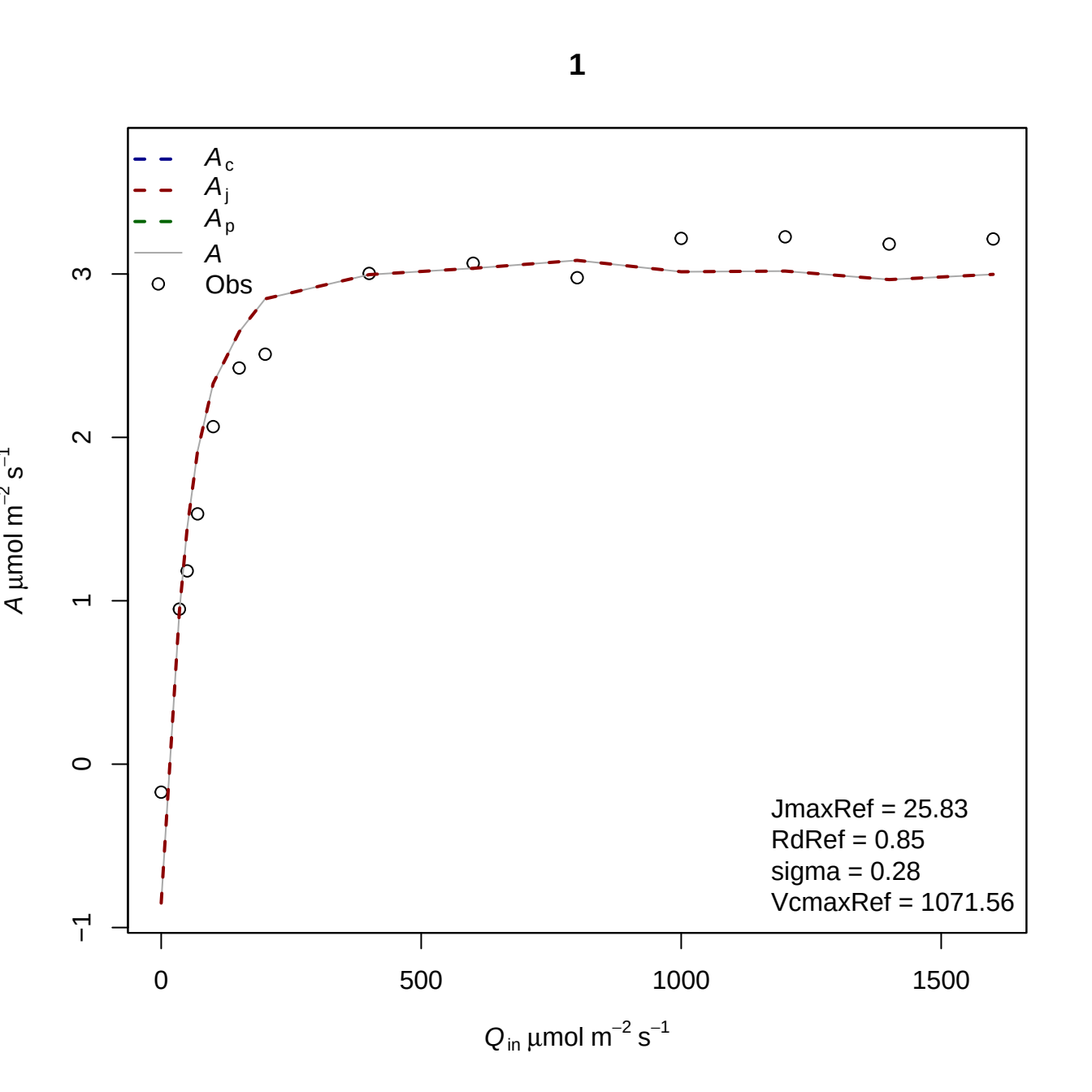
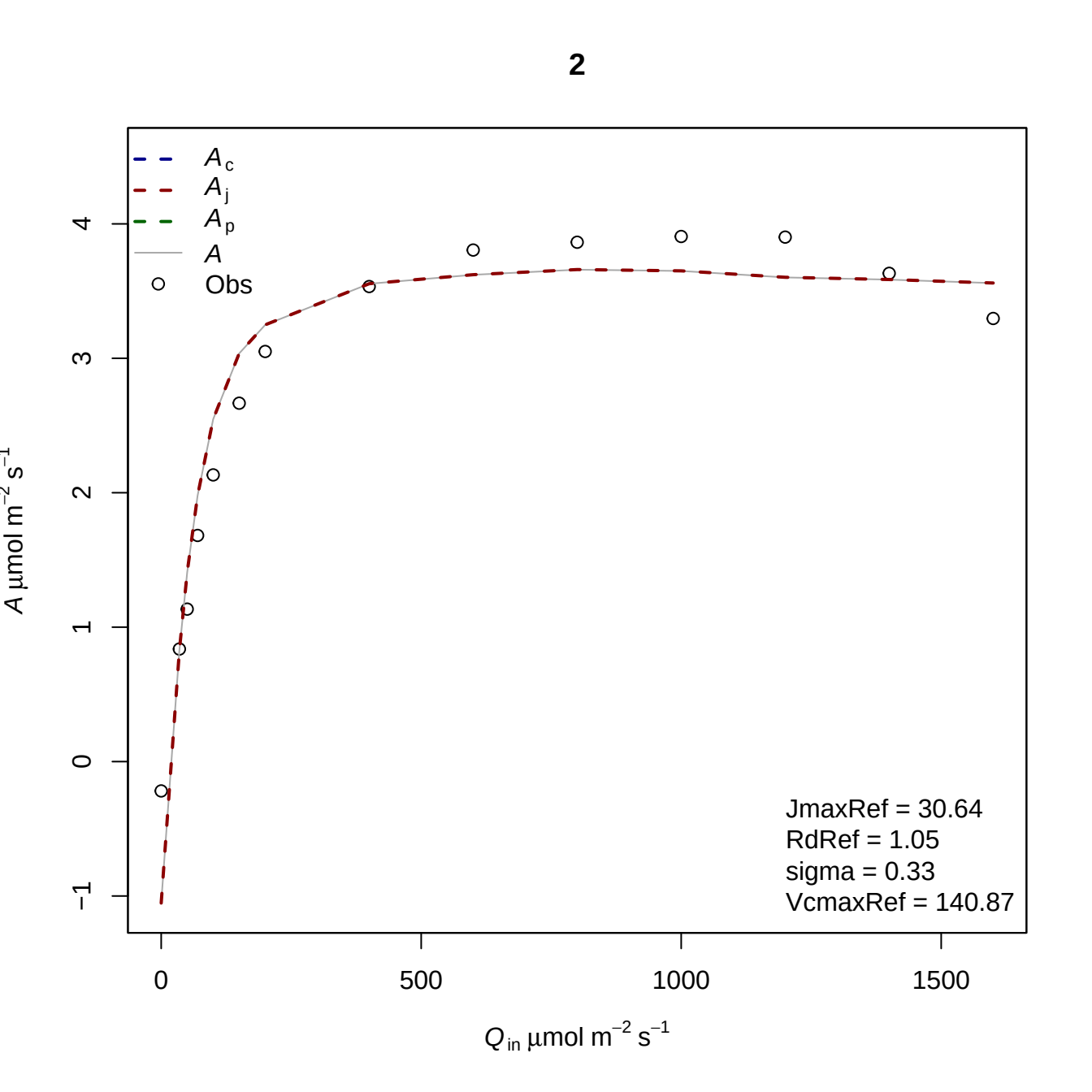
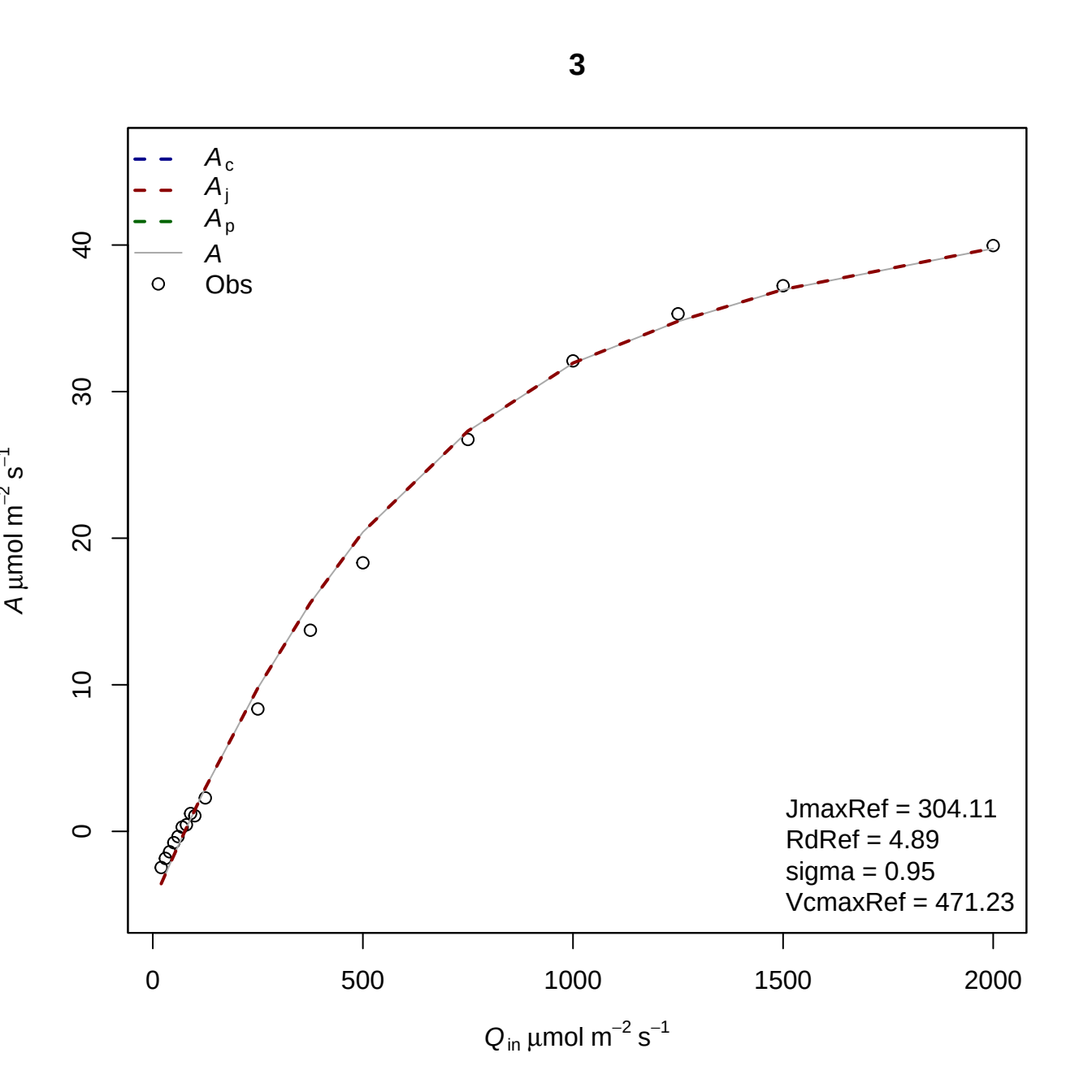
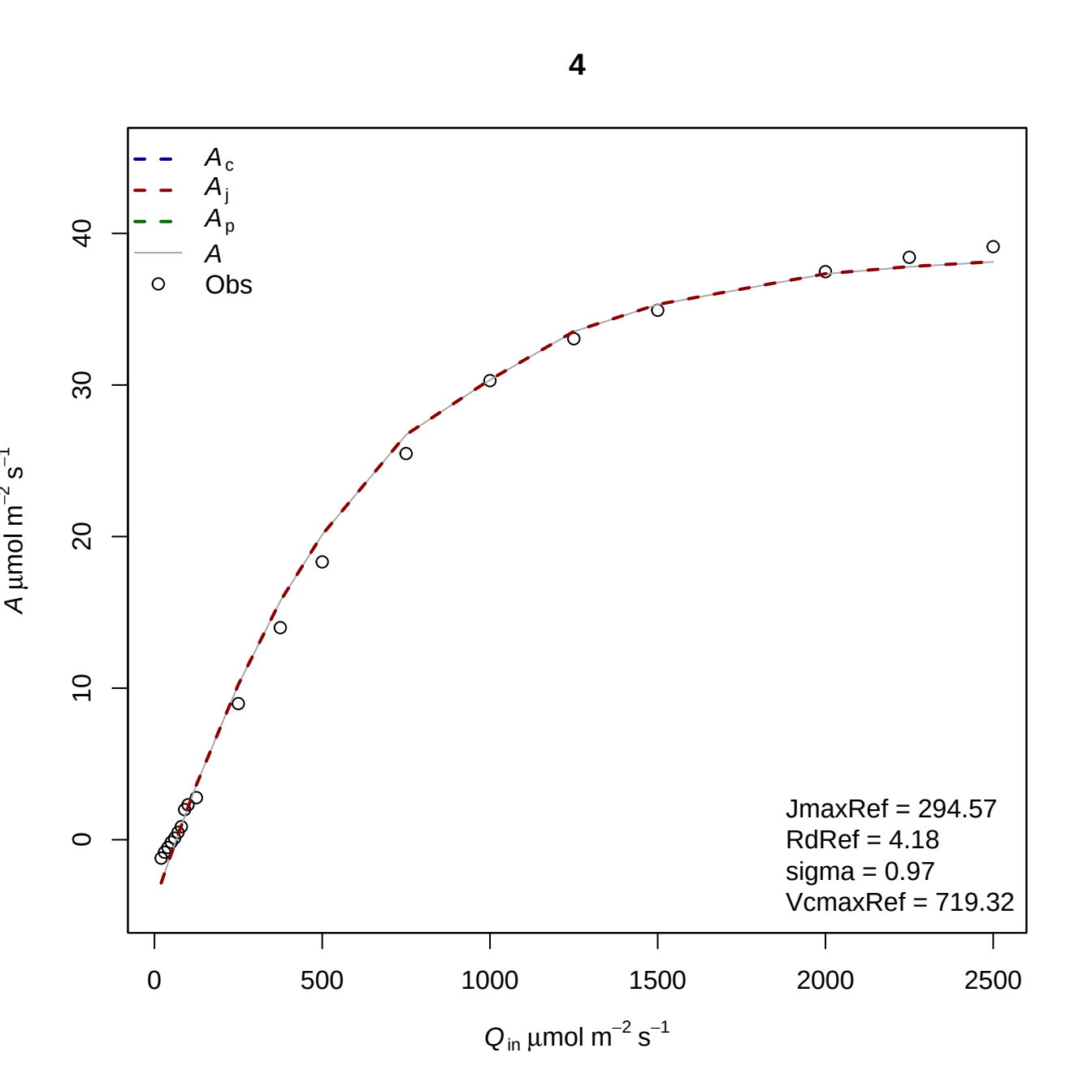
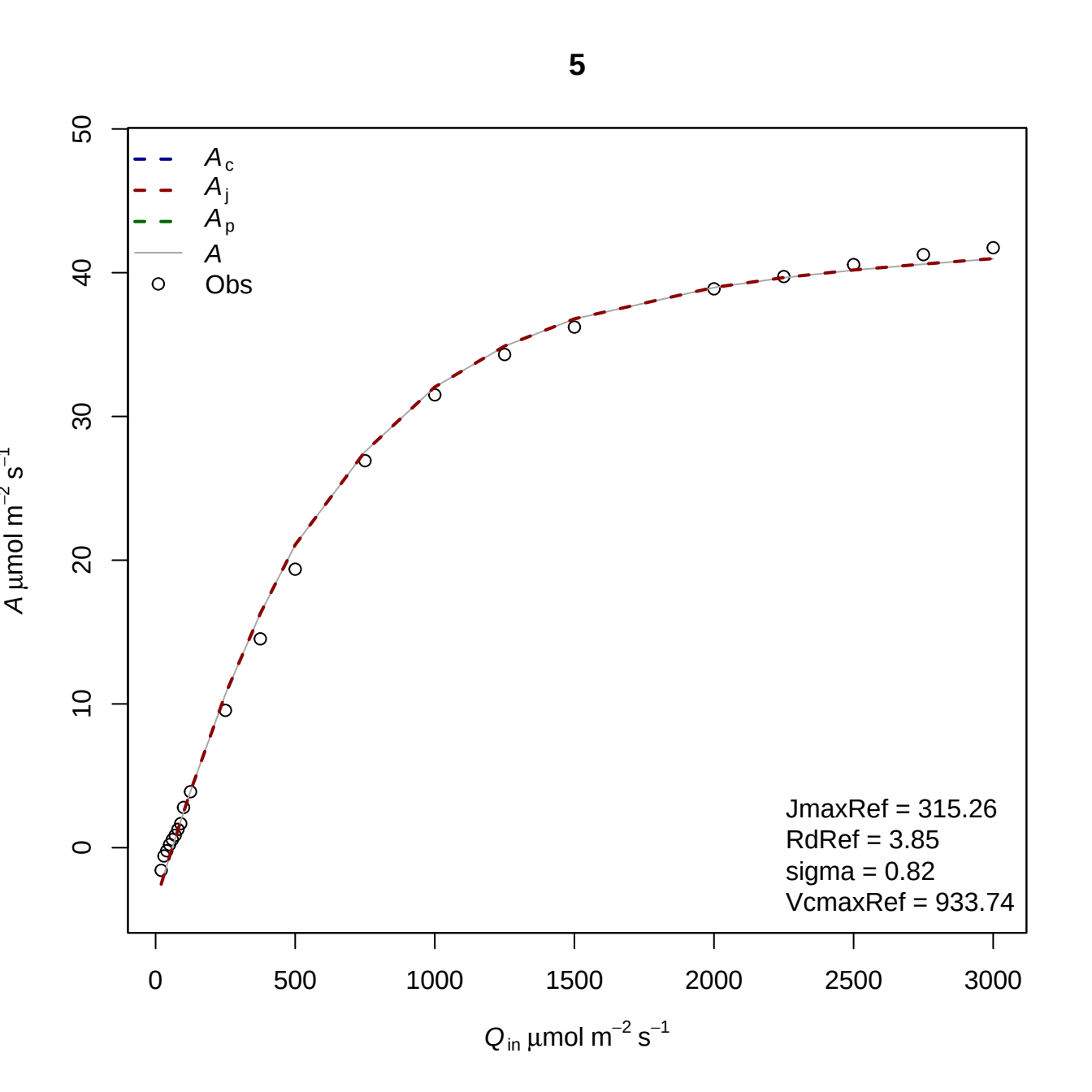
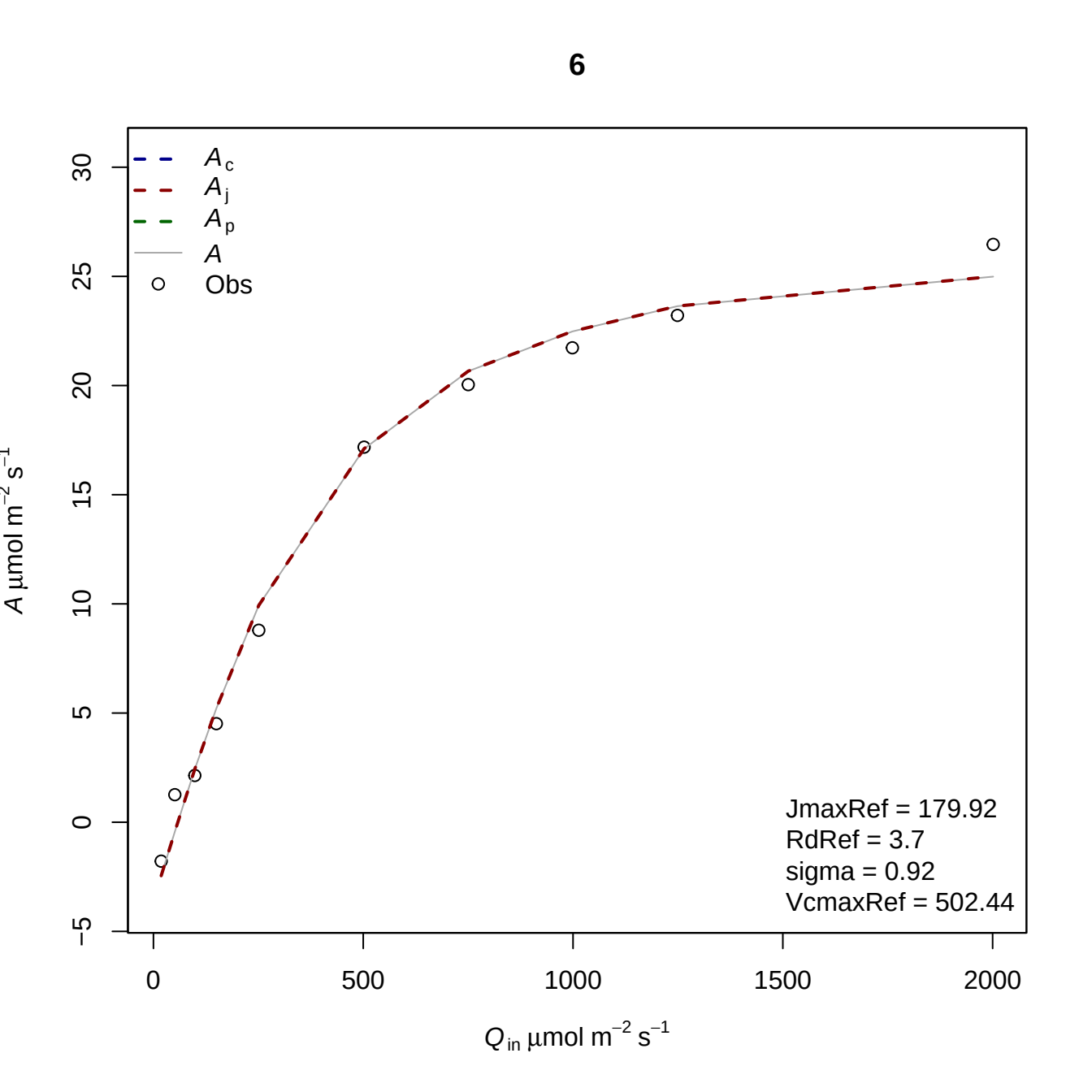

1
A_c
A_j
A_p
A
Obs
JmaxRef = 25.83
RdRef = 0.85
sigma = 0.28
VcmaxRef = 1071.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

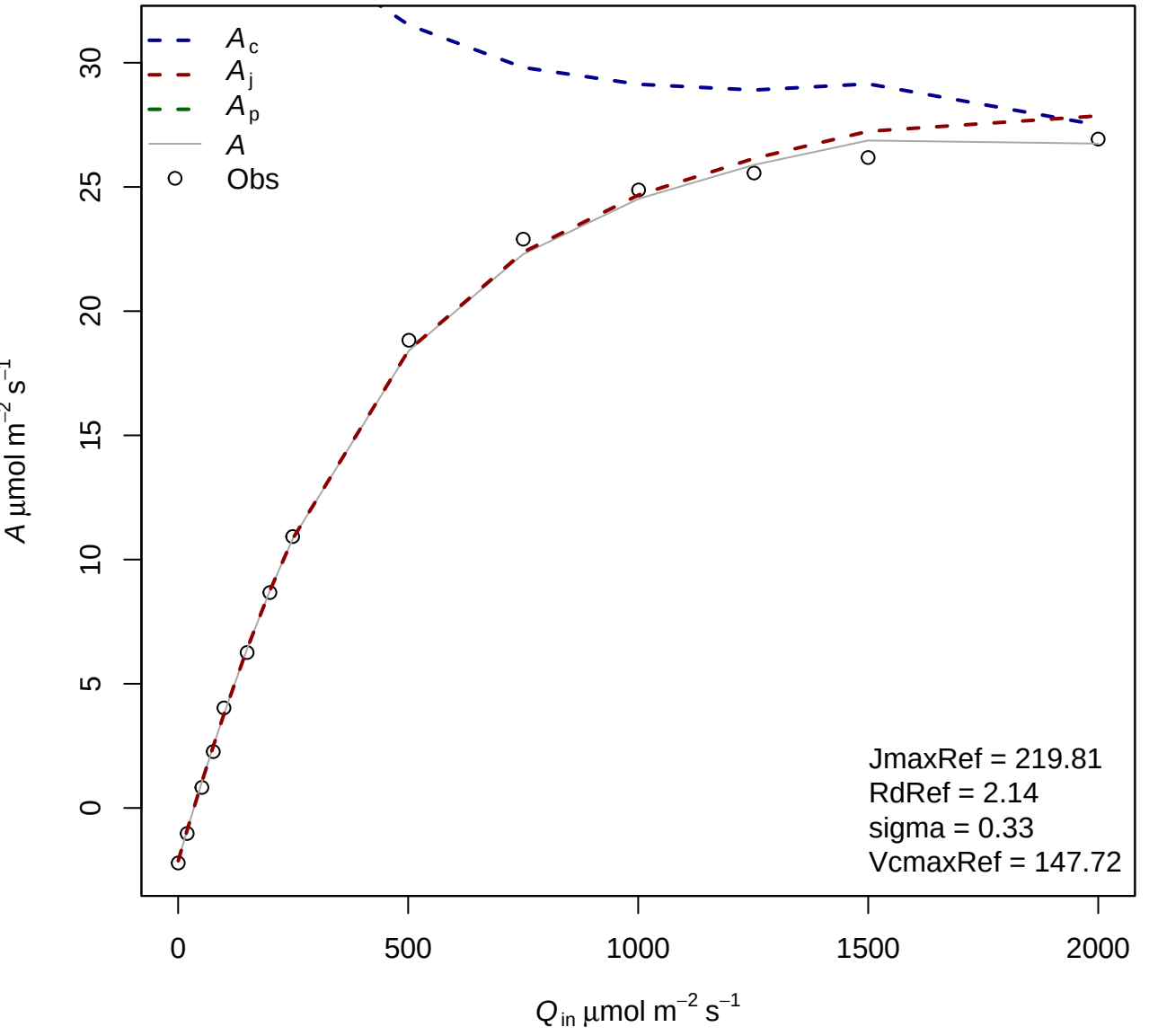

2
A_c
A_j
A_p
A
Obs
JmaxRef = 30.64
RdRef = 1.05
sigma = 0.33
VcmaxRef = 140.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

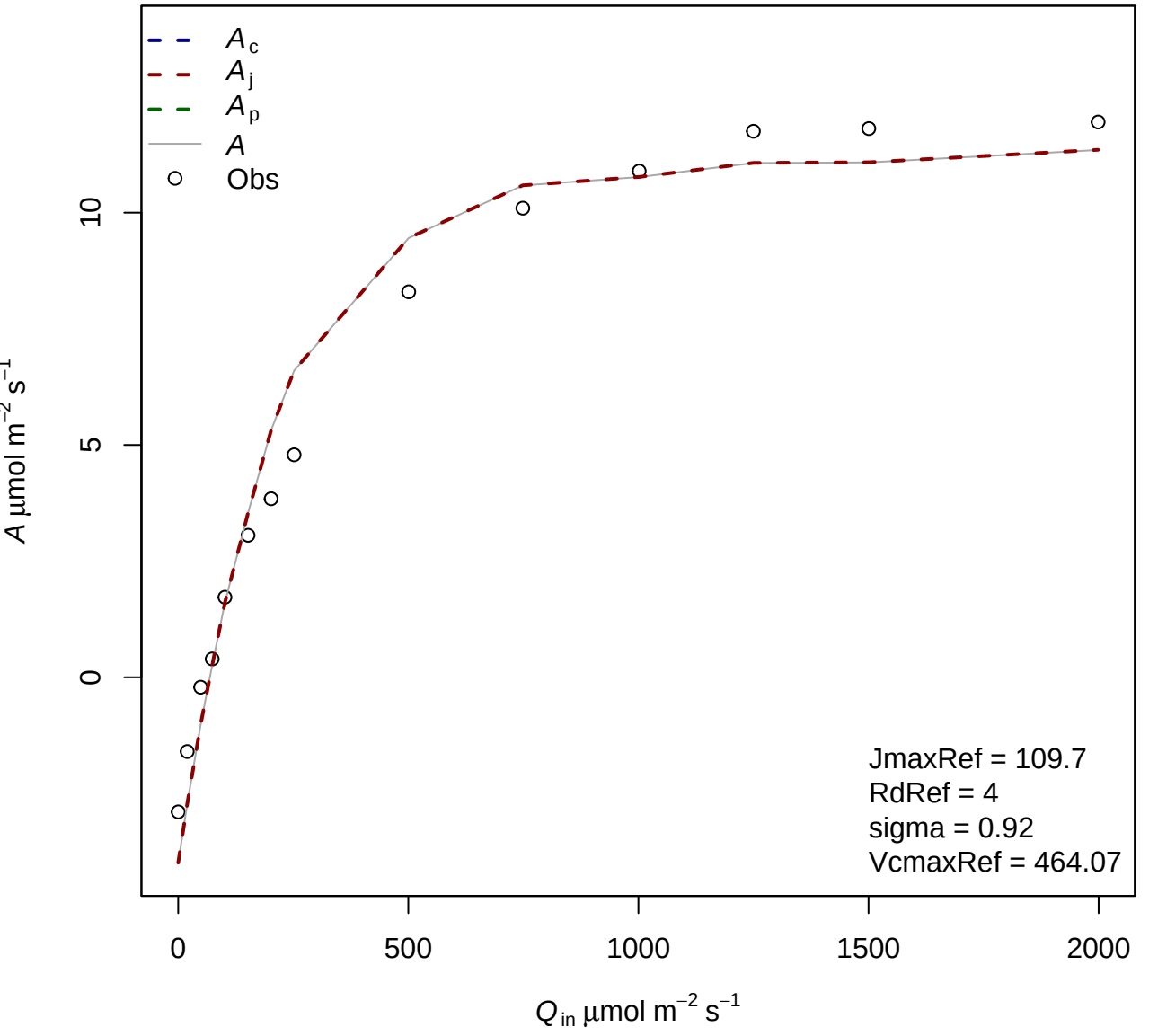

3
A_c
A_j
A_p
A
Obs
JmaxRef = 304.11
RdRef = 4.89
sigma = 0.95
VcmaxRef = 471.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

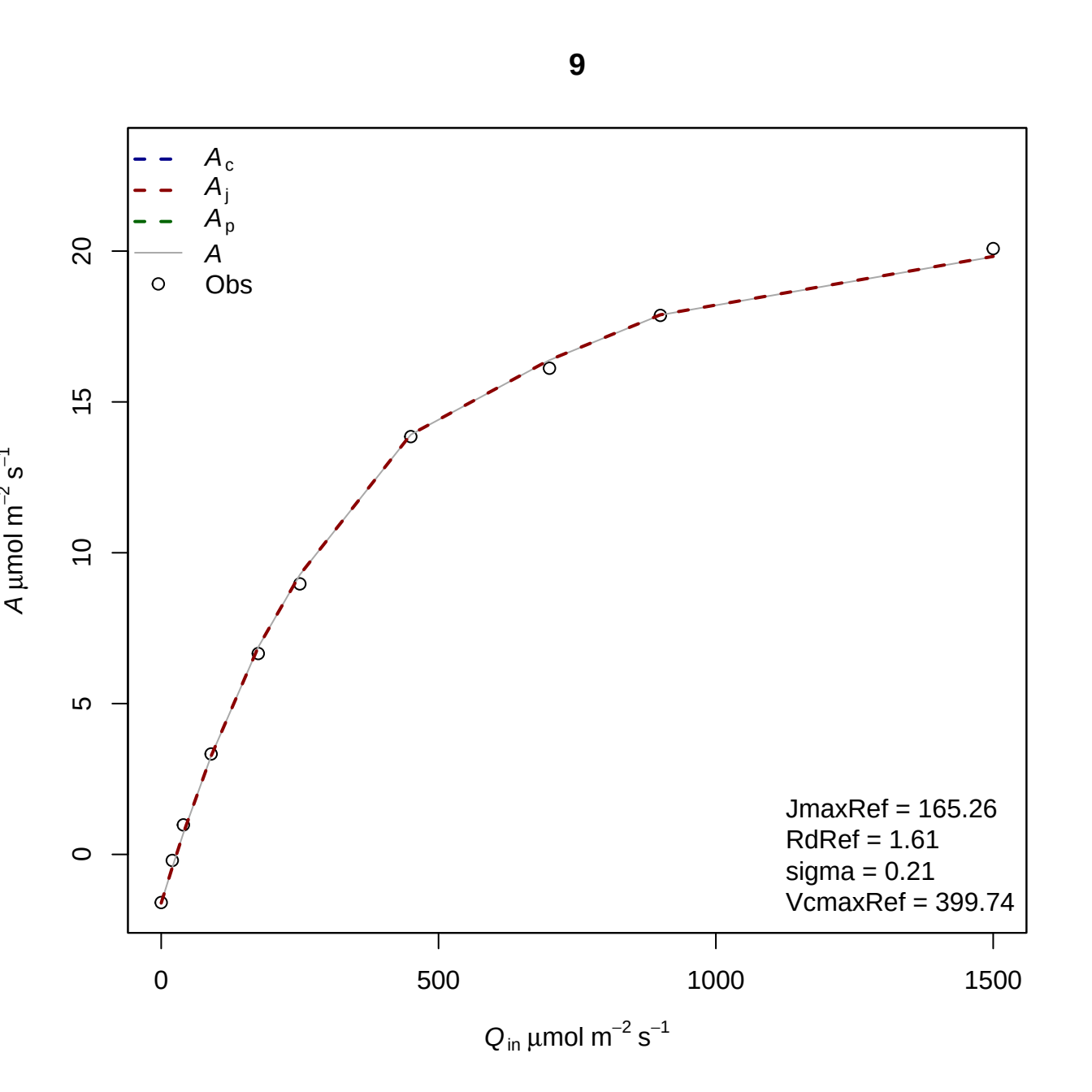

4
A_c
A_j
A_p
A
Obs
JmaxRef = 294.57
RdRef = 4.18
sigma = 0.97
VcmaxRef = 719.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000
2500

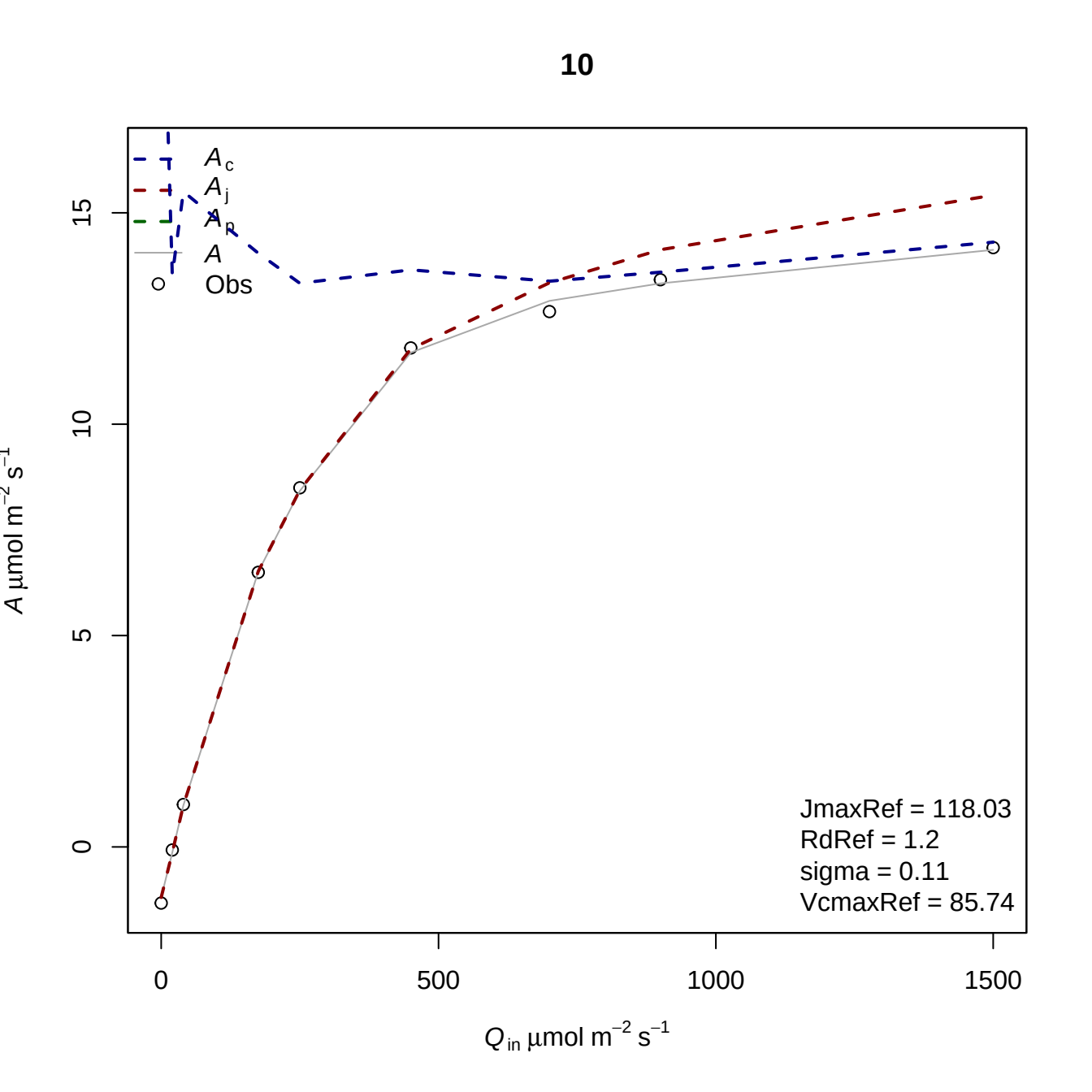

5
A_c
A_j
A_p
A
Obs
JmaxRef = 315.26
RdRef = 3.85
sigma = 0.82
VcmaxRef = 933.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
50
0
500
1000
1500
2000
2500
3000

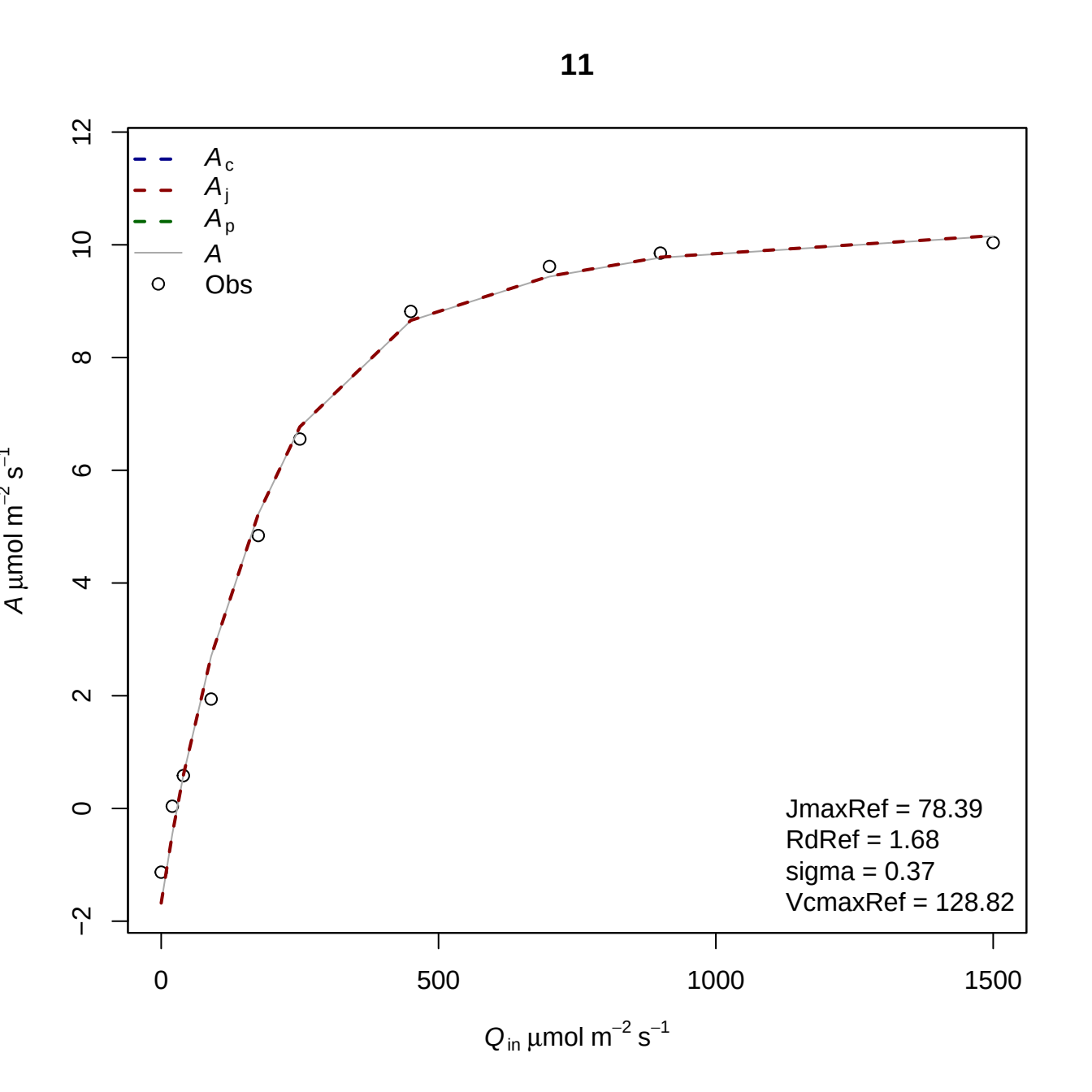

6
A_c
A_j
A_p
A
Obs
JmaxRef = 179.92
RdRef = 3.7
sigma = 0.92
VcmaxRef = 502.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
30
0
500
1000
1500
2000

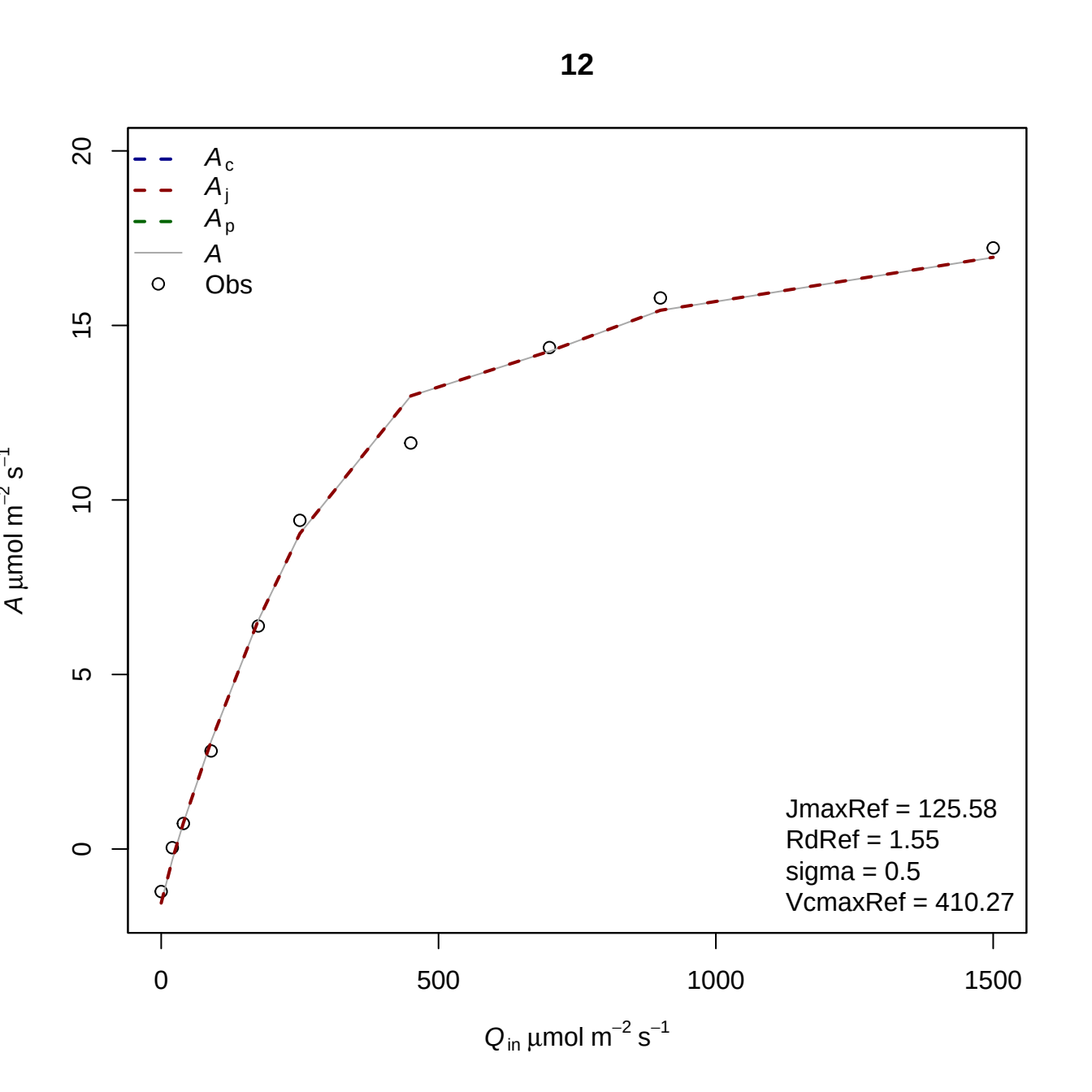

A_c
A_j
A_p
A
Obs
JmaxRef = 219.81
RdRef = 2.14
sigma = 0.33
VcmaxRef = 147.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

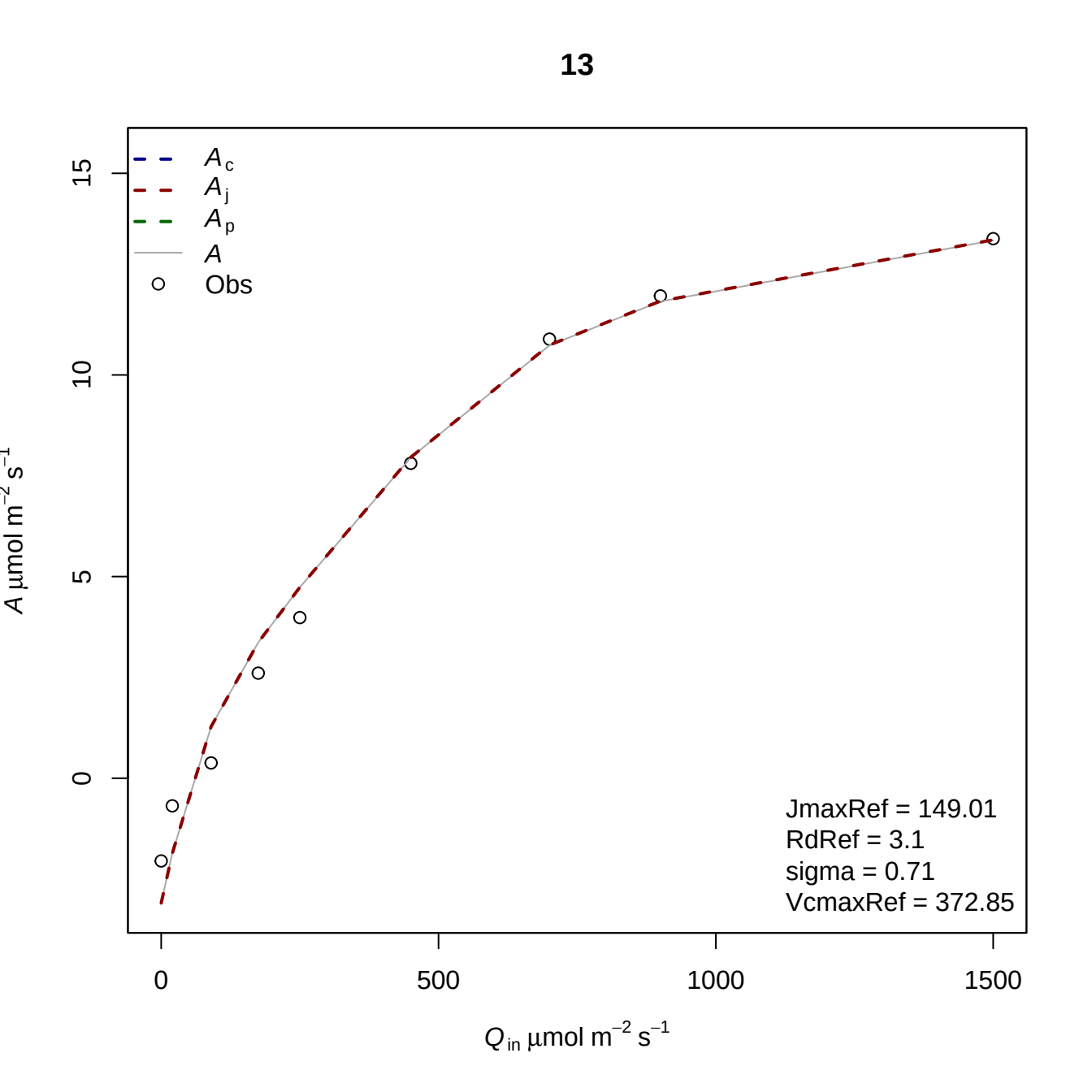

A_c
A_j
A_p
A
Obs
JmaxRef = 109.7
RdRef = 4
sigma = 0.92
VcmaxRef = 464.07
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

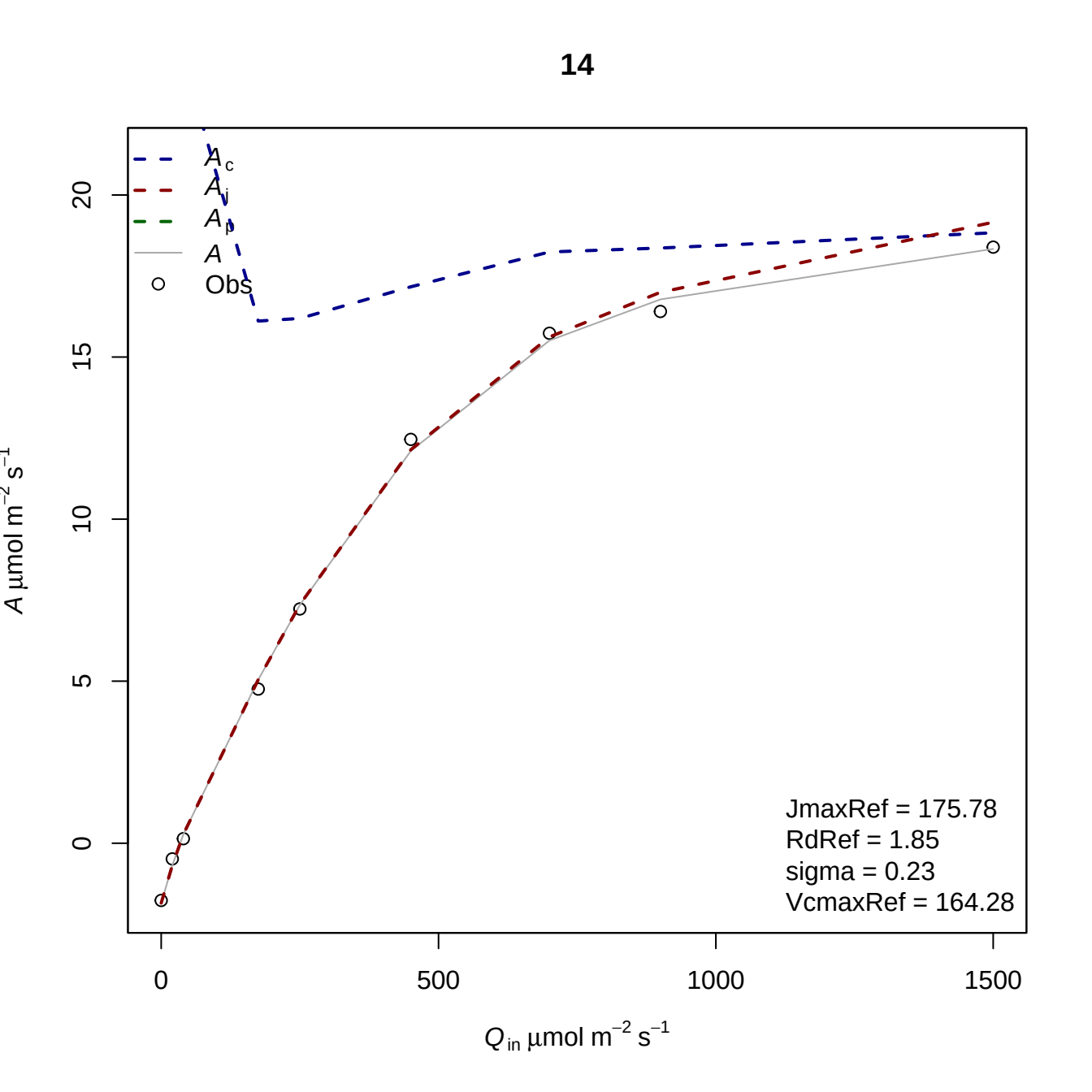

9
A_c
A_j
A_p
A
Obs
JmaxRef = 165.26
RdRef = 1.61
sigma = 0.21
VcmaxRef = 399.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

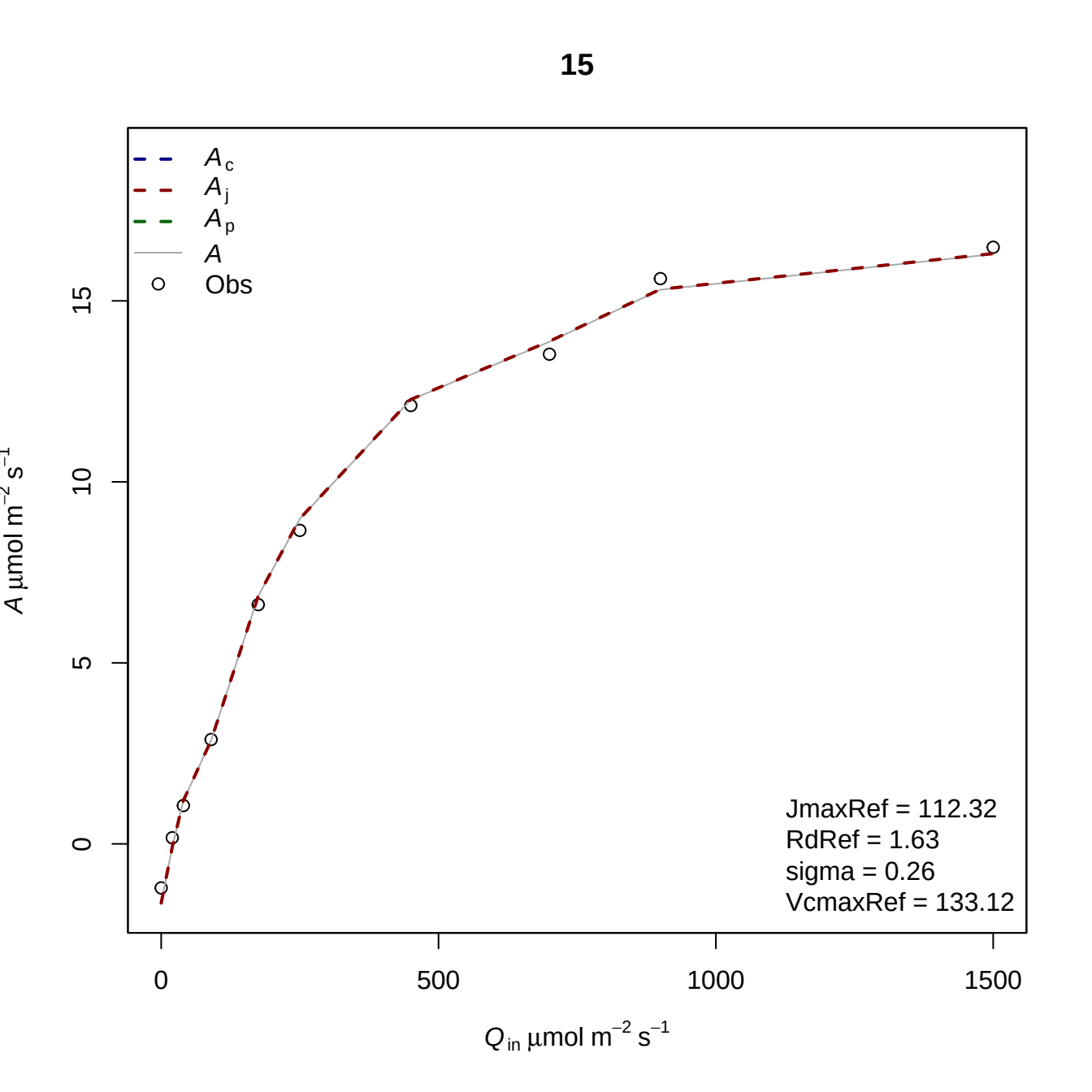

10
A_c
A_j
A_p
A
Obs
JmaxRef = 118.03
RdRef = 1.2
sigma = 0.11
VcmaxRef = 85.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

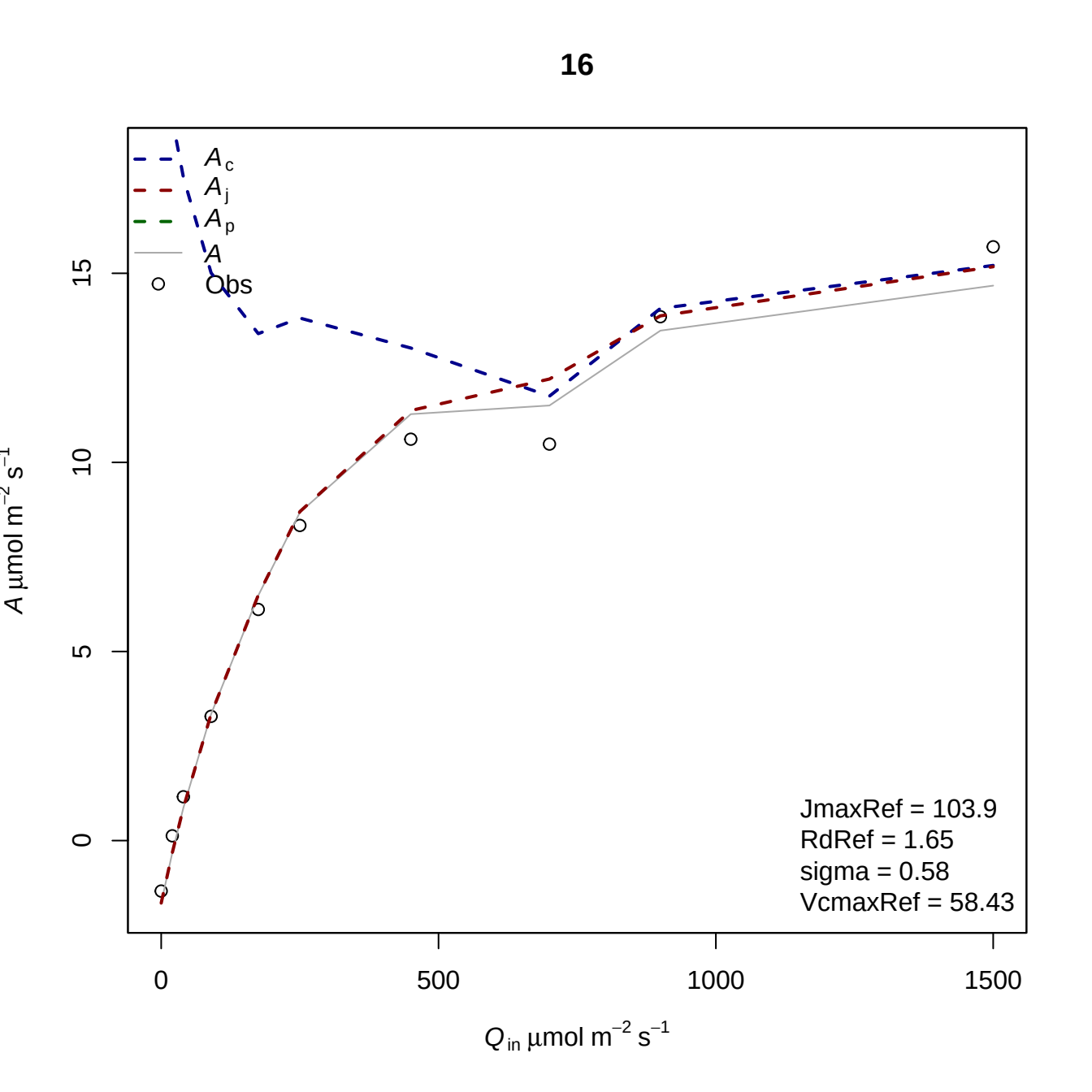

11
A_c
A_j
A_p
A
Obs
JmaxRef = 78.39
RdRef = 1.68
sigma = 0.37
VcmaxRef = 128.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

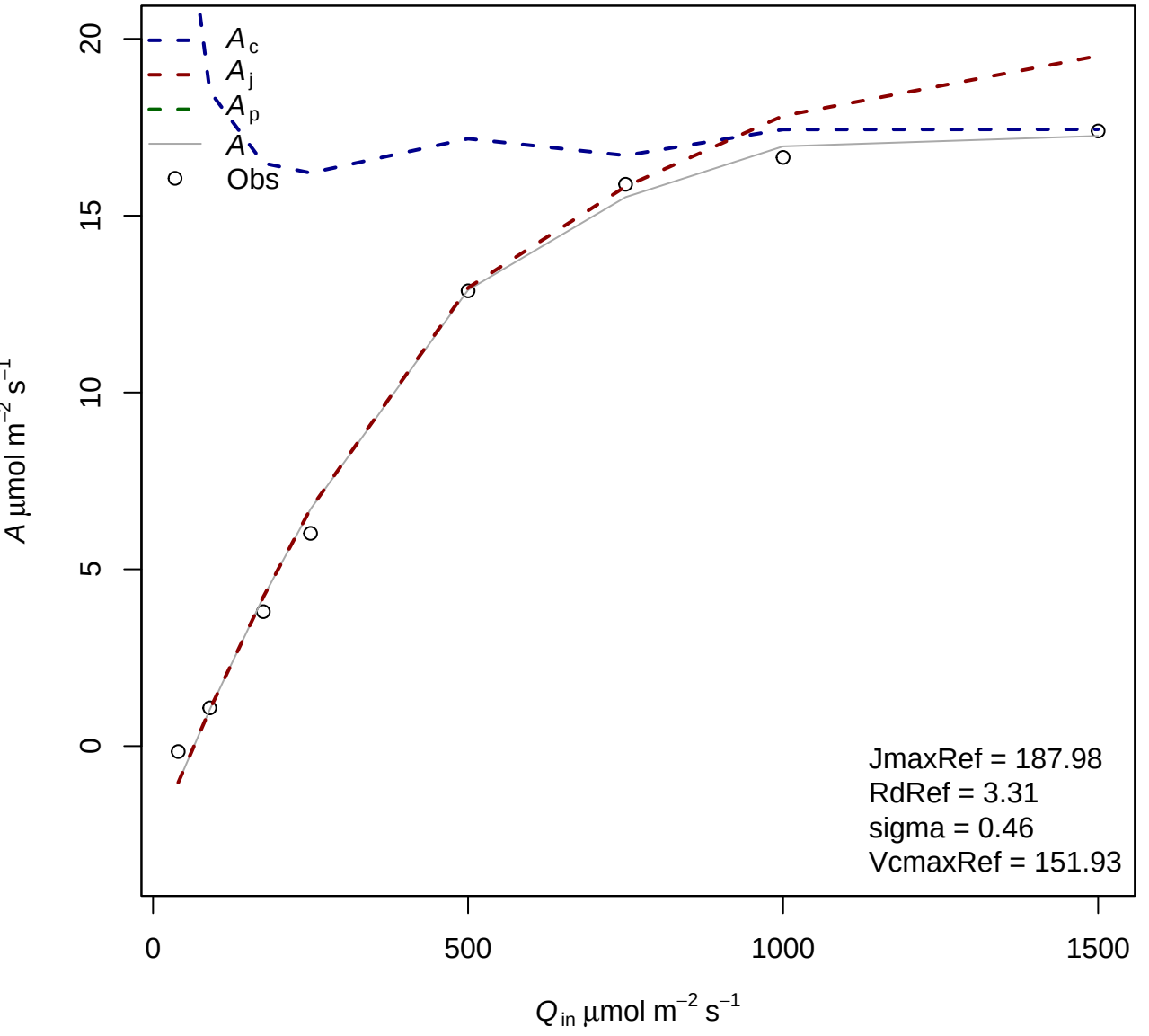

12
A_c
A_j
A_p
A
Obs
JmaxRef = 125.58
RdRef = 1.55
sigma = 0.5
VcmaxRef = 410.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

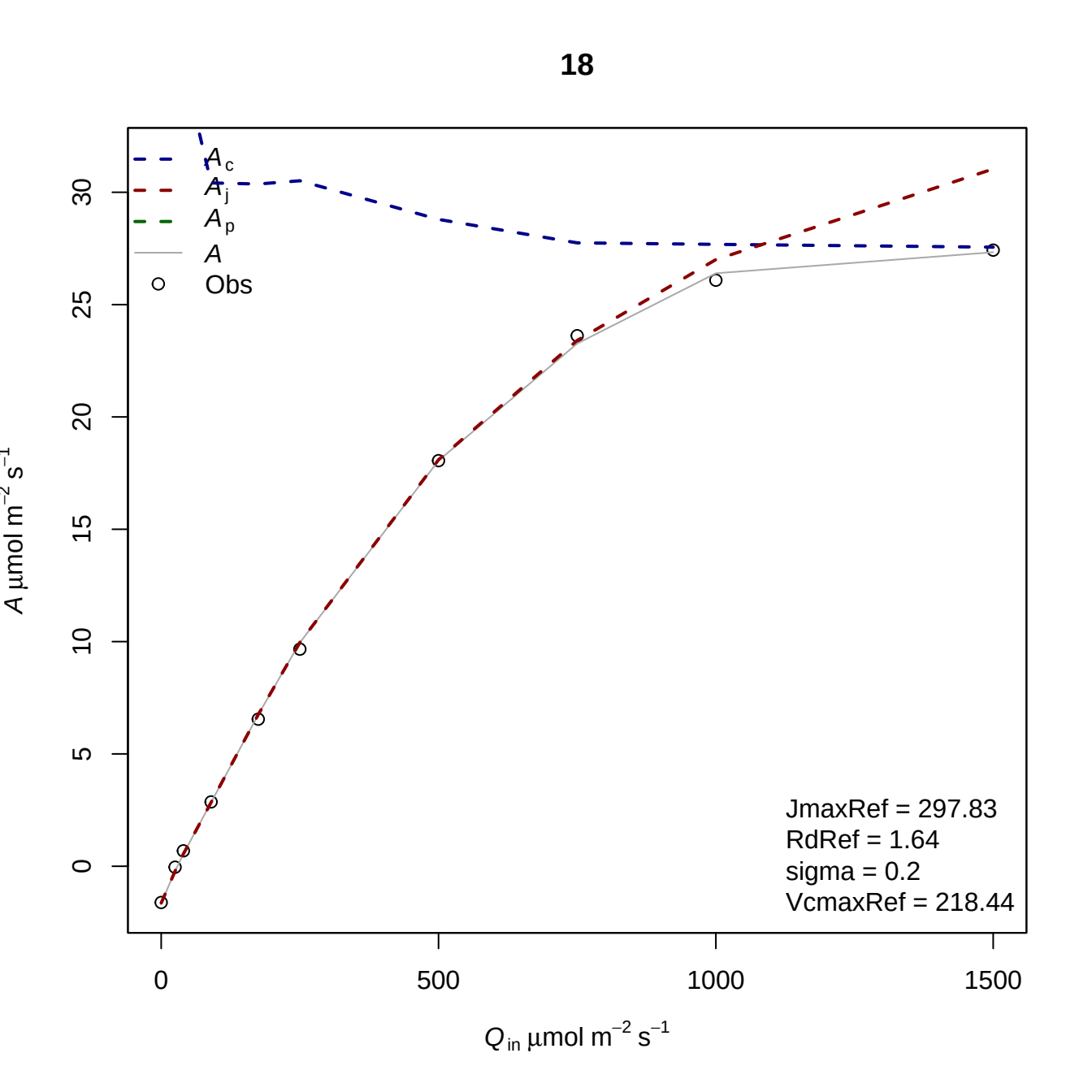

13
A_c
A_j
A_p
A
Obs
JmaxRef = 149.01
RdRef = 3.1
sigma = 0.71
VcmaxRef = 372.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

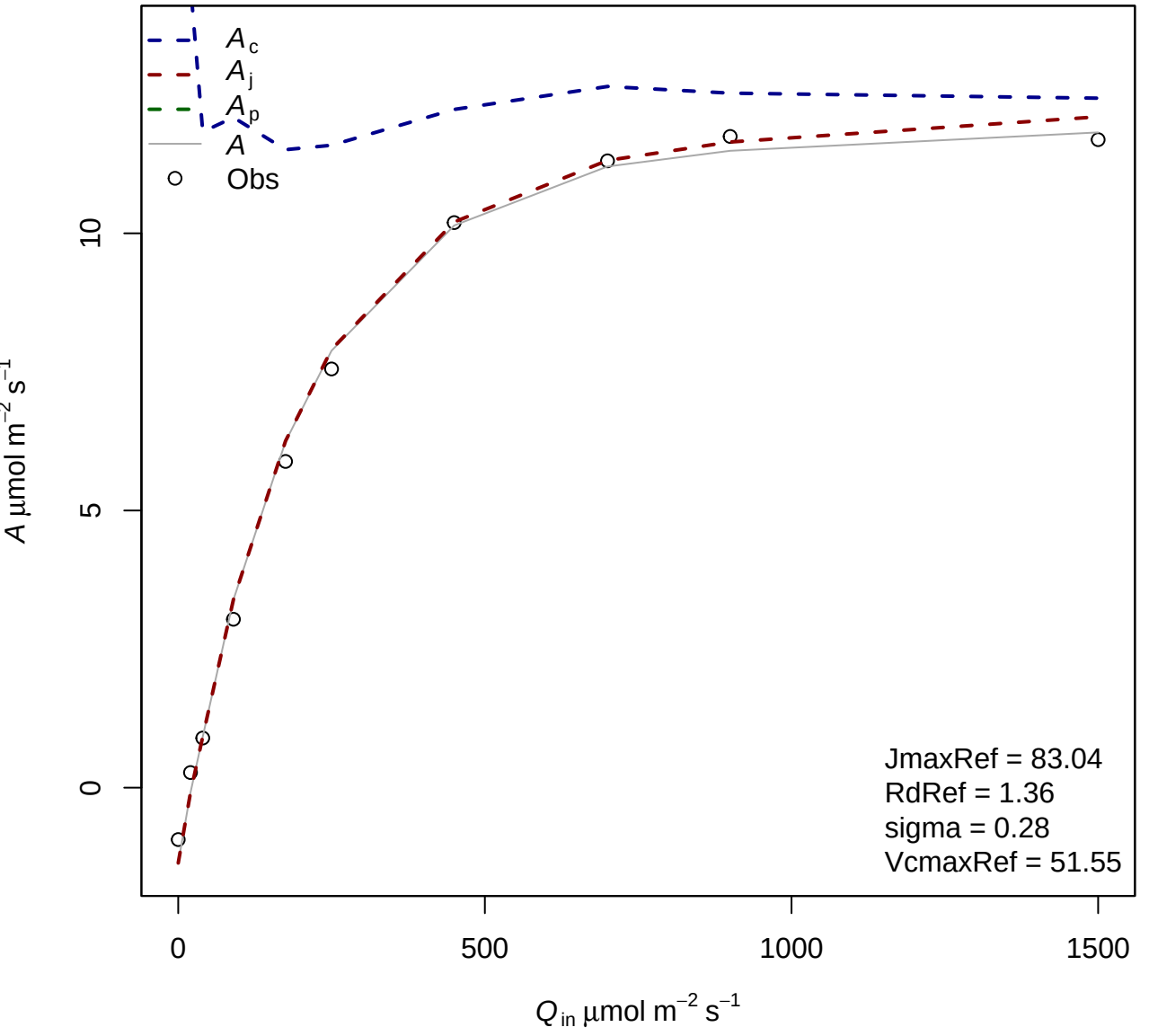

14
A_c
A_j
A_p
A
Obs
JmaxRef = 175.78
RdRef = 1.85
sigma = 0.23
VcmaxRef = 164.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

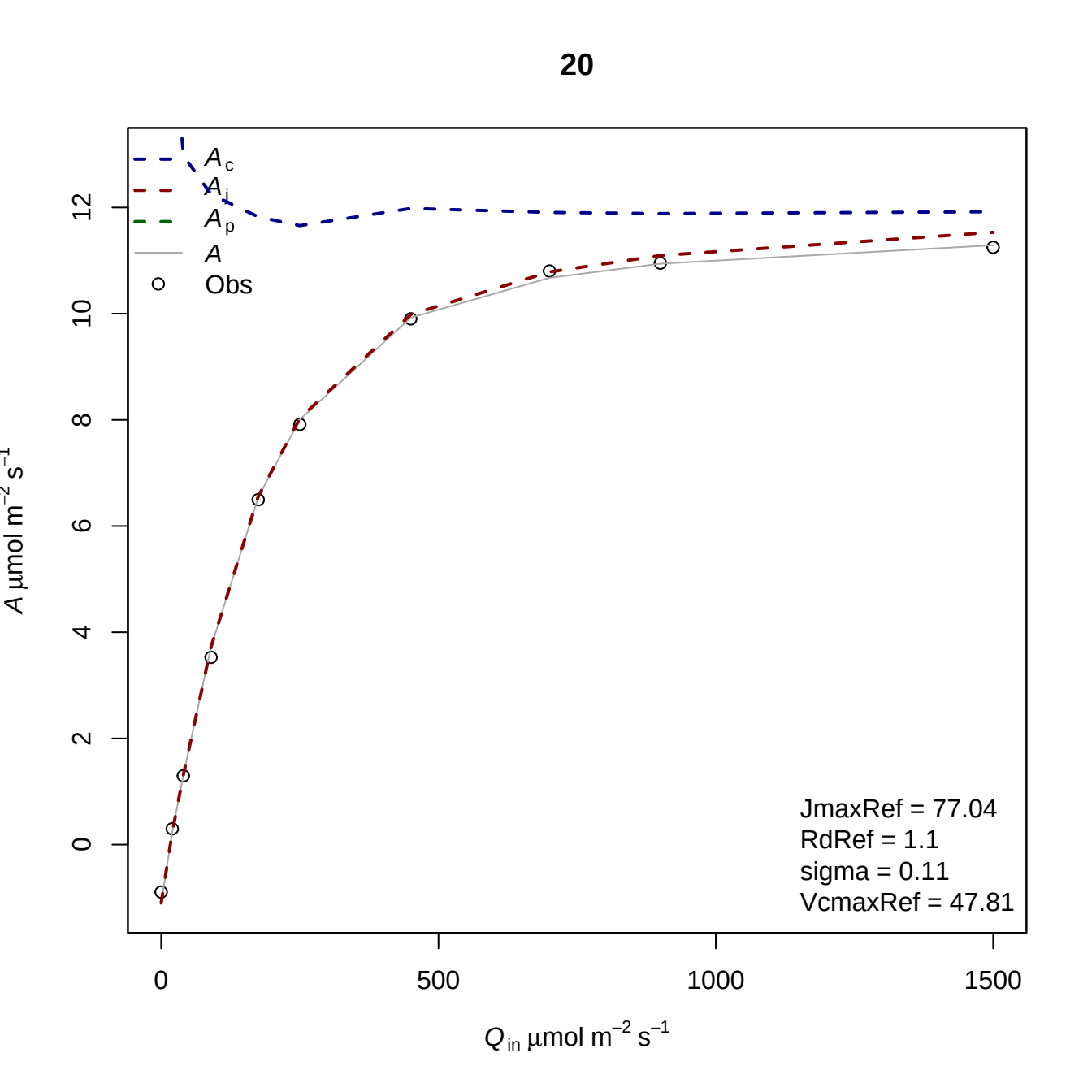

15
A_c
A_j
A_p
A
Obs
JmaxRef = 112.32
RdRef = 1.63
sigma = 0.26
VcmaxRef = 133.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

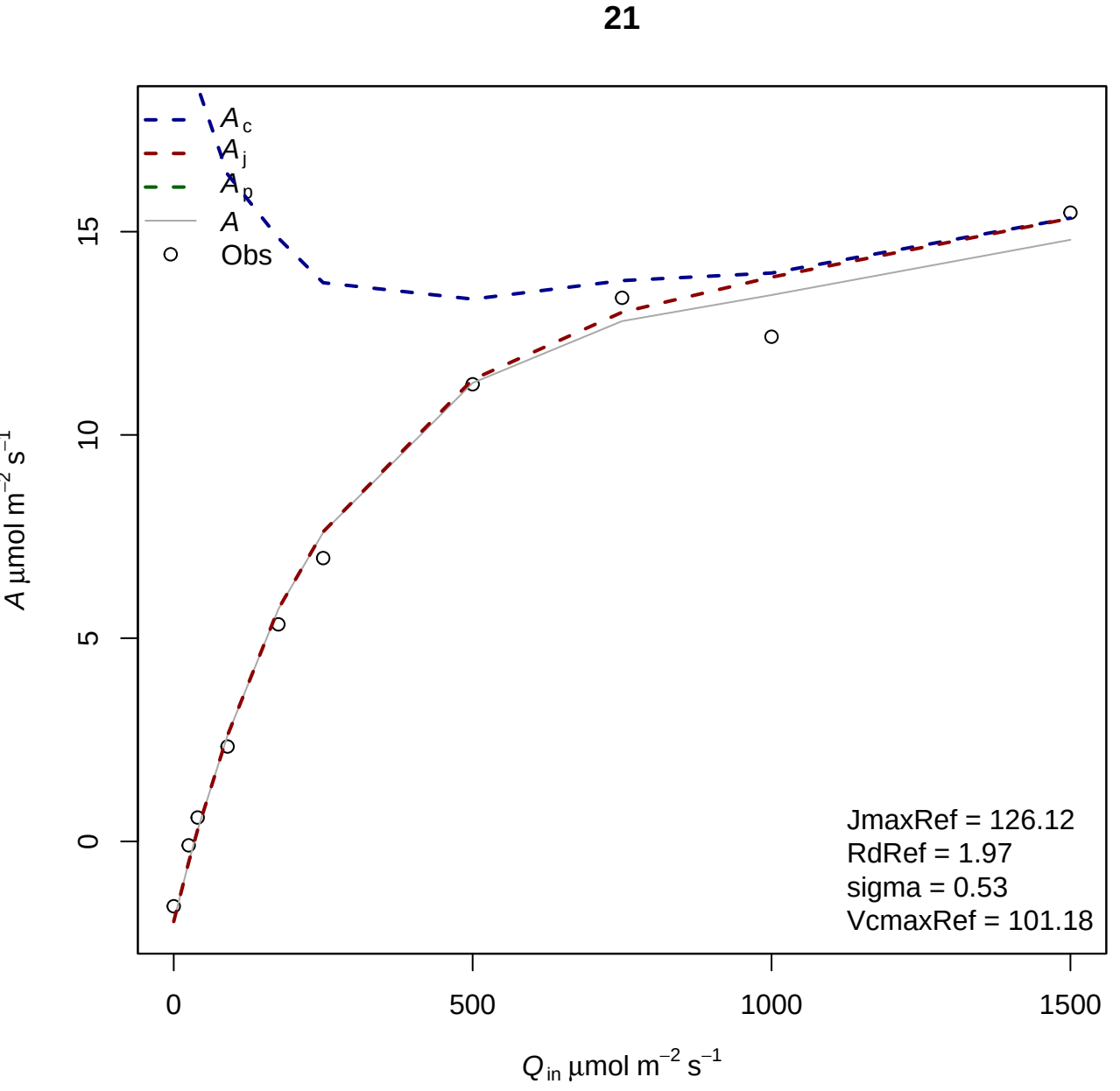

16
A_c
A_j
A_p
A
Obs
JmaxRef = 103.9
RdRef = 1.65
sigma = 0.58
VcmaxRef = 58.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

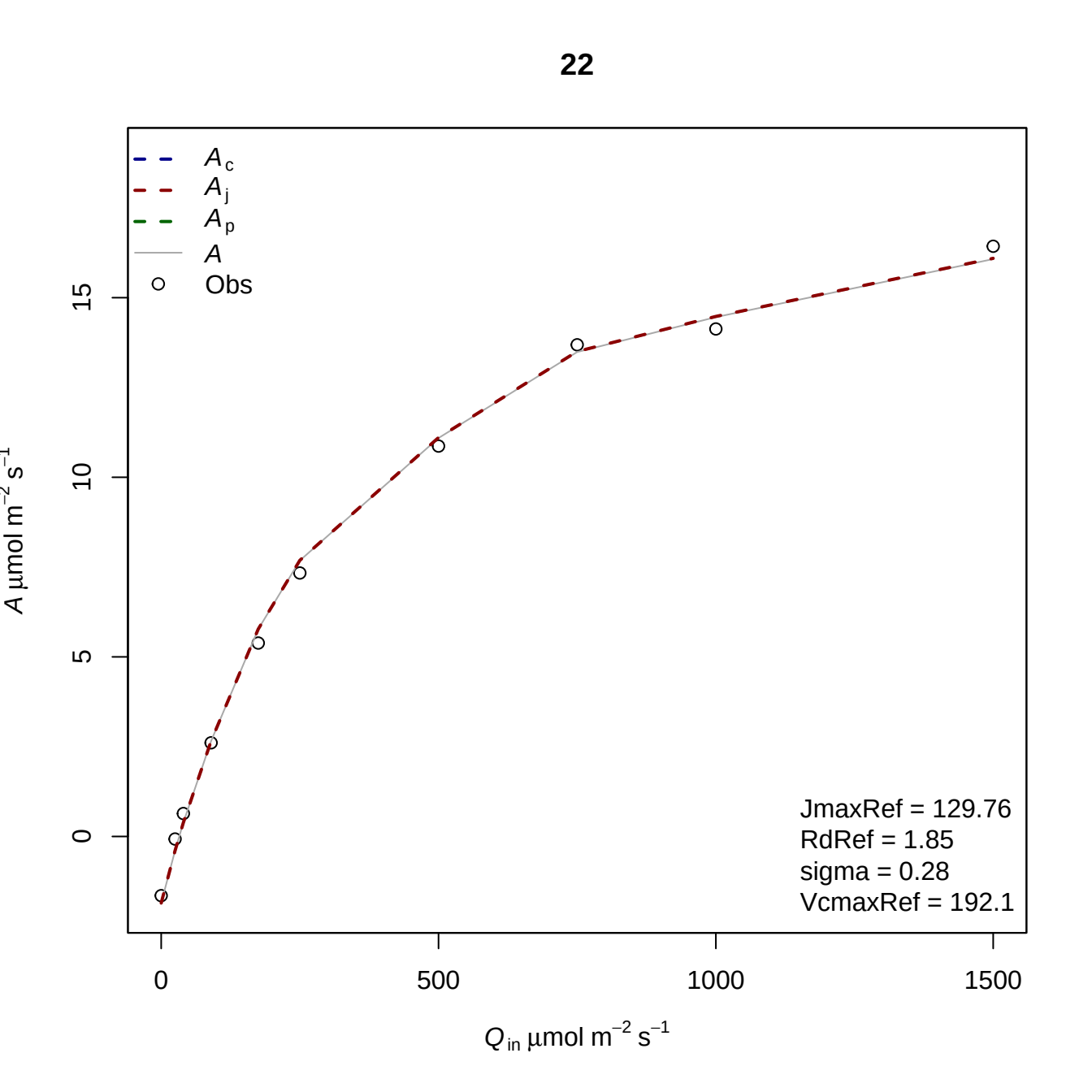

A_c
A_j
A_p
A
Obs
JmaxRef = 187.98
RdRef = 3.31
sigma = 0.46
VcmaxRef = 151.93
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

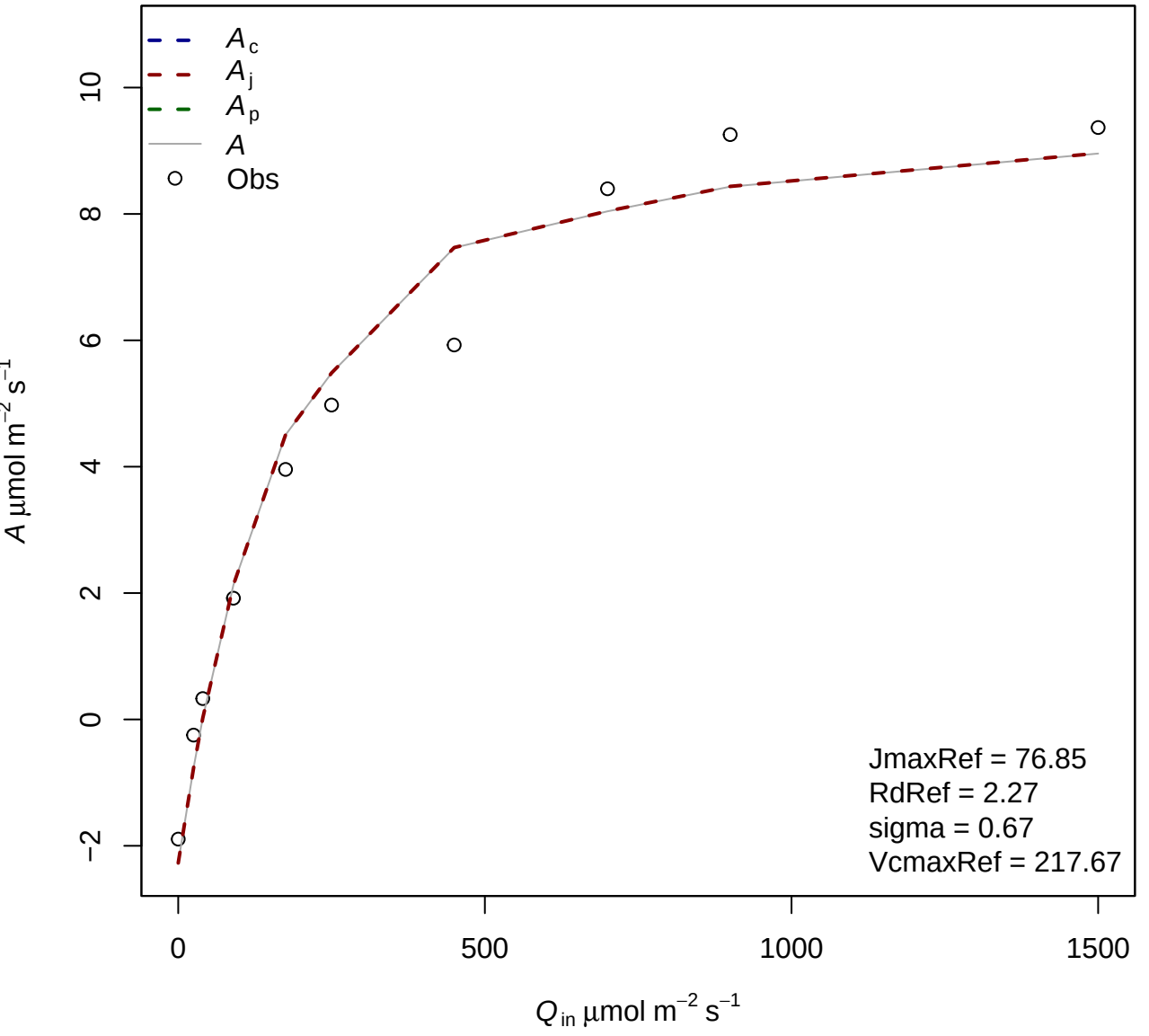

18
A_c
A_j
A_p
A
Obs
JmaxRef = 297.83
RdRef = 1.64
sigma = 0.2
VcmaxRef = 218.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500

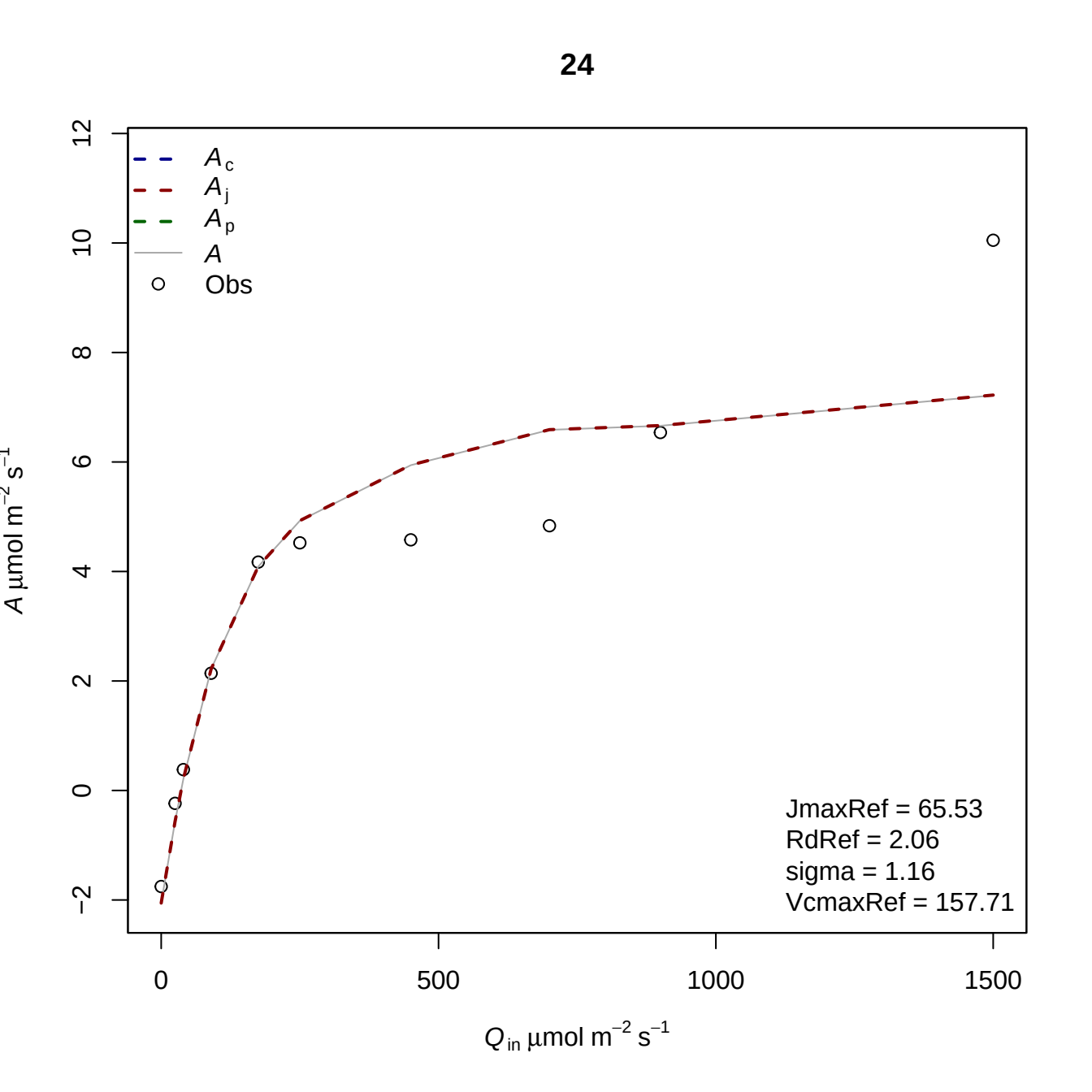

A_c
A_j
A_p
A
Obs
JmaxRef = 83.04
RdRef = 1.36
sigma = 0.28
VcmaxRef = 51.55
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

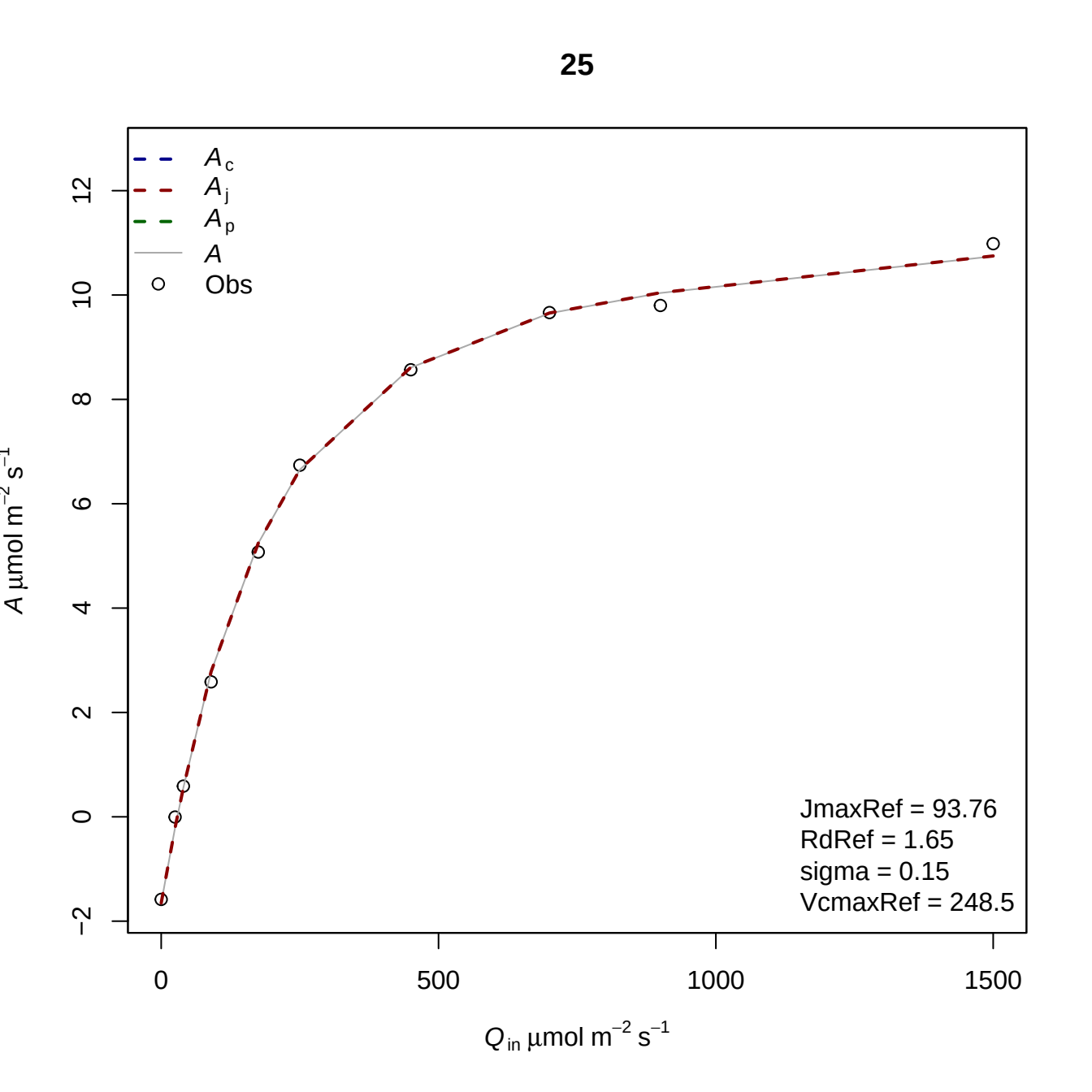

20
A_c
A_j
A_p
A
Obs
JmaxRef = 77.04
RdRef = 1.1
sigma = 0.11
VcmaxRef = 47.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

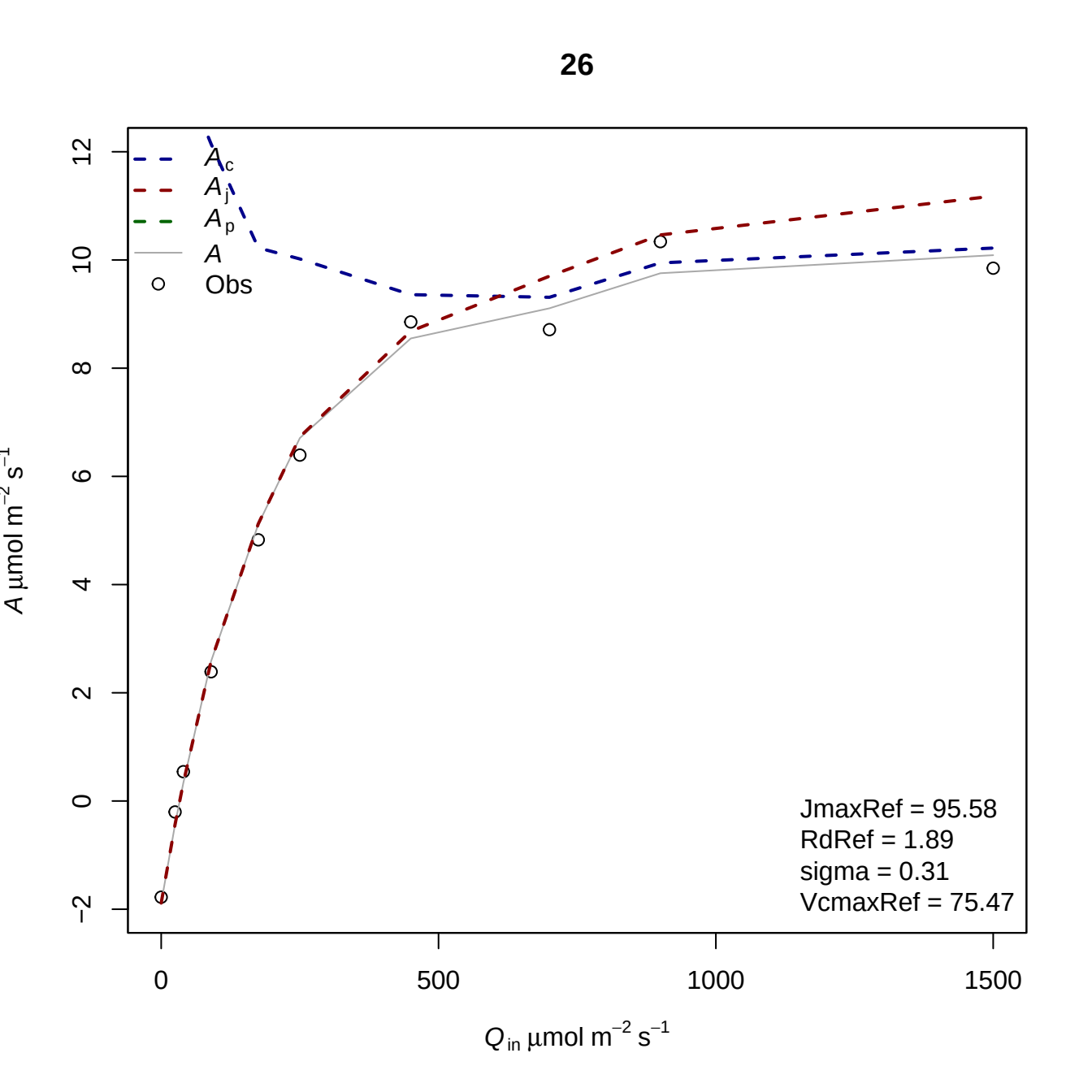

21
A_c
A_j
A_p
A
Obs
JmaxRef = 126.12
RdRef = 1.97
sigma = 0.53
VcmaxRef = 101.18
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

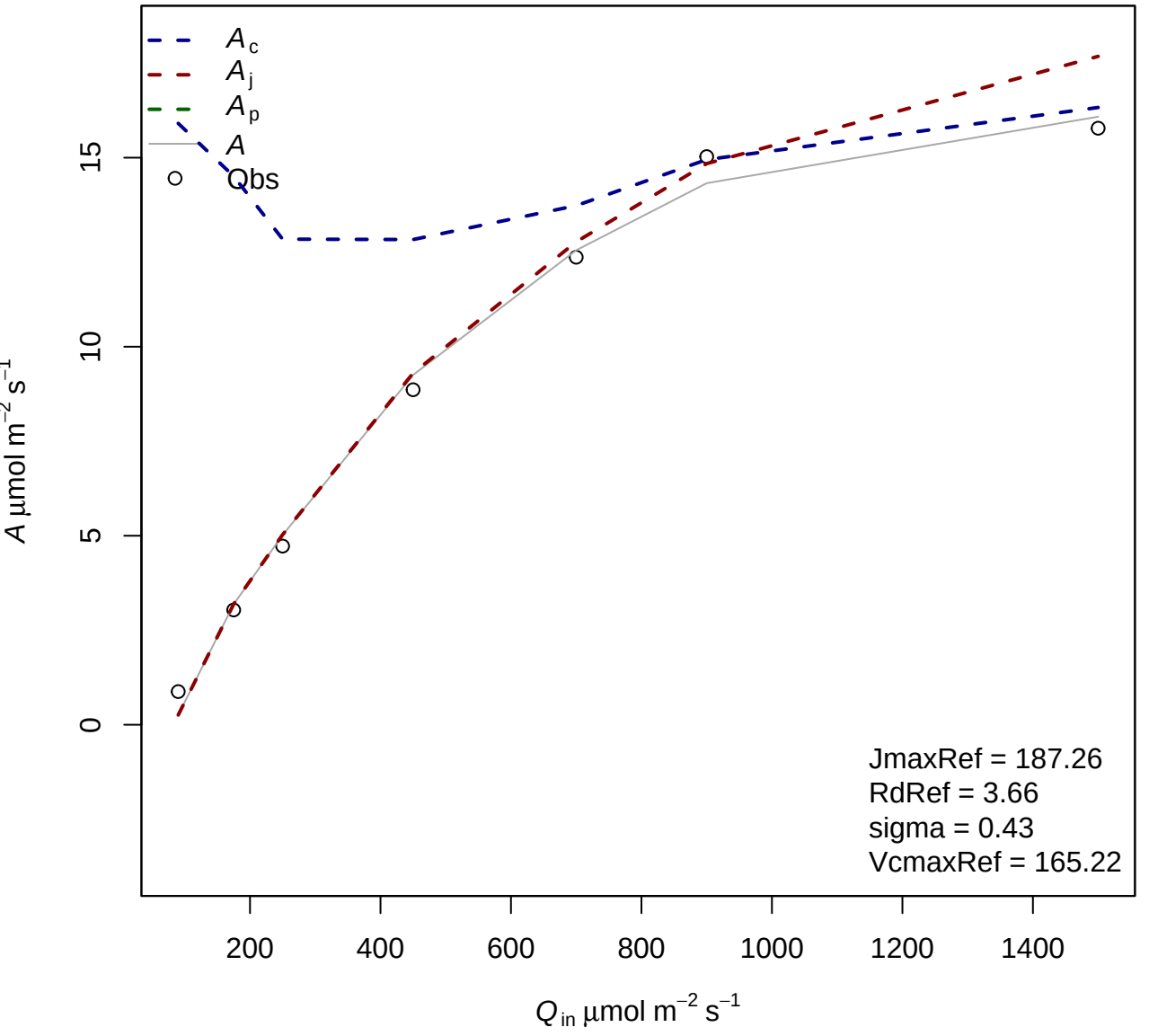

22
A_c
A_j
A_p
A
Obs
JmaxRef = 129.76
RdRef = 1.85
sigma = 0.28
VcmaxRef = 192.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

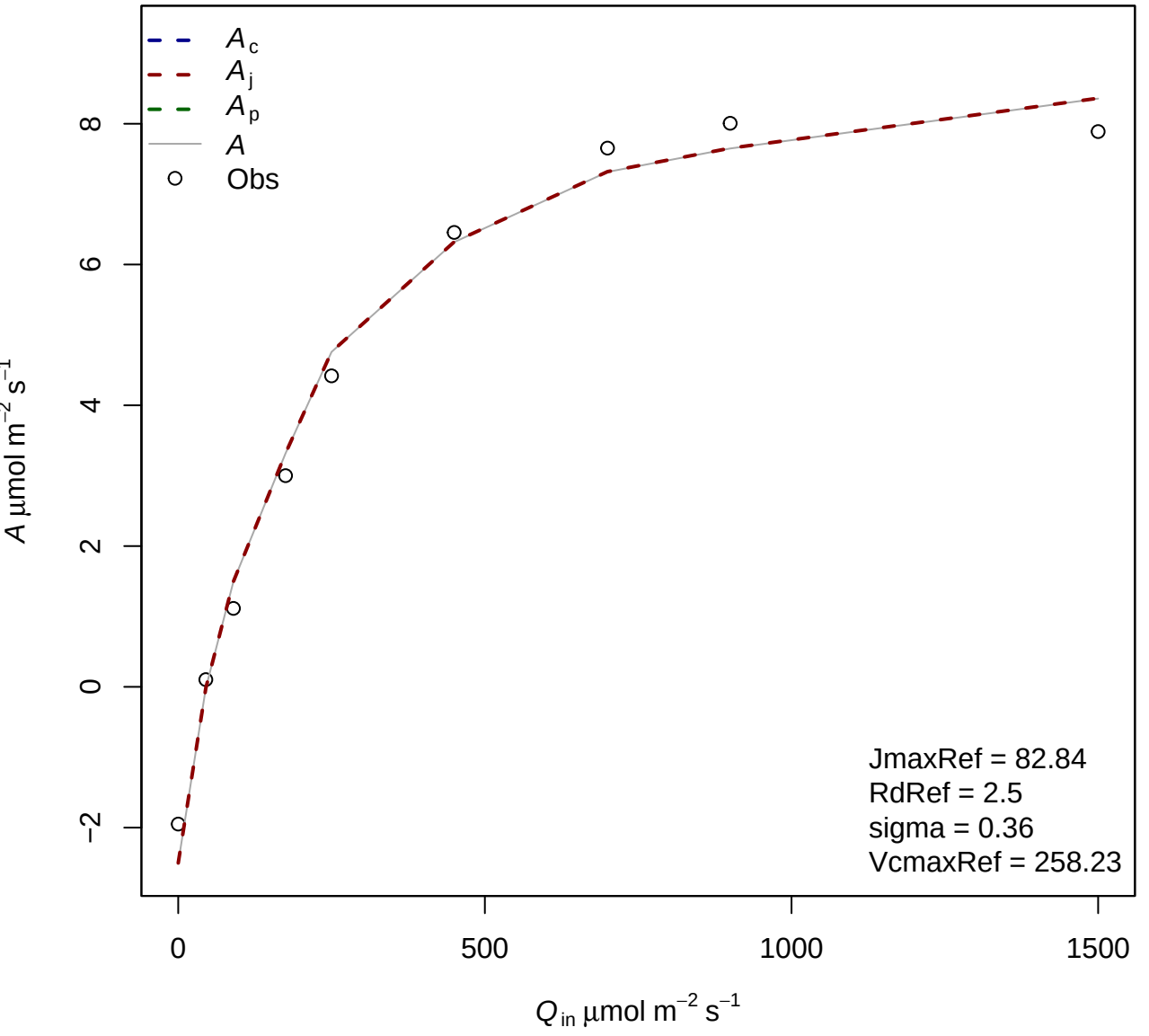

A_c
A_j
A_p
A
Obs
JmaxRef = 76.85
RdRef = 2.27
sigma = 0.67
VcmaxRef = 217.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

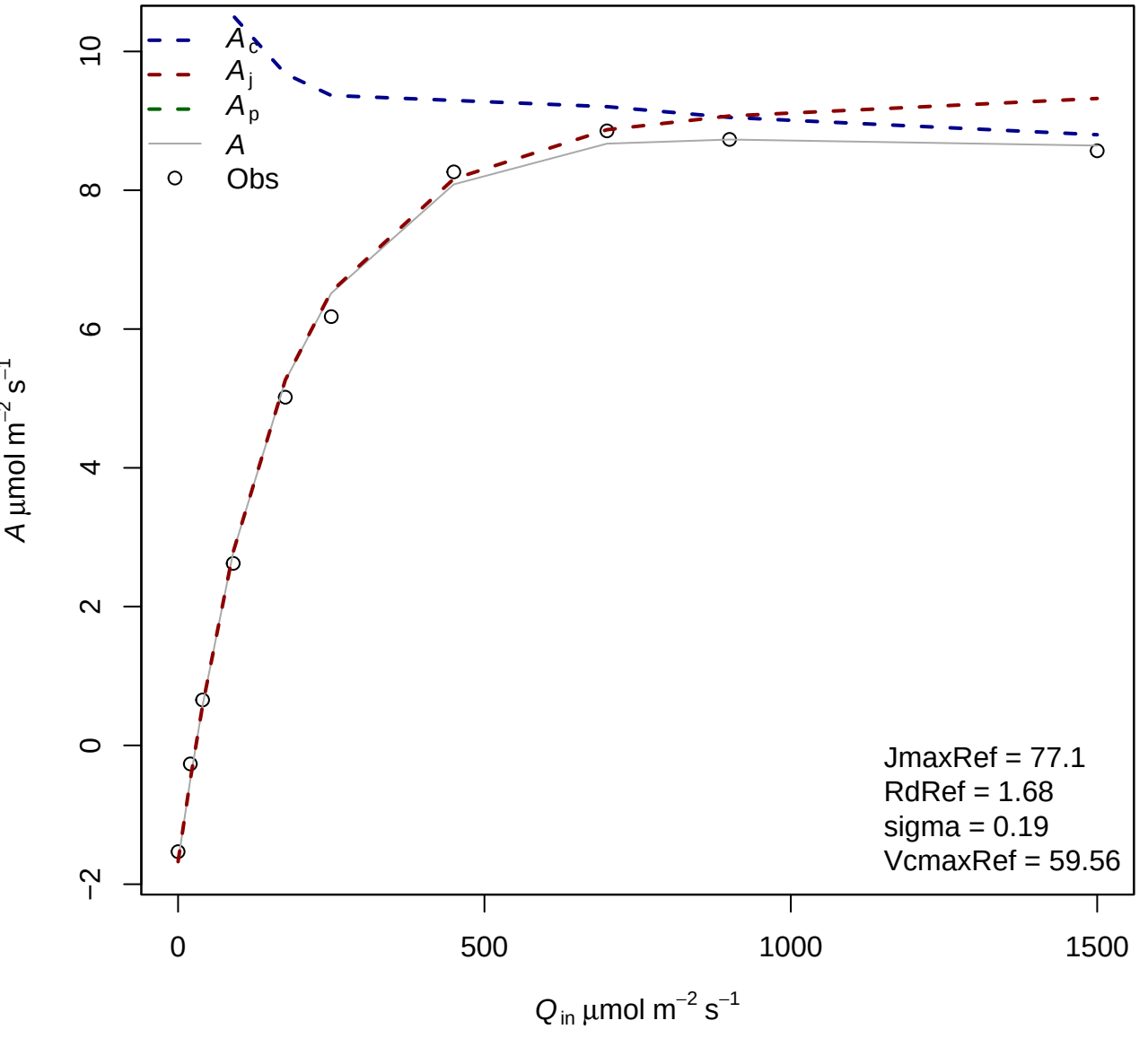

24
A_c
A_j
A_p
A
Obs
JmaxRef = 65.53
RdRef = 2.06
sigma = 1.16
VcmaxRef = 157.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

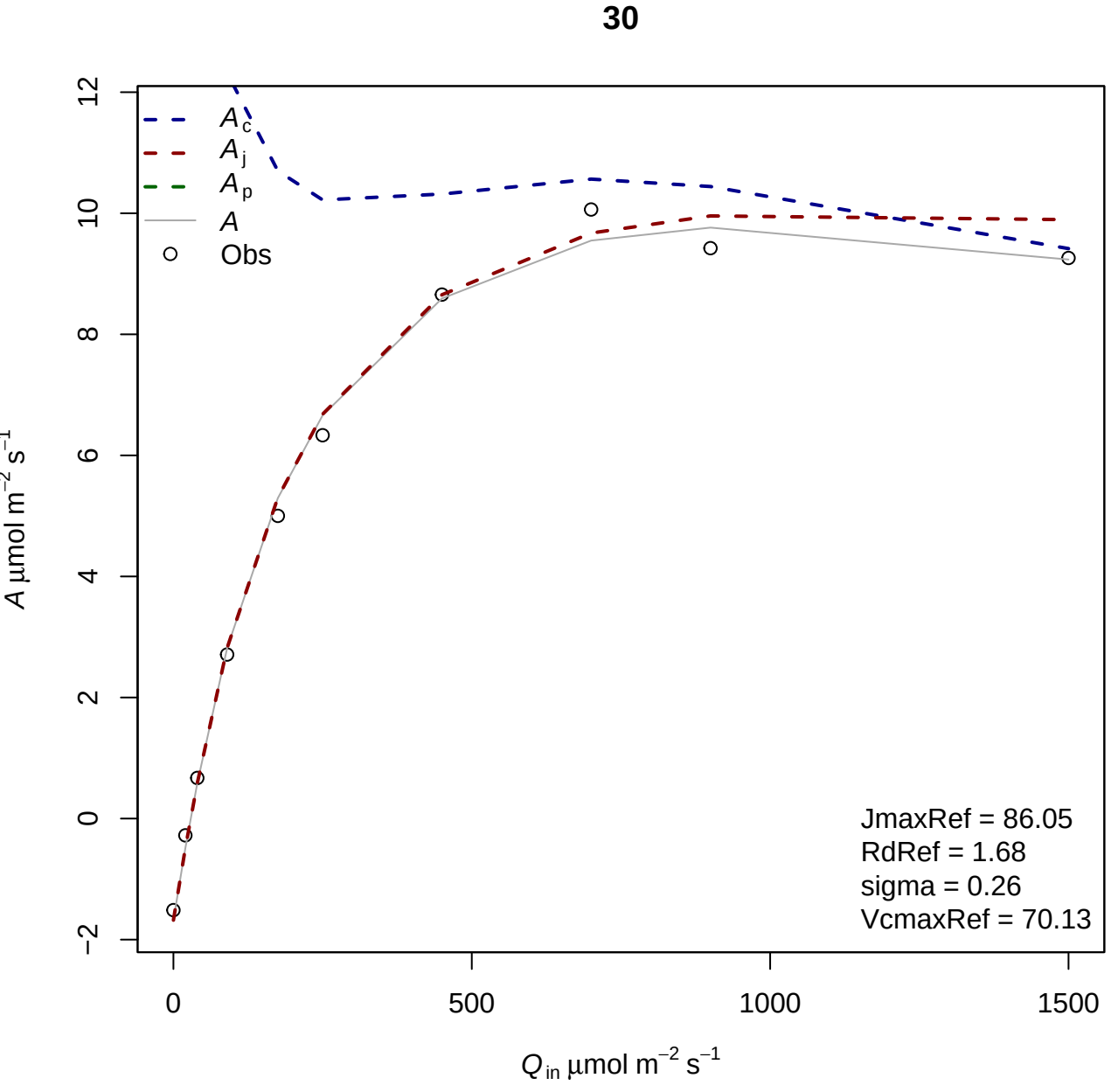

25
A_c
A_j
A_p
A
Obs
JmaxRef = 93.76
RdRef = 1.65
sigma = 0.15
VcmaxRef = 248.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

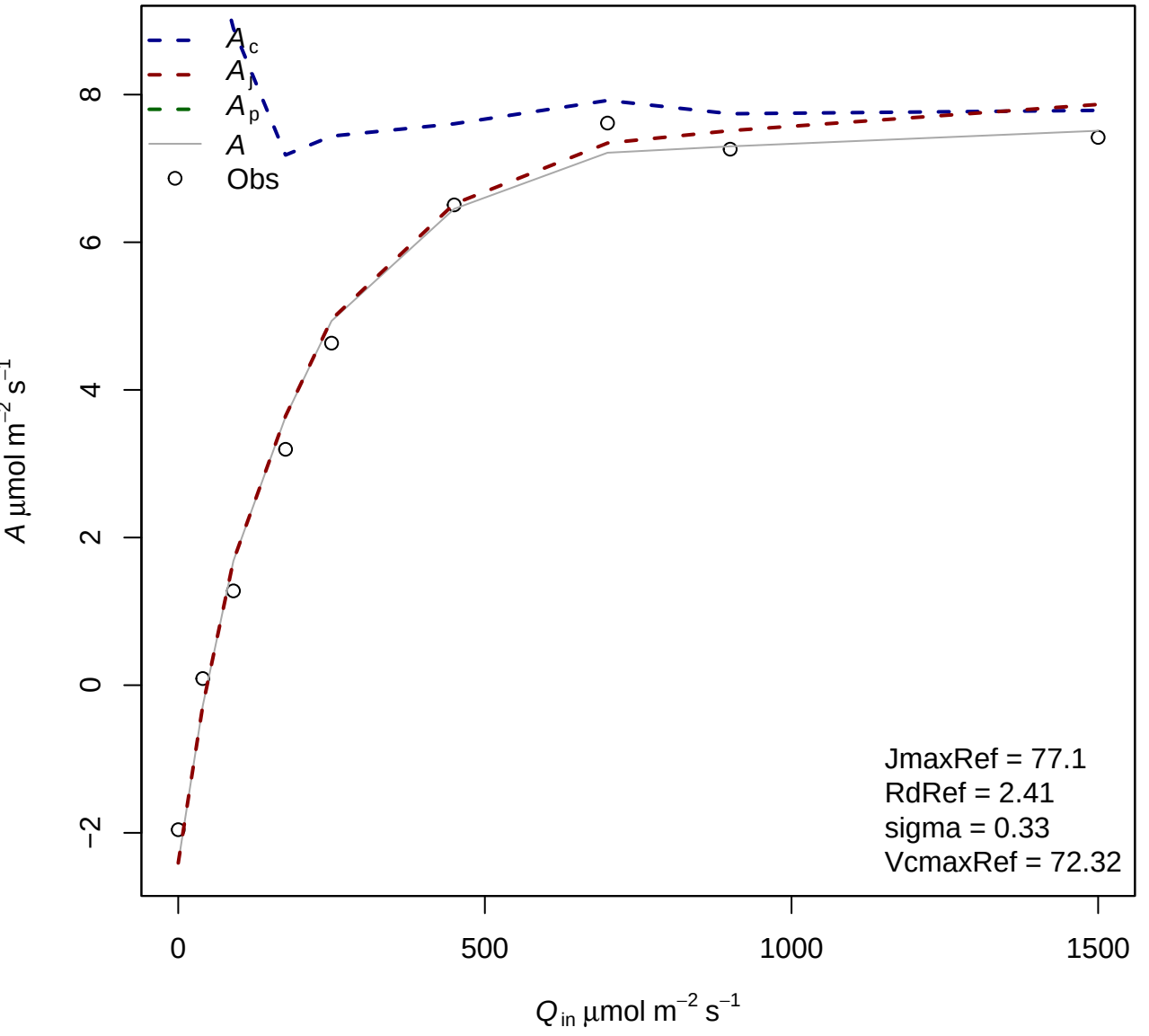

26
A_c
A_j
A_p
A
Obs
JmaxRef = 95.58
RdRef = 1.89
sigma = 0.31
VcmaxRef = 75.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

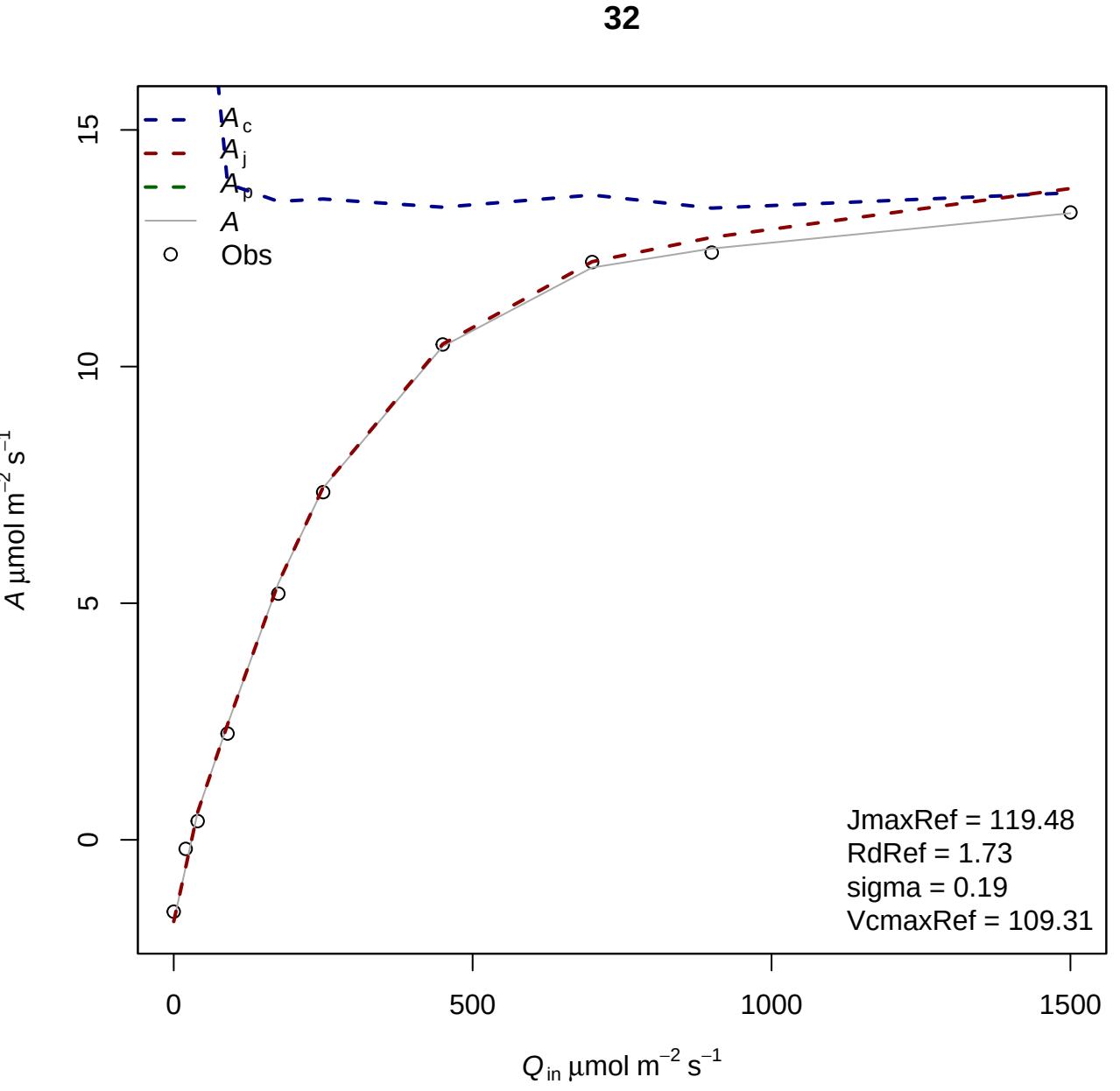

A_c
A_j
A_p
A
Qbs
JmaxRef = 187.26
RdRef = 3.66
sigma = 0.43
VcmaxRef = 165.22
0
5
10
15
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

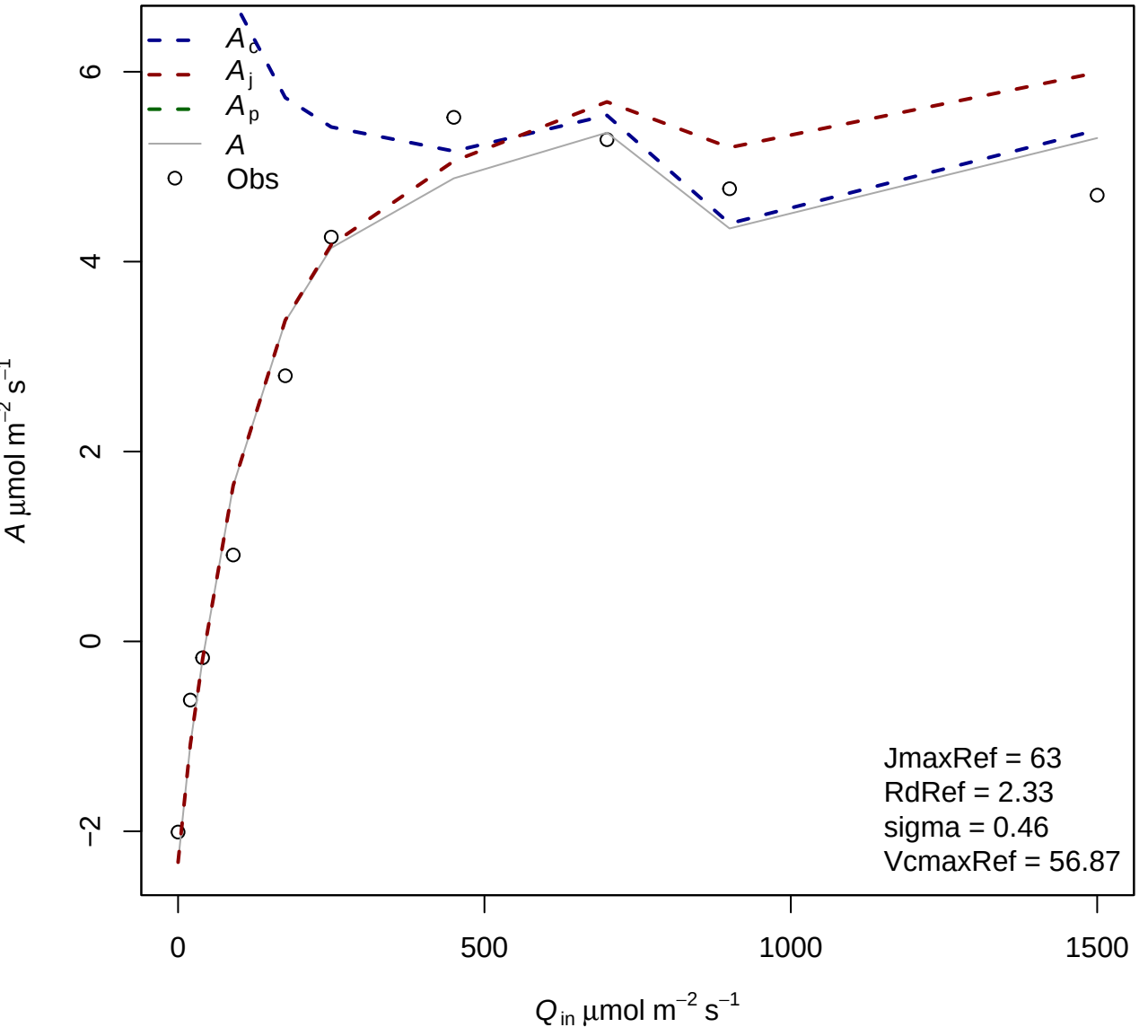

A_c
A_j
A_p
A
Obs
JmaxRef = 82.84
RdRef = 2.5
sigma = 0.36
VcmaxRef = 258.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

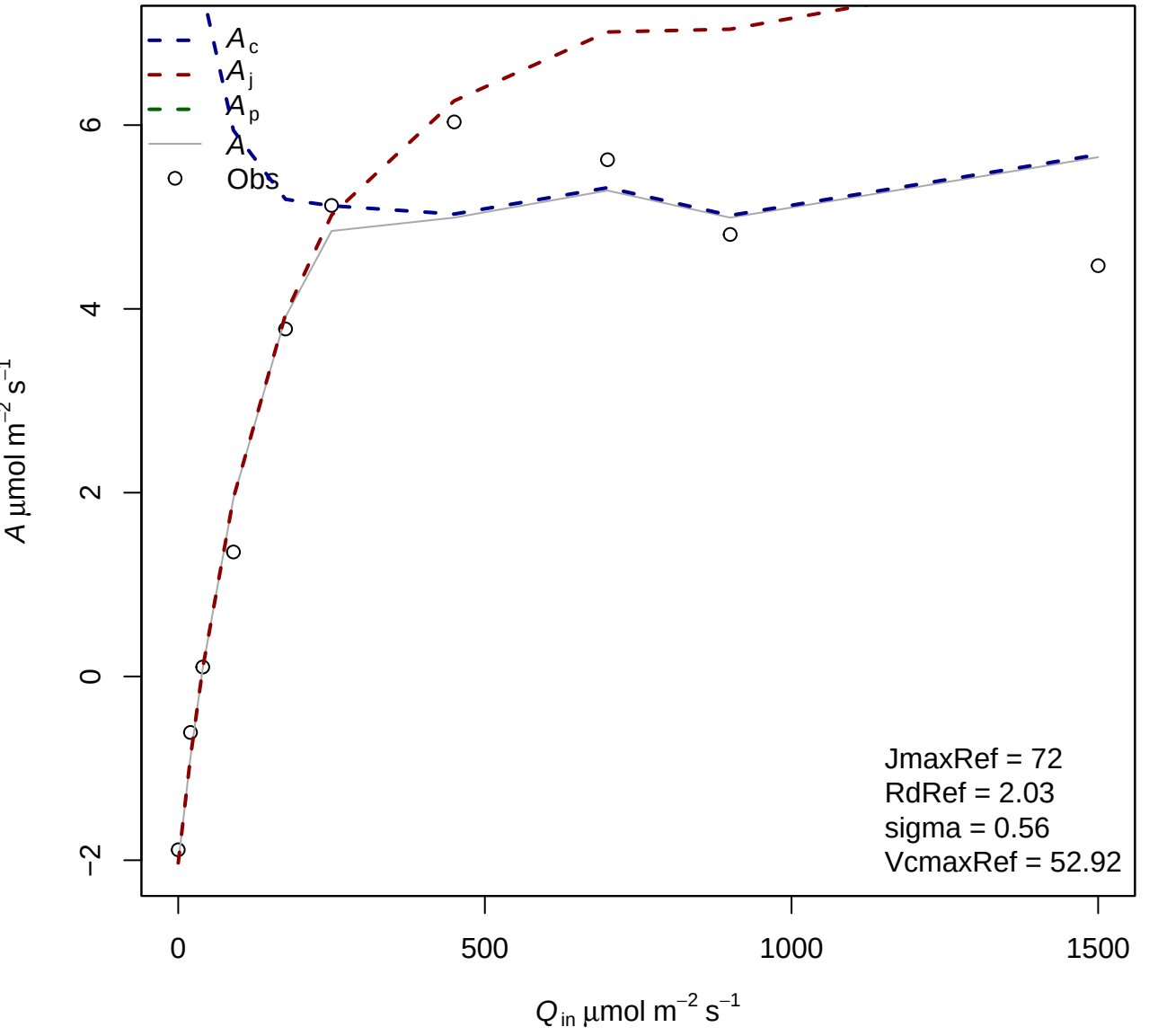

A_c
A_j
A_p
A
Obs
JmaxRef = 77.1
RdRef = 1.68
sigma = 0.19
VcmaxRef = 59.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

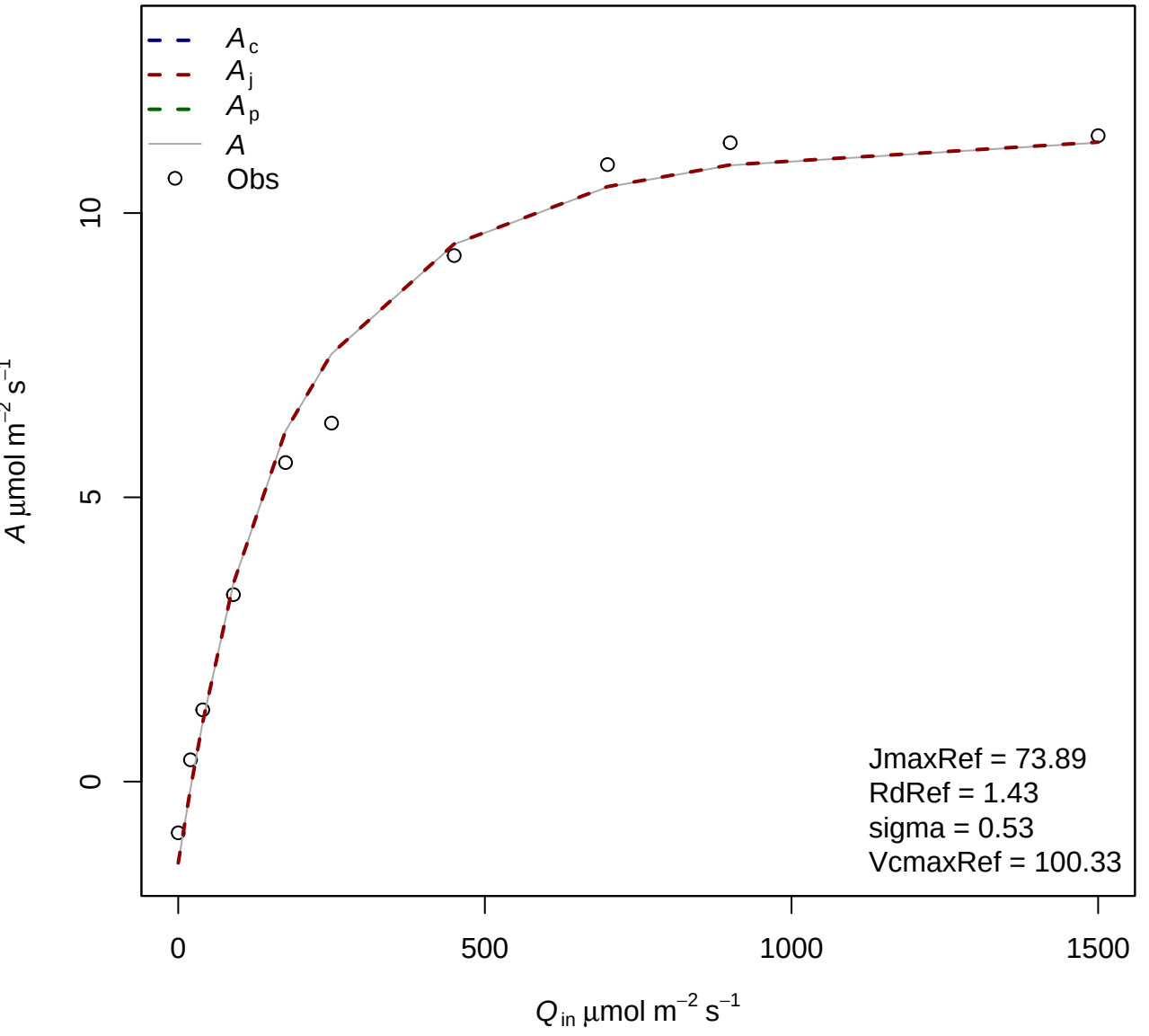

30
A_c
A_j
A_p
A
Obs
JmaxRef = 86.05
RdRef = 1.68
sigma = 0.26
VcmaxRef = 70.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

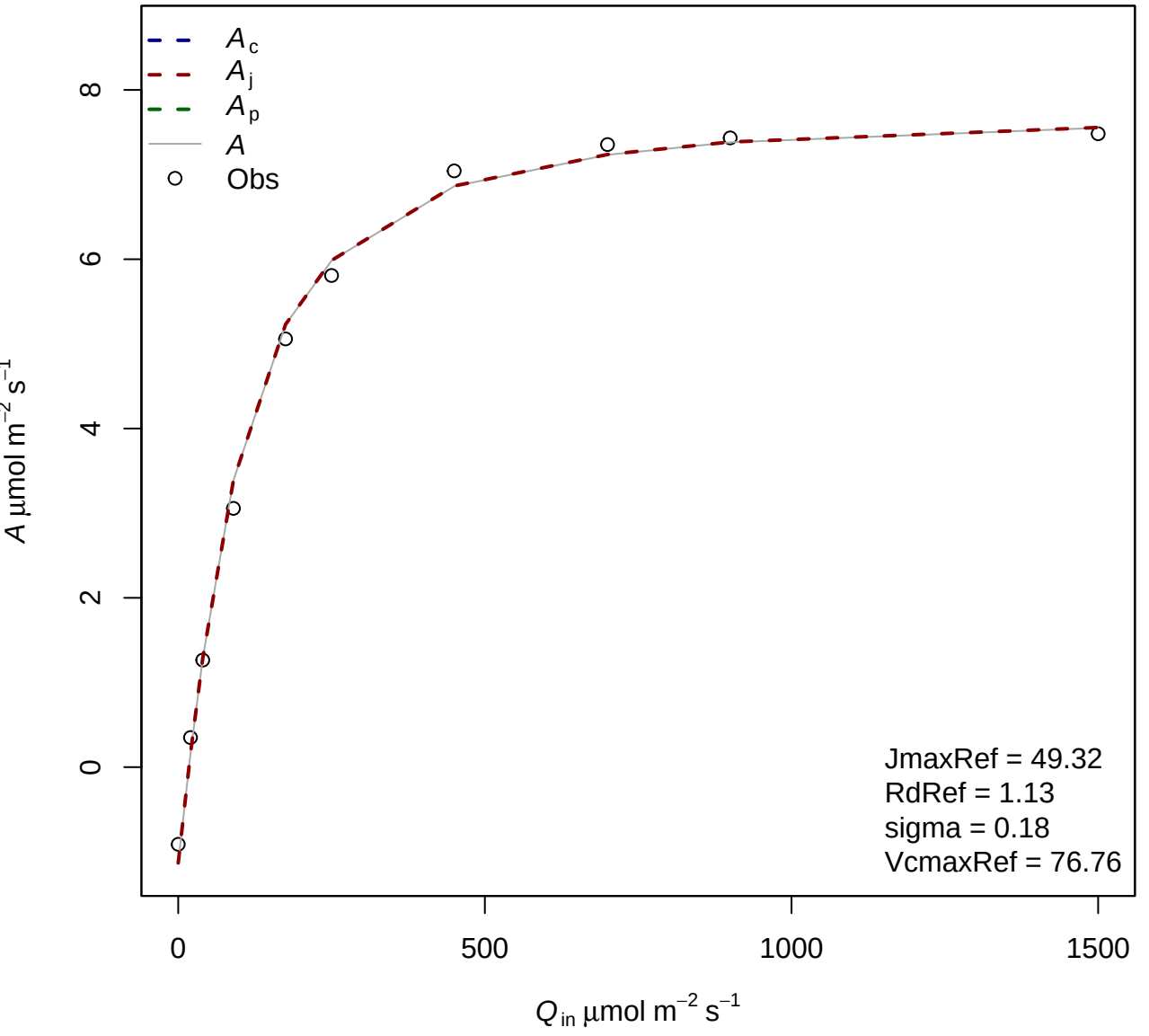

A_c
A_j
A_p
A
Obs
JmaxRef = 77.1
RdRef = 2.41
sigma = 0.33
VcmaxRef = 72.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

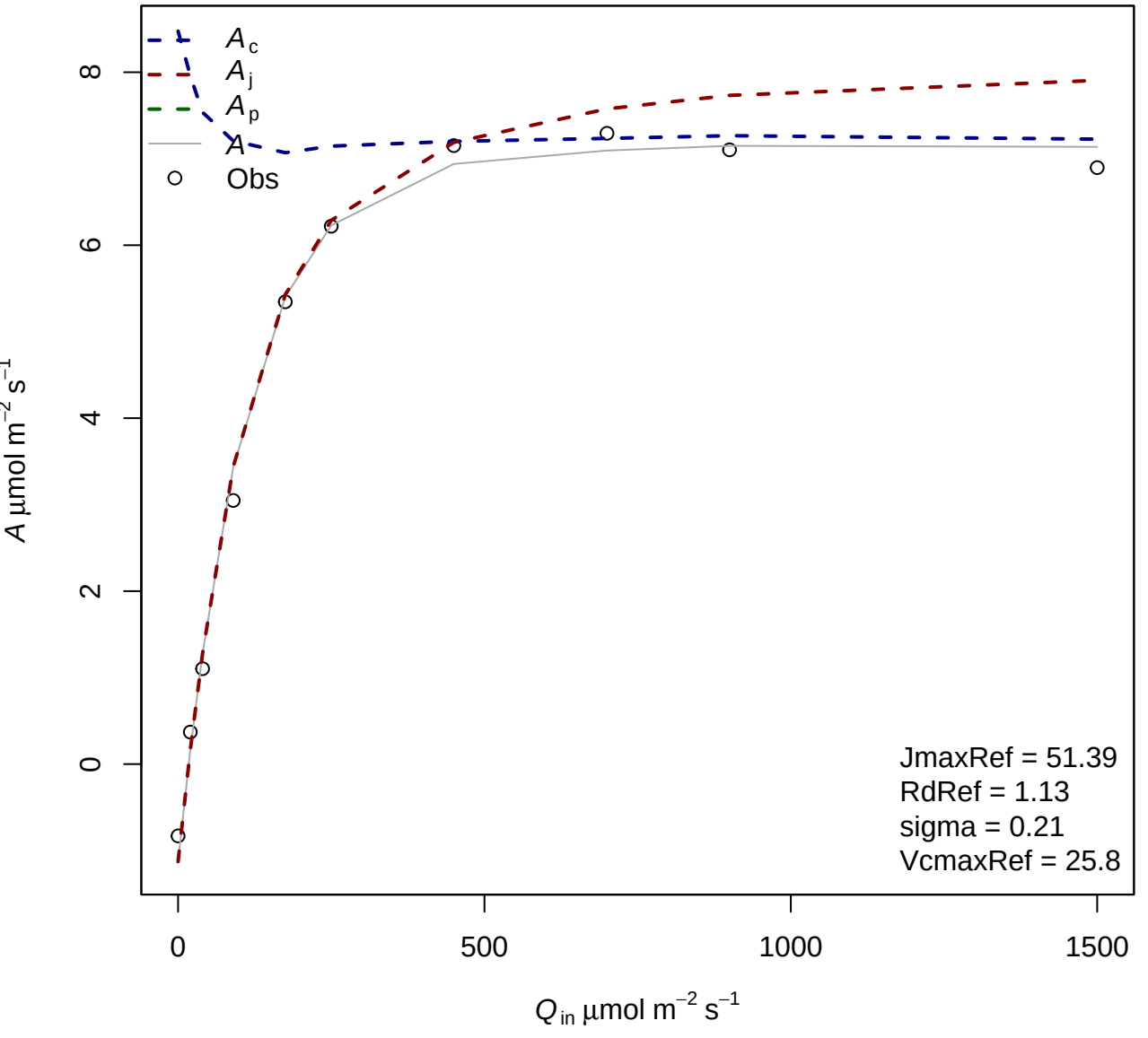

32
A_c
A_j
A_p
A
Obs
JmaxRef = 119.48
RdRef = 1.73
sigma = 0.19
VcmaxRef = 109.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

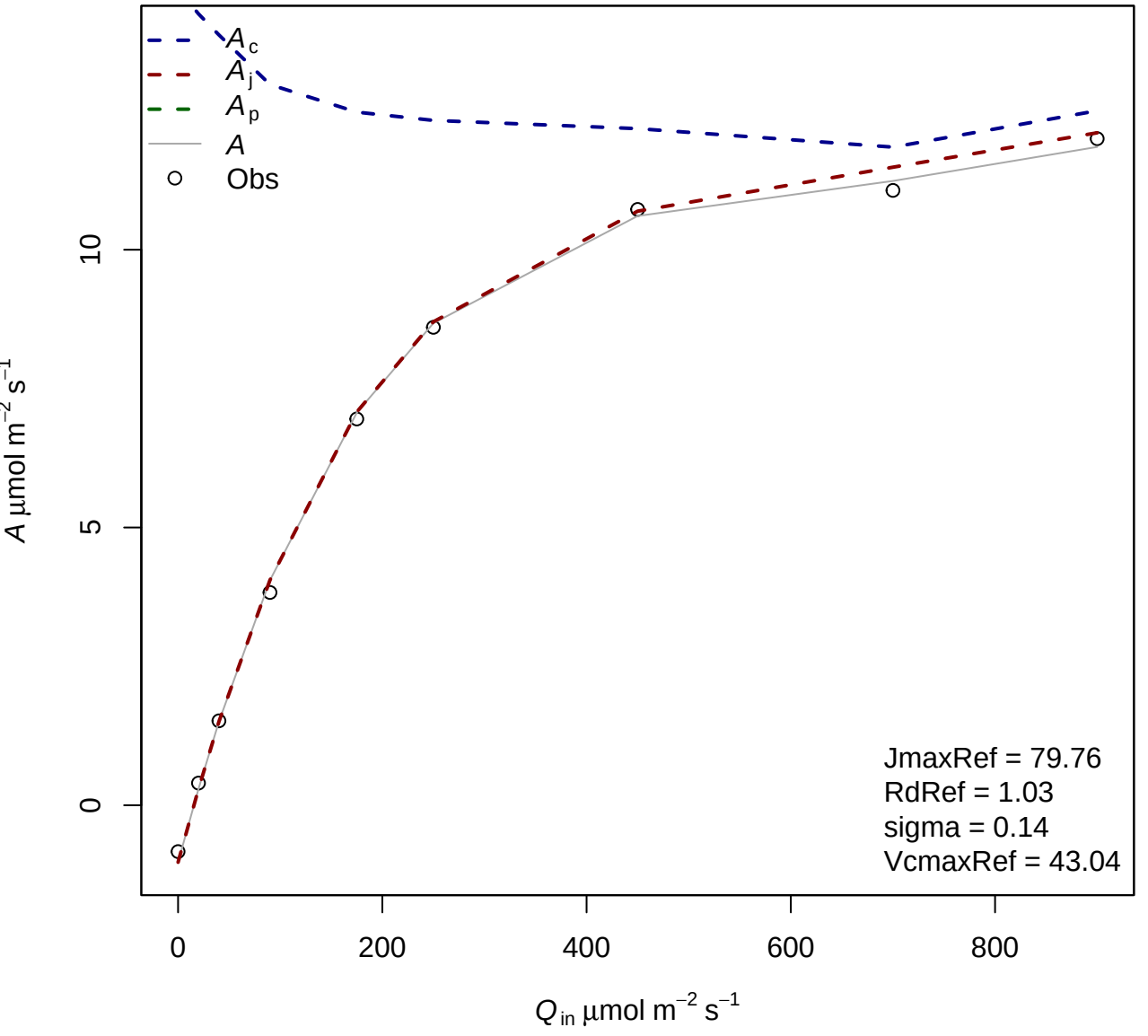

A_c
A_j
A_p
A
Obs
JmaxRef = 63
RdRef = 2.33
sigma = 0.46
VcmaxRef = 56.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

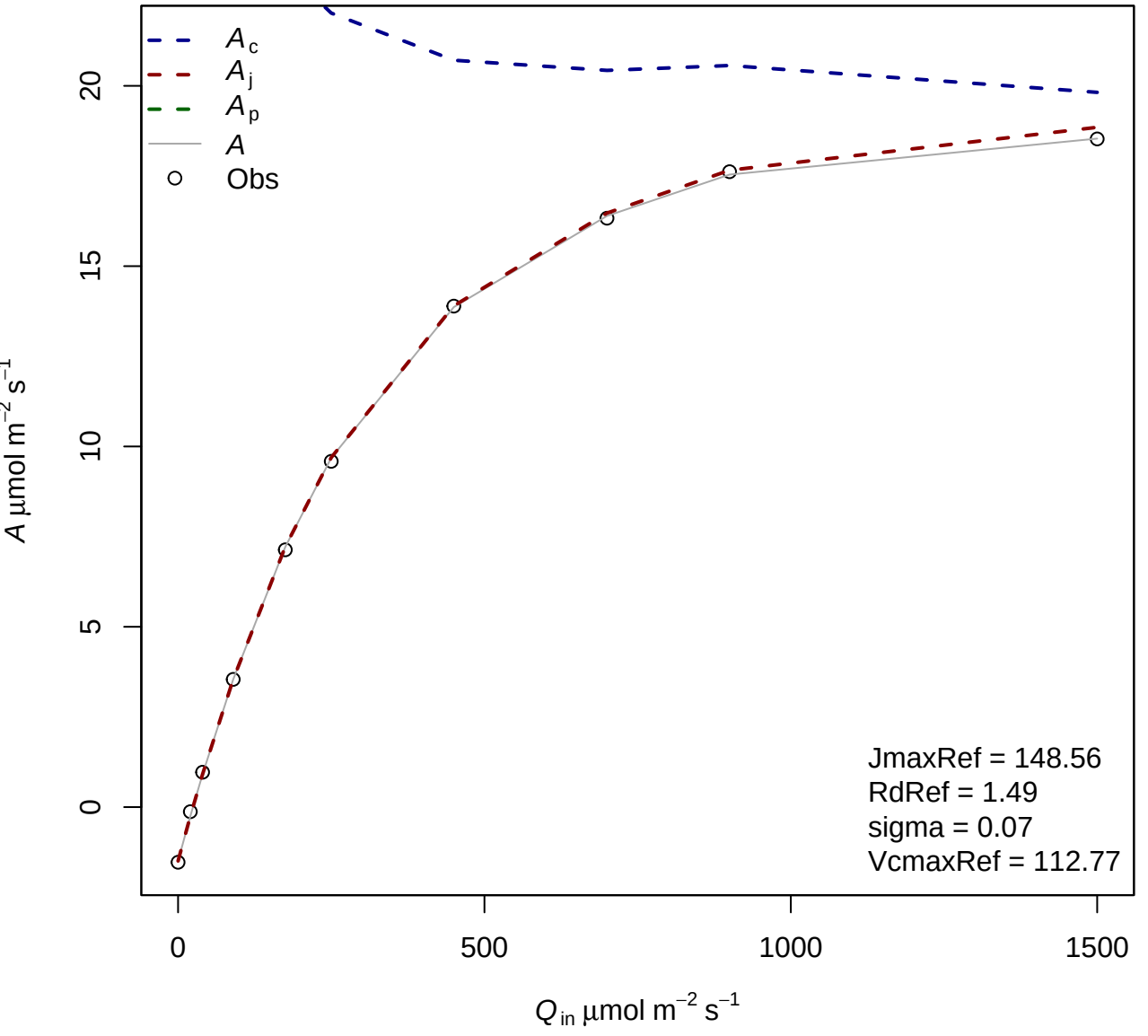

A_c
A_j
A_p
A
Obs
JmaxRef = 72
RdRef = 2.03
sigma = 0.56
VcmaxRef = 52.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

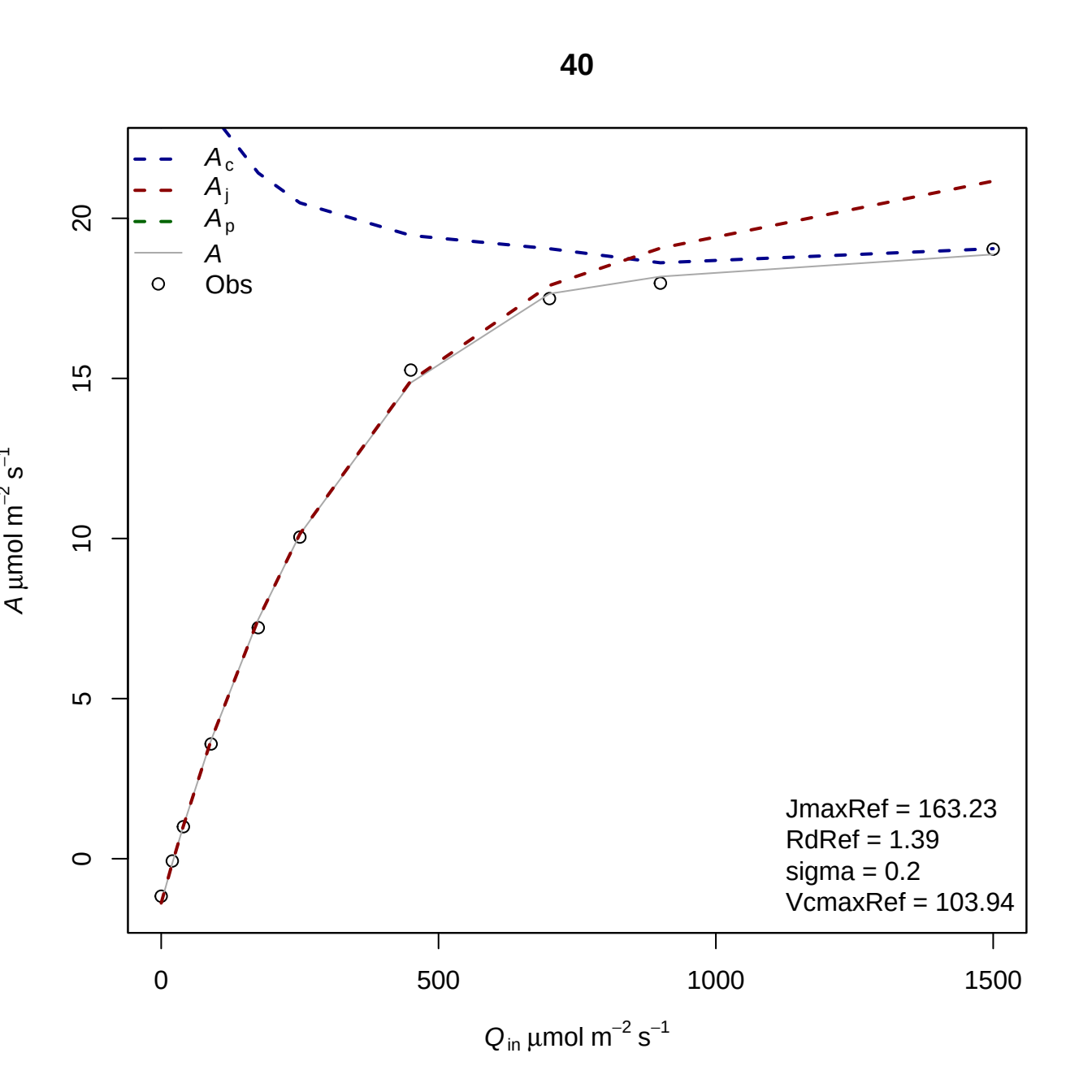

A_c
A_j
A_p
A
Obs
JmaxRef = 73.89
RdRef = 1.43
sigma = 0.53
VcmaxRef = 100.33
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

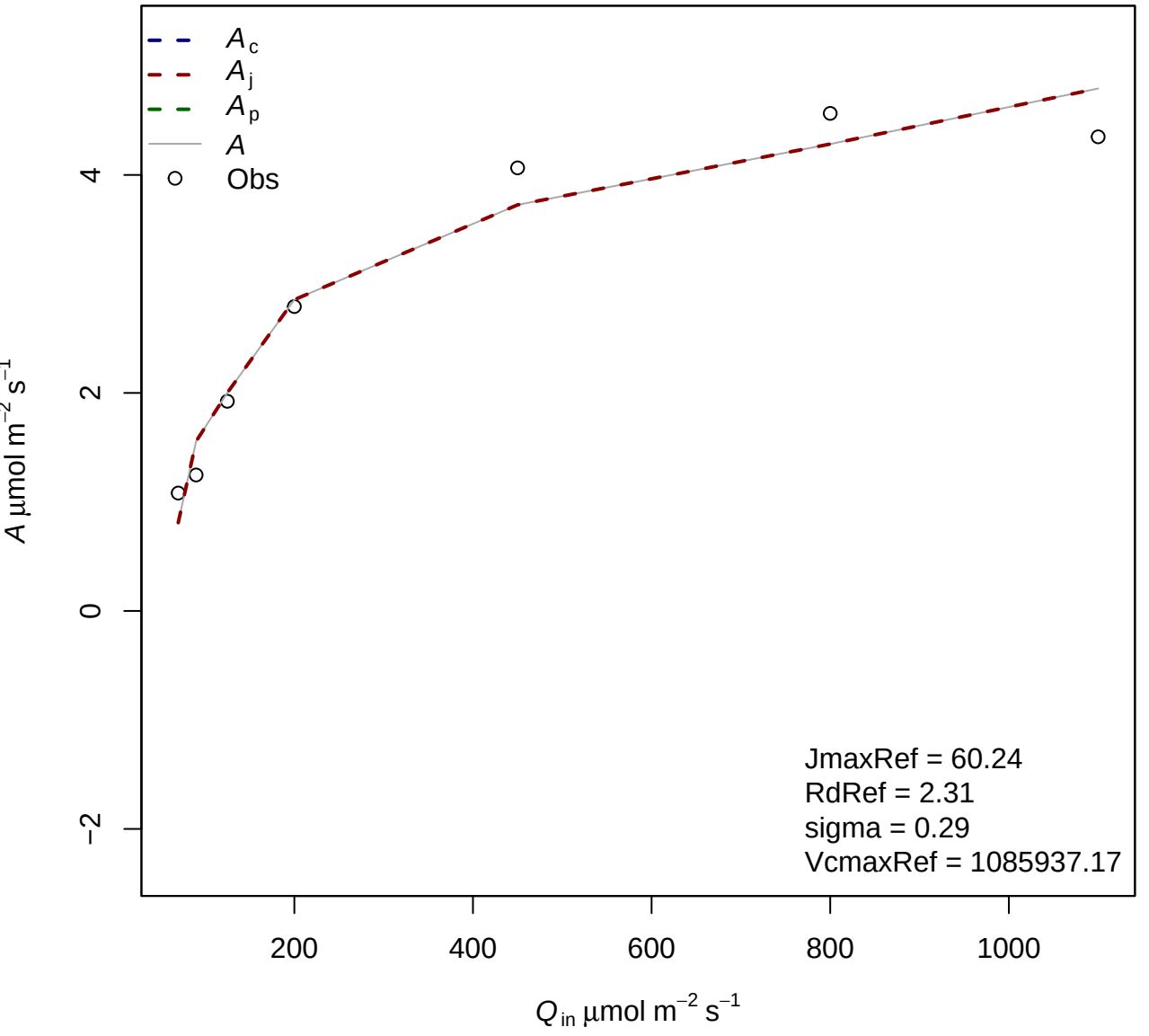

A_c
A_j
A_p
A
Obs
JmaxRef = 49.32
RdRef = 1.13
sigma = 0.18
VcmaxRef = 76.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

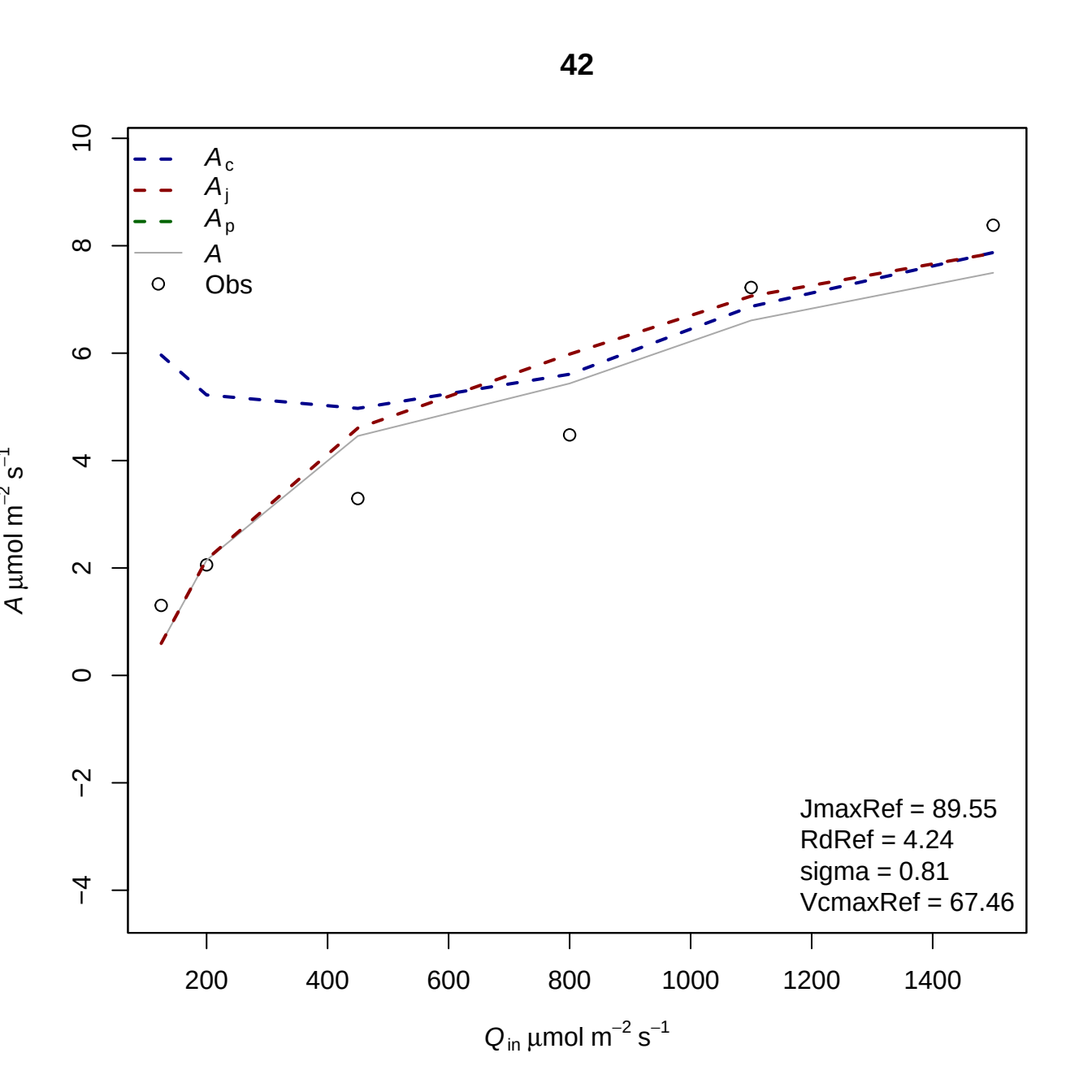

A_c
A_j
A_p
A
Obs
JmaxRef = 51.39
RdRef = 1.13
sigma = 0.21
VcmaxRef = 25.8
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

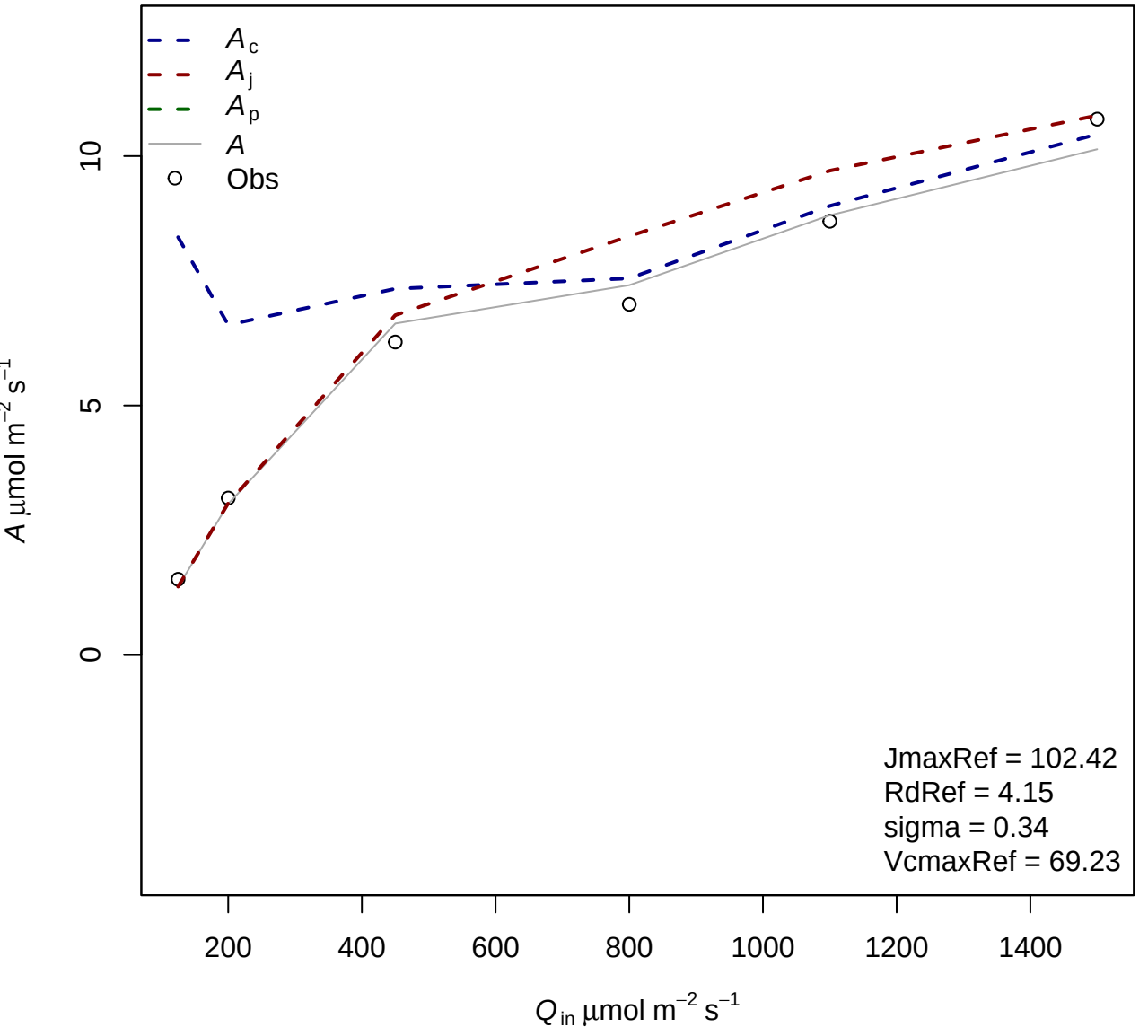

A_c
A_j
A_p
A
Obs
JmaxRef = 79.76
RdRef = 1.03
sigma = 0.14
VcmaxRef = 43.04
0
5
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
Q_{in} µmol m^{-2} s^{-1}

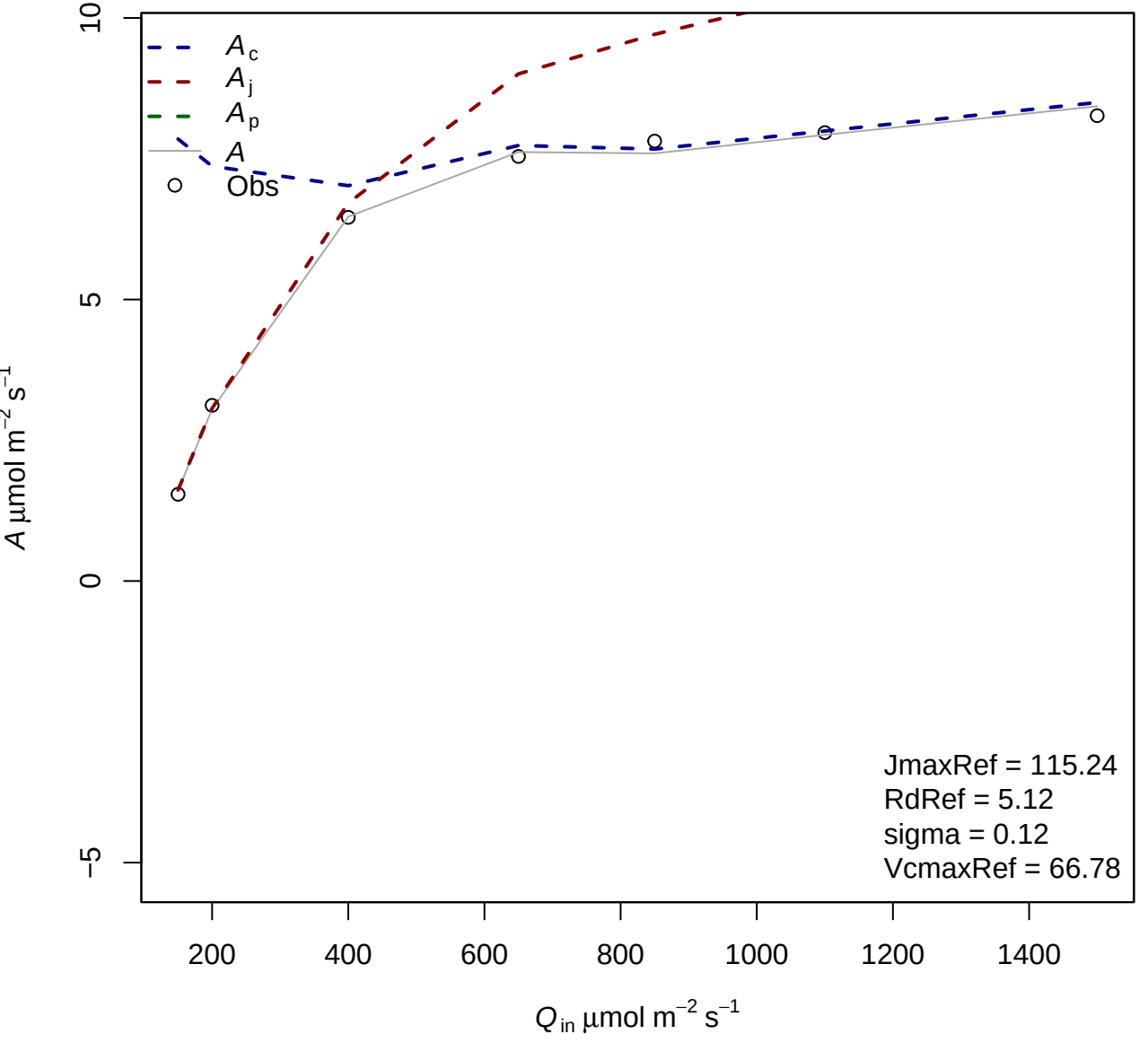

A_c
A_j
A_p
A
Obs
JmaxRef = 148.56
RdRef = 1.49
sigma = 0.07
VcmaxRef = 112.77
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

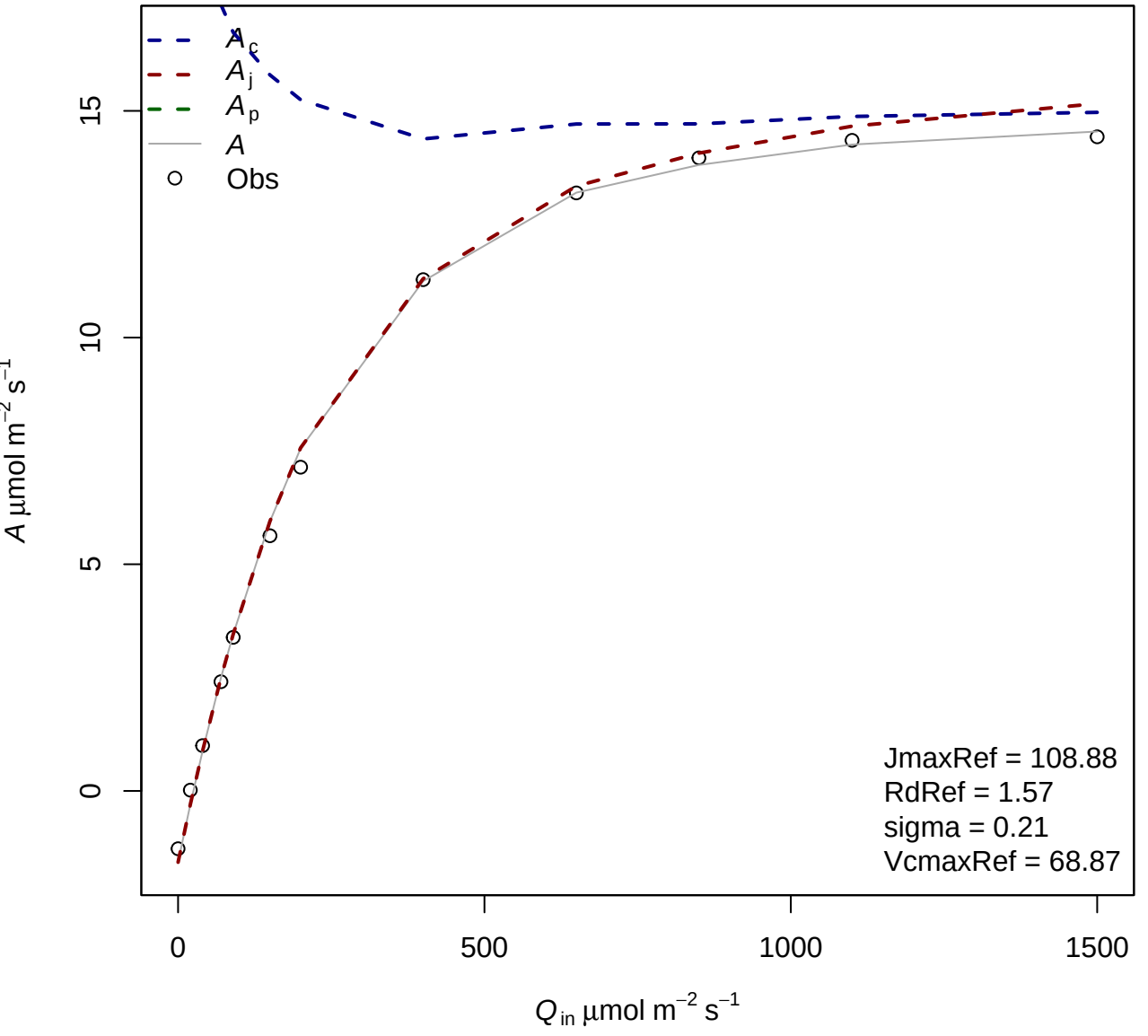

40
A_c
A_j
A_p
A
Obs
JmaxRef = 163.23
RdRef = 1.39
sigma = 0.2
VcmaxRef = 103.94
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

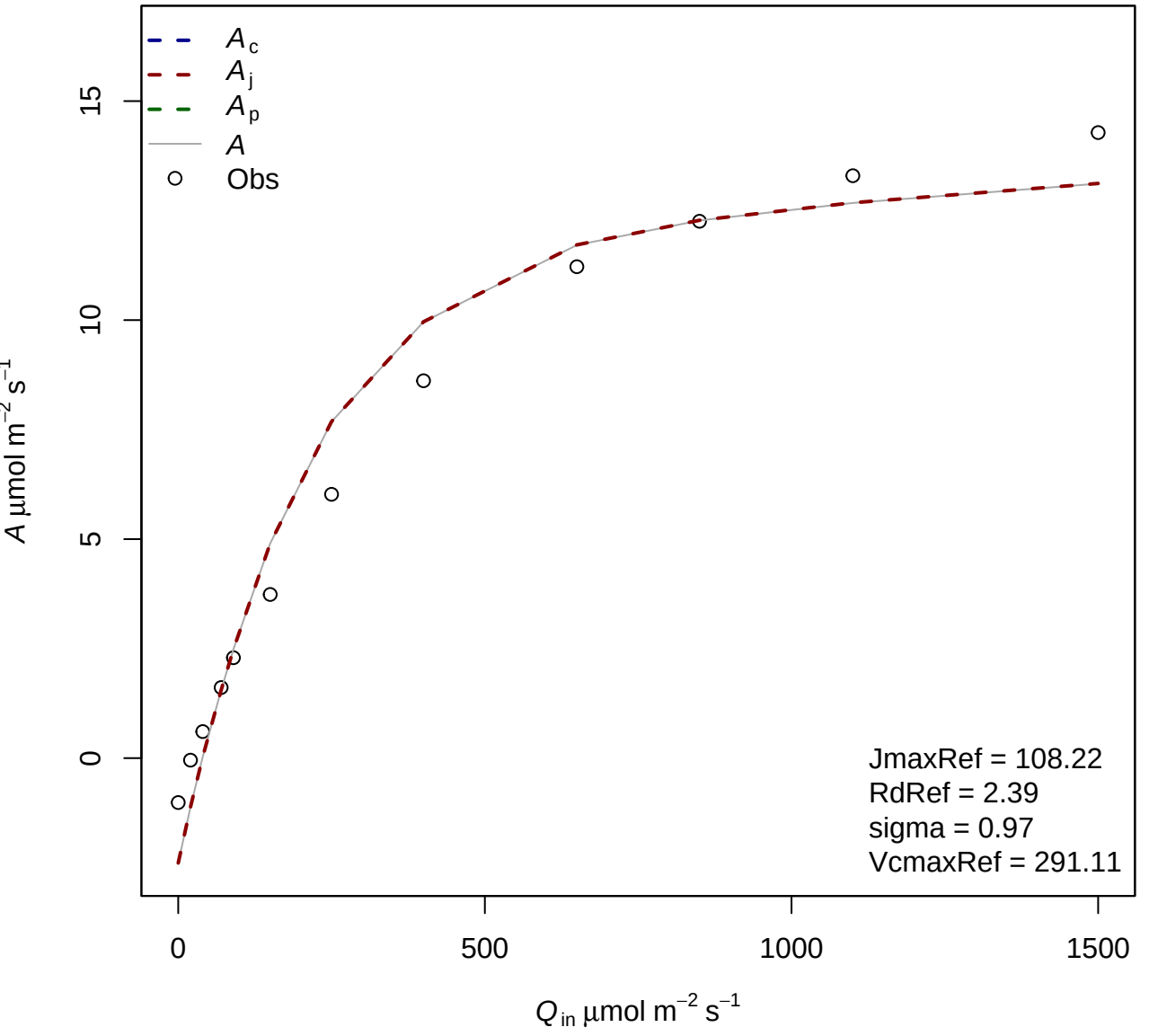

A_c
A_j
A_p
A
Obs
JmaxRef = 60.24
RdRef = 2.31
sigma = 0.29
VcmaxRef = 1085937.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
200
400
600
800
1000

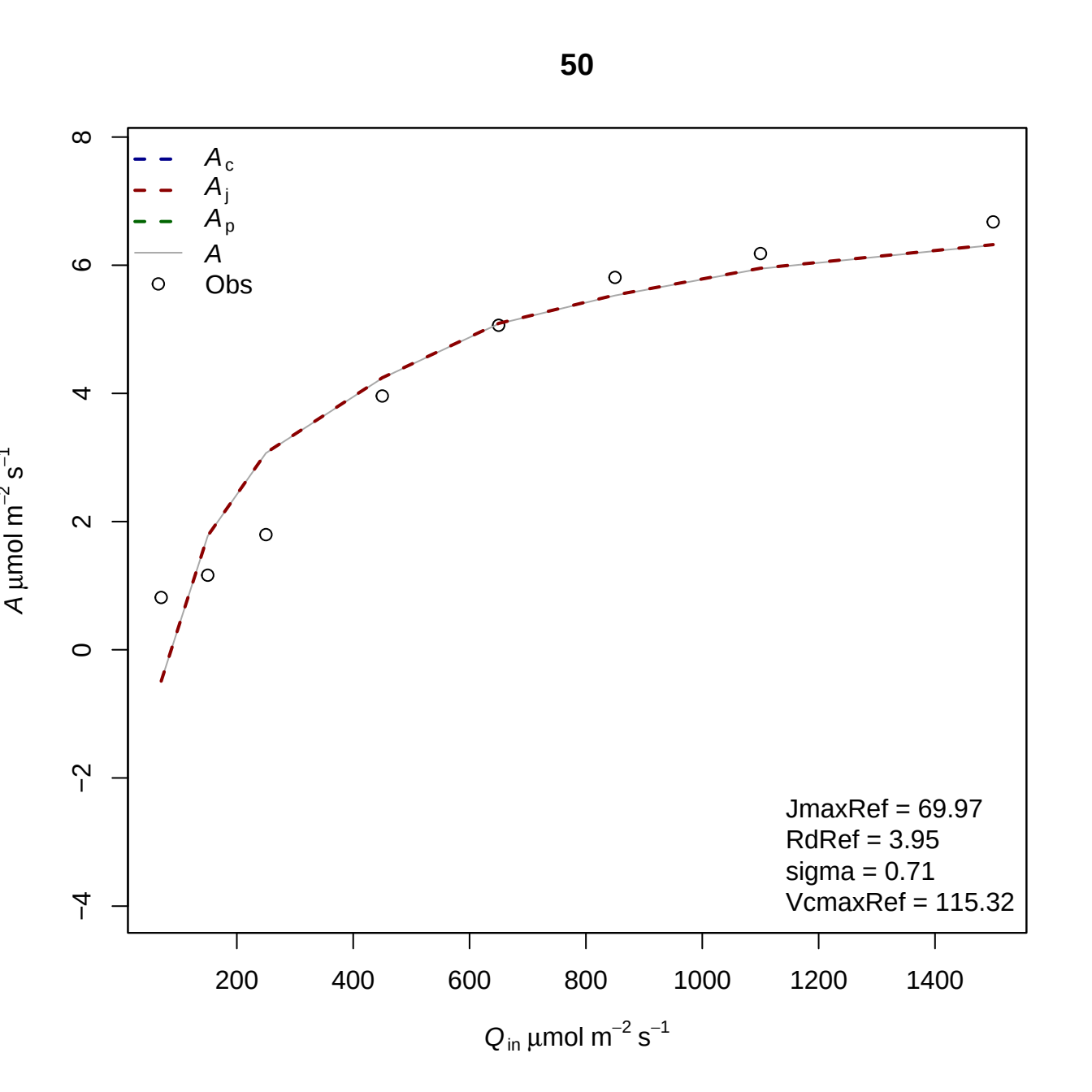

42
A_c
A_j
A_p
A
Obs
JmaxRef = 89.55
RdRef = 4.24
sigma = 0.81
VcmaxRef = 67.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

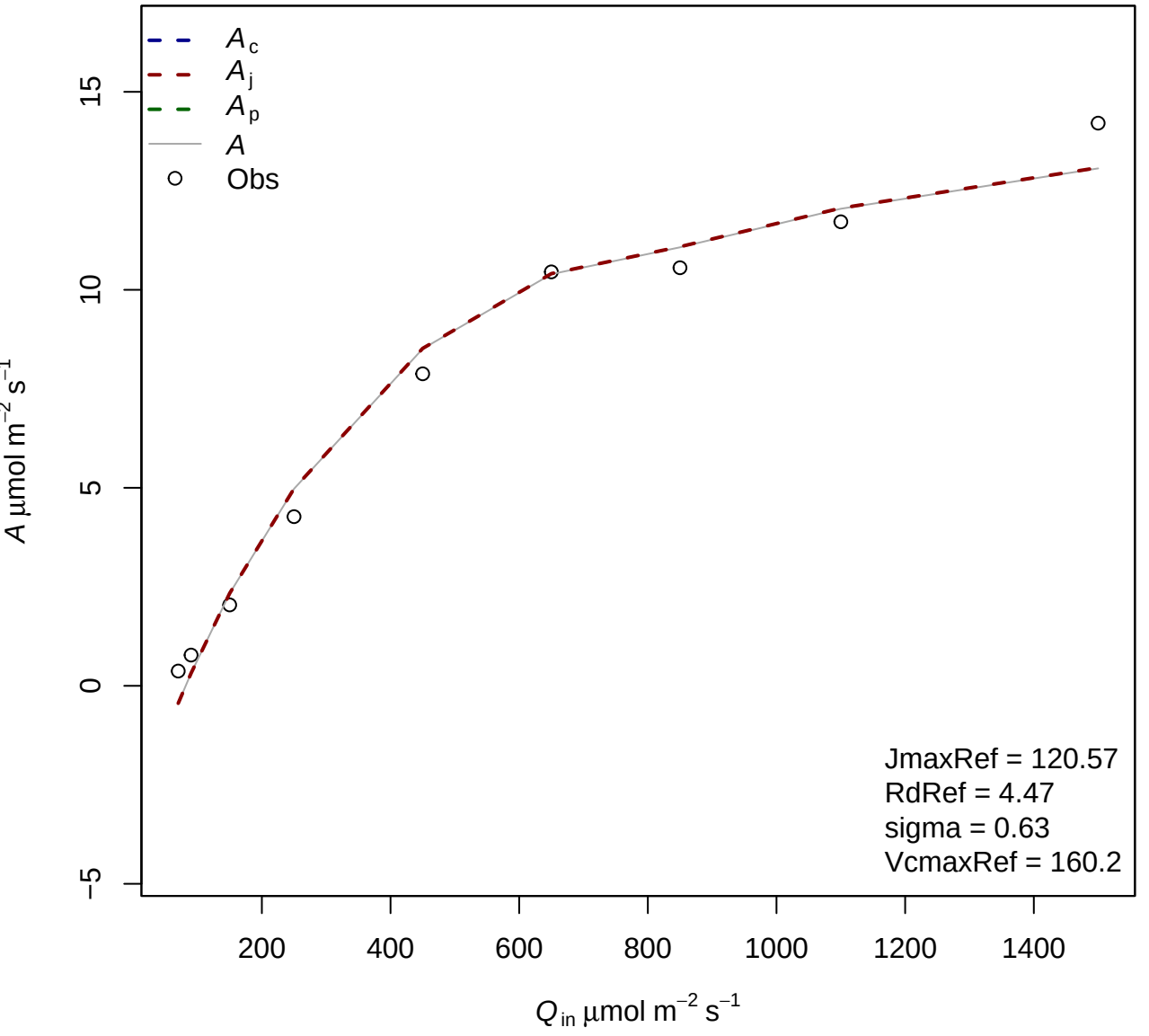

A_c
A_j
A_p
A
Obs
0
5
10
A µmol m^{-2} s^{-1}
200
400
600
800
1000
1200
1400
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 102.42
RdRef = 4.15
sigma = 0.34
VcmaxRef = 69.23

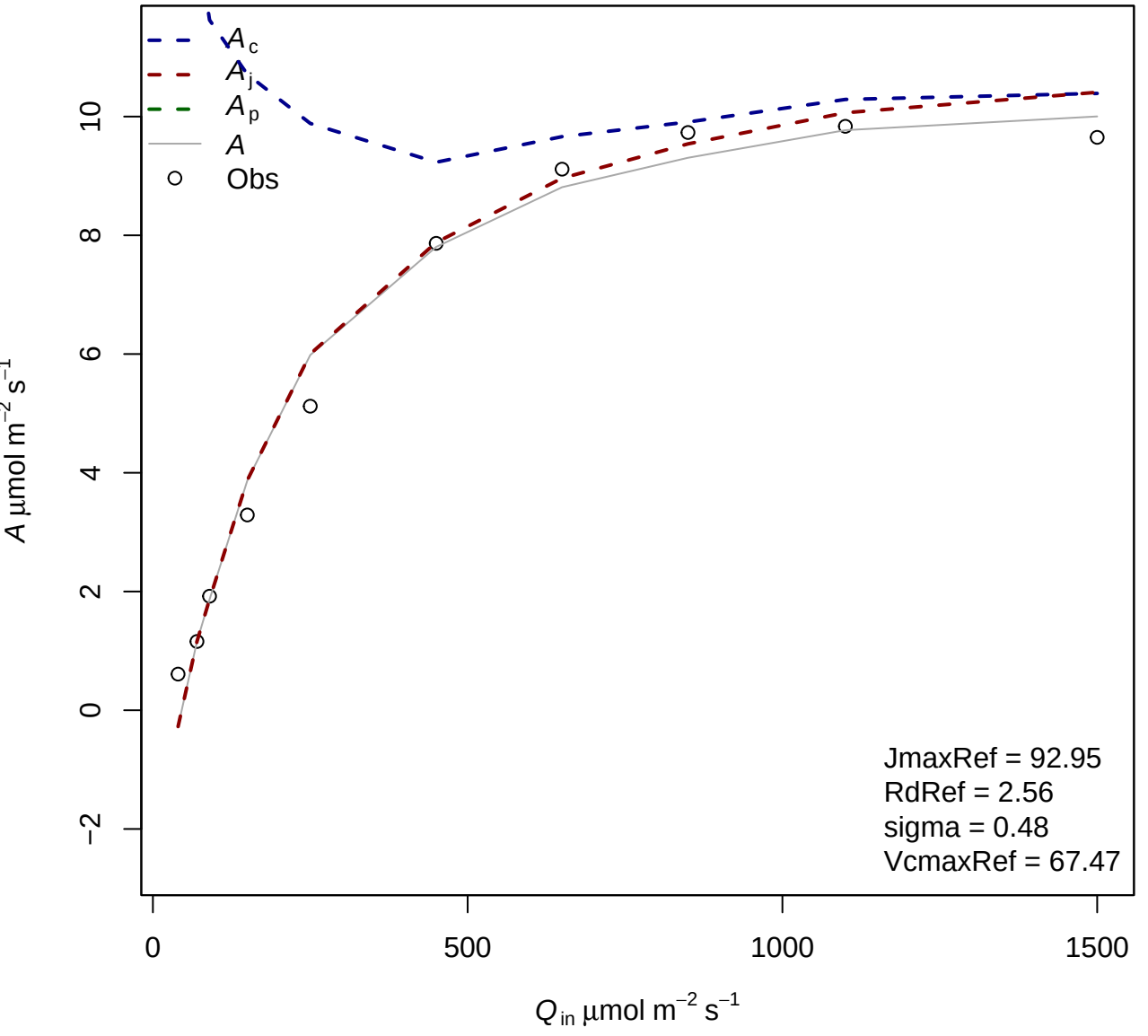

A_c
A_j
A_p
A
Obs
JmaxRef = 115.24
RdRef = 5.12
sigma = 0.12
VcmaxRef = 66.78
10
5
0
−5
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

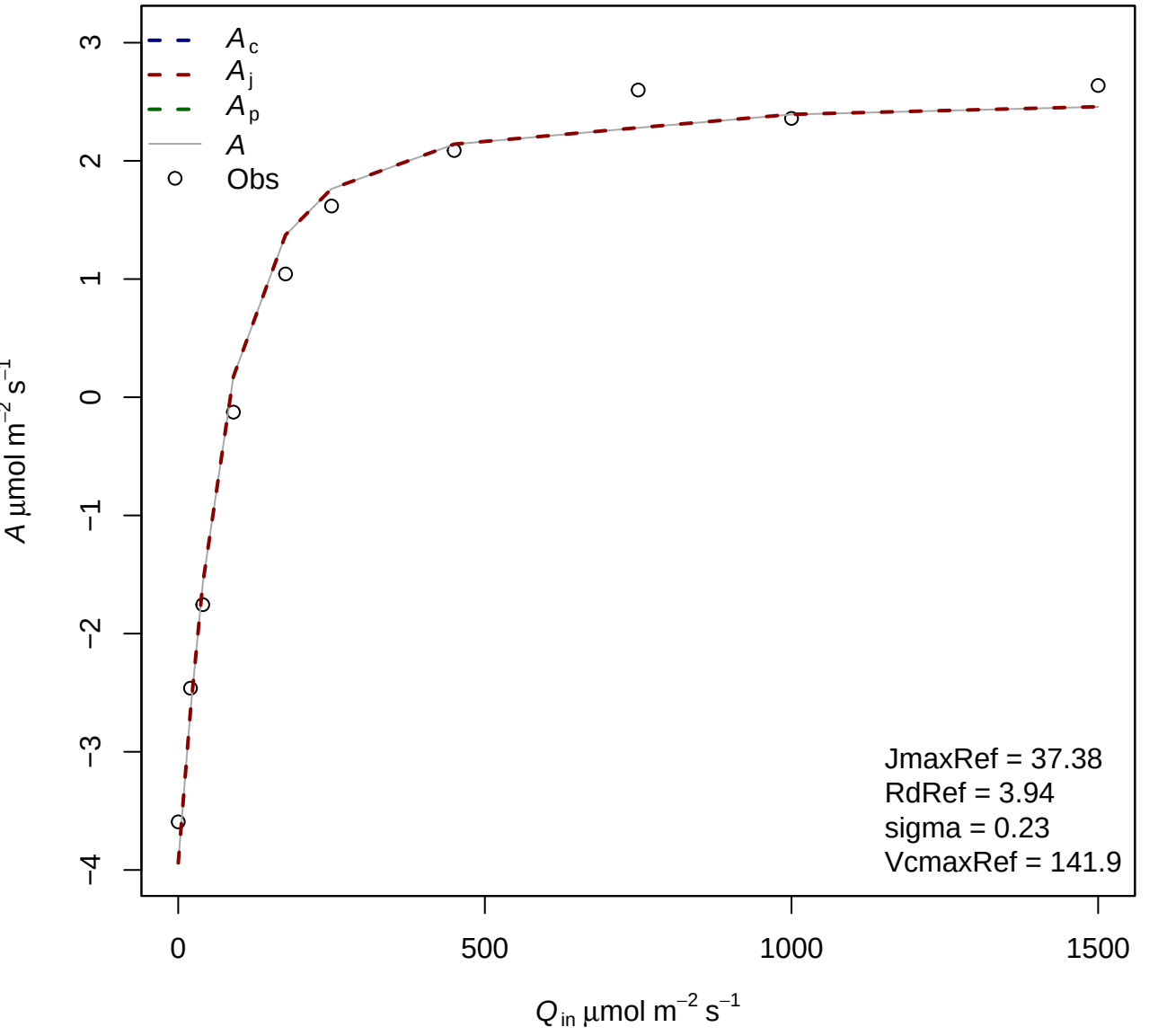

A_c
A_j
A_p
A
Obs
JmaxRef = 108.88
RdRef = 1.57
sigma = 0.21
VcmaxRef = 68.87
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

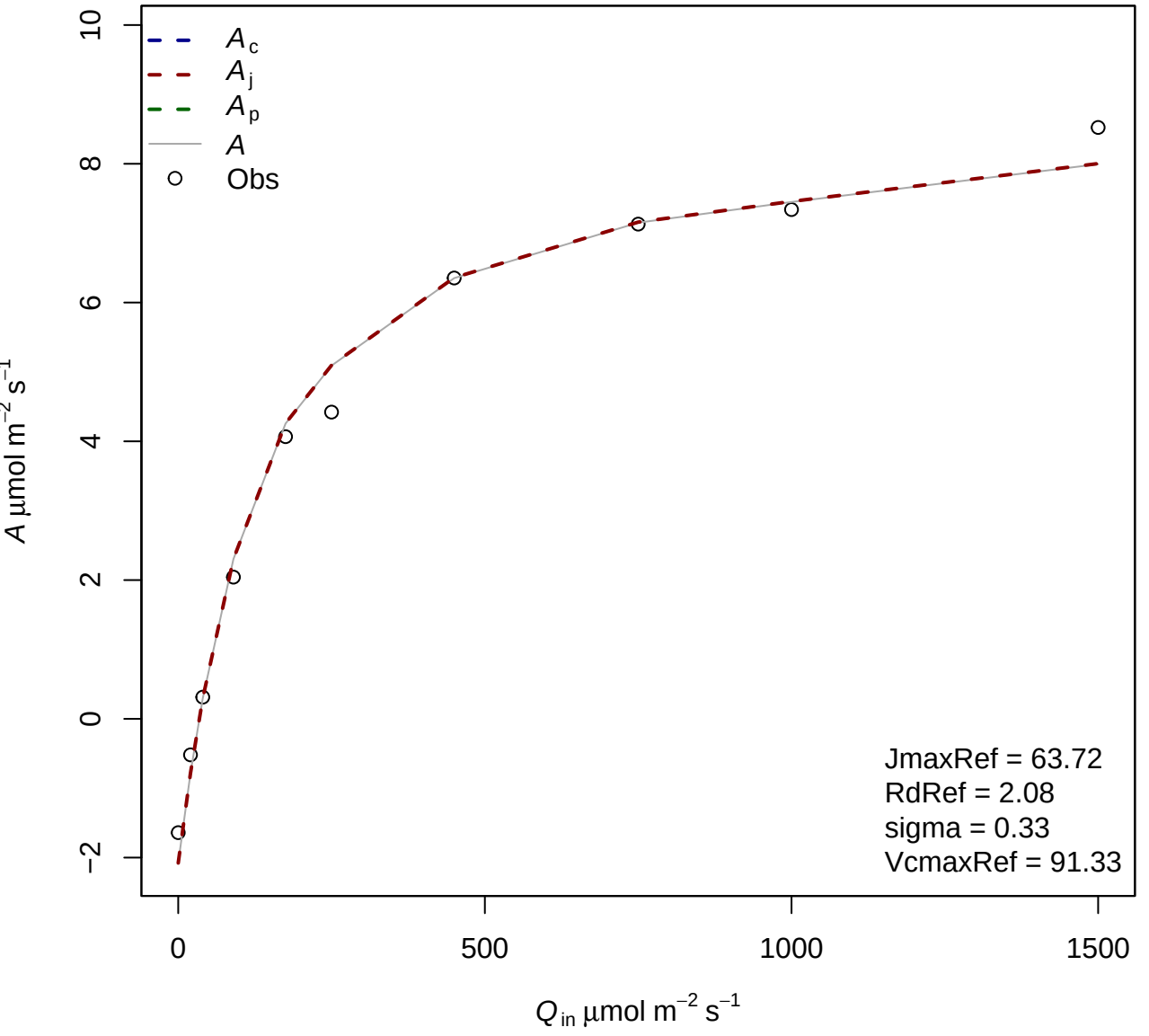

A_c
A_j
A_p
A
Obs
JmaxRef = 108.22
RdRef = 2.39
sigma = 0.97
VcmaxRef = 291.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

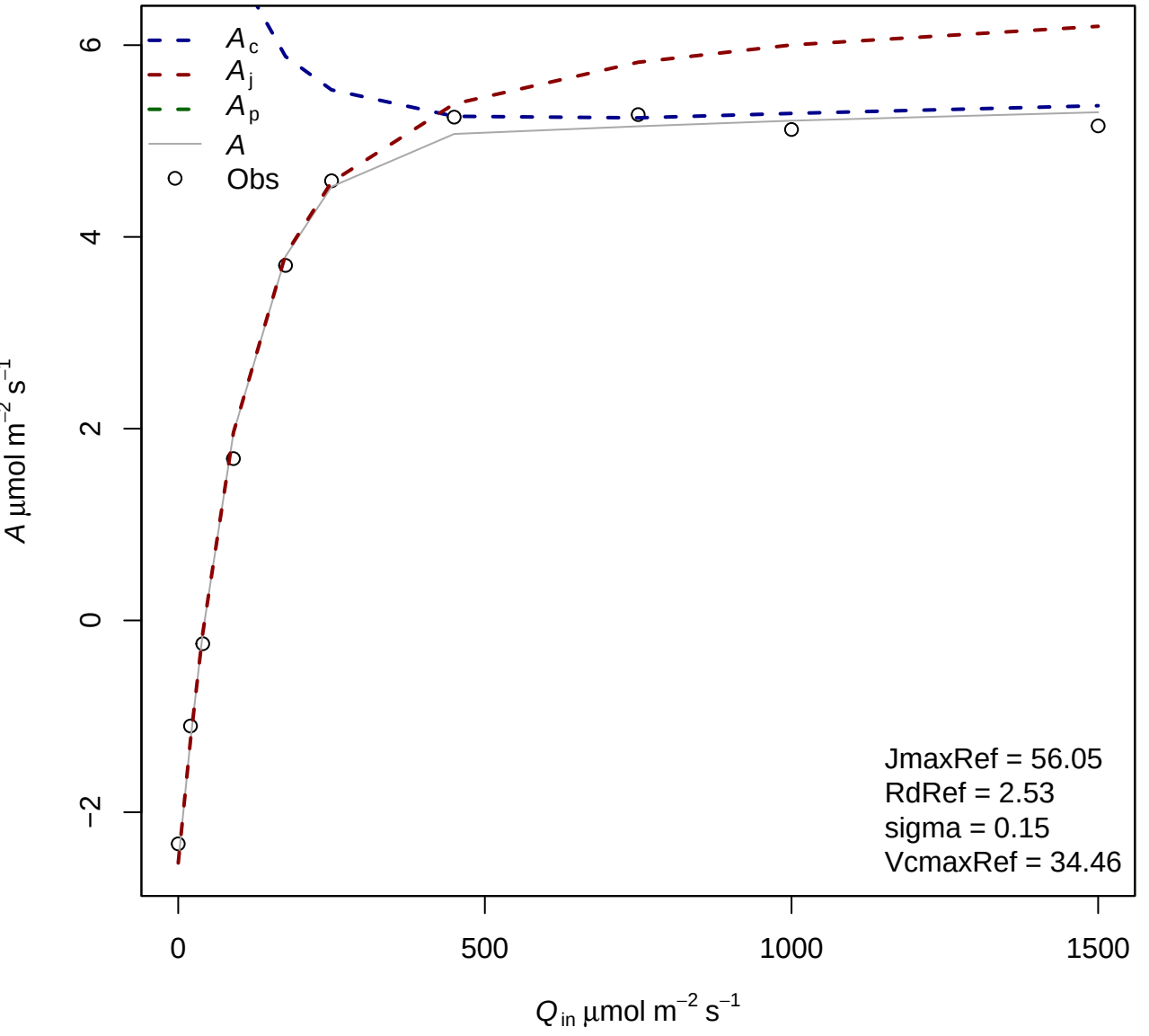

50
A_c
A_j
A_p
A
Obs
JmaxRef = 69.97
RdRef = 3.95
sigma = 0.71
VcmaxRef = 115.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

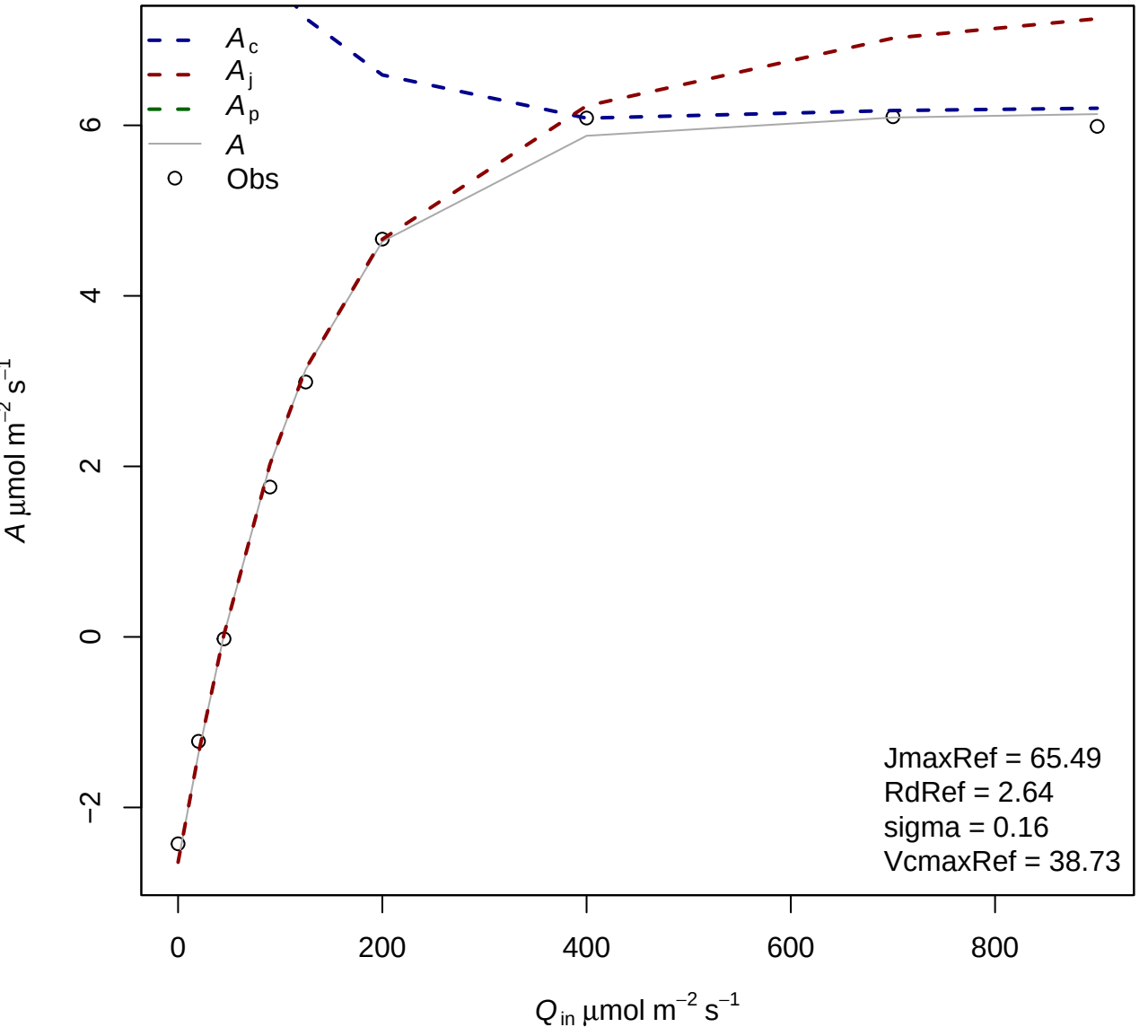

A_c
A_j
A_p
A
Obs
JmaxRef = 120.57
RdRef = 4.47
sigma = 0.63
VcmaxRef = 160.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
200
400
600
800
1000
1200
1400

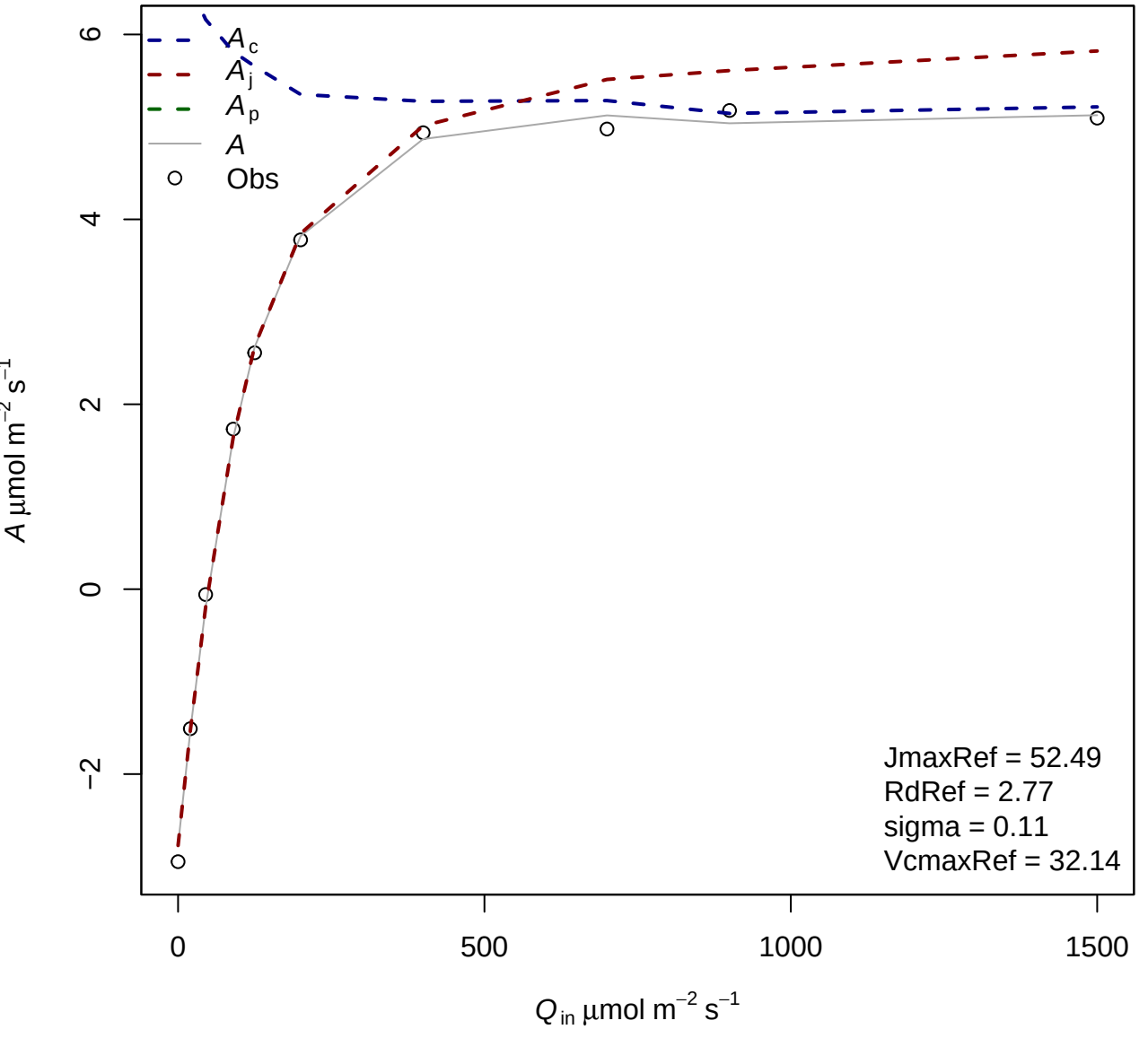

A_c
A_j
A_p
A
Obs
JmaxRef = 92.95
RdRef = 2.56
sigma = 0.48
VcmaxRef = 67.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

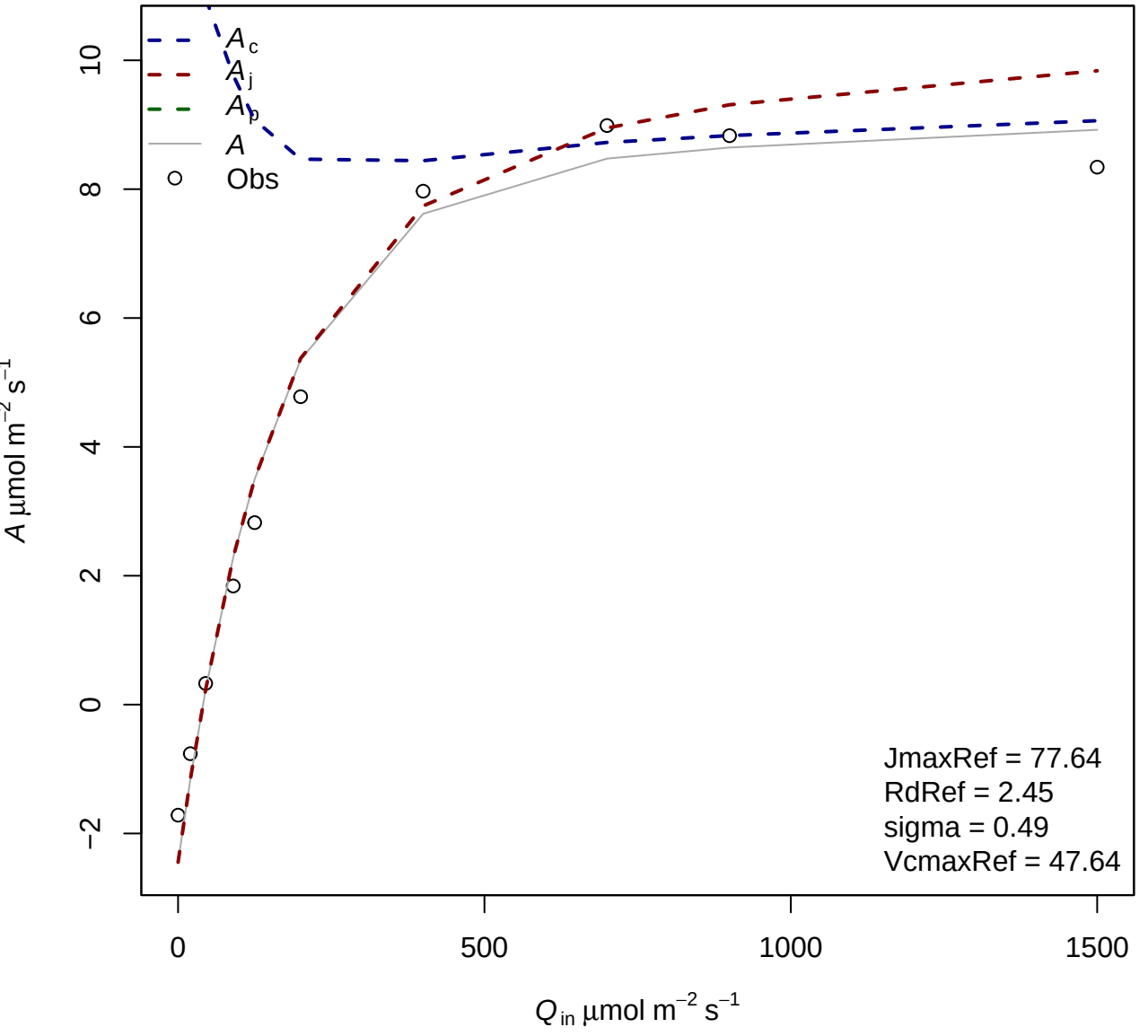

A_c
A_j
A_p
A
Obs
JmaxRef = 37.38
RdRef = 3.94
sigma = 0.23
VcmaxRef = 141.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
−3
−4
0
500
1000
1500

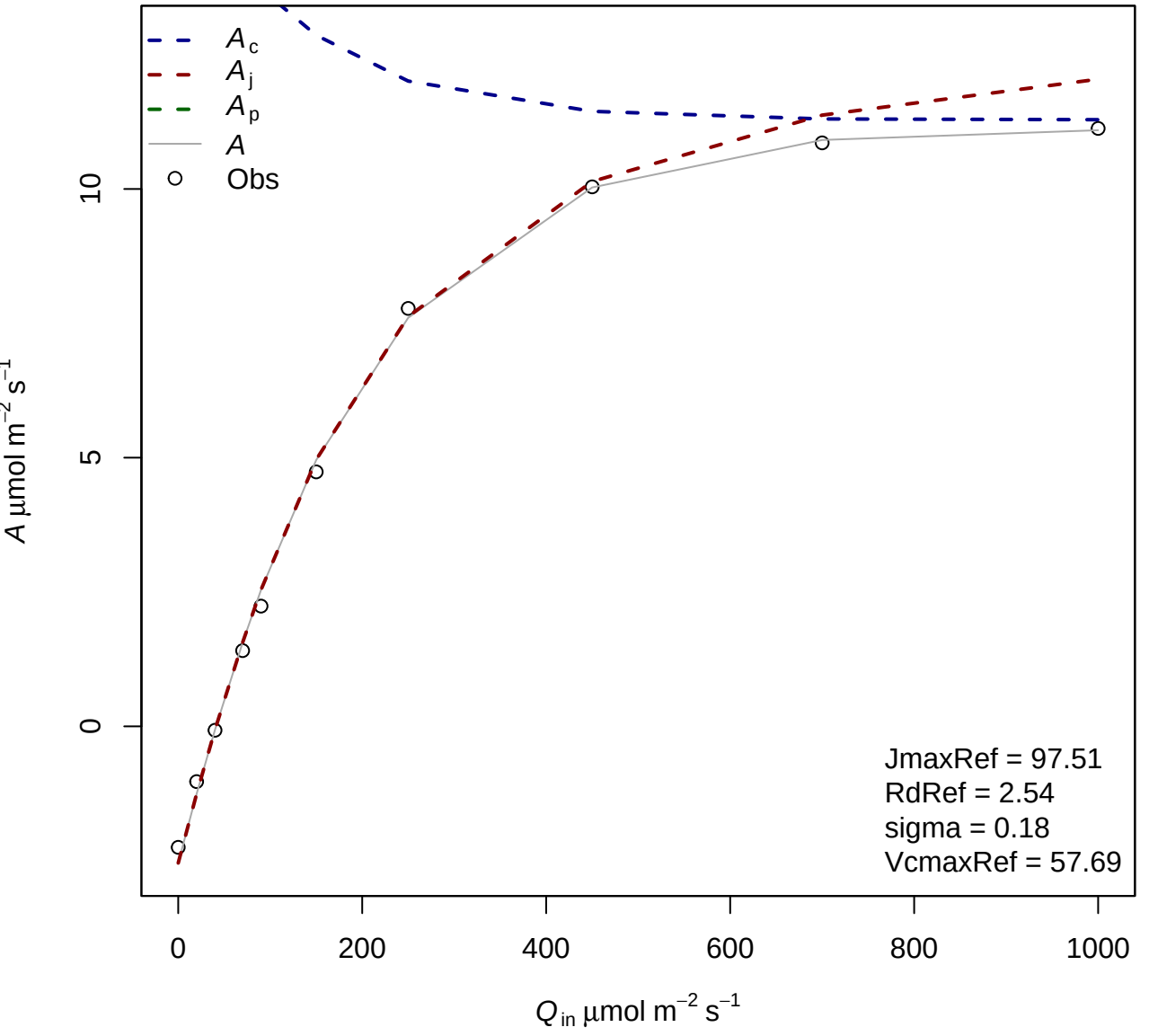

A_c
A_j
A_p
A
Obs
JmaxRef = 63.72
RdRef = 2.08
sigma = 0.33
VcmaxRef = 91.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

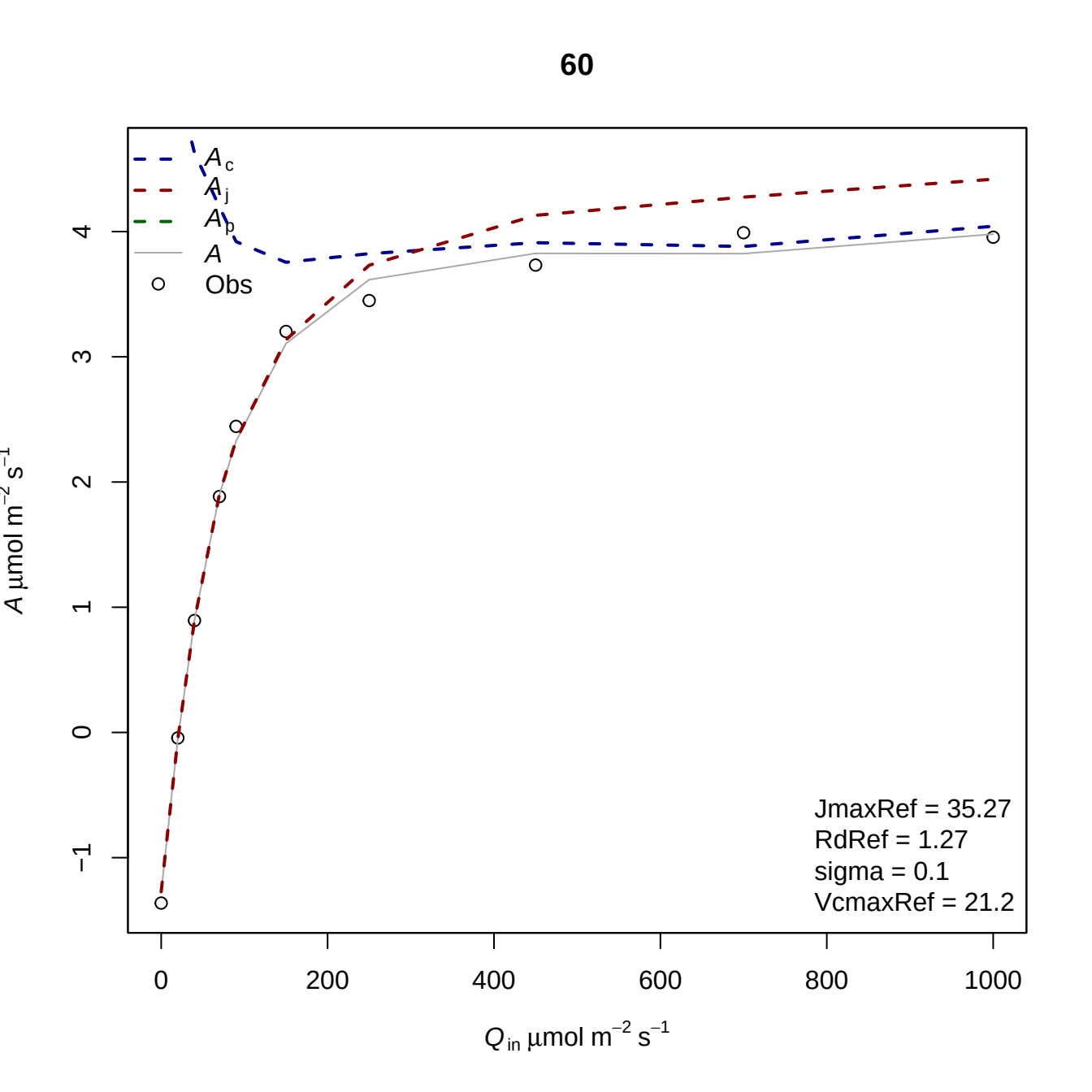

A_c
A_j
A_p
A
Obs
JmaxRef = 56.05
RdRef = 2.53
sigma = 0.15
VcmaxRef = 34.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

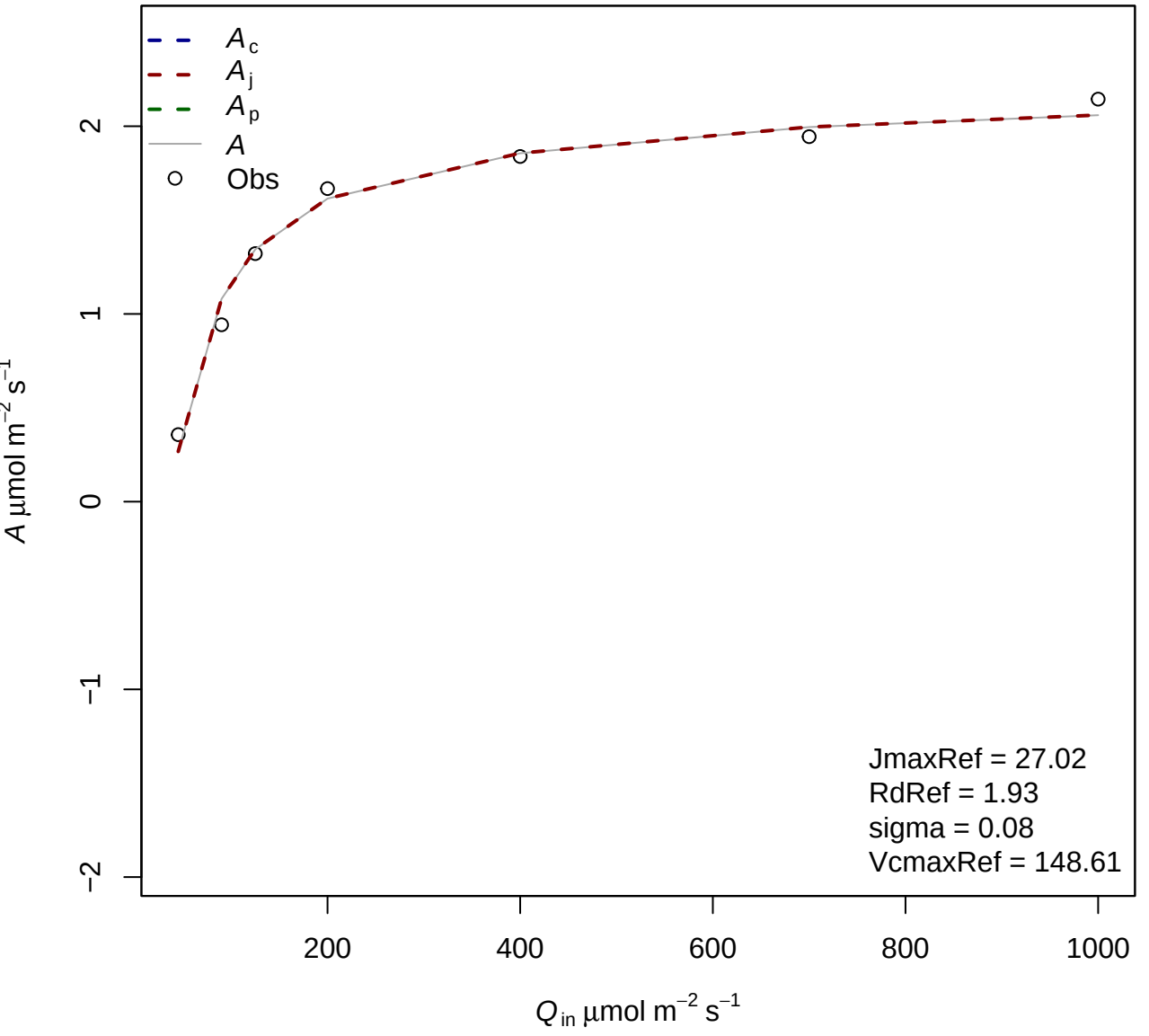

A_c
A_j
A_p
A
Obs
JmaxRef = 65.49
RdRef = 2.64
sigma = 0.16
VcmaxRef = 38.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

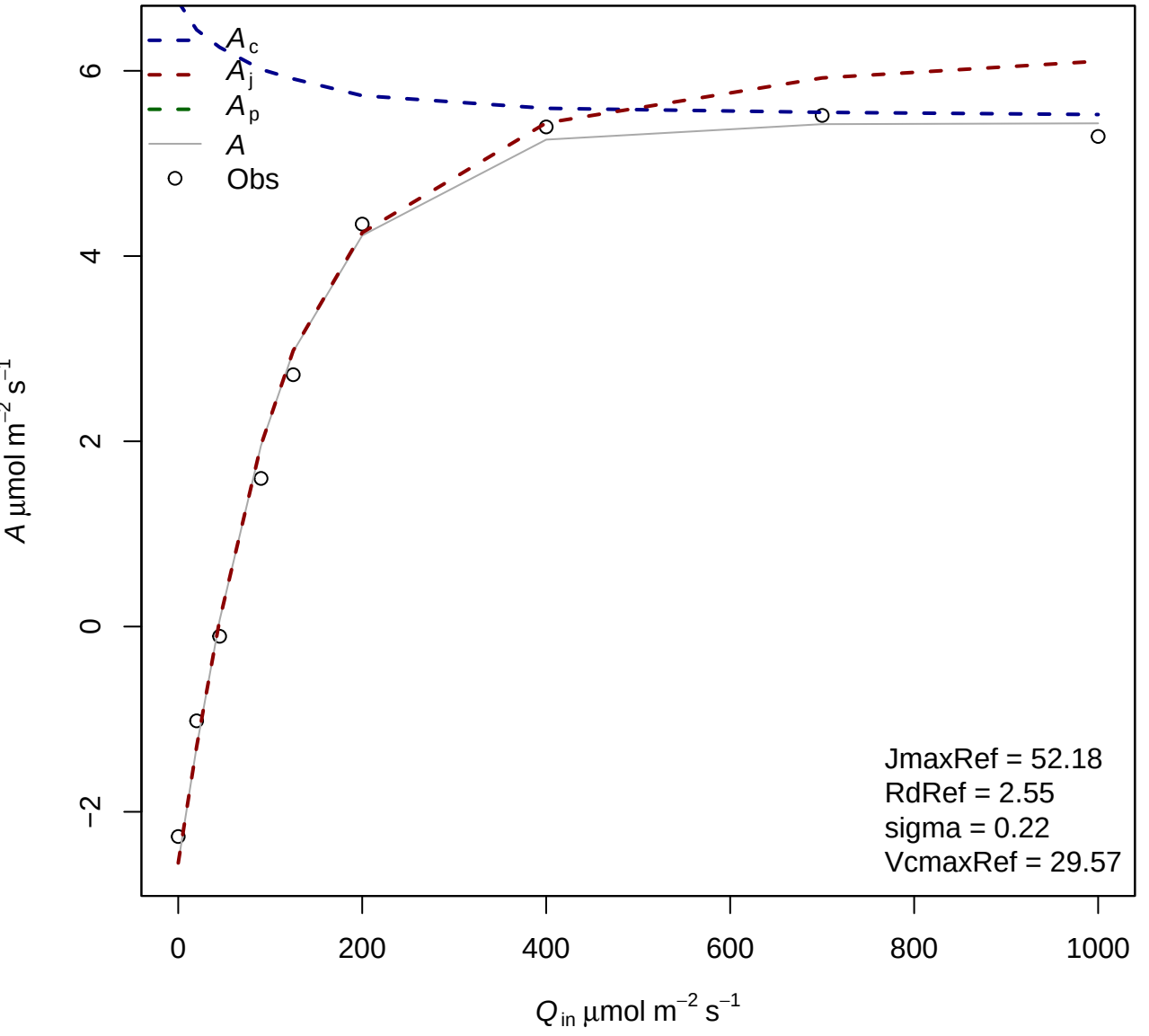

A_c
A_j
A_p
A
Obs
JmaxRef = 52.49
RdRef = 2.77
sigma = 0.11
VcmaxRef = 32.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

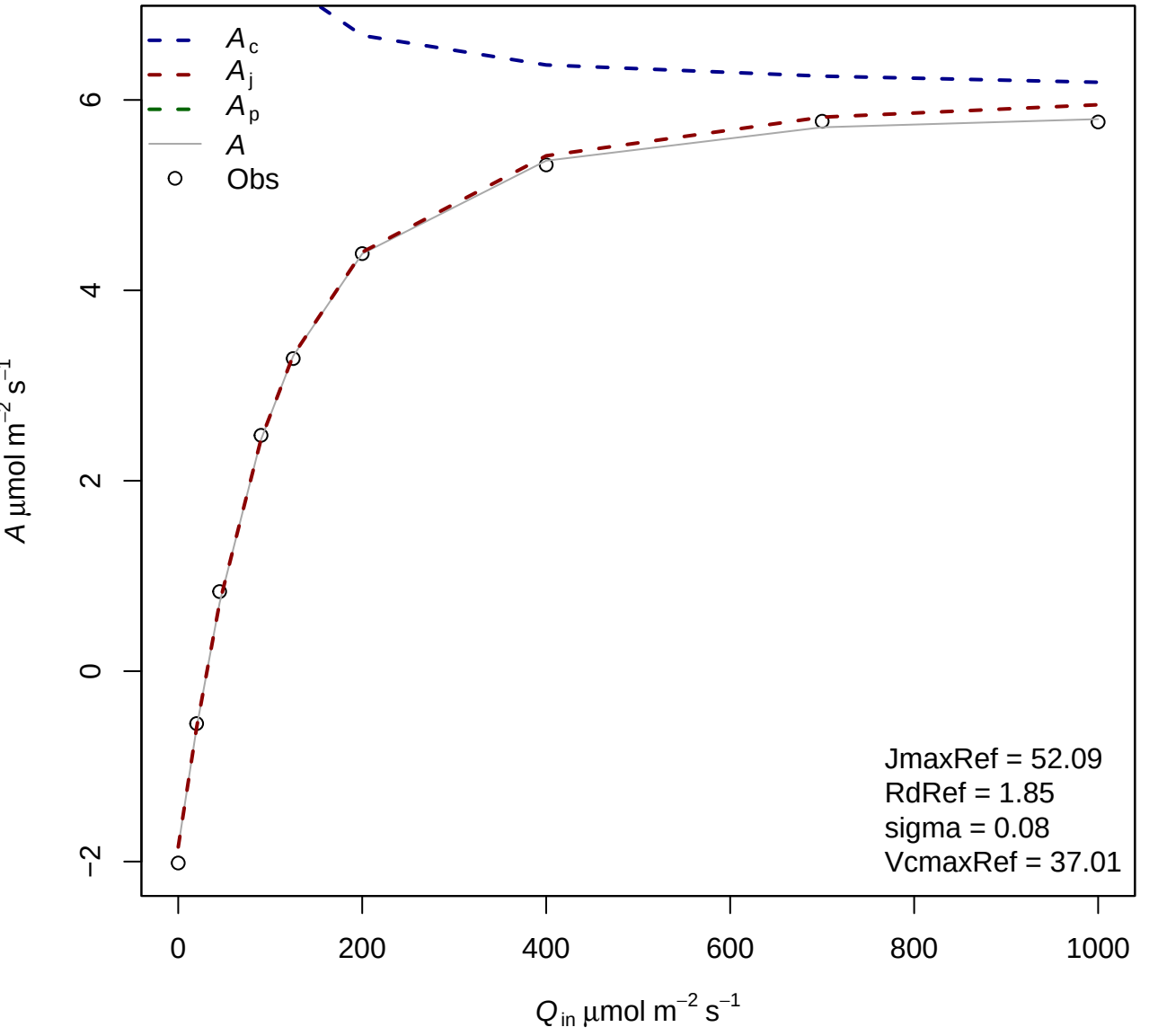

A_c
A_j
A_p
A
Obs
JmaxRef = 77.64
RdRef = 2.45
sigma = 0.49
VcmaxRef = 47.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

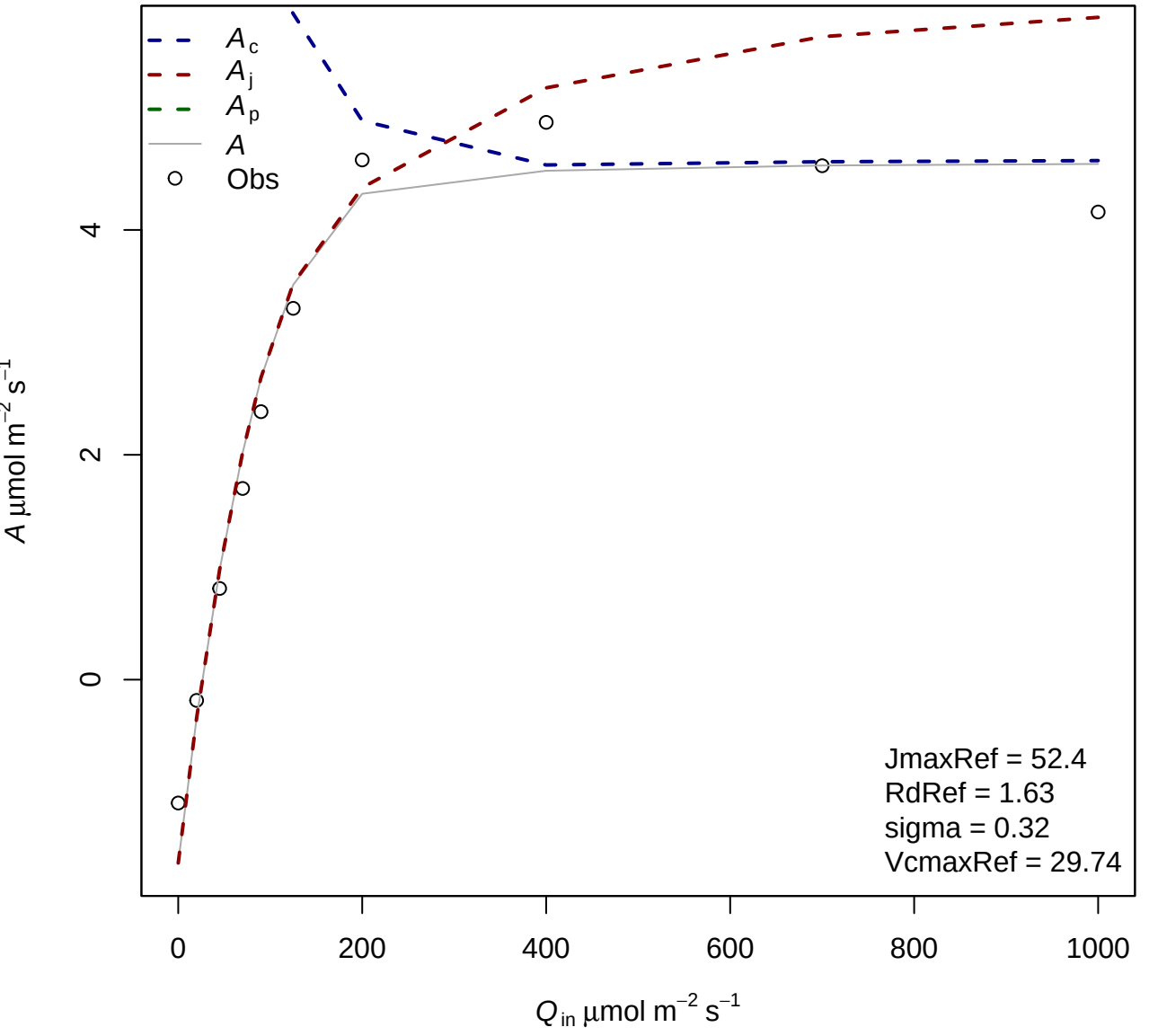

A_c
A_j
A_p
A
Obs
JmaxRef = 97.51
RdRef = 2.54
sigma = 0.18
VcmaxRef = 57.69
A µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
Q_{in} µmol m^{-2} s^{-1}

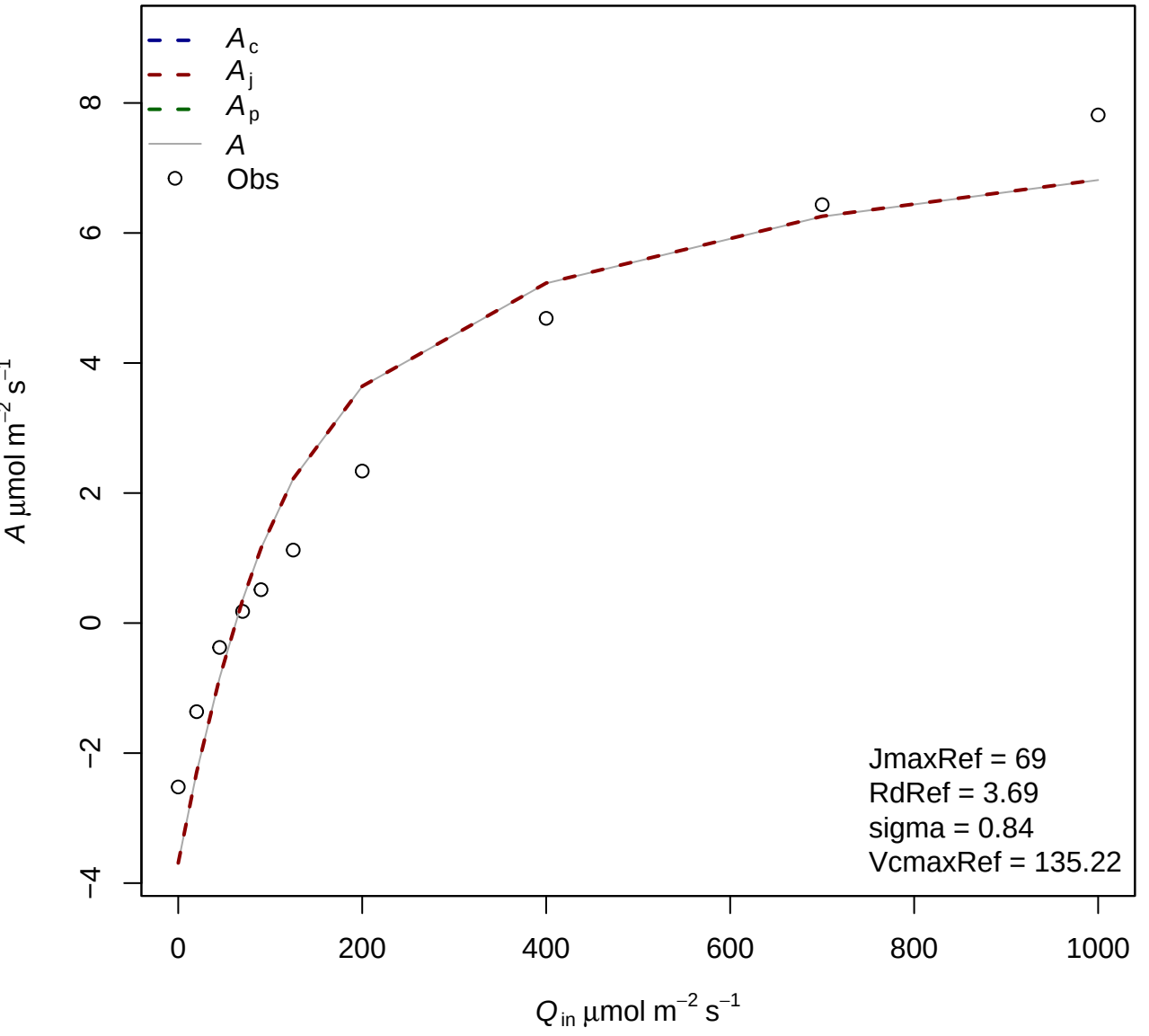

60
A_c
A_j
A_p
A
Obs
JmaxRef = 35.27
RdRef = 1.27
sigma = 0.1
VcmaxRef = 21.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

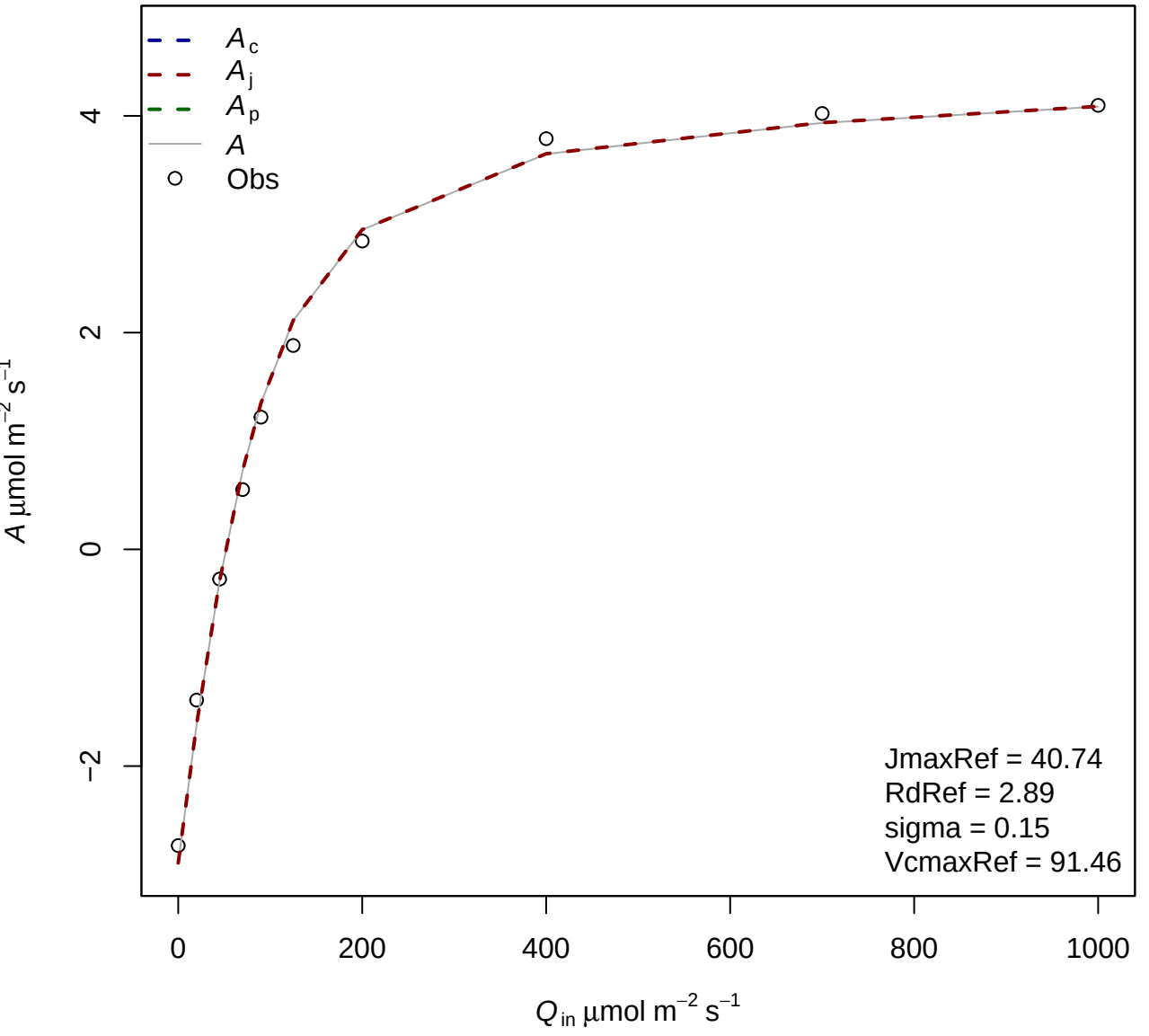

A_c
A_j
A_p
A
Obs
JmaxRef = 27.02
RdRef = 1.93
sigma = 0.08
VcmaxRef = 148.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
200
400
600
800
1000

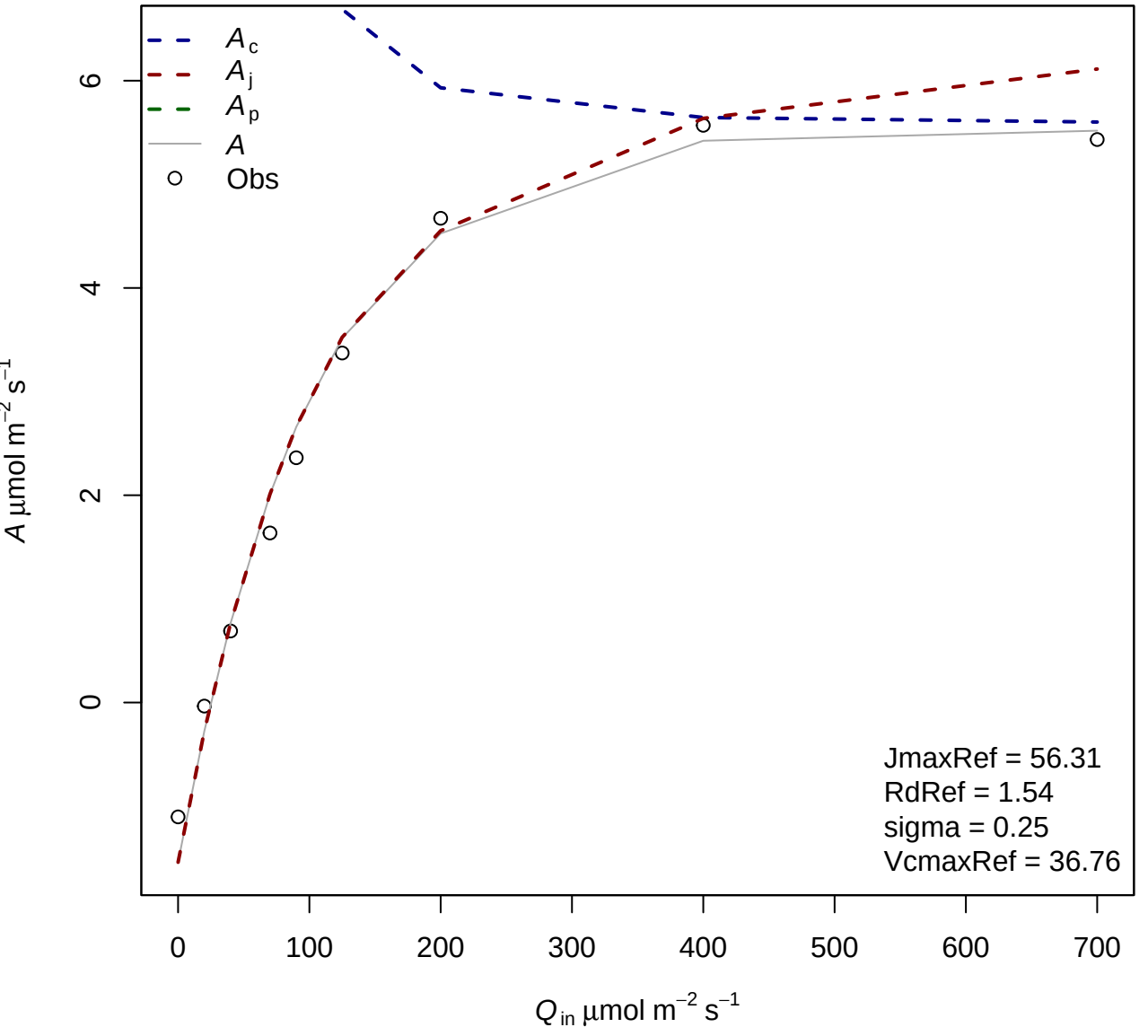

A_c
A_j
A_p
A
Obs
JmaxRef = 52.18
RdRef = 2.55
sigma = 0.22
VcmaxRef = 29.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

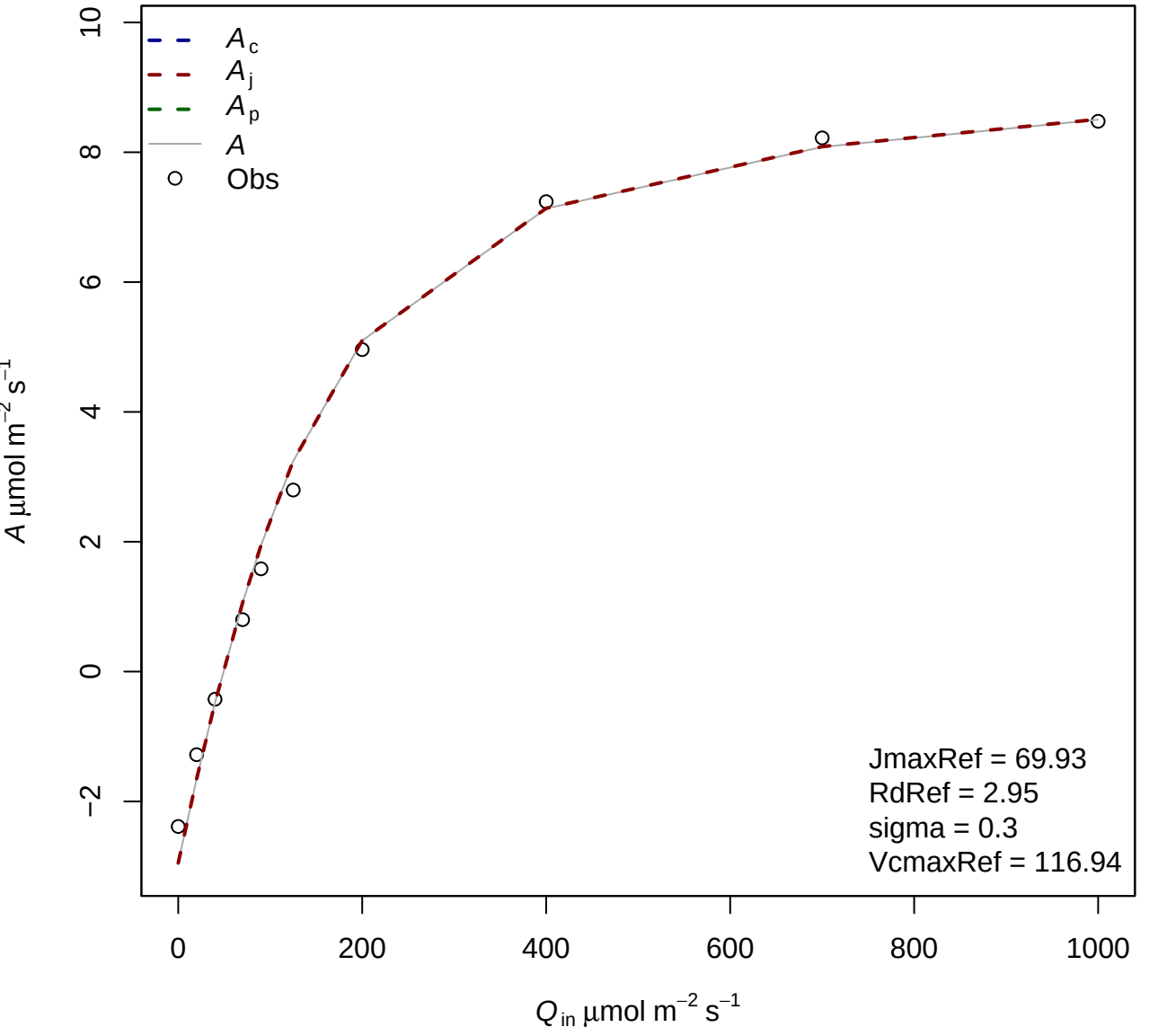

A_c
A_j
A_p
A
Obs
JmaxRef = 52.09
RdRef = 1.85
sigma = 0.08
VcmaxRef = 37.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

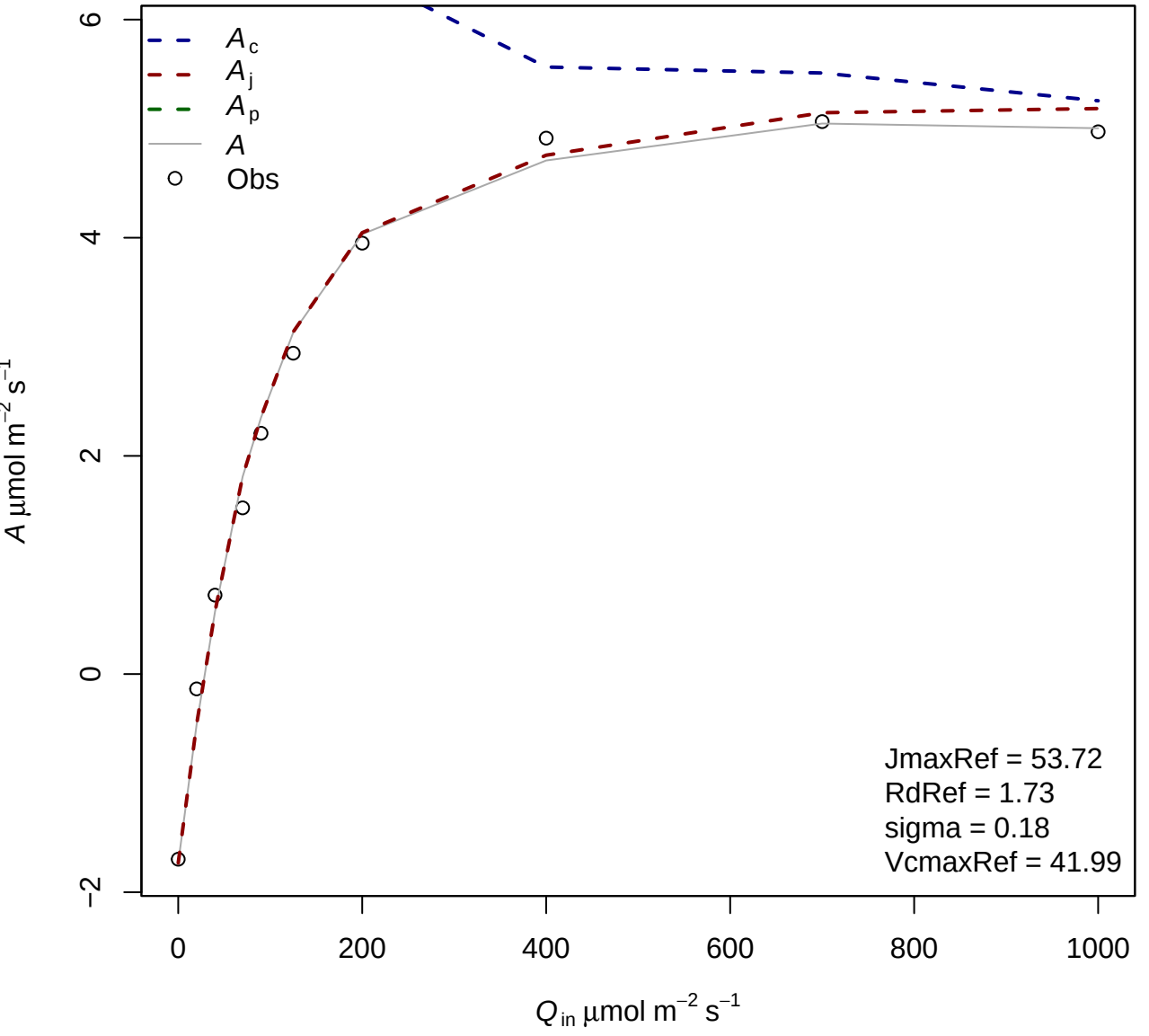

A_c
A_j
A_p
A
Obs
JmaxRef = 52.4
RdRef = 1.63
sigma = 0.32
VcmaxRef = 29.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
0
200
400
600
800
1000

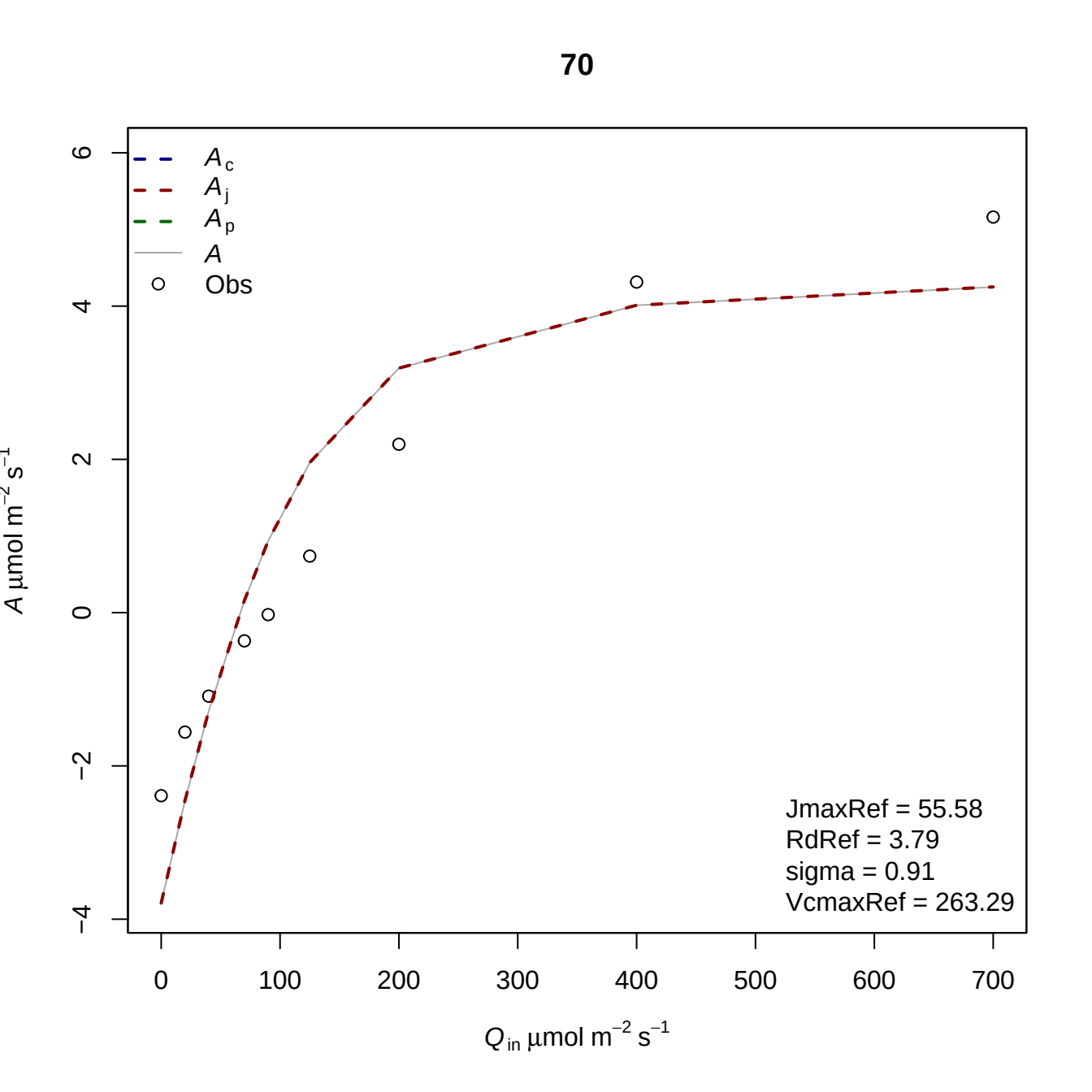

A_c
A_j
A_p
A
Obs
JmaxRef = 69
RdRef = 3.69
sigma = 0.84
VcmaxRef = 135.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000

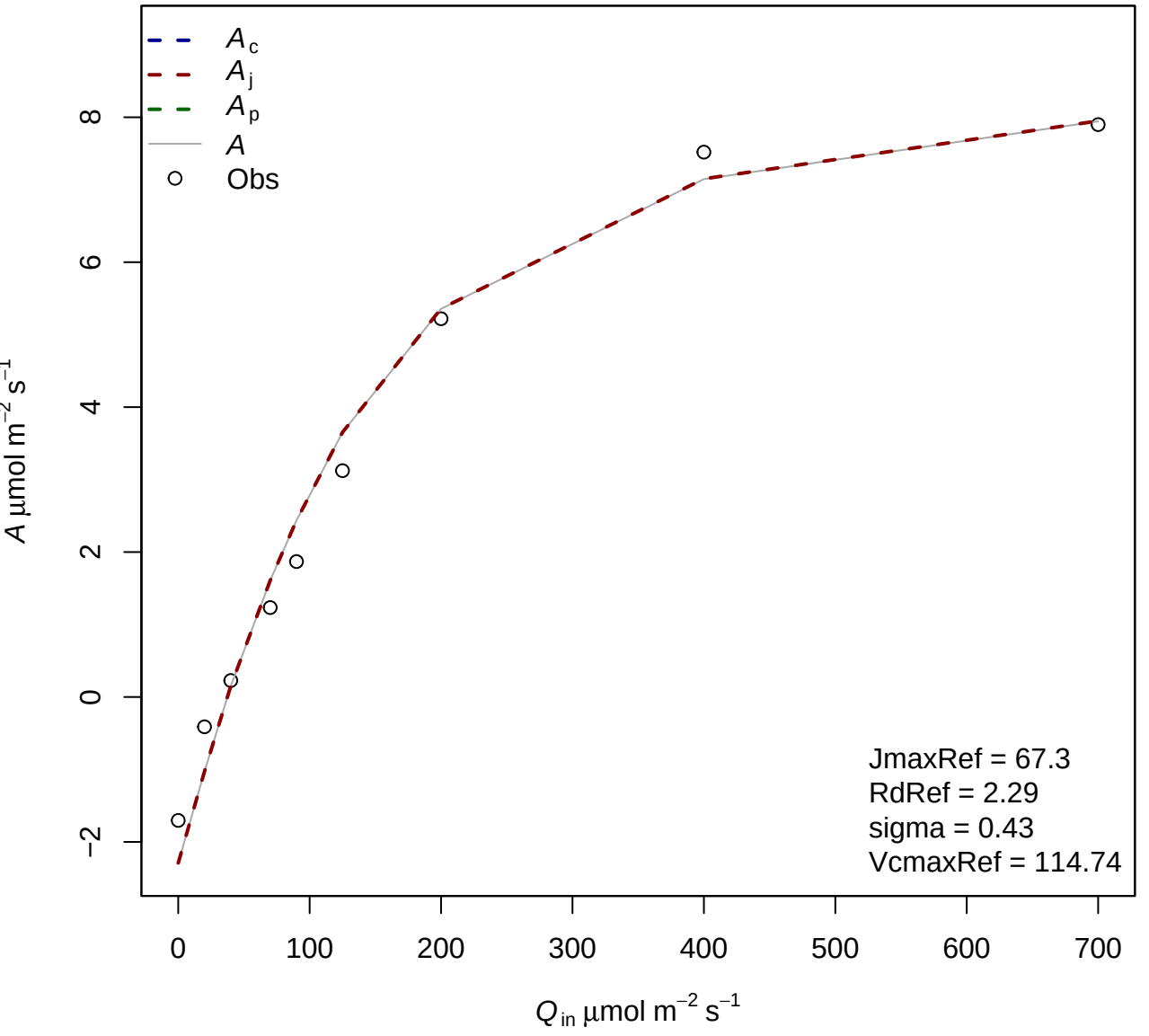

A_c
A_j
A_p
A
Obs
JmaxRef = 40.74
RdRef = 2.89
sigma = 0.15
VcmaxRef = 91.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

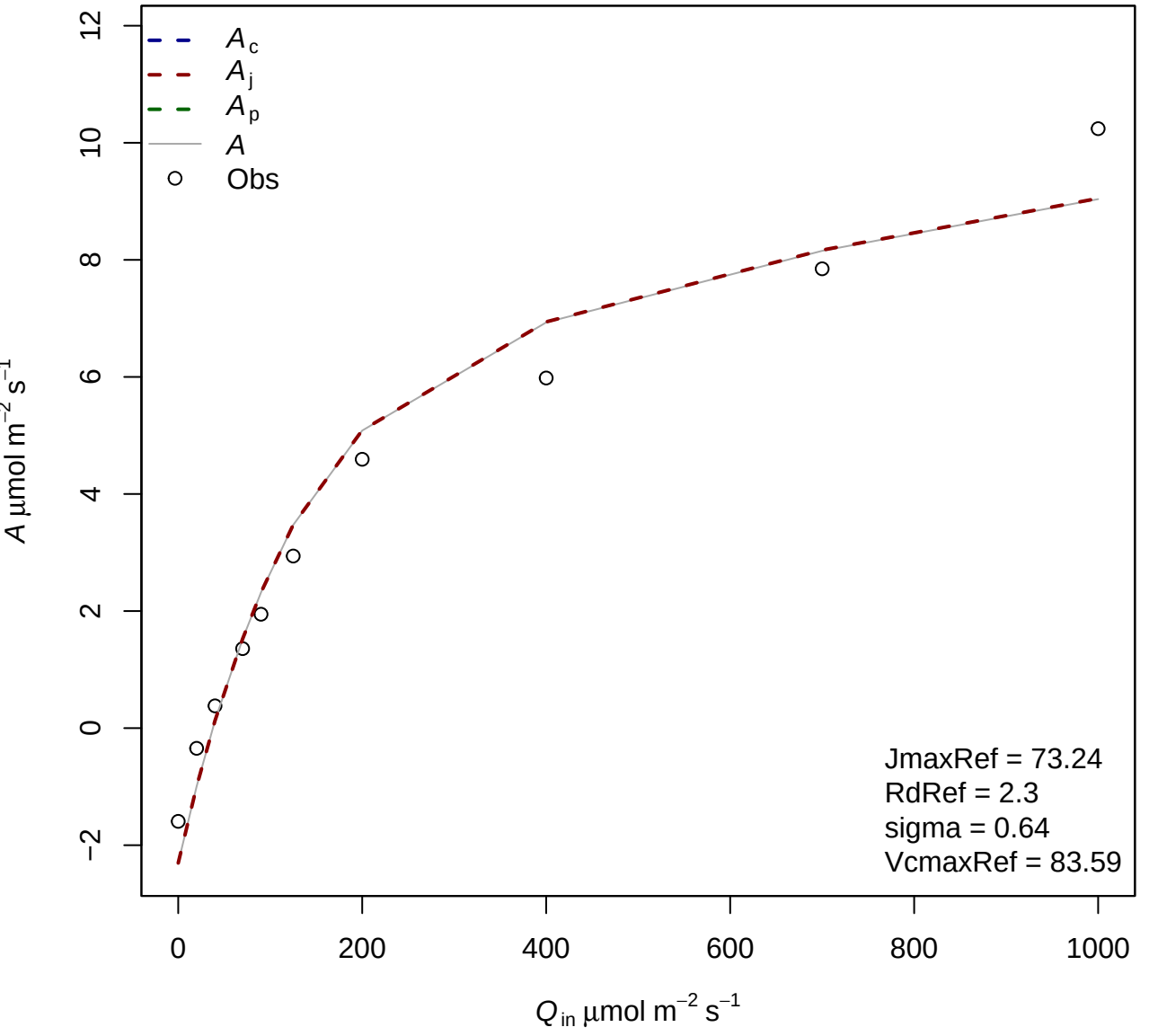

A_c
A_j
A_p
A
Obs
JmaxRef = 56.31
RdRef = 1.54
sigma = 0.25
VcmaxRef = 36.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
100
200
300
400
500
600
700

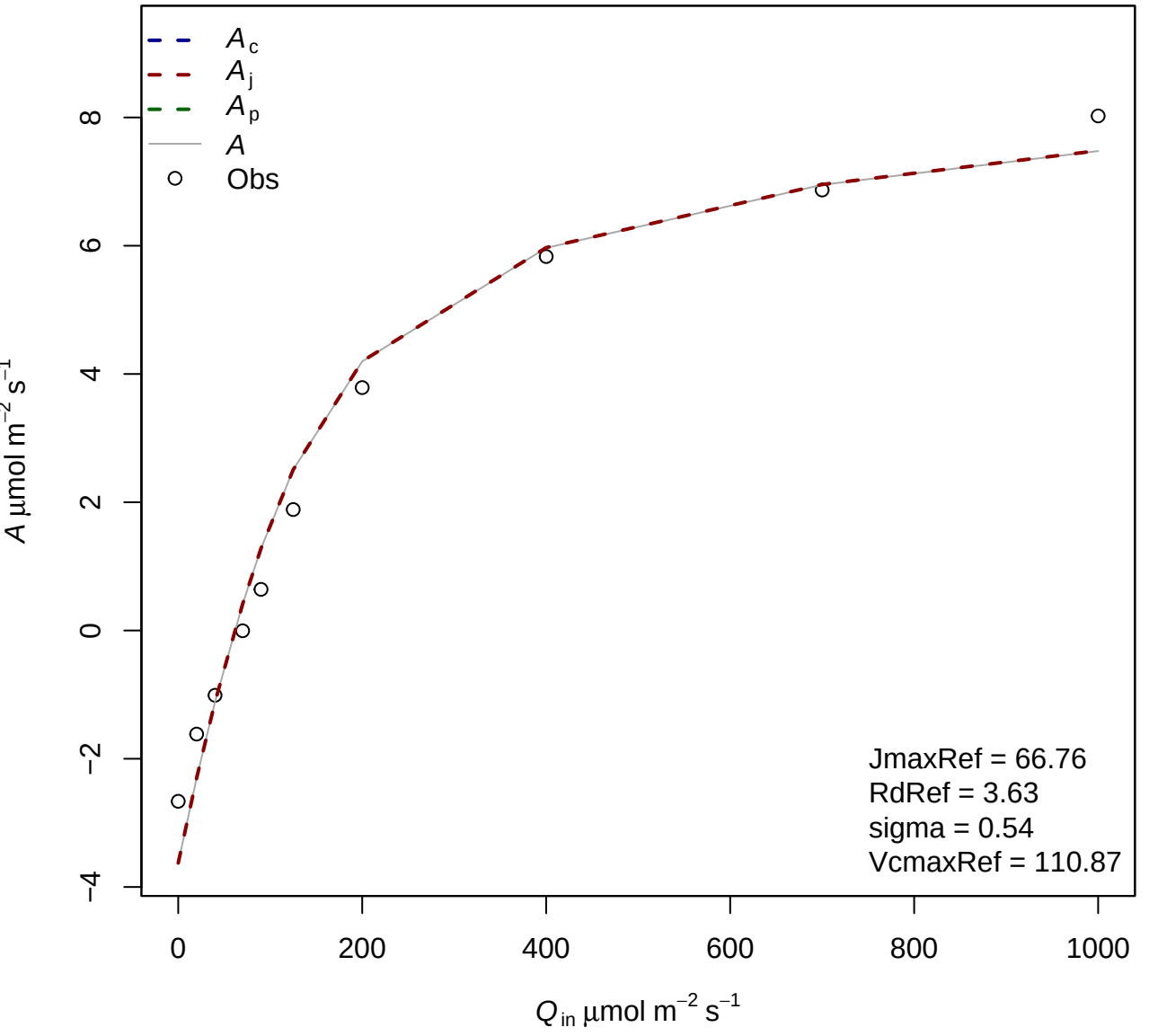

A_c
A_j
A_p
A
Obs
JmaxRef = 69.93
RdRef = 2.95
sigma = 0.3
VcmaxRef = 116.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
200
400
600
800
1000

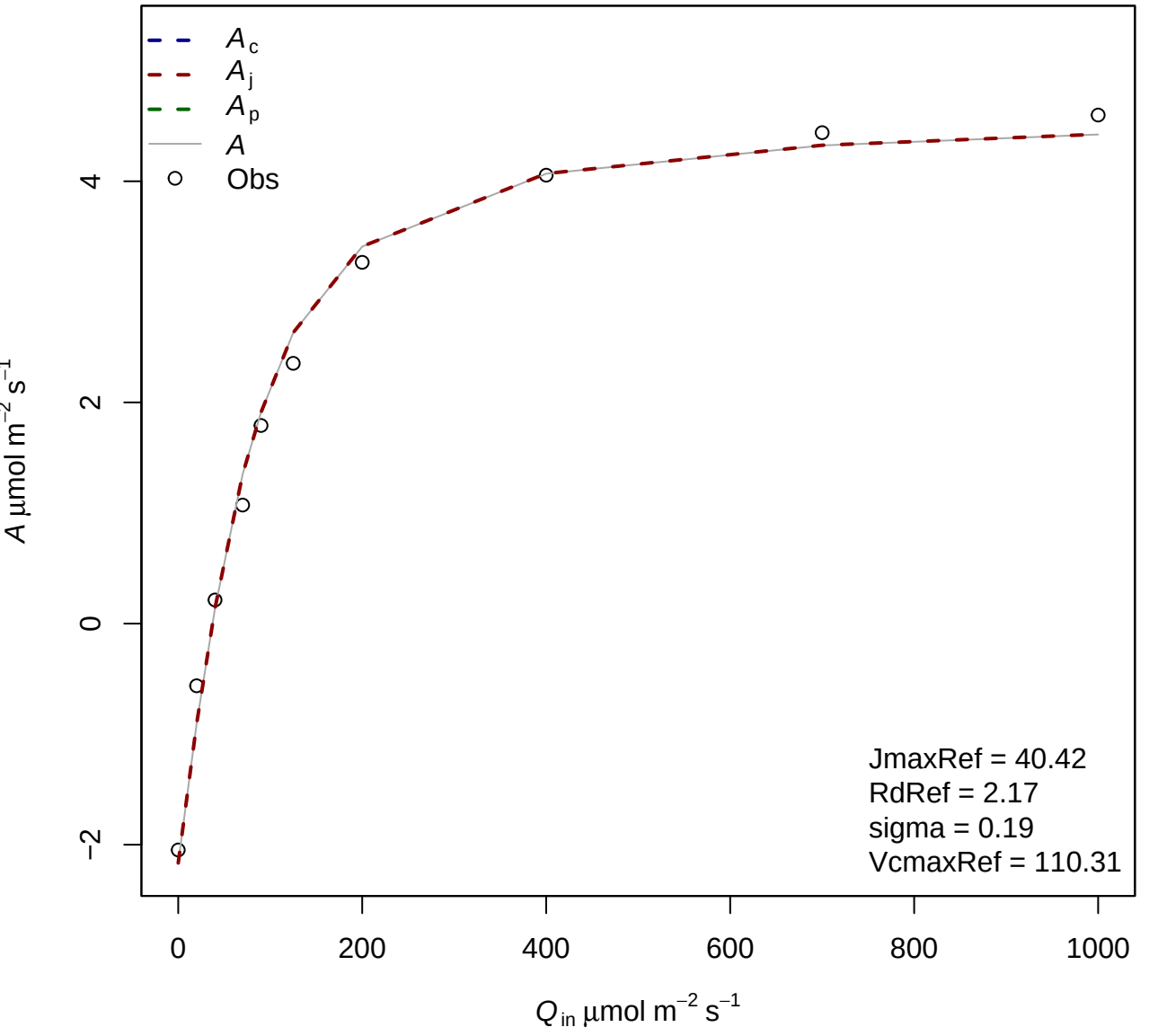

A_c
A_j
A_p
A
Obs
JmaxRef = 53.72
RdRef = 1.73
sigma = 0.18
VcmaxRef = 41.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

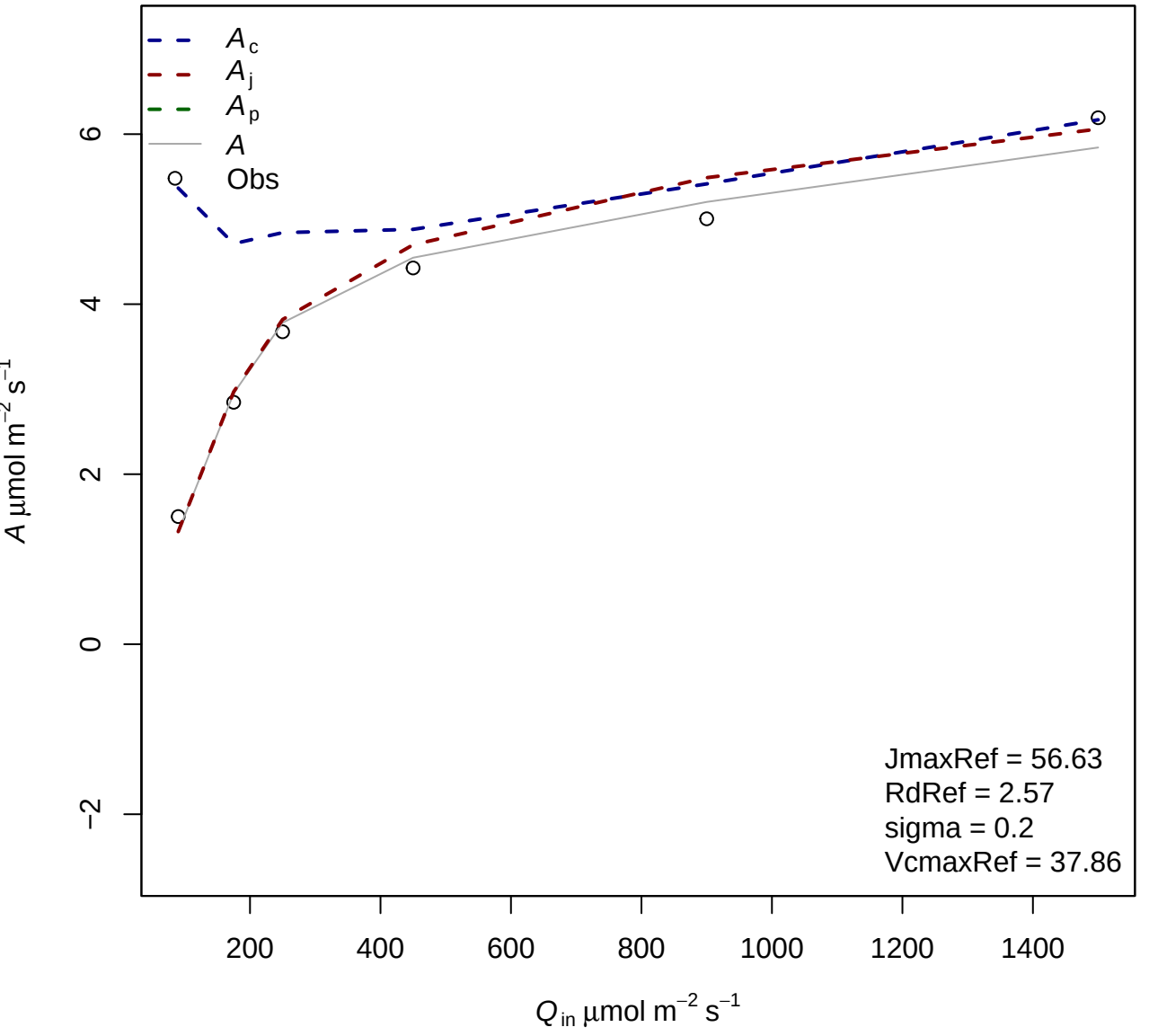

70
A_c
A_j
A_p
A
Obs
JmaxRef = 55.58
RdRef = 3.79
sigma = 0.91
VcmaxRef = 263.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
100
200
300
400
500
600
700

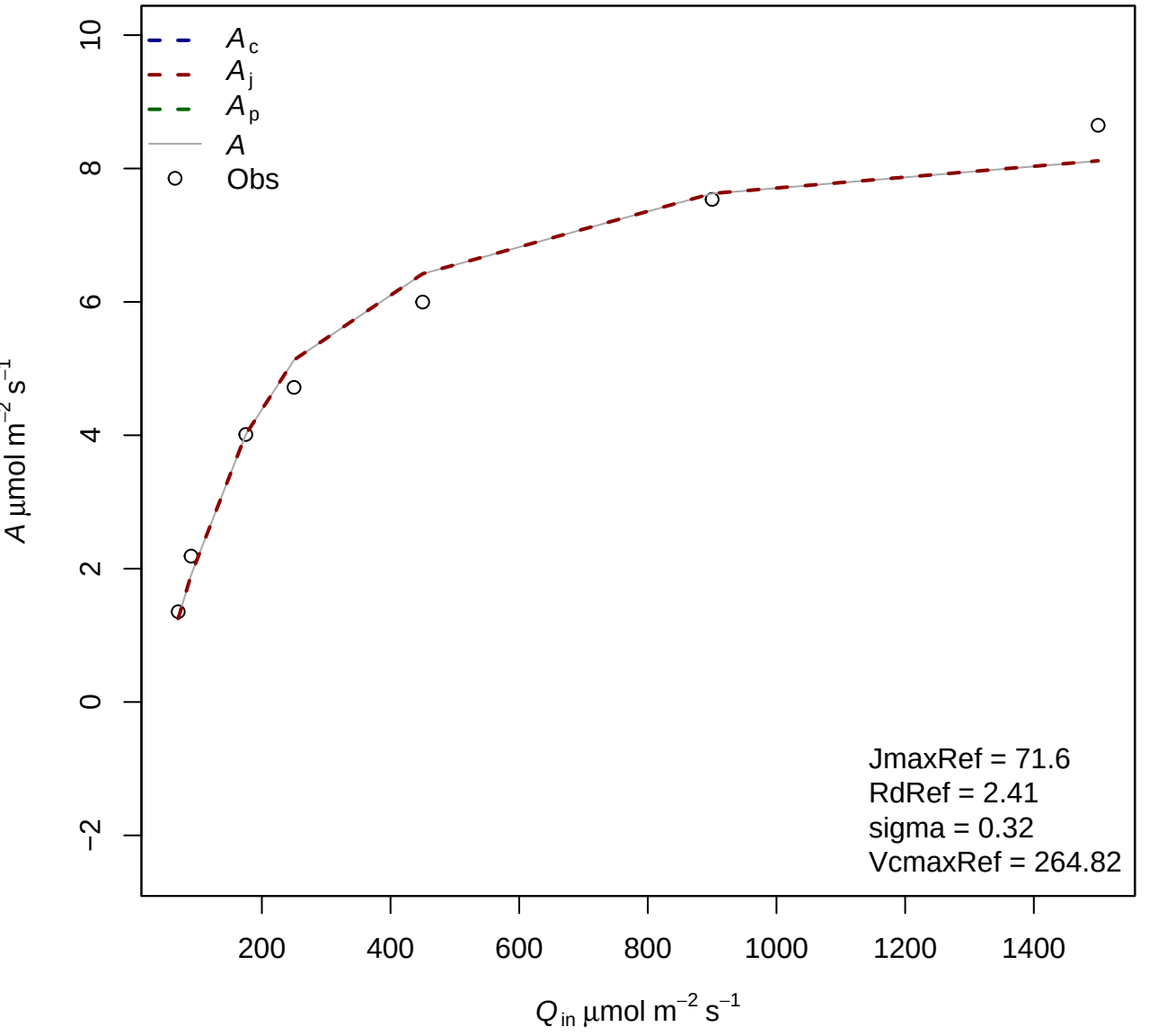

A_c
A_j
A_p
A
Obs
JmaxRef = 67.3
RdRef = 2.29
sigma = 0.43
VcmaxRef = 114.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
100
200
300
400
500
600
700

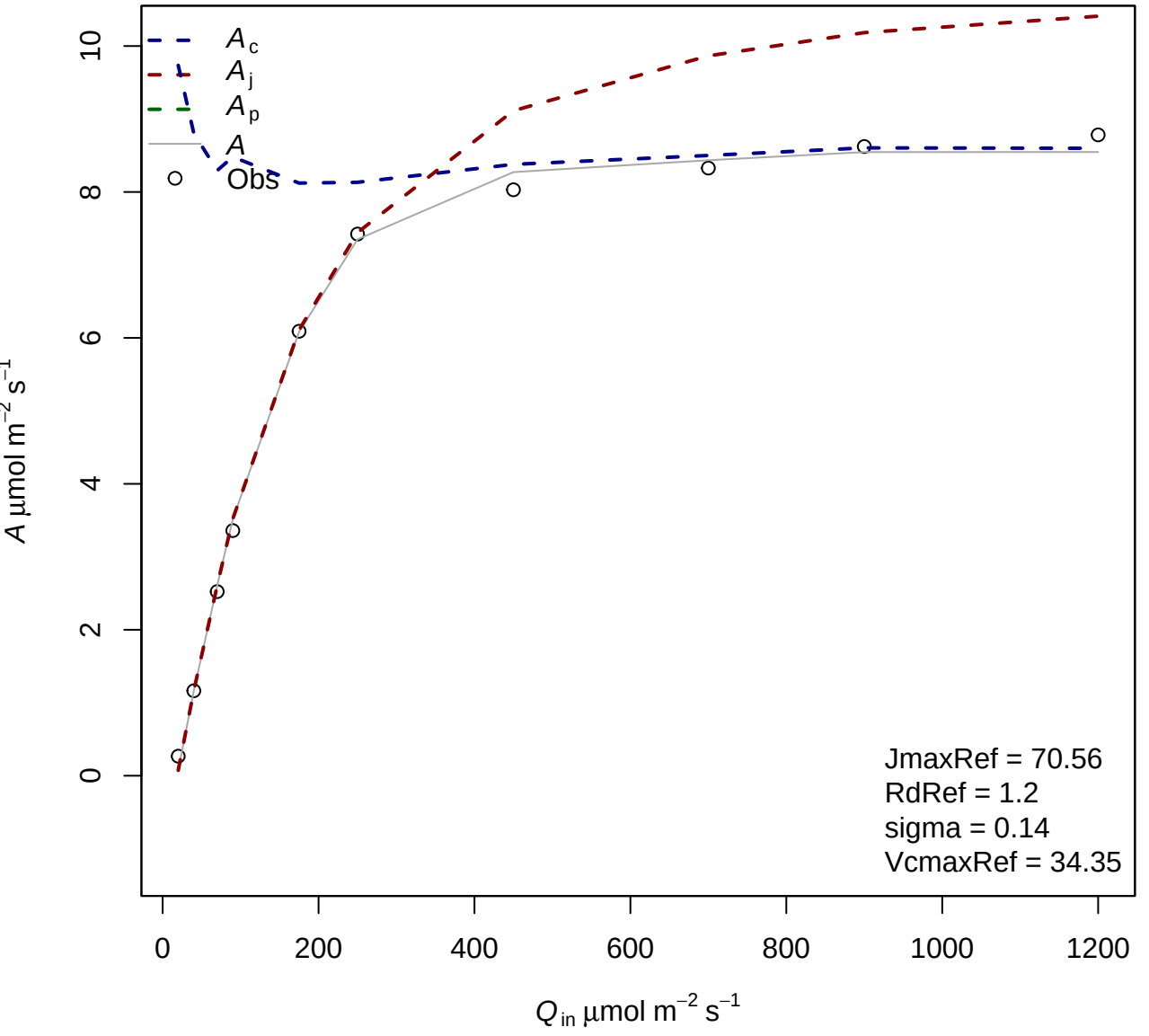

A_c
A_j
A_p
A
Obs
JmaxRef = 73.24
RdRef = 2.3
sigma = 0.64
VcmaxRef = 83.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
200
400
600
800
1000

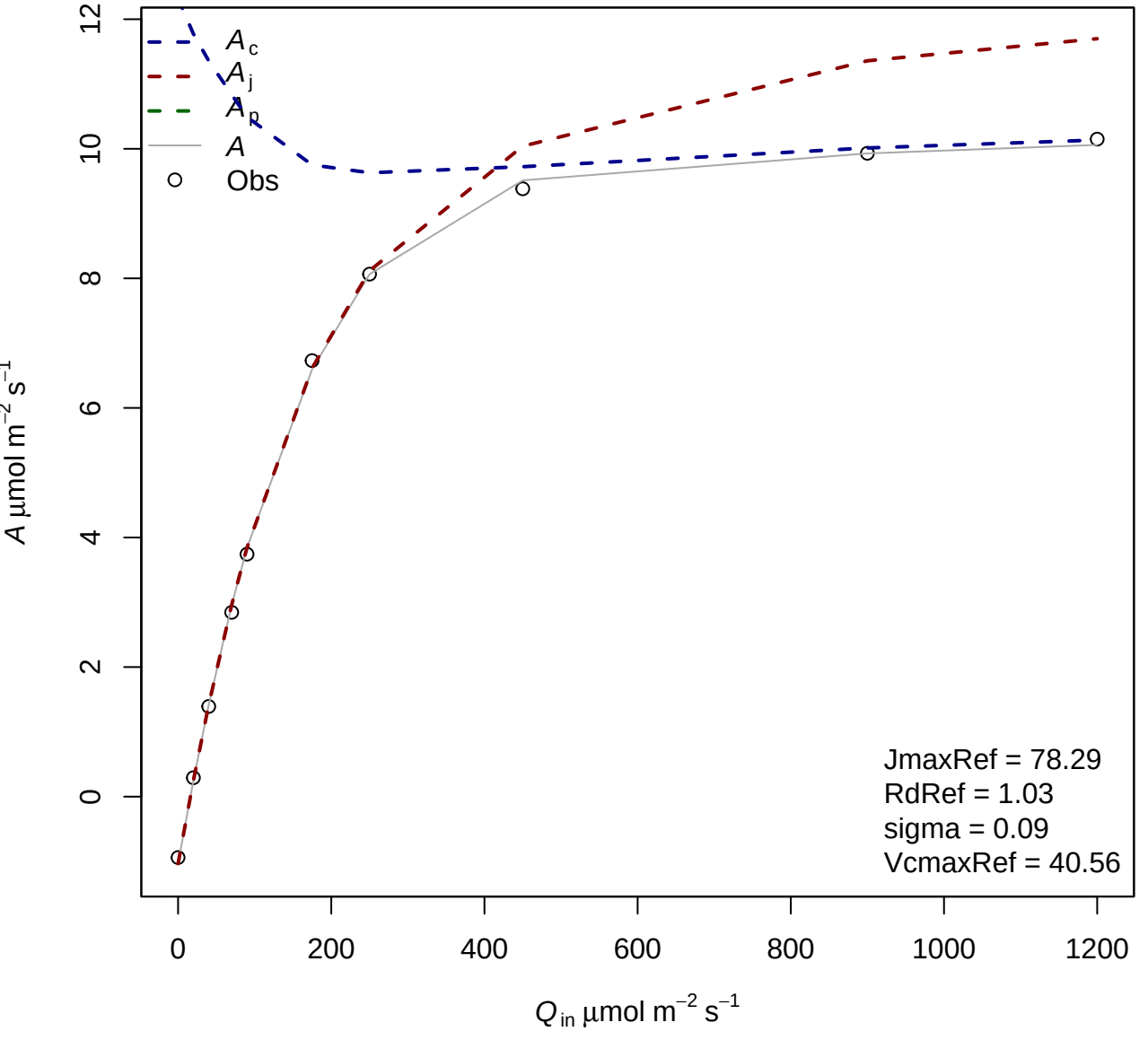

A_c
A_j
A_p
A
Obs
JmaxRef = 66.76
RdRef = 3.63
sigma = 0.54
VcmaxRef = 110.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000

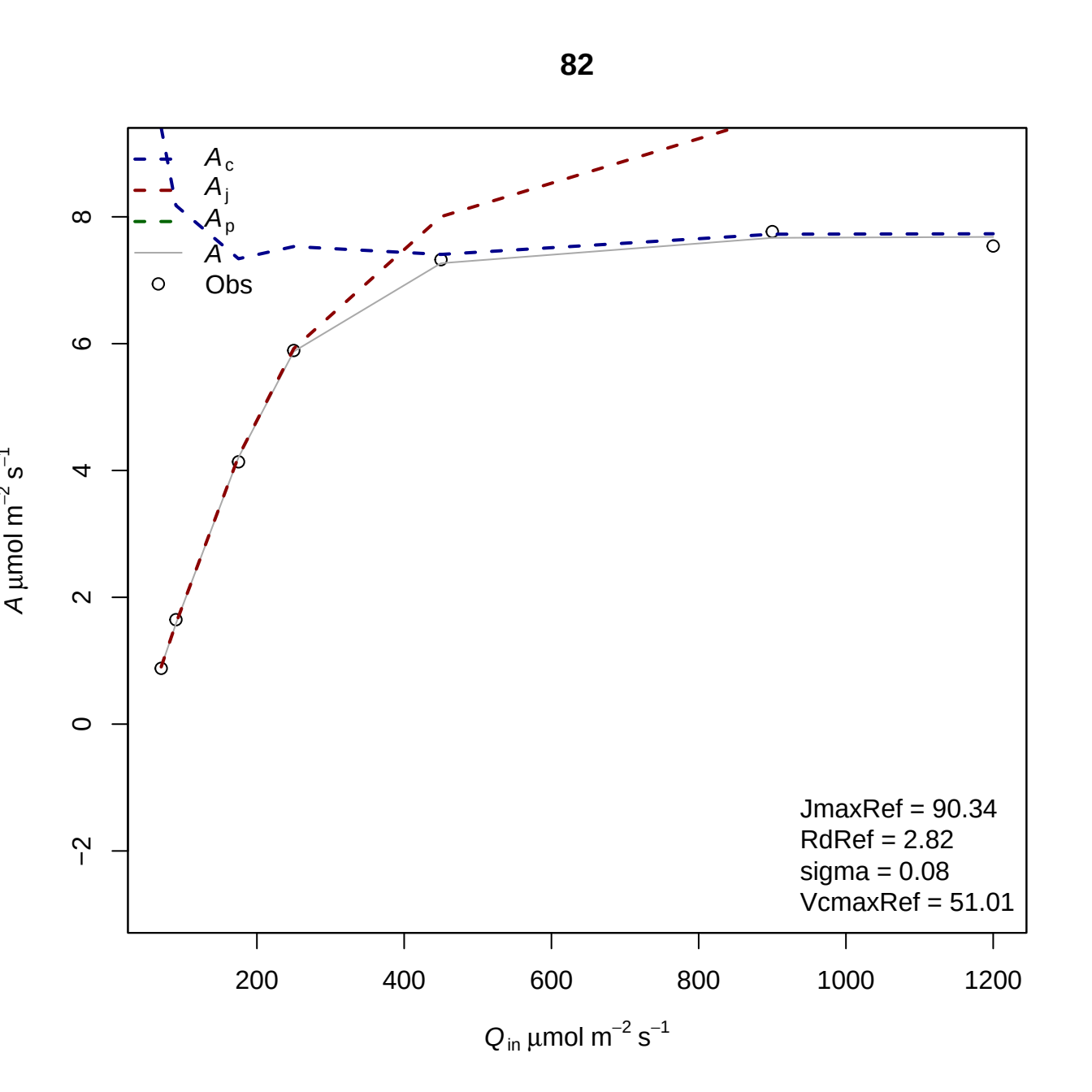

A_c
A_j
A_p
A
Obs
JmaxRef = 40.42
RdRef = 2.17
sigma = 0.19
VcmaxRef = 110.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

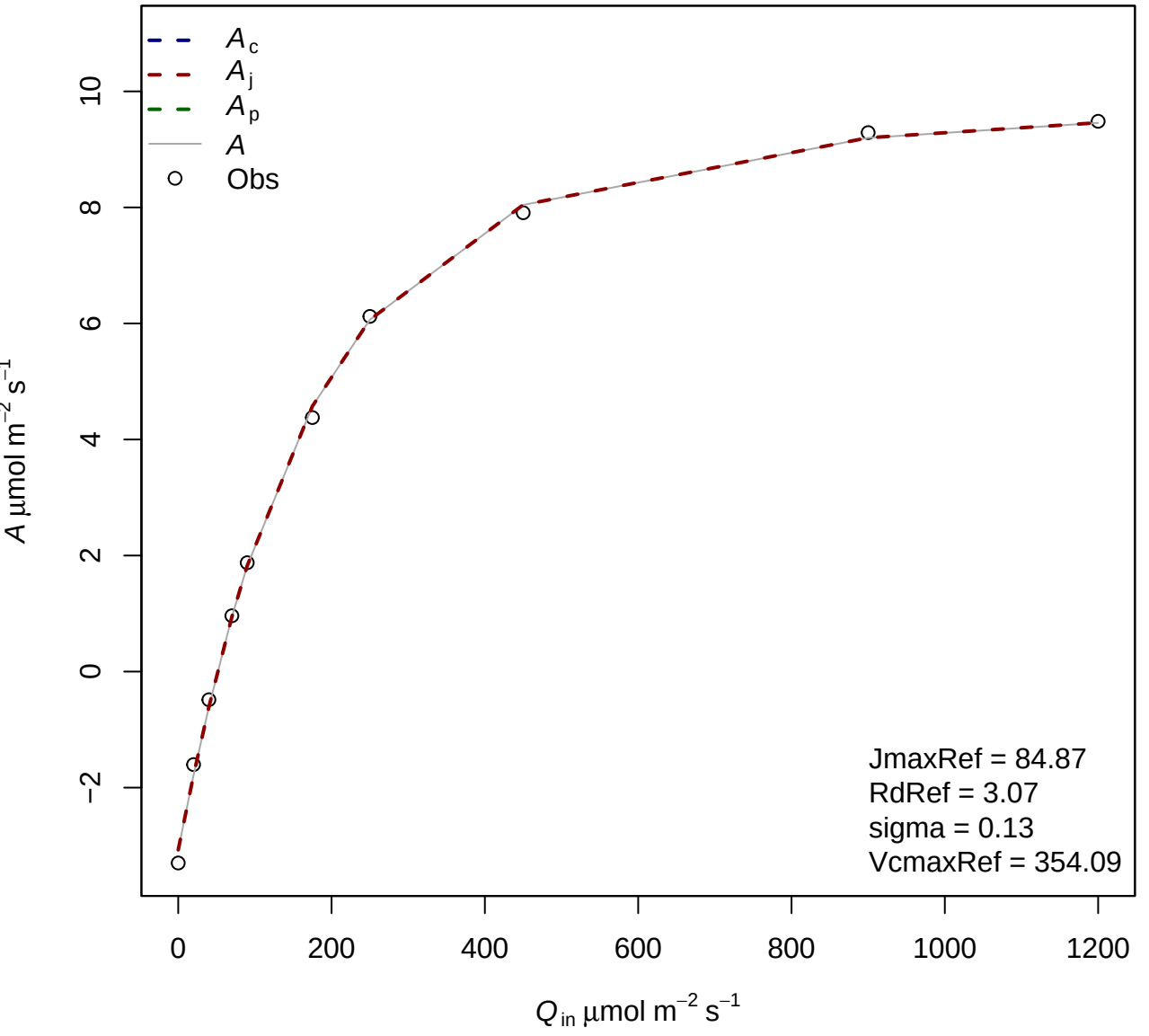

A_c
A_j
A_p
A
Obs
JmaxRef = 56.63
RdRef = 2.57
sigma = 0.2
VcmaxRef = 37.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
200
400
600
800
1000
1200
1400

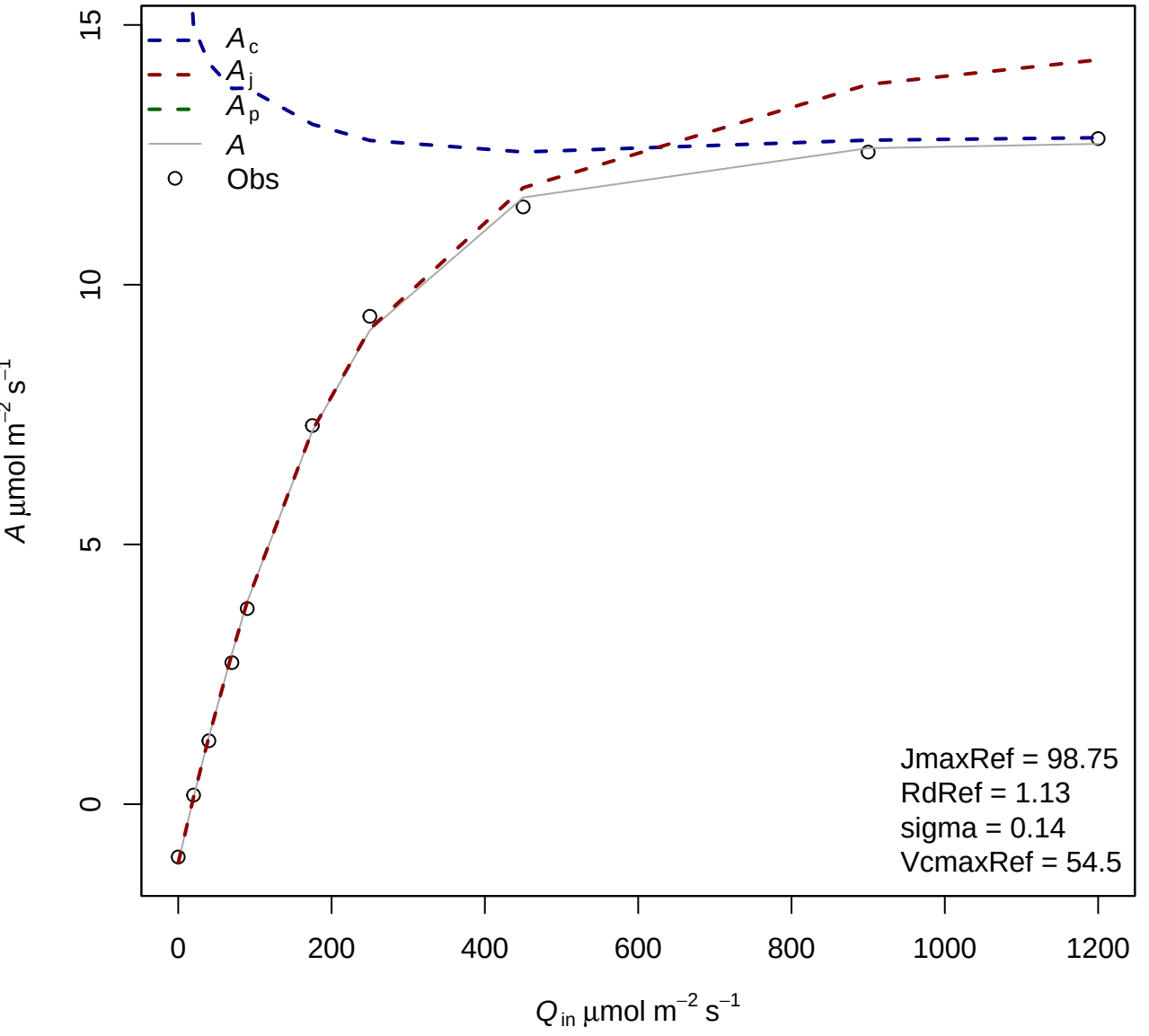

A_c
A_j
A_p
A
Obs
JmaxRef = 71.6
RdRef = 2.41
sigma = 0.32
VcmaxRef = 264.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400

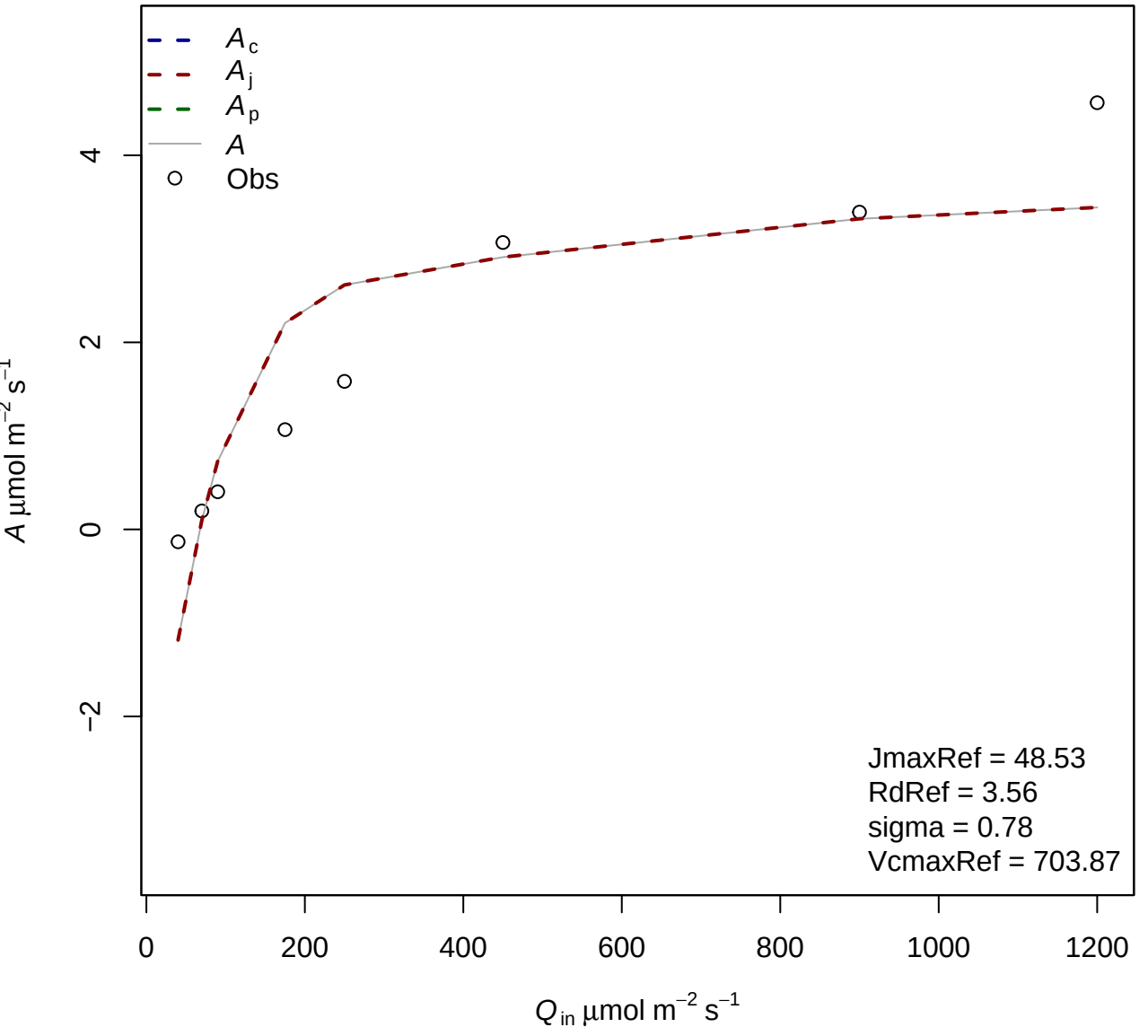

A_c
A_j
A_p
A
Obs
JmaxRef = 70.56
RdRef = 1.2
sigma = 0.14
VcmaxRef = 34.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

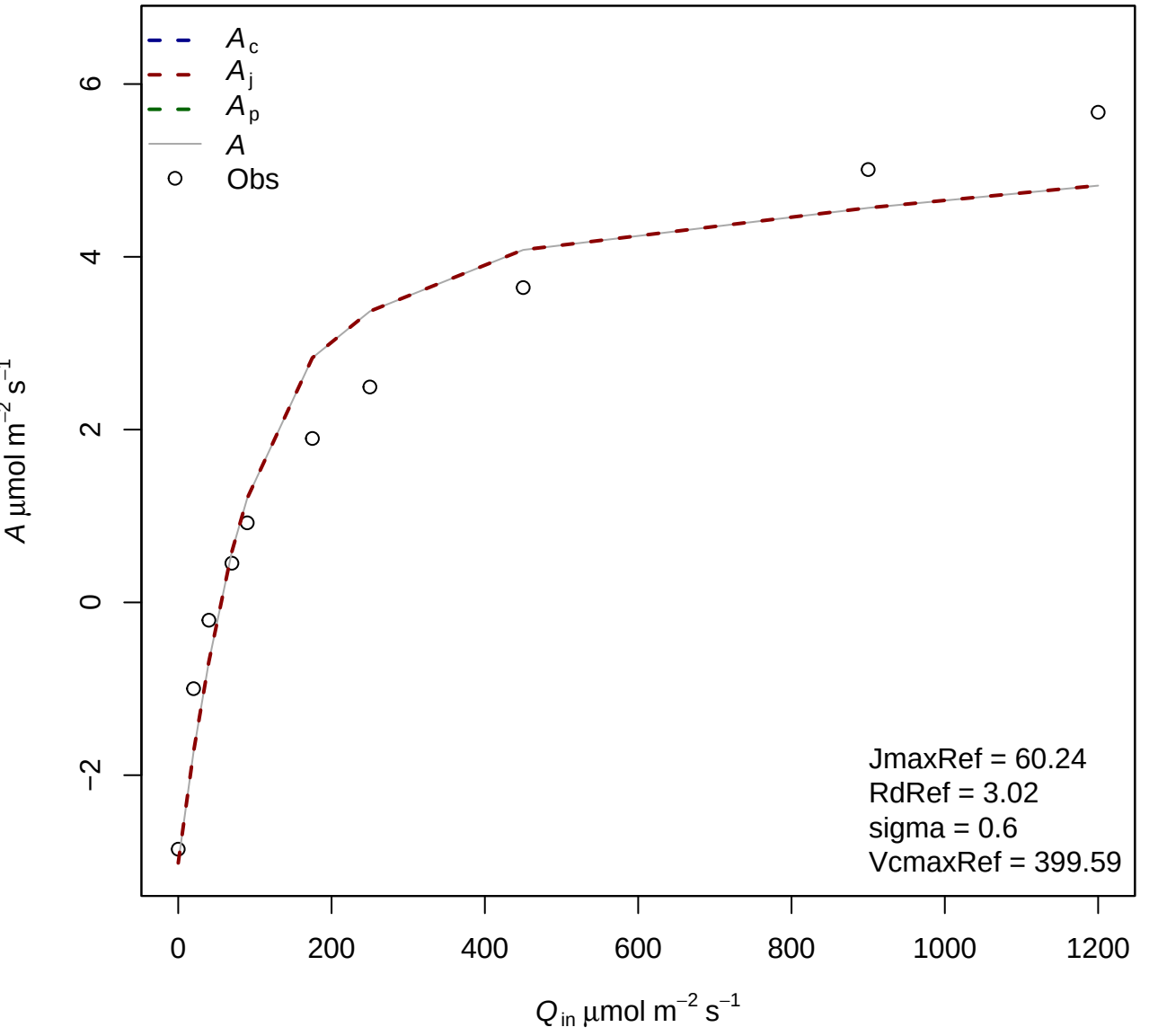

A_c
A_j
A_p
A
Obs
JmaxRef = 78.29
RdRef = 1.03
sigma = 0.09
VcmaxRef = 40.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

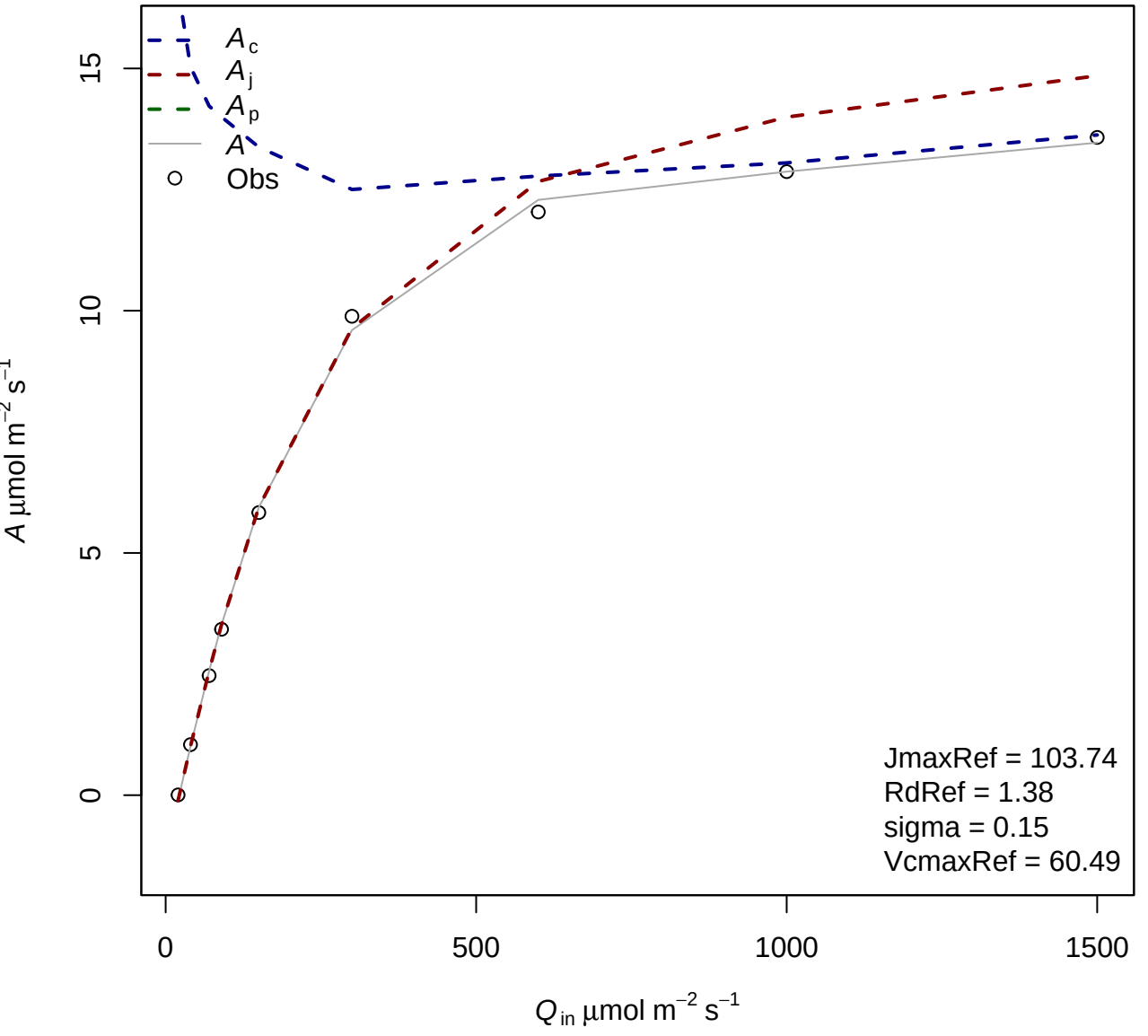

82
A_c
A_j
A_p
A
Obs
JmaxRef = 90.34
RdRef = 2.82
sigma = 0.08
VcmaxRef = 51.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
200
400
600
800
1000
1200

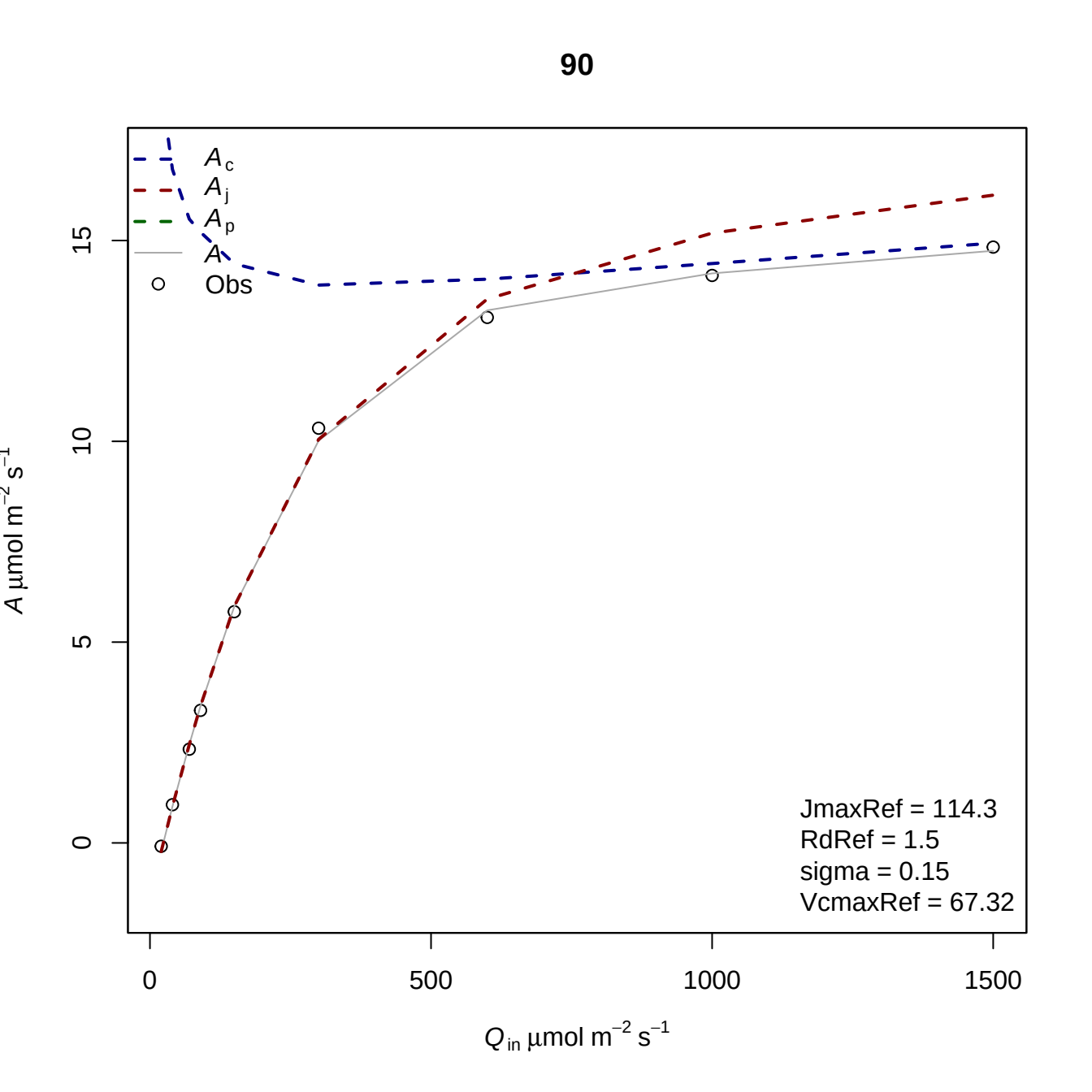

A_c
A_j
A_p
A
Obs
JmaxRef = 84.87
RdRef = 3.07
sigma = 0.13
VcmaxRef = 354.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

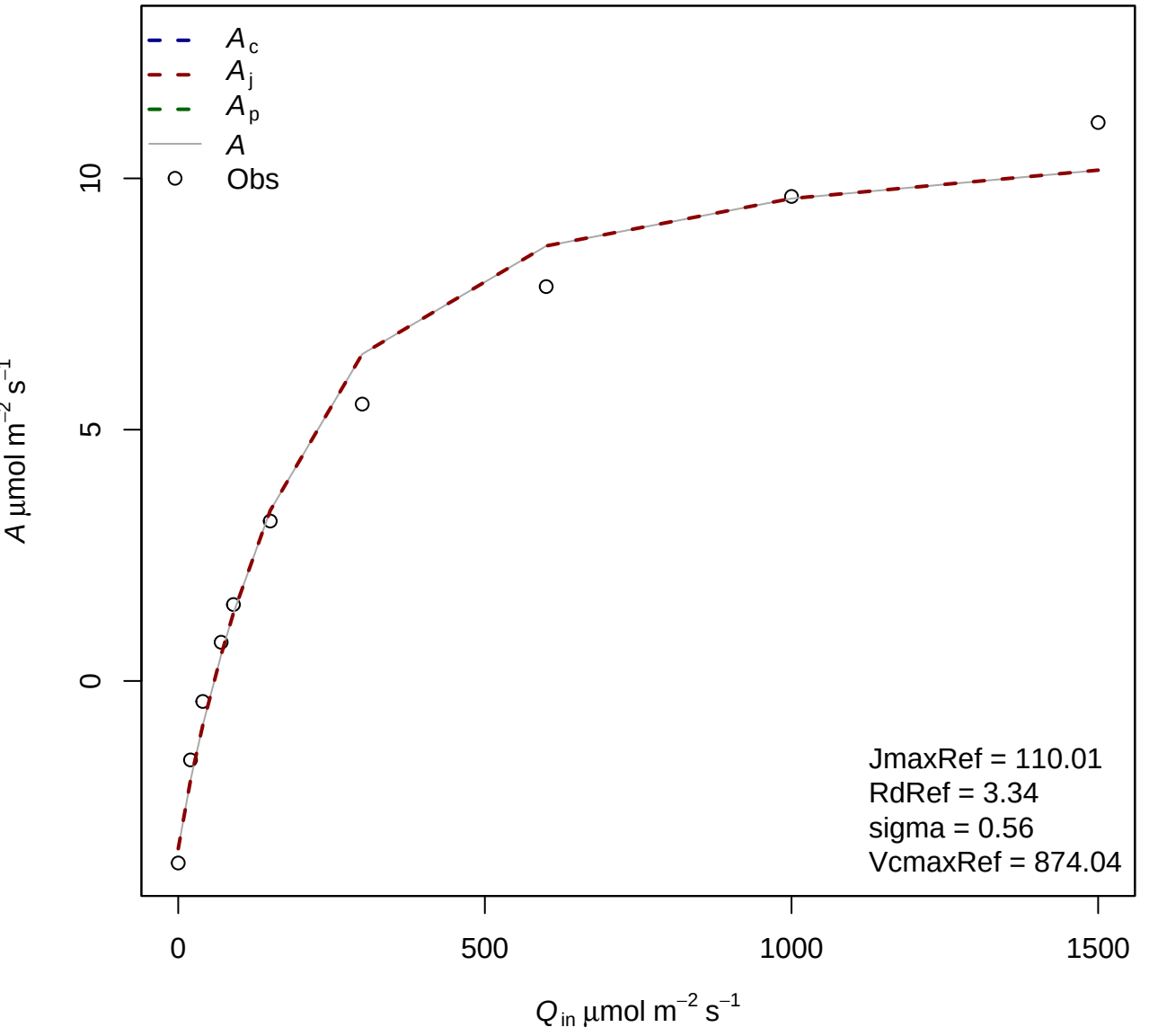

A_c
A_j
A_p
A
Obs
JmaxRef = 98.75
RdRef = 1.13
sigma = 0.14
VcmaxRef = 54.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200

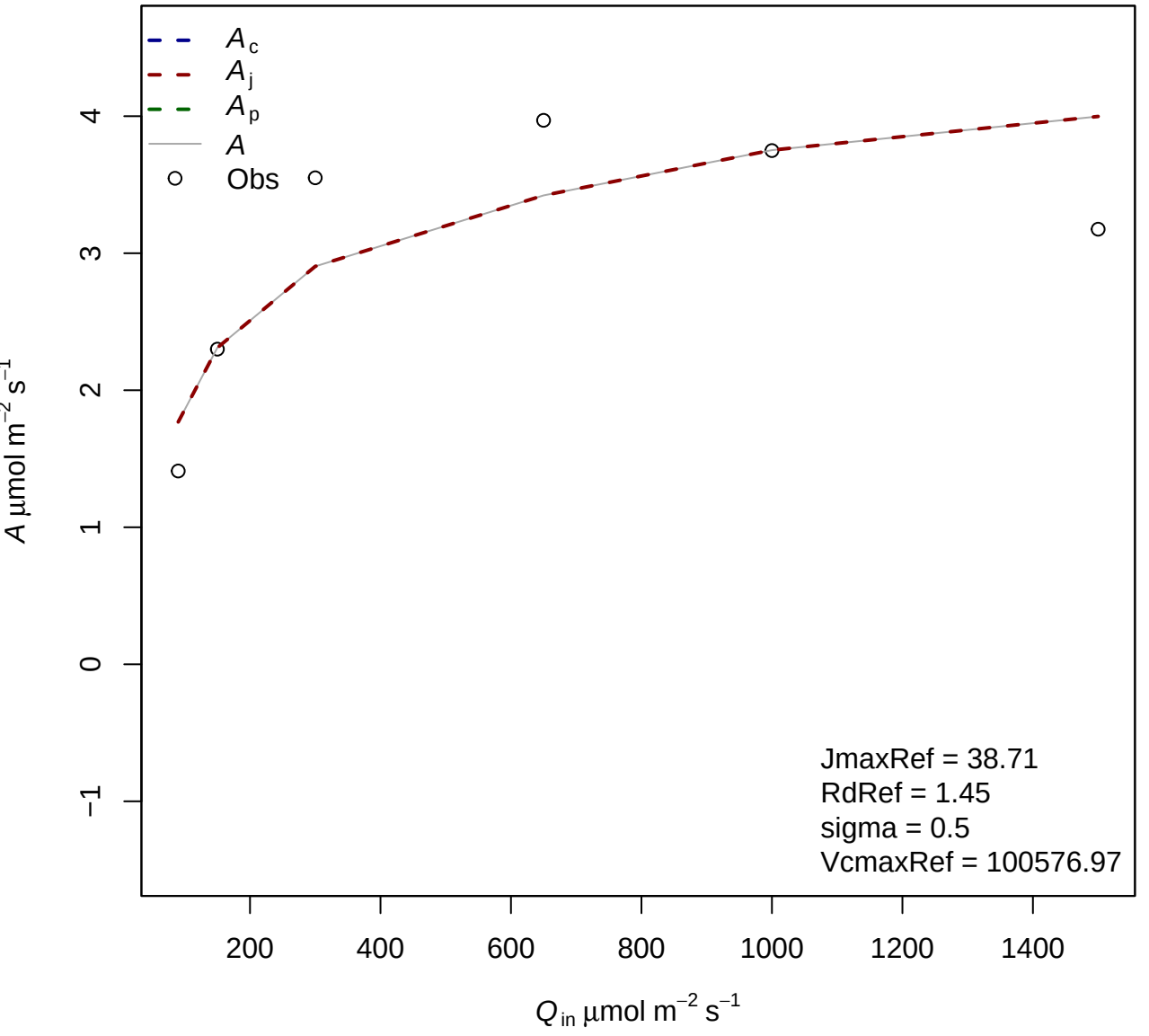

A_c
A_j
A_p
A
Obs
JmaxRef = 48.53
RdRef = 3.56
sigma = 0.78
VcmaxRef = 703.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200

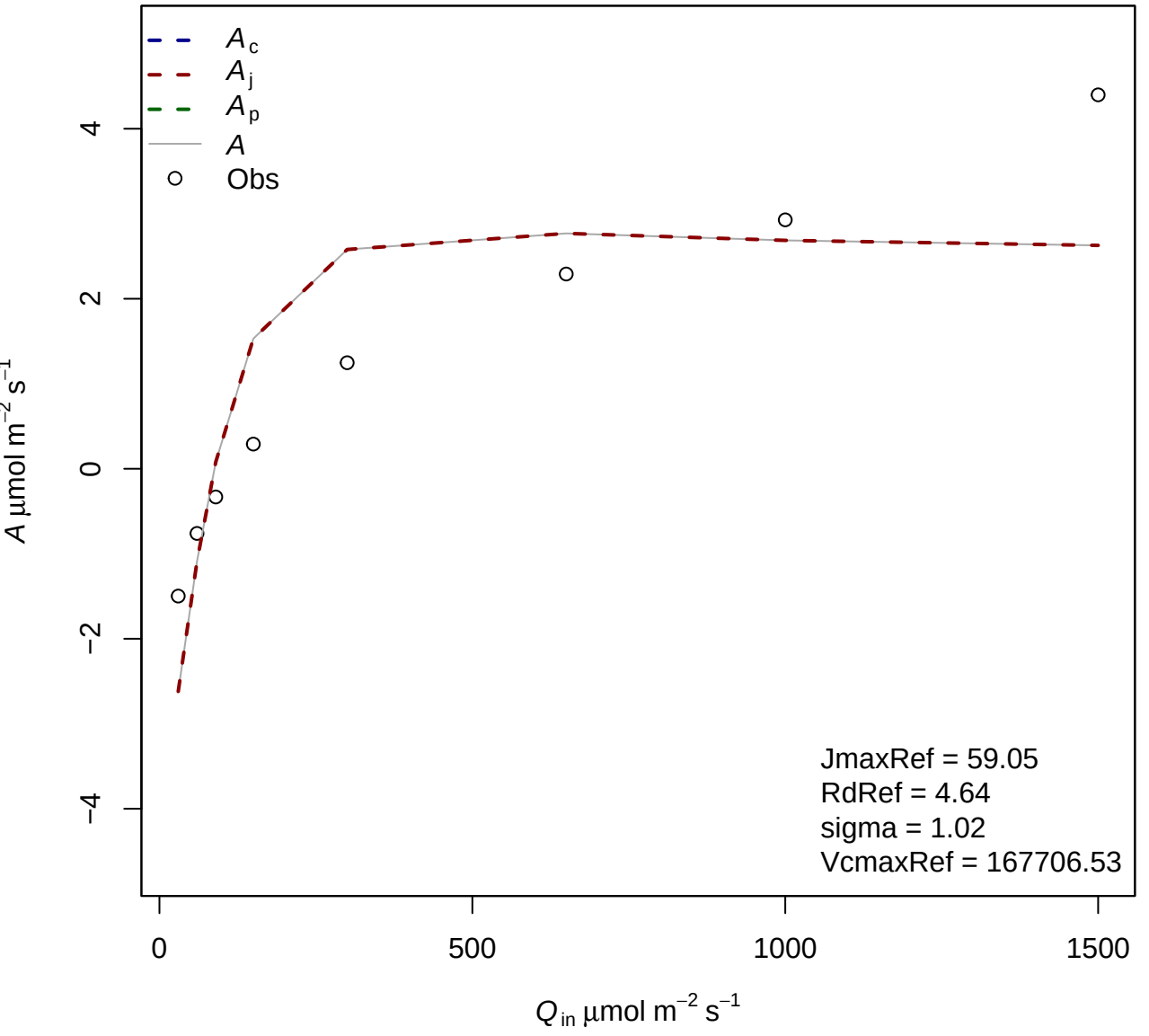

A_c
A_j
A_p
A
Obs
JmaxRef = 60.24
RdRef = 3.02
sigma = 0.6
VcmaxRef = 399.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

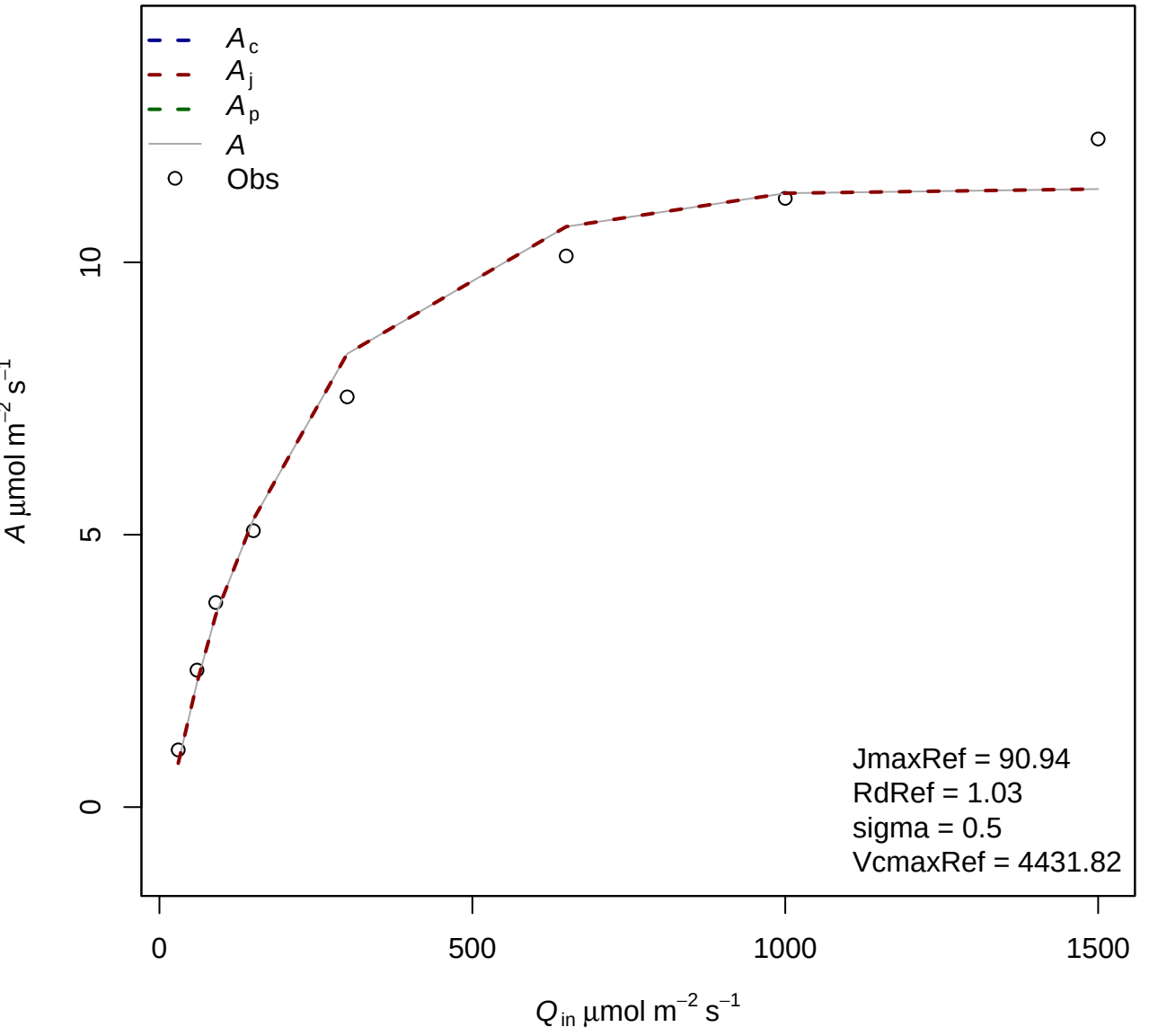

A_c
A_j
A_p
A
Obs
JmaxRef = 103.74
RdRef = 1.38
sigma = 0.15
VcmaxRef = 60.49
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

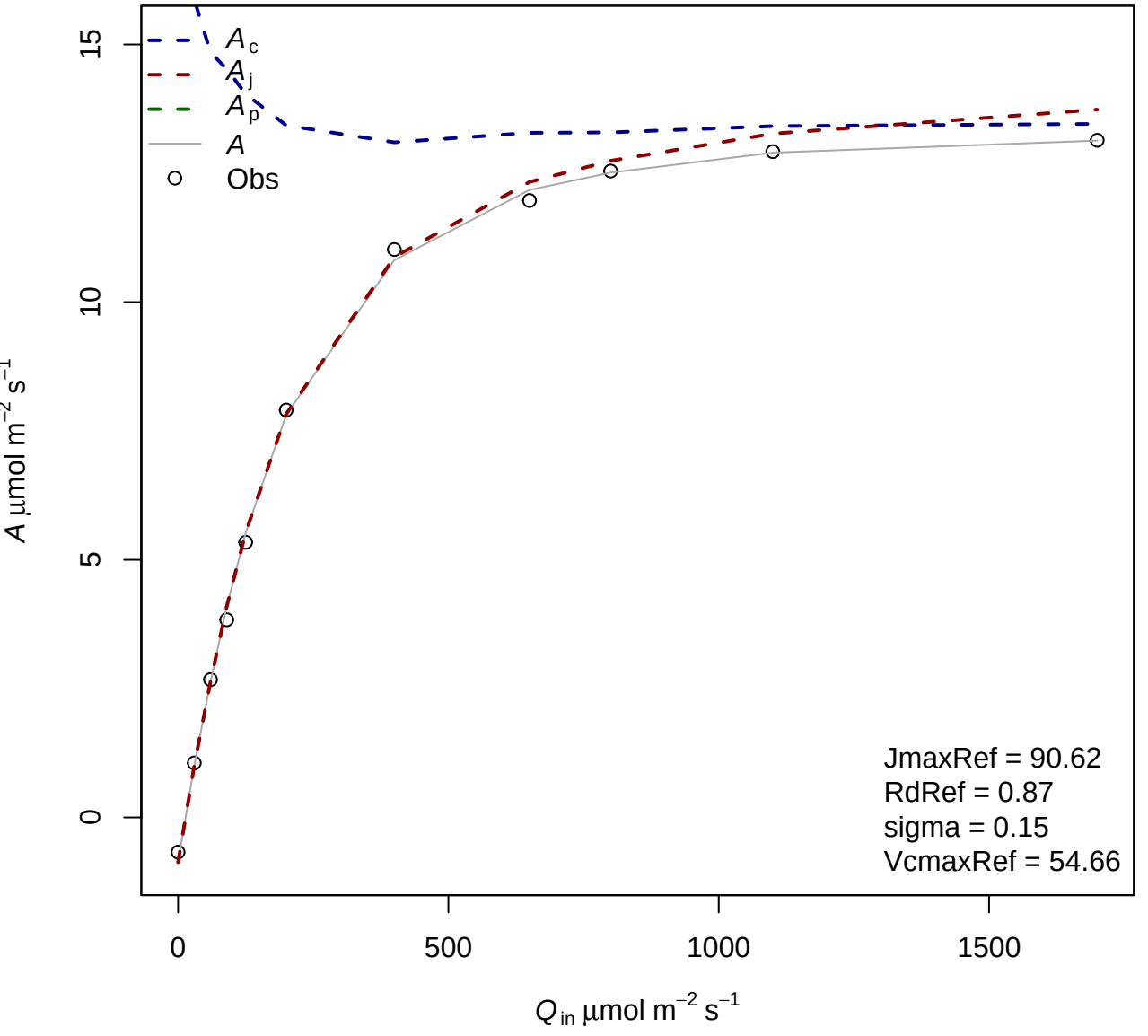

90
A_c
A_j
A_p
A
Obs
JmaxRef = 114.3
RdRef = 1.5
sigma = 0.15
VcmaxRef = 67.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

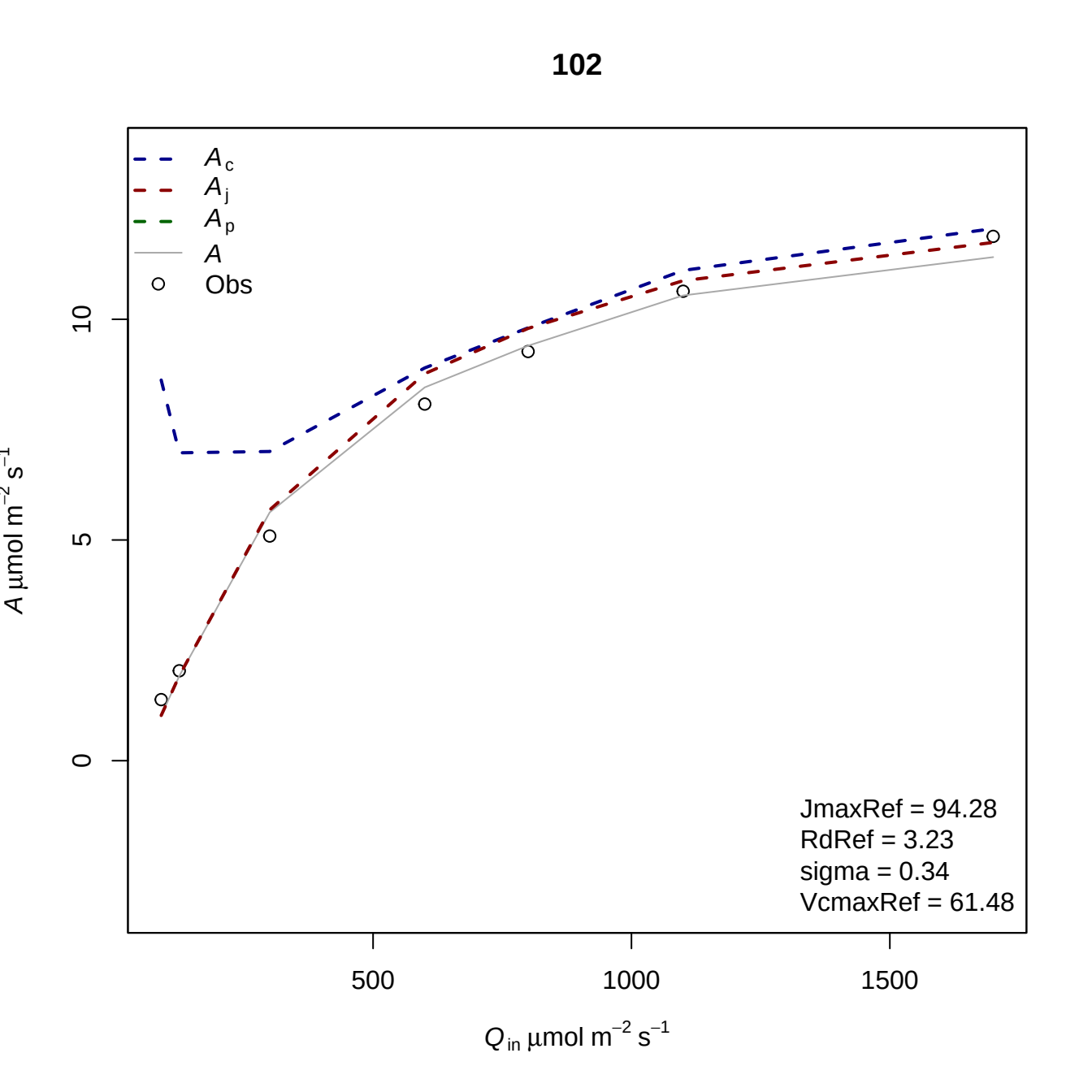

A_c
A_j
A_p
A
Obs
JmaxRef = 110.01
RdRef = 3.34
sigma = 0.56
VcmaxRef = 874.04
0
5
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

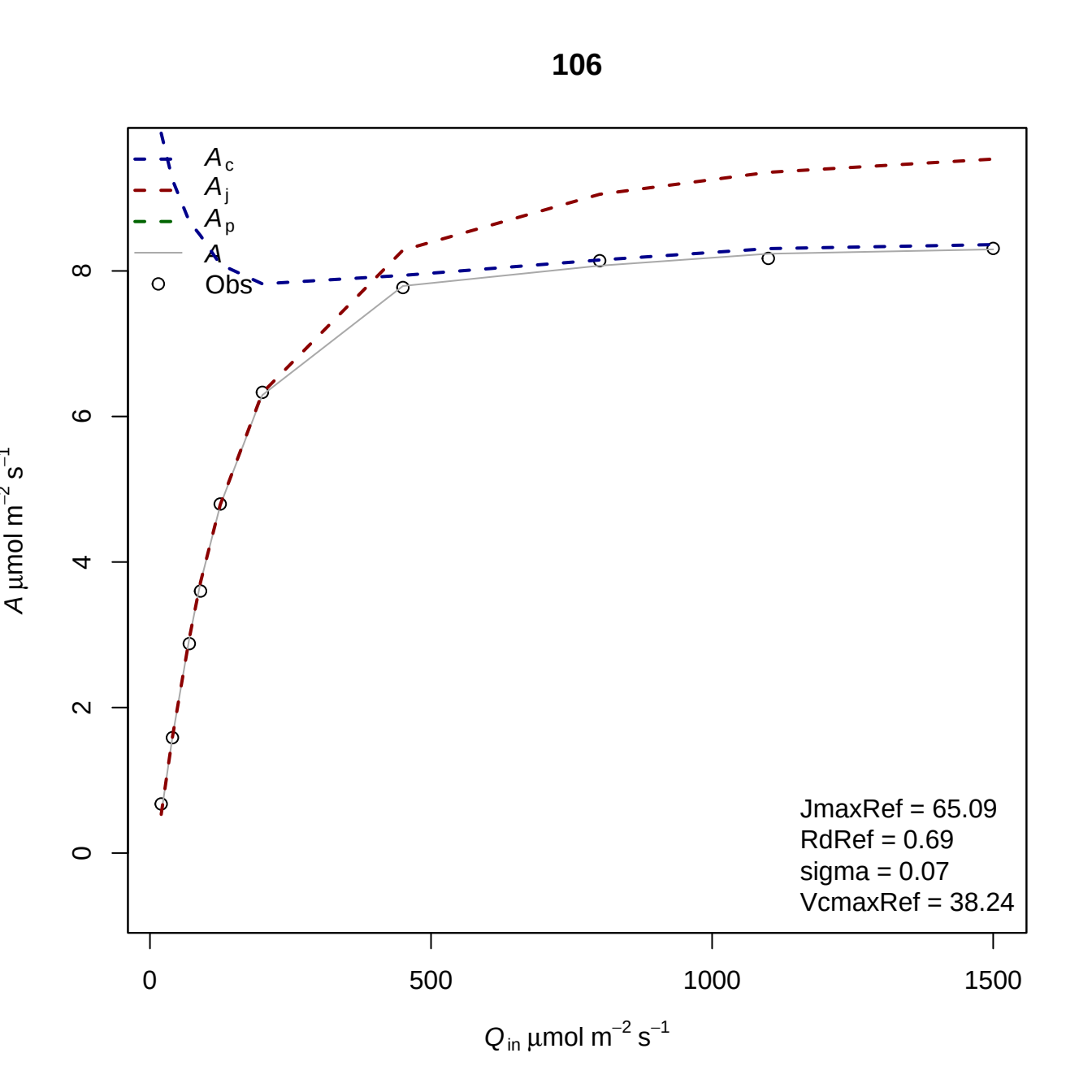

A_c
A_j
A_p
A
Obs
JmaxRef = 38.71
RdRef = 1.45
sigma = 0.5
VcmaxRef = 100576.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
200
400
600
800
1000
1200
1400

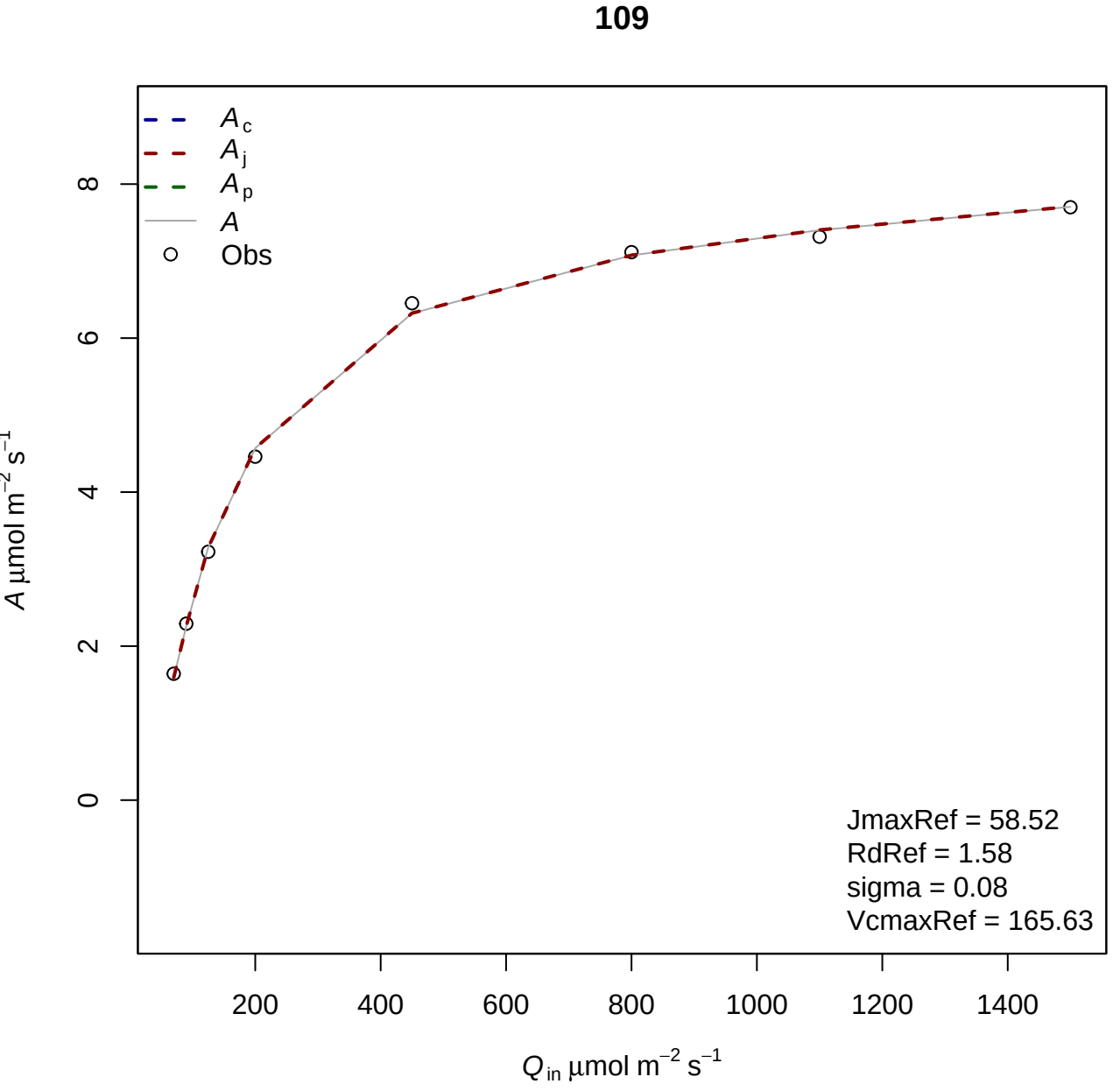

A_c
A_j
A_p
A
Obs
JmaxRef = 59.05
RdRef = 4.64
sigma = 1.02
VcmaxRef = 167706.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

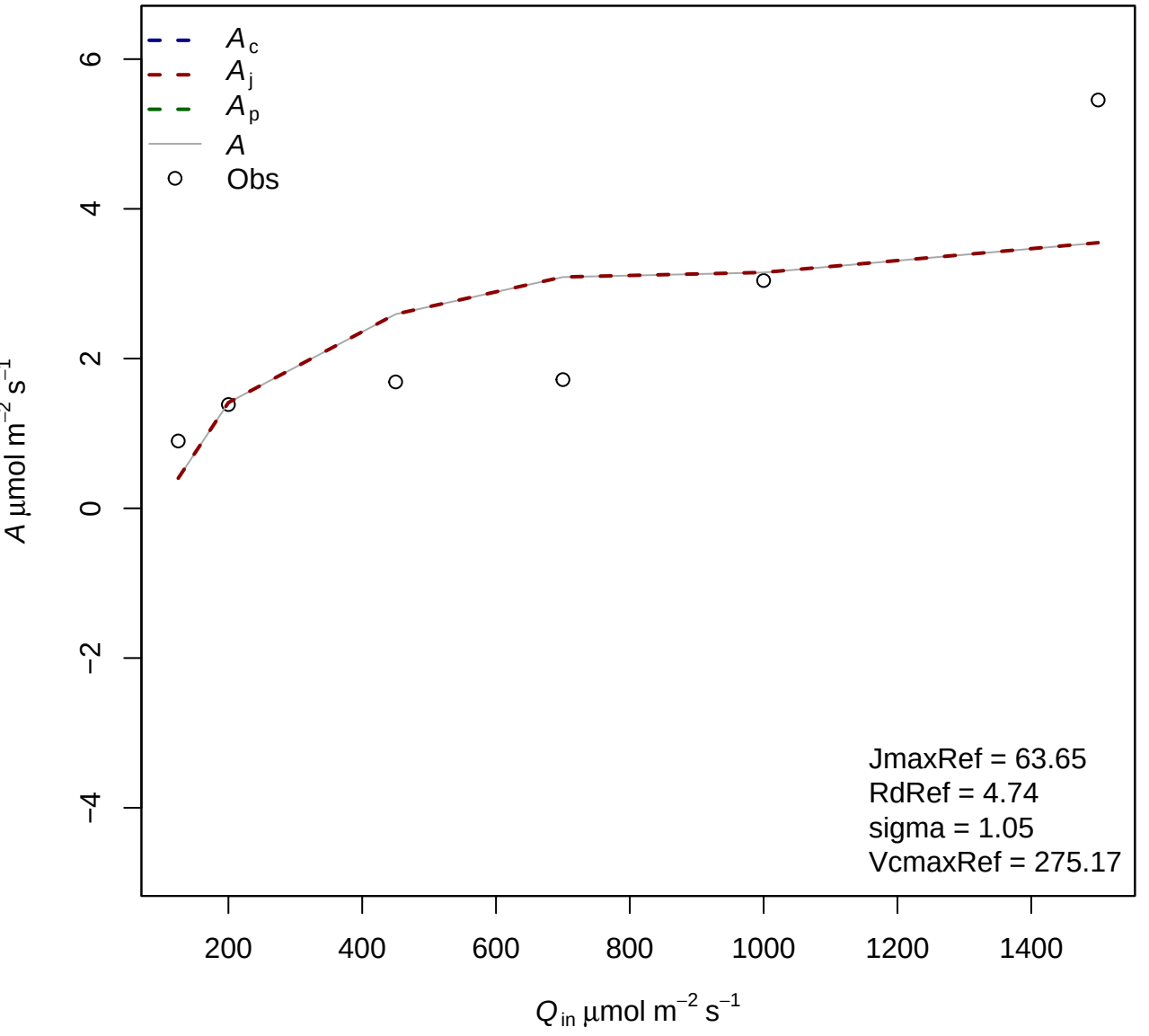

A_c
A_j
A_p
A
Obs
JmaxRef = 90.94
RdRef = 1.03
sigma = 0.5
VcmaxRef = 4431.82
0
5
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

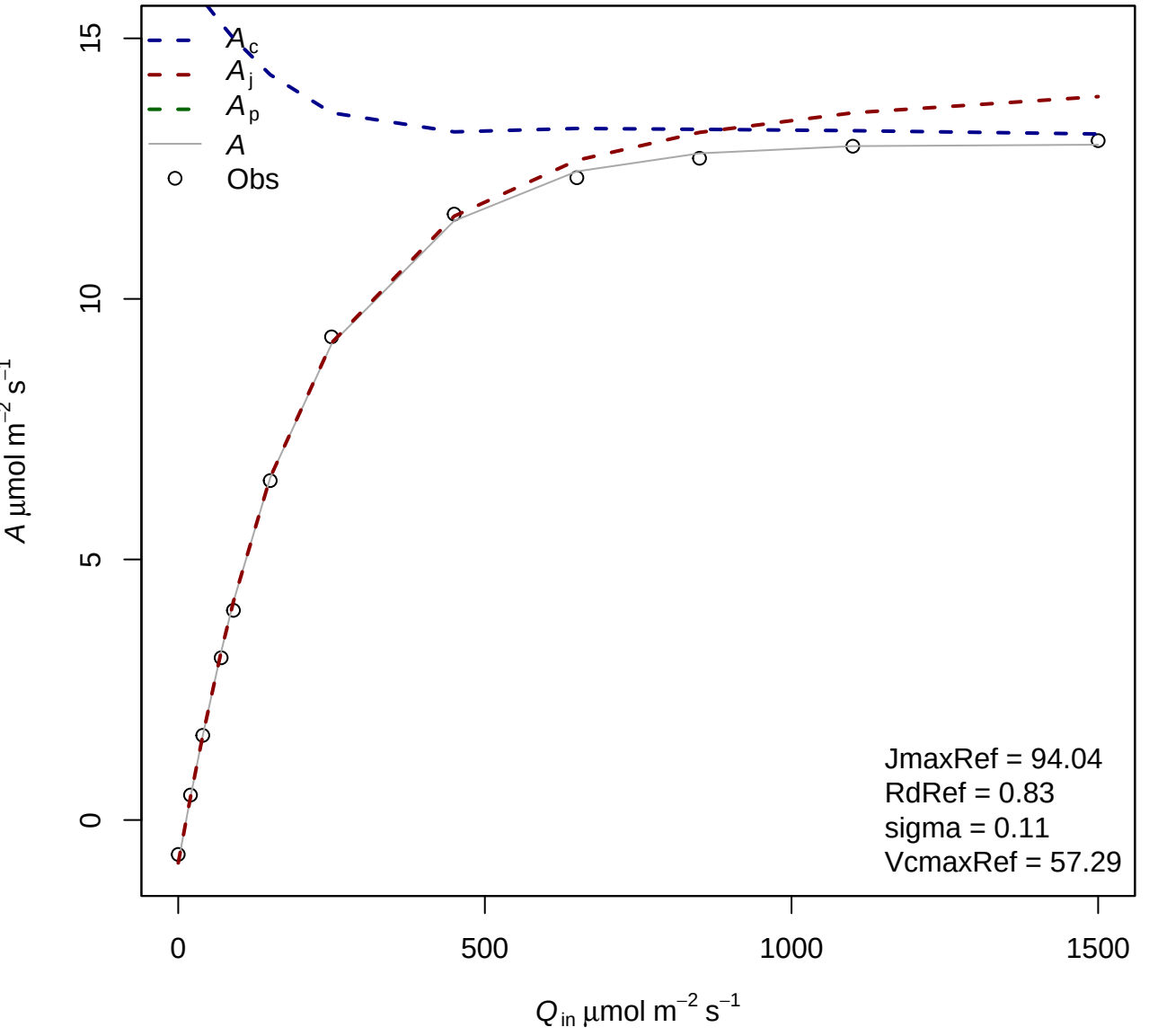

A_c
A_j
A_p
A
Obs
JmaxRef = 90.62
RdRef = 0.87
sigma = 0.15
VcmaxRef = 54.66
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

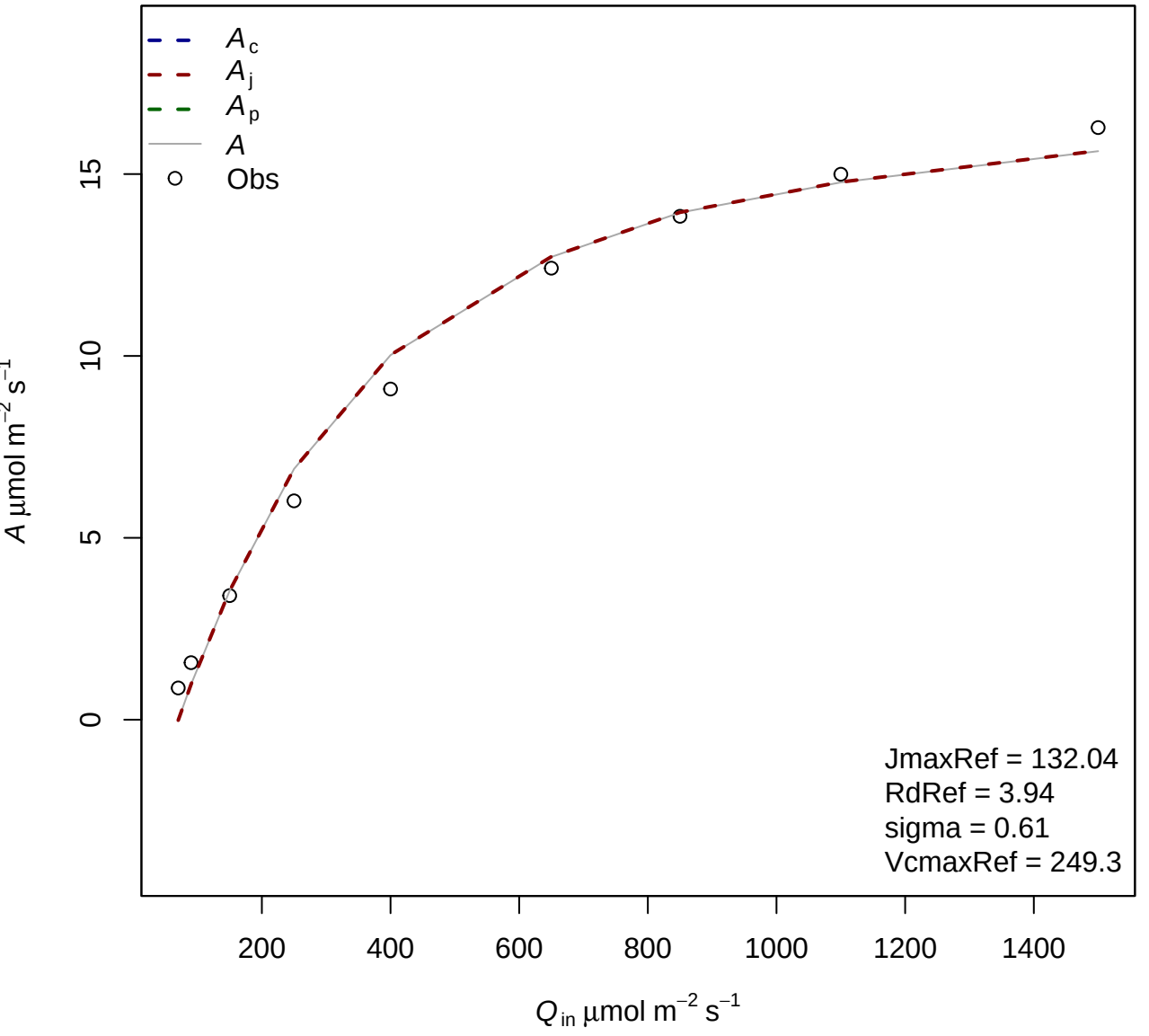

102
A_c
A_j
A_p
A
Obs
JmaxRef = 94.28
RdRef = 3.23
sigma = 0.34
VcmaxRef = 61.48
0
5
10
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

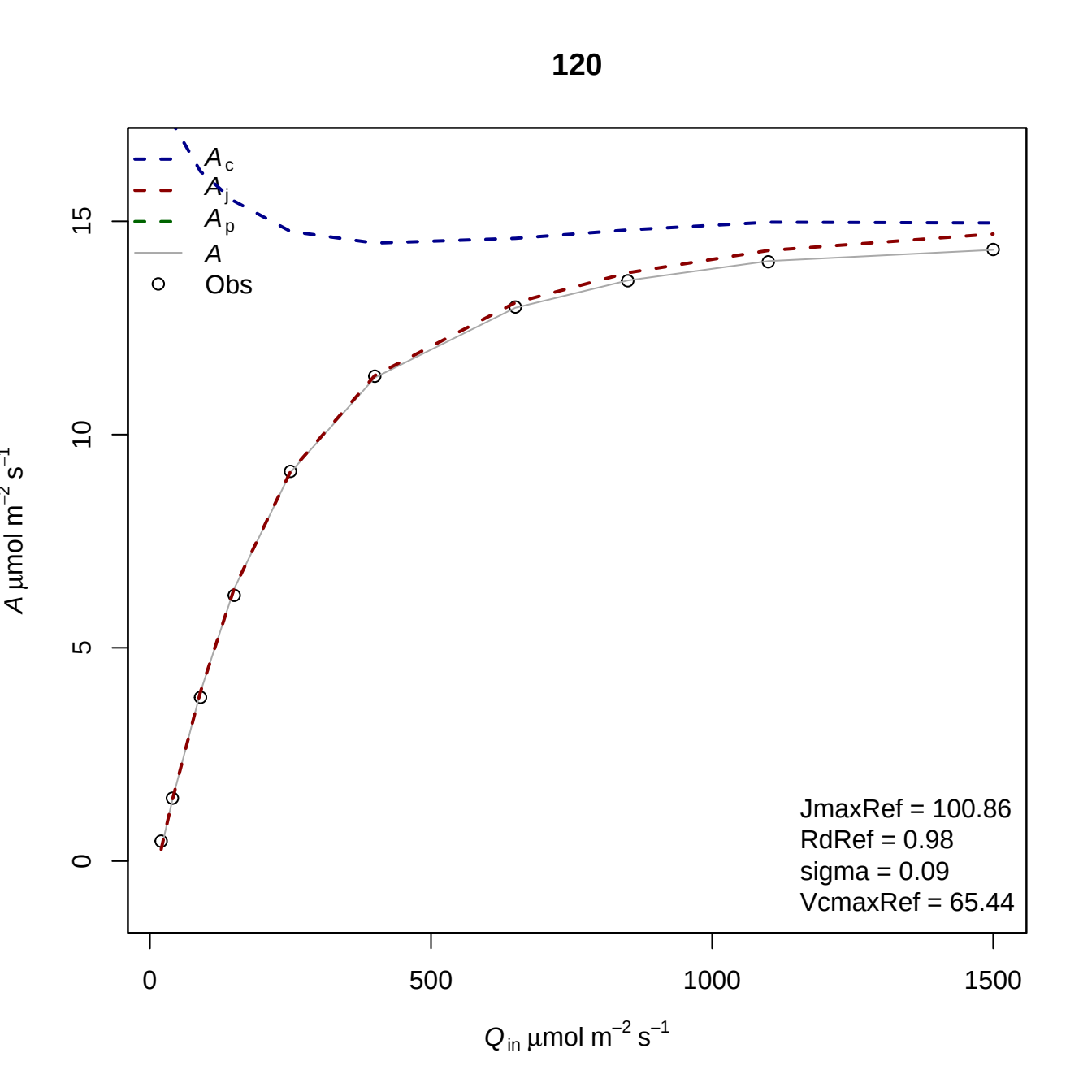

106
A_c
A_j
A_p
A
Obs
JmaxRef = 65.09
RdRef = 0.69
sigma = 0.07
VcmaxRef = 38.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

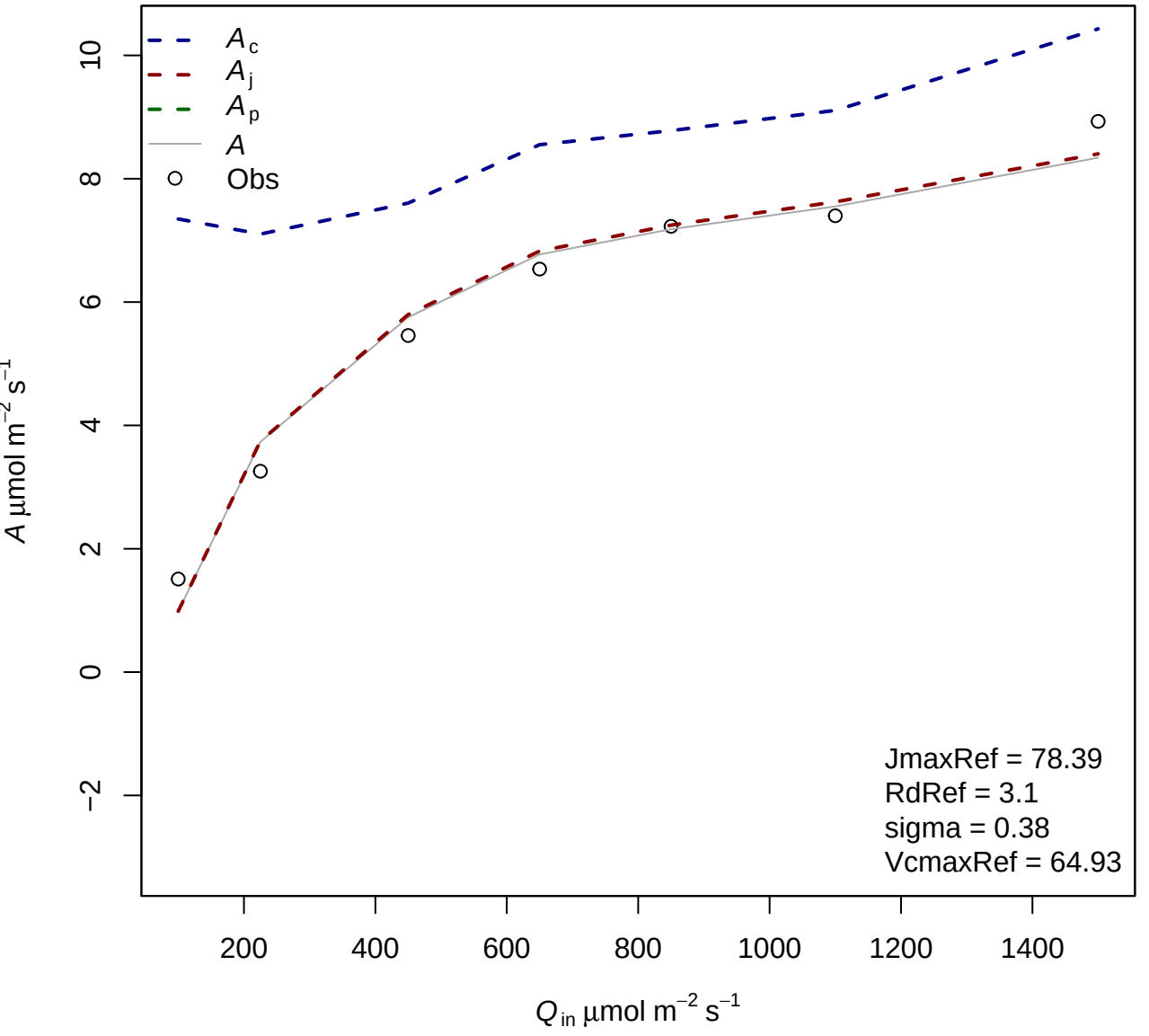

109
A_c
A_j
A_p
A
Obs
JmaxRef = 58.52
RdRef = 1.58
sigma = 0.08
VcmaxRef = 165.63
0
2
4
6
8
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

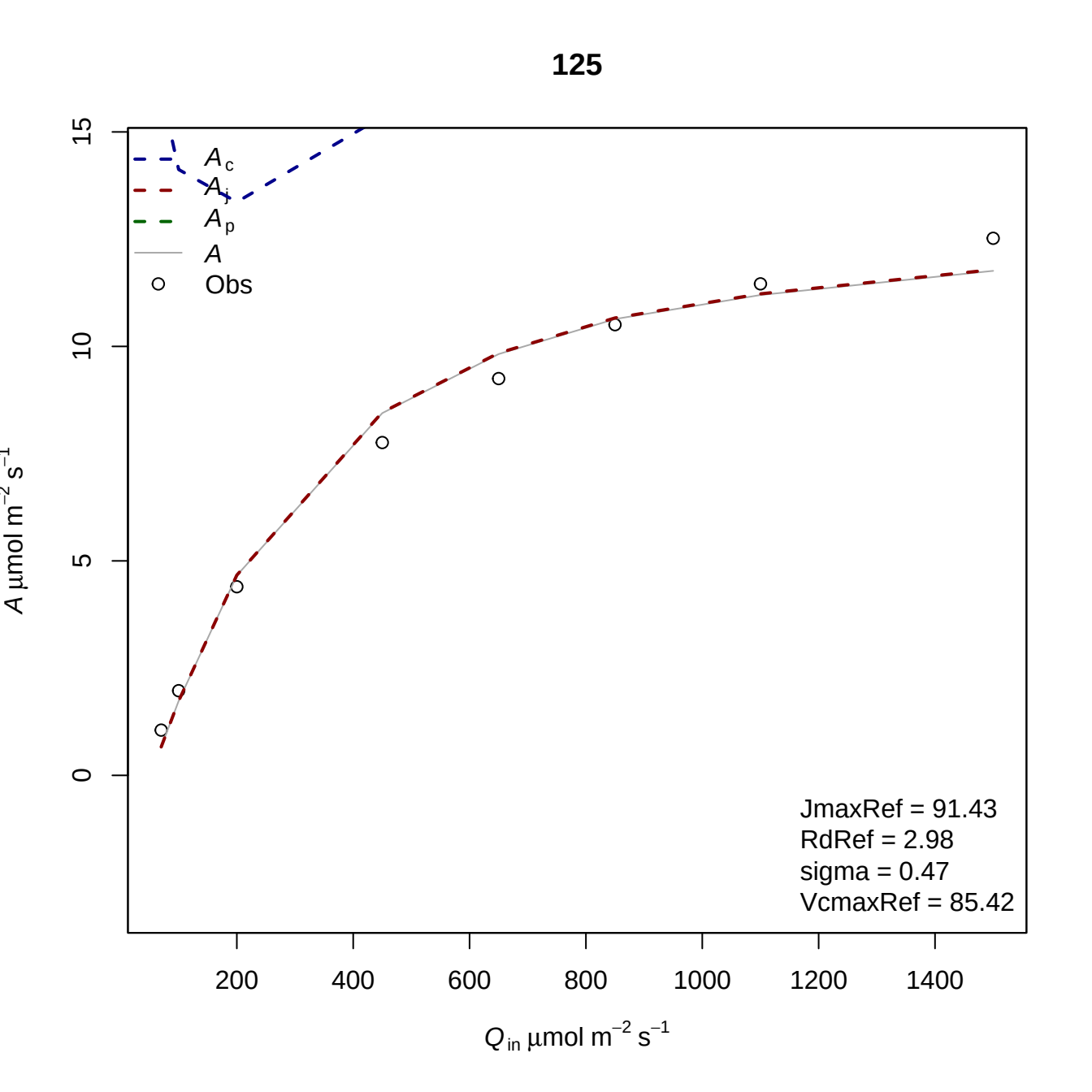

A_c
A_j
A_p
A
Obs
JmaxRef = 63.65
RdRef = 4.74
sigma = 1.05
VcmaxRef = 275.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

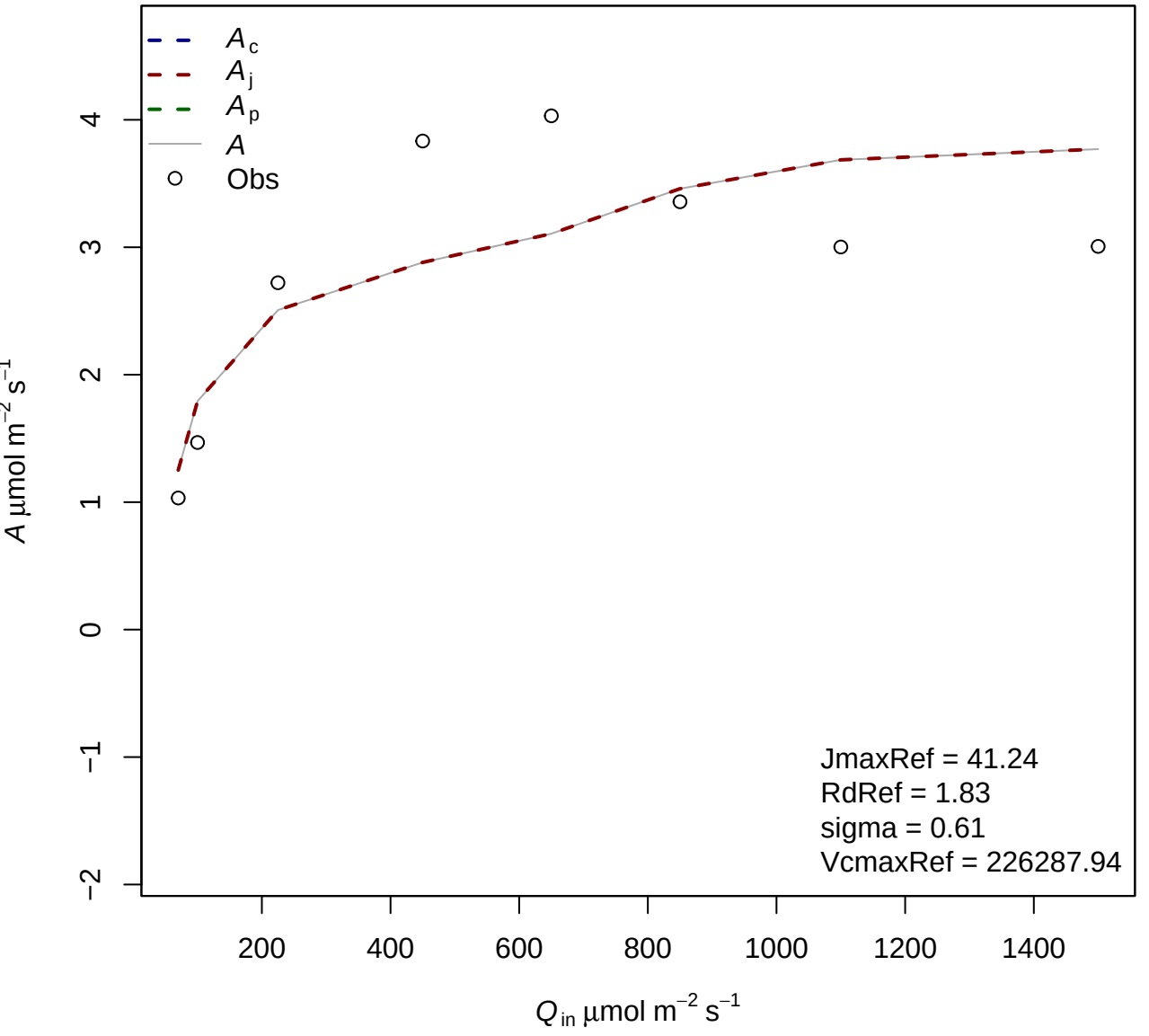

A_c
A_j
A_p
A
Obs
JmaxRef = 94.04
RdRef = 0.83
sigma = 0.11
VcmaxRef = 57.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

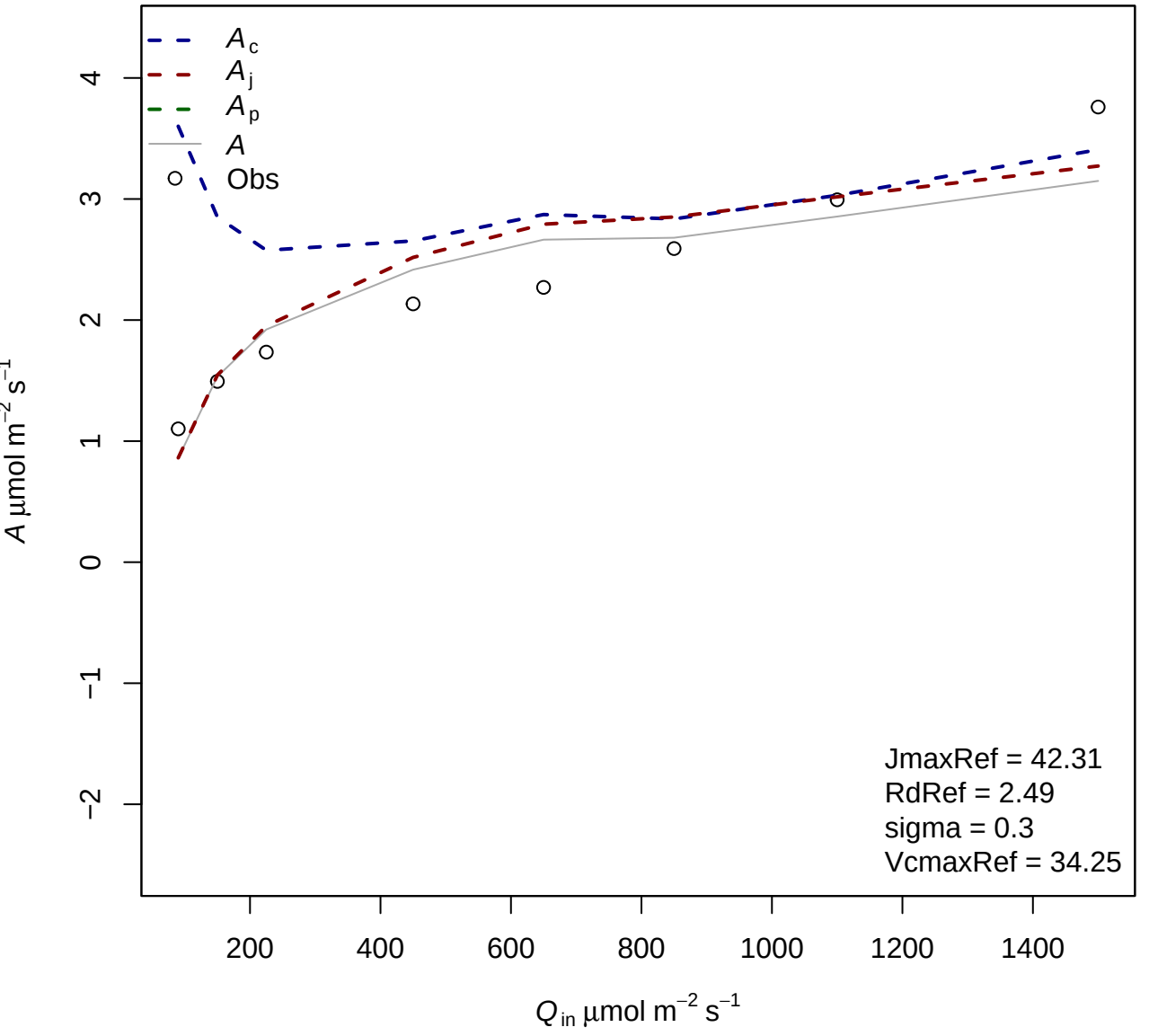

A_c
A_j
A_p
A
Obs
JmaxRef = 132.04
RdRef = 3.94
sigma = 0.61
VcmaxRef = 249.3
0
5
10
15
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

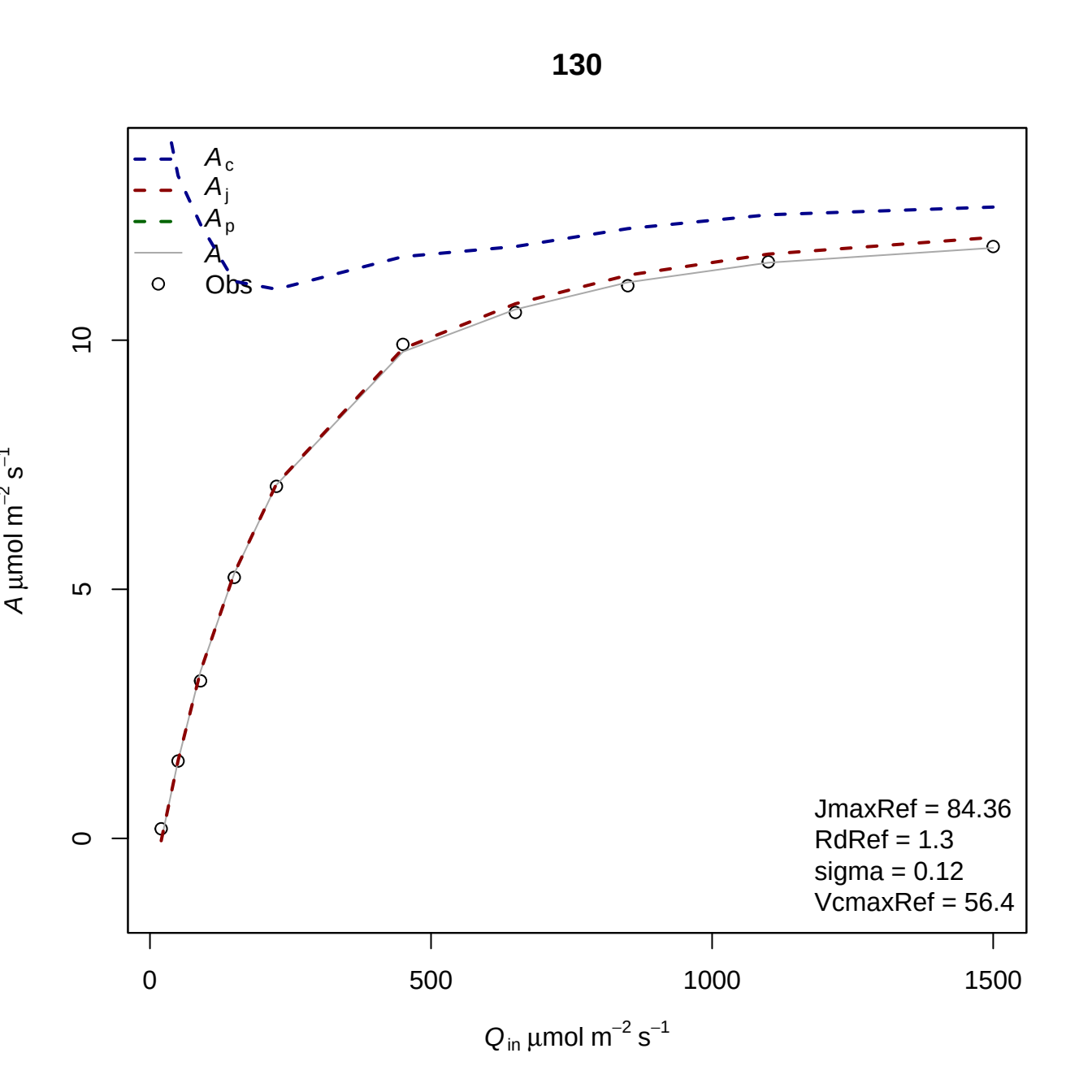

120
A_c
A_j
A_p
A
Obs
JmaxRef = 100.86
RdRef = 0.98
sigma = 0.09
VcmaxRef = 65.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

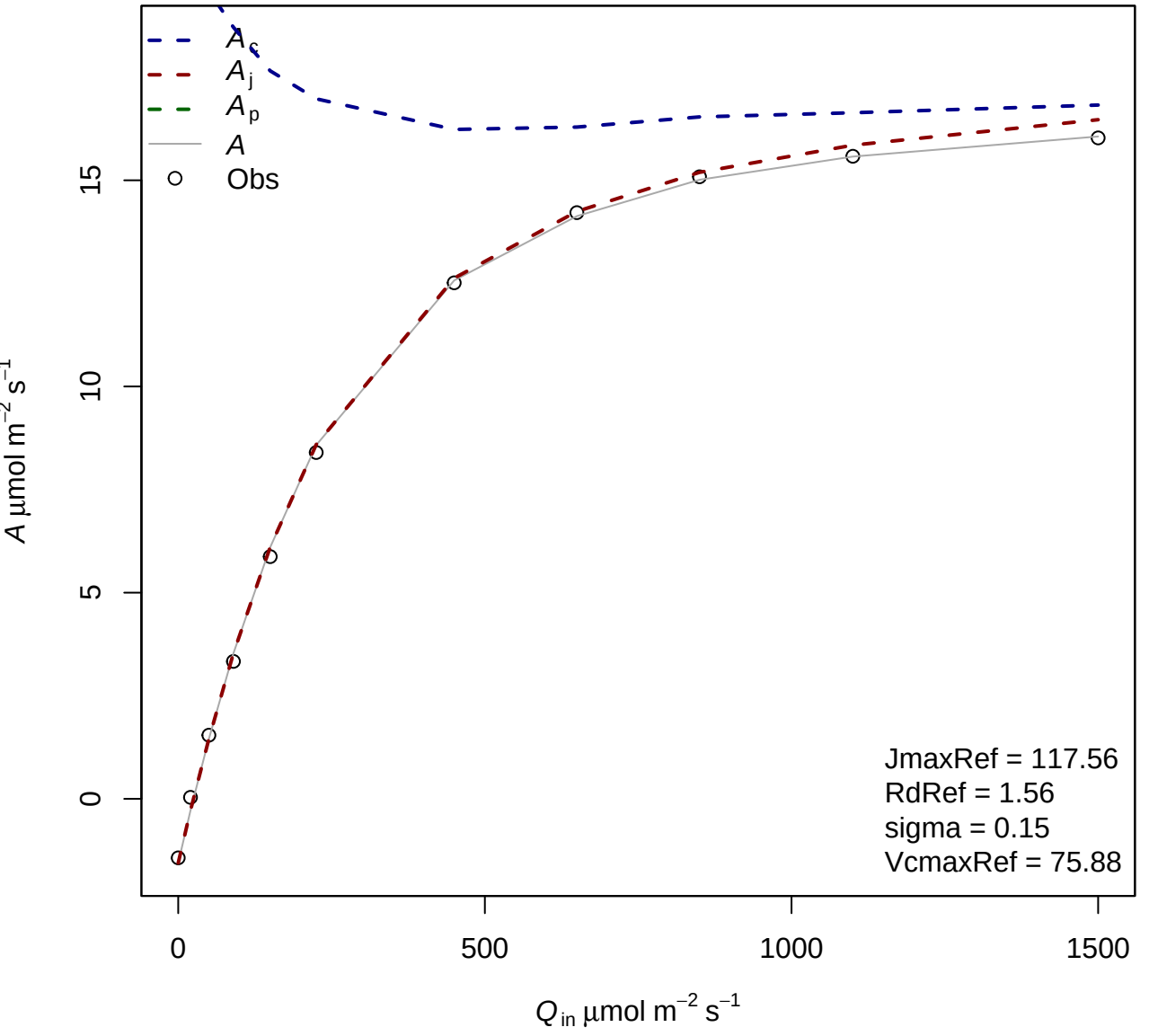

A_c
A_j
A_p
A
Obs
JmaxRef = 78.39
RdRef = 3.1
sigma = 0.38
VcmaxRef = 64.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
200
400
600
800
1000
1200
1400

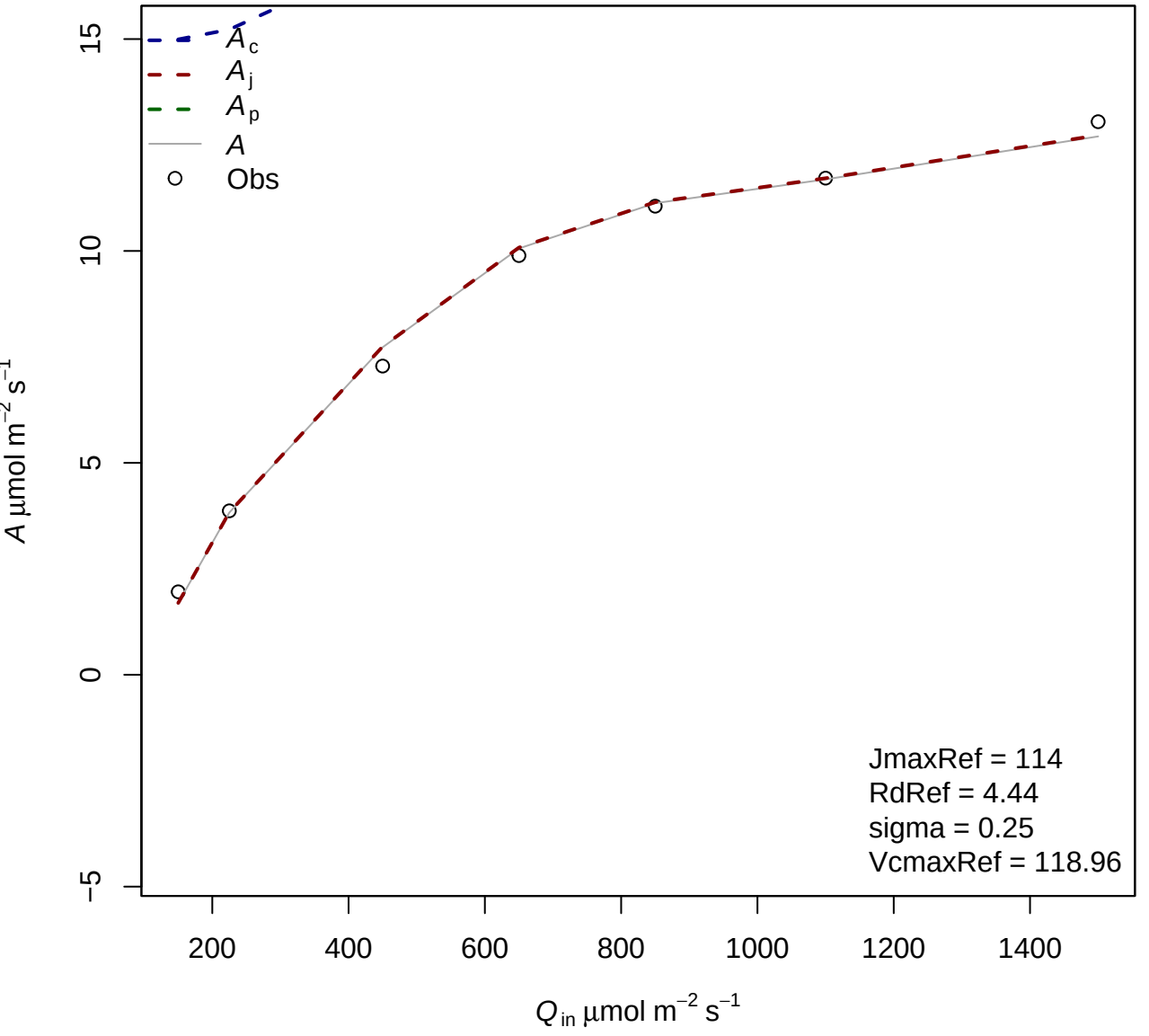

125
A_c
A_j
A_p
A
Obs
JmaxRef = 91.43
RdRef = 2.98
sigma = 0.47
VcmaxRef = 85.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
200
400
600
800
1000
1200
1400

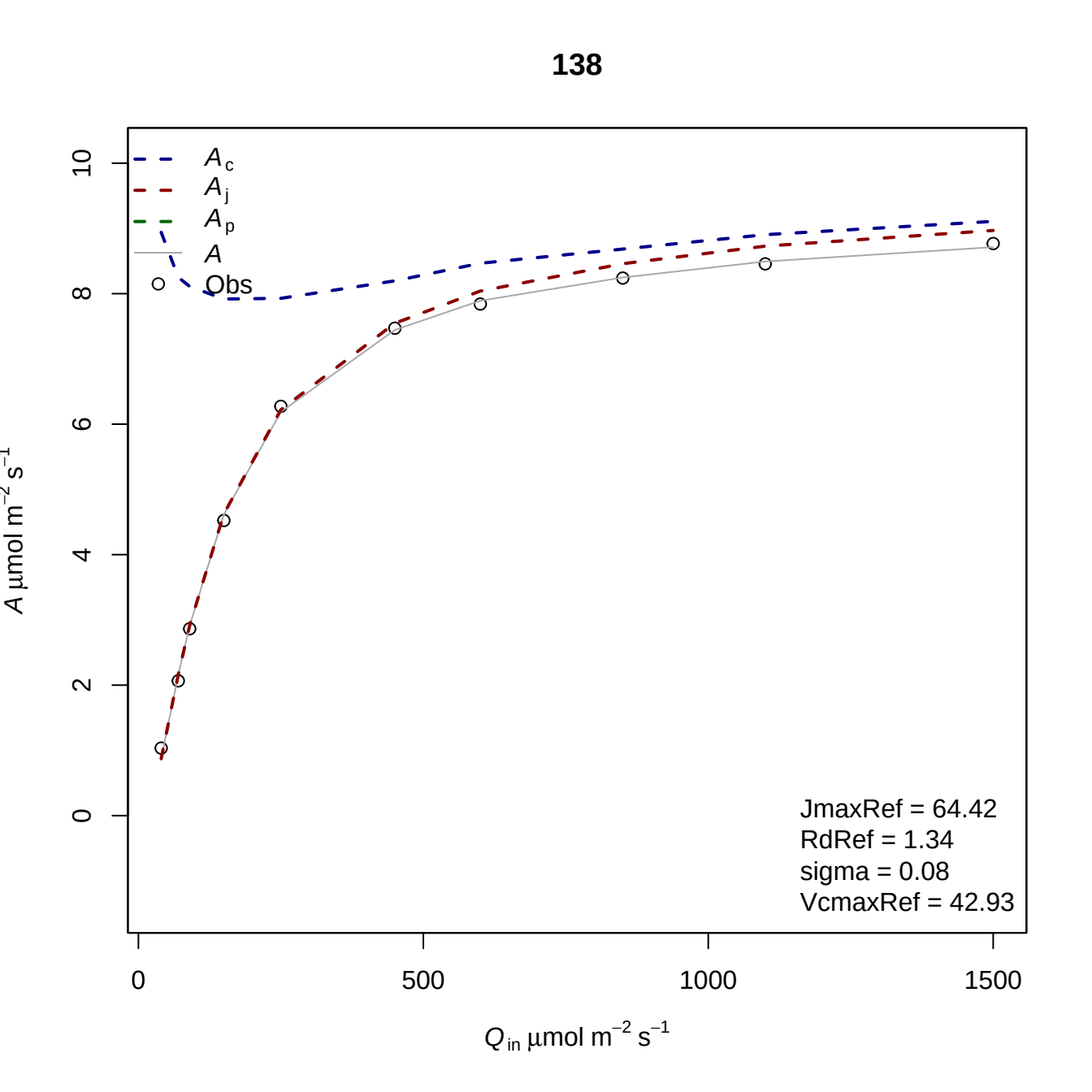

A_c
A_j
A_p
A
Obs
JmaxRef = 41.24
RdRef = 1.83
sigma = 0.61
VcmaxRef = 226287.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
200
400
600
800
1000
1200
1400

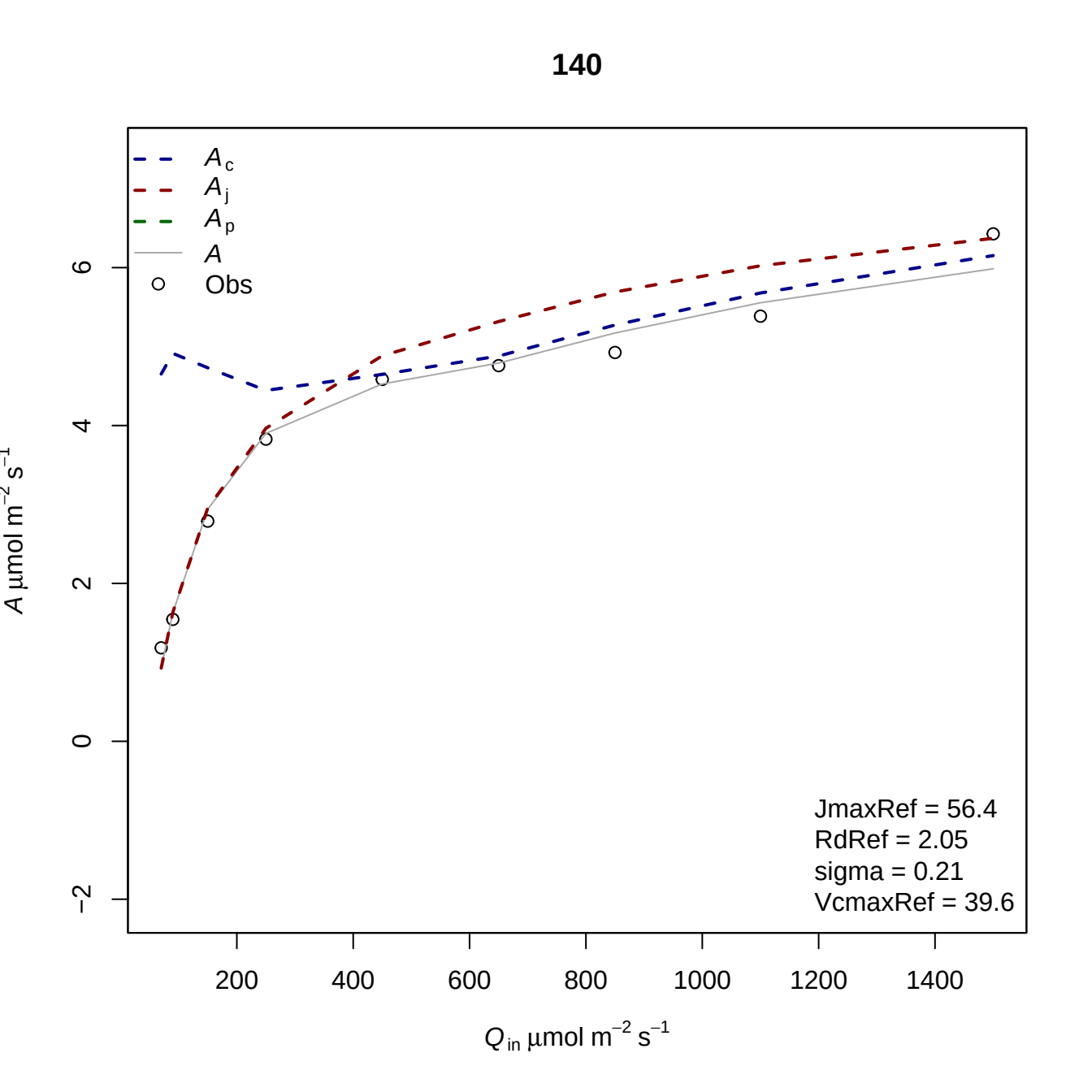

A_c
A_j
A_p
A
Obs
JmaxRef = 42.31
RdRef = 2.49
sigma = 0.3
VcmaxRef = 34.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
200
400
600
800
1000
1200
1400

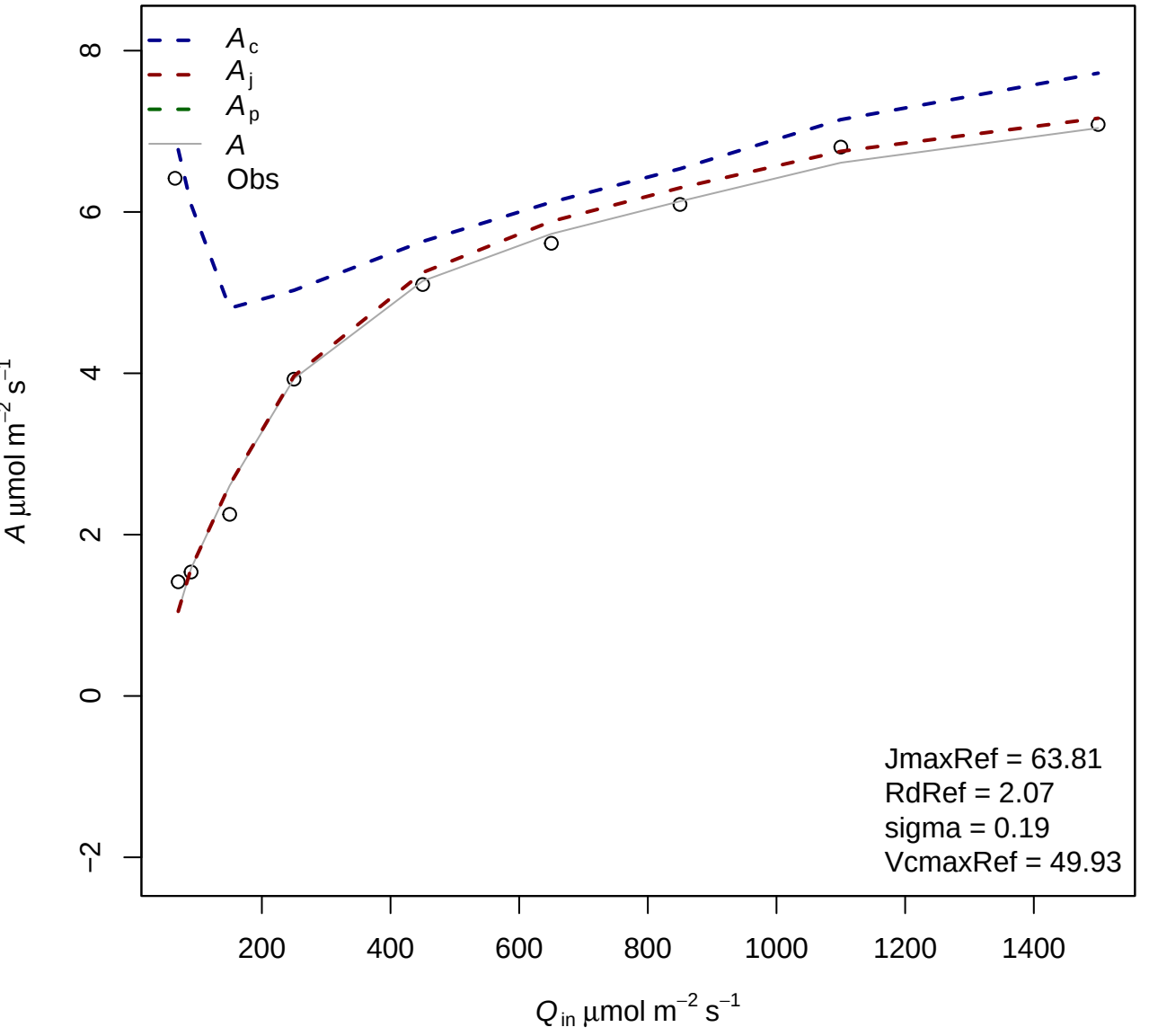

130
A_c
A_j
A_p
A
Obs
JmaxRef = 84.36
RdRef = 1.3
sigma = 0.12
VcmaxRef = 56.4
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

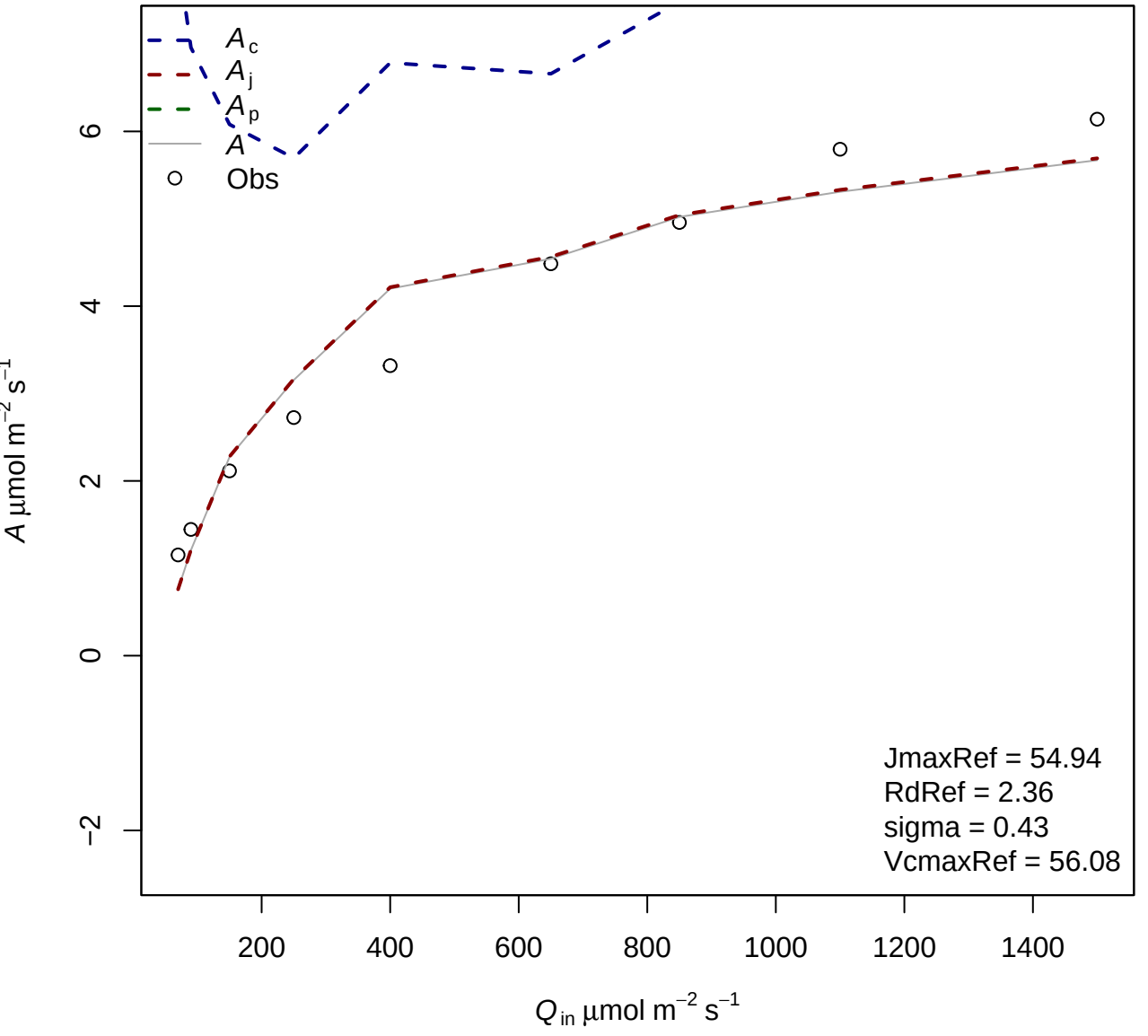

A_c
A_j
A_p
A
Obs
JmaxRef = 117.56
RdRef = 1.56
sigma = 0.15
VcmaxRef = 75.88
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

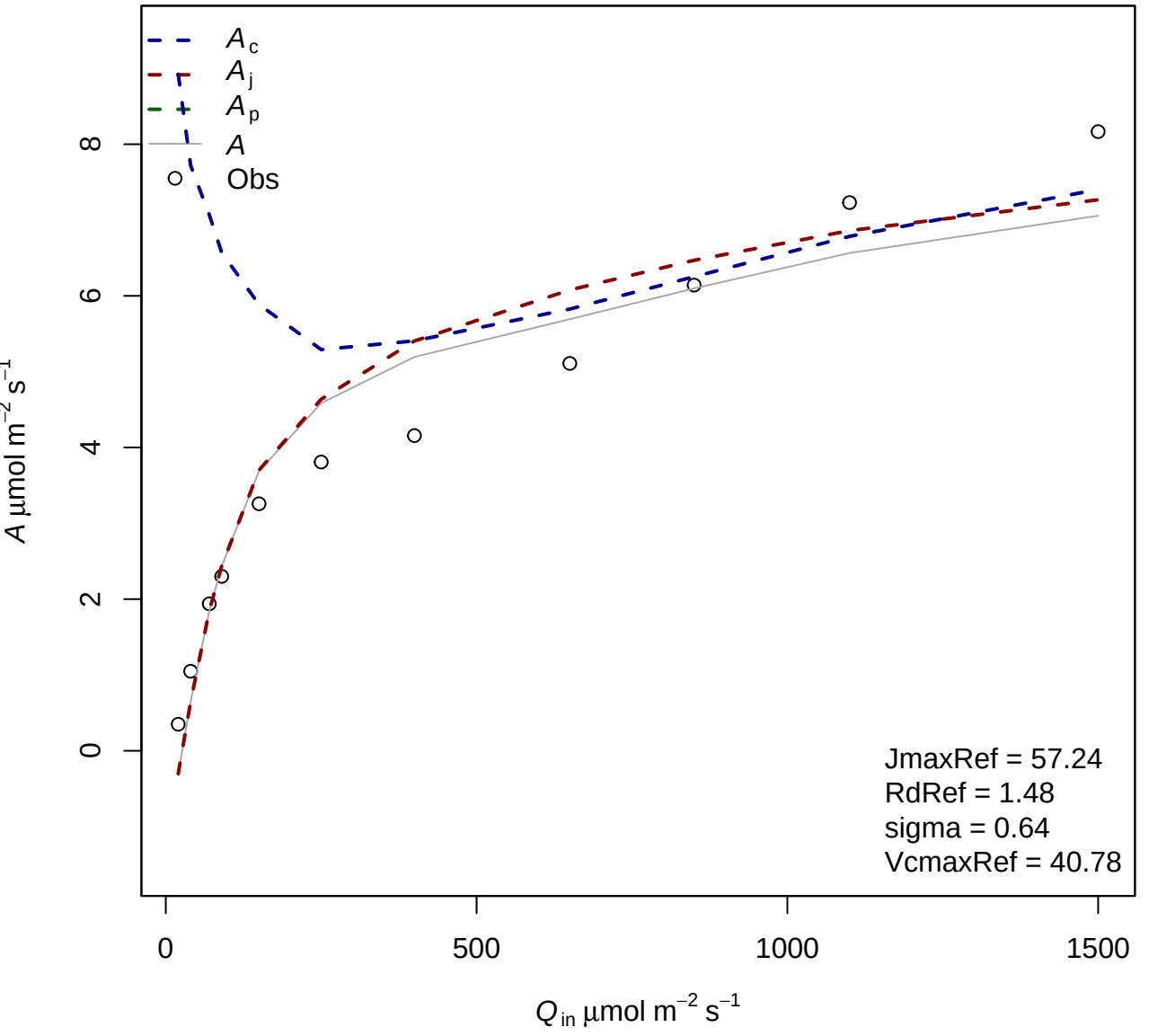

A_c
A_j
A_p
A
Obs
15
10
5
0
−5
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 114
RdRef = 4.44
sigma = 0.25
VcmaxRef = 118.96

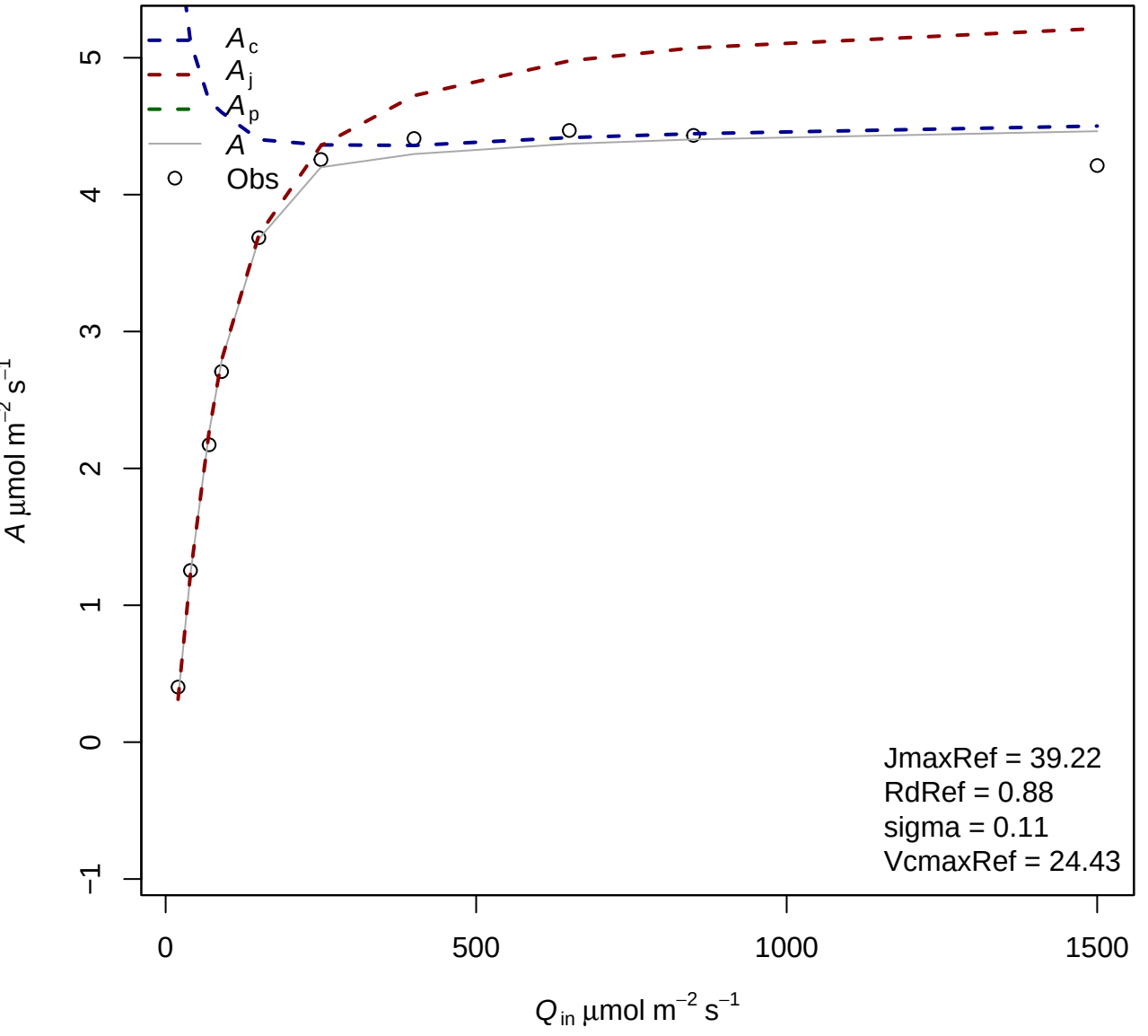

138
A_c
A_j
A_p
A
Obs
JmaxRef = 64.42
RdRef = 1.34
sigma = 0.08
VcmaxRef = 42.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

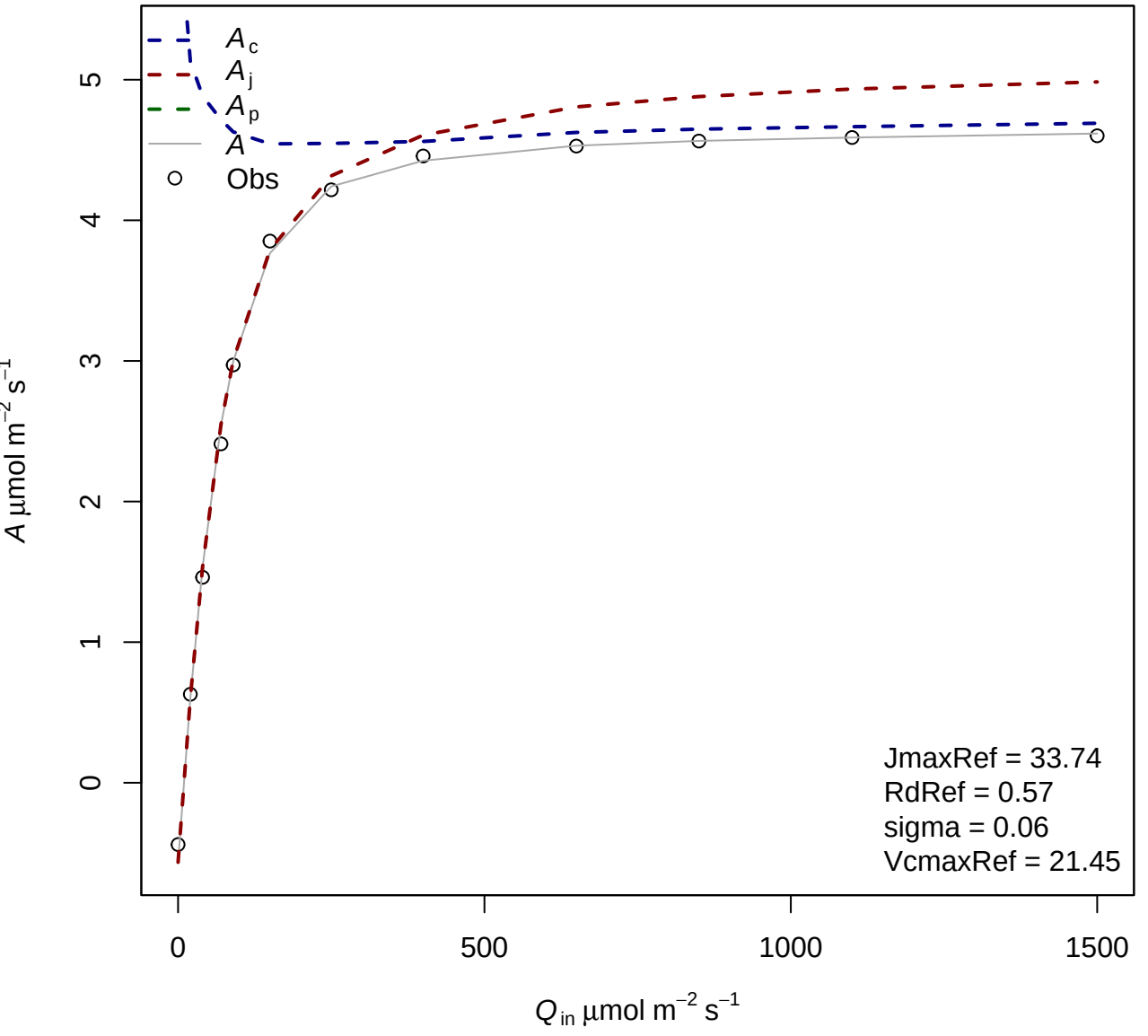

140
A_c
A_j
A_p
A
Obs
JmaxRef = 56.4
RdRef = 2.05
sigma = 0.21
VcmaxRef = 39.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
200
400
600
800
1000
1200
1400

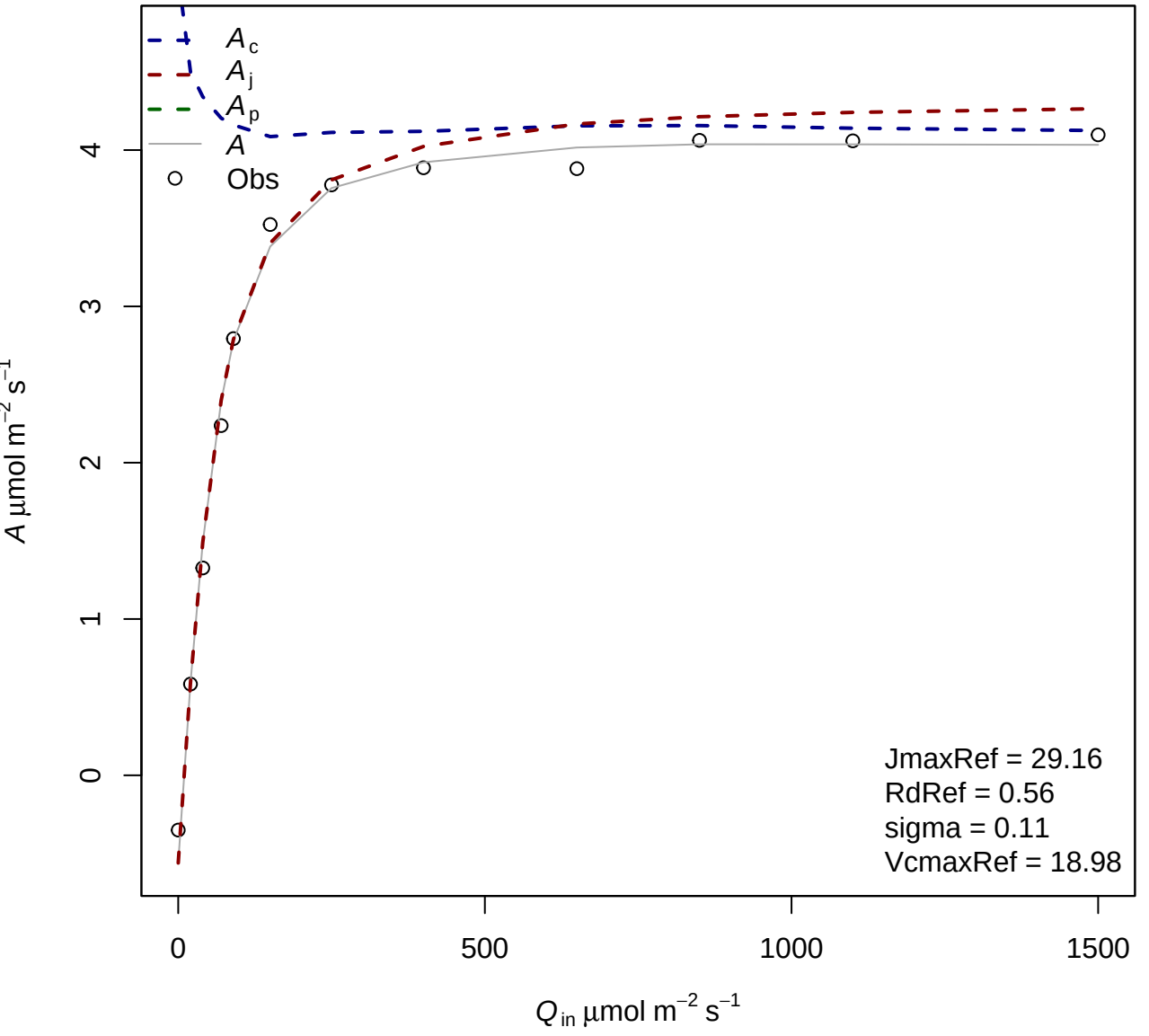

A_c
A_j
A_p
A
Obs
JmaxRef = 63.81
RdRef = 2.07
sigma = 0.19
VcmaxRef = 49.93
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

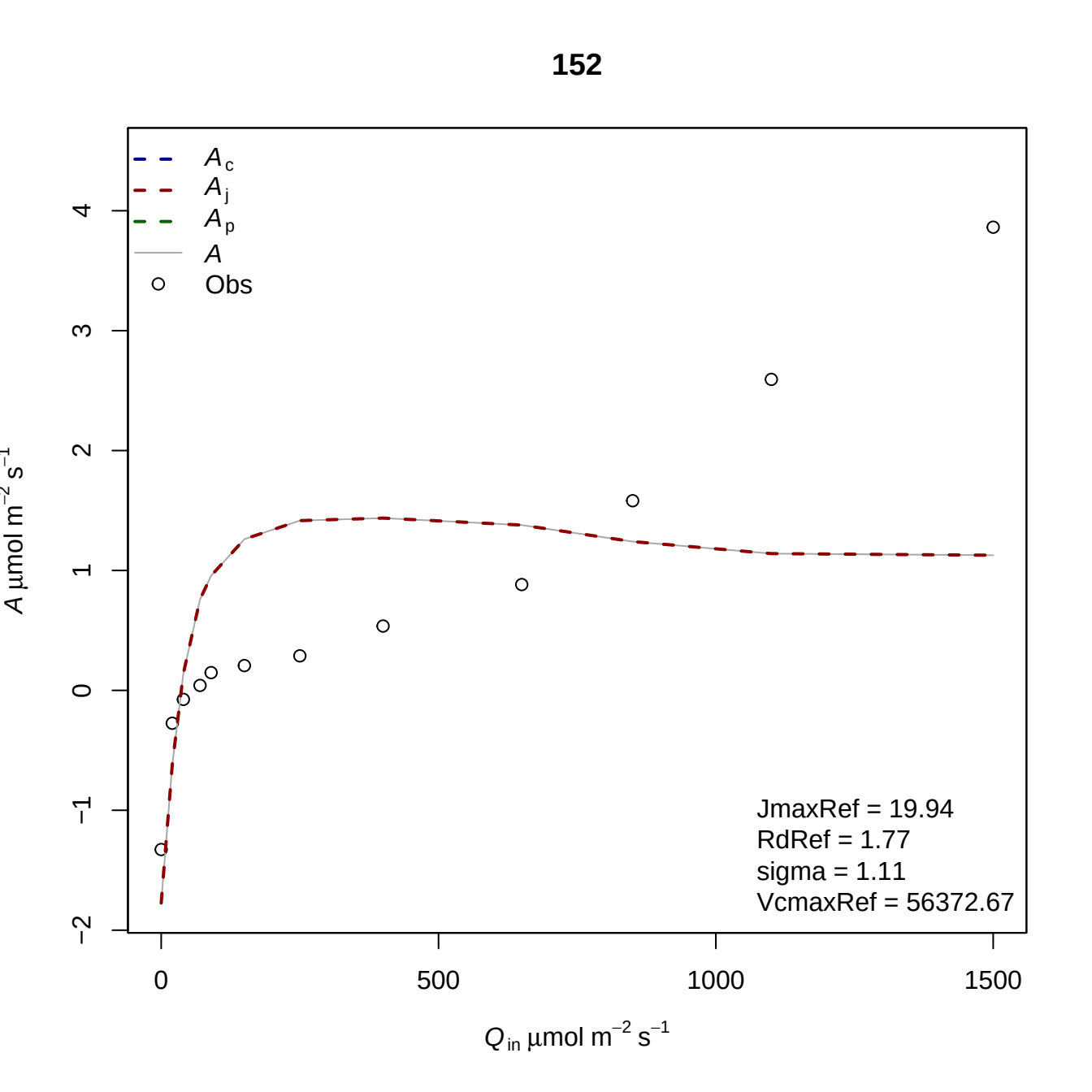

A_c
A_j
A_p
A
Obs
JmaxRef = 54.94
RdRef = 2.36
sigma = 0.43
VcmaxRef = 56.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
200
400
600
800
1000
1200
1400

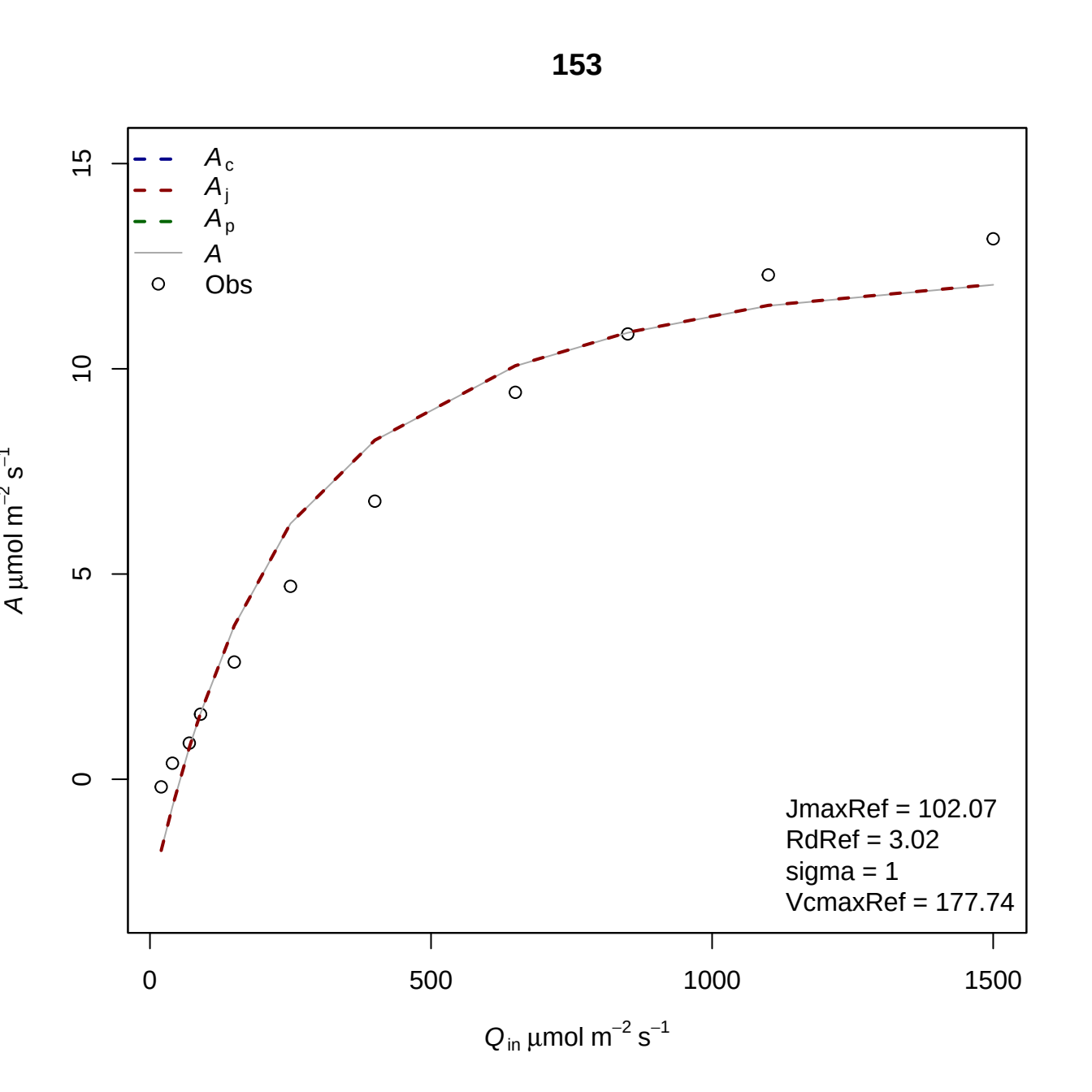

A_c
A_j
A_p
A
Obs
JmaxRef = 57.24
RdRef = 1.48
sigma = 0.64
VcmaxRef = 40.78
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

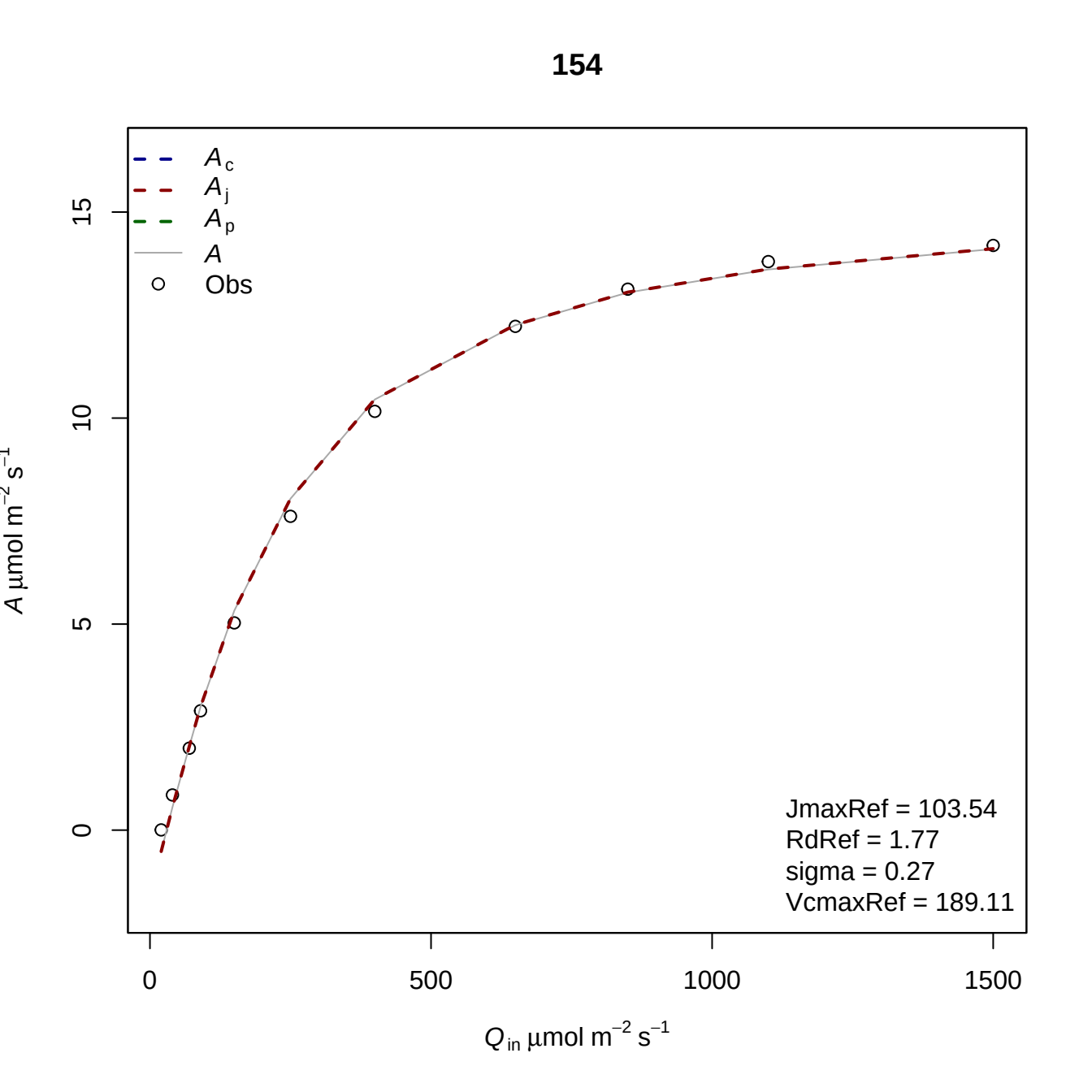

A_c
A_j
A_p
A
Obs
JmaxRef = 39.22
RdRef = 0.88
sigma = 0.11
VcmaxRef = 24.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

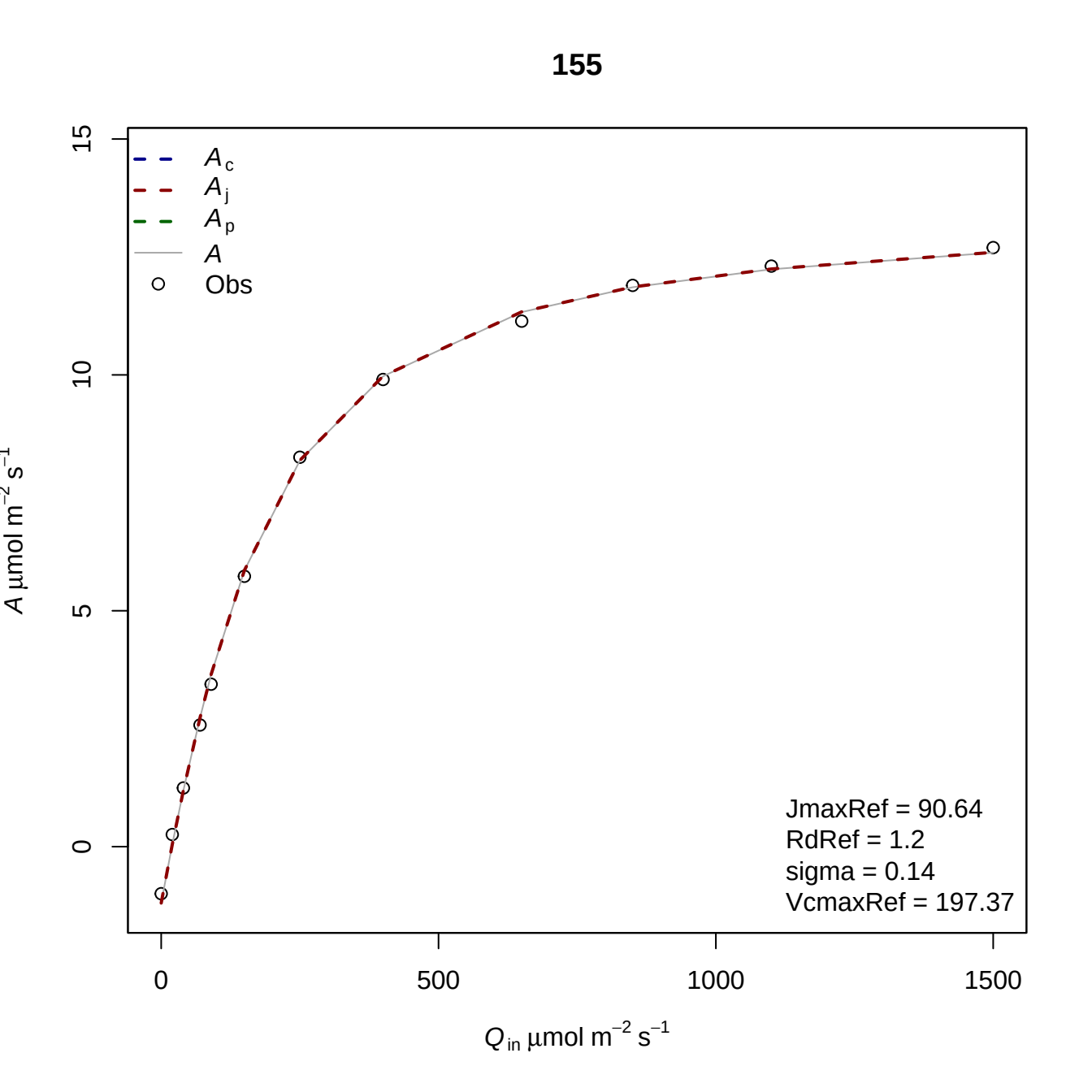

A_c
A_j
A_p
A
Obs
JmaxRef = 33.74
RdRef = 0.57
sigma = 0.06
VcmaxRef = 21.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

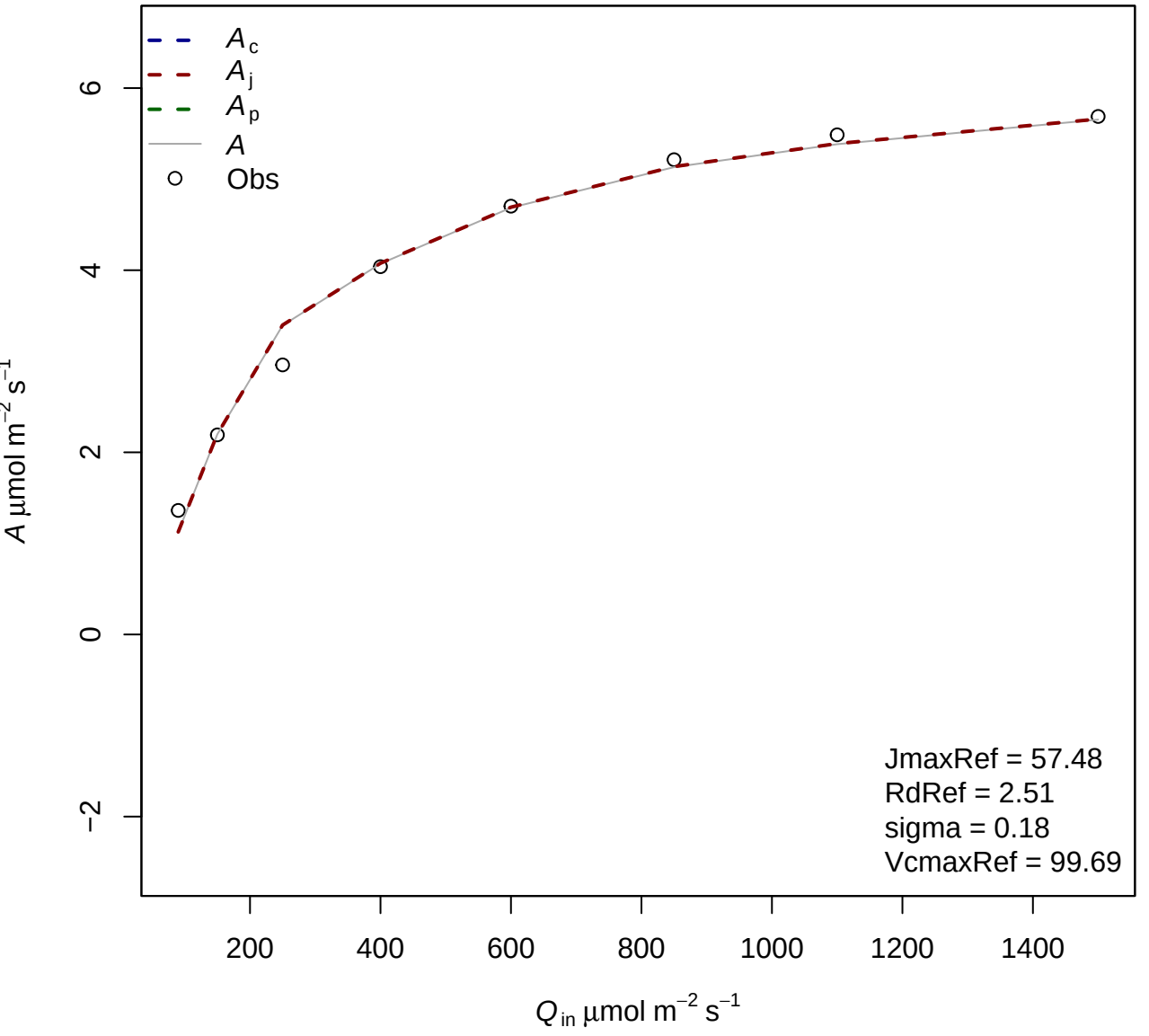

A_c
A_j
A_p
A
Obs
JmaxRef = 29.16
RdRef = 0.56
sigma = 0.11
VcmaxRef = 18.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

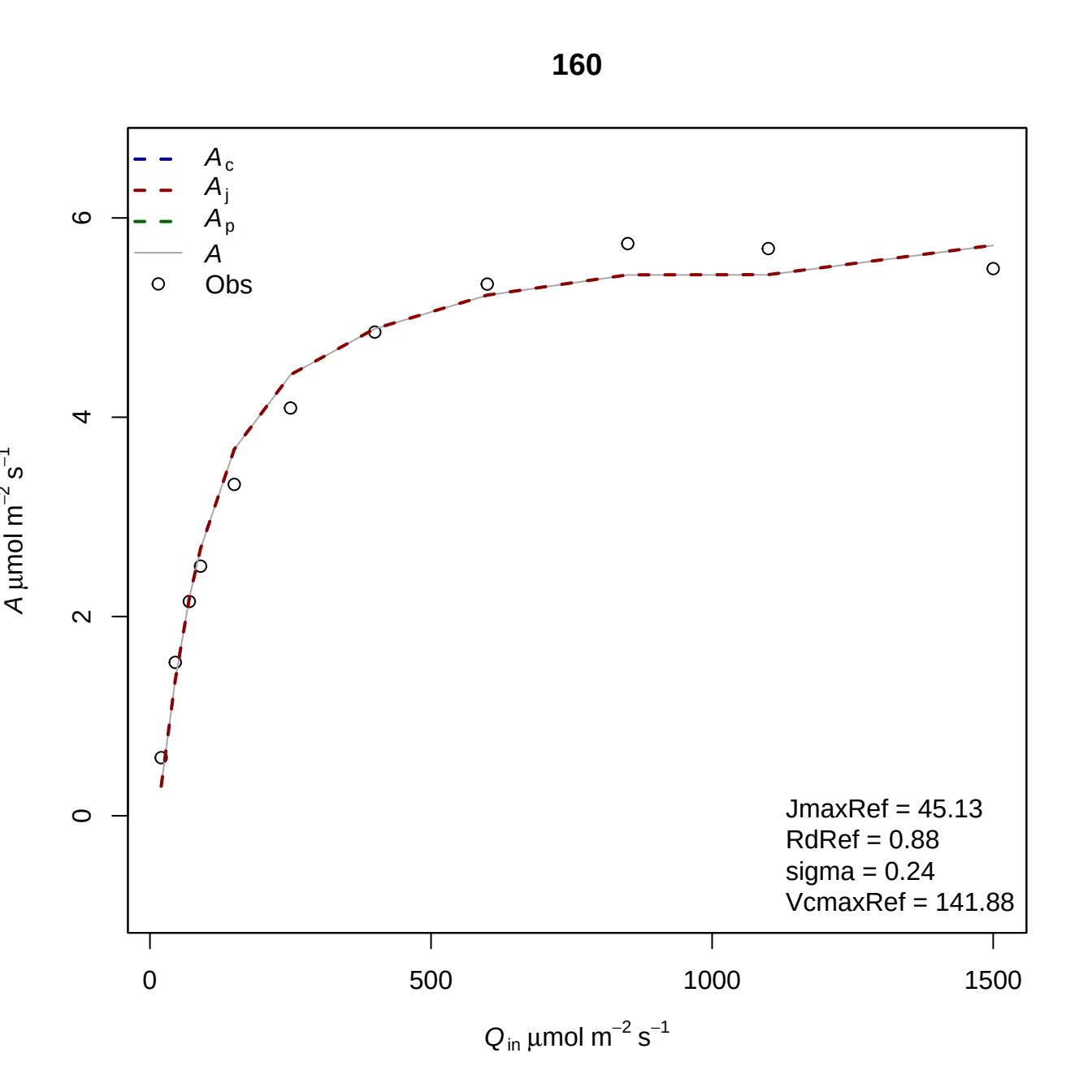

152
A_c
A_j
A_p
A
Obs
JmaxRef = 19.94
RdRef = 1.77
sigma = 1.11
VcmaxRef = 56372.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
500
1000
1500

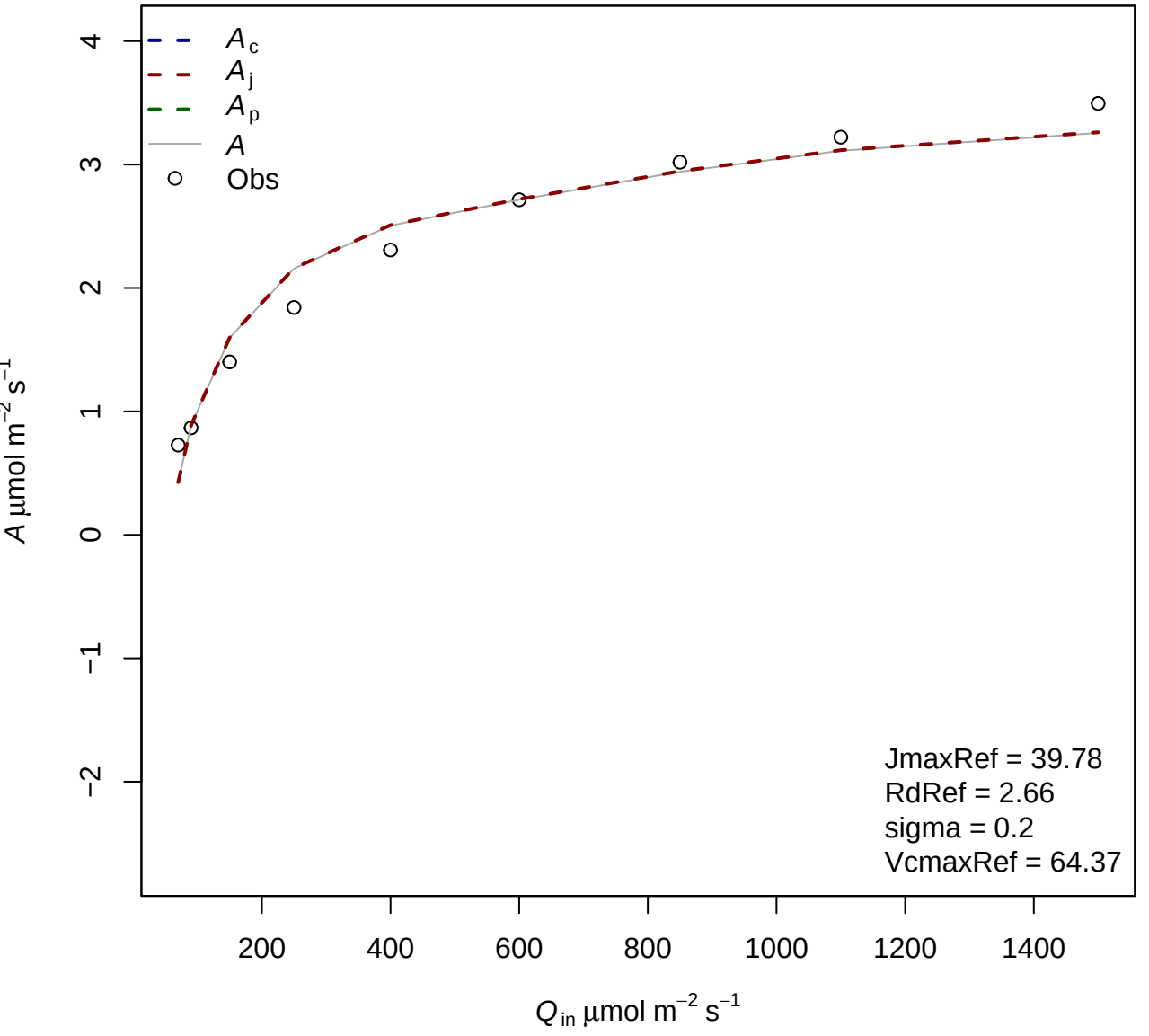

153
A_c
A_j
A_p
A
Obs
JmaxRef = 102.07
RdRef = 3.02
sigma = 1
VcmaxRef = 177.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

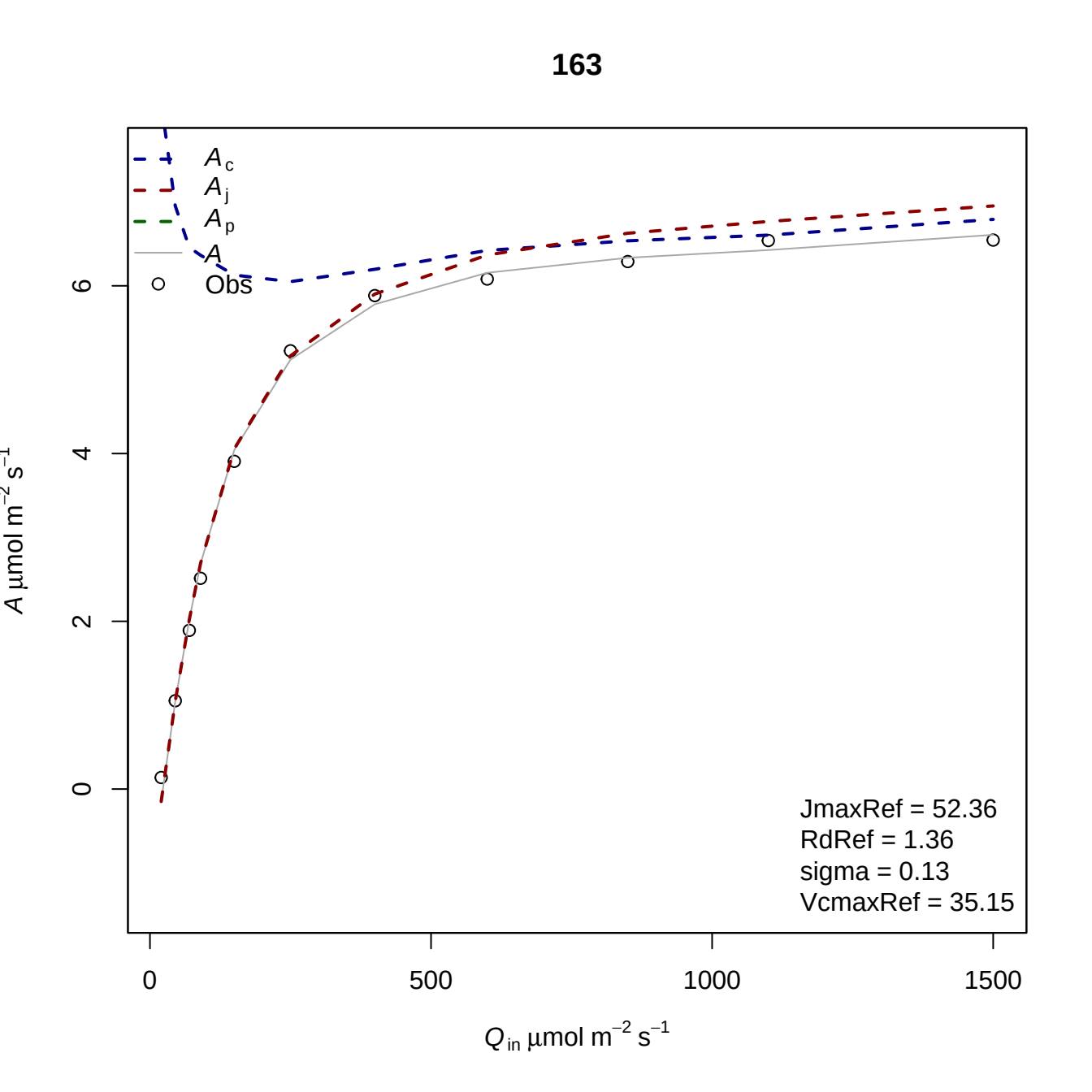

154
A_c
A_j
A_p
A
Obs
JmaxRef = 103.54
RdRef = 1.77
sigma = 0.27
VcmaxRef = 189.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

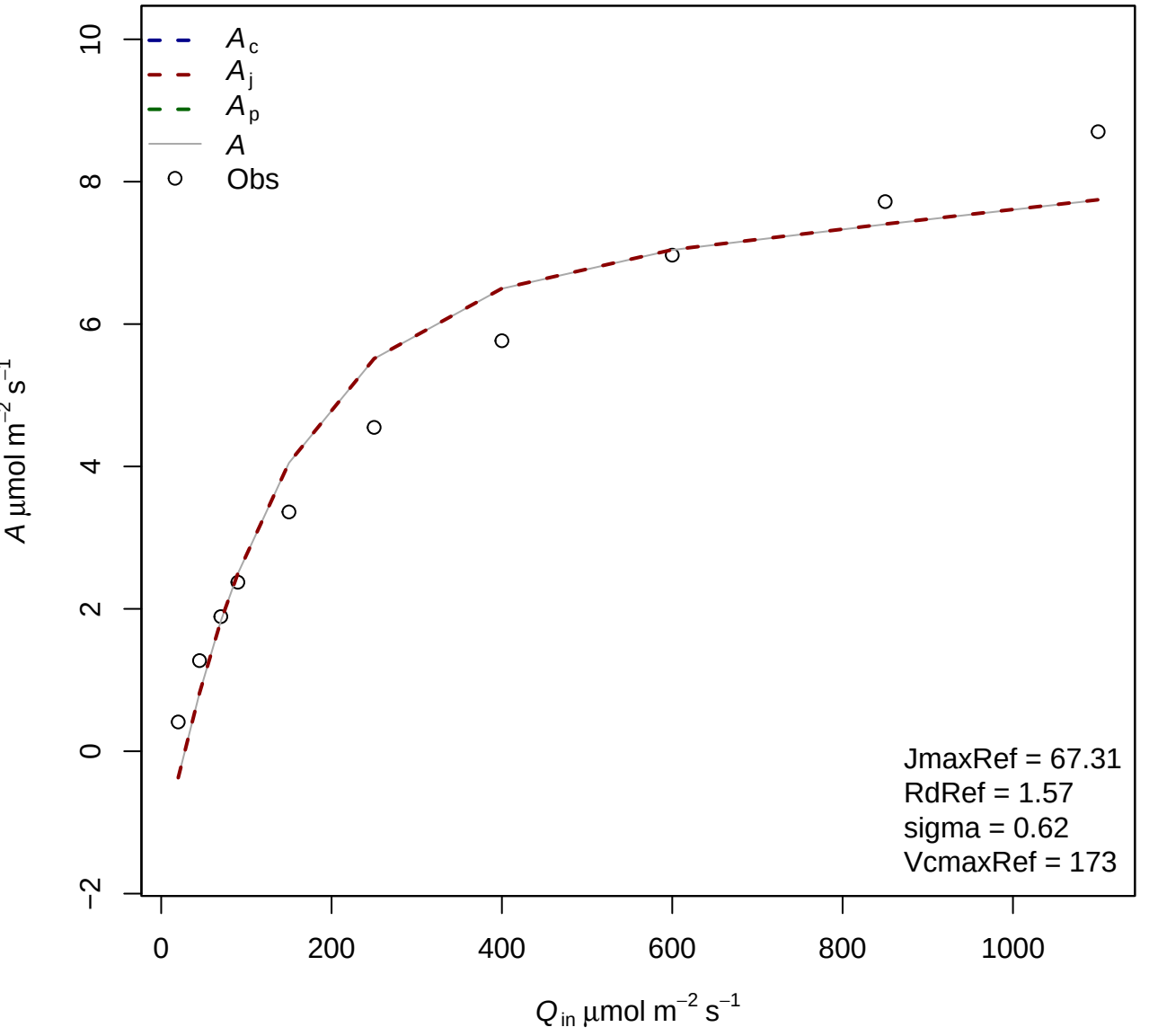

155
A_c
A_j
A_p
A
Obs
JmaxRef = 90.64
RdRef = 1.2
sigma = 0.14
VcmaxRef = 197.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

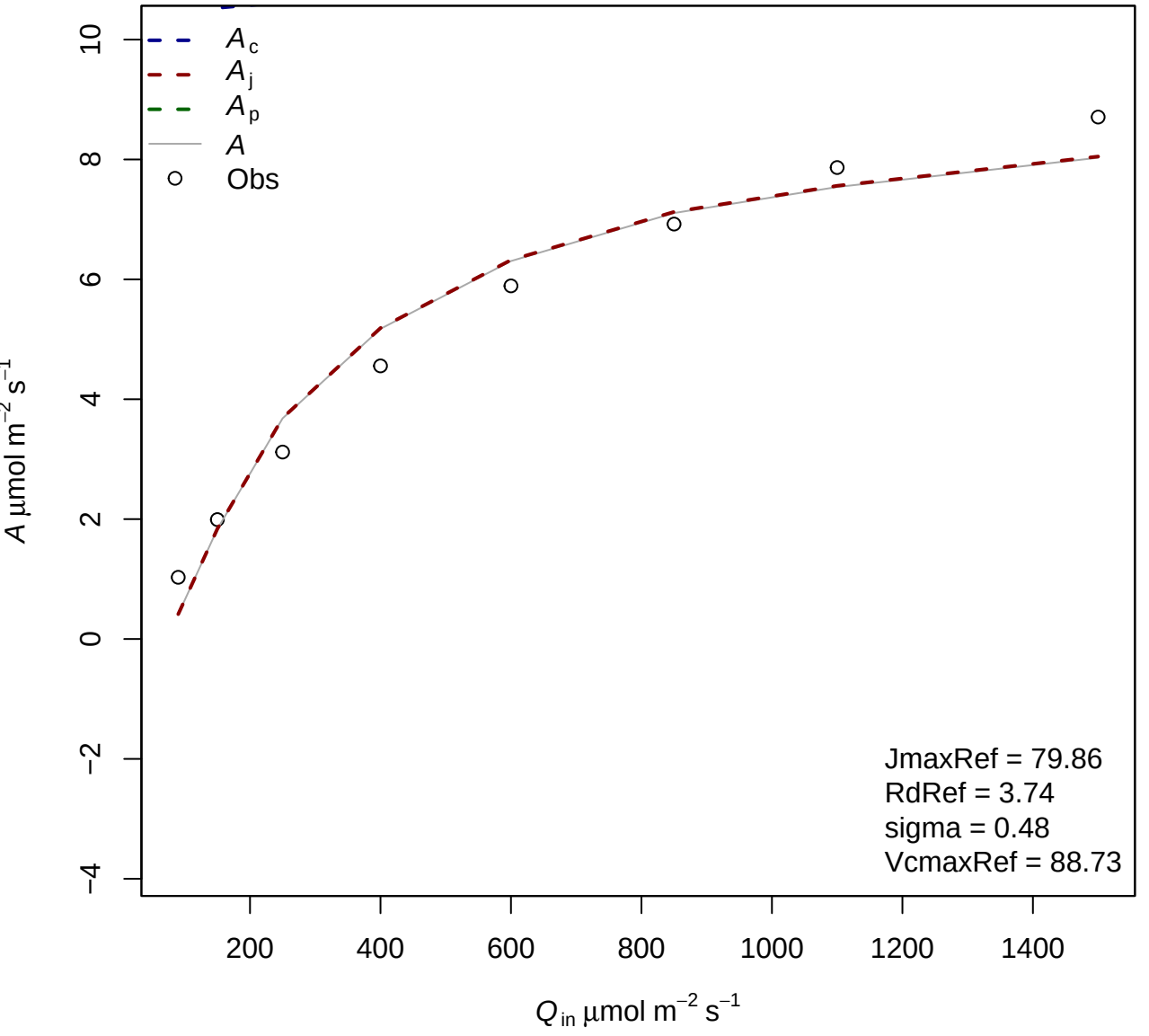

A_c
A_j
A_p
A
Obs
JmaxRef = 57.48
RdRef = 2.51
sigma = 0.18
VcmaxRef = 99.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
200
400
600
800
1000
1200
1400

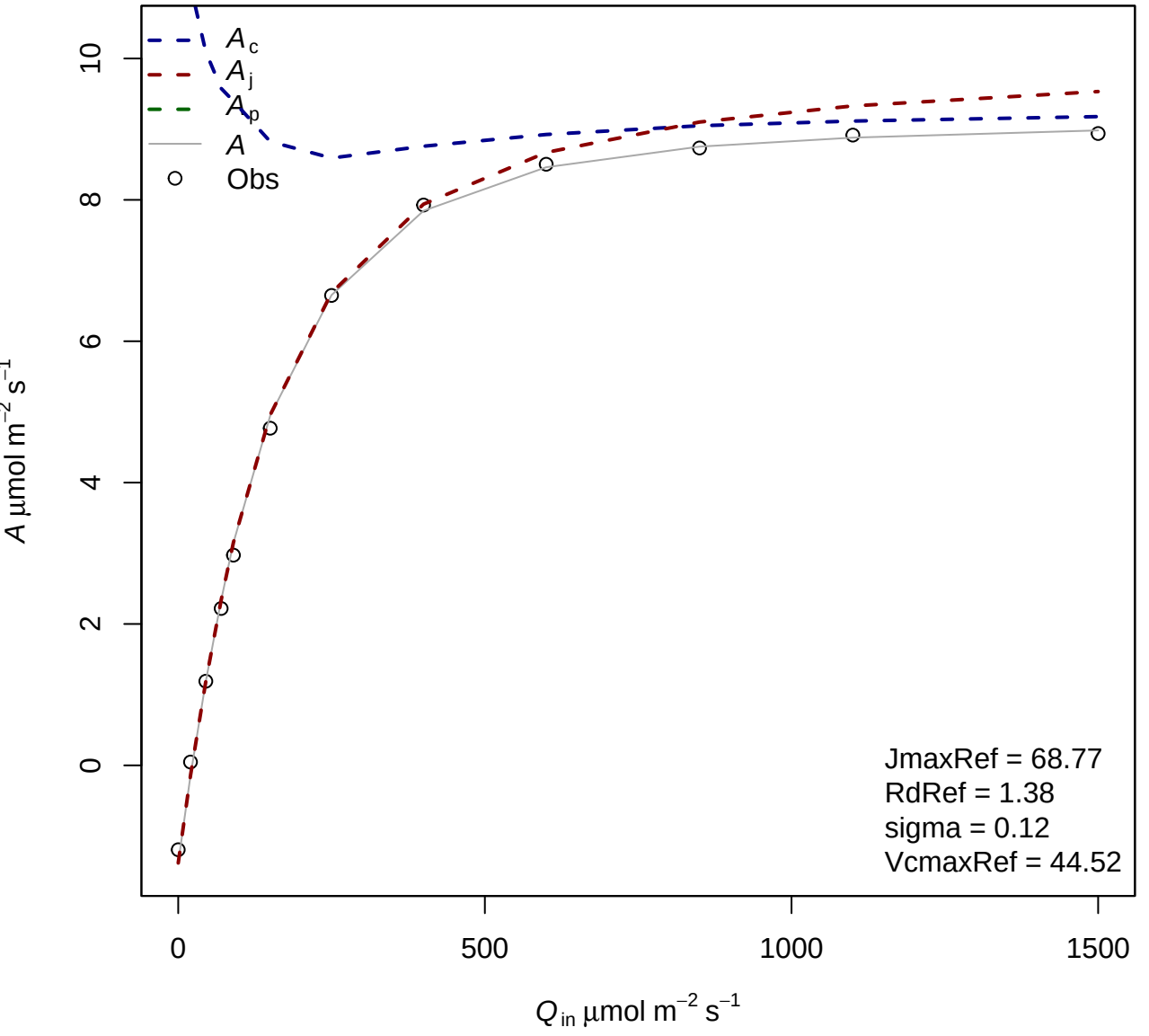

160
A_c
A_j
A_p
A
Obs
JmaxRef = 45.13
RdRef = 0.88
sigma = 0.24
VcmaxRef = 141.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

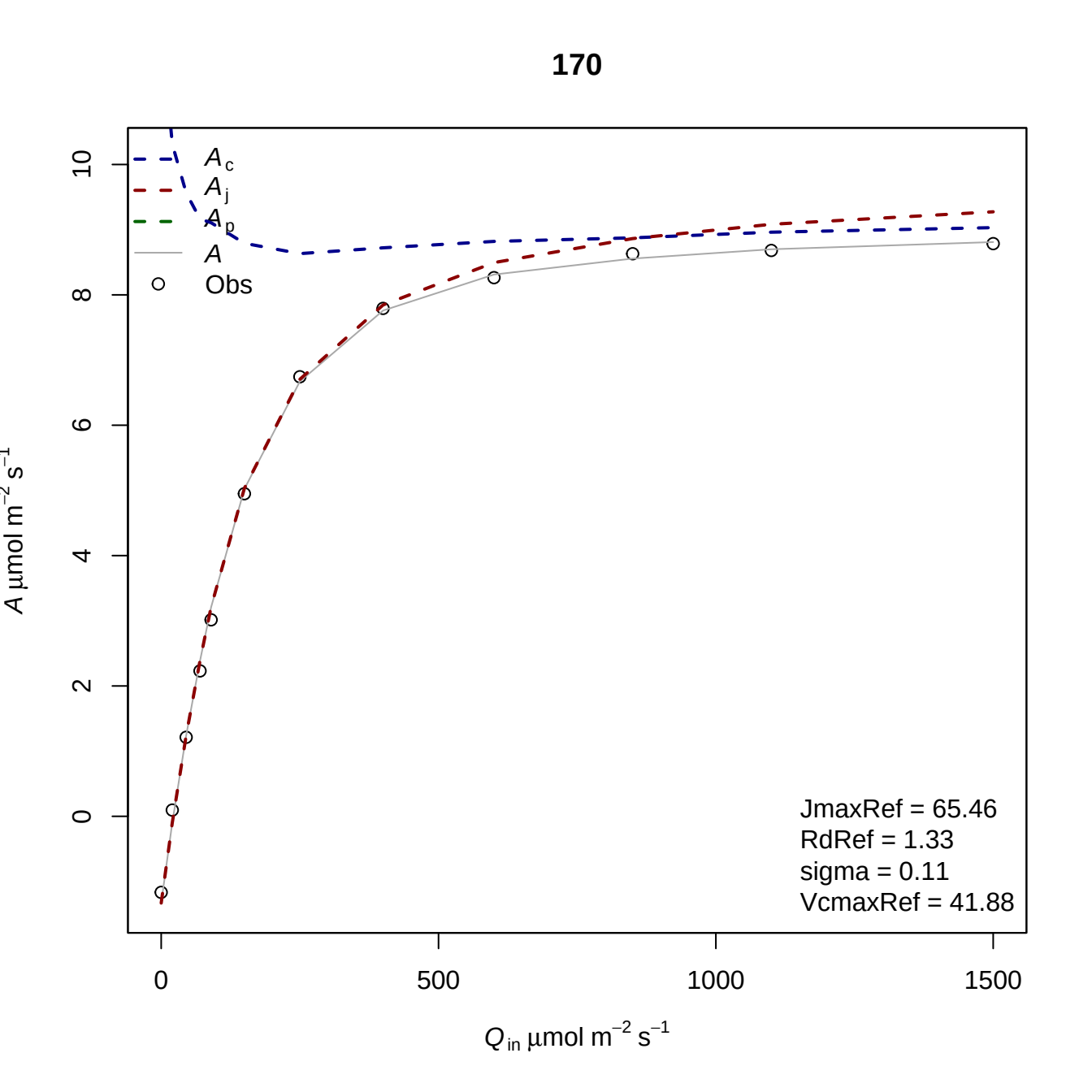

A_c
A_j
A_p
A
Obs
JmaxRef = 39.78
RdRef = 2.66
sigma = 0.2
VcmaxRef = 64.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
200
400
600
800
1000
1200
1400

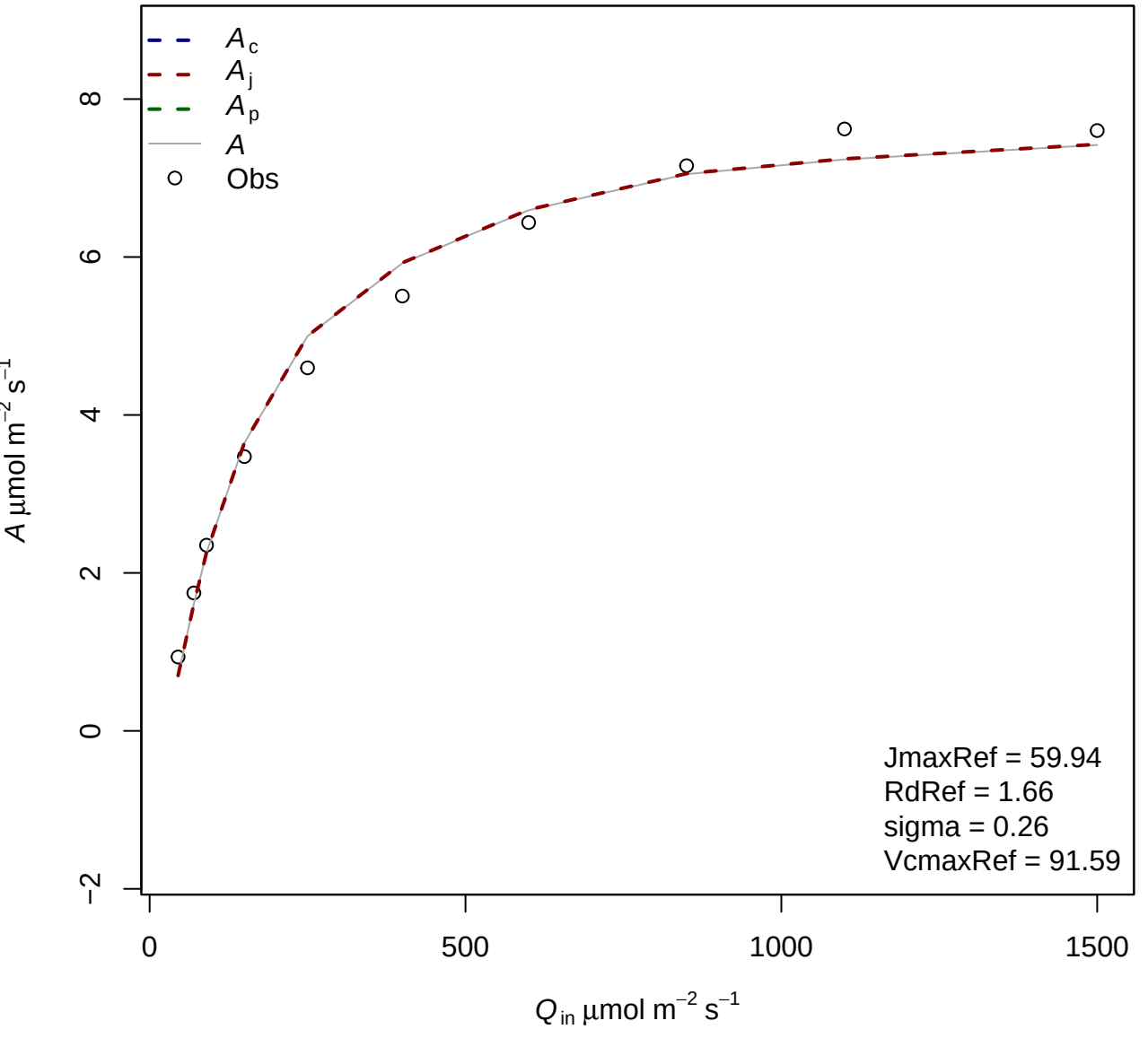

163
A_c
A_j
A_p
A
Obs
JmaxRef = 52.36
RdRef = 1.36
sigma = 0.13
VcmaxRef = 35.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

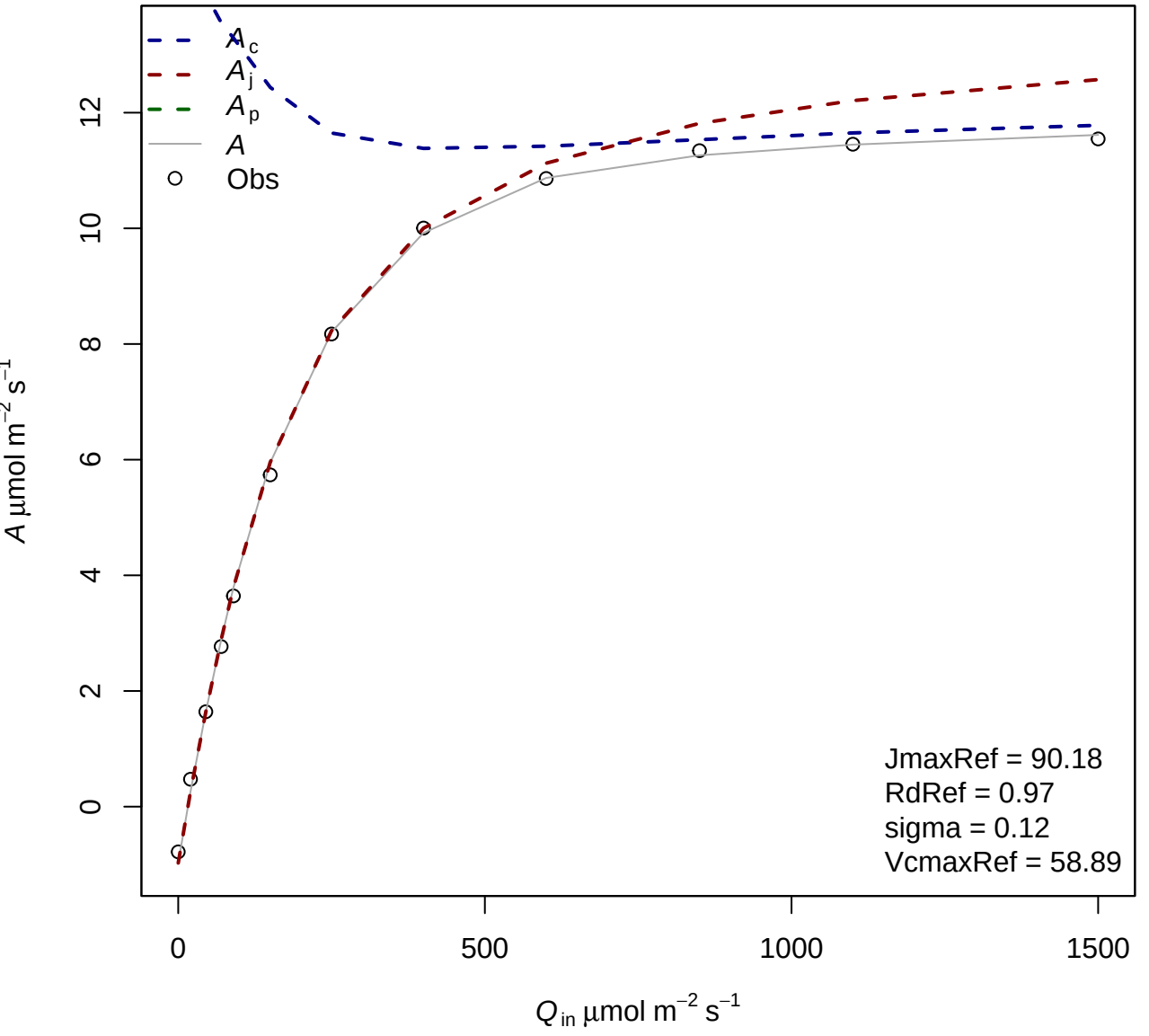

A_c
A_j
A_p
A
Obs
JmaxRef = 67.31
RdRef = 1.57
sigma = 0.62
VcmaxRef = 173
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000

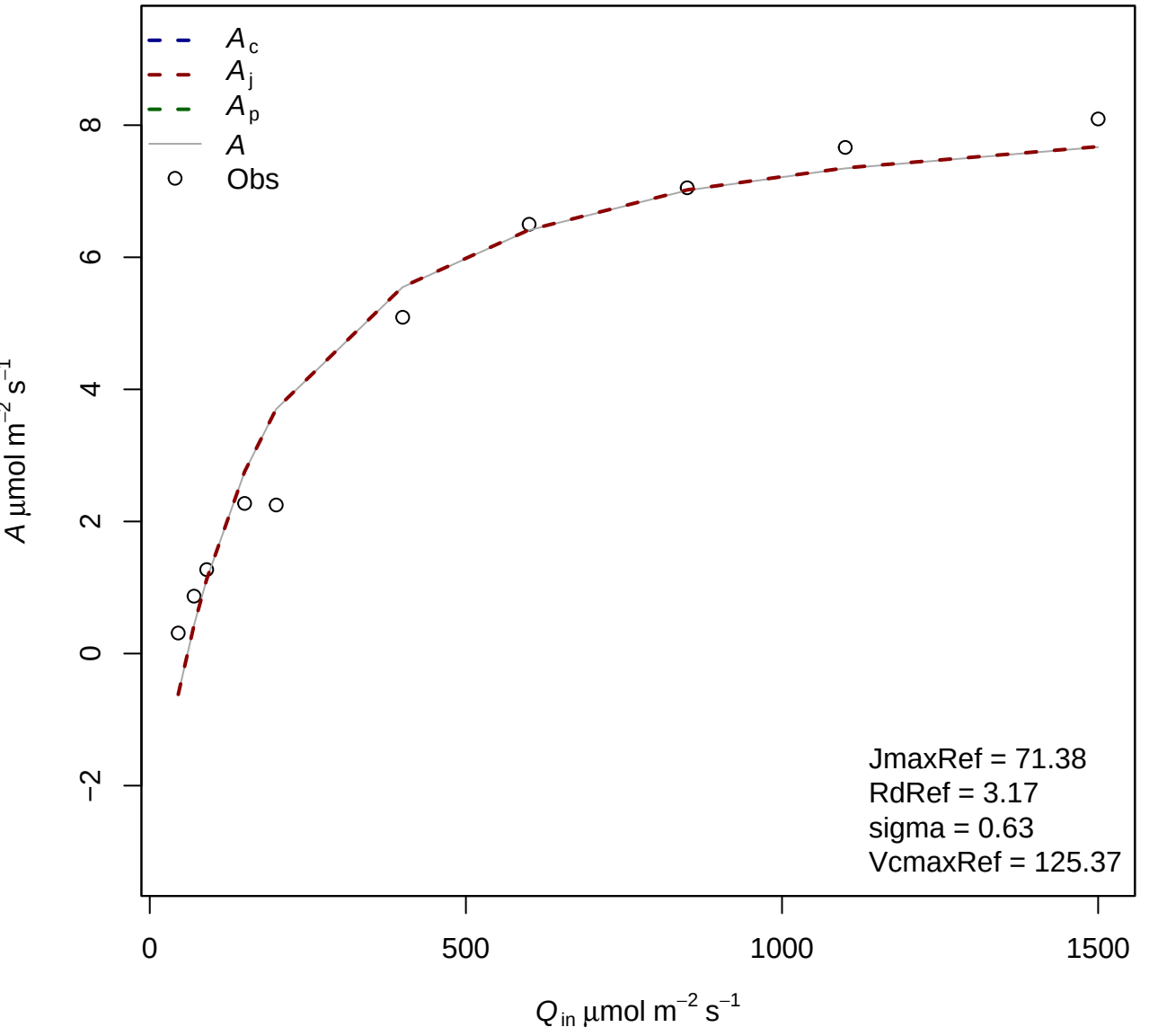

A_c
A_j
A_p
A
Obs
JmaxRef = 79.86
RdRef = 3.74
sigma = 0.48
VcmaxRef = 88.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

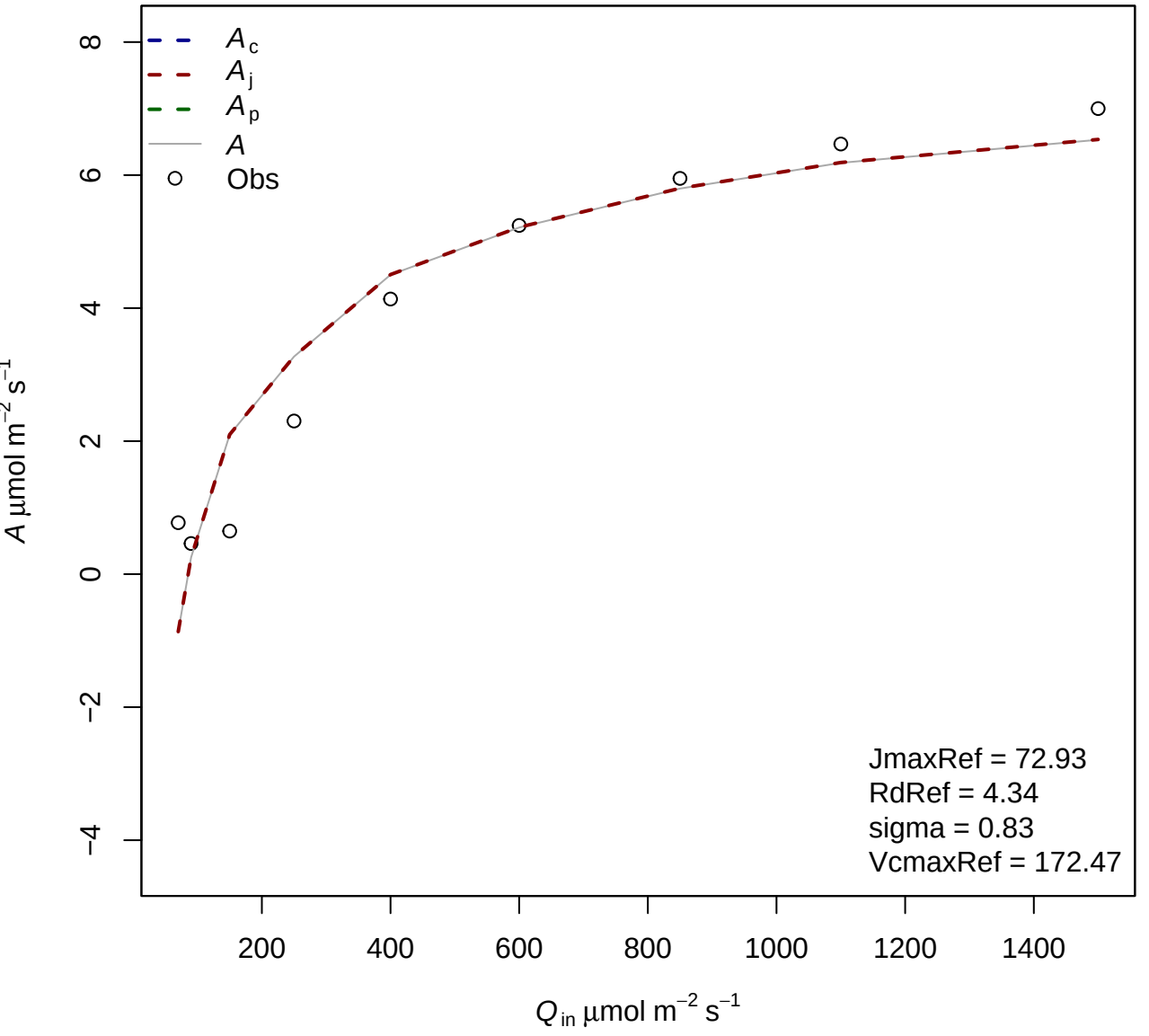

A_c
A_j
A_p
A
Obs
JmaxRef = 68.77
RdRef = 1.38
sigma = 0.12
VcmaxRef = 44.52
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

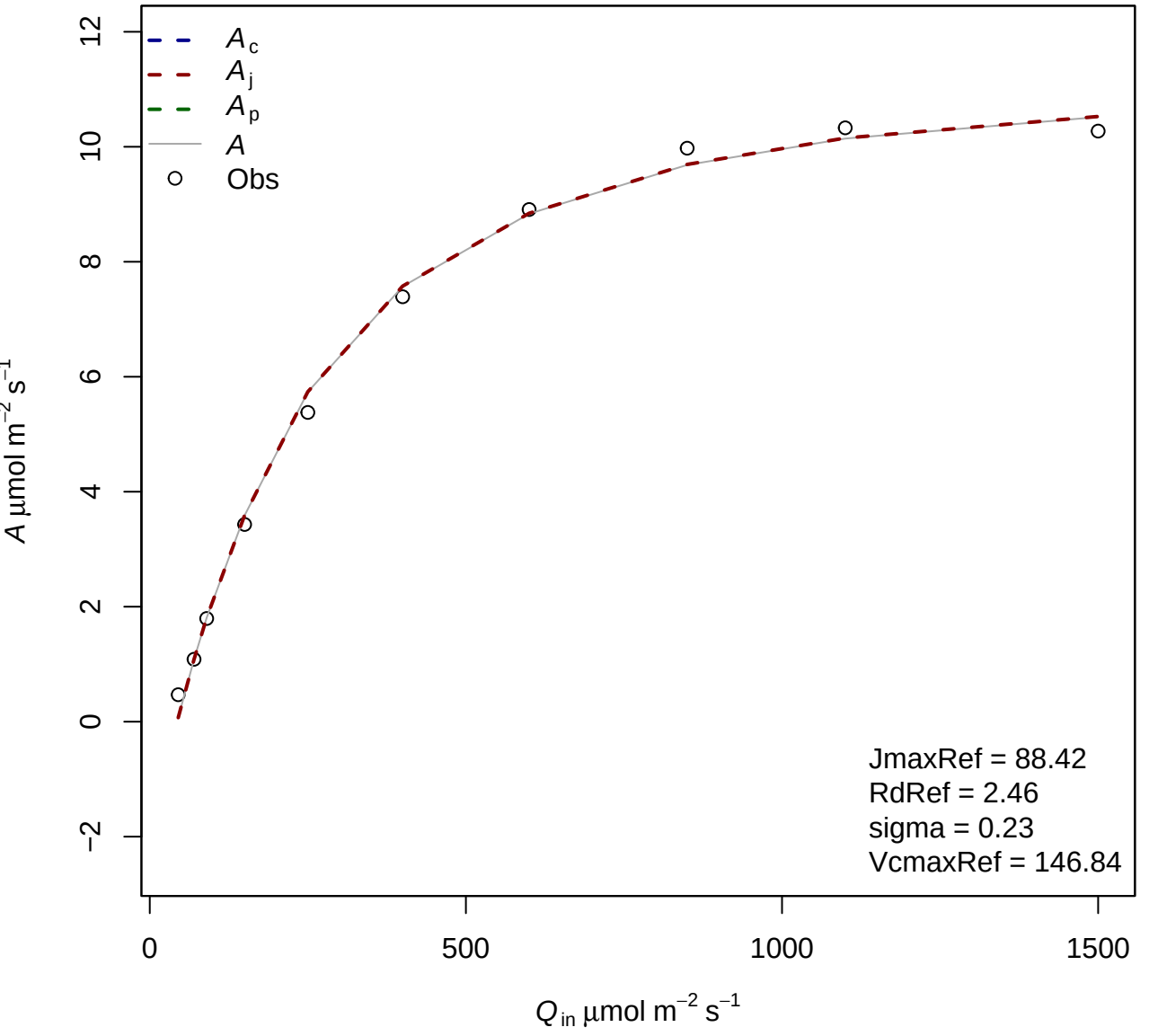

170
A_c
A_j
A_p
A
Obs
JmaxRef = 65.46
RdRef = 1.33
sigma = 0.11
VcmaxRef = 41.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

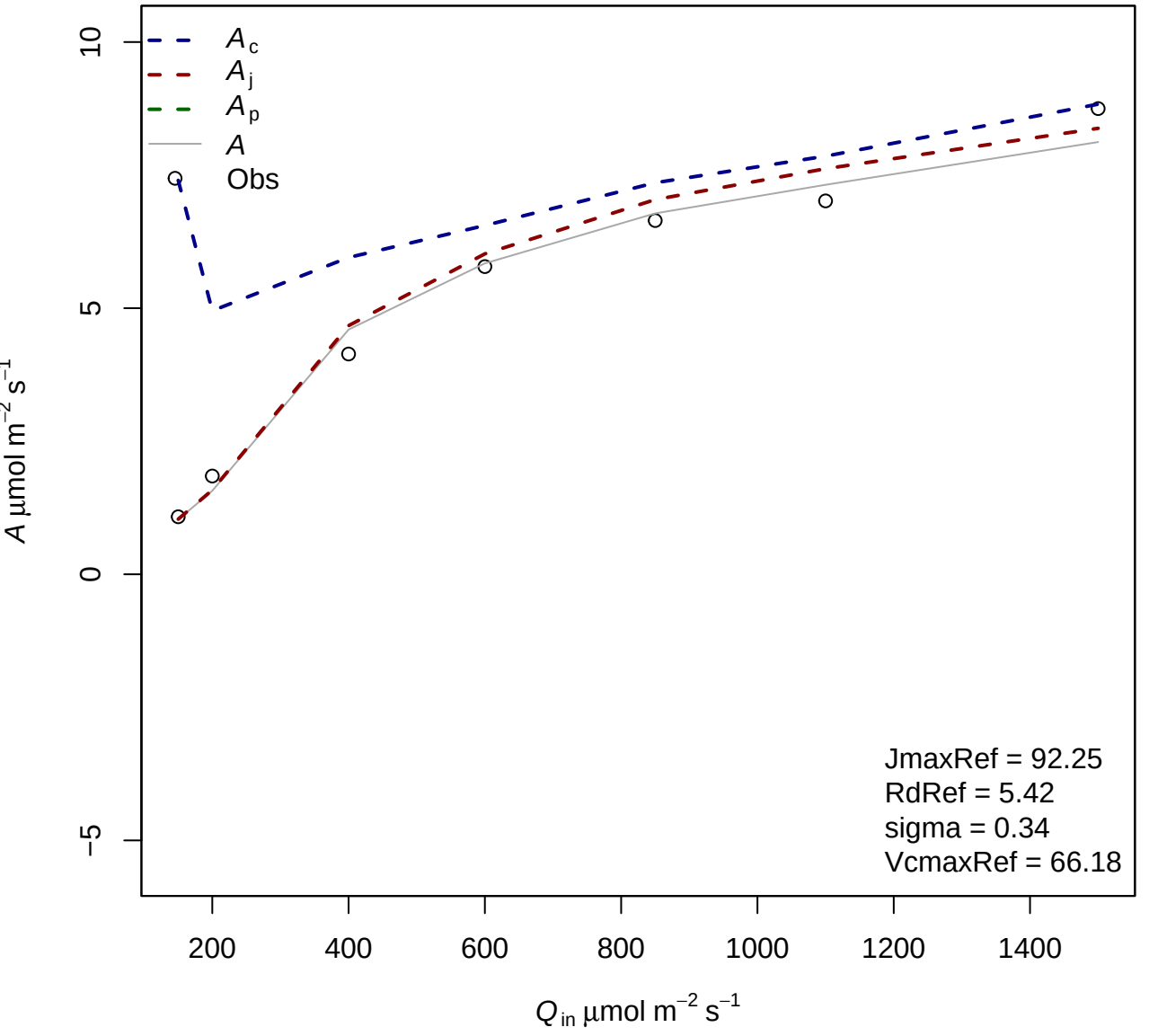

A_c
A_j
A_p
A
Obs
JmaxRef = 59.94
RdRef = 1.66
sigma = 0.26
VcmaxRef = 91.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

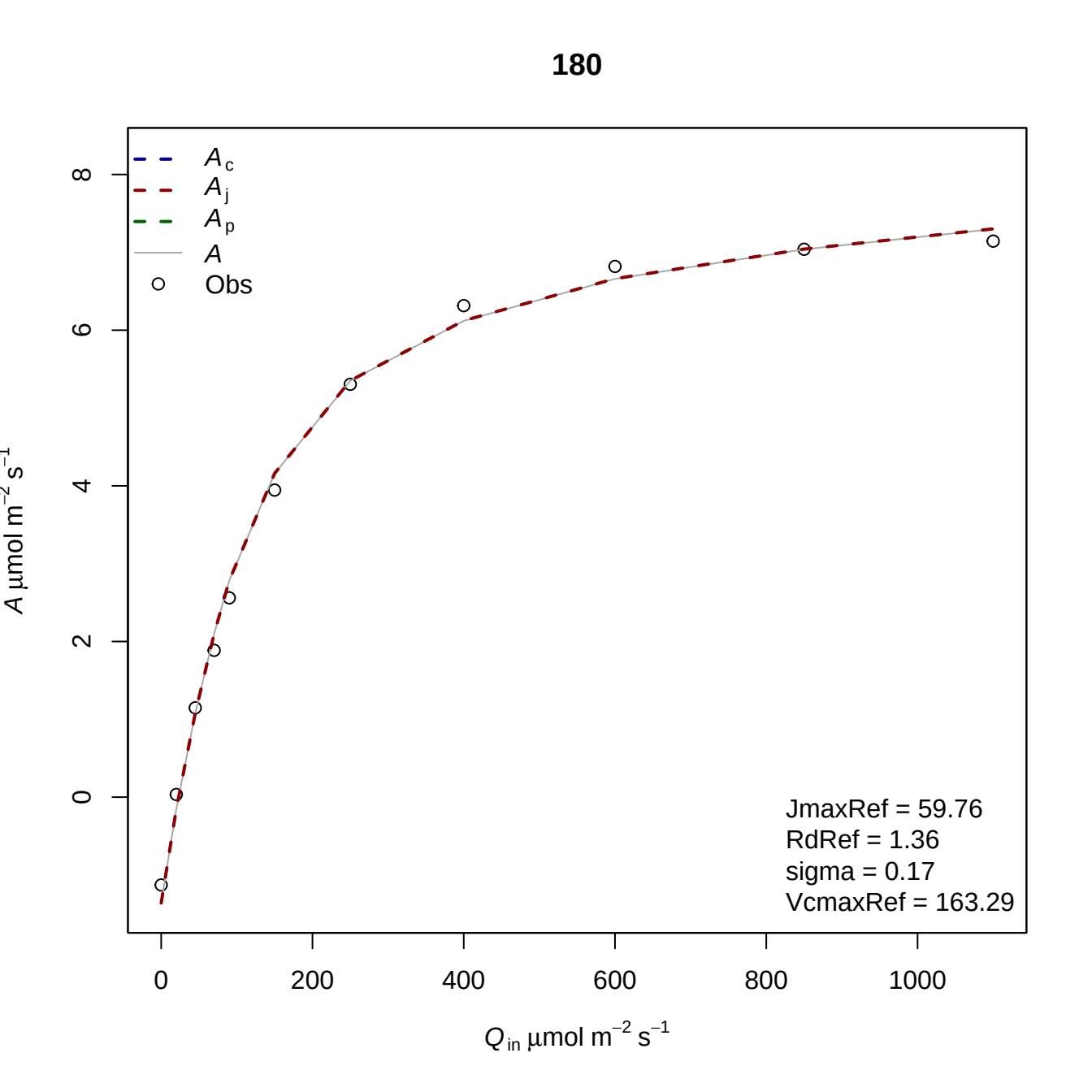

A_c
A_j
A_p
A
Obs
JmaxRef = 90.18
RdRef = 0.97
sigma = 0.12
VcmaxRef = 58.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

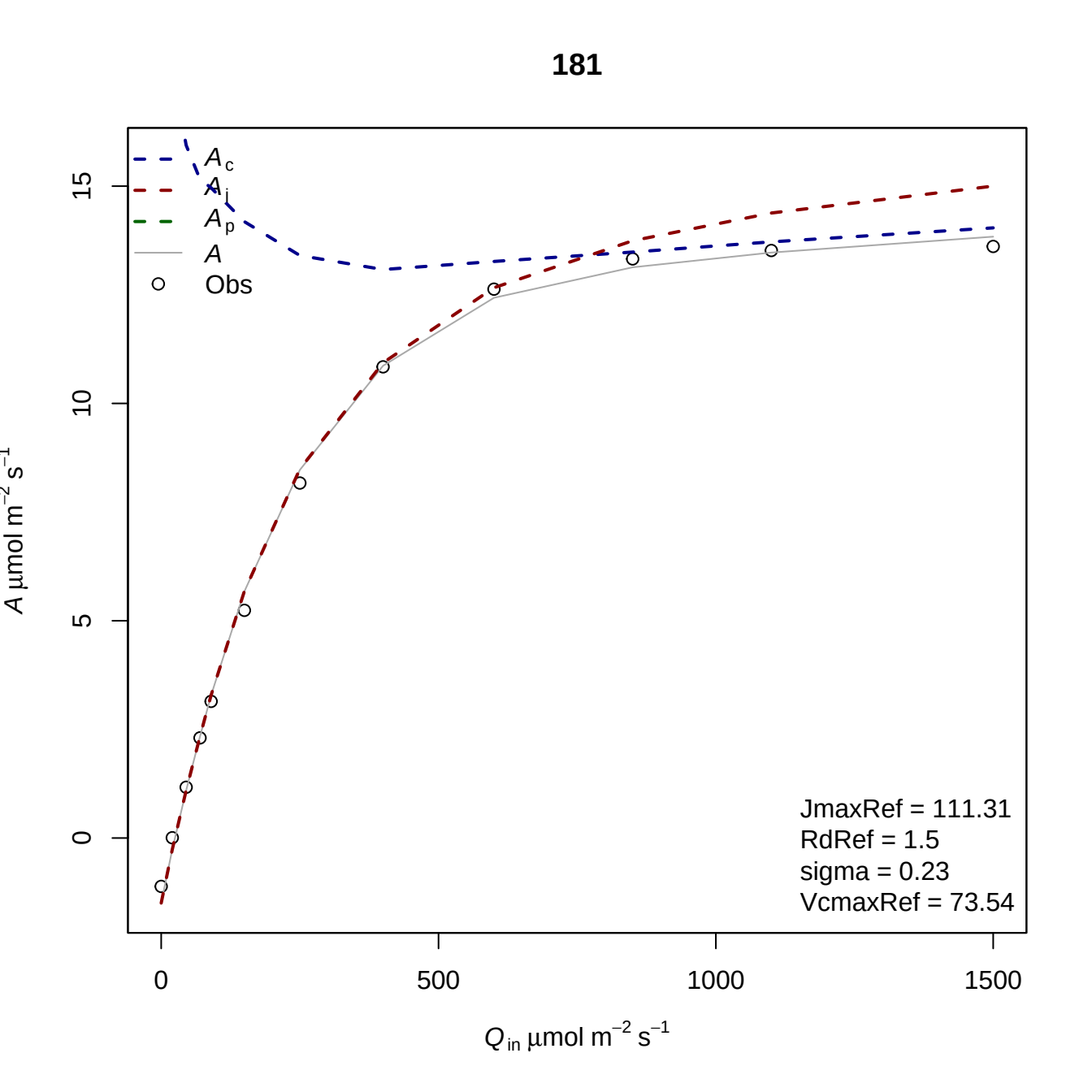

A_c
A_j
A_p
A
Obs
JmaxRef = 71.38
RdRef = 3.17
sigma = 0.63
VcmaxRef = 125.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

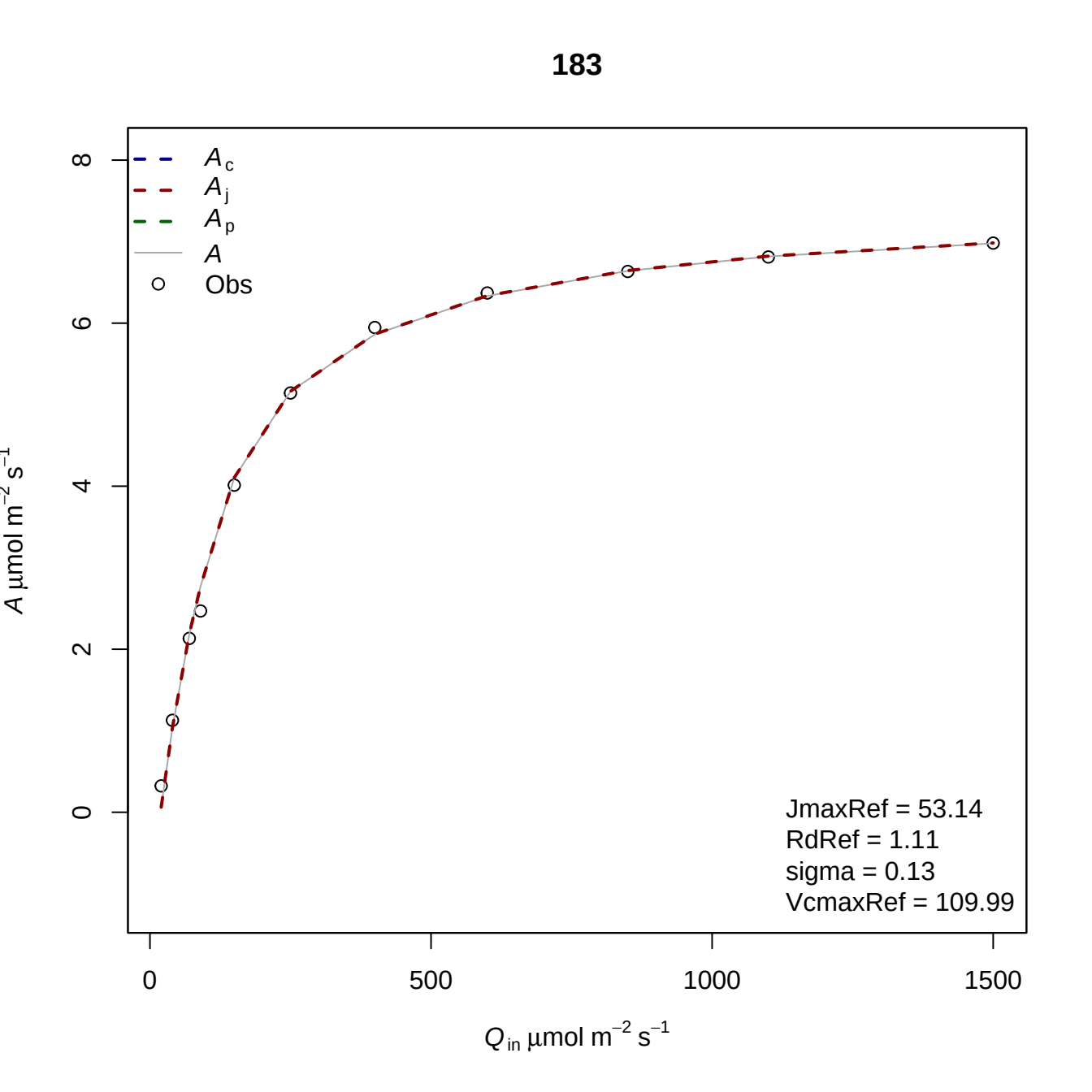

A_c
A_j
A_p
A
Obs
JmaxRef = 72.93
RdRef = 4.34
sigma = 0.83
VcmaxRef = 172.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

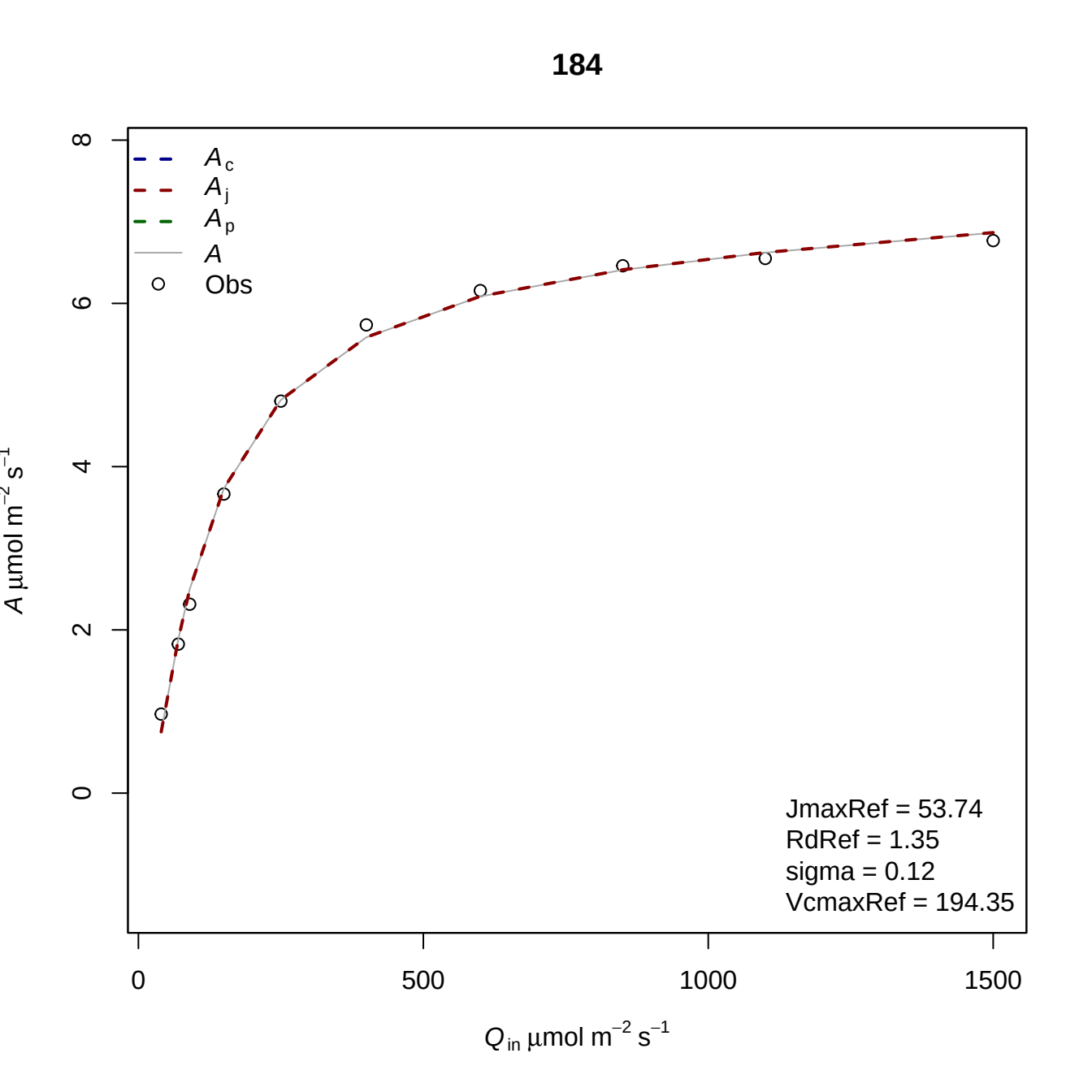

A_c
A_j
A_p
A
Obs
JmaxRef = 88.42
RdRef = 2.46
sigma = 0.23
VcmaxRef = 146.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

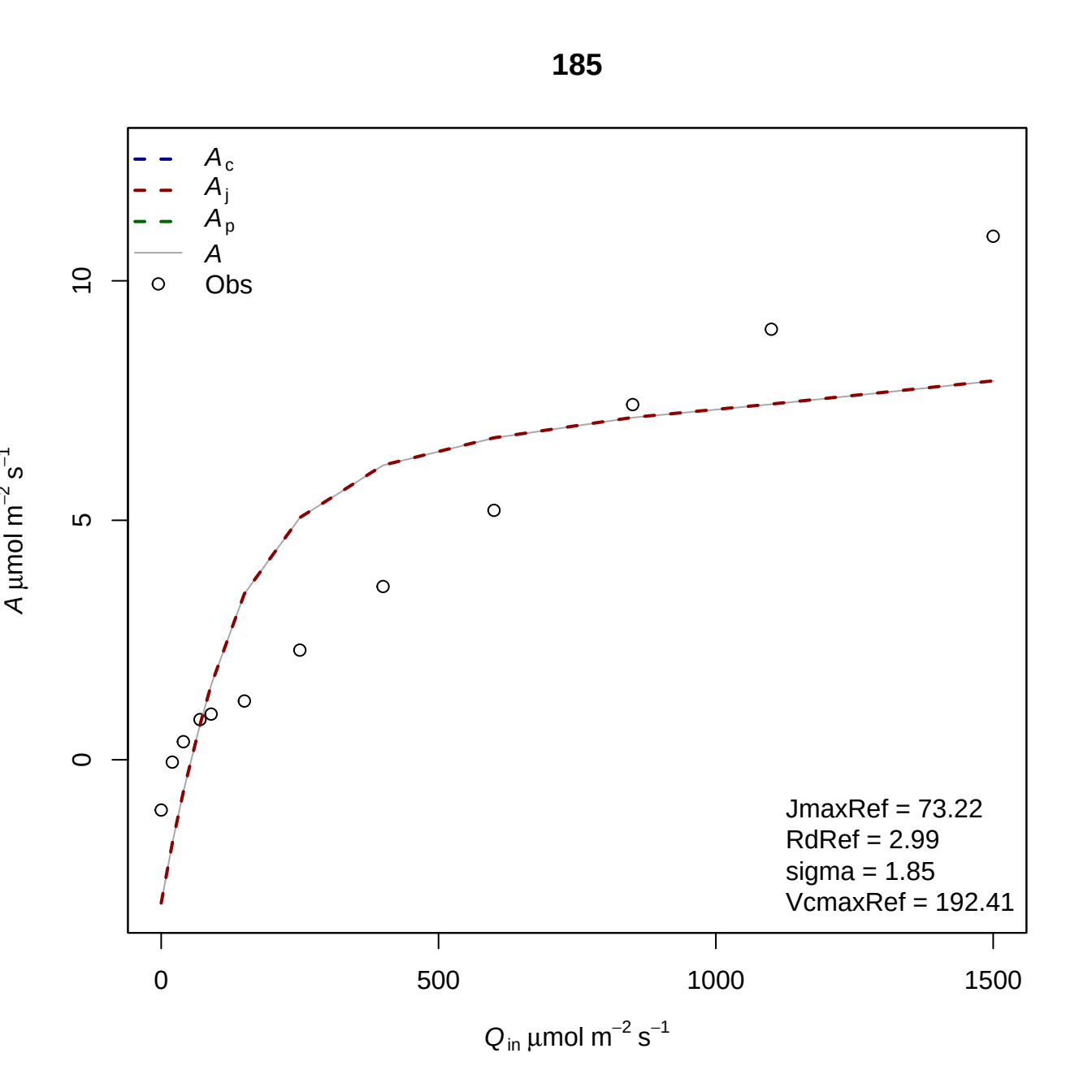

A_c
A_j
A_p
A
Obs
JmaxRef = 92.25
RdRef = 5.42
sigma = 0.34
VcmaxRef = 66.18
10
5
0
−5
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

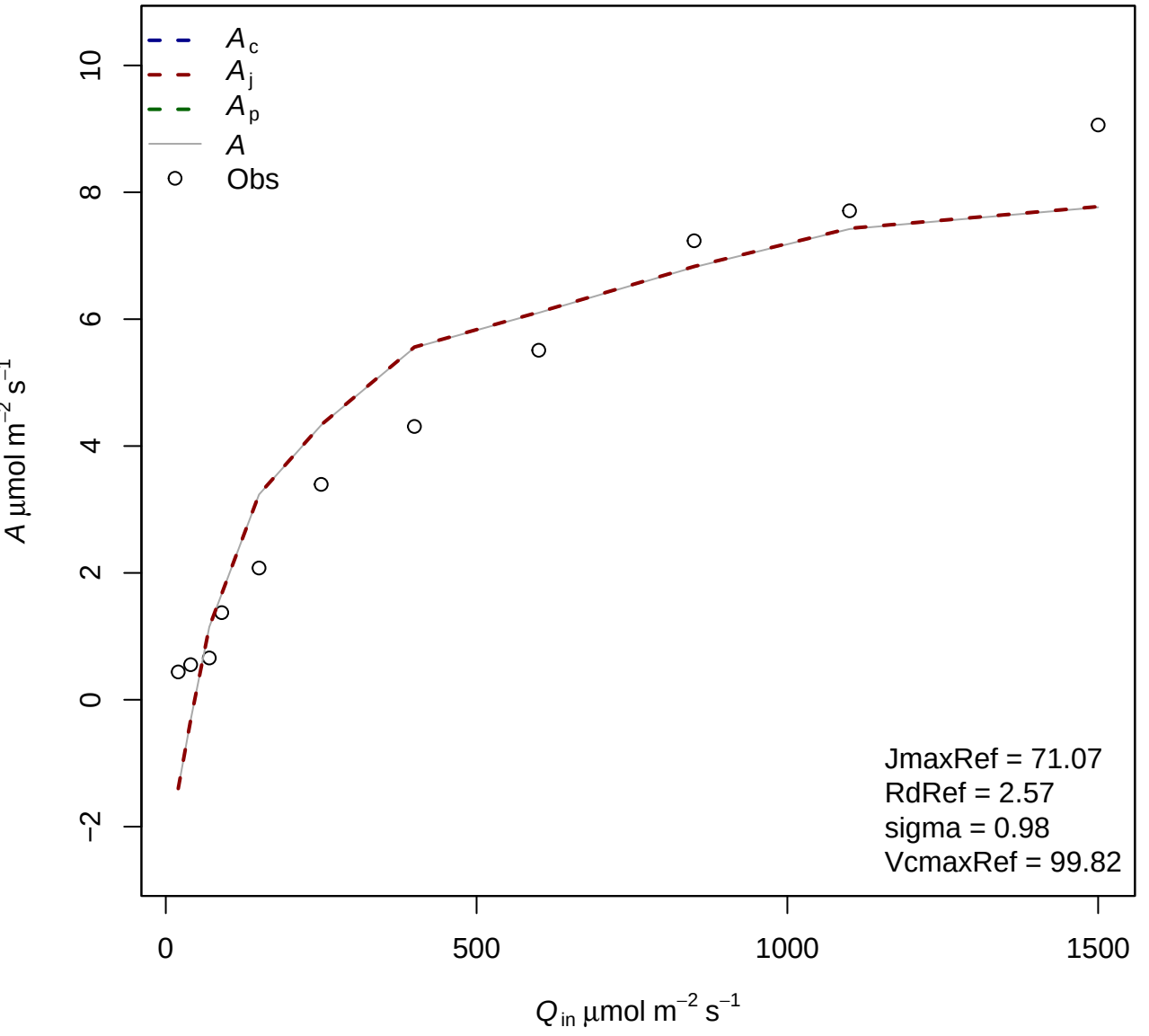

180
A_c
A_j
A_p
A
Obs
JmaxRef = 59.76
RdRef = 1.36
sigma = 0.17
VcmaxRef = 163.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
200
400
600
800
1000

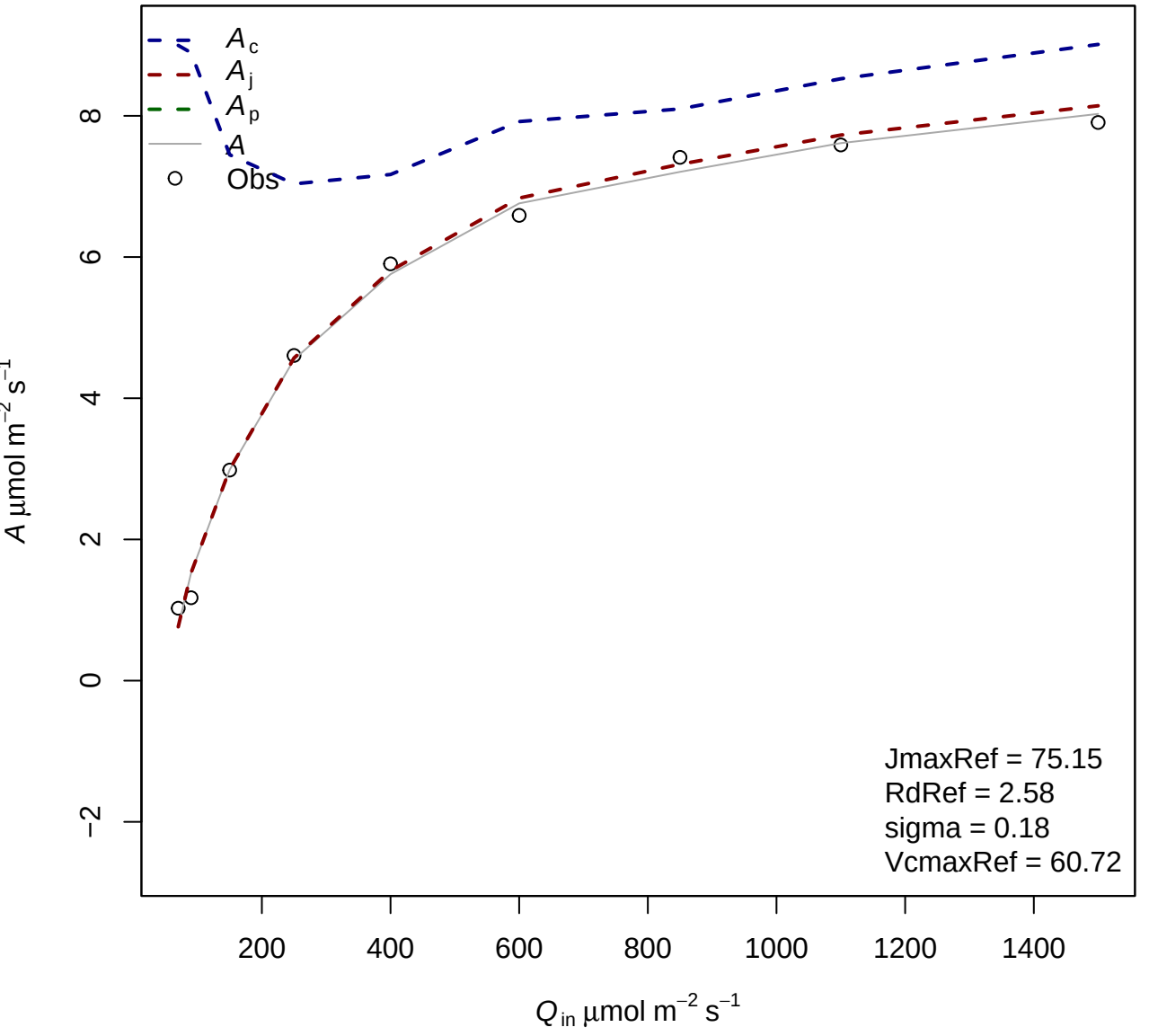

181
A_c
A_j
A_p
A
Obs
JmaxRef = 111.31
RdRef = 1.5
sigma = 0.23
VcmaxRef = 73.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

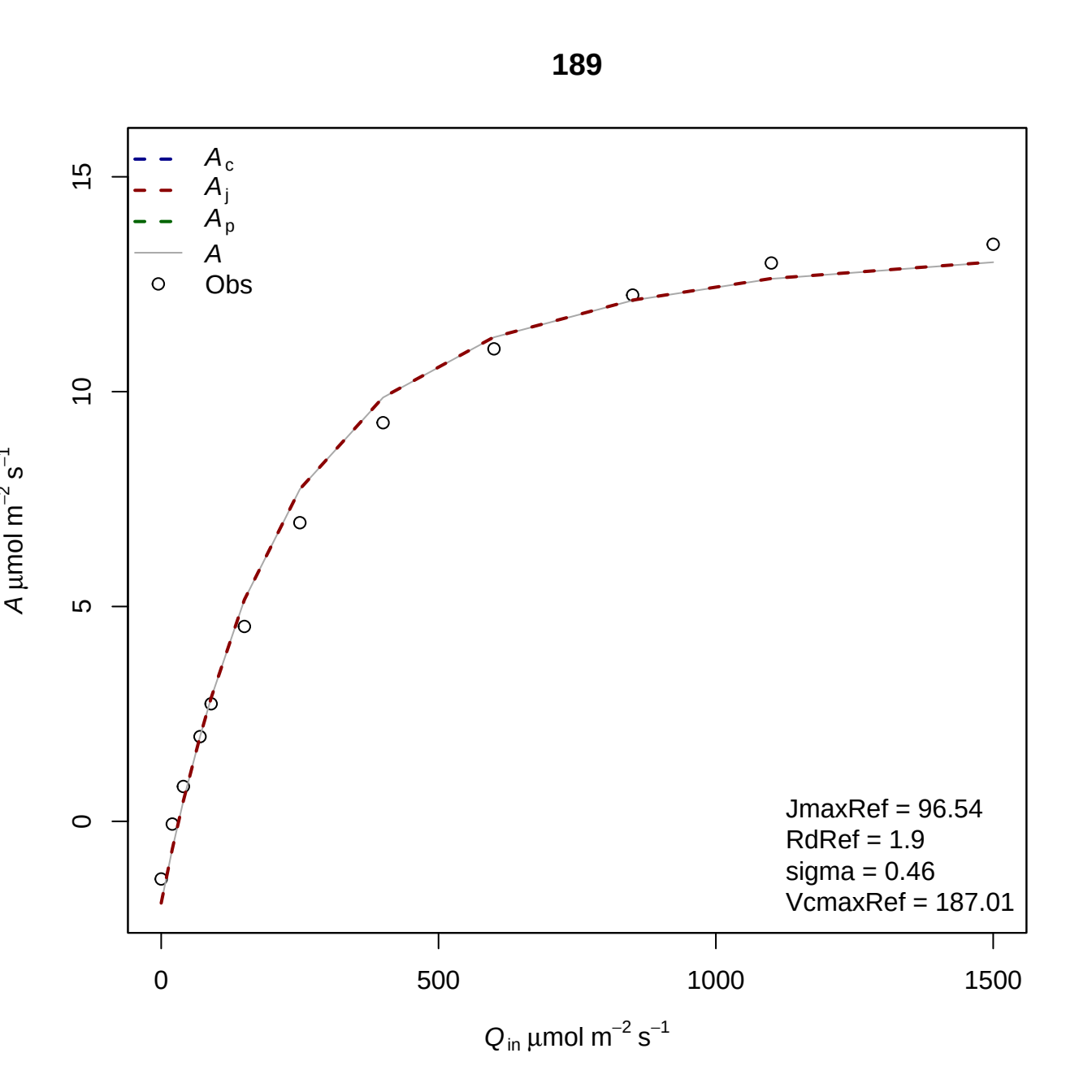

183
A_c
A_j
A_p
A
Obs
JmaxRef = 53.14
RdRef = 1.11
sigma = 0.13
VcmaxRef = 109.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

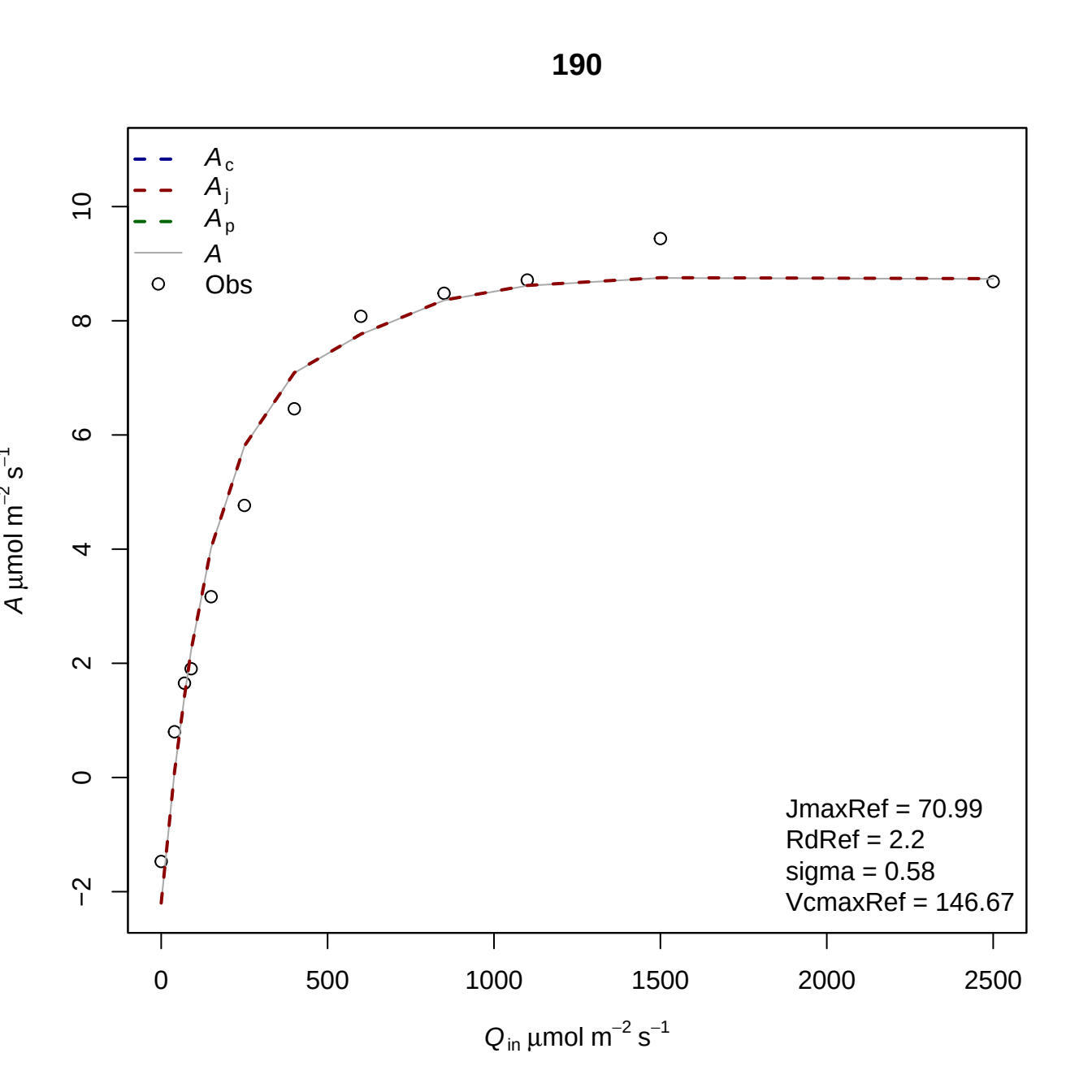

184
A_c
A_j
A_p
A
Obs
JmaxRef = 53.74
RdRef = 1.35
sigma = 0.12
VcmaxRef = 194.35
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

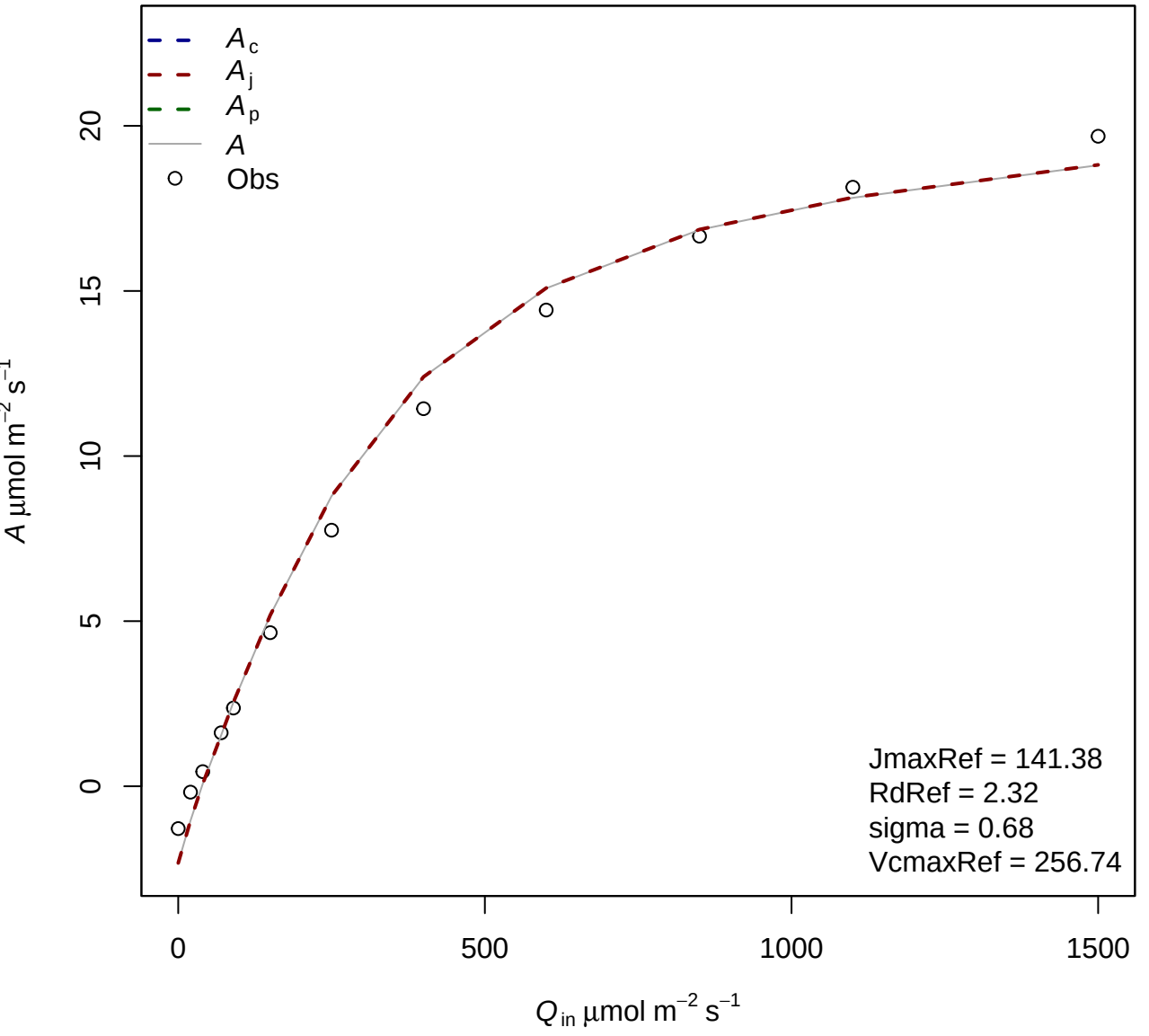

185
A_c
A_j
A_p
A
Obs
JmaxRef = 73.22
RdRef = 2.99
sigma = 1.85
VcmaxRef = 192.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

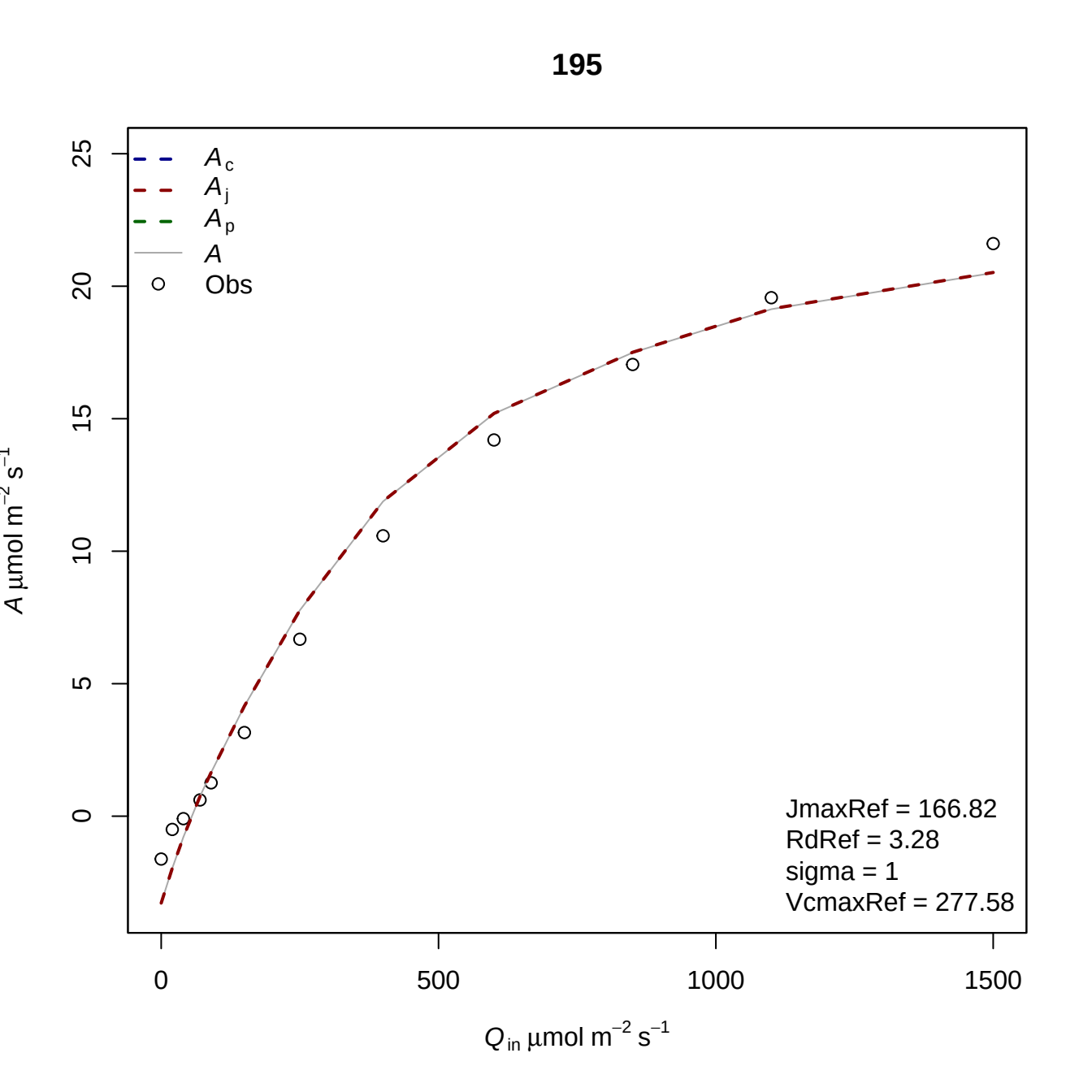

A_c
A_j
A_p
A
Obs
JmaxRef = 71.07
RdRef = 2.57
sigma = 0.98
VcmaxRef = 99.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

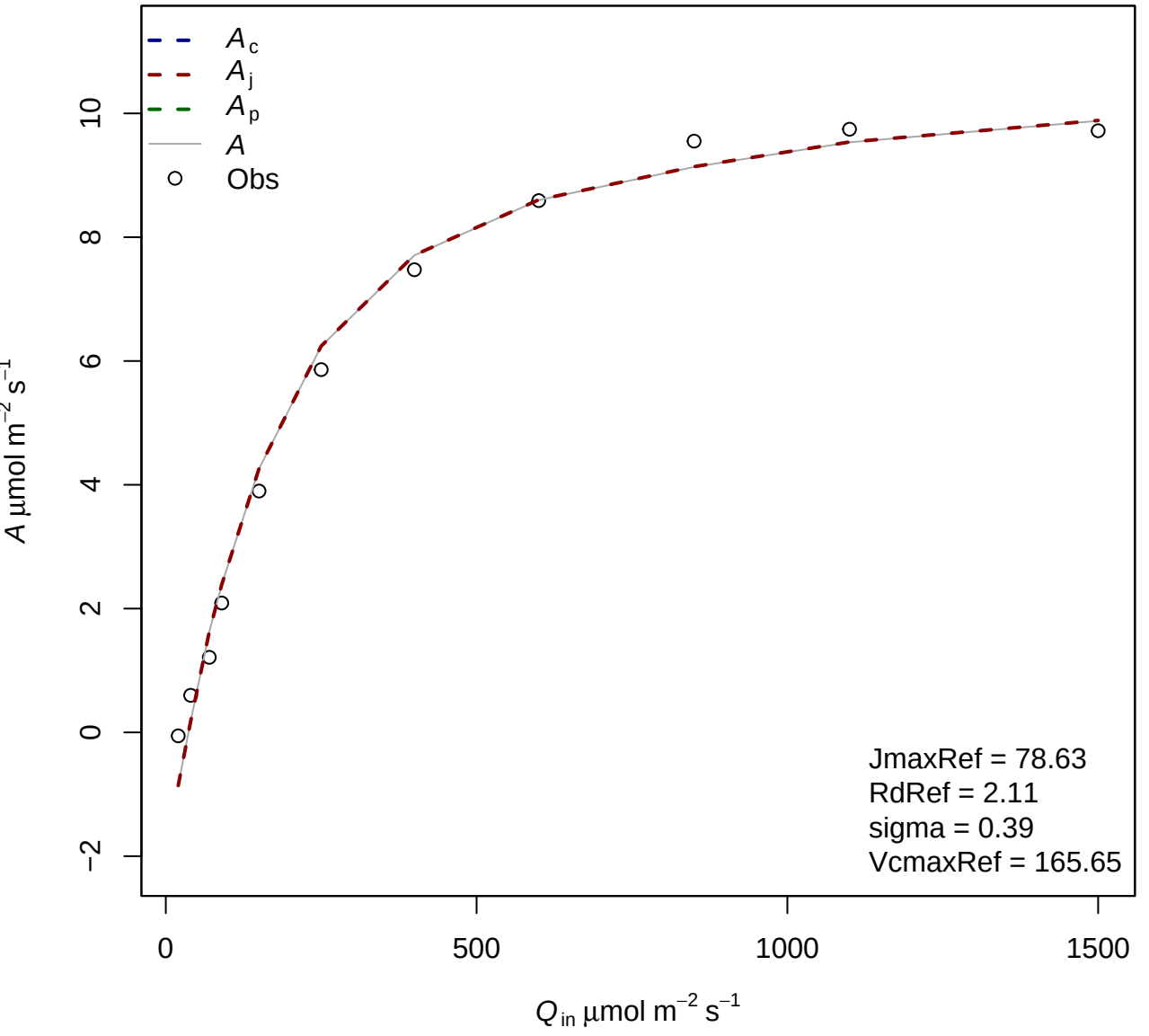

A_c
A_j
A_p
A
Obs
JmaxRef = 75.15
RdRef = 2.58
sigma = 0.18
VcmaxRef = 60.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
200
400
600
800
1000
1200
1400

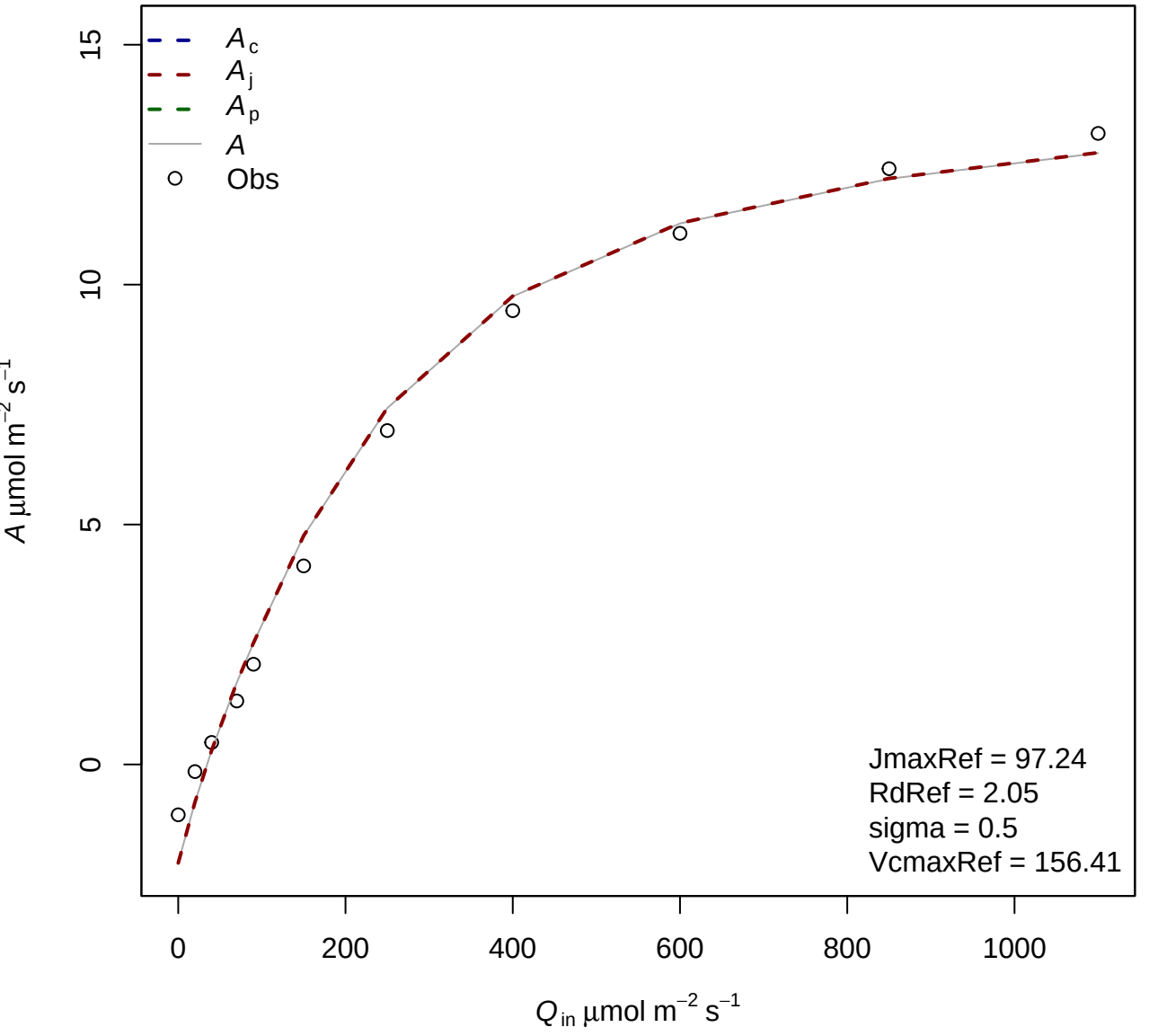

189
A_c
A_j
A_p
A
Obs
JmaxRef = 96.54
RdRef = 1.9
sigma = 0.46
VcmaxRef = 187.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

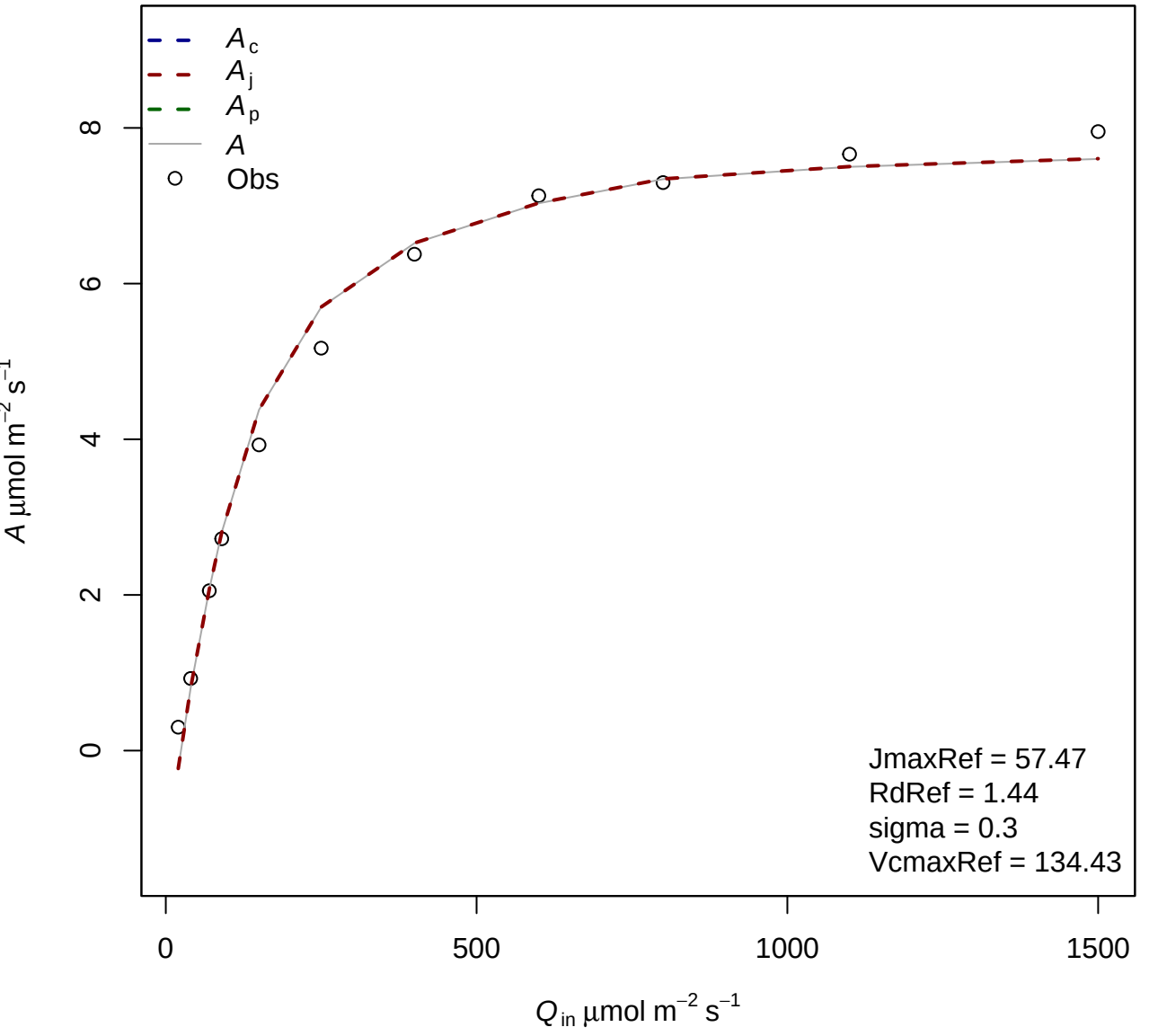

190
A_c
A_j
A_p
A
Obs
JmaxRef = 70.99
RdRef = 2.2
sigma = 0.58
VcmaxRef = 146.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000
2500

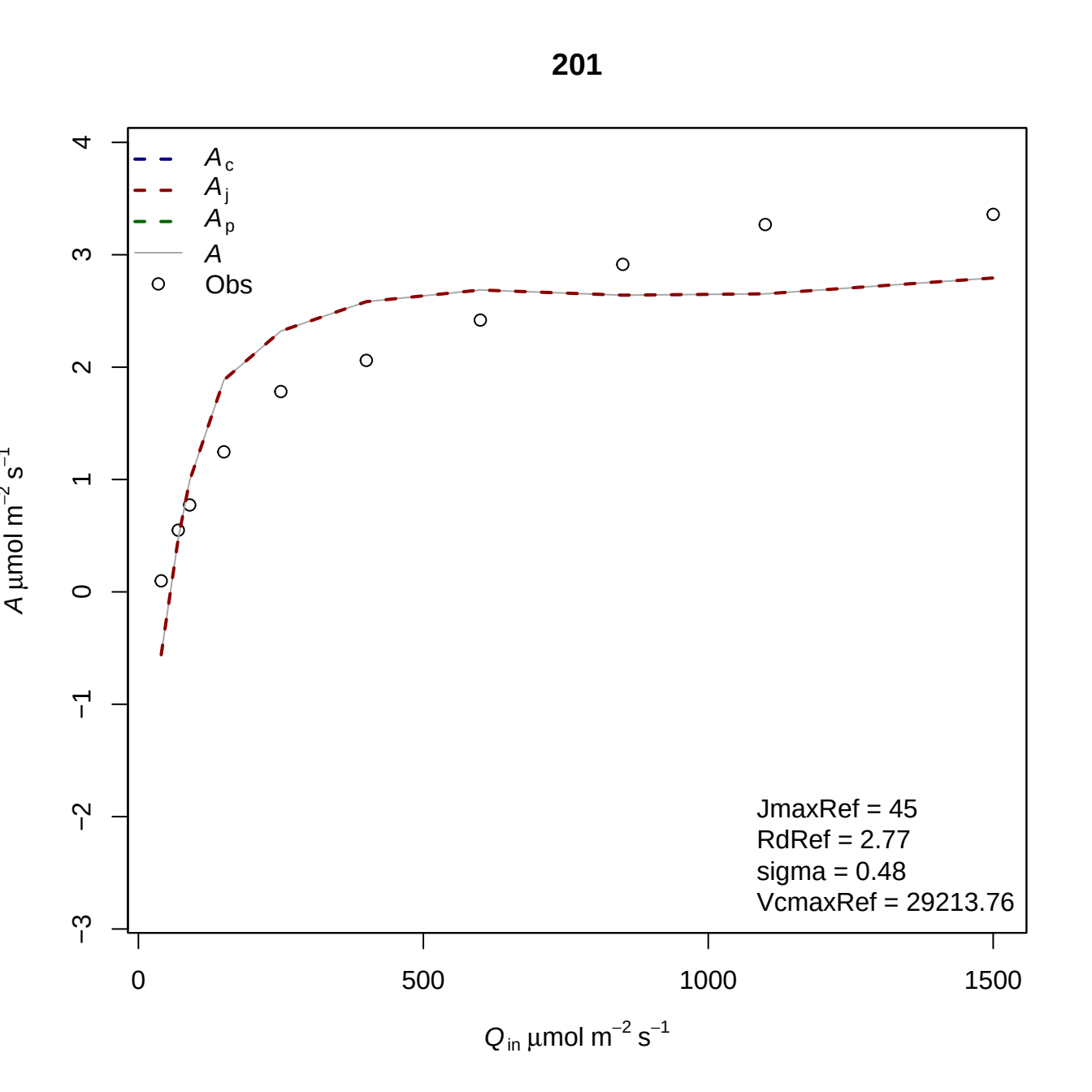

A_c
A_j
A_p
A
Obs
JmaxRef = 141.38
RdRef = 2.32
sigma = 0.68
VcmaxRef = 256.74
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

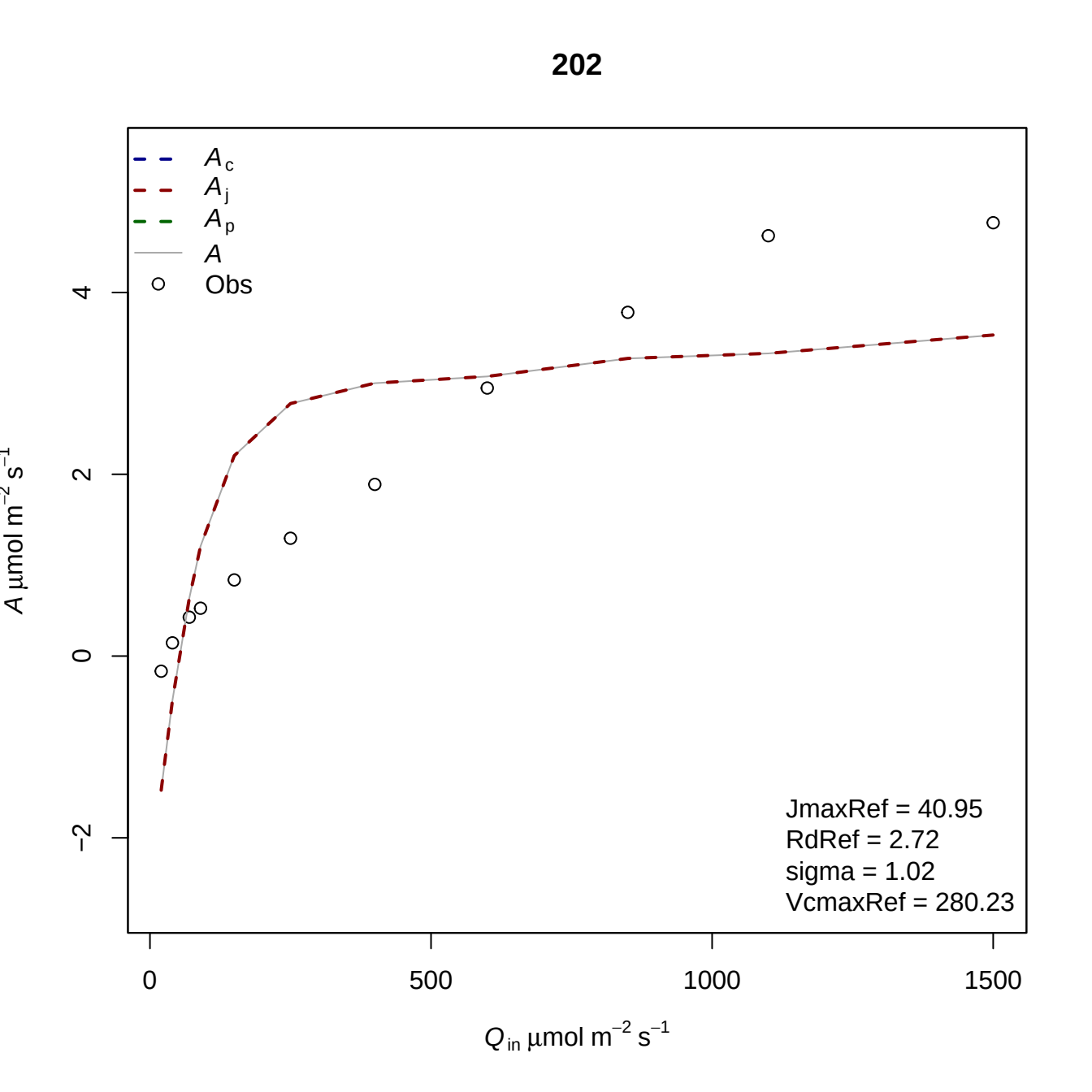

195
A_c
A_j
A_p
A
Obs
JmaxRef = 166.82
RdRef = 3.28
sigma = 1
VcmaxRef = 277.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

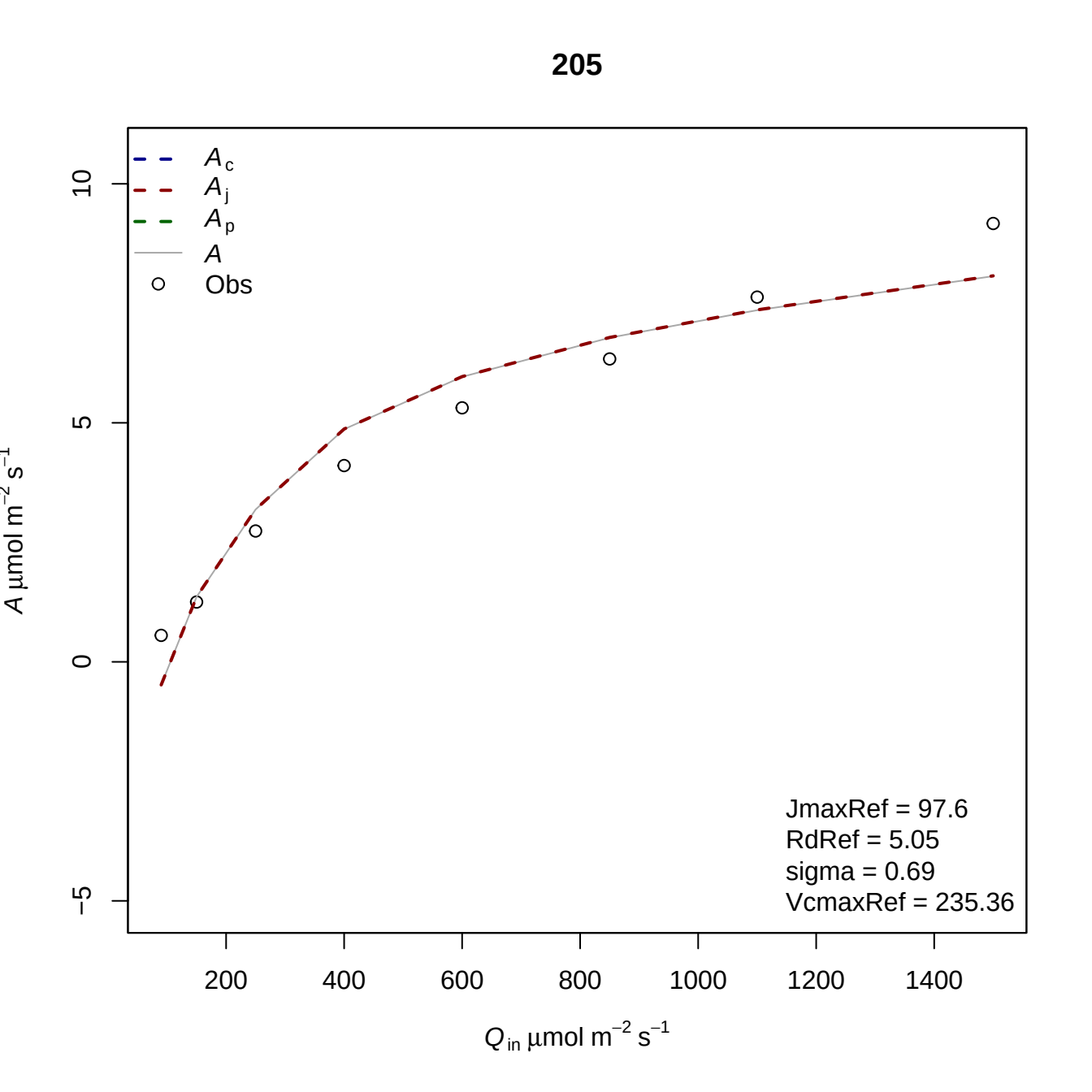

A_c
A_j
A_p
A
Obs
JmaxRef = 78.63
RdRef = 2.11
sigma = 0.39
VcmaxRef = 165.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

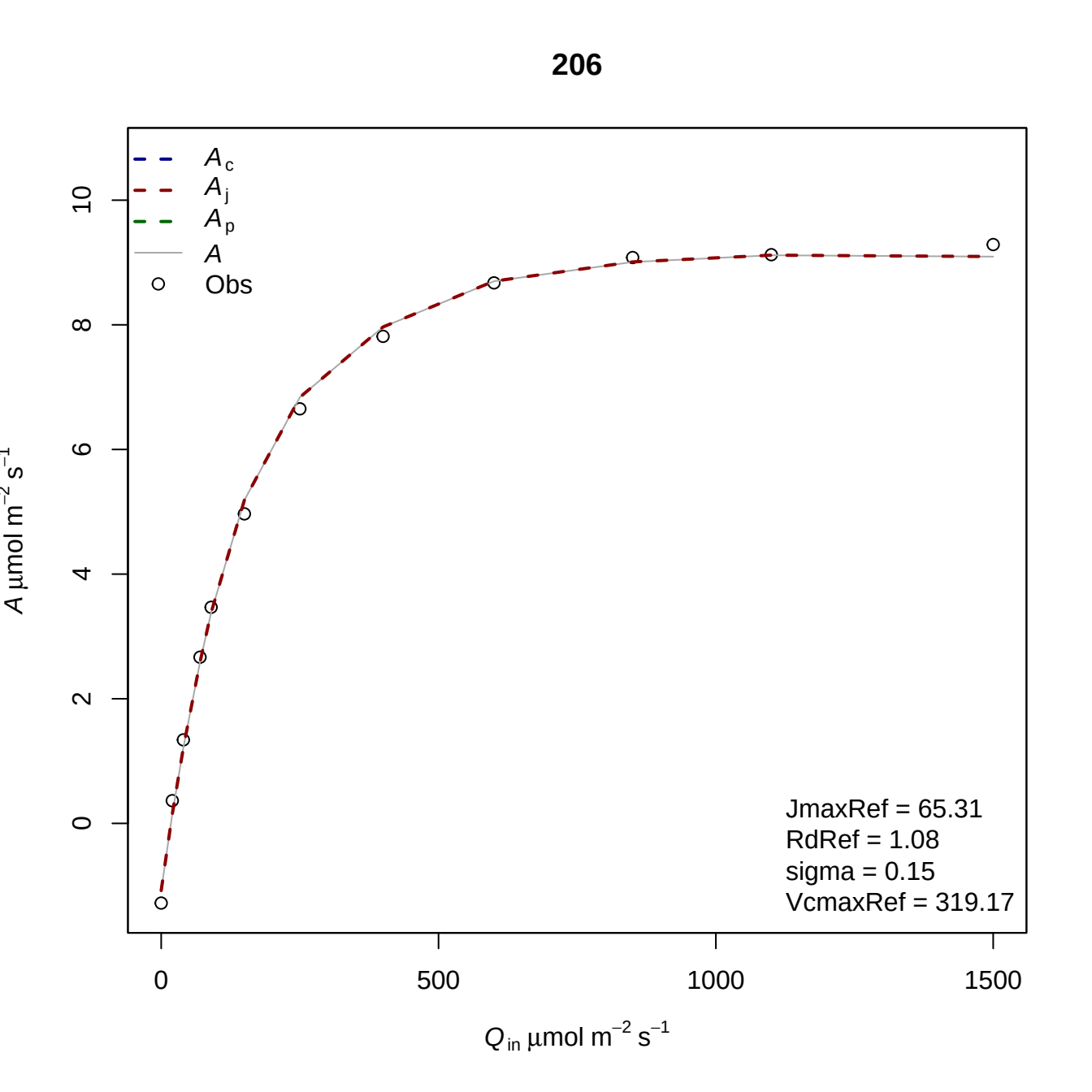

A_c
A_j
A_p
A
Obs
JmaxRef = 97.24
RdRef = 2.05
sigma = 0.5
VcmaxRef = 156.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000

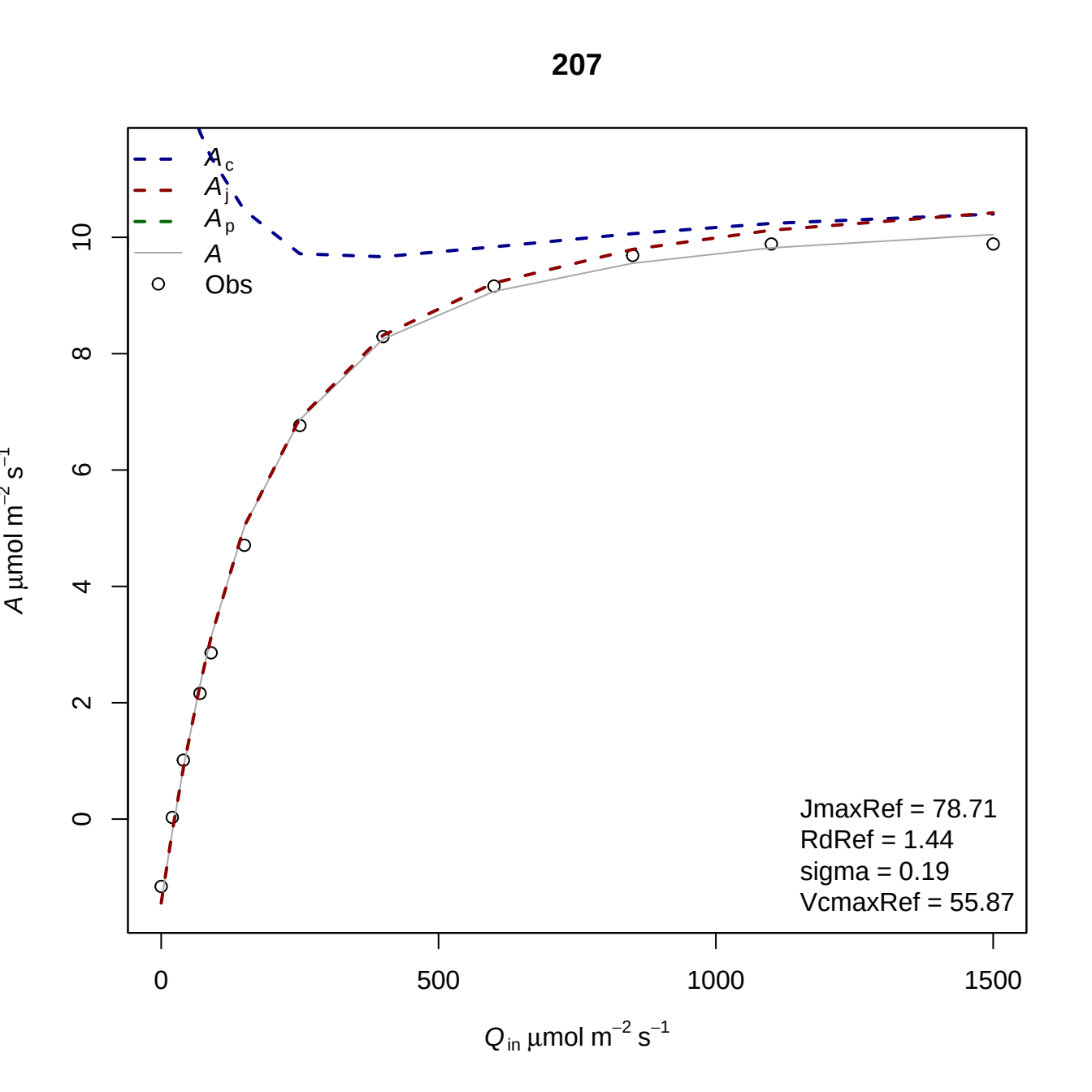

A_c
A_j
A_p
A
Obs
JmaxRef = 57.47
RdRef = 1.44
sigma = 0.3
VcmaxRef = 134.43
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

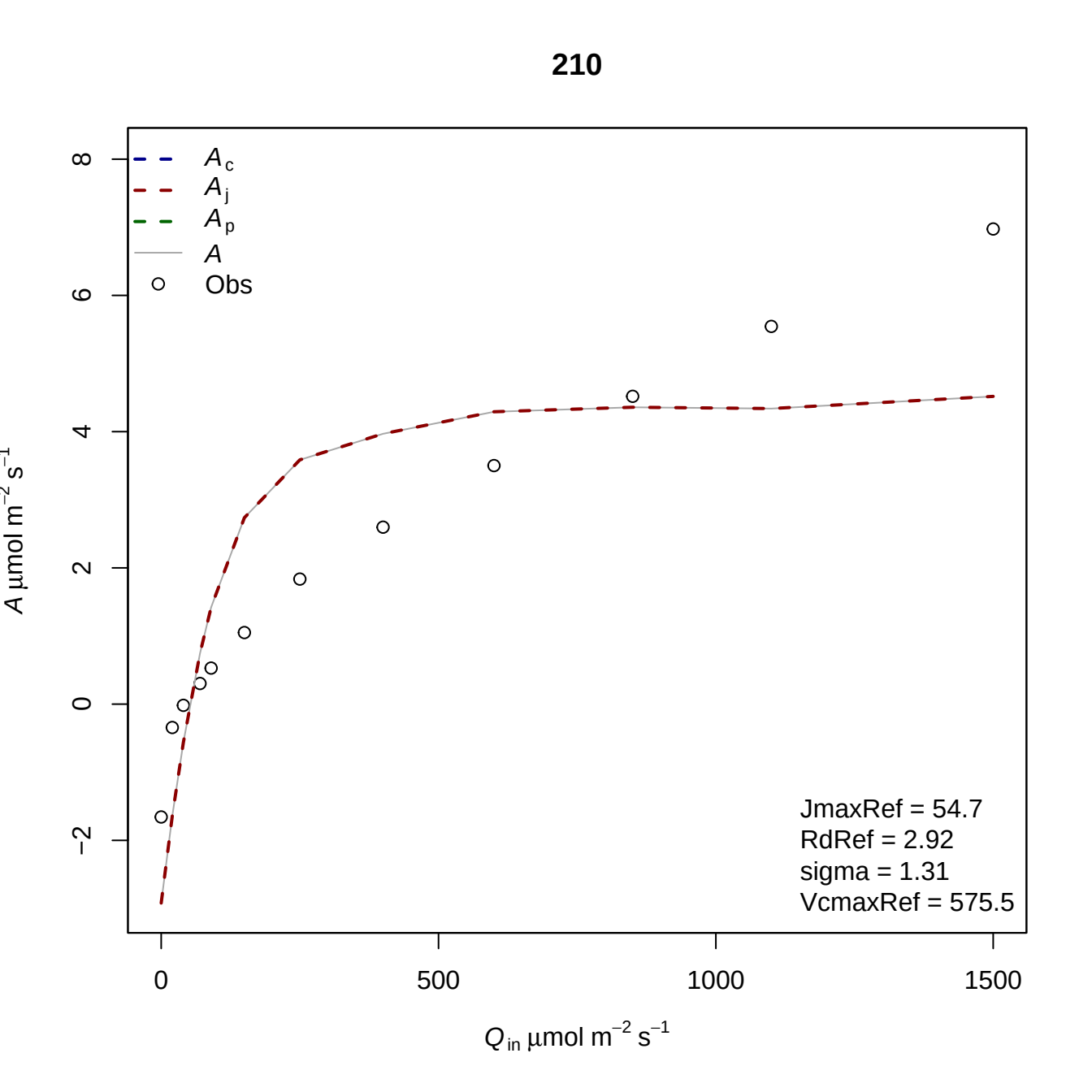

201
A_c
A_j
A_p
A
Obs
JmaxRef = 45
RdRef = 2.77
sigma = 0.48
VcmaxRef = 29213.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500

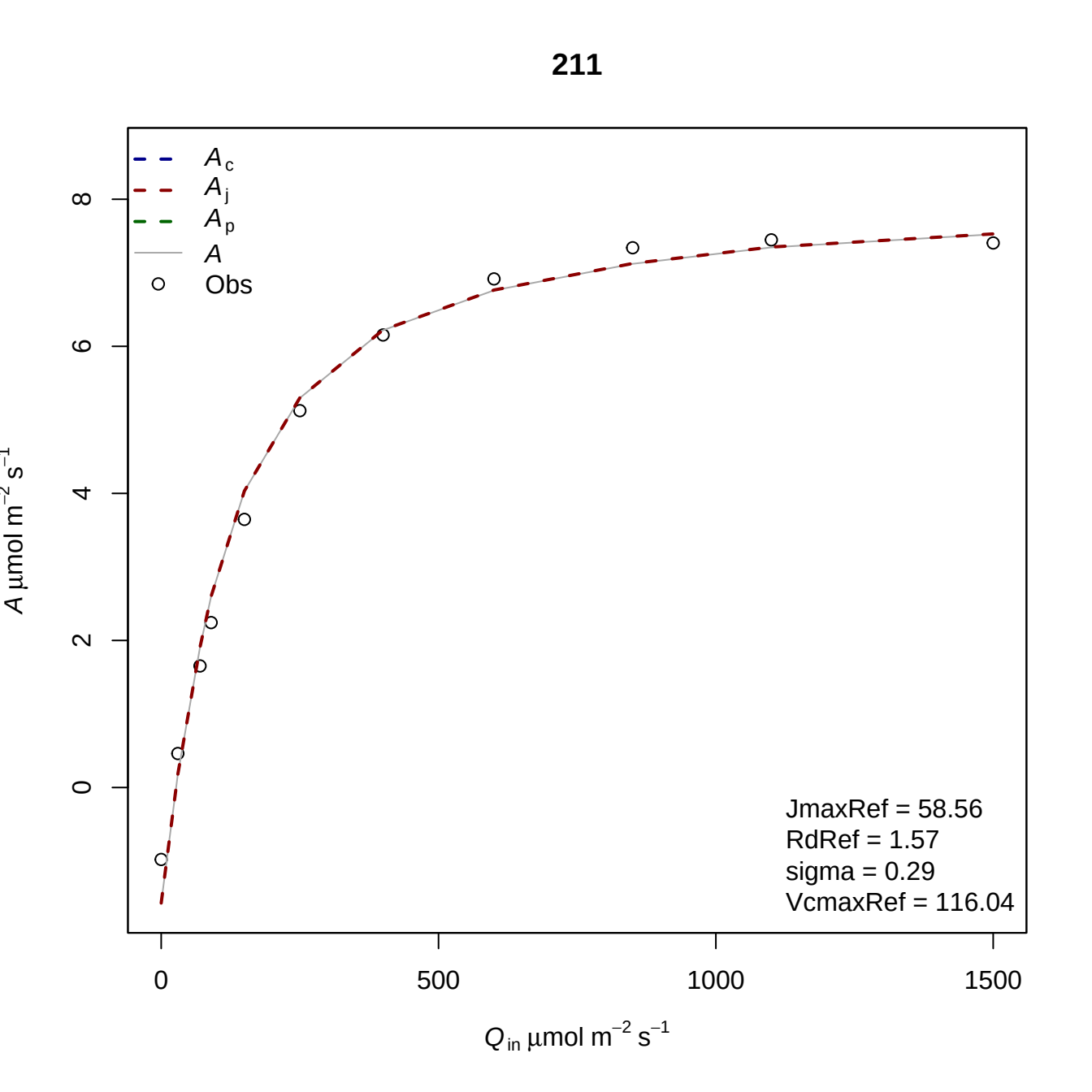

202
A_c
A_j
A_p
A
Obs
JmaxRef = 40.95
RdRef = 2.72
sigma = 1.02
VcmaxRef = 280.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

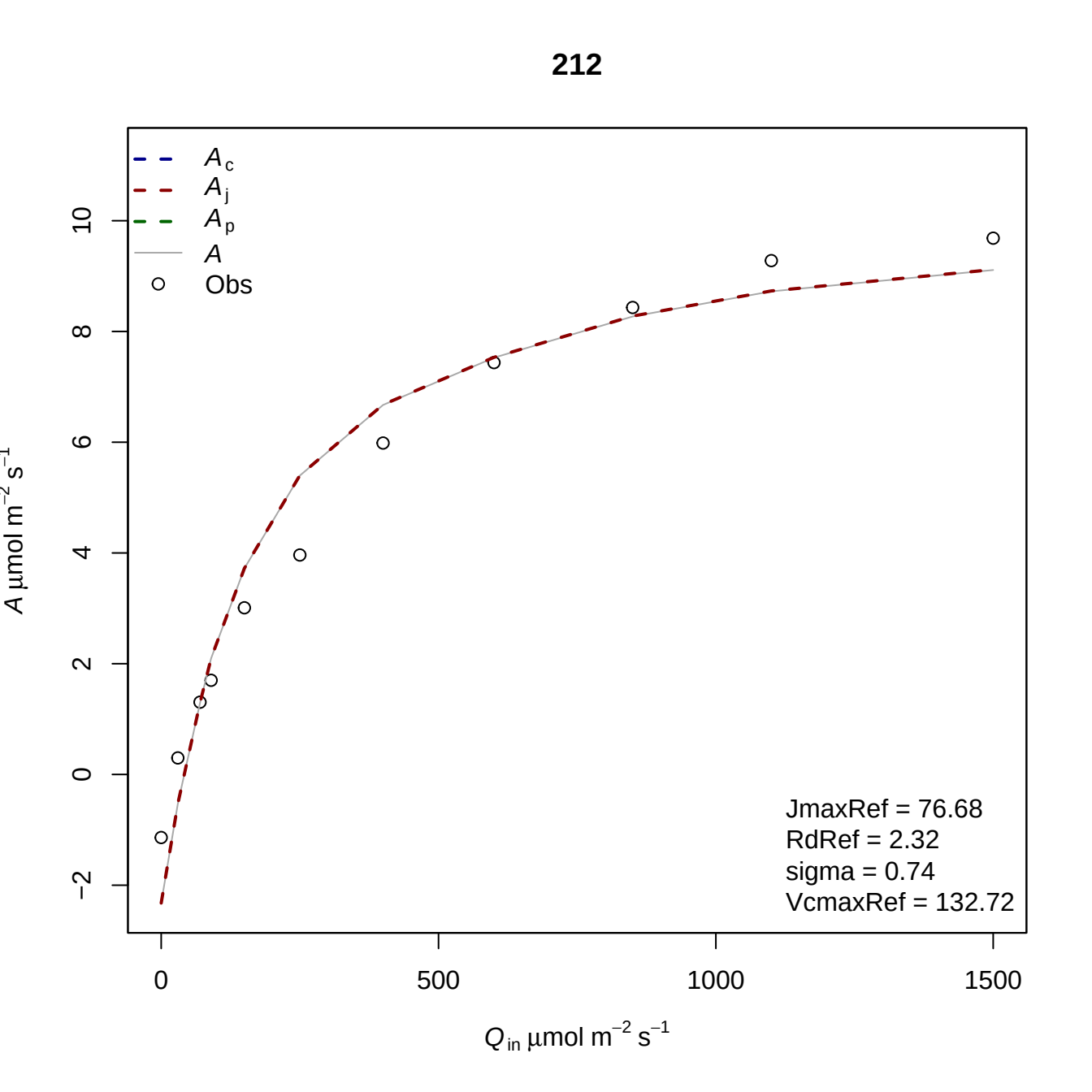

205
A_c
A_j
A_p
A
Obs
JmaxRef = 97.6
RdRef = 5.05
sigma = 0.69
VcmaxRef = 235.36
10
5
0
−5
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

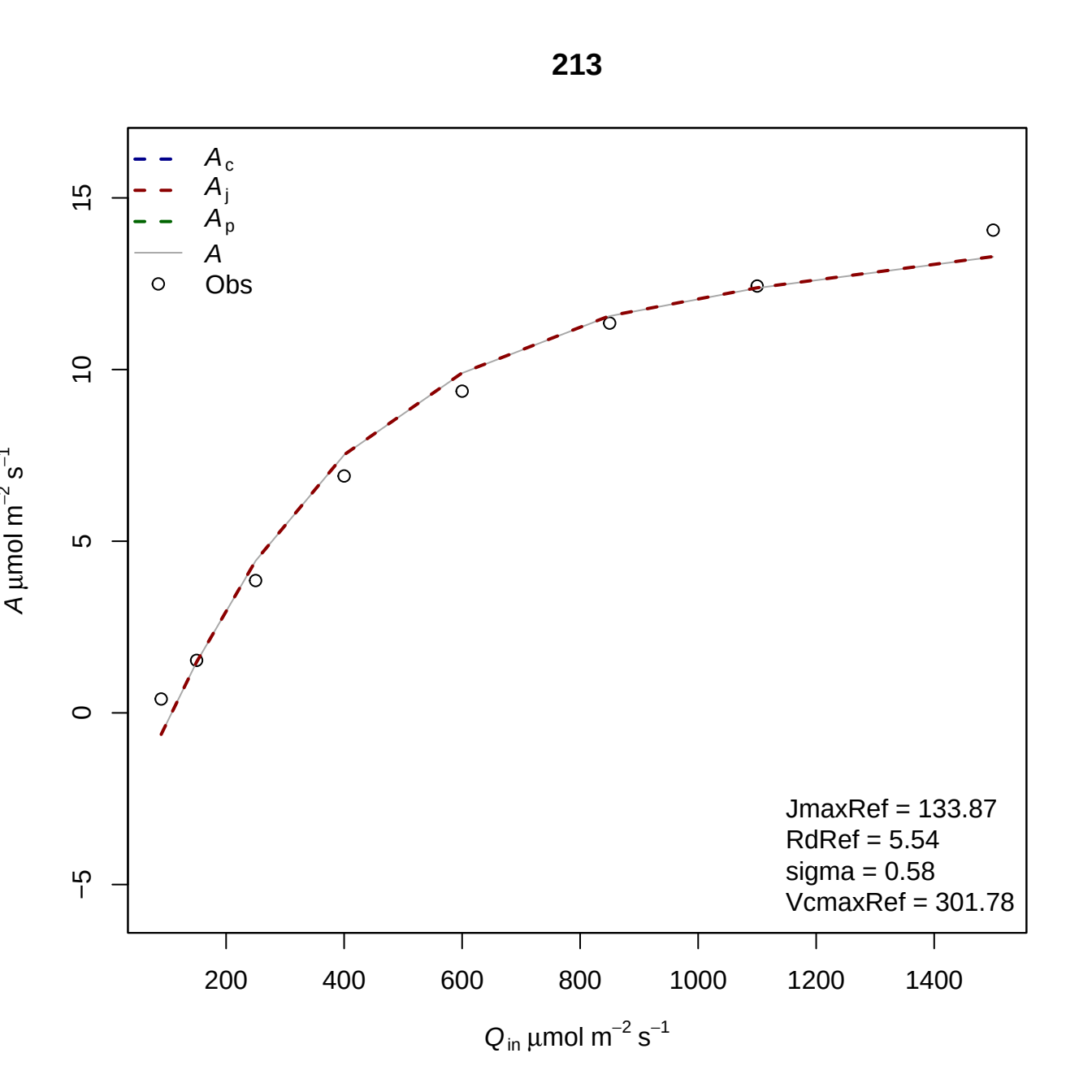

206
A_c
A_j
A_p
A
Obs
JmaxRef = 65.31
RdRef = 1.08
sigma = 0.15
VcmaxRef = 319.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

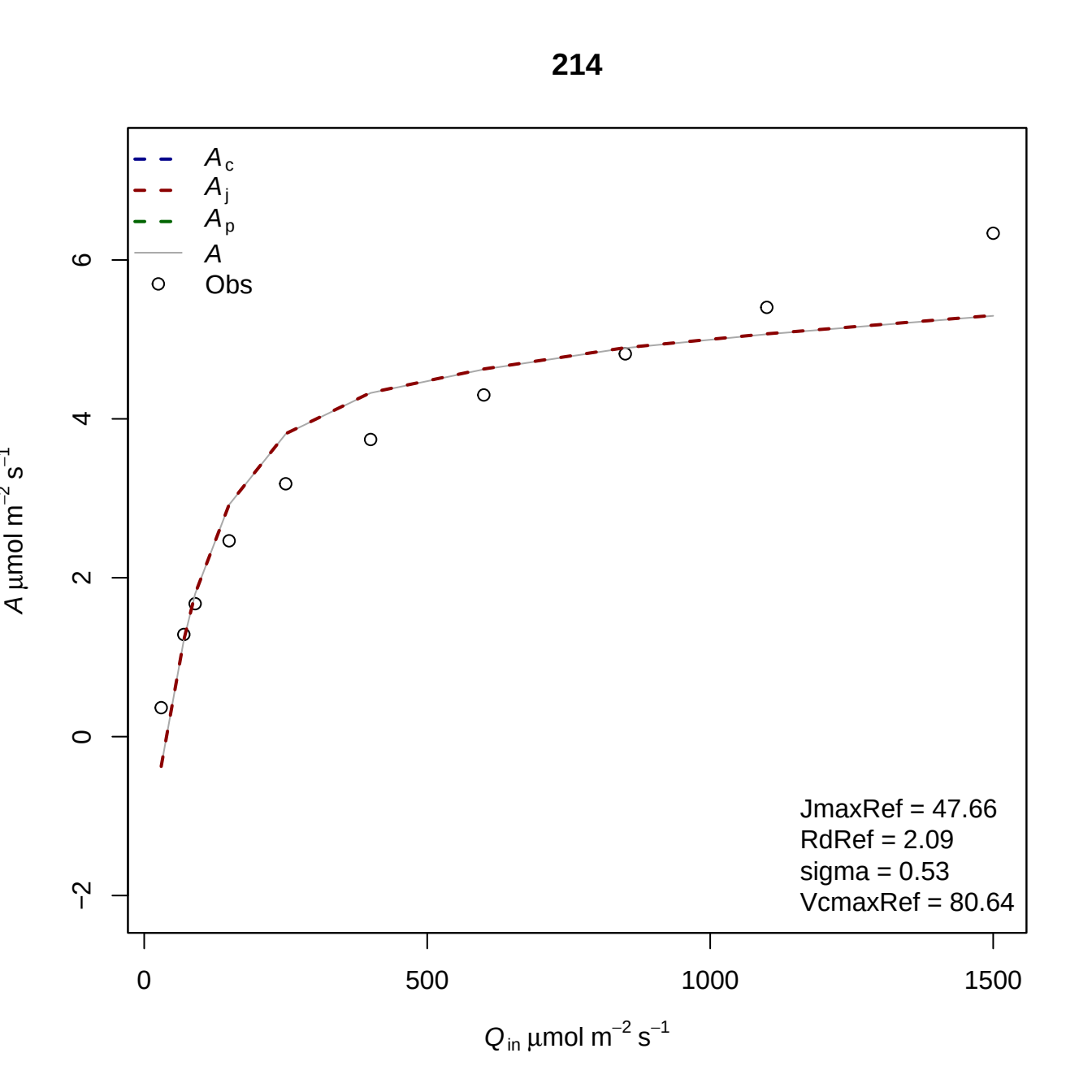

207
A_c
A_j
A_p
A
Obs
JmaxRef = 78.71
RdRef = 1.44
sigma = 0.19
VcmaxRef = 55.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

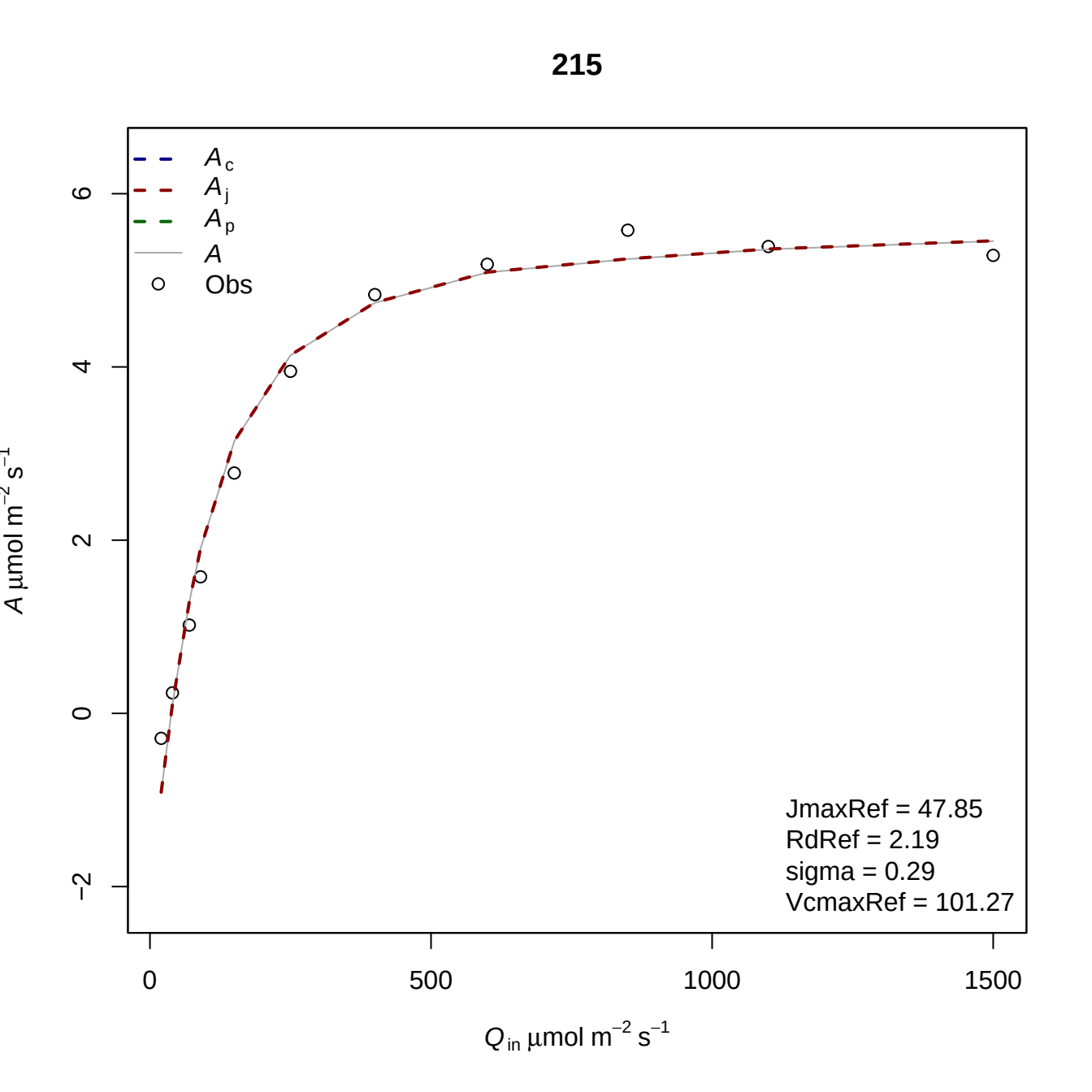

210
A_c
A_j
A_p
A
Obs
JmaxRef = 54.7
RdRef = 2.92
sigma = 1.31
VcmaxRef = 575.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

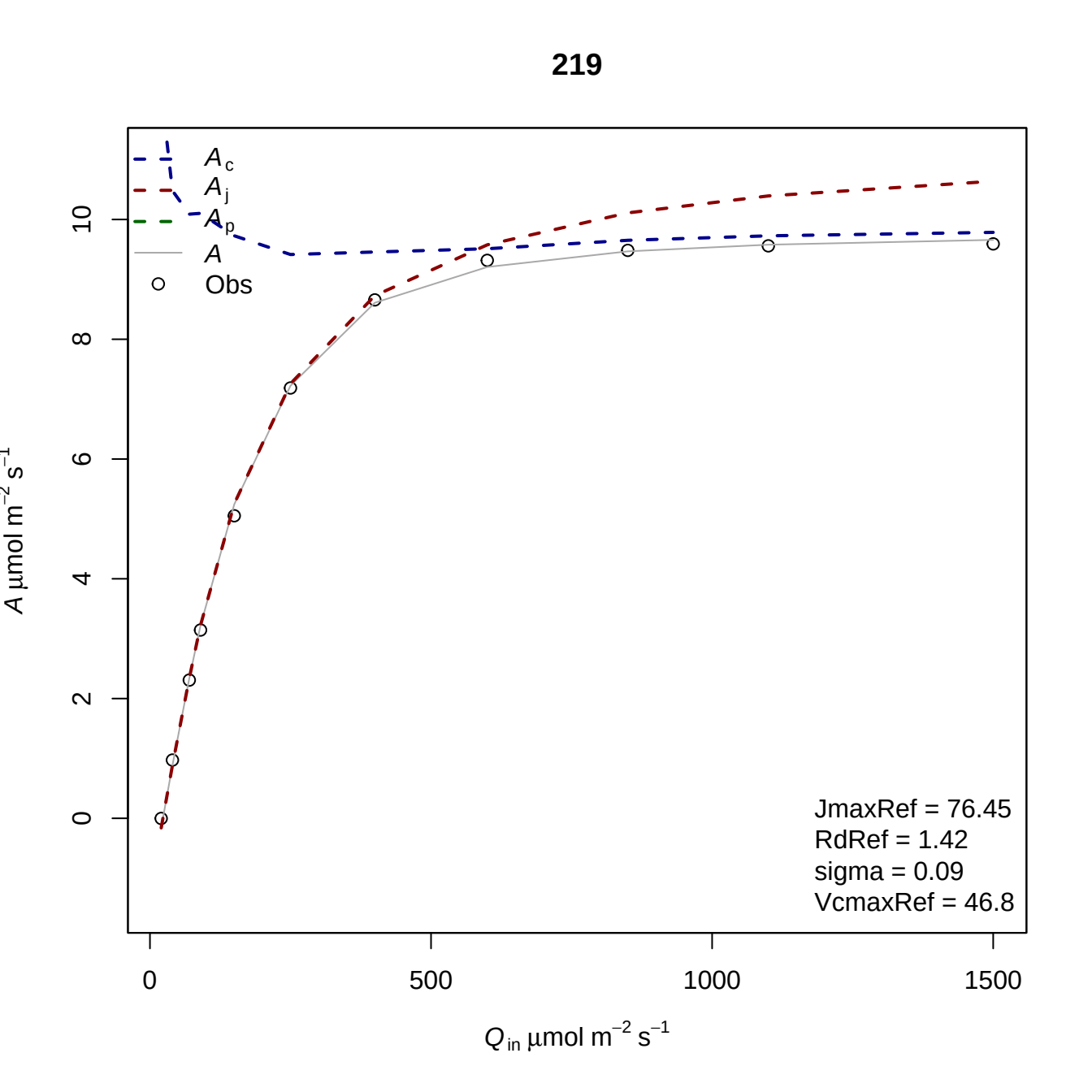

211
A_c
A_j
A_p
A
Obs
JmaxRef = 58.56
RdRef = 1.57
sigma = 0.29
VcmaxRef = 116.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

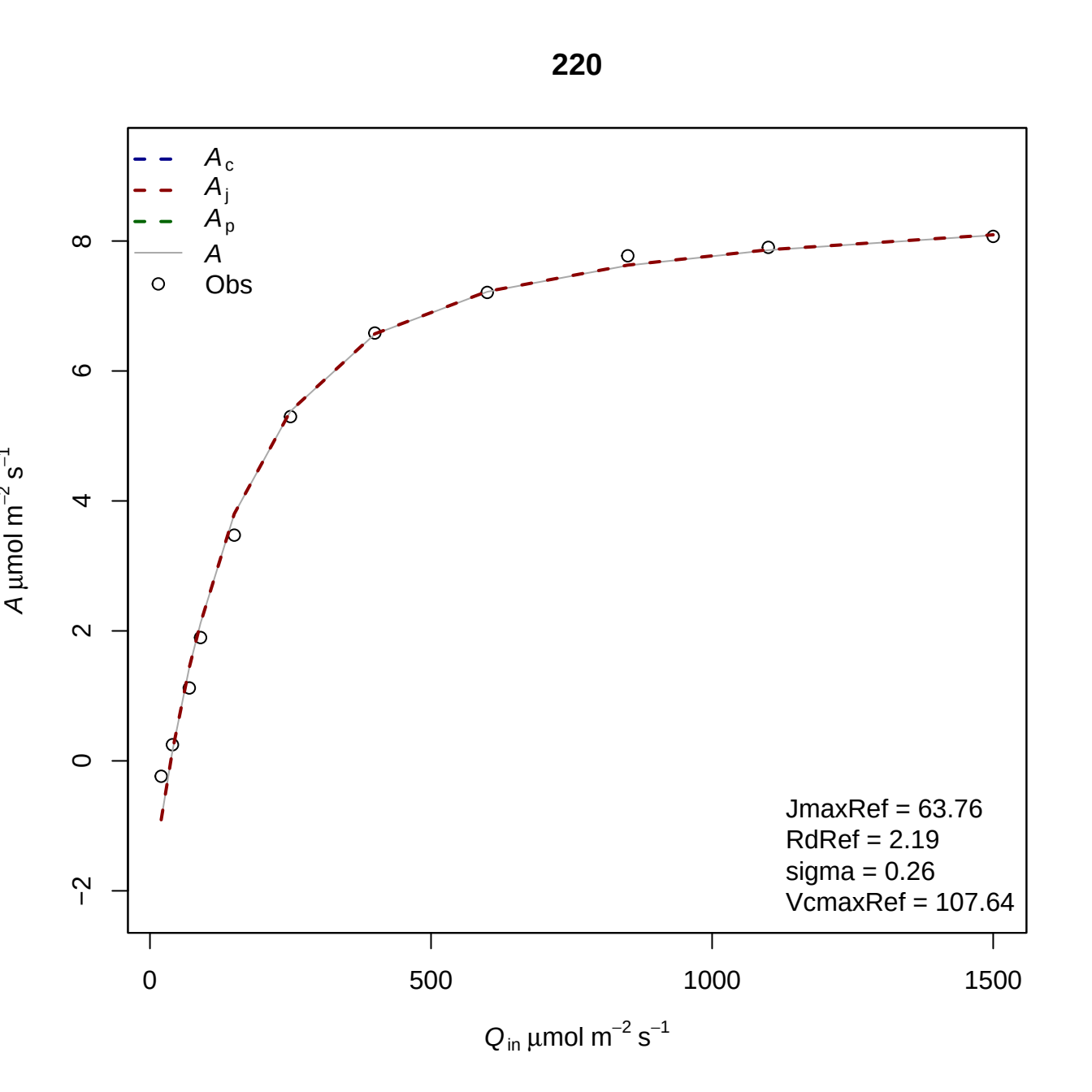

212
A_c
A_j
A_p
A
Obs
JmaxRef = 76.68
RdRef = 2.32
sigma = 0.74
VcmaxRef = 132.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

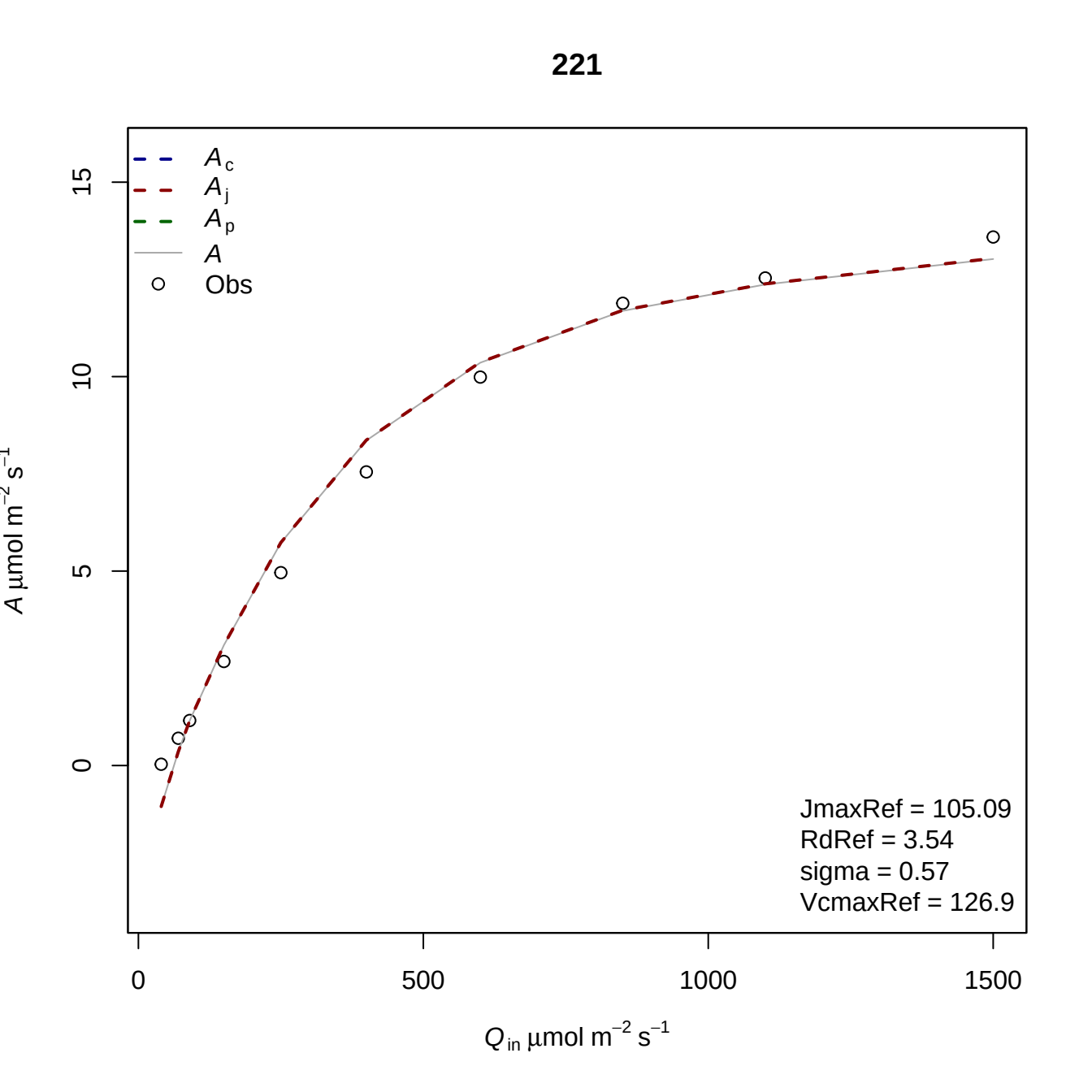

213
A_c
A_j
A_p
A
Obs
JmaxRef = 133.87
RdRef = 5.54
sigma = 0.58
VcmaxRef = 301.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
200
400
600
800
1000
1200
1400

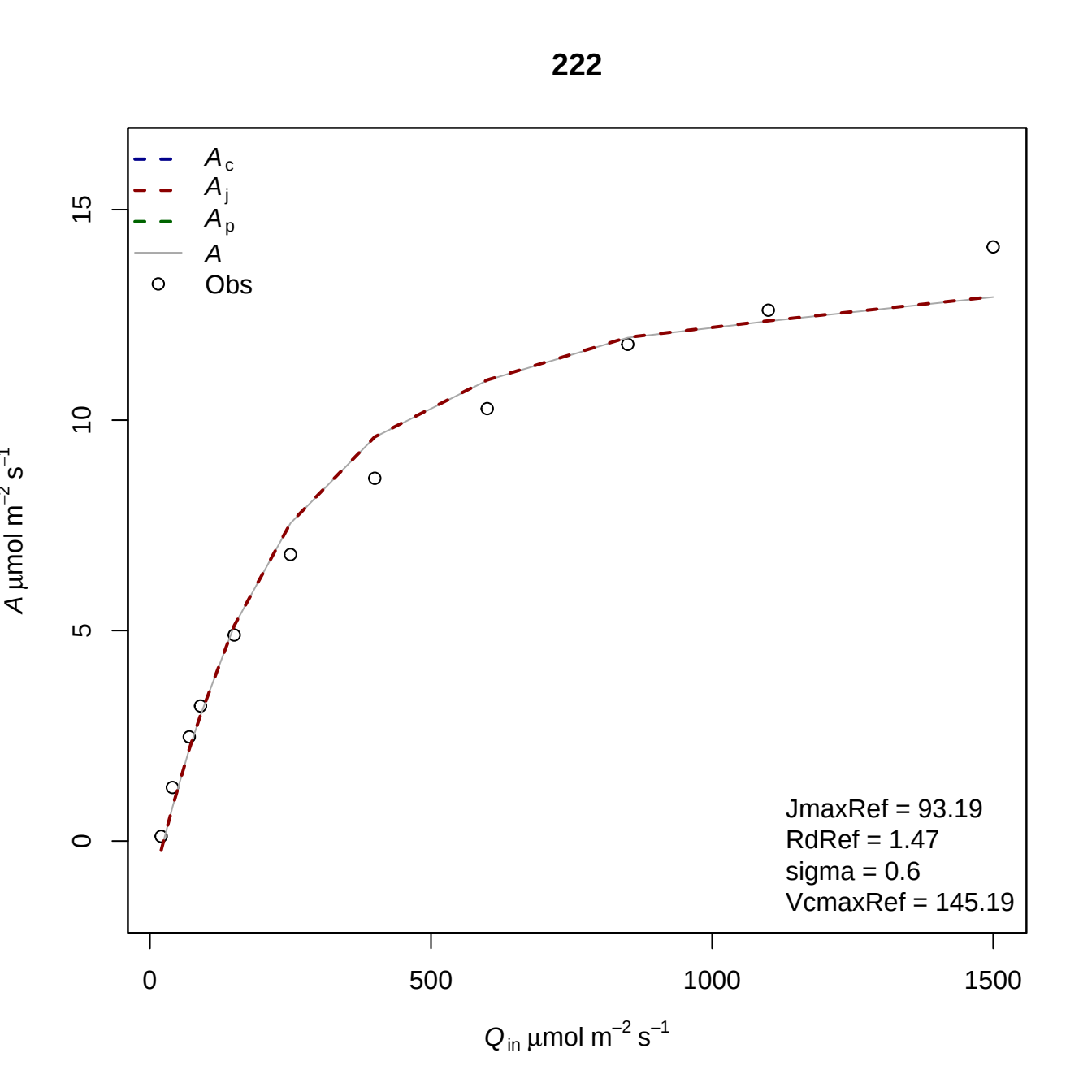

214
A_c
A_j
A_p
A
Obs
JmaxRef = 47.66
RdRef = 2.09
sigma = 0.53
VcmaxRef = 80.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

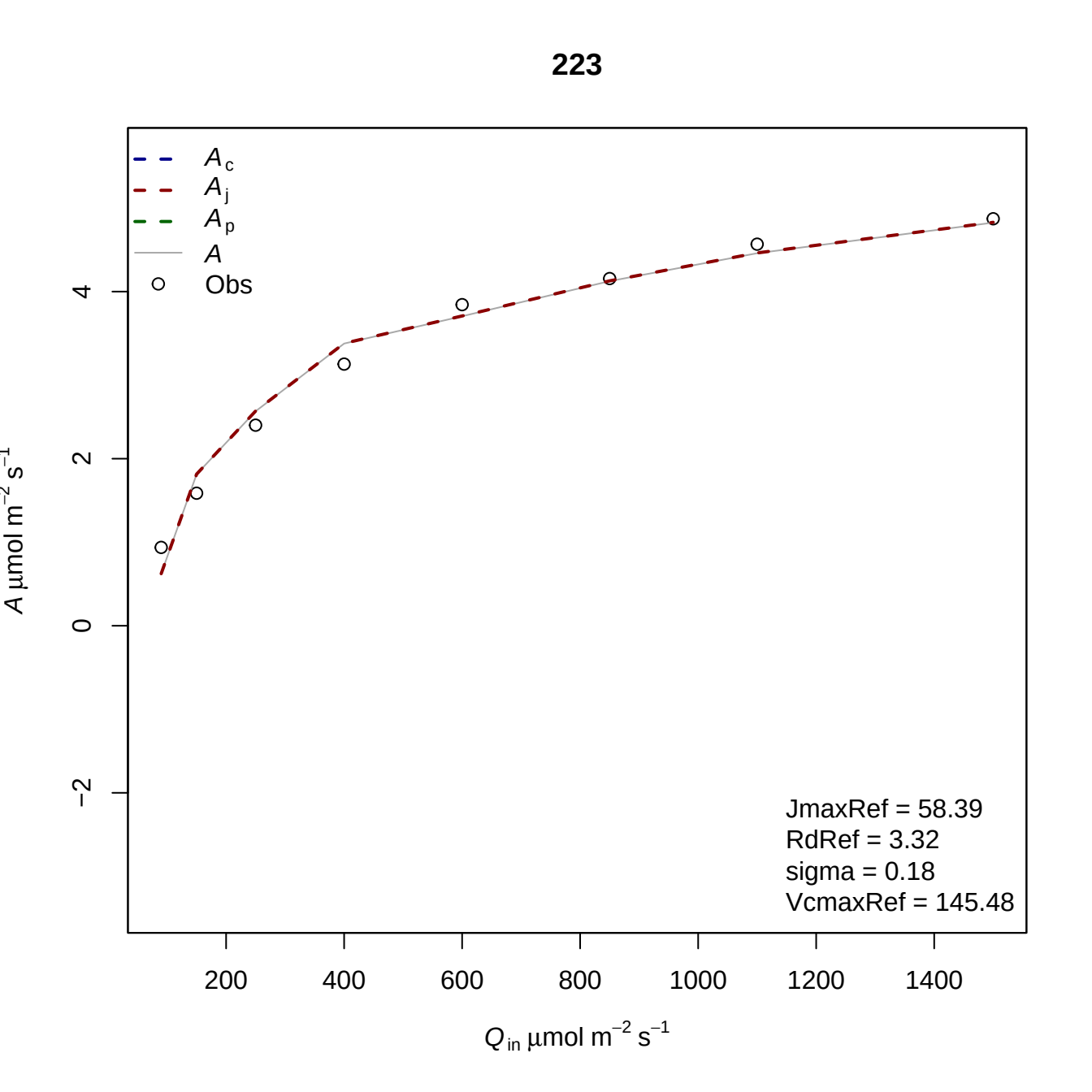

215
A_c
A_j
A_p
A
Obs
JmaxRef = 47.85
RdRef = 2.19
sigma = 0.29
VcmaxRef = 101.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

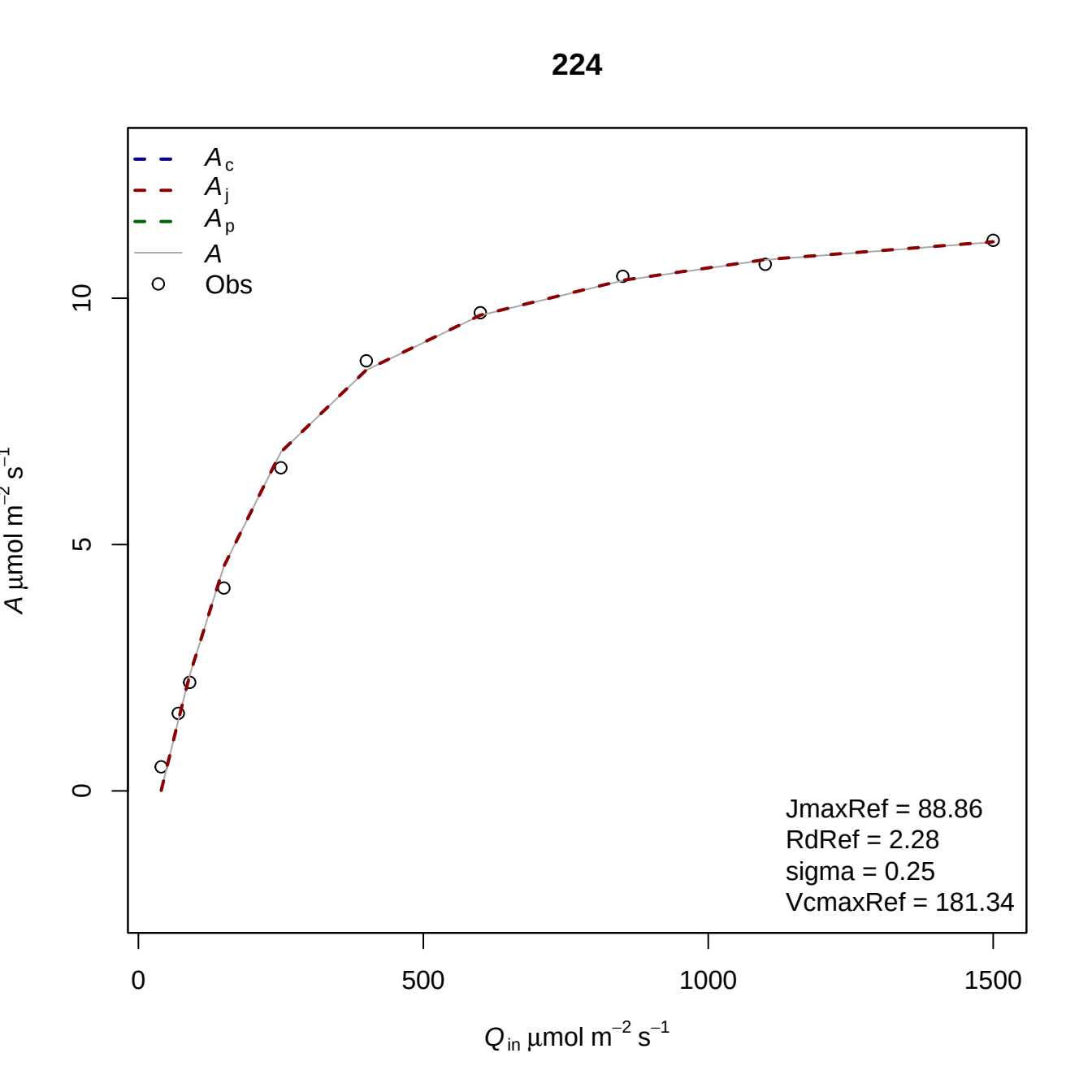

219
A_c
A_j
A_p
A
Obs
JmaxRef = 76.45
RdRef = 1.42
sigma = 0.09
VcmaxRef = 46.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

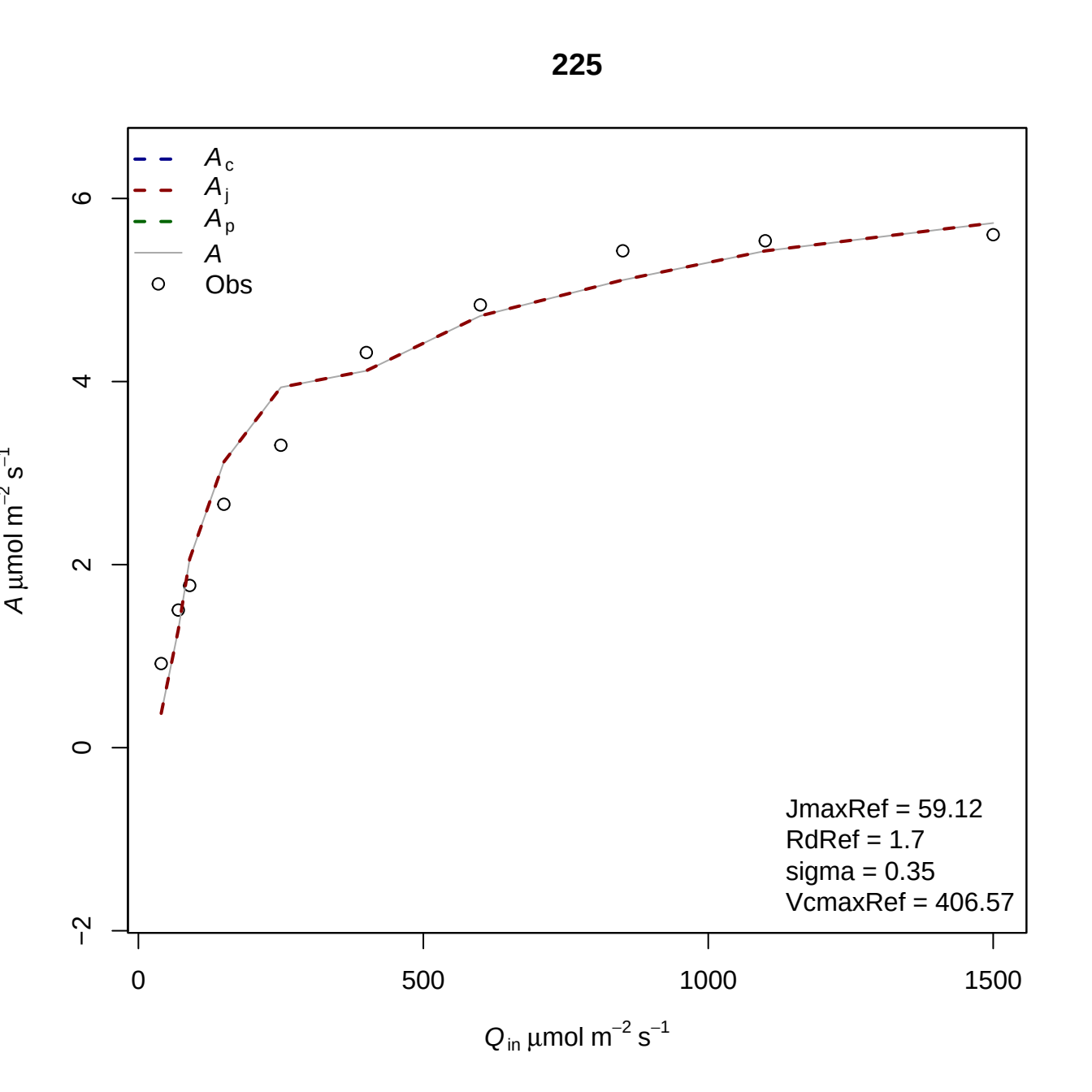

220
A_c
A_j
A_p
A
Obs
JmaxRef = 63.76
RdRef = 2.19
sigma = 0.26
VcmaxRef = 107.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

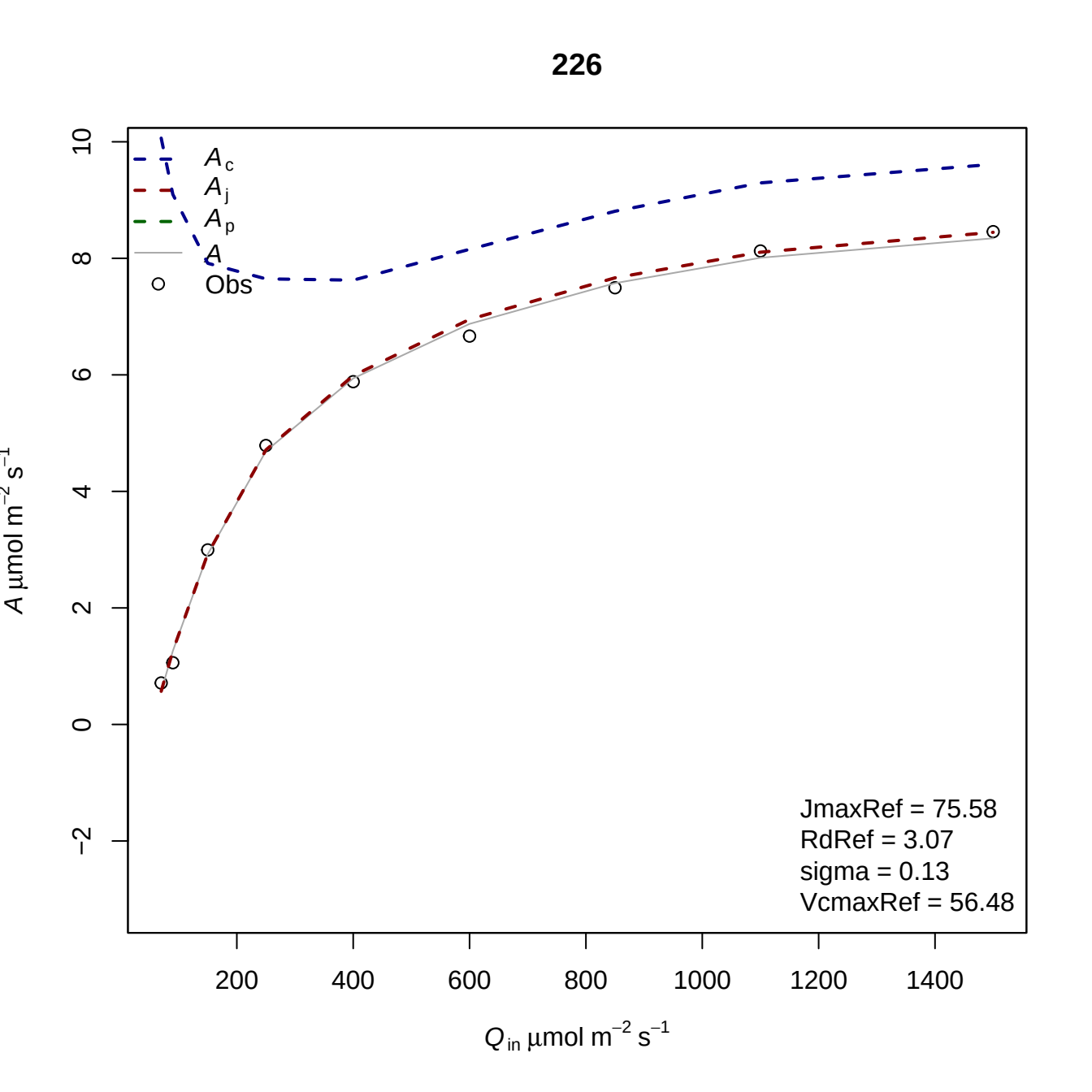

221
A_c
A_j
A_p
A
Obs
JmaxRef = 105.09
RdRef = 3.54
sigma = 0.57
VcmaxRef = 126.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

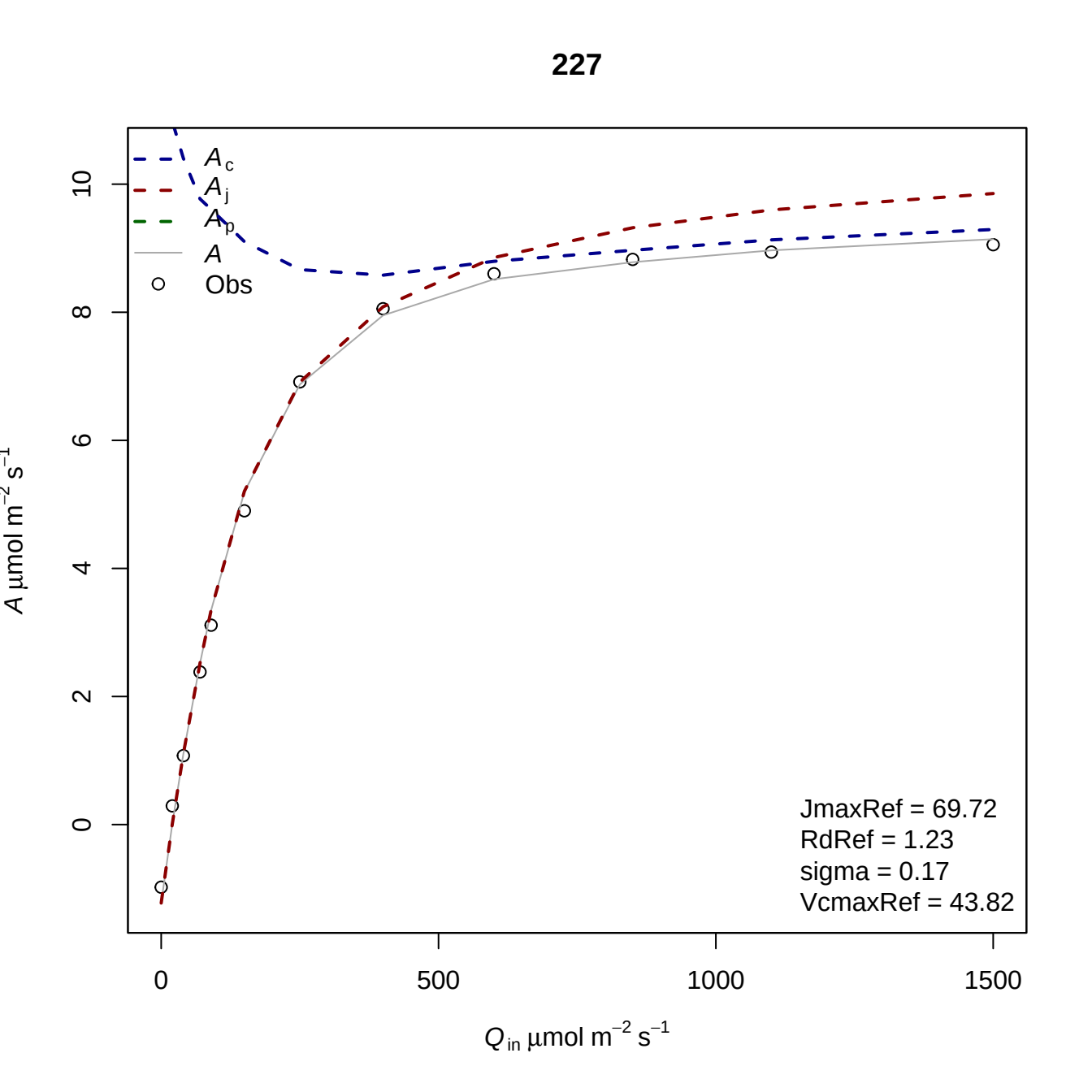

222
A_c
A_j
A_p
A
Obs
JmaxRef = 93.19
RdRef = 1.47
sigma = 0.6
VcmaxRef = 145.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

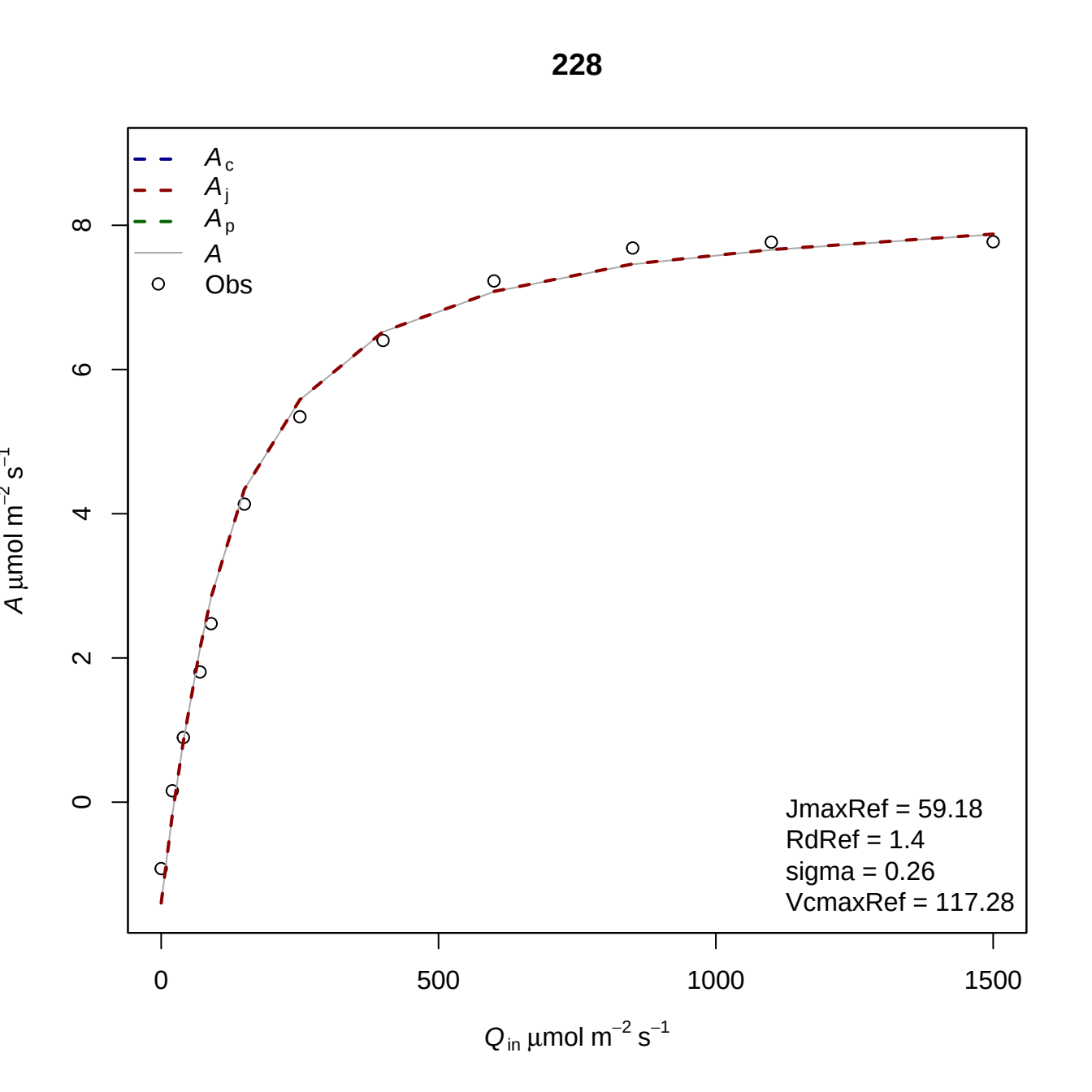

223
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
4
2
0
−2
200
400
600
800
1000
1200
1400
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 58.39
RdRef = 3.32
sigma = 0.18
VcmaxRef = 145.48

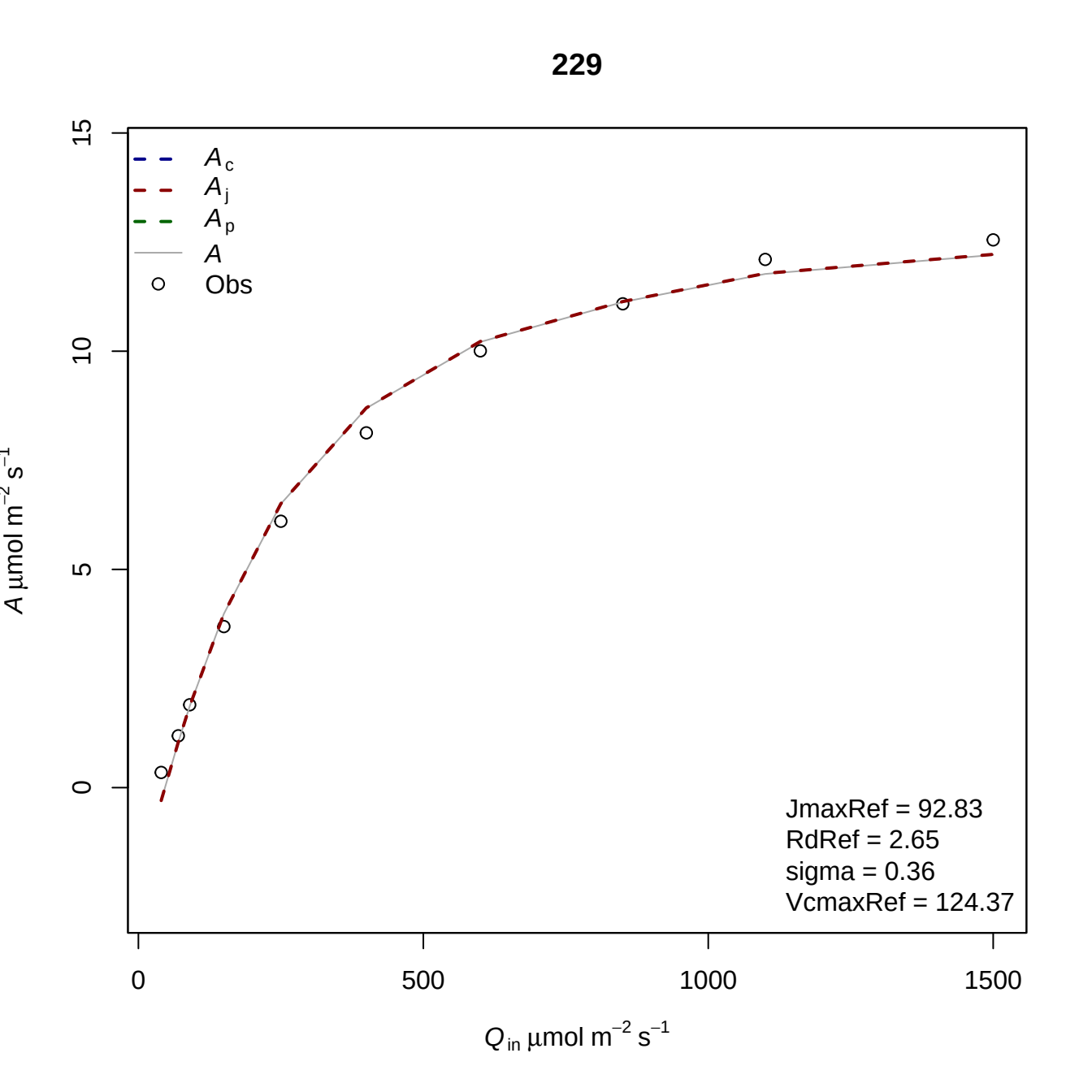

224
A_c
A_j
A_p
A
Obs
JmaxRef = 88.86
RdRef = 2.28
sigma = 0.25
VcmaxRef = 181.34
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

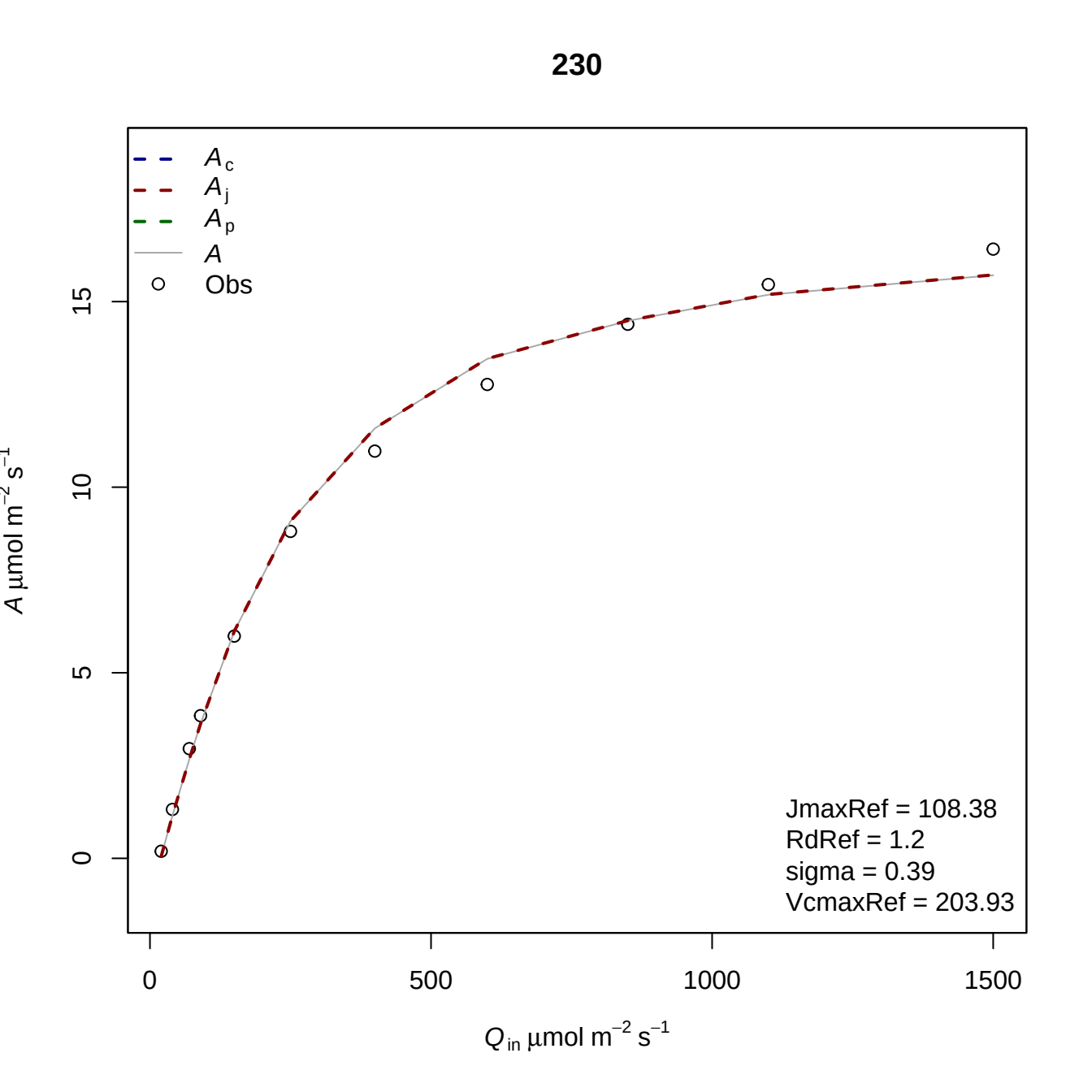

225
A_c
A_j
A_p
A
Obs
JmaxRef = 59.12
RdRef = 1.7
sigma = 0.35
VcmaxRef = 406.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

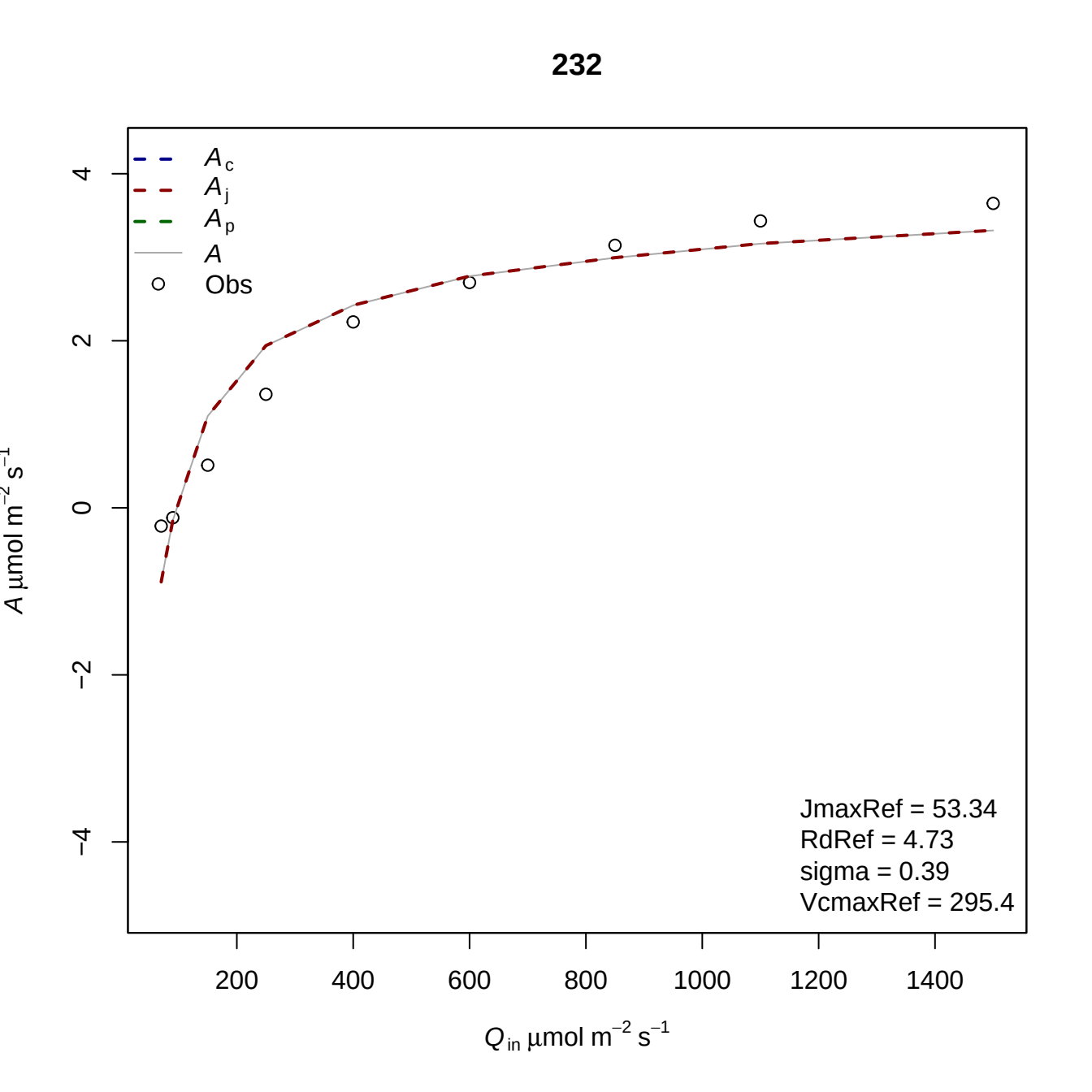

226
A_c
A_j
A_p
A
Obs
JmaxRef = 75.58
RdRef = 3.07
sigma = 0.13
VcmaxRef = 56.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400

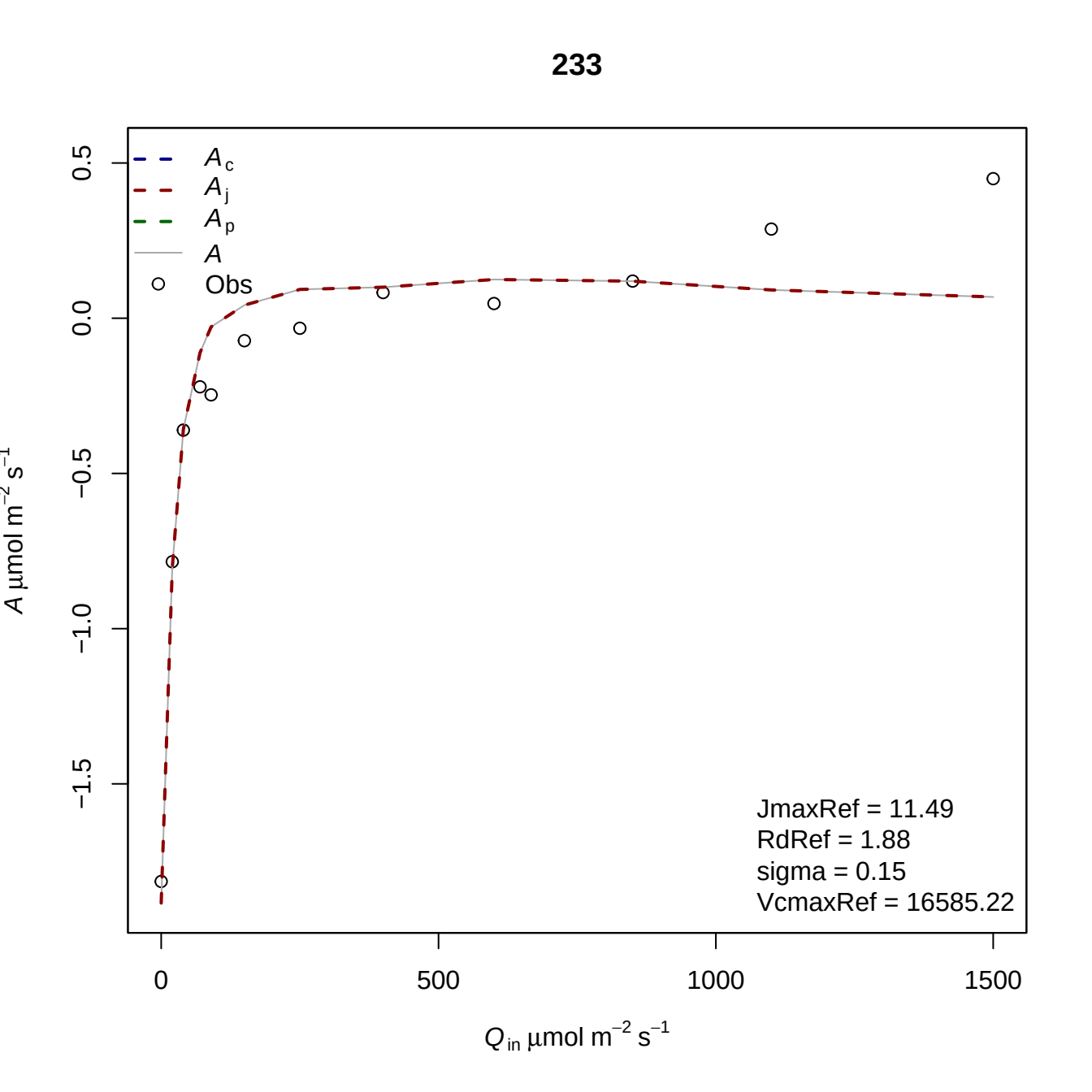

227
A_c
A_j
A_p
A
Obs
JmaxRef = 69.72
RdRef = 1.23
sigma = 0.17
VcmaxRef = 43.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

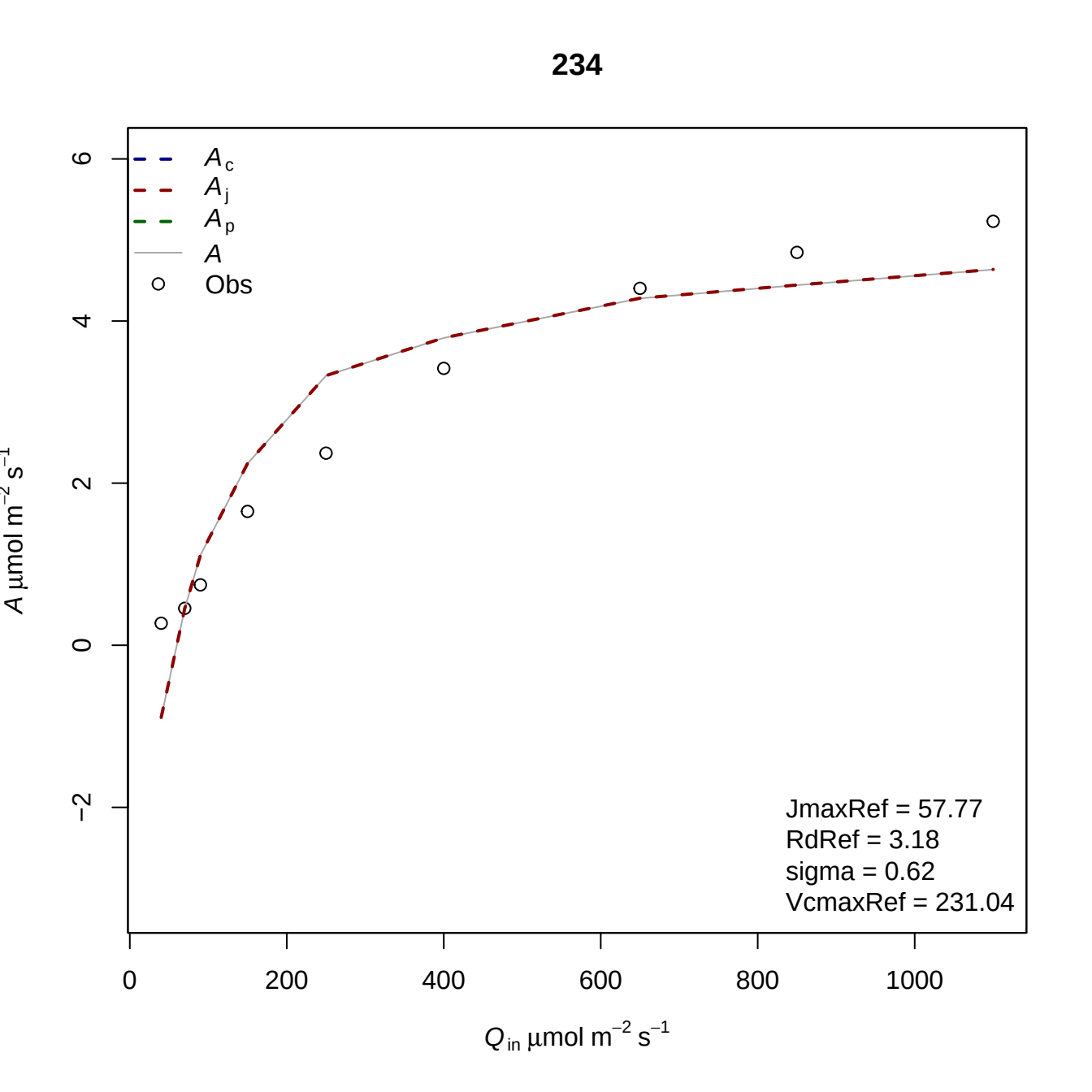

228
A_c
A_j
A_p
A
Obs
JmaxRef = 59.18
RdRef = 1.4
sigma = 0.26
VcmaxRef = 117.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

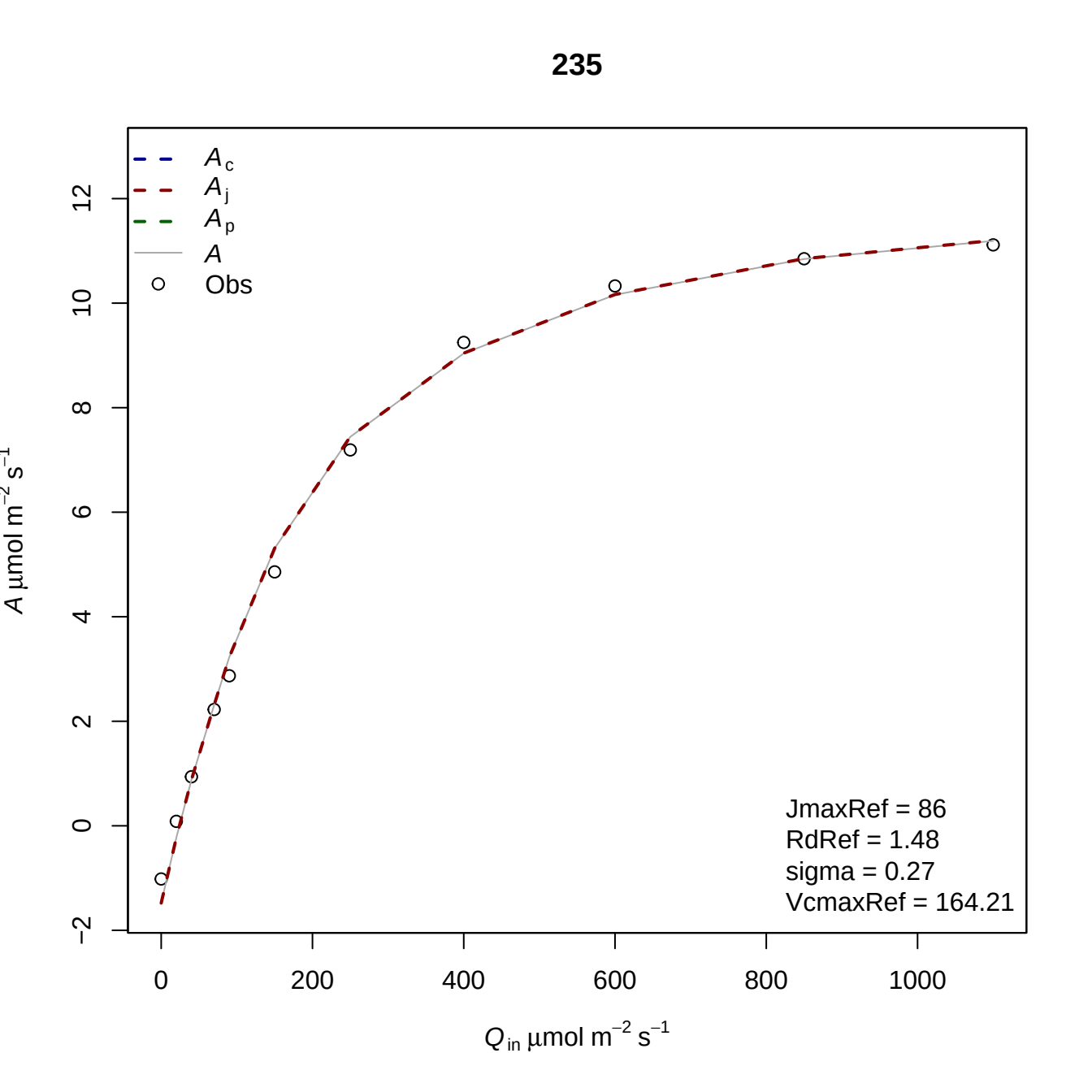

229
A_c
A_j
A_p
A
Obs
JmaxRef = 92.83
RdRef = 2.65
sigma = 0.36
VcmaxRef = 124.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

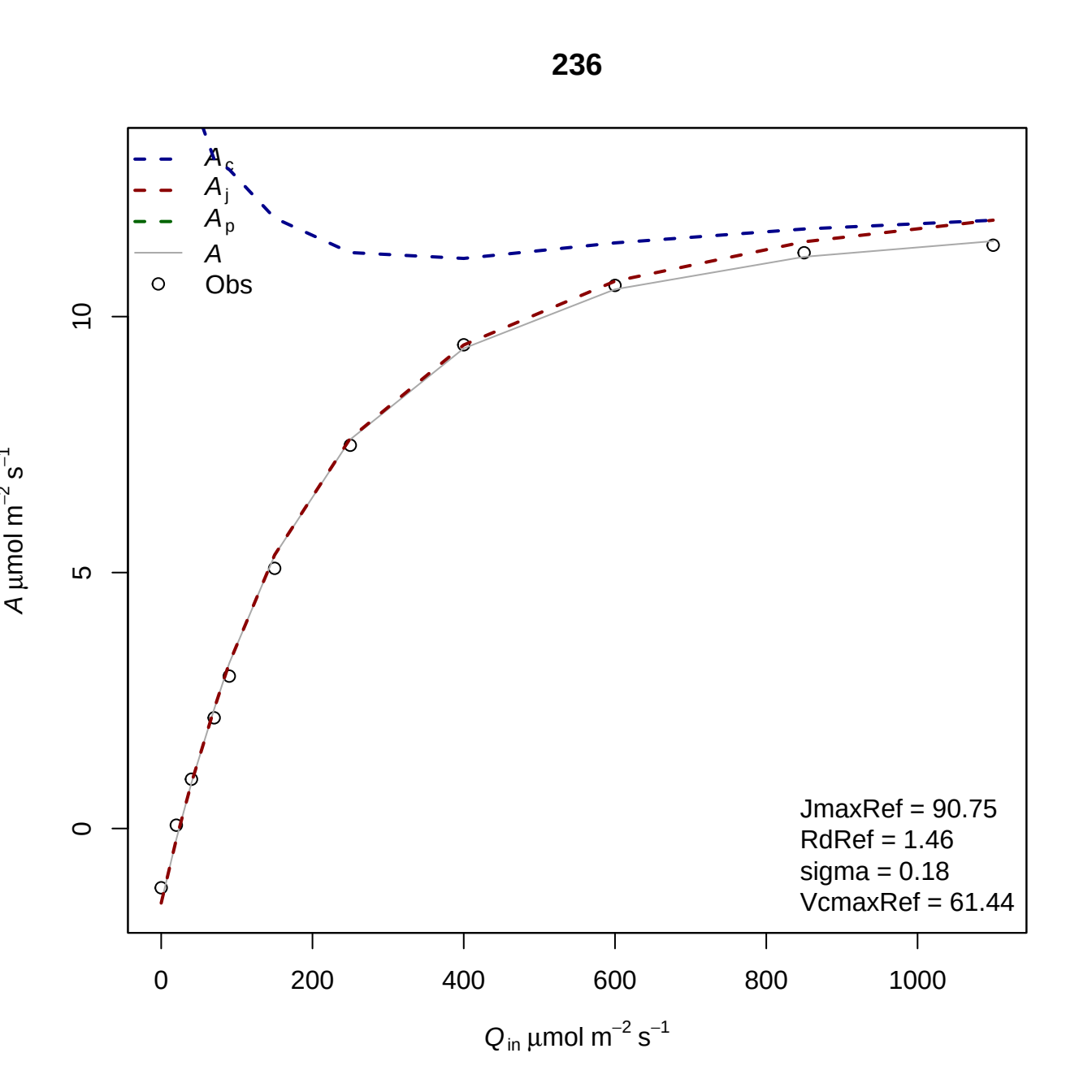

230
A_c
A_j
A_p
A
Obs
JmaxRef = 108.38
RdRef = 1.2
sigma = 0.39
VcmaxRef = 203.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

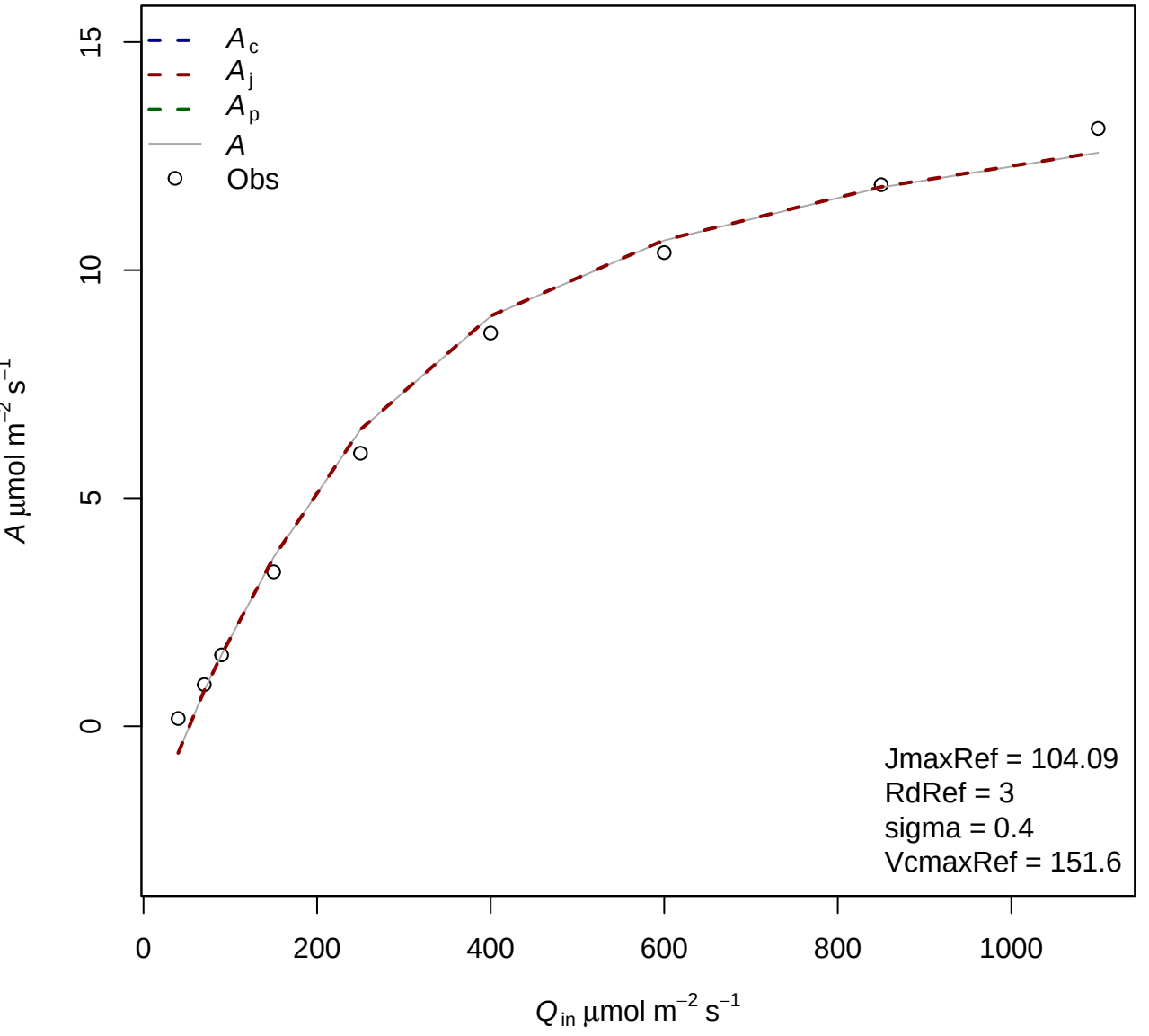

232
A_c
A_j
A_p
A
Obs
JmaxRef = 53.34
RdRef = 4.73
sigma = 0.39
VcmaxRef = 295.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

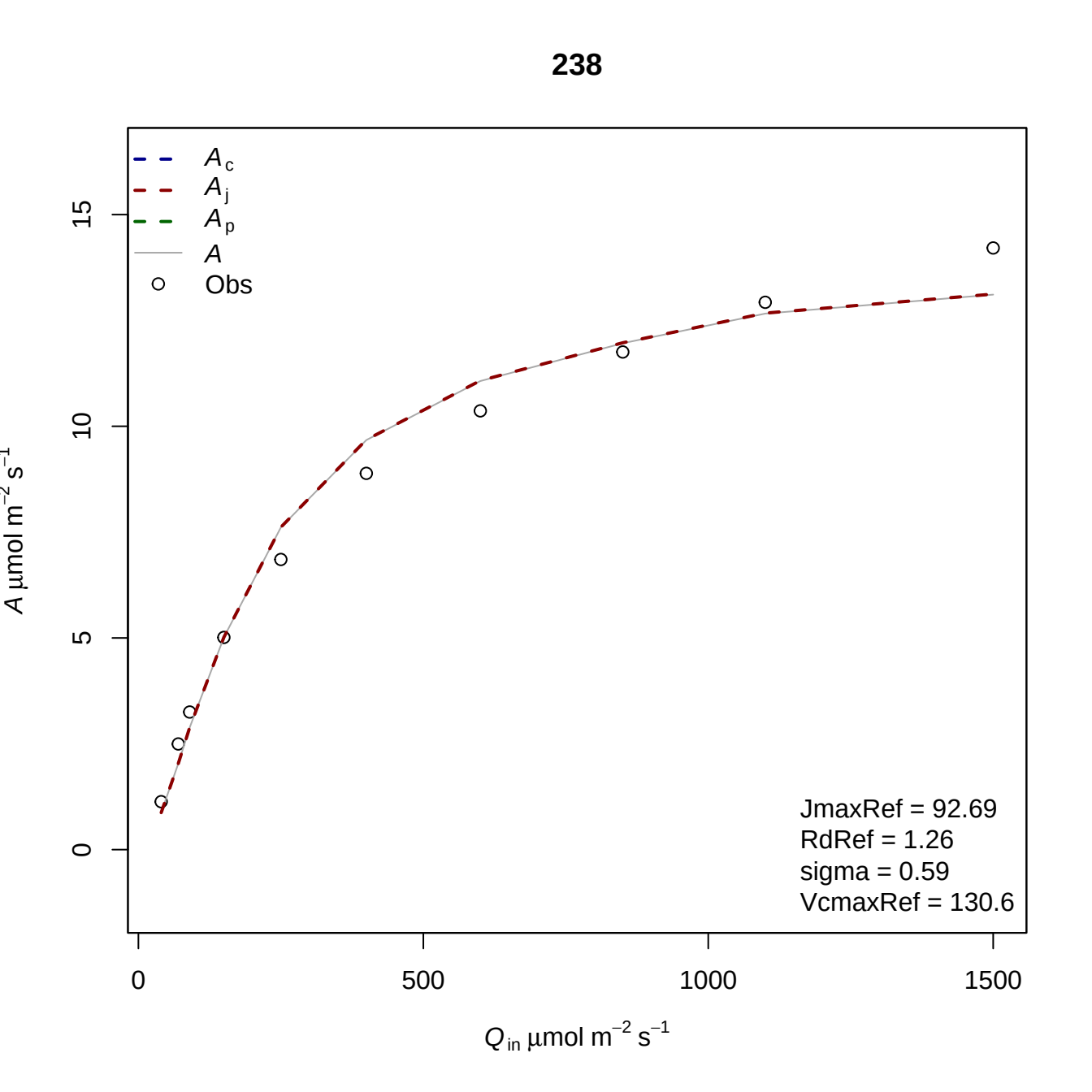

233
A_c
A_j
A_p
A
Obs
JmaxRef = 11.49
RdRef = 1.88
sigma = 0.15
VcmaxRef = 16585.22
0.5
0.0
−0.5
−1.0
−1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

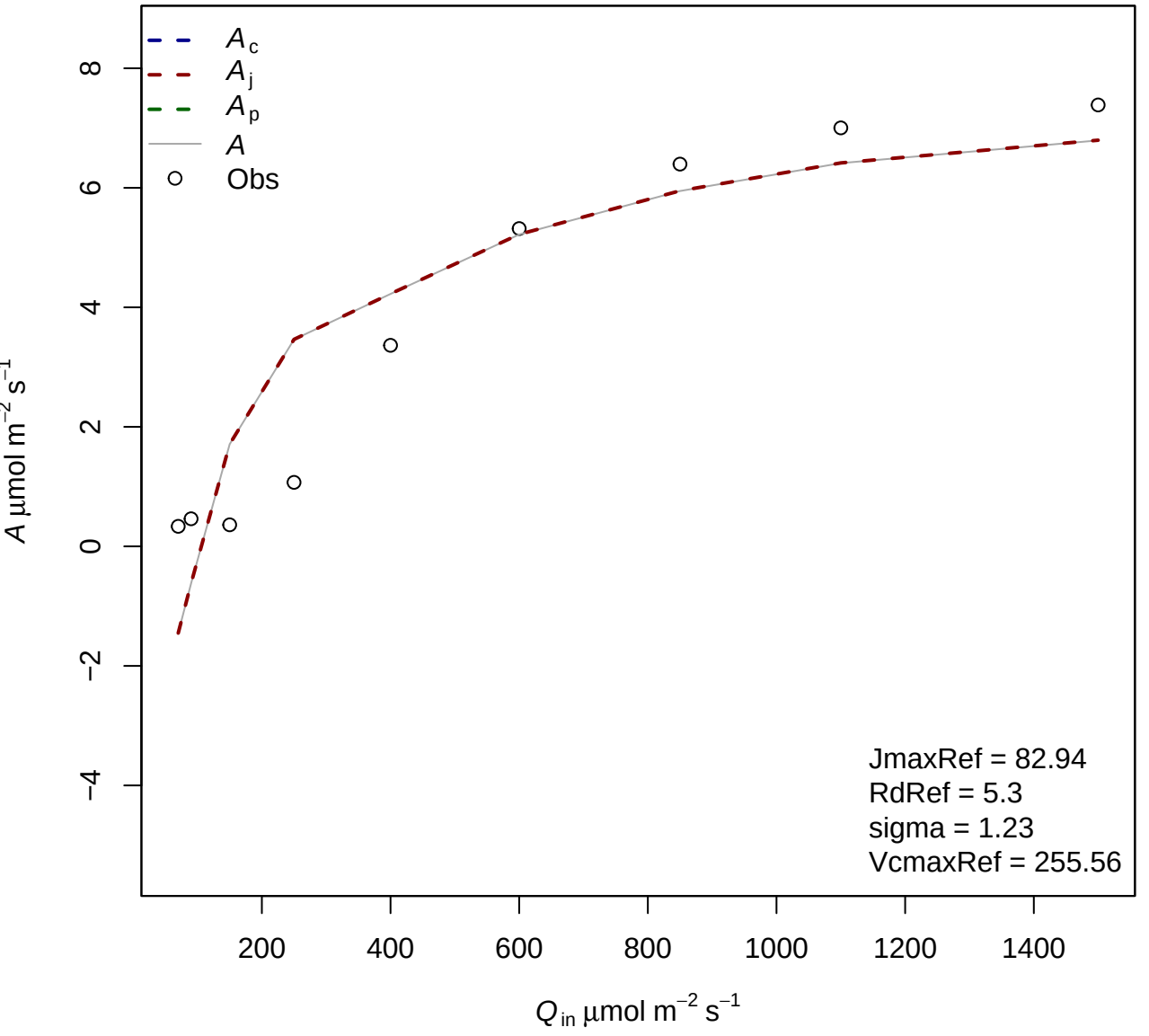

234
A_c
A_j
A_p
A
Obs
JmaxRef = 57.77
RdRef = 3.18
sigma = 0.62
VcmaxRef = 231.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

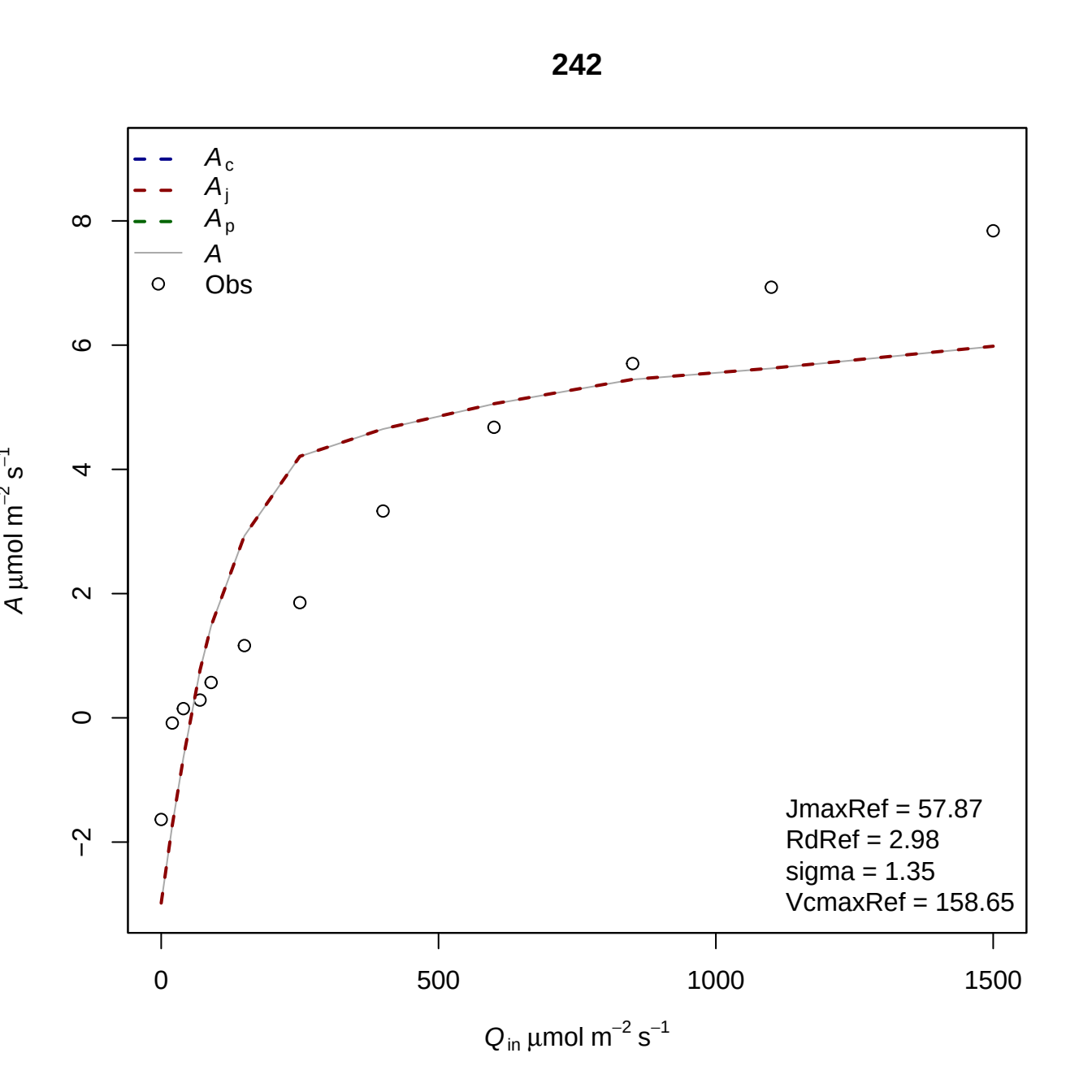

235
A_c
A_j
A_p
A
Obs
JmaxRef = 86
RdRef = 1.48
sigma = 0.27
VcmaxRef = 164.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000

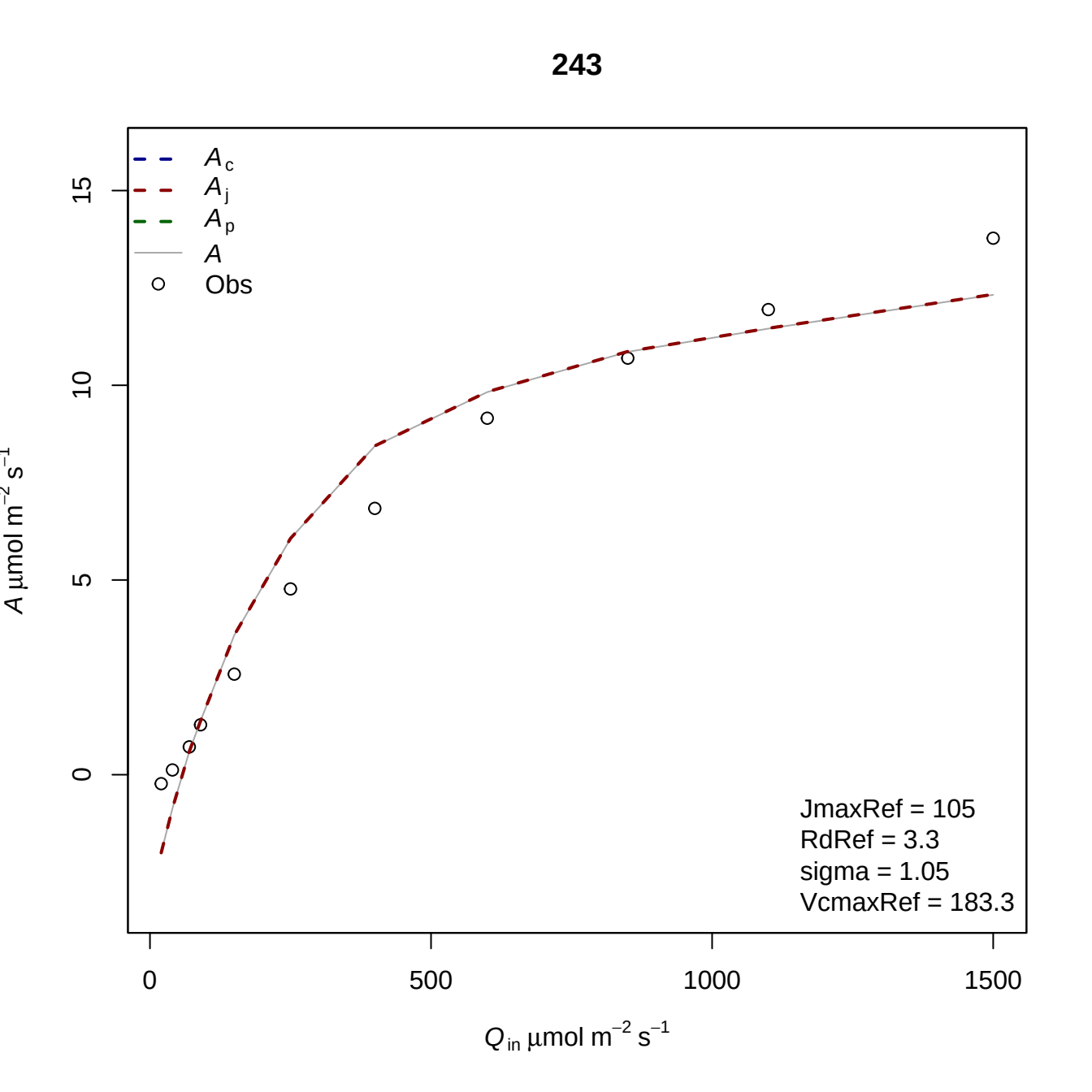

236
A_c
A_j
A_p
A
Obs
JmaxRef = 90.75
RdRef = 1.46
sigma = 0.18
VcmaxRef = 61.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

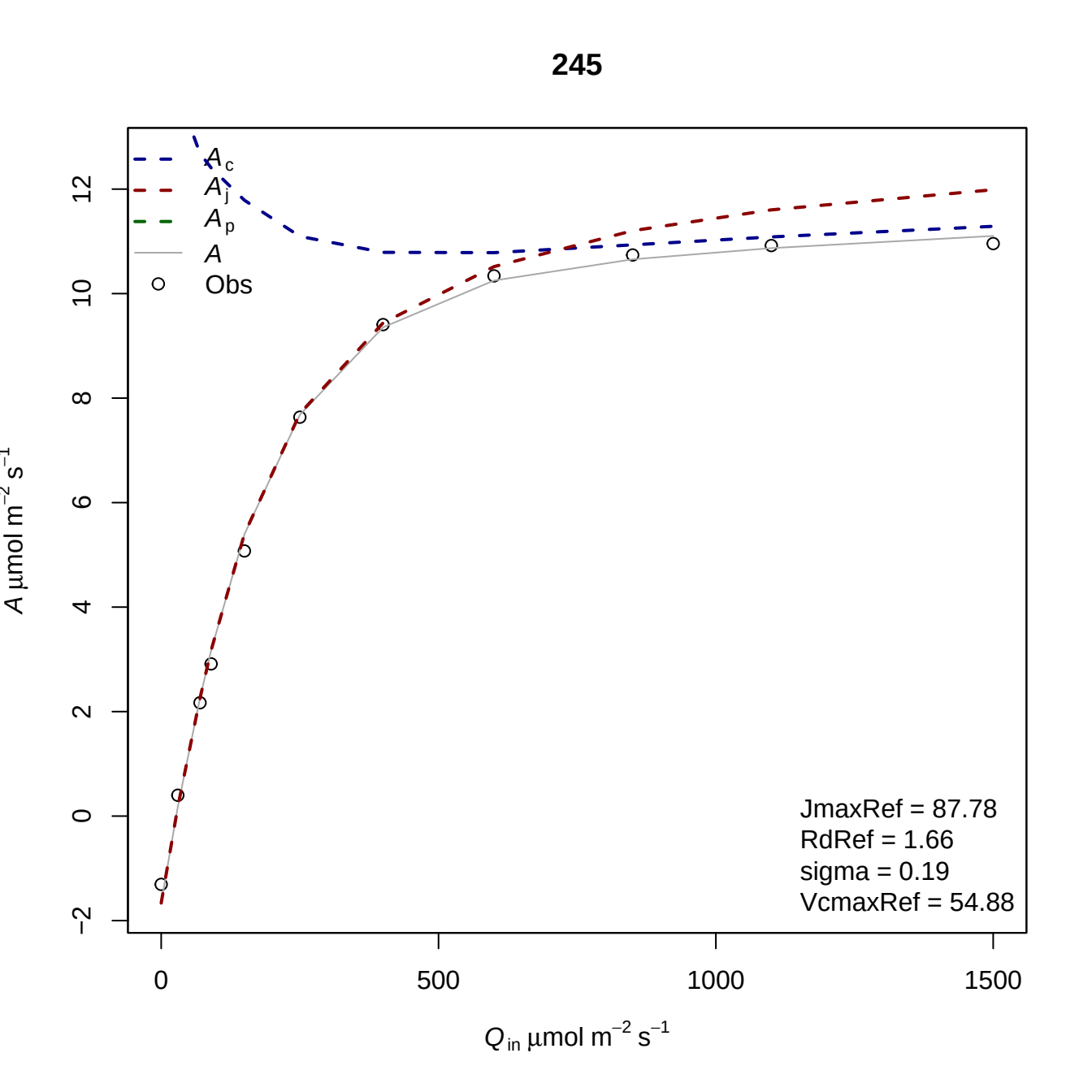

A_c
A_j
A_p
A
Obs
JmaxRef = 104.09
RdRef = 3
sigma = 0.4
VcmaxRef = 151.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000

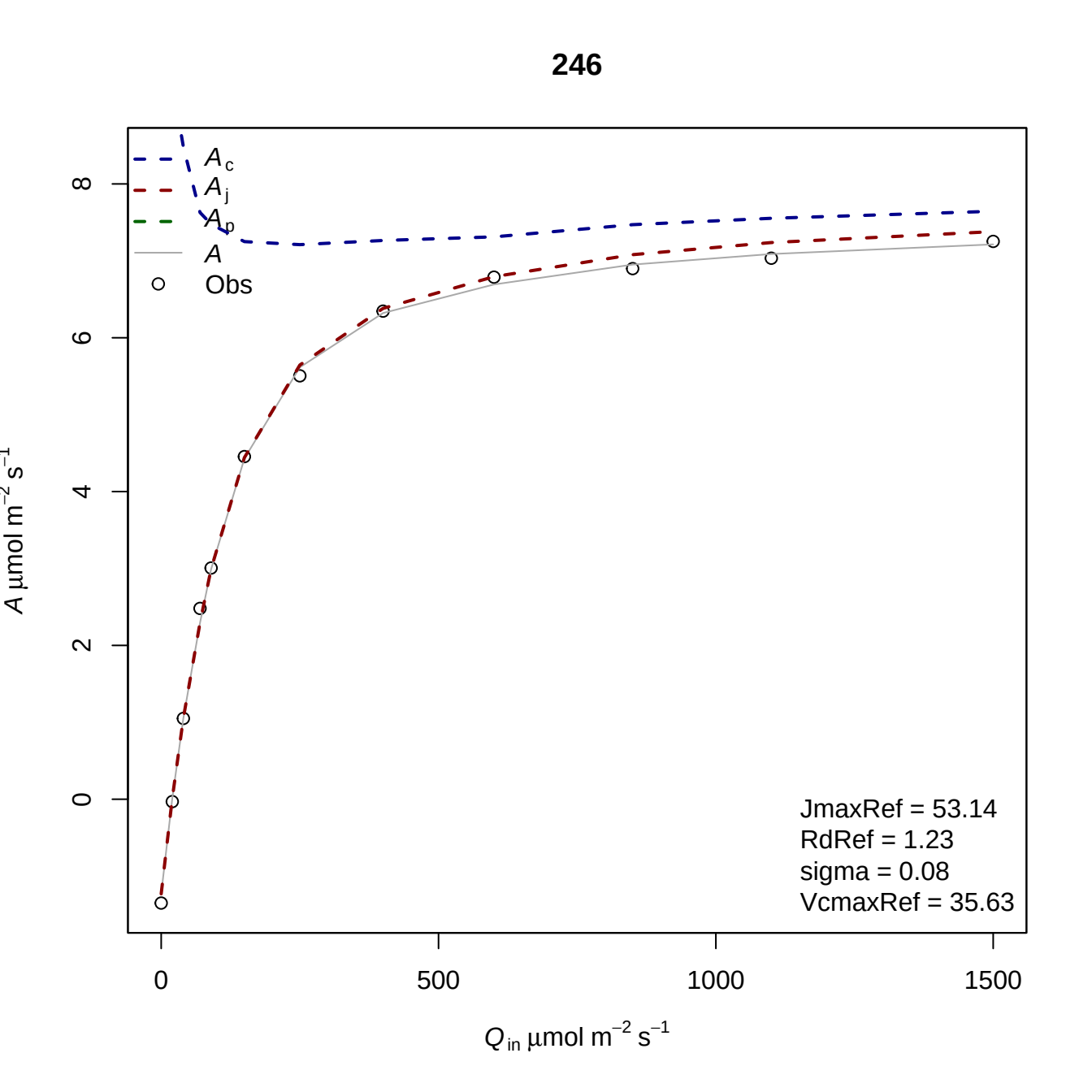

238
A_c
A_j
A_p
A
Obs
JmaxRef = 92.69
RdRef = 1.26
sigma = 0.59
VcmaxRef = 130.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

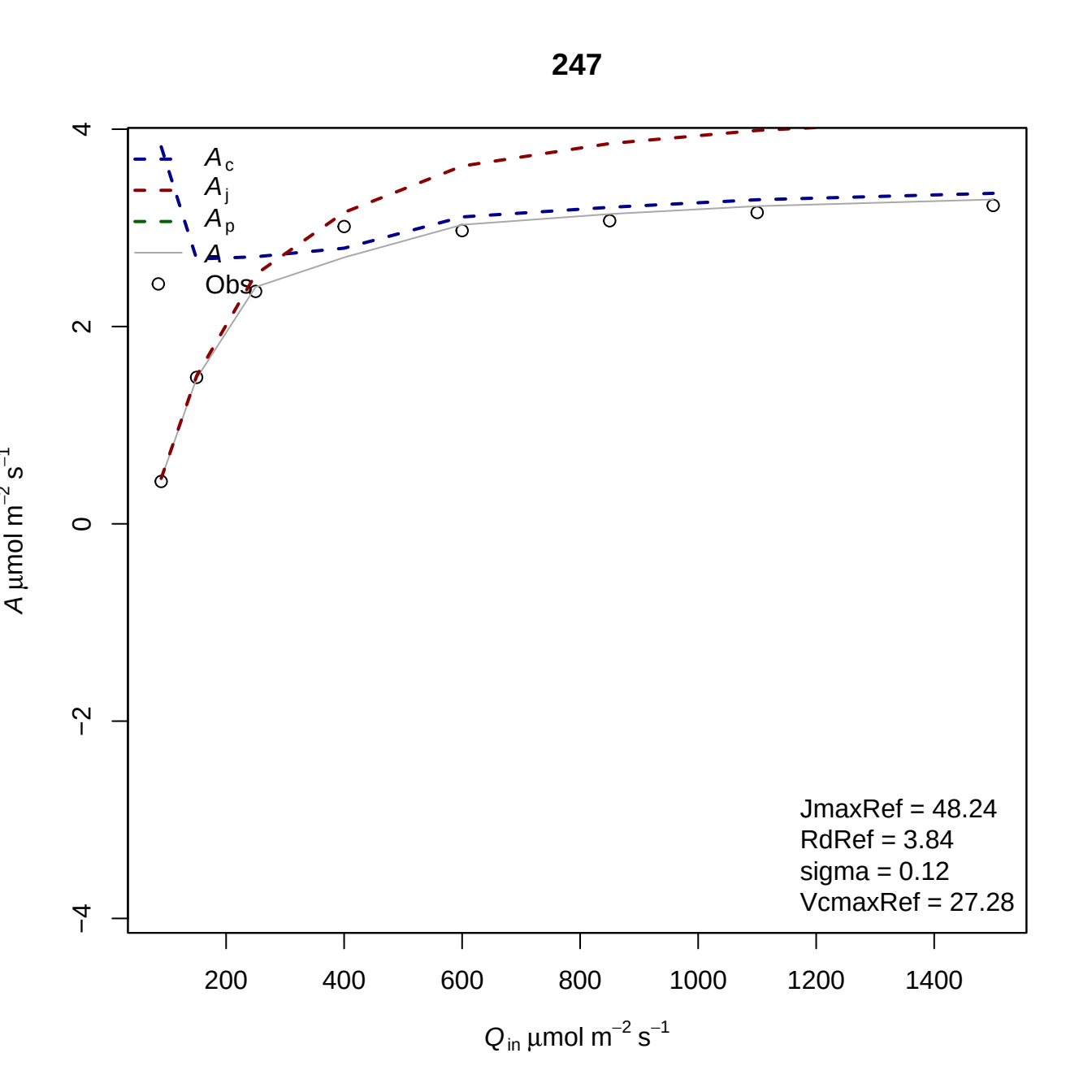

A_c
A_j
A_p
A
Obs
JmaxRef = 82.94
RdRef = 5.3
sigma = 1.23
VcmaxRef = 255.56
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

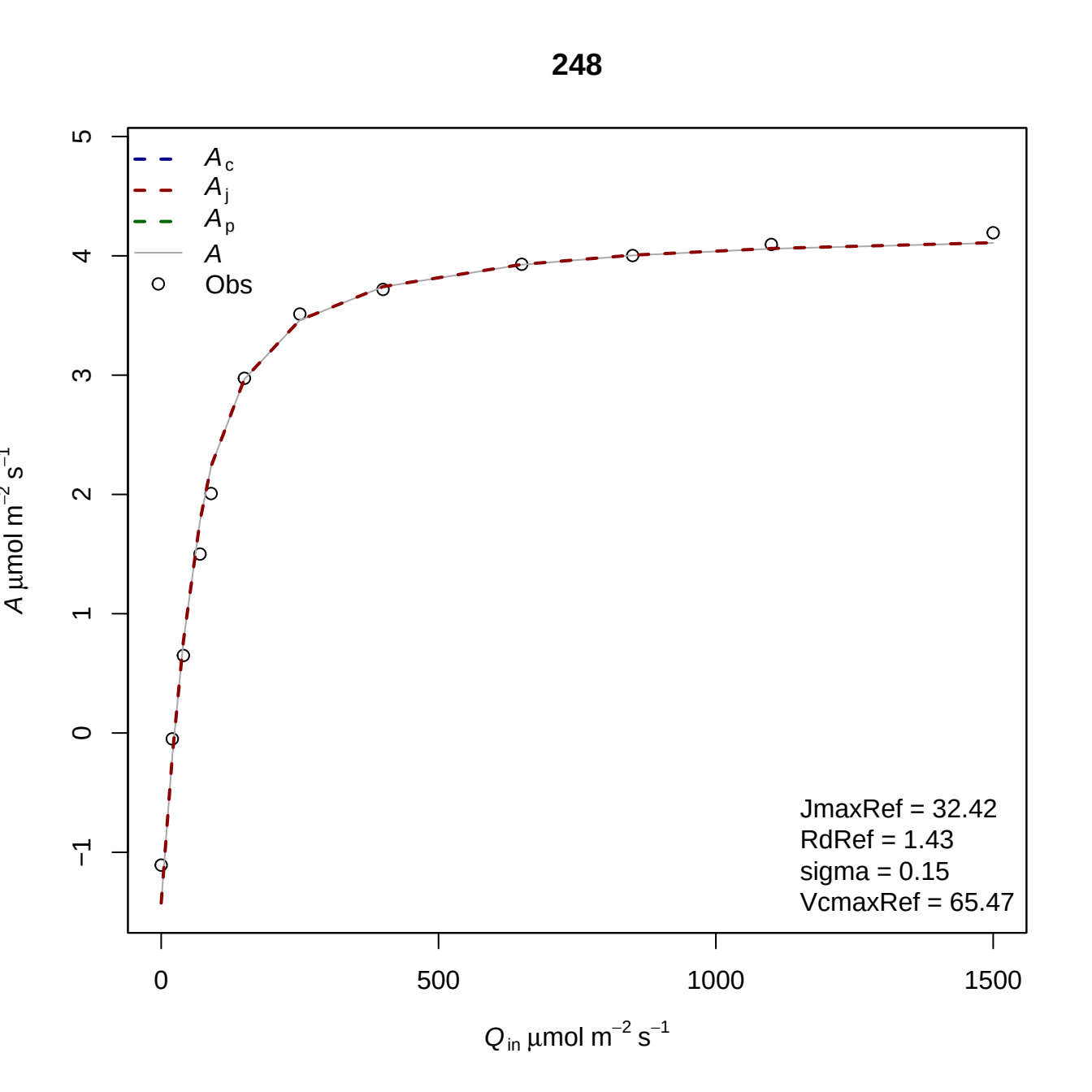

242
A_c
A_j
A_p
A
Obs
JmaxRef = 57.87
RdRef = 2.98
sigma = 1.35
VcmaxRef = 158.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

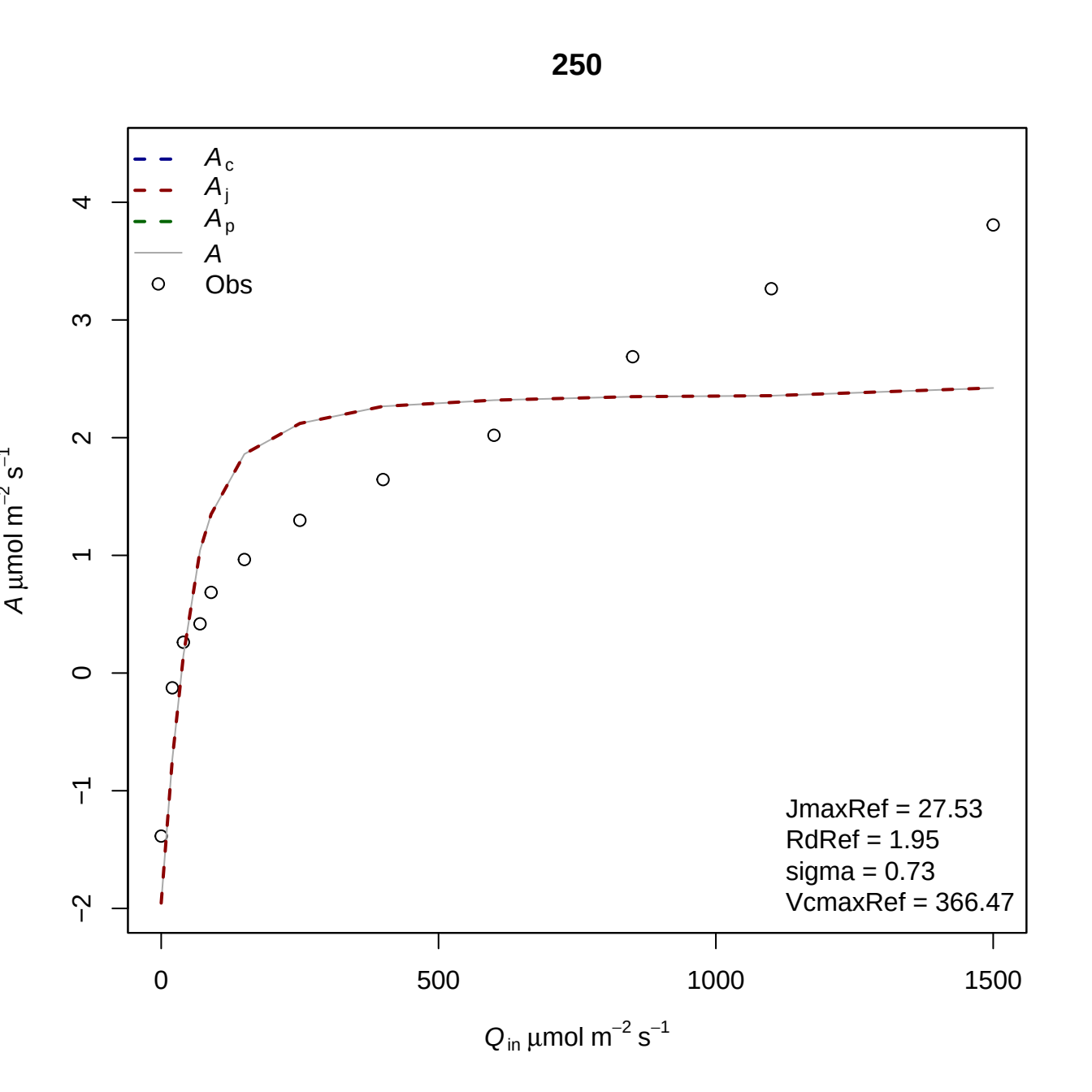

243
A_c
A_j
A_p
A
Obs
JmaxRef = 105
RdRef = 3.3
sigma = 1.05
VcmaxRef = 183.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

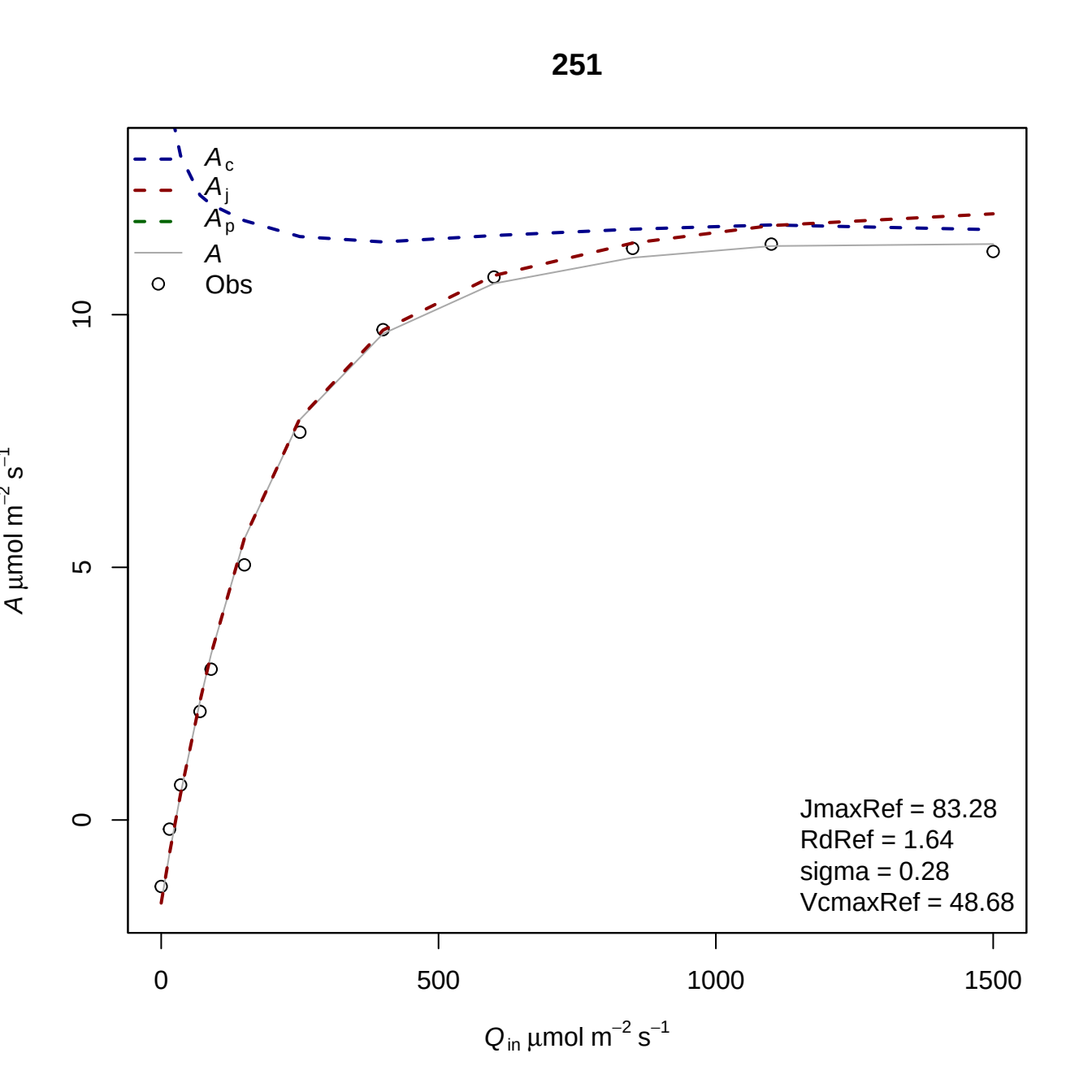

245
A_c
A_j
A_p
A
Obs
JmaxRef = 87.78
RdRef = 1.66
sigma = 0.19
VcmaxRef = 54.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

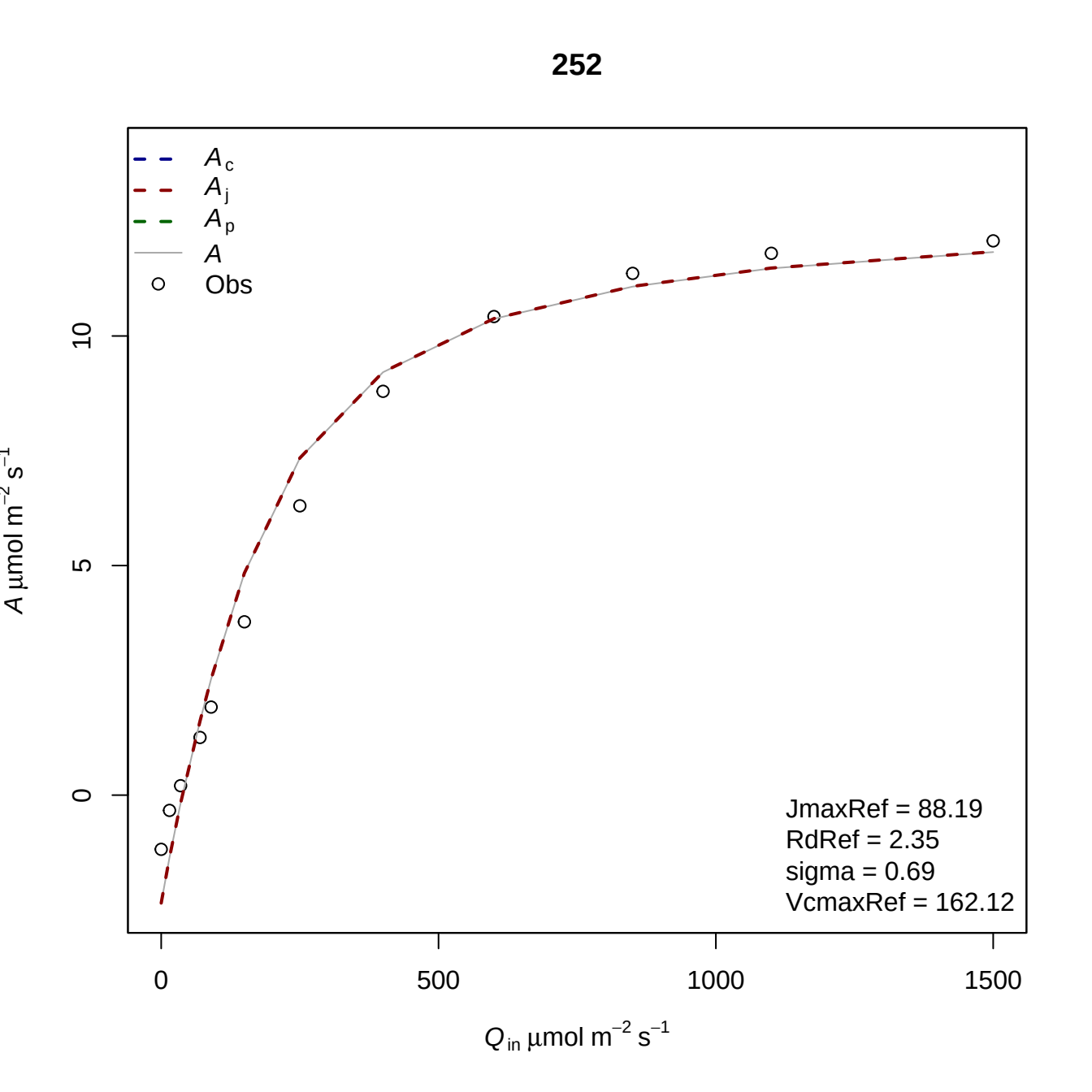

246
A_c
A_j
A_p
A
Obs
JmaxRef = 53.14
RdRef = 1.23
sigma = 0.08
VcmaxRef = 35.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

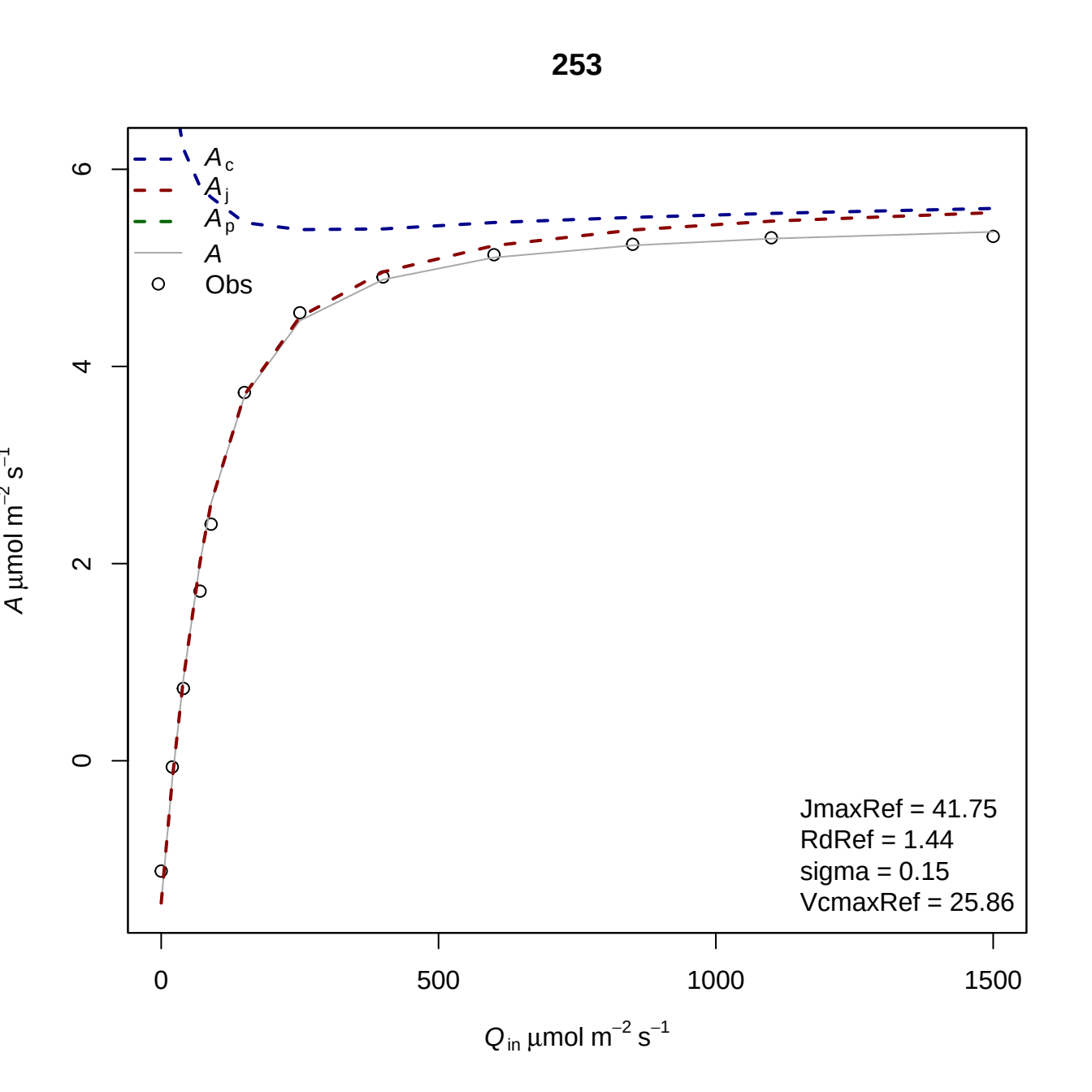

247
A_c
A_j
A_p
A
Obs
JmaxRef = 48.24
RdRef = 3.84
sigma = 0.12
VcmaxRef = 27.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

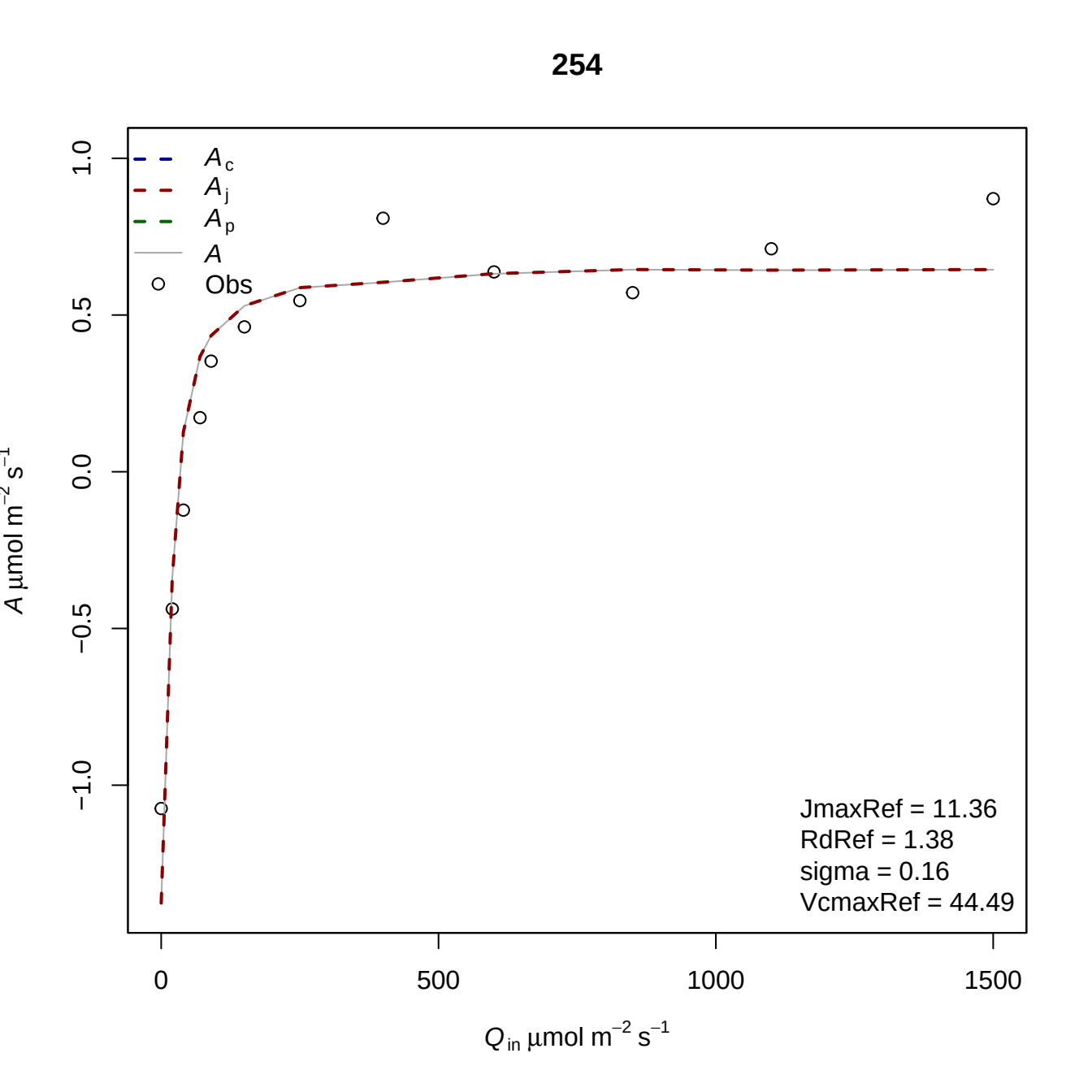

248
A_c
A_j
A_p
A
Obs
JmaxRef = 32.42
RdRef = 1.43
sigma = 0.15
VcmaxRef = 65.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
0
500
1000
1500

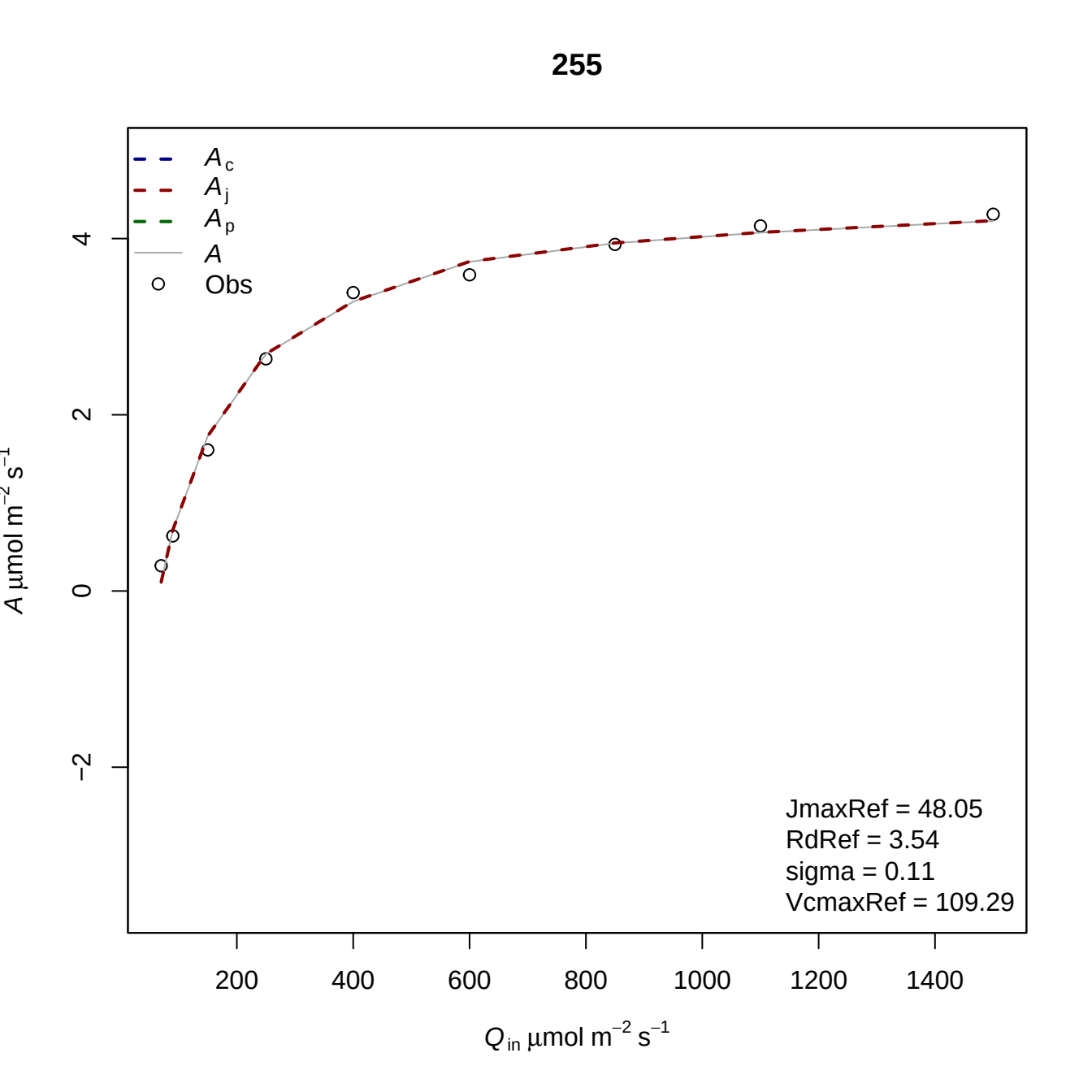

250
A_c
A_j
A_p
A
Obs
JmaxRef = 27.53
RdRef = 1.95
sigma = 0.73
VcmaxRef = 366.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
500
1000
1500

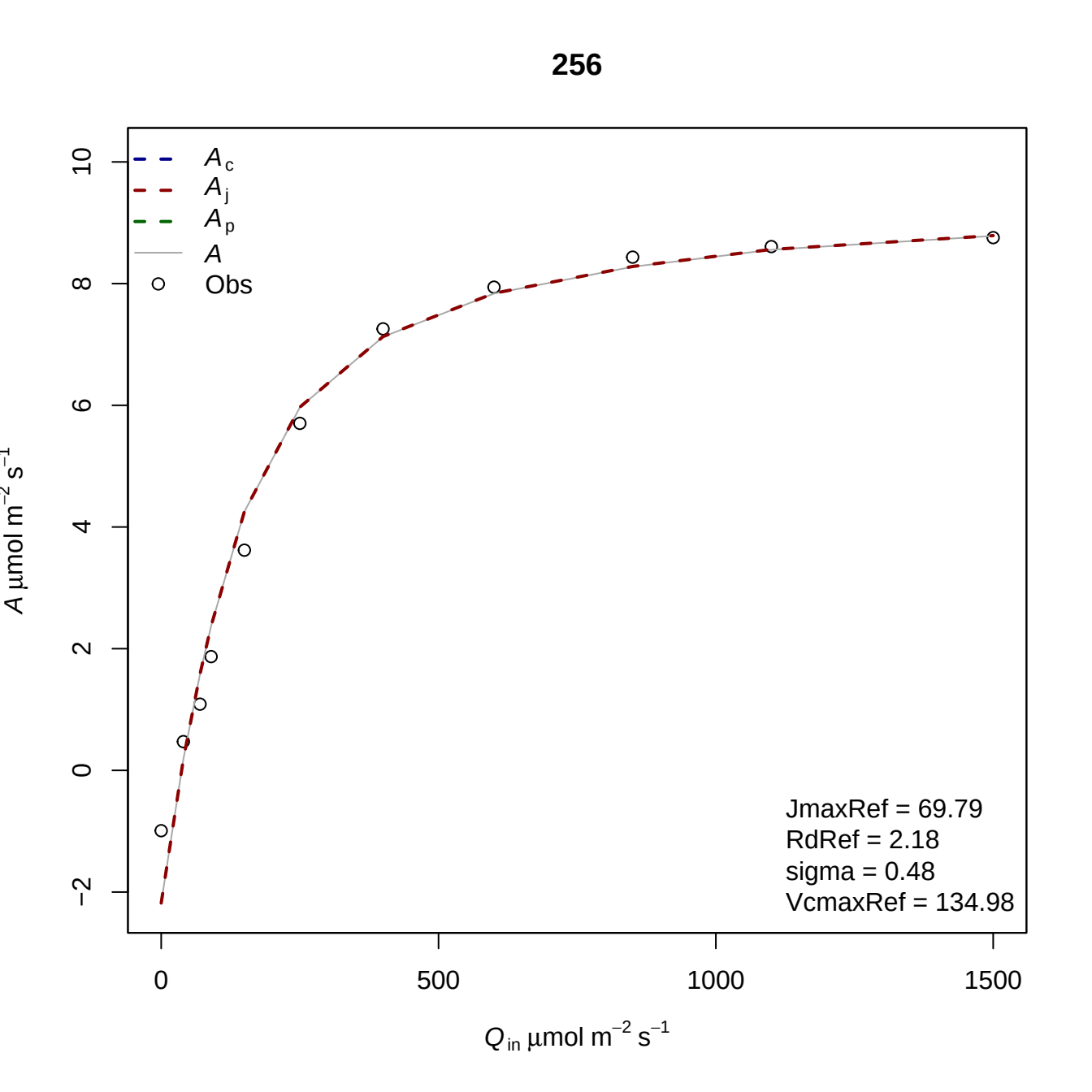

251
A_c
A_j
A_p
A
Obs
JmaxRef = 83.28
RdRef = 1.64
sigma = 0.28
VcmaxRef = 48.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

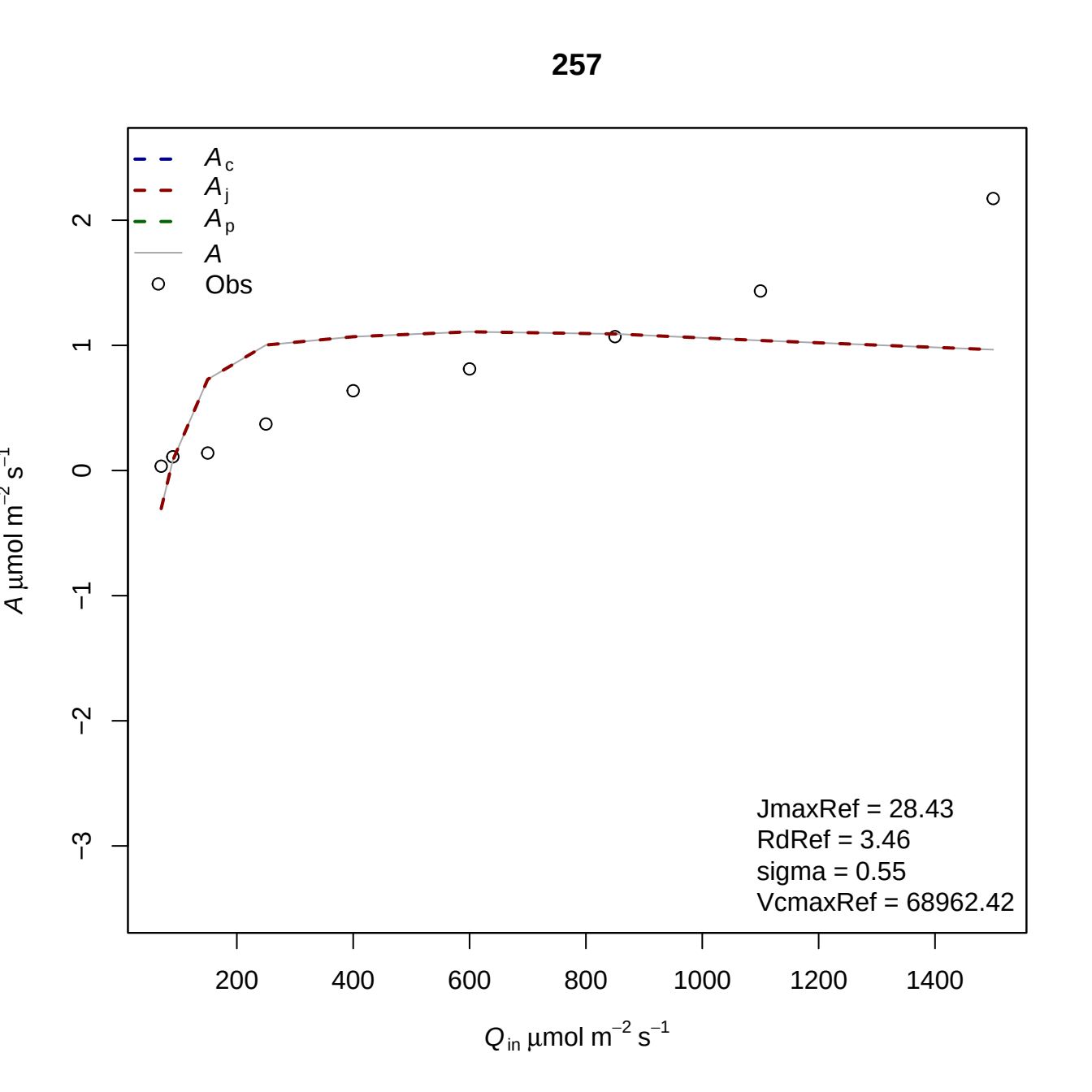

252
A_c
A_j
A_p
A
Obs
JmaxRef = 88.19
RdRef = 2.35
sigma = 0.69
VcmaxRef = 162.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

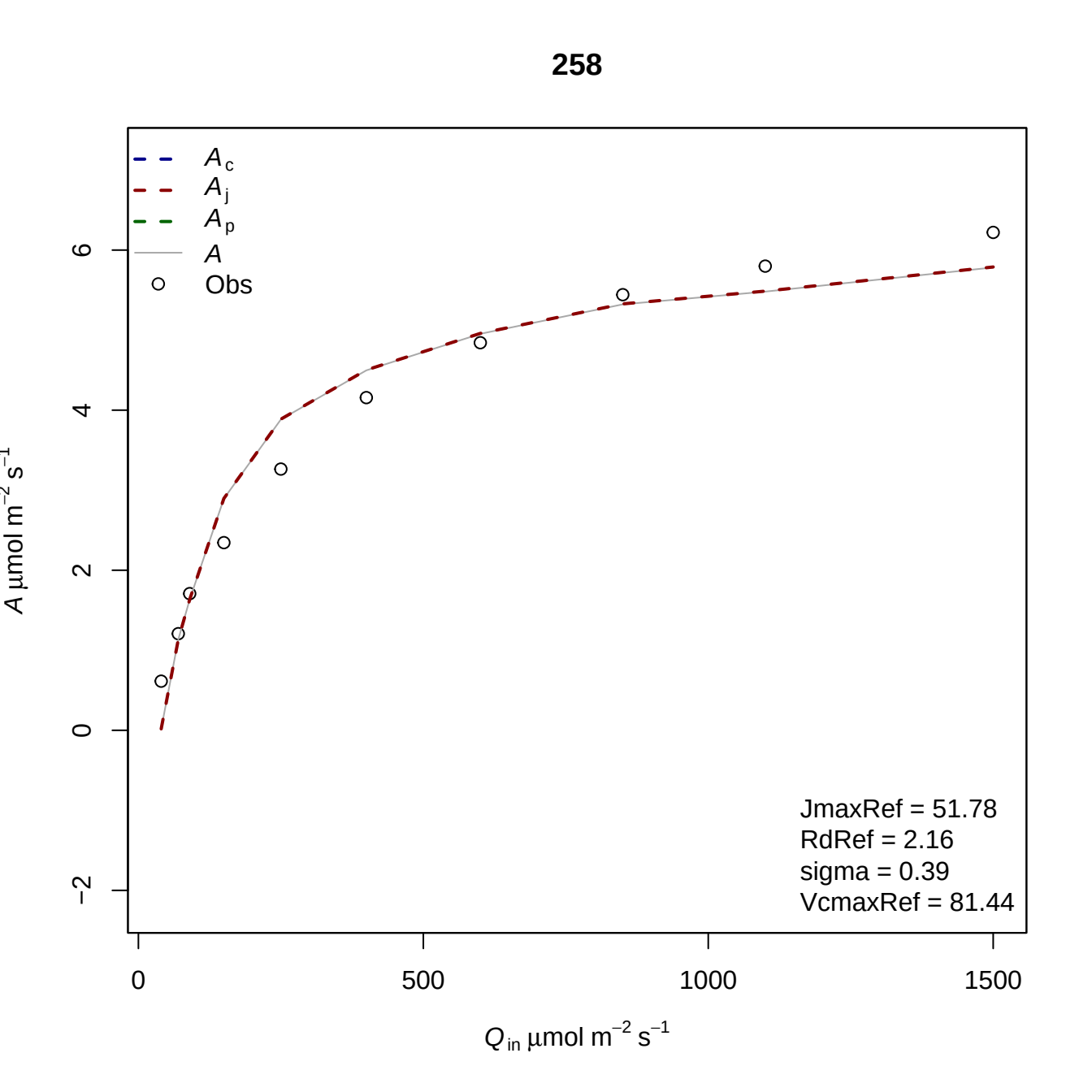

253
A_c
A_j
A_p
A
Obs
JmaxRef = 41.75
RdRef = 1.44
sigma = 0.15
VcmaxRef = 25.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

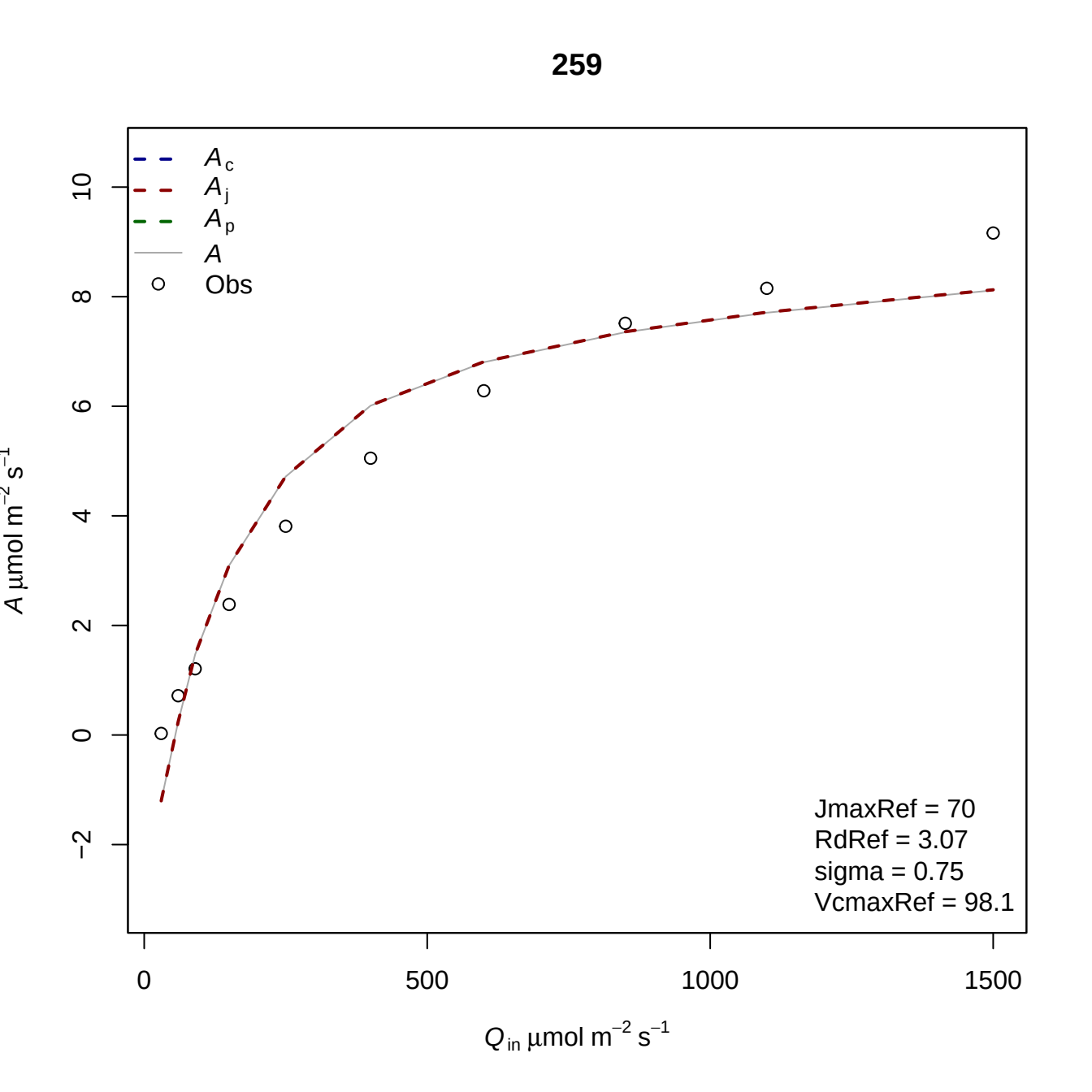

254
A_c
A_j
A_p
A
Obs
1.0
0.5
0.0
−0.5
−1.0
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 11.36
RdRef = 1.38
sigma = 0.16
VcmaxRef = 44.49

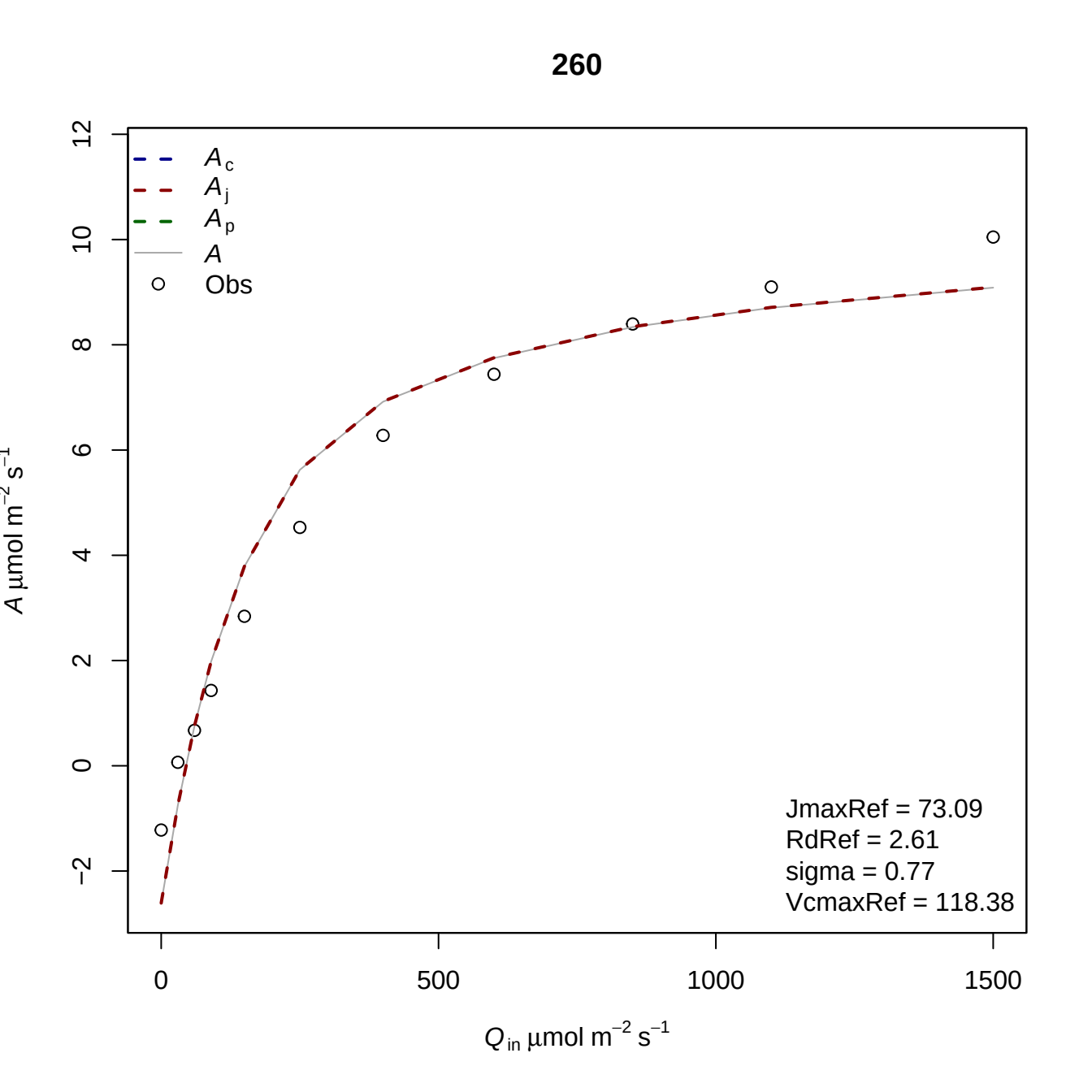

255
A_c
A_j
A_p
A
Obs
4
2
0
−2
A µmol m^{-2} s^{-1}
200
400
600
800
1000
1200
1400
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 48.05
RdRef = 3.54
sigma = 0.11
VcmaxRef = 109.29

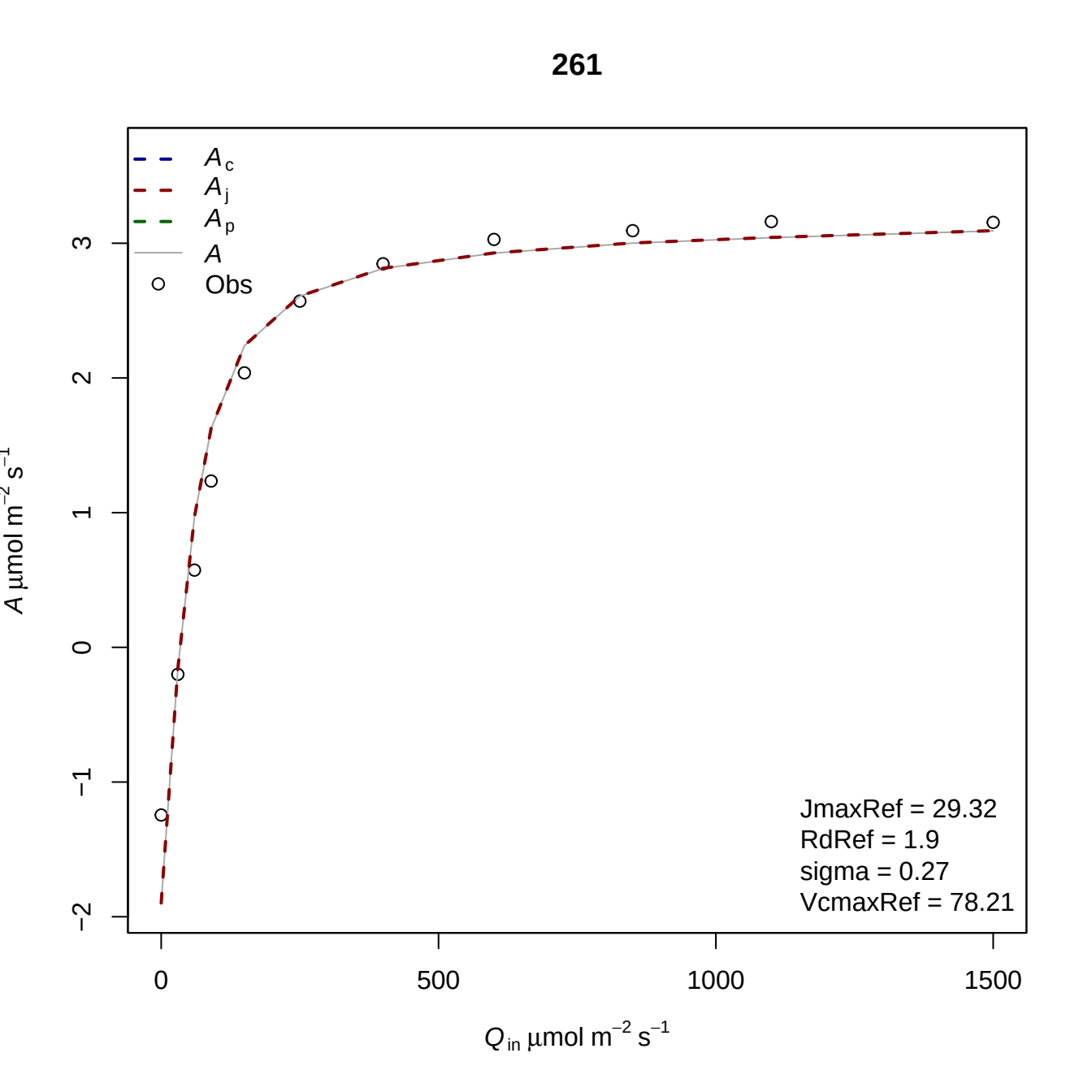

256
A_c
A_j
A_p
A
Obs
JmaxRef = 69.79
RdRef = 2.18
sigma = 0.48
VcmaxRef = 134.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

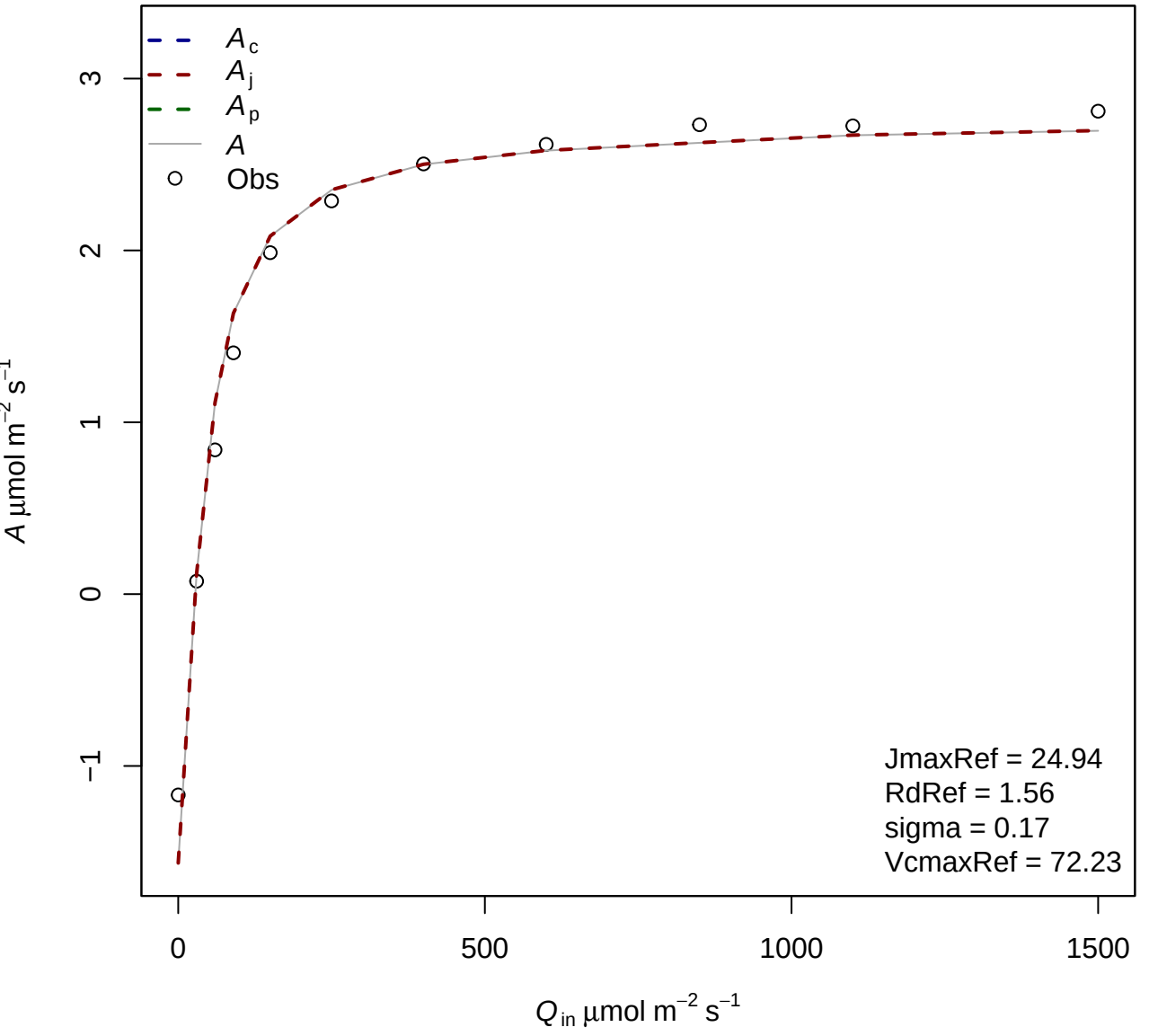

257
A_c
A_j
A_p
A
Obs
JmaxRef = 28.43
RdRef = 3.46
sigma = 0.55
VcmaxRef = 68962.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
−3
200
400
600
800
1000
1200
1400

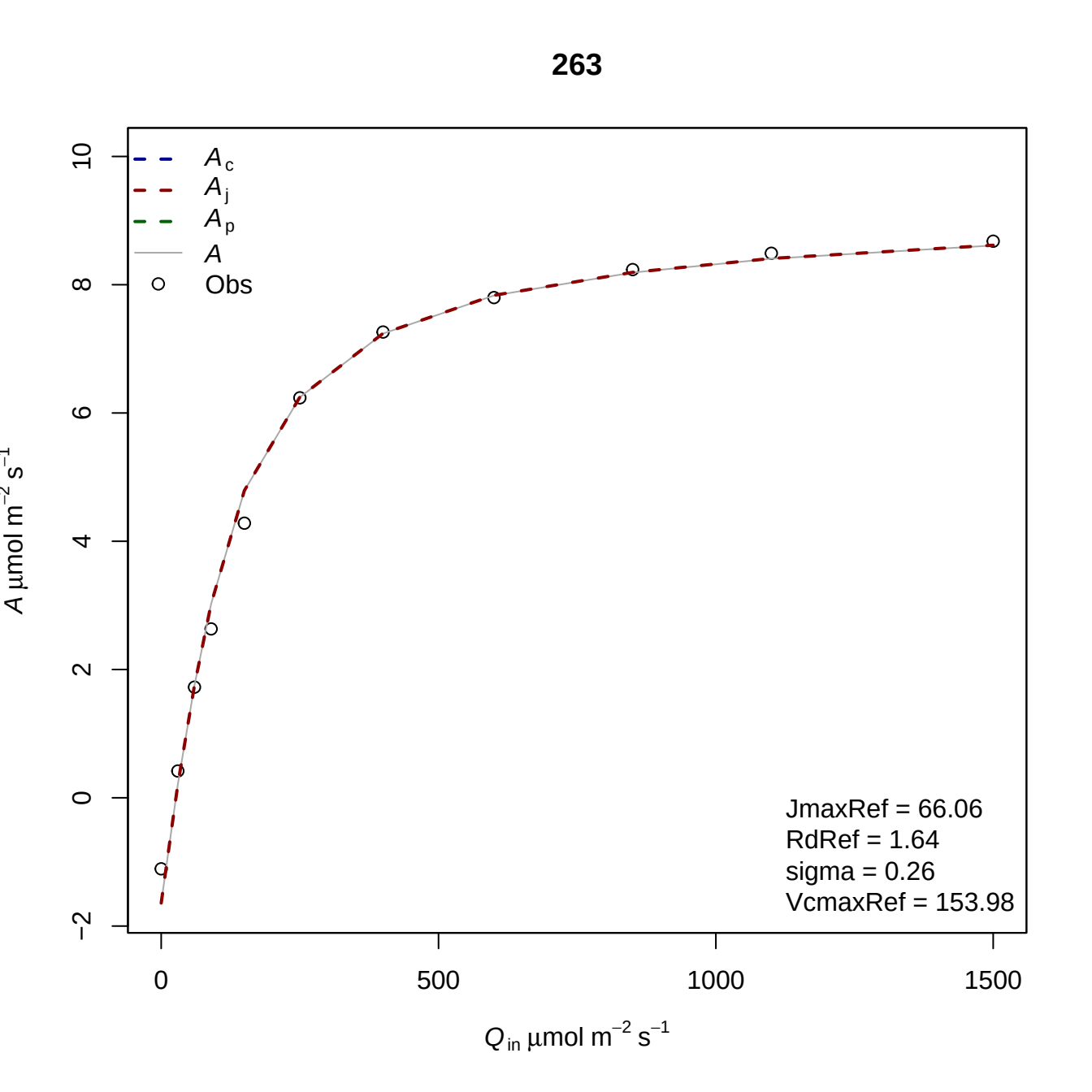

258
A_c
A_j
A_p
A
Obs
JmaxRef = 51.78
RdRef = 2.16
sigma = 0.39
VcmaxRef = 81.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

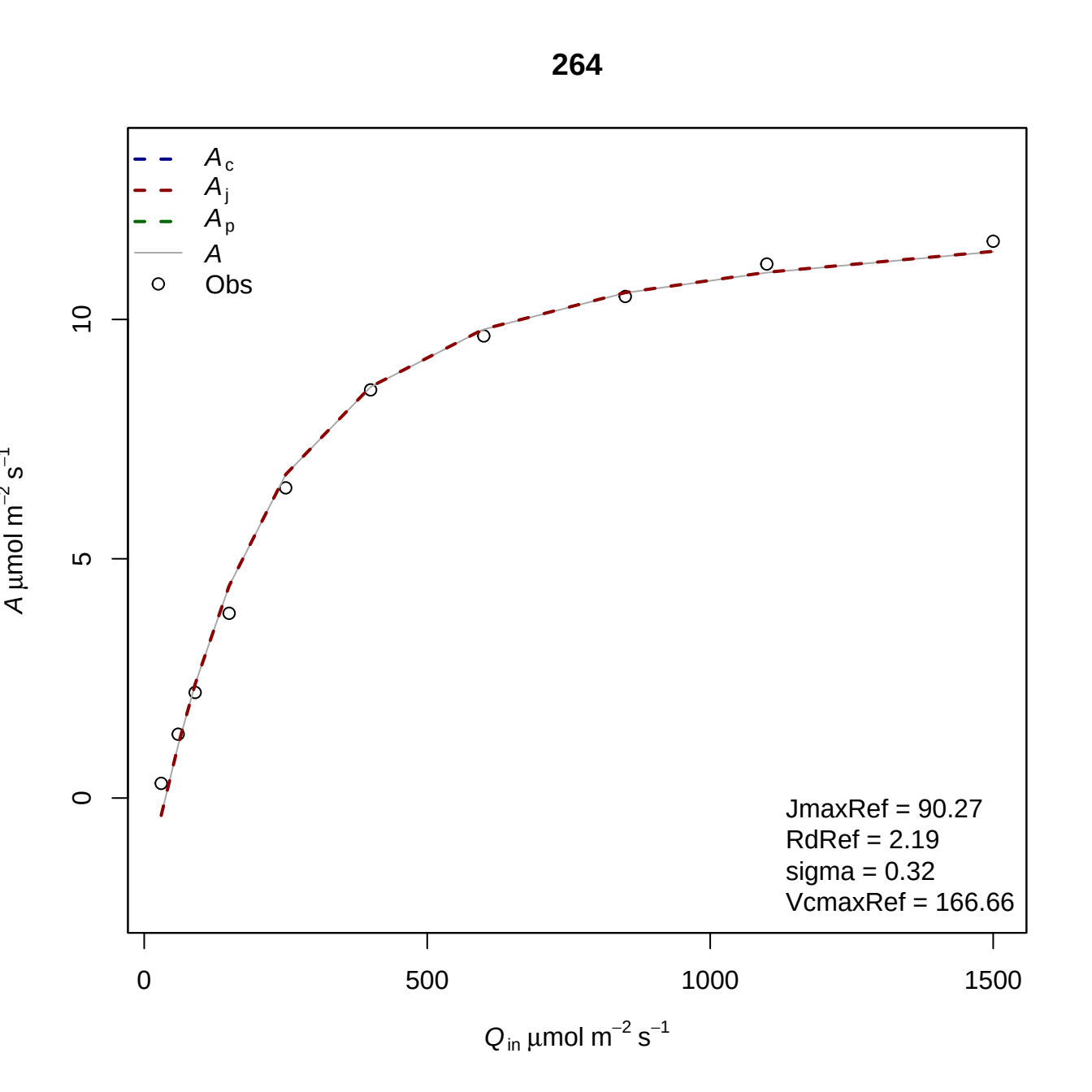

259
A_c
A_j
A_p
A
Obs
JmaxRef = 70
RdRef = 3.07
sigma = 0.75
VcmaxRef = 98.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

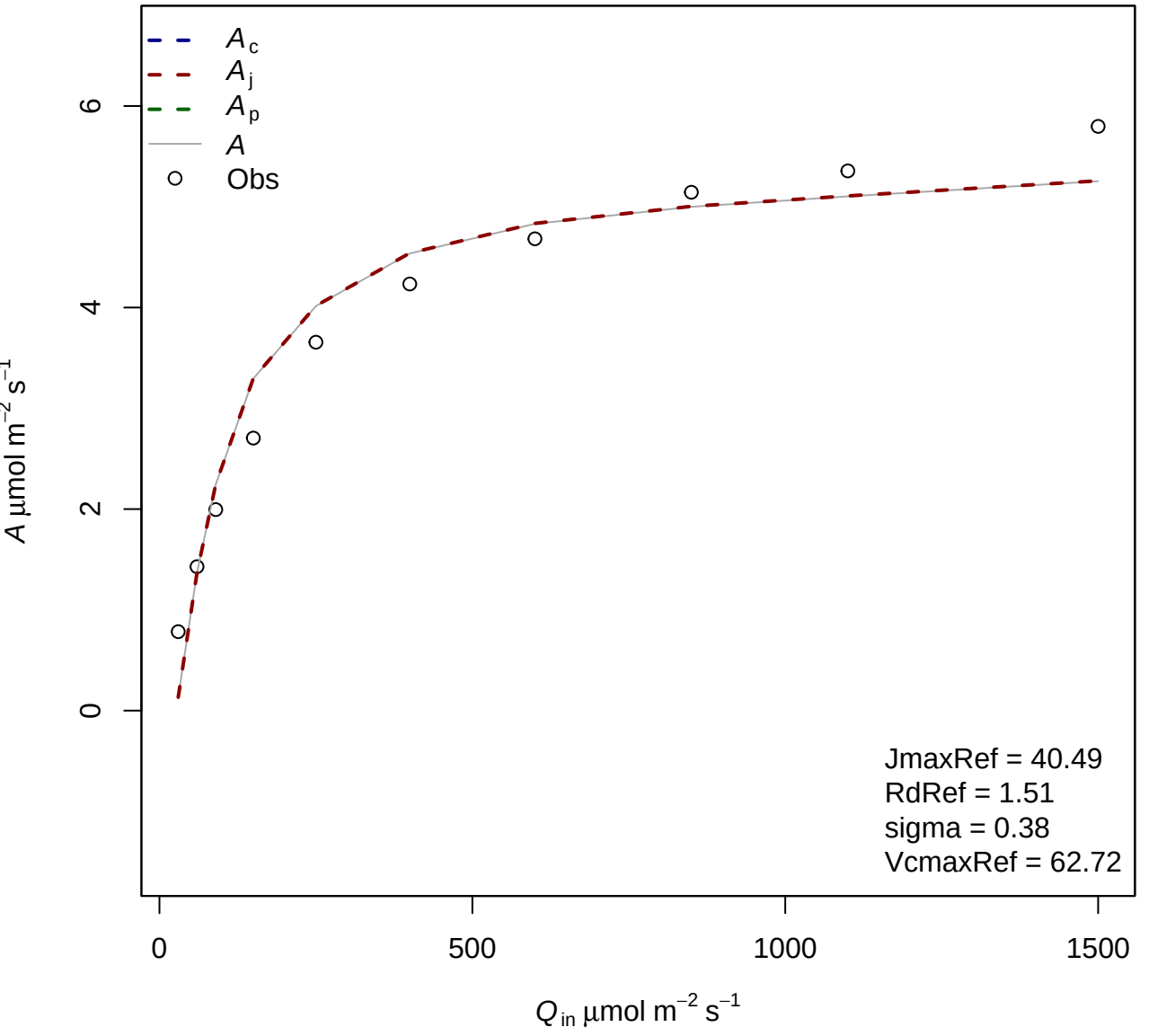

260
A_c
A_j
A_p
A
Obs
JmaxRef = 73.09
RdRef = 2.61
sigma = 0.77
VcmaxRef = 118.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

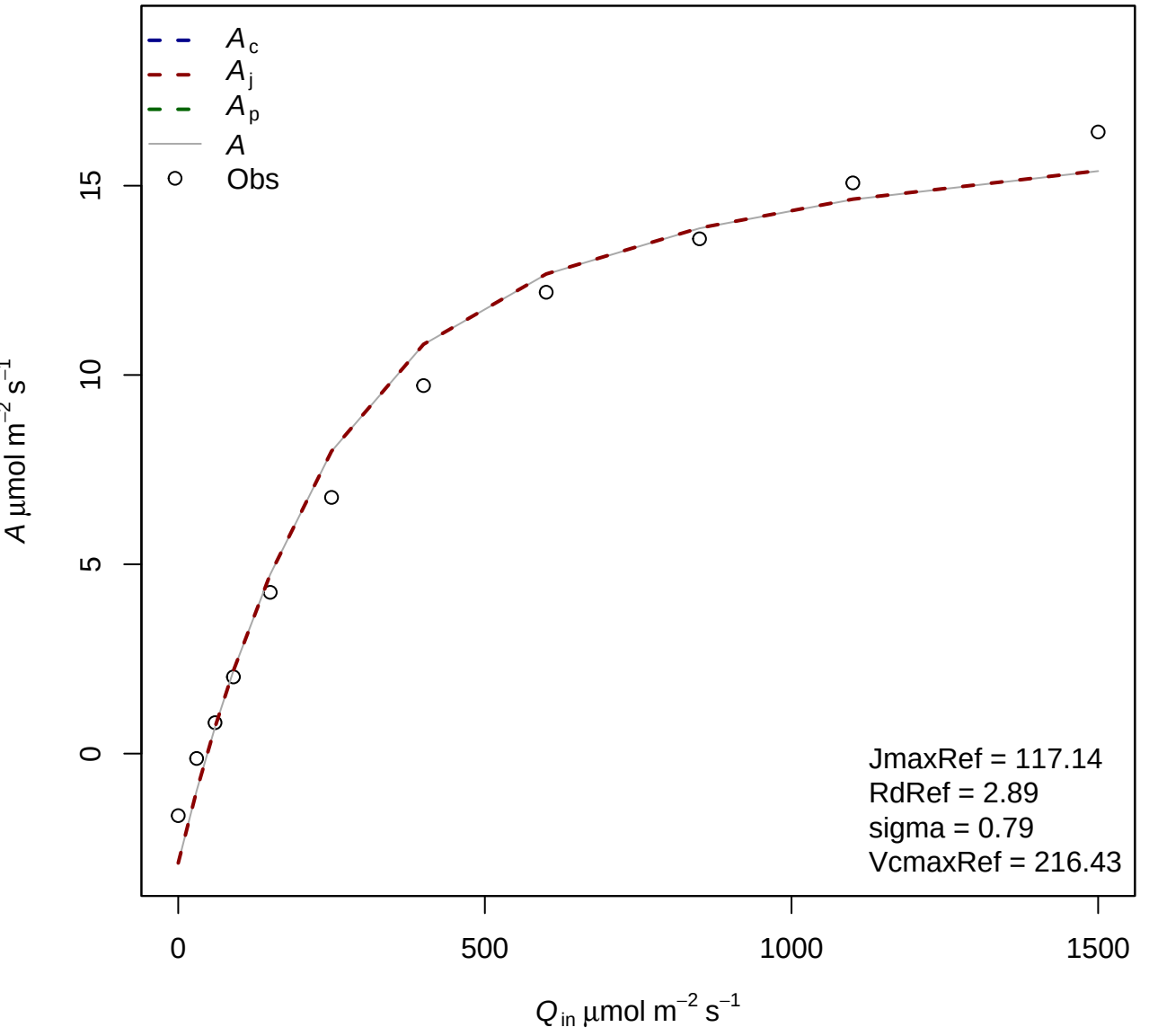

261
A_c
A_j
A_p
A
Obs
JmaxRef = 29.32
RdRef = 1.9
sigma = 0.27
VcmaxRef = 78.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
500
1000
1500

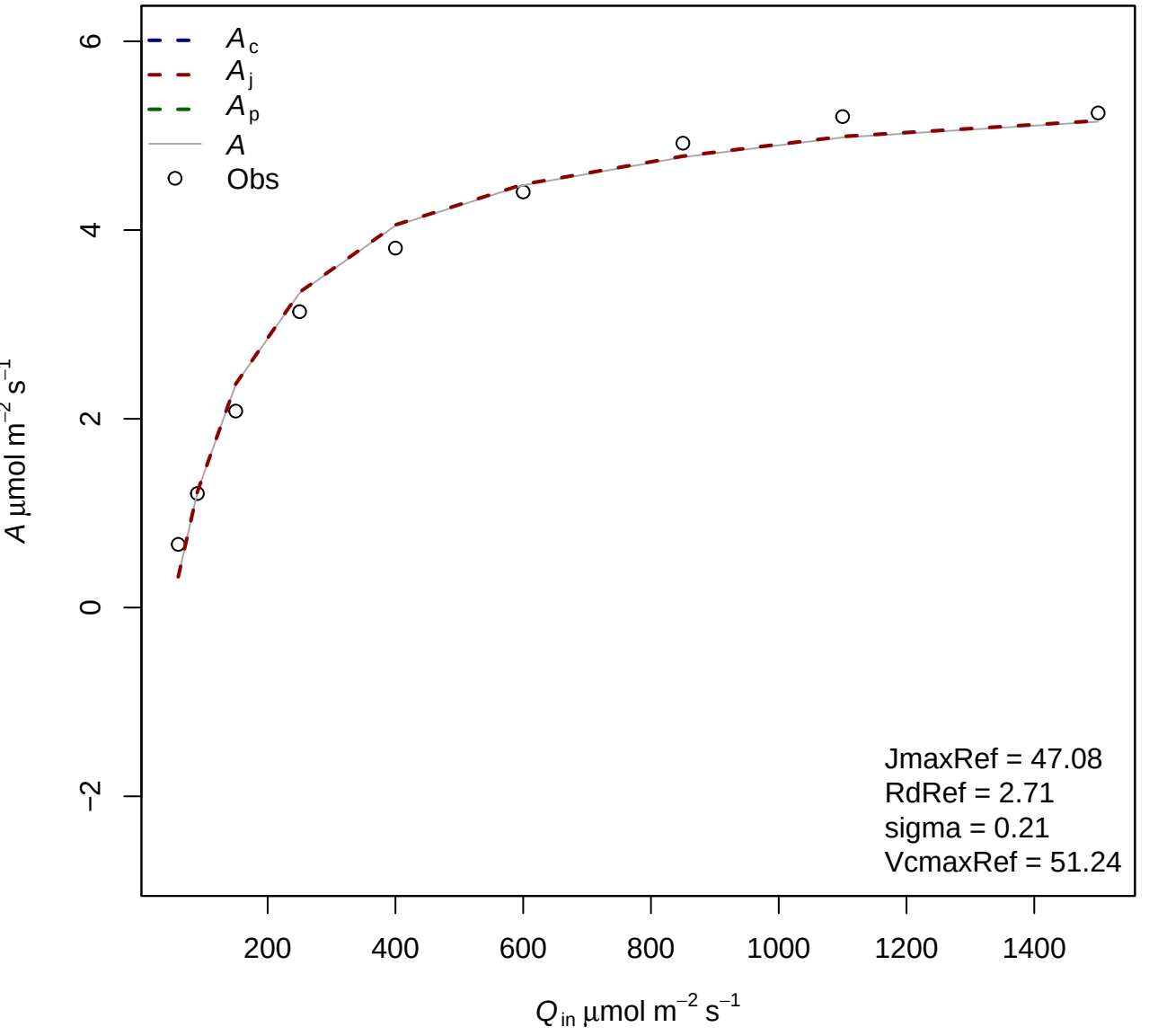

A_c
A_j
A_p
A
Obs
JmaxRef = 24.94
RdRef = 1.56
sigma = 0.17
VcmaxRef = 72.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

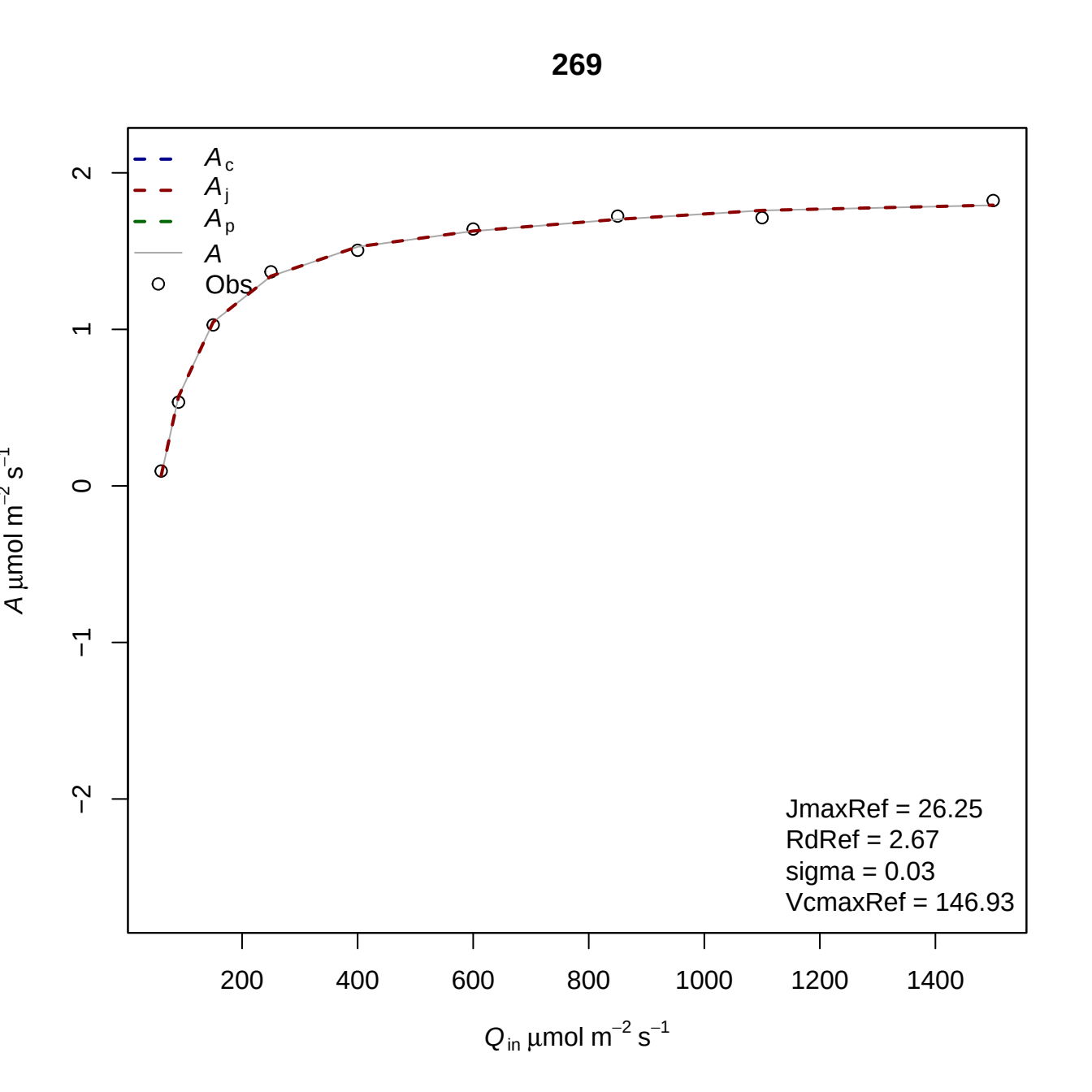

263
A_c
A_j
A_p
A
Obs
JmaxRef = 66.06
RdRef = 1.64
sigma = 0.26
VcmaxRef = 153.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

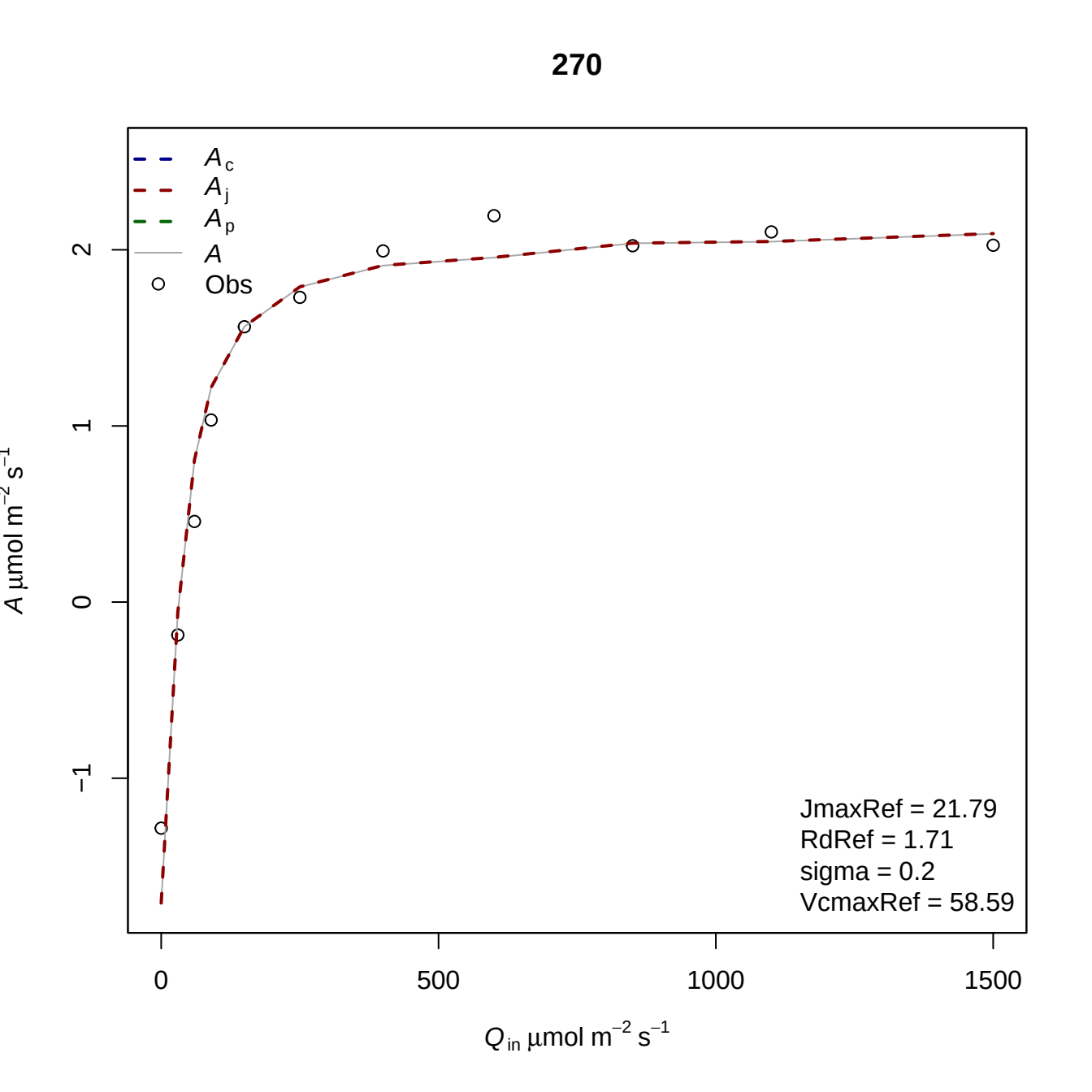

264
A_c
A_j
A_p
A
Obs
JmaxRef = 90.27
RdRef = 2.19
sigma = 0.32
VcmaxRef = 166.66
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

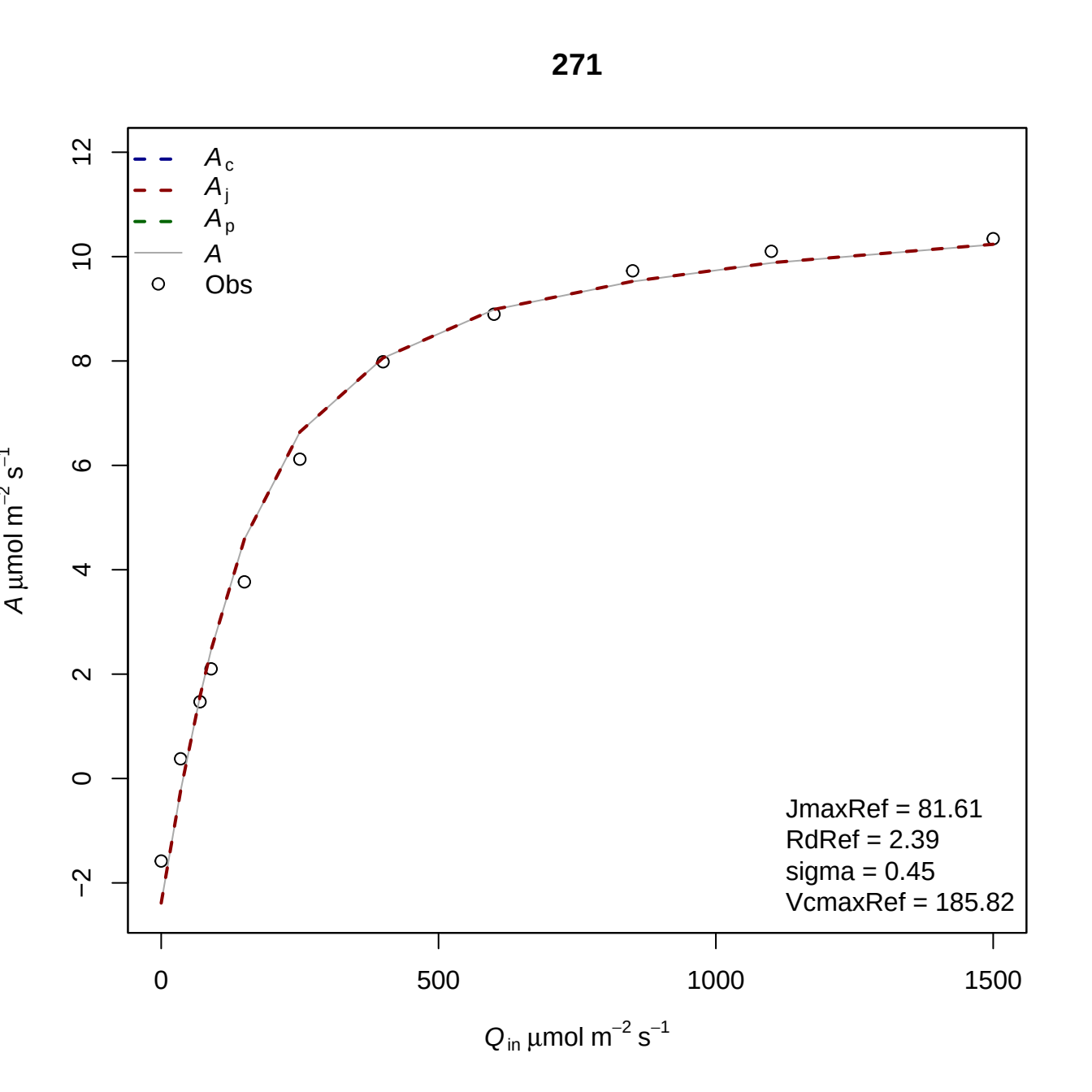

A_c
A_j
A_p
A
Obs
JmaxRef = 40.49
RdRef = 1.51
sigma = 0.38
VcmaxRef = 62.72
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

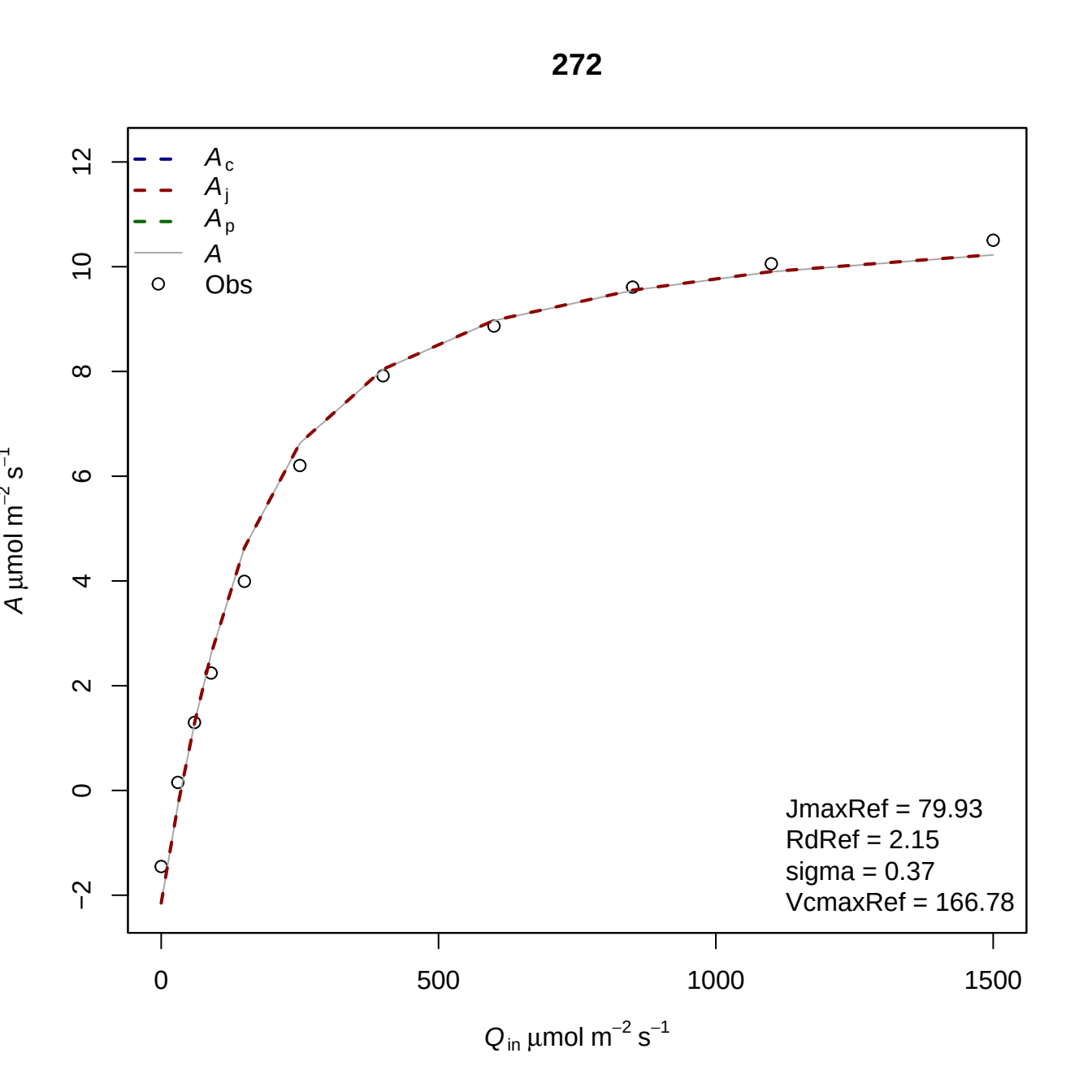

A_c
A_j
A_p
A
Obs
JmaxRef = 117.14
RdRef = 2.89
sigma = 0.79
VcmaxRef = 216.43
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

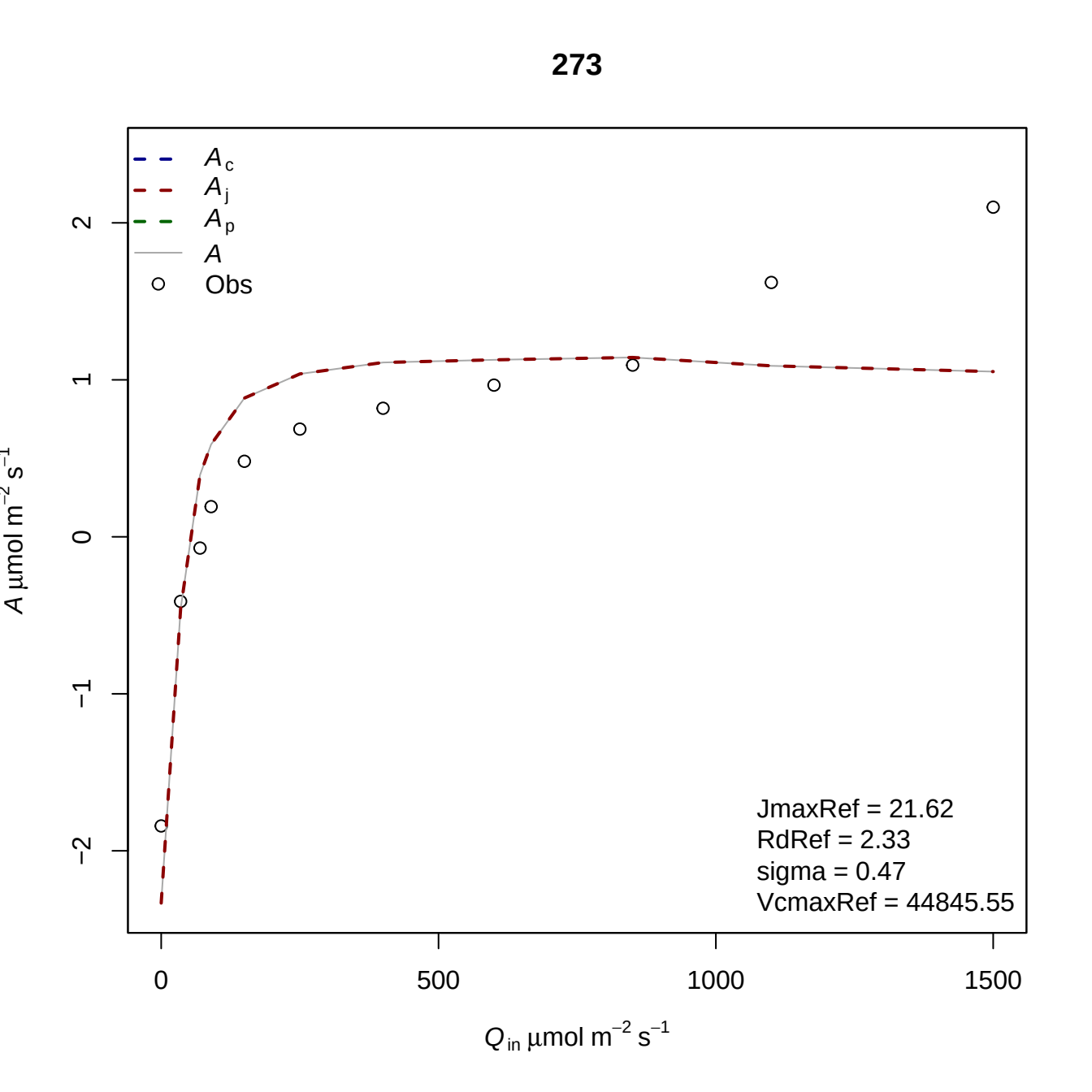

A_c
A_j
A_p
A
Obs
JmaxRef = 47.08
RdRef = 2.71
sigma = 0.21
VcmaxRef = 51.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
200
400
600
800
1000
1200
1400

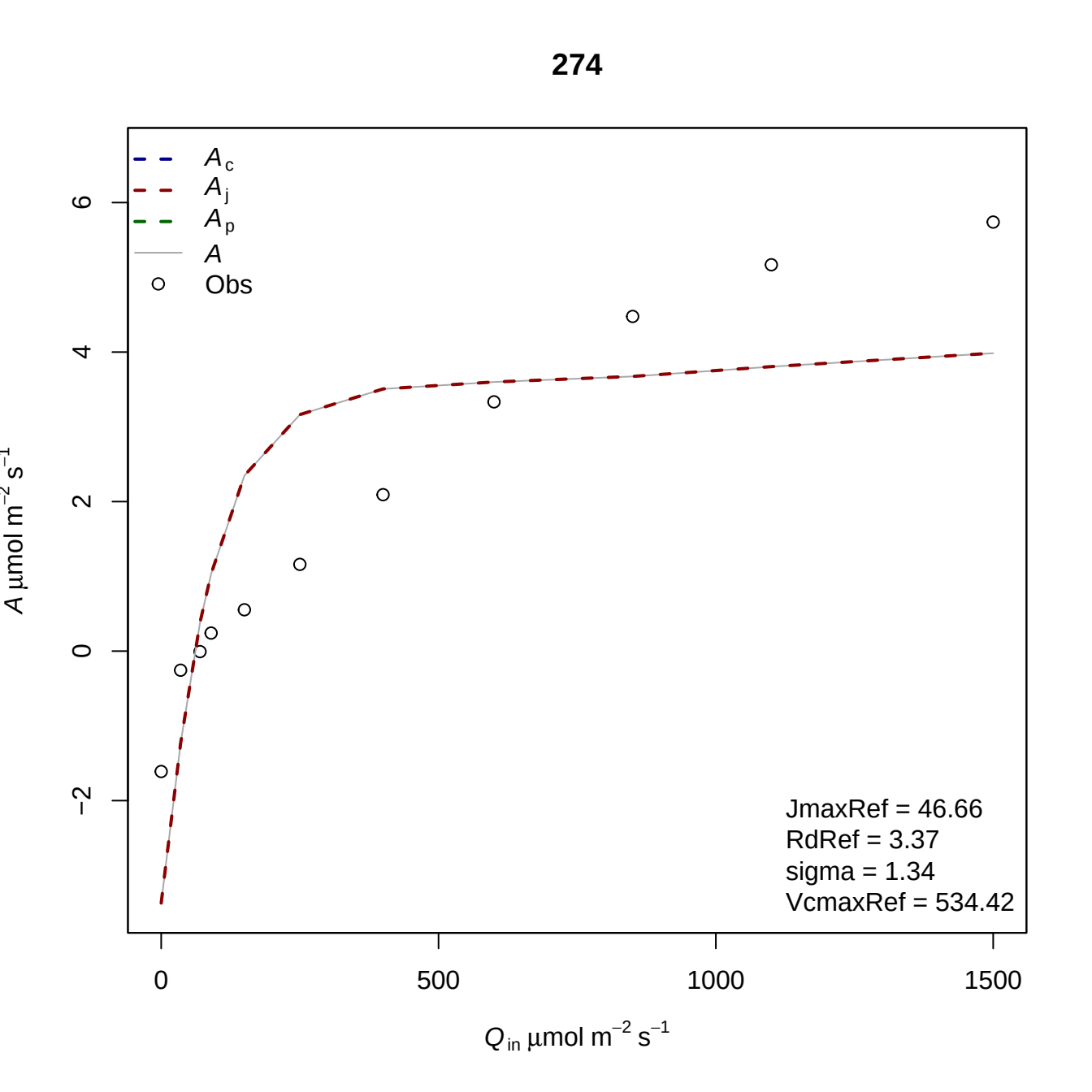

269
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
2
1
0
−1
−2
200
400
600
800
1000
1200
1400
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 26.25
RdRef = 2.67
sigma = 0.03
VcmaxRef = 146.93

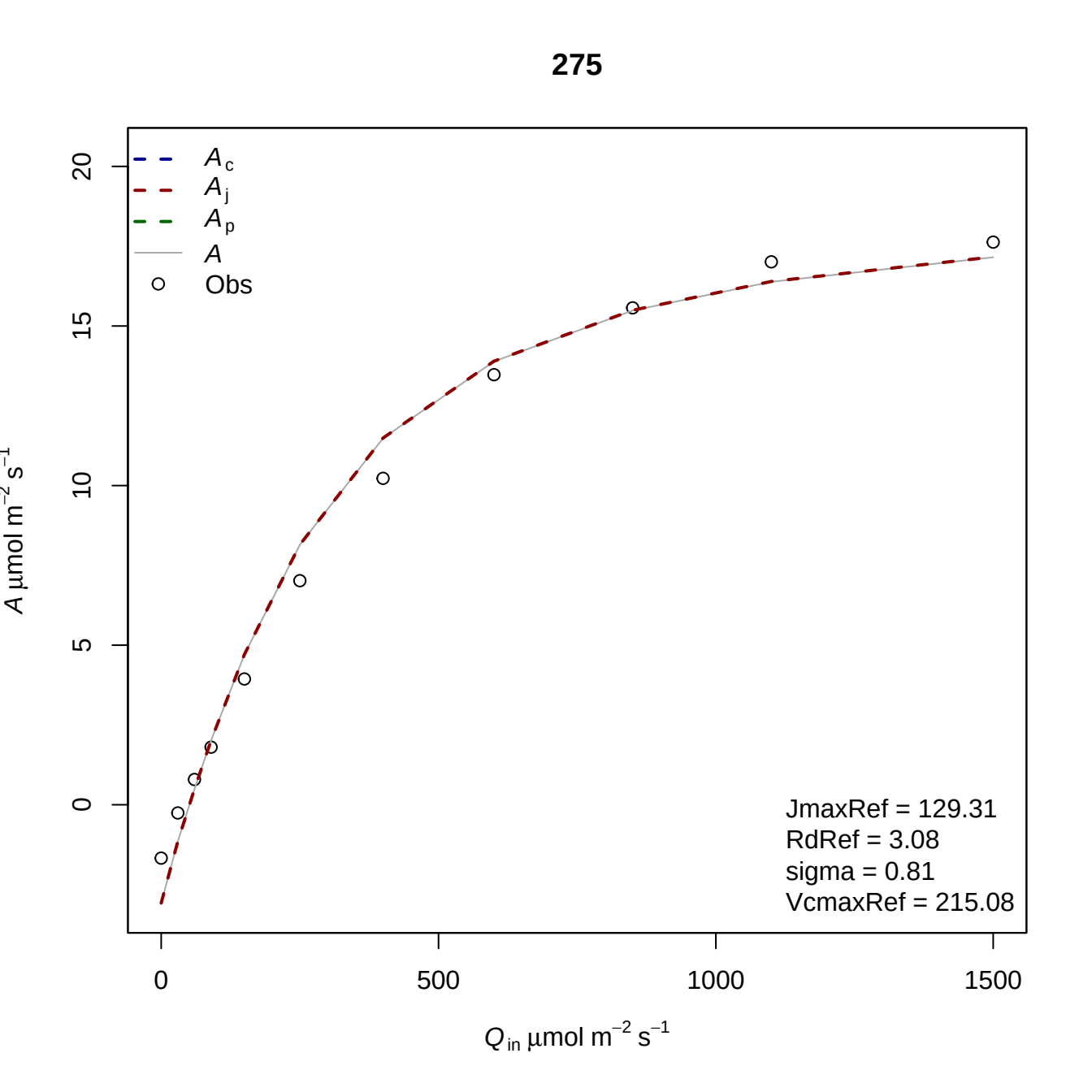

270
A_c
A_j
A_p
A
Obs
JmaxRef = 21.79
RdRef = 1.71
sigma = 0.2
VcmaxRef = 58.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

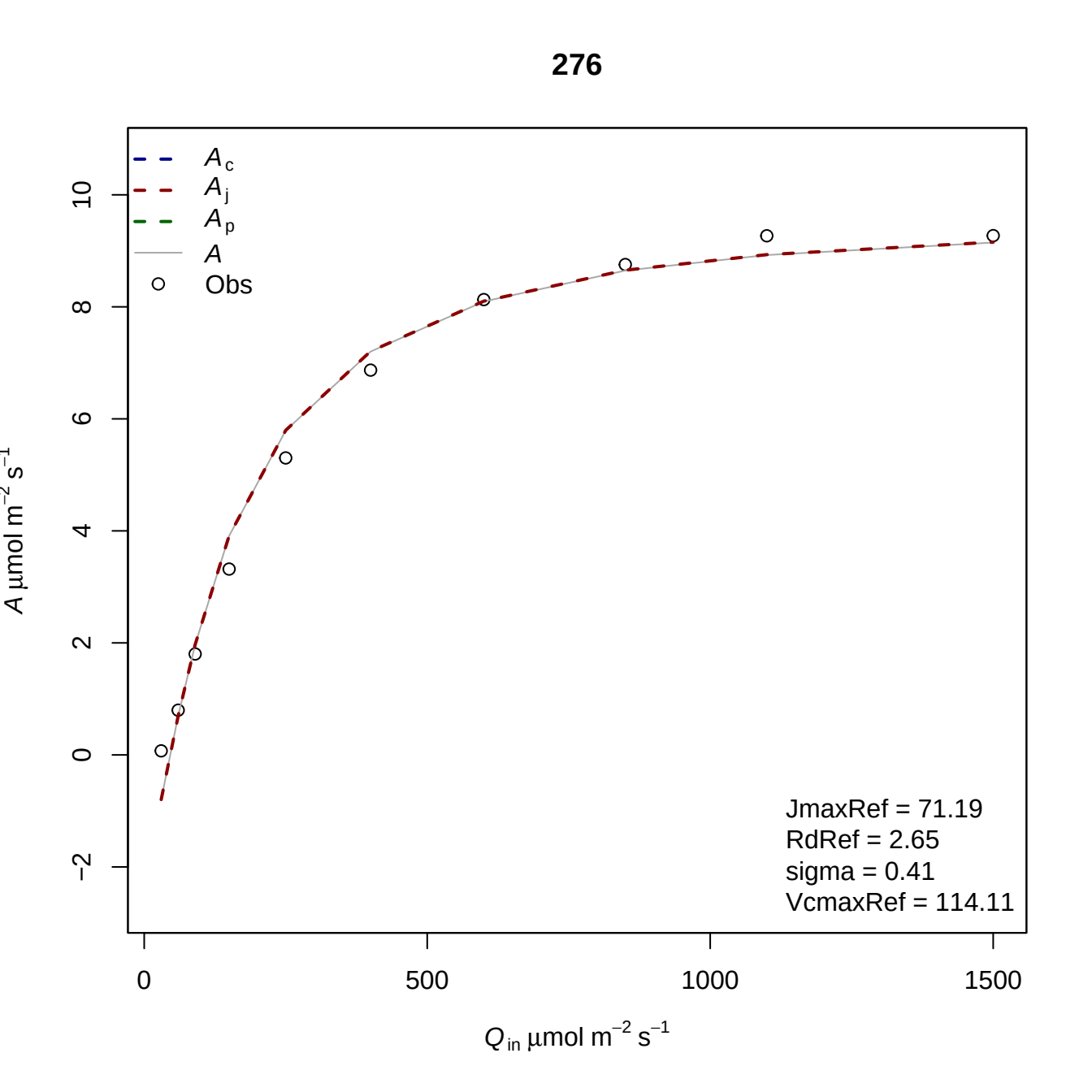

271
A_c
A_j
A_p
A
Obs
JmaxRef = 81.61
RdRef = 2.39
sigma = 0.45
VcmaxRef = 185.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

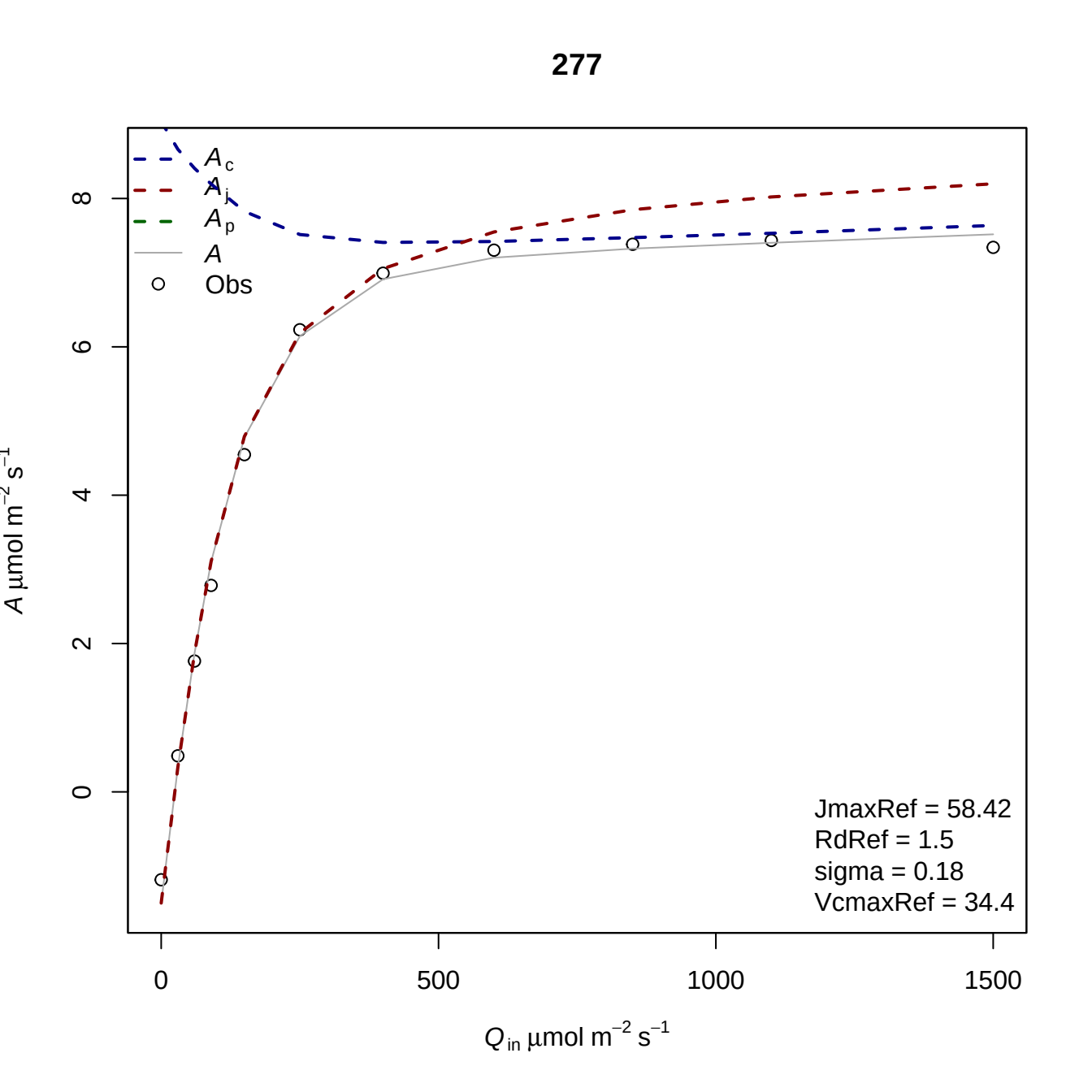

272
A_c
A_j
A_p
A
Obs
JmaxRef = 79.93
RdRef = 2.15
sigma = 0.37
VcmaxRef = 166.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

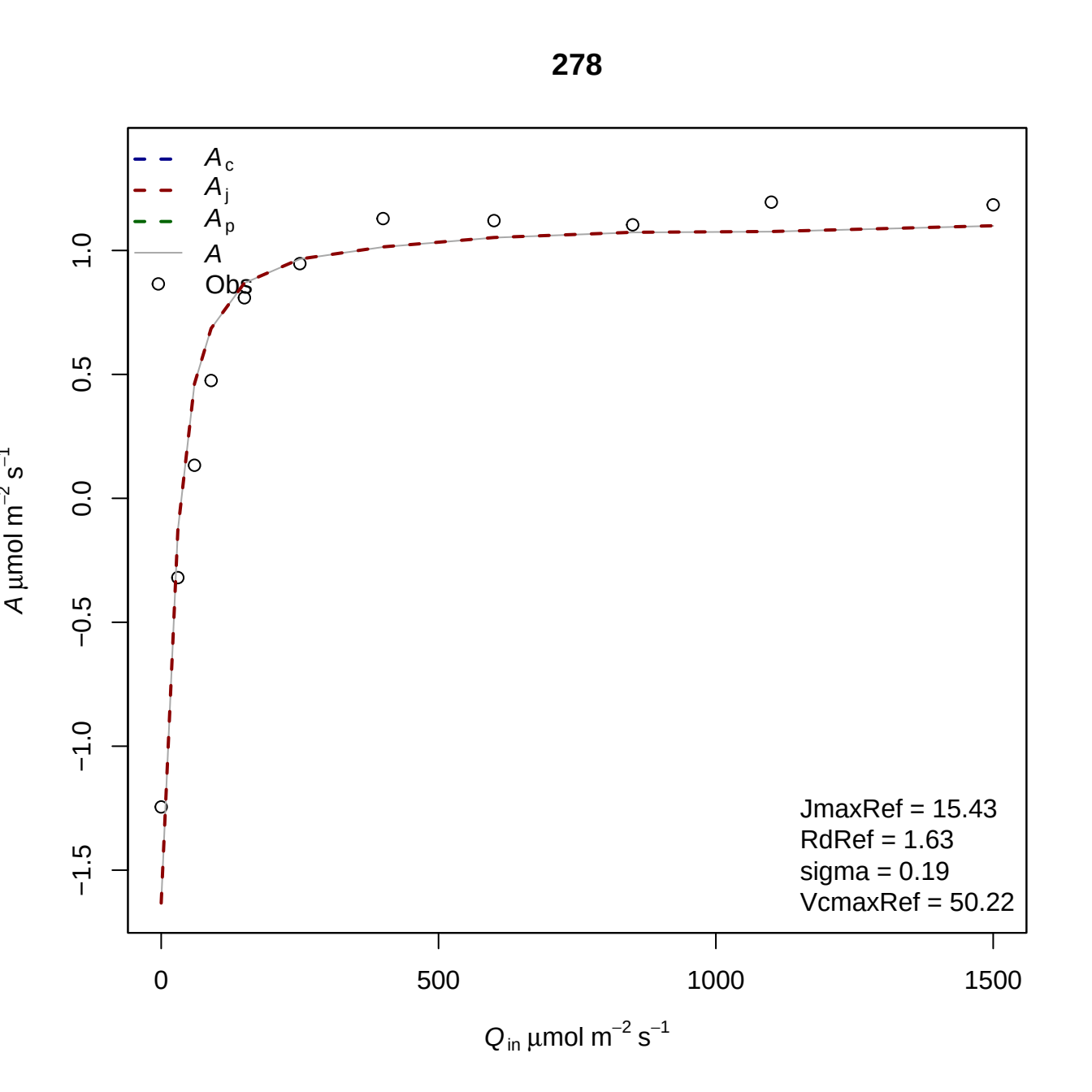

273
A_c
A_j
A_p
A
Obs
JmaxRef = 21.62
RdRef = 2.33
sigma = 0.47
VcmaxRef = 44845.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
500
1000
1500

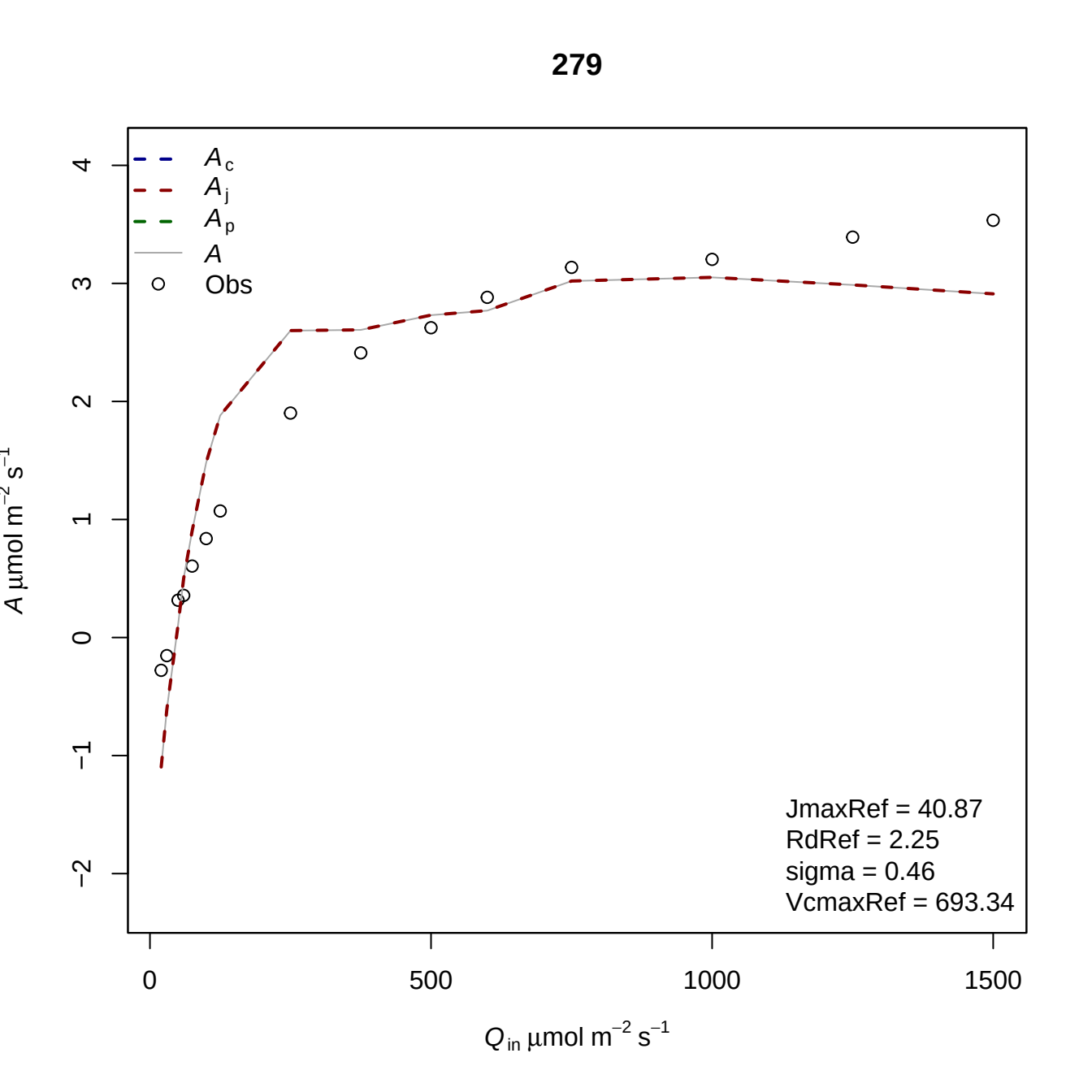

274
A_c
A_j
A_p
A
Obs
JmaxRef = 46.66
RdRef = 3.37
sigma = 1.34
VcmaxRef = 534.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

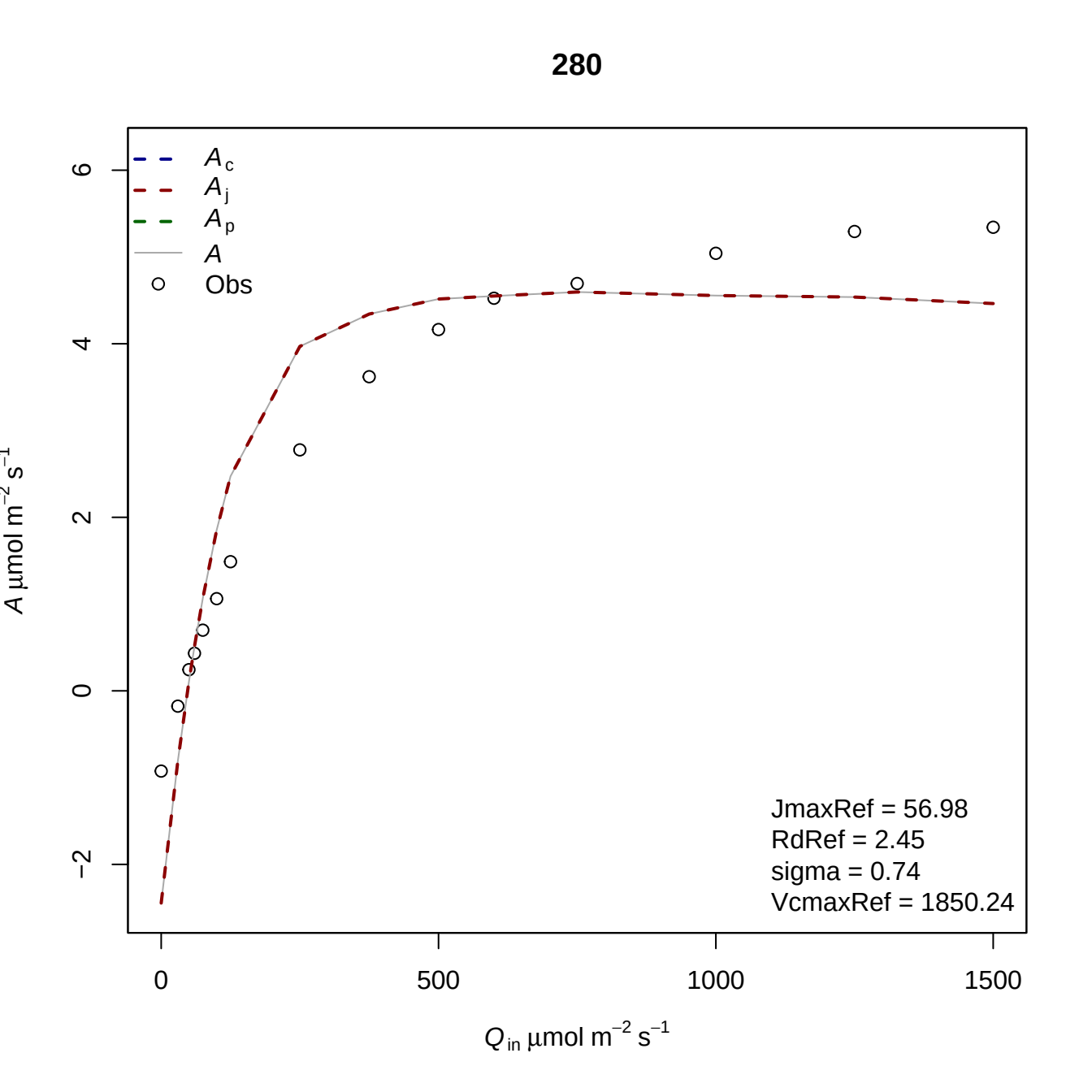

275
A_c
A_j
A_p
A
Obs
JmaxRef = 129.31
RdRef = 3.08
sigma = 0.81
VcmaxRef = 215.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

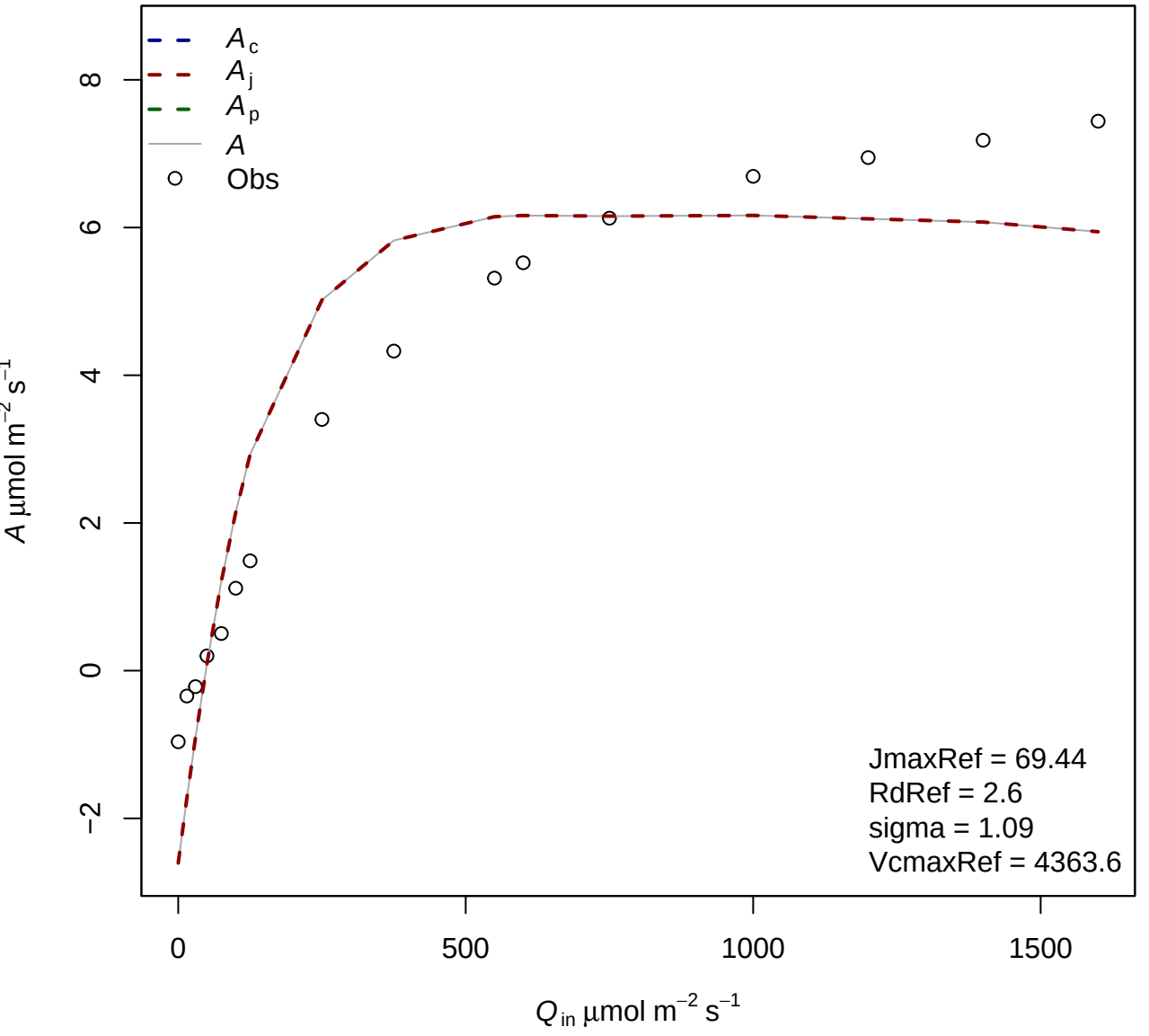

276
A_c
A_j
A_p
A
Obs
JmaxRef = 71.19
RdRef = 2.65
sigma = 0.41
VcmaxRef = 114.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

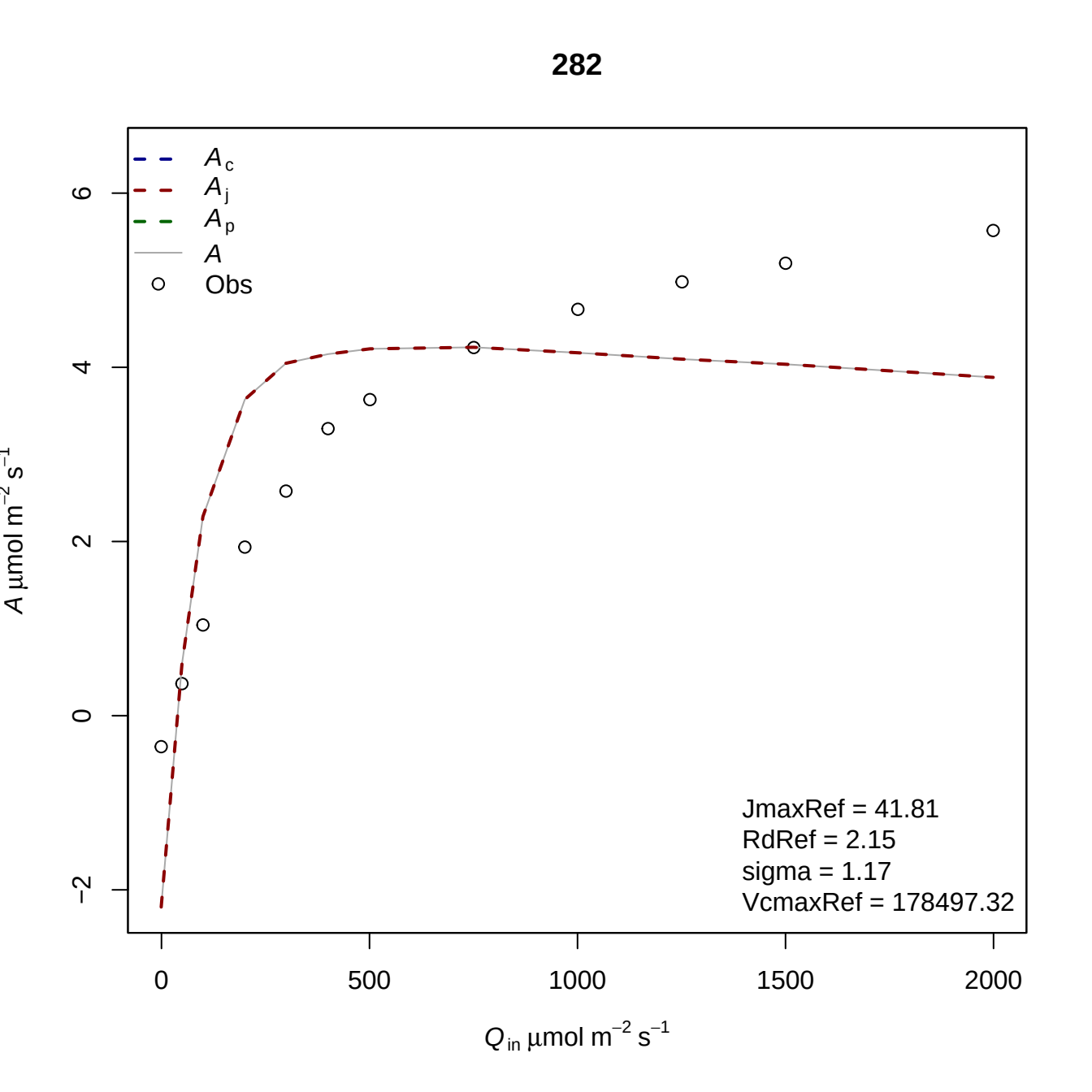

277
A_c
A_j
A_p
A
Obs
JmaxRef = 58.42
RdRef = 1.5
sigma = 0.18
VcmaxRef = 34.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

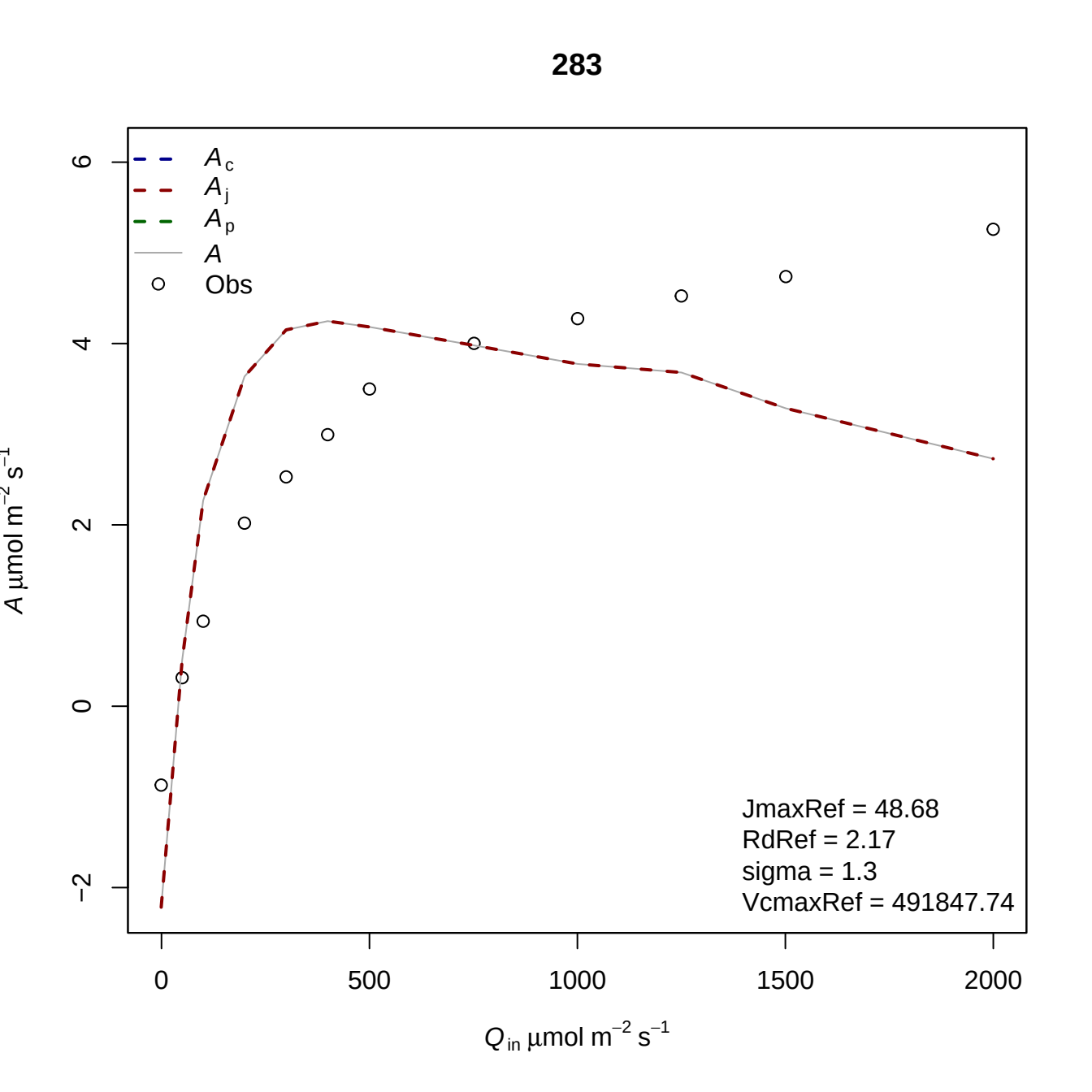

278
A_c
A_j
A_p
A
Obs
JmaxRef = 15.43
RdRef = 1.63
sigma = 0.19
VcmaxRef = 50.22
A µmol m^{-2} s^{-1}
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

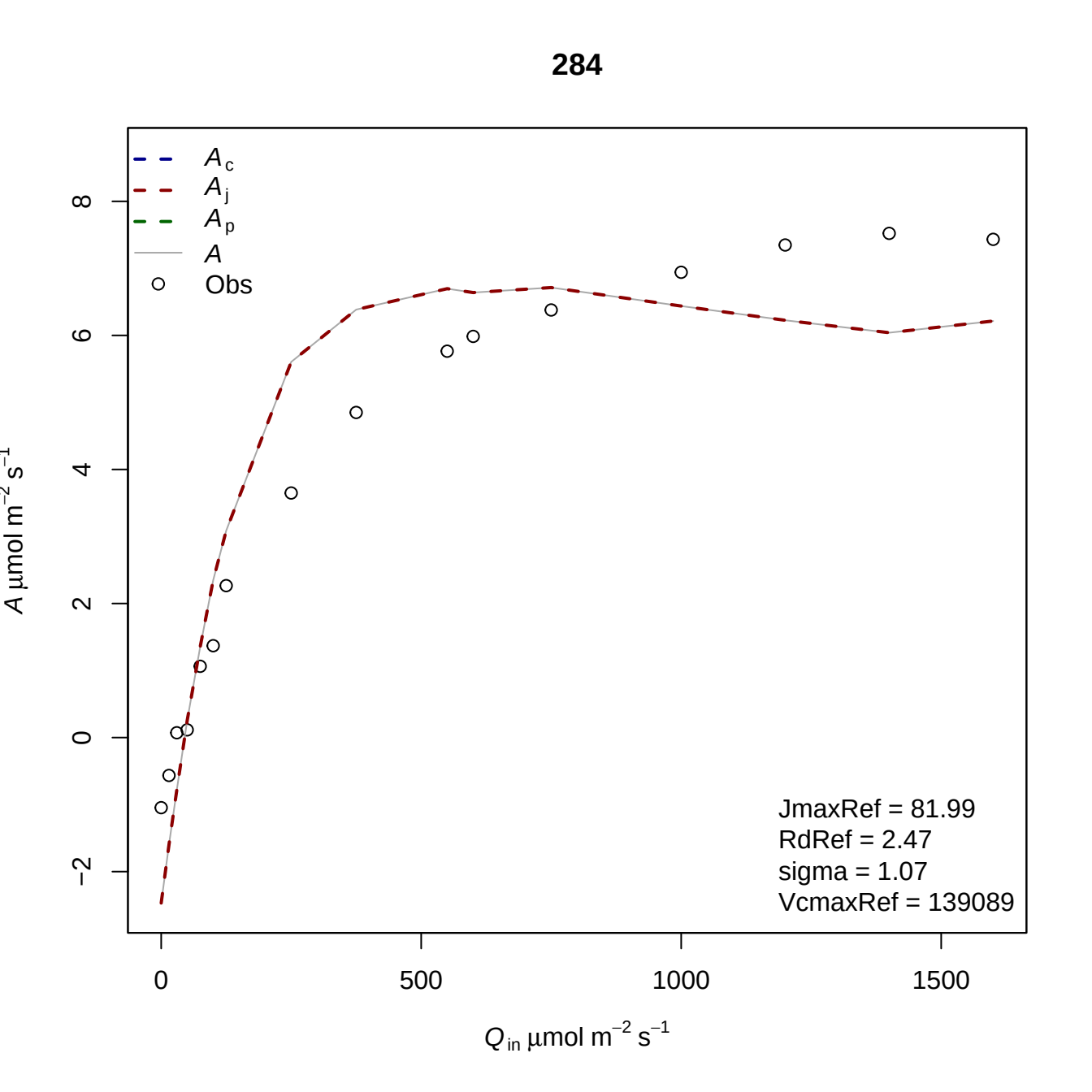

279
A_c
A_j
A_p
A
Obs
JmaxRef = 40.87
RdRef = 2.25
sigma = 0.46
VcmaxRef = 693.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

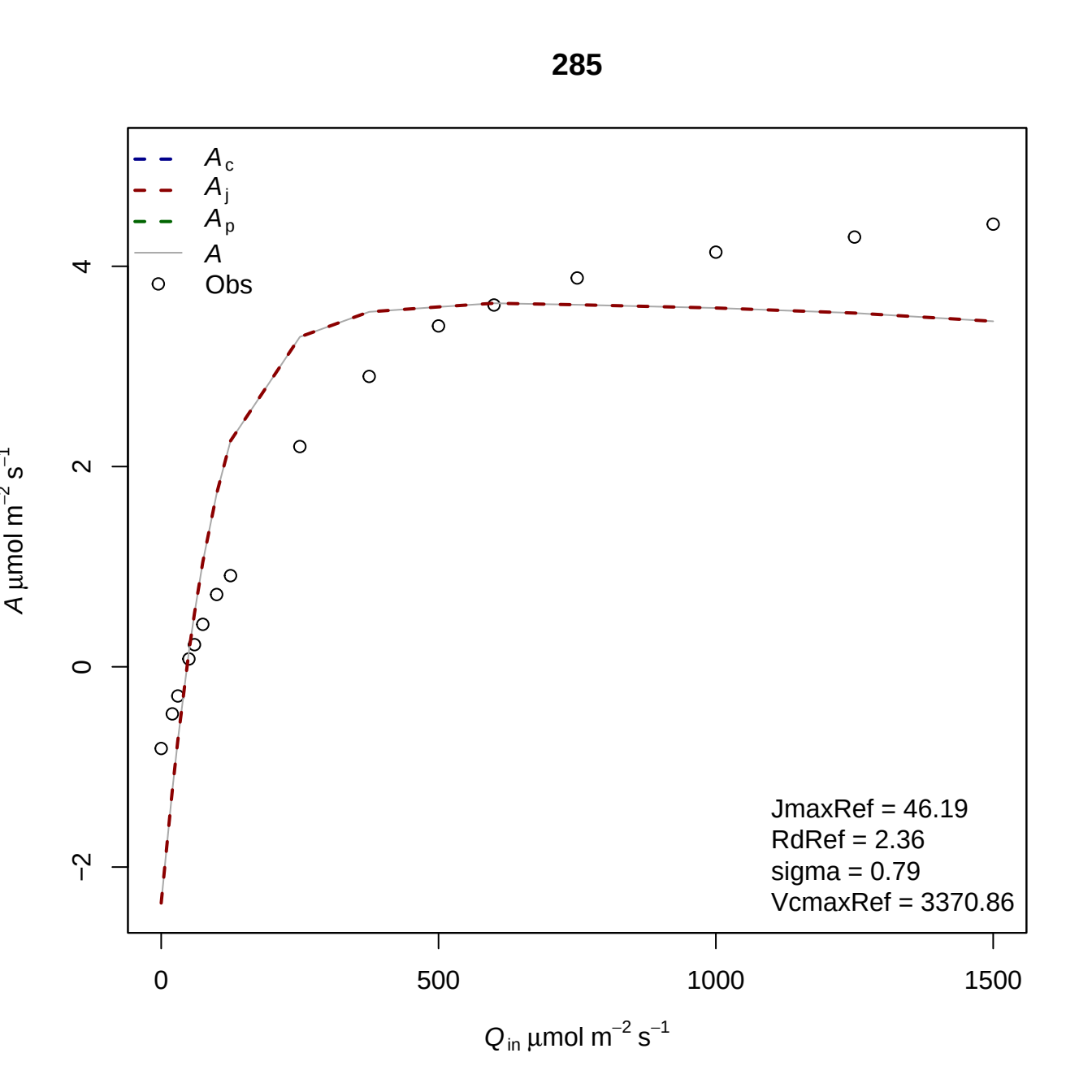

280
A_c
A_j
A_p
A
Obs
JmaxRef = 56.98
RdRef = 2.45
sigma = 0.74
VcmaxRef = 1850.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

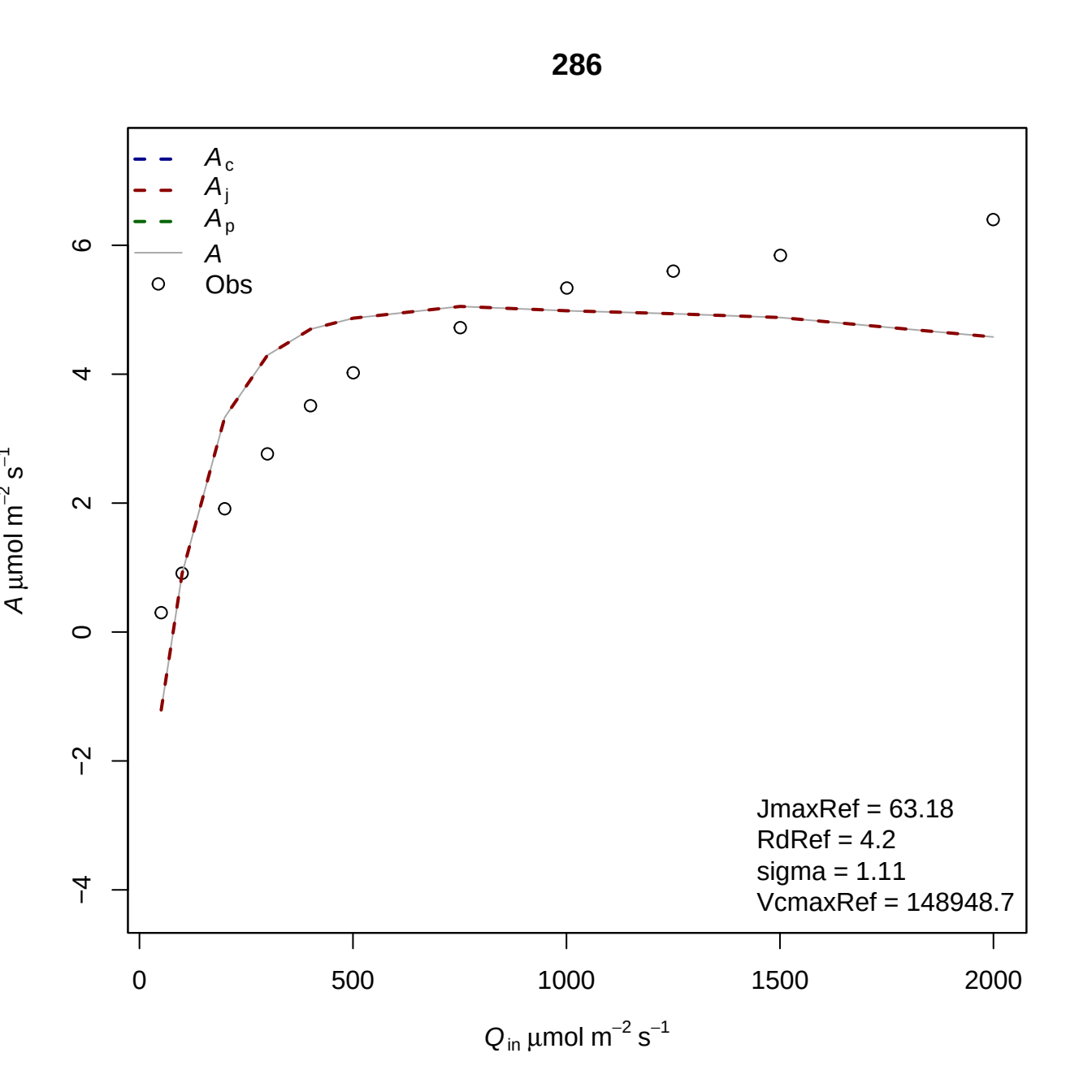

A_c
A_j
A_p
A
Obs
JmaxRef = 69.44
RdRef = 2.6
sigma = 1.09
VcmaxRef = 4363.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

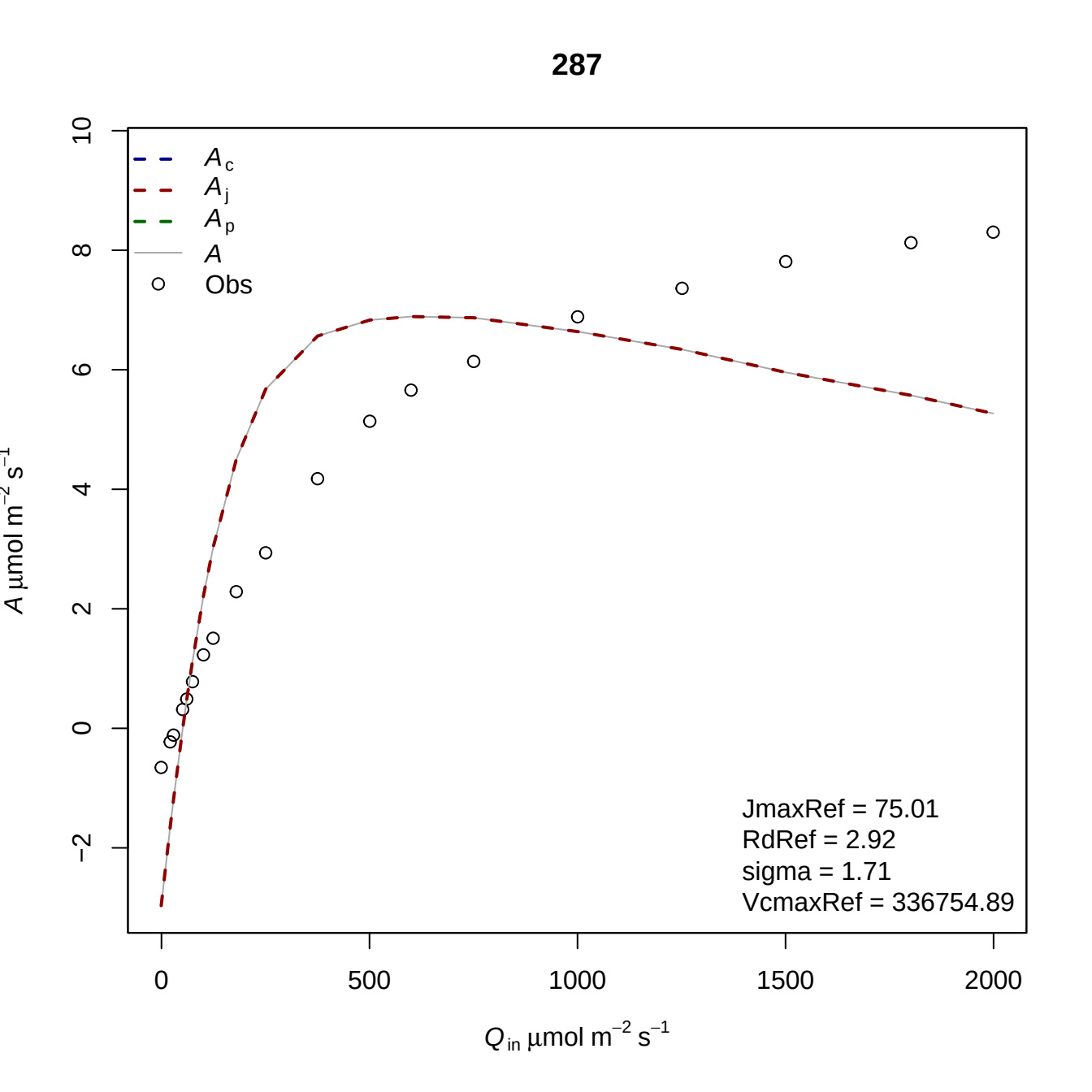

282
A_c
A_j
A_p
A
Obs
JmaxRef = 41.81
RdRef = 2.15
sigma = 1.17
VcmaxRef = 178497.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

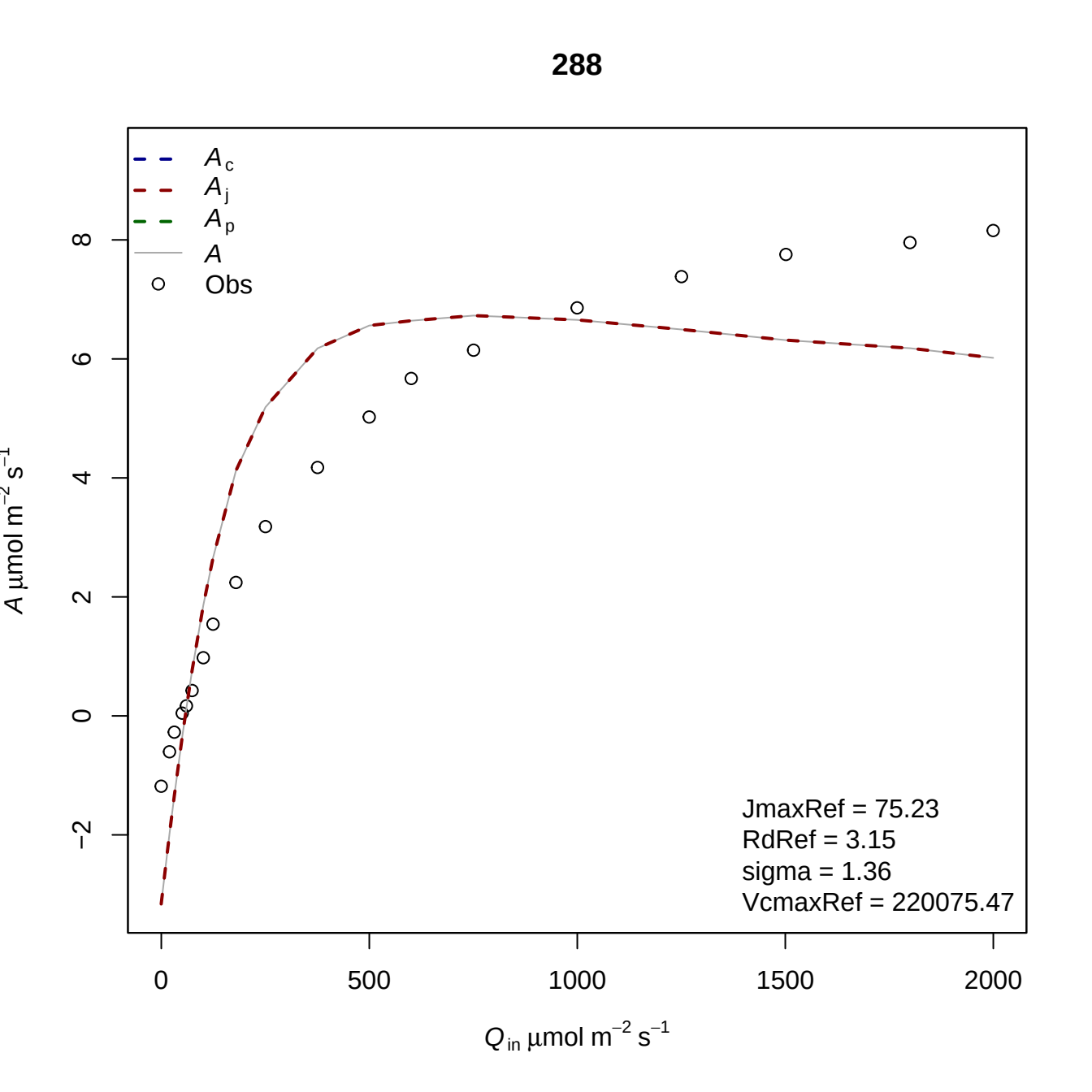

283
A_c
A_j
A_p
A
Obs
JmaxRef = 48.68
RdRef = 2.17
sigma = 1.3
VcmaxRef = 491847.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

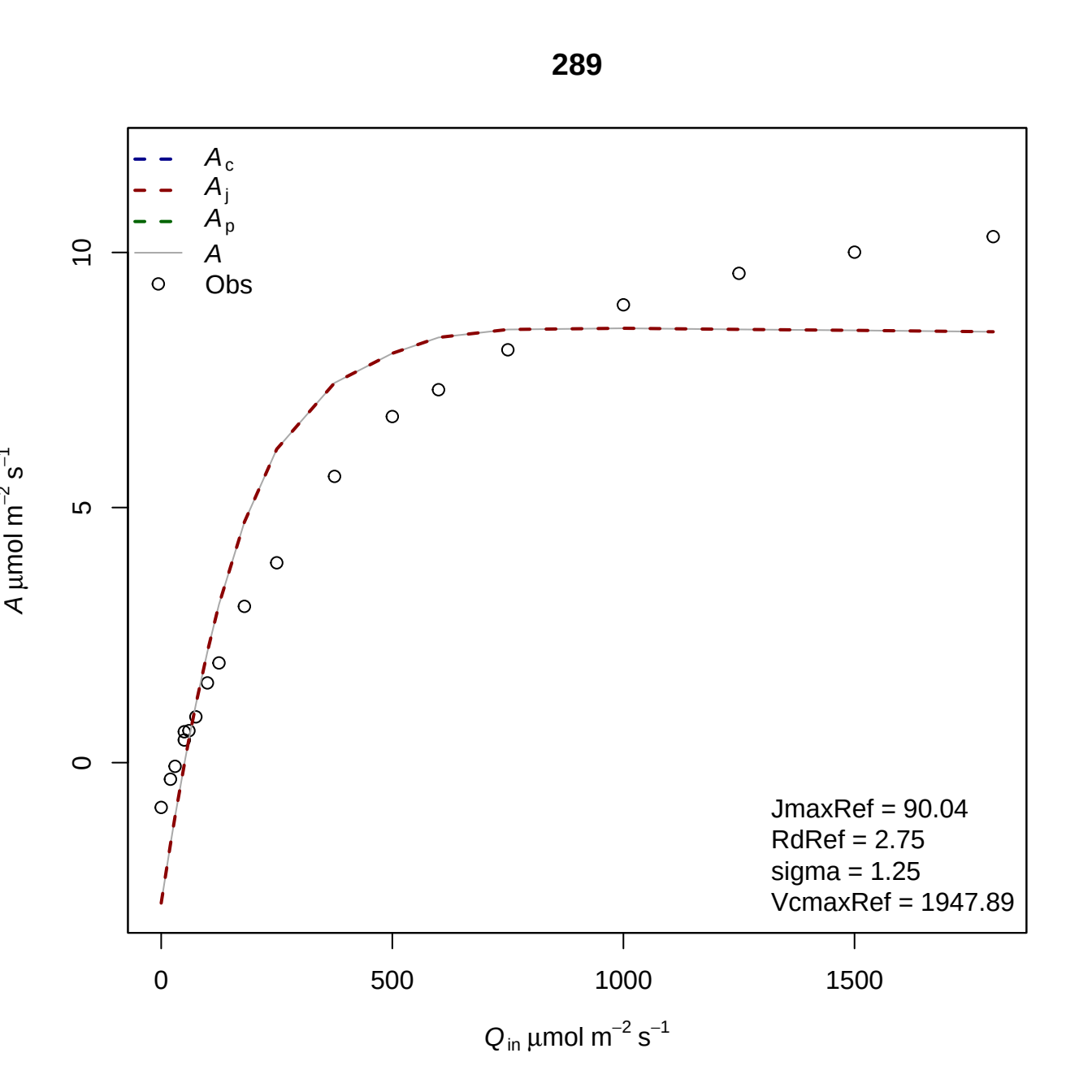

284
A_c
A_j
A_p
A
Obs
JmaxRef = 81.99
RdRef = 2.47
sigma = 1.07
VcmaxRef = 139089
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

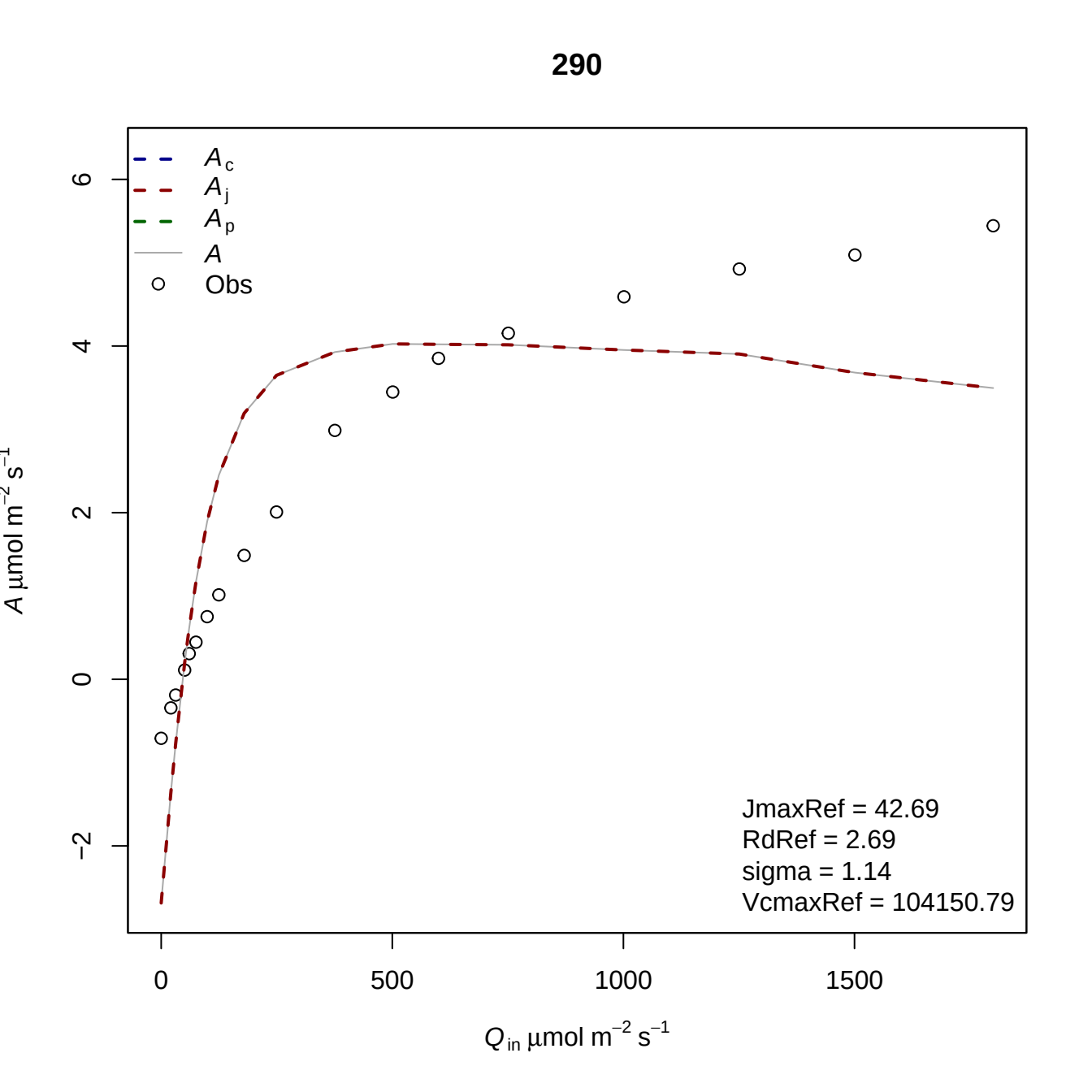

285
A_c
A_j
A_p
A
Obs
JmaxRef = 46.19
RdRef = 2.36
sigma = 0.79
VcmaxRef = 3370.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

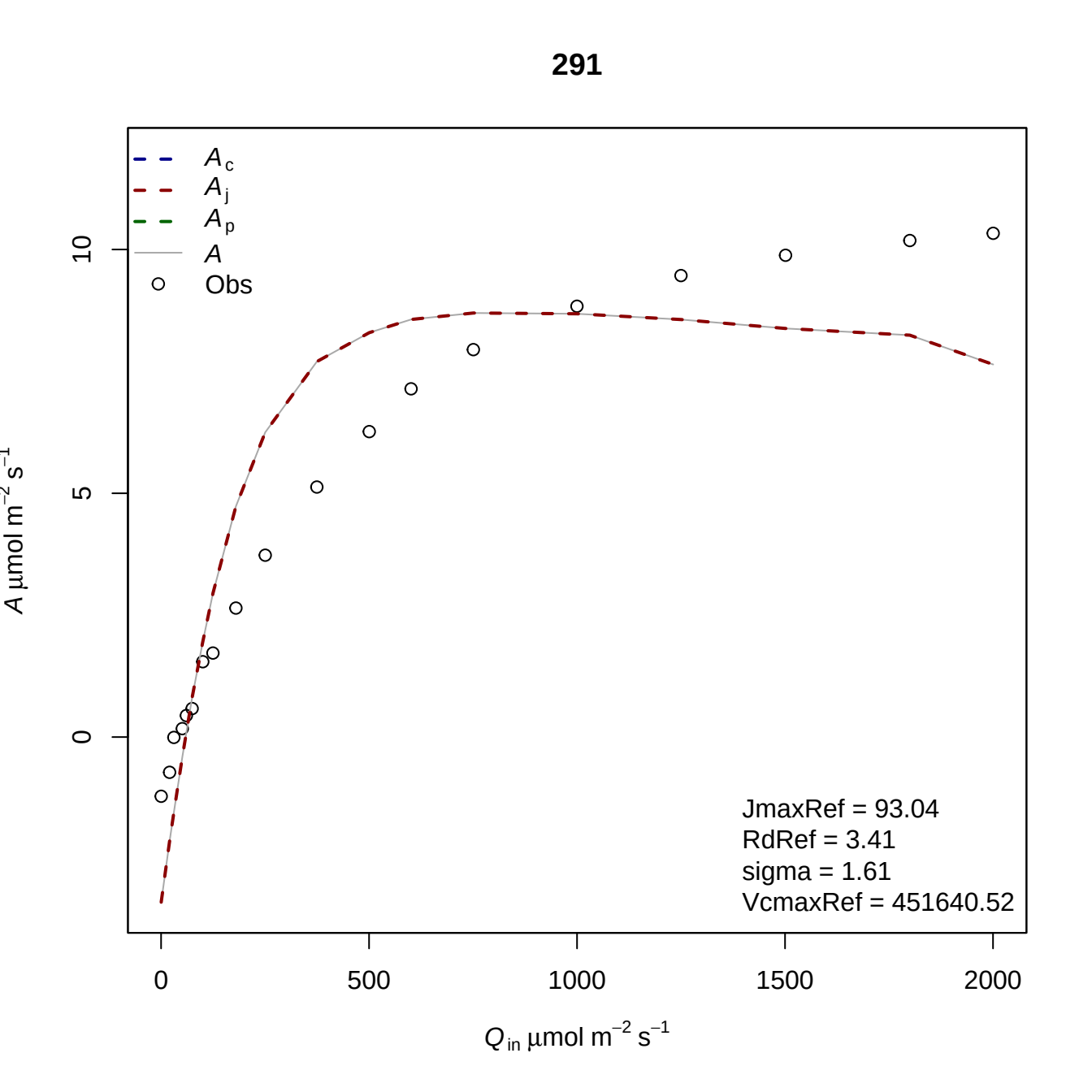

286
A_c
A_j
A_p
A
Obs
JmaxRef = 63.18
RdRef = 4.2
sigma = 1.11
VcmaxRef = 148948.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

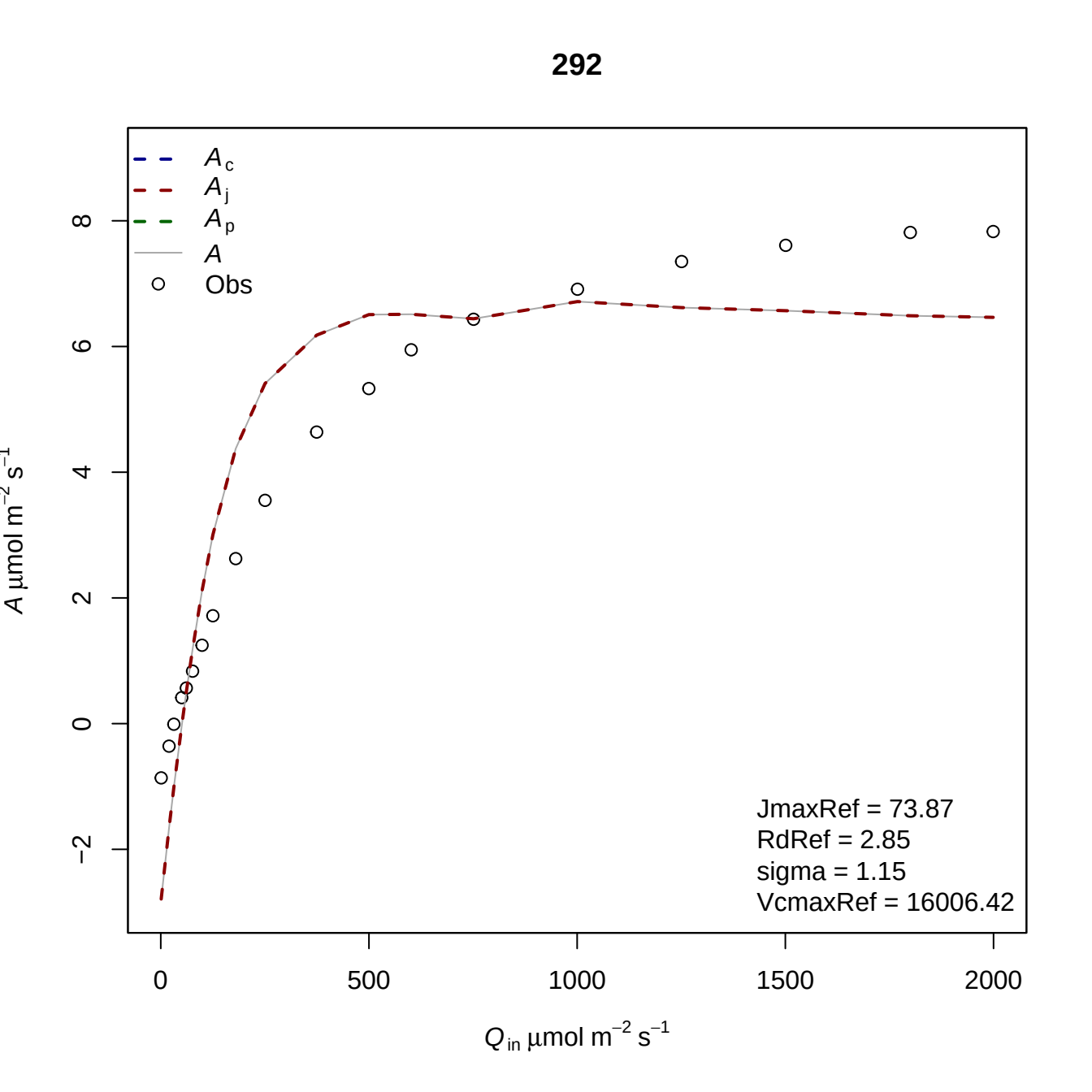

287
A_c
A_j
A_p
A
Obs
JmaxRef = 75.01
RdRef = 2.92
sigma = 1.71
VcmaxRef = 336754.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

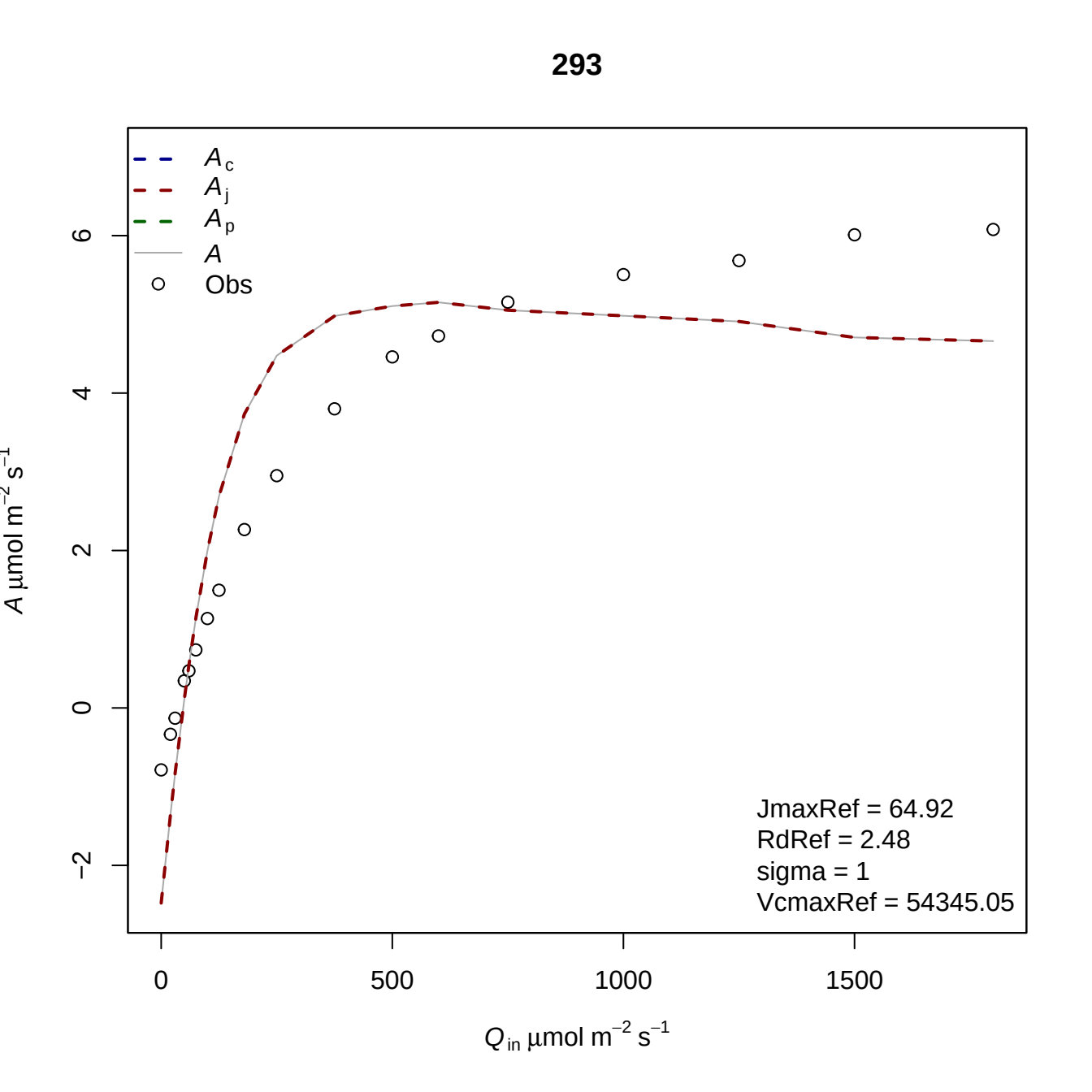

288
A_c
A_j
A_p
A
Obs
JmaxRef = 75.23
RdRef = 3.15
sigma = 1.36
VcmaxRef = 220075.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

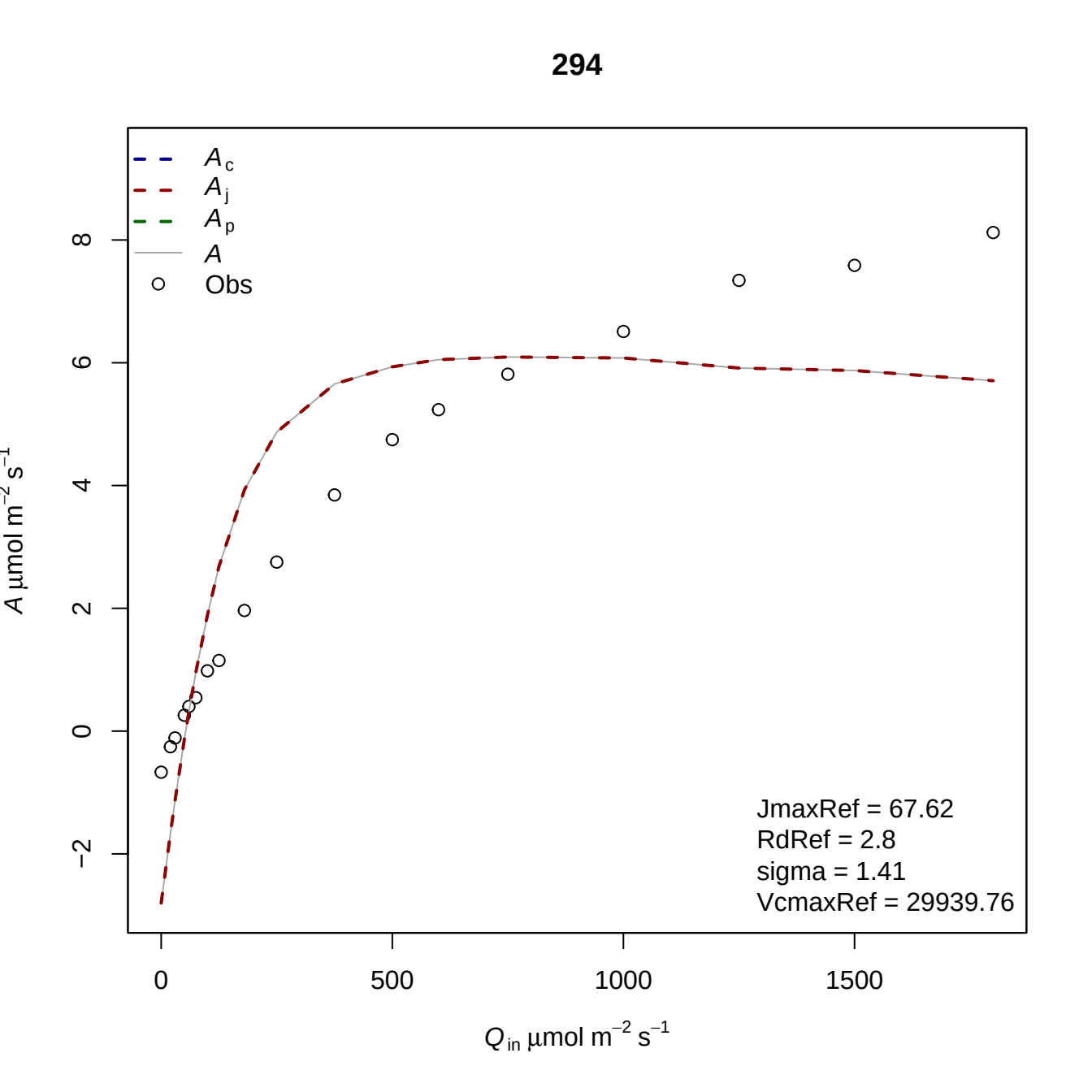

289
A_c
A_j
A_p
A
Obs
JmaxRef = 90.04
RdRef = 2.75
sigma = 1.25
VcmaxRef = 1947.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

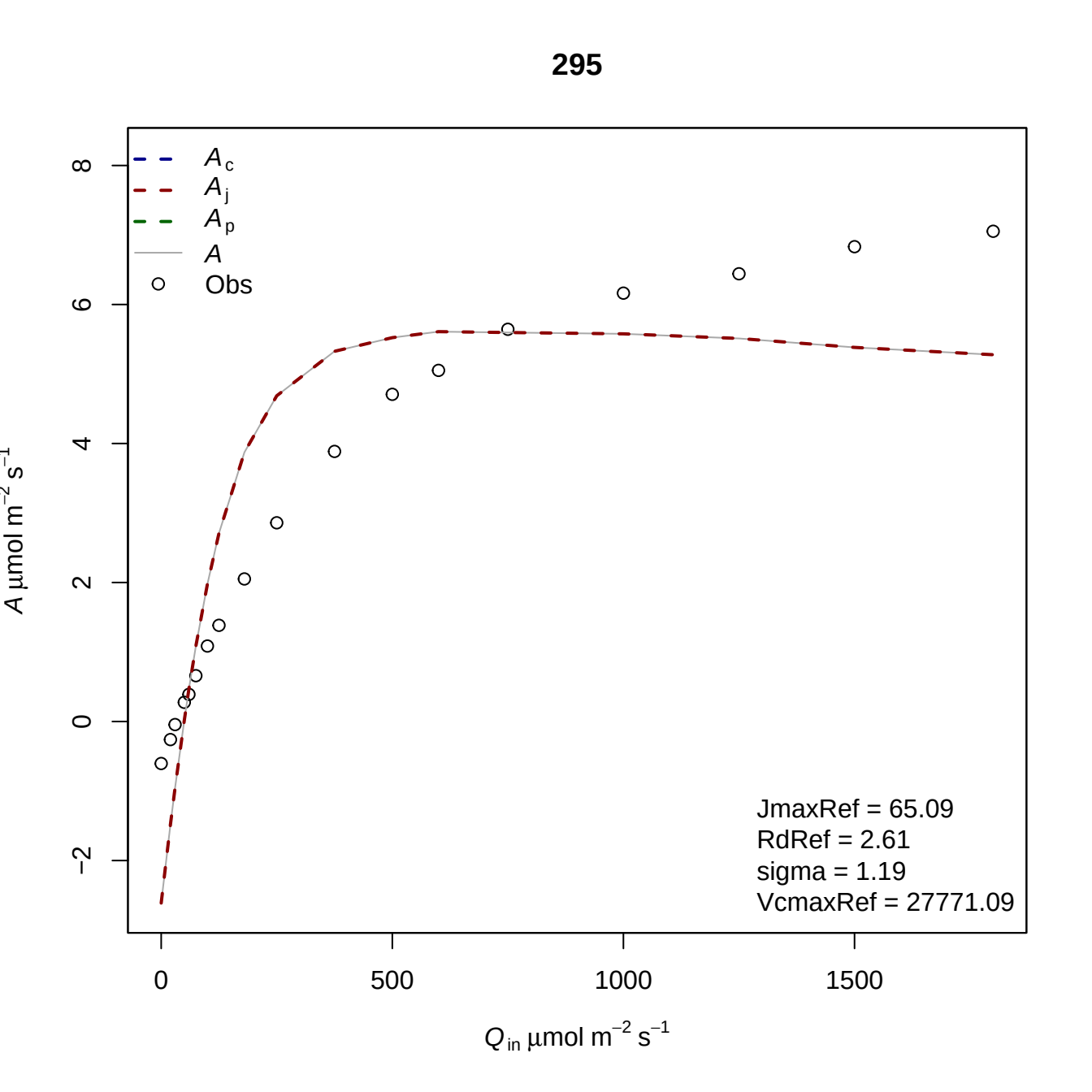

290
A_c
A_j
A_p
A
Obs
JmaxRef = 42.69
RdRef = 2.69
sigma = 1.14
VcmaxRef = 104150.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

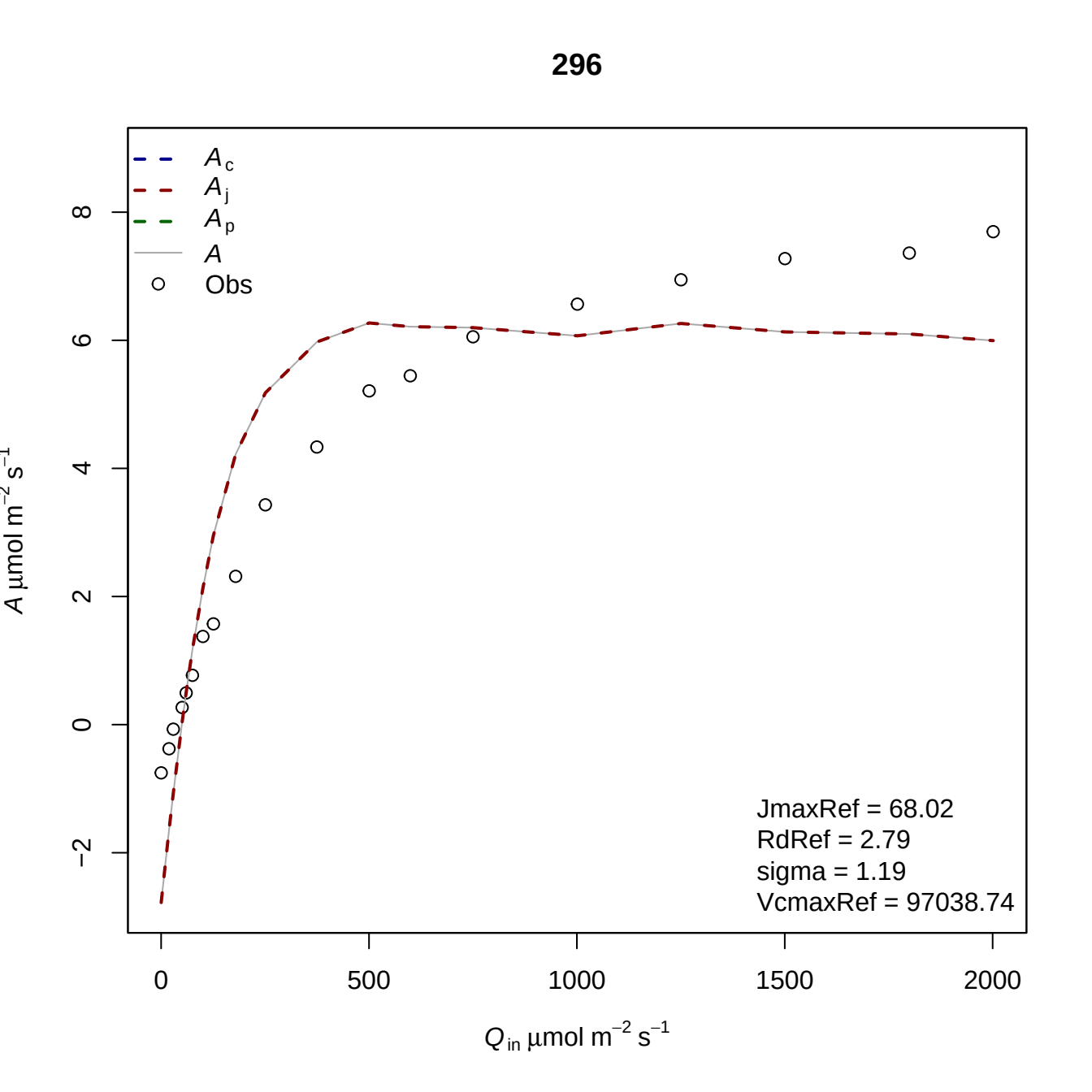

291
A_c
A_j
A_p
A
Obs
JmaxRef = 93.04
RdRef = 3.41
sigma = 1.61
VcmaxRef = 451640.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

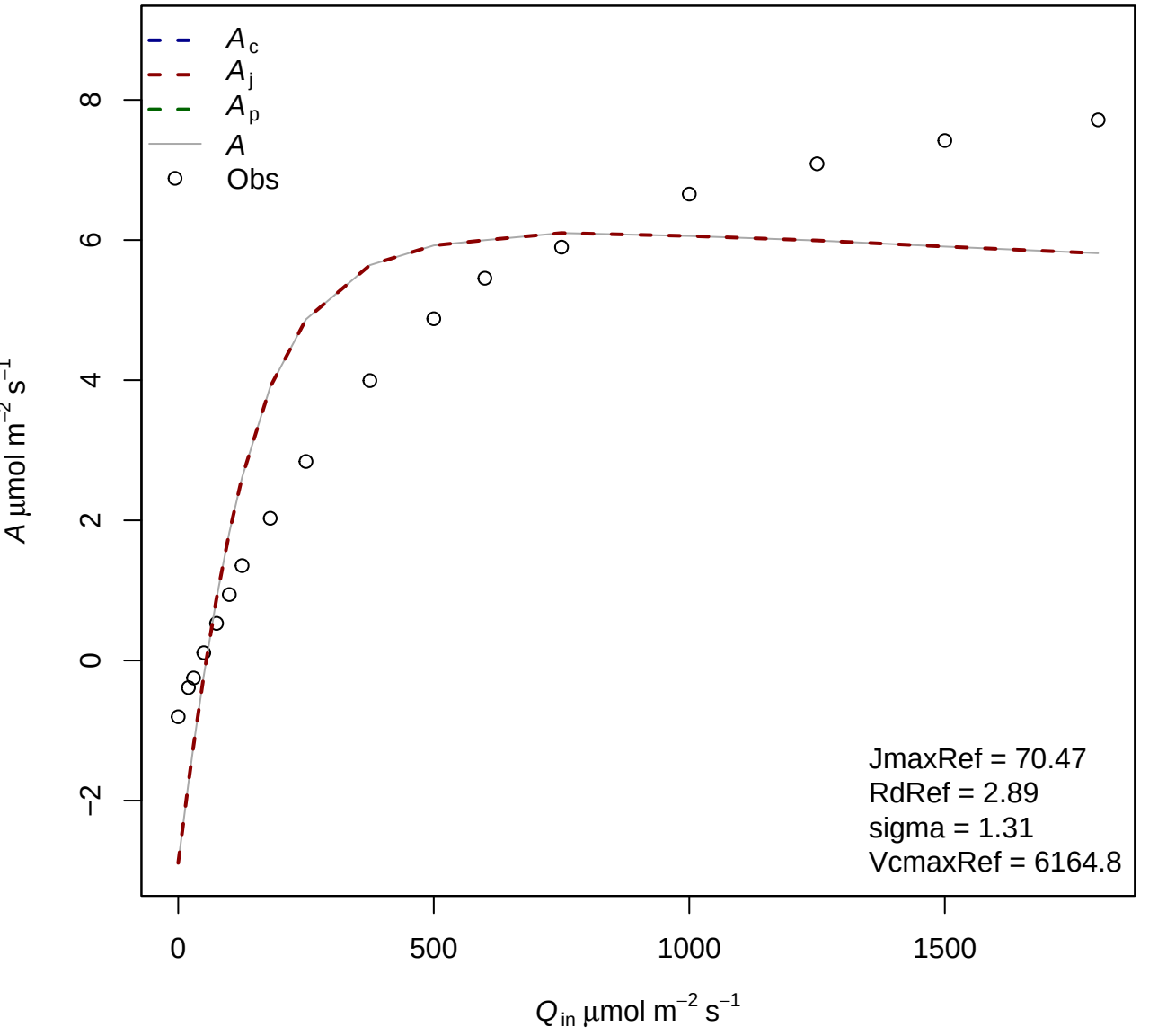

292
A_c
A_j
A_p
A
Obs
JmaxRef = 73.87
RdRef = 2.85
sigma = 1.15
VcmaxRef = 16006.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

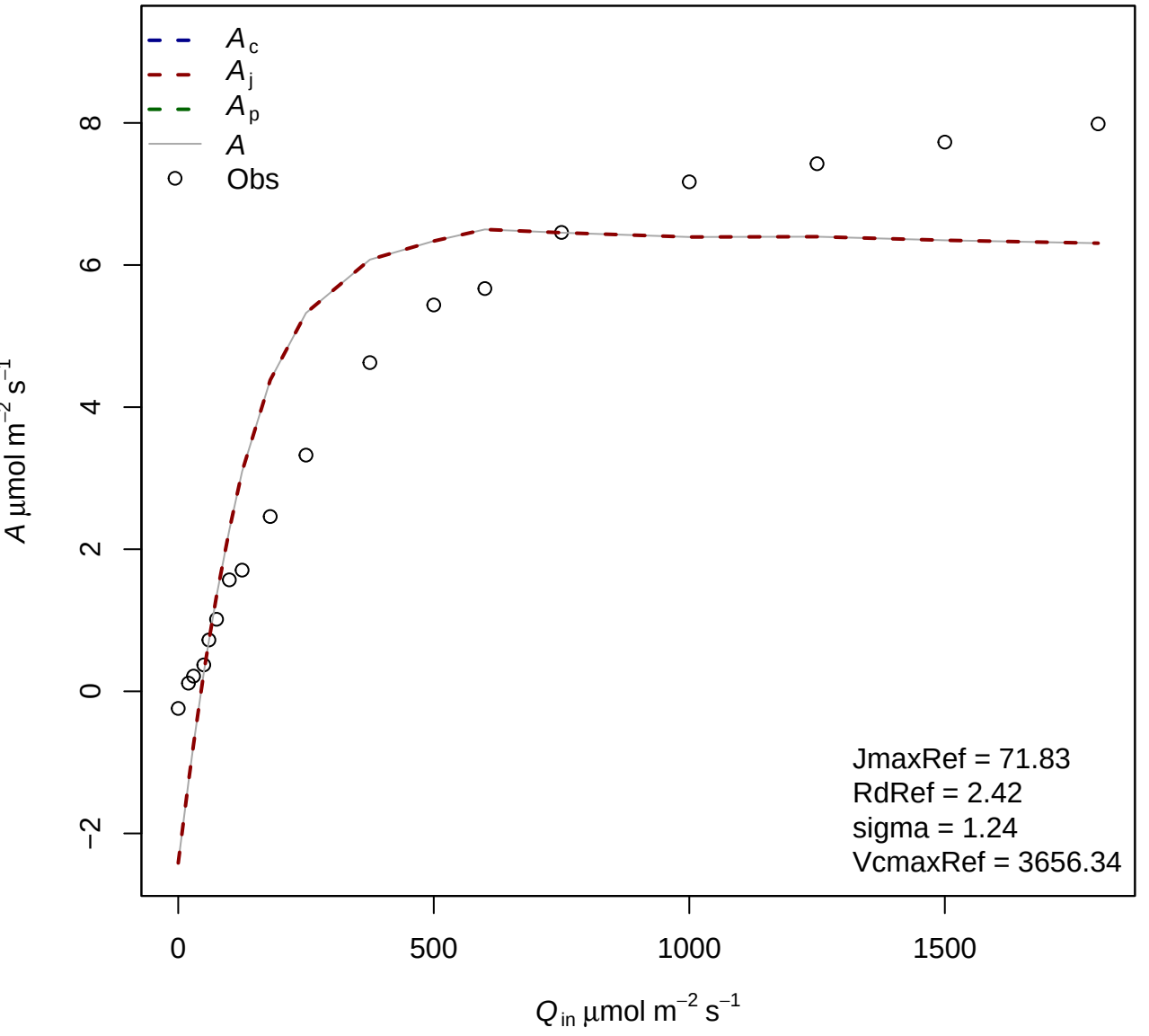

293
A_c
A_j
A_p
A
Obs
JmaxRef = 64.92
RdRef = 2.48
sigma = 1
VcmaxRef = 54345.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

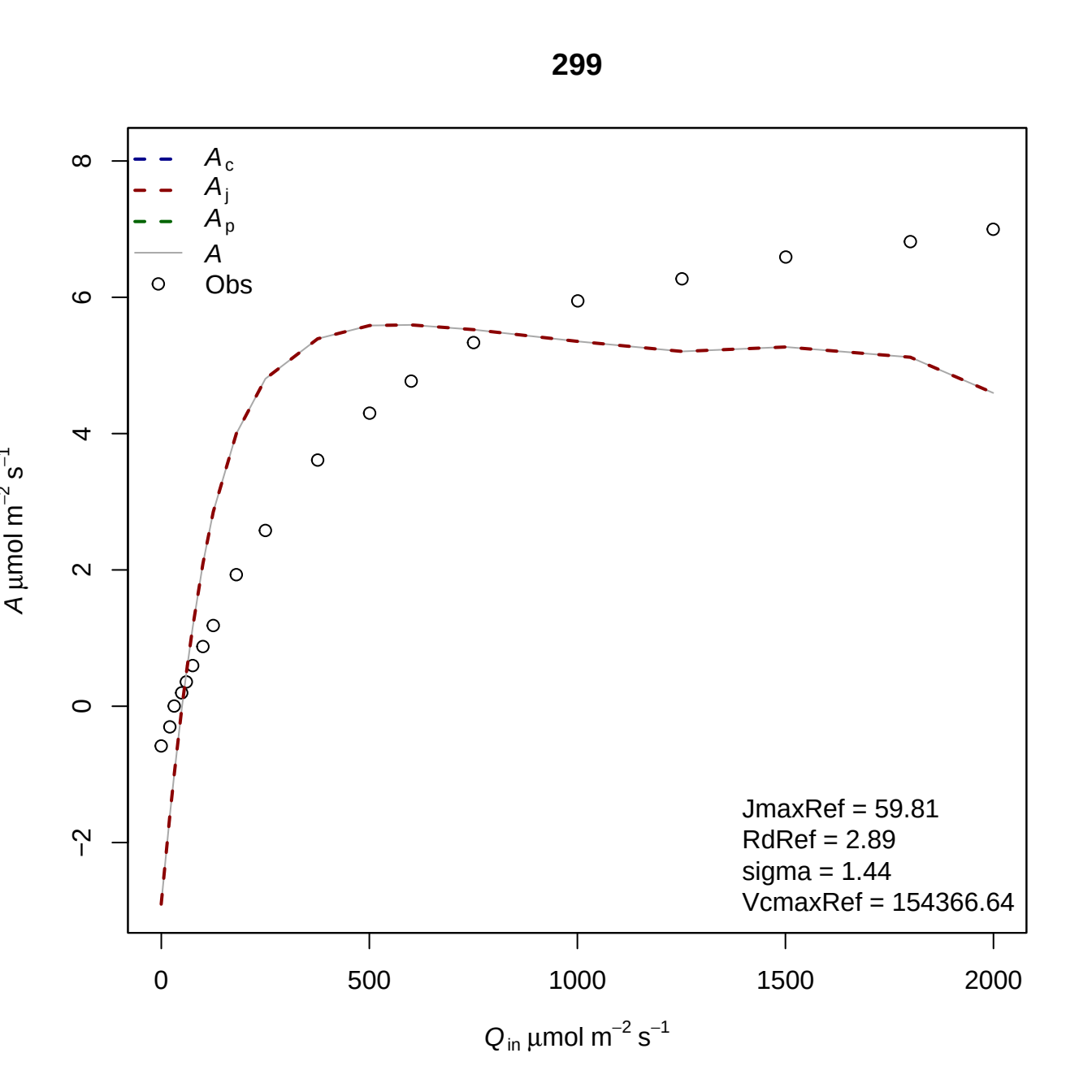

294
A_c
A_j
A_p
A
Obs
JmaxRef = 67.62
RdRef = 2.8
sigma = 1.41
VcmaxRef = 29939.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

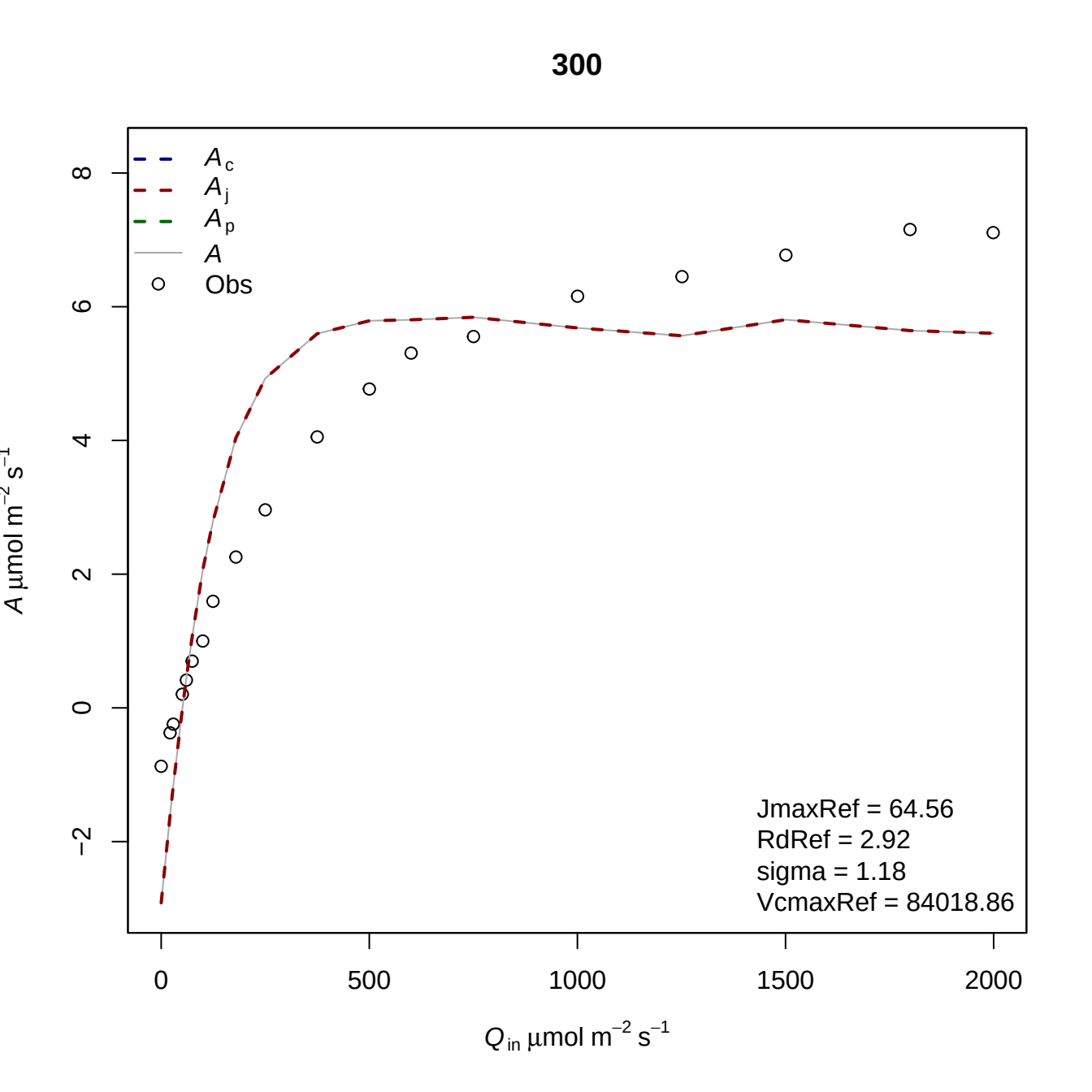

295
A_c
A_j
A_p
A
Obs
JmaxRef = 65.09
RdRef = 2.61
sigma = 1.19
VcmaxRef = 27771.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

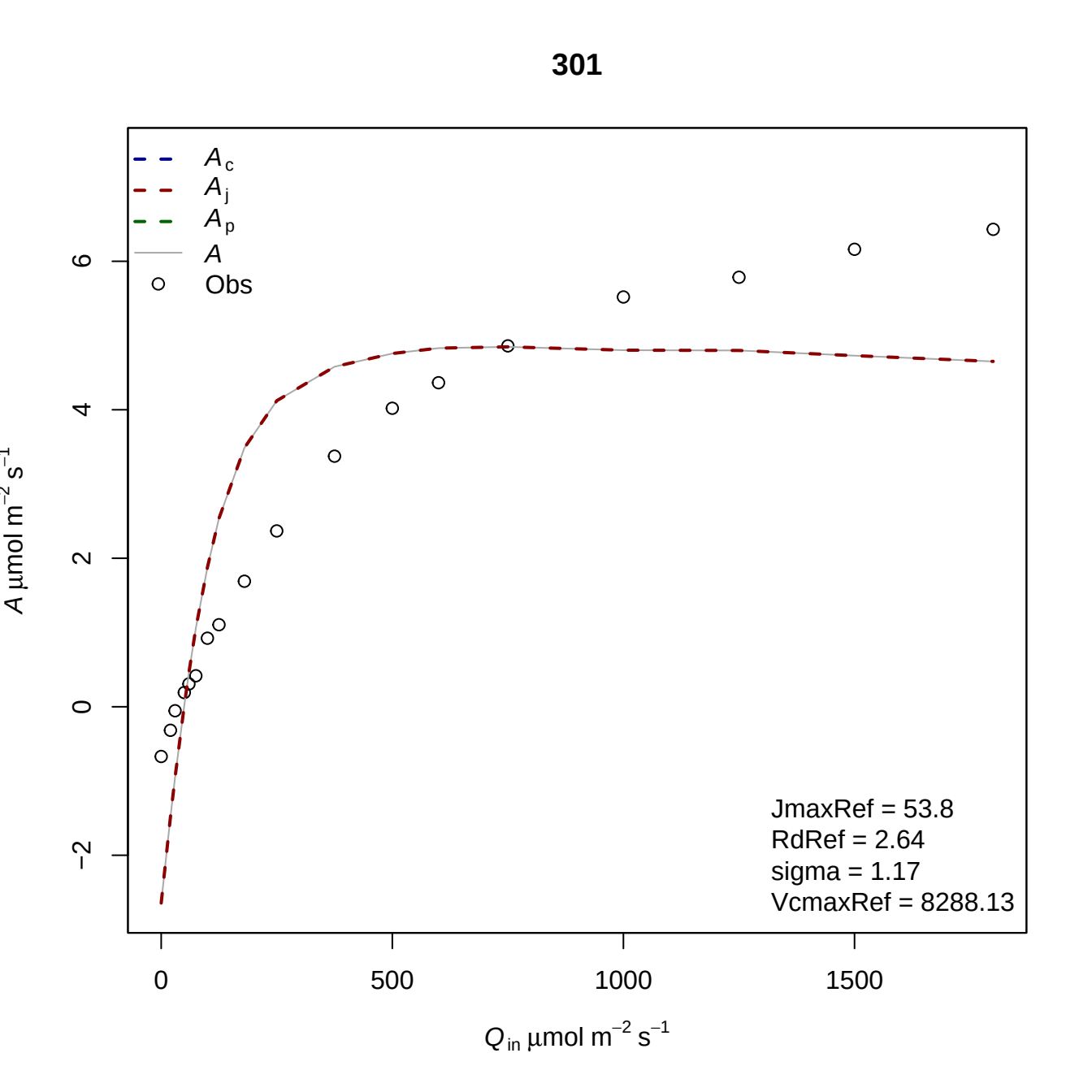

296
A_c
A_j
A_p
A
Obs
JmaxRef = 68.02
RdRef = 2.79
sigma = 1.19
VcmaxRef = 97038.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

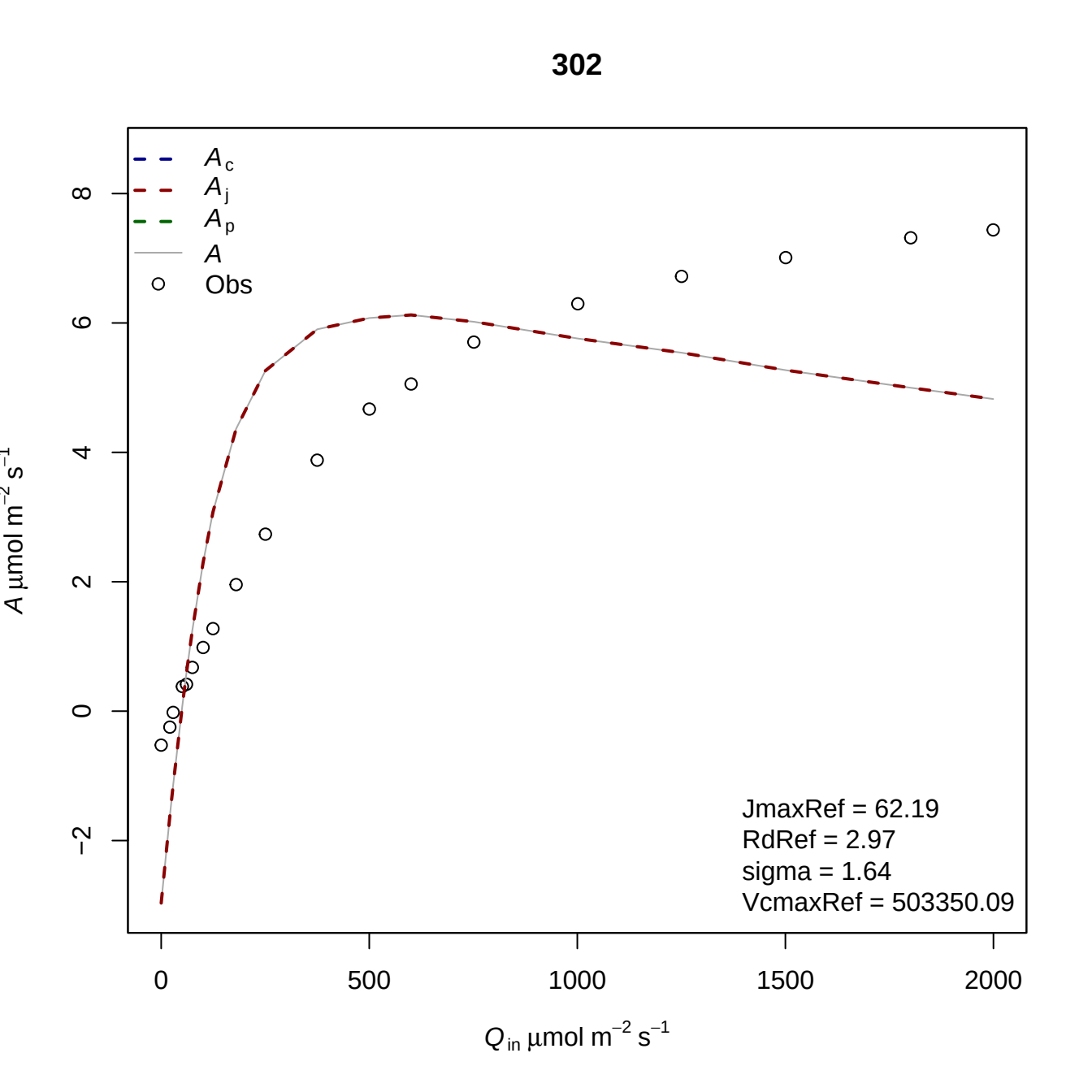

A_c
A_j
A_p
A
Obs
JmaxRef = 70.47
RdRef = 2.89
sigma = 1.31
VcmaxRef = 6164.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

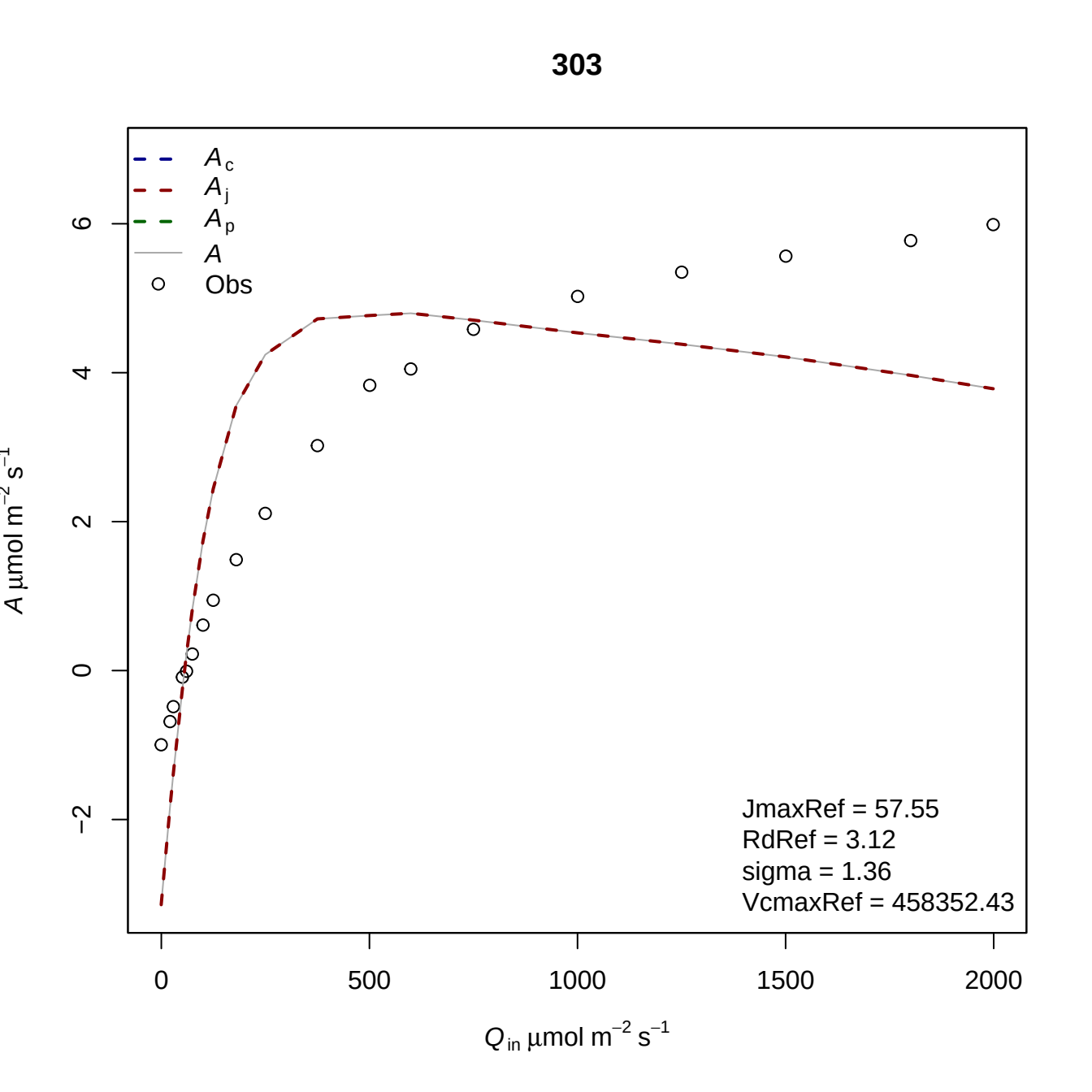

A_c
A_j
A_p
A
Obs
JmaxRef = 71.83
RdRef = 2.42
sigma = 1.24
VcmaxRef = 3656.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

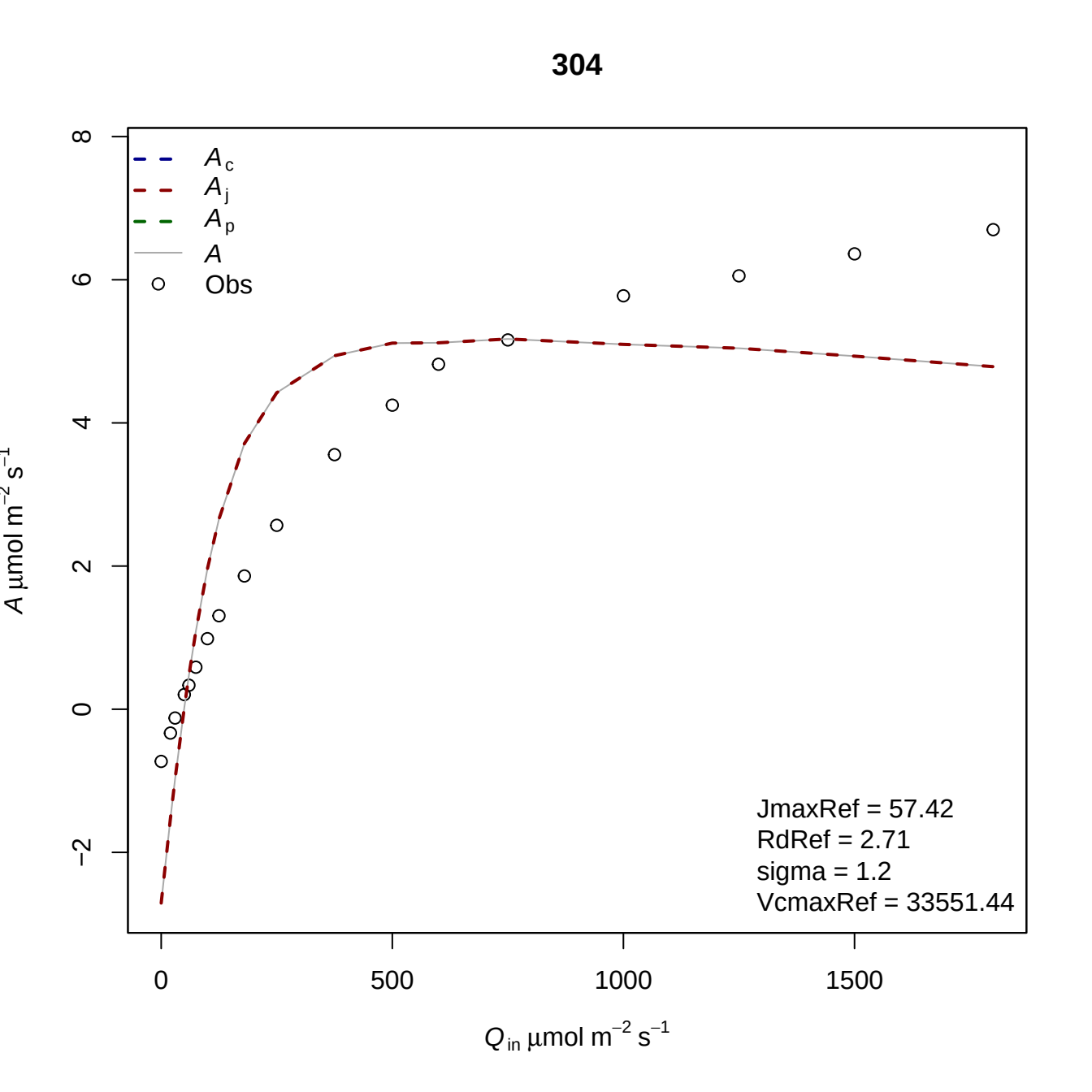

299
A_c
A_j
A_p
A
Obs
JmaxRef = 59.81
RdRef = 2.89
sigma = 1.44
VcmaxRef = 154366.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

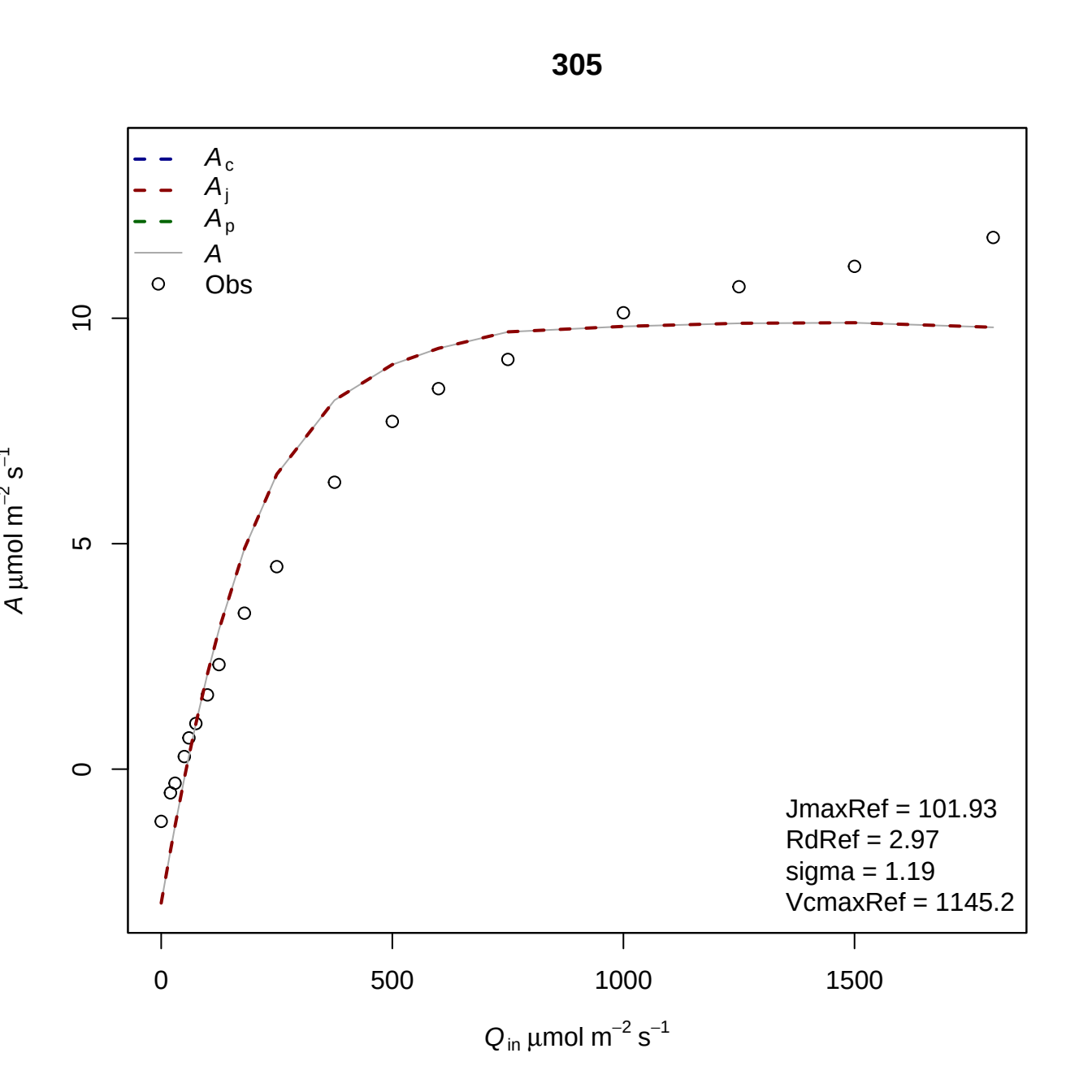

300
A_c
A_j
A_p
A
Obs
JmaxRef = 64.56
RdRef = 2.92
sigma = 1.18
VcmaxRef = 84018.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

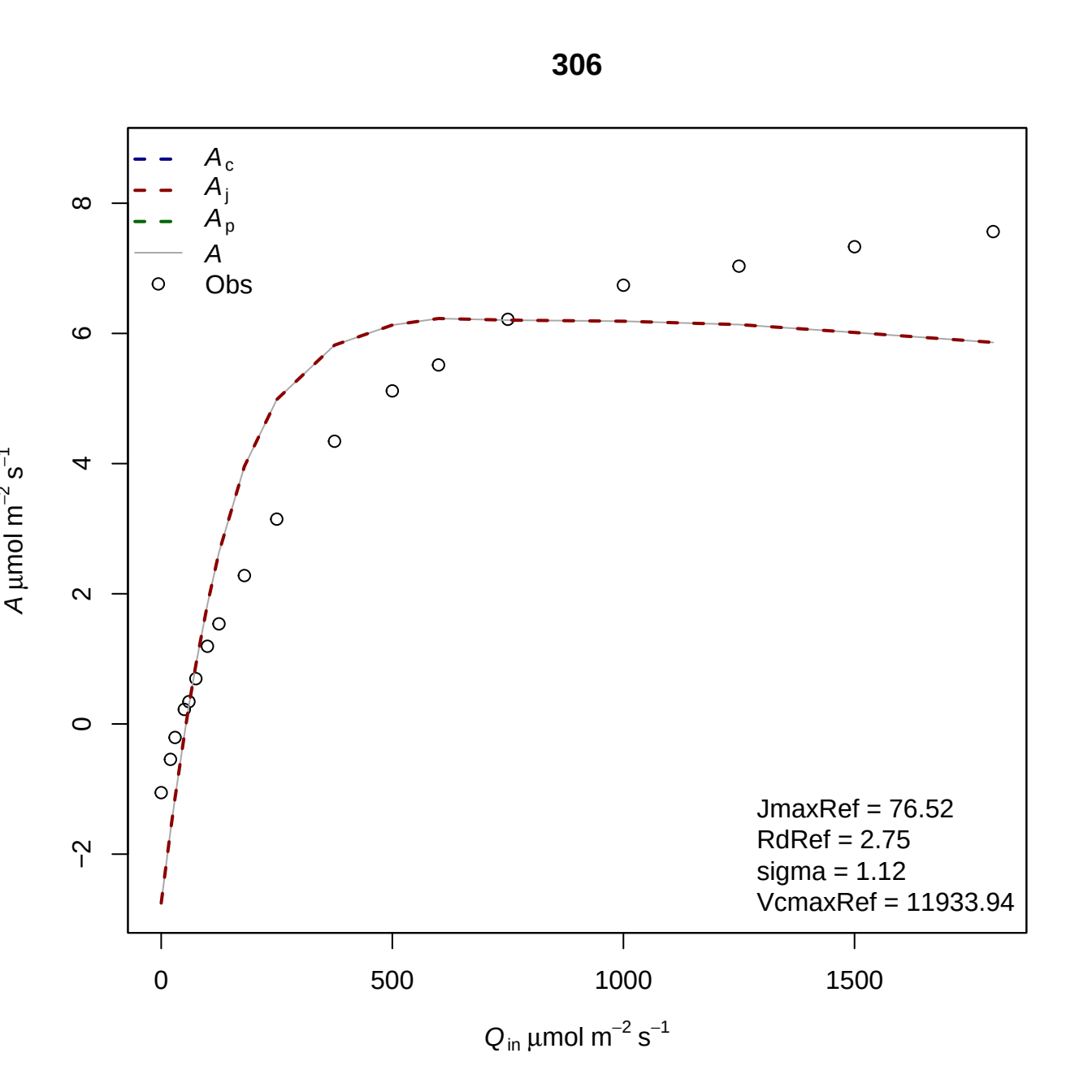

301
A_c
A_j
A_p
A
Obs
JmaxRef = 53.8
RdRef = 2.64
sigma = 1.17
VcmaxRef = 8288.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

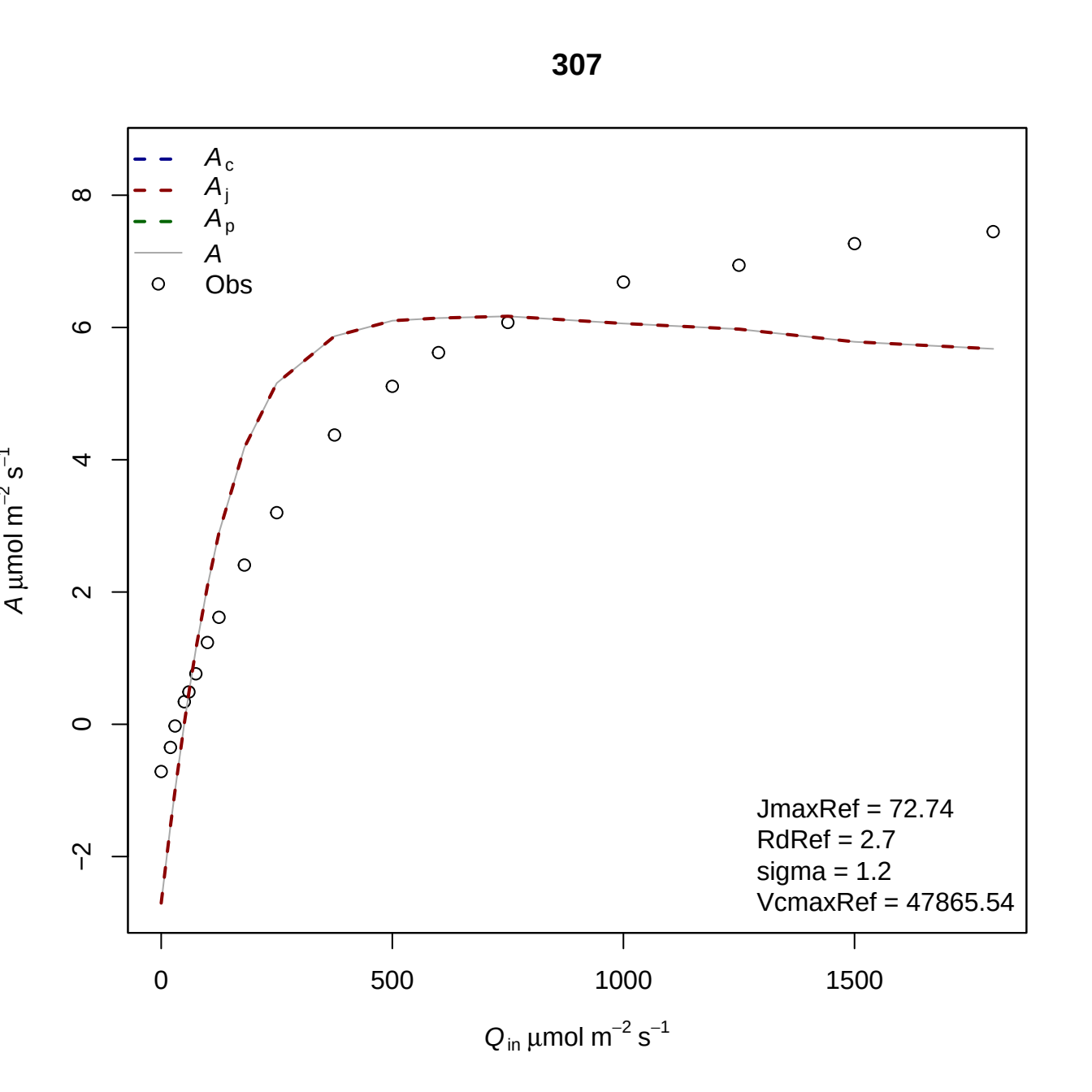

302
A_c
A_j
A_p
A
Obs
JmaxRef = 62.19
RdRef = 2.97
sigma = 1.64
VcmaxRef = 503350.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

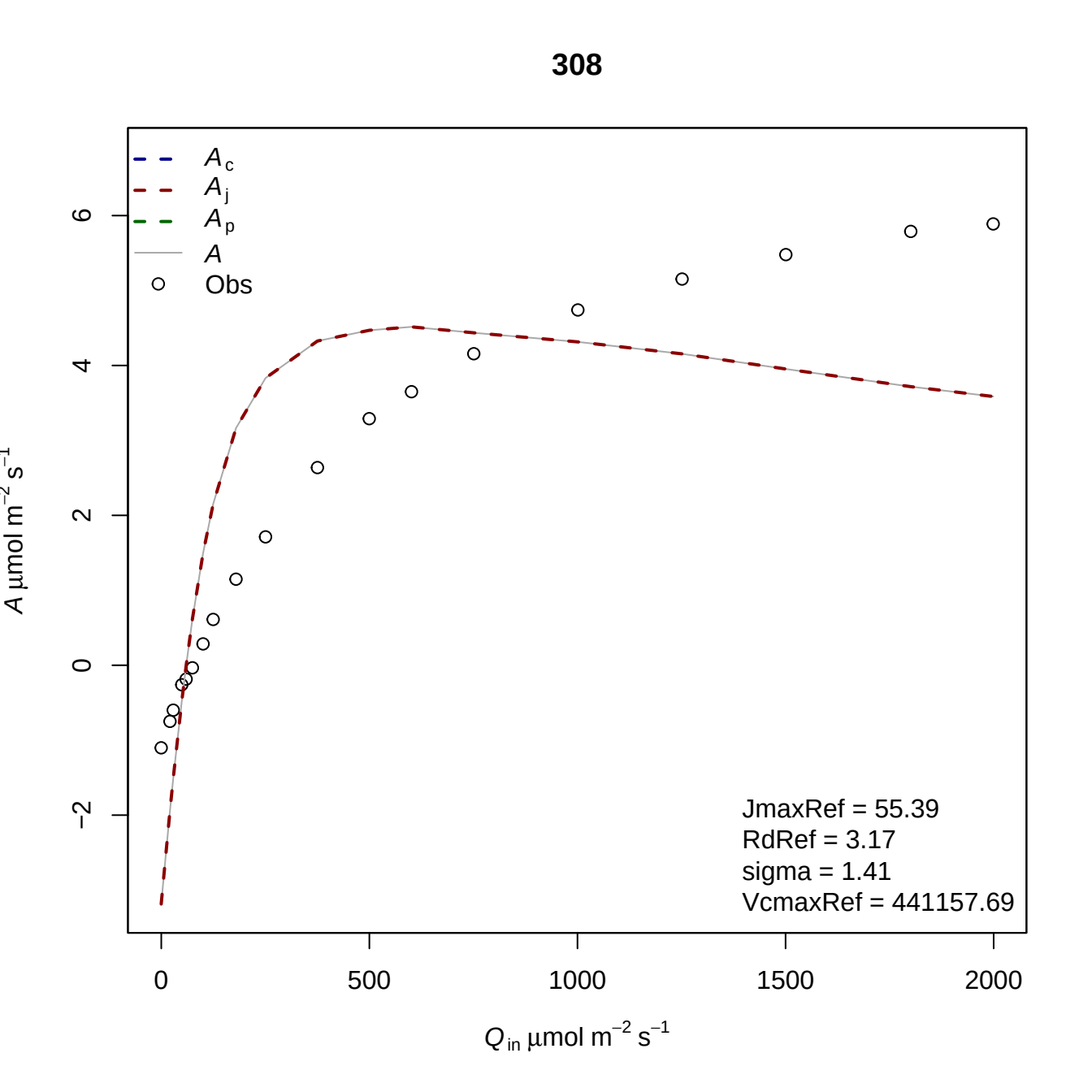

303
A_c
A_j
A_p
A
Obs
JmaxRef = 57.55
RdRef = 3.12
sigma = 1.36
VcmaxRef = 458352.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

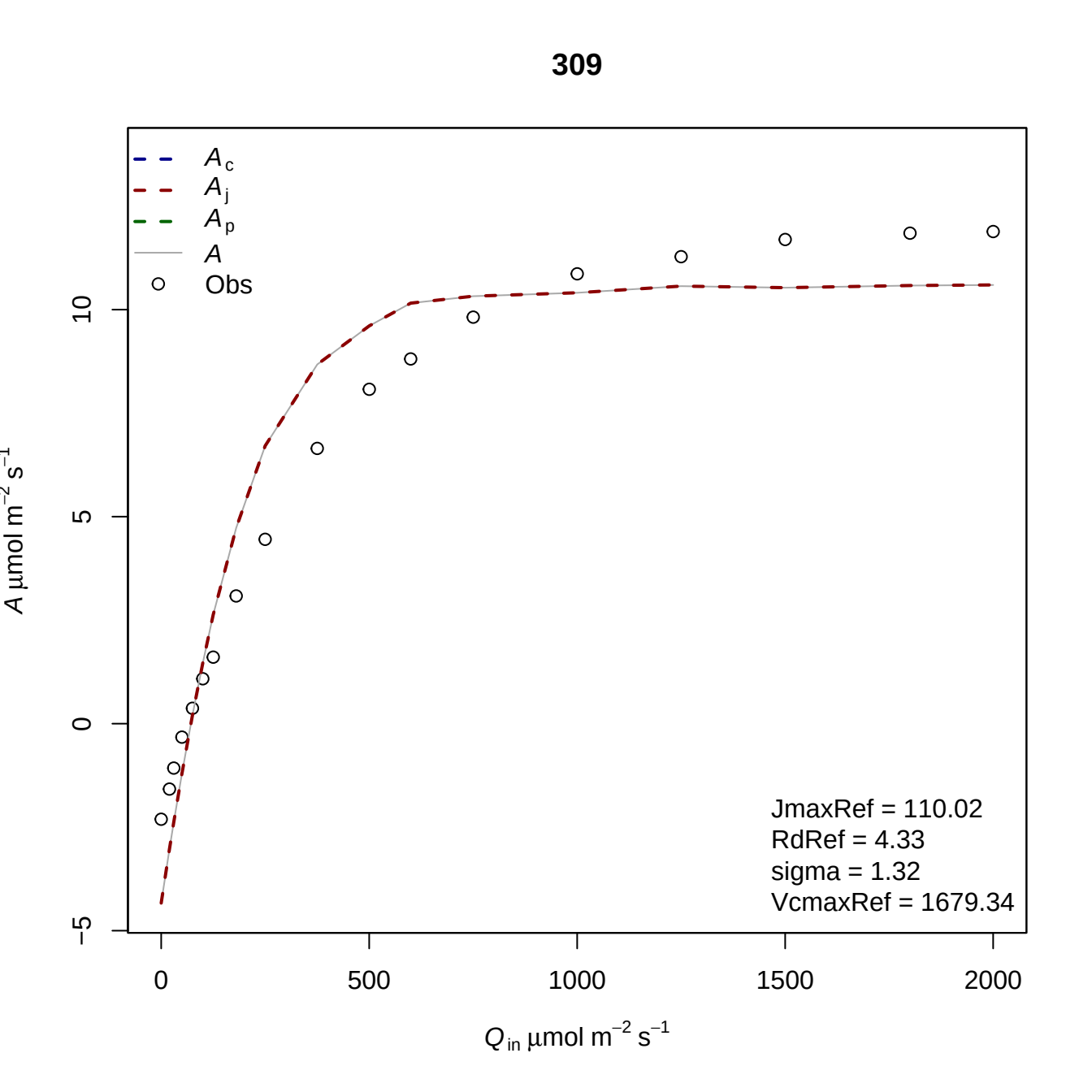

304
A_c
A_j
A_p
A
Obs
JmaxRef = 57.42
RdRef = 2.71
sigma = 1.2
VcmaxRef = 33551.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

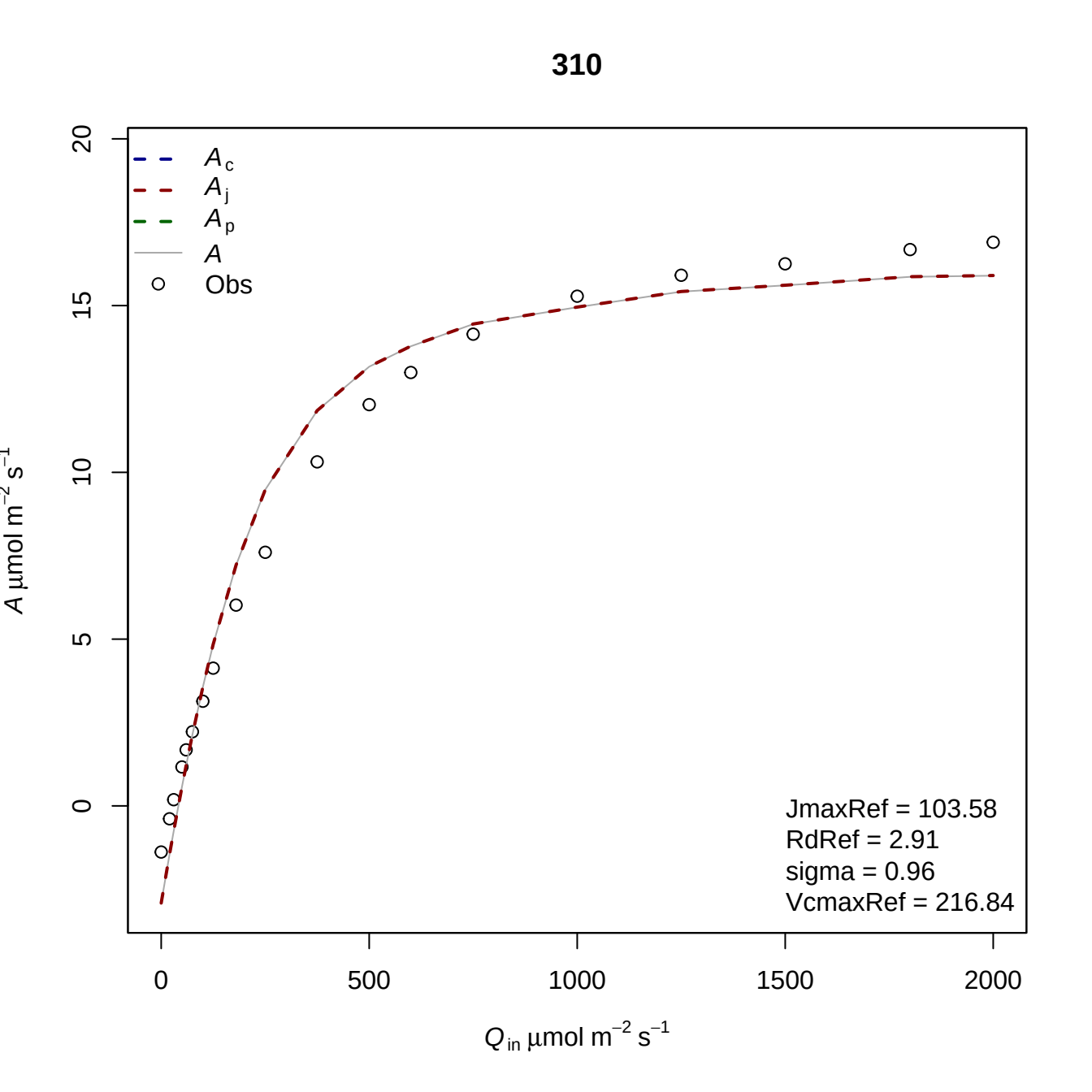

305
A_c
A_j
A_p
A
Obs
JmaxRef = 101.93
RdRef = 2.97
sigma = 1.19
VcmaxRef = 1145.2
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

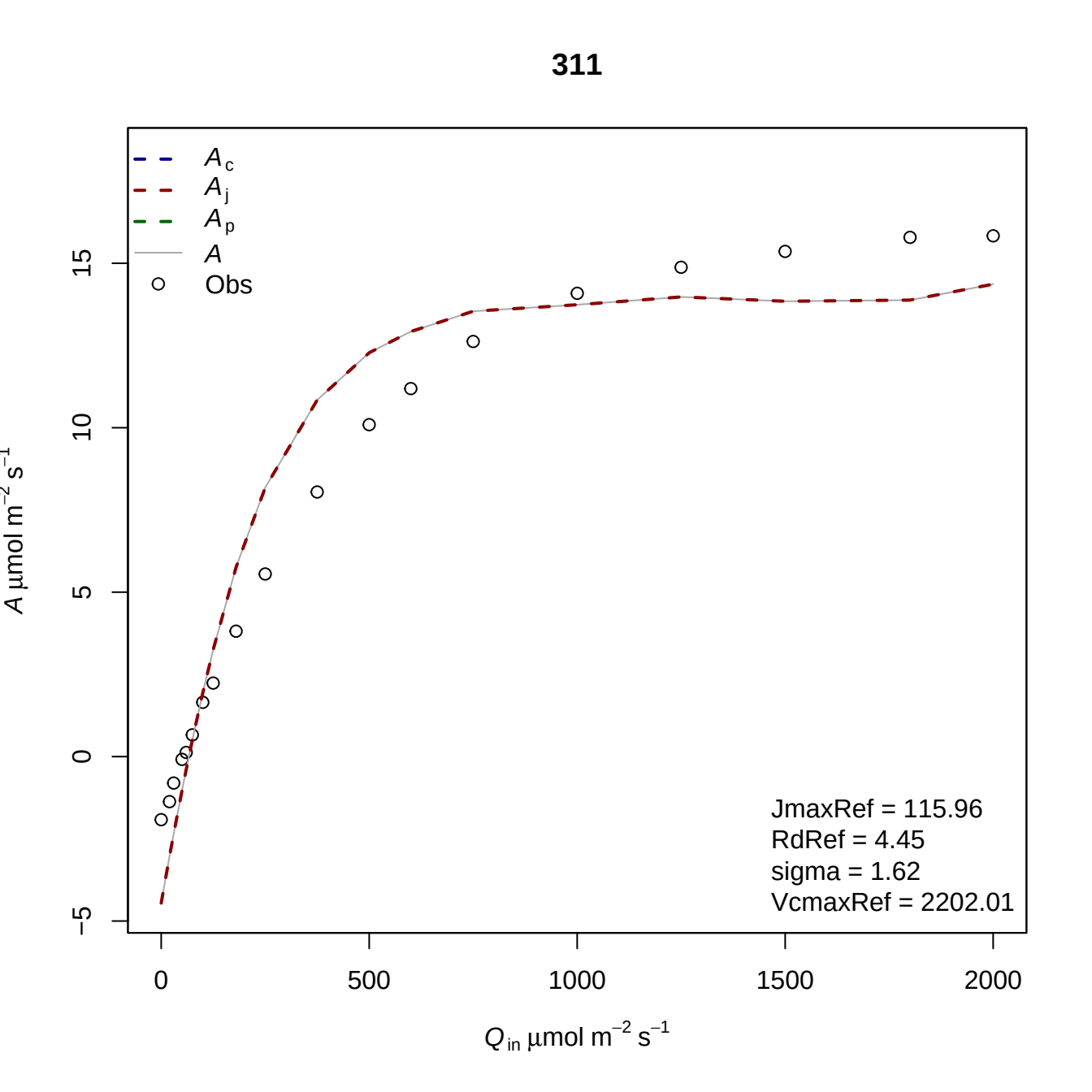

306
A_c
A_j
A_p
A
Obs
JmaxRef = 76.52
RdRef = 2.75
sigma = 1.12
VcmaxRef = 11933.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

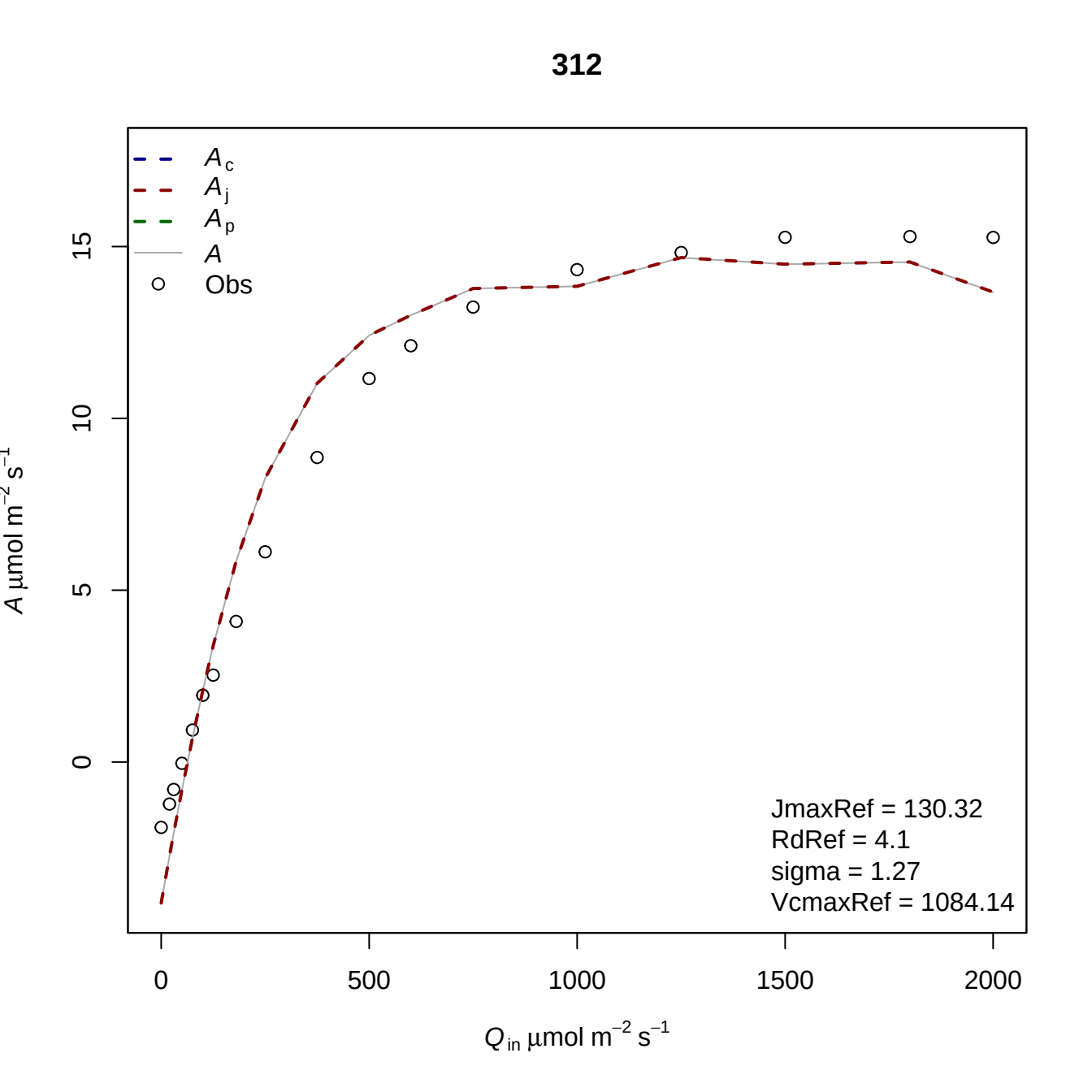

307
A_c
A_j
A_p
A
Obs
JmaxRef = 72.74
RdRef = 2.7
sigma = 1.2
VcmaxRef = 47865.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

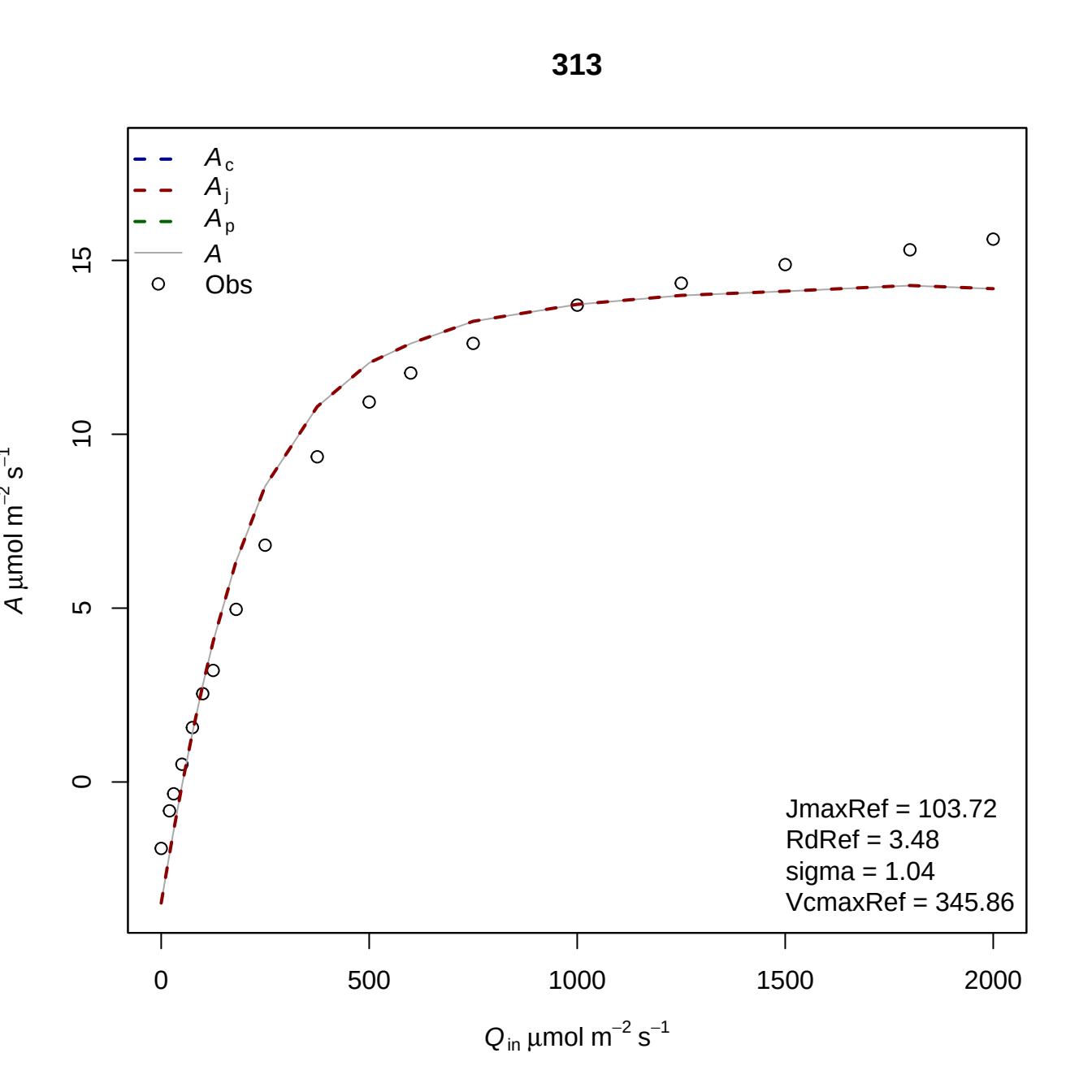

308
A_c
A_j
A_p
A
Obs
JmaxRef = 55.39
RdRef = 3.17
sigma = 1.41
VcmaxRef = 441157.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

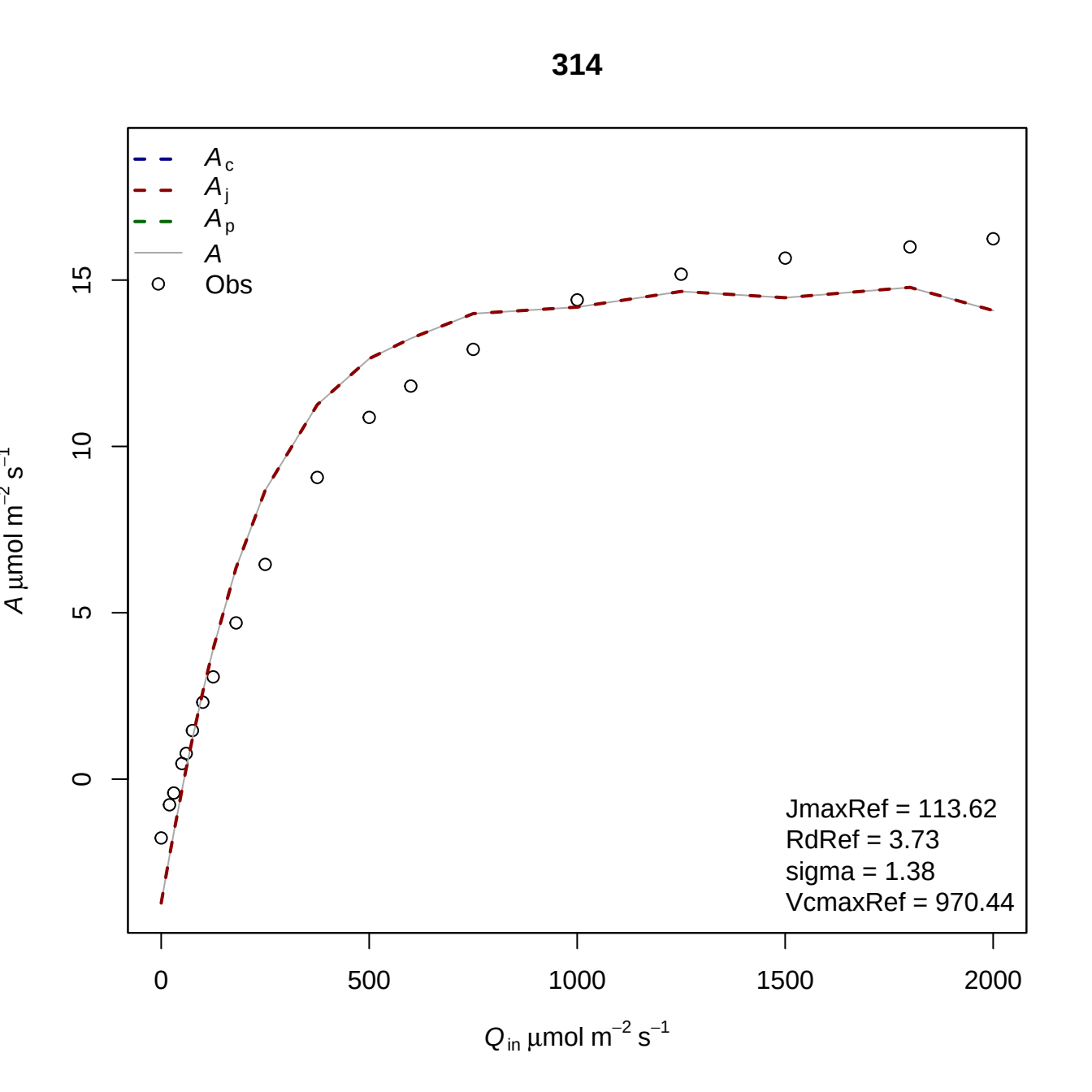

309
A_c
A_j
A_p
A
Obs
JmaxRef = 110.02
RdRef = 4.33
sigma = 1.32
VcmaxRef = 1679.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

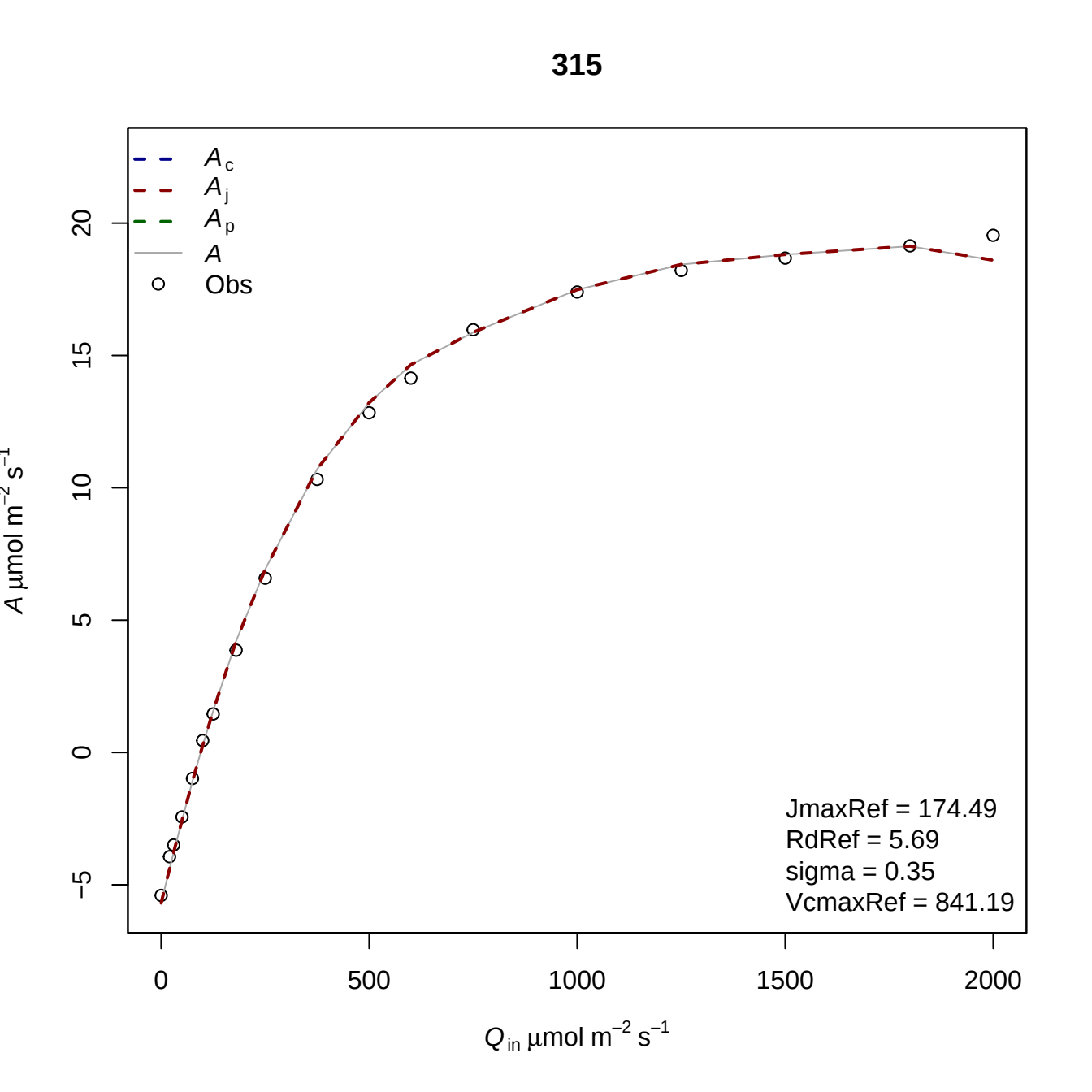

310
A_c
A_j
A_p
A
Obs
JmaxRef = 103.58
RdRef = 2.91
sigma = 0.96
VcmaxRef = 216.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

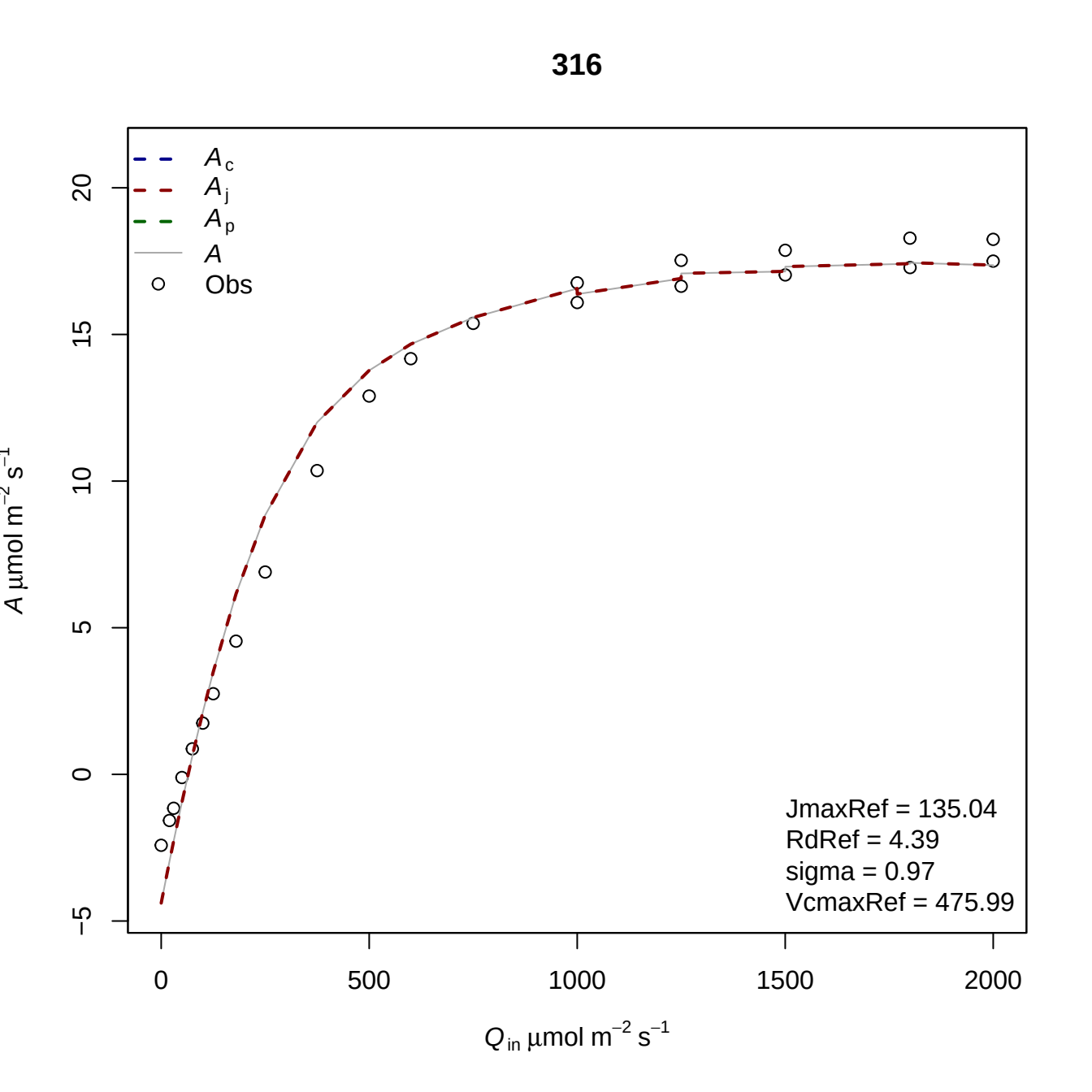

311
A_c
A_j
A_p
A
Obs
JmaxRef = 115.96
RdRef = 4.45
sigma = 1.62
VcmaxRef = 2202.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

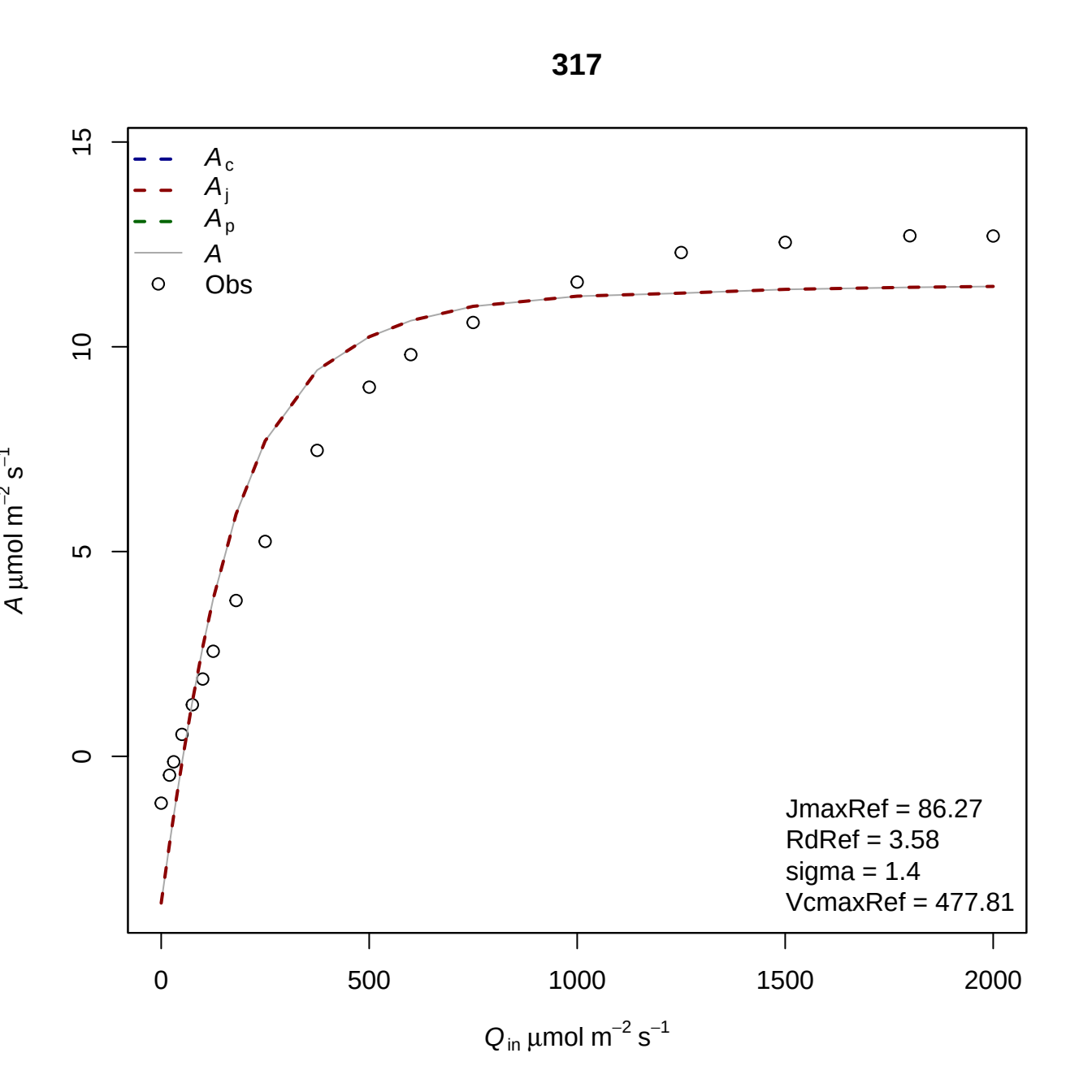

312
A_c
A_j
A_p
A
Obs
JmaxRef = 130.32
RdRef = 4.1
sigma = 1.27
VcmaxRef = 1084.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

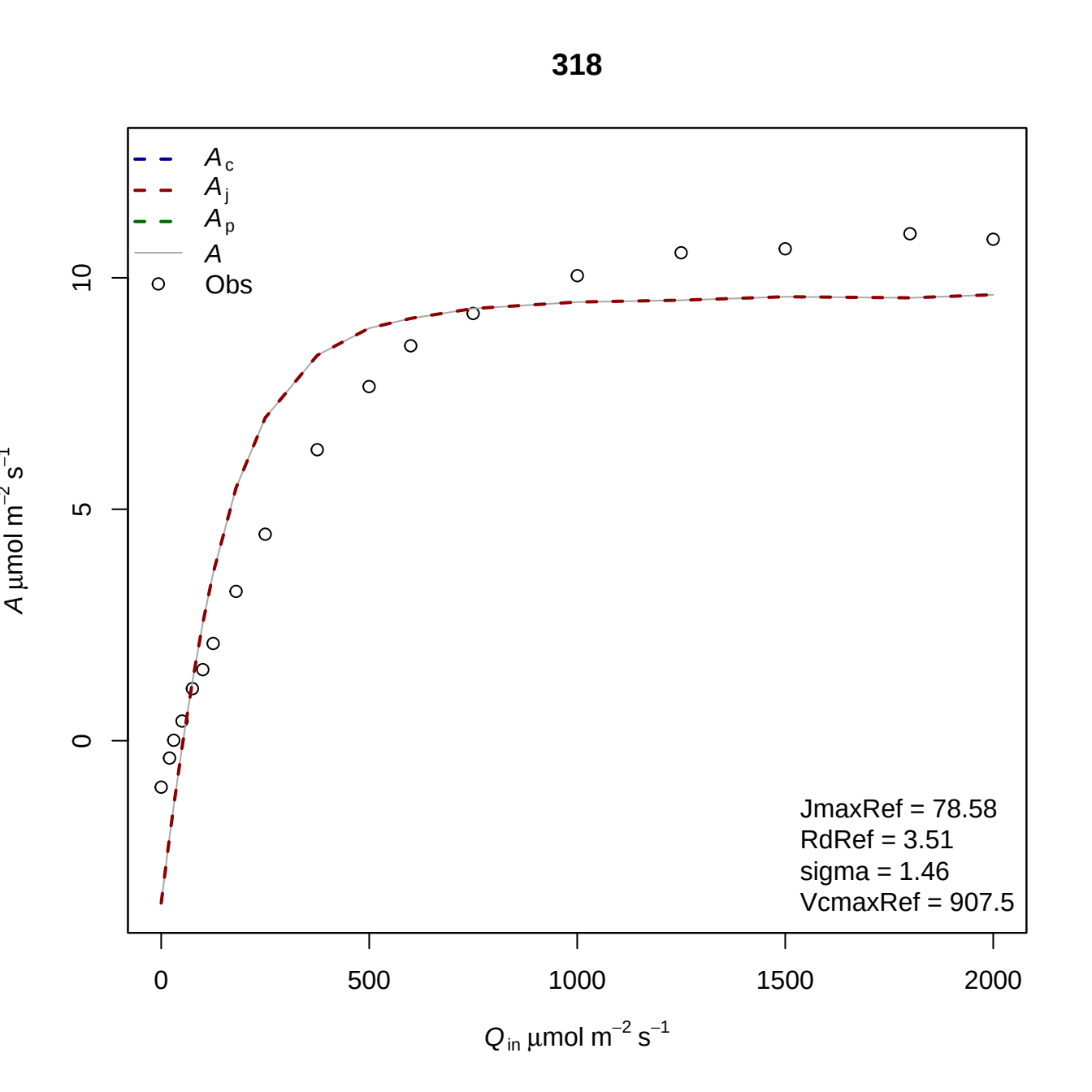

313
A_c
A_j
A_p
A
Obs
JmaxRef = 103.72
RdRef = 3.48
sigma = 1.04
VcmaxRef = 345.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

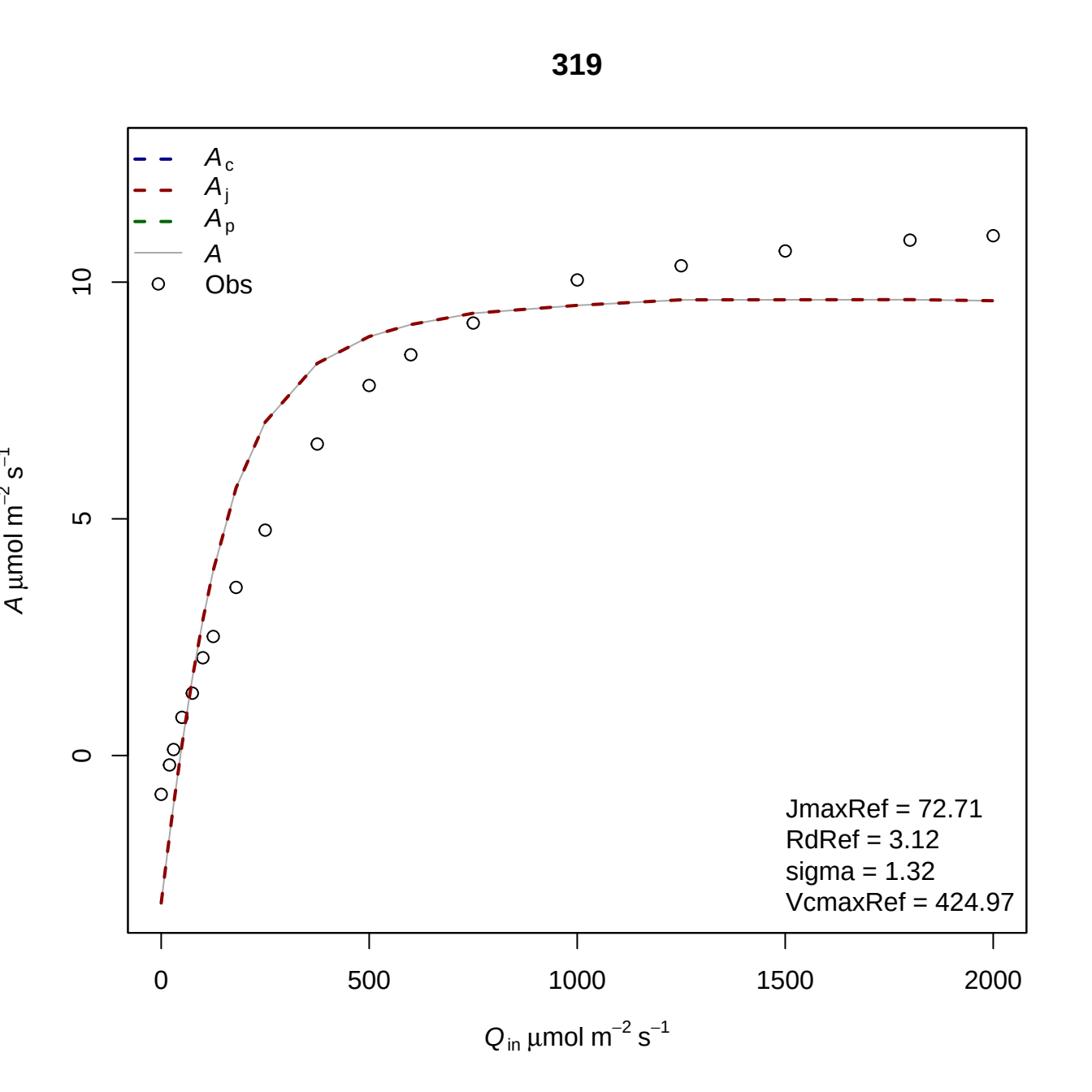

314
A_c
A_j
A_p
A
Obs
JmaxRef = 113.62
RdRef = 3.73
sigma = 1.38
VcmaxRef = 970.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

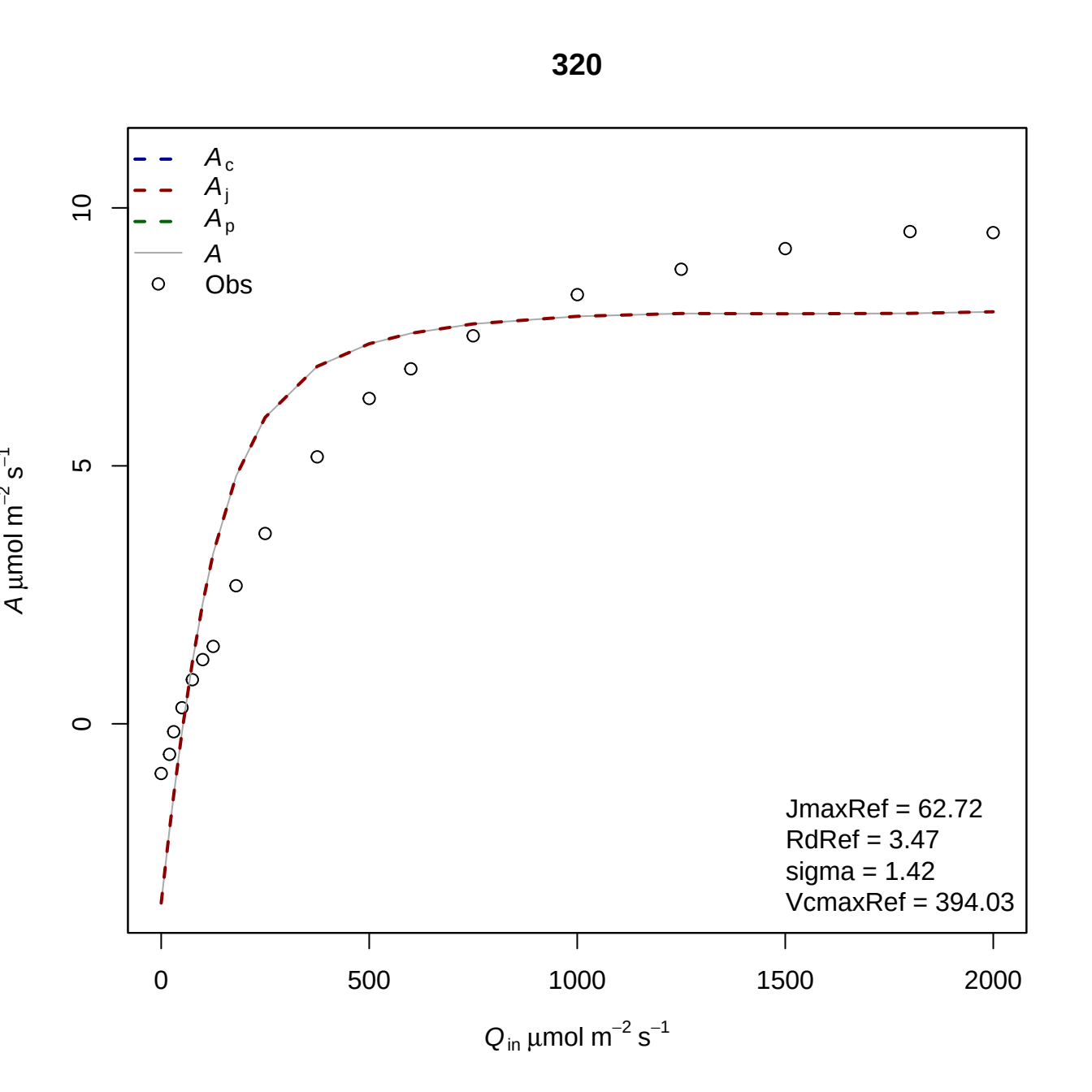

315
A_c
A_j
A_p
A
Obs
JmaxRef = 174.49
RdRef = 5.69
sigma = 0.35
VcmaxRef = 841.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

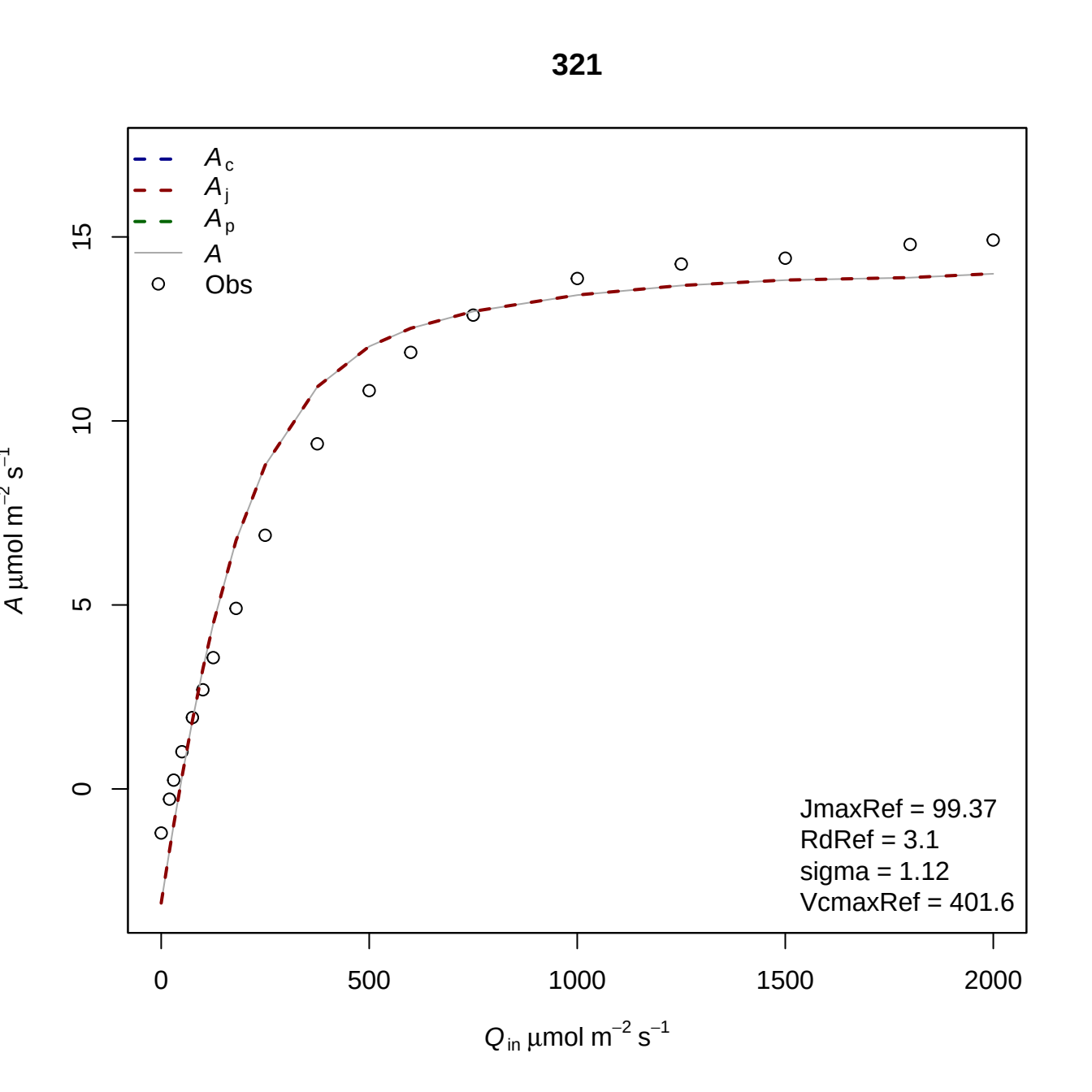

316
A_c
A_j
A_p
A
Obs
JmaxRef = 135.04
RdRef = 4.39
sigma = 0.97
VcmaxRef = 475.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

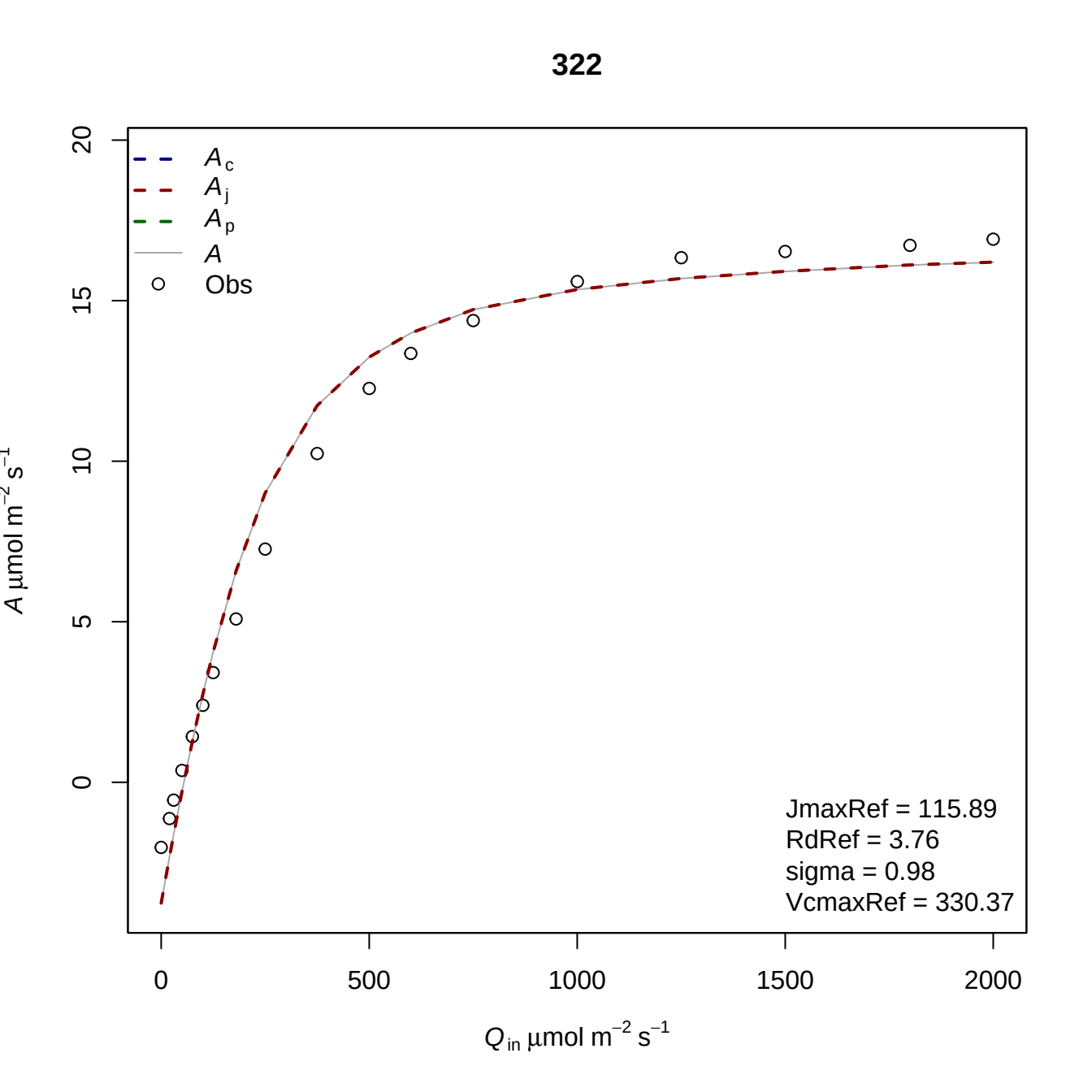

317
A_c
A_j
A_p
A
Obs
JmaxRef = 86.27
RdRef = 3.58
sigma = 1.4
VcmaxRef = 477.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

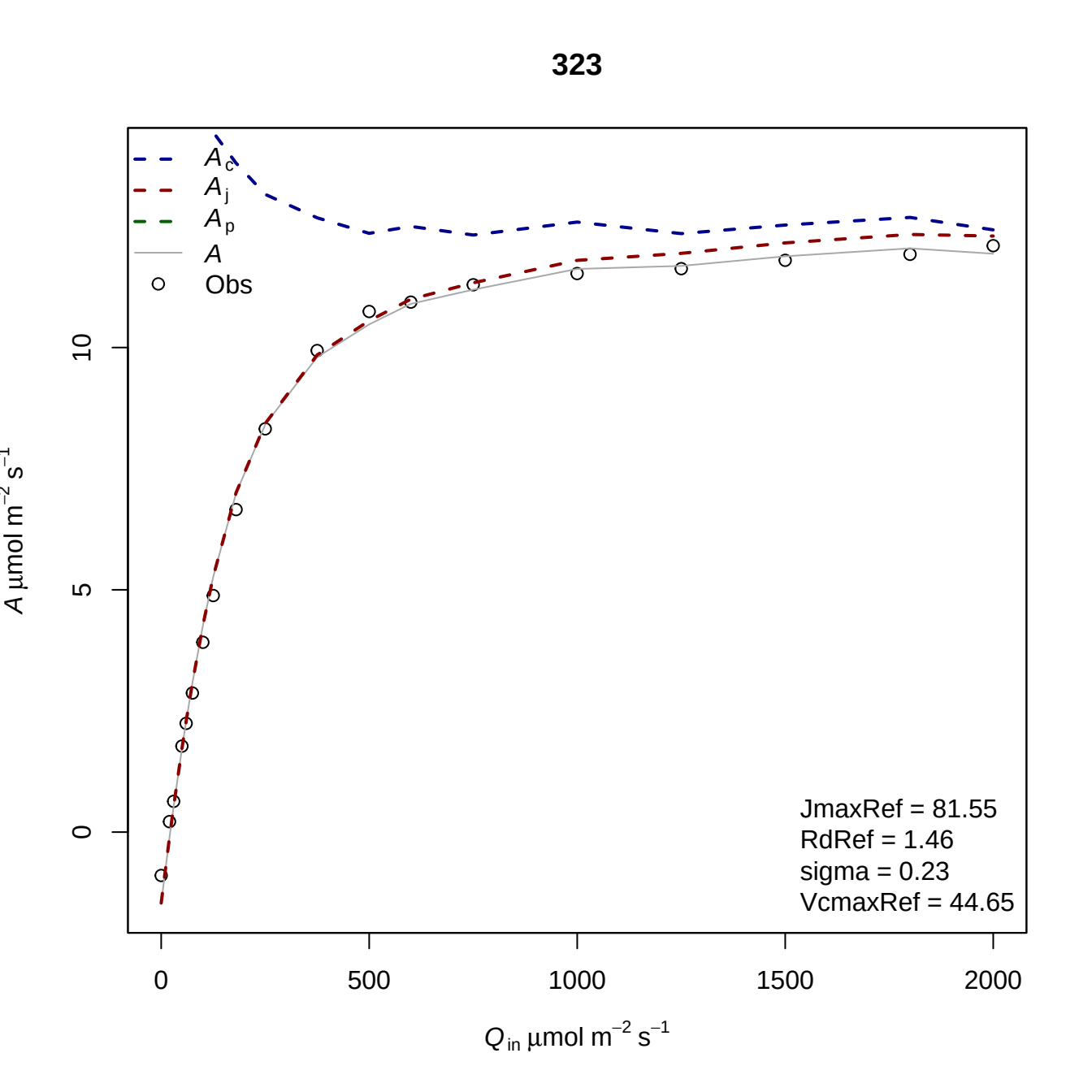

318
A_c
A_j
A_p
A
Obs
JmaxRef = 78.58
RdRef = 3.51
sigma = 1.46
VcmaxRef = 907.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

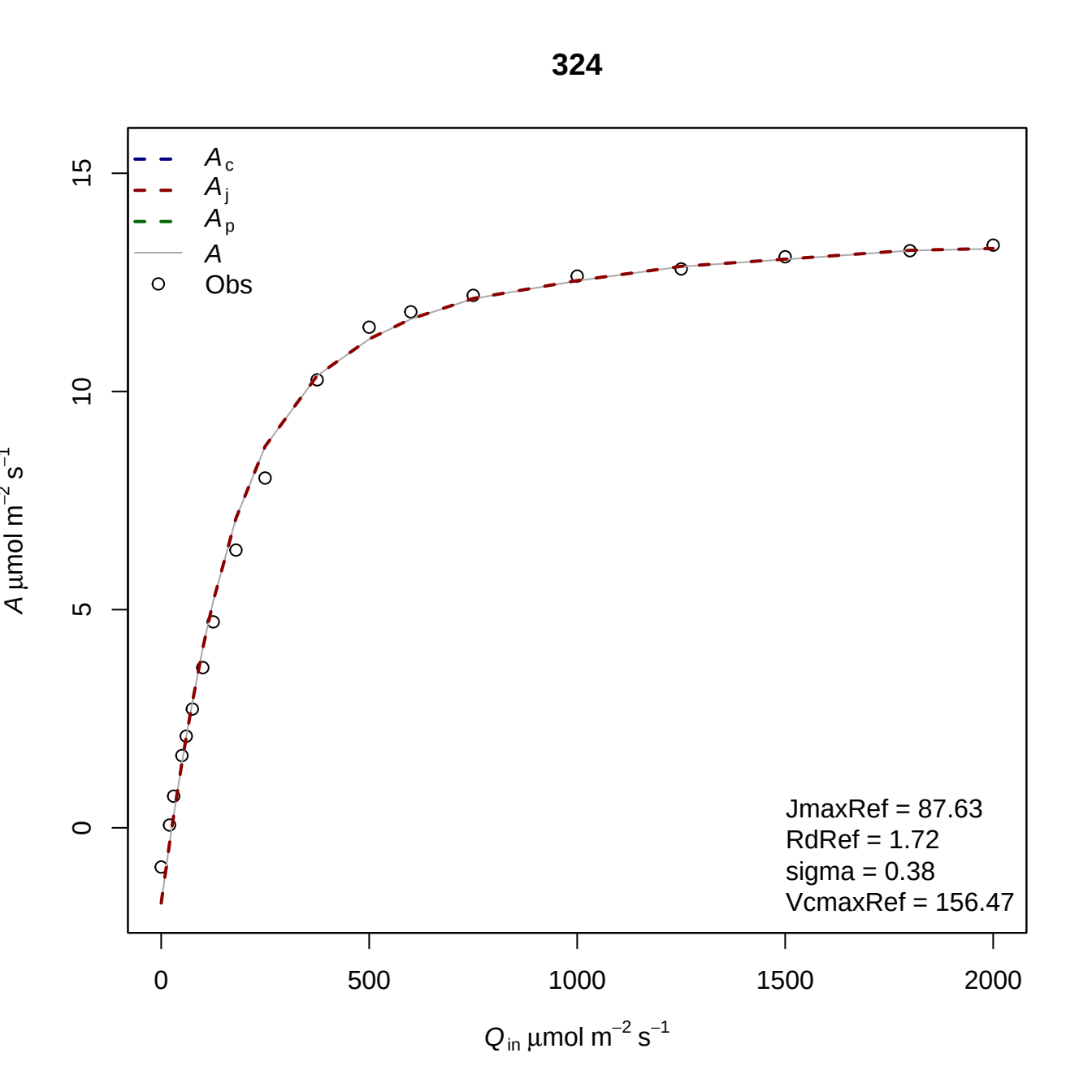

319
A_c
A_j
A_p
A
Obs
JmaxRef = 72.71
RdRef = 3.12
sigma = 1.32
VcmaxRef = 424.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

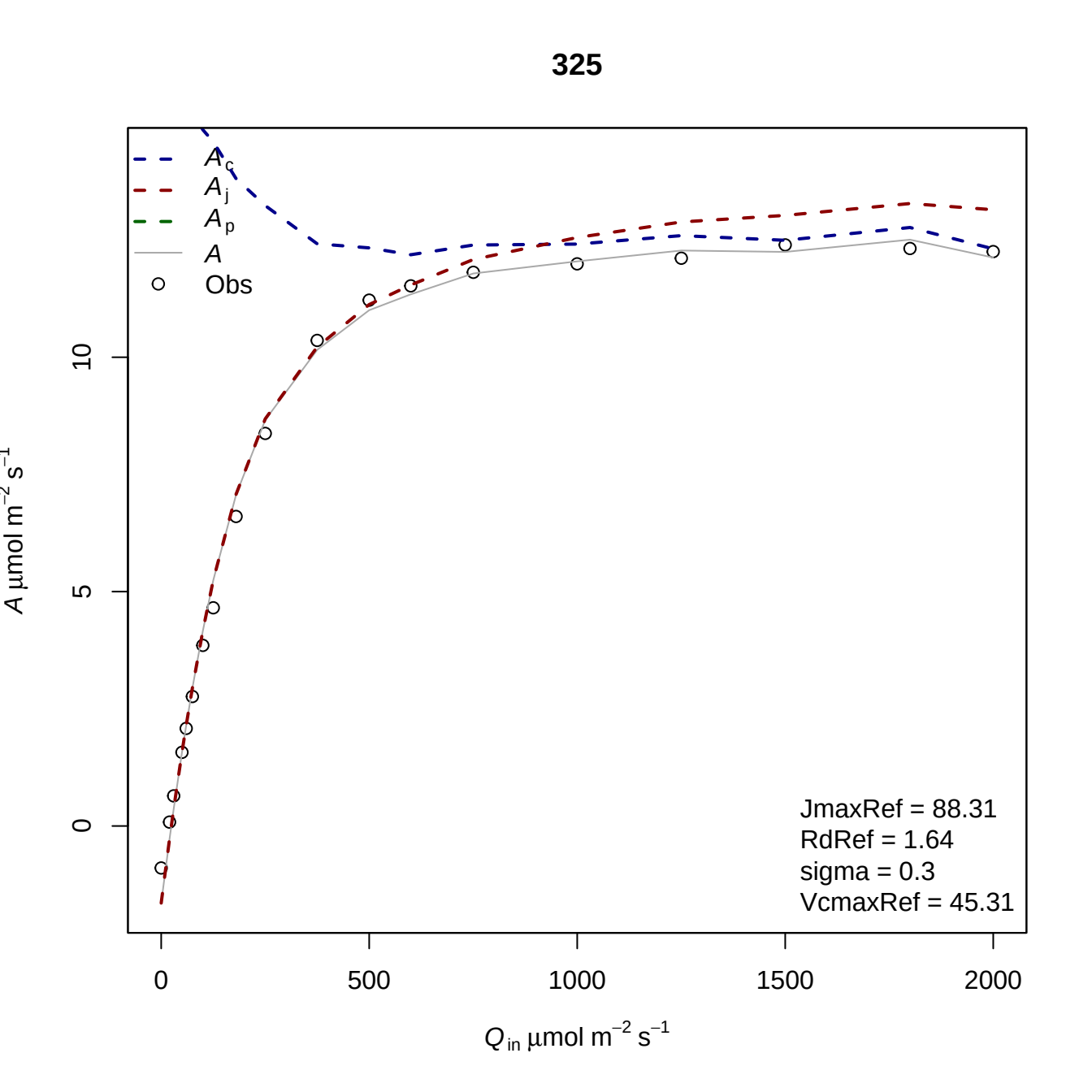

320
A_c
A_j
A_p
A
Obs
JmaxRef = 62.72
RdRef = 3.47
sigma = 1.42
VcmaxRef = 394.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

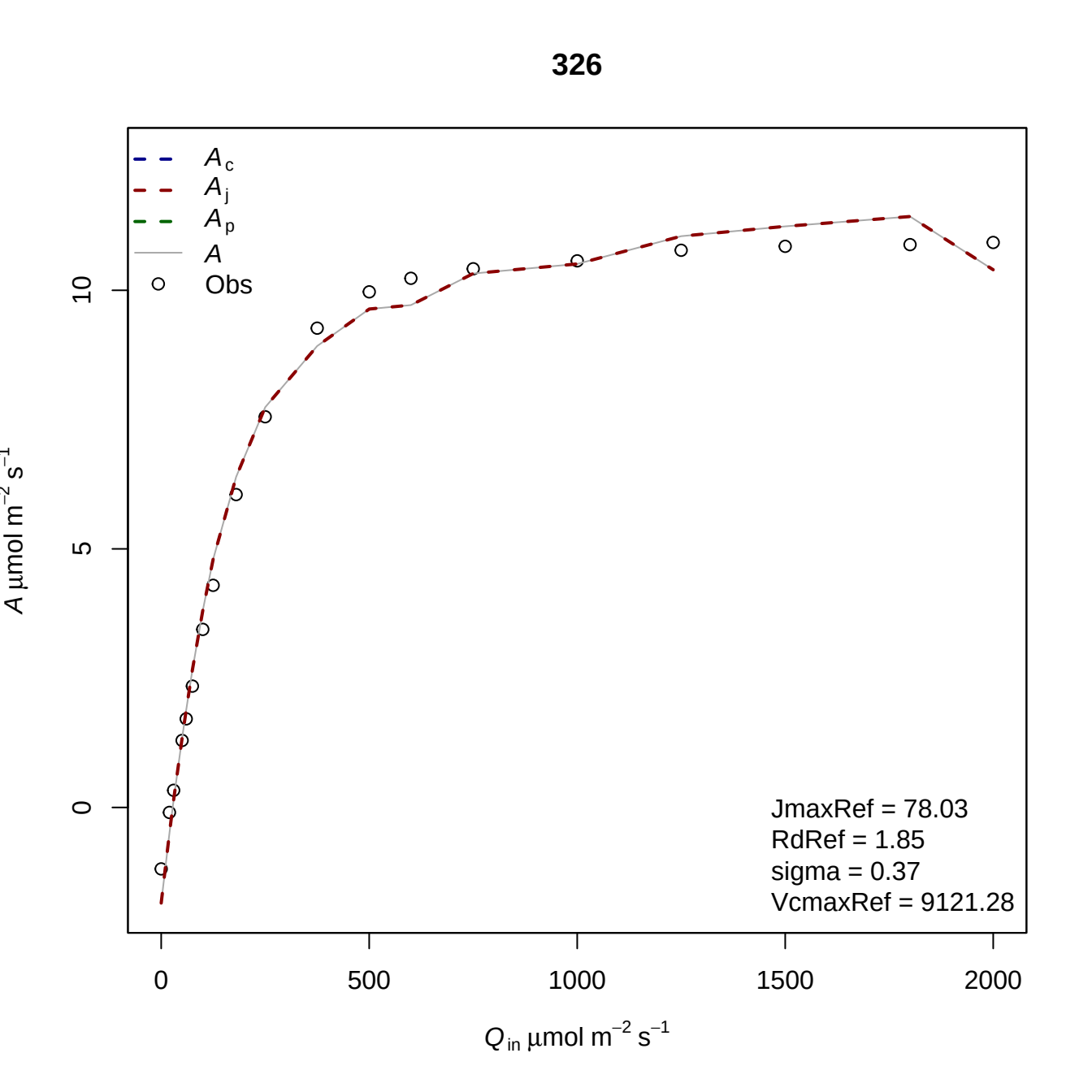

321
A_c
A_j
A_p
A
Obs
JmaxRef = 99.37
RdRef = 3.1
sigma = 1.12
VcmaxRef = 401.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

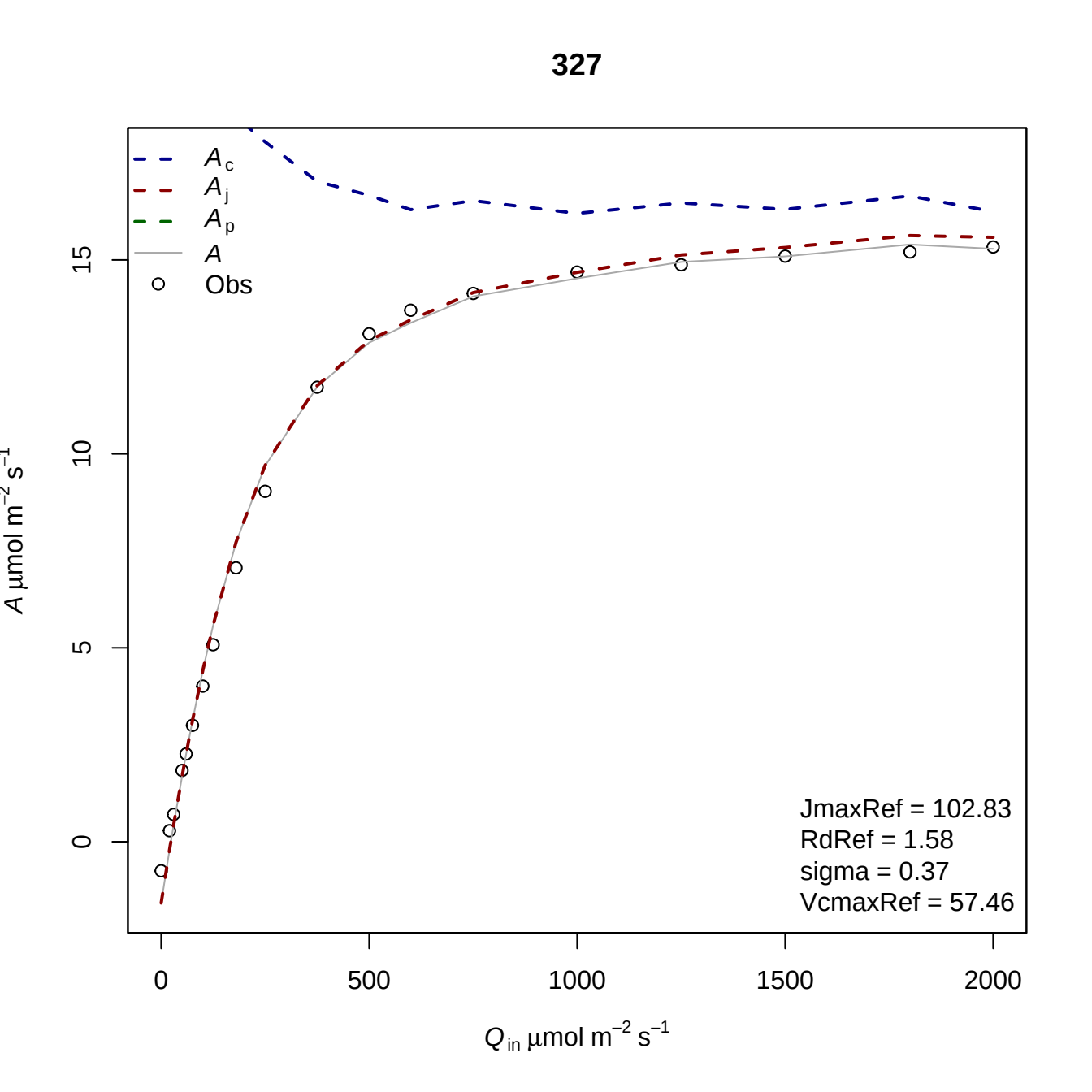

322
A_c
A_j
A_p
A
Obs
JmaxRef = 115.89
RdRef = 3.76
sigma = 0.98
VcmaxRef = 330.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

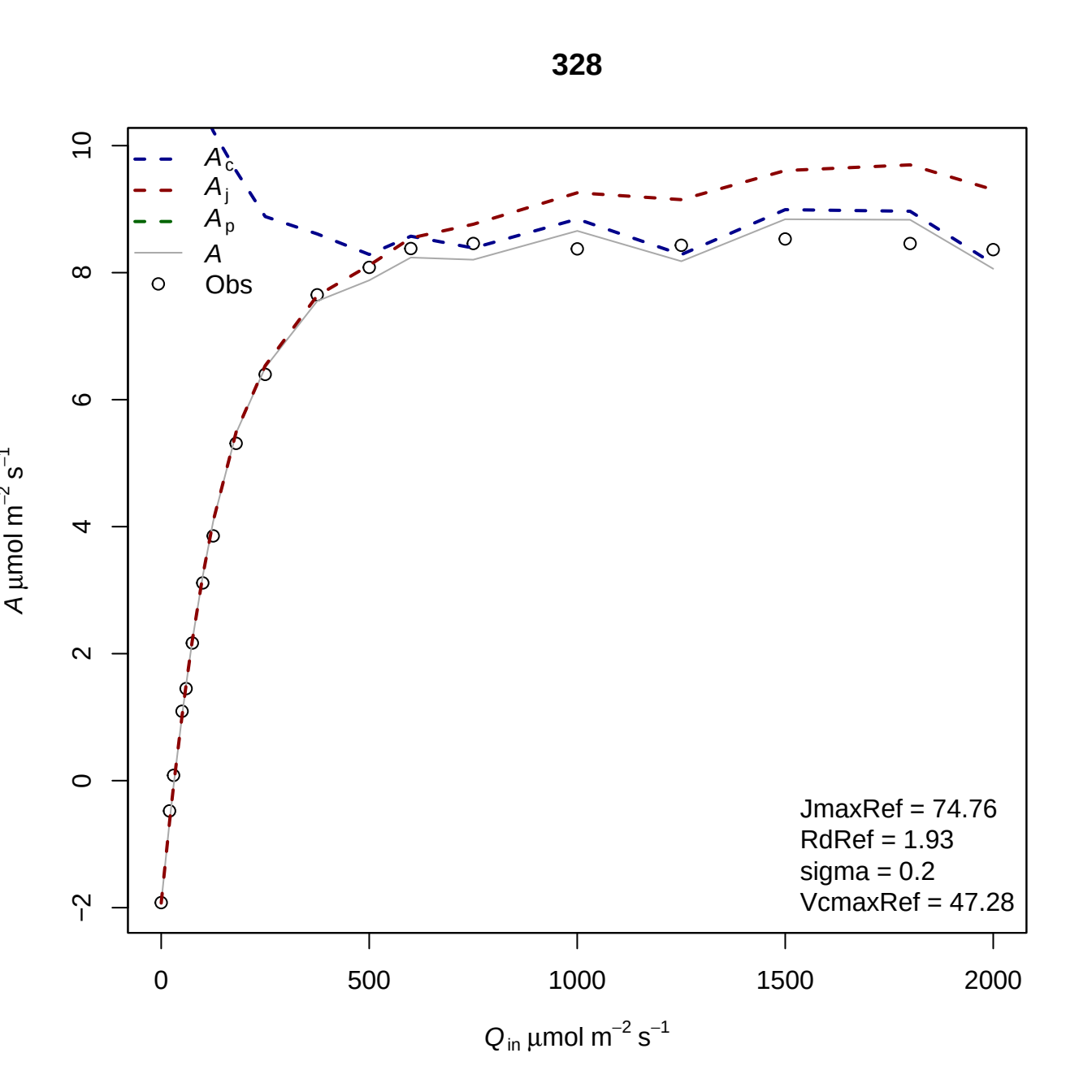

323
A_c
A_j
A_p
A
Obs
JmaxRef = 81.55
RdRef = 1.46
sigma = 0.23
VcmaxRef = 44.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

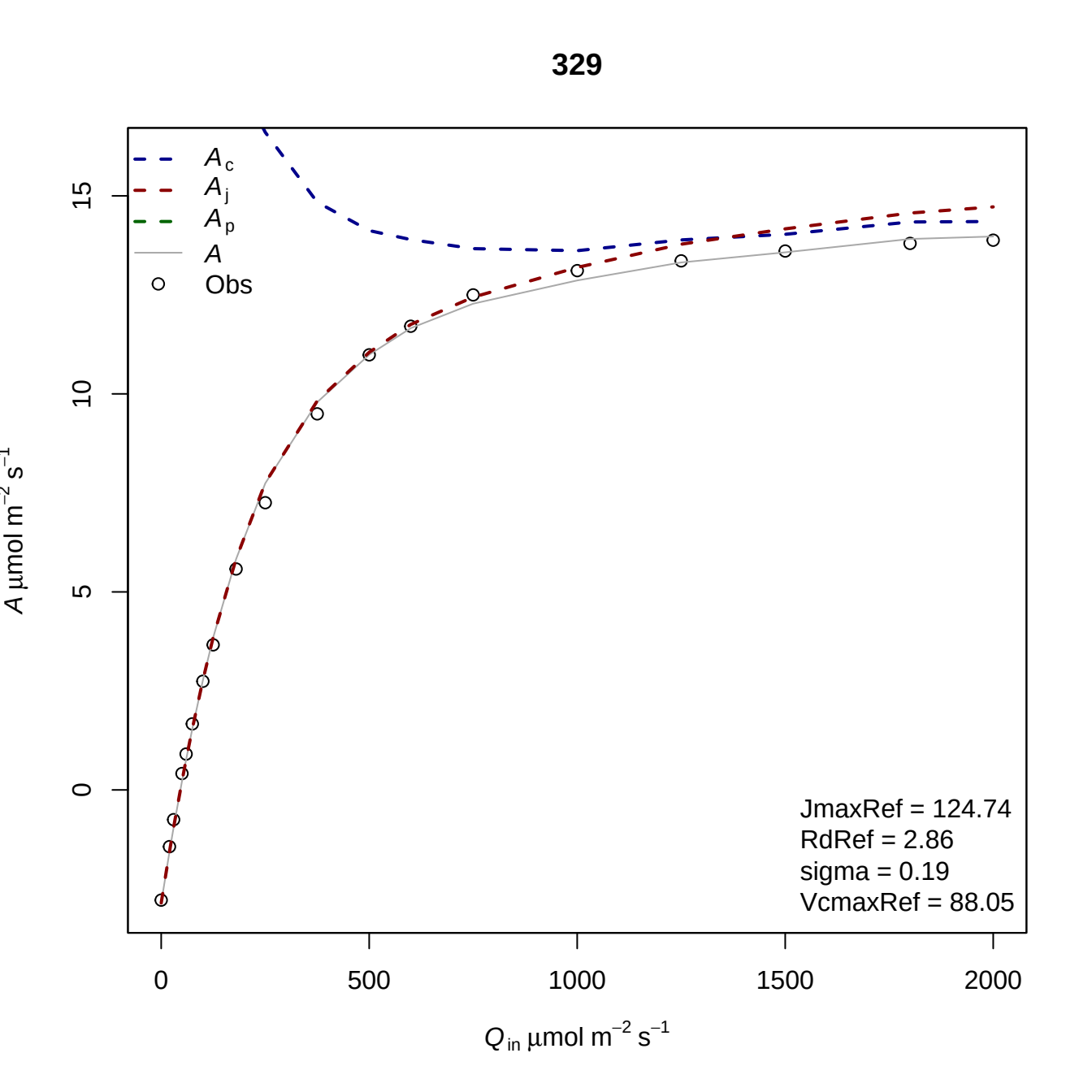

324
A_c
A_j
A_p
A
Obs
JmaxRef = 87.63
RdRef = 1.72
sigma = 0.38
VcmaxRef = 156.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

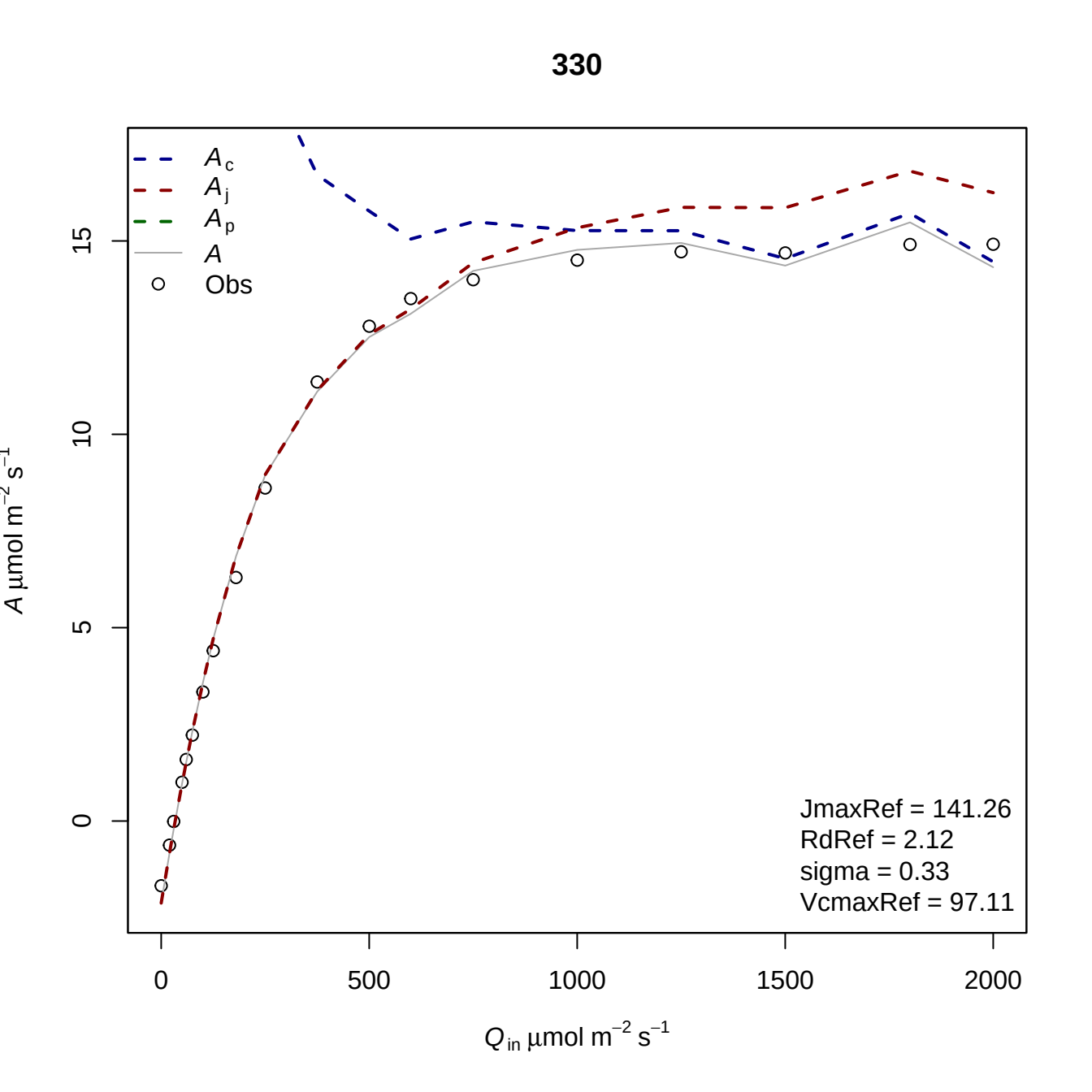

325
A_c
A_j
A_p
A
Obs
JmaxRef = 88.31
RdRef = 1.64
sigma = 0.3
VcmaxRef = 45.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

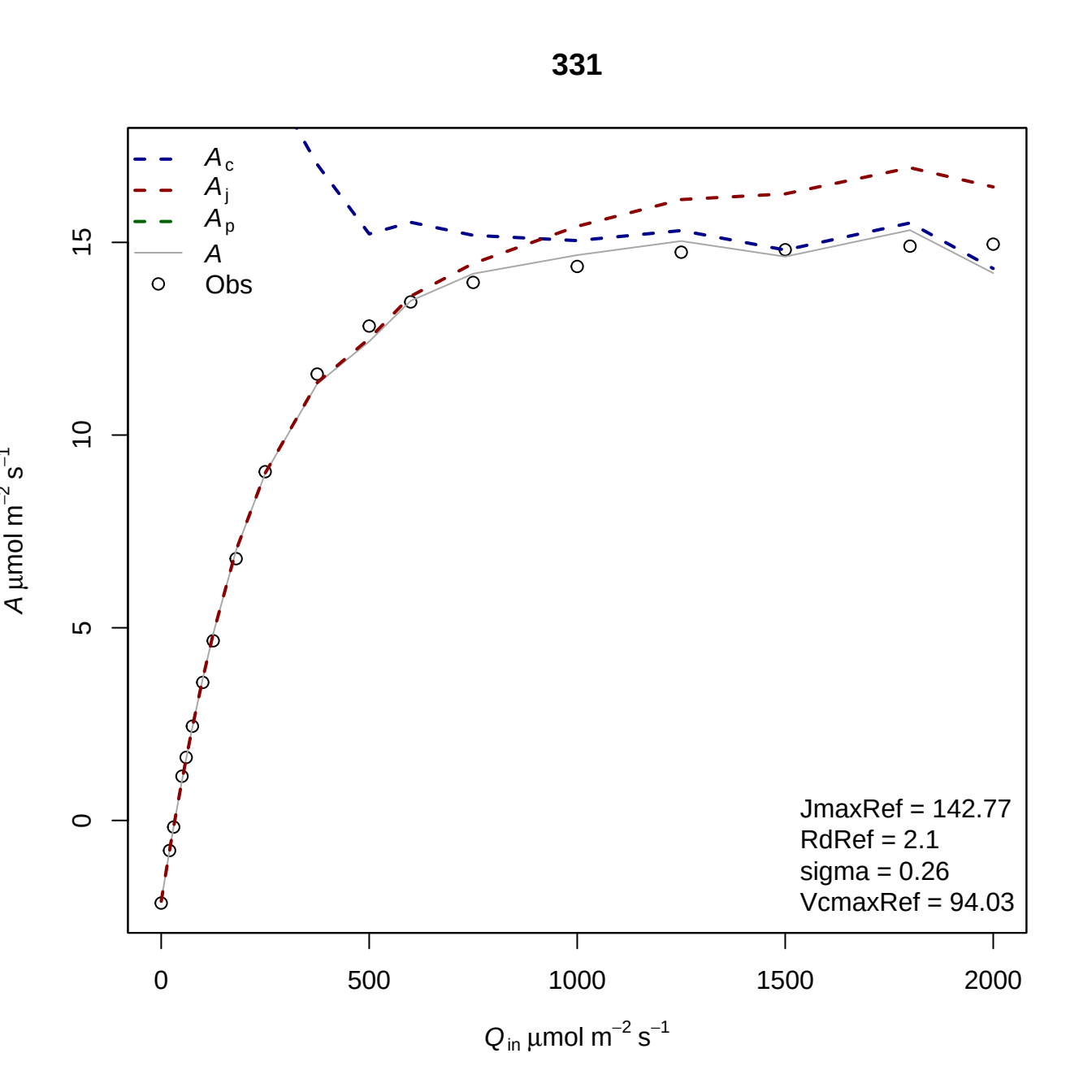

326
A_c
A_j
A_p
A
Obs
JmaxRef = 78.03
RdRef = 1.85
sigma = 0.37
VcmaxRef = 9121.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

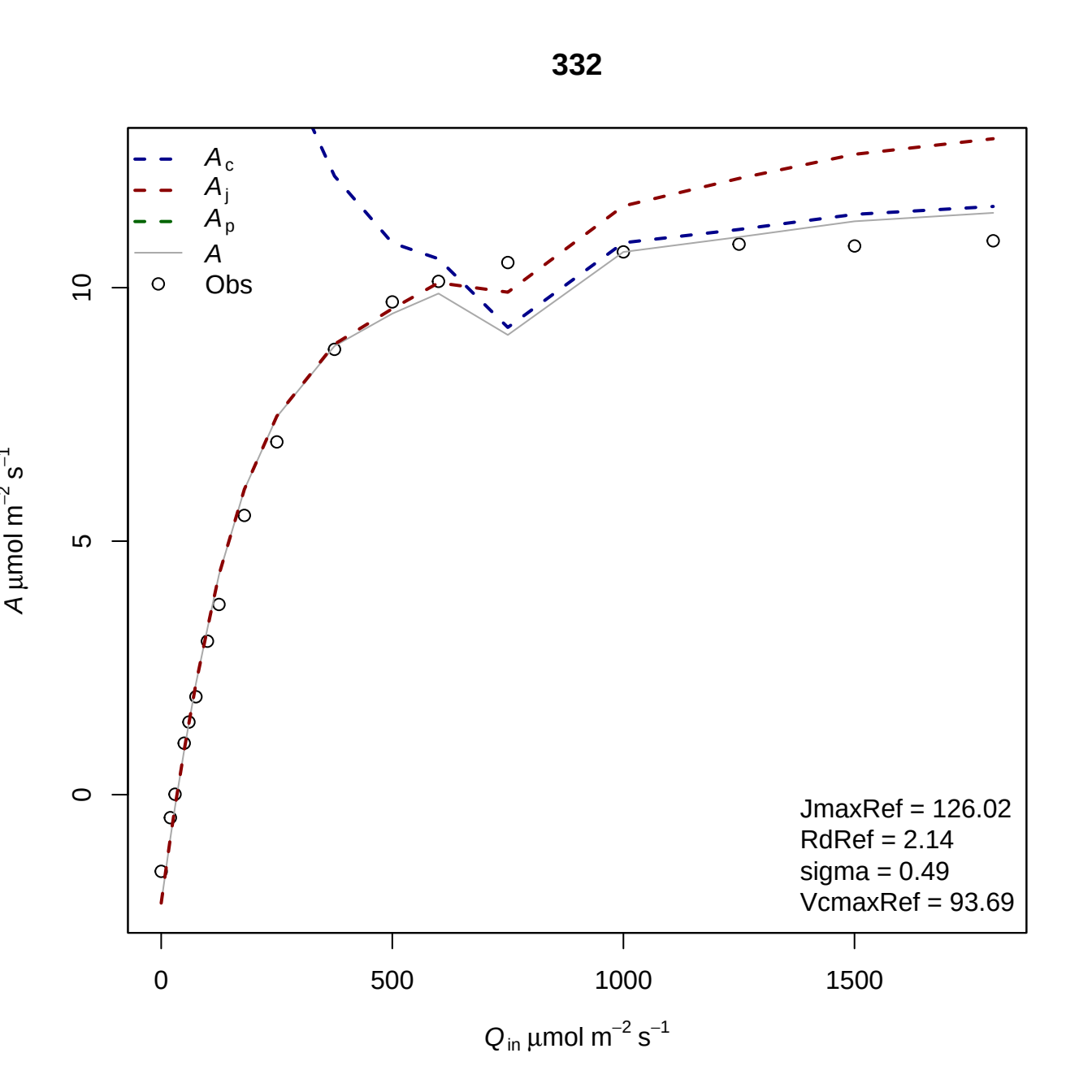

327
A_c
A_j
A_p
A
Obs
JmaxRef = 102.83
RdRef = 1.58
sigma = 0.37
VcmaxRef = 57.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

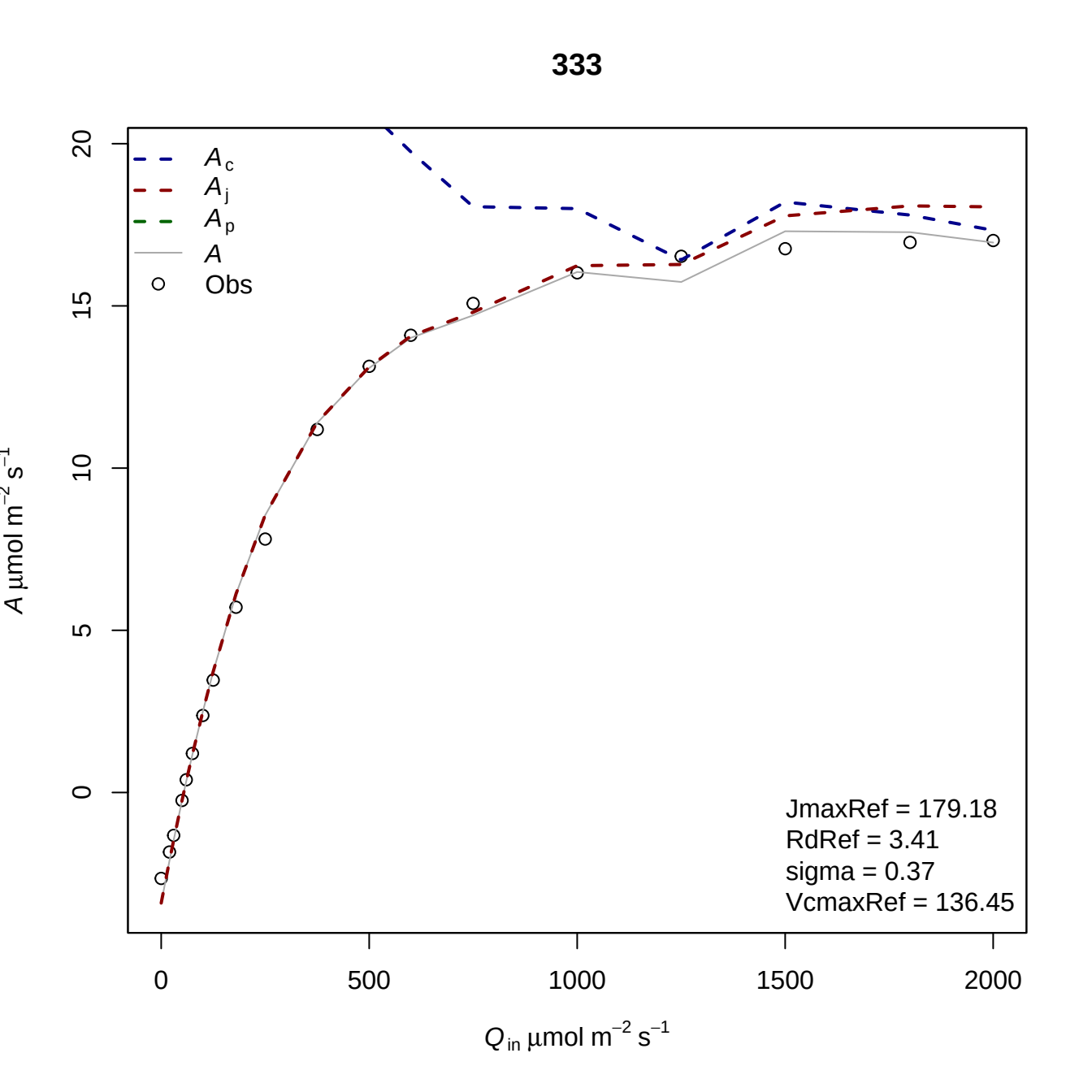

328
A_c
A_j
A_p
A
Obs
JmaxRef = 74.76
RdRef = 1.93
sigma = 0.2
VcmaxRef = 47.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

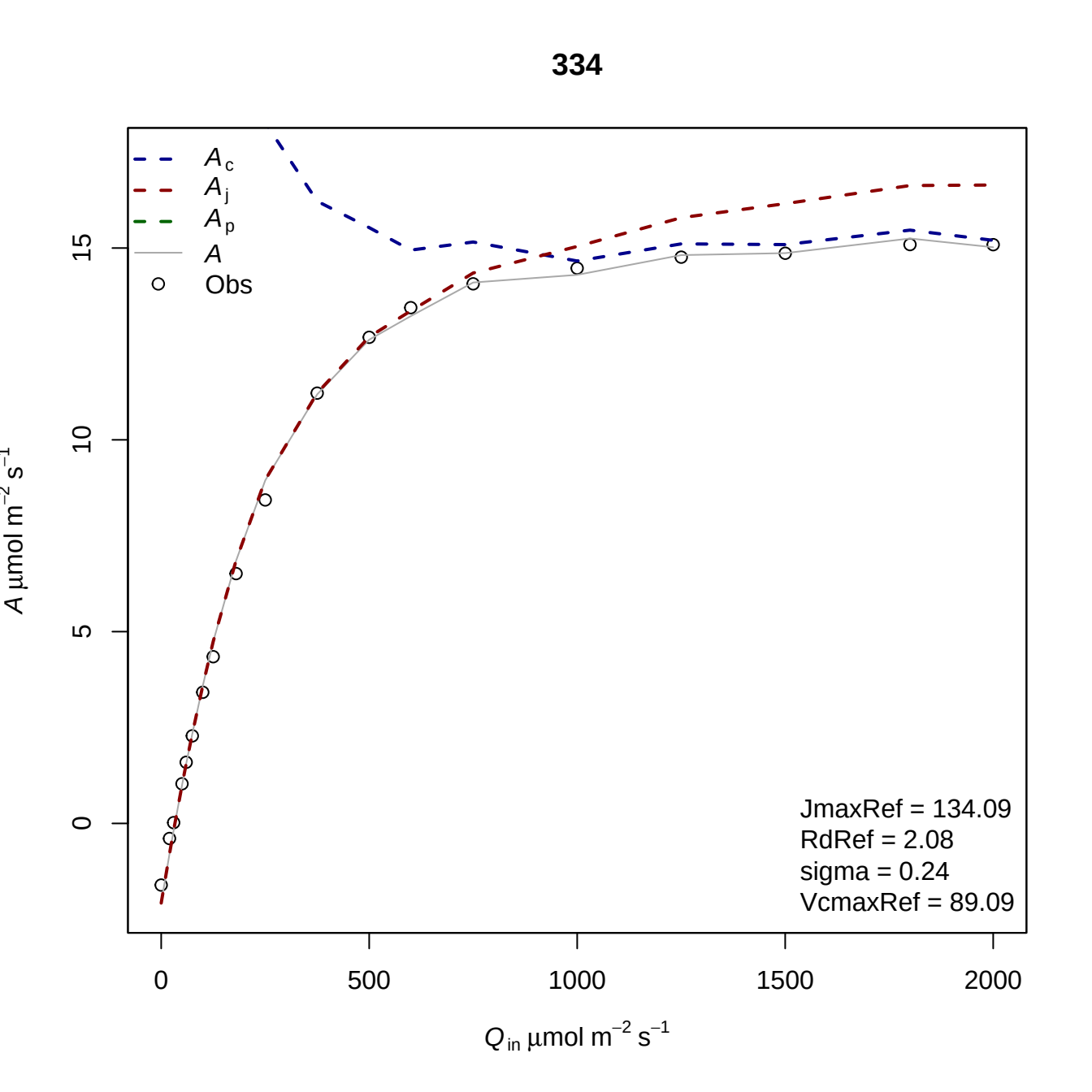

329
A_c
A_j
A_p
A
Obs
JmaxRef = 124.74
RdRef = 2.86
sigma = 0.19
VcmaxRef = 88.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

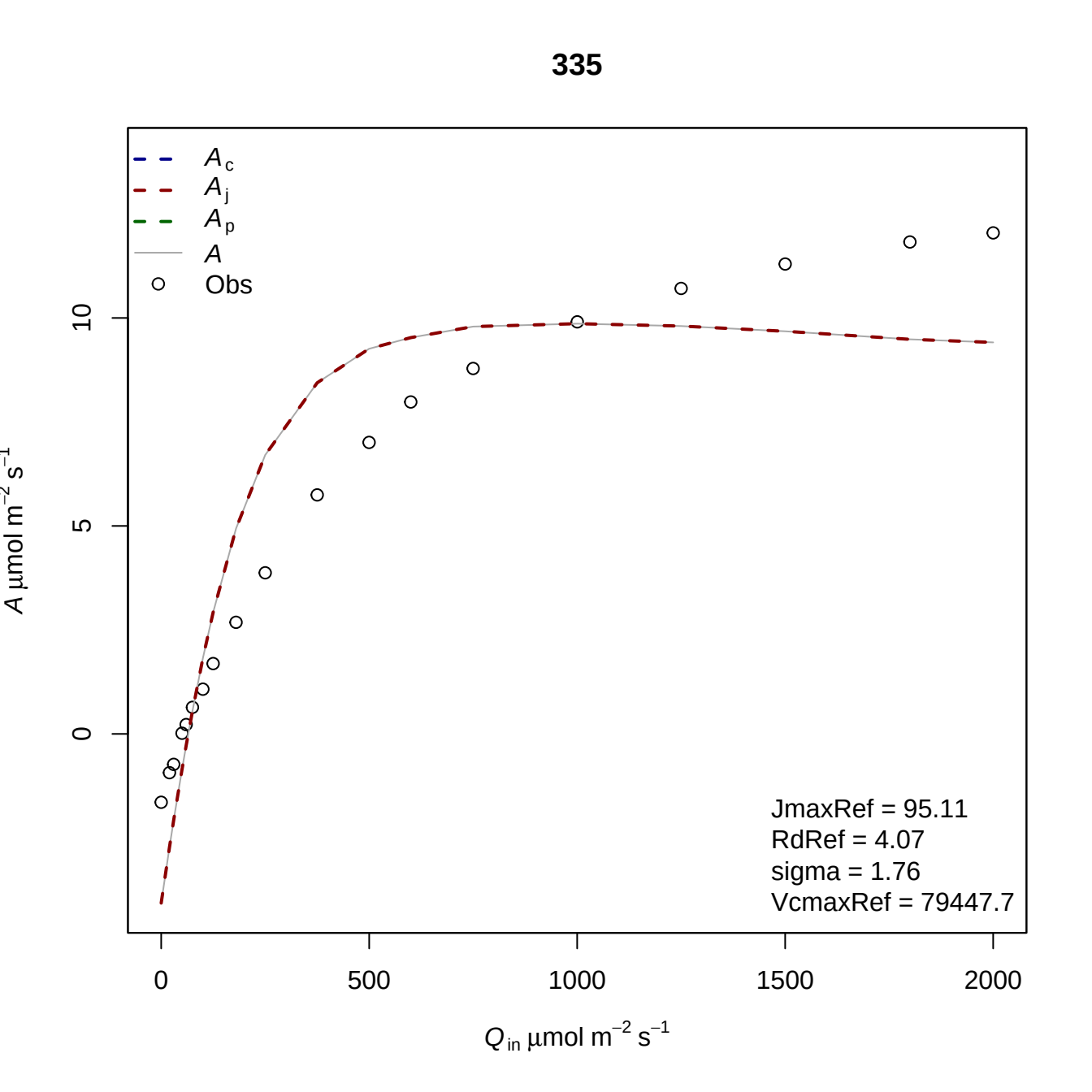

330
A_c
A_j
A_p
A
Obs
JmaxRef = 141.26
RdRef = 2.12
sigma = 0.33
VcmaxRef = 97.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

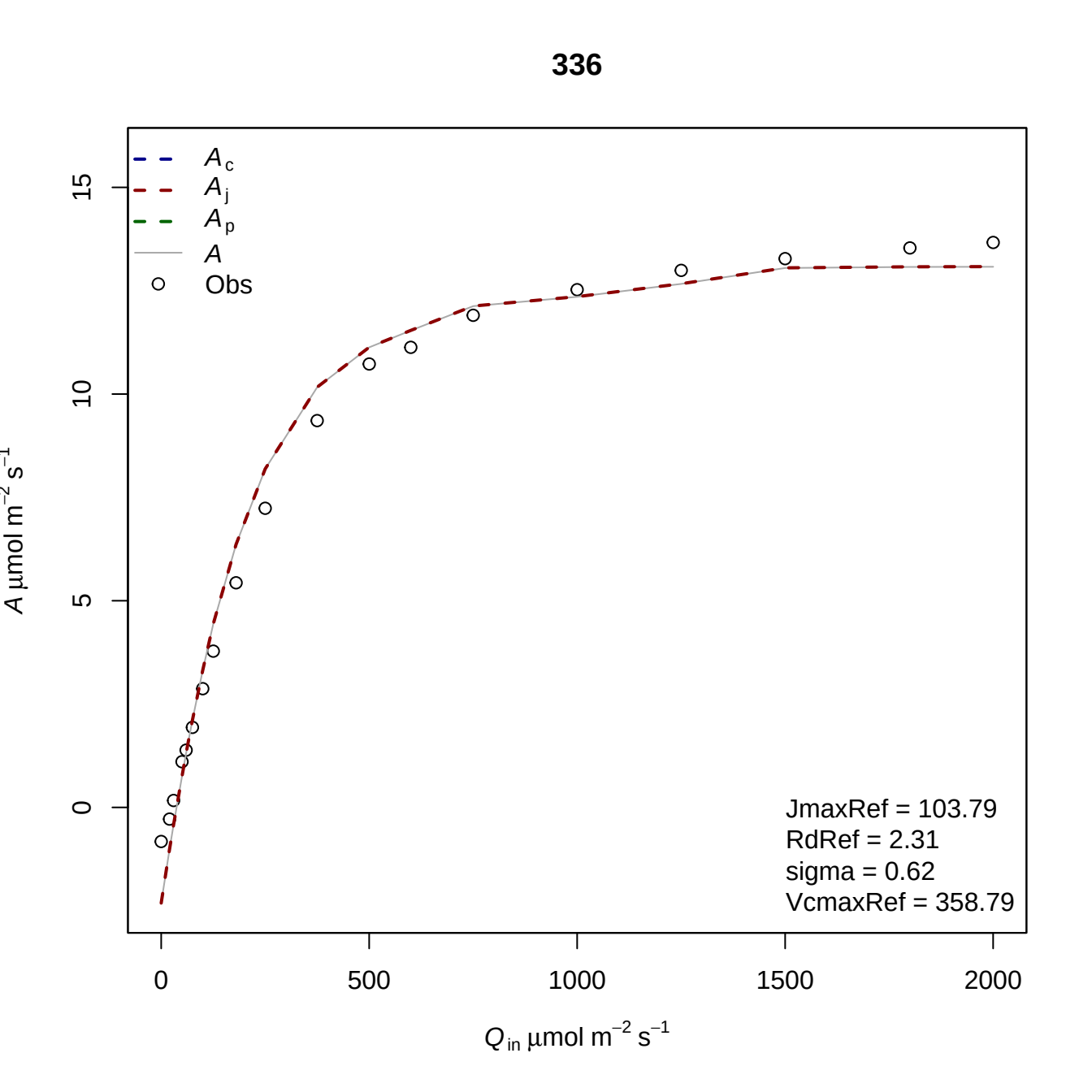

331
A_c
A_j
A_p
A
Obs
JmaxRef = 142.77
RdRef = 2.1
sigma = 0.26
VcmaxRef = 94.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

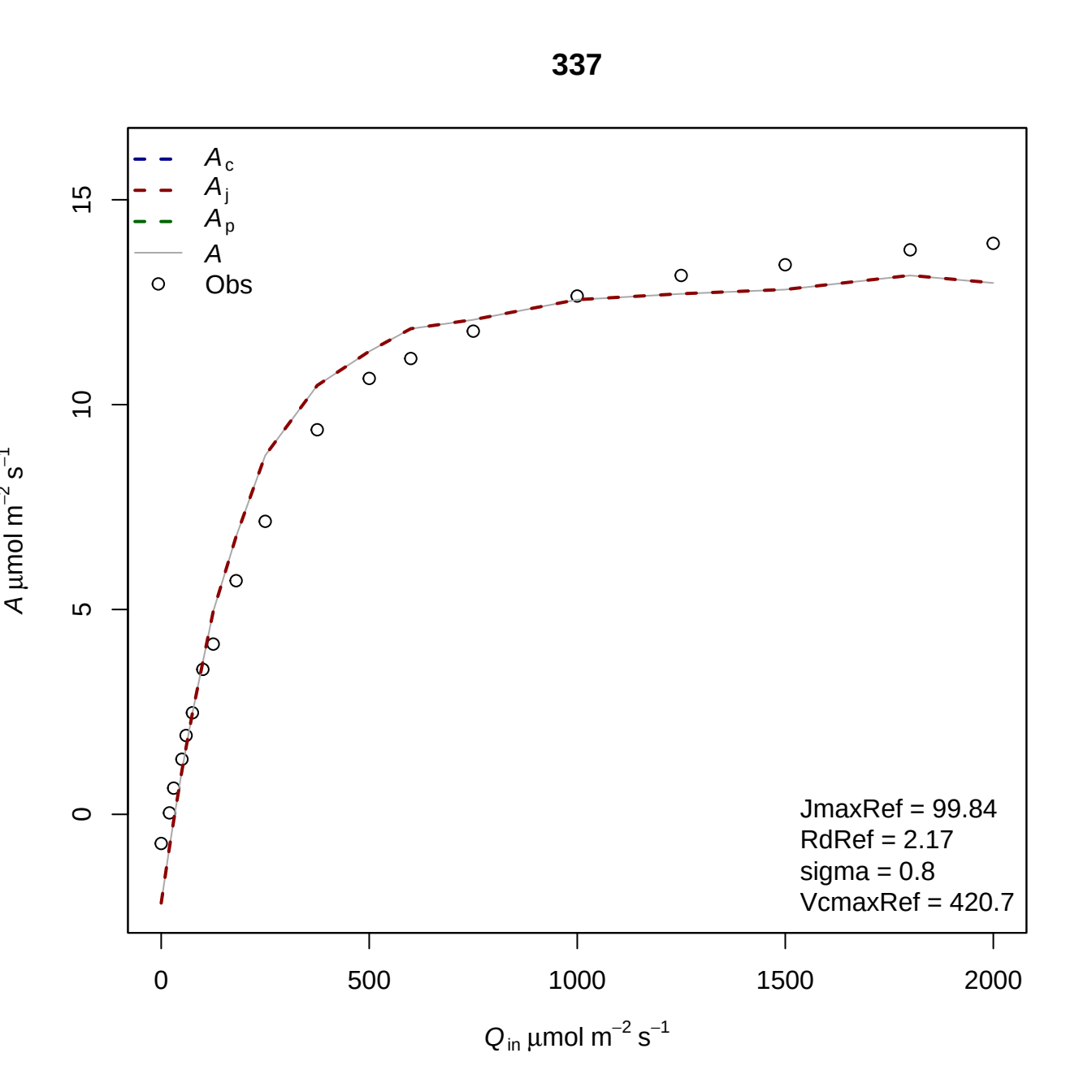

332
A_c
A_j
A_p
A
Obs
JmaxRef = 126.02
RdRef = 2.14
sigma = 0.49
VcmaxRef = 93.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

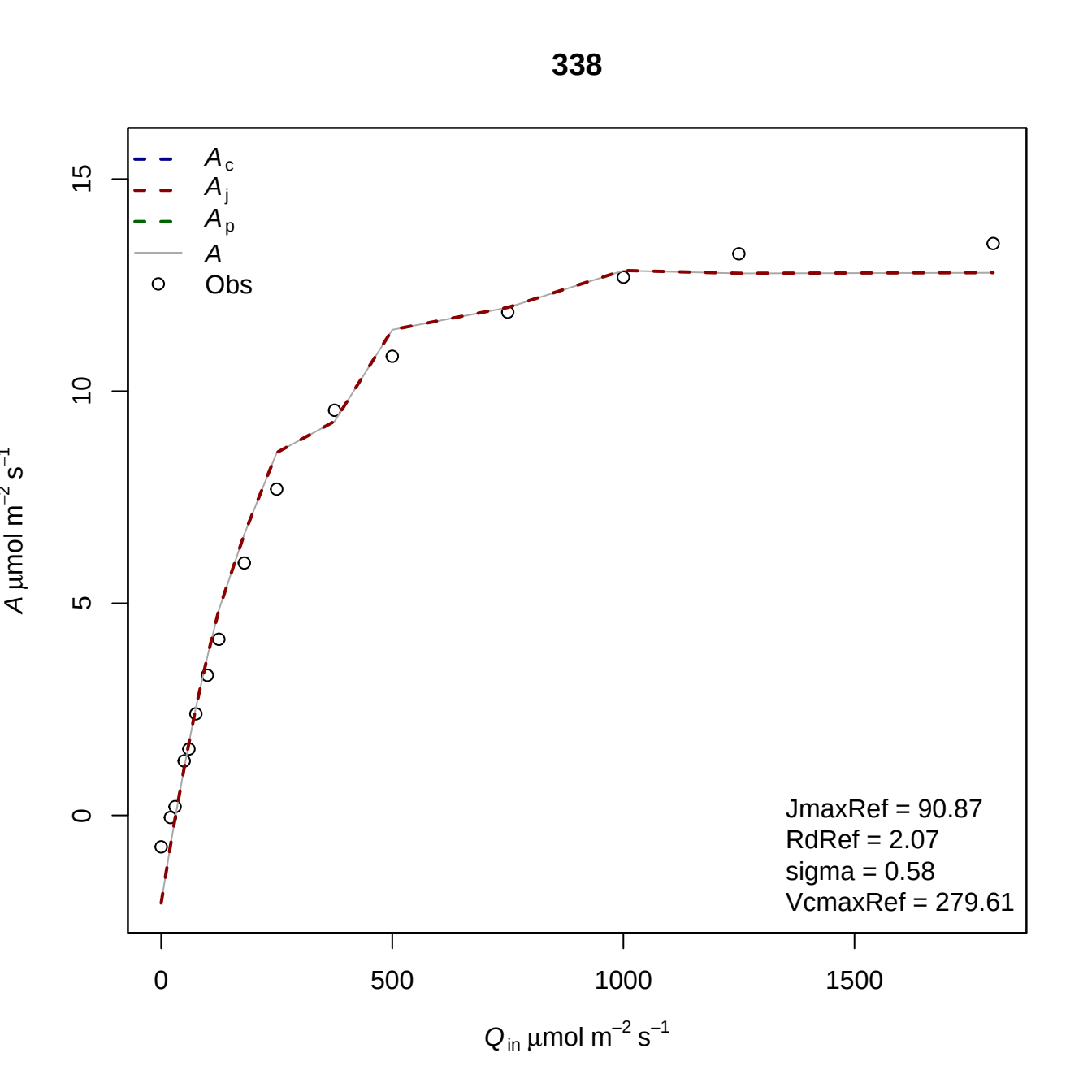

333
A_c
A_j
A_p
A
Obs
JmaxRef = 179.18
RdRef = 3.41
sigma = 0.37
VcmaxRef = 136.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

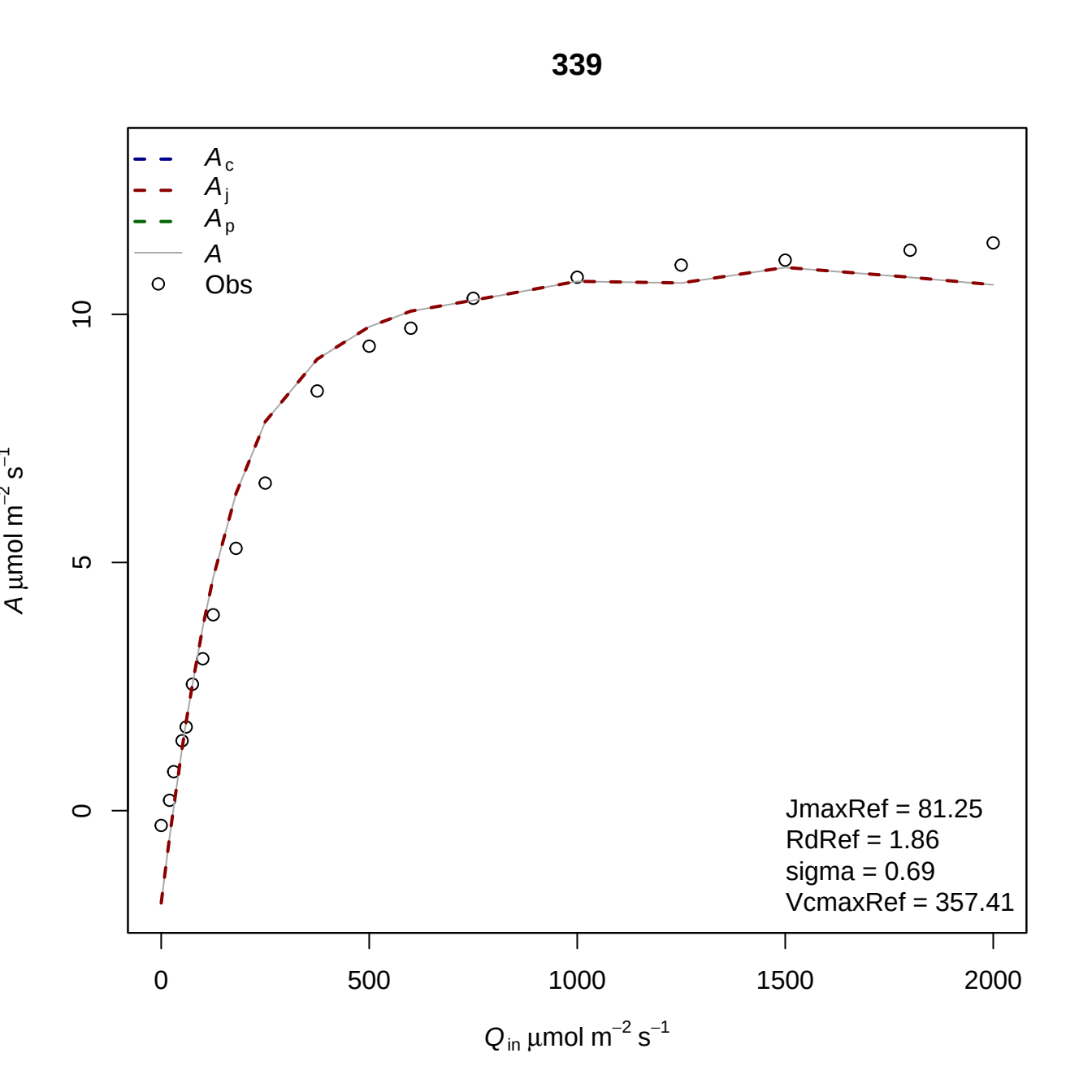

334
A_c
A_j
A_p
A
Obs
JmaxRef = 134.09
RdRef = 2.08
sigma = 0.24
VcmaxRef = 89.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

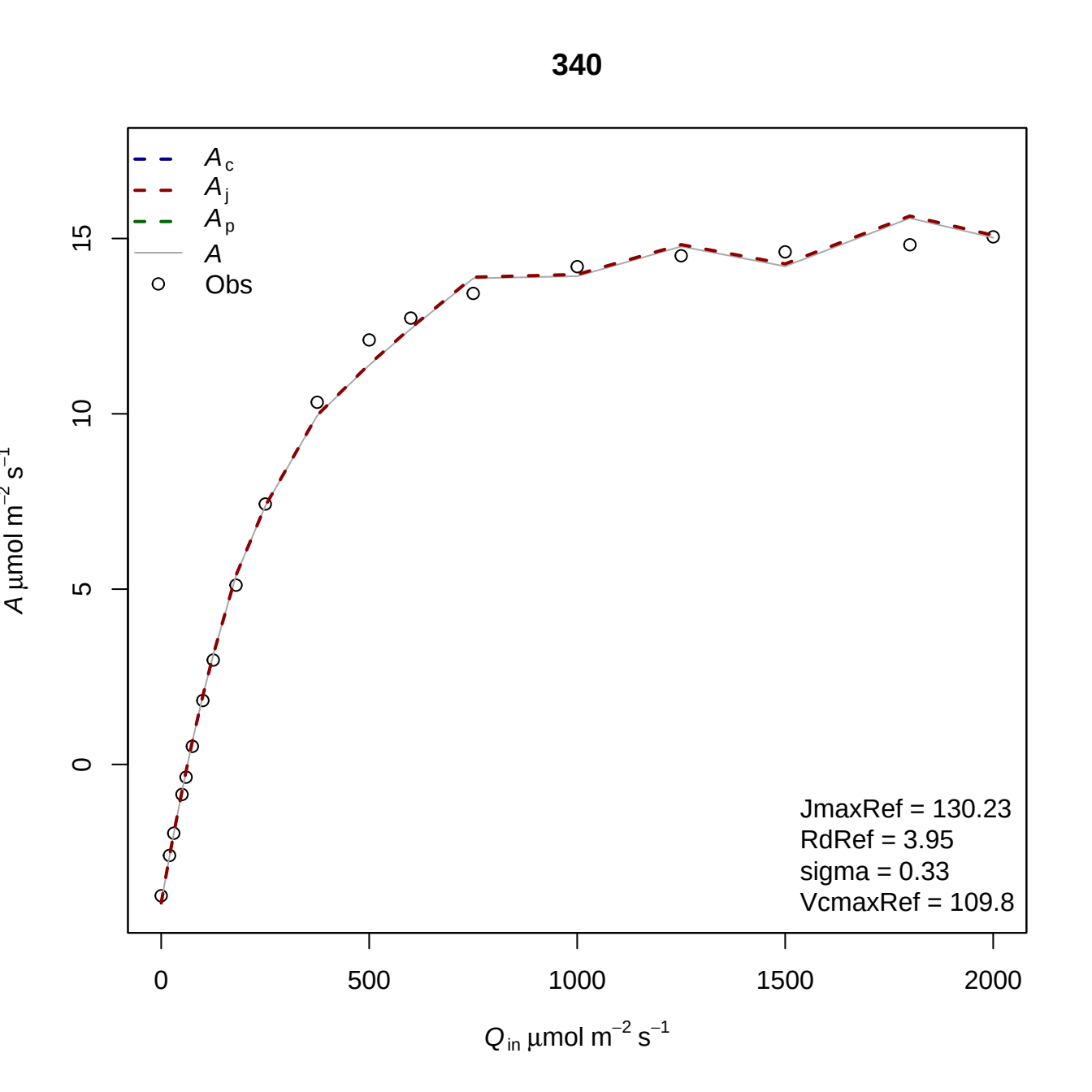

335
A_c
A_j
A_p
A
Obs
JmaxRef = 95.11
RdRef = 4.07
sigma = 1.76
VcmaxRef = 79447.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

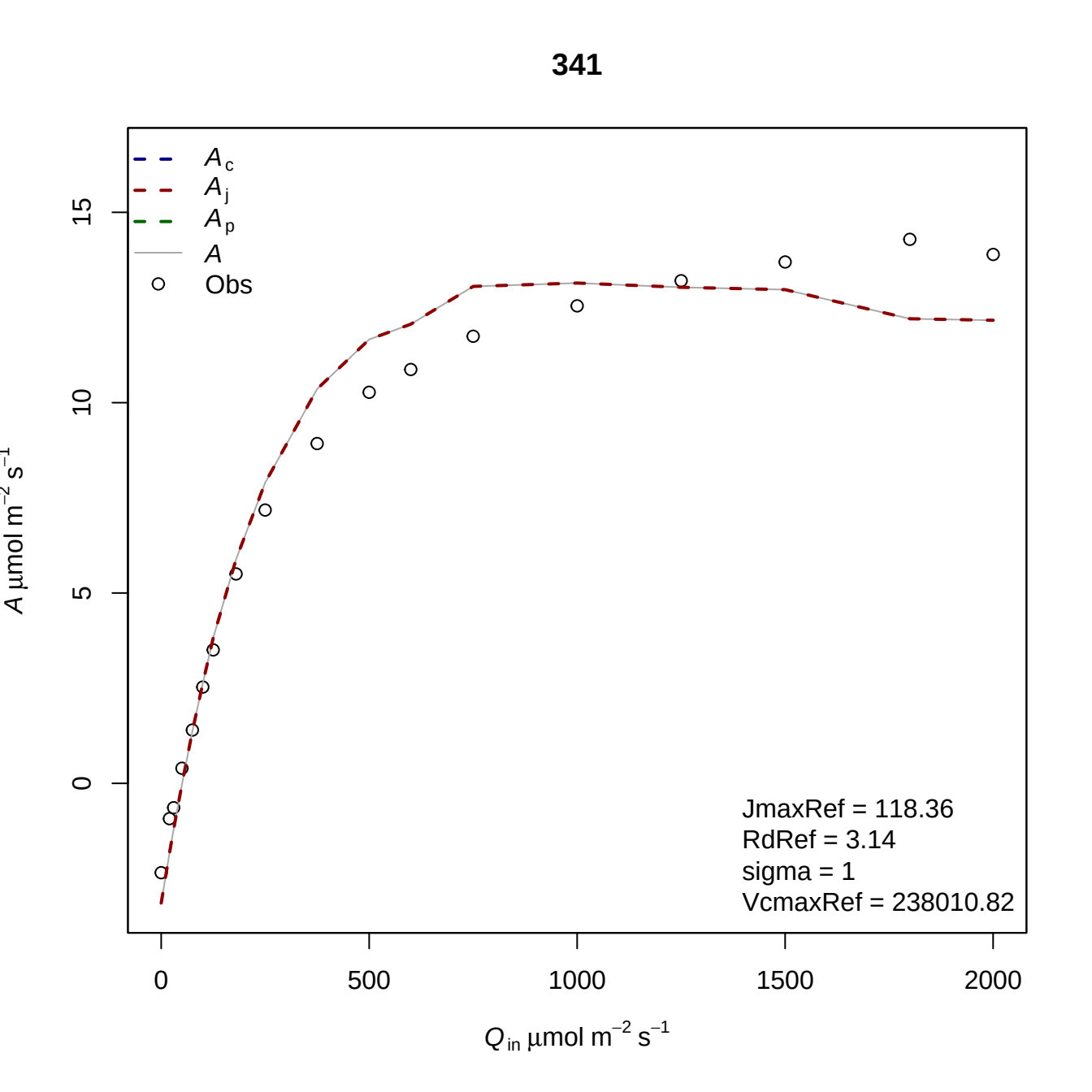

336
A_c
A_j
A_p
A
Obs
JmaxRef = 103.79
RdRef = 2.31
sigma = 0.62
VcmaxRef = 358.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

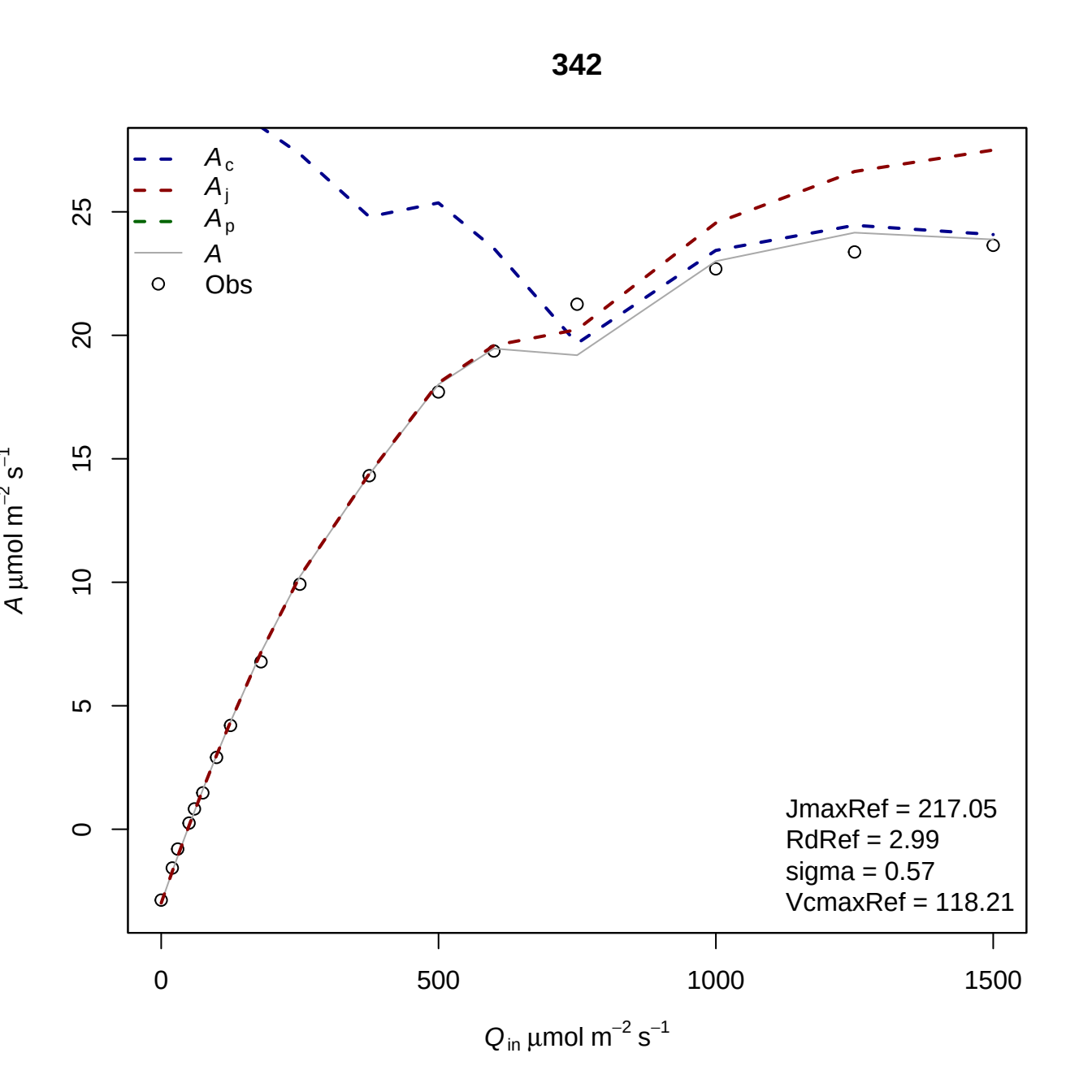

337
A_c
A_j
A_p
A
Obs
JmaxRef = 99.84
RdRef = 2.17
sigma = 0.8
VcmaxRef = 420.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

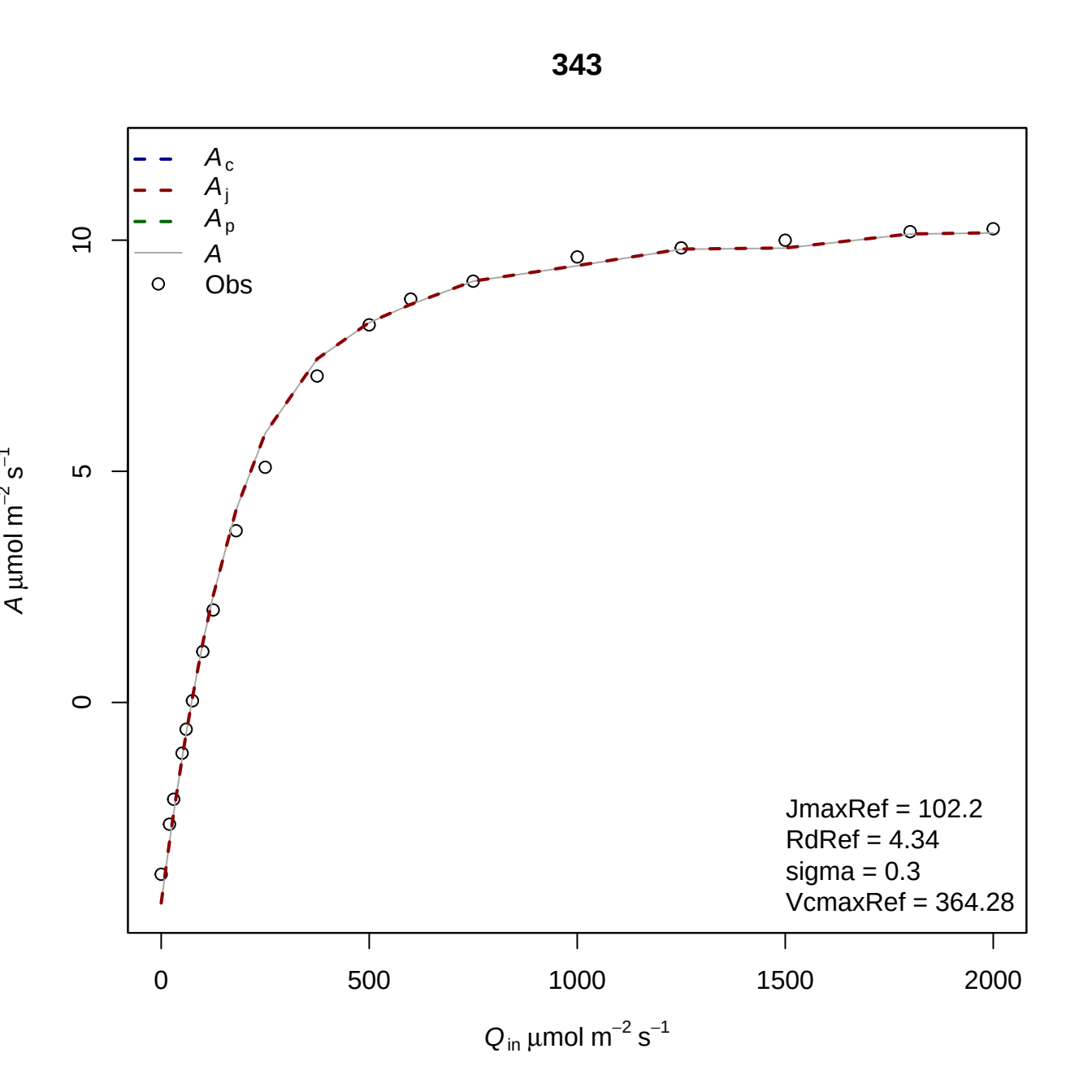

338
A_c
A_j
A_p
A
Obs
JmaxRef = 90.87
RdRef = 2.07
sigma = 0.58
VcmaxRef = 279.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

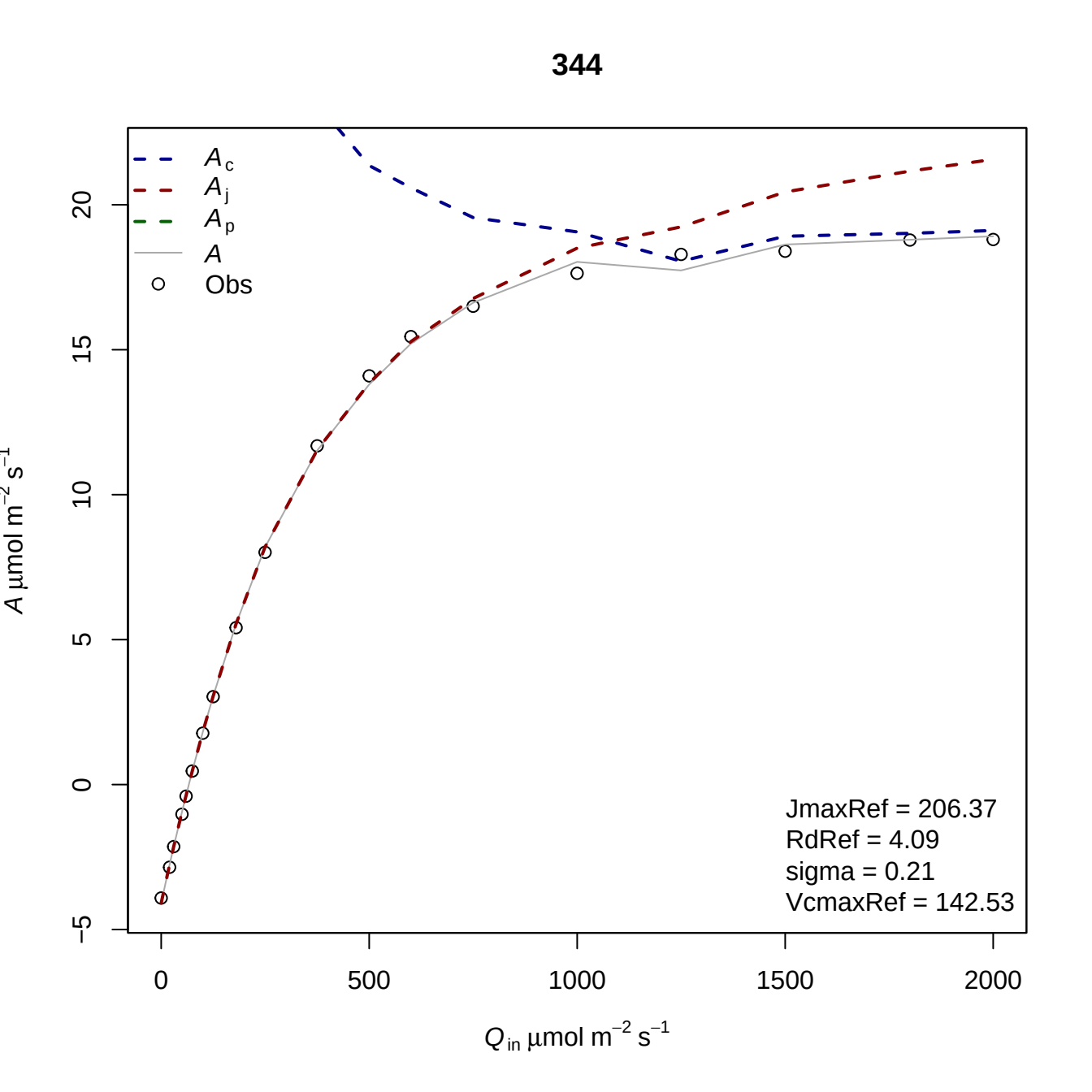

339
A_c
A_j
A_p
A
Obs
JmaxRef = 81.25
RdRef = 1.86
sigma = 0.69
VcmaxRef = 357.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

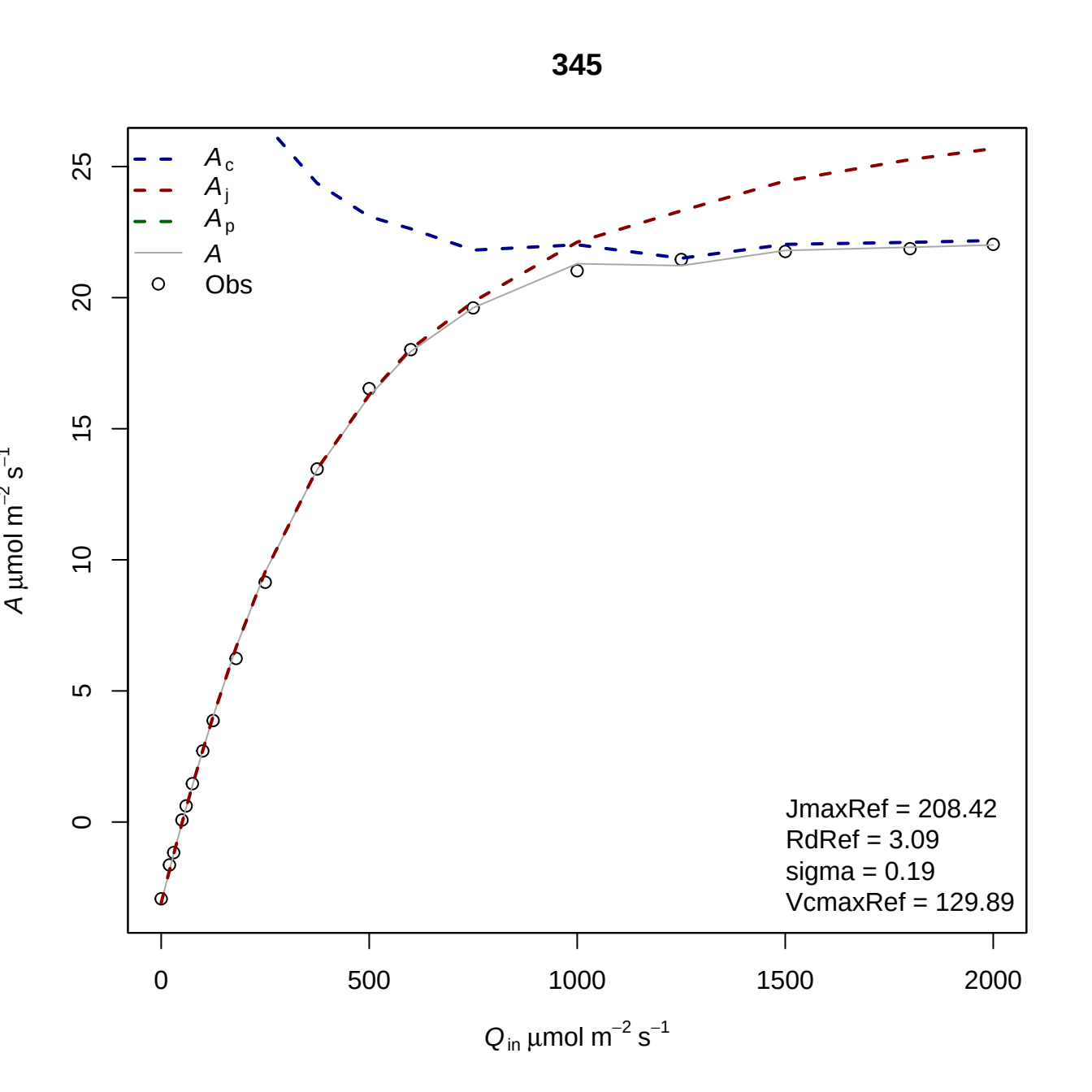

340
A_c
A_j
A_p
A
Obs
JmaxRef = 130.23
RdRef = 3.95
sigma = 0.33
VcmaxRef = 109.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

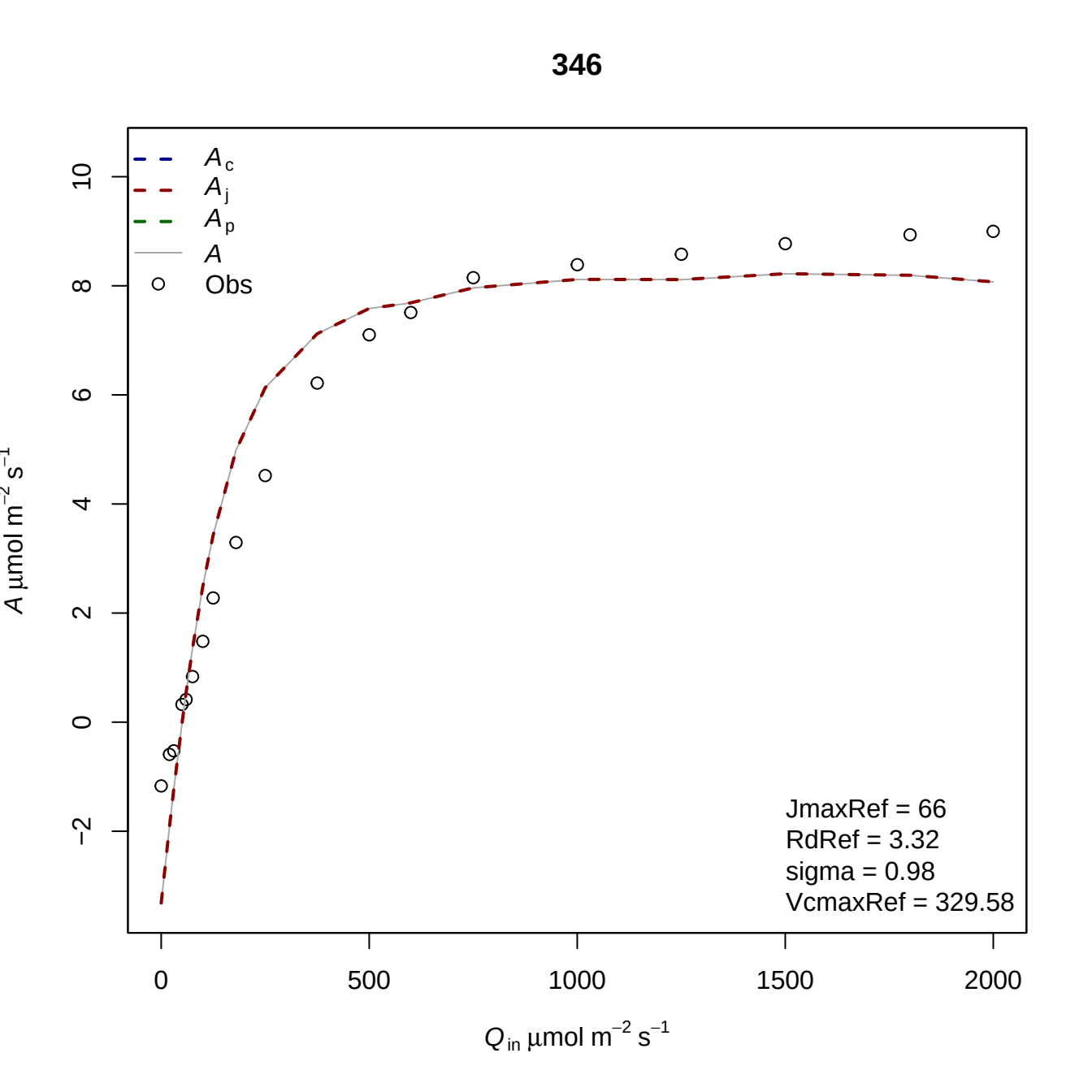

341
A_c
A_j
A_p
A
Obs
JmaxRef = 118.36
RdRef = 3.14
sigma = 1
VcmaxRef = 238010.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

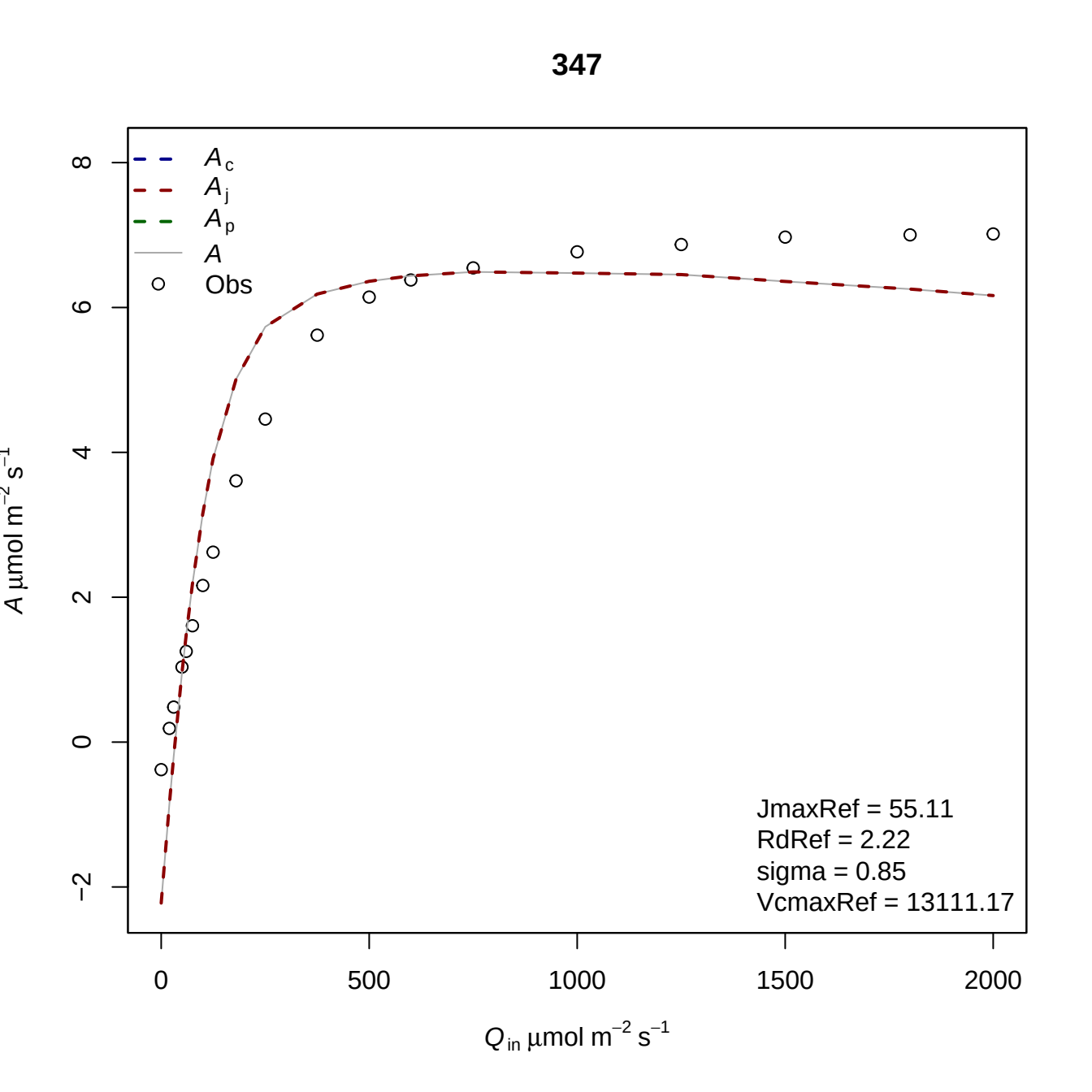

342
A_c
A_j
A_p
A
Obs
JmaxRef = 217.05
RdRef = 2.99
sigma = 0.57
VcmaxRef = 118.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

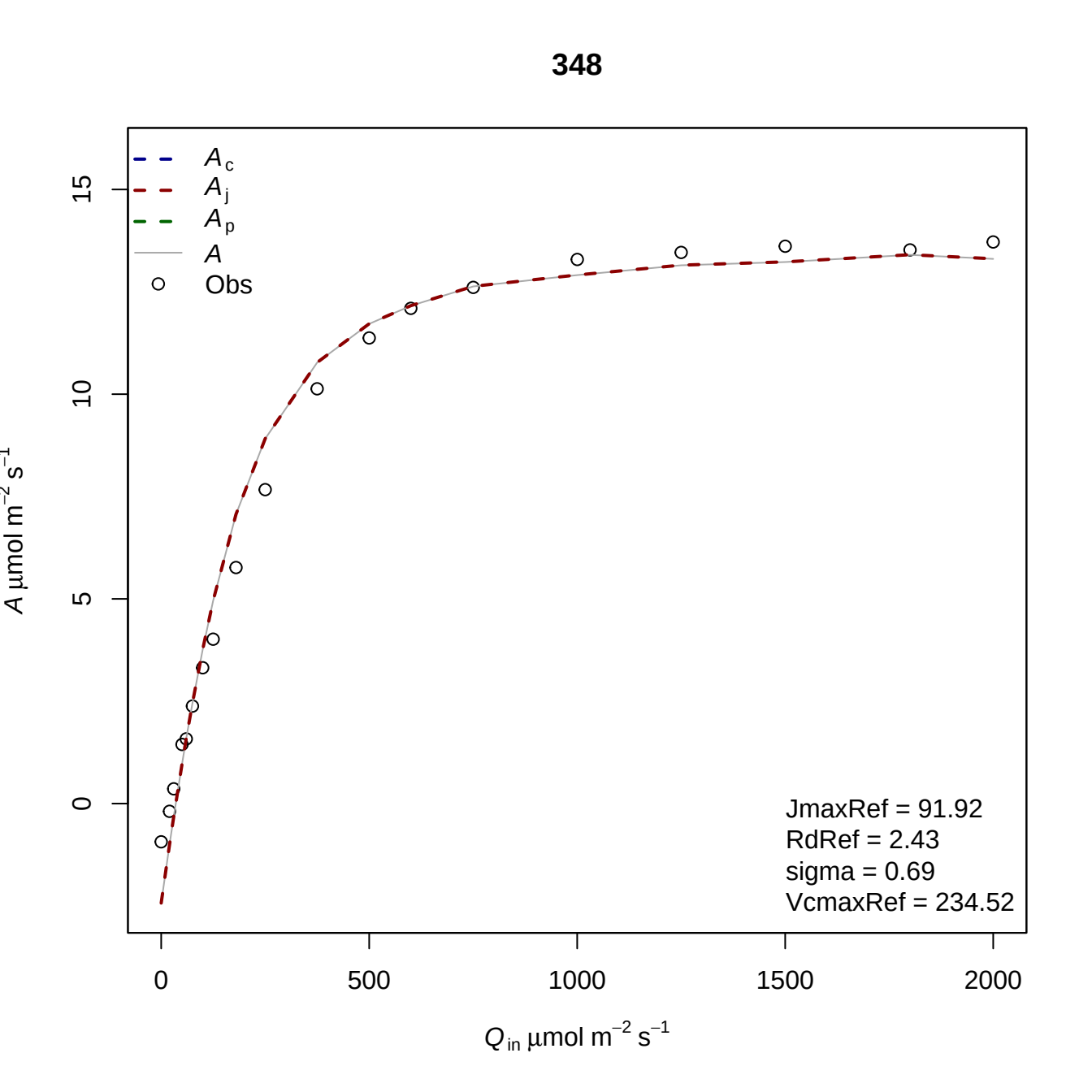

343
A_c
A_j
A_p
A
Obs
JmaxRef = 102.2
RdRef = 4.34
sigma = 0.3
VcmaxRef = 364.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

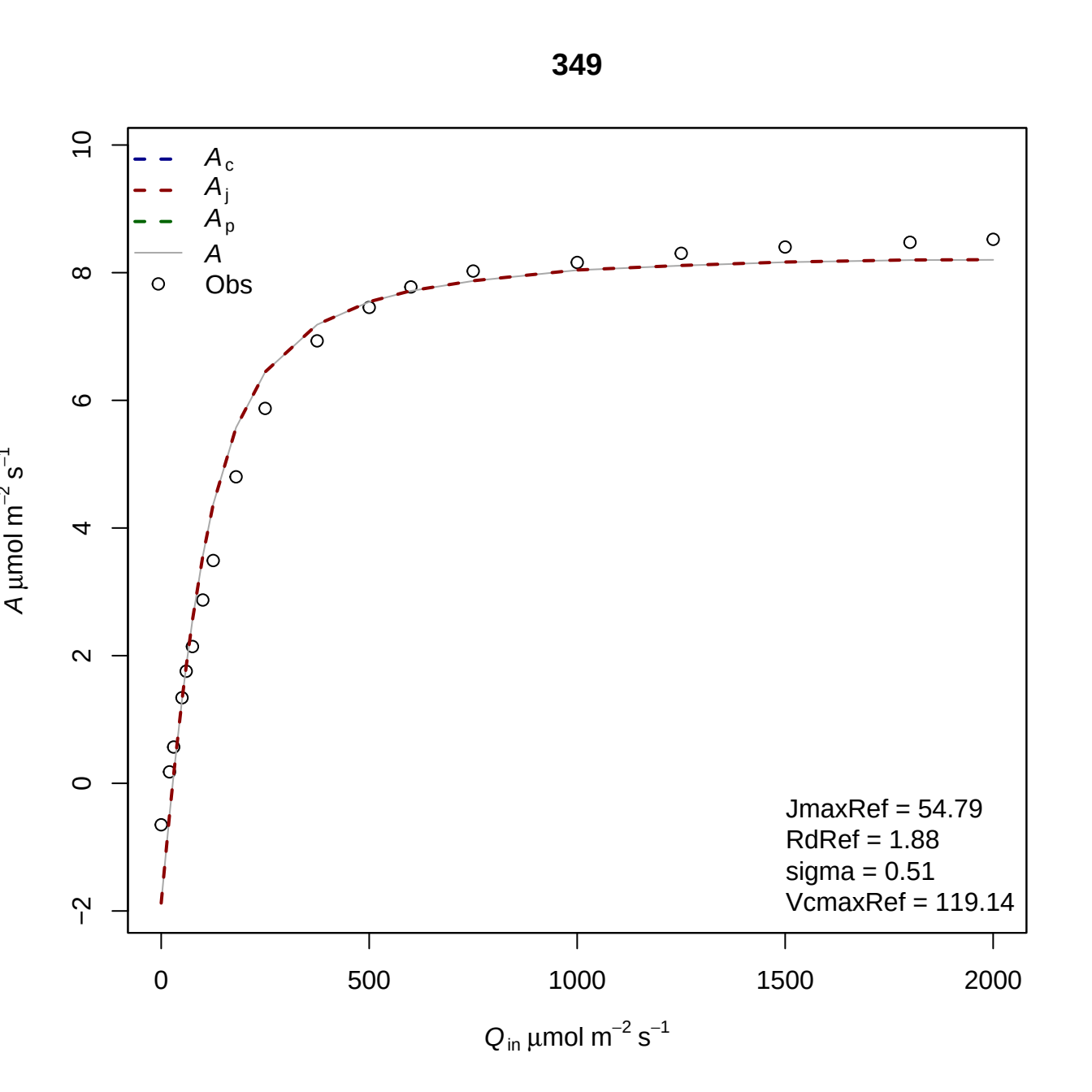

344
A_c
A_j
A_p
A
Obs
JmaxRef = 206.37
RdRef = 4.09
sigma = 0.21
VcmaxRef = 142.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

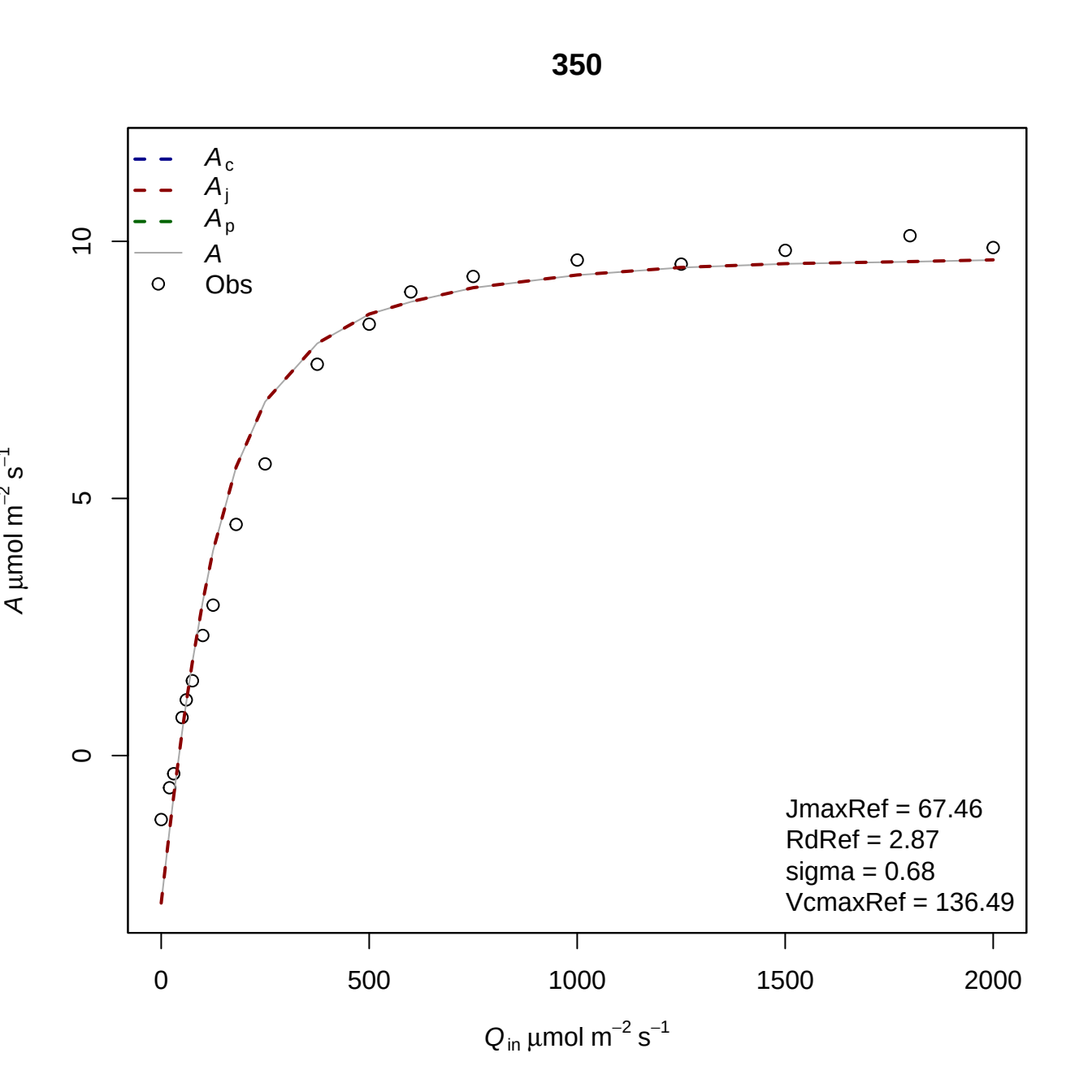

345
A_c
A_j
A_p
A
Obs
JmaxRef = 208.42
RdRef = 3.09
sigma = 0.19
VcmaxRef = 129.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

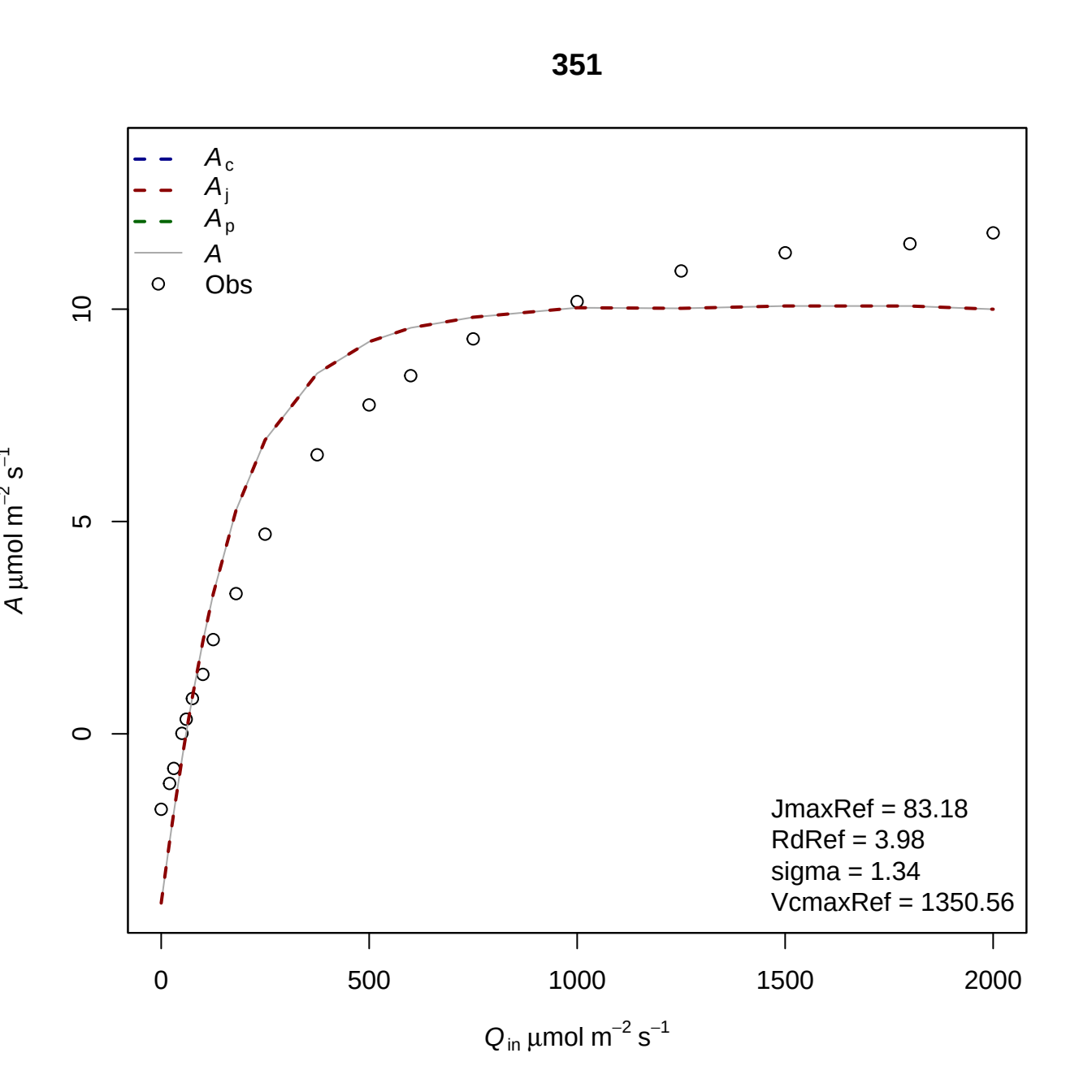

346
A_c
A_j
A_p
A
Obs
JmaxRef = 66
RdRef = 3.32
sigma = 0.98
VcmaxRef = 329.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

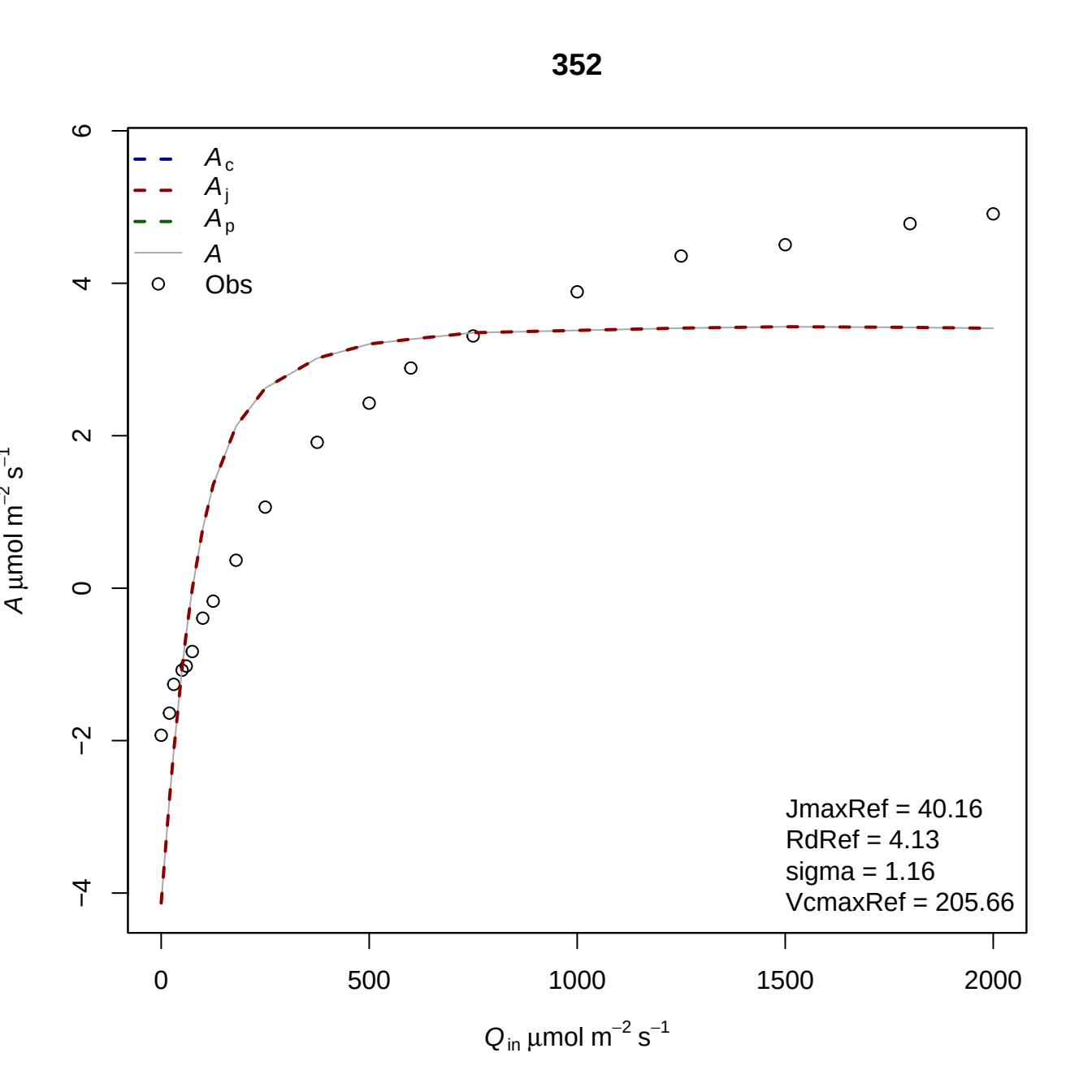

347
A_c
A_j
A_p
A
Obs
JmaxRef = 55.11
RdRef = 2.22
sigma = 0.85
VcmaxRef = 13111.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

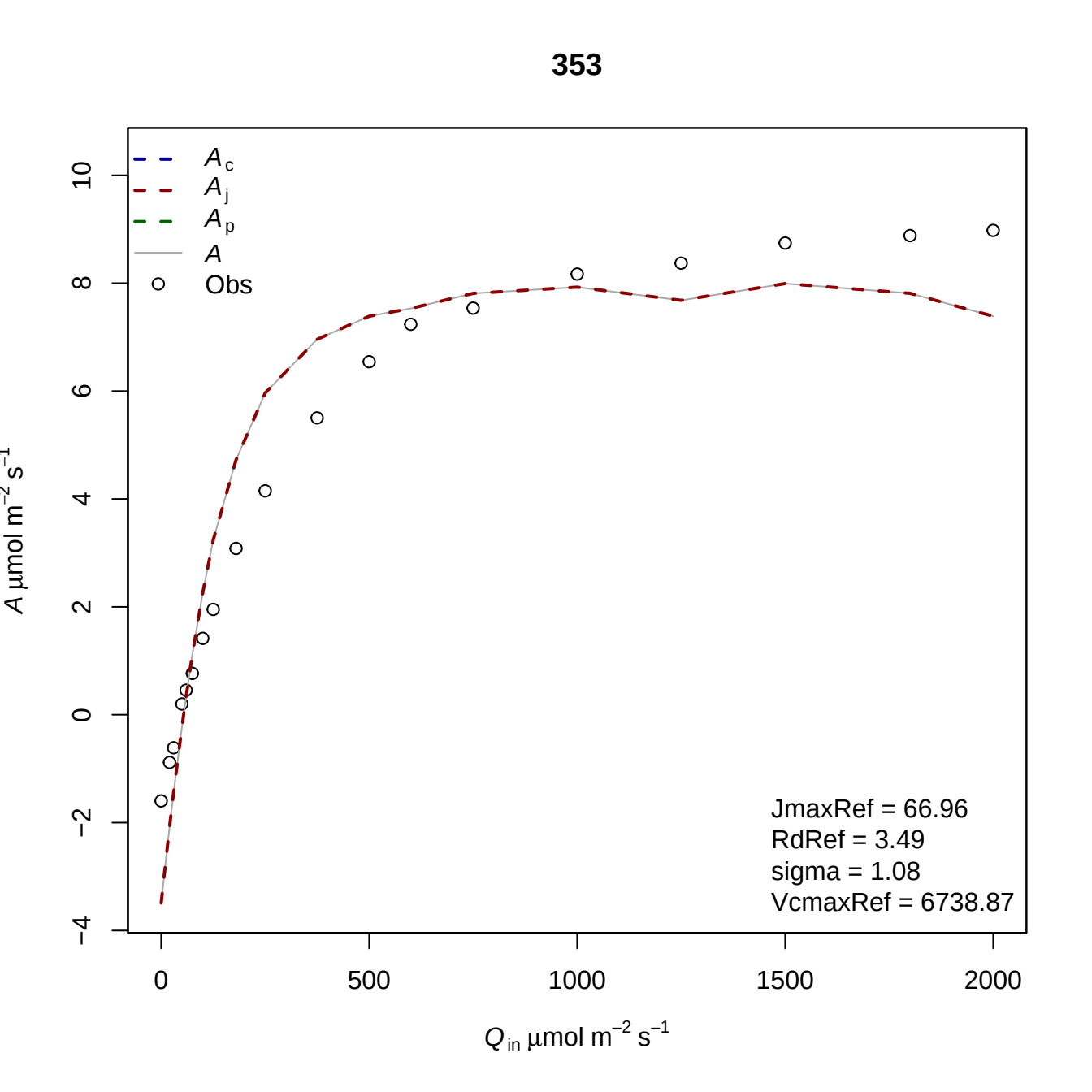

348
A_c
A_j
A_p
A
Obs
JmaxRef = 91.92
RdRef = 2.43
sigma = 0.69
VcmaxRef = 234.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

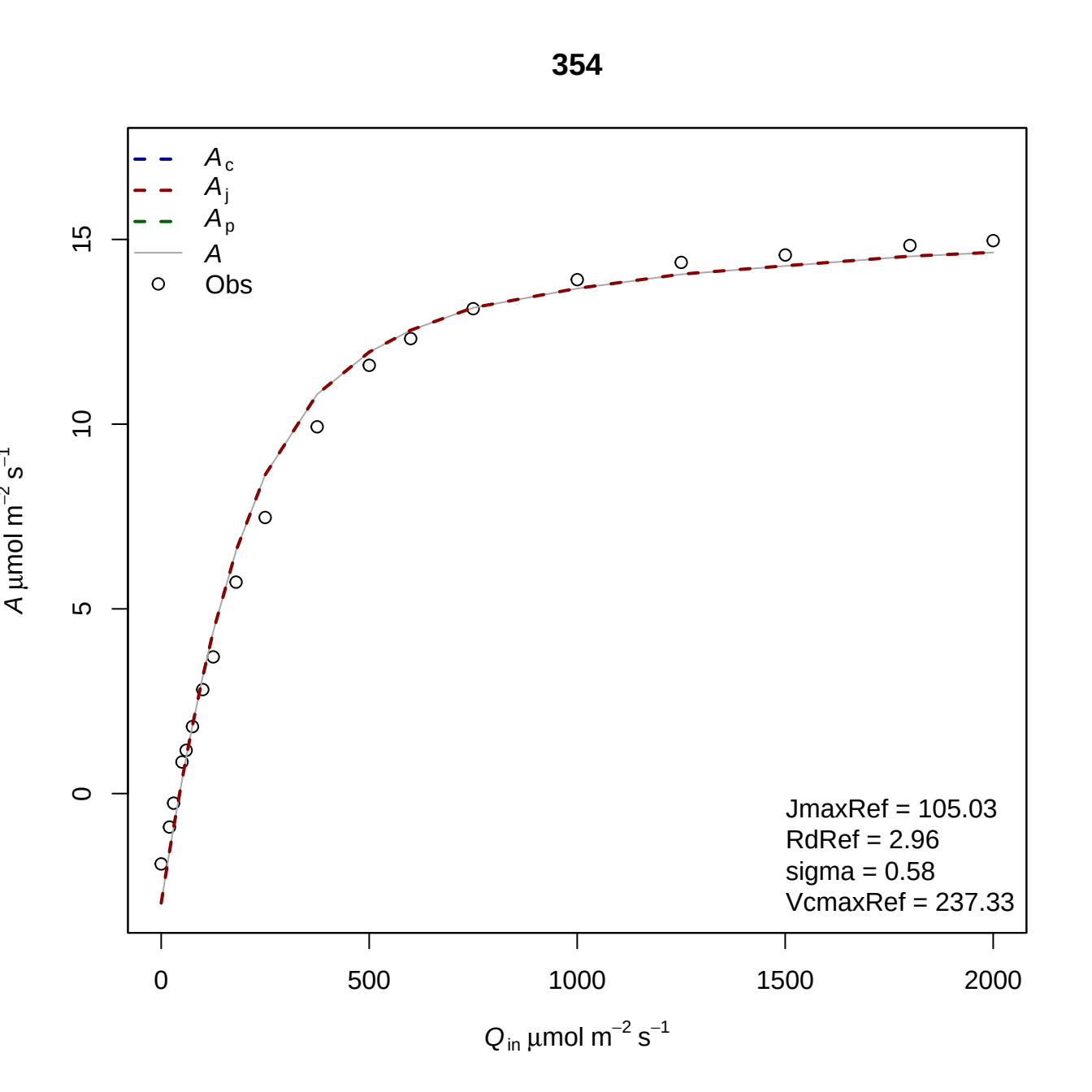

349
A_c
A_j
A_p
A
Obs
JmaxRef = 54.79
RdRef = 1.88
sigma = 0.51
VcmaxRef = 119.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

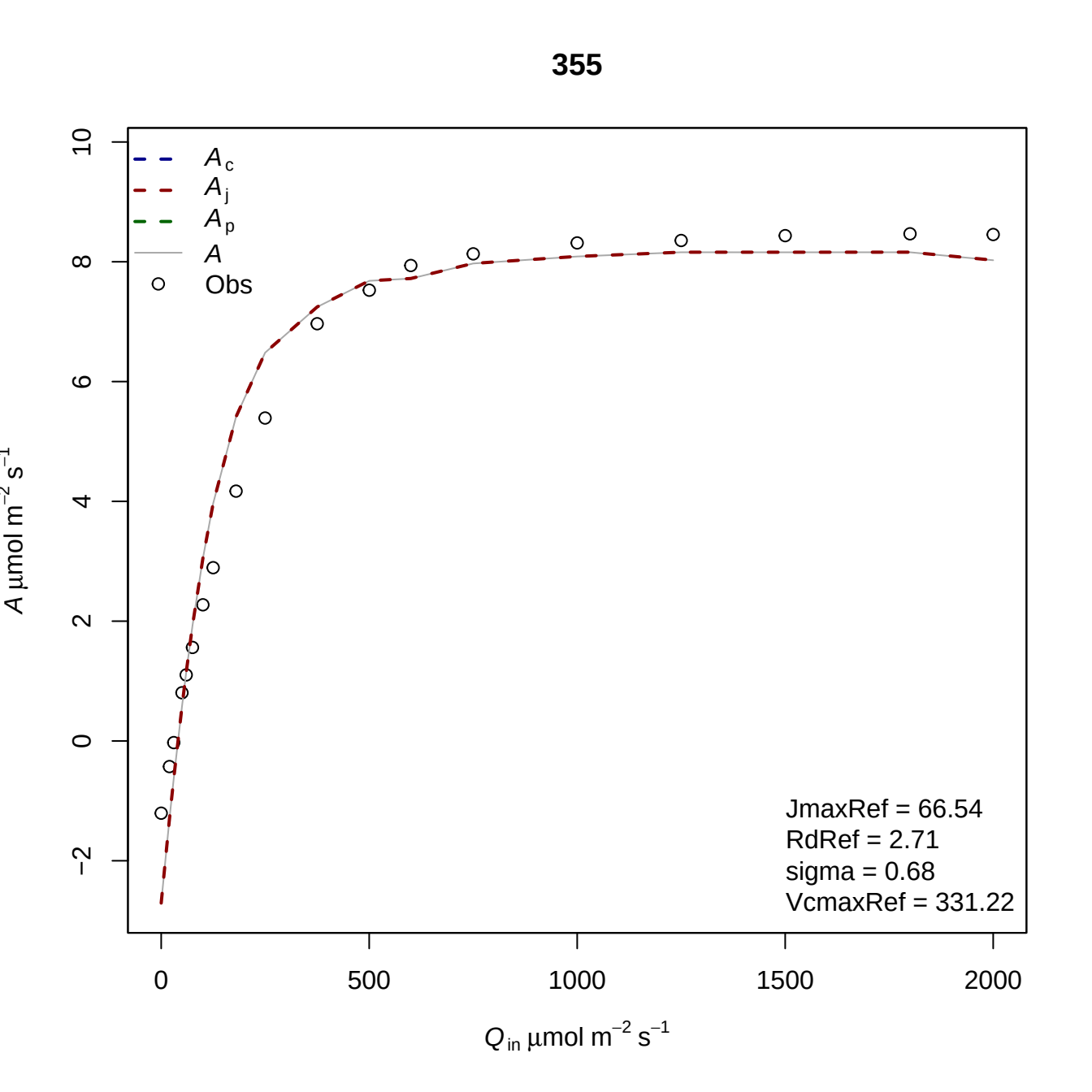

350
A_c
A_j
A_p
A
Obs
JmaxRef = 67.46
RdRef = 2.87
sigma = 0.68
VcmaxRef = 136.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

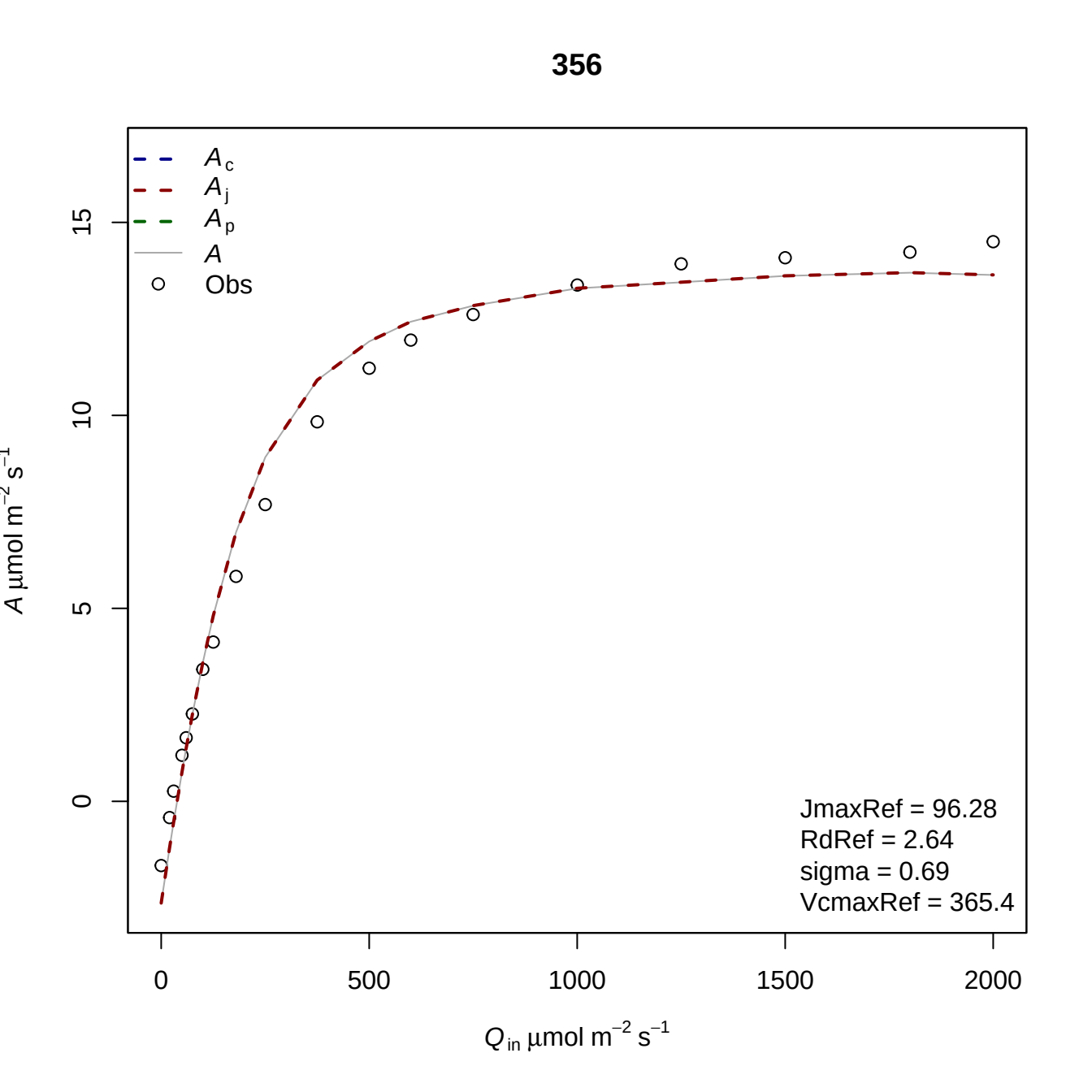

351
A_c
A_j
A_p
A
Obs
JmaxRef = 83.18
RdRef = 3.98
sigma = 1.34
VcmaxRef = 1350.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

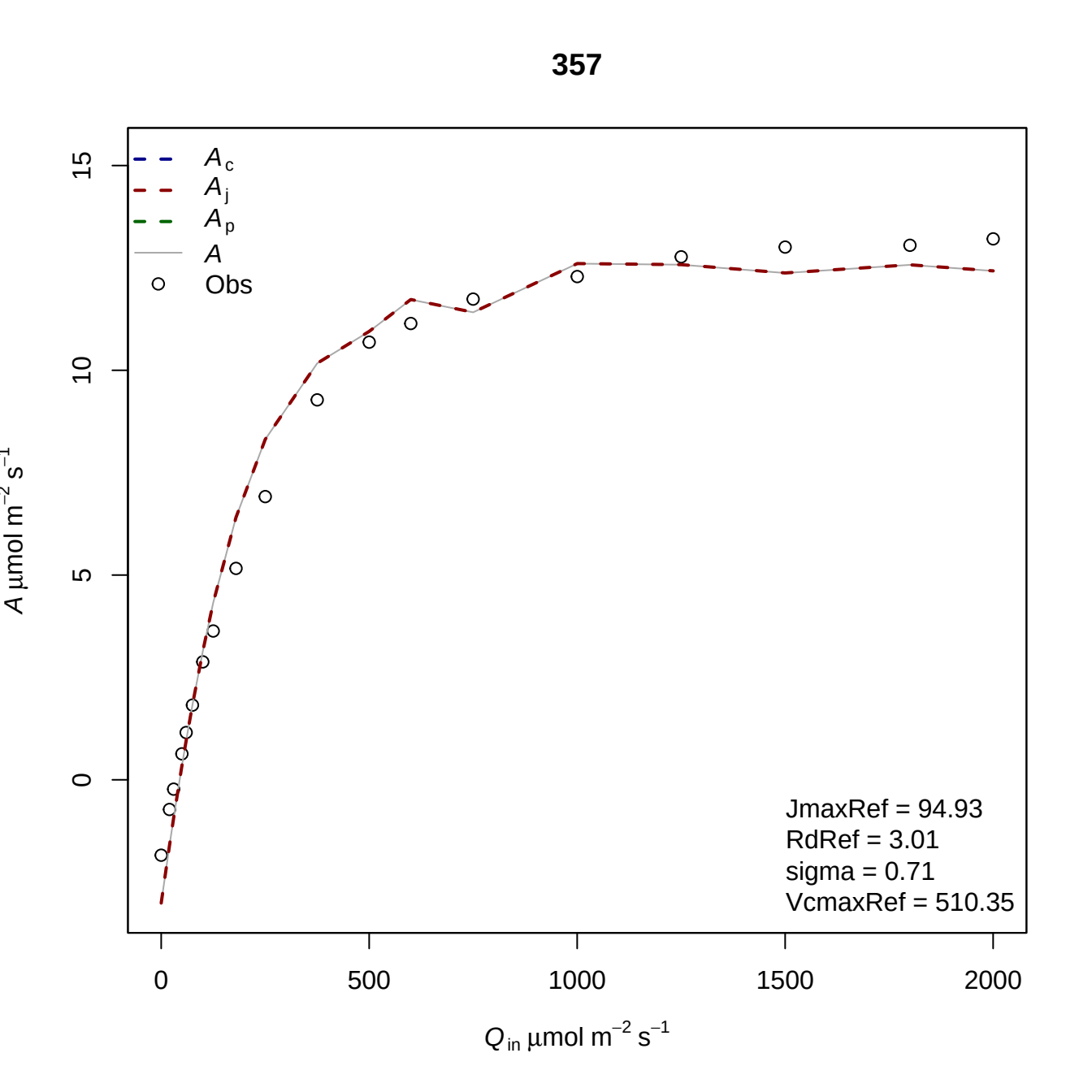

352
A_c
A_j
A_p
A
Obs
JmaxRef = 40.16
RdRef = 4.13
sigma = 1.16
VcmaxRef = 205.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

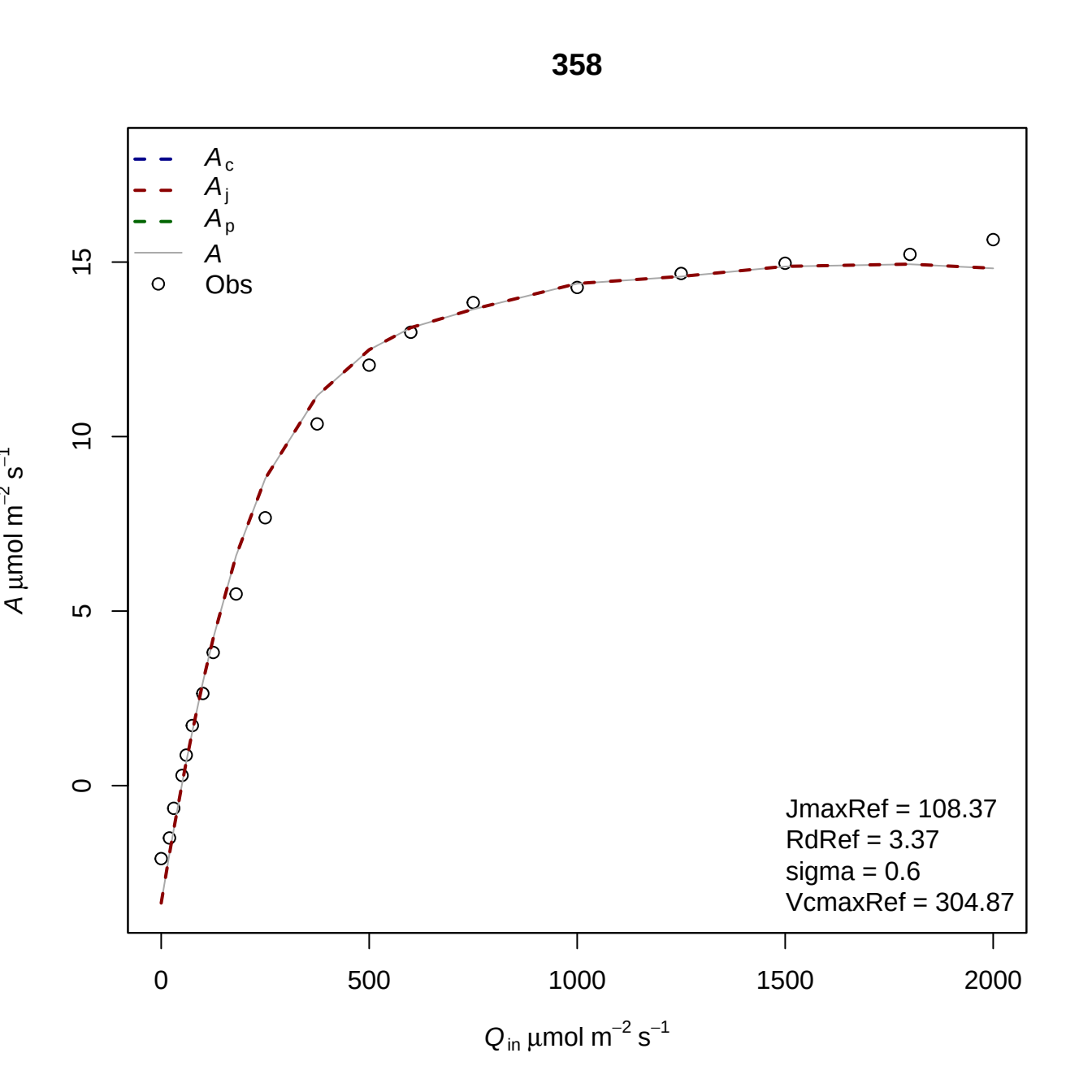

353
A_c
A_j
A_p
A
Obs
JmaxRef = 66.96
RdRef = 3.49
sigma = 1.08
VcmaxRef = 6738.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

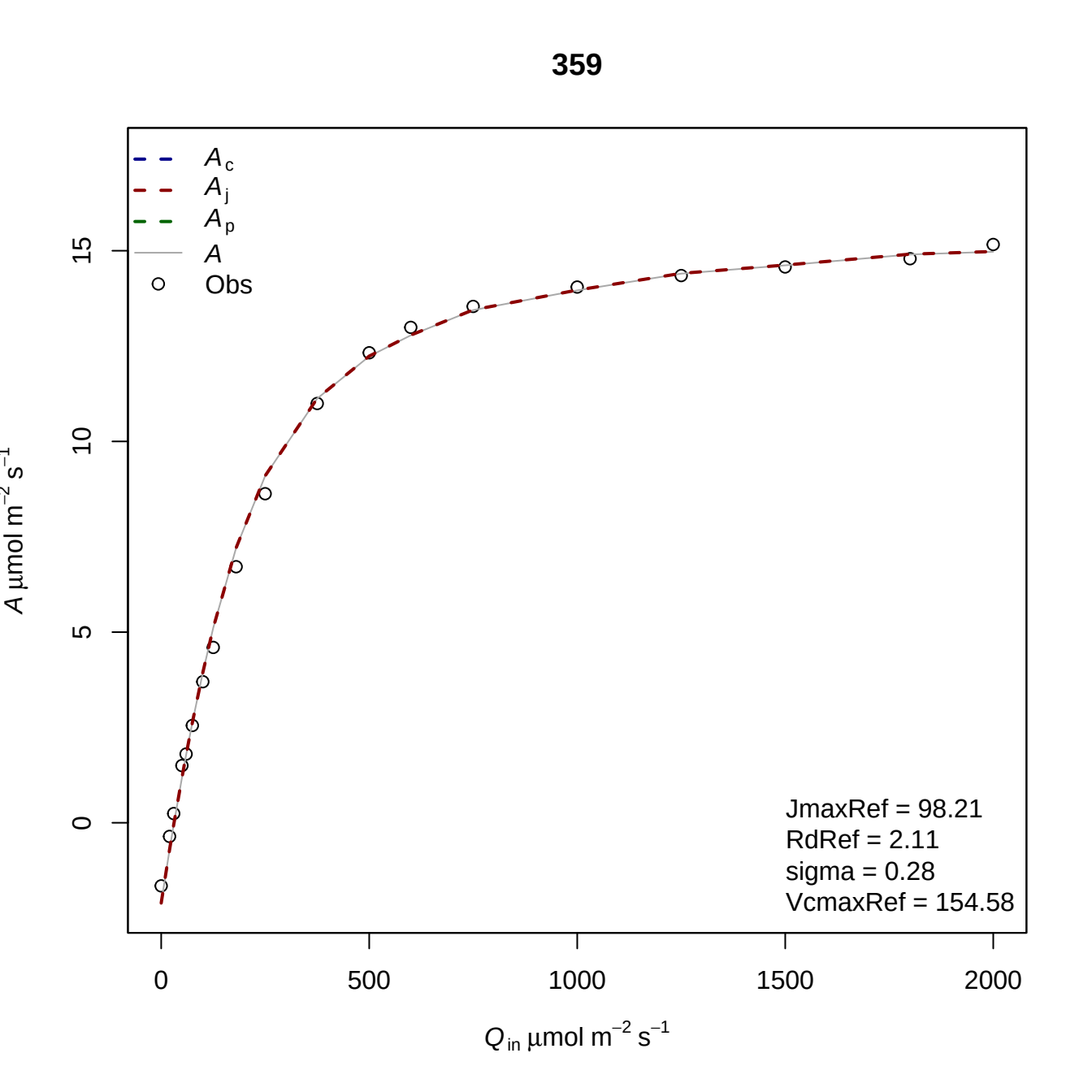

354
A_c
A_j
A_p
A
Obs
JmaxRef = 105.03
RdRef = 2.96
sigma = 0.58
VcmaxRef = 237.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

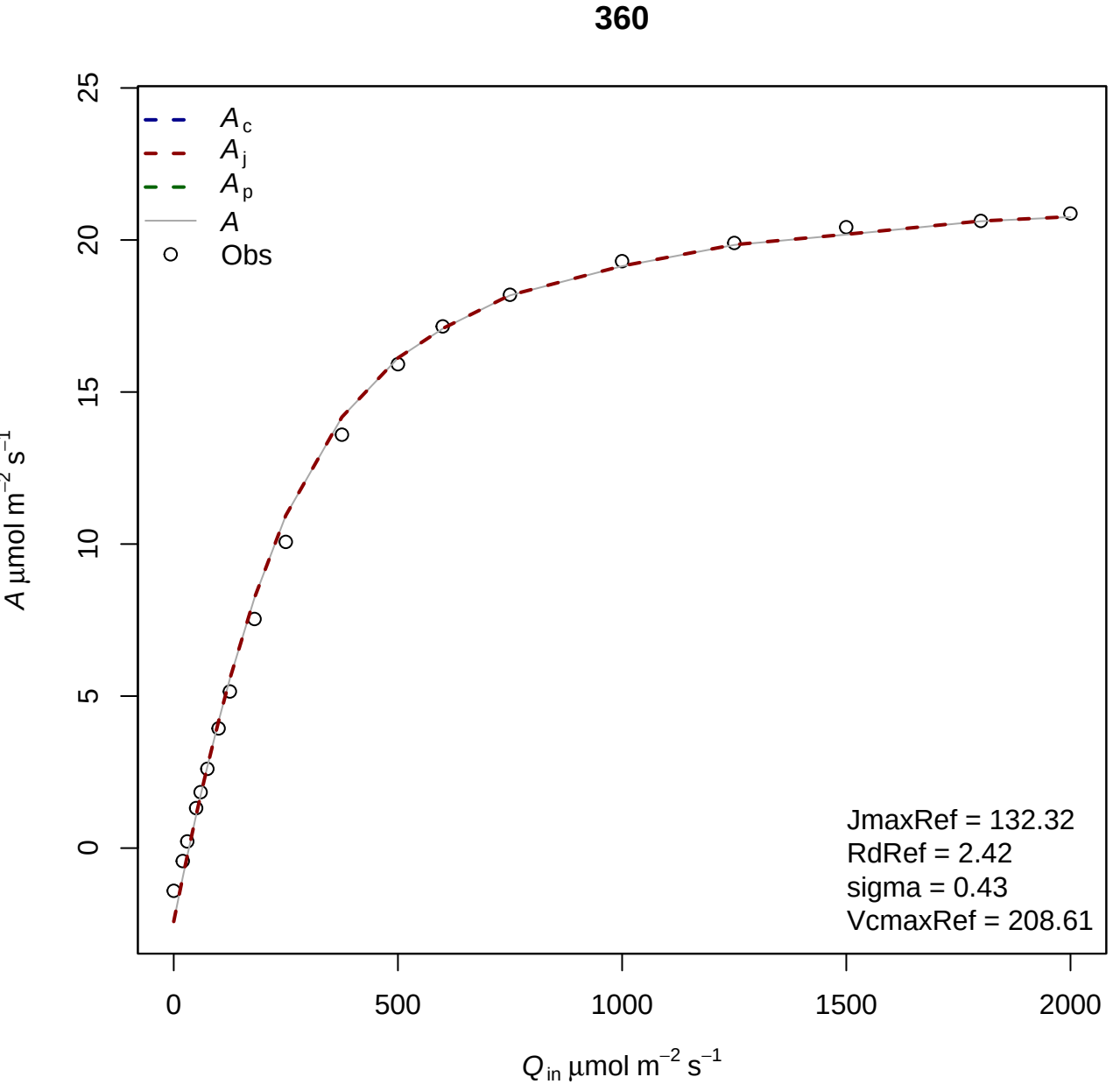

355
A_c
A_j
A_p
A
Obs
JmaxRef = 66.54
RdRef = 2.71
sigma = 0.68
VcmaxRef = 331.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

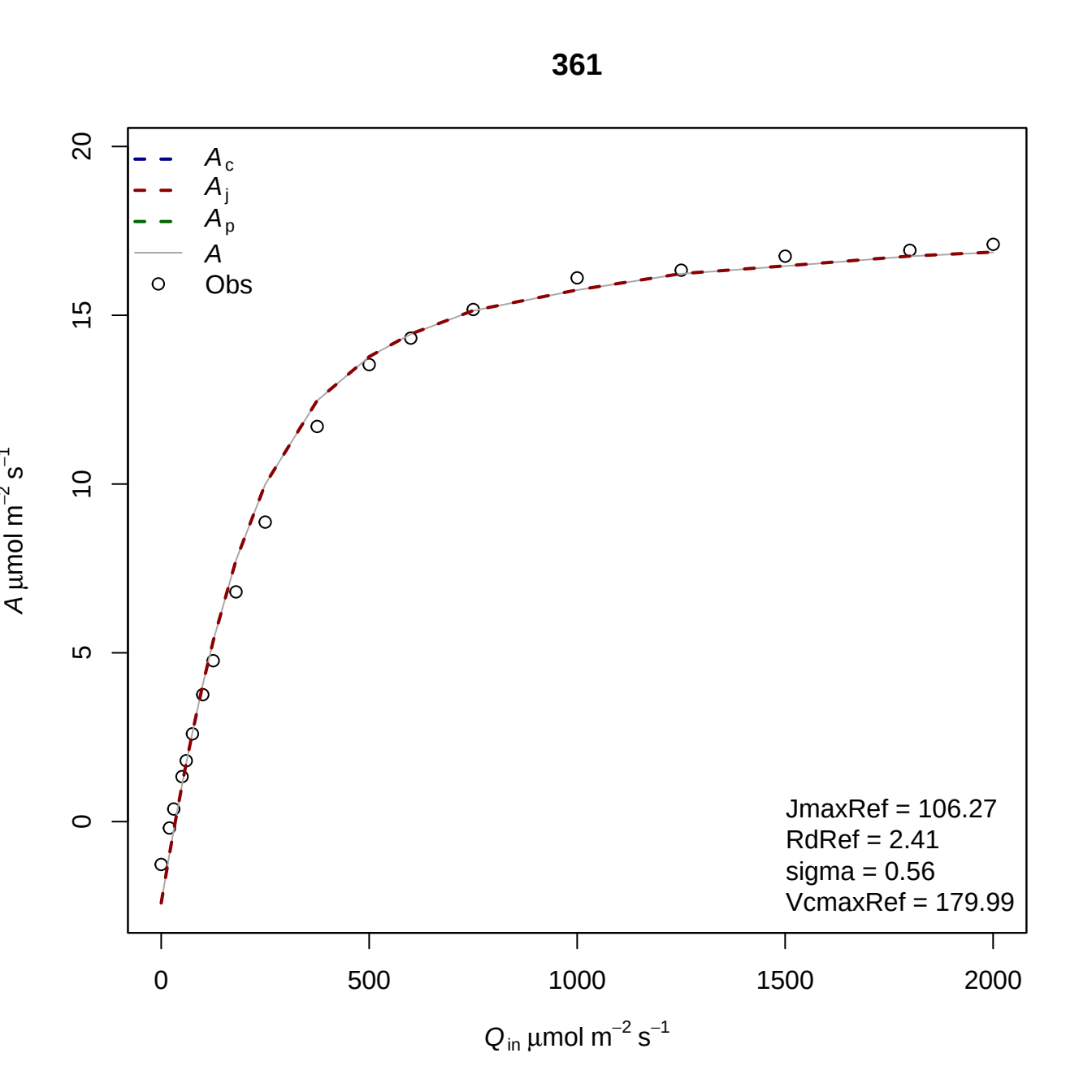

356
A_c
A_j
A_p
A
Obs
JmaxRef = 96.28
RdRef = 2.64
sigma = 0.69
VcmaxRef = 365.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

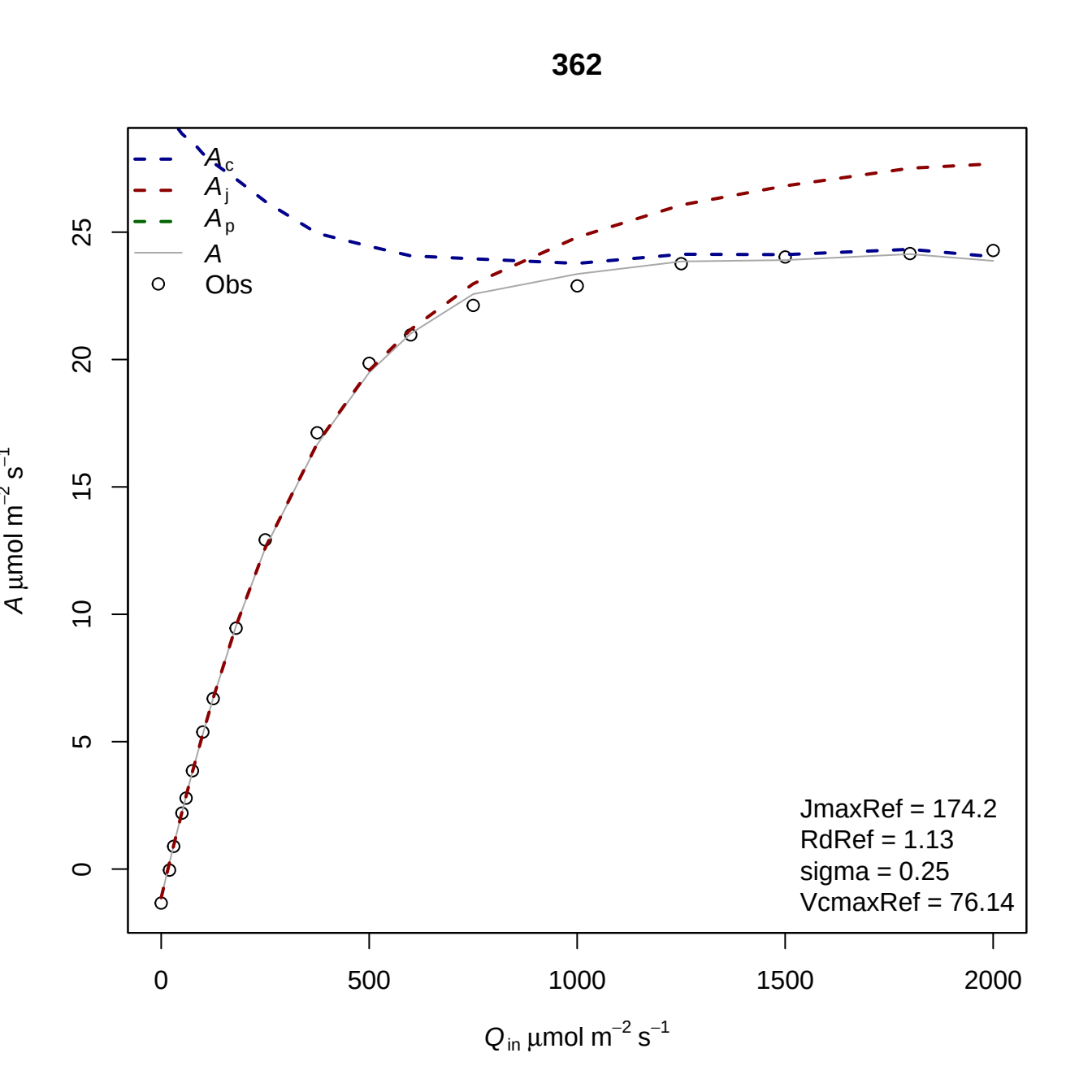

357
A_c
A_j
A_p
A
Obs
JmaxRef = 94.93
RdRef = 3.01
sigma = 0.71
VcmaxRef = 510.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

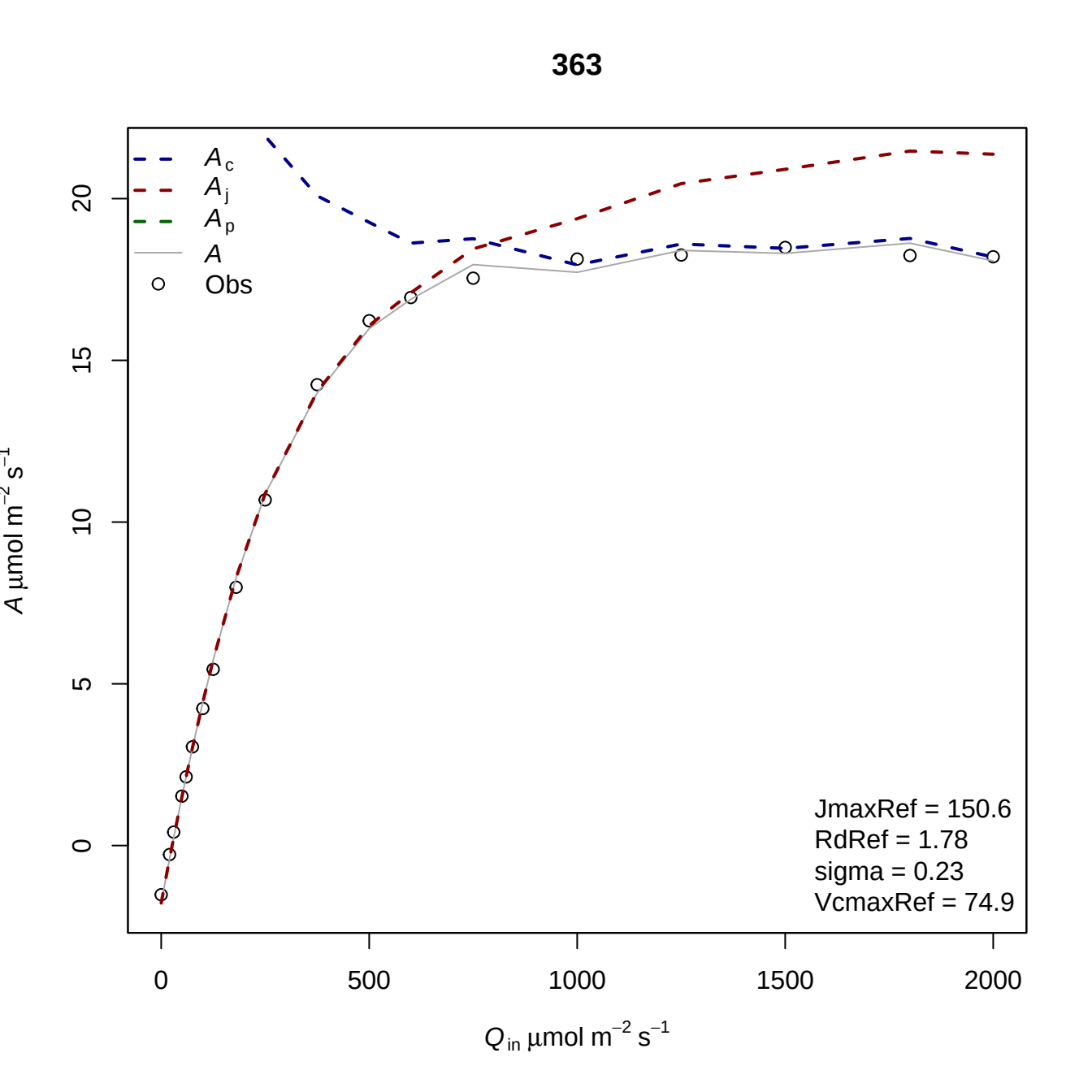

358
A_c
A_j
A_p
A
Obs
JmaxRef = 108.37
RdRef = 3.37
sigma = 0.6
VcmaxRef = 304.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

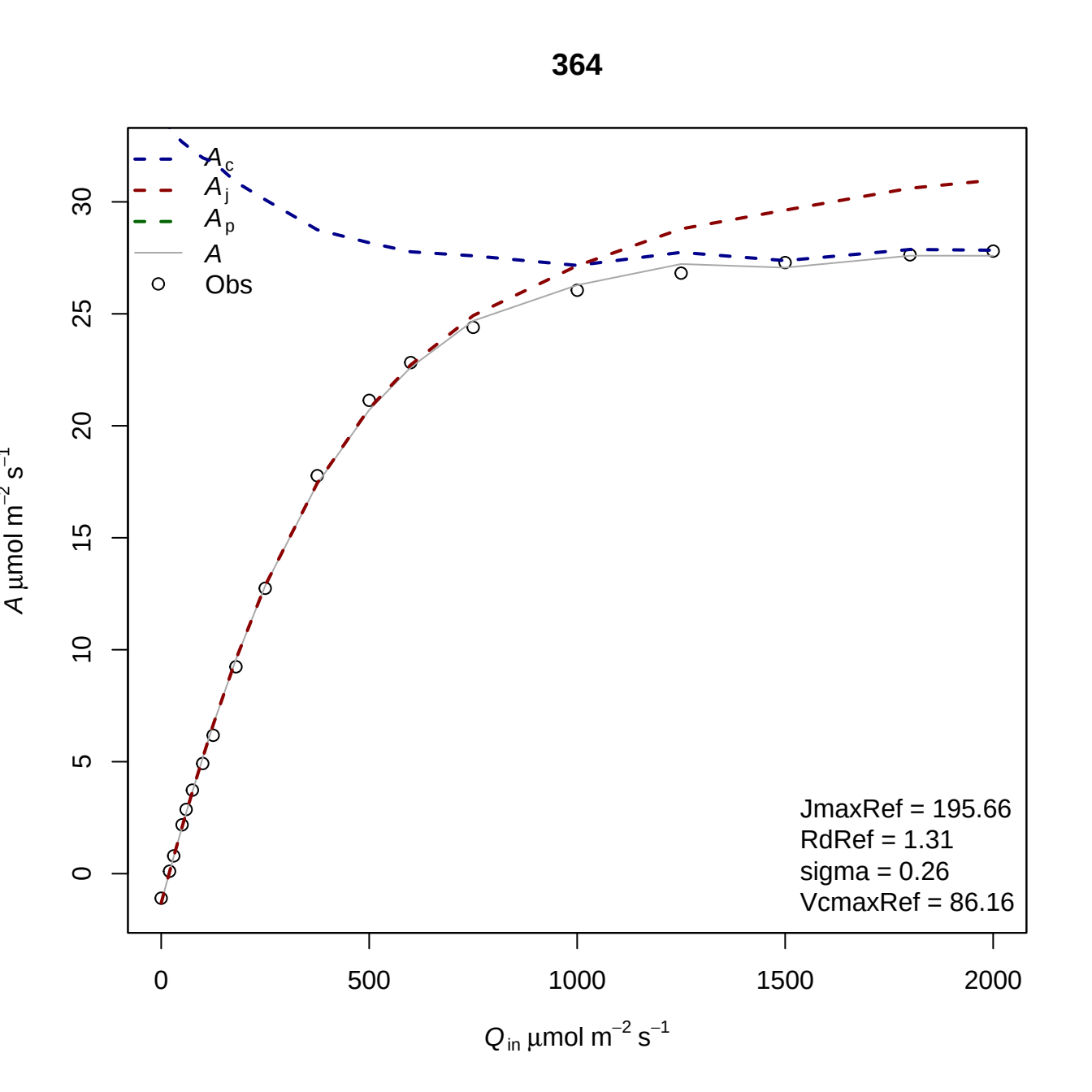

359
A_c
A_j
A_p
A
Obs
JmaxRef = 98.21
RdRef = 2.11
sigma = 0.28
VcmaxRef = 154.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

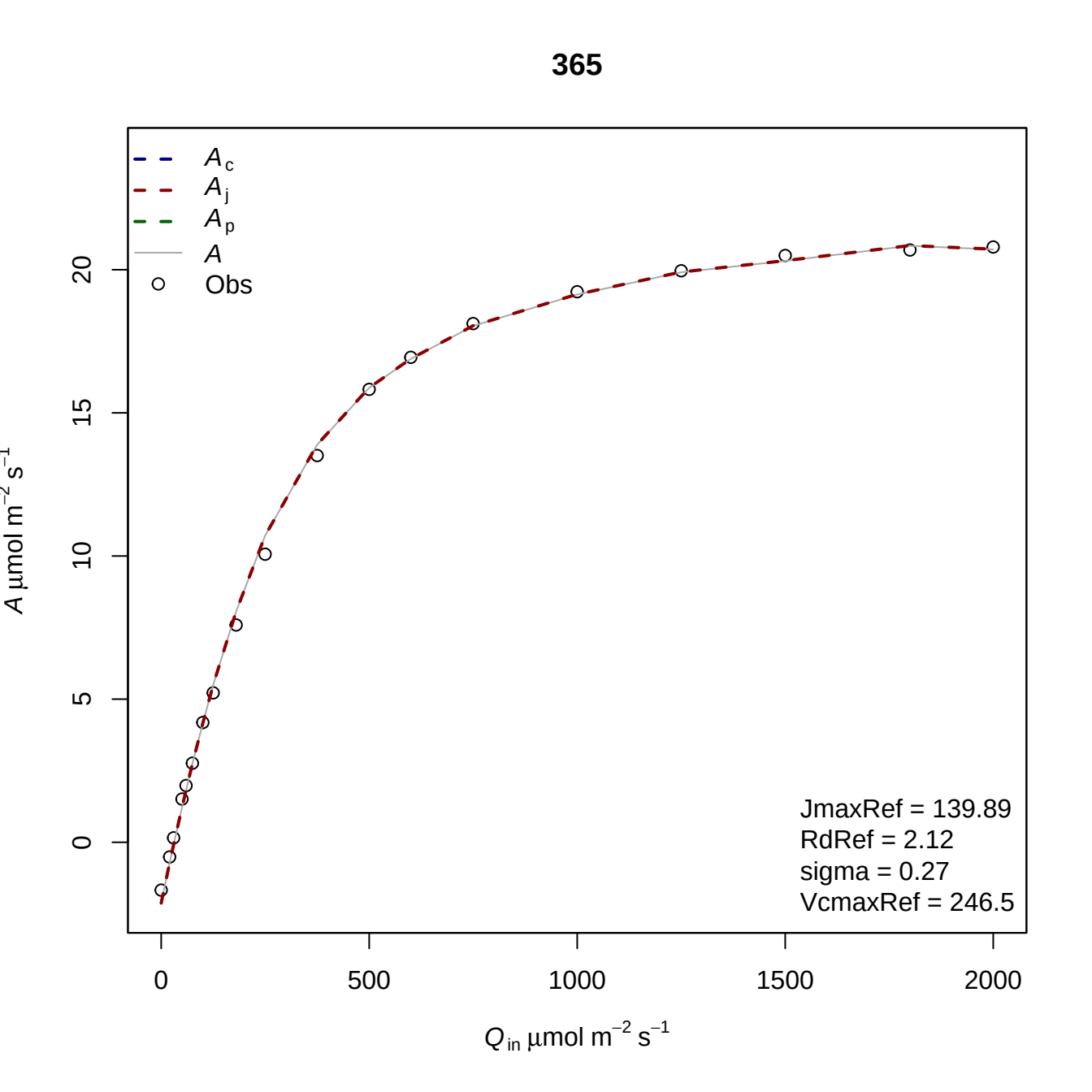

360
A_c
A_j
A_p
A
Obs
JmaxRef = 132.32
RdRef = 2.42
sigma = 0.43
VcmaxRef = 208.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

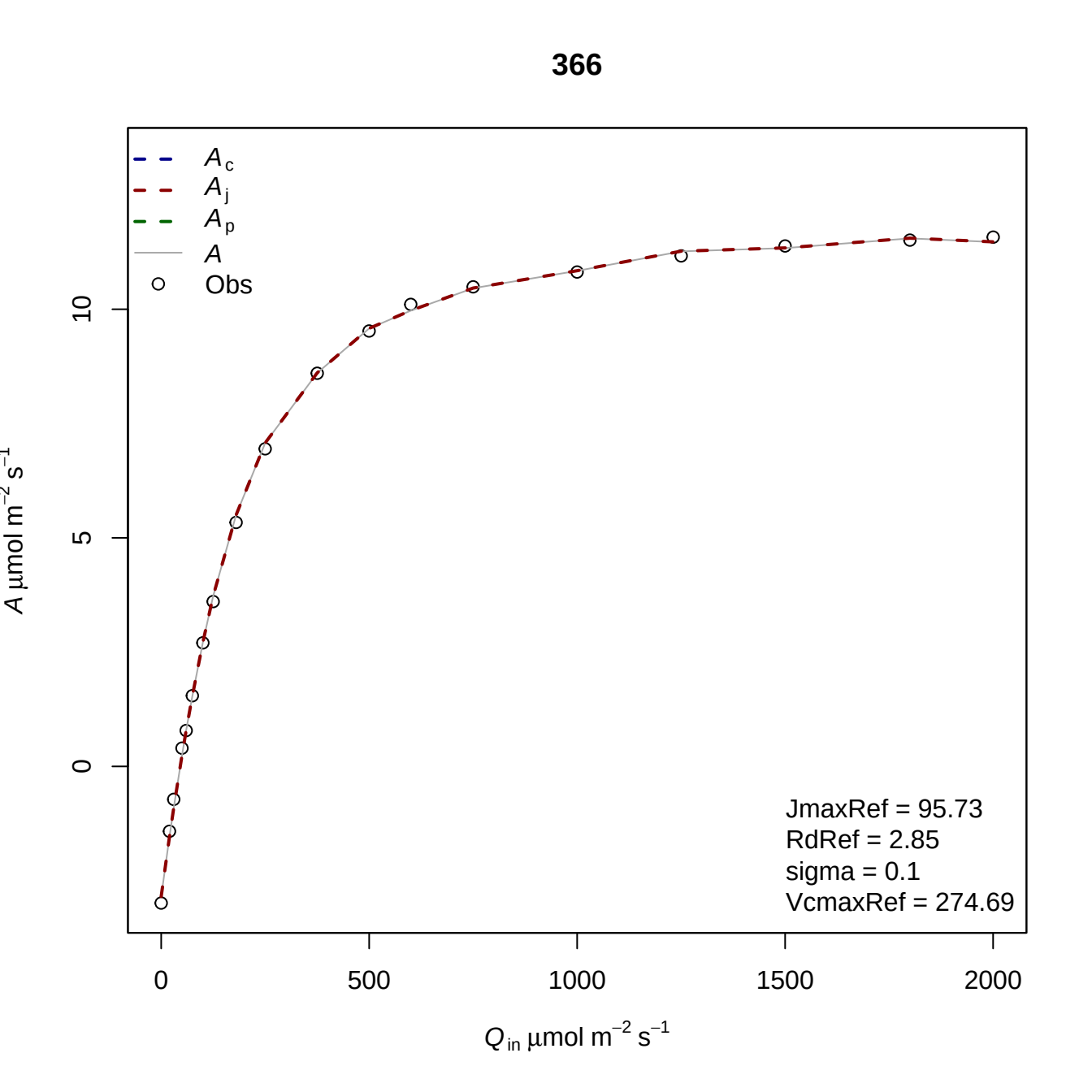

361
A_c
A_j
A_p
A
Obs
JmaxRef = 106.27
RdRef = 2.41
sigma = 0.56
VcmaxRef = 179.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

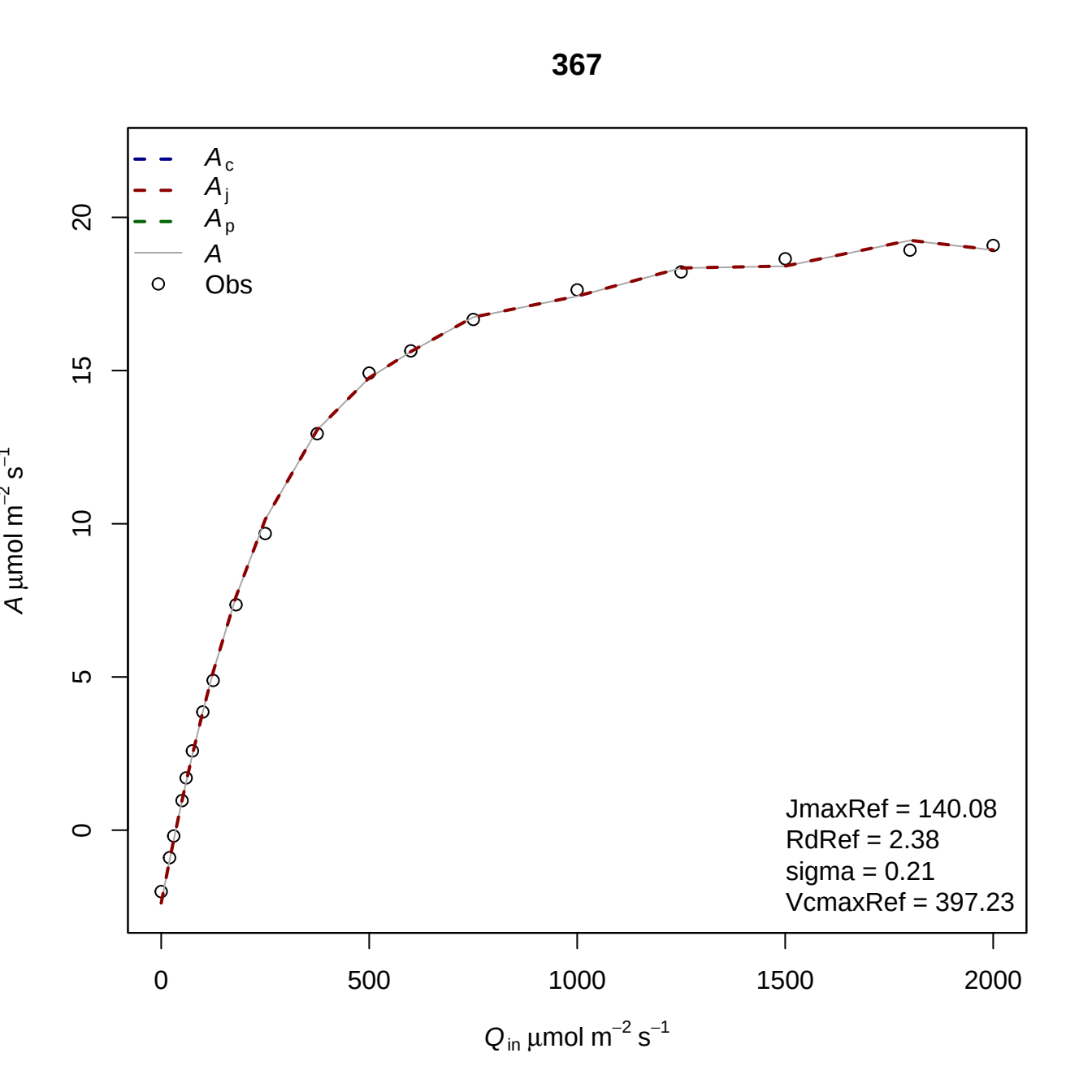

362
A_c
A_j
A_p
A
Obs
JmaxRef = 174.2
RdRef = 1.13
sigma = 0.25
VcmaxRef = 76.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

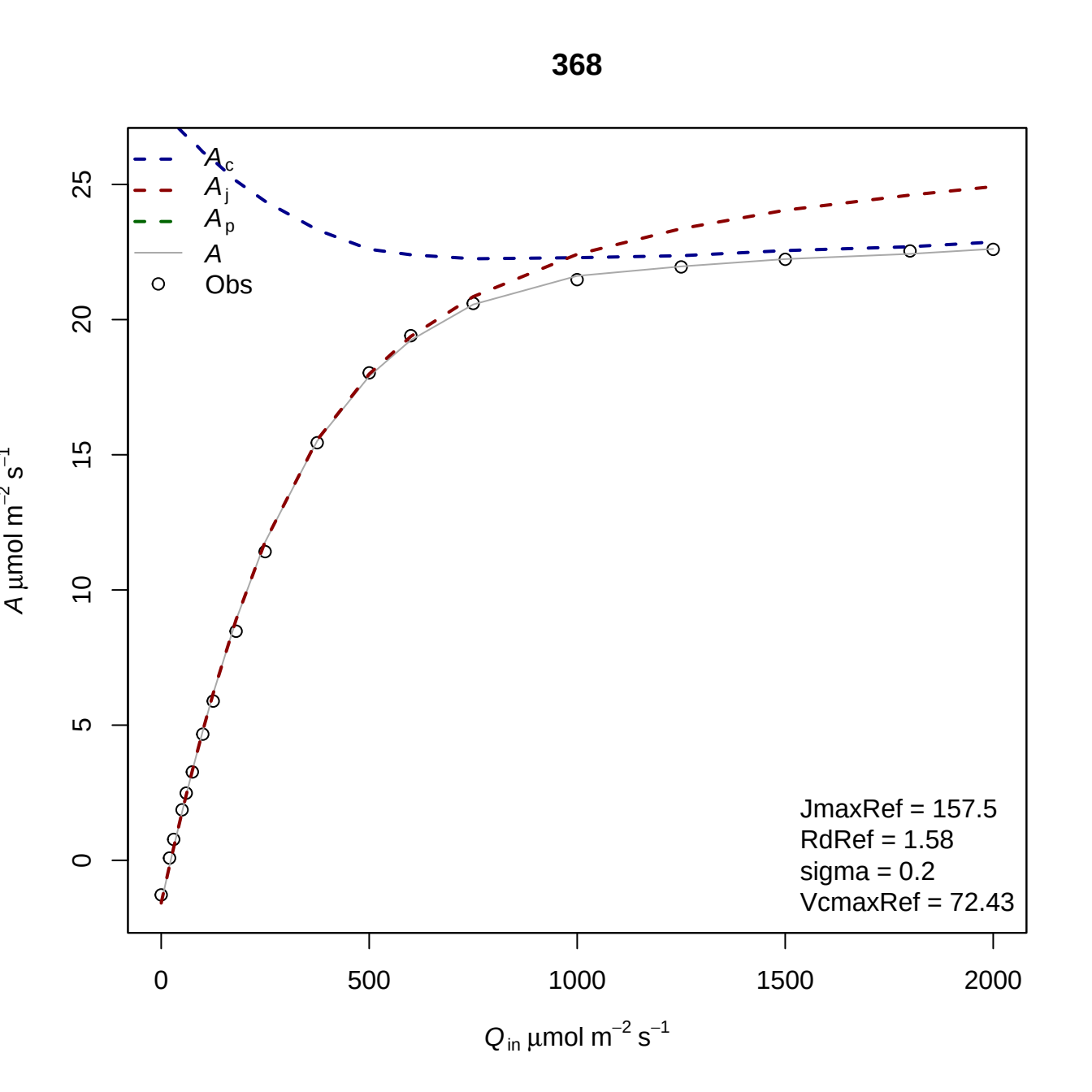

363
A_c
A_j
A_p
A
Obs
JmaxRef = 150.6
RdRef = 1.78
sigma = 0.23
VcmaxRef = 74.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

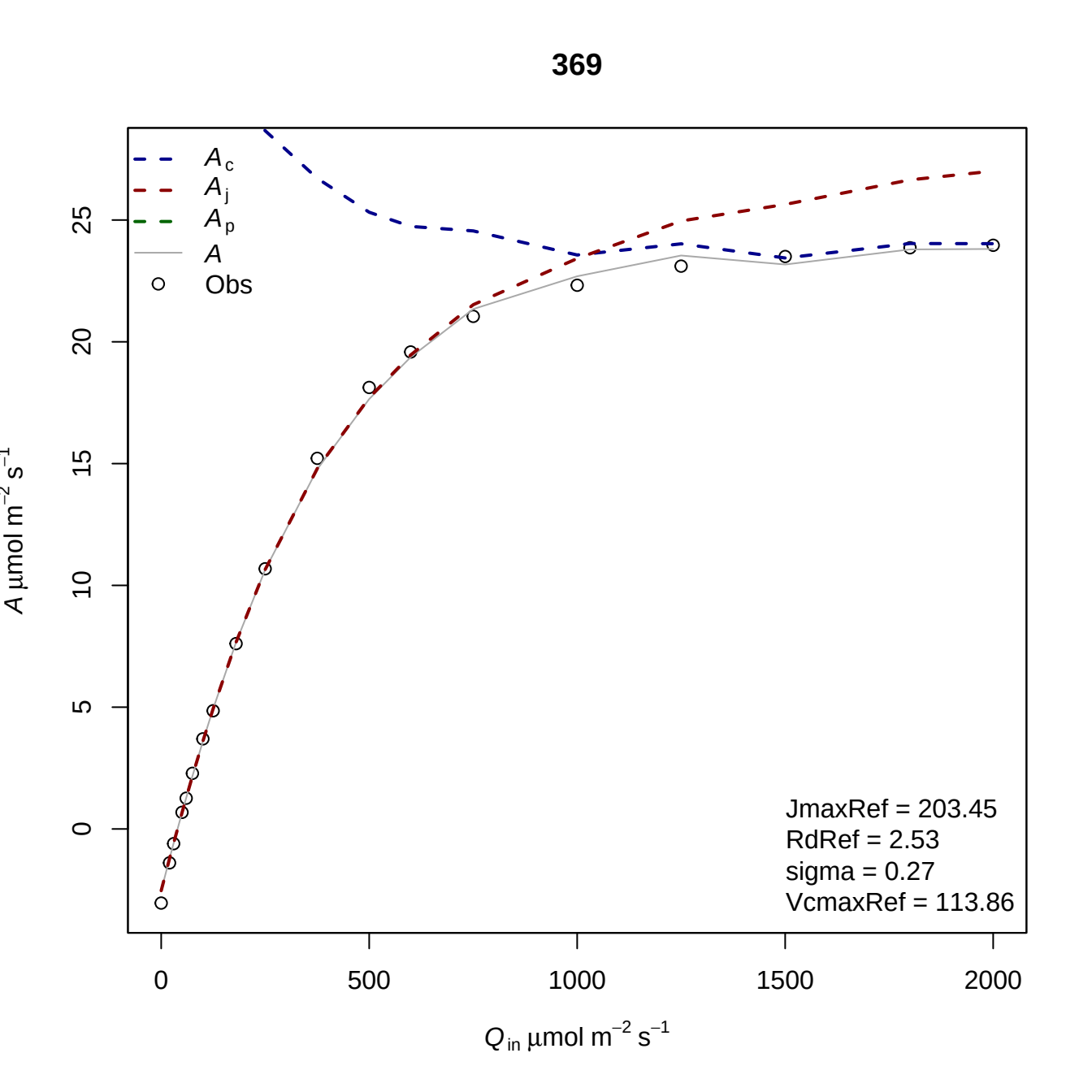

364
A_c
A_j
A_p
A
Obs
JmaxRef = 195.66
RdRef = 1.31
sigma = 0.26
VcmaxRef = 86.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

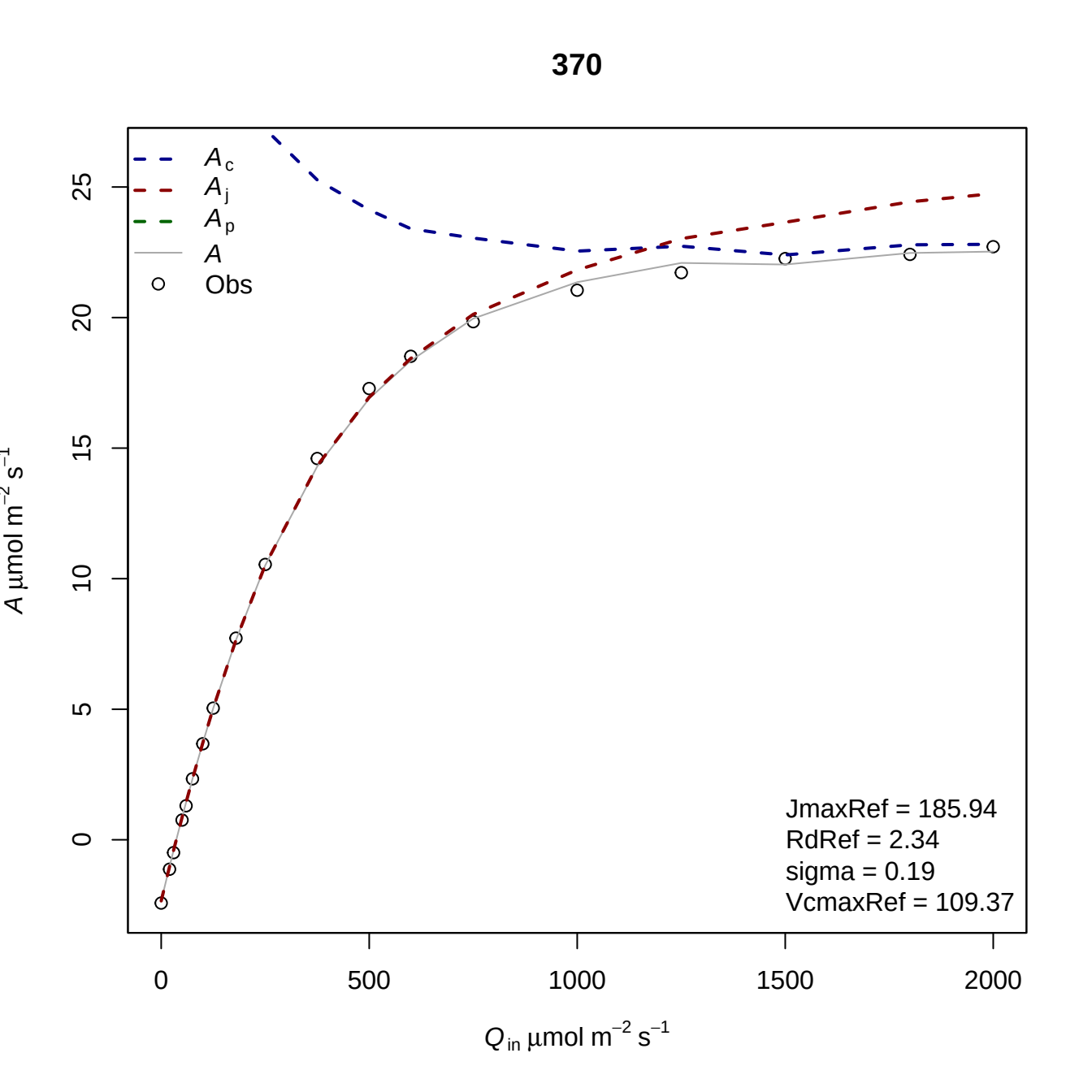

365
A_c
A_j
A_p
A
Obs
JmaxRef = 139.89
RdRef = 2.12
sigma = 0.27
VcmaxRef = 246.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

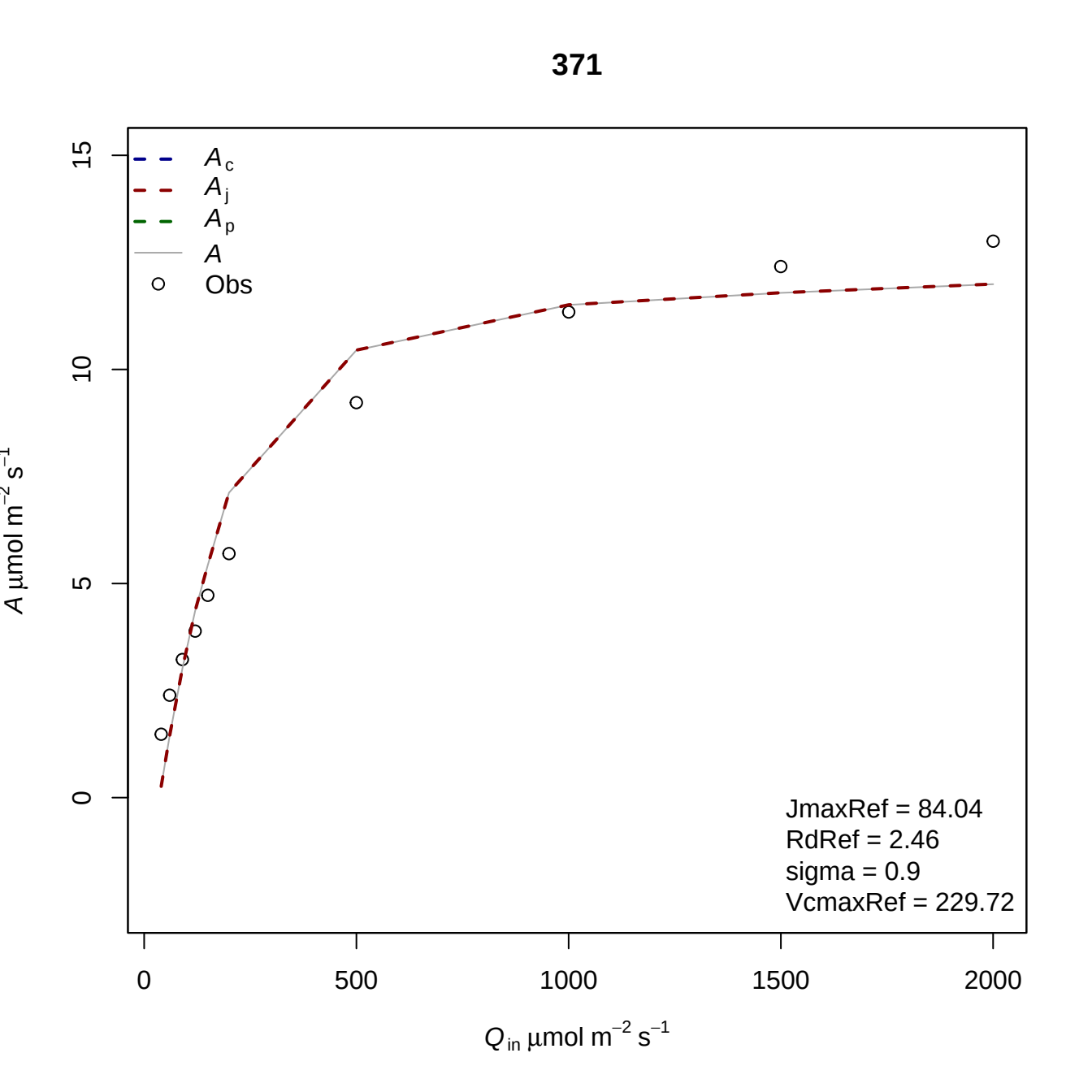

366
A_c
A_j
A_p
A
Obs
JmaxRef = 95.73
RdRef = 2.85
sigma = 0.1
VcmaxRef = 274.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

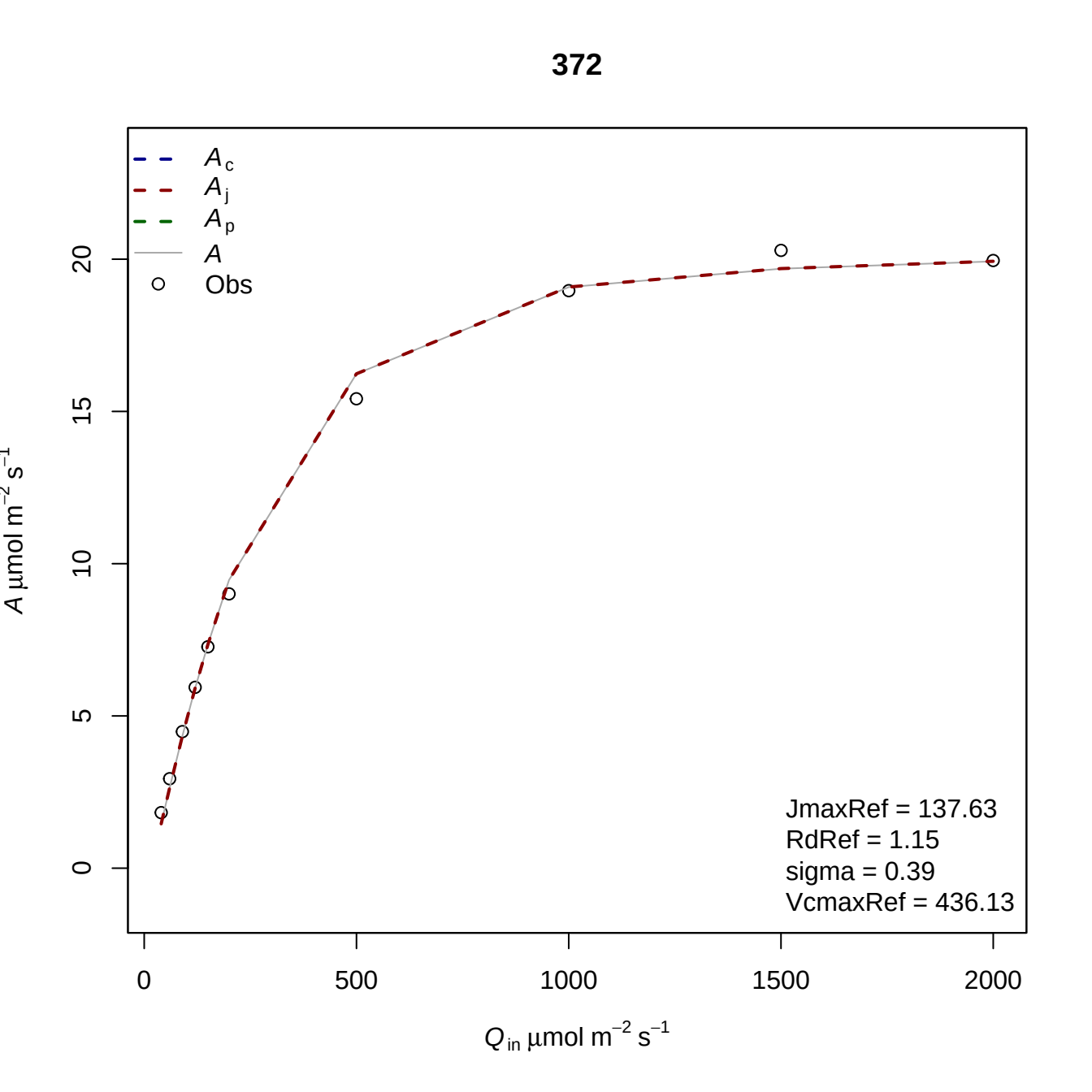

367
A_c
A_j
A_p
A
Obs
JmaxRef = 140.08
RdRef = 2.38
sigma = 0.21
VcmaxRef = 397.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

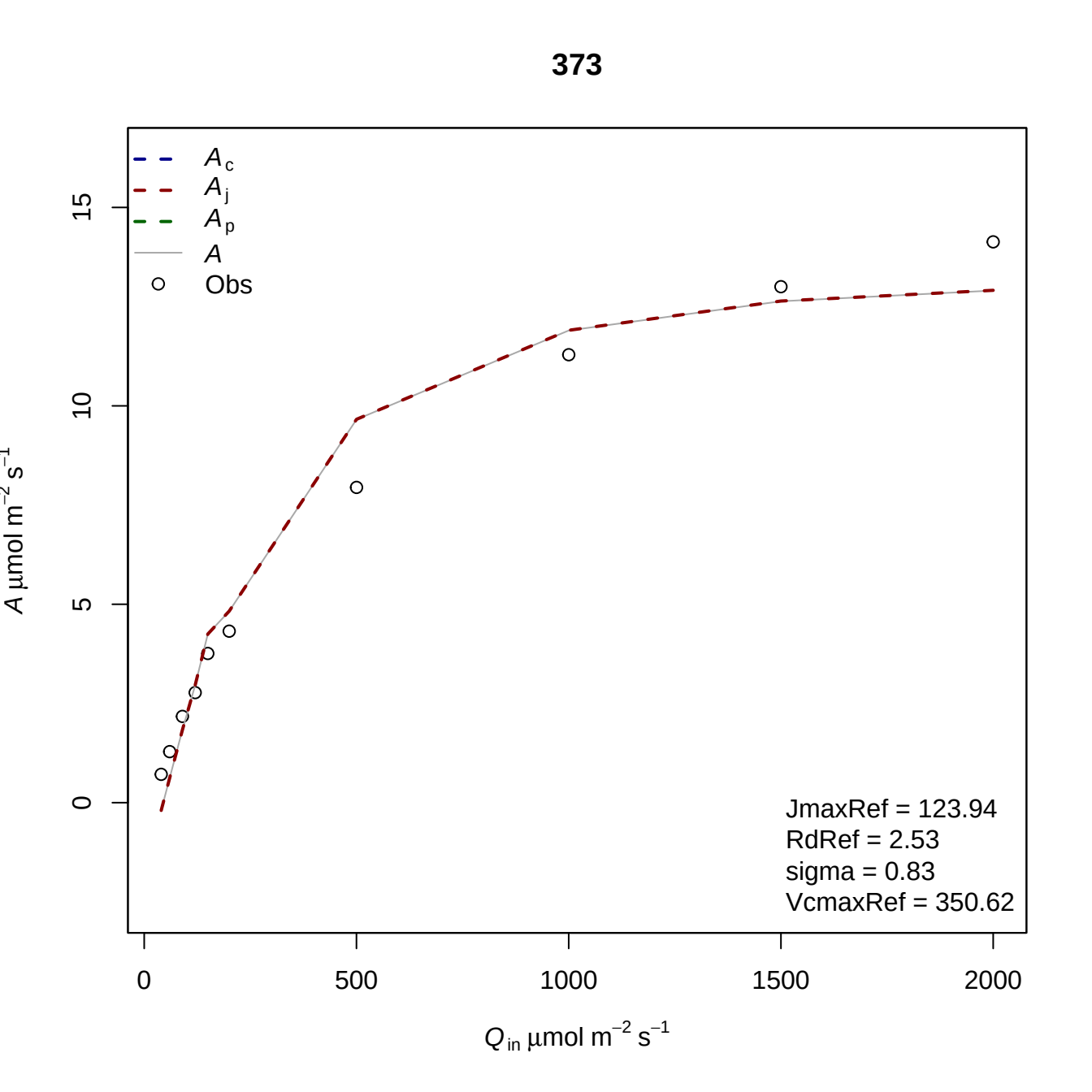

368
A_c
A_j
A_p
A
Obs
JmaxRef = 157.5
RdRef = 1.58
sigma = 0.2
VcmaxRef = 72.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

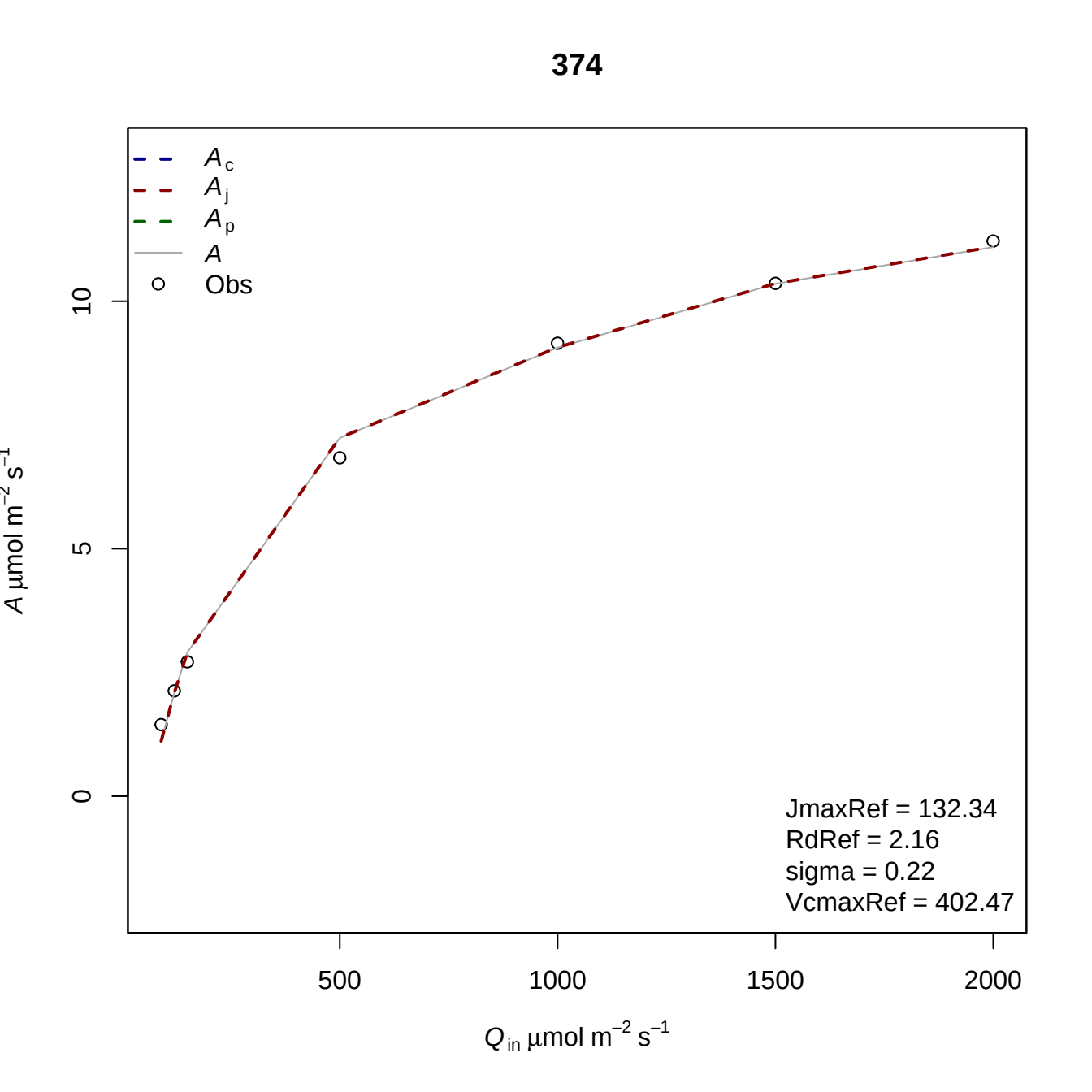

369
A_c
A_j
A_p
A
Obs
JmaxRef = 203.45
RdRef = 2.53
sigma = 0.27
VcmaxRef = 113.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

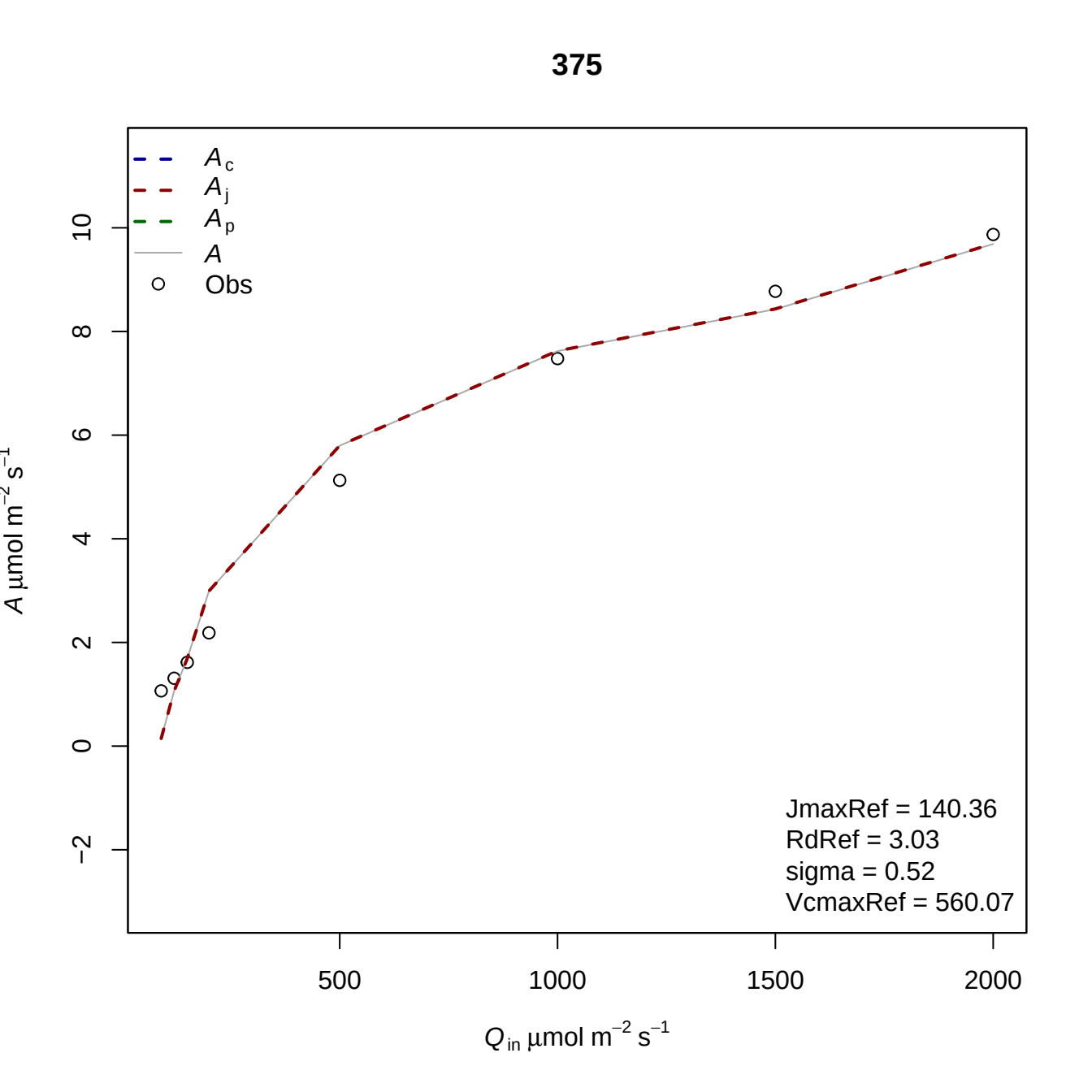

370
A_c
A_j
A_p
A
Obs
JmaxRef = 185.94
RdRef = 2.34
sigma = 0.19
VcmaxRef = 109.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

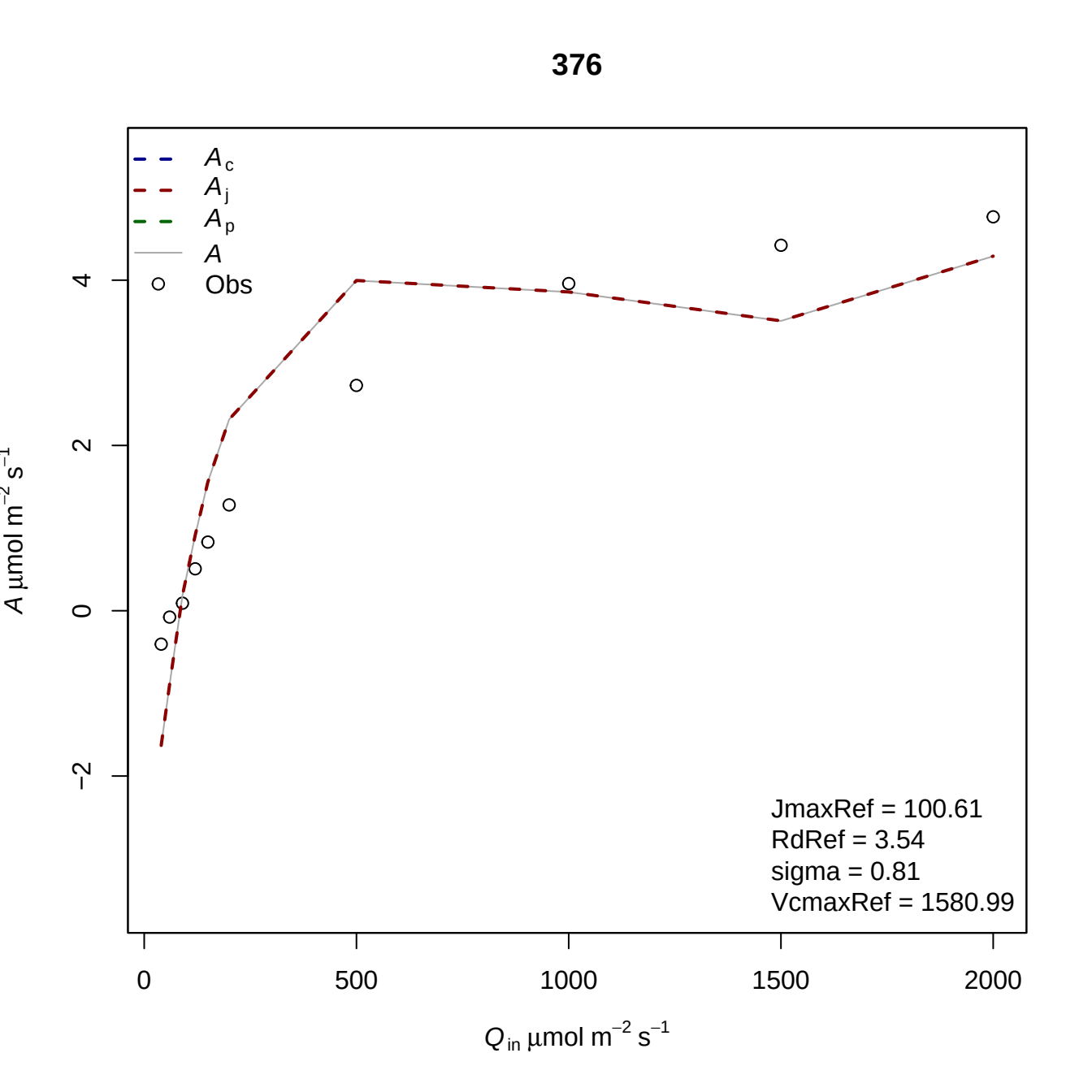

371
A_c
A_j
A_p
A
Obs
JmaxRef = 84.04
RdRef = 2.46
sigma = 0.9
VcmaxRef = 229.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

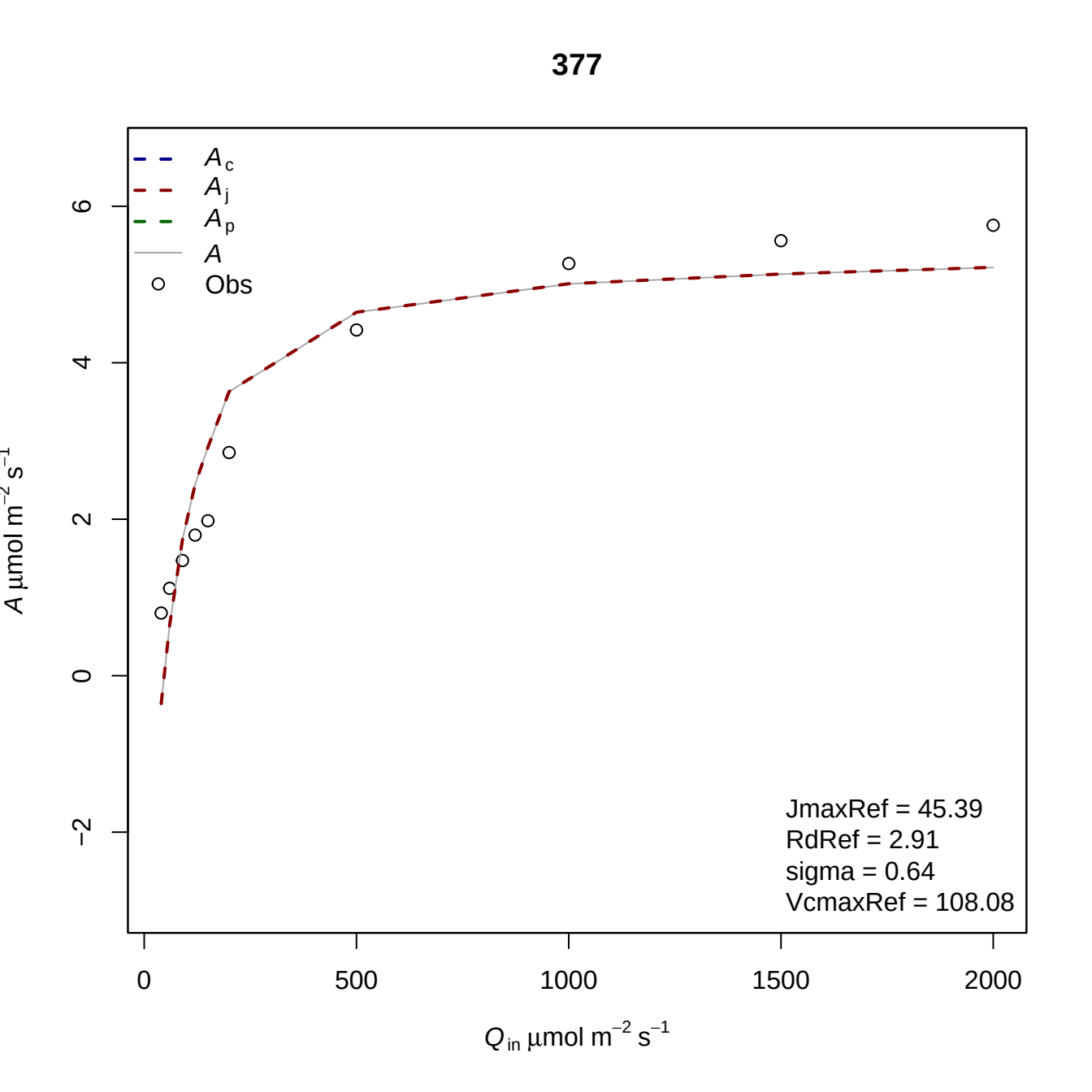

372
A_c
A_j
A_p
A
Obs
JmaxRef = 137.63
RdRef = 1.15
sigma = 0.39
VcmaxRef = 436.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

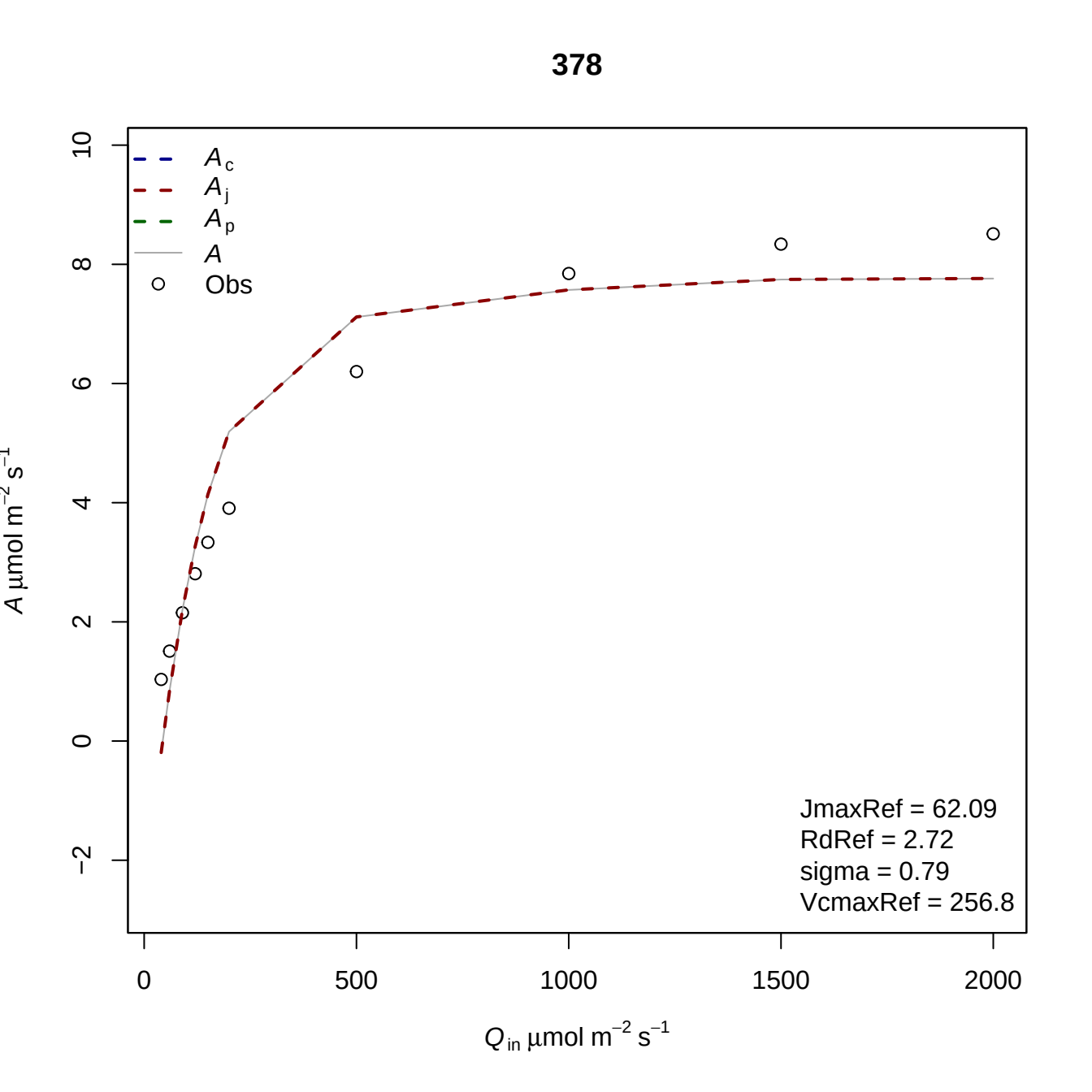

373
A_c
A_j
A_p
A
Obs
JmaxRef = 123.94
RdRef = 2.53
sigma = 0.83
VcmaxRef = 350.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

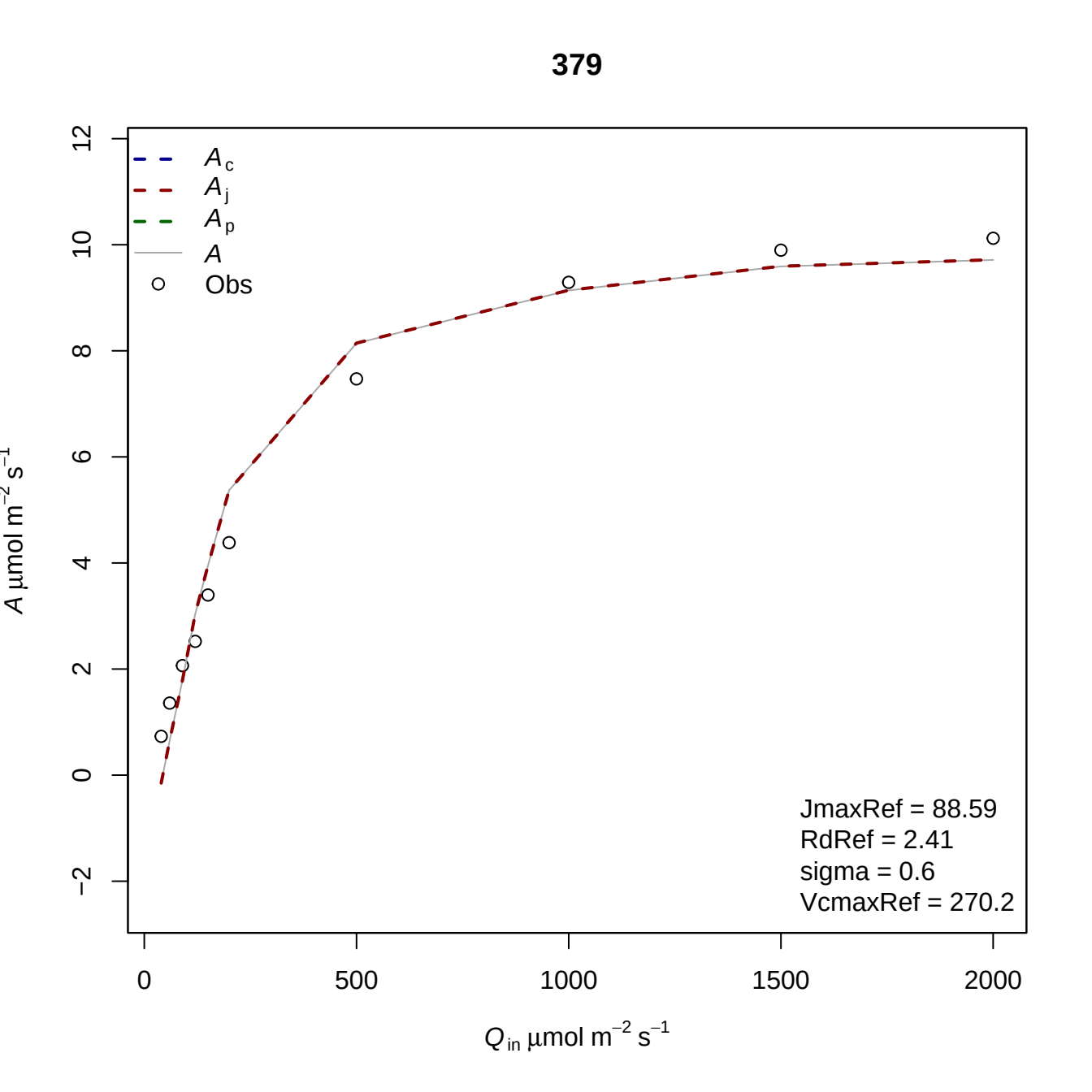

374
A_c
A_j
A_p
A
Obs
JmaxRef = 132.34
RdRef = 2.16
sigma = 0.22
VcmaxRef = 402.47
0
5
10
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

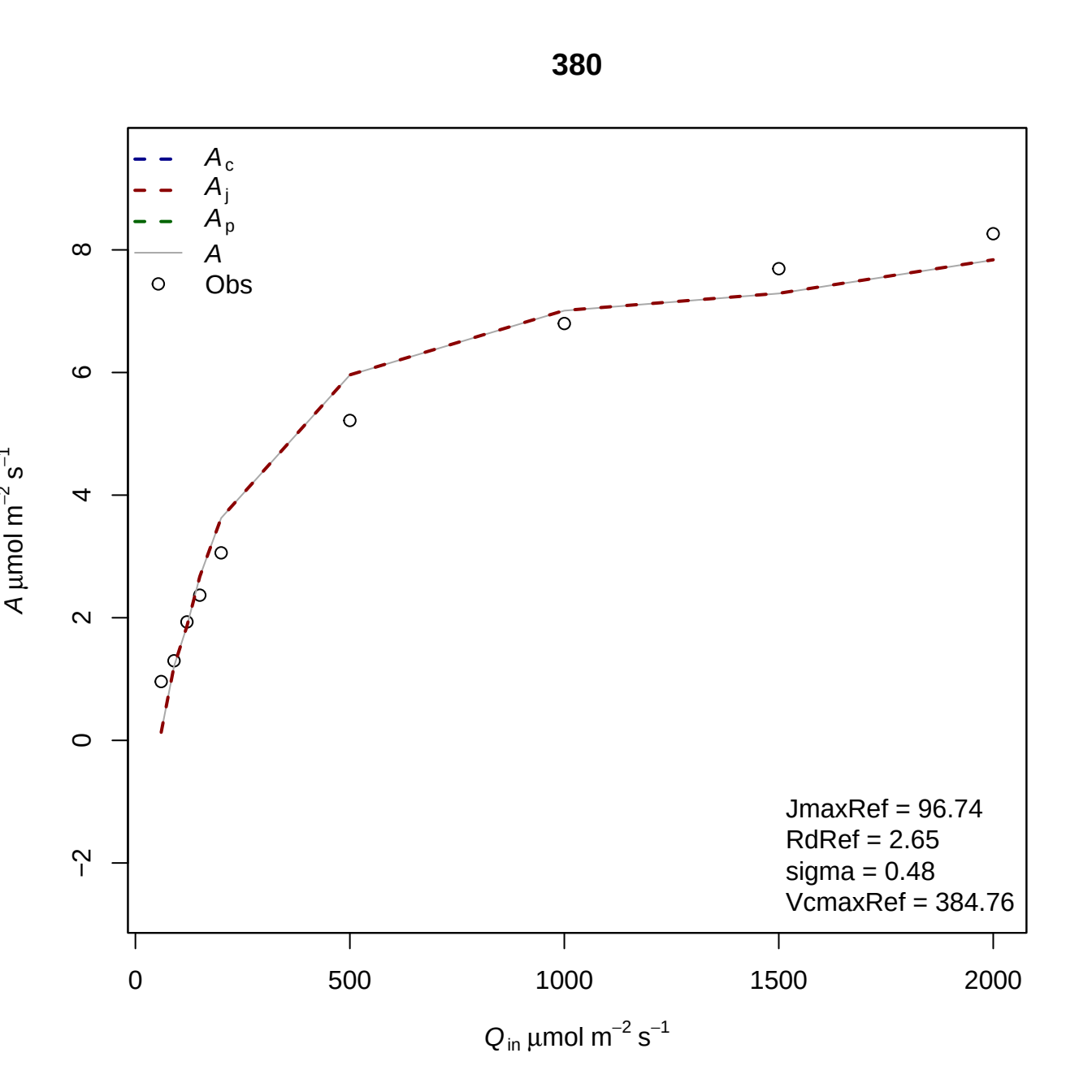

375
A_c
A_j
A_p
A
Obs
JmaxRef = 140.36
RdRef = 3.03
sigma = 0.52
VcmaxRef = 560.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
500
1000
1500
2000

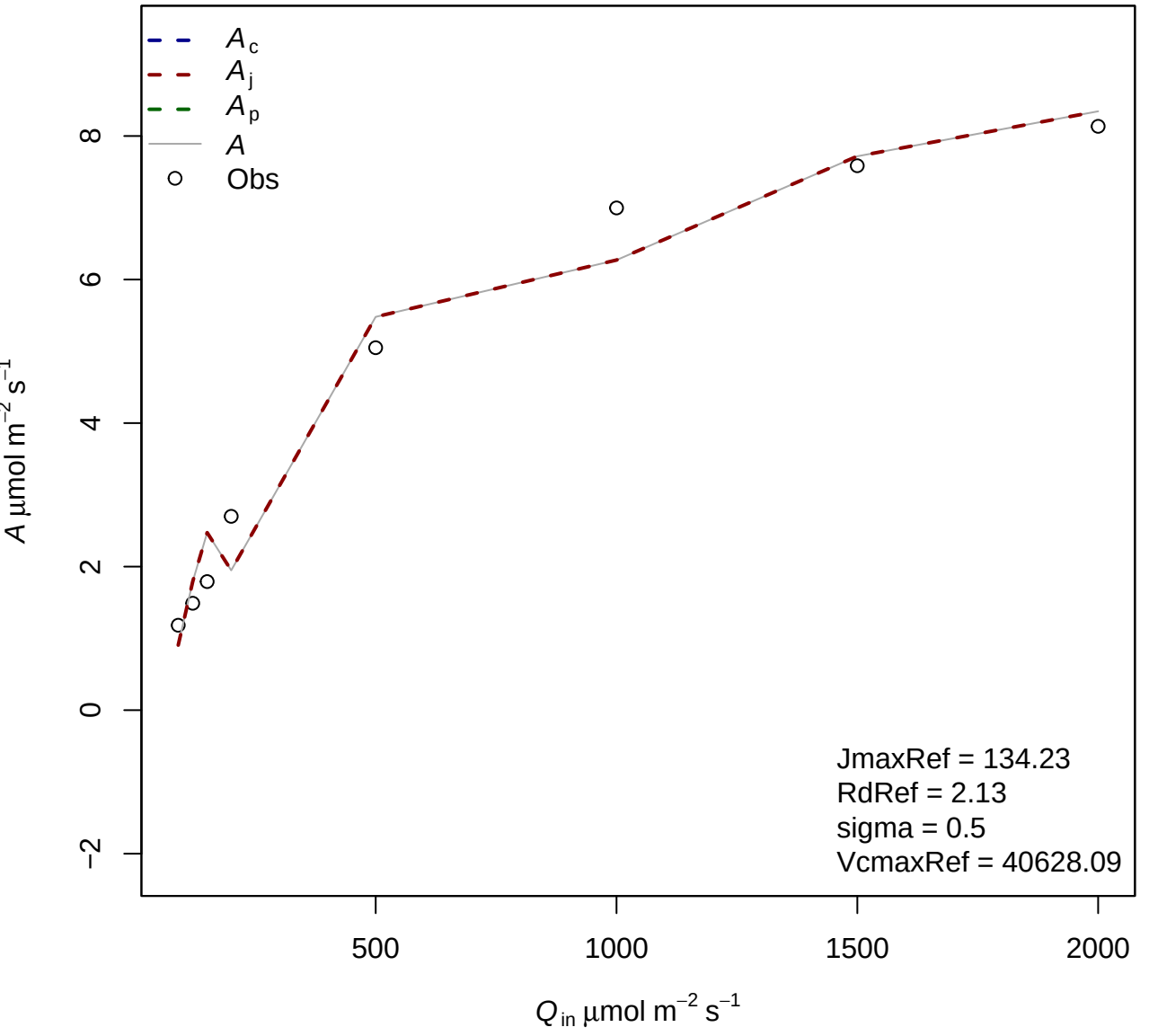

376
A_c
A_j
A_p
A
Obs
JmaxRef = 100.61
RdRef = 3.54
sigma = 0.81
VcmaxRef = 1580.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

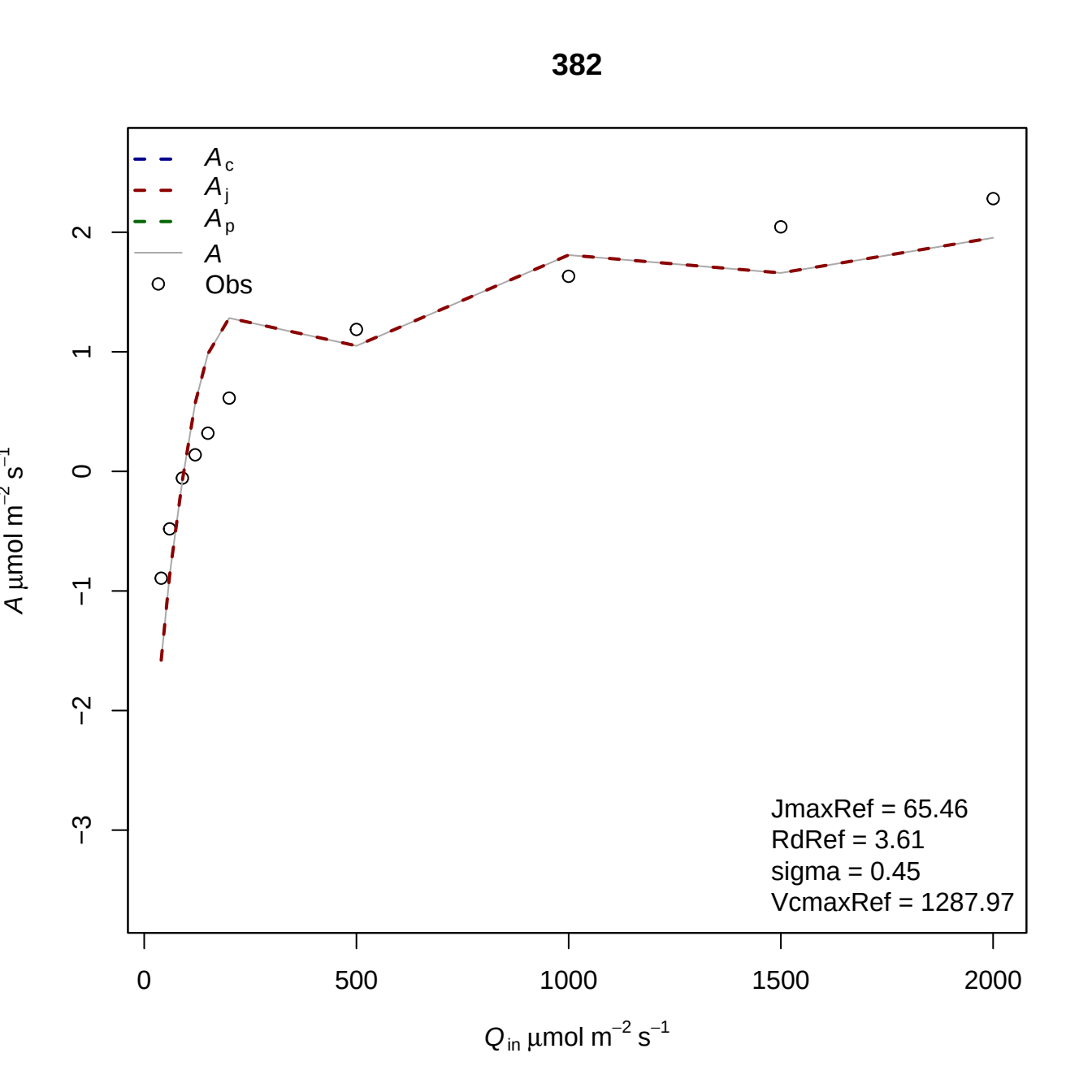

377
A_c
A_j
A_p
A
Obs
JmaxRef = 45.39
RdRef = 2.91
sigma = 0.64
VcmaxRef = 108.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

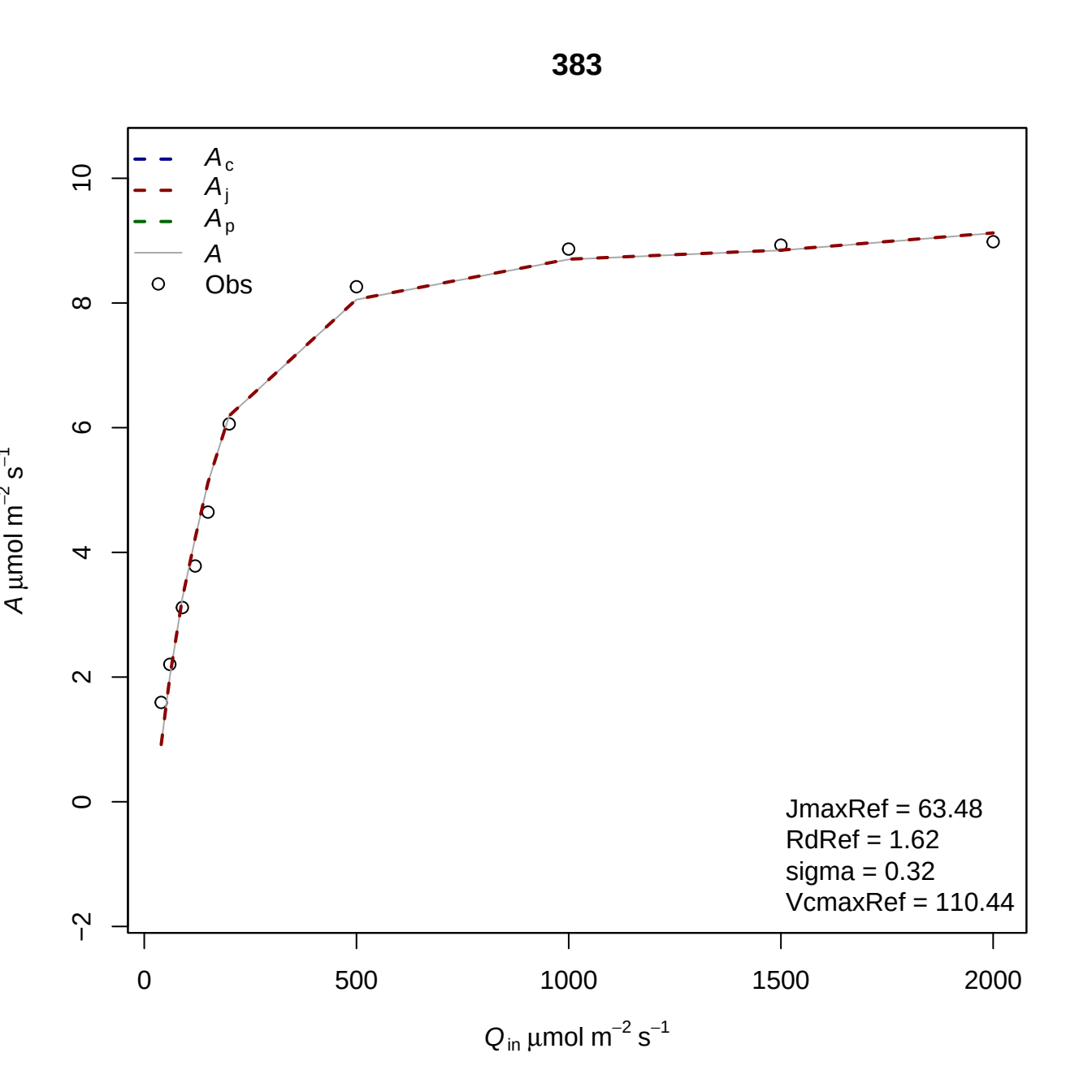

378
A_c
A_j
A_p
A
Obs
JmaxRef = 62.09
RdRef = 2.72
sigma = 0.79
VcmaxRef = 256.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

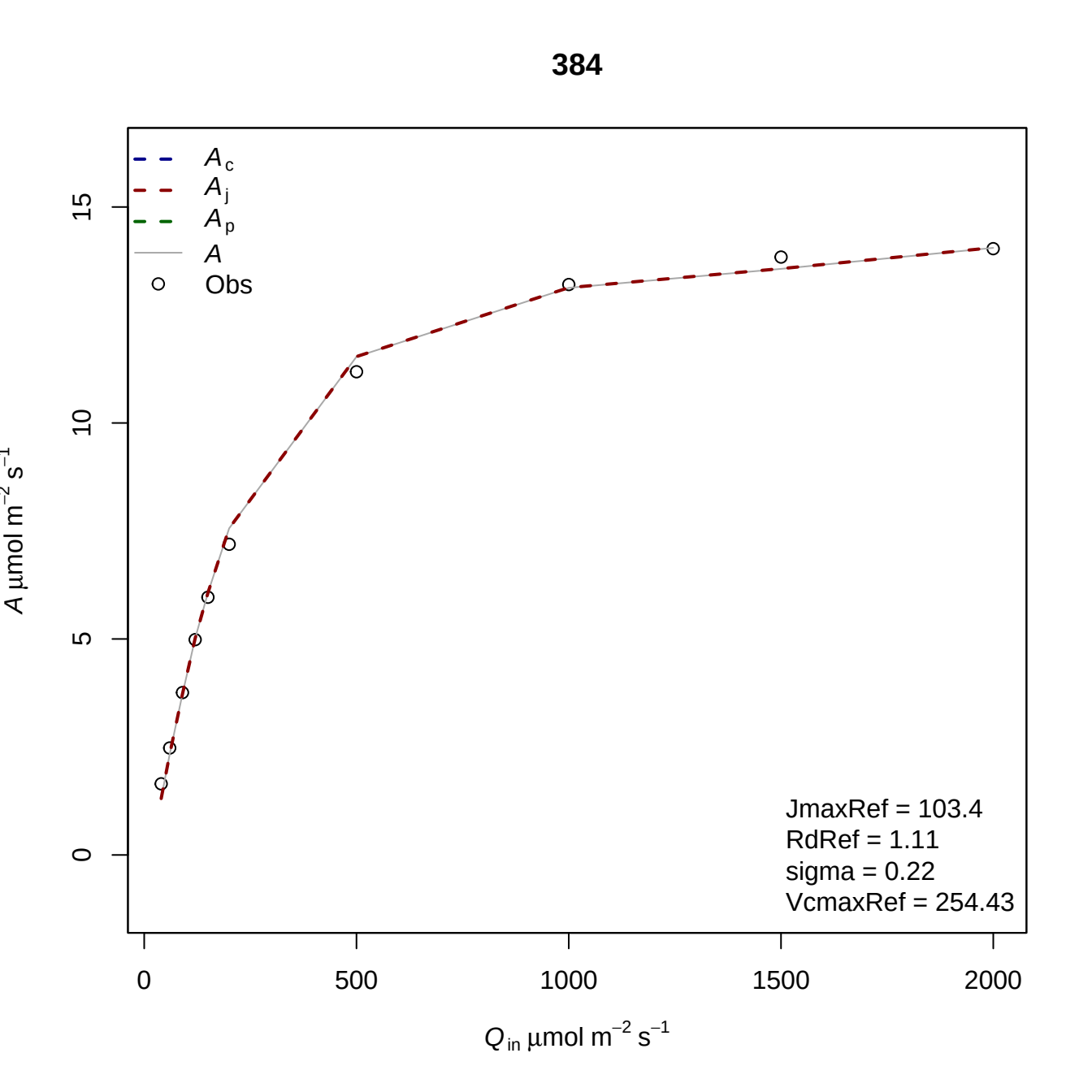

379
A_c
A_j
A_p
A
Obs
JmaxRef = 88.59
RdRef = 2.41
sigma = 0.6
VcmaxRef = 270.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

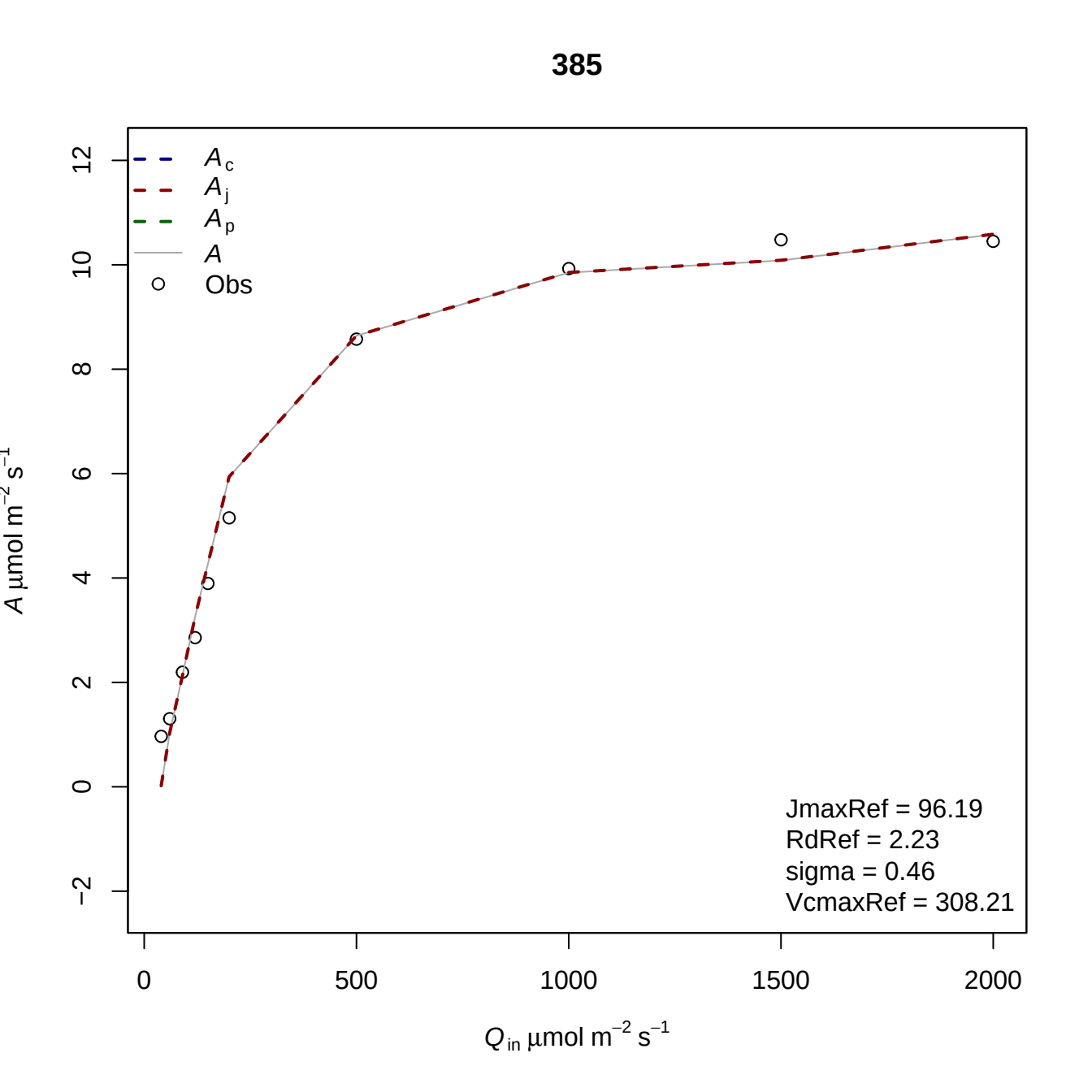

380
A_c
A_j
A_p
A
Obs
JmaxRef = 96.74
RdRef = 2.65
sigma = 0.48
VcmaxRef = 384.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

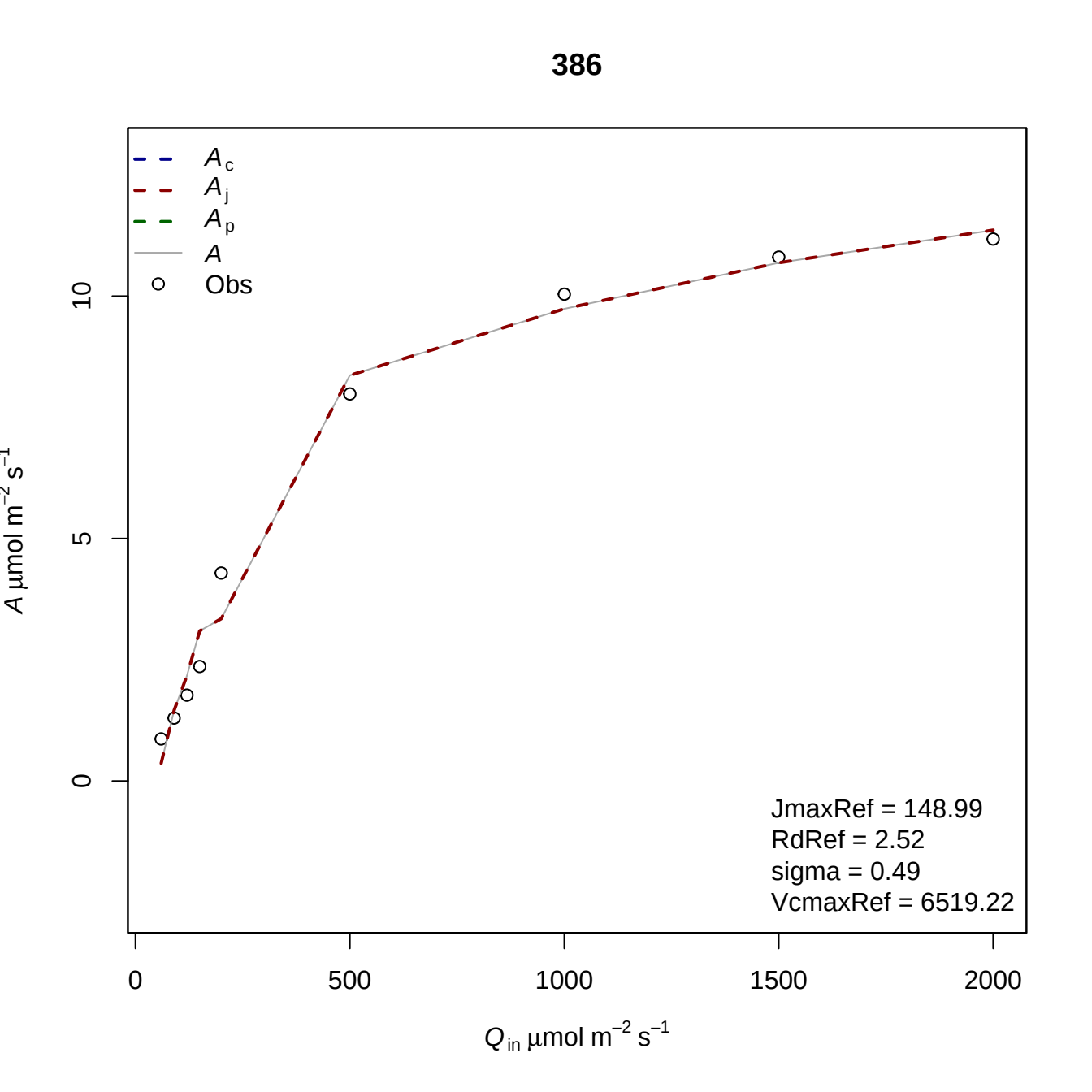

A_c
A_j
A_p
A
Obs
JmaxRef = 134.23
RdRef = 2.13
sigma = 0.5
VcmaxRef = 40628.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
500
1000
1500
2000

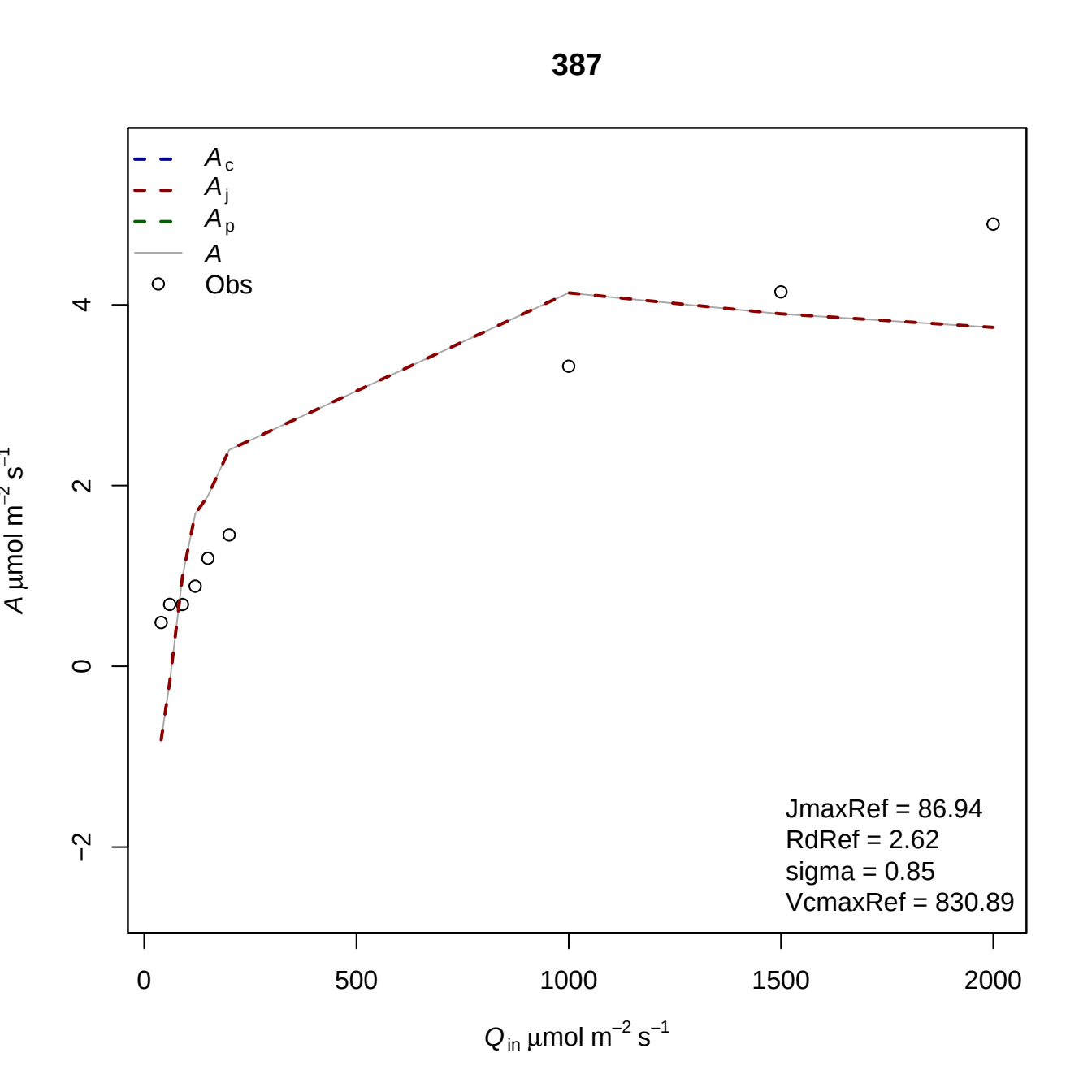

382
A_c
A_j
A_p
A
Obs
JmaxRef = 65.46
RdRef = 3.61
sigma = 0.45
VcmaxRef = 1287.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
−3
0
500
1000
1500
2000

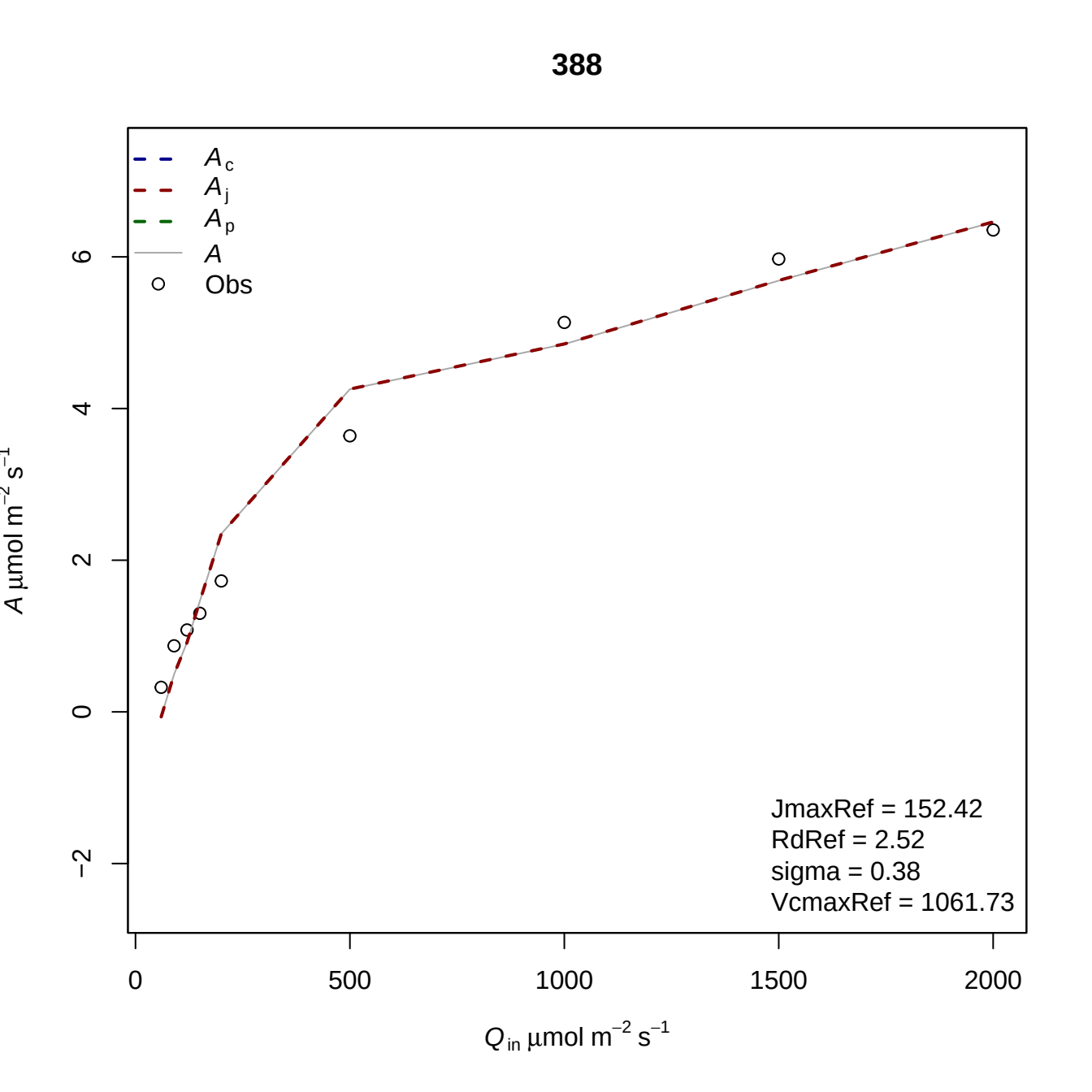

383
A_c
A_j
A_p
A
Obs
JmaxRef = 63.48
RdRef = 1.62
sigma = 0.32
VcmaxRef = 110.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

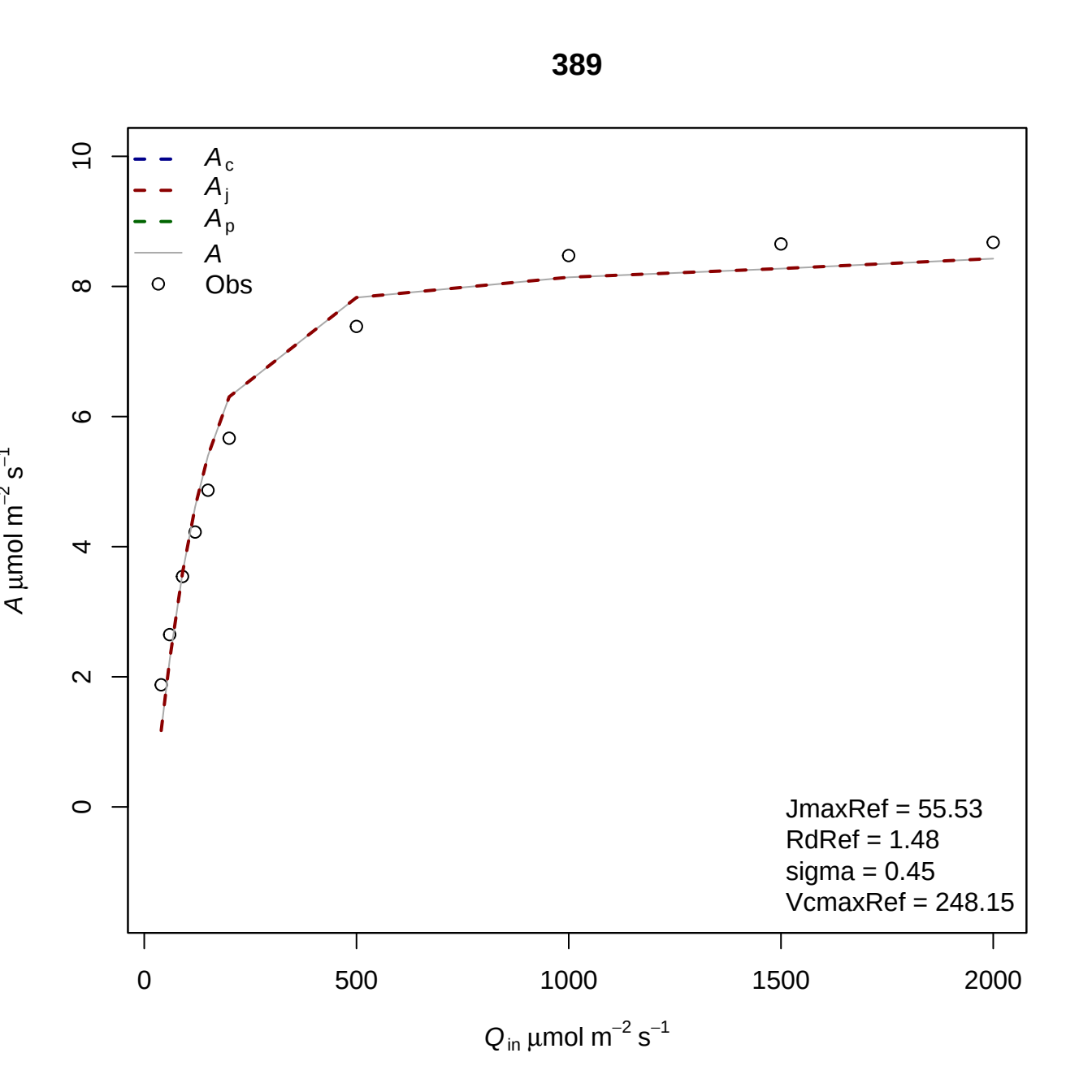

384
A_c
A_j
A_p
A
Obs
JmaxRef = 103.4
RdRef = 1.11
sigma = 0.22
VcmaxRef = 254.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

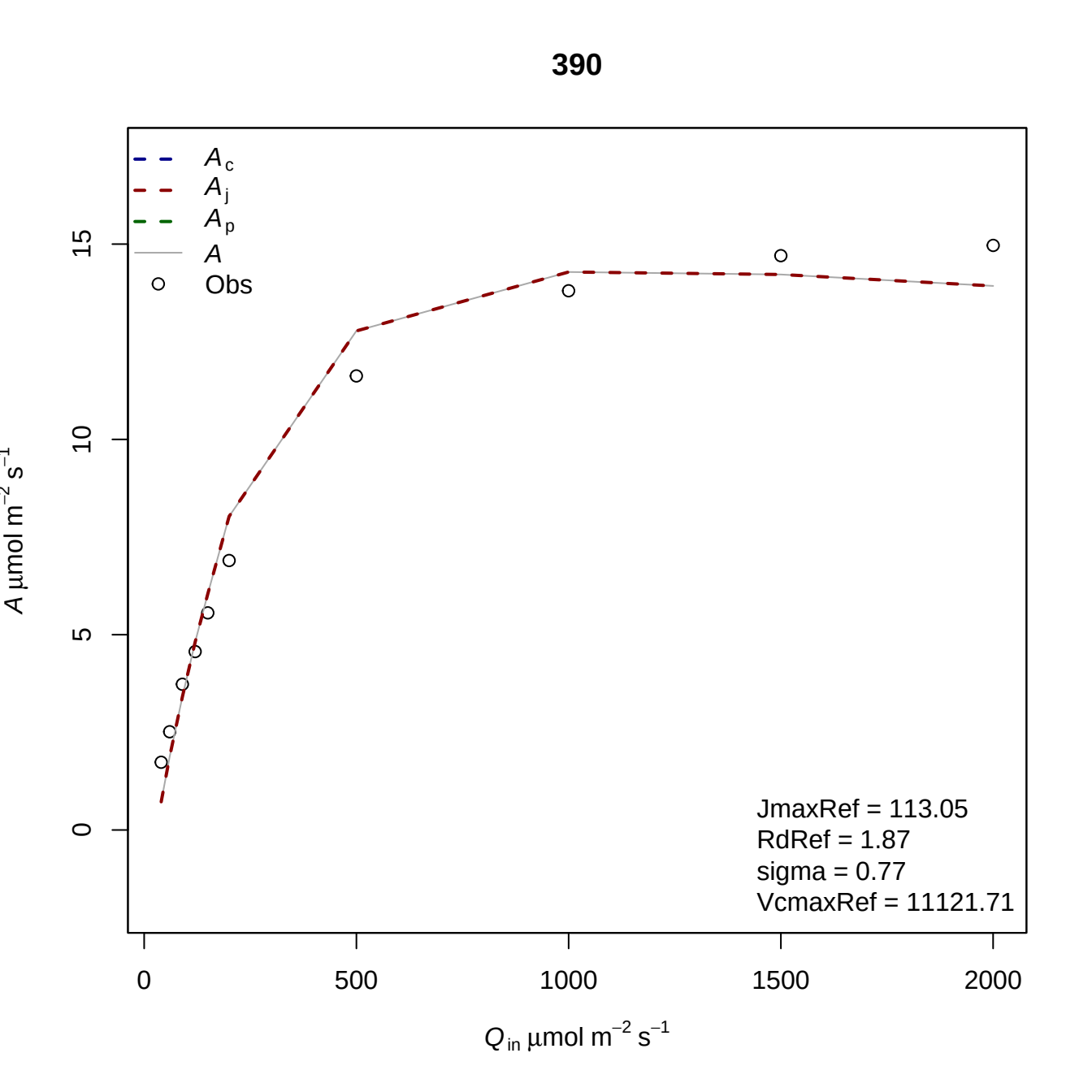

385
A_c
A_j
A_p
A
Obs
JmaxRef = 96.19
RdRef = 2.23
sigma = 0.46
VcmaxRef = 308.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

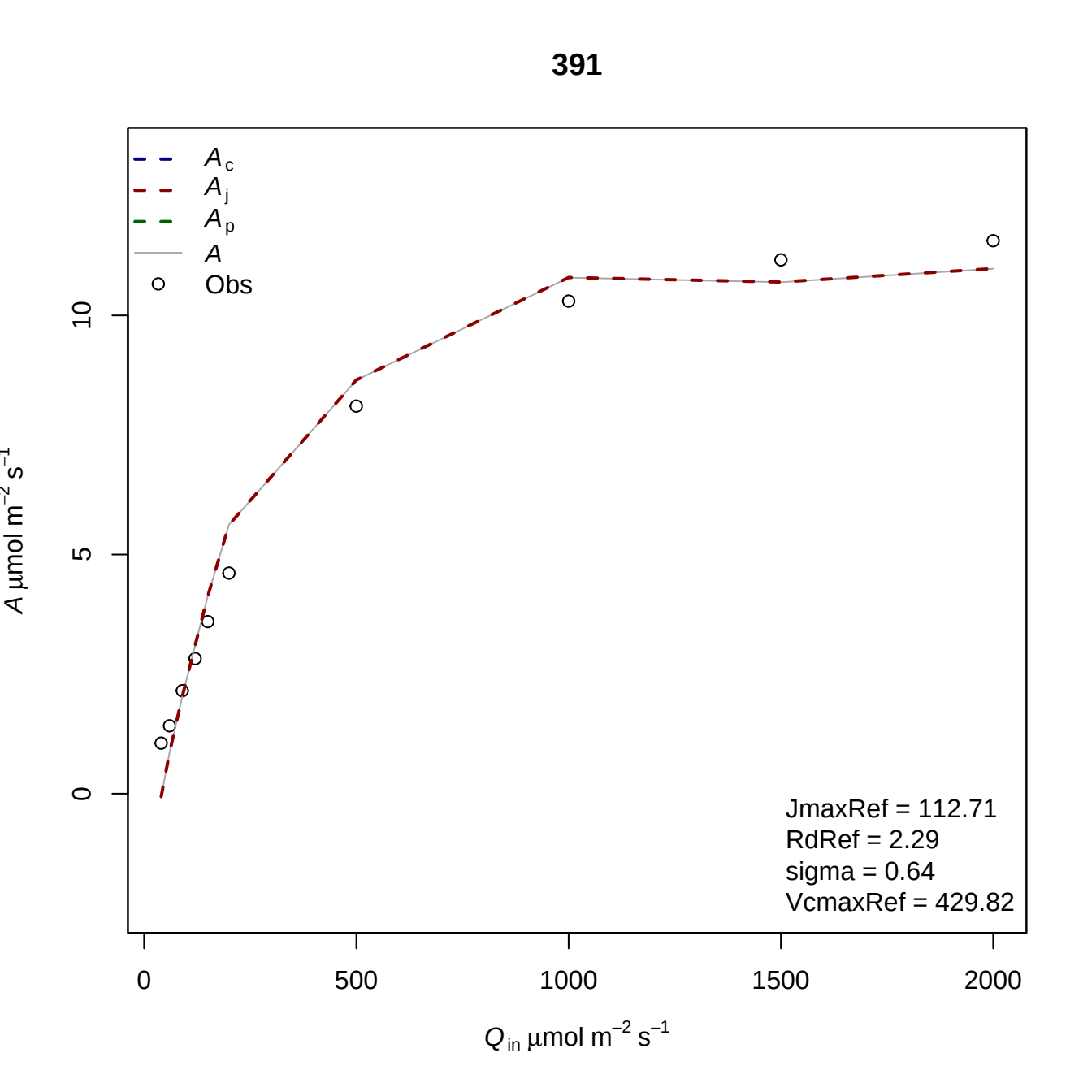

386
A_c
A_j
A_p
A
Obs
JmaxRef = 148.99
RdRef = 2.52
sigma = 0.49
VcmaxRef = 6519.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

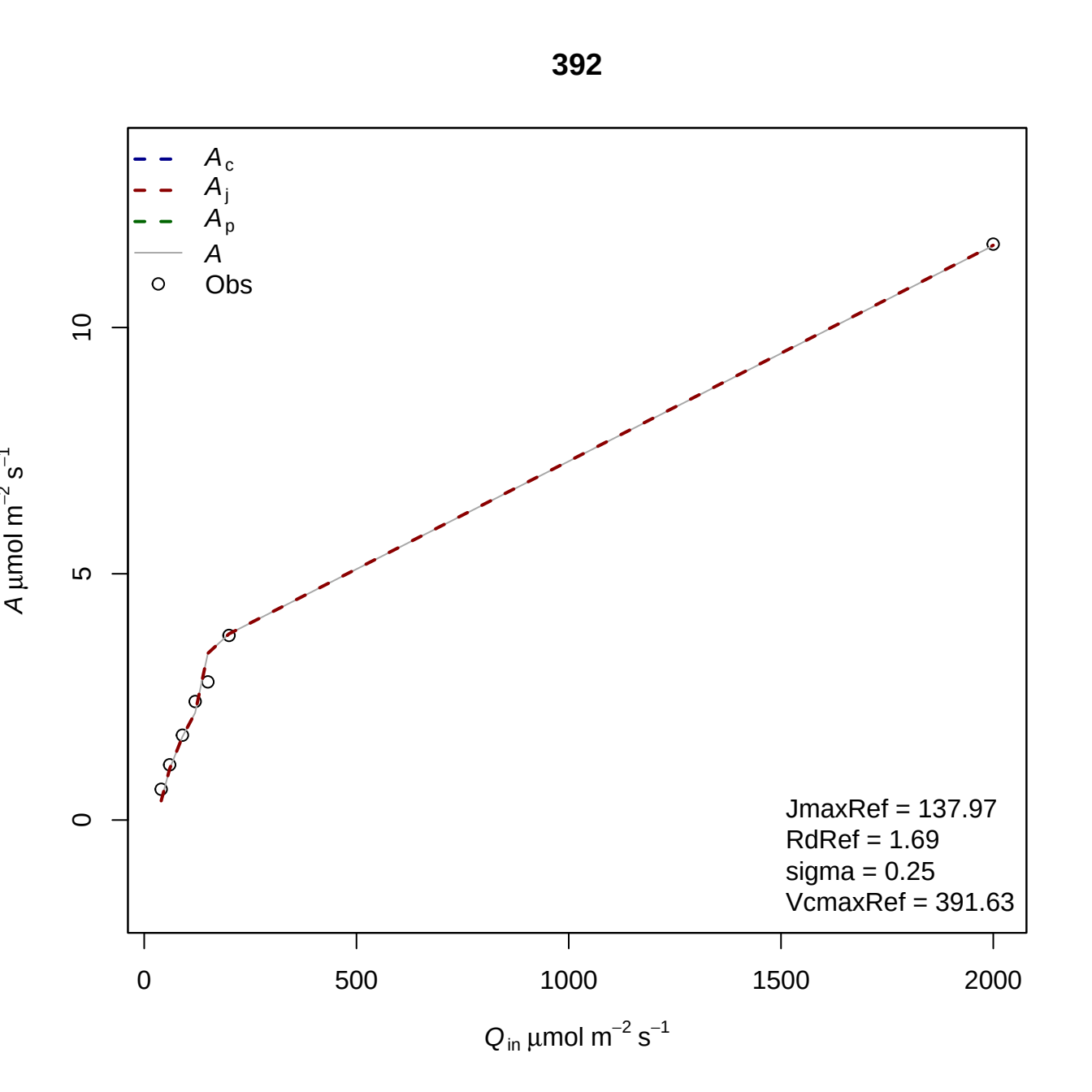

387
A_c
A_j
A_p
A
Obs
JmaxRef = 86.94
RdRef = 2.62
sigma = 0.85
VcmaxRef = 830.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

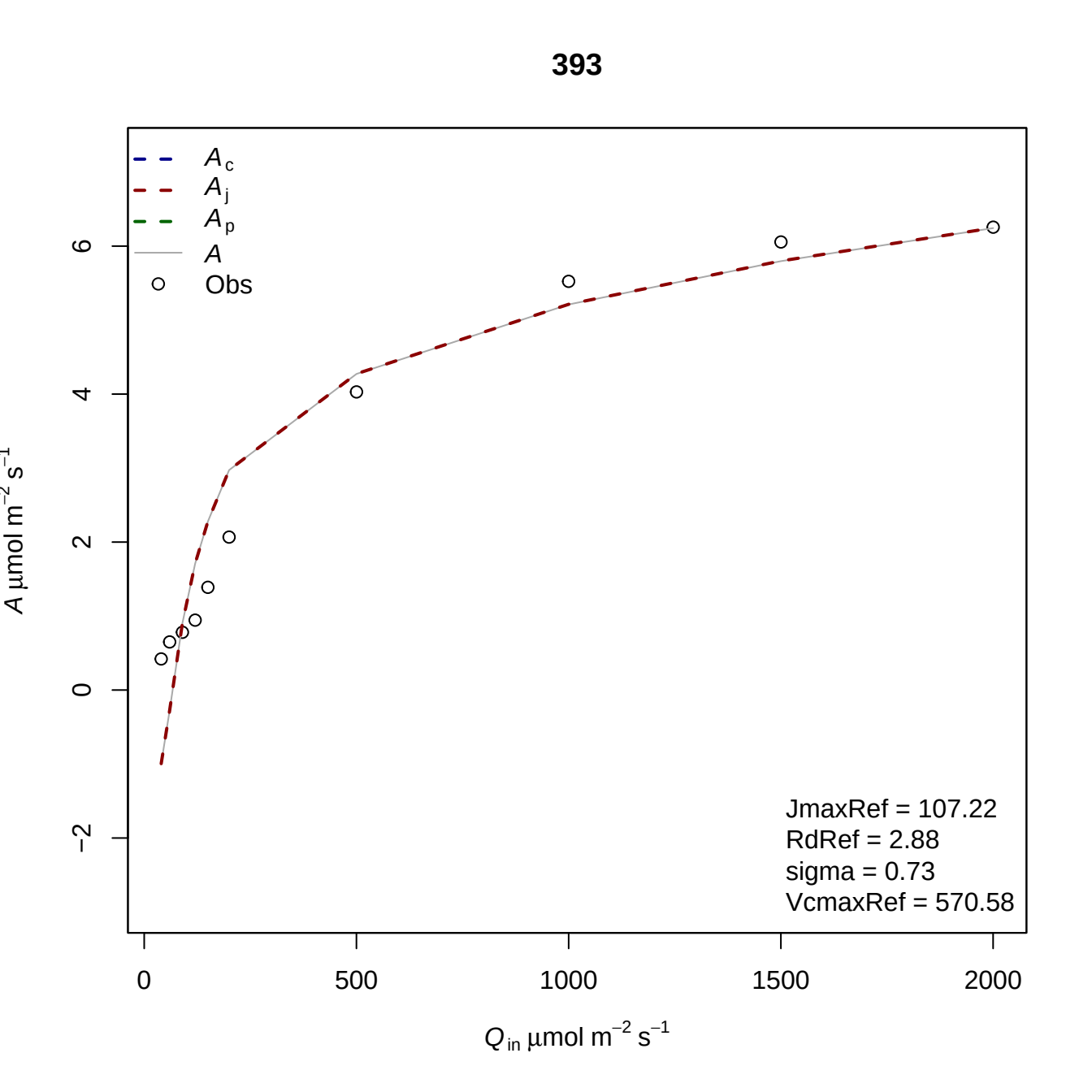

388
A_c
A_j
A_p
A
Obs
JmaxRef = 152.42
RdRef = 2.52
sigma = 0.38
VcmaxRef = 1061.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

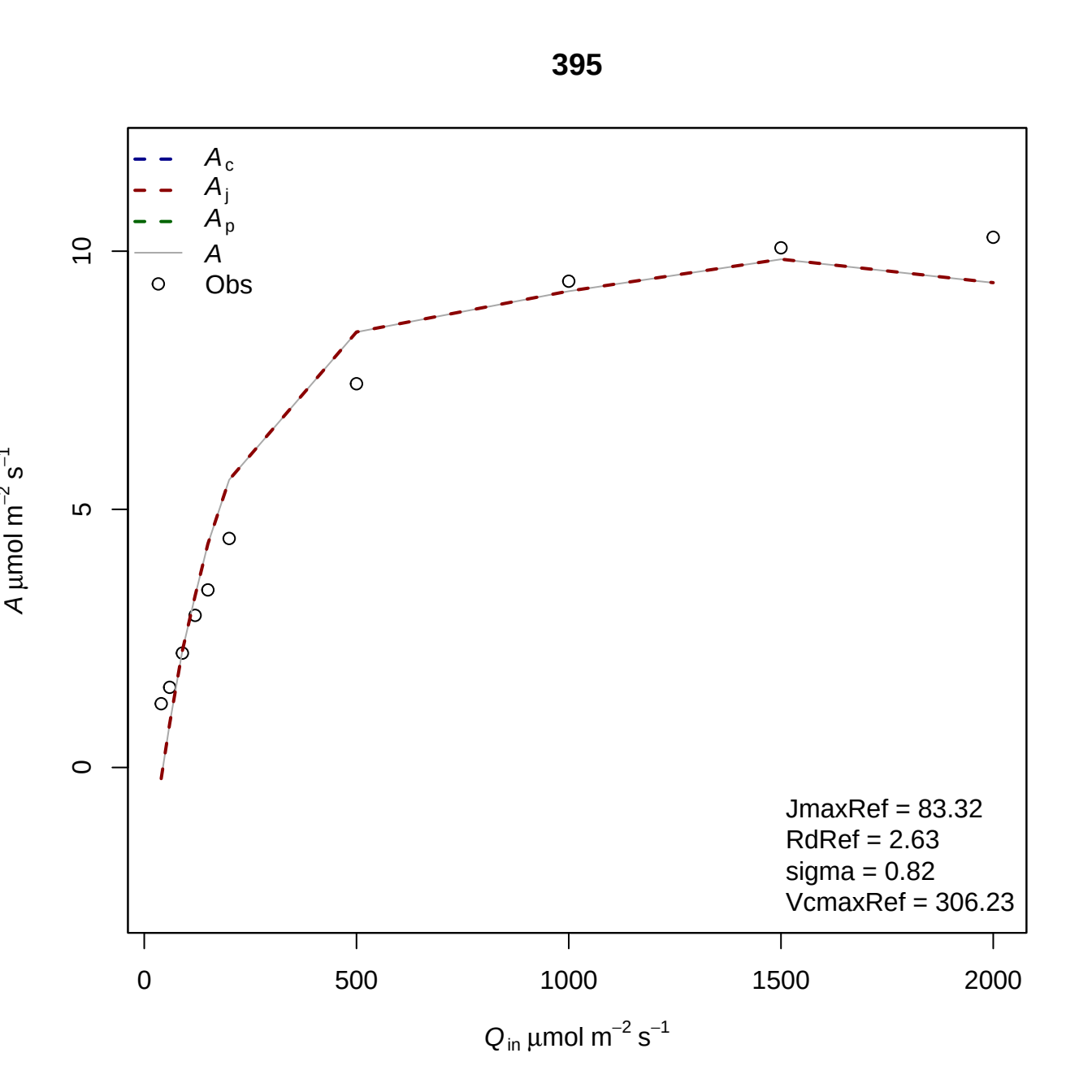

389
A_c
A_j
A_p
A
Obs
JmaxRef = 55.53
RdRef = 1.48
sigma = 0.45
VcmaxRef = 248.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

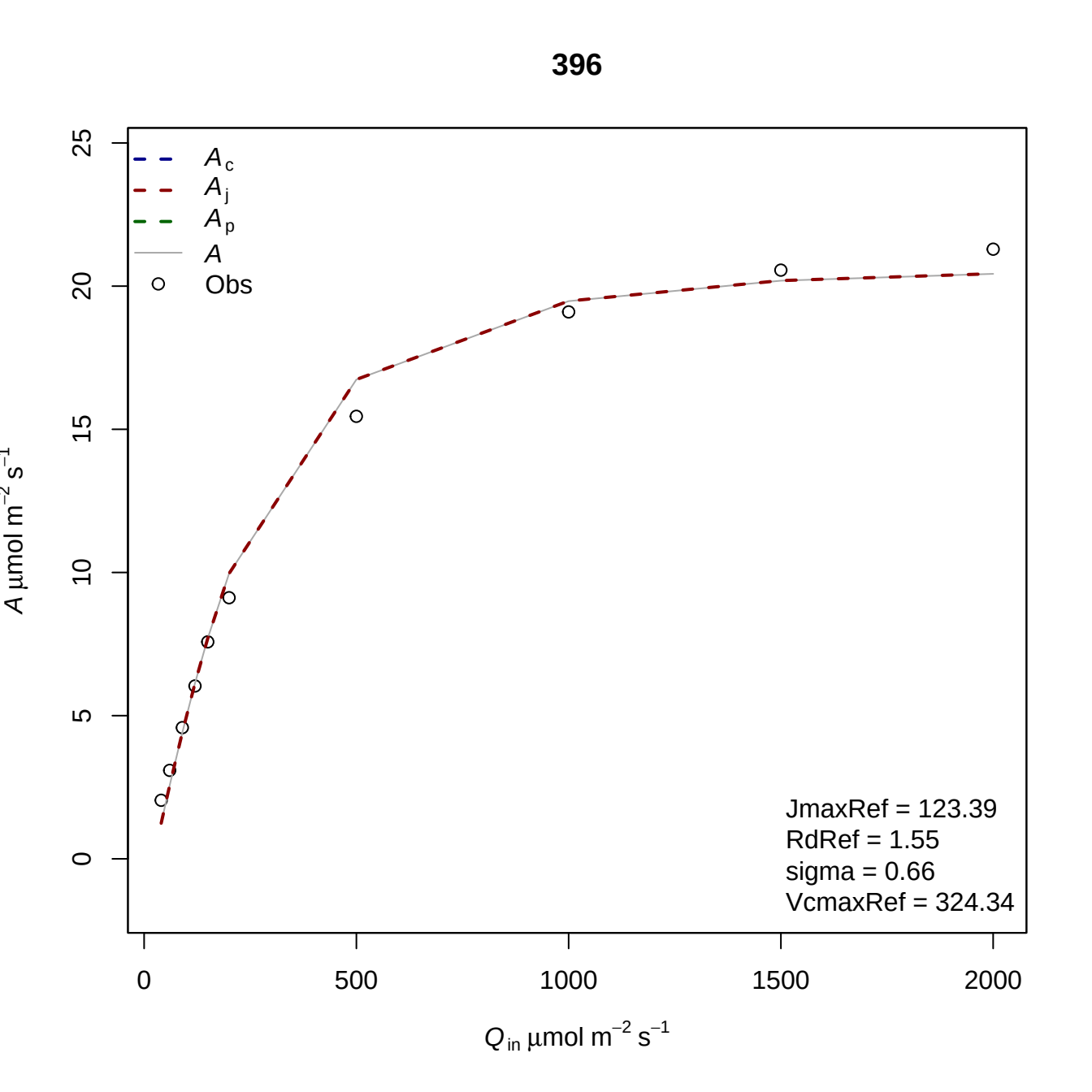

390
A_c
A_j
A_p
A
Obs
JmaxRef = 113.05
RdRef = 1.87
sigma = 0.77
VcmaxRef = 11121.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

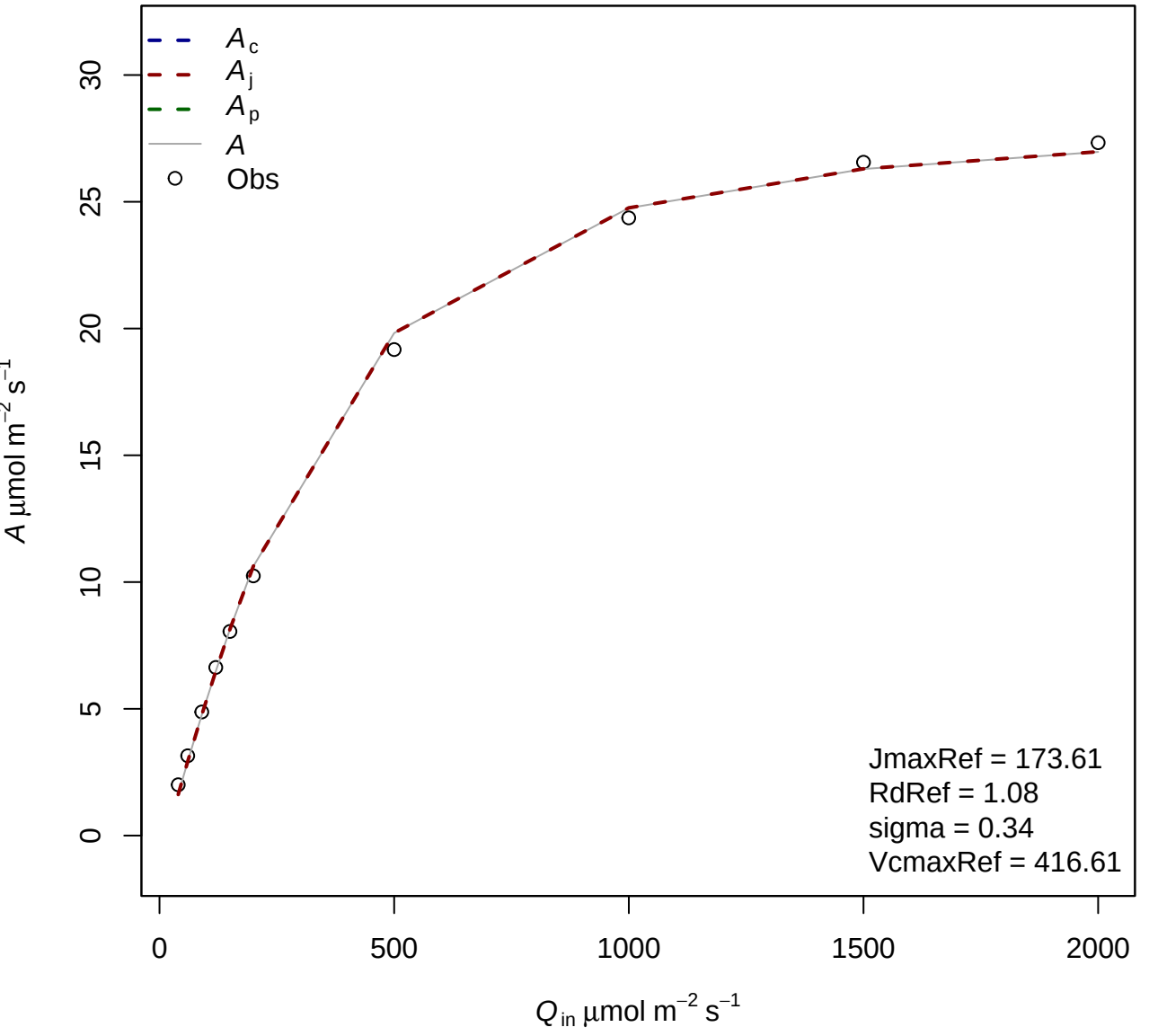

391
A_c
A_j
A_p
A
Obs
JmaxRef = 112.71
RdRef = 2.29
sigma = 0.64
VcmaxRef = 429.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

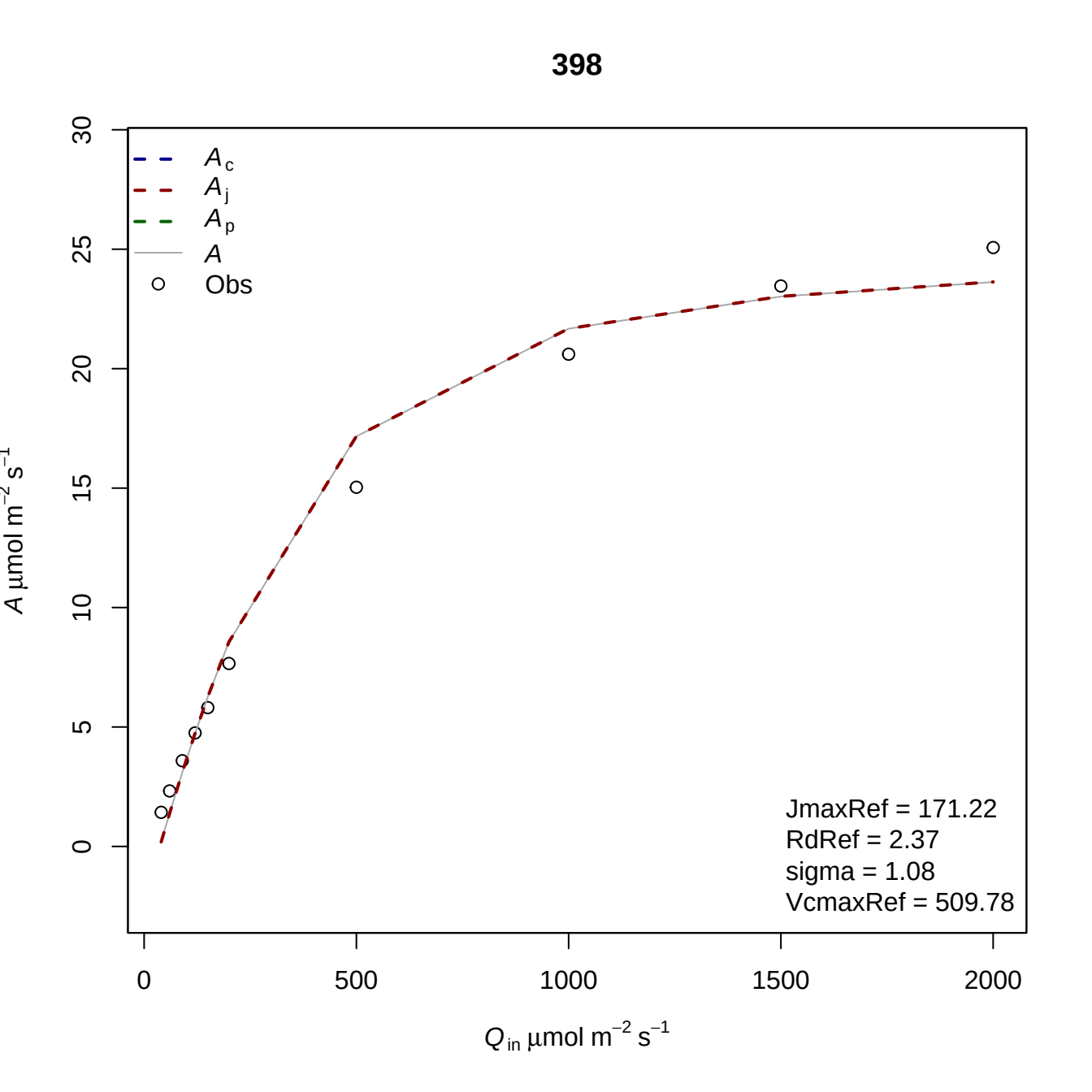

392
A_c
A_j
A_p
A
Obs
JmaxRef = 137.97
RdRef = 1.69
sigma = 0.25
VcmaxRef = 391.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

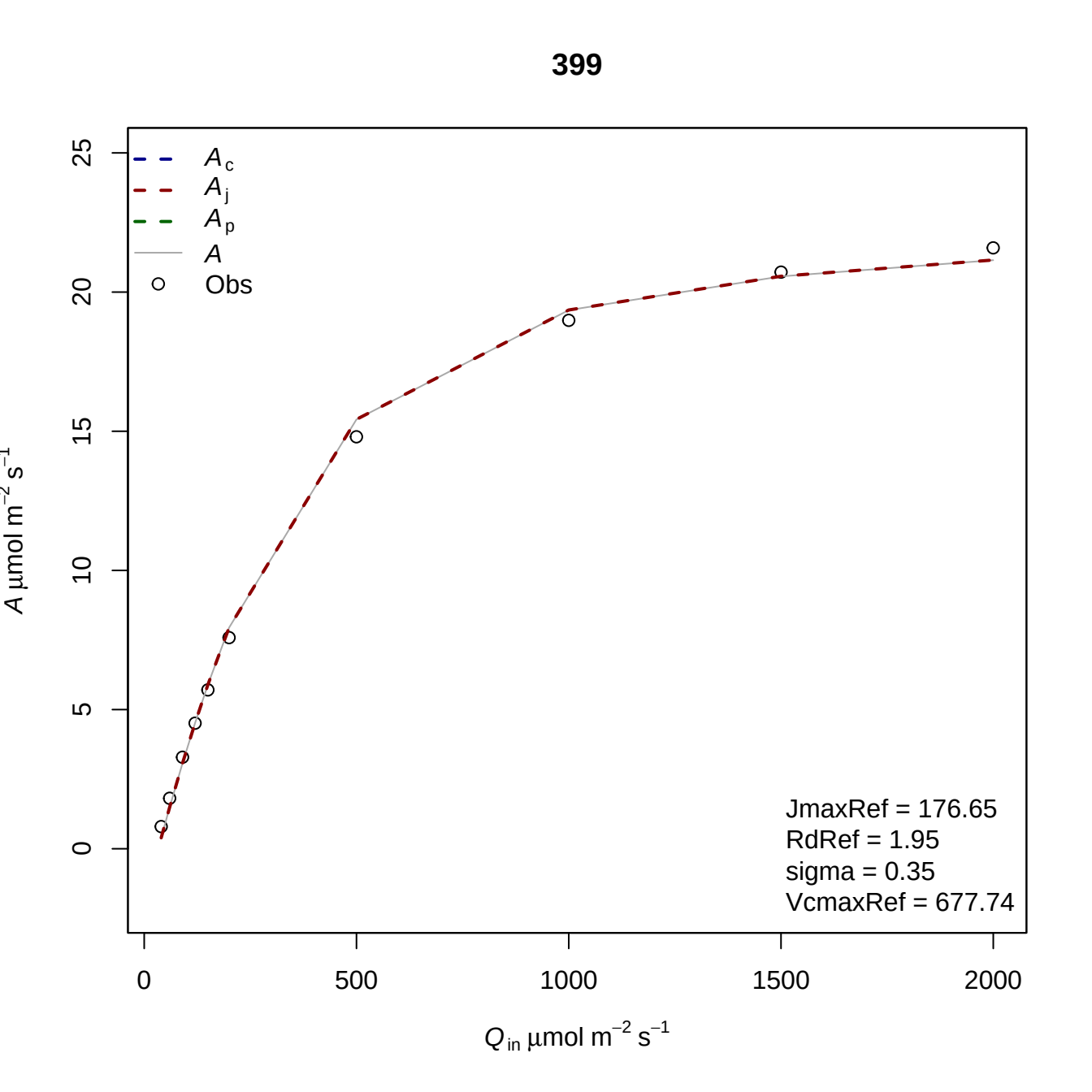

393
A_c
A_j
A_p
A
Obs
JmaxRef = 107.22
RdRef = 2.88
sigma = 0.73
VcmaxRef = 570.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

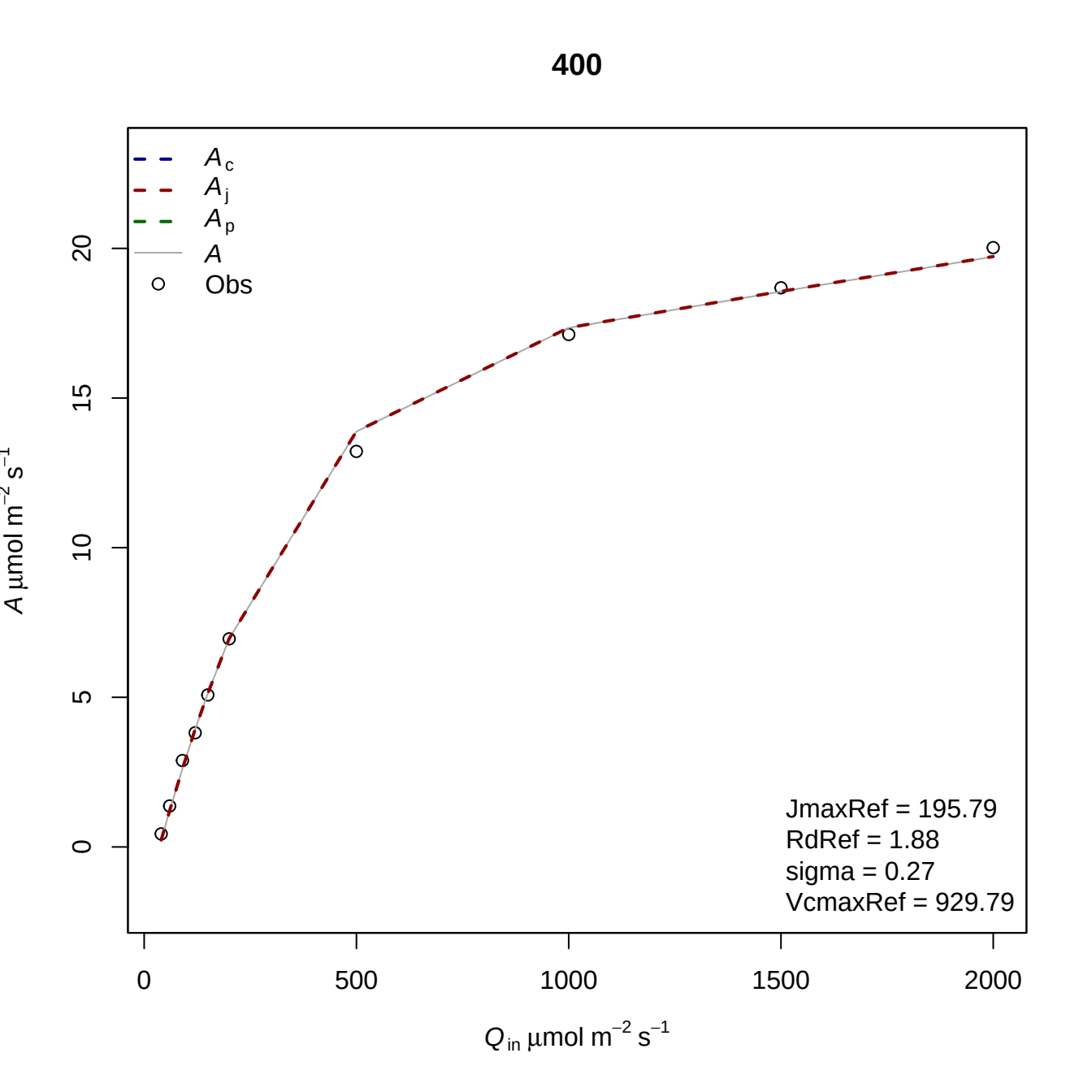

395
A_c
A_j
A_p
A
Obs
JmaxRef = 83.32
RdRef = 2.63
sigma = 0.82
VcmaxRef = 306.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

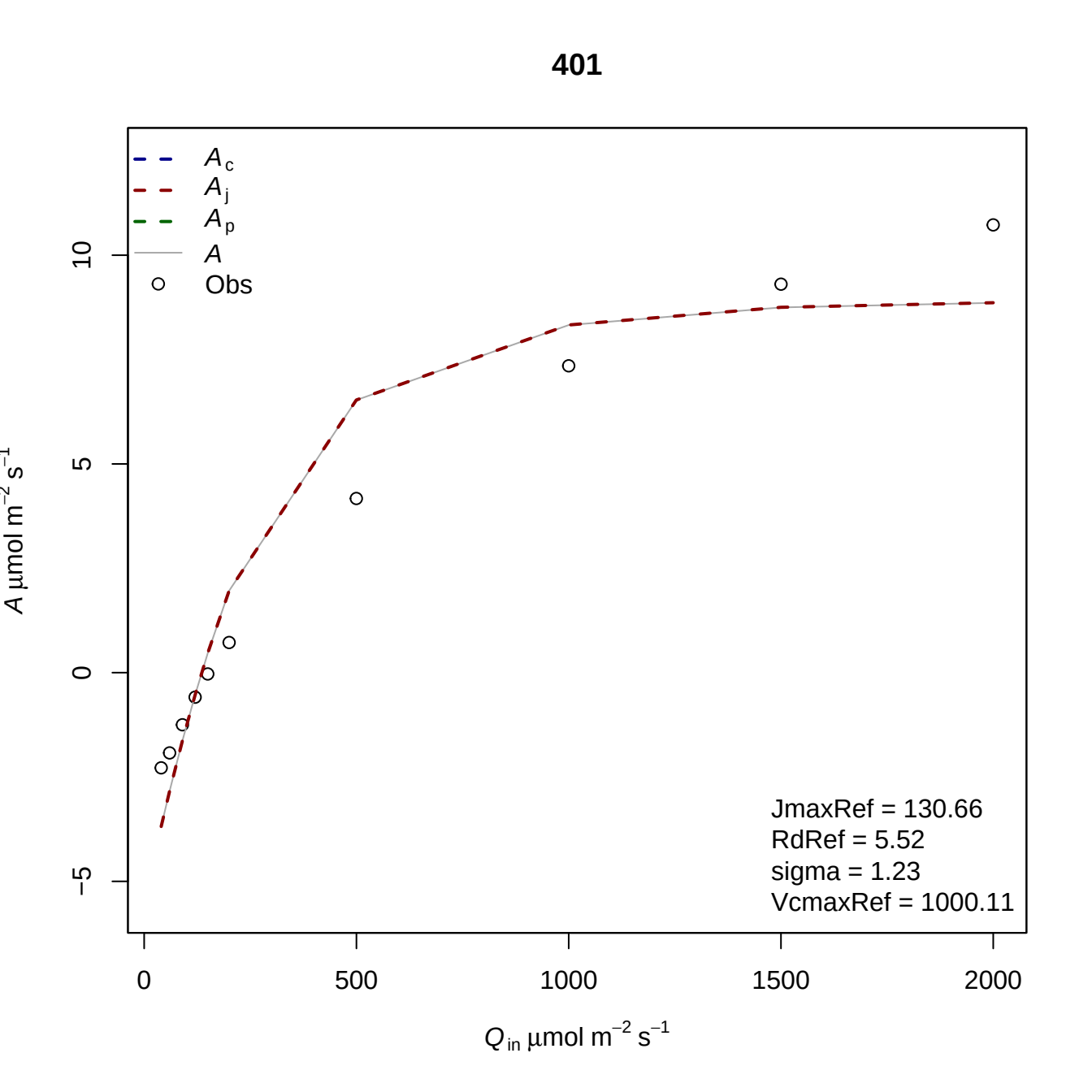

396
A_c
A_j
A_p
A
Obs
JmaxRef = 123.39
RdRef = 1.55
sigma = 0.66
VcmaxRef = 324.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

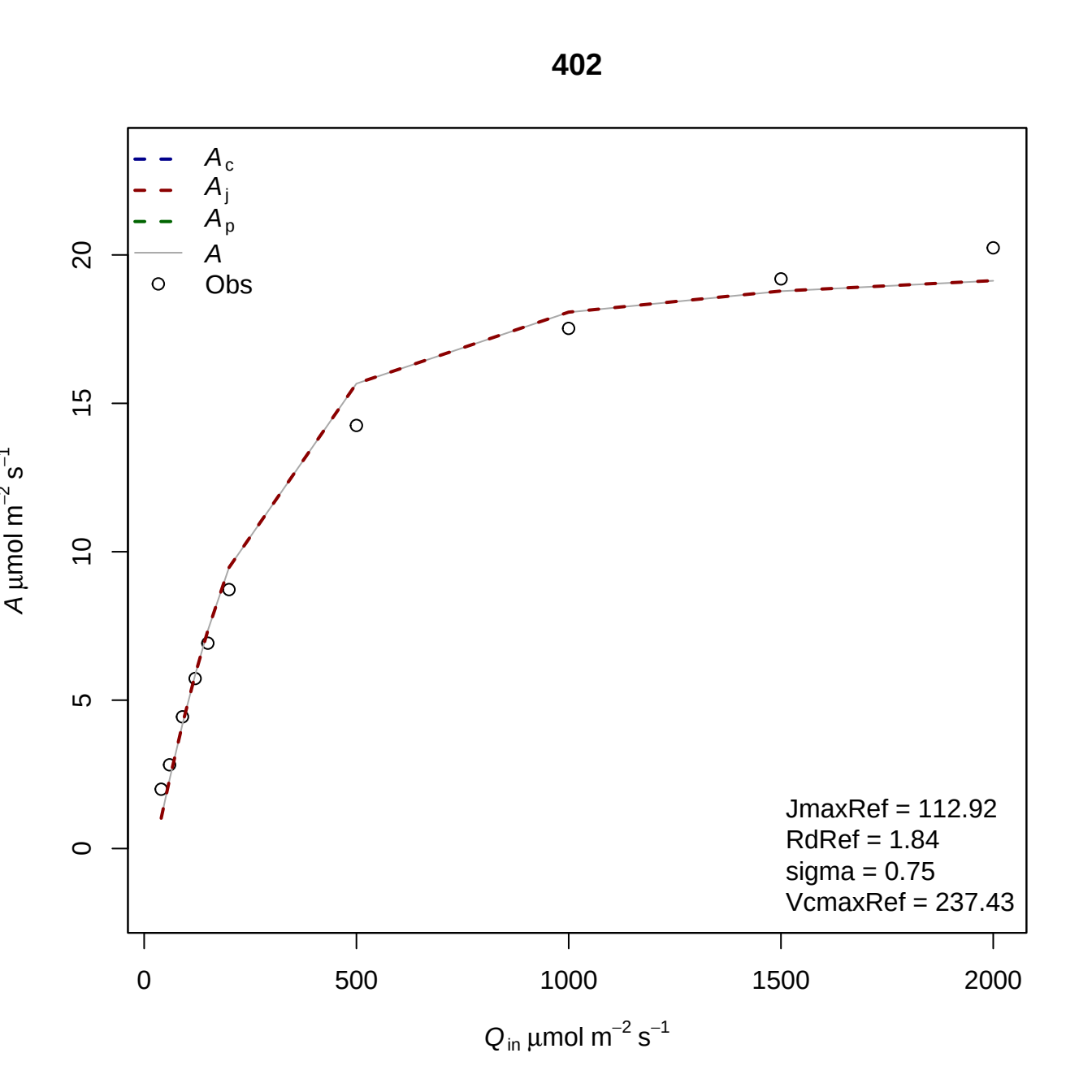

A_c
A_j
A_p
A
Obs
JmaxRef = 173.61
RdRef = 1.08
sigma = 0.34
VcmaxRef = 416.61
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

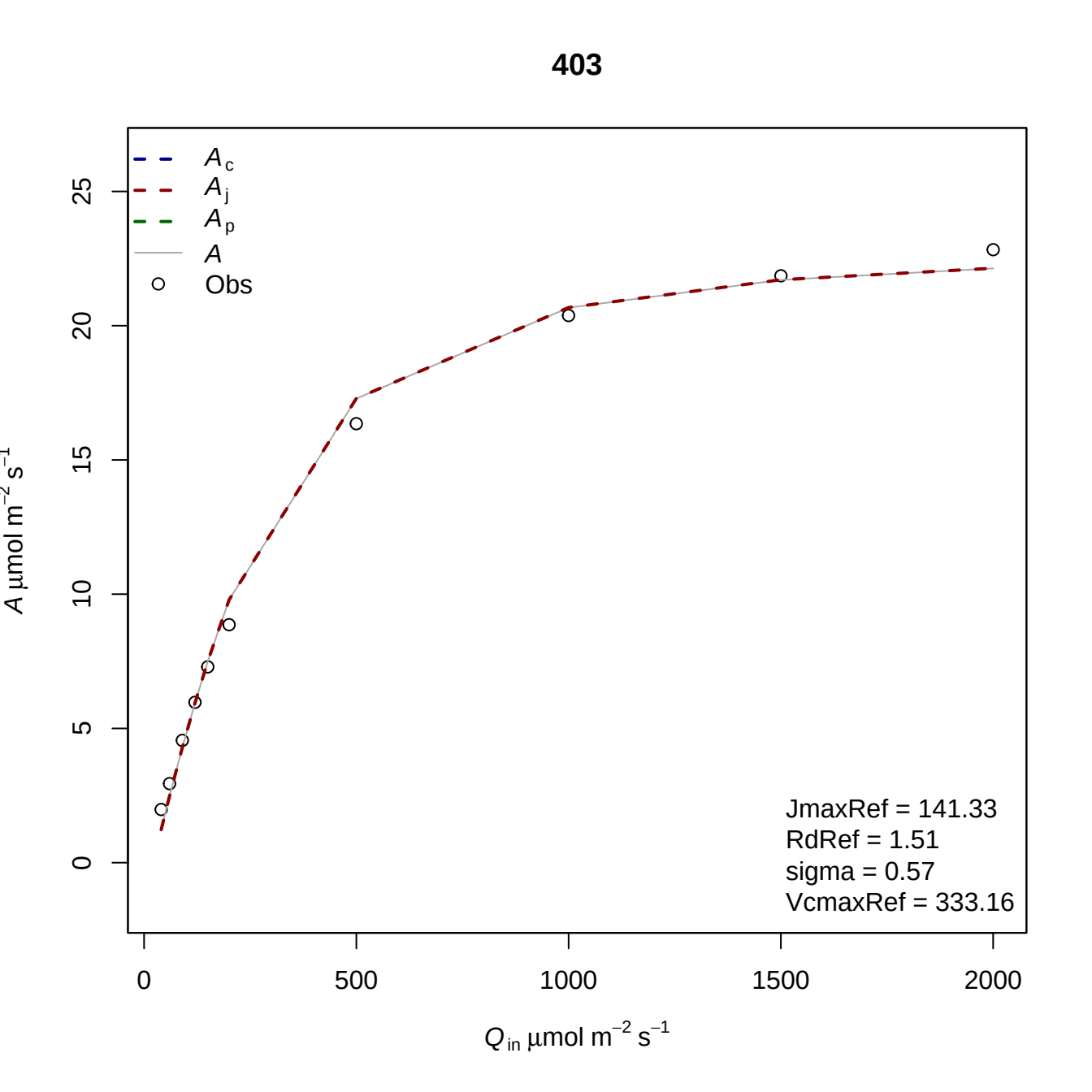

398
A_c
A_j
A_p
A
Obs
JmaxRef = 171.22
RdRef = 2.37
sigma = 1.08
VcmaxRef = 509.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

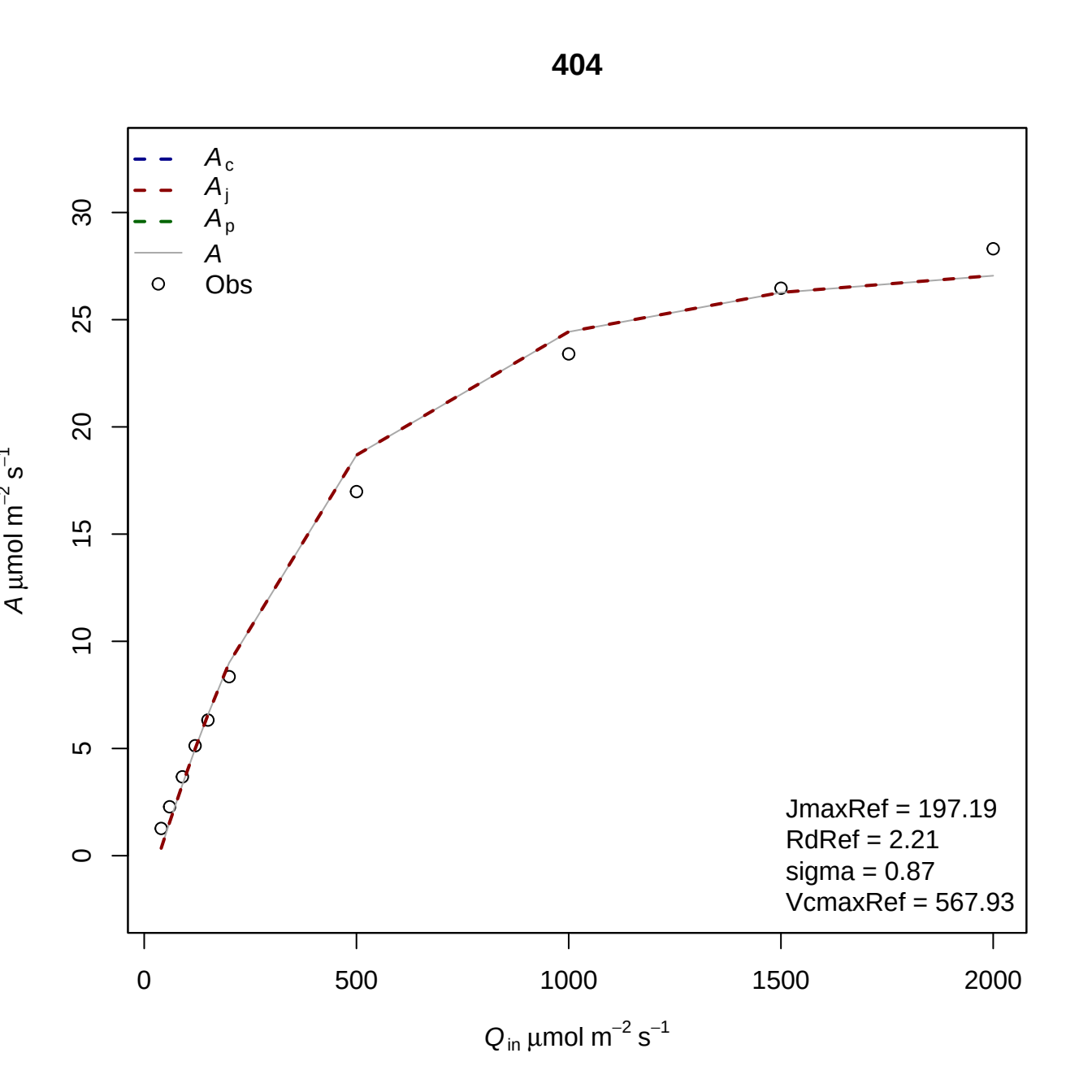

399
A_c
A_j
A_p
A
Obs
JmaxRef = 176.65
RdRef = 1.95
sigma = 0.35
VcmaxRef = 677.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

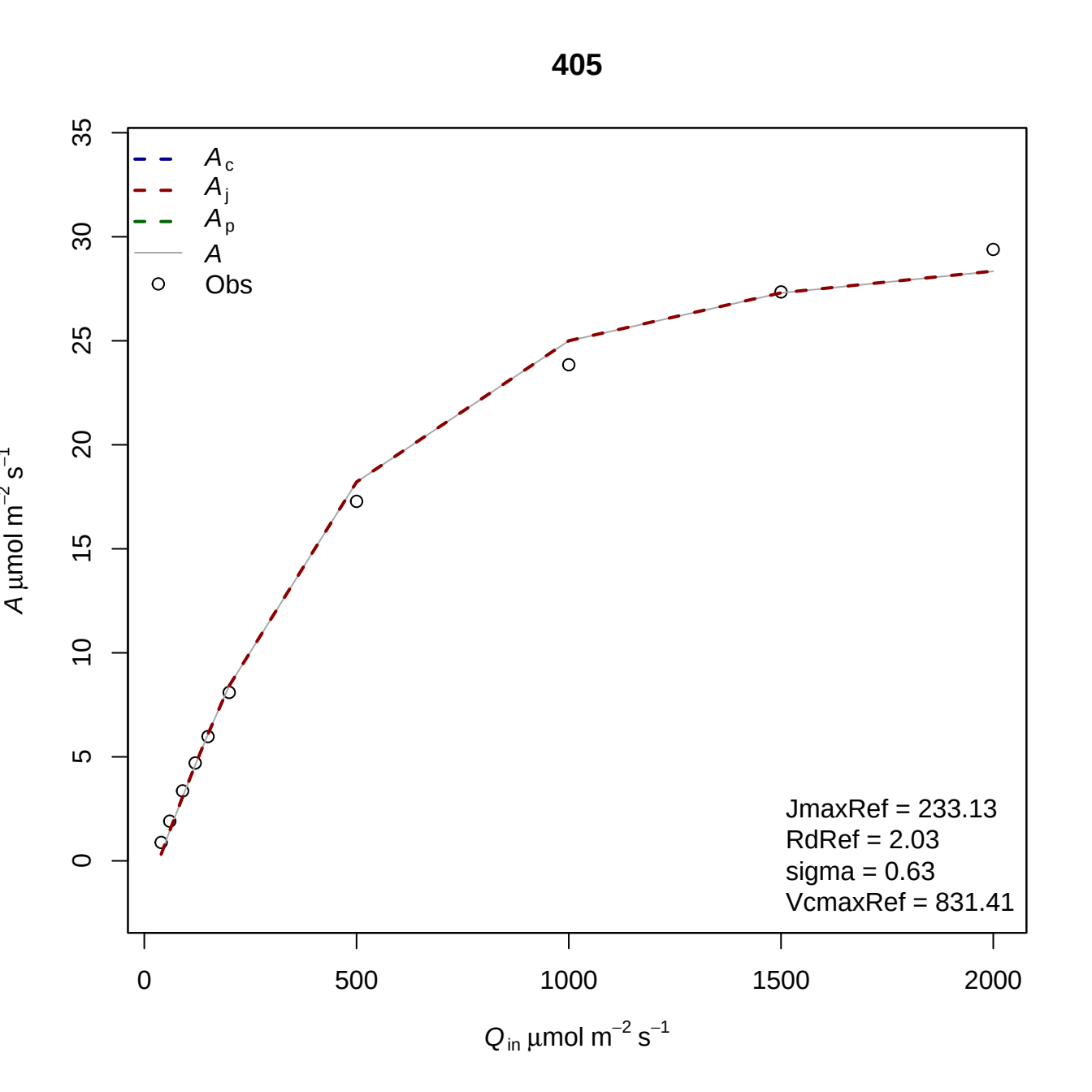

400
A_c
A_j
A_p
A
Obs
JmaxRef = 195.79
RdRef = 1.88
sigma = 0.27
VcmaxRef = 929.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

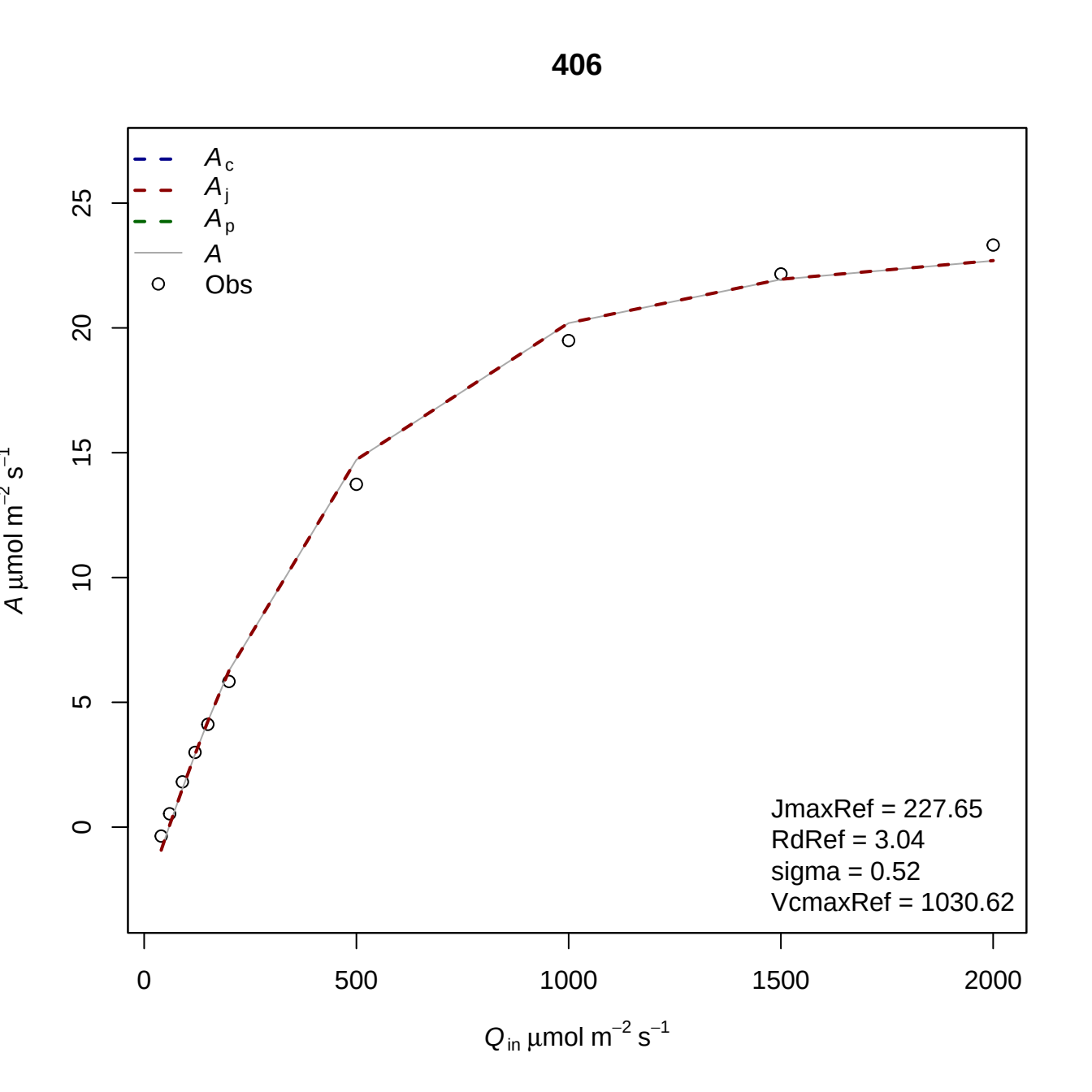

401
A_c
A_j
A_p
A
Obs
JmaxRef = 130.66
RdRef = 5.52
sigma = 1.23
VcmaxRef = 1000.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

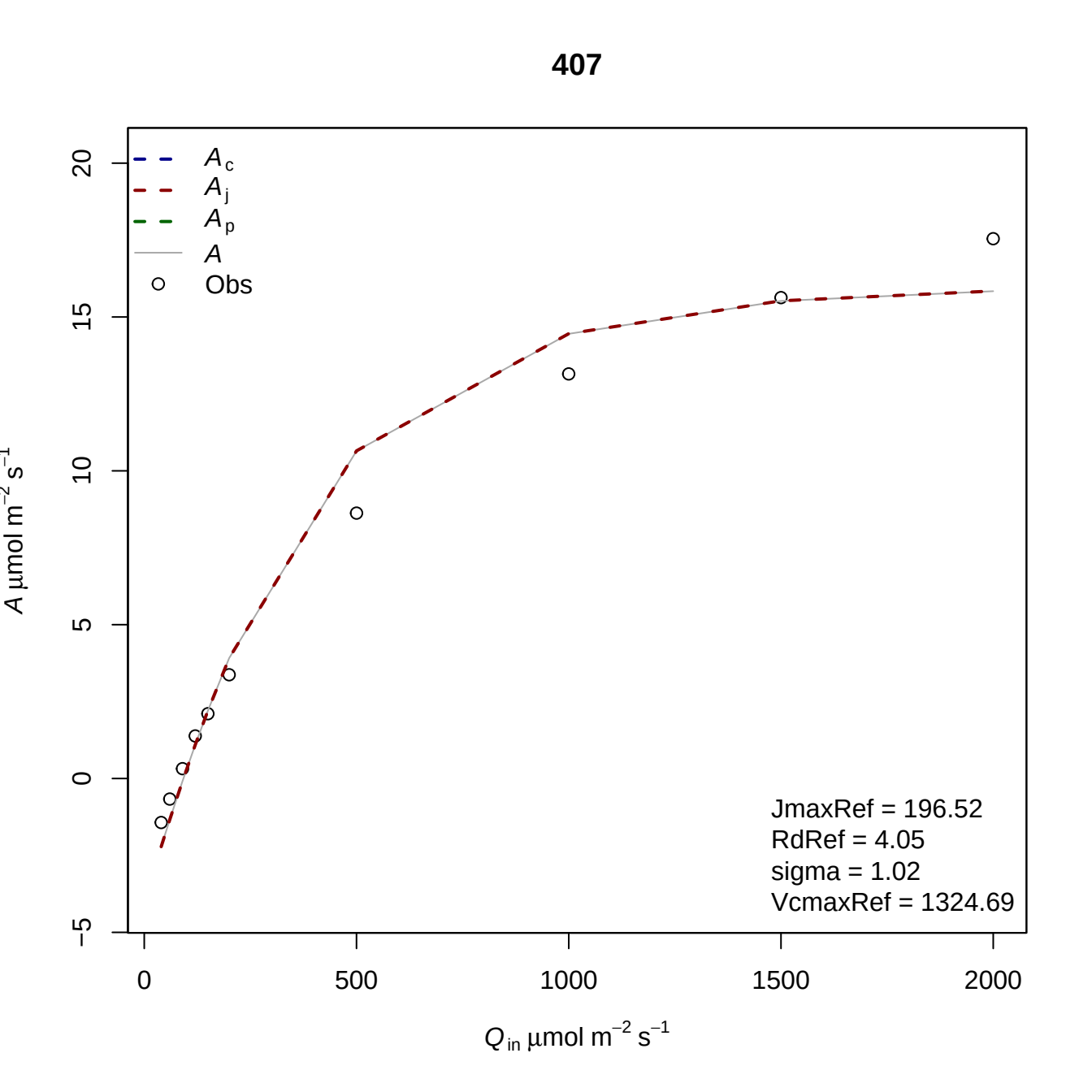

402
A_c
A_j
A_p
A
Obs
JmaxRef = 112.92
RdRef = 1.84
sigma = 0.75
VcmaxRef = 237.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

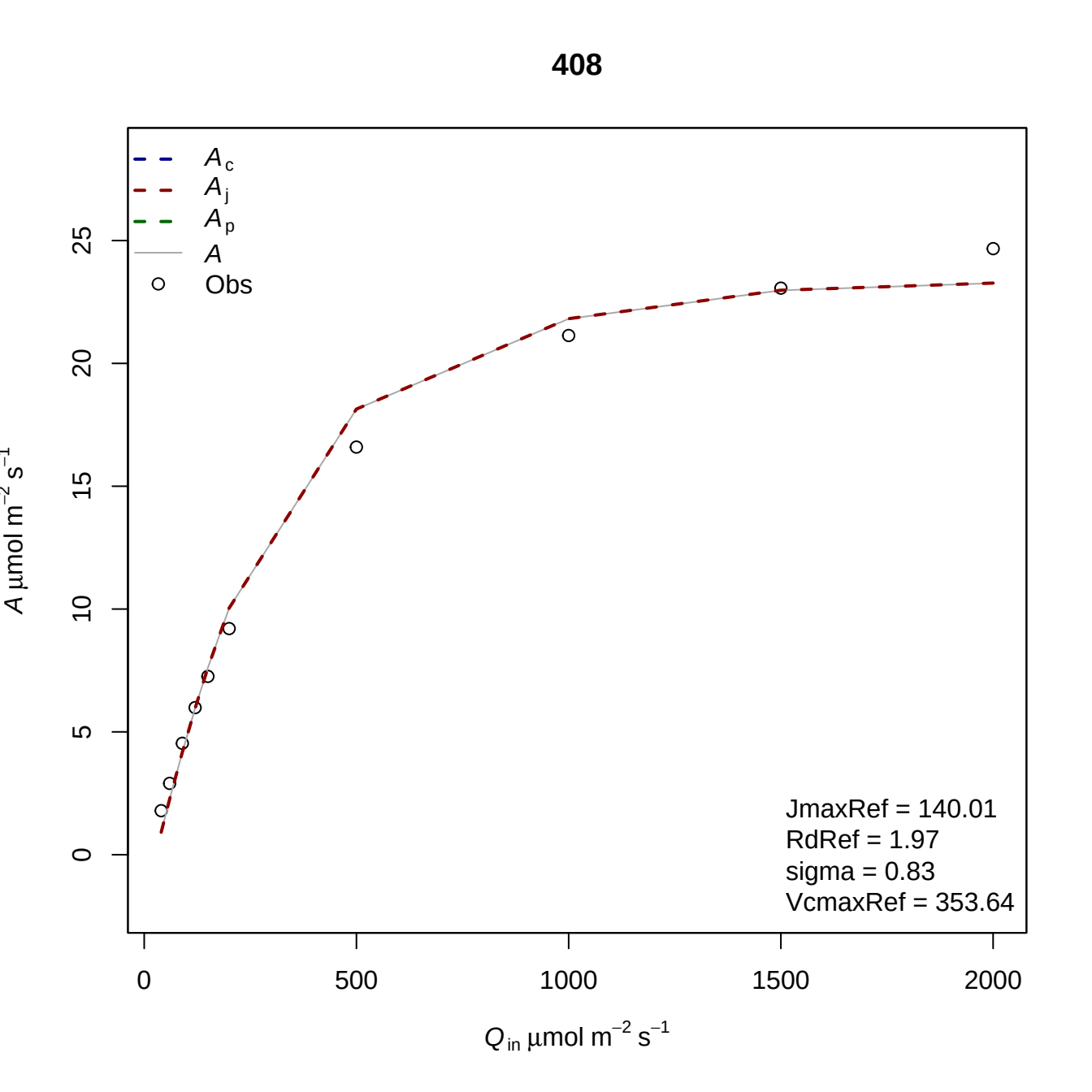

403
A_c
A_j
A_p
A
Obs
JmaxRef = 141.33
RdRef = 1.51
sigma = 0.57
VcmaxRef = 333.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

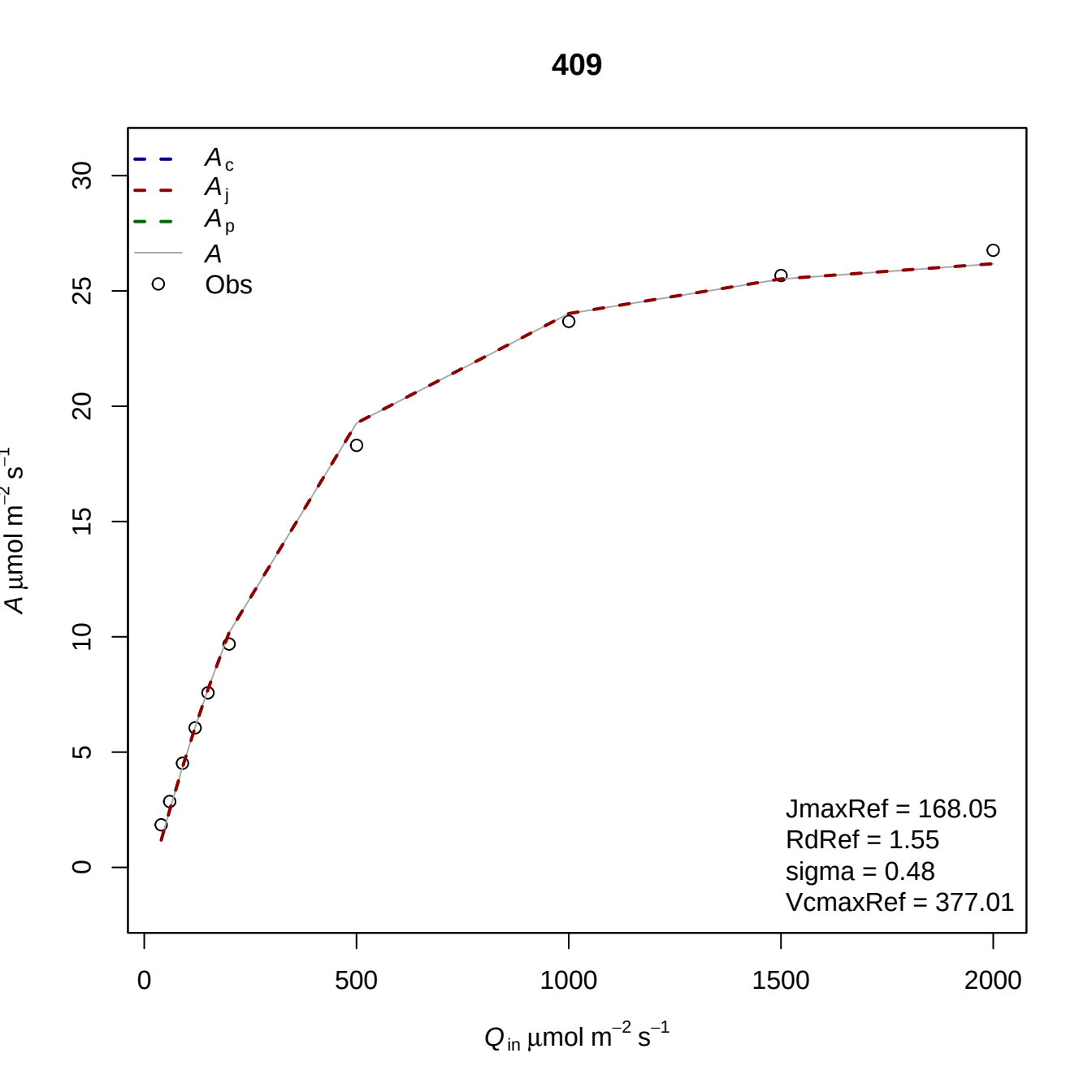

404
A_c
A_j
A_p
A
Obs
JmaxRef = 197.19
RdRef = 2.21
sigma = 0.87
VcmaxRef = 567.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

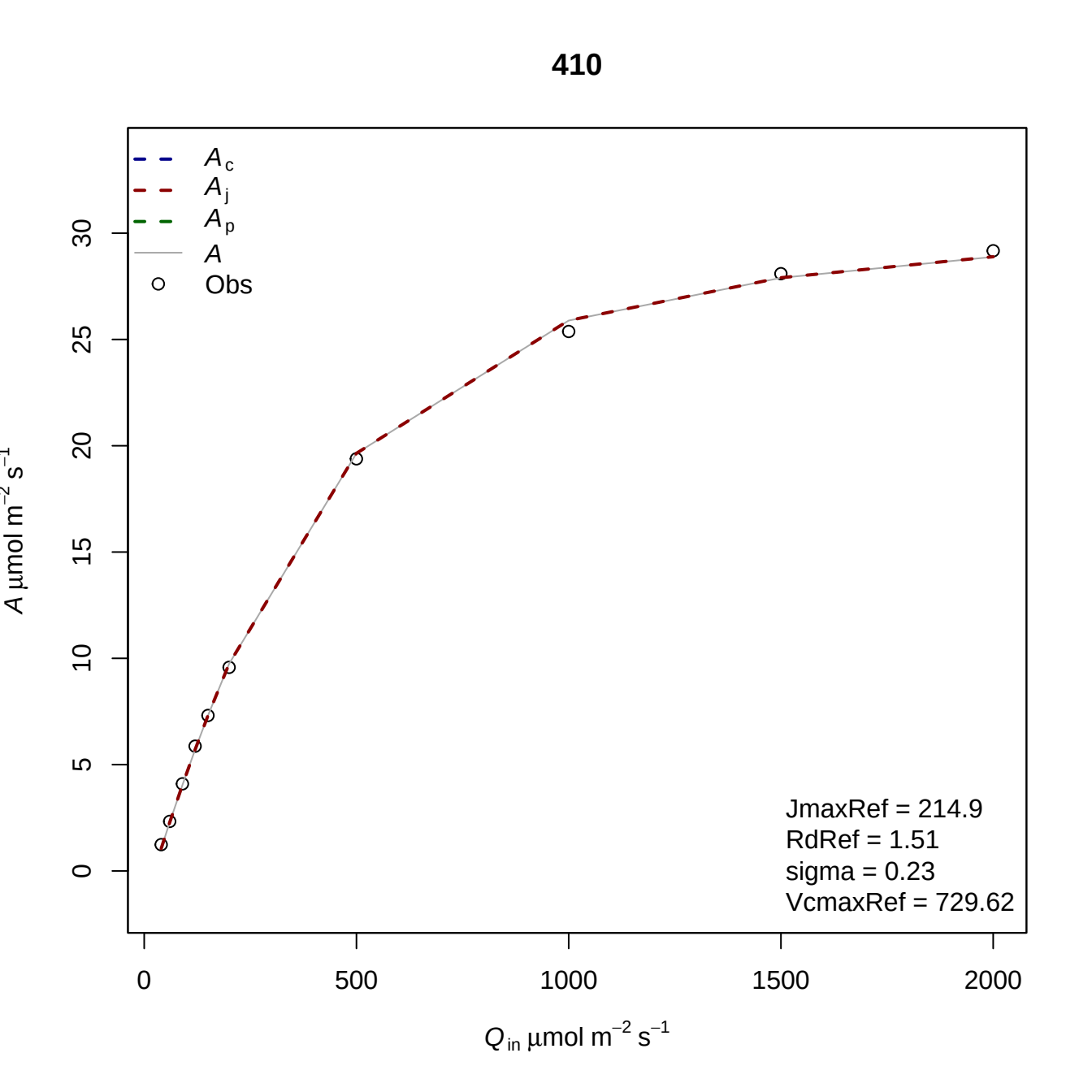

405
A_c
A_j
A_p
A
Obs
JmaxRef = 233.13
RdRef = 2.03
sigma = 0.63
VcmaxRef = 831.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

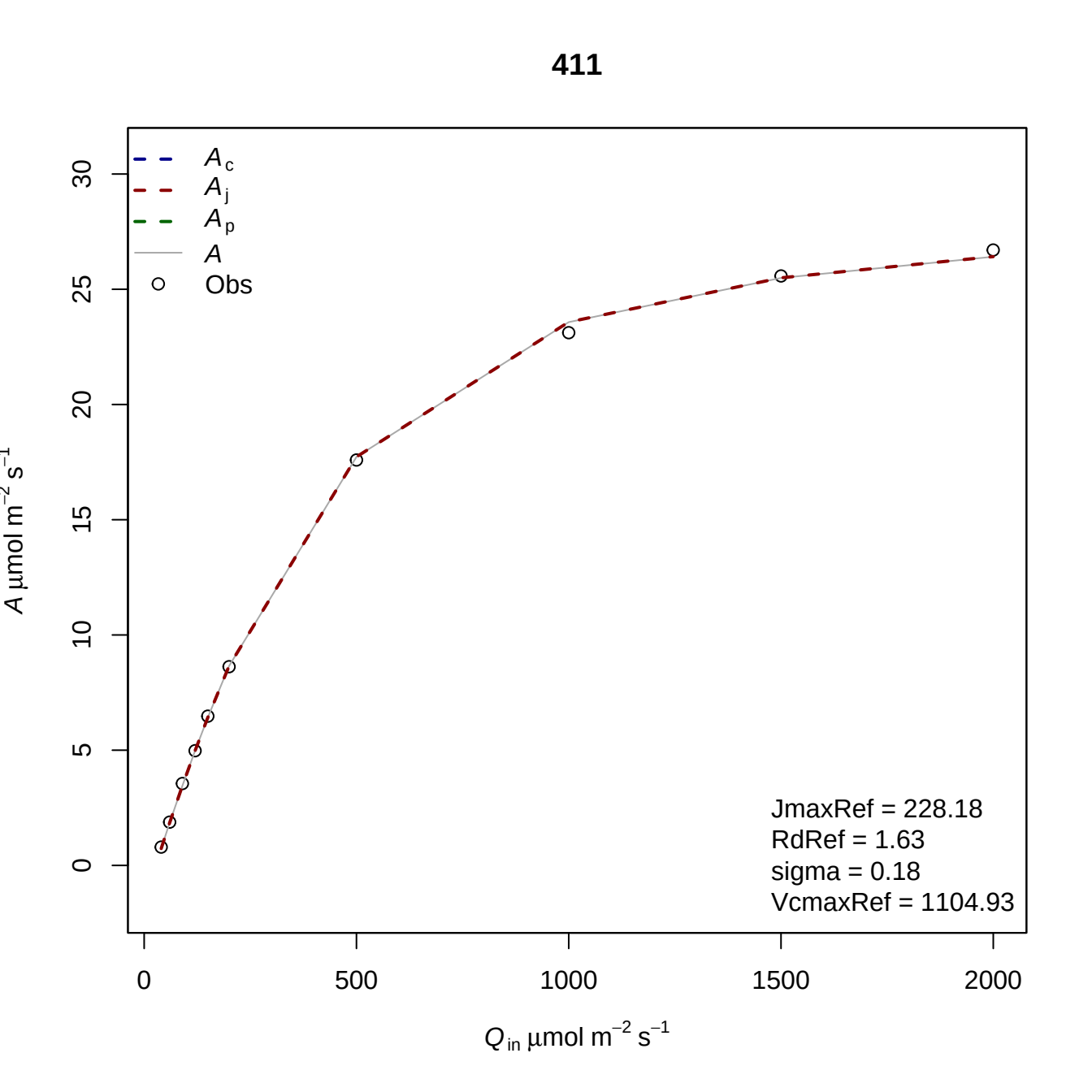

406
A_c
A_j
A_p
A
Obs
JmaxRef = 227.65
RdRef = 3.04
sigma = 0.52
VcmaxRef = 1030.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

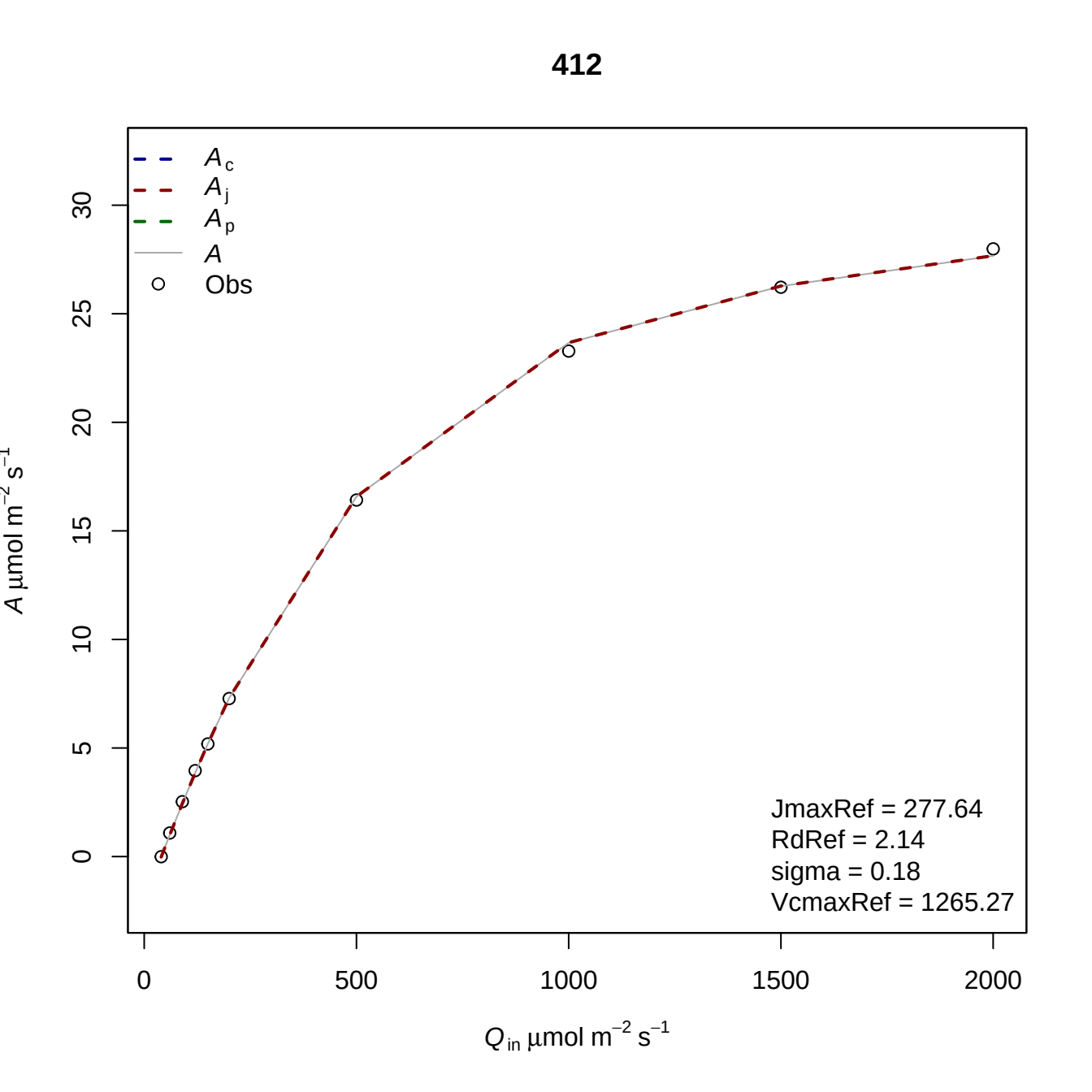

407
A_c
A_j
A_p
A
Obs
JmaxRef = 196.52
RdRef = 4.05
sigma = 1.02
VcmaxRef = 1324.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

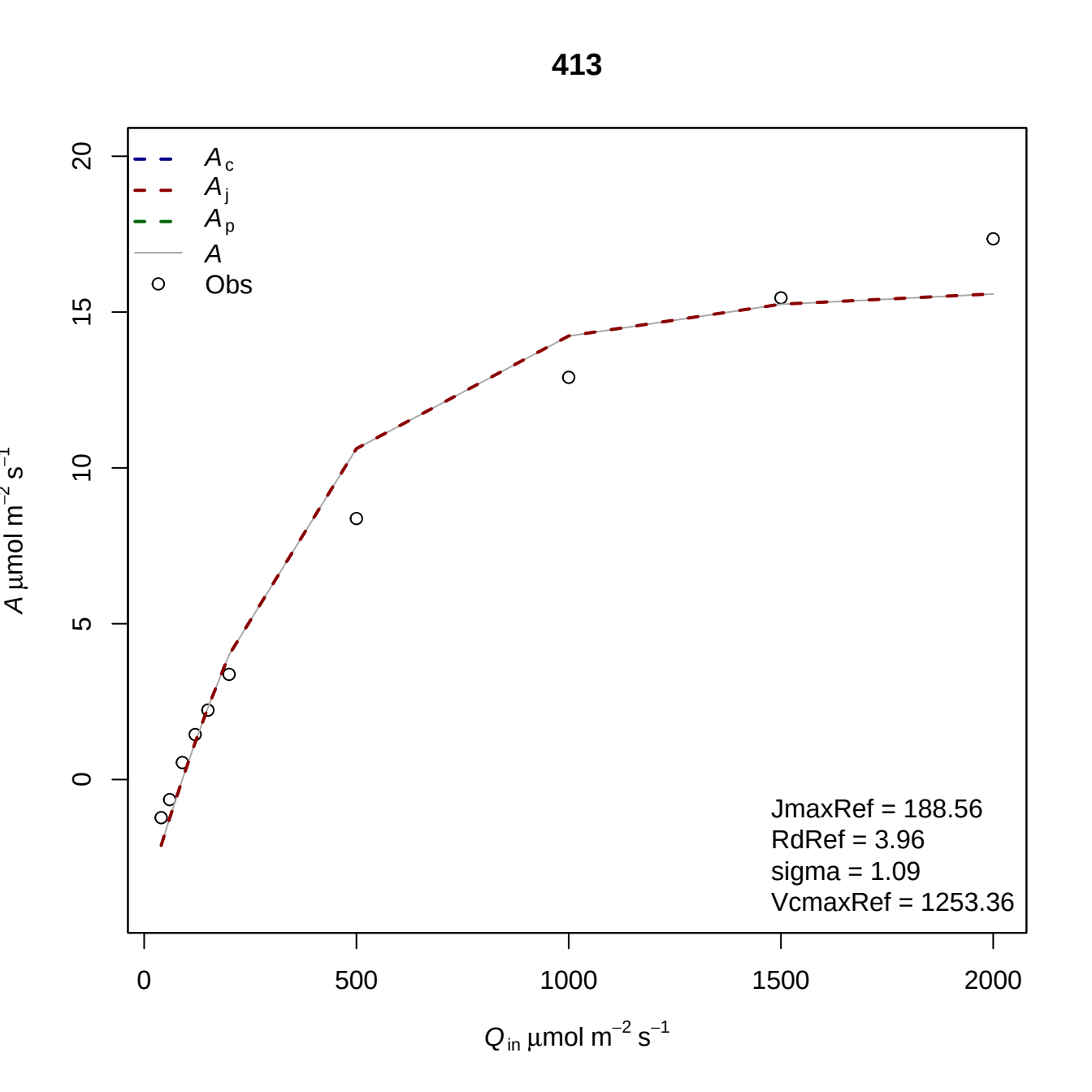

408
A_c
A_j
A_p
A
Obs
JmaxRef = 140.01
RdRef = 1.97
sigma = 0.83
VcmaxRef = 353.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

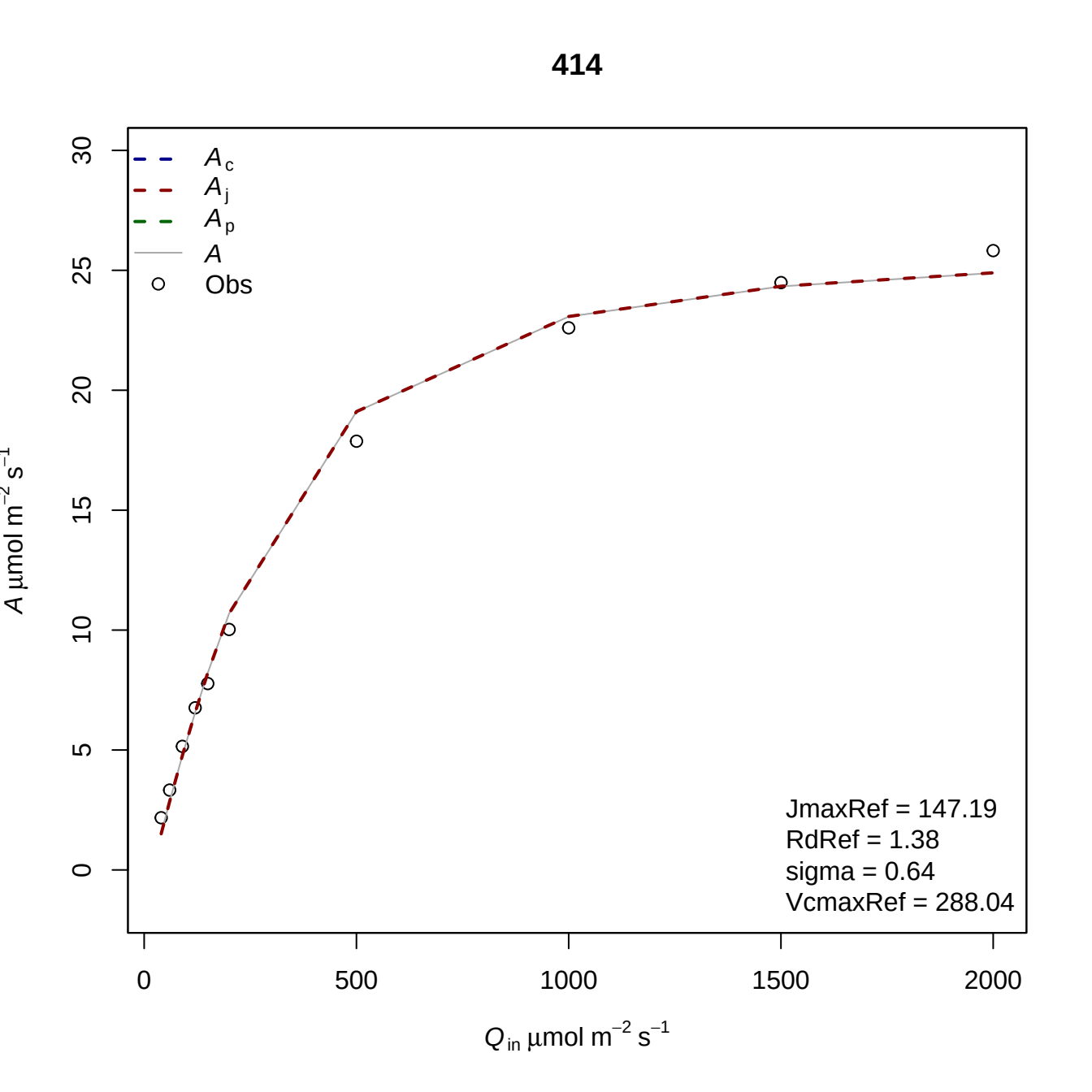

409
A_c
A_j
A_p
A
Obs
JmaxRef = 168.05
RdRef = 1.55
sigma = 0.48
VcmaxRef = 377.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

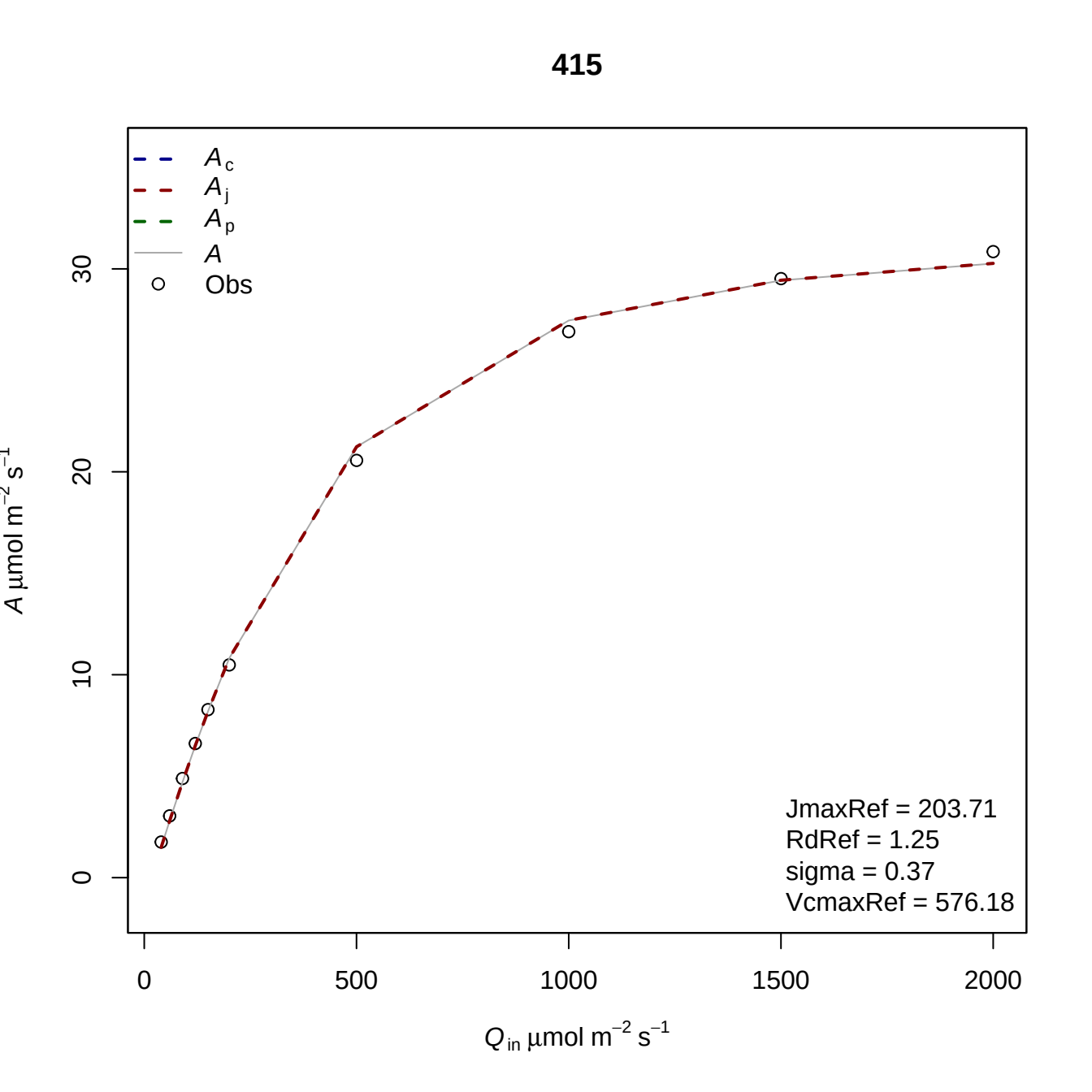

410
A_c
A_j
A_p
A
Obs
JmaxRef = 214.9
RdRef = 1.51
sigma = 0.23
VcmaxRef = 729.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

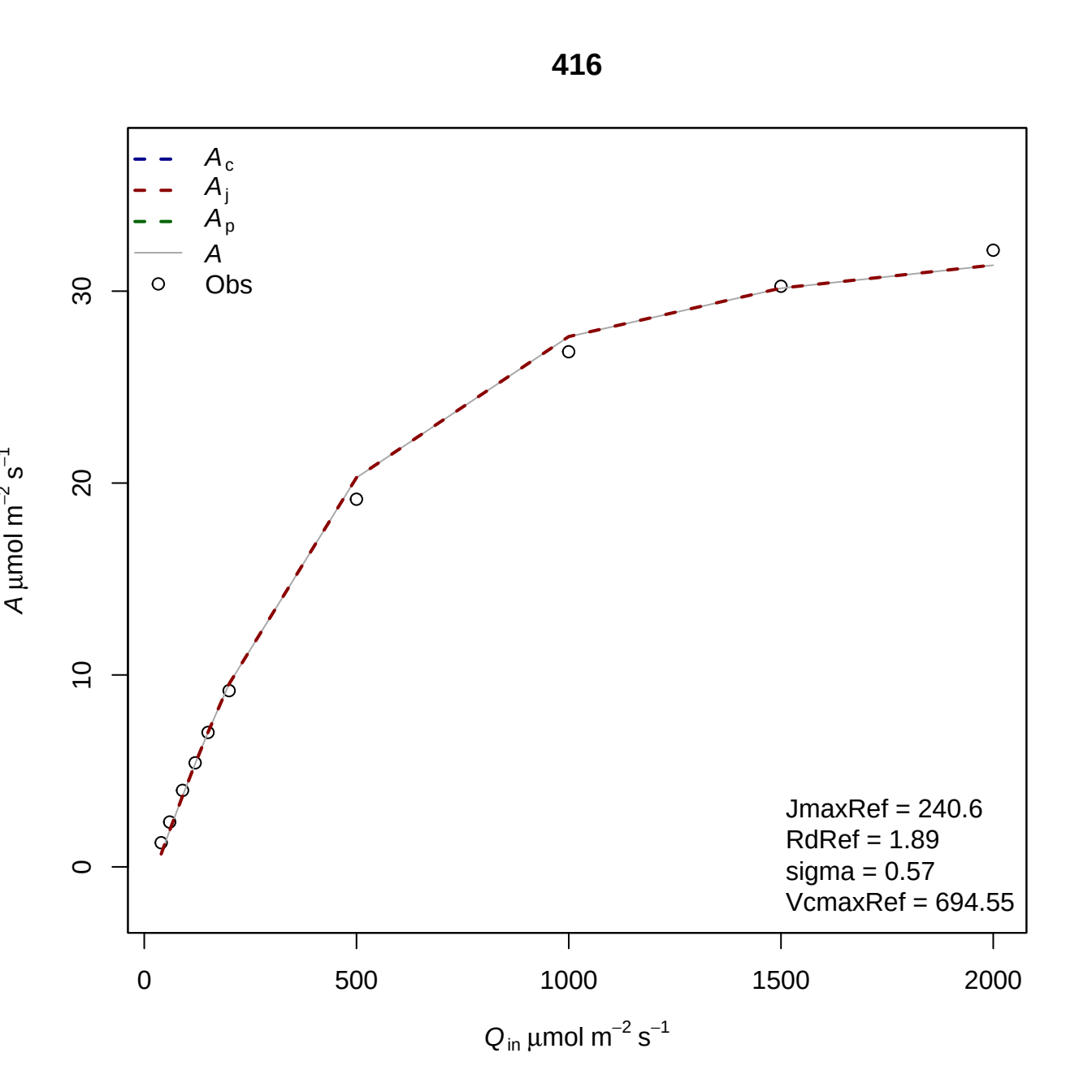

411
A_c
A_j
A_p
A
Obs
JmaxRef = 228.18
RdRef = 1.63
sigma = 0.18
VcmaxRef = 1104.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

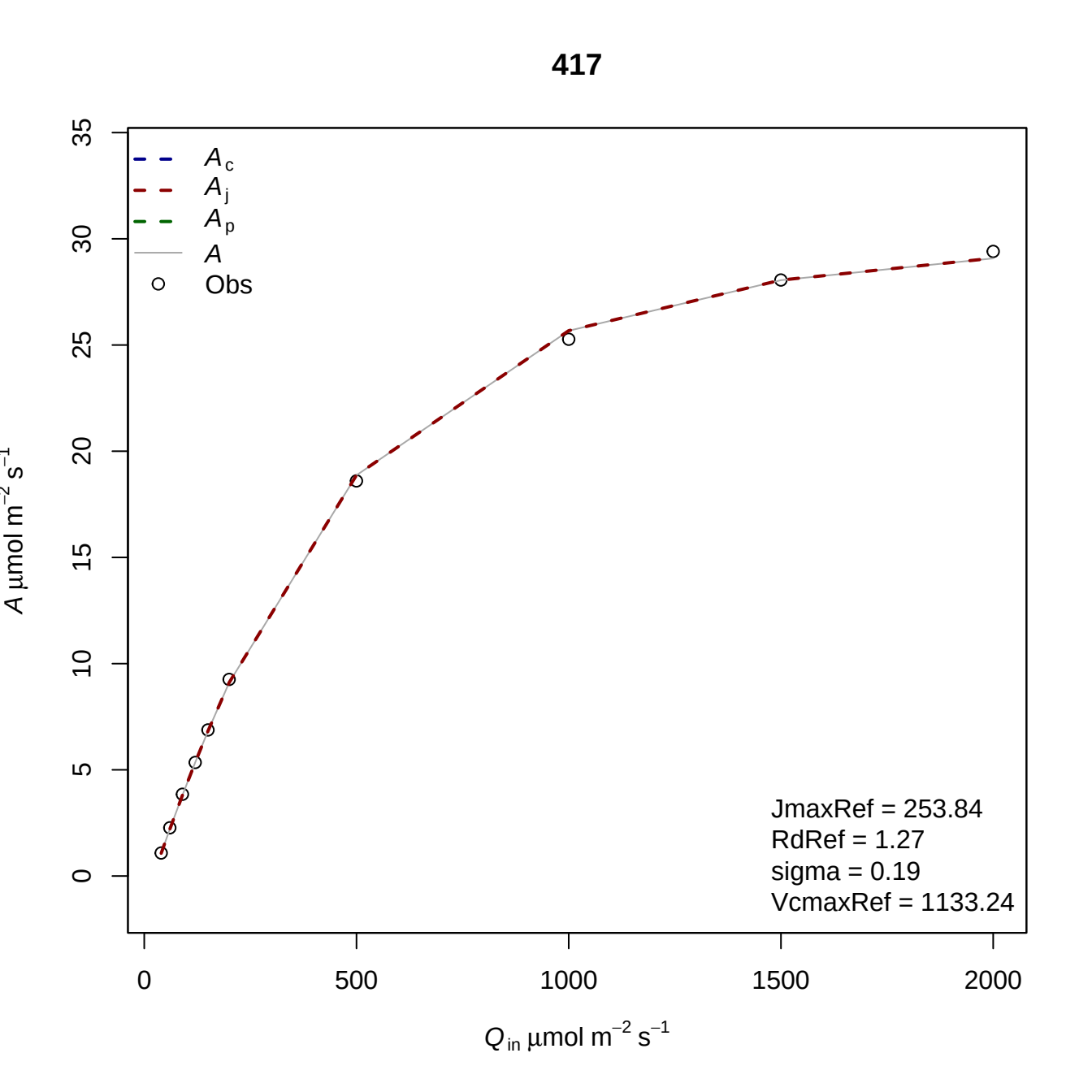

412
A_c
A_j
A_p
A
Obs
JmaxRef = 277.64
RdRef = 2.14
sigma = 0.18
VcmaxRef = 1265.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

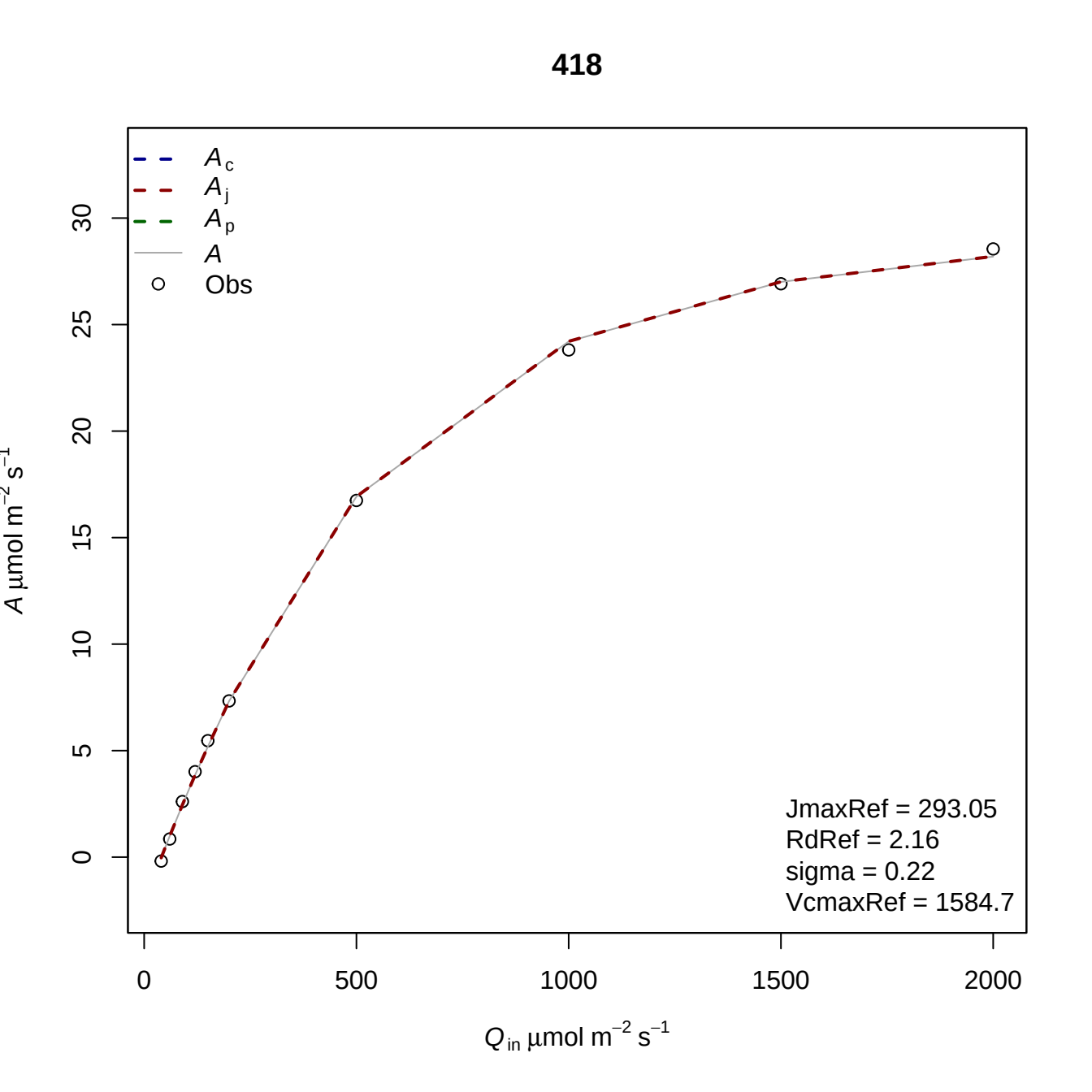

413
A_c
A_j
A_p
A
Obs
JmaxRef = 188.56
RdRef = 3.96
sigma = 1.09
VcmaxRef = 1253.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

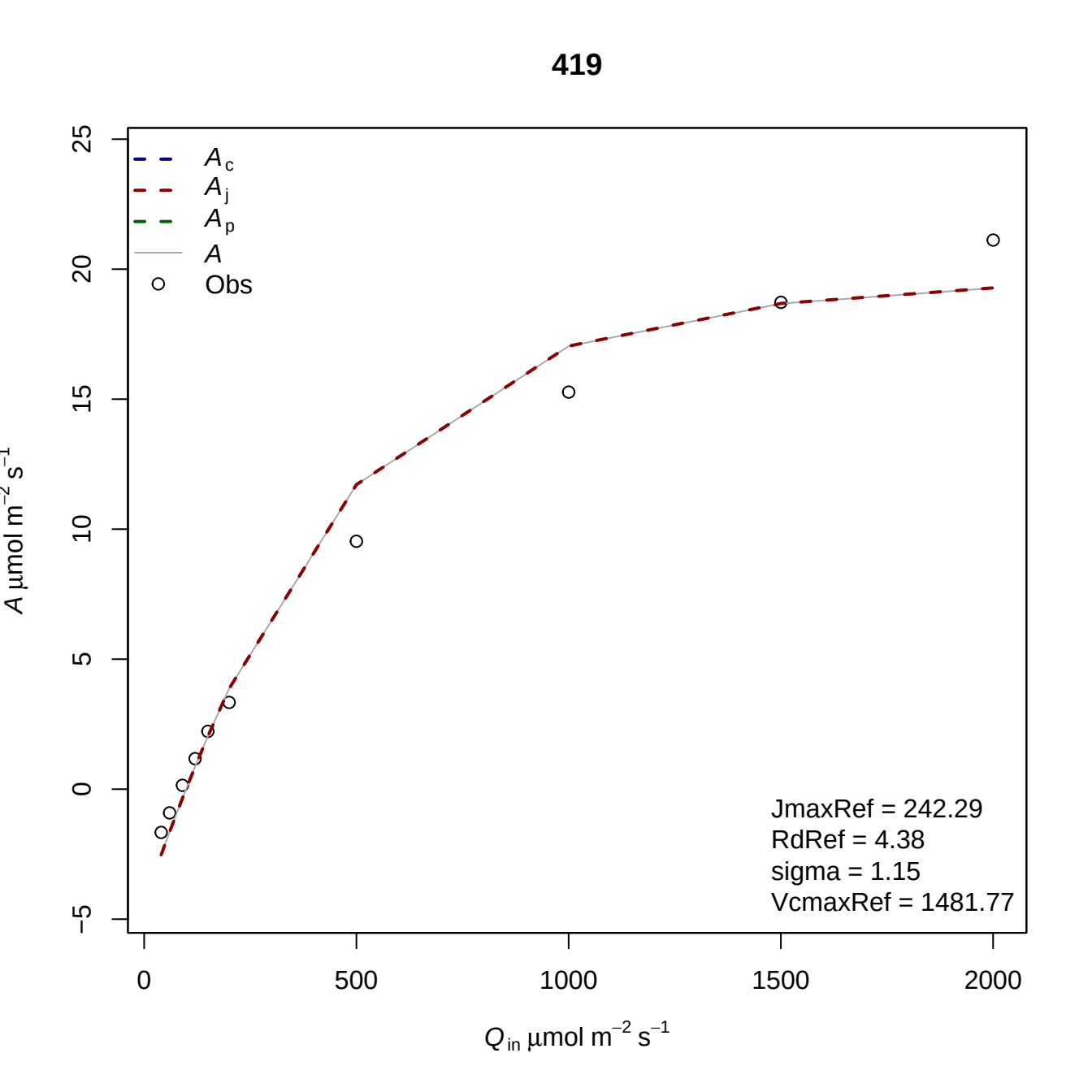

414
A_c
A_j
A_p
A
Obs
JmaxRef = 147.19
RdRef = 1.38
sigma = 0.64
VcmaxRef = 288.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

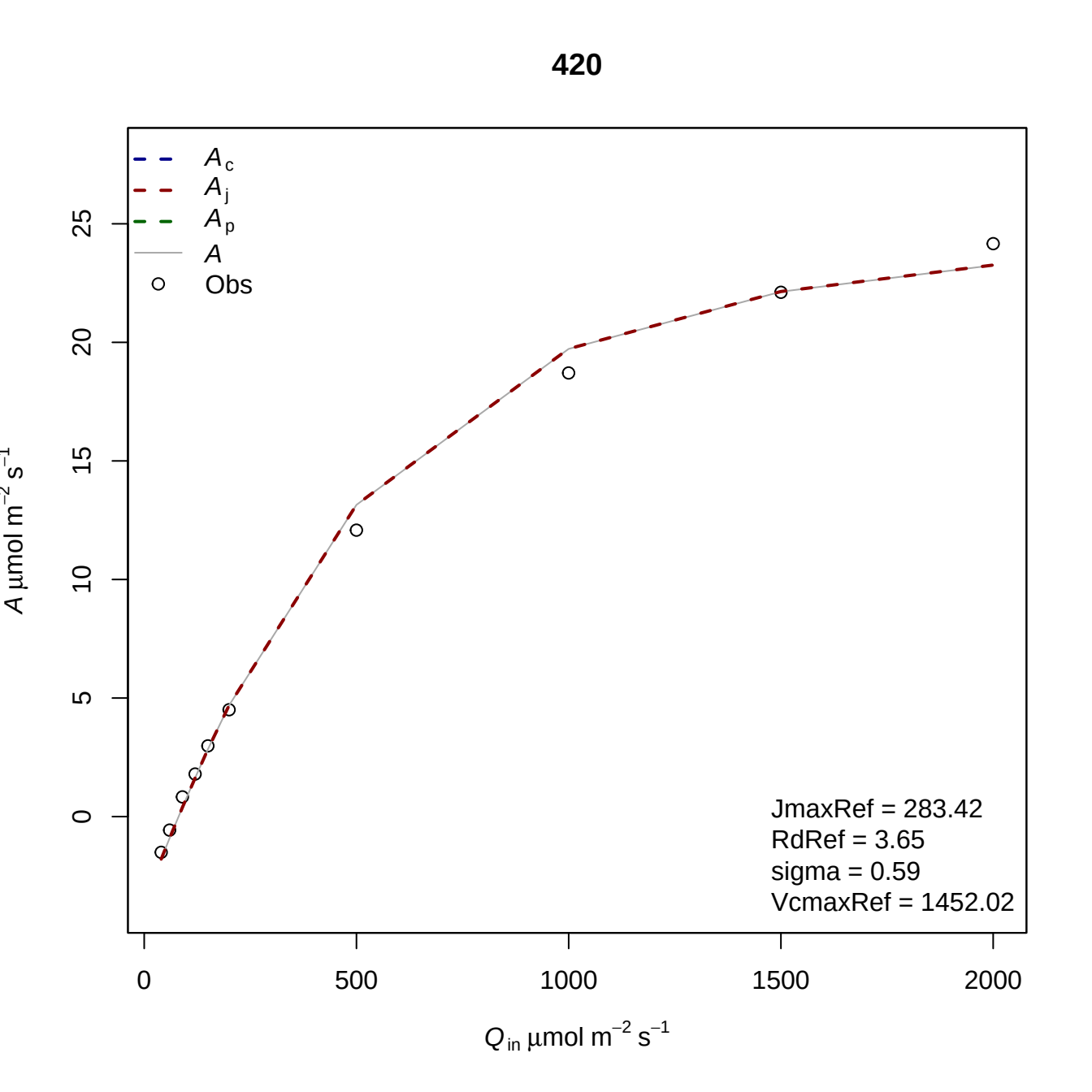

415
A_c
A_j
A_p
A
Obs
JmaxRef = 203.71
RdRef = 1.25
sigma = 0.37
VcmaxRef = 576.18
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

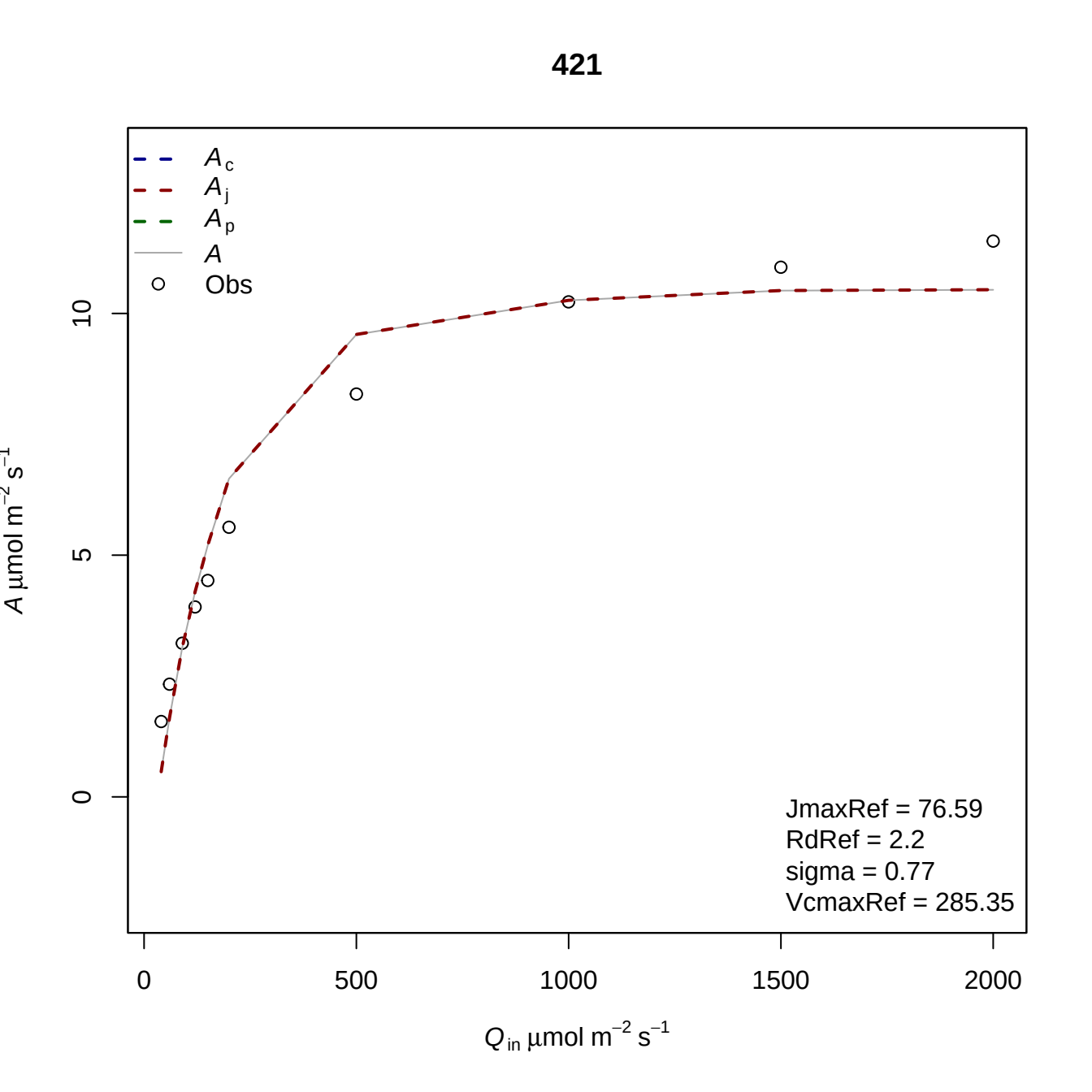

416
A_c
A_j
A_p
A
Obs
JmaxRef = 240.6
RdRef = 1.89
sigma = 0.57
VcmaxRef = 694.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

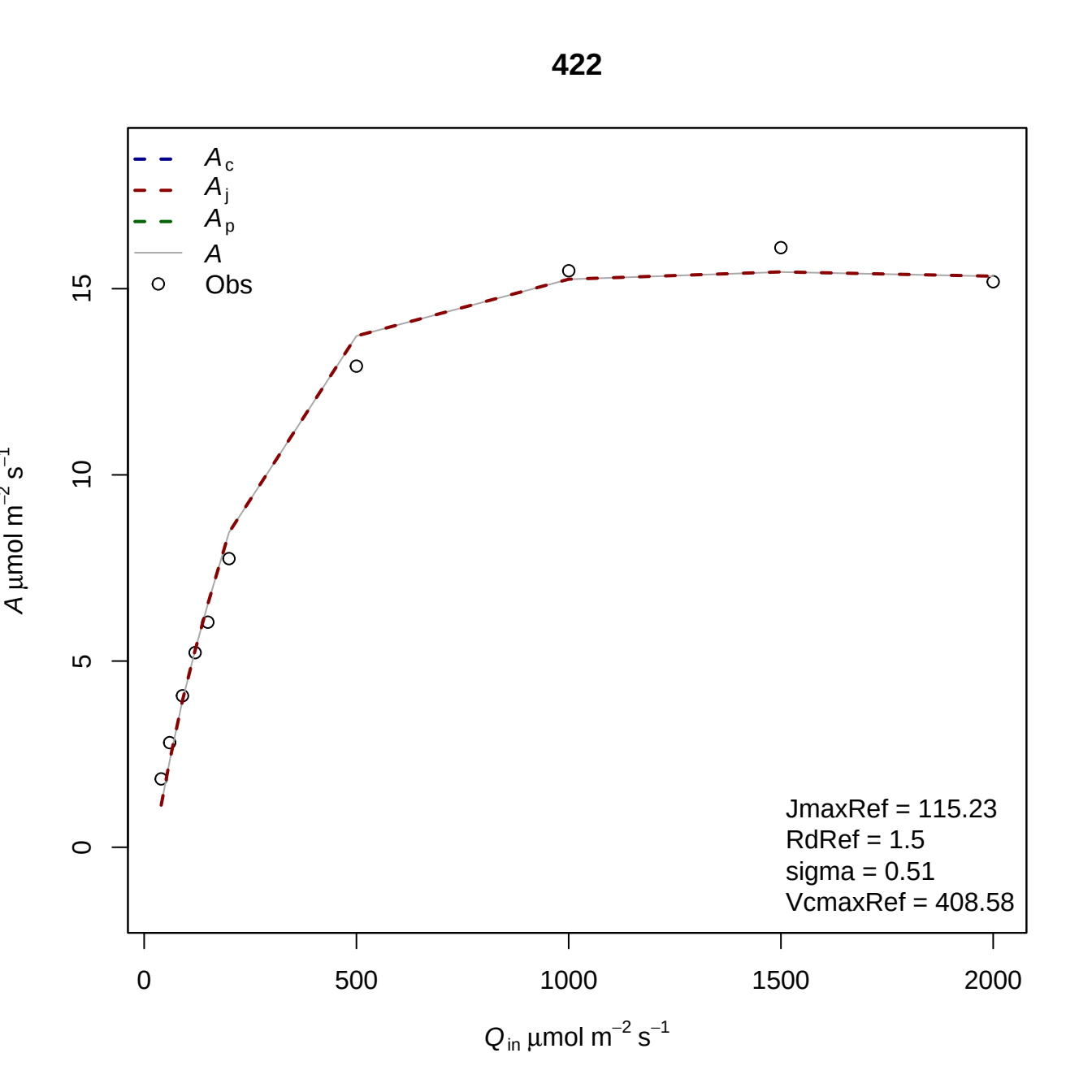

417
A_c
A_j
A_p
A
Obs
JmaxRef = 253.84
RdRef = 1.27
sigma = 0.19
VcmaxRef = 1133.24
0
5
10
15
20
25
30
35
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

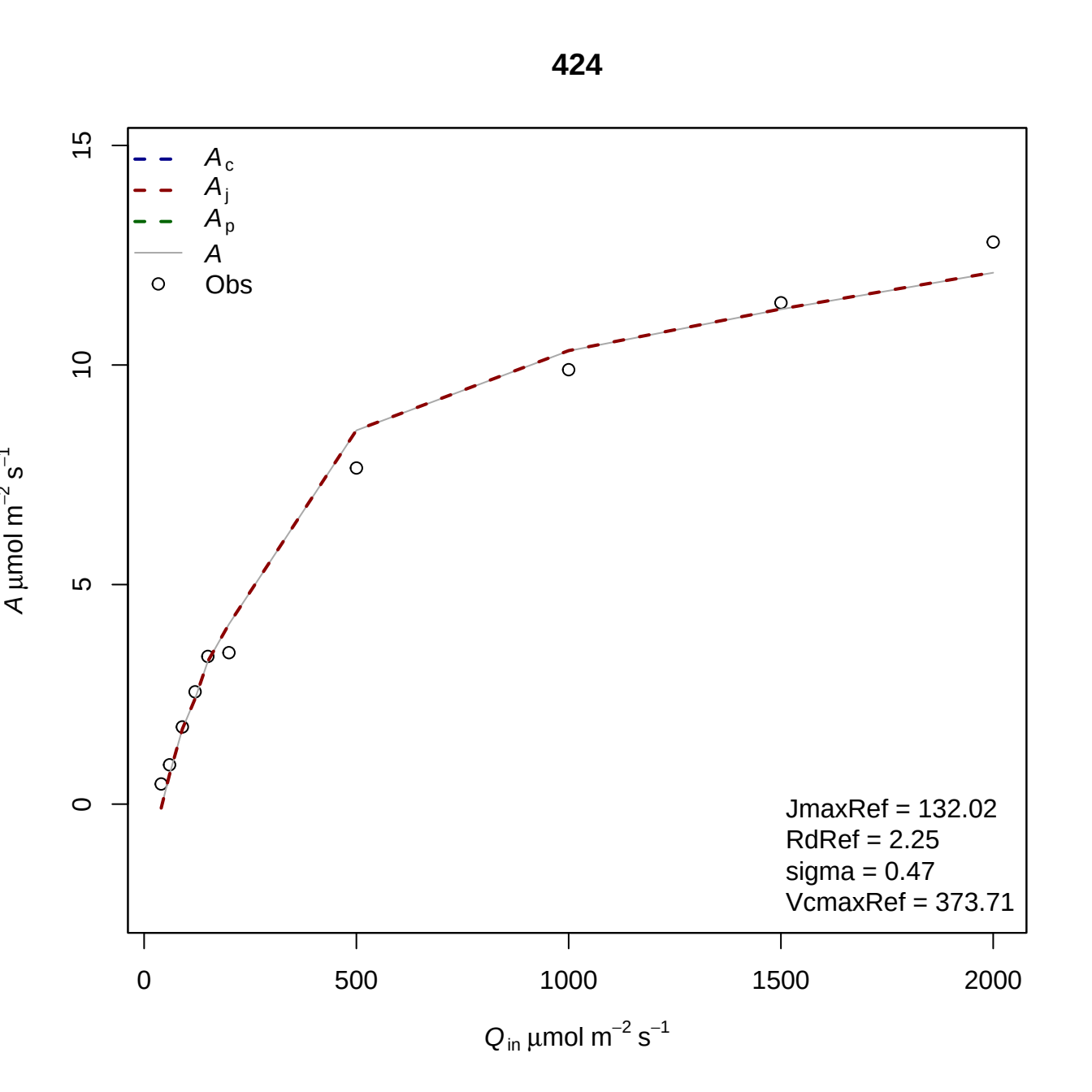

418
A_c
A_j
A_p
A
Obs
JmaxRef = 293.05
RdRef = 2.16
sigma = 0.22
VcmaxRef = 1584.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

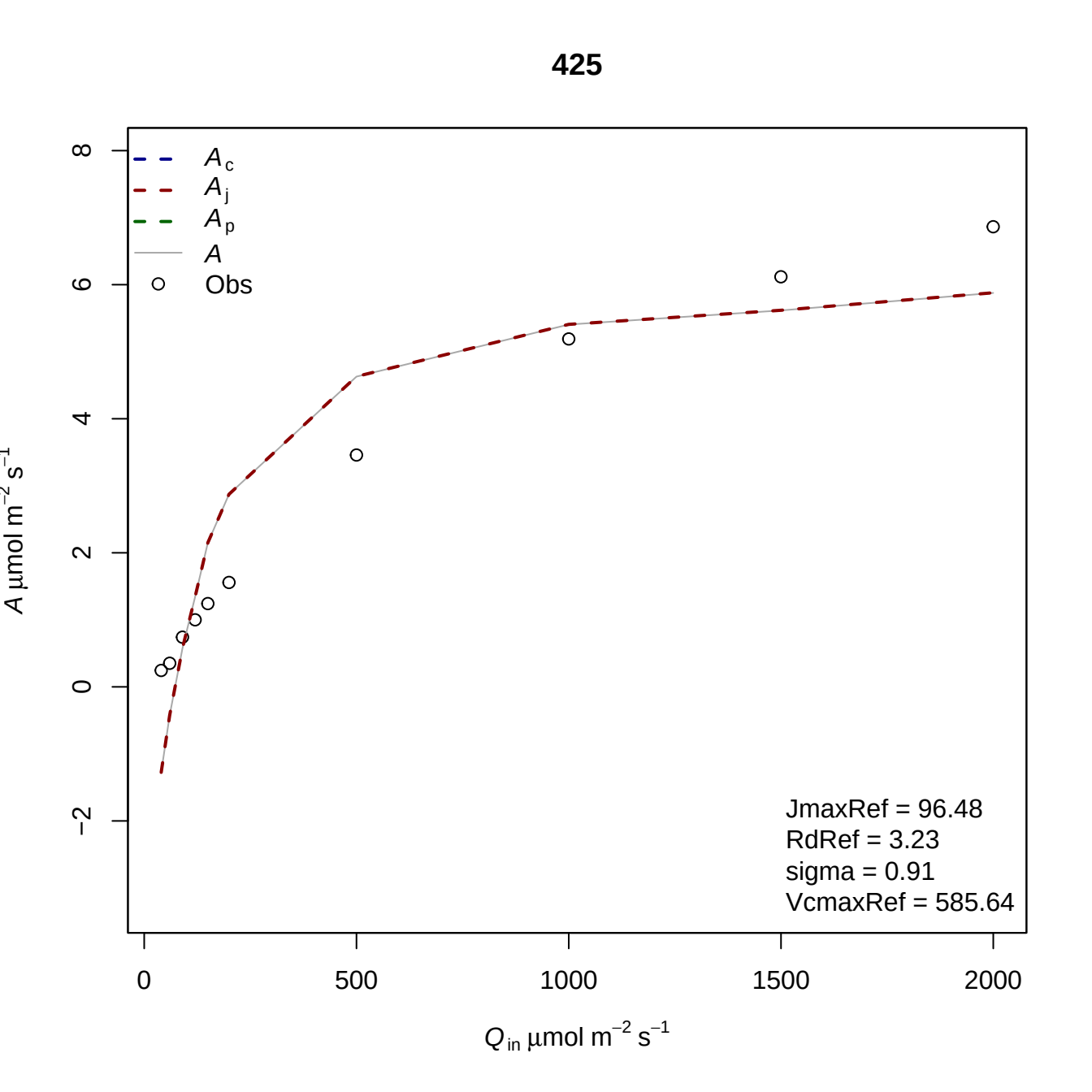

419
A_c
A_j
A_p
A
Obs
JmaxRef = 242.29
RdRef = 4.38
sigma = 1.15
VcmaxRef = 1481.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

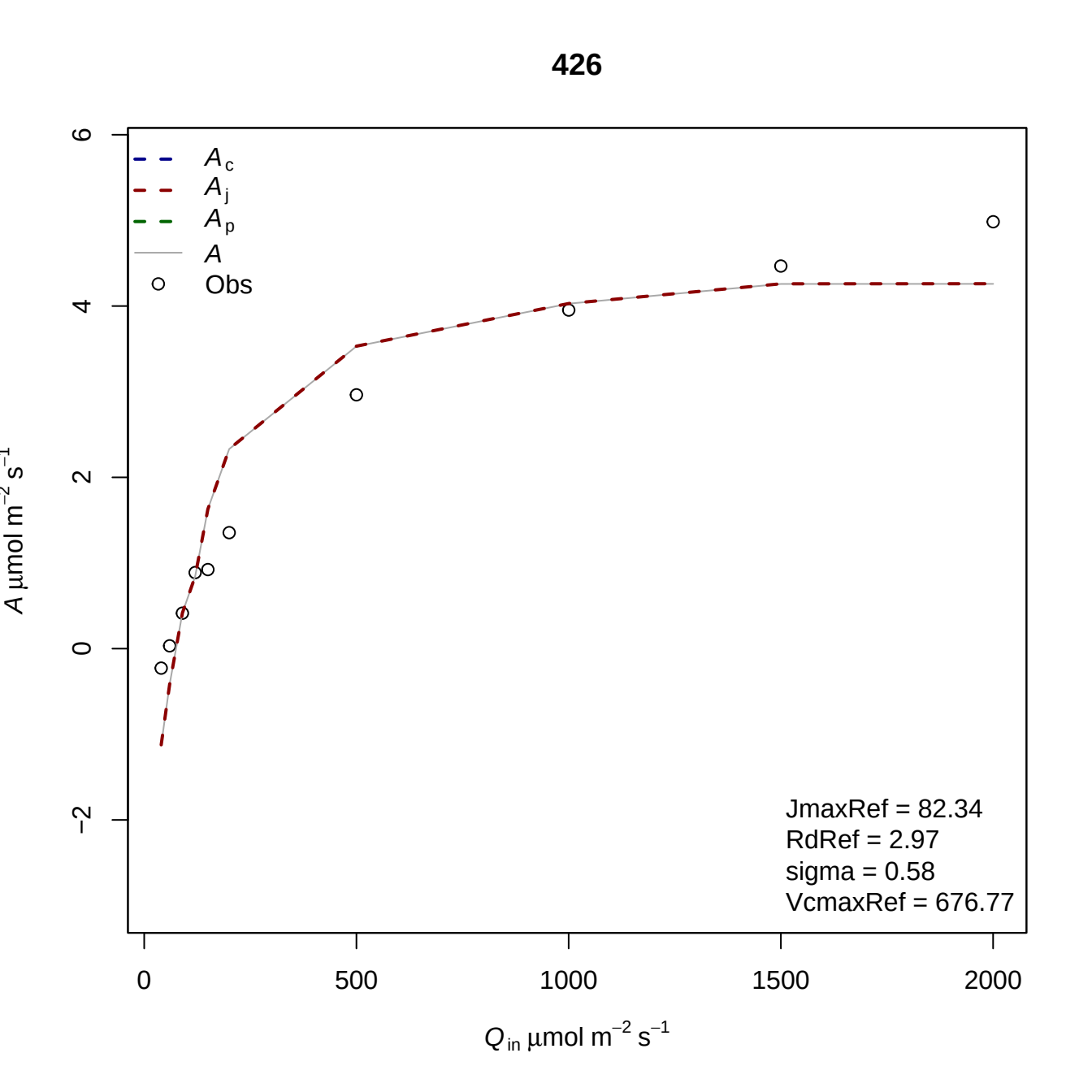

420
A_c
A_j
A_p
A
Obs
JmaxRef = 283.42
RdRef = 3.65
sigma = 0.59
VcmaxRef = 1452.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

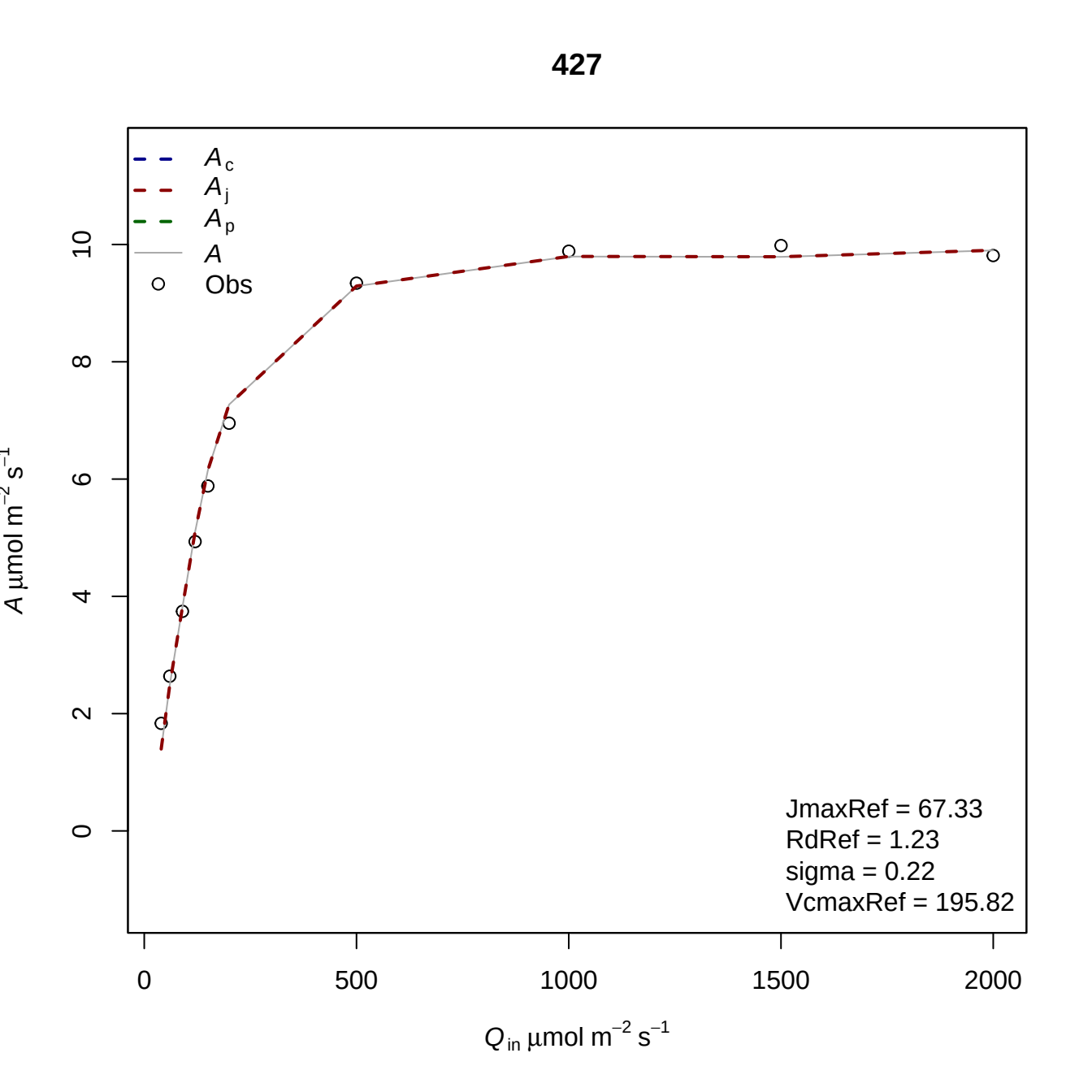

421
A_c
A_j
A_p
A
Obs
JmaxRef = 76.59
RdRef = 2.2
sigma = 0.77
VcmaxRef = 285.35
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

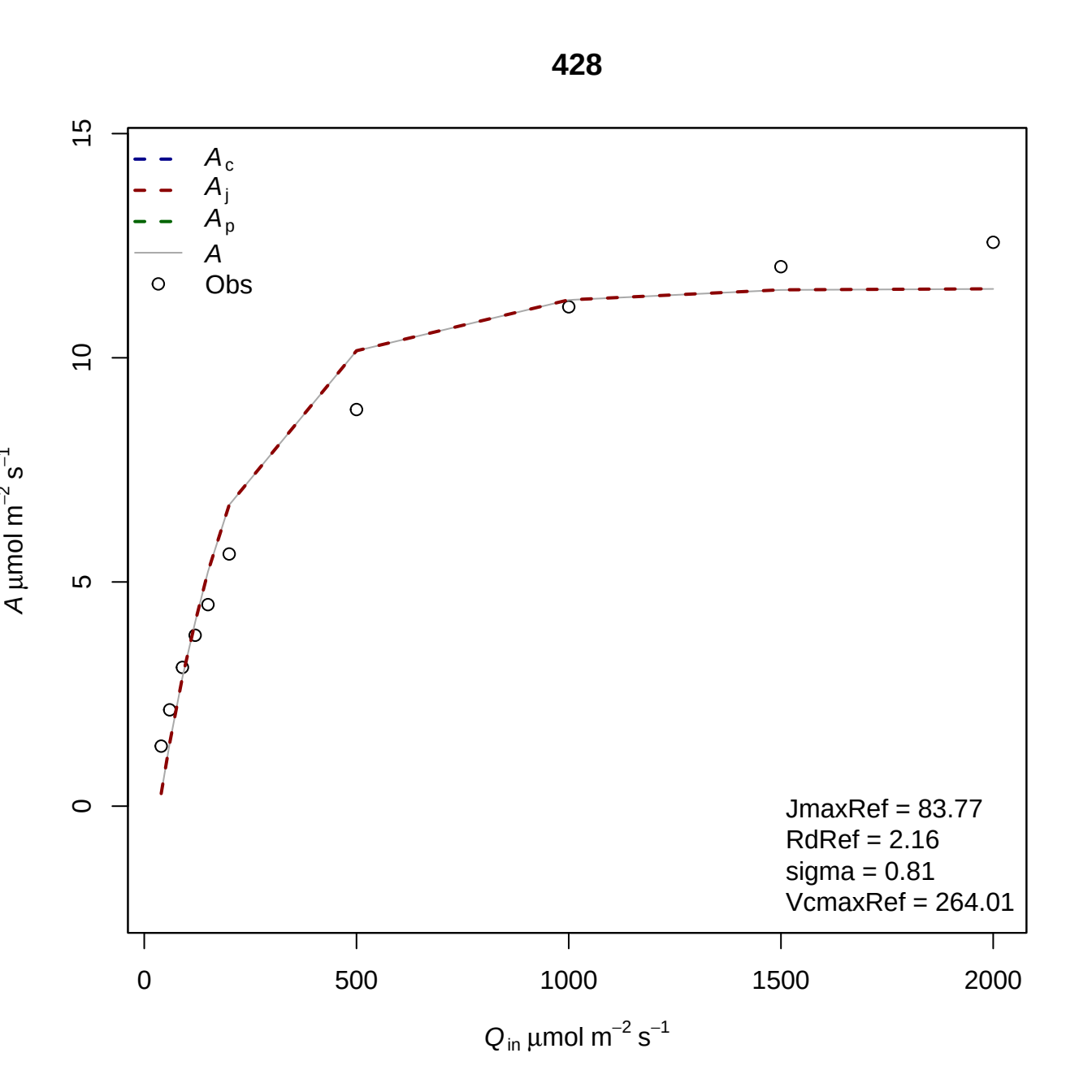

422
A_c
A_j
A_p
A
Obs
JmaxRef = 115.23
RdRef = 1.5
sigma = 0.51
VcmaxRef = 408.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

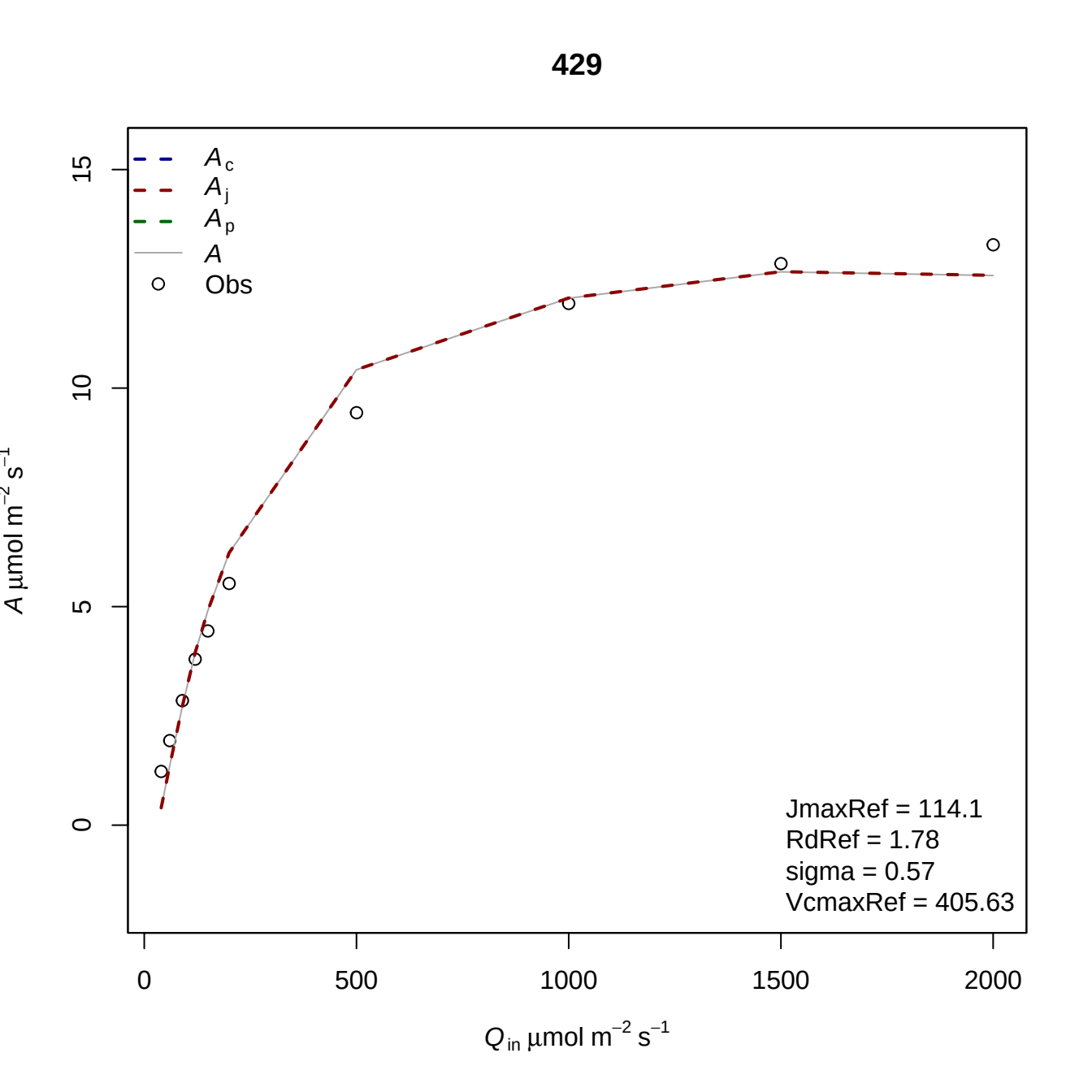

424
A_c
A_j
A_p
A
Obs
JmaxRef = 132.02
RdRef = 2.25
sigma = 0.47
VcmaxRef = 373.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

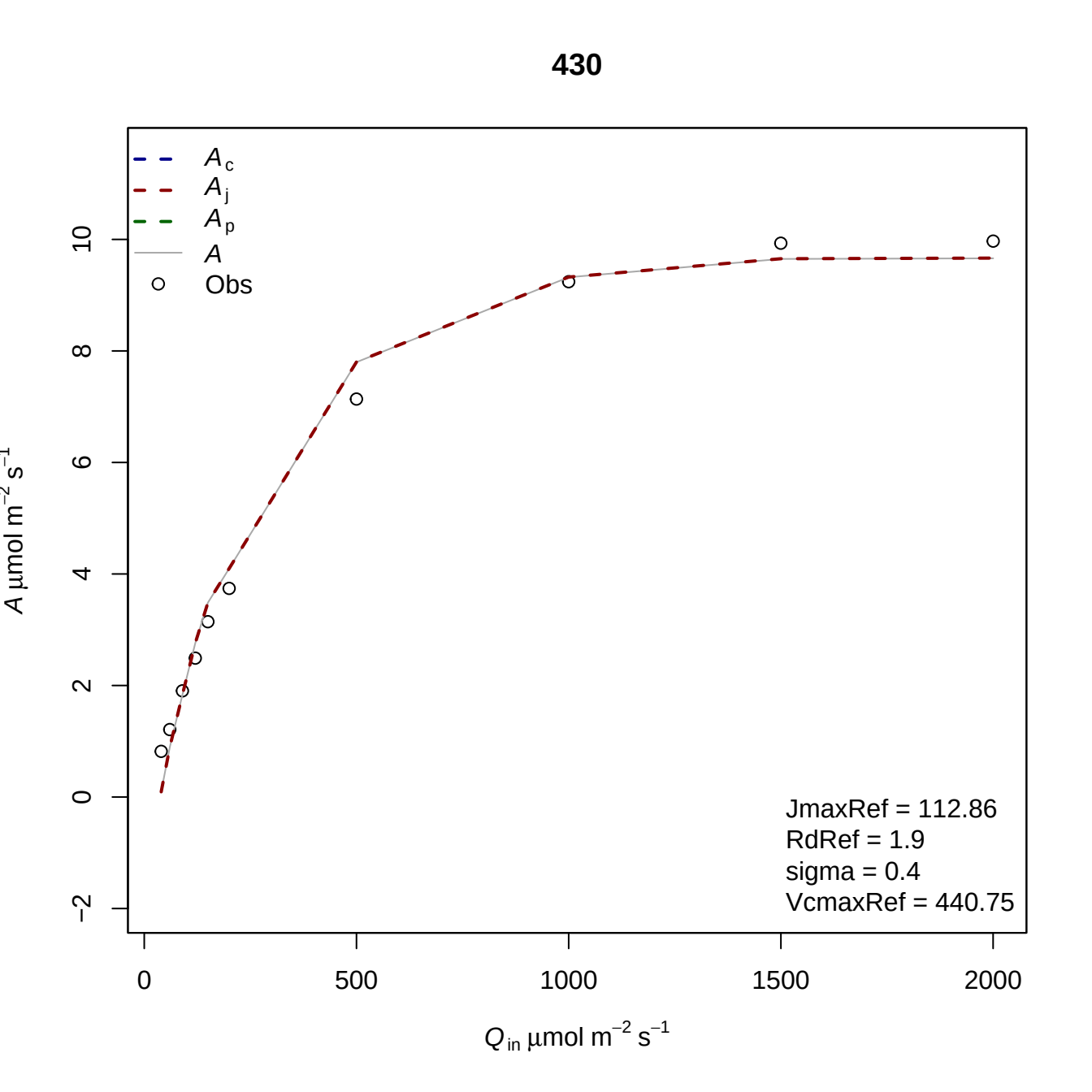

425
A_c
A_j
A_p
A
Obs
JmaxRef = 96.48
RdRef = 3.23
sigma = 0.91
VcmaxRef = 585.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

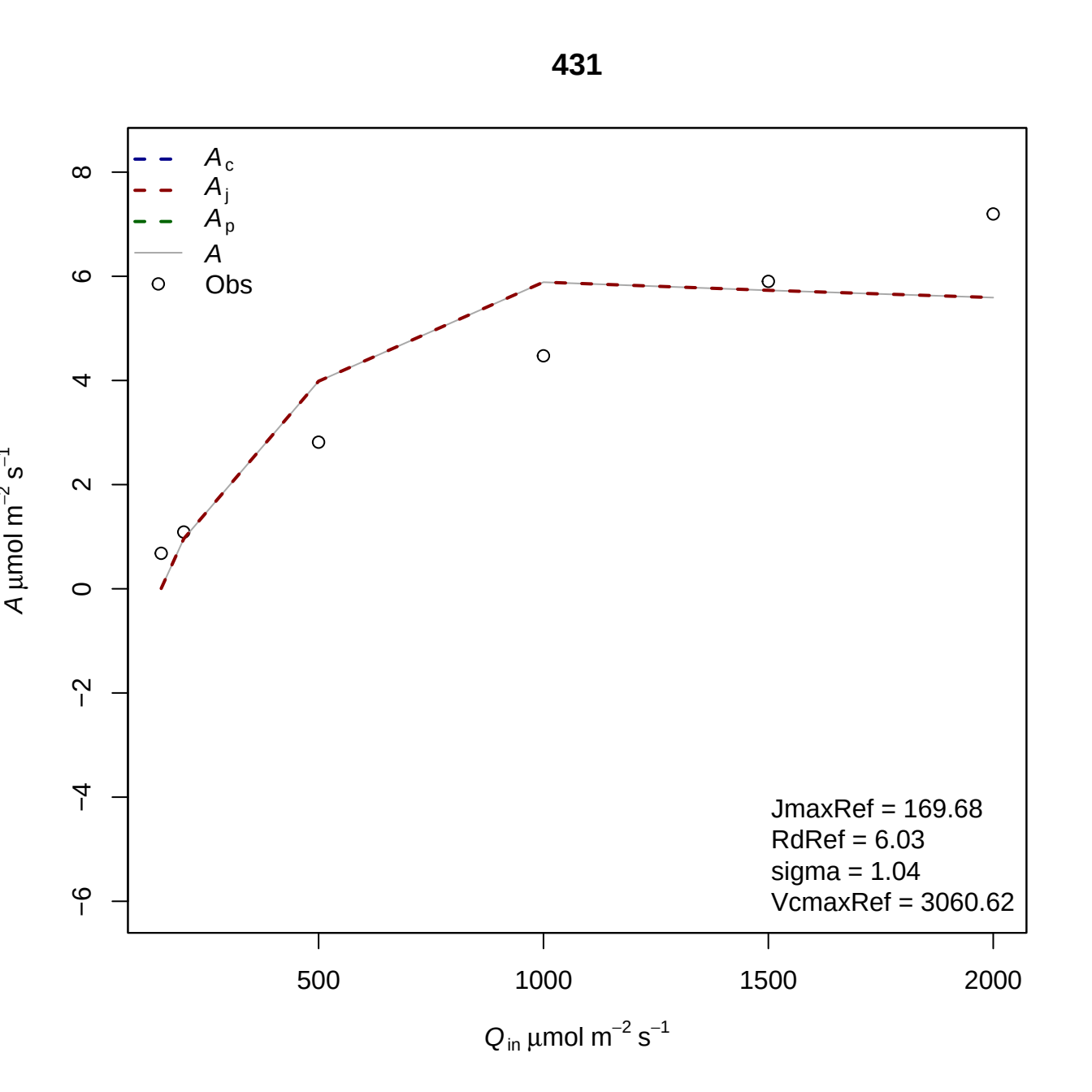

426
A_c
A_j
A_p
A
Obs
JmaxRef = 82.34
RdRef = 2.97
sigma = 0.58
VcmaxRef = 676.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

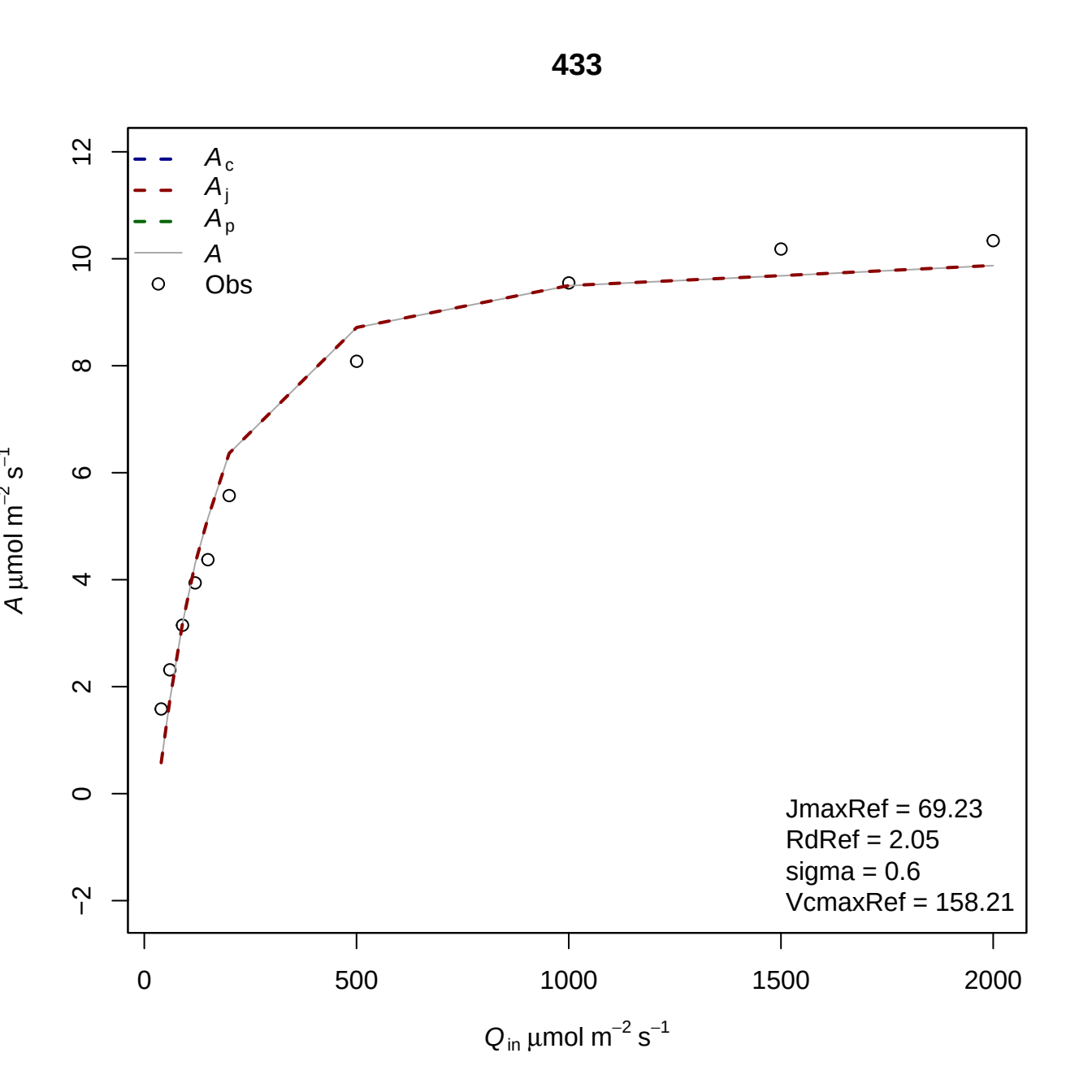

427
A_c
A_j
A_p
A
Obs
JmaxRef = 67.33
RdRef = 1.23
sigma = 0.22
VcmaxRef = 195.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

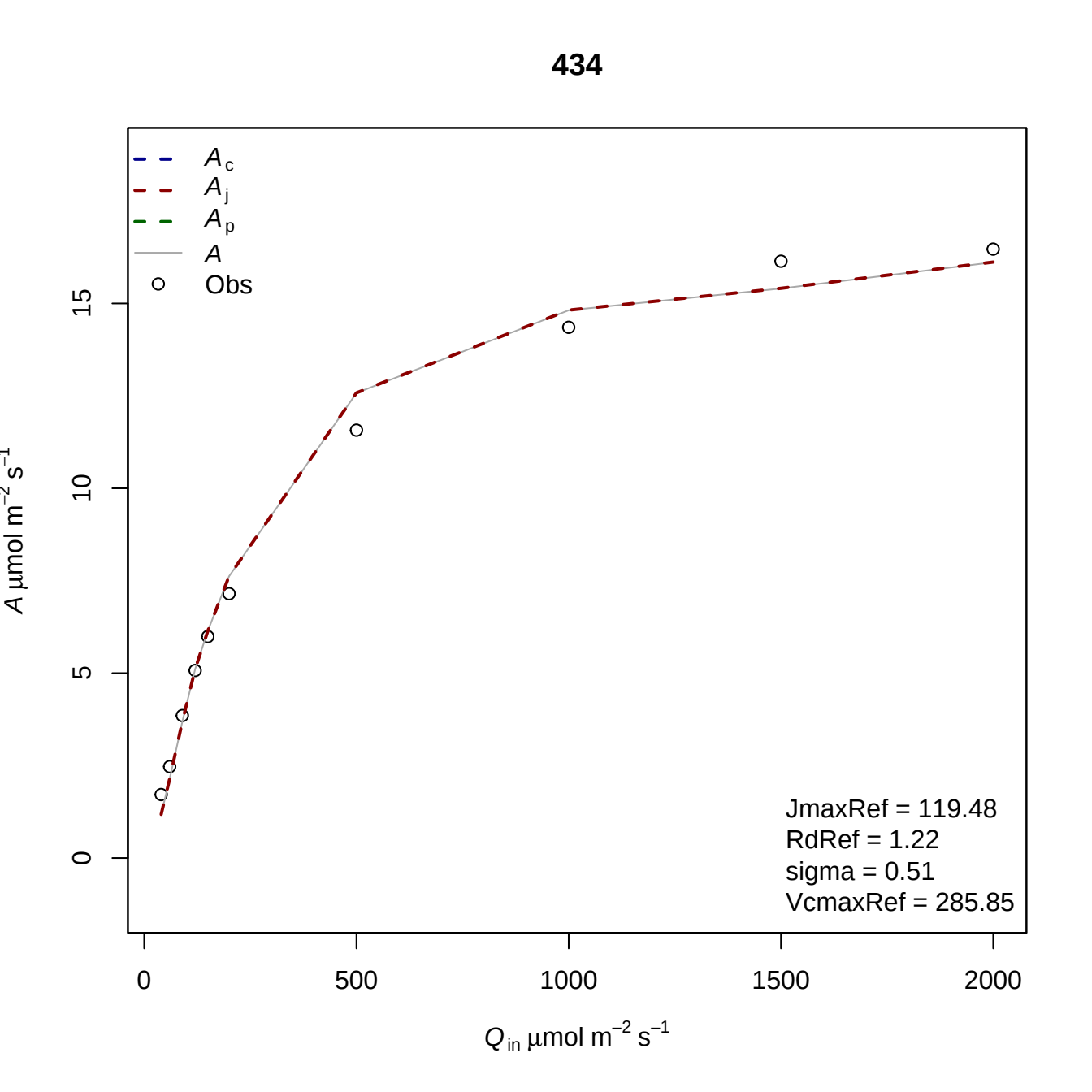

428
A_c
A_j
A_p
A
Obs
JmaxRef = 83.77
RdRef = 2.16
sigma = 0.81
VcmaxRef = 264.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

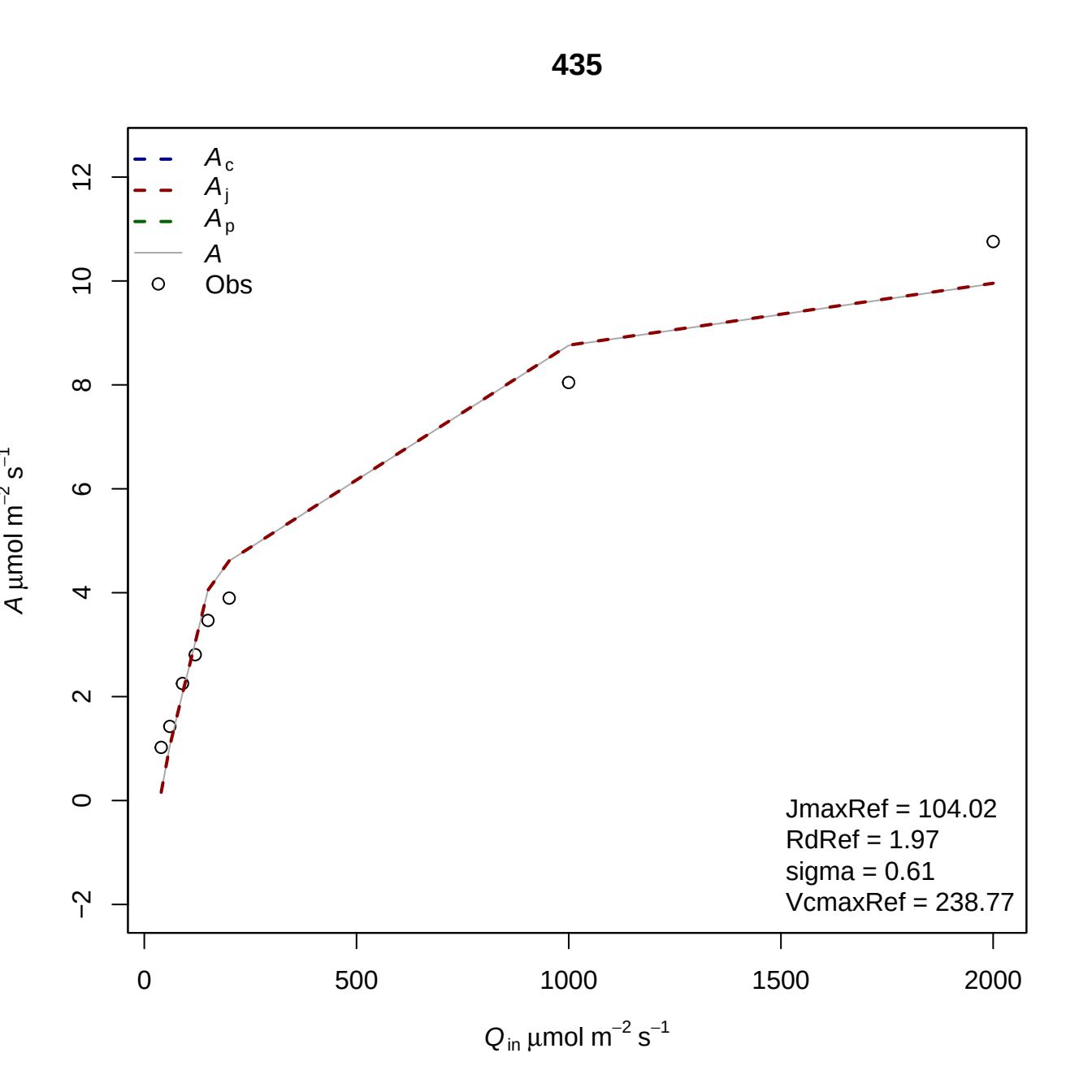

429
A_c
A_j
A_p
A
Obs
JmaxRef = 114.1
RdRef = 1.78
sigma = 0.57
VcmaxRef = 405.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

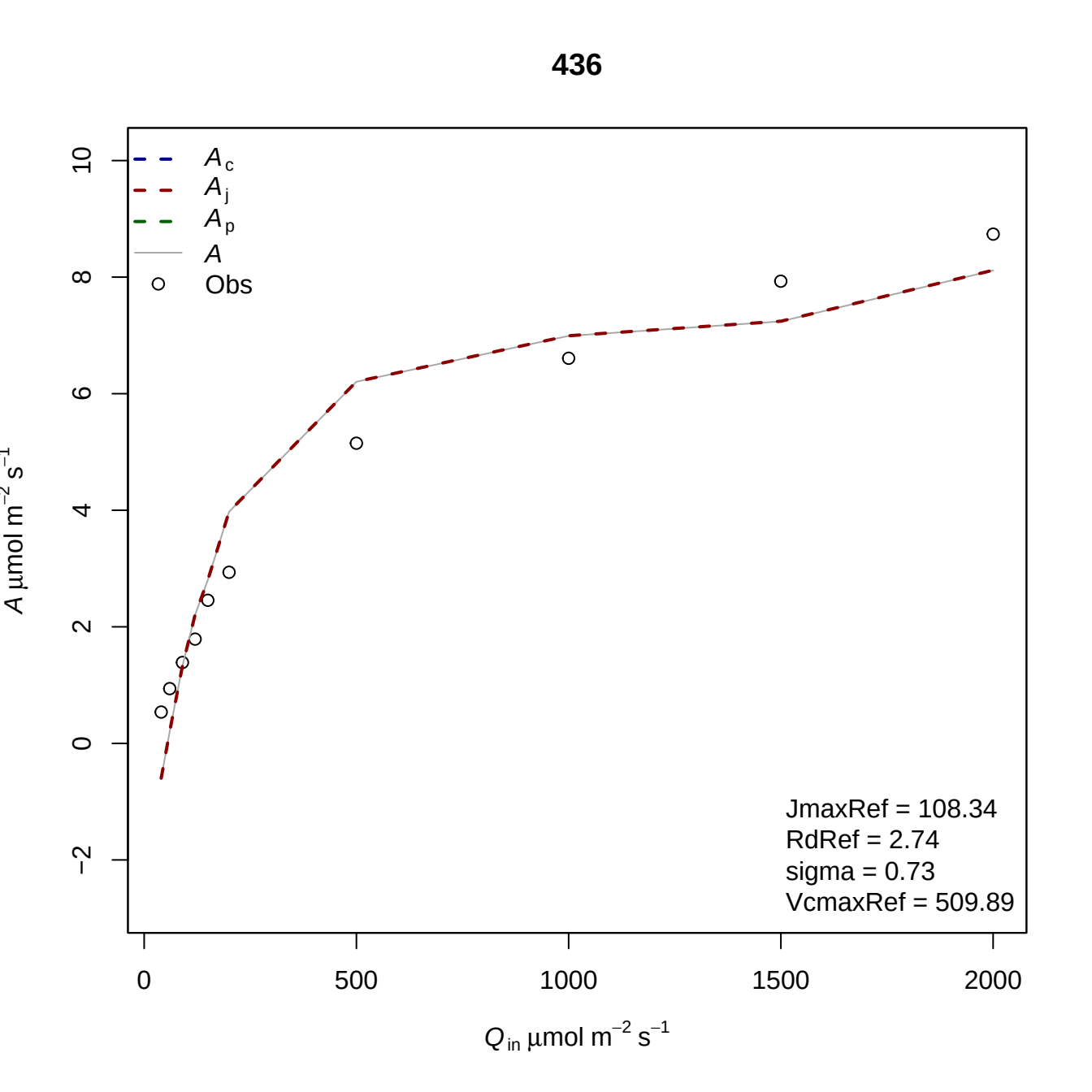

430
A_c
A_j
A_p
A
Obs
JmaxRef = 112.86
RdRef = 1.9
sigma = 0.4
VcmaxRef = 440.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

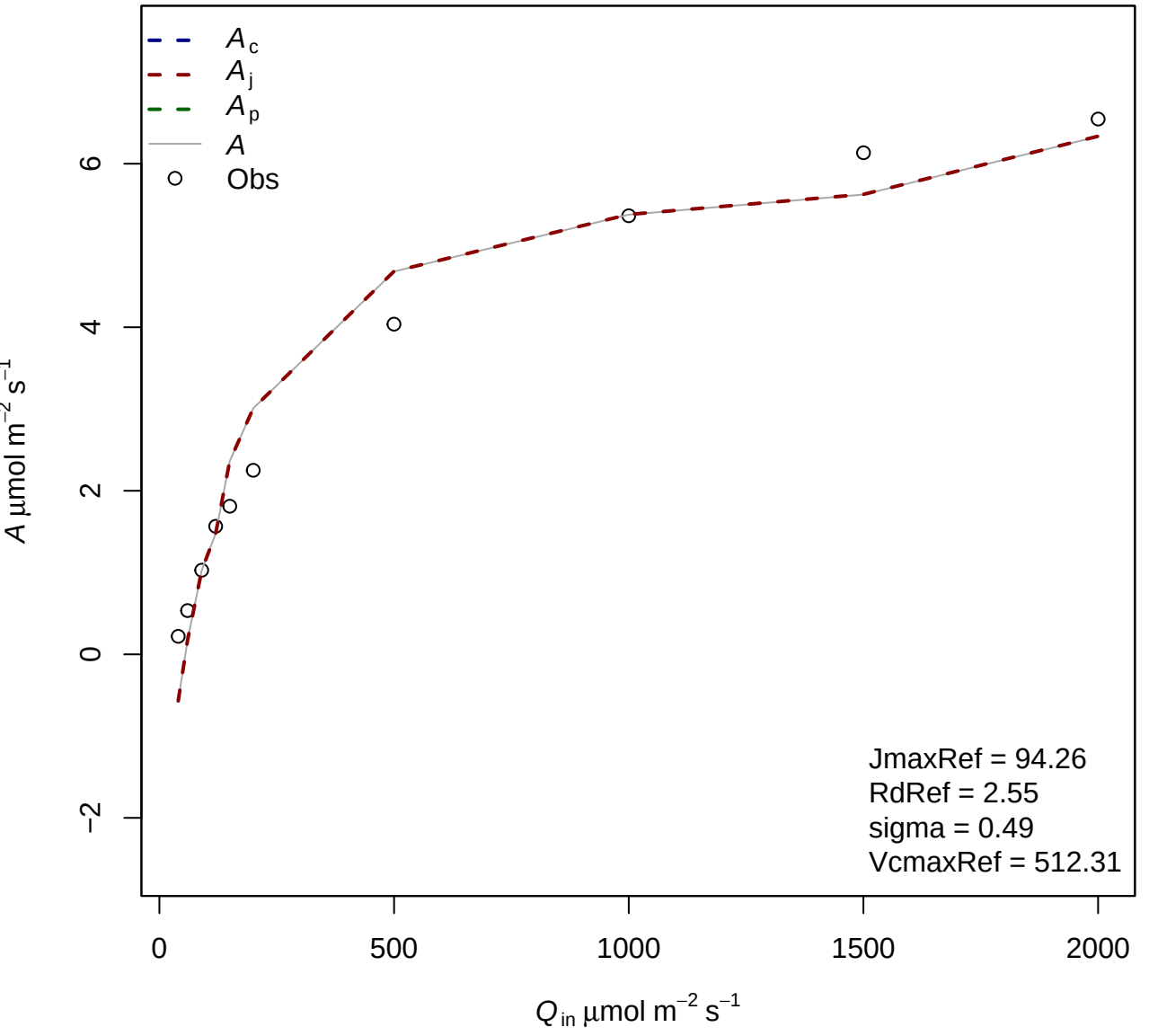

431
A_c
A_j
A_p
A
Obs
JmaxRef = 169.68
RdRef = 6.03
sigma = 1.04
VcmaxRef = 3060.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
−6
500
1000
1500
2000

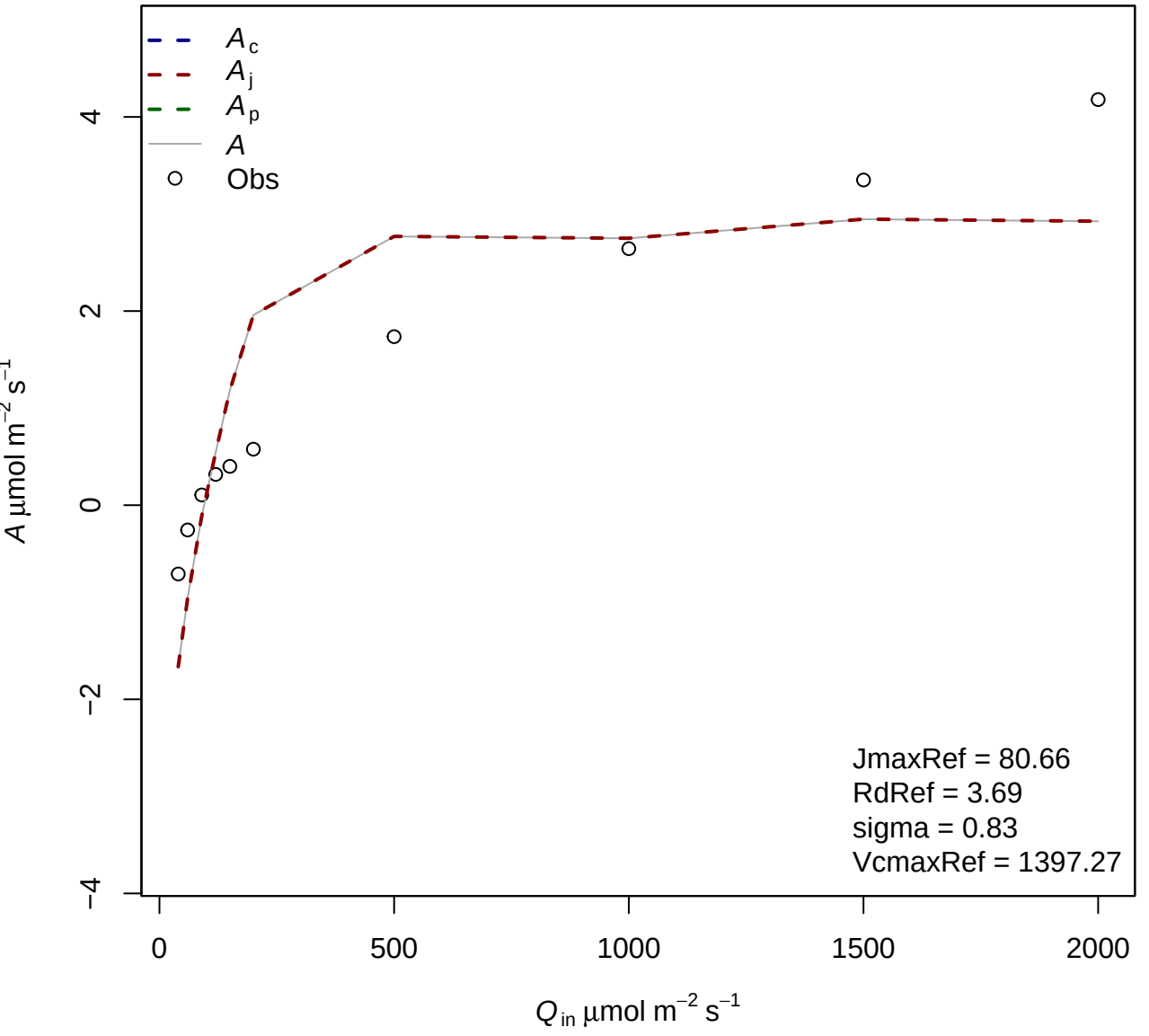

433
A_c
A_j
A_p
A
Obs
JmaxRef = 69.23
RdRef = 2.05
sigma = 0.6
VcmaxRef = 158.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

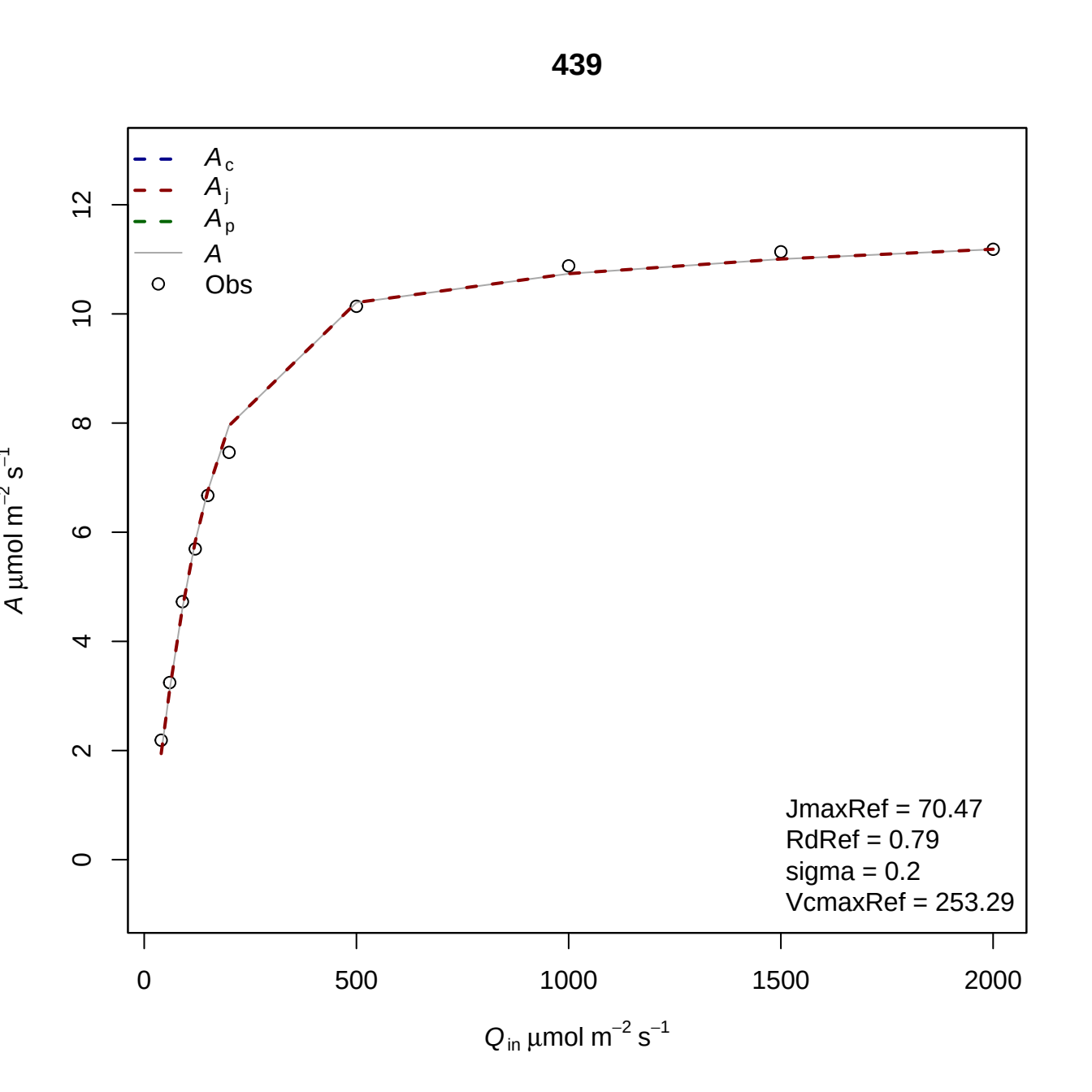

434
A_c
A_j
A_p
A
Obs
JmaxRef = 119.48
RdRef = 1.22
sigma = 0.51
VcmaxRef = 285.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

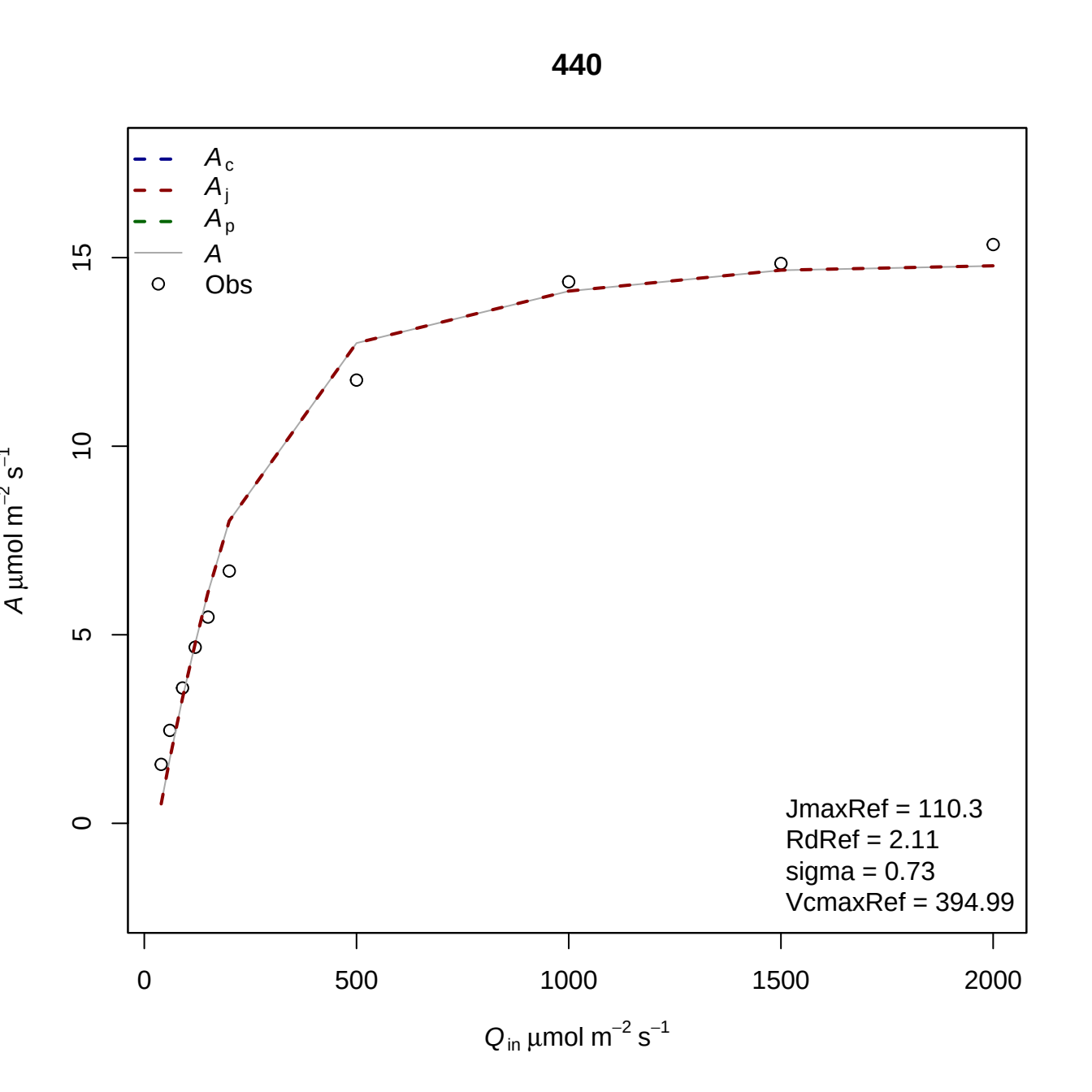

435
A_c
A_j
A_p
A
Obs
JmaxRef = 104.02
RdRef = 1.97
sigma = 0.61
VcmaxRef = 238.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

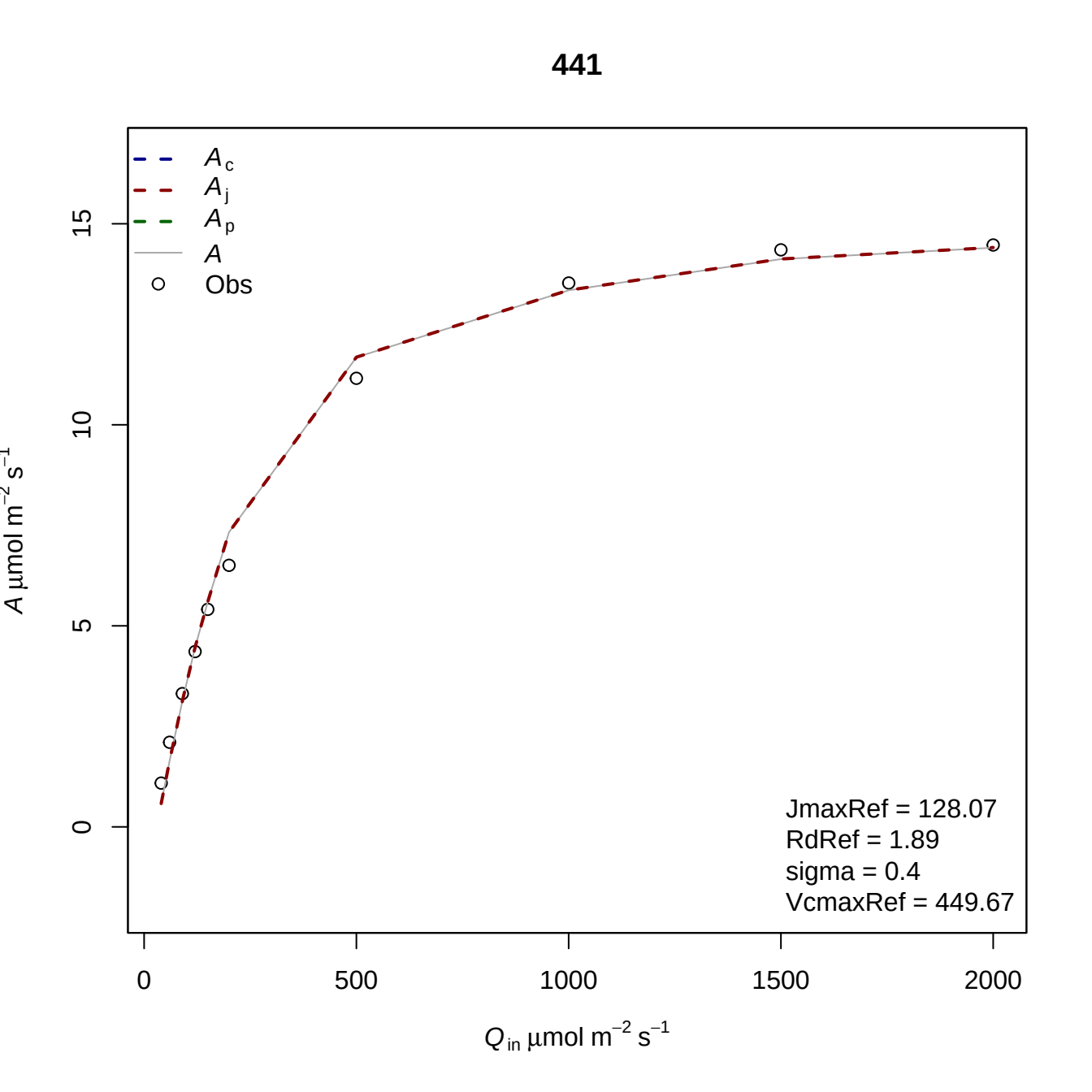

436
A_c
A_j
A_p
A
Obs
JmaxRef = 108.34
RdRef = 2.74
sigma = 0.73
VcmaxRef = 509.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

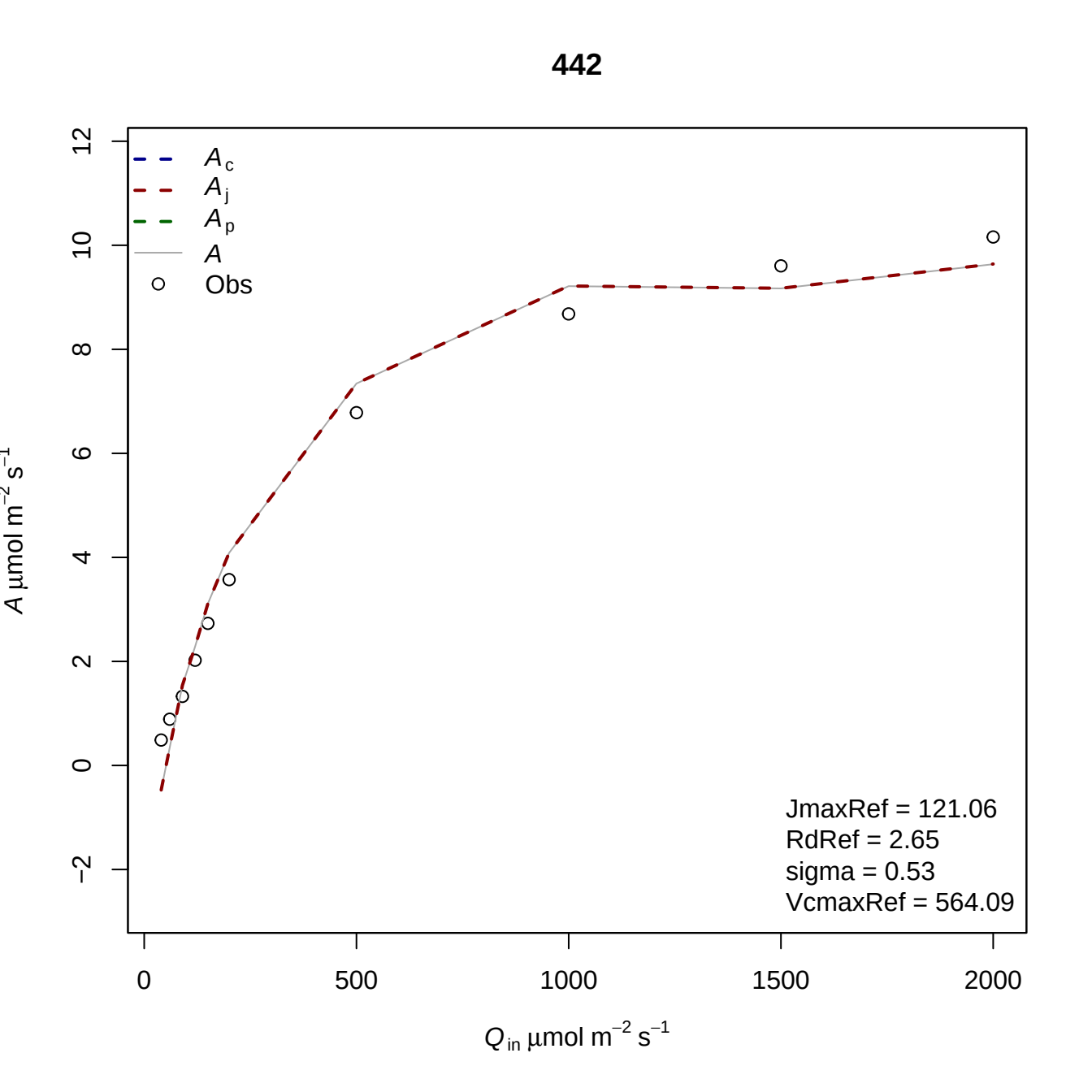

A_c
A_j
A_p
A
Obs
JmaxRef = 94.26
RdRef = 2.55
sigma = 0.49
VcmaxRef = 512.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

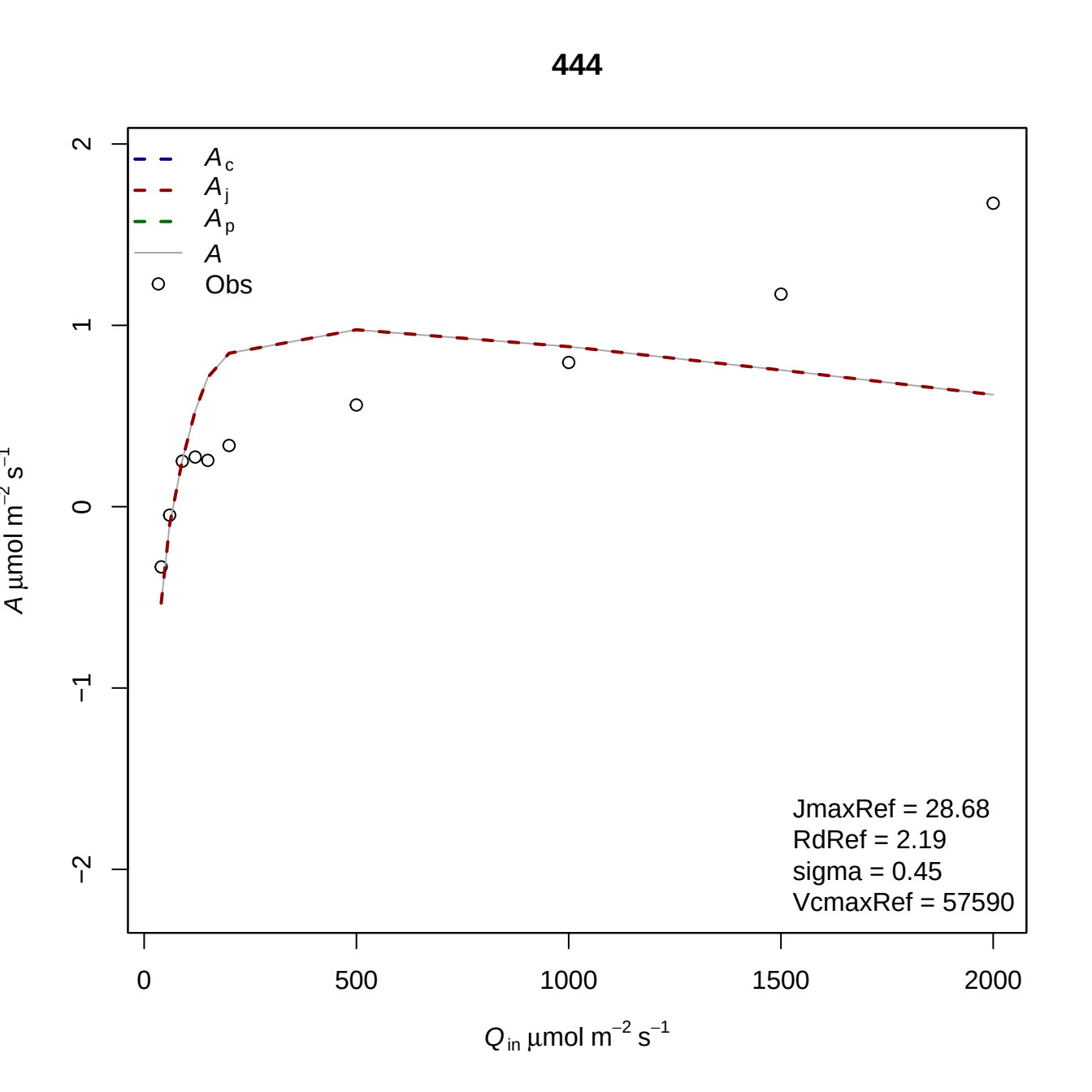

A_c
A_j
A_p
A
Obs
JmaxRef = 80.66
RdRef = 3.69
sigma = 0.83
VcmaxRef = 1397.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500
2000

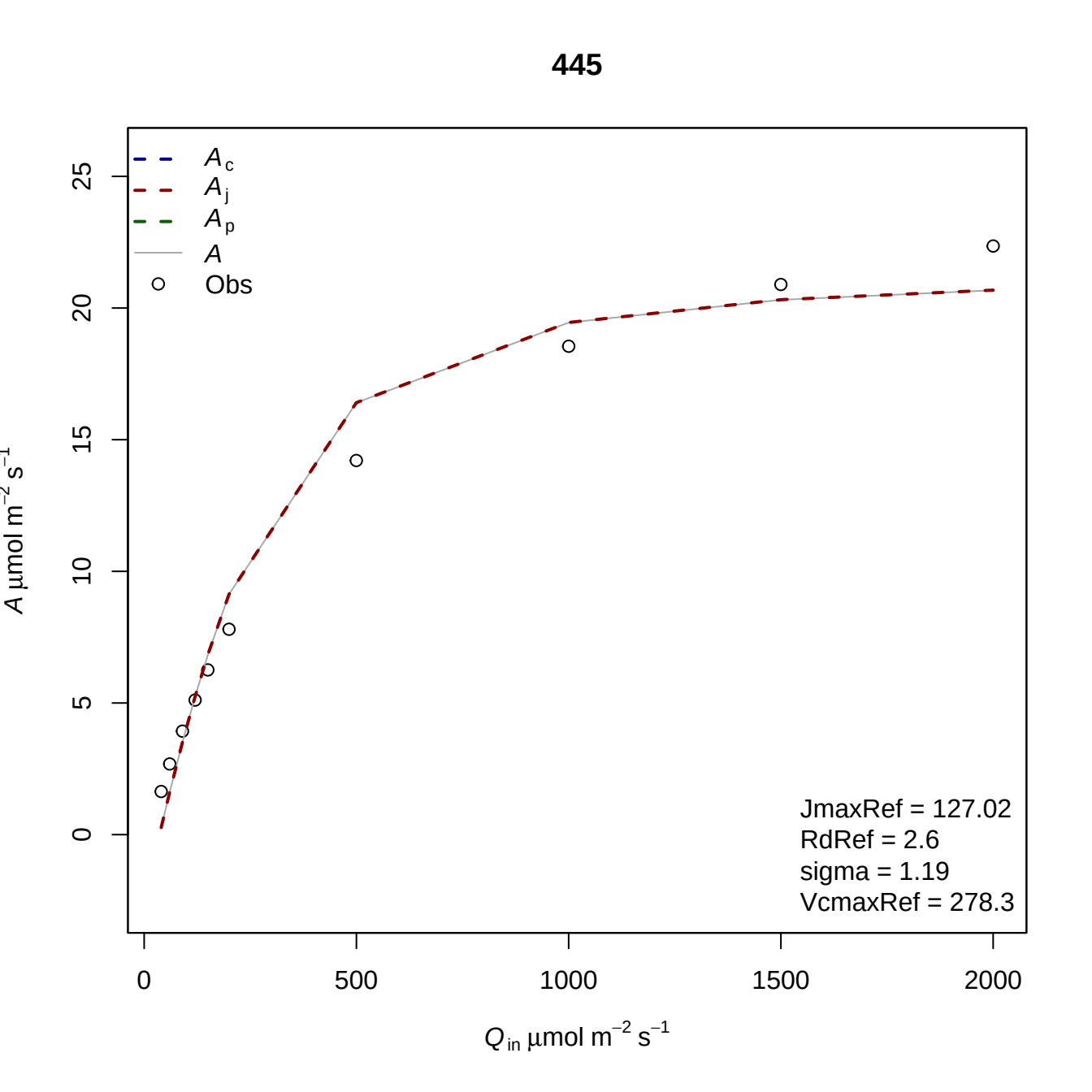

439
A_c
A_j
A_p
A
Obs
JmaxRef = 70.47
RdRef = 0.79
sigma = 0.2
VcmaxRef = 253.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

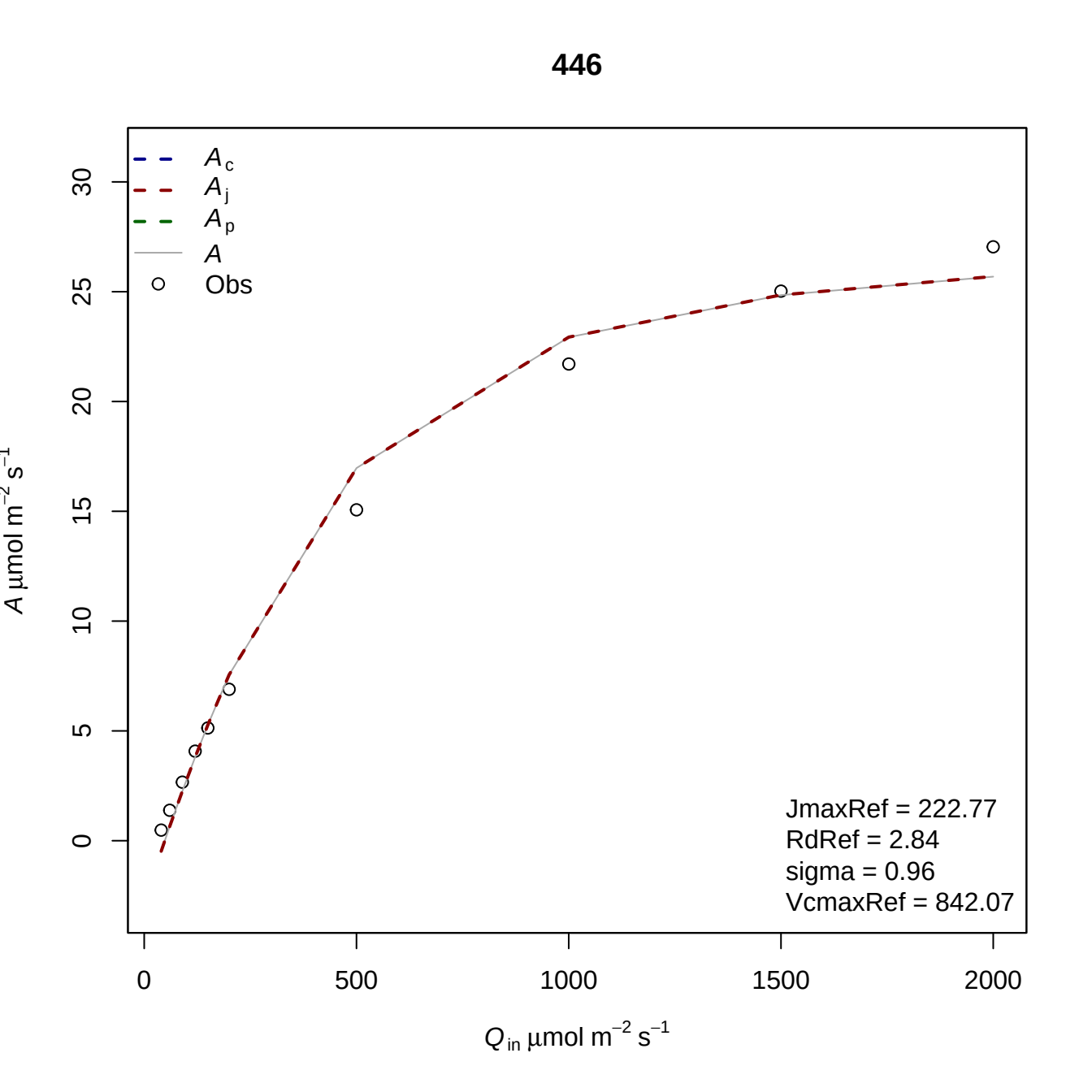

440
A_c
A_j
A_p
A
Obs
JmaxRef = 110.3
RdRef = 2.11
sigma = 0.73
VcmaxRef = 394.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

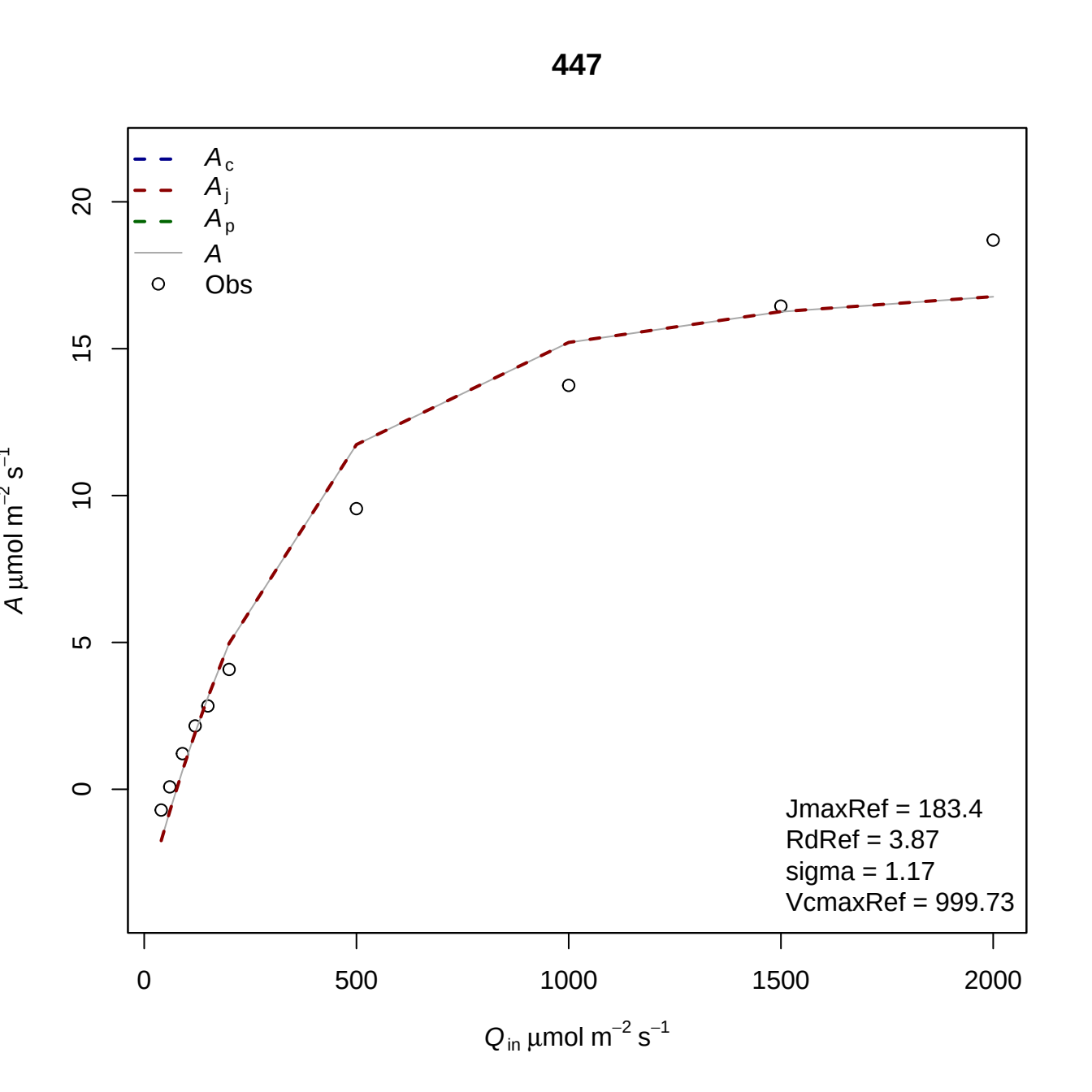

441
A_c
A_j
A_p
A
Obs
JmaxRef = 128.07
RdRef = 1.89
sigma = 0.4
VcmaxRef = 449.67
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

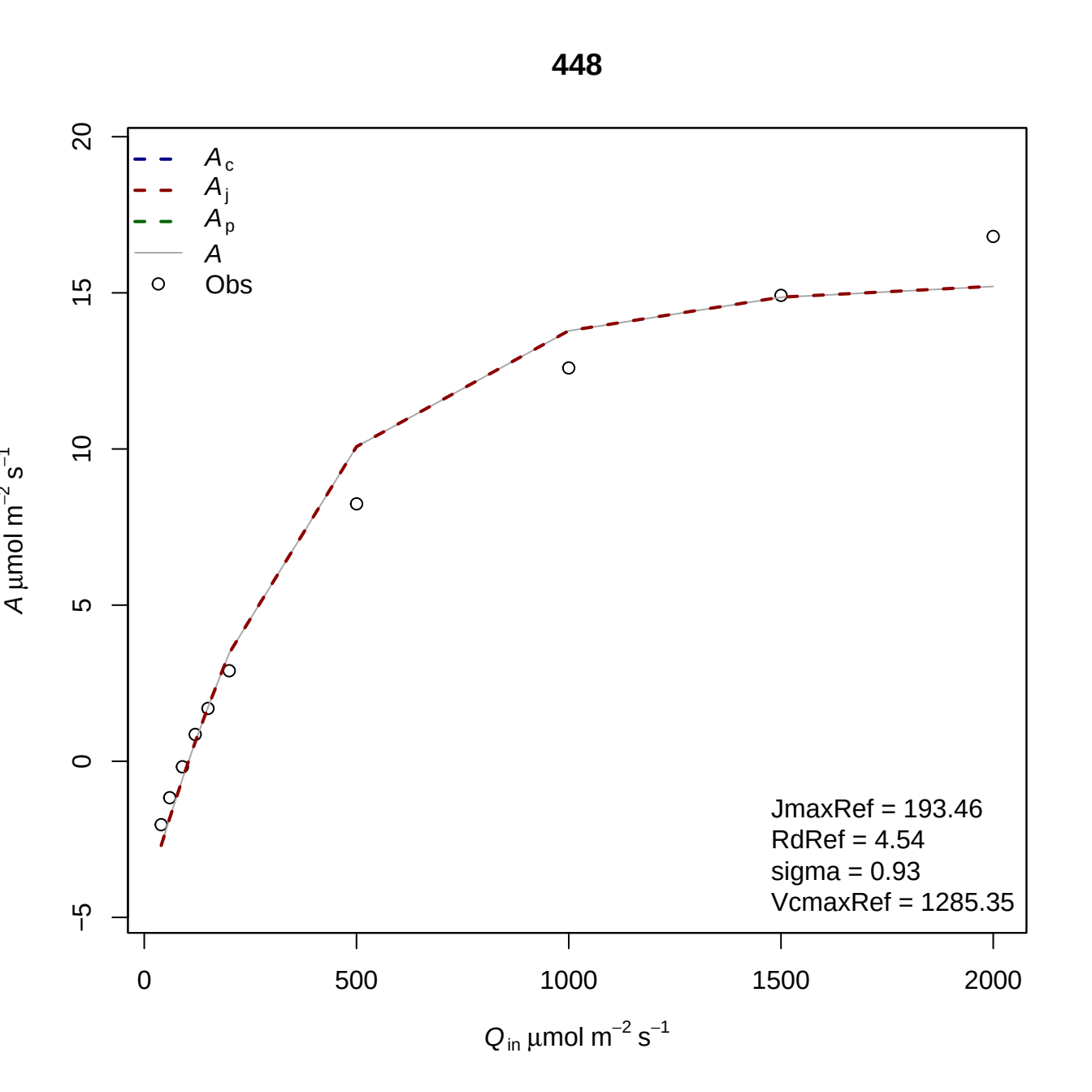

442
A_c
A_j
A_p
A
Obs
JmaxRef = 121.06
RdRef = 2.65
sigma = 0.53
VcmaxRef = 564.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

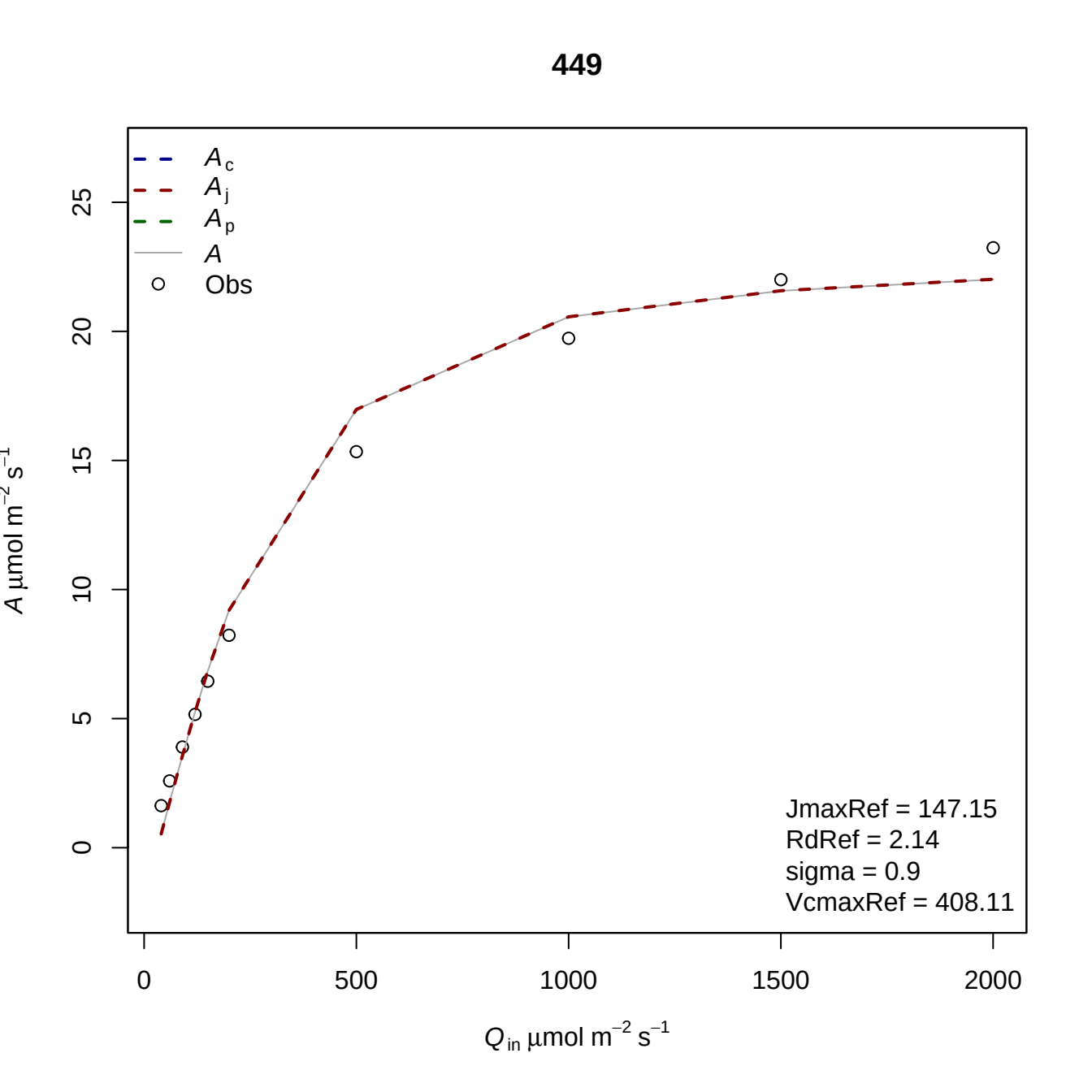

444
A_c
A_j
A_p
A
Obs
JmaxRef = 28.68
RdRef = 2.19
sigma = 0.45
VcmaxRef = 57590
2
1
0
−1
−2
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

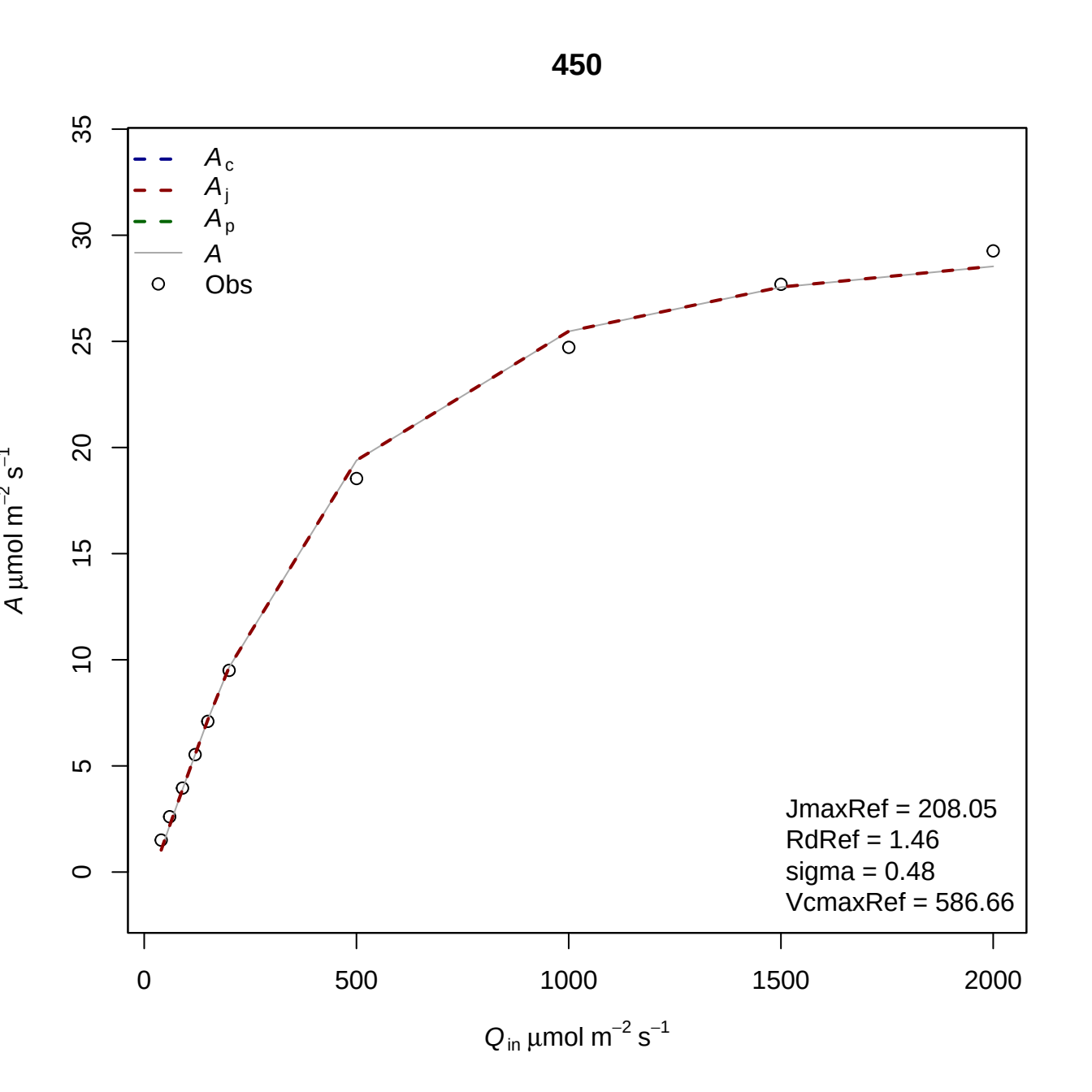

445
A_c
A_j
A_p
A
Obs
JmaxRef = 127.02
RdRef = 2.6
sigma = 1.19
VcmaxRef = 278.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

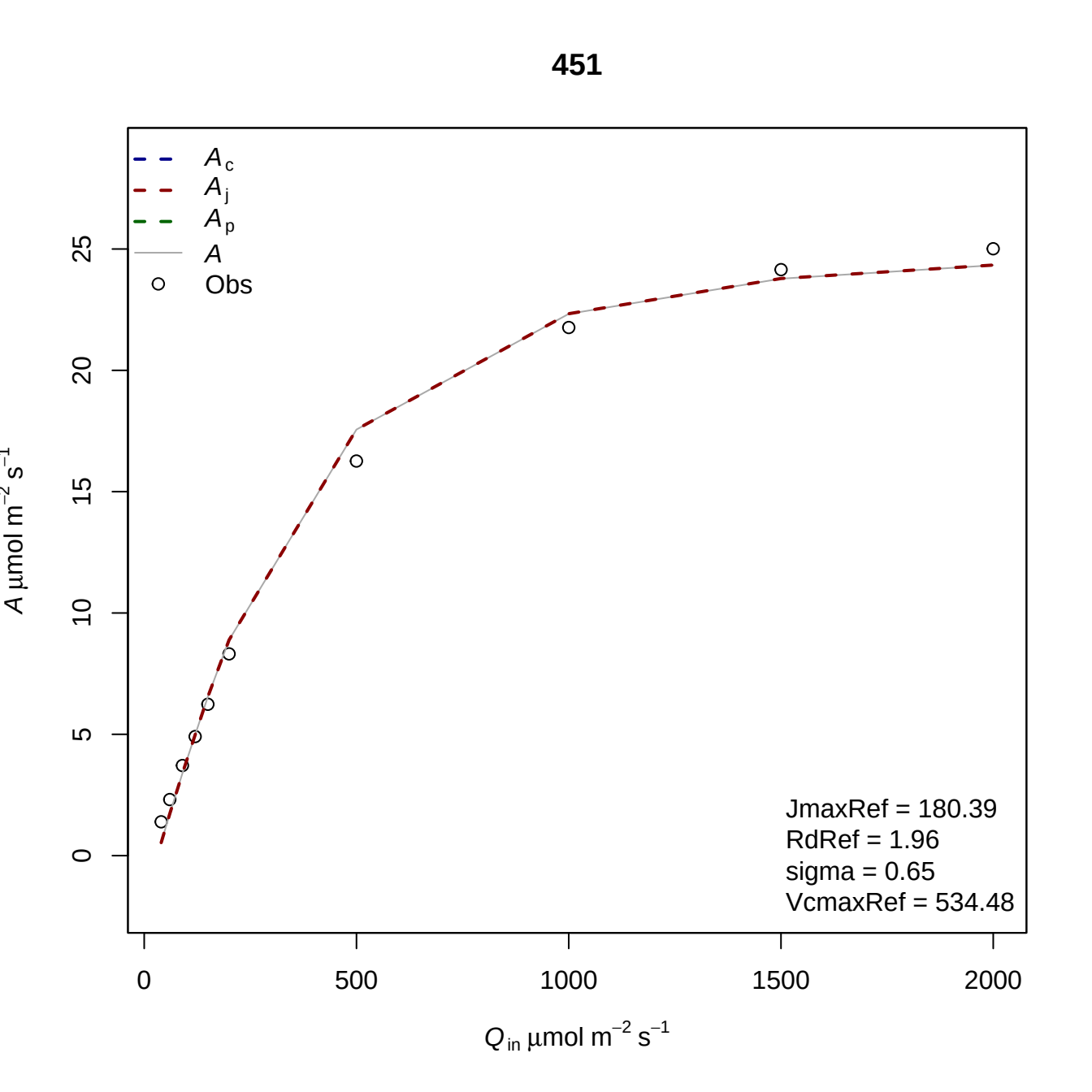

446
A_c
A_j
A_p
A
Obs
JmaxRef = 222.77
RdRef = 2.84
sigma = 0.96
VcmaxRef = 842.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

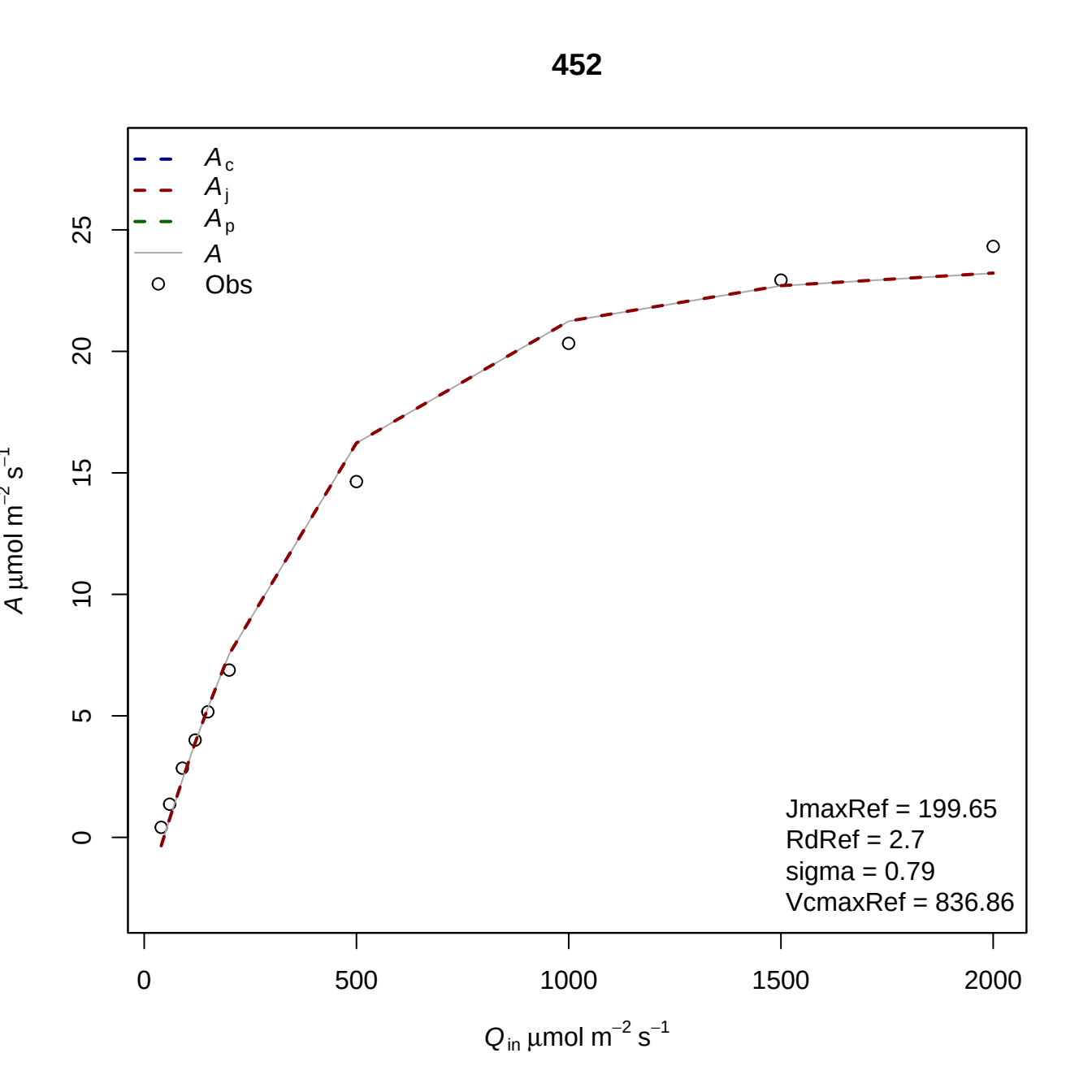

447
A_c
A_j
A_p
A
Obs
JmaxRef = 183.4
RdRef = 3.87
sigma = 1.17
VcmaxRef = 999.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

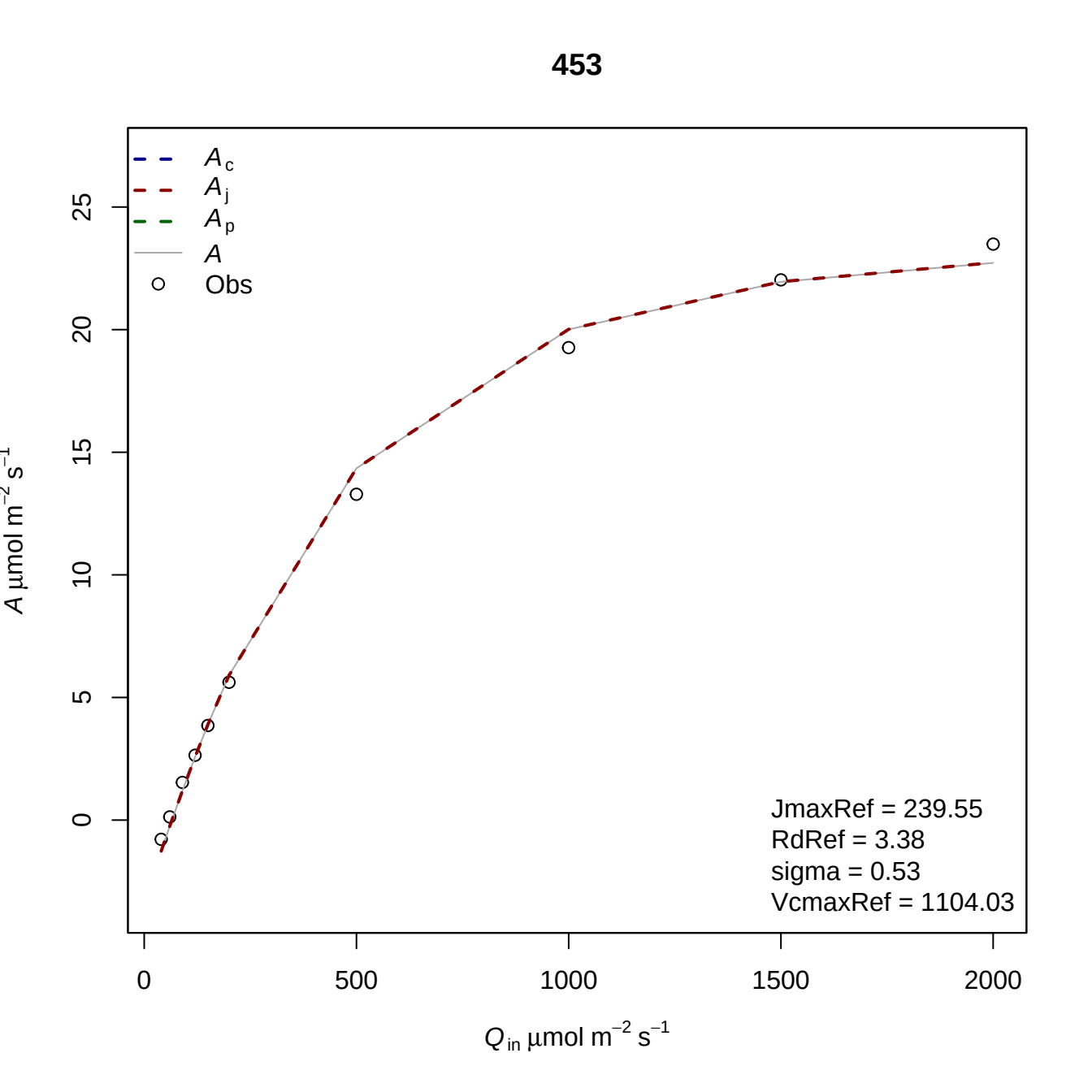

448
A_c
A_j
A_p
A
Obs
JmaxRef = 193.46
RdRef = 4.54
sigma = 0.93
VcmaxRef = 1285.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

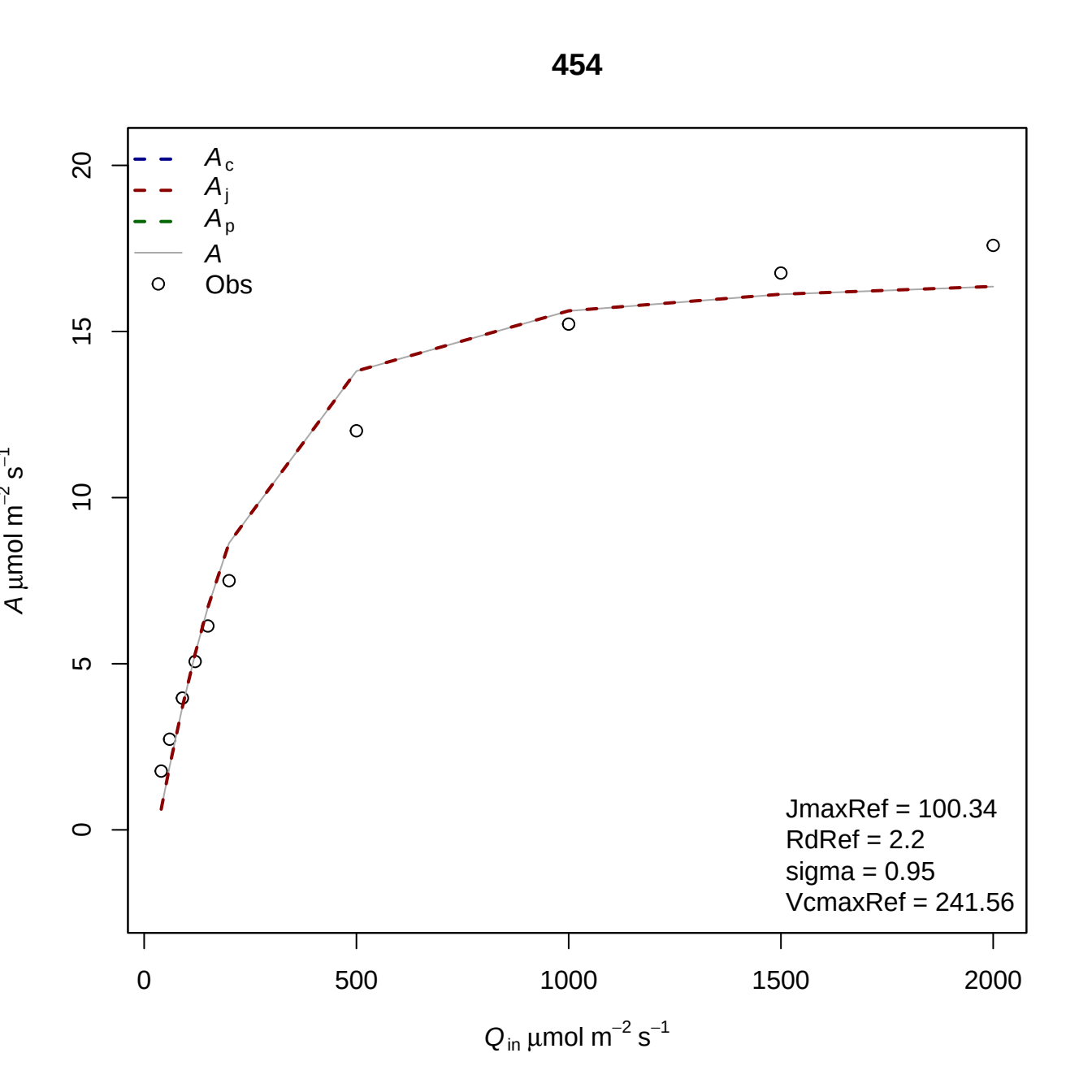

449
A_c
A_j
A_p
A
Obs
JmaxRef = 147.15
RdRef = 2.14
sigma = 0.9
VcmaxRef = 408.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

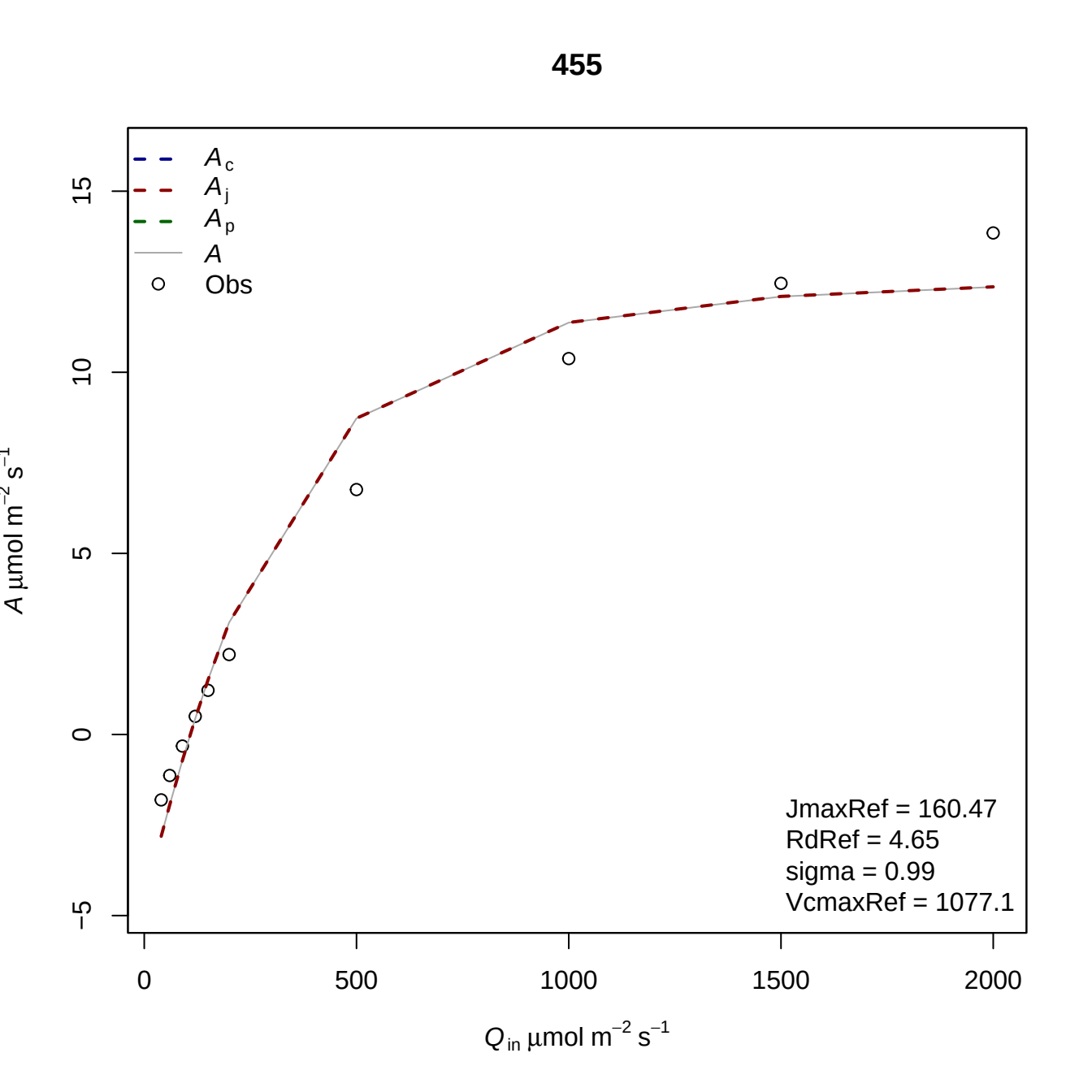

450
A_c
A_j
A_p
A
Obs
JmaxRef = 208.05
RdRef = 1.46
sigma = 0.48
VcmaxRef = 586.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

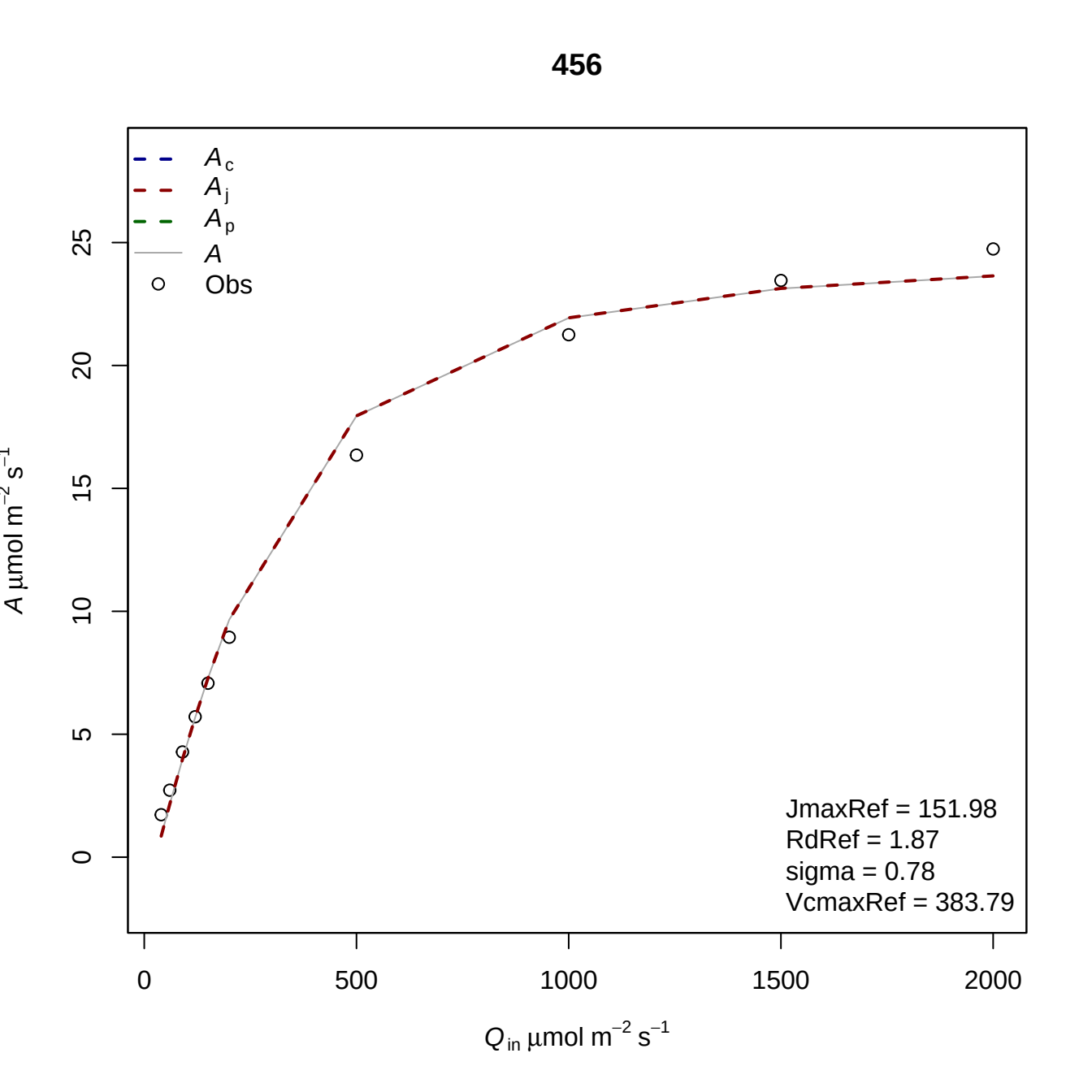

451
A_c
A_j
A_p
A
Obs
JmaxRef = 180.39
RdRef = 1.96
sigma = 0.65
VcmaxRef = 534.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

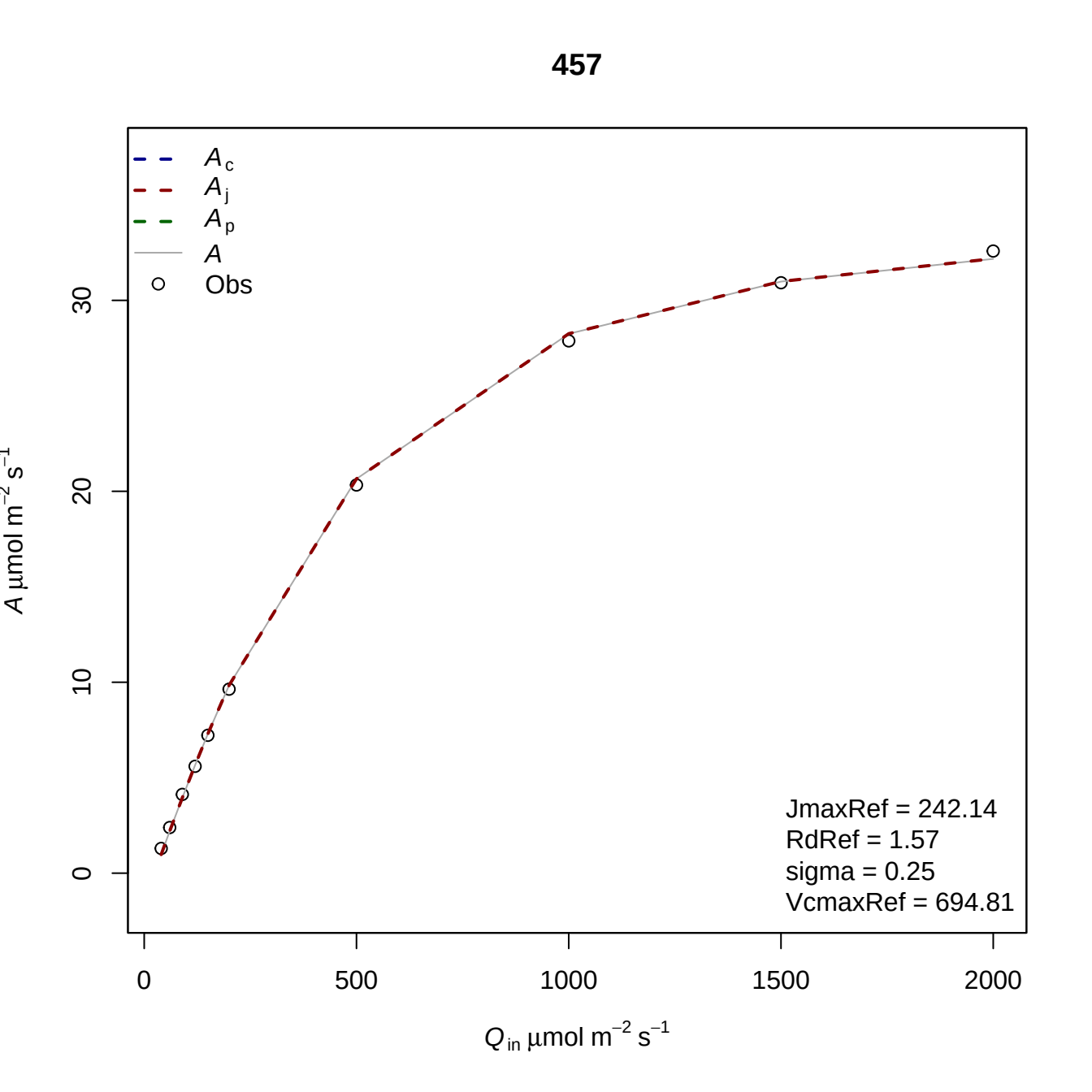

452
A_c
A_j
A_p
A
Obs
JmaxRef = 199.65
RdRef = 2.7
sigma = 0.79
VcmaxRef = 836.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

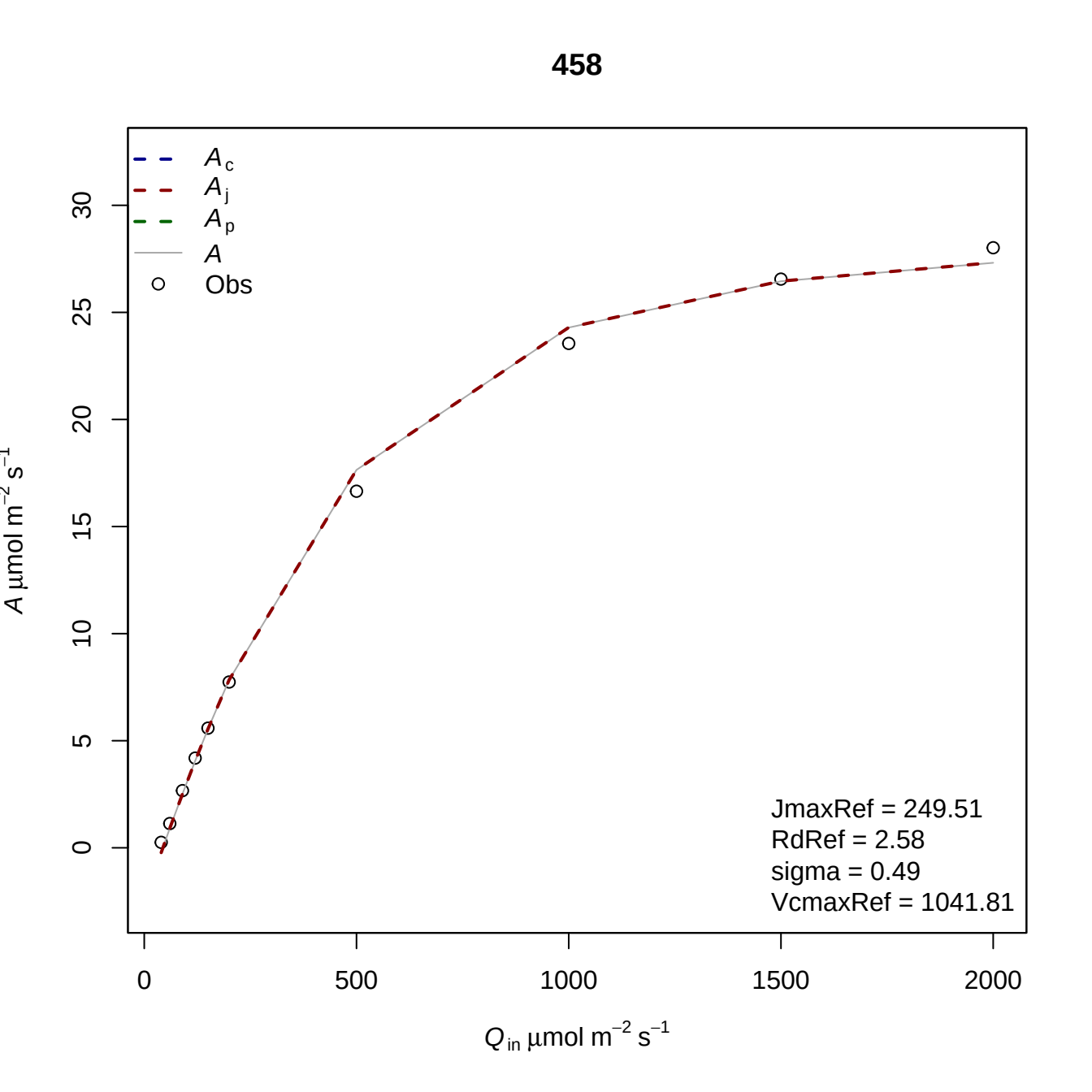

453
A_c
A_j
A_p
A
Obs
JmaxRef = 239.55
RdRef = 3.38
sigma = 0.53
VcmaxRef = 1104.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

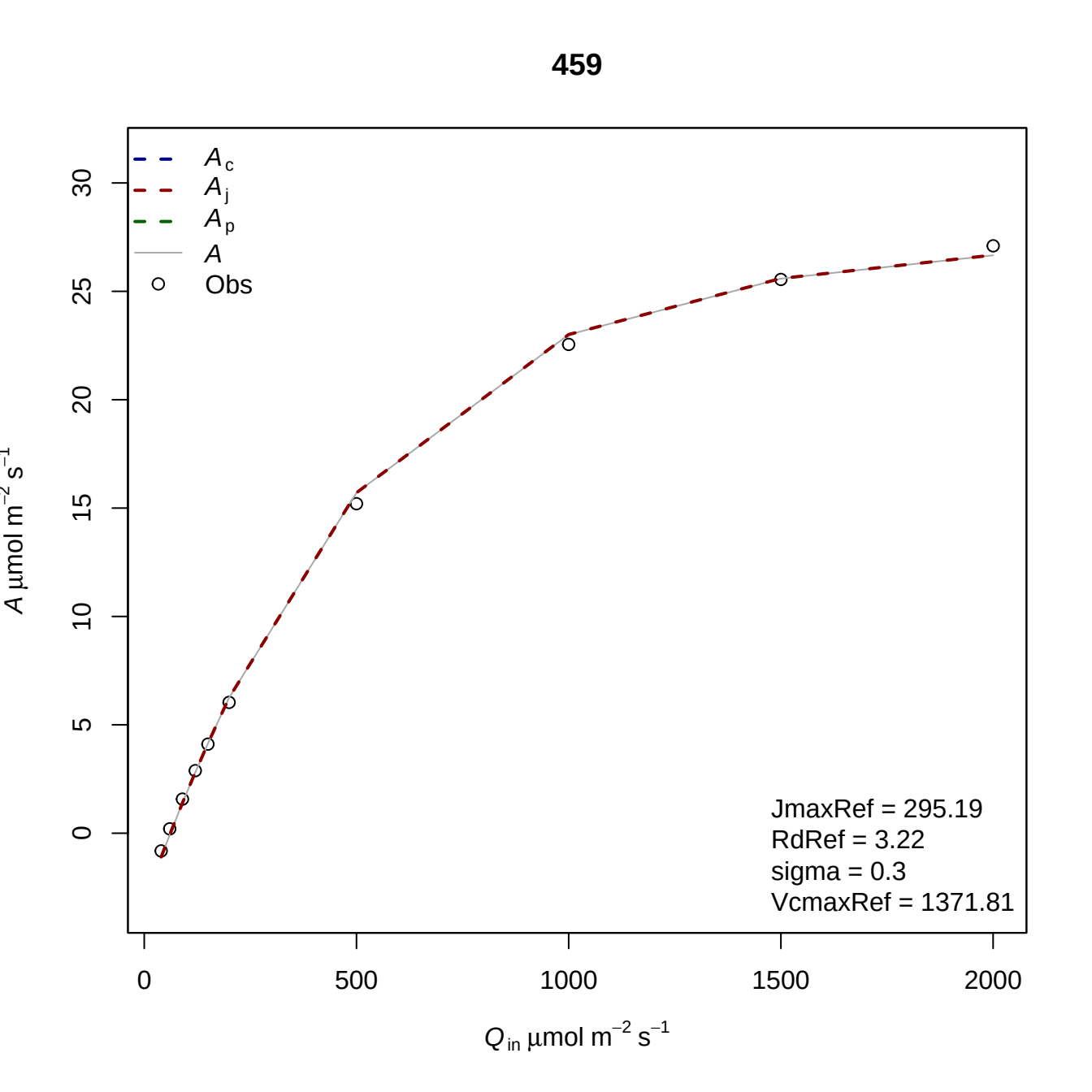

454
A_c
A_j
A_p
A
Obs
JmaxRef = 100.34
RdRef = 2.2
sigma = 0.95
VcmaxRef = 241.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

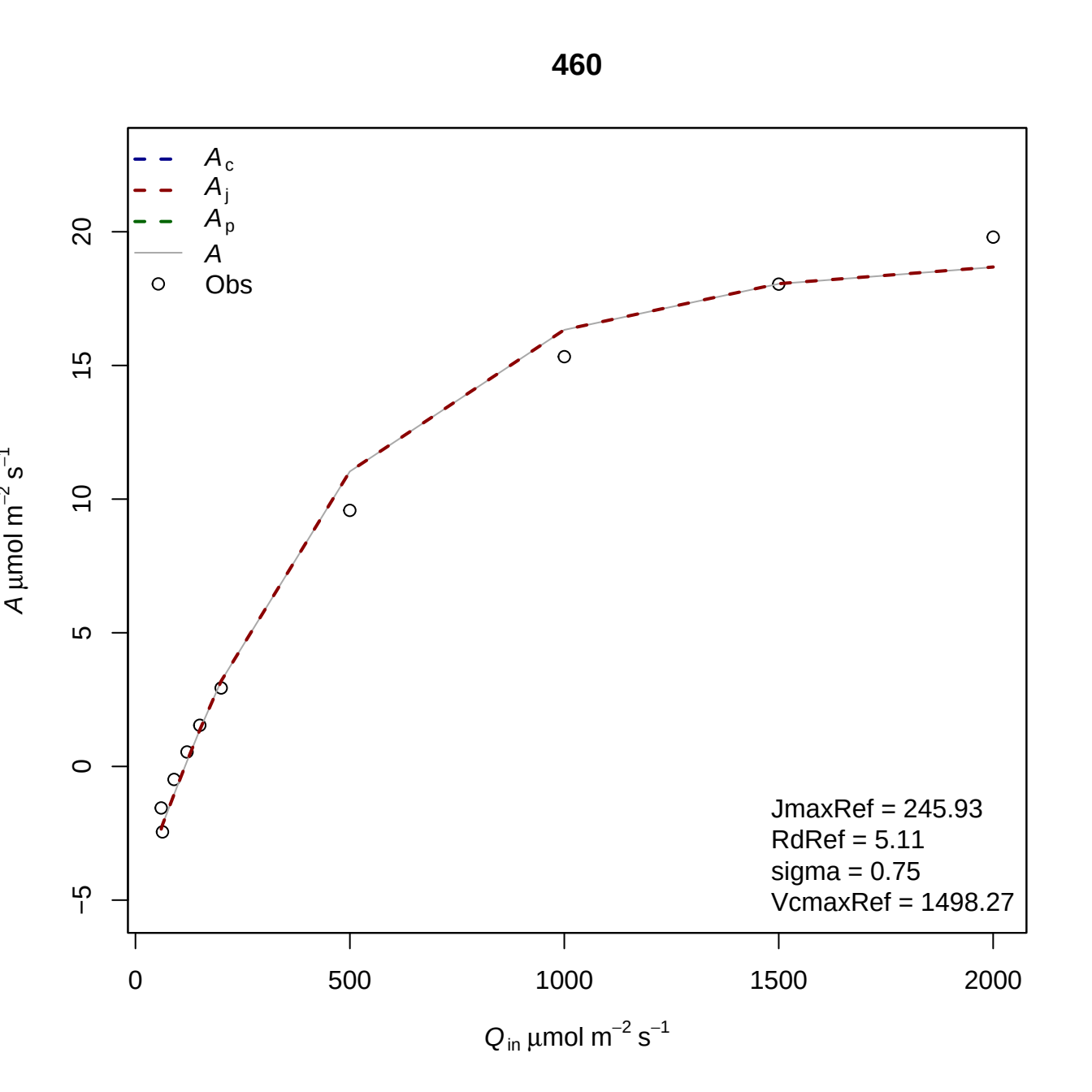

455
A_c
A_j
A_p
A
Obs
JmaxRef = 160.47
RdRef = 4.65
sigma = 0.99
VcmaxRef = 1077.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

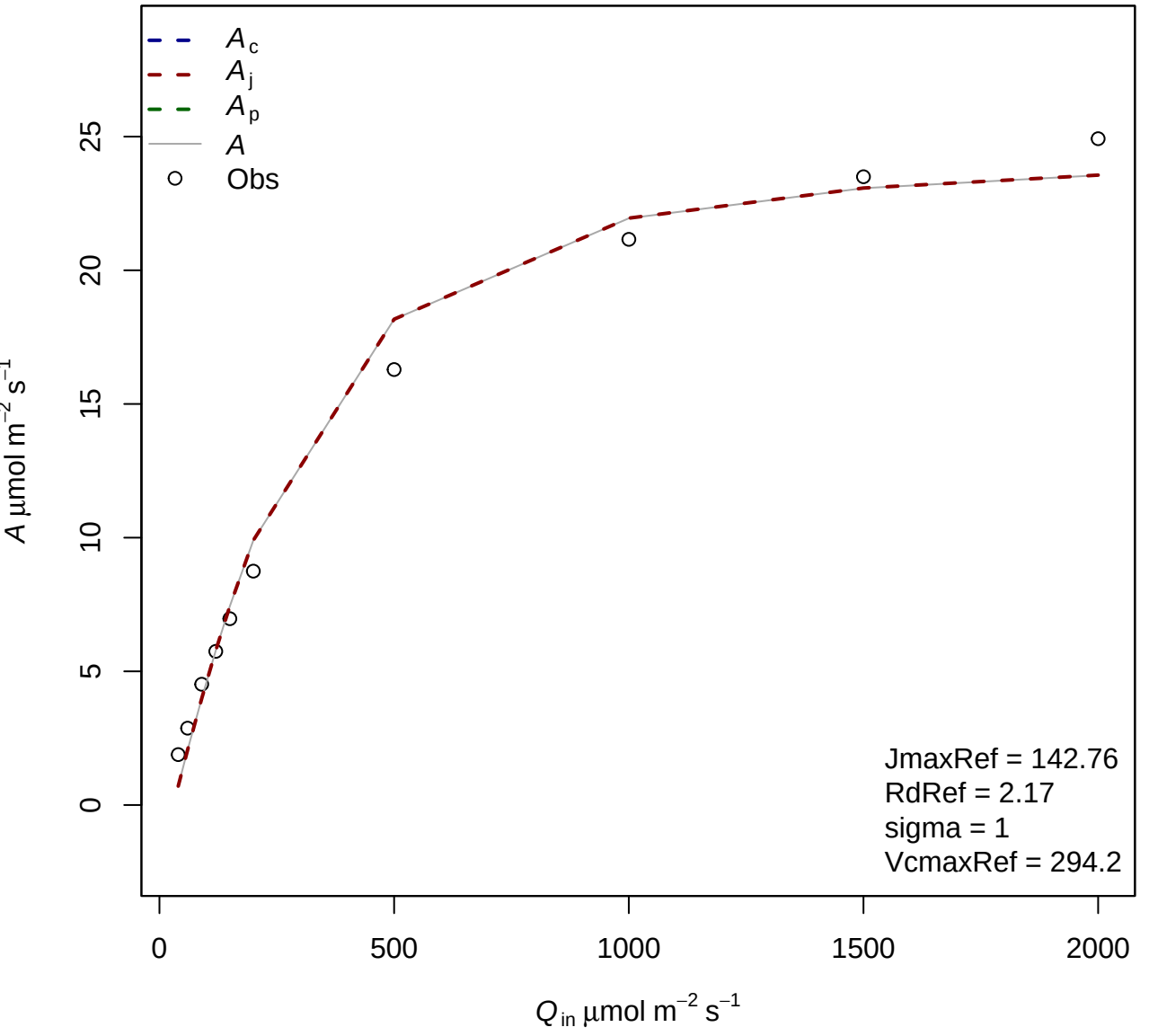

456
A_c
A_j
A_p
A
Obs
JmaxRef = 151.98
RdRef = 1.87
sigma = 0.78
VcmaxRef = 383.79
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

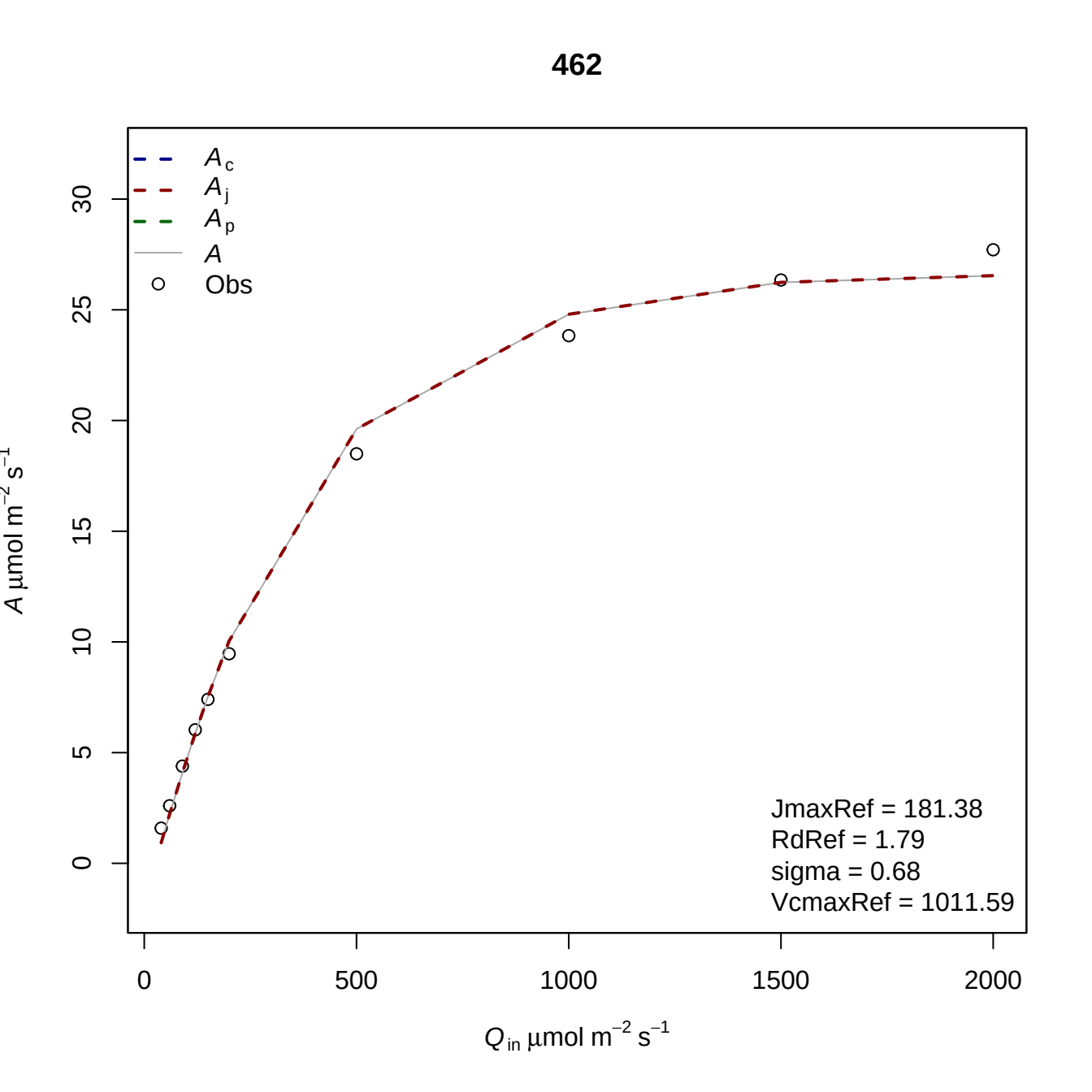

457
A_c
A_j
A_p
A
Obs
JmaxRef = 242.14
RdRef = 1.57
sigma = 0.25
VcmaxRef = 694.81
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

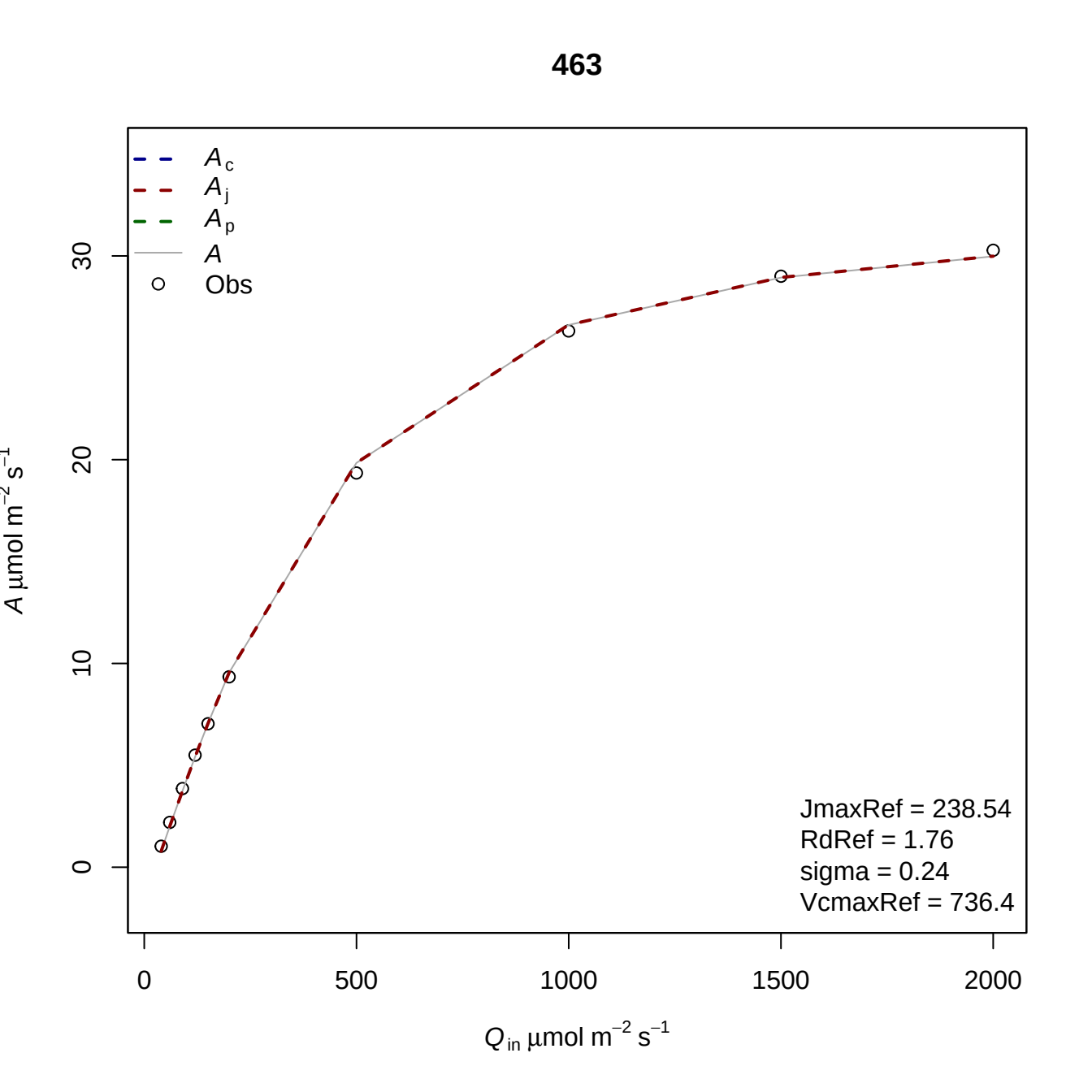

458
A_c
A_j
A_p
A
Obs
JmaxRef = 249.51
RdRef = 2.58
sigma = 0.49
VcmaxRef = 1041.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

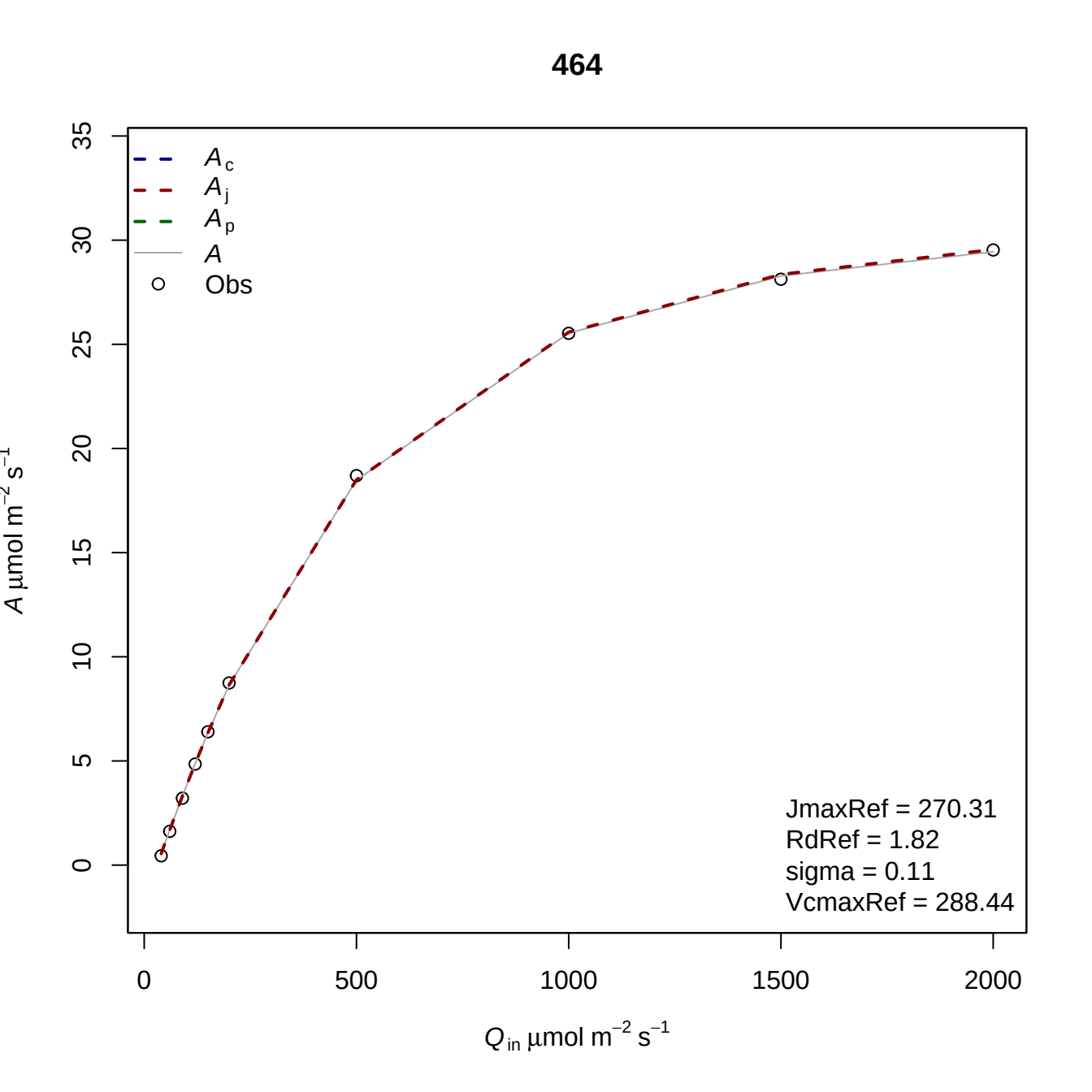

459
A_c
A_j
A_p
A
Obs
JmaxRef = 295.19
RdRef = 3.22
sigma = 0.3
VcmaxRef = 1371.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

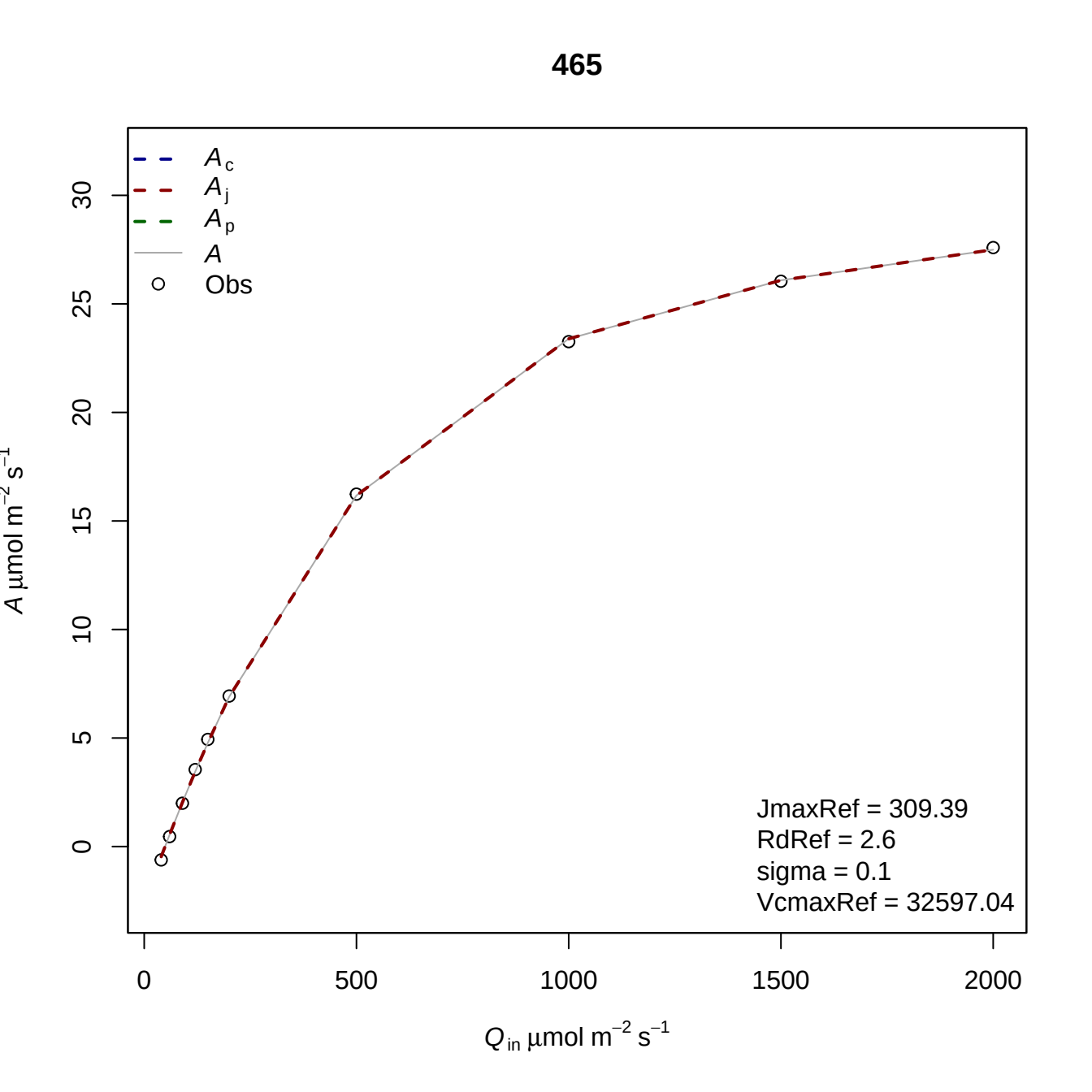

460
A_c
A_j
A_p
A
Obs
JmaxRef = 245.93
RdRef = 5.11
sigma = 0.75
VcmaxRef = 1498.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

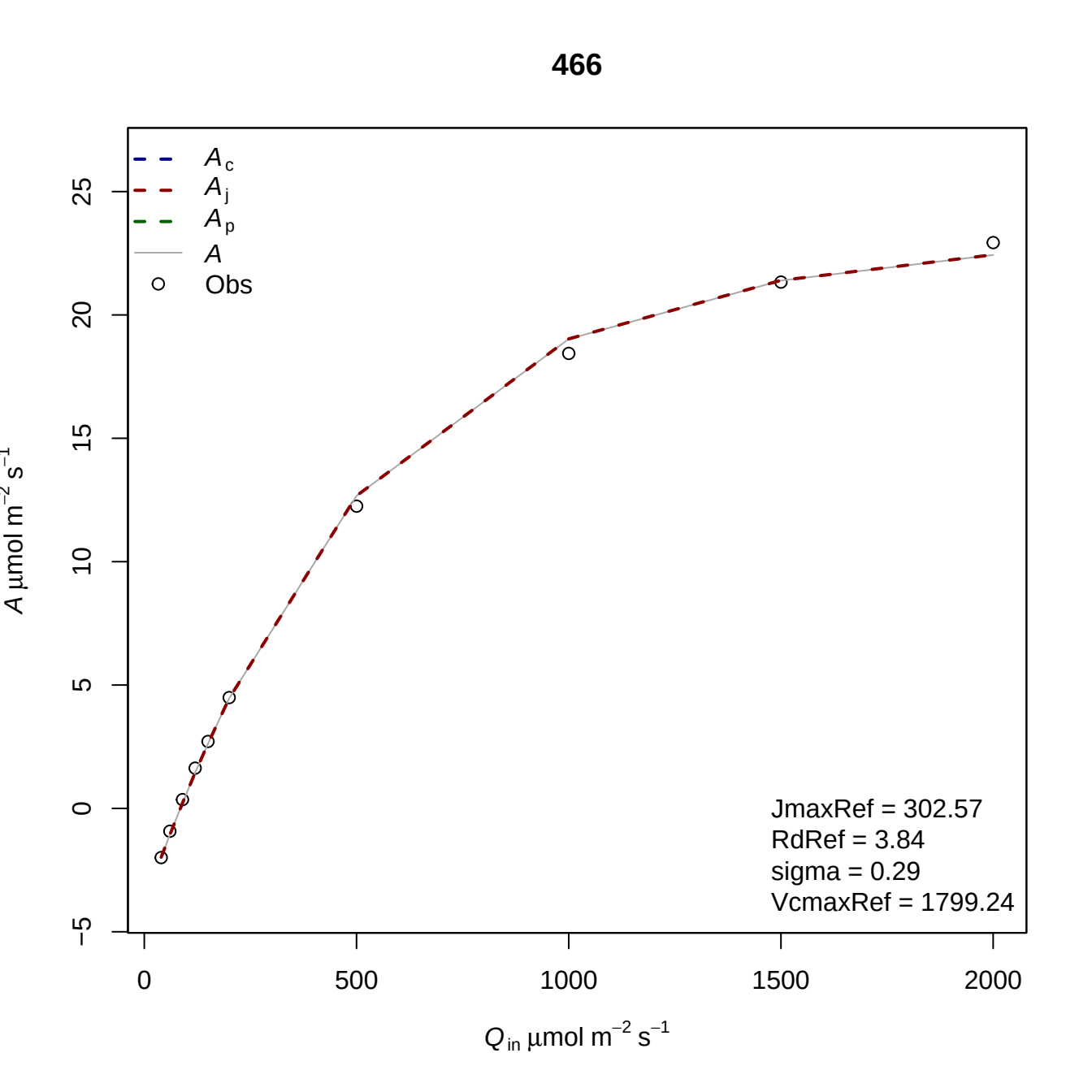

A_c
A_j
A_p
A
Obs
JmaxRef = 142.76
RdRef = 2.17
sigma = 1
VcmaxRef = 294.2
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

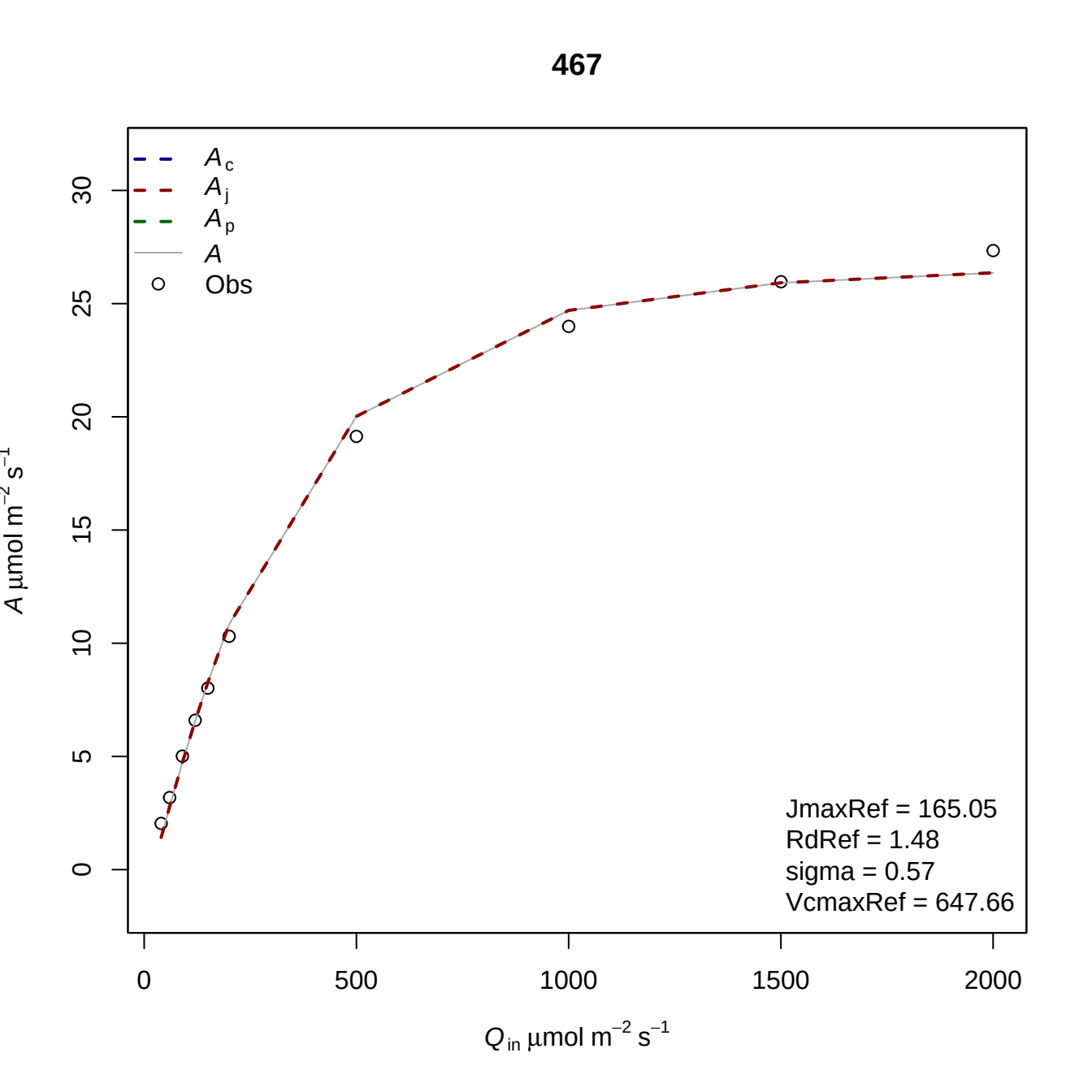

462
A_c
A_j
A_p
A
Obs
JmaxRef = 181.38
RdRef = 1.79
sigma = 0.68
VcmaxRef = 1011.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

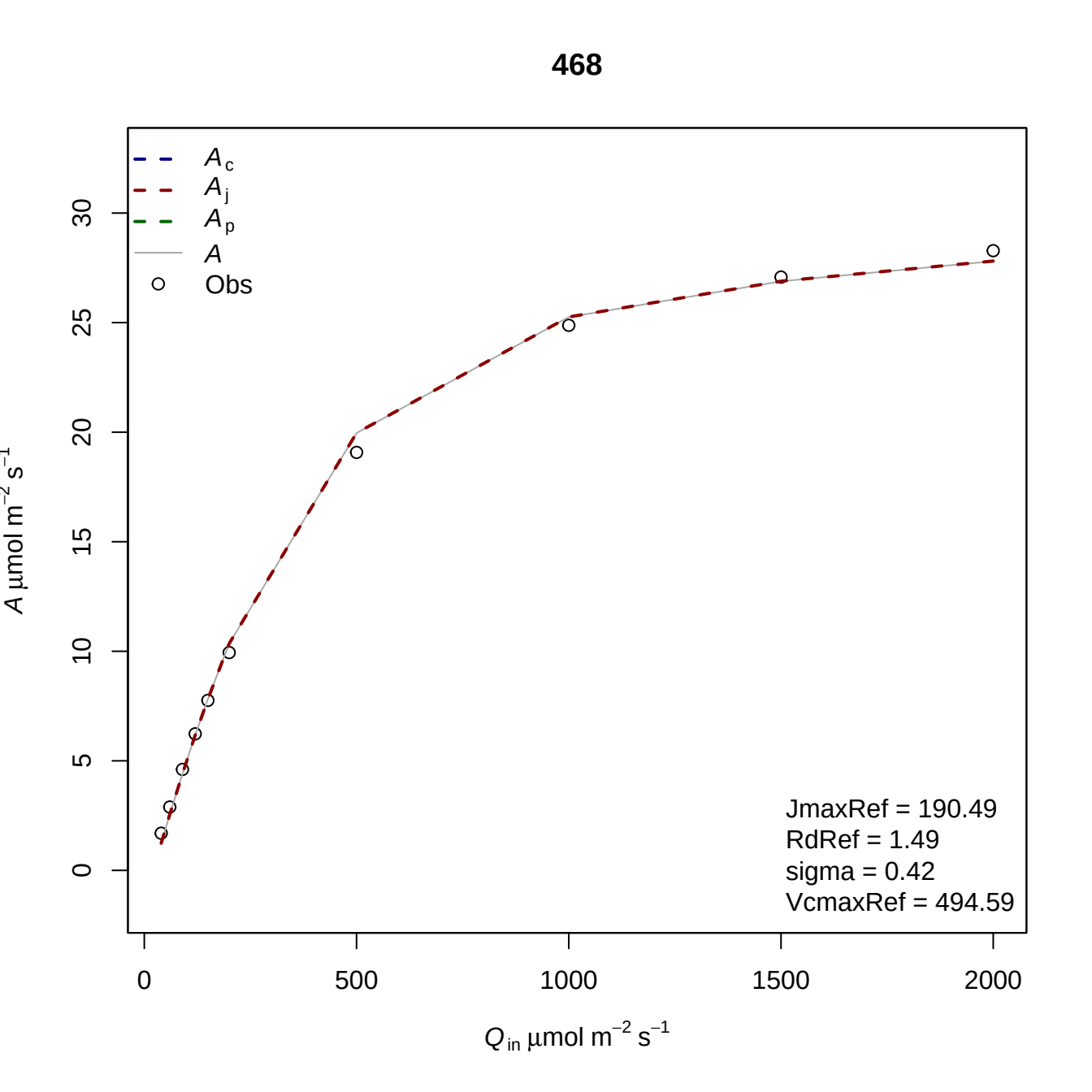

463
A_c
A_j
A_p
A
Obs
JmaxRef = 238.54
RdRef = 1.76
sigma = 0.24
VcmaxRef = 736.4
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

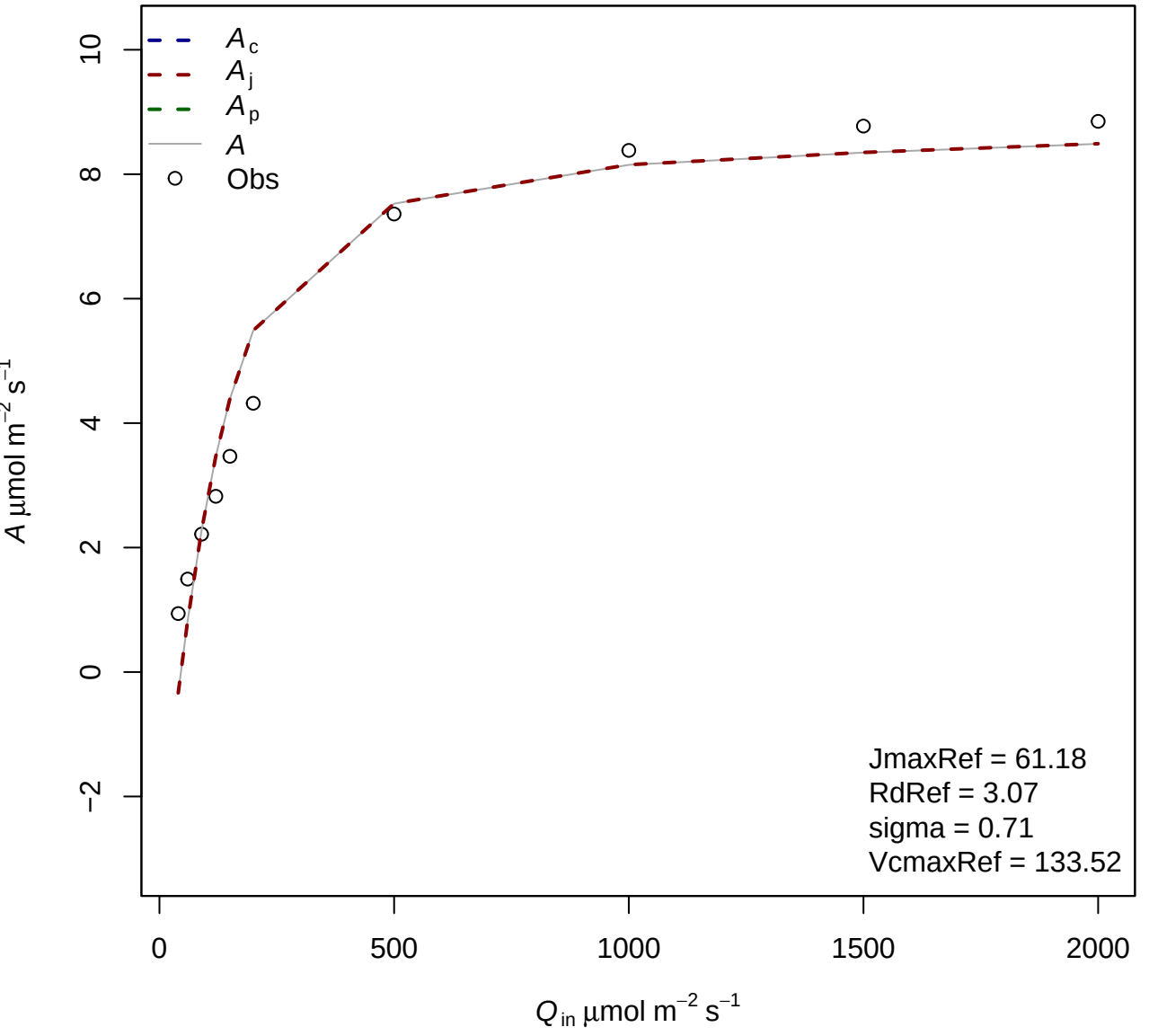

464
A_c
A_j
A_p
A
Obs
JmaxRef = 270.31
RdRef = 1.82
sigma = 0.11
VcmaxRef = 288.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

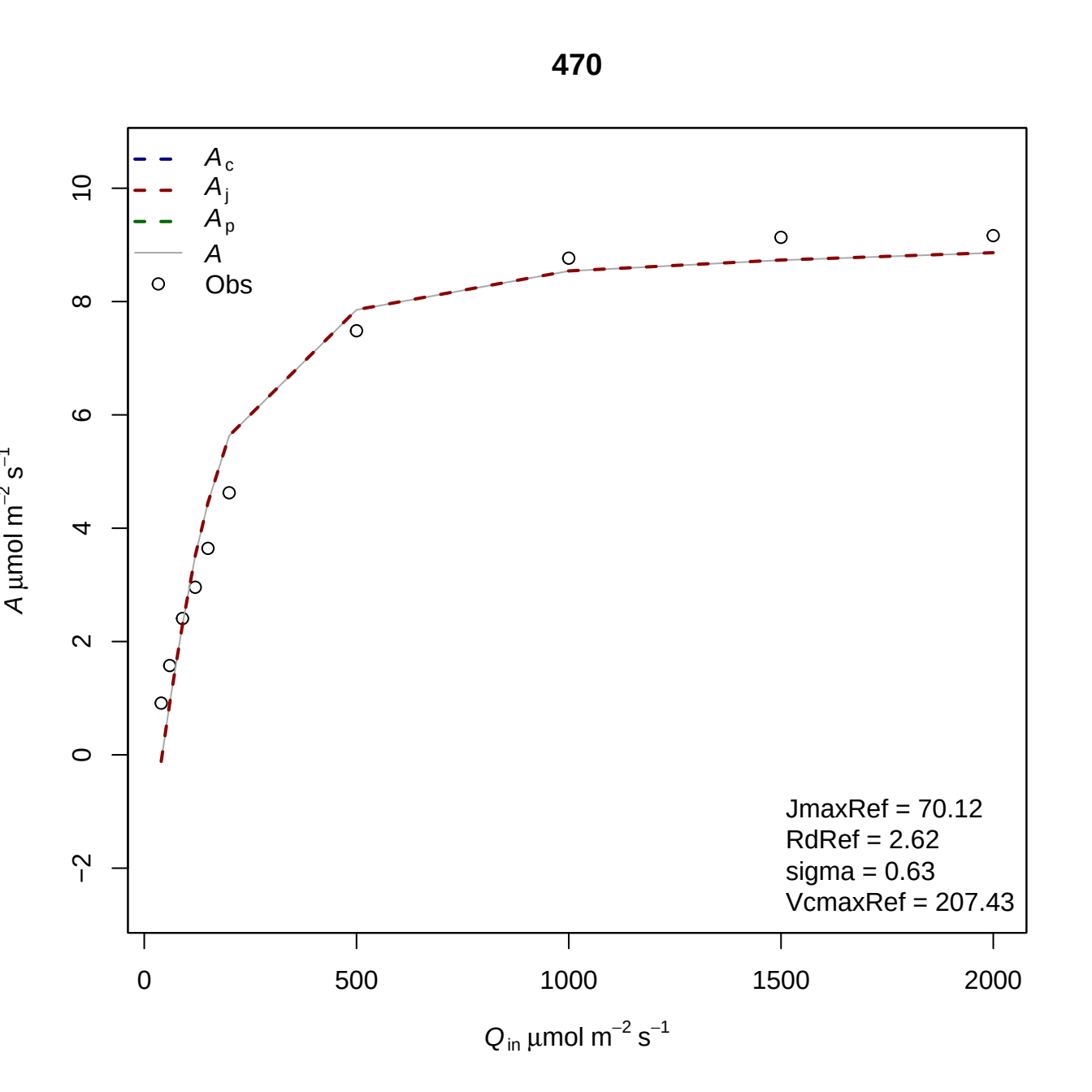

465
A_c
A_j
A_p
A
Obs
JmaxRef = 309.39
RdRef = 2.6
sigma = 0.1
VcmaxRef = 32597.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

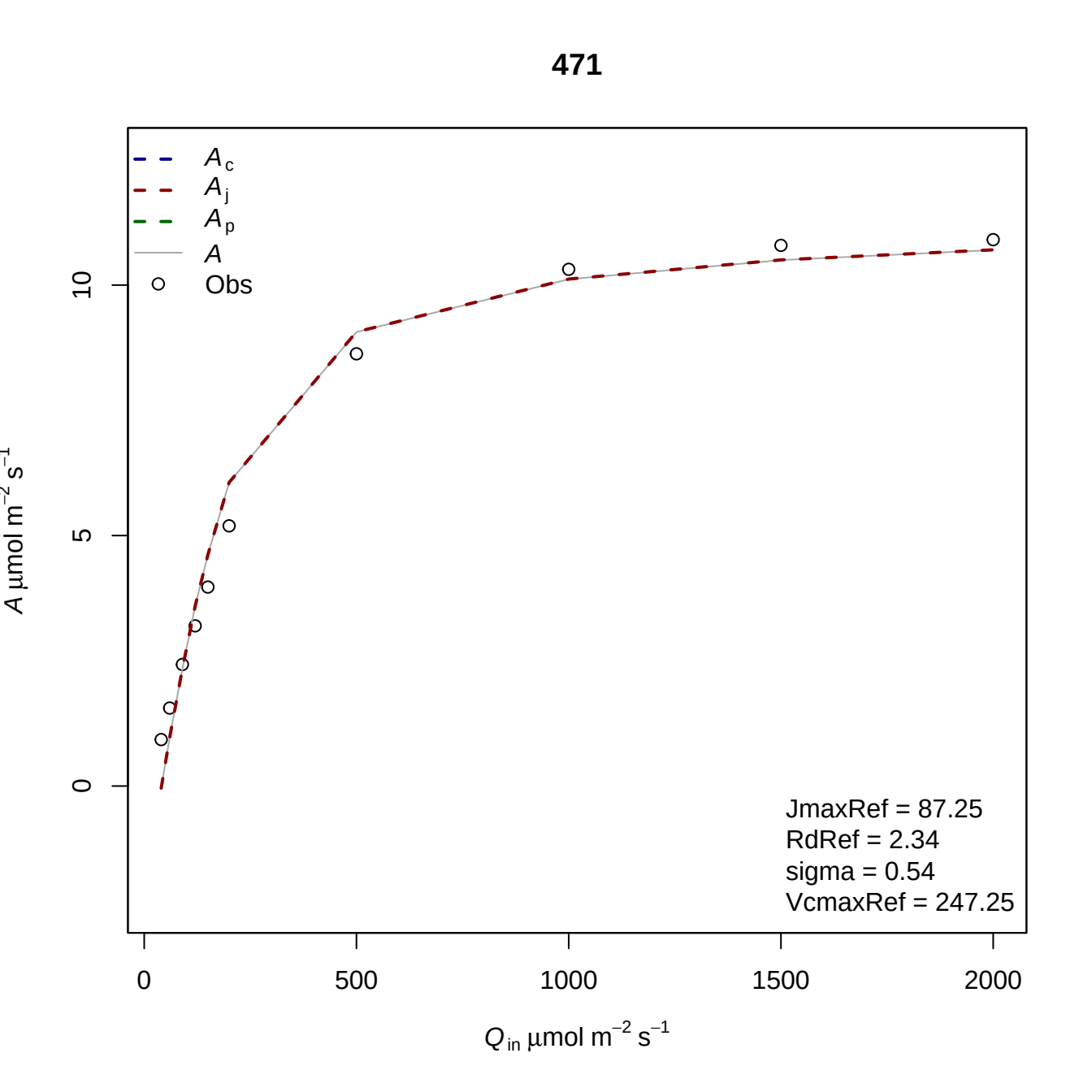

466
A_c
A_j
A_p
A
Obs
JmaxRef = 302.57
RdRef = 3.84
sigma = 0.29
VcmaxRef = 1799.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500
2000

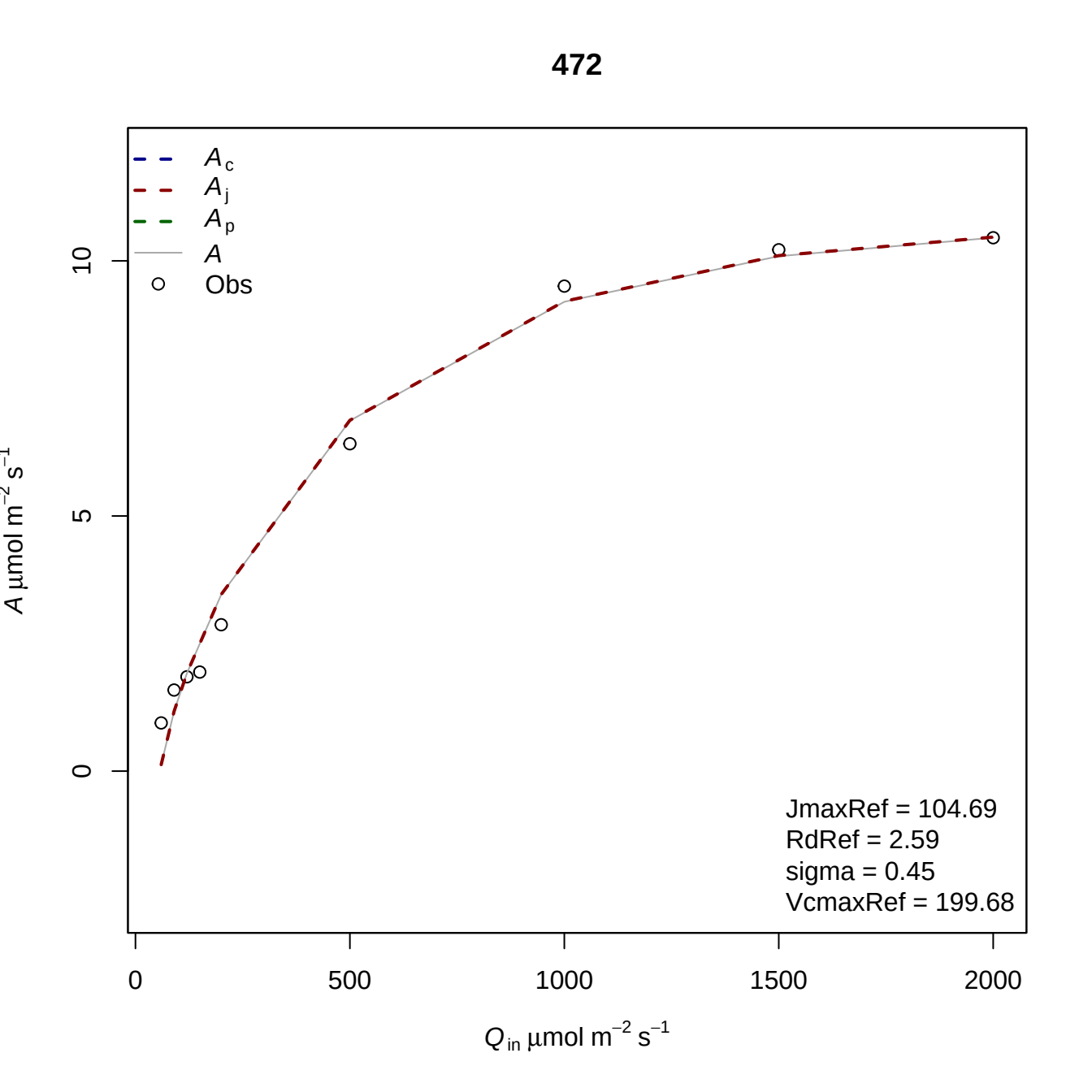

467
A_c
A_j
A_p
A
Obs
JmaxRef = 165.05
RdRef = 1.48
sigma = 0.57
VcmaxRef = 647.66
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

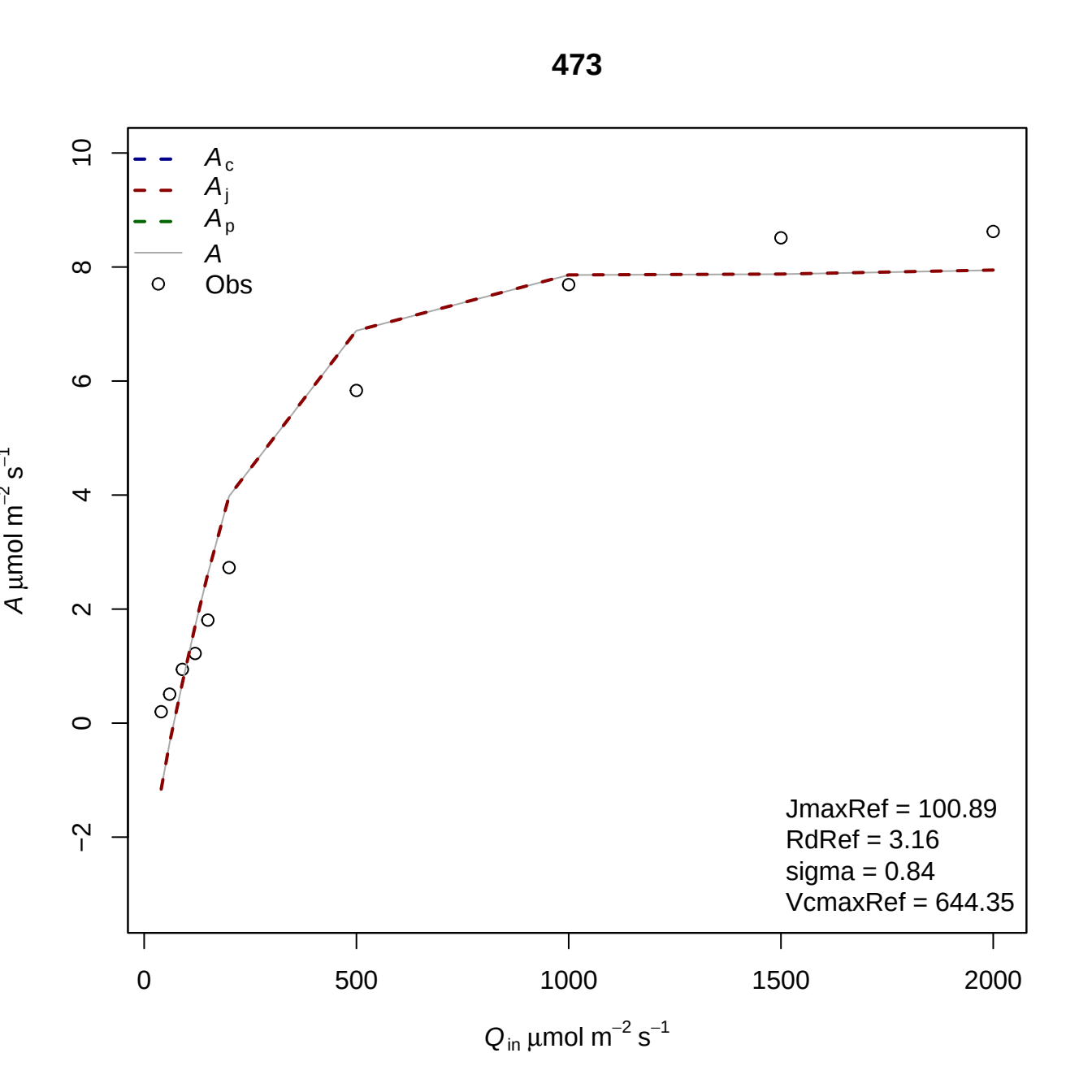

468
A_c
A_j
A_p
A
Obs
JmaxRef = 190.49
RdRef = 1.49
sigma = 0.42
VcmaxRef = 494.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

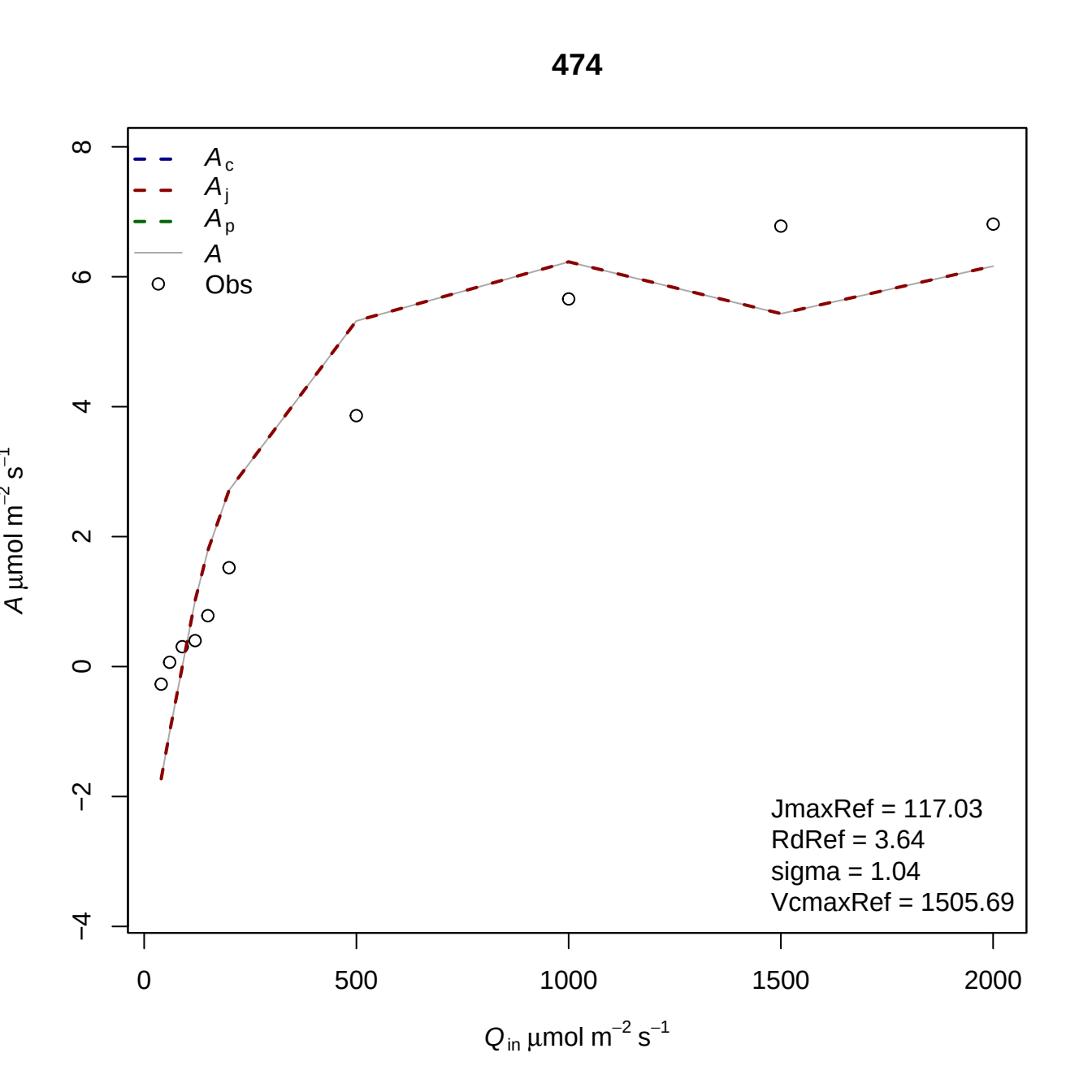

A_c
A_j
A_p
A
Obs
JmaxRef = 61.18
RdRef = 3.07
sigma = 0.71
VcmaxRef = 133.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

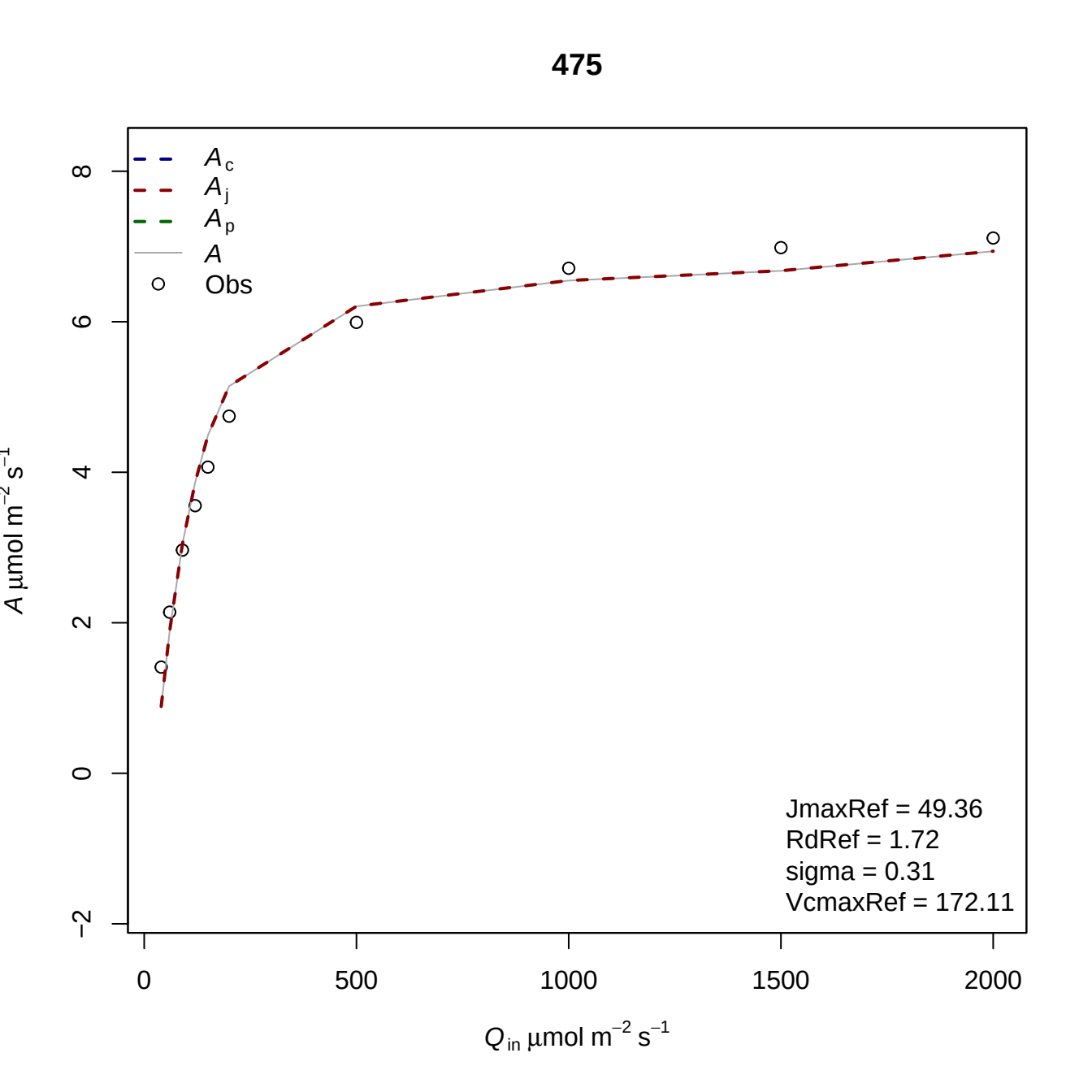

470
A_c
A_j
A_p
A
Obs
JmaxRef = 70.12
RdRef = 2.62
sigma = 0.63
VcmaxRef = 207.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

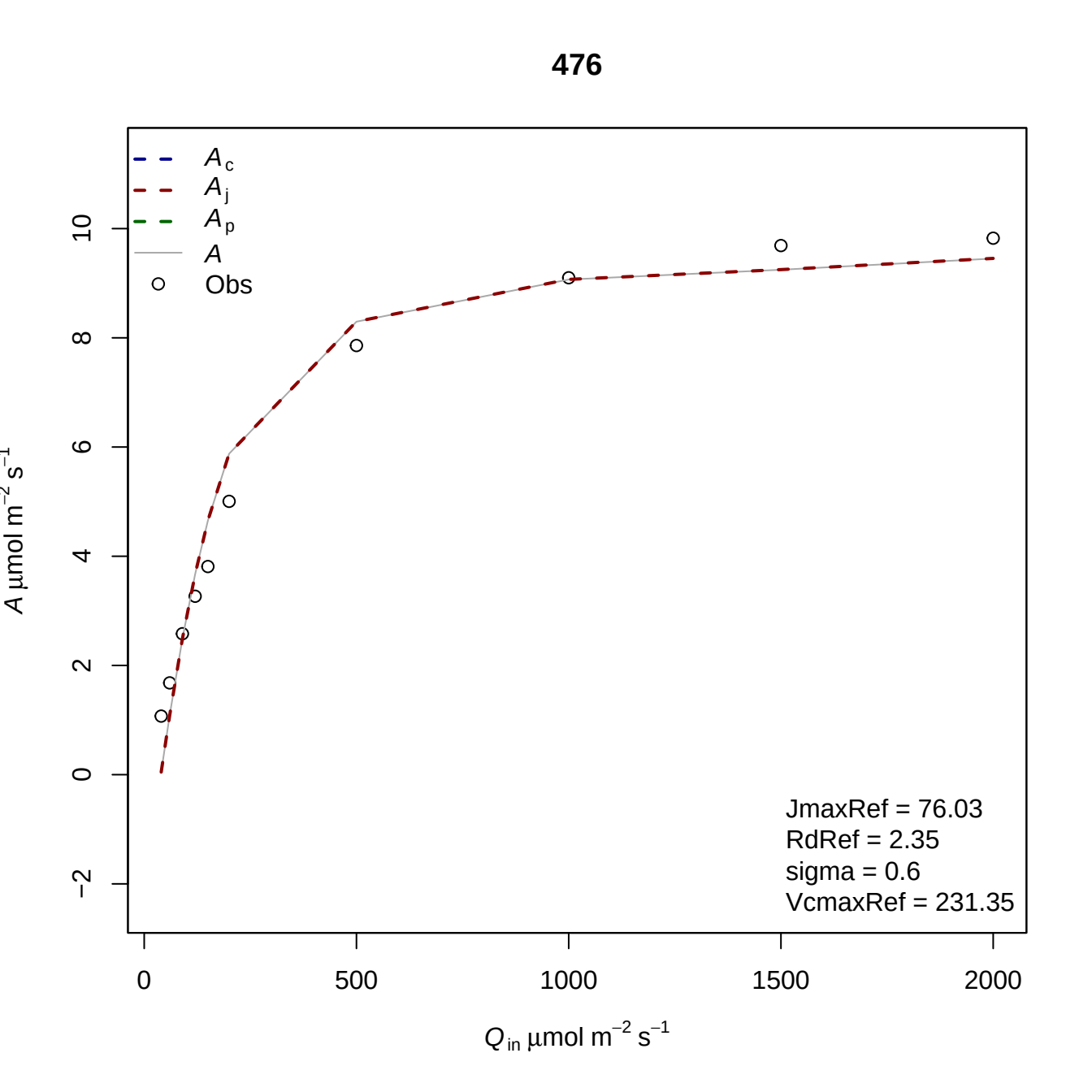

471
A_c
A_j
A_p
A
Obs
JmaxRef = 87.25
RdRef = 2.34
sigma = 0.54
VcmaxRef = 247.25
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

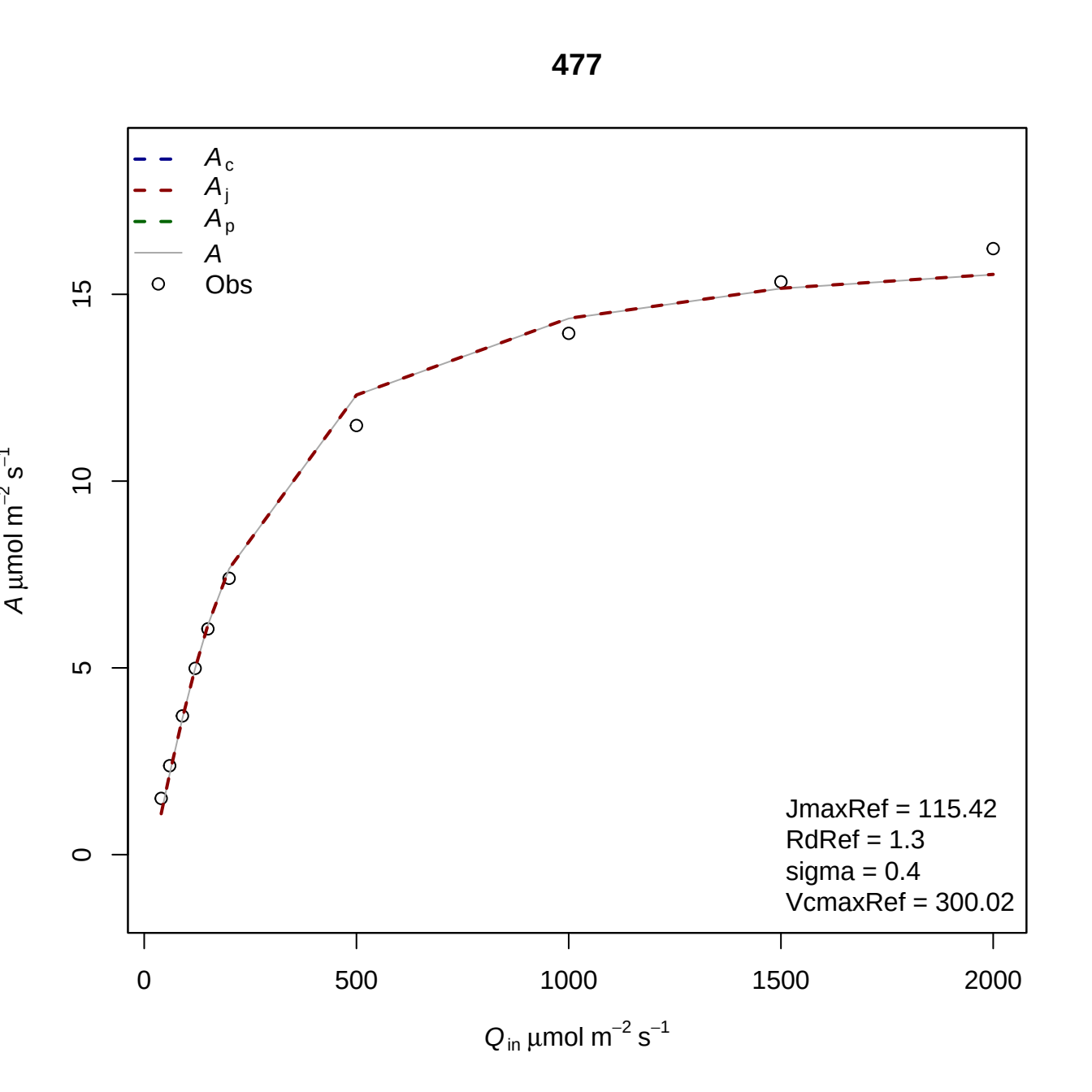

472
A_c
A_j
A_p
A
Obs
JmaxRef = 104.69
RdRef = 2.59
sigma = 0.45
VcmaxRef = 199.68
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

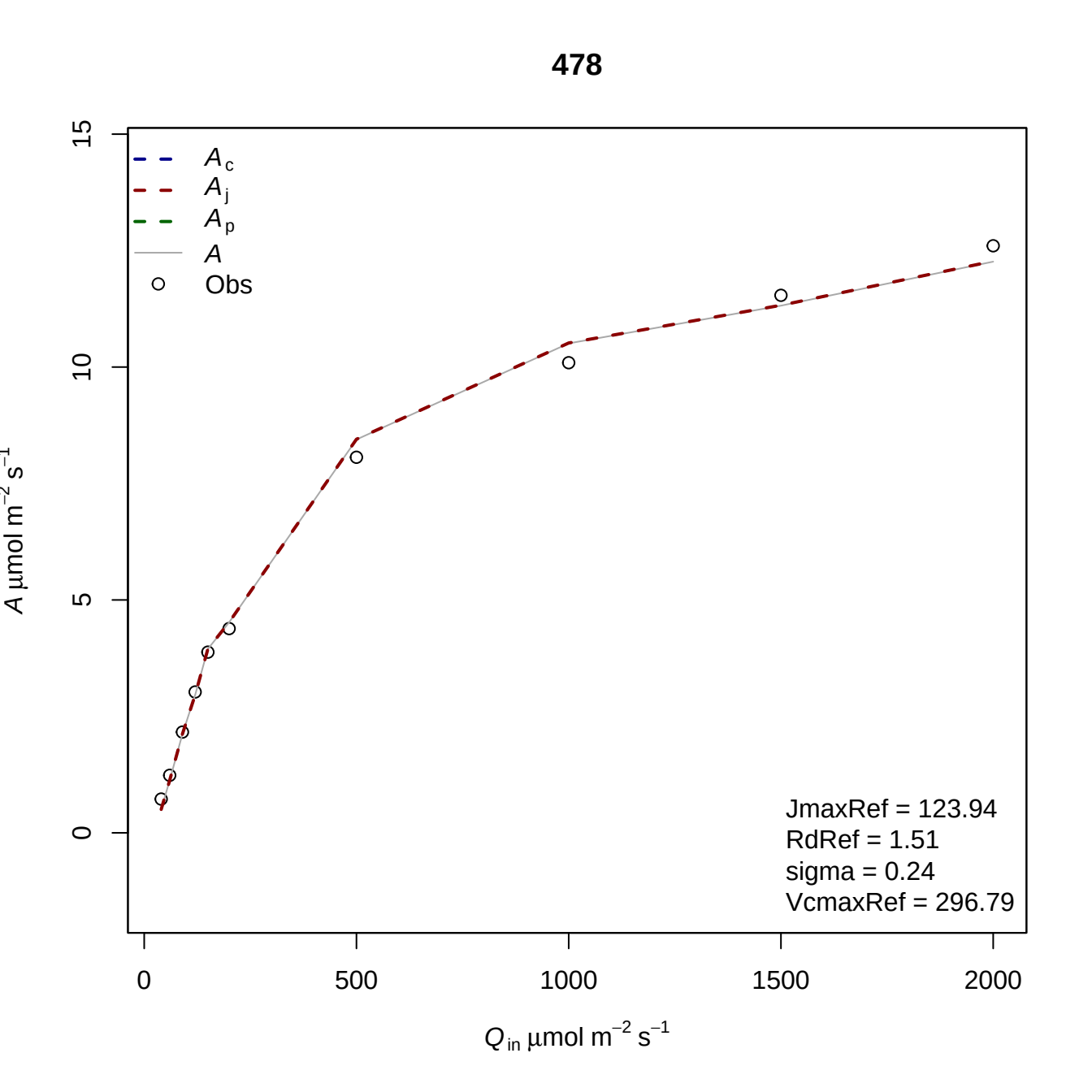

473
A_c
A_j
A_p
A
Obs
JmaxRef = 100.89
RdRef = 3.16
sigma = 0.84
VcmaxRef = 644.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

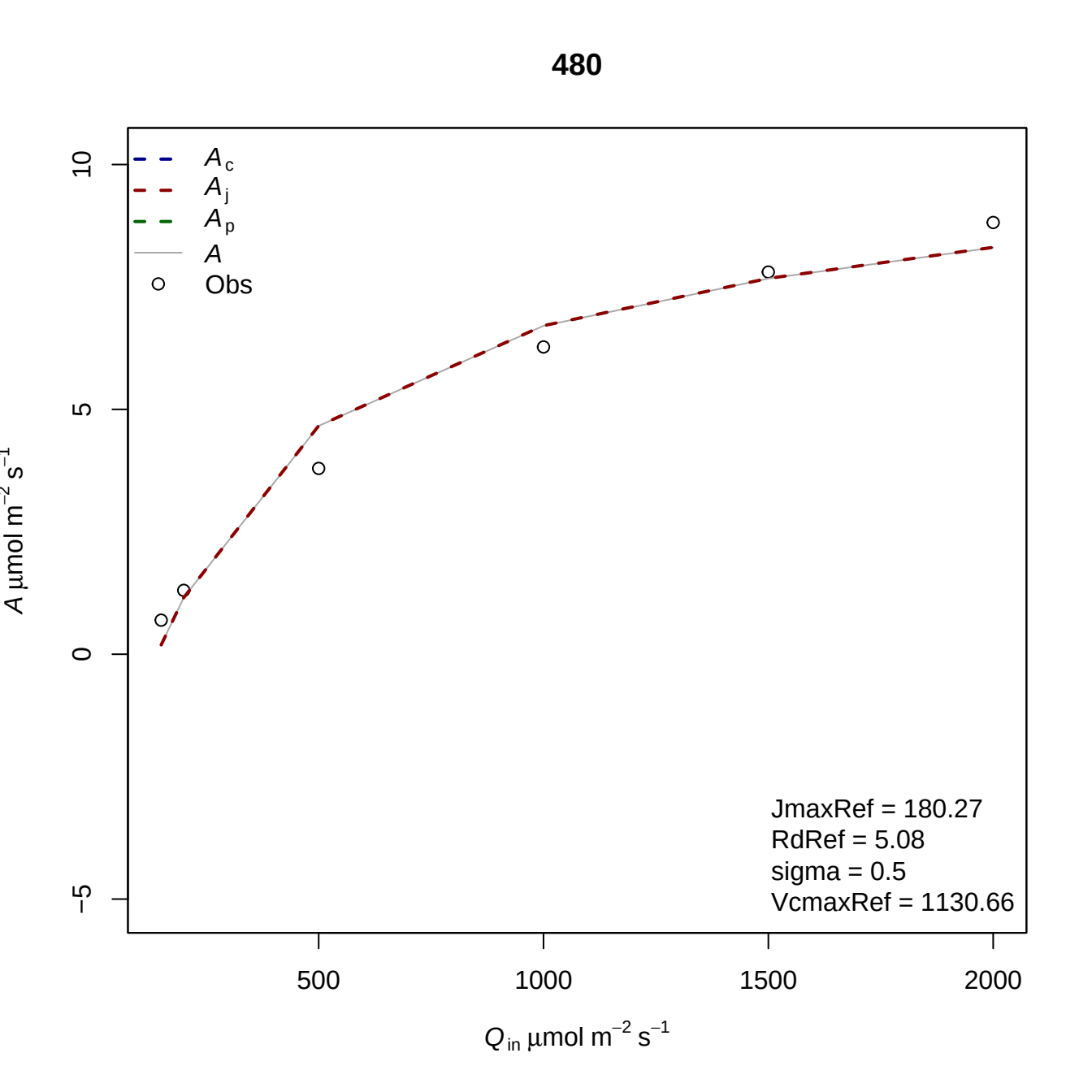

474
A_c
A_j
A_p
A
Obs
JmaxRef = 117.03
RdRef = 3.64
sigma = 1.04
VcmaxRef = 1505.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

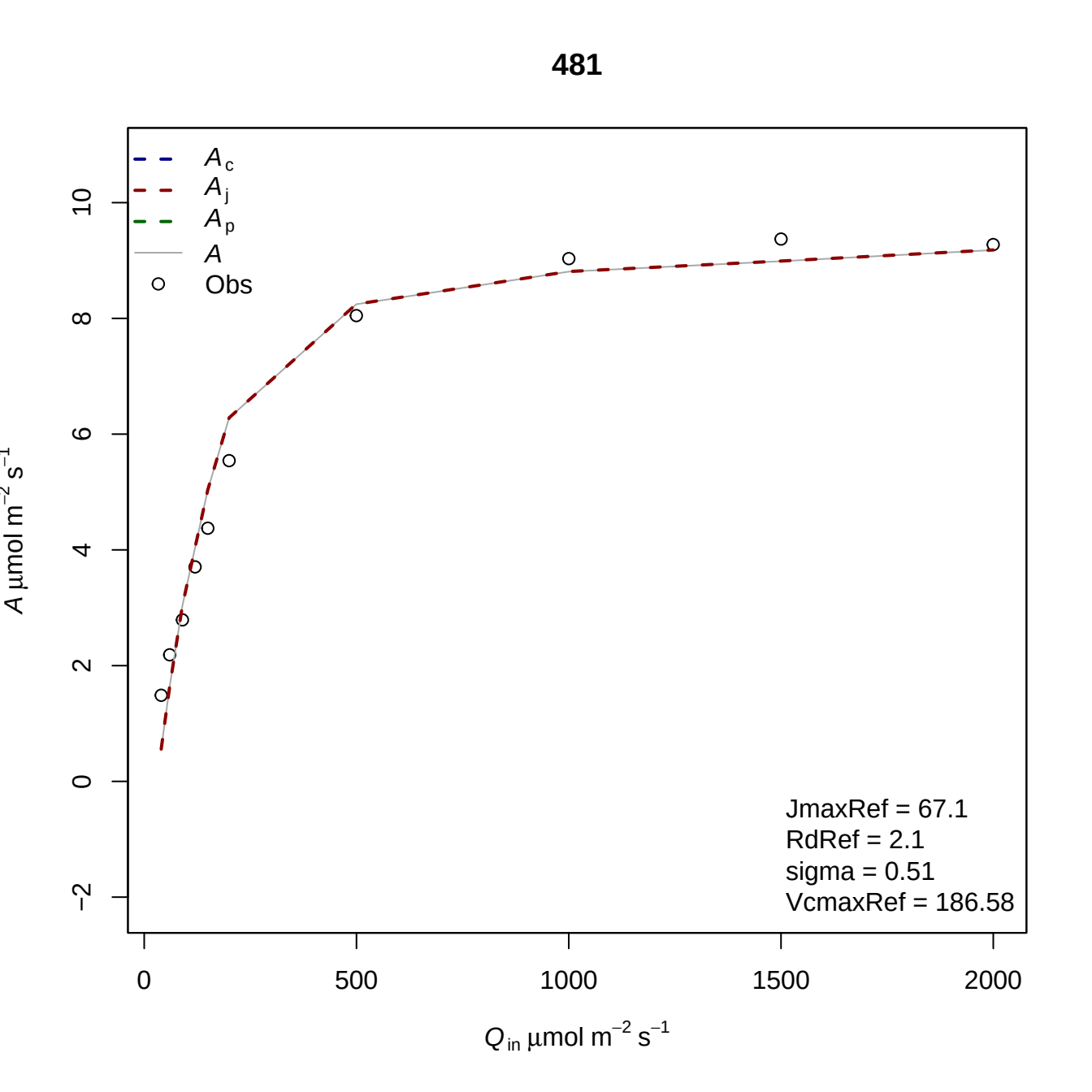

475
A_c
A_j
A_p
A
Obs
JmaxRef = 49.36
RdRef = 1.72
sigma = 0.31
VcmaxRef = 172.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

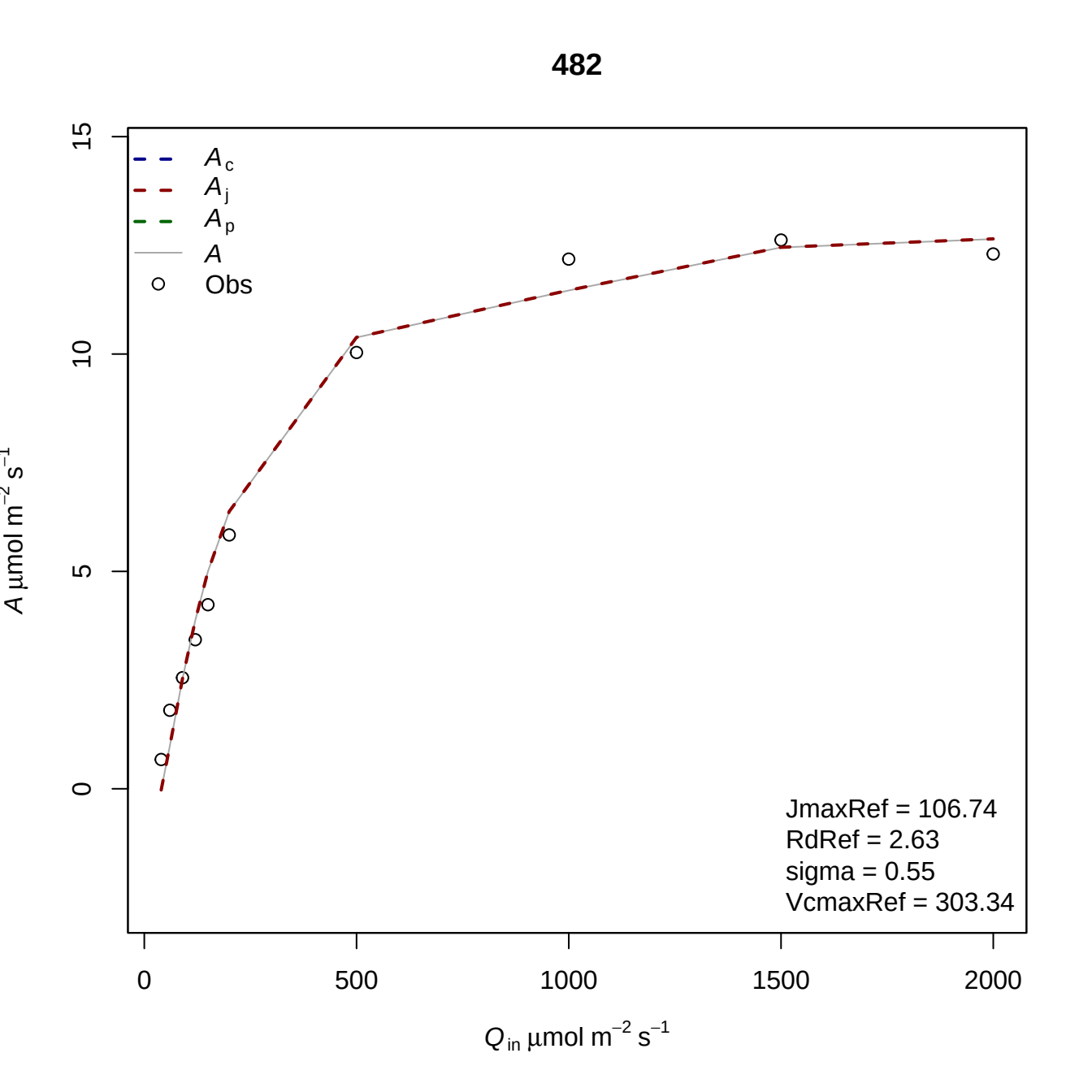

476
A_c
A_j
A_p
A
Obs
JmaxRef = 76.03
RdRef = 2.35
sigma = 0.6
VcmaxRef = 231.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

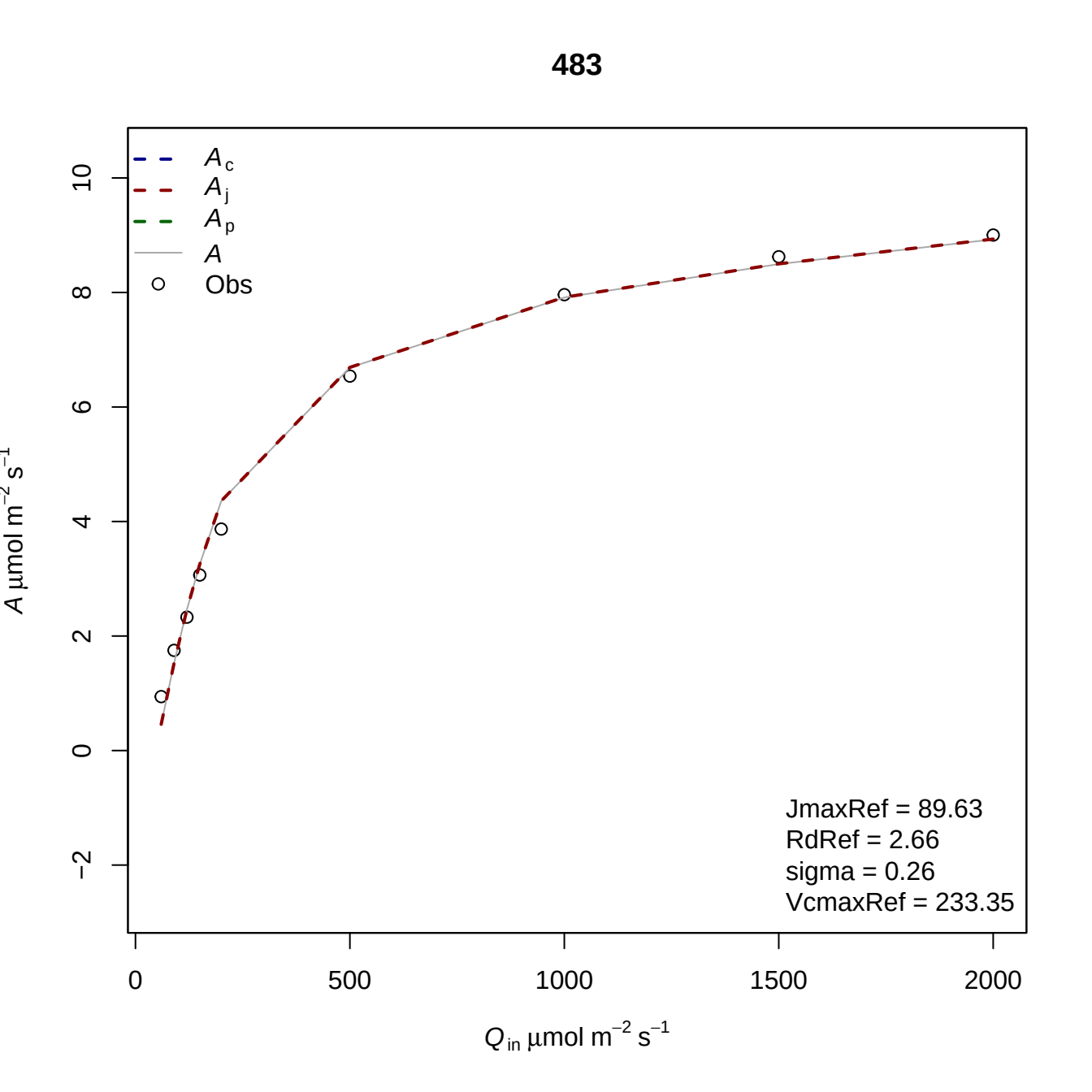

477
A_c
A_j
A_p
A
Obs
JmaxRef = 115.42
RdRef = 1.3
sigma = 0.4
VcmaxRef = 300.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

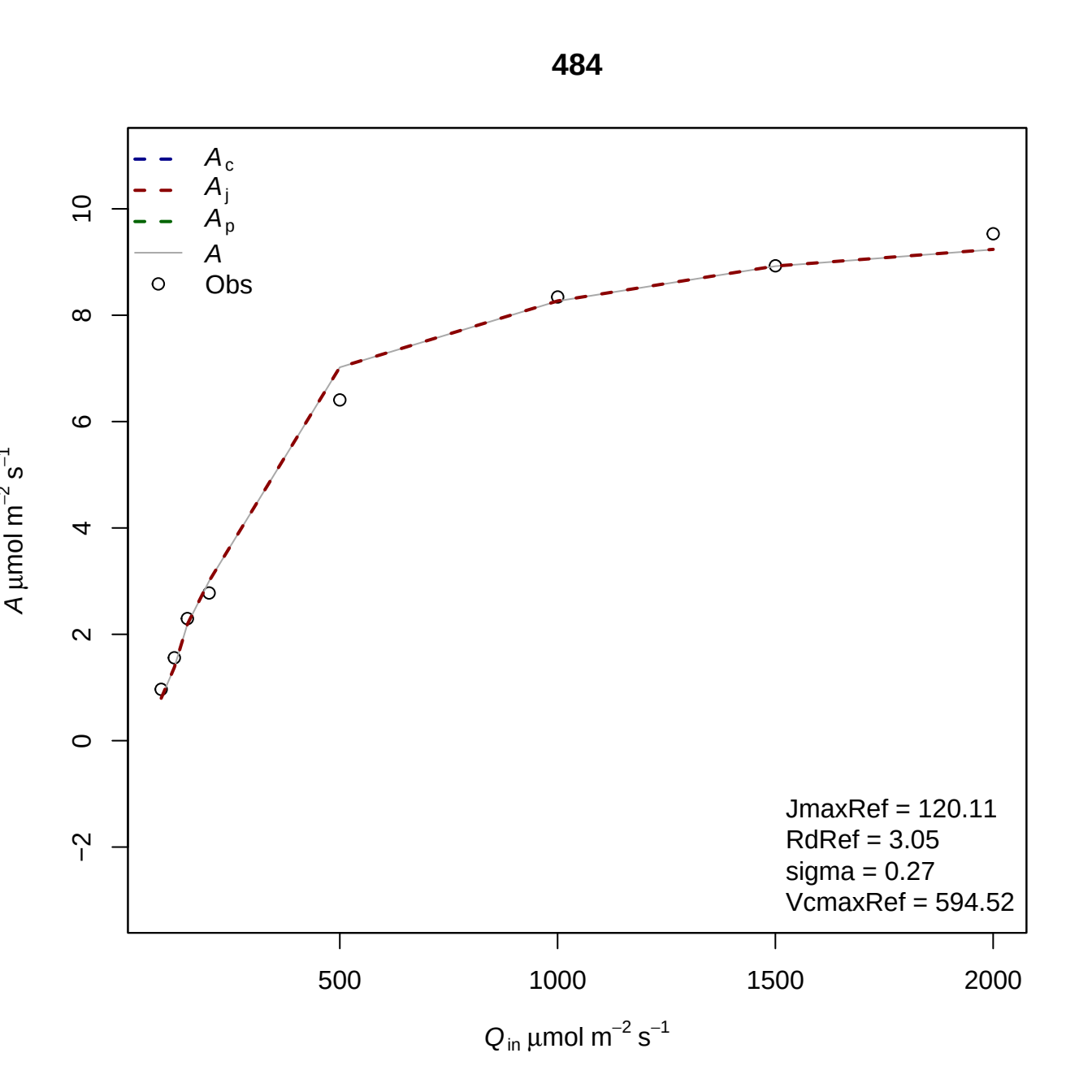

478
A_c
A_j
A_p
A
Obs
JmaxRef = 123.94
RdRef = 1.51
sigma = 0.24
VcmaxRef = 296.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

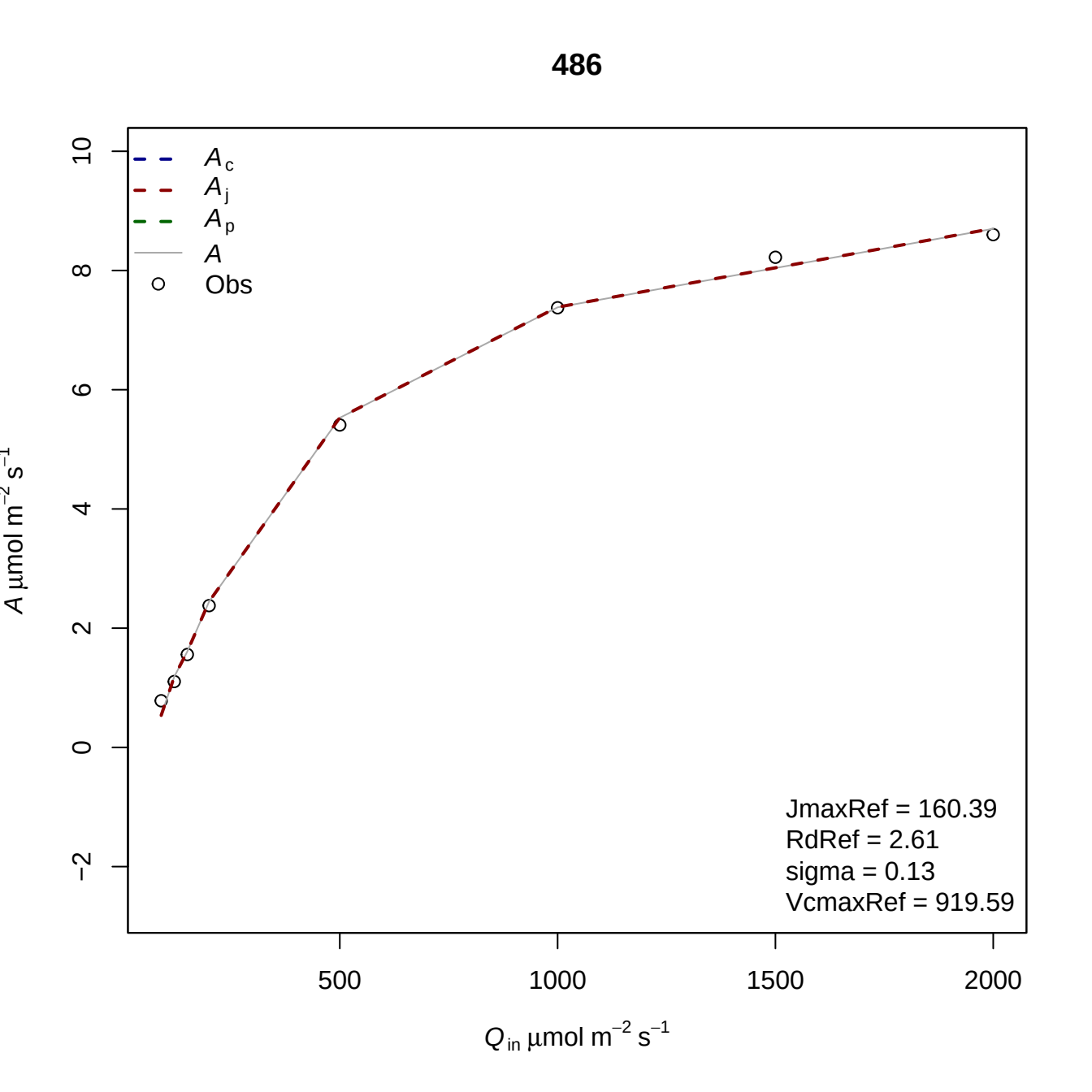

480
A_c
A_j
A_p
A
Obs
10
5
0
−5
A µmol m^{-2} s^{-1}
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 180.27
RdRef = 5.08
sigma = 0.5
VcmaxRef = 1130.66

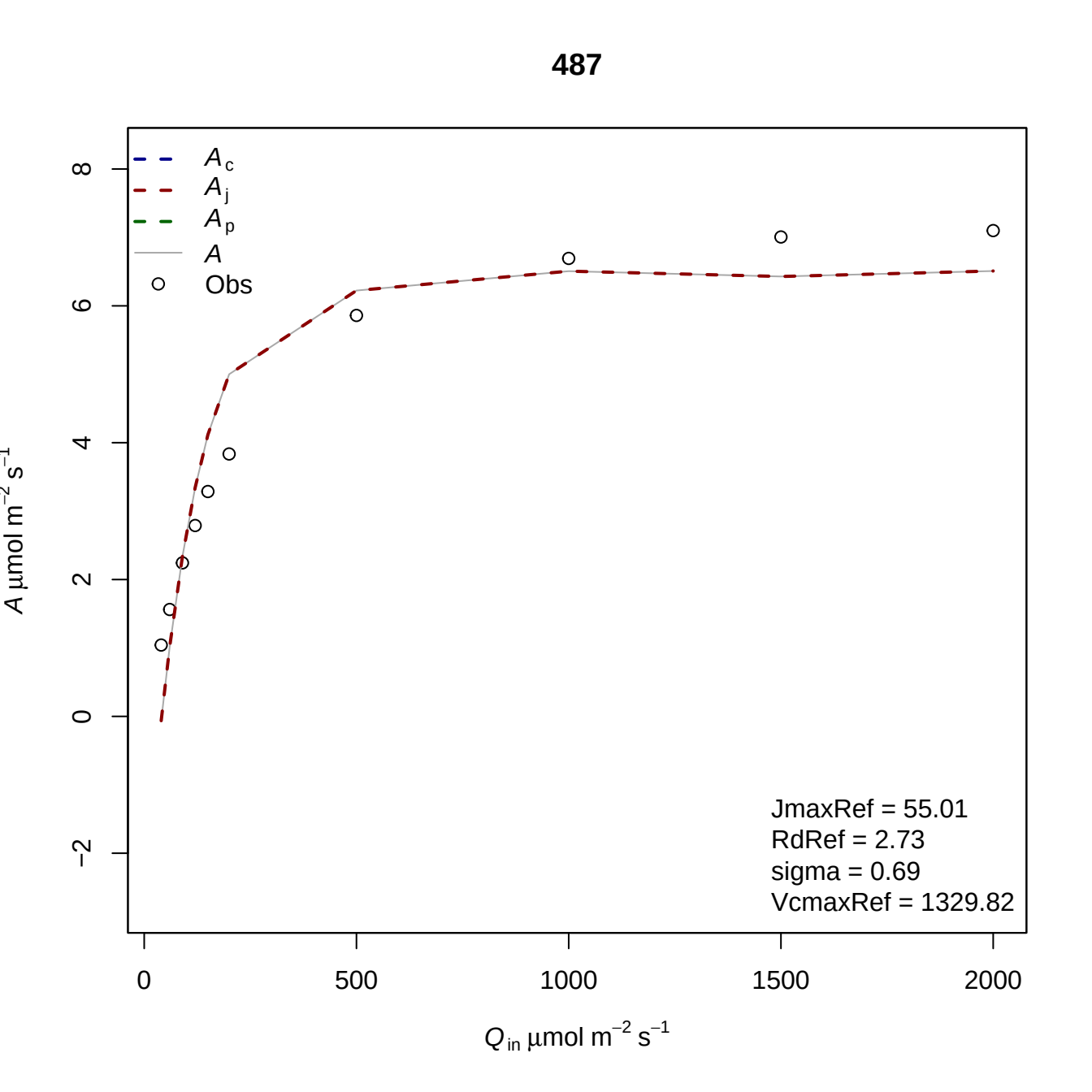

481
A_c
A_j
A_p
A
Obs
JmaxRef = 67.1
RdRef = 2.1
sigma = 0.51
VcmaxRef = 186.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

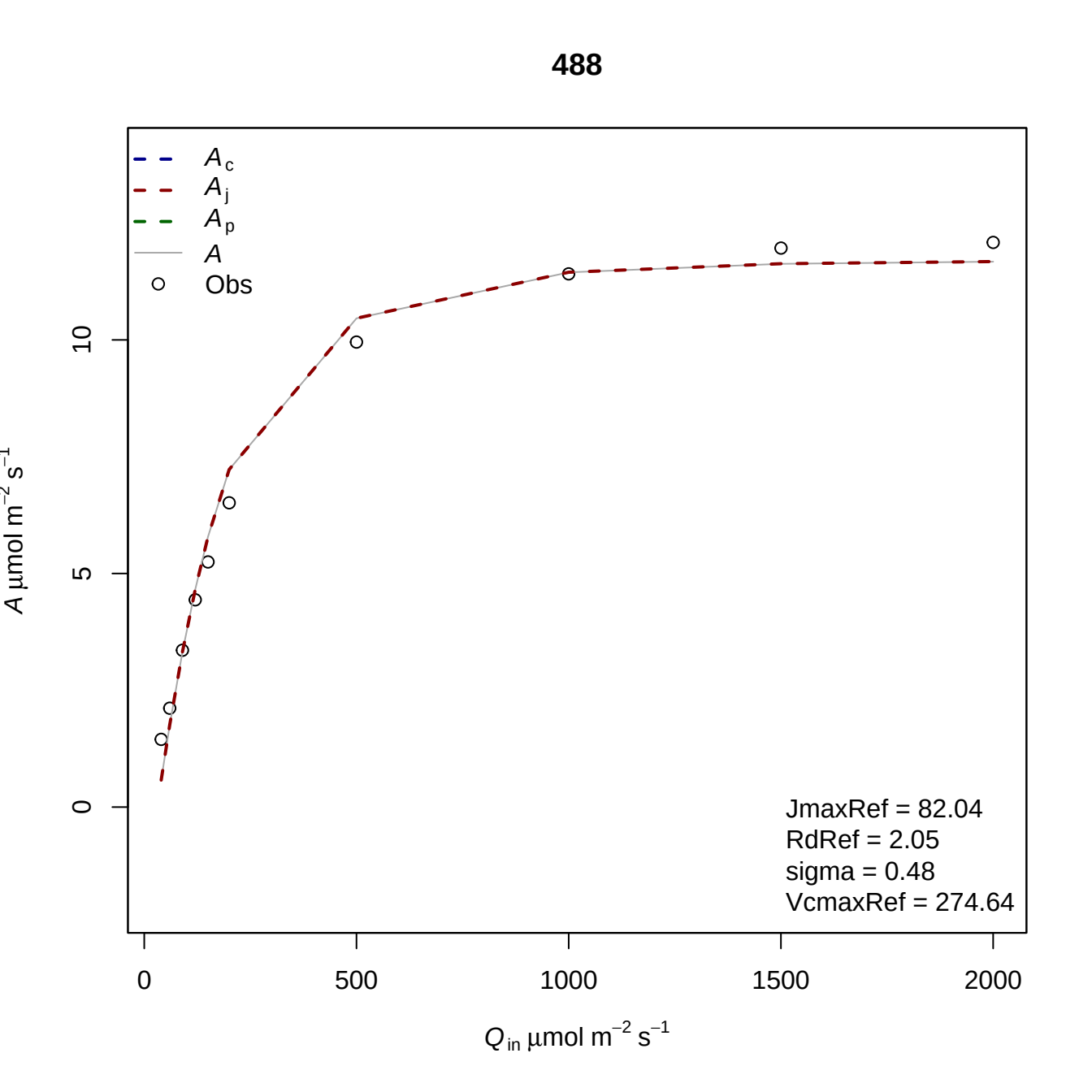

482
A_c
A_j
A_p
A
Obs
JmaxRef = 106.74
RdRef = 2.63
sigma = 0.55
VcmaxRef = 303.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

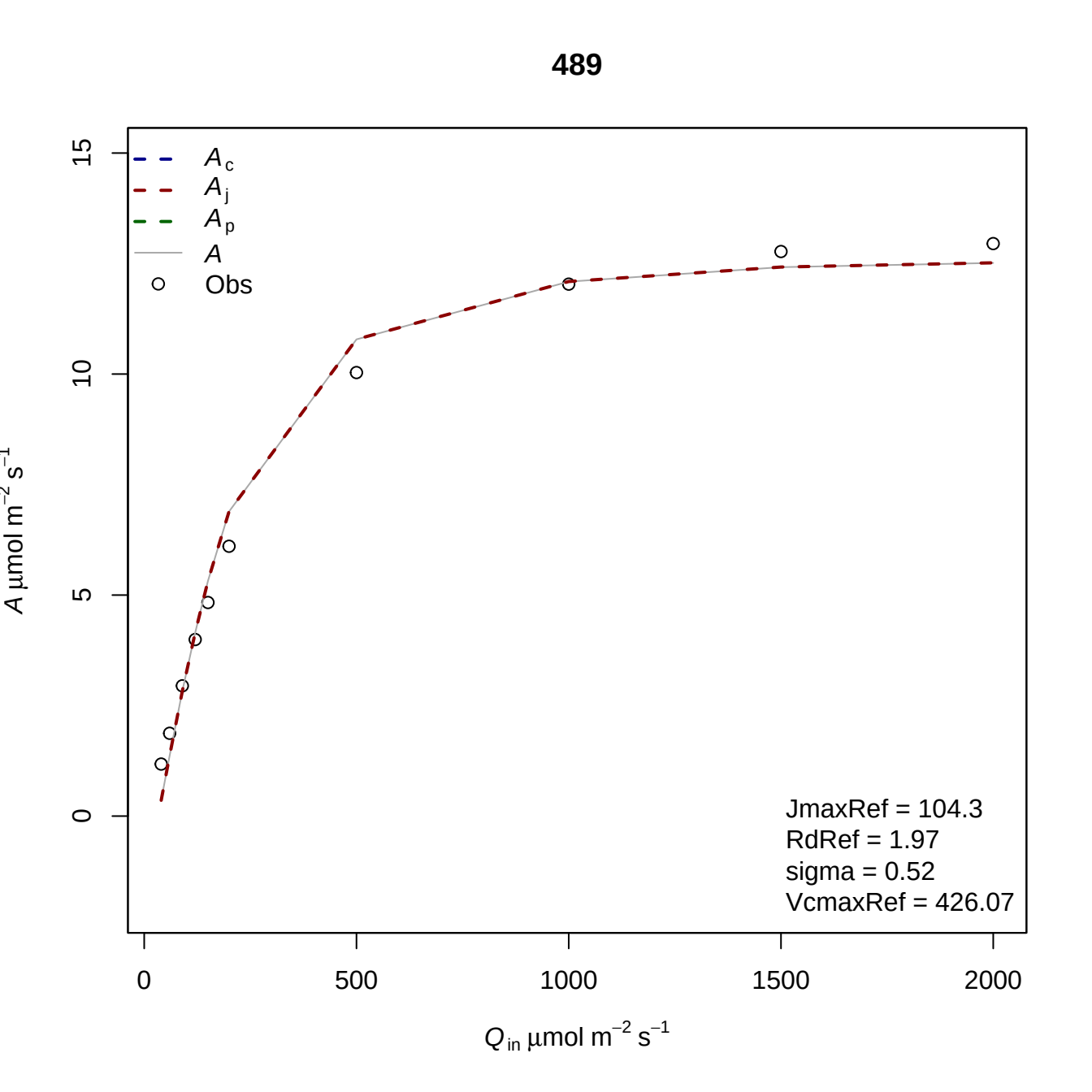

483
A_c
A_j
A_p
A
Obs
JmaxRef = 89.63
RdRef = 2.66
sigma = 0.26
VcmaxRef = 233.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

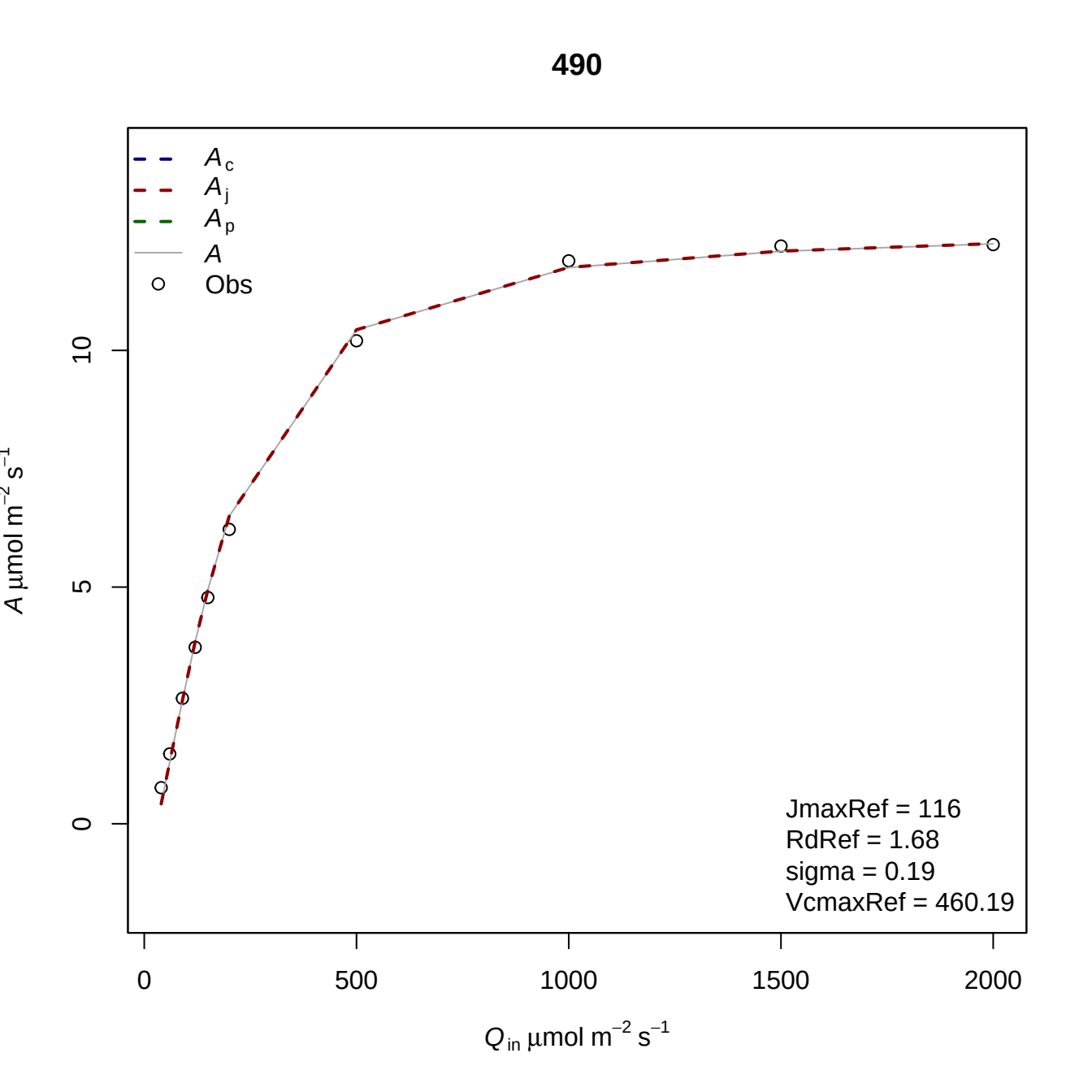

484
A_c
A_j
A_p
A
Obs
JmaxRef = 120.11
RdRef = 3.05
sigma = 0.27
VcmaxRef = 594.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
500
1000
1500
2000

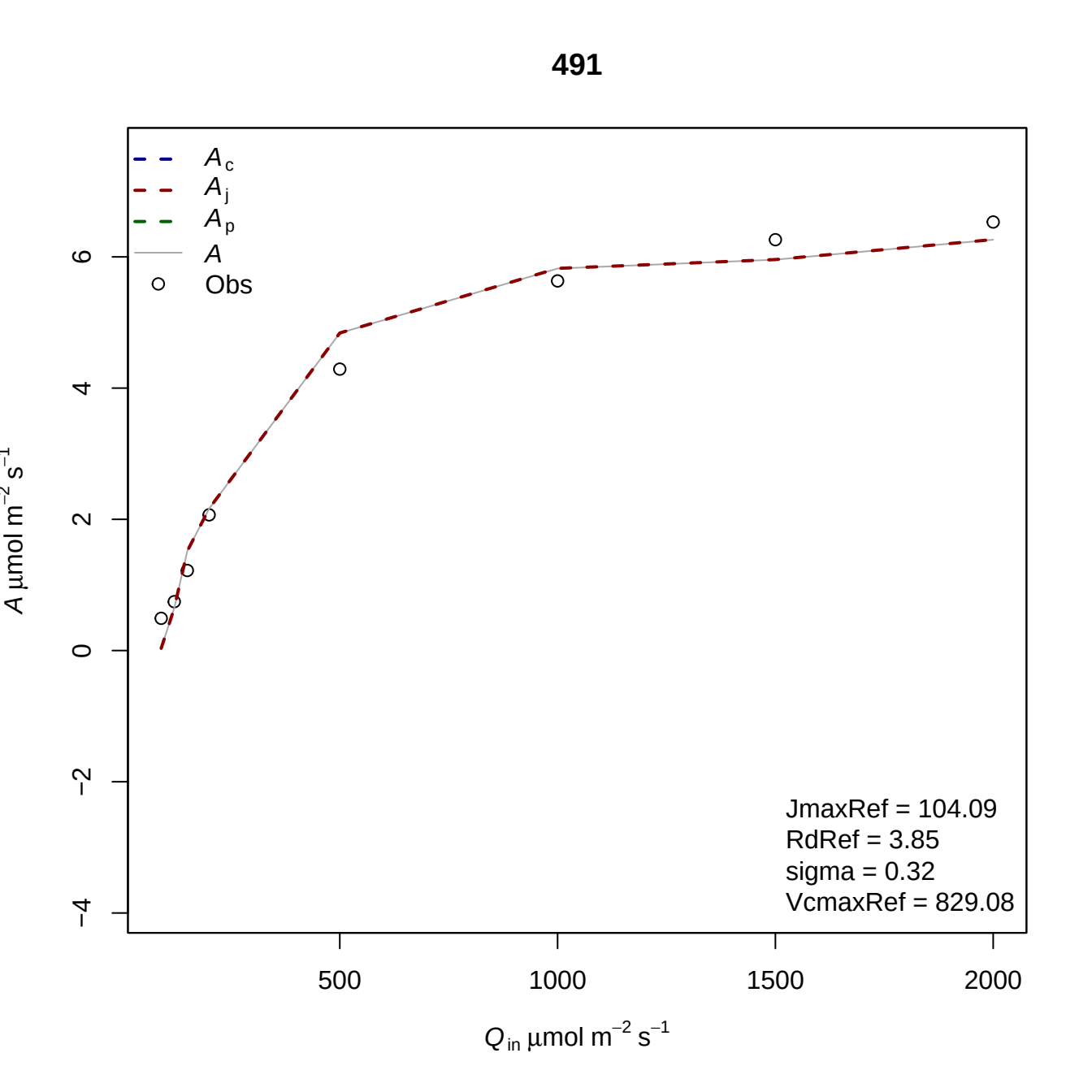

486
A_c
A_j
A_p
A
Obs
JmaxRef = 160.39
RdRef = 2.61
sigma = 0.13
VcmaxRef = 919.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
500
1000
1500
2000

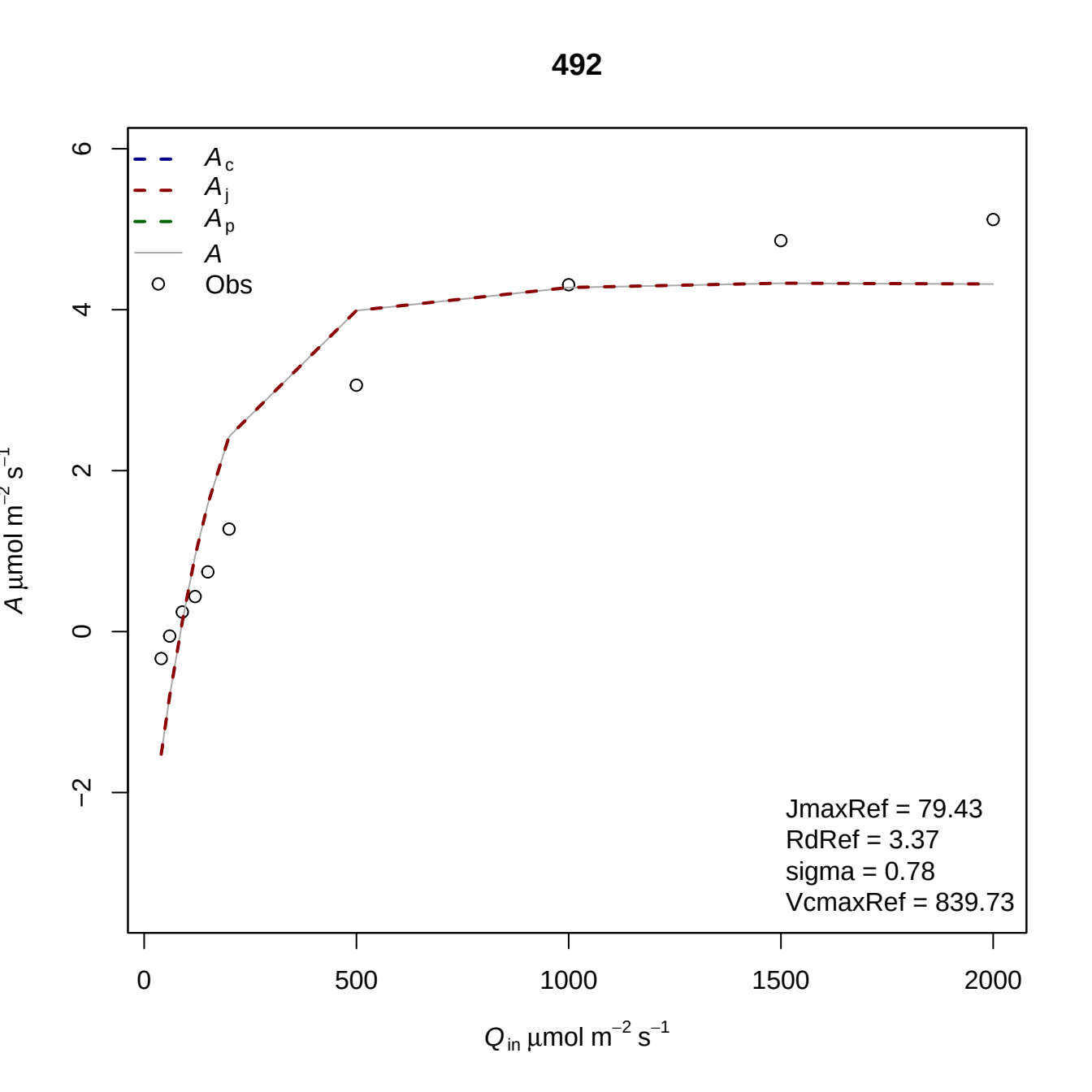

487
A_c
A_j
A_p
A
Obs
JmaxRef = 55.01
RdRef = 2.73
sigma = 0.69
VcmaxRef = 1329.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

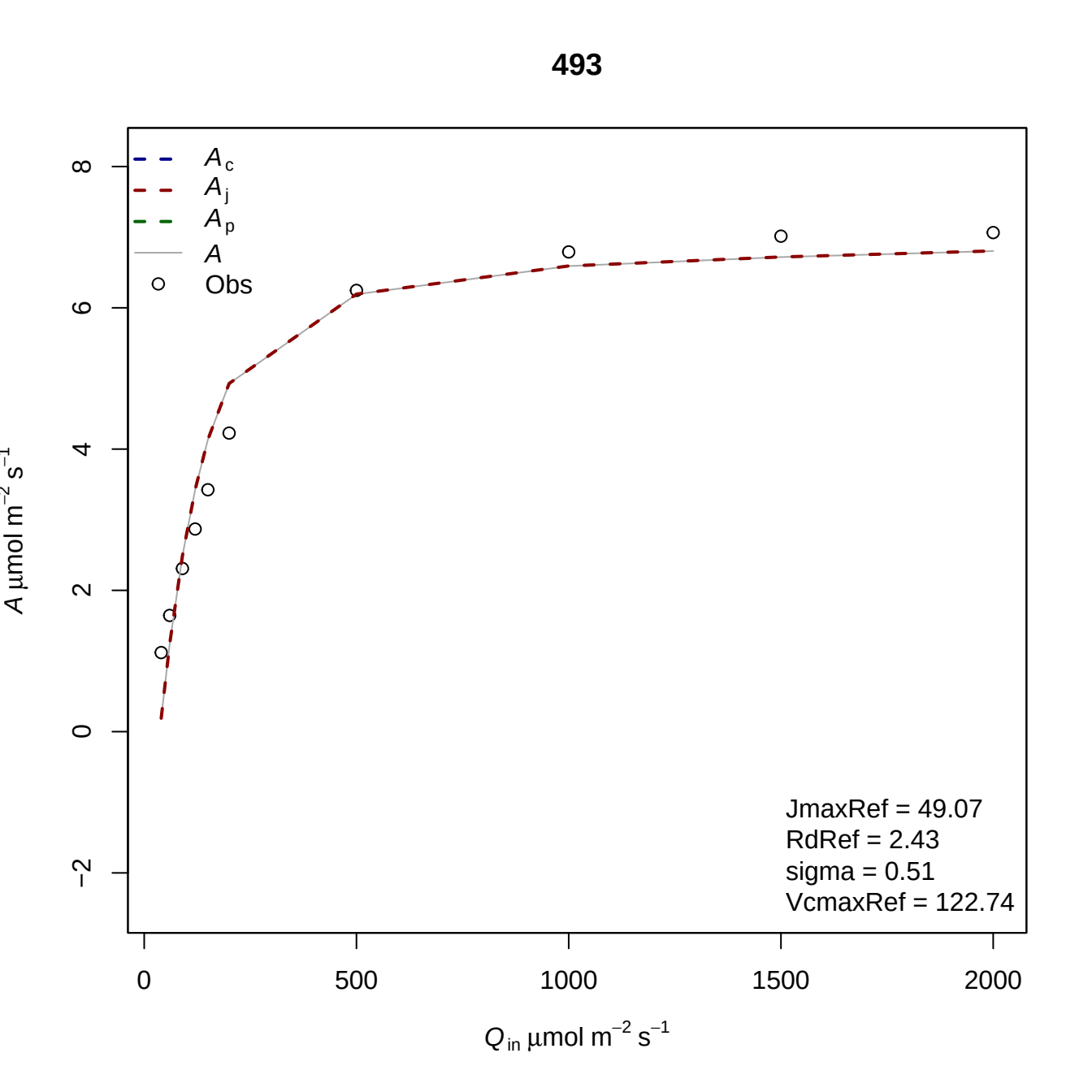

488
A_c
A_j
A_p
A
Obs
JmaxRef = 82.04
RdRef = 2.05
sigma = 0.48
VcmaxRef = 274.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

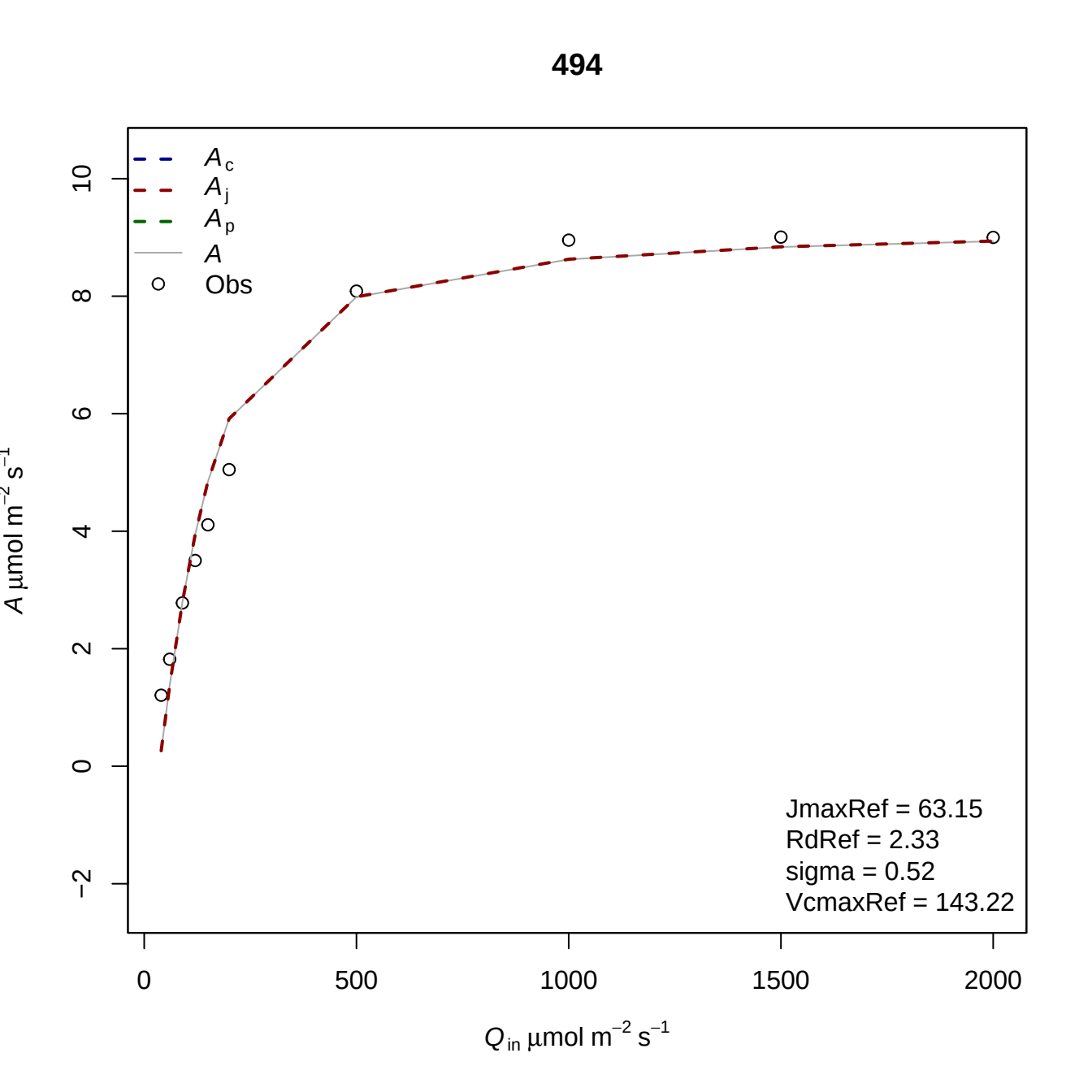

489
A_c
A_j
A_p
A
Obs
JmaxRef = 104.3
RdRef = 1.97
sigma = 0.52
VcmaxRef = 426.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

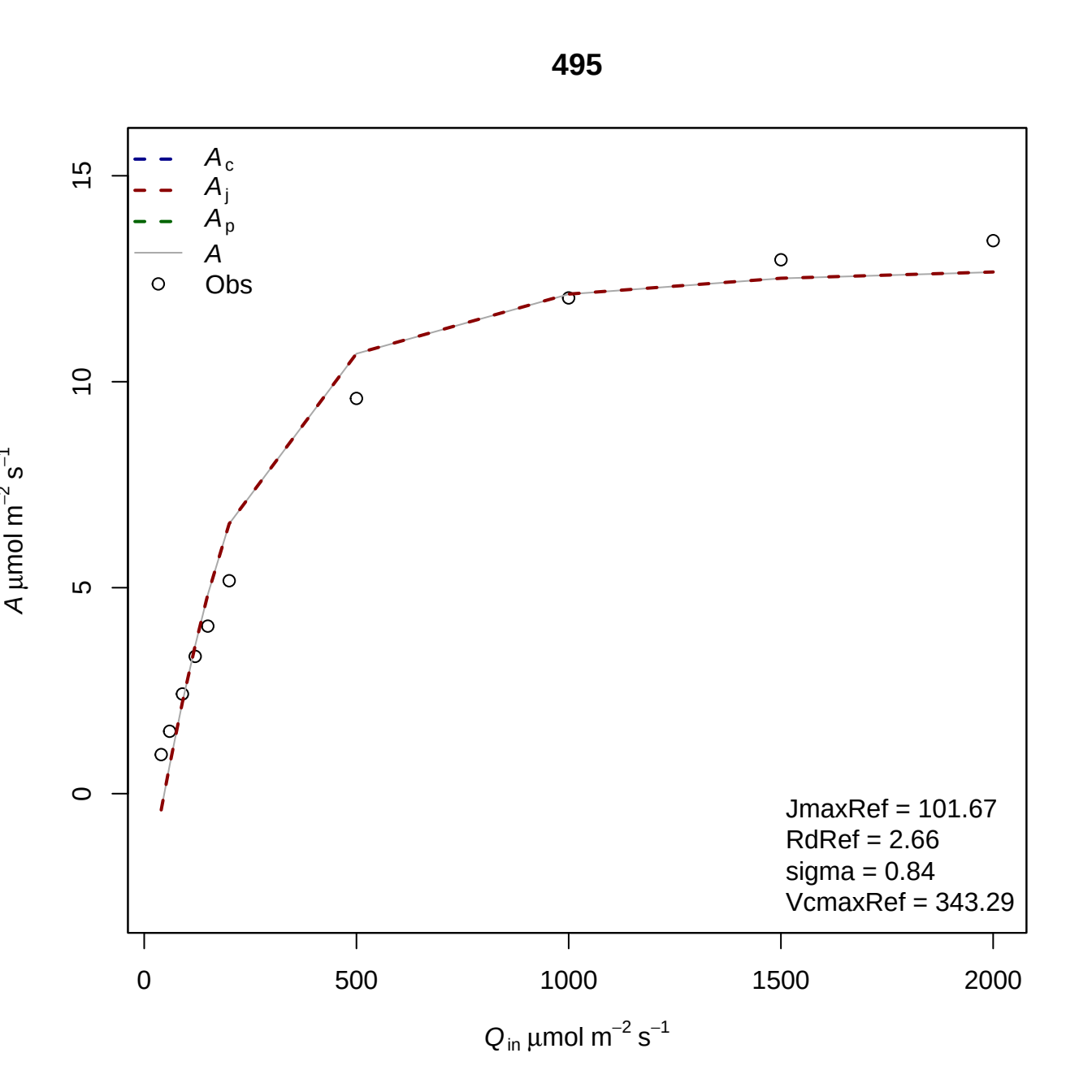

490
A_c
A_j
A_p
A
Obs
JmaxRef = 116
RdRef = 1.68
sigma = 0.19
VcmaxRef = 460.19
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

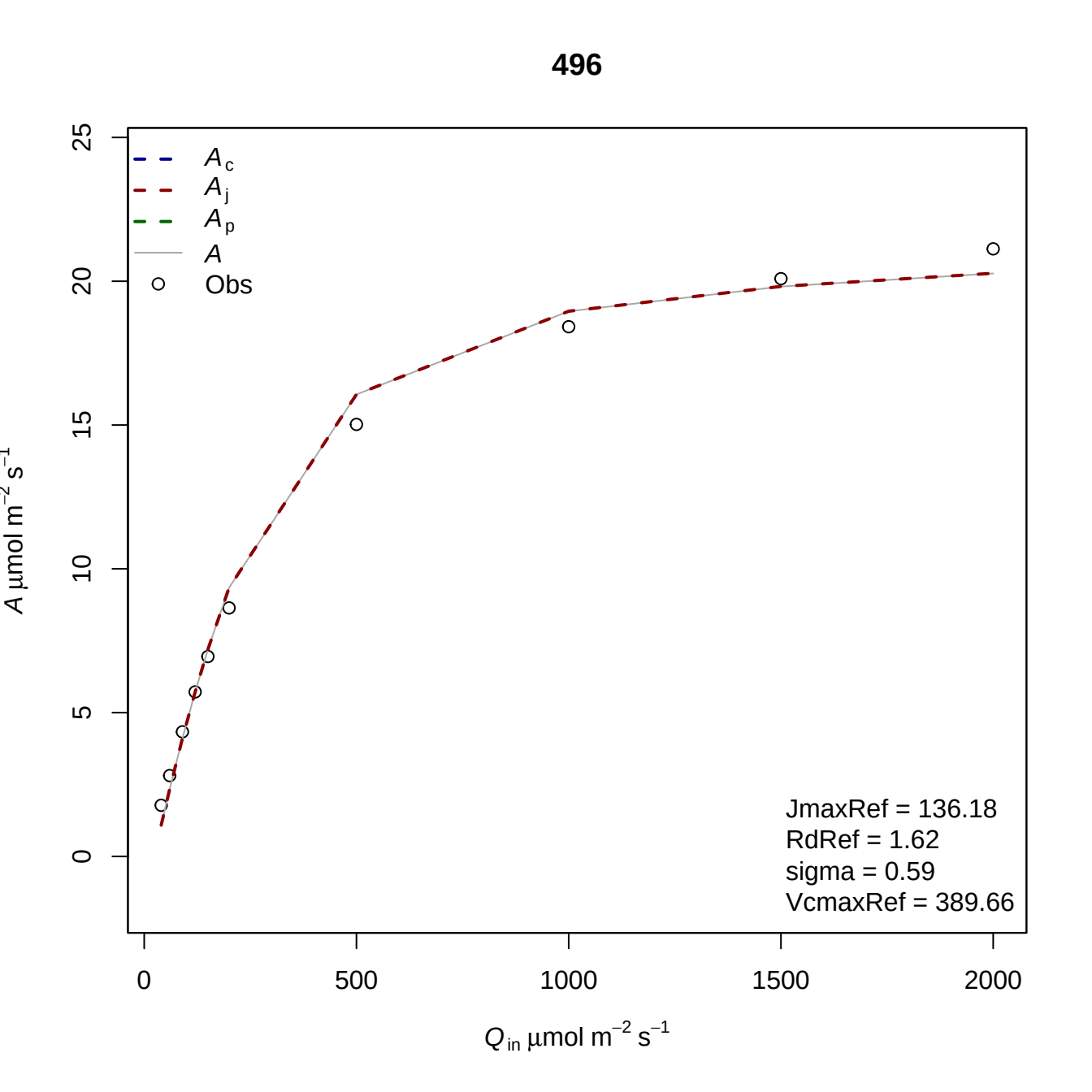

491
A_c
A_j
A_p
A
Obs
JmaxRef = 104.09
RdRef = 3.85
sigma = 0.32
VcmaxRef = 829.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
6
500
1000
1500
2000

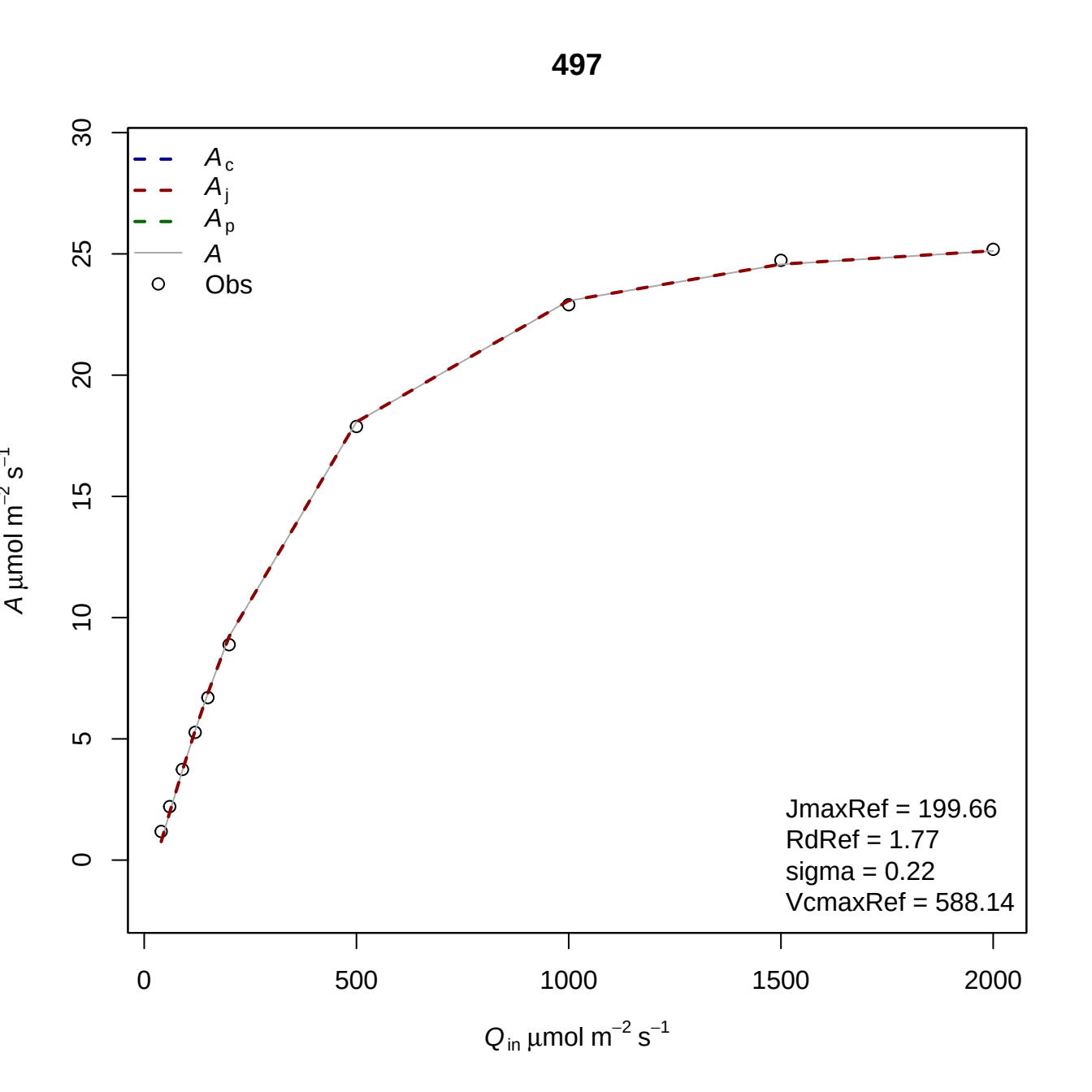

492
A_c
A_j
A_p
A
Obs
JmaxRef = 79.43
RdRef = 3.37
sigma = 0.78
VcmaxRef = 839.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

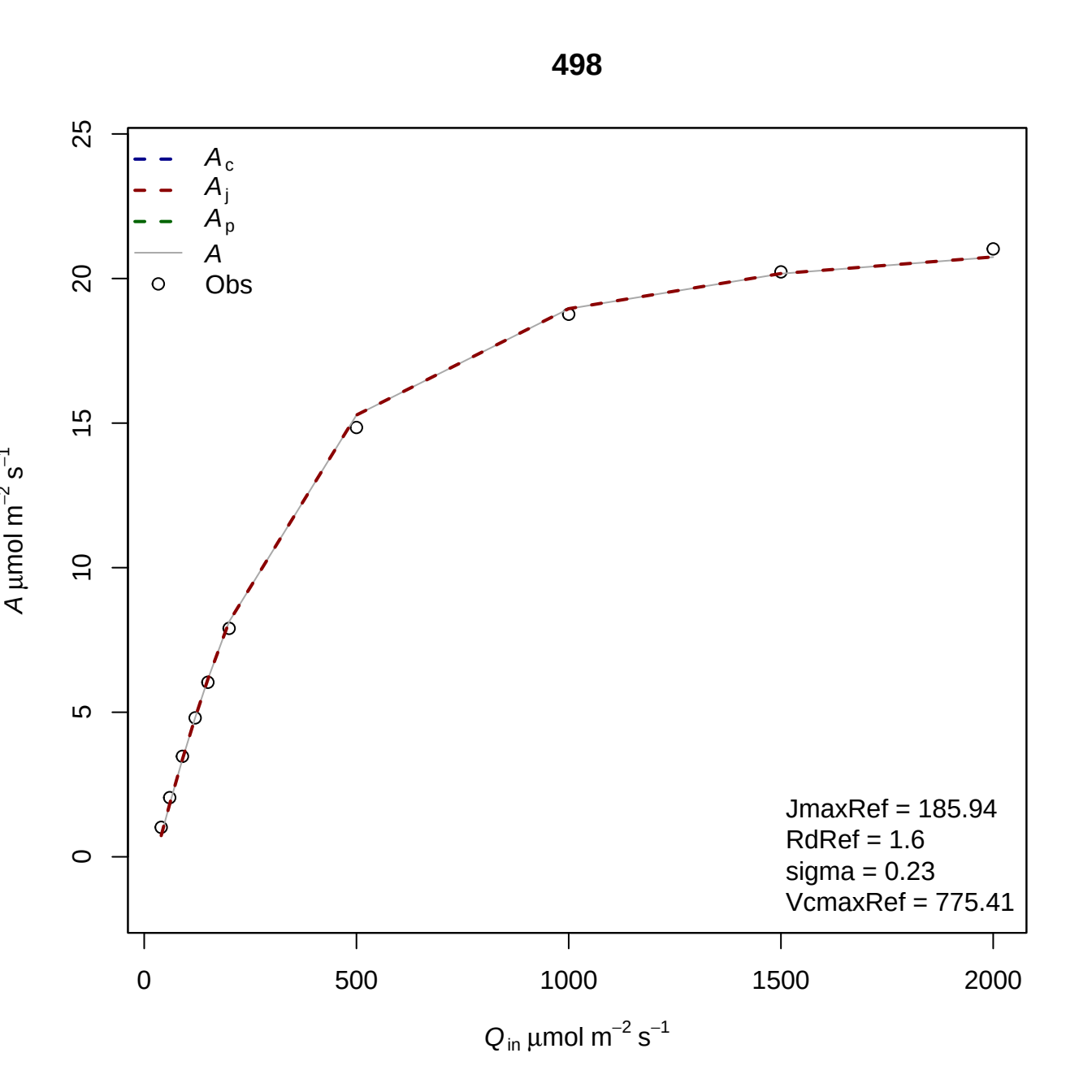

493
A_c
A_j
A_p
A
Obs
JmaxRef = 49.07
RdRef = 2.43
sigma = 0.51
VcmaxRef = 122.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

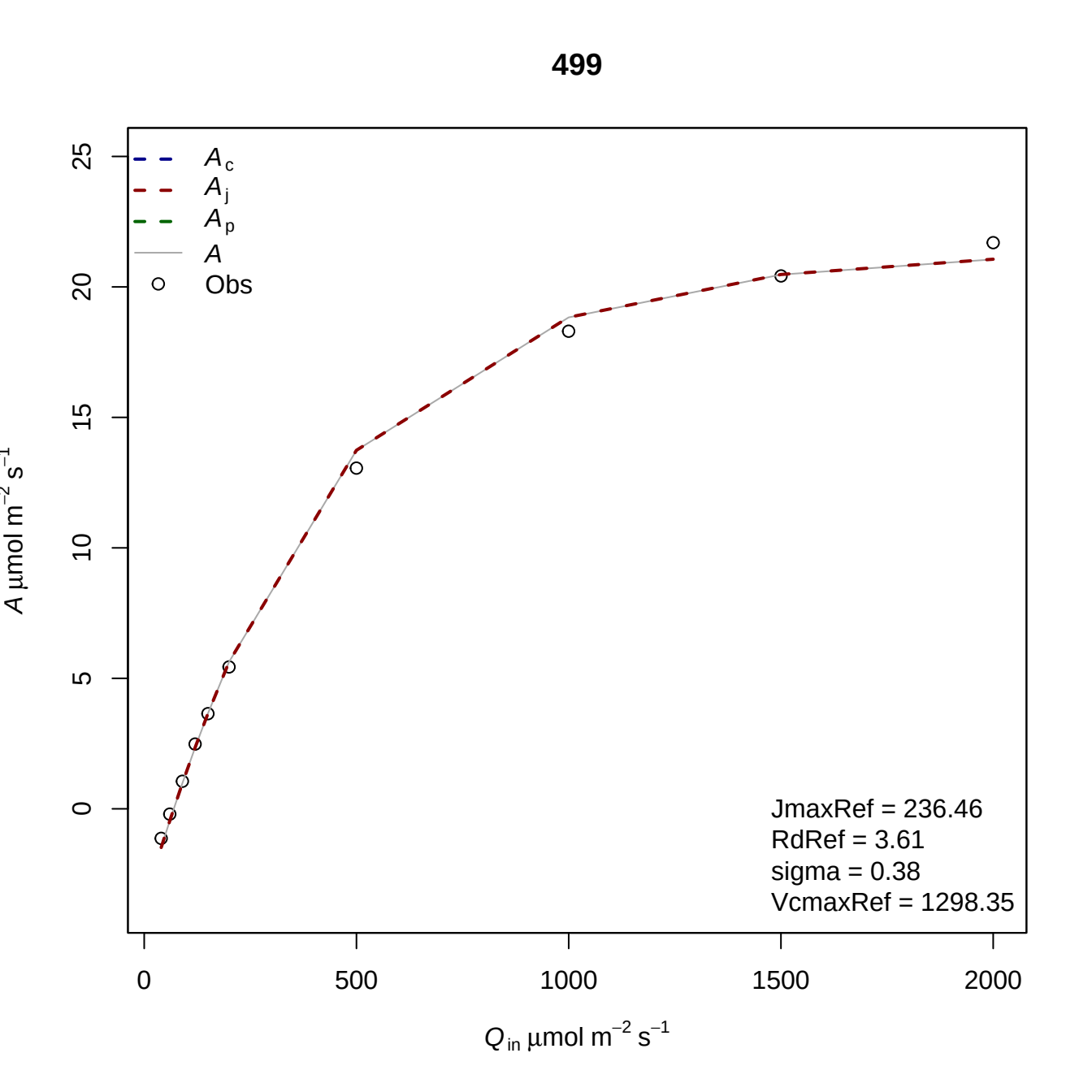

494
A_c
A_j
A_p
A
Obs
JmaxRef = 63.15
RdRef = 2.33
sigma = 0.52
VcmaxRef = 143.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

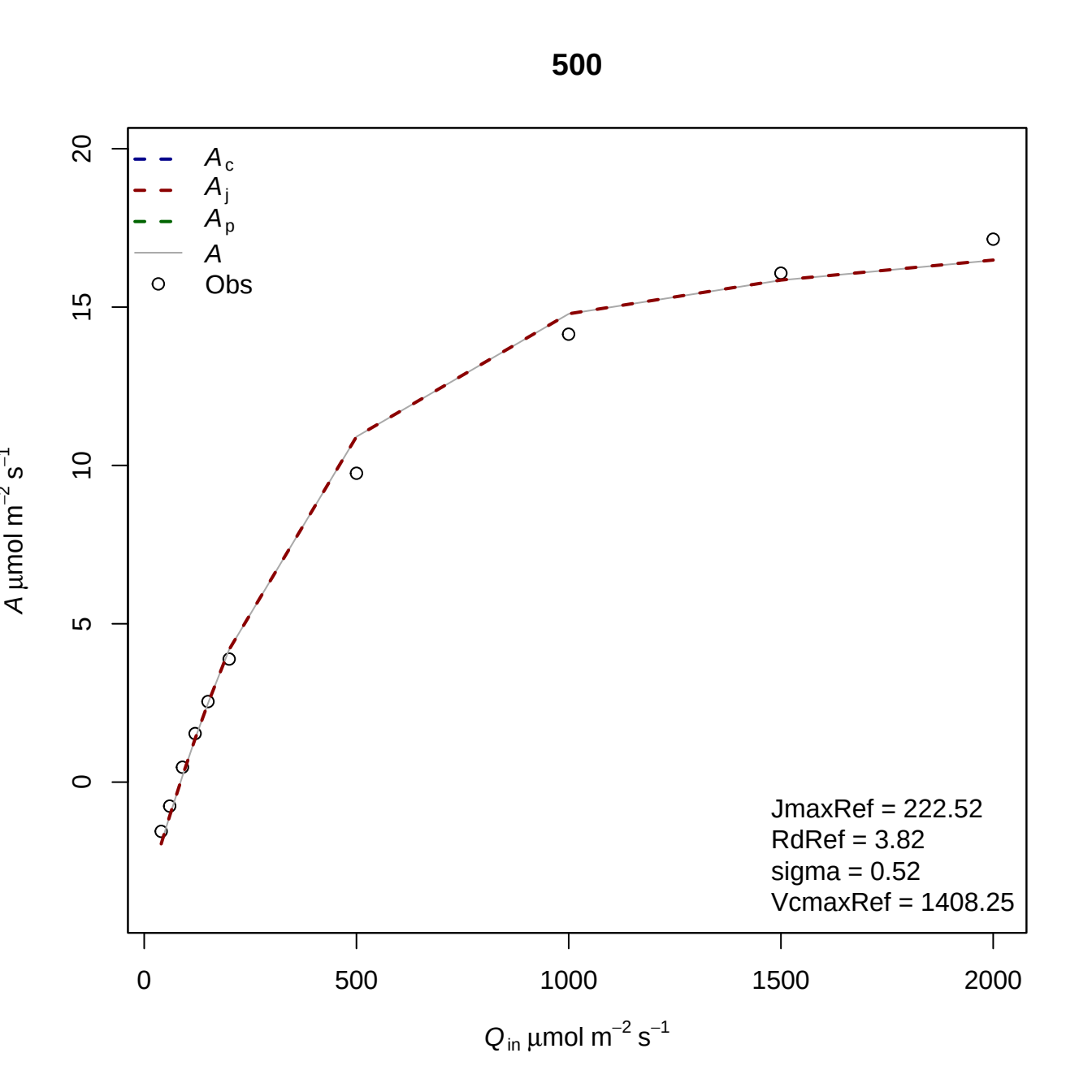

495
A_c
A_j
A_p
A
Obs
JmaxRef = 101.67
RdRef = 2.66
sigma = 0.84
VcmaxRef = 343.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

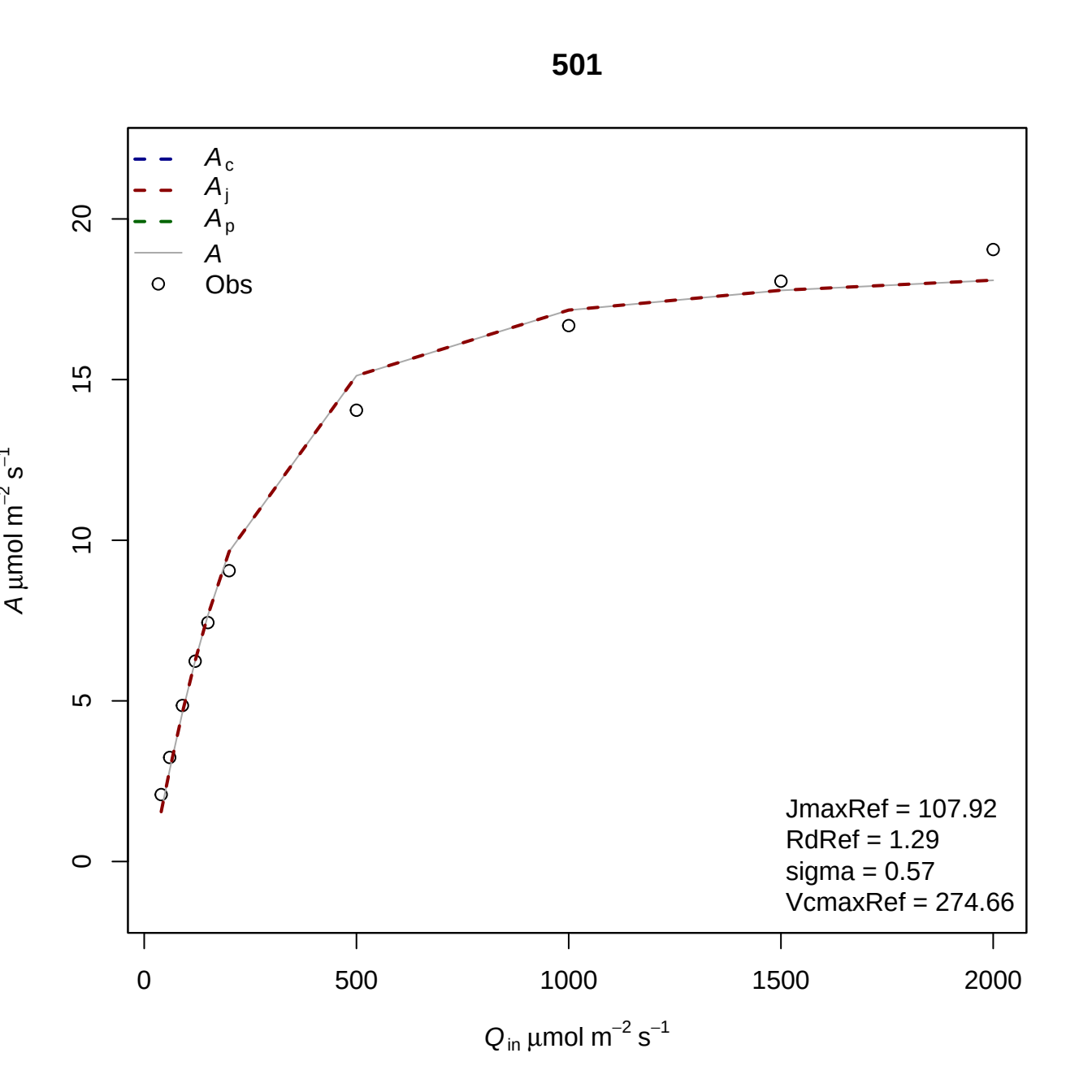

496
A_c
A_j
A_p
A
Obs
JmaxRef = 136.18
RdRef = 1.62
sigma = 0.59
VcmaxRef = 389.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

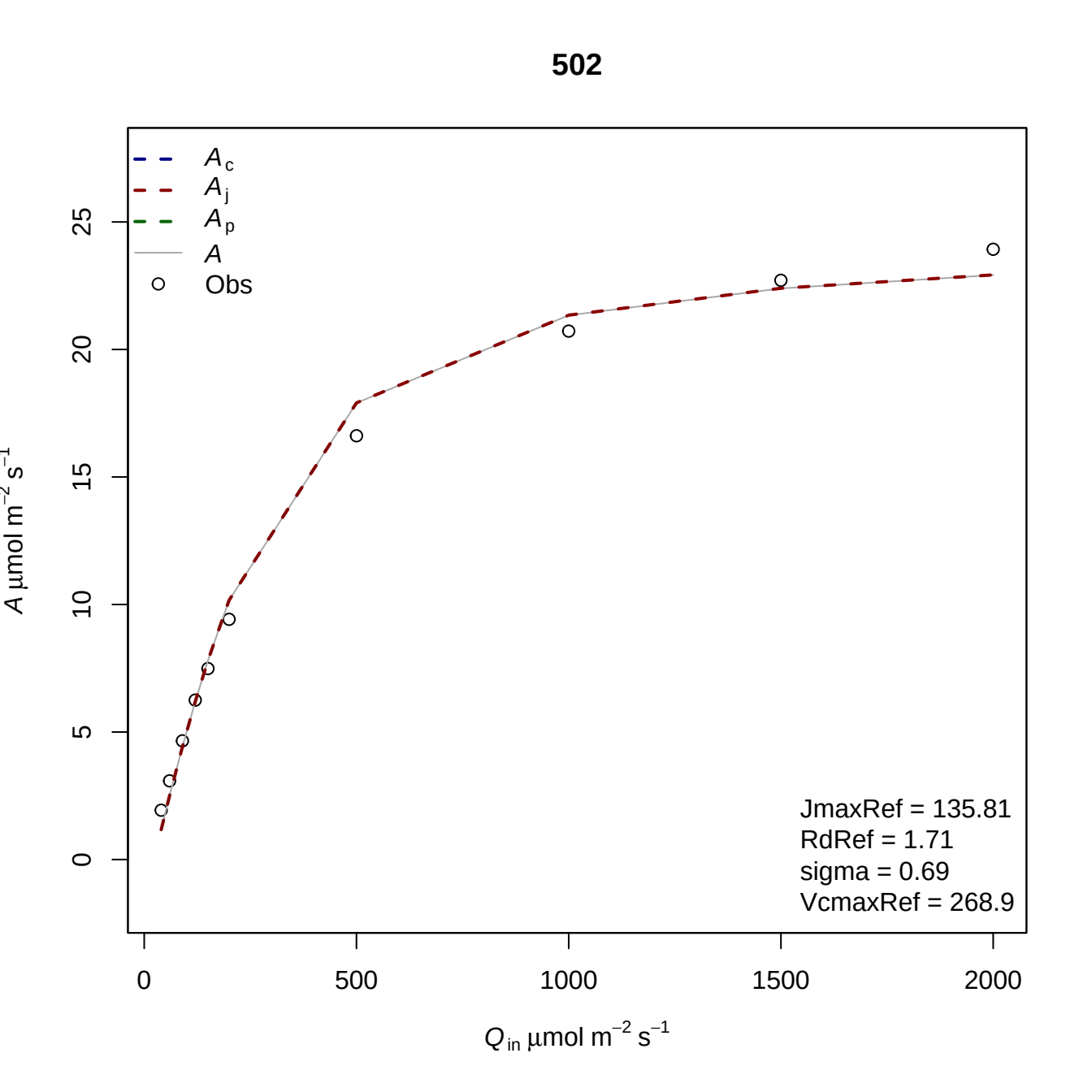

497
A_c
A_j
A_p
A
Obs
JmaxRef = 199.66
RdRef = 1.77
sigma = 0.22
VcmaxRef = 588.14
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

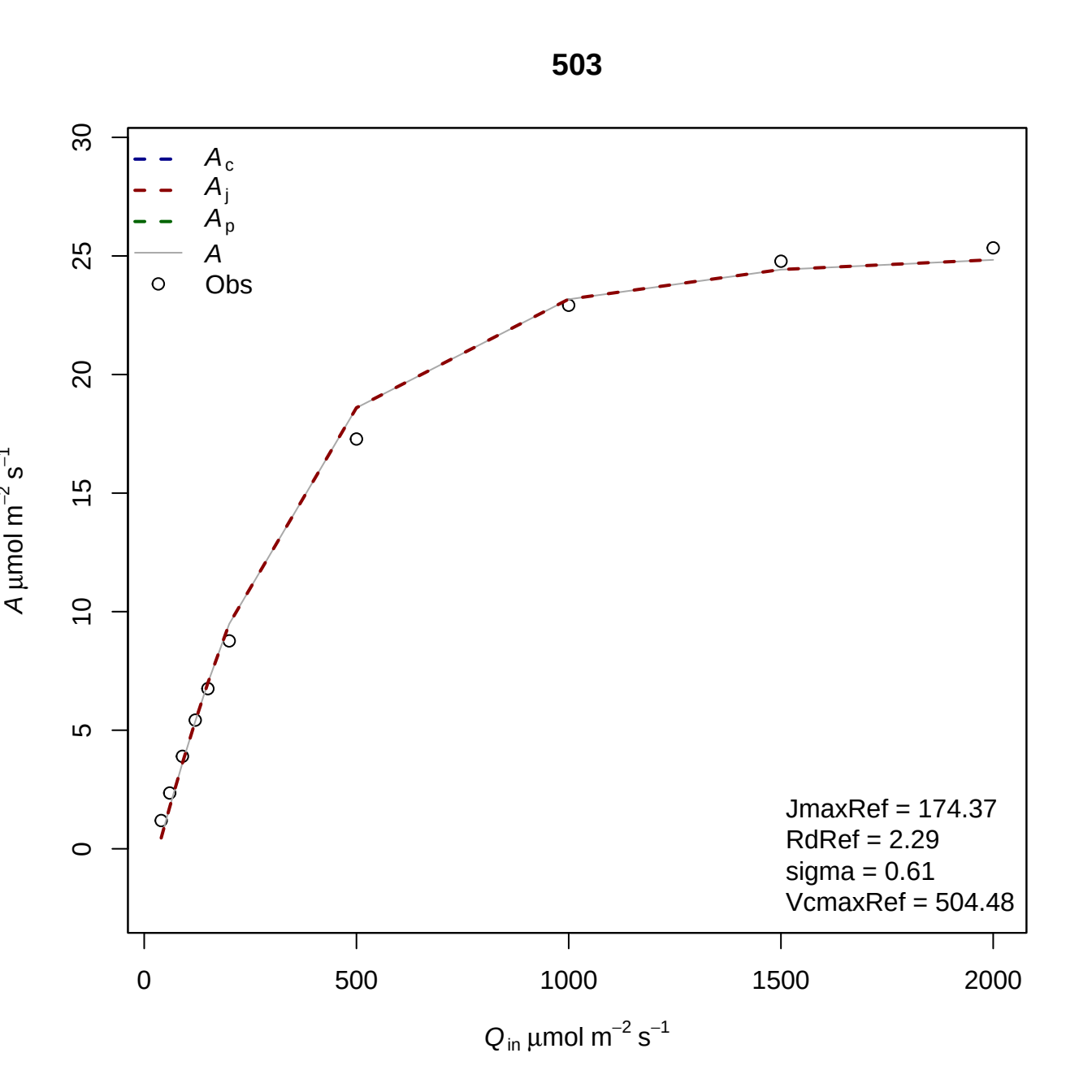

498
A_c
A_j
A_p
A
Obs
JmaxRef = 185.94
RdRef = 1.6
sigma = 0.23
VcmaxRef = 775.41
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

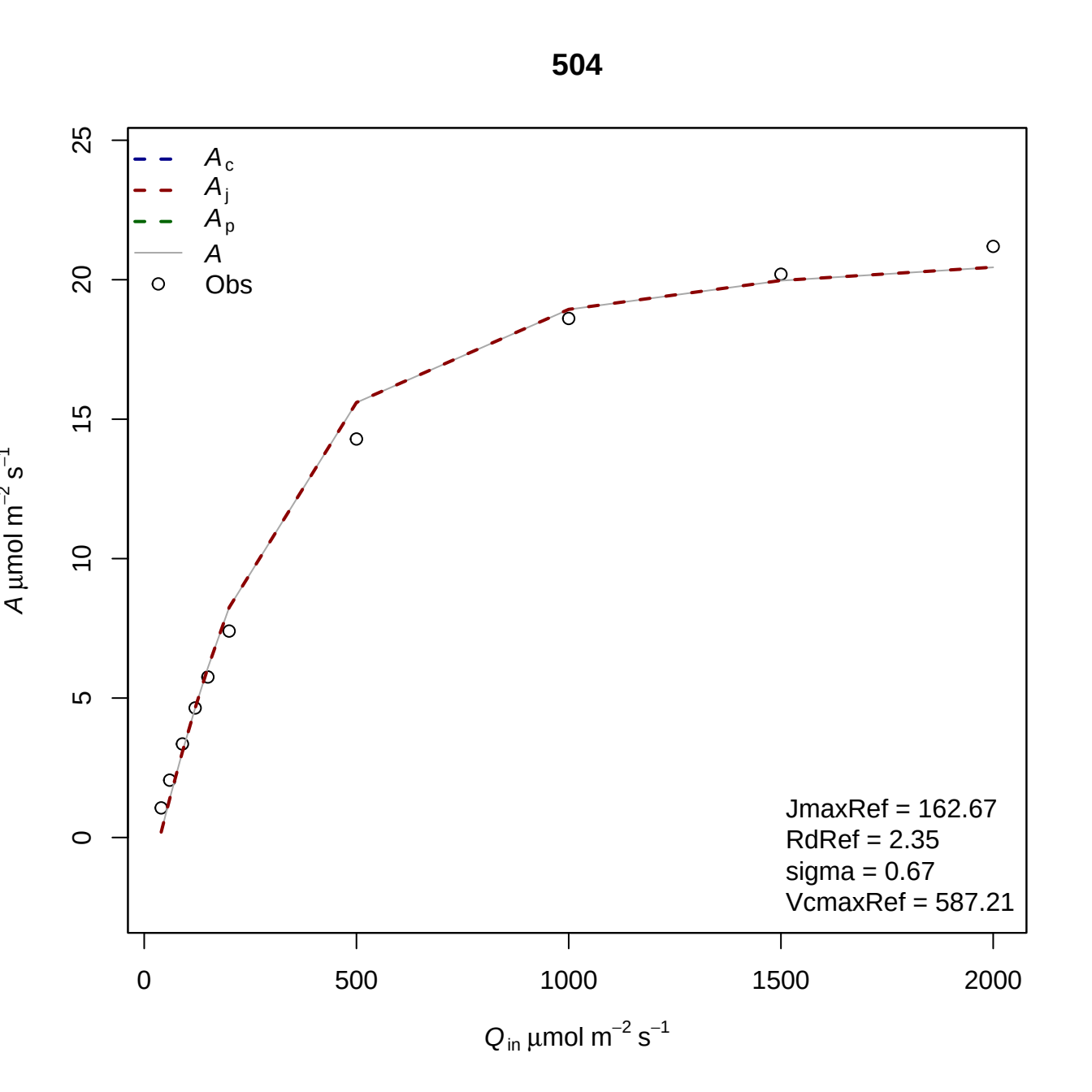

499
A_c
A_j
A_p
A
Obs
JmaxRef = 236.46
RdRef = 3.61
sigma = 0.38
VcmaxRef = 1298.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

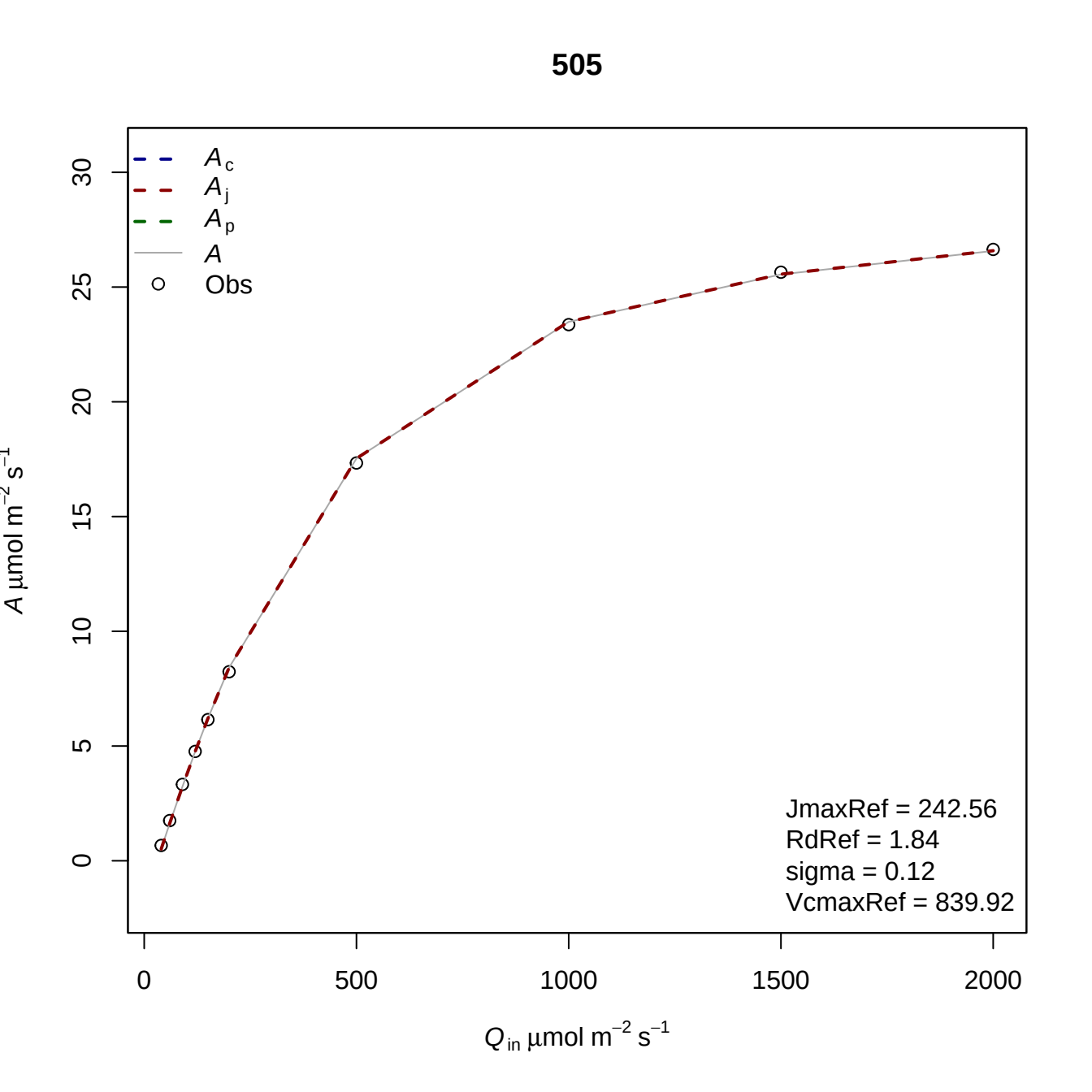

500
A_c
A_j
A_p
A
Obs
JmaxRef = 222.52
RdRef = 3.82
sigma = 0.52
VcmaxRef = 1408.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

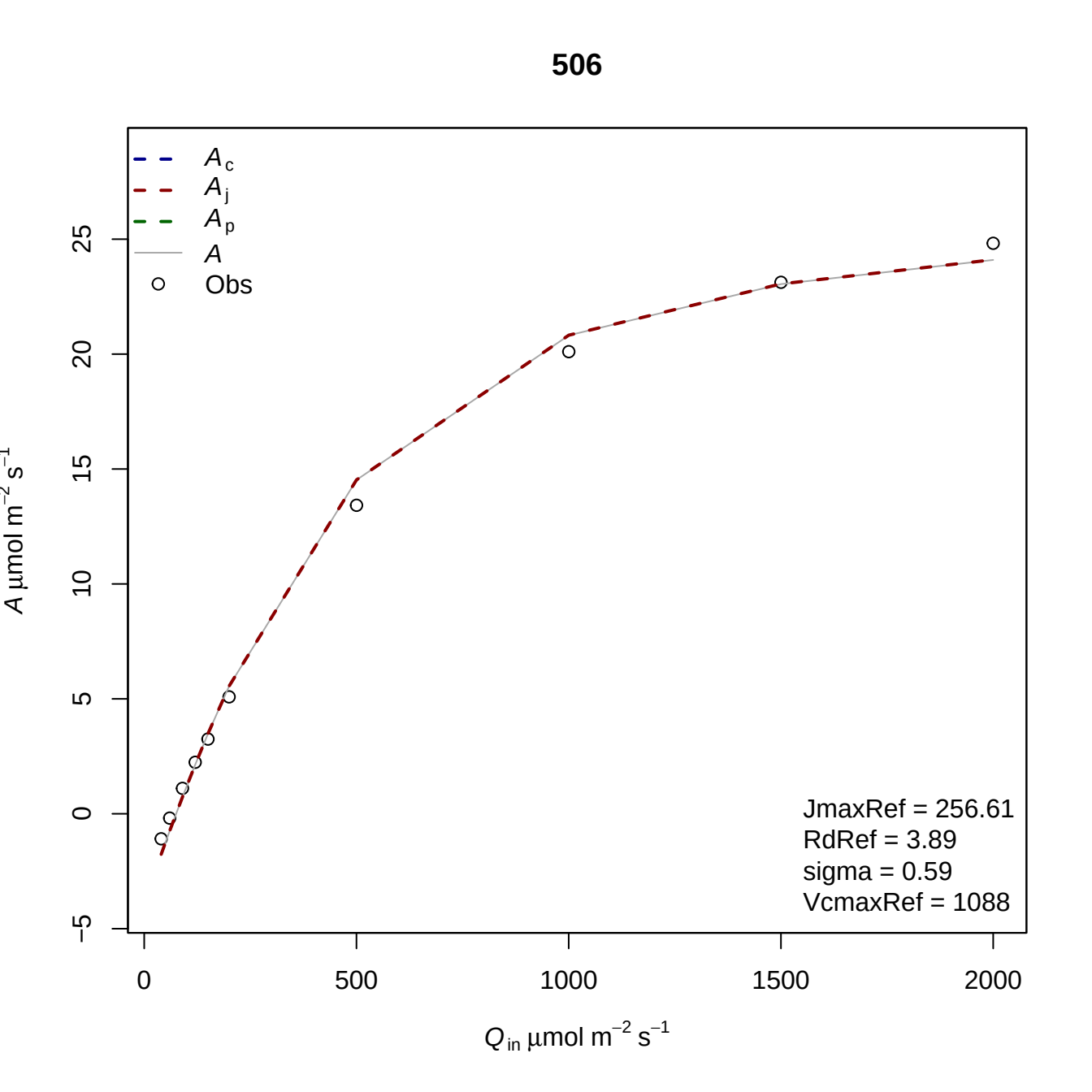

501
A_c
A_j
A_p
A
Obs
JmaxRef = 107.92
RdRef = 1.29
sigma = 0.57
VcmaxRef = 274.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

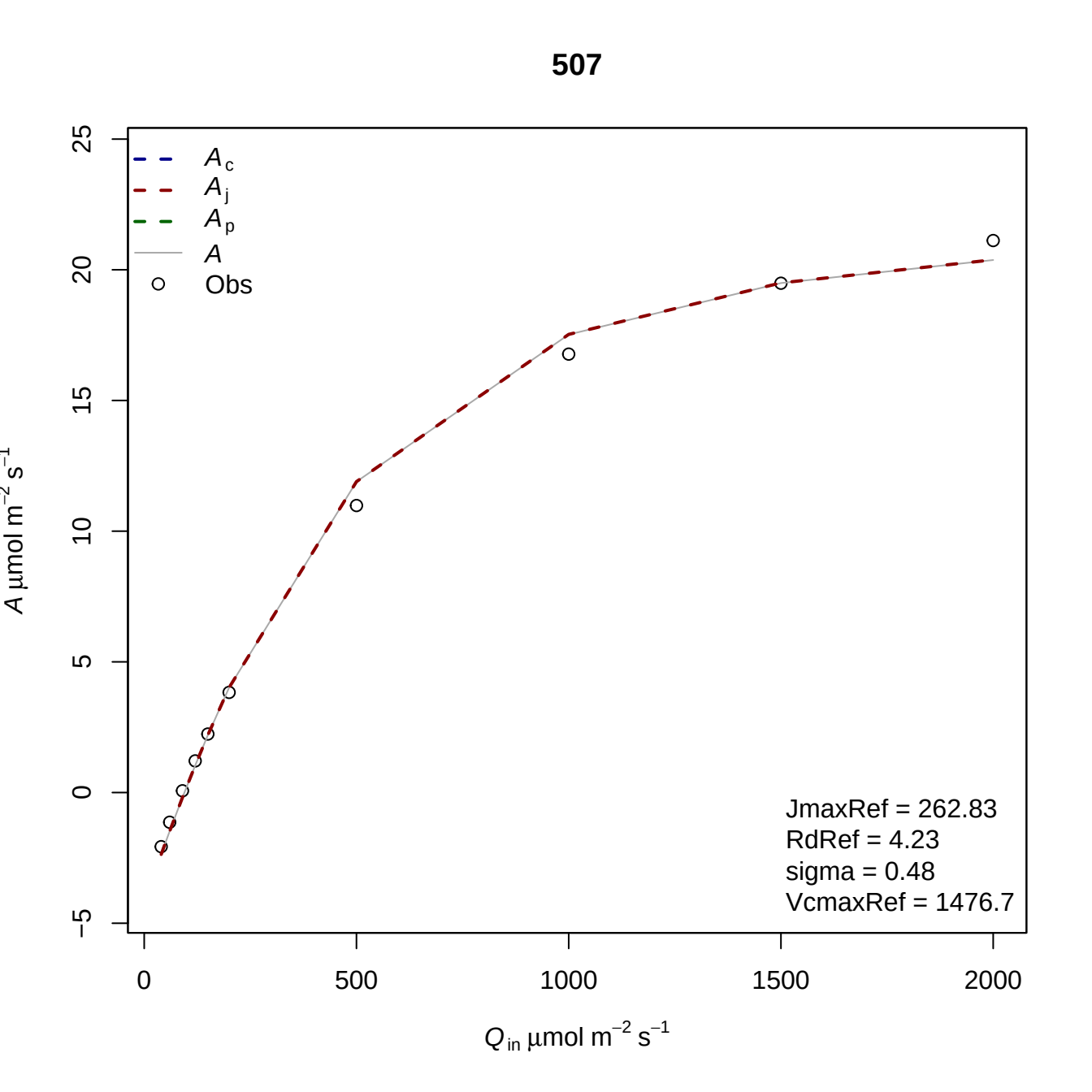

502
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 135.81
RdRef = 1.71
sigma = 0.69
VcmaxRef = 268.9

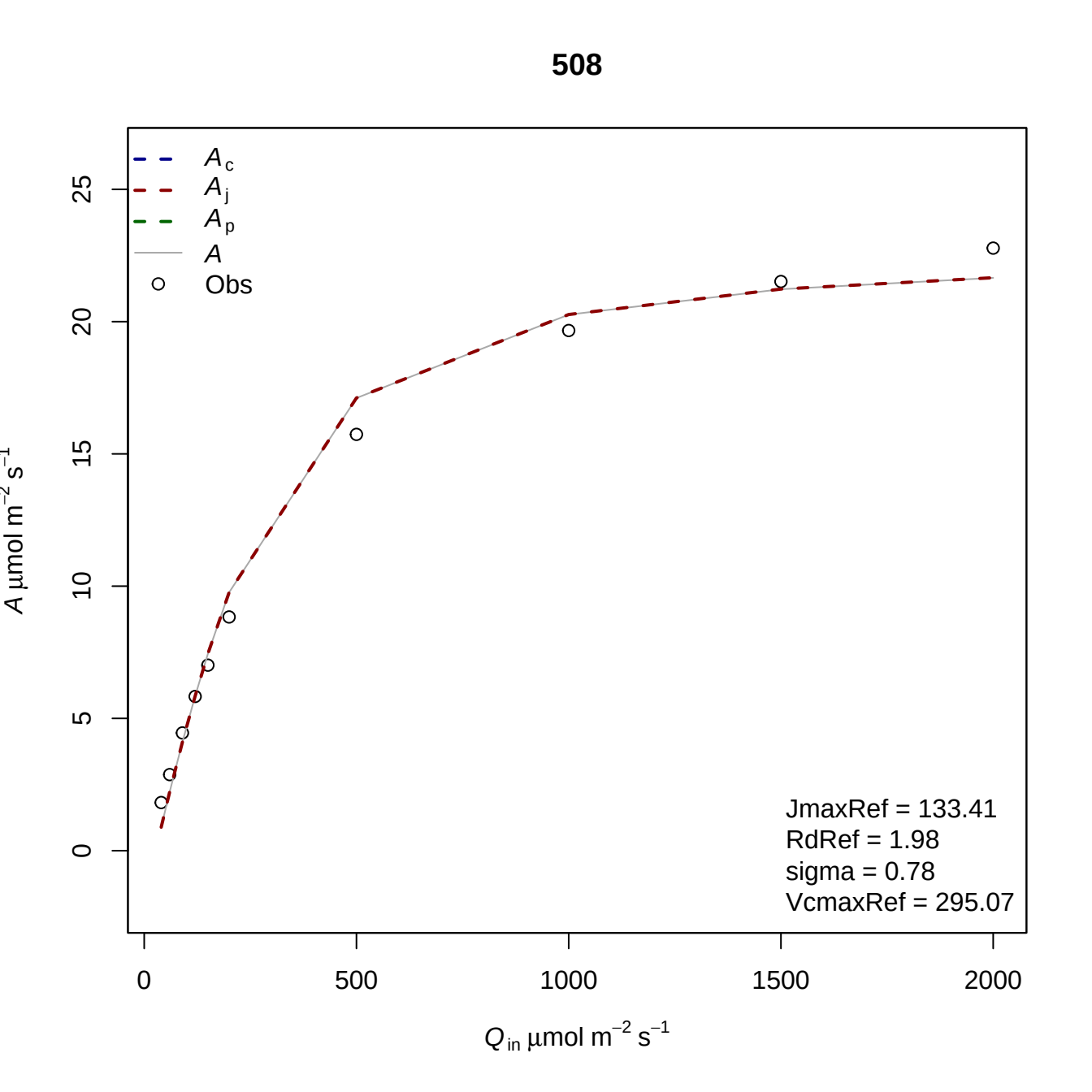

503
A_c
A_j
A_p
A
Obs
JmaxRef = 174.37
RdRef = 2.29
sigma = 0.61
VcmaxRef = 504.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

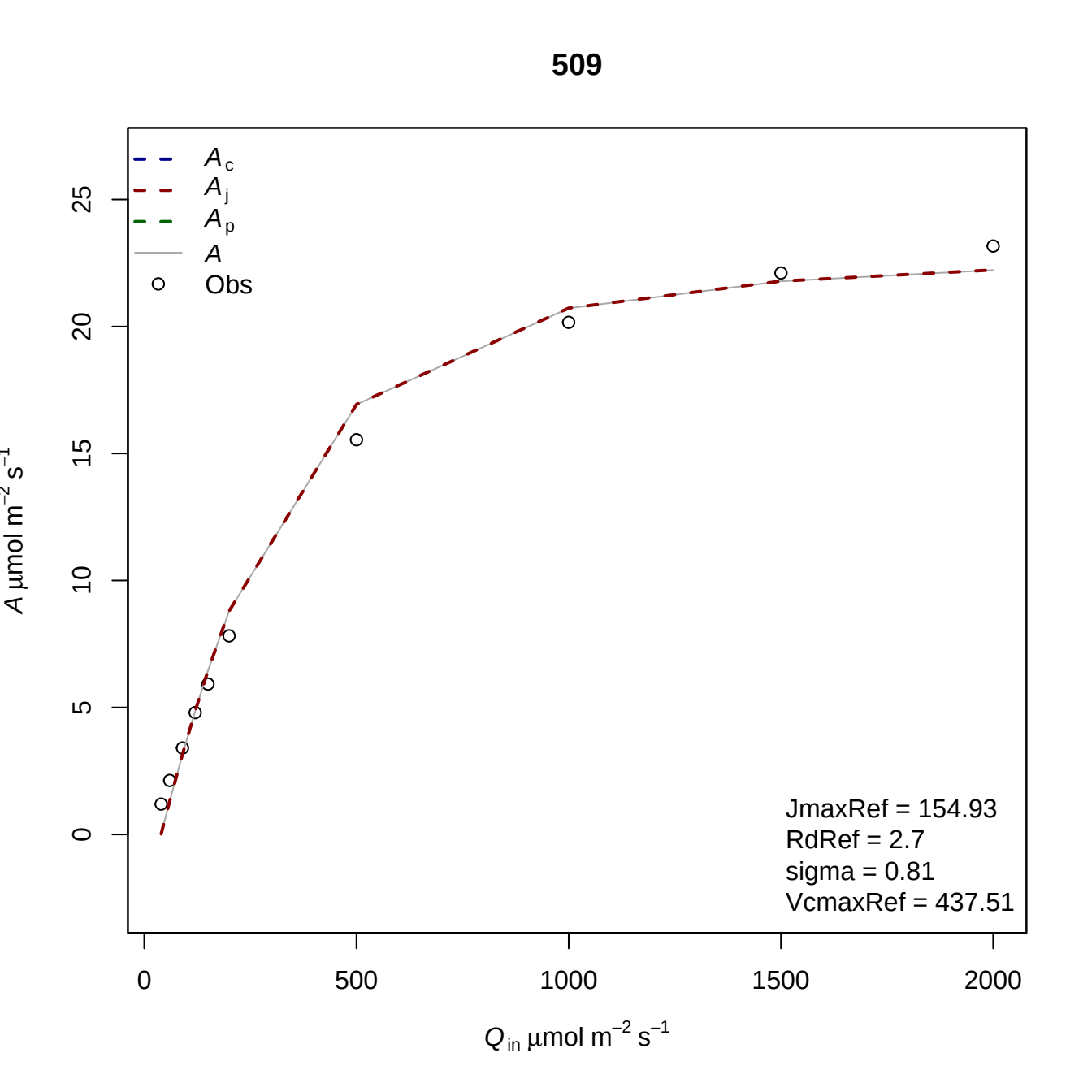

504
A_c
A_j
A_p
A
Obs
JmaxRef = 162.67
RdRef = 2.35
sigma = 0.67
VcmaxRef = 587.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

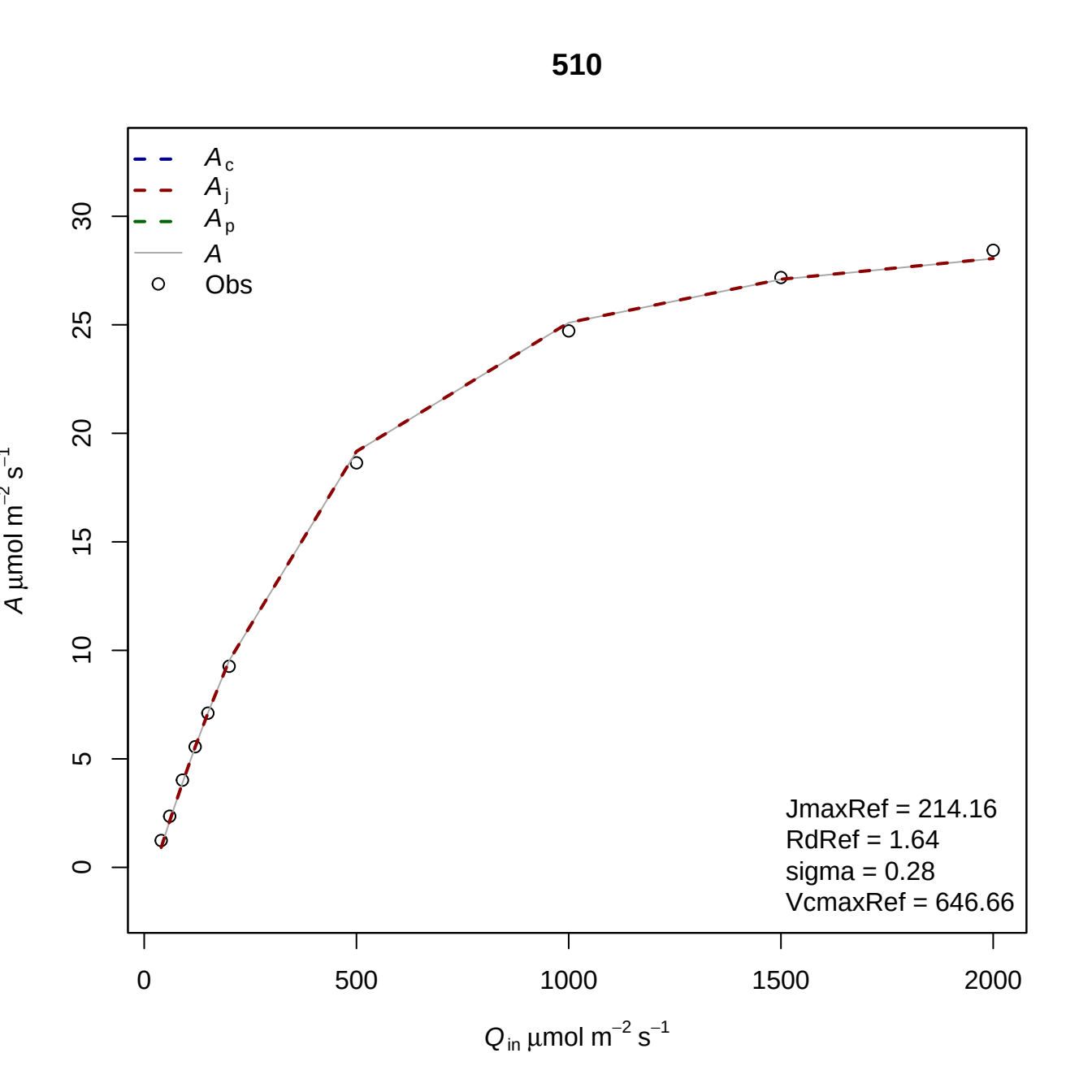

505
A_c
A_j
A_p
A
Obs
JmaxRef = 242.56
RdRef = 1.84
sigma = 0.12
VcmaxRef = 839.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

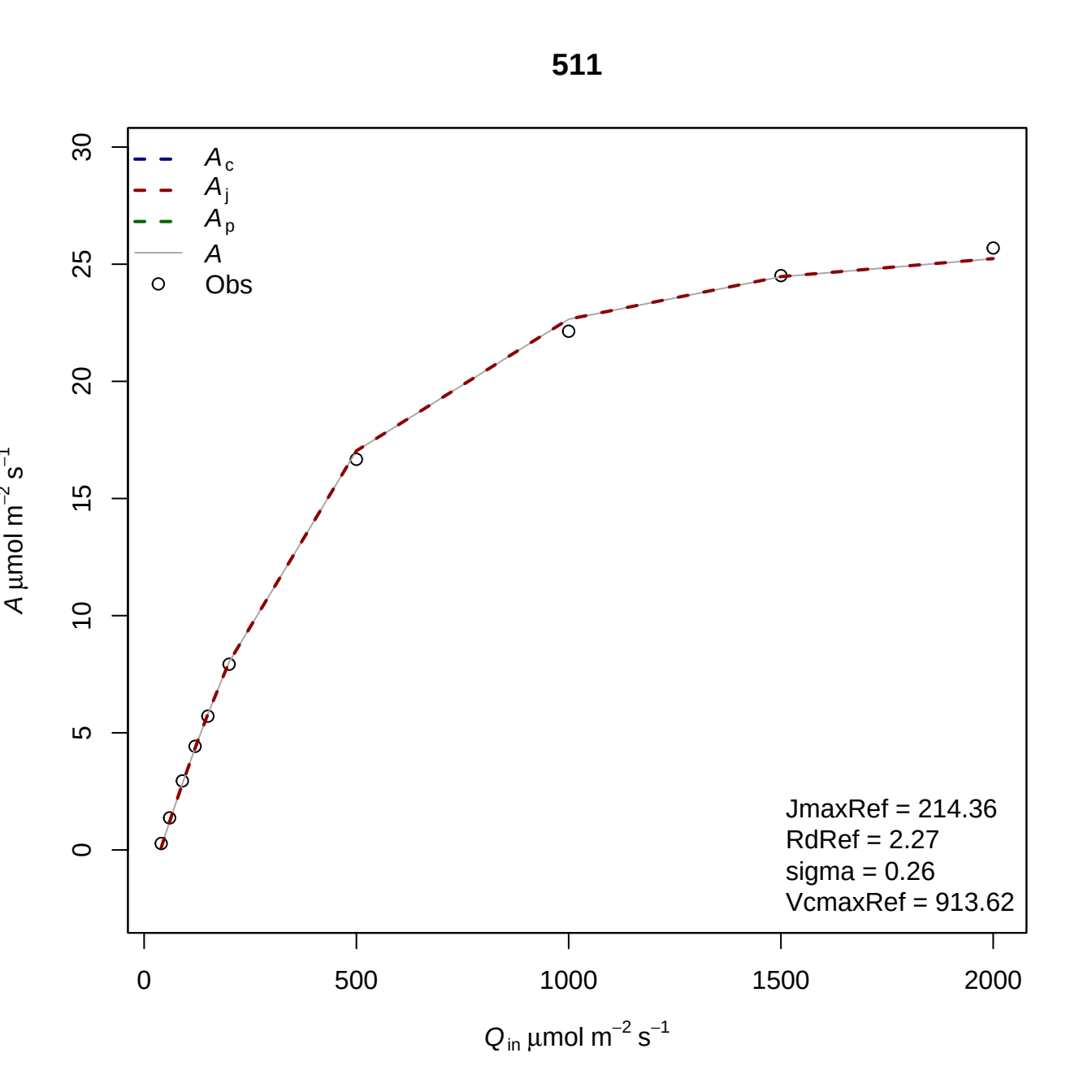

506
A_c
A_j
A_p
A
Obs
JmaxRef = 256.61
RdRef = 3.89
sigma = 0.59
VcmaxRef = 1088
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500
2000

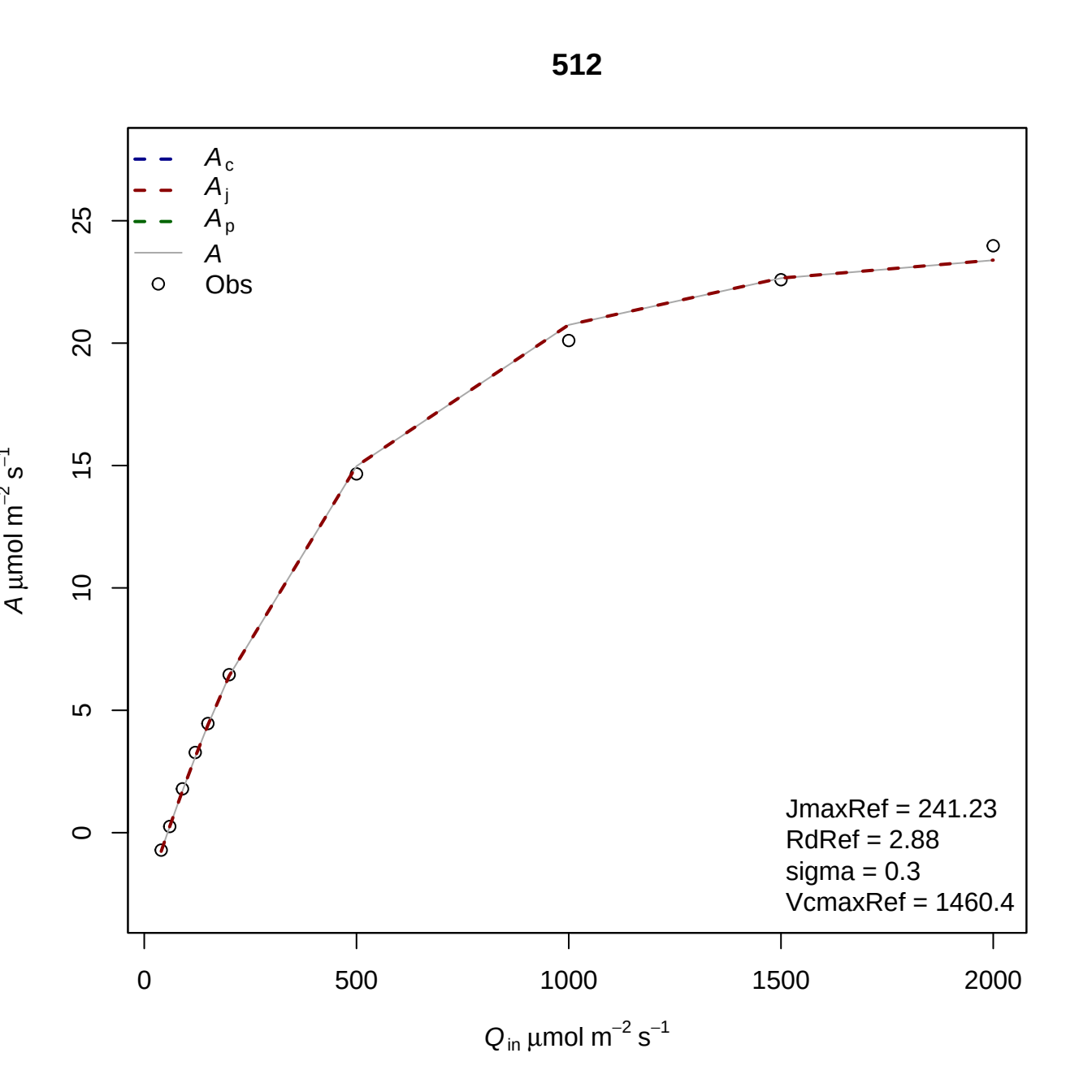

507
A_c
A_j
A_p
A
Obs
JmaxRef = 262.83
RdRef = 4.23
sigma = 0.48
VcmaxRef = 1476.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

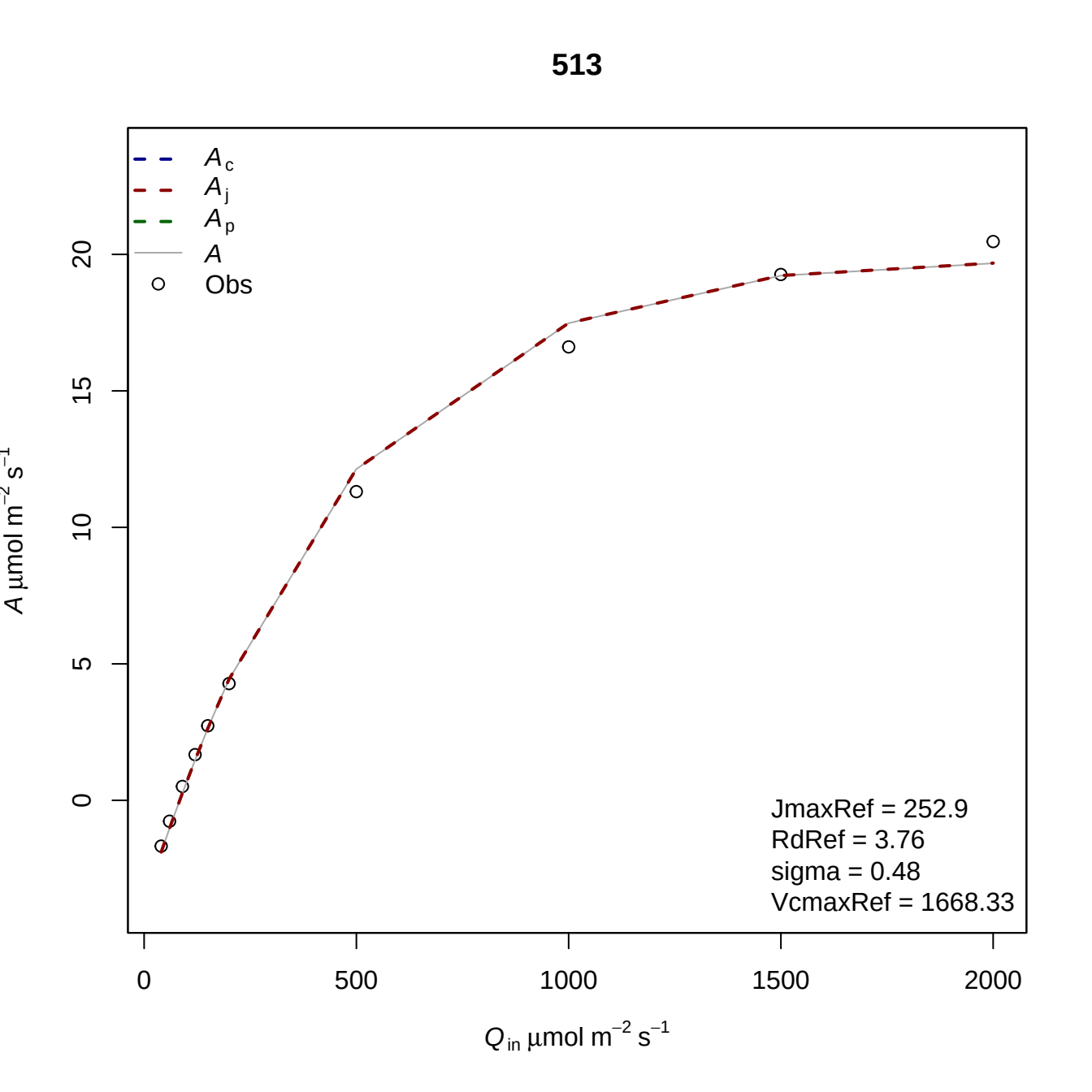

508
A_c
A_j
A_p
A
Obs
JmaxRef = 133.41
RdRef = 1.98
sigma = 0.78
VcmaxRef = 295.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

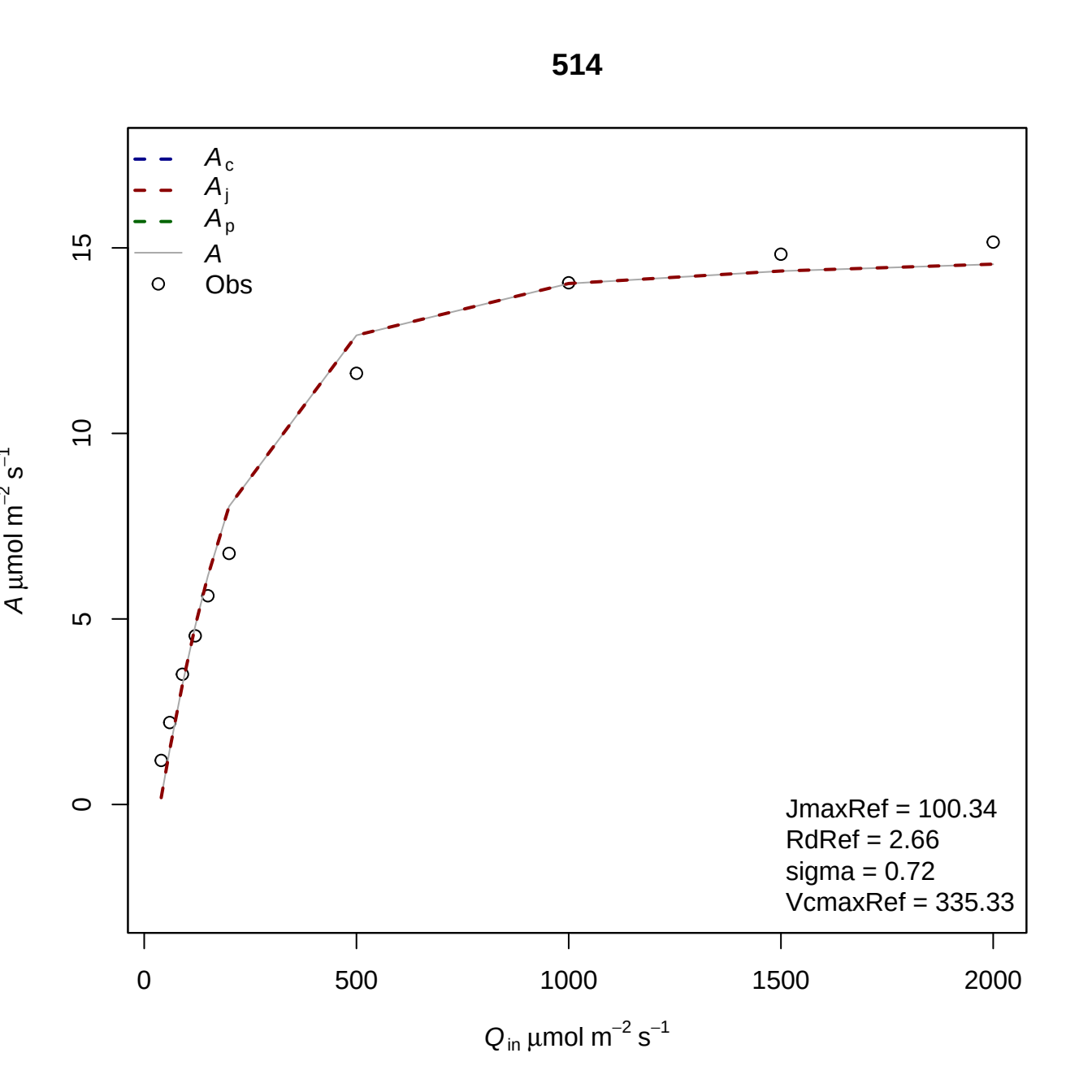

509
A_c
A_j
A_p
A
Obs
JmaxRef = 154.93
RdRef = 2.7
sigma = 0.81
VcmaxRef = 437.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

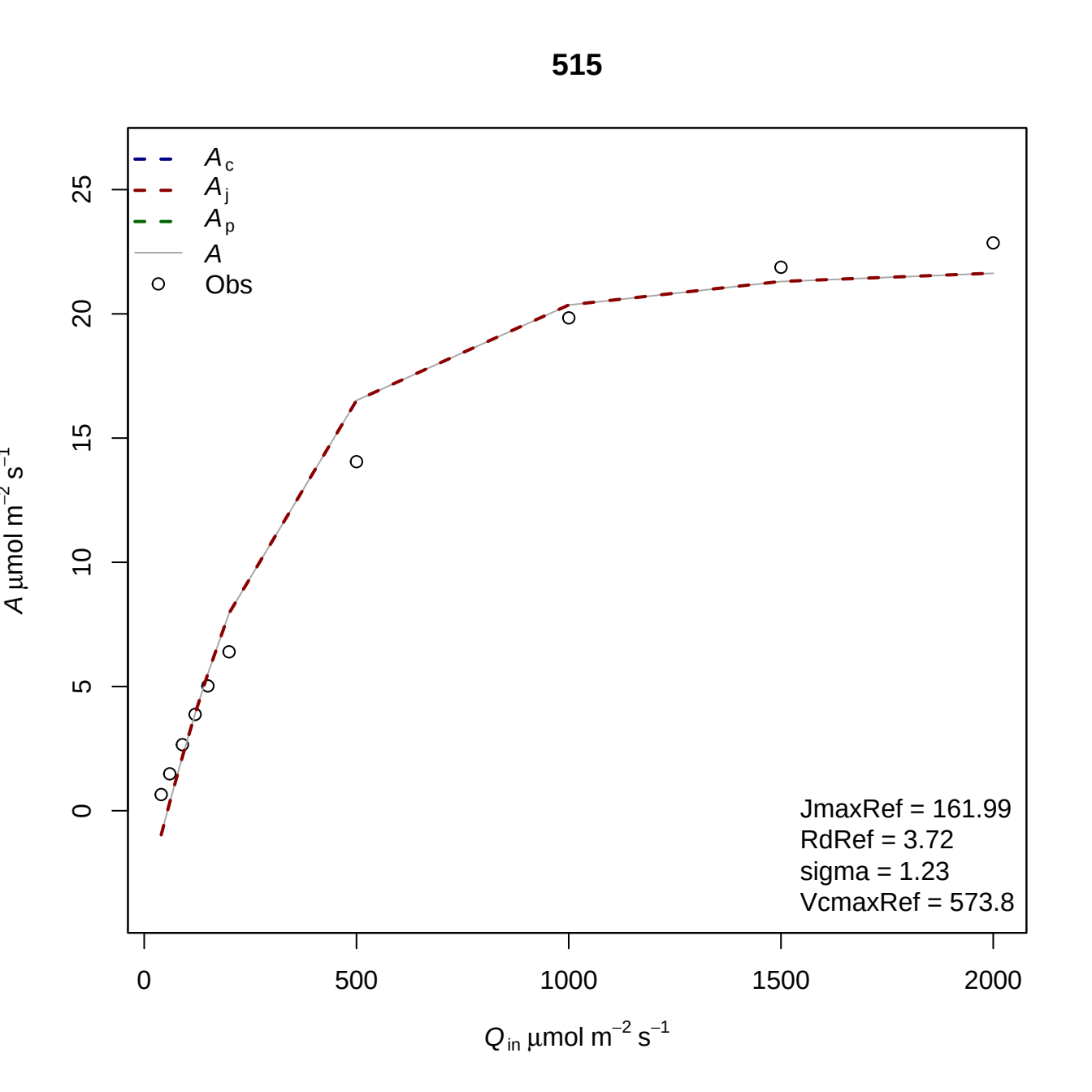

510
A_c
A_j
A_p
A
Obs
JmaxRef = 214.16
RdRef = 1.64
sigma = 0.28
VcmaxRef = 646.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

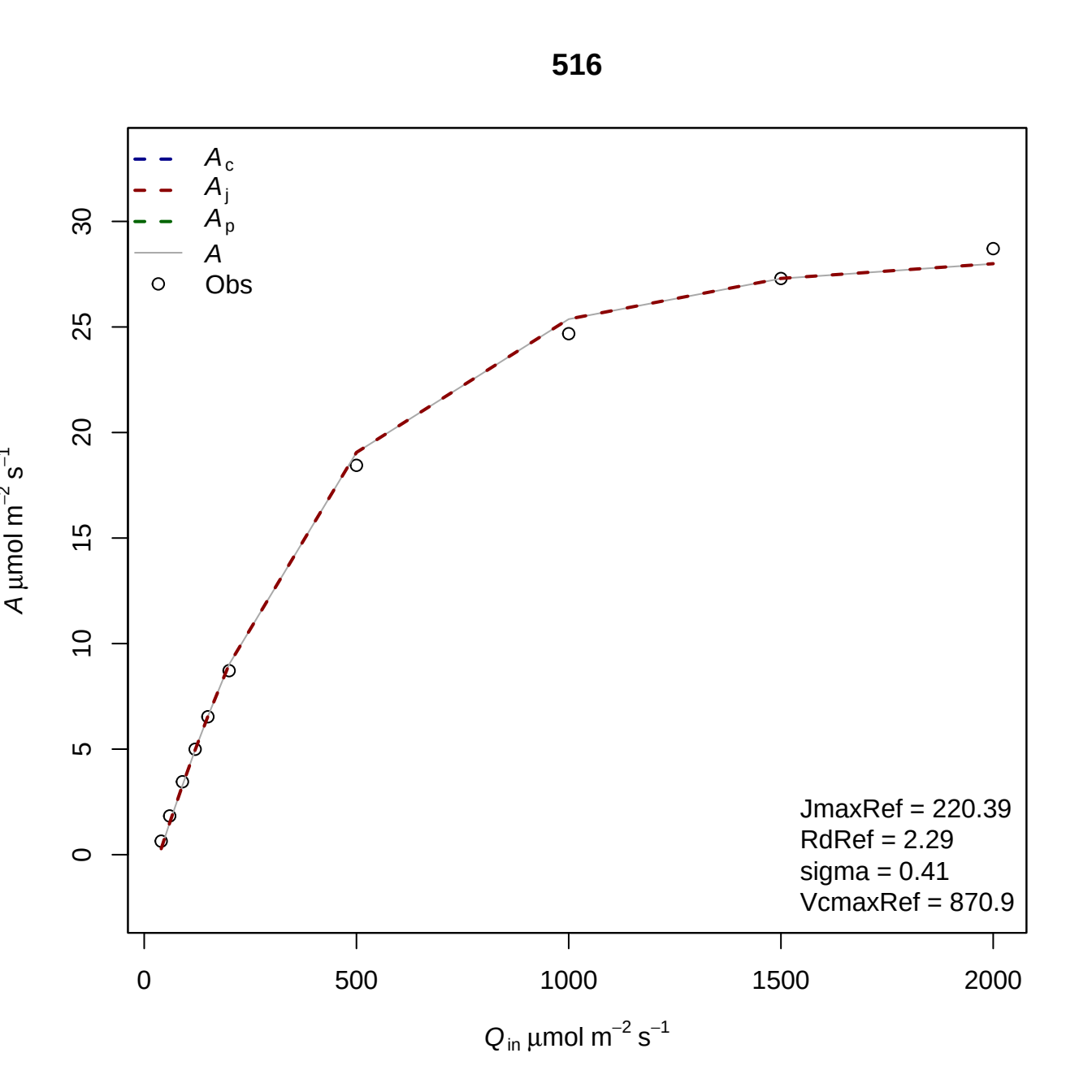

511
A_c
A_j
A_p
A
Obs
JmaxRef = 214.36
RdRef = 2.27
sigma = 0.26
VcmaxRef = 913.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

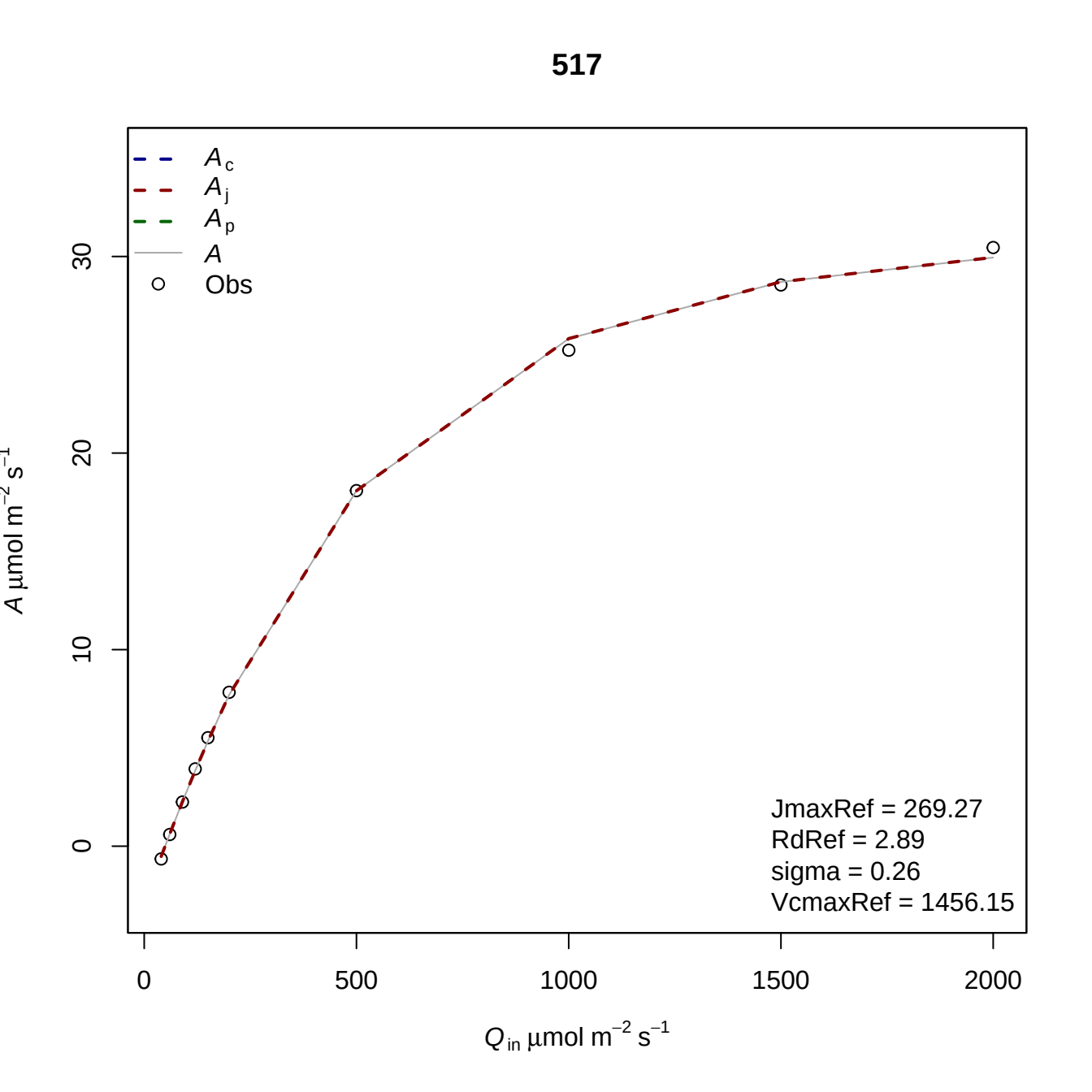

512
A_c
A_j
A_p
A
Obs
JmaxRef = 241.23
RdRef = 2.88
sigma = 0.3
VcmaxRef = 1460.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

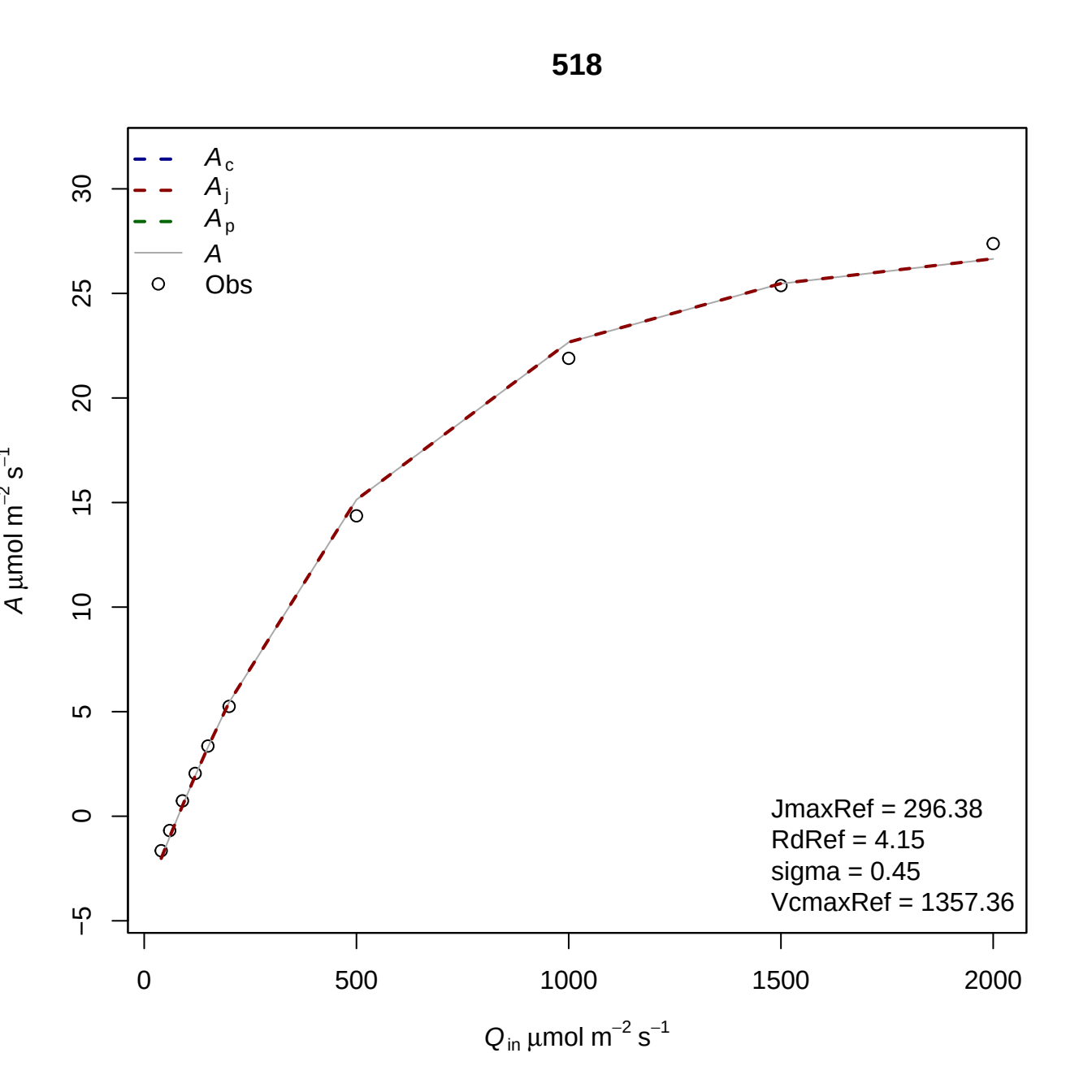

513
A_c
A_j
A_p
A
Obs
JmaxRef = 252.9
RdRef = 3.76
sigma = 0.48
VcmaxRef = 1668.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

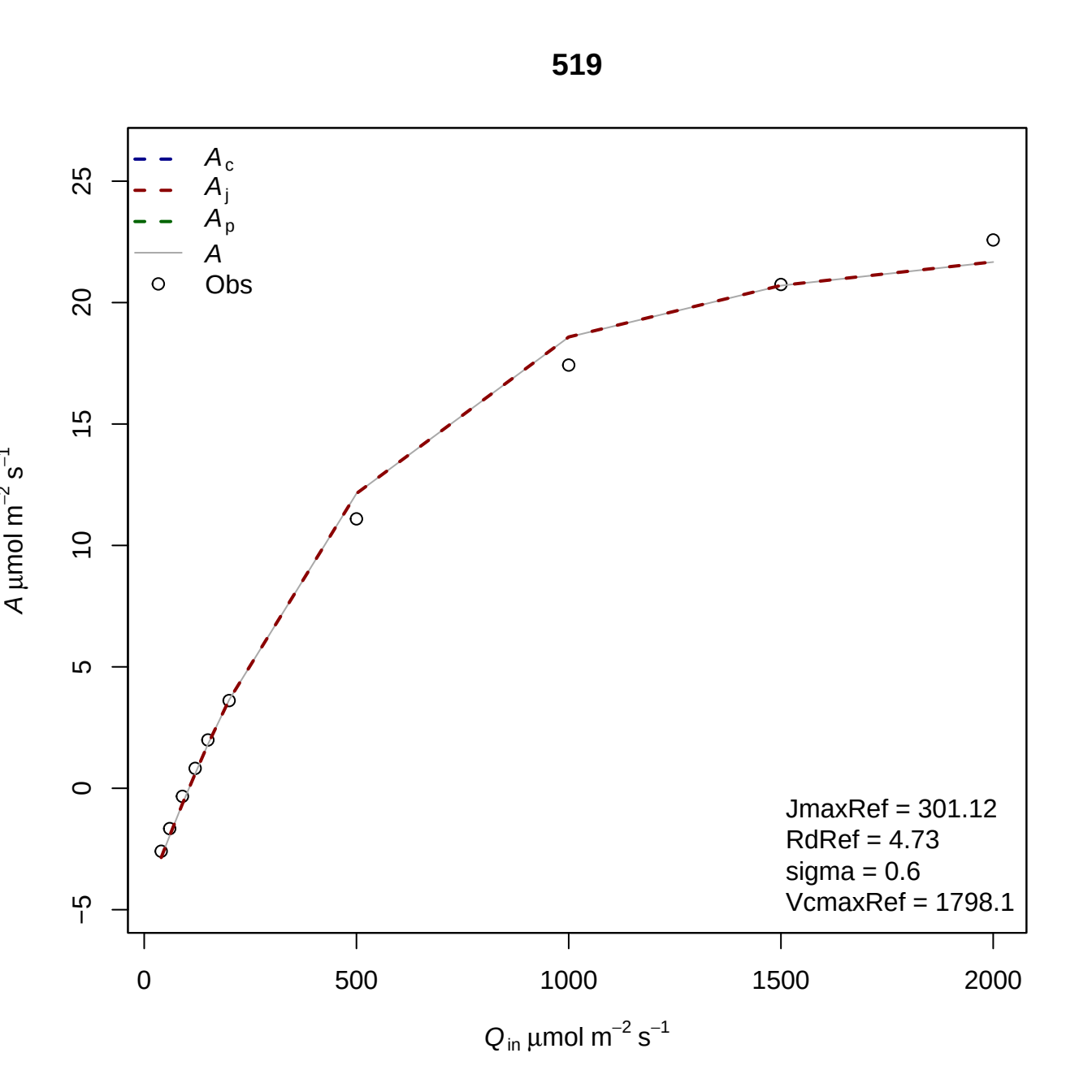

514
A_c
A_j
A_p
A
Obs
JmaxRef = 100.34
RdRef = 2.66
sigma = 0.72
VcmaxRef = 335.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

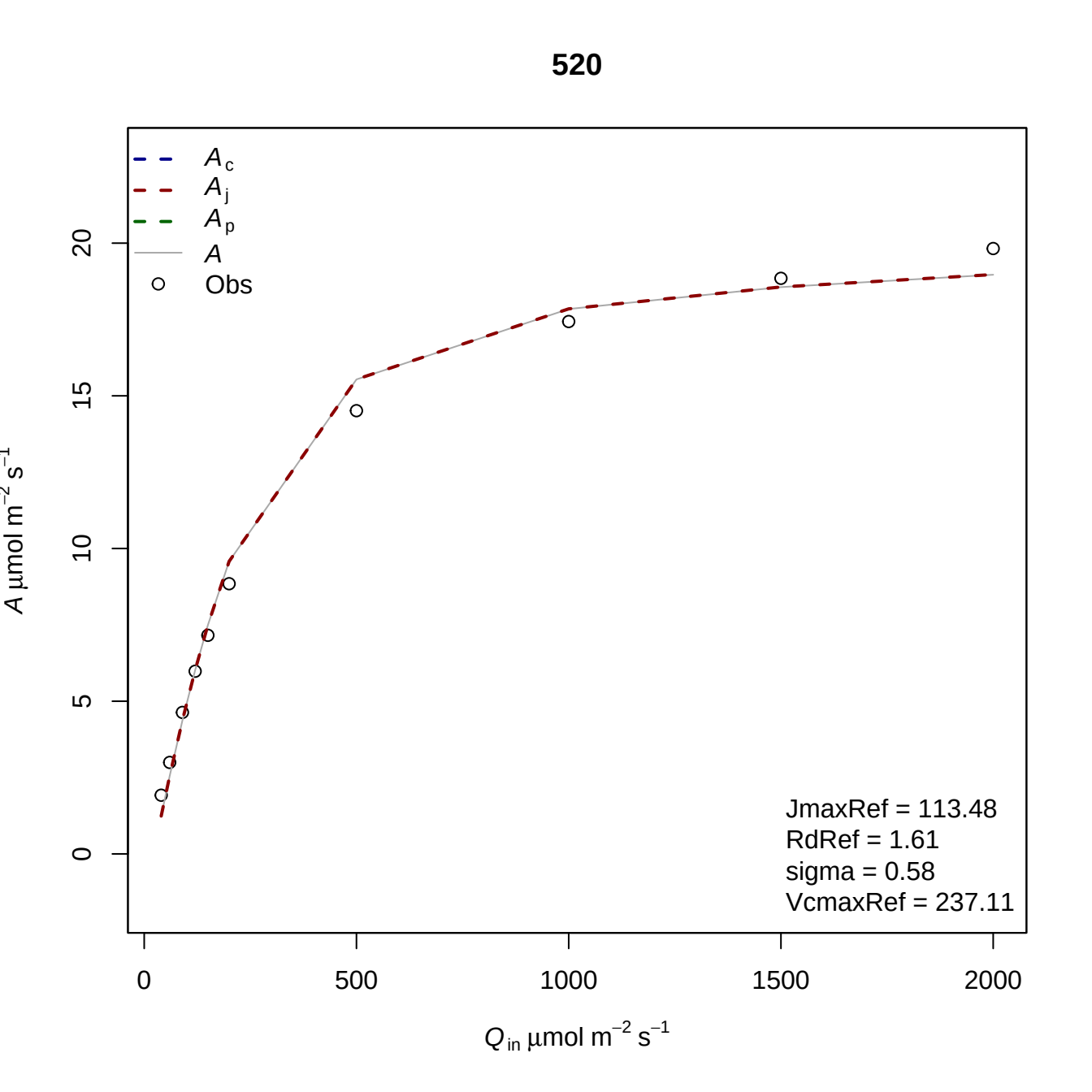

515
A_c
A_j
A_p
A
Obs
JmaxRef = 161.99
RdRef = 3.72
sigma = 1.23
VcmaxRef = 573.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

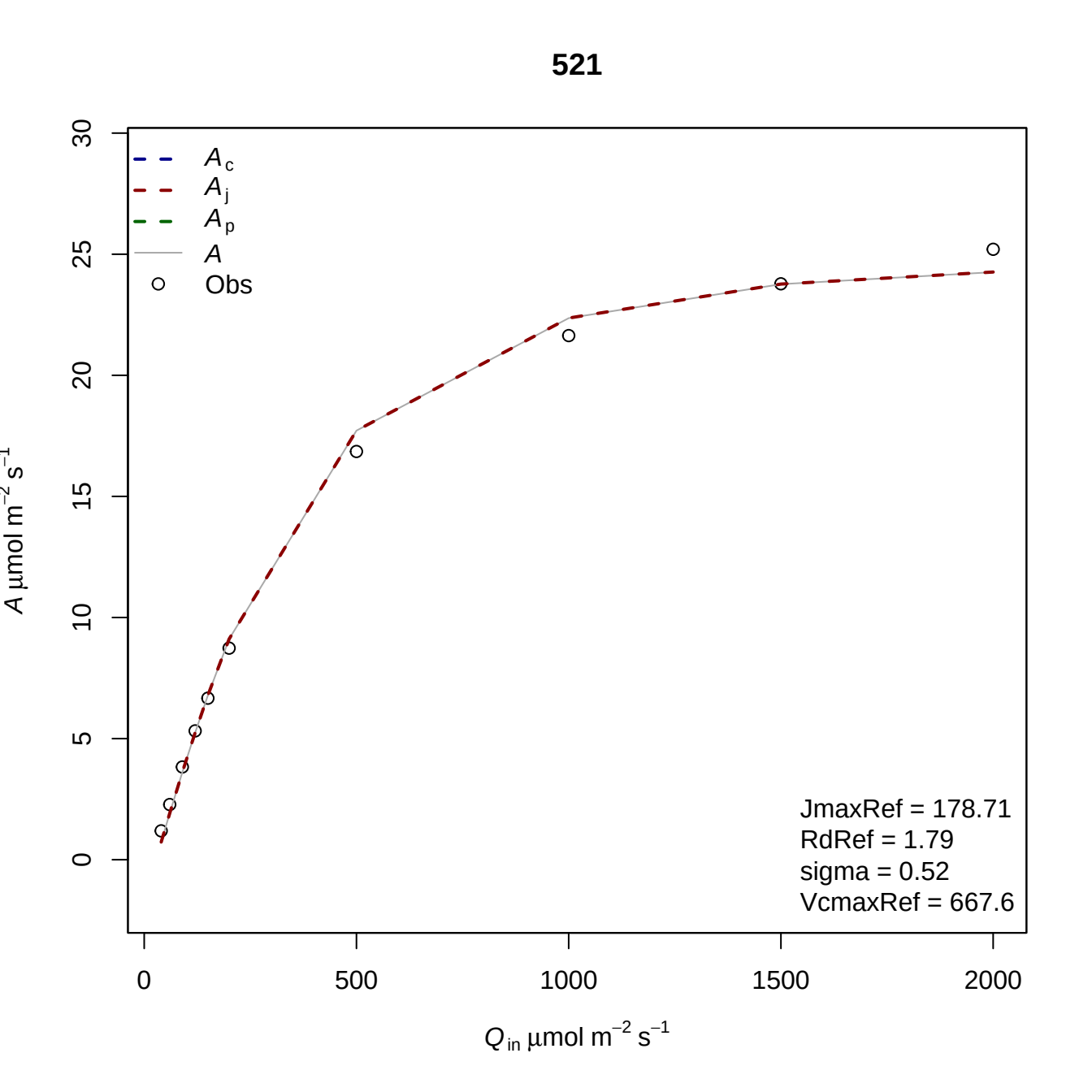

516
A_c
A_j
A_p
A
Obs
JmaxRef = 220.39
RdRef = 2.29
sigma = 0.41
VcmaxRef = 870.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

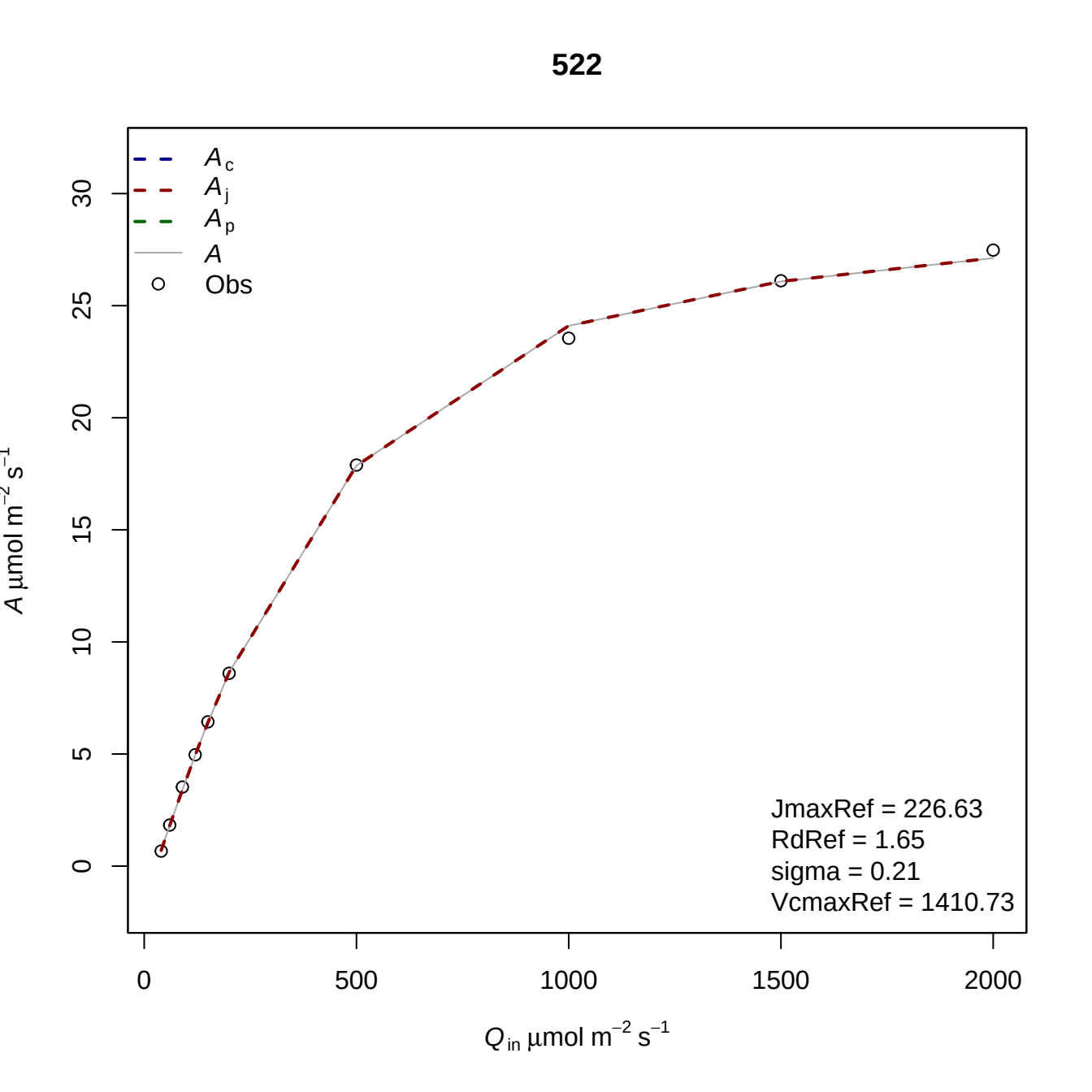

517
A_c
A_j
A_p
A
Obs
JmaxRef = 269.27
RdRef = 2.89
sigma = 0.26
VcmaxRef = 1456.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

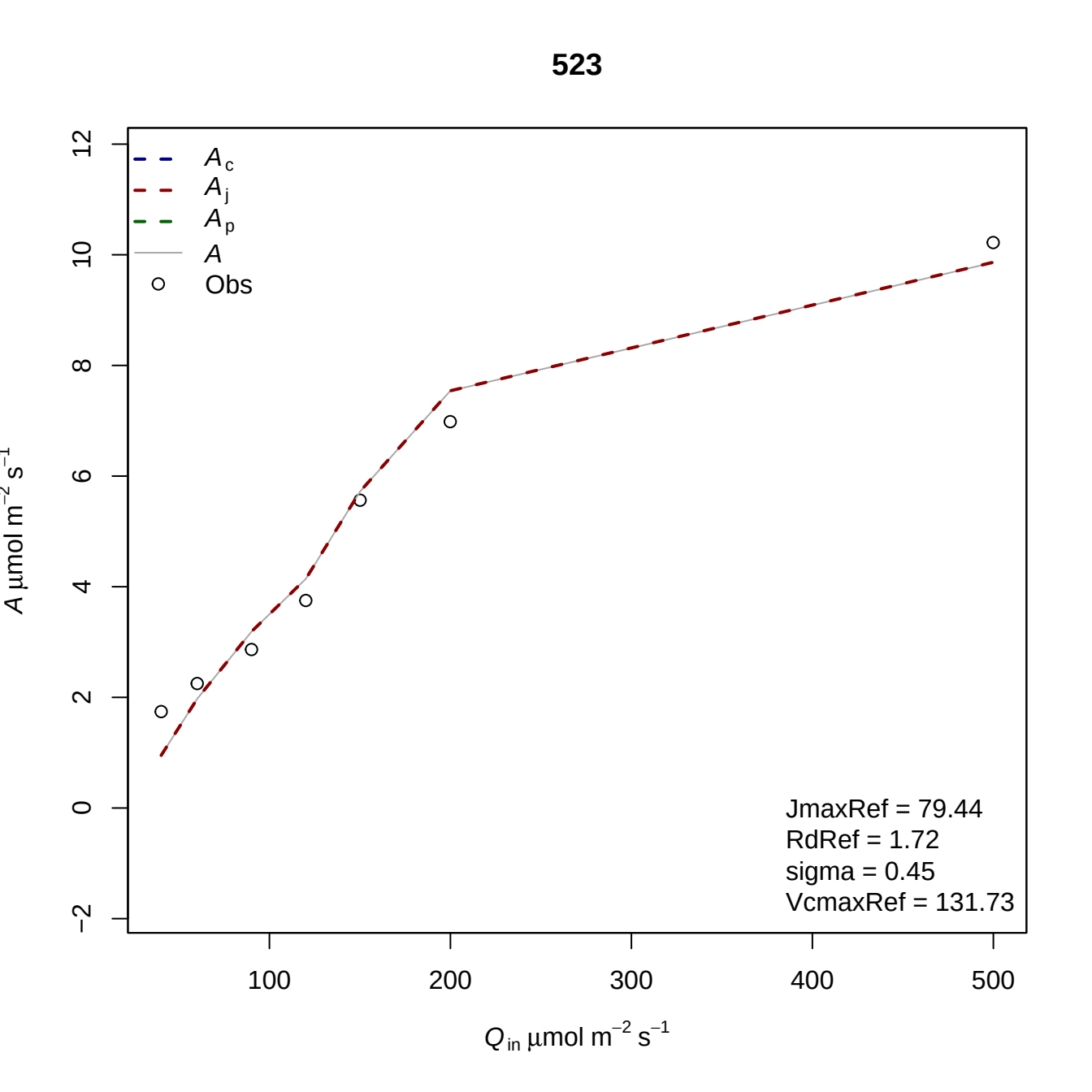

518
A_c
A_j
A_p
A
Obs
JmaxRef = 296.38
RdRef = 4.15
sigma = 0.45
VcmaxRef = 1357.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
30
0
500
1000
1500
2000

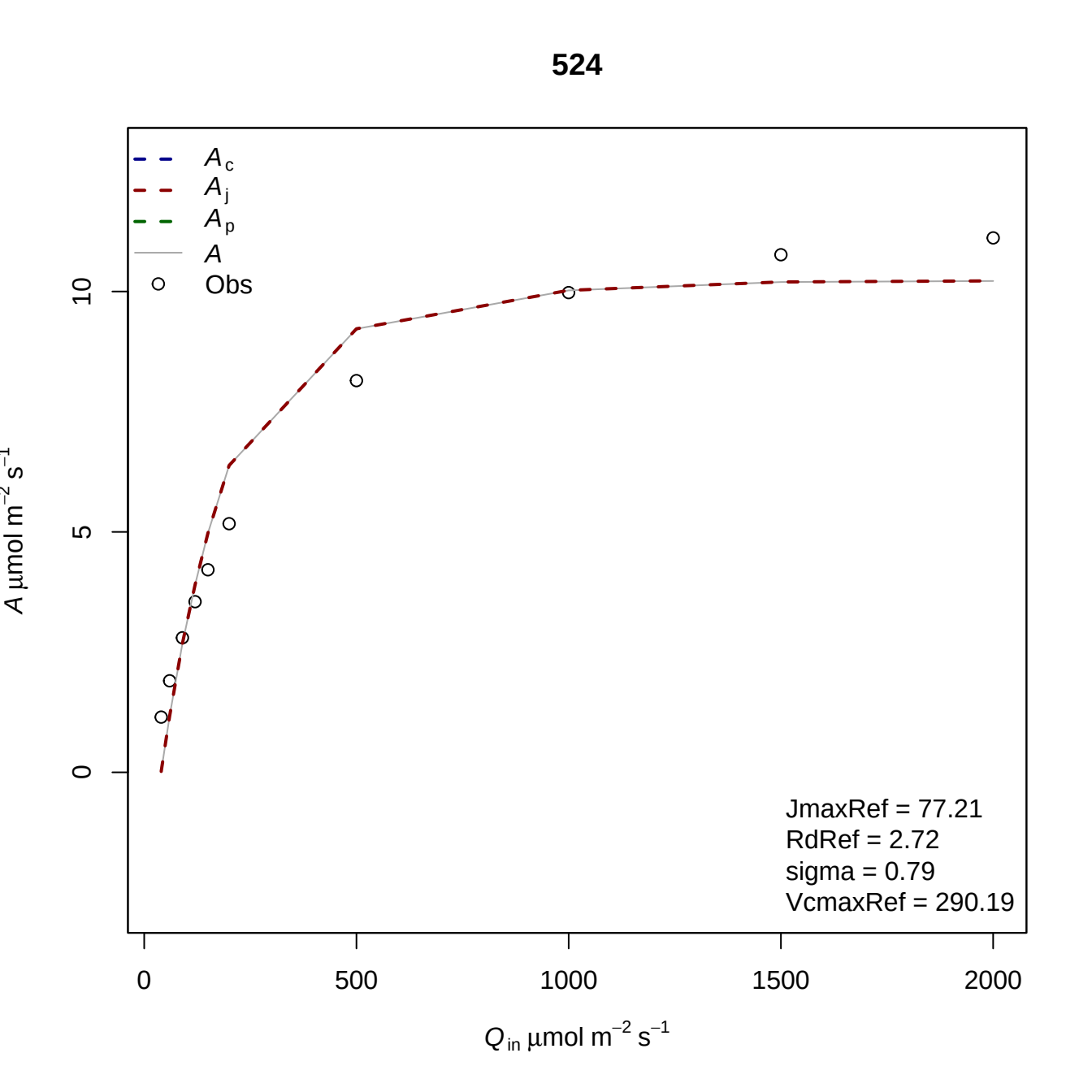

519
A_c
A_j
A_p
A
Obs
JmaxRef = 301.12
RdRef = 4.73
sigma = 0.6
VcmaxRef = 1798.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

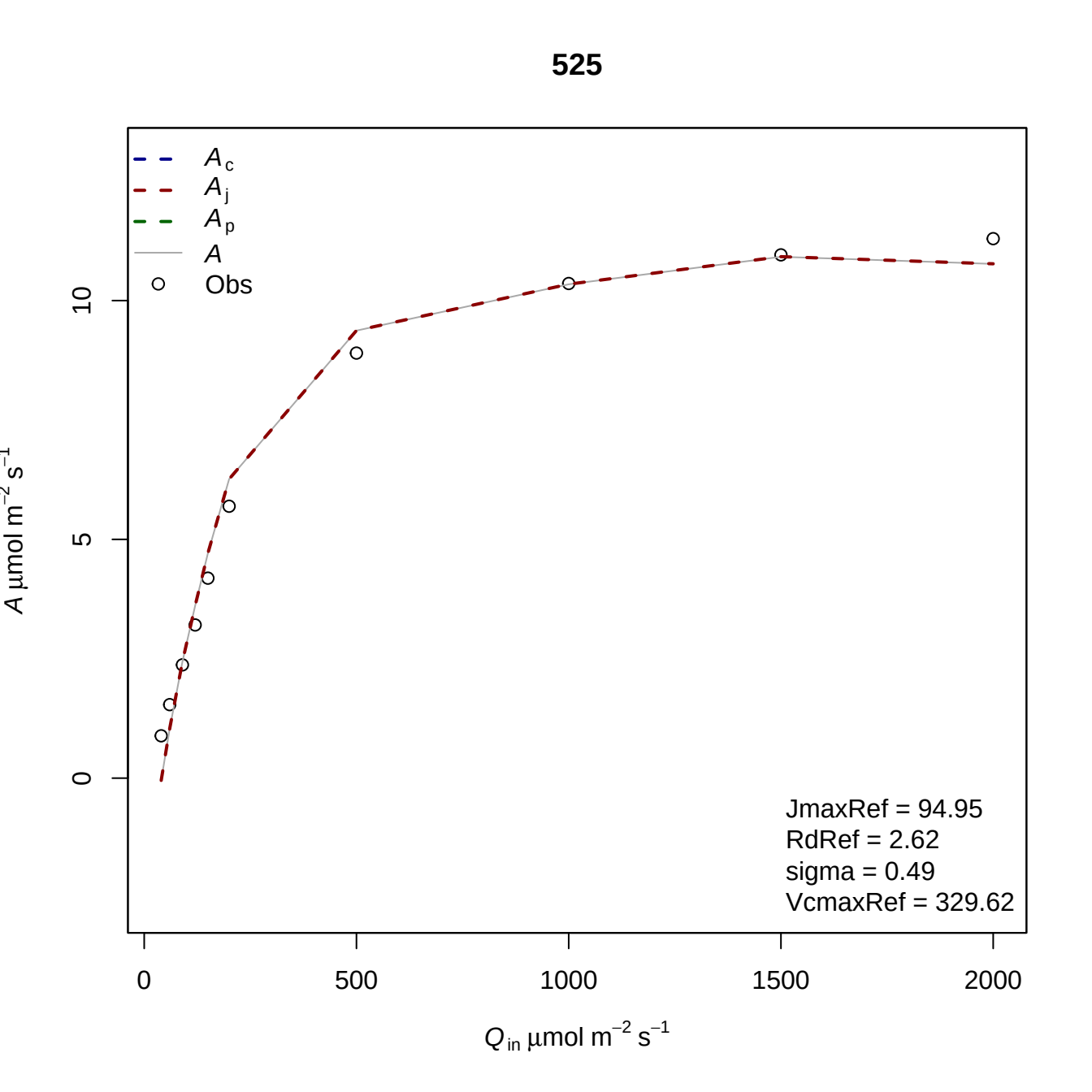

520
A_c
A_j
A_p
A
Obs
JmaxRef = 113.48
RdRef = 1.61
sigma = 0.58
VcmaxRef = 237.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

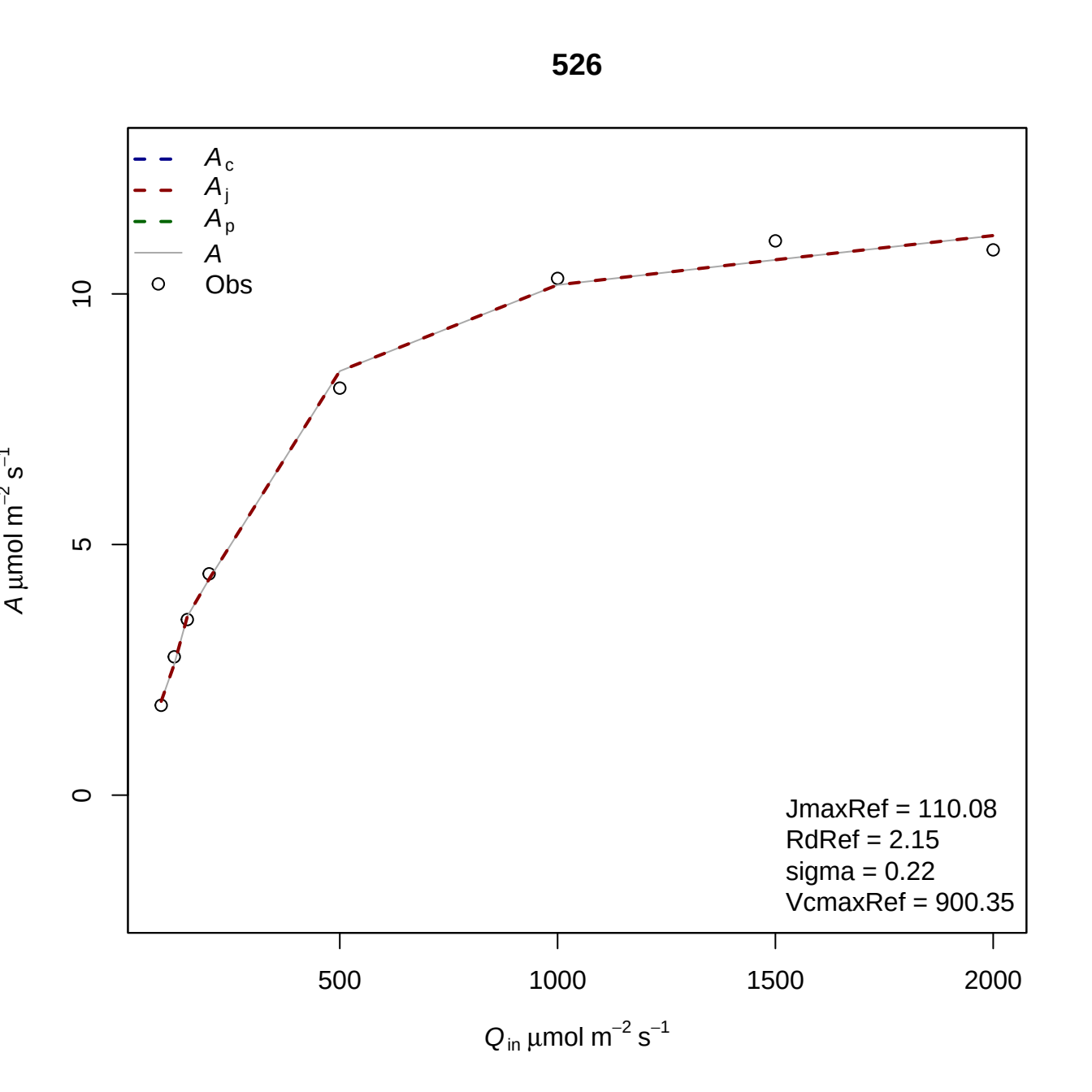

521
A_c
A_j
A_p
A
Obs
JmaxRef = 178.71
RdRef = 1.79
sigma = 0.52
VcmaxRef = 667.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

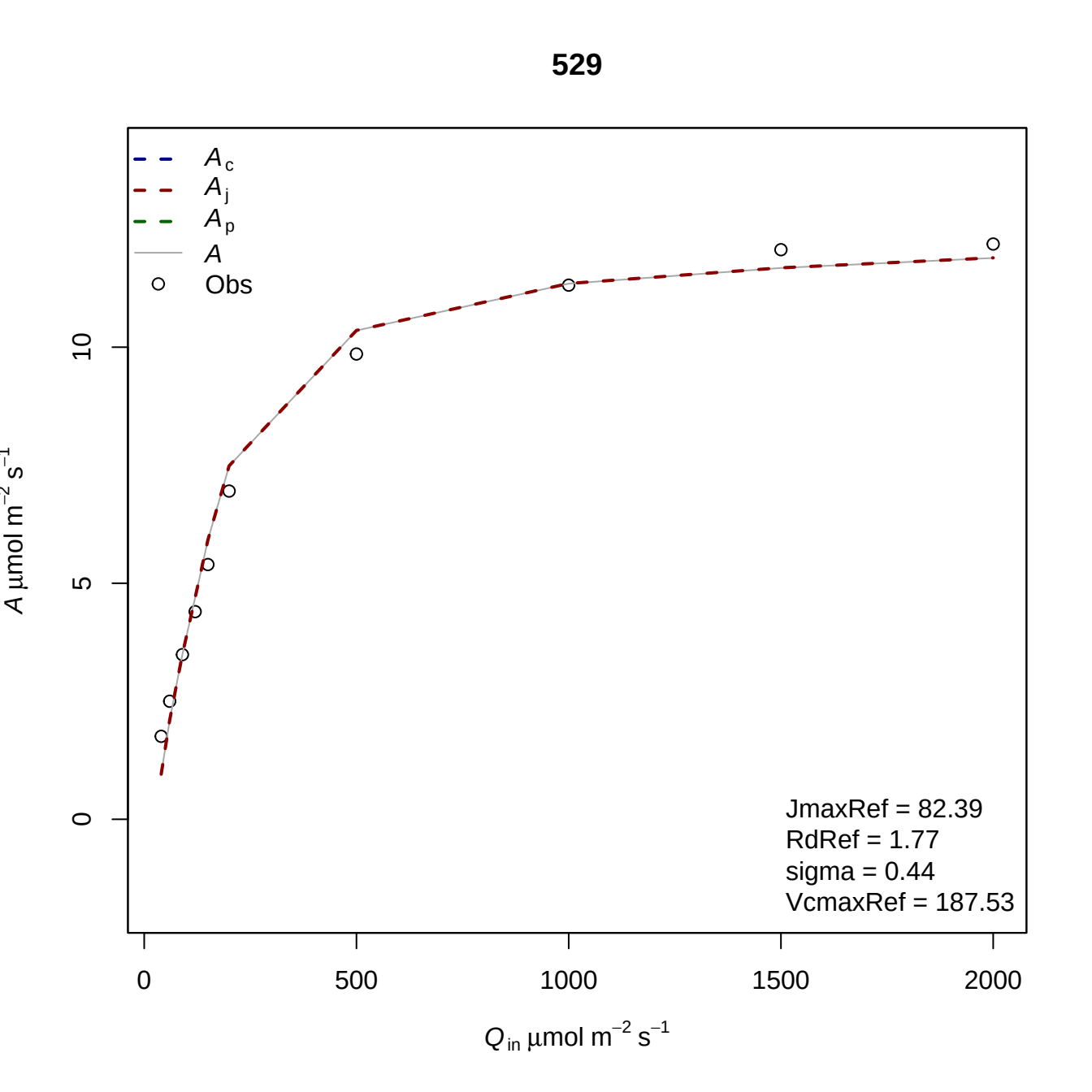

522
A_c
A_j
A_p
A
Obs
JmaxRef = 226.63
RdRef = 1.65
sigma = 0.21
VcmaxRef = 1410.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

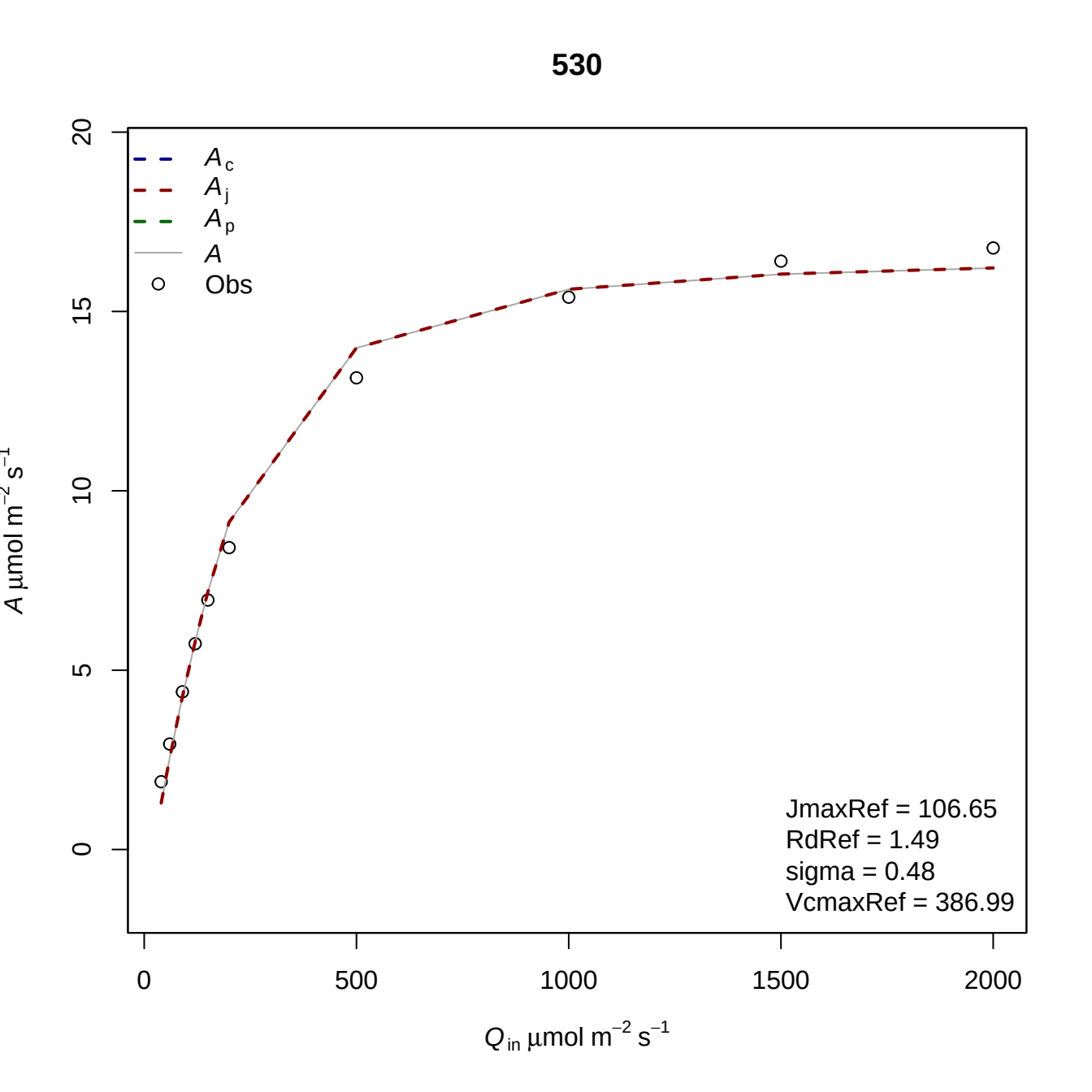

523
A_c
A_j
A_p
A
Obs
JmaxRef = 79.44
RdRef = 1.72
sigma = 0.45
VcmaxRef = 131.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
100
200
300
400
500

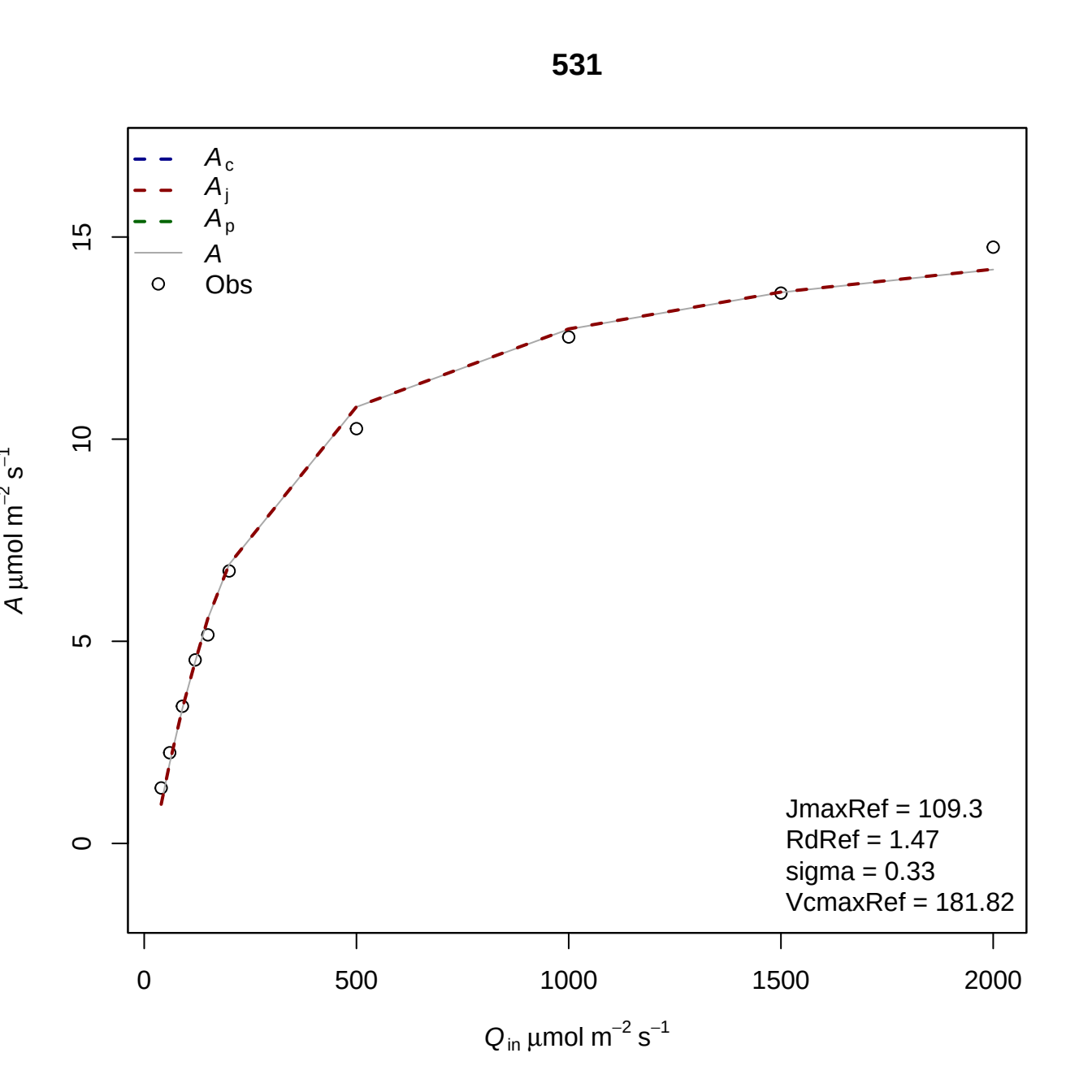

524
A_c
A_j
A_p
A
Obs
JmaxRef = 77.21
RdRef = 2.72
sigma = 0.79
VcmaxRef = 290.19
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

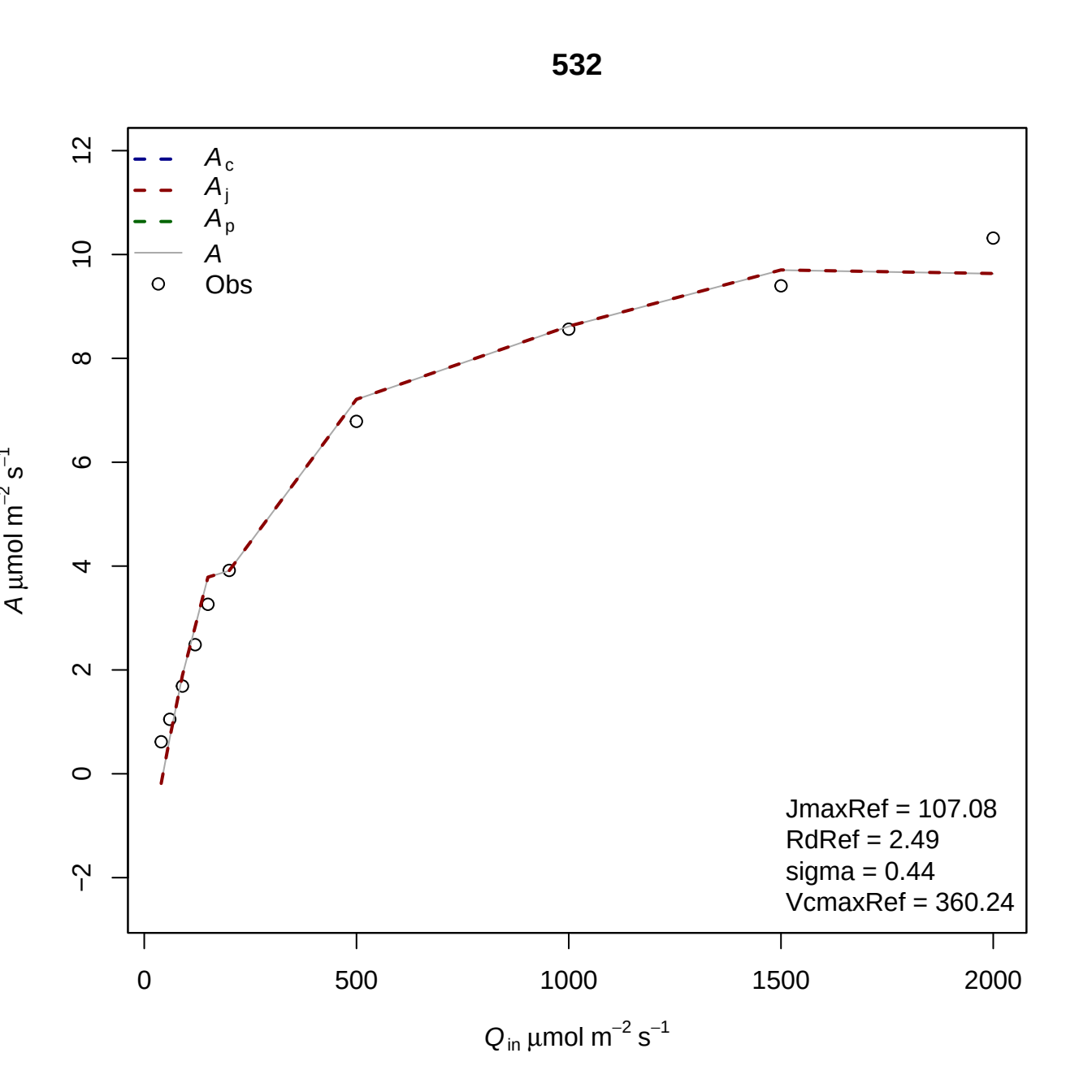

525
A_c
A_j
A_p
A
Obs
JmaxRef = 94.95
RdRef = 2.62
sigma = 0.49
VcmaxRef = 329.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

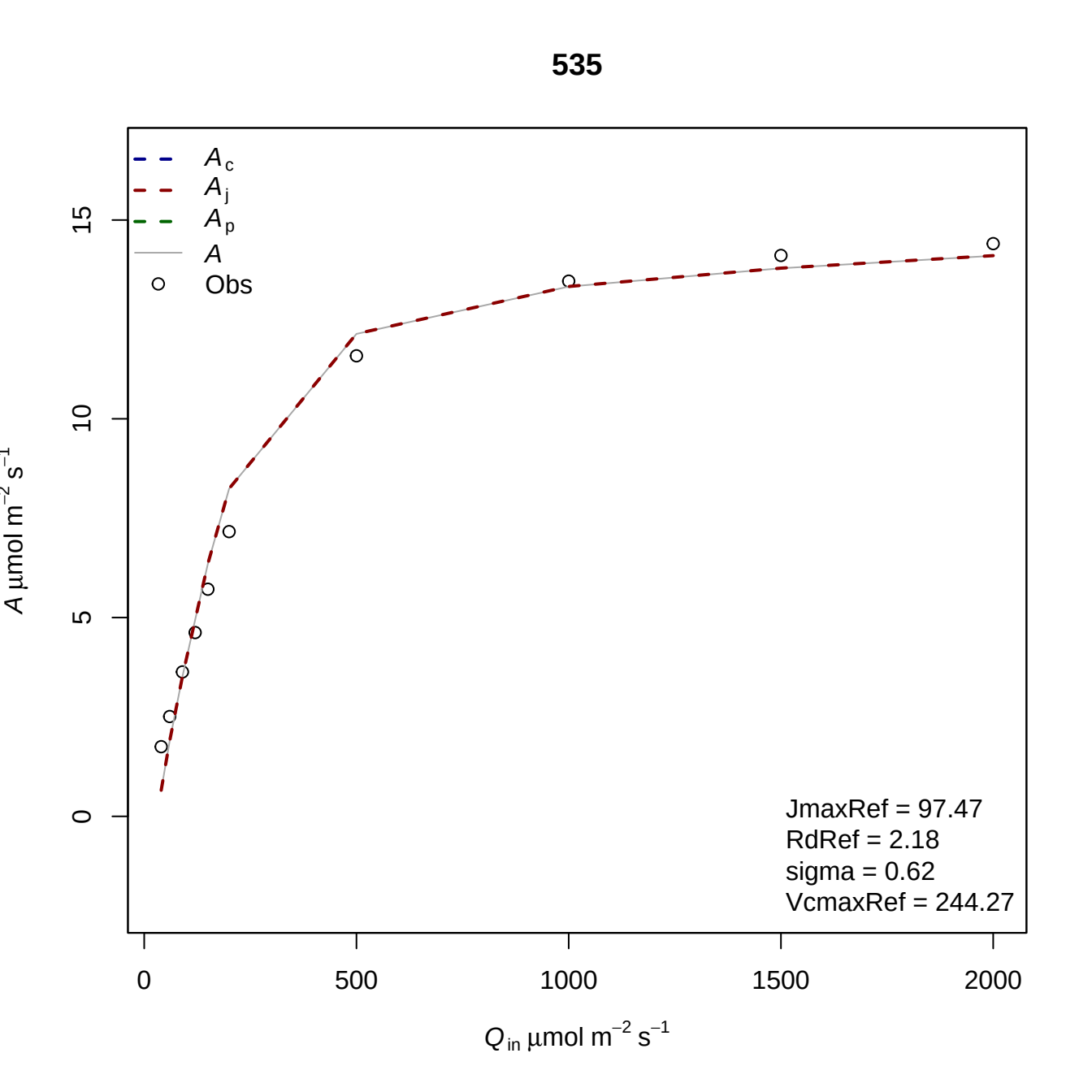

526
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
5
10
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 110.08
RdRef = 2.15
sigma = 0.22
VcmaxRef = 900.35

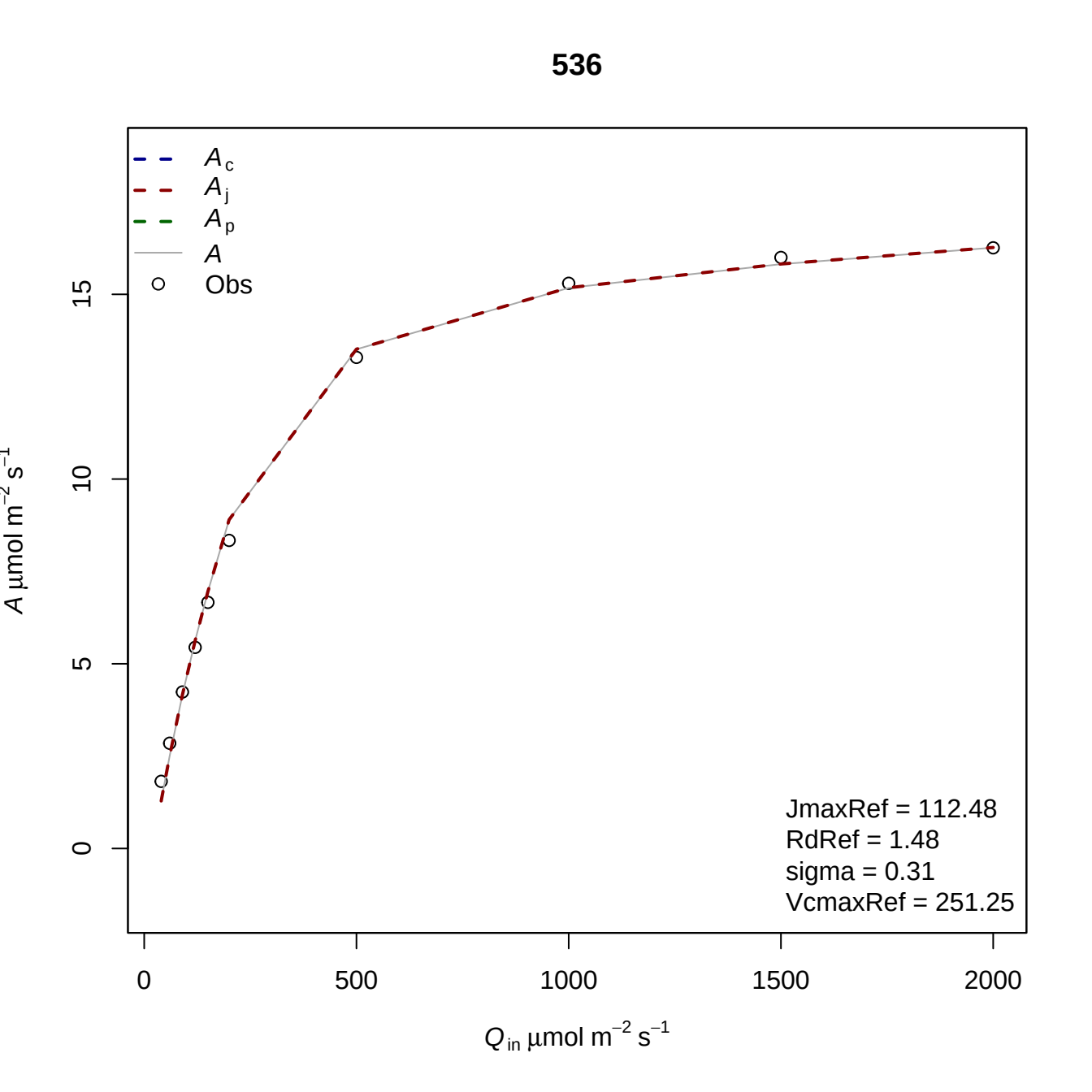

529
A_c
A_j
A_p
A
Obs
JmaxRef = 82.39
RdRef = 1.77
sigma = 0.44
VcmaxRef = 187.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

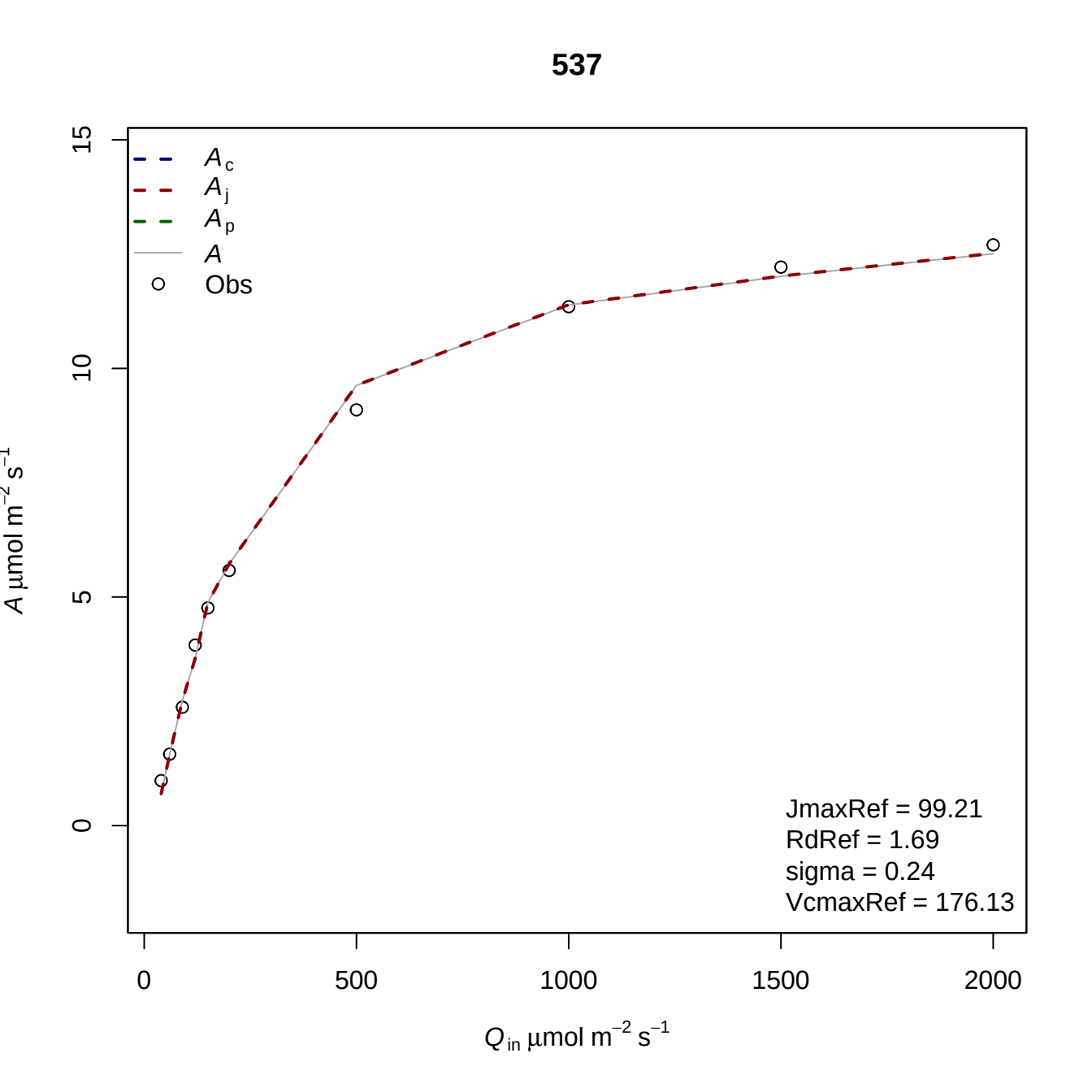

530
A_c
A_j
A_p
A
Obs
JmaxRef = 106.65
RdRef = 1.49
sigma = 0.48
VcmaxRef = 386.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

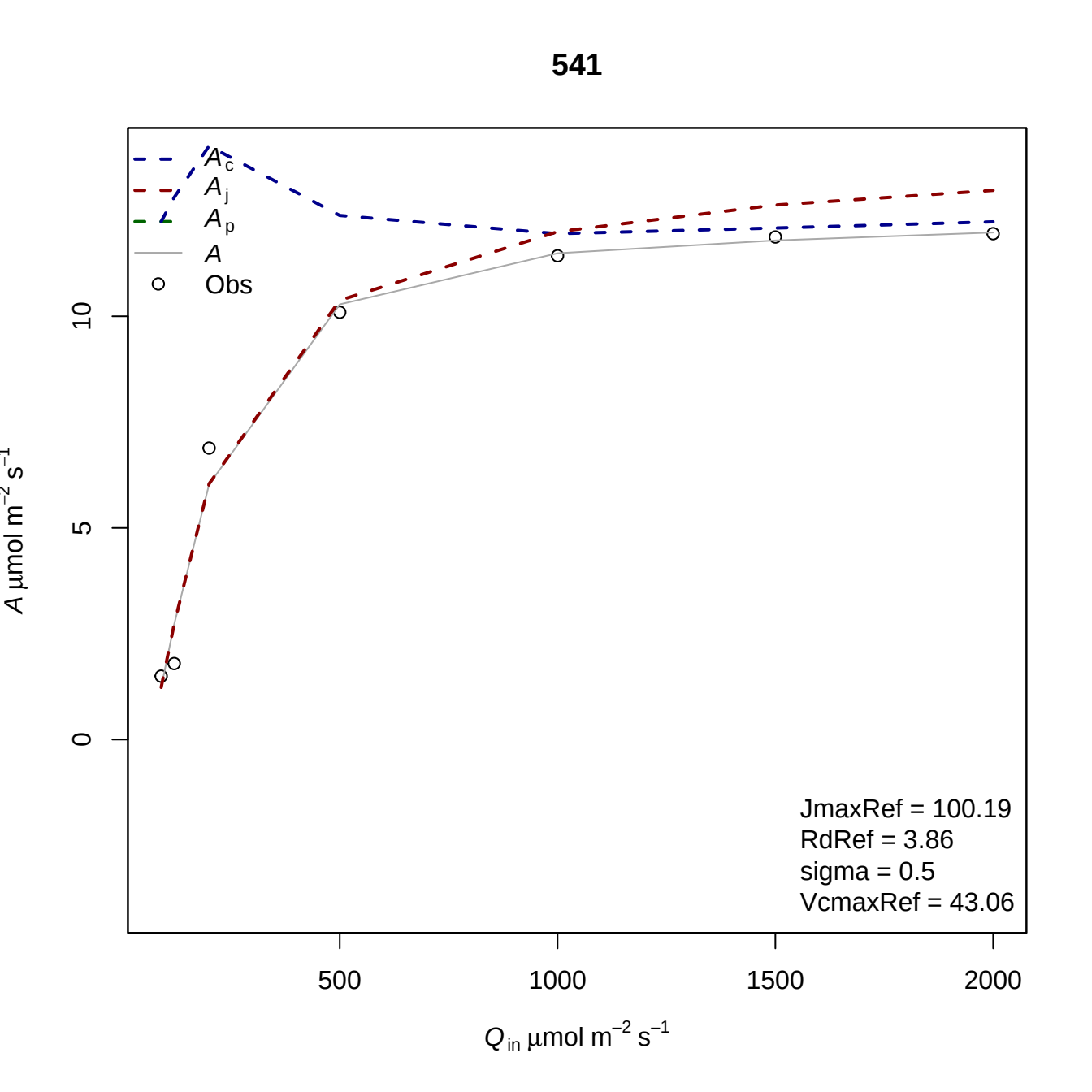

531
A_c
A_j
A_p
A
Obs
JmaxRef = 109.3
RdRef = 1.47
sigma = 0.33
VcmaxRef = 181.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

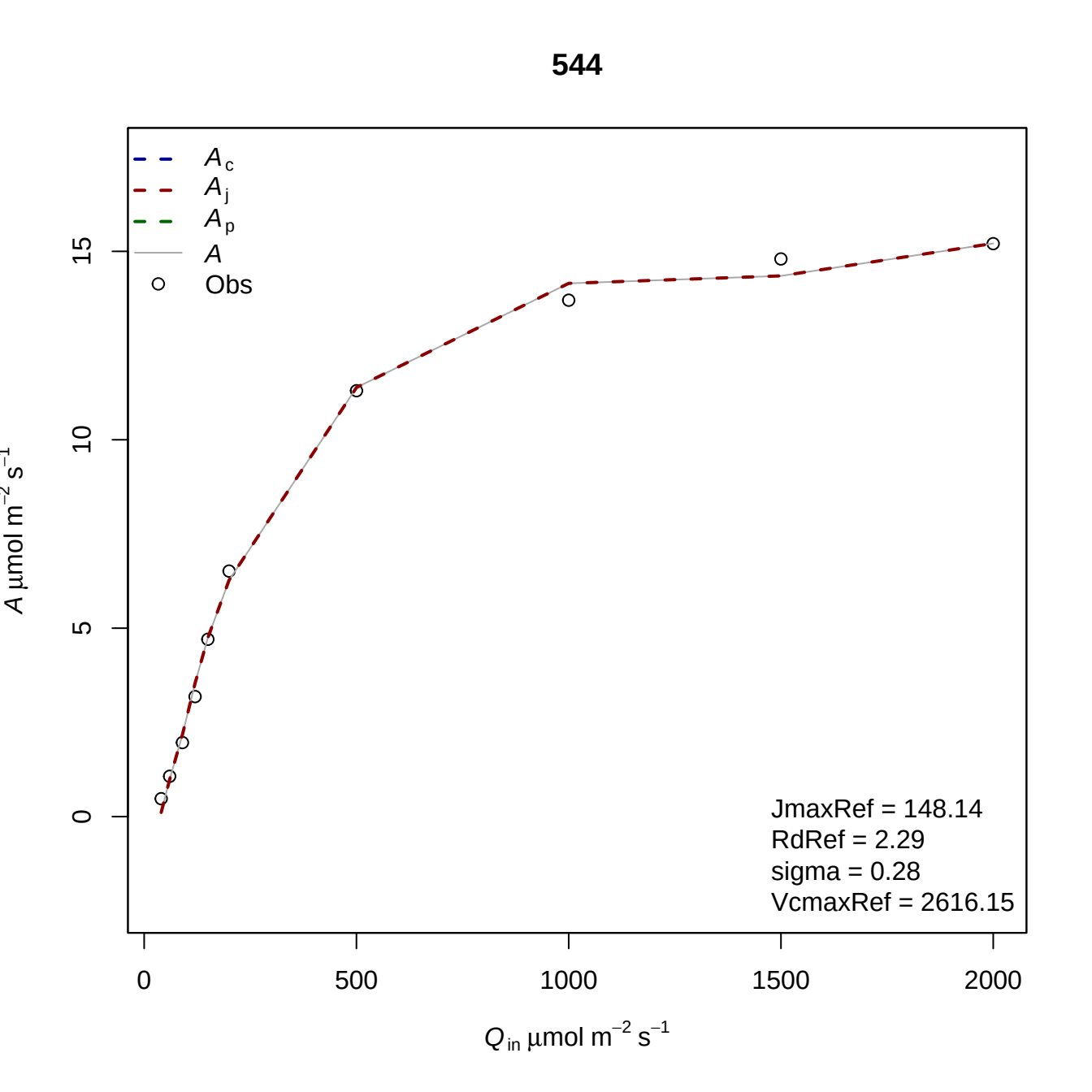

532
A_c
A_j
A_p
A
Obs
JmaxRef = 107.08
RdRef = 2.49
sigma = 0.44
VcmaxRef = 360.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

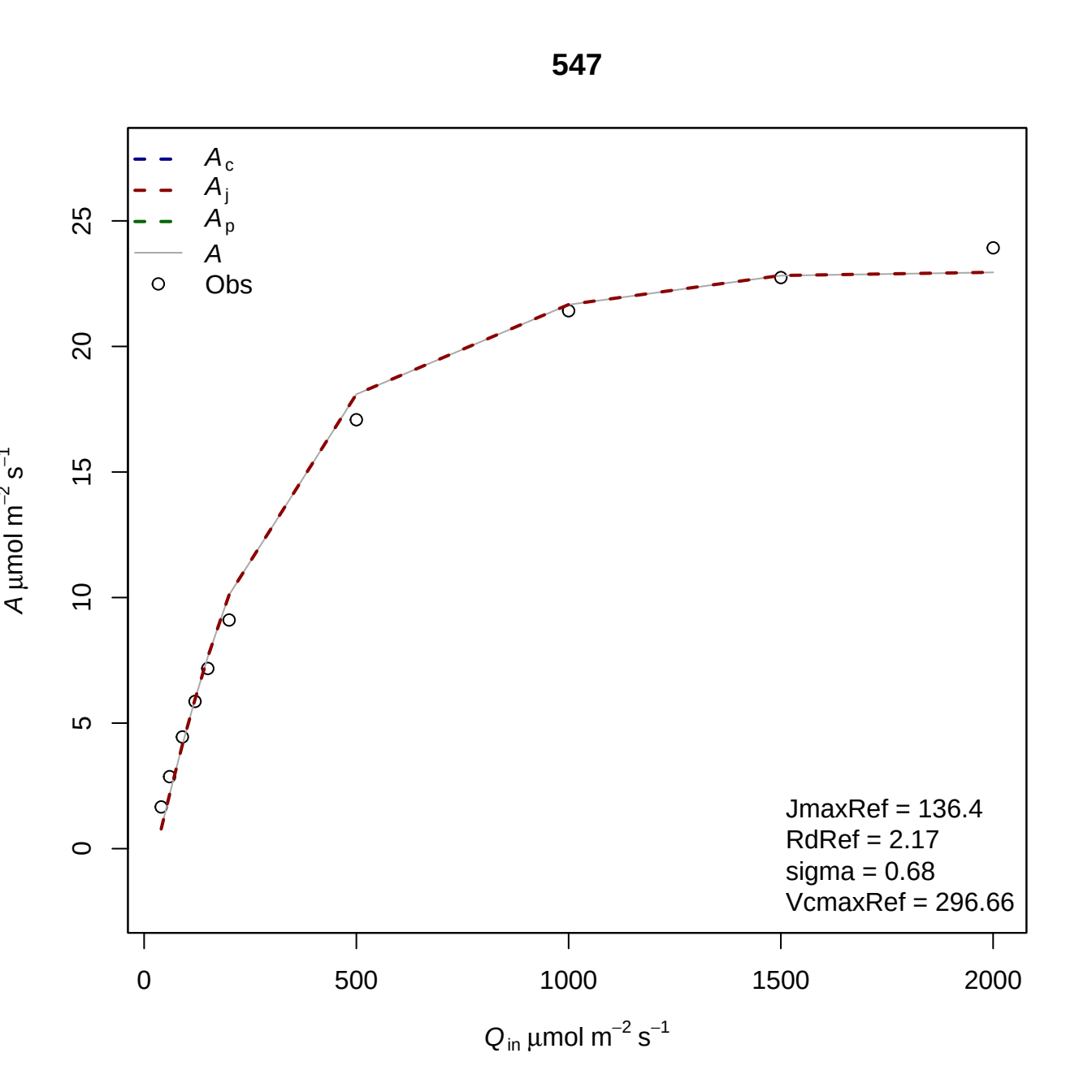

535
A_c
A_j
A_p
A
Obs
JmaxRef = 97.47
RdRef = 2.18
sigma = 0.62
VcmaxRef = 244.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

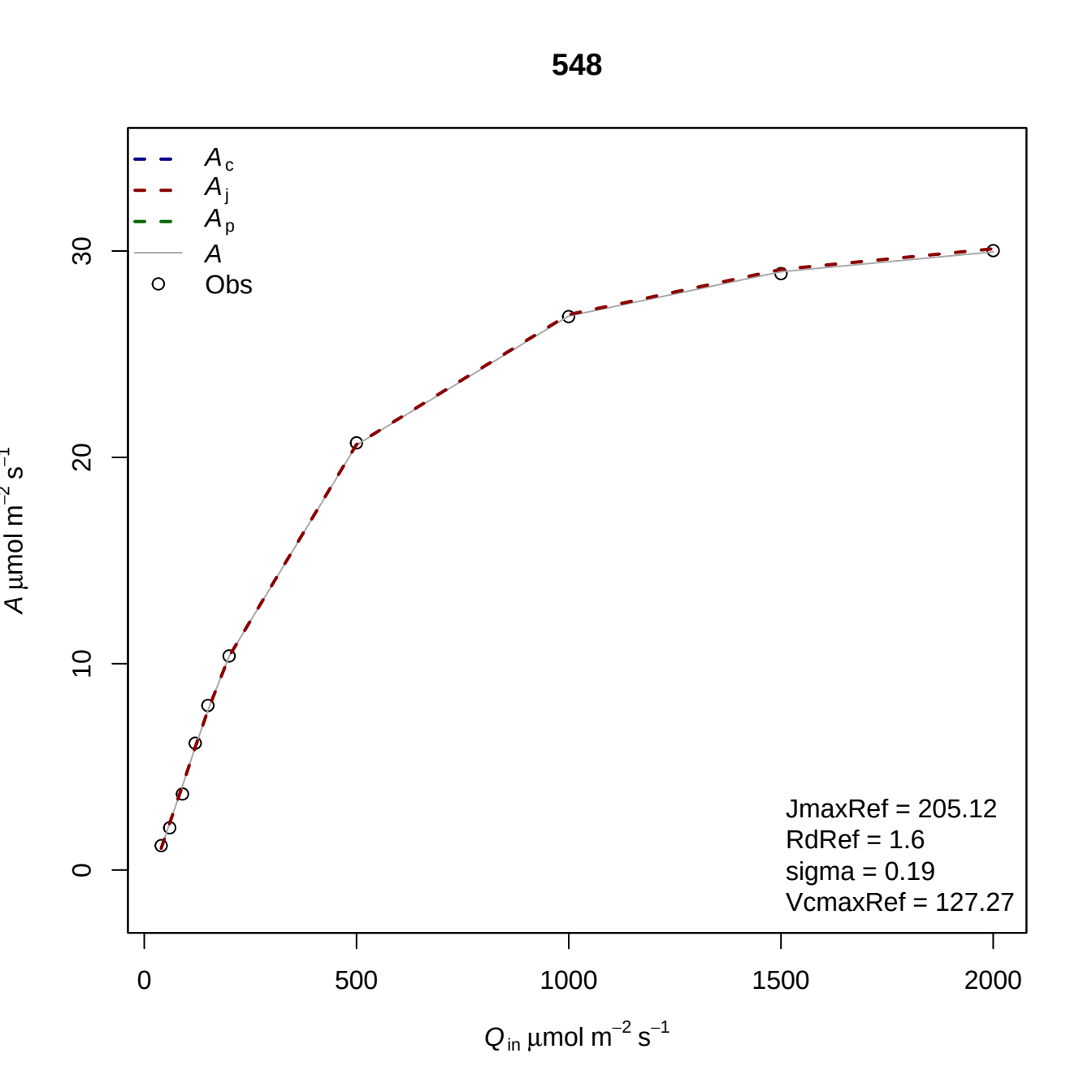

536
A_c
A_j
A_p
A
Obs
JmaxRef = 112.48
RdRef = 1.48
sigma = 0.31
VcmaxRef = 251.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

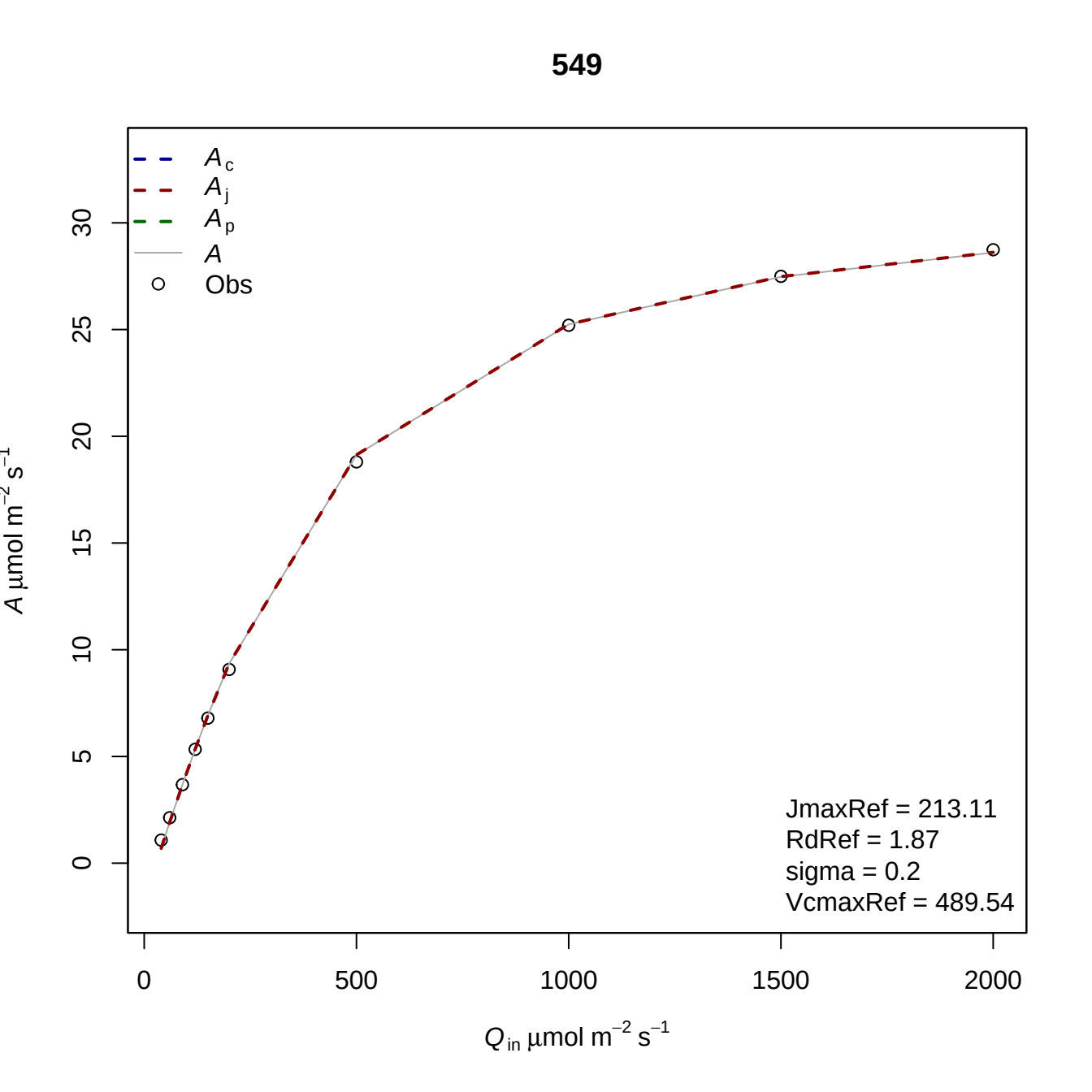

537
A_c
A_j
A_p
A
Obs
JmaxRef = 99.21
RdRef = 1.69
sigma = 0.24
VcmaxRef = 176.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

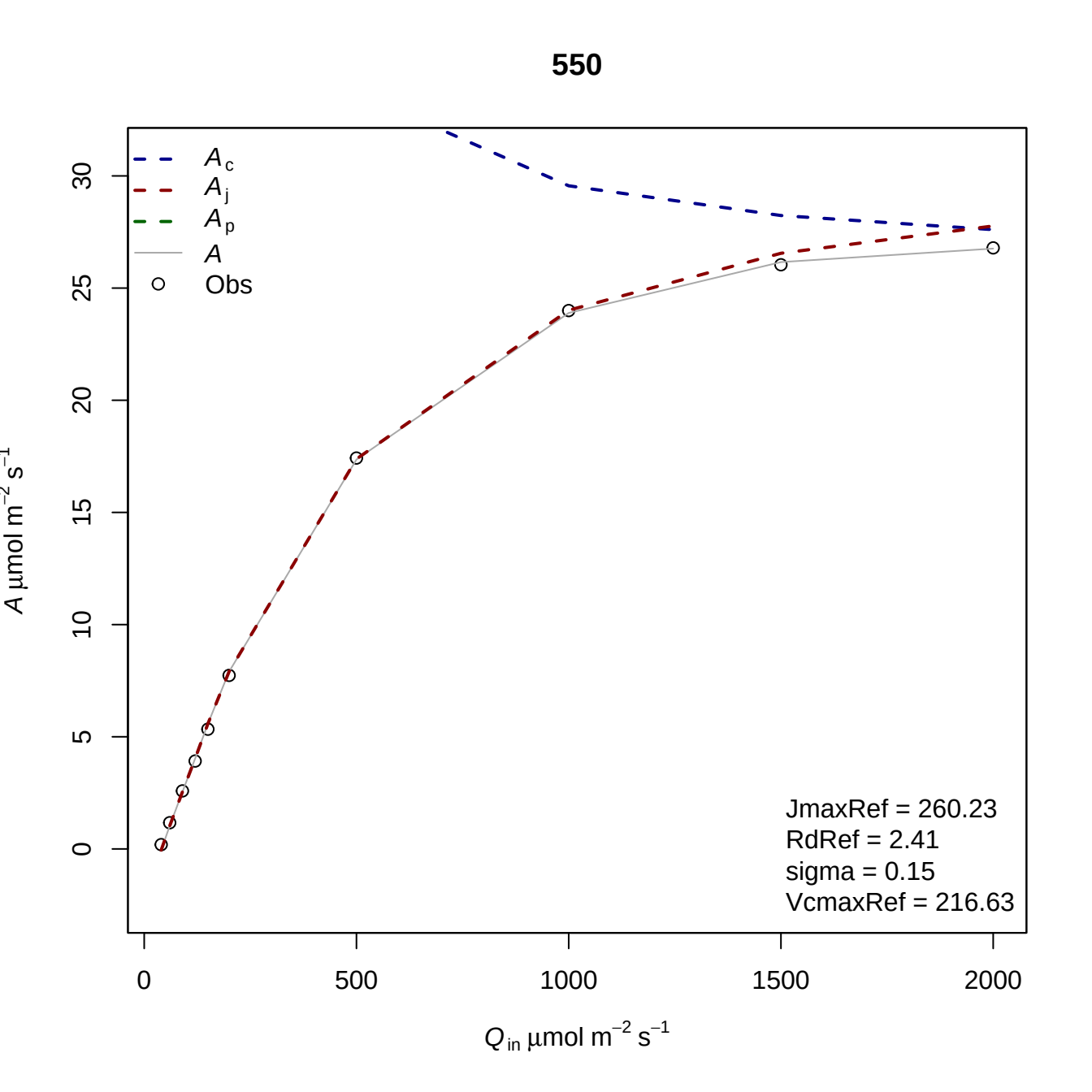

541
A_c
A_j
A_p
A
Obs
JmaxRef = 100.19
RdRef = 3.86
sigma = 0.5
VcmaxRef = 43.06
0
5
10
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

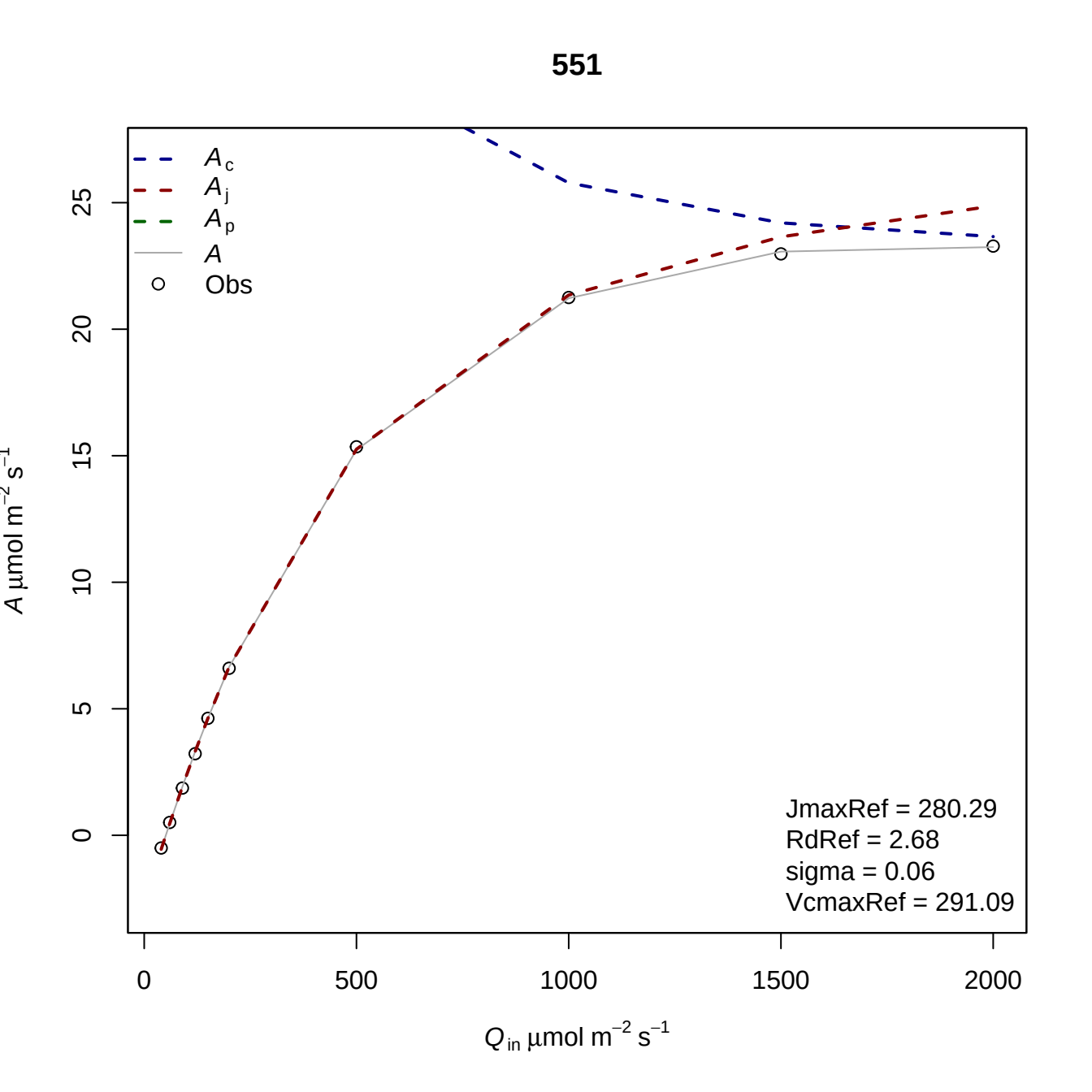

544
A_c
A_j
A_p
A
Obs
JmaxRef = 148.14
RdRef = 2.29
sigma = 0.28
VcmaxRef = 2616.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

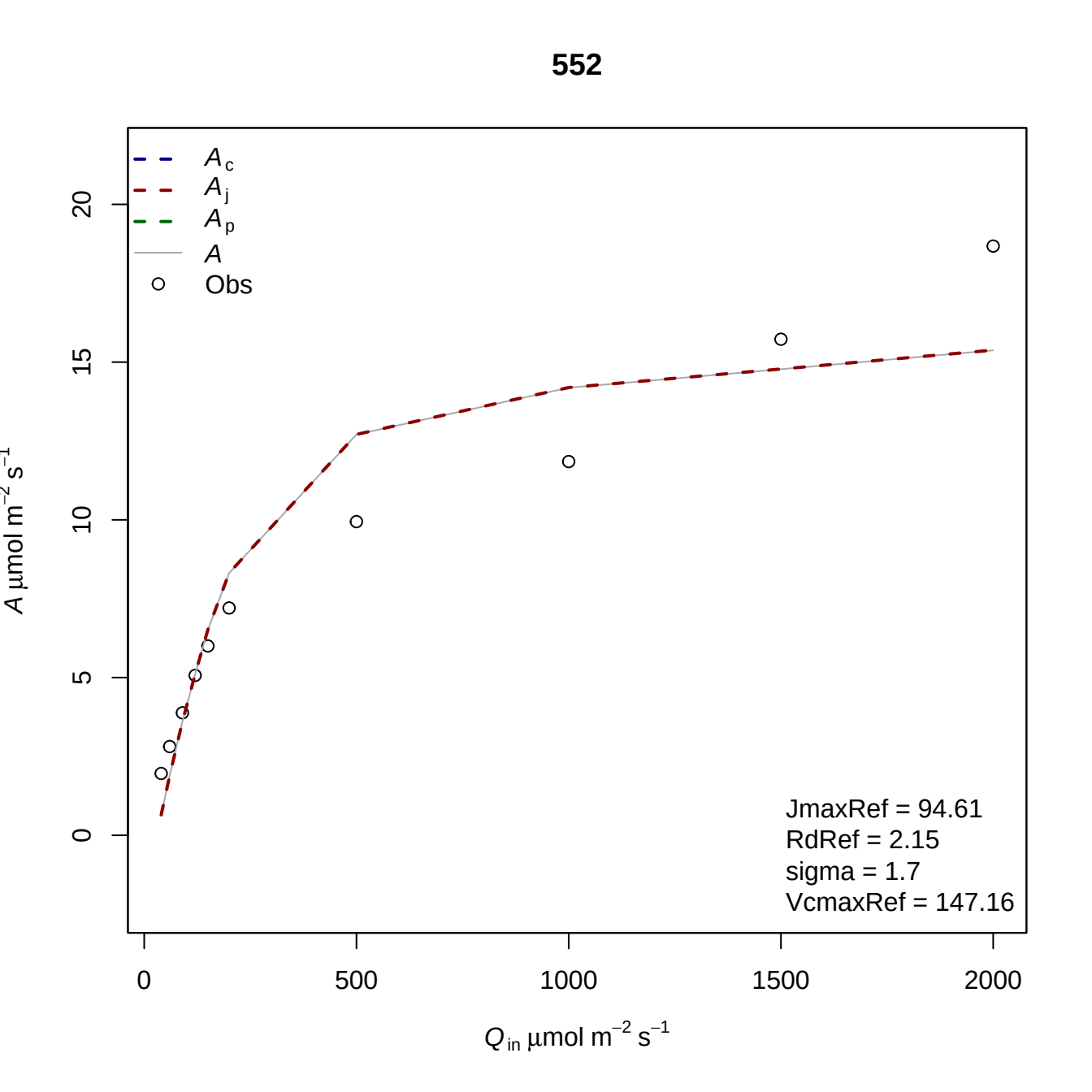

547
A_c
A_j
A_p
A
Obs
JmaxRef = 136.4
RdRef = 2.17
sigma = 0.68
VcmaxRef = 296.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

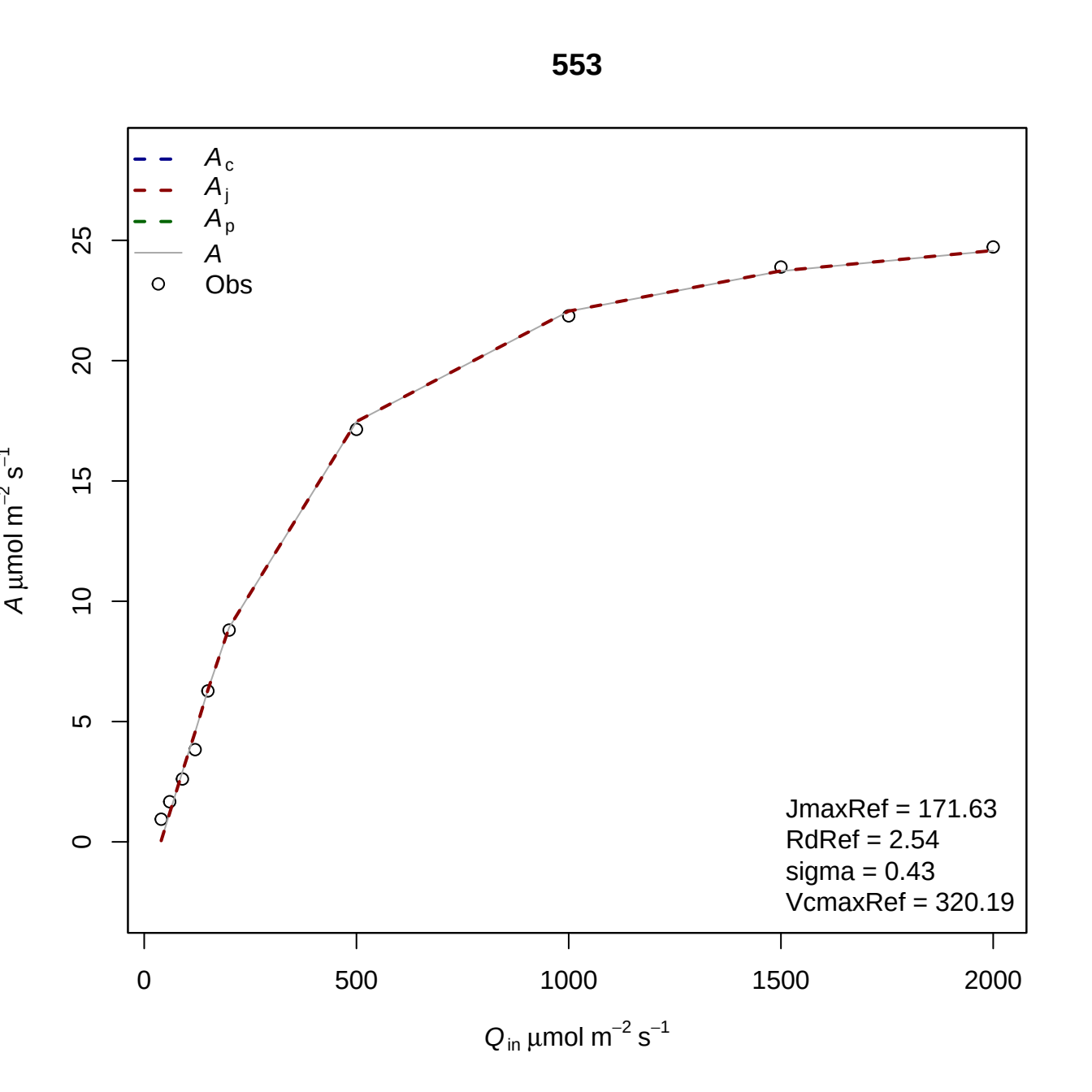

548
A_c
A_j
A_p
A
Obs
JmaxRef = 205.12
RdRef = 1.6
sigma = 0.19
VcmaxRef = 127.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

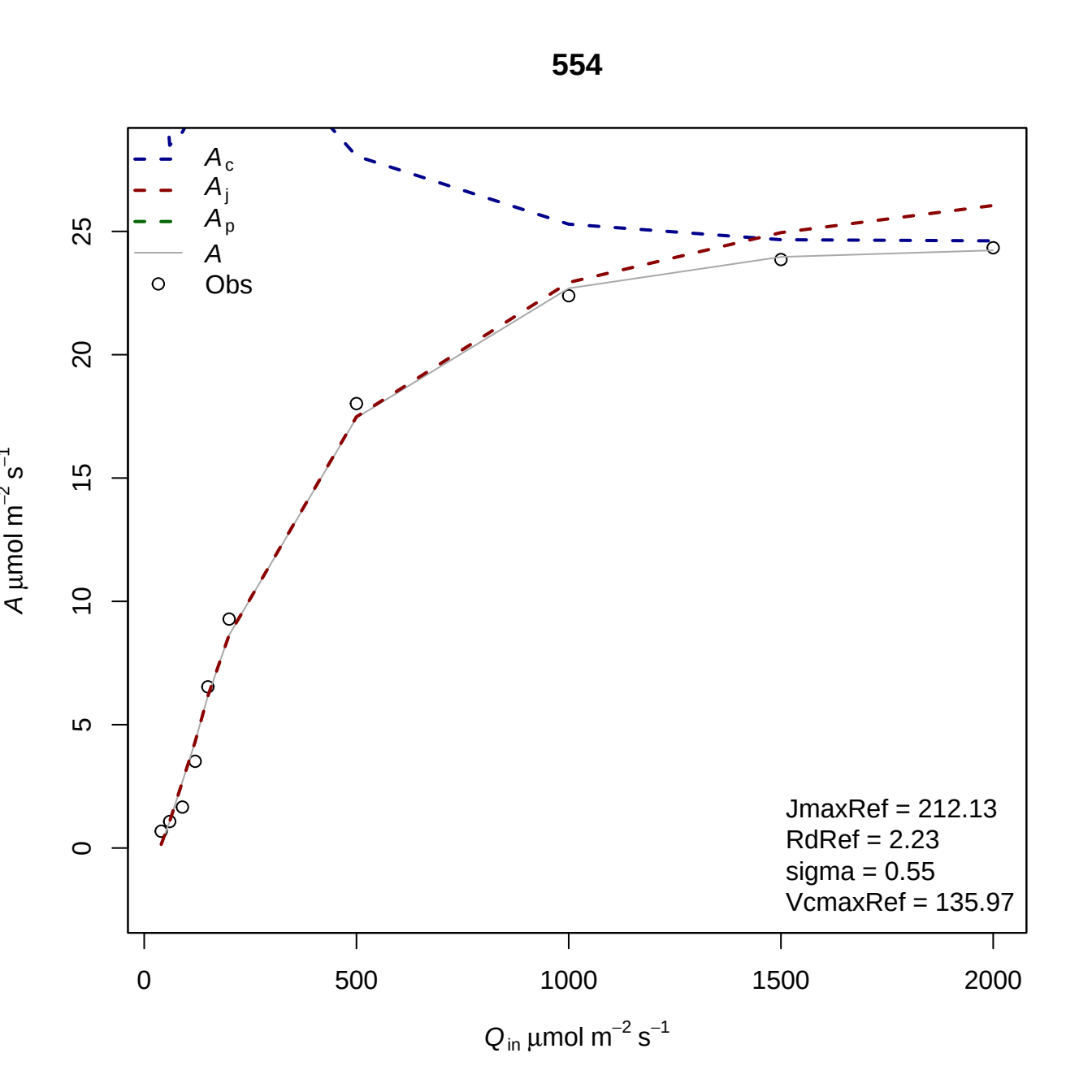

549
A_c
A_j
A_p
A
Obs
JmaxRef = 213.11
RdRef = 1.87
sigma = 0.2
VcmaxRef = 489.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

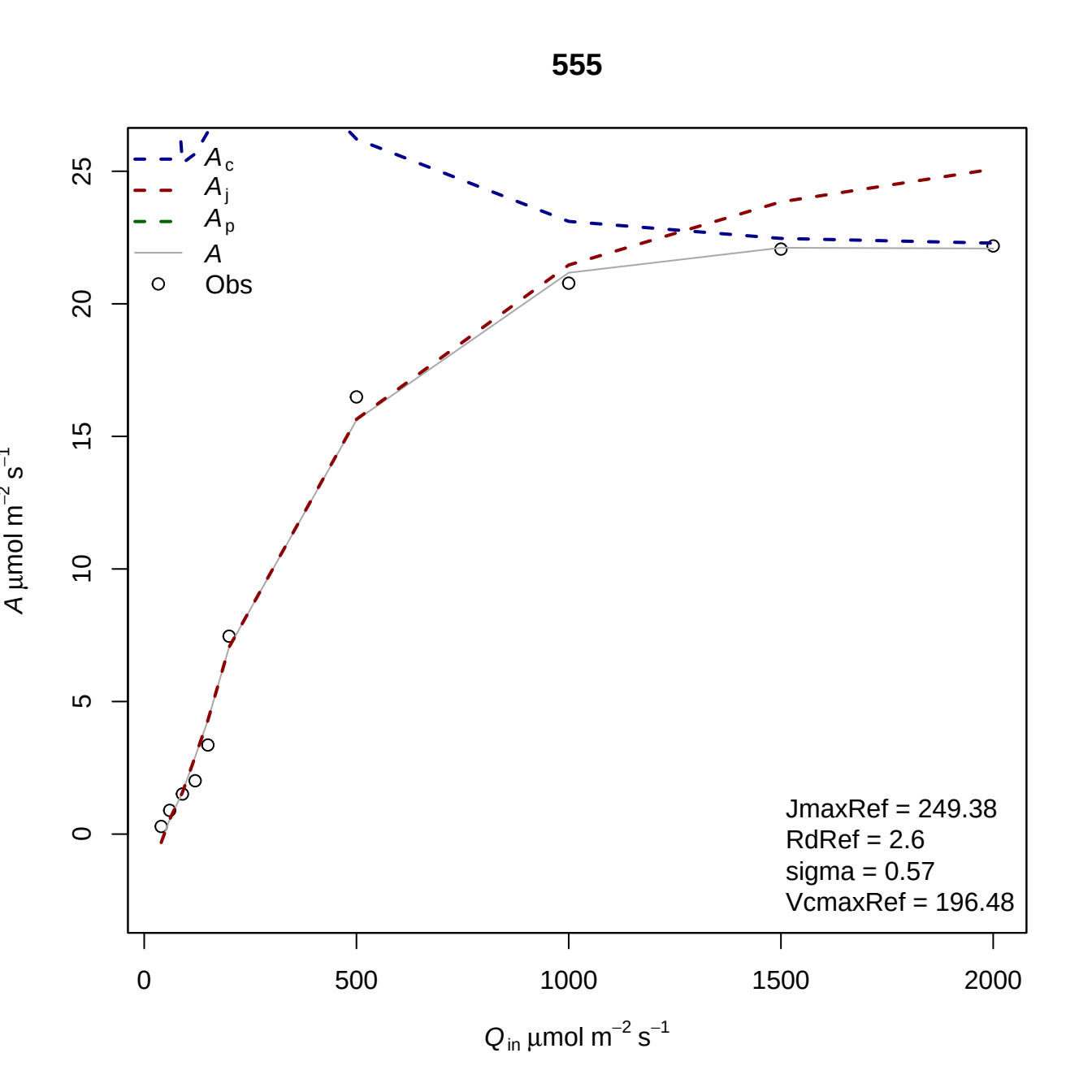

550
A_c
A_j
A_p
A
Obs
JmaxRef = 260.23
RdRef = 2.41
sigma = 0.15
VcmaxRef = 216.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

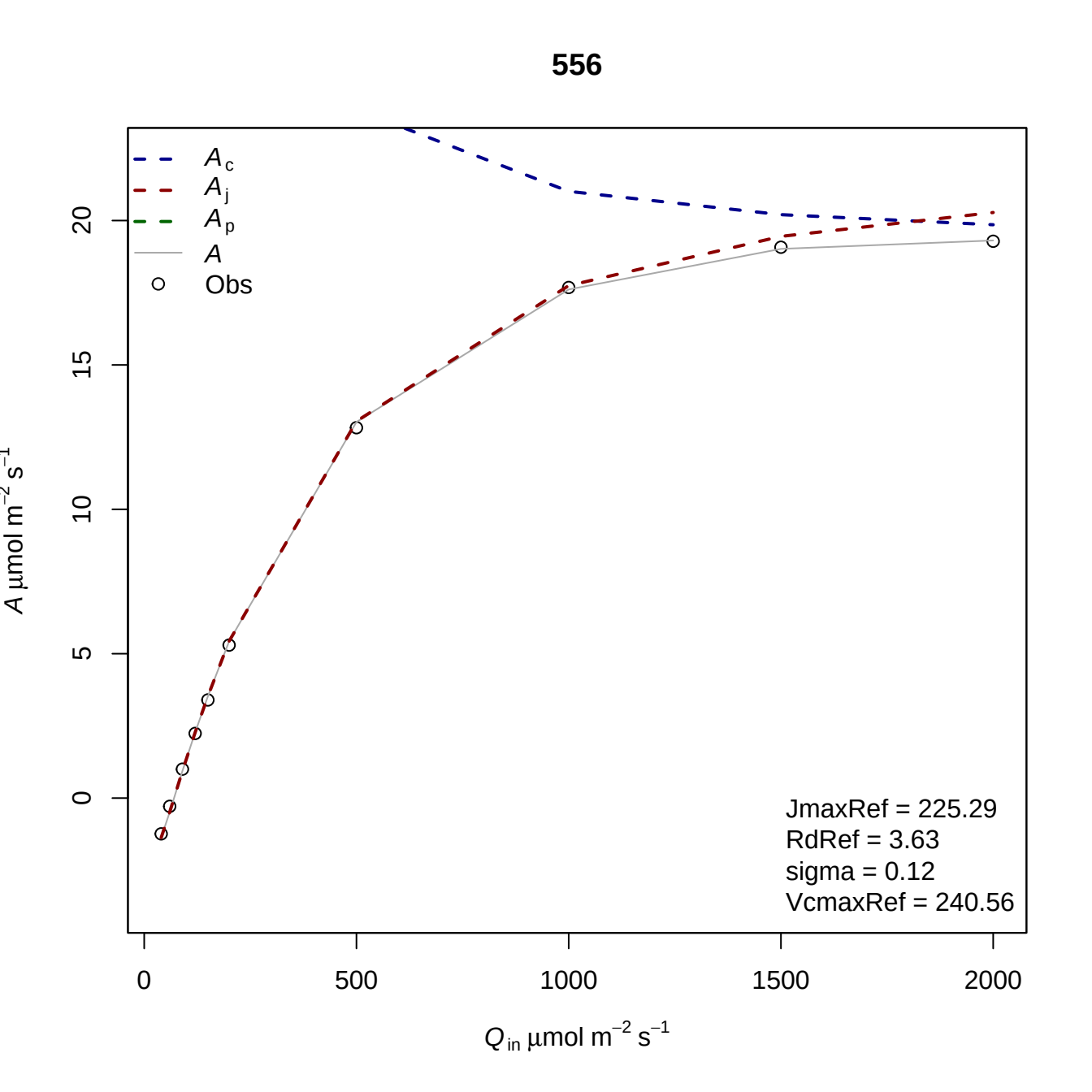

551
A_c
A_j
A_p
A
Obs
JmaxRef = 280.29
RdRef = 2.68
sigma = 0.06
VcmaxRef = 291.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

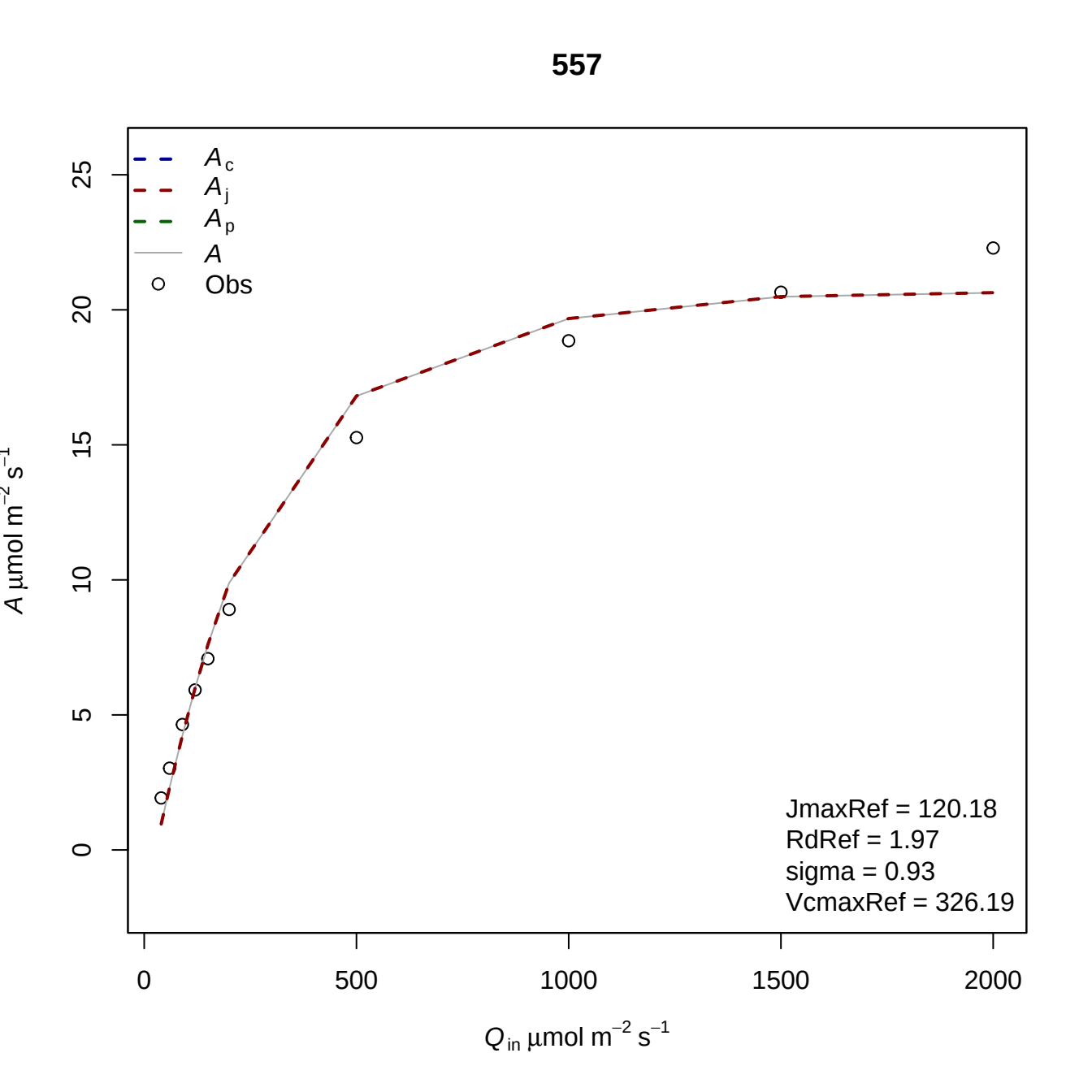

552
A_c
A_j
A_p
A
Obs
JmaxRef = 94.61
RdRef = 2.15
sigma = 1.7
VcmaxRef = 147.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

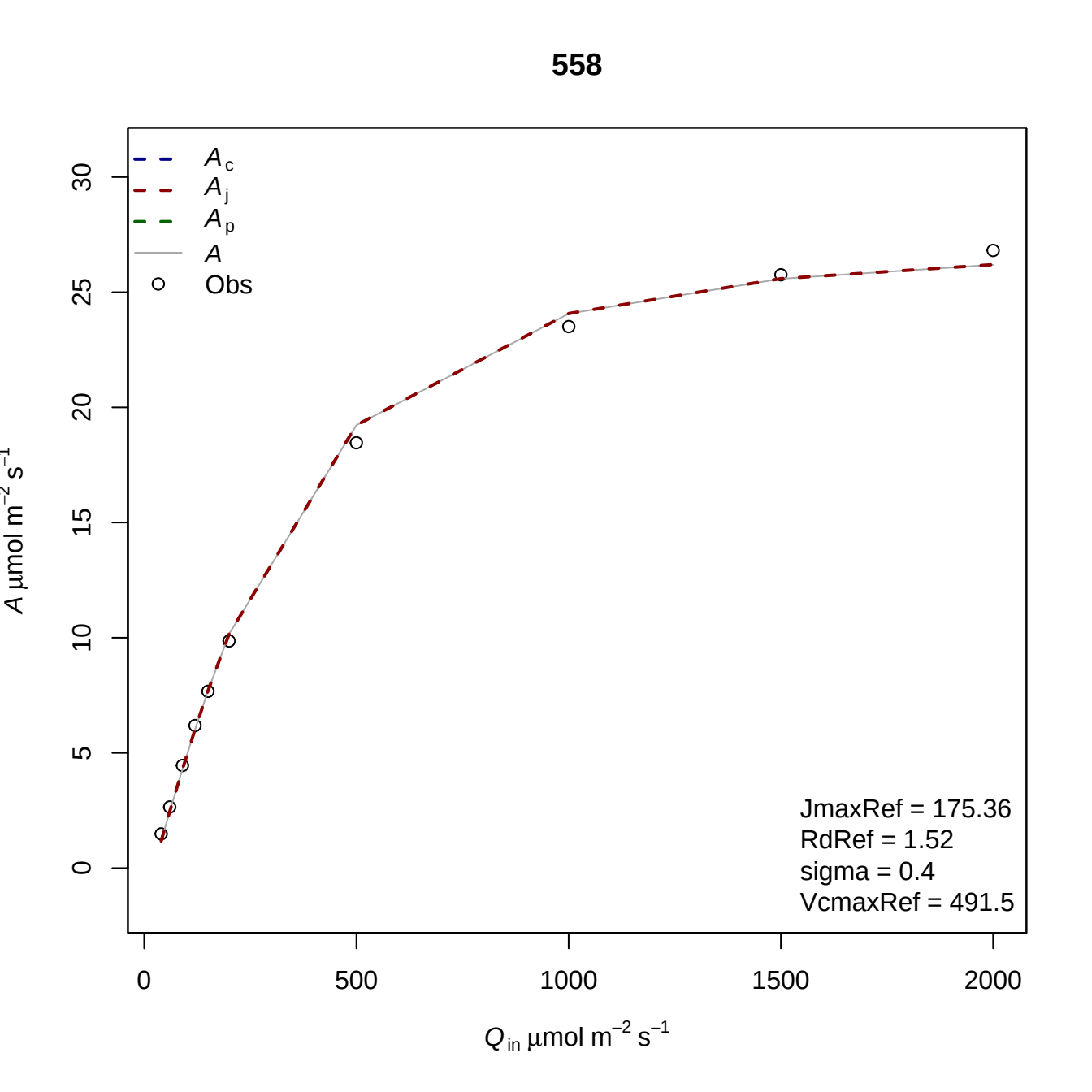

553
A_c
A_j
A_p
A
Obs
JmaxRef = 171.63
RdRef = 2.54
sigma = 0.43
VcmaxRef = 320.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

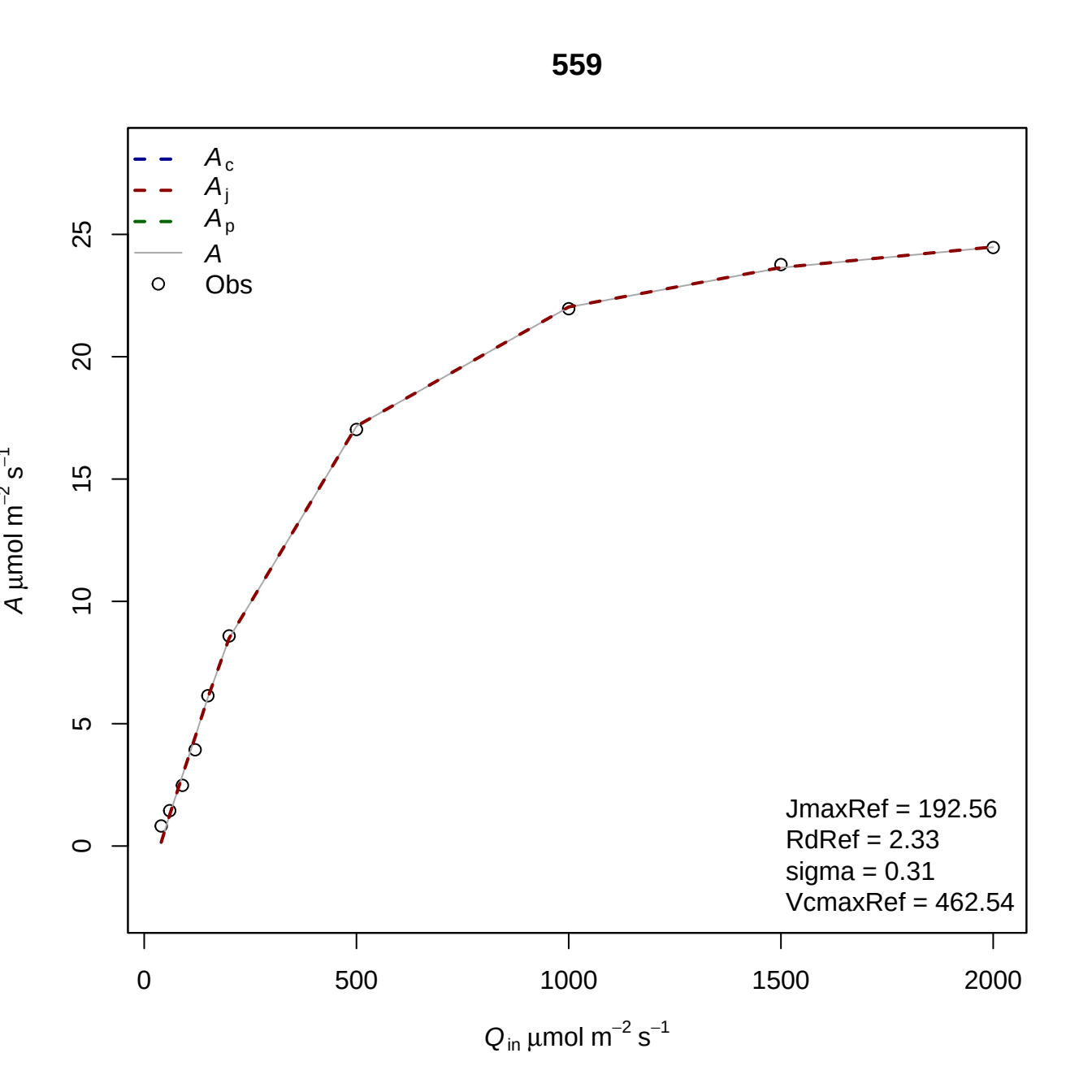

554
A_c
A_j
A_p
A
Obs
JmaxRef = 212.13
RdRef = 2.23
sigma = 0.55
VcmaxRef = 135.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

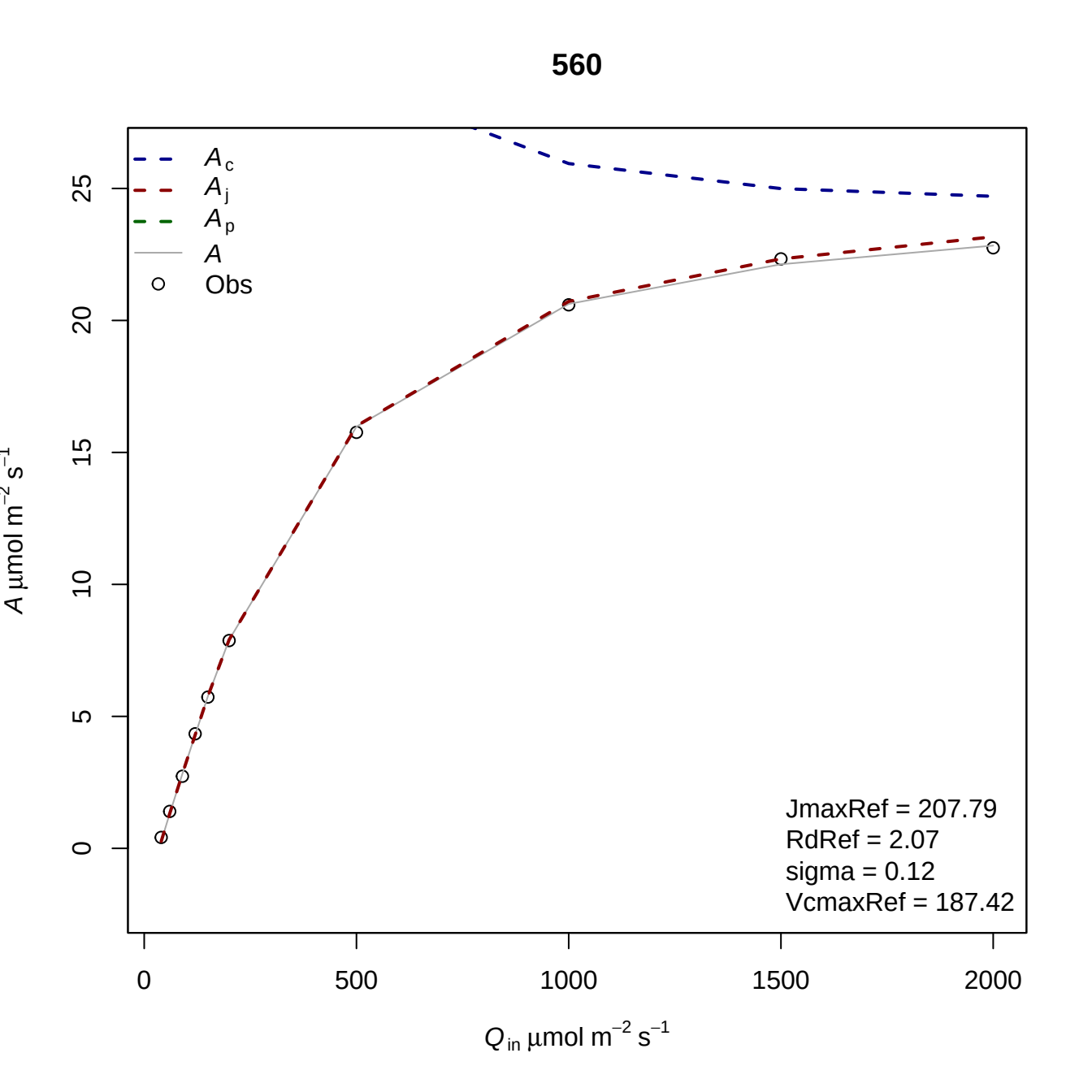

555
A_c
A_j
A_p
A
Obs
JmaxRef = 249.38
RdRef = 2.6
sigma = 0.57
VcmaxRef = 196.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

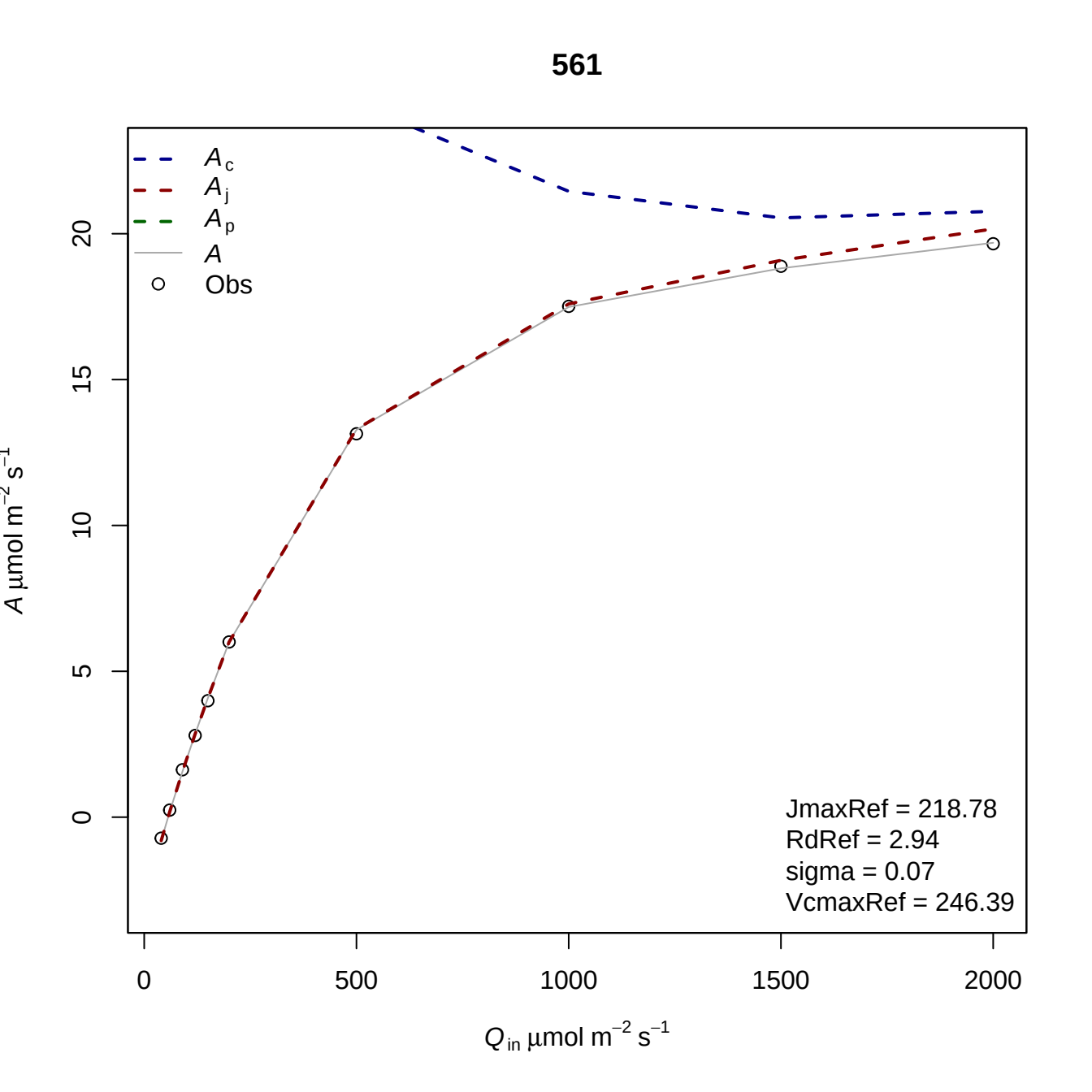

556
A_c
A_j
A_p
A
Obs
JmaxRef = 225.29
RdRef = 3.63
sigma = 0.12
VcmaxRef = 240.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

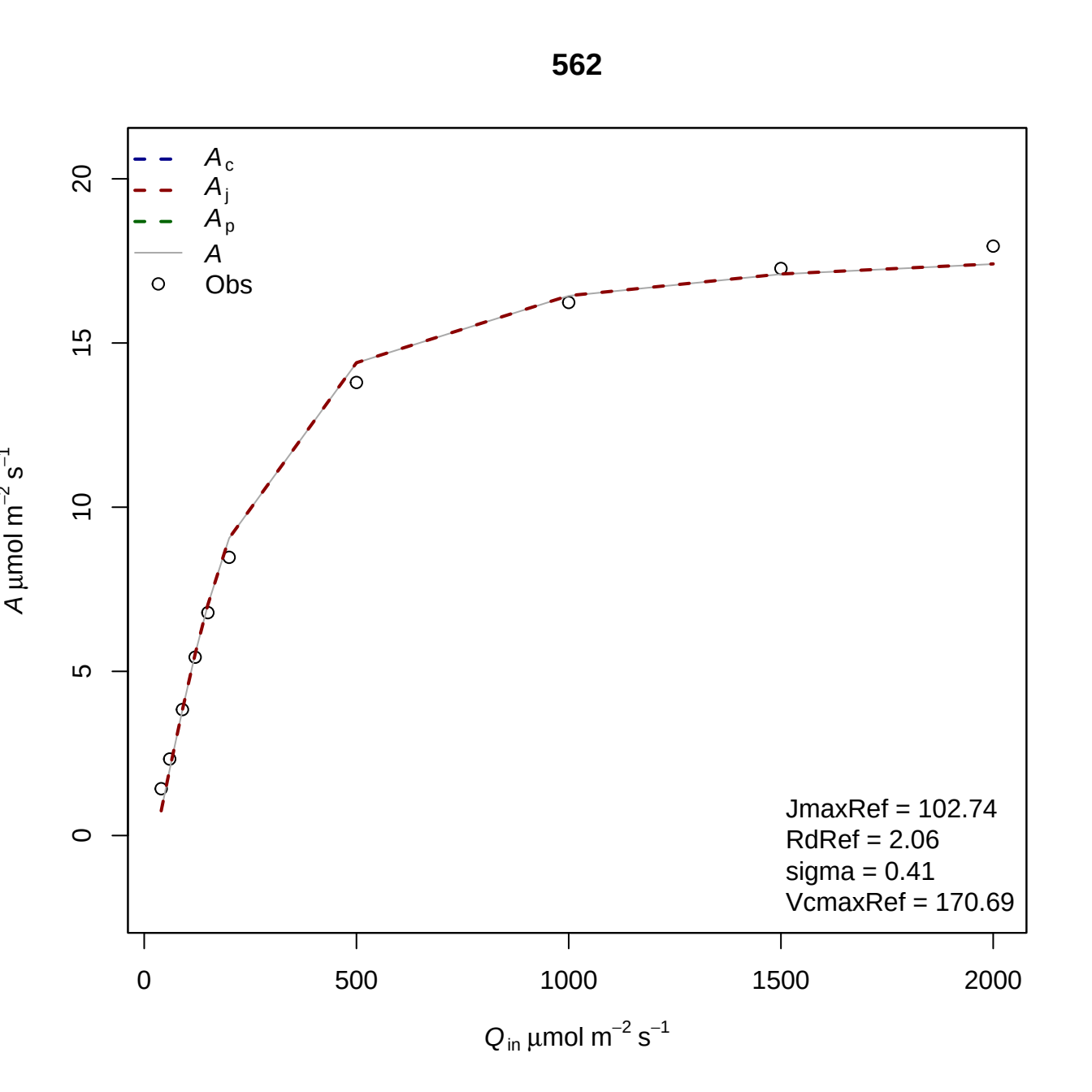

557
A_c
A_j
A_p
A
Obs
JmaxRef = 120.18
RdRef = 1.97
sigma = 0.93
VcmaxRef = 326.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

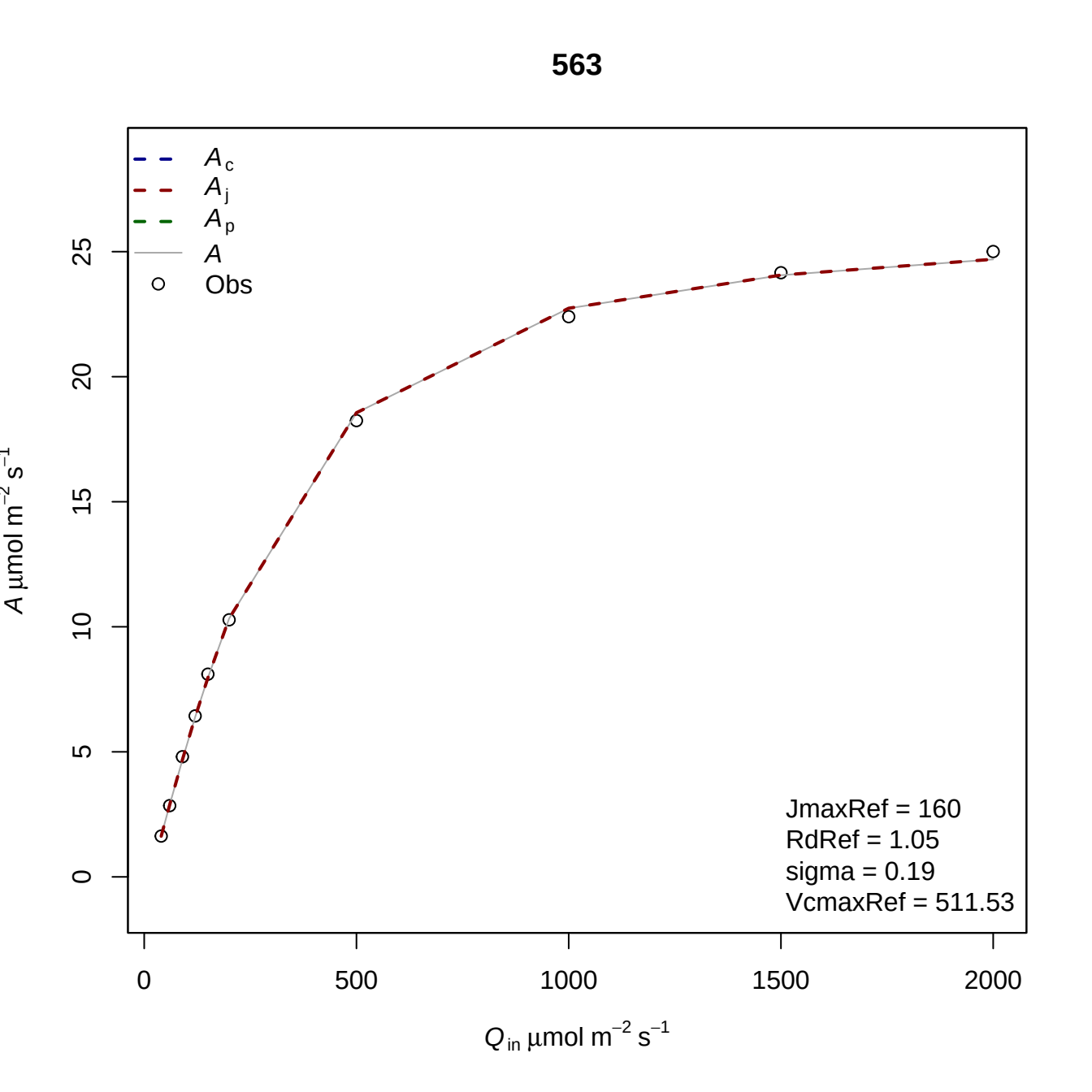

558
A_c
A_j
A_p
A
Obs
JmaxRef = 175.36
RdRef = 1.52
sigma = 0.4
VcmaxRef = 491.5
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

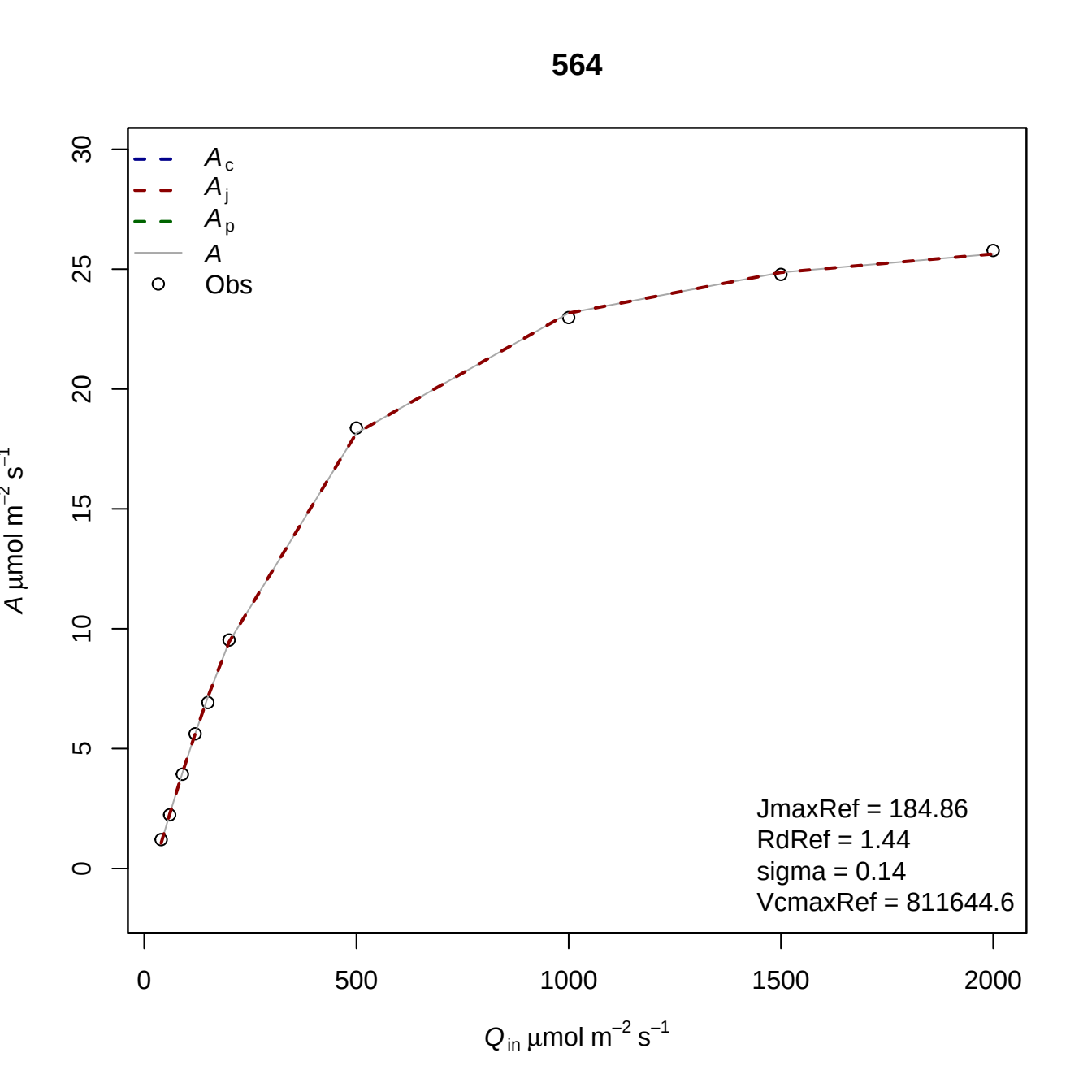

559
A_c
A_j
A_p
A
Obs
JmaxRef = 192.56
RdRef = 2.33
sigma = 0.31
VcmaxRef = 462.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

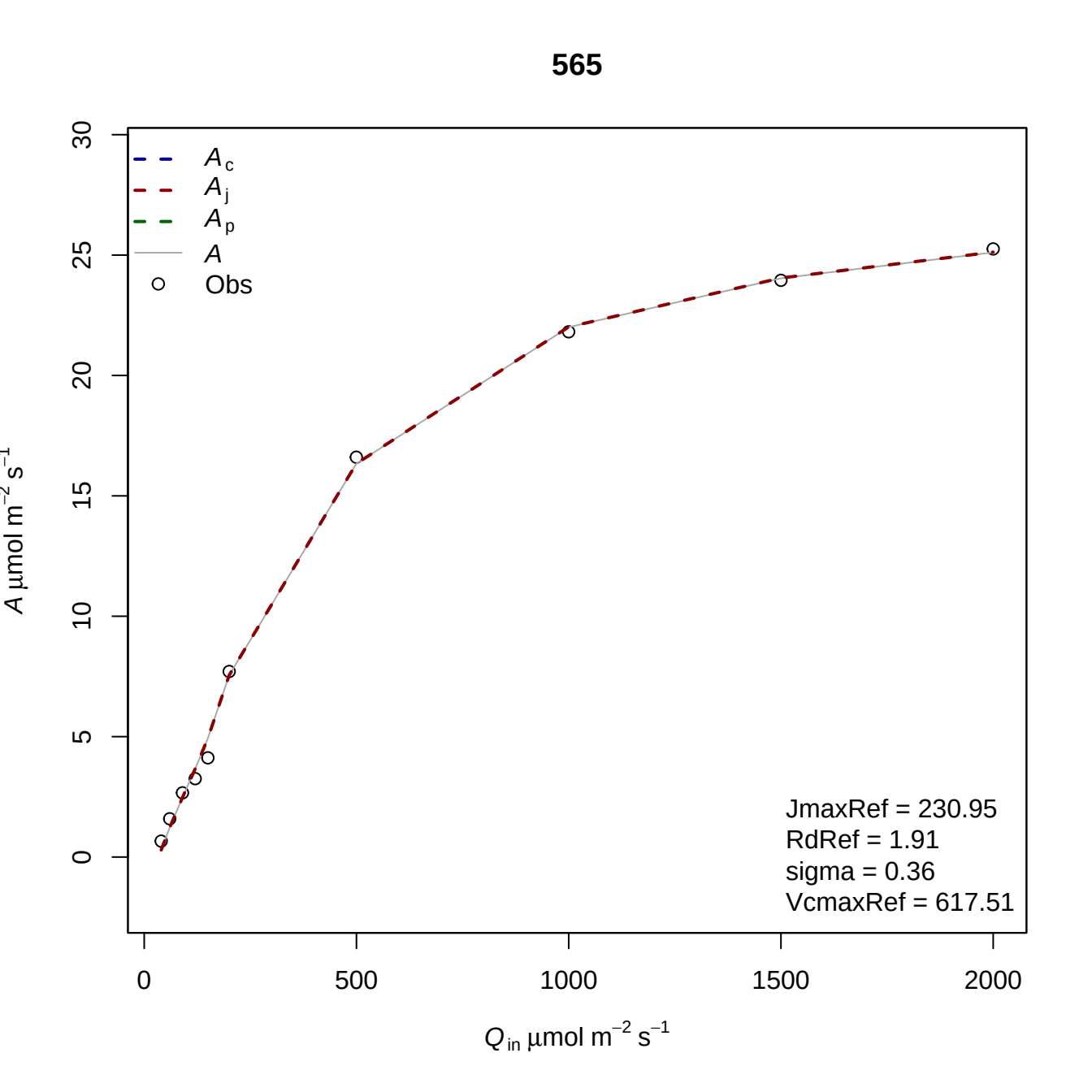

560
A_c
A_j
A_p
A
Obs
JmaxRef = 207.79
RdRef = 2.07
sigma = 0.12
VcmaxRef = 187.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

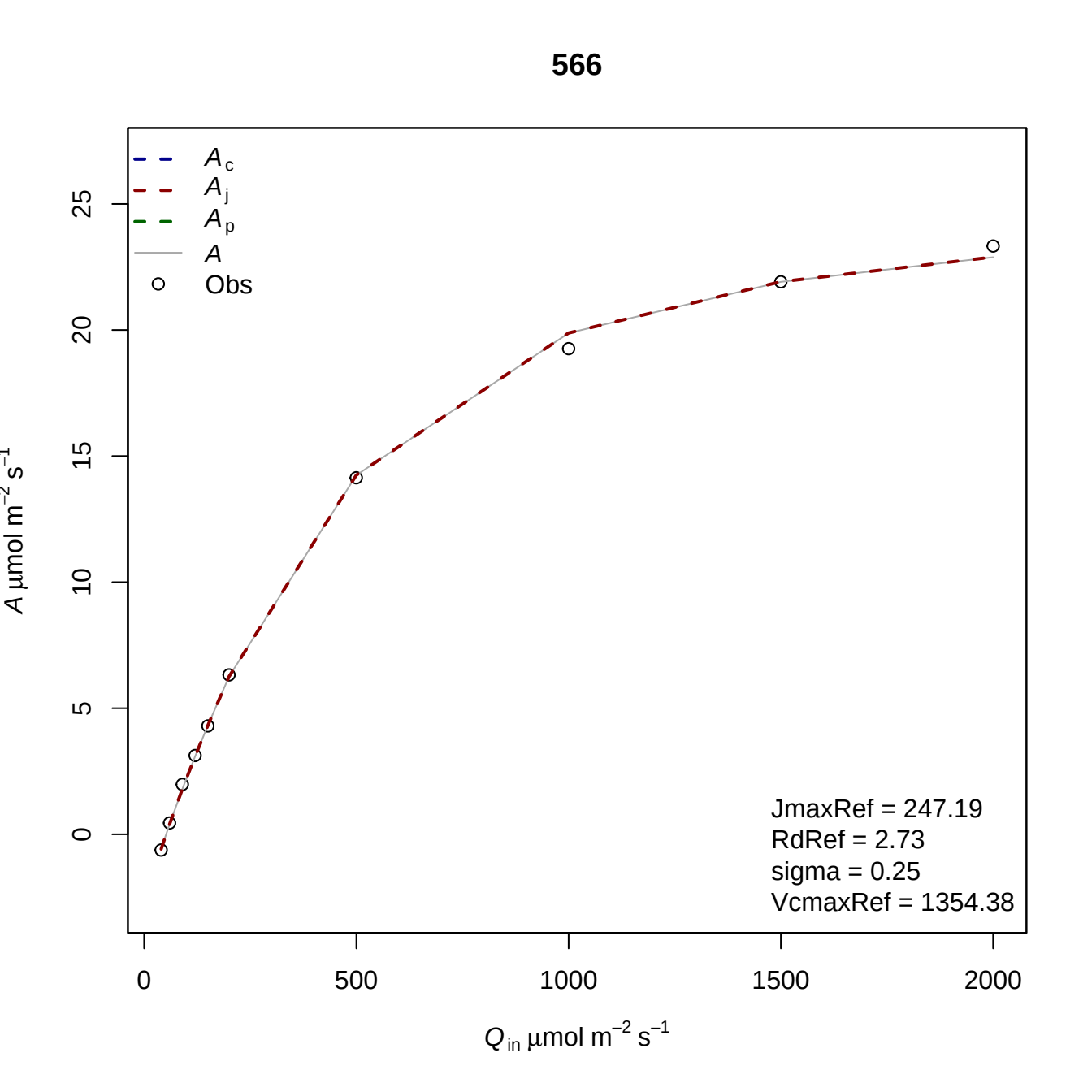

561
A_c
A_j
A_p
A
Obs
JmaxRef = 218.78
RdRef = 2.94
sigma = 0.07
VcmaxRef = 246.39
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

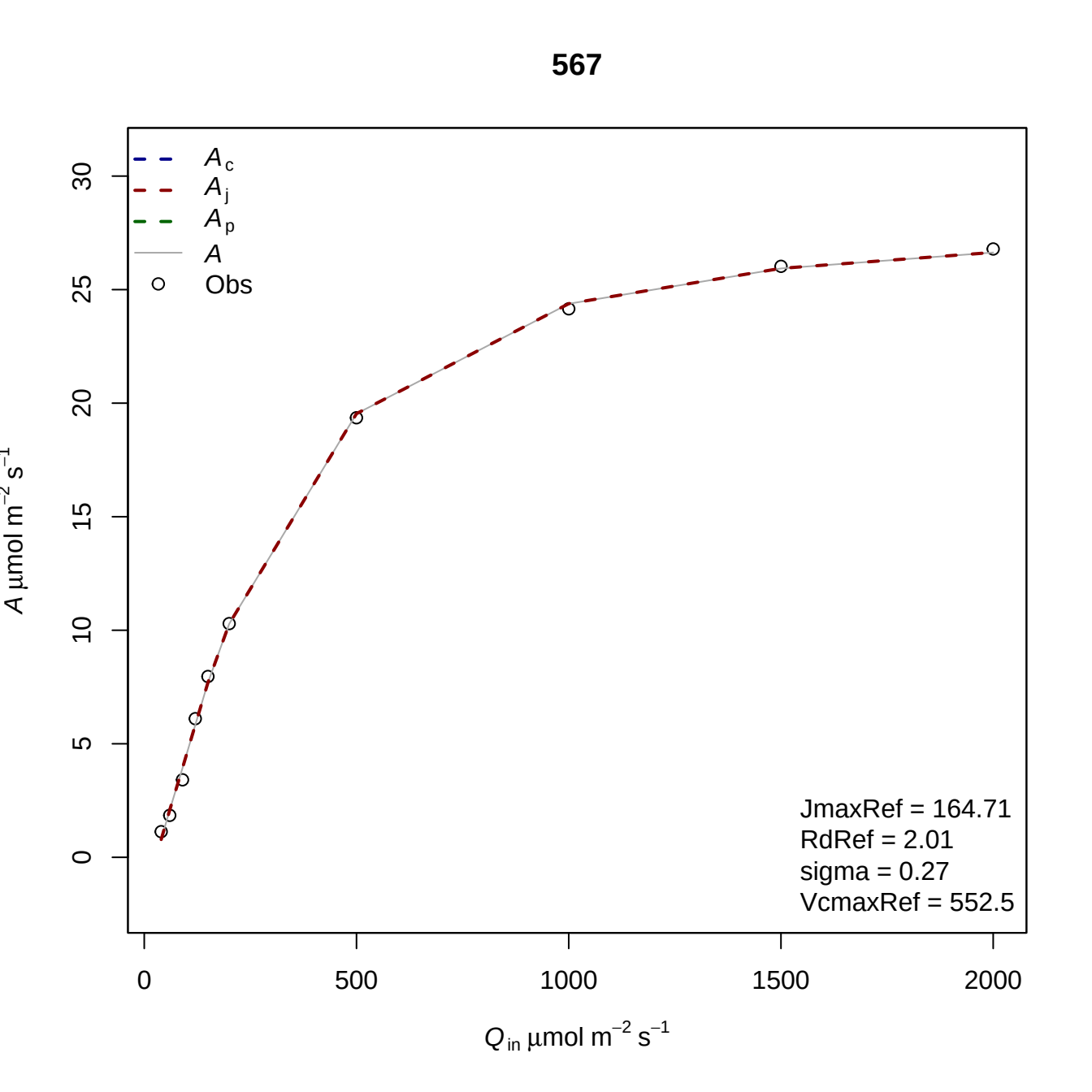

562
A_c
A_j
A_p
A
Obs
JmaxRef = 102.74
RdRef = 2.06
sigma = 0.41
VcmaxRef = 170.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

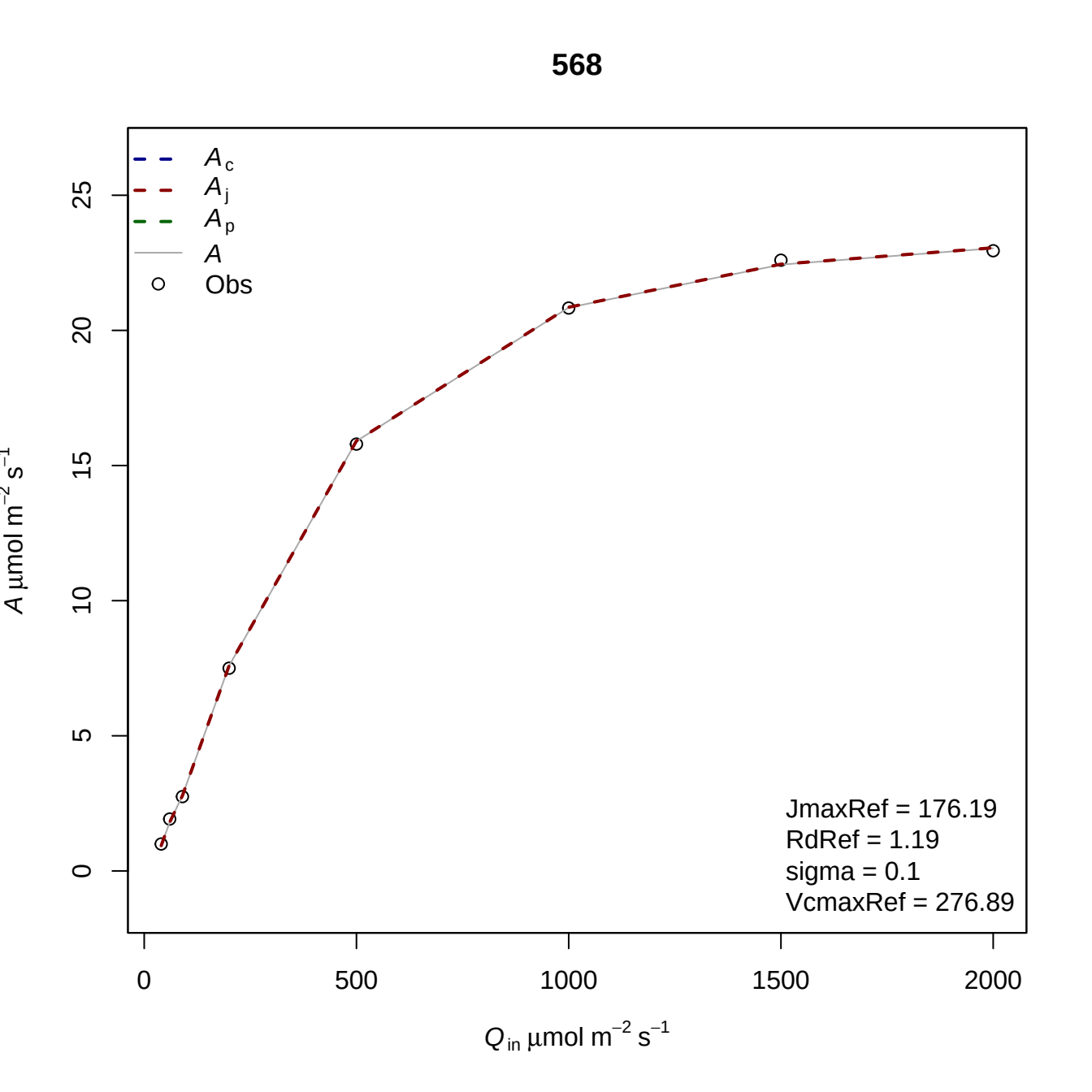

563
A_c
A_j
A_p
A
Obs
JmaxRef = 160
RdRef = 1.05
sigma = 0.19
VcmaxRef = 511.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

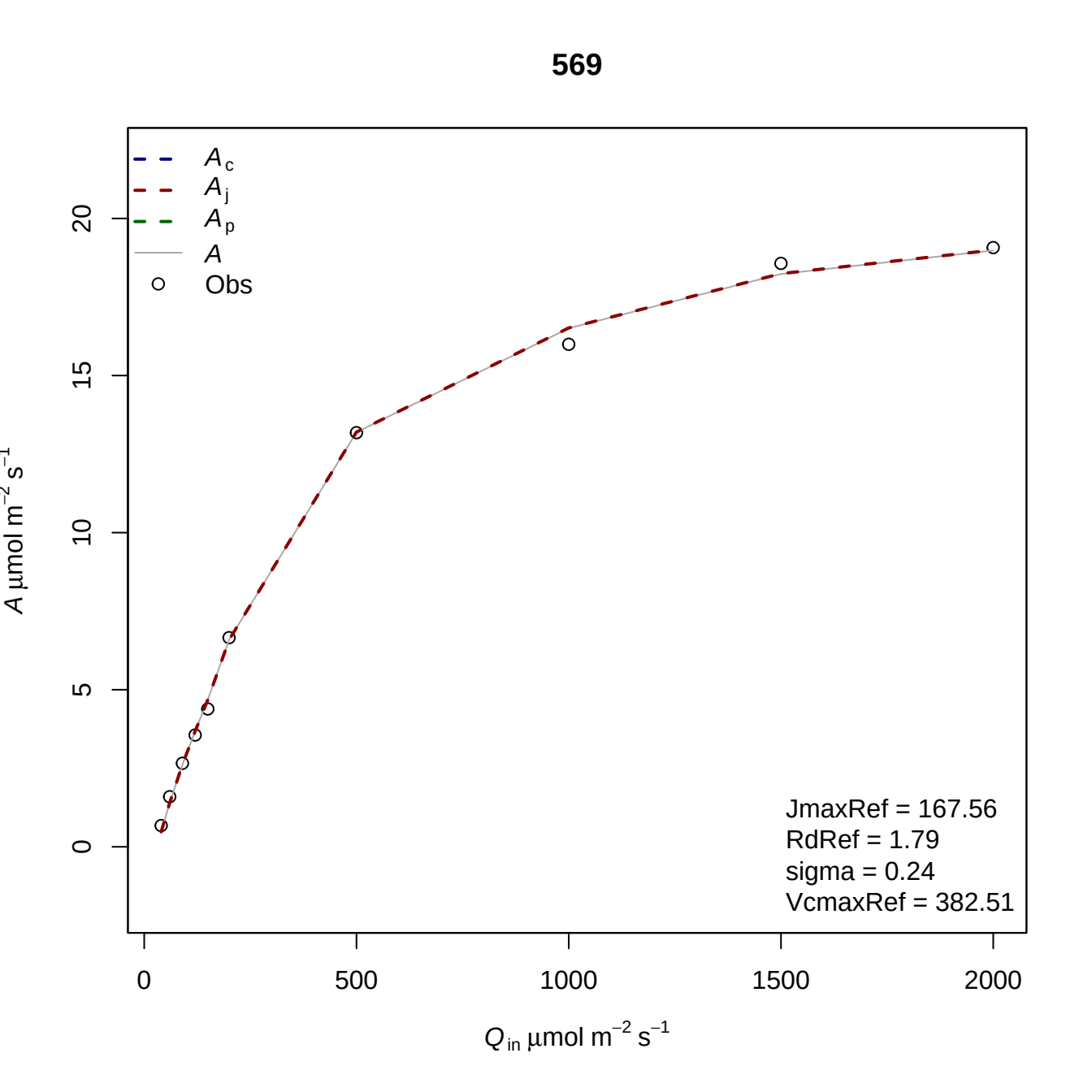

564
A_c
A_j
A_p
A
Obs
JmaxRef = 184.86
RdRef = 1.44
sigma = 0.14
VcmaxRef = 811644.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

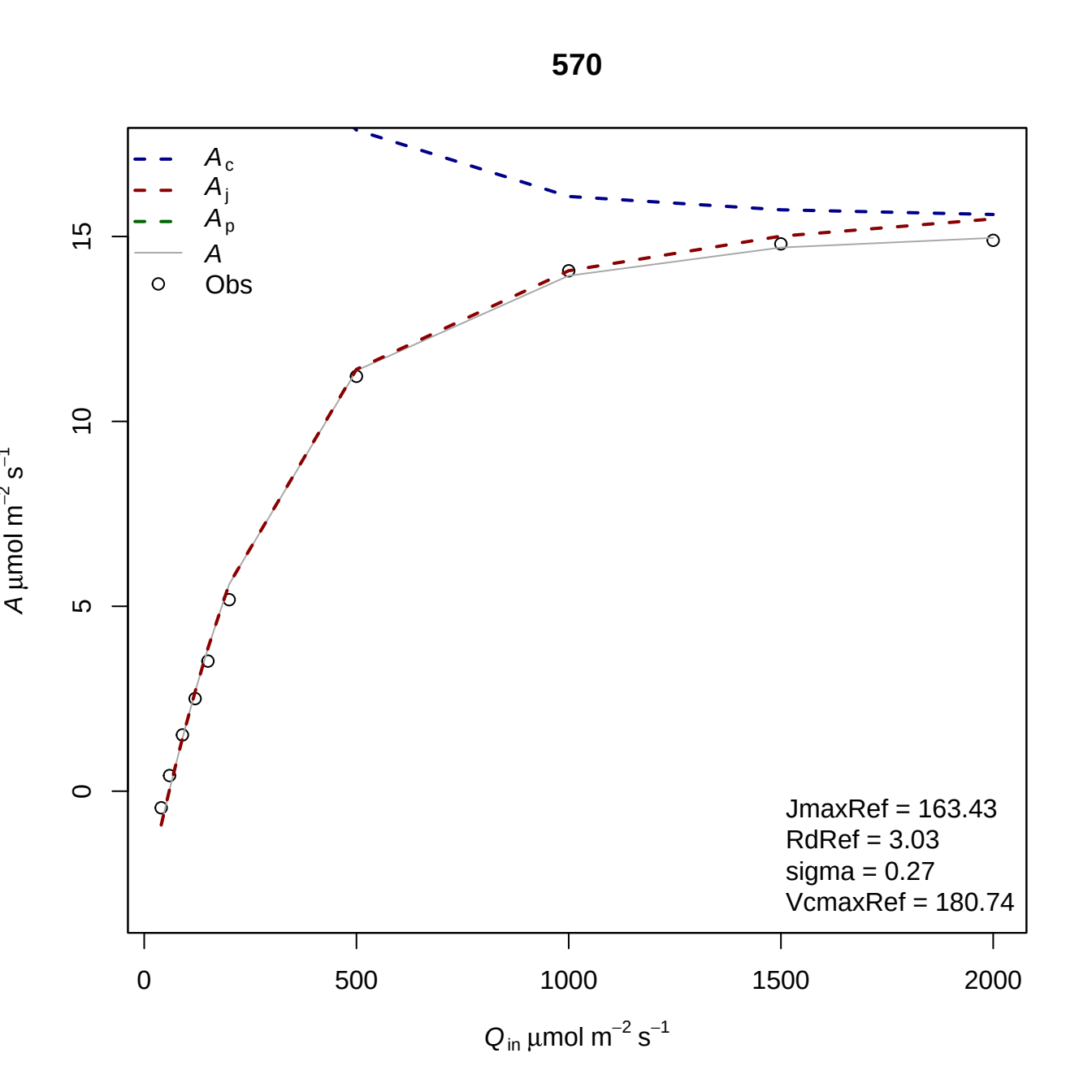

565
A_c
A_j
A_p
A
Obs
JmaxRef = 230.95
RdRef = 1.91
sigma = 0.36
VcmaxRef = 617.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

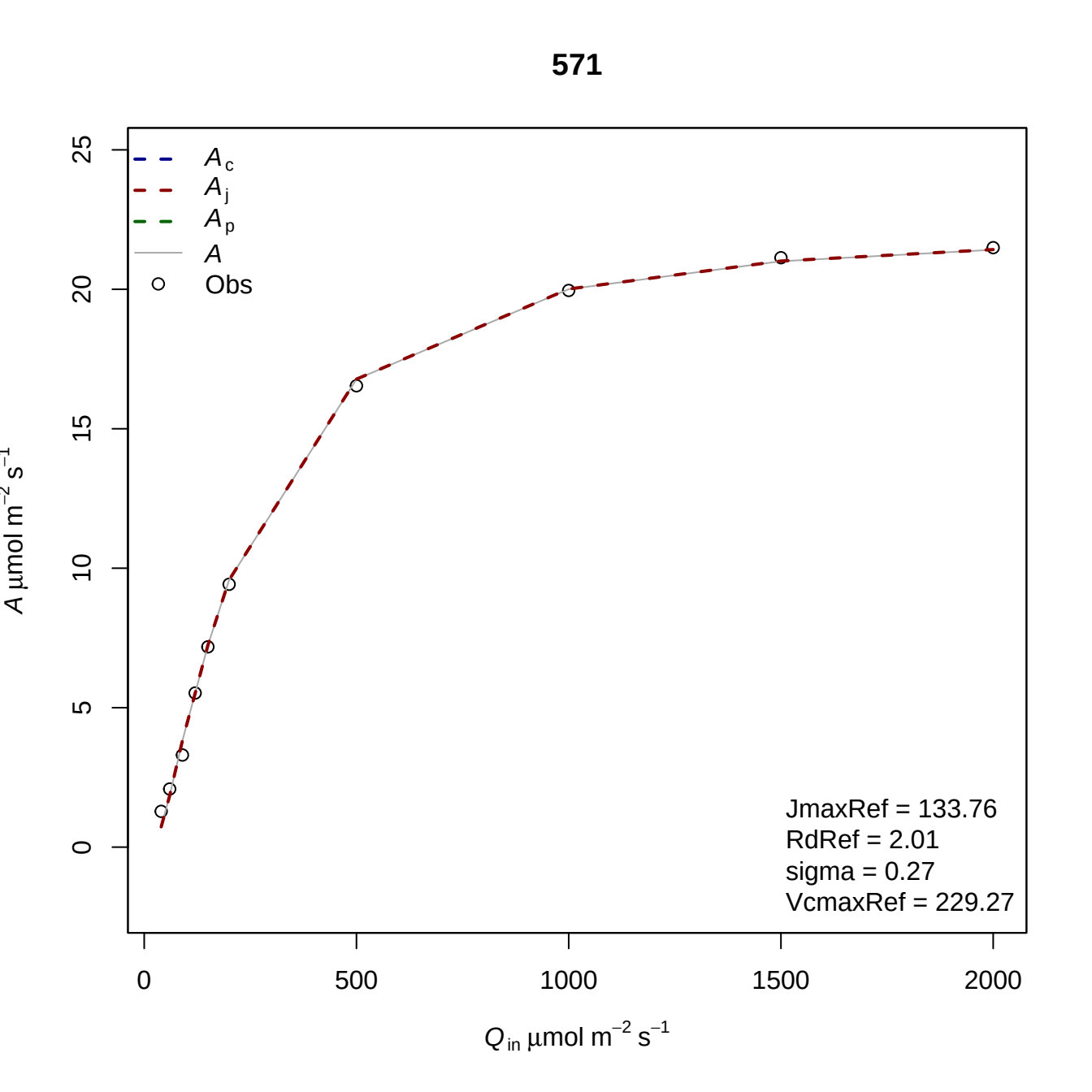

566
A_c
A_j
A_p
A
Obs
JmaxRef = 247.19
RdRef = 2.73
sigma = 0.25
VcmaxRef = 1354.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

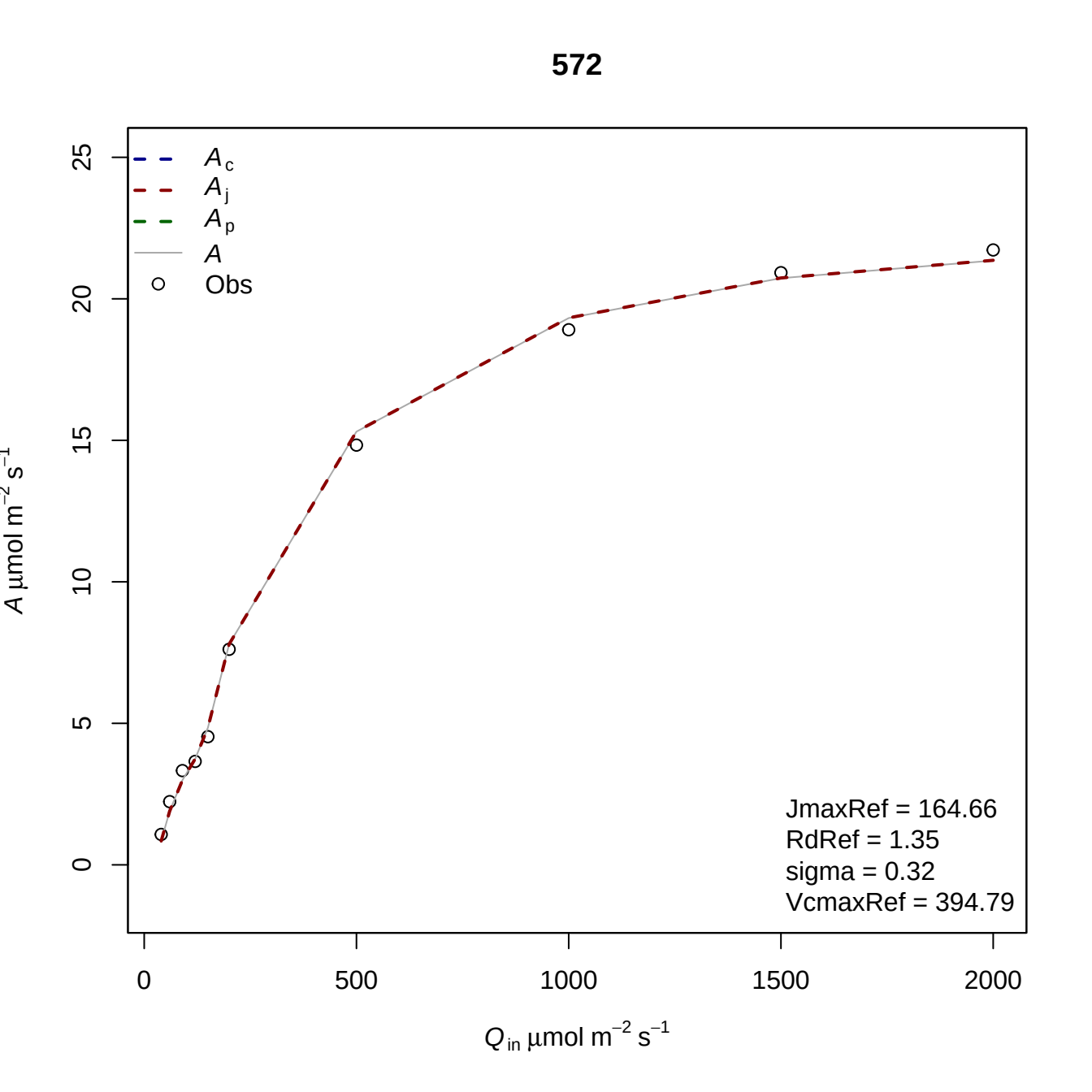

567
A_c
A_j
A_p
A
Obs
JmaxRef = 164.71
RdRef = 2.01
sigma = 0.27
VcmaxRef = 552.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

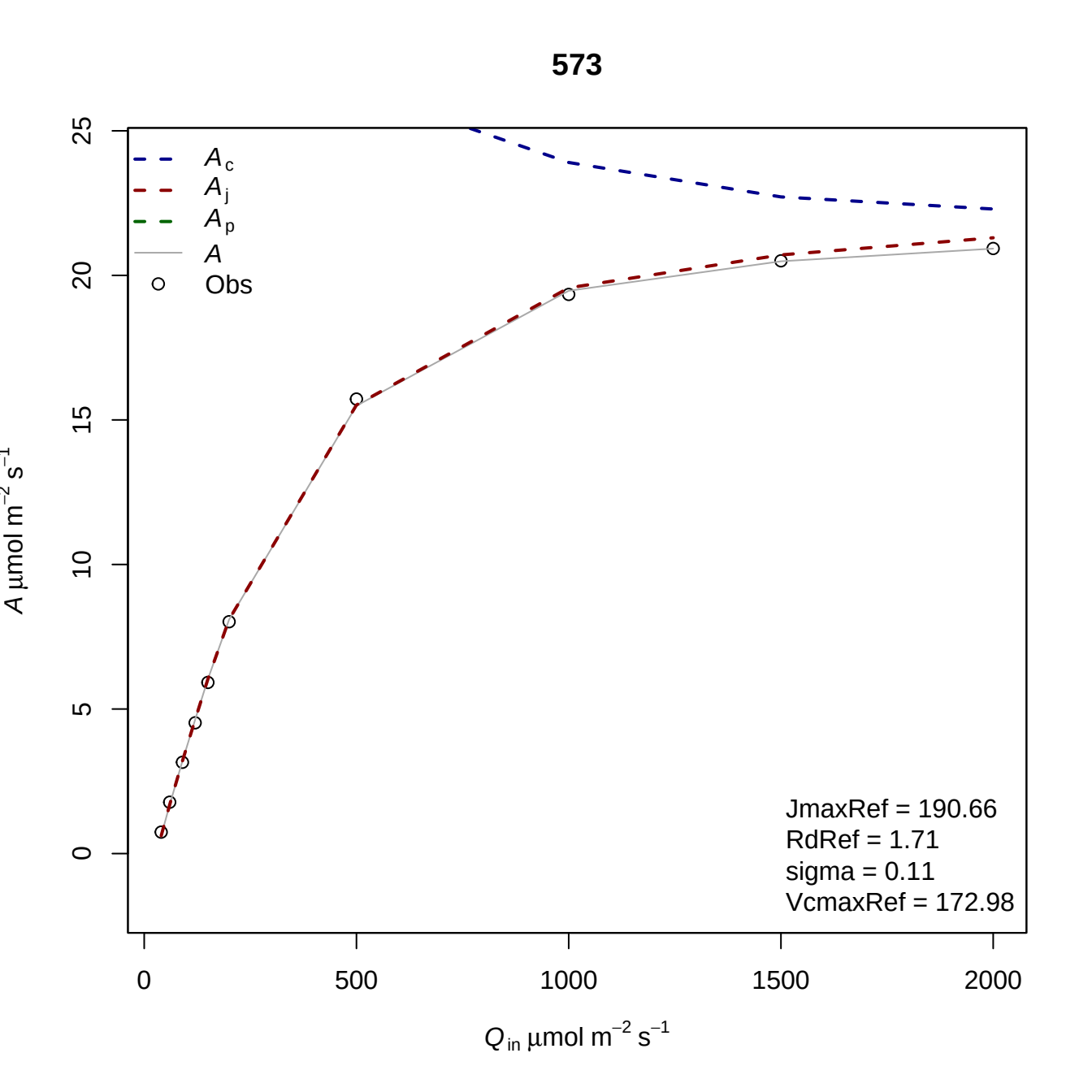

568
A_c
A_j
A_p
A
Obs
JmaxRef = 176.19
RdRef = 1.19
sigma = 0.1
VcmaxRef = 276.89
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

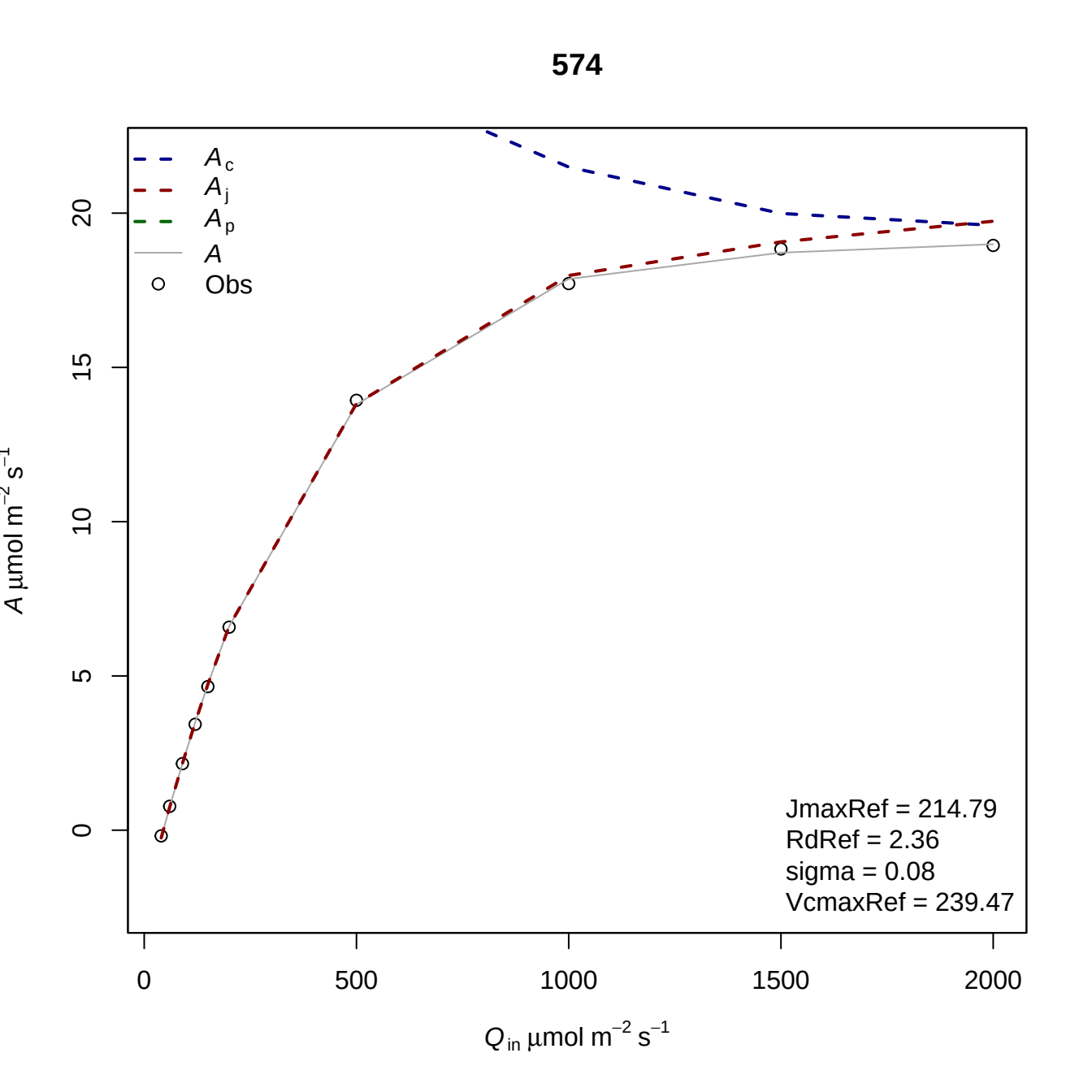

569
A_c
A_j
A_p
A
Obs
JmaxRef = 167.56
RdRef = 1.79
sigma = 0.24
VcmaxRef = 382.51
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

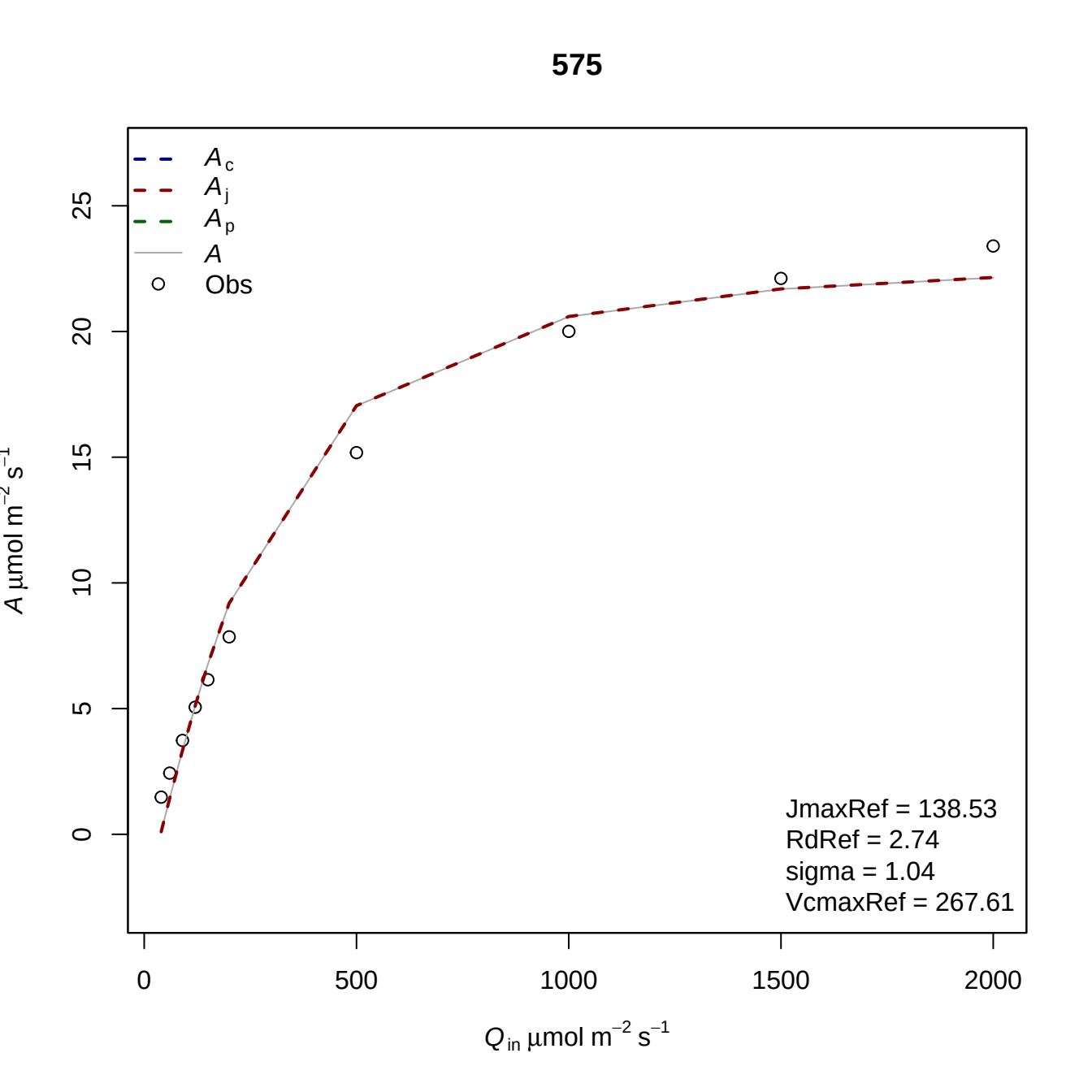

570
A_c
A_j
A_p
A
Obs
JmaxRef = 163.43
RdRef = 3.03
sigma = 0.27
VcmaxRef = 180.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

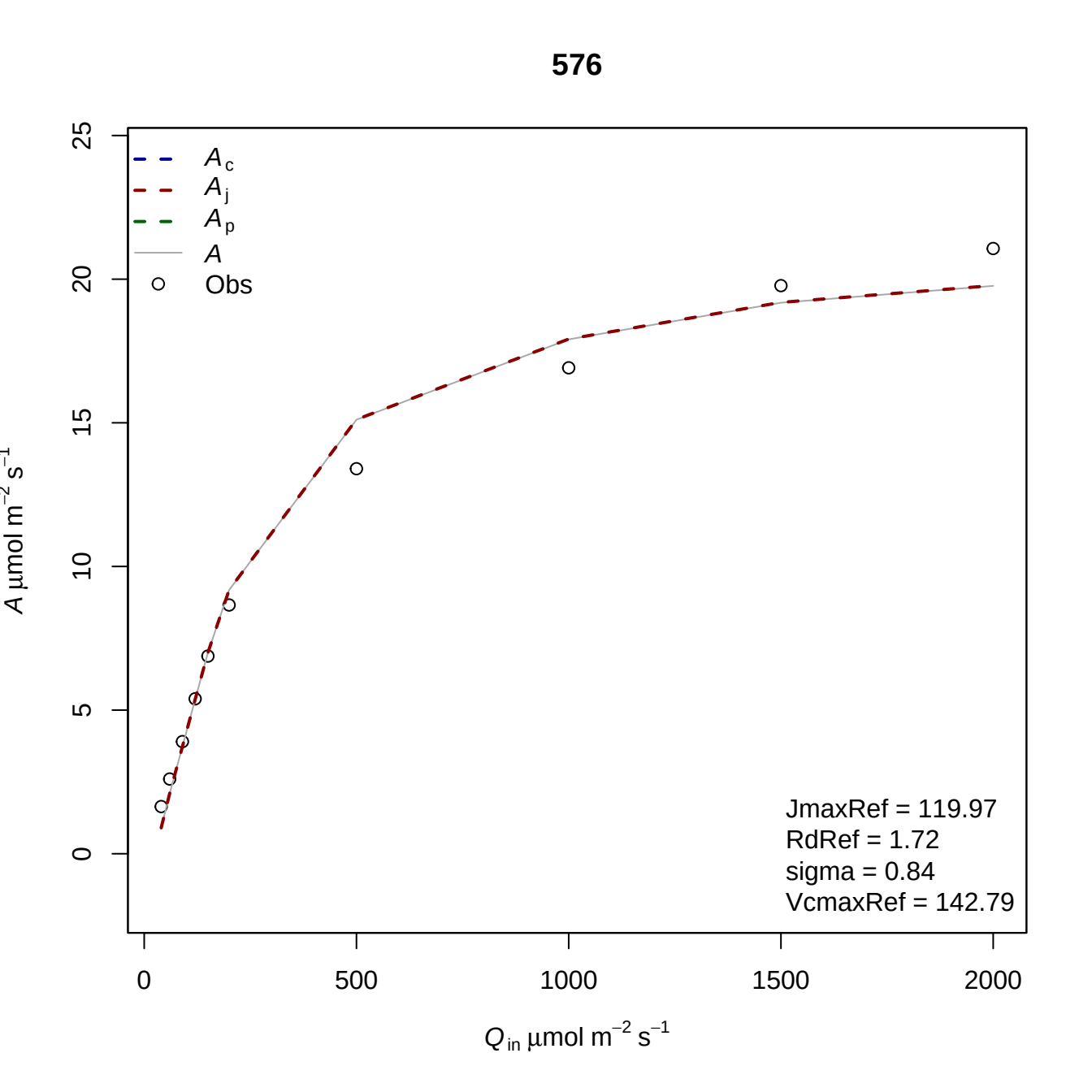

571
A_c
A_j
A_p
A
Obs
JmaxRef = 133.76
RdRef = 2.01
sigma = 0.27
VcmaxRef = 229.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

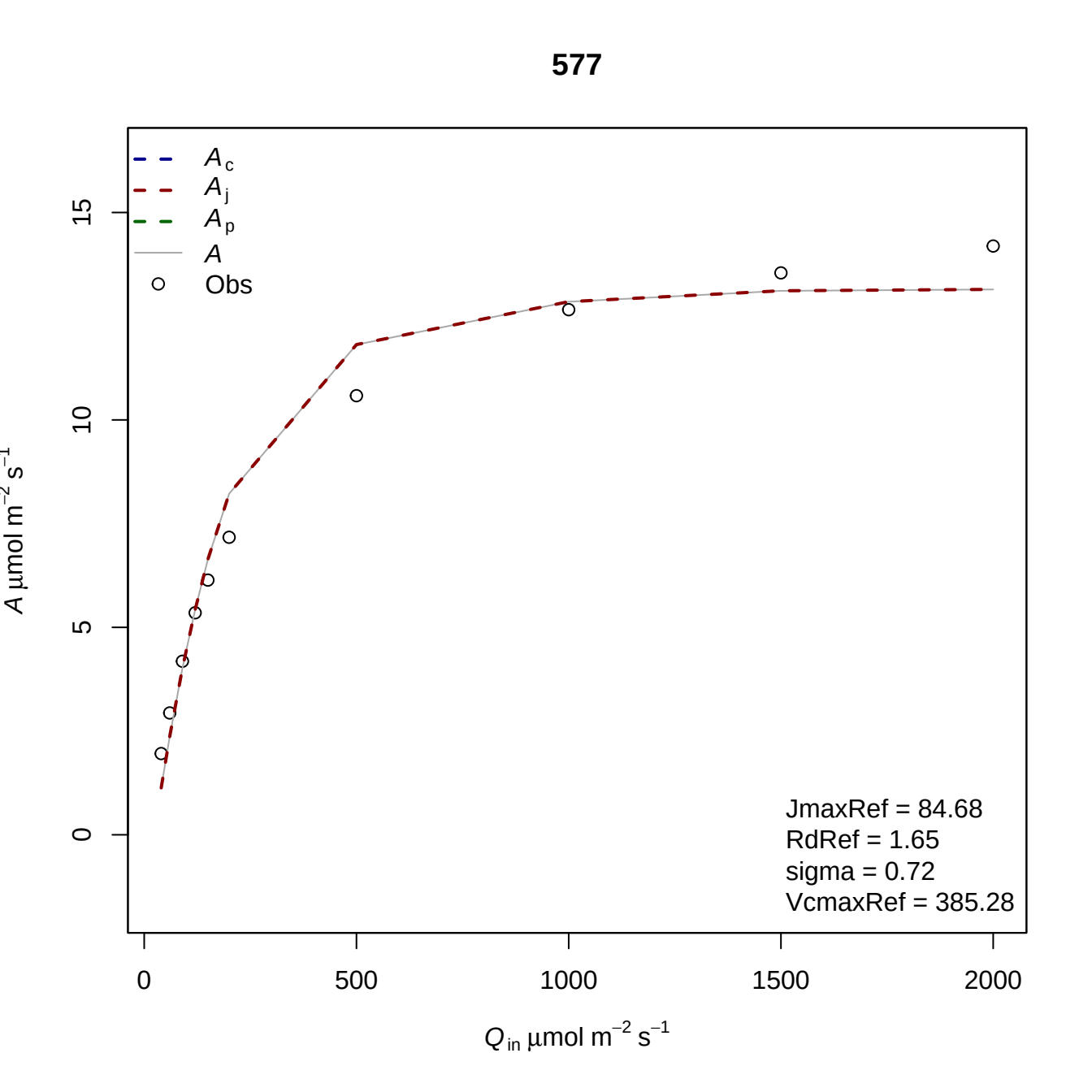

572
A_c
A_j
A_p
A
Obs
JmaxRef = 164.66
RdRef = 1.35
sigma = 0.32
VcmaxRef = 394.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

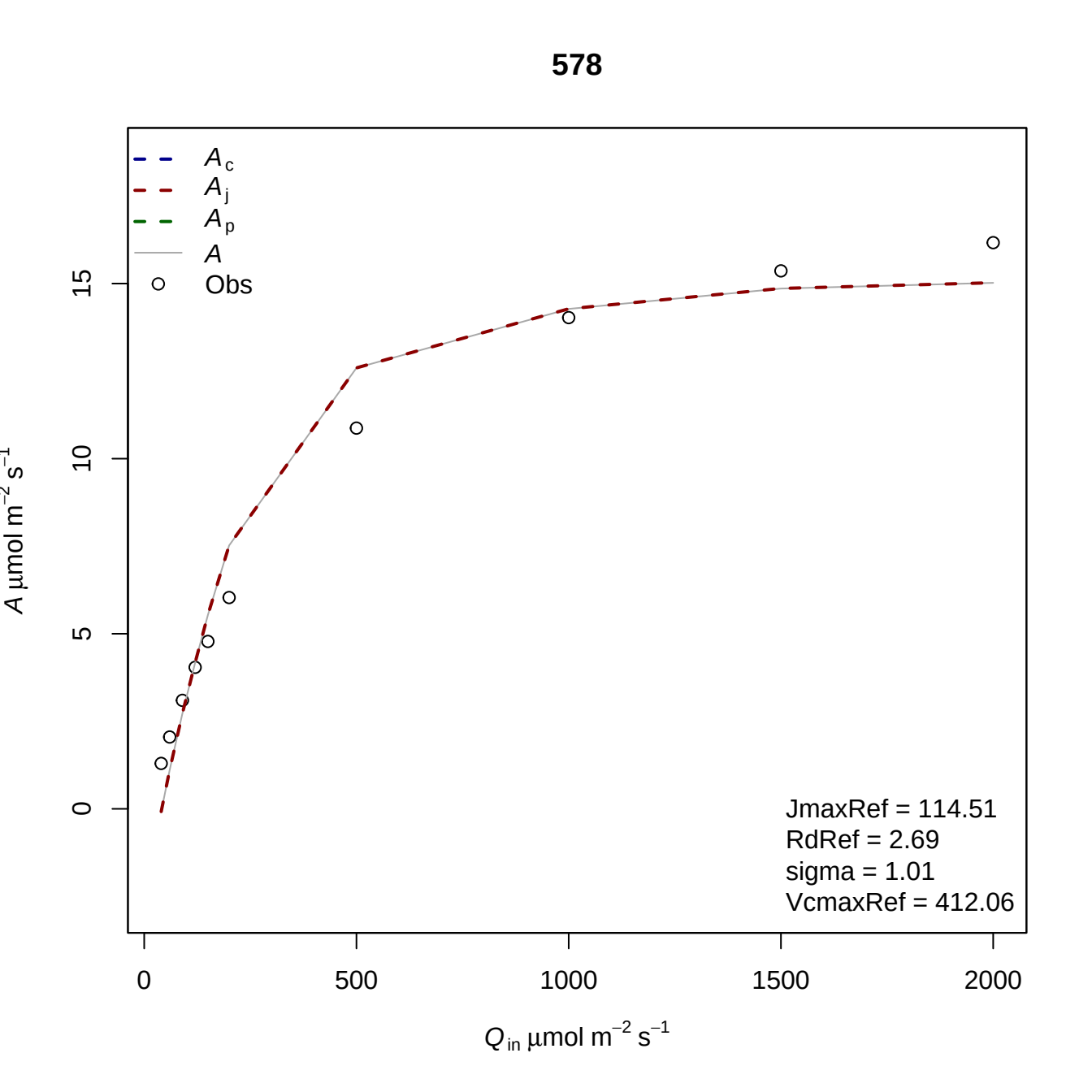

573
A_c
A_j
A_p
A
Obs
JmaxRef = 190.66
RdRef = 1.71
sigma = 0.11
VcmaxRef = 172.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

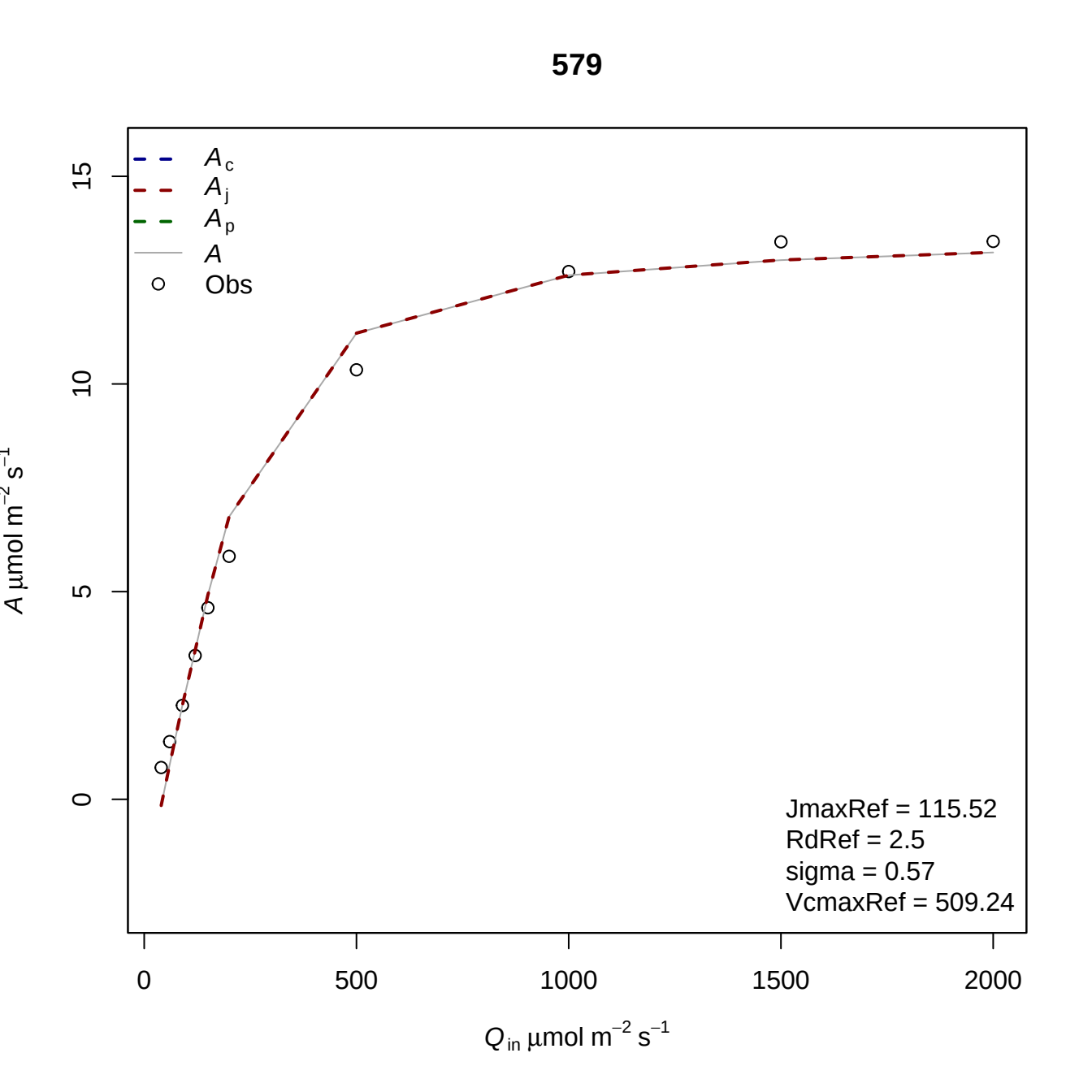

574
A_c
A_j
A_p
A
Obs
JmaxRef = 214.79
RdRef = 2.36
sigma = 0.08
VcmaxRef = 239.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

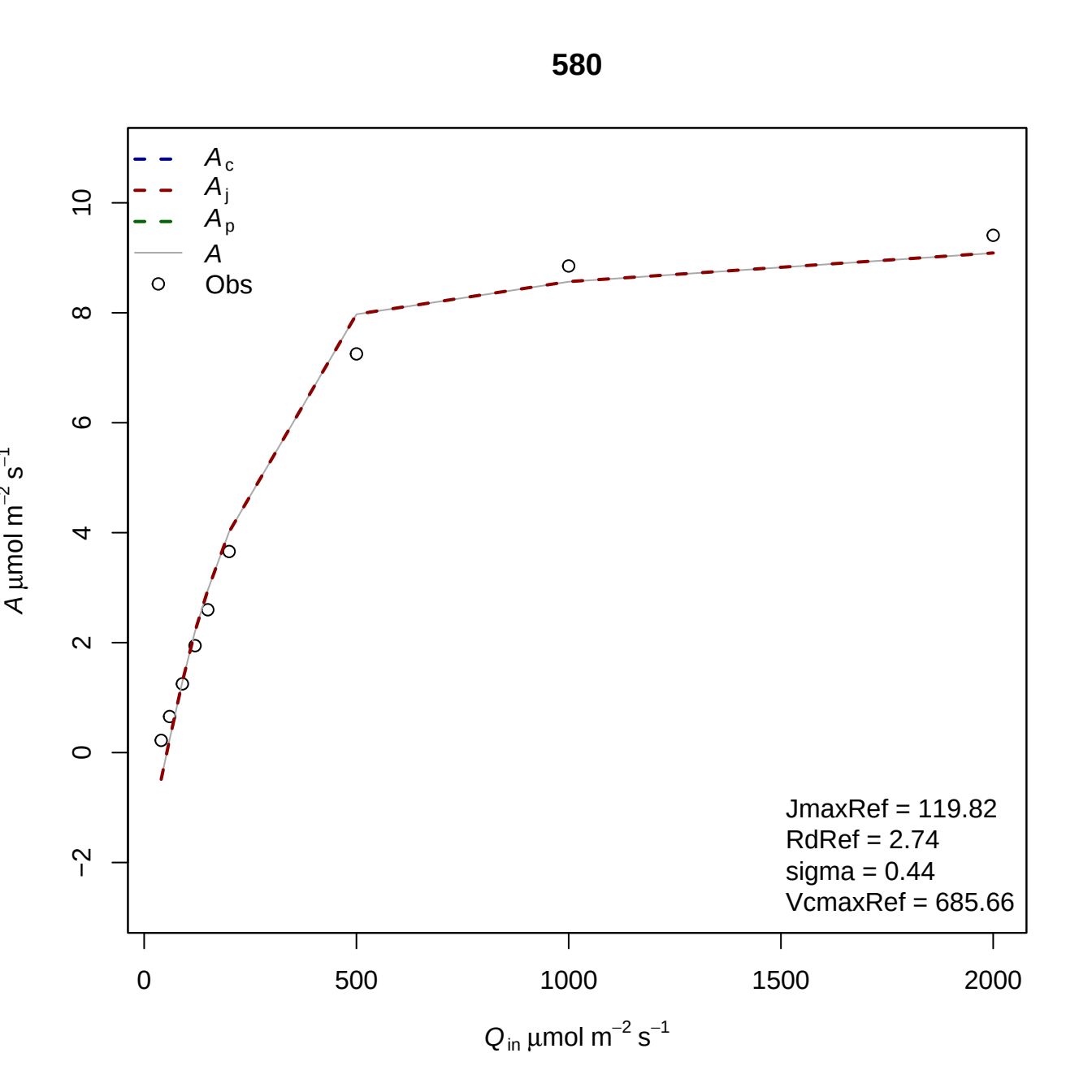

575
A_c
A_j
A_p
A
Obs
JmaxRef = 138.53
RdRef = 2.74
sigma = 1.04
VcmaxRef = 267.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

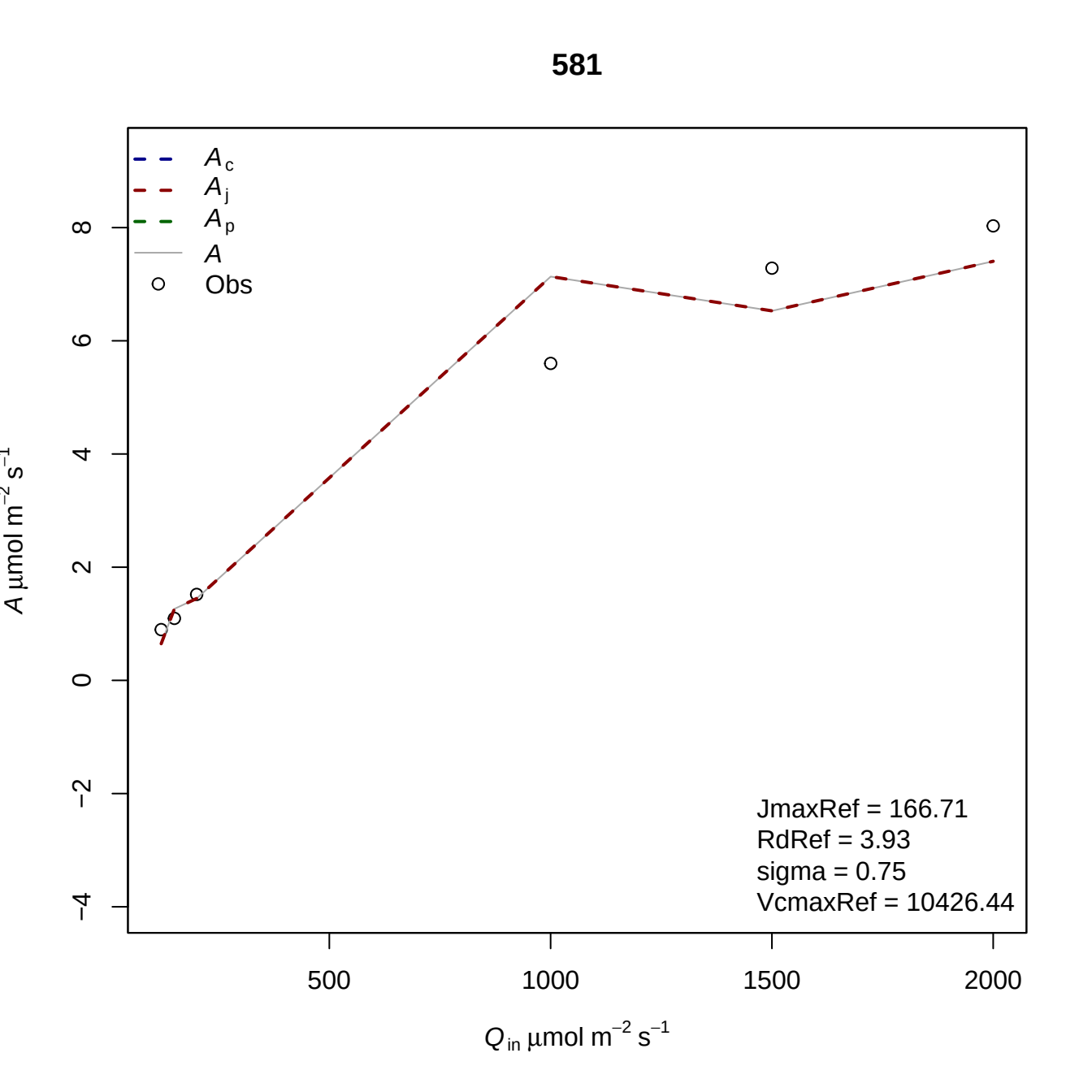

576
A_c
A_j
A_p
A
Obs
JmaxRef = 119.97
RdRef = 1.72
sigma = 0.84
VcmaxRef = 142.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

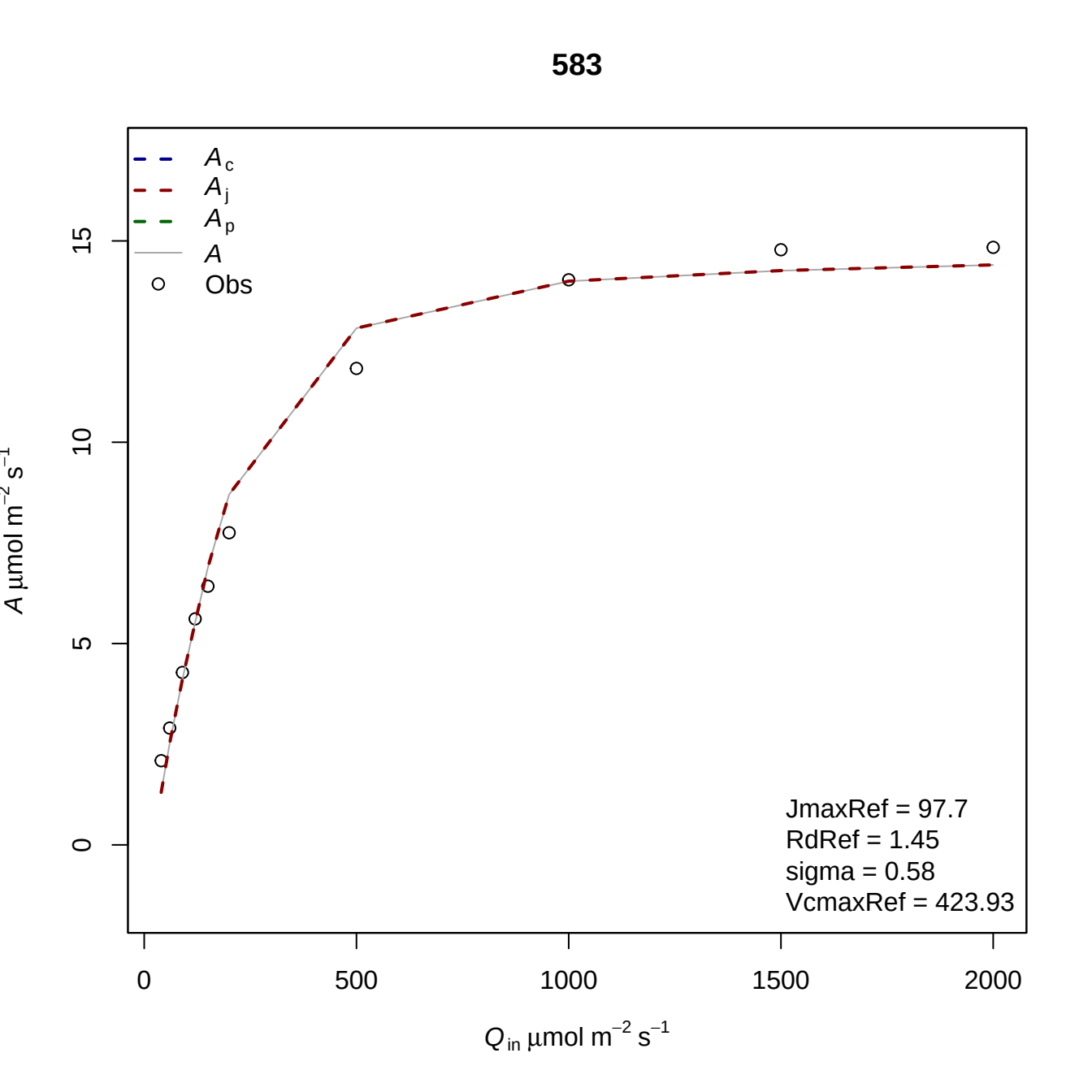

577
A_c
A_j
A_p
A
Obs
JmaxRef = 84.68
RdRef = 1.65
sigma = 0.72
VcmaxRef = 385.28
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

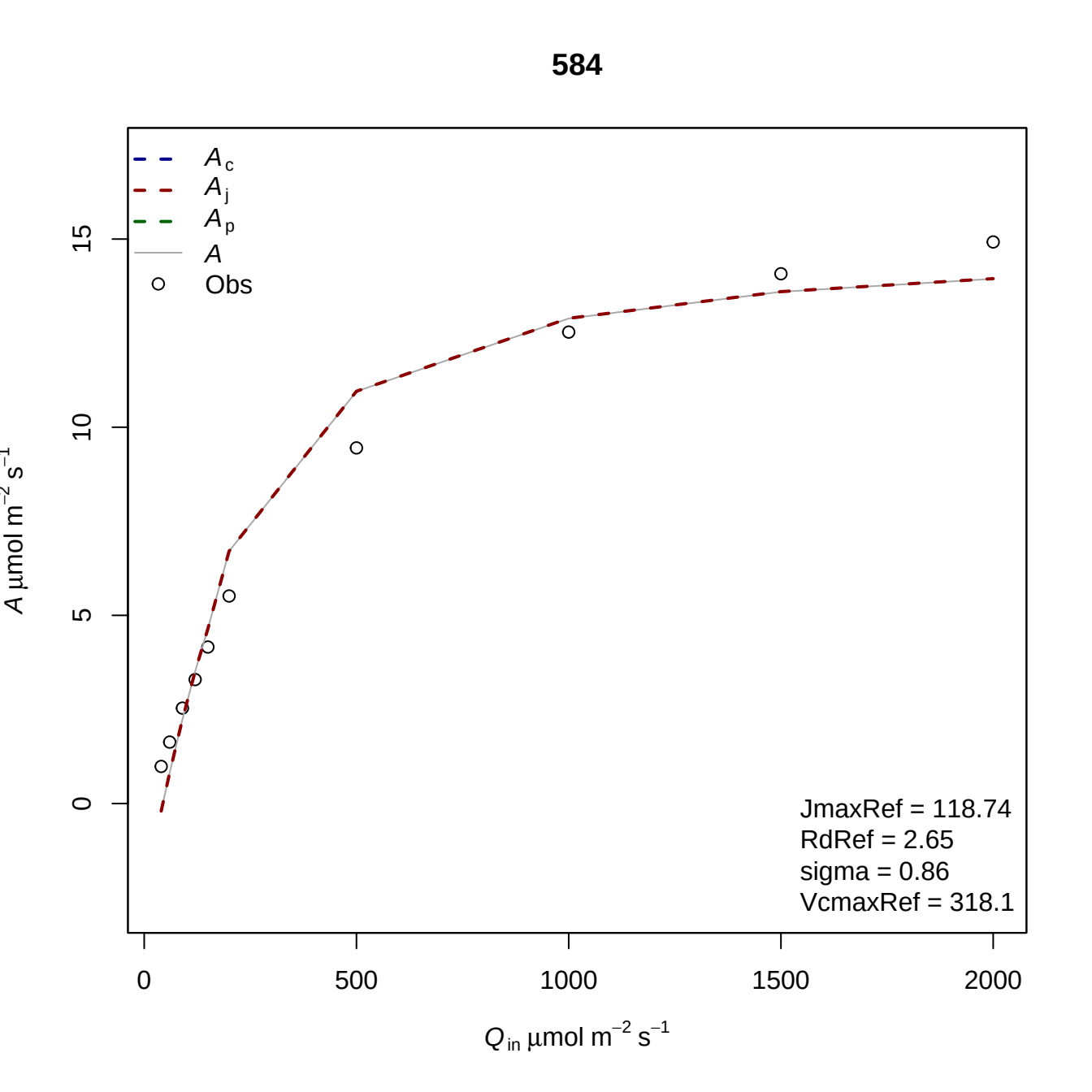

578
A_c
A_j
A_p
A
Obs
JmaxRef = 114.51
RdRef = 2.69
sigma = 1.01
VcmaxRef = 412.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

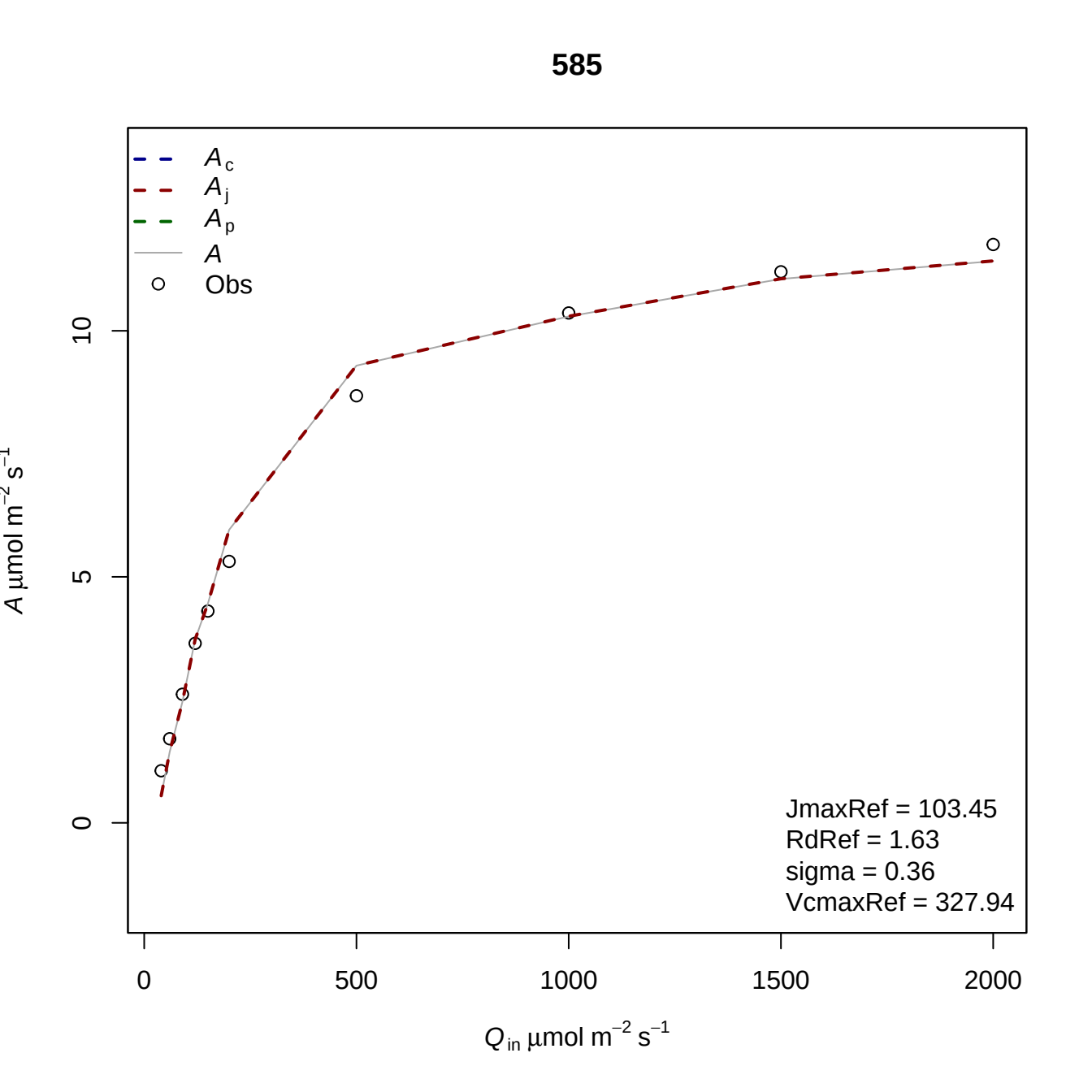

579
A_c
A_j
A_p
A
Obs
JmaxRef = 115.52
RdRef = 2.5
sigma = 0.57
VcmaxRef = 509.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

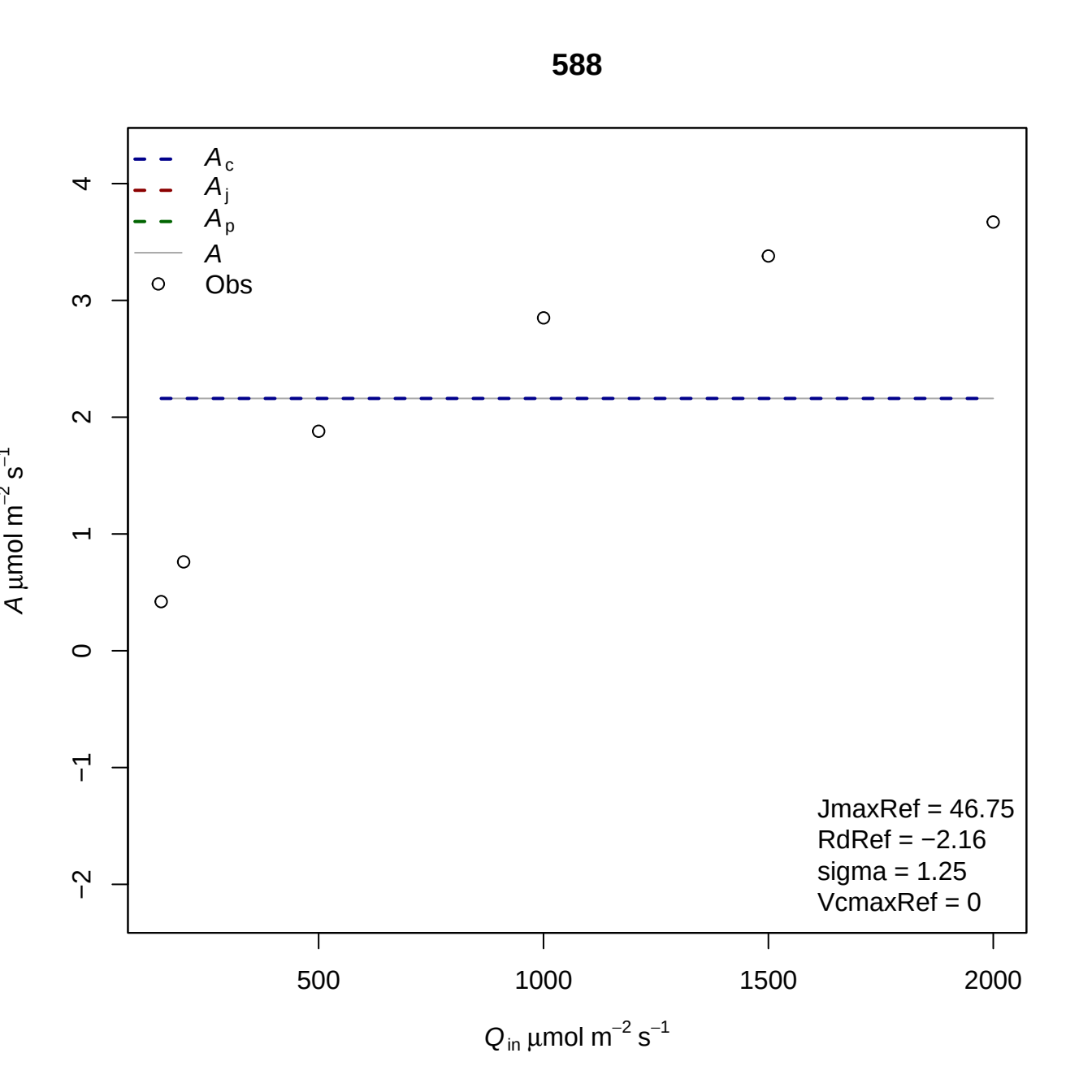

580
A_c
A_j
A_p
A
Obs
JmaxRef = 119.82
RdRef = 2.74
sigma = 0.44
VcmaxRef = 685.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

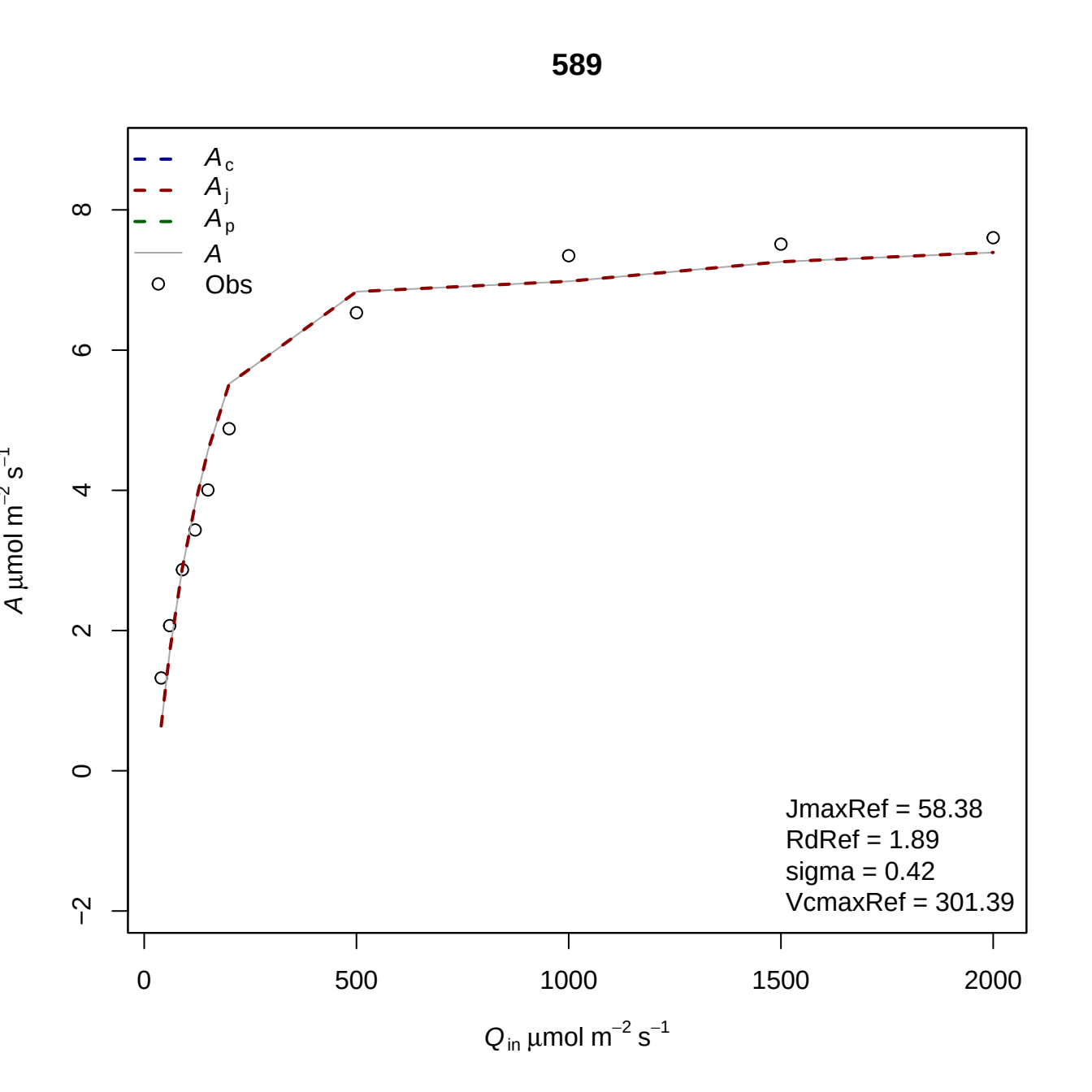

581
A_c
A_j
A_p
A
Obs
JmaxRef = 166.71
RdRef = 3.93
sigma = 0.75
VcmaxRef = 10426.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
500
1000
1500
2000

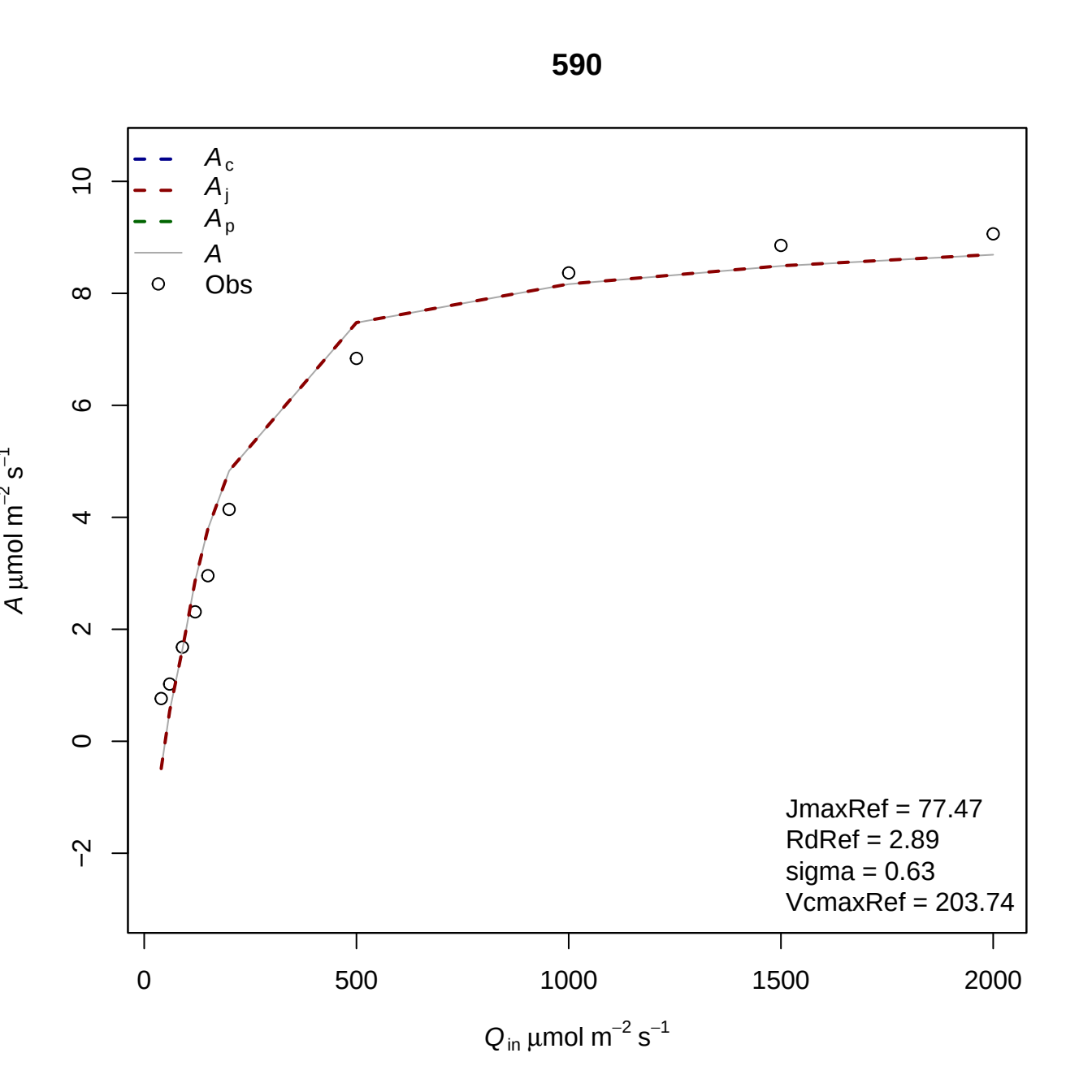

583
A_c
A_j
A_p
A
Obs
JmaxRef = 97.7
RdRef = 1.45
sigma = 0.58
VcmaxRef = 423.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

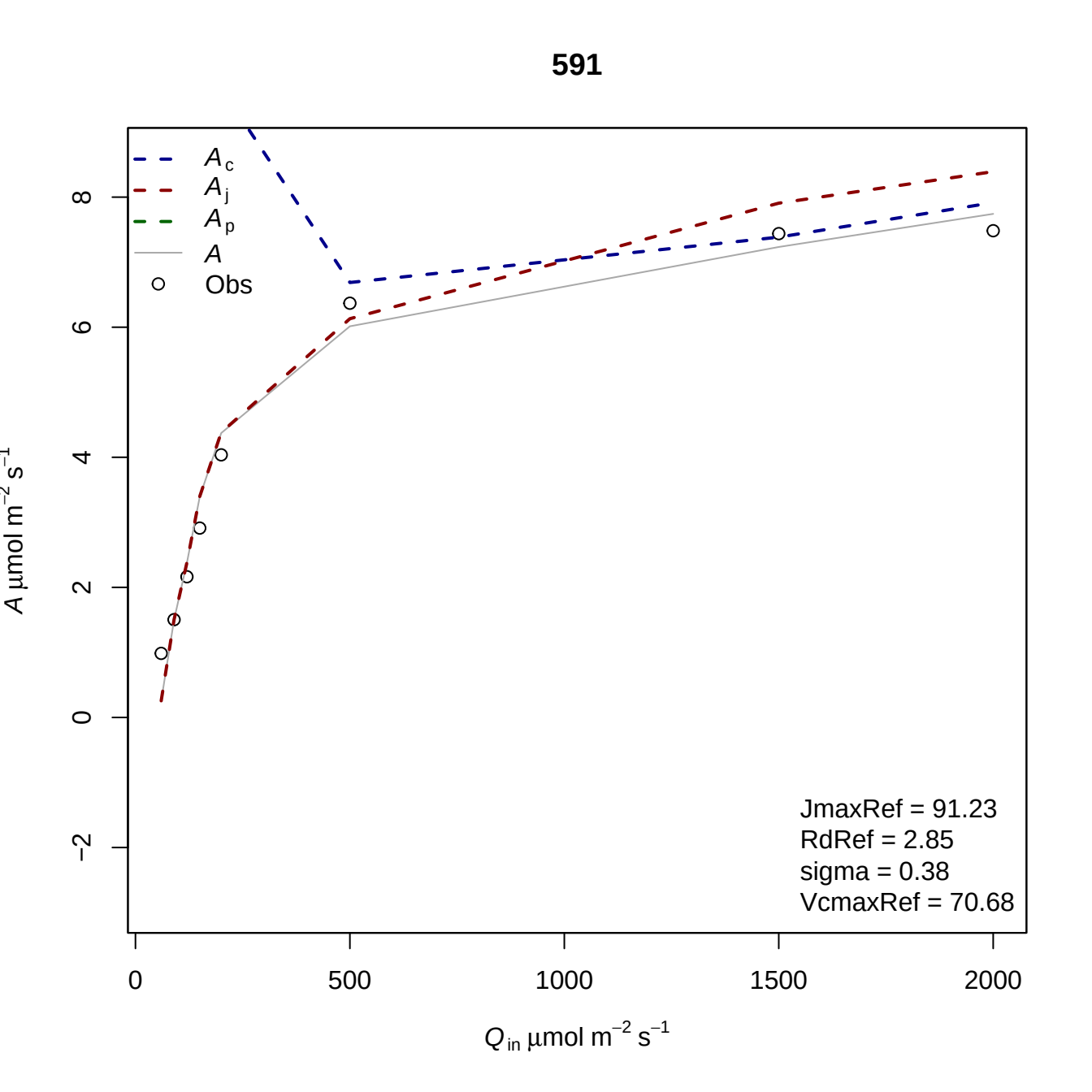

584
A_c
A_j
A_p
A
Obs
JmaxRef = 118.74
RdRef = 2.65
sigma = 0.86
VcmaxRef = 318.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

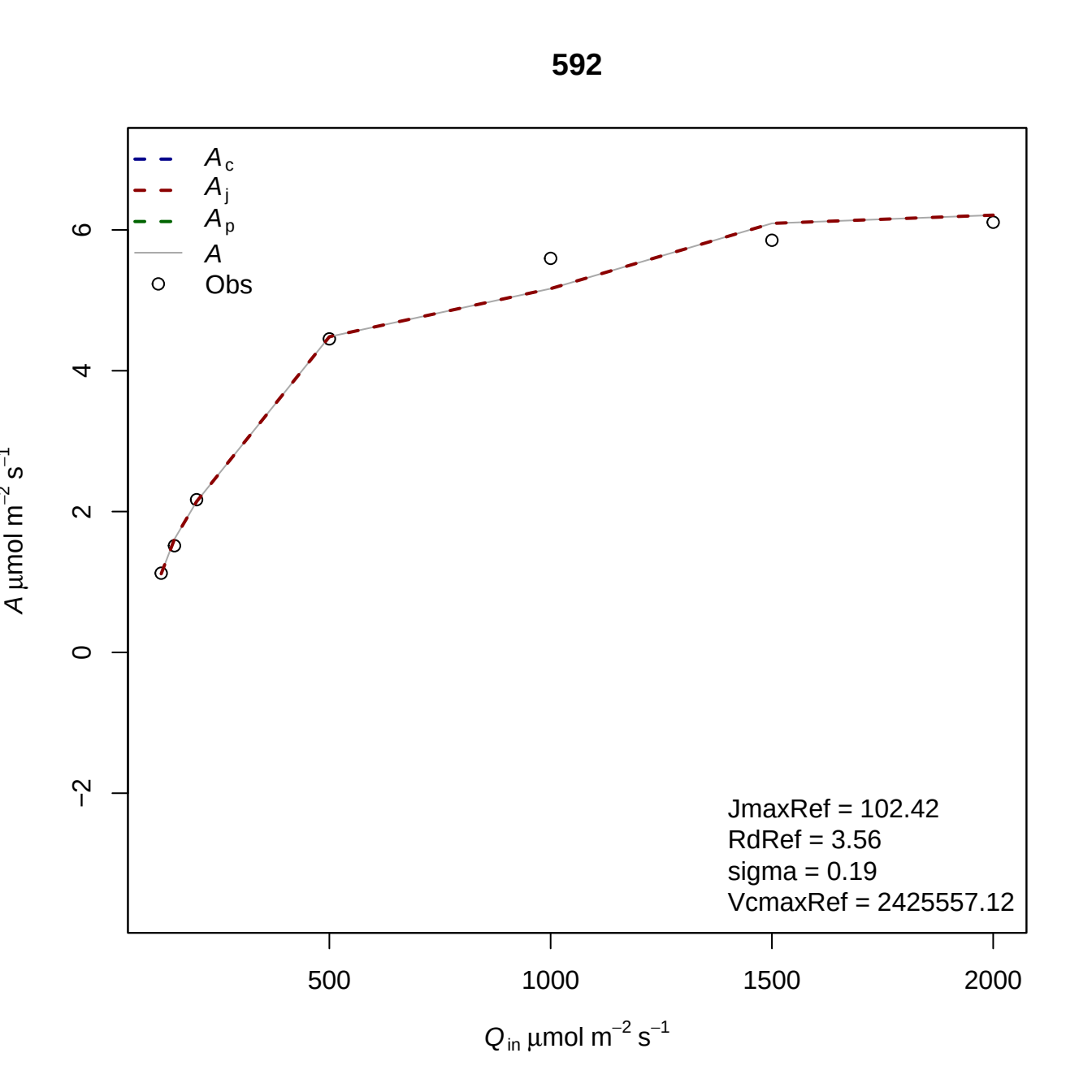

585
A_c
A_j
A_p
A
Obs
JmaxRef = 103.45
RdRef = 1.63
sigma = 0.36
VcmaxRef = 327.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

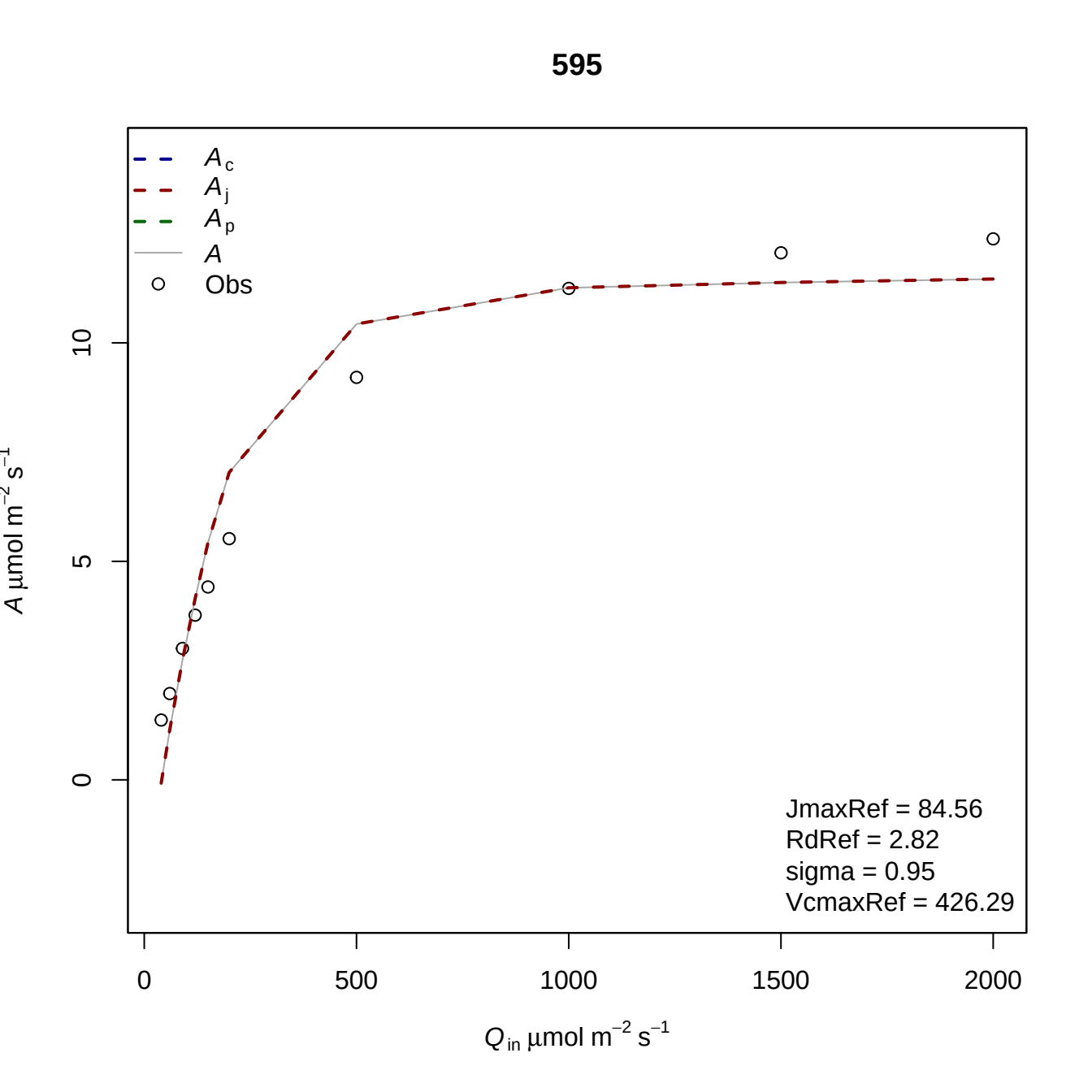

588
A_c
A_j
A_p
A
Obs
JmaxRef = 46.75
RdRef = −2.16
sigma = 1.25
VcmaxRef = 0
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
500
1000
1500
2000

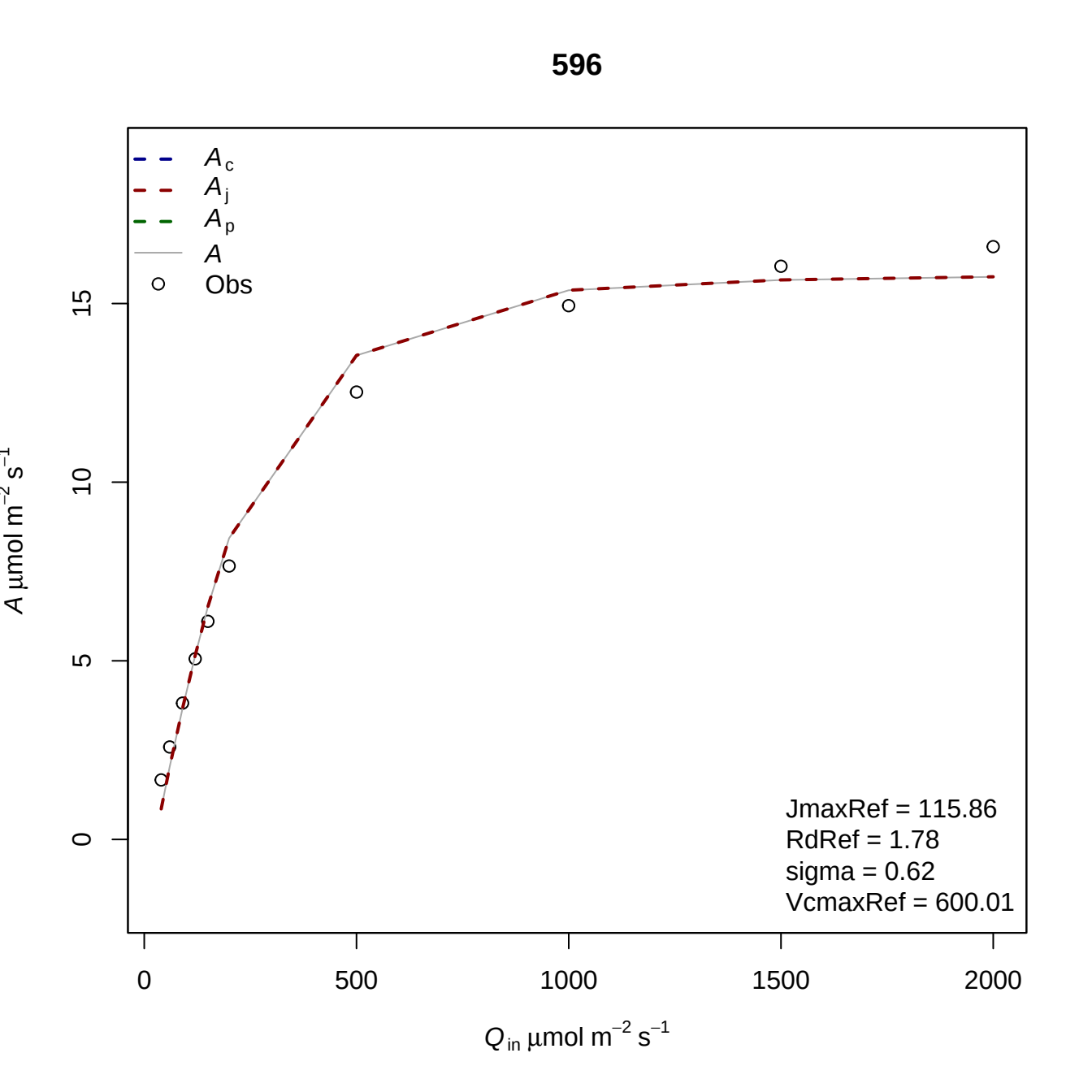

589
A_c
A_j
A_p
A
Obs
JmaxRef = 58.38
RdRef = 1.89
sigma = 0.42
VcmaxRef = 301.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

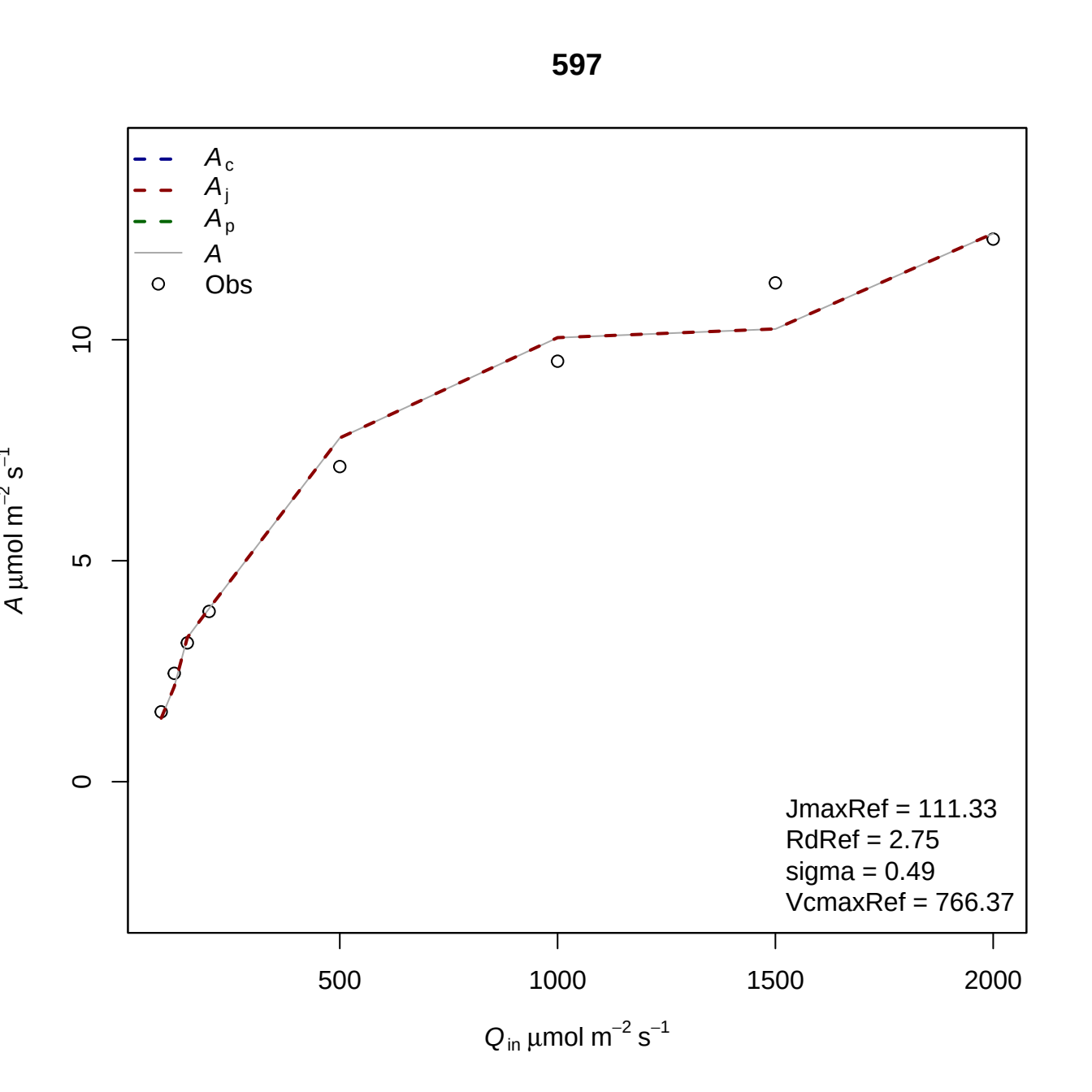

590
A_c
A_j
A_p
A
Obs
JmaxRef = 77.47
RdRef = 2.89
sigma = 0.63
VcmaxRef = 203.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

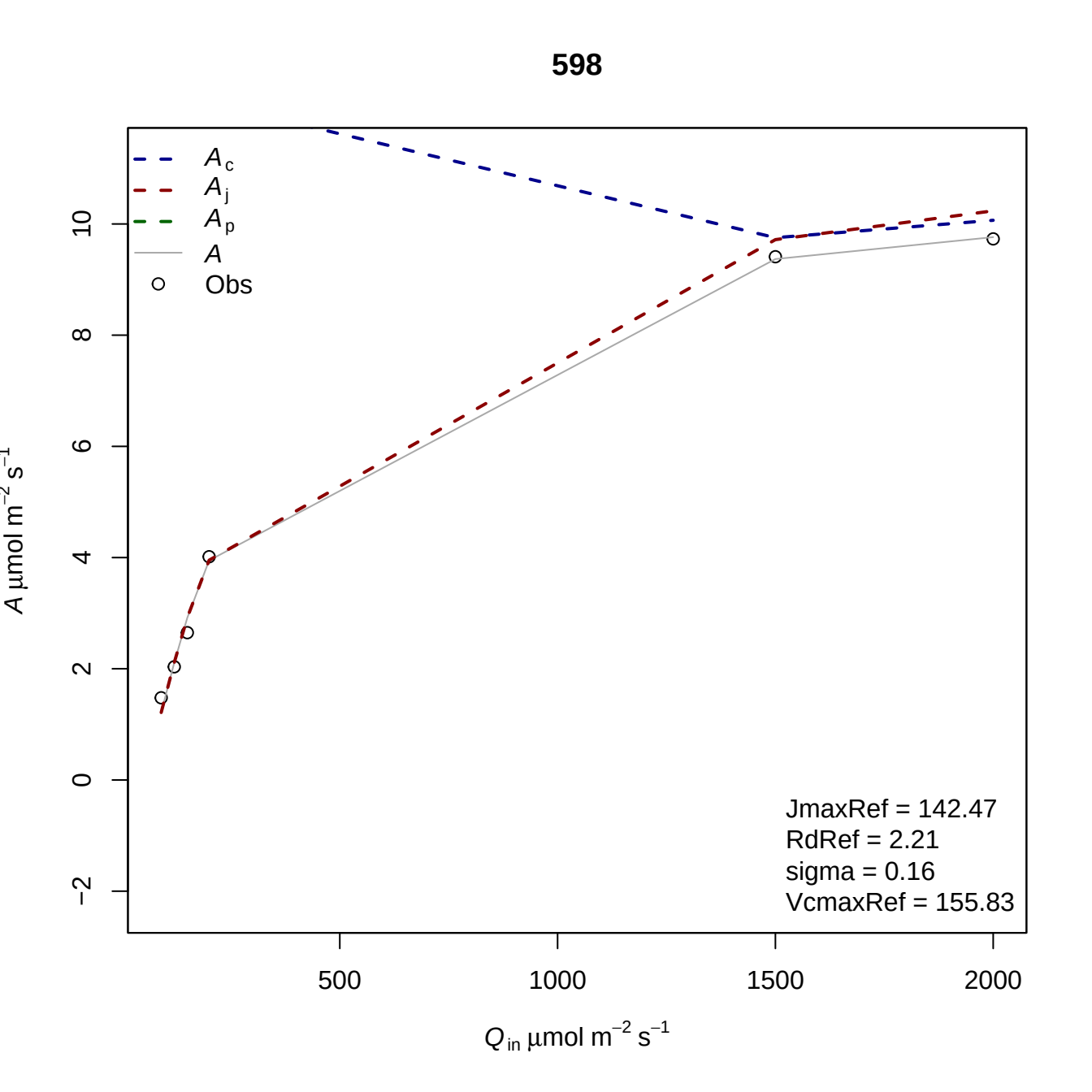

591
A_c
A_j
A_p
A
Obs
JmaxRef = 91.23
RdRef = 2.85
sigma = 0.38
VcmaxRef = 70.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

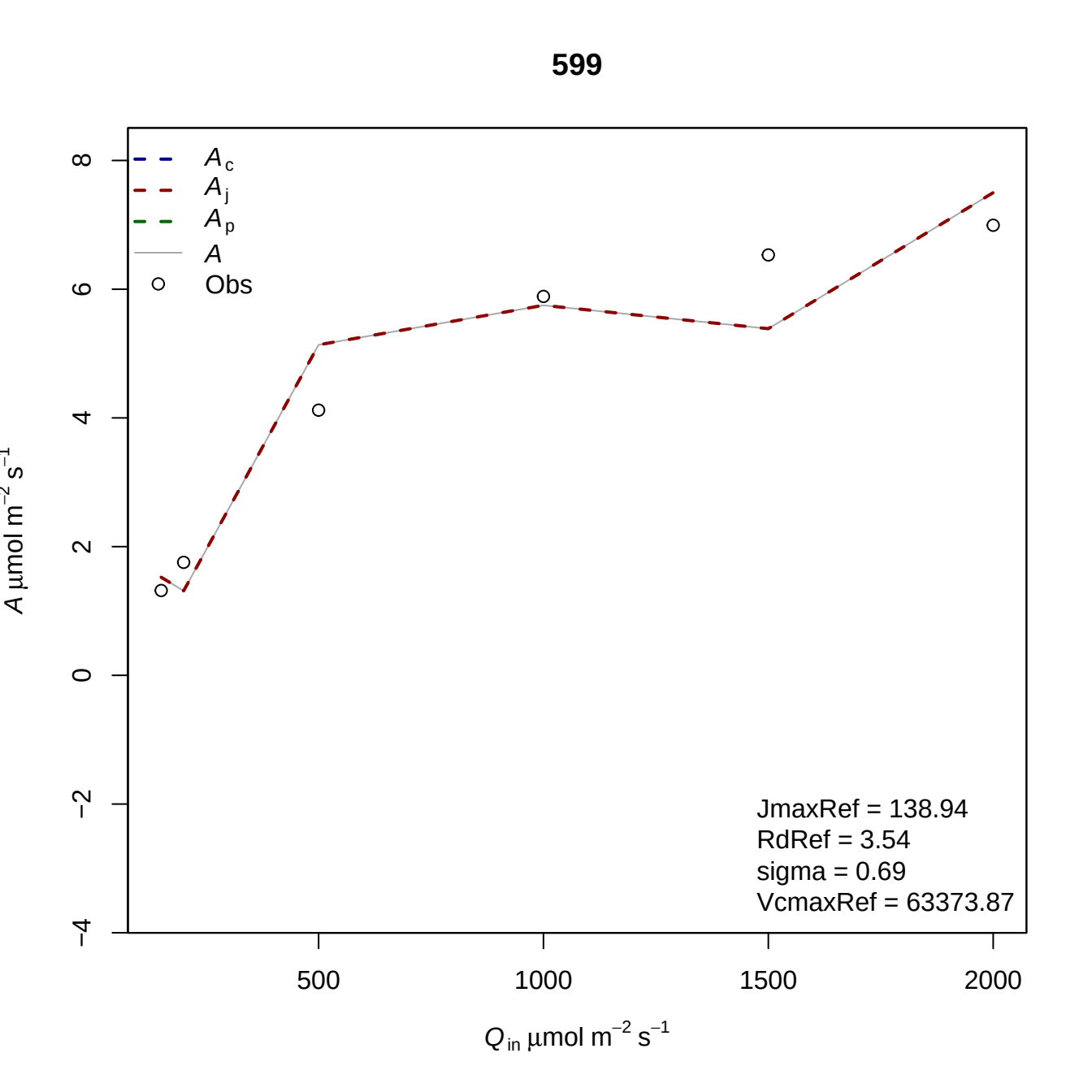

592
A_c
A_j
A_p
A
Obs
JmaxRef = 102.42
RdRef = 3.56
sigma = 0.19
VcmaxRef = 2425557.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
500
1000
1500
2000

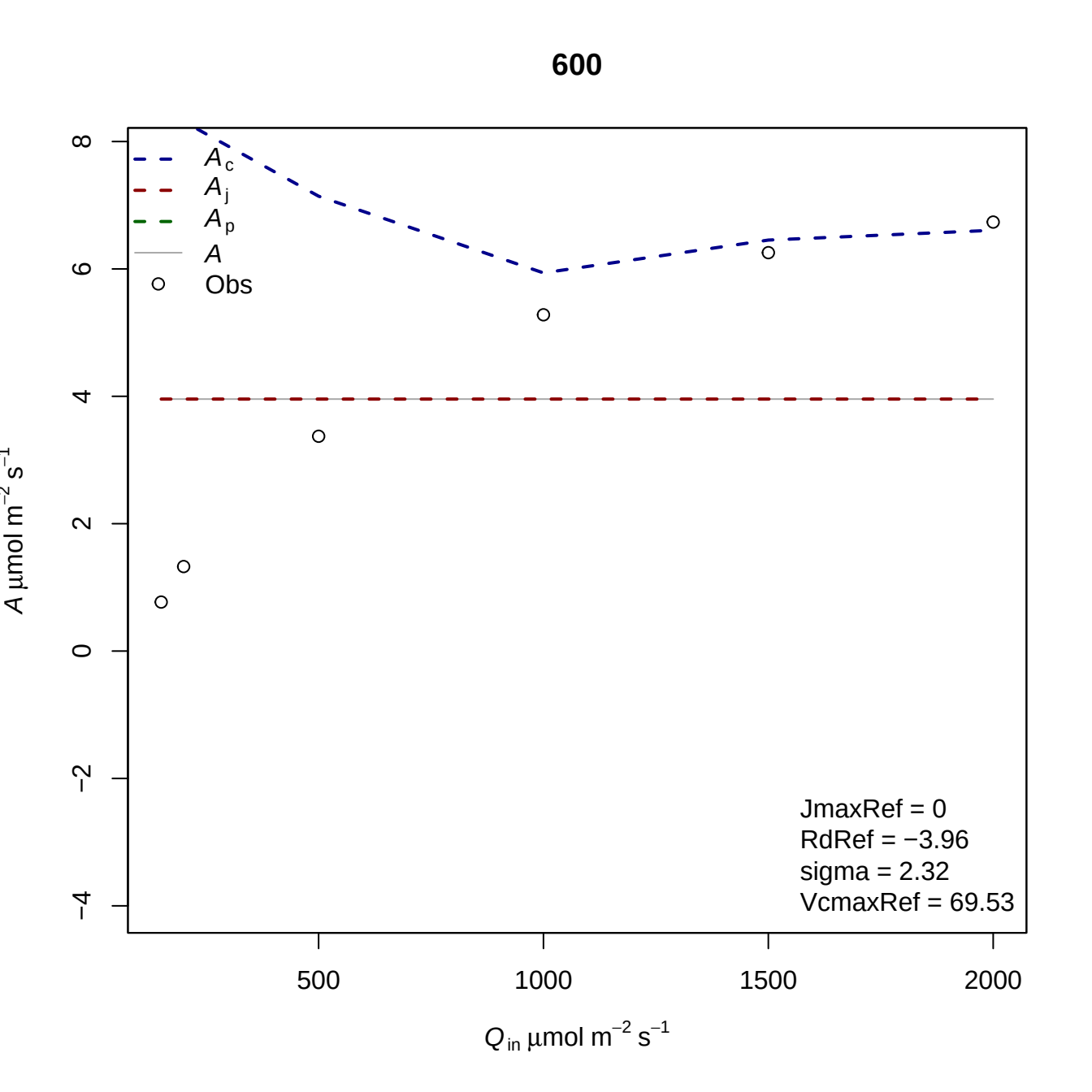

595
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 84.56
RdRef = 2.82
sigma = 0.95
VcmaxRef = 426.29

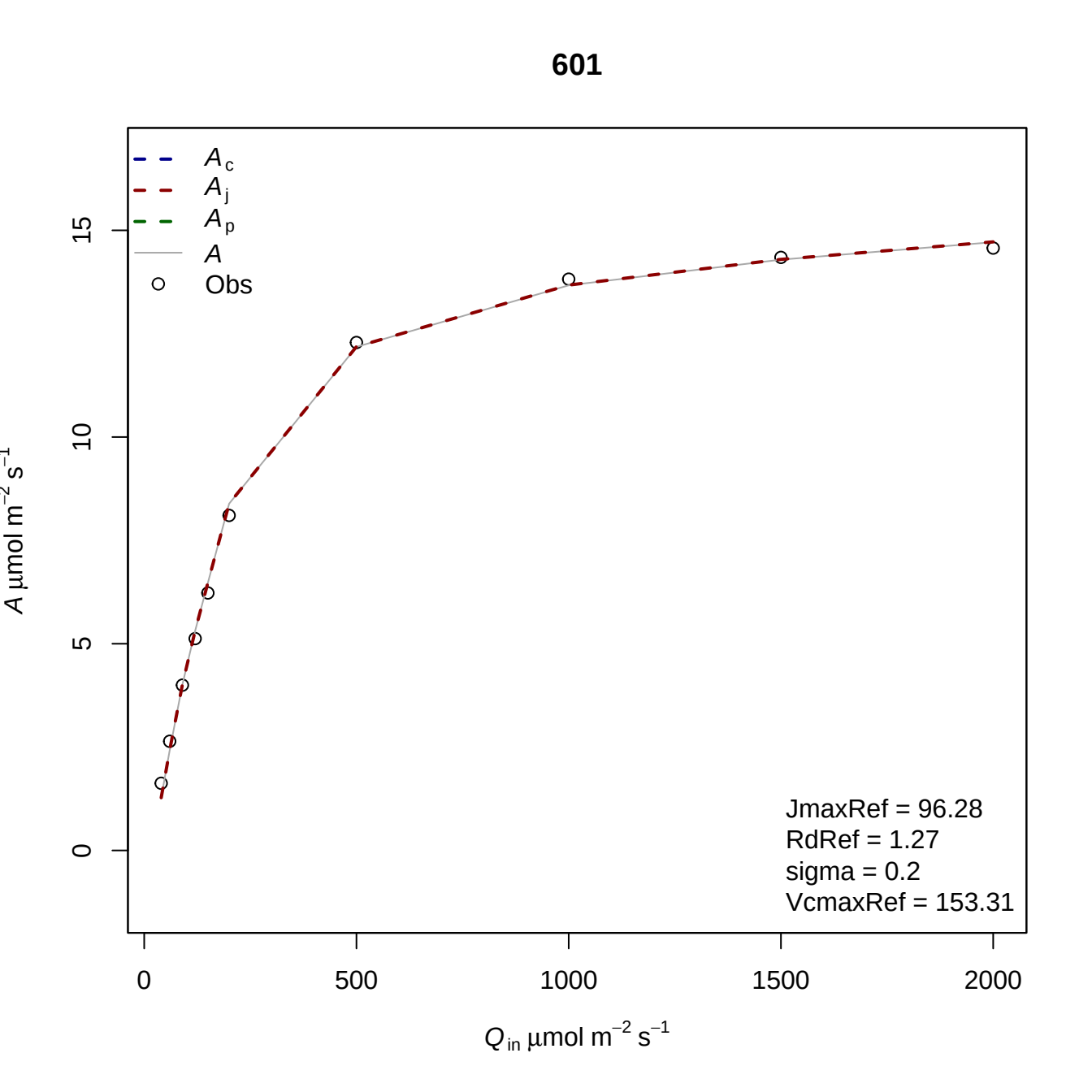

596
A_c
A_j
A_p
A
Obs
JmaxRef = 115.86
RdRef = 1.78
sigma = 0.62
VcmaxRef = 600.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

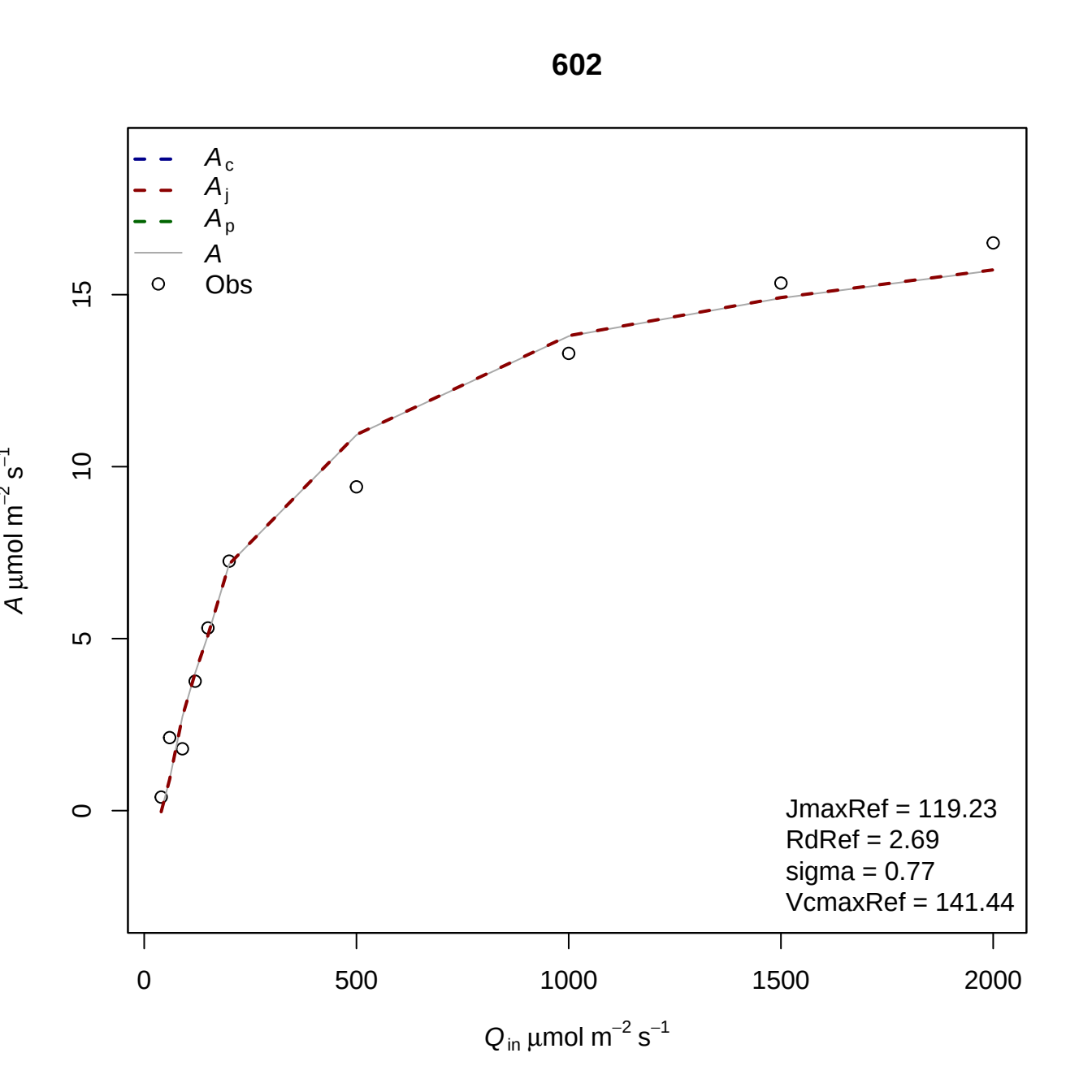

597
A_c
A_j
A_p
A
Obs
JmaxRef = 111.33
RdRef = 2.75
sigma = 0.49
VcmaxRef = 766.37
0
5
10
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

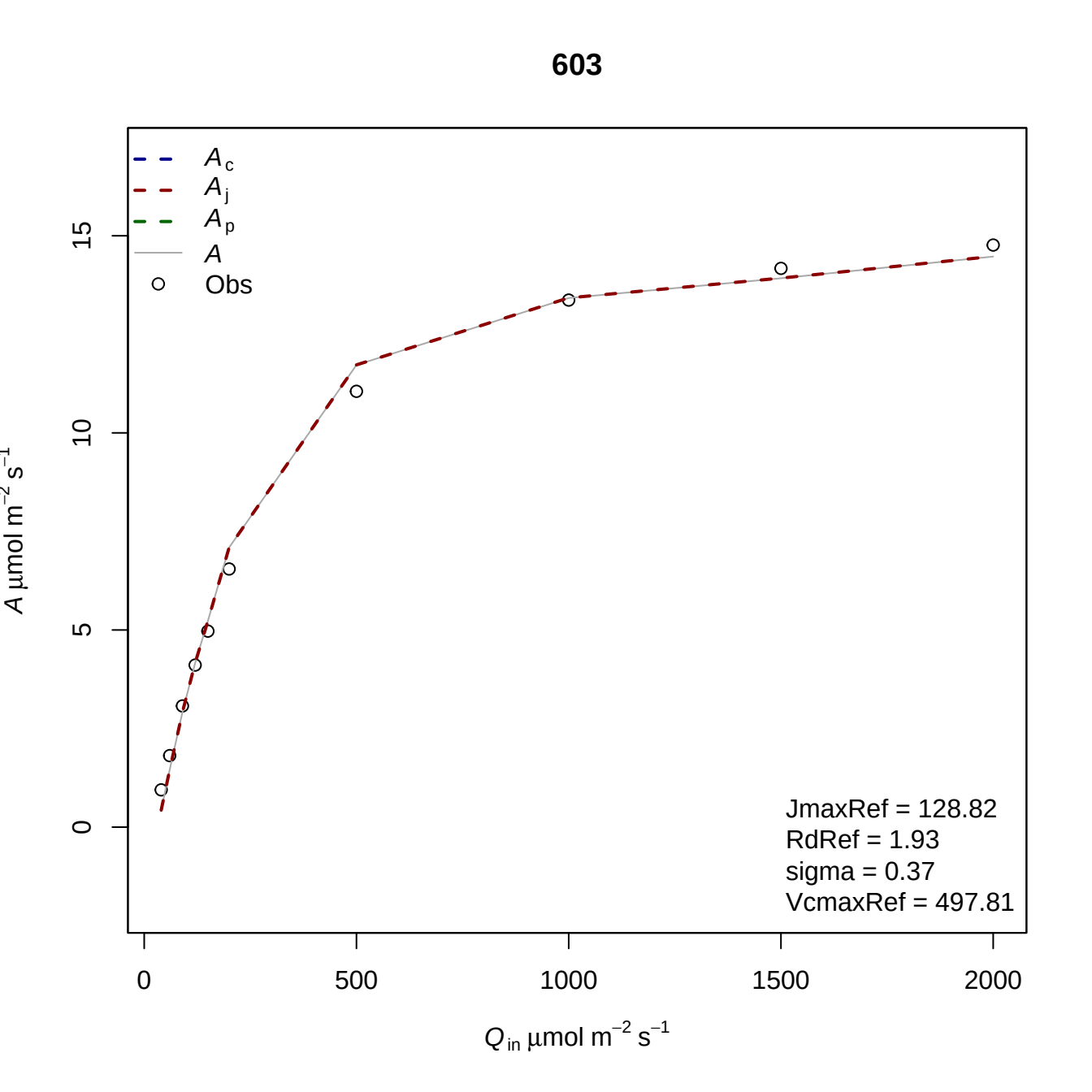

598
A_c
A_j
A_p
A
Obs
JmaxRef = 142.47
RdRef = 2.21
sigma = 0.16
VcmaxRef = 155.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
500
1000
1500
2000

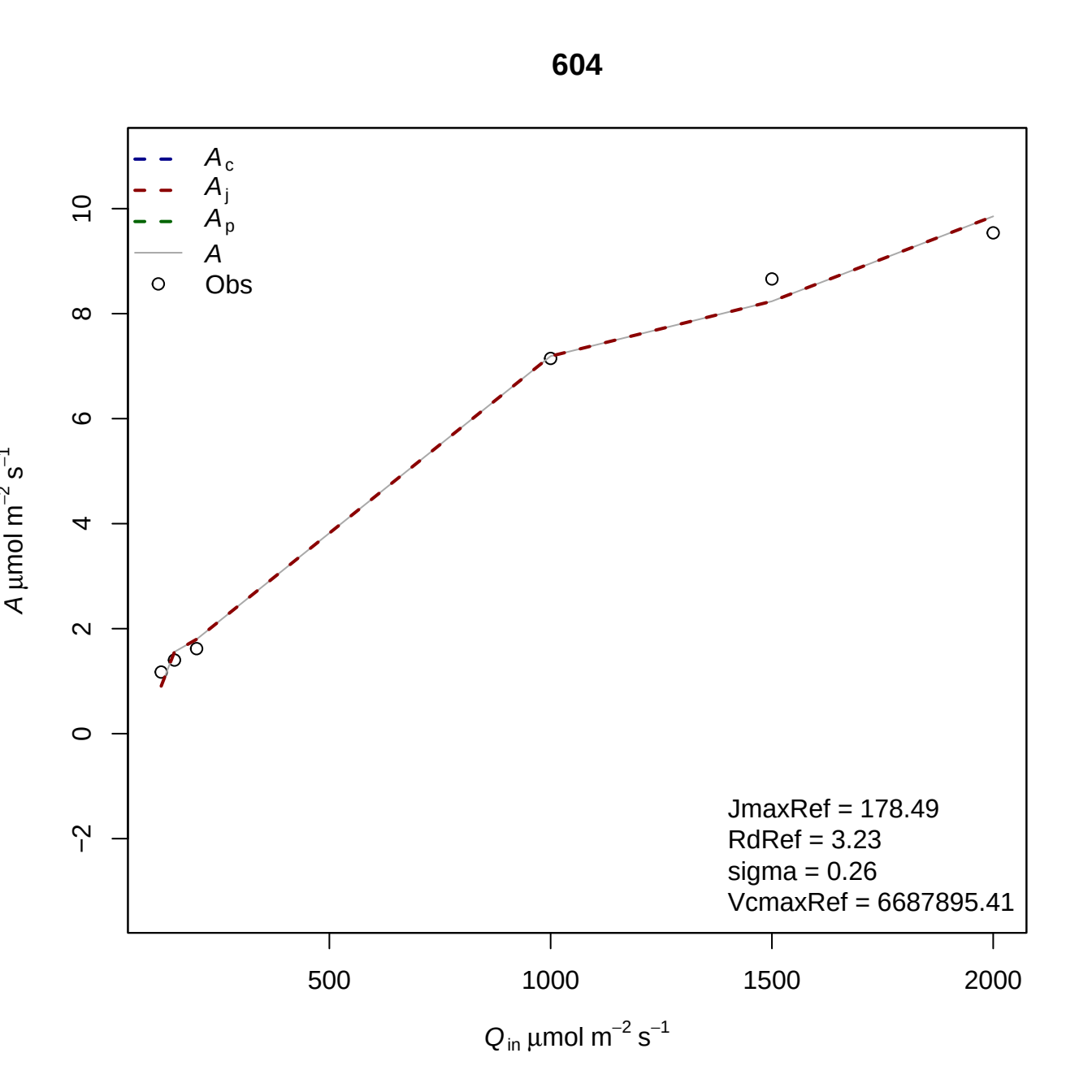

599
A_c
A_j
A_p
A
Obs
JmaxRef = 138.94
RdRef = 3.54
sigma = 0.69
VcmaxRef = 63373.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
500
1000
1500
2000

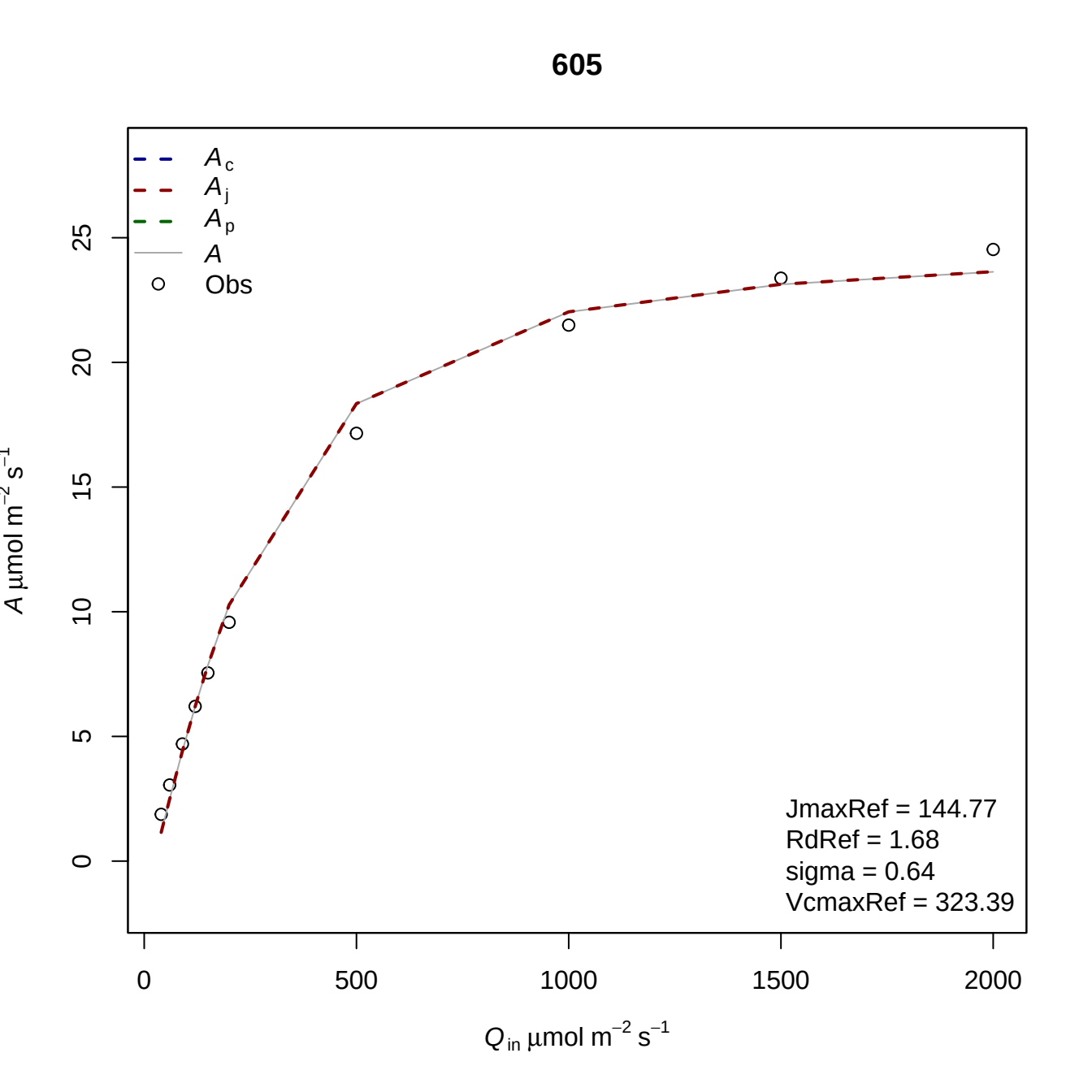

600
A_c
A_j
A_p
A
Obs
JmaxRef = 0
RdRef = −3.96
sigma = 2.32
VcmaxRef = 69.53
8
6
4
2
0
−2
−4
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

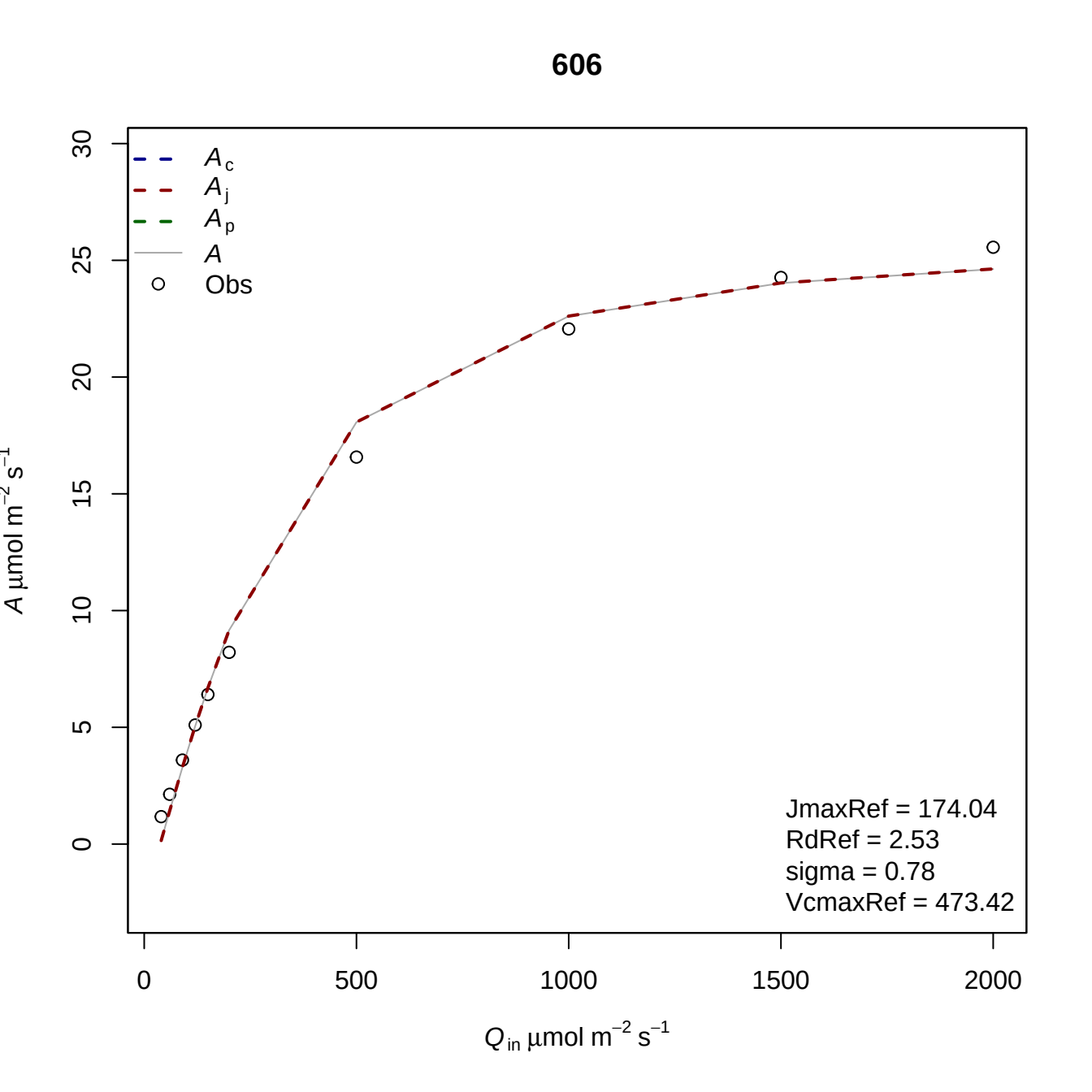

601
A_c
A_j
A_p
A
Obs
JmaxRef = 96.28
RdRef = 1.27
sigma = 0.2
VcmaxRef = 153.31
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

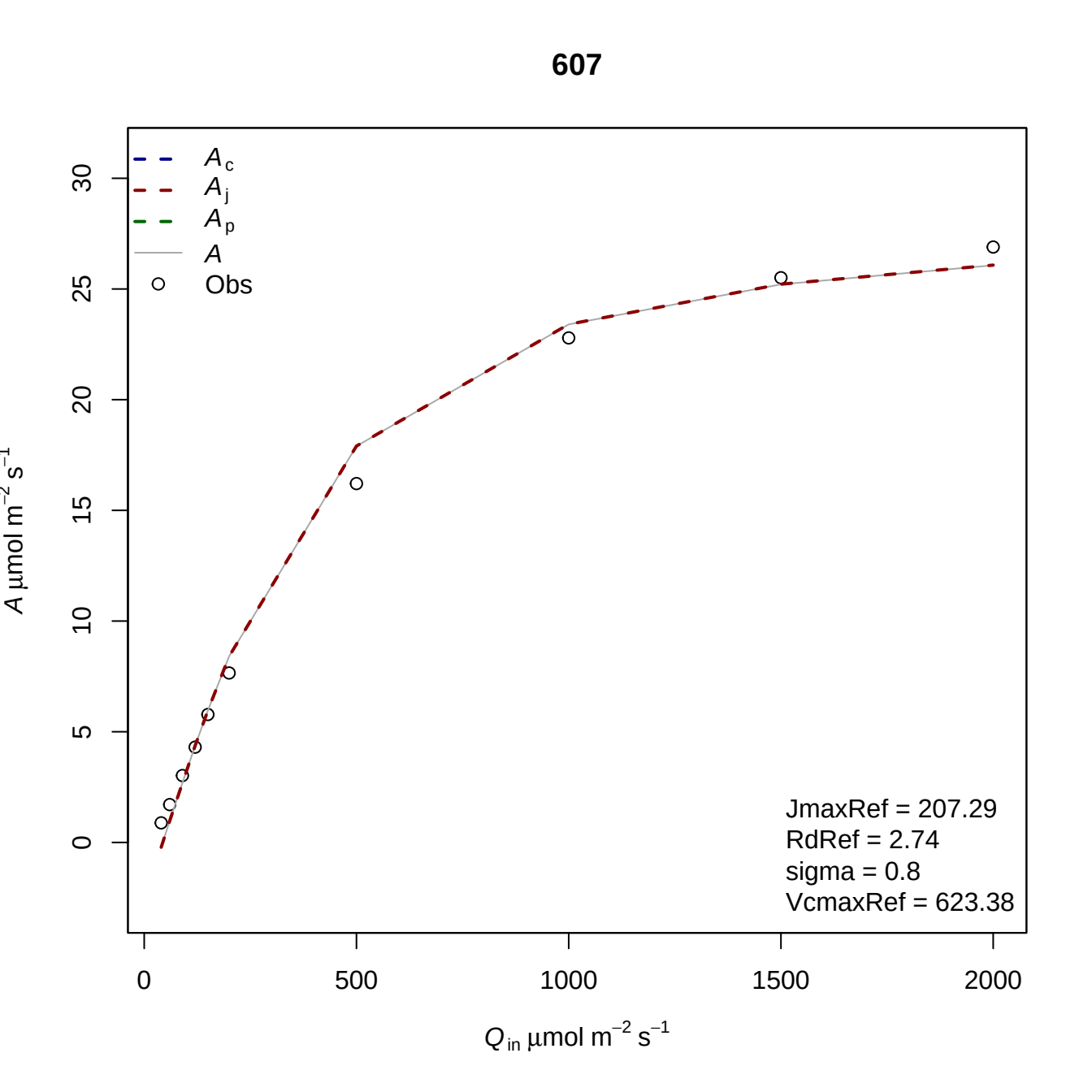

602
A_c
A_j
A_p
A
Obs
JmaxRef = 119.23
RdRef = 2.69
sigma = 0.77
VcmaxRef = 141.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

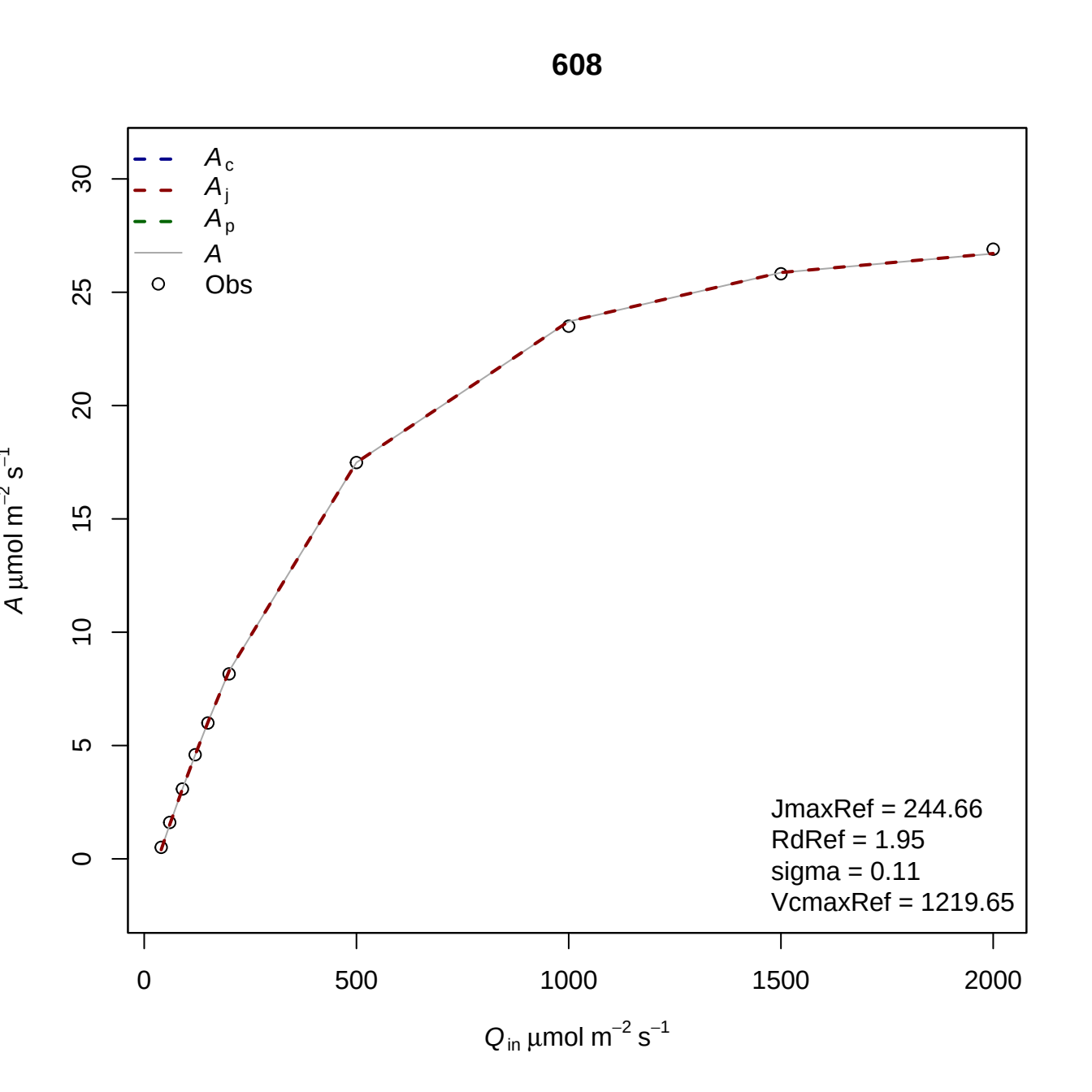

603
A_c
A_j
A_p
A
Obs
JmaxRef = 128.82
RdRef = 1.93
sigma = 0.37
VcmaxRef = 497.81
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

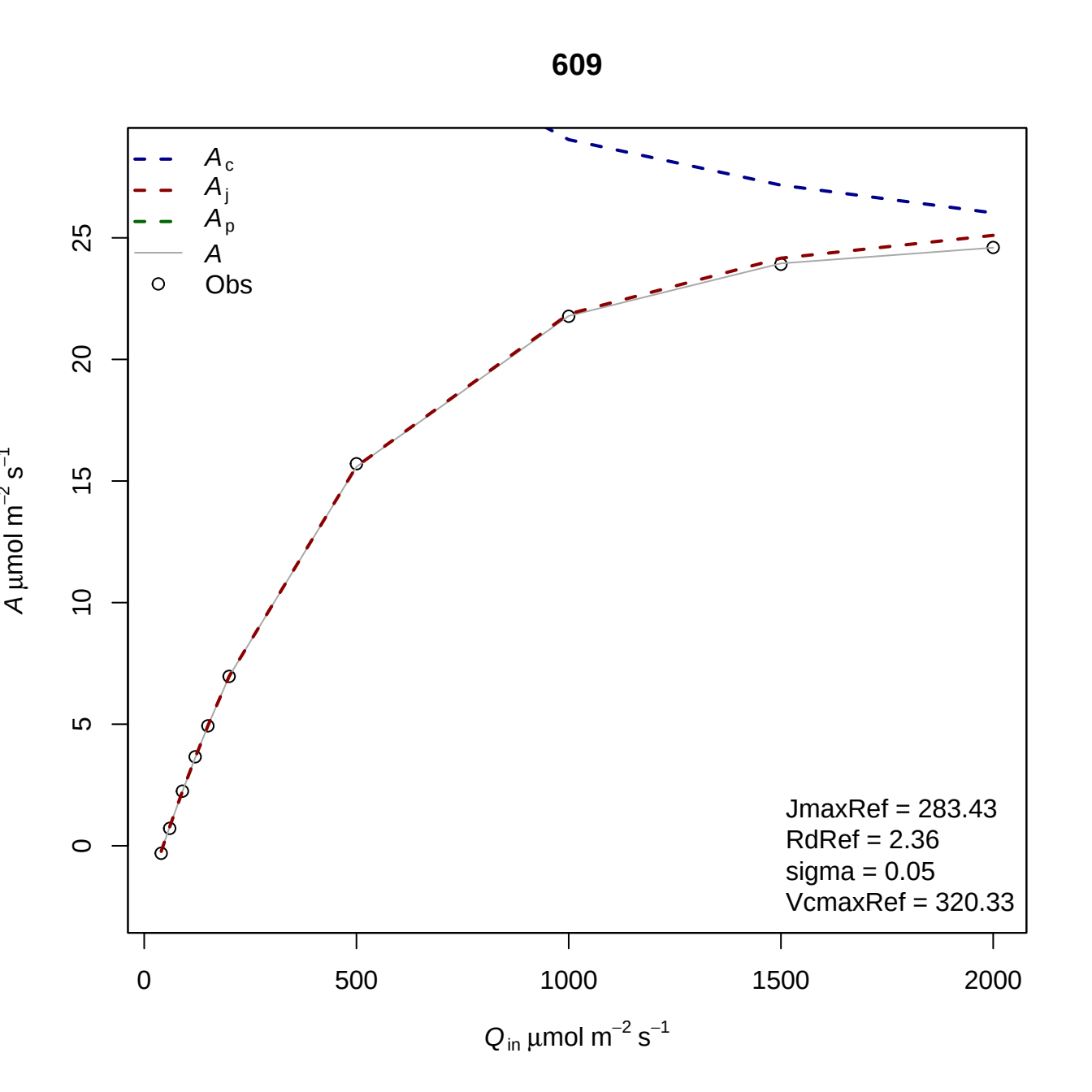

604
A_c
A_j
A_p
A
Obs
JmaxRef = 178.49
RdRef = 3.23
sigma = 0.26
VcmaxRef = 6687895.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
500
1000
1500
2000

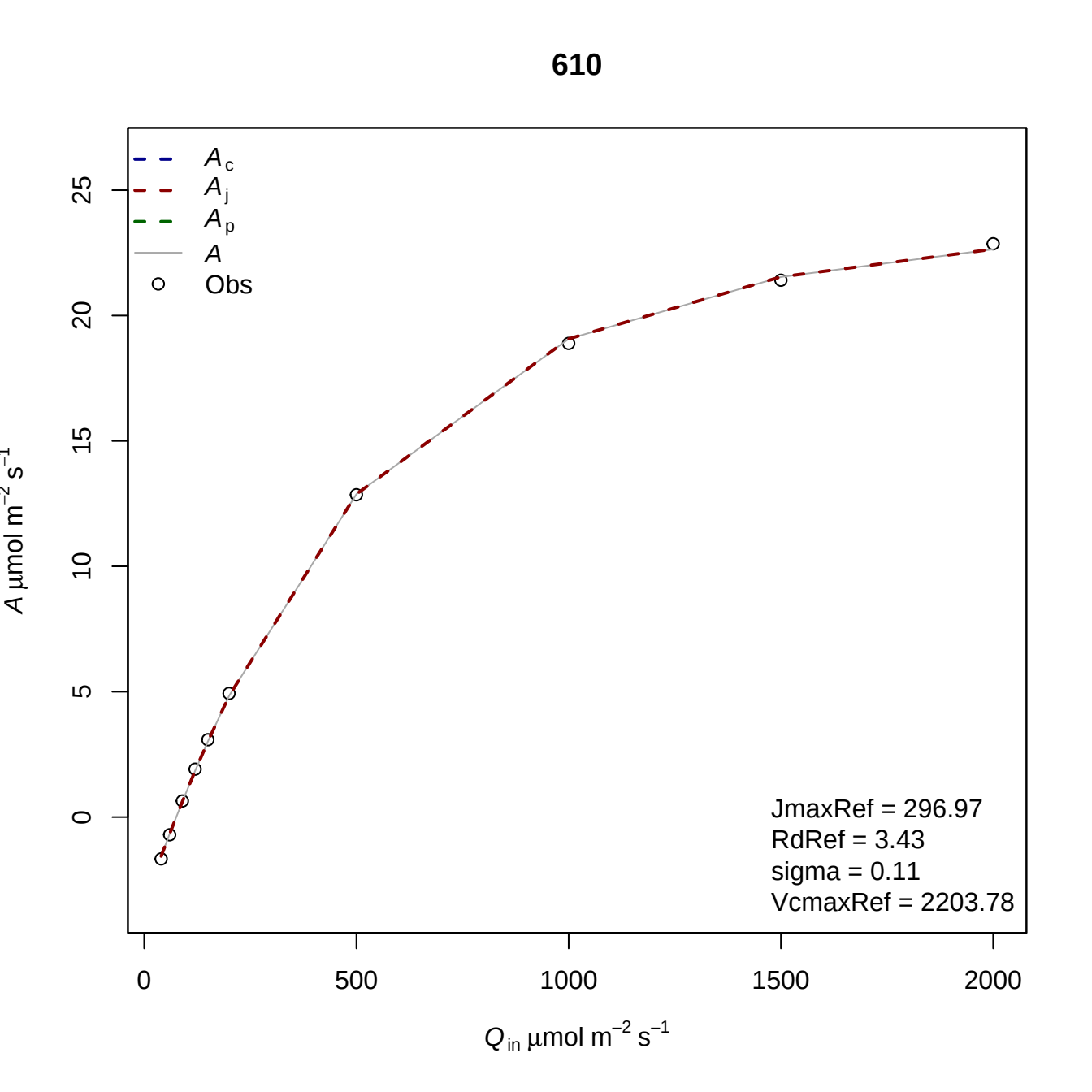

605
A_c
A_j
A_p
A
Obs
JmaxRef = 144.77
RdRef = 1.68
sigma = 0.64
VcmaxRef = 323.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

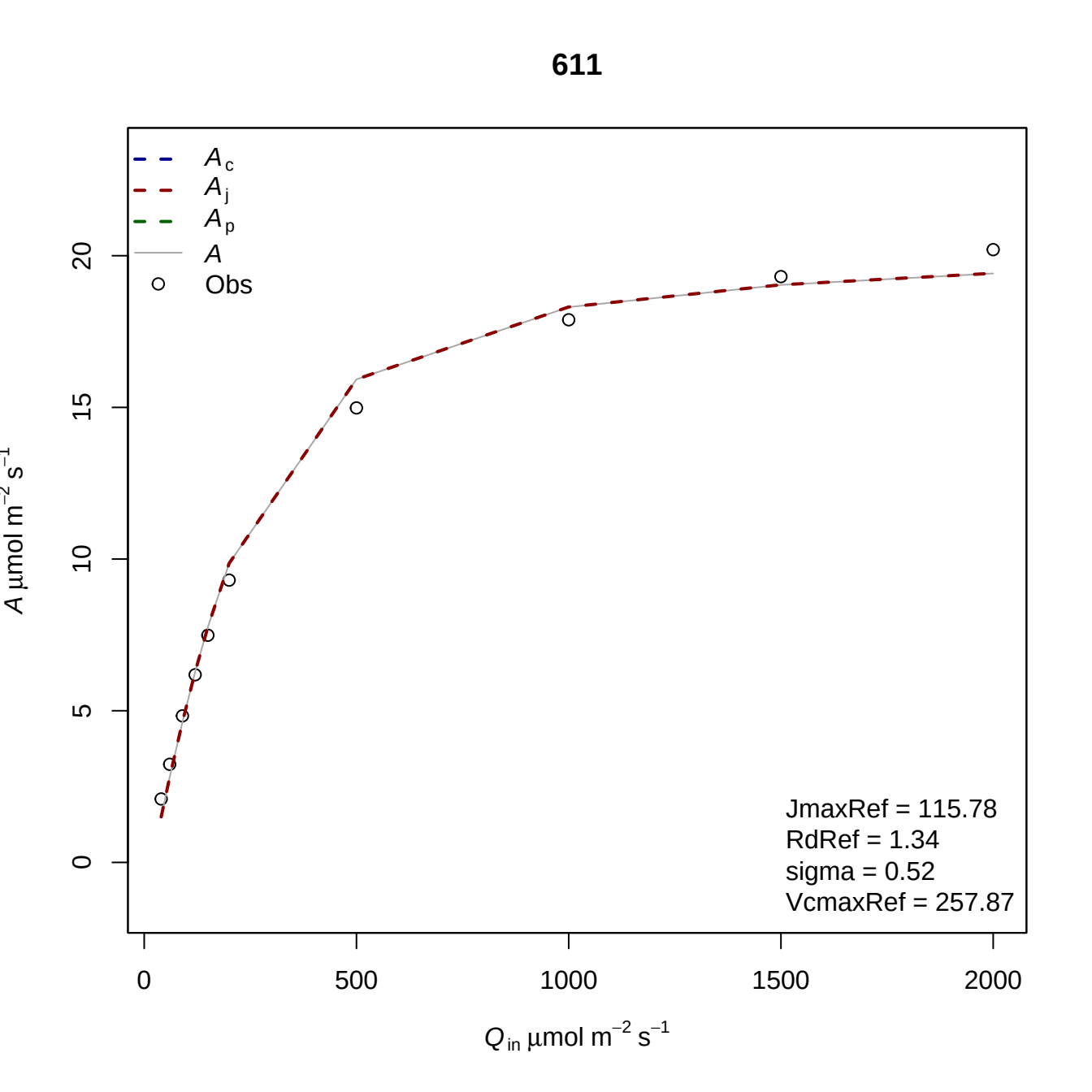

606
A_c
A_j
A_p
A
Obs
JmaxRef = 174.04
RdRef = 2.53
sigma = 0.78
VcmaxRef = 473.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

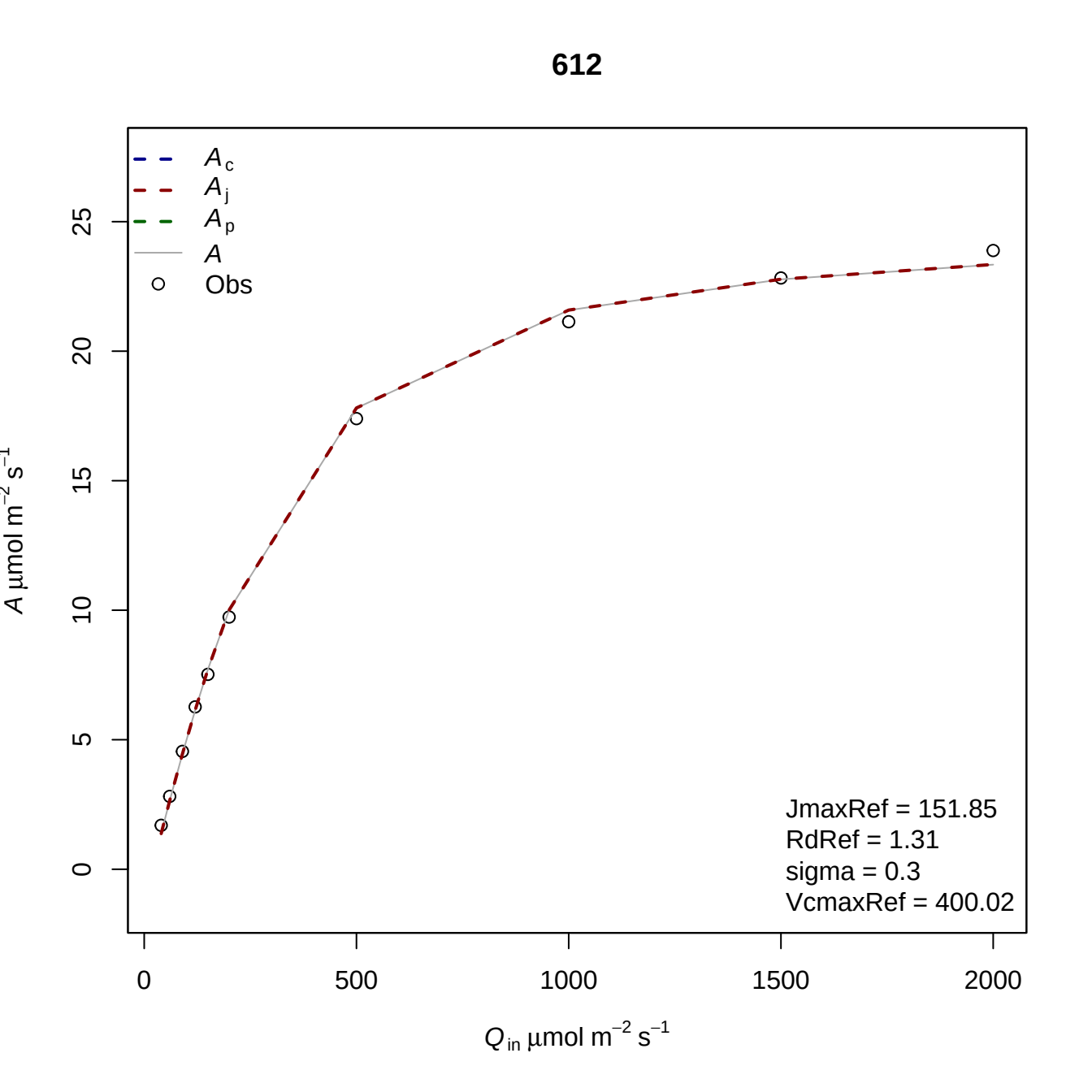

607
A_c
A_j
A_p
A
Obs
JmaxRef = 207.29
RdRef = 2.74
sigma = 0.8
VcmaxRef = 623.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

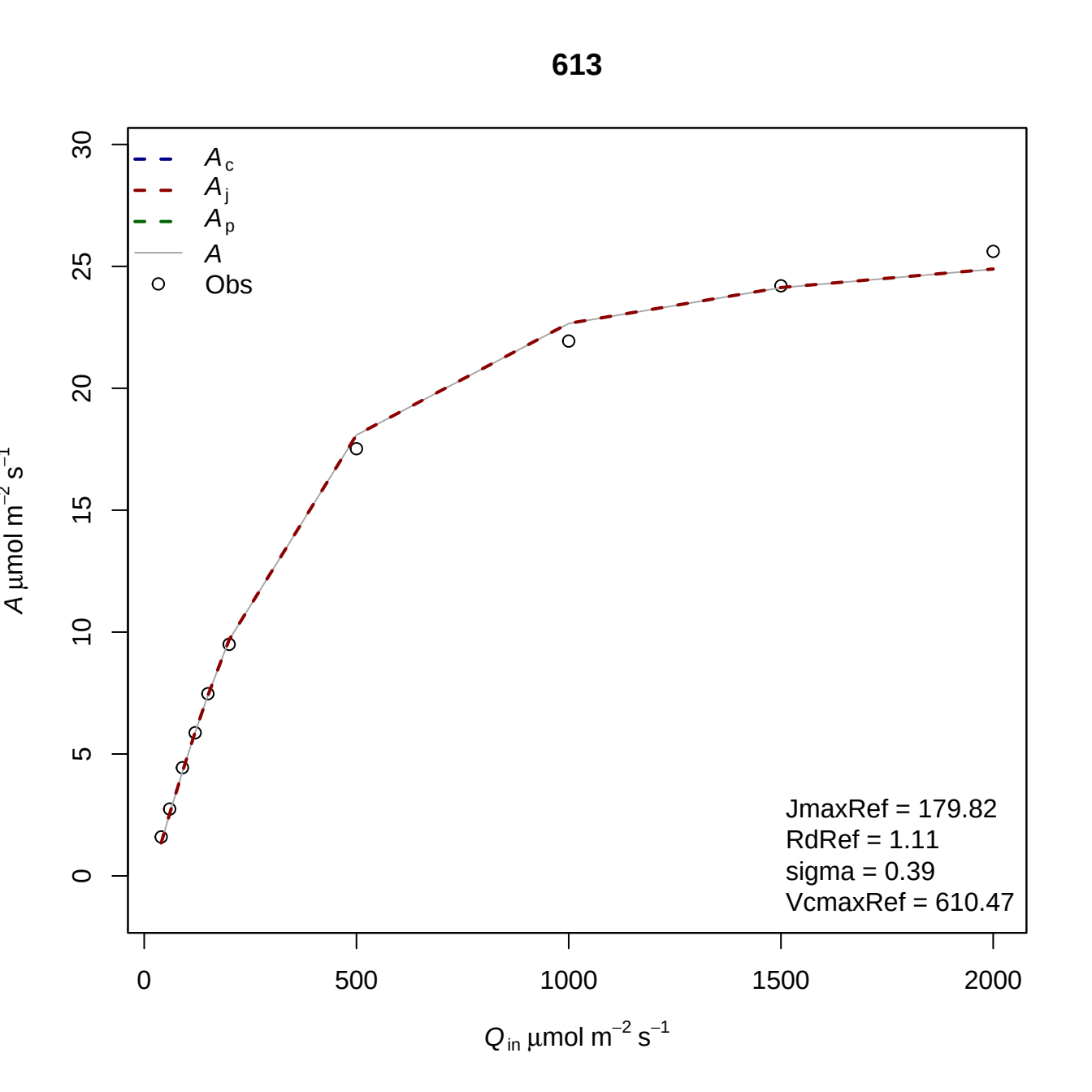

608
A_c
A_j
A_p
A
Obs
JmaxRef = 244.66
RdRef = 1.95
sigma = 0.11
VcmaxRef = 1219.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

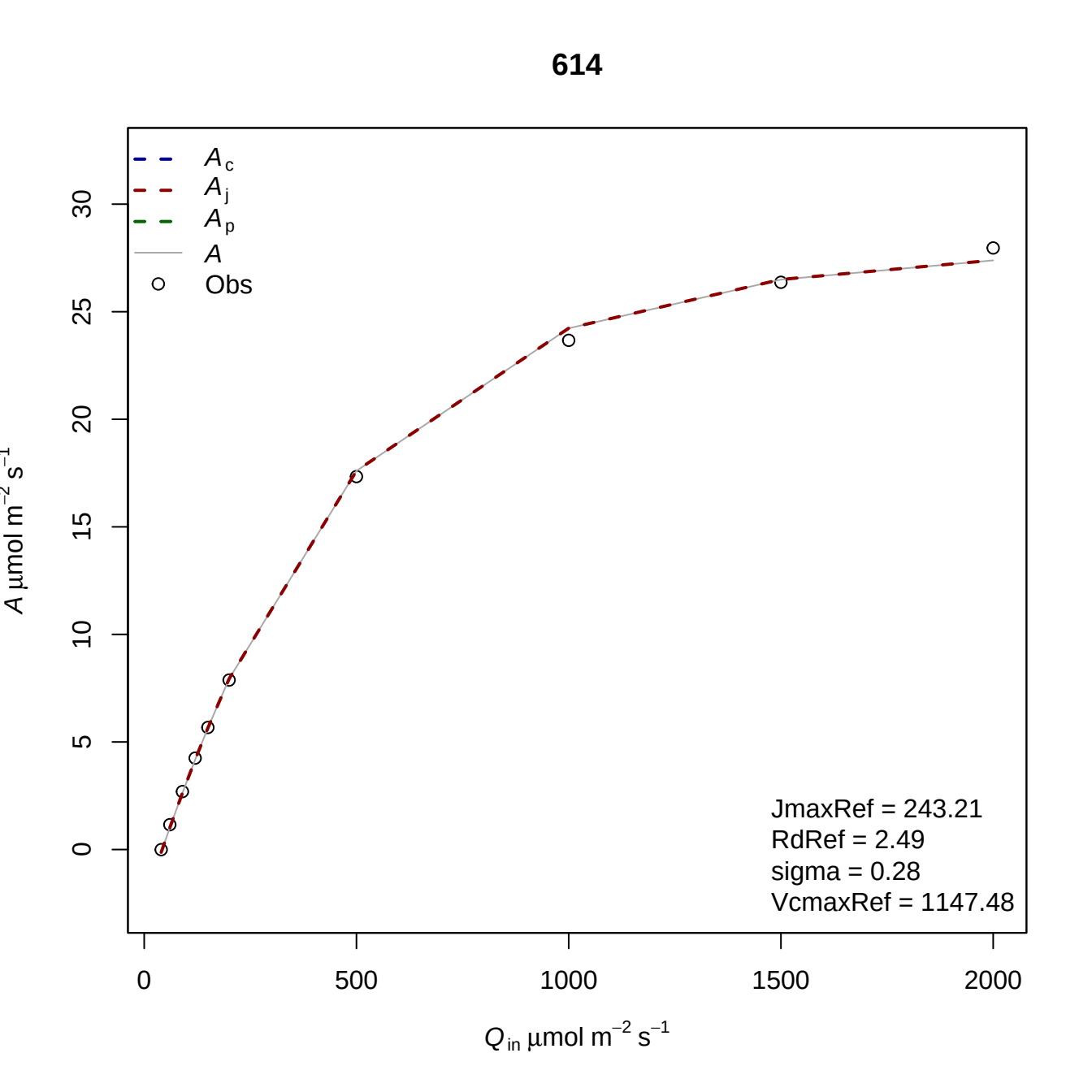

609
A_c
A_j
A_p
A
Obs
JmaxRef = 283.43
RdRef = 2.36
sigma = 0.05
VcmaxRef = 320.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

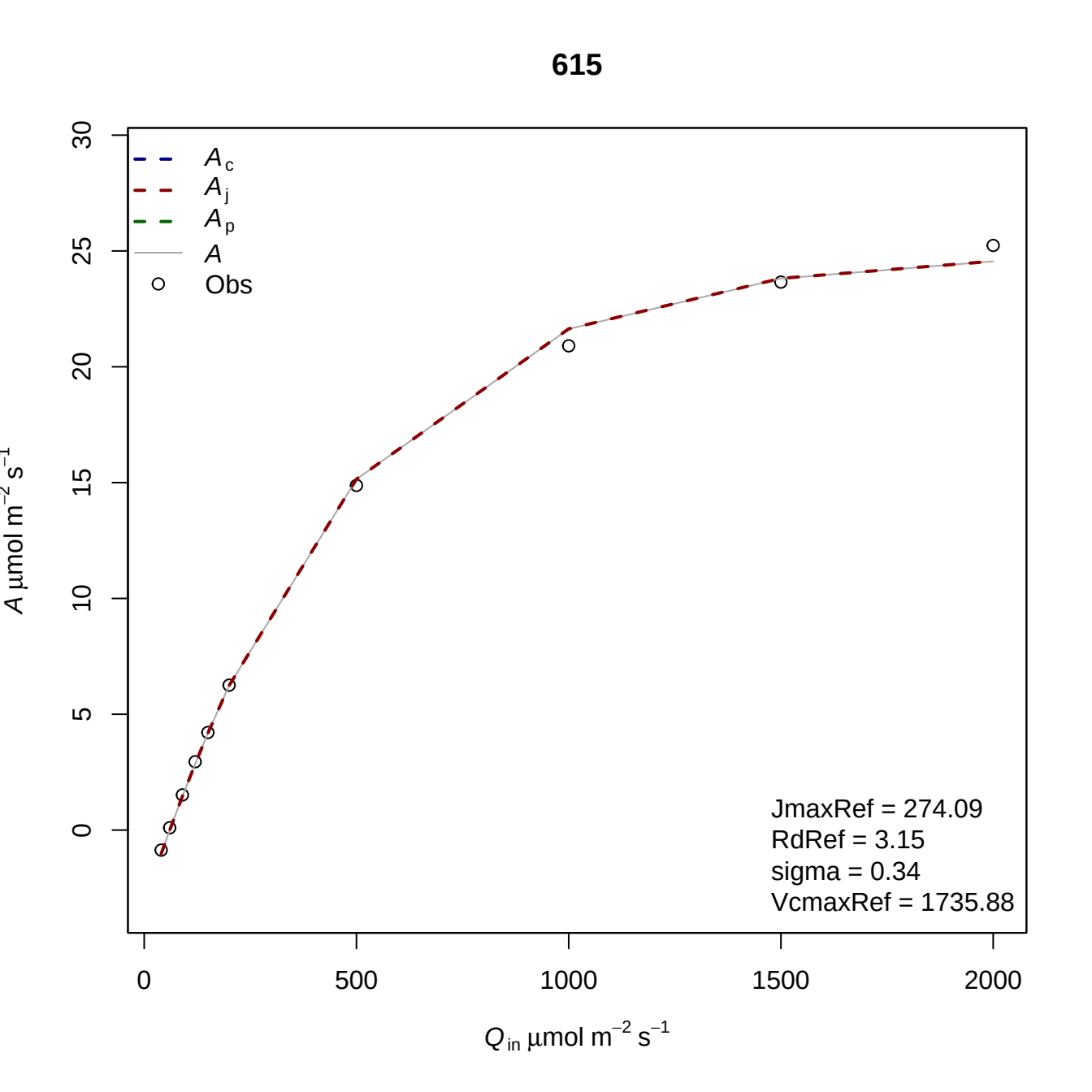

610
A_c
A_j
A_p
A
Obs
JmaxRef = 296.97
RdRef = 3.43
sigma = 0.11
VcmaxRef = 2203.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

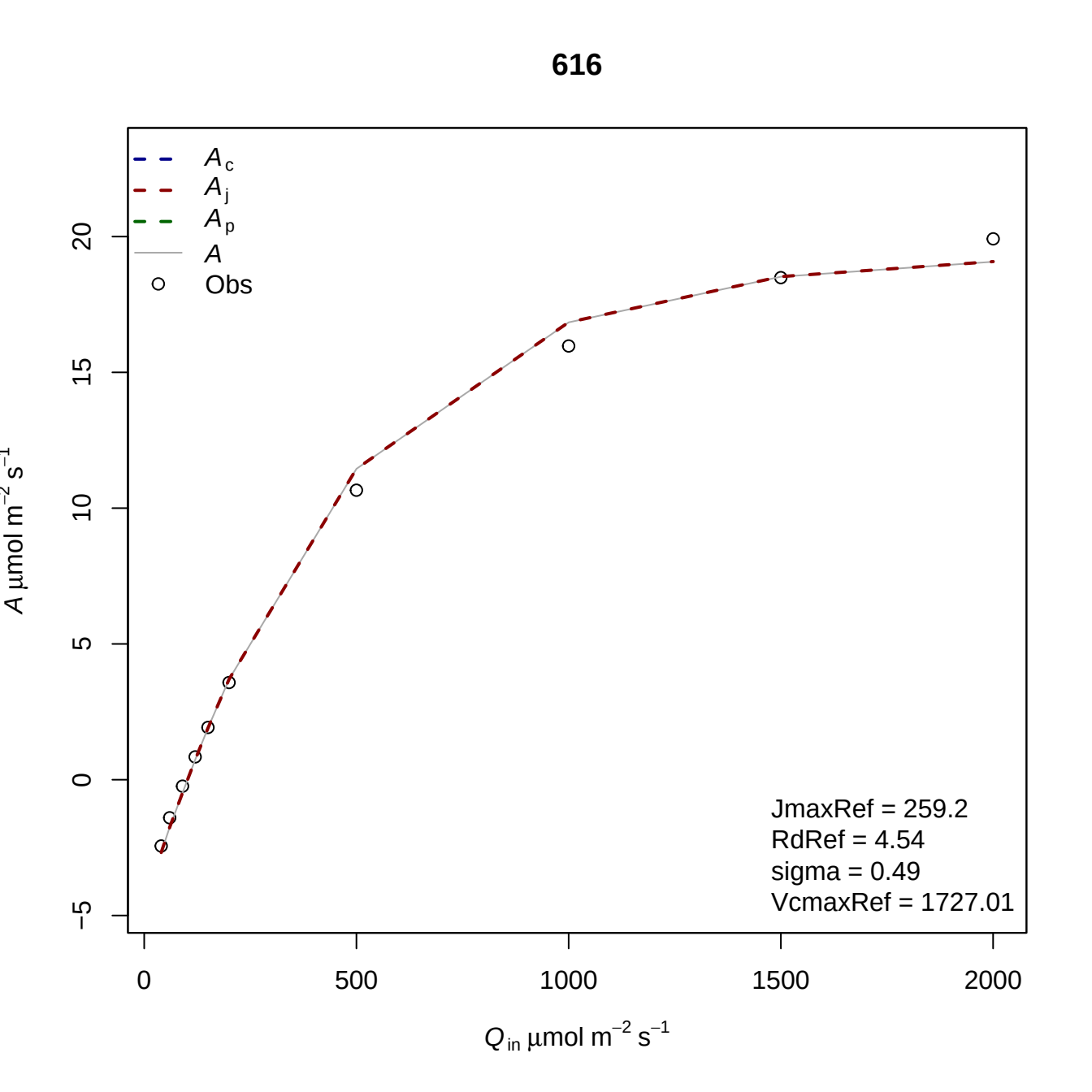

611
A_c
A_j
A_p
A
Obs
JmaxRef = 115.78
RdRef = 1.34
sigma = 0.52
VcmaxRef = 257.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

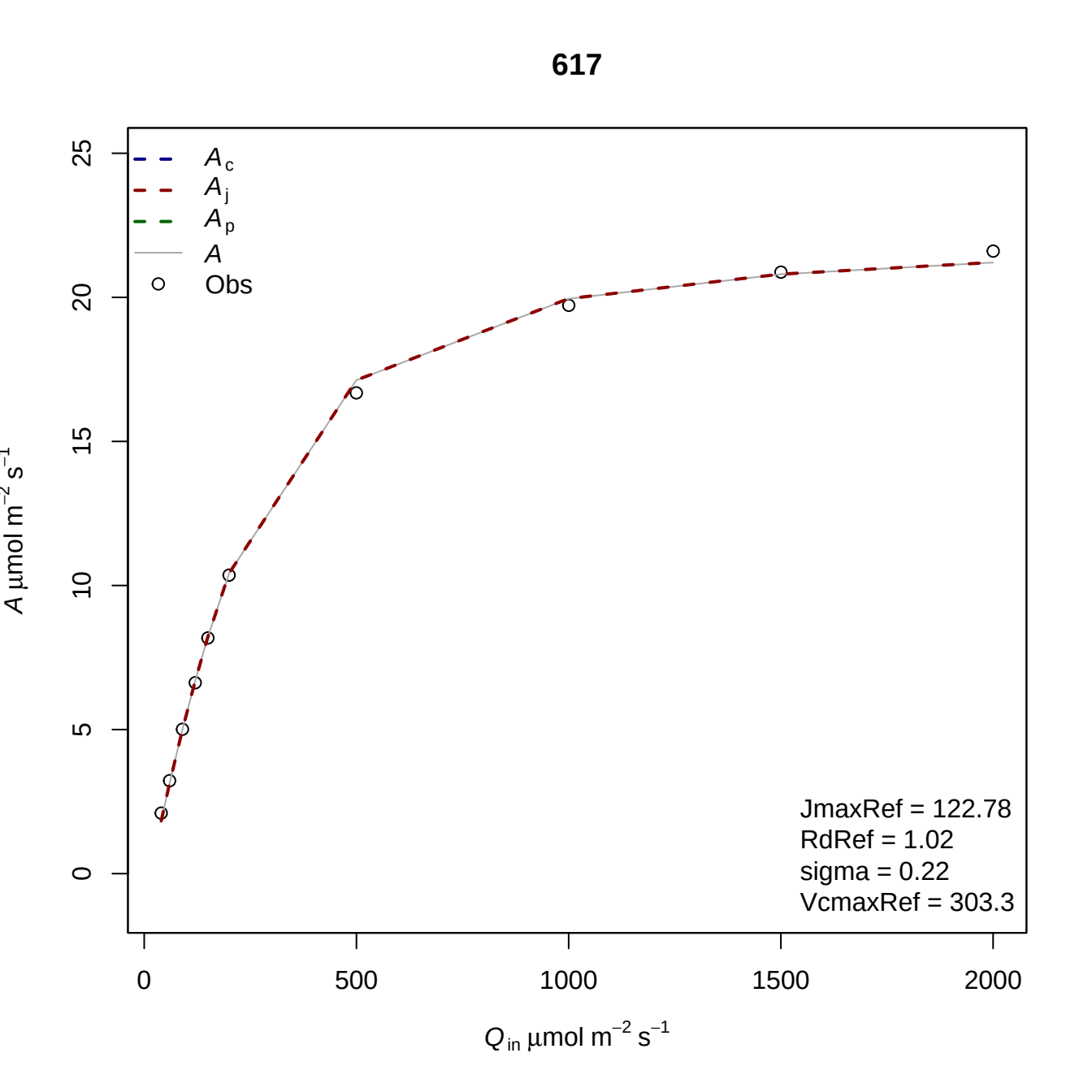

612
A_c
A_j
A_p
A
Obs
JmaxRef = 151.85
RdRef = 1.31
sigma = 0.3
VcmaxRef = 400.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

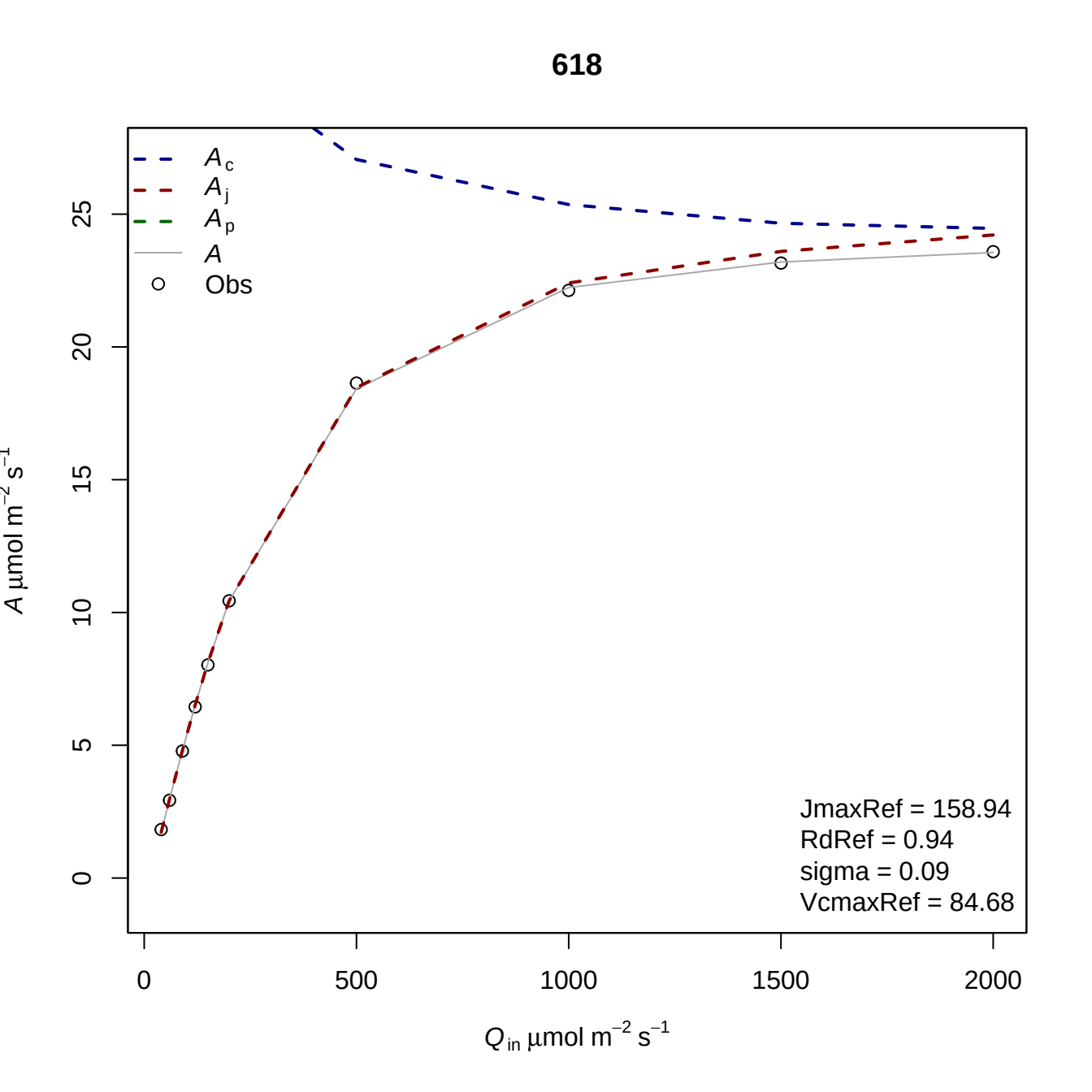

613
A_c
A_j
A_p
A
Obs
JmaxRef = 179.82
RdRef = 1.11
sigma = 0.39
VcmaxRef = 610.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

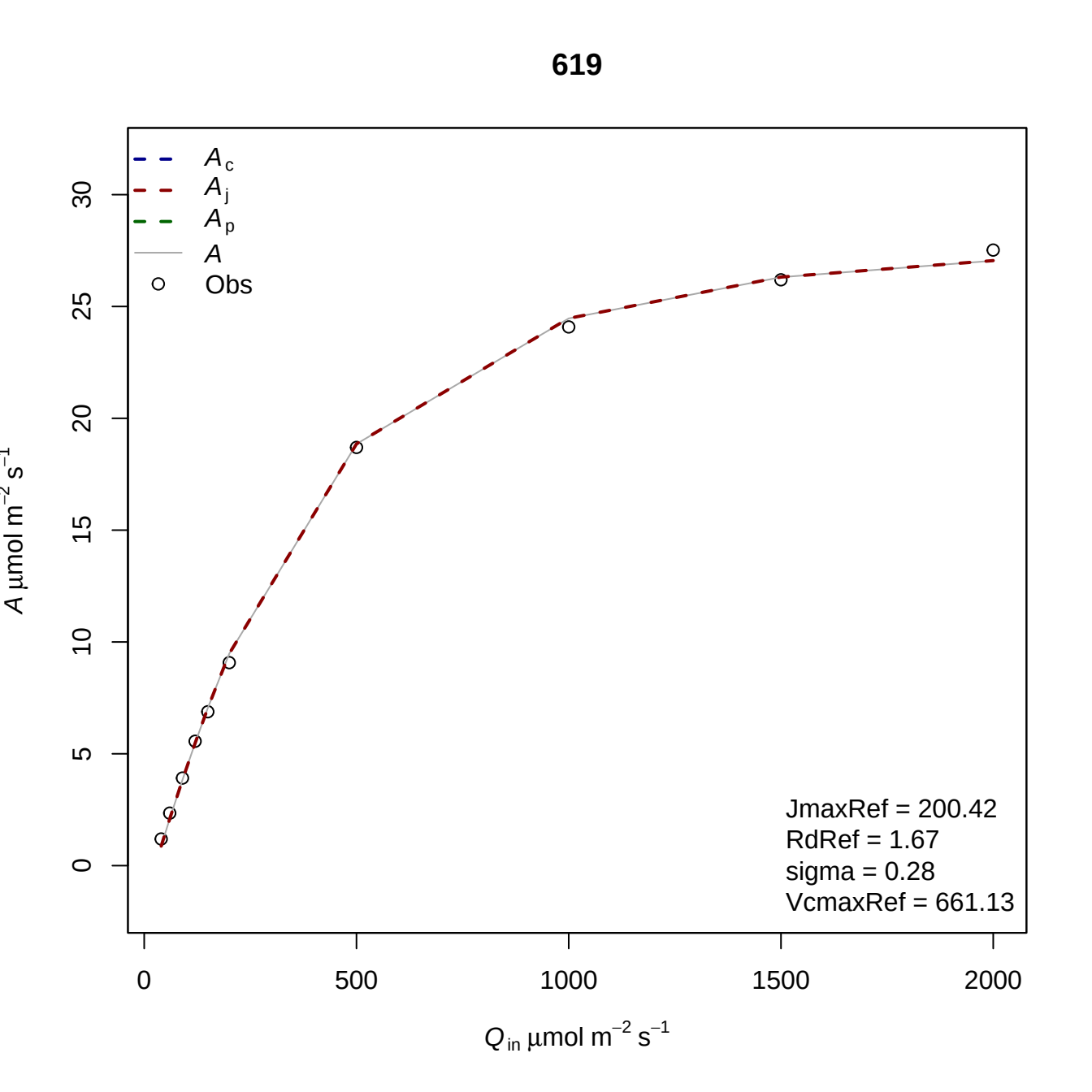

614
A_c
A_j
A_p
A
Obs
JmaxRef = 243.21
RdRef = 2.49
sigma = 0.28
VcmaxRef = 1147.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

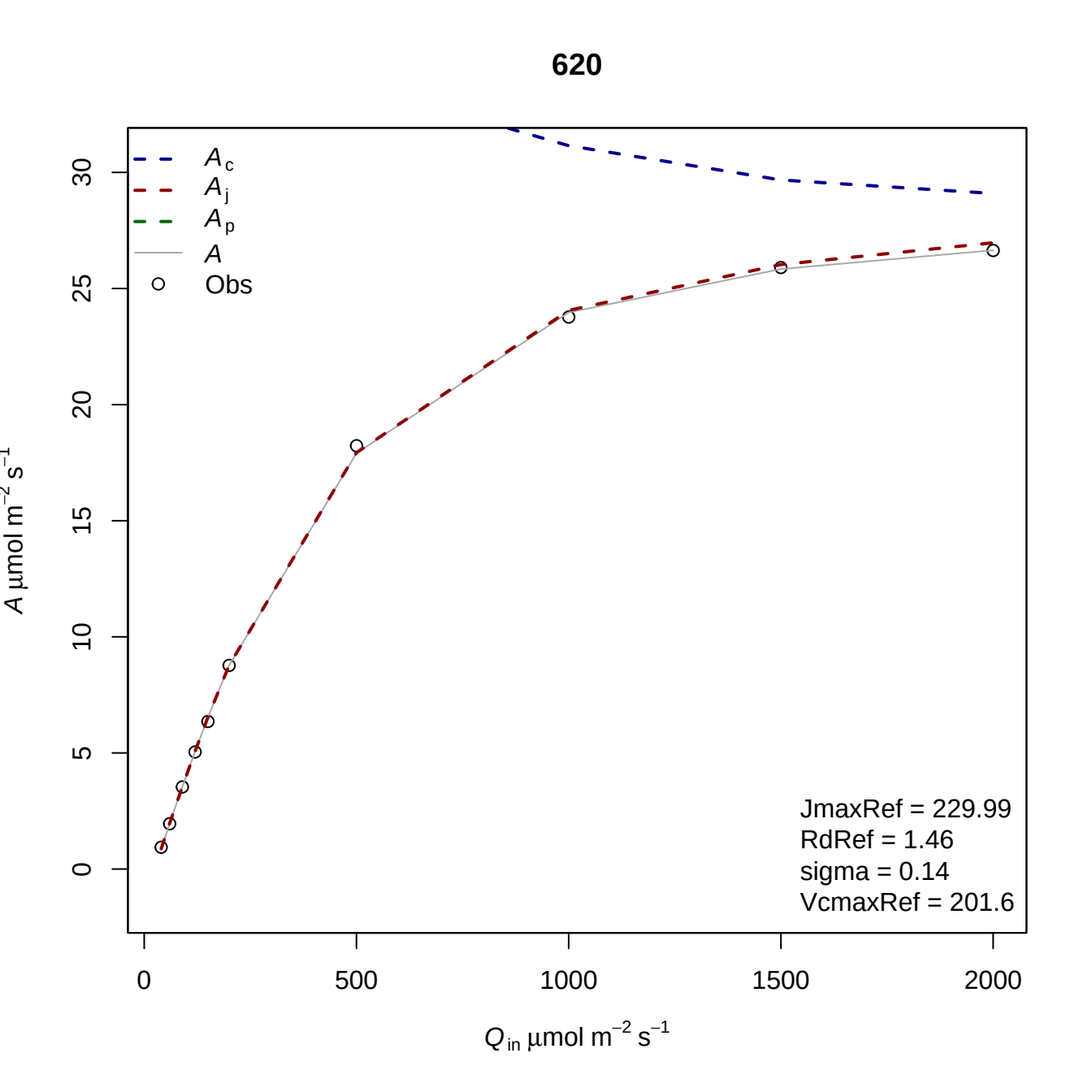

615
A_c
A_j
A_p
A
Obs
JmaxRef = 274.09
RdRef = 3.15
sigma = 0.34
VcmaxRef = 1735.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

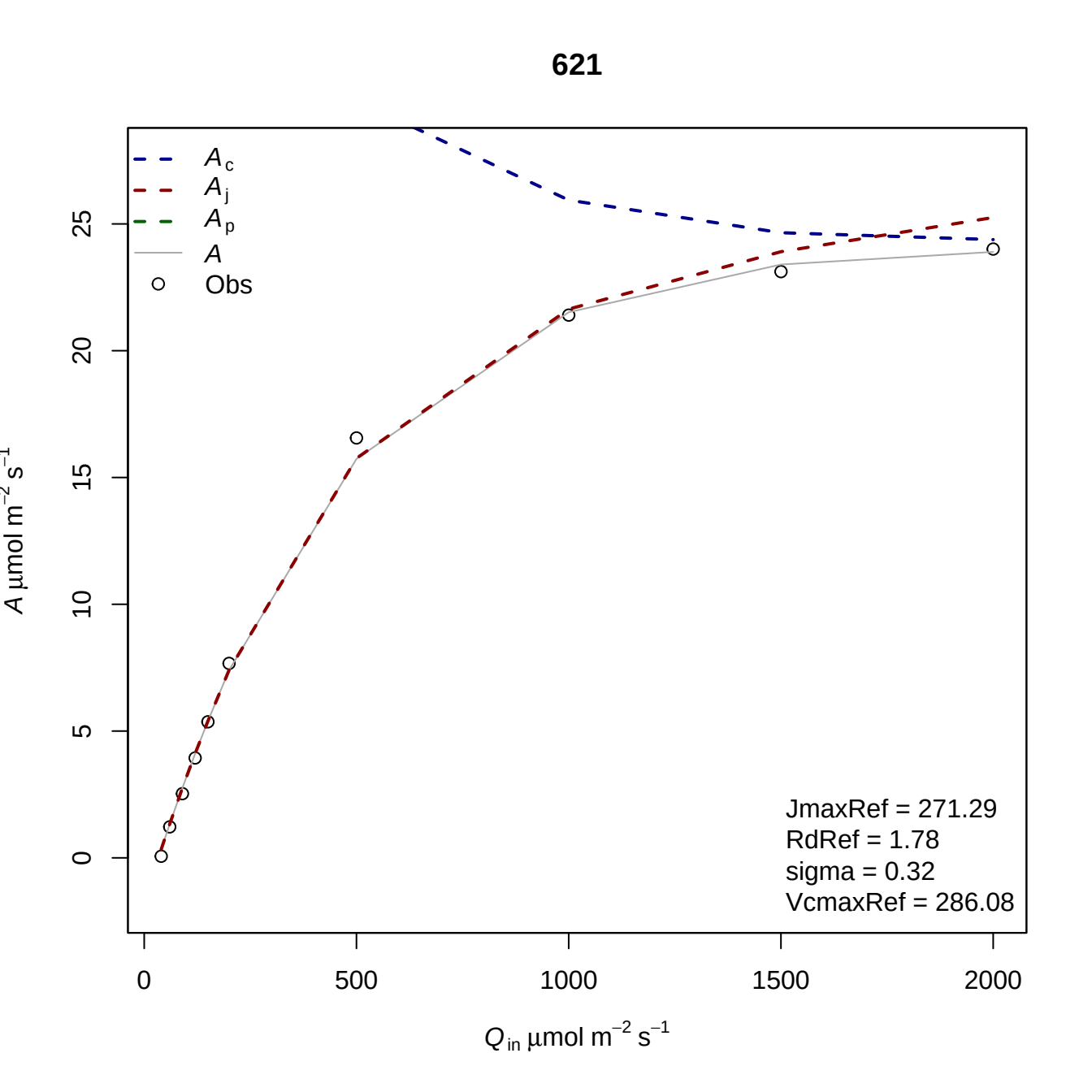

616
A_c
A_j
A_p
A
Obs
JmaxRef = 259.2
RdRef = 4.54
sigma = 0.49
VcmaxRef = 1727.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

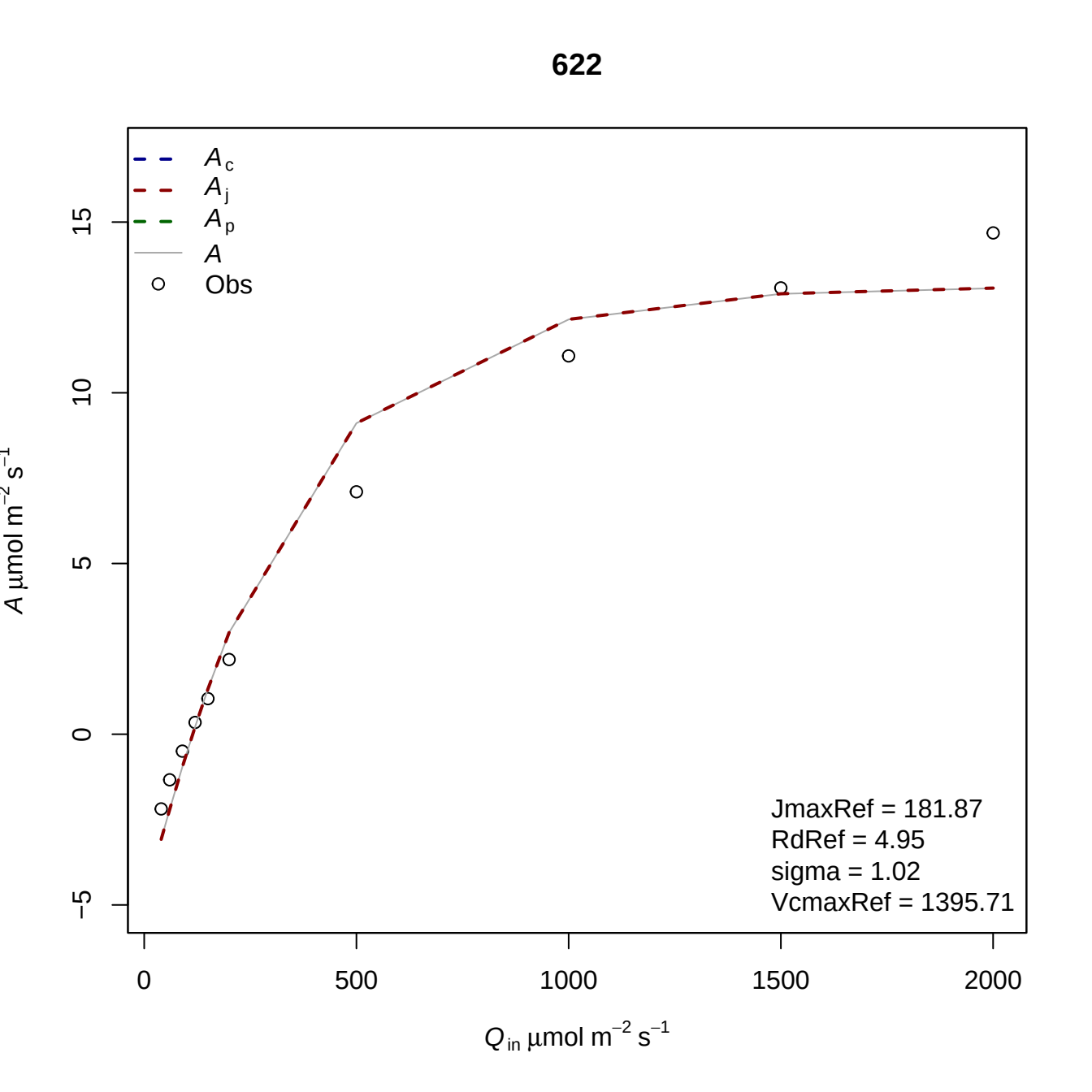

617
A_c
A_j
A_p
A
Obs
JmaxRef = 122.78
RdRef = 1.02
sigma = 0.22
VcmaxRef = 303.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

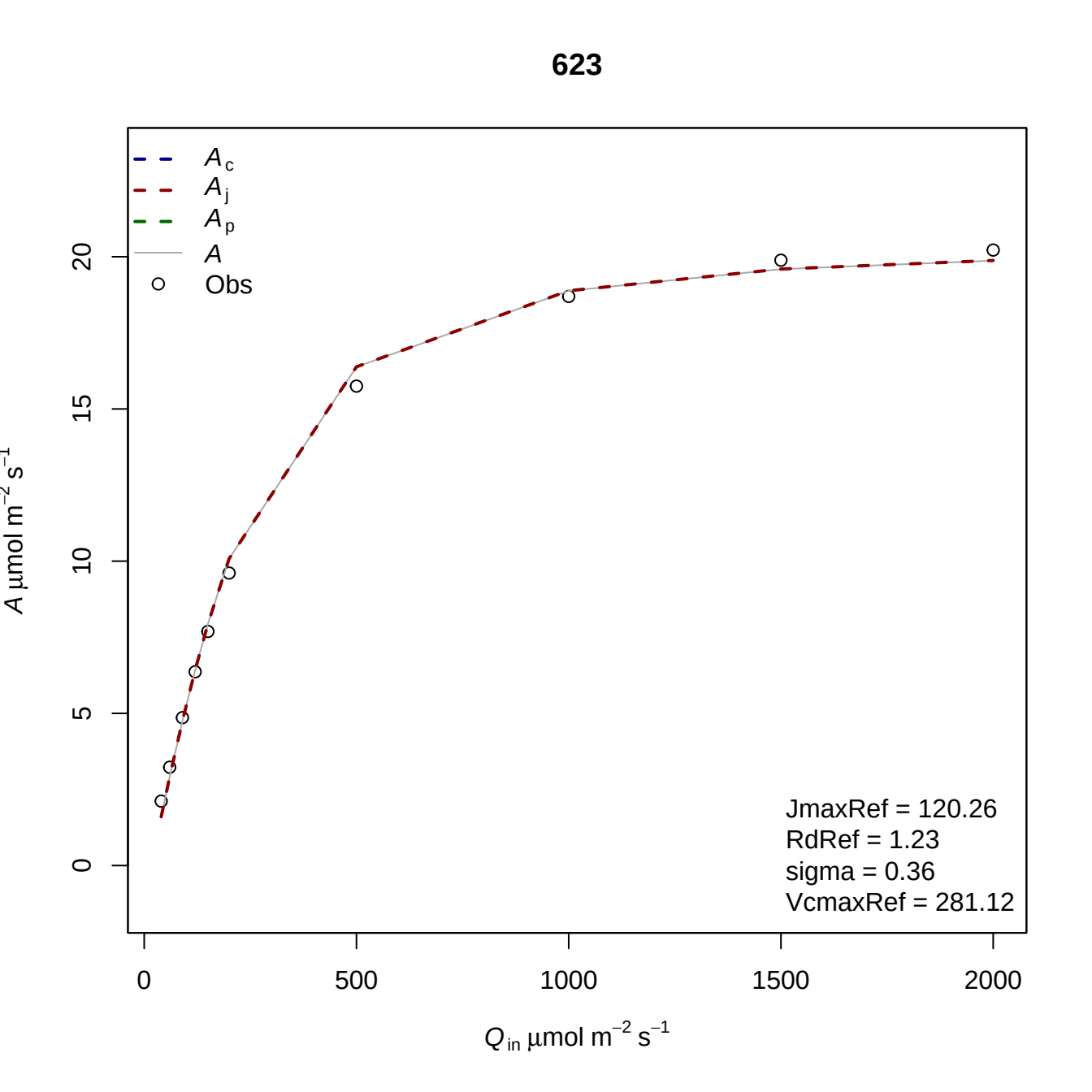

618
A_c
A_j
A_p
A
Obs
JmaxRef = 158.94
RdRef = 0.94
sigma = 0.09
VcmaxRef = 84.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

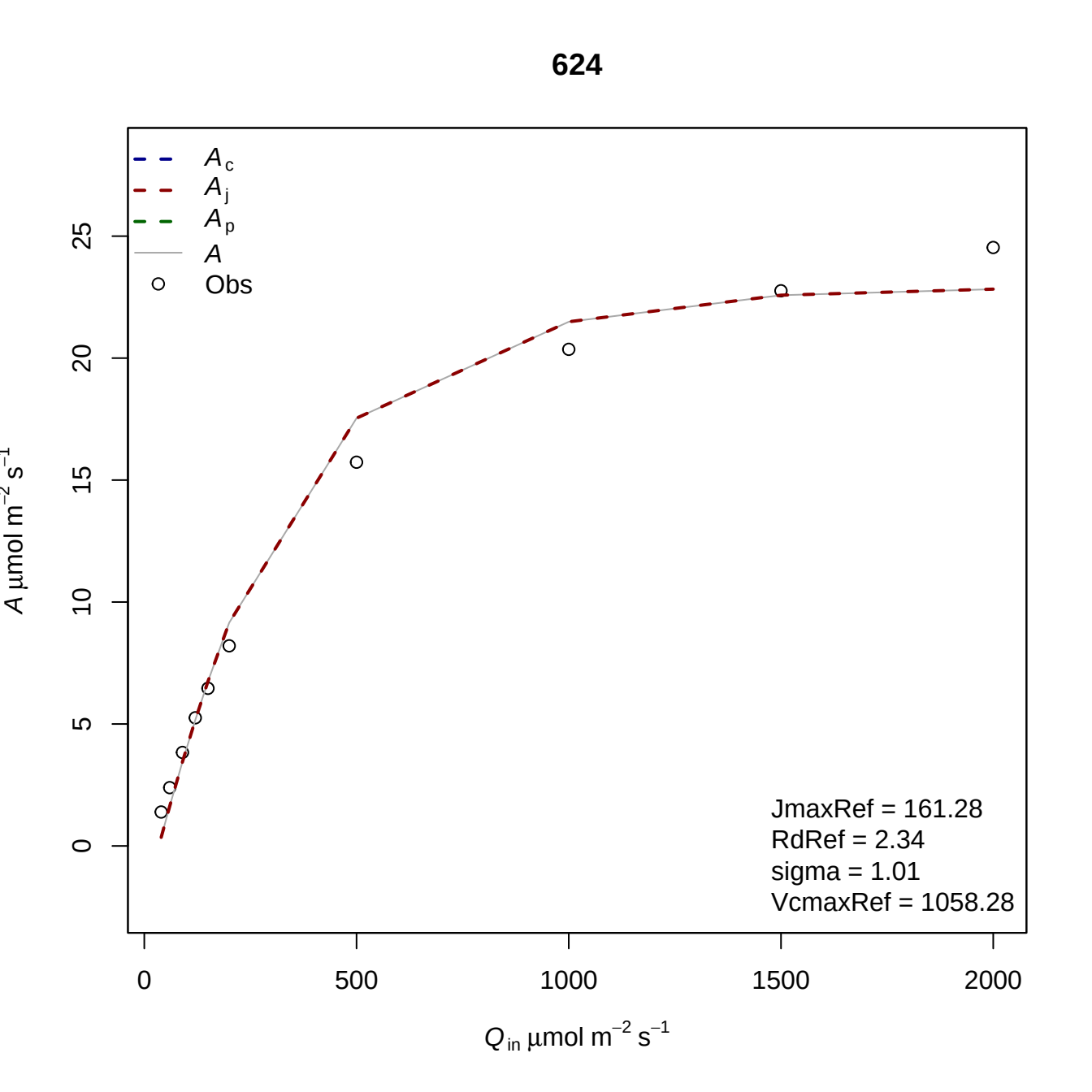

619
A_c
A_j
A_p
A
Obs
JmaxRef = 200.42
RdRef = 1.67
sigma = 0.28
VcmaxRef = 661.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

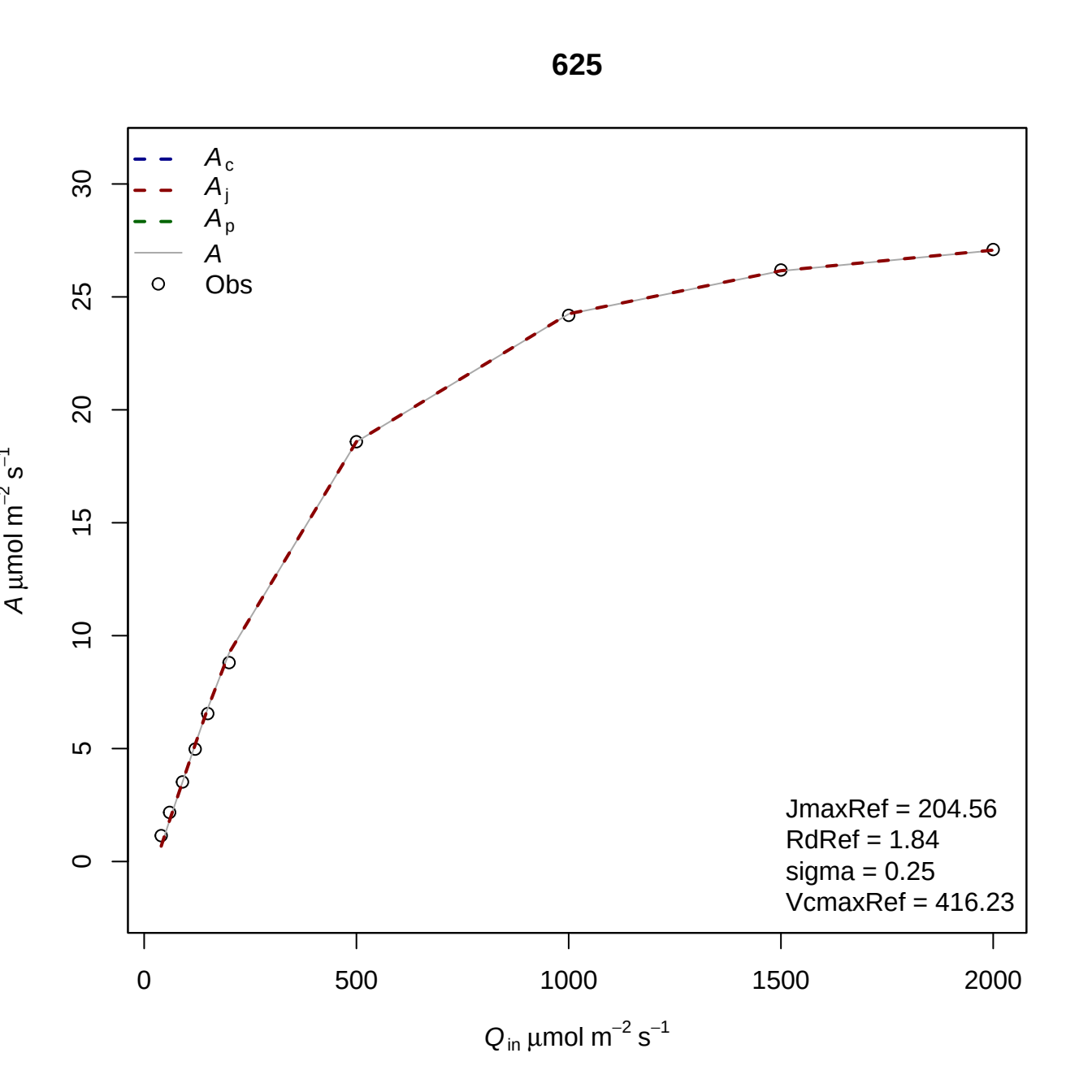

620
A_c
A_j
A_p
A
Obs
JmaxRef = 229.99
RdRef = 1.46
sigma = 0.14
VcmaxRef = 201.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

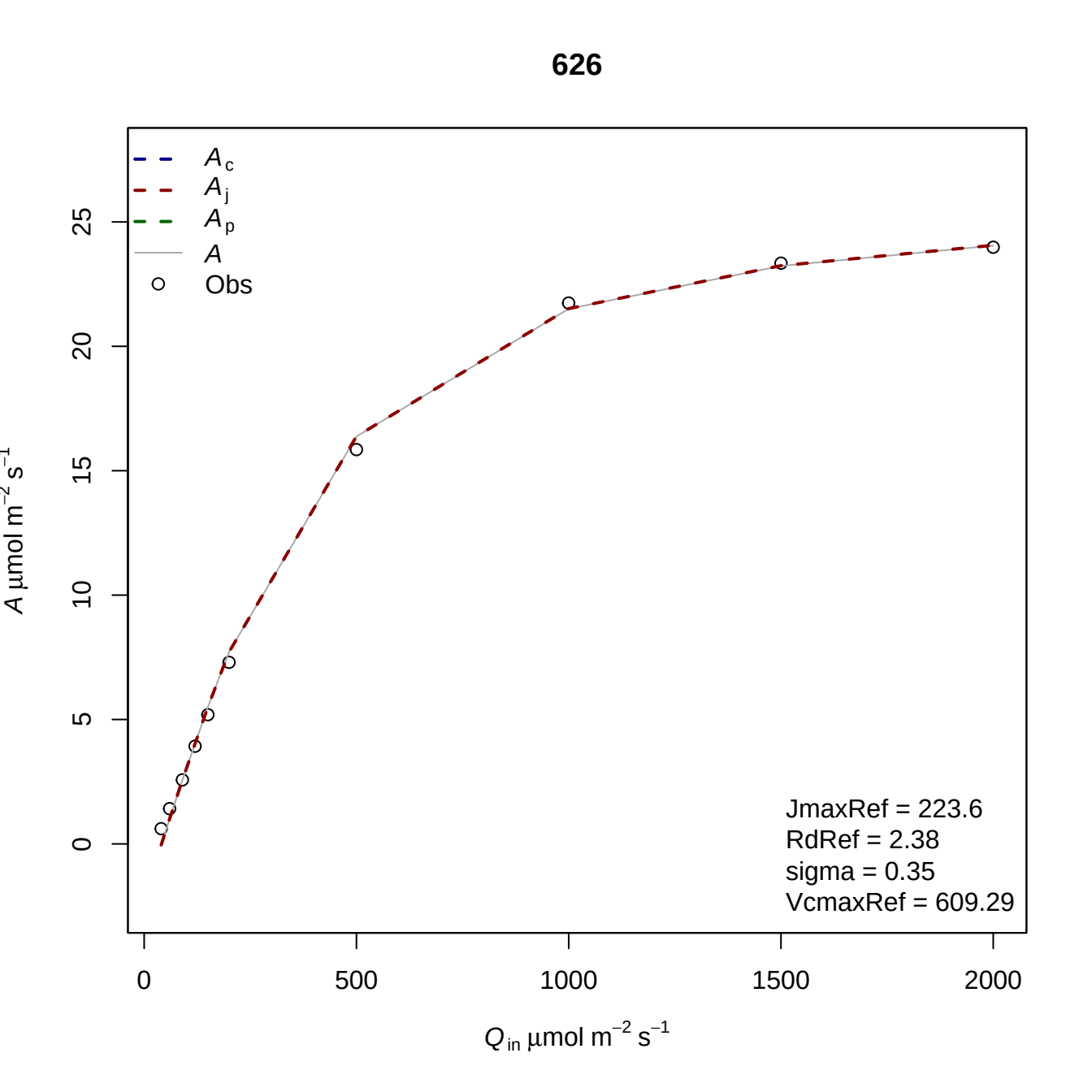

621
A_c
A_j
A_p
A
Obs
JmaxRef = 271.29
RdRef = 1.78
sigma = 0.32
VcmaxRef = 286.08
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

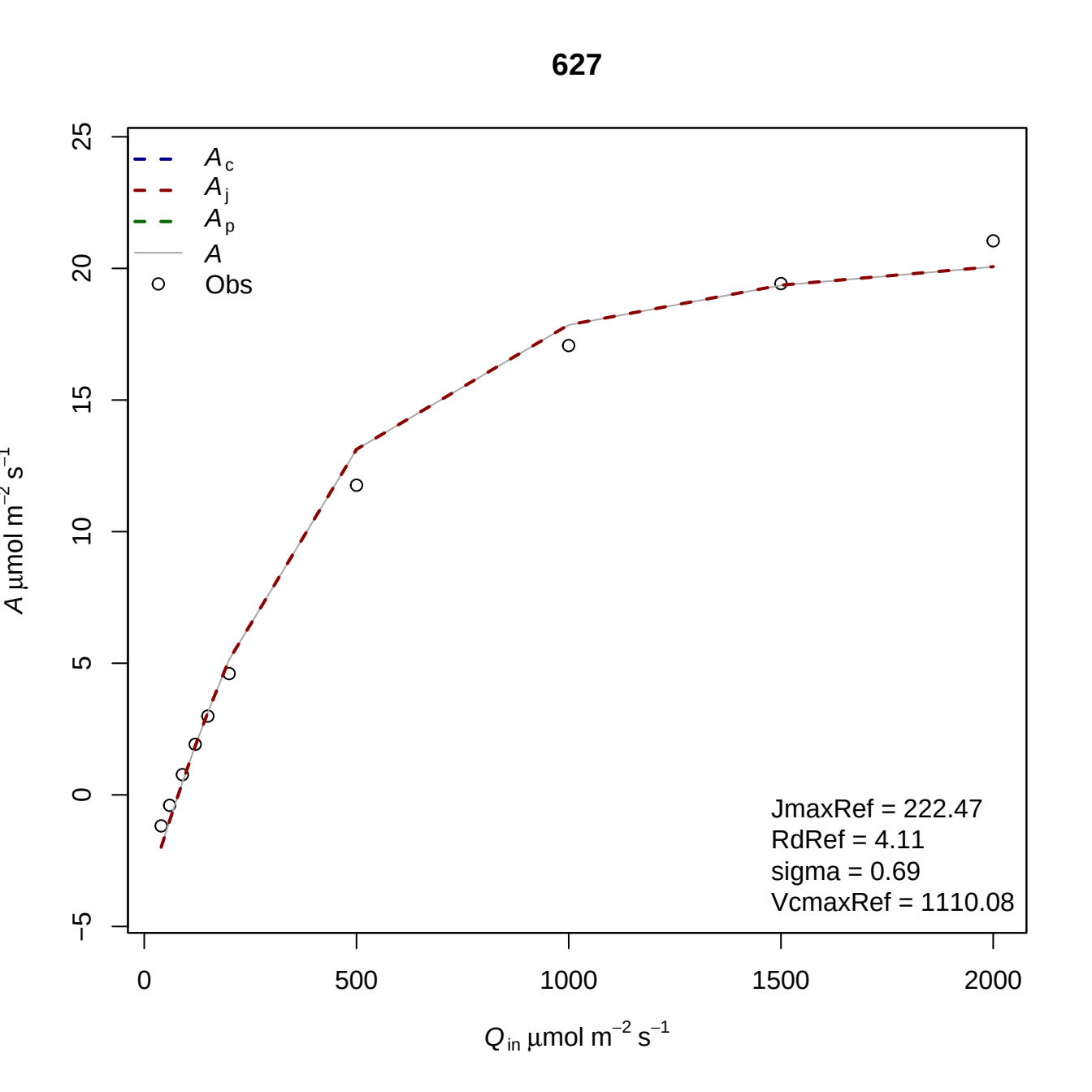

622
A_c
A_j
A_p
A
Obs
JmaxRef = 181.87
RdRef = 4.95
sigma = 1.02
VcmaxRef = 1395.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

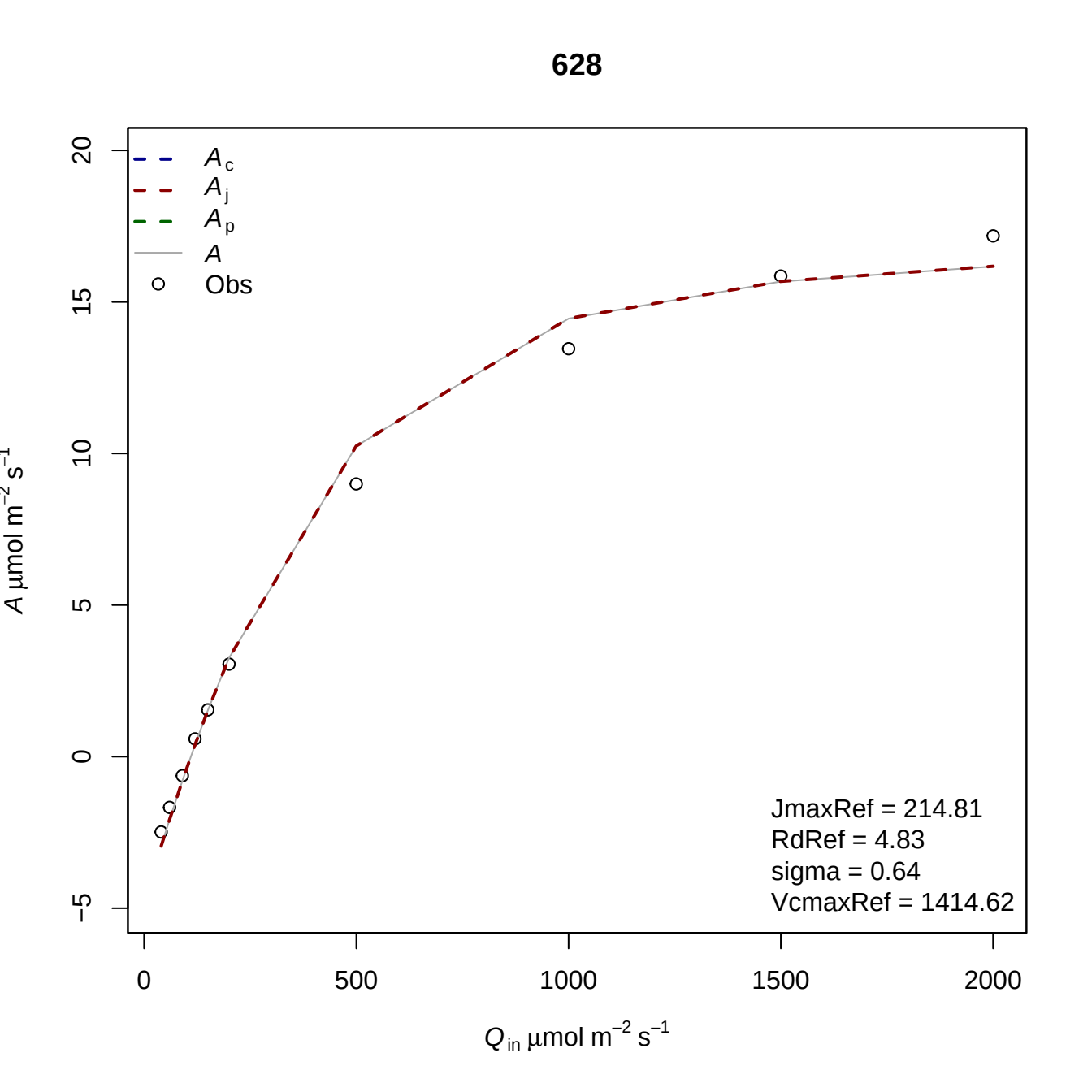

623
A_c
A_j
A_p
A
Obs
JmaxRef = 120.26
RdRef = 1.23
sigma = 0.36
VcmaxRef = 281.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

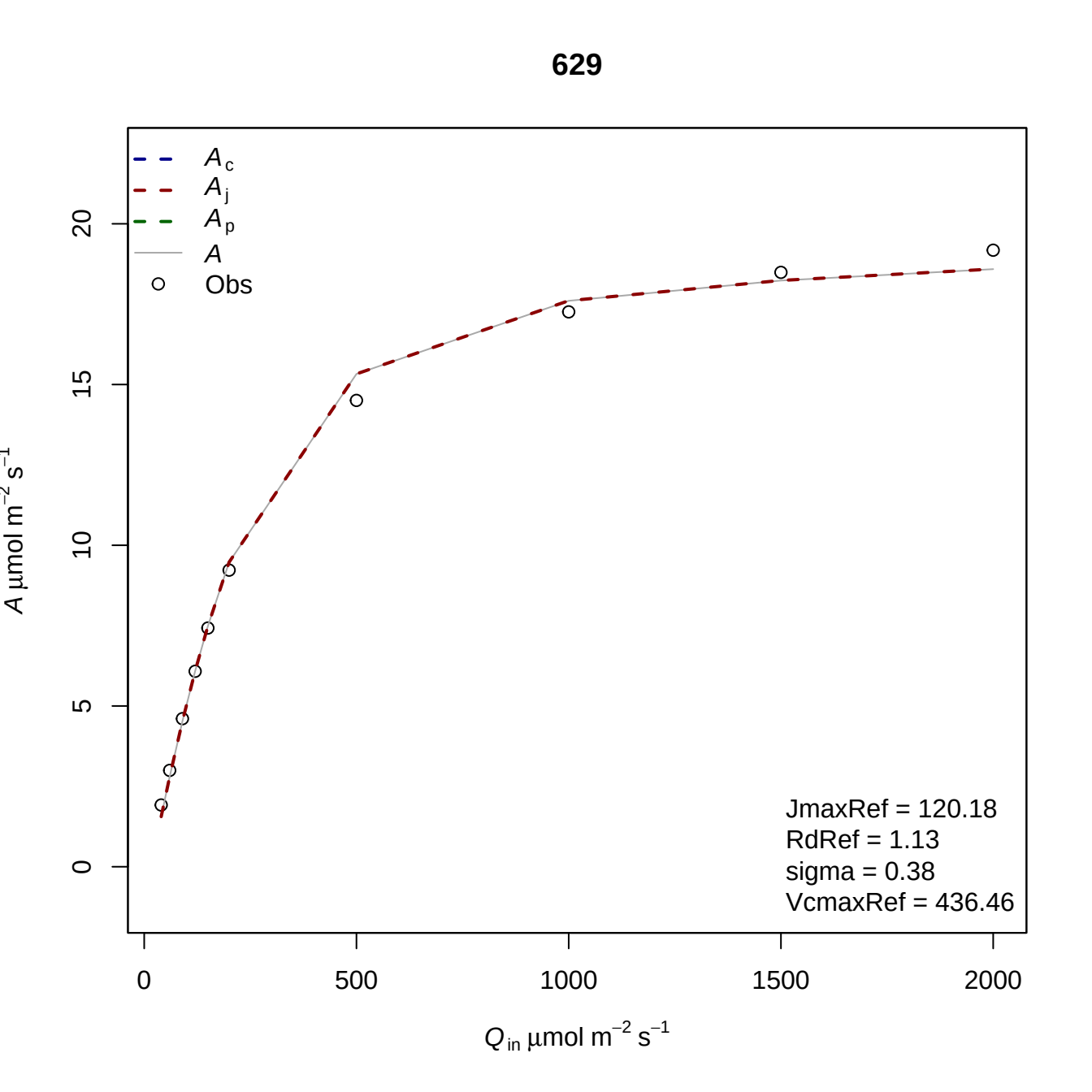

624
A_c
A_j
A_p
A
Obs
JmaxRef = 161.28
RdRef = 2.34
sigma = 1.01
VcmaxRef = 1058.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

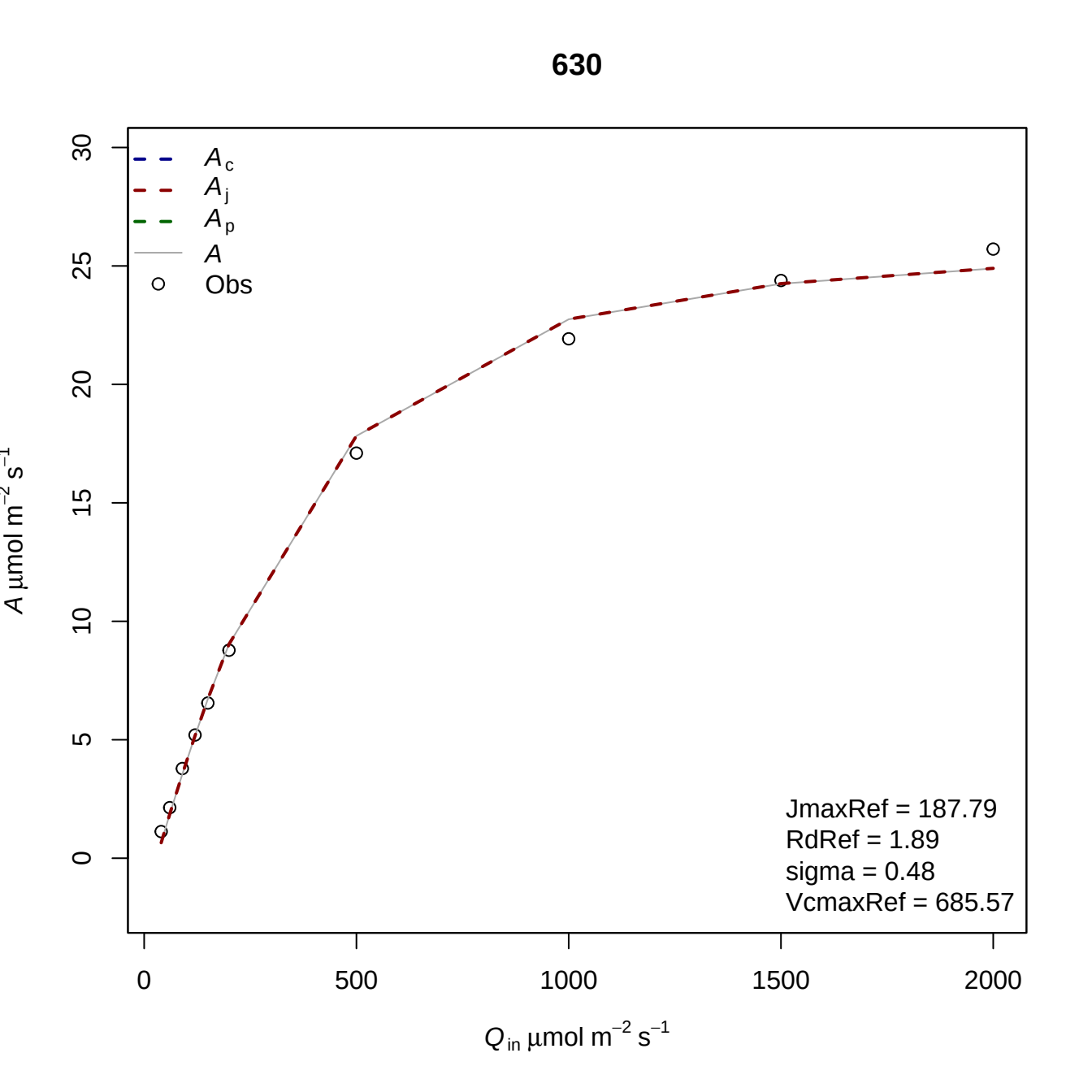

625
A_c
A_j
A_p
A
Obs
JmaxRef = 204.56
RdRef = 1.84
sigma = 0.25
VcmaxRef = 416.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

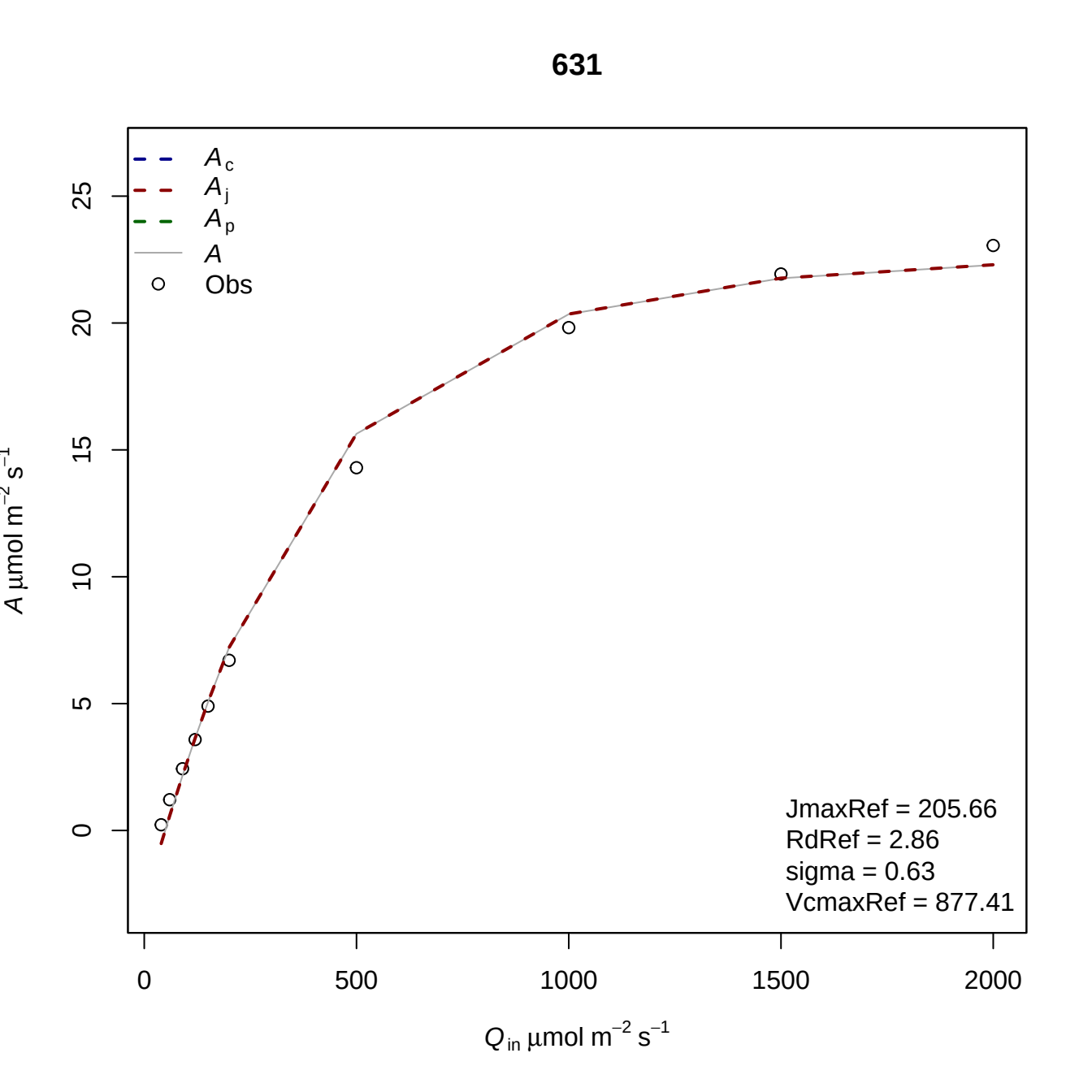

626
A_c
A_j
A_p
A
Obs
JmaxRef = 223.6
RdRef = 2.38
sigma = 0.35
VcmaxRef = 609.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

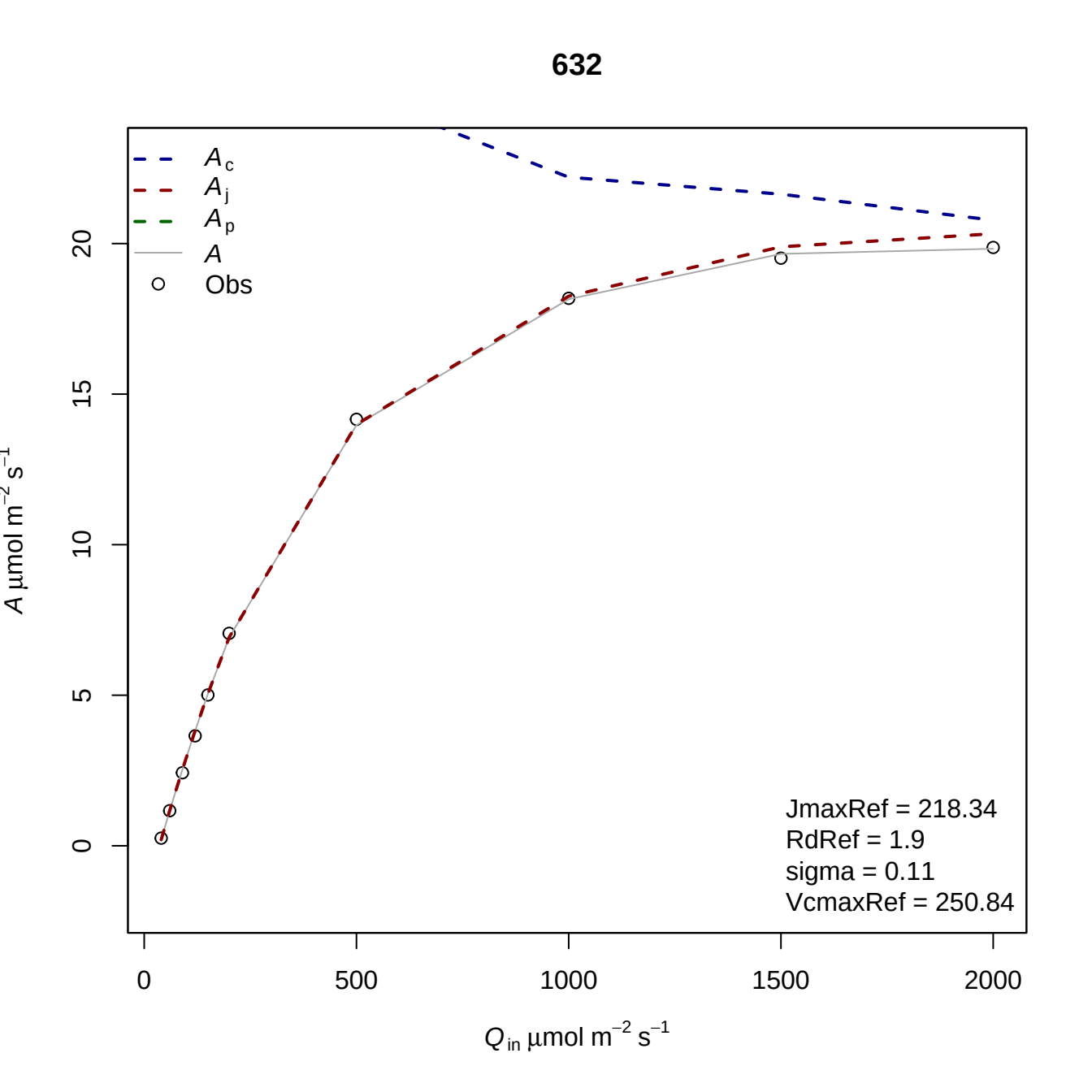

627
A_c
A_j
A_p
A
Obs
JmaxRef = 222.47
RdRef = 4.11
sigma = 0.69
VcmaxRef = 1110.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

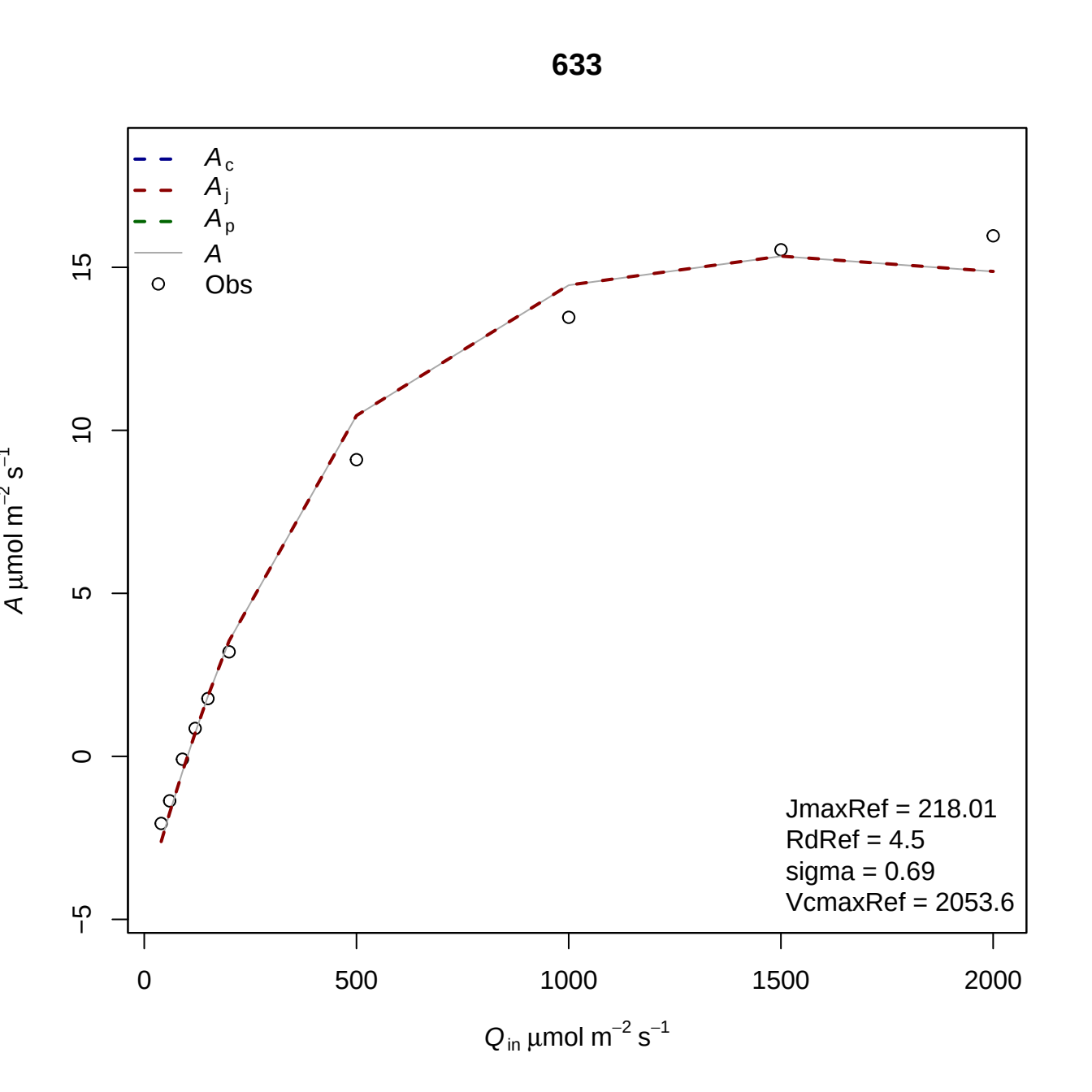

628
A_c
A_j
A_p
A
Obs
JmaxRef = 214.81
RdRef = 4.83
sigma = 0.64
VcmaxRef = 1414.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

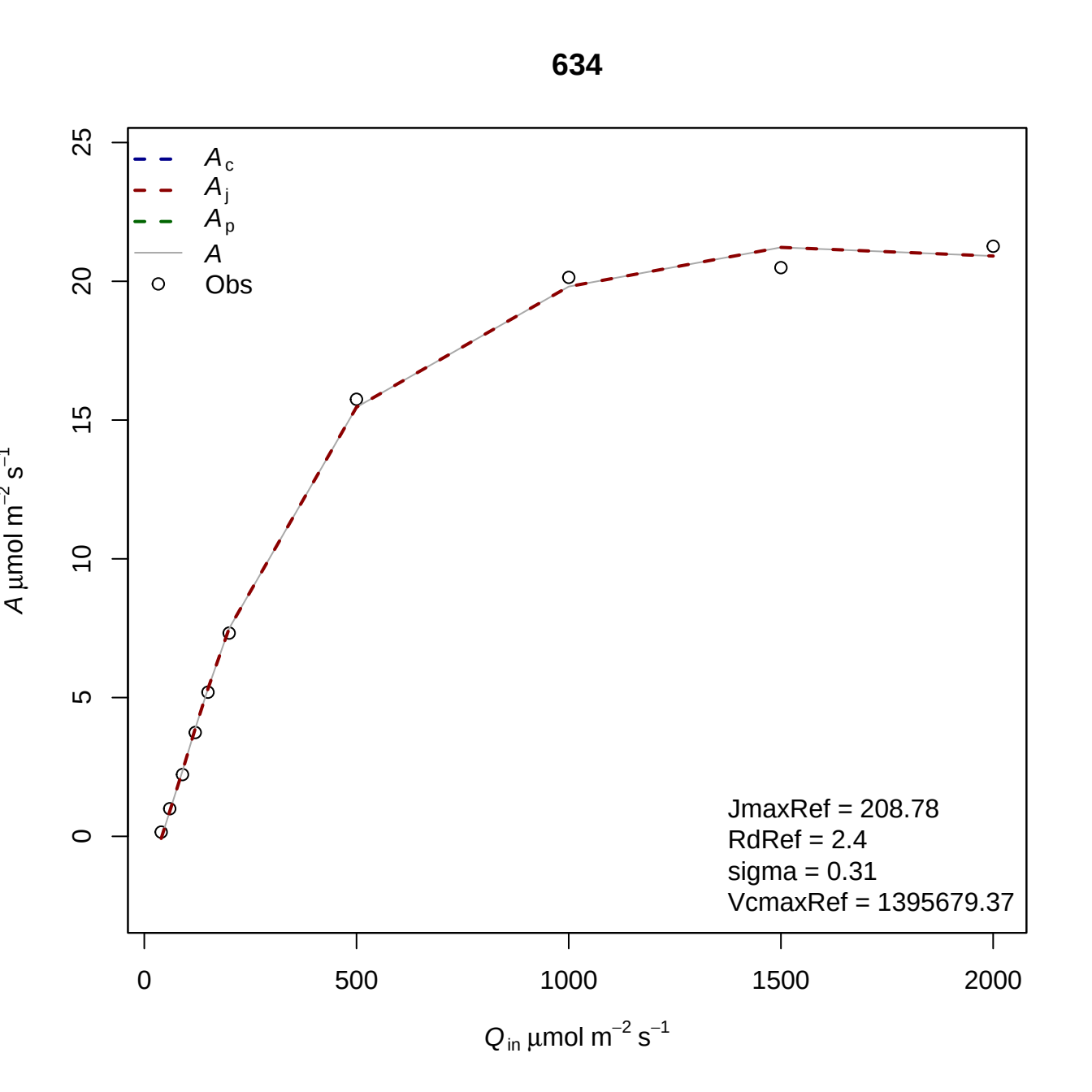

629
A_c
A_j
A_p
A
Obs
JmaxRef = 120.18
RdRef = 1.13
sigma = 0.38
VcmaxRef = 436.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

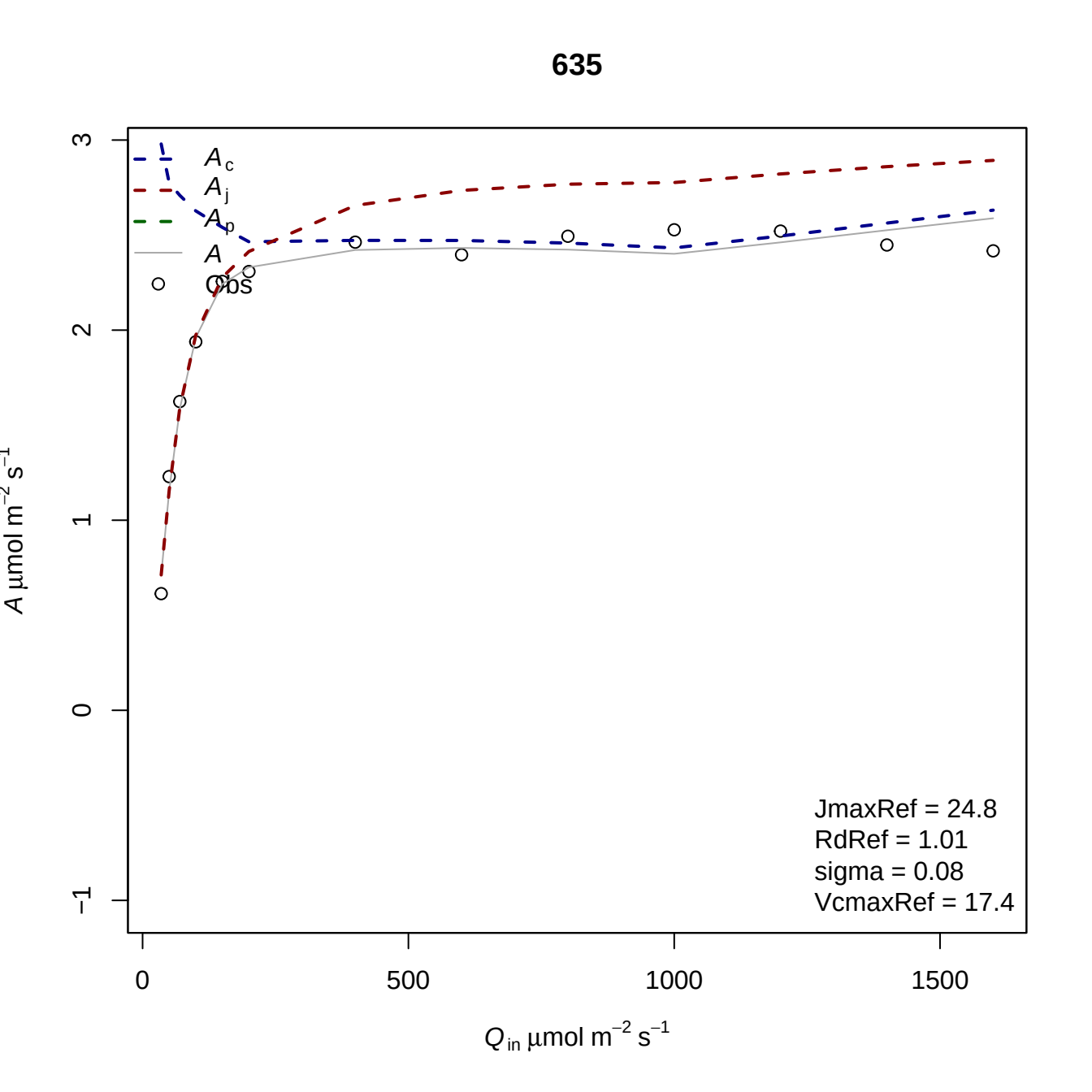

630
A_c
A_j
A_p
A
Obs
JmaxRef = 187.79
RdRef = 1.89
sigma = 0.48
VcmaxRef = 685.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

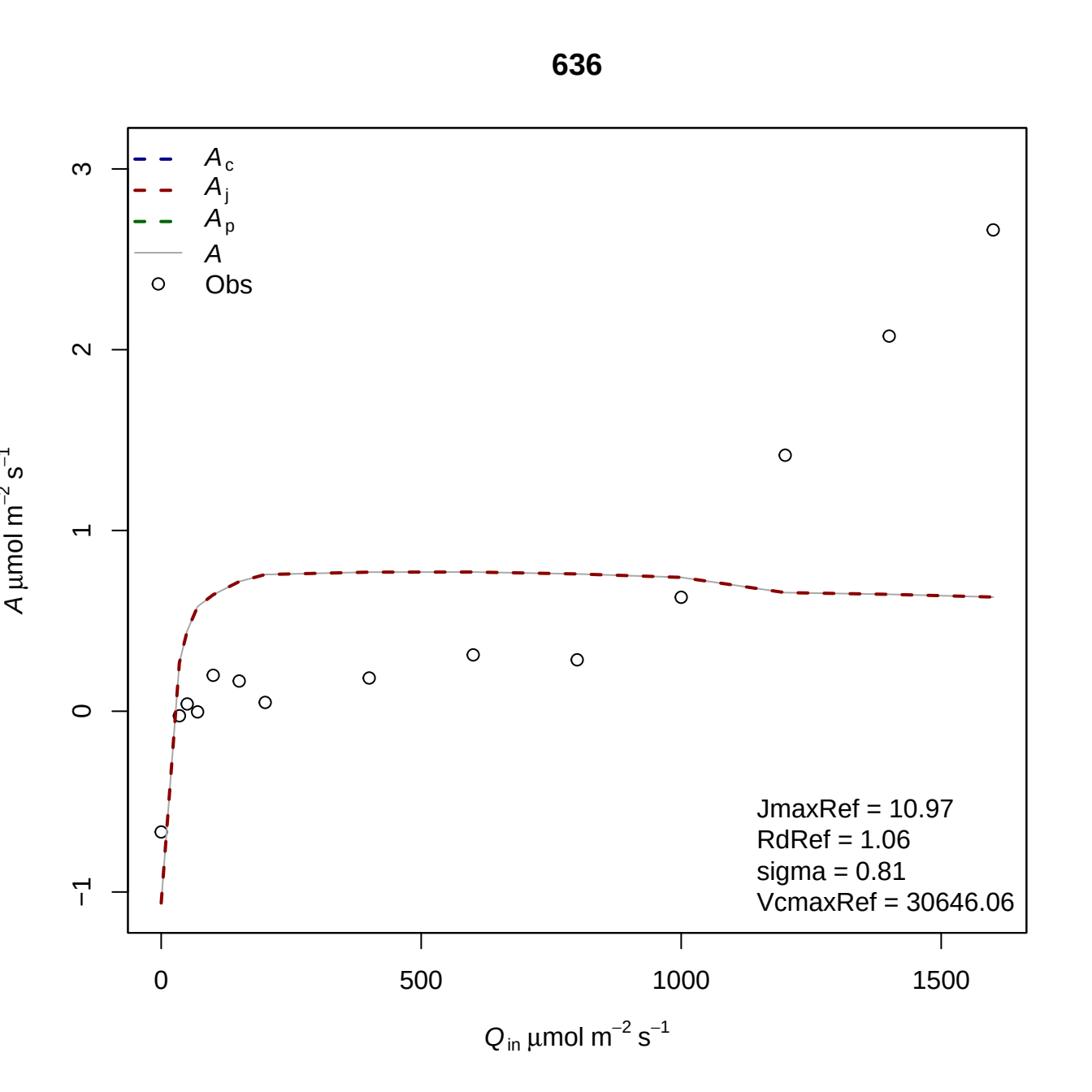

631
A_c
A_j
A_p
A
Obs
JmaxRef = 205.66
RdRef = 2.86
sigma = 0.63
VcmaxRef = 877.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

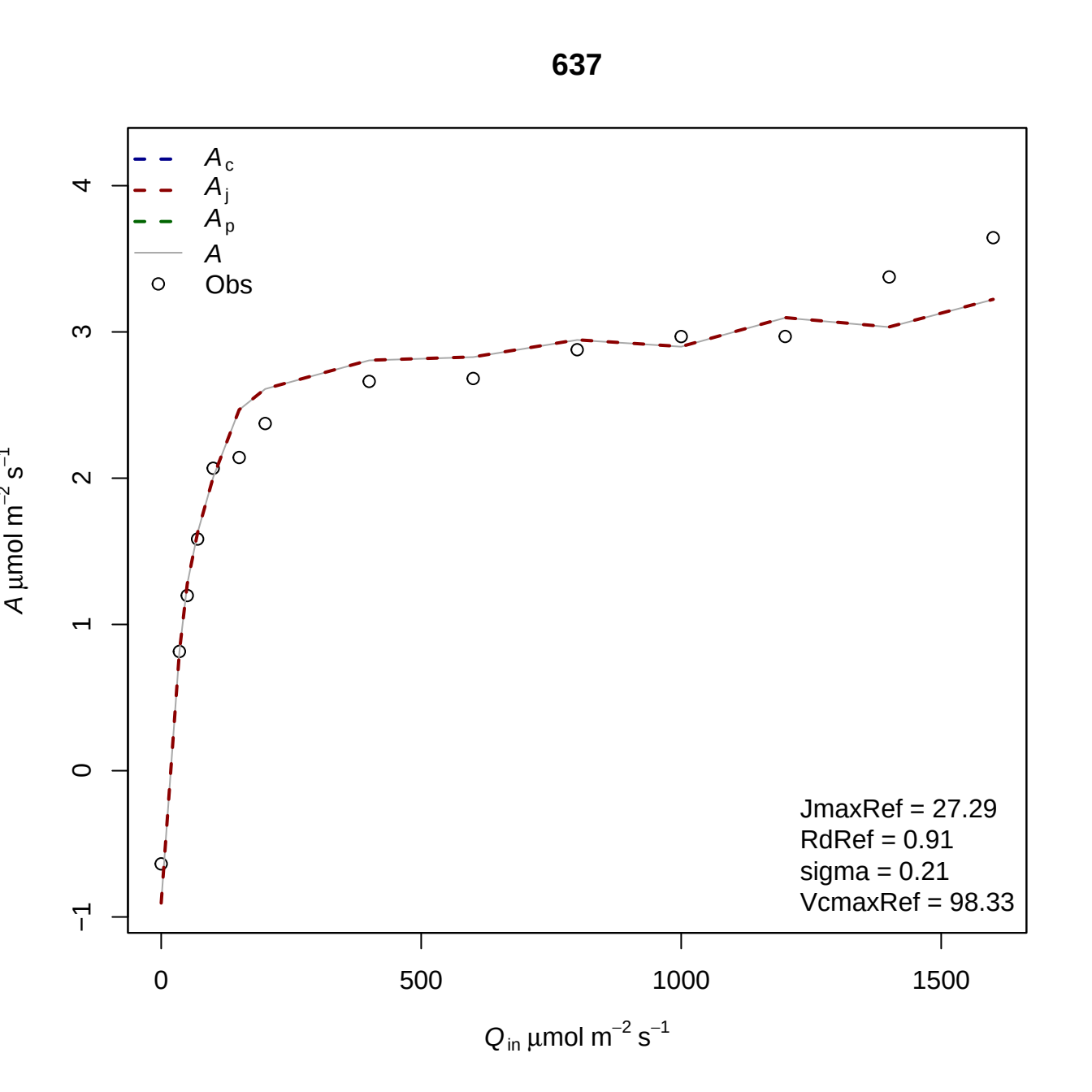

632
A_c
A_j
A_p
A
Obs
JmaxRef = 218.34
RdRef = 1.9
sigma = 0.11
VcmaxRef = 250.84
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

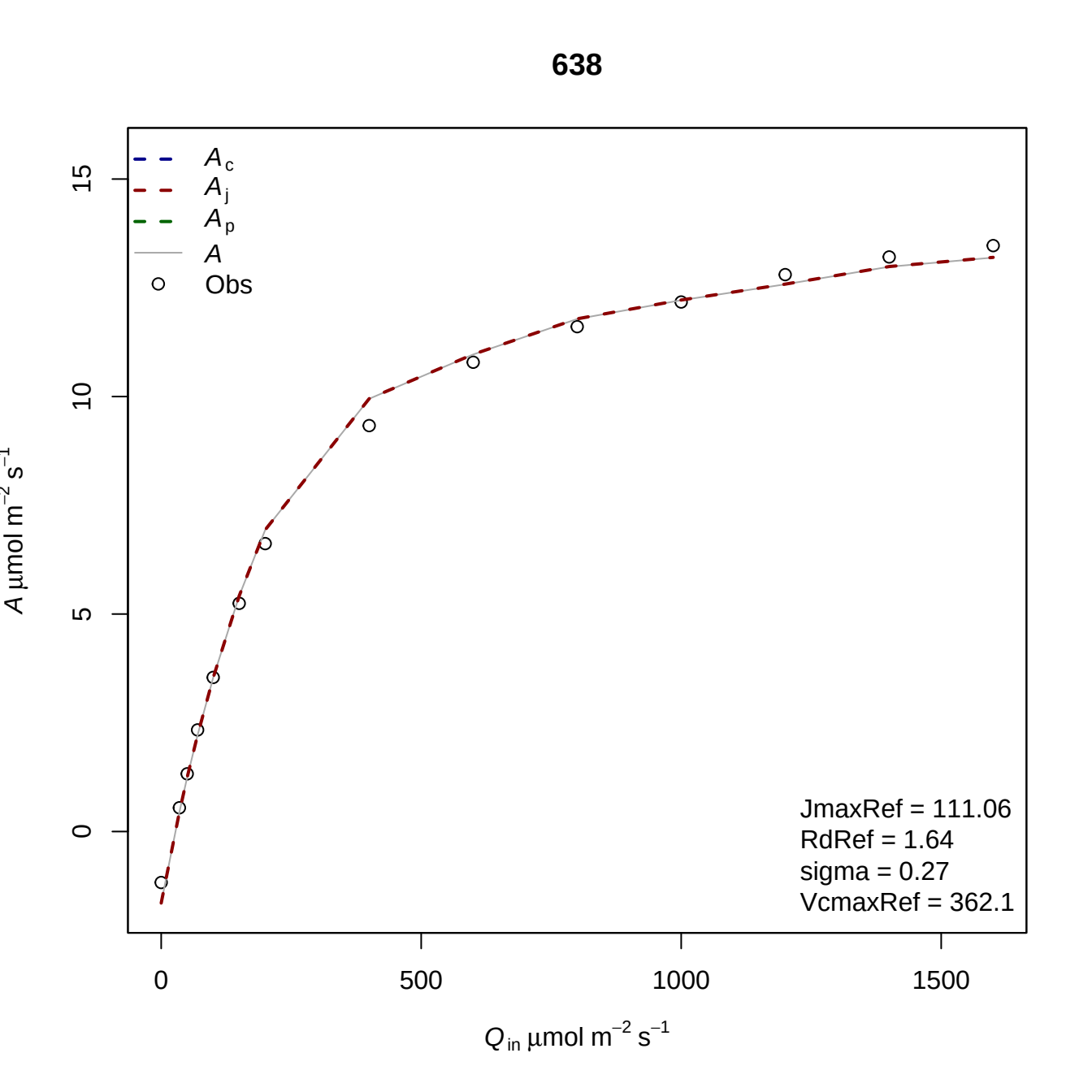

633
A_c
A_j
A_p
A
Obs
JmaxRef = 218.01
RdRef = 4.5
sigma = 0.69
VcmaxRef = 2053.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

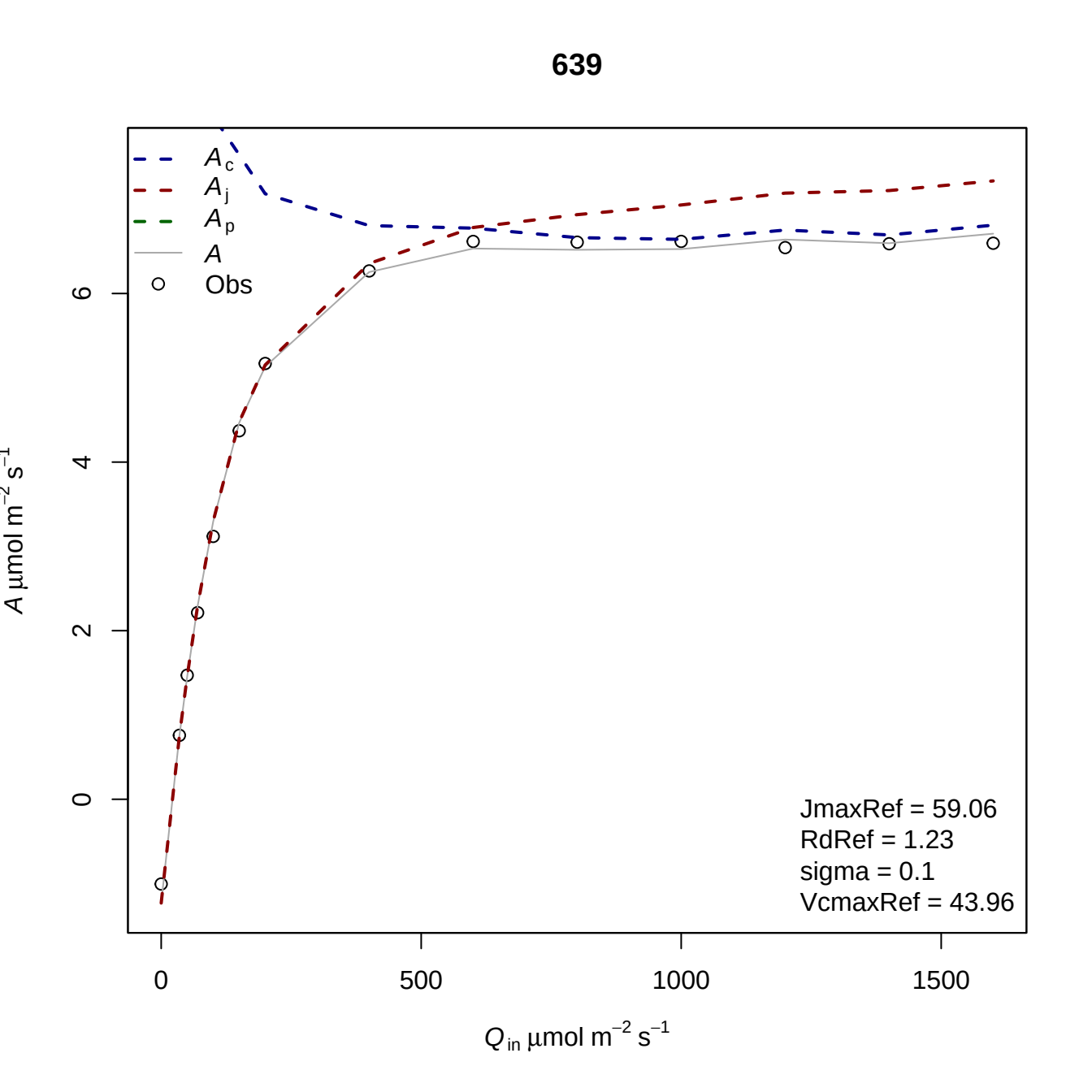

634
A_c
A_j
A_p
A
Obs
JmaxRef = 208.78
RdRef = 2.4
sigma = 0.31
VcmaxRef = 1395679.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

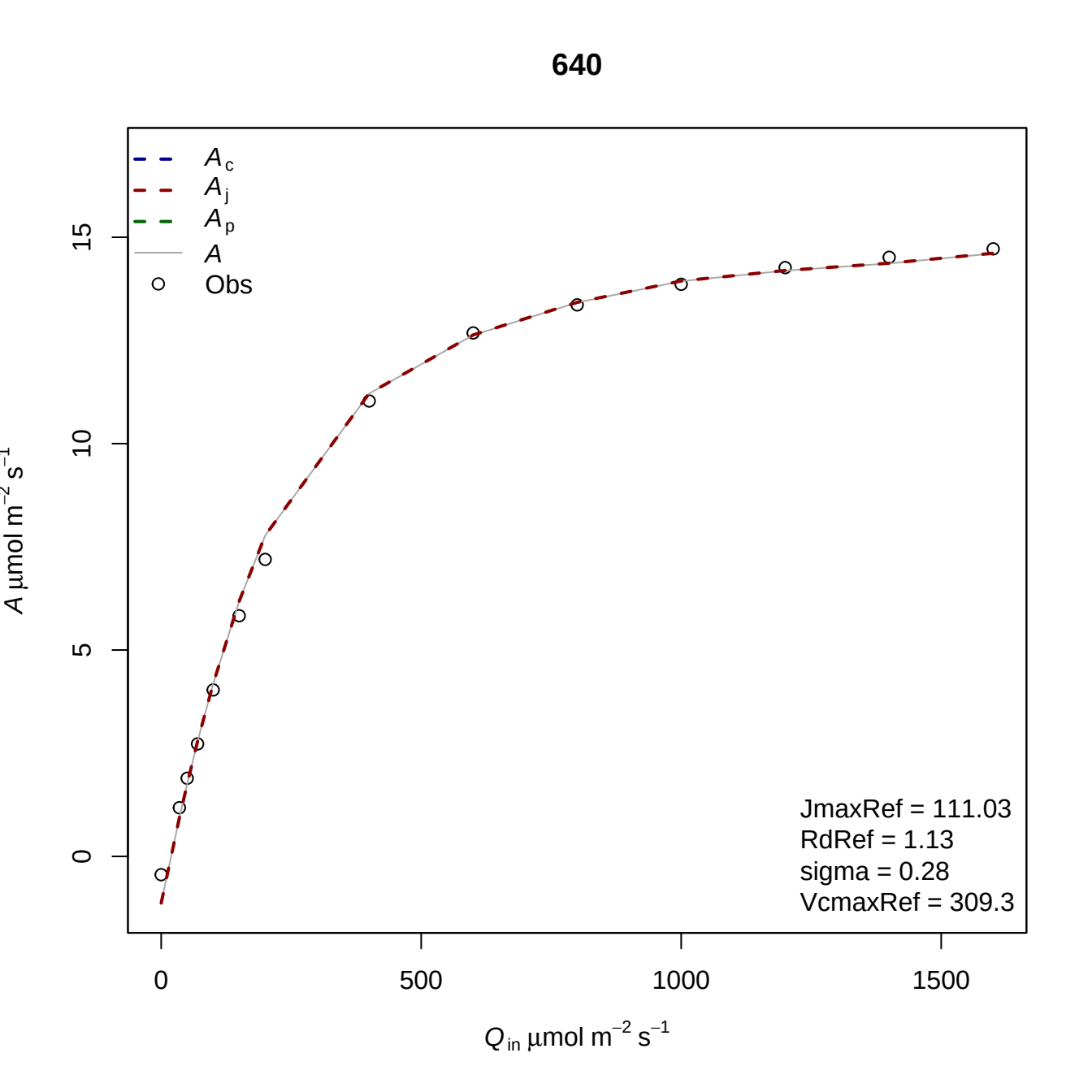

635
A_c
A_j
A_p
A
Obs
JmaxRef = 24.8
RdRef = 1.01
sigma = 0.08
VcmaxRef = 17.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

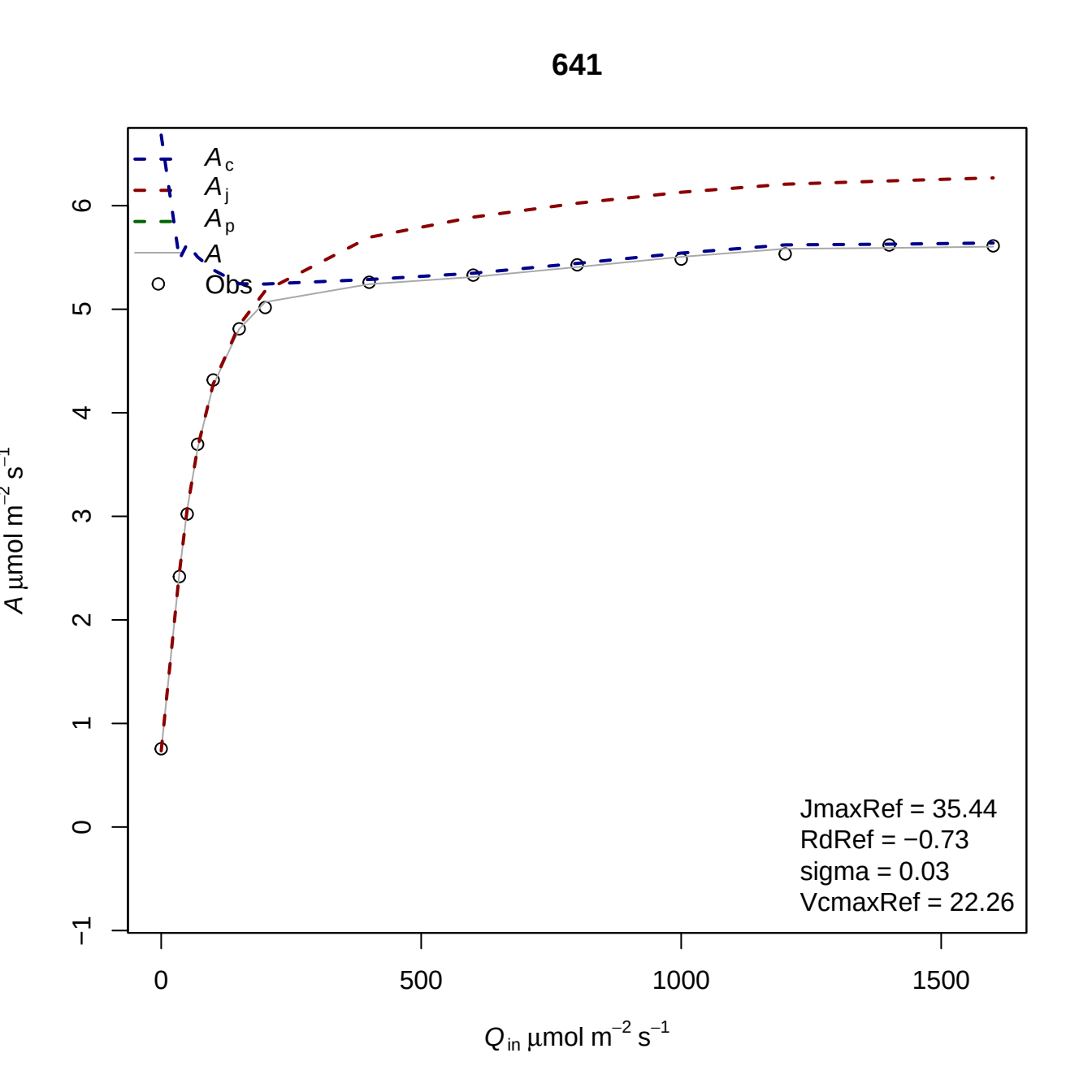

636
A_c
A_j
A_p
A
Obs
JmaxRef = 10.97
RdRef = 1.06
sigma = 0.81
VcmaxRef = 30646.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

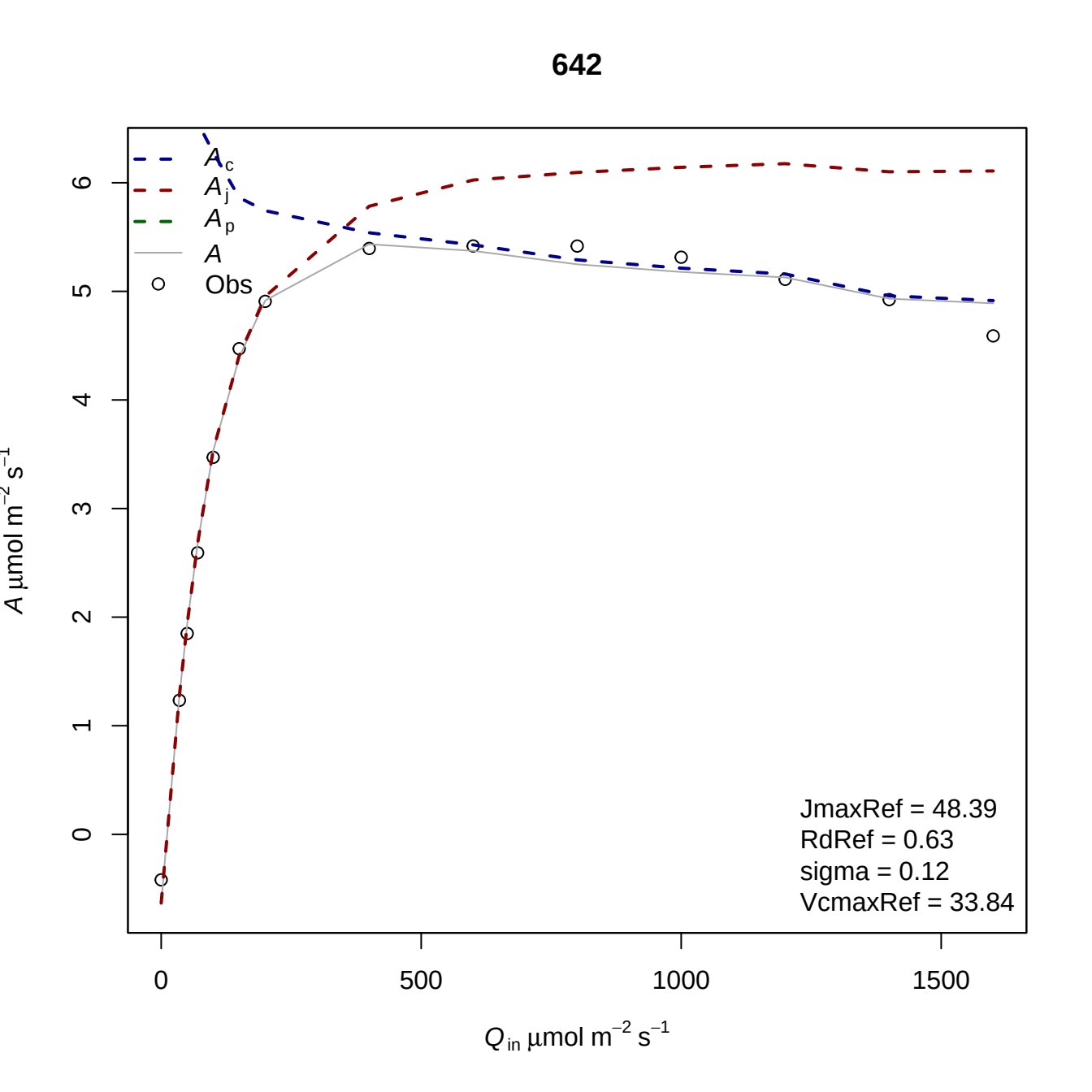

637
A_c
A_j
A_p
A
Obs
JmaxRef = 27.29
RdRef = 0.91
sigma = 0.21
VcmaxRef = 98.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

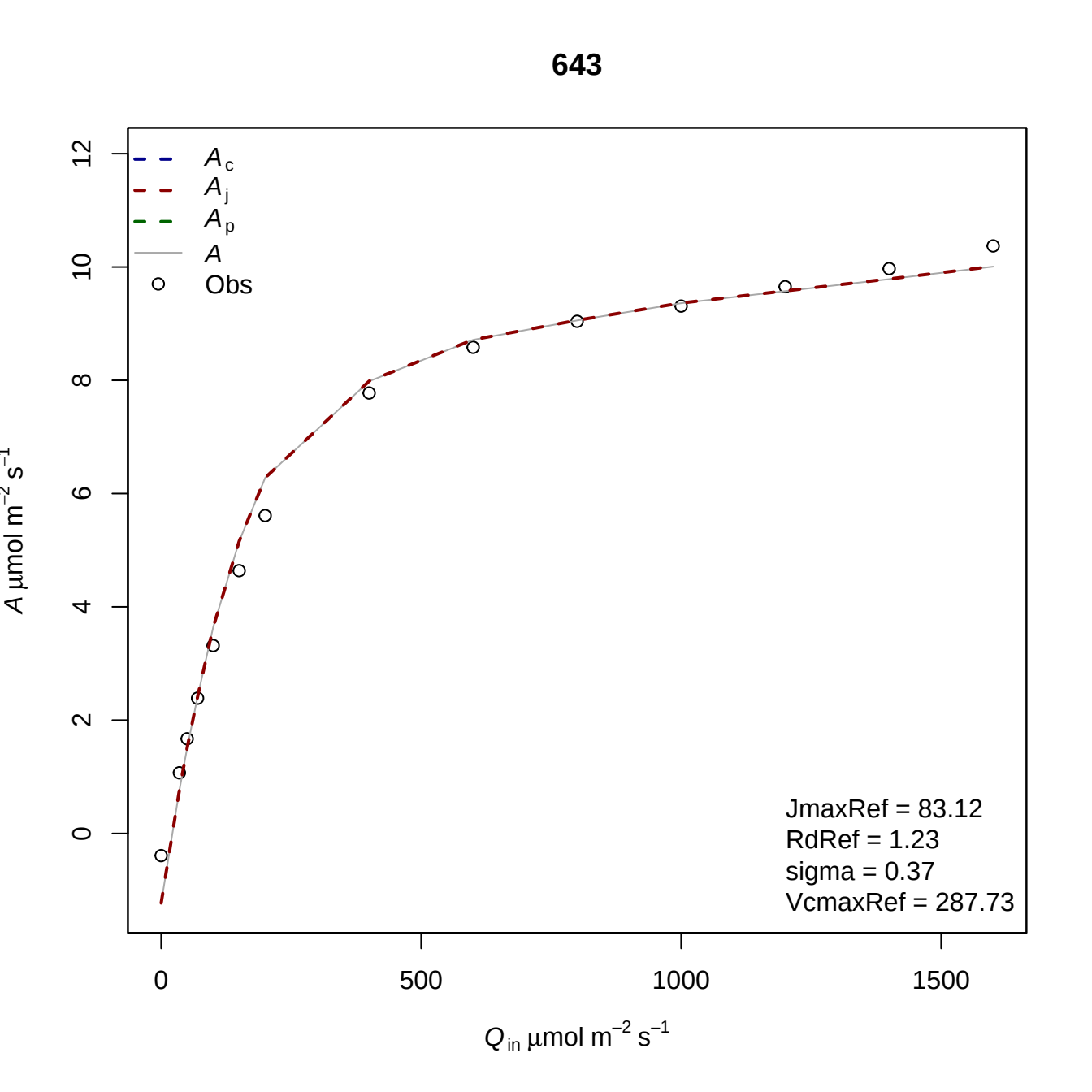

638
A_c
A_j
A_p
A
Obs
JmaxRef = 111.06
RdRef = 1.64
sigma = 0.27
VcmaxRef = 362.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

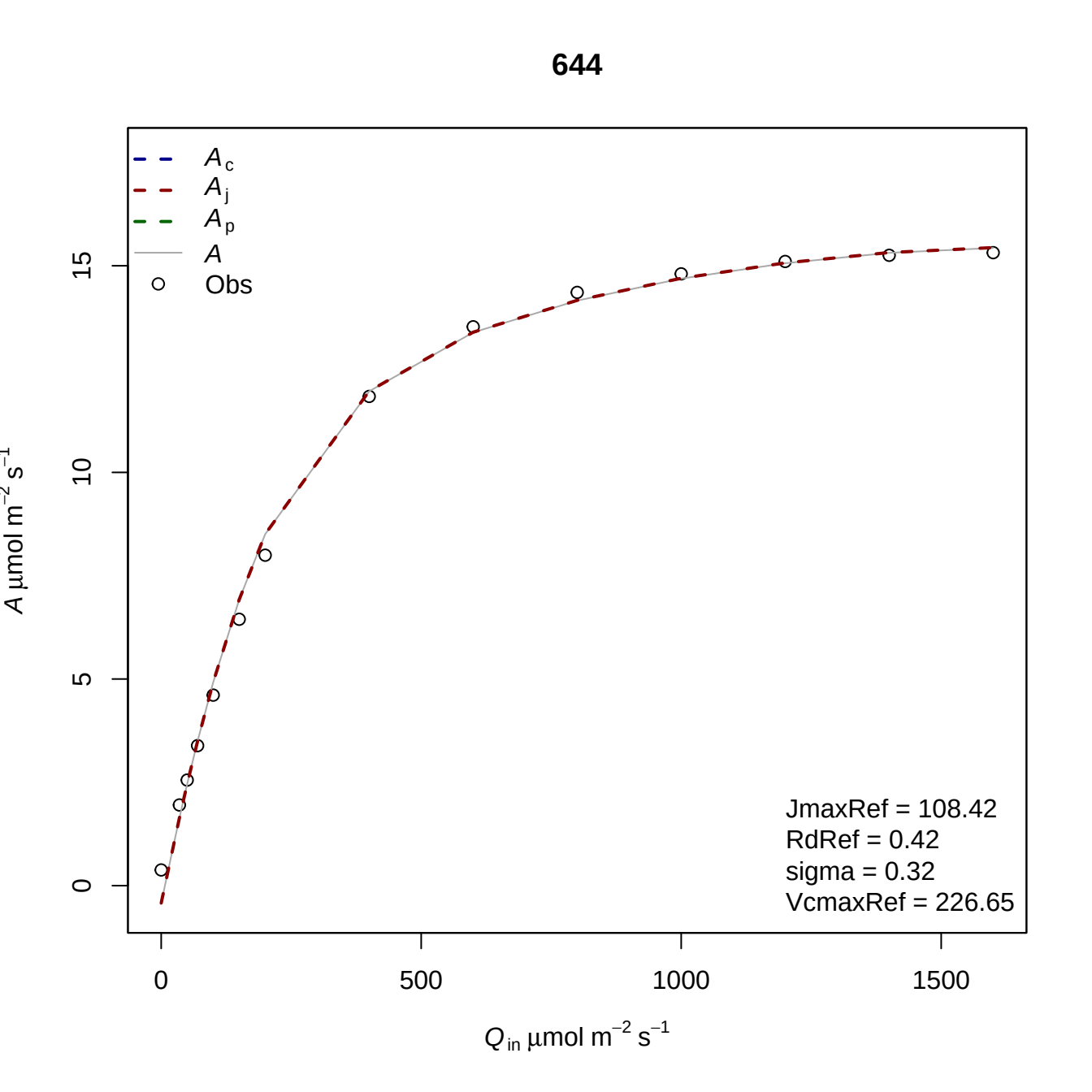

639
A_c
A_j
A_p
A
Obs
JmaxRef = 59.06
RdRef = 1.23
sigma = 0.1
VcmaxRef = 43.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

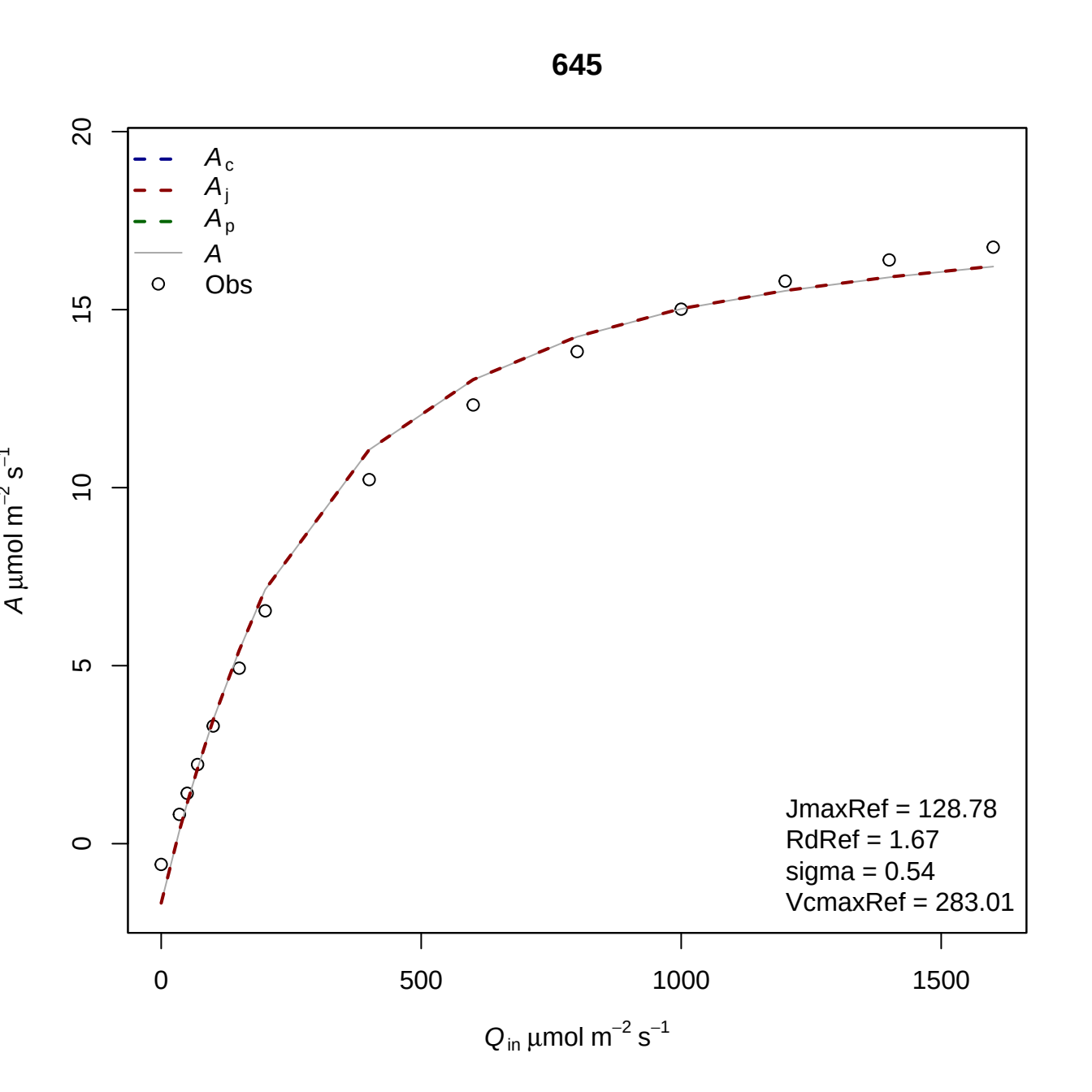

640
A_c
A_j
A_p
A
Obs
JmaxRef = 111.03
RdRef = 1.13
sigma = 0.28
VcmaxRef = 309.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

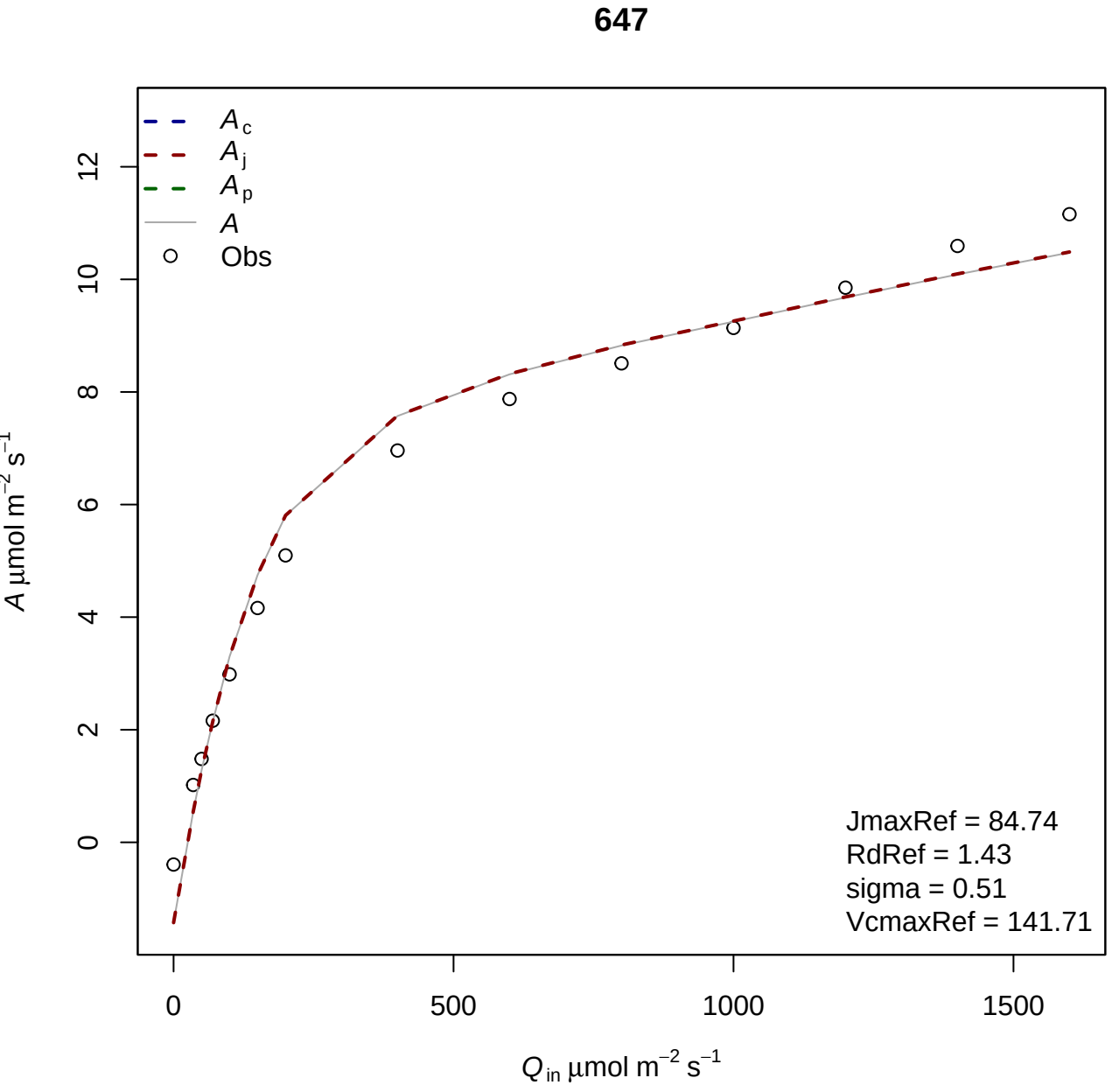

641
A_c
A_j
A_p
A
Obs
JmaxRef = 35.44
RdRef = −0.73
sigma = 0.03
VcmaxRef = 22.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

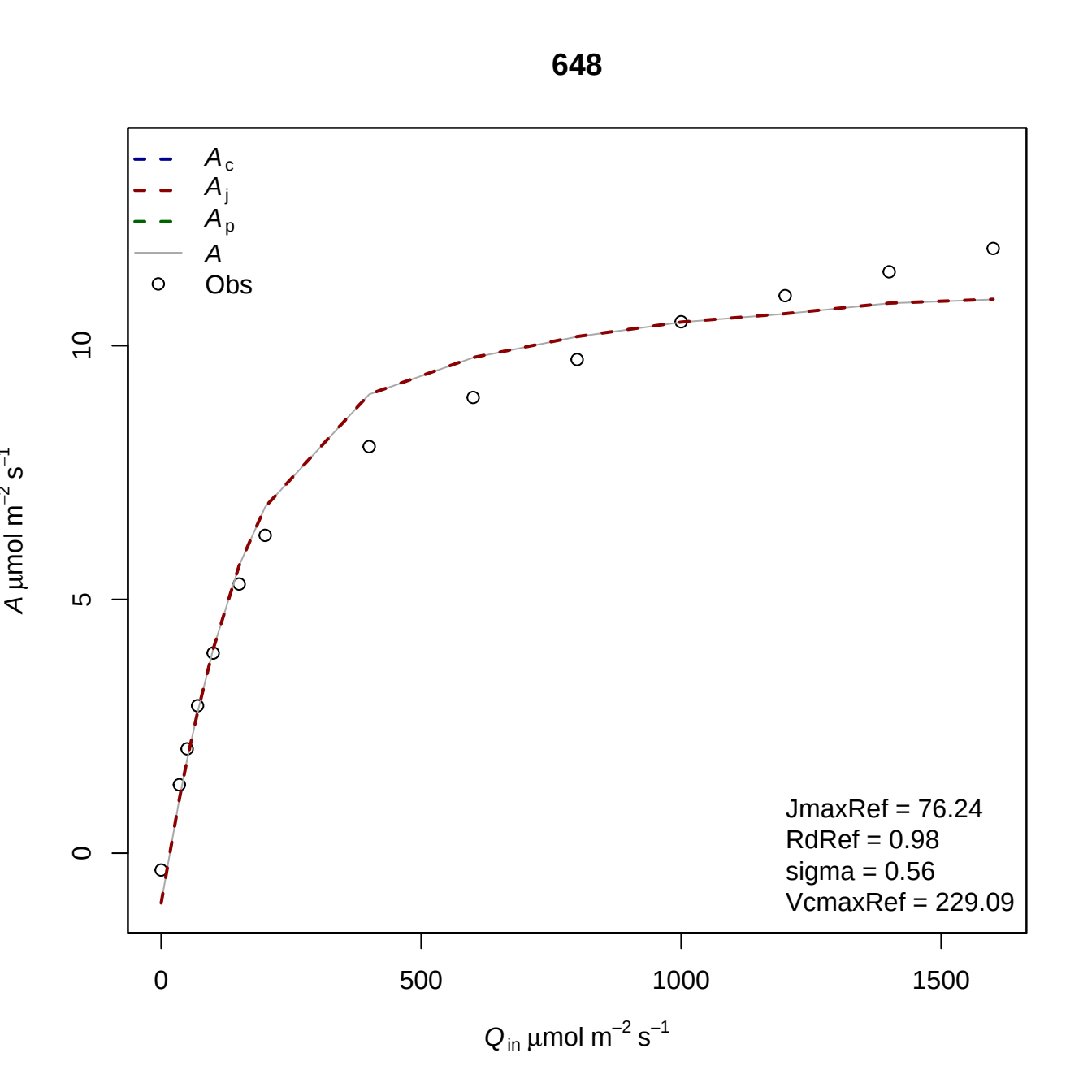

642
A_c
A_j
A_p
A
Obs
JmaxRef = 48.39
RdRef = 0.63
sigma = 0.12
VcmaxRef = 33.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

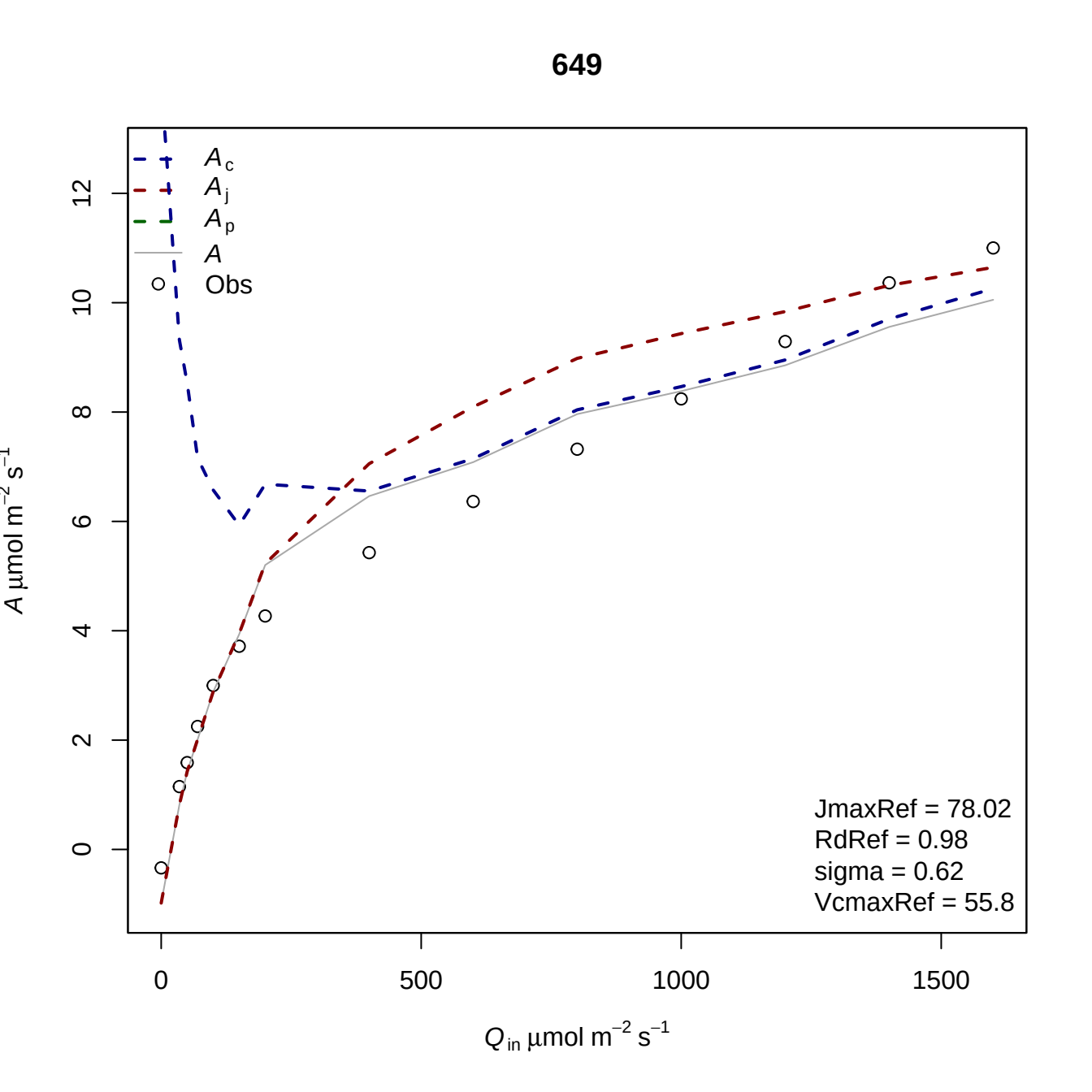

643
A_c
A_j
A_p
A
Obs
JmaxRef = 83.12
RdRef = 1.23
sigma = 0.37
VcmaxRef = 287.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

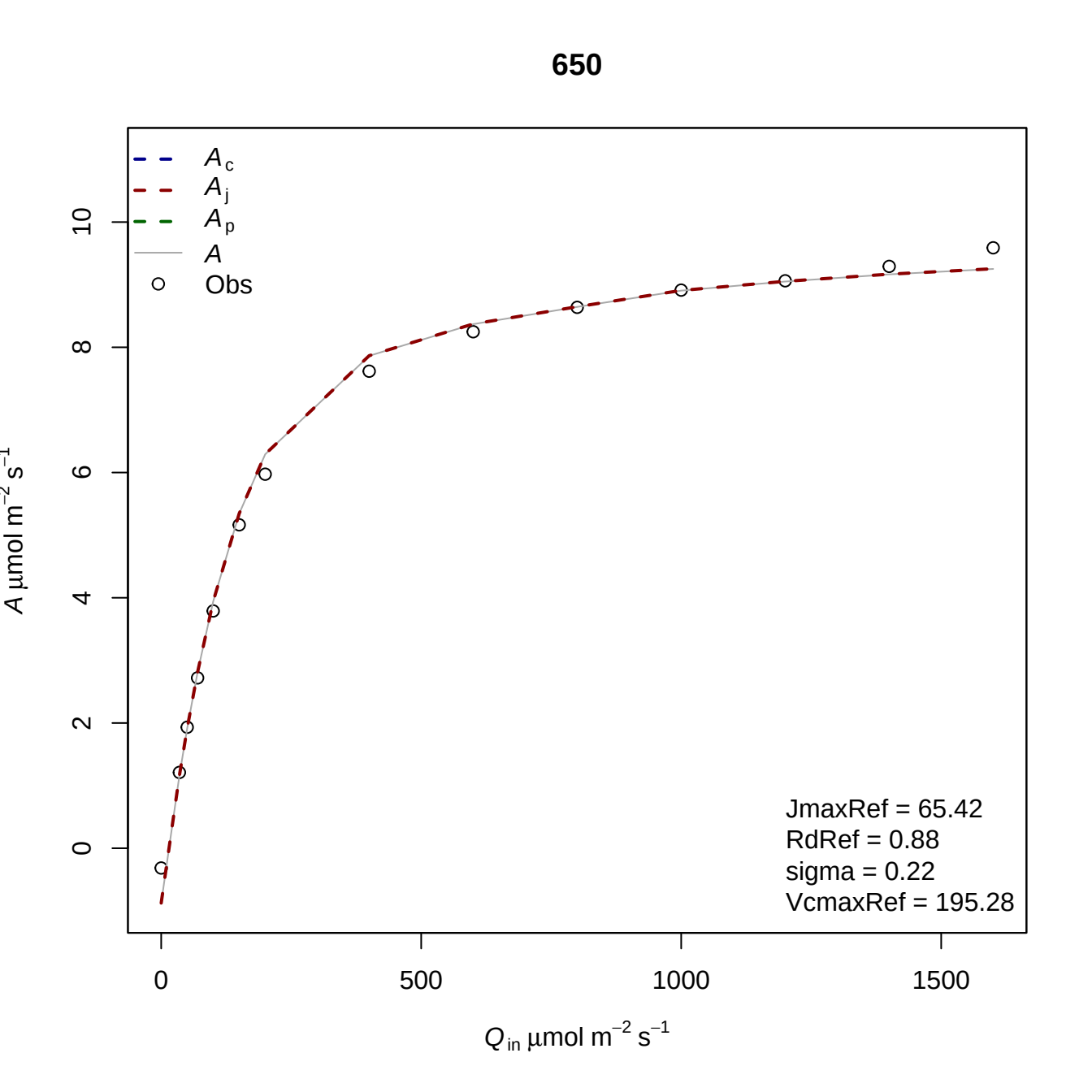

644
A_c
A_j
A_p
A
Obs
JmaxRef = 108.42
RdRef = 0.42
sigma = 0.32
VcmaxRef = 226.65
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

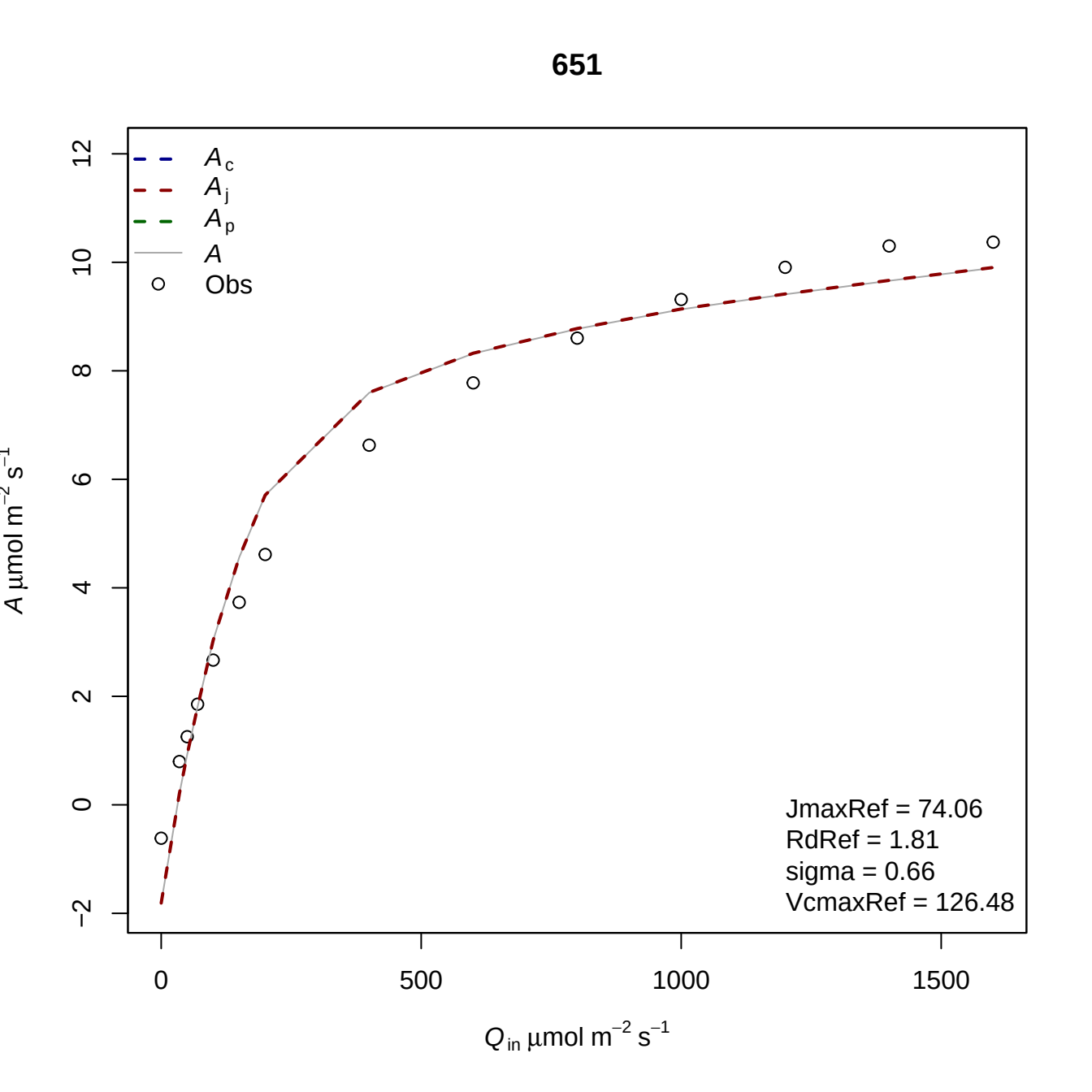

645
A_c
A_j
A_p
A
Obs
JmaxRef = 128.78
RdRef = 1.67
sigma = 0.54
VcmaxRef = 283.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

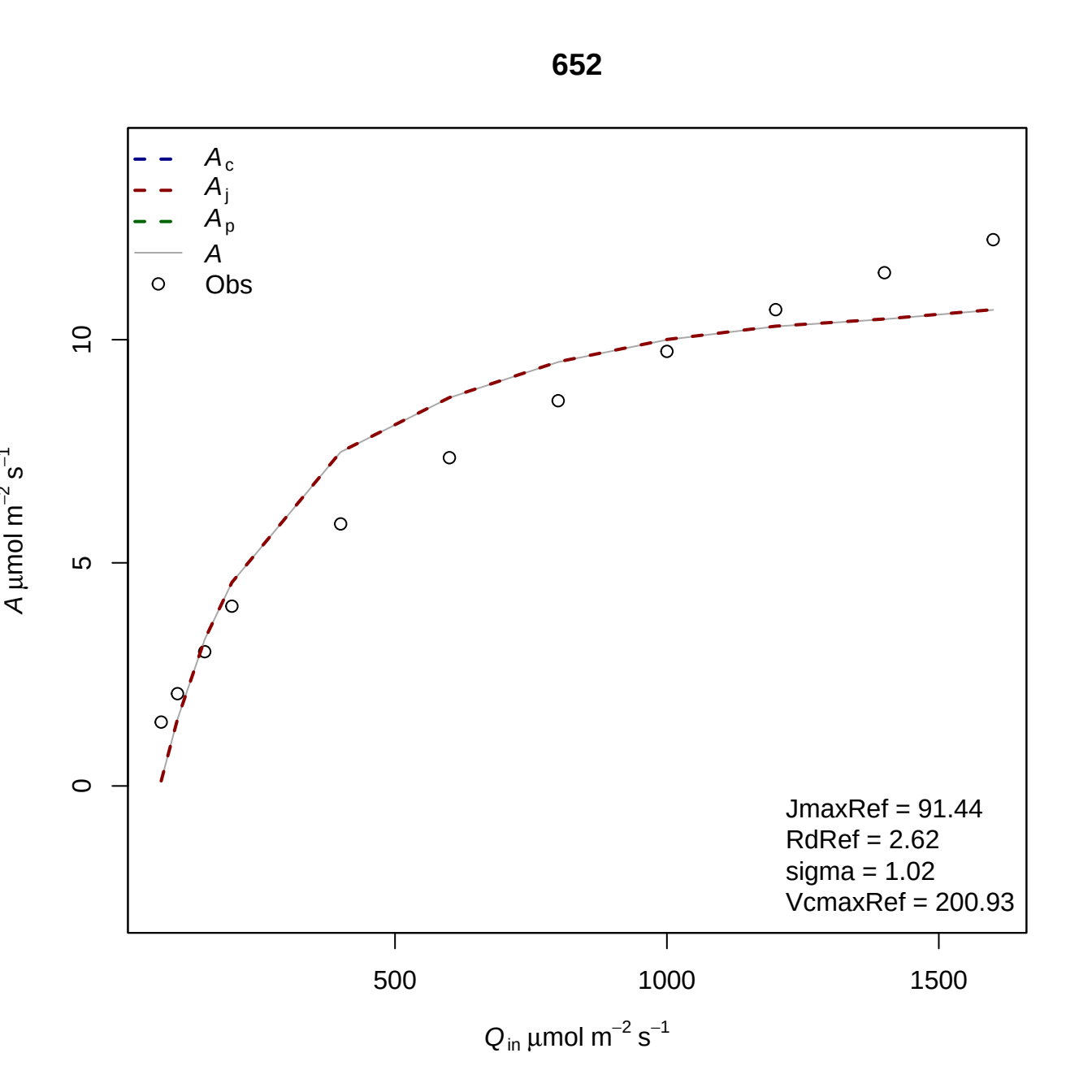

647
A_c
A_j
A_p
A
Obs
JmaxRef = 84.74
RdRef = 1.43
sigma = 0.51
VcmaxRef = 141.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

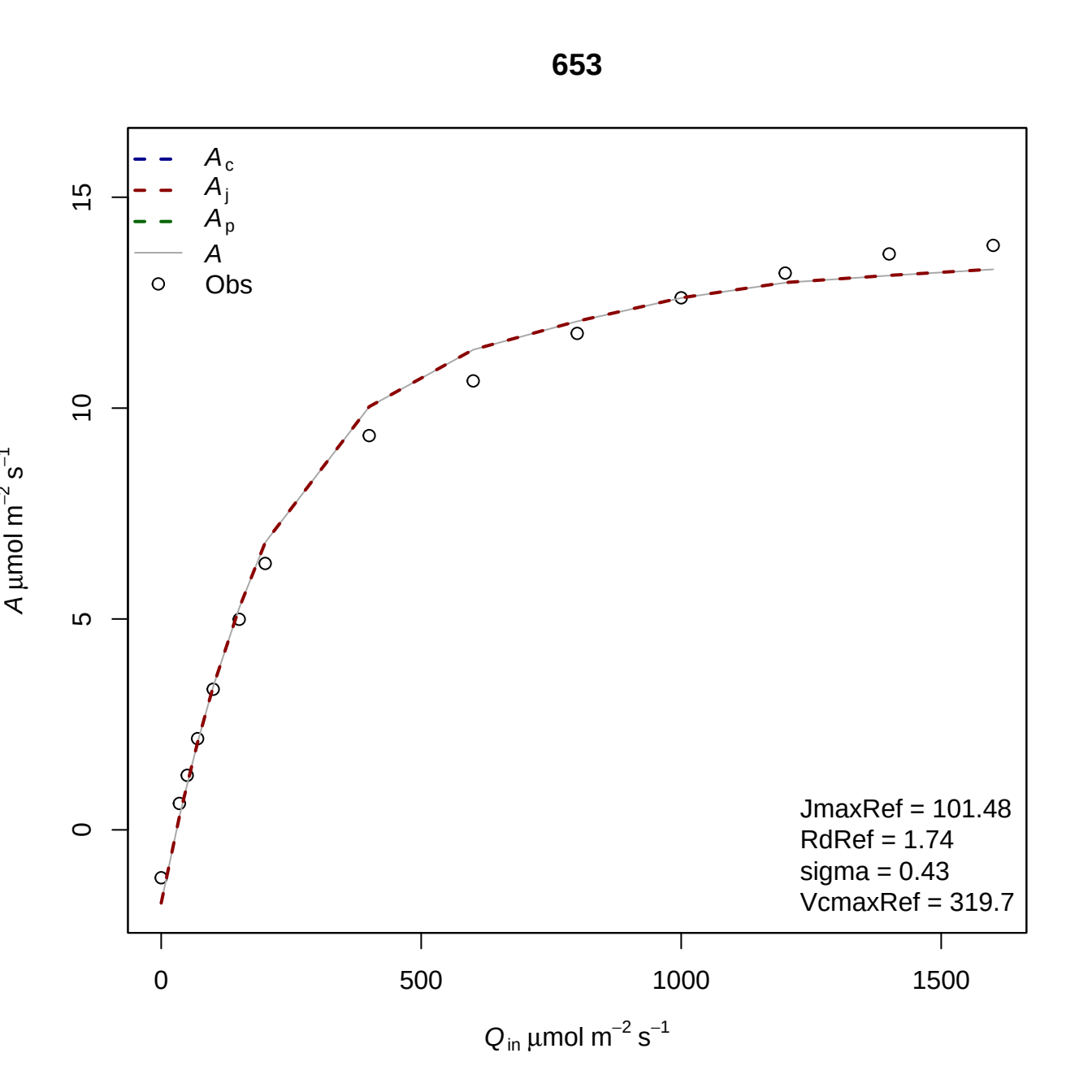

648
A_c
A_j
A_p
A
Obs
JmaxRef = 76.24
RdRef = 0.98
sigma = 0.56
VcmaxRef = 229.09
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

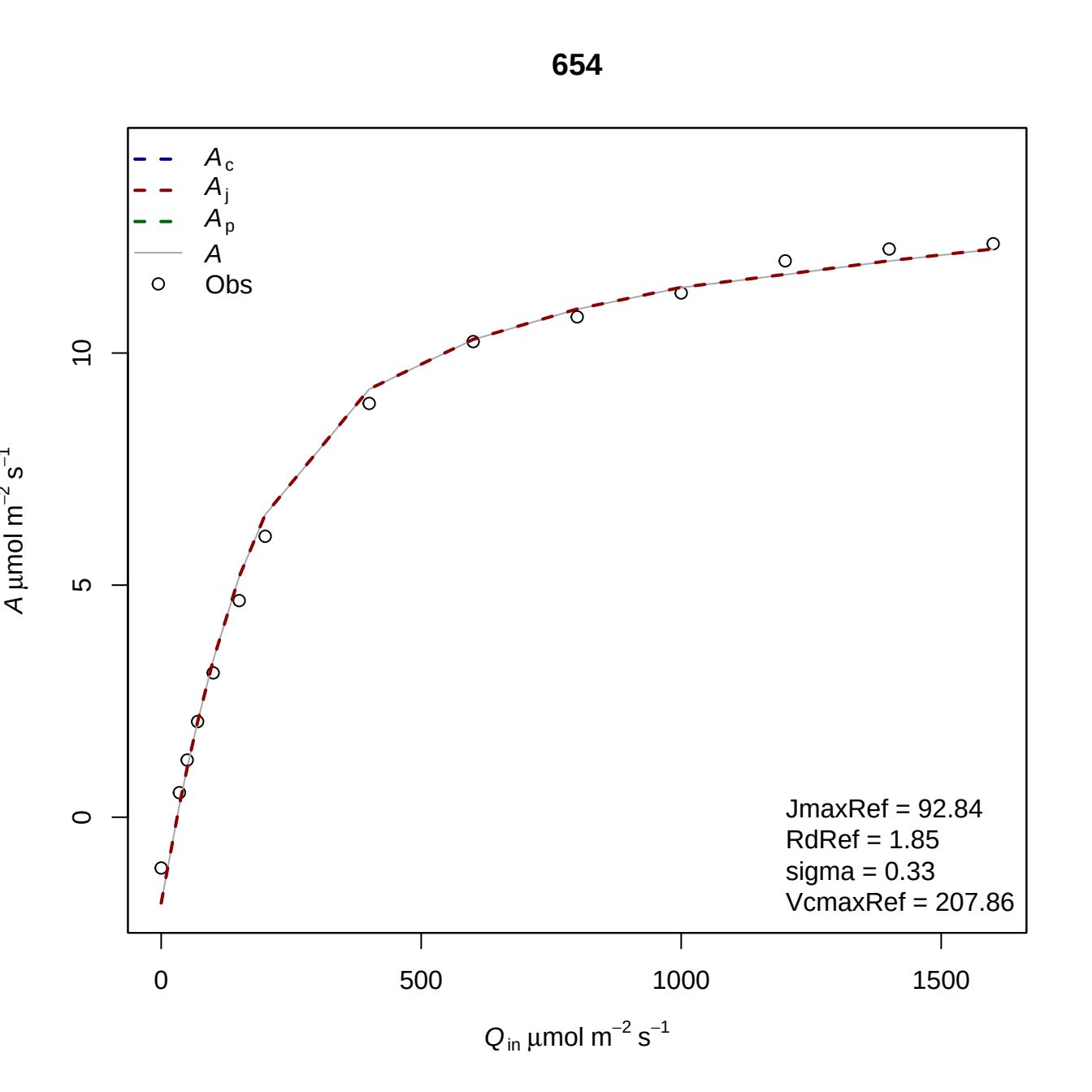

649
A_c
A_j
A_p
A
Obs
JmaxRef = 78.02
RdRef = 0.98
sigma = 0.62
VcmaxRef = 55.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

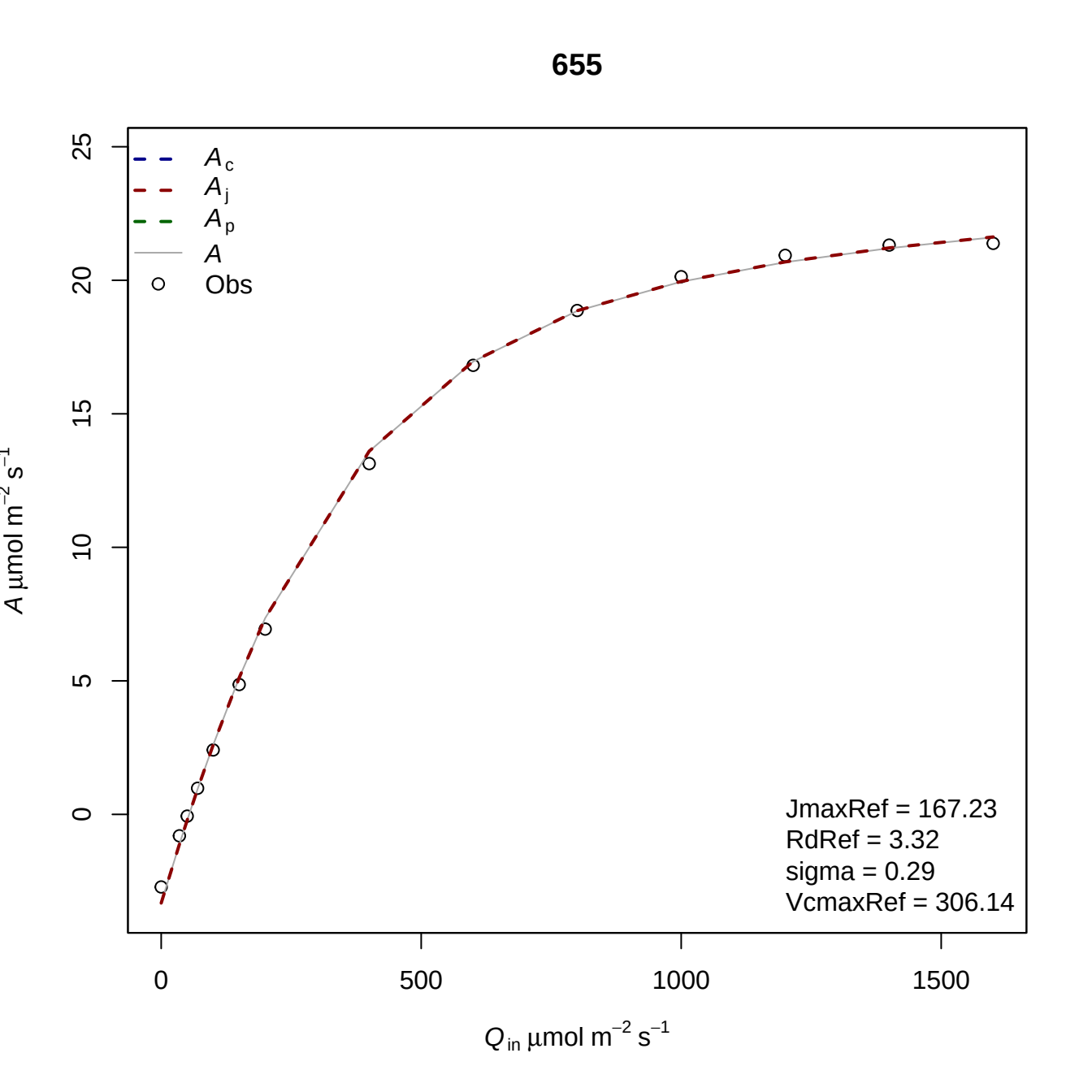

650
A_c
A_j
A_p
A
Obs
JmaxRef = 65.42
RdRef = 0.88
sigma = 0.22
VcmaxRef = 195.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

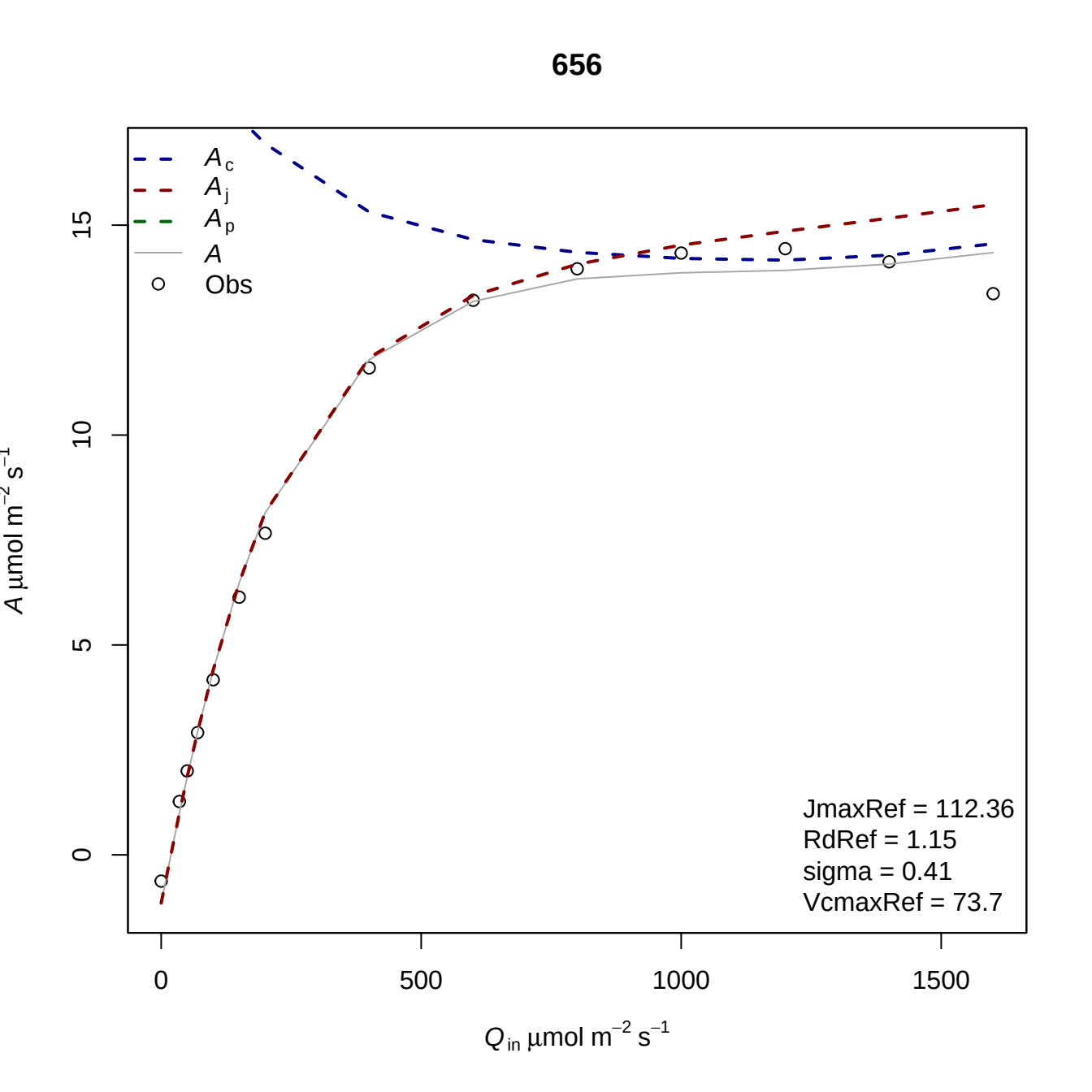

651
A_c
A_j
A_p
A
Obs
JmaxRef = 74.06
RdRef = 1.81
sigma = 0.66
VcmaxRef = 126.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

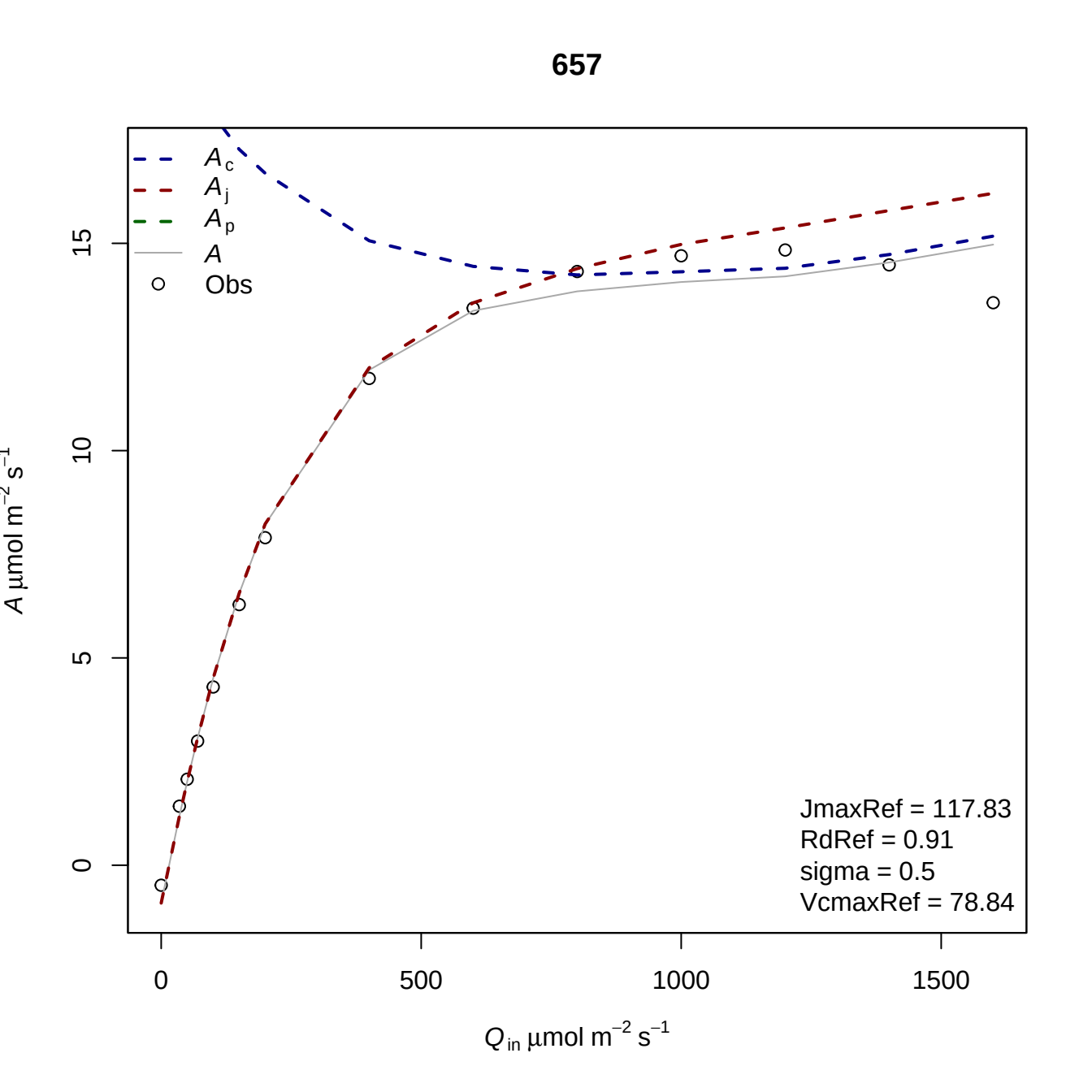

652
A_c
A_j
A_p
A
Obs
JmaxRef = 91.44
RdRef = 2.62
sigma = 1.02
VcmaxRef = 200.93
0
5
10
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

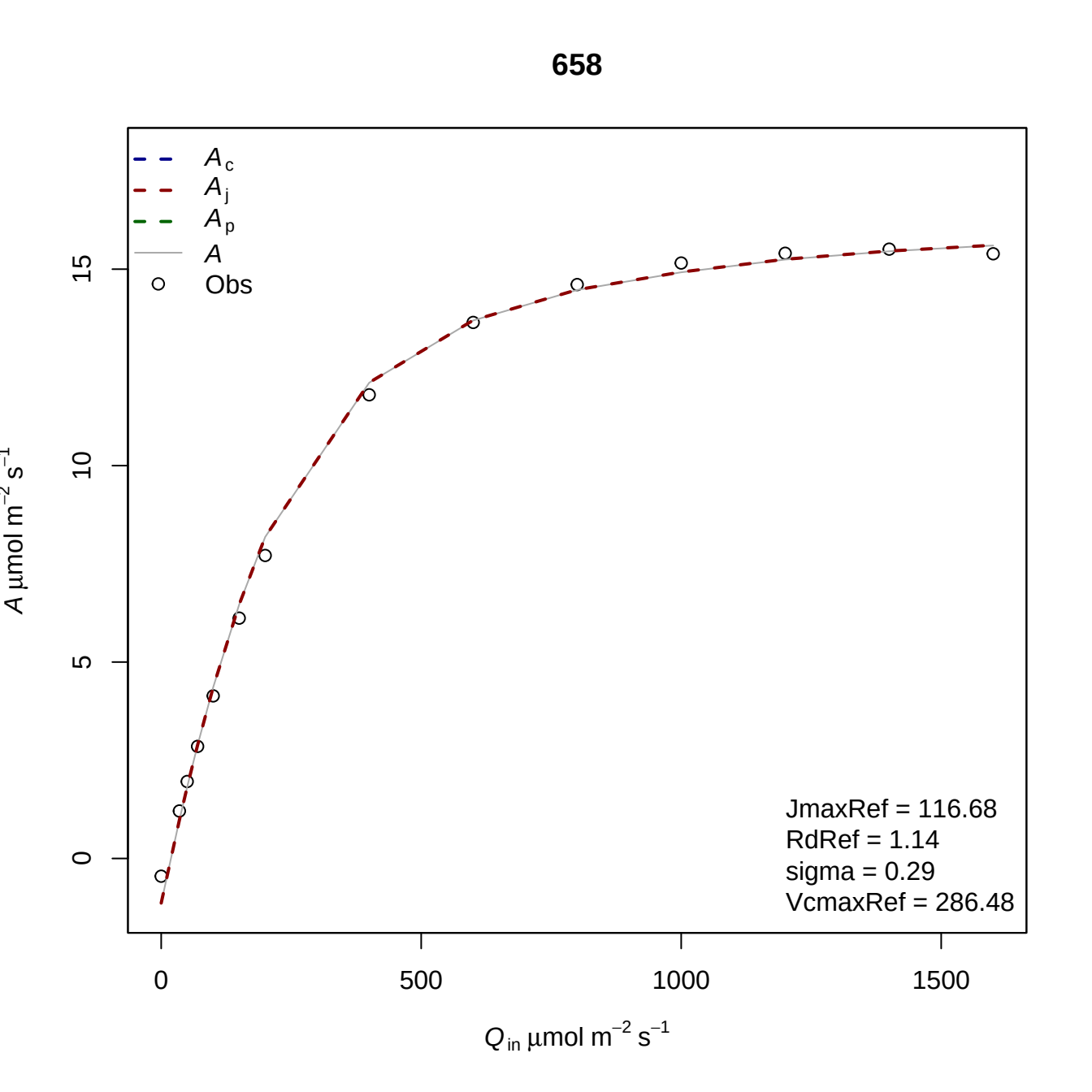

653
A_c
A_j
A_p
A
Obs
JmaxRef = 101.48
RdRef = 1.74
sigma = 0.43
VcmaxRef = 319.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

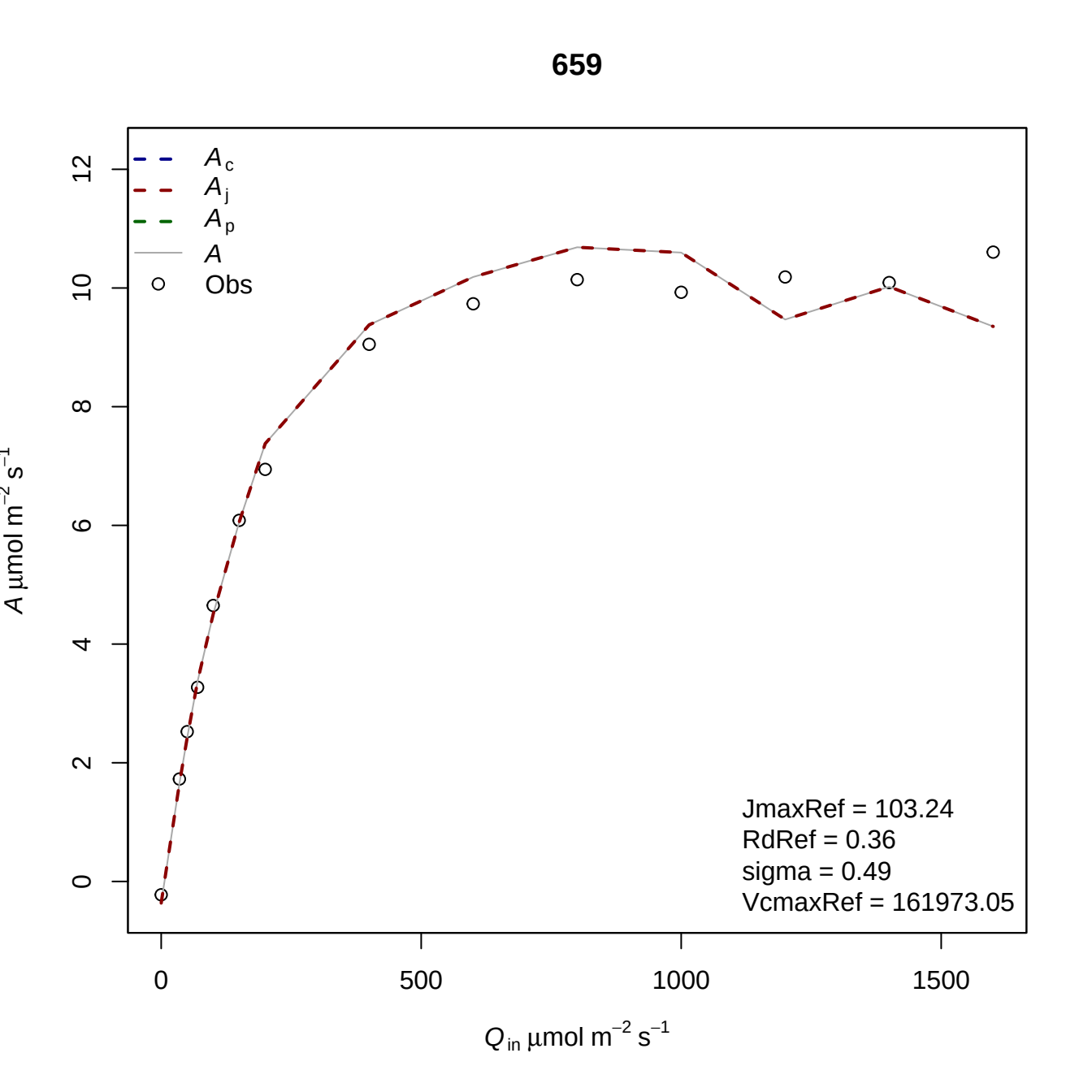

654
A_c
A_j
A_p
A
Obs
JmaxRef = 92.84
RdRef = 1.85
sigma = 0.33
VcmaxRef = 207.86
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

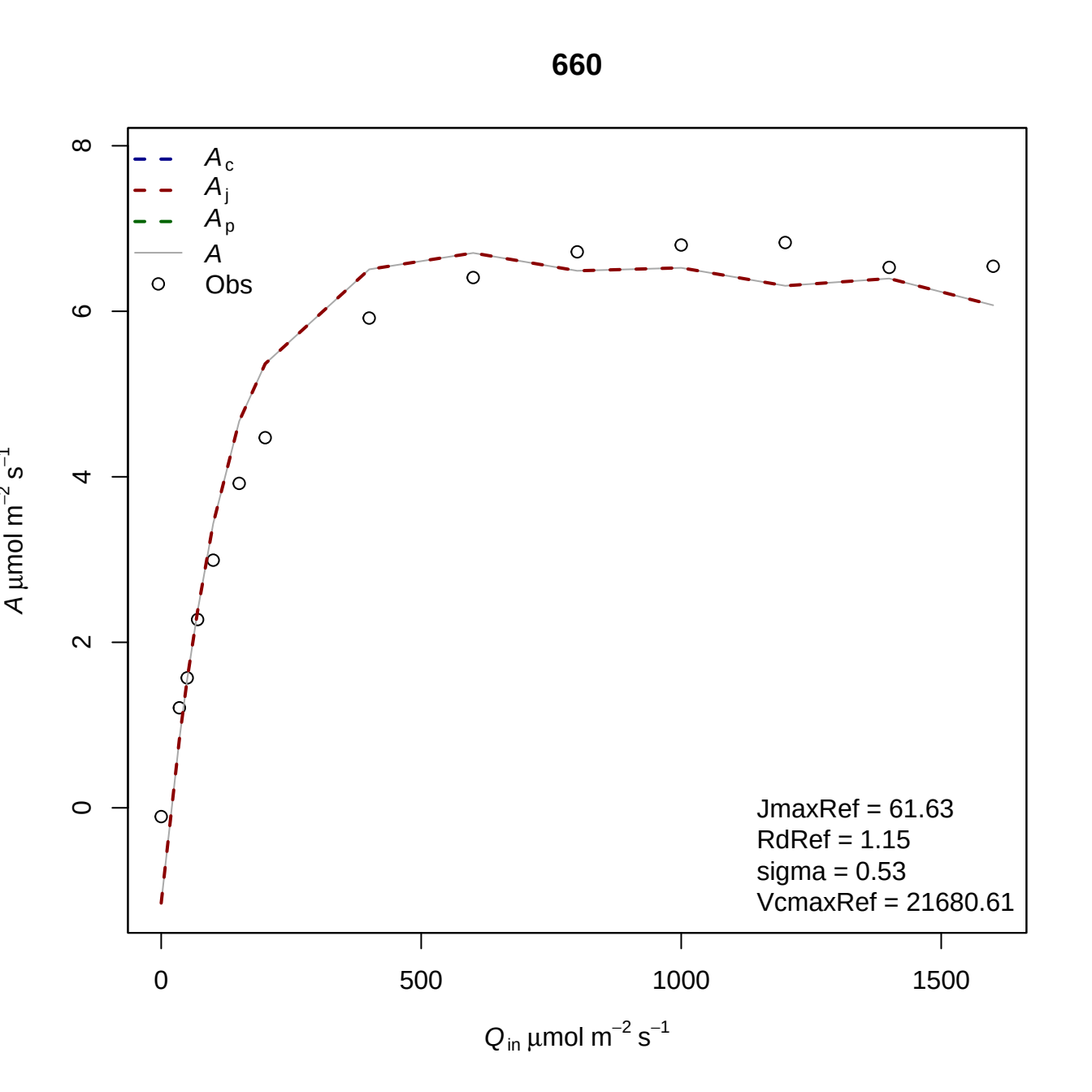

655
A_c
A_j
A_p
A
Obs
JmaxRef = 167.23
RdRef = 3.32
sigma = 0.29
VcmaxRef = 306.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

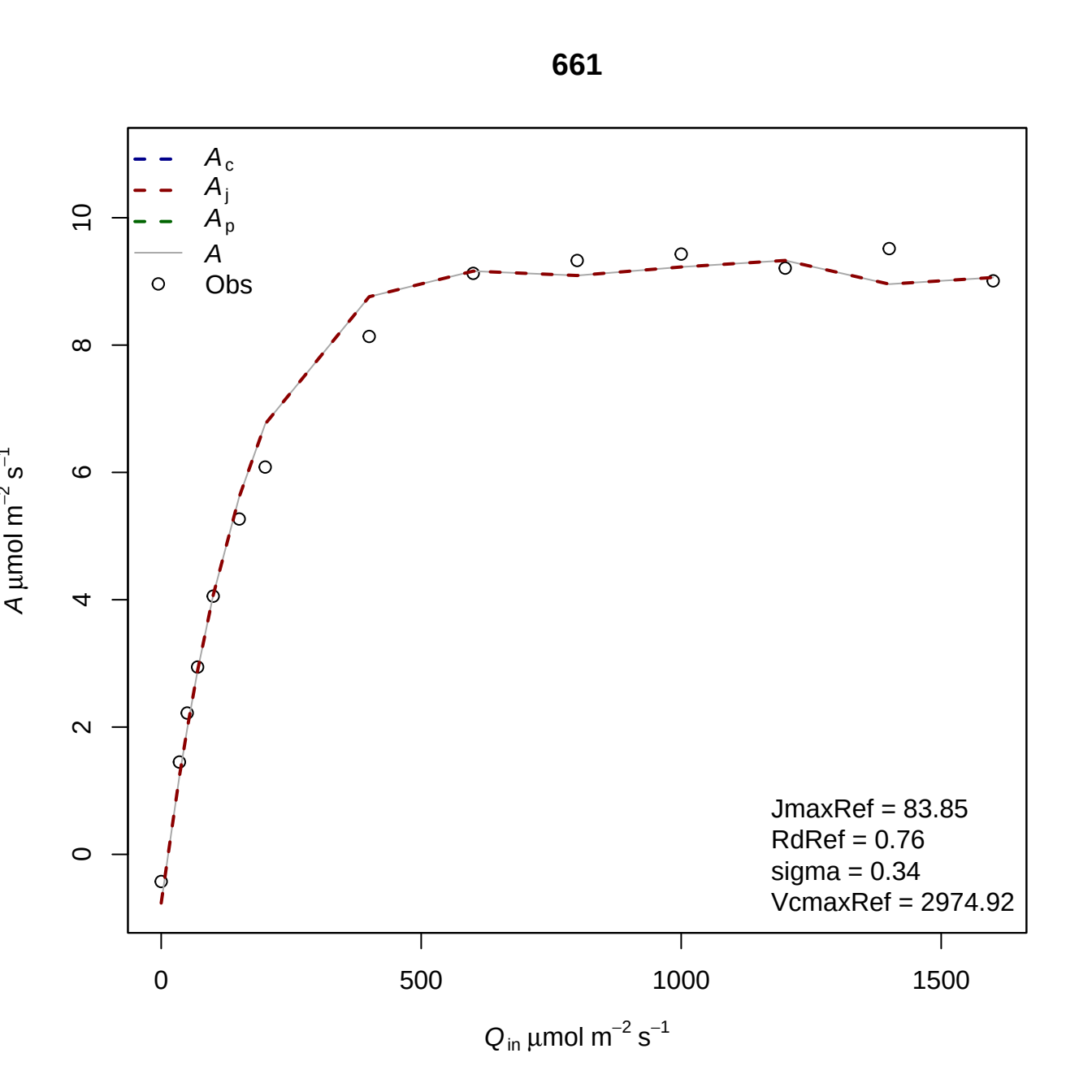

656
A_c
A_j
A_p
A
Obs
JmaxRef = 112.36
RdRef = 1.15
sigma = 0.41
VcmaxRef = 73.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

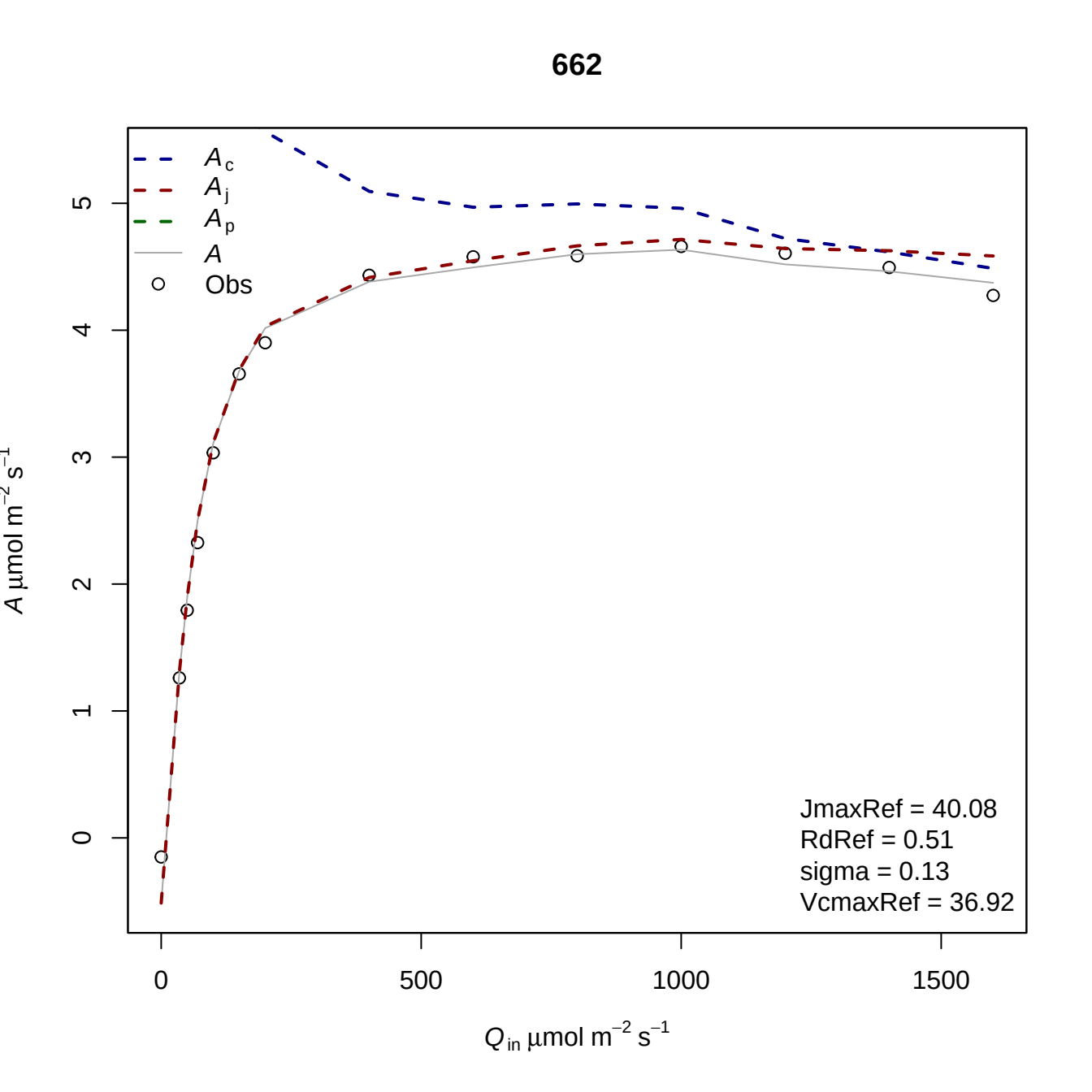

657
A_c
A_j
A_p
A
Obs
JmaxRef = 117.83
RdRef = 0.91
sigma = 0.5
VcmaxRef = 78.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

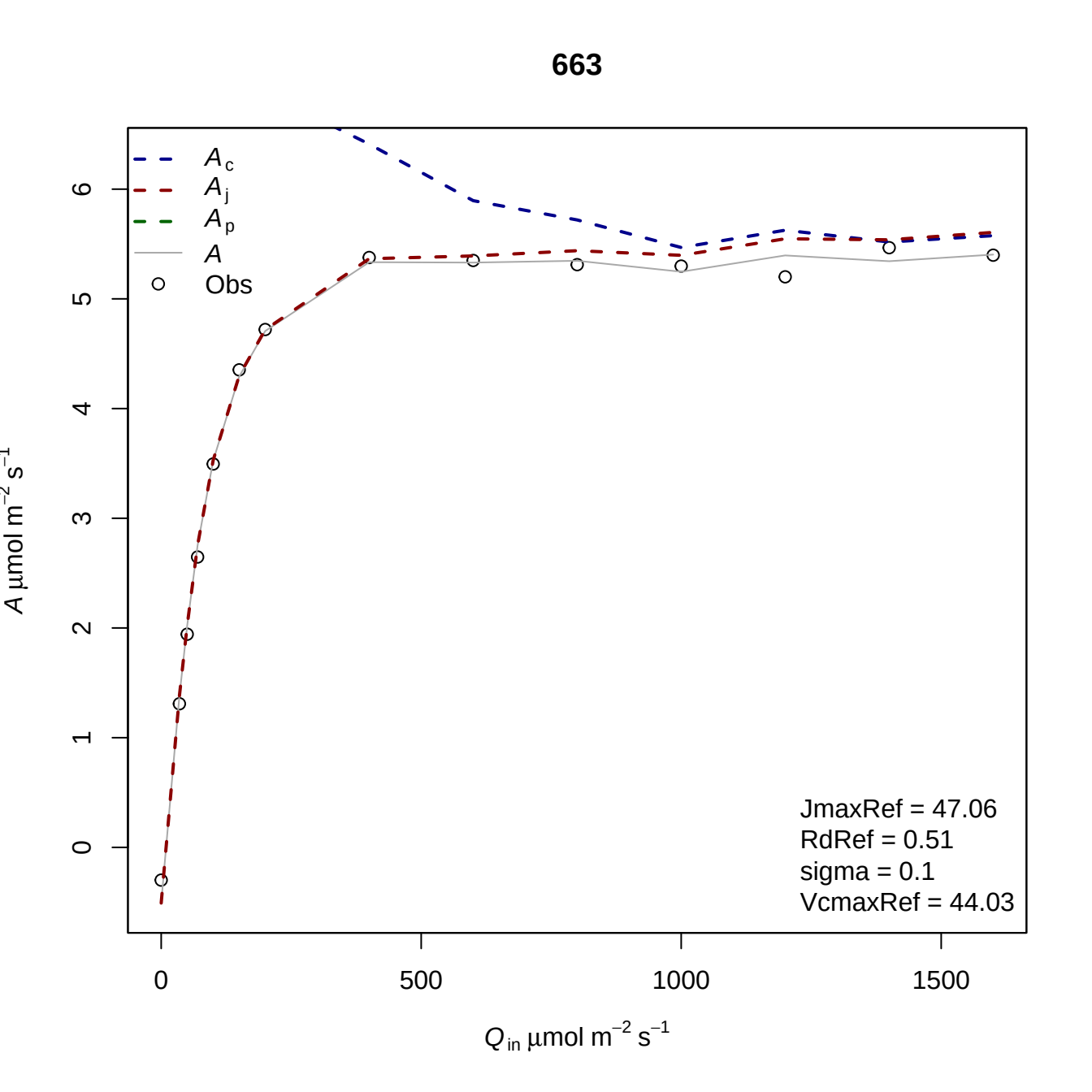

658
A_c
A_j
A_p
A
Obs
JmaxRef = 116.68
RdRef = 1.14
sigma = 0.29
VcmaxRef = 286.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

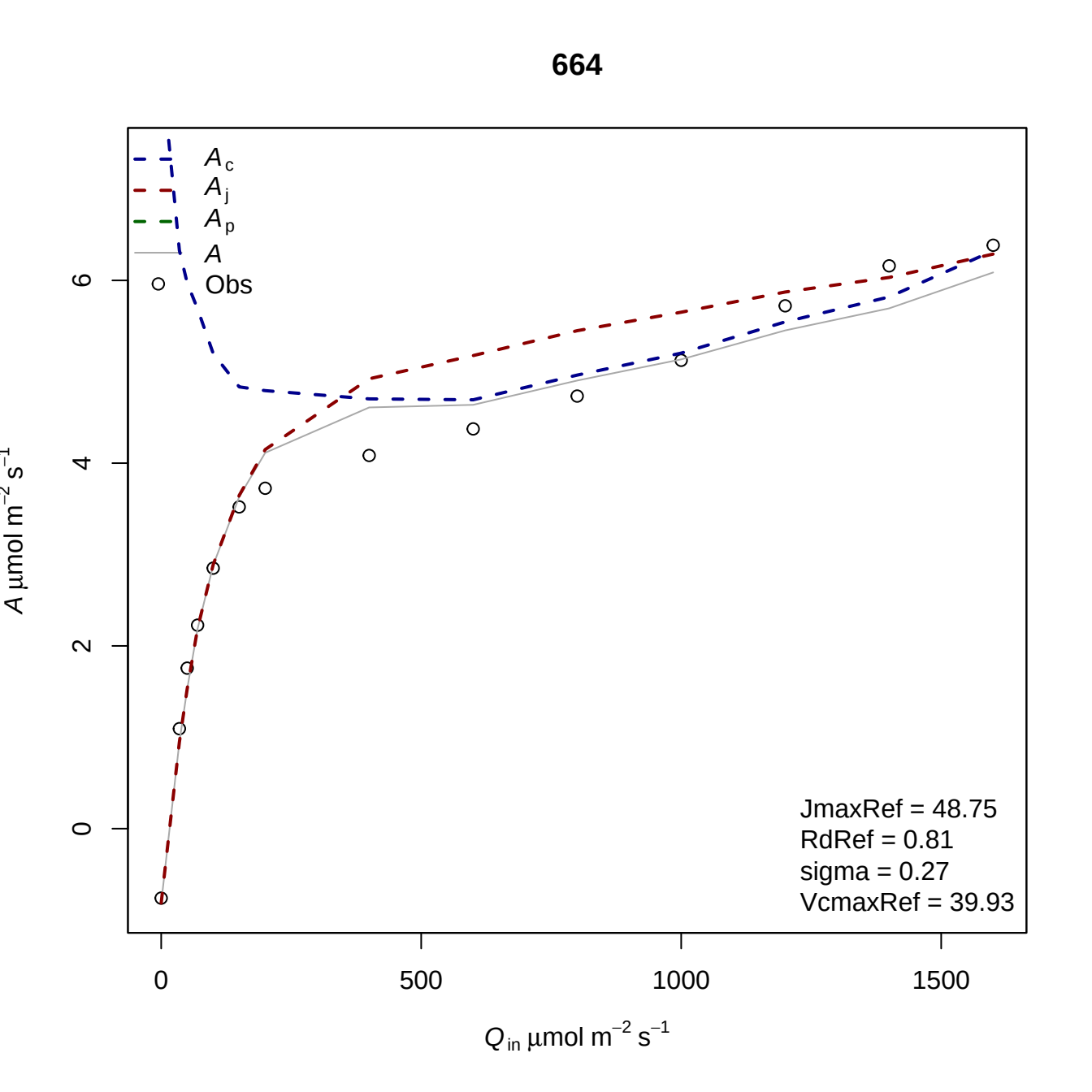

659
A_c
A_j
A_p
A
Obs
JmaxRef = 103.24
RdRef = 0.36
sigma = 0.49
VcmaxRef = 161973.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

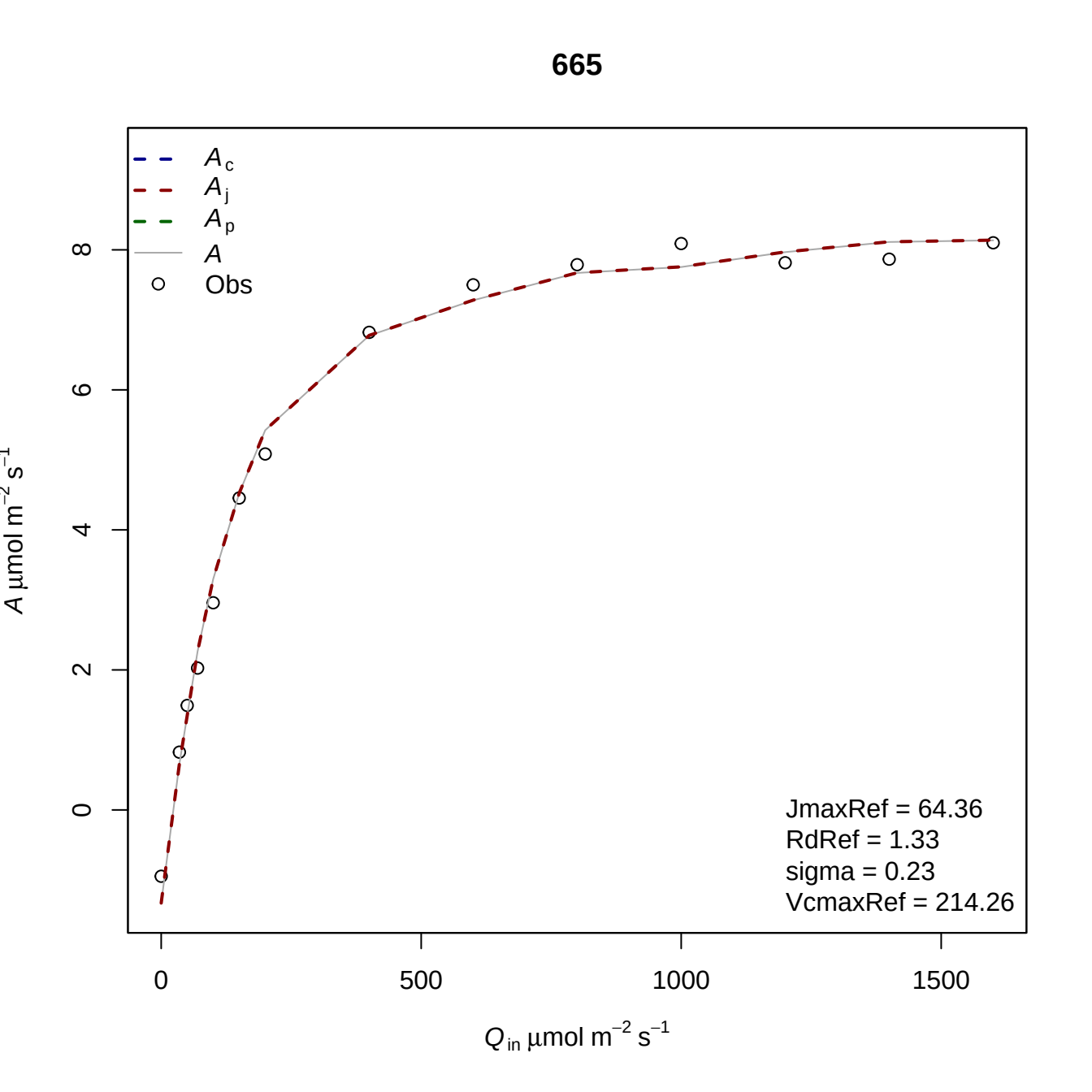

660
A_c
A_j
A_p
A
Obs
JmaxRef = 61.63
RdRef = 1.15
sigma = 0.53
VcmaxRef = 21680.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

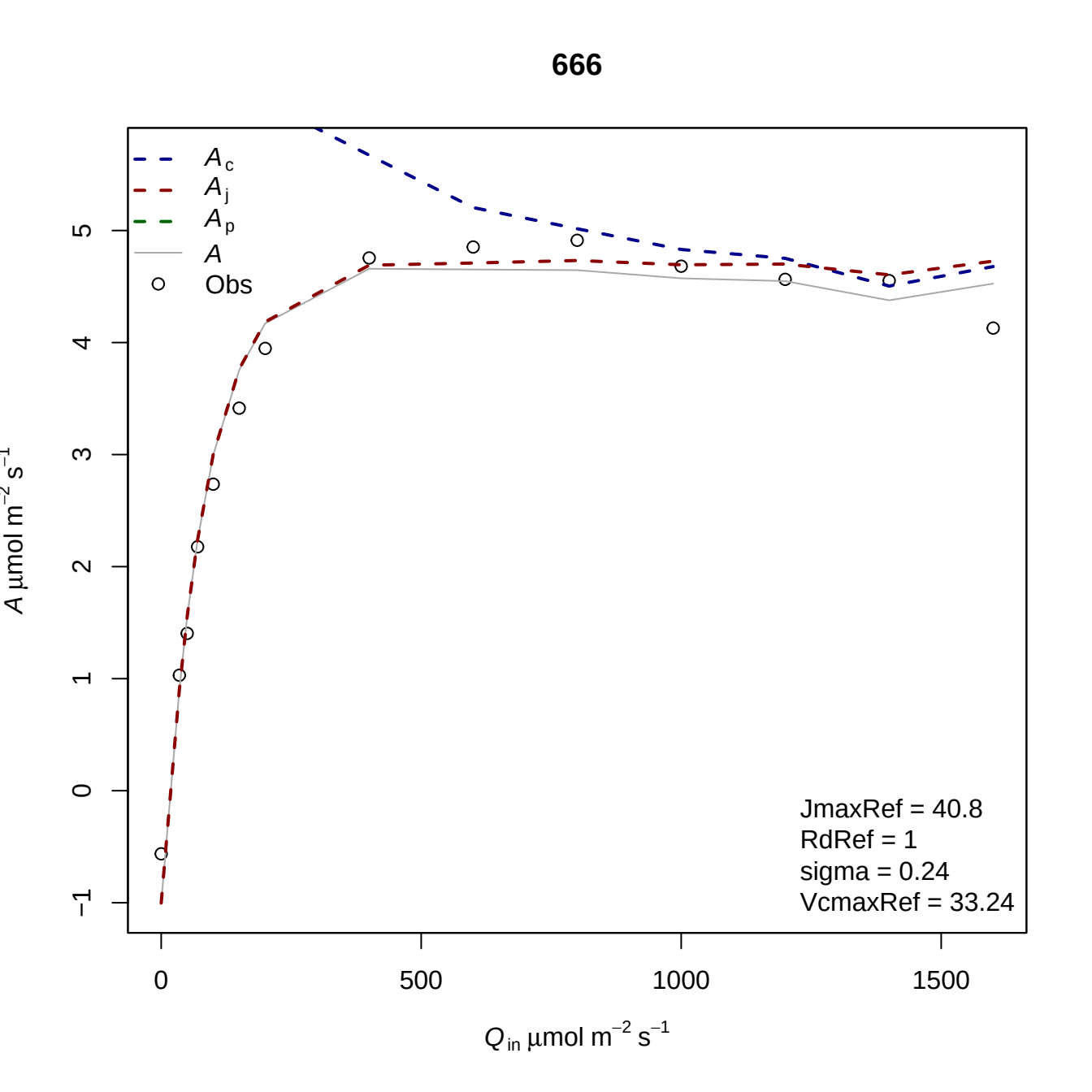

661
A_c
A_j
A_p
A
Obs
JmaxRef = 83.85
RdRef = 0.76
sigma = 0.34
VcmaxRef = 2974.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

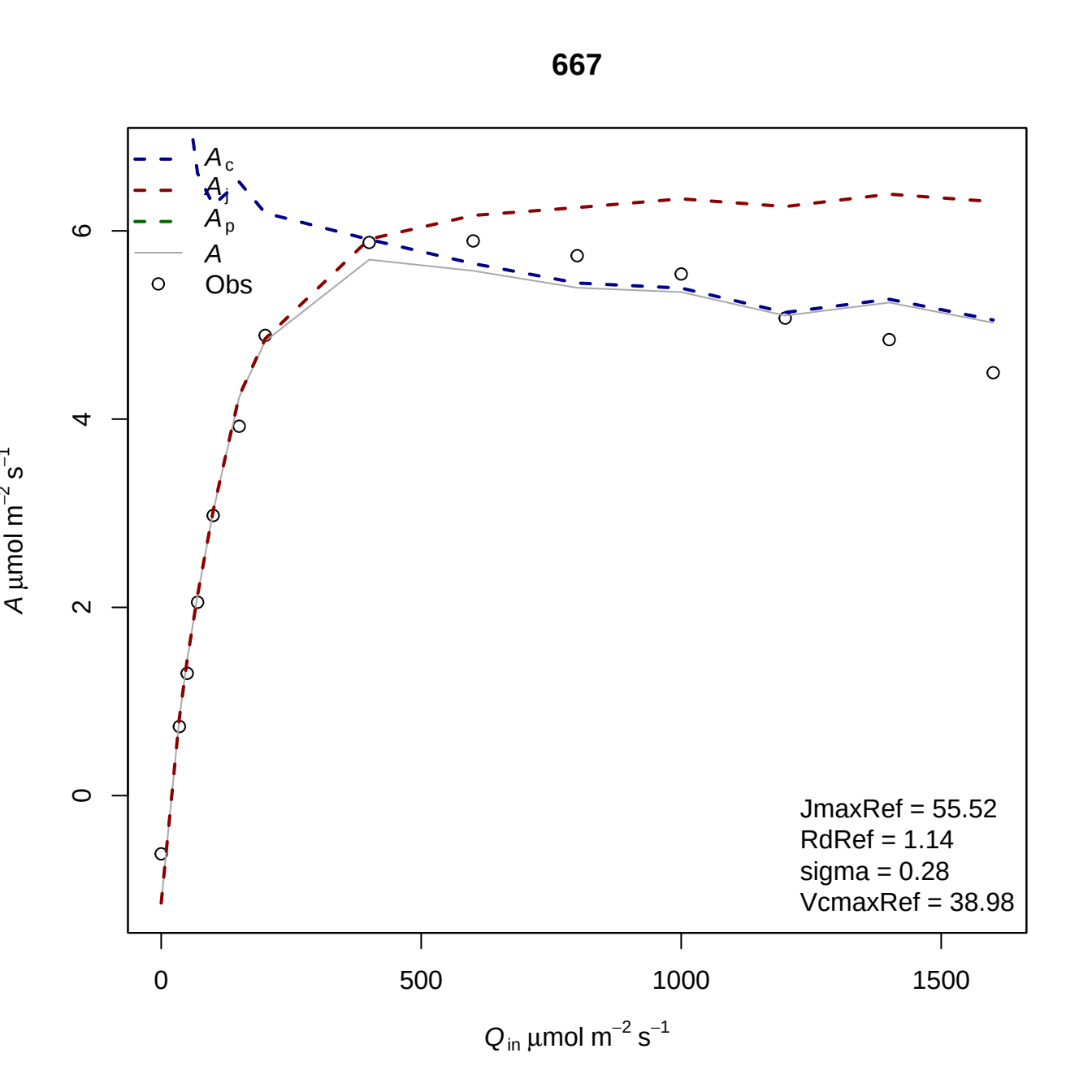

662
A_c
A_j
A_p
A
Obs
JmaxRef = 40.08
RdRef = 0.51
sigma = 0.13
VcmaxRef = 36.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

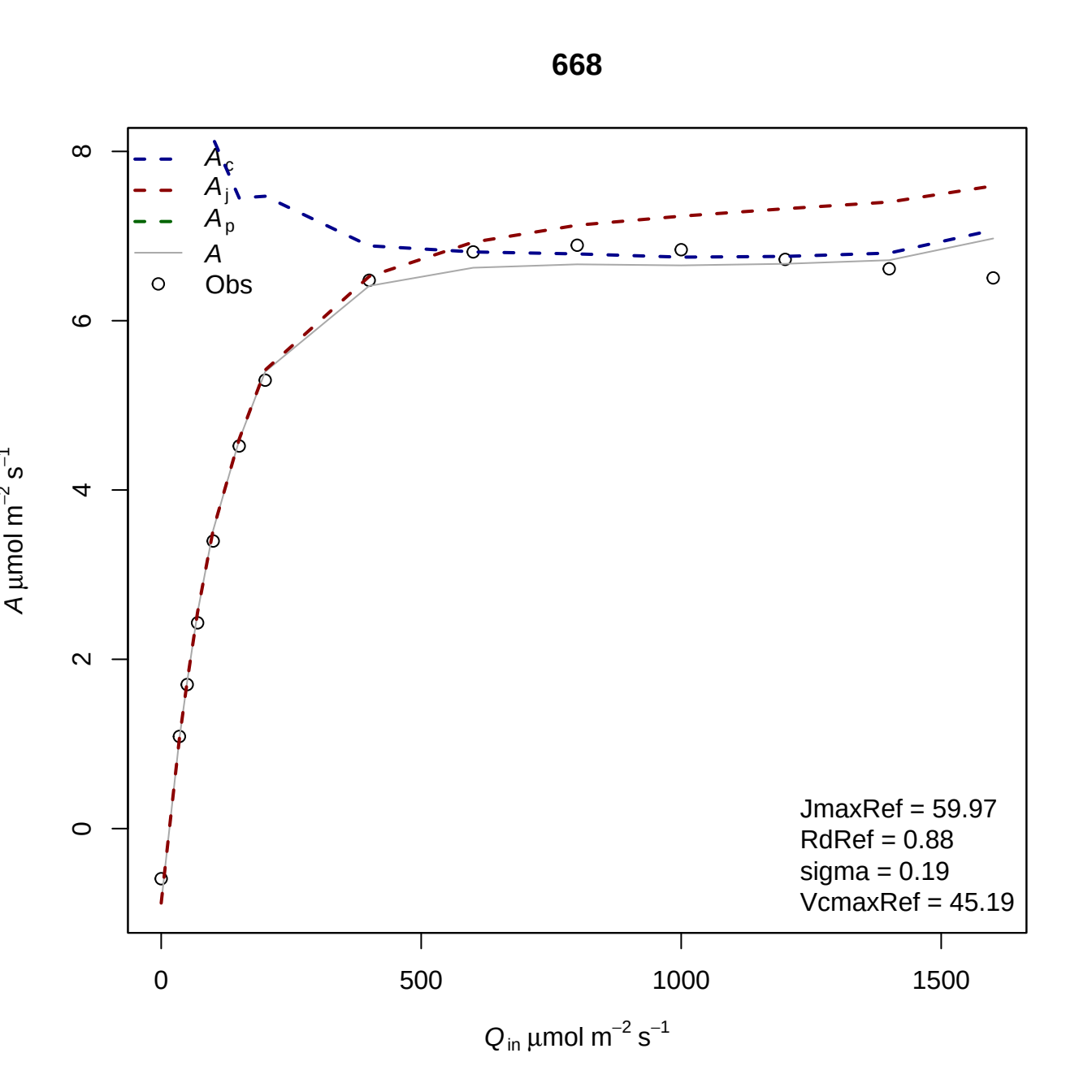

663
A_c
A_j
A_p
A
Obs
JmaxRef = 47.06
RdRef = 0.51
sigma = 0.1
VcmaxRef = 44.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

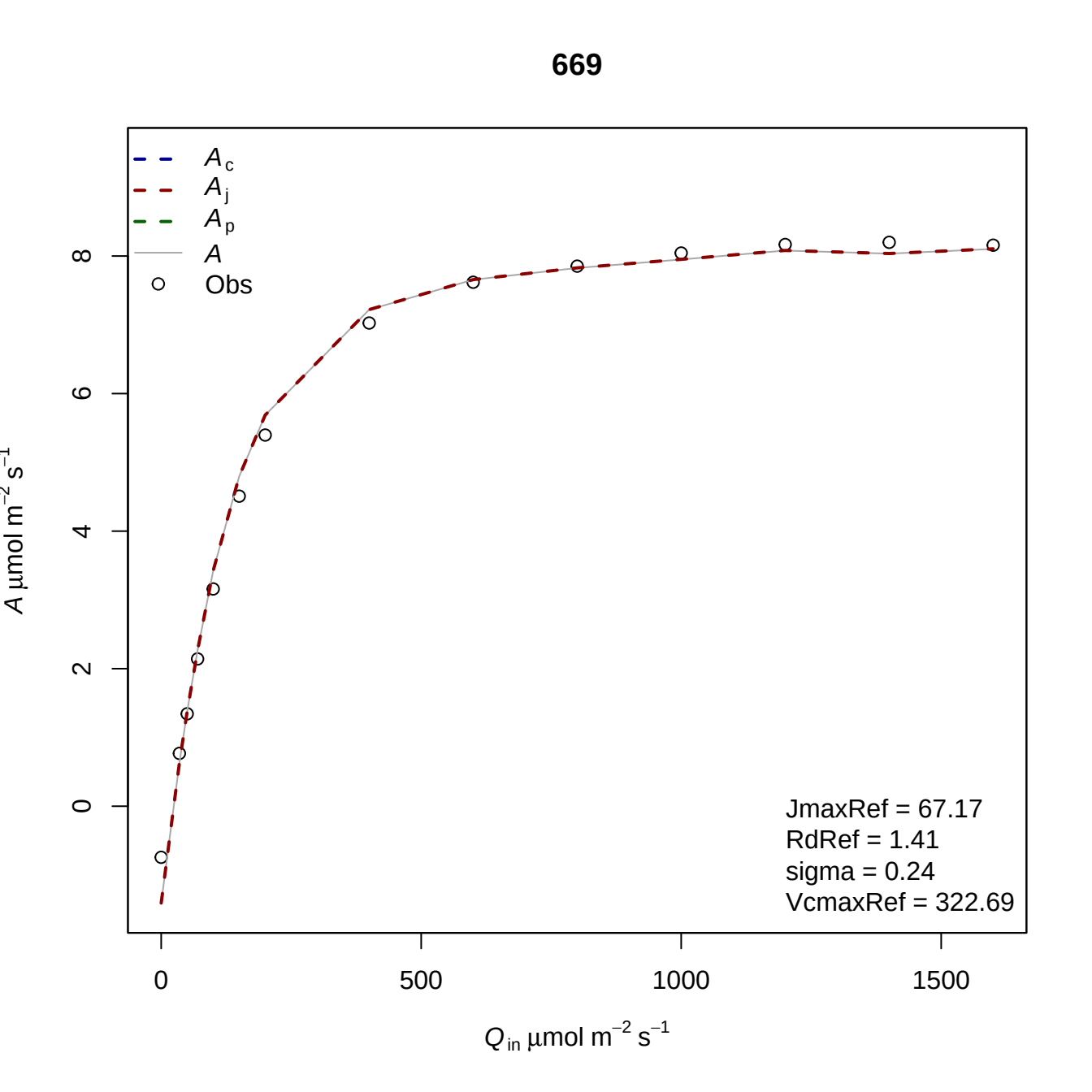

664
A_c
A_j
A_p
A
Obs
JmaxRef = 48.75
RdRef = 0.81
sigma = 0.27
VcmaxRef = 39.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

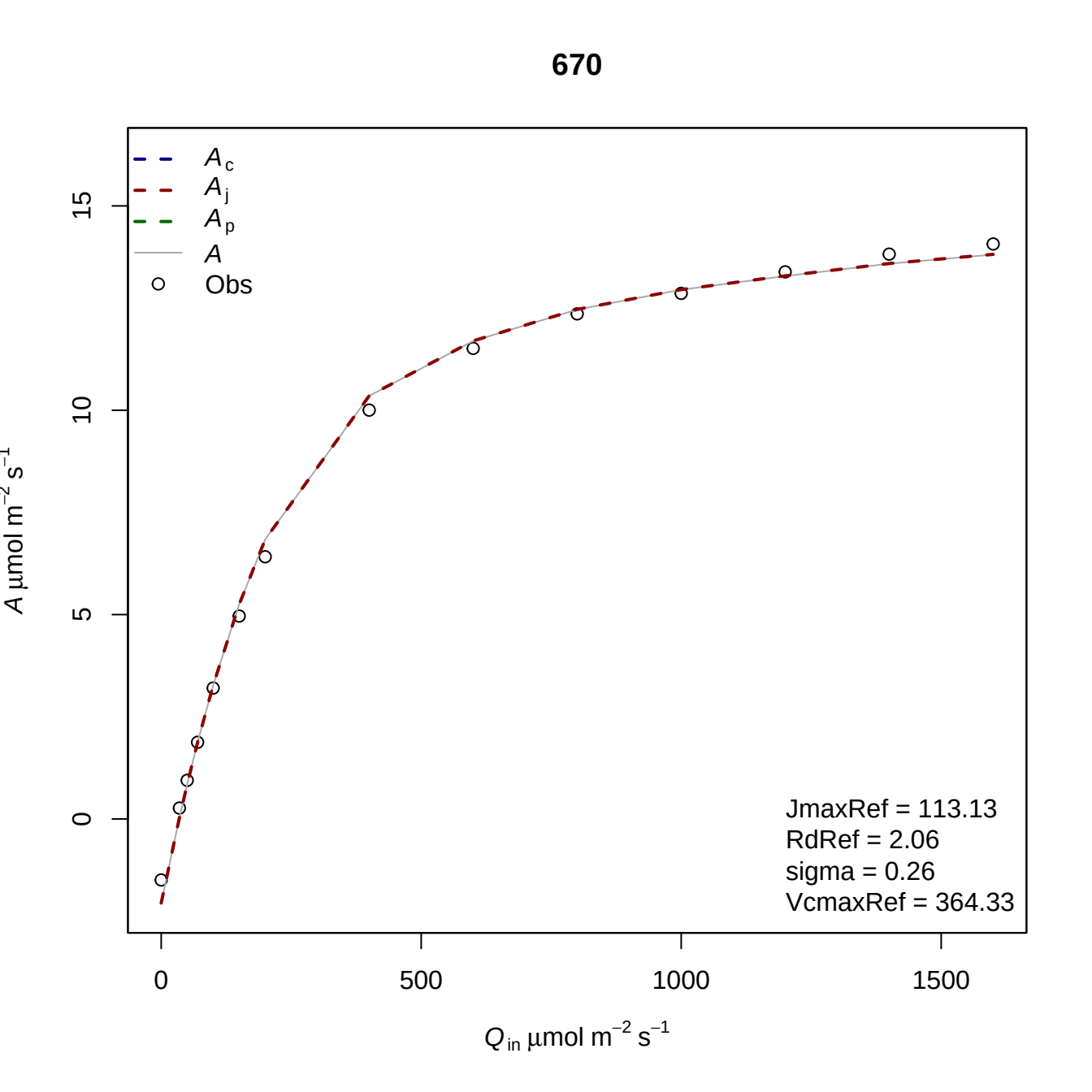

665
A_c
A_j
A_p
A
Obs
JmaxRef = 64.36
RdRef = 1.33
sigma = 0.23
VcmaxRef = 214.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

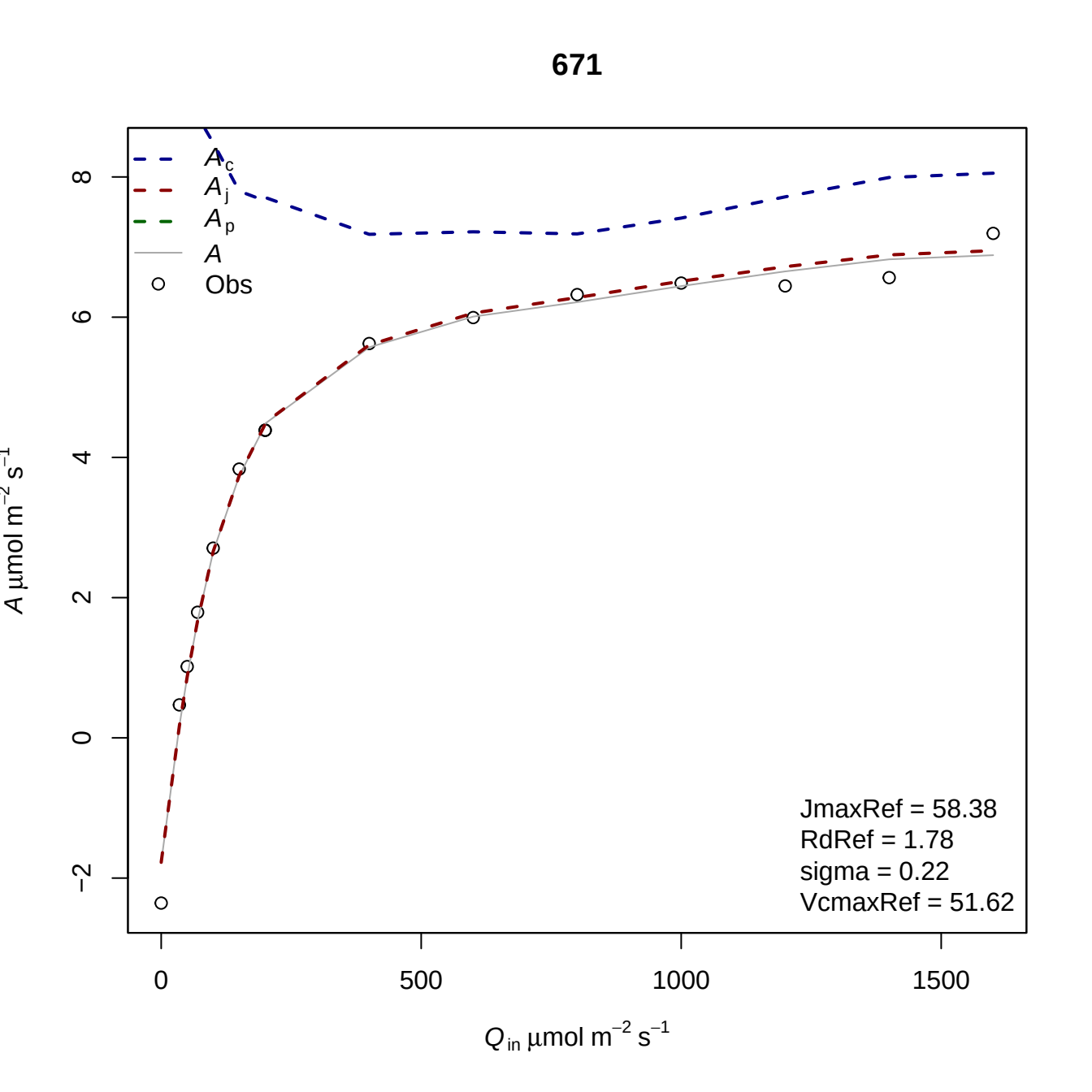

666
A_c
A_j
A_p
A
Obs
JmaxRef = 40.8
RdRef = 1
sigma = 0.24
VcmaxRef = 33.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

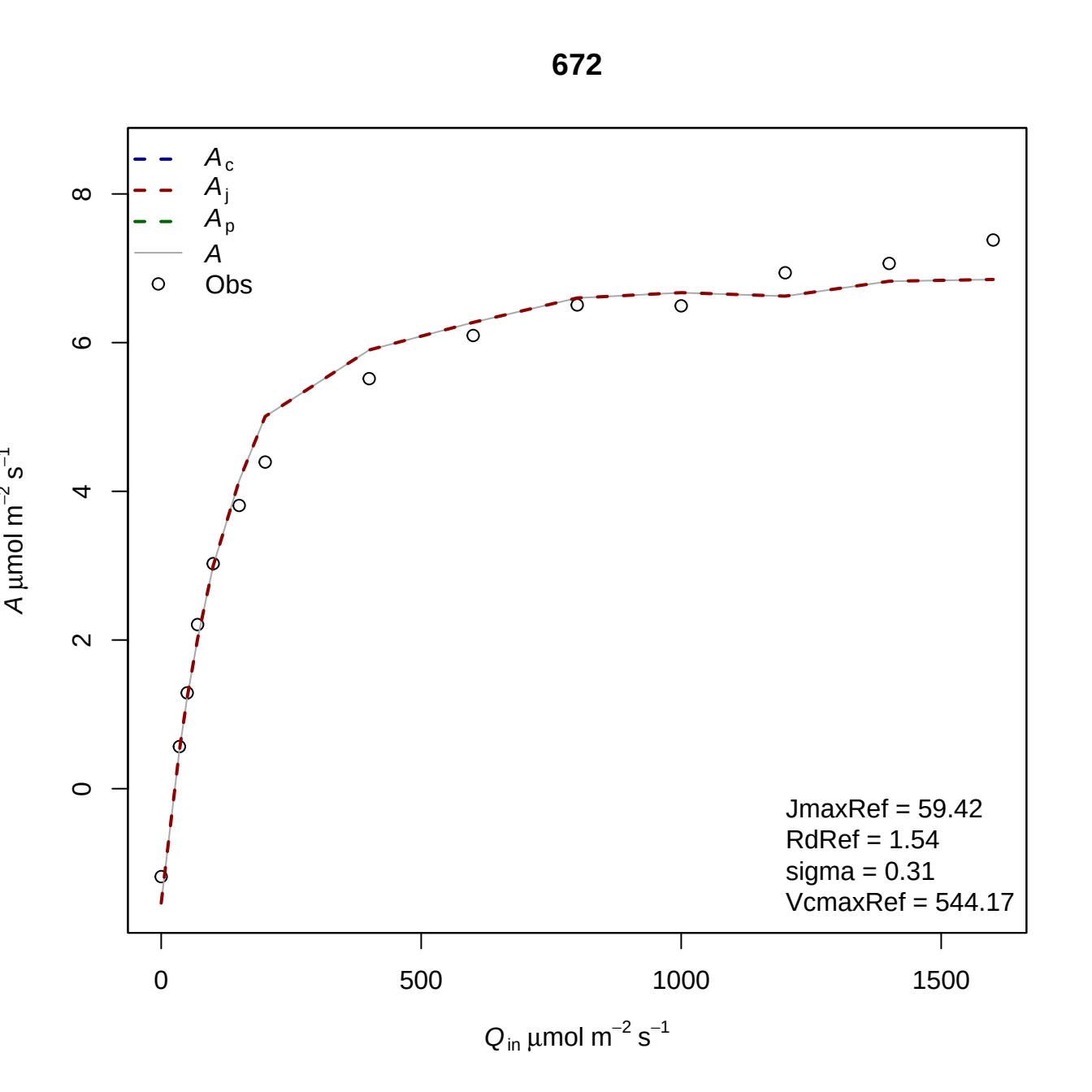

667
A_c
A_j
A_p
A
Obs
JmaxRef = 55.52
RdRef = 1.14
sigma = 0.28
VcmaxRef = 38.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

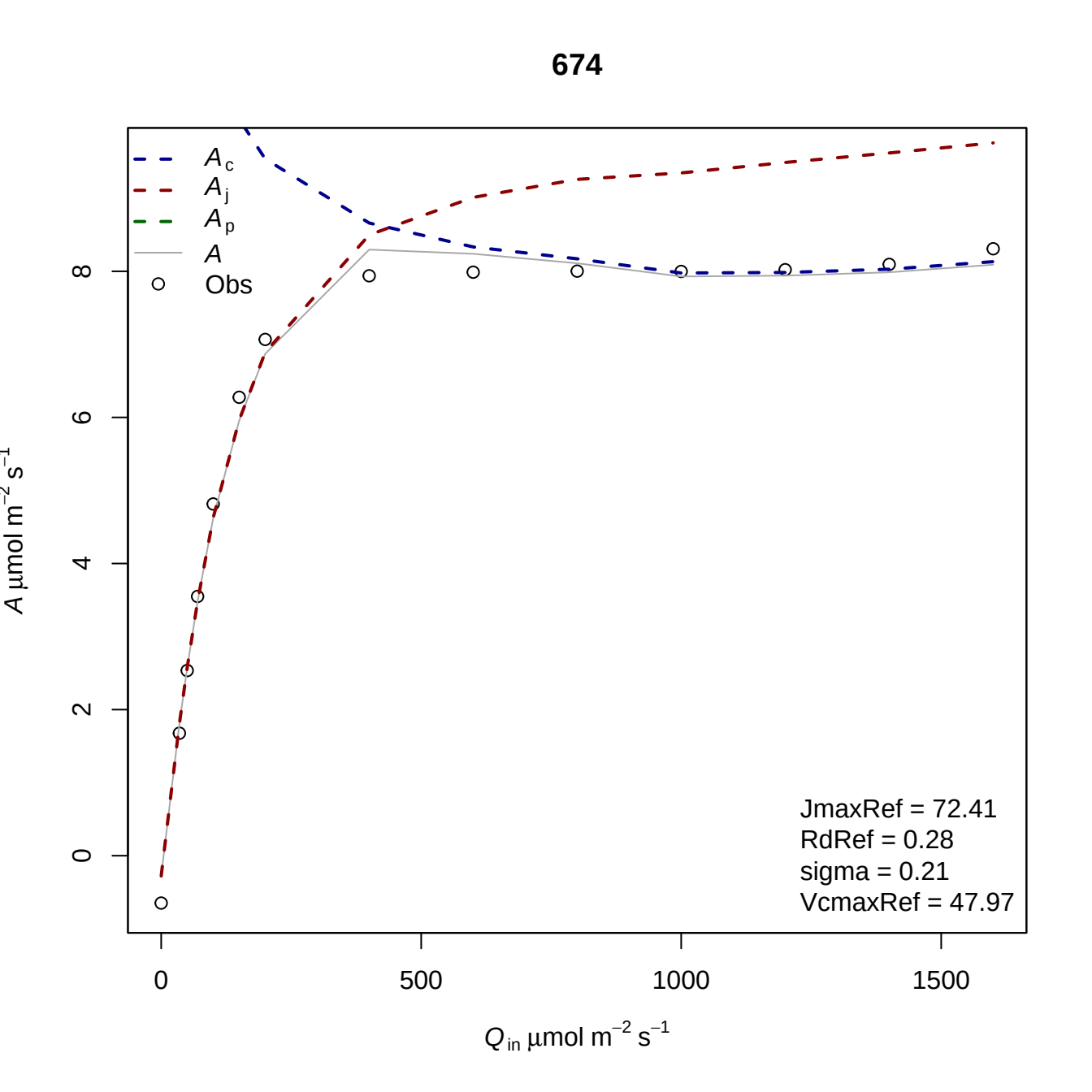

668
A_c
A_j
A_p
A
Obs
JmaxRef = 59.97
RdRef = 0.88
sigma = 0.19
VcmaxRef = 45.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

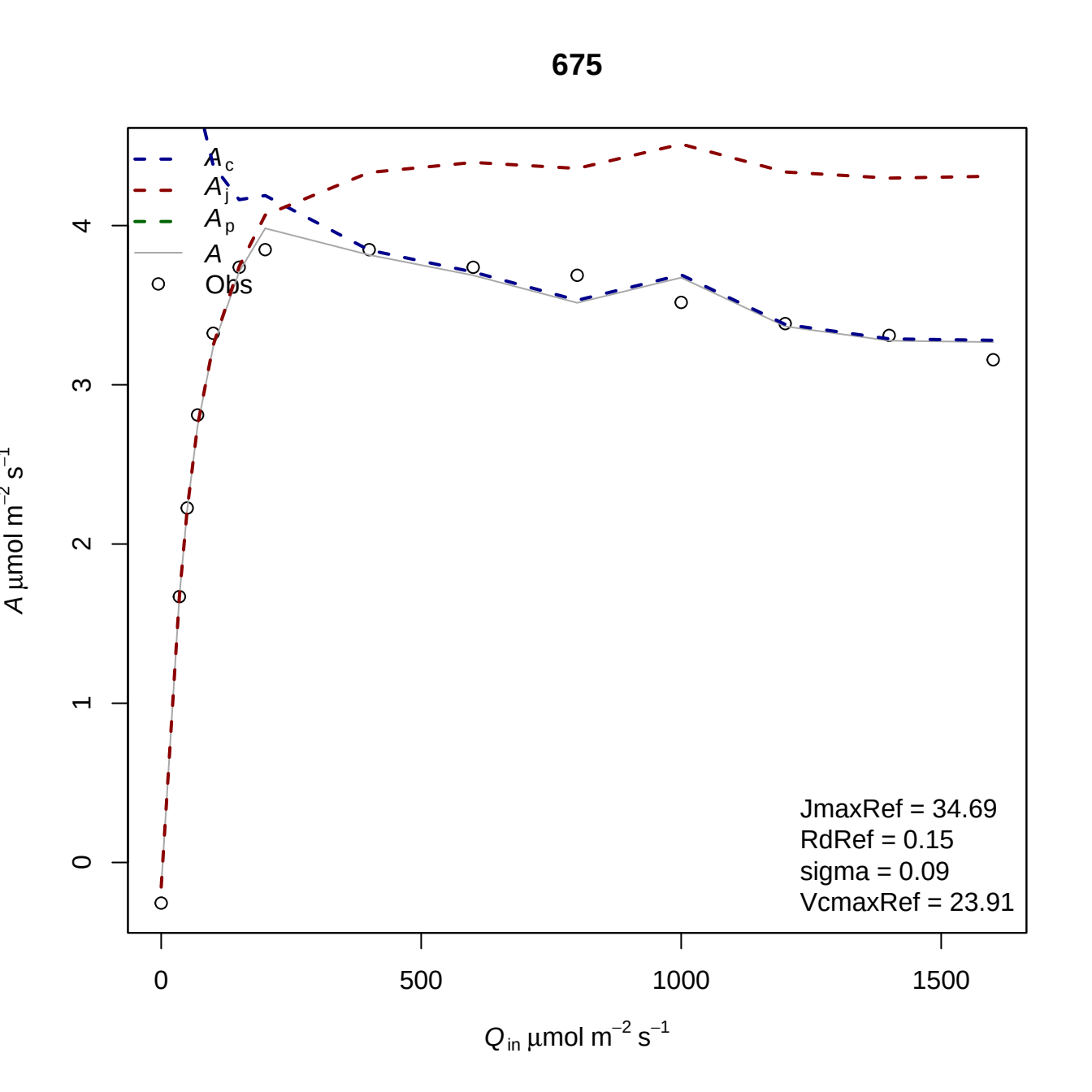

669
A_c
A_j
A_p
A
Obs
JmaxRef = 67.17
RdRef = 1.41
sigma = 0.24
VcmaxRef = 322.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

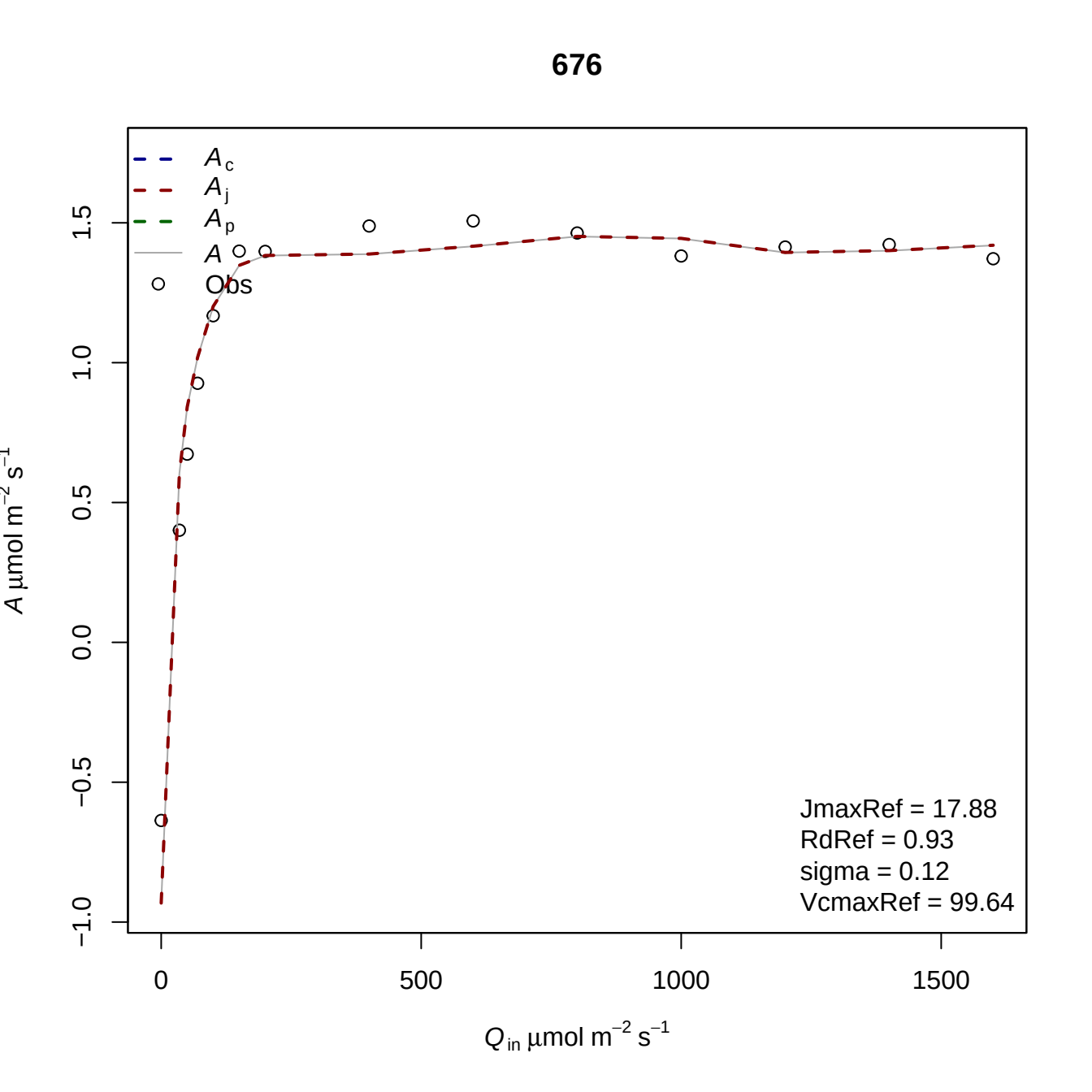

670
A_c
A_j
A_p
A
Obs
JmaxRef = 113.13
RdRef = 2.06
sigma = 0.26
VcmaxRef = 364.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

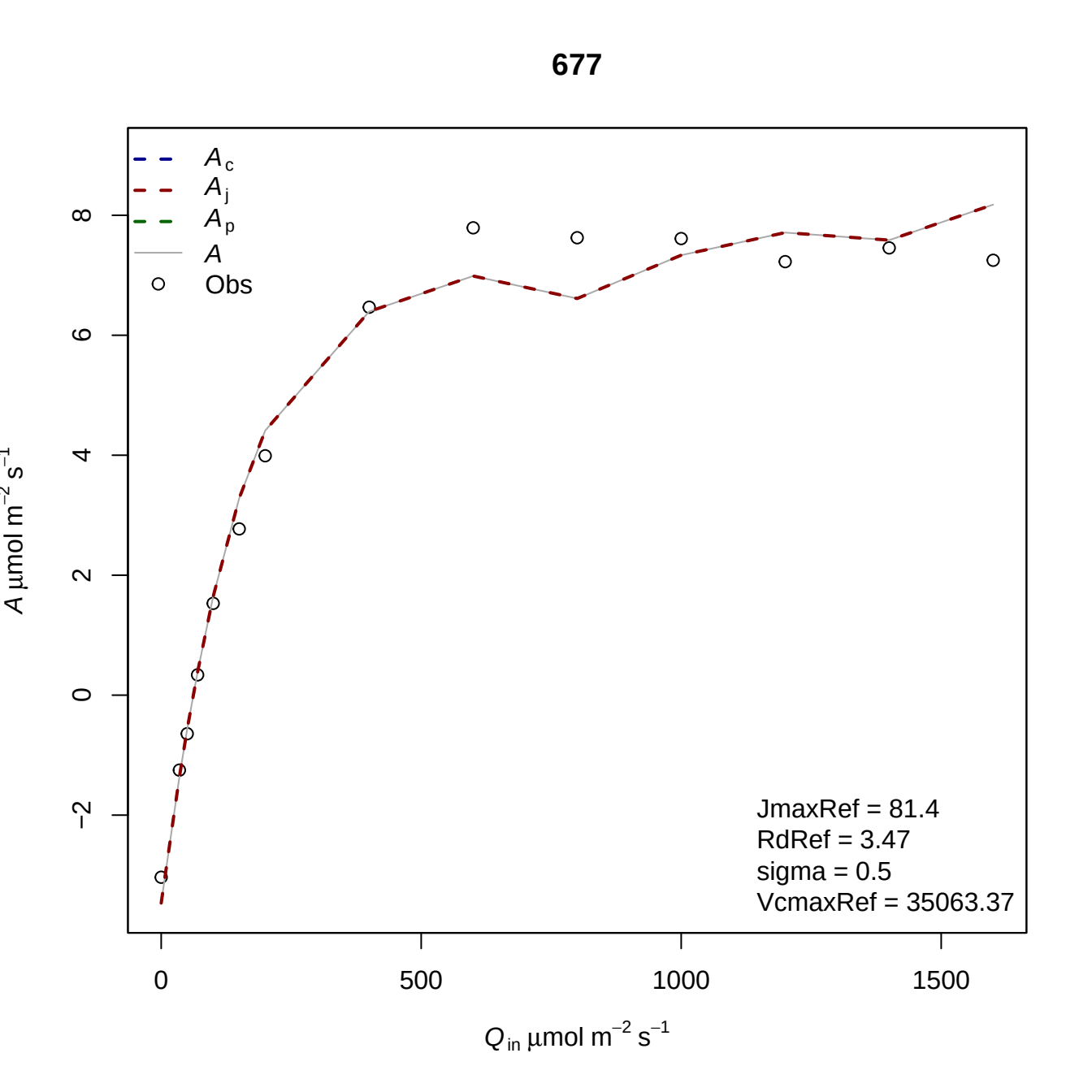

671
A_c
A_j
A_p
A
Obs
JmaxRef = 58.38
RdRef = 1.78
sigma = 0.22
VcmaxRef = 51.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

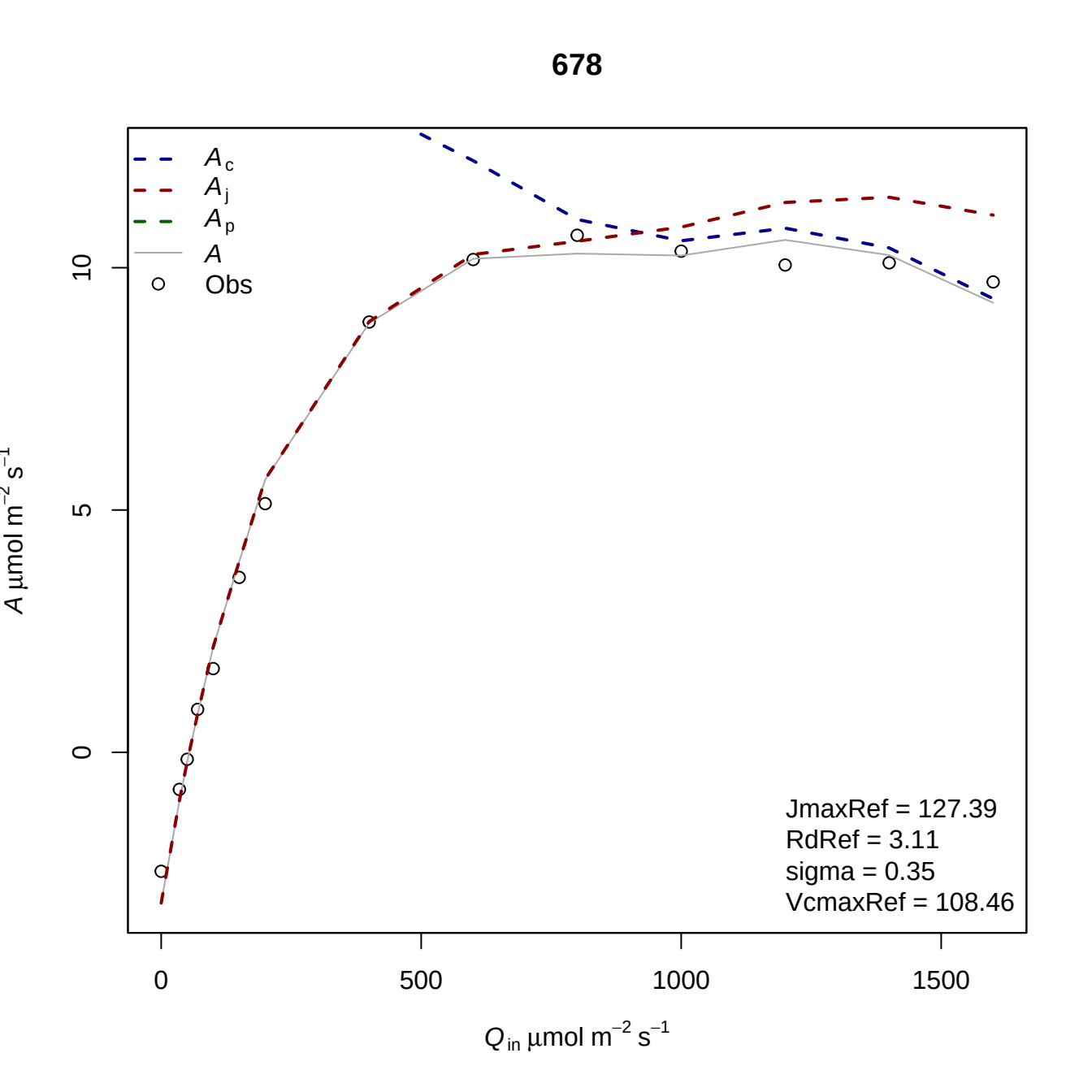

672
A_c
A_j
A_p
A
Obs
JmaxRef = 59.42
RdRef = 1.54
sigma = 0.31
VcmaxRef = 544.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

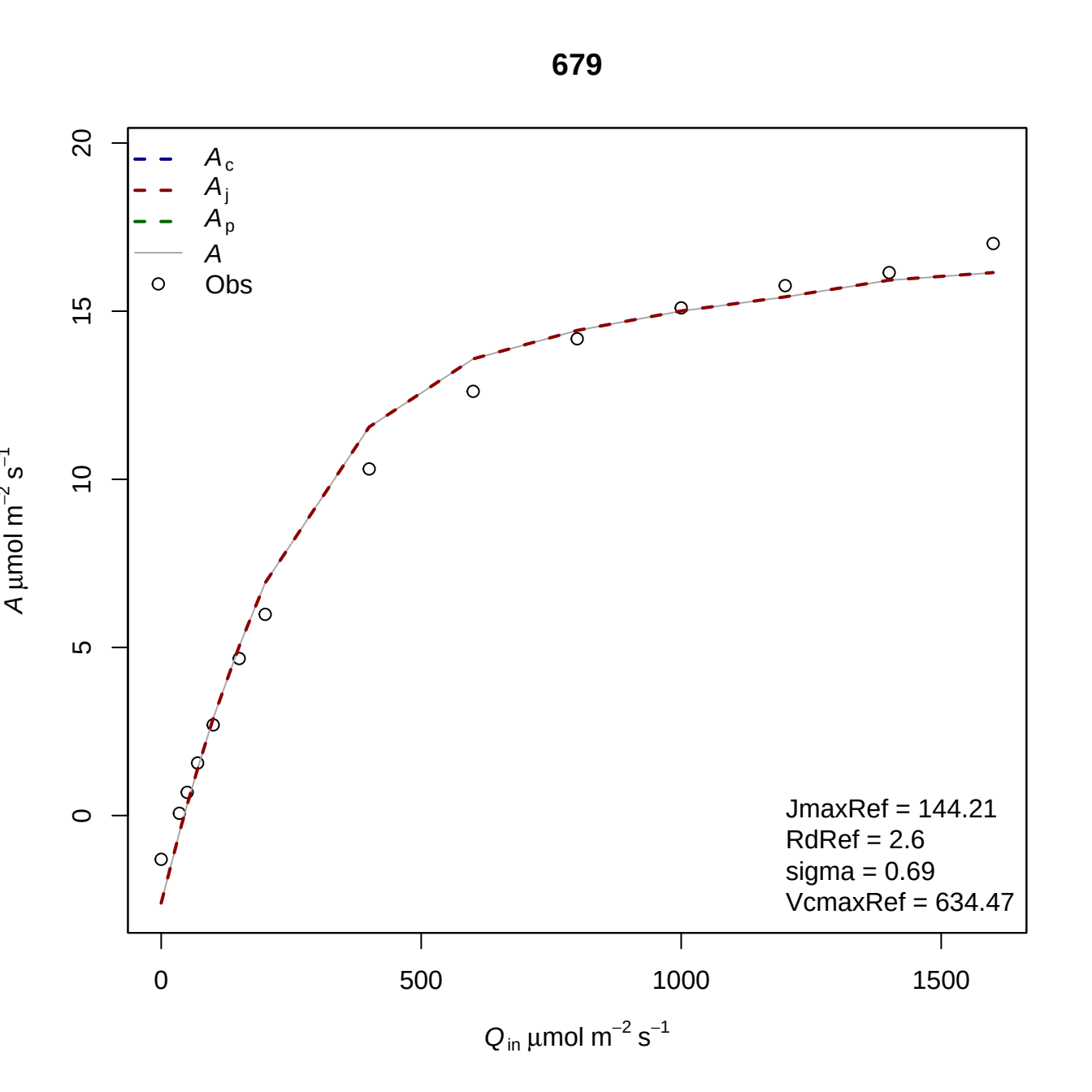

674
A_c
A_j
A_p
A
Obs
JmaxRef = 72.41
RdRef = 0.28
sigma = 0.21
VcmaxRef = 47.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

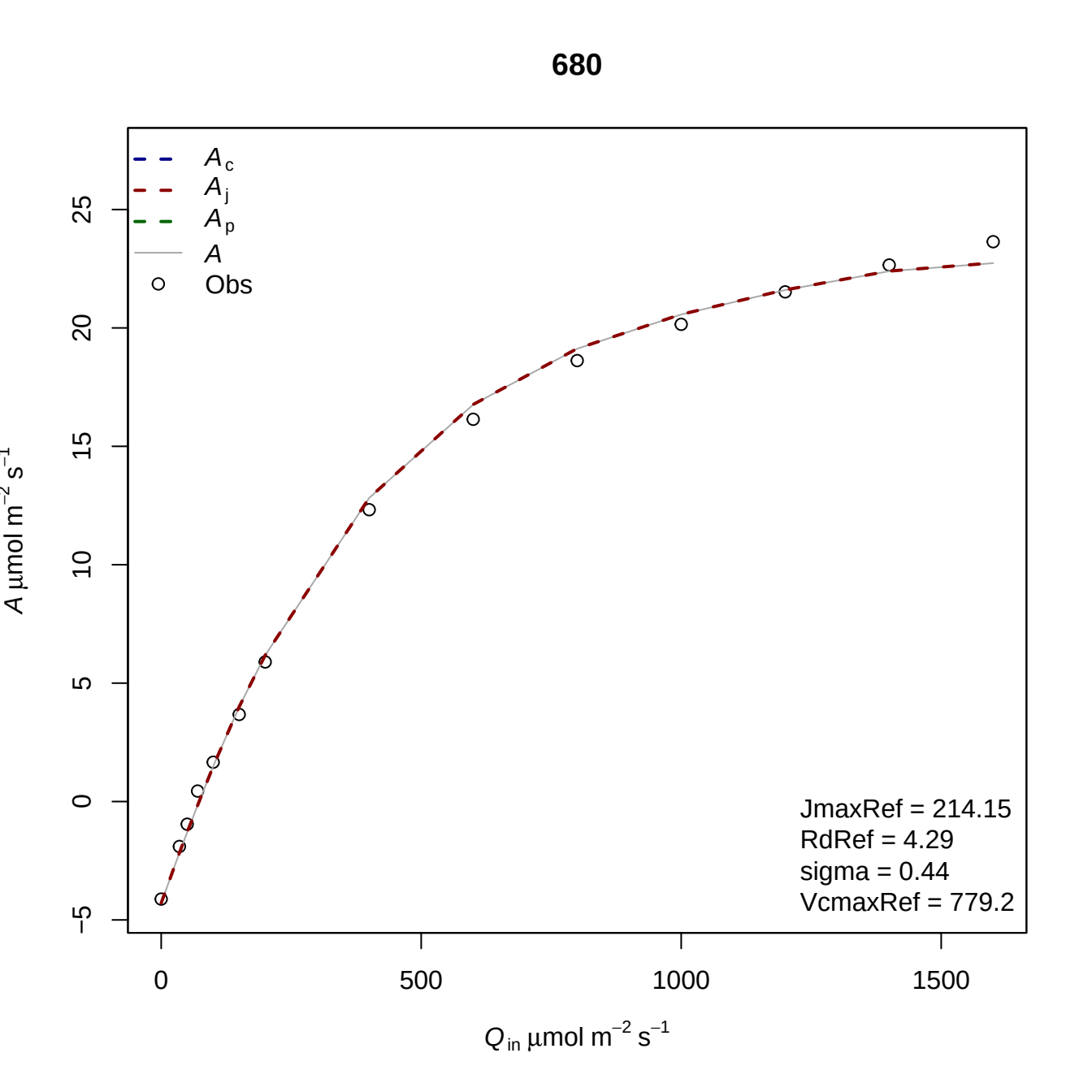

675
A_c
A_j
A_p
A
Obs
JmaxRef = 34.69
RdRef = 0.15
sigma = 0.09
VcmaxRef = 23.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

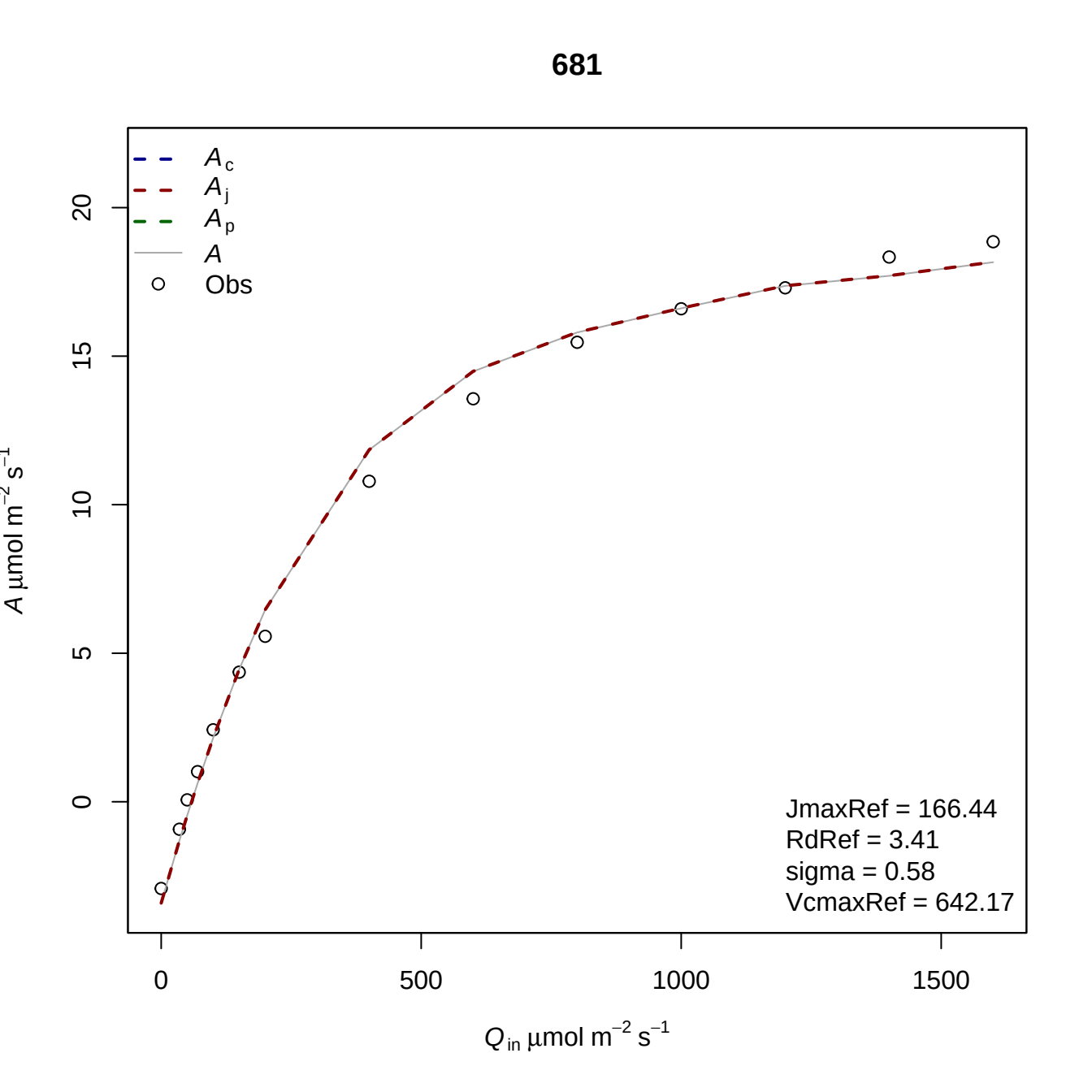

676
A_c
A_j
A_p
A
Obs
JmaxRef = 17.88
RdRef = 0.93
sigma = 0.12
VcmaxRef = 99.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1.0
−0.5
0.0
0.5
1.0
1.5
0
500
1000
1500

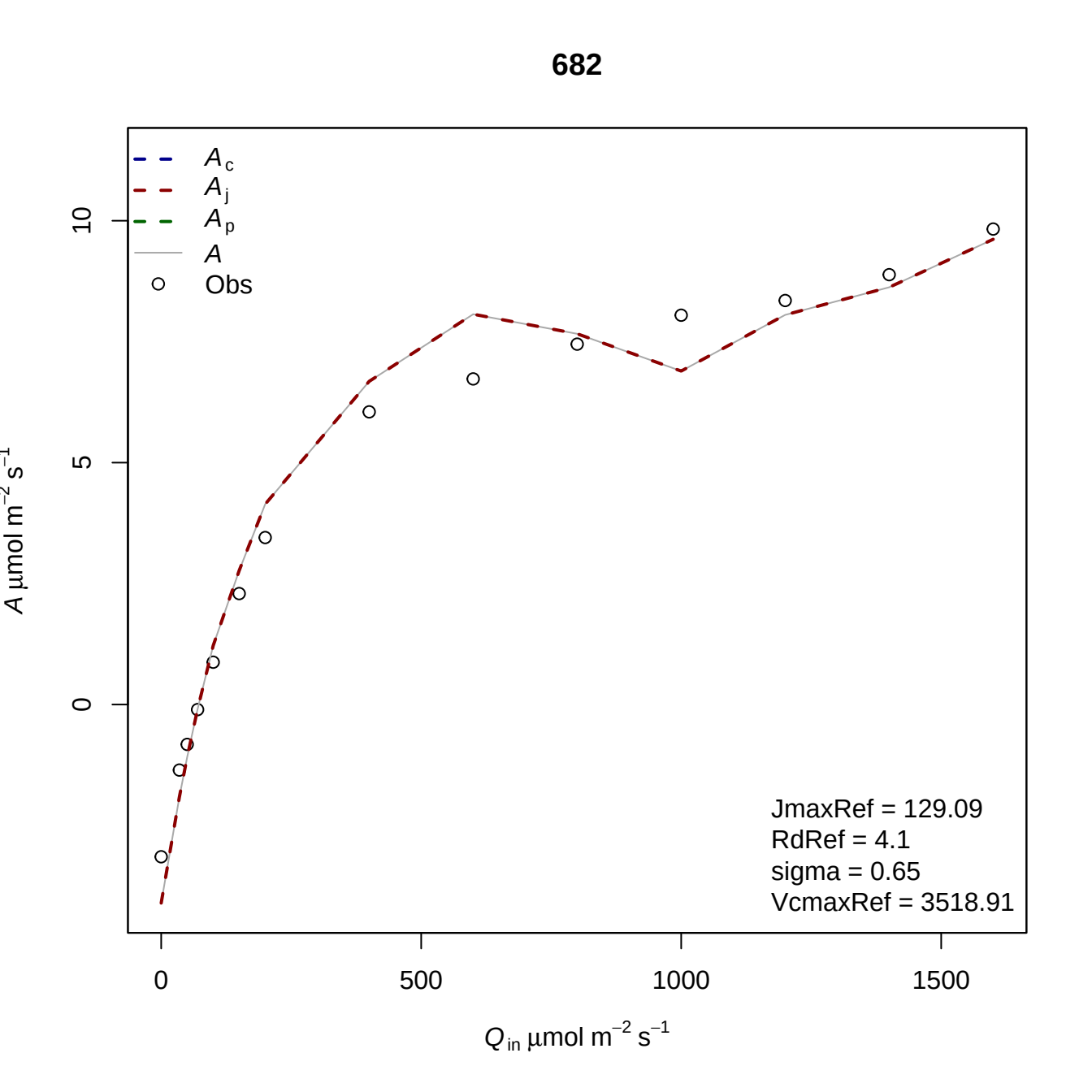

677
A_c
A_j
A_p
A
Obs
JmaxRef = 81.4
RdRef = 3.47
sigma = 0.5
VcmaxRef = 35063.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

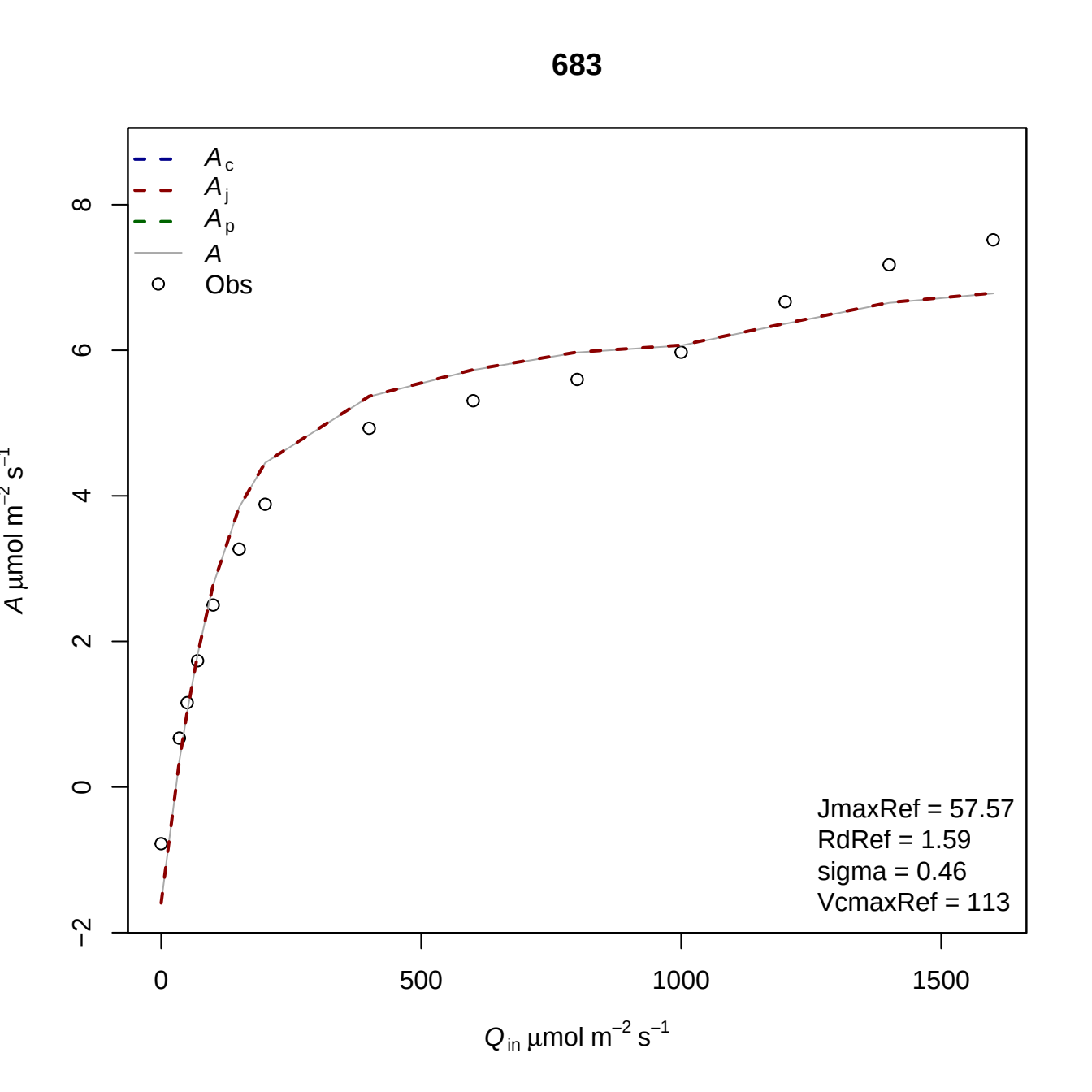

678
A_c
A_j
A_p
A
Obs
JmaxRef = 127.39
RdRef = 3.11
sigma = 0.35
VcmaxRef = 108.46
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

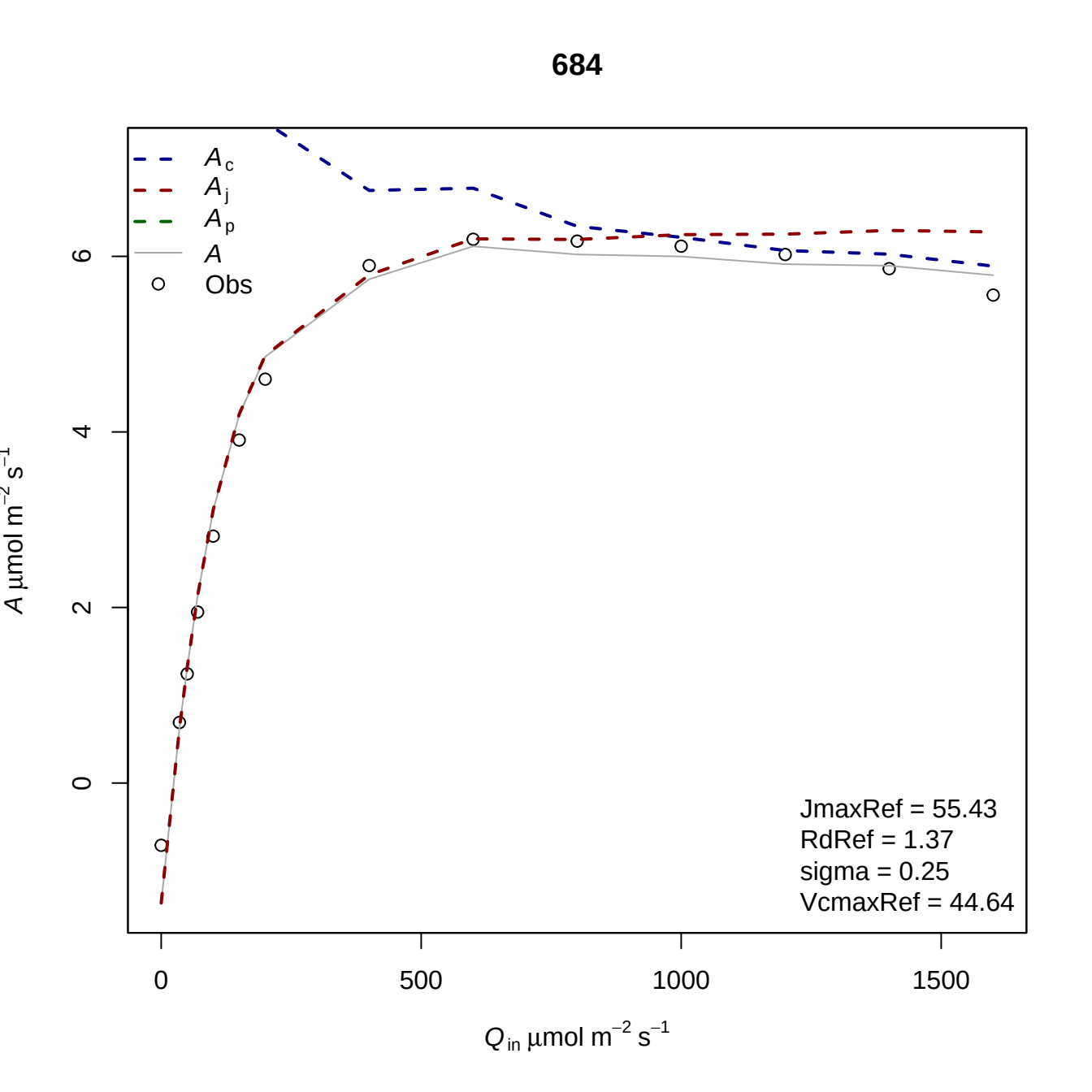

679
A_c
A_j
A_p
A
Obs
JmaxRef = 144.21
RdRef = 2.6
sigma = 0.69
VcmaxRef = 634.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

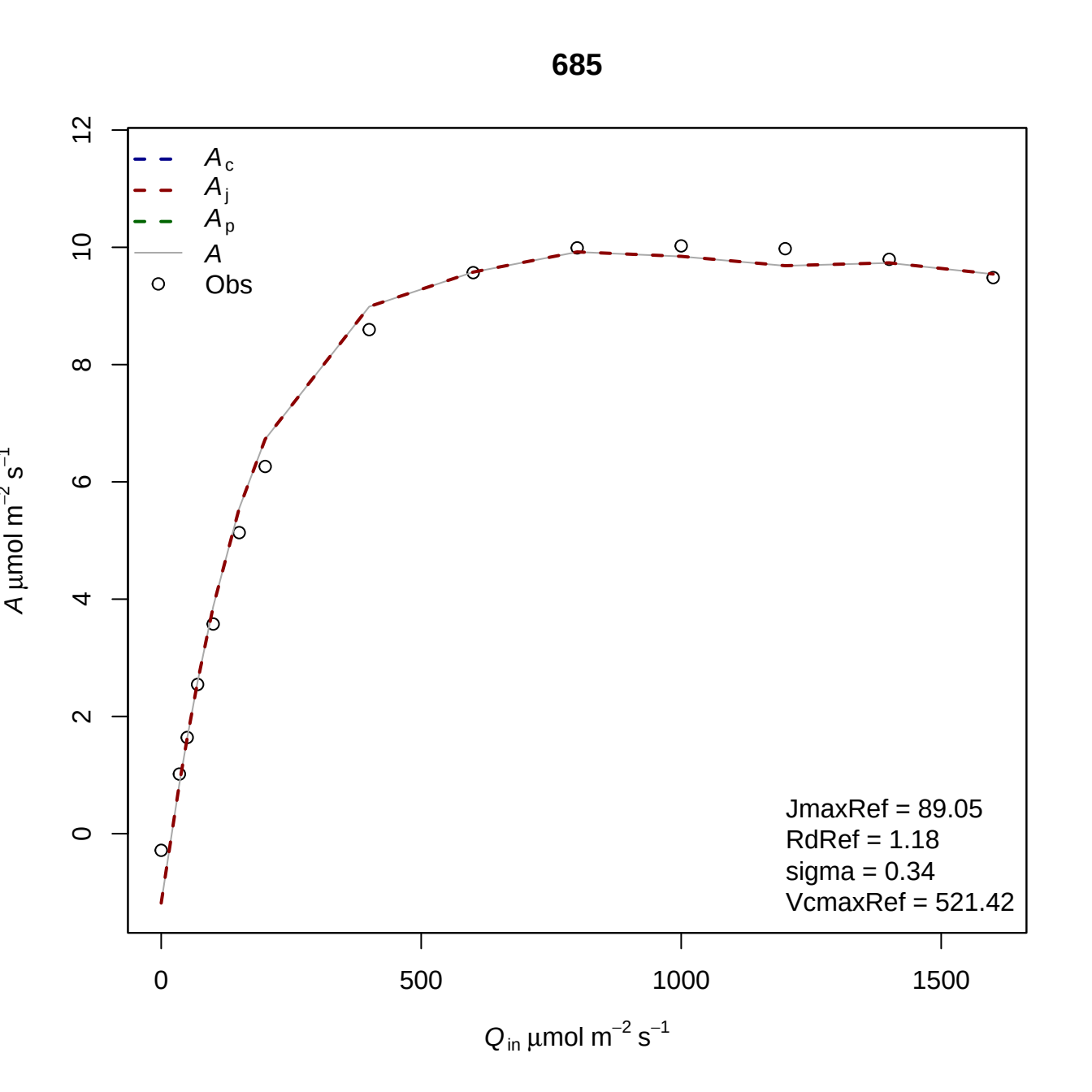

680
A_c
A_j
A_p
A
Obs
JmaxRef = 214.15
RdRef = 4.29
sigma = 0.44
VcmaxRef = 779.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500

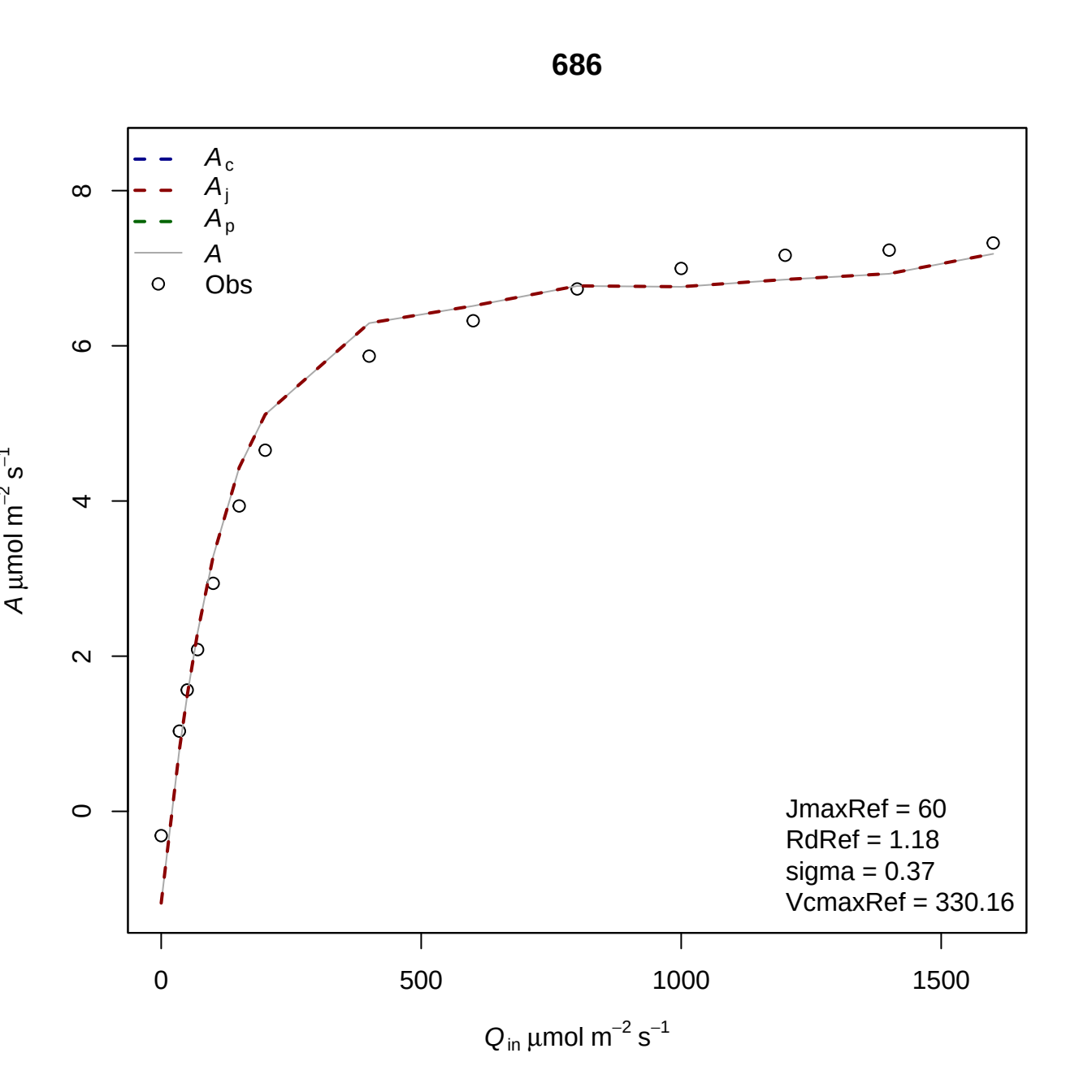

681
A_c
A_j
A_p
A
Obs
JmaxRef = 166.44
RdRef = 3.41
sigma = 0.58
VcmaxRef = 642.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

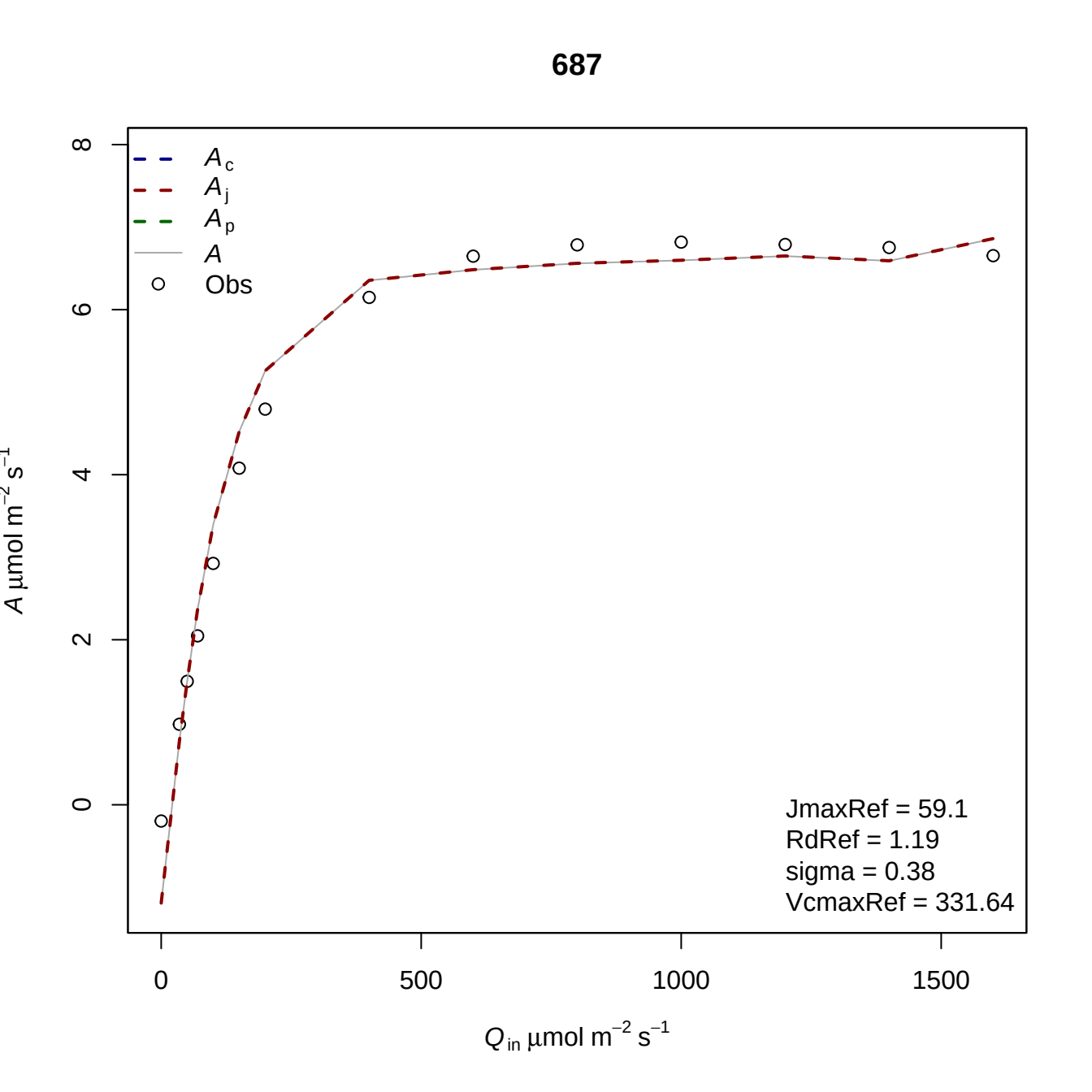

682
A_c
A_j
A_p
A
Obs
JmaxRef = 129.09
RdRef = 4.1
sigma = 0.65
VcmaxRef = 3518.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

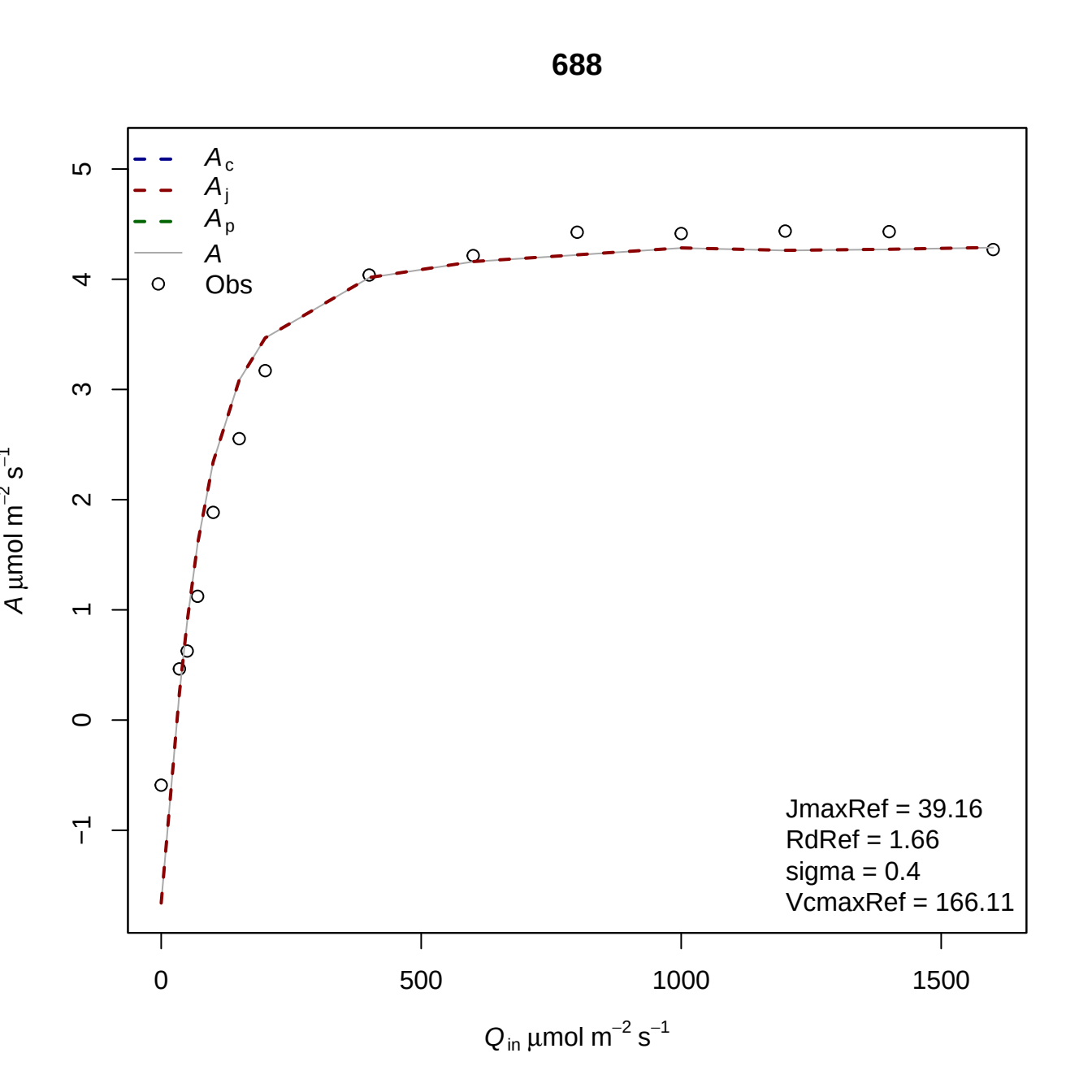

683
A_c
A_j
A_p
A
Obs
JmaxRef = 57.57
RdRef = 1.59
sigma = 0.46
VcmaxRef = 113
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

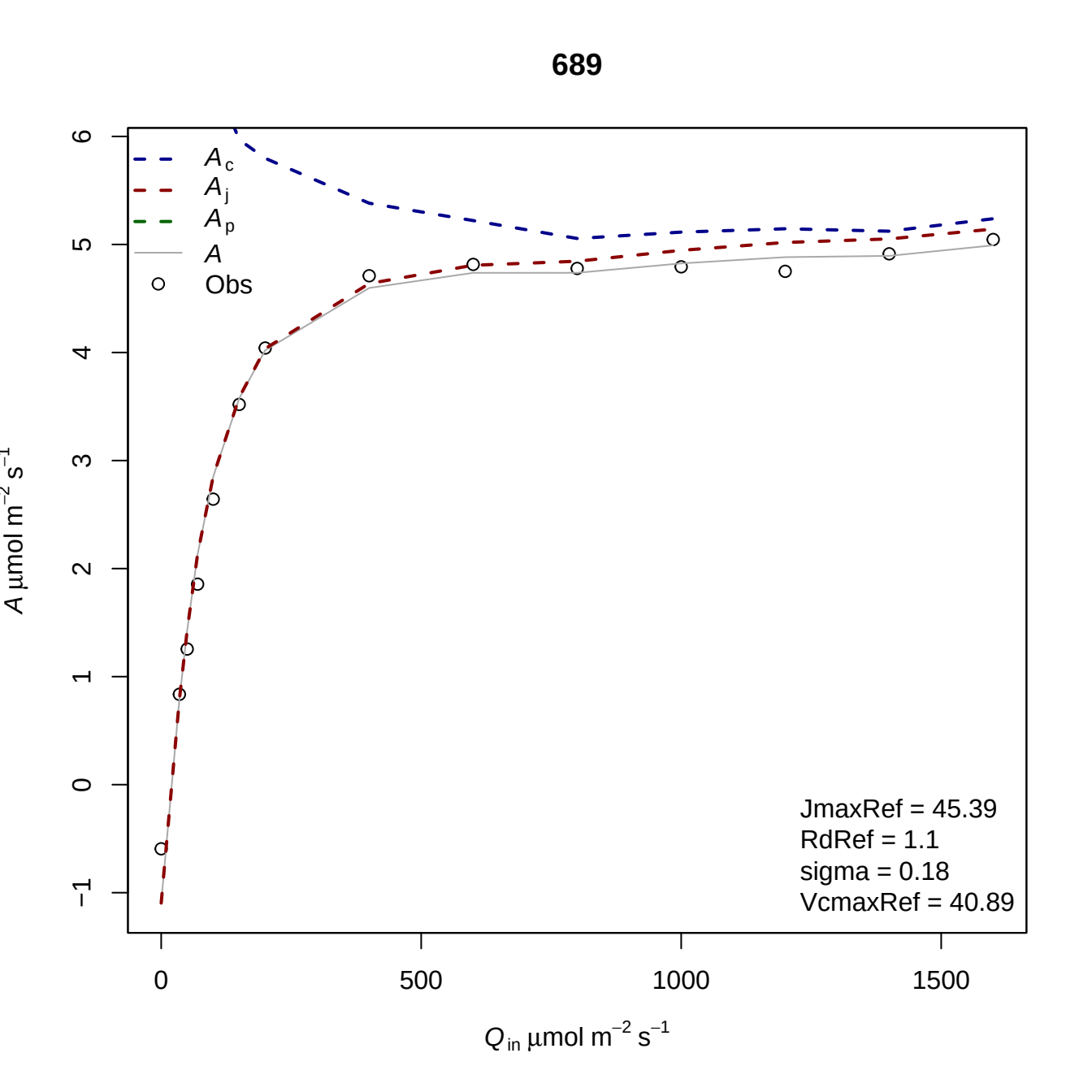

684
A_c
A_j
A_p
A
Obs
JmaxRef = 55.43
RdRef = 1.37
sigma = 0.25
VcmaxRef = 44.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

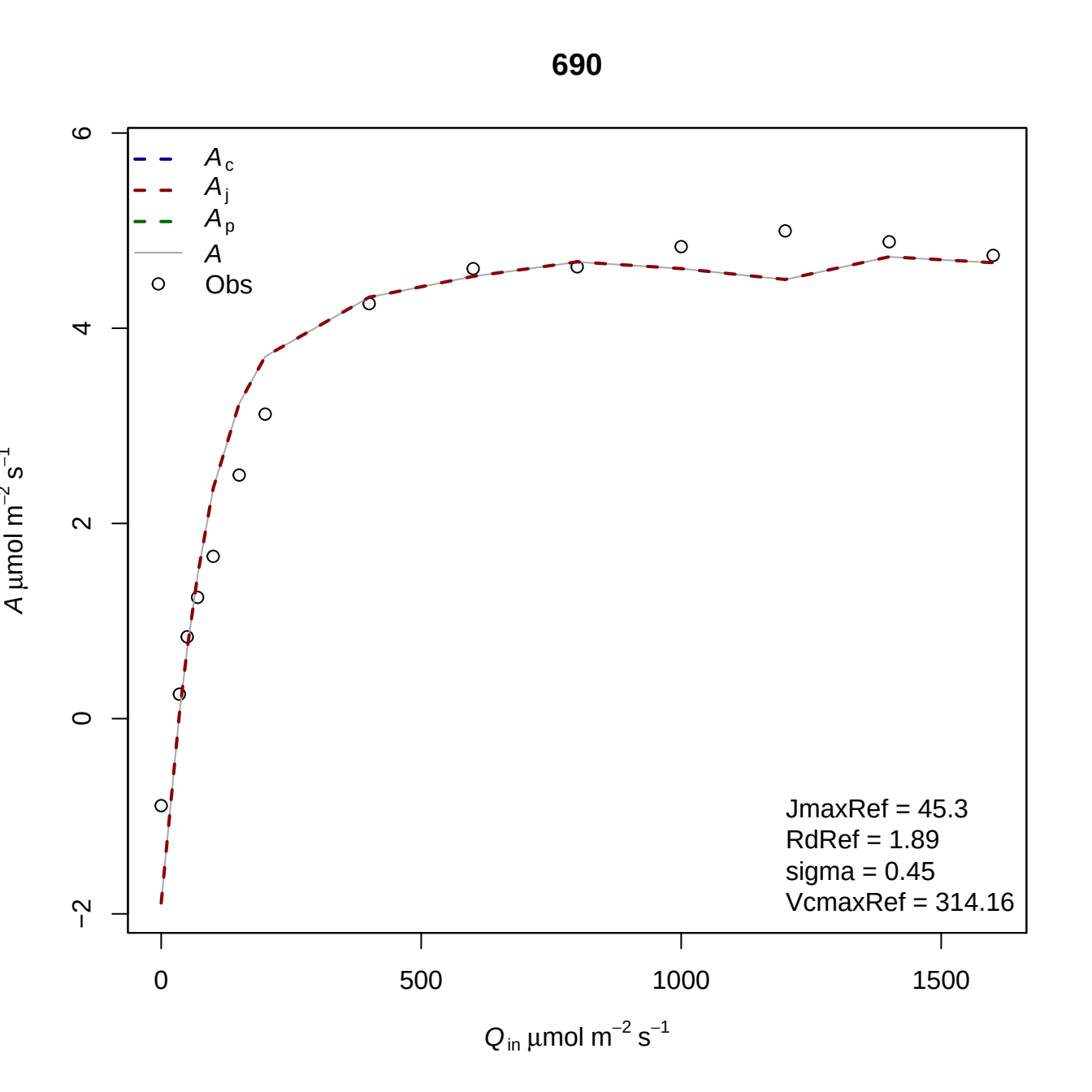

685
A_c
A_j
A_p
A
Obs
JmaxRef = 89.05
RdRef = 1.18
sigma = 0.34
VcmaxRef = 521.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

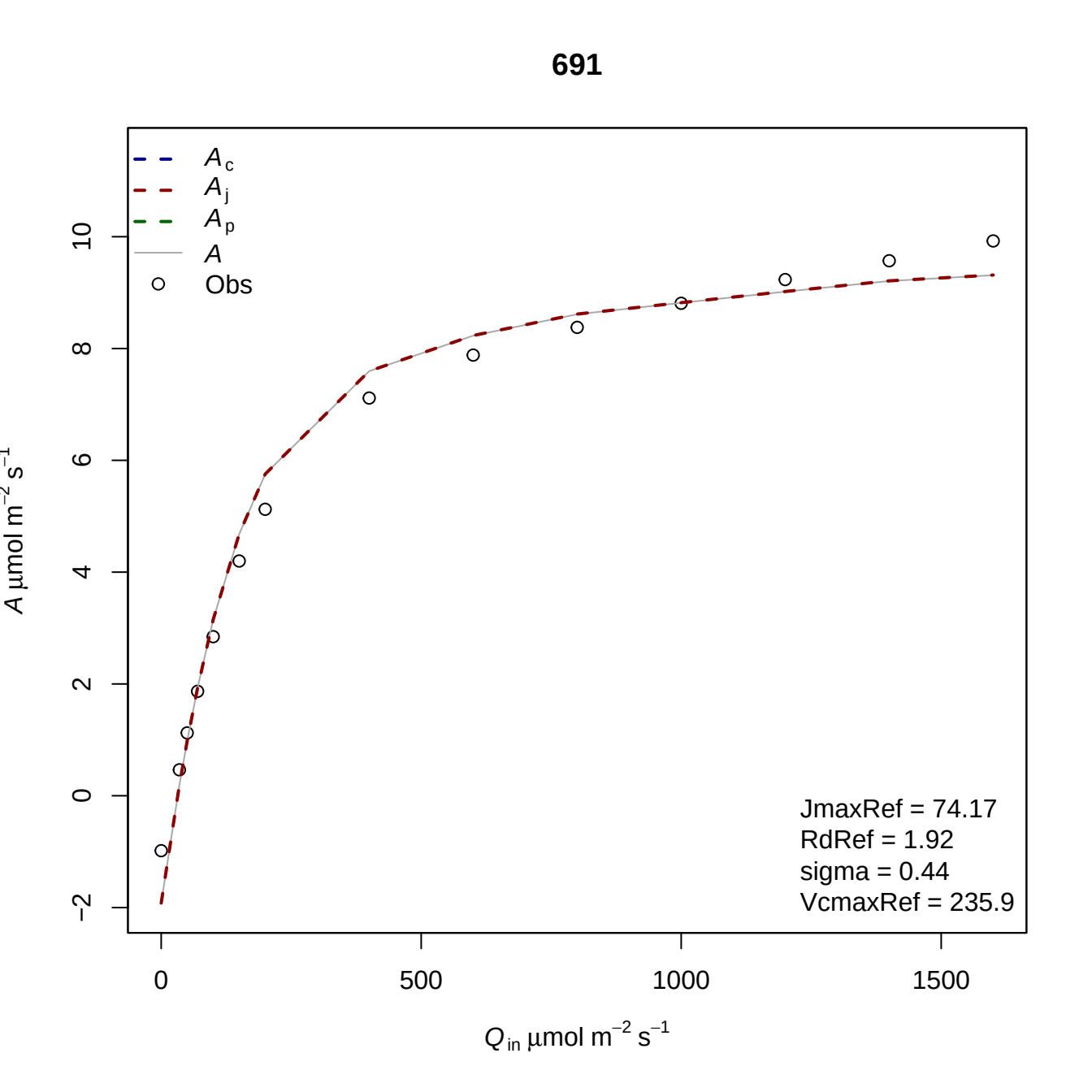

686
A_c
A_j
A_p
A
Obs
JmaxRef = 60
RdRef = 1.18
sigma = 0.37
VcmaxRef = 330.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

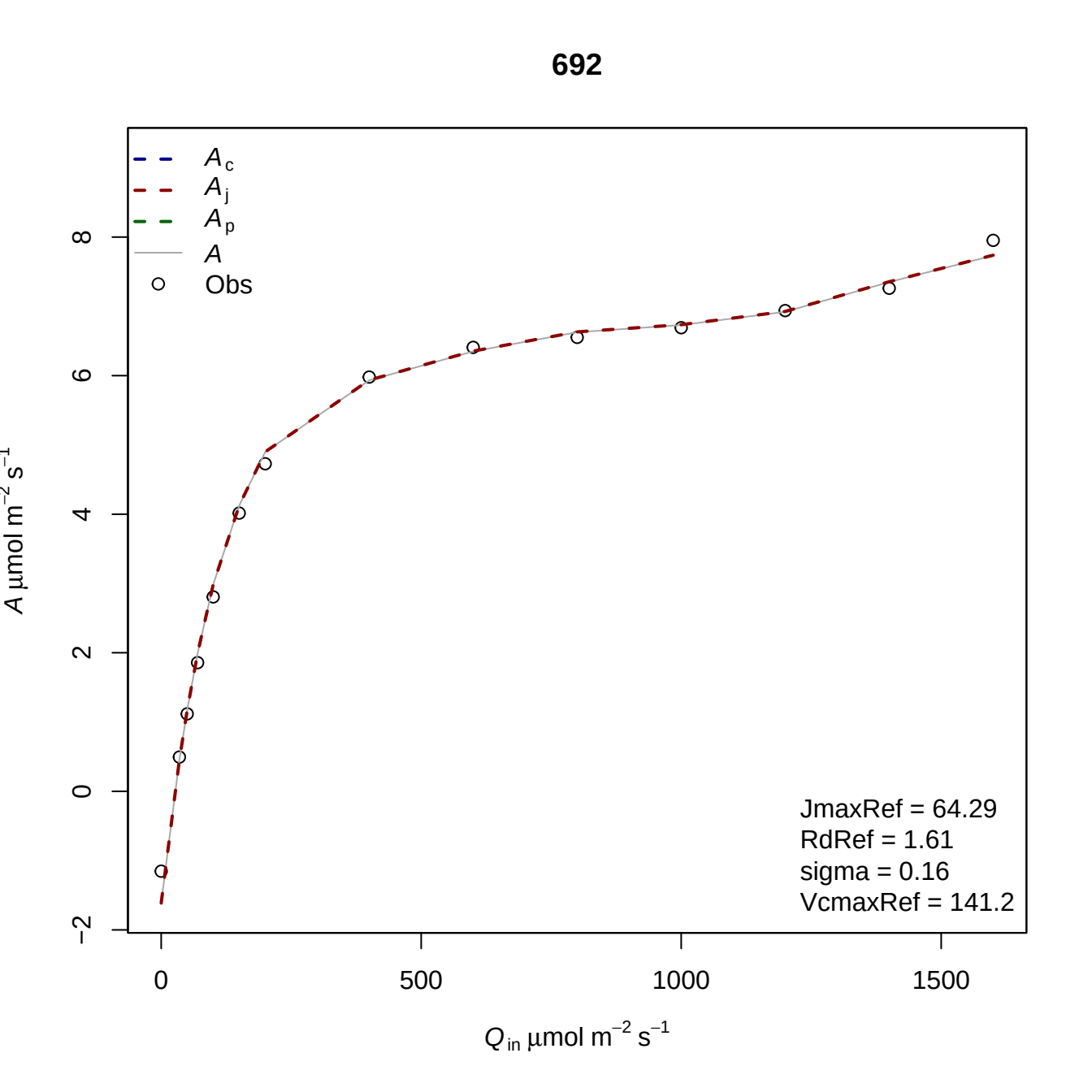

687
A_c
A_j
A_p
A
Obs
JmaxRef = 59.1
RdRef = 1.19
sigma = 0.38
VcmaxRef = 331.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

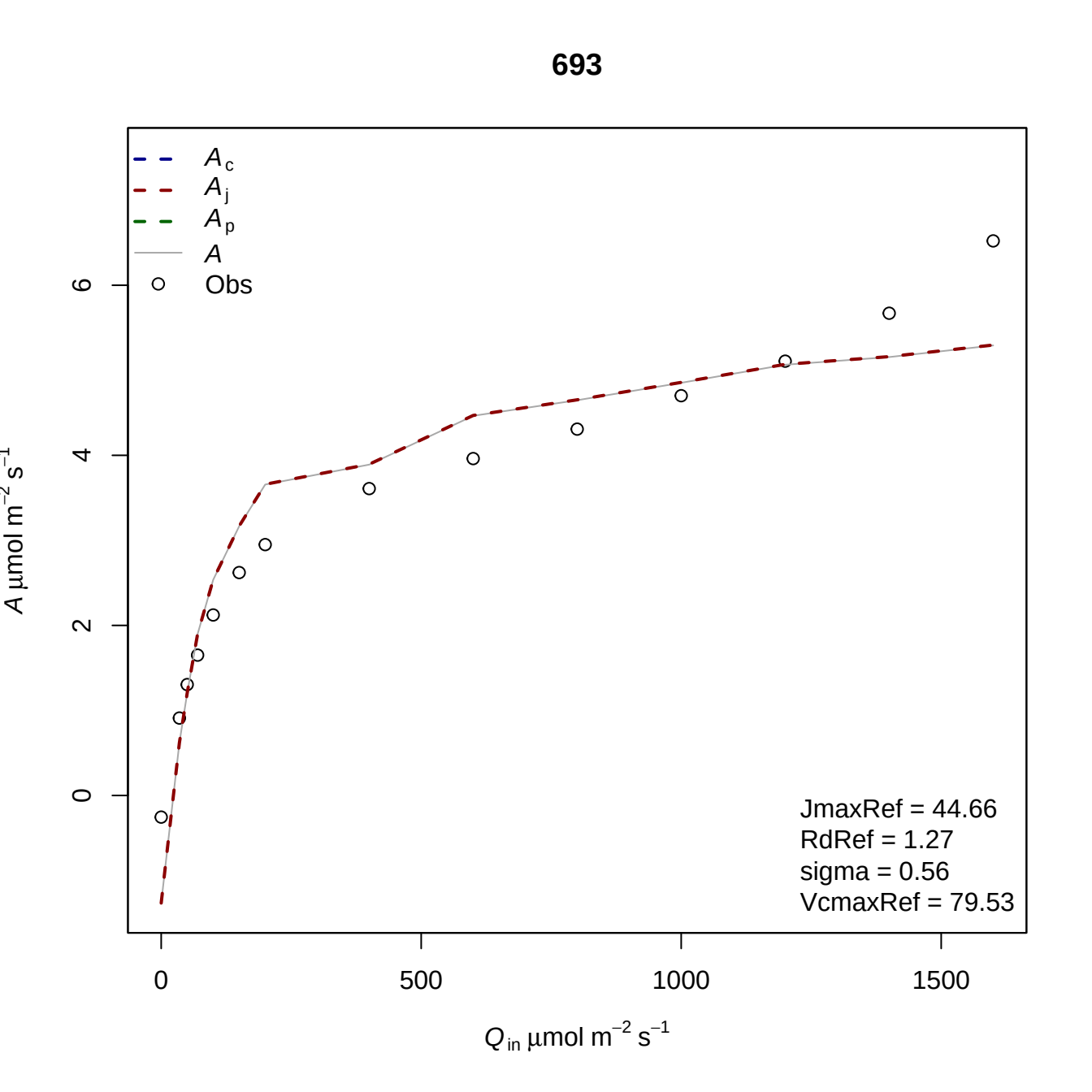

688
A_c
A_j
A_p
A
Obs
JmaxRef = 39.16
RdRef = 1.66
sigma = 0.4
VcmaxRef = 166.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

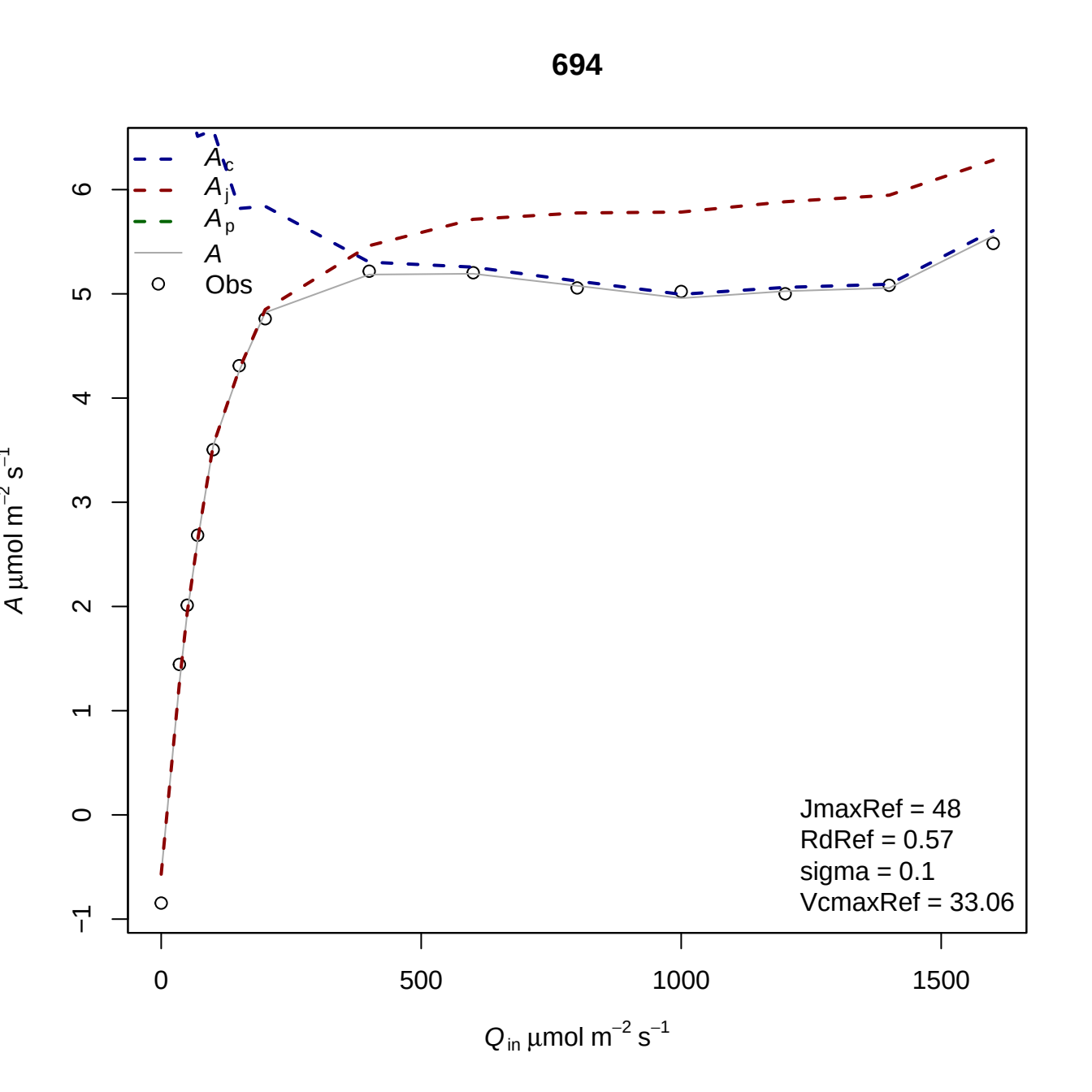

689
A_c
A_j
A_p
A
Obs
JmaxRef = 45.39
RdRef = 1.1
sigma = 0.18
VcmaxRef = 40.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

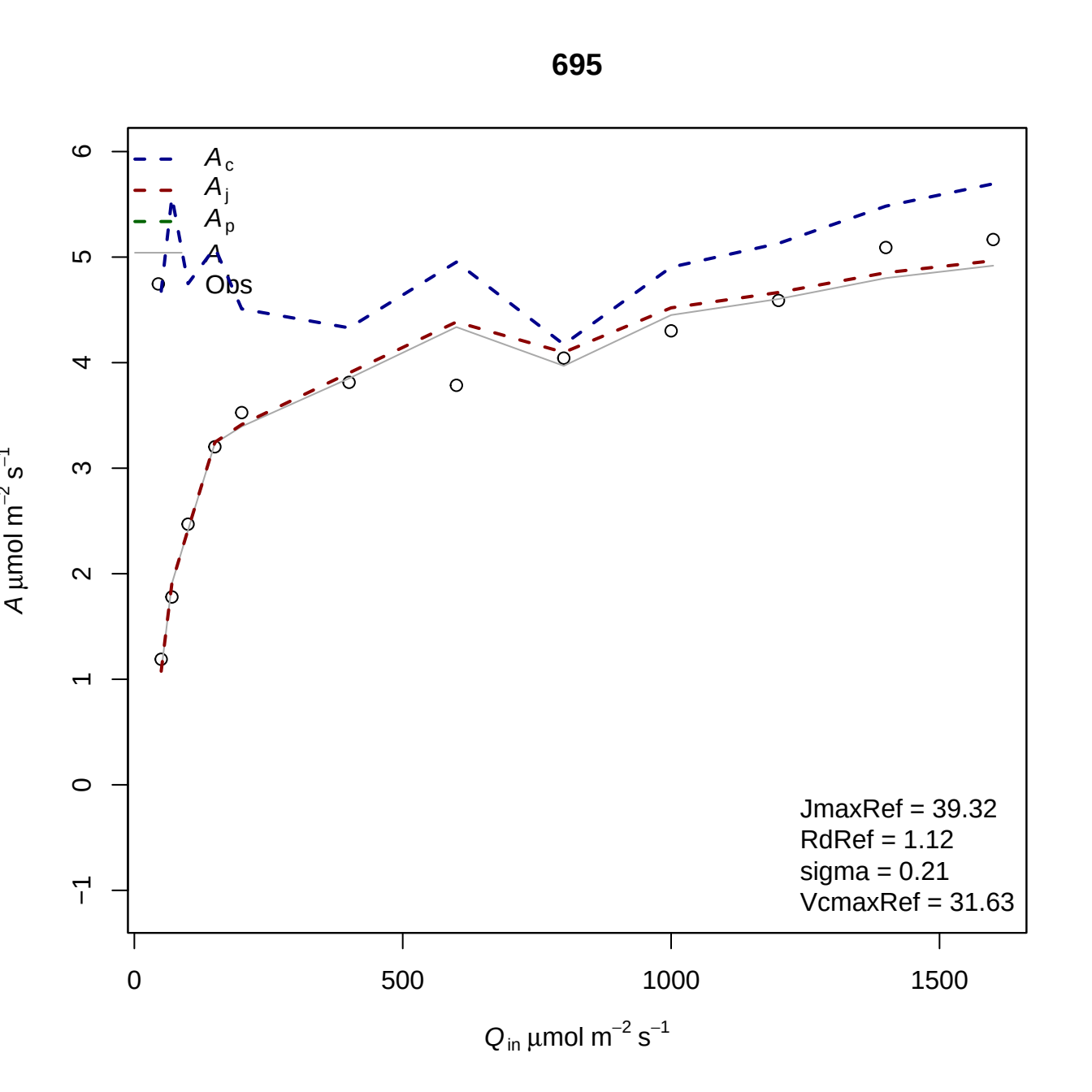

690
A_c
A_j
A_p
A
Obs
JmaxRef = 45.3
RdRef = 1.89
sigma = 0.45
VcmaxRef = 314.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

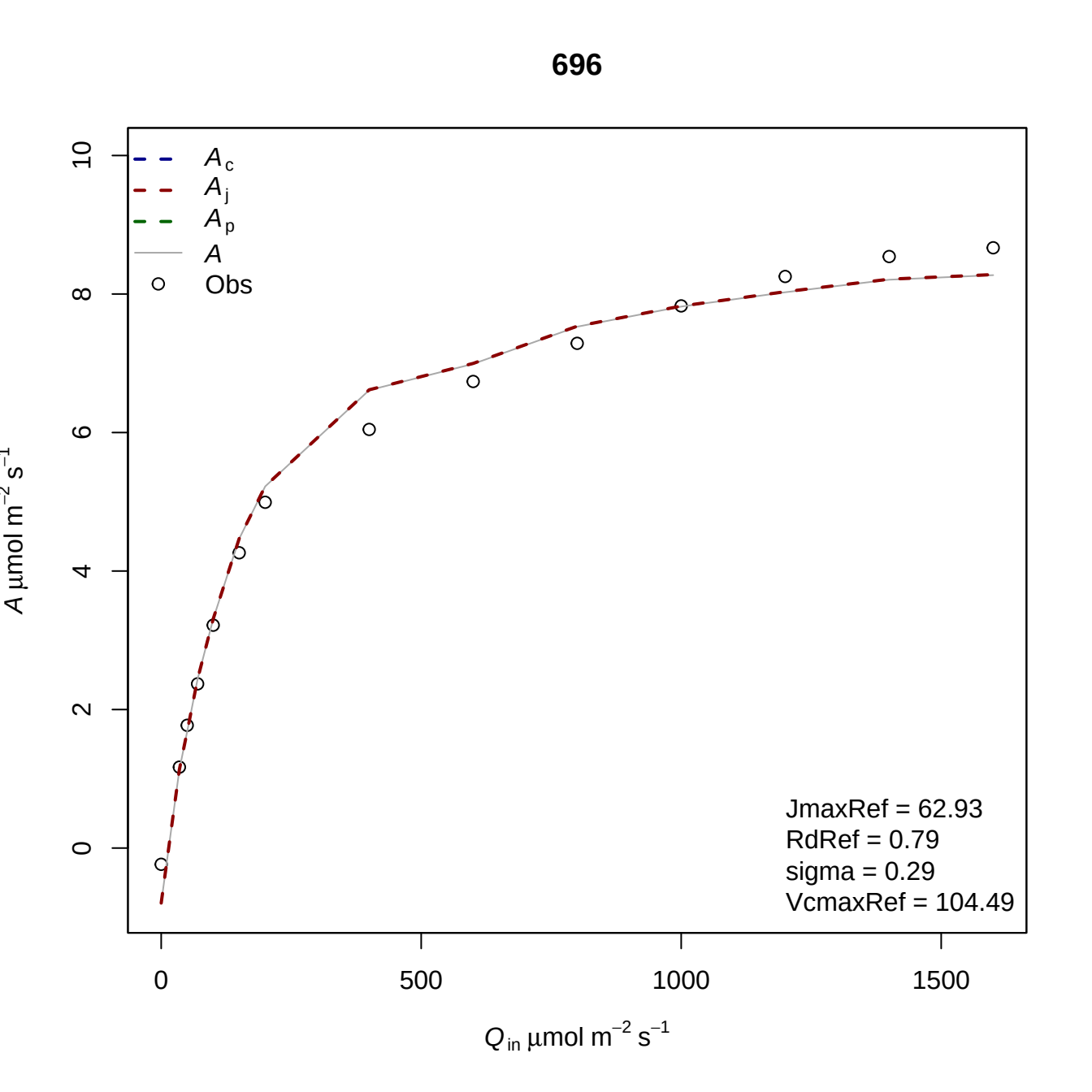

691
A_c
A_j
A_p
A
Obs
JmaxRef = 74.17
RdRef = 1.92
sigma = 0.44
VcmaxRef = 235.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

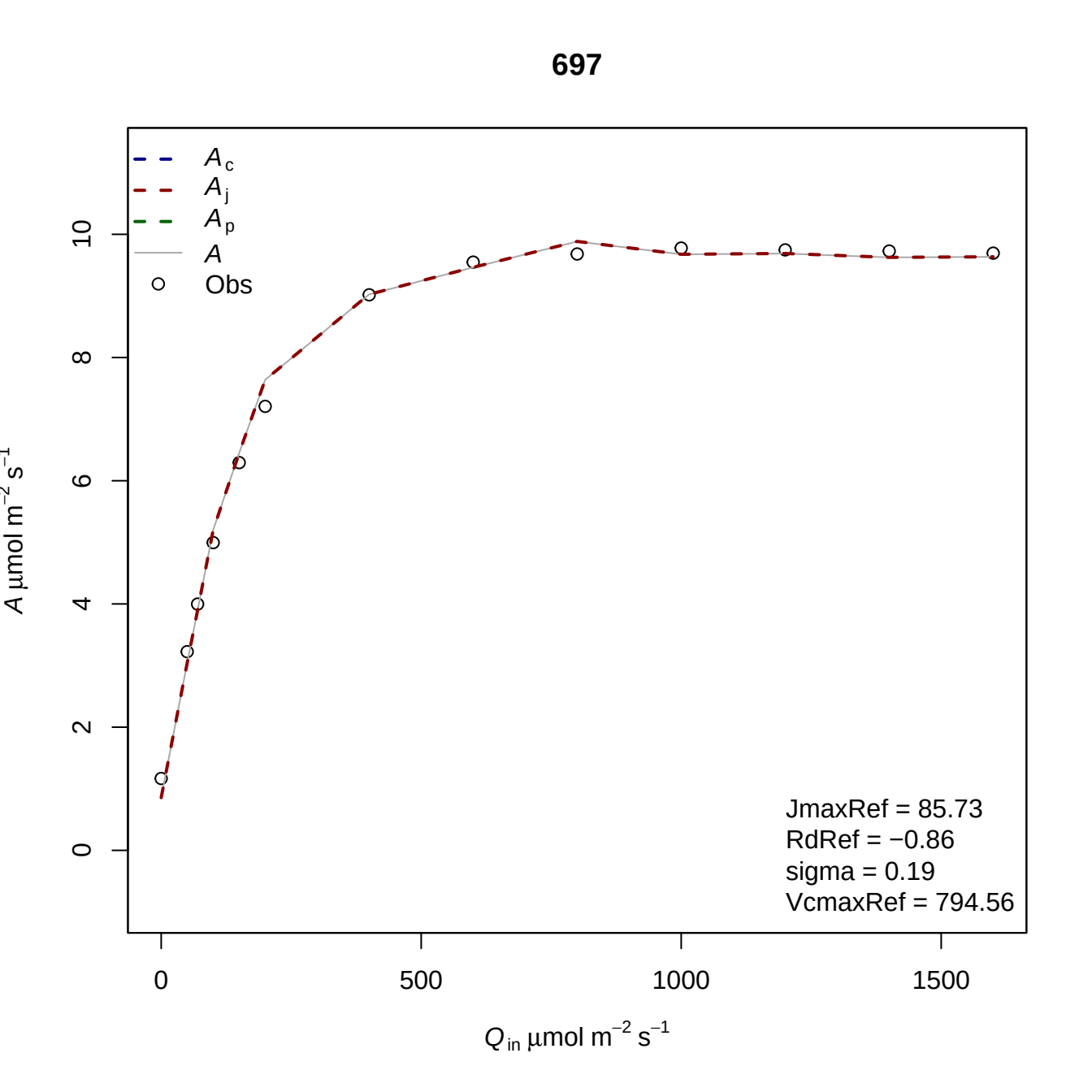

692
A_c
A_j
A_p
A
Obs
JmaxRef = 64.29
RdRef = 1.61
sigma = 0.16
VcmaxRef = 141.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

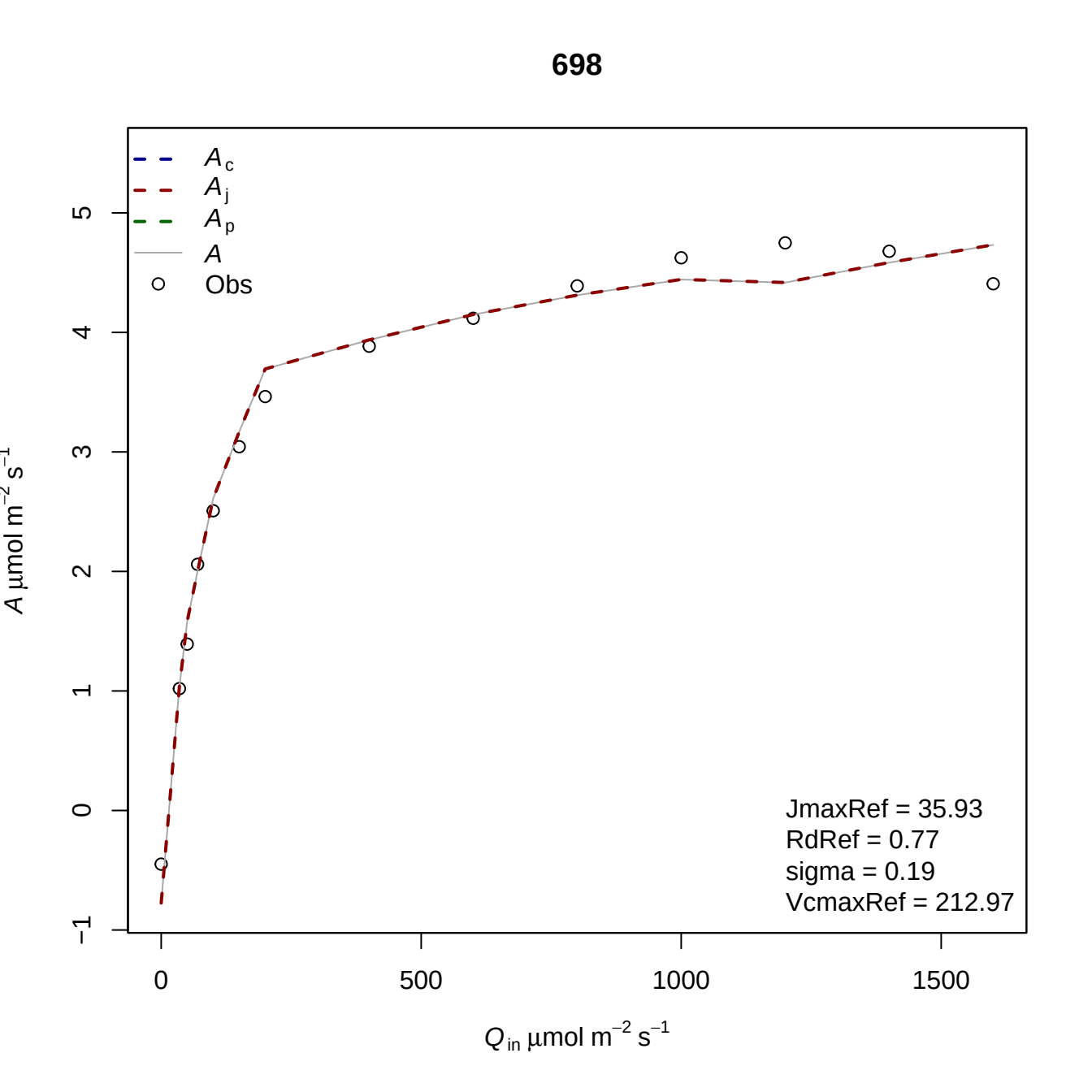

693
A_c
A_j
A_p
A
Obs
JmaxRef = 44.66
RdRef = 1.27
sigma = 0.56
VcmaxRef = 79.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

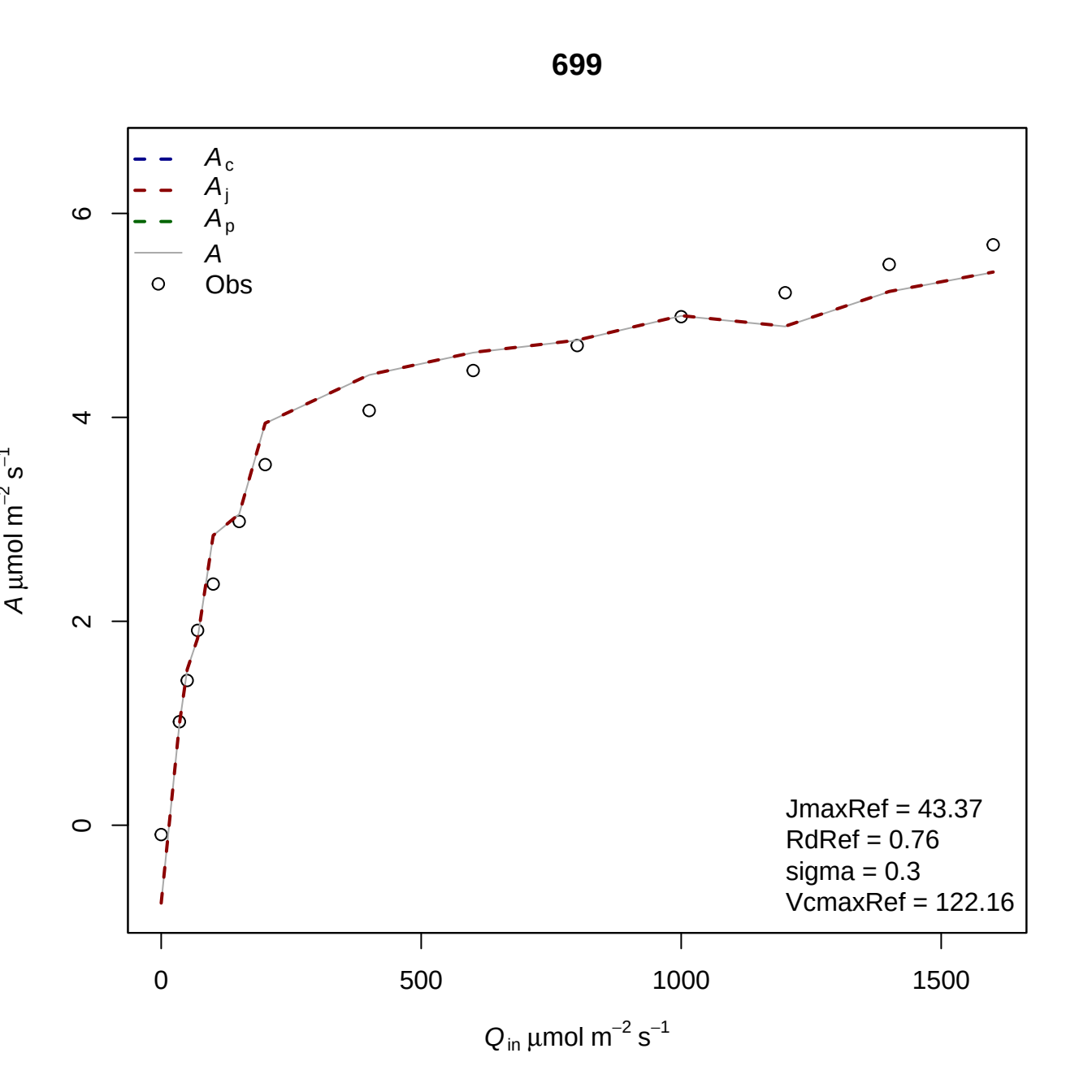

694
A_c
A_j
A_p
A
Obs
JmaxRef = 48
RdRef = 0.57
sigma = 0.1
VcmaxRef = 33.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

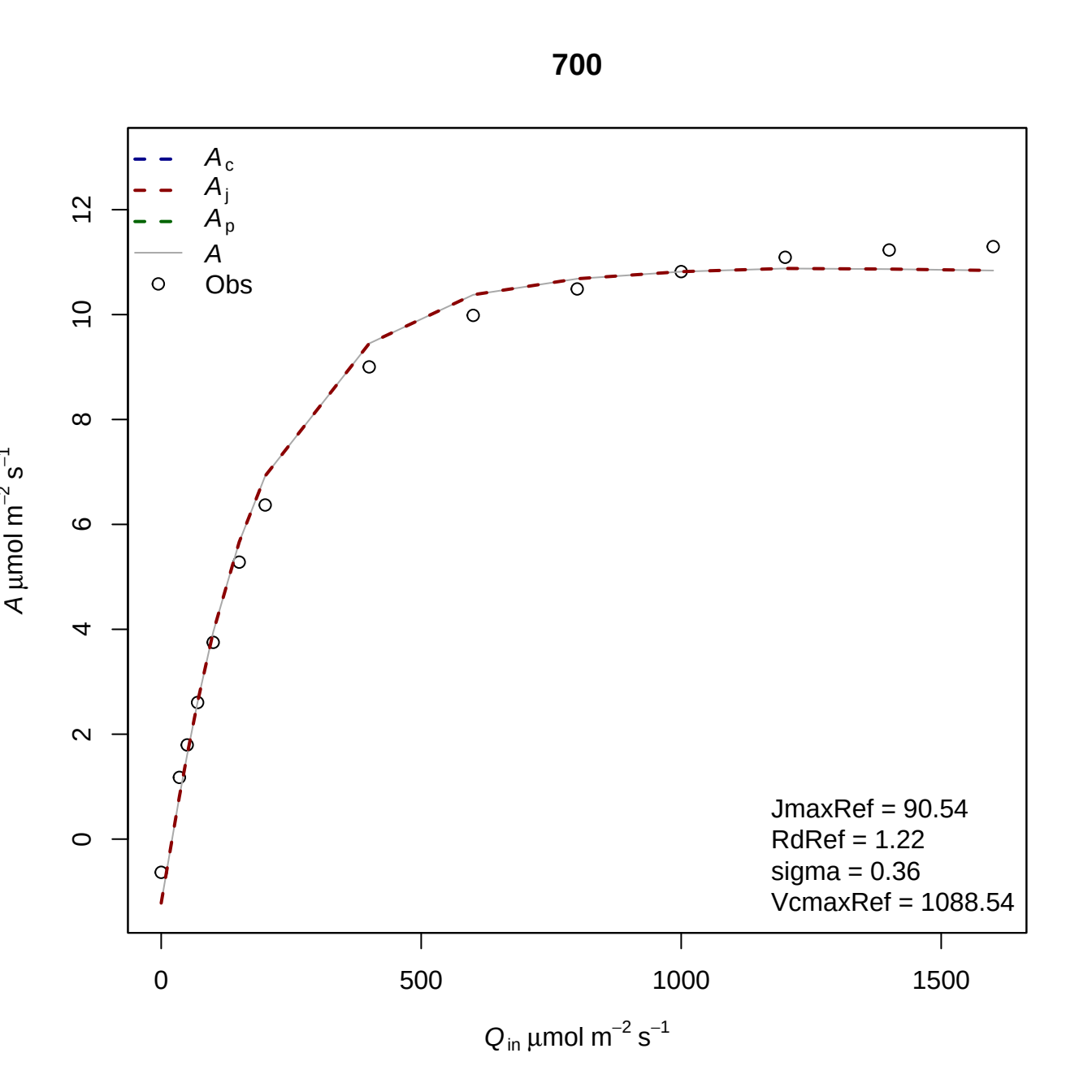

695
A_c
A_j
A_p
A
Obs
JmaxRef = 39.32
RdRef = 1.12
sigma = 0.21
VcmaxRef = 31.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

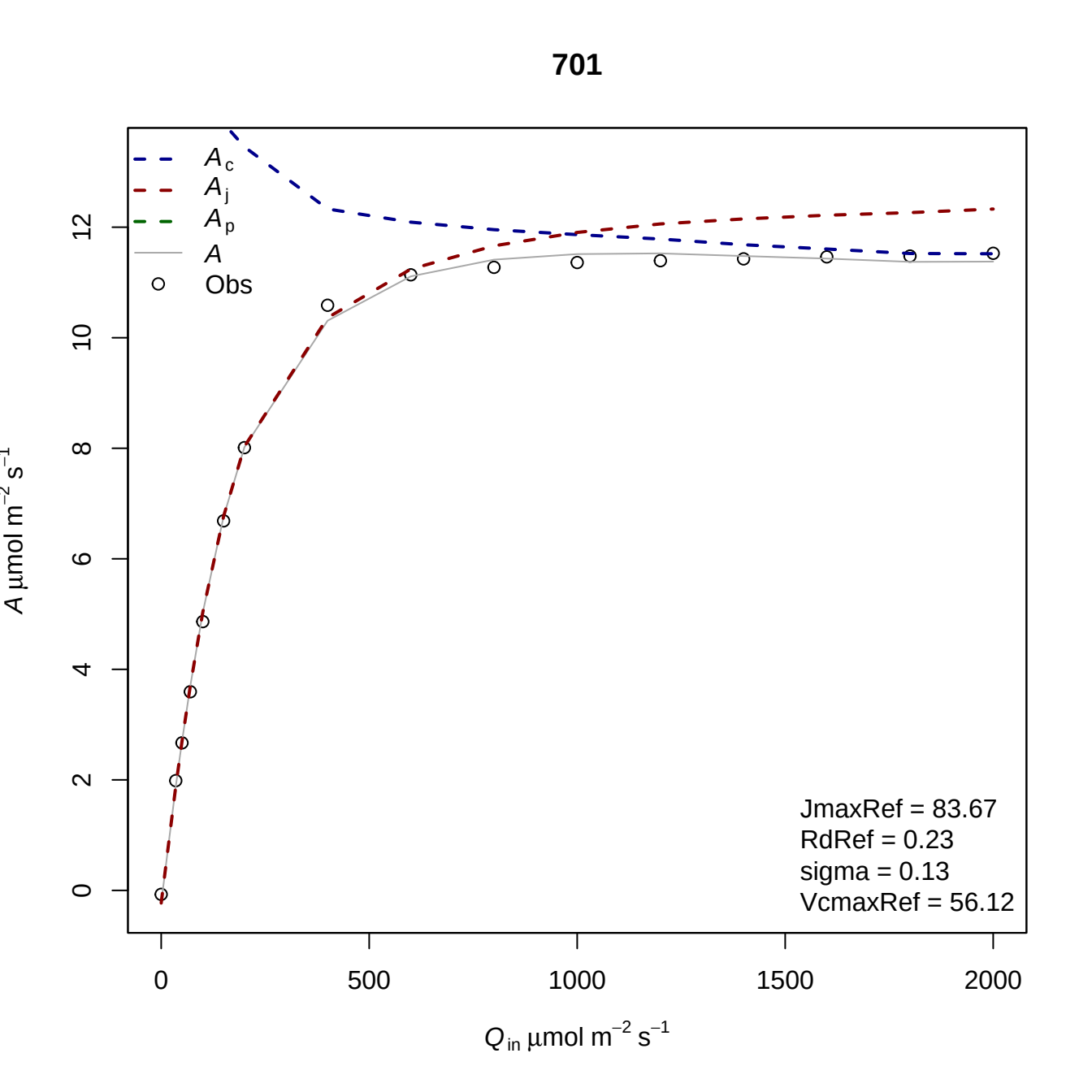

696
A_c
A_j
A_p
A
Obs
JmaxRef = 62.93
RdRef = 0.79
sigma = 0.29
VcmaxRef = 104.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

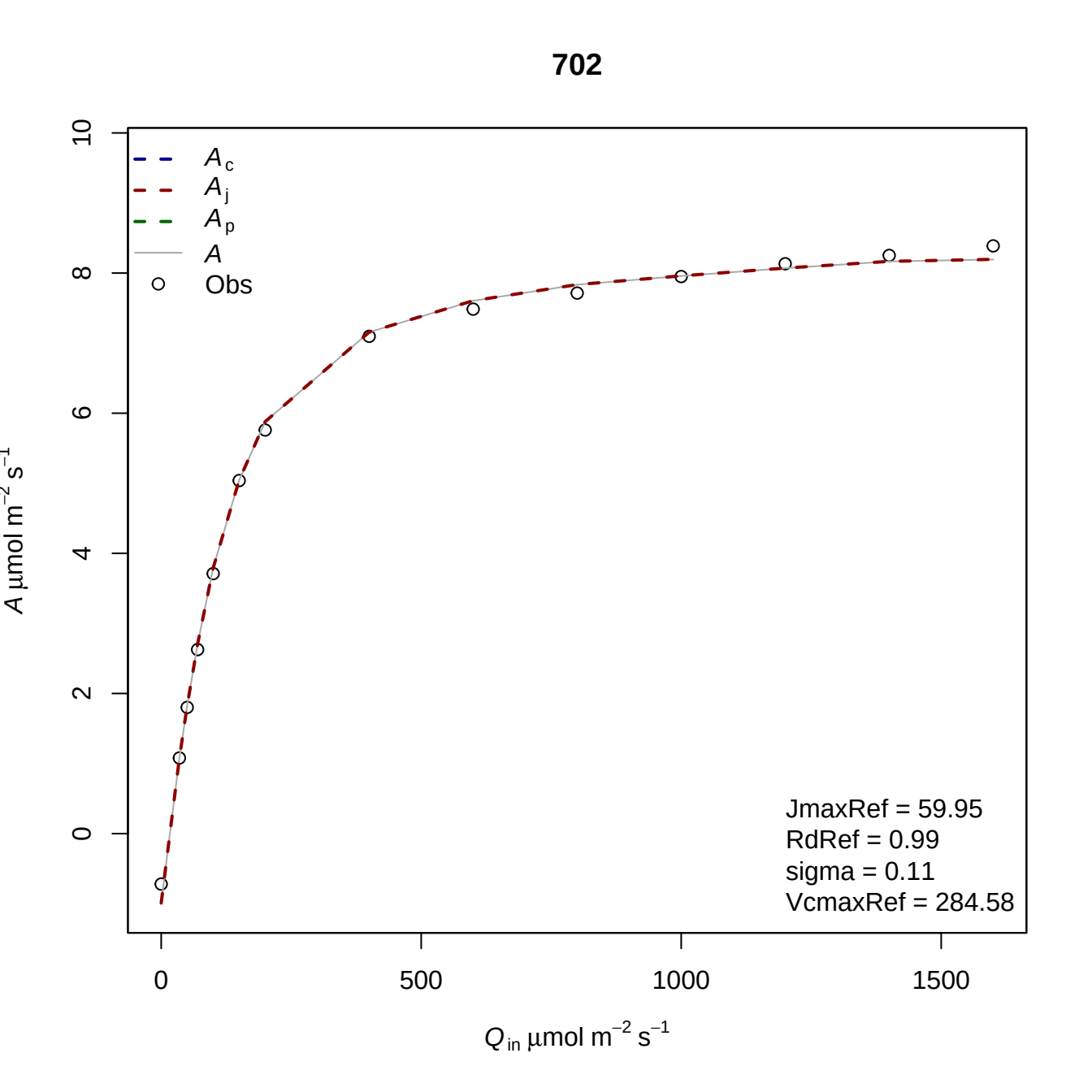

697
A_c
A_j
A_p
A
Obs
JmaxRef = 85.73
RdRef = −0.86
sigma = 0.19
VcmaxRef = 794.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

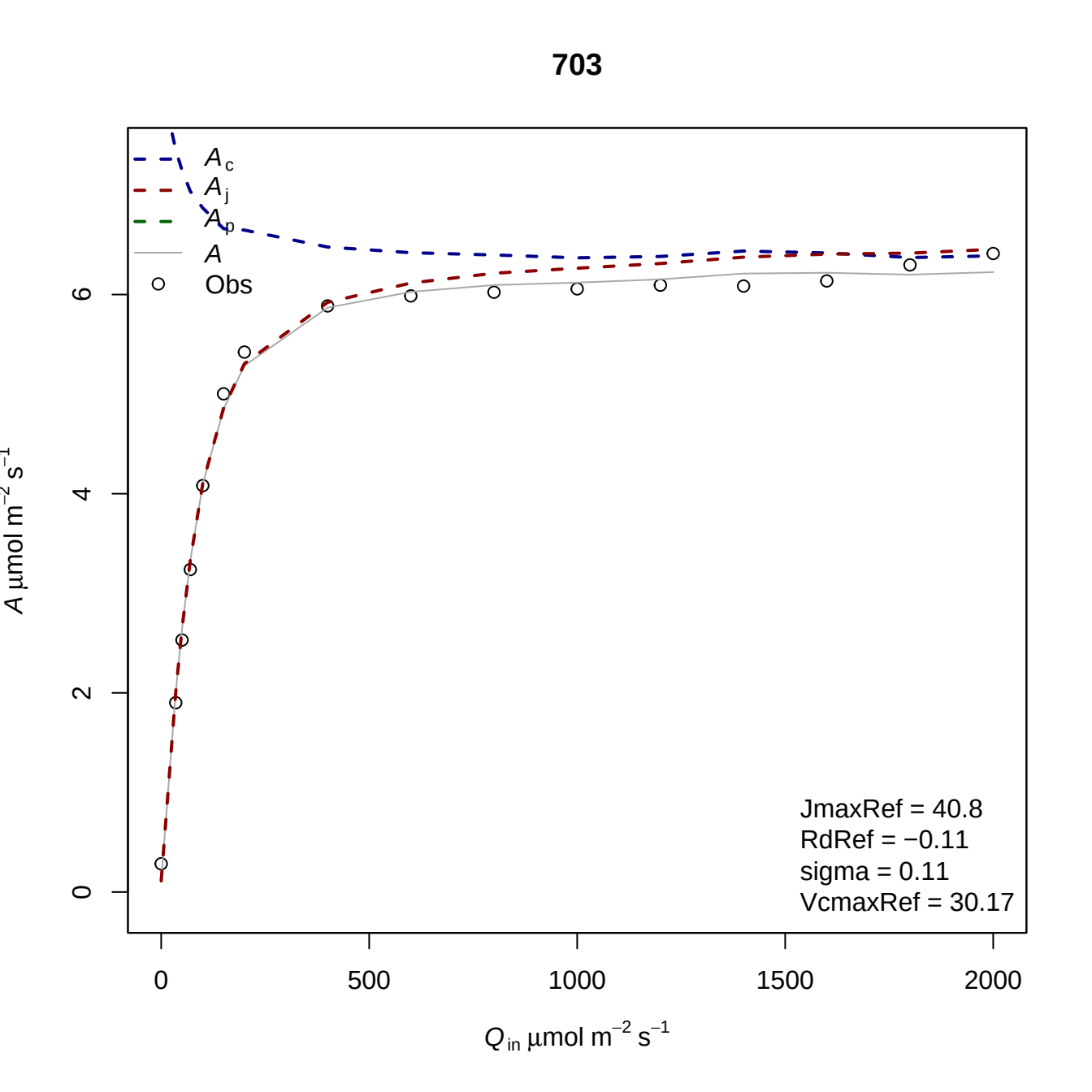

698
A_c
A_j
A_p
A
Obs
JmaxRef = 35.93
RdRef = 0.77
sigma = 0.19
VcmaxRef = 212.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

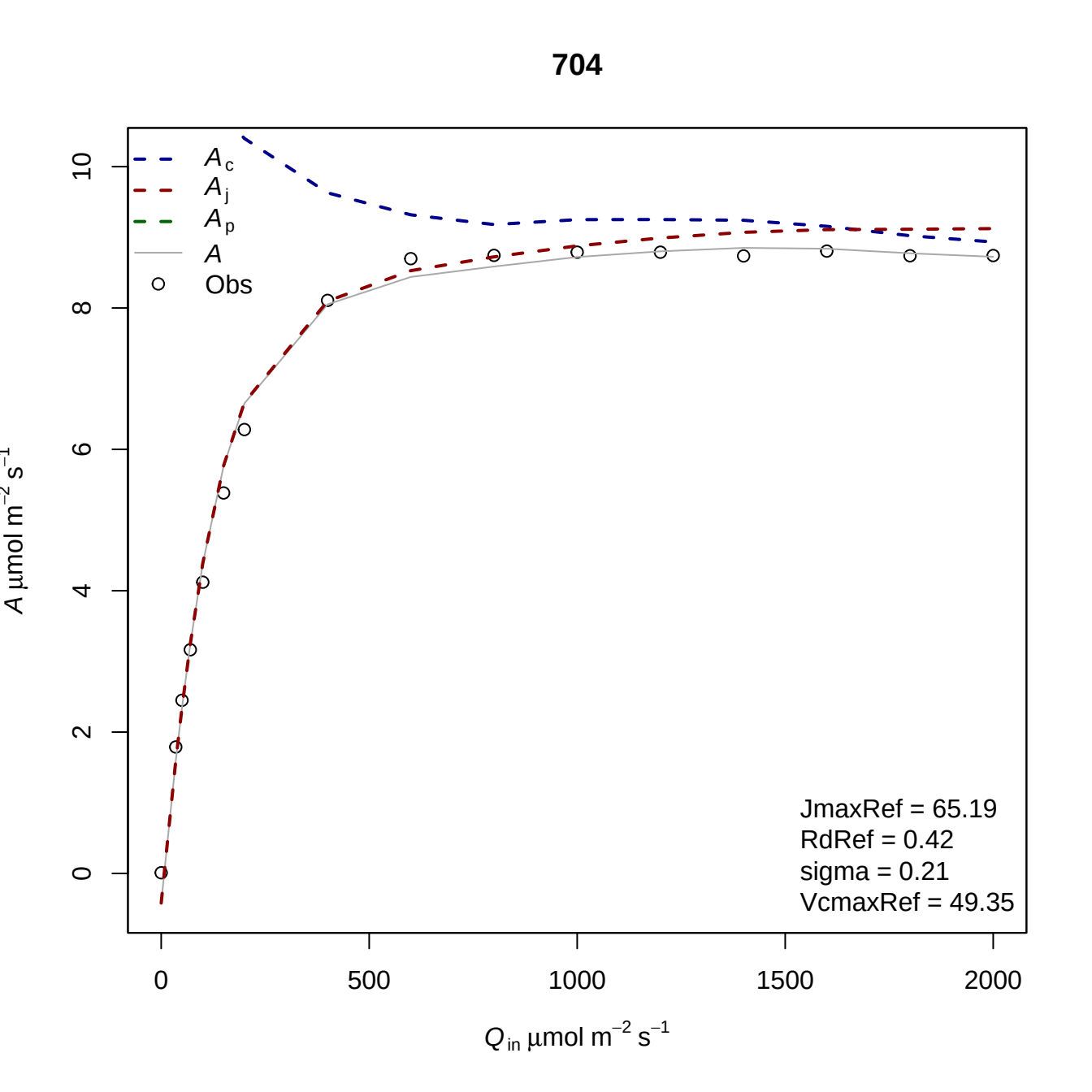

699
A_c
A_j
A_p
A
Obs
JmaxRef = 43.37
RdRef = 0.76
sigma = 0.3
VcmaxRef = 122.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

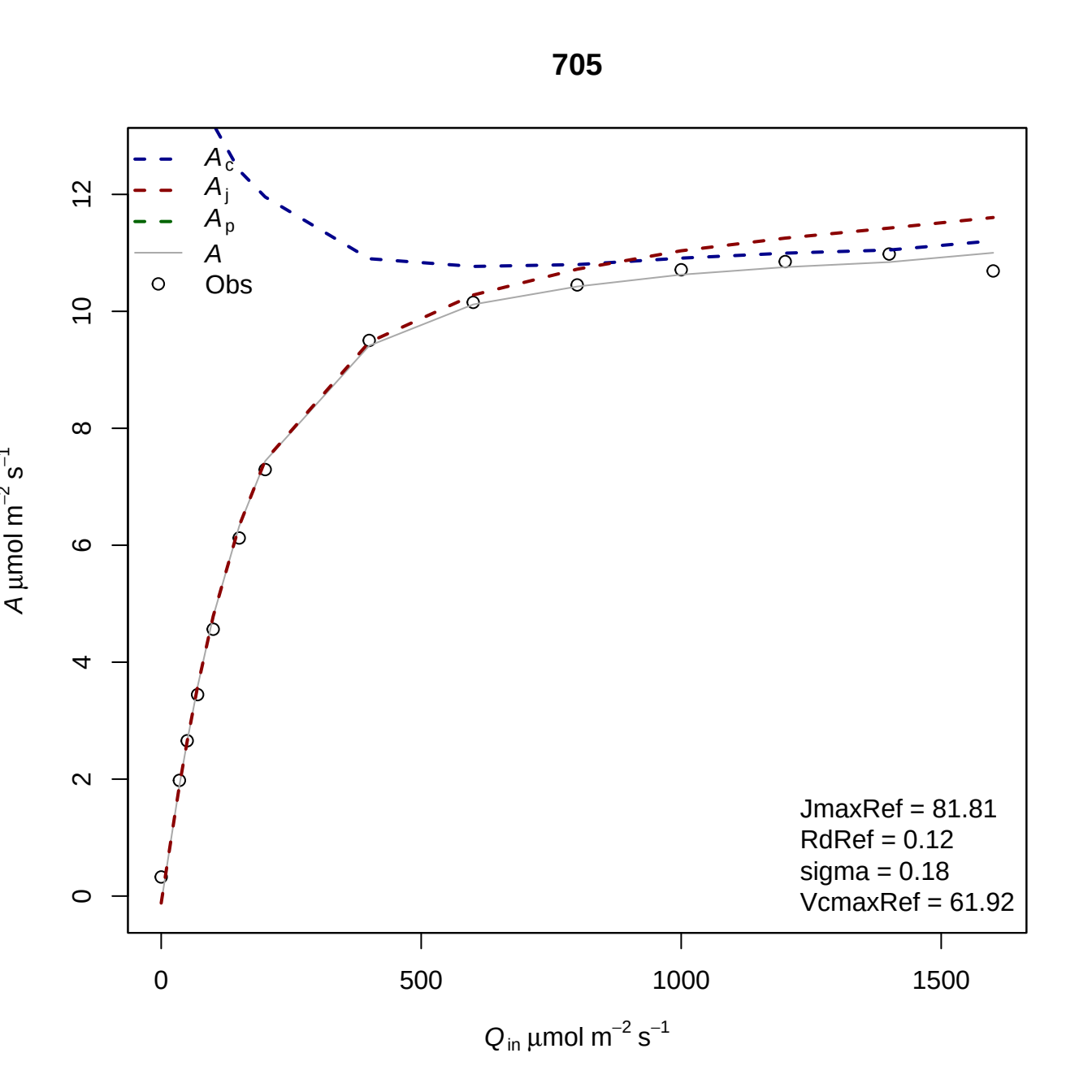

700
A_c
A_j
A_p
A
Obs
JmaxRef = 90.54
RdRef = 1.22
sigma = 0.36
VcmaxRef = 1088.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

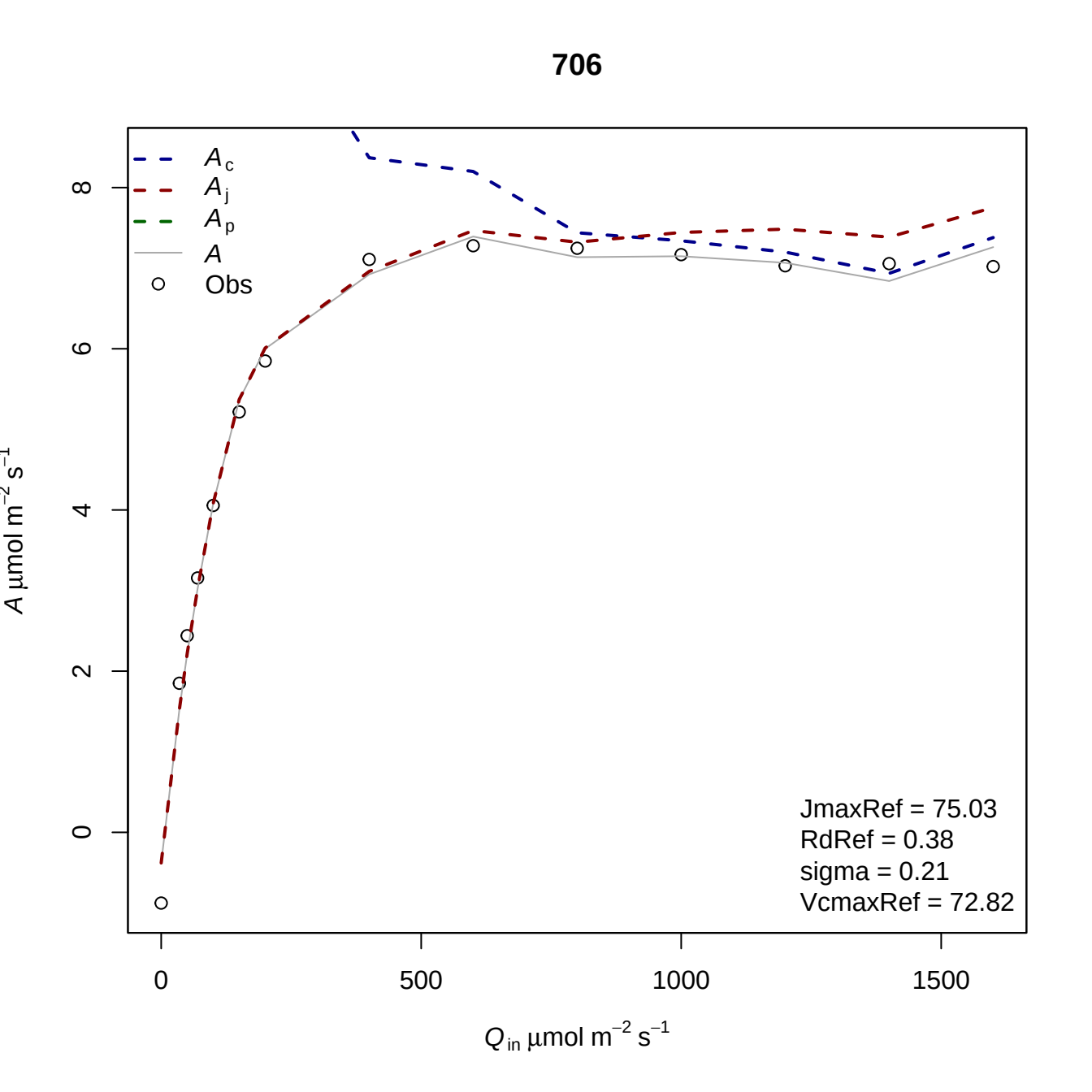

701
A_c
A_j
A_p
A
Obs
JmaxRef = 83.67
RdRef = 0.23
sigma = 0.13
VcmaxRef = 56.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

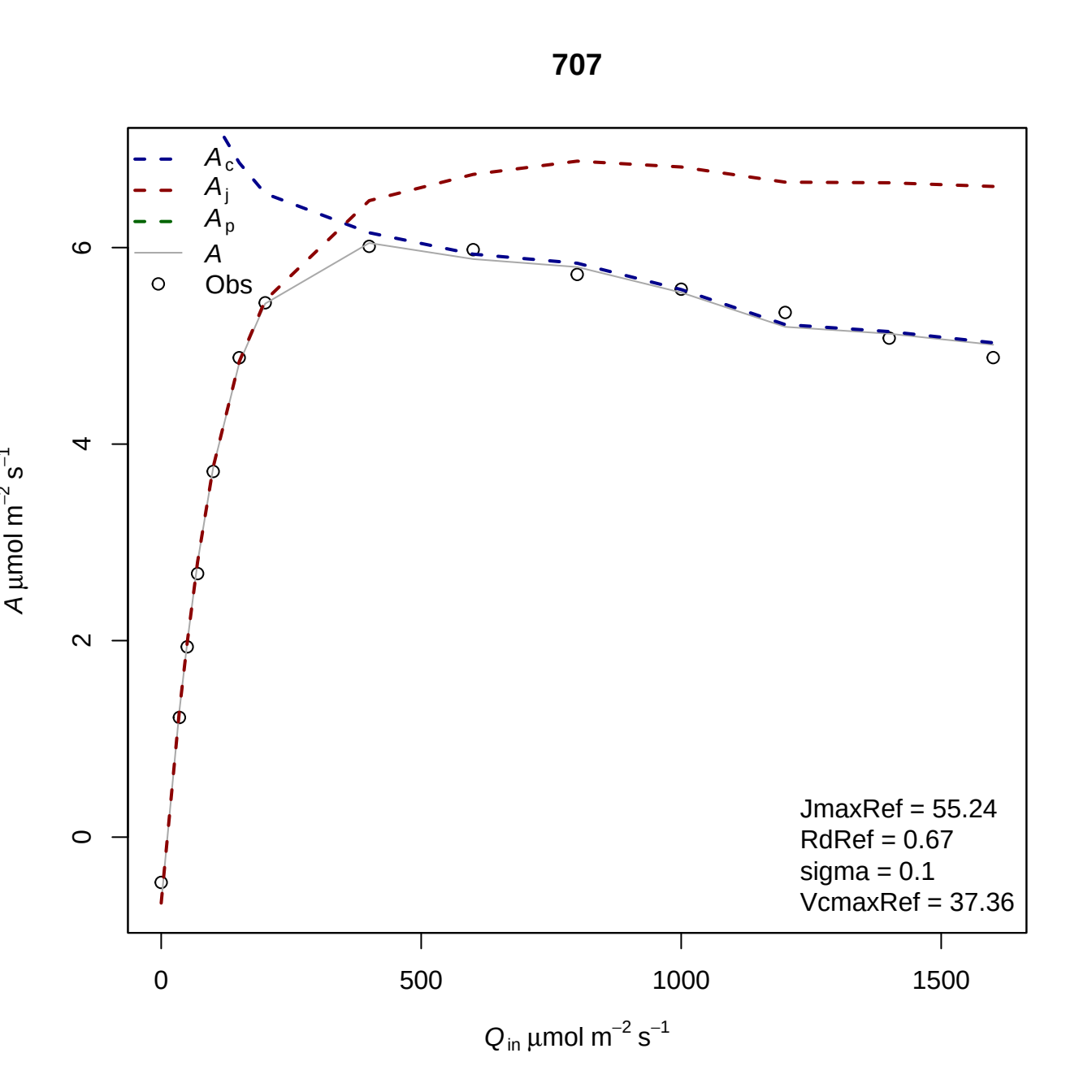

702
A_c
A_j
A_p
A
Obs
JmaxRef = 59.95
RdRef = 0.99
sigma = 0.11
VcmaxRef = 284.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

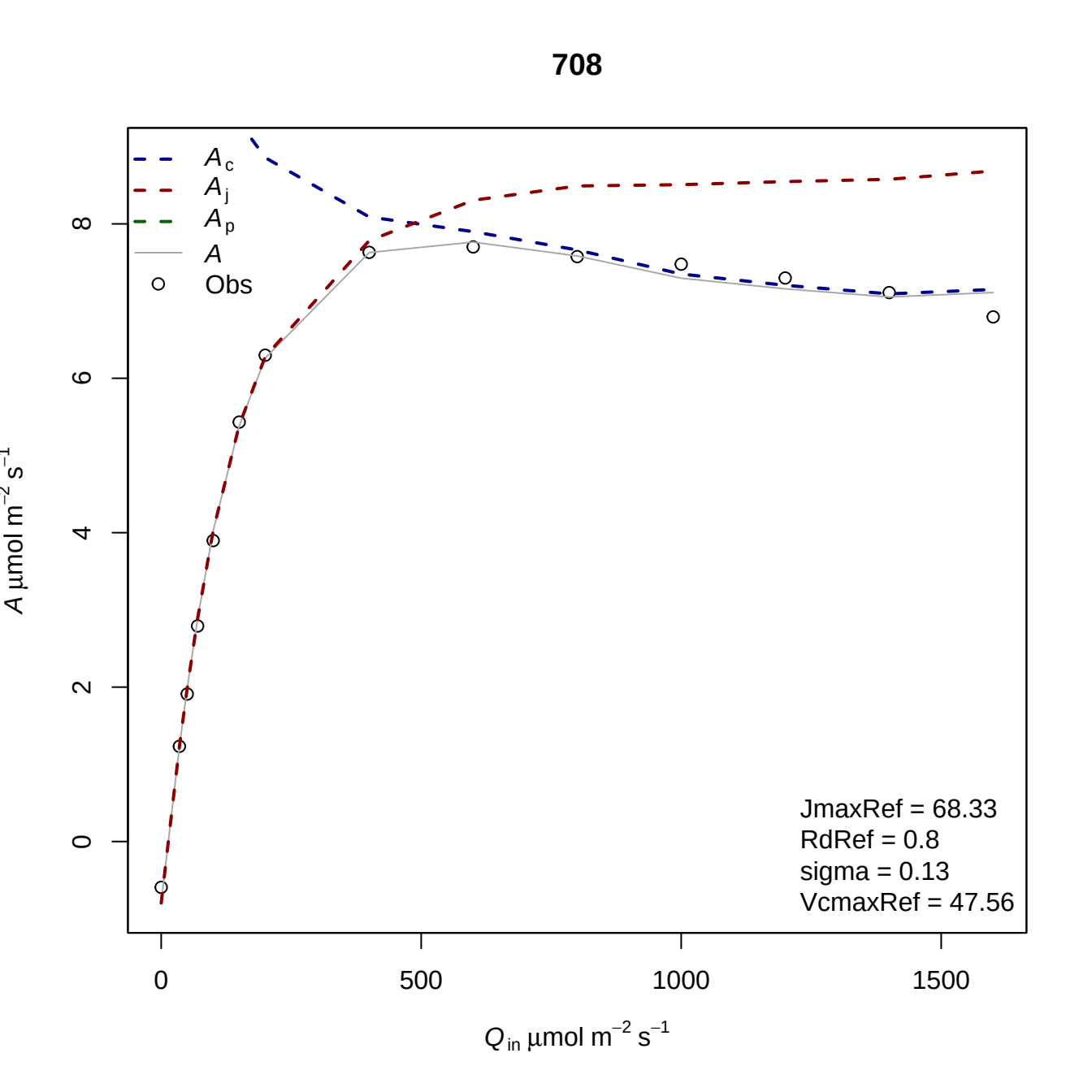

703
A_c
A_j
A_p
A
Obs
JmaxRef = 40.8
RdRef = −0.11
sigma = 0.11
VcmaxRef = 30.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

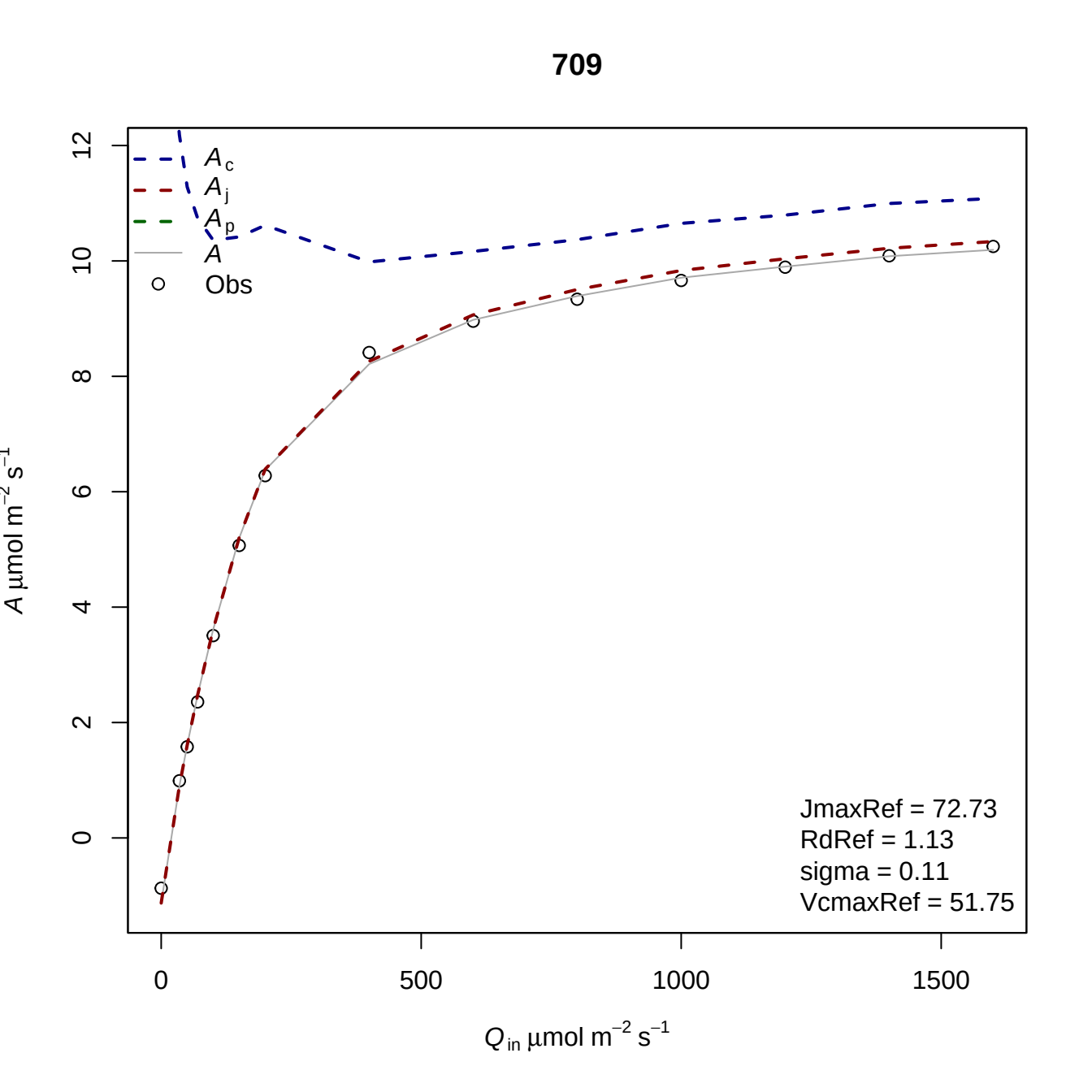

704
A_c
A_j
A_p
A
Obs
JmaxRef = 65.19
RdRef = 0.42
sigma = 0.21
VcmaxRef = 49.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

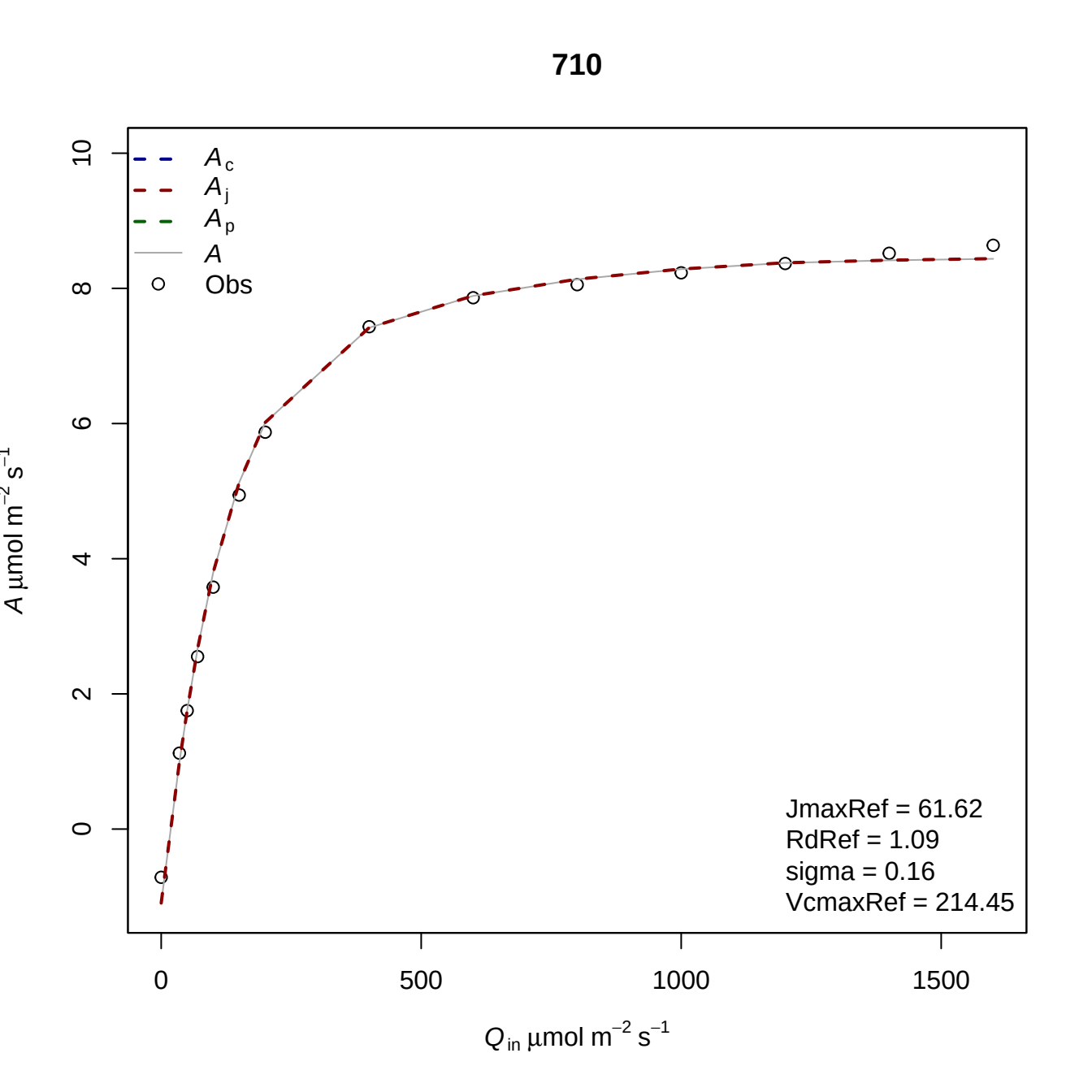

705
A_c
A_j
A_p
A
Obs
JmaxRef = 81.81
RdRef = 0.12
sigma = 0.18
VcmaxRef = 61.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

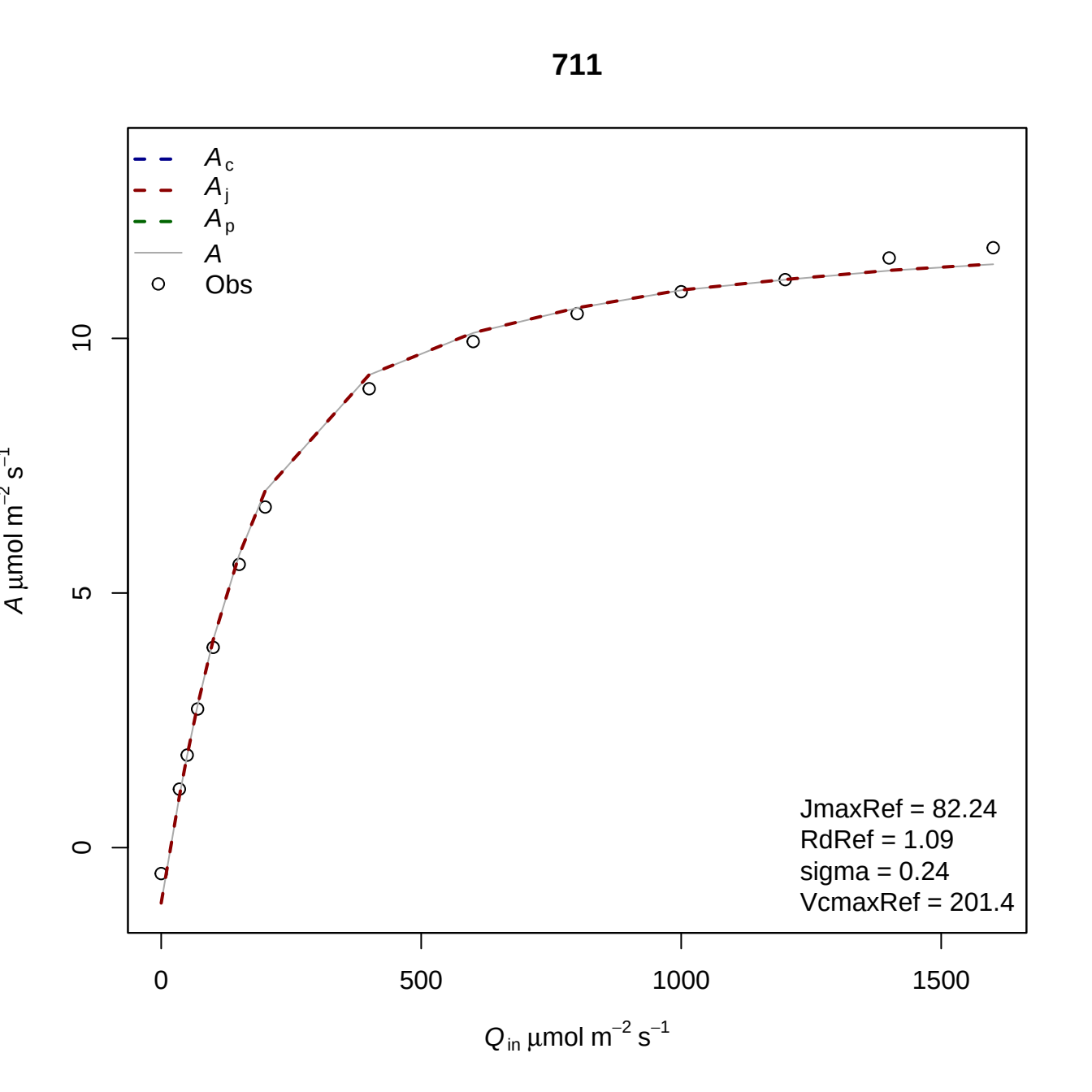

706
A_c
A_j
A_p
A
Obs
JmaxRef = 75.03
RdRef = 0.38
sigma = 0.21
VcmaxRef = 72.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

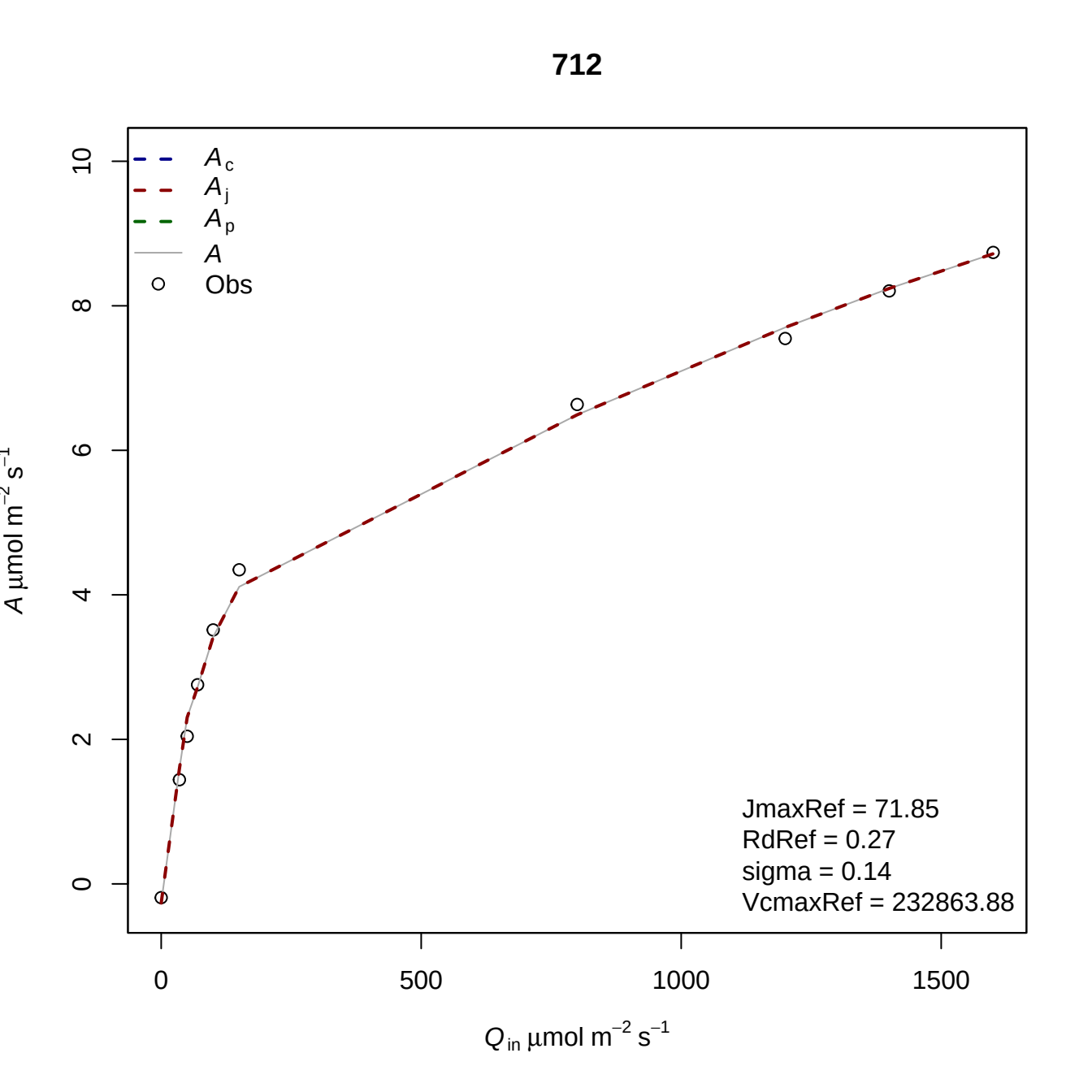

707
A_c
A_j
A_p
A
Obs
JmaxRef = 55.24
RdRef = 0.67
sigma = 0.1
VcmaxRef = 37.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

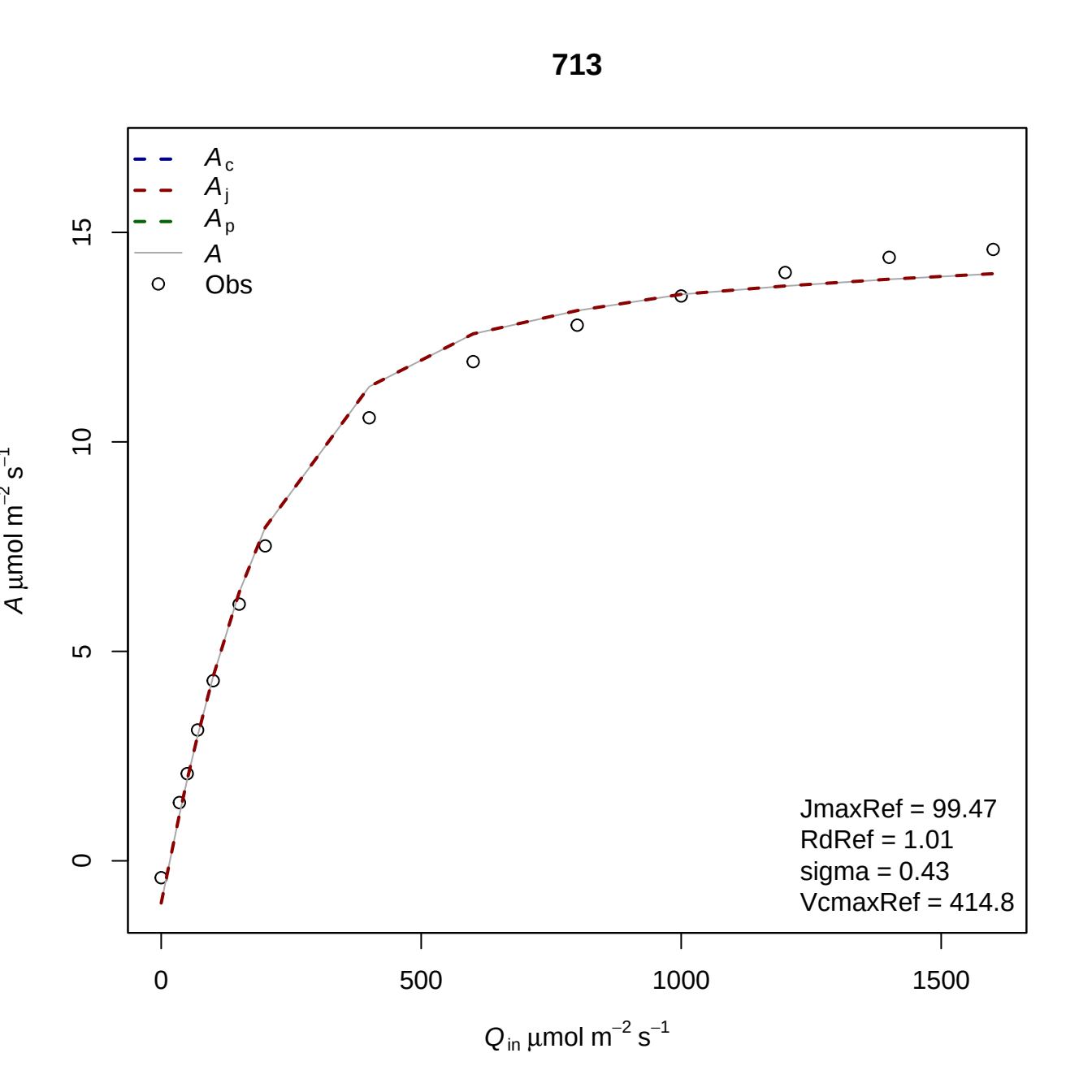

708
A_c
A_j
A_p
A
Obs
JmaxRef = 68.33
RdRef = 0.8
sigma = 0.13
VcmaxRef = 47.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

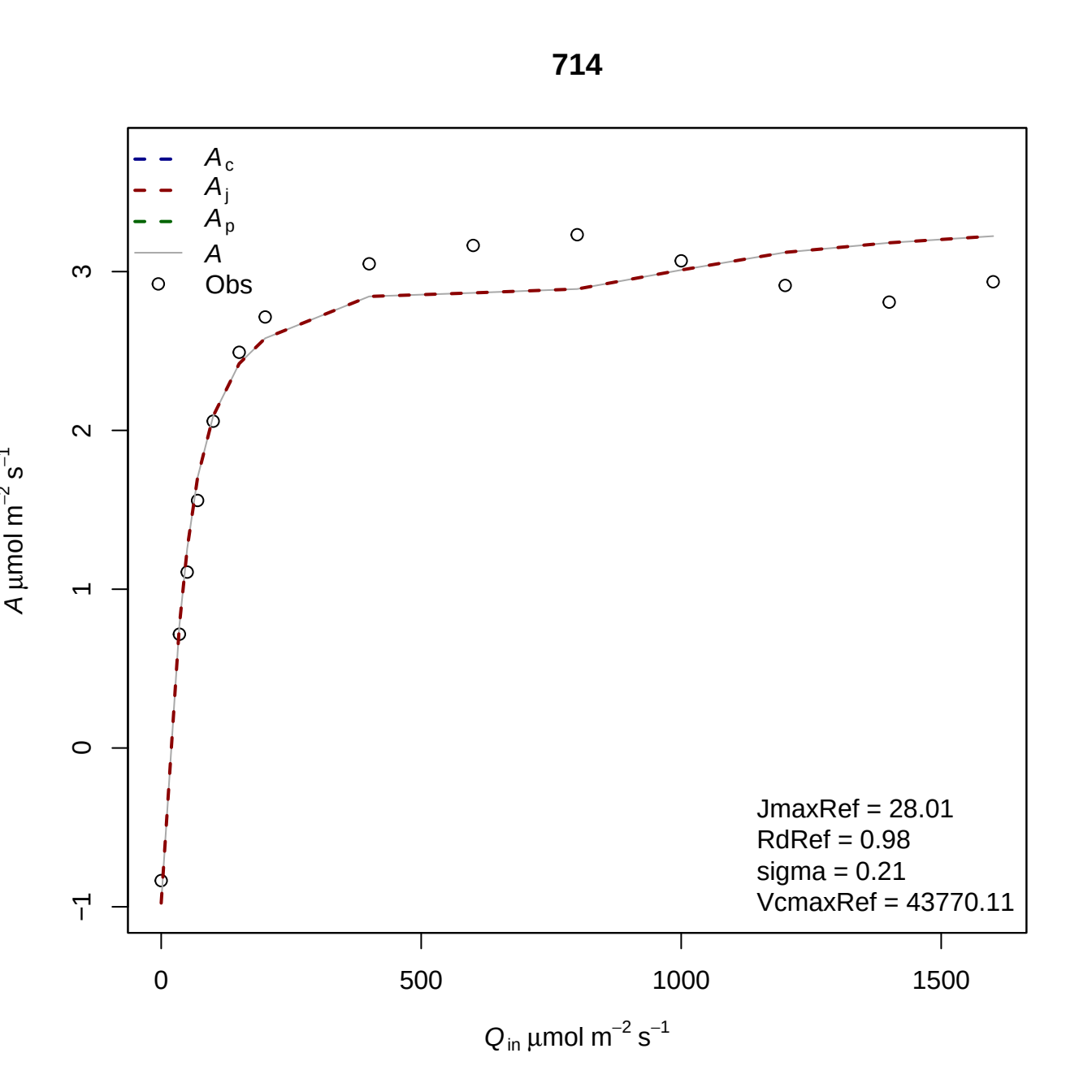

709
A_c
A_j
A_p
A
Obs
JmaxRef = 72.73
RdRef = 1.13
sigma = 0.11
VcmaxRef = 51.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

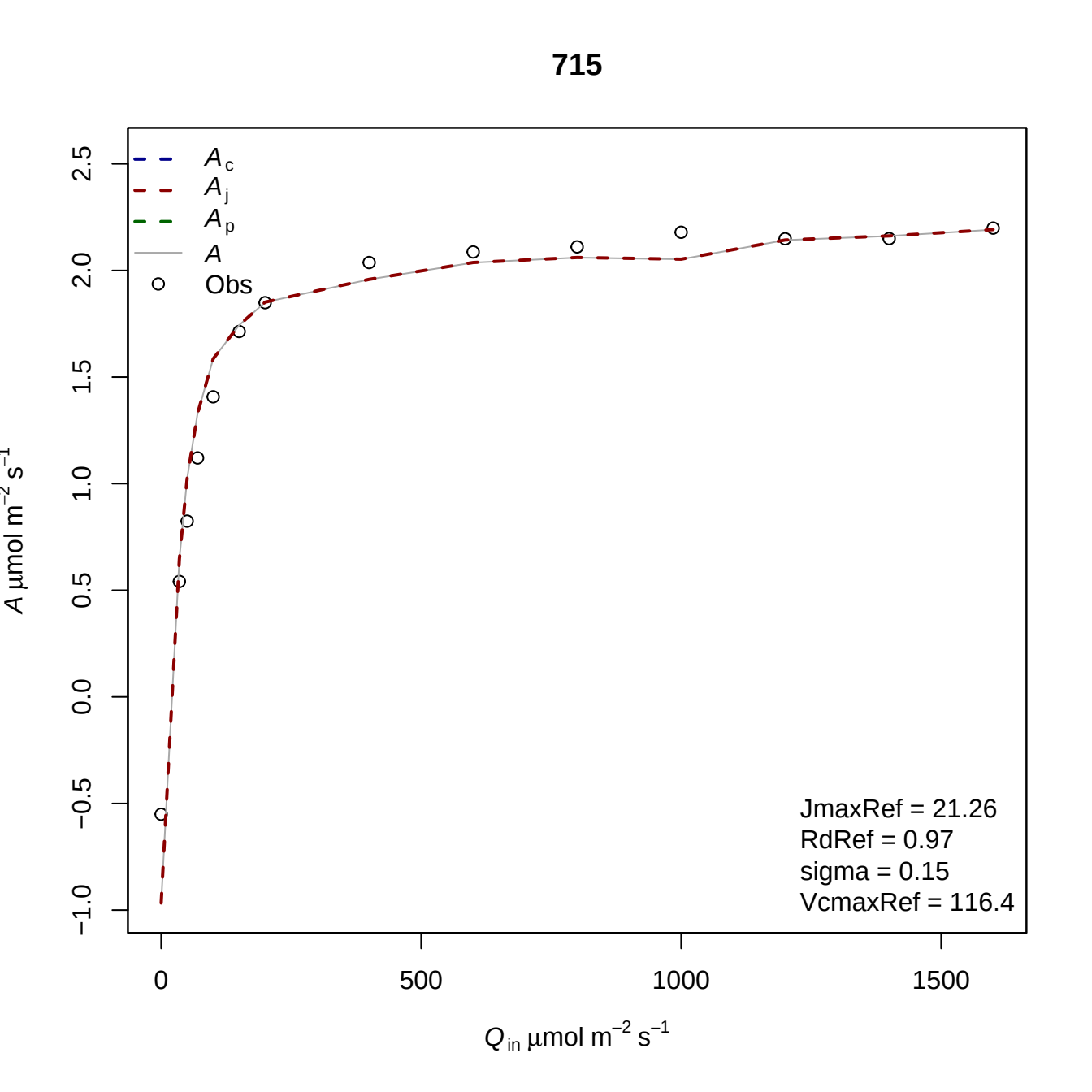

710
A_c
A_j
A_p
A
Obs
JmaxRef = 61.62
RdRef = 1.09
sigma = 0.16
VcmaxRef = 214.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

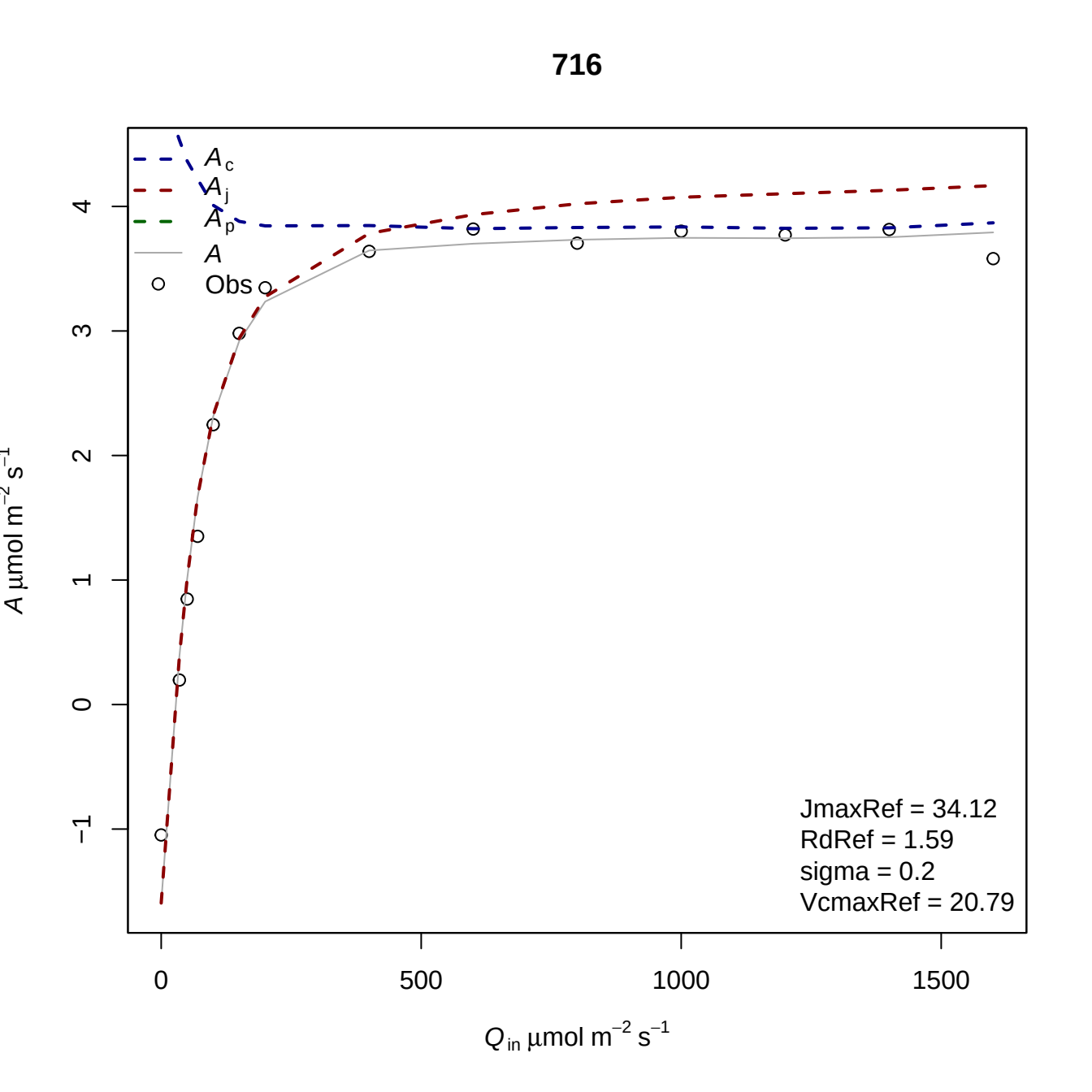

711
A_c
A_j
A_p
A
Obs
JmaxRef = 82.24
RdRef = 1.09
sigma = 0.24
VcmaxRef = 201.4
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

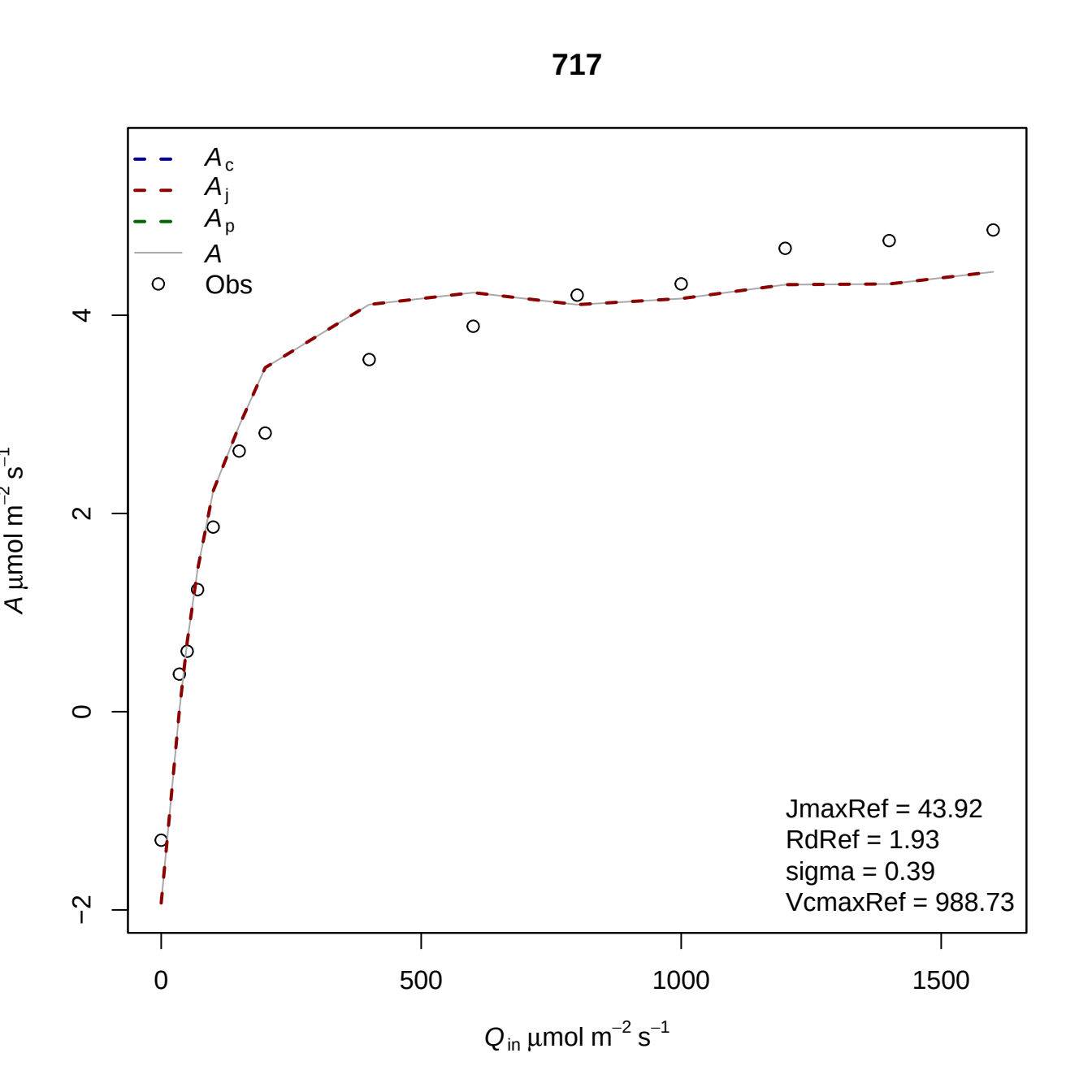

712
A_c
A_j
A_p
A
Obs
JmaxRef = 71.85
RdRef = 0.27
sigma = 0.14
VcmaxRef = 232863.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

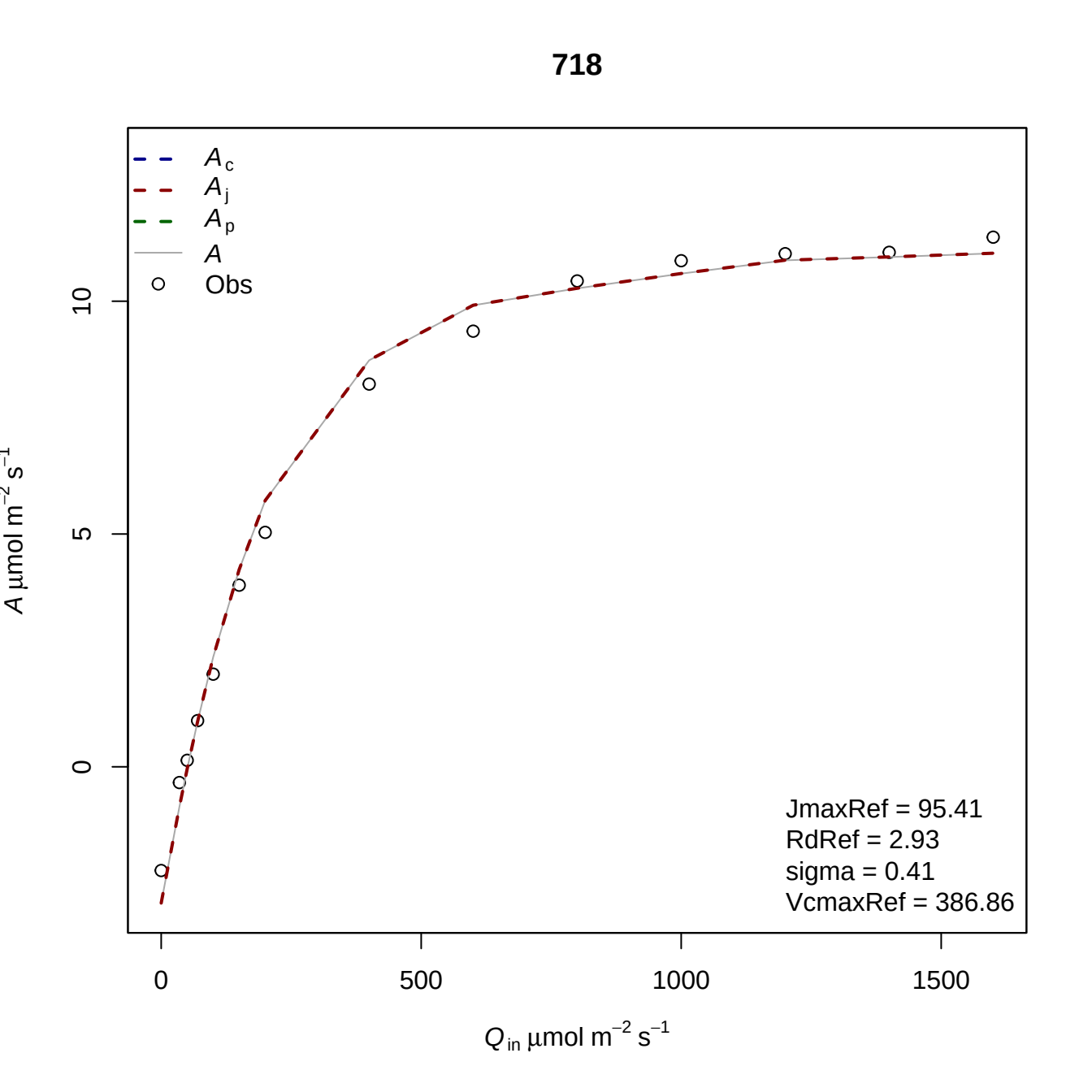

713
A_c
A_j
A_p
A
Obs
JmaxRef = 99.47
RdRef = 1.01
sigma = 0.43
VcmaxRef = 414.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

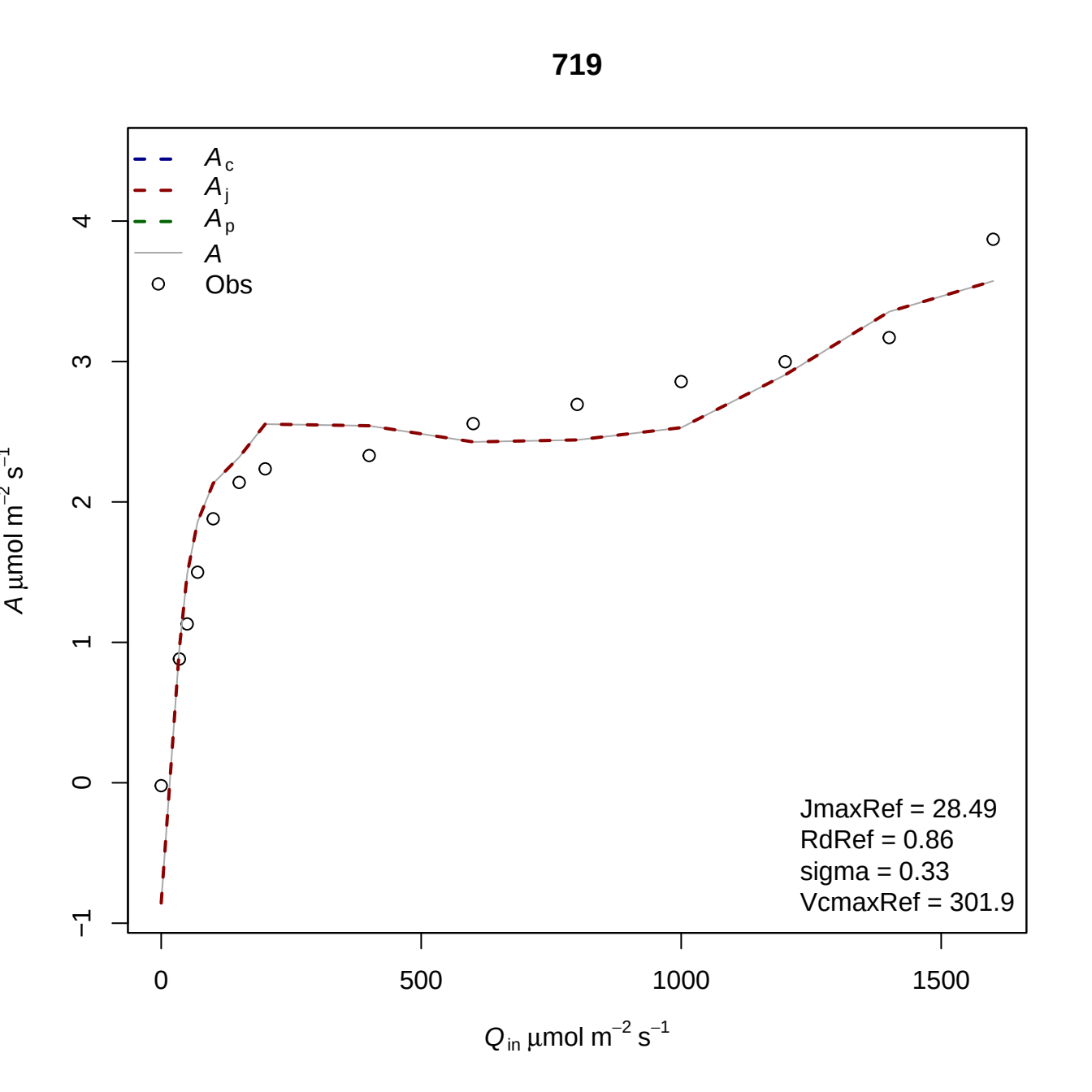

714
A_c
A_j
A_p
A
Obs
JmaxRef = 28.01
RdRef = 0.98
sigma = 0.21
VcmaxRef = 43770.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

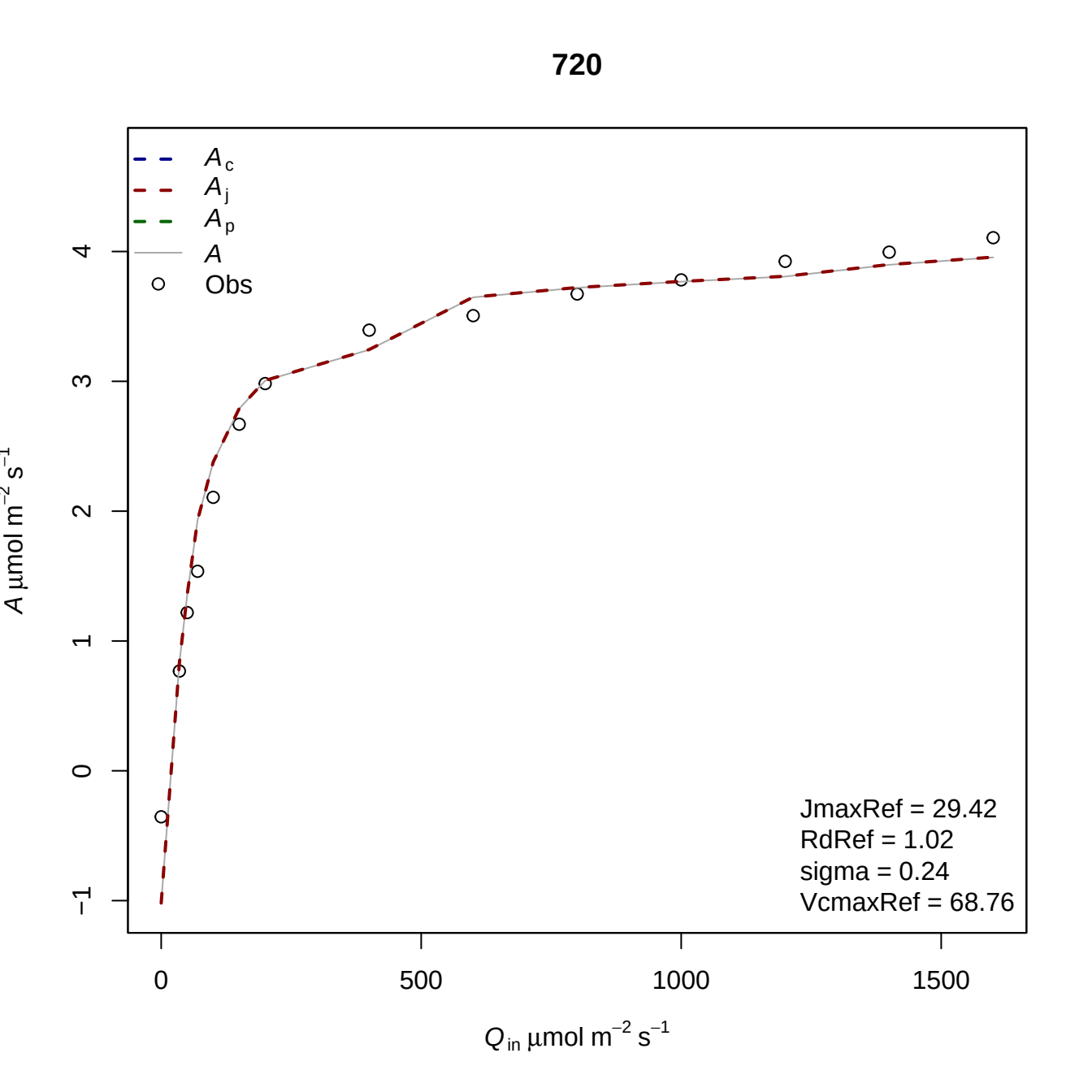

715
A_c
A_j
A_p
A
Obs
JmaxRef = 21.26
RdRef = 0.97
sigma = 0.15
VcmaxRef = 116.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500

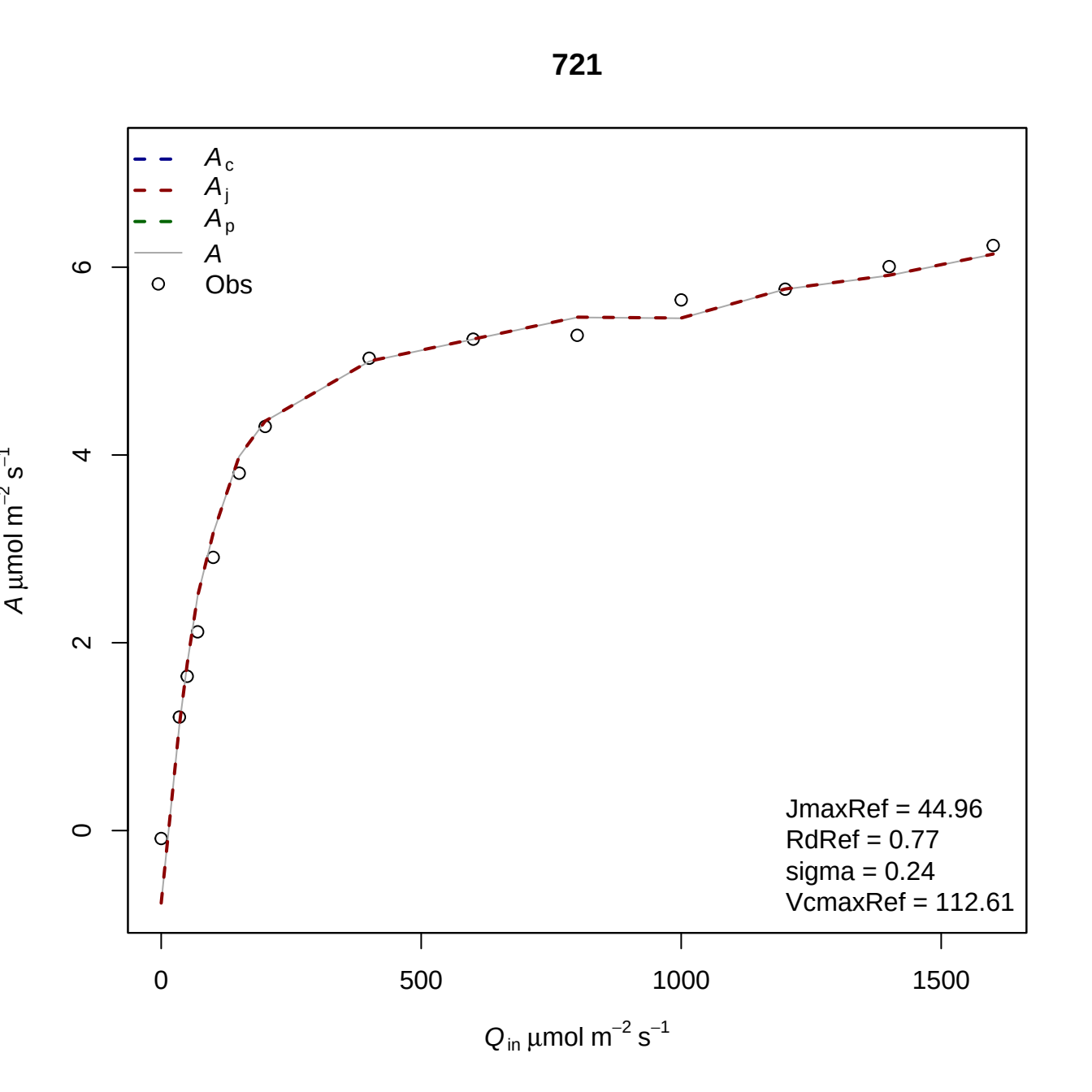

716
A_c
A_j
A_p
A
Obs
JmaxRef = 34.12
RdRef = 1.59
sigma = 0.2
VcmaxRef = 20.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

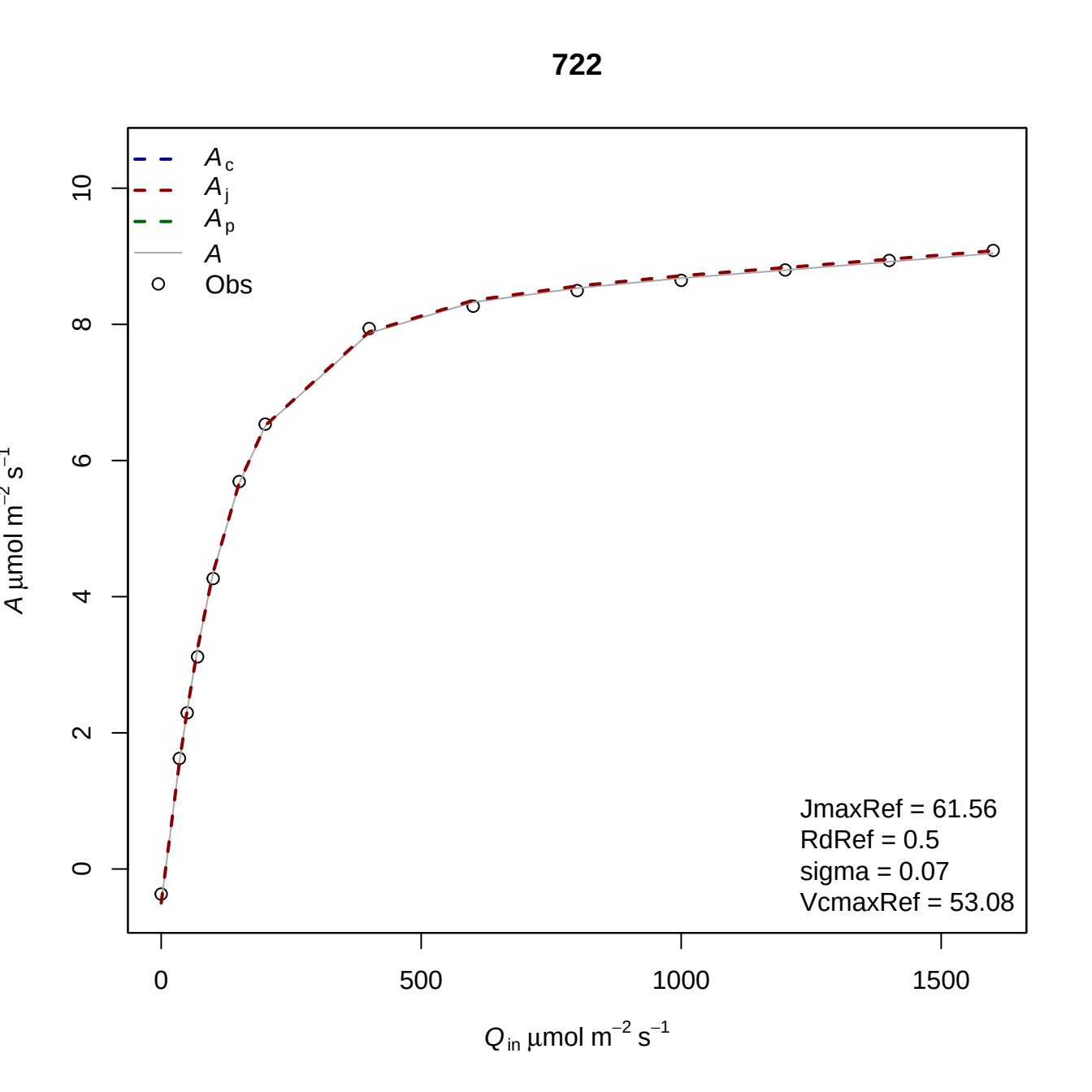

717
A_c
A_j
A_p
A
Obs
JmaxRef = 43.92
RdRef = 1.93
sigma = 0.39
VcmaxRef = 988.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

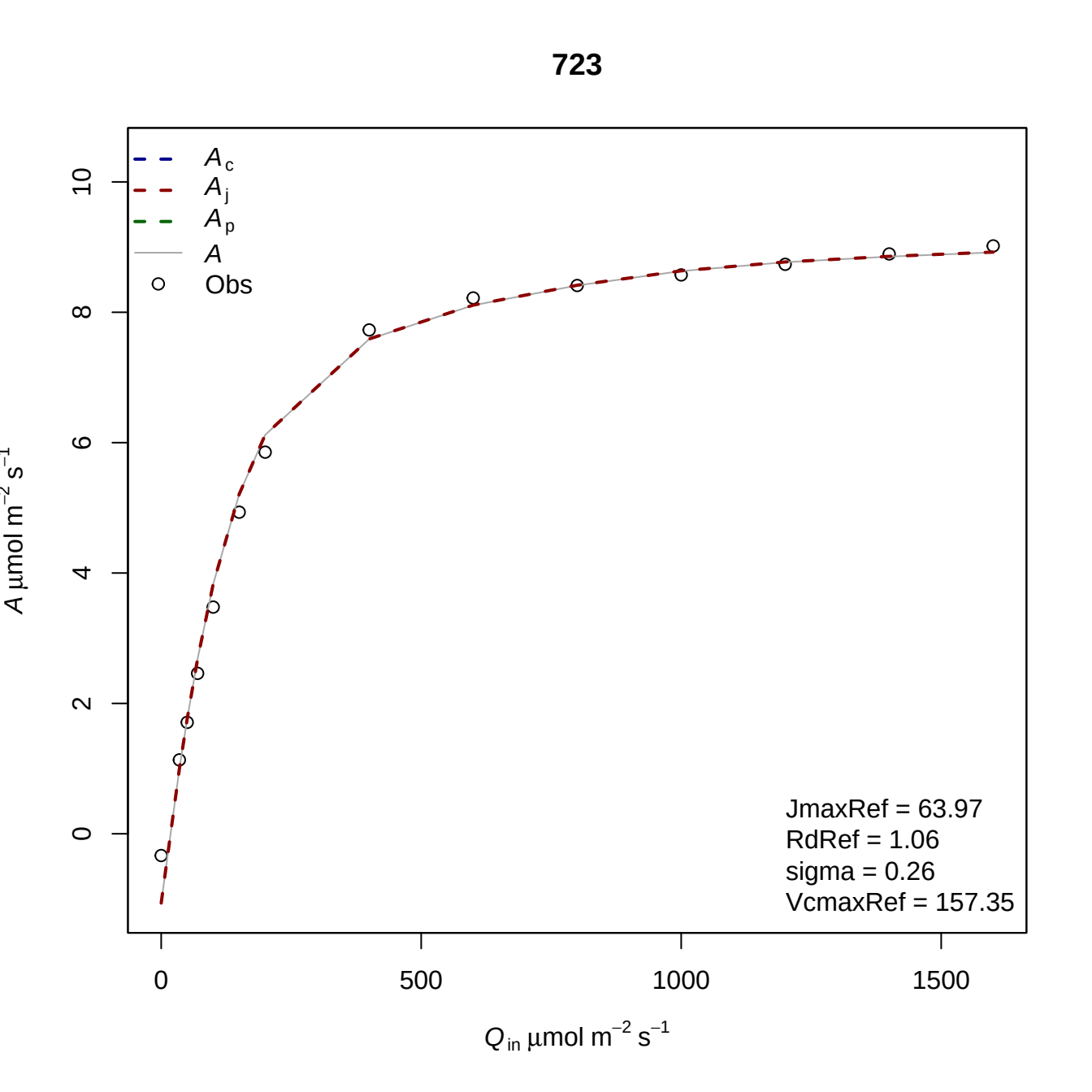

718
A_c
A_j
A_p
A
Obs
JmaxRef = 95.41
RdRef = 2.93
sigma = 0.41
VcmaxRef = 386.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

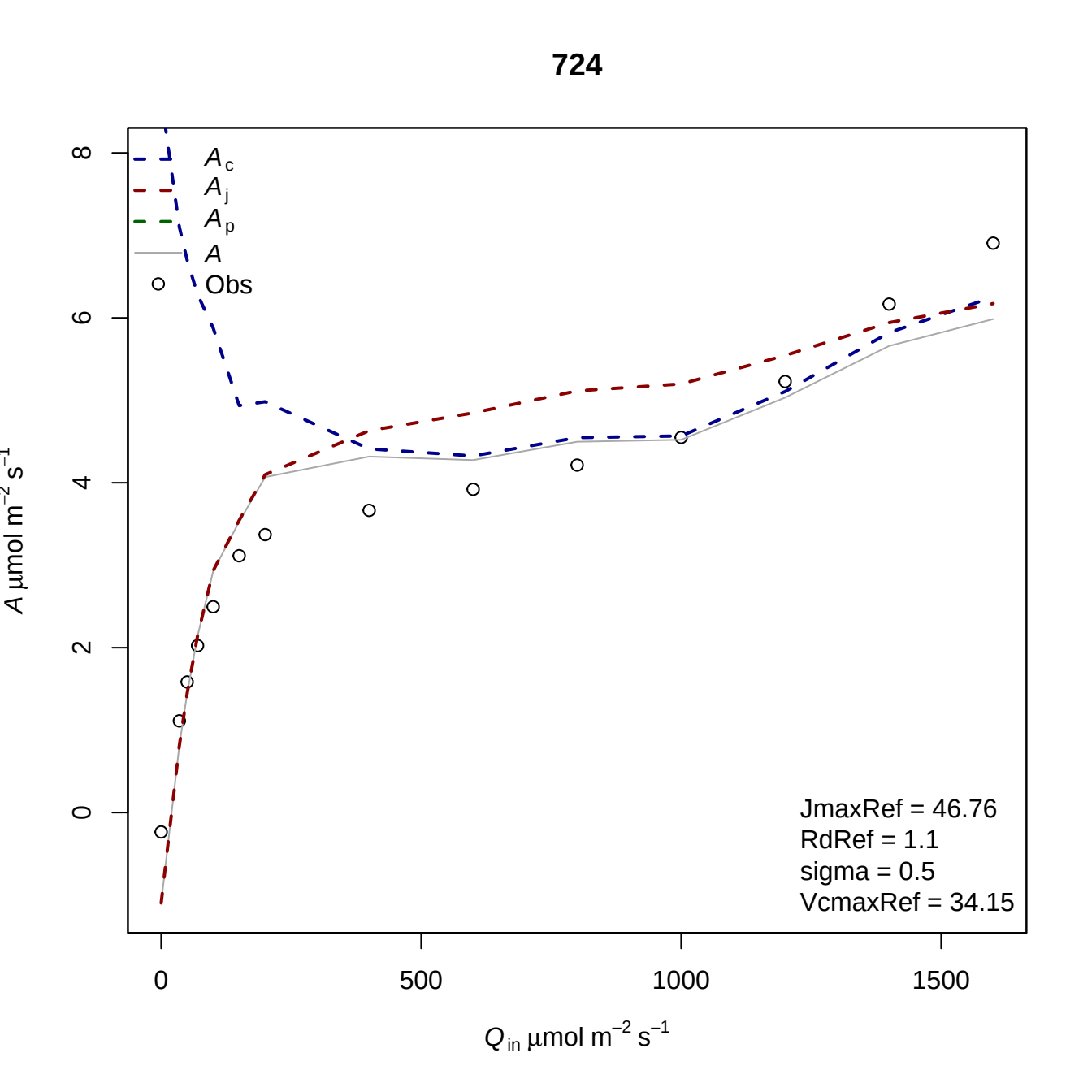

719
A_c
A_j
A_p
A
Obs
JmaxRef = 28.49
RdRef = 0.86
sigma = 0.33
VcmaxRef = 301.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

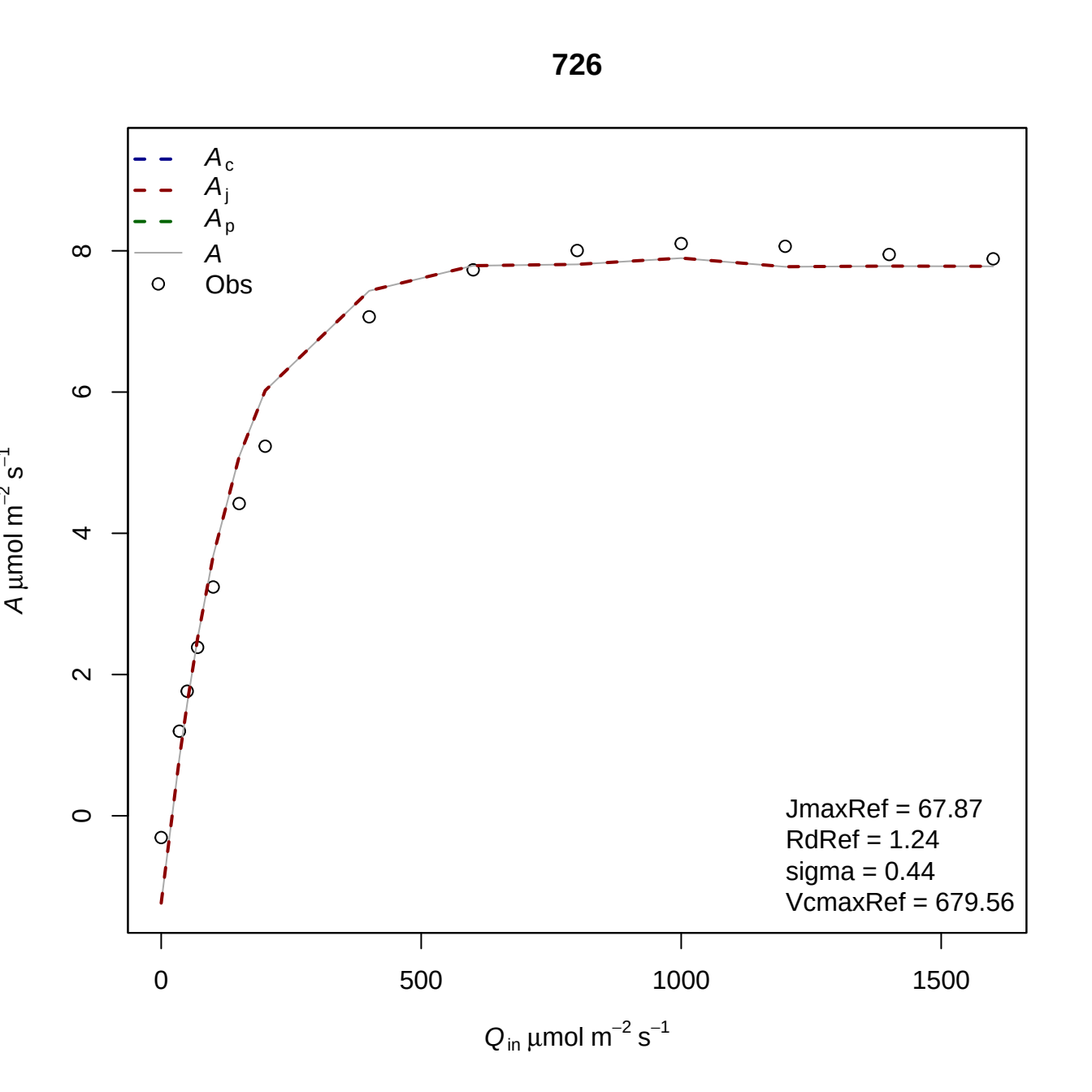

720
A_c
A_j
A_p
A
Obs
JmaxRef = 29.42
RdRef = 1.02
sigma = 0.24
VcmaxRef = 68.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

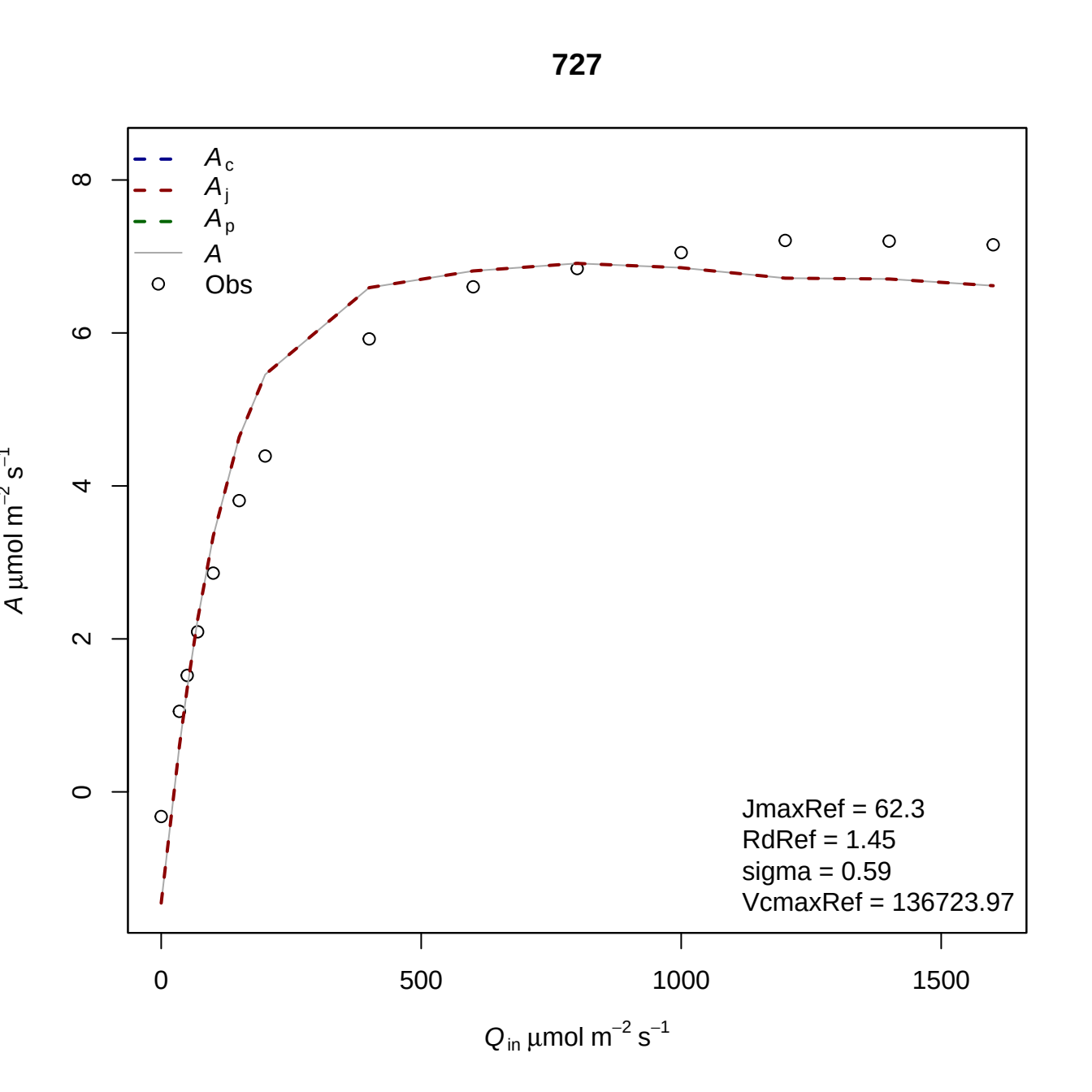

721
A_c
A_j
A_p
A
Obs
JmaxRef = 44.96
RdRef = 0.77
sigma = 0.24
VcmaxRef = 112.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

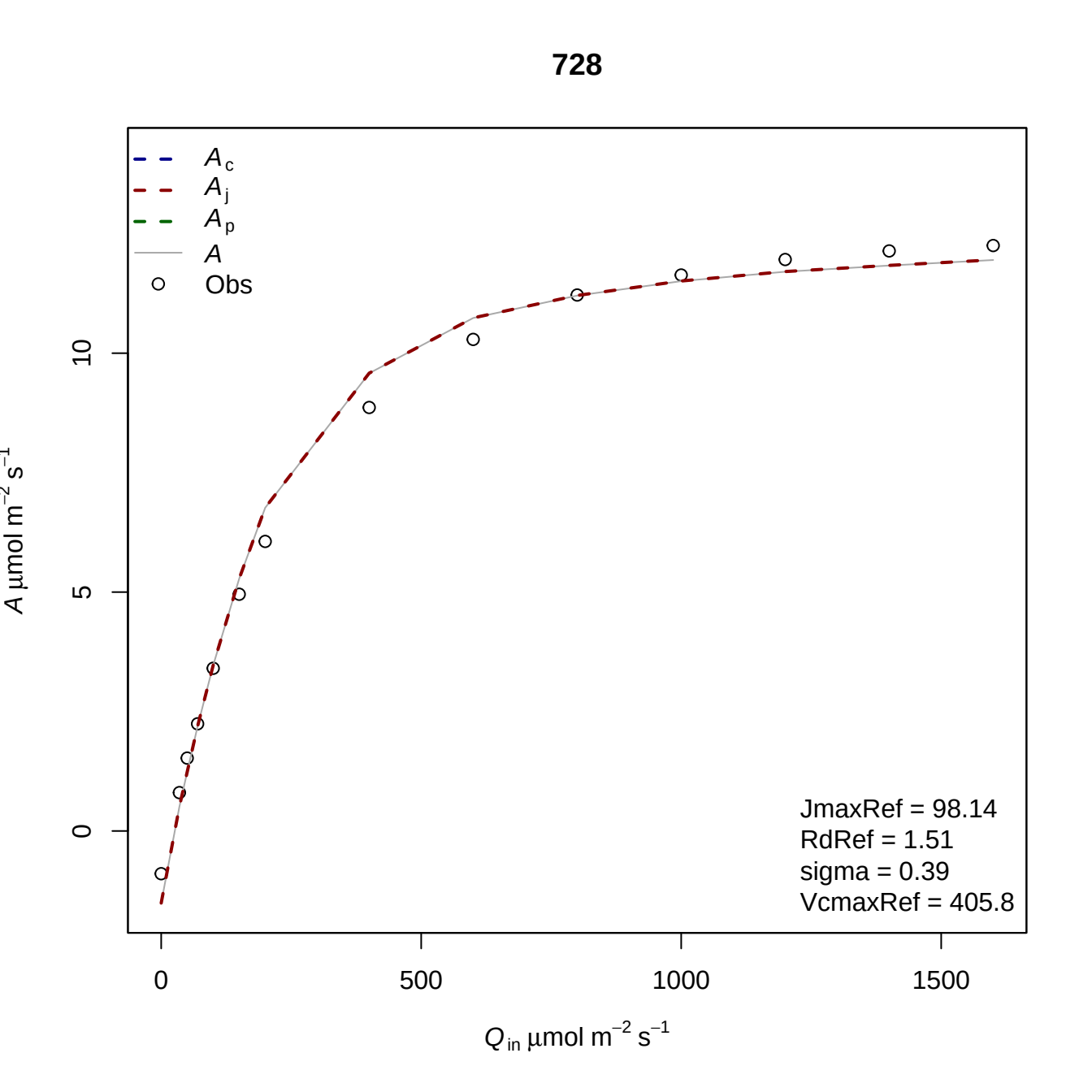

722
A_c
A_j
A_p
A
Obs
JmaxRef = 61.56
RdRef = 0.5
sigma = 0.07
VcmaxRef = 53.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

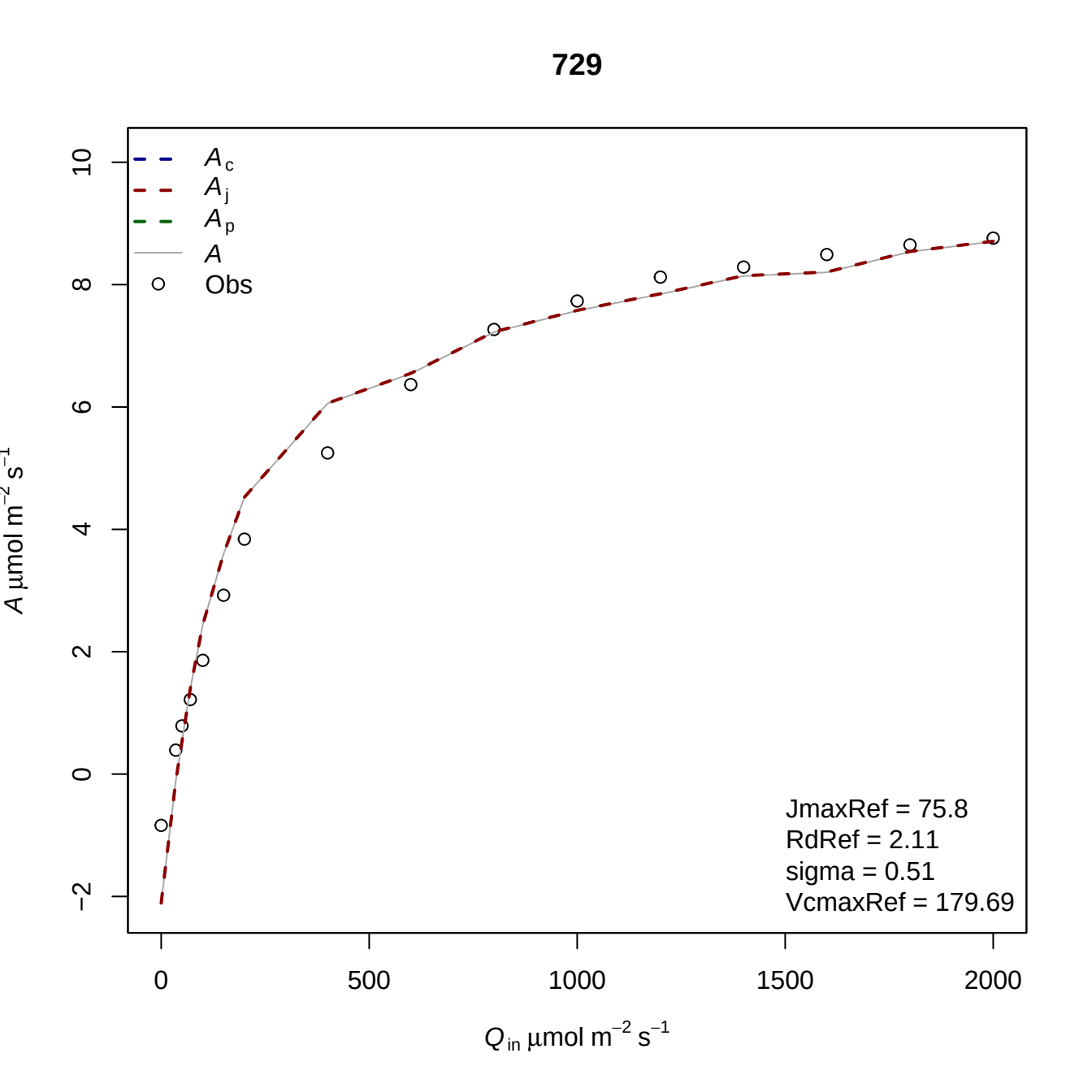

723
A_c
A_j
A_p
A
Obs
JmaxRef = 63.97
RdRef = 1.06
sigma = 0.26
VcmaxRef = 157.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

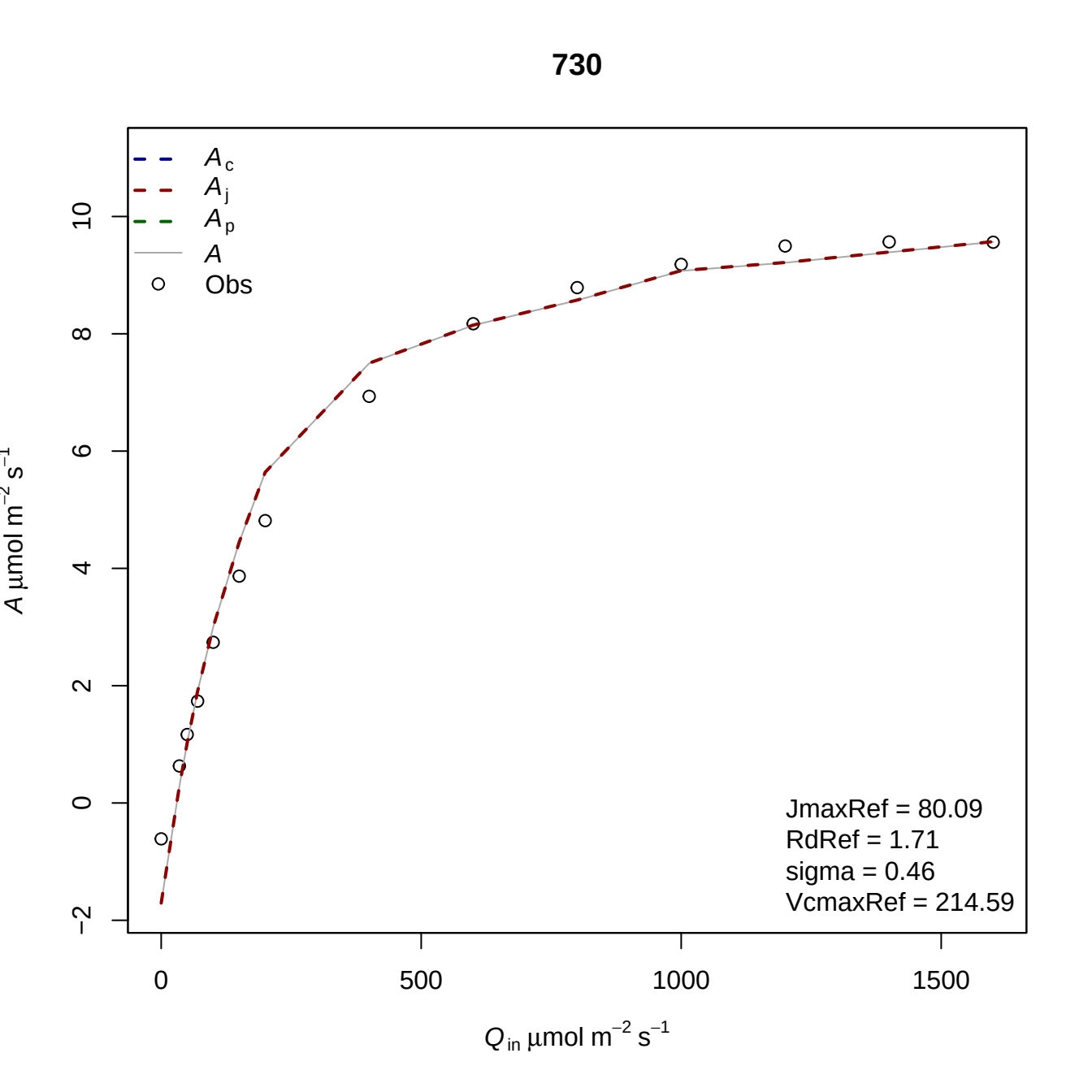

724
A_c
A_j
A_p
A
Obs
JmaxRef = 46.76
RdRef = 1.1
sigma = 0.5
VcmaxRef = 34.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

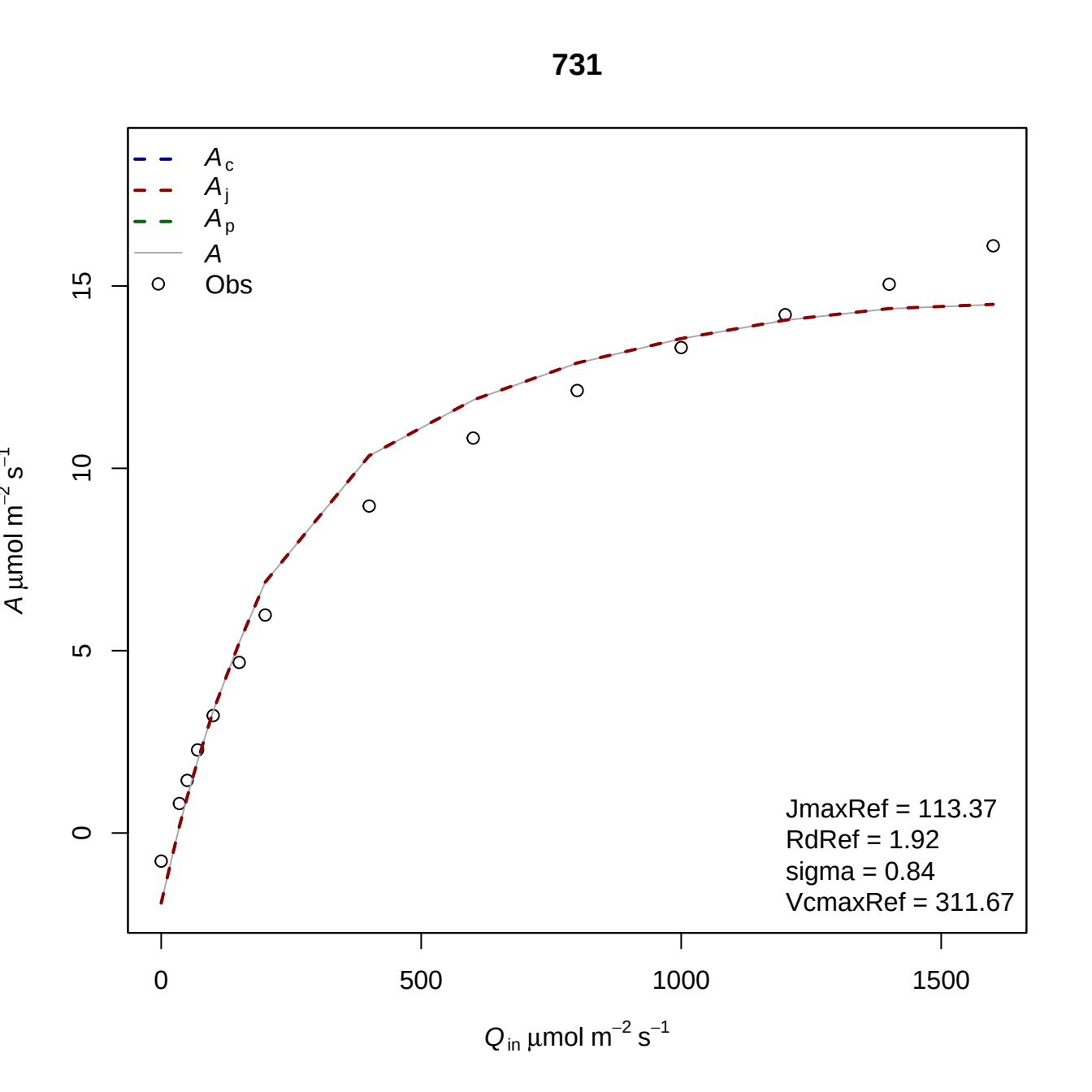

726
A_c
A_j
A_p
A
Obs
JmaxRef = 67.87
RdRef = 1.24
sigma = 0.44
VcmaxRef = 679.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

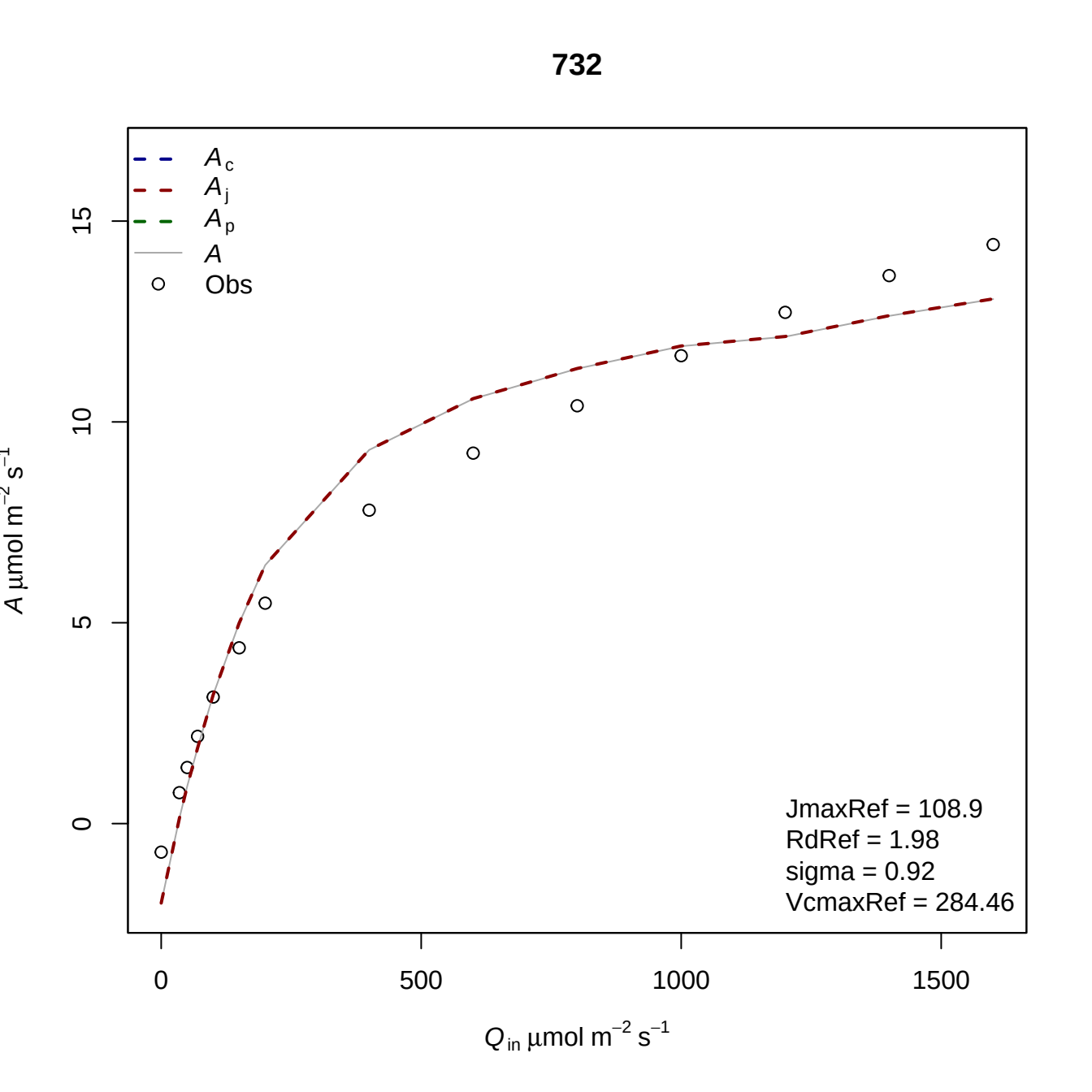

727
A_c
A_j
A_p
A
Obs
JmaxRef = 62.3
RdRef = 1.45
sigma = 0.59
VcmaxRef = 136723.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

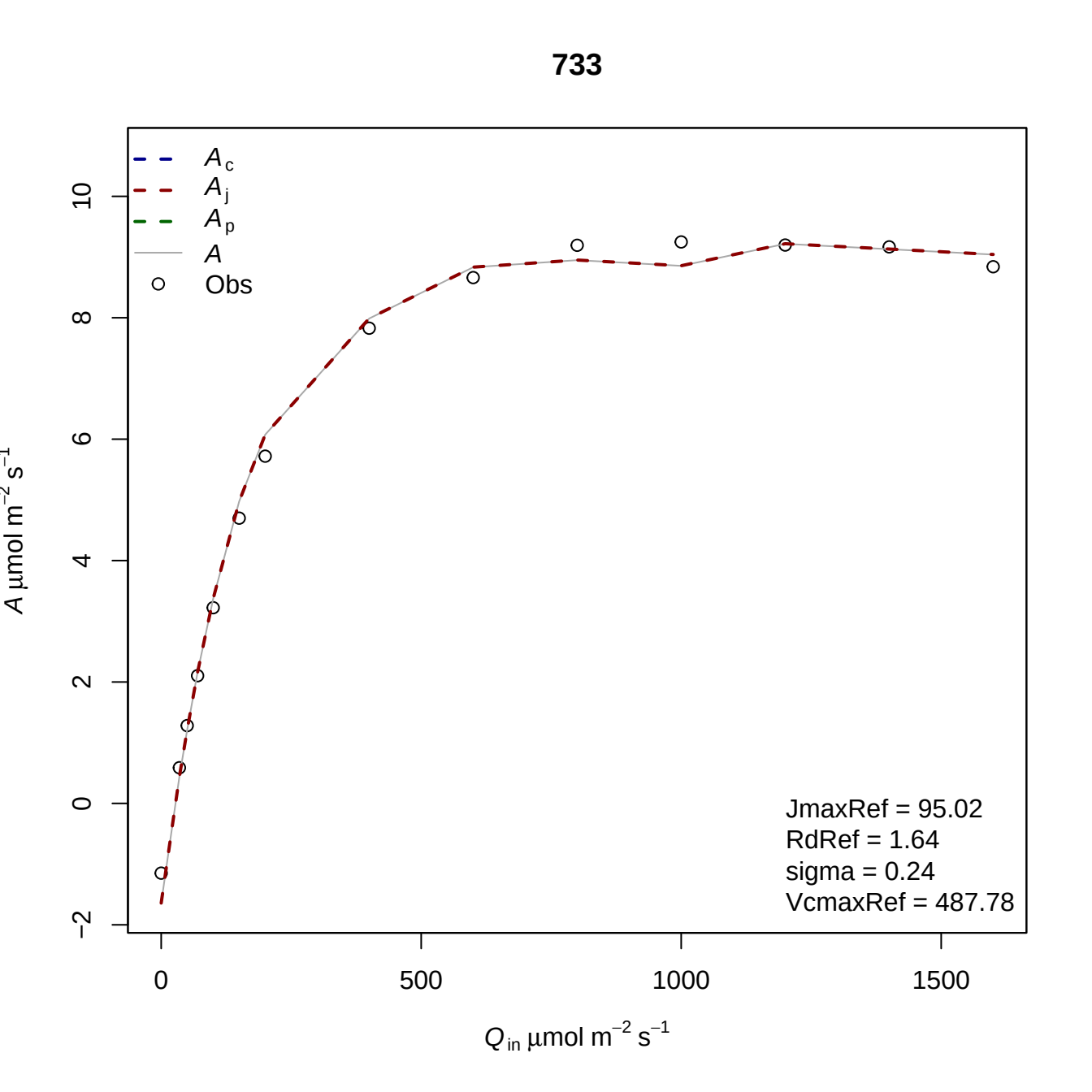

728
A_c
A_j
A_p
A
Obs
JmaxRef = 98.14
RdRef = 1.51
sigma = 0.39
VcmaxRef = 405.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

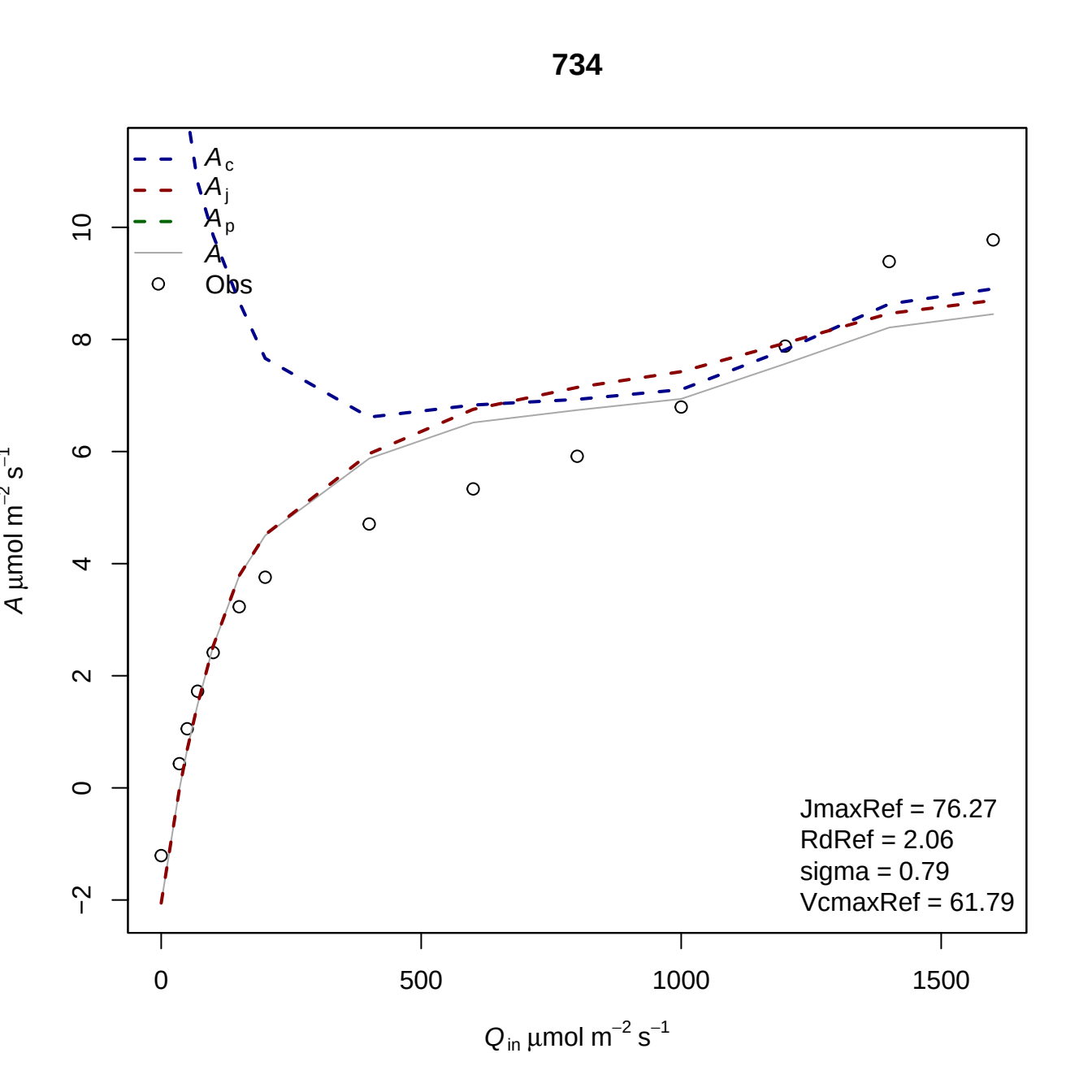

729
A_c
A_j
A_p
A
Obs
JmaxRef = 75.8
RdRef = 2.11
sigma = 0.51
VcmaxRef = 179.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

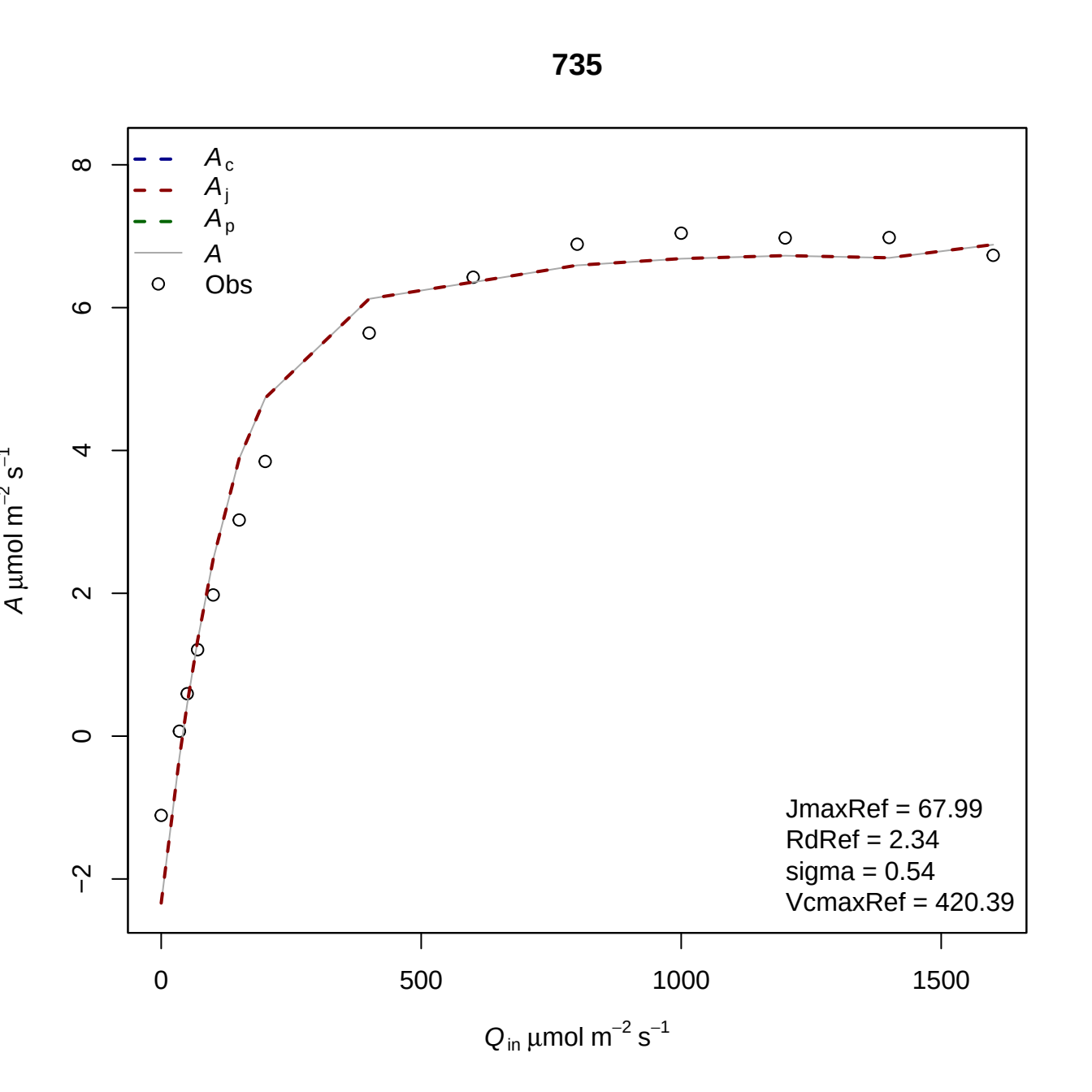

730
A_c
A_j
A_p
A
Obs
JmaxRef = 80.09
RdRef = 1.71
sigma = 0.46
VcmaxRef = 214.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

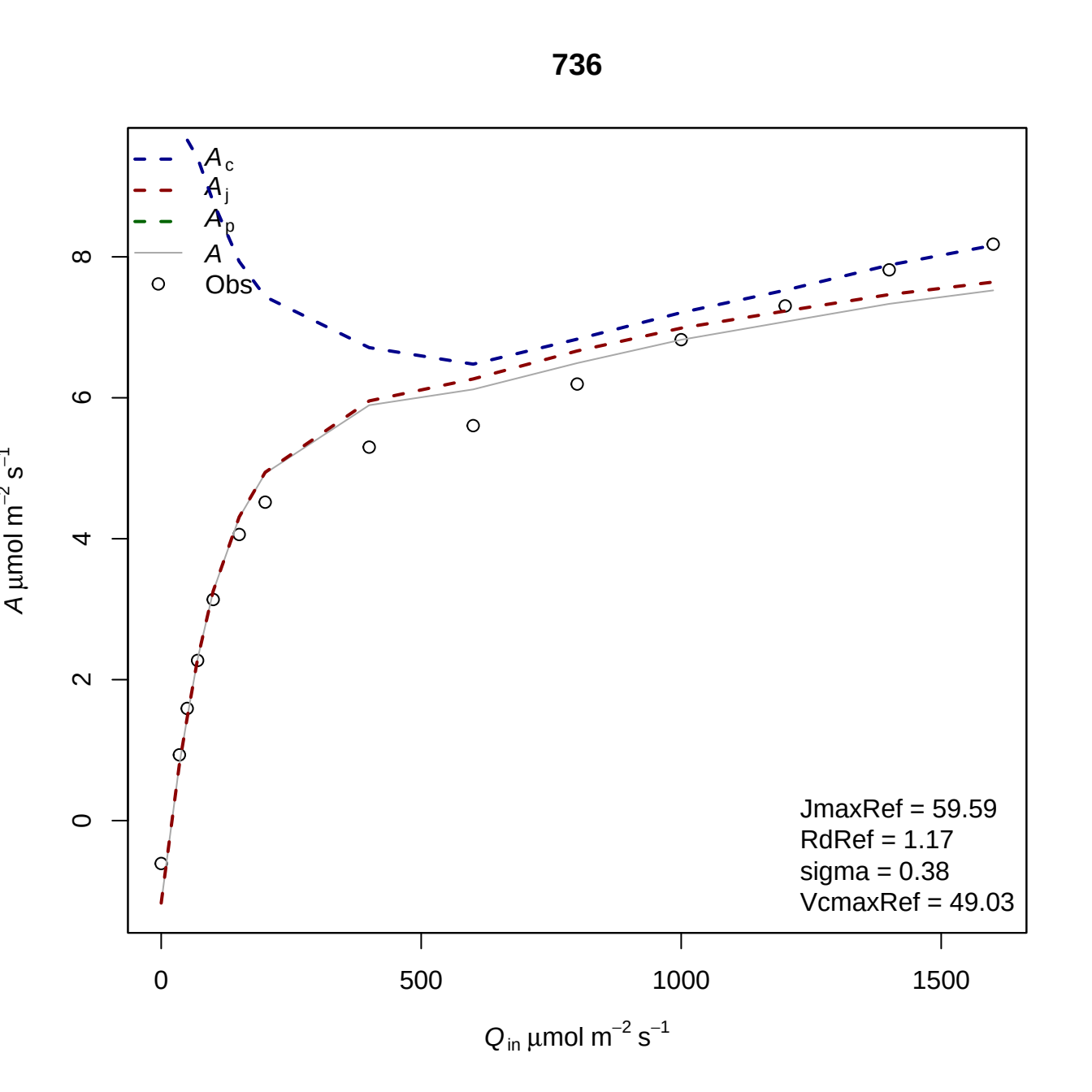

731
A_c
A_j
A_p
A
Obs
JmaxRef = 113.37
RdRef = 1.92
sigma = 0.84
VcmaxRef = 311.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

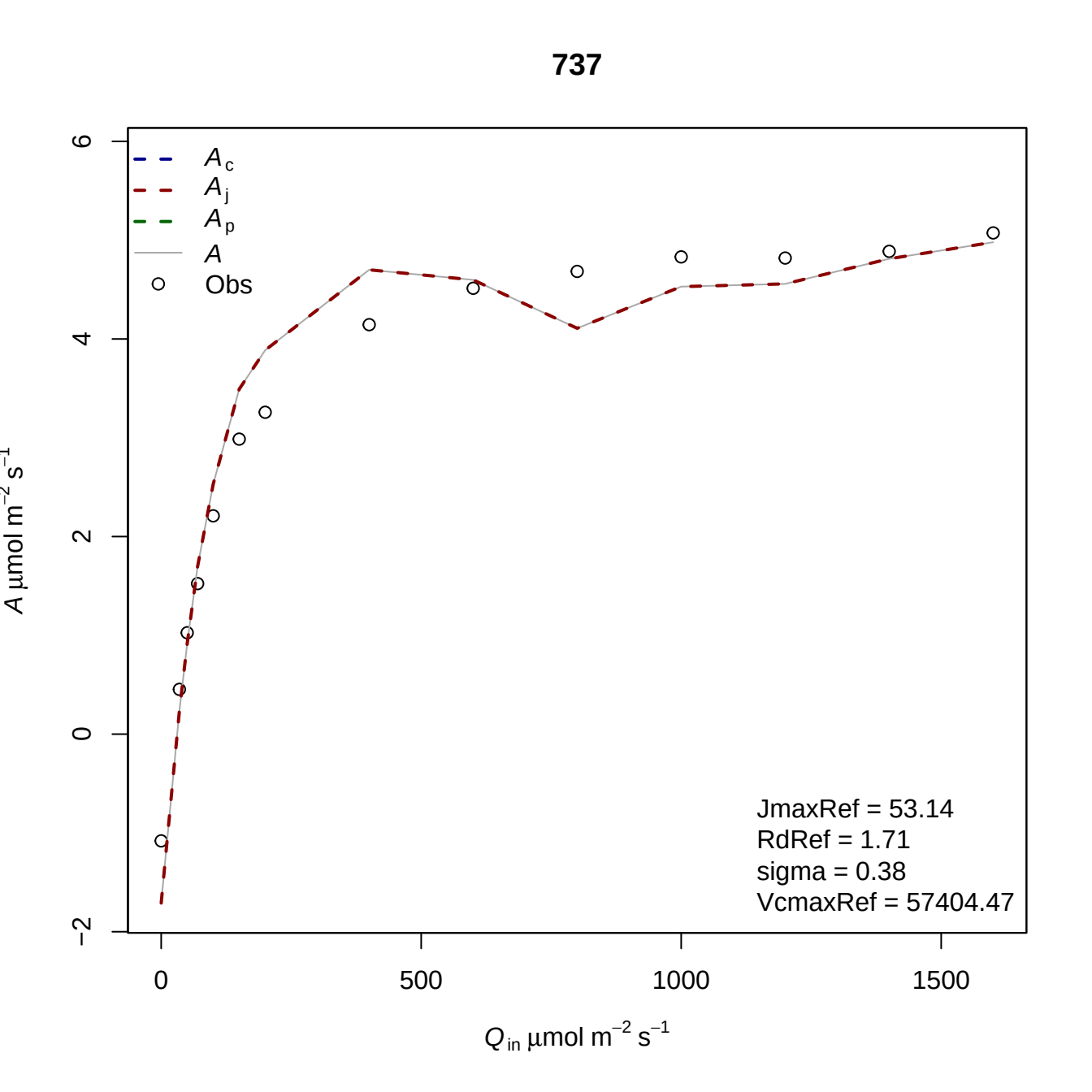

732
A_c
A_j
A_p
A
Obs
JmaxRef = 108.9
RdRef = 1.98
sigma = 0.92
VcmaxRef = 284.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

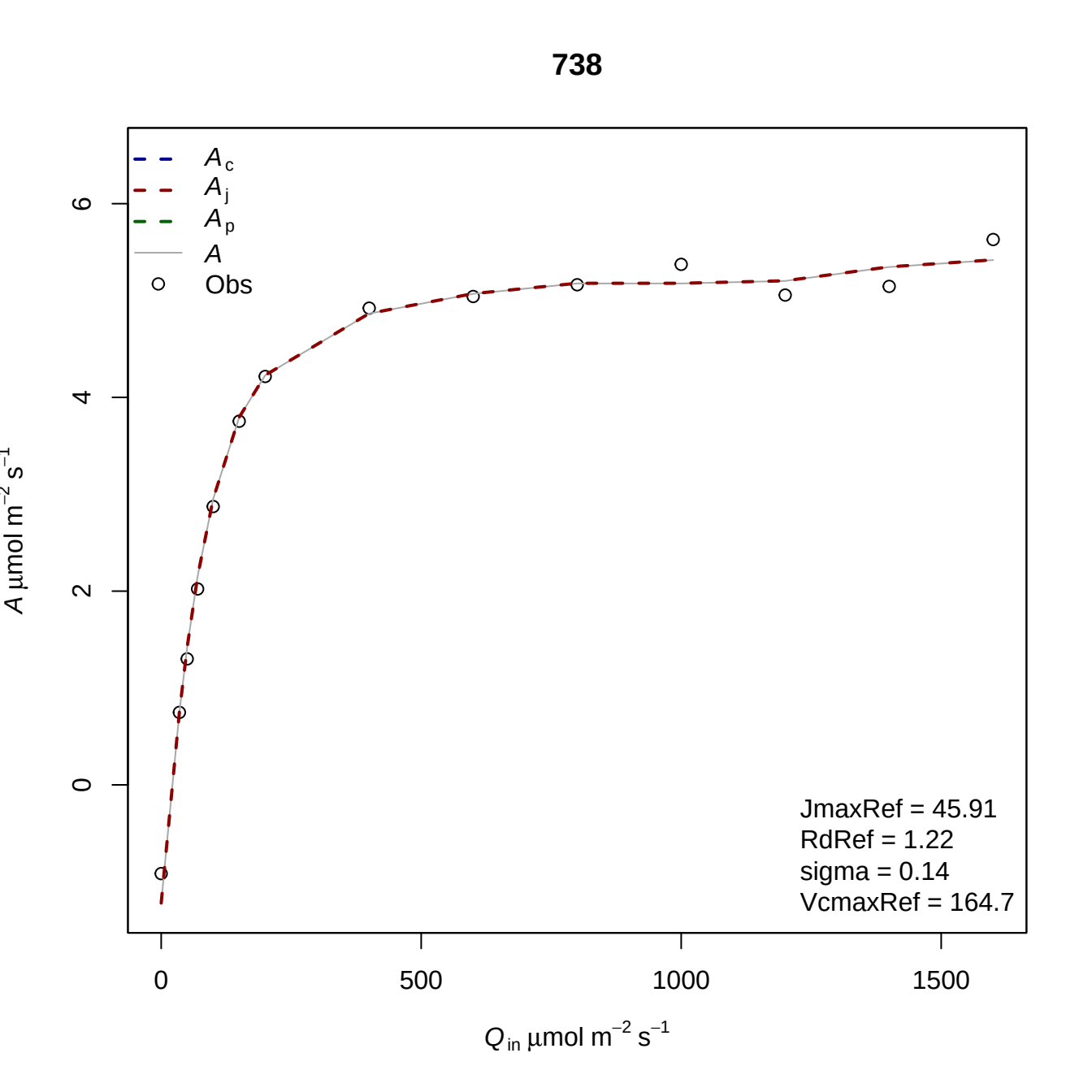

733
A_c
A_j
A_p
A
Obs
JmaxRef = 95.02
RdRef = 1.64
sigma = 0.24
VcmaxRef = 487.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

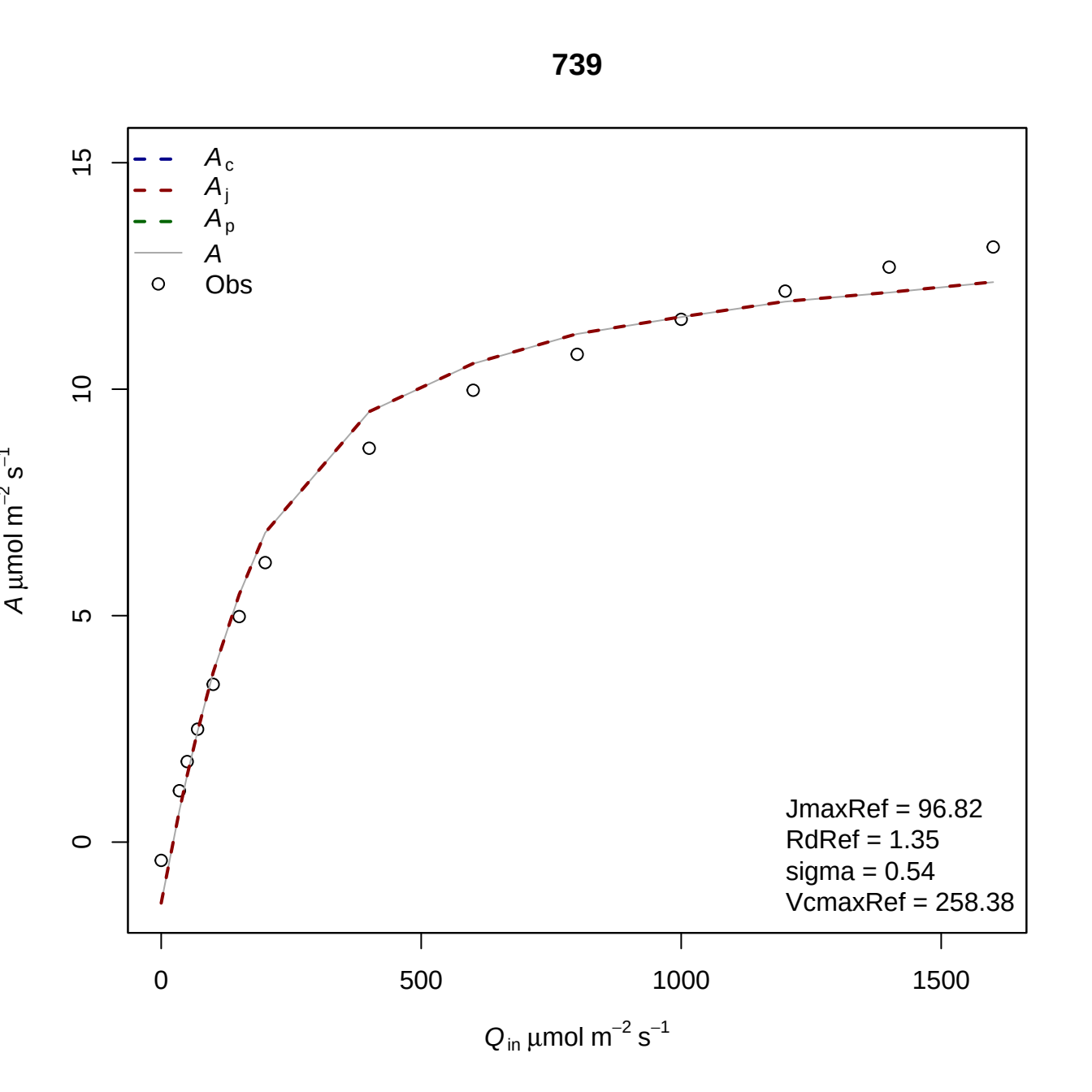

734
A_c
A_j
A_p
A_n
Obs
JmaxRef = 76.27
RdRef = 2.06
sigma = 0.79
VcmaxRef = 61.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

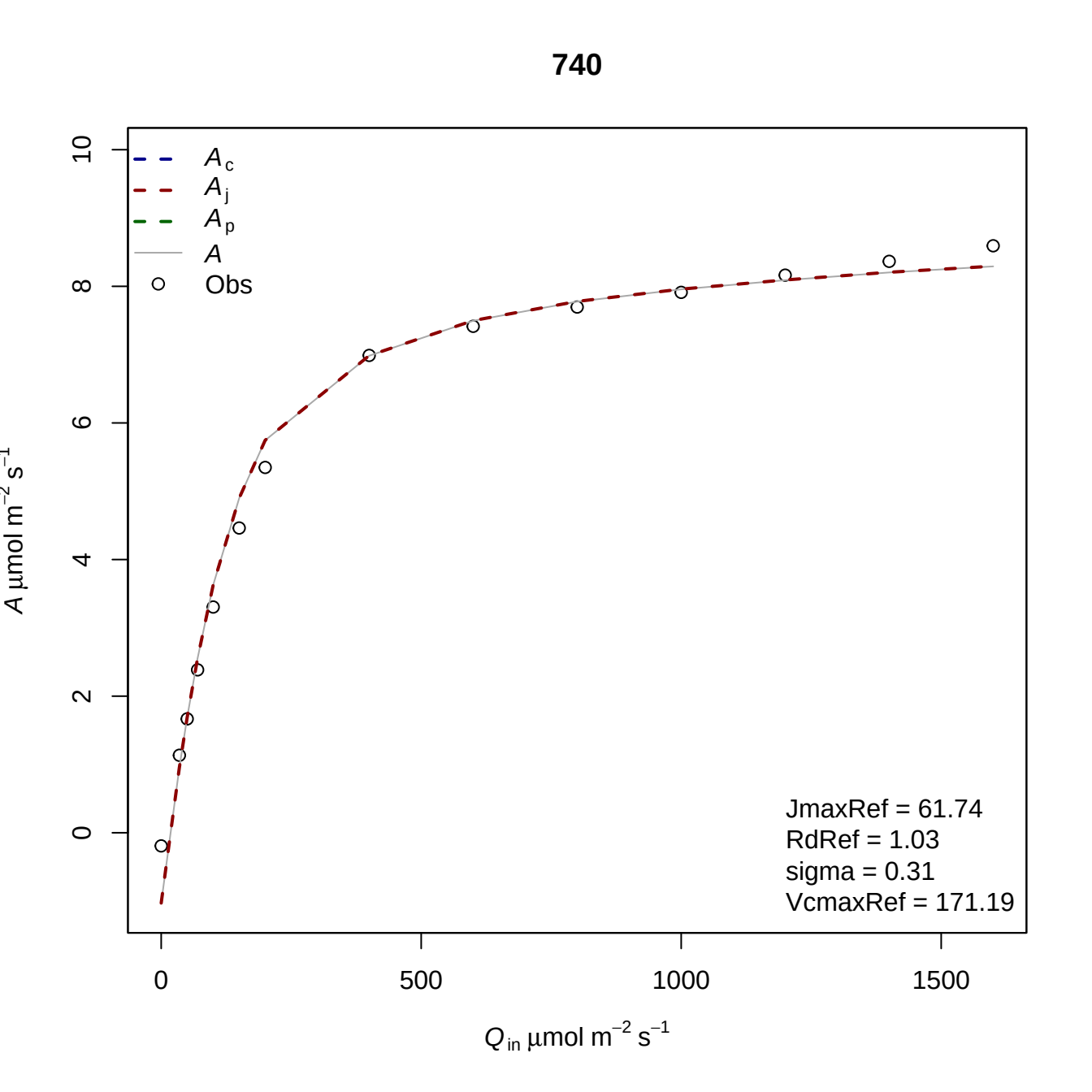

735
A_c
A_j
A_p
A
Obs
JmaxRef = 67.99
RdRef = 2.34
sigma = 0.54
VcmaxRef = 420.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

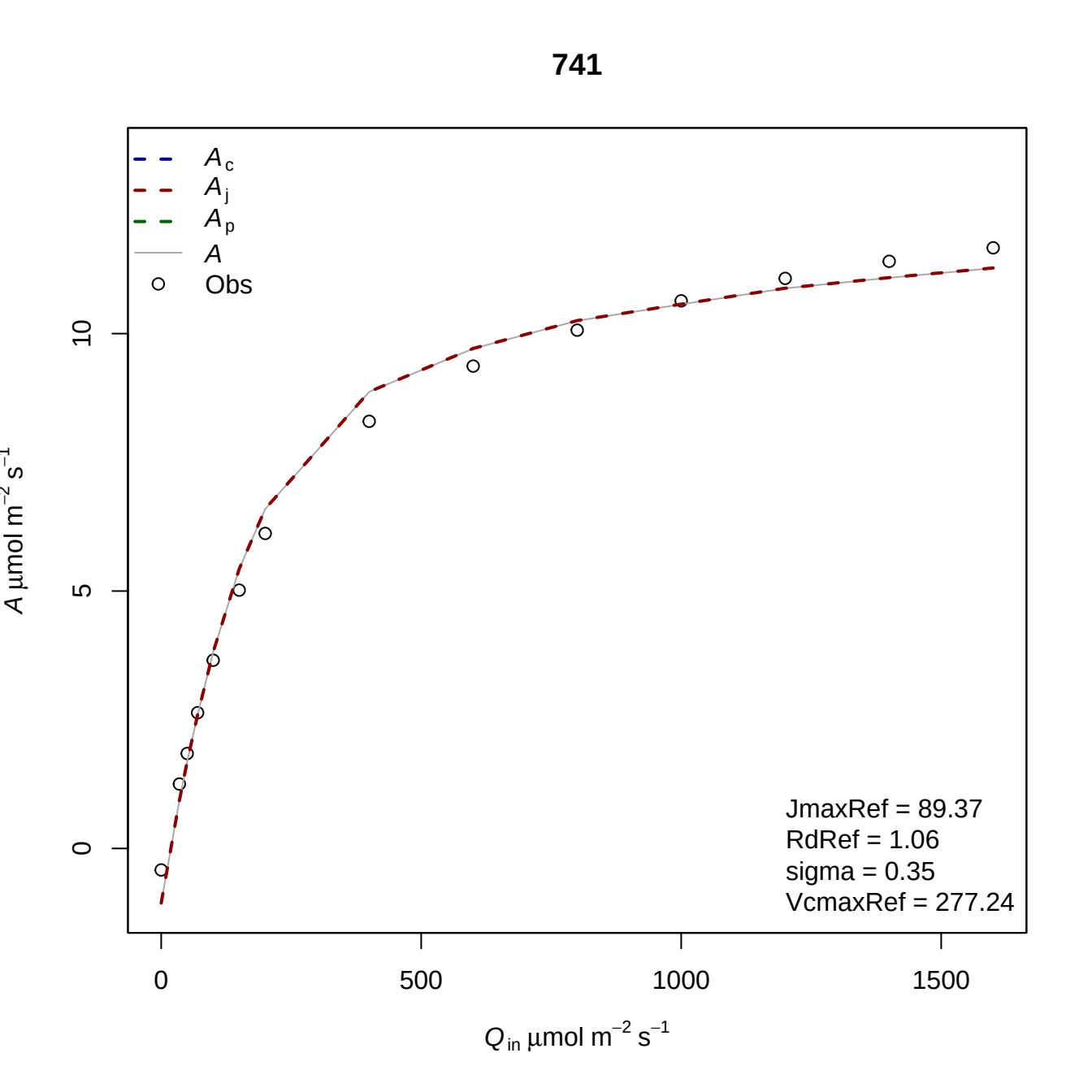

736
A_c
A_j
A_p
A
Obs
JmaxRef = 59.59
RdRef = 1.17
sigma = 0.38
VcmaxRef = 49.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

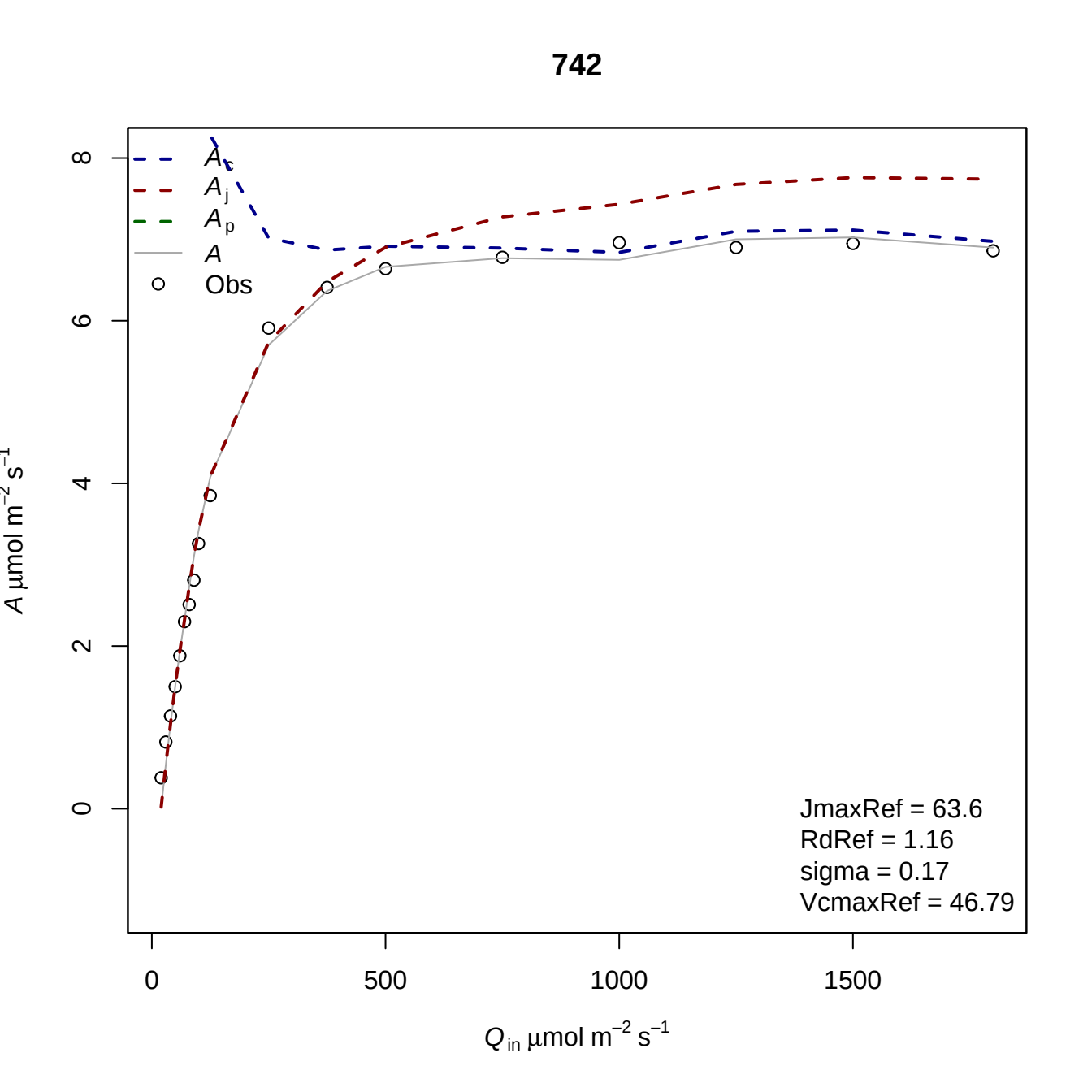

737
A_c
A_j
A_p
A
Obs
JmaxRef = 53.14
RdRef = 1.71
sigma = 0.38
VcmaxRef = 57404.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

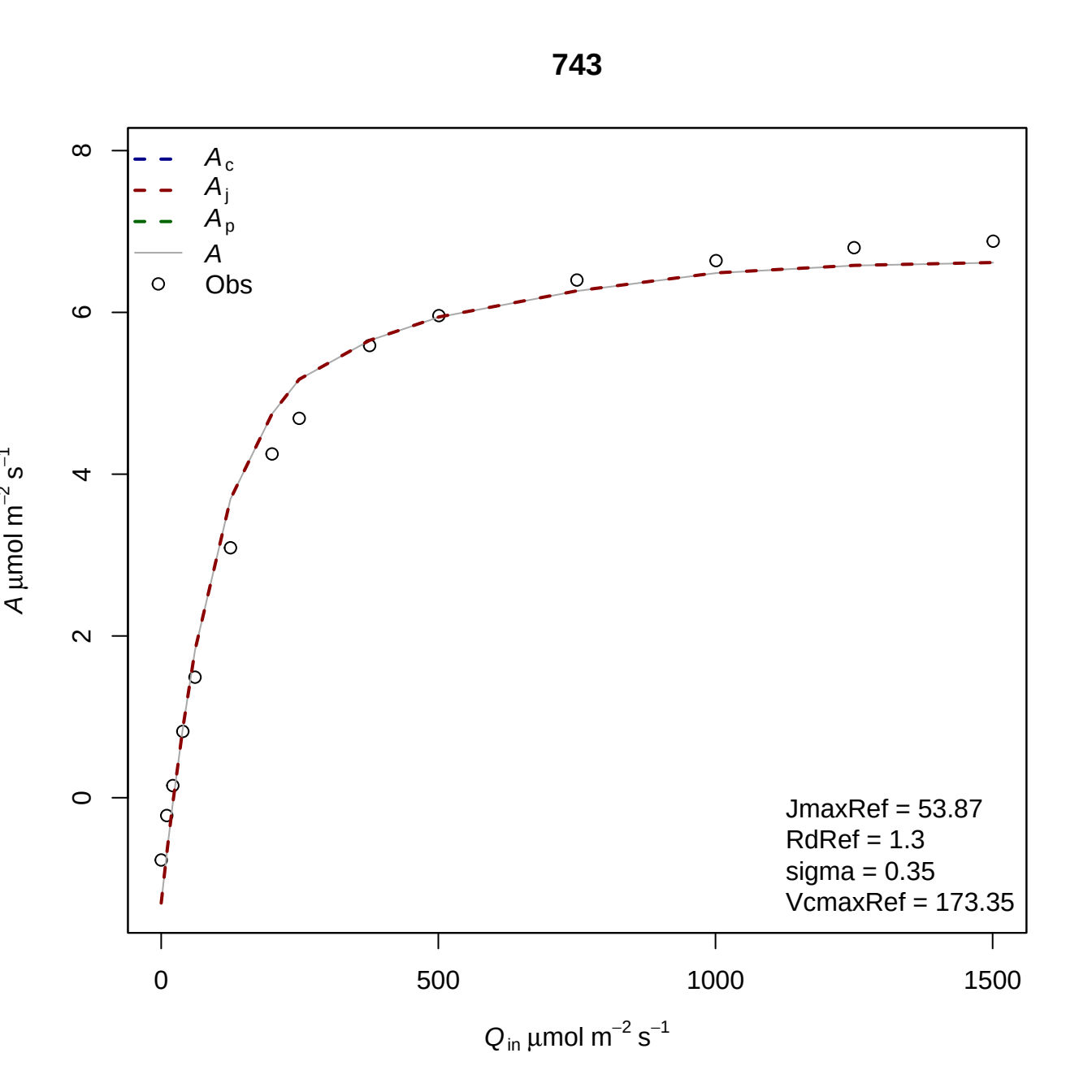

738
A_c
A_j
A_p
A
Obs
JmaxRef = 45.91
RdRef = 1.22
sigma = 0.14
VcmaxRef = 164.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

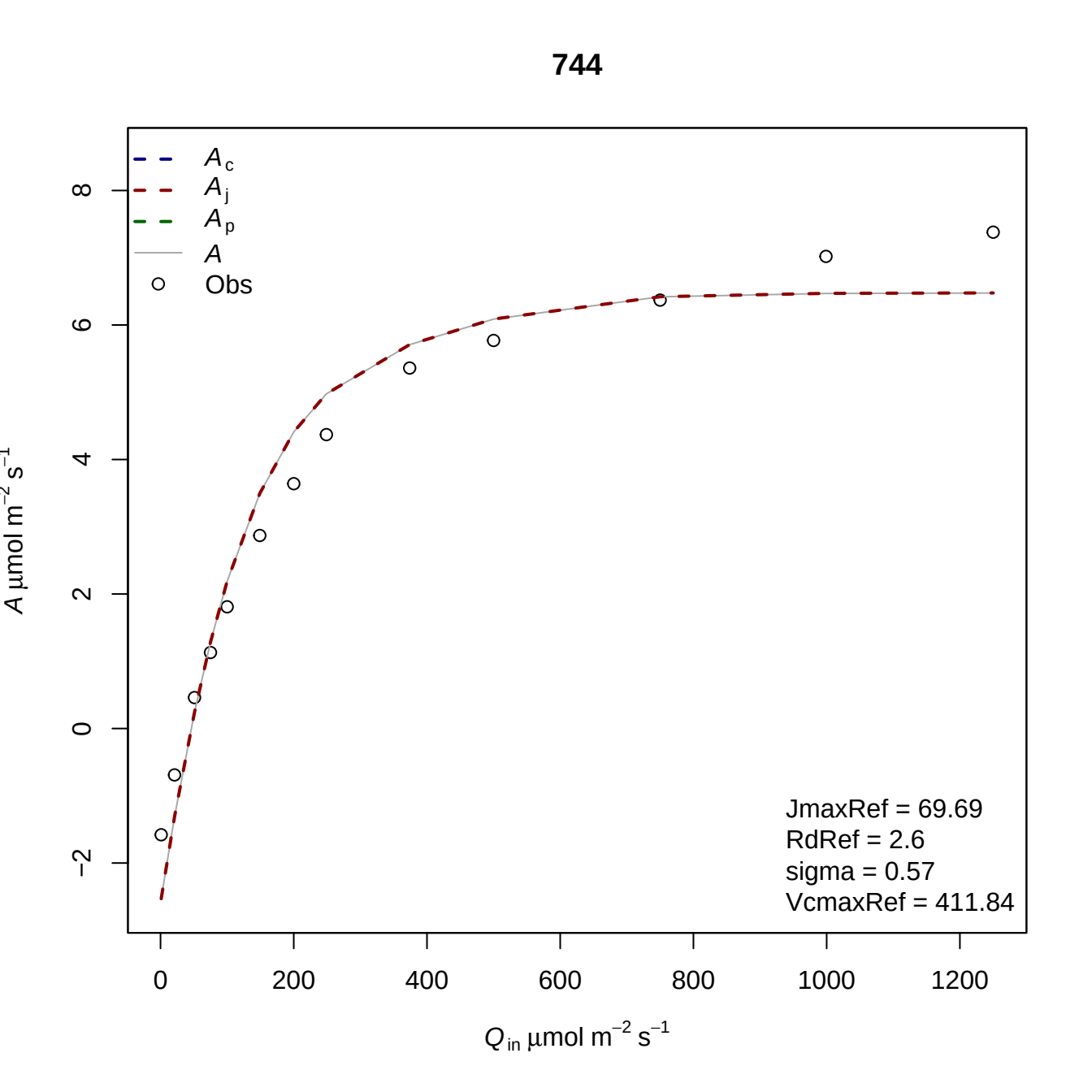

739
A_c
A_j
A_p
A
Obs
JmaxRef = 96.82
RdRef = 1.35
sigma = 0.54
VcmaxRef = 258.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

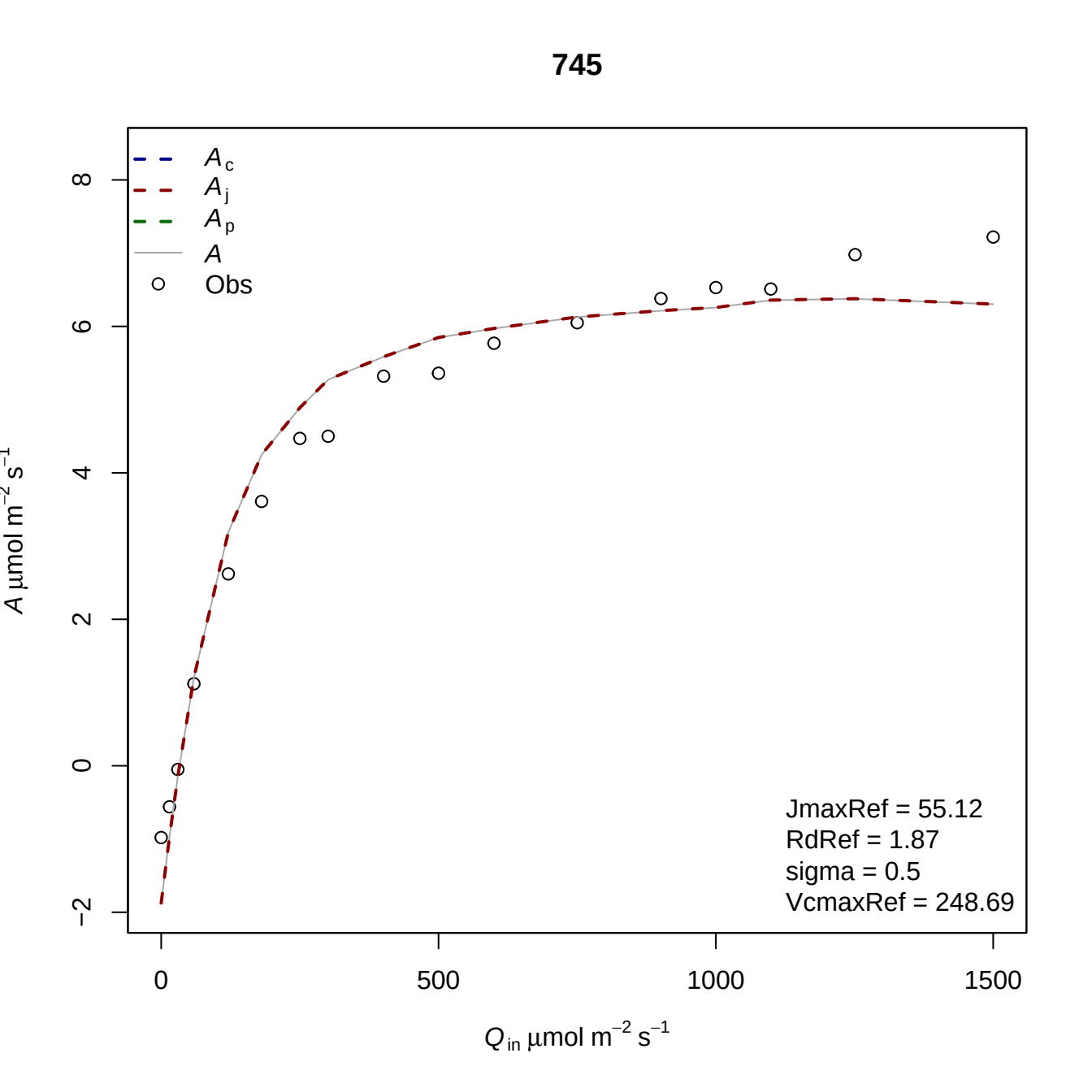

740
A_c
A_j
A_p
A
Obs
JmaxRef = 61.74
RdRef = 1.03
sigma = 0.31
VcmaxRef = 171.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

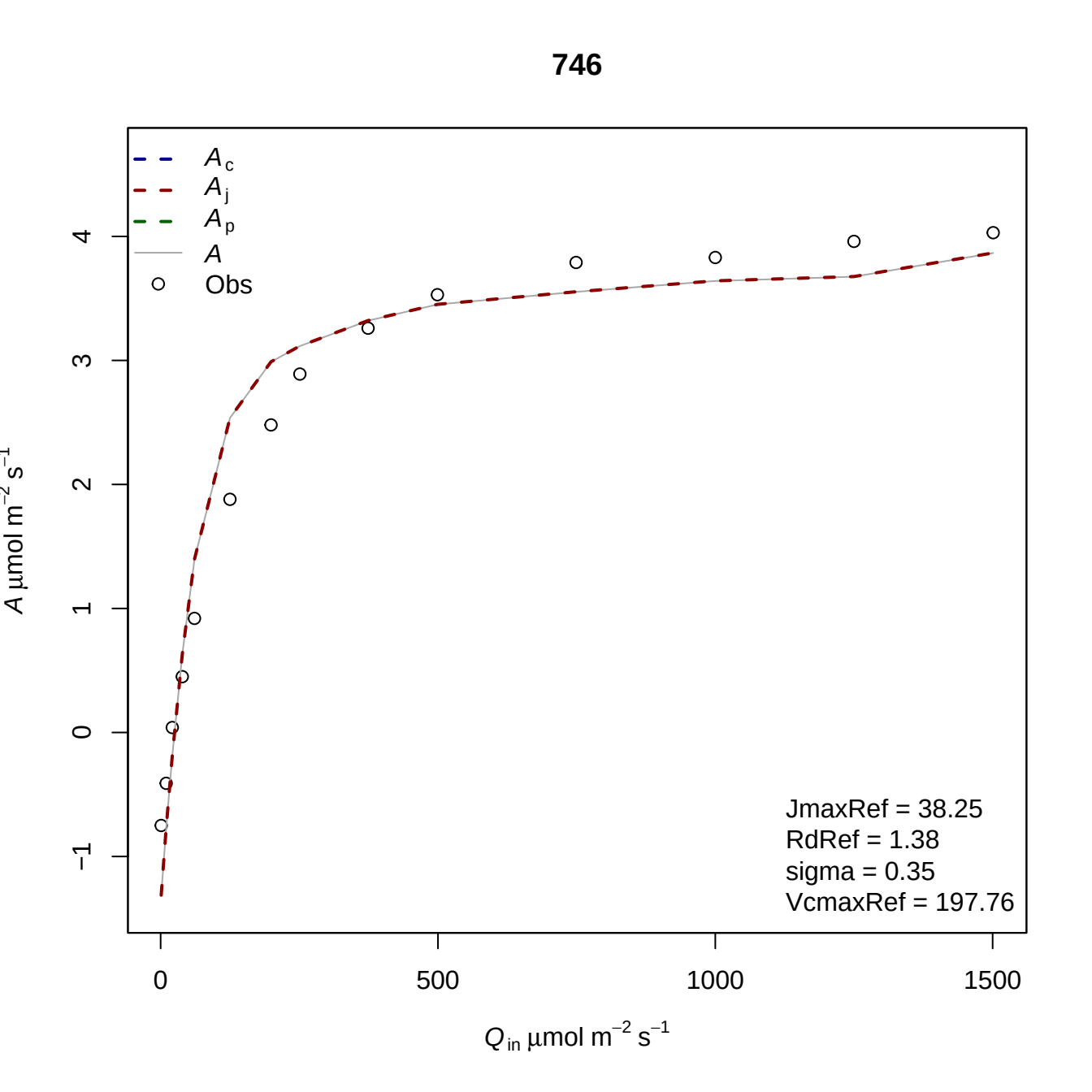

741
A_c
A_j
A_p
A
Obs
JmaxRef = 89.37
RdRef = 1.06
sigma = 0.35
VcmaxRef = 277.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

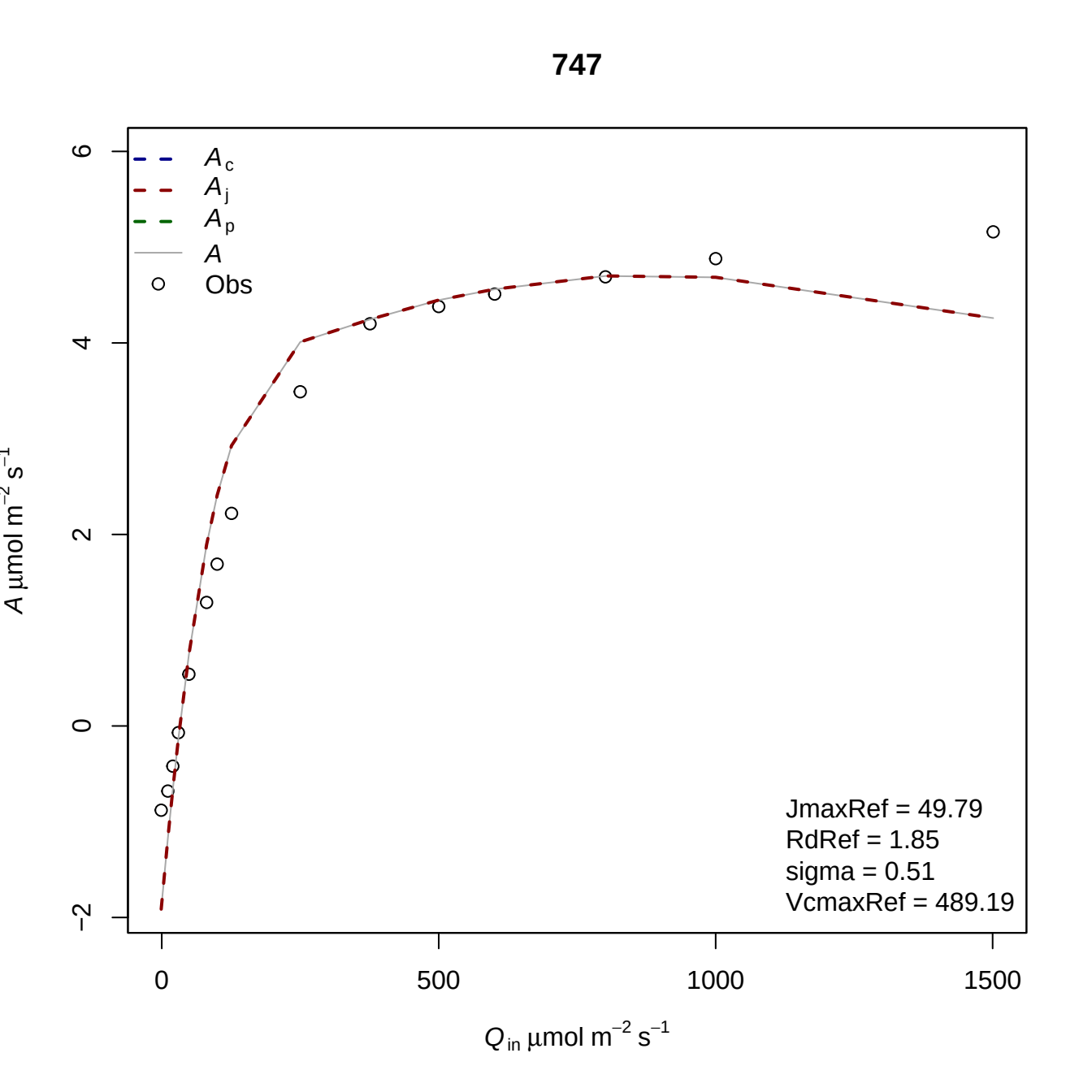

742
A_c
A_j
A_p
A
Obs
JmaxRef = 63.6
RdRef = 1.16
sigma = 0.17
VcmaxRef = 46.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

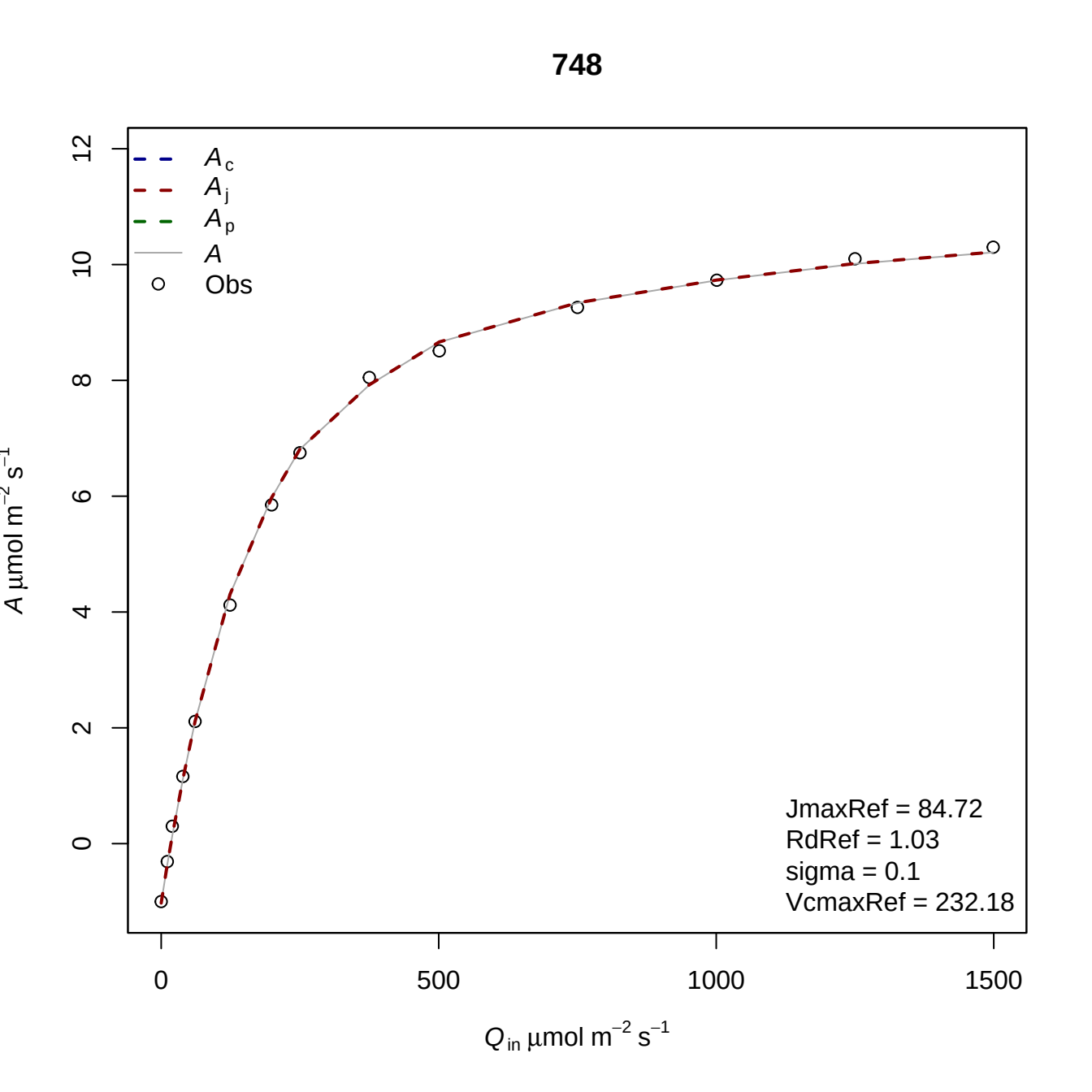

743
A_c
A_j
A_p
A
Obs
JmaxRef = 53.87
RdRef = 1.3
sigma = 0.35
VcmaxRef = 173.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

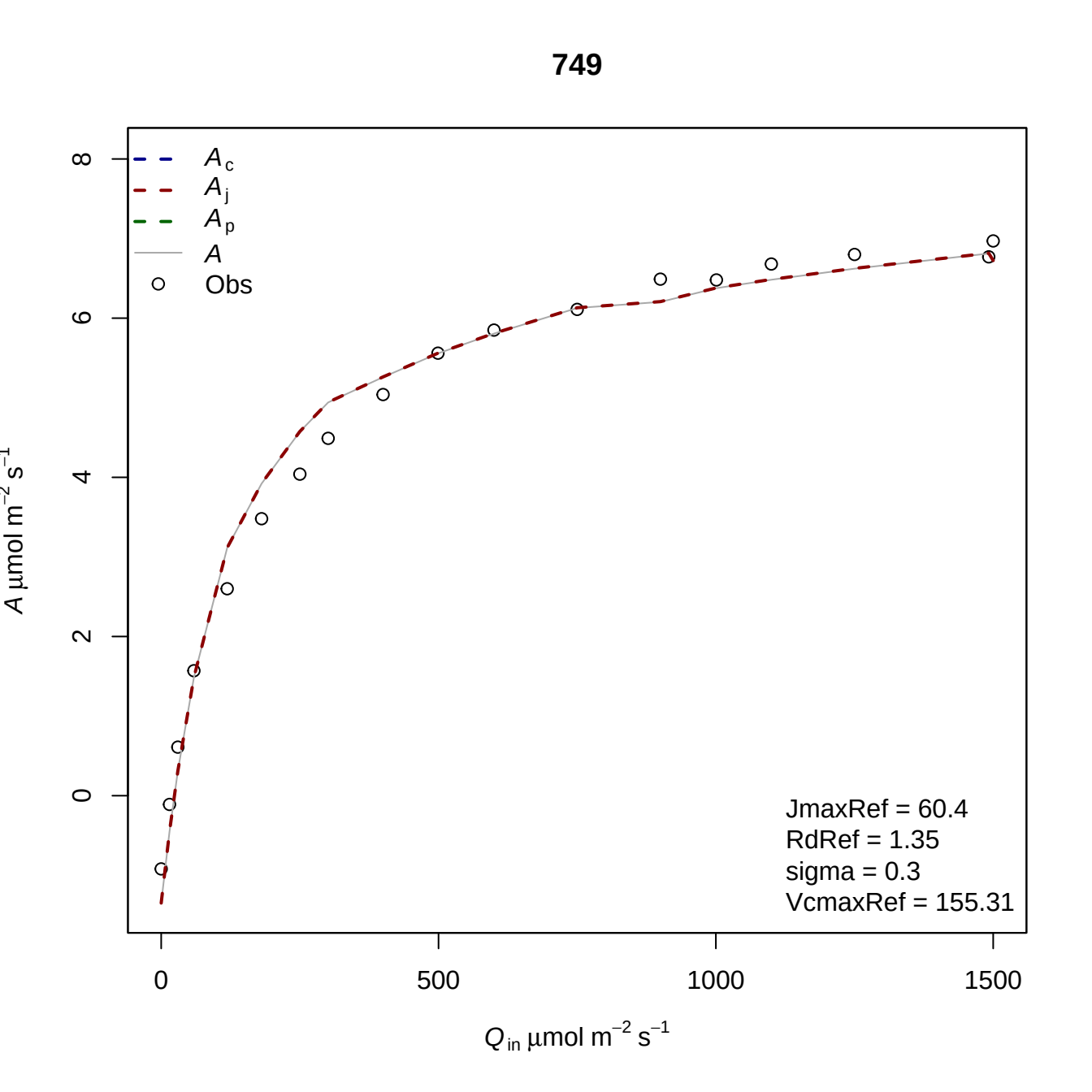

744
A_c
A_j
A_p
A
Obs
JmaxRef = 69.69
RdRef = 2.6
sigma = 0.57
VcmaxRef = 411.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

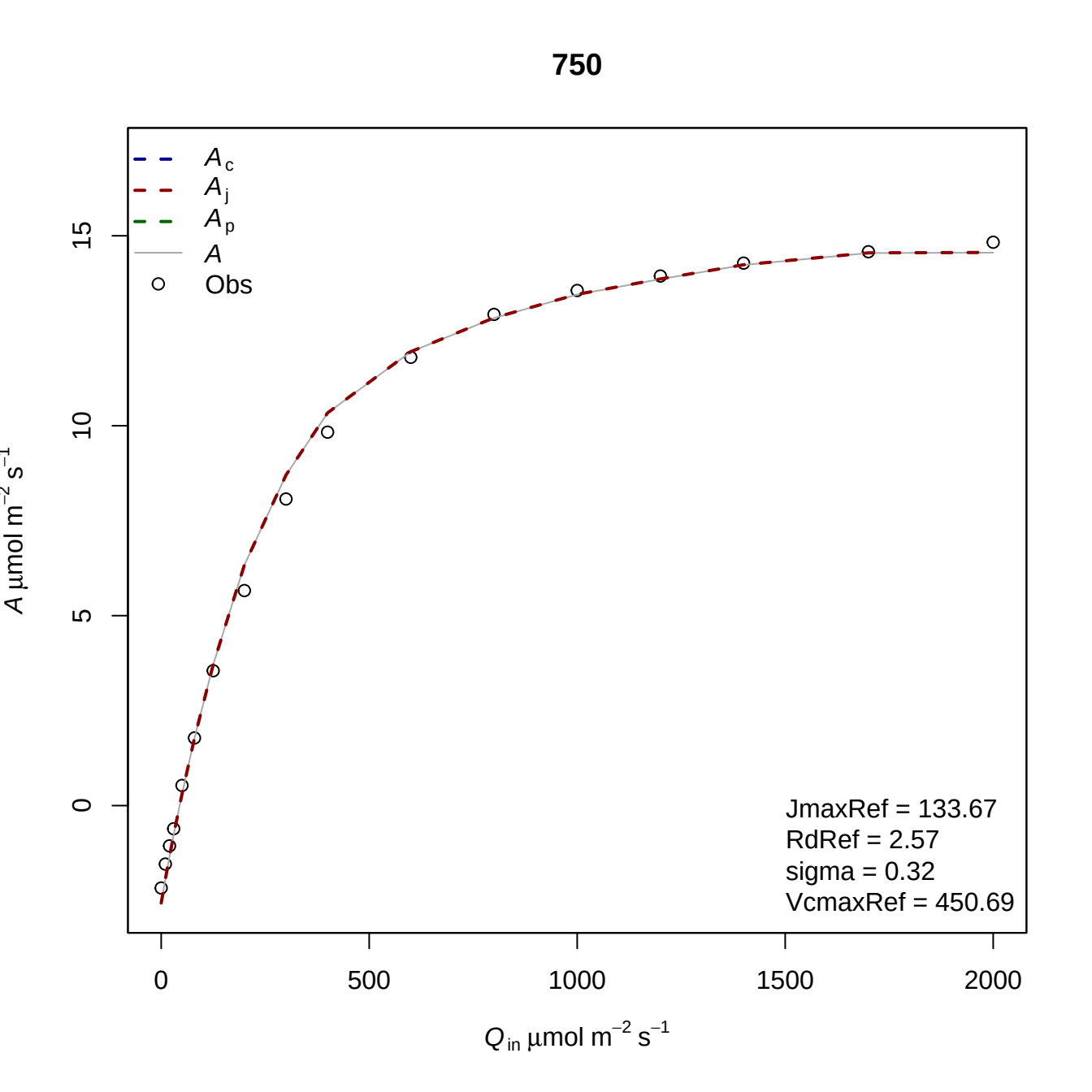

745
A_c
A_j
A_p
A
Obs
JmaxRef = 55.12
RdRef = 1.87
sigma = 0.5
VcmaxRef = 248.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

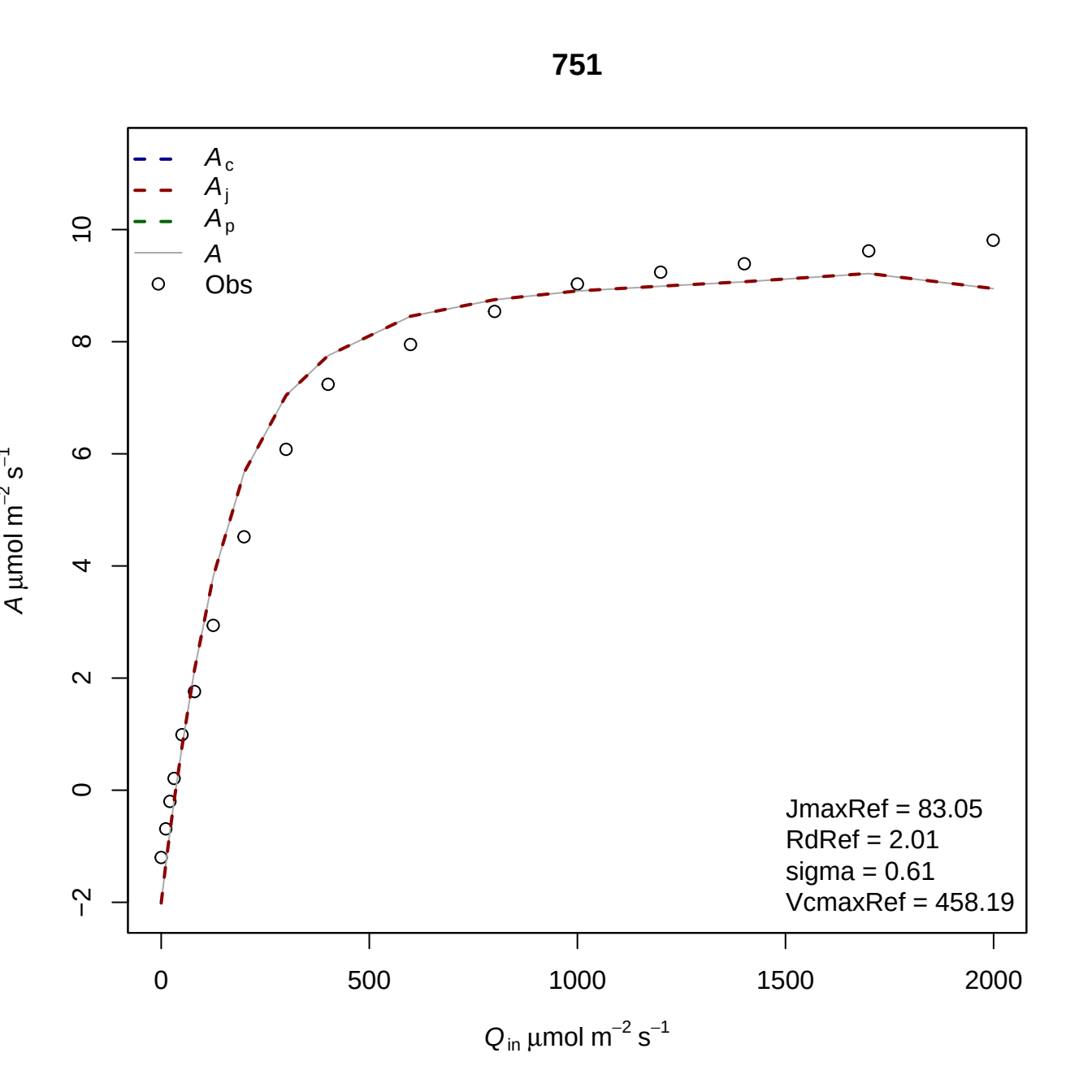

746
A_c
A_j
A_p
A
Obs
JmaxRef = 38.25
RdRef = 1.38
sigma = 0.35
VcmaxRef = 197.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

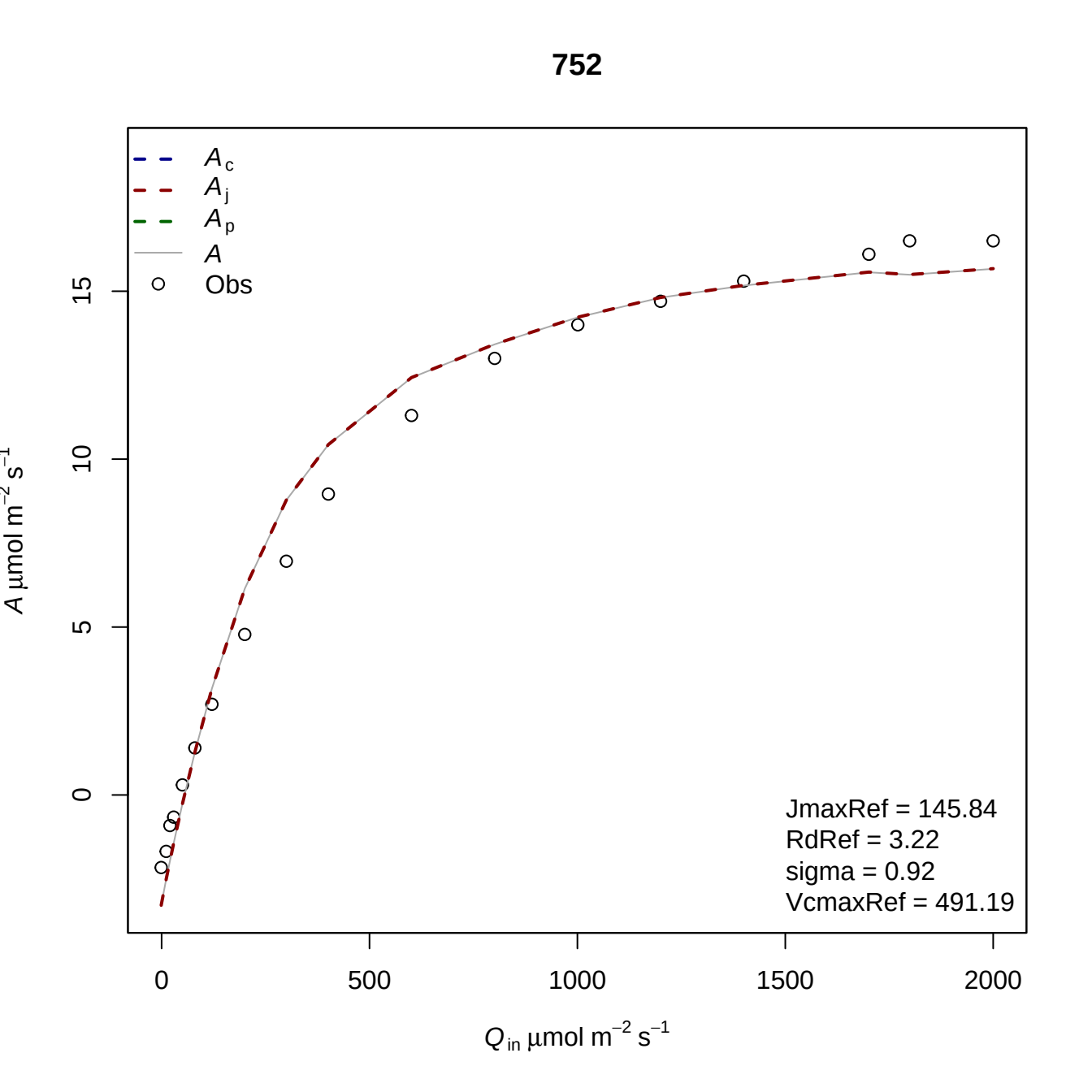

747
A_c
A_j
A_p
A
Obs
JmaxRef = 49.79
RdRef = 1.85
sigma = 0.51
VcmaxRef = 489.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

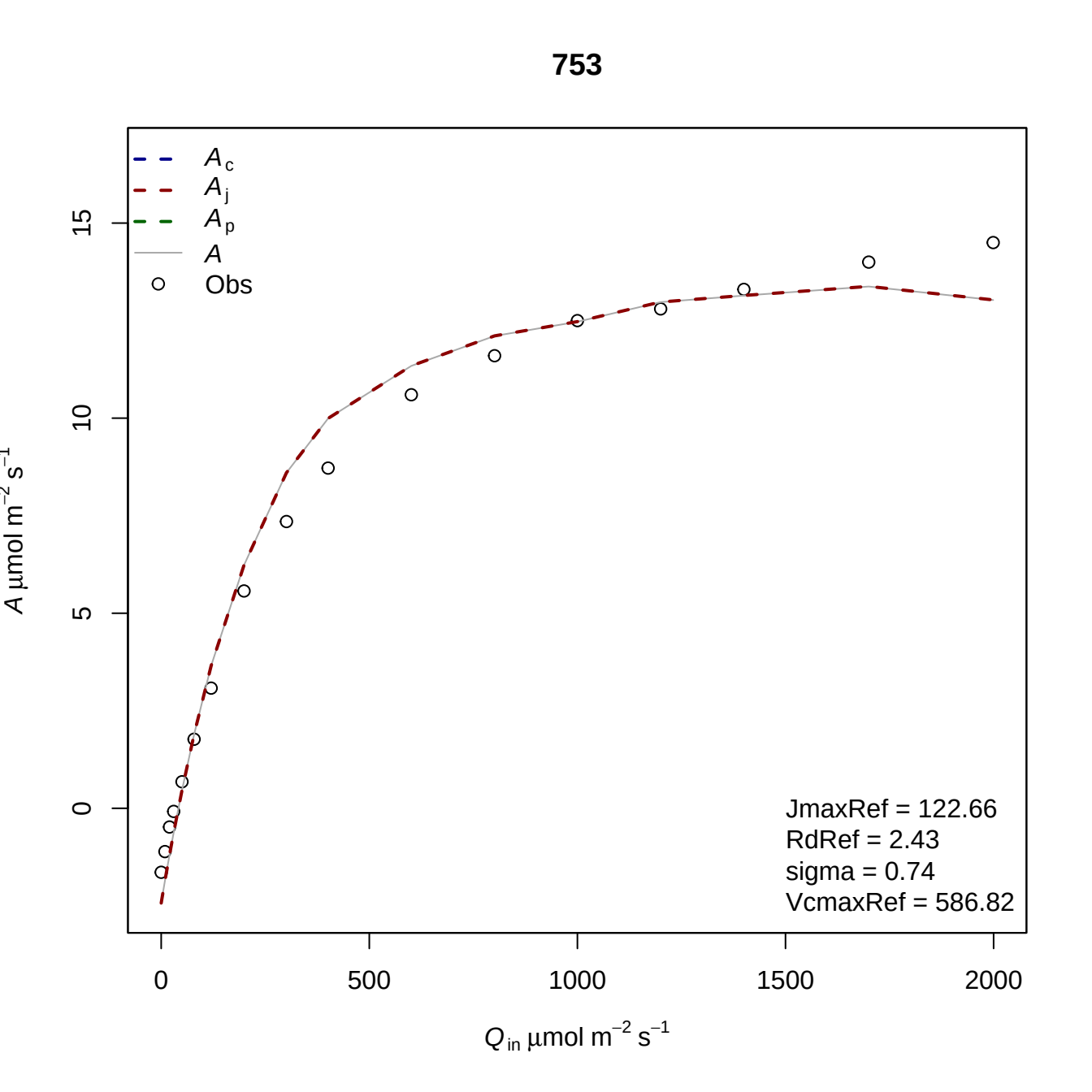

748
A_c
A_j
A_p
A
Obs
JmaxRef = 84.72
RdRef = 1.03
sigma = 0.1
VcmaxRef = 232.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

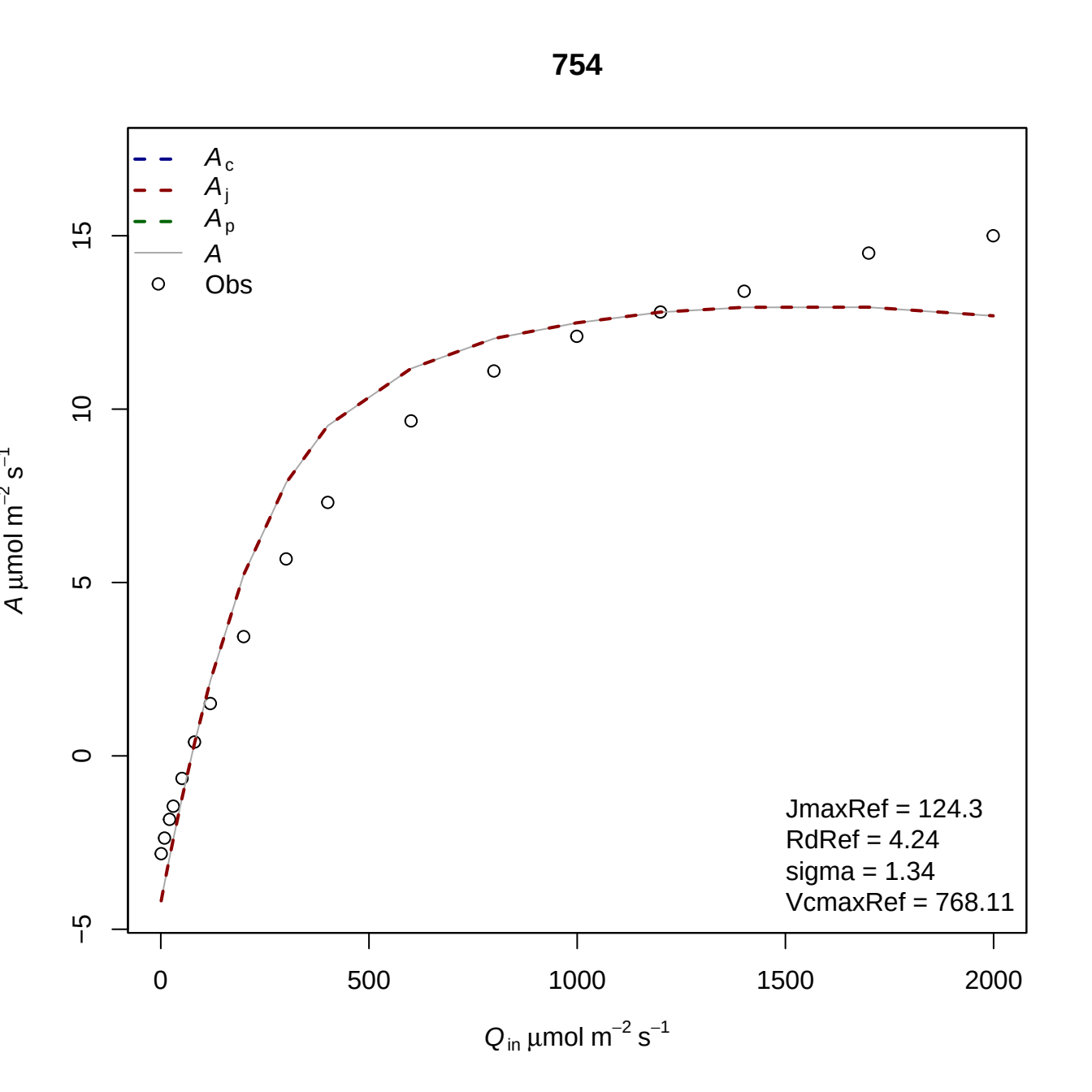

749
A_c
A_j
A_p
A
Obs
JmaxRef = 60.4
RdRef = 1.35
sigma = 0.3
VcmaxRef = 155.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

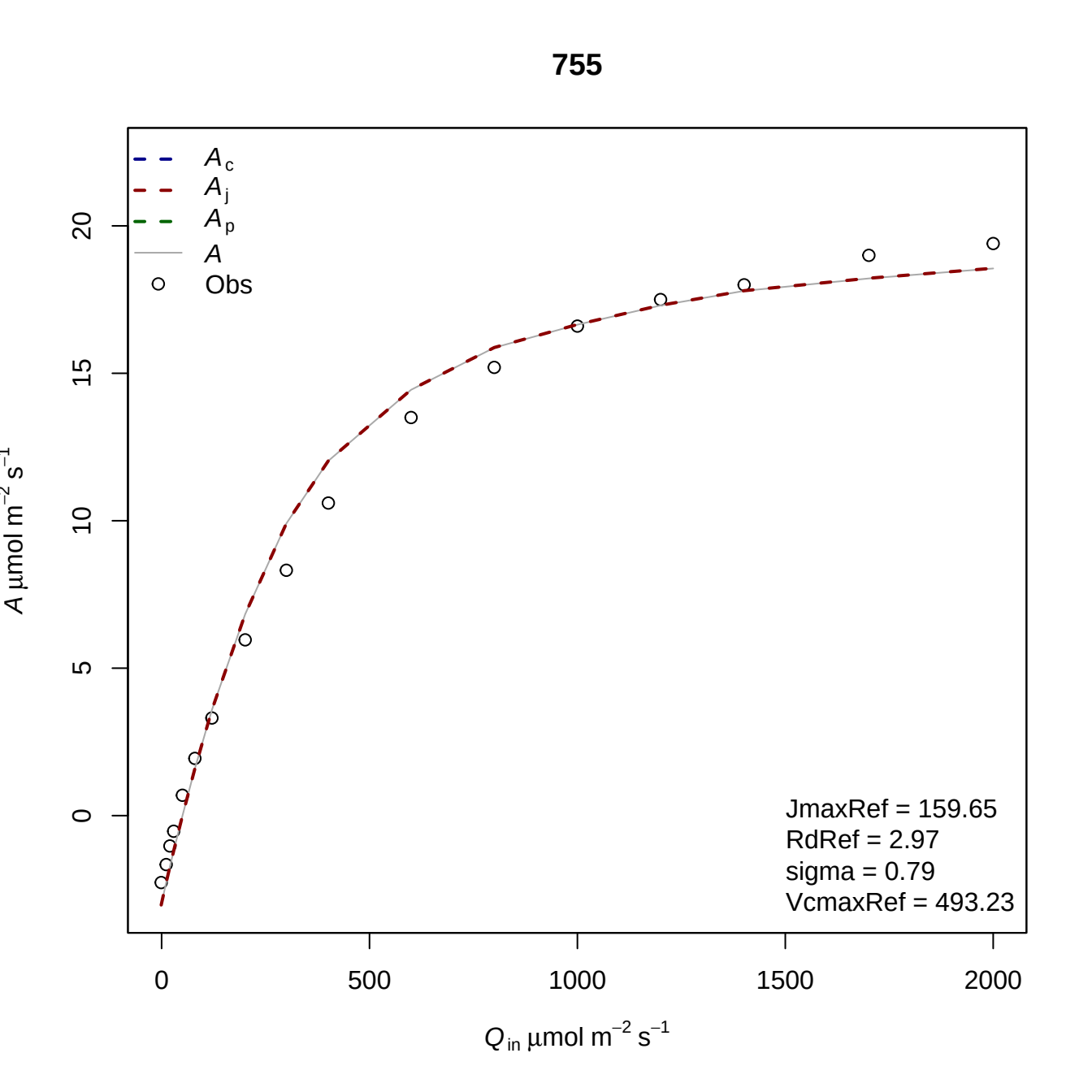

750
A_c
A_j
A_p
A
Obs
JmaxRef = 133.67
RdRef = 2.57
sigma = 0.32
VcmaxRef = 450.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

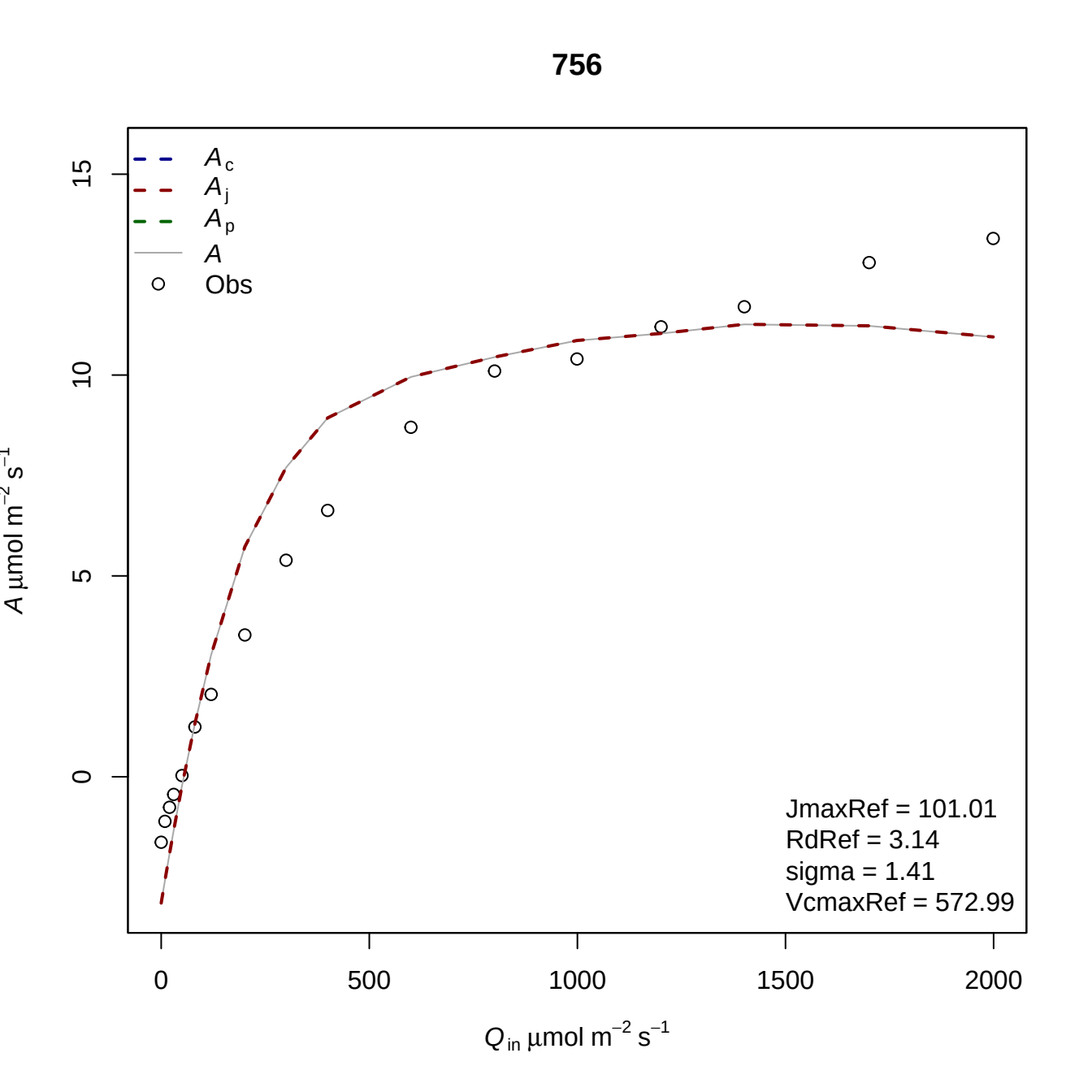

751
A_c
A_j
A_p
A
Obs
JmaxRef = 83.05
RdRef = 2.01
sigma = 0.61
VcmaxRef = 458.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

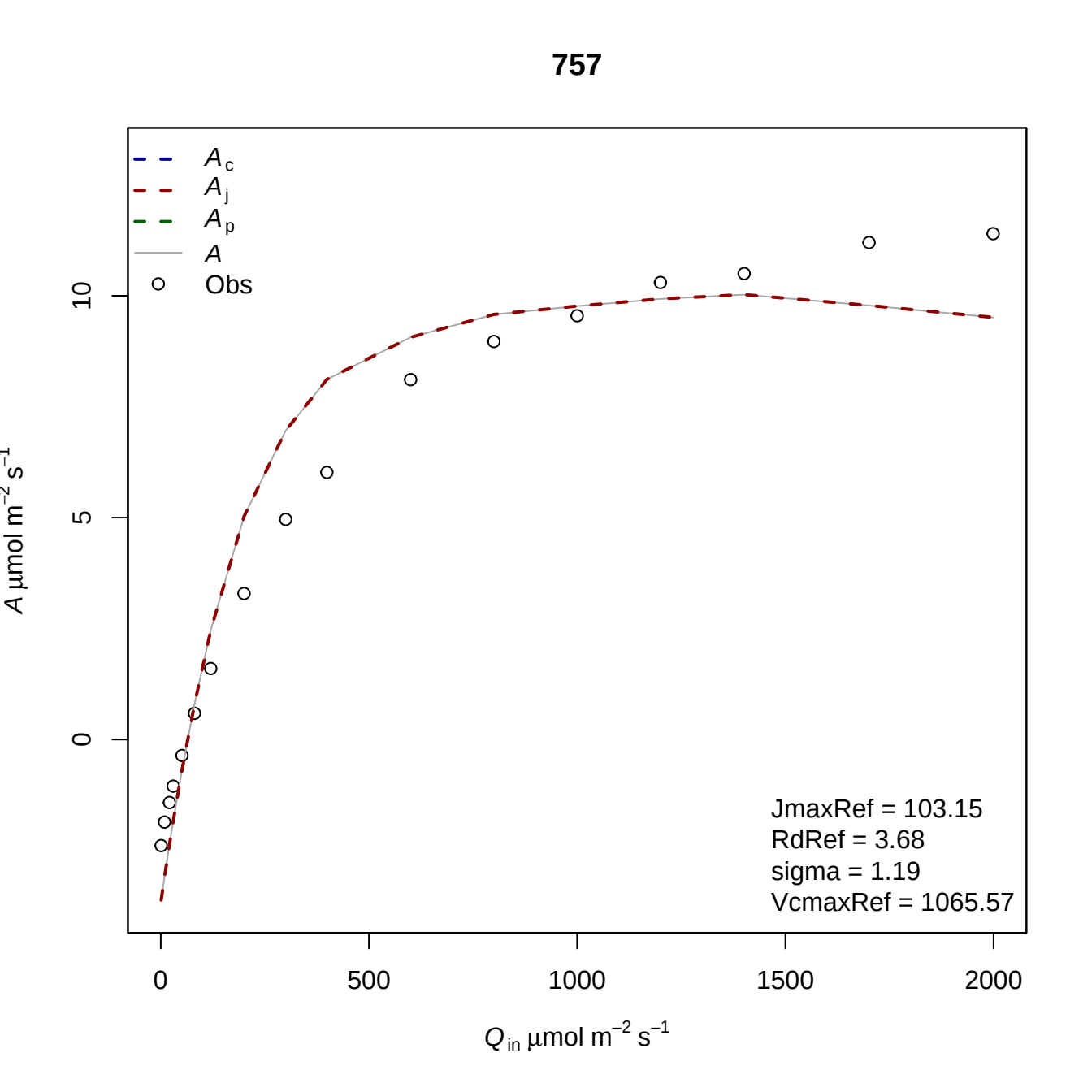

752
A_c
A_j
A_p
A
Obs
JmaxRef = 145.84
RdRef = 3.22
sigma = 0.92
VcmaxRef = 491.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

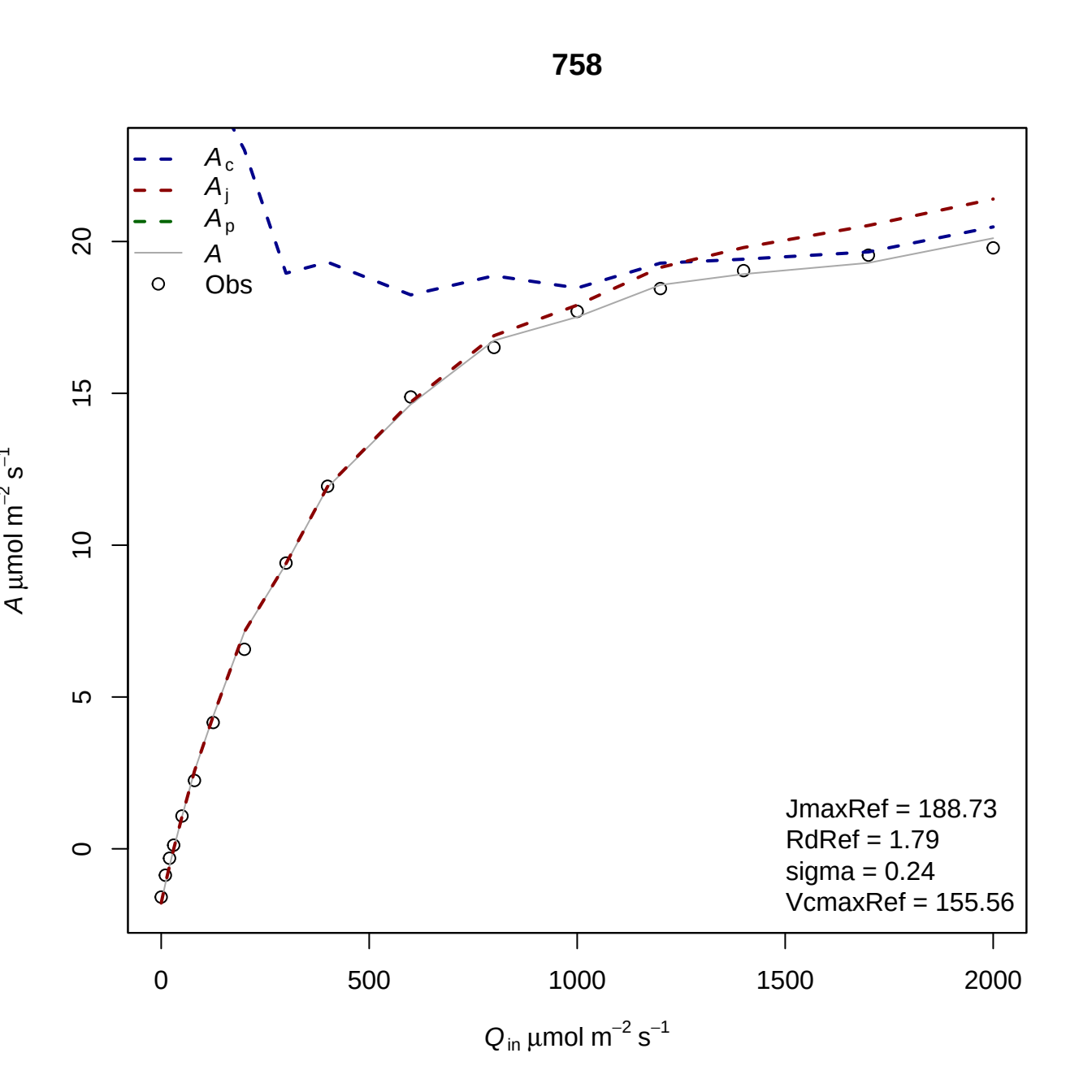

753
A_c
A_j
A_p
A
Obs
JmaxRef = 122.66
RdRef = 2.43
sigma = 0.74
VcmaxRef = 586.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

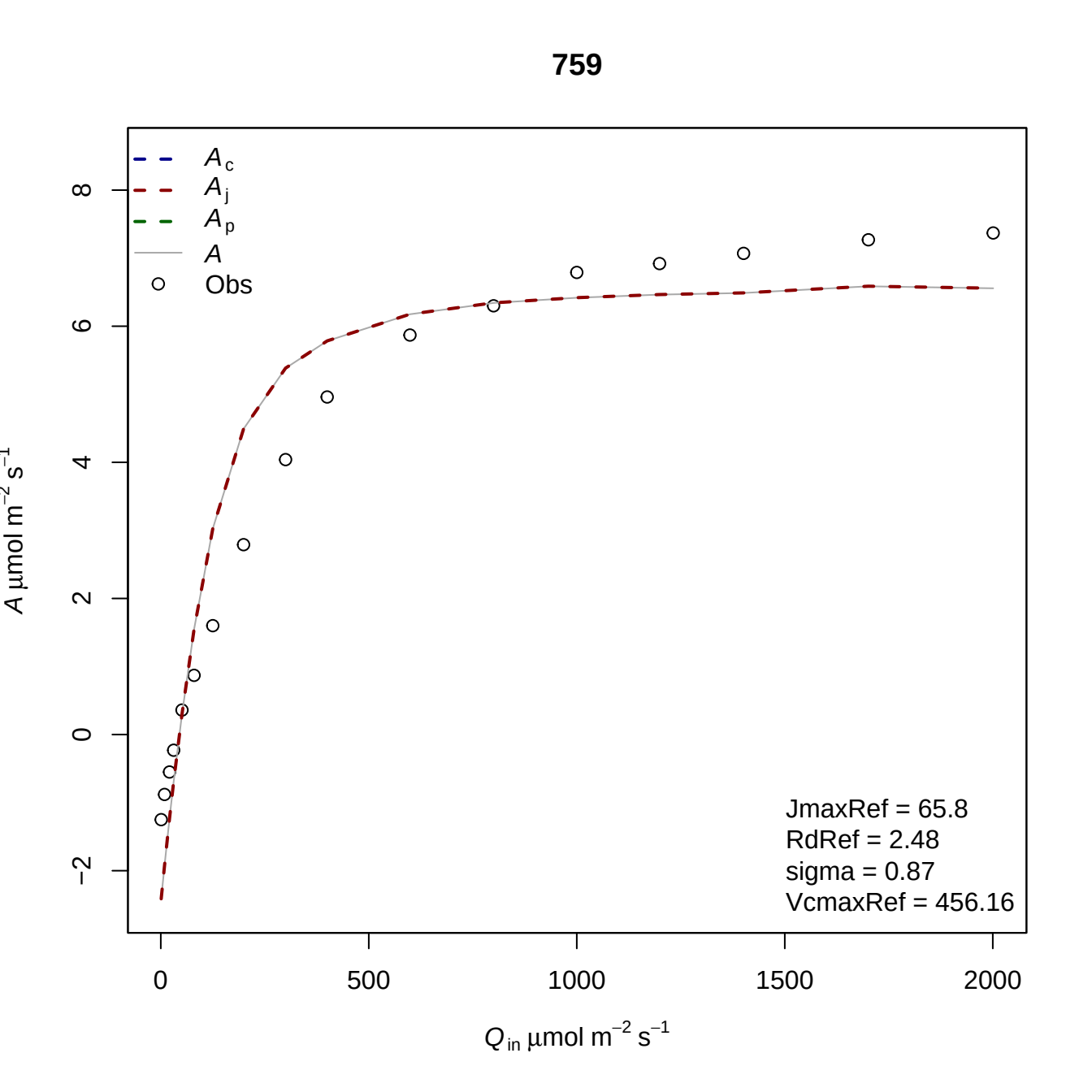

754
A_c
A_j
A_p
A
Obs
JmaxRef = 124.3
RdRef = 4.24
sigma = 1.34
VcmaxRef = 768.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

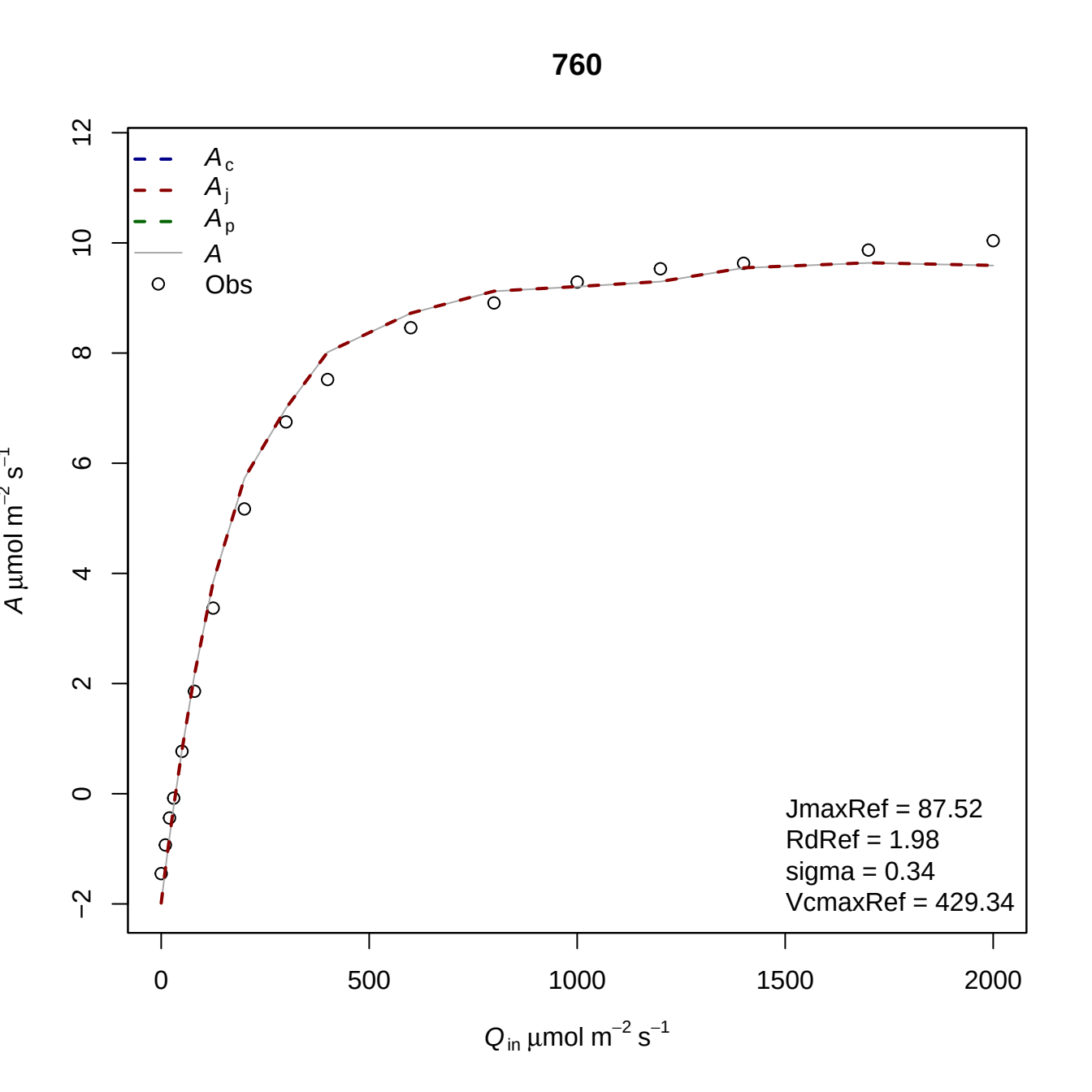

755
A_c
A_j
A_p
A
Obs
JmaxRef = 159.65
RdRef = 2.97
sigma = 0.79
VcmaxRef = 493.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

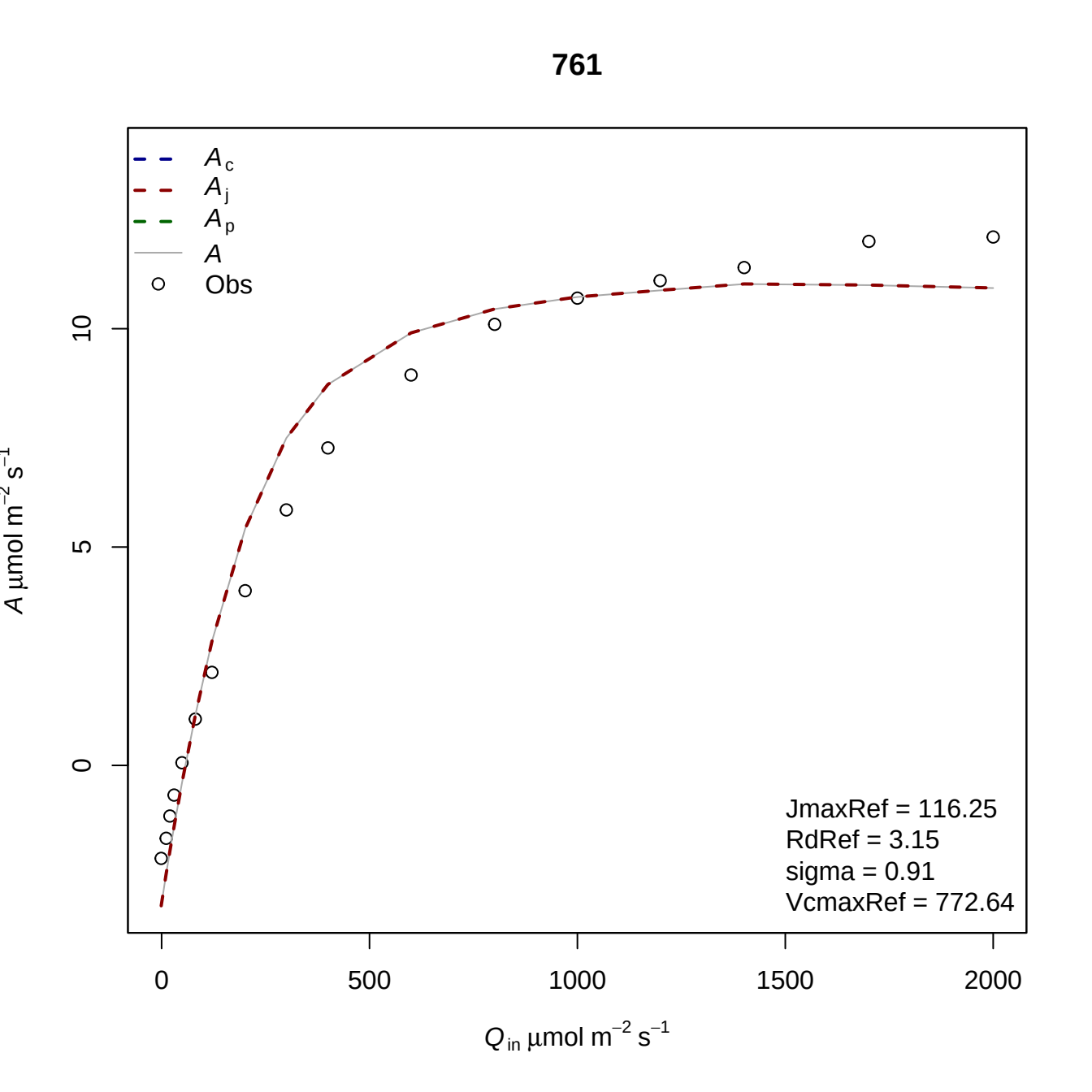

756
A_c
A_j
A_p
A
Obs
JmaxRef = 101.01
RdRef = 3.14
sigma = 1.41
VcmaxRef = 572.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

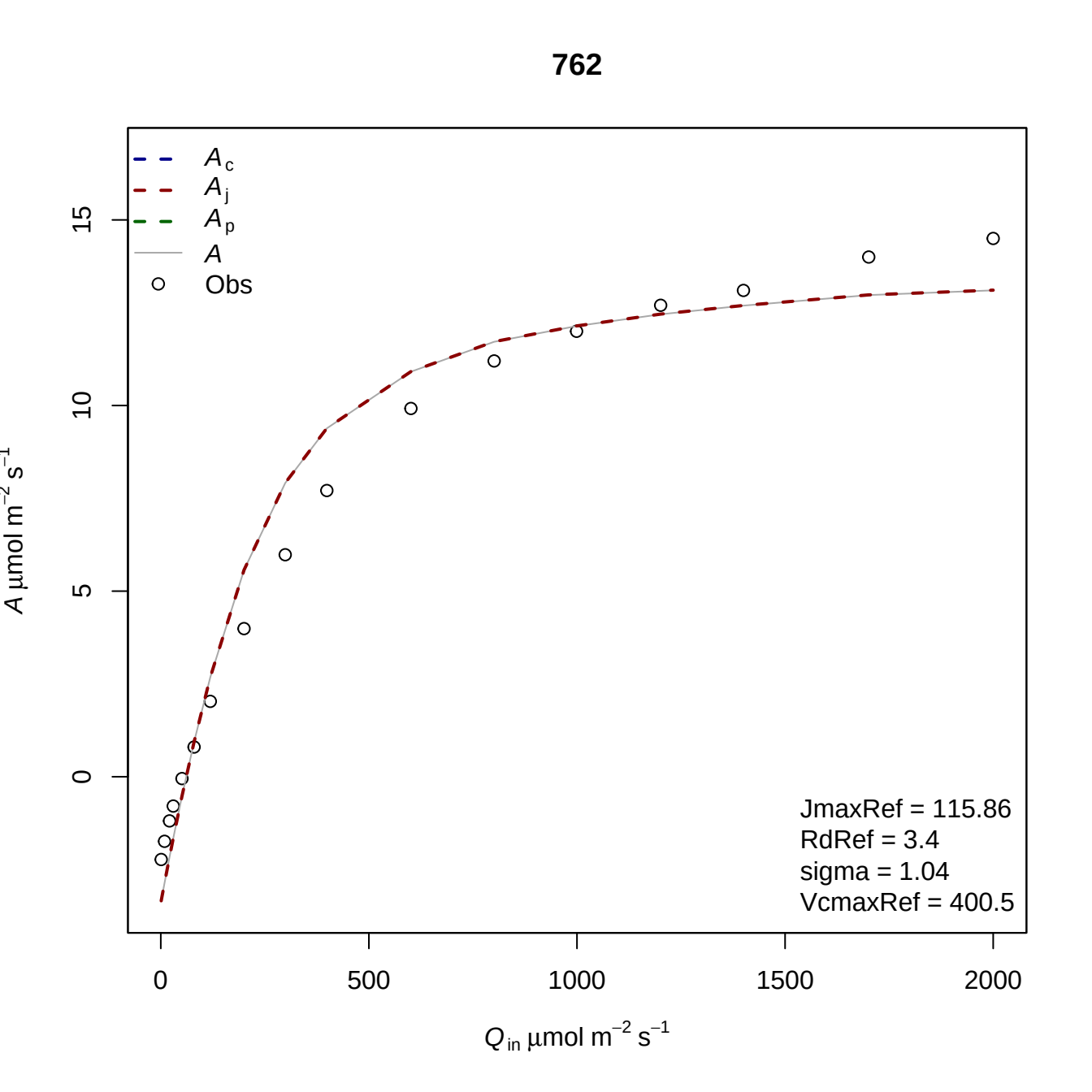

757
A_c
A_j
A_p
A
Obs
JmaxRef = 103.15
RdRef = 3.68
sigma = 1.19
VcmaxRef = 1065.57
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

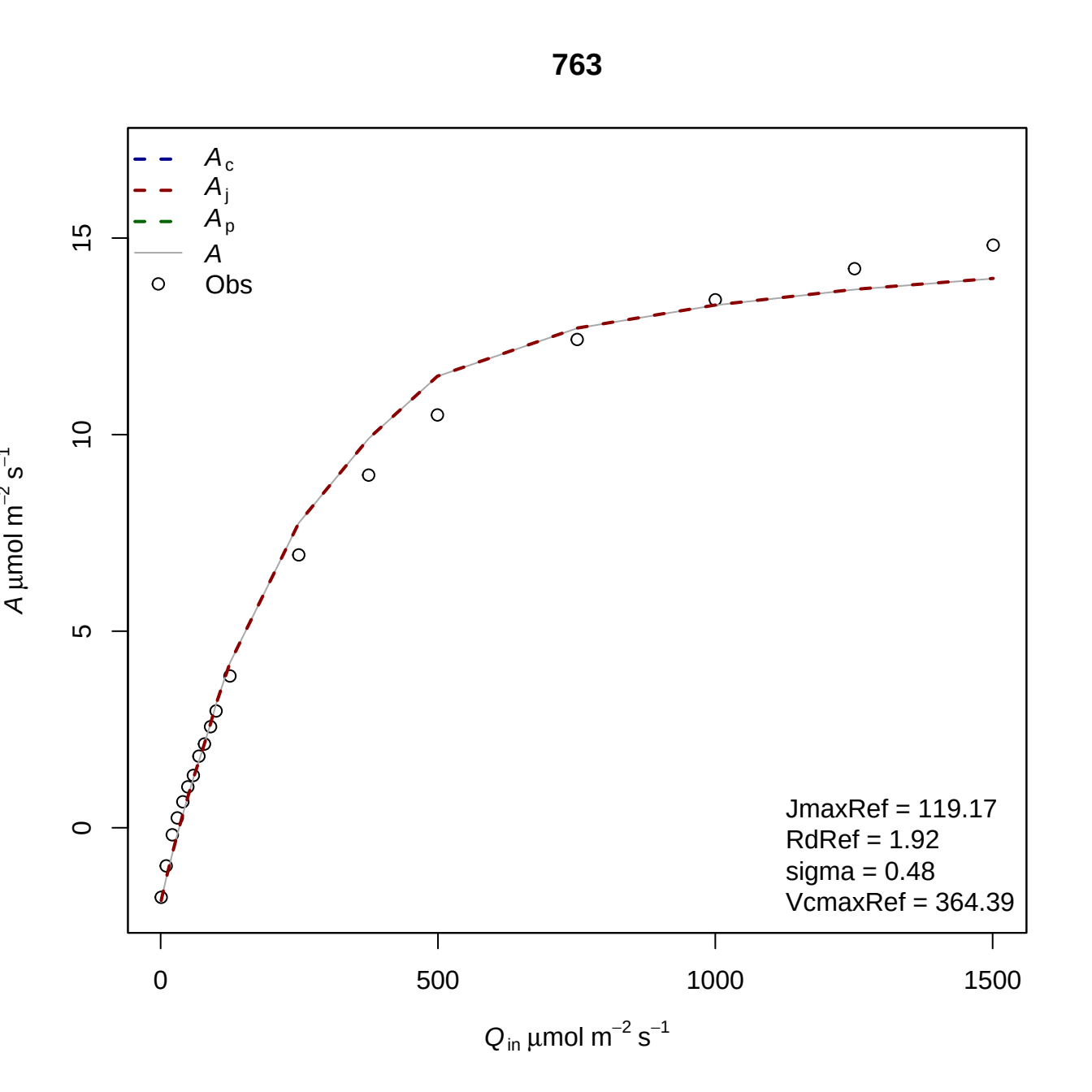

758
A_c
A_j
A_p
A
Obs
JmaxRef = 188.73
RdRef = 1.79
sigma = 0.24
VcmaxRef = 155.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

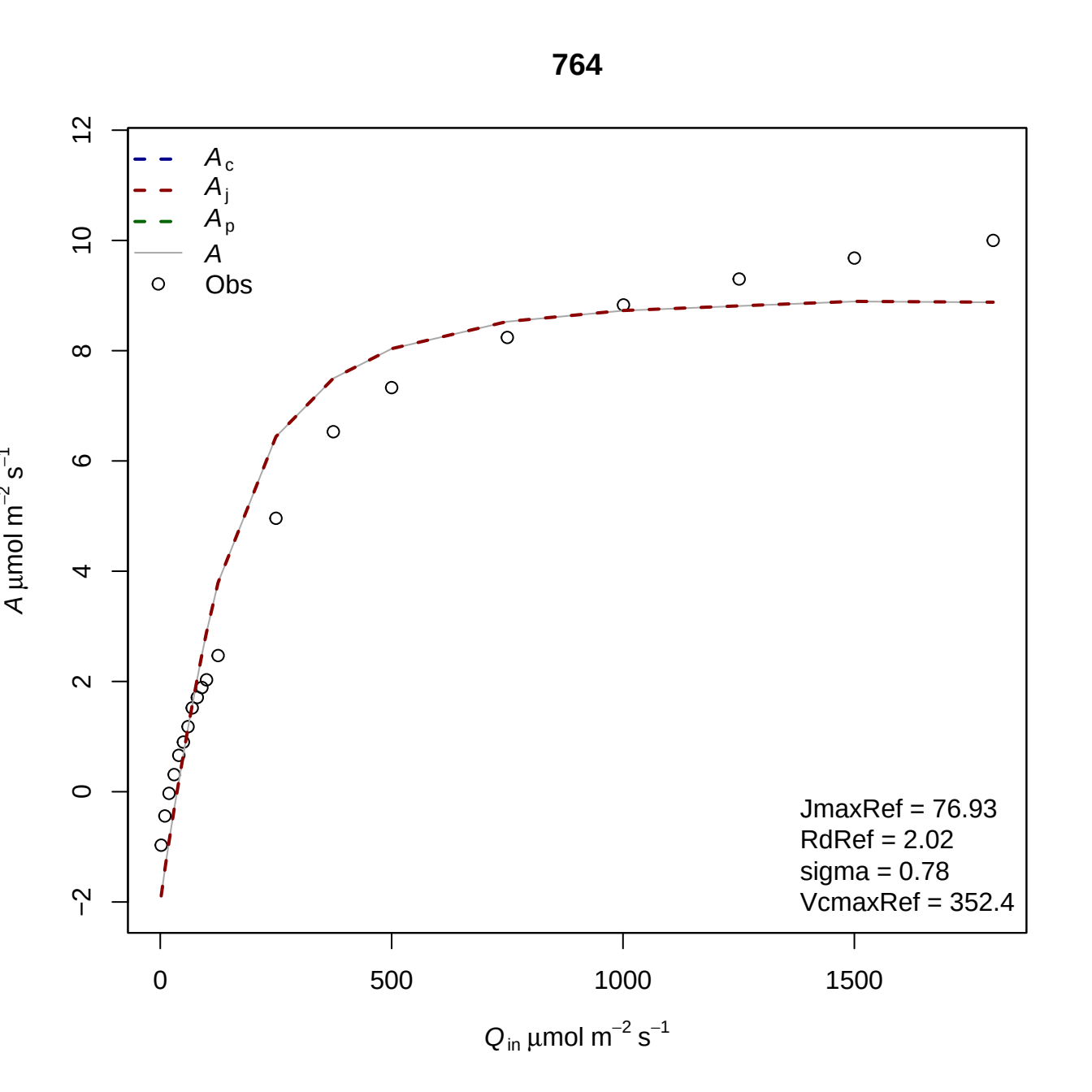

759
A_c
A_j
A_p
A
Obs
JmaxRef = 65.8
RdRef = 2.48
sigma = 0.87
VcmaxRef = 456.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

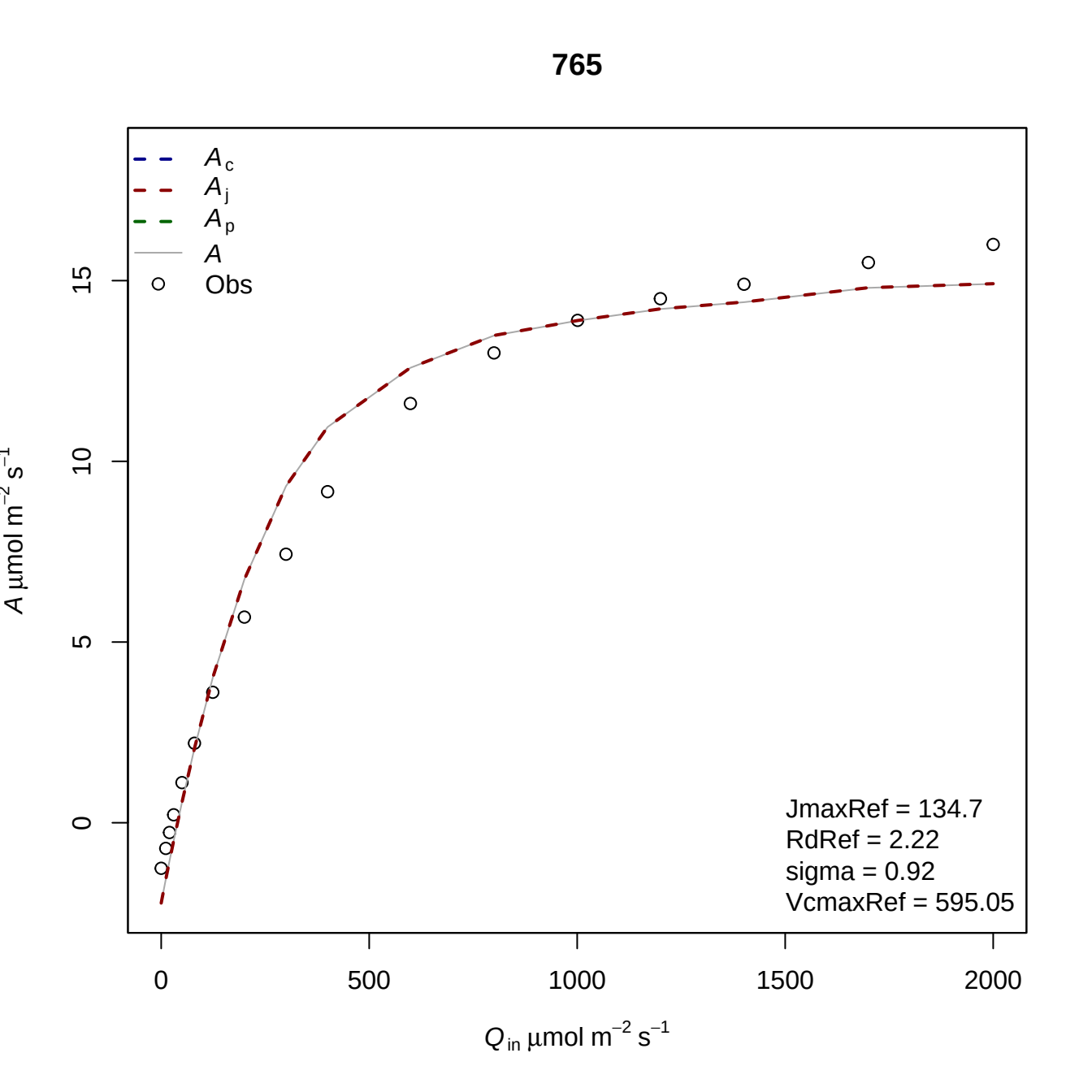

760
A_c
A_j
A_p
A
Obs
JmaxRef = 87.52
RdRef = 1.98
sigma = 0.34
VcmaxRef = 429.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

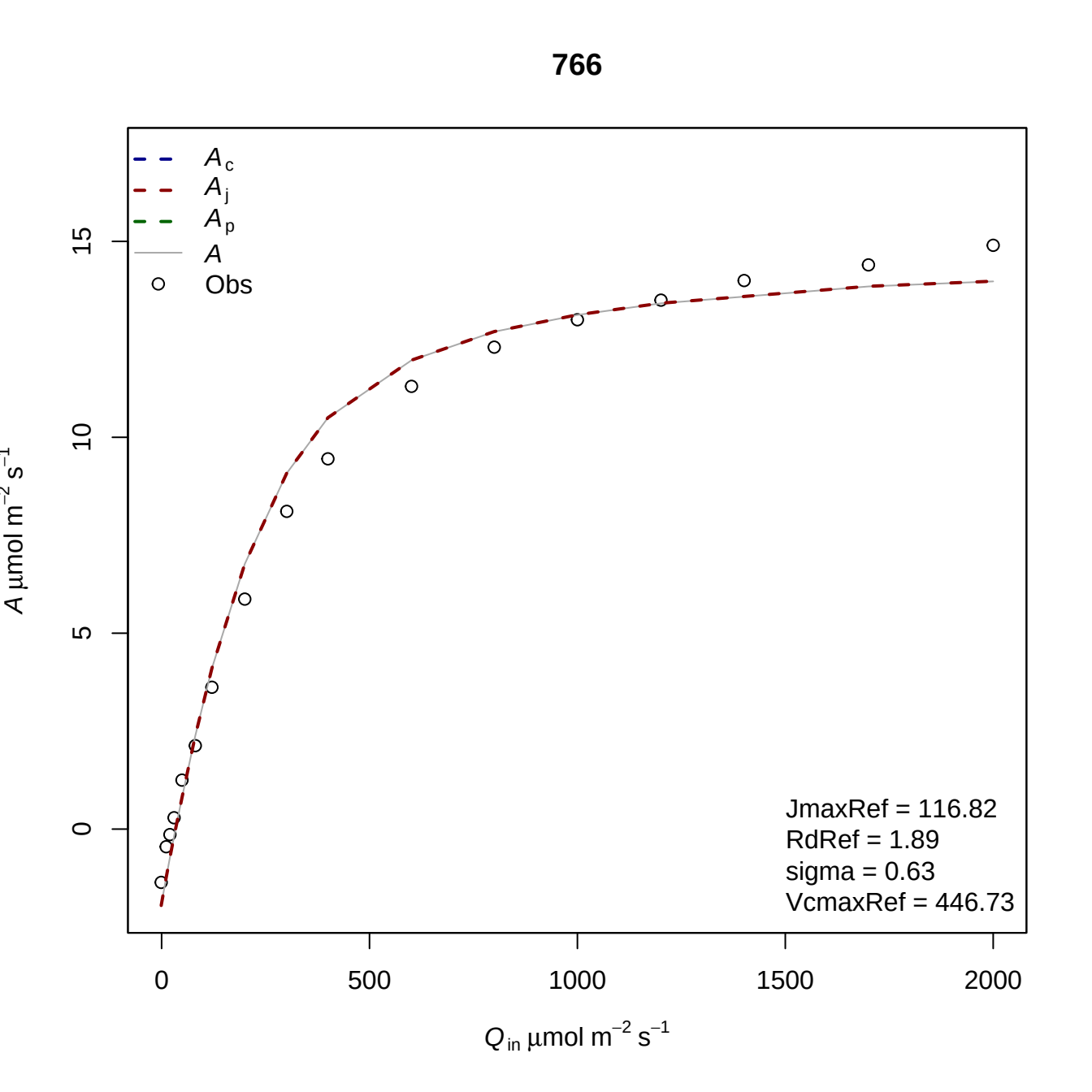

761
A_c
A_j
A_p
A
Obs
JmaxRef = 116.25
RdRef = 3.15
sigma = 0.91
VcmaxRef = 772.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

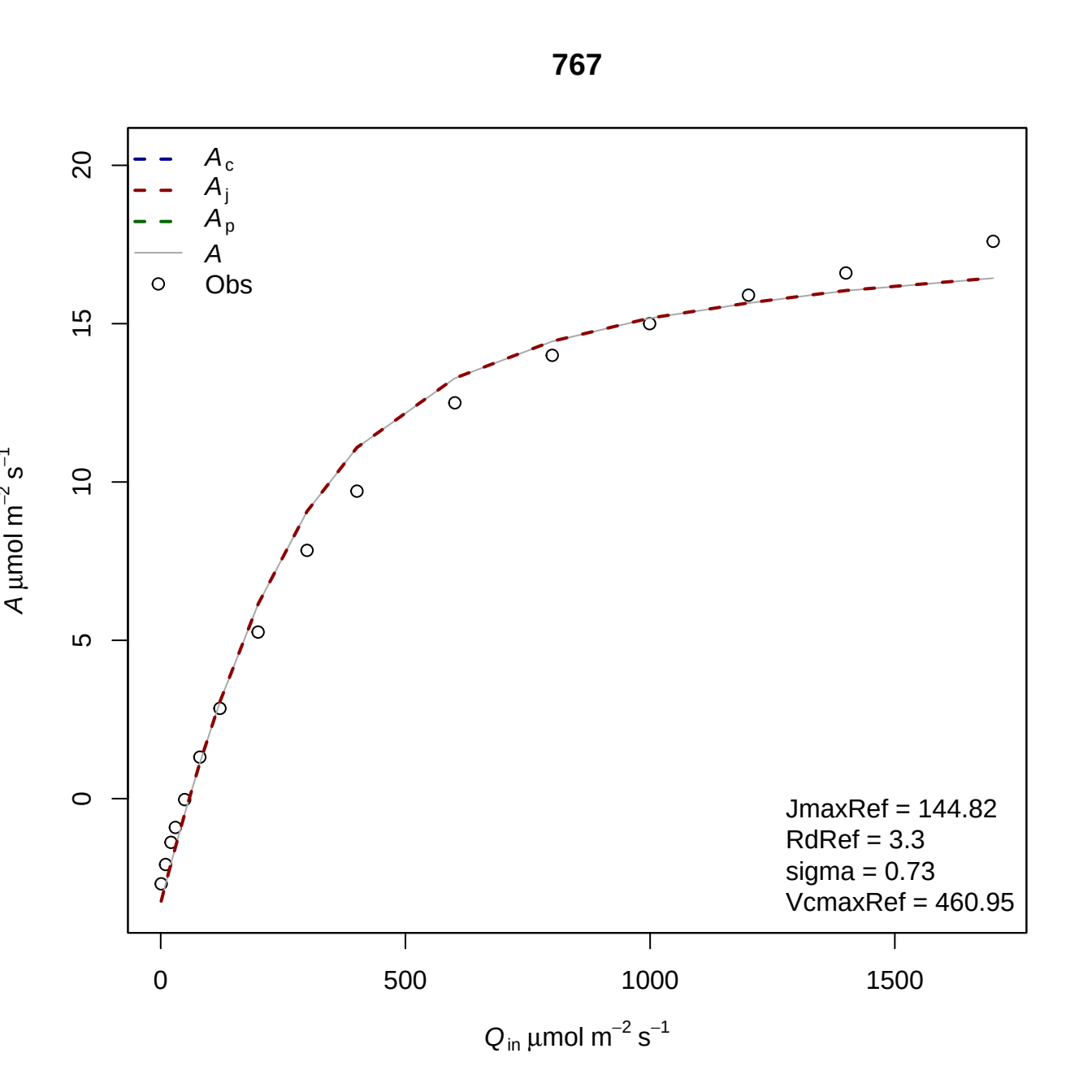

762
A_c
A_j
A_p
A
Obs
JmaxRef = 115.86
RdRef = 3.4
sigma = 1.04
VcmaxRef = 400.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

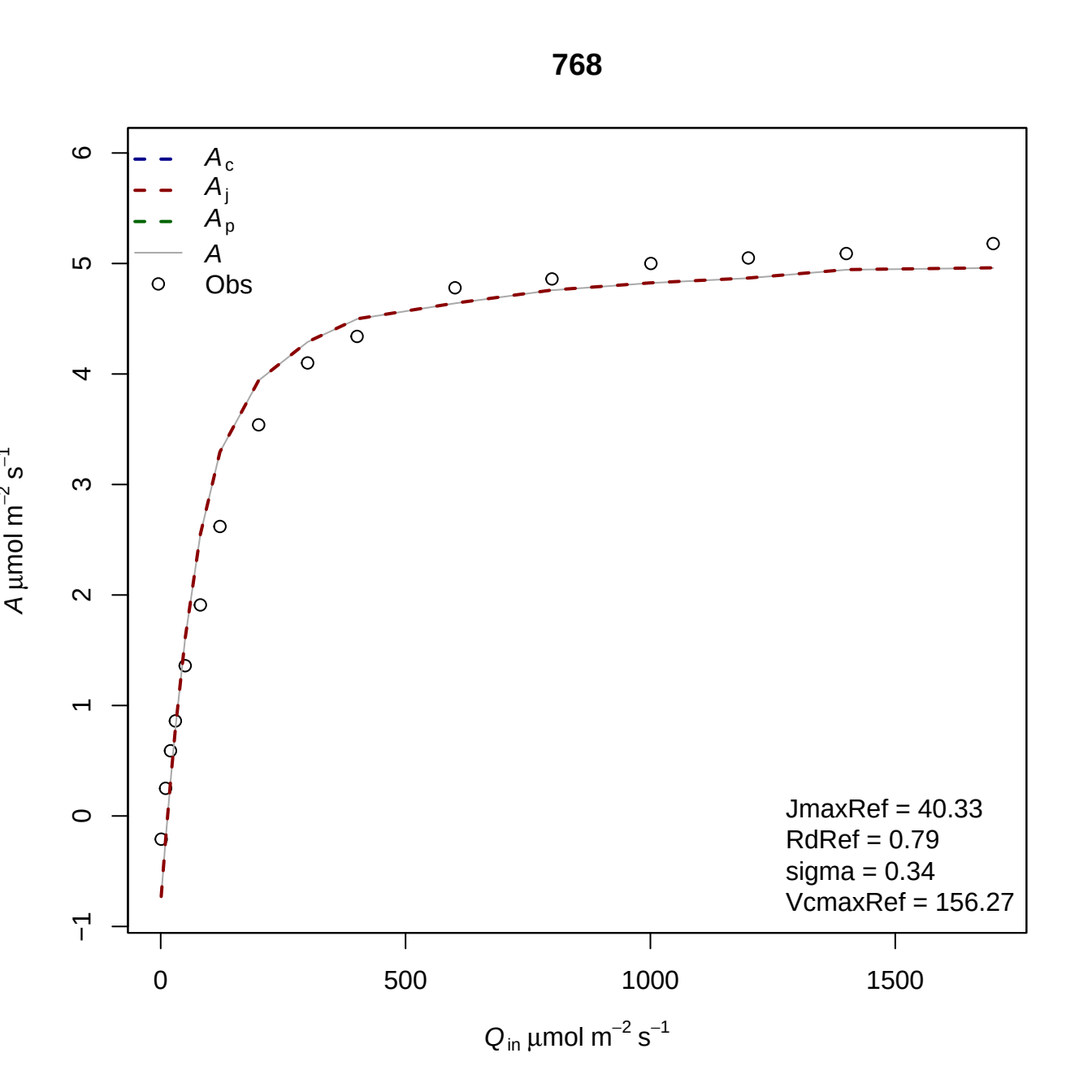

763
A_c
A_j
A_p
A
Obs
JmaxRef = 119.17
RdRef = 1.92
sigma = 0.48
VcmaxRef = 364.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

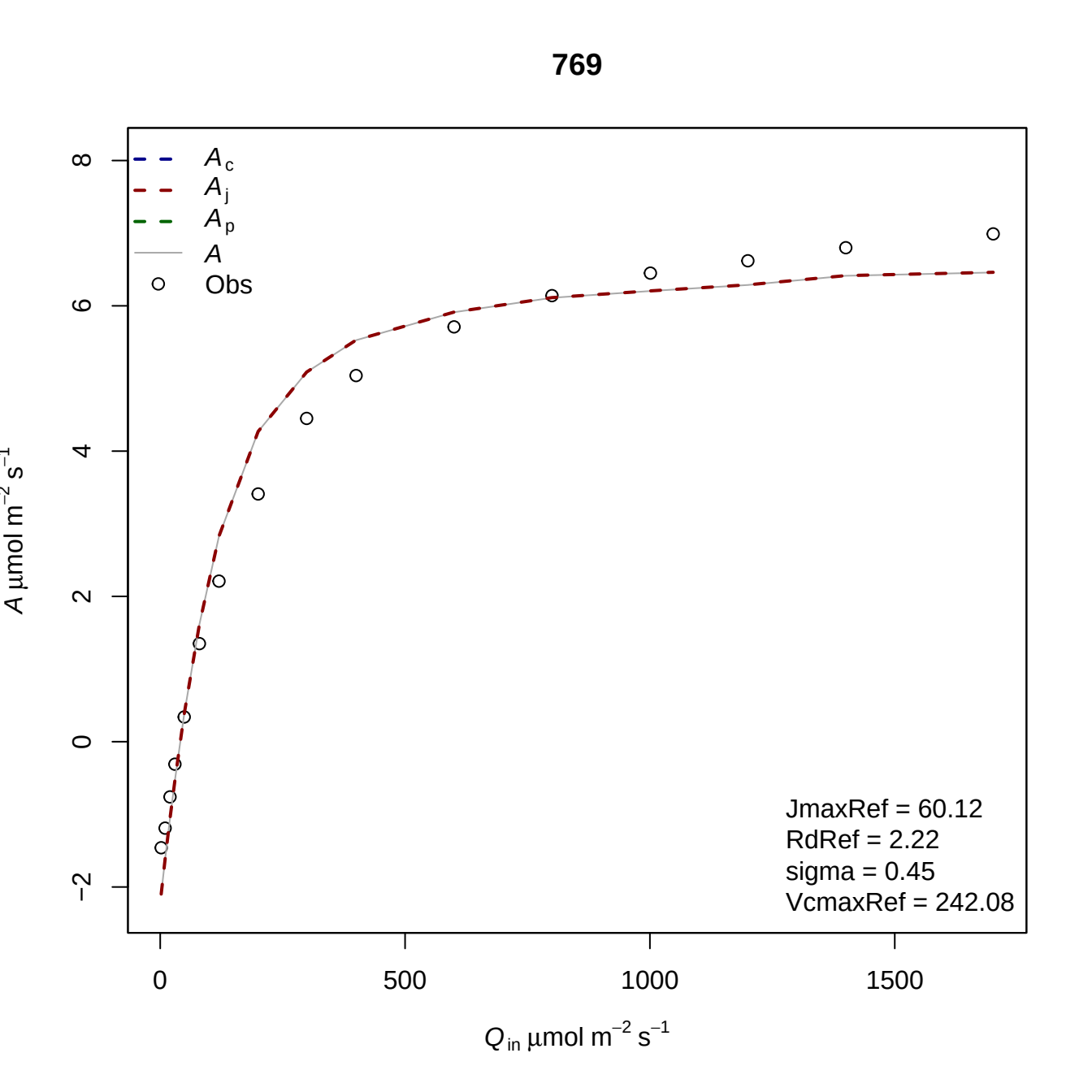

764
A_c
A_j
A_p
A
Obs
JmaxRef = 76.93
RdRef = 2.02
sigma = 0.78
VcmaxRef = 352.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

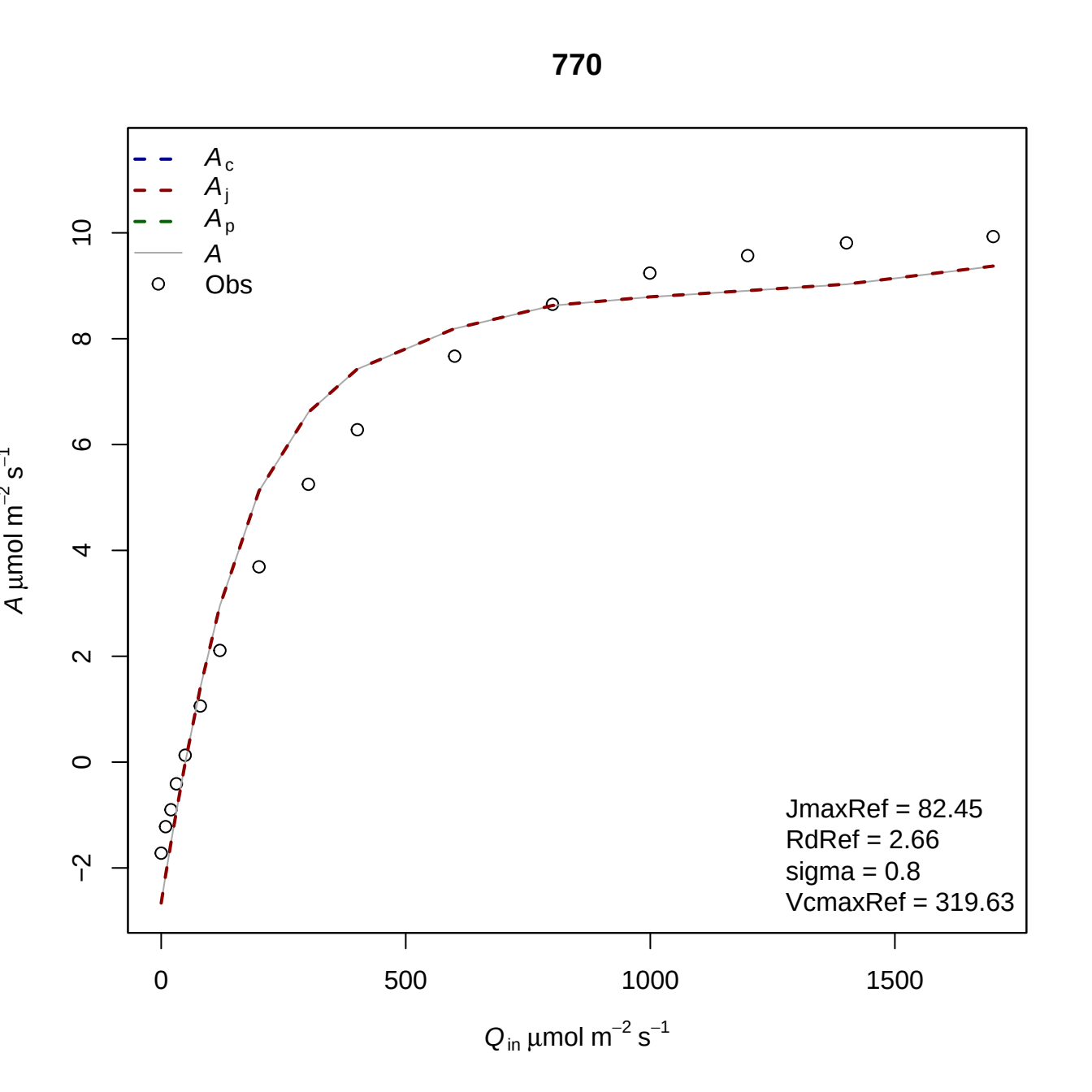

765
A_c
A_j
A_p
A
Obs
JmaxRef = 134.7
RdRef = 2.22
sigma = 0.92
VcmaxRef = 595.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

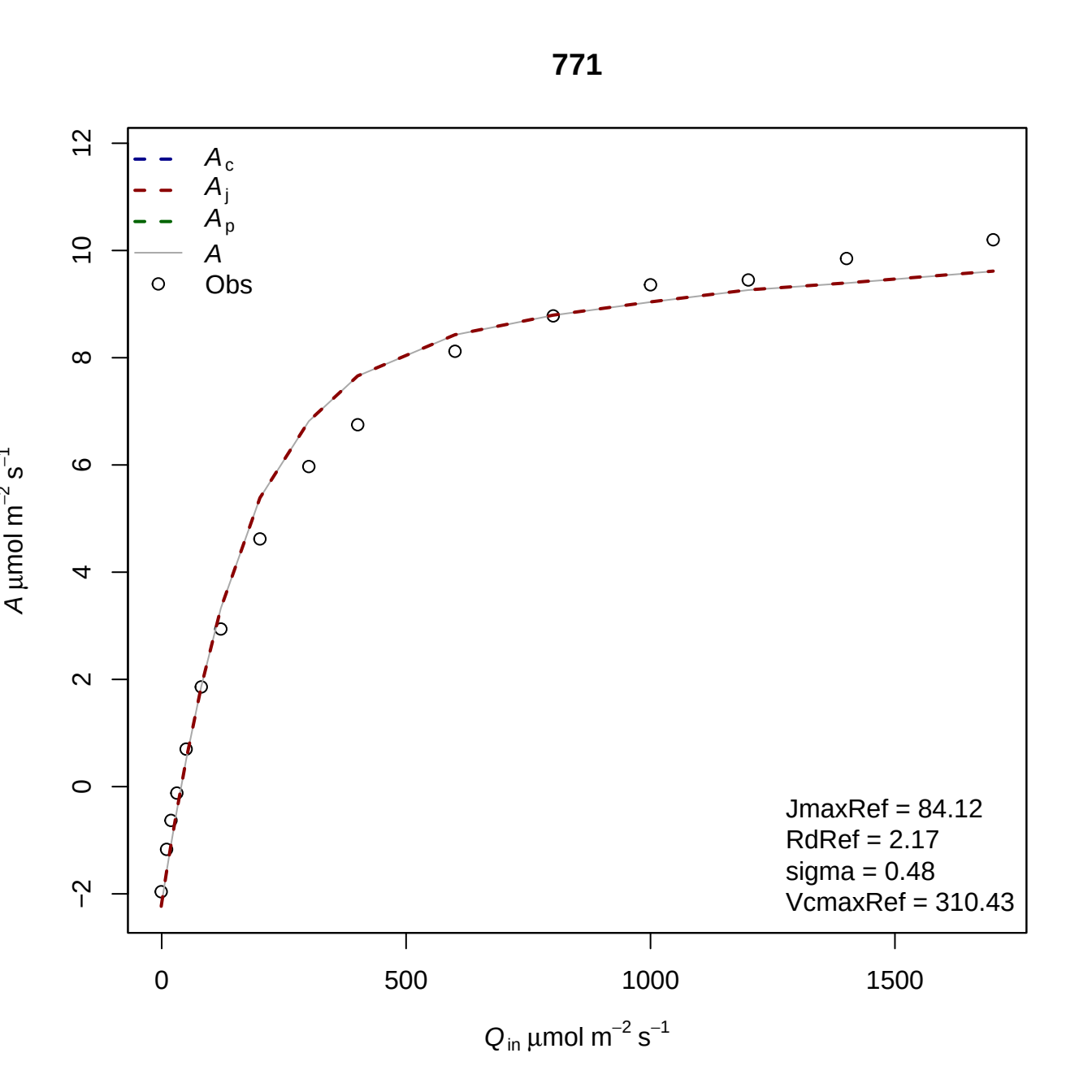

766
A_c
A_j
A_p
A
Obs
JmaxRef = 116.82
RdRef = 1.89
sigma = 0.63
VcmaxRef = 446.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

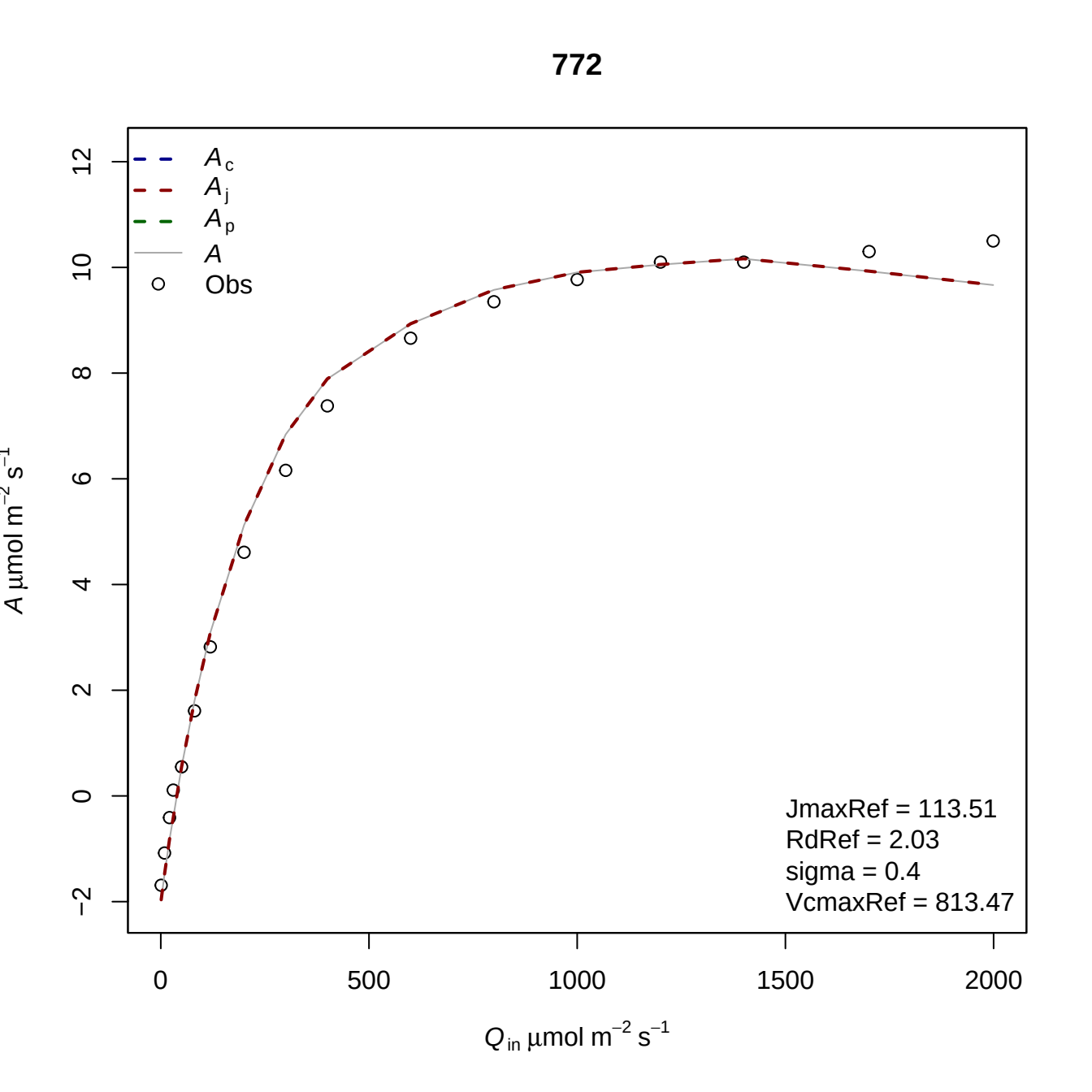

767
A_c
A_j
A_p
A
Obs
JmaxRef = 144.82
RdRef = 3.3
sigma = 0.73
VcmaxRef = 460.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

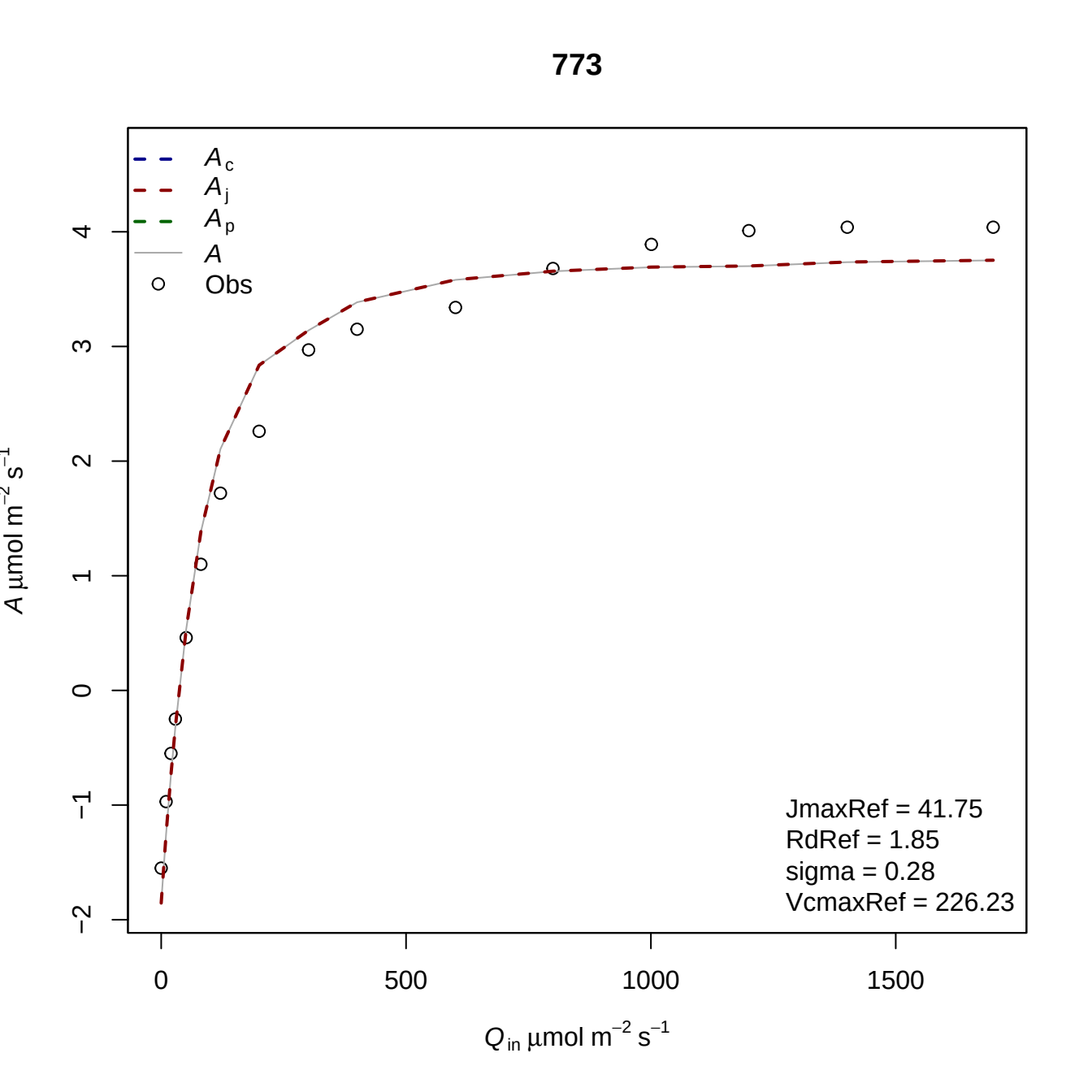

768
A_c
A_j
A_p
A
Obs
JmaxRef = 40.33
RdRef = 0.79
sigma = 0.34
VcmaxRef = 156.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

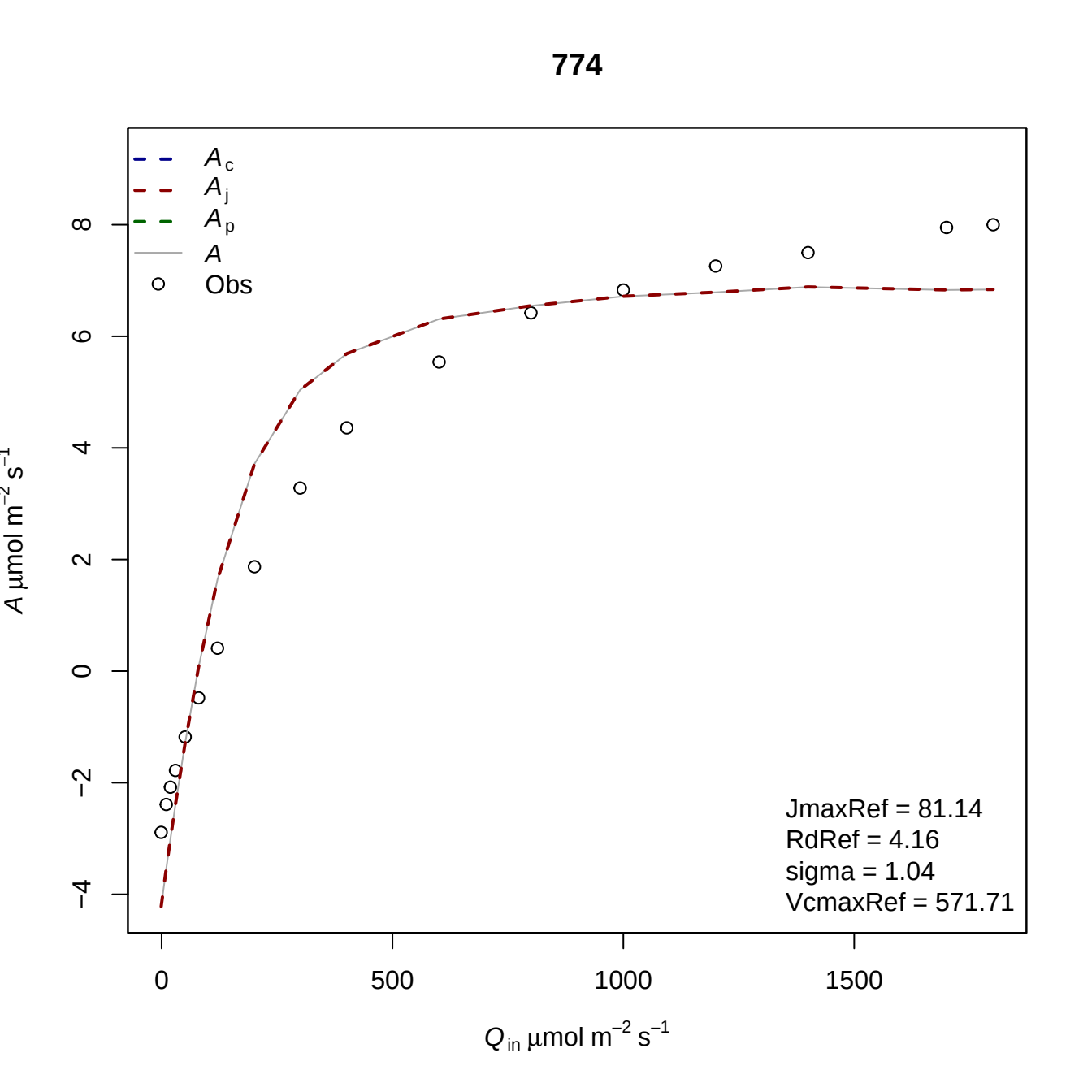

769
A_c
A_j
A_p
A
Obs
JmaxRef = 60.12
RdRef = 2.22
sigma = 0.45
VcmaxRef = 242.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

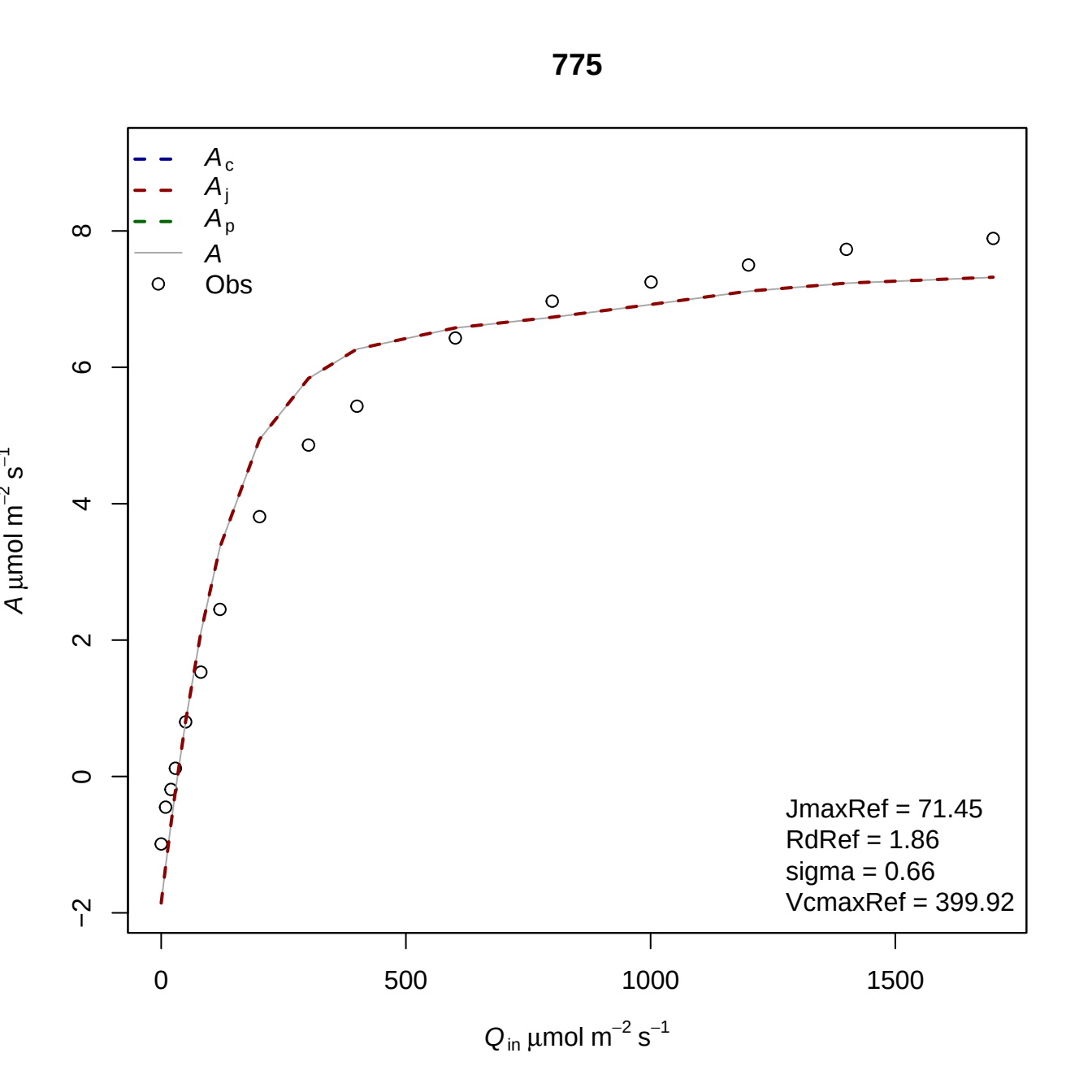

770
A_c
A_j
A_p
A
Obs
JmaxRef = 82.45
RdRef = 2.66
sigma = 0.8
VcmaxRef = 319.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

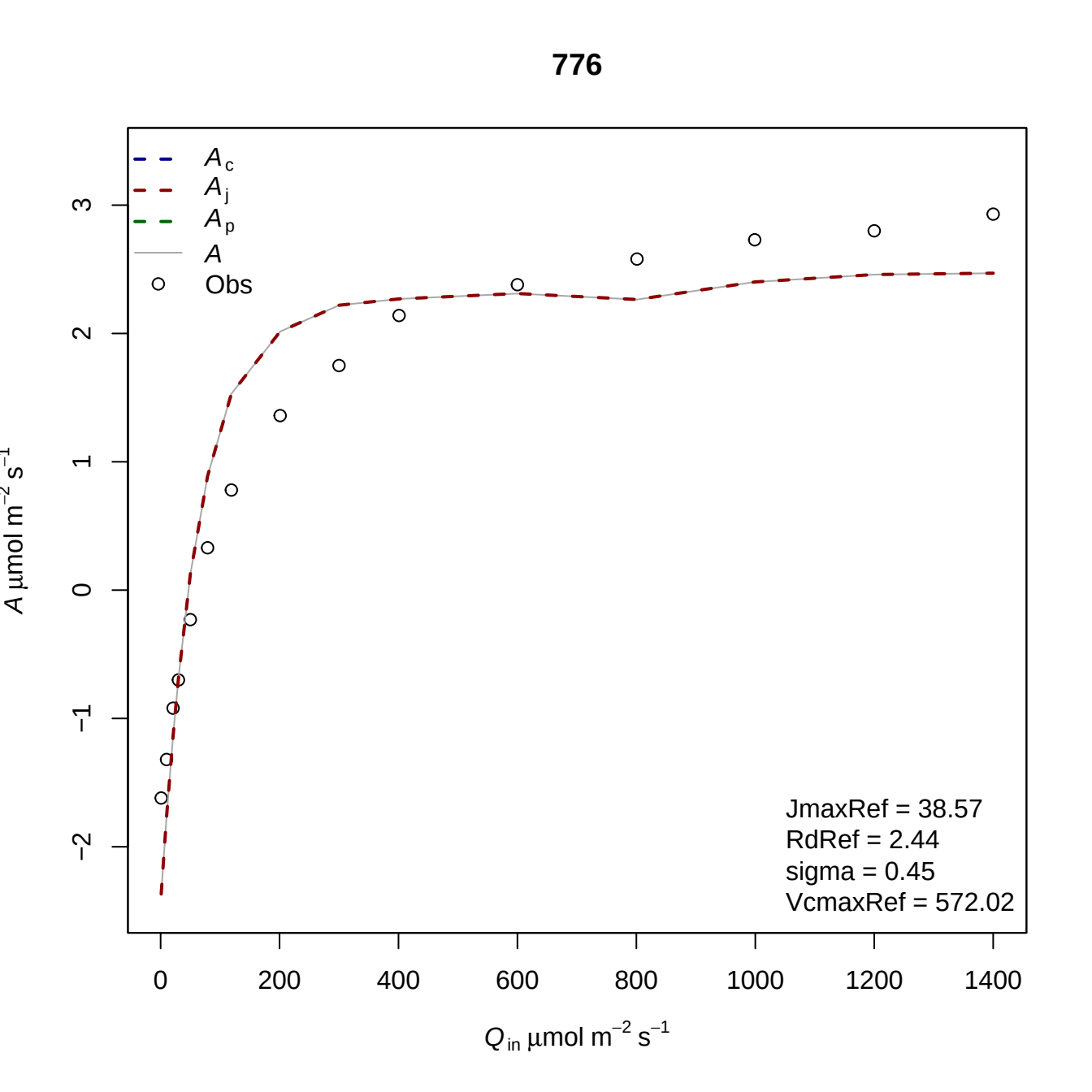

771
A_c
A_j
A_p
A
Obs
JmaxRef = 84.12
RdRef = 2.17
sigma = 0.48
VcmaxRef = 310.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

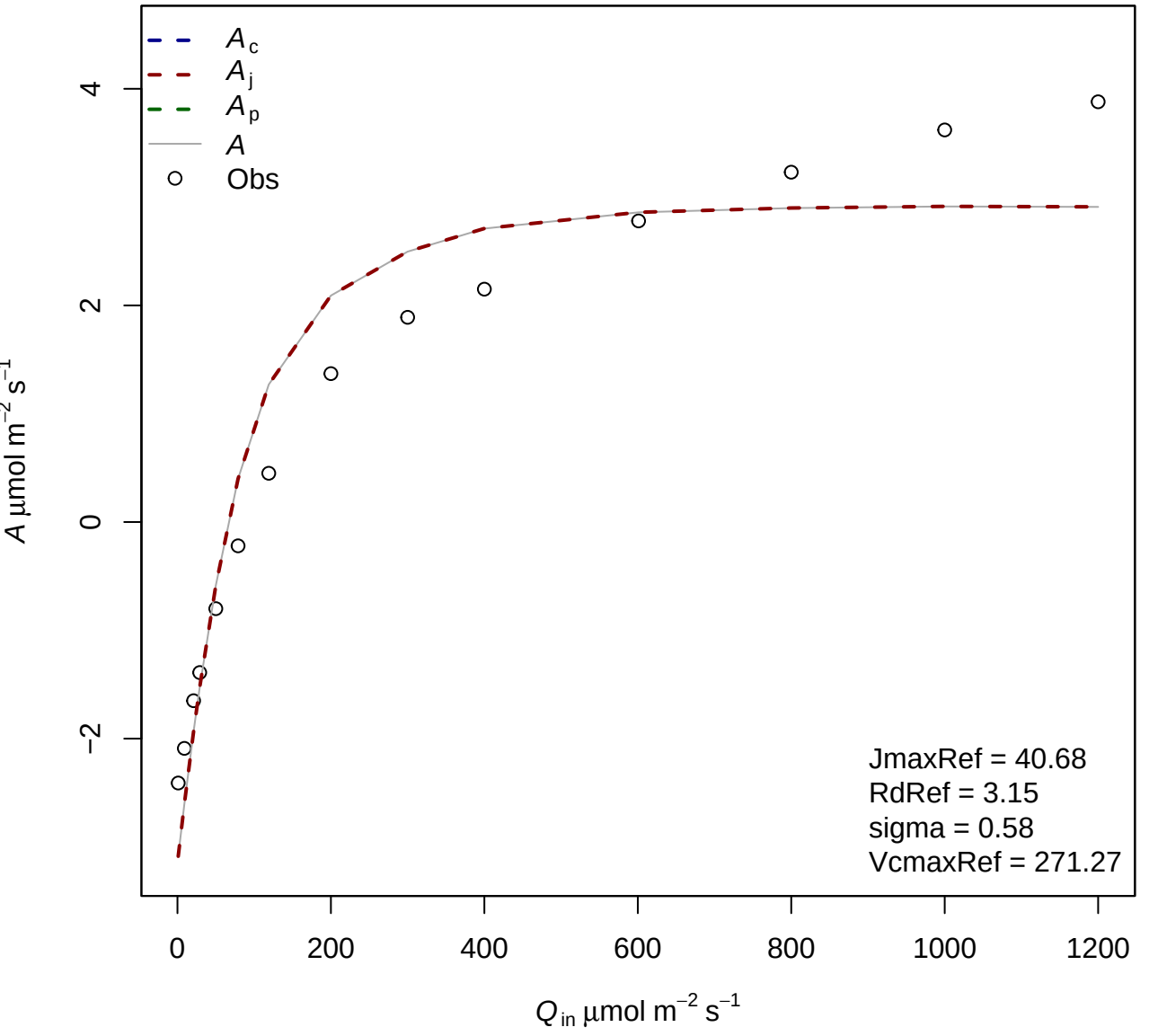

772
A_c
A_j
A_p
A
Obs
JmaxRef = 113.51
RdRef = 2.03
sigma = 0.4
VcmaxRef = 813.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

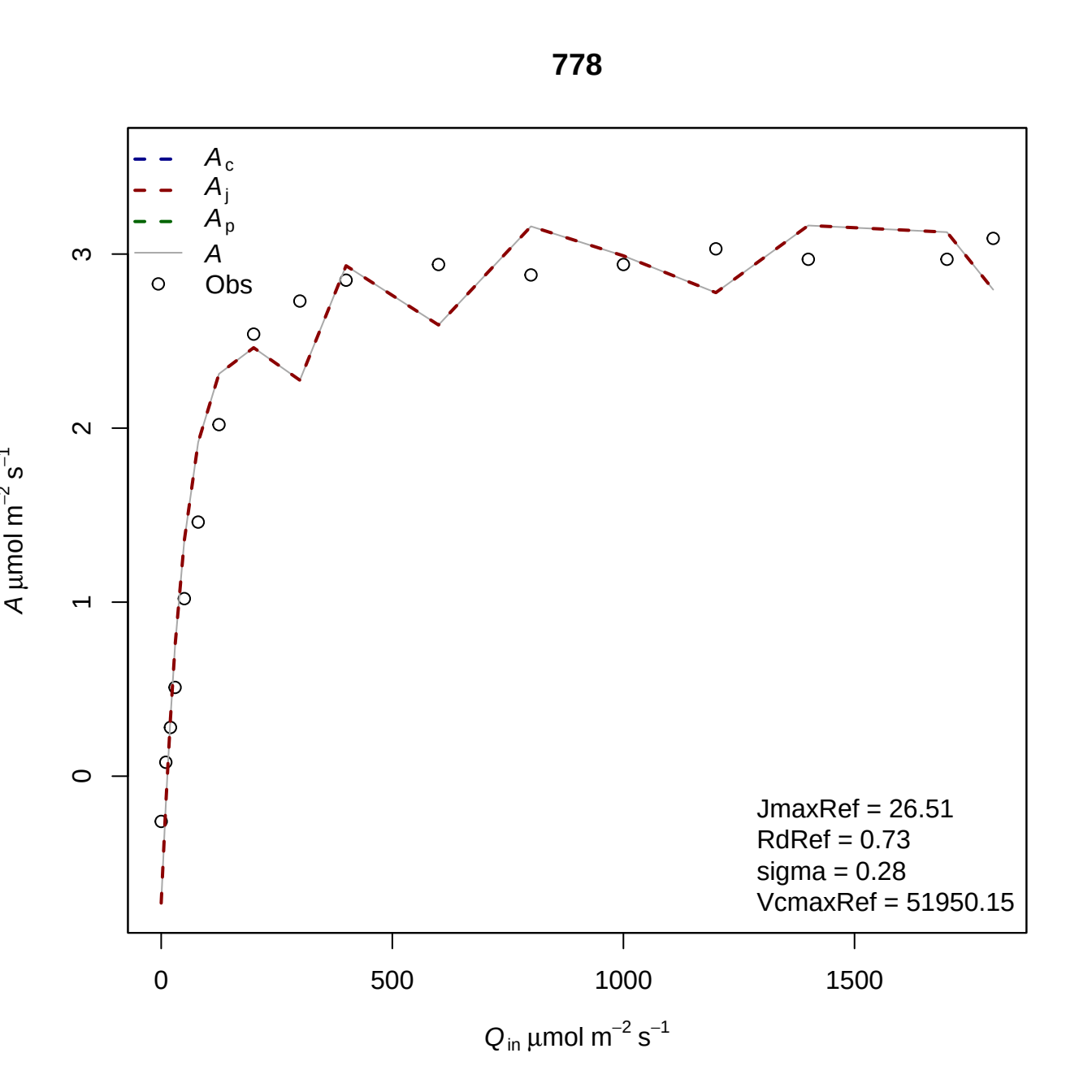

773
A_c
A_j
A_p
A
Obs
JmaxRef = 41.75
RdRef = 1.85
sigma = 0.28
VcmaxRef = 226.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
500
1000
1500

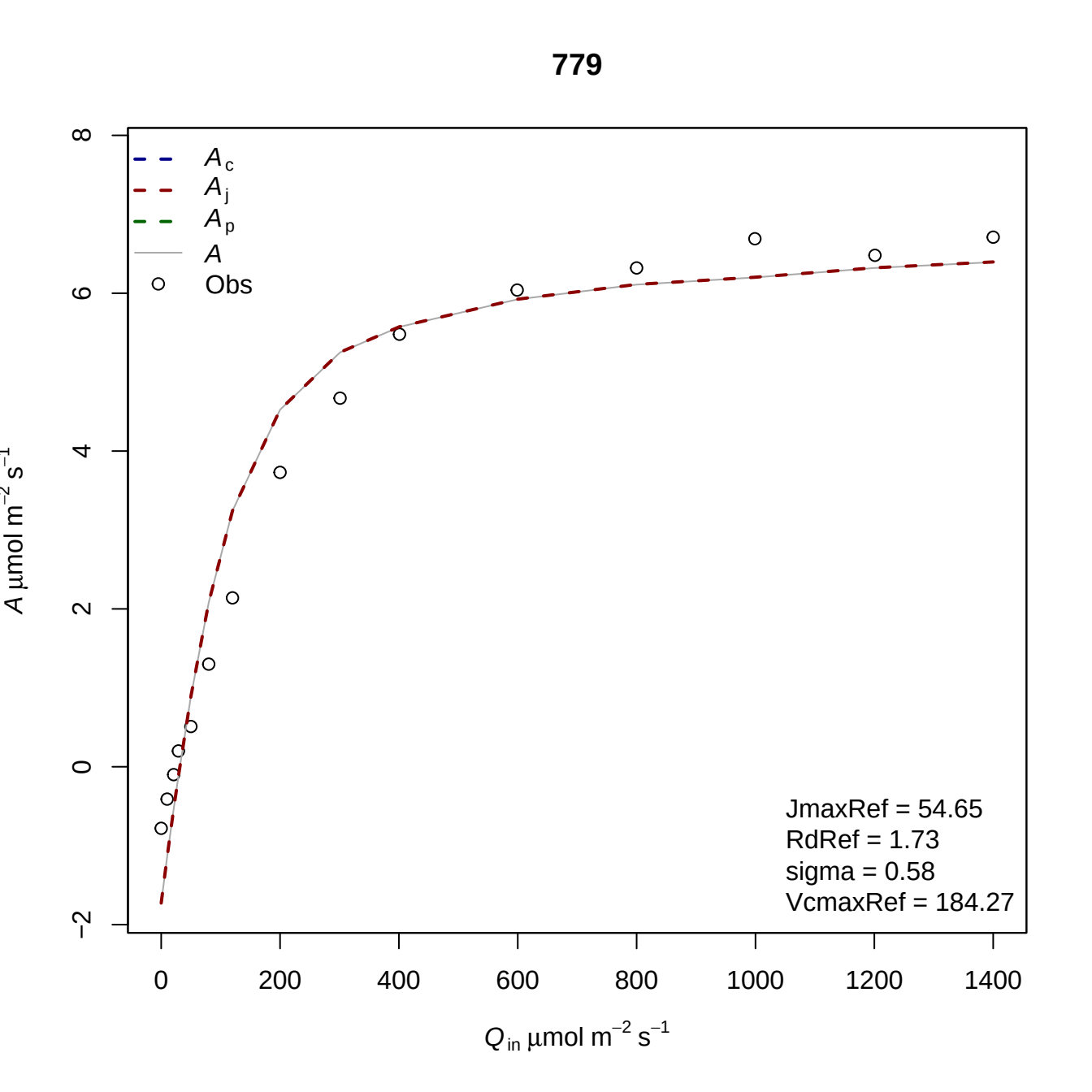

774
A_c
A_j
A_p
A
Obs
JmaxRef = 81.14
RdRef = 4.16
sigma = 1.04
VcmaxRef = 571.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
6
8
0
500
1000
1500

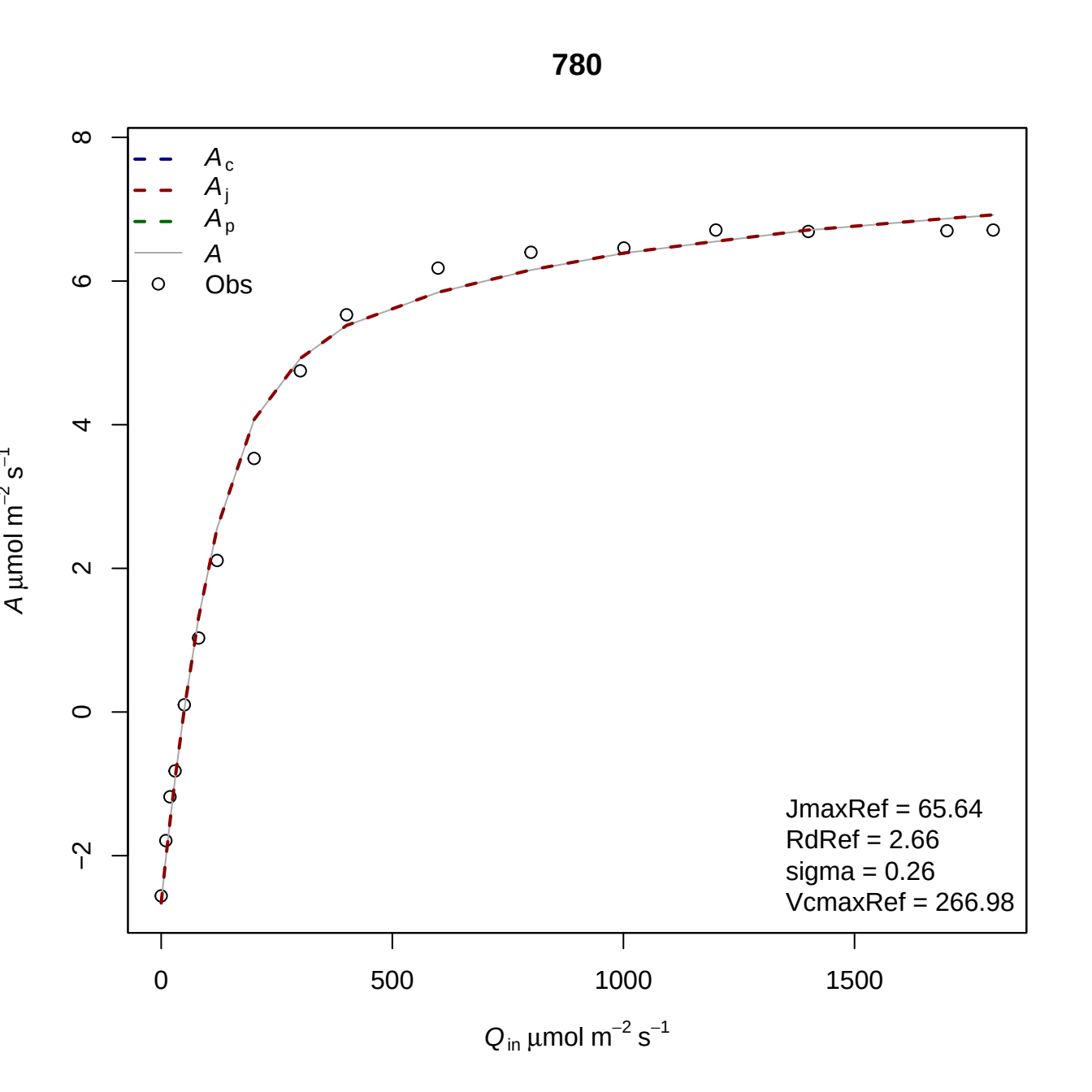

775
A_c
A_j
A_p
A
Obs
JmaxRef = 71.45
RdRef = 1.86
sigma = 0.66
VcmaxRef = 399.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

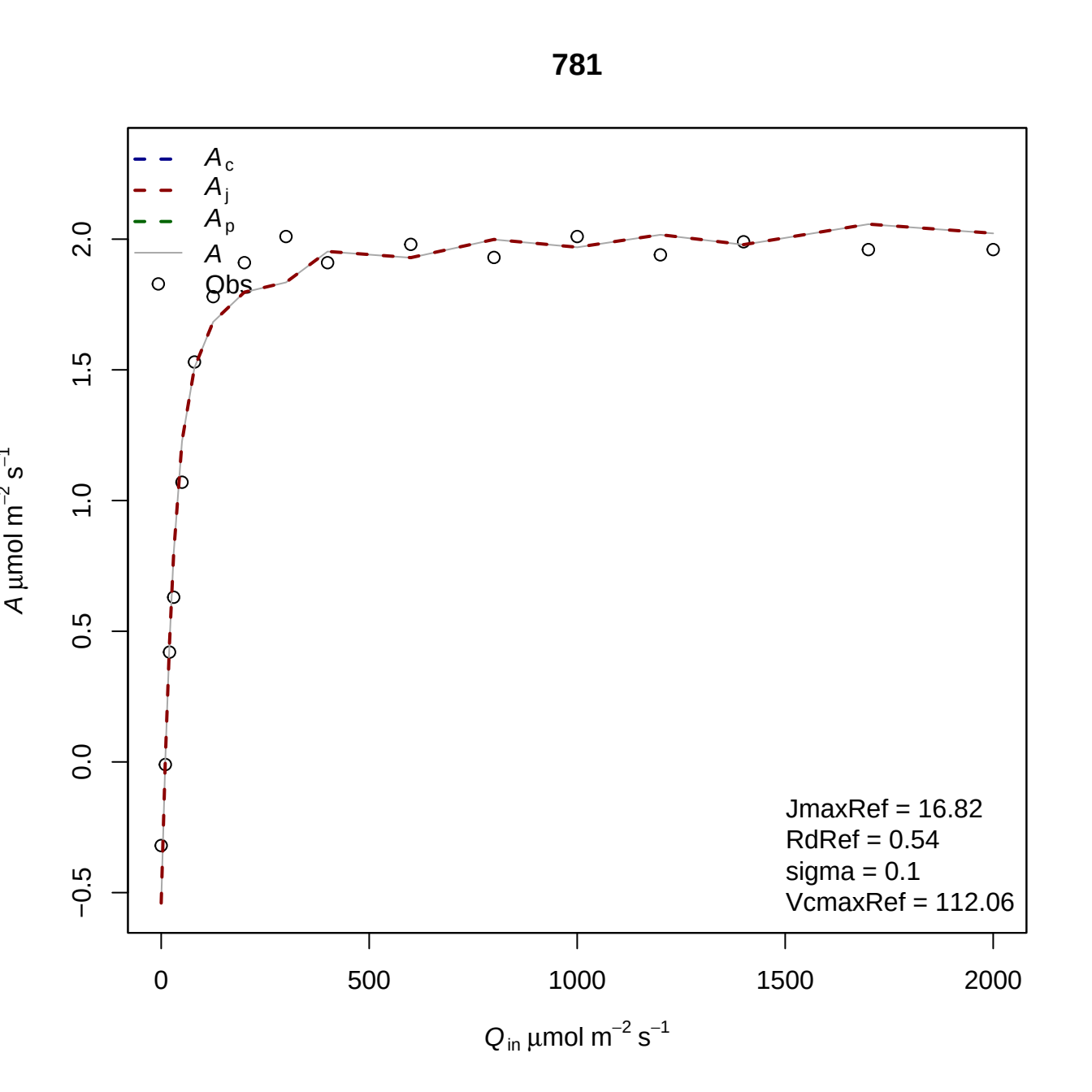

776
A_c
A_j
A_p
A
Obs
JmaxRef = 38.57
RdRef = 2.44
sigma = 0.45
VcmaxRef = 572.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

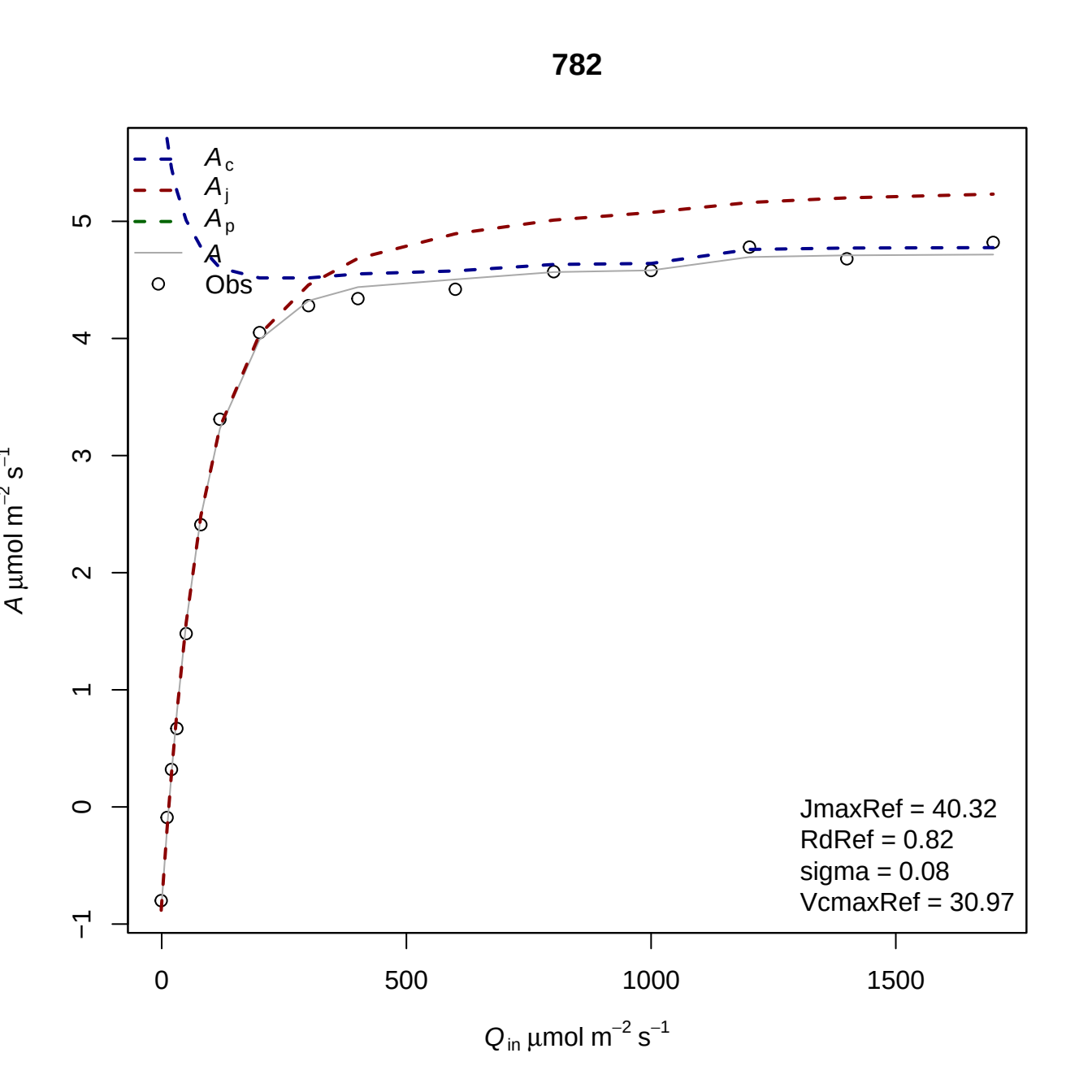

A_c
A_j
A_p
A
Obs
JmaxRef = 40.68
RdRef = 3.15
sigma = 0.58
VcmaxRef = 271.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200

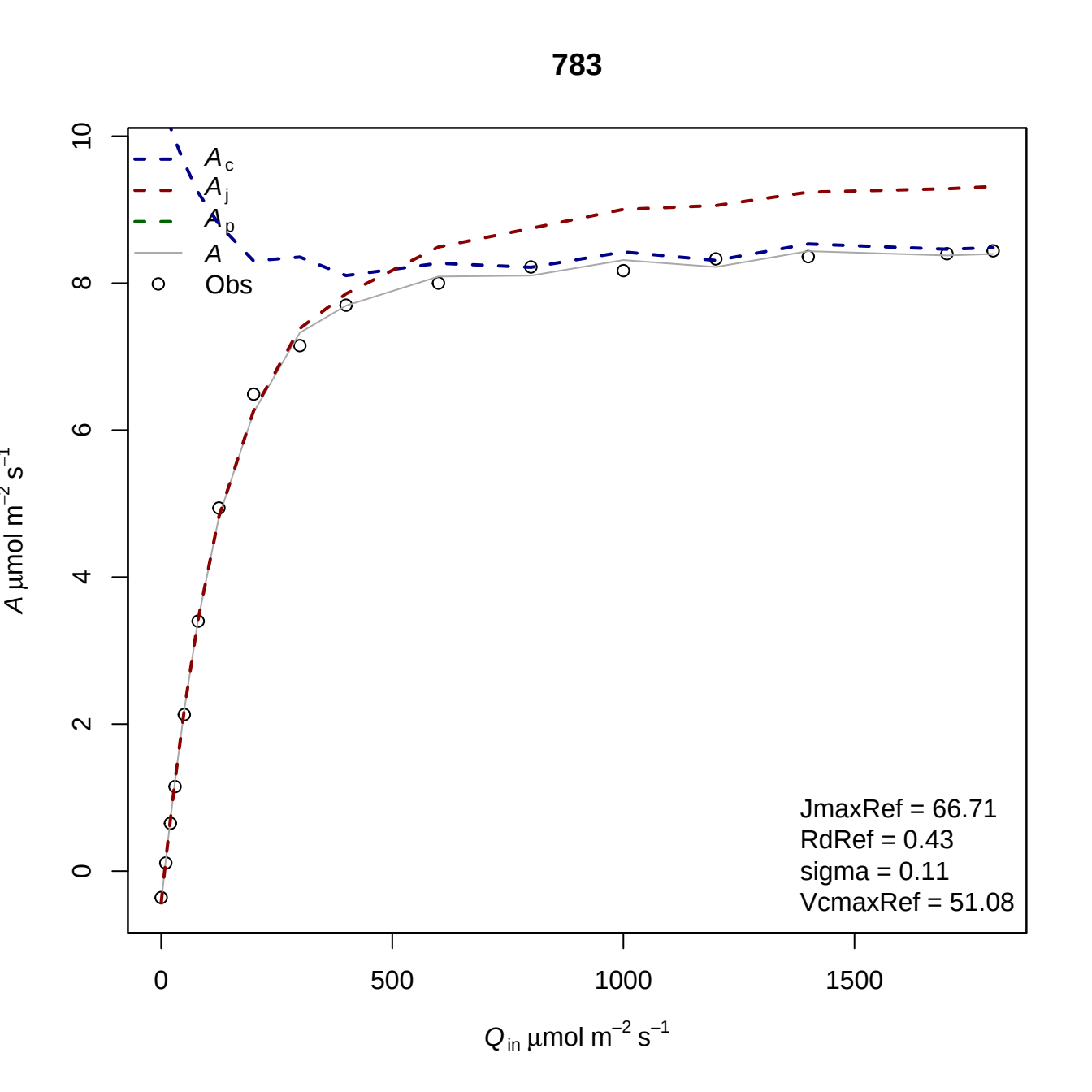

778
A_c
A_j
A_p
A
Obs
JmaxRef = 26.51
RdRef = 0.73
sigma = 0.28
VcmaxRef = 51950.15
0
1
2
3
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

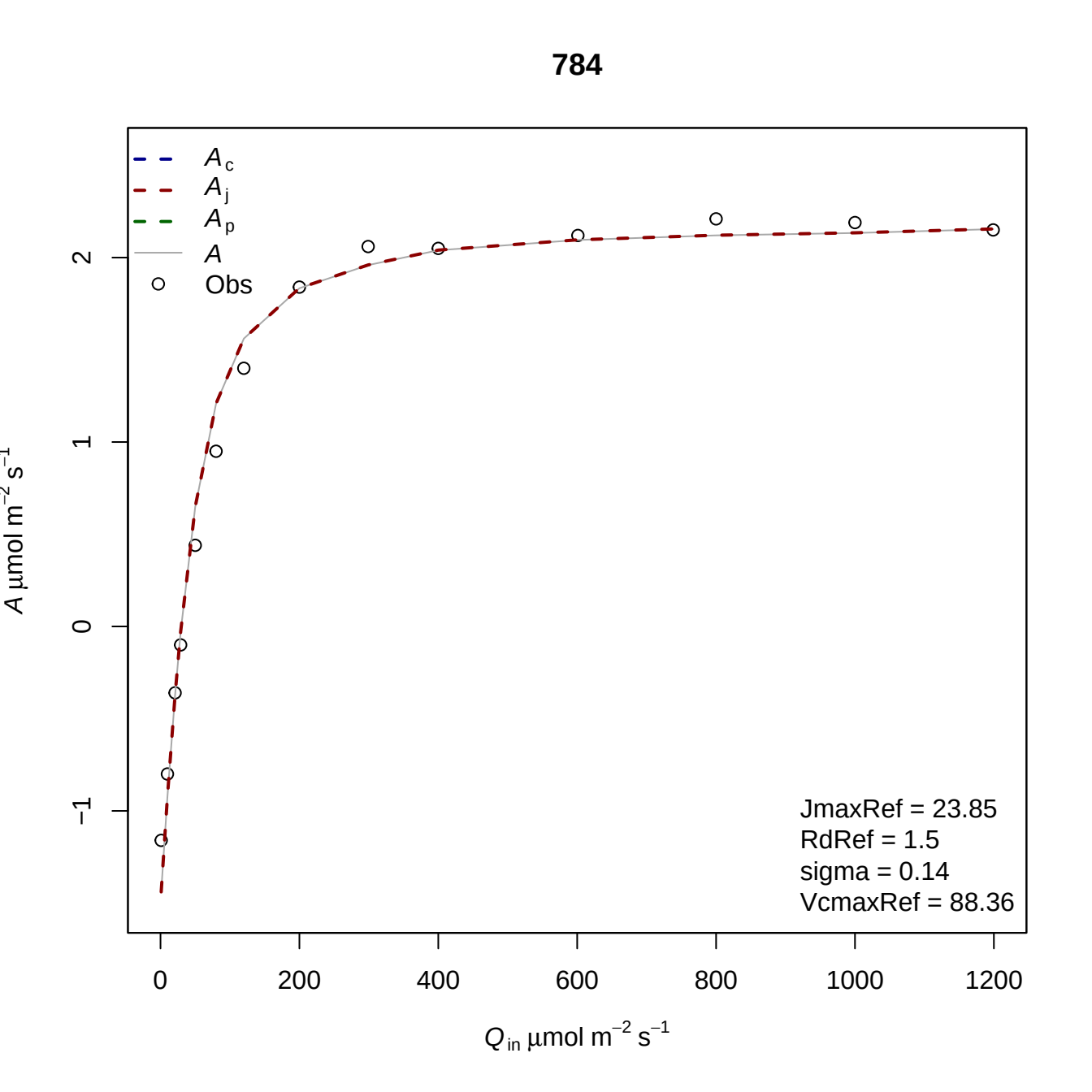

779
A_c
A_j
A_p
A
Obs
JmaxRef = 54.65
RdRef = 1.73
sigma = 0.58
VcmaxRef = 184.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

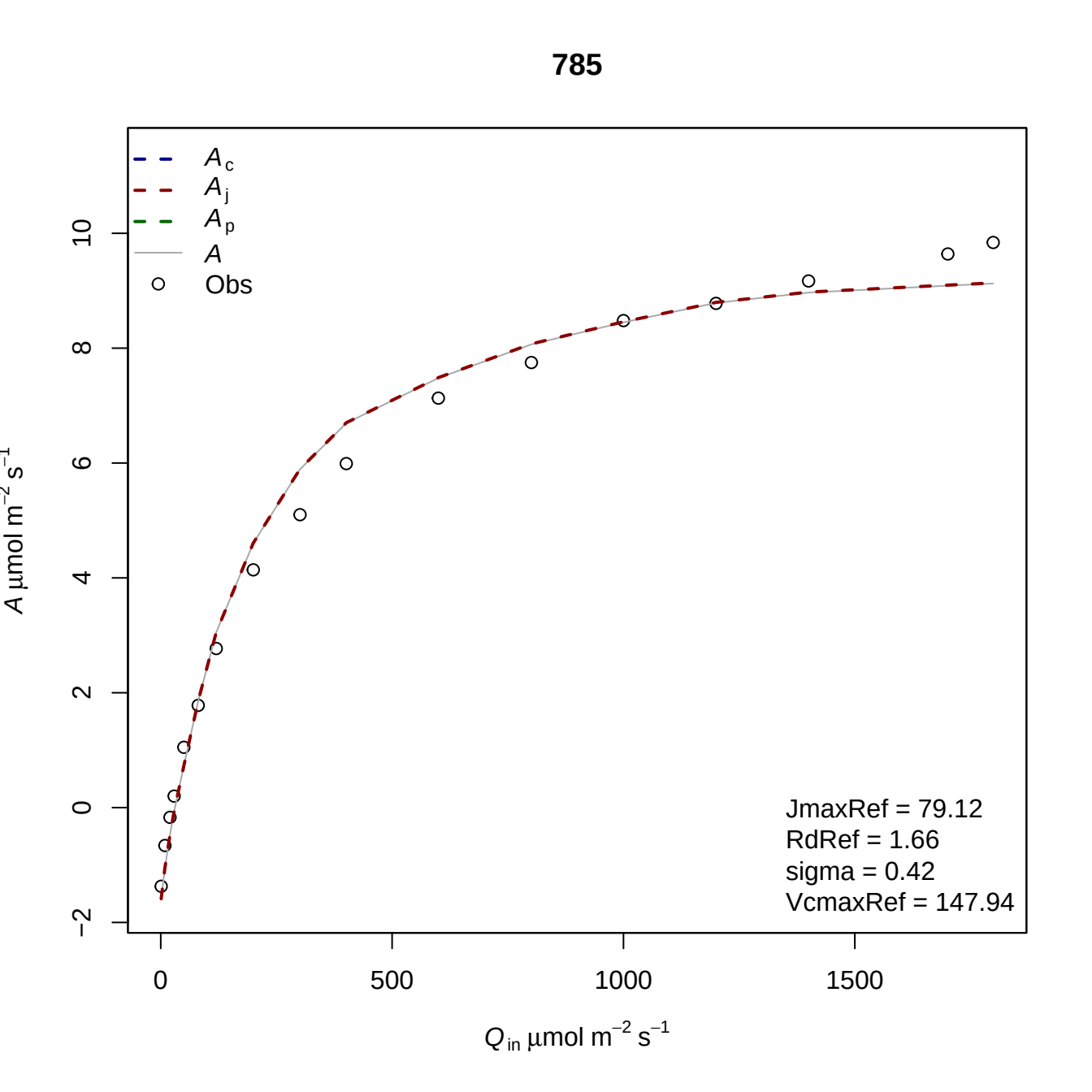

780
A_c
A_j
A_p
A
Obs
JmaxRef = 65.64
RdRef = 2.66
sigma = 0.26
VcmaxRef = 266.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

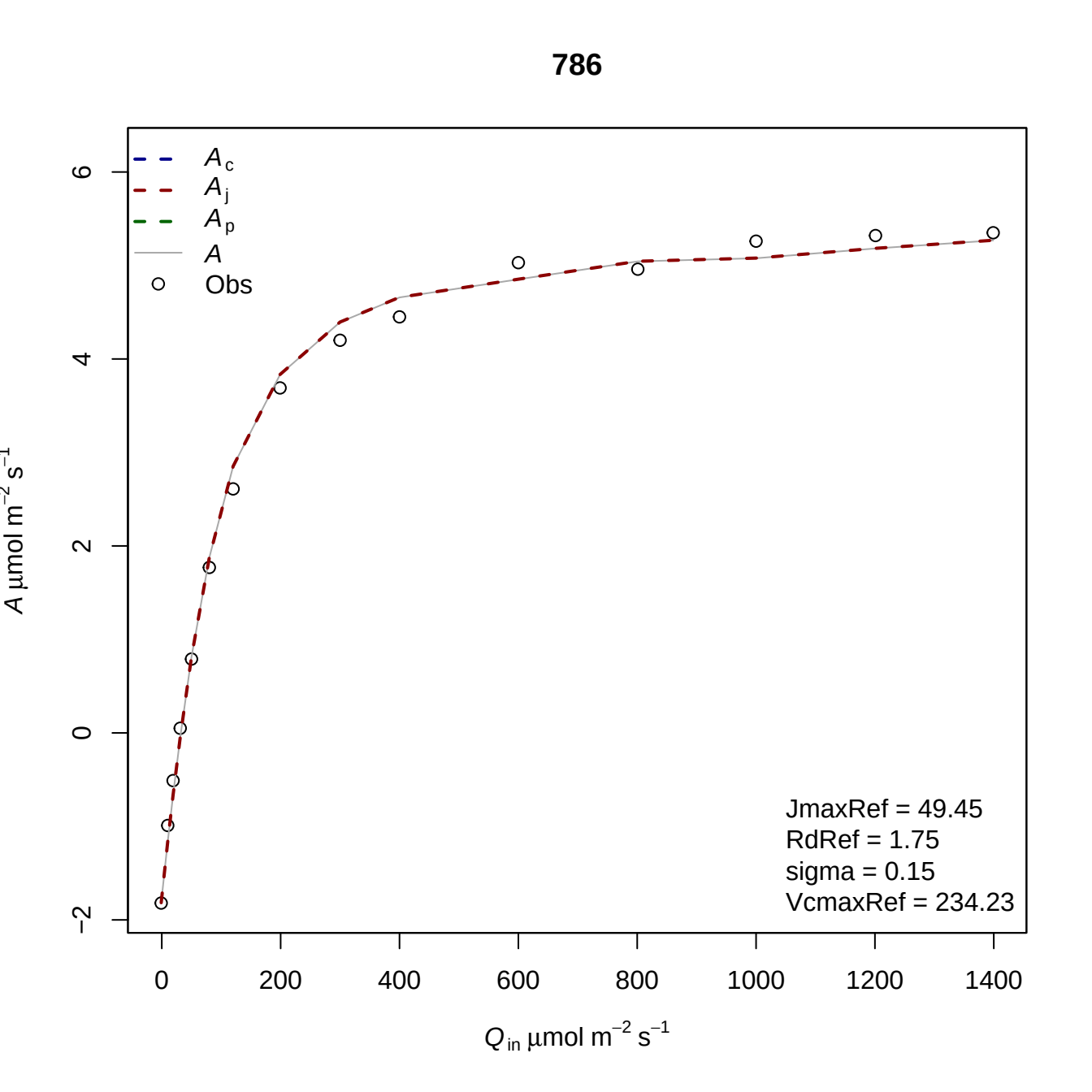

781
A_c
A_j
A_p
A
Obs
JmaxRef = 16.82
RdRef = 0.54
sigma = 0.1
VcmaxRef = 112.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
2000

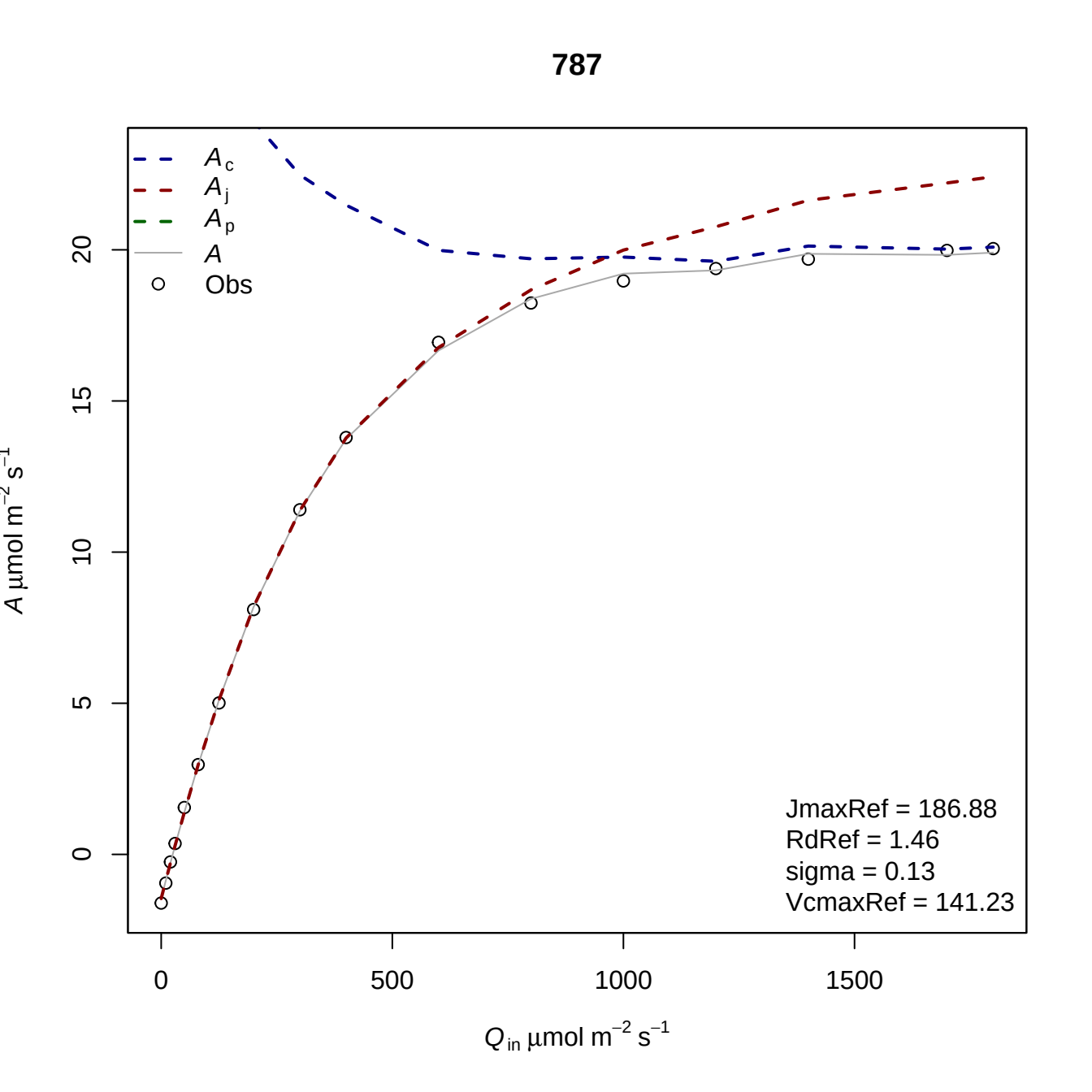

782
A_c
A_j
A_p
A
Obs
JmaxRef = 40.32
RdRef = 0.82
sigma = 0.08
VcmaxRef = 30.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

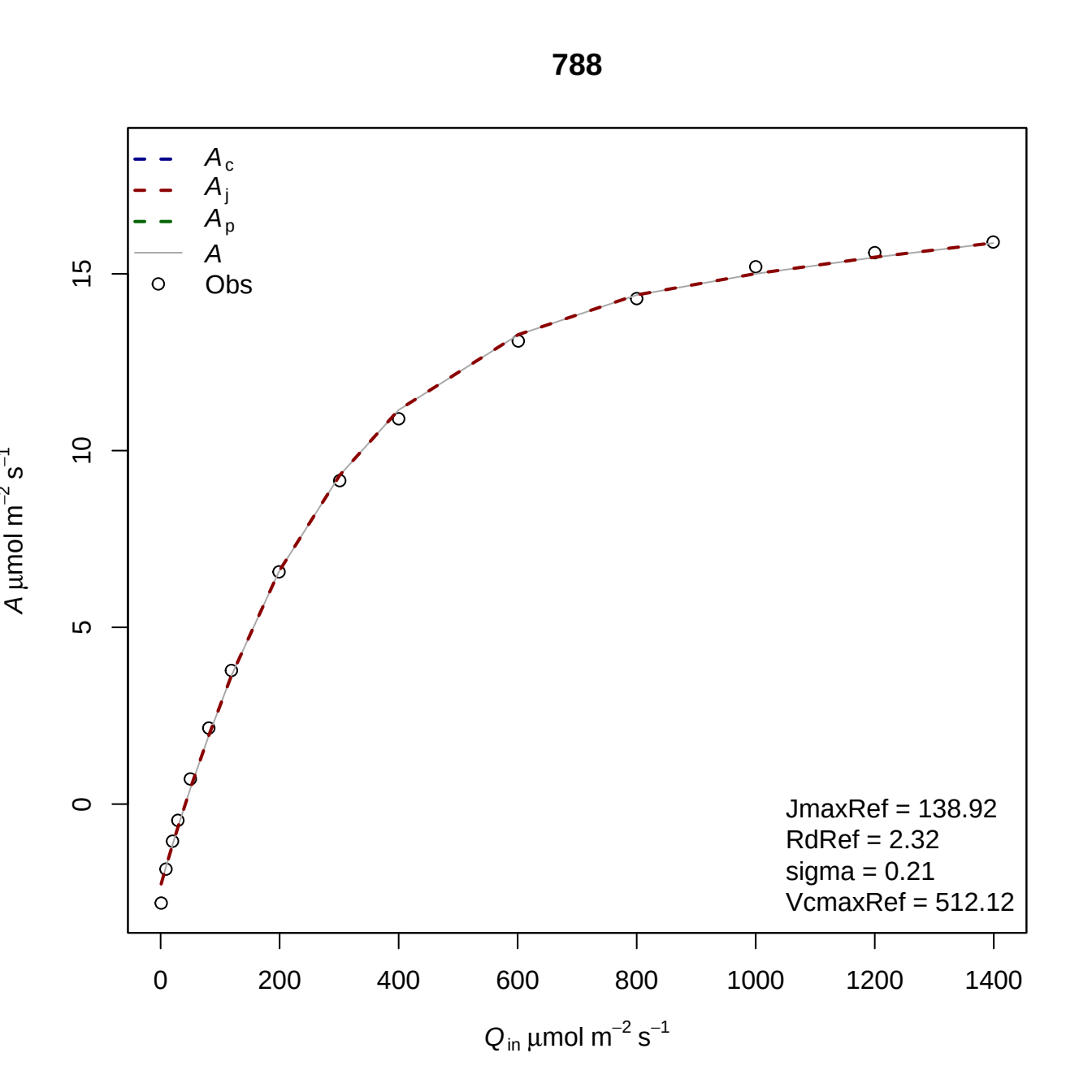

783
A_c
A_j
A_p
A
Obs
JmaxRef = 66.71
RdRef = 0.43
sigma = 0.11
VcmaxRef = 51.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

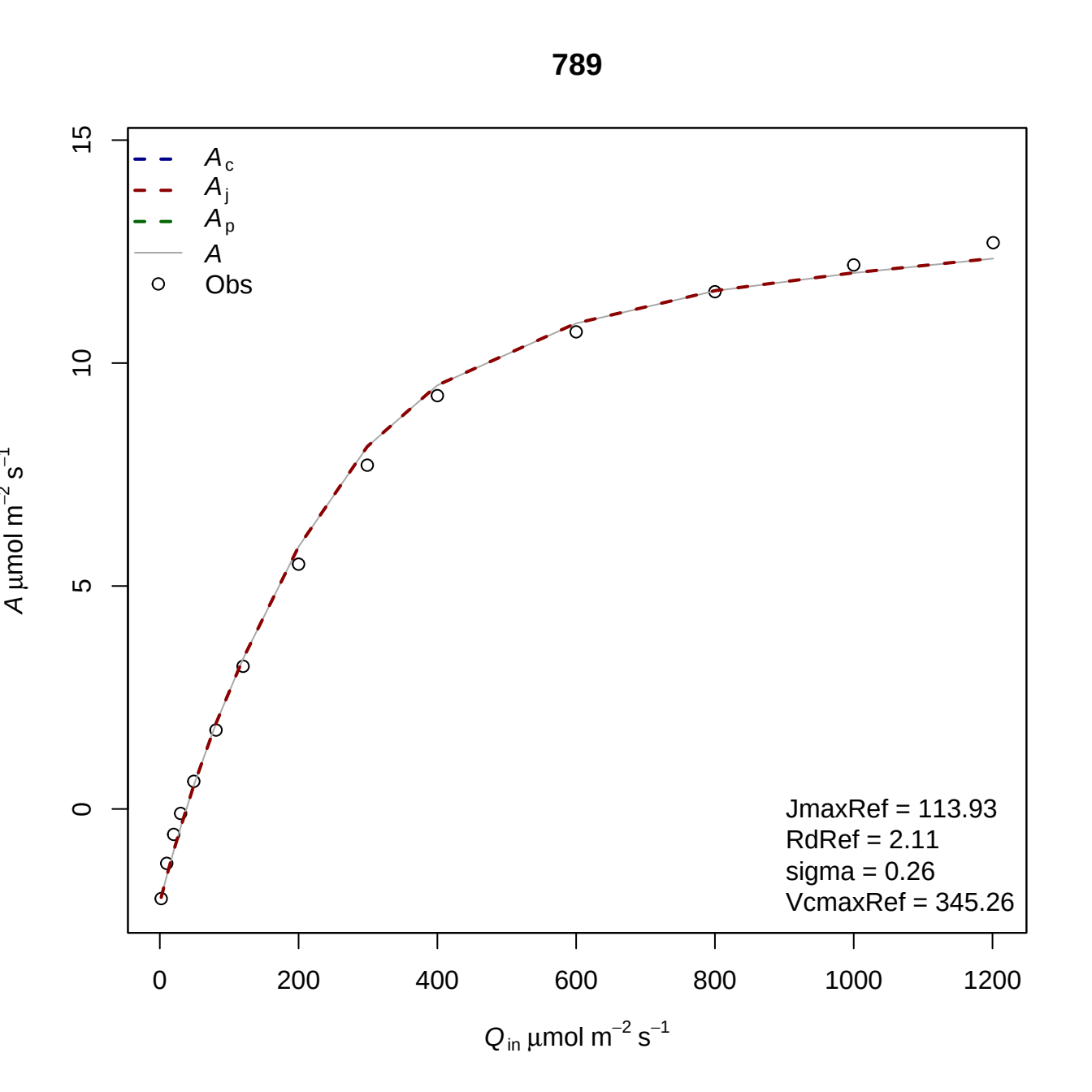

784
A_c
A_j
A_p
A
Obs
JmaxRef = 23.85
RdRef = 1.5
sigma = 0.14
VcmaxRef = 88.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
200
400
600
800
1000
1200

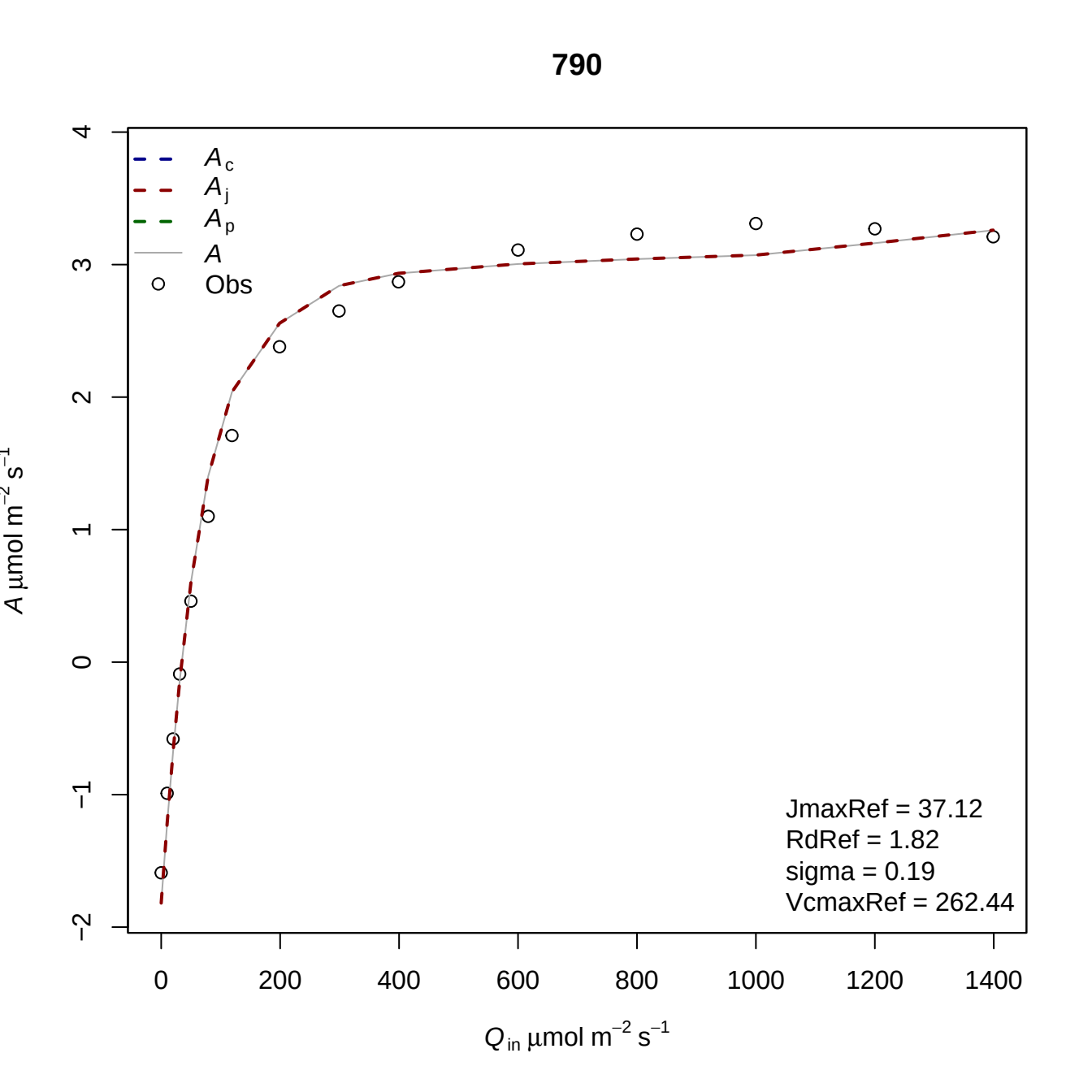

785
A_c
A_j
A_p
A
Obs
JmaxRef = 79.12
RdRef = 1.66
sigma = 0.42
VcmaxRef = 147.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

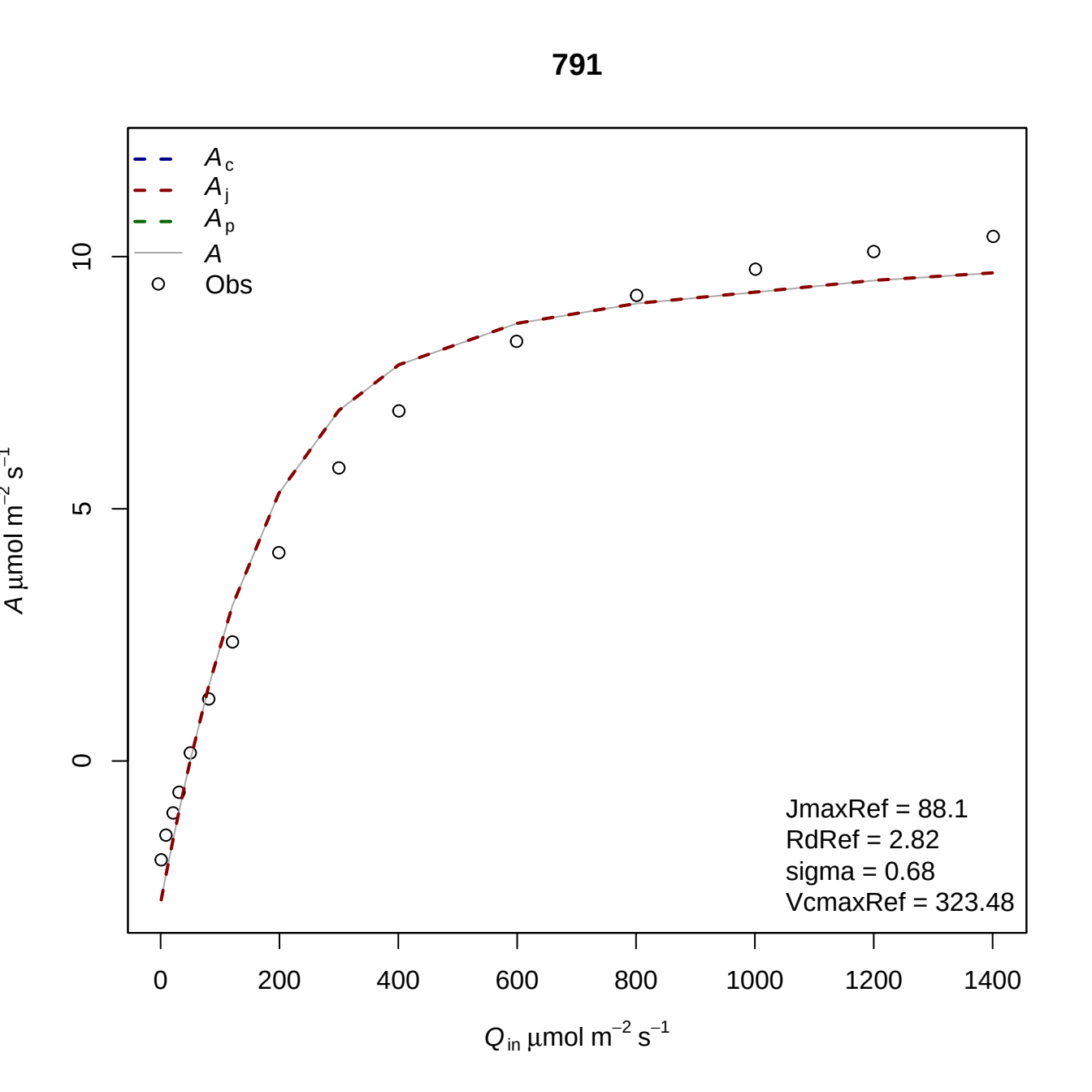

786
A_c
A_j
A_p
A
Obs
JmaxRef = 49.45
RdRef = 1.75
sigma = 0.15
VcmaxRef = 234.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

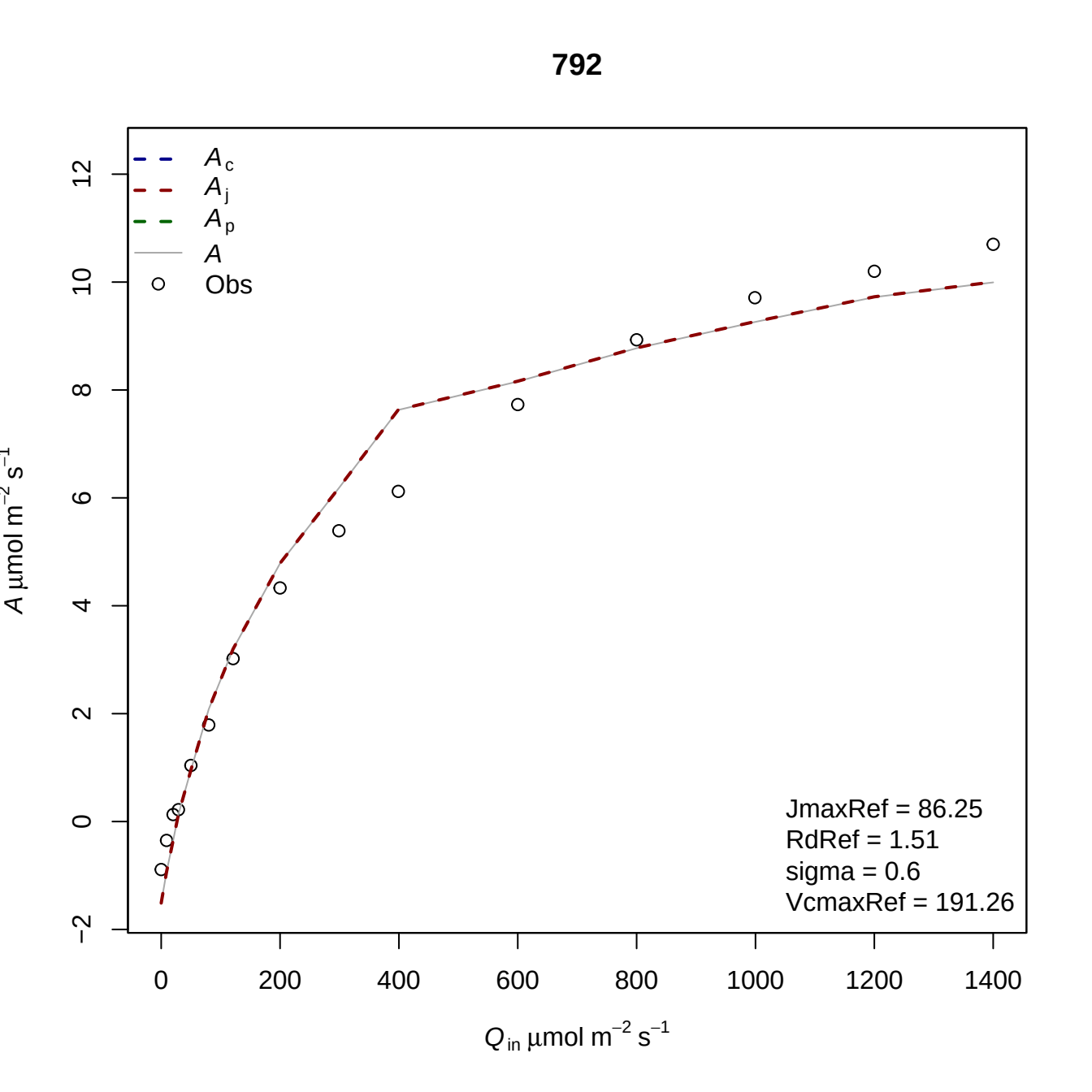

787
A_c
A_j
A_p
A
Obs
JmaxRef = 186.88
RdRef = 1.46
sigma = 0.13
VcmaxRef = 141.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

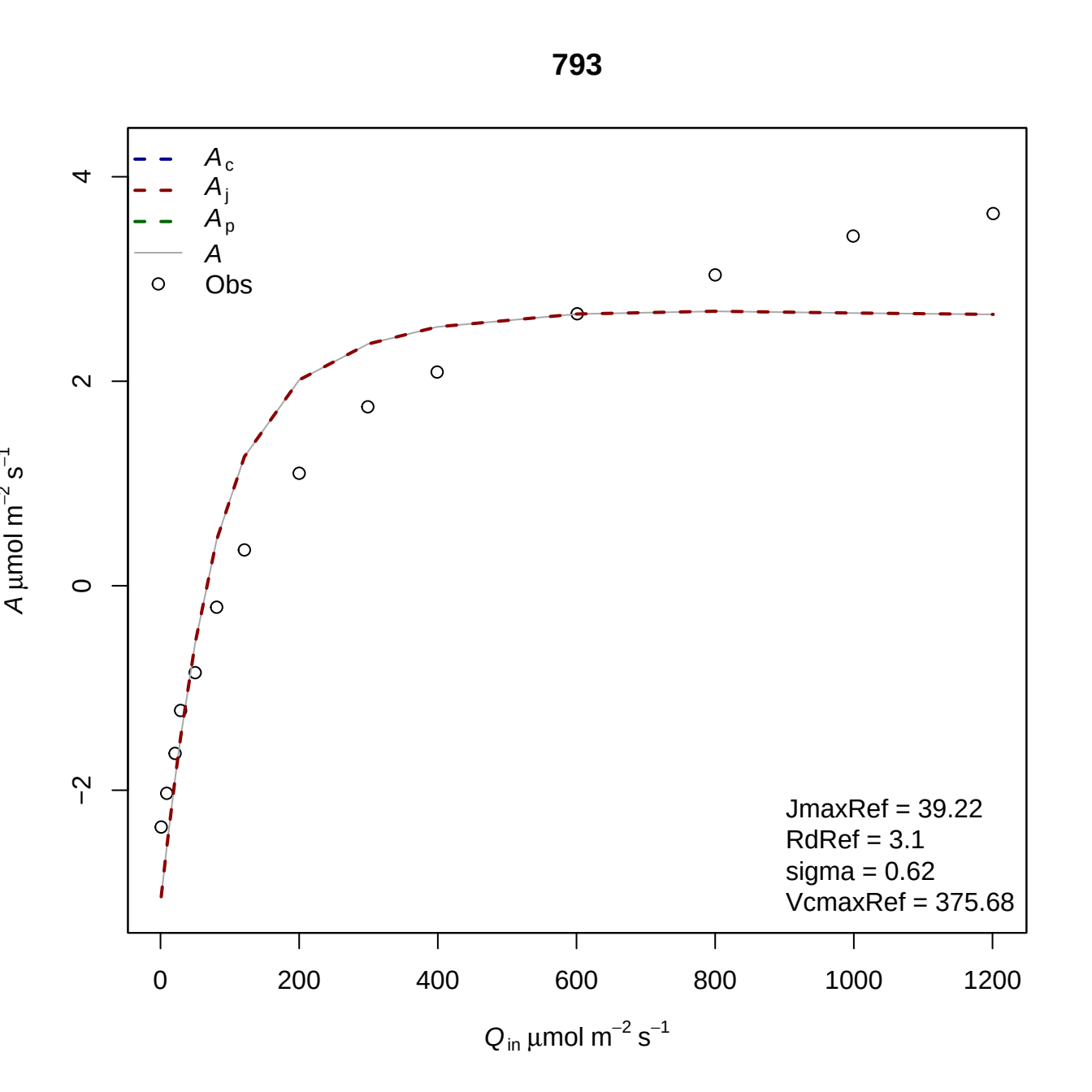

788
A_c
A_j
A_p
A
Obs
JmaxRef = 138.92
RdRef = 2.32
sigma = 0.21
VcmaxRef = 512.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

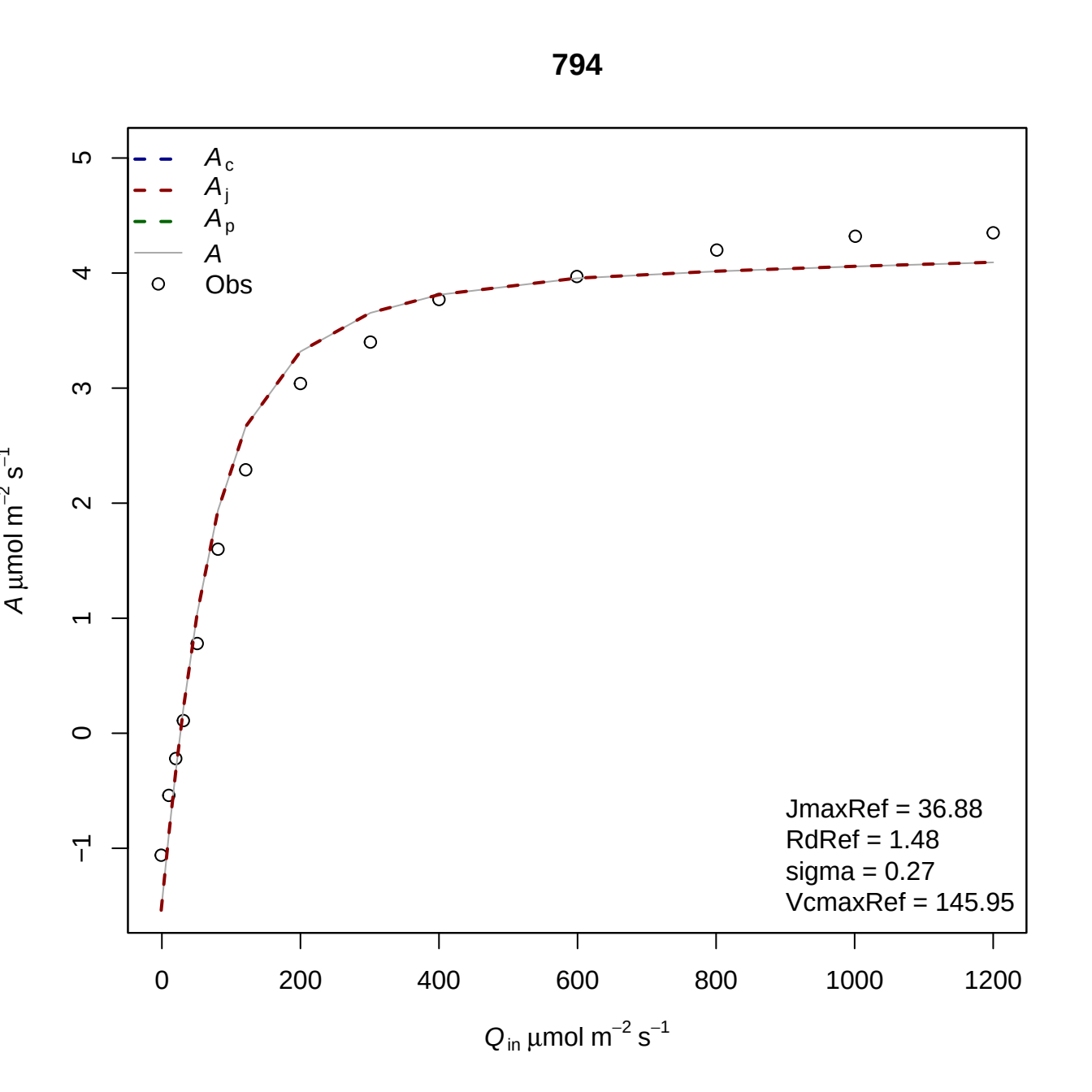

789
A_c
A_j
A_p
A
Obs
JmaxRef = 113.93
RdRef = 2.11
sigma = 0.26
VcmaxRef = 345.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200

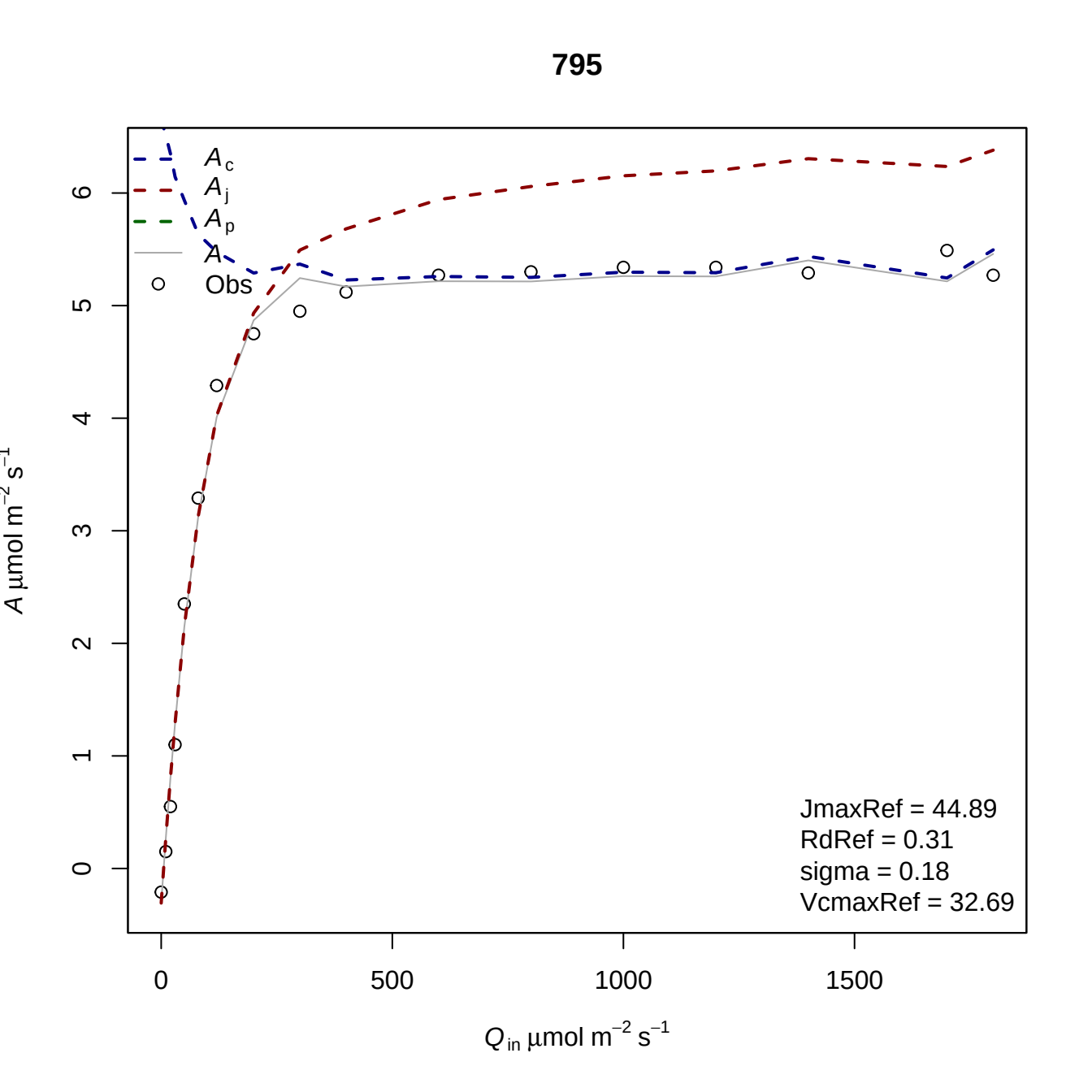

790
A_c
A_j
A_p
A
Obs
JmaxRef = 37.12
RdRef = 1.82
sigma = 0.19
VcmaxRef = 262.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400

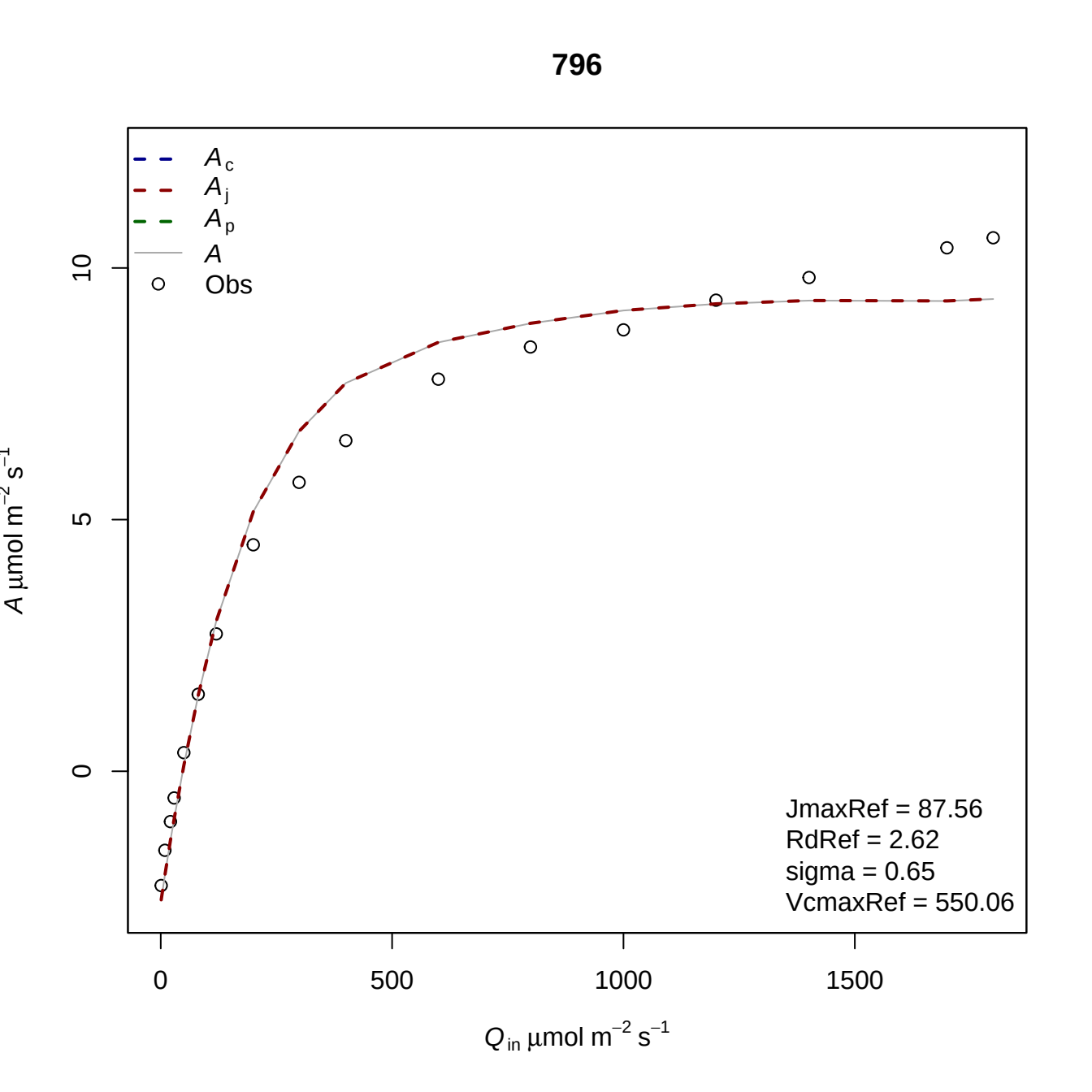

791
A_c
A_j
A_p
A
Obs
JmaxRef = 88.1
RdRef = 2.82
sigma = 0.68
VcmaxRef = 323.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200
1400

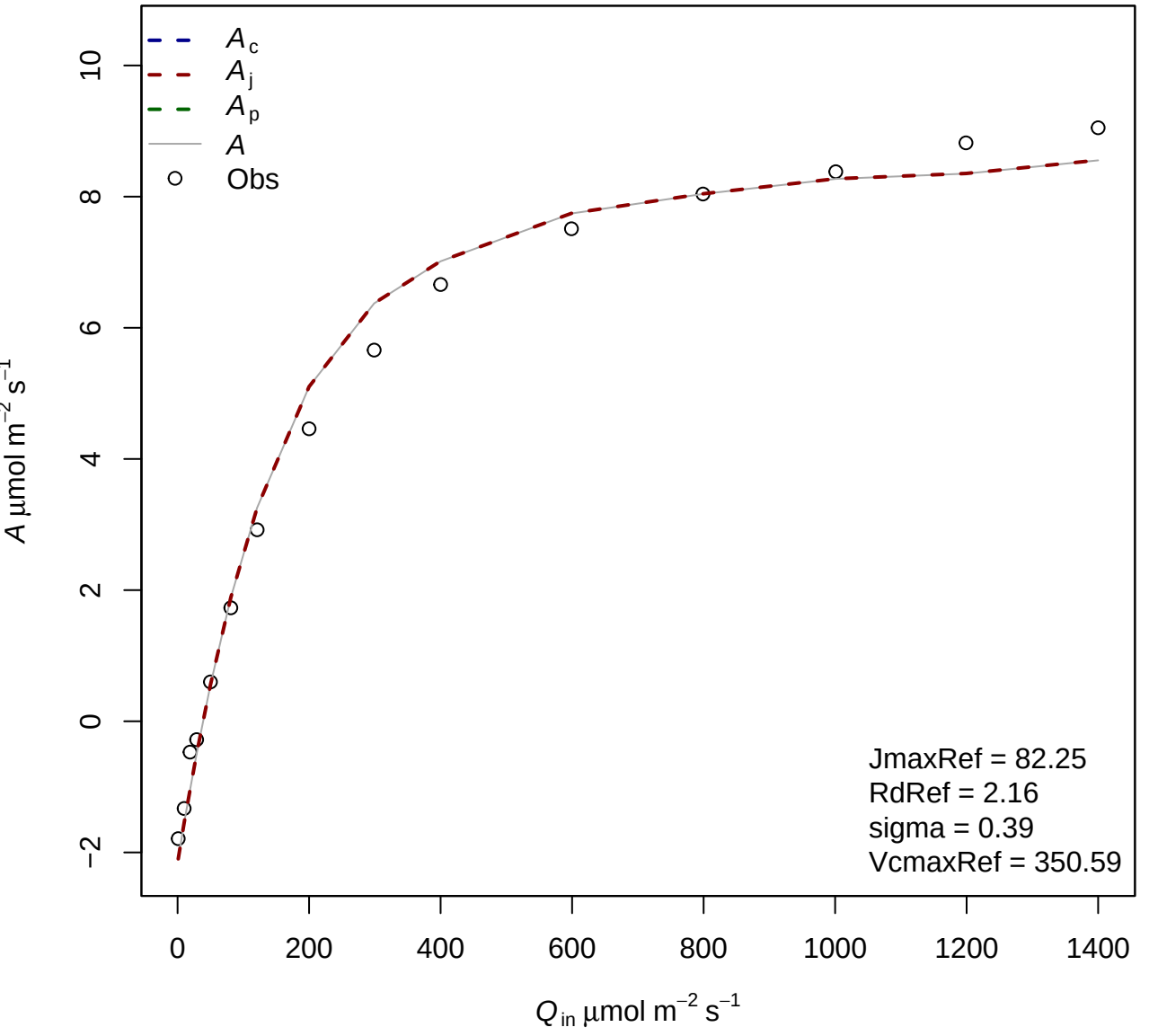

792
A_c
A_j
A_p
A
Obs
JmaxRef = 86.25
RdRef = 1.51
sigma = 0.6
VcmaxRef = 191.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

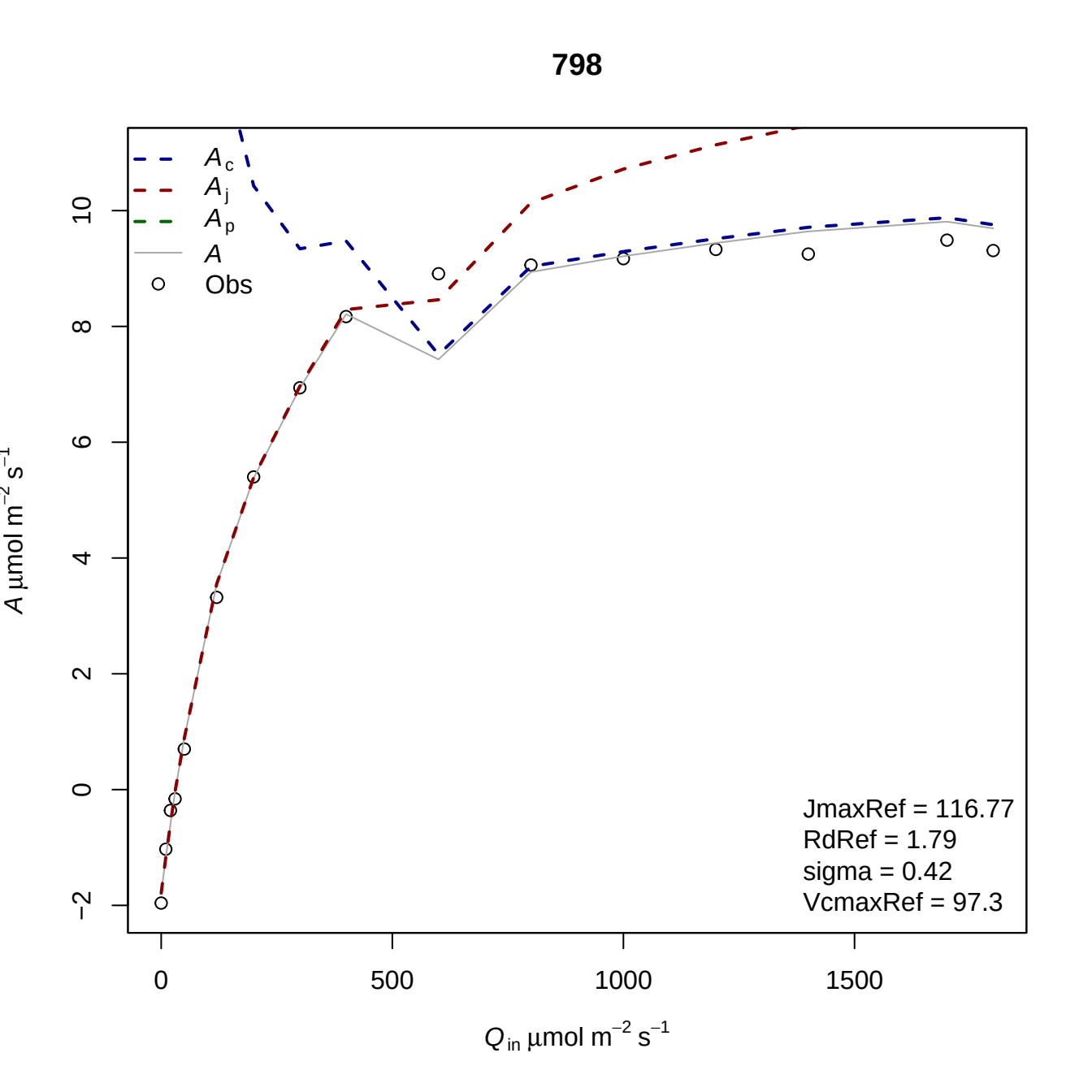

793
A_c
A_j
A_p
A
Obs
JmaxRef = 39.22
RdRef = 3.1
sigma = 0.62
VcmaxRef = 375.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
0
200
400
600
800
1000
1200

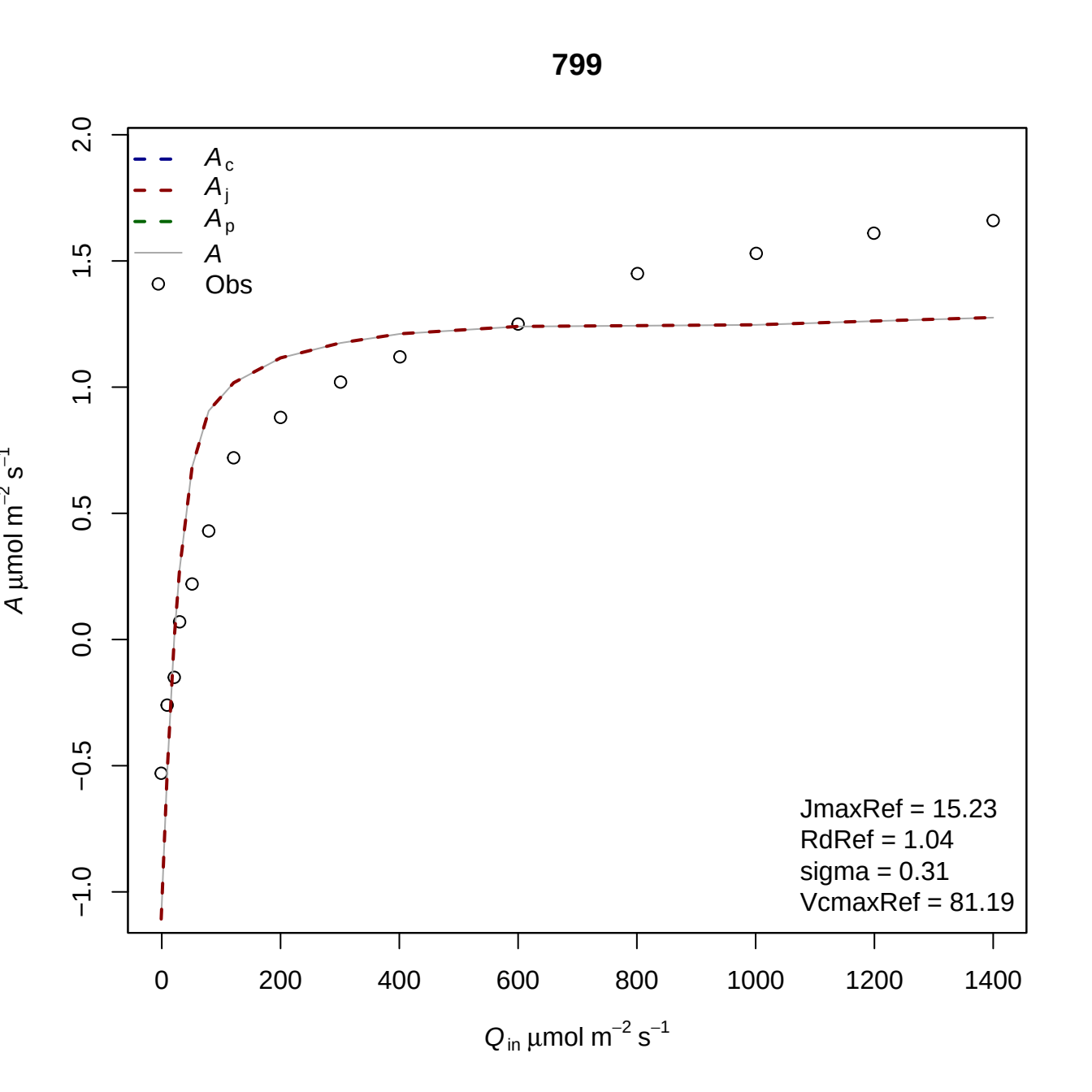

794
A_c
A_j
A_p
A
Obs
JmaxRef = 36.88
RdRef = 1.48
sigma = 0.27
VcmaxRef = 145.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
0
200
400
600
800
1000
1200

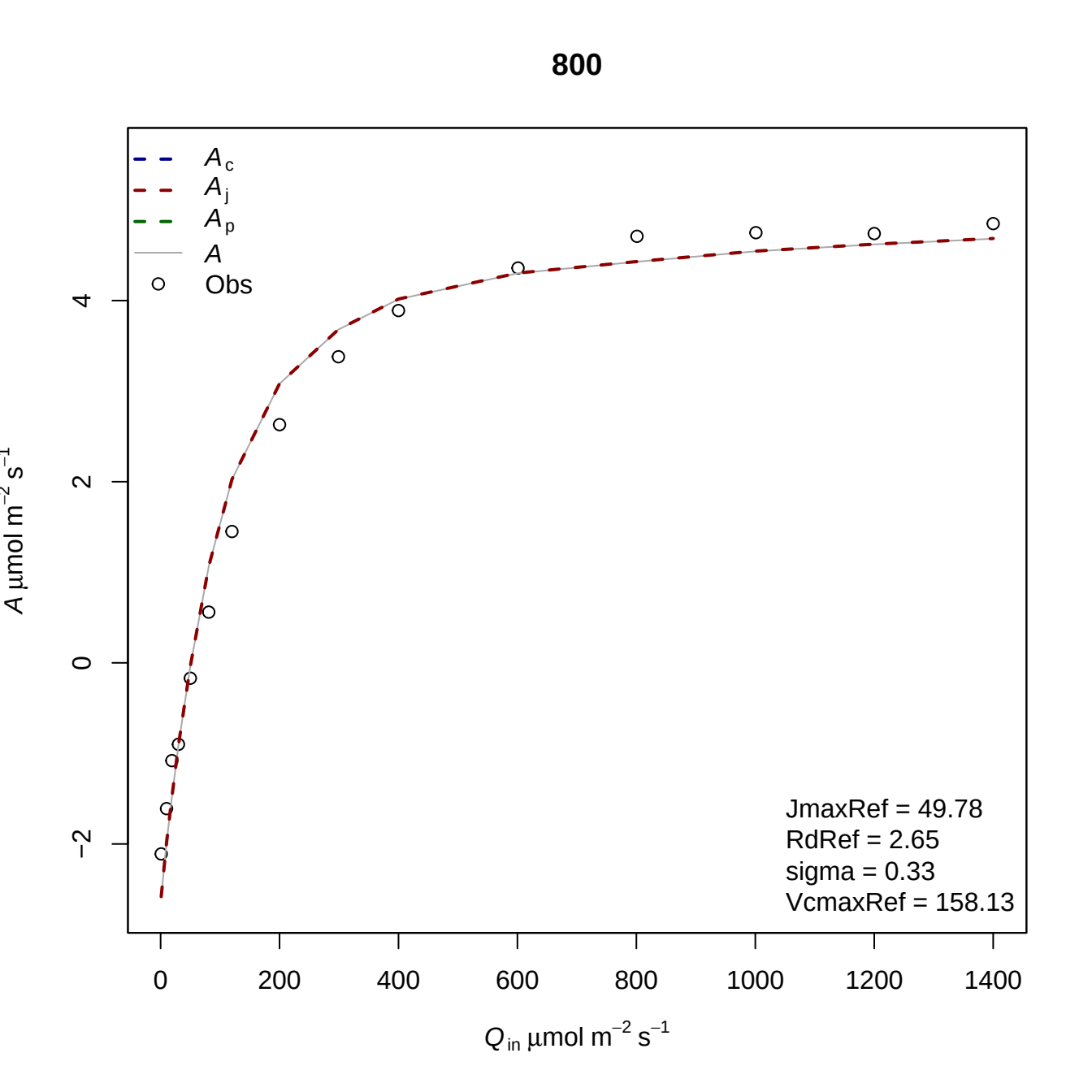

795
A_c
A_j
A_p
A
Obs
JmaxRef = 44.89
RdRef = 0.31
sigma = 0.18
VcmaxRef = 32.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

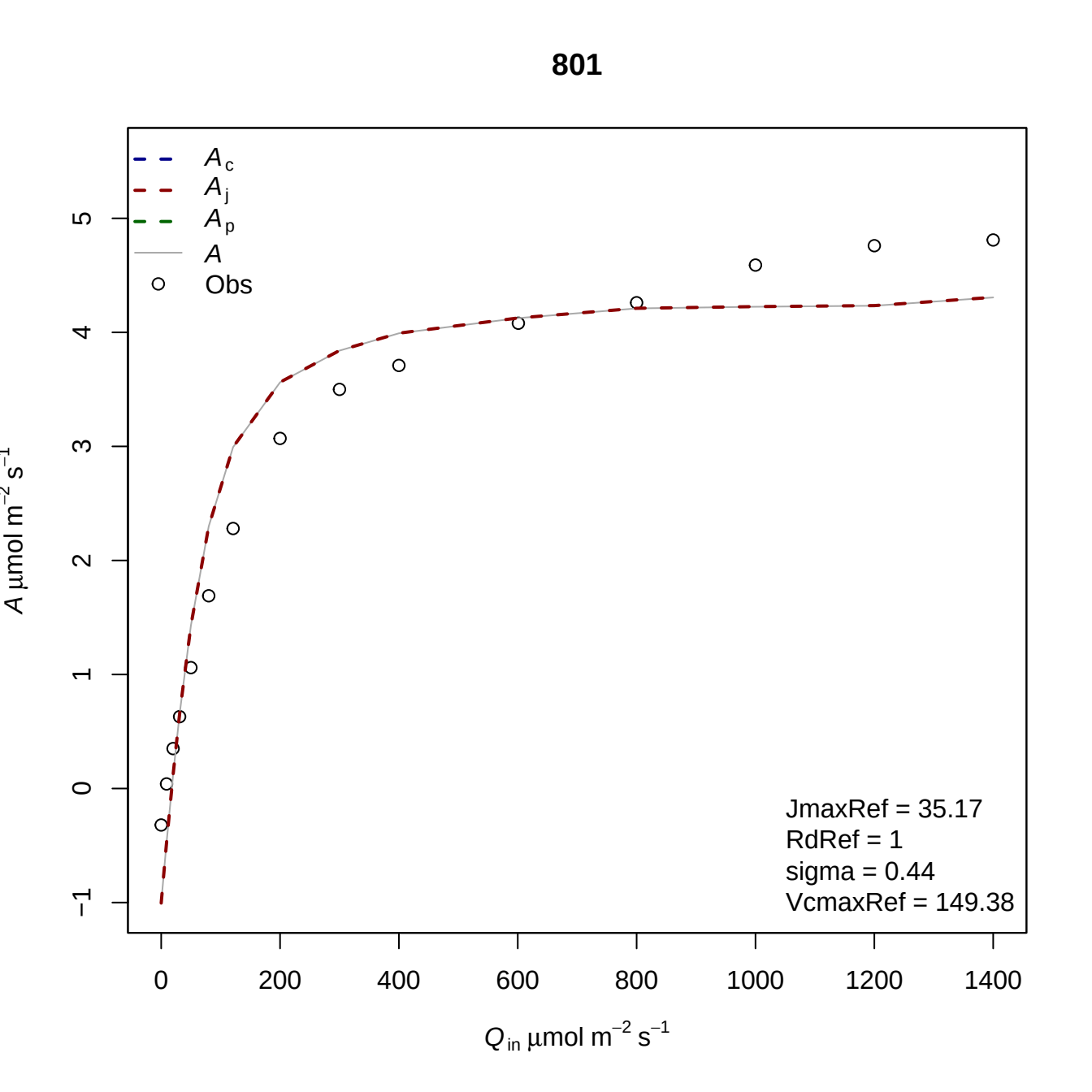

796
A_c
A_j
A_p
A
Obs
JmaxRef = 87.56
RdRef = 2.62
sigma = 0.65
VcmaxRef = 550.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

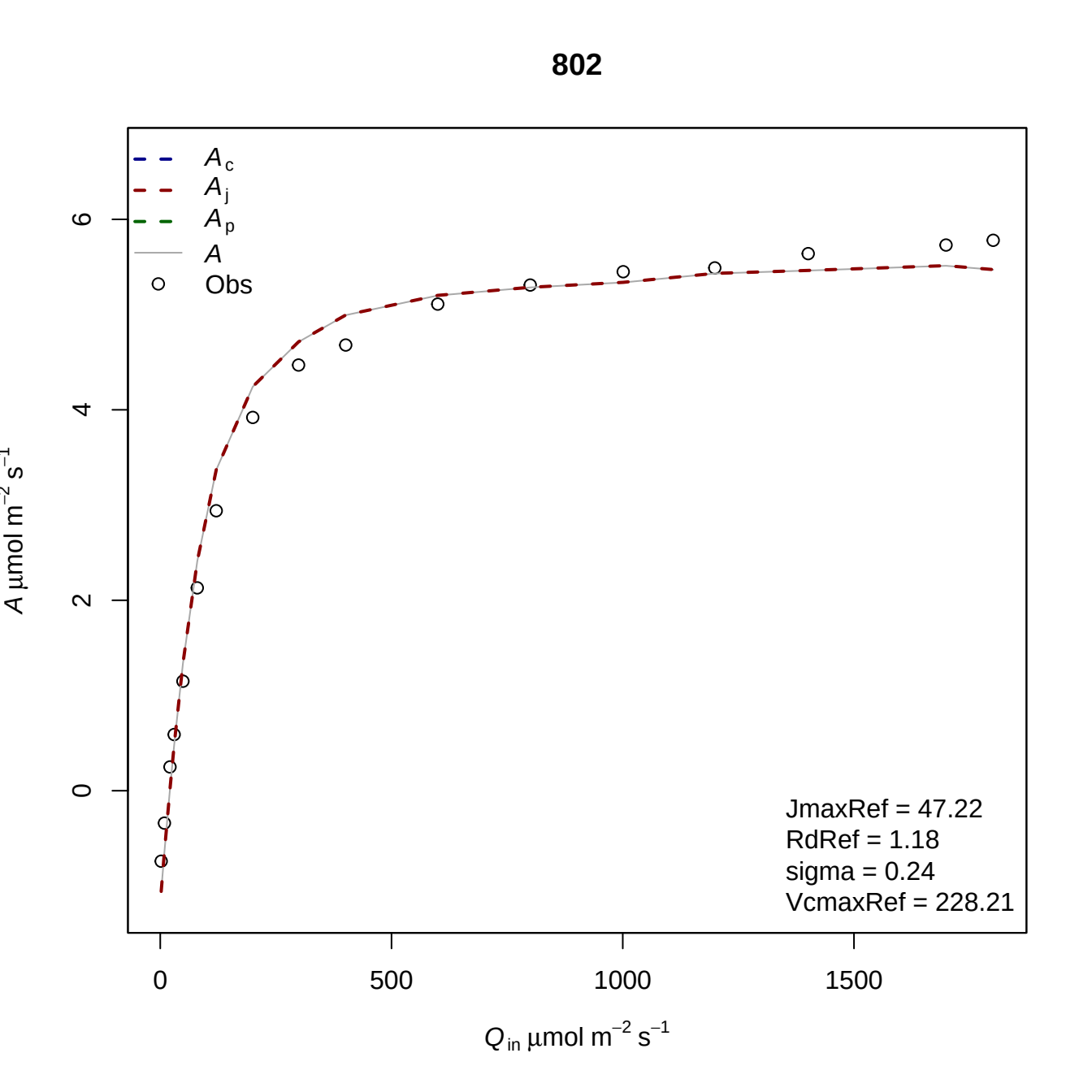

A_c
A_j
A_p
A
Obs
JmaxRef = 82.25
RdRef = 2.16
sigma = 0.39
VcmaxRef = 350.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

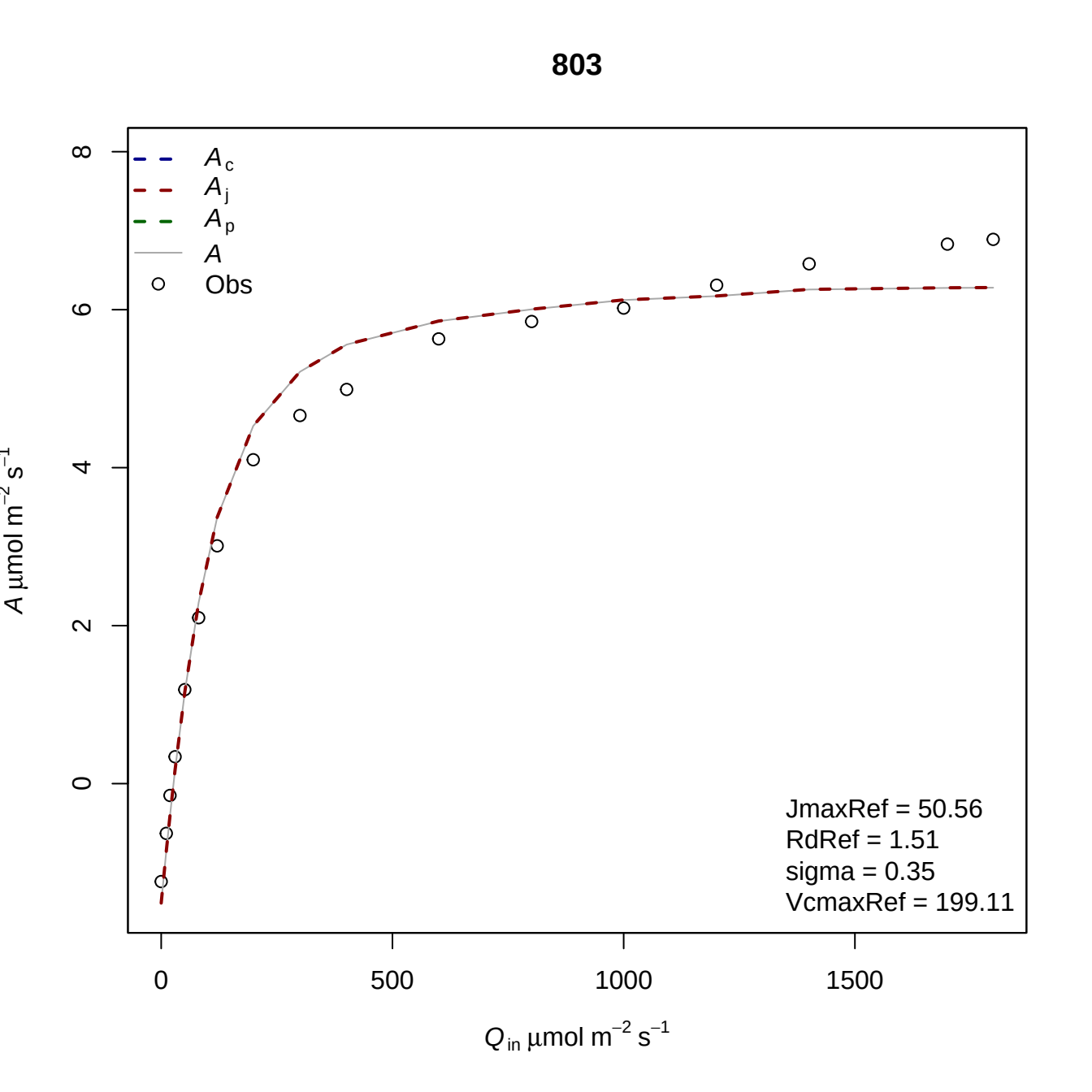

798
A_c
A_j
A_p
A
Obs
JmaxRef = 116.77
RdRef = 1.79
sigma = 0.42
VcmaxRef = 97.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

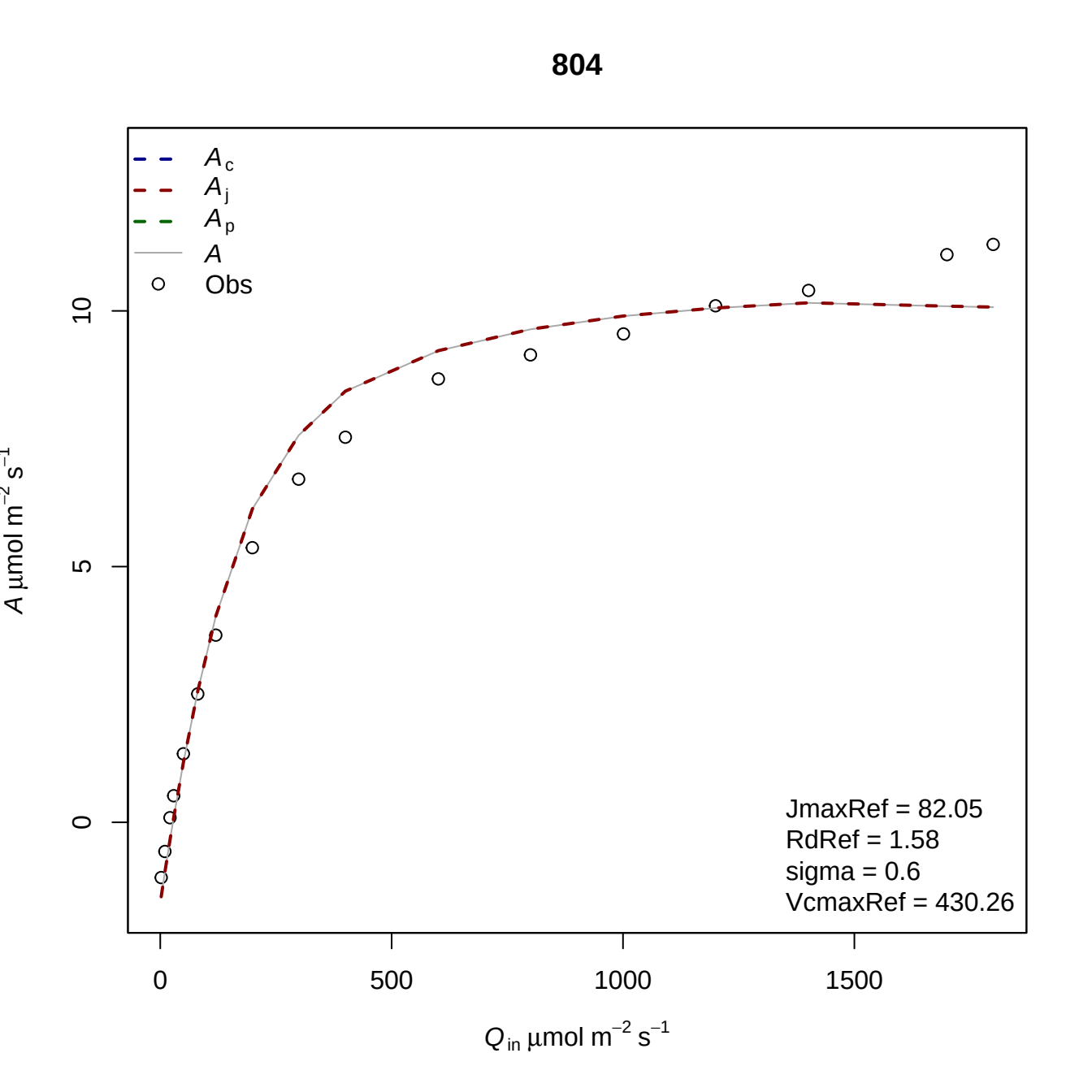

799
A_c
A_j
A_p
A
Obs
JmaxRef = 15.23
RdRef = 1.04
sigma = 0.31
VcmaxRef = 81.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
200
400
600
800
1000
1200
1400

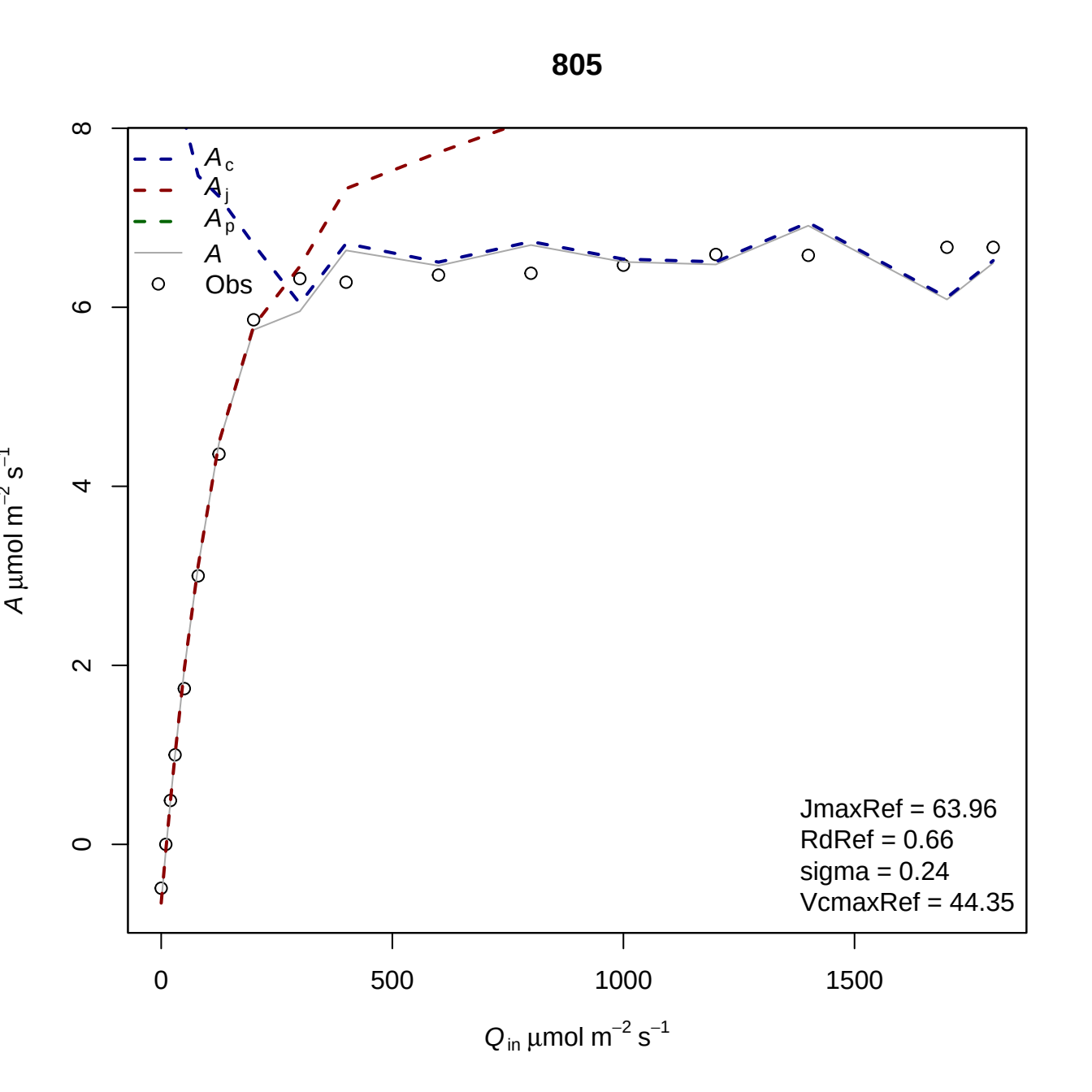

800
A_c
A_j
A_p
A
Obs
JmaxRef = 49.78
RdRef = 2.65
sigma = 0.33
VcmaxRef = 158.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200
1400

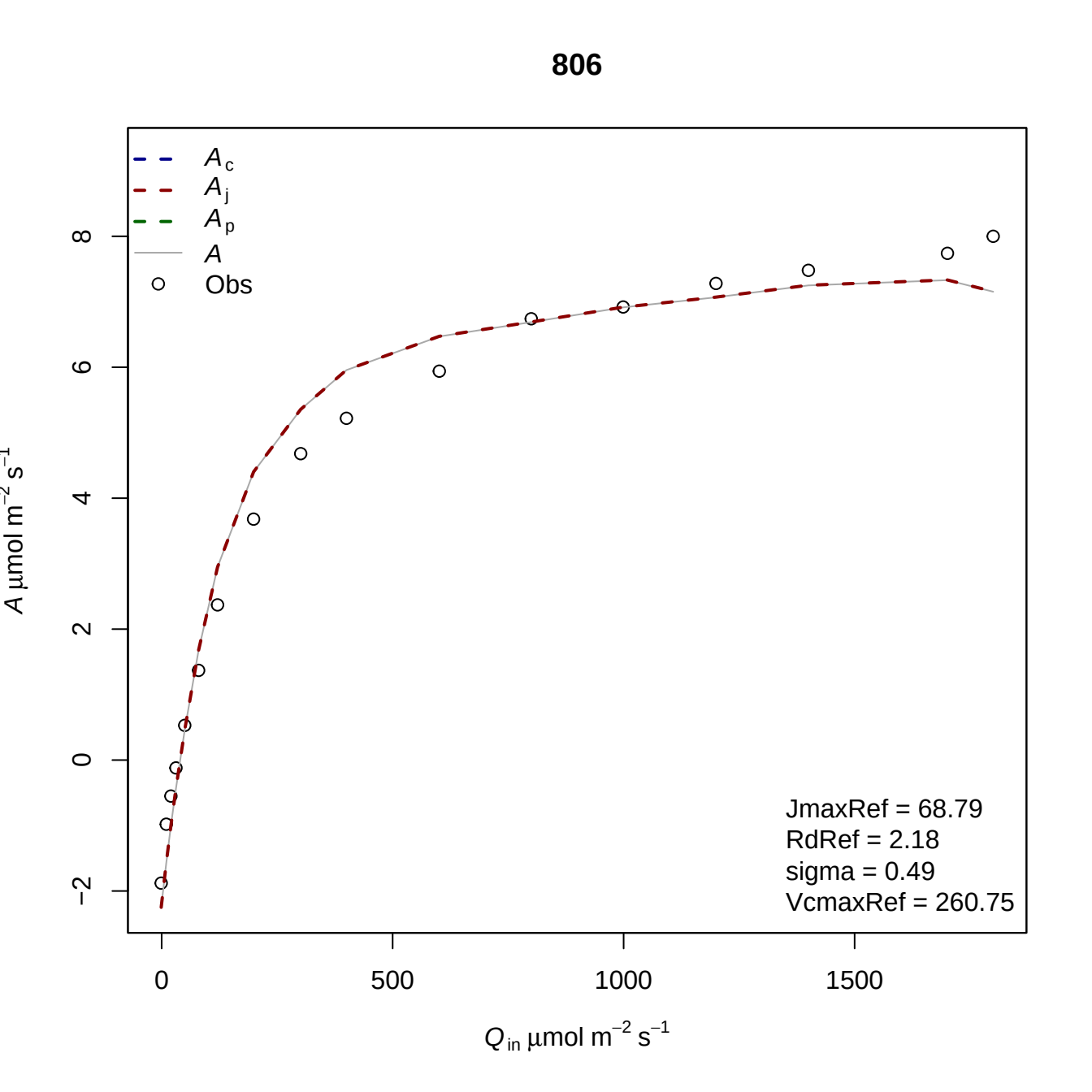

801
A_c
A_j
A_p
A
Obs
JmaxRef = 35.17
RdRef = 1
sigma = 0.44
VcmaxRef = 149.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000
1200
1400

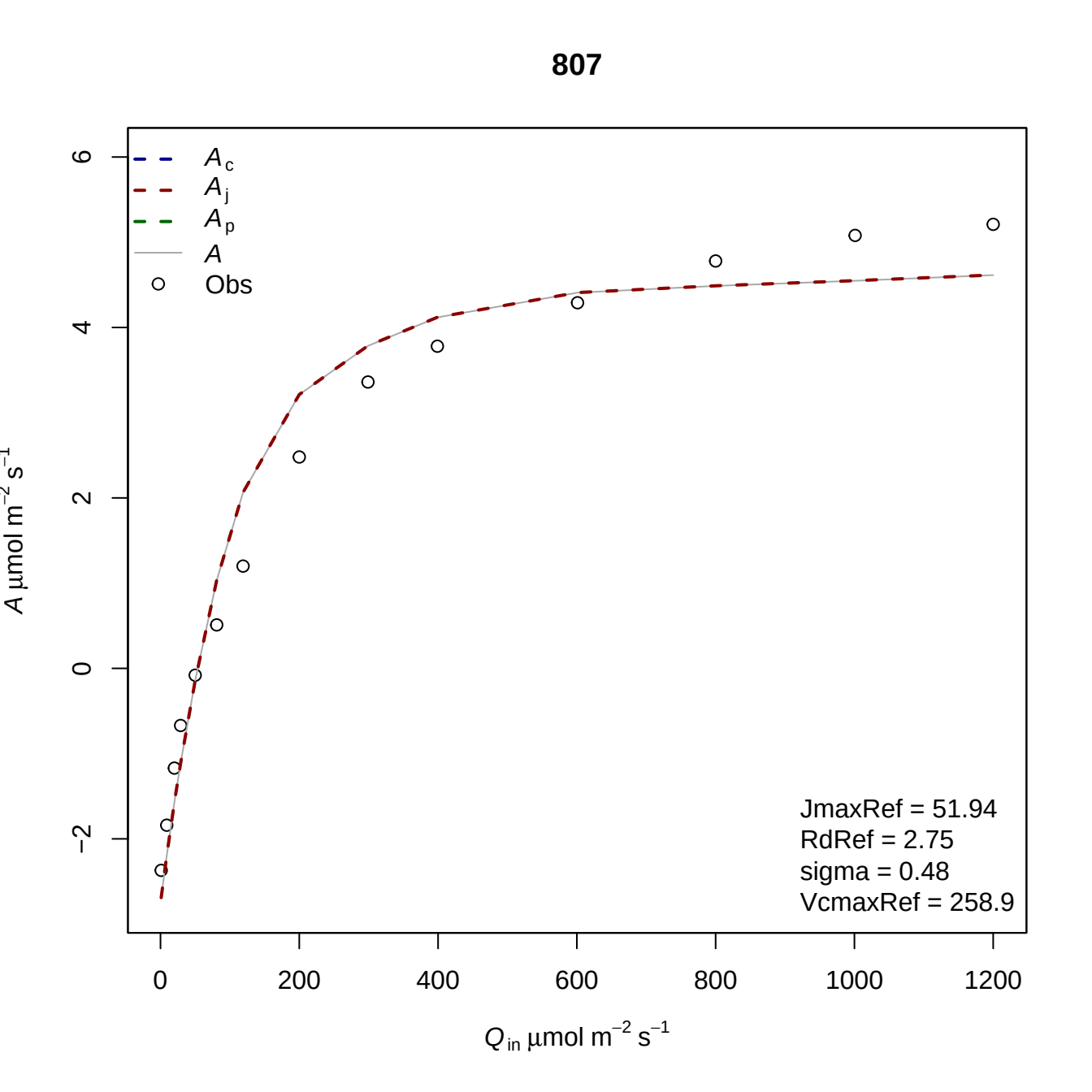

802
A_c
A_j
A_p
A
Obs
JmaxRef = 47.22
RdRef = 1.18
sigma = 0.24
VcmaxRef = 228.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

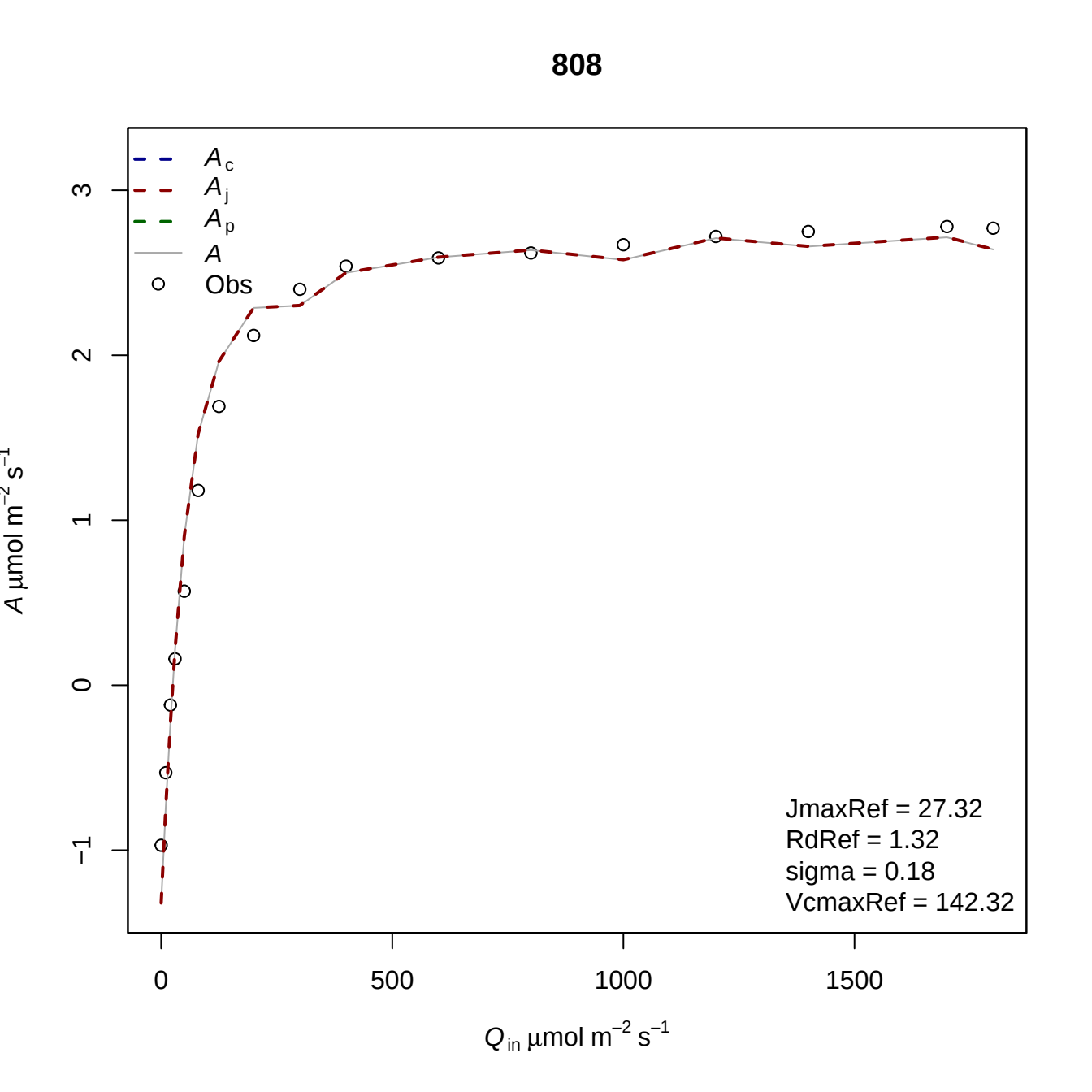

803
A_c
A_j
A_p
A
Obs
JmaxRef = 50.56
RdRef = 1.51
sigma = 0.35
VcmaxRef = 199.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

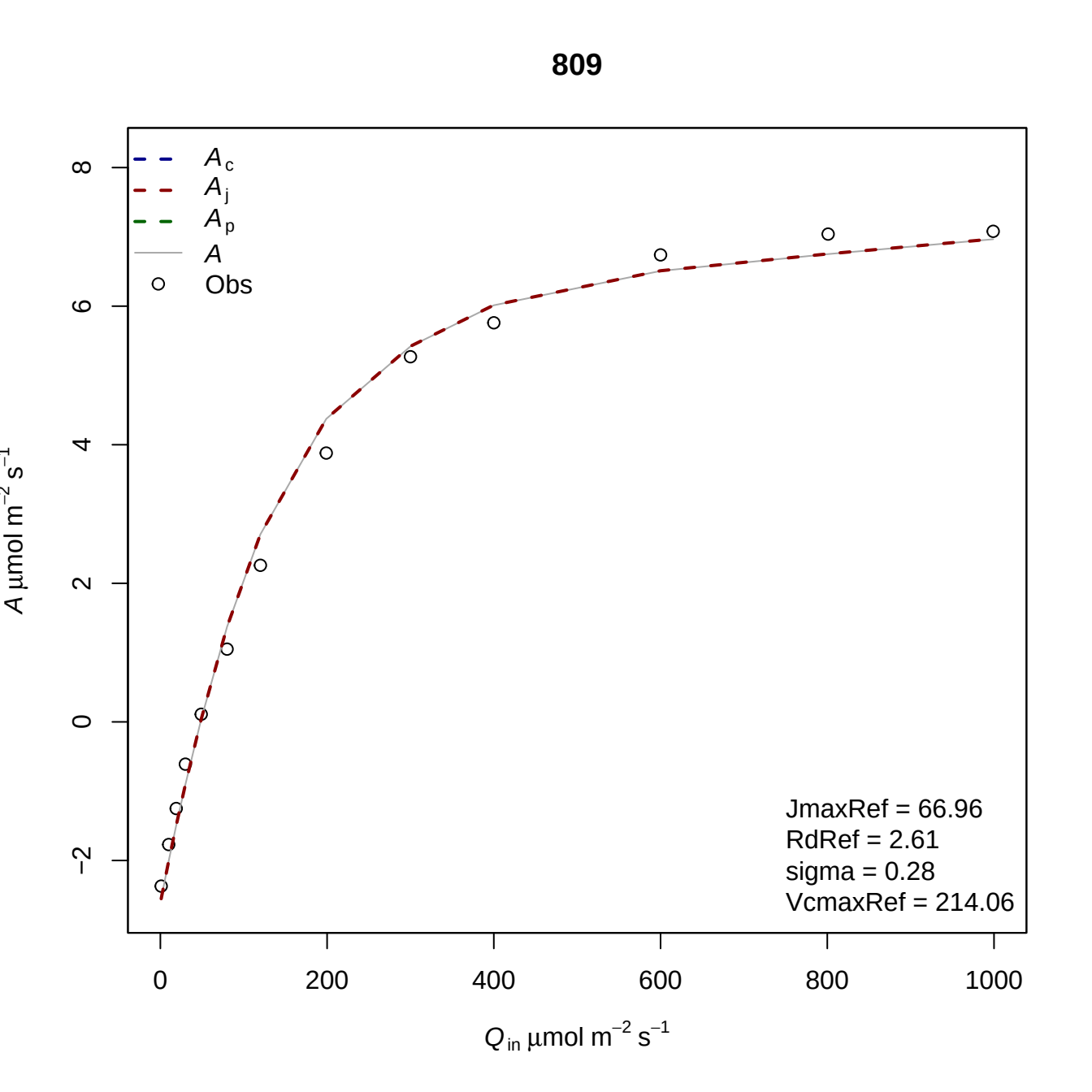

804
A_c
A_j
A_p
A
Obs
JmaxRef = 82.05
RdRef = 1.58
sigma = 0.6
VcmaxRef = 430.26
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

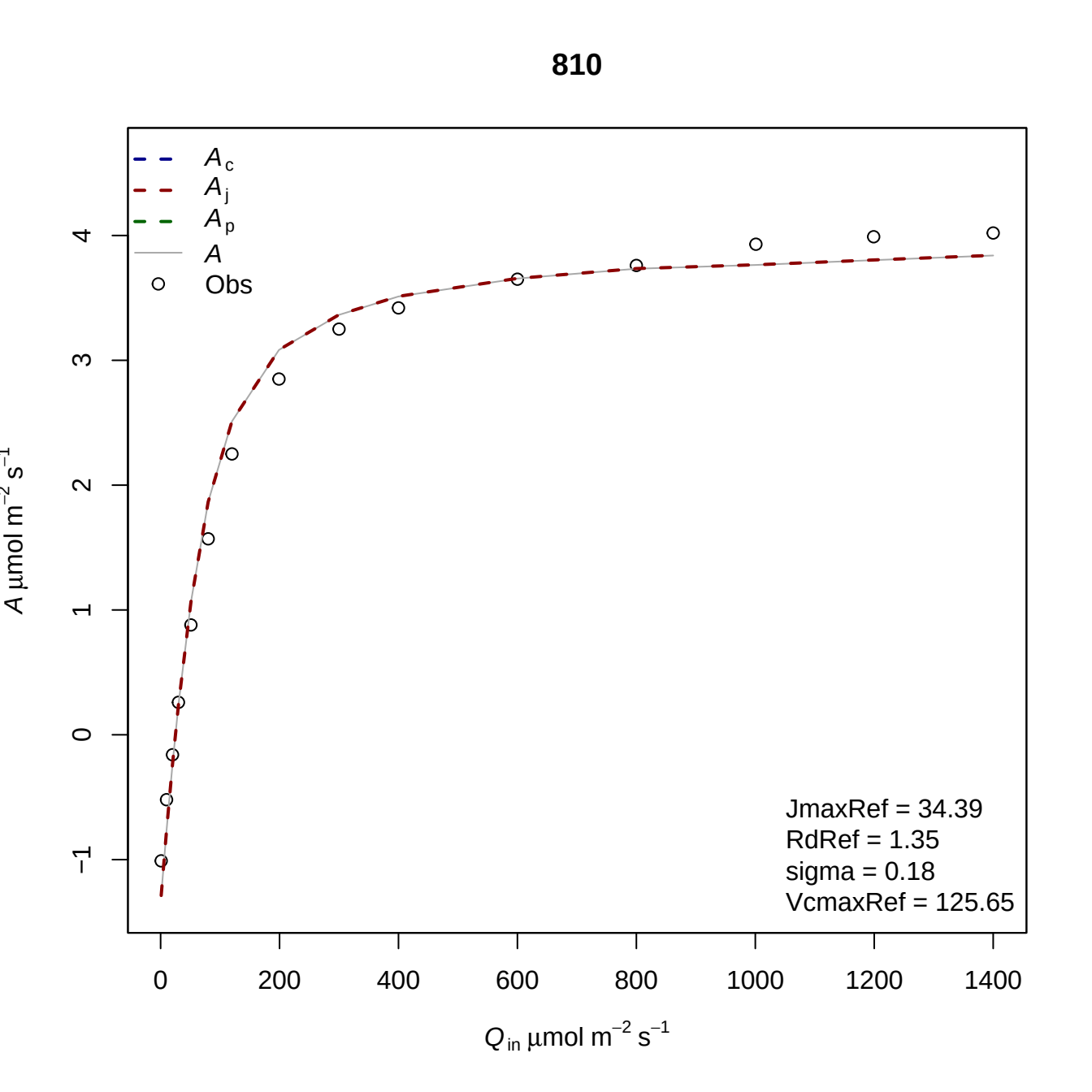

805
A_c
A_j
A_p
A
Obs
JmaxRef = 63.96
RdRef = 0.66
sigma = 0.24
VcmaxRef = 44.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

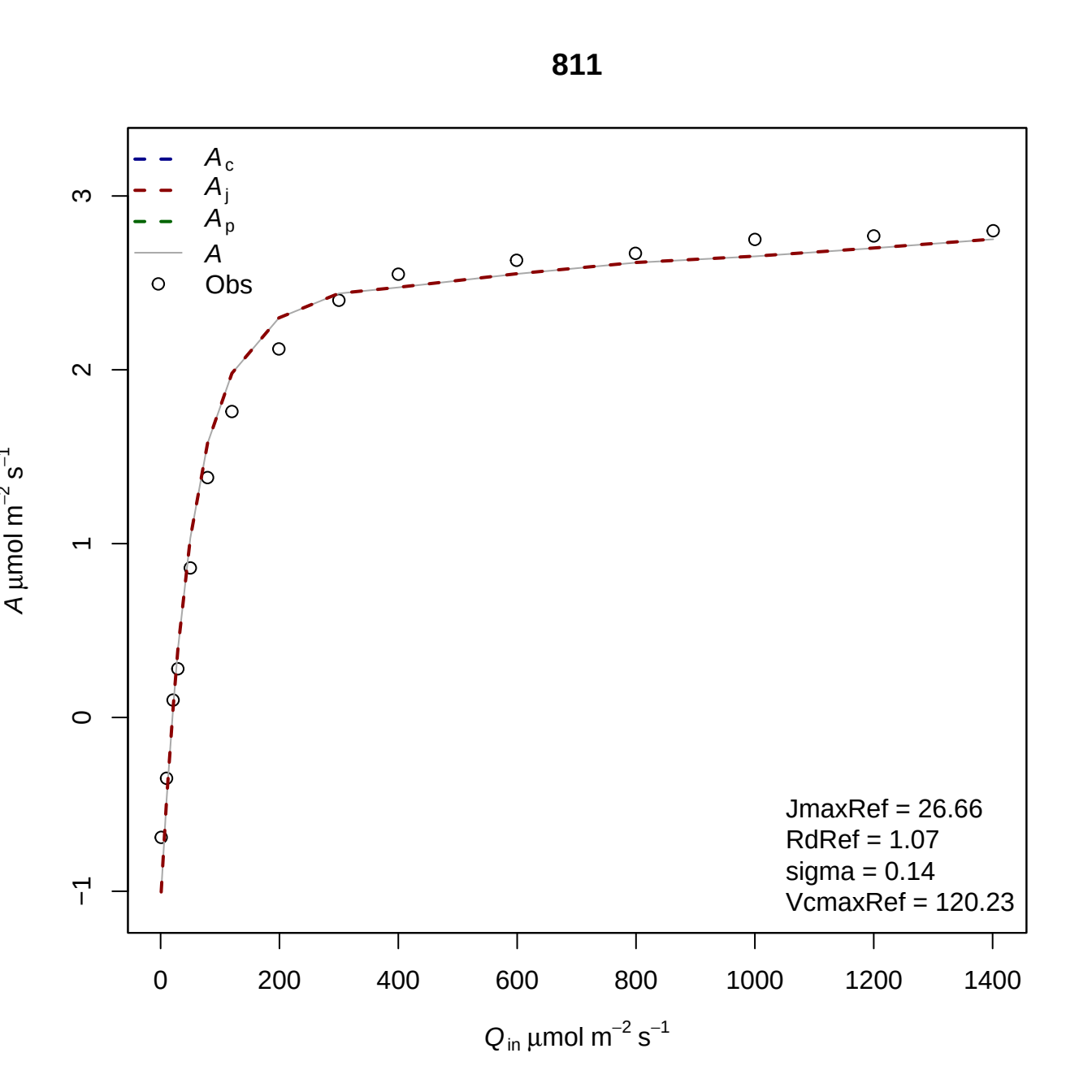

806
A_c
A_j
A_p
A
Obs
JmaxRef = 68.79
RdRef = 2.18
sigma = 0.49
VcmaxRef = 260.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

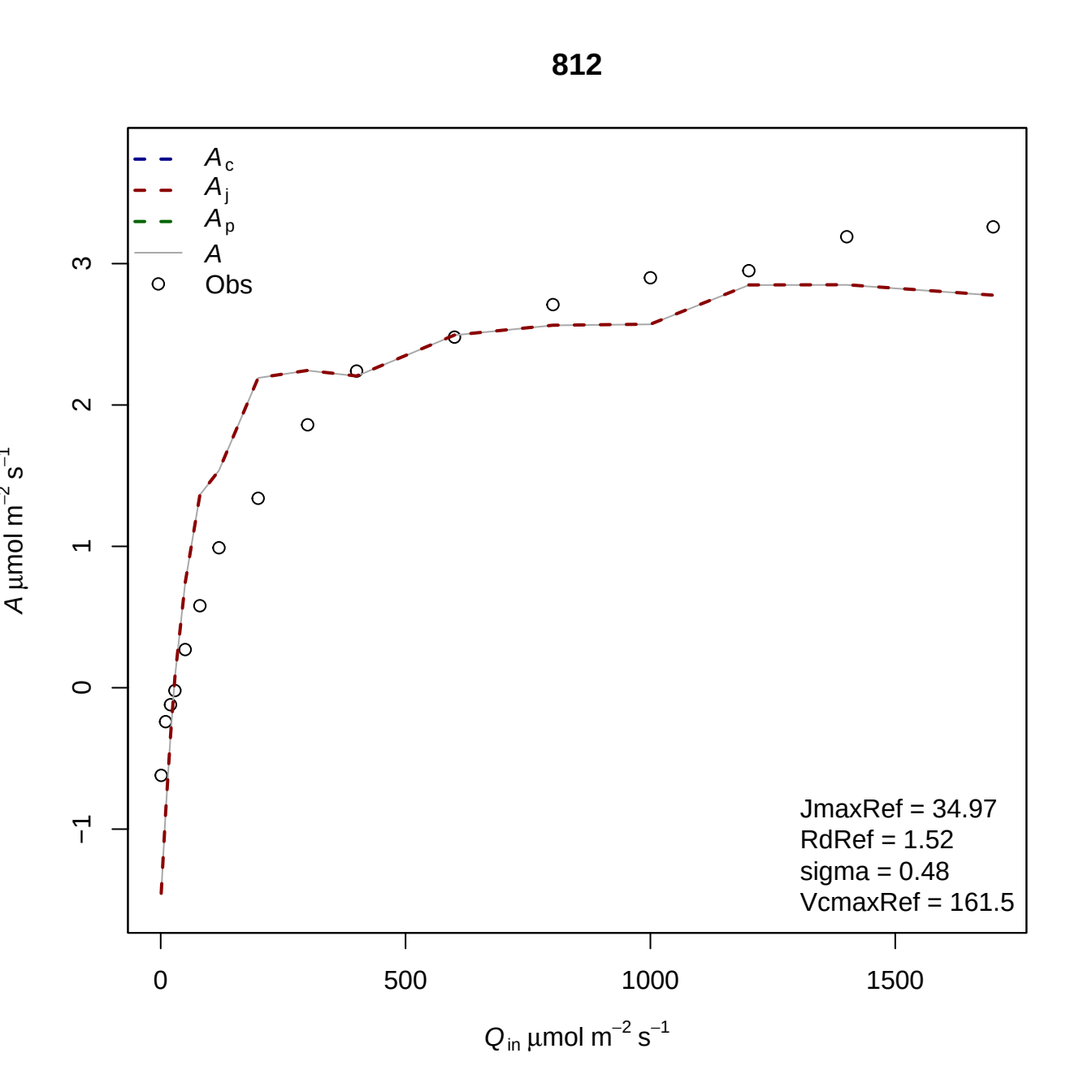

807
A_c
A_j
A_p
A
Obs
JmaxRef = 51.94
RdRef = 2.75
sigma = 0.48
VcmaxRef = 258.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

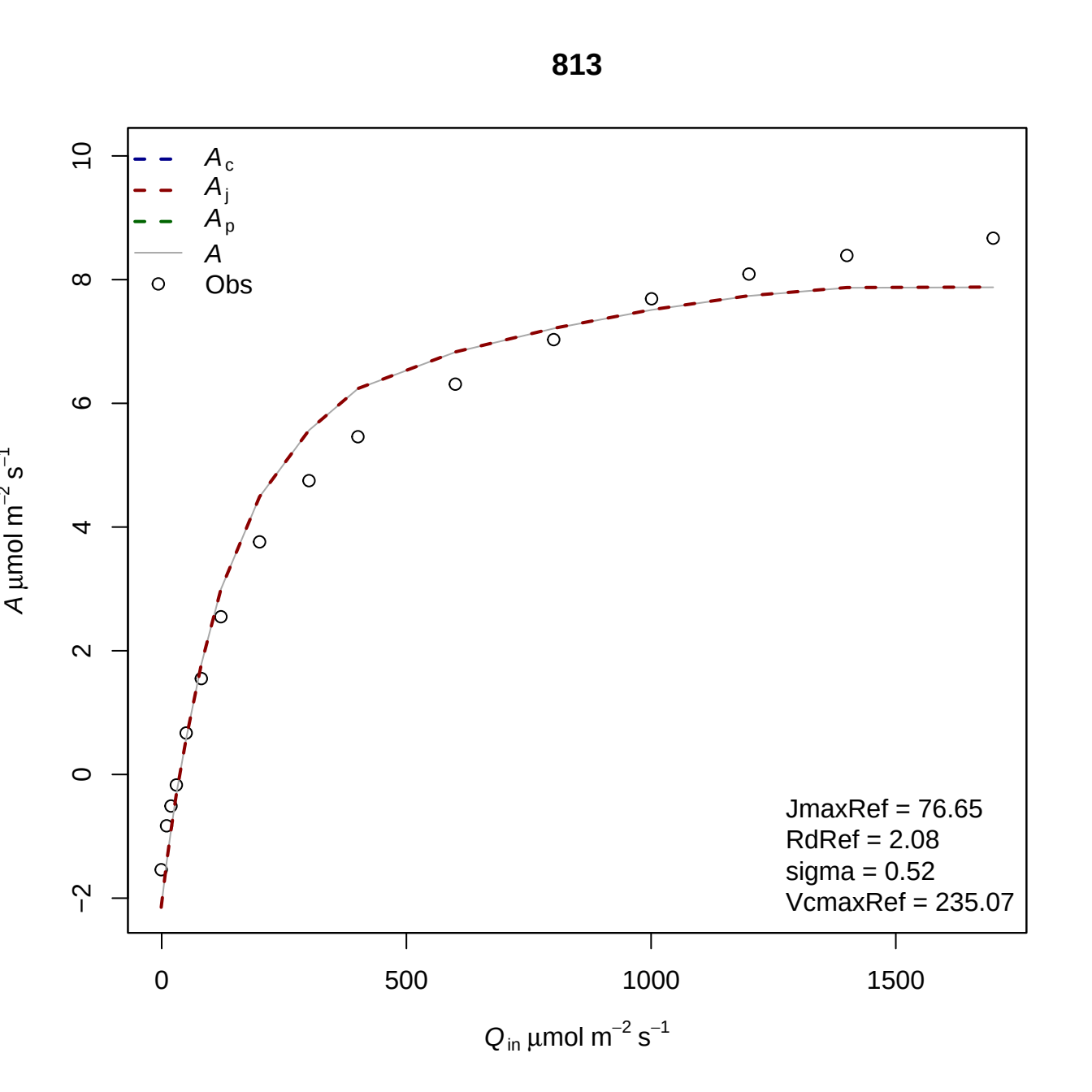

808
A_c
A_j
A_p
A
Obs
JmaxRef = 27.32
RdRef = 1.32
sigma = 0.18
VcmaxRef = 142.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

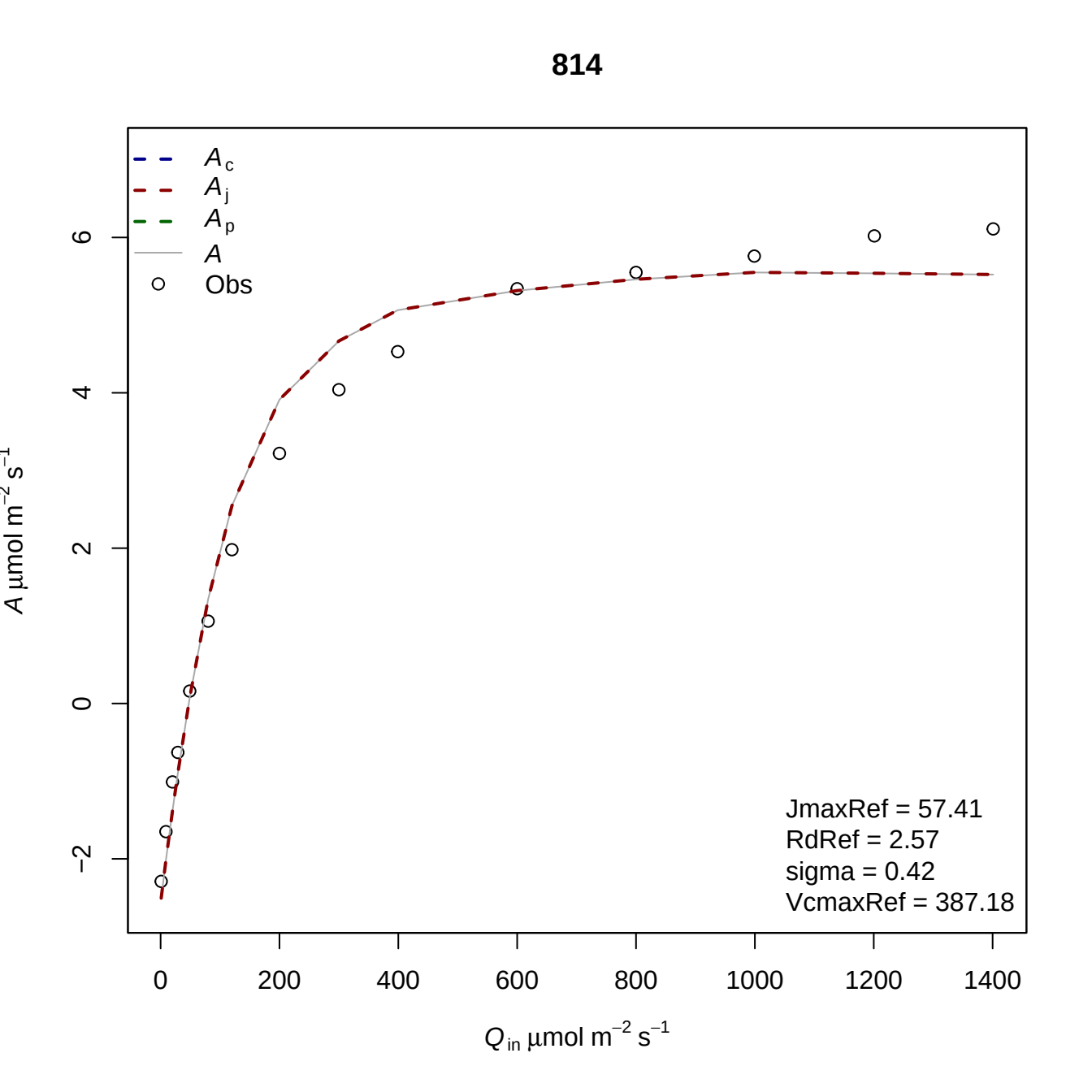

809
A_c
A_j
A_p
A
Obs
JmaxRef = 66.96
RdRef = 2.61
sigma = 0.28
VcmaxRef = 214.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
200
400
600
800
1000

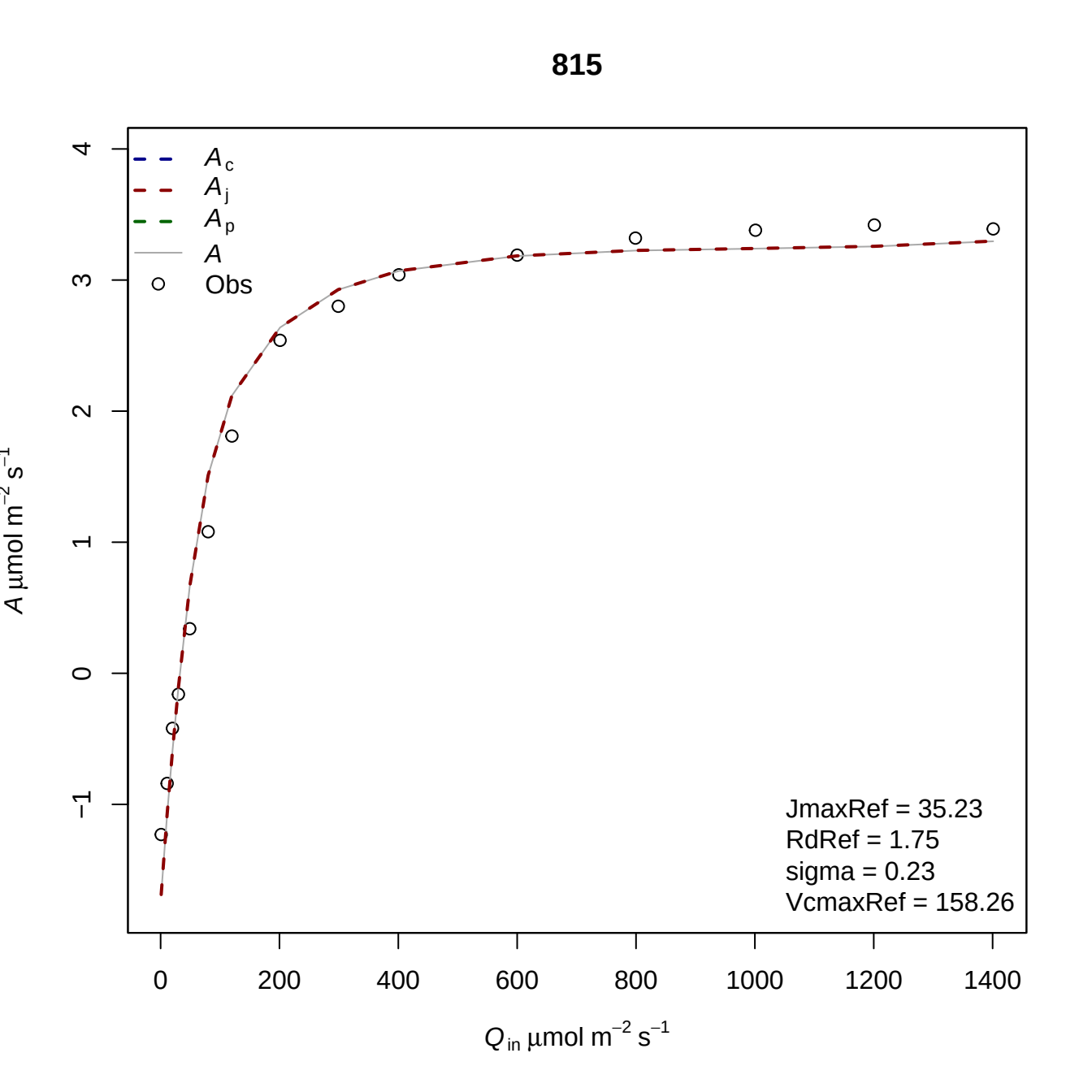

810
A_c
A_j
A_p
A
Obs
JmaxRef = 34.39
RdRef = 1.35
sigma = 0.18
VcmaxRef = 125.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

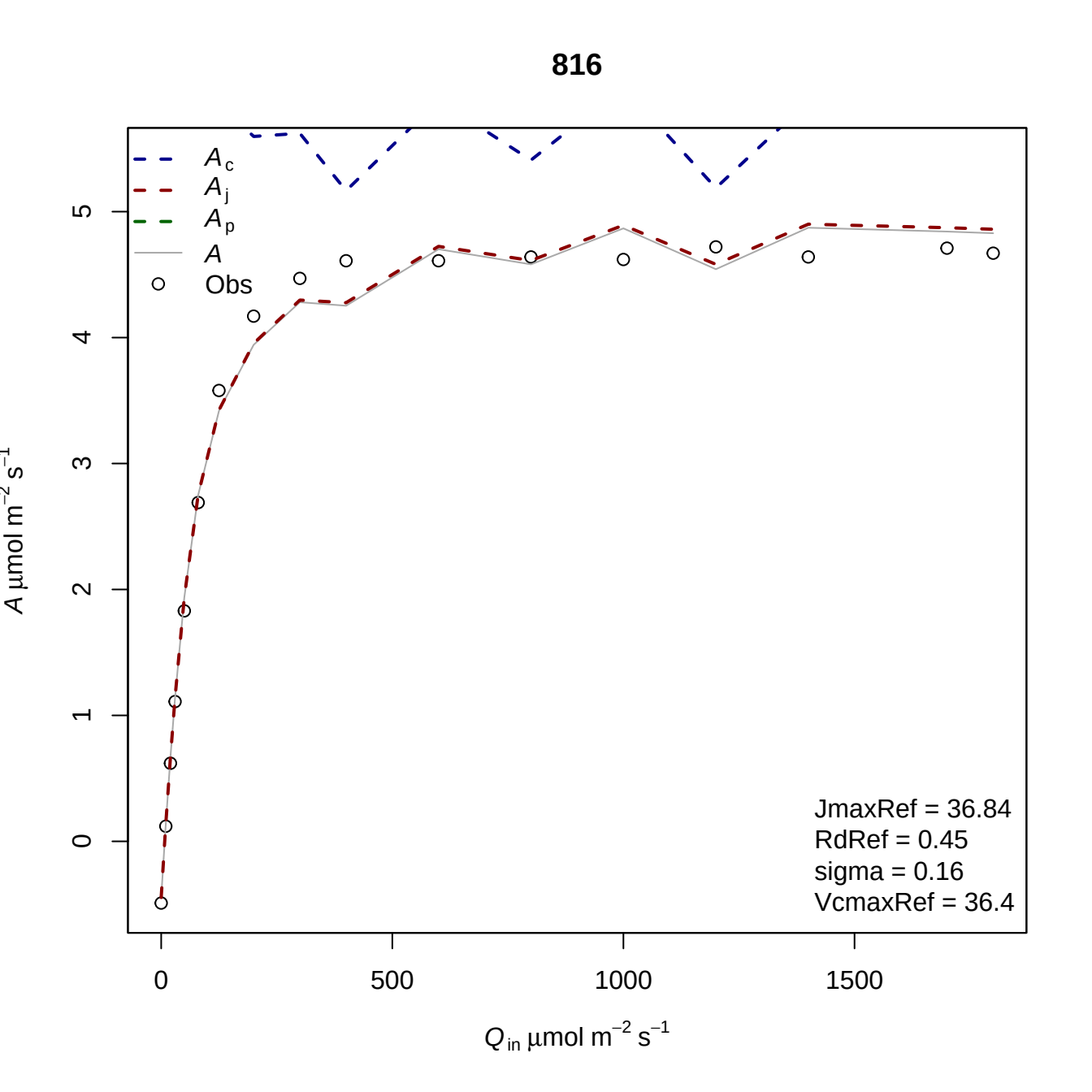

811
A_c
A_j
A_p
A
Obs
JmaxRef = 26.66
RdRef = 1.07
sigma = 0.14
VcmaxRef = 120.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

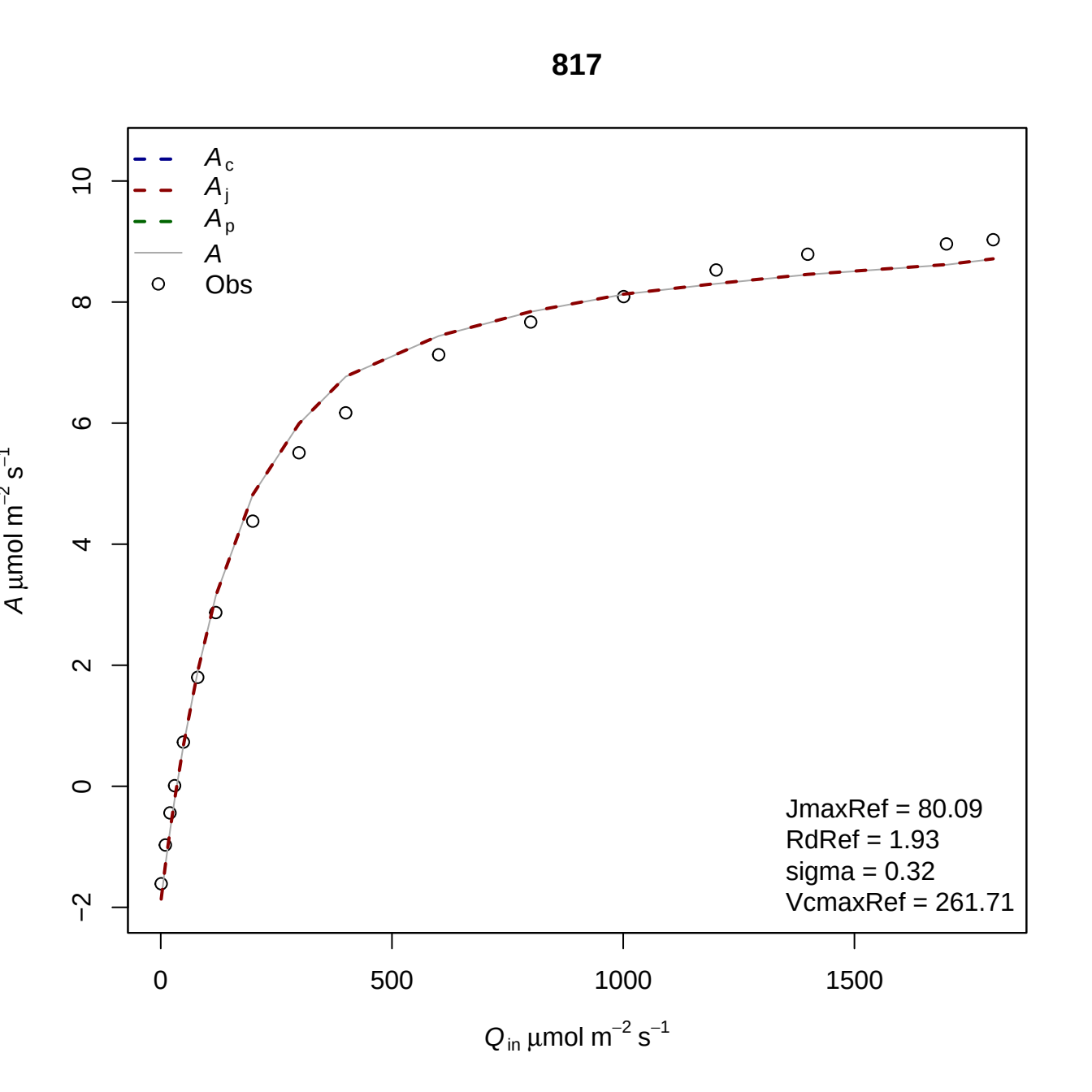

812
A_c
A_j
A_p
A
Obs
JmaxRef = 34.97
RdRef = 1.52
sigma = 0.48
VcmaxRef = 161.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

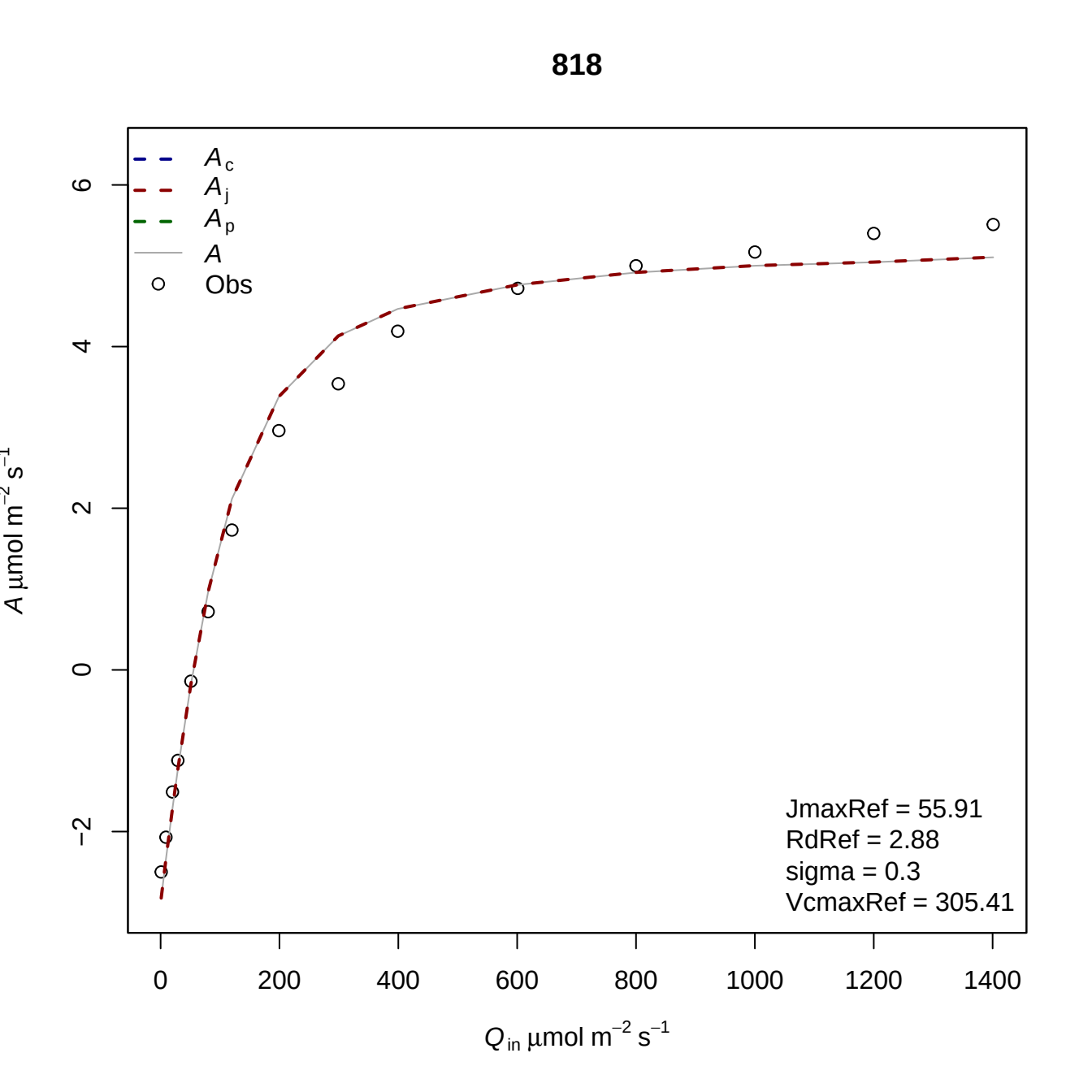

813
A_c
A_j
A_p
A
Obs
JmaxRef = 76.65
RdRef = 2.08
sigma = 0.52
VcmaxRef = 235.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

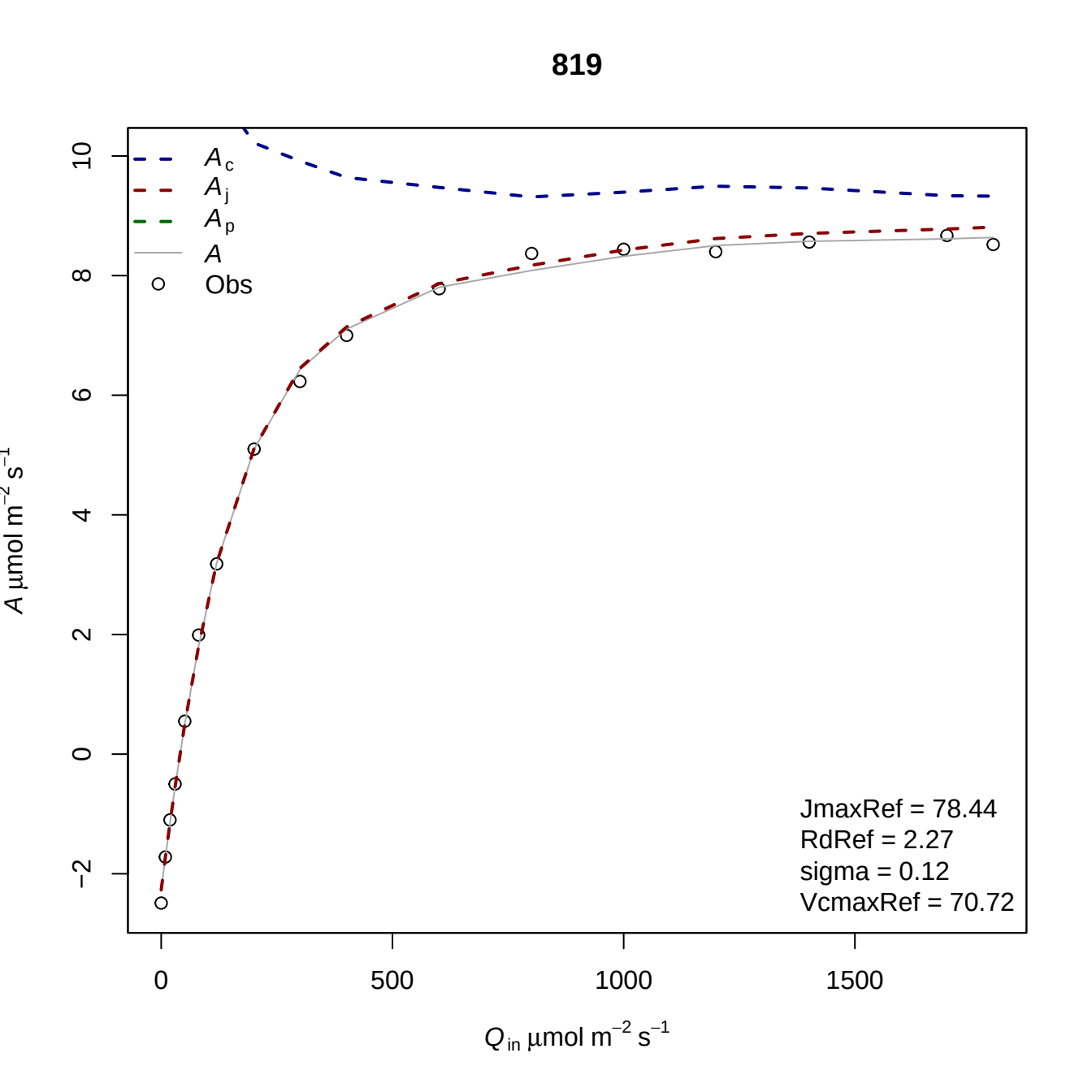

814
A_c
A_j
A_p
A
Obs
JmaxRef = 57.41
RdRef = 2.57
sigma = 0.42
VcmaxRef = 387.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

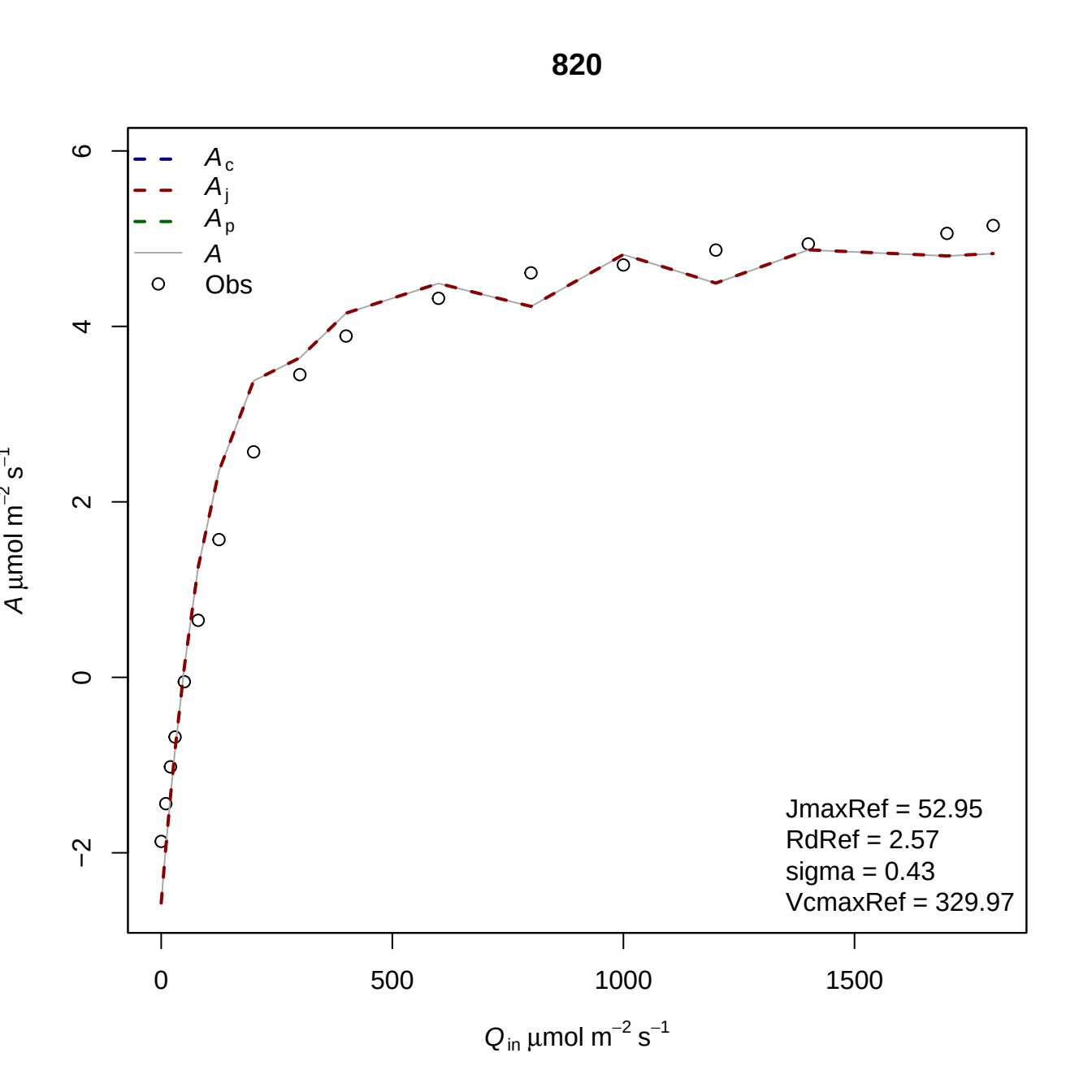

815
A_c
A_j
A_p
A
Obs
JmaxRef = 35.23
RdRef = 1.75
sigma = 0.23
VcmaxRef = 158.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

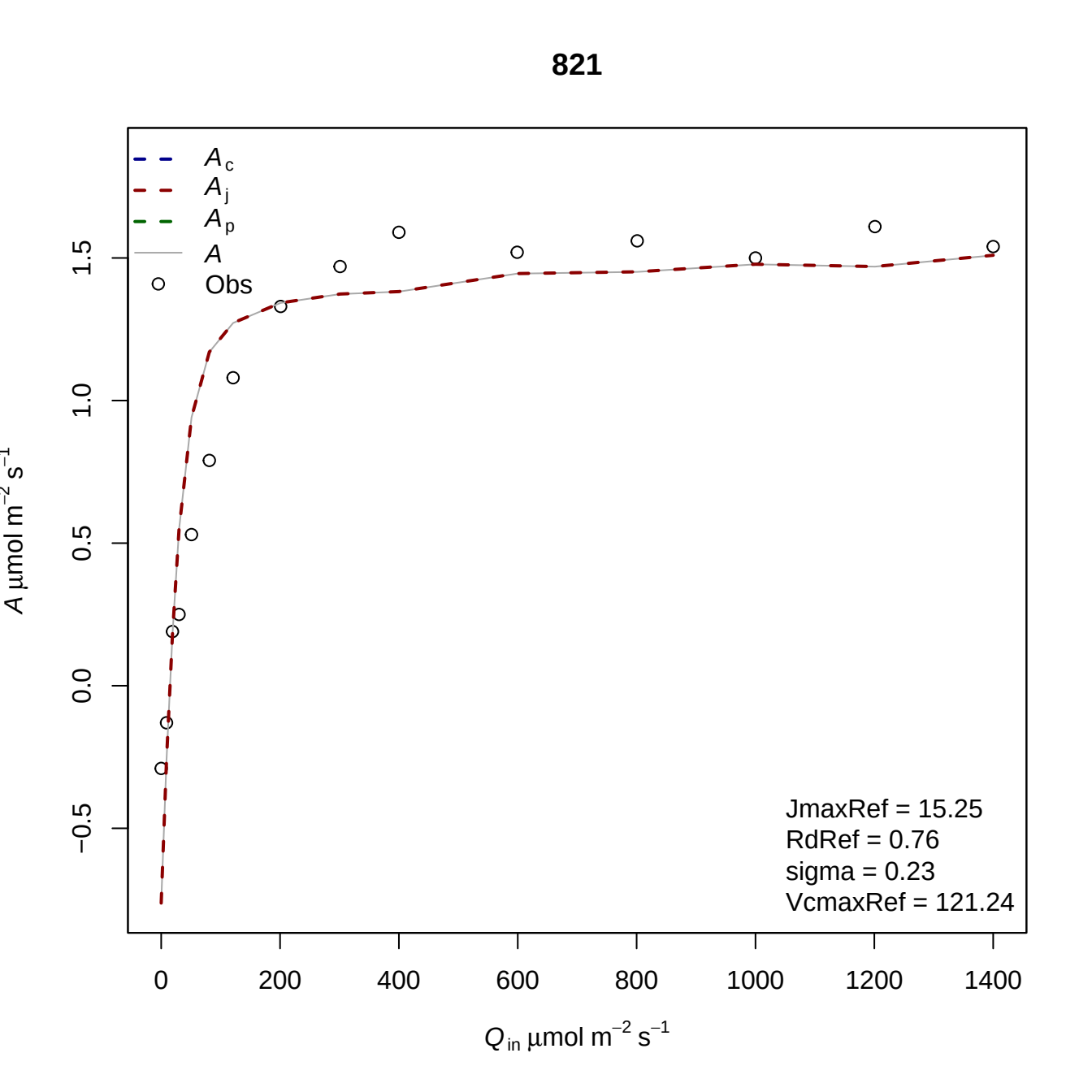

816
A_c
A_j
A_p
A
Obs
JmaxRef = 36.84
RdRef = 0.45
sigma = 0.16
VcmaxRef = 36.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

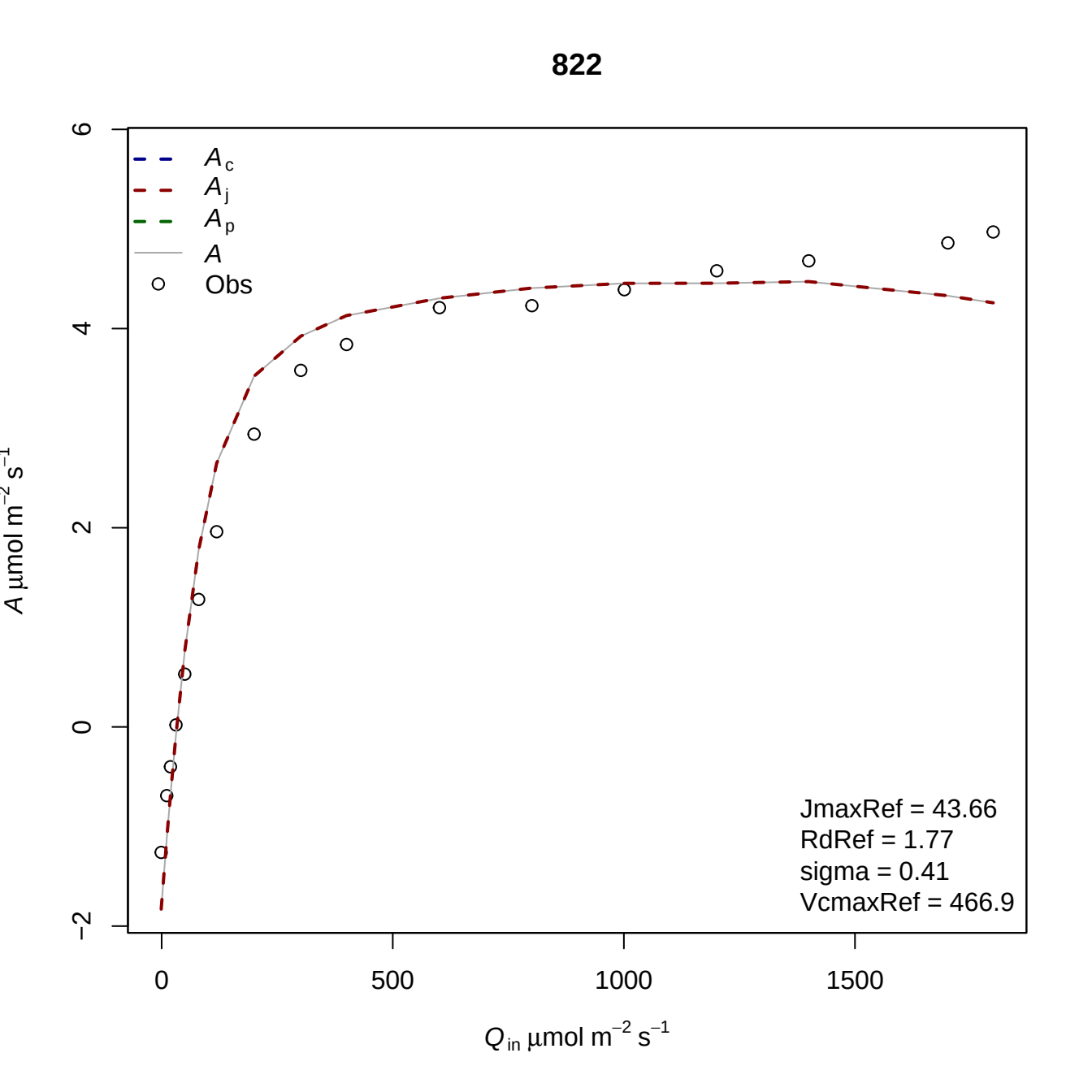

817
A_c
A_j
A_p
A
Obs
JmaxRef = 80.09
RdRef = 1.93
sigma = 0.32
VcmaxRef = 261.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

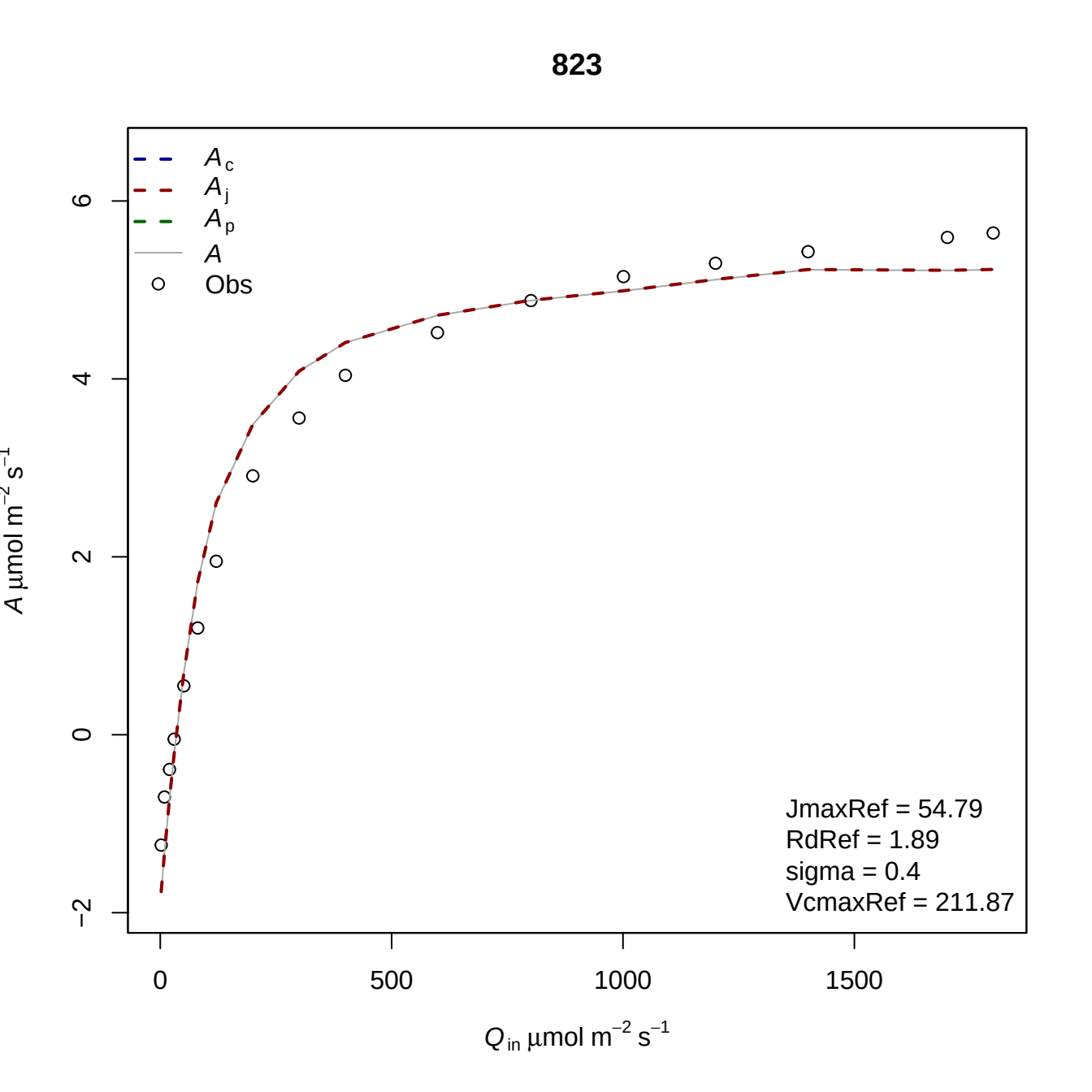

818
A_c
A_j
A_p
A
Obs
JmaxRef = 55.91
RdRef = 2.88
sigma = 0.3
VcmaxRef = 305.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200
1400

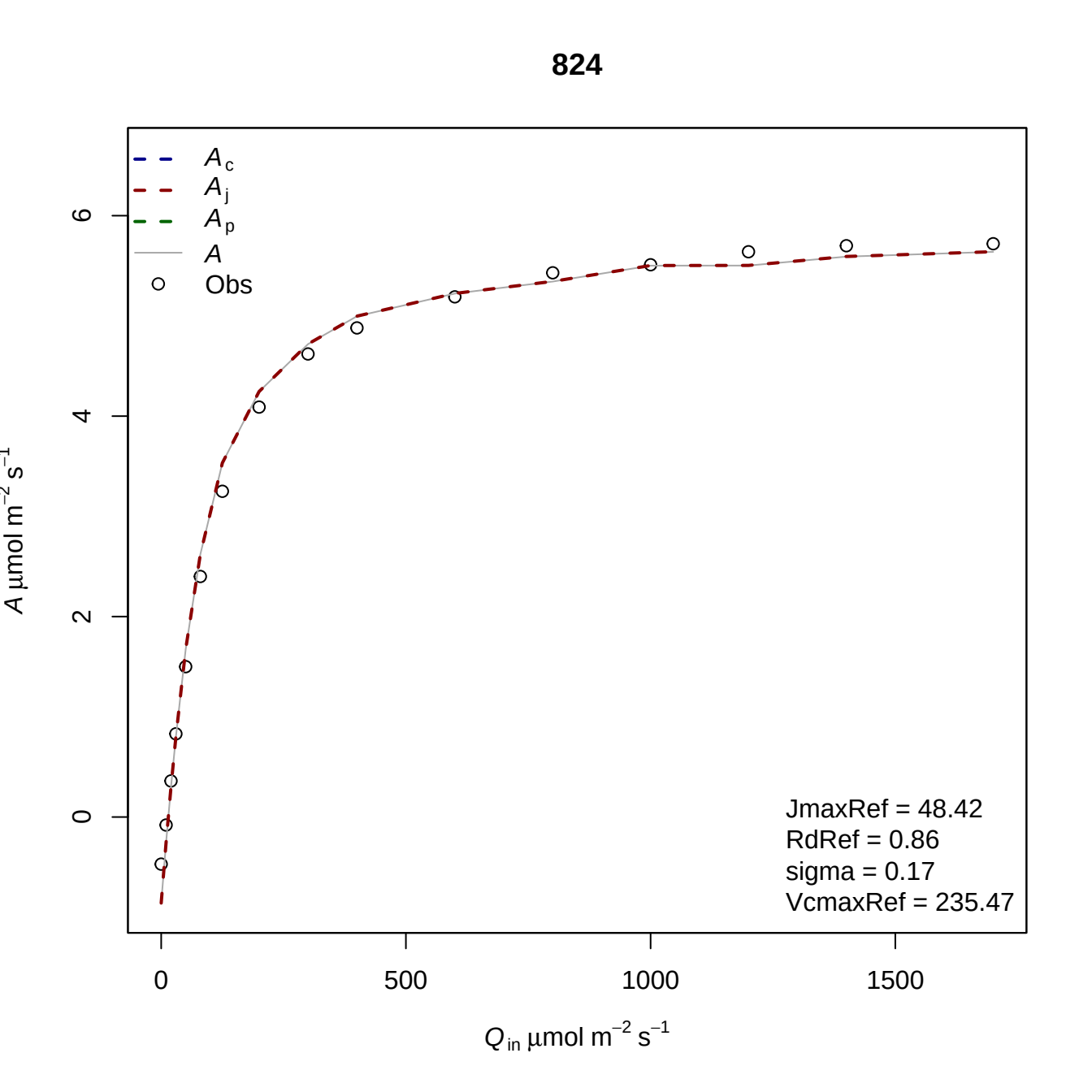

819
A_c
A_j
A_p
A
Obs
JmaxRef = 78.44
RdRef = 2.27
sigma = 0.12
VcmaxRef = 70.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

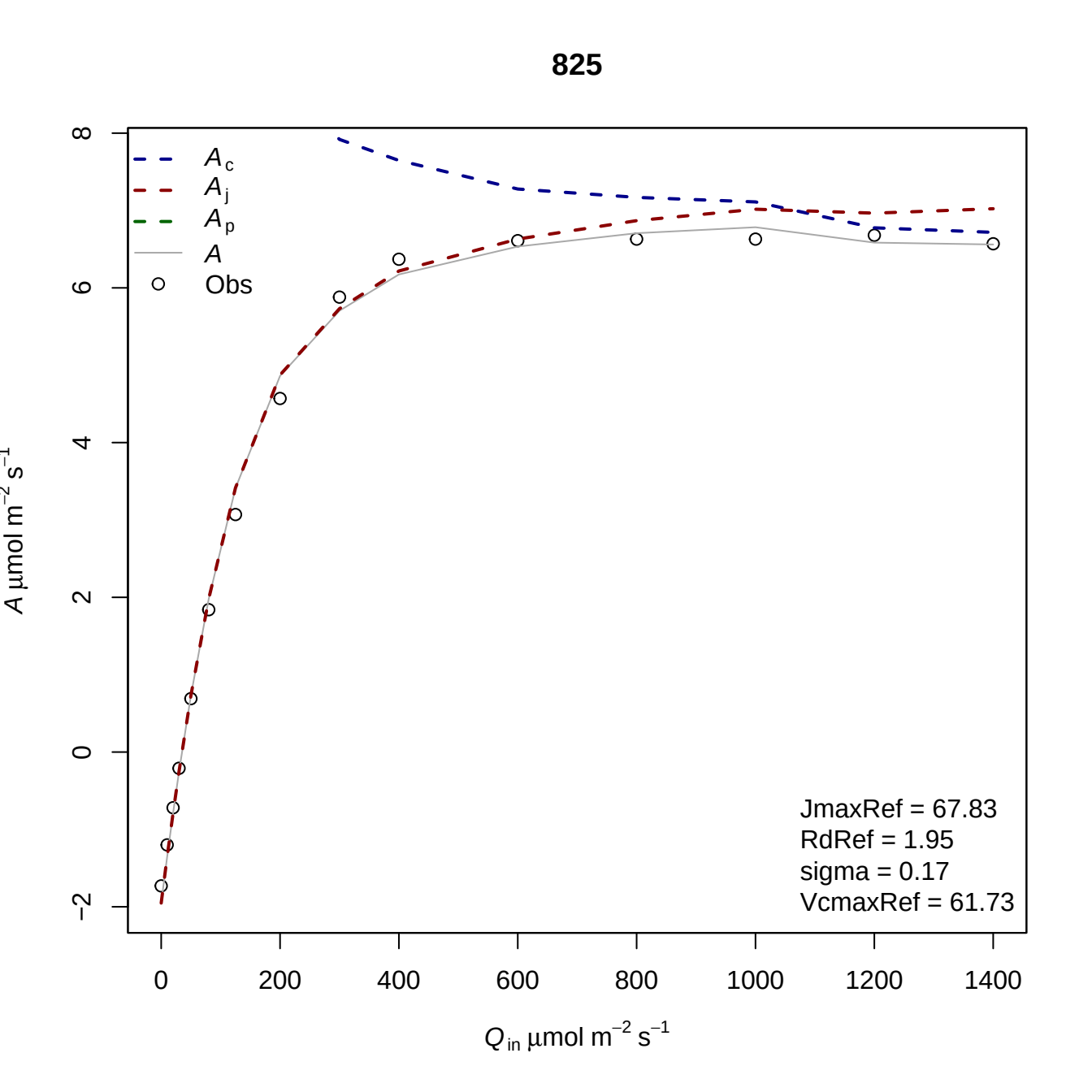

820
A_c
A_j
A_p
A
Obs
JmaxRef = 52.95
RdRef = 2.57
sigma = 0.43
VcmaxRef = 329.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

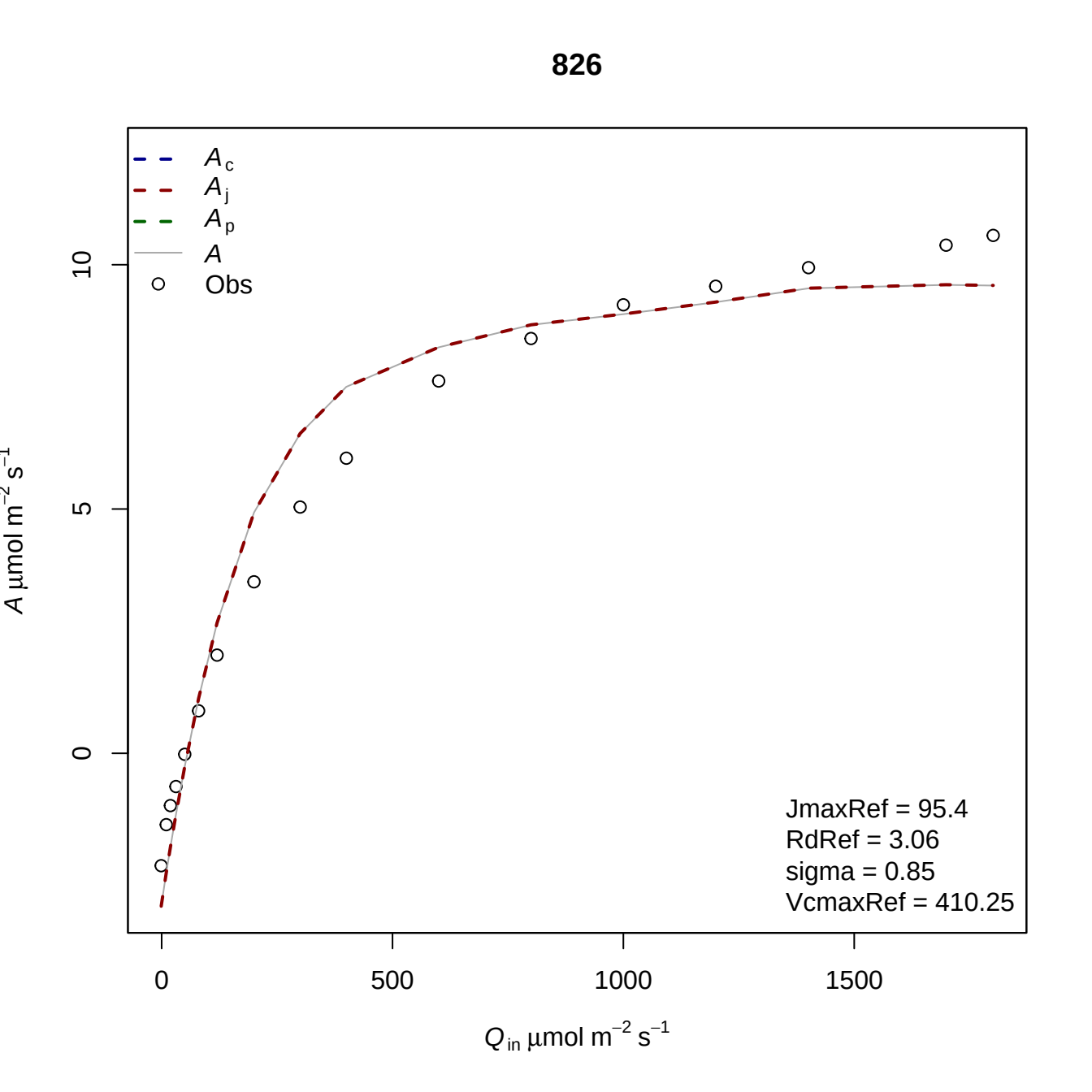

821
A_c
A_j
A_p
A
Obs
JmaxRef = 15.25
RdRef = 0.76
sigma = 0.23
VcmaxRef = 121.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
0
200
400
600
800
1000
1200
1400

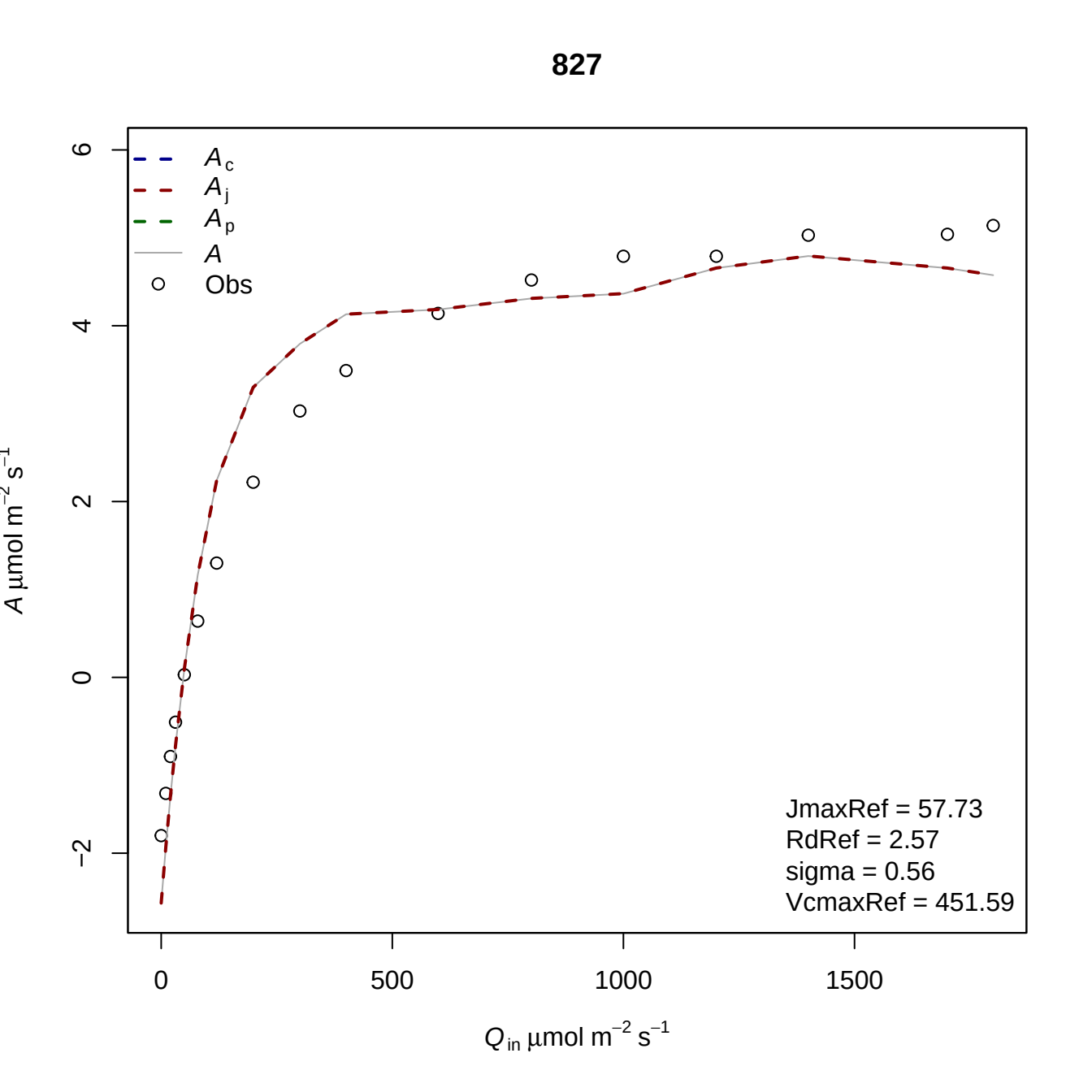

822
A_c
A_j
A_p
A
Obs
JmaxRef = 43.66
RdRef = 1.77
sigma = 0.41
VcmaxRef = 466.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

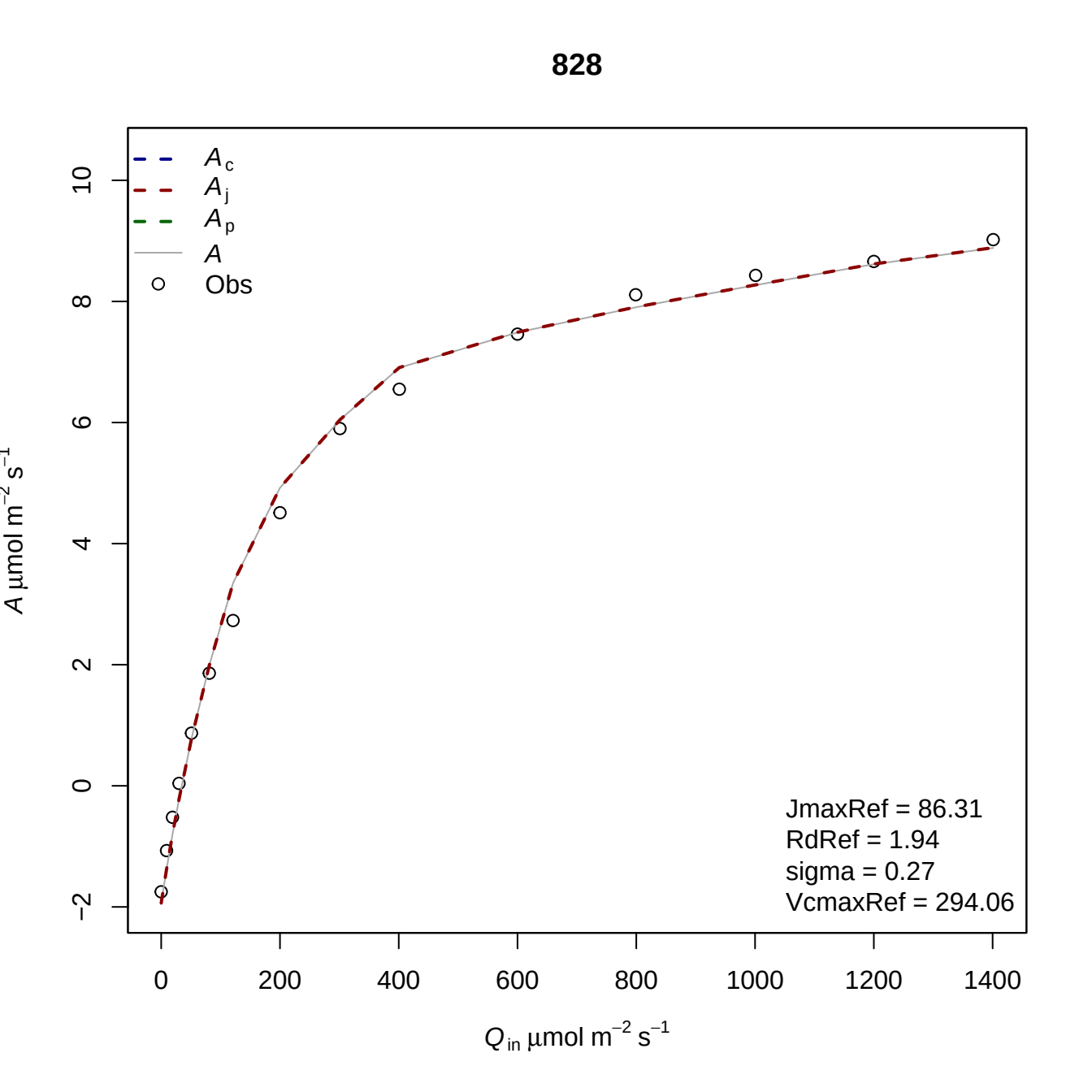

823
A_c
A_j
A_p
A
Obs
JmaxRef = 54.79
RdRef = 1.89
sigma = 0.4
VcmaxRef = 211.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

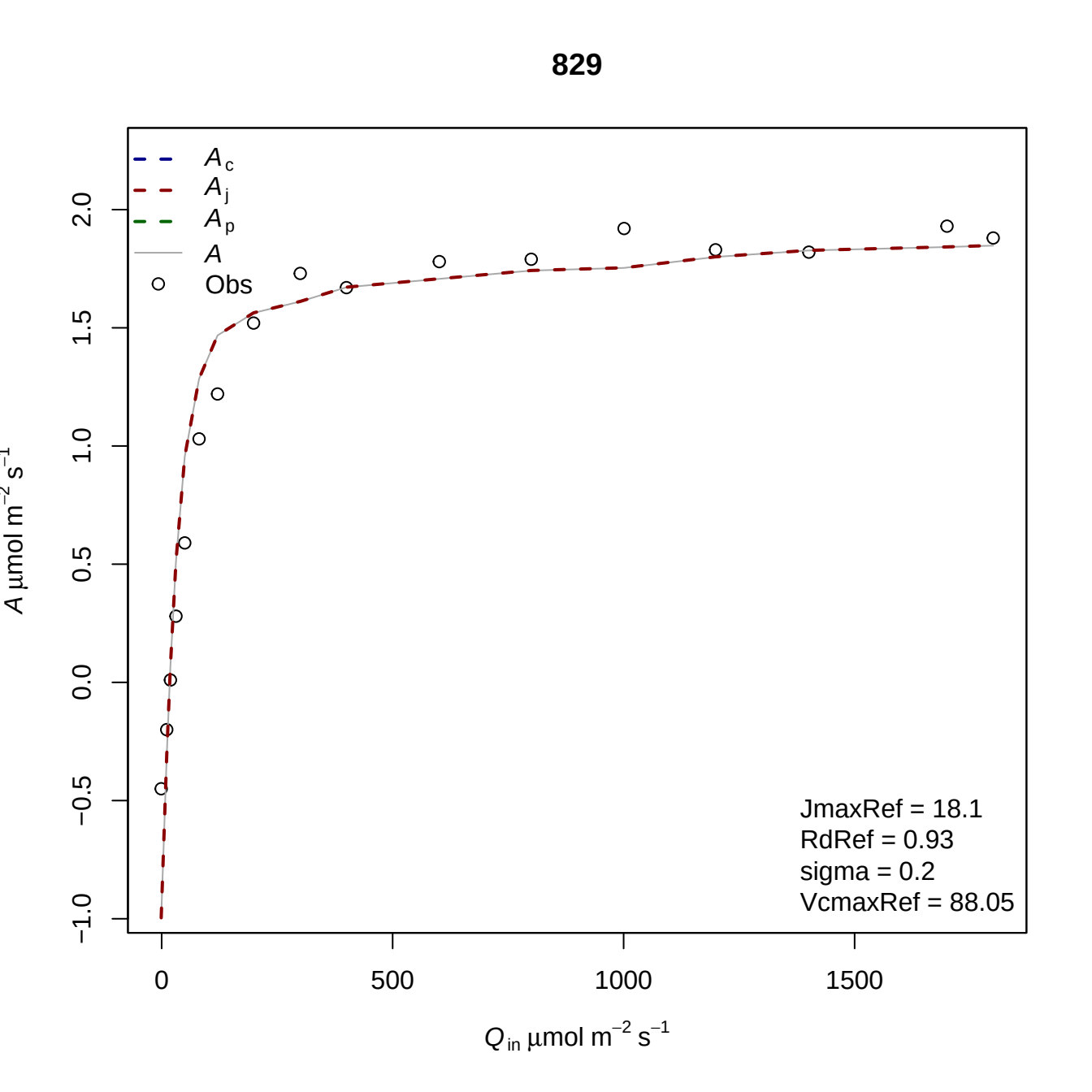

824
A_c
A_j
A_p
A
Obs
JmaxRef = 48.42
RdRef = 0.86
sigma = 0.17
VcmaxRef = 235.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

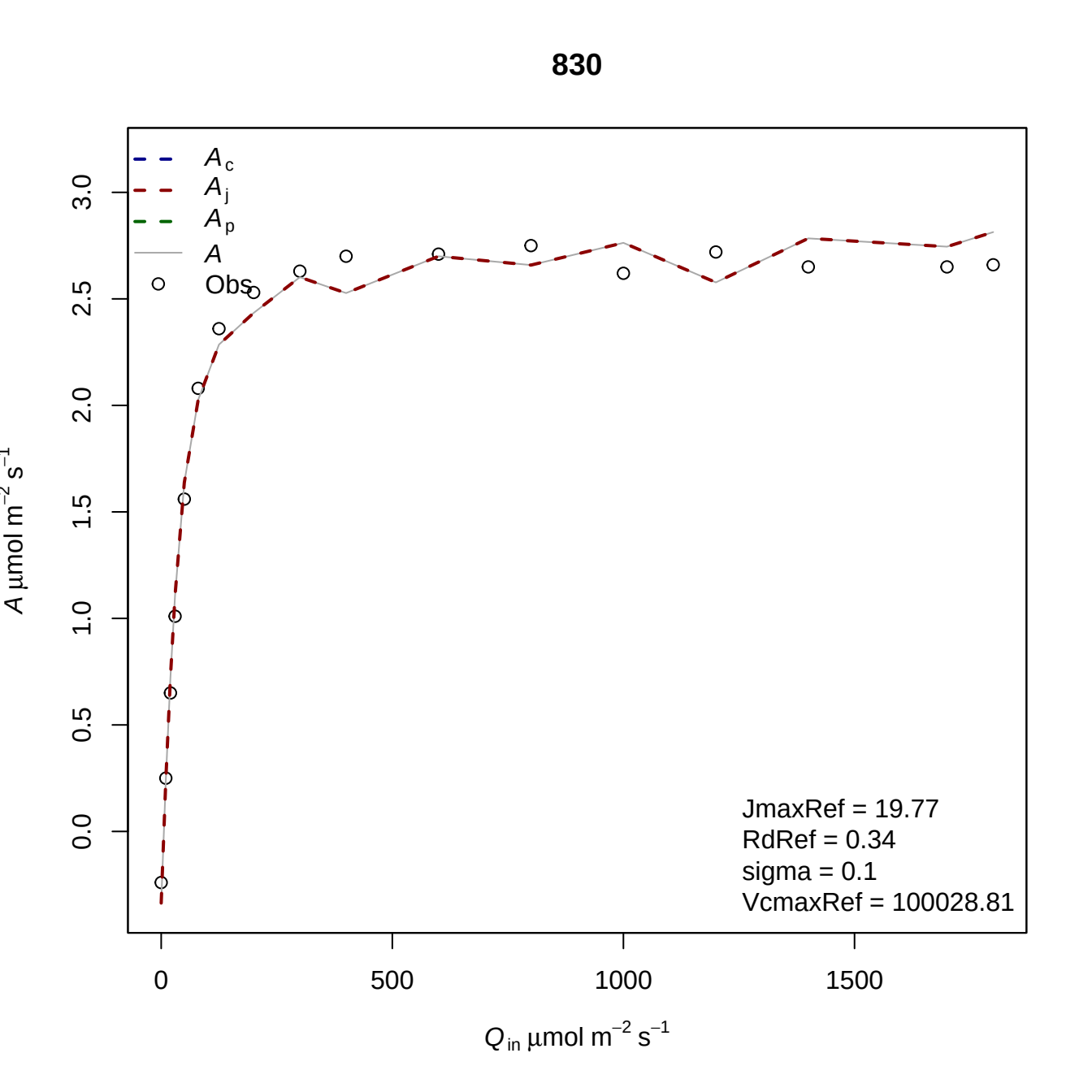

825
A_c
A_j
A_p
A
Obs
JmaxRef = 67.83
RdRef = 1.95
sigma = 0.17
VcmaxRef = 61.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
200
400
600
800
1000
1200
1400

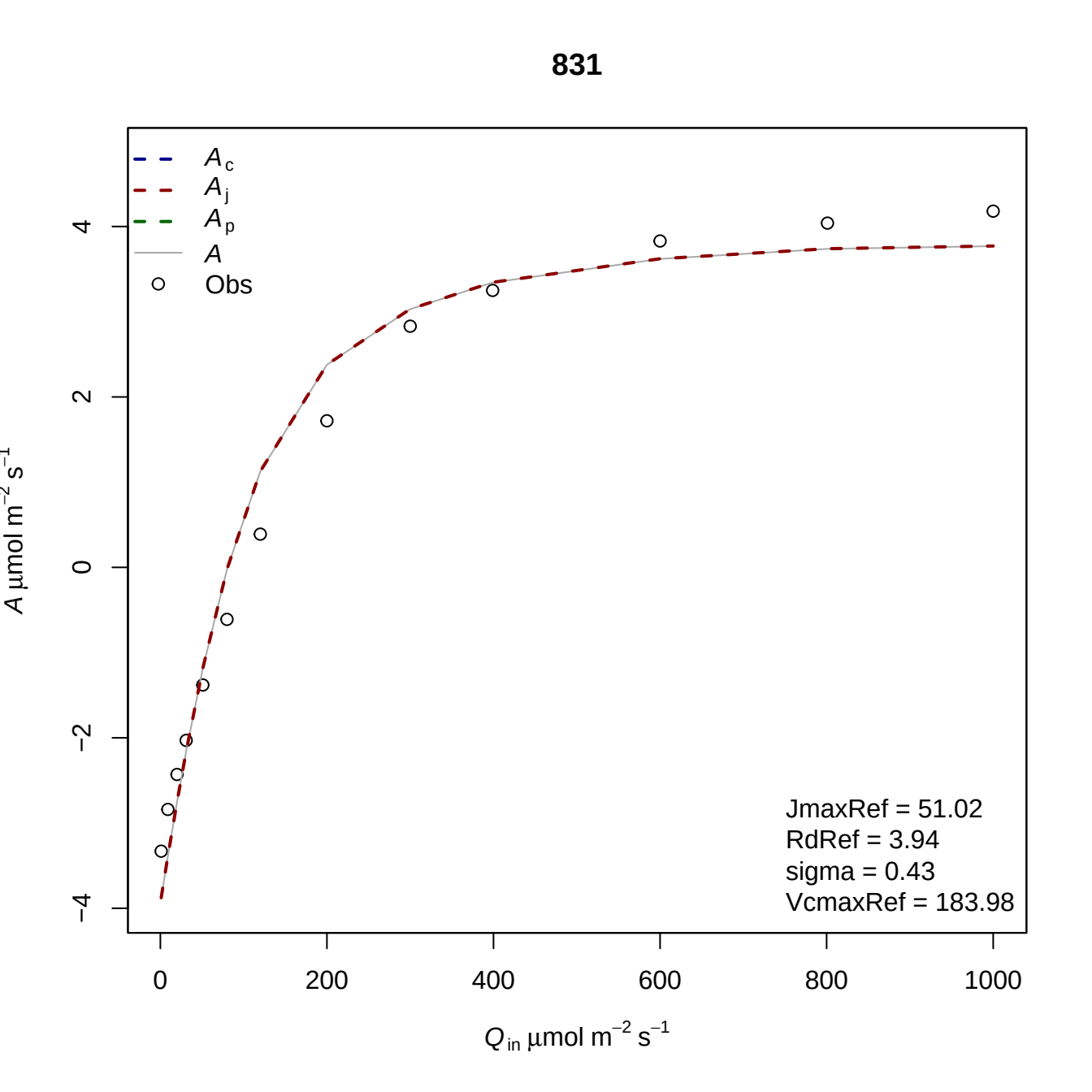

826
A_c
A_j
A_p
A
Obs
JmaxRef = 95.4
RdRef = 3.06
sigma = 0.85
VcmaxRef = 410.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

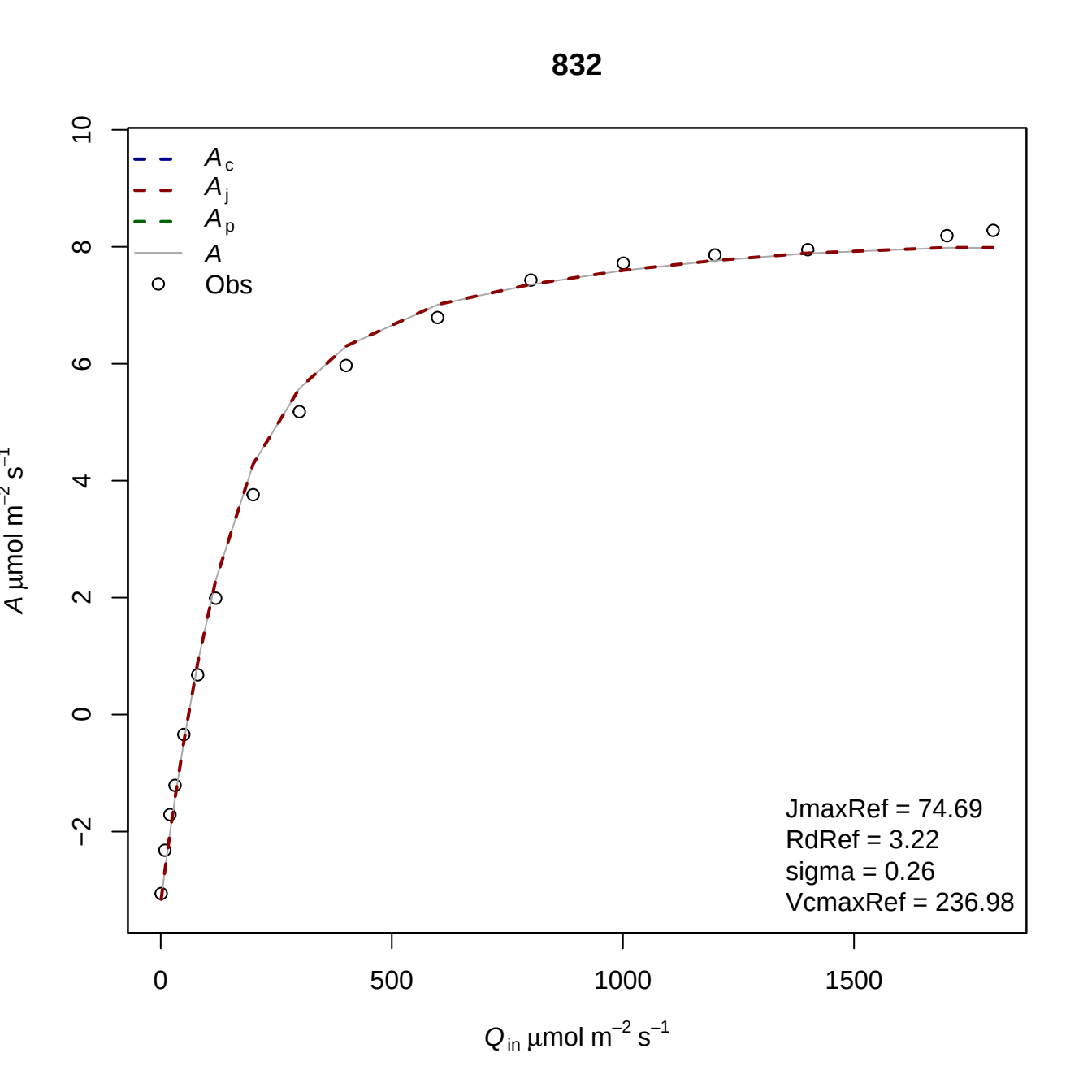

827
A_c
A_j
A_p
A
Obs
JmaxRef = 57.73
RdRef = 2.57
sigma = 0.56
VcmaxRef = 451.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

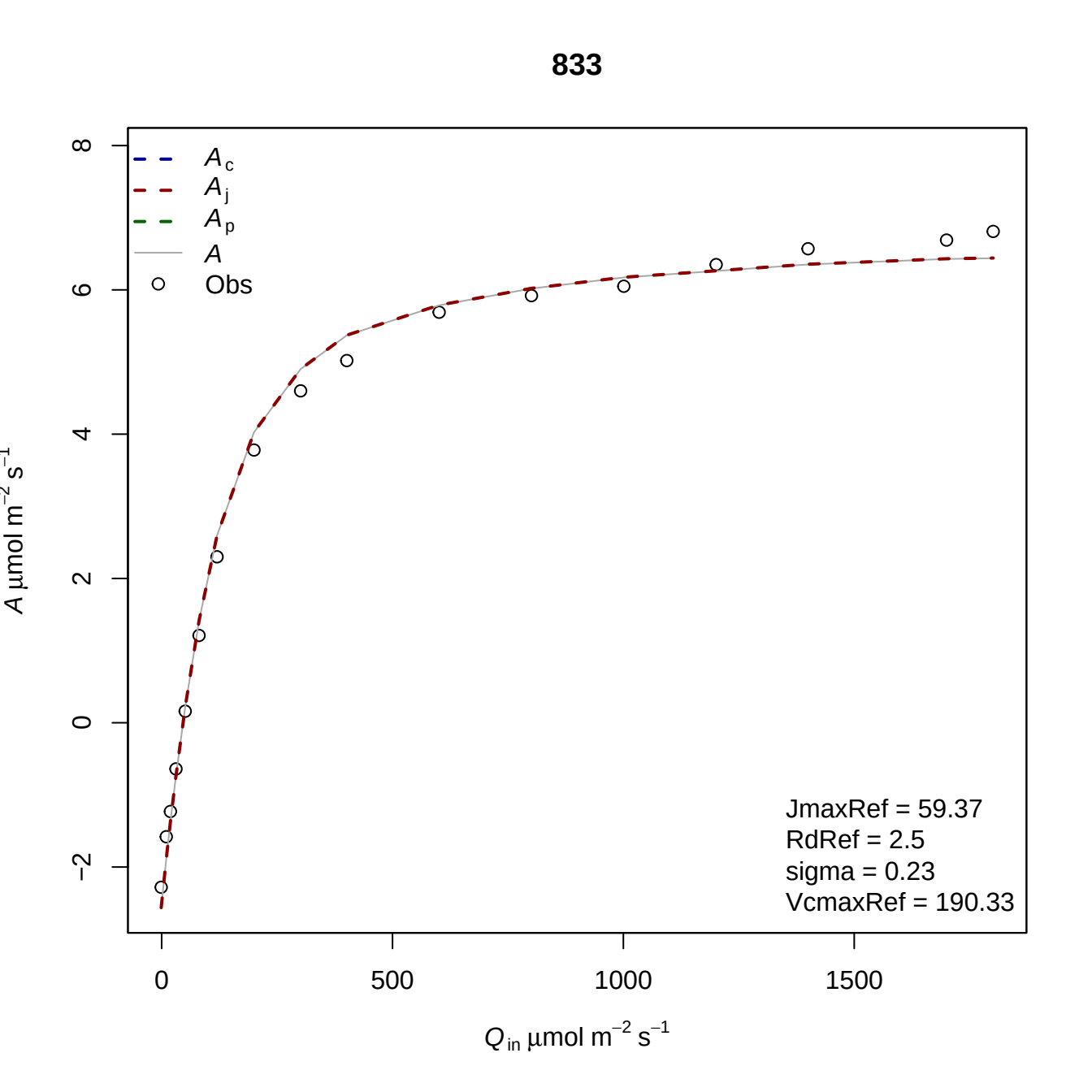

828
A_c
A_j
A_p
A
Obs
JmaxRef = 86.31
RdRef = 1.94
sigma = 0.27
VcmaxRef = 294.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

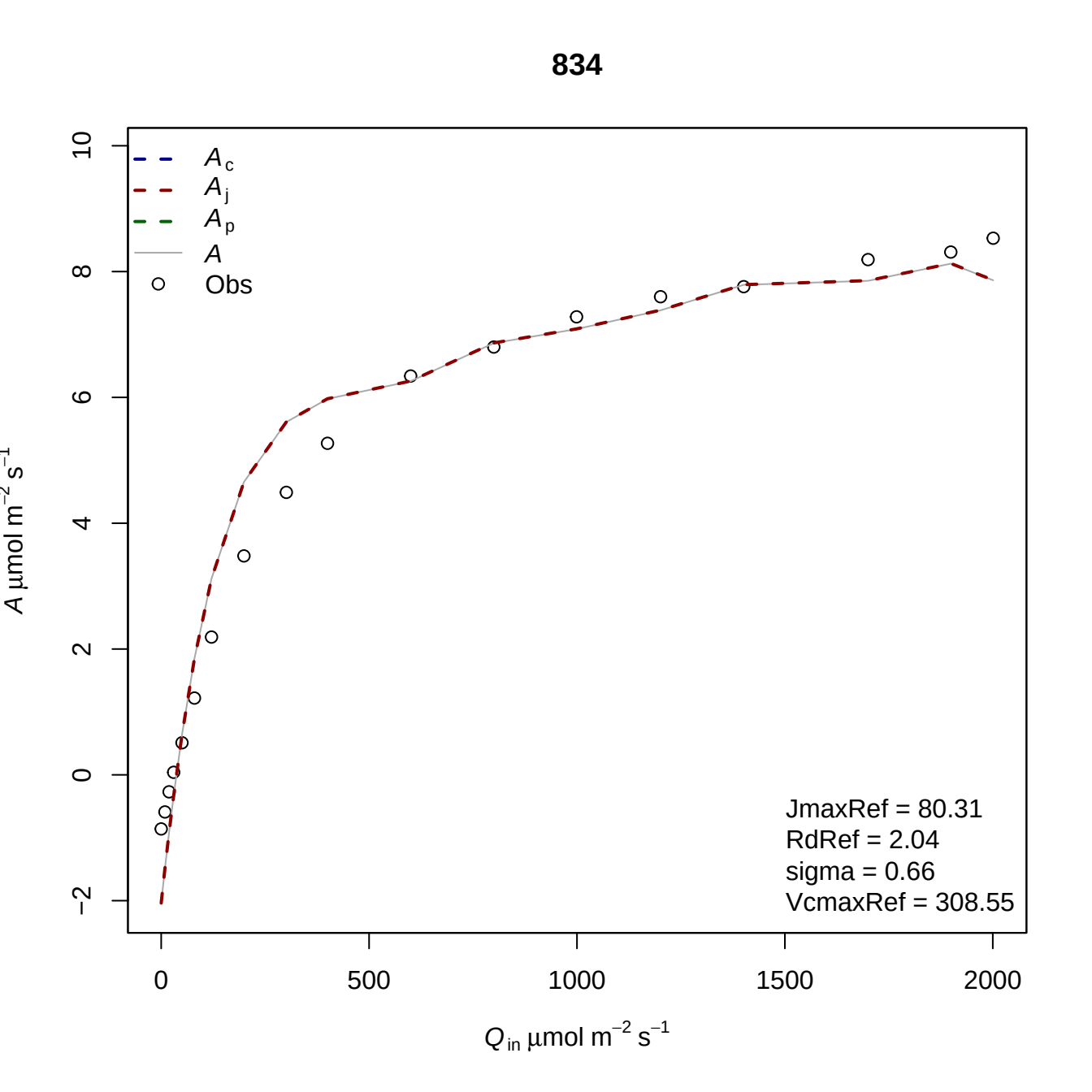

829
A_c
A_j
A_p
A
Obs
JmaxRef = 18.1
RdRef = 0.93
sigma = 0.2
VcmaxRef = 88.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500

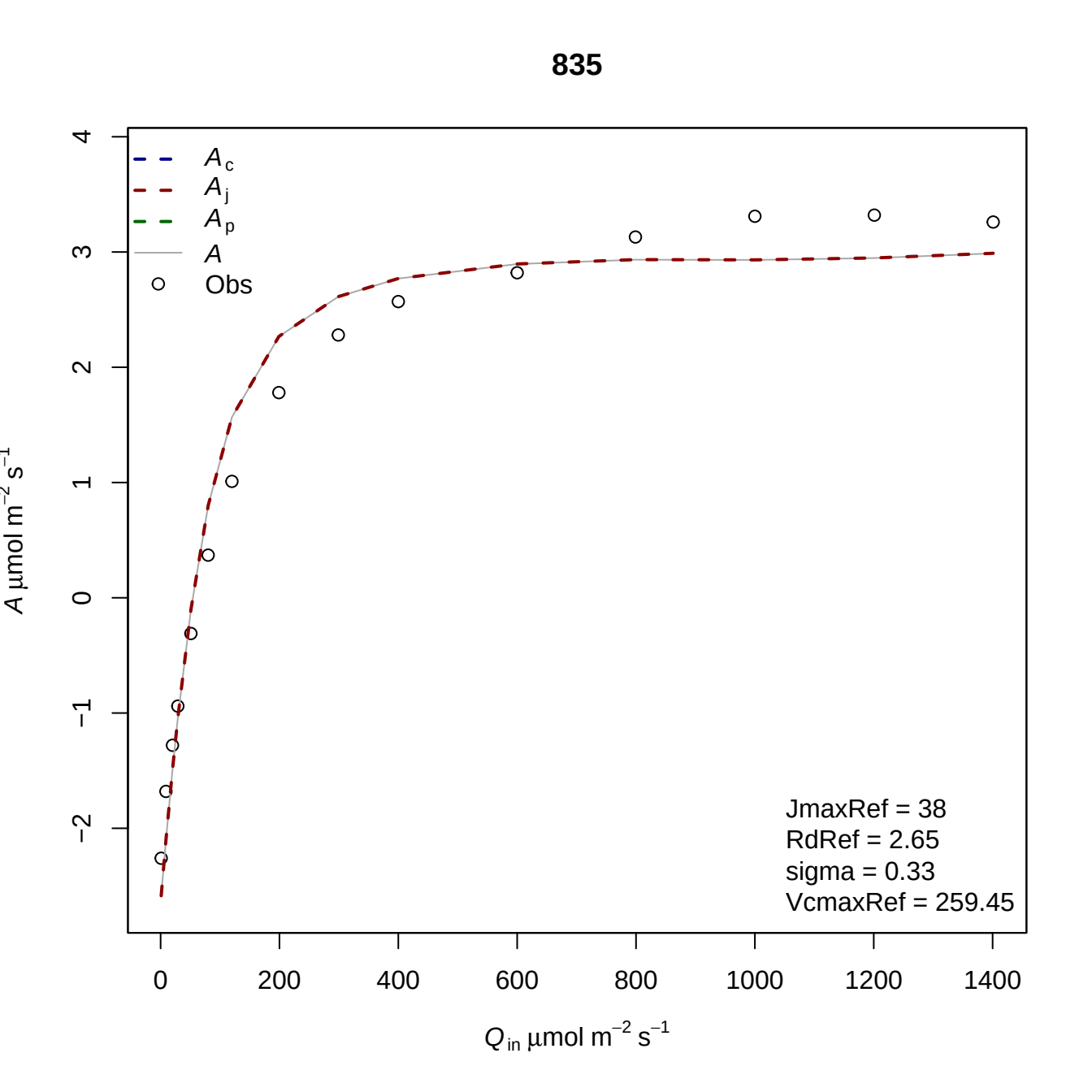

830
A_c
A_j
A_p
A
Obs
JmaxRef = 19.77
RdRef = 0.34
sigma = 0.1
VcmaxRef = 100028.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500

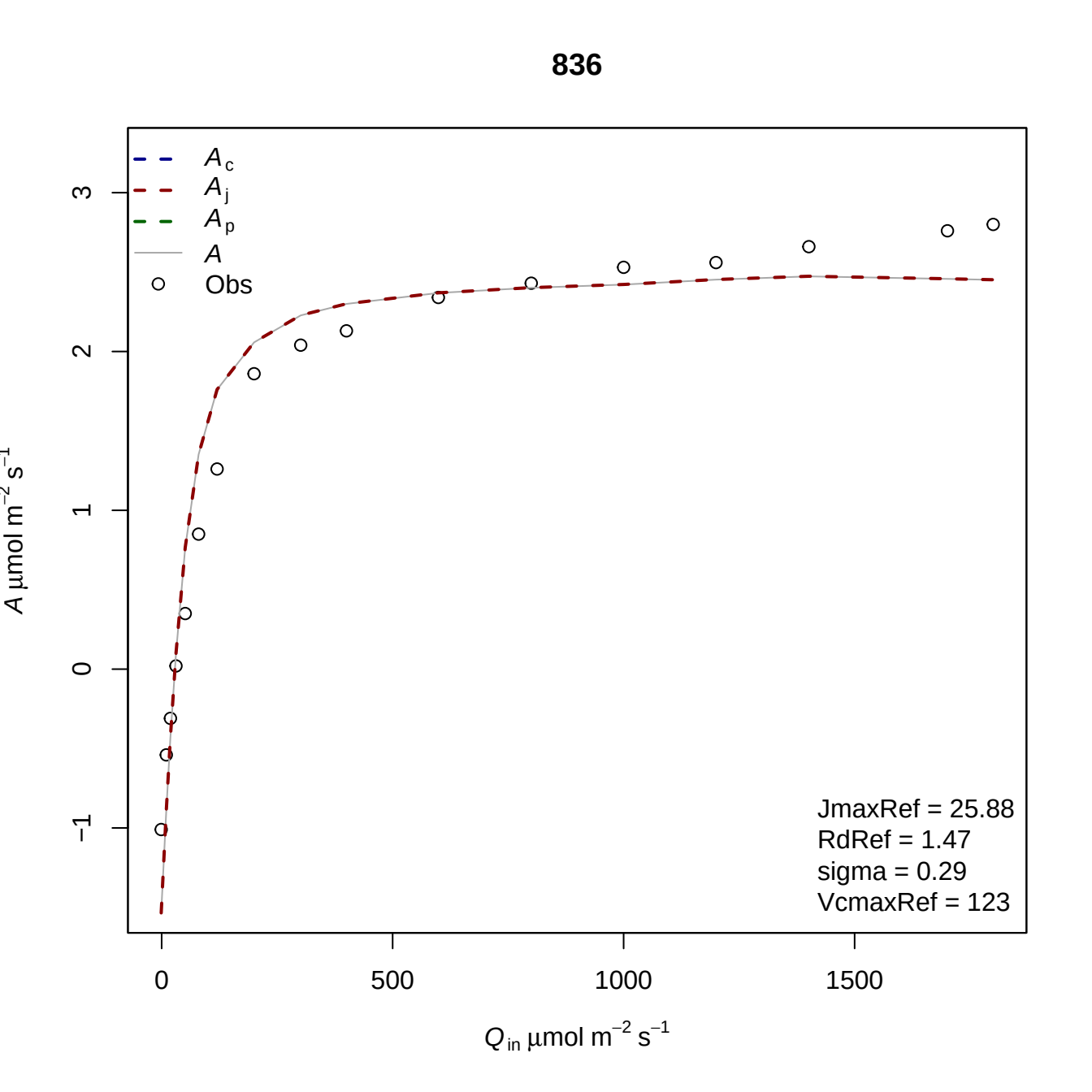

831
A_c
A_j
A_p
A
Obs
JmaxRef = 51.02
RdRef = 3.94
sigma = 0.43
VcmaxRef = 183.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
200
400
600
800
1000

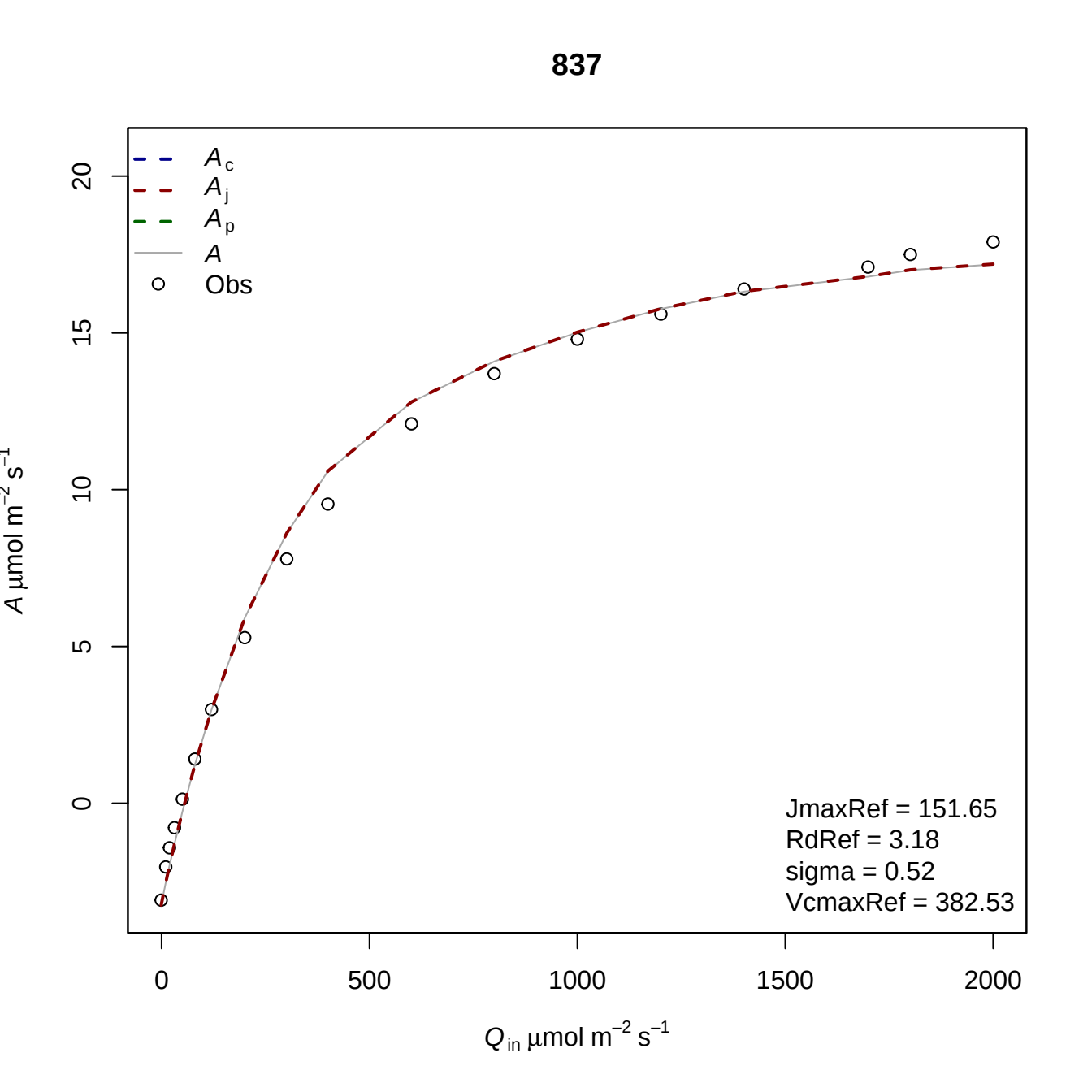

832
A_c
A_j
A_p
A
Obs
JmaxRef = 74.69
RdRef = 3.22
sigma = 0.26
VcmaxRef = 236.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

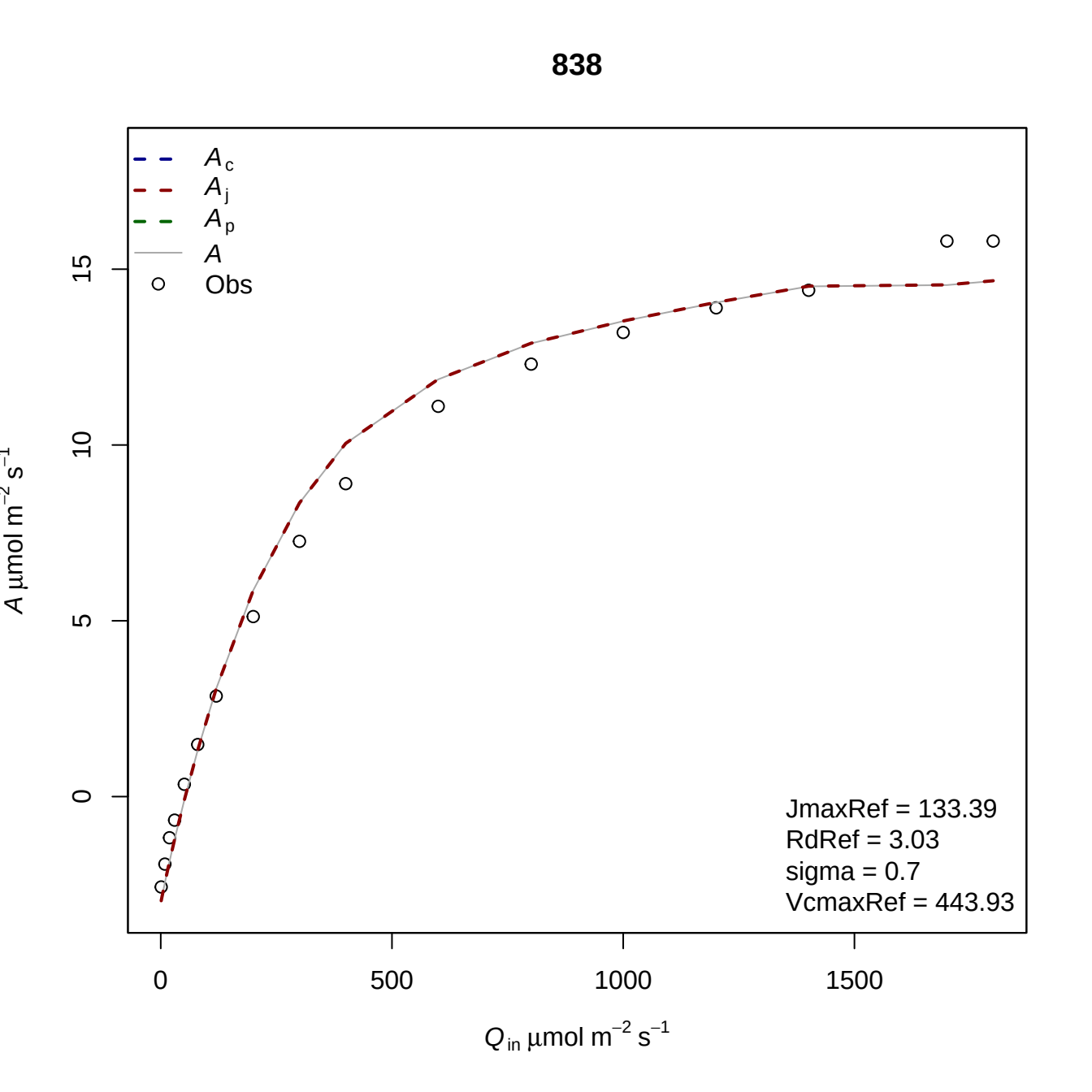

833
A_c
A_j
A_p
A
Obs
JmaxRef = 59.37
RdRef = 2.5
sigma = 0.23
VcmaxRef = 190.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

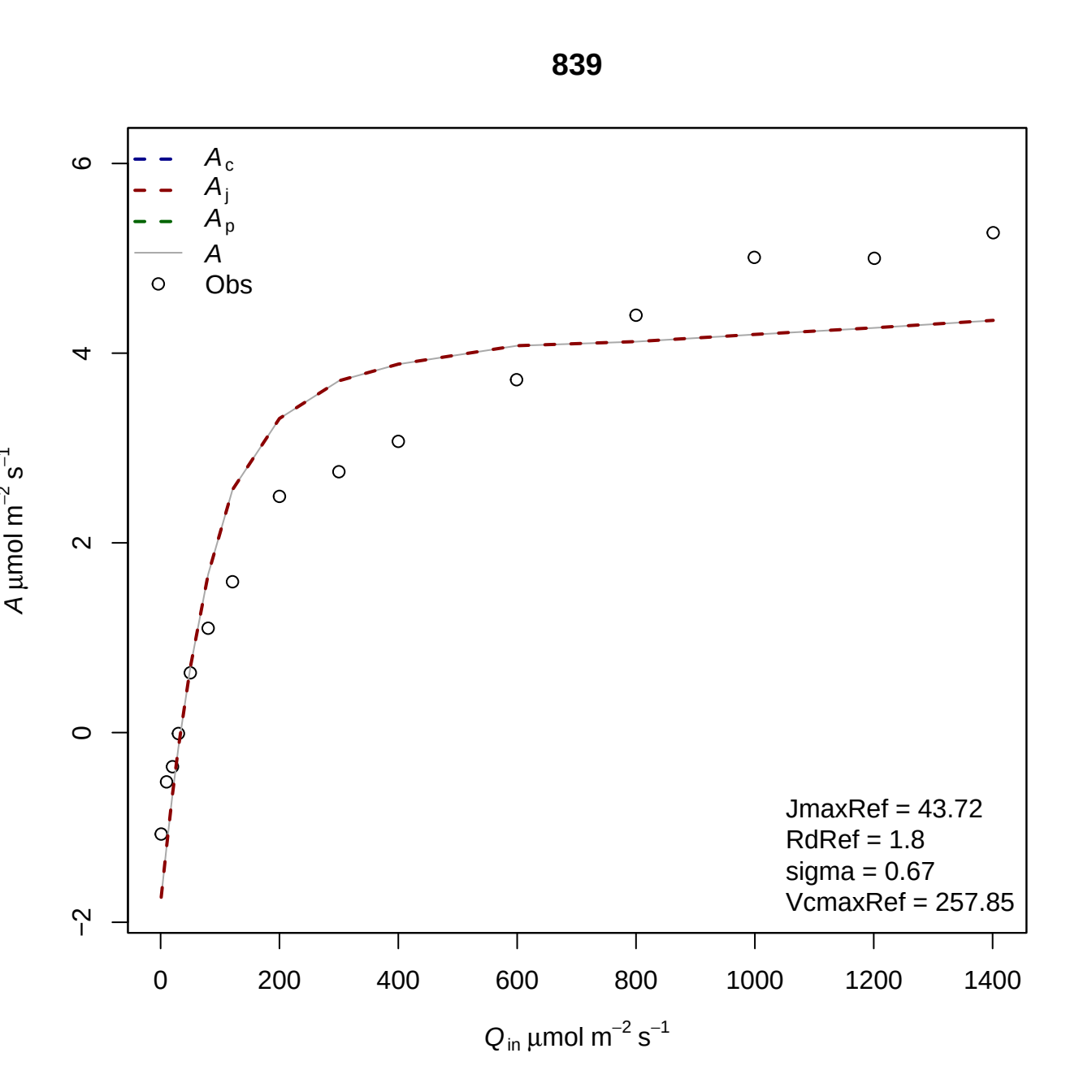

834
A_c
A_j
A_p
A
Obs
JmaxRef = 80.31
RdRef = 2.04
sigma = 0.66
VcmaxRef = 308.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

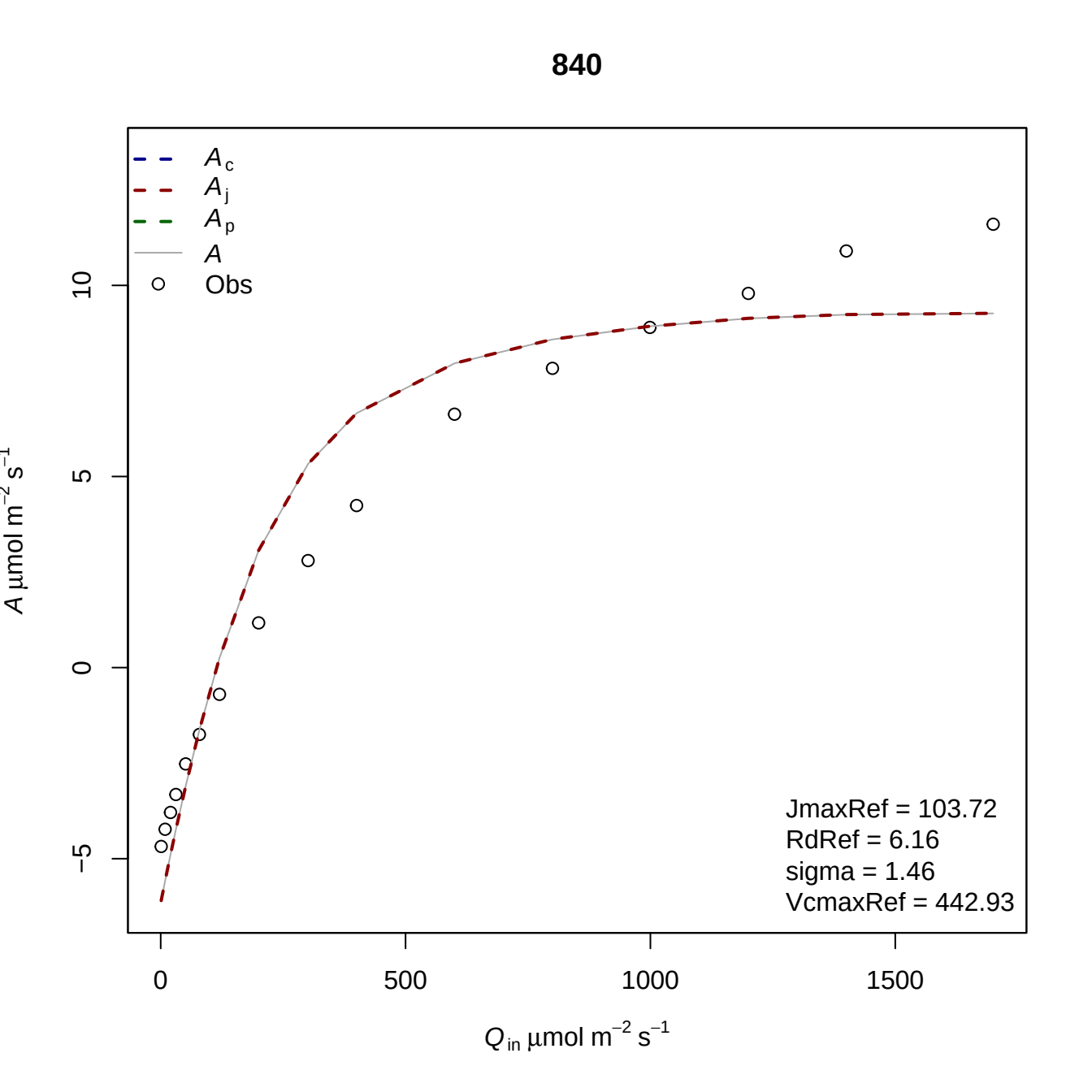

835
A_c
A_j
A_p
A
Obs
JmaxRef = 38
RdRef = 2.65
sigma = 0.33
VcmaxRef = 259.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400

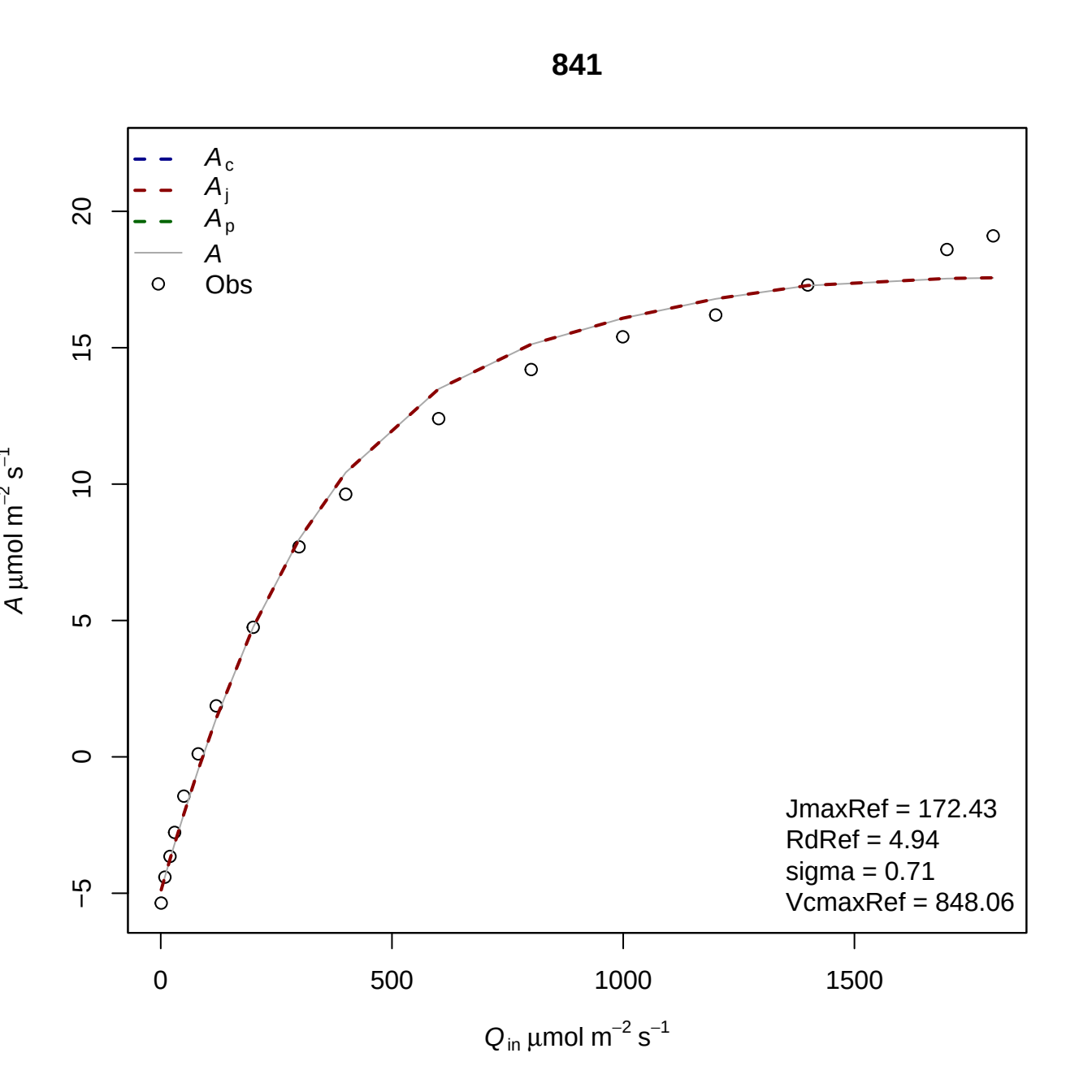

836
A_c
A_j
A_p
A
Obs
JmaxRef = 25.88
RdRef = 1.47
sigma = 0.29
VcmaxRef = 123
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

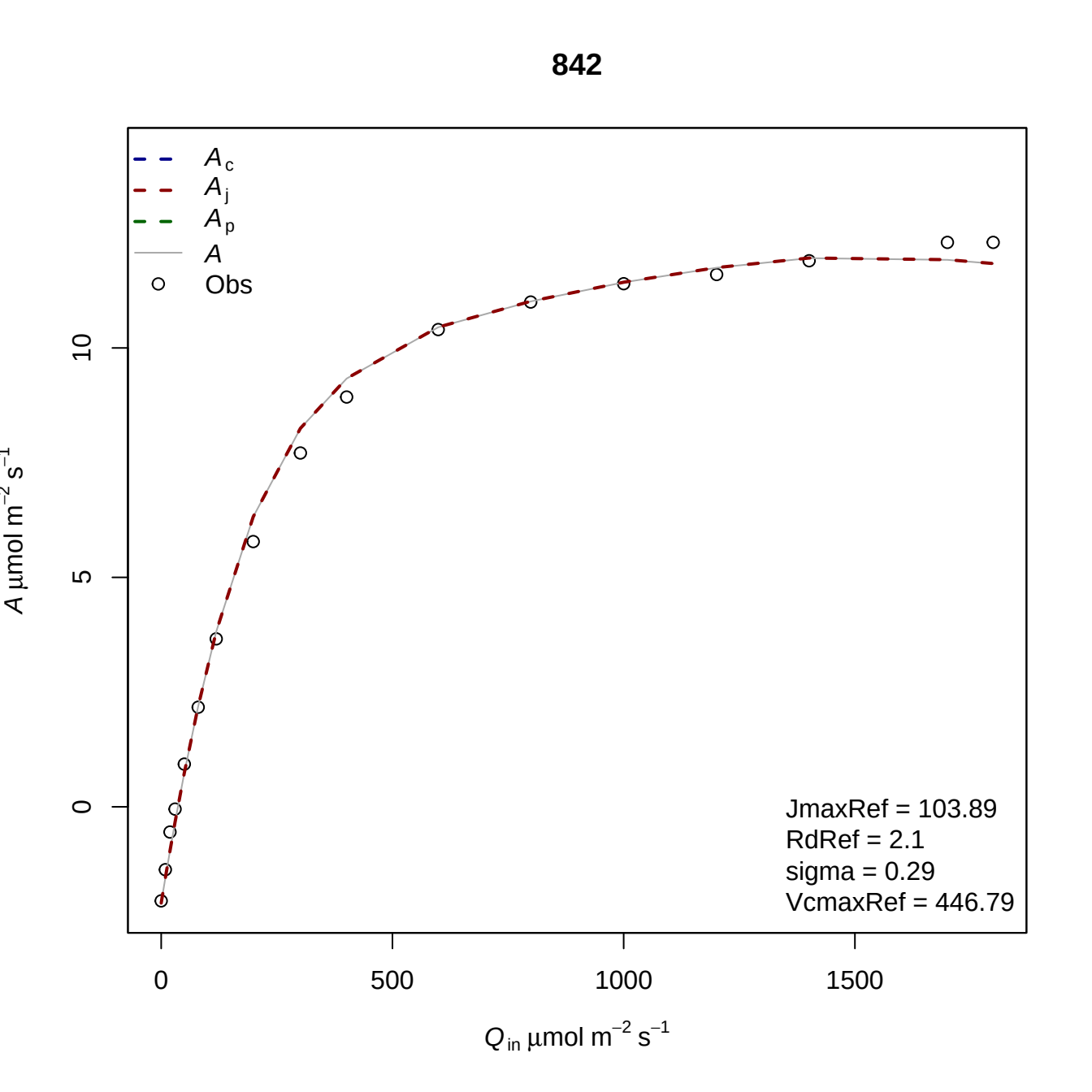

837
A_c
A_j
A_p
A
Obs
JmaxRef = 151.65
RdRef = 3.18
sigma = 0.52
VcmaxRef = 382.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

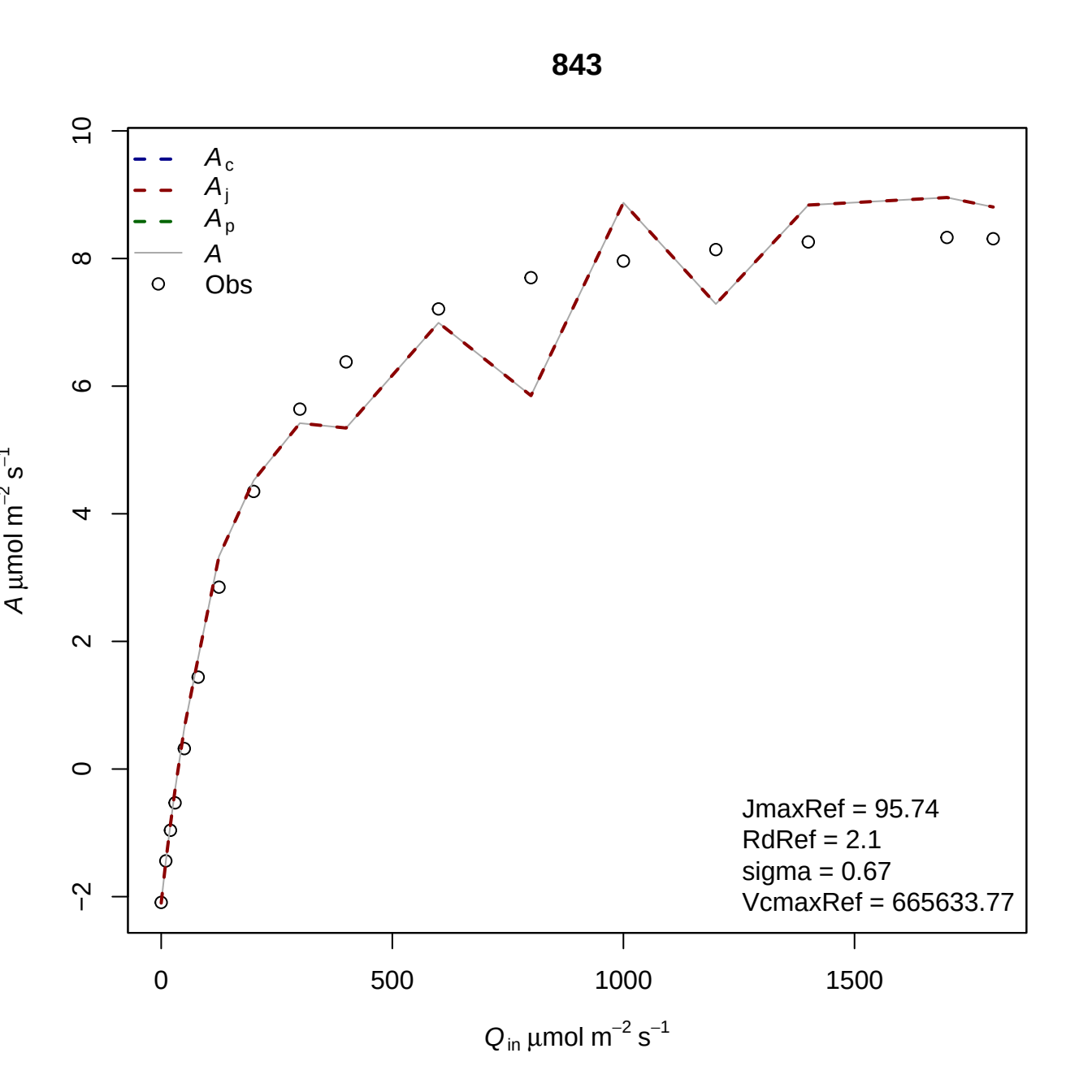

838
A_c
A_j
A_p
A
Obs
JmaxRef = 133.39
RdRef = 3.03
sigma = 0.7
VcmaxRef = 443.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

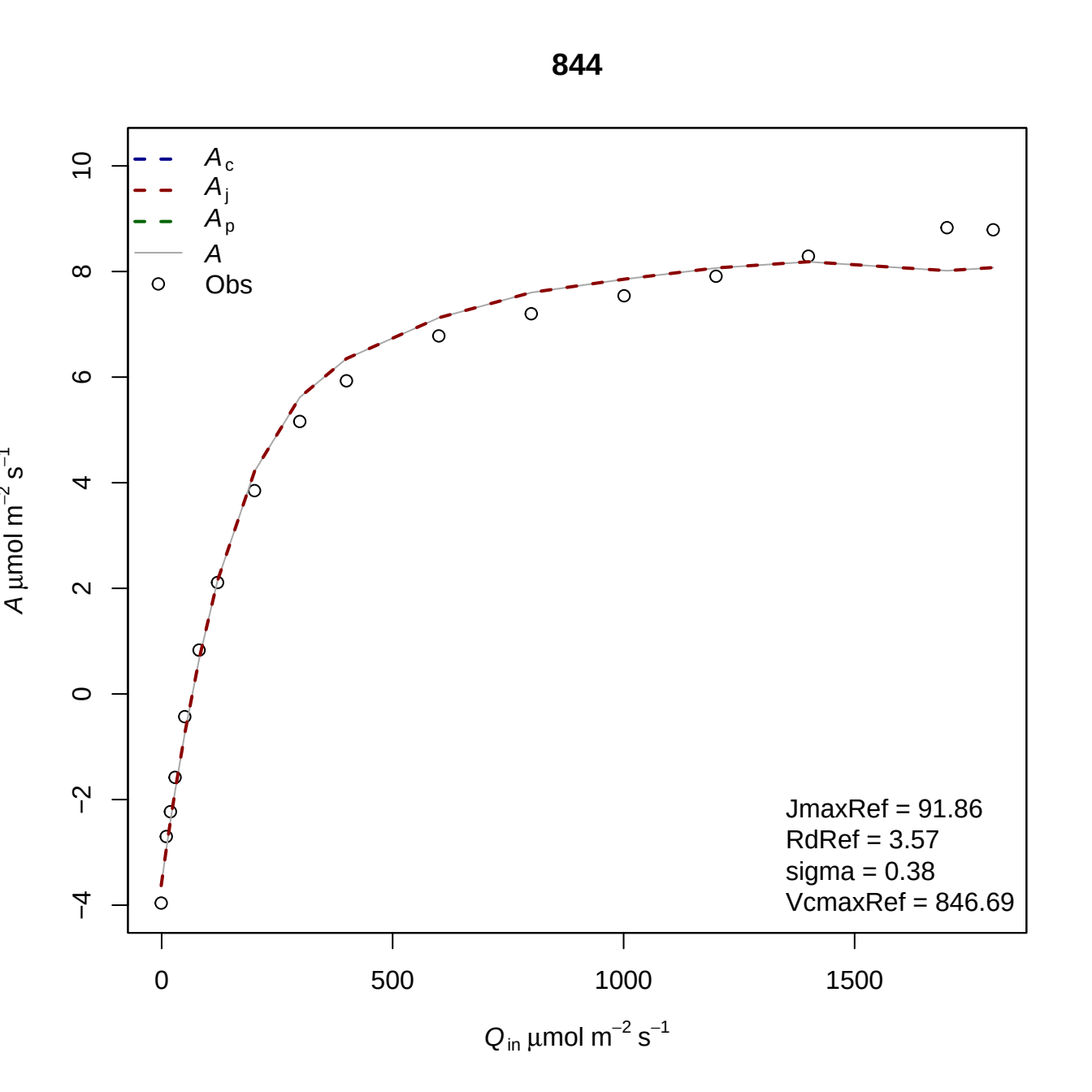

839
A_c
A_j
A_p
A
Obs
JmaxRef = 43.72
RdRef = 1.8
sigma = 0.67
VcmaxRef = 257.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

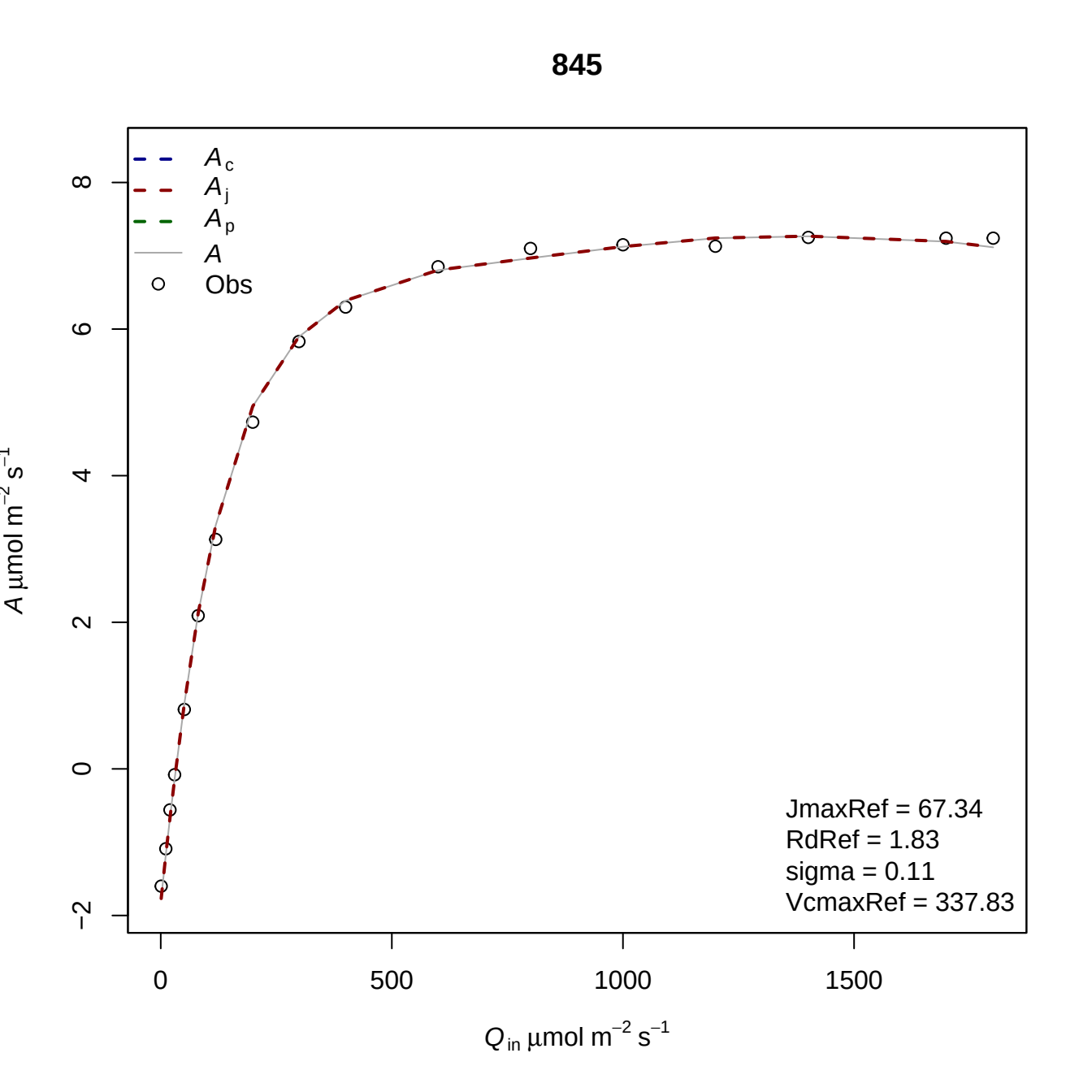

840
A_c
A_j
A_p
A
Obs
JmaxRef = 103.72
RdRef = 6.16
sigma = 1.46
VcmaxRef = 442.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

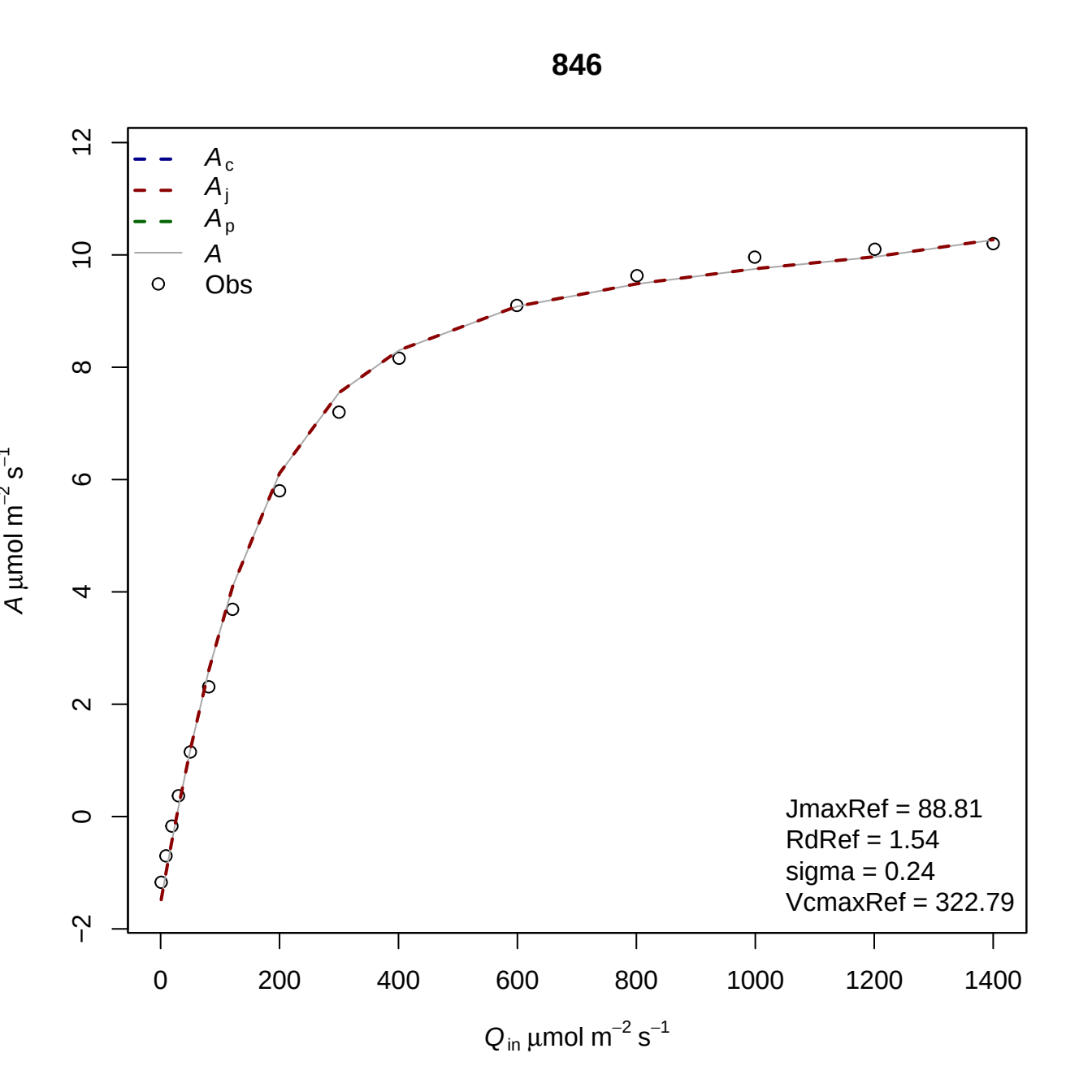

841
A_c
A_j
A_p
A
Obs
JmaxRef = 172.43
RdRef = 4.94
sigma = 0.71
VcmaxRef = 848.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500

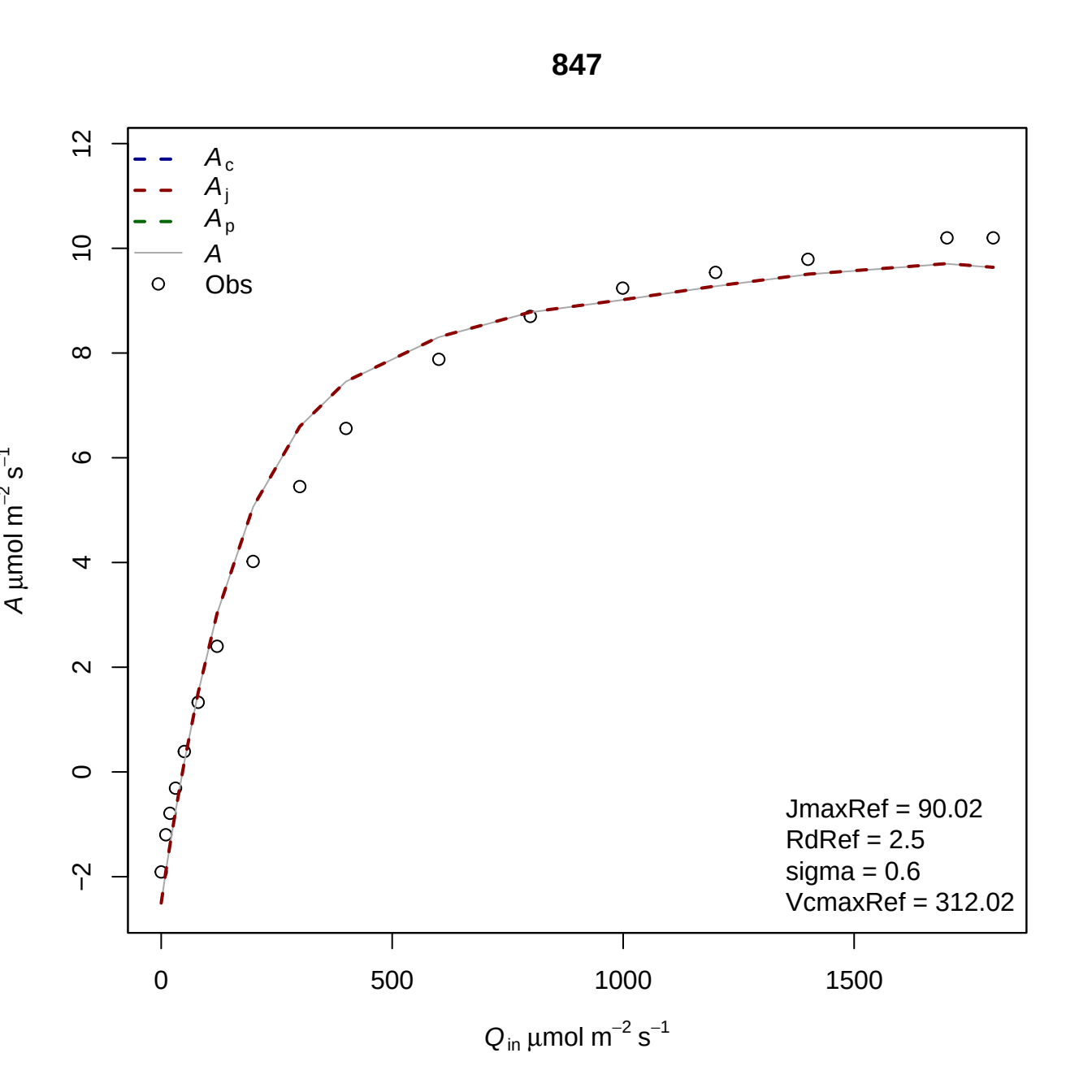

842
A_c
A_j
A_p
A
Obs
JmaxRef = 103.89
RdRef = 2.1
sigma = 0.29
VcmaxRef = 446.79
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

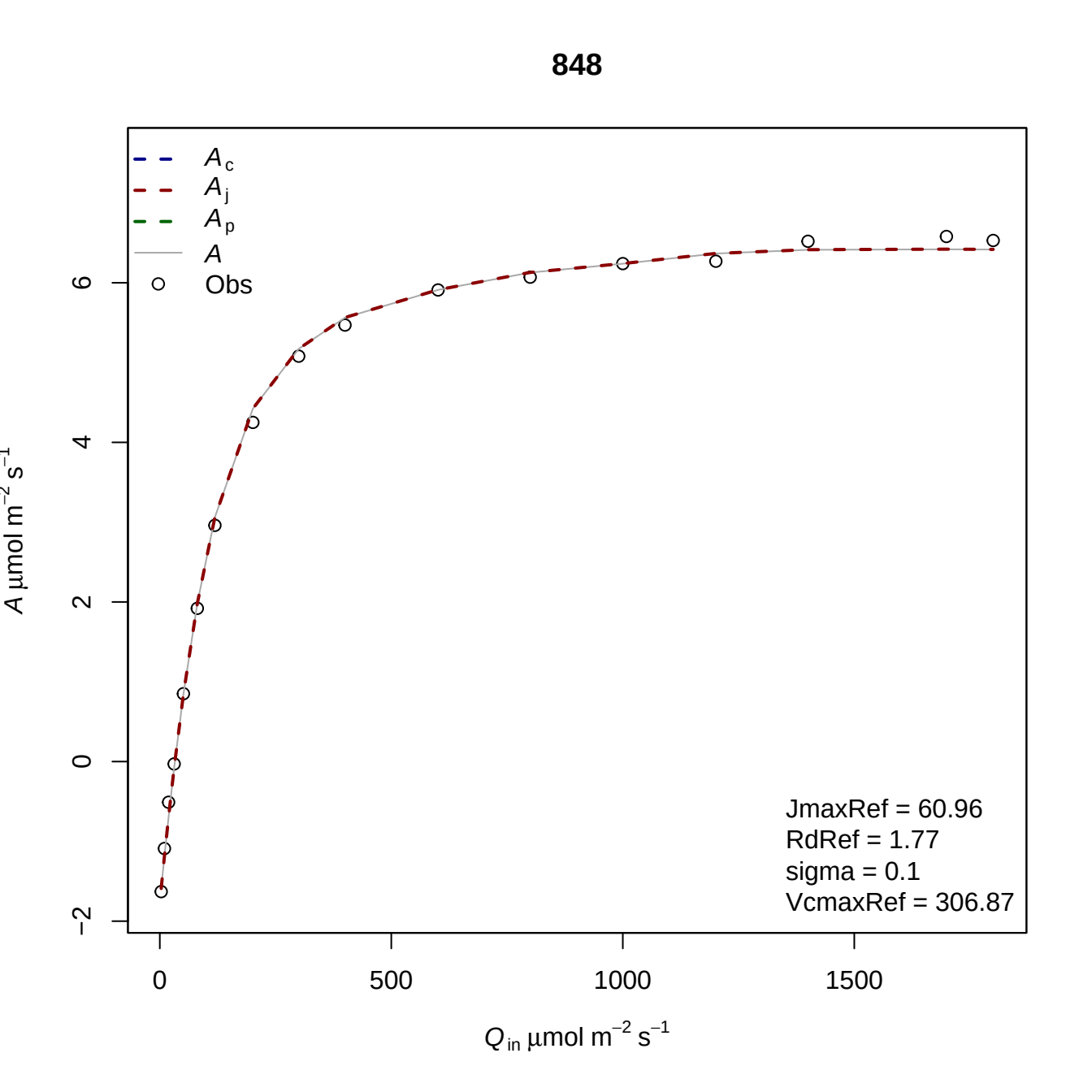

843
A_c
A_j
A_p
A
Obs
JmaxRef = 95.74
RdRef = 2.1
sigma = 0.67
VcmaxRef = 665633.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

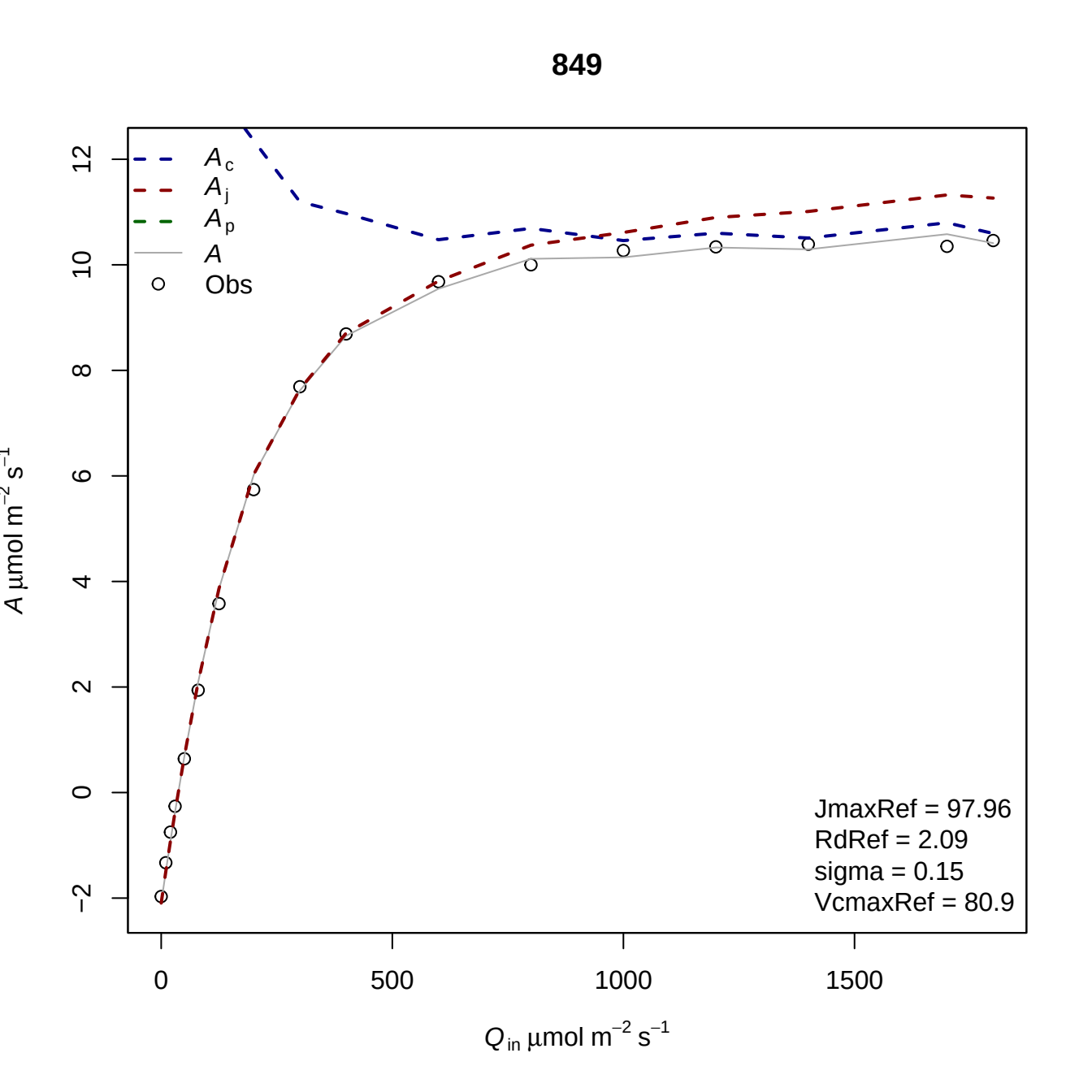

844
A_c
A_j
A_p
A
Obs
JmaxRef = 91.86
RdRef = 3.57
sigma = 0.38
VcmaxRef = 846.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

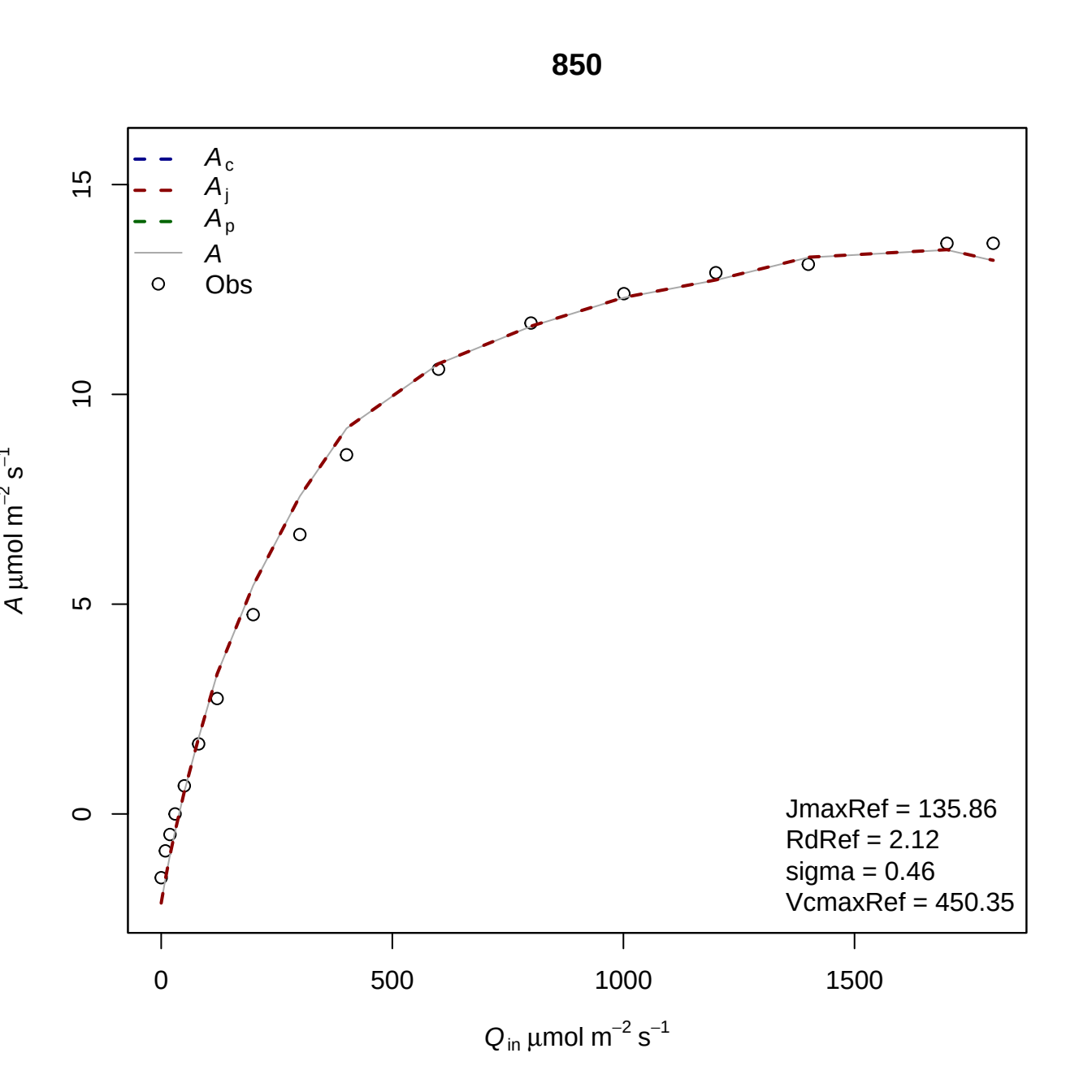

845
A_c
A_j
A_p
A
Obs
JmaxRef = 67.34
RdRef = 1.83
sigma = 0.11
VcmaxRef = 337.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

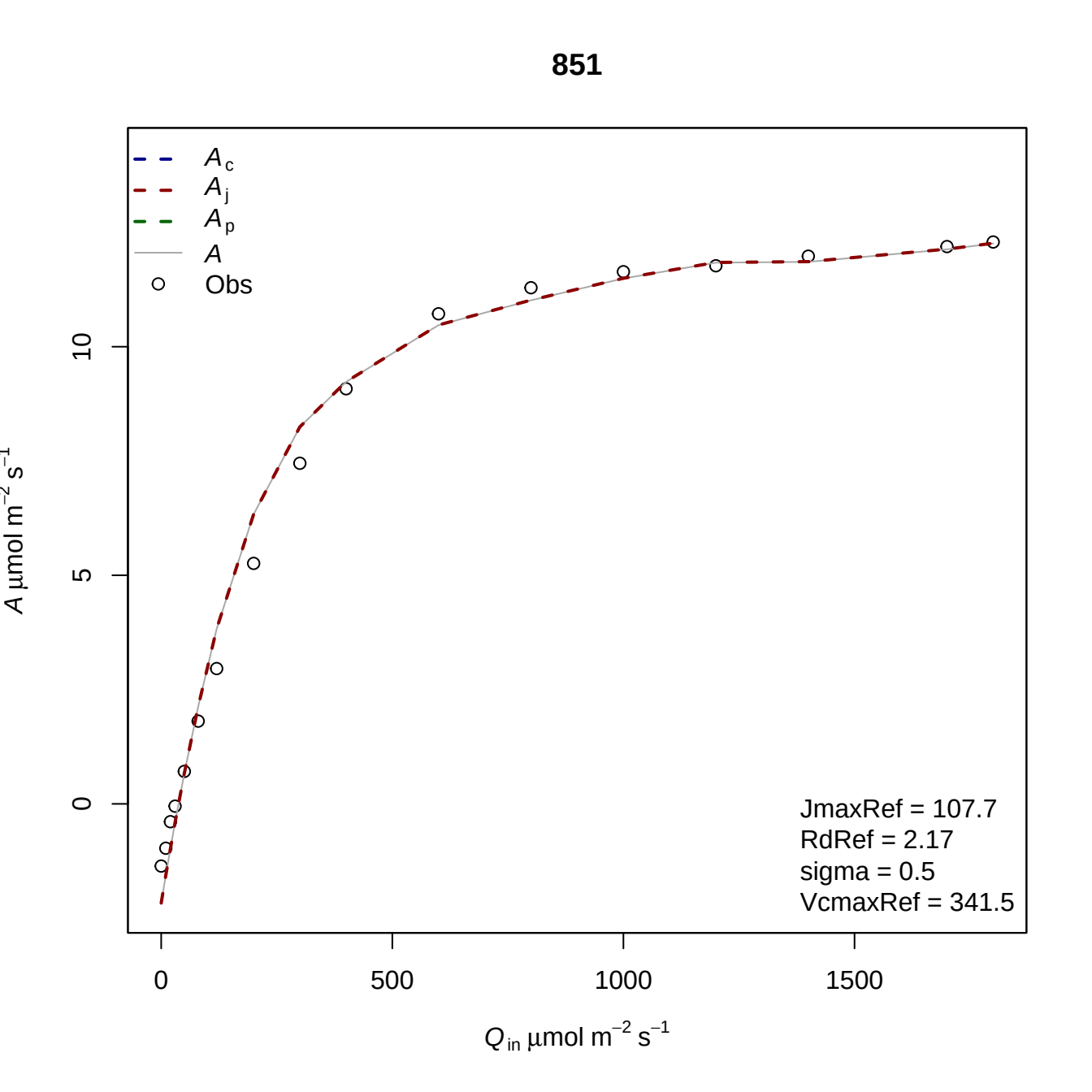

846
A_c
A_j
A_p
A
Obs
JmaxRef = 88.81
RdRef = 1.54
sigma = 0.24
VcmaxRef = 322.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

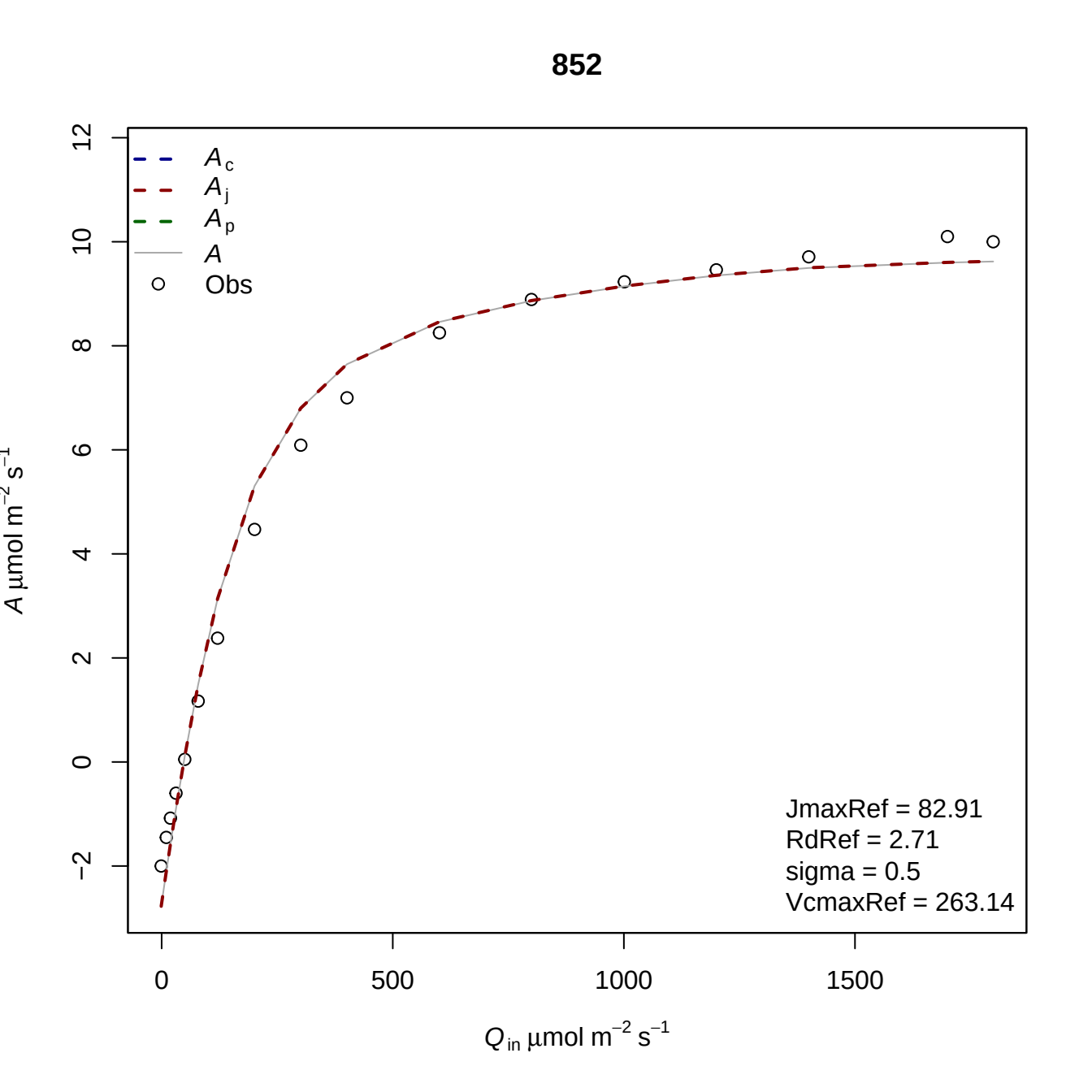

847
A_c
A_j
A_p
A
Obs
JmaxRef = 90.02
RdRef = 2.5
sigma = 0.6
VcmaxRef = 312.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

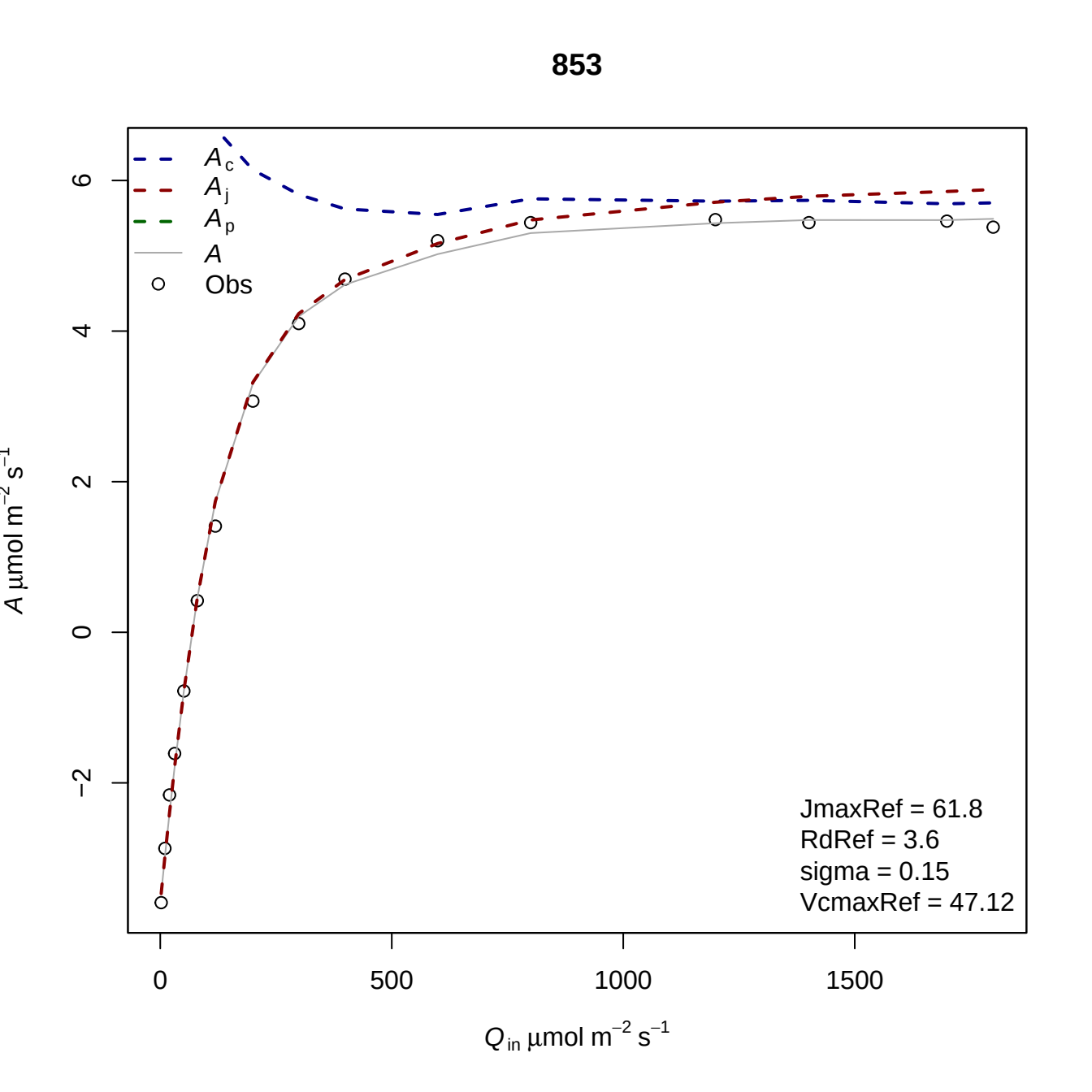

848
A_c
A_j
A_p
A
Obs
JmaxRef = 60.96
RdRef = 1.77
sigma = 0.1
VcmaxRef = 306.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

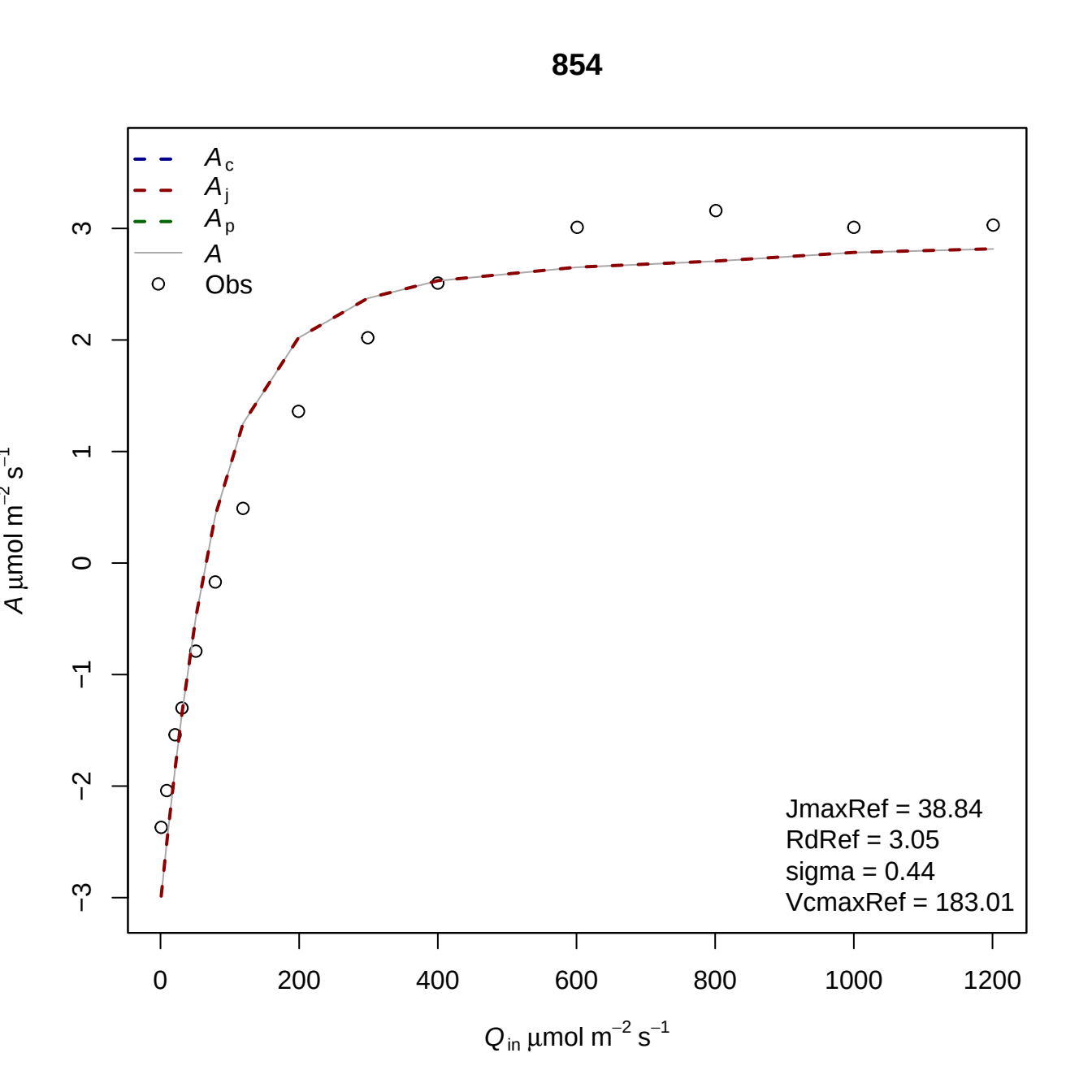

849
A_c
A_j
A_p
A
Obs
JmaxRef = 97.96
RdRef = 2.09
sigma = 0.15
VcmaxRef = 80.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

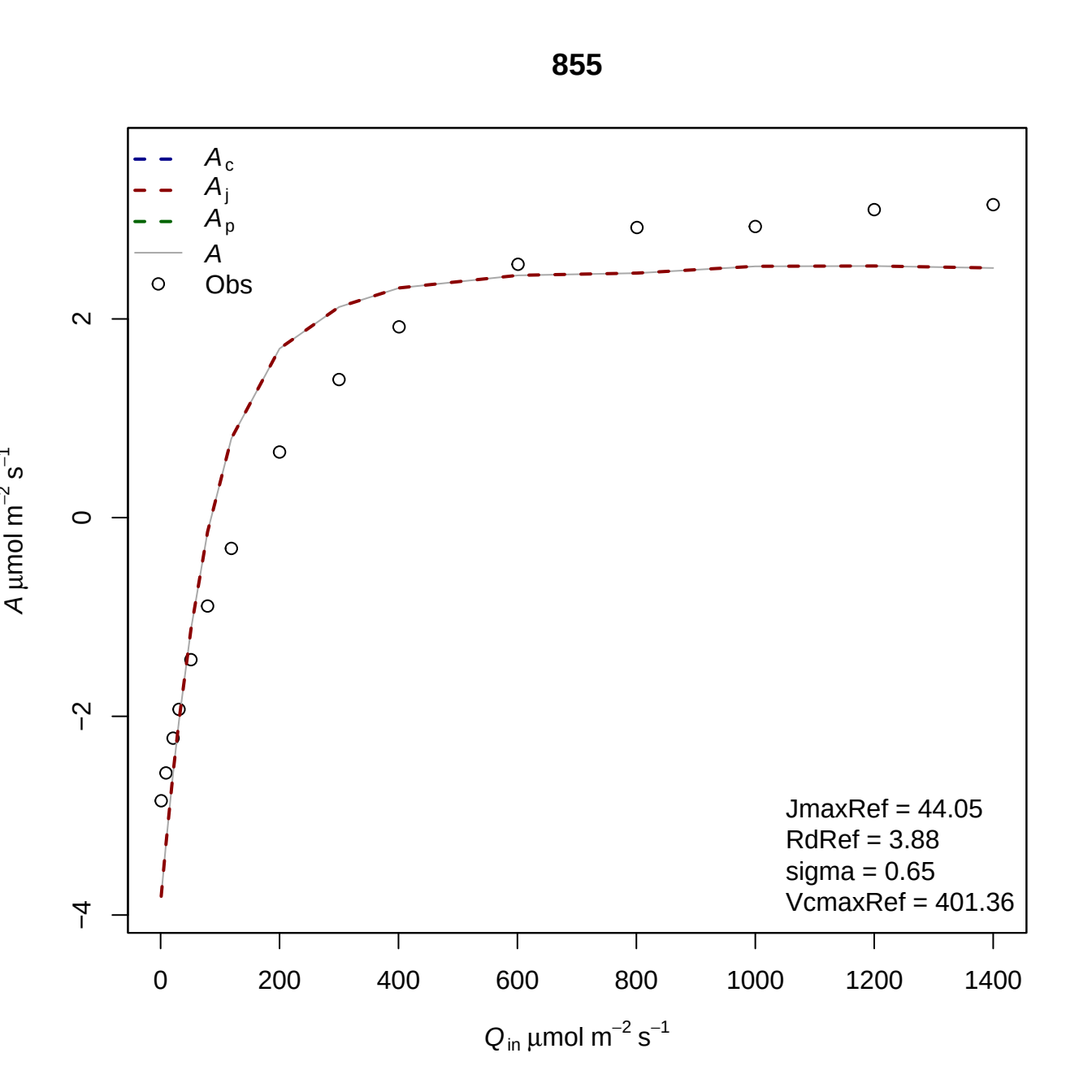

850
A_c
A_j
A_p
A
Obs
JmaxRef = 135.86
RdRef = 2.12
sigma = 0.46
VcmaxRef = 450.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

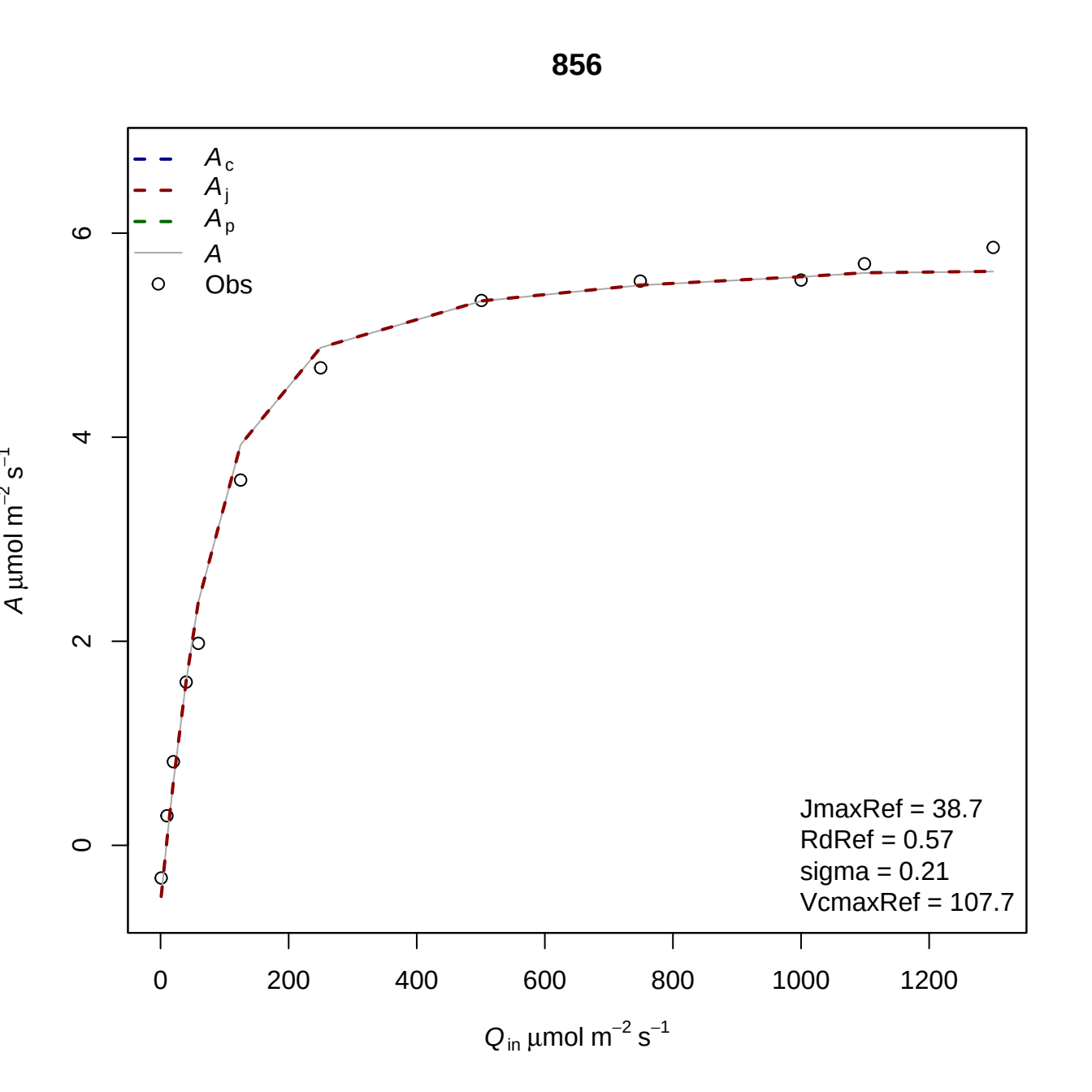

851
A_c
A_j
A_p
A
Obs
JmaxRef = 107.7
RdRef = 2.17
sigma = 0.5
VcmaxRef = 341.5
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

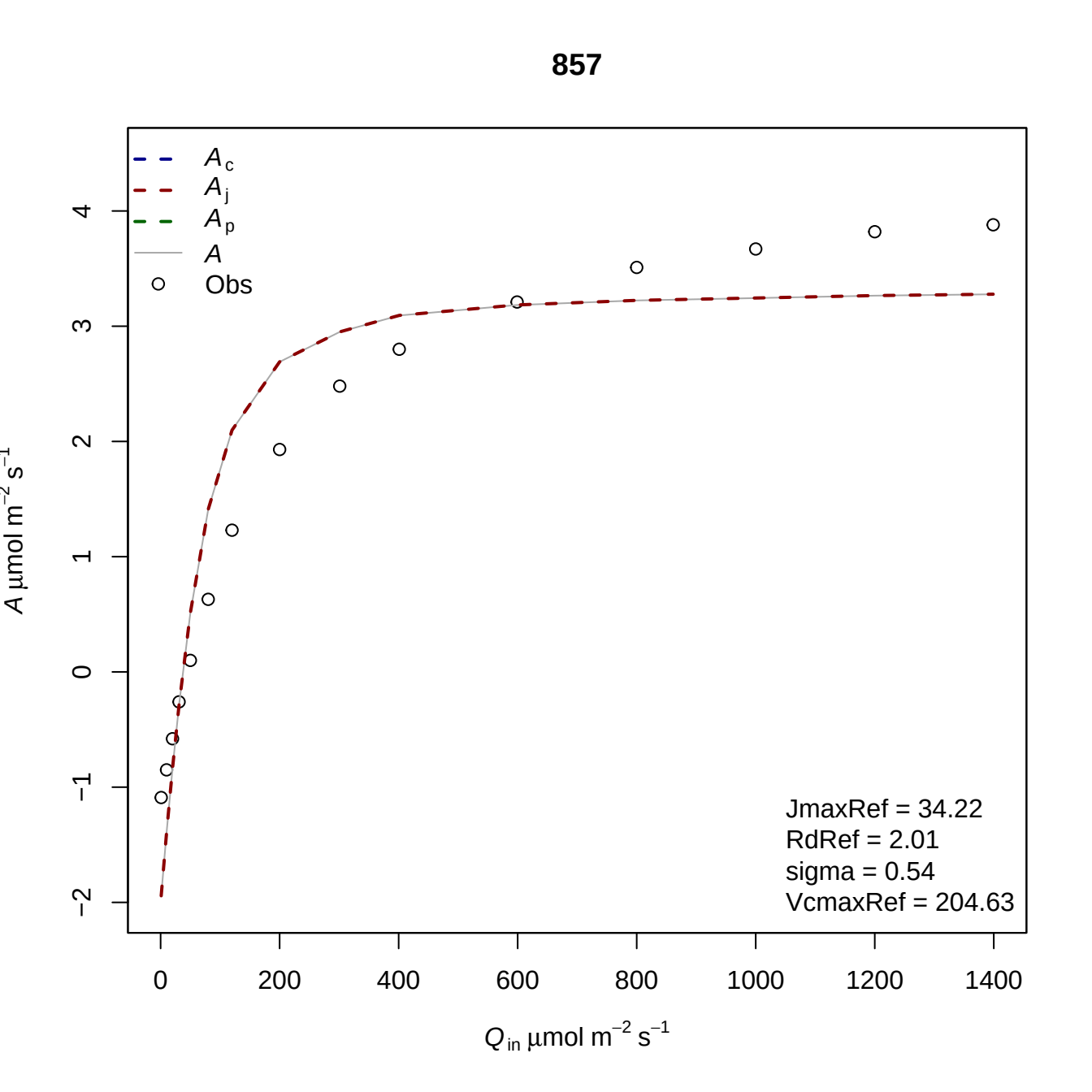

852
A_c
A_j
A_p
A
Obs
JmaxRef = 82.91
RdRef = 2.71
sigma = 0.5
VcmaxRef = 263.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

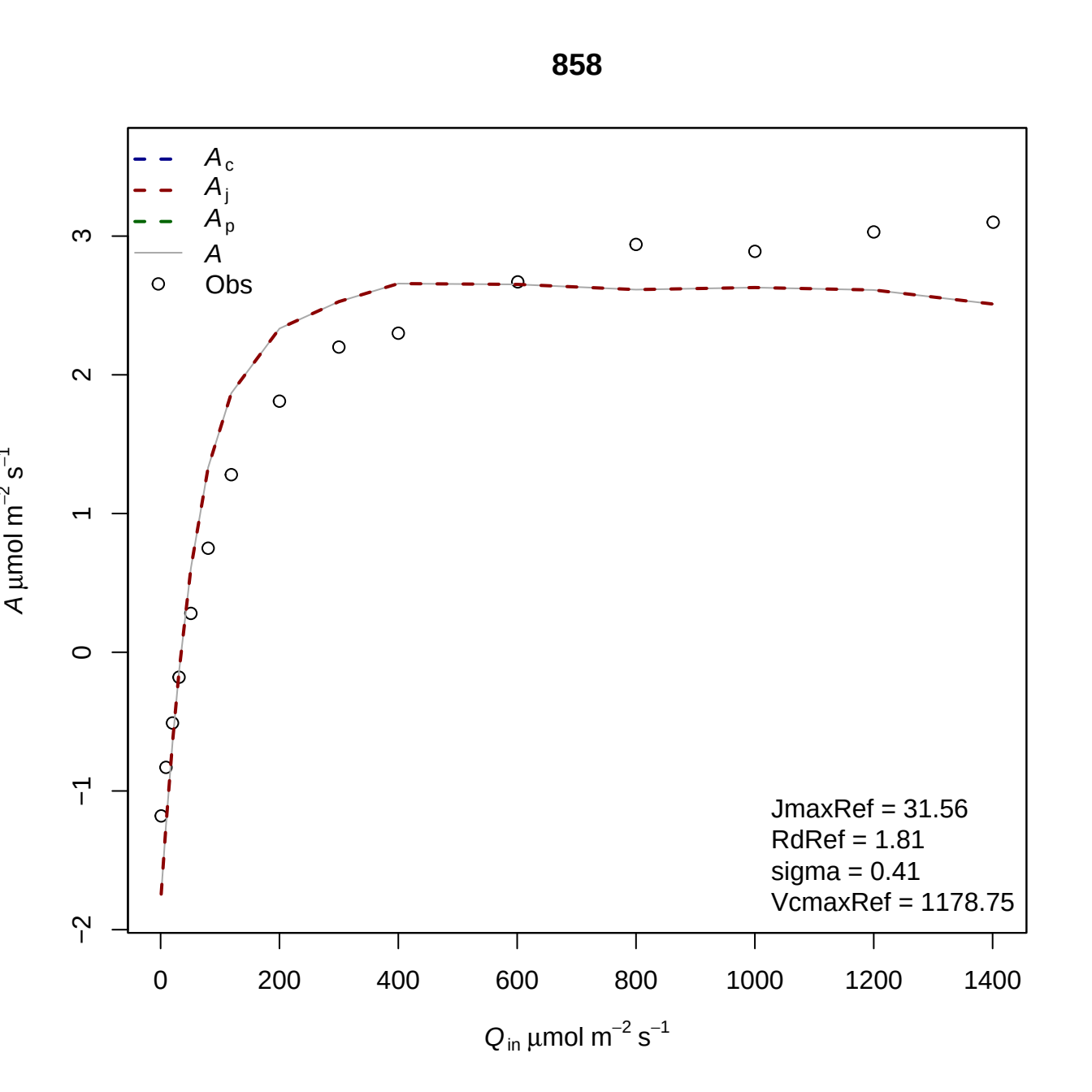

853
A_c
A_j
A_p
A
Obs
JmaxRef = 61.8
RdRef = 3.6
sigma = 0.15
VcmaxRef = 47.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

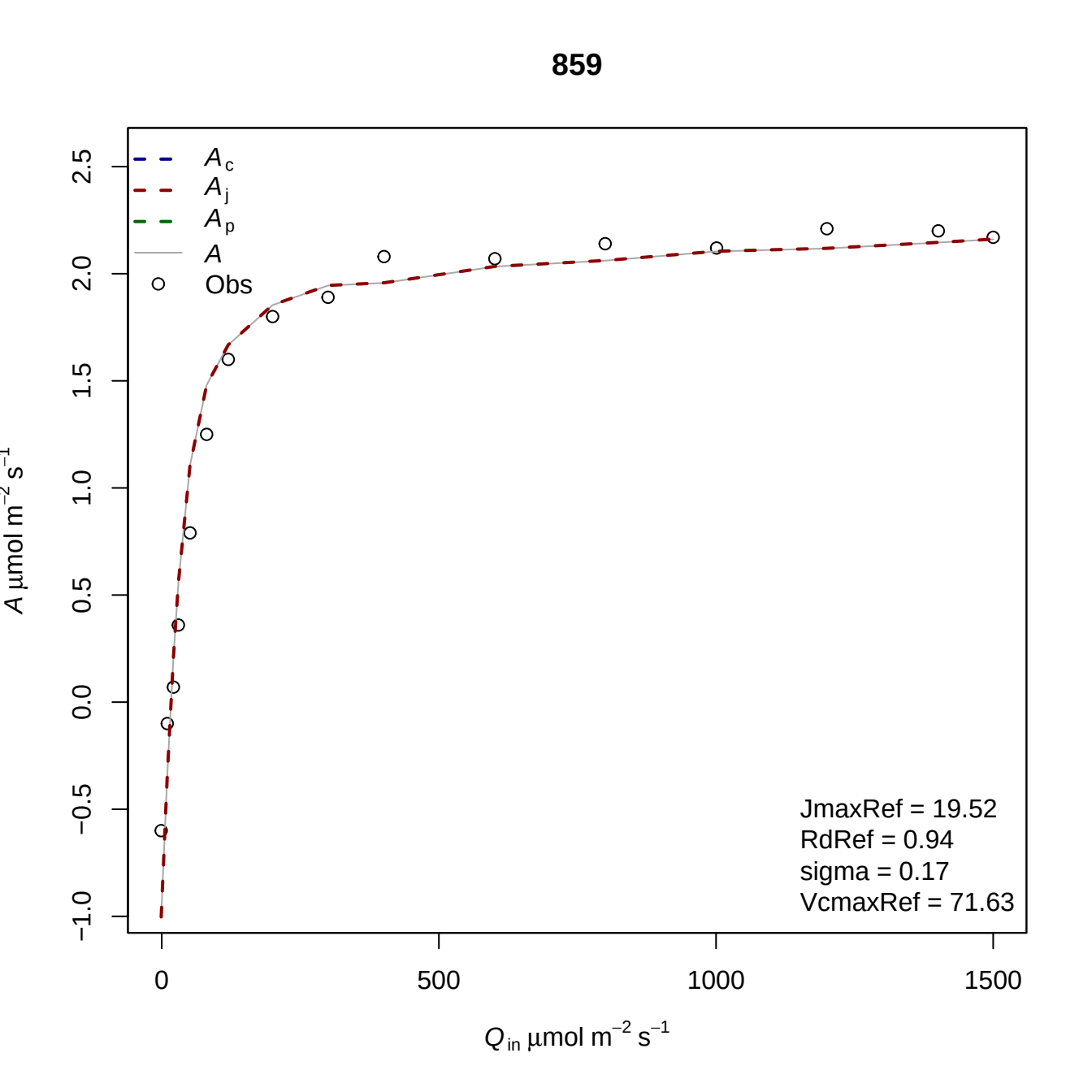

854
A_c
A_j
A_p
A
Obs
JmaxRef = 38.84
RdRef = 3.05
sigma = 0.44
VcmaxRef = 183.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−3
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200

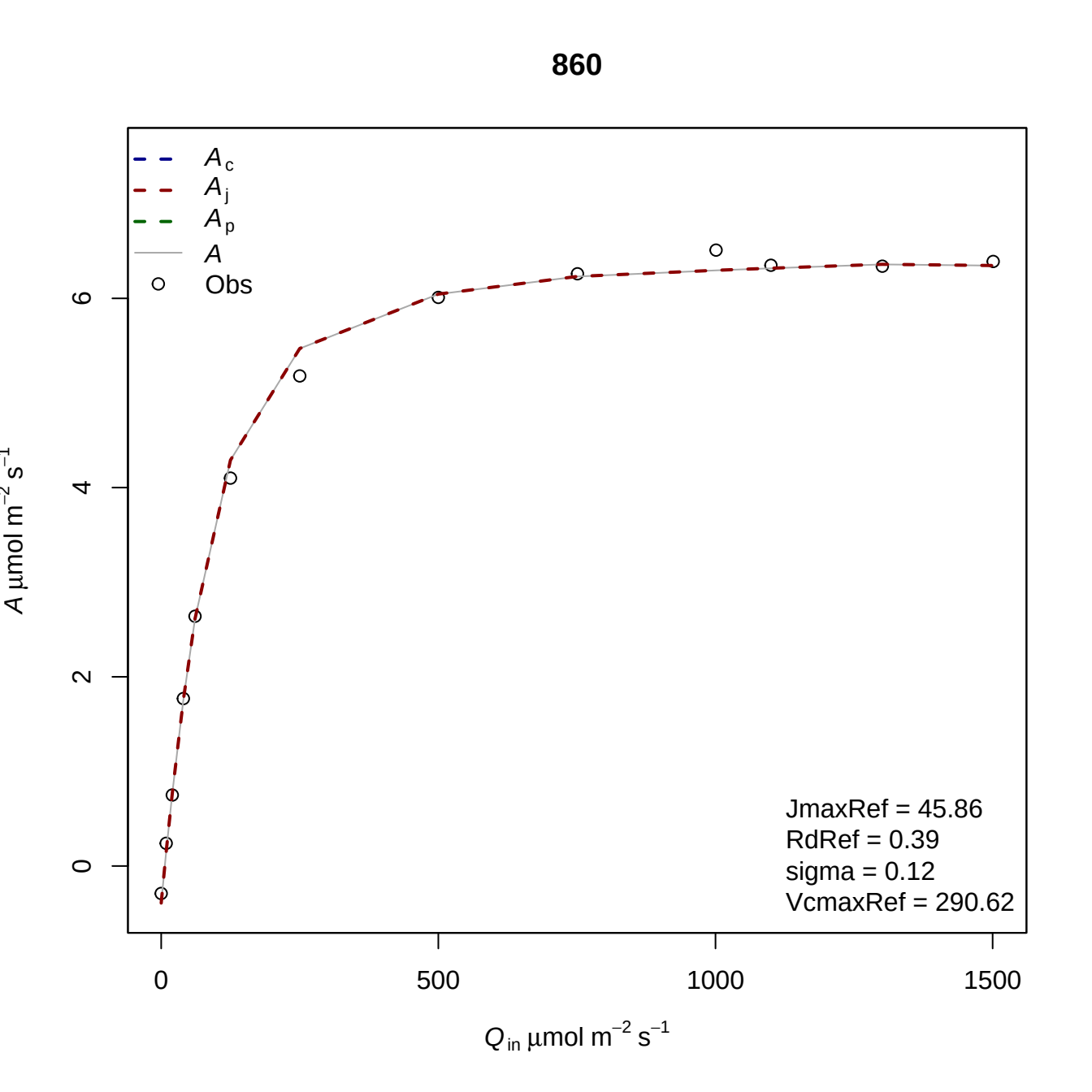

855
A_c
A_j
A_p
A
Obs
JmaxRef = 44.05
RdRef = 3.88
sigma = 0.65
VcmaxRef = 401.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
0
200
400
600
800
1000
1200
1400

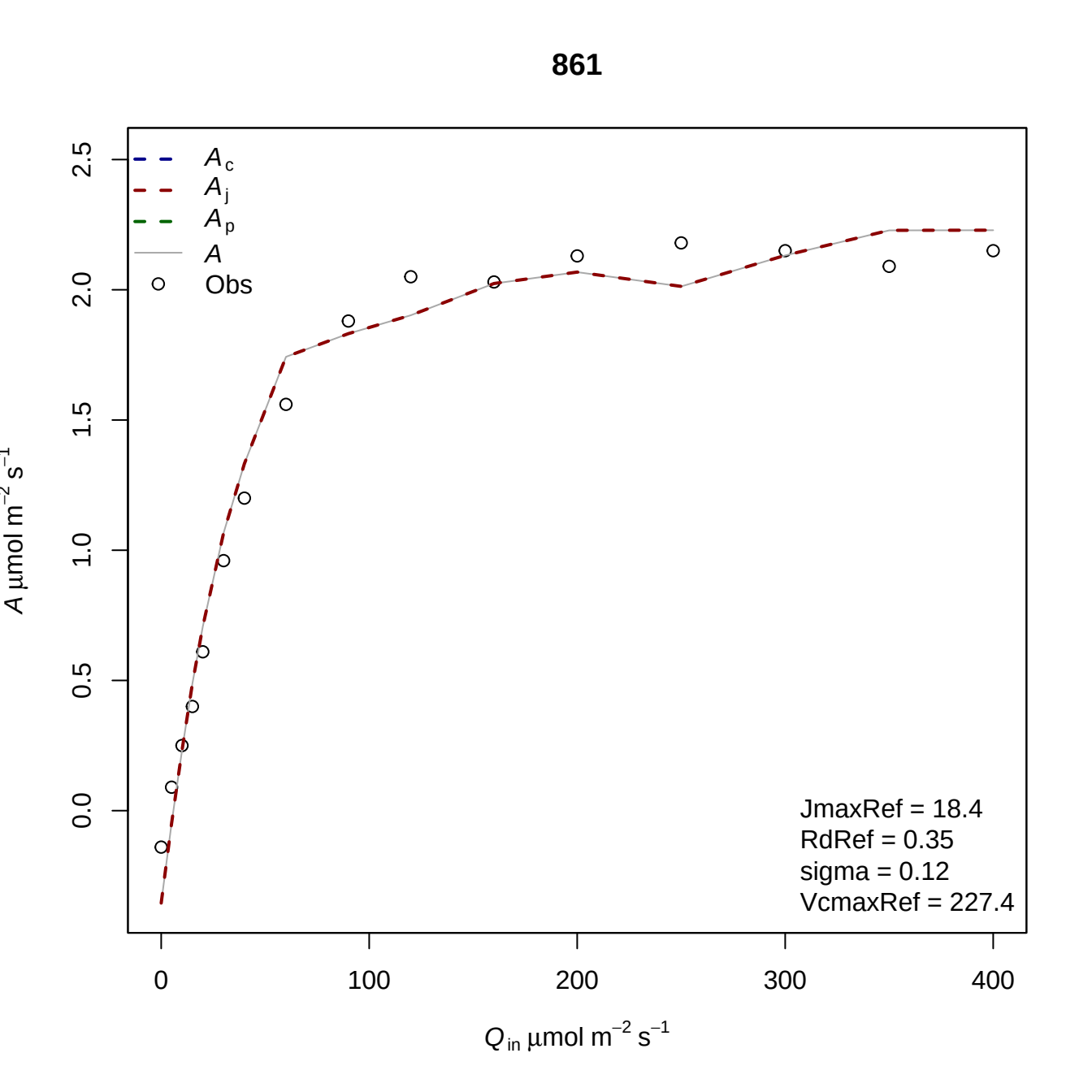

856
A_c
A_j
A_p
A
Obs
JmaxRef = 38.7
RdRef = 0.57
sigma = 0.21
VcmaxRef = 107.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000
1200

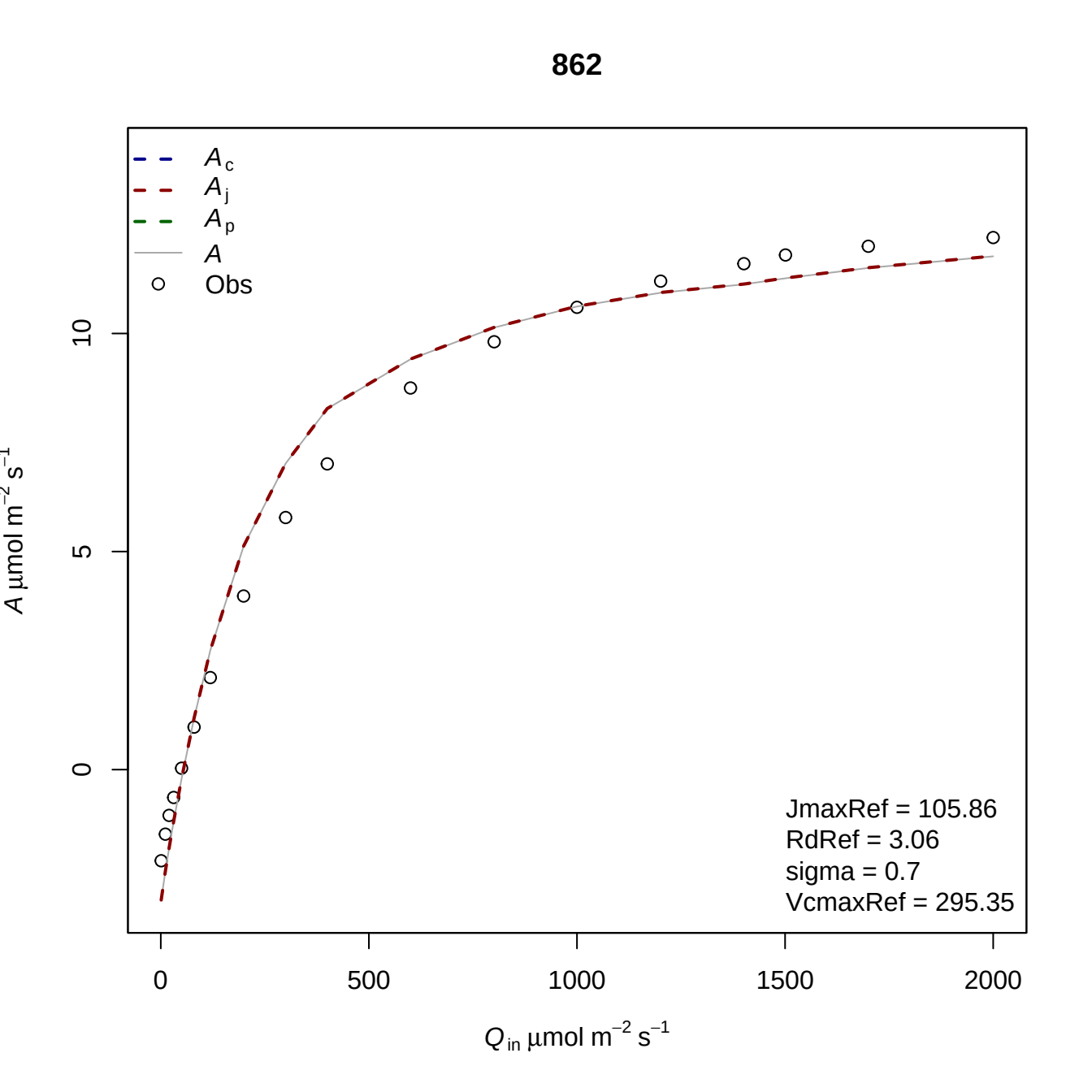

857
A_c
A_j
A_p
A
Obs
JmaxRef = 34.22
RdRef = 2.01
sigma = 0.54
VcmaxRef = 204.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

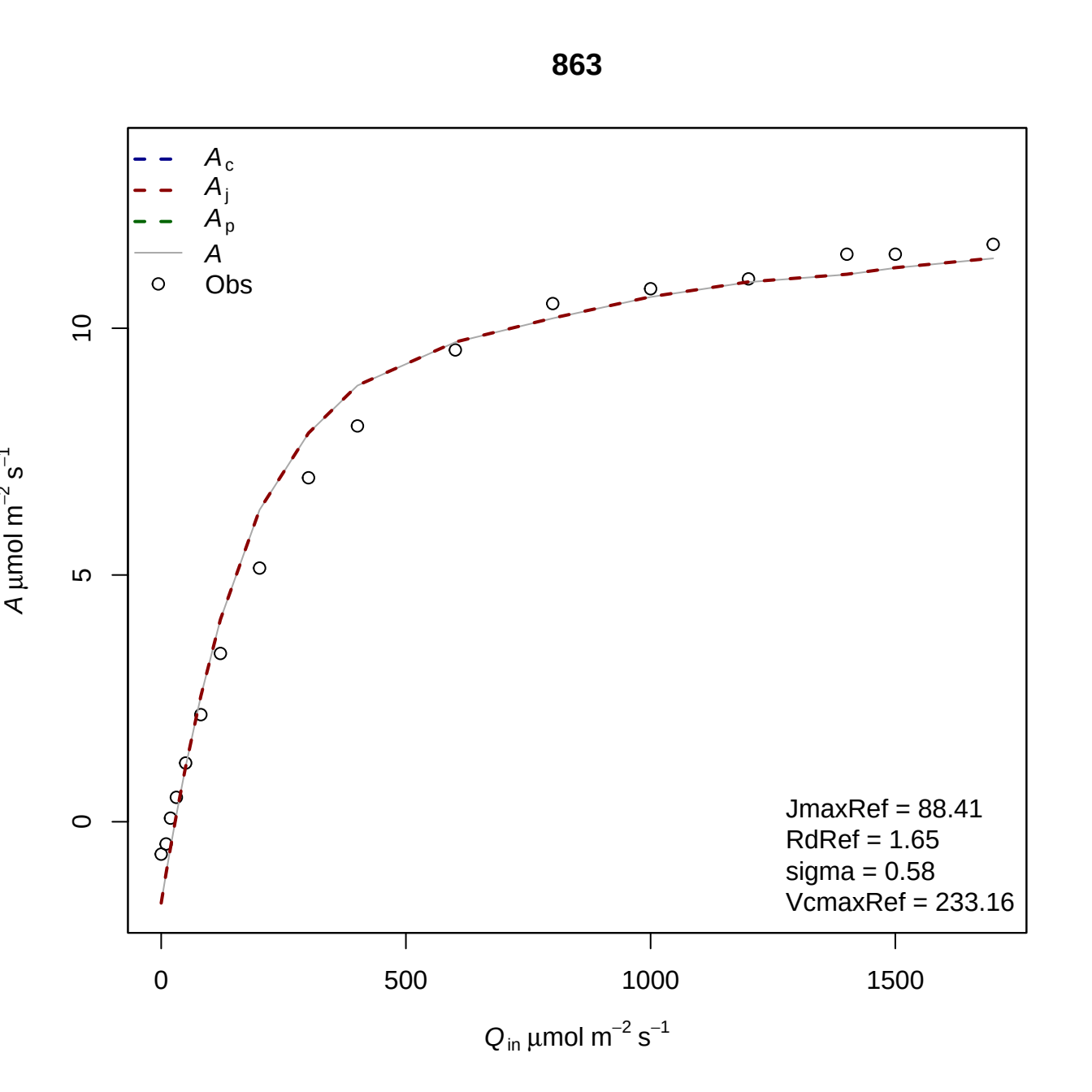

858
A_c
A_j
A_p
A
Obs
JmaxRef = 31.56
RdRef = 1.81
sigma = 0.41
VcmaxRef = 1178.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

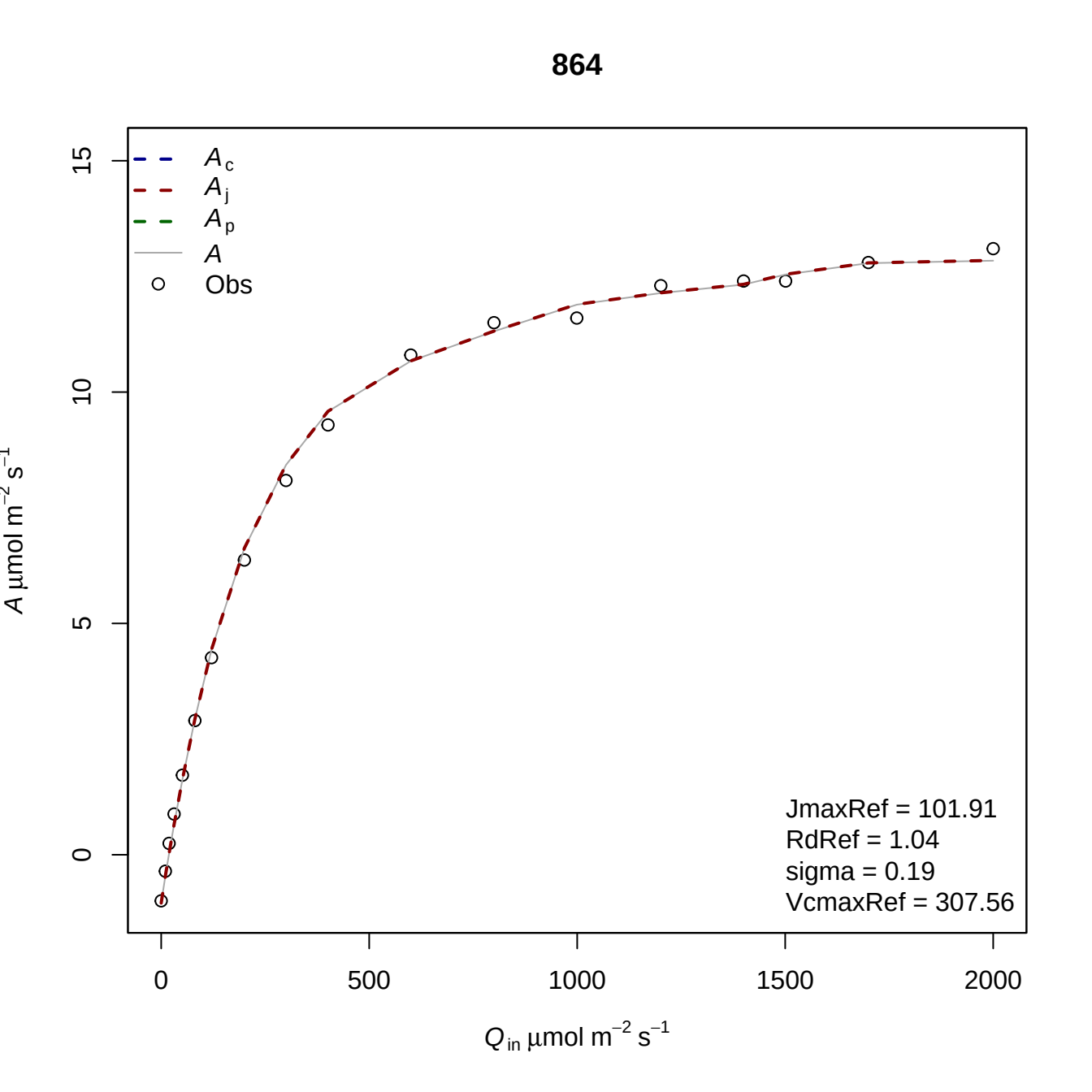

859
A_c
A_j
A_p
A
Obs
JmaxRef = 19.52
RdRef = 0.94
sigma = 0.17
VcmaxRef = 71.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500

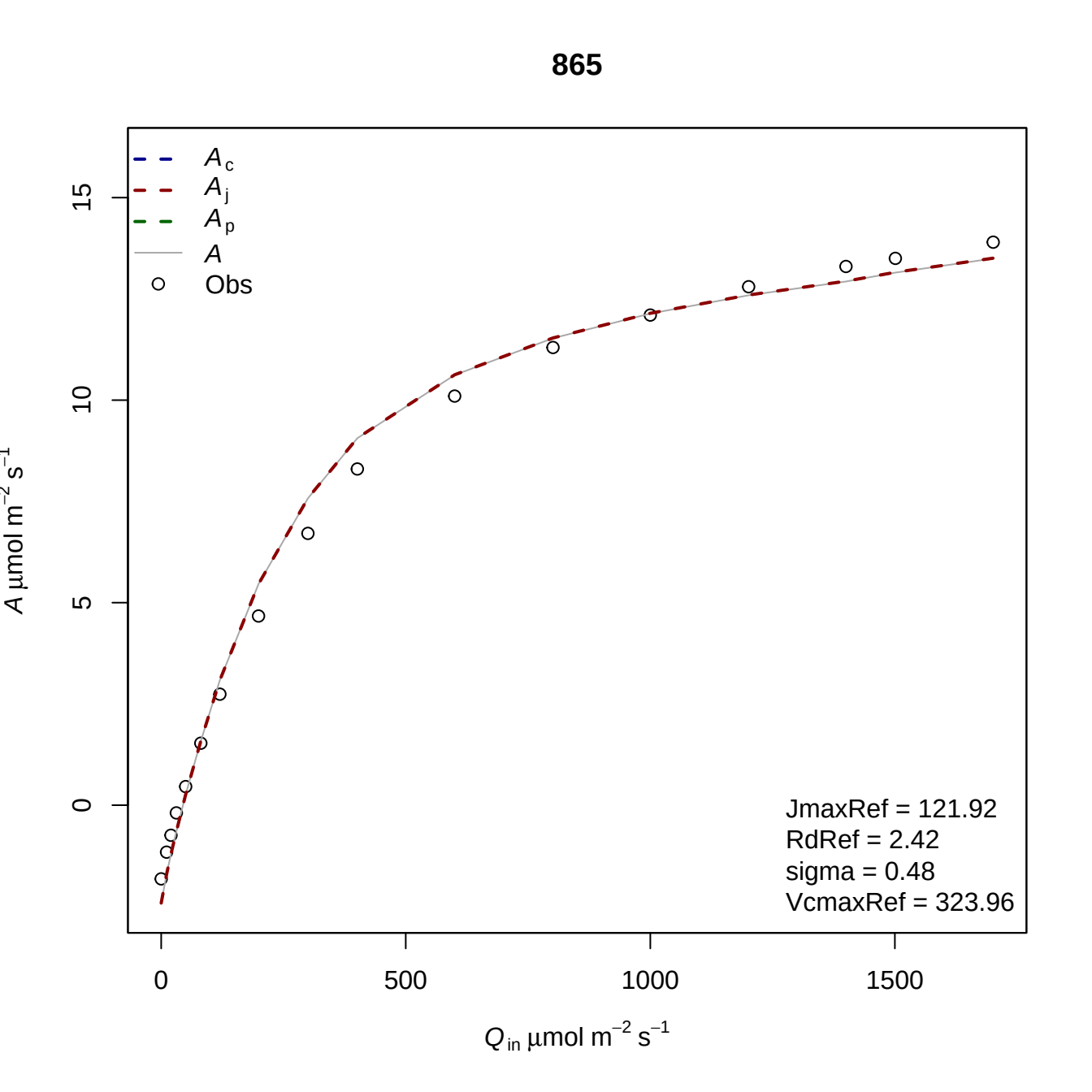

860
A_c
A_j
A_p
A
Obs
JmaxRef = 45.86
RdRef = 0.39
sigma = 0.12
VcmaxRef = 290.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

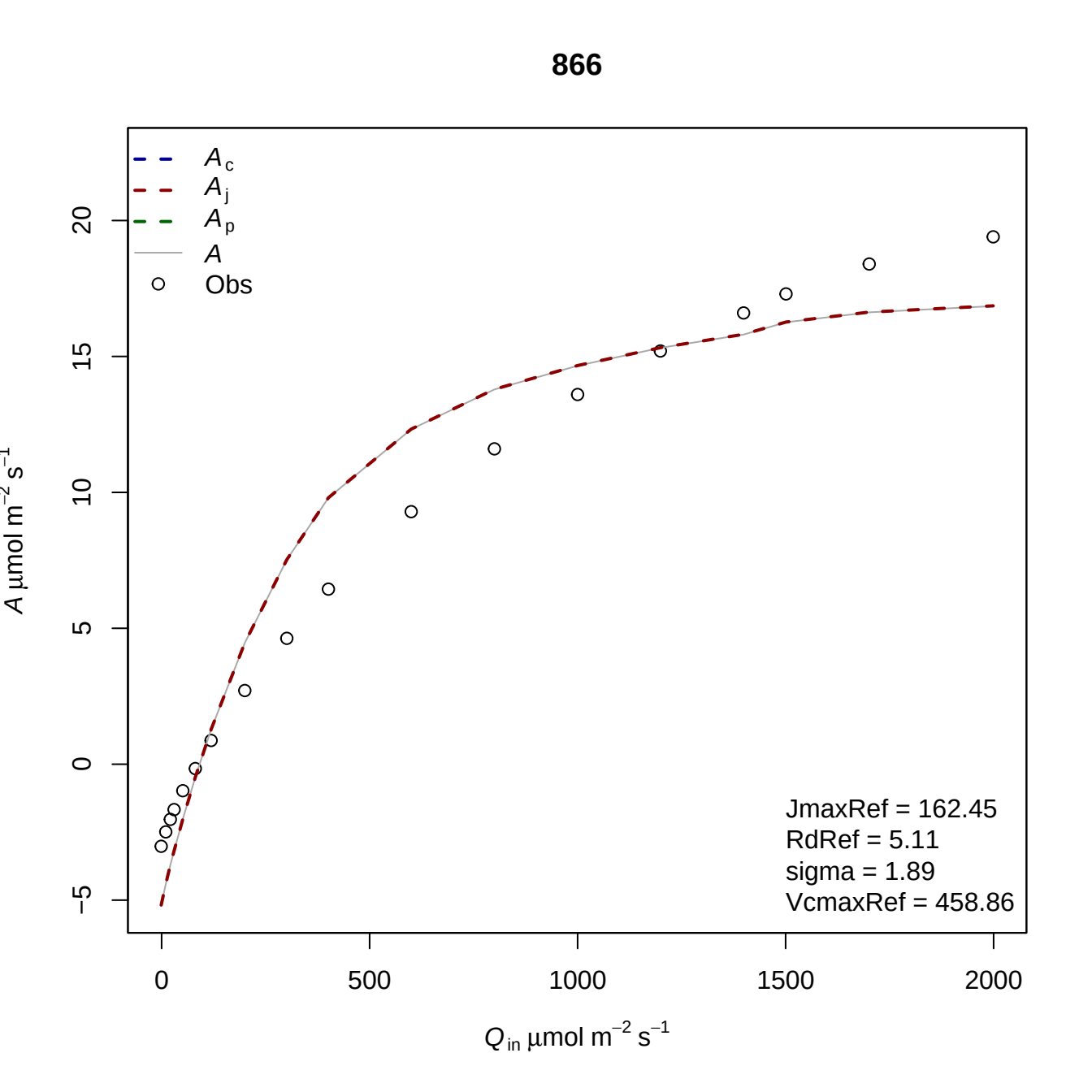

861
A_c
A_j
A_p
A
Obs
JmaxRef = 18.4
RdRef = 0.35
sigma = 0.12
VcmaxRef = 227.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
0
100
200
300
400

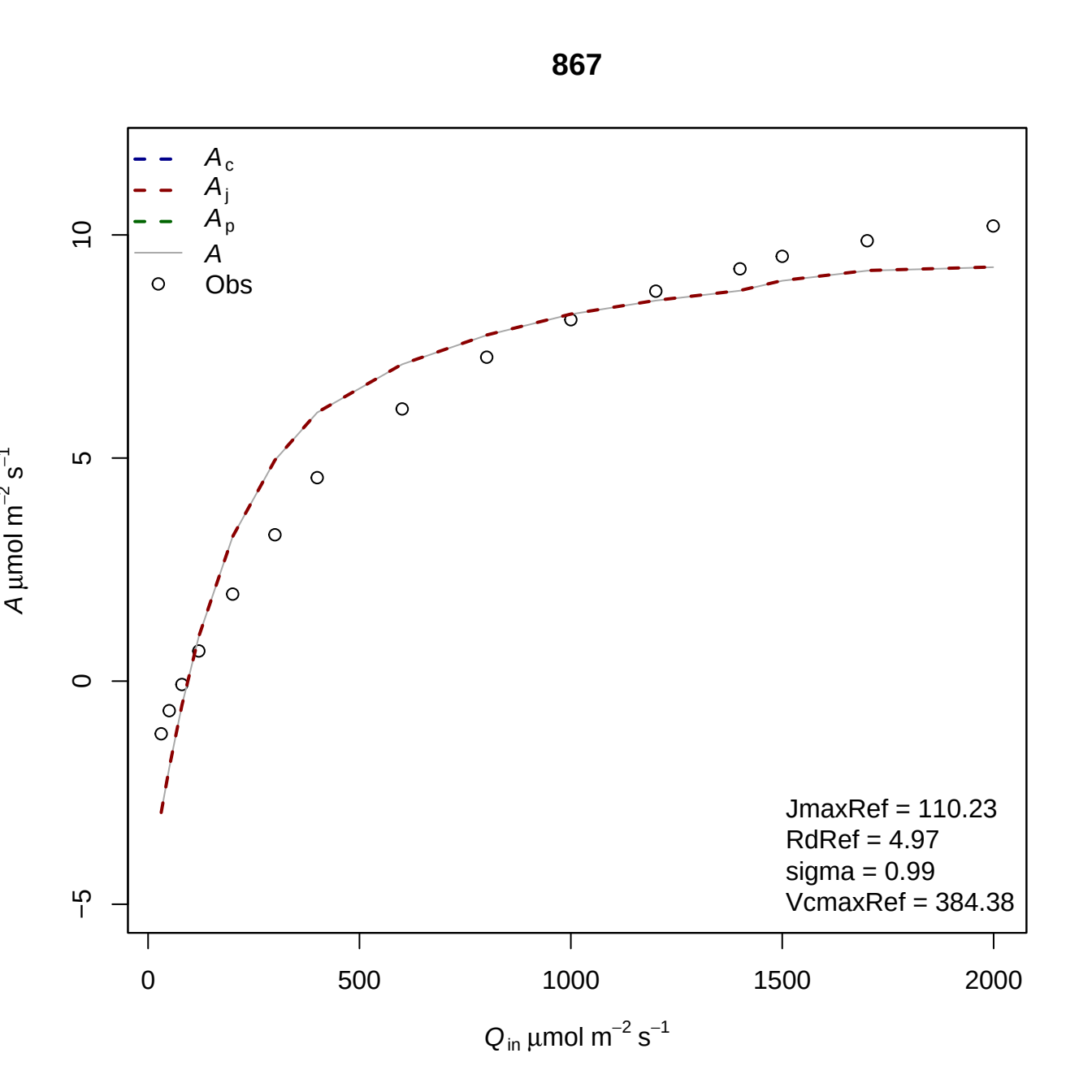

862
A_c
A_j
A_p
A
Obs
JmaxRef = 105.86
RdRef = 3.06
sigma = 0.7
VcmaxRef = 295.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

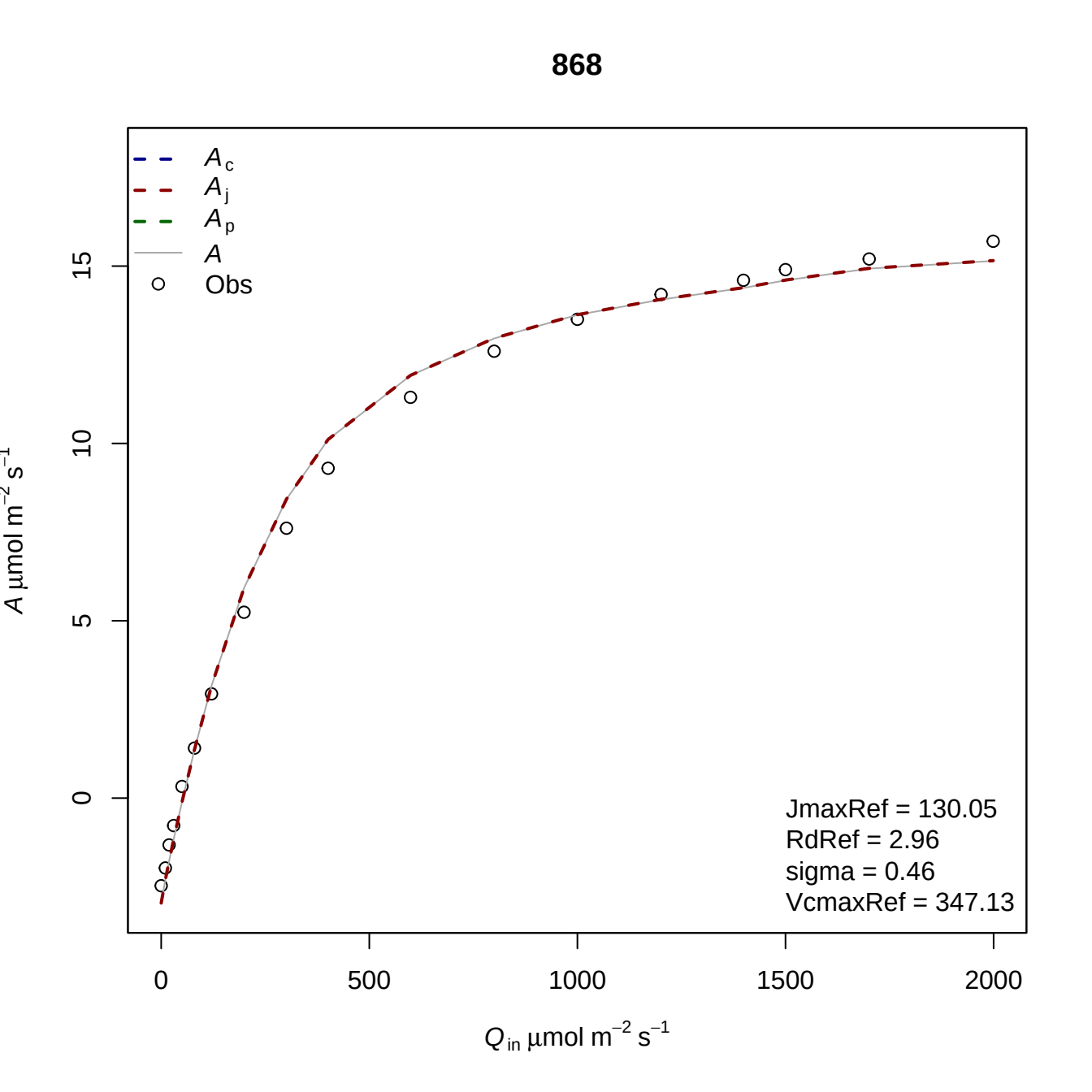

863
A_c
A_j
A_p
A
Obs
JmaxRef = 88.41
RdRef = 1.65
sigma = 0.58
VcmaxRef = 233.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

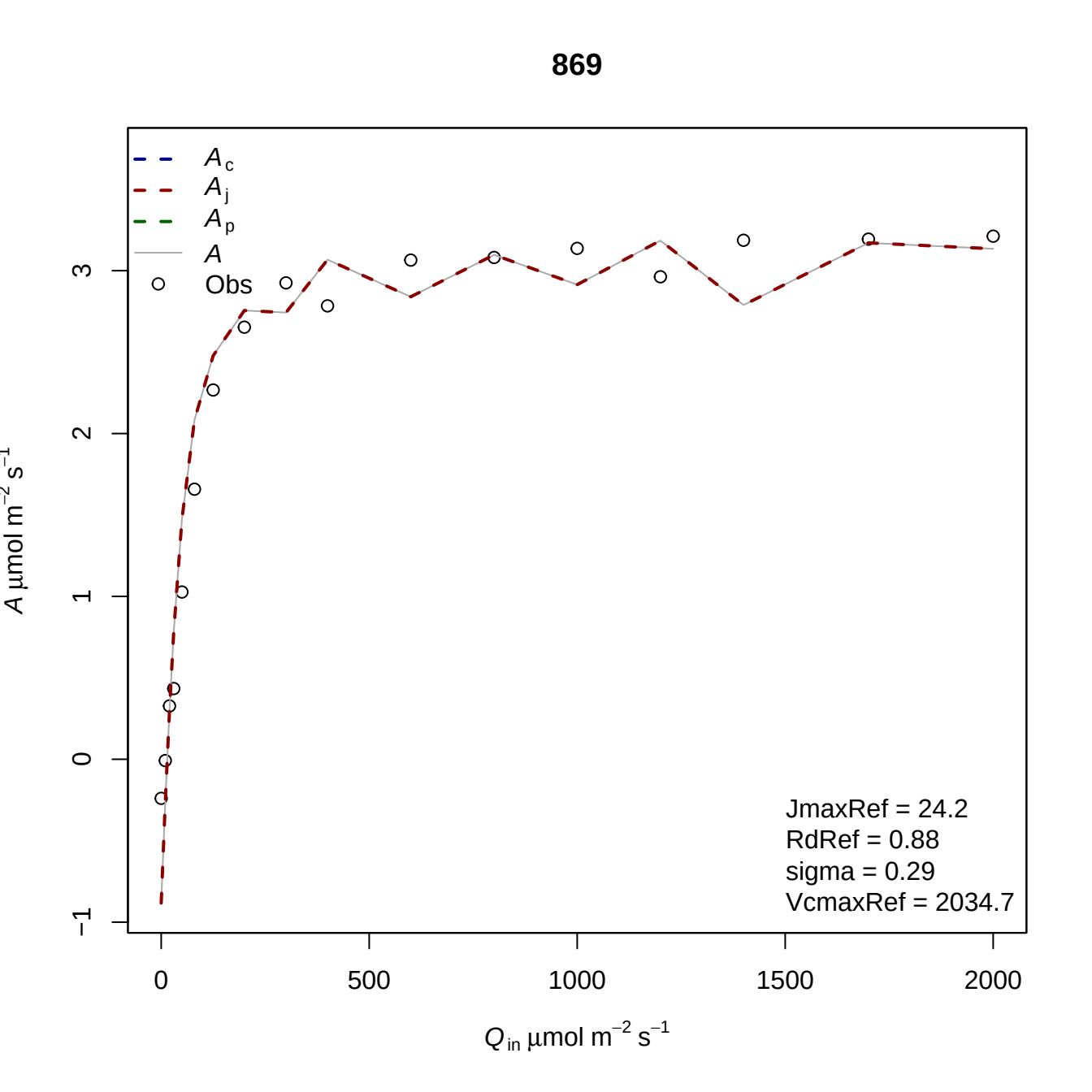

864
A_c
A_j
A_p
A
Obs
JmaxRef = 101.91
RdRef = 1.04
sigma = 0.19
VcmaxRef = 307.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

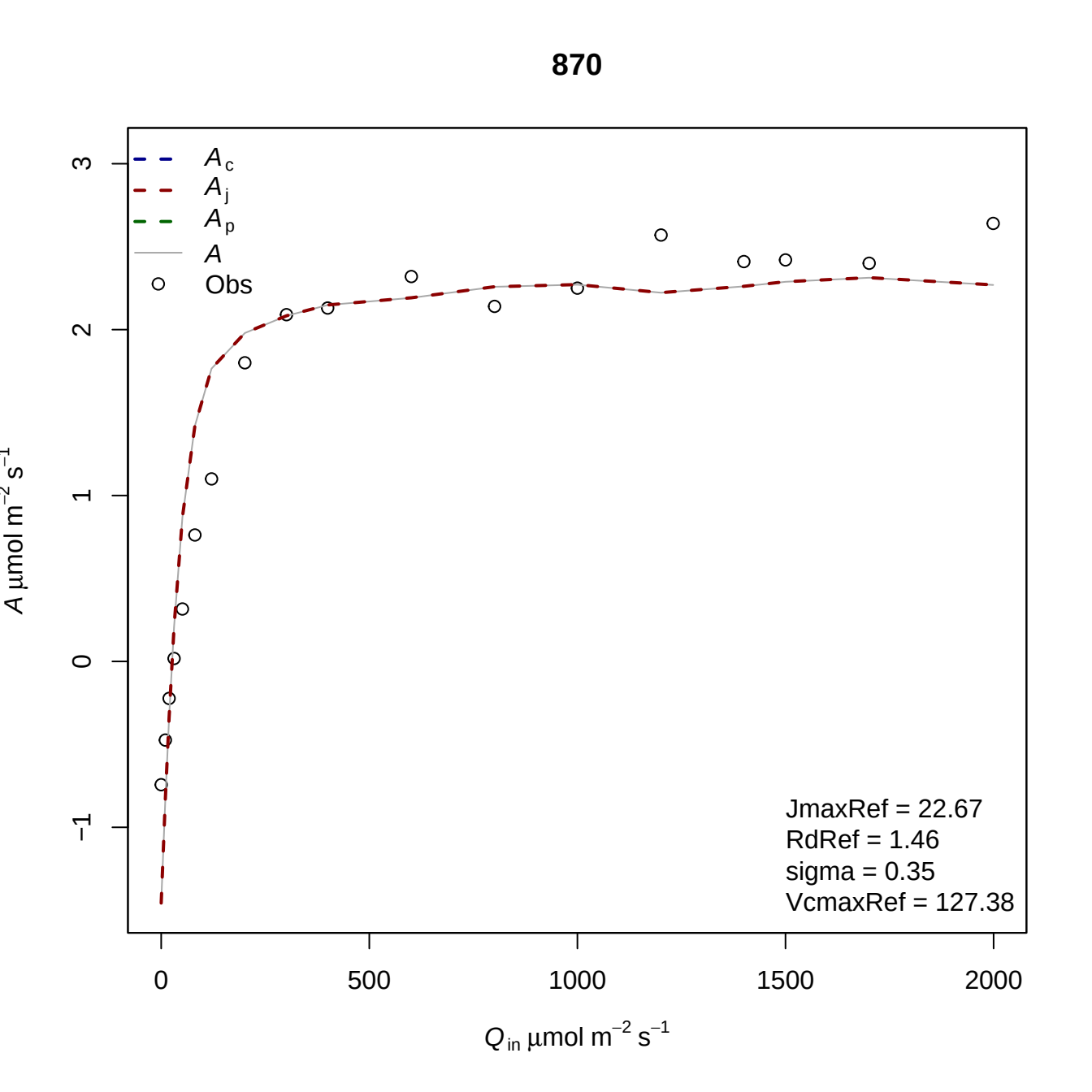

865
A_c
A_j
A_p
A
Obs
JmaxRef = 121.92
RdRef = 2.42
sigma = 0.48
VcmaxRef = 323.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

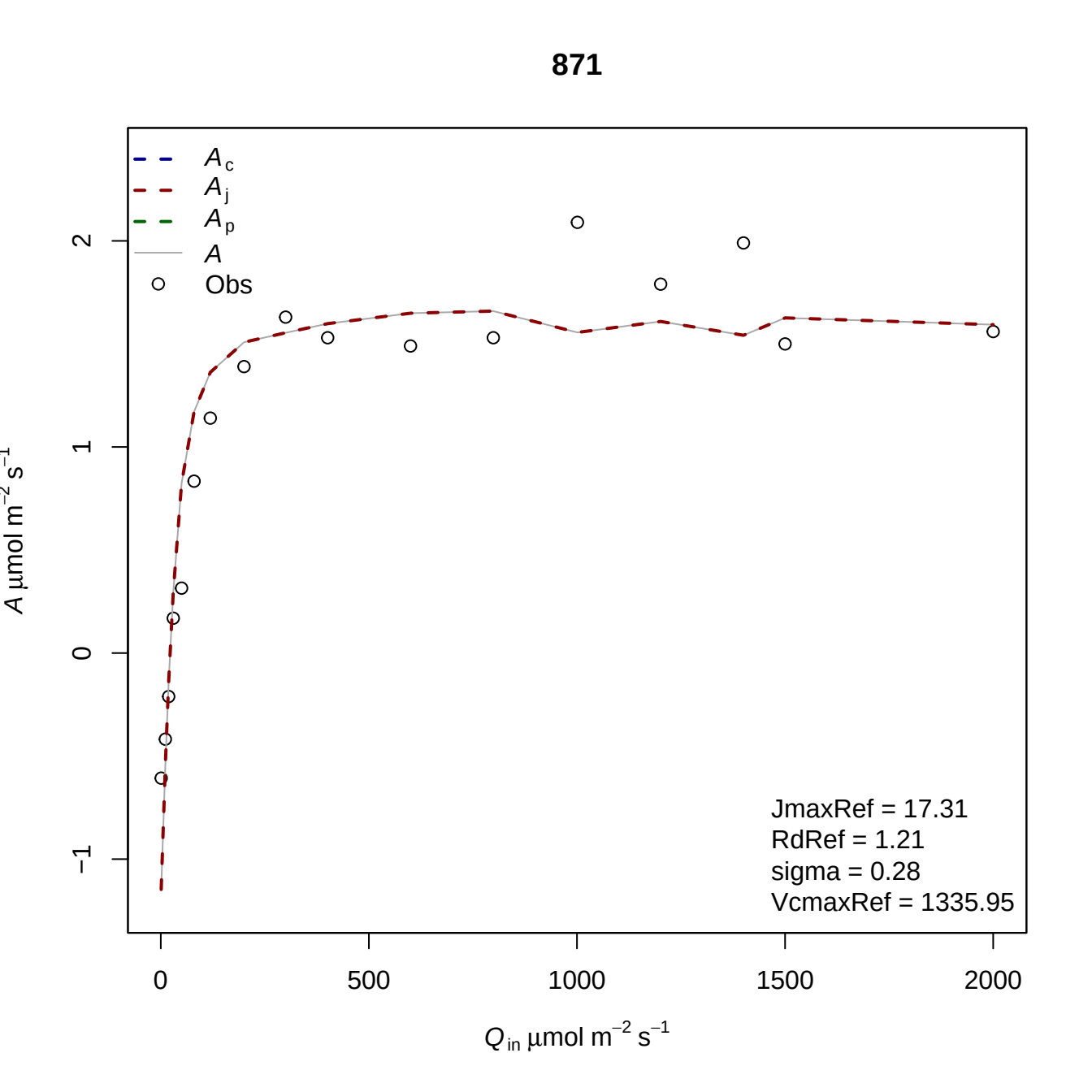

866
A_c
A_j
A_p
A
Obs
JmaxRef = 162.45
RdRef = 5.11
sigma = 1.89
VcmaxRef = 458.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

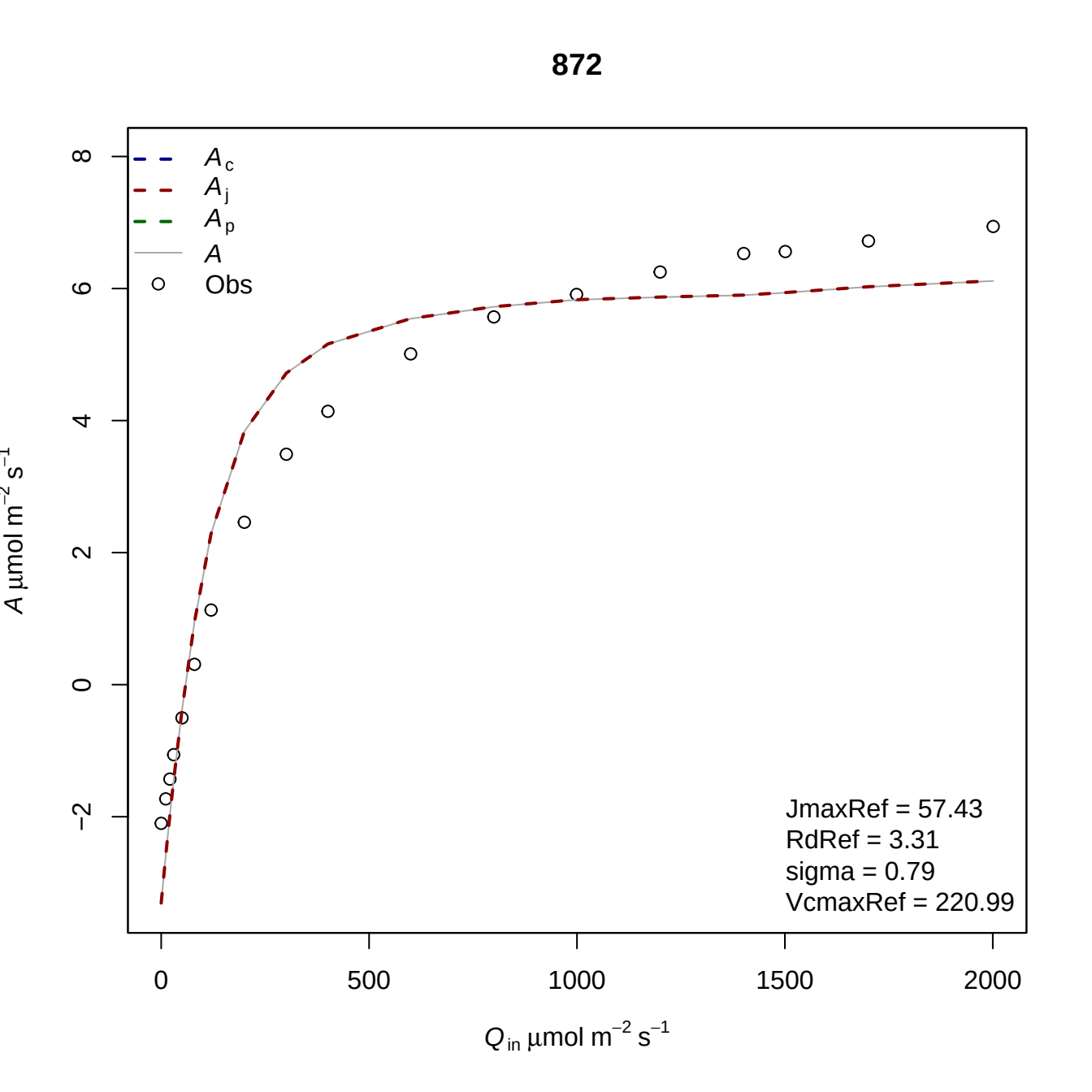

867
A_c
A_j
A_p
A
Obs
JmaxRef = 110.23
RdRef = 4.97
sigma = 0.99
VcmaxRef = 384.38
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

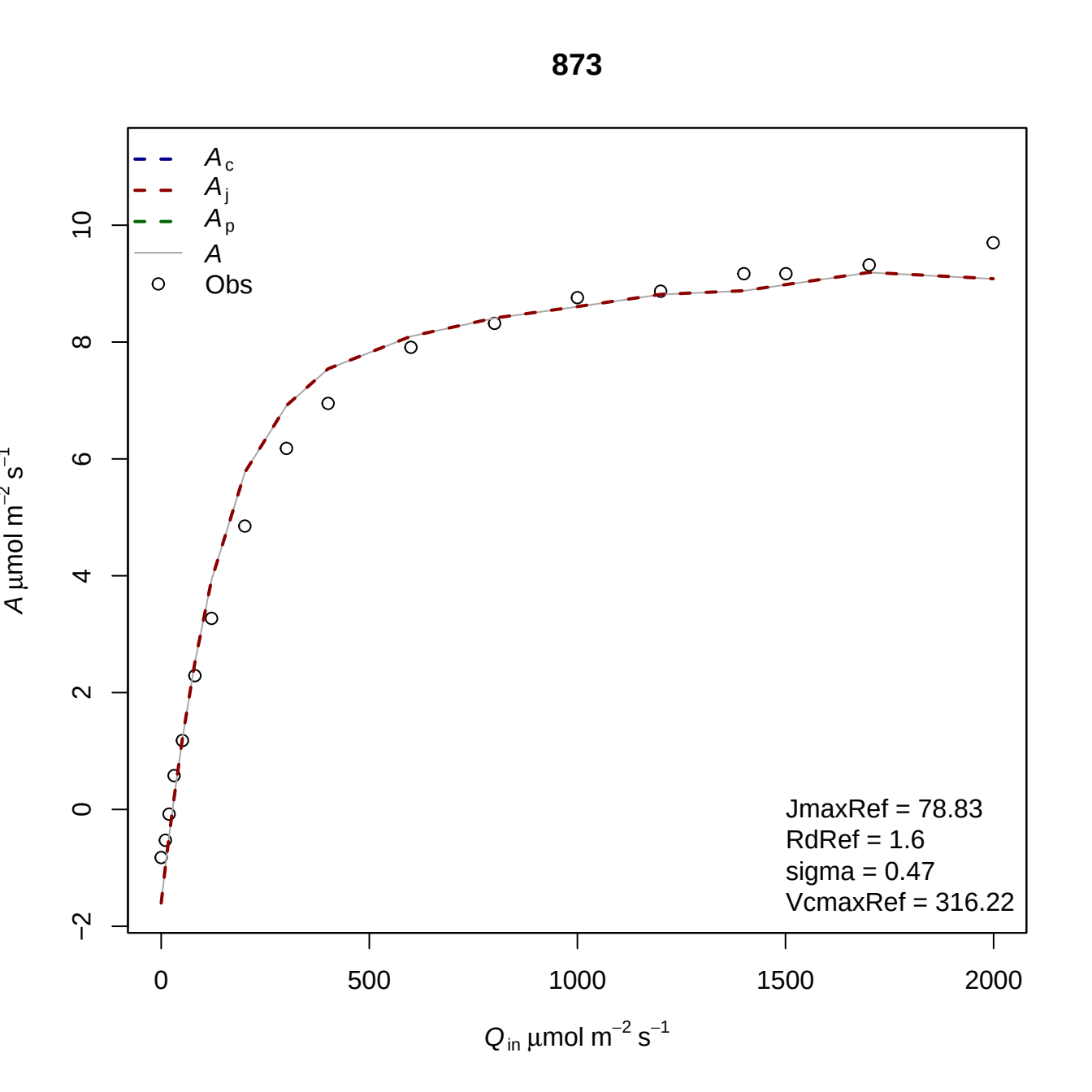

868
A_c
A_j
A_p
A
Obs
JmaxRef = 130.05
RdRef = 2.96
sigma = 0.46
VcmaxRef = 347.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

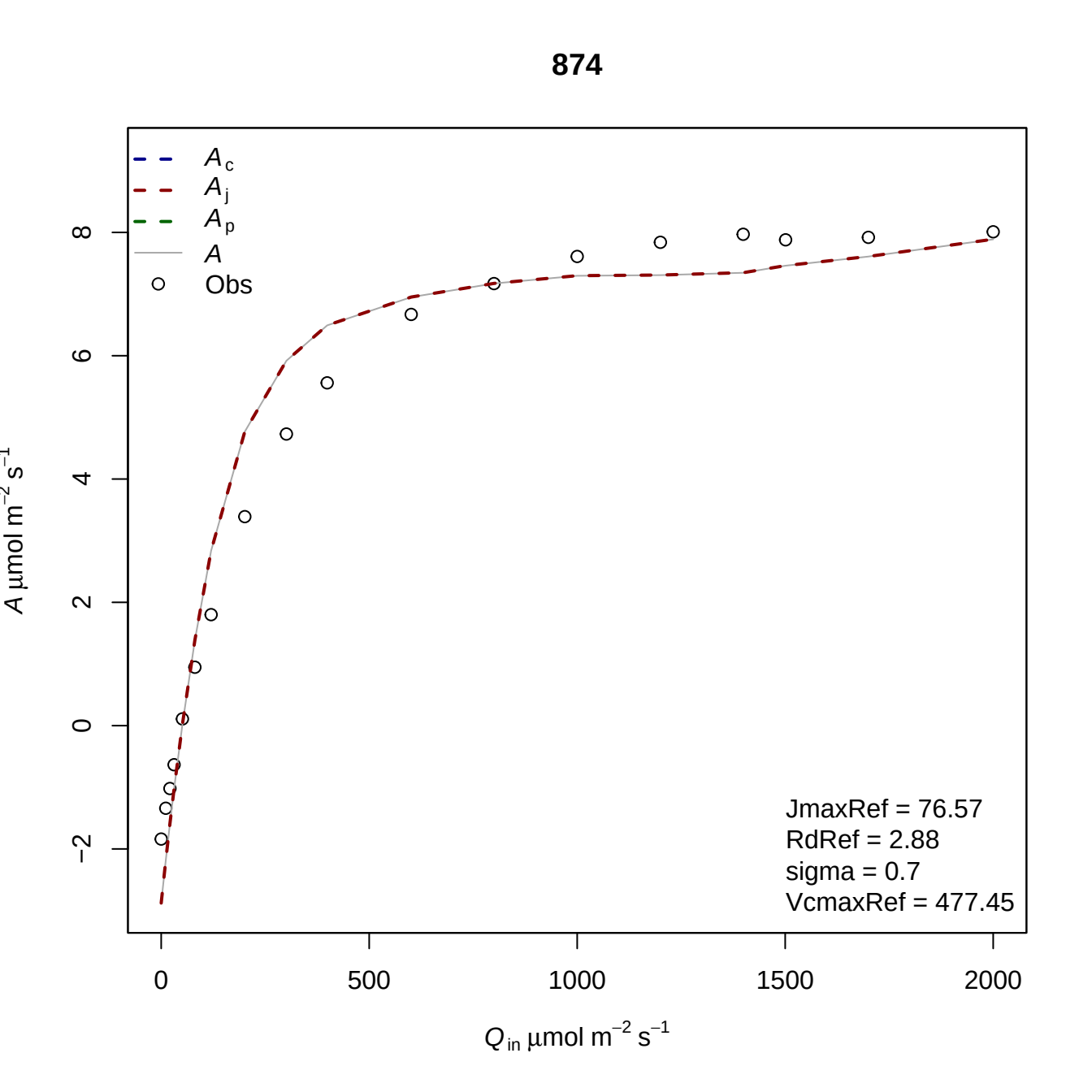

869
A_c
A_j
A_p
A
Obs
JmaxRef = 24.2
RdRef = 0.88
sigma = 0.29
VcmaxRef = 2034.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

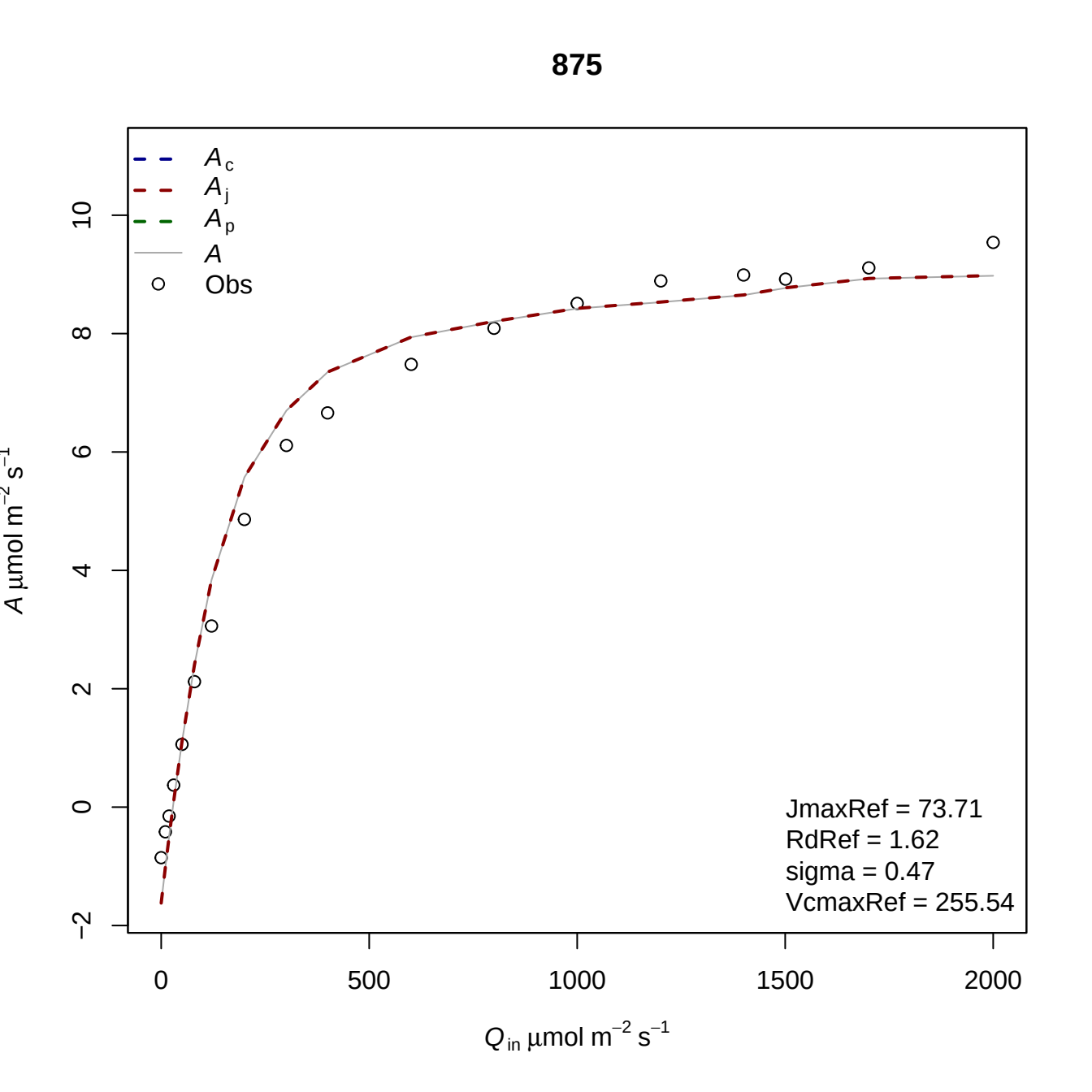

870
A_c
A_j
A_p
A
Obs
JmaxRef = 22.67
RdRef = 1.46
sigma = 0.35
VcmaxRef = 127.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500
2000

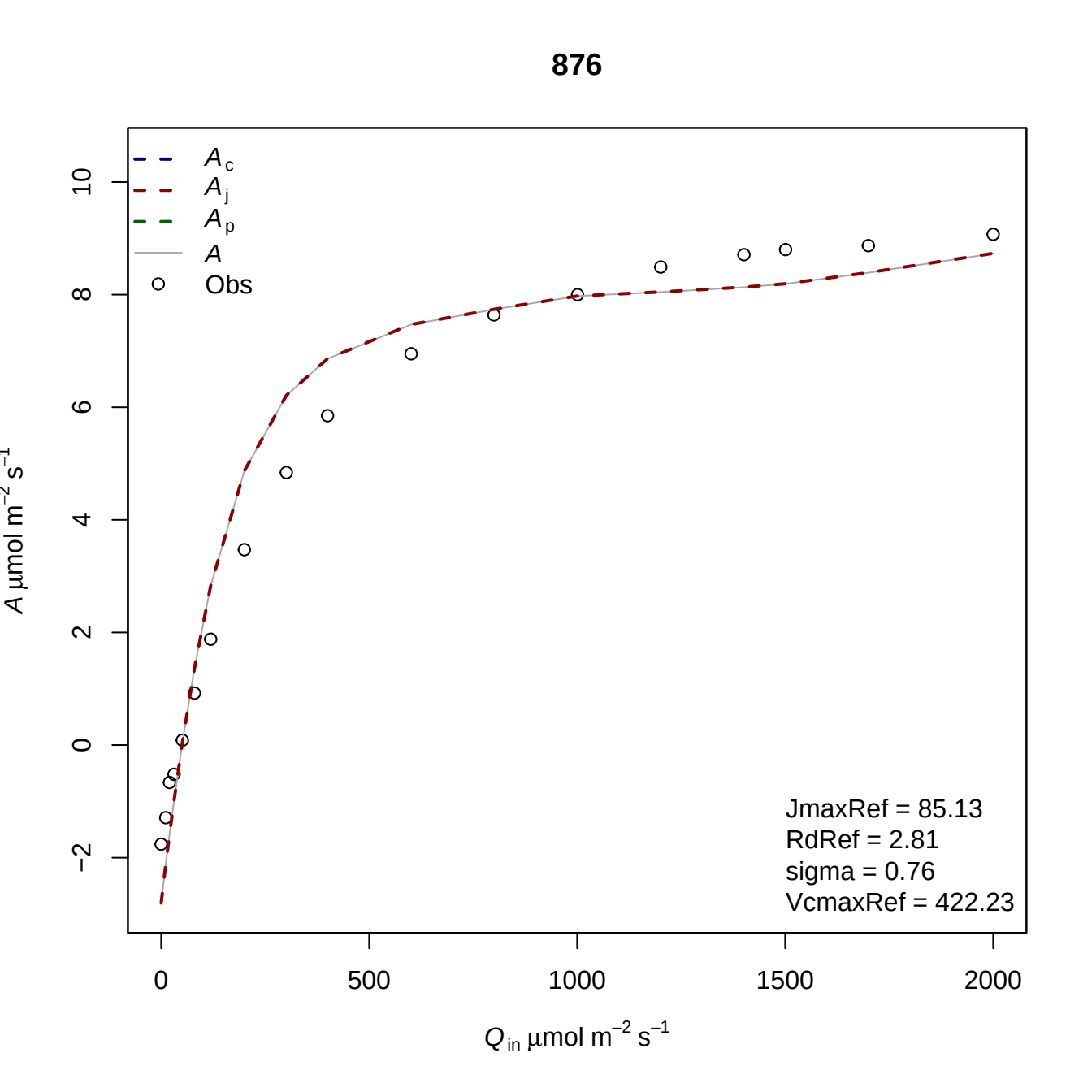

871
A_c
A_j
A_p
A
Obs
JmaxRef = 17.31
RdRef = 1.21
sigma = 0.28
VcmaxRef = 1335.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500
2000

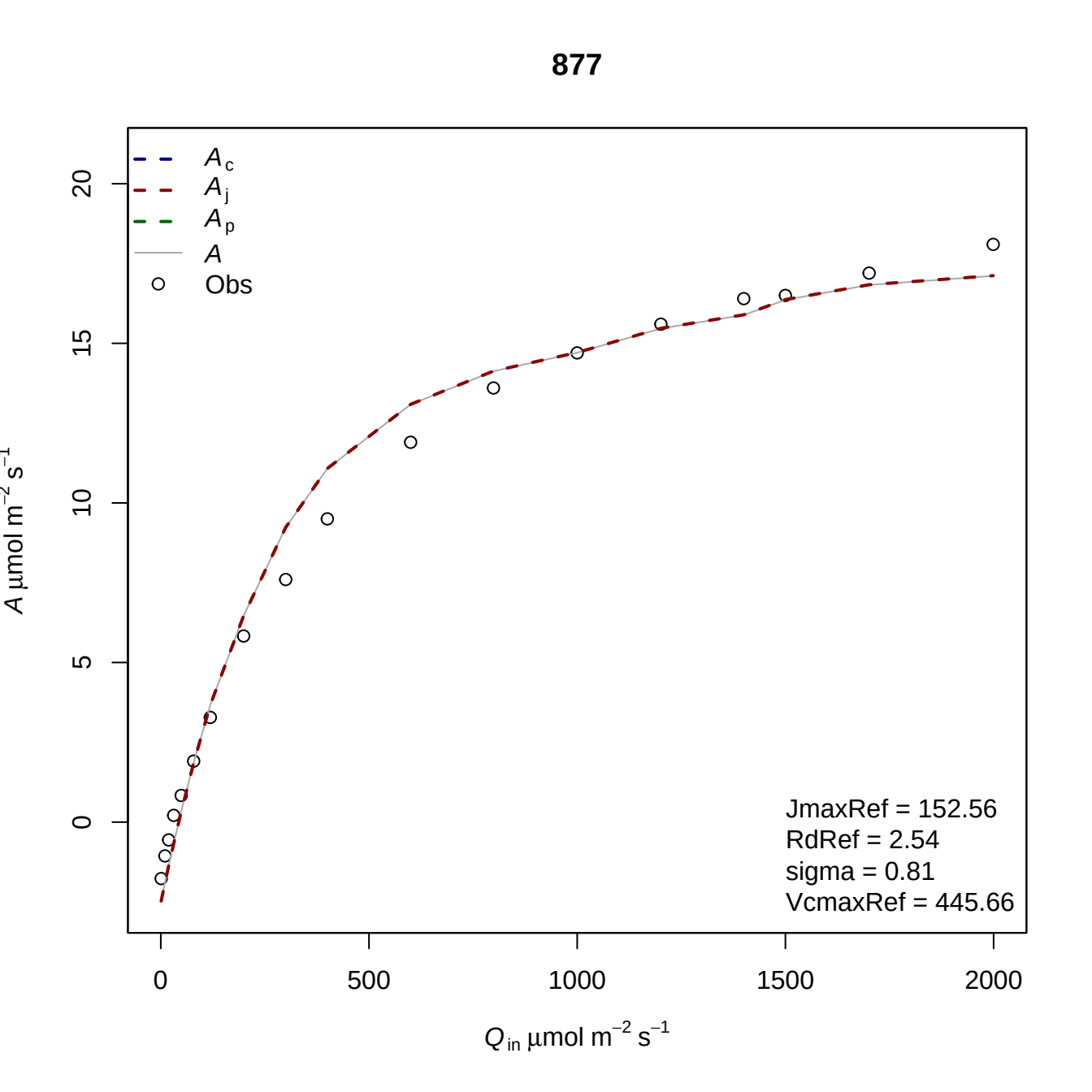

872
A_c
A_j
A_p
A
Obs
JmaxRef = 57.43
RdRef = 3.31
sigma = 0.79
VcmaxRef = 220.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

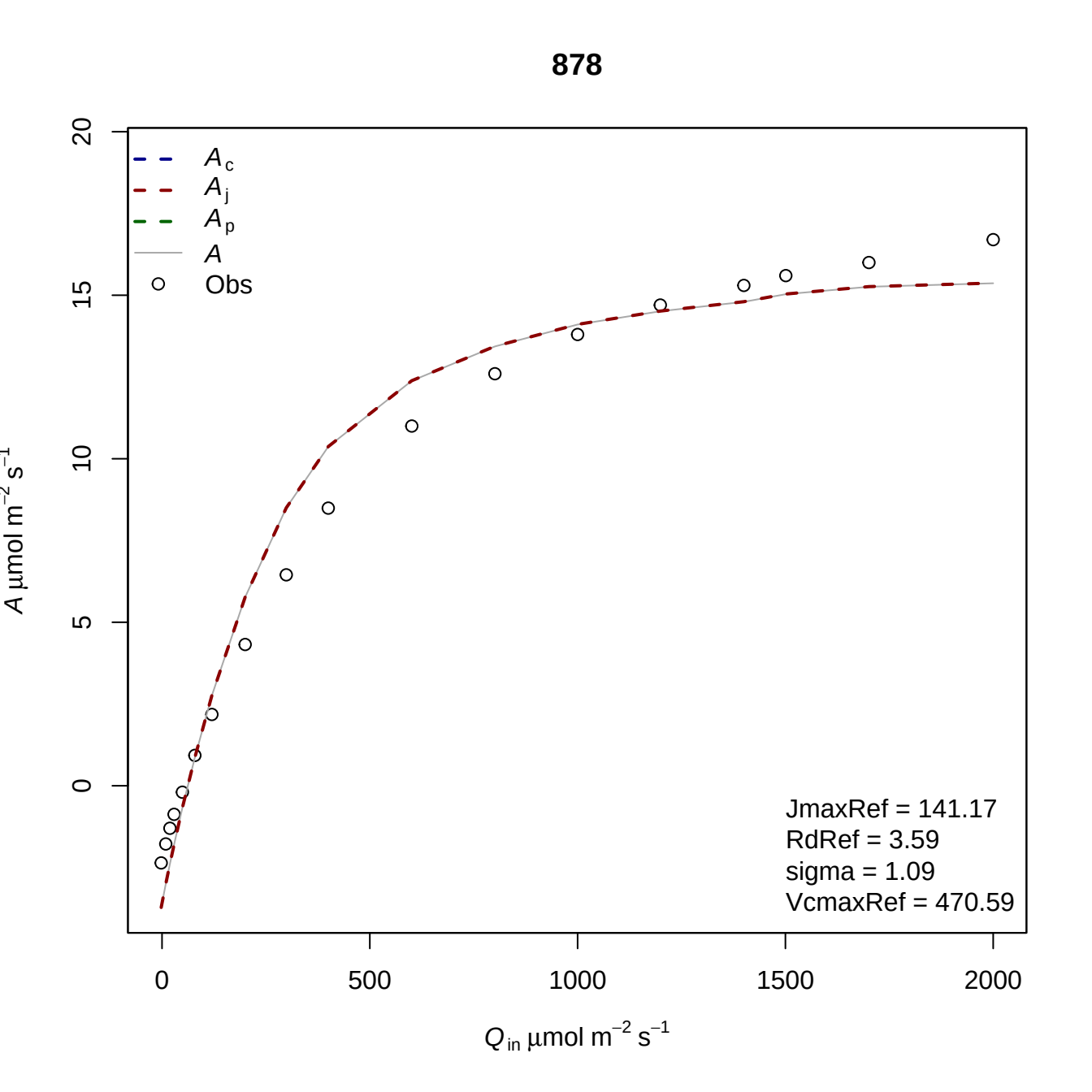

873
A_c
A_j
A_p
A
Obs
JmaxRef = 78.83
RdRef = 1.6
sigma = 0.47
VcmaxRef = 316.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

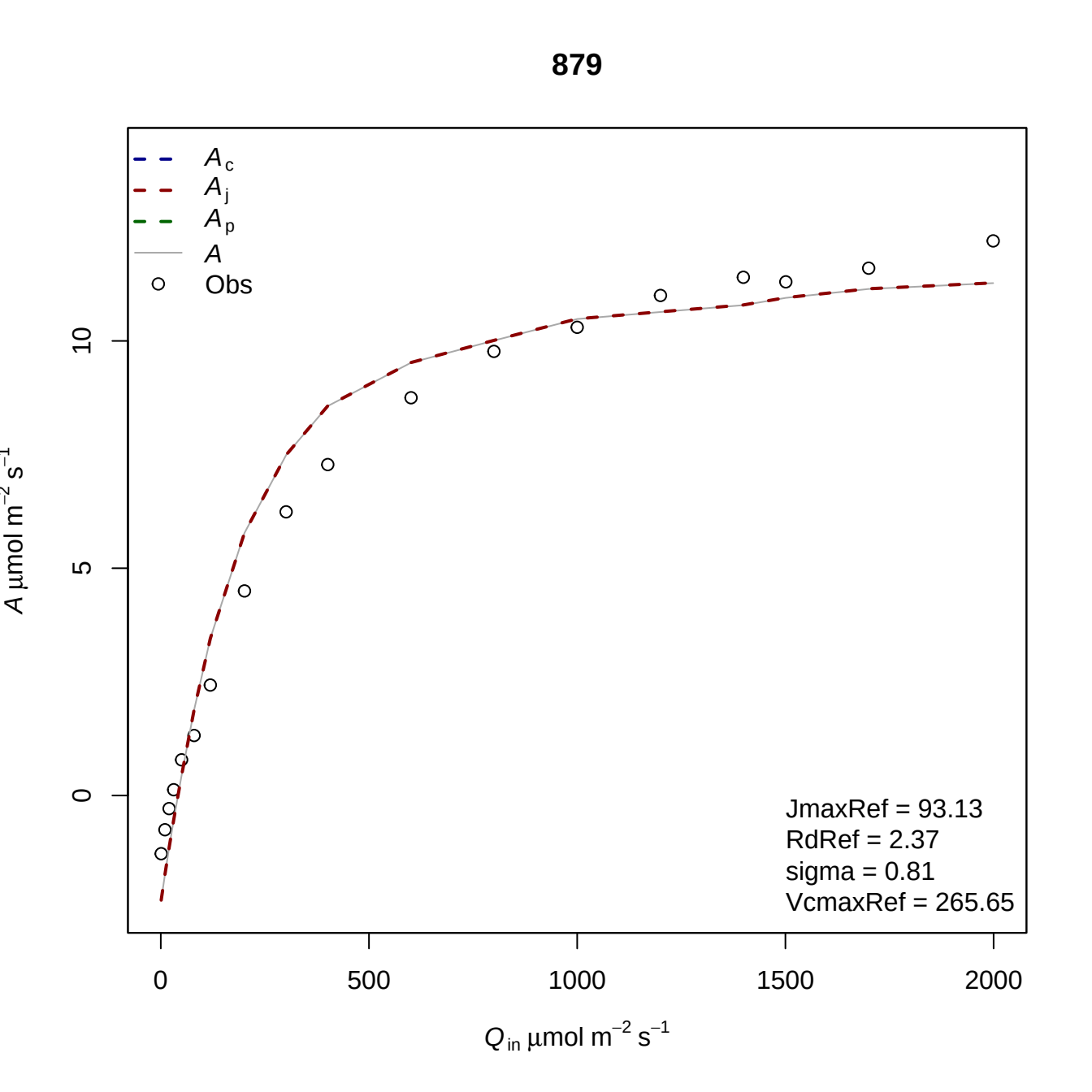

874
A_c
A_j
A_p
A
Obs
JmaxRef = 76.57
RdRef = 2.88
sigma = 0.7
VcmaxRef = 477.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

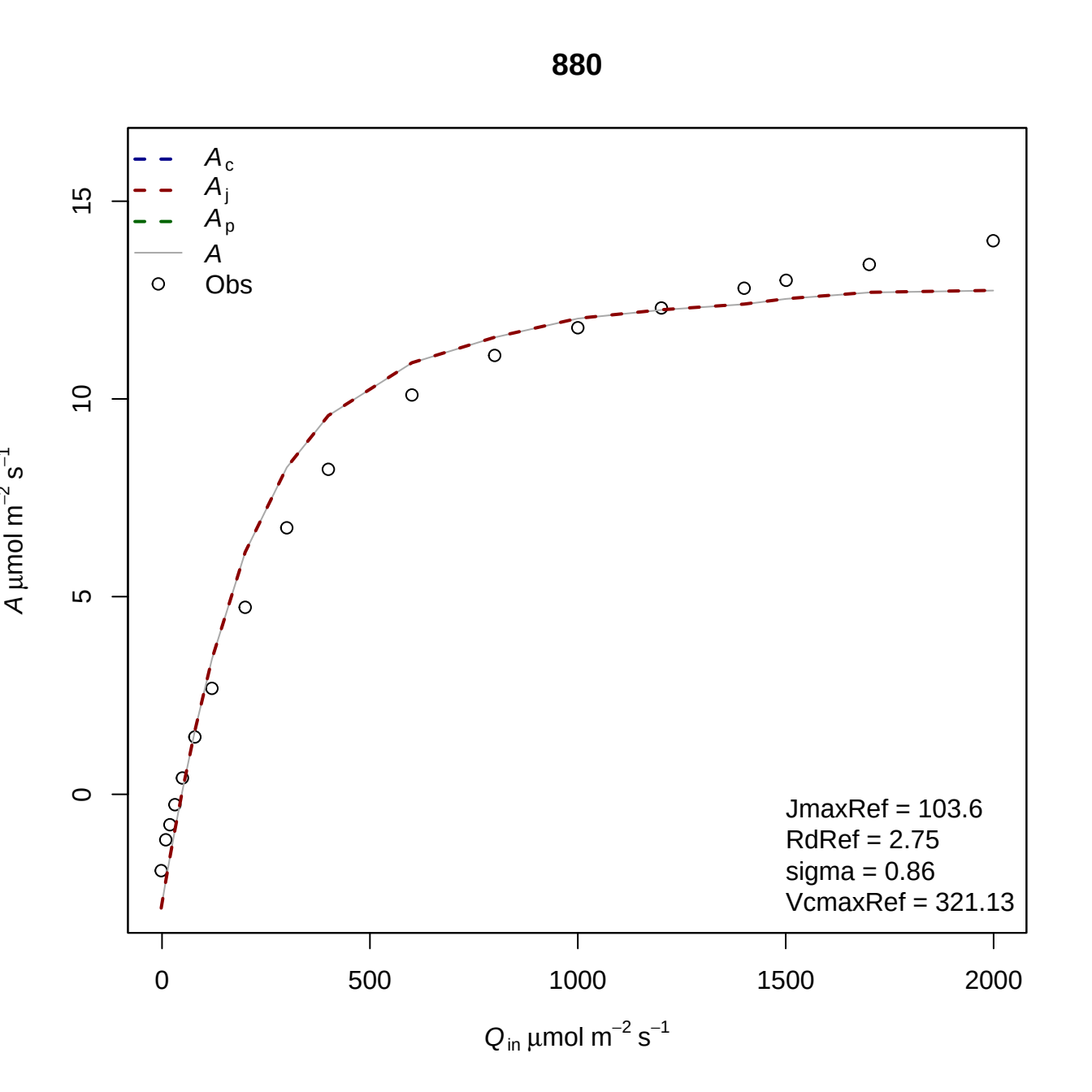

875
A_c
A_j
A_p
A
Obs
JmaxRef = 73.71
RdRef = 1.62
sigma = 0.47
VcmaxRef = 255.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

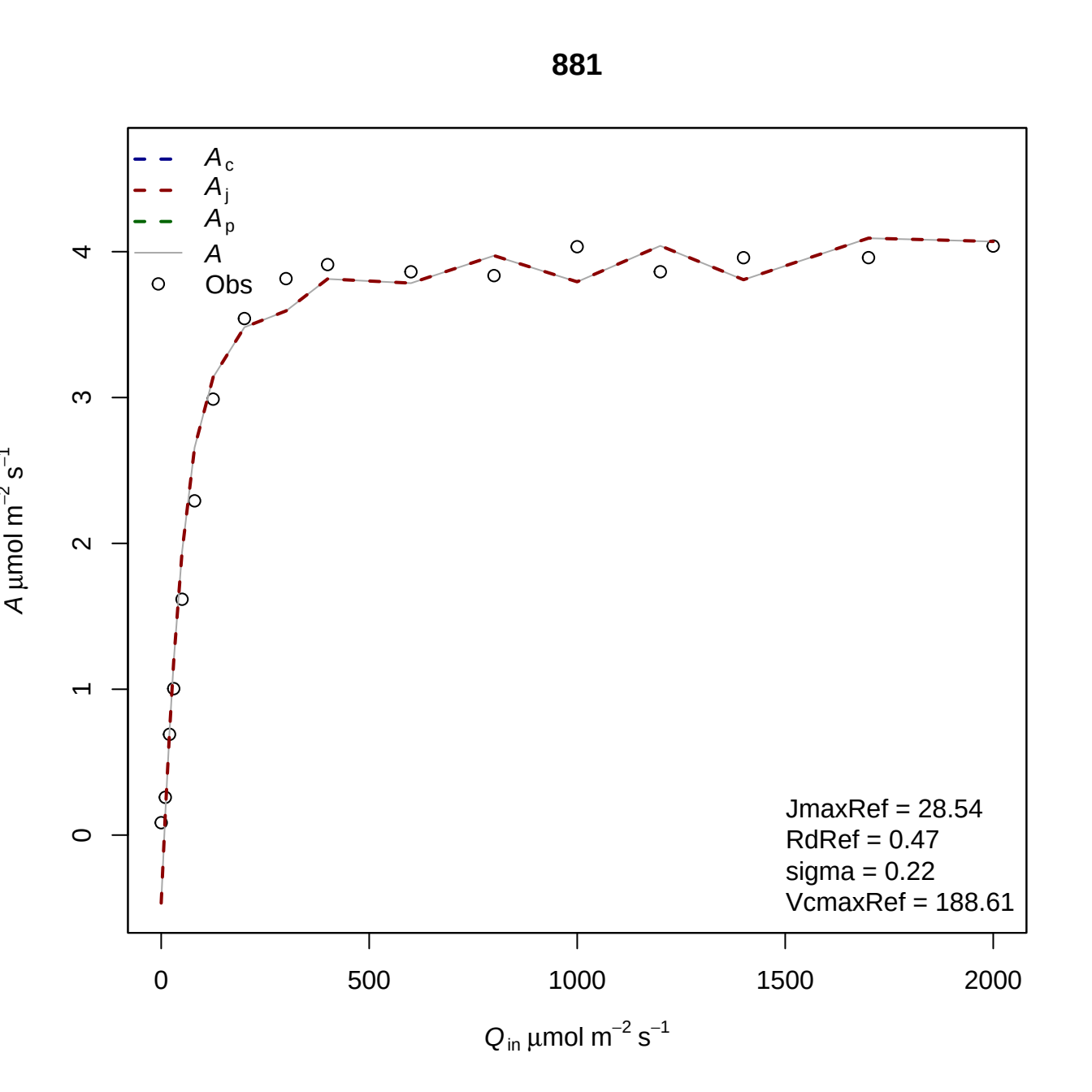

876
A_c
A_j
A_p
A
Obs
JmaxRef = 85.13
RdRef = 2.81
sigma = 0.76
VcmaxRef = 422.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

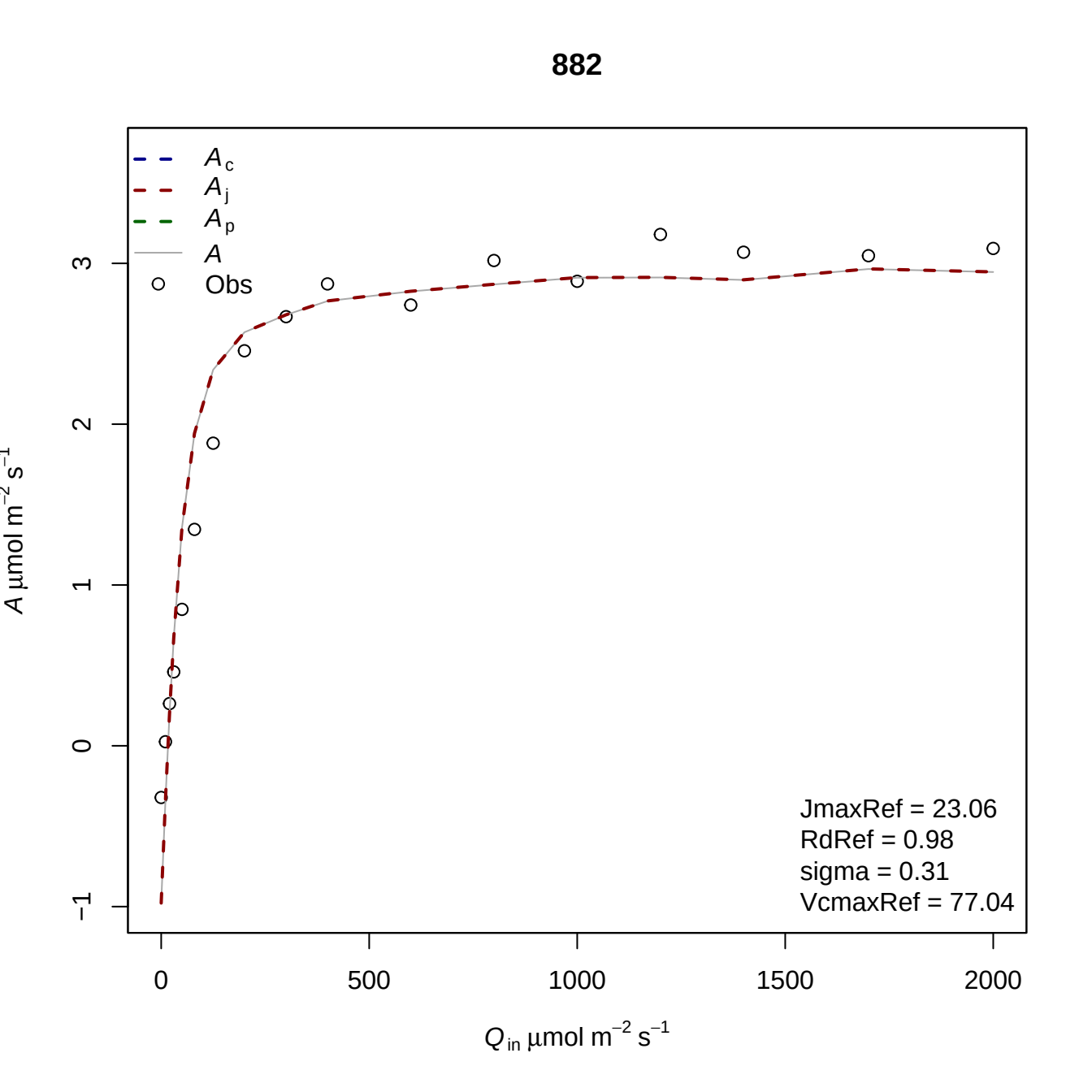

877
A_c
A_j
A_p
A
Obs
JmaxRef = 152.56
RdRef = 2.54
sigma = 0.81
VcmaxRef = 445.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

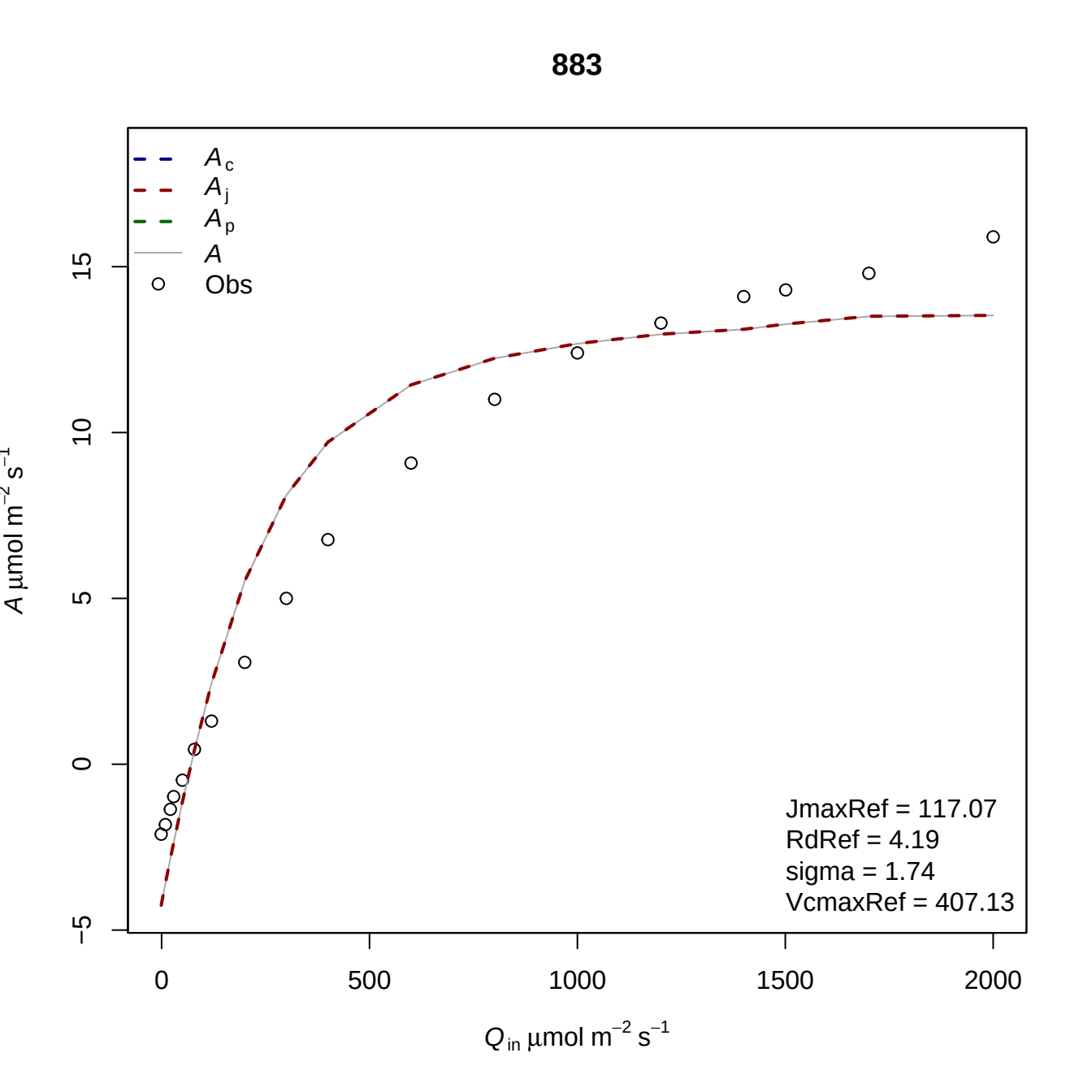

878
A_c
A_j
A_p
A
Obs
JmaxRef = 141.17
RdRef = 3.59
sigma = 1.09
VcmaxRef = 470.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

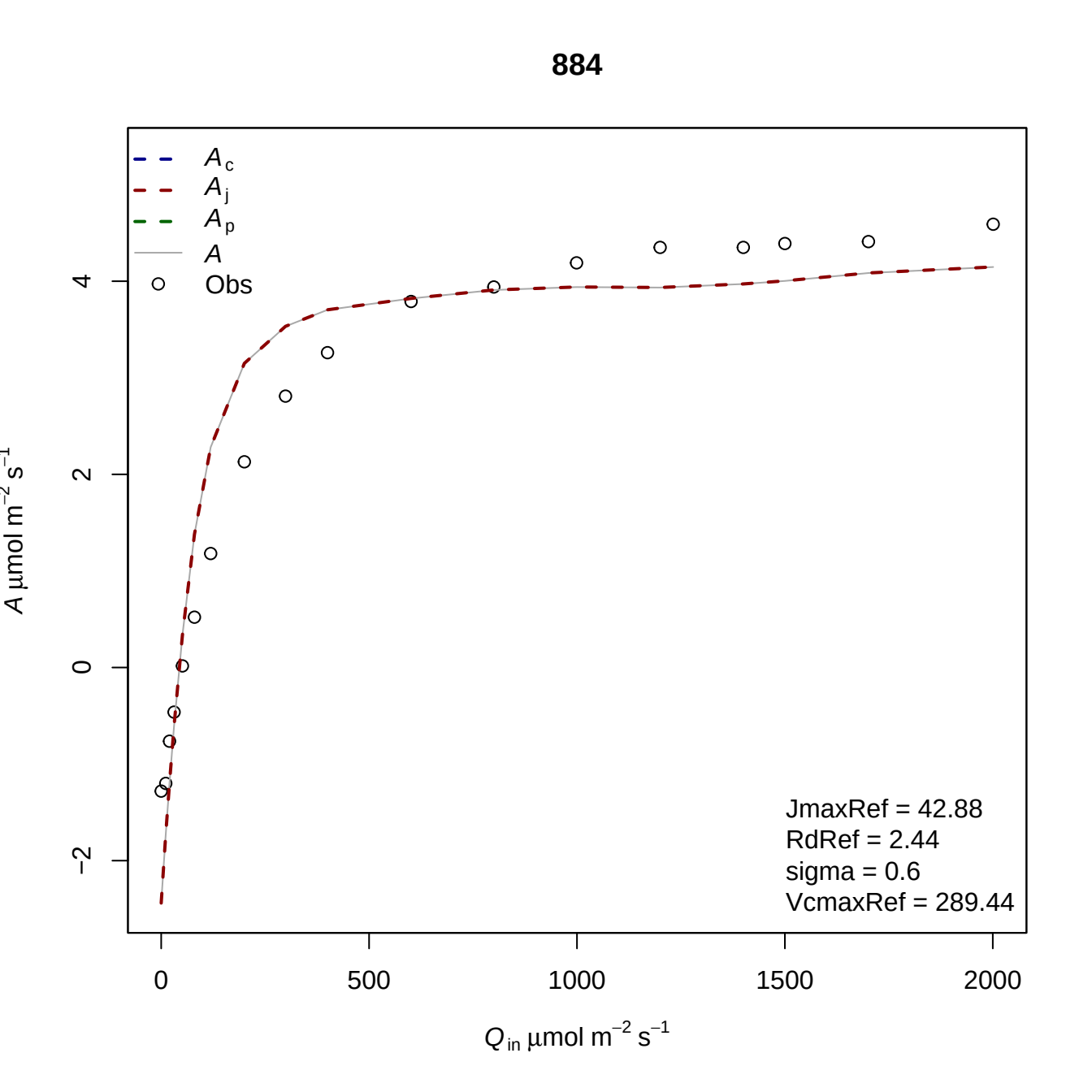

879
A_c
A_j
A_p
A
Obs
JmaxRef = 93.13
RdRef = 2.37
sigma = 0.81
VcmaxRef = 265.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

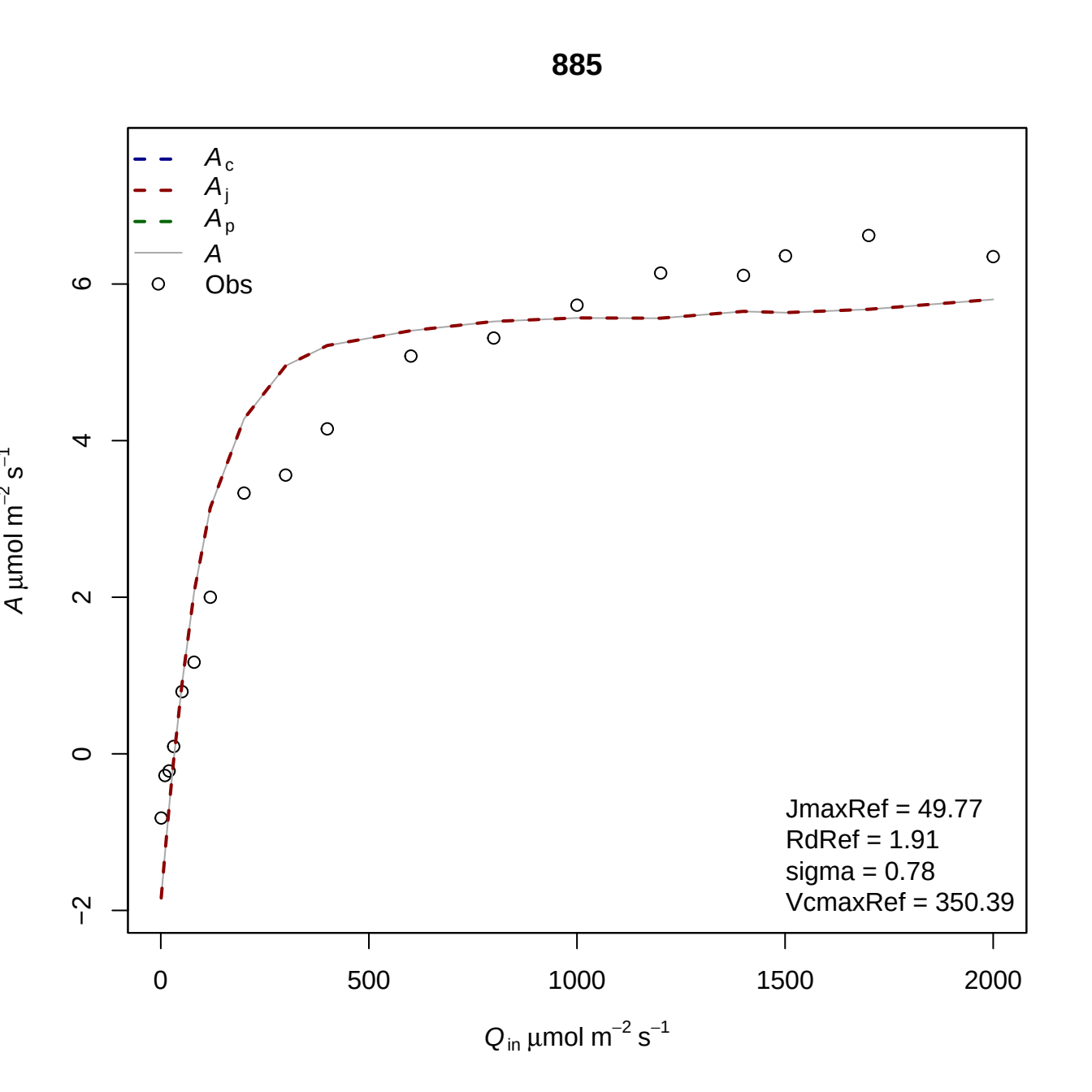

880
A_c
A_j
A_p
A
Obs
JmaxRef = 103.6
RdRef = 2.75
sigma = 0.86
VcmaxRef = 321.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

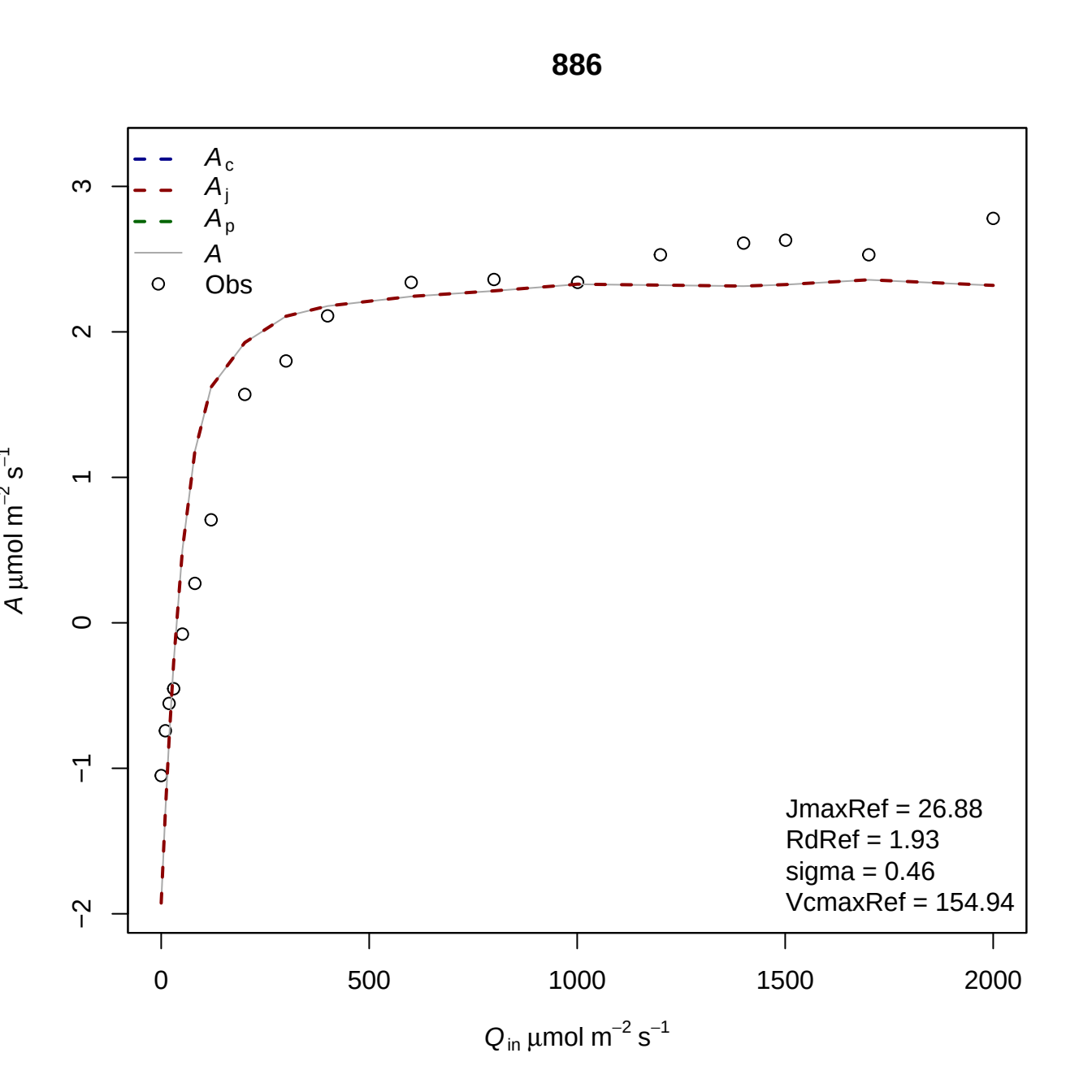

881
A_c
A_j
A_p
A
Obs
JmaxRef = 28.54
RdRef = 0.47
sigma = 0.22
VcmaxRef = 188.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

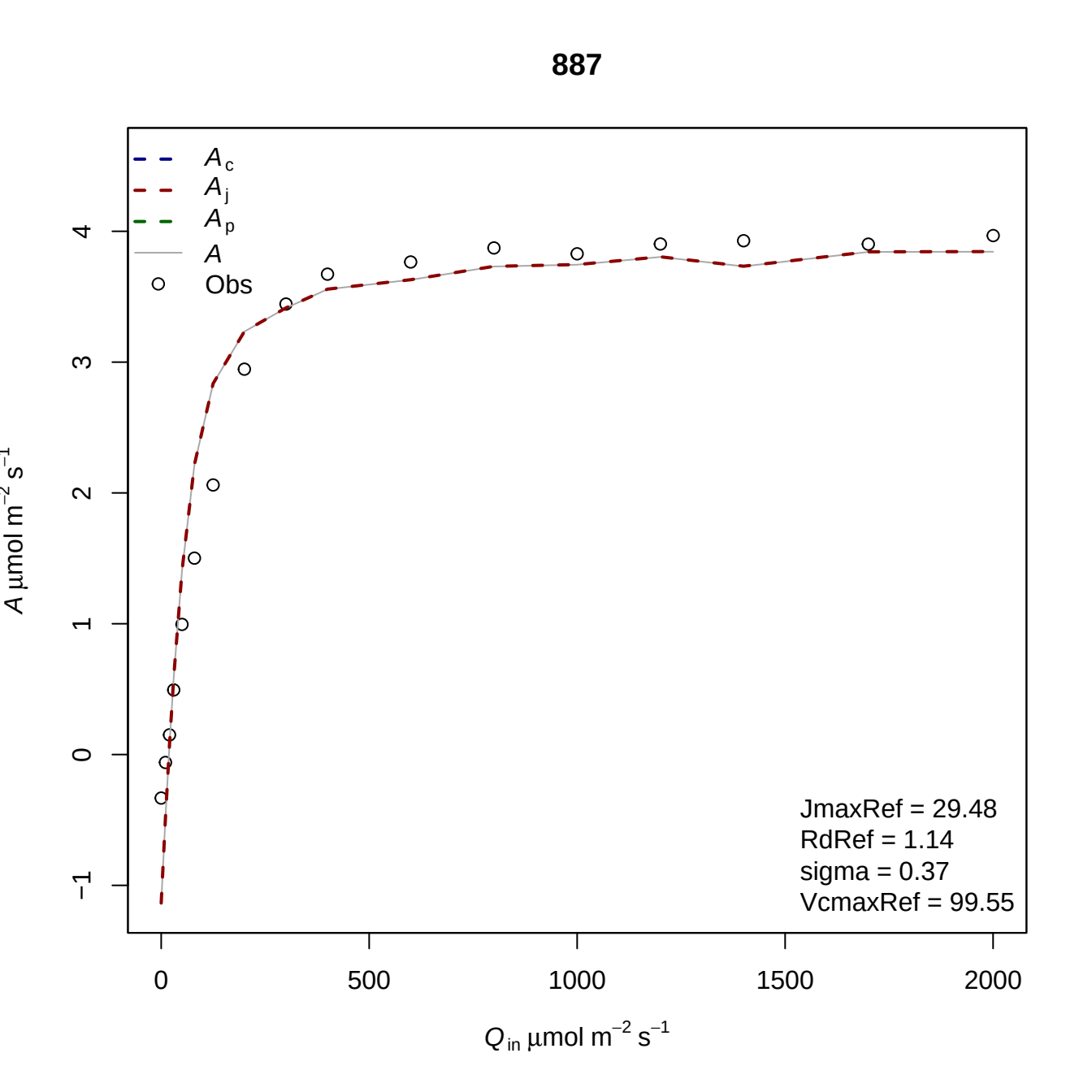

882
A_c
A_j
A_p
A
Obs
JmaxRef = 23.06
RdRef = 0.98
sigma = 0.31
VcmaxRef = 77.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

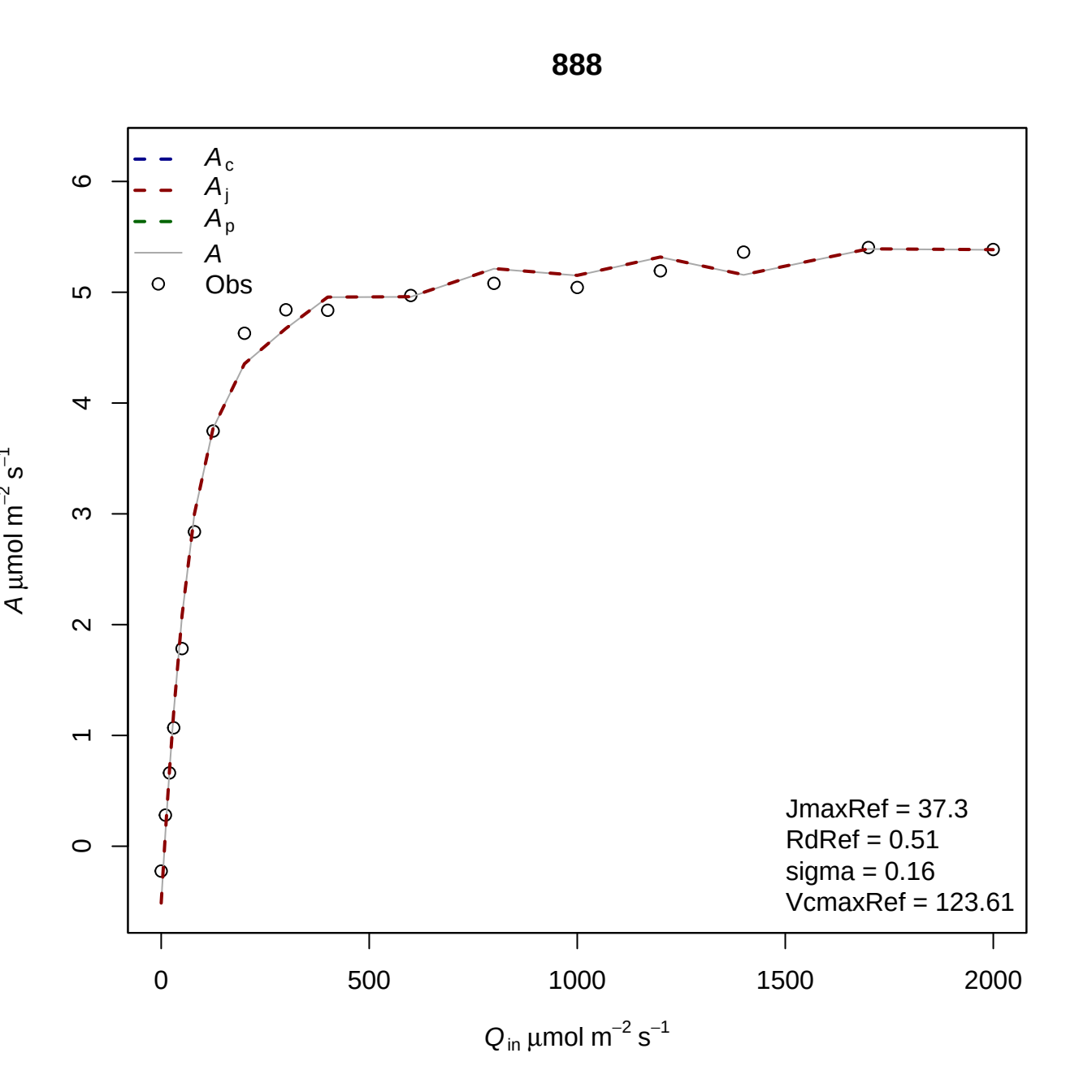

883
A_c
A_j
A_p
A
Obs
JmaxRef = 117.07
RdRef = 4.19
sigma = 1.74
VcmaxRef = 407.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

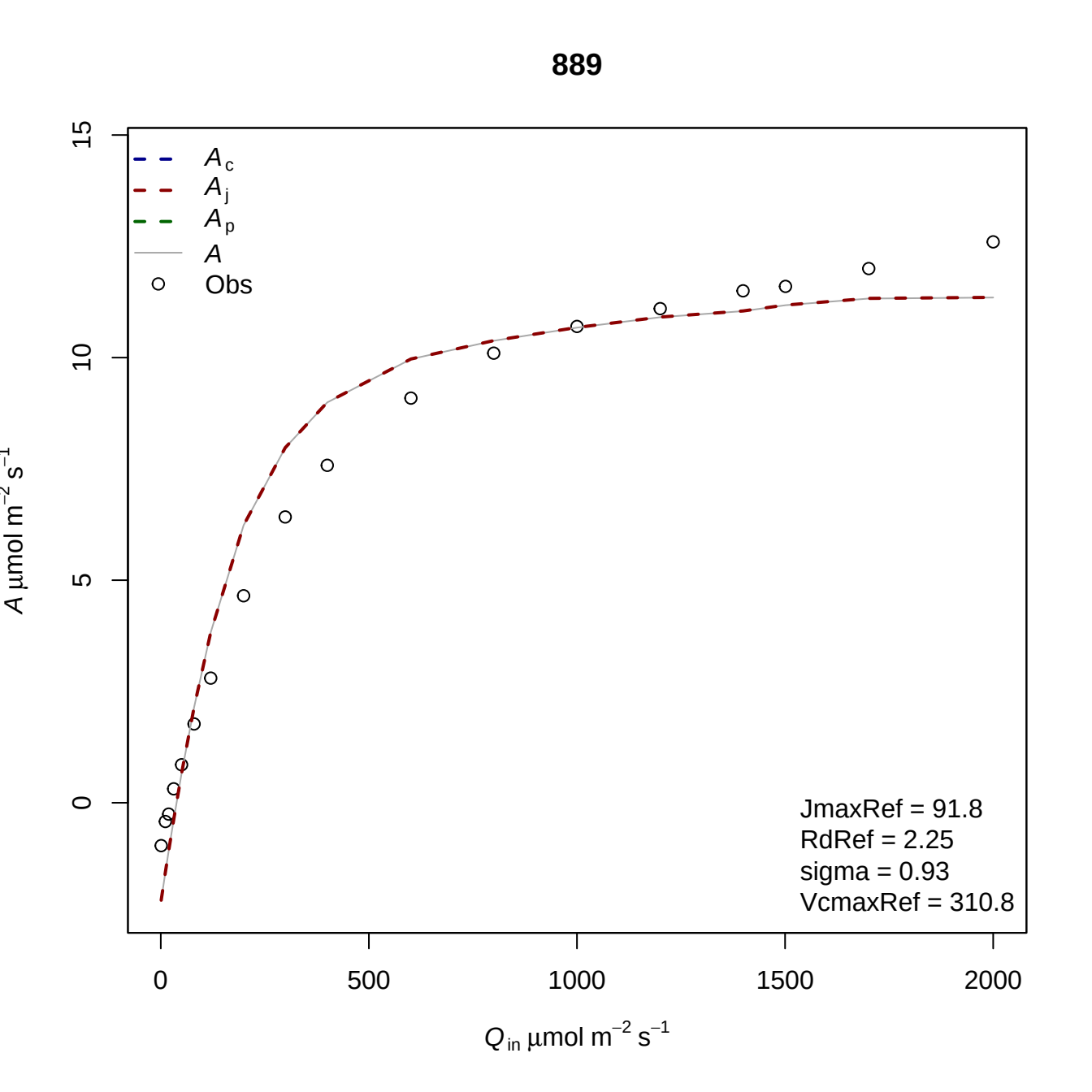

884
A_c
A_j
A_p
A
Obs
JmaxRef = 42.88
RdRef = 2.44
sigma = 0.6
VcmaxRef = 289.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

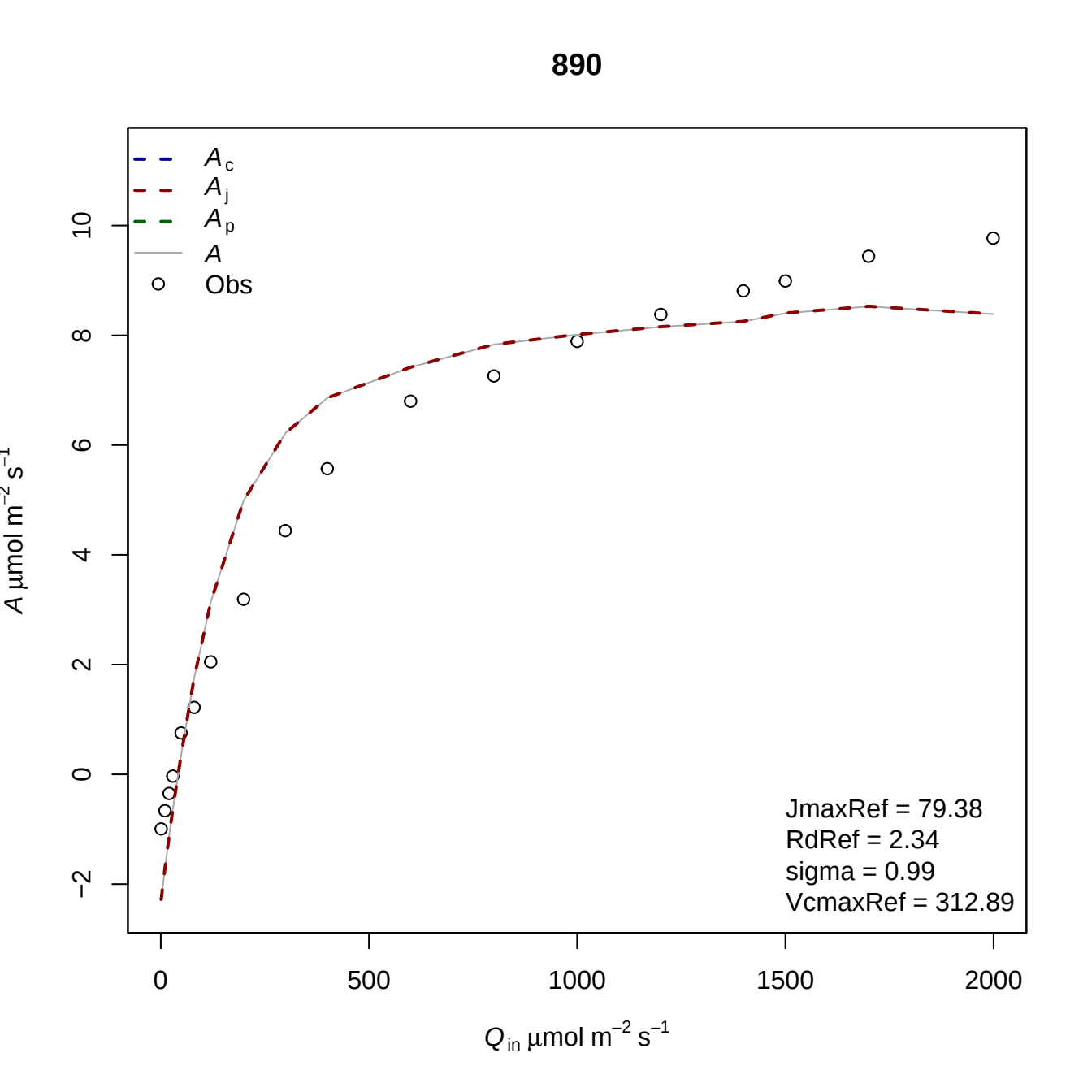

885
A_c
A_j
A_p
A
Obs
JmaxRef = 49.77
RdRef = 1.91
sigma = 0.78
VcmaxRef = 350.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

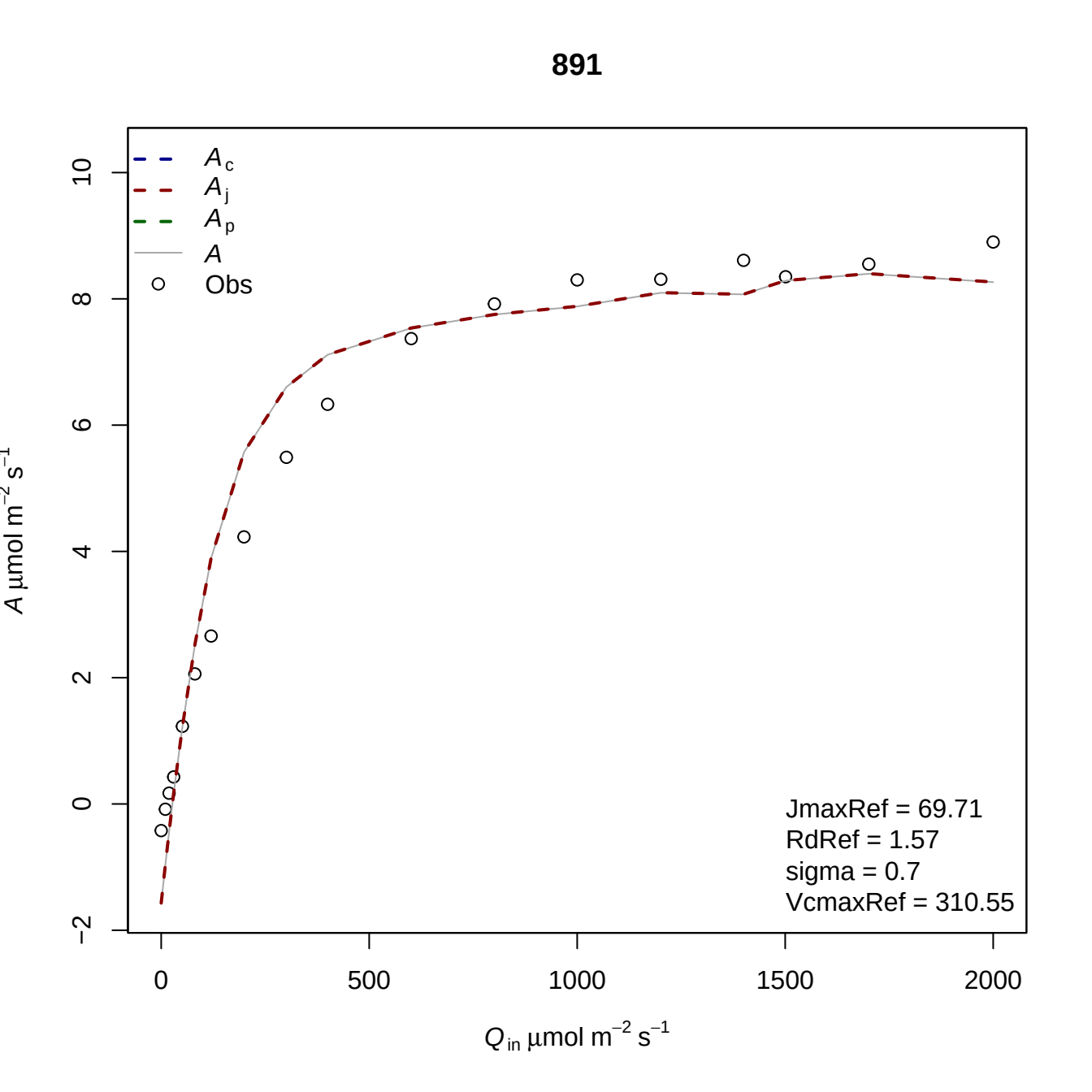

886
A_c
A_j
A_p
A
Obs
JmaxRef = 26.88
RdRef = 1.93
sigma = 0.46
VcmaxRef = 154.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
500
1000
1500
2000

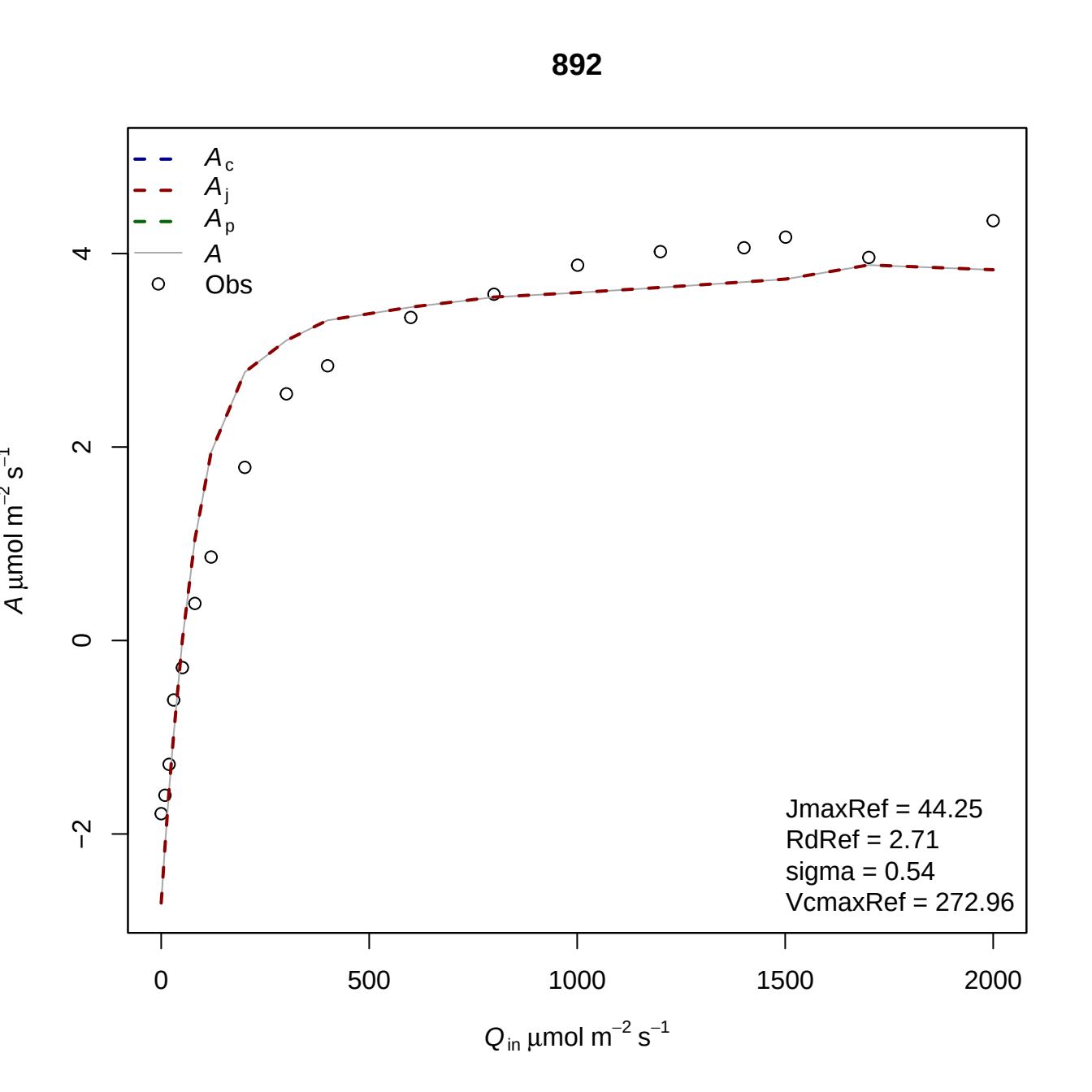

887
A_c
A_j
A_p
A
Obs
JmaxRef = 29.48
RdRef = 1.14
sigma = 0.37
VcmaxRef = 99.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

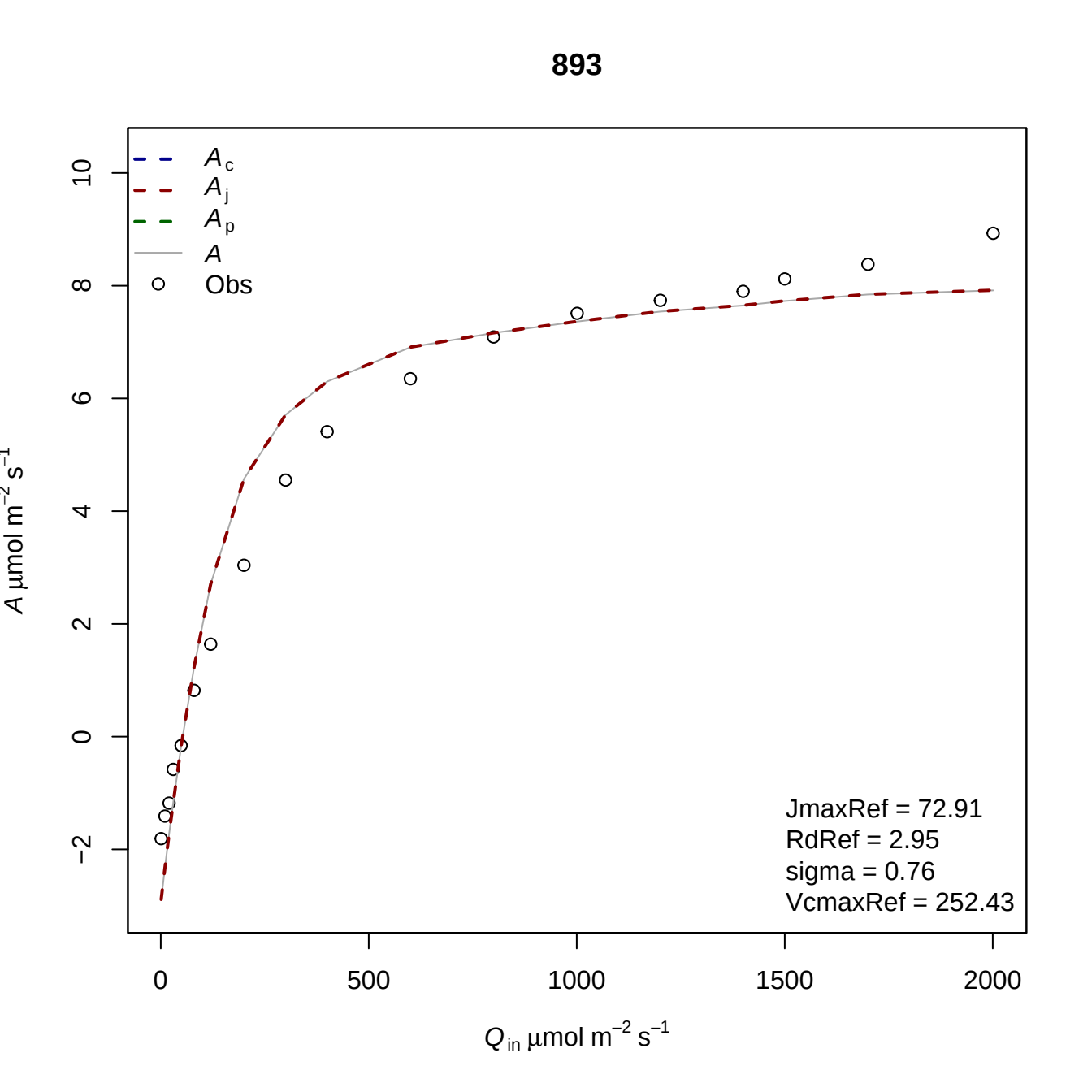

888
A_c
A_j
A_p
A
Obs
JmaxRef = 37.3
RdRef = 0.51
sigma = 0.16
VcmaxRef = 123.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500
2000

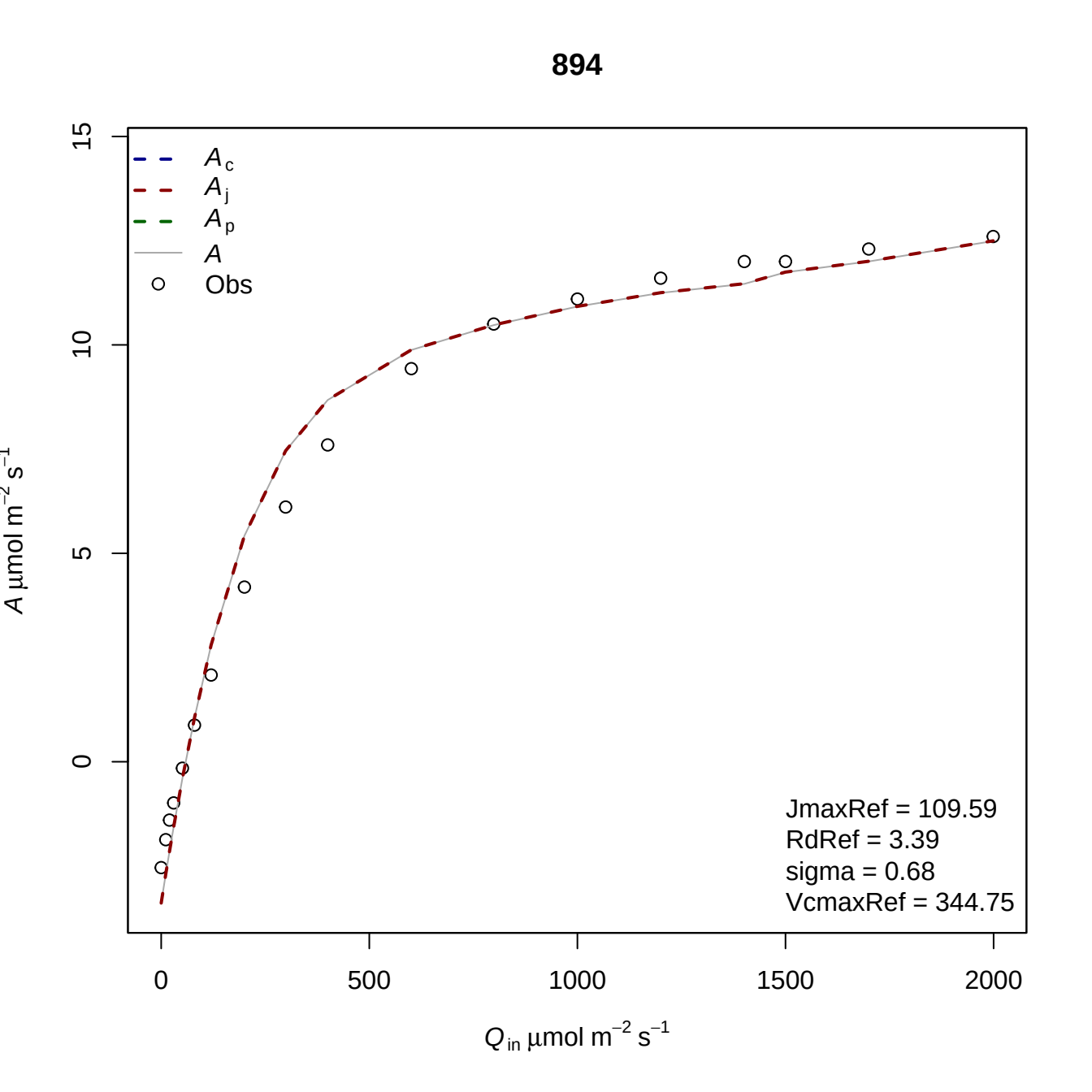

889
A_c
A_j
A_p
A
Obs
JmaxRef = 91.8
RdRef = 2.25
sigma = 0.93
VcmaxRef = 310.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

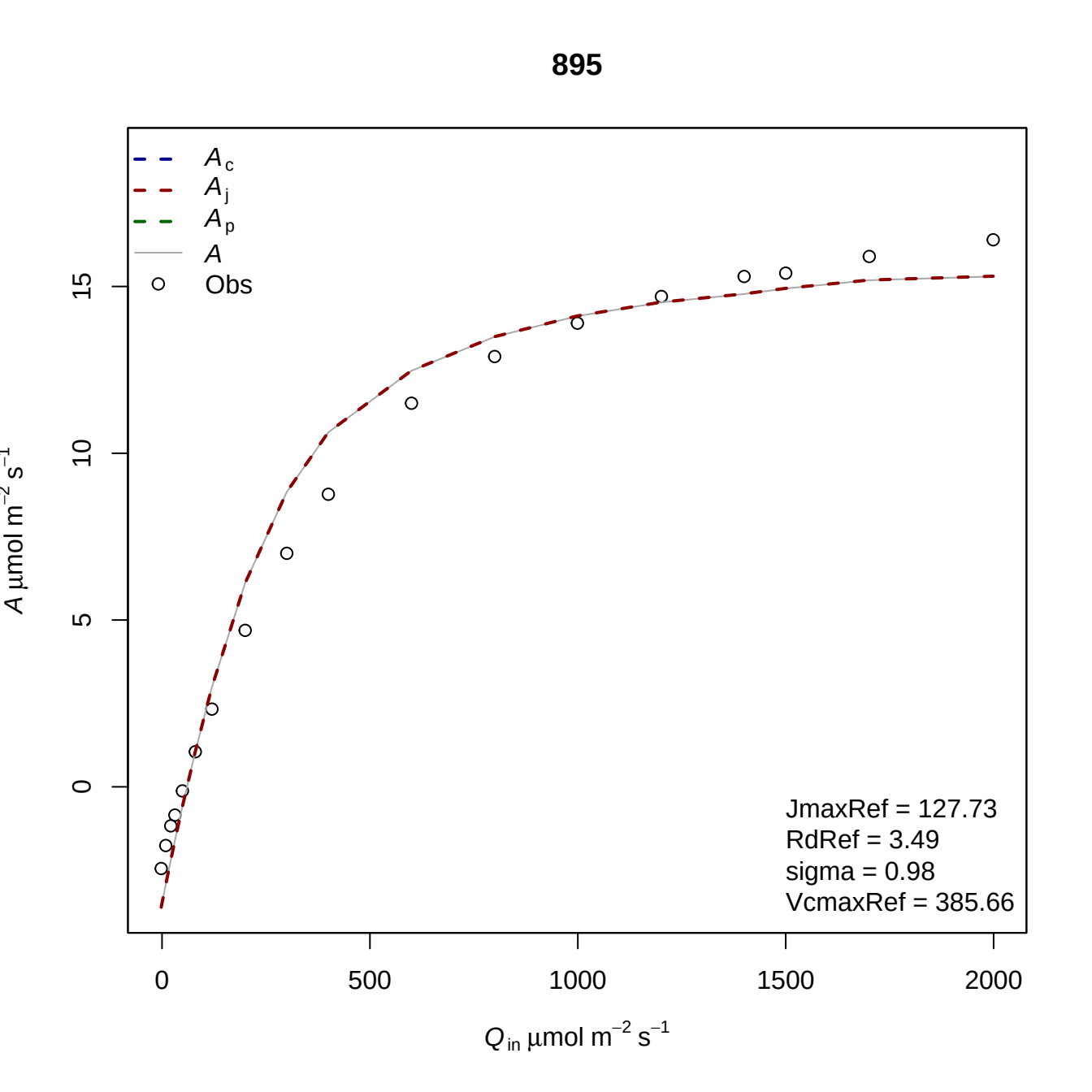

890
A_c
A_j
A_p
A
Obs
JmaxRef = 79.38
RdRef = 2.34
sigma = 0.99
VcmaxRef = 312.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

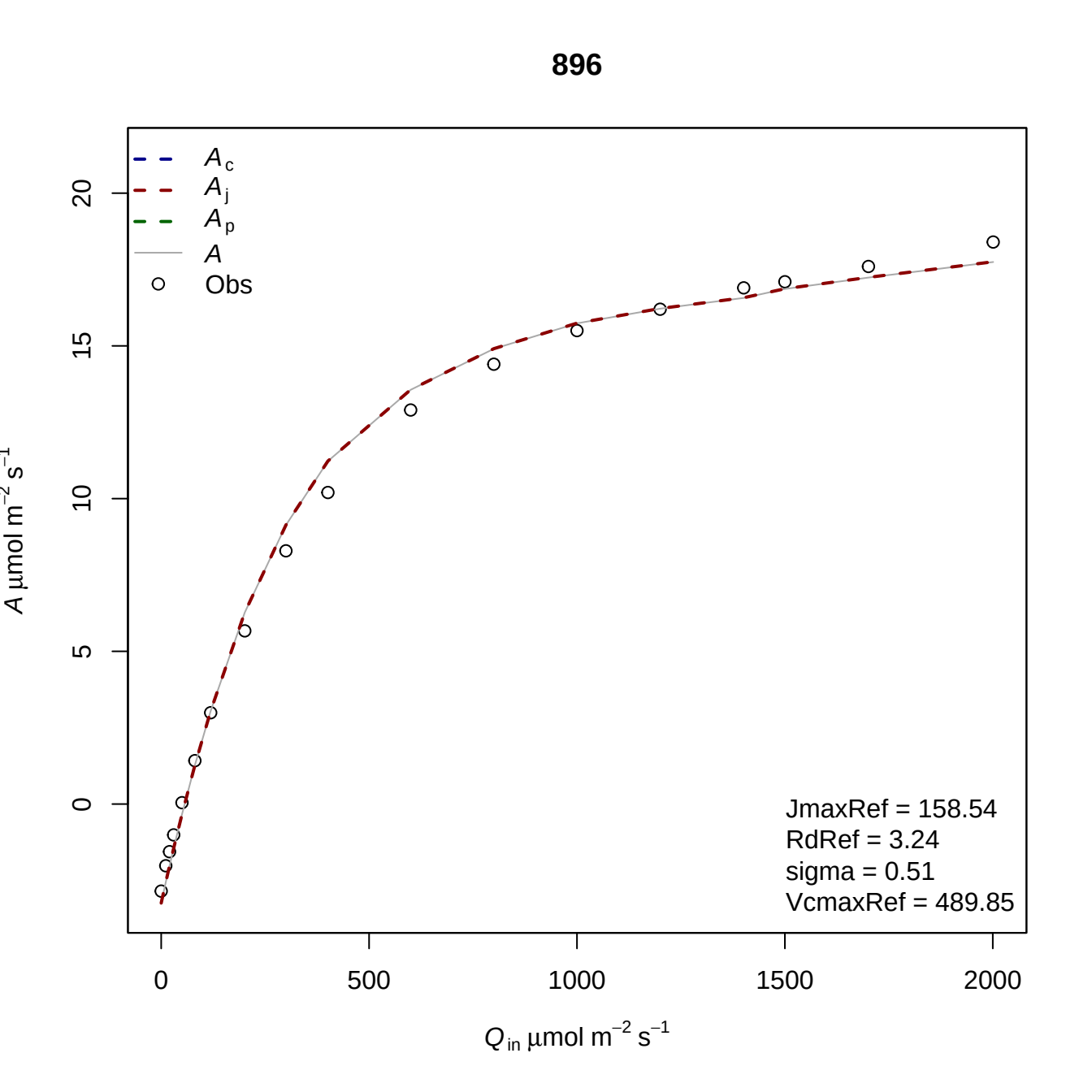

891
A_c
A_j
A_p
A
Obs
JmaxRef = 69.71
RdRef = 1.57
sigma = 0.7
VcmaxRef = 310.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

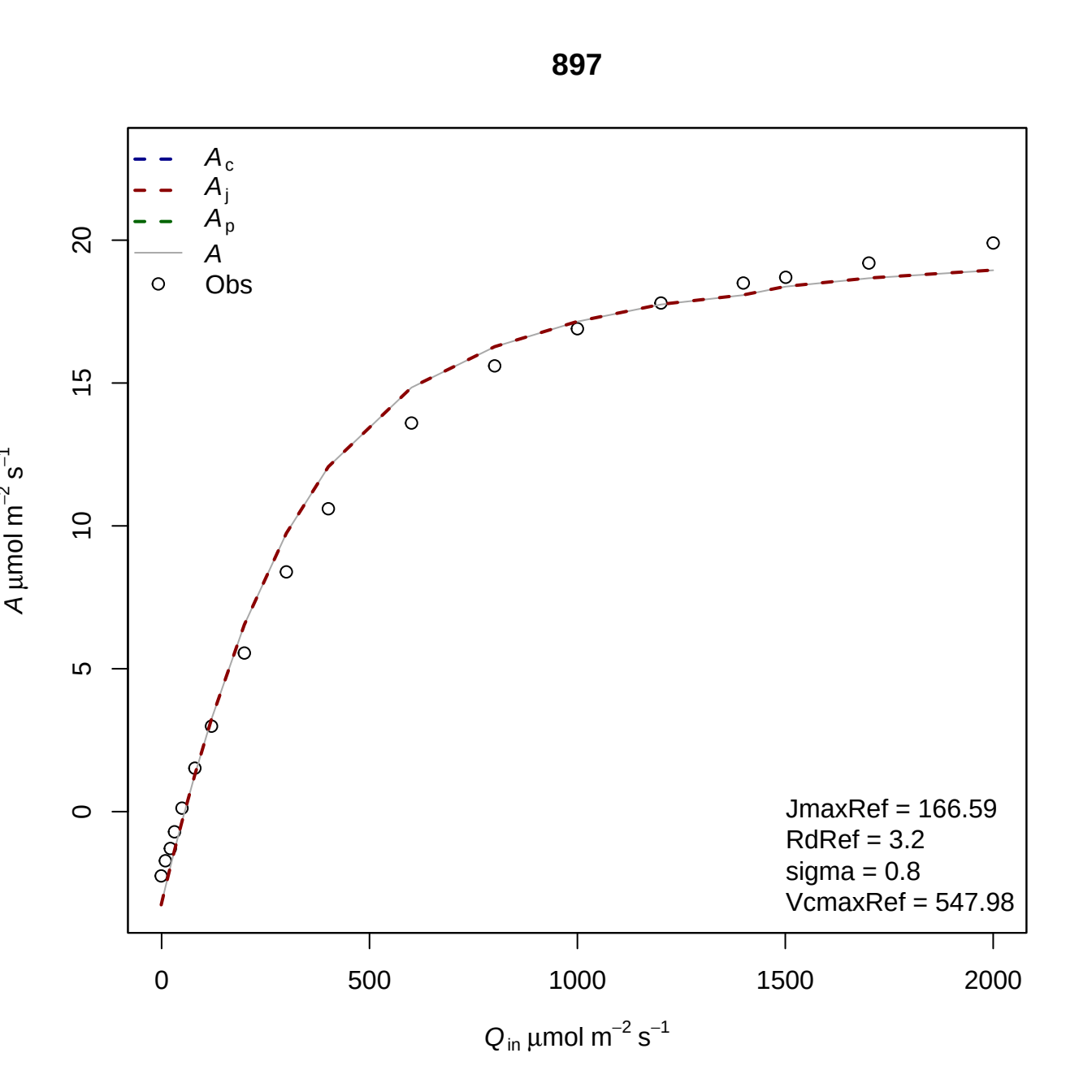

892
A_c
A_j
A_p
A
Obs
JmaxRef = 44.25
RdRef = 2.71
sigma = 0.54
VcmaxRef = 272.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

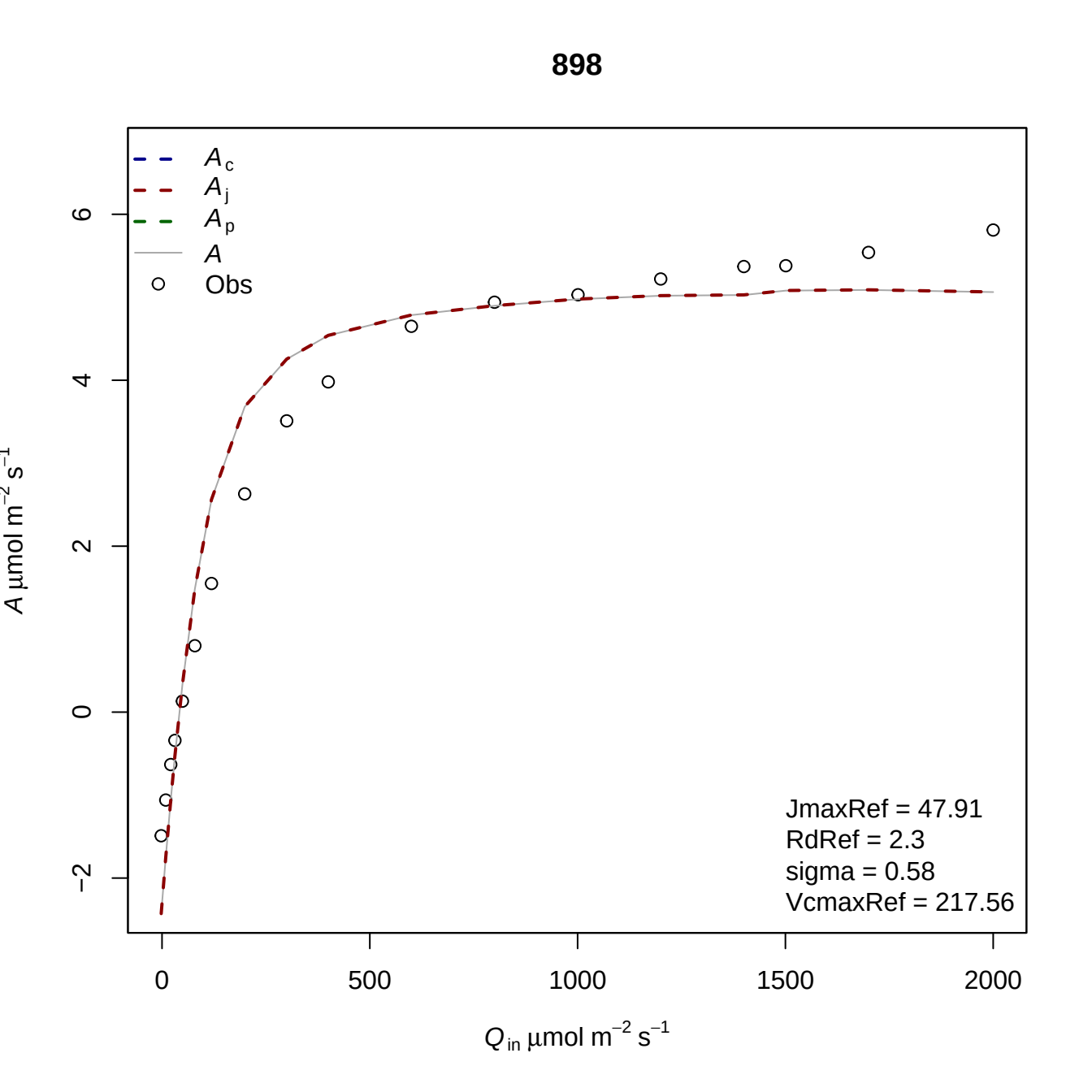

893
A_c
A_j
A_p
A
Obs
JmaxRef = 72.91
RdRef = 2.95
sigma = 0.76
VcmaxRef = 252.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

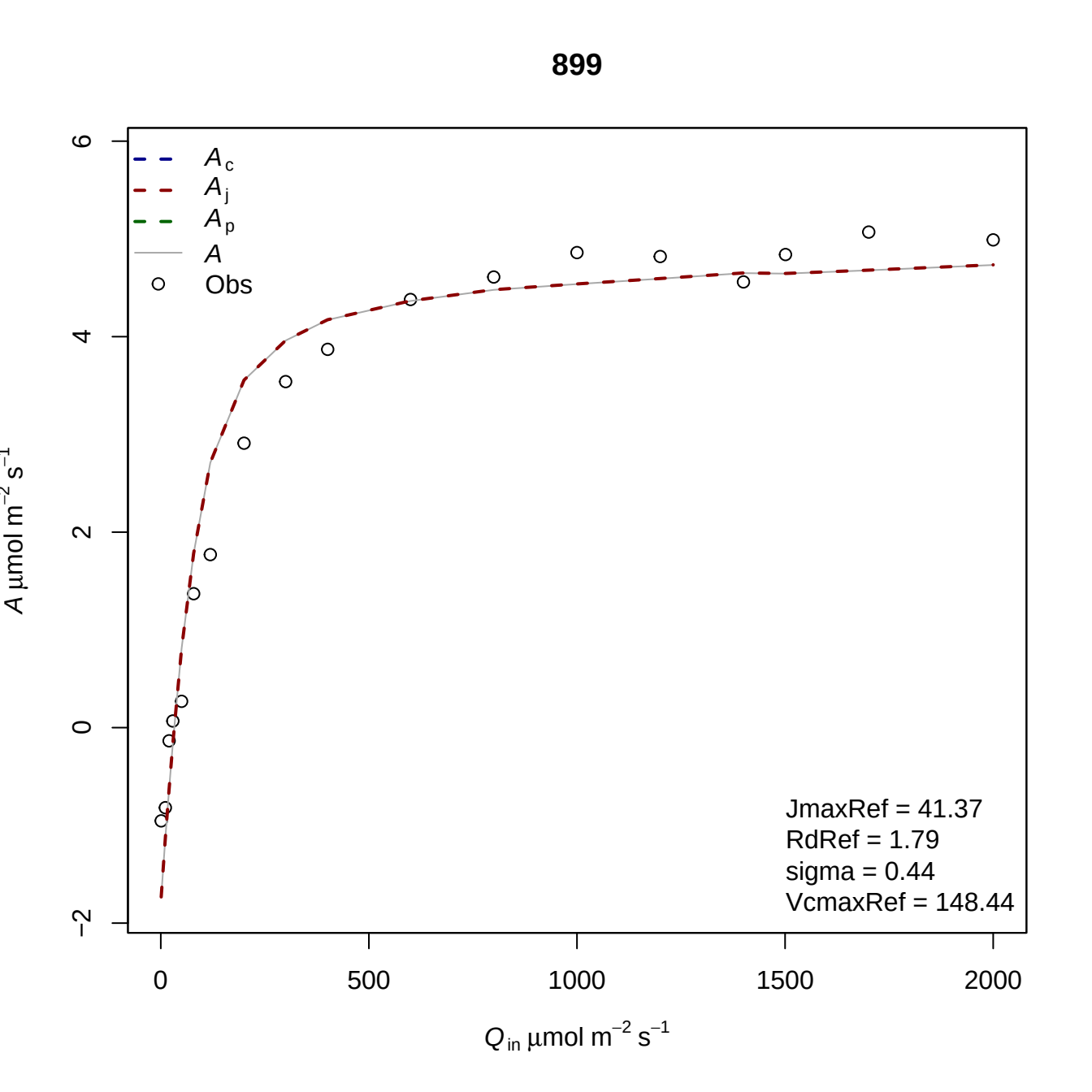

894
A_c
A_j
A_p
A
Obs
JmaxRef = 109.59
RdRef = 3.39
sigma = 0.68
VcmaxRef = 344.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

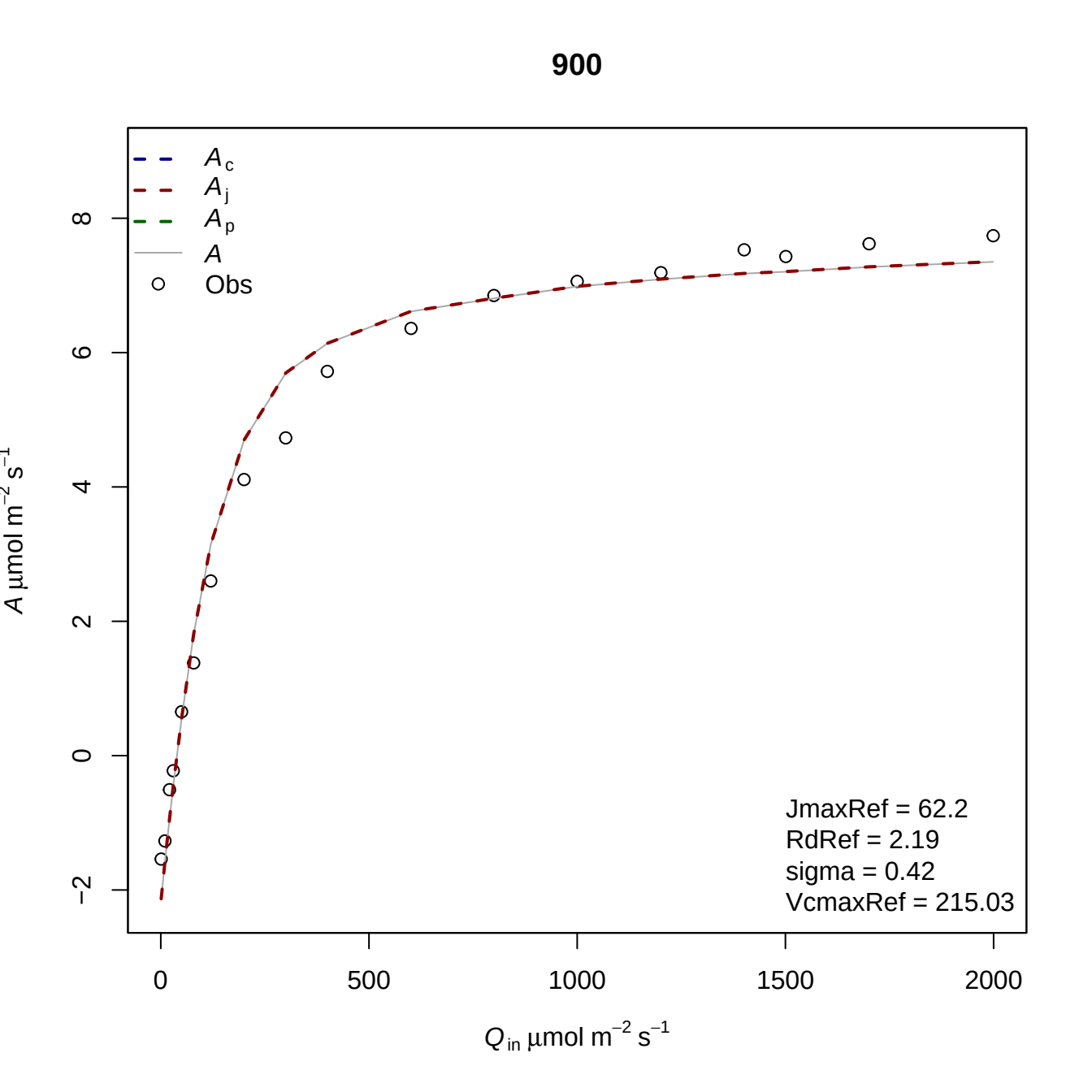

895
A_c
A_j
A_p
A
Obs
JmaxRef = 127.73
RdRef = 3.49
sigma = 0.98
VcmaxRef = 385.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

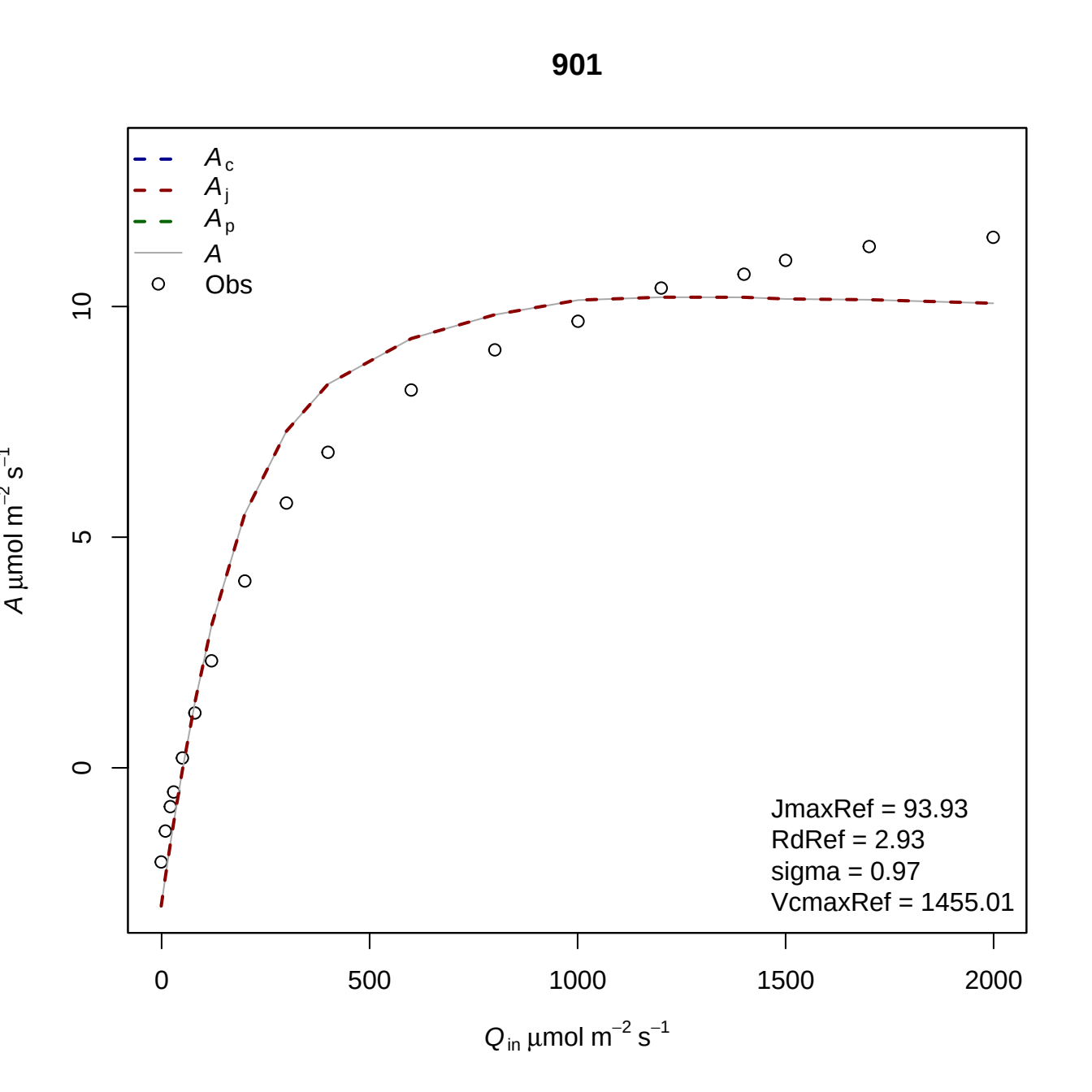

896
A_c
A_j
A_p
A
Obs
JmaxRef = 158.54
RdRef = 3.24
sigma = 0.51
VcmaxRef = 489.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

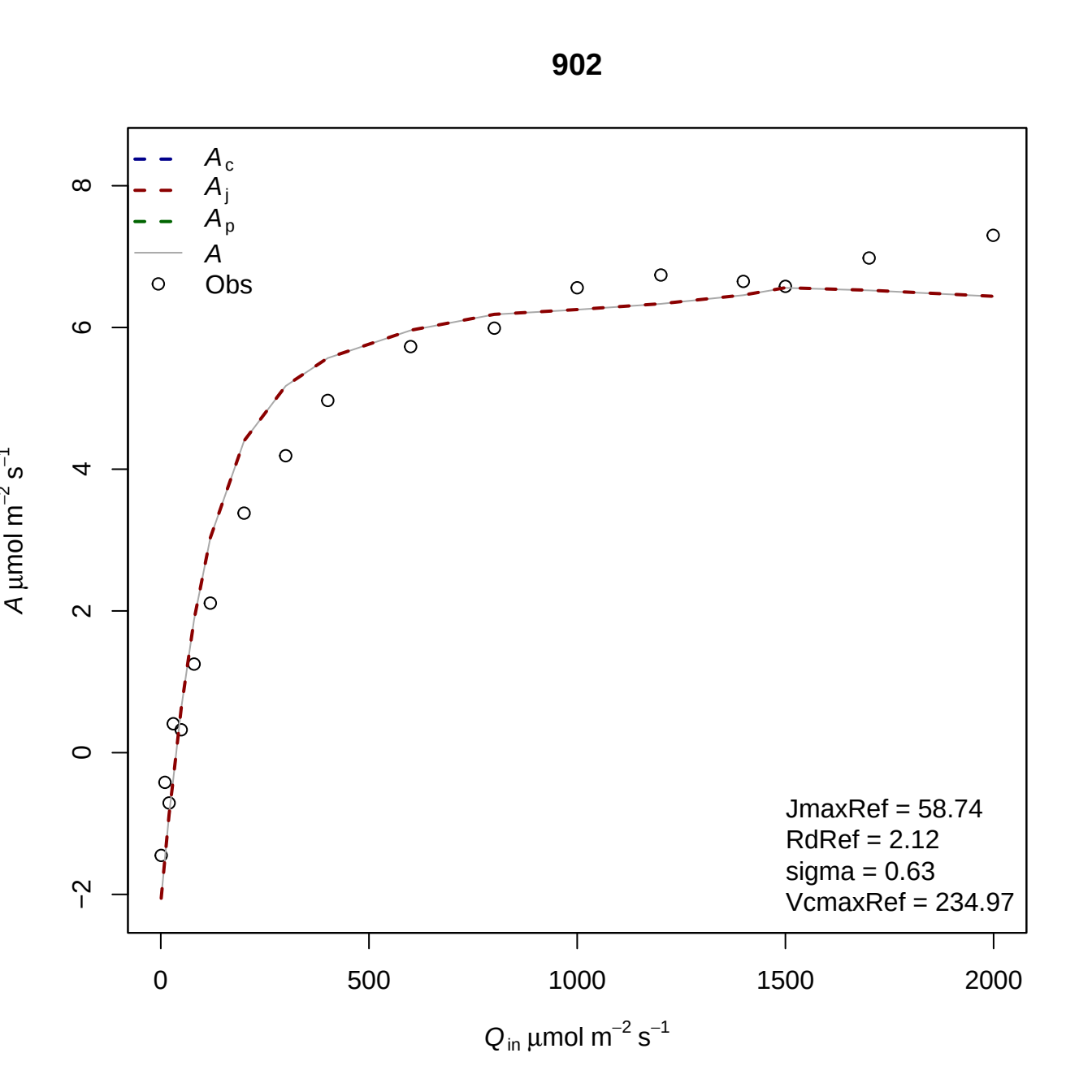

897
A_c
A_j
A_p
A
Obs
JmaxRef = 166.59
RdRef = 3.2
sigma = 0.8
VcmaxRef = 547.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

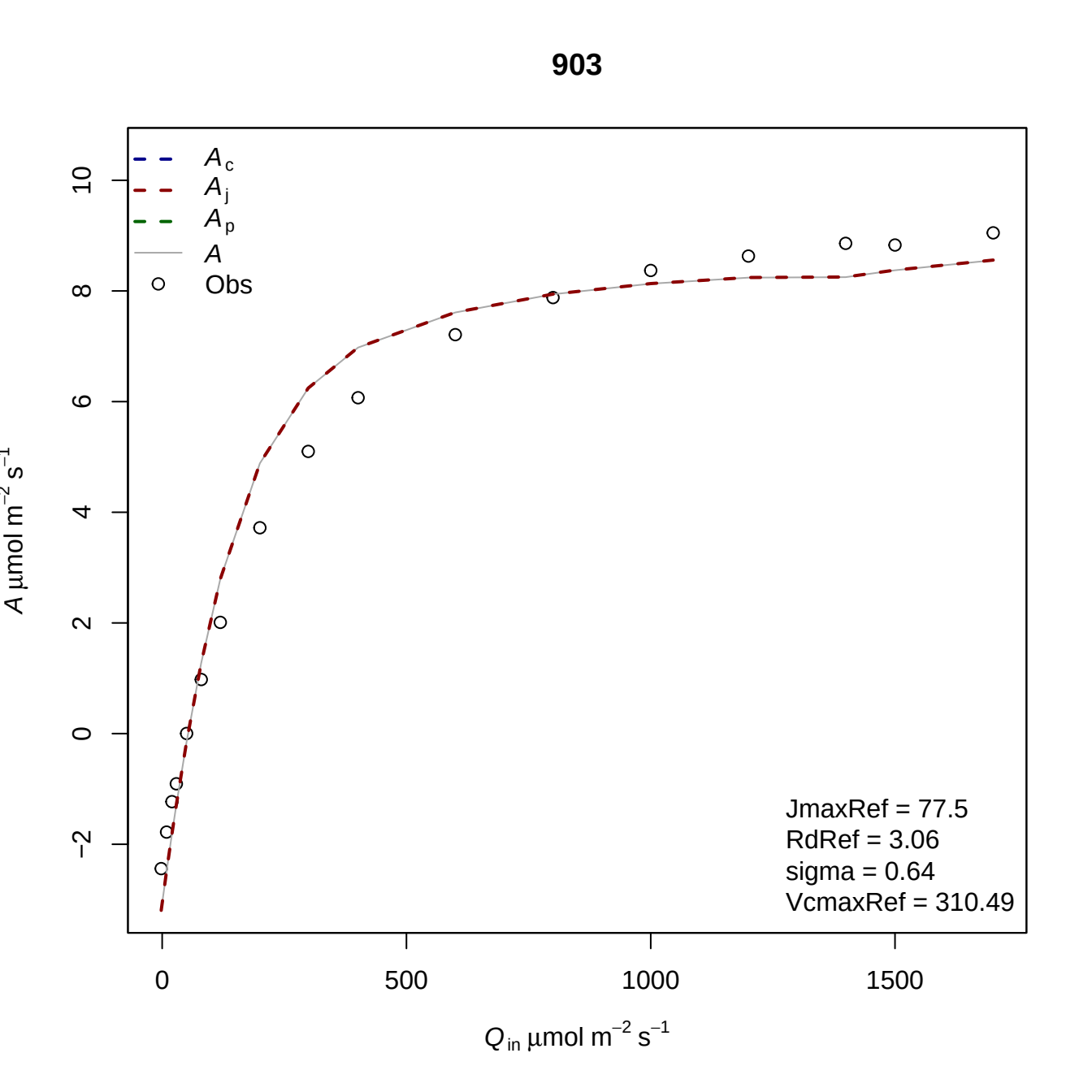

898
A_c
A_j
A_p
A
Obs
JmaxRef = 47.91
RdRef = 2.3
sigma = 0.58
VcmaxRef = 217.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

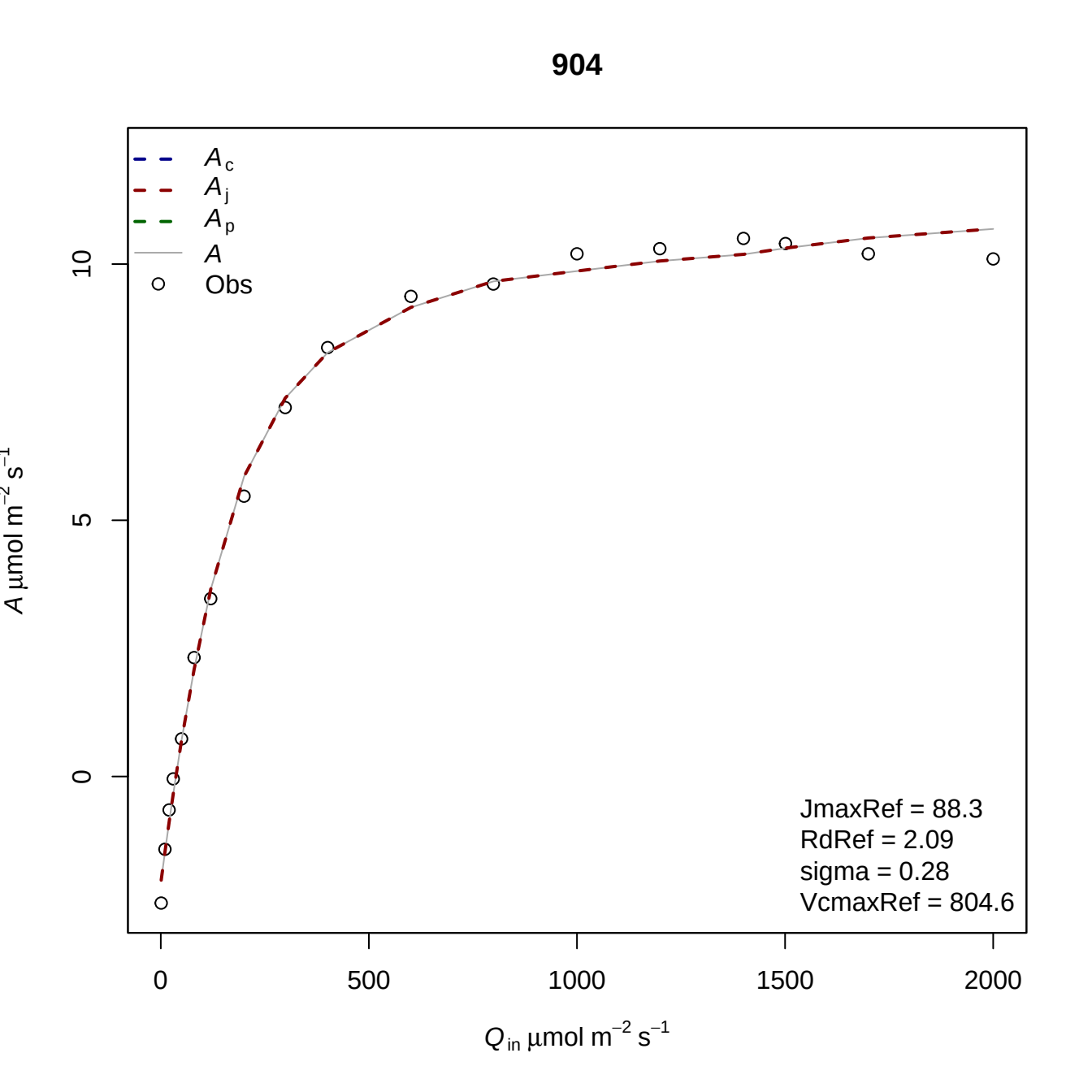

899
A_c
A_j
A_p
A
Obs
JmaxRef = 41.37
RdRef = 1.79
sigma = 0.44
VcmaxRef = 148.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

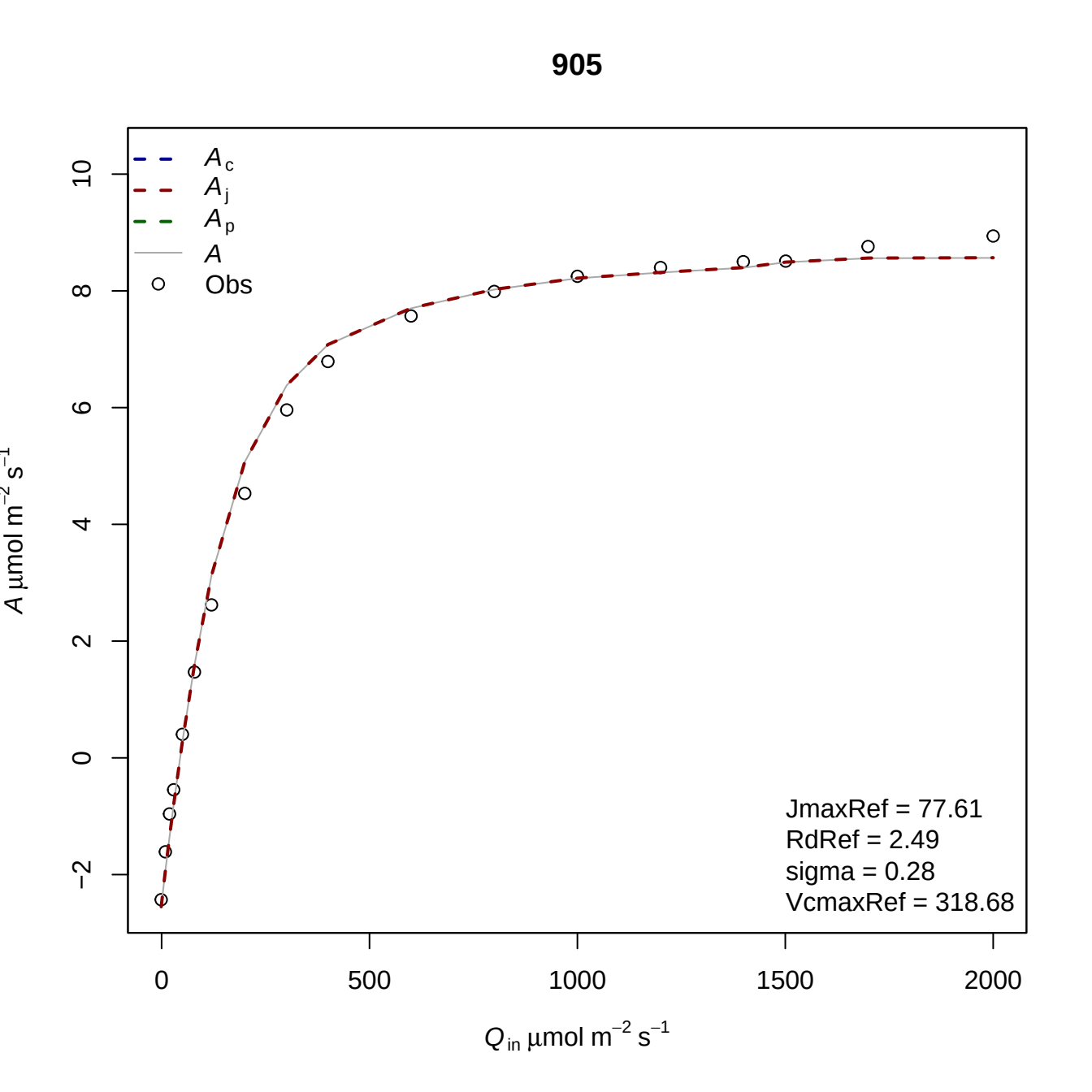

900
A_c
A_j
A_p
A
Obs
JmaxRef = 62.2
RdRef = 2.19
sigma = 0.42
VcmaxRef = 215.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

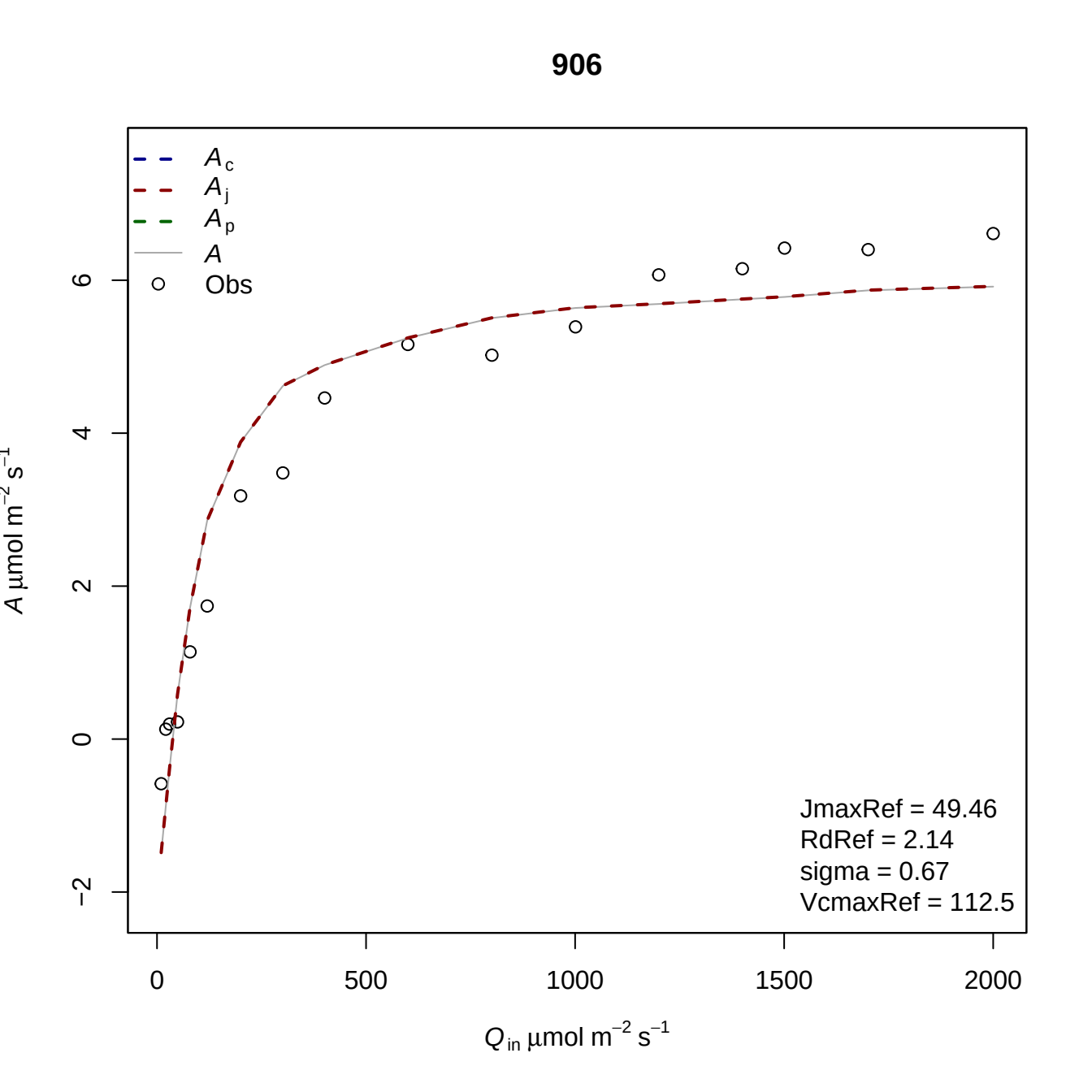

901
A_c
A_j
A_p
A
Obs
JmaxRef = 93.93
RdRef = 2.93
sigma = 0.97
VcmaxRef = 1455.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

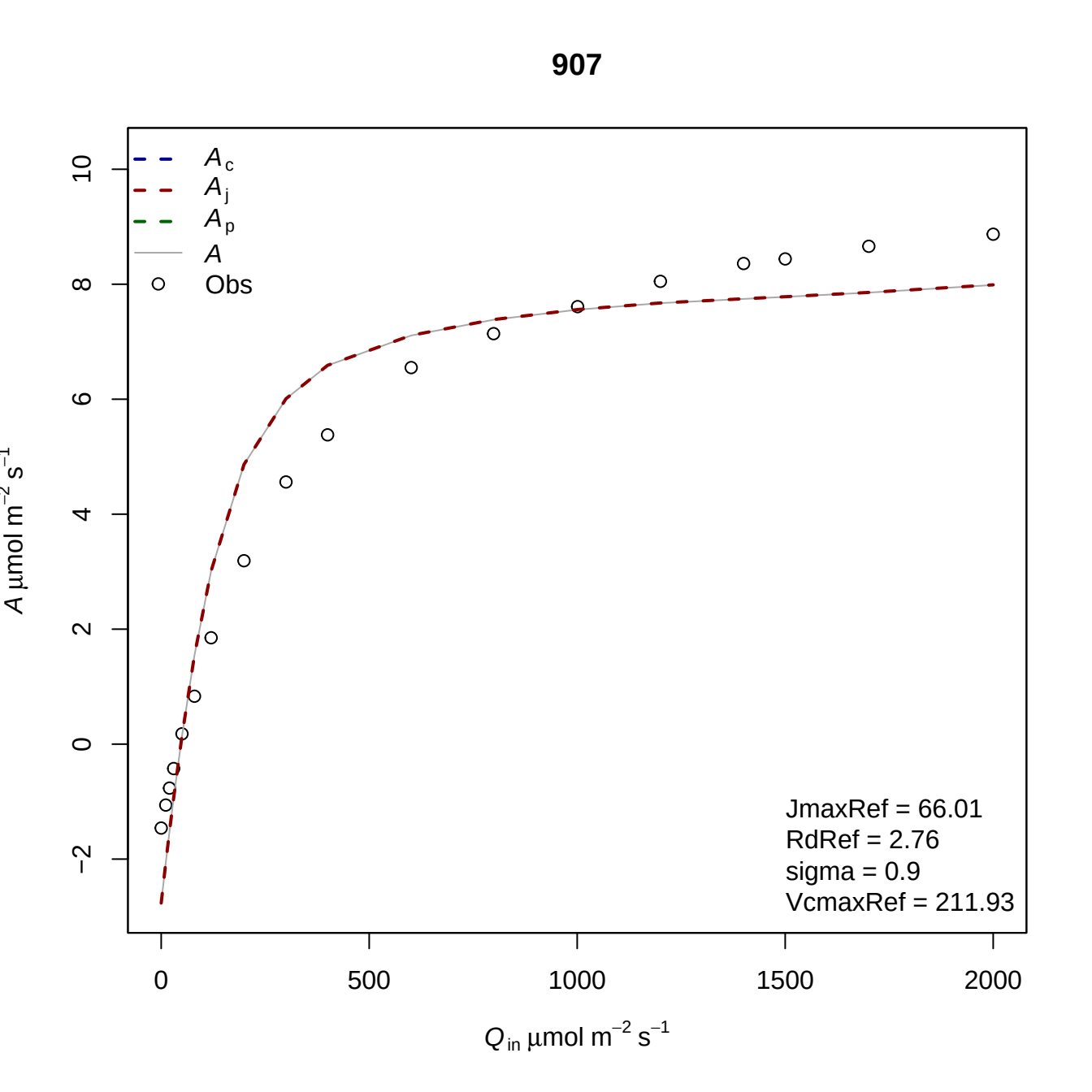

902
A_c
A_j
A_p
A
Obs
JmaxRef = 58.74
RdRef = 2.12
sigma = 0.63
VcmaxRef = 234.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

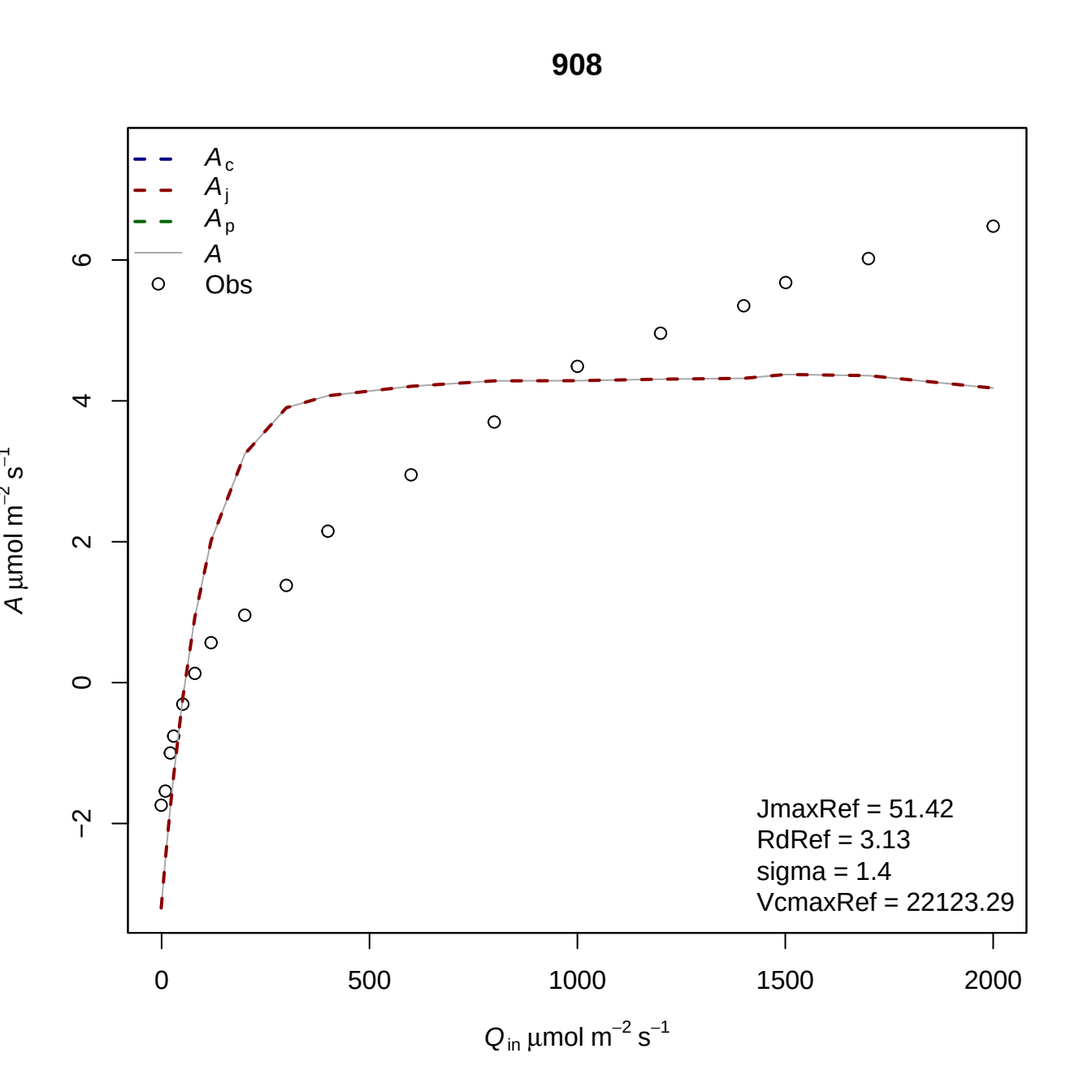

903
A_c
A_j
A_p
A
Obs
JmaxRef = 77.5
RdRef = 3.06
sigma = 0.64
VcmaxRef = 310.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

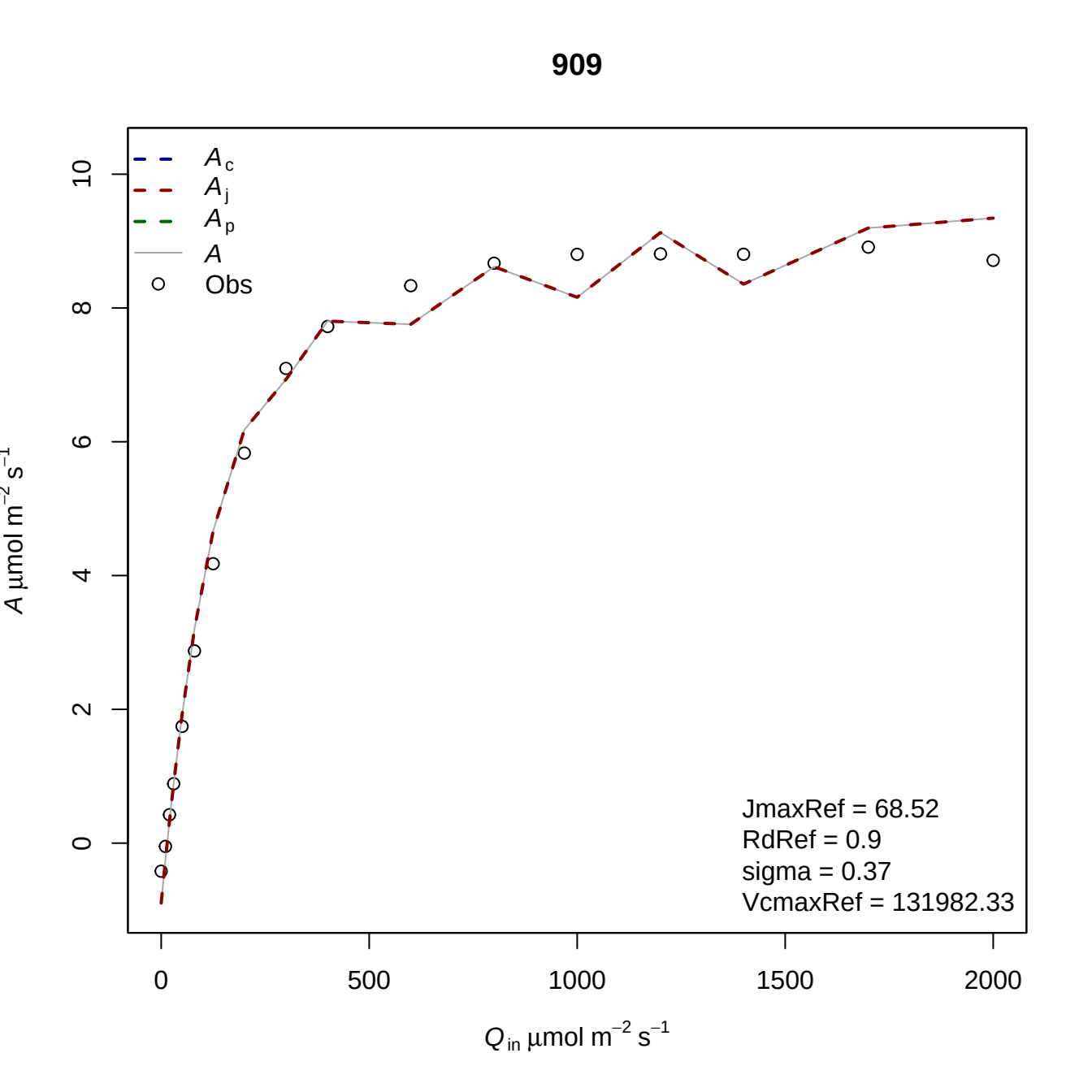

904
A_c
A_j
A_p
A
Obs
JmaxRef = 88.3
RdRef = 2.09
sigma = 0.28
VcmaxRef = 804.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

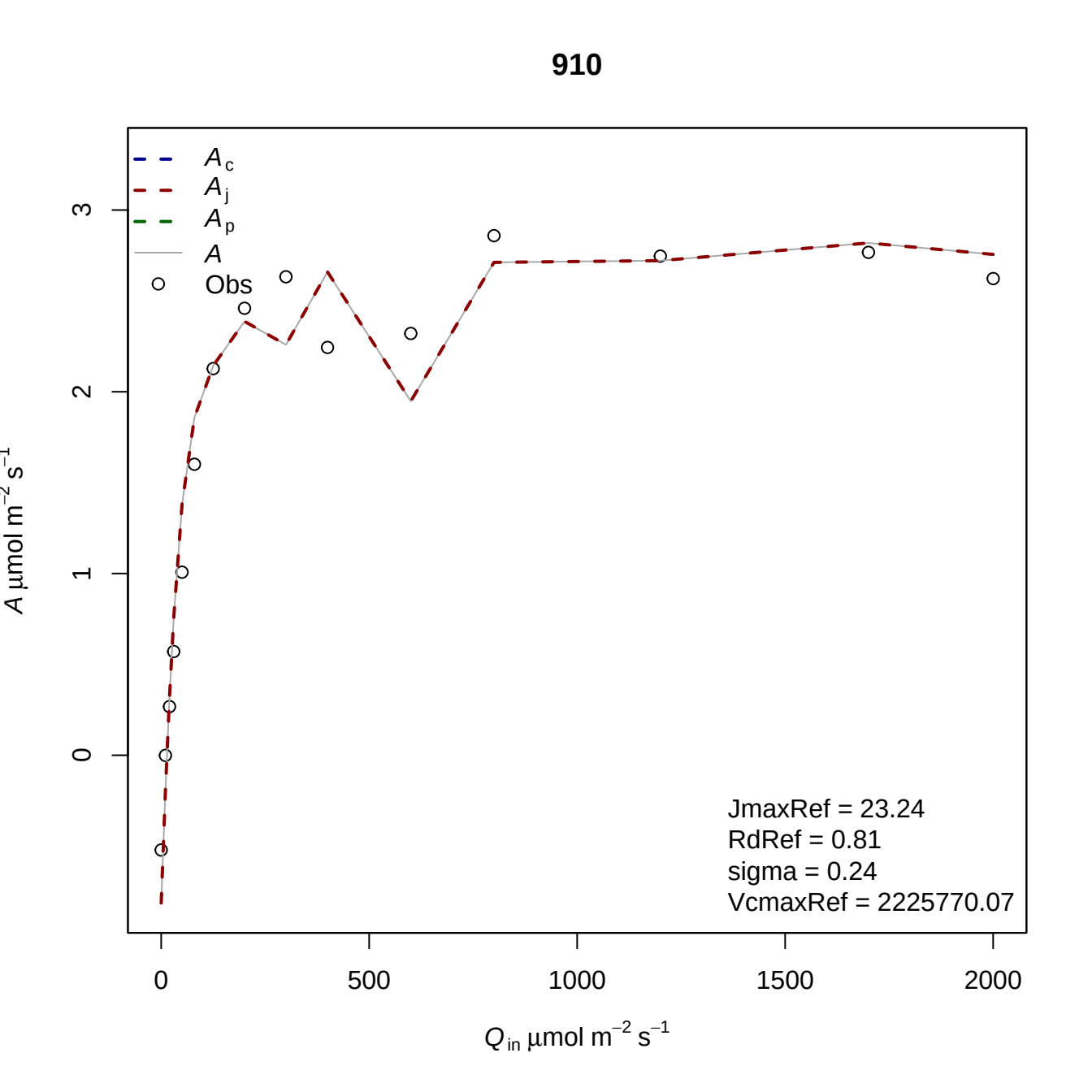

905
A_c
A_j
A_p
A
Obs
JmaxRef = 77.61
RdRef = 2.49
sigma = 0.28
VcmaxRef = 318.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

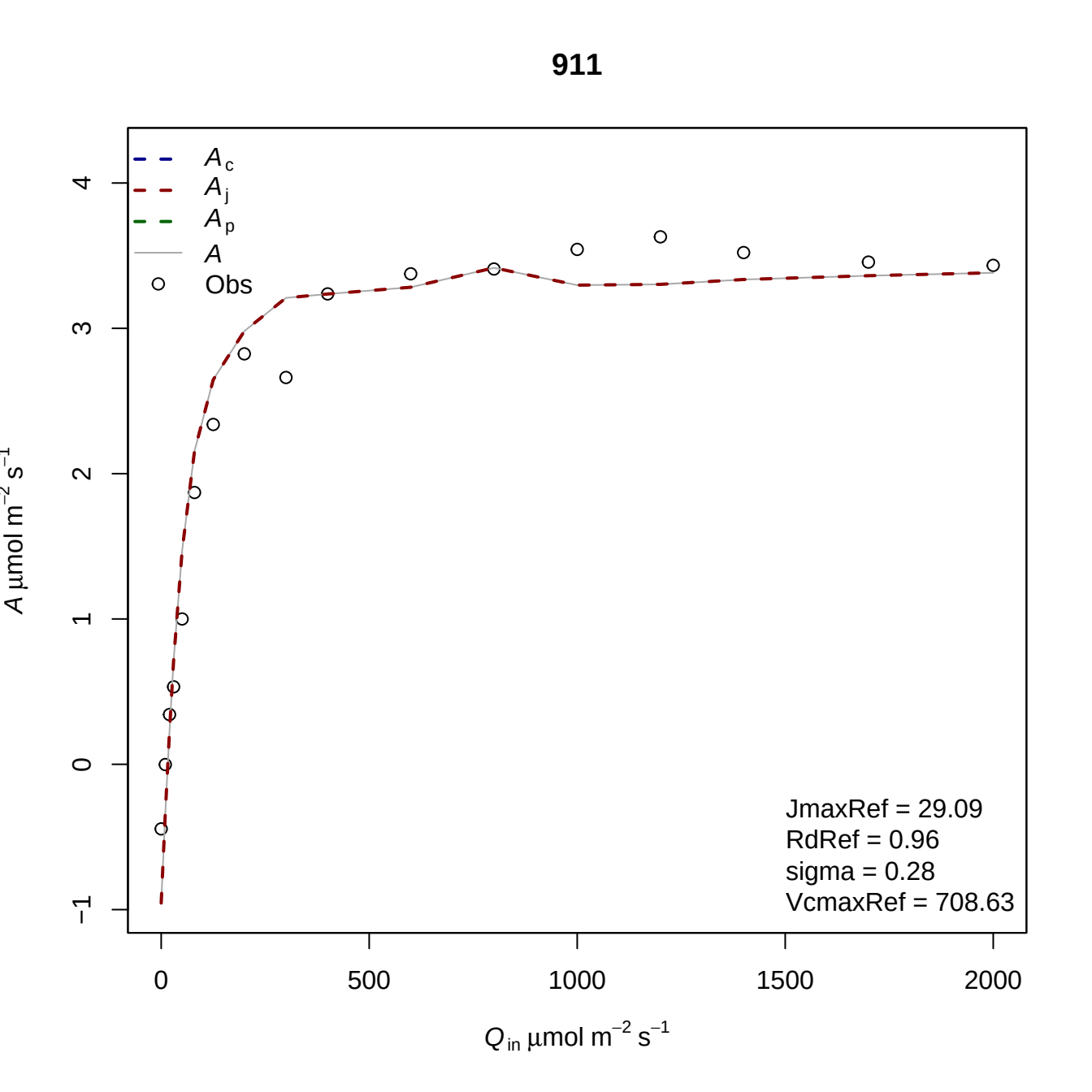

906
A_c
A_j
A_p
A
Obs
JmaxRef = 49.46
RdRef = 2.14
sigma = 0.67
VcmaxRef = 112.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

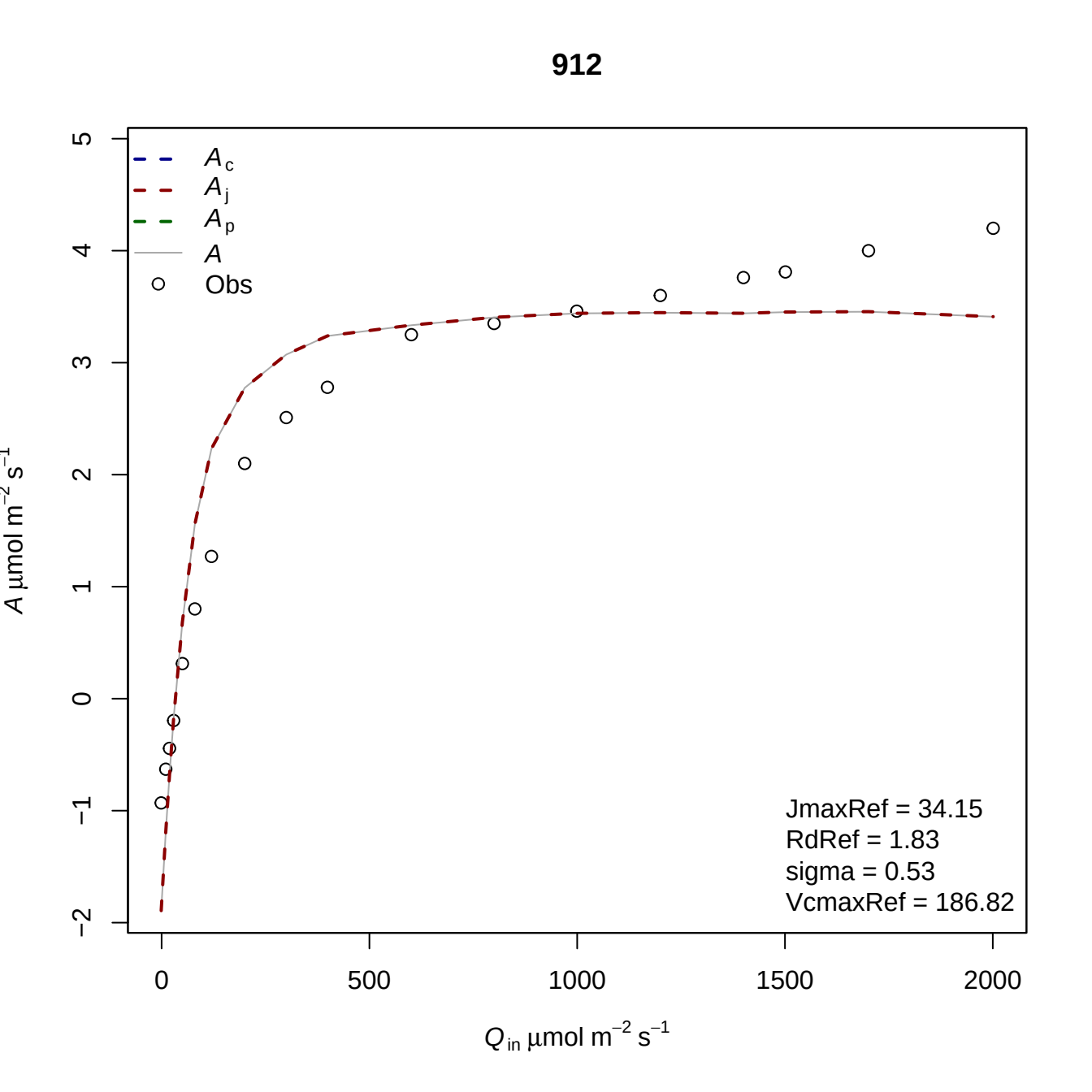

907
A_c
A_j
A_p
A
Obs
JmaxRef = 66.01
RdRef = 2.76
sigma = 0.9
VcmaxRef = 211.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

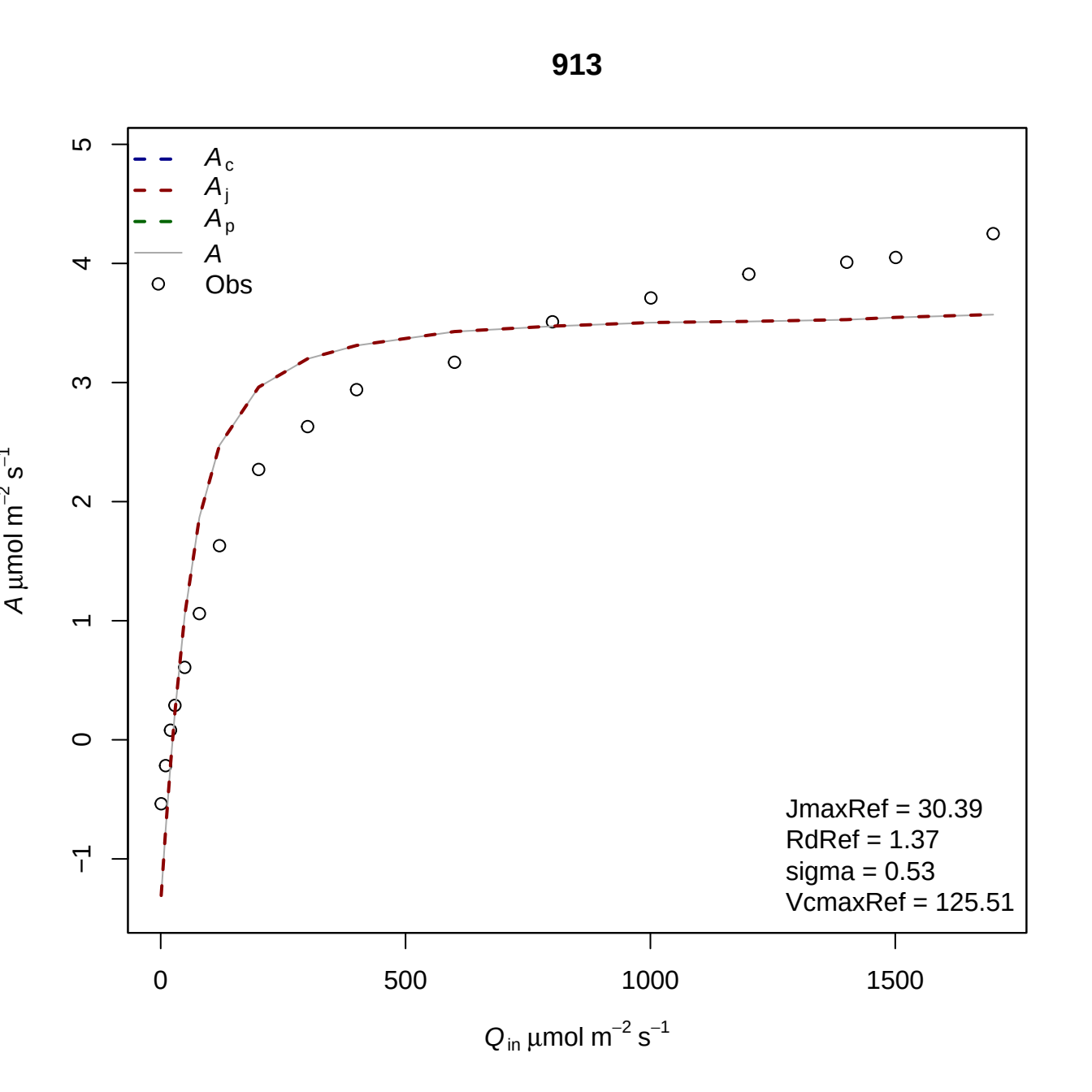

908
A_c
A_j
A_p
A
Obs
JmaxRef = 51.42
RdRef = 3.13
sigma = 1.4
VcmaxRef = 22123.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

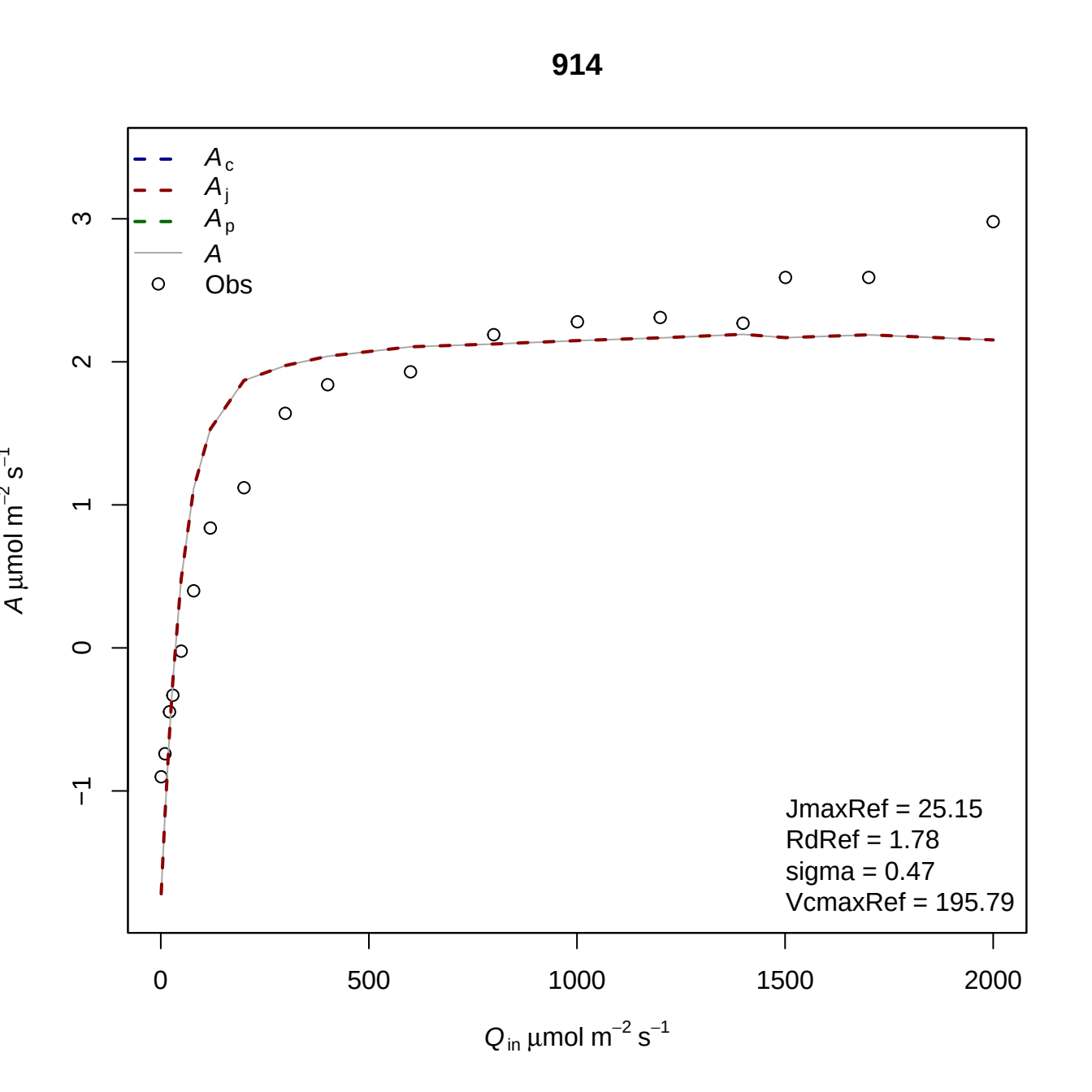

909
A_c
A_j
A_p
A
Obs
JmaxRef = 68.52
RdRef = 0.9
sigma = 0.37
VcmaxRef = 131982.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

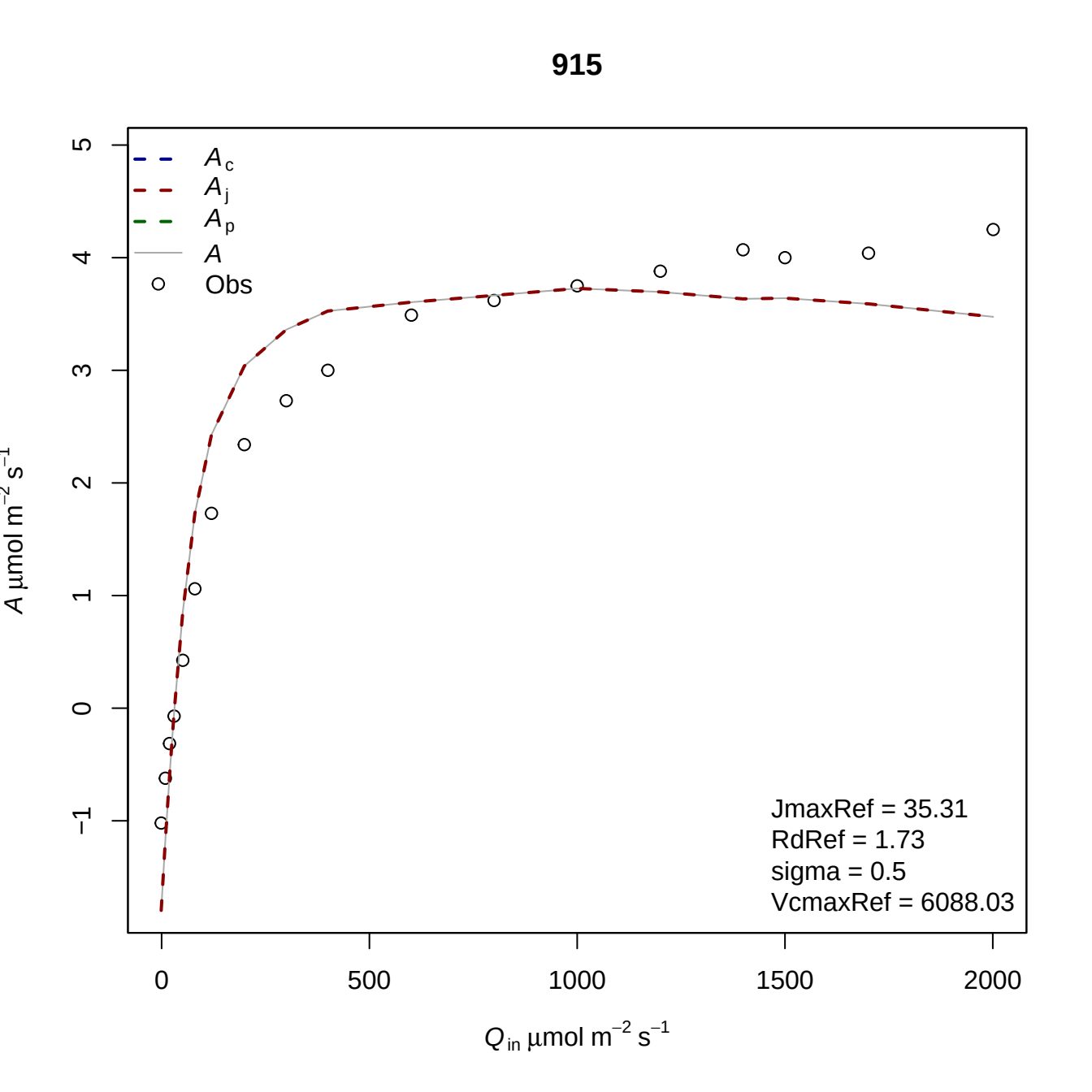

910
A_c
A_j
A_p
A
Obs
JmaxRef = 23.24
RdRef = 0.81
sigma = 0.24
VcmaxRef = 2225770.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

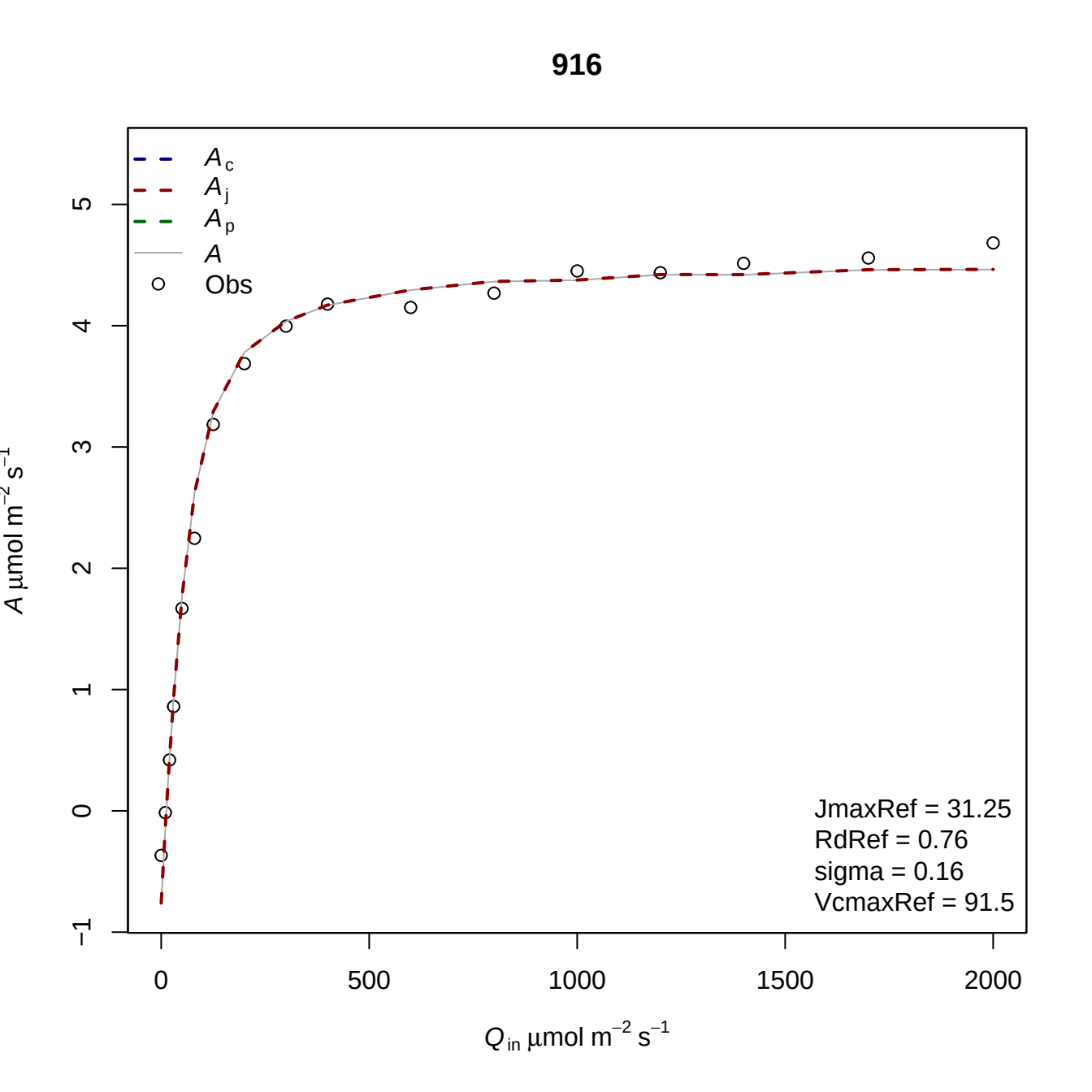

911
A_c
A_j
A_p
A
Obs
JmaxRef = 29.09
RdRef = 0.96
sigma = 0.28
VcmaxRef = 708.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

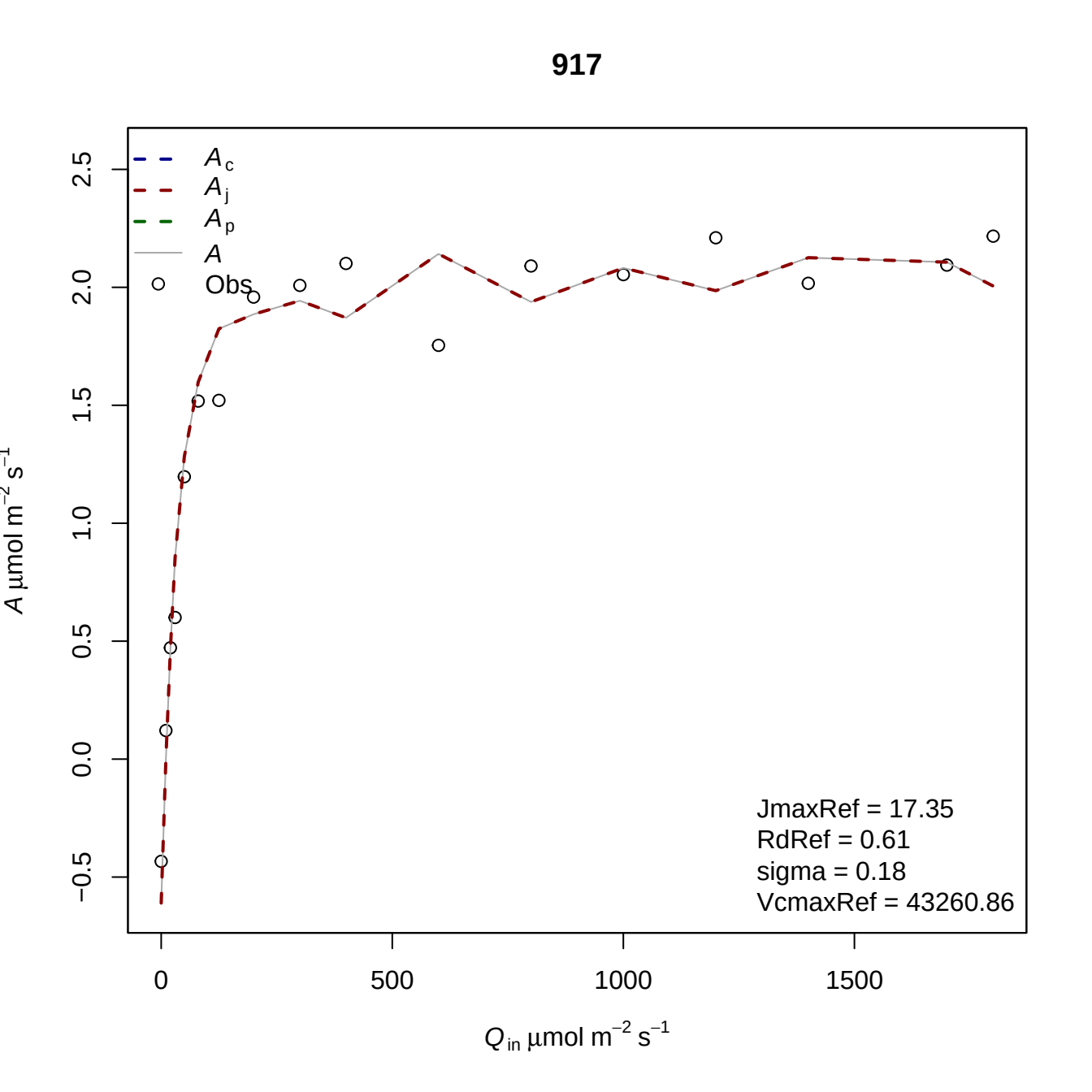

912
A_c
A_j
A_p
A
Obs
JmaxRef = 34.15
RdRef = 1.83
sigma = 0.53
VcmaxRef = 186.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

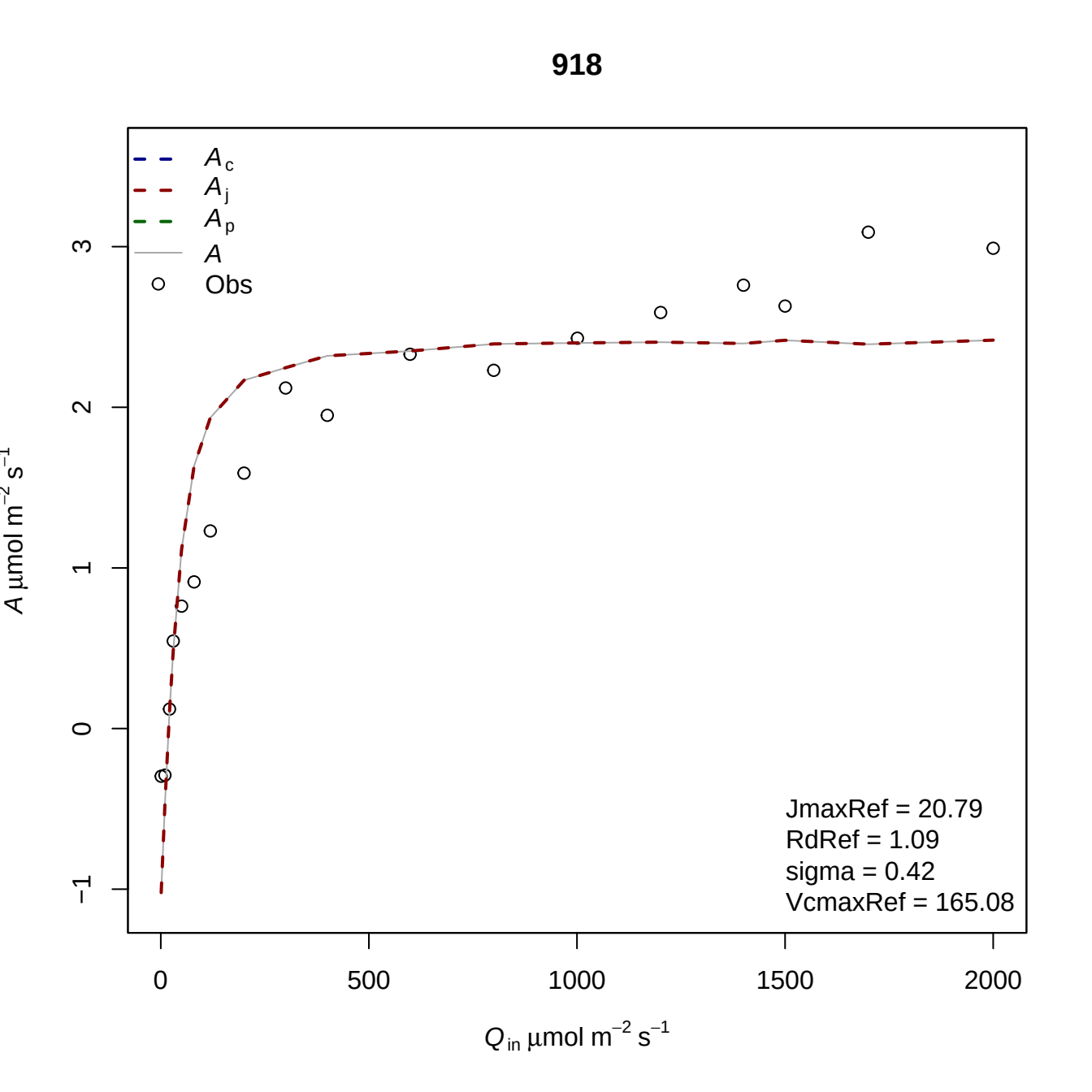

913
A_c
A_j
A_p
A
Obs
JmaxRef = 30.39
RdRef = 1.37
sigma = 0.53
VcmaxRef = 125.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

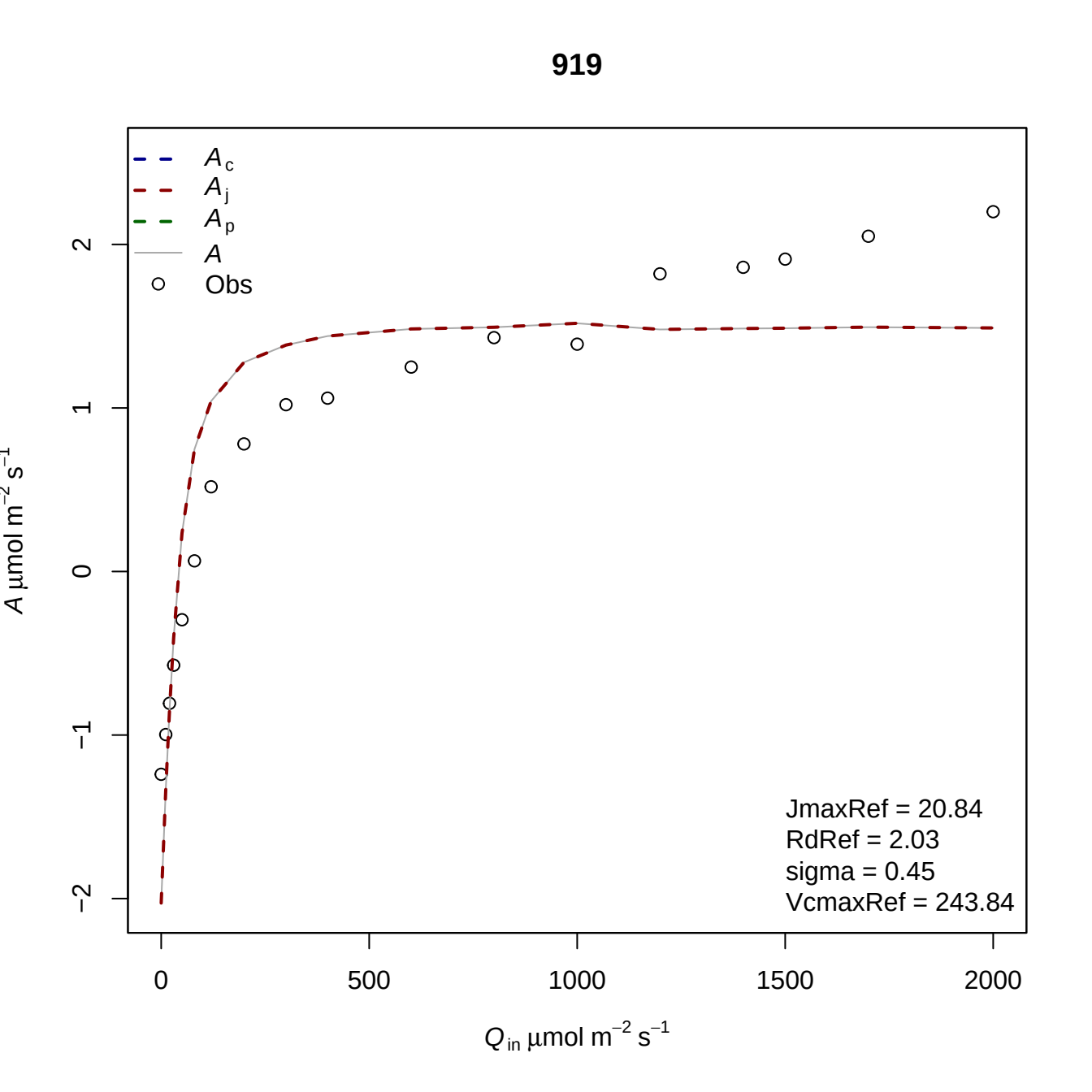

914
A_c
A_j
A_p
A
Obs
JmaxRef = 25.15
RdRef = 1.78
sigma = 0.47
VcmaxRef = 195.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

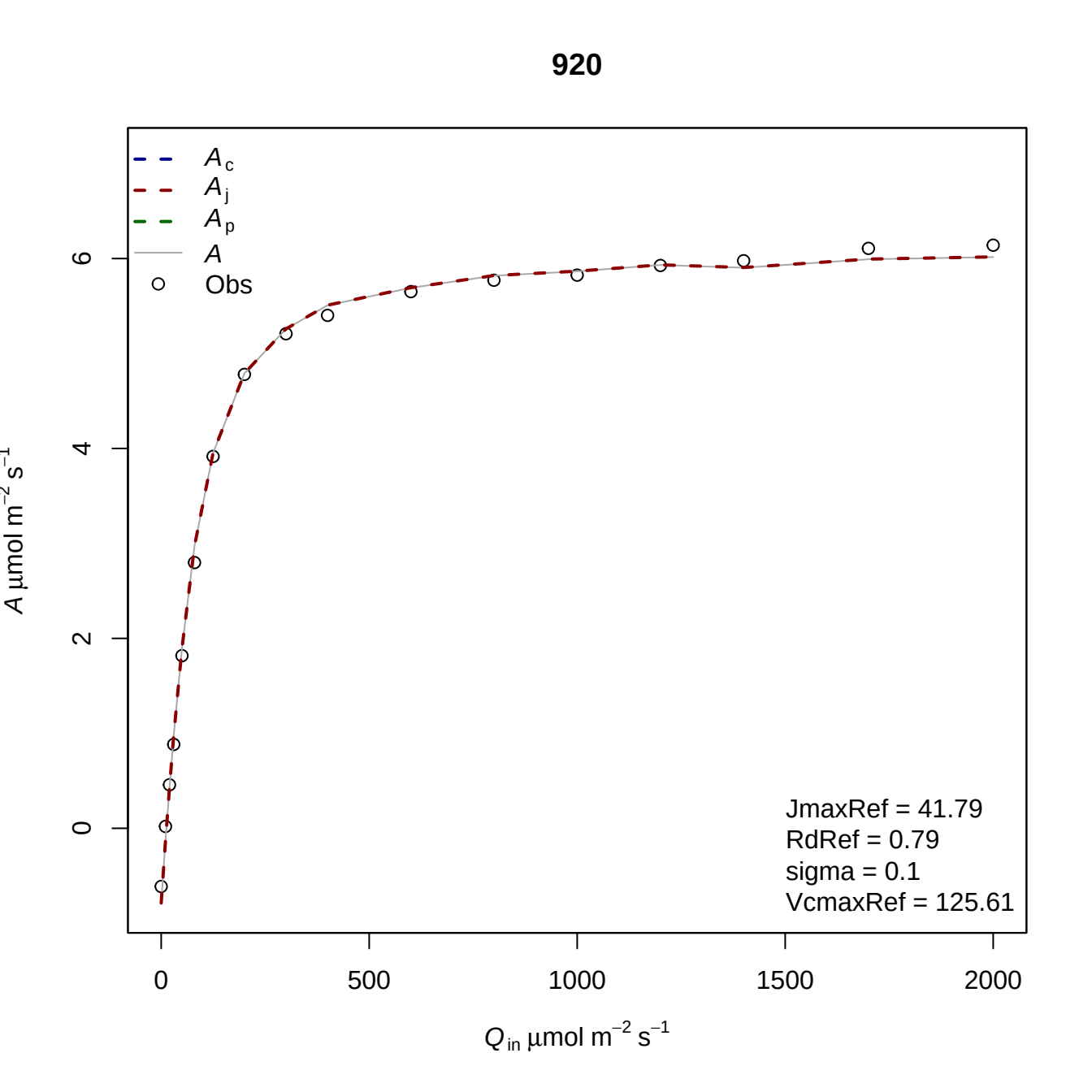

915
A_c
A_j
A_p
A
Obs
JmaxRef = 35.31
RdRef = 1.73
sigma = 0.5
VcmaxRef = 6088.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

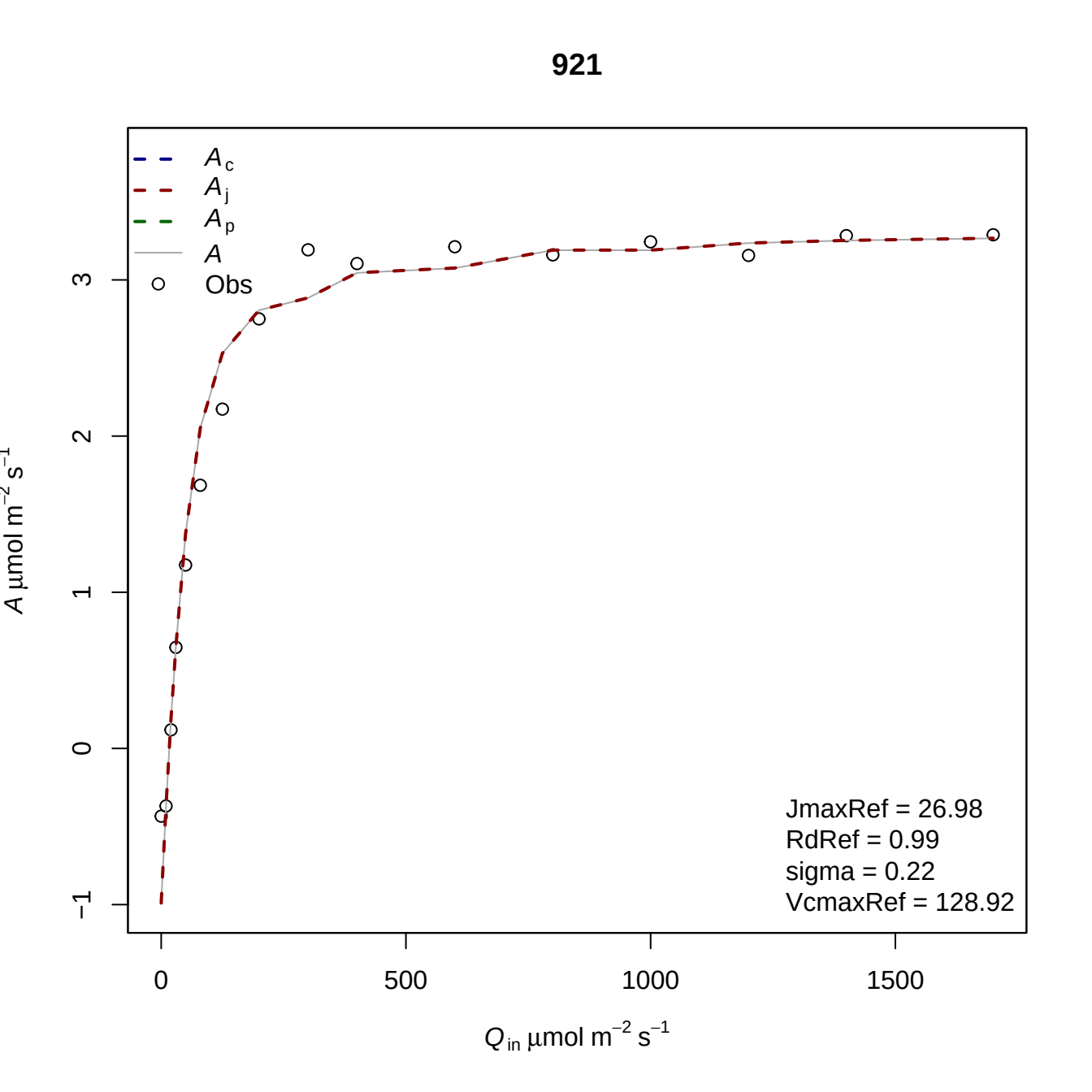

916
A_c
A_j
A_p
A
Obs
JmaxRef = 31.25
RdRef = 0.76
sigma = 0.16
VcmaxRef = 91.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

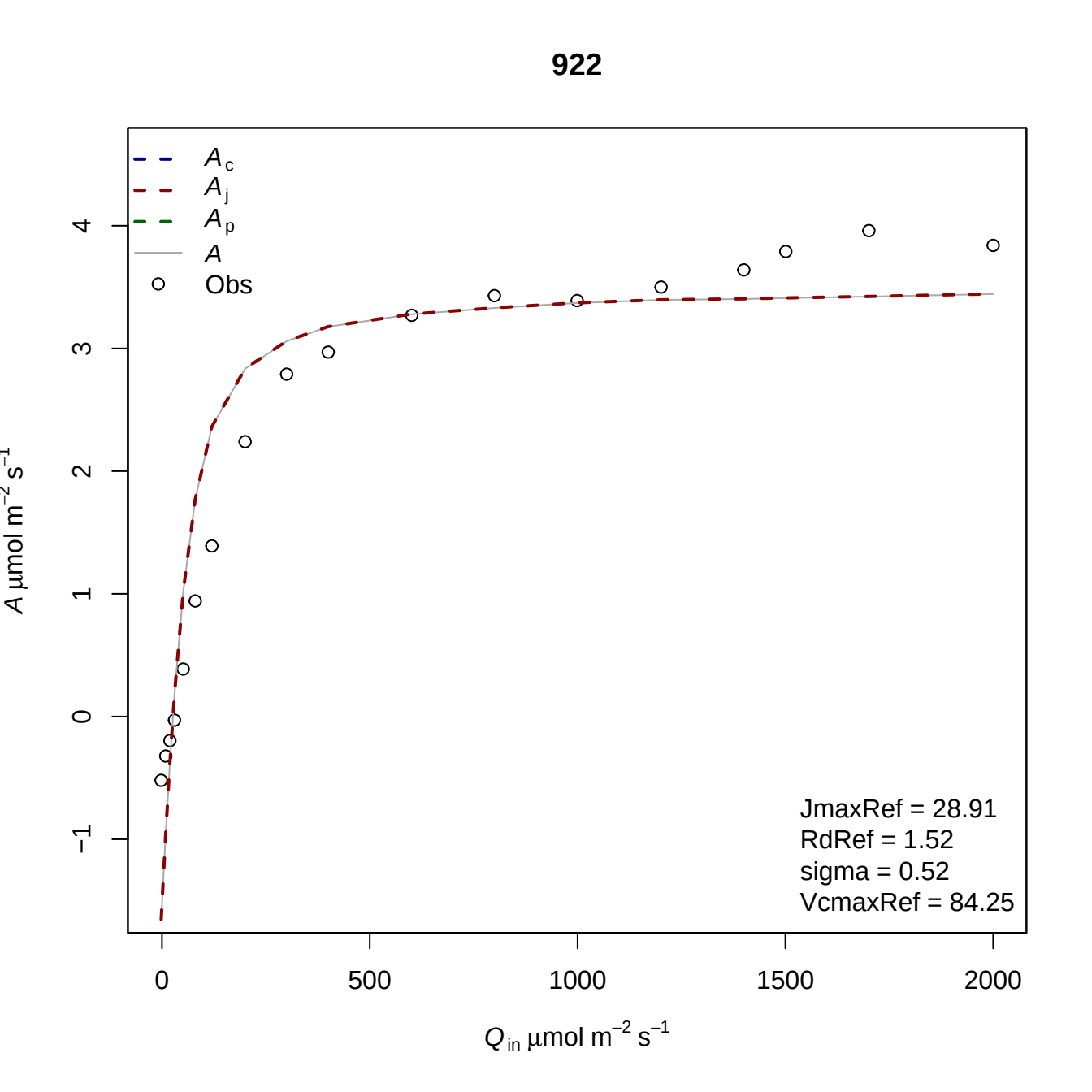

917
A_c
A_j
A_p
A
Obs
JmaxRef = 17.35
RdRef = 0.61
sigma = 0.18
VcmaxRef = 43260.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500

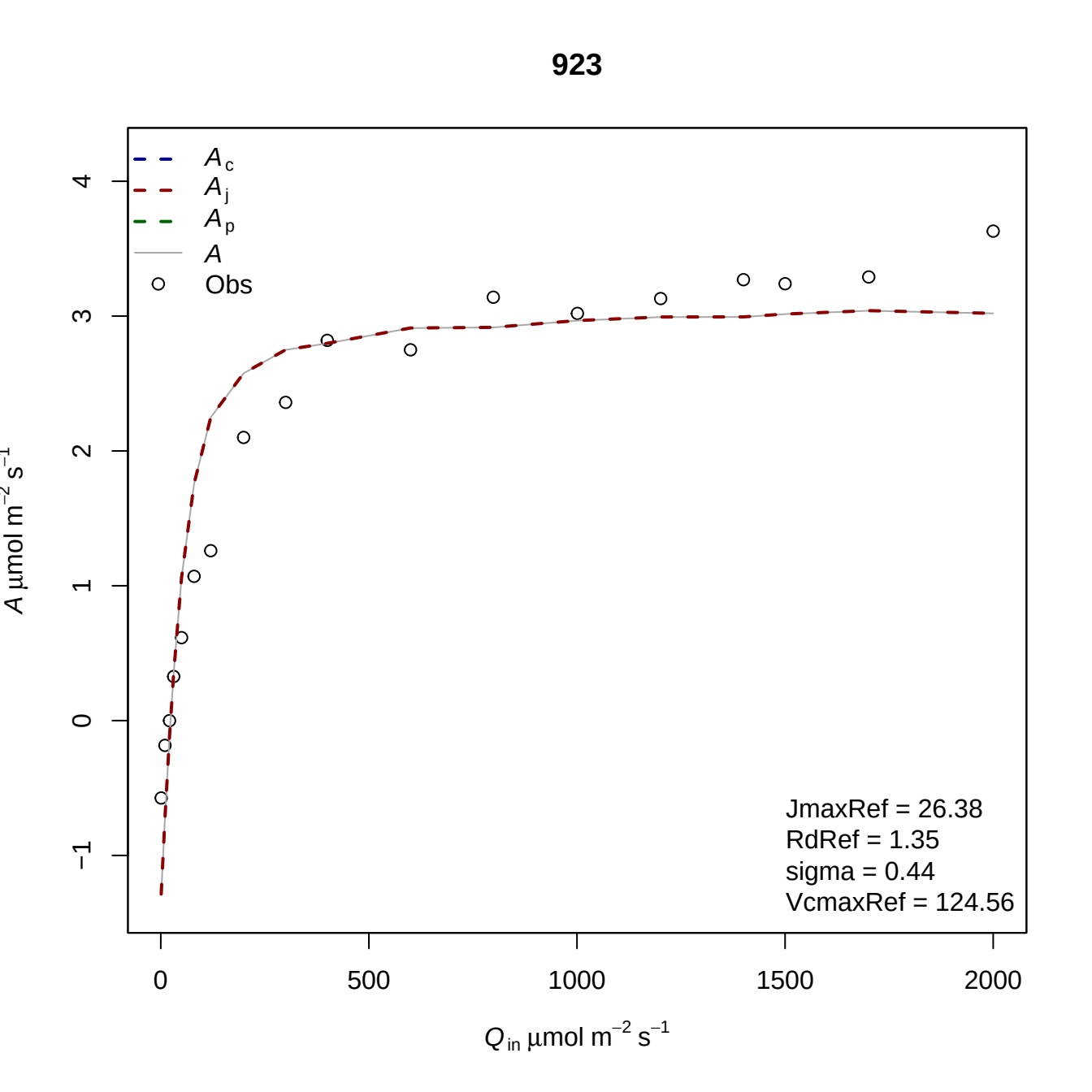

918
A_c
A_j
A_p
A
Obs
JmaxRef = 20.79
RdRef = 1.09
sigma = 0.42
VcmaxRef = 165.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

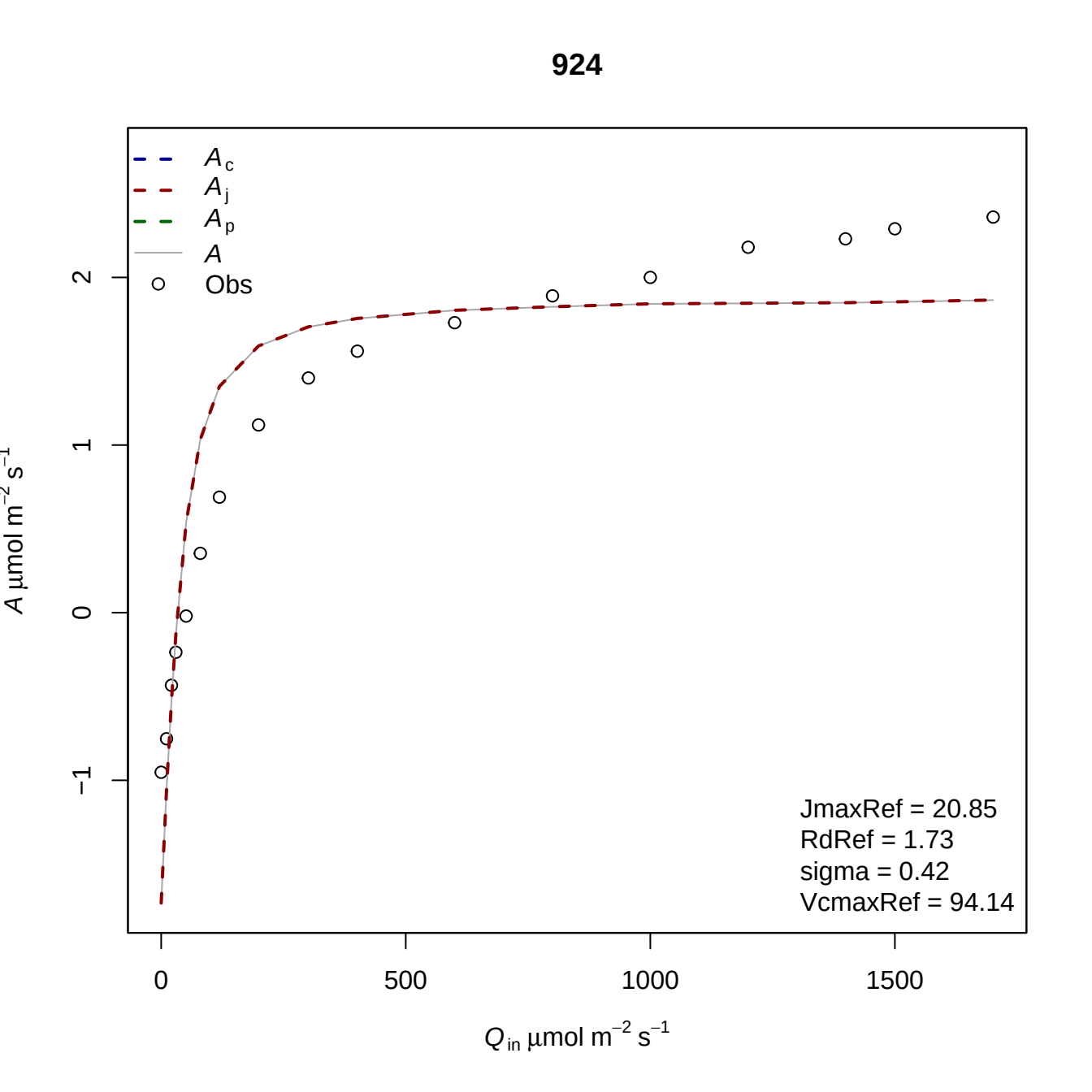

919
A_c
A_j
A_p
A
Obs
JmaxRef = 20.84
RdRef = 2.03
sigma = 0.45
VcmaxRef = 243.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
0
500
1000
1500
2000

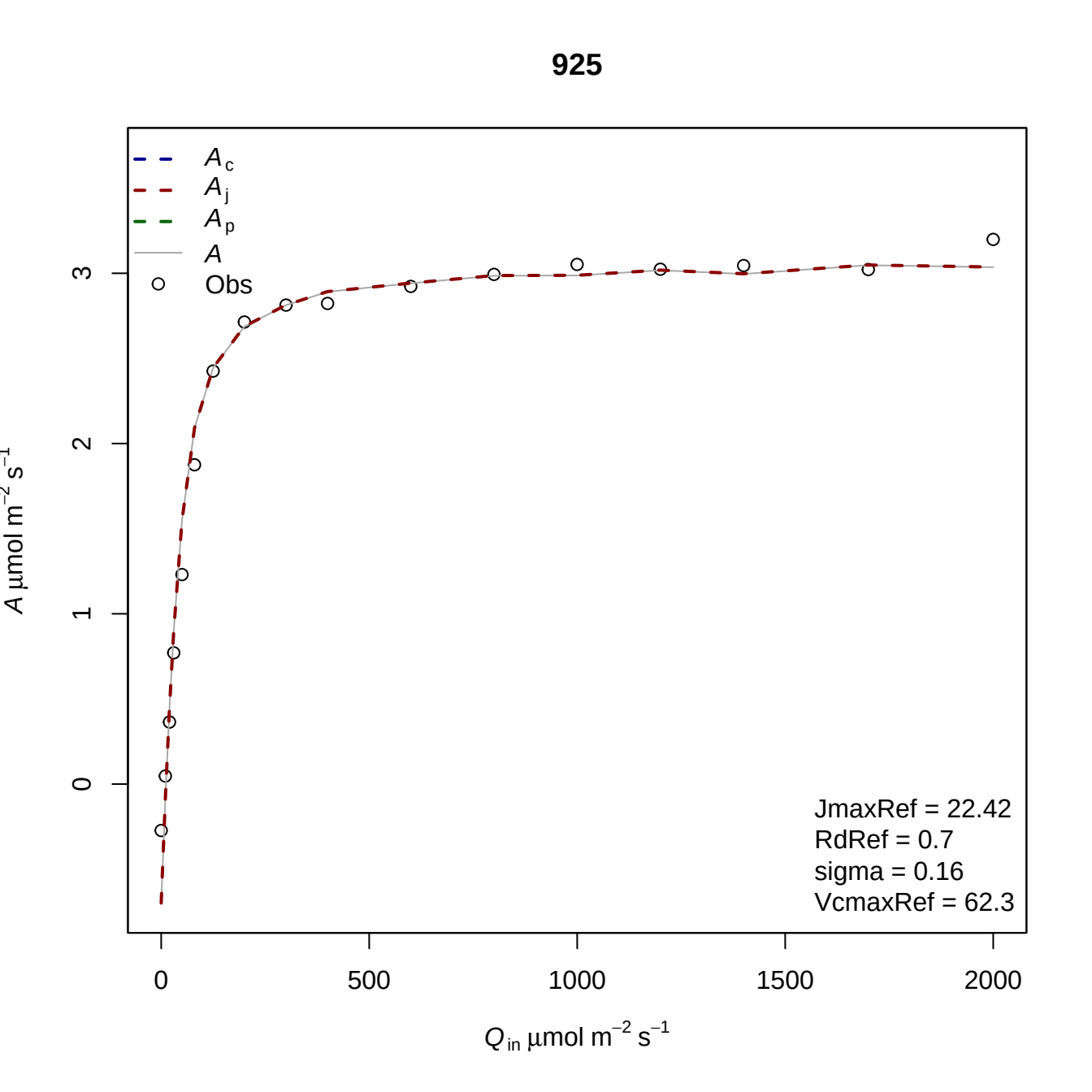

920
A_c
A_j
A_p
A
Obs
JmaxRef = 41.79
RdRef = 0.79
sigma = 0.1
VcmaxRef = 125.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

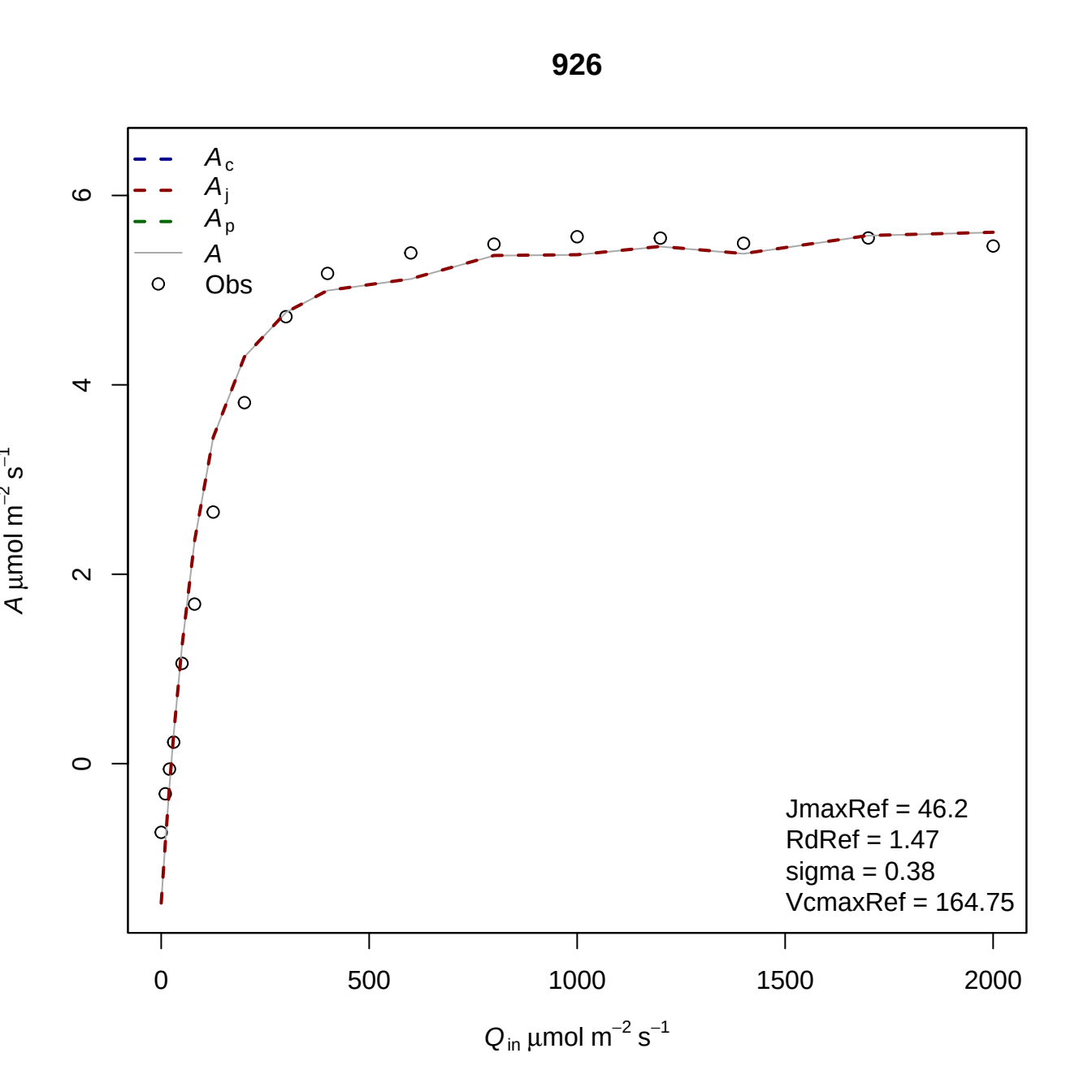

921
A_c
A_j
A_p
A
Obs
JmaxRef = 26.98
RdRef = 0.99
sigma = 0.22
VcmaxRef = 128.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

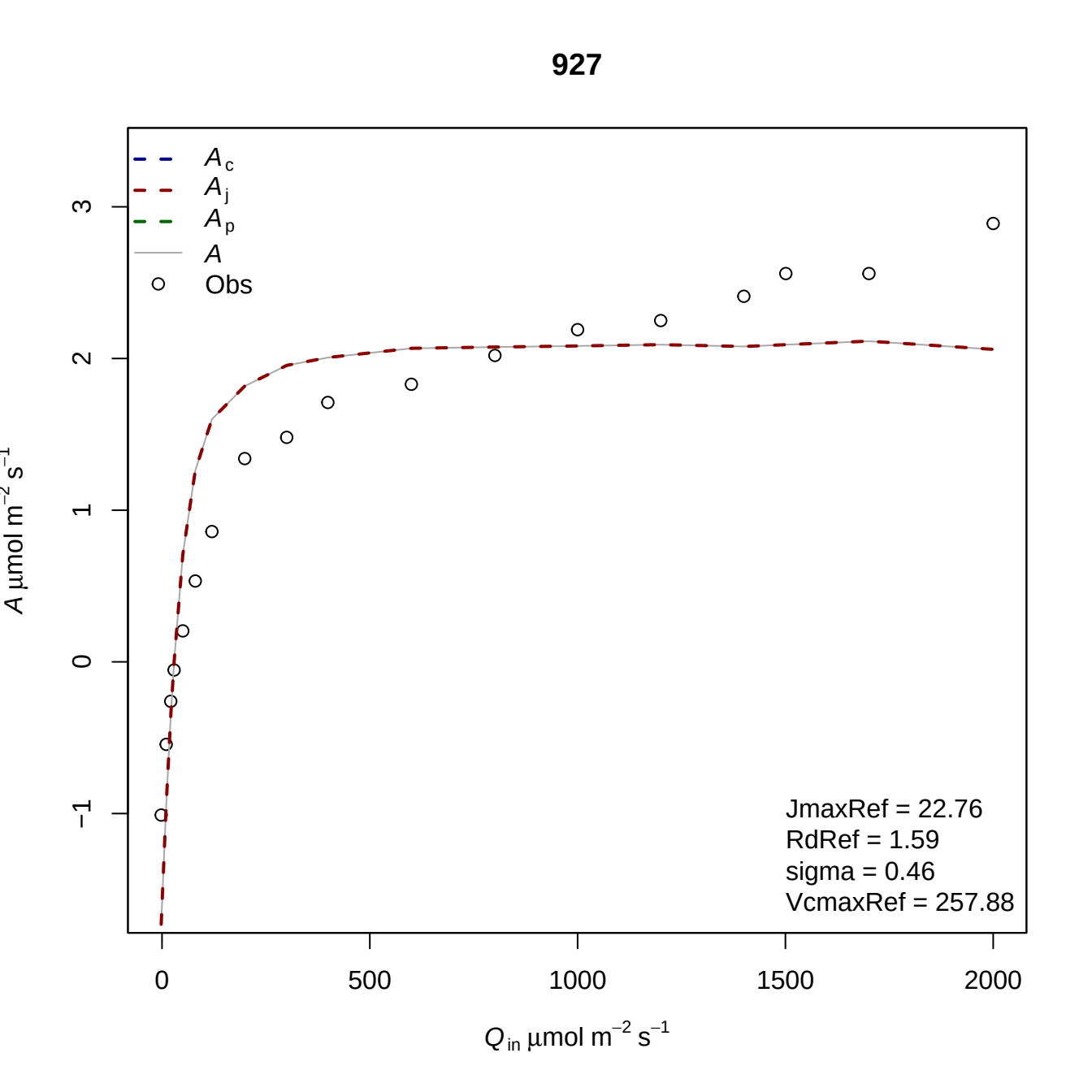

922
A_c
A_j
A_p
A
Obs
JmaxRef = 28.91
RdRef = 1.52
sigma = 0.52
VcmaxRef = 84.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

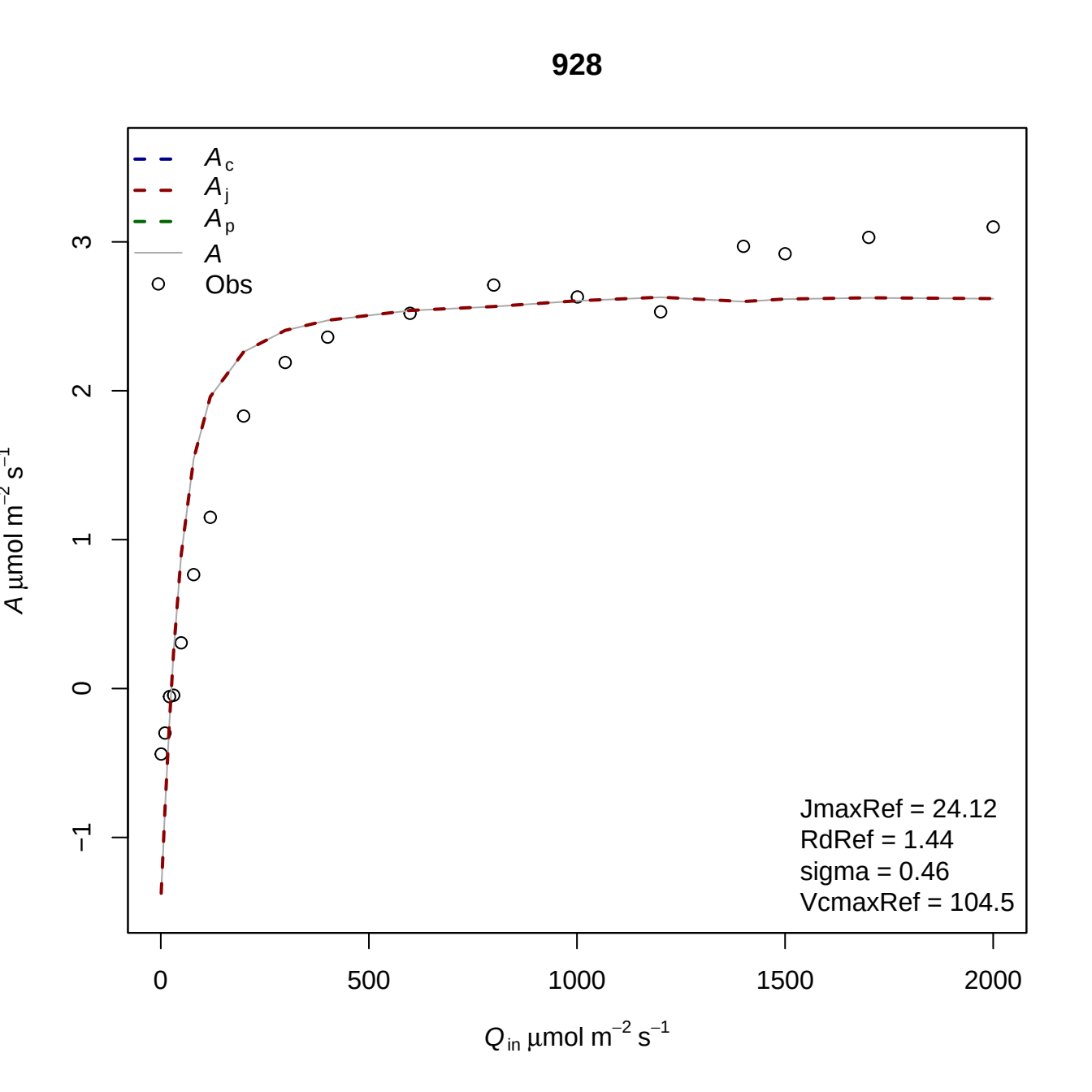

923
A_c
A_j
A_p
A
Obs
JmaxRef = 26.38
RdRef = 1.35
sigma = 0.44
VcmaxRef = 124.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

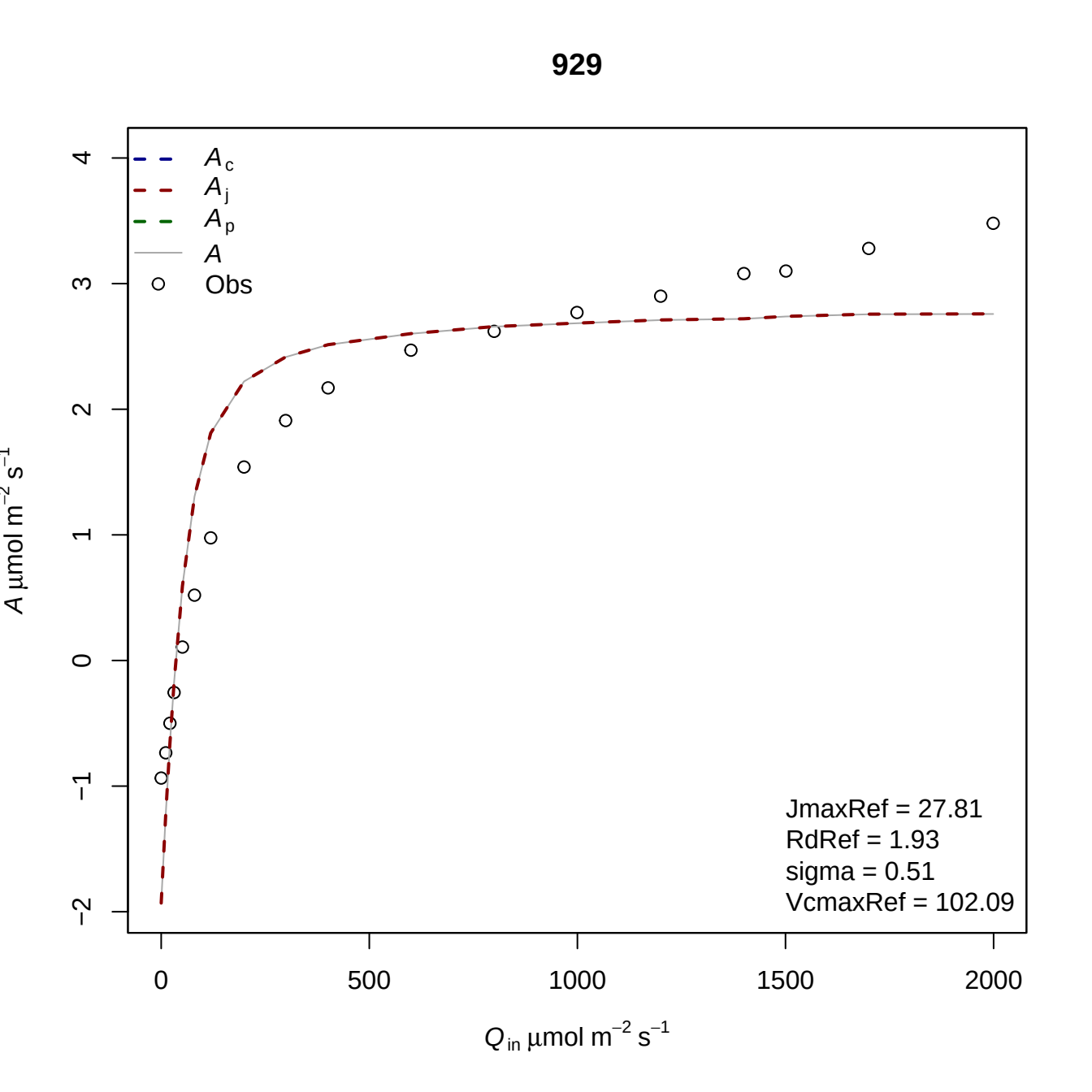

924
A_c
A_j
A_p
A
Obs
JmaxRef = 20.85
RdRef = 1.73
sigma = 0.42
VcmaxRef = 94.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

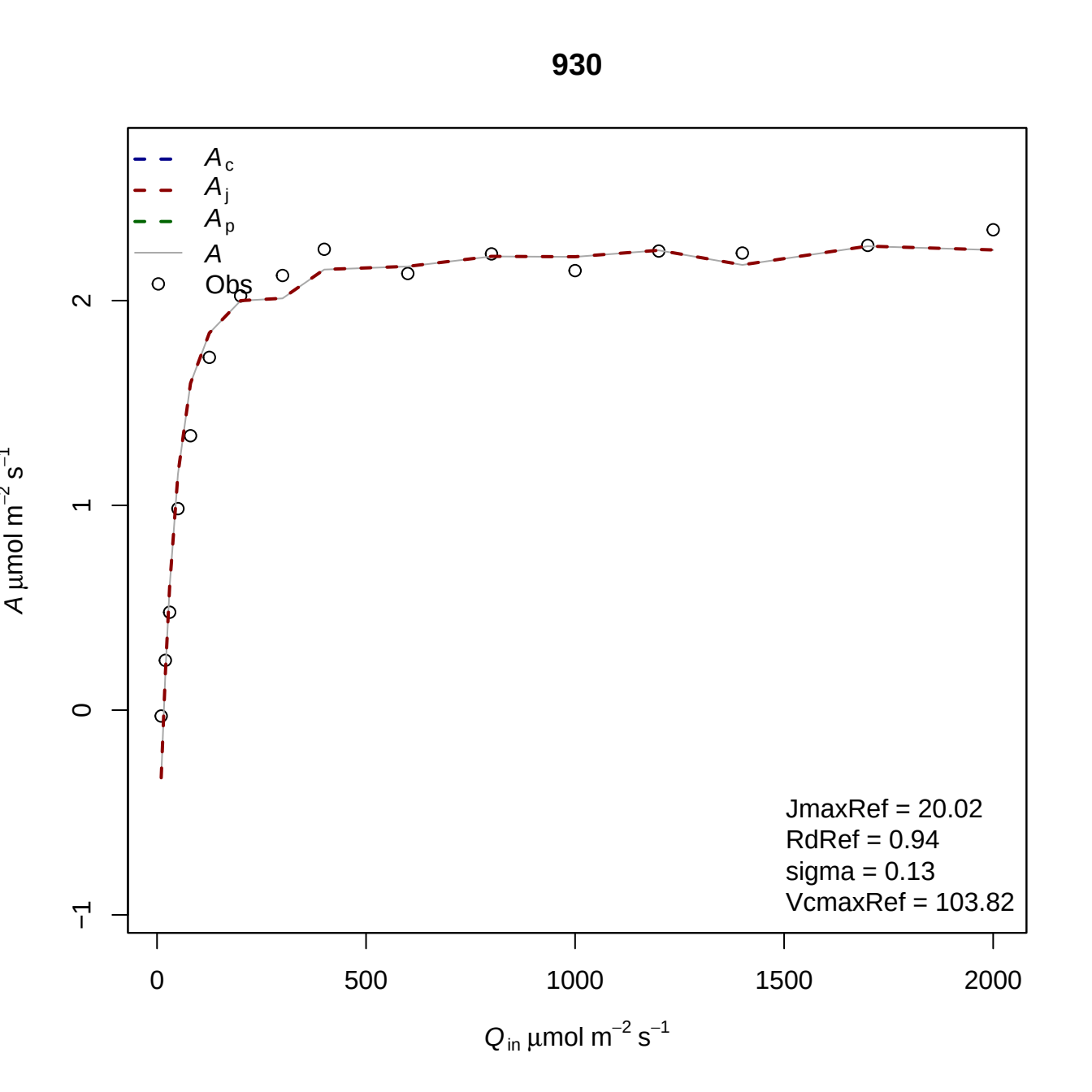

925
A_c
A_j
A_p
A
Obs
JmaxRef = 22.42
RdRef = 0.7
sigma = 0.16
VcmaxRef = 62.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

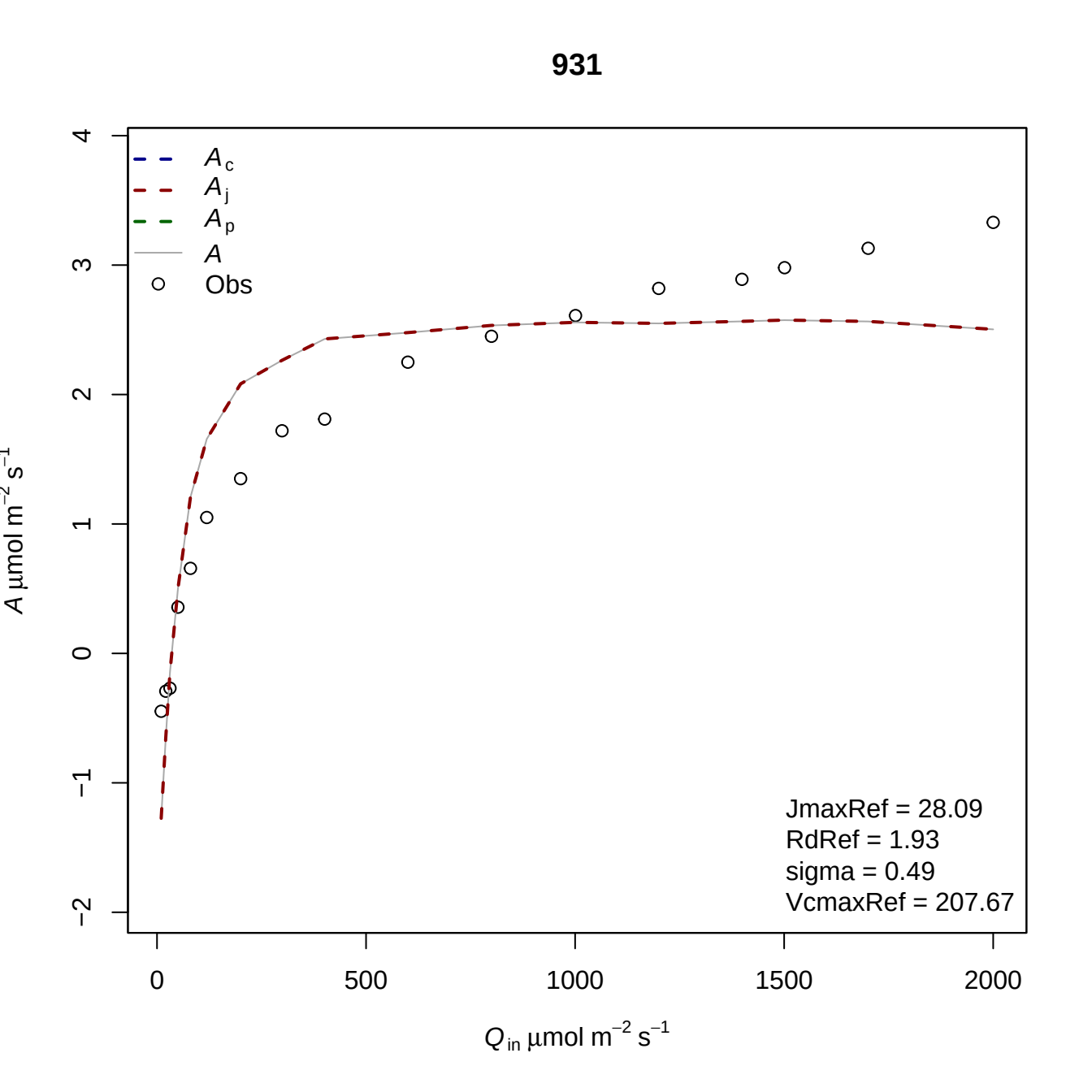

926
A_c
A_j
A_p
A
Obs
JmaxRef = 46.2
RdRef = 1.47
sigma = 0.38
VcmaxRef = 164.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

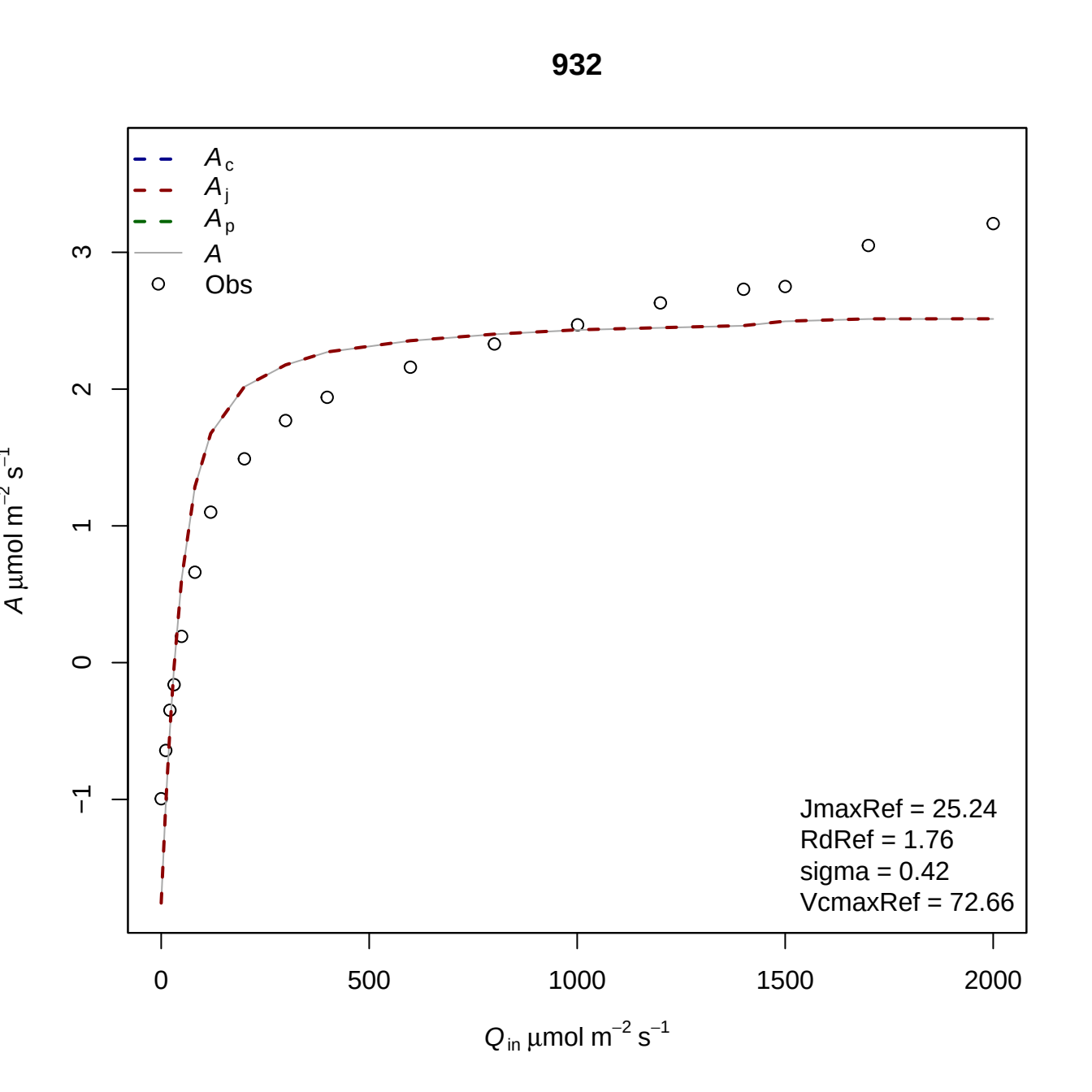

927
A_c
A_j
A_p
A
Obs
JmaxRef = 22.76
RdRef = 1.59
sigma = 0.46
VcmaxRef = 257.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

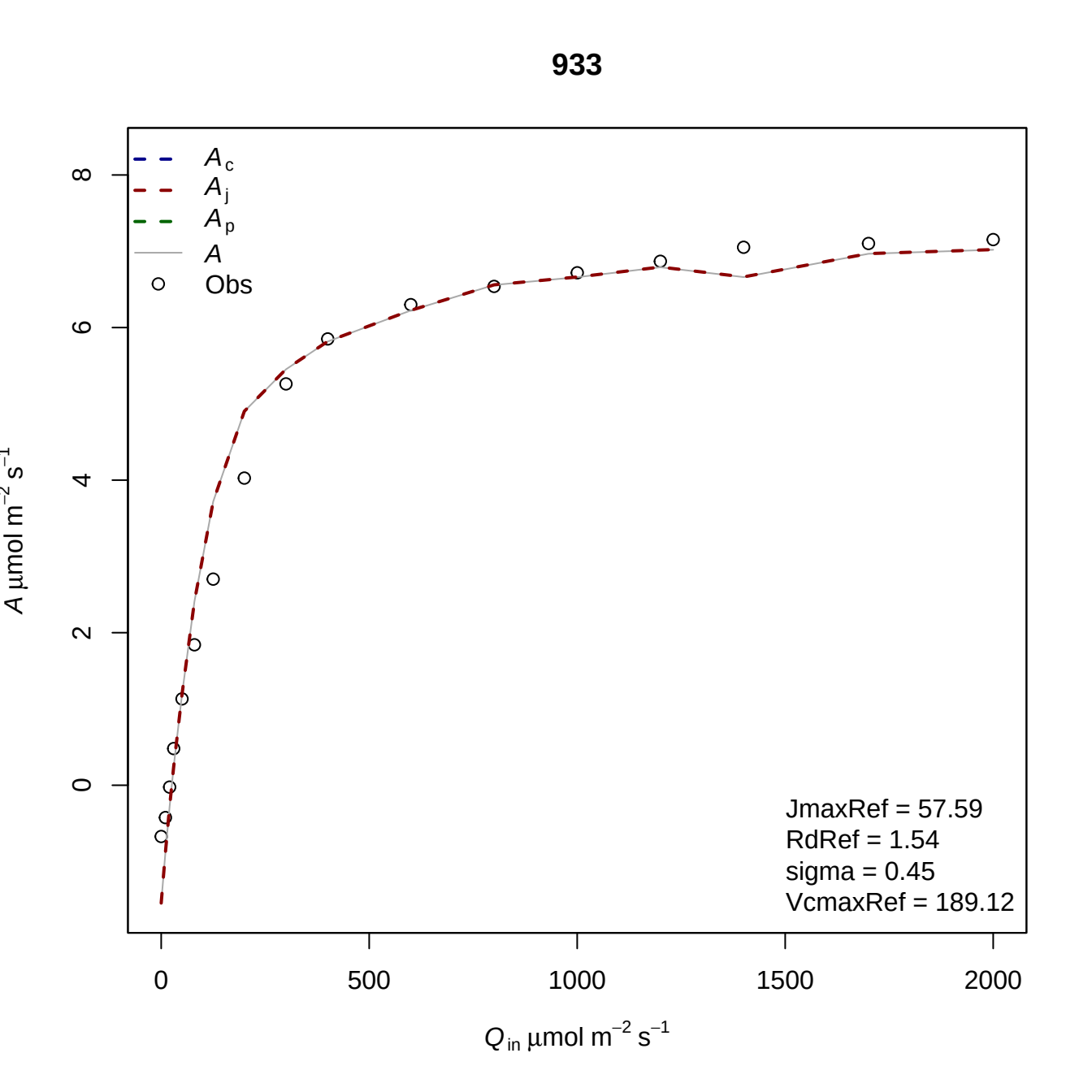

928
A_c
A_j
A_p
A
Obs
JmaxRef = 24.12
RdRef = 1.44
sigma = 0.46
VcmaxRef = 104.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

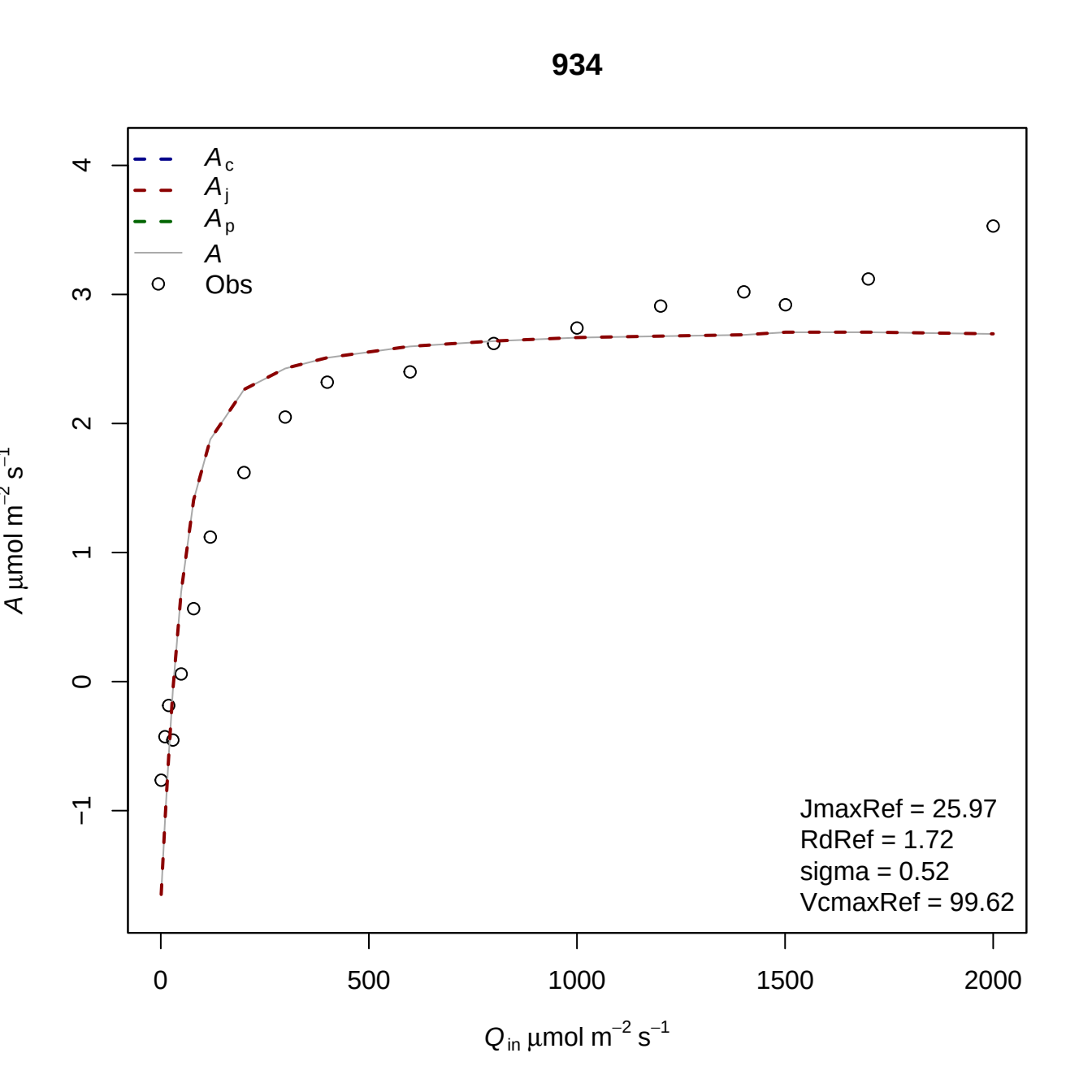

929
A_c
A_j
A_p
A
Obs
JmaxRef = 27.81
RdRef = 1.93
sigma = 0.51
VcmaxRef = 102.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

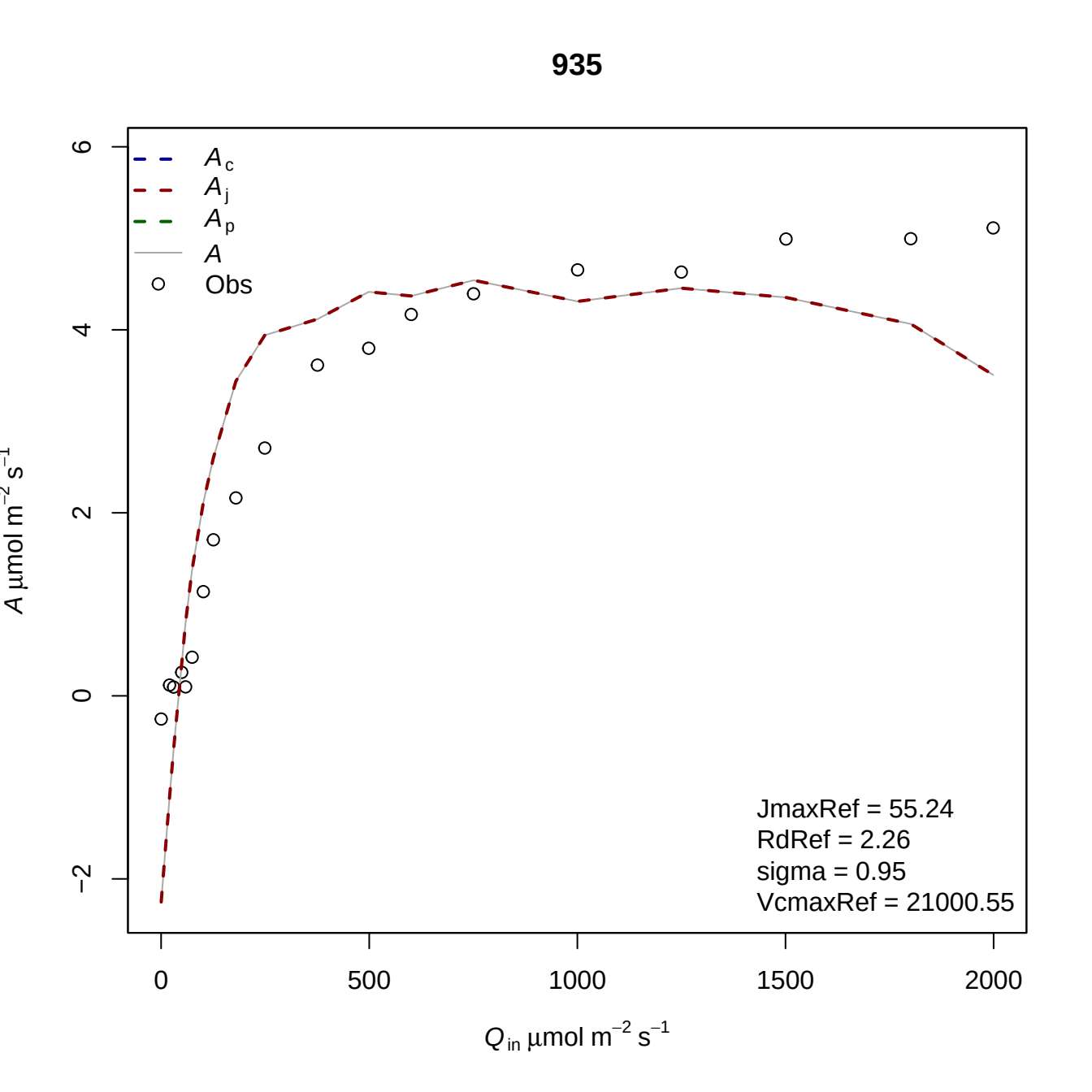

930
A_c
A_j
A_p
A
Obs
JmaxRef = 20.02
RdRef = 0.94
sigma = 0.13
VcmaxRef = 103.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500
2000

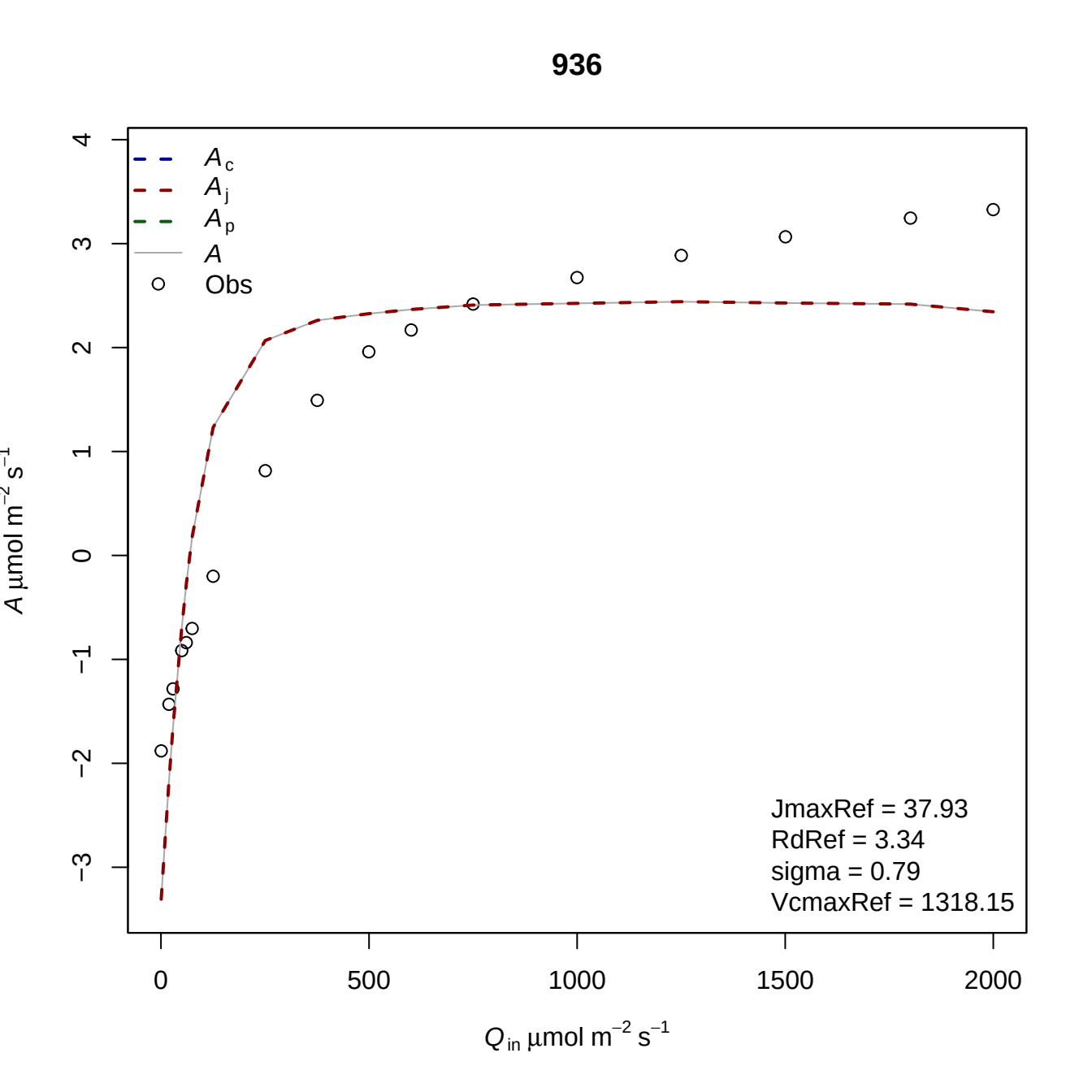

931
A_c
A_j
A_p
A
Obs
JmaxRef = 28.09
RdRef = 1.93
sigma = 0.49
VcmaxRef = 207.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

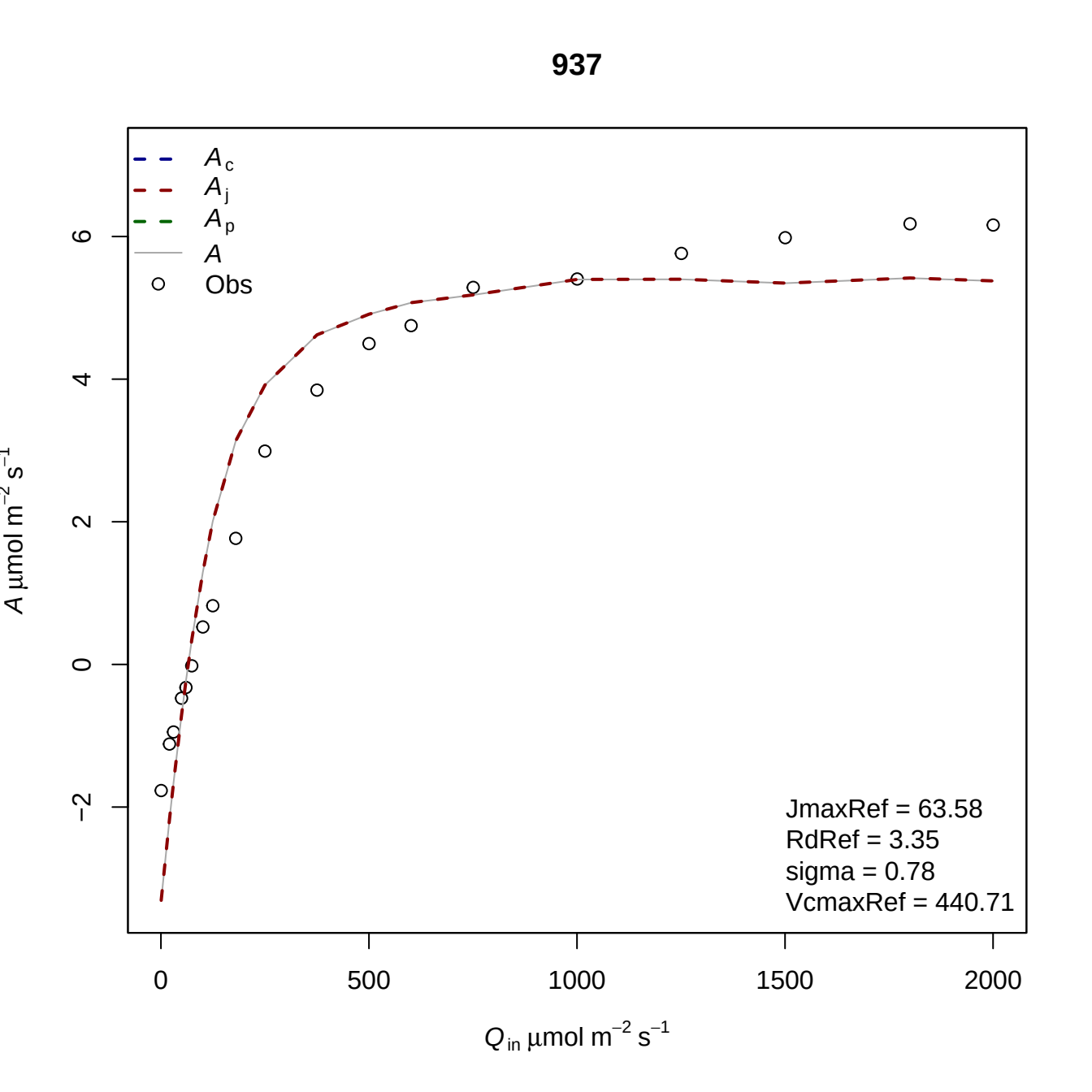

932
A_c
A_j
A_p
A
Obs
JmaxRef = 25.24
RdRef = 1.76
sigma = 0.42
VcmaxRef = 72.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

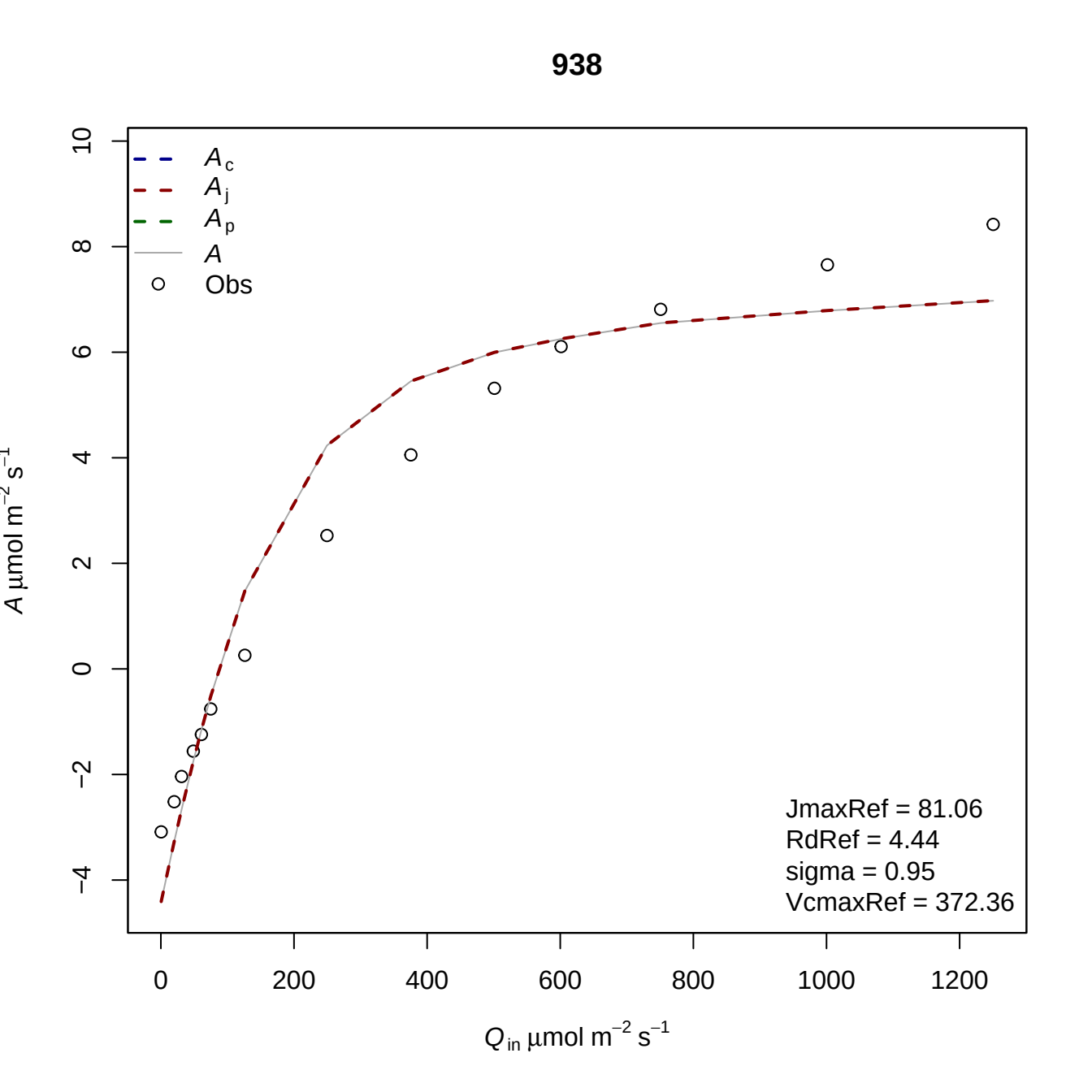

933
A_c
A_j
A_p
A
Obs
JmaxRef = 57.59
RdRef = 1.54
sigma = 0.45
VcmaxRef = 189.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

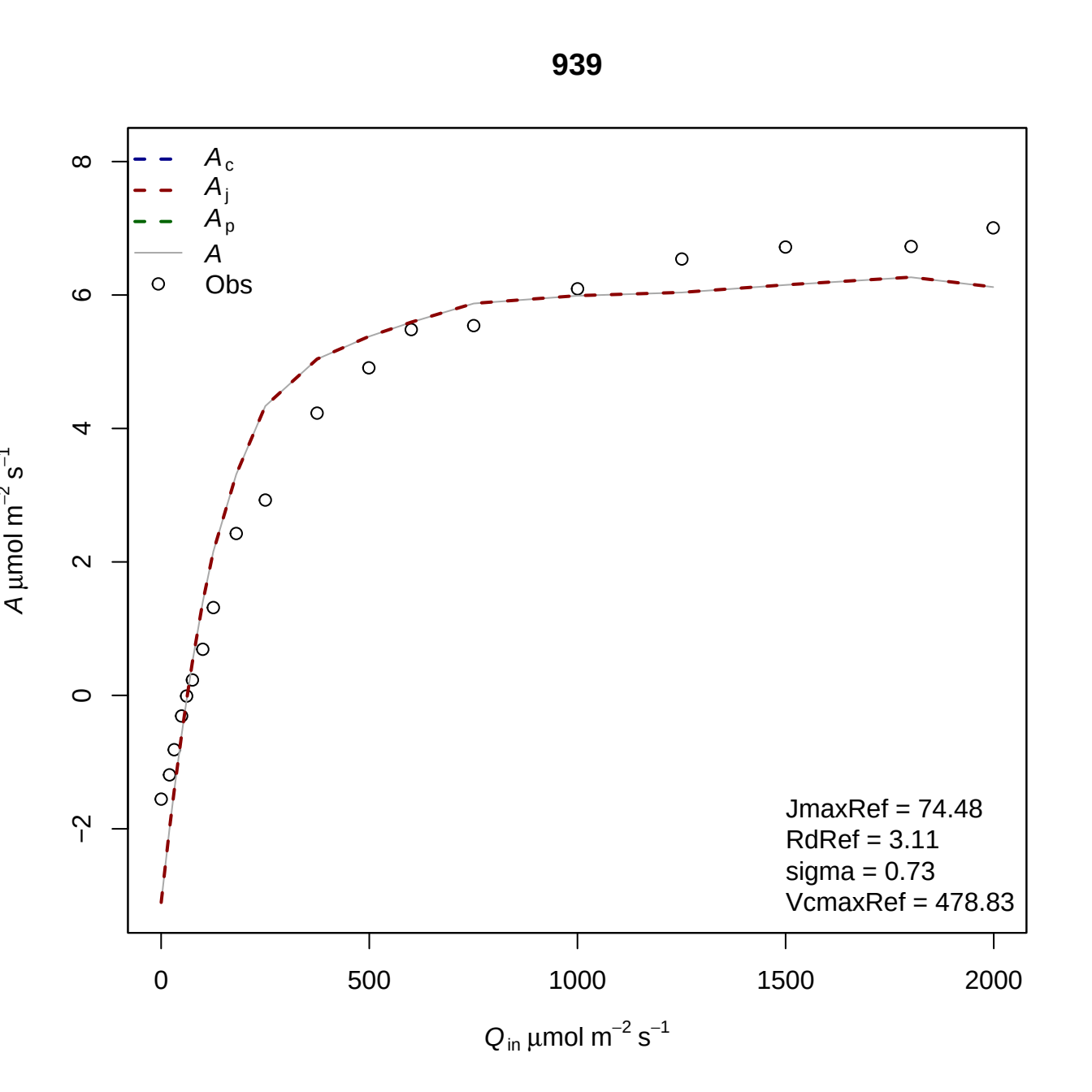

934
A_c
A_j
A_p
A
Obs
JmaxRef = 25.97
RdRef = 1.72
sigma = 0.52
VcmaxRef = 99.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

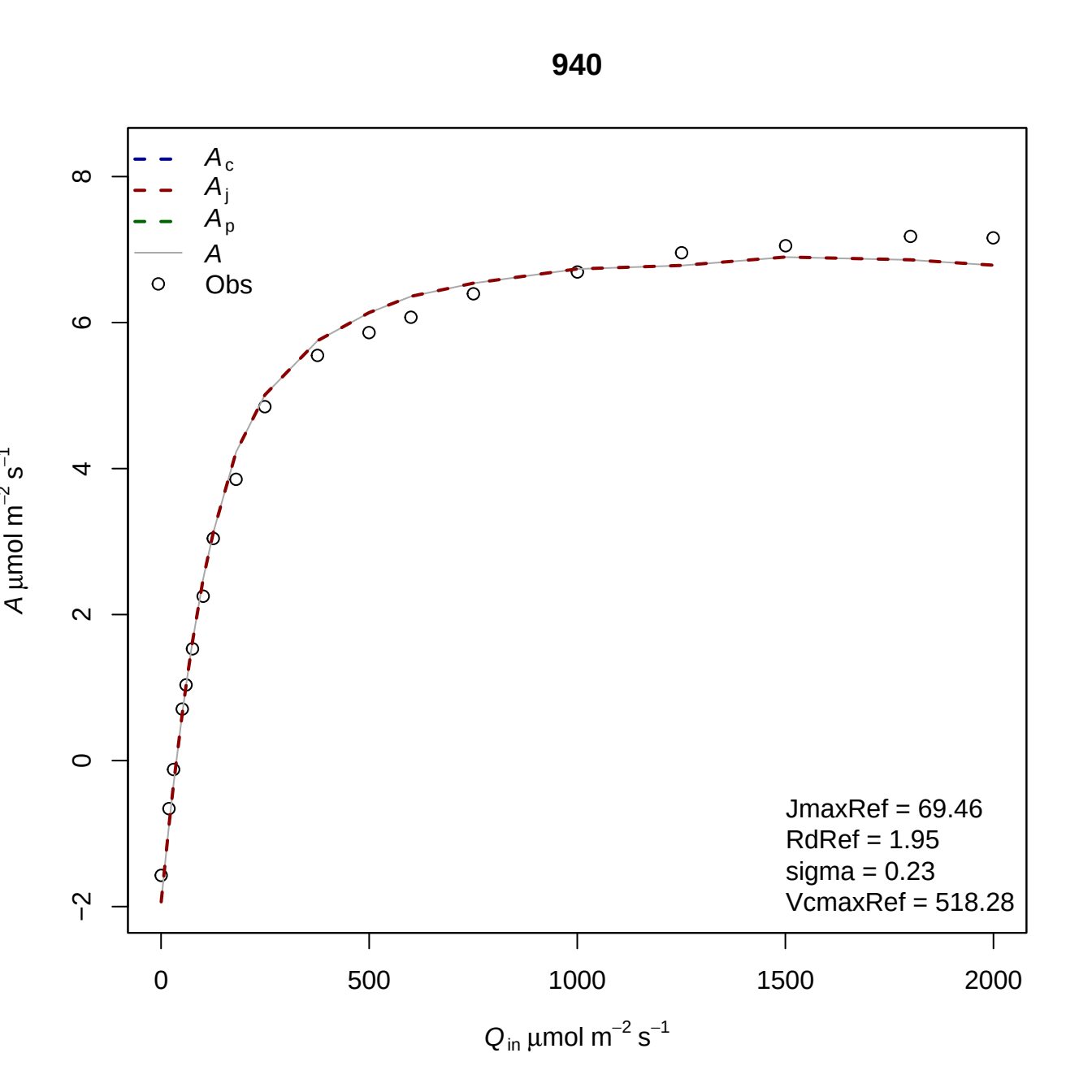

935
A_c
A_j
A_p
A
Obs
JmaxRef = 55.24
RdRef = 2.26
sigma = 0.95
VcmaxRef = 21000.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

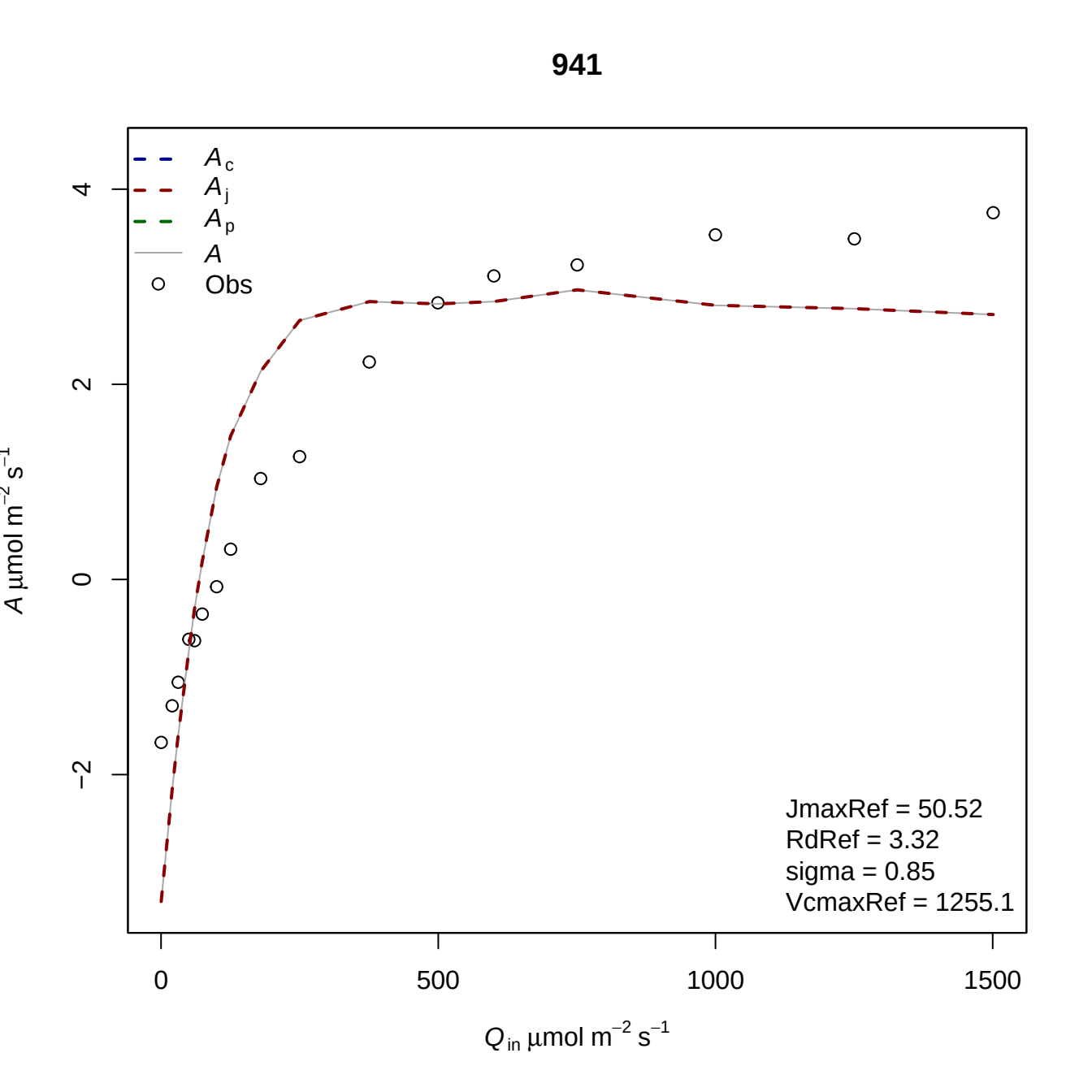

936
A_c
A_j
A_p
A
Obs
JmaxRef = 37.93
RdRef = 3.34
sigma = 0.79
VcmaxRef = 1318.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500
2000

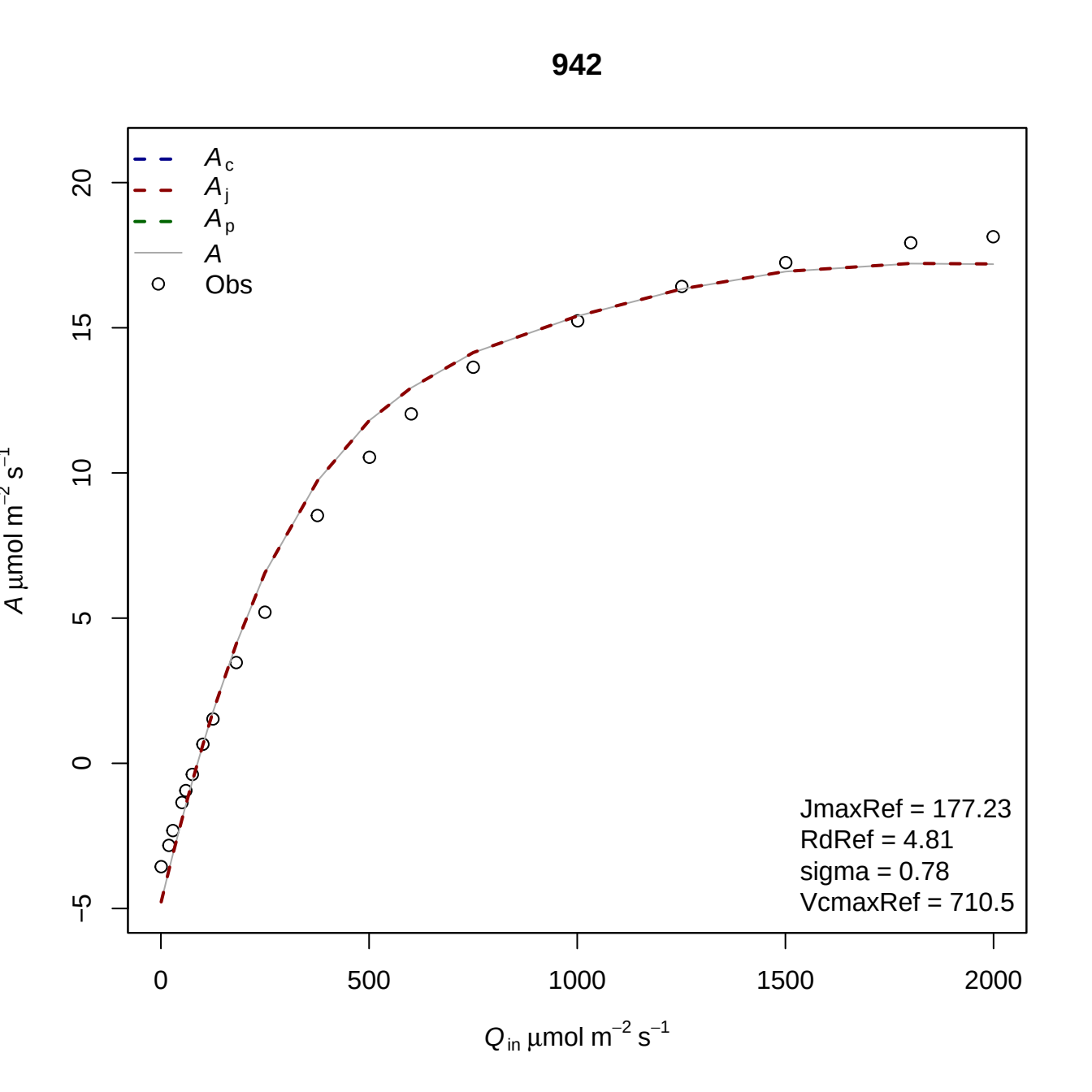

937
A_c
A_j
A_p
A
Obs
JmaxRef = 63.58
RdRef = 3.35
sigma = 0.78
VcmaxRef = 440.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

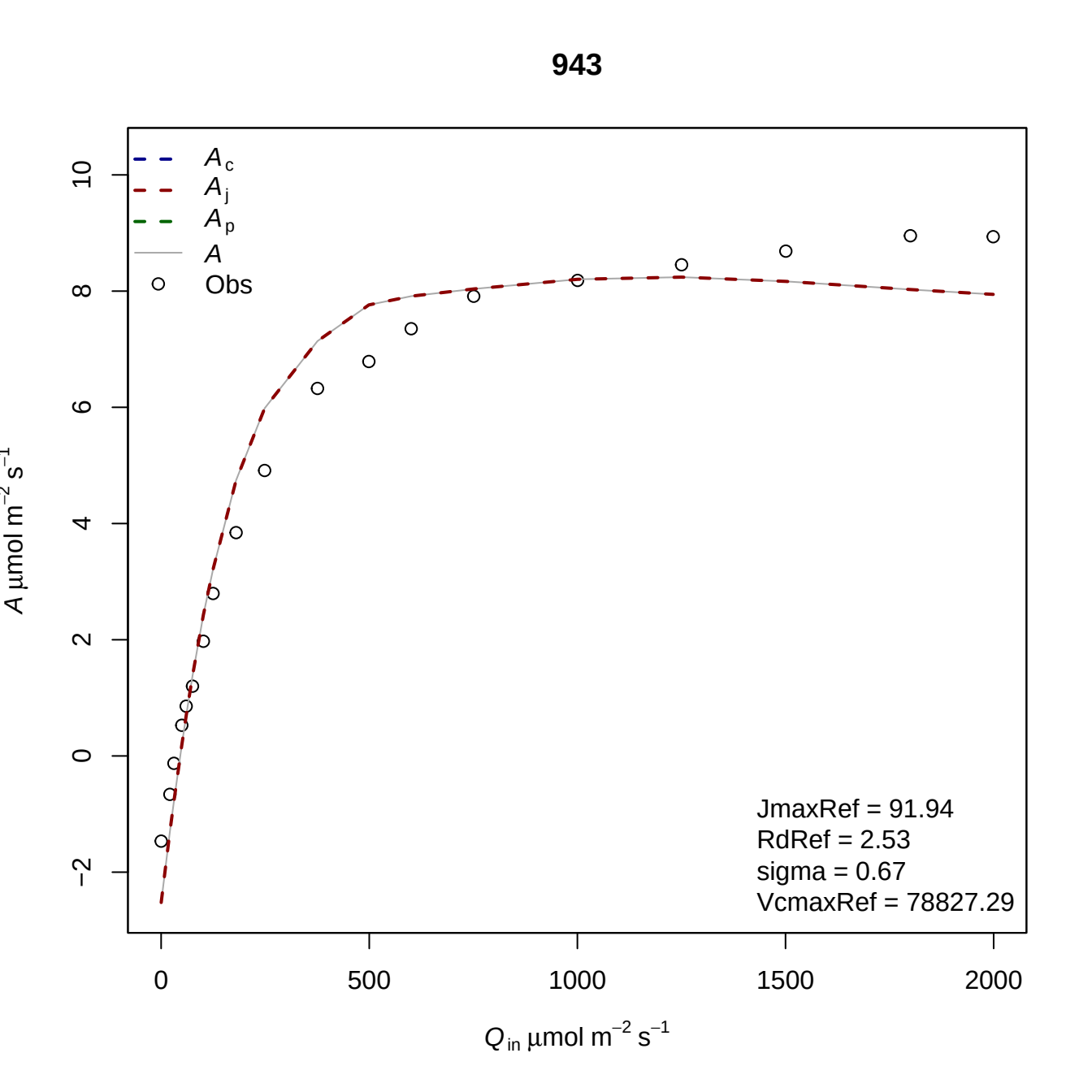

938
A_c
A_j
A_p
A
Obs
JmaxRef = 81.06
RdRef = 4.44
sigma = 0.95
VcmaxRef = 372.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

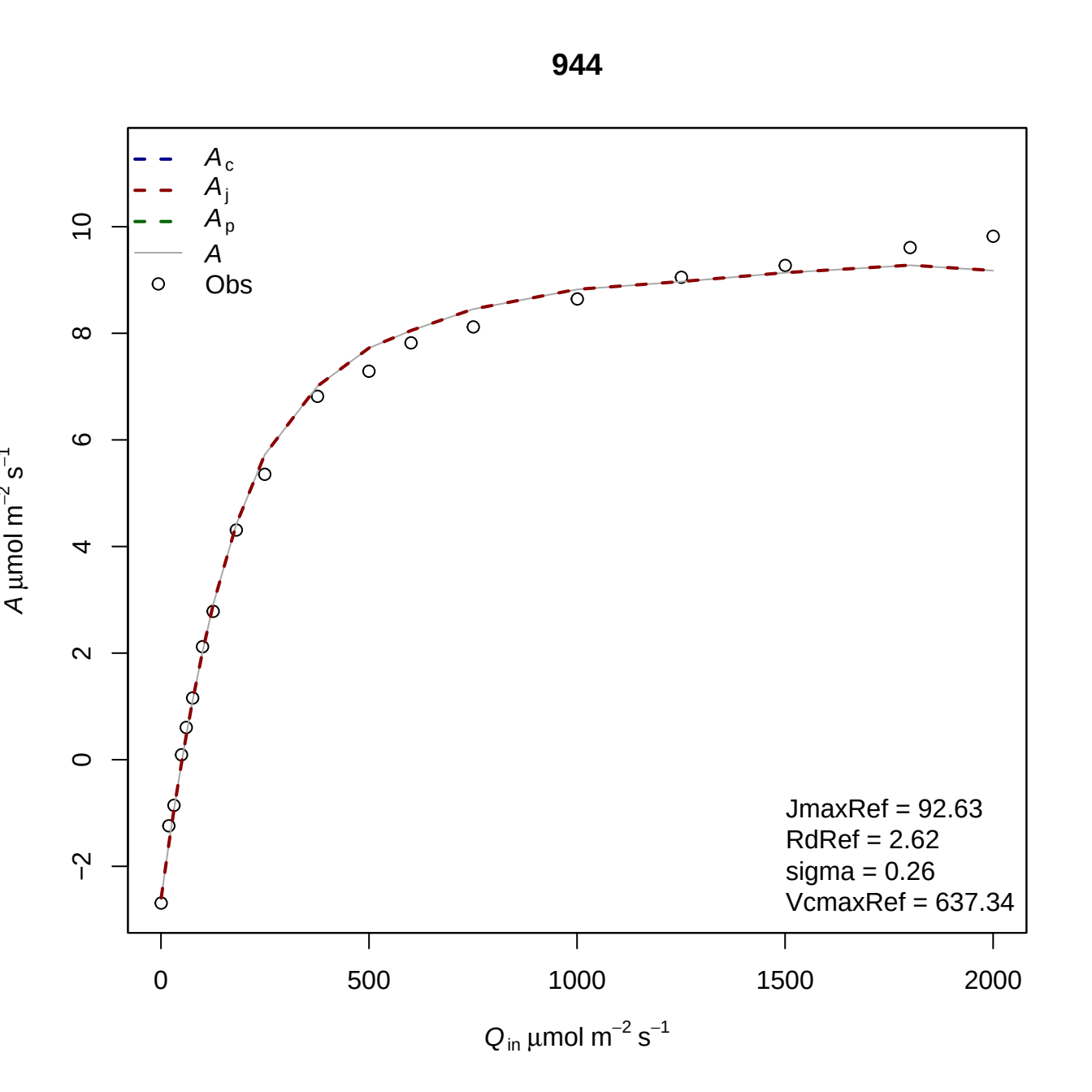

939
A_c
A_j
A_p
A
Obs
JmaxRef = 74.48
RdRef = 3.11
sigma = 0.73
VcmaxRef = 478.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

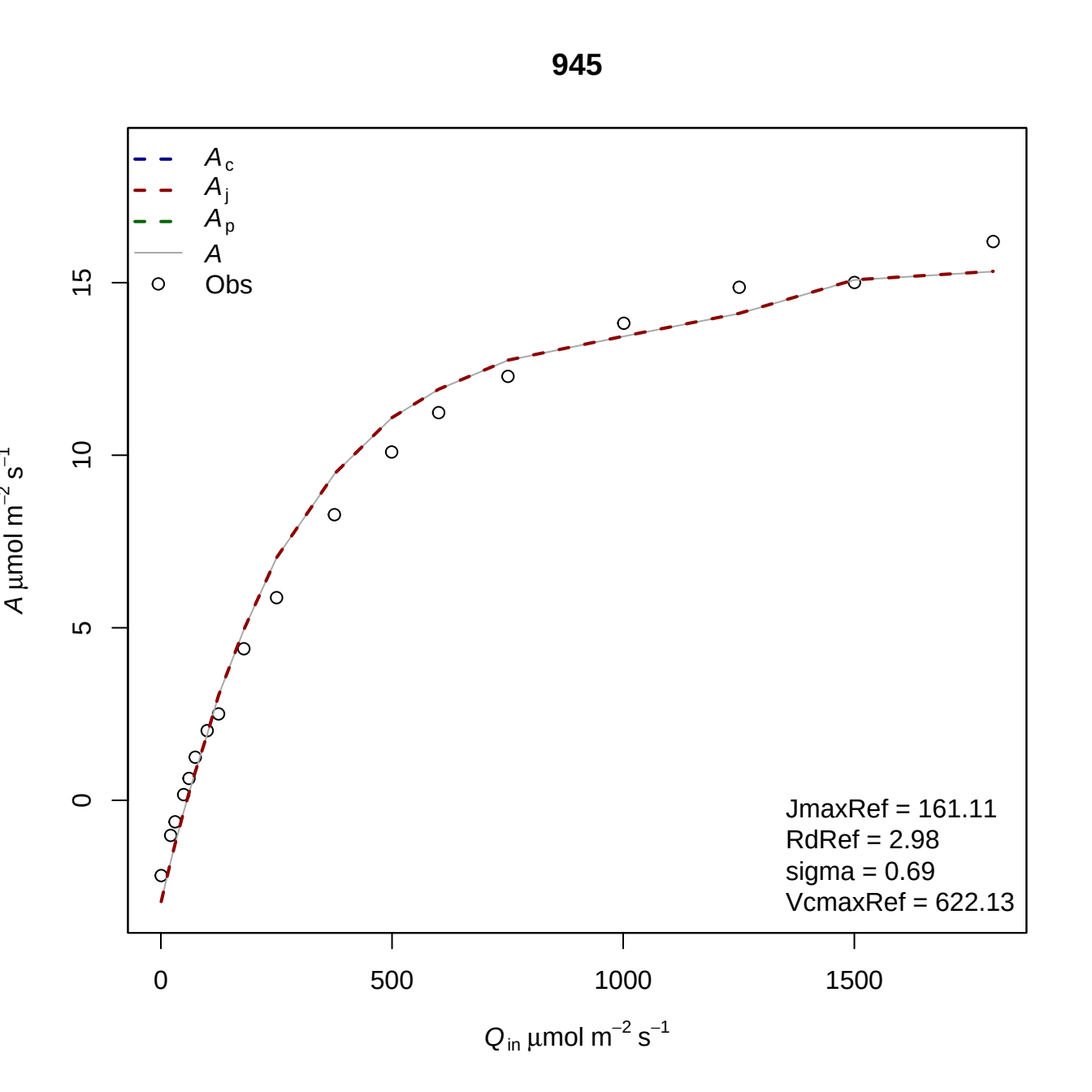

940
A_c
A_j
A_p
A
Obs
JmaxRef = 69.46
RdRef = 1.95
sigma = 0.23
VcmaxRef = 518.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

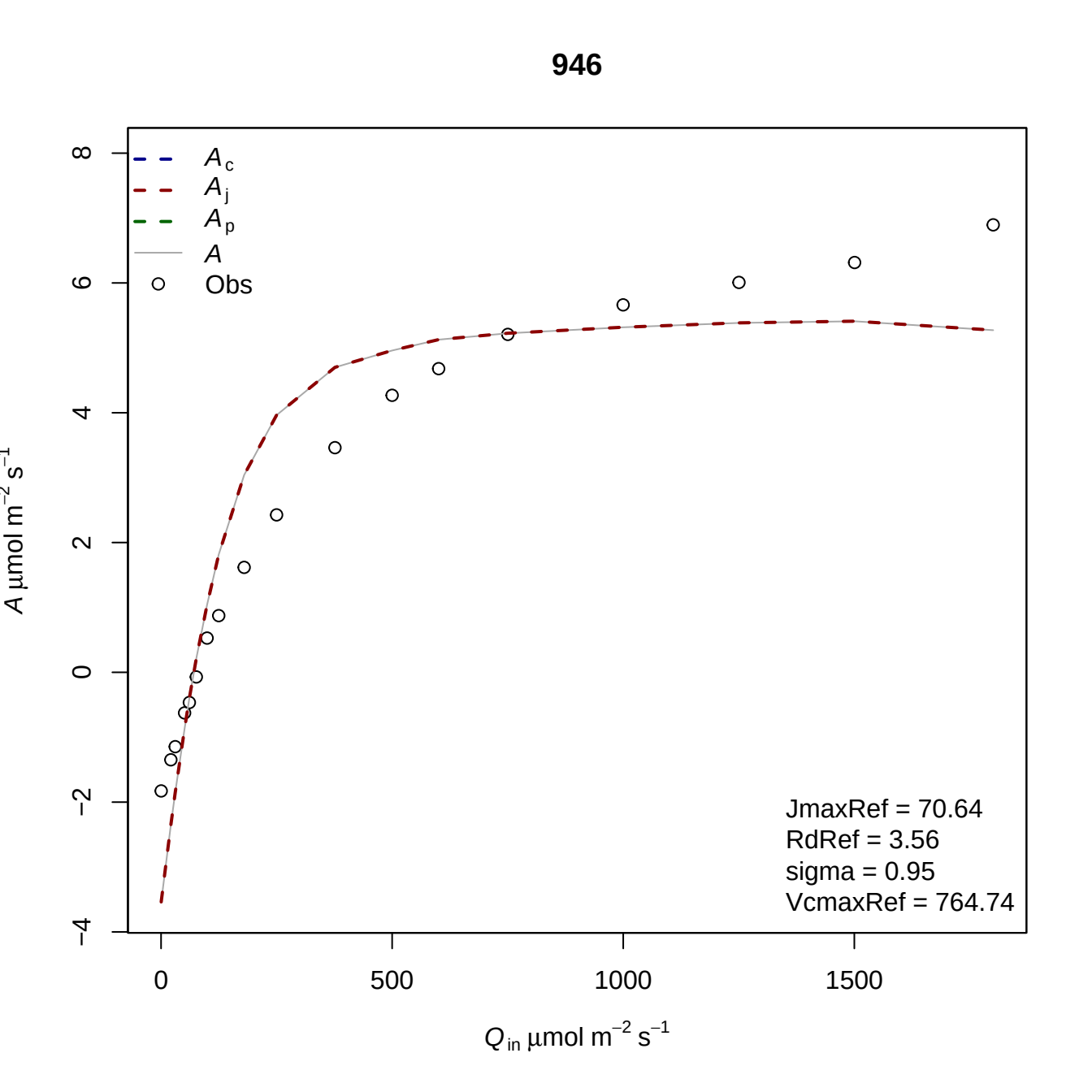

941
A_c
A_j
A_p
A
Obs
JmaxRef = 50.52
RdRef = 3.32
sigma = 0.85
VcmaxRef = 1255.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
0
500
1000
1500

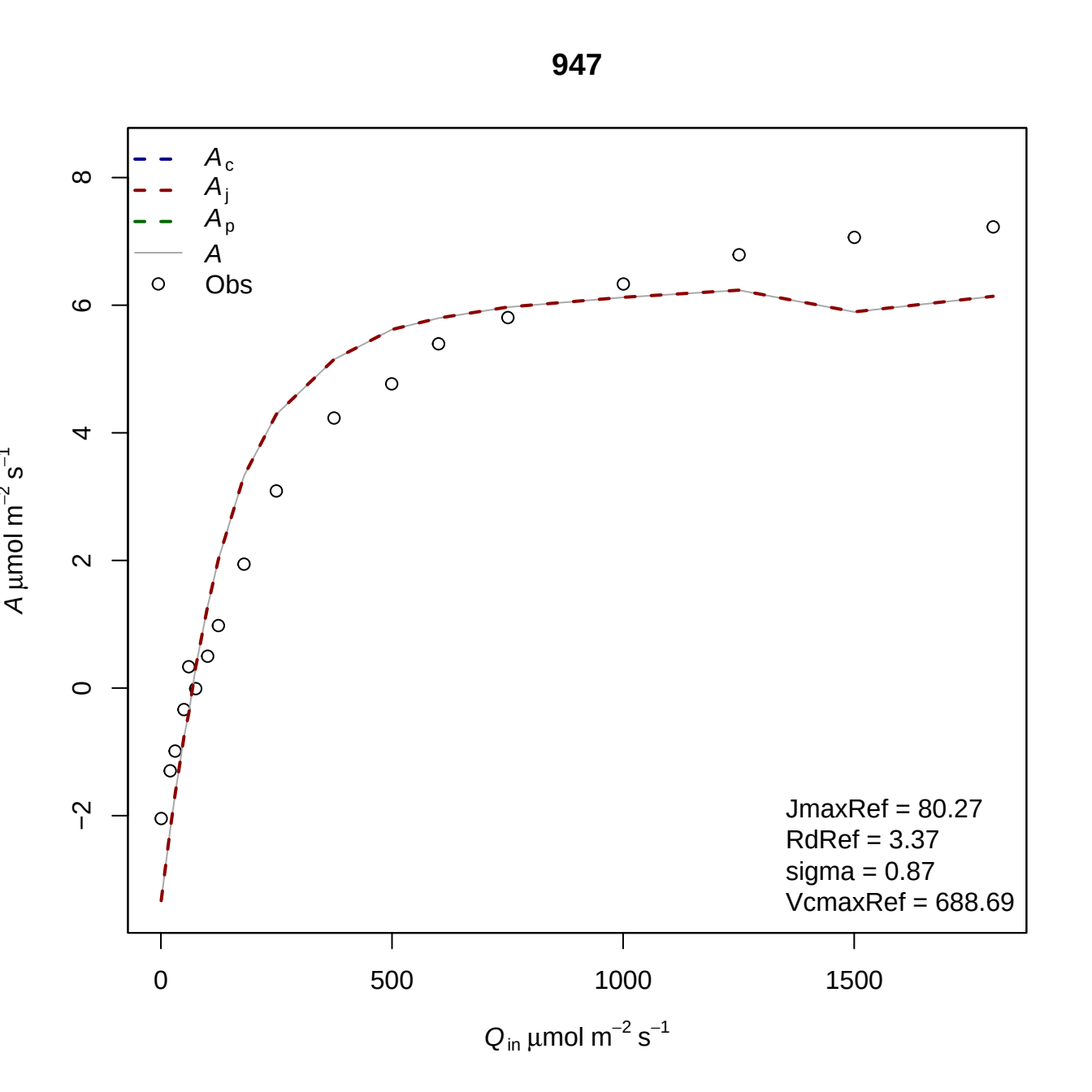

942
A_c
A_j
A_p
A
Obs
JmaxRef = 177.23
RdRef = 4.81
sigma = 0.78
VcmaxRef = 710.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

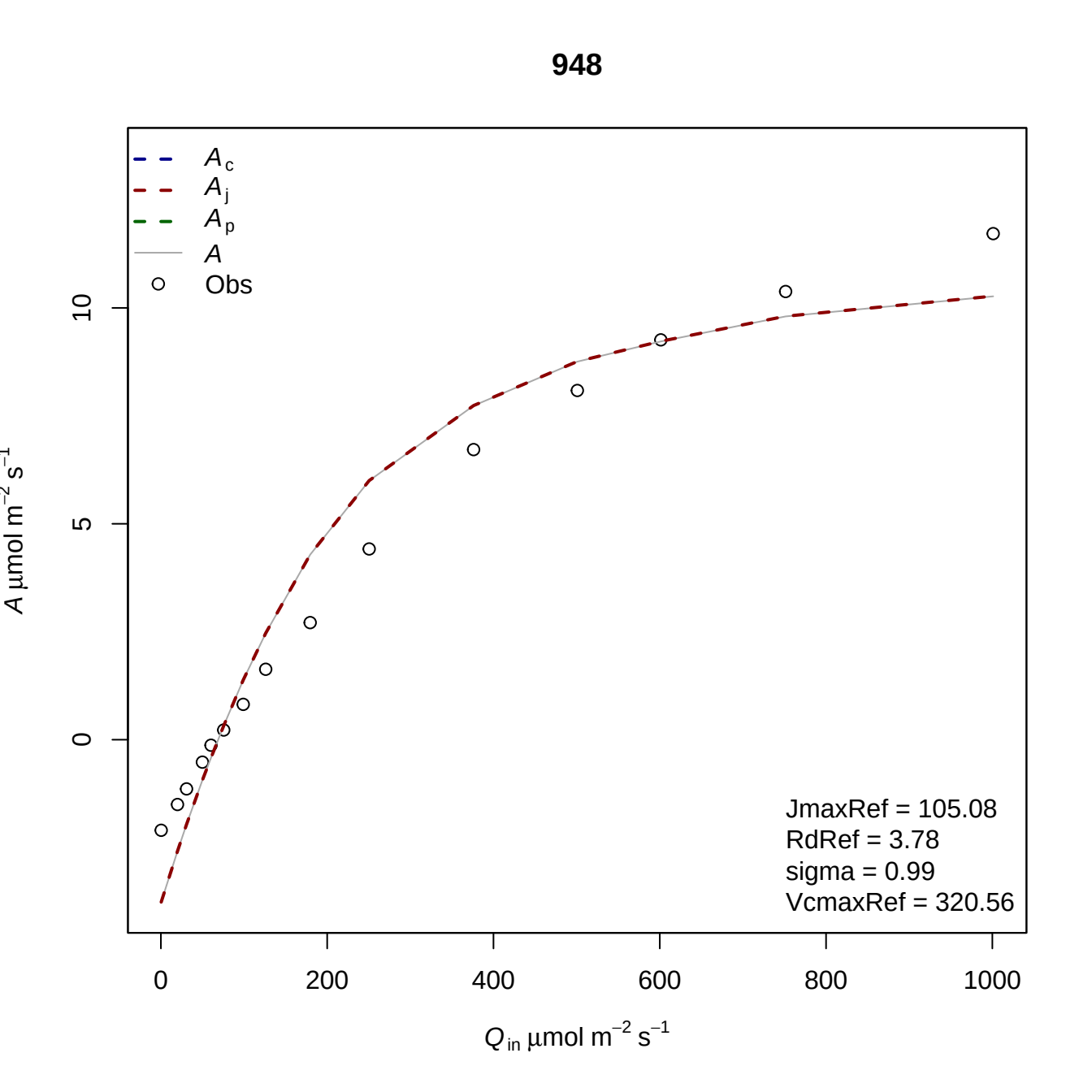

943
A_c
A_j
A_p
A
Obs
JmaxRef = 91.94
RdRef = 2.53
sigma = 0.67
VcmaxRef = 78827.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

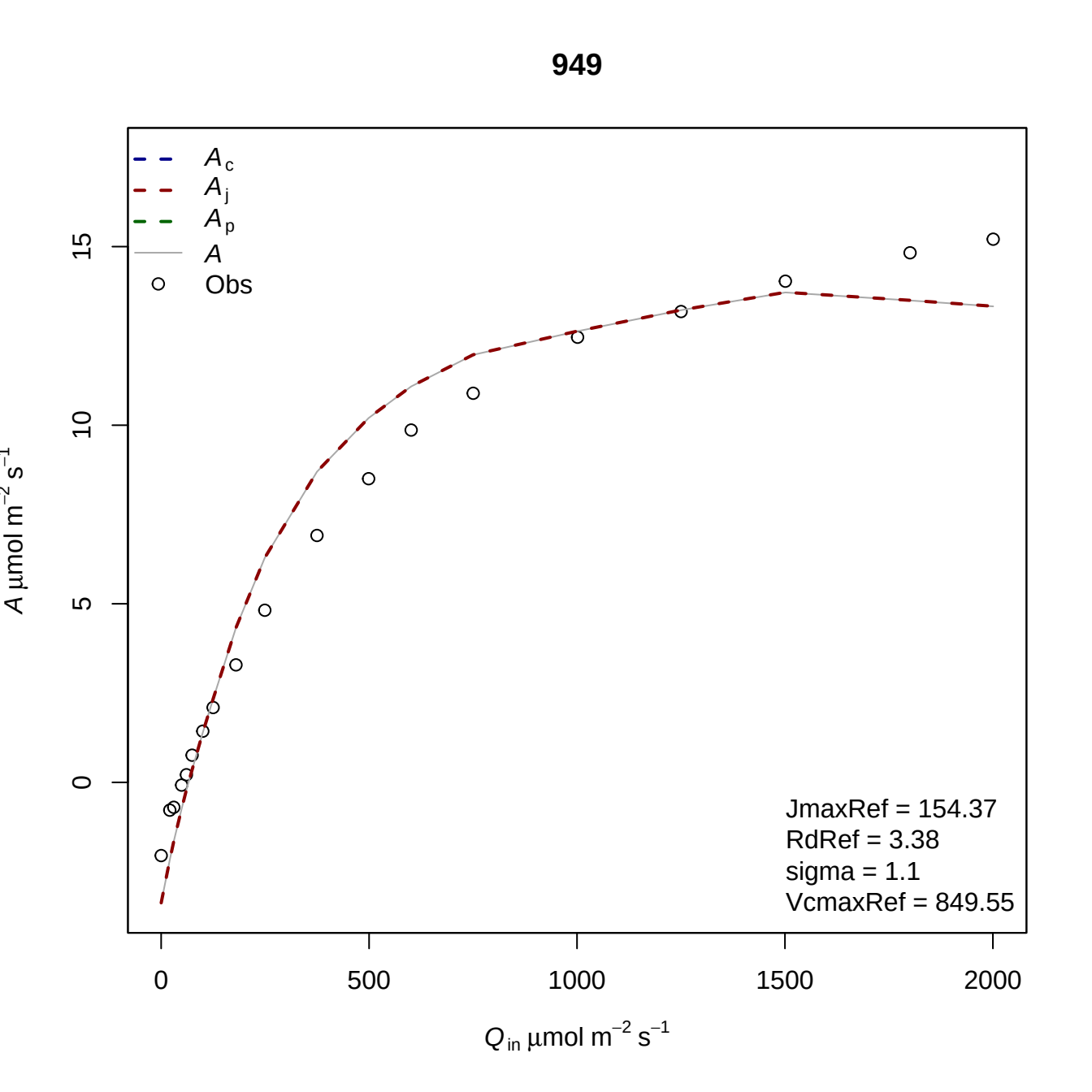

944
A_c
A_j
A_p
A
Obs
JmaxRef = 92.63
RdRef = 2.62
sigma = 0.26
VcmaxRef = 637.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

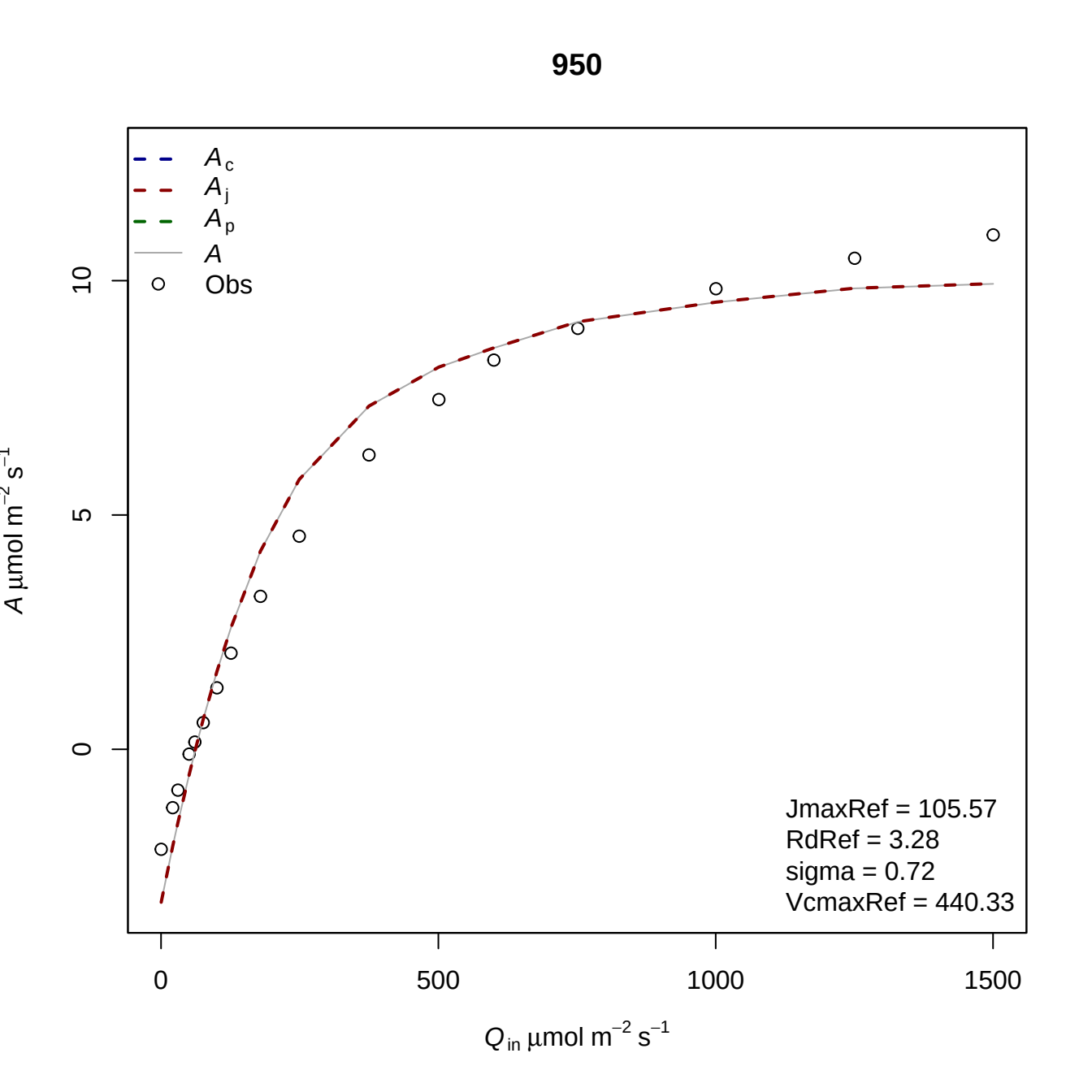

945
A_c
A_j
A_p
A
Obs
JmaxRef = 161.11
RdRef = 2.98
sigma = 0.69
VcmaxRef = 622.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

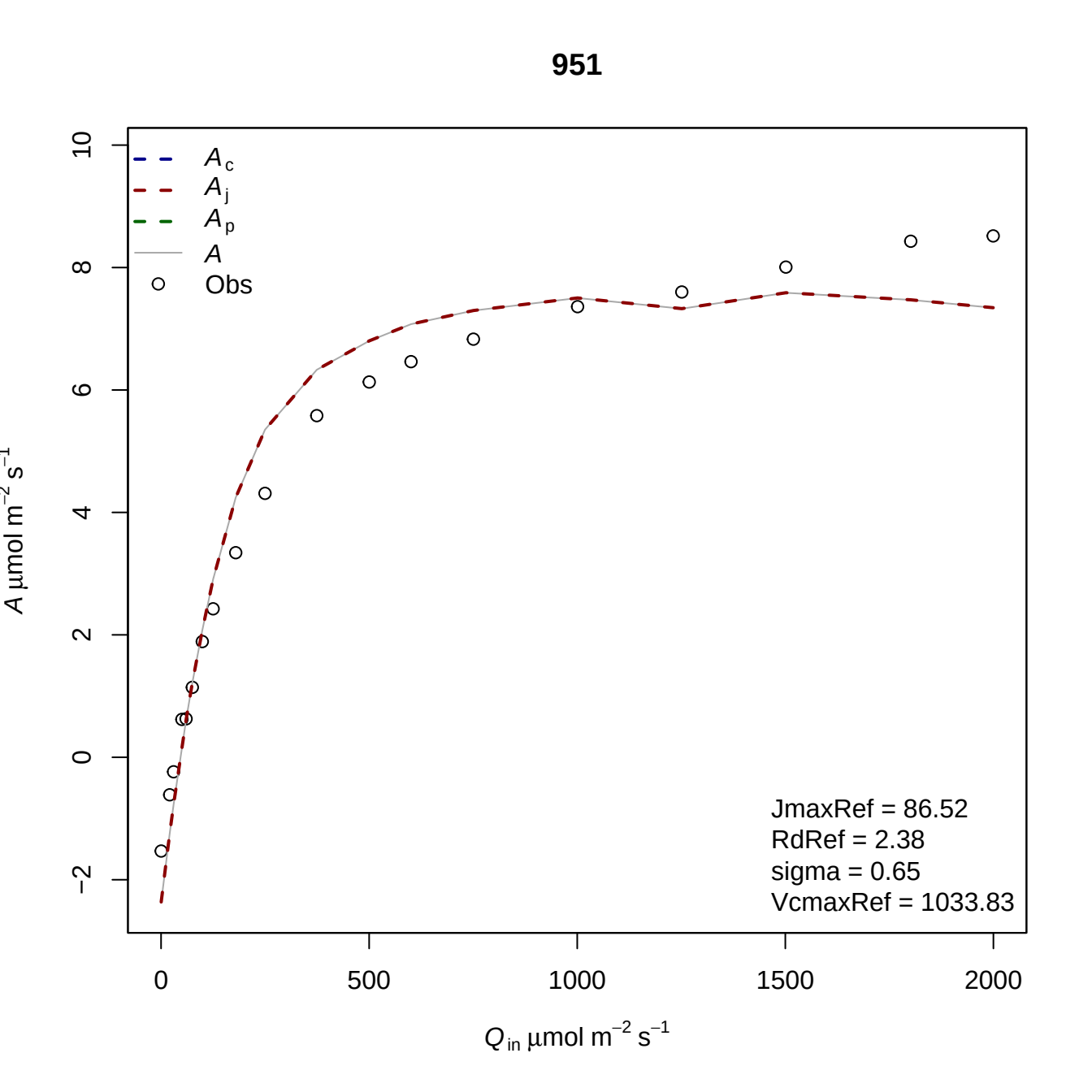

946
A_c
A_j
A_p
A
Obs
JmaxRef = 70.64
RdRef = 3.56
sigma = 0.95
VcmaxRef = 764.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

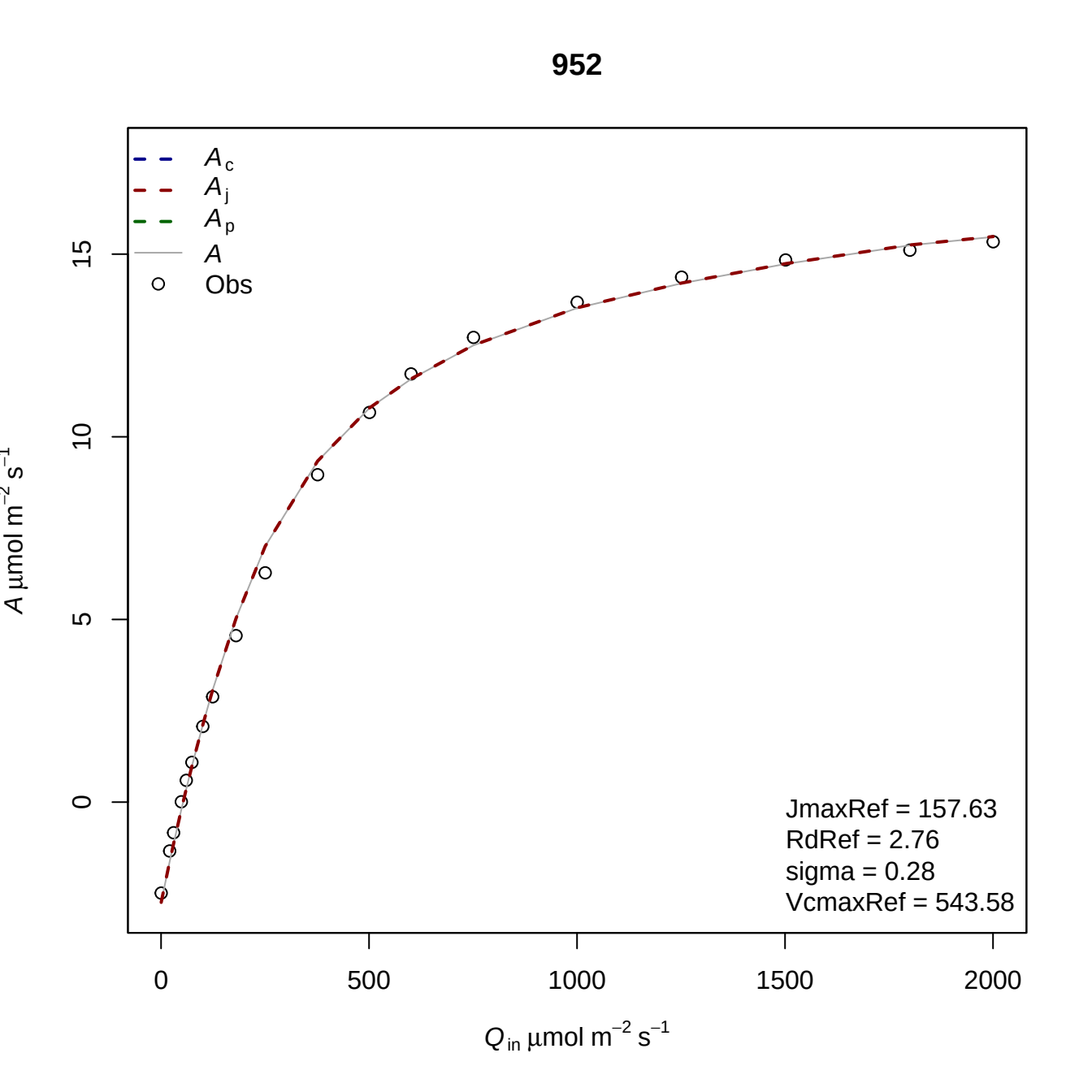

947
A_c
A_j
A_p
A
Obs
JmaxRef = 80.27
RdRef = 3.37
sigma = 0.87
VcmaxRef = 688.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

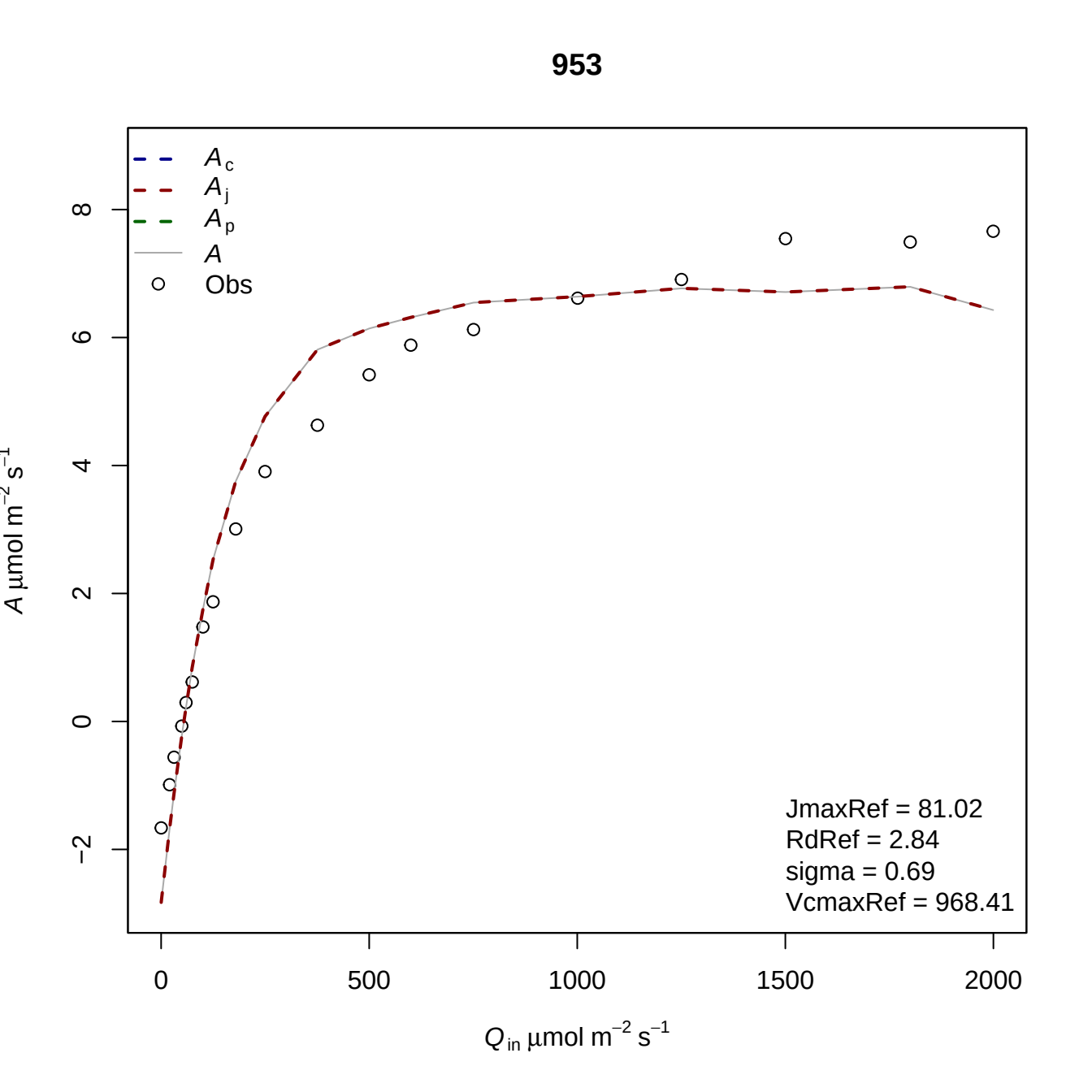

948
A_c
A_j
A_p
A
Obs
JmaxRef = 105.08
RdRef = 3.78
sigma = 0.99
VcmaxRef = 320.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

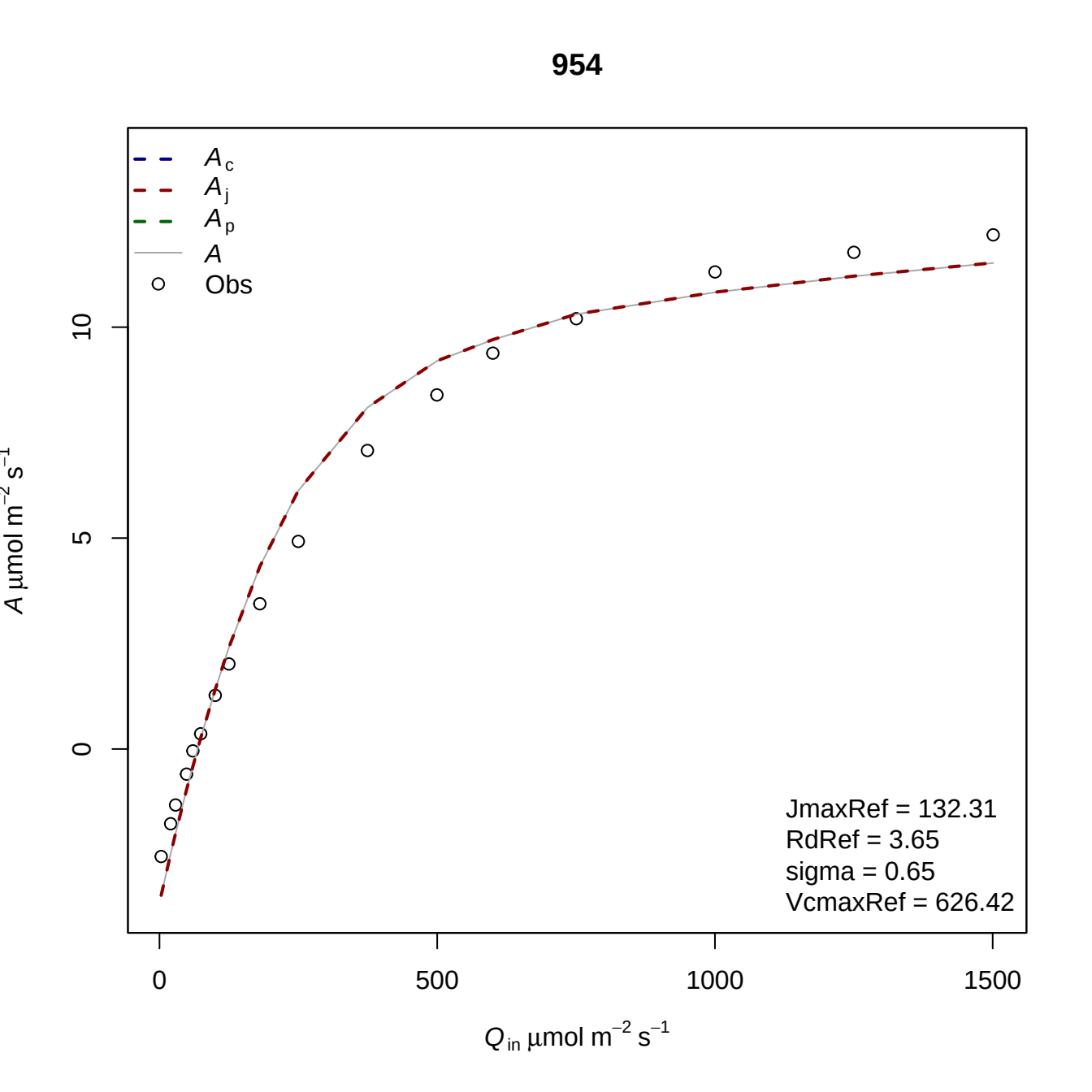

949
A_c
A_j
A_p
A
Obs
JmaxRef = 154.37
RdRef = 3.38
sigma = 1.1
VcmaxRef = 849.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

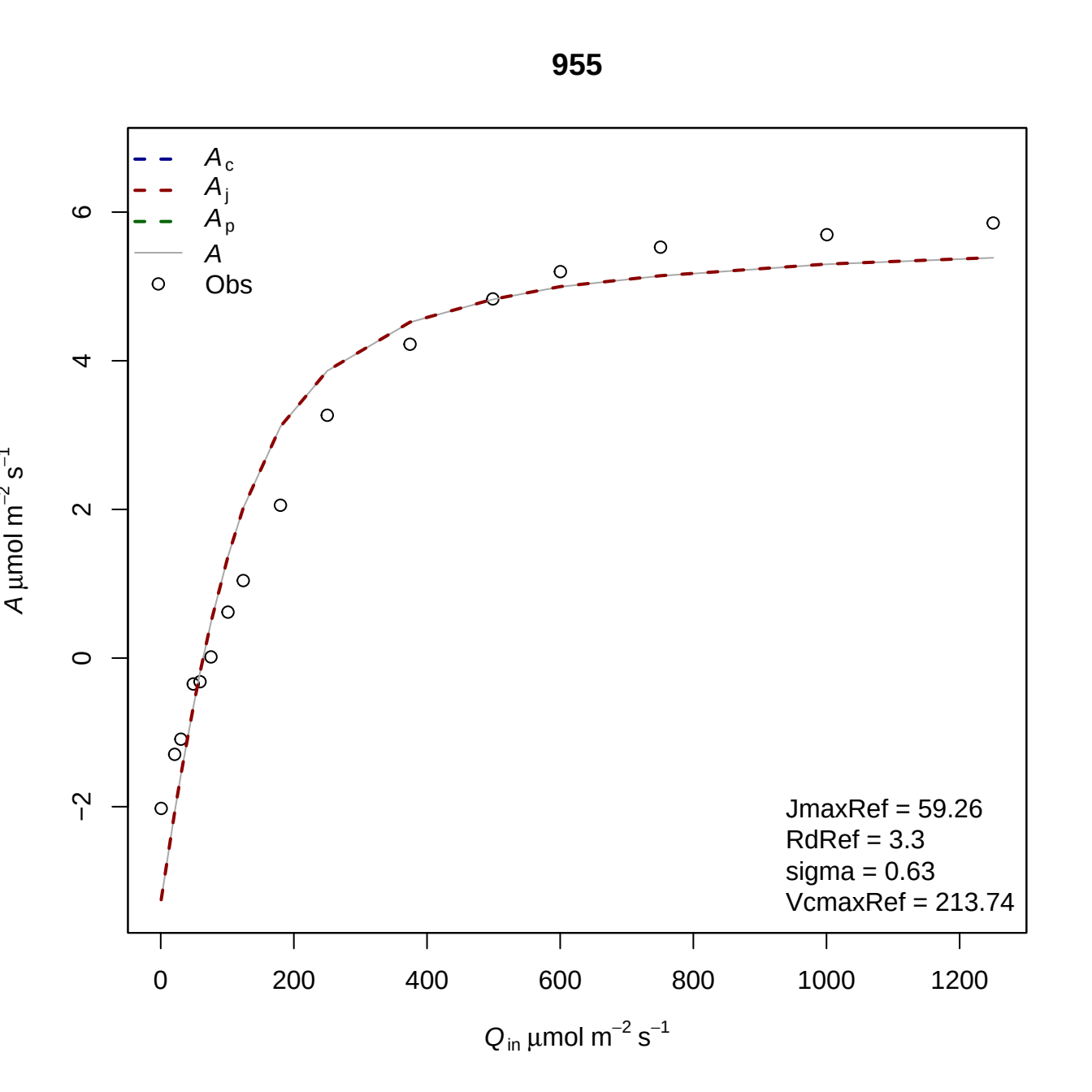

950
A_c
A_j
A_p
A
Obs
JmaxRef = 105.57
RdRef = 3.28
sigma = 0.72
VcmaxRef = 440.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

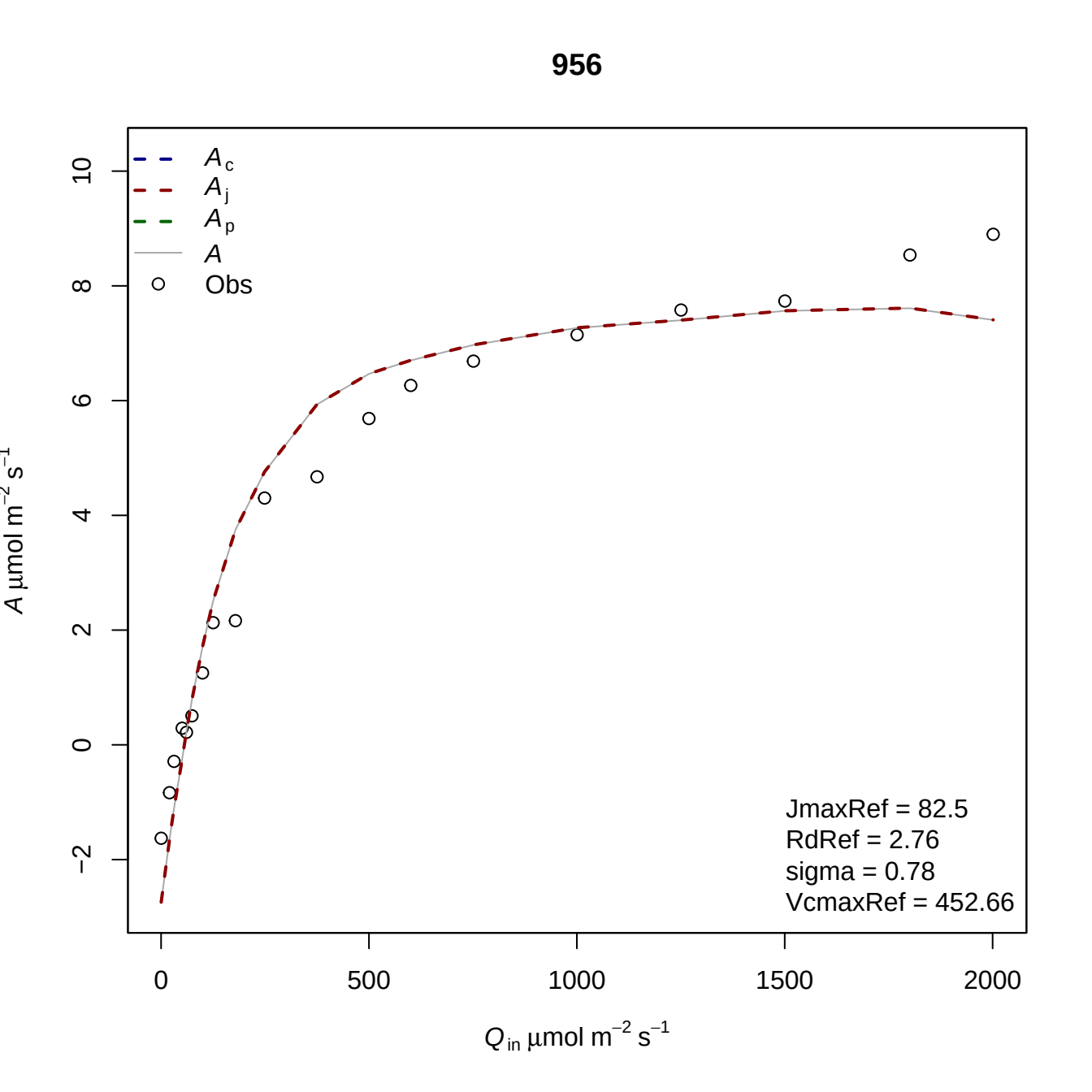

951
A_c
A_j
A_p
A
Obs
JmaxRef = 86.52
RdRef = 2.38
sigma = 0.65
VcmaxRef = 1033.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

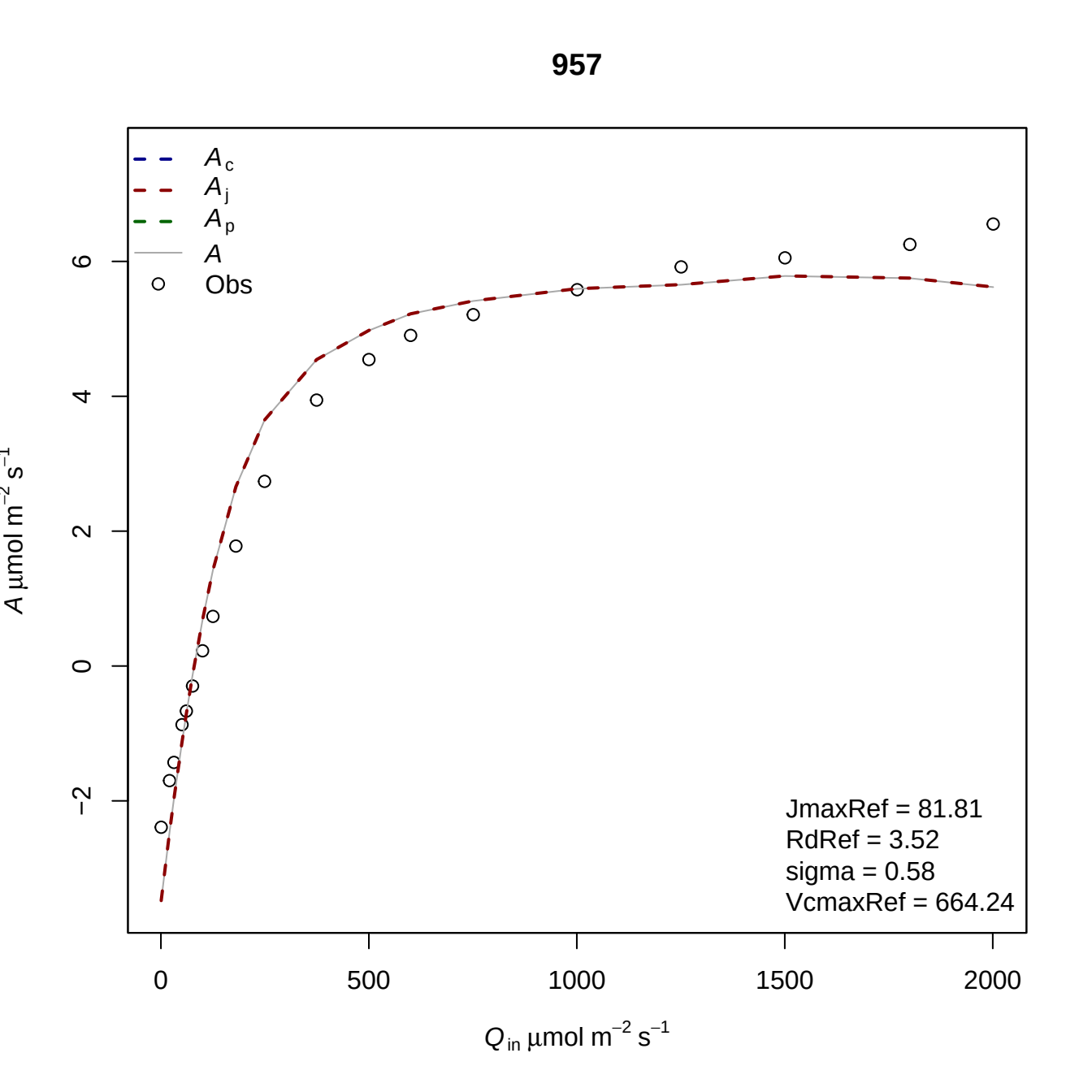

952
A_c
A_j
A_p
A
Obs
JmaxRef = 157.63
RdRef = 2.76
sigma = 0.28
VcmaxRef = 543.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

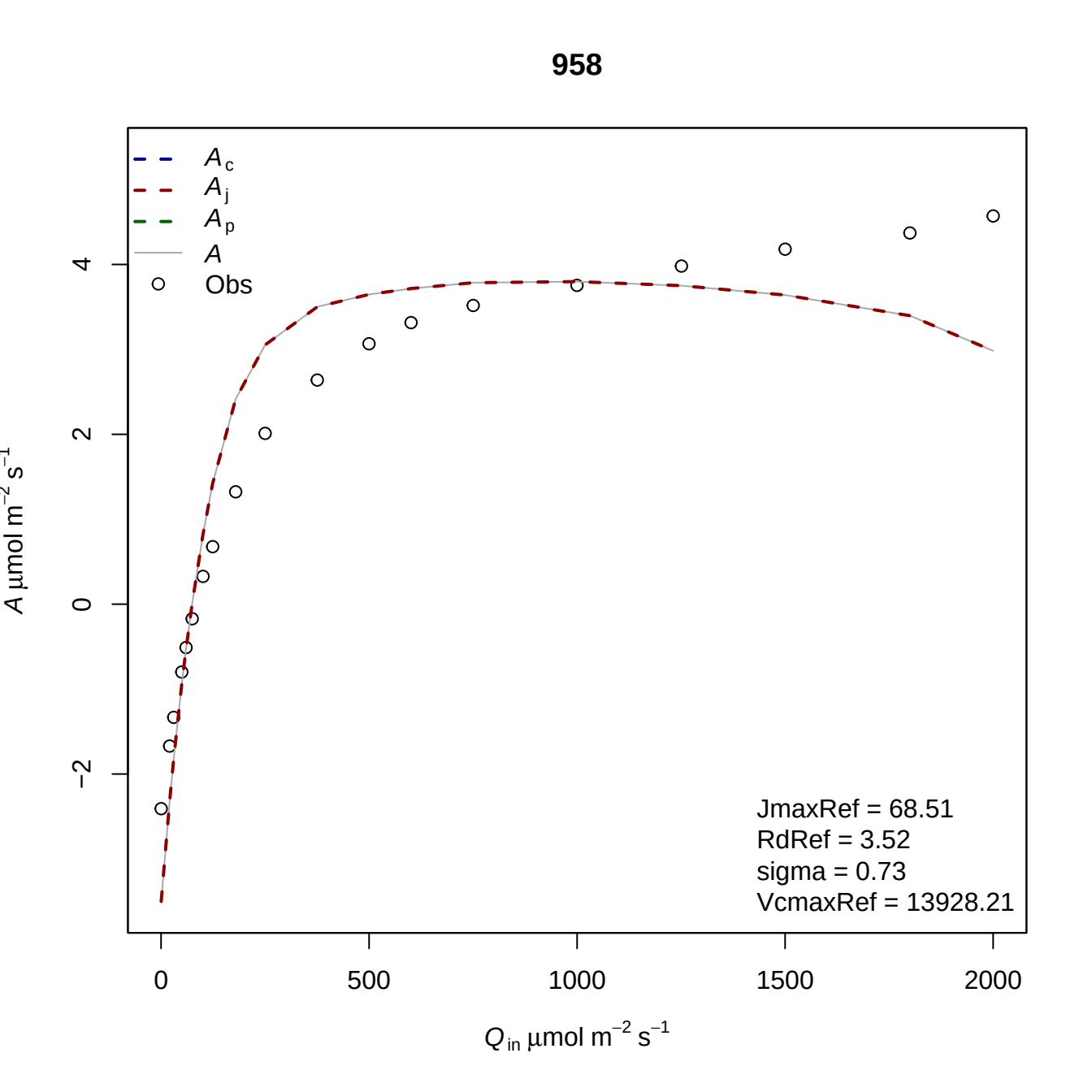

953
A_c
A_j
A_p
A
Obs
JmaxRef = 81.02
RdRef = 2.84
sigma = 0.69
VcmaxRef = 968.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

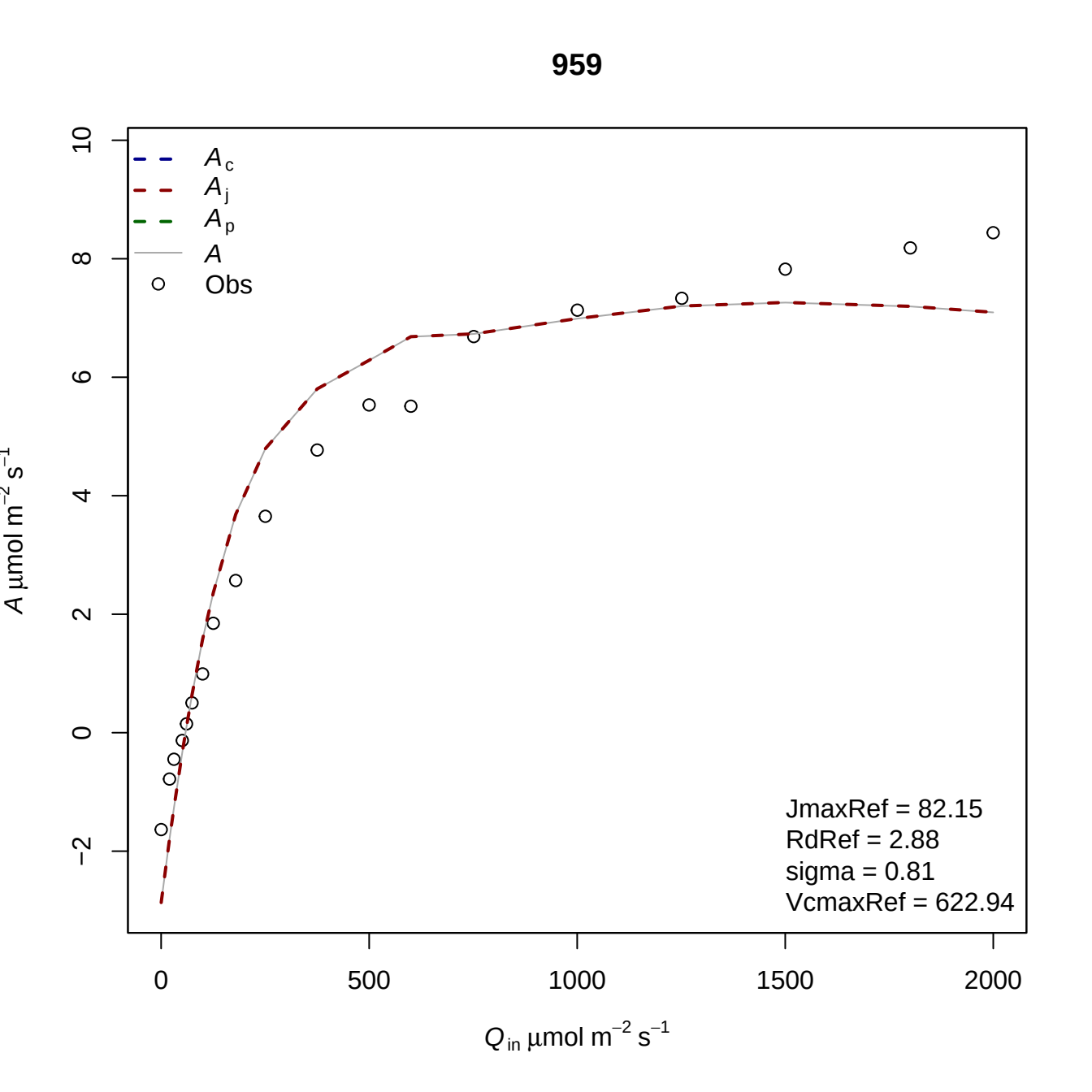

954
A_c
A_j
A_p
A
Obs
JmaxRef = 132.31
RdRef = 3.65
sigma = 0.65
VcmaxRef = 626.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

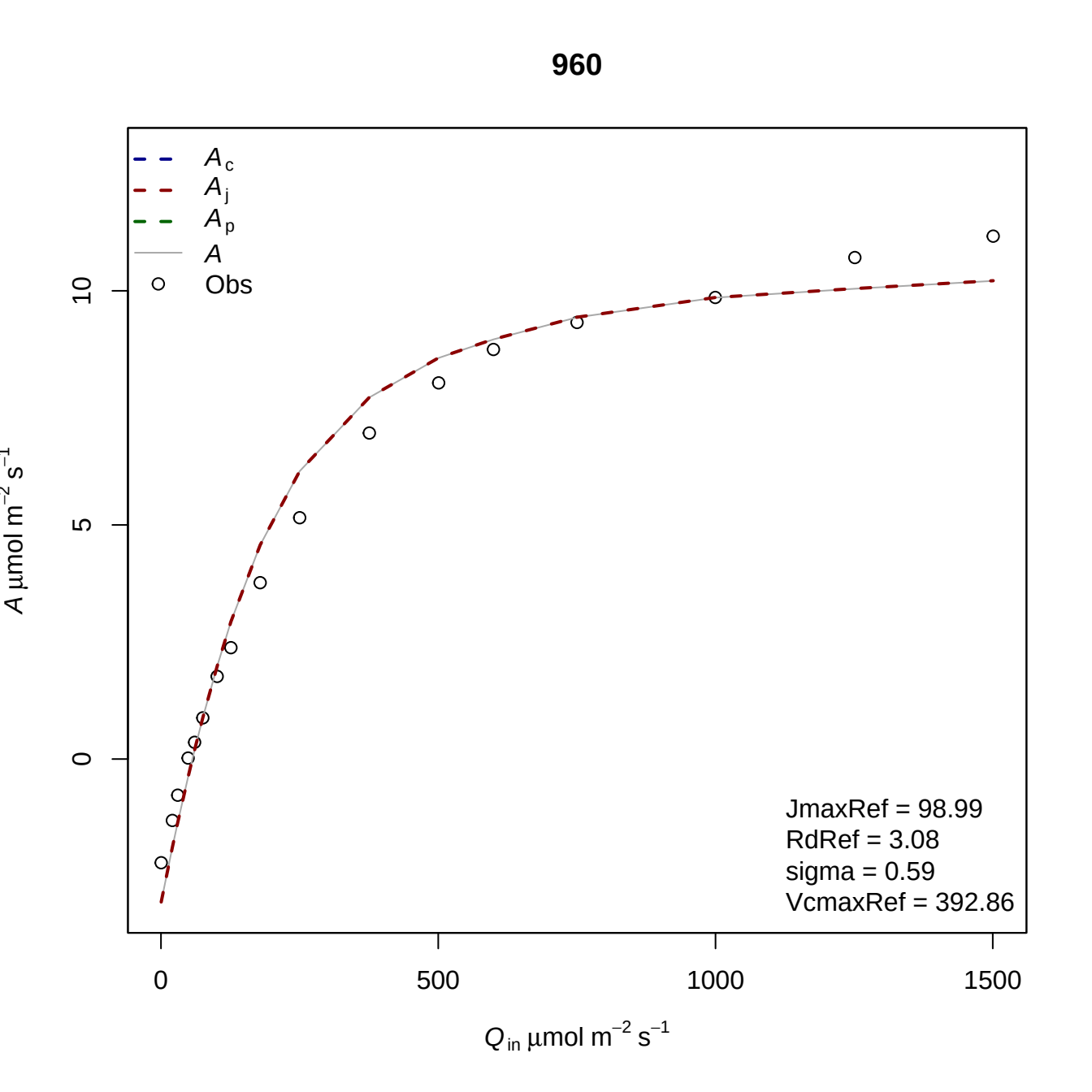

955
A_c
A_j
A_p
A
Obs
JmaxRef = 59.26
RdRef = 3.3
sigma = 0.63
VcmaxRef = 213.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

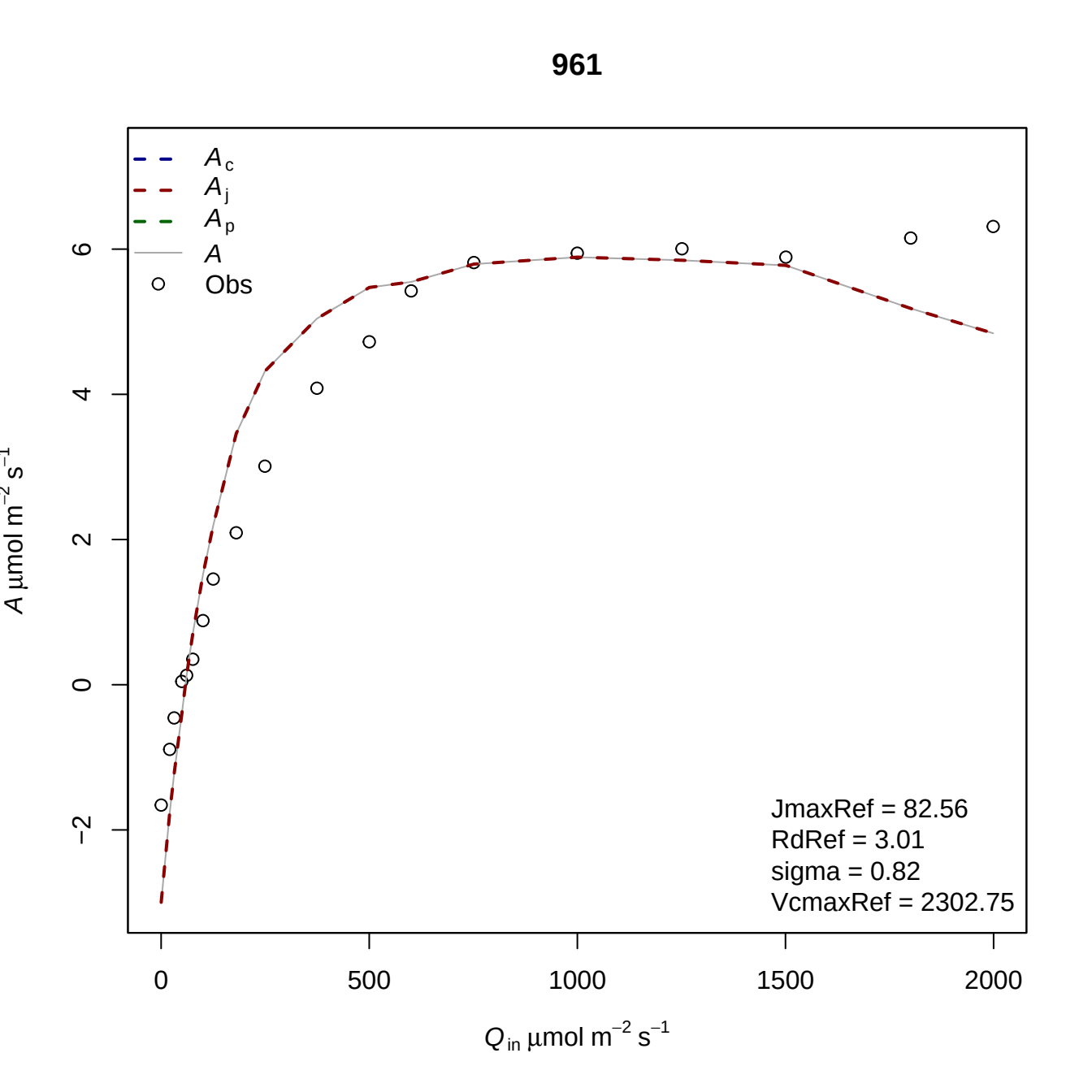

956
A_c
A_j
A_p
A
Obs
JmaxRef = 82.5
RdRef = 2.76
sigma = 0.78
VcmaxRef = 452.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

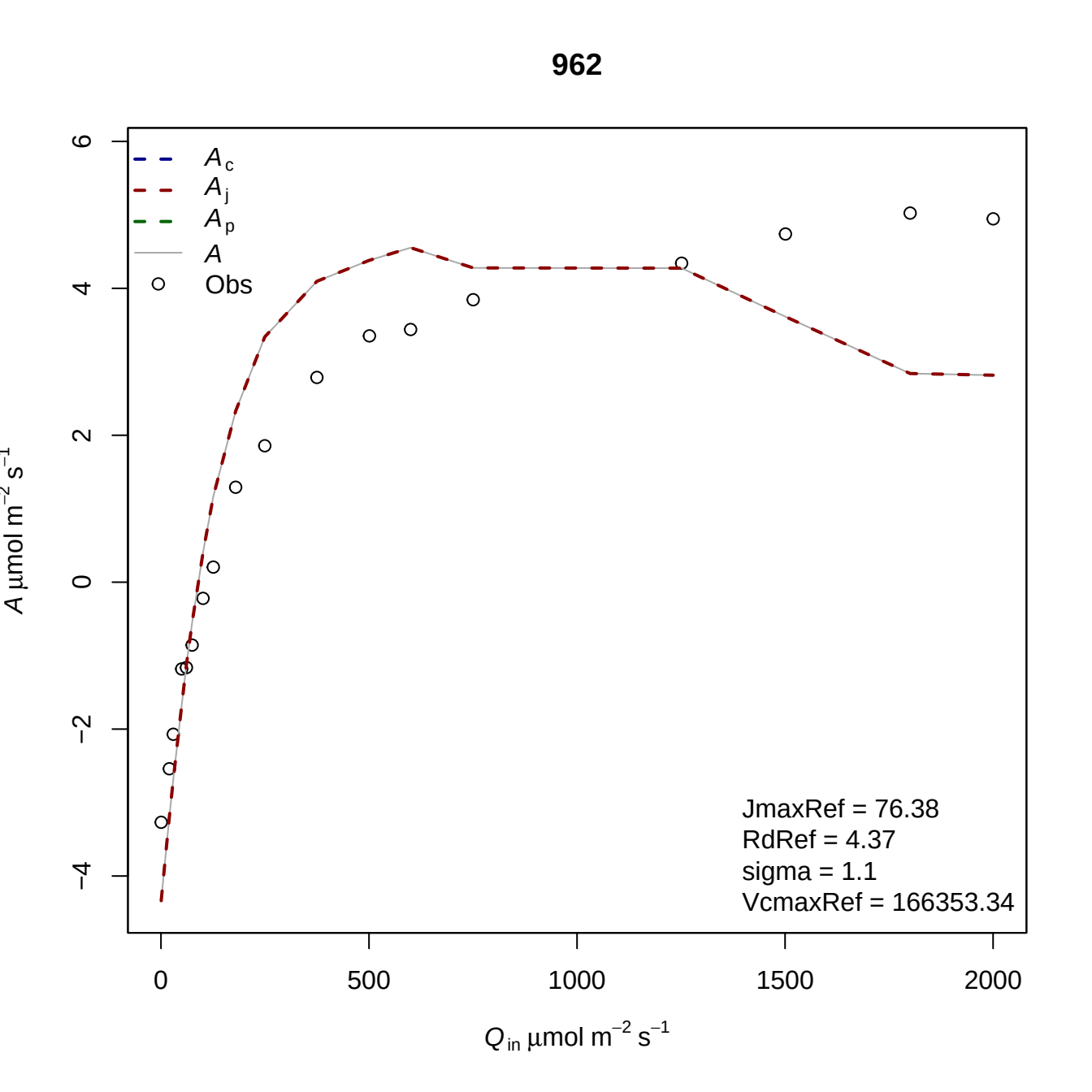

957
A_c
A_j
A_p
A
Obs
JmaxRef = 81.81
RdRef = 3.52
sigma = 0.58
VcmaxRef = 664.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

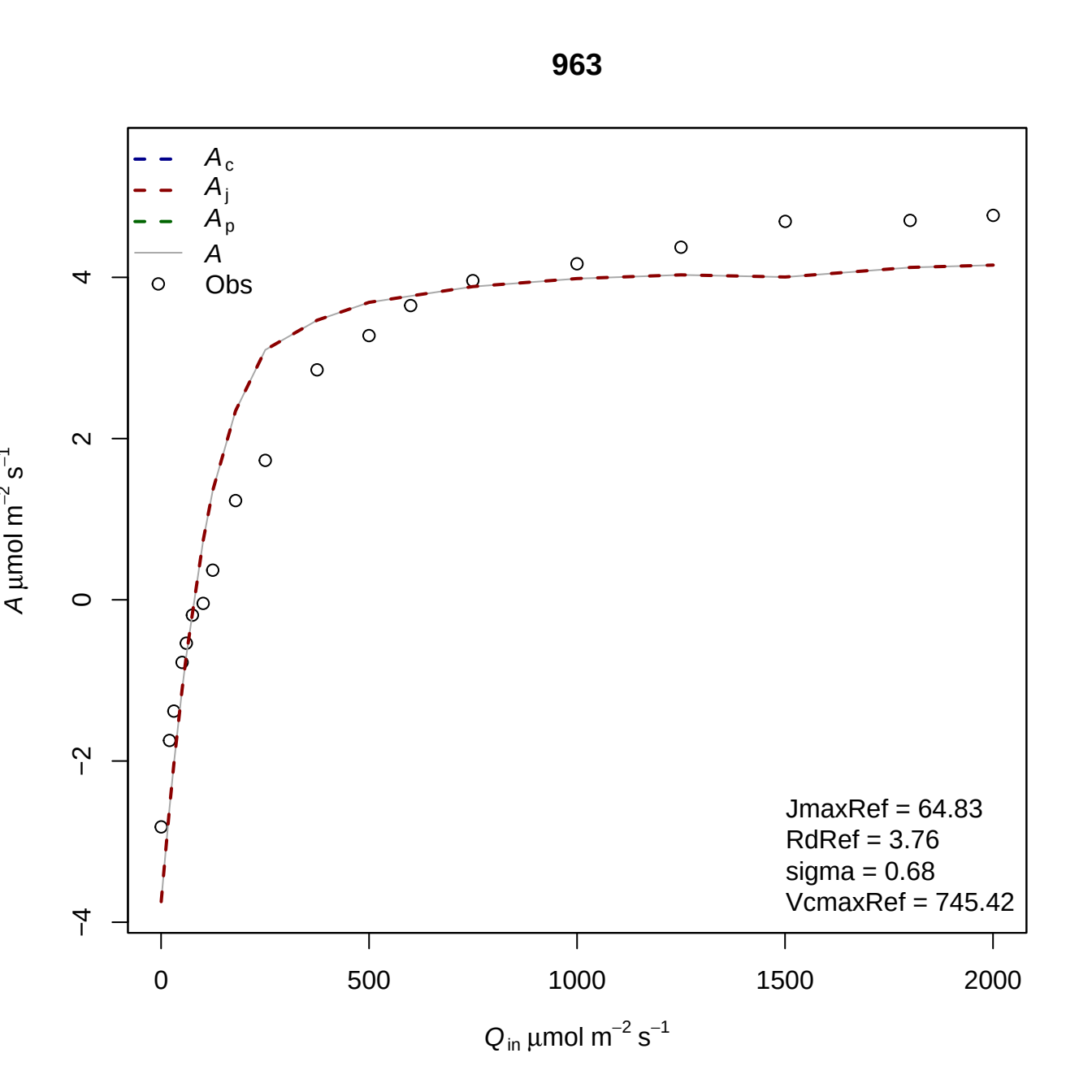

958
A_c
A_j
A_p
A
Obs
JmaxRef = 68.51
RdRef = 3.52
sigma = 0.73
VcmaxRef = 13928.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

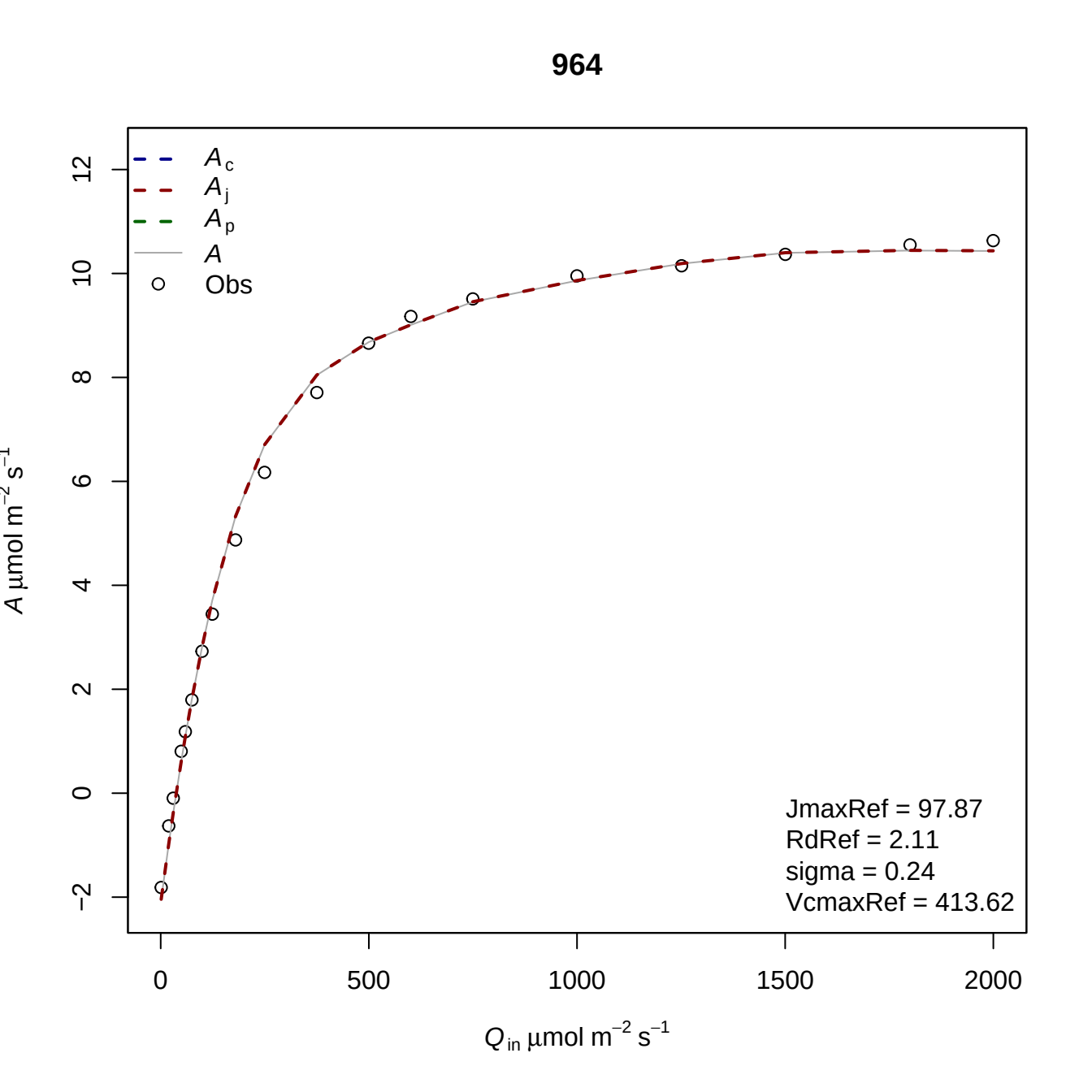

959
A_c
A_j
A_p
A
Obs
JmaxRef = 82.15
RdRef = 2.88
sigma = 0.81
VcmaxRef = 622.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

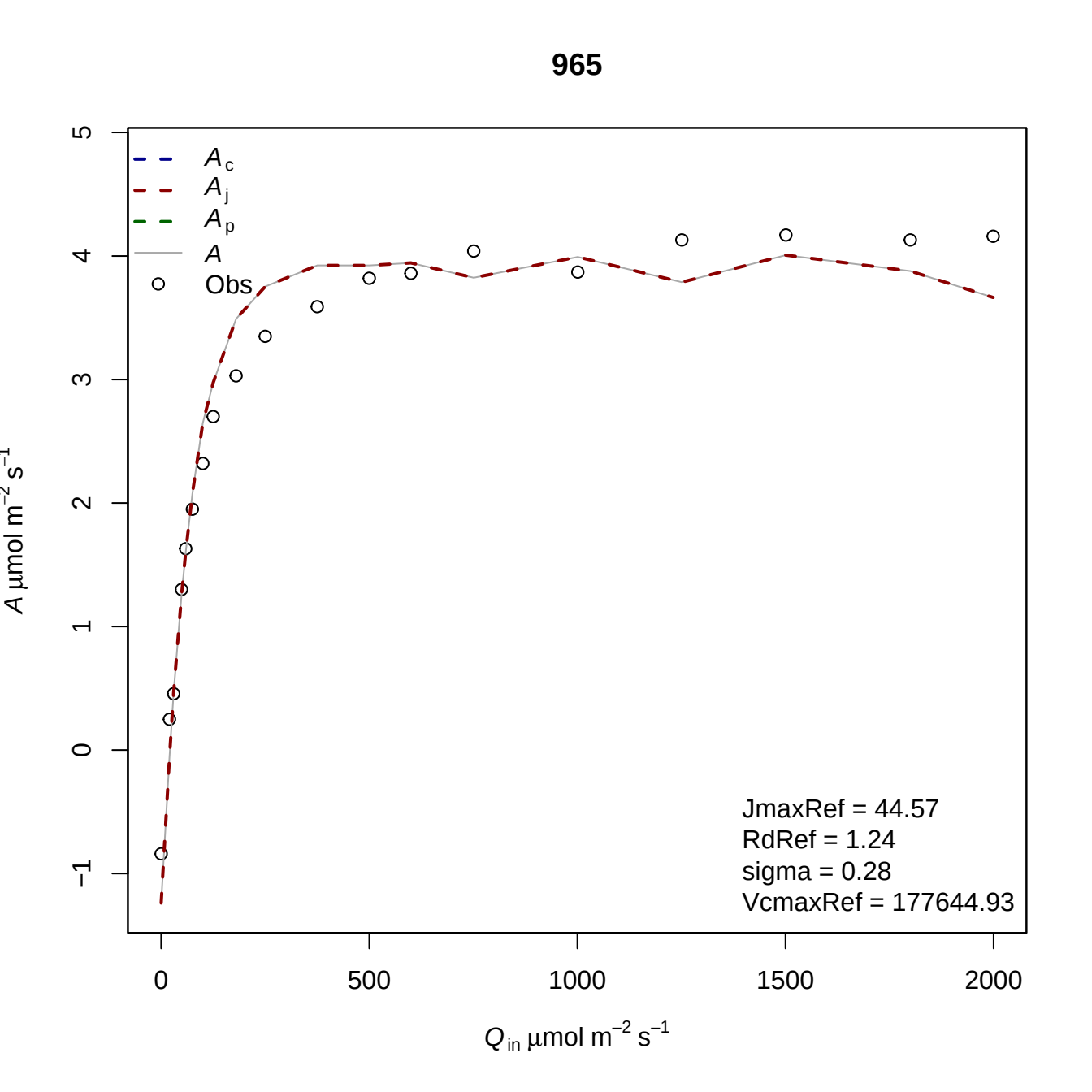

960
A_c
A_j
A_p
A
Obs
JmaxRef = 98.99
RdRef = 3.08
sigma = 0.59
VcmaxRef = 392.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

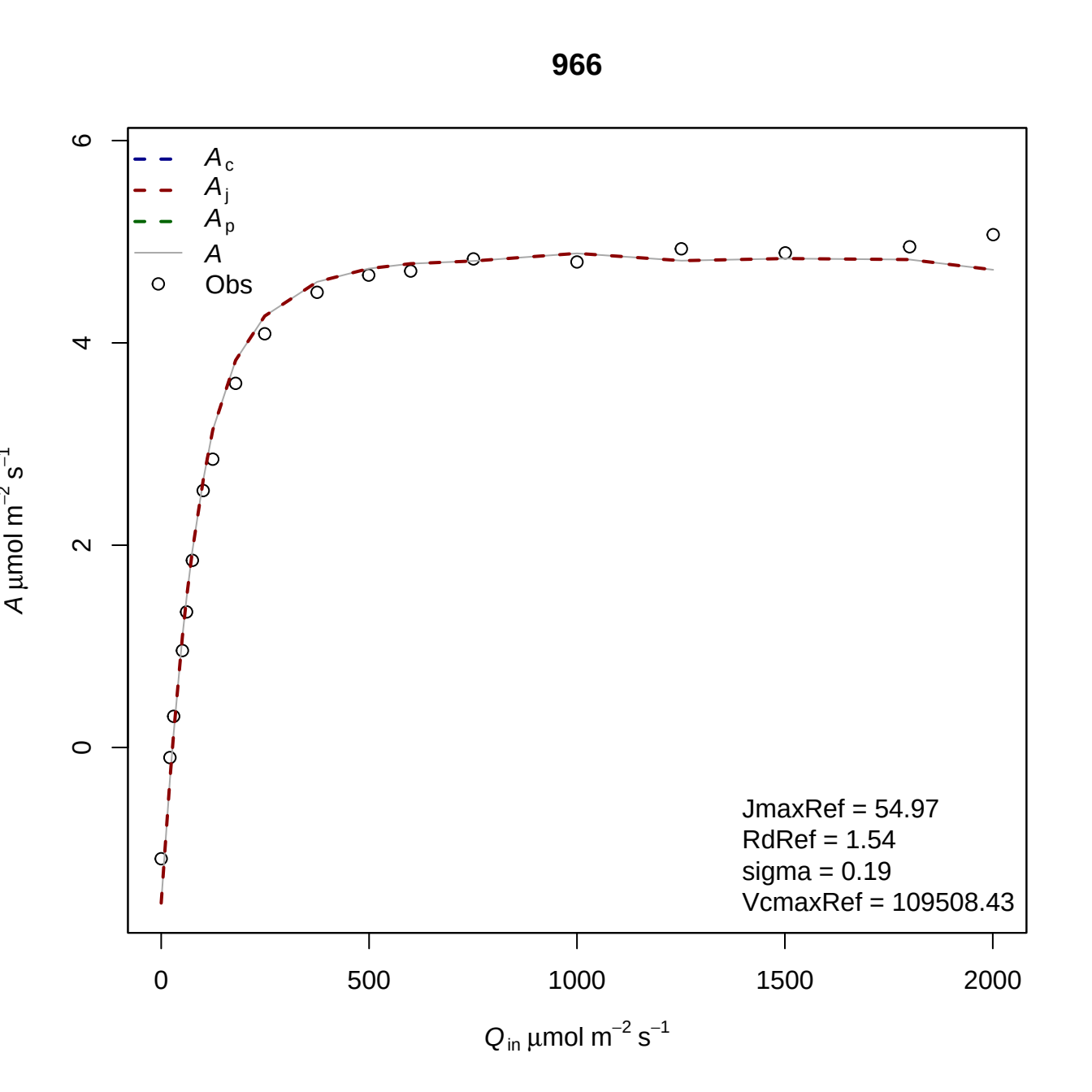

961
A_c
A_j
A_p
A
Obs
JmaxRef = 82.56
RdRef = 3.01
sigma = 0.82
VcmaxRef = 2302.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

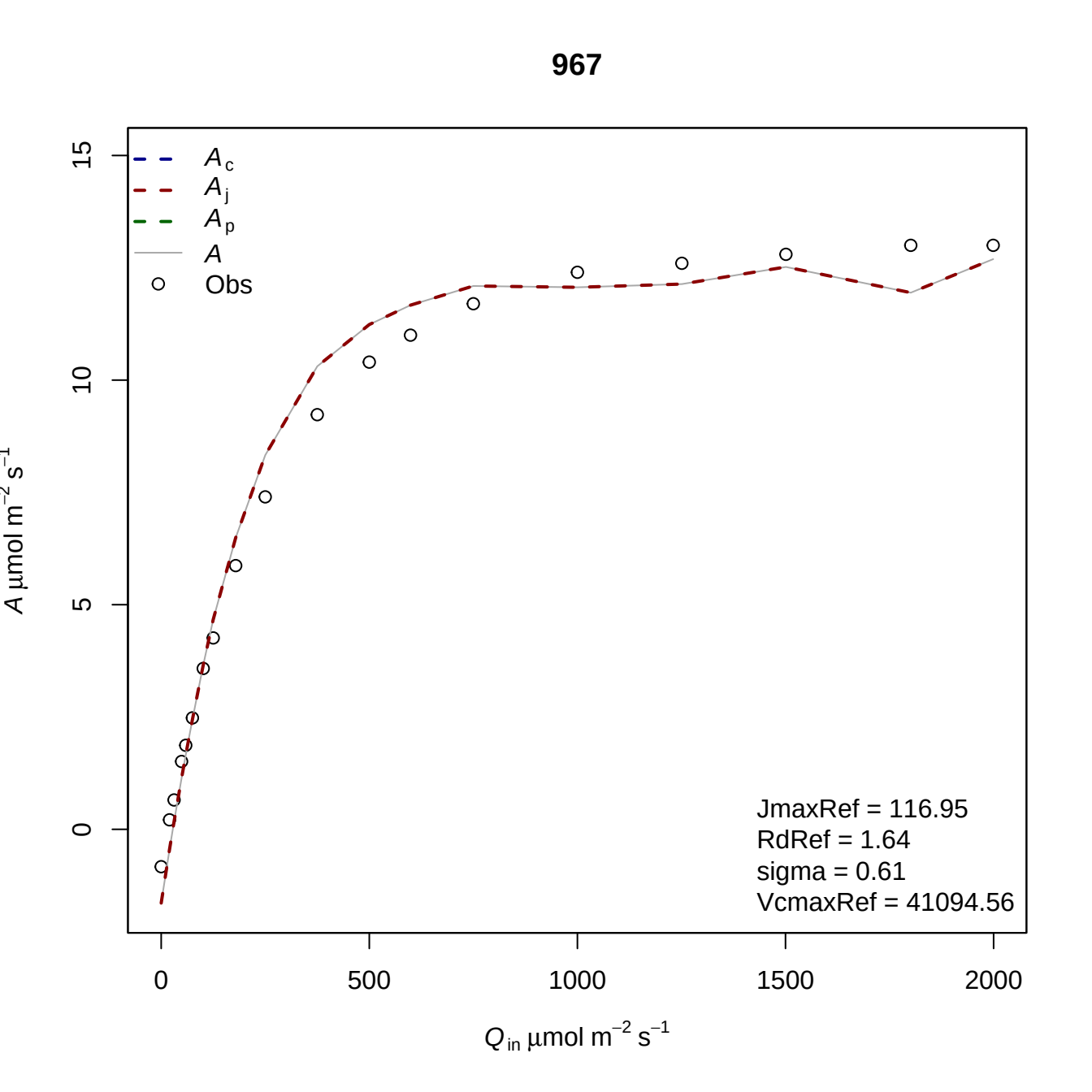

962
A_c
A_j
A_p
A
Obs
JmaxRef = 76.38
RdRef = 4.37
sigma = 1.1
VcmaxRef = 166353.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

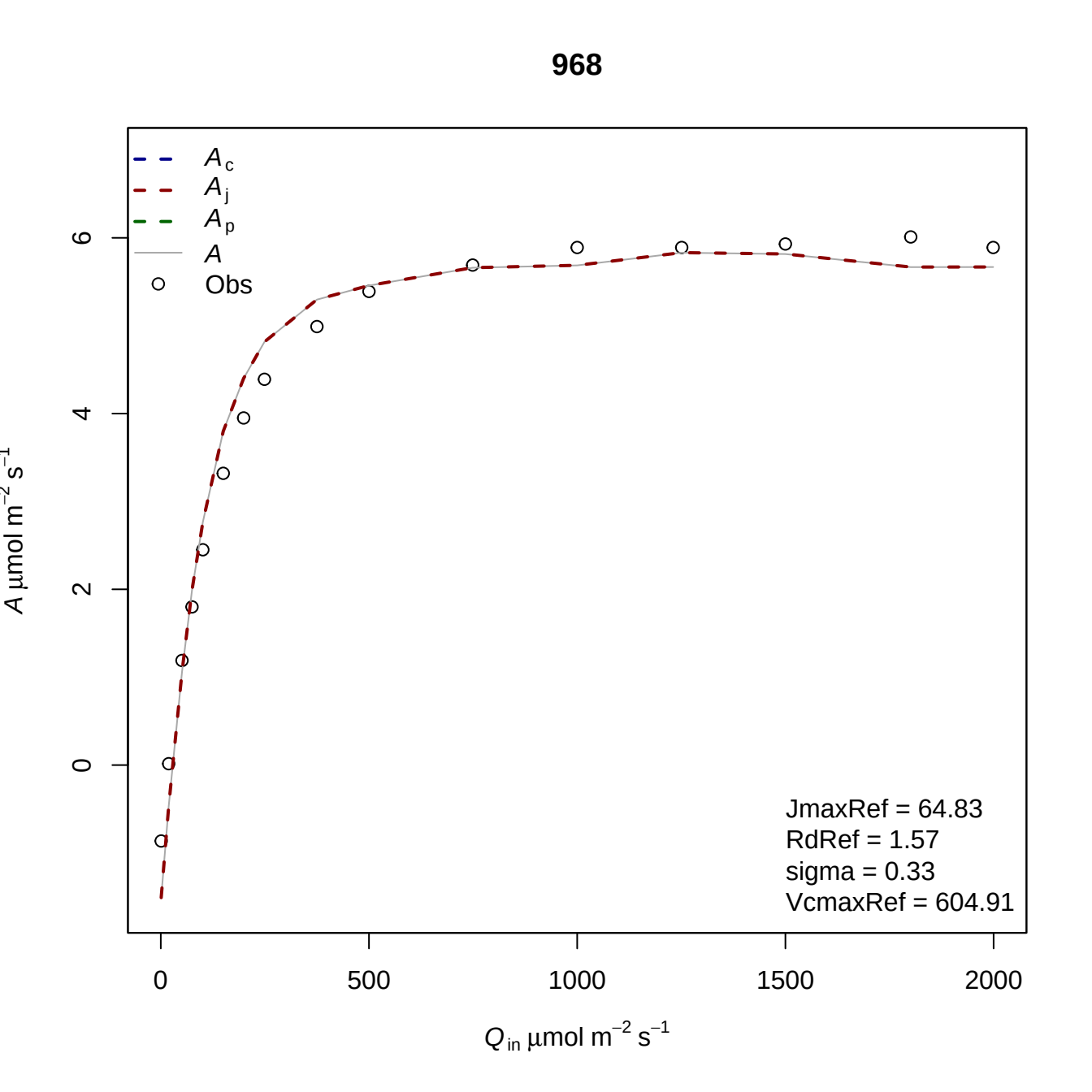

963
A_c
A_j
A_p
A
Obs
JmaxRef = 64.83
RdRef = 3.76
sigma = 0.68
VcmaxRef = 745.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
0
500
1000
1500
2000

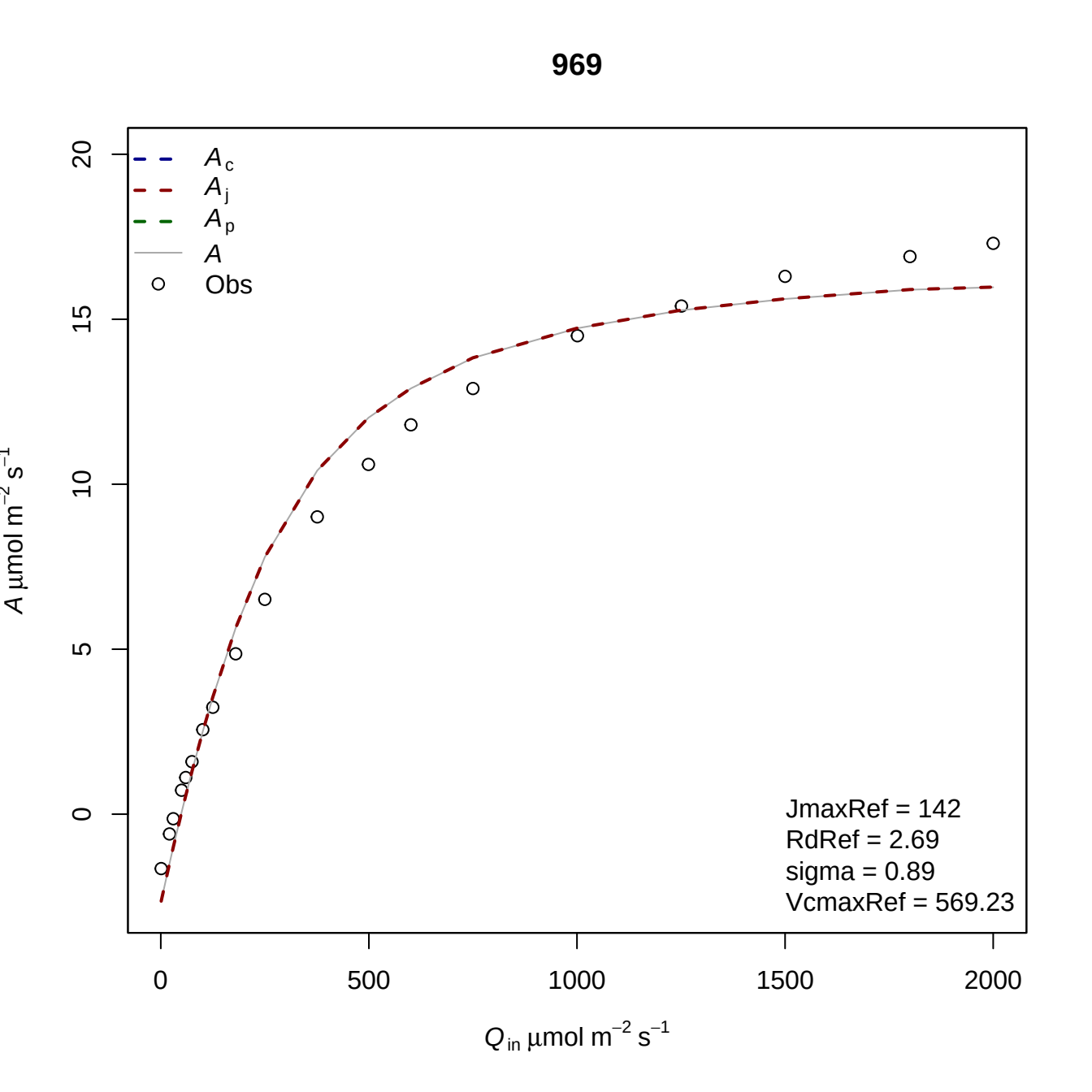

964
A_c
A_j
A_p
A
Obs
JmaxRef = 97.87
RdRef = 2.11
sigma = 0.24
VcmaxRef = 413.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

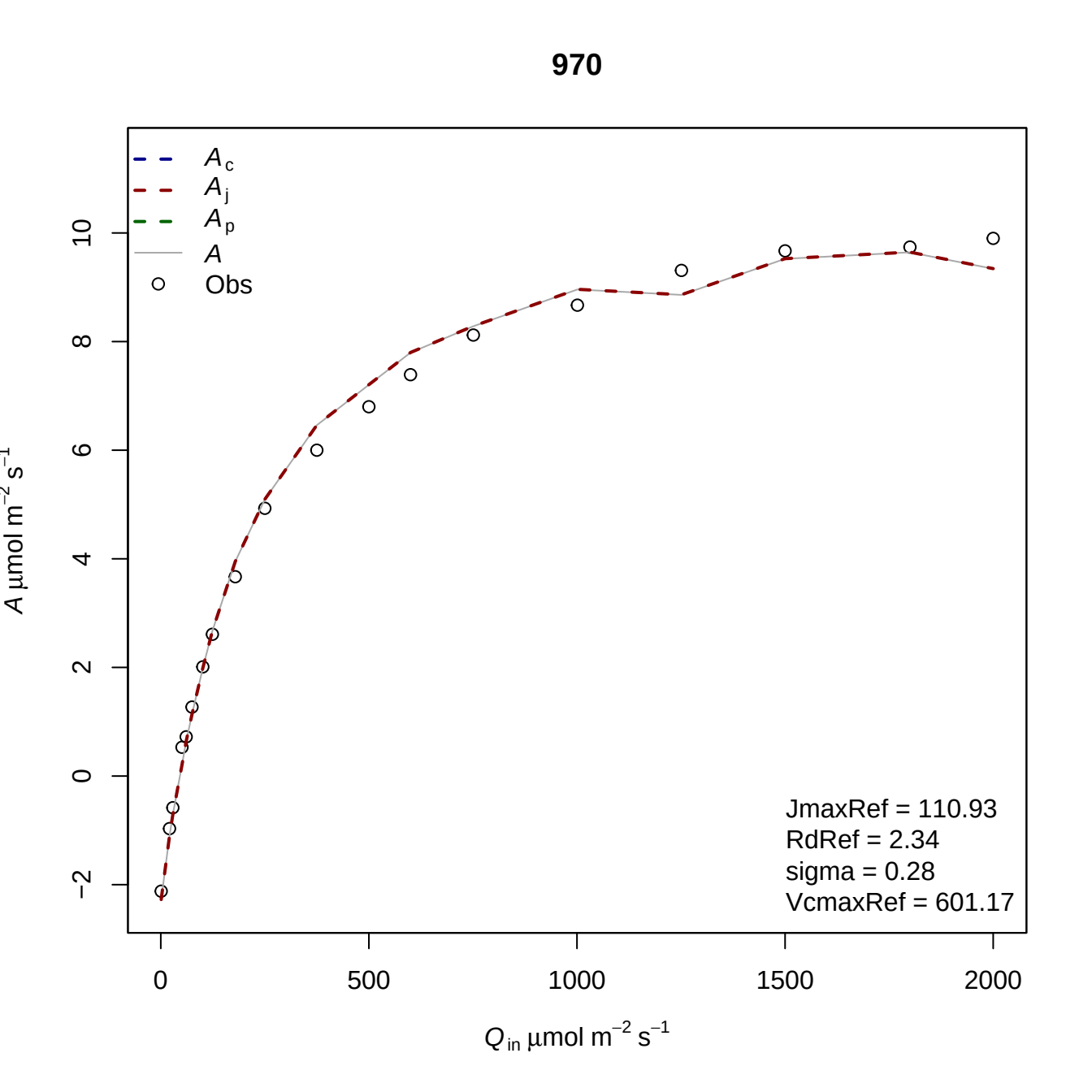

965
A_c
A_j
A_p
A
Obs
JmaxRef = 44.57
RdRef = 1.24
sigma = 0.28
VcmaxRef = 177644.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
0
500
1000
1500
2000

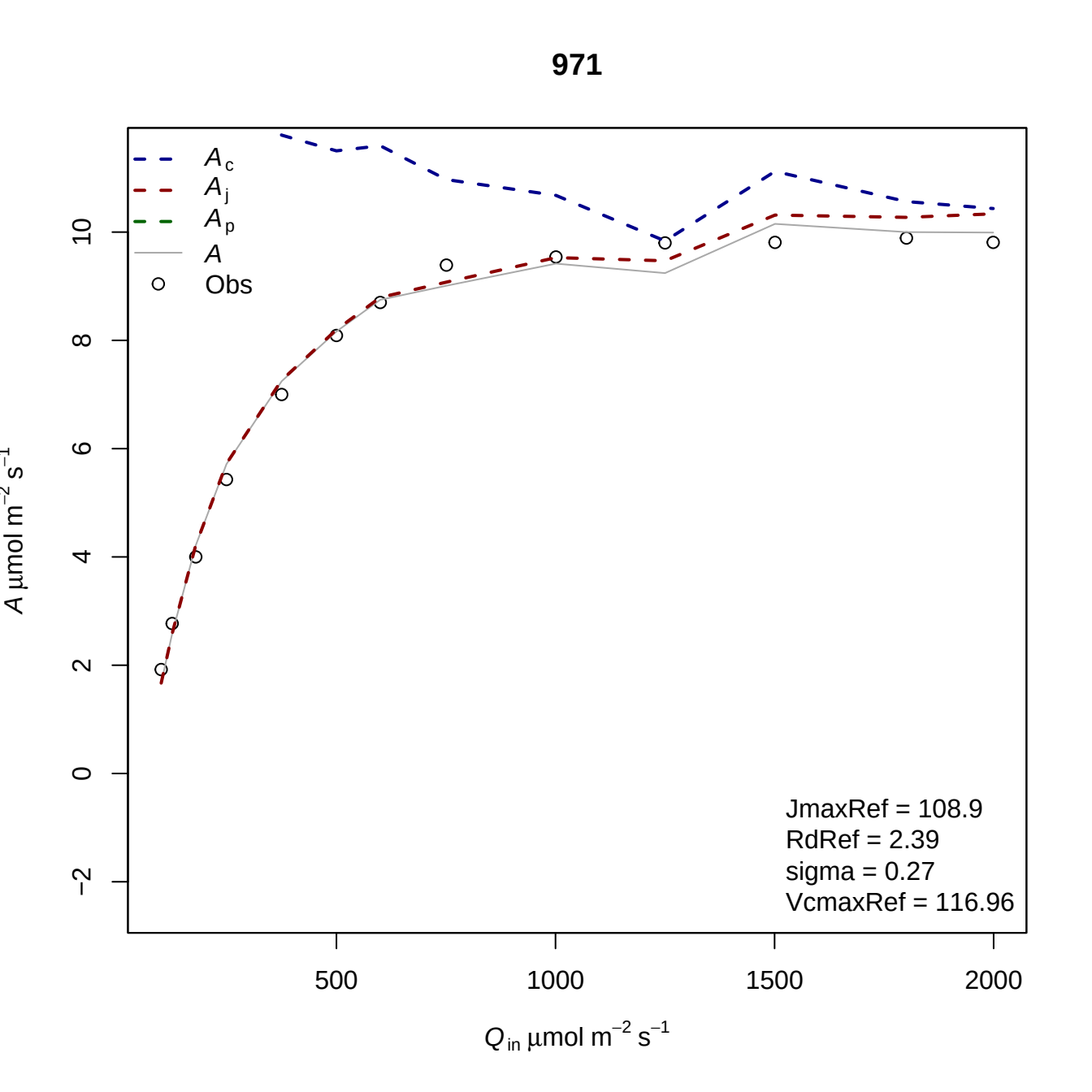

966
A_c
A_j
A_p
A
Obs
JmaxRef = 54.97
RdRef = 1.54
sigma = 0.19
VcmaxRef = 109508.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

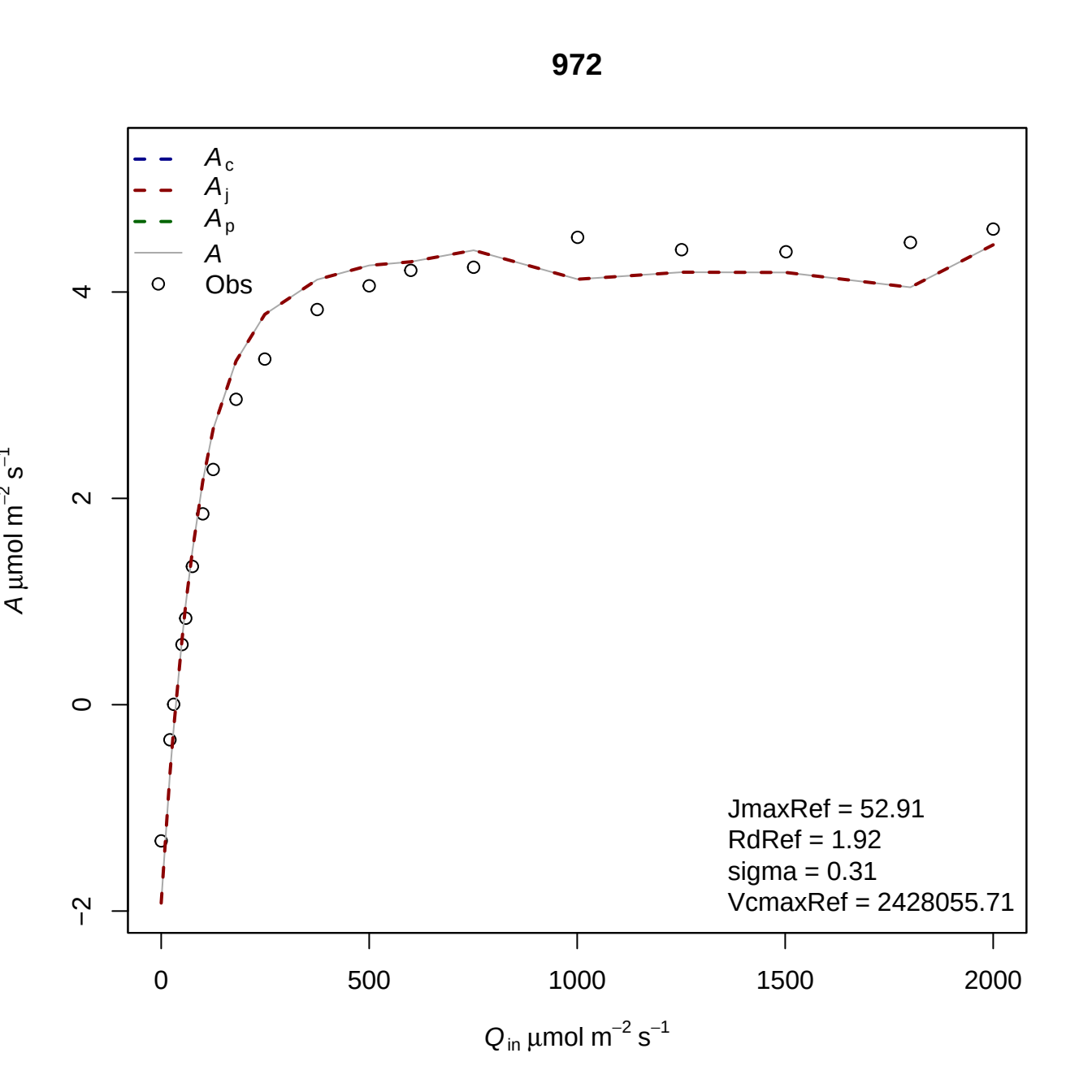

967
A_c
A_j
A_p
A
Obs
JmaxRef = 116.95
RdRef = 1.64
sigma = 0.61
VcmaxRef = 41094.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

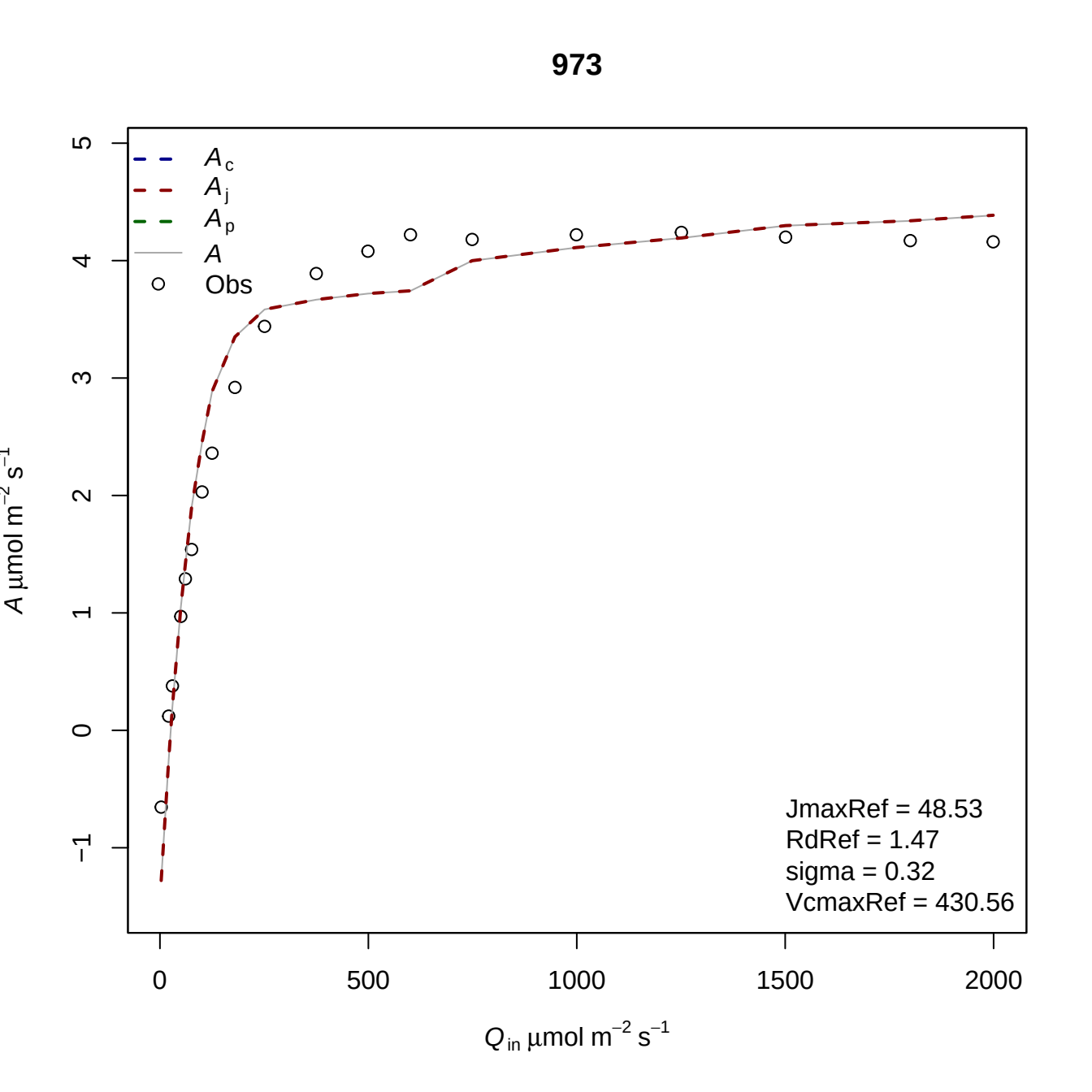

968
A_c
A_j
A_p
A
Obs
JmaxRef = 64.83
RdRef = 1.57
sigma = 0.33
VcmaxRef = 604.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

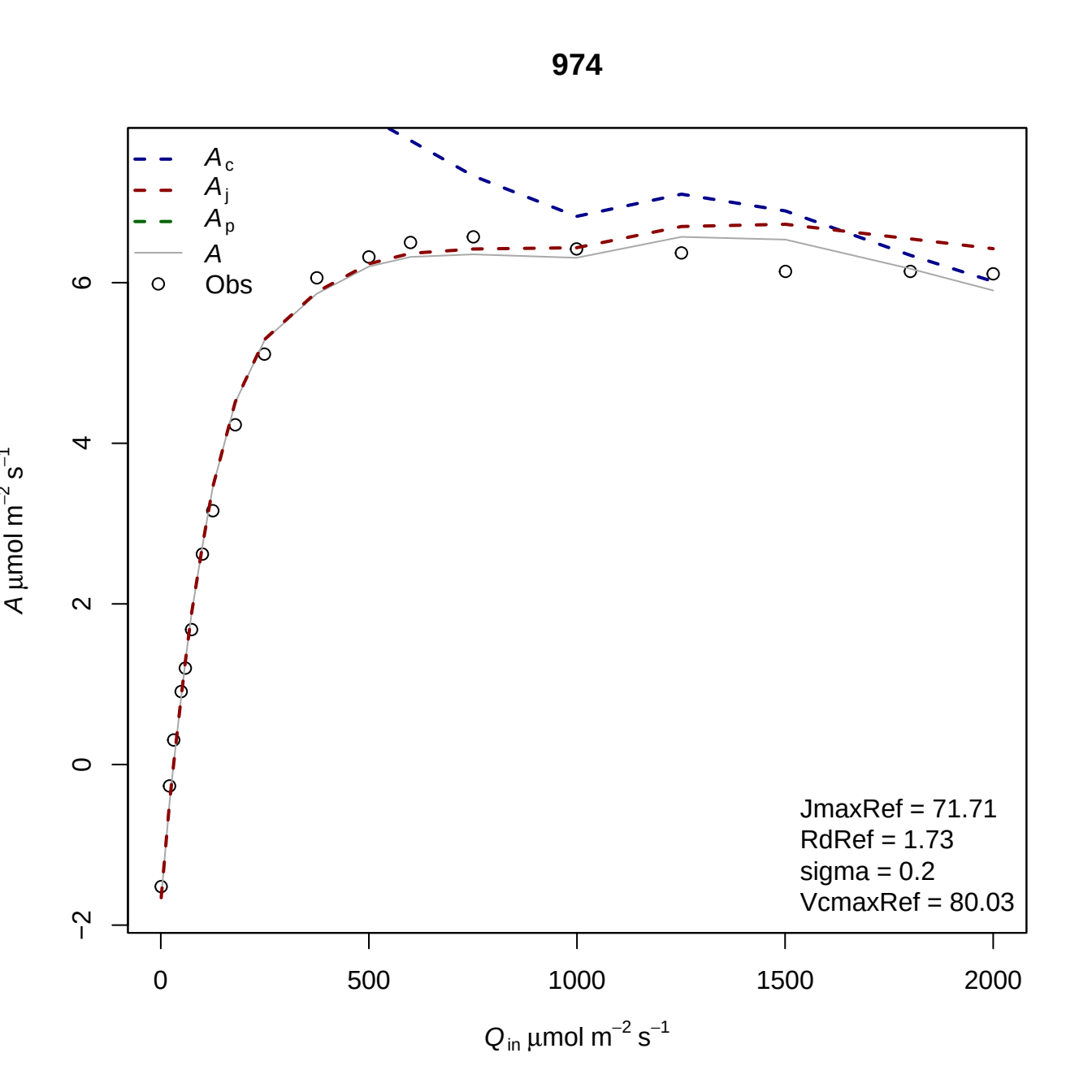

969
A_c
A_j
A_p
A
Obs
JmaxRef = 142
RdRef = 2.69
sigma = 0.89
VcmaxRef = 569.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

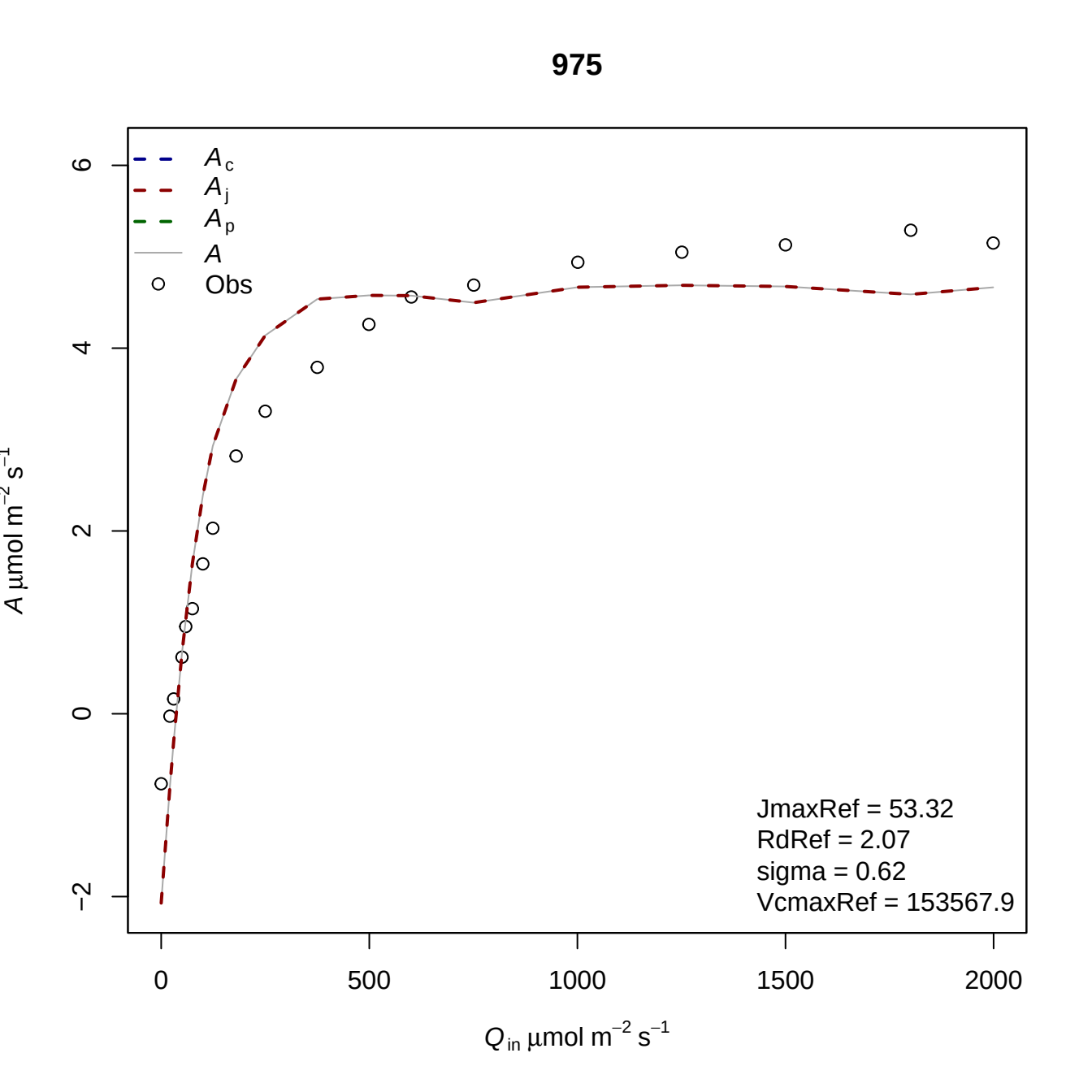

970
A_c
A_j
A_p
A
Obs
JmaxRef = 110.93
RdRef = 2.34
sigma = 0.28
VcmaxRef = 601.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

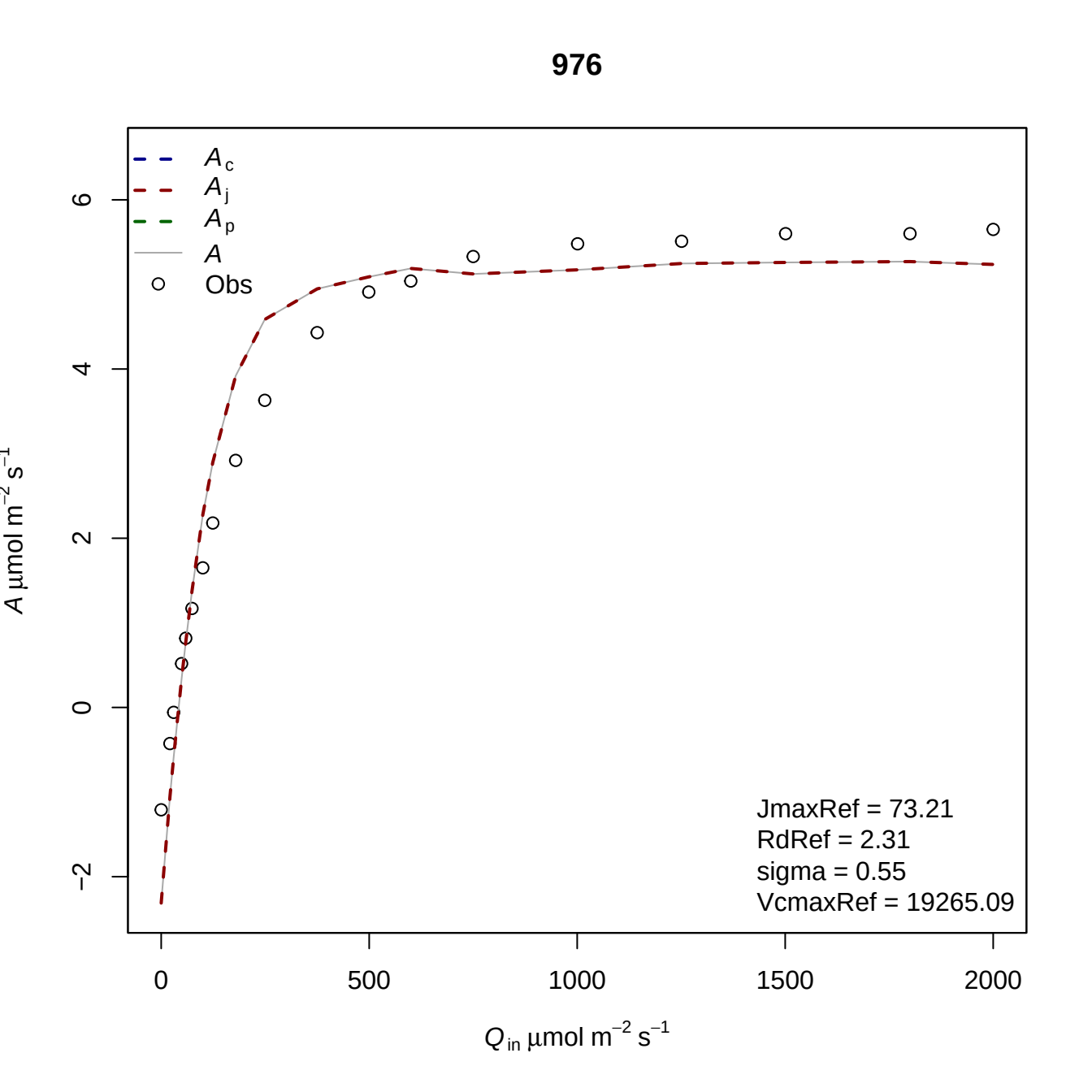

971
A_c
A_j
A_p
A
Obs
JmaxRef = 108.9
RdRef = 2.39
sigma = 0.27
VcmaxRef = 116.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
500
1000
1500
2000

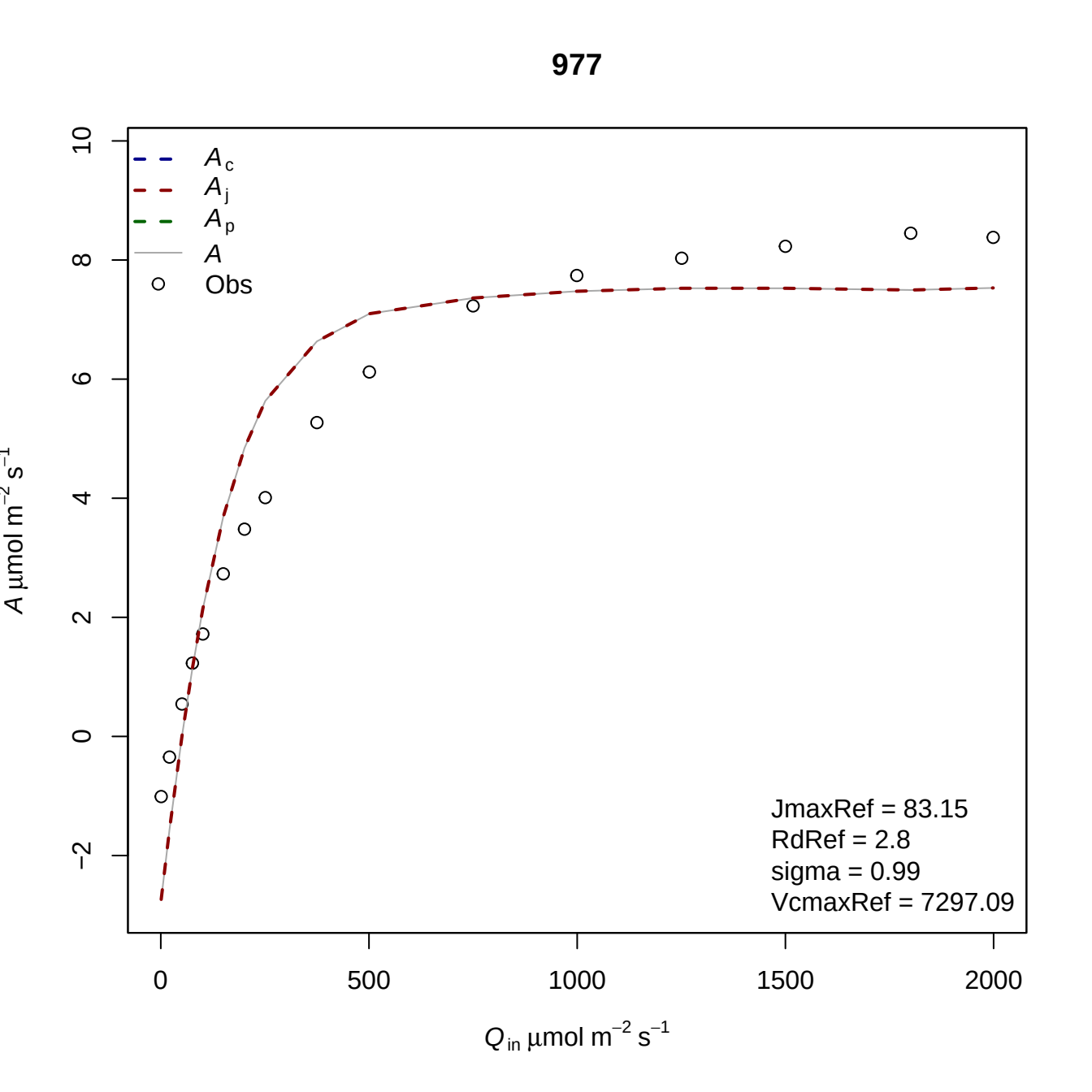

972
A_c
A_j
A_p
A
Obs
JmaxRef = 52.91
RdRef = 1.92
sigma = 0.31
VcmaxRef = 2428055.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

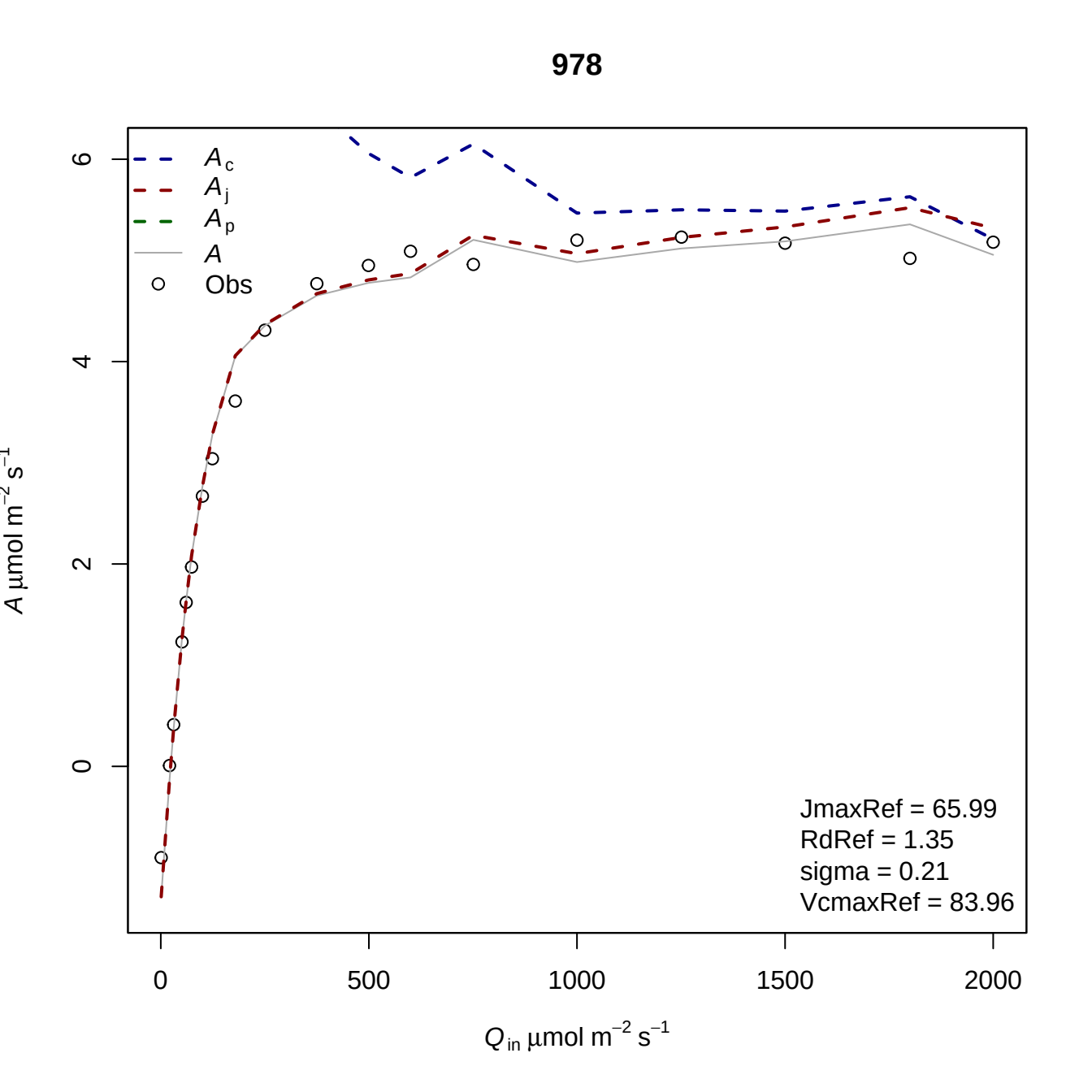

973
A_c
A_j
A_p
A
Obs
JmaxRef = 48.53
RdRef = 1.47
sigma = 0.32
VcmaxRef = 430.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

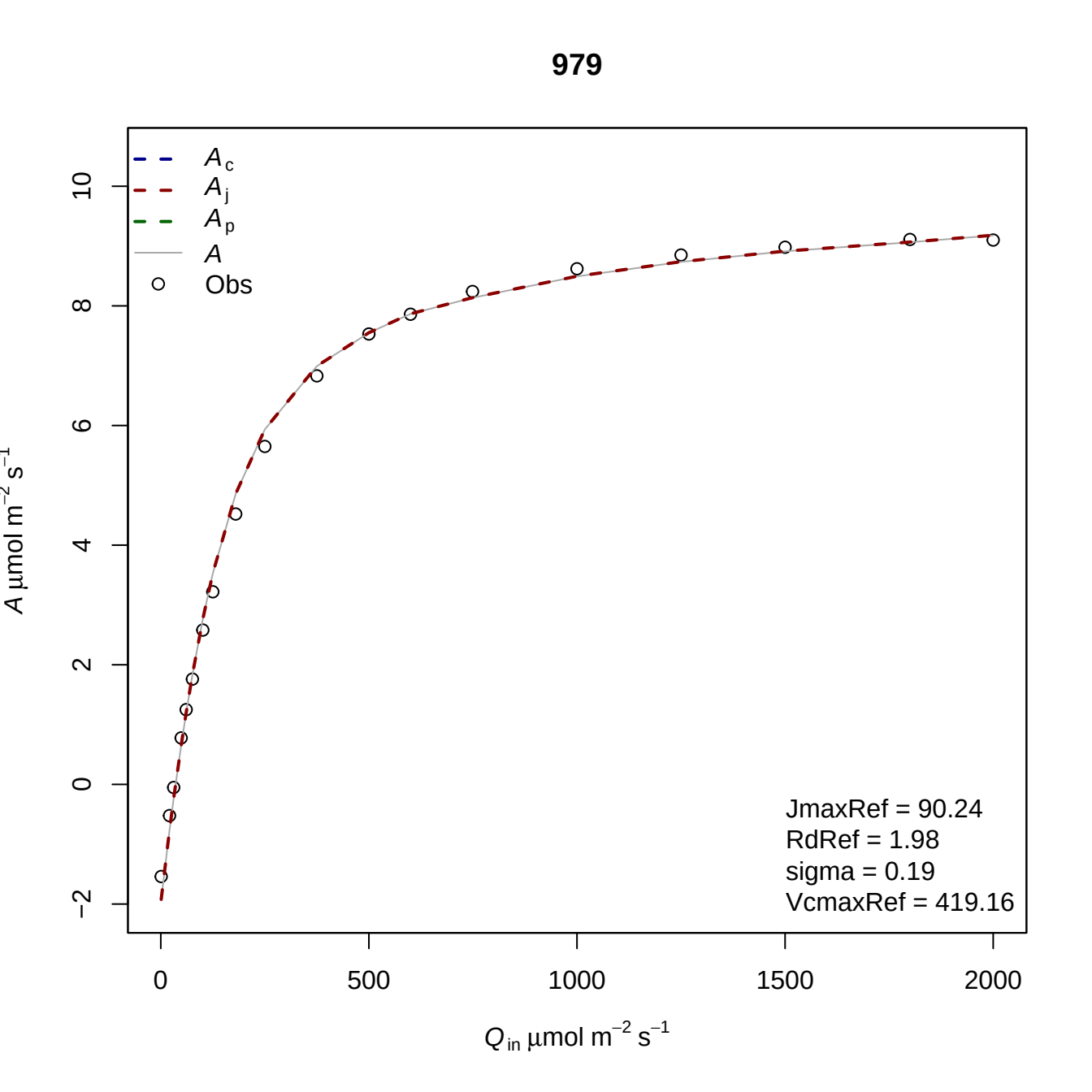

974
A_c
A_j
A_p
A
Obs
JmaxRef = 71.71
RdRef = 1.73
sigma = 0.2
VcmaxRef = 80.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

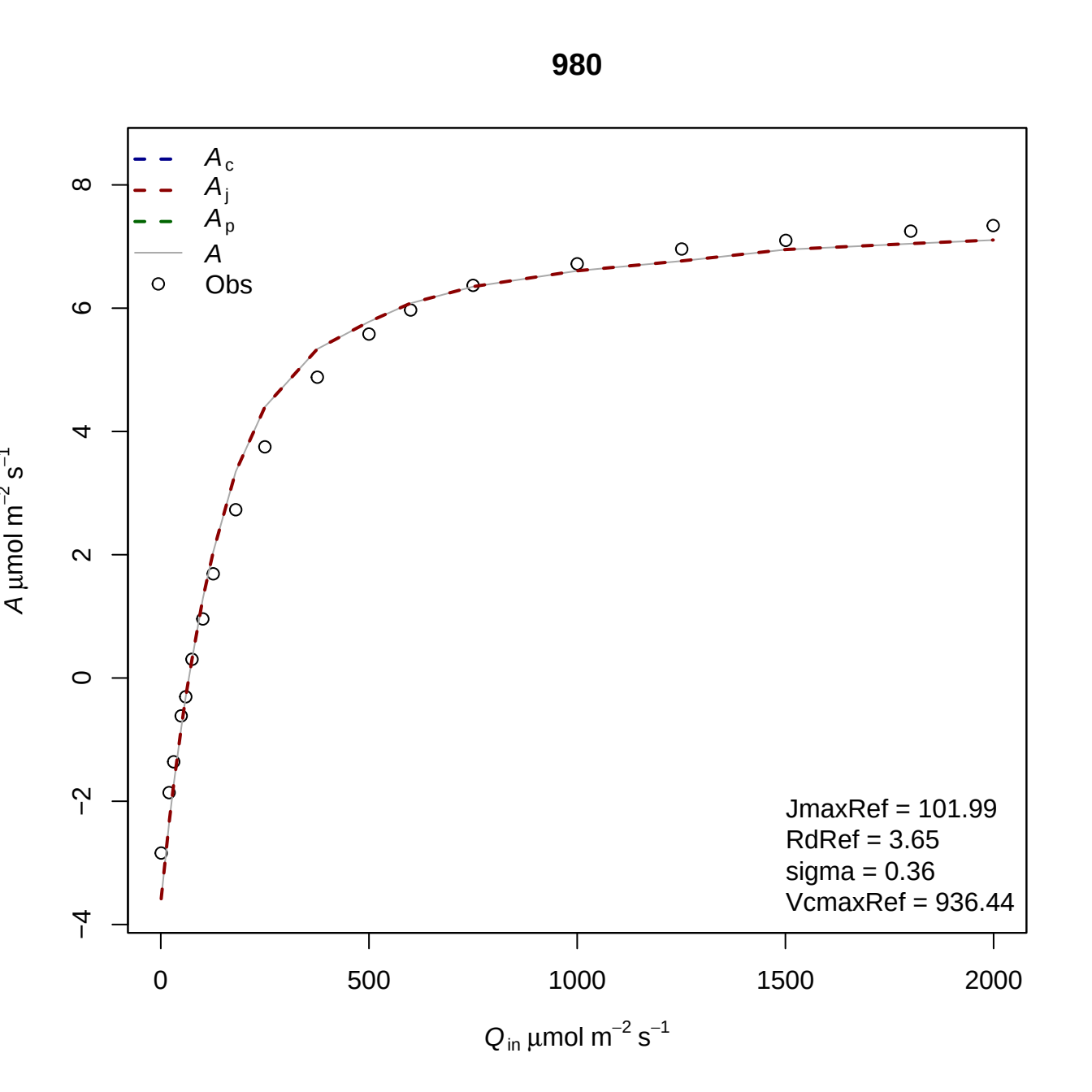

975
A_c
A_j
A_p
A
Obs
JmaxRef = 53.32
RdRef = 2.07
sigma = 0.62
VcmaxRef = 153567.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

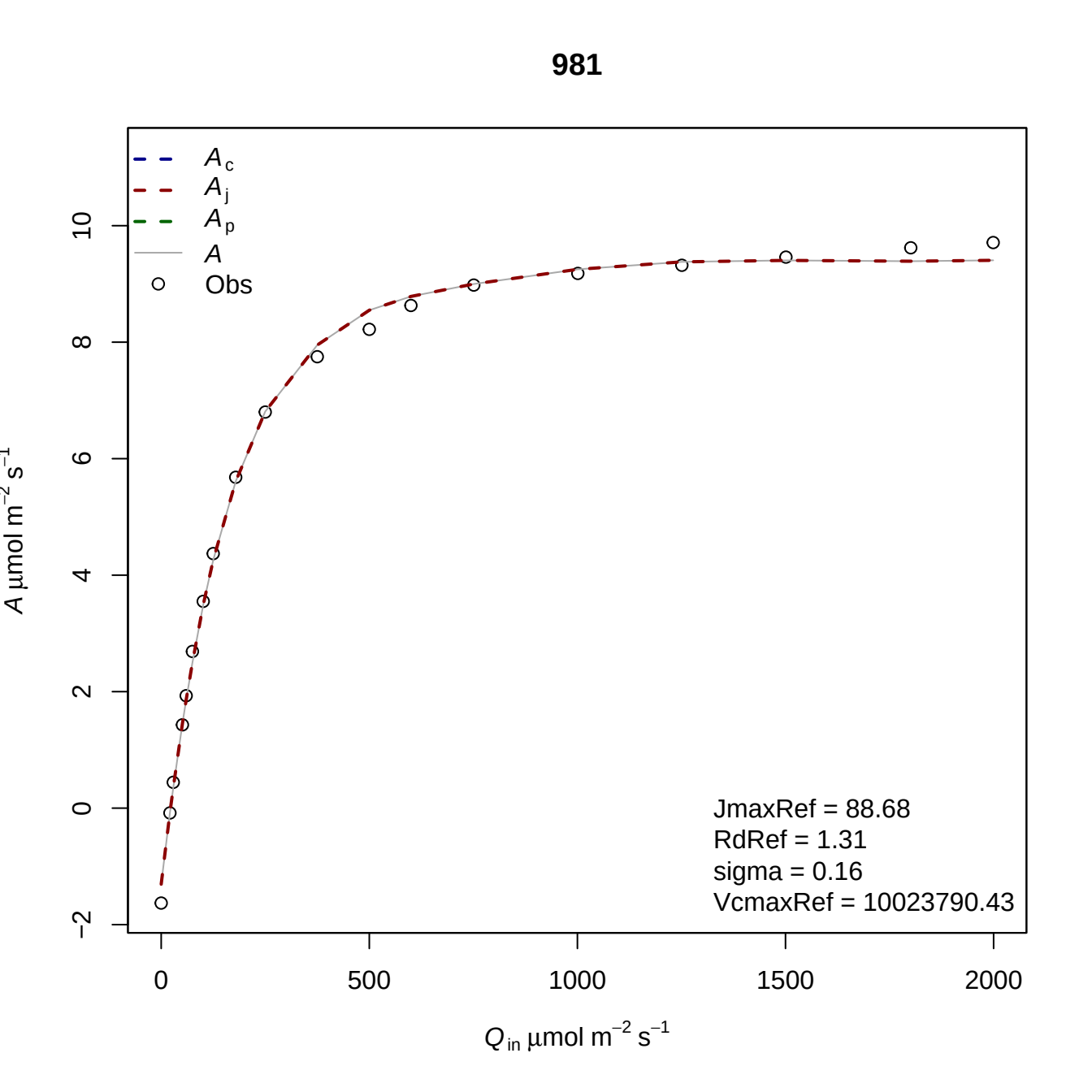

976
A_c
A_j
A_p
A
Obs
JmaxRef = 73.21
RdRef = 2.31
sigma = 0.55
VcmaxRef = 19265.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

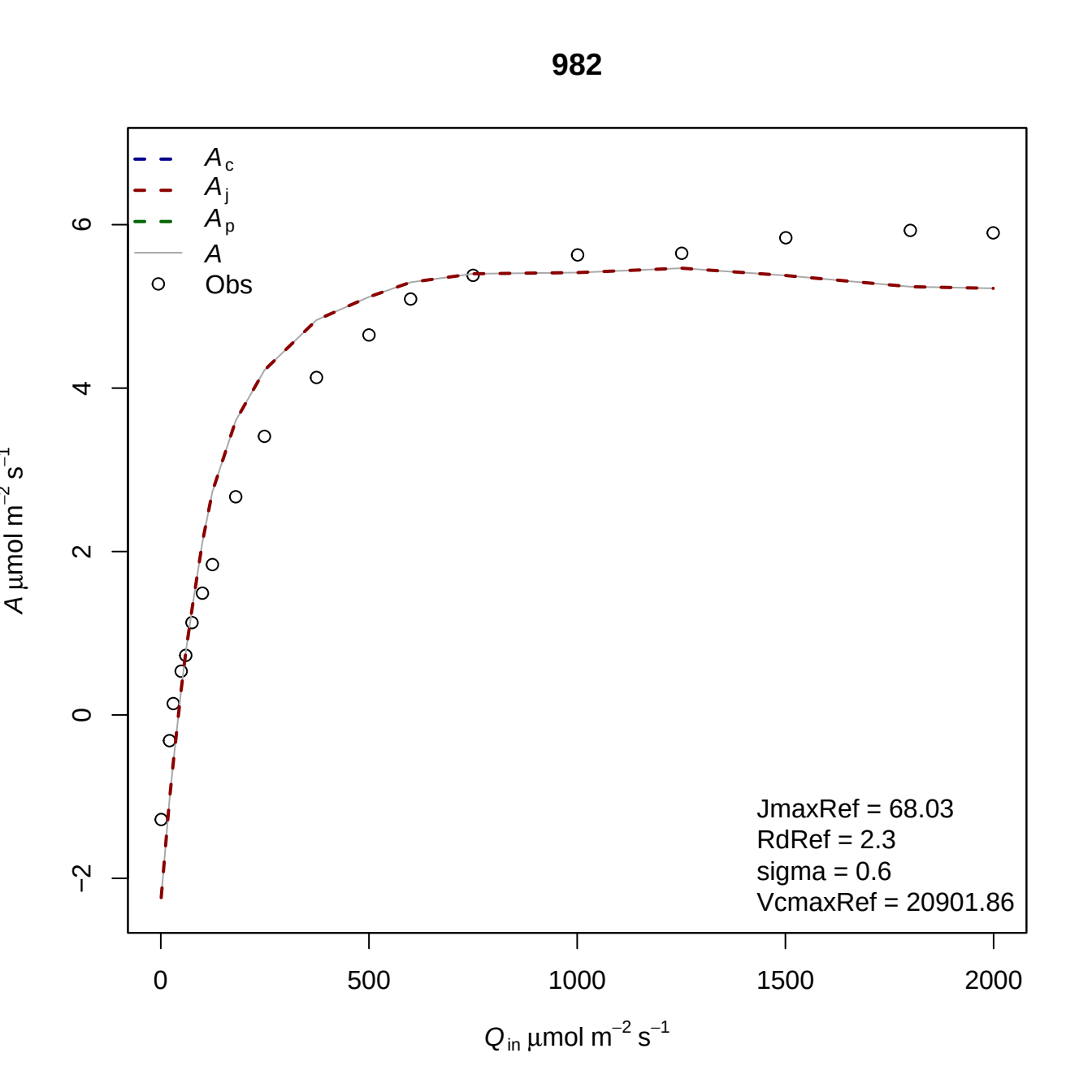

977
A_c
A_j
A_p
A
Obs
JmaxRef = 83.15
RdRef = 2.8
sigma = 0.99
VcmaxRef = 7297.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

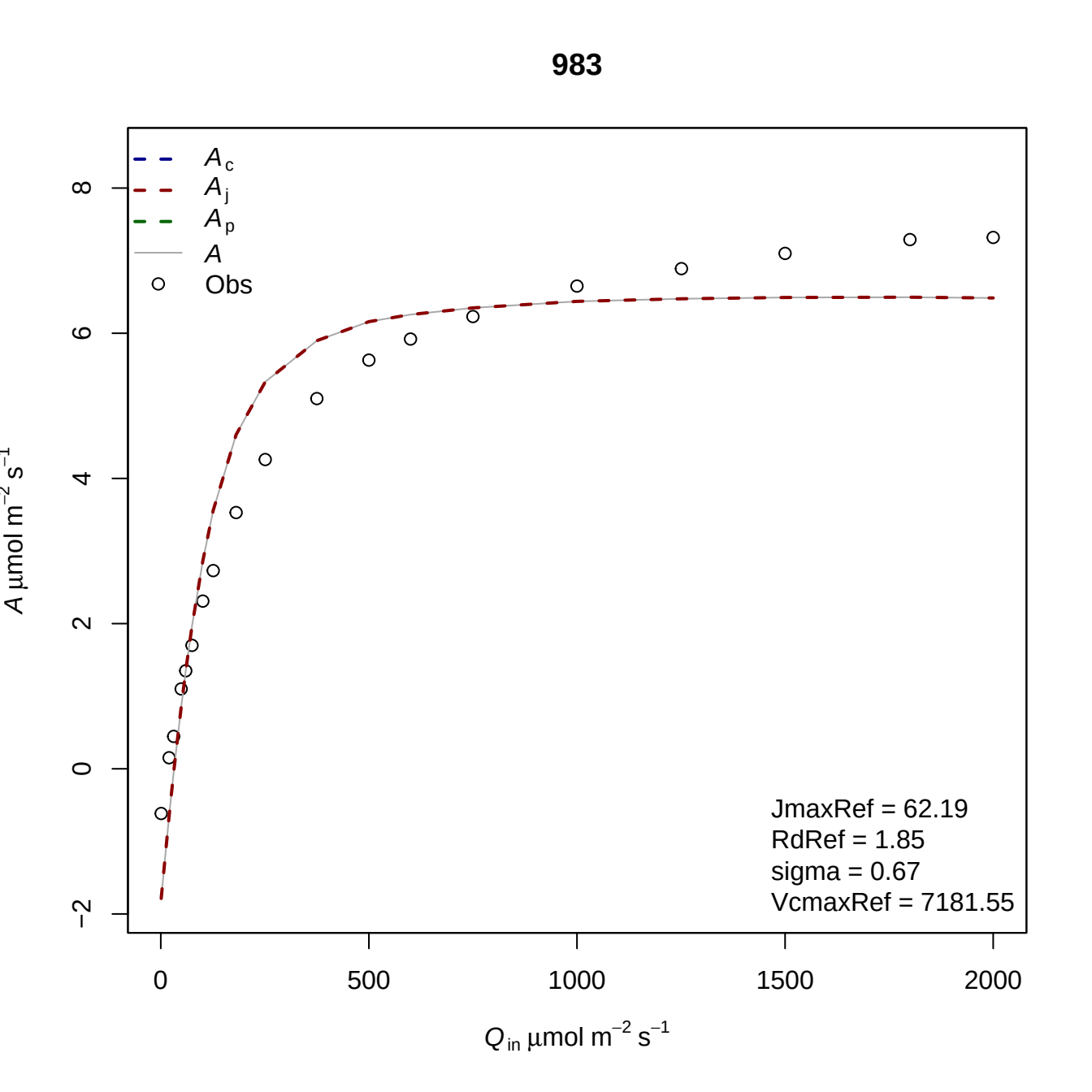

978
A_c
A_j
A_p
A
Obs
JmaxRef = 65.99
RdRef = 1.35
sigma = 0.21
VcmaxRef = 83.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

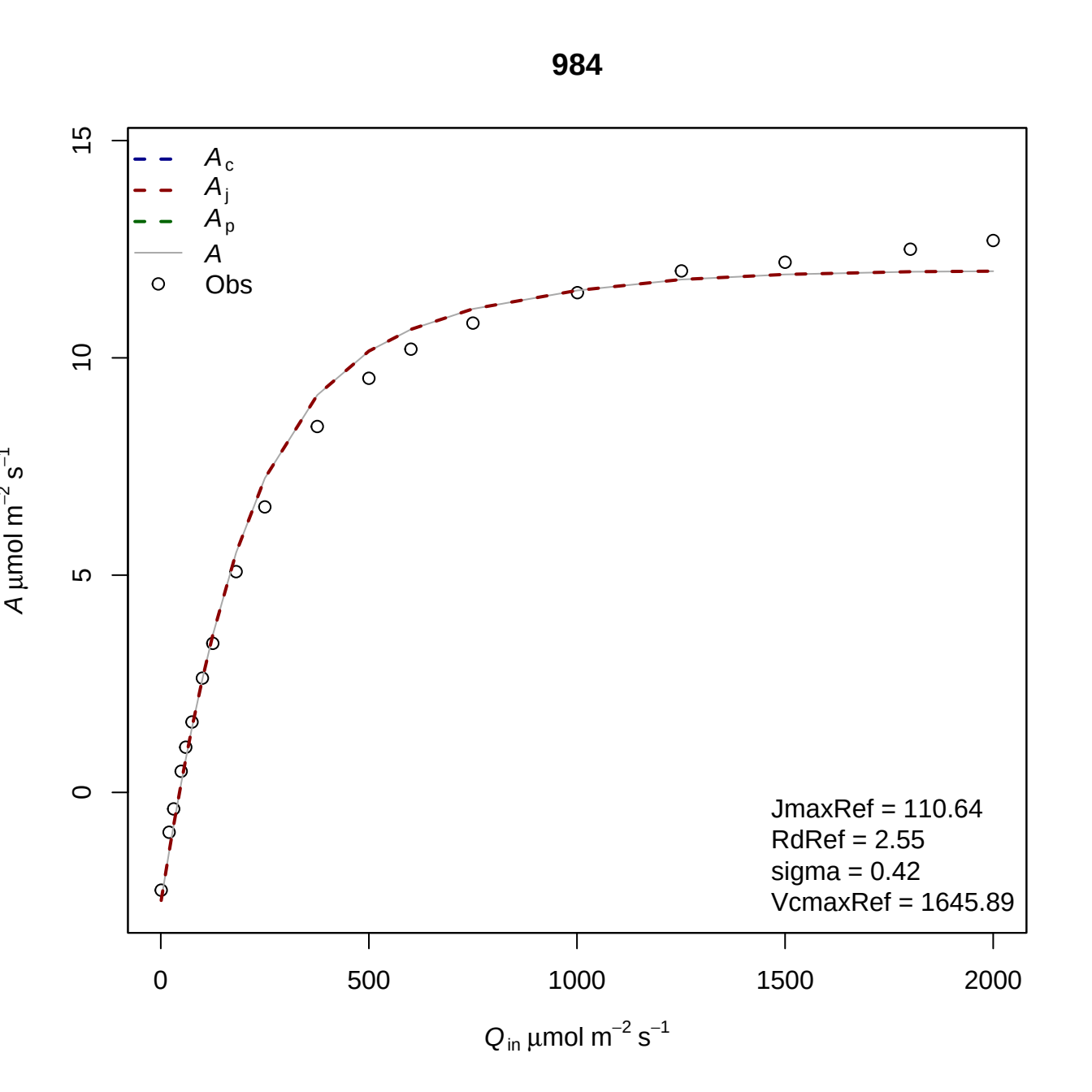

979
A_c
A_j
A_p
A
Obs
JmaxRef = 90.24
RdRef = 1.98
sigma = 0.19
VcmaxRef = 419.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

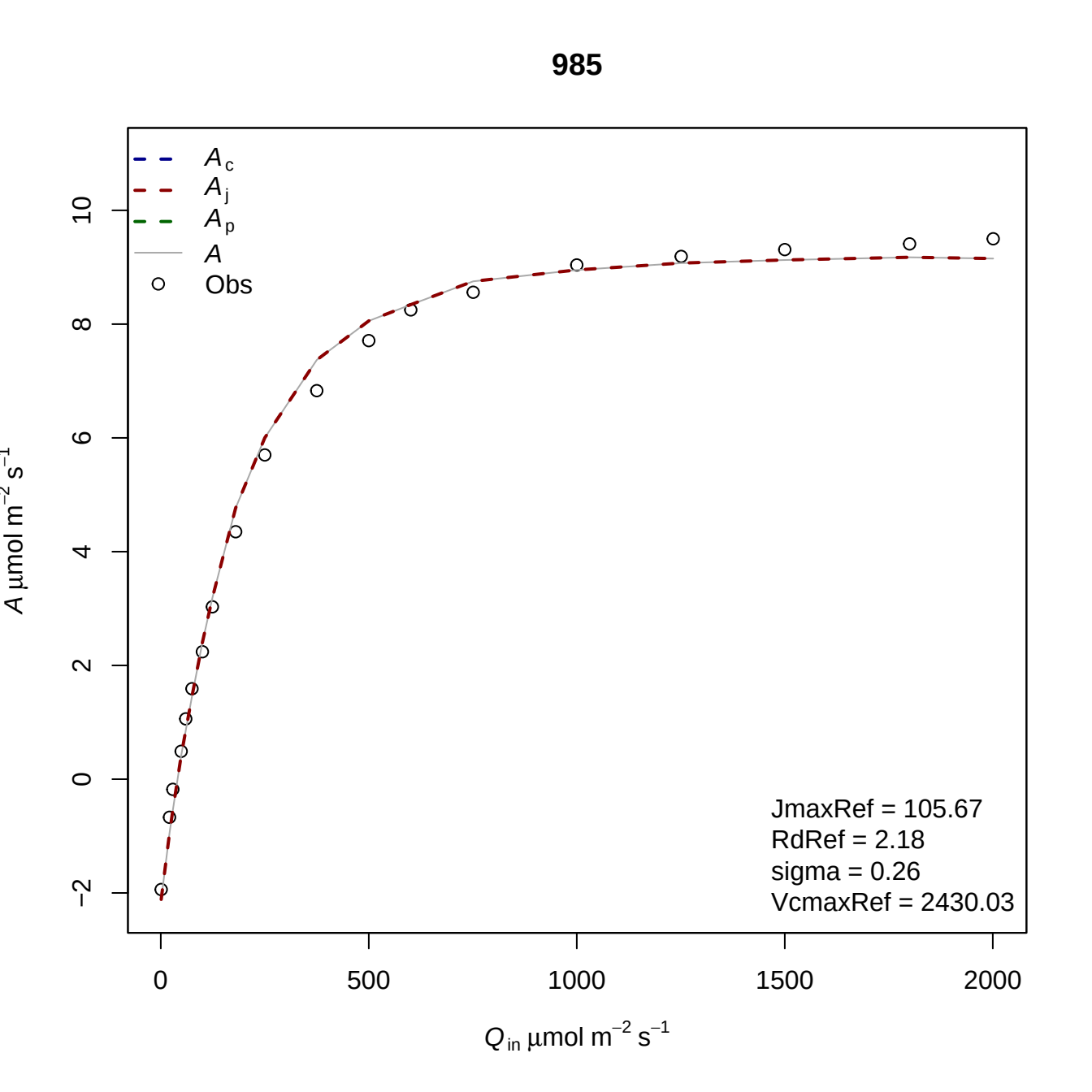

980
A_c
A_j
A_p
A
Obs
JmaxRef = 101.99
RdRef = 3.65
sigma = 0.36
VcmaxRef = 936.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

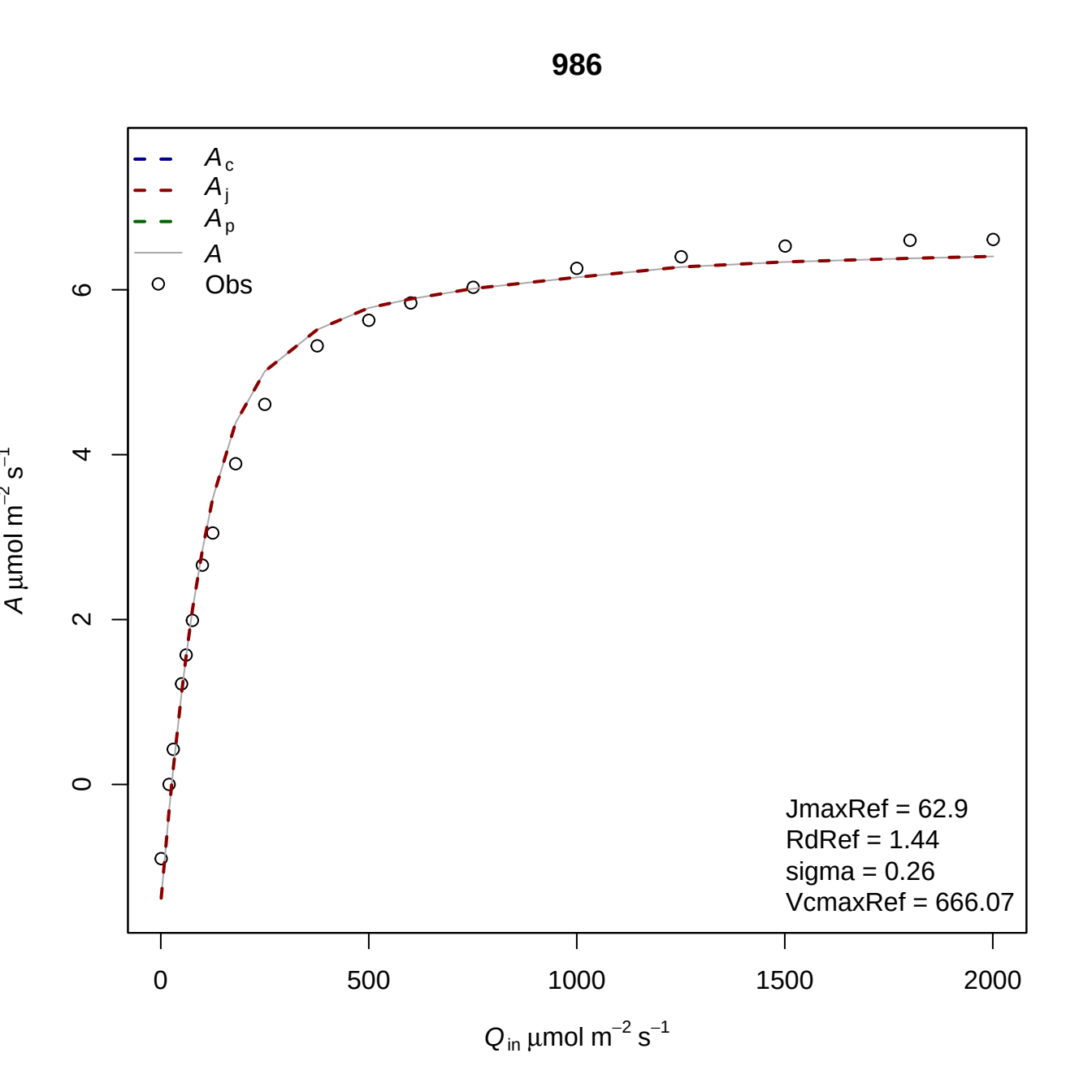

981
A_c
A_j
A_p
A
Obs
JmaxRef = 88.68
RdRef = 1.31
sigma = 0.16
VcmaxRef = 10023790.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

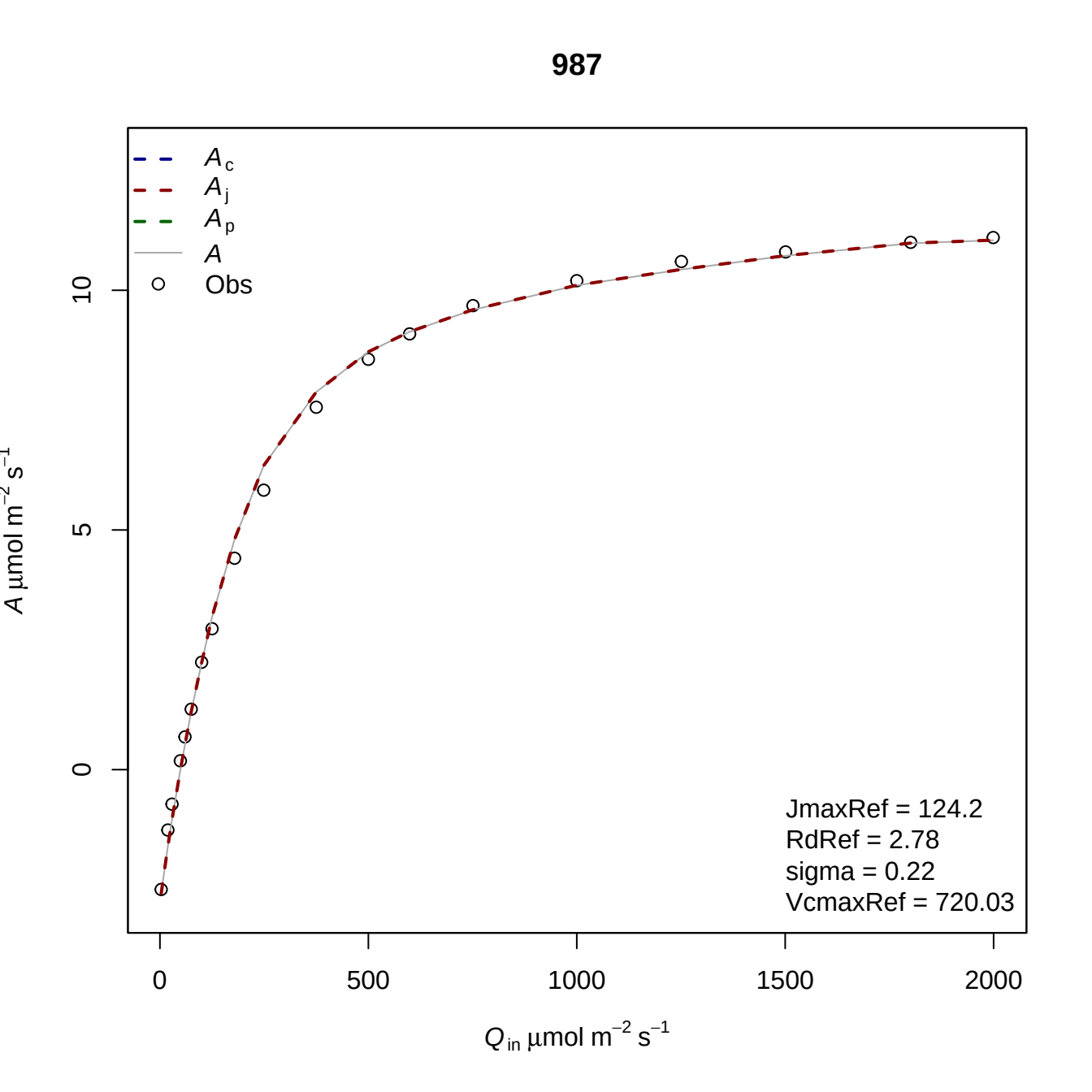

982
A_c
A_j
A_p
A
Obs
JmaxRef = 68.03
RdRef = 2.3
sigma = 0.6
VcmaxRef = 20901.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

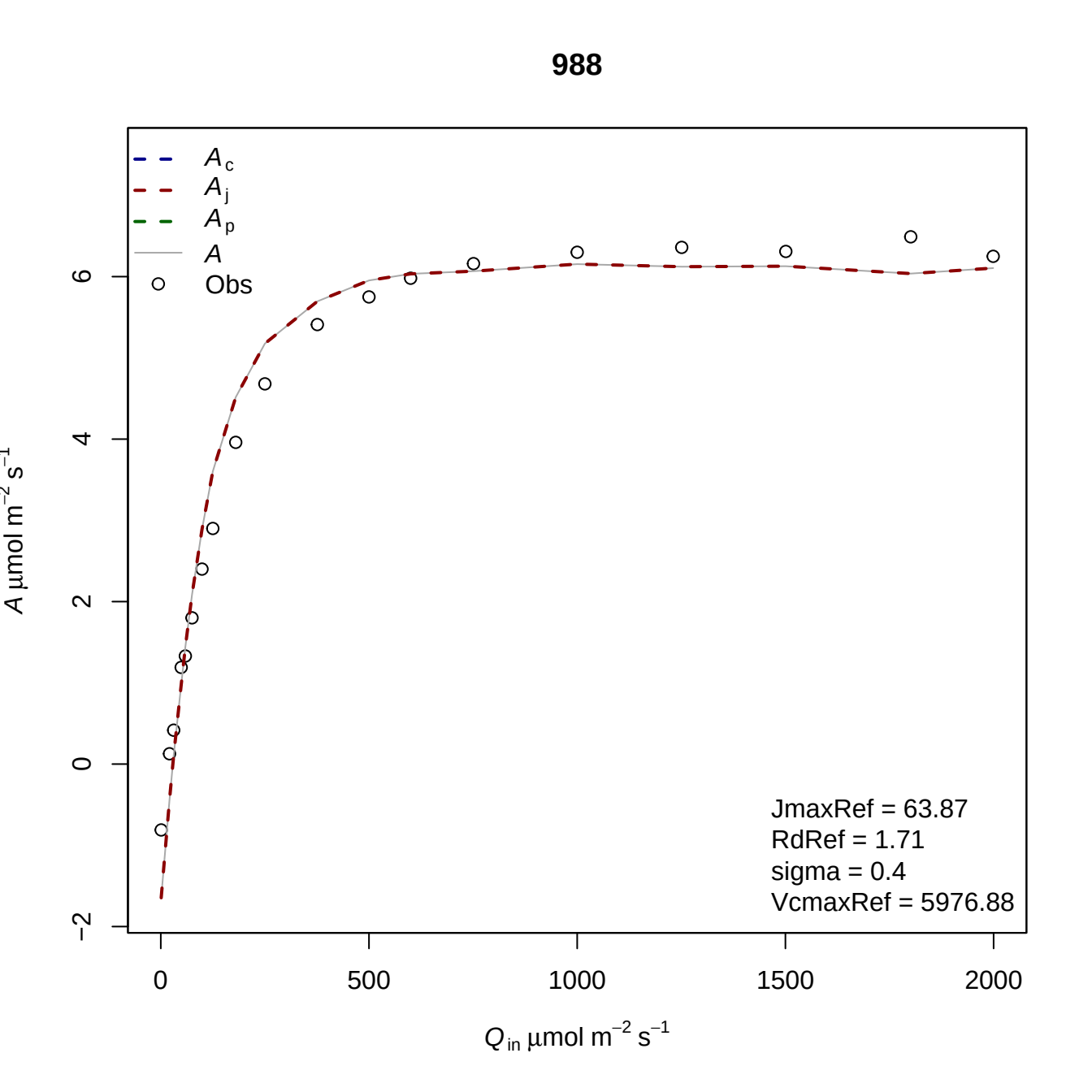

983
A_c
A_j
A_p
A
Obs
JmaxRef = 62.19
RdRef = 1.85
sigma = 0.67
VcmaxRef = 7181.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

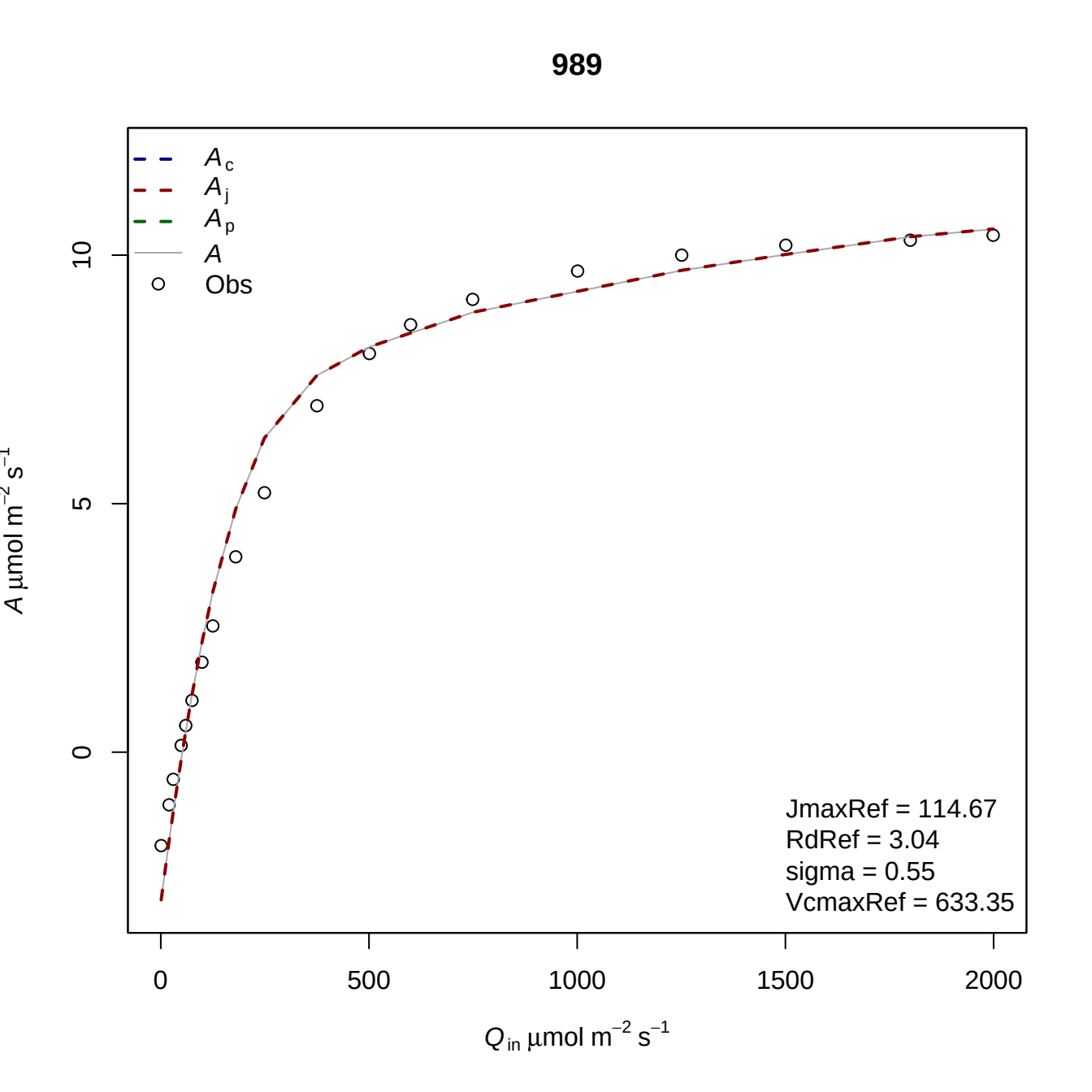

984
A_c
A_j
A_p
A
Obs
JmaxRef = 110.64
RdRef = 2.55
sigma = 0.42
VcmaxRef = 1645.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

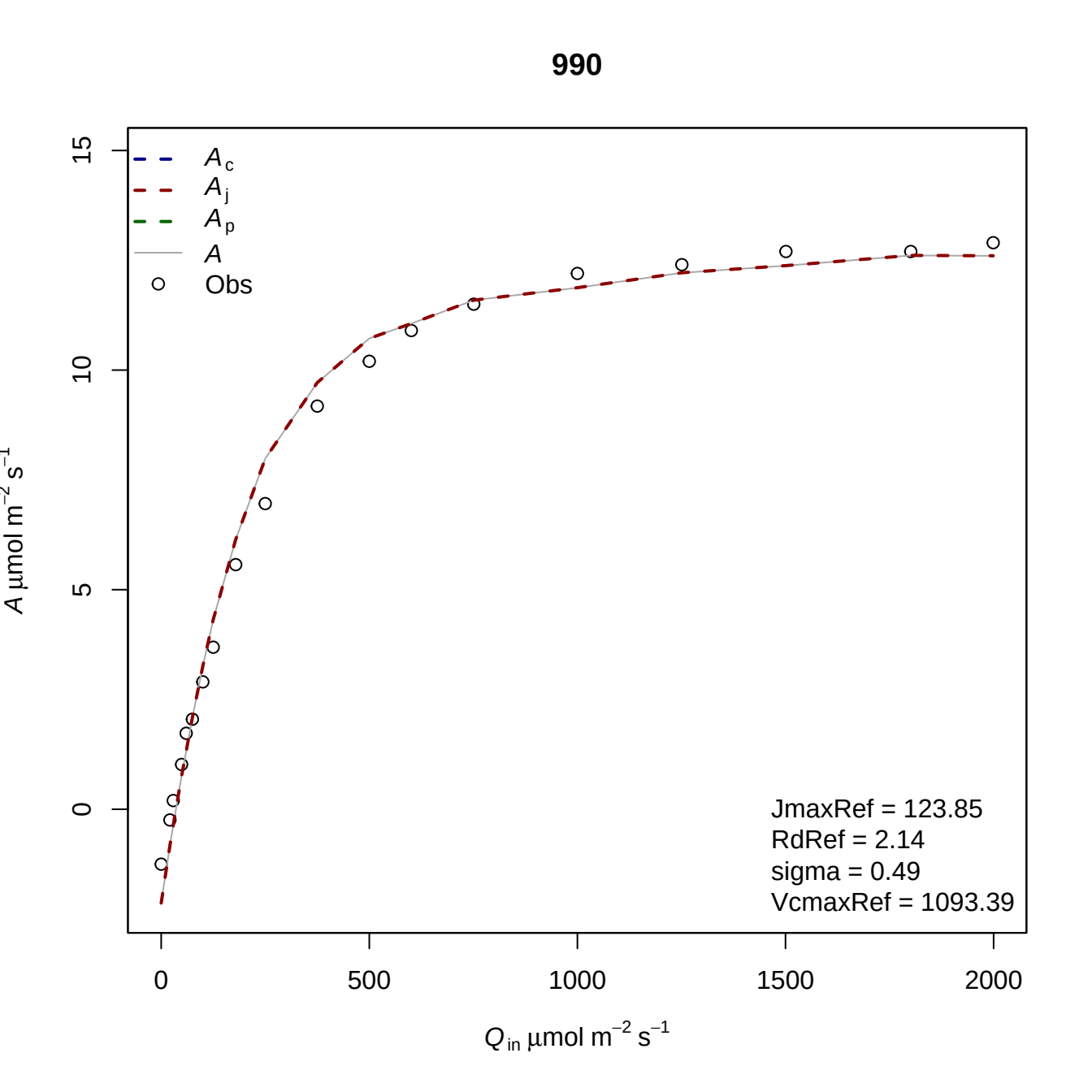

985
A_c
A_j
A_p
A
Obs
JmaxRef = 105.67
RdRef = 2.18
sigma = 0.26
VcmaxRef = 2430.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

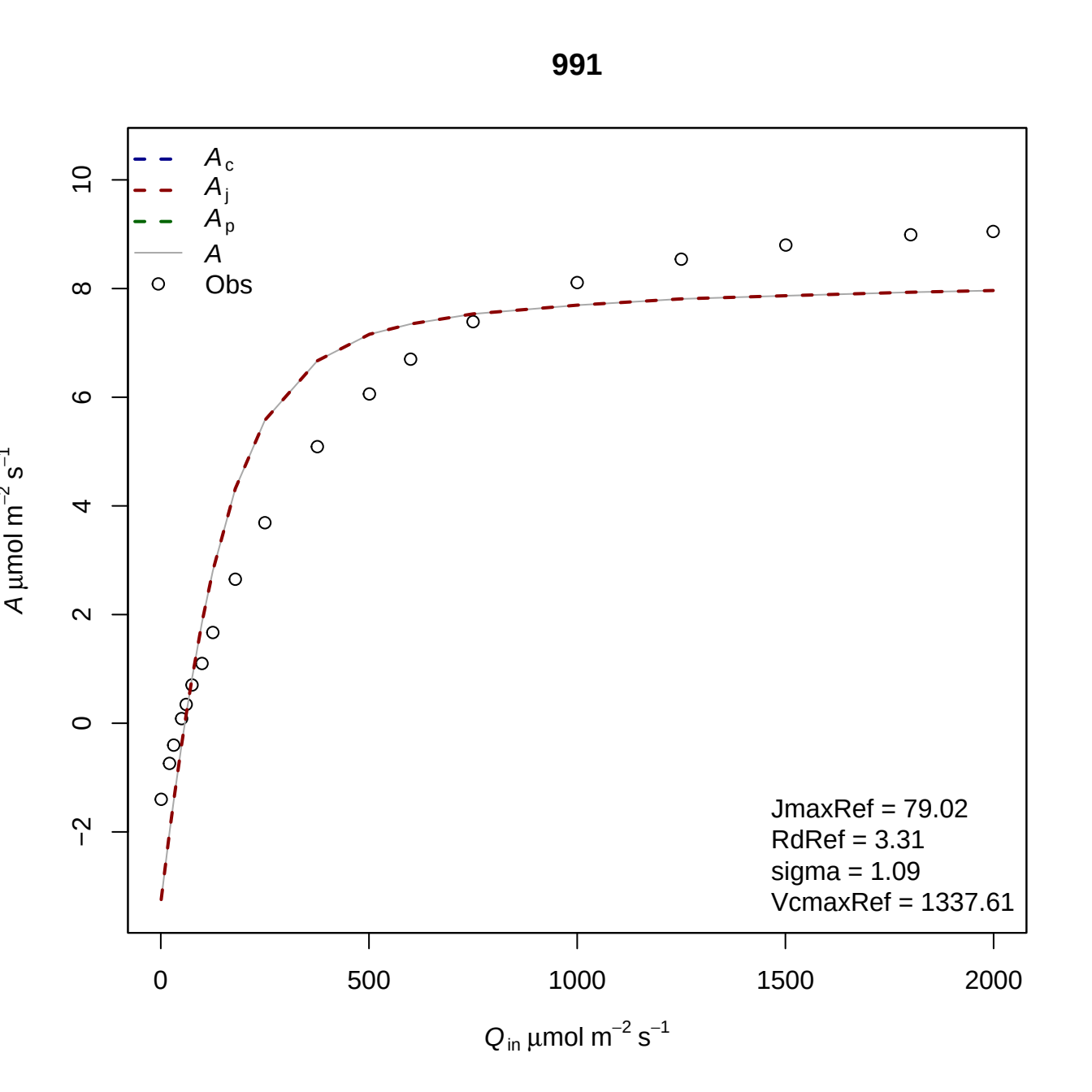

986
A_c
A_j
A_p
A
Obs
JmaxRef = 62.9
RdRef = 1.44
sigma = 0.26
VcmaxRef = 666.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

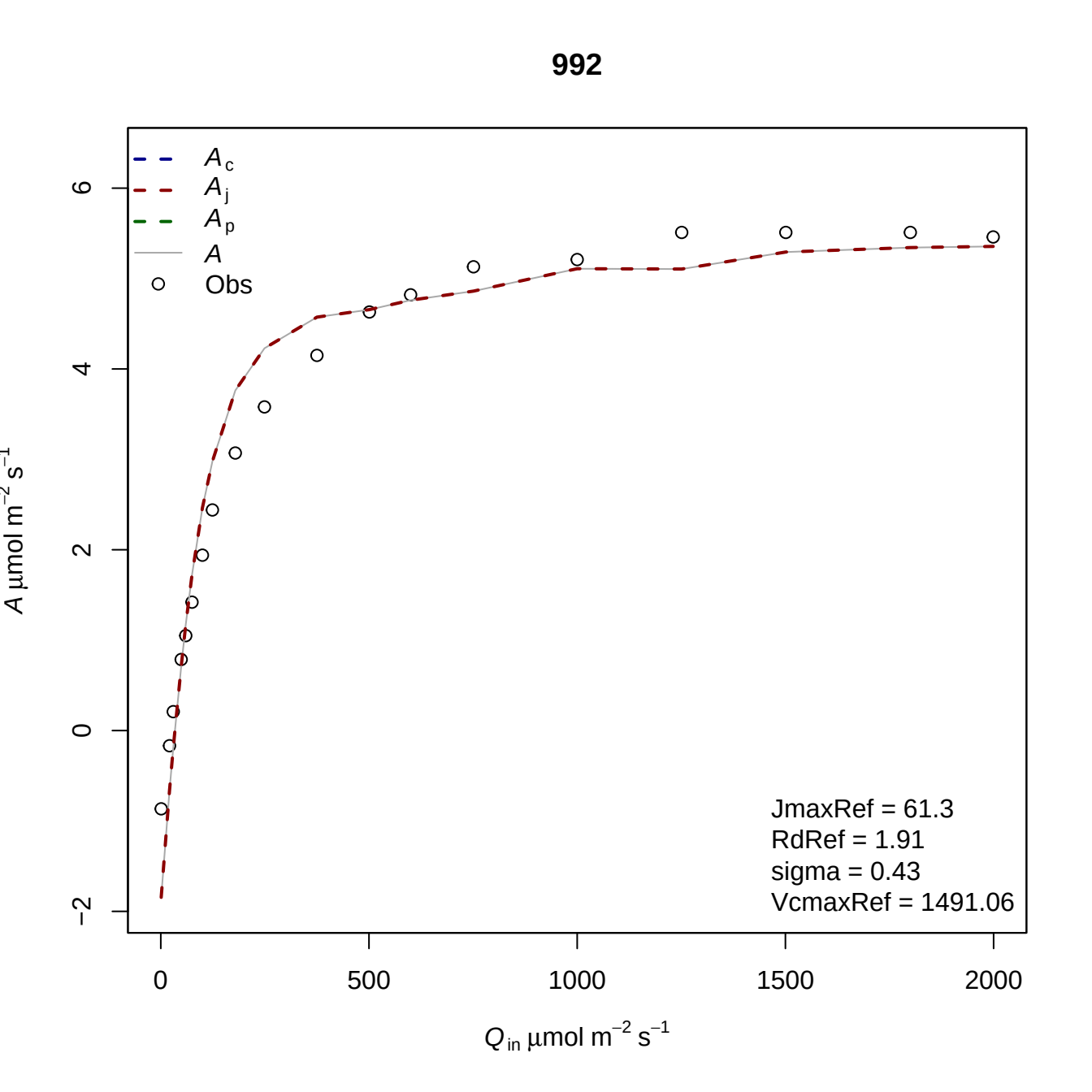

987
A_c
A_j
A_p
A
Obs
JmaxRef = 124.2
RdRef = 2.78
sigma = 0.22
VcmaxRef = 720.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

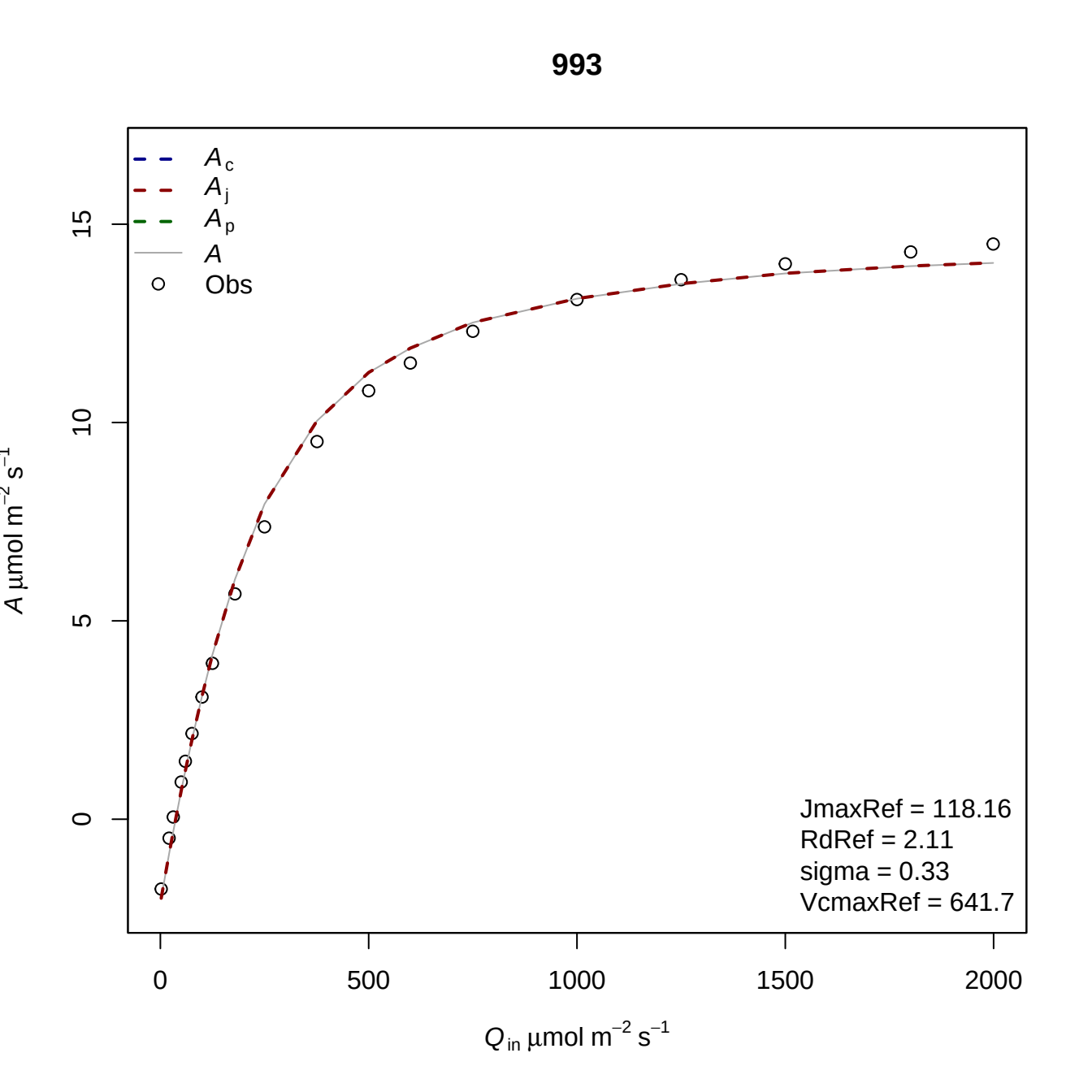

988
A_c
A_j
A_p
A
Obs
JmaxRef = 63.87
RdRef = 1.71
sigma = 0.4
VcmaxRef = 5976.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

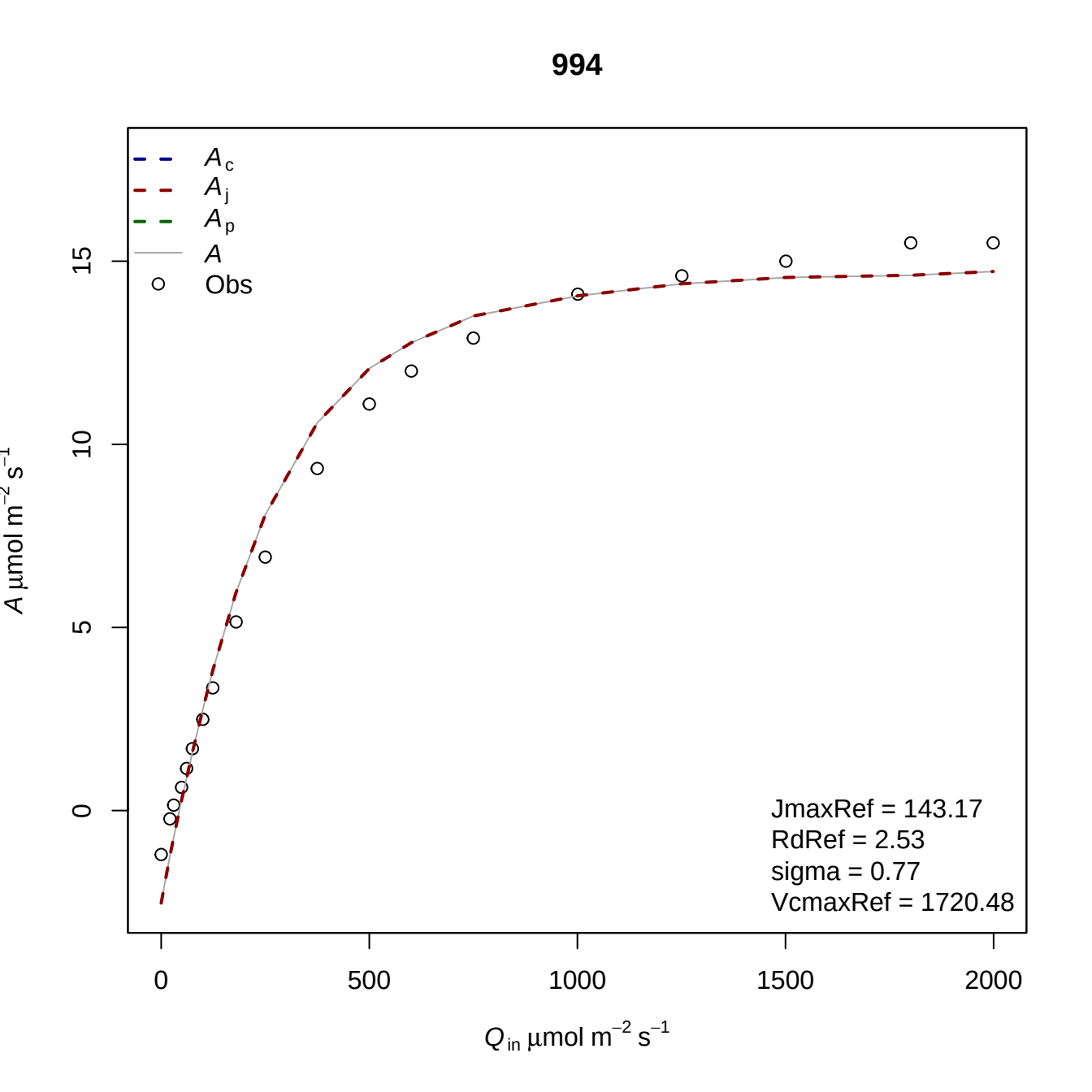

989
A_c
A_j
A_p
A
Obs
JmaxRef = 114.67
RdRef = 3.04
sigma = 0.55
VcmaxRef = 633.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

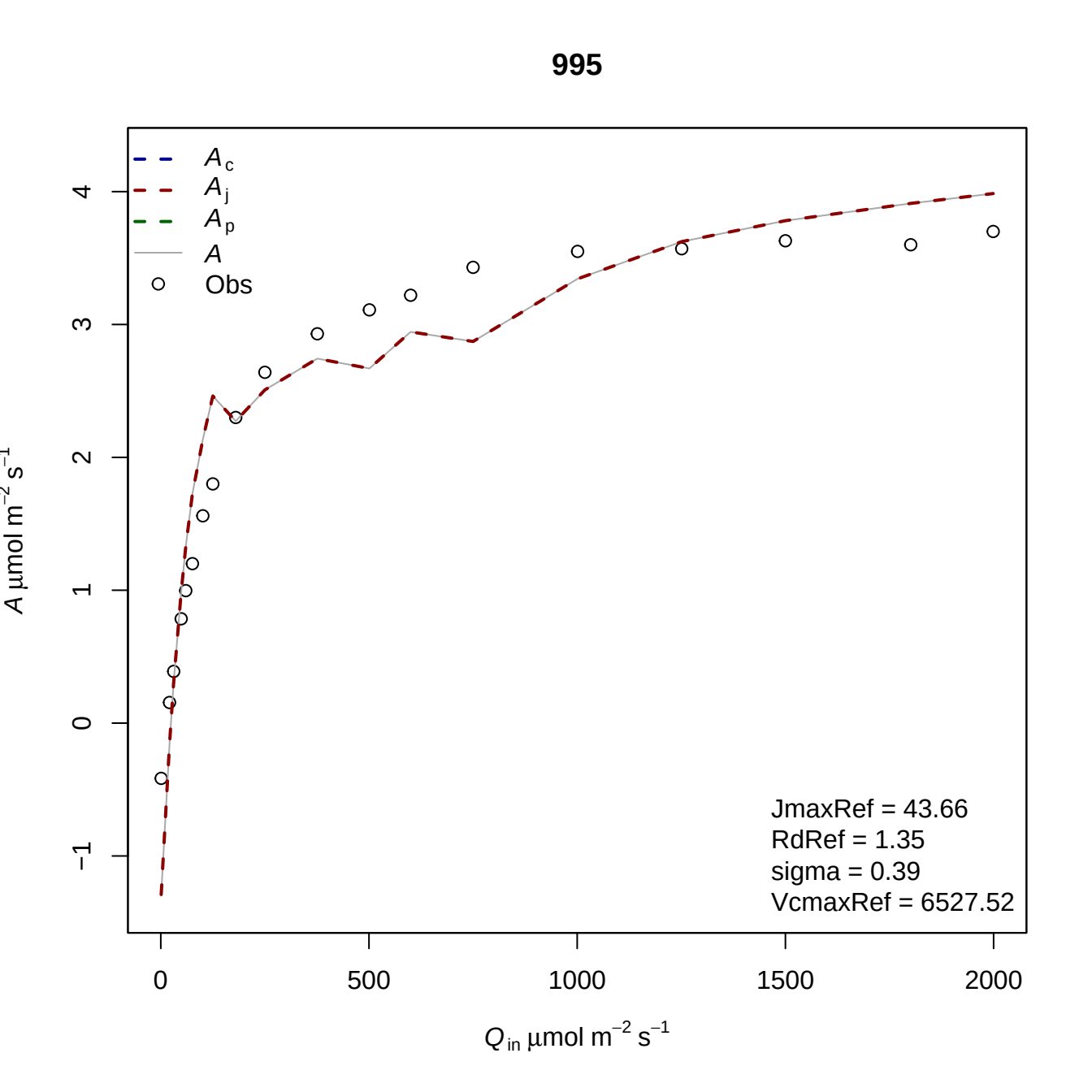

990
A_c
A_j
A_p
A
Obs
JmaxRef = 123.85
RdRef = 2.14
sigma = 0.49
VcmaxRef = 1093.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

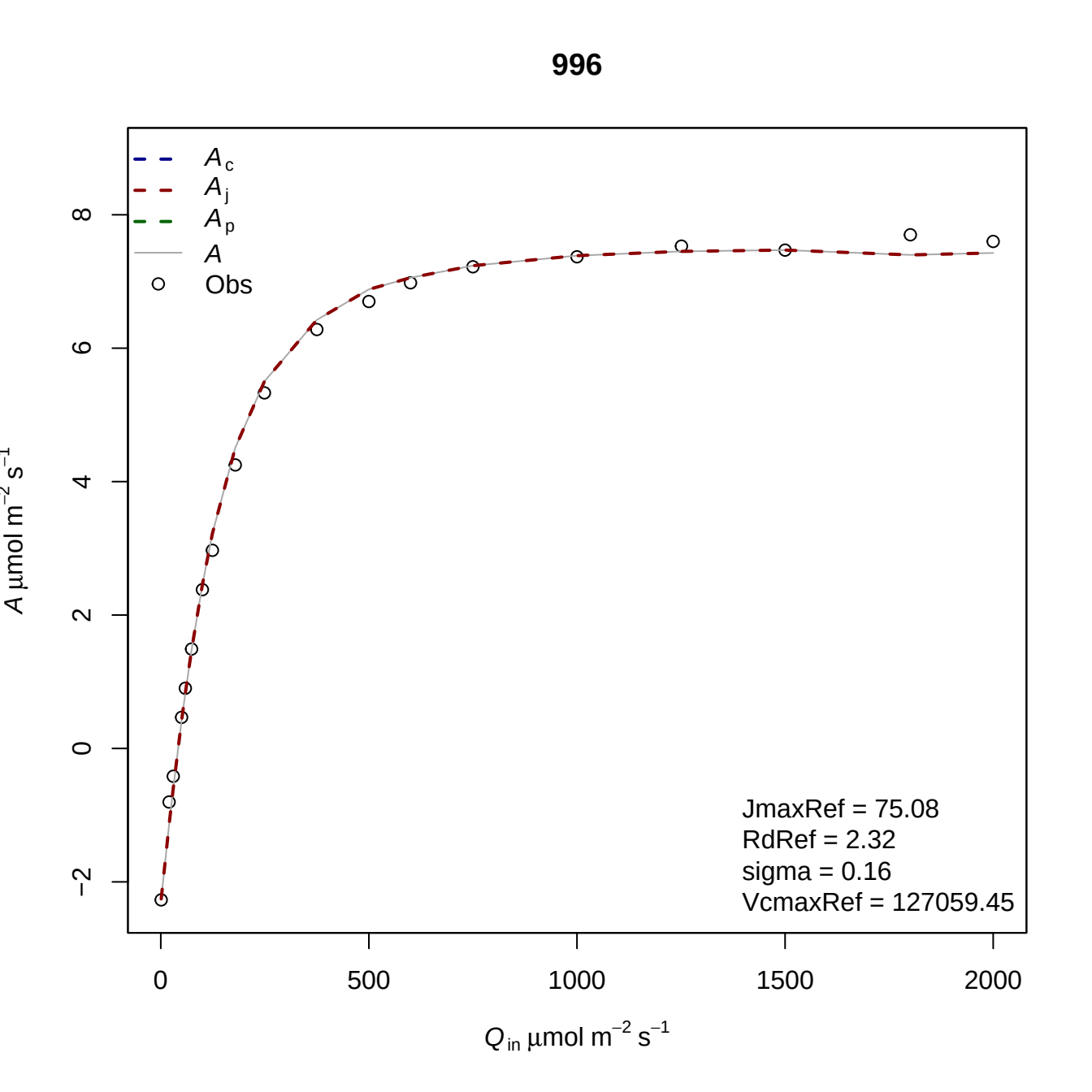

991
A_c
A_j
A_p
A
Obs
JmaxRef = 79.02
RdRef = 3.31
sigma = 1.09
VcmaxRef = 1337.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

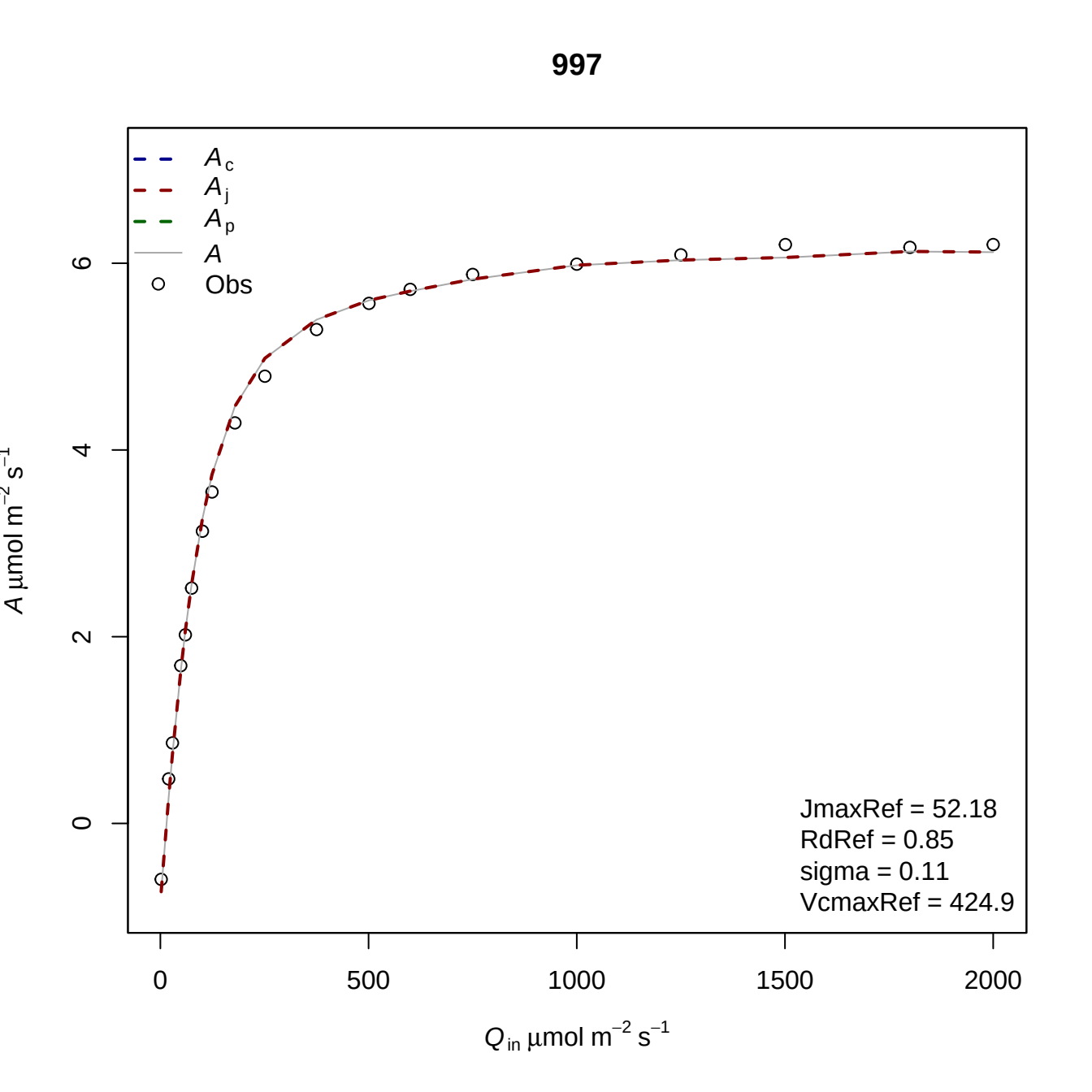

992
A_c
A_j
A_p
A
Obs
JmaxRef = 61.3
RdRef = 1.91
sigma = 0.43
VcmaxRef = 1491.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

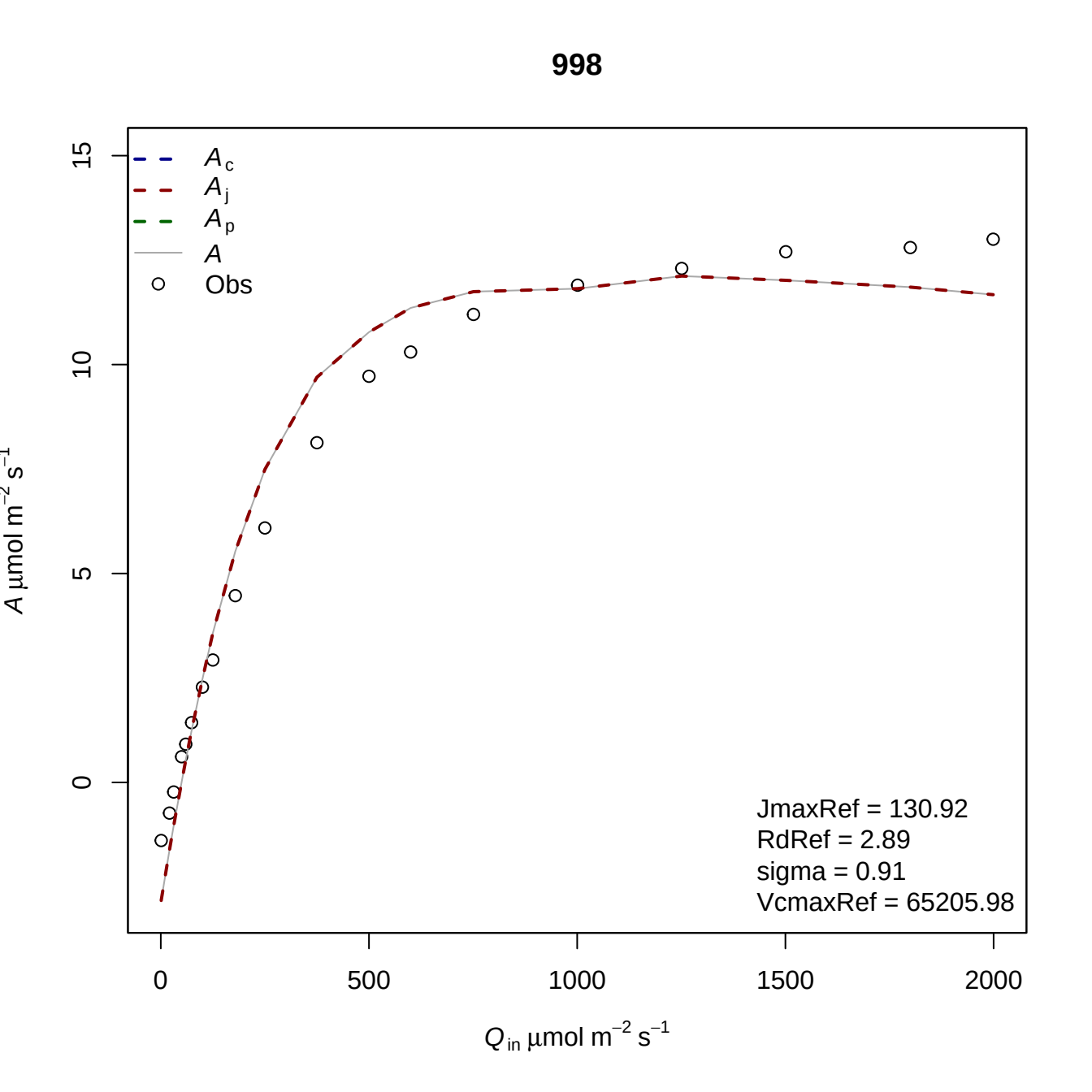

993
A_c
A_j
A_p
A
Obs
JmaxRef = 118.16
RdRef = 2.11
sigma = 0.33
VcmaxRef = 641.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

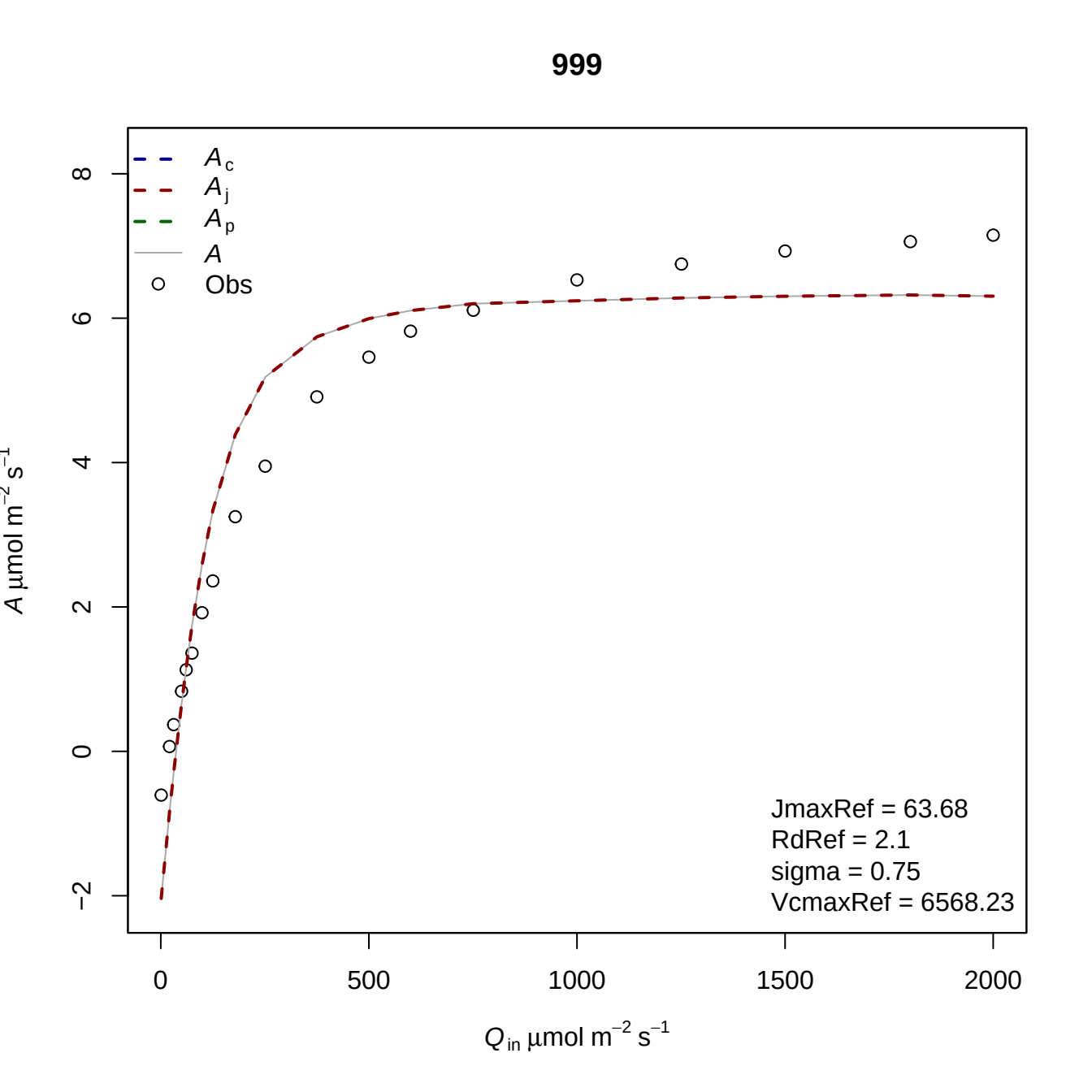

994
A_c
A_j
A_p
A
Obs
JmaxRef = 143.17
RdRef = 2.53
sigma = 0.77
VcmaxRef = 1720.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

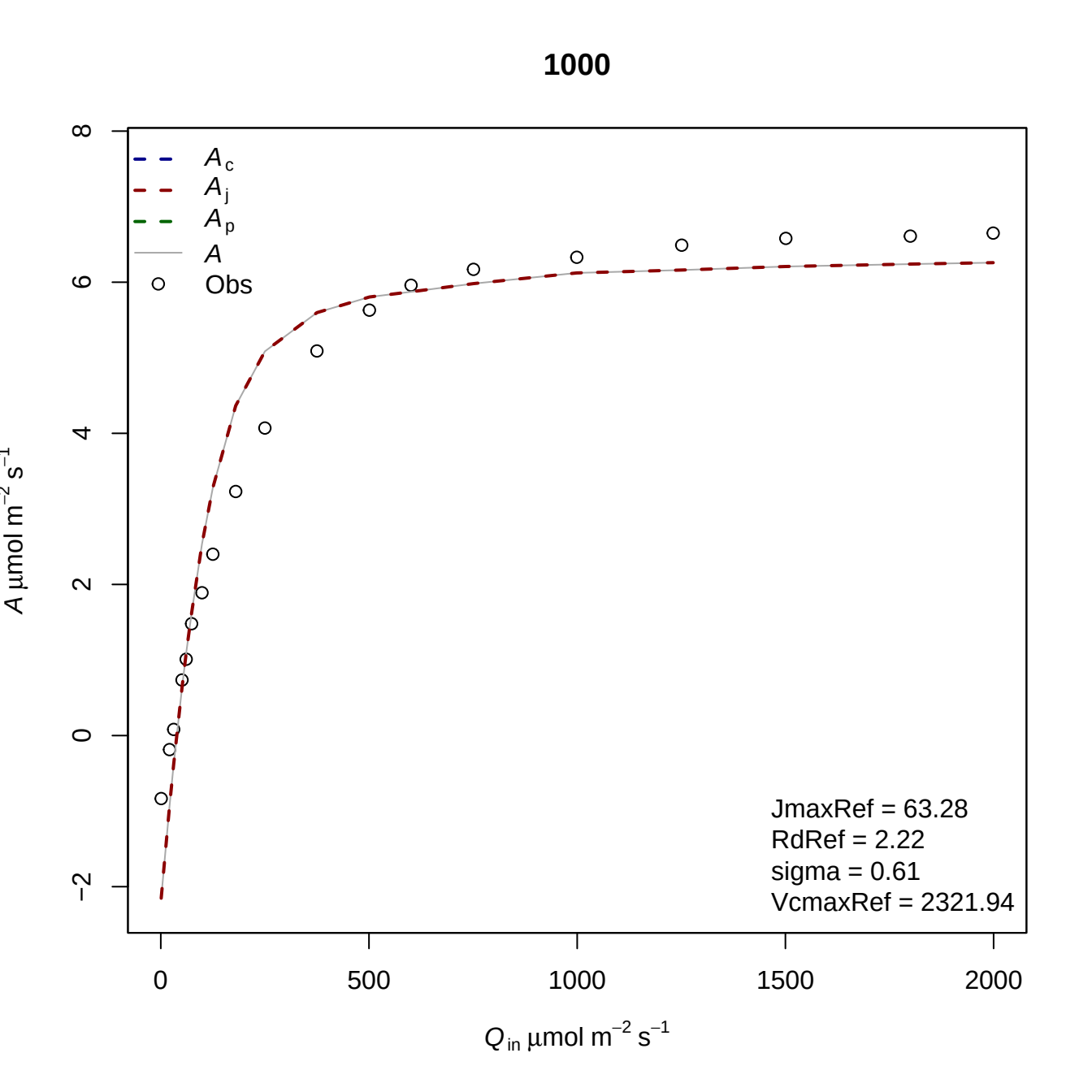

995
A_c
A_j
A_p
A
Obs
JmaxRef = 43.66
RdRef = 1.35
sigma = 0.39
VcmaxRef = 6527.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

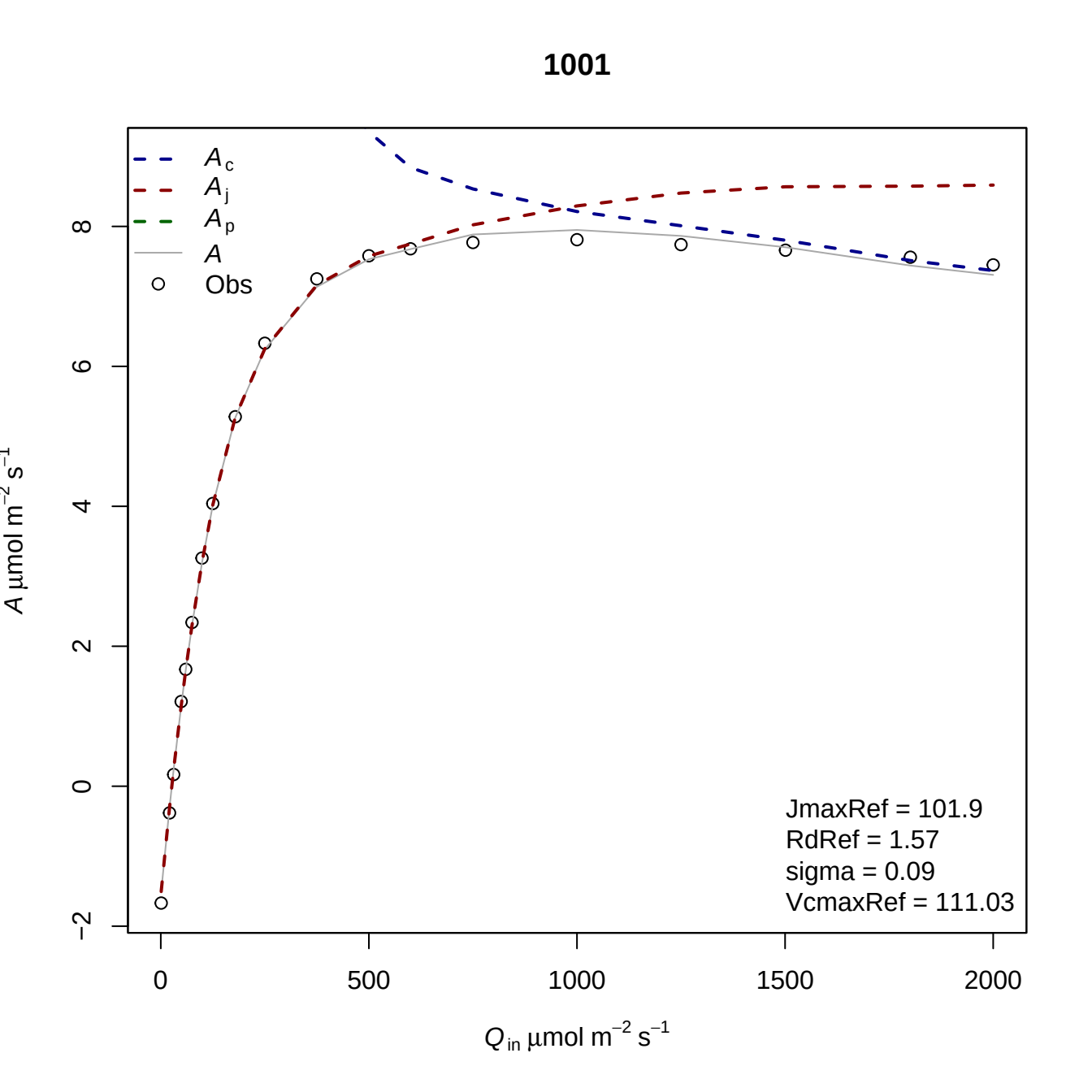

996
A_c
A_j
A_p
A
Obs
JmaxRef = 75.08
RdRef = 2.32
sigma = 0.16
VcmaxRef = 127059.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

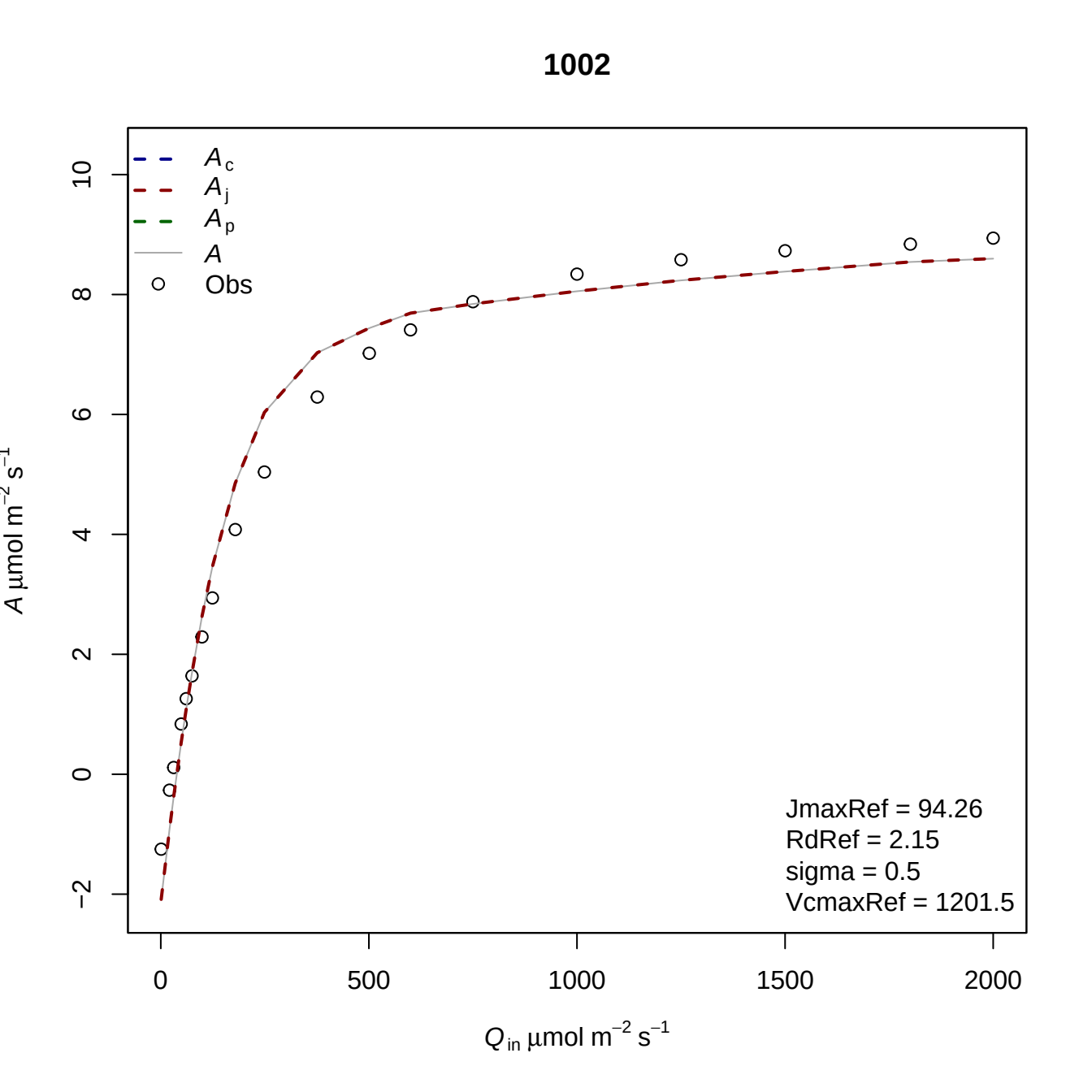

997
A_c
A_j
A_p
A
Obs
JmaxRef = 52.18
RdRef = 0.85
sigma = 0.11
VcmaxRef = 424.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

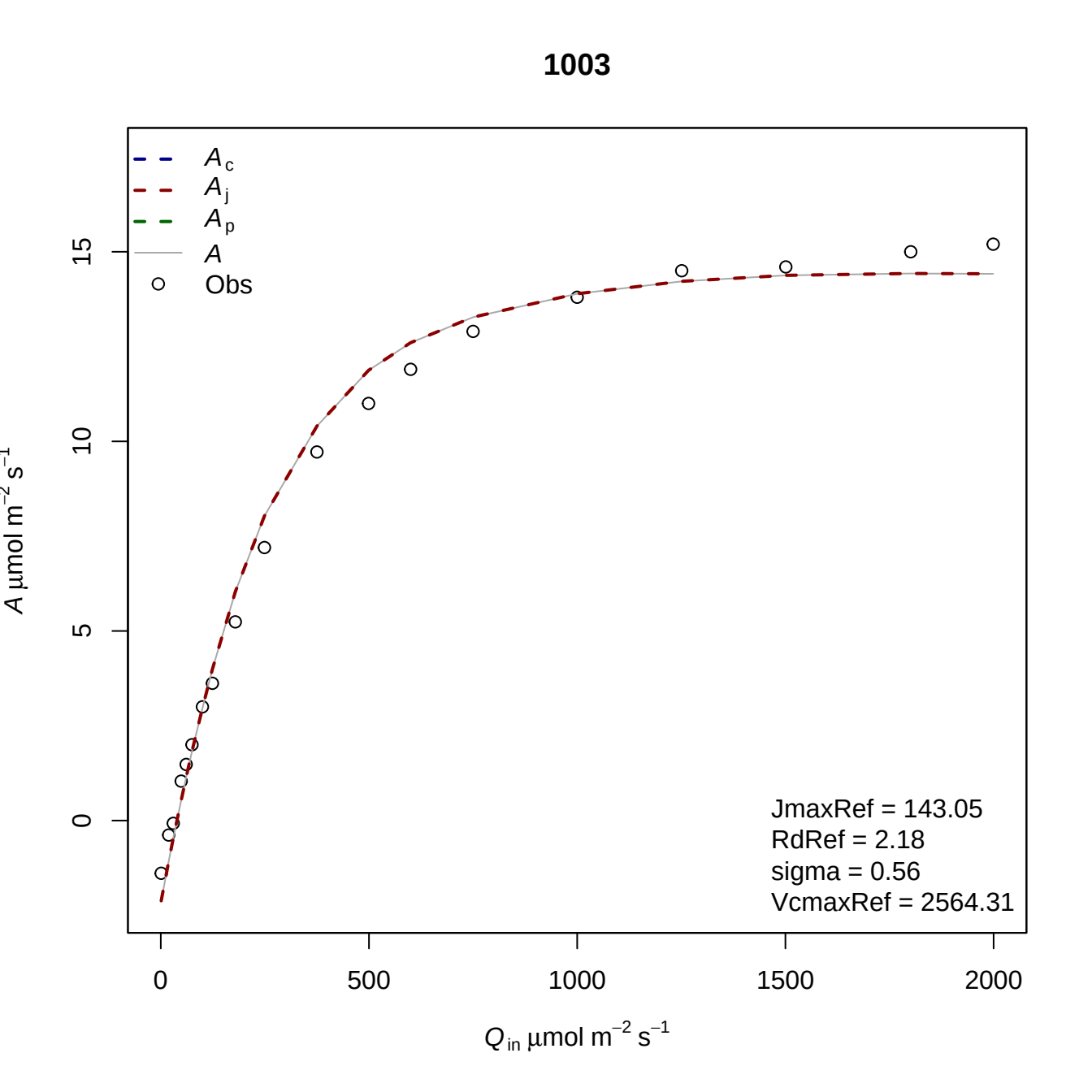

998
A_c
A_j
A_p
A
Obs
JmaxRef = 130.92
RdRef = 2.89
sigma = 0.91
VcmaxRef = 65205.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

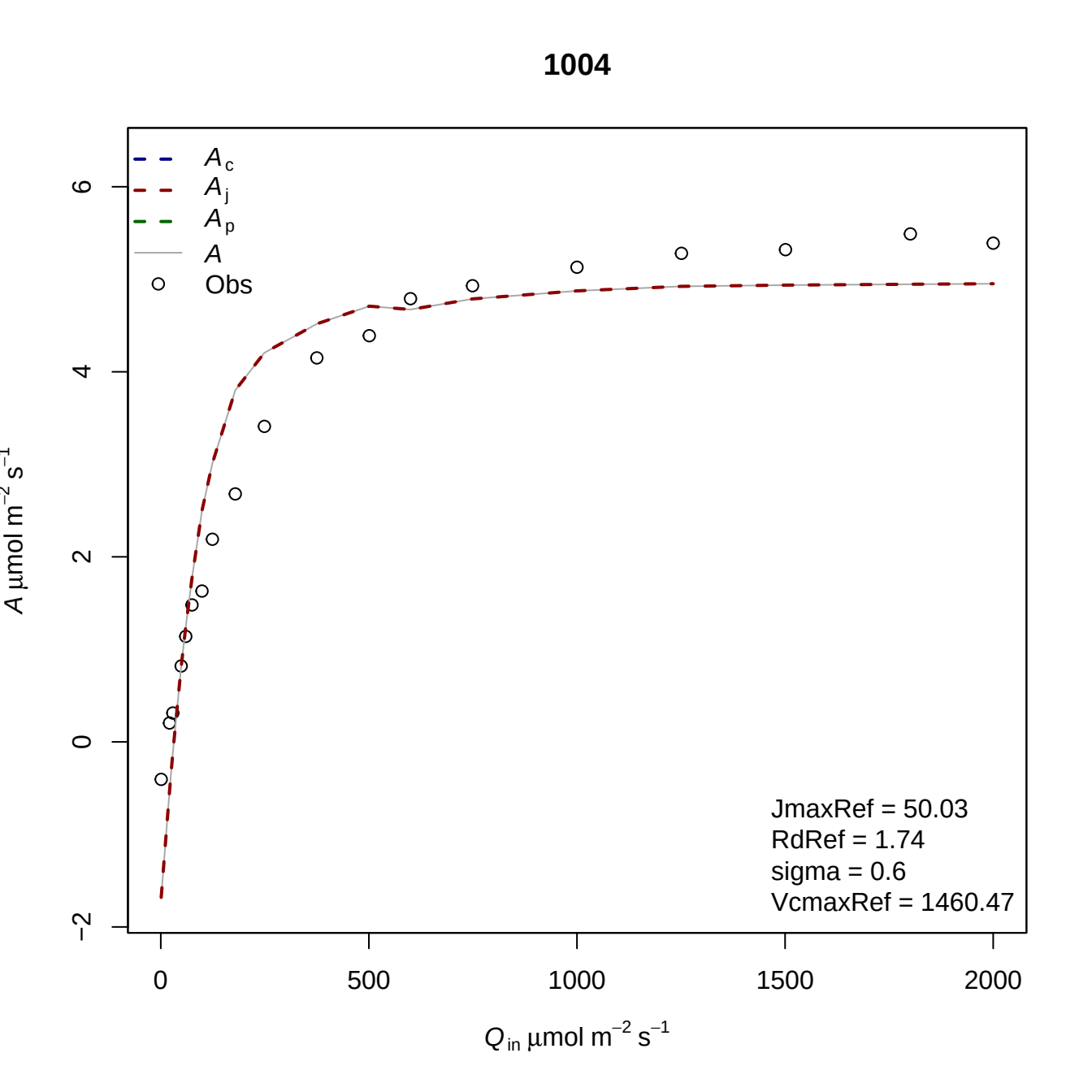

999
A_c
A_j
A_p
A
Obs
JmaxRef = 63.68
RdRef = 2.1
sigma = 0.75
VcmaxRef = 6568.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

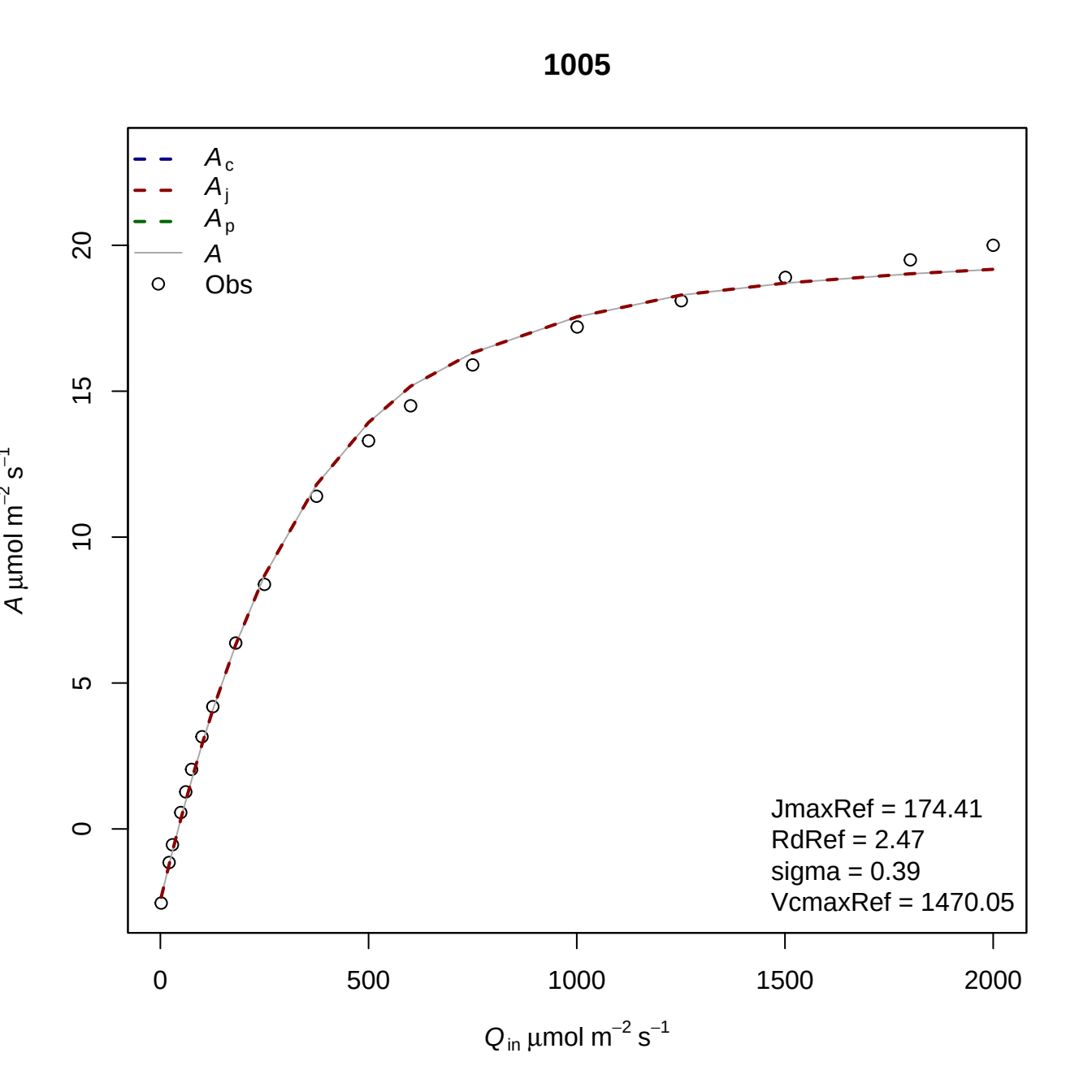

1000
A_c
A_j
A_p
A
Obs
JmaxRef = 63.28
RdRef = 2.22
sigma = 0.61
VcmaxRef = 2321.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

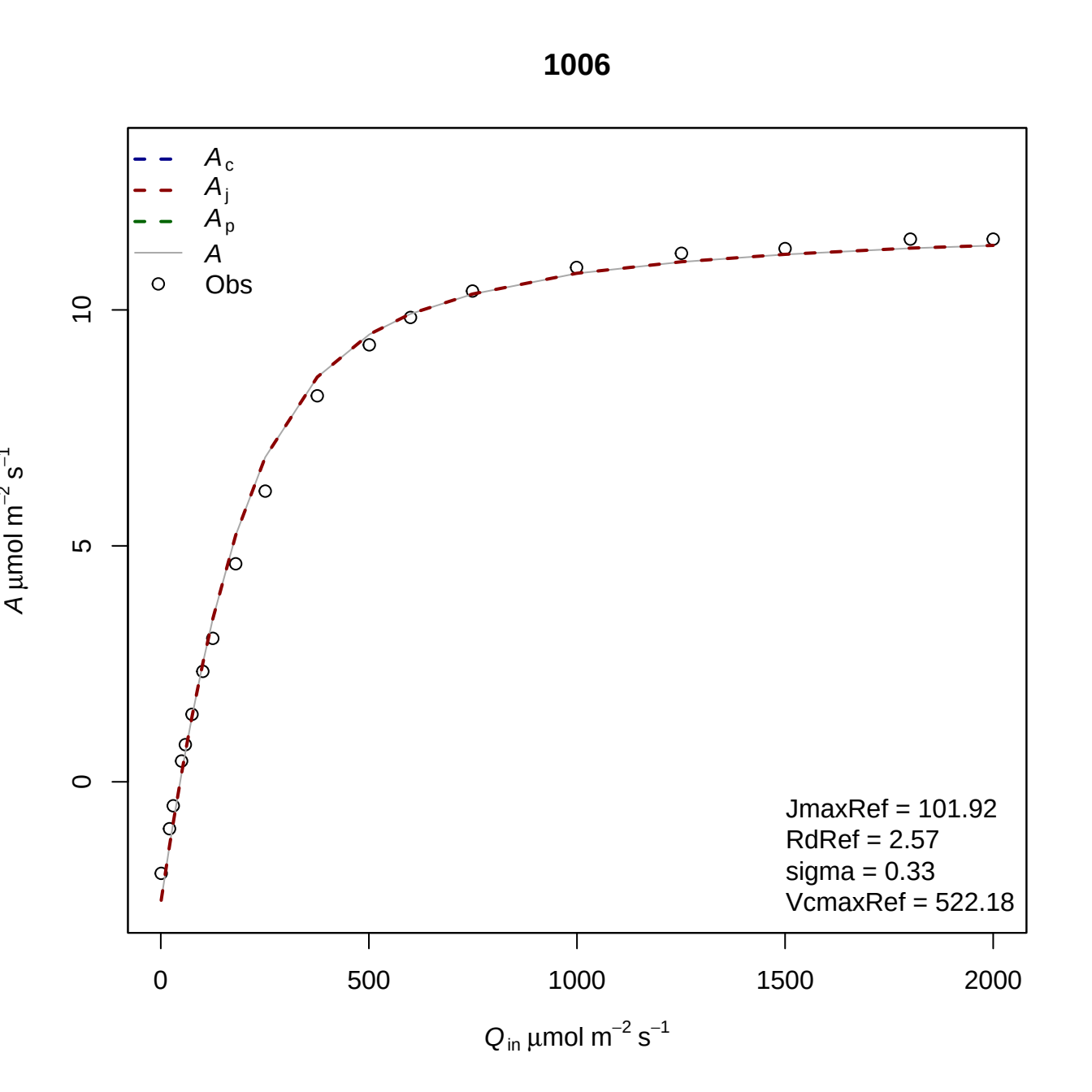

1001
A_c
A_j
A_p
A
Obs
JmaxRef = 101.9
RdRef = 1.57
sigma = 0.09
VcmaxRef = 111.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

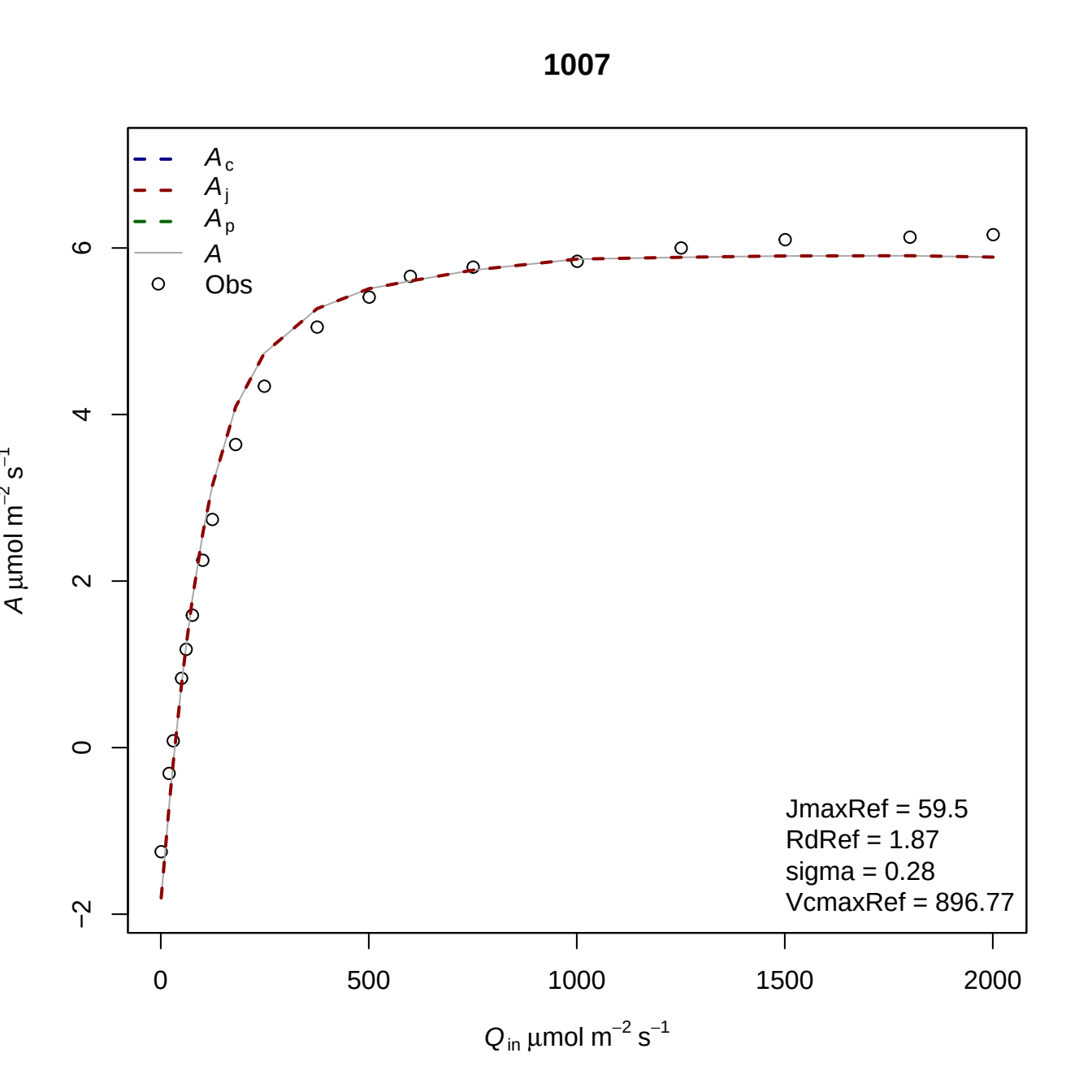

1002
A_c
A_j
A_p
A
Obs
JmaxRef = 94.26
RdRef = 2.15
sigma = 0.5
VcmaxRef = 1201.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

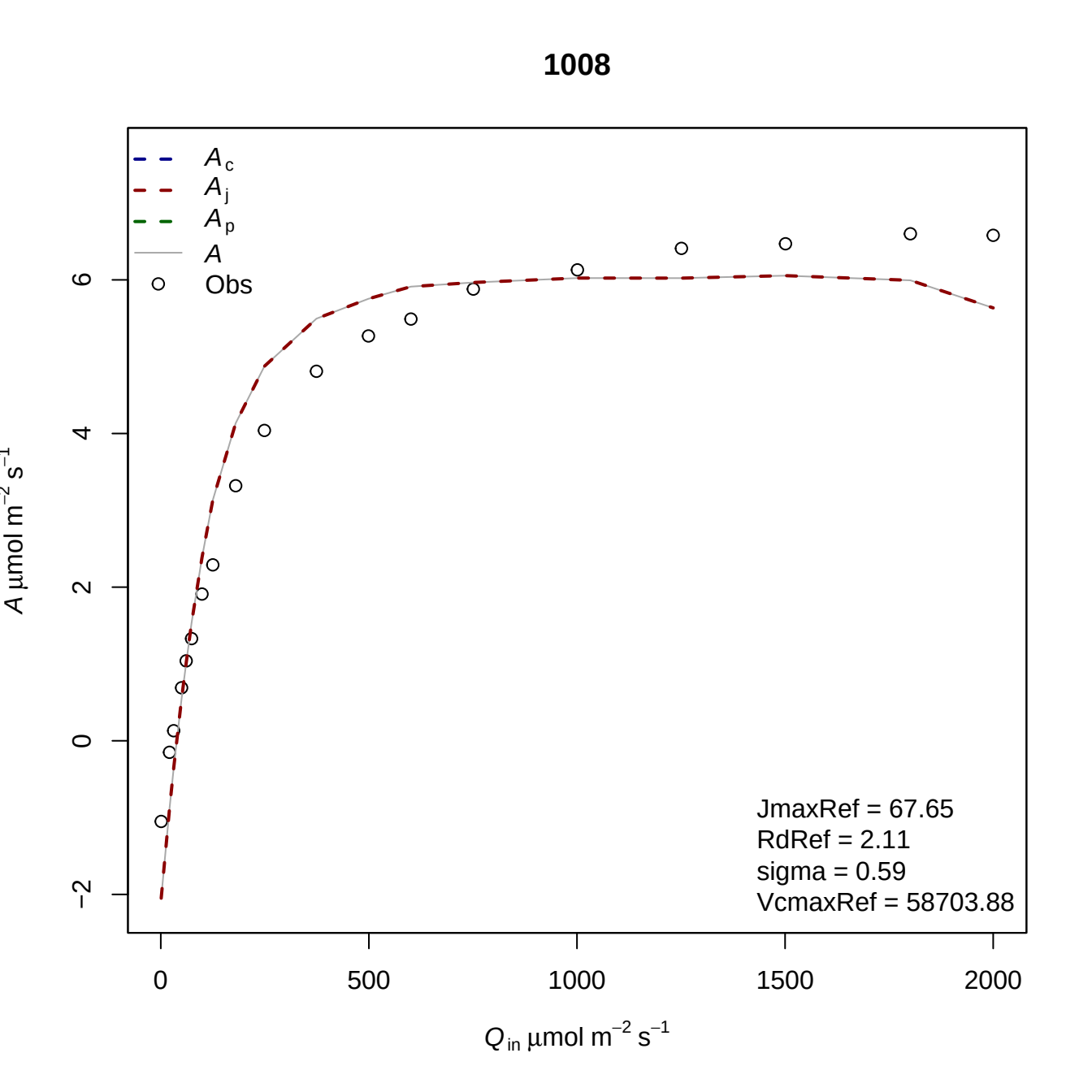

1003
A_c
A_j
A_p
A
Obs
JmaxRef = 143.05
RdRef = 2.18
sigma = 0.56
VcmaxRef = 2564.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

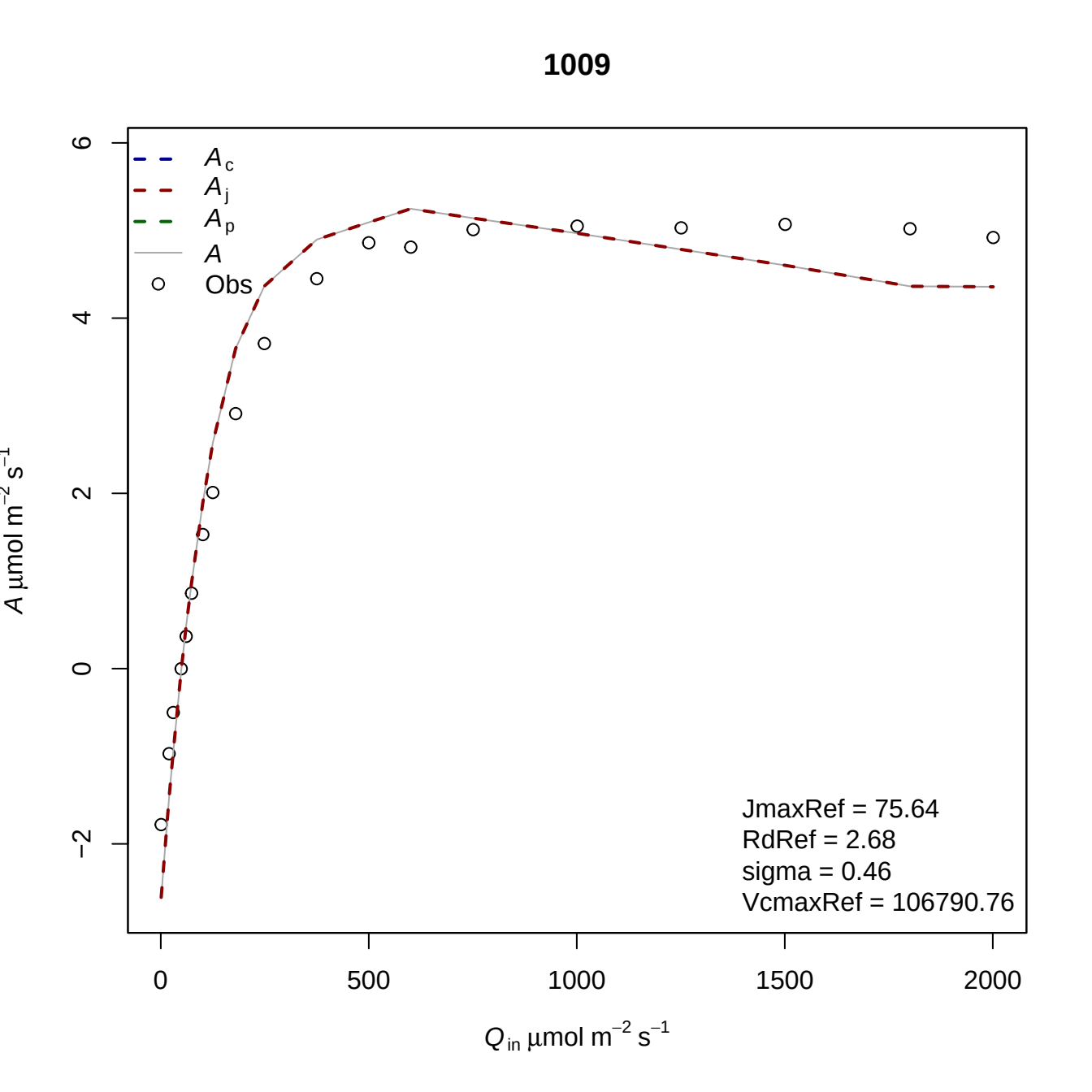

1004
A_c
A_j
A_p
A
Obs
JmaxRef = 50.03
RdRef = 1.74
sigma = 0.6
VcmaxRef = 1460.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

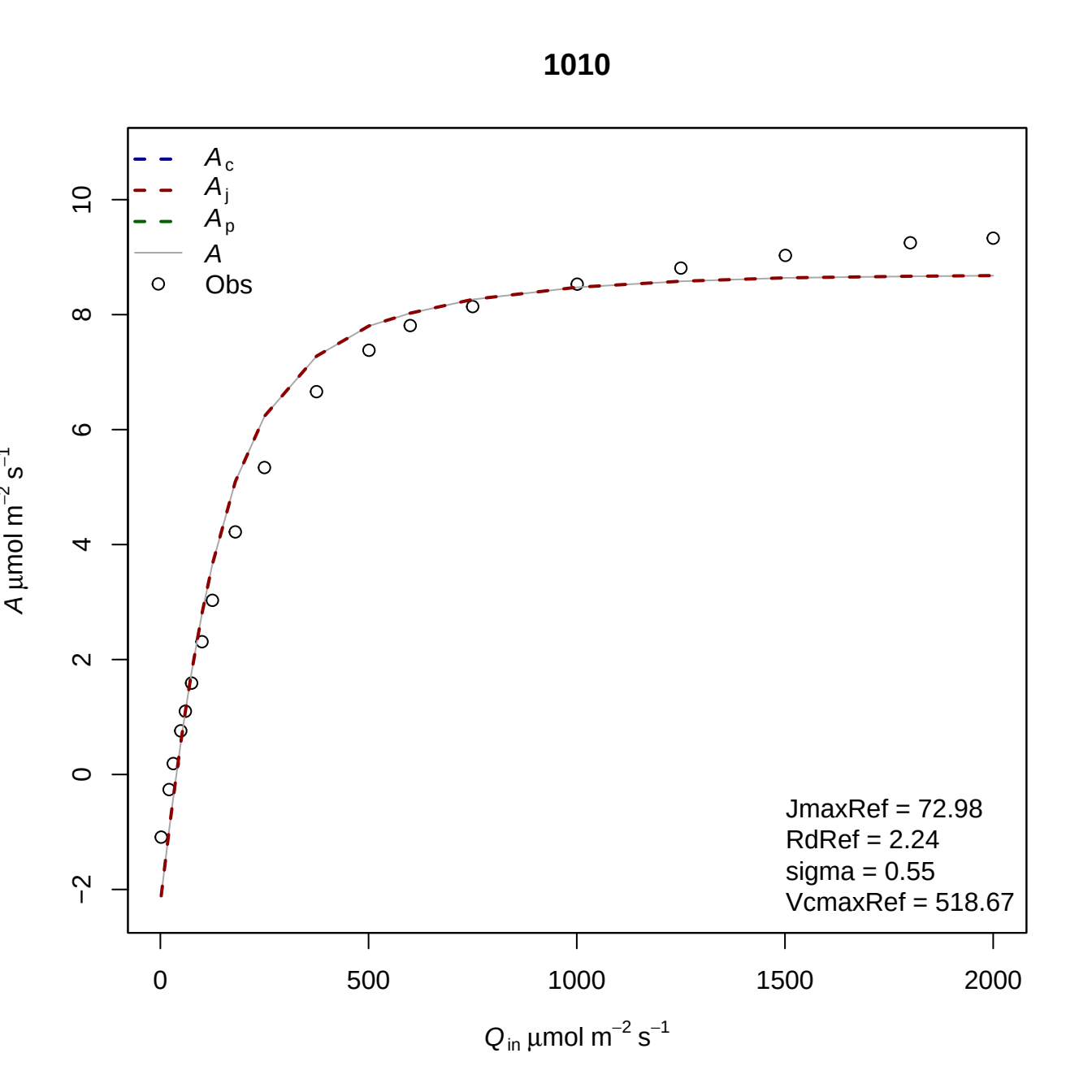

1005
A_c
A_j
A_p
A
Obs
JmaxRef = 174.41
RdRef = 2.47
sigma = 0.39
VcmaxRef = 1470.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

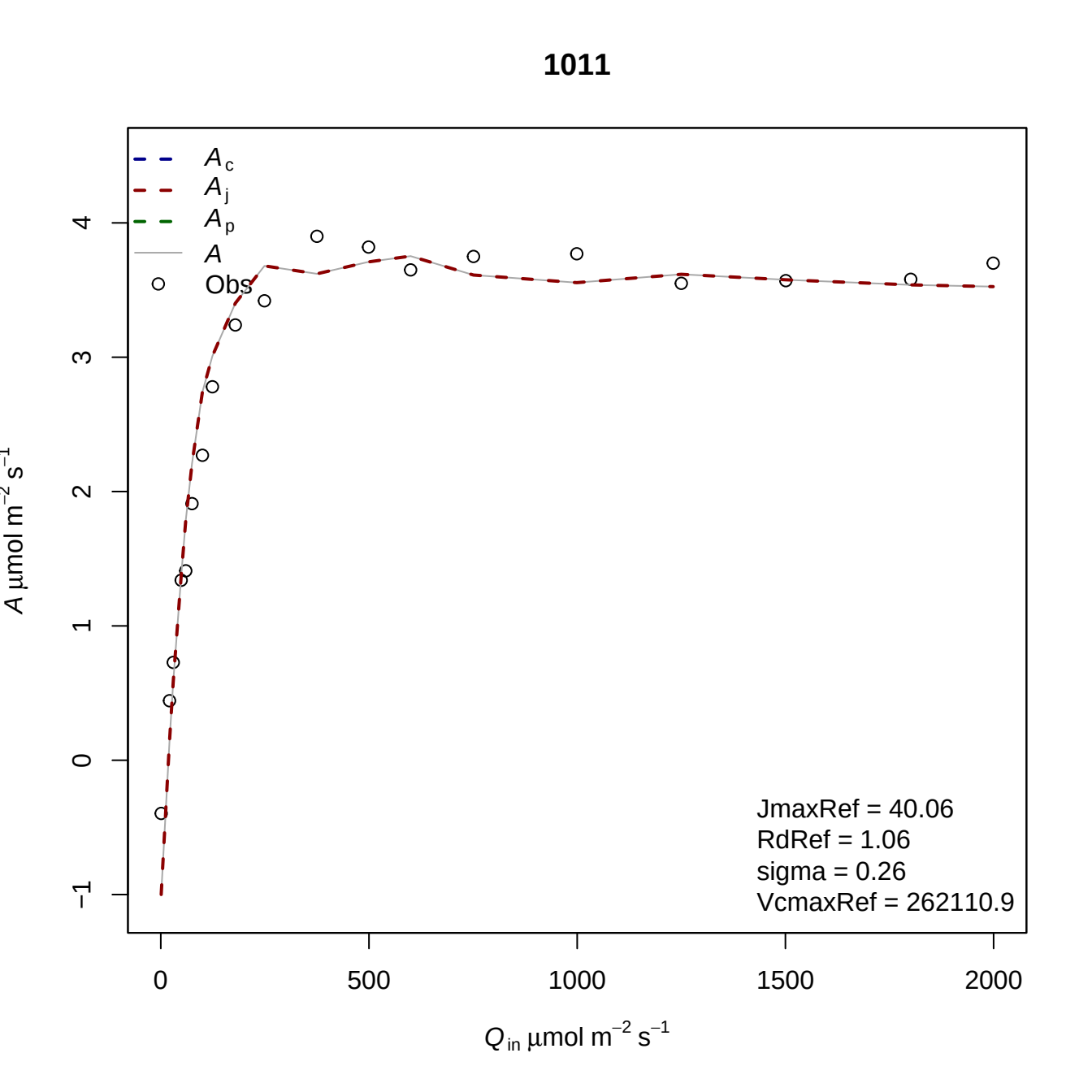

1006
A_c
A_j
A_p
A
Obs
JmaxRef = 101.92
RdRef = 2.57
sigma = 0.33
VcmaxRef = 522.18
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

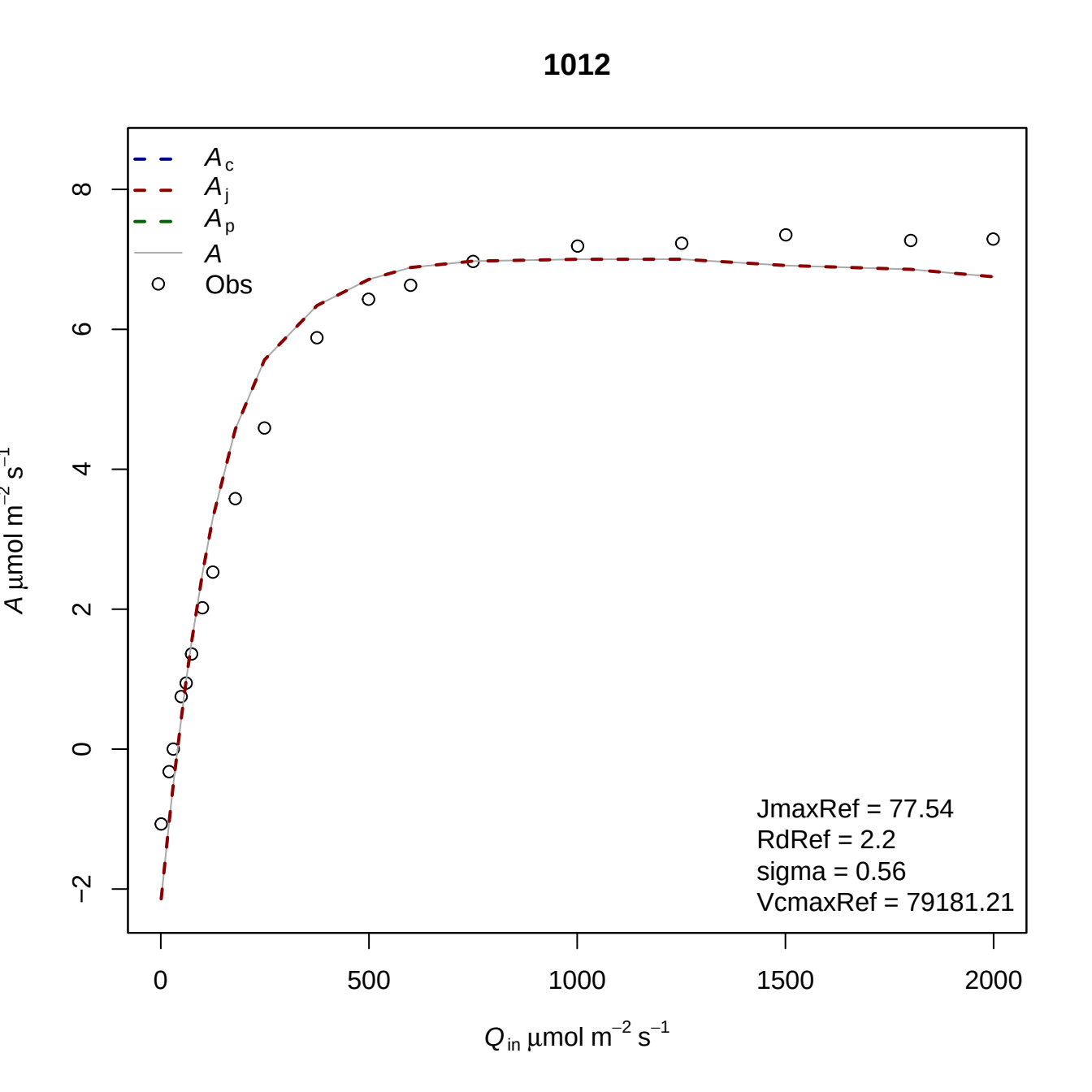

1007
A_c
A_j
A_p
A
Obs
JmaxRef = 59.5
RdRef = 1.87
sigma = 0.28
VcmaxRef = 896.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

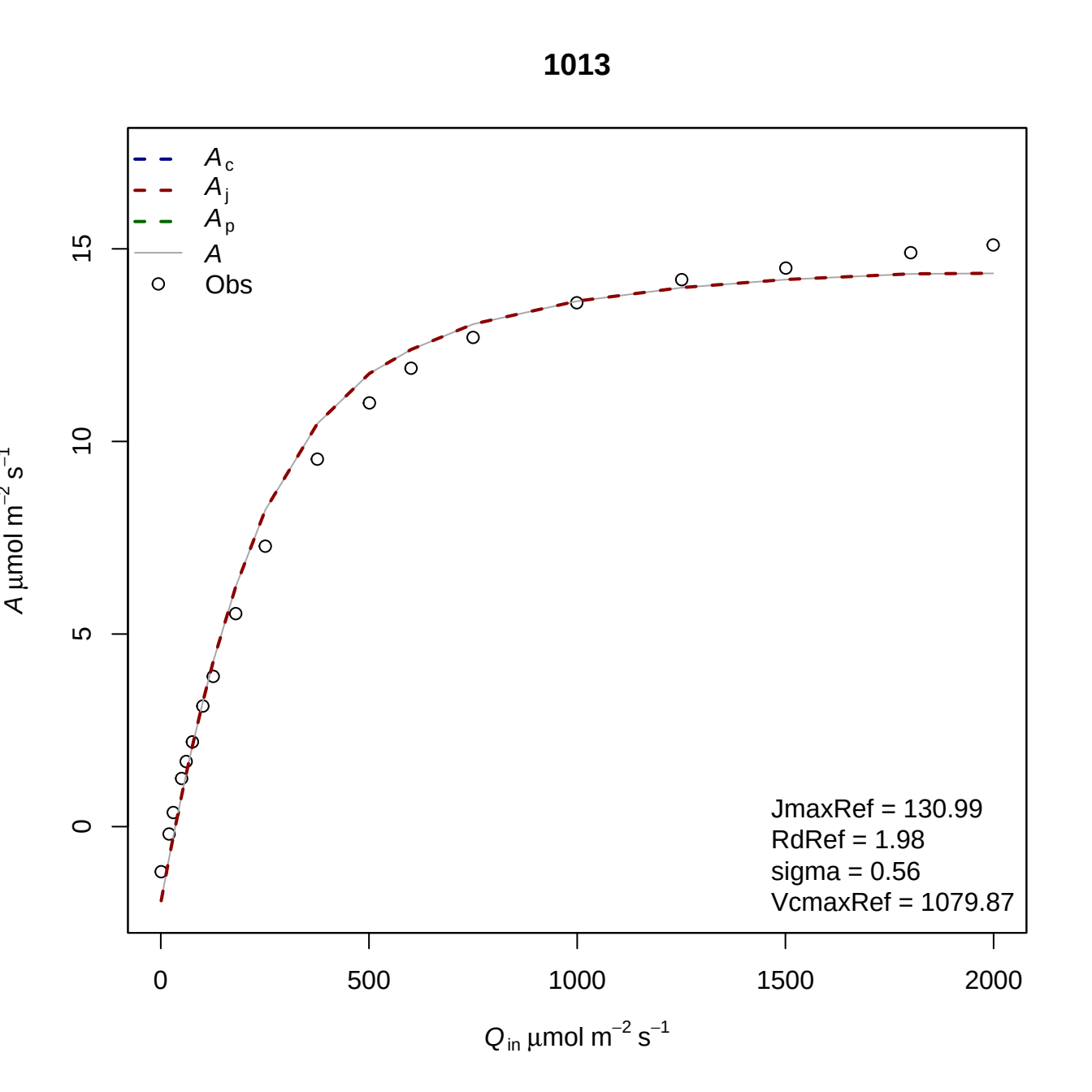

1008
A_c
A_j
A_p
A
Obs
JmaxRef = 67.65
RdRef = 2.11
sigma = 0.59
VcmaxRef = 58703.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

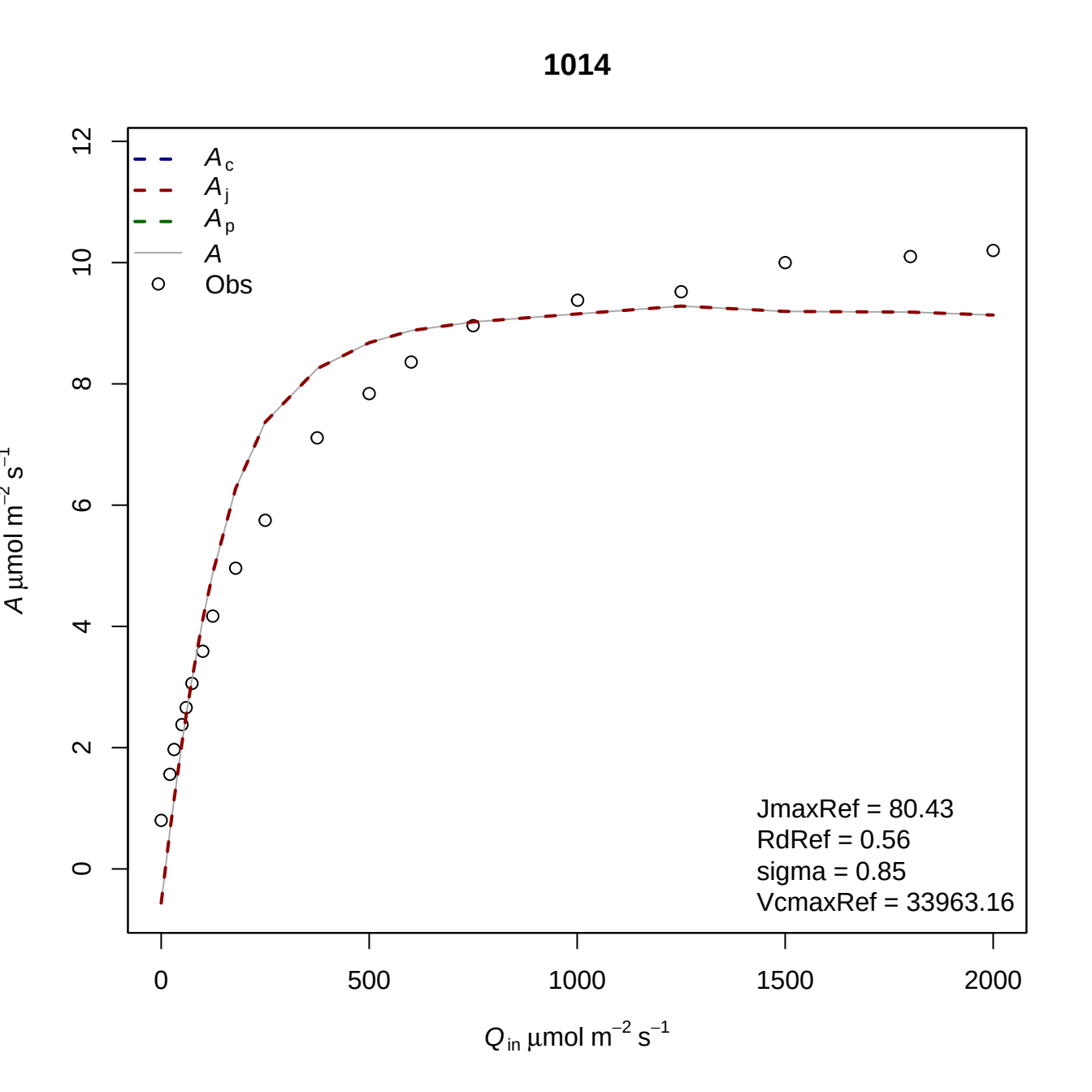

1009
A_c
A_j
A_p
A
Obs
JmaxRef = 75.64
RdRef = 2.68
sigma = 0.46
VcmaxRef = 106790.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

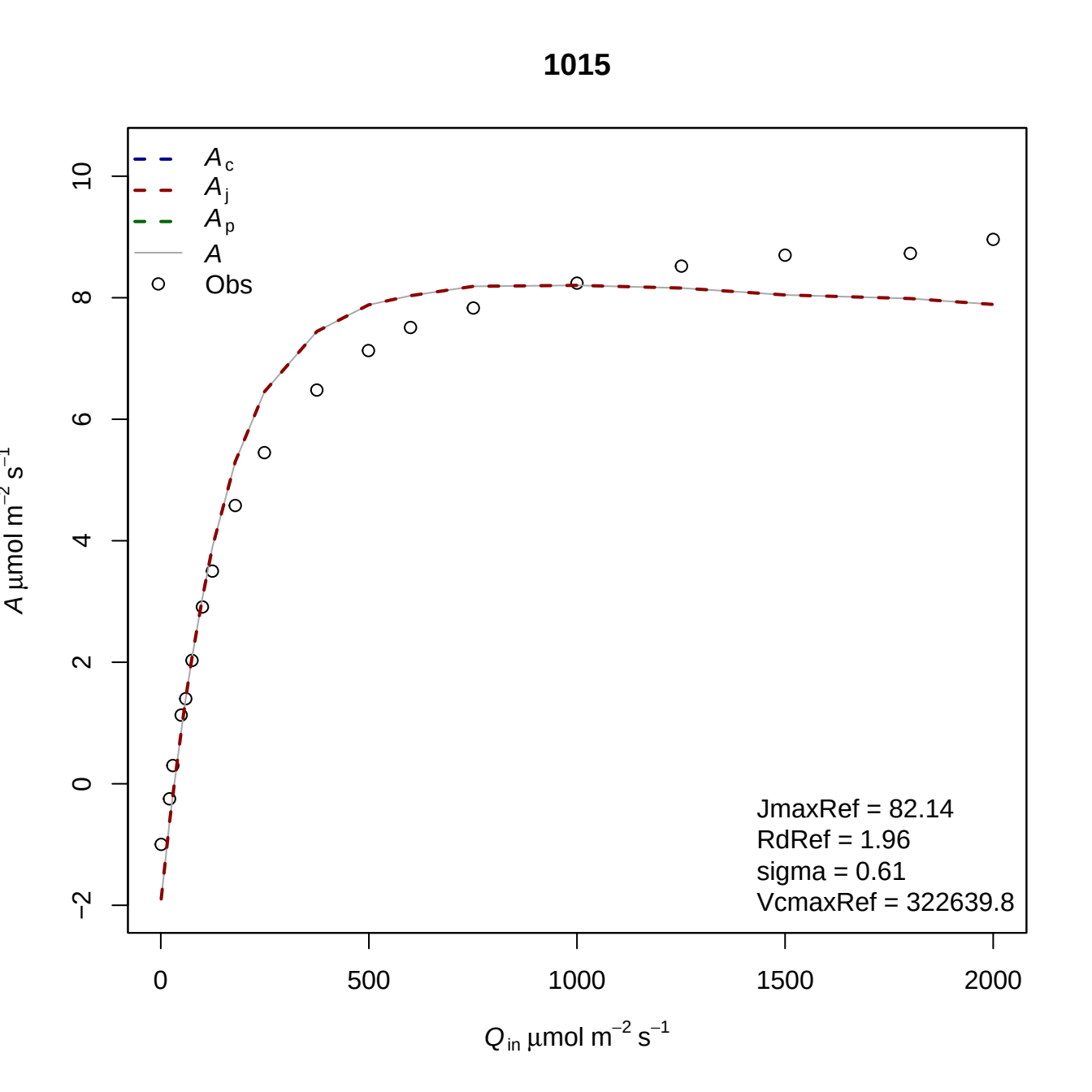

1010
A_c
A_j
A_p
A
Obs
JmaxRef = 72.98
RdRef = 2.24
sigma = 0.55
VcmaxRef = 518.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

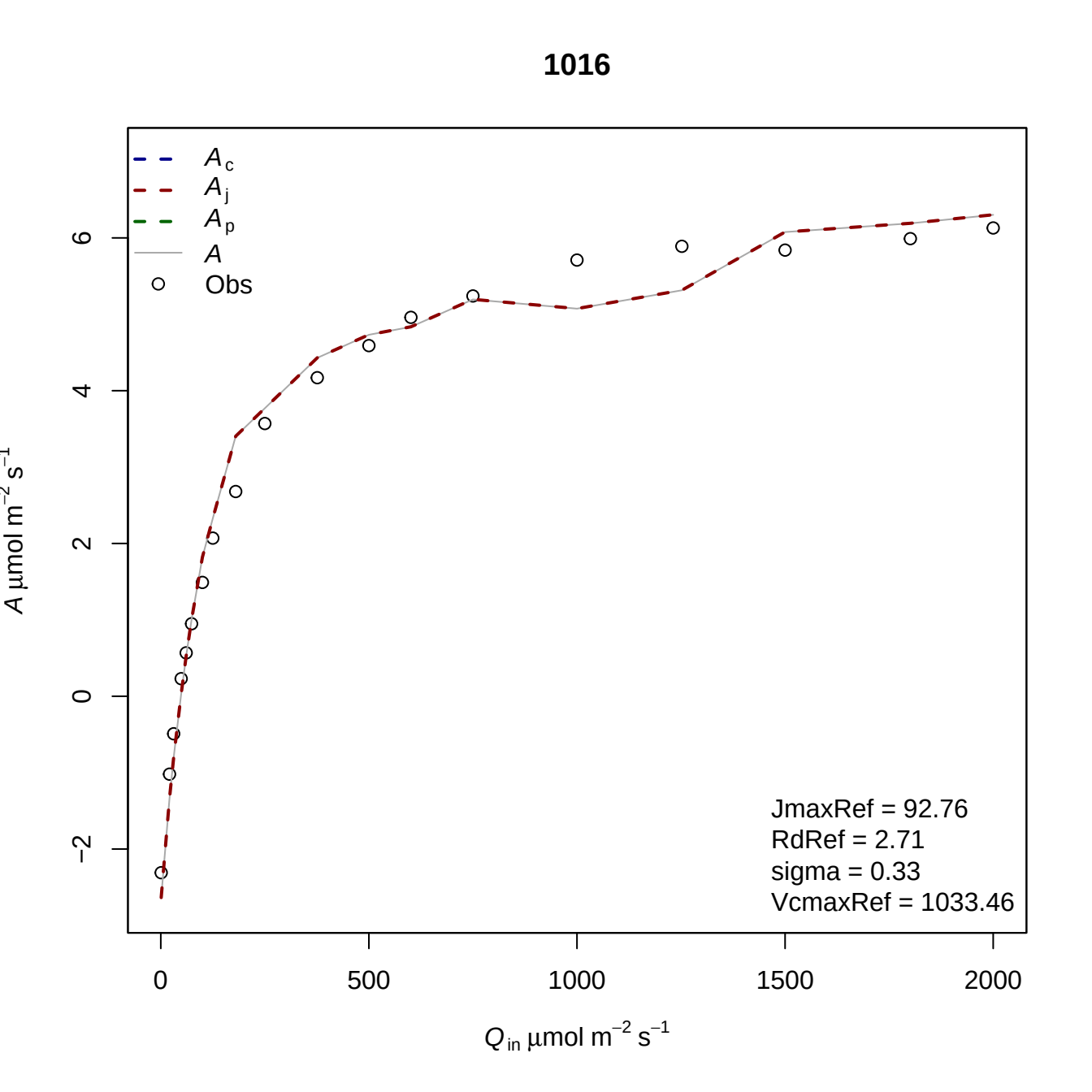

1011
A_c
A_j
A_p
A
Obs
JmaxRef = 40.06
RdRef = 1.06
sigma = 0.26
VcmaxRef = 262110.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

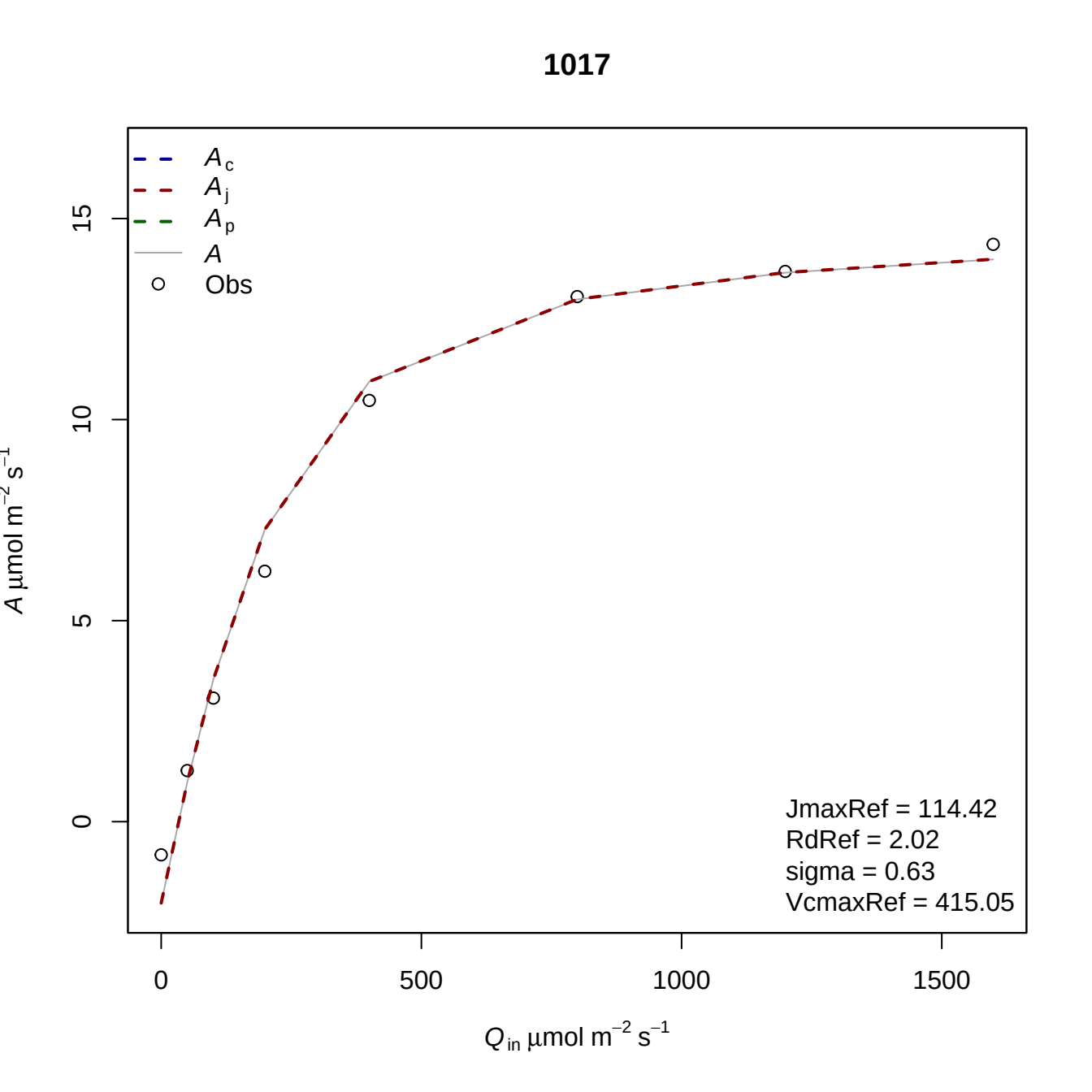

1012
A_c
A_j
A_p
A
Obs
JmaxRef = 77.54
RdRef = 2.2
sigma = 0.56
VcmaxRef = 79181.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

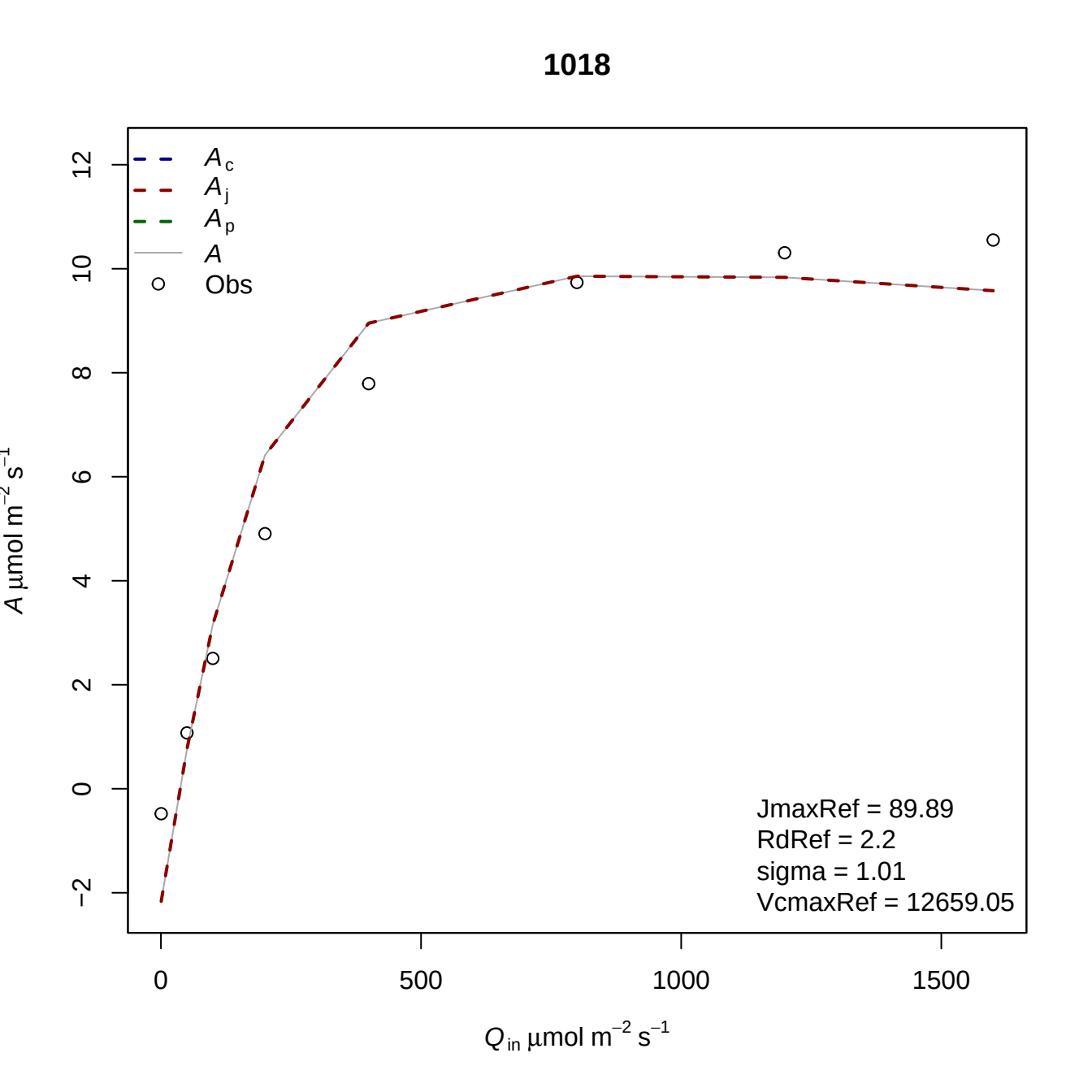

1013
A_c
A_j
A_p
A
Obs
JmaxRef = 130.99
RdRef = 1.98
sigma = 0.56
VcmaxRef = 1079.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

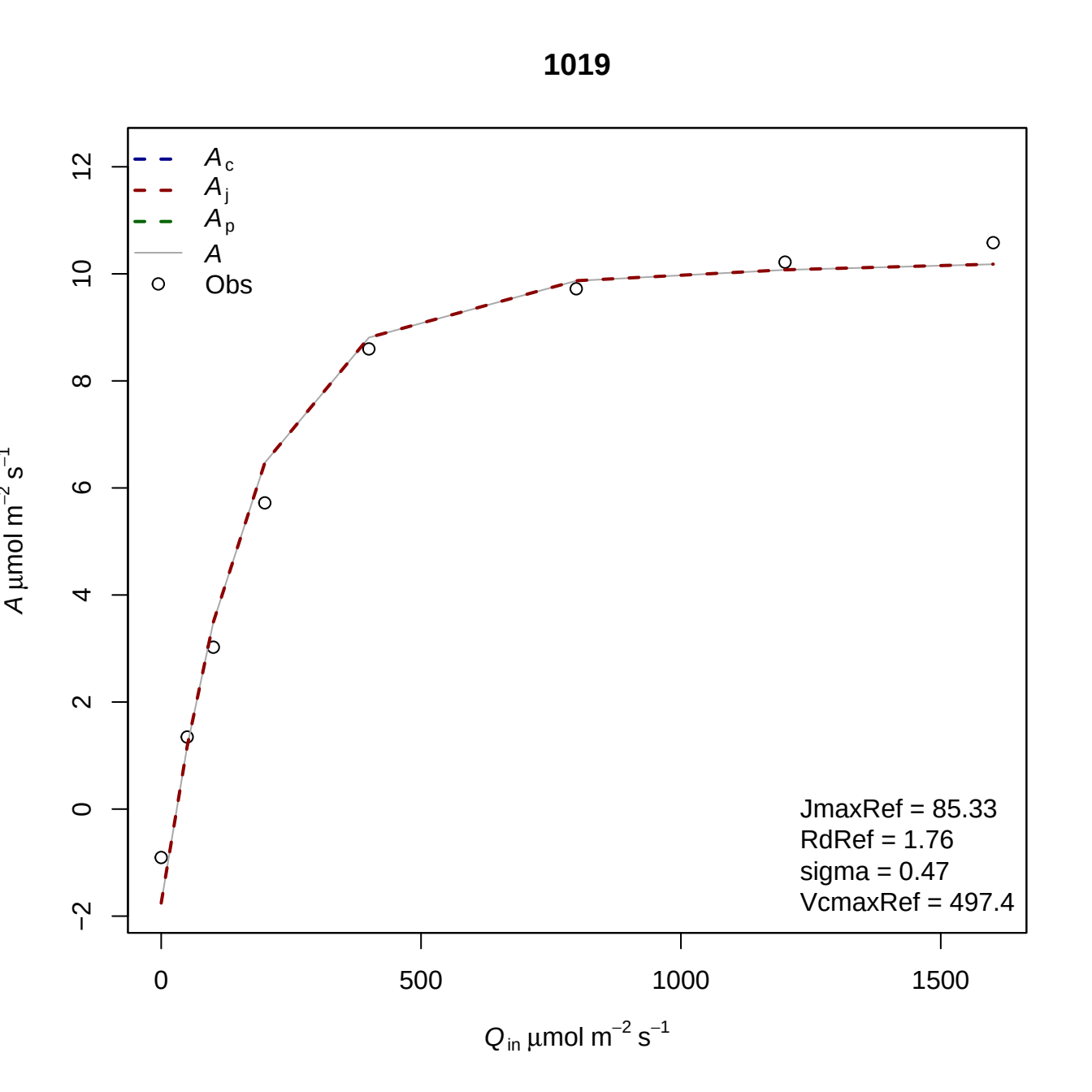

1014
A_c
A_j
A_p
A
Obs
JmaxRef = 80.43
RdRef = 0.56
sigma = 0.85
VcmaxRef = 33963.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

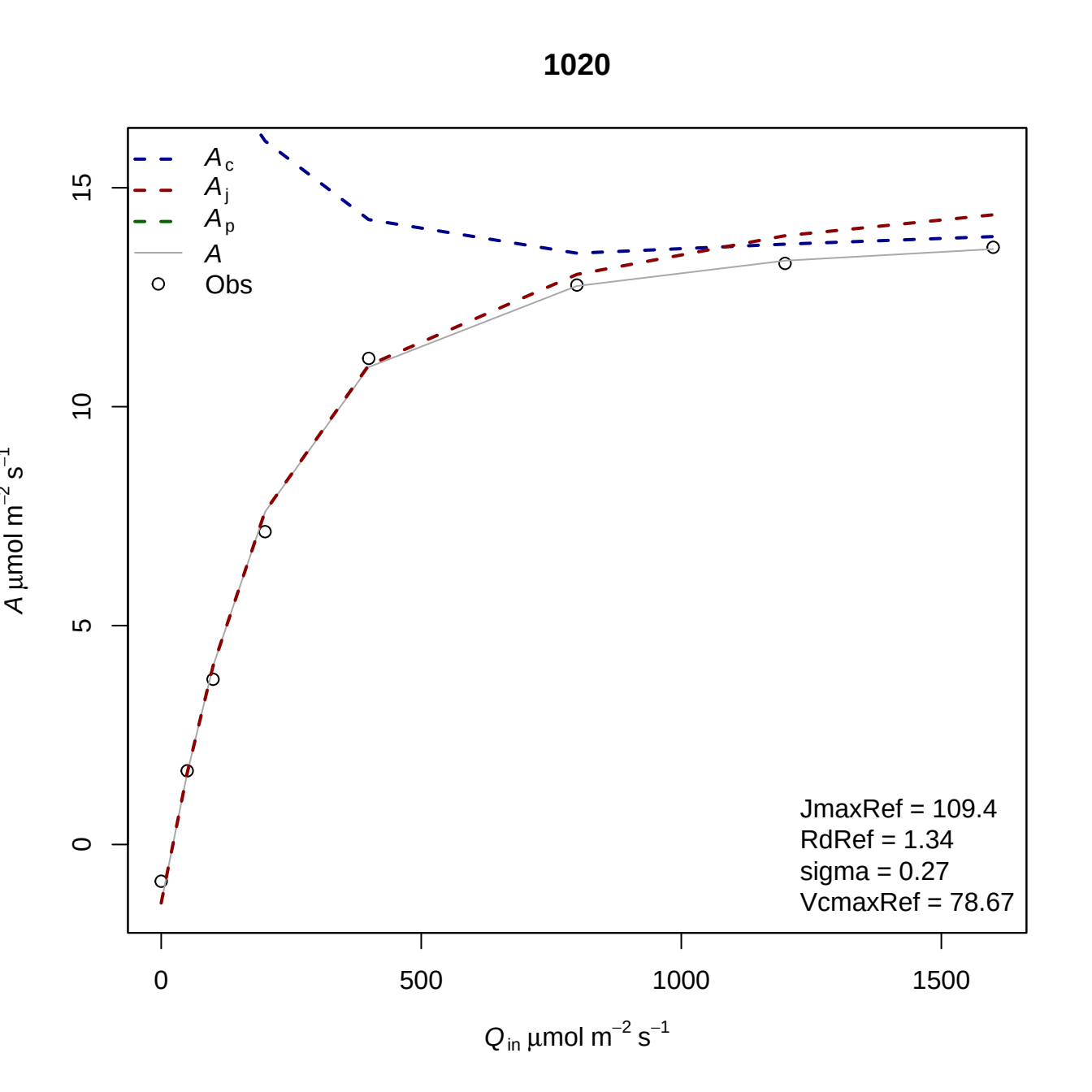

1015
A_c
A_j
A_p
A
Obs
JmaxRef = 82.14
RdRef = 1.96
sigma = 0.61
VcmaxRef = 322639.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

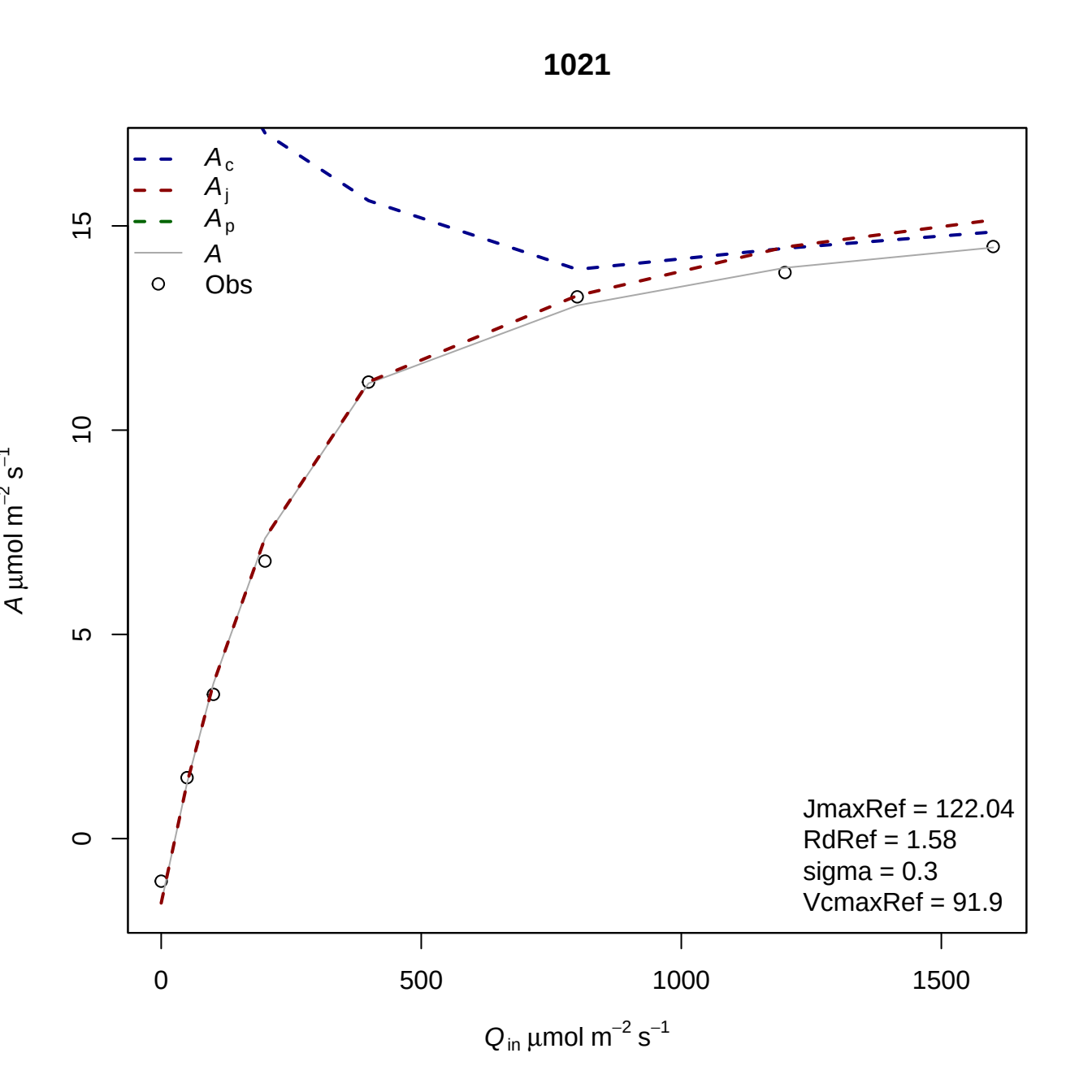

1016
A_c
A_j
A_p
A
Obs
JmaxRef = 92.76
RdRef = 2.71
sigma = 0.33
VcmaxRef = 1033.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

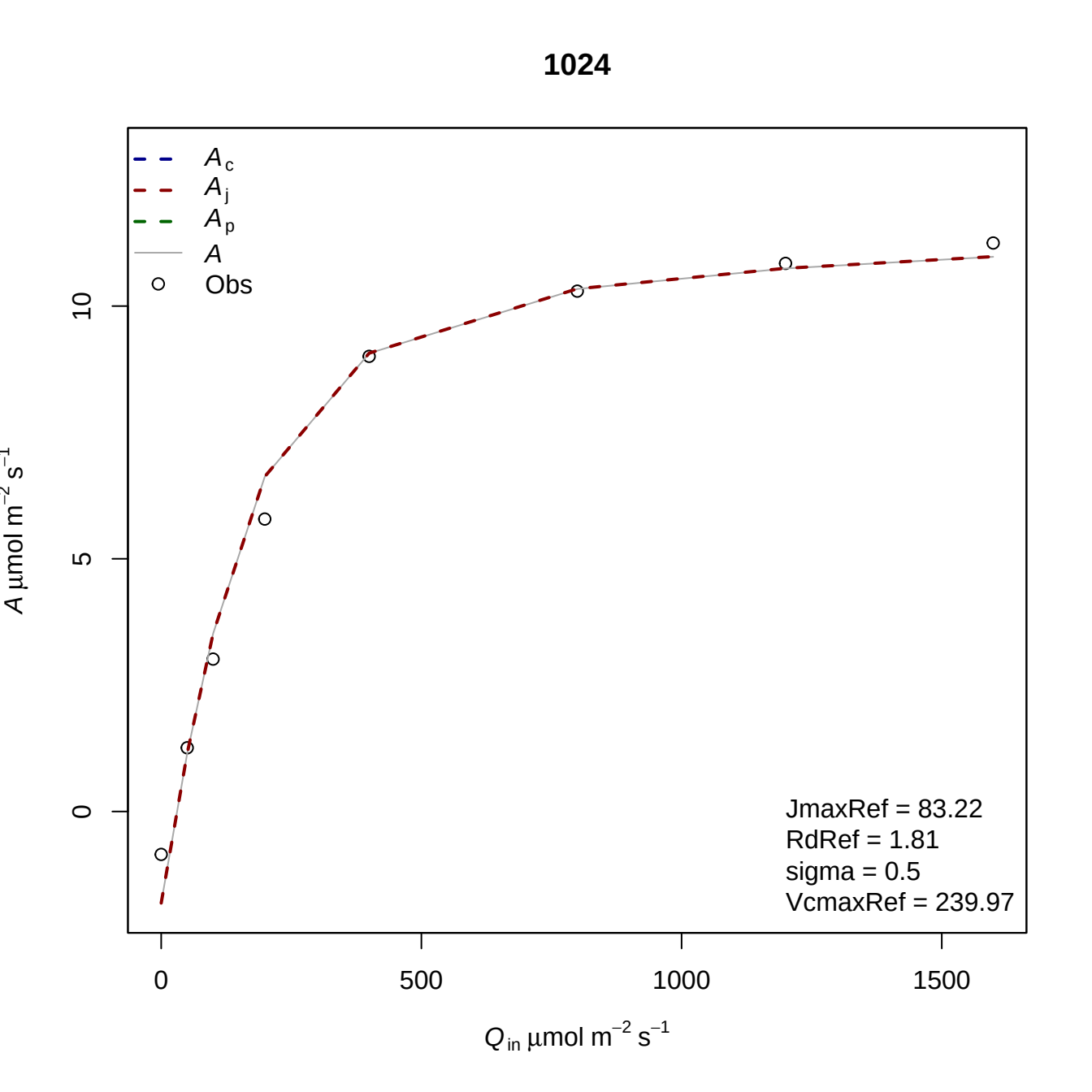

1017
A_c
A_j
A_p
A
Obs
JmaxRef = 114.42
RdRef = 2.02
sigma = 0.63
VcmaxRef = 415.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

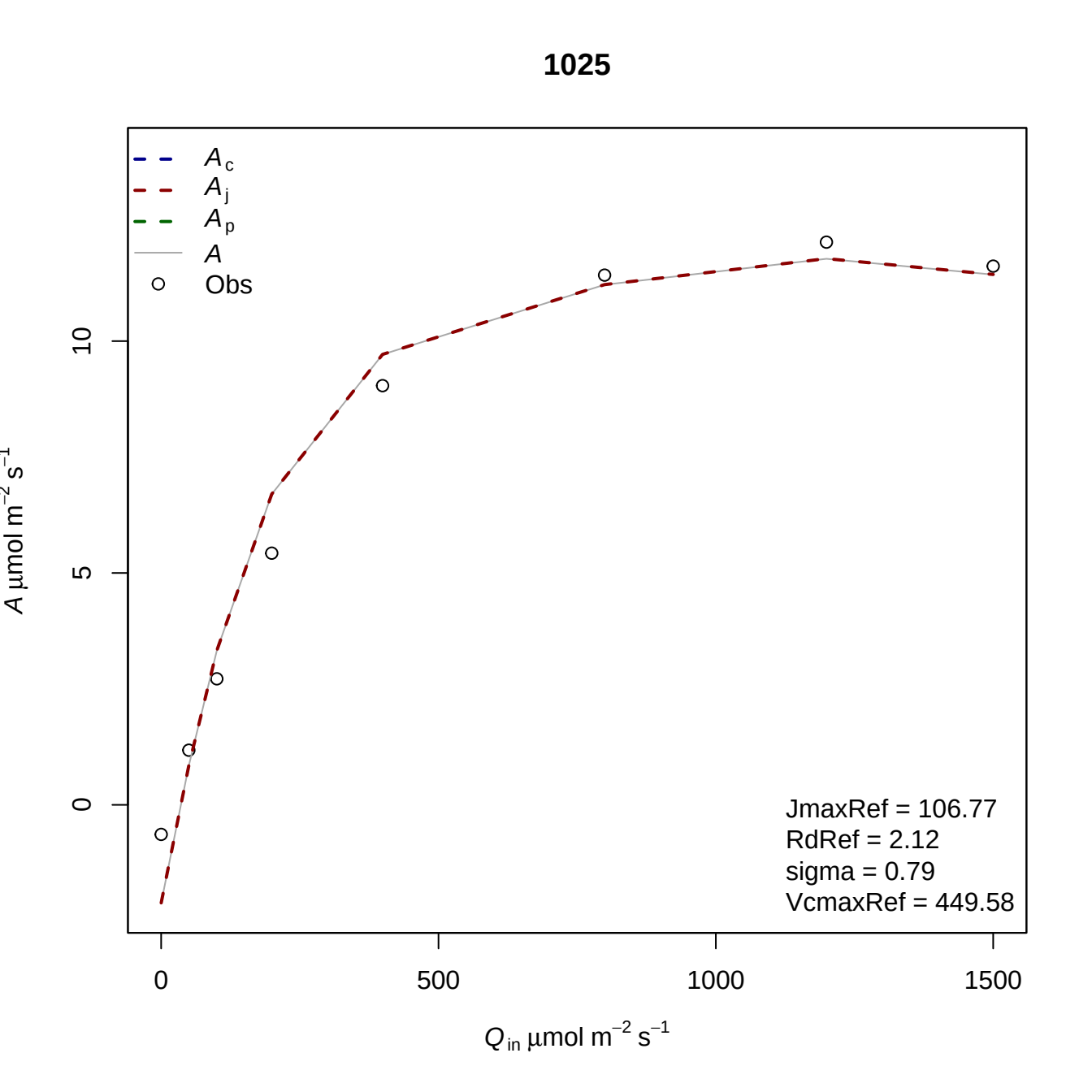

1018
A_c
A_j
A_p
A
Obs
JmaxRef = 89.89
RdRef = 2.2
sigma = 1.01
VcmaxRef = 12659.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

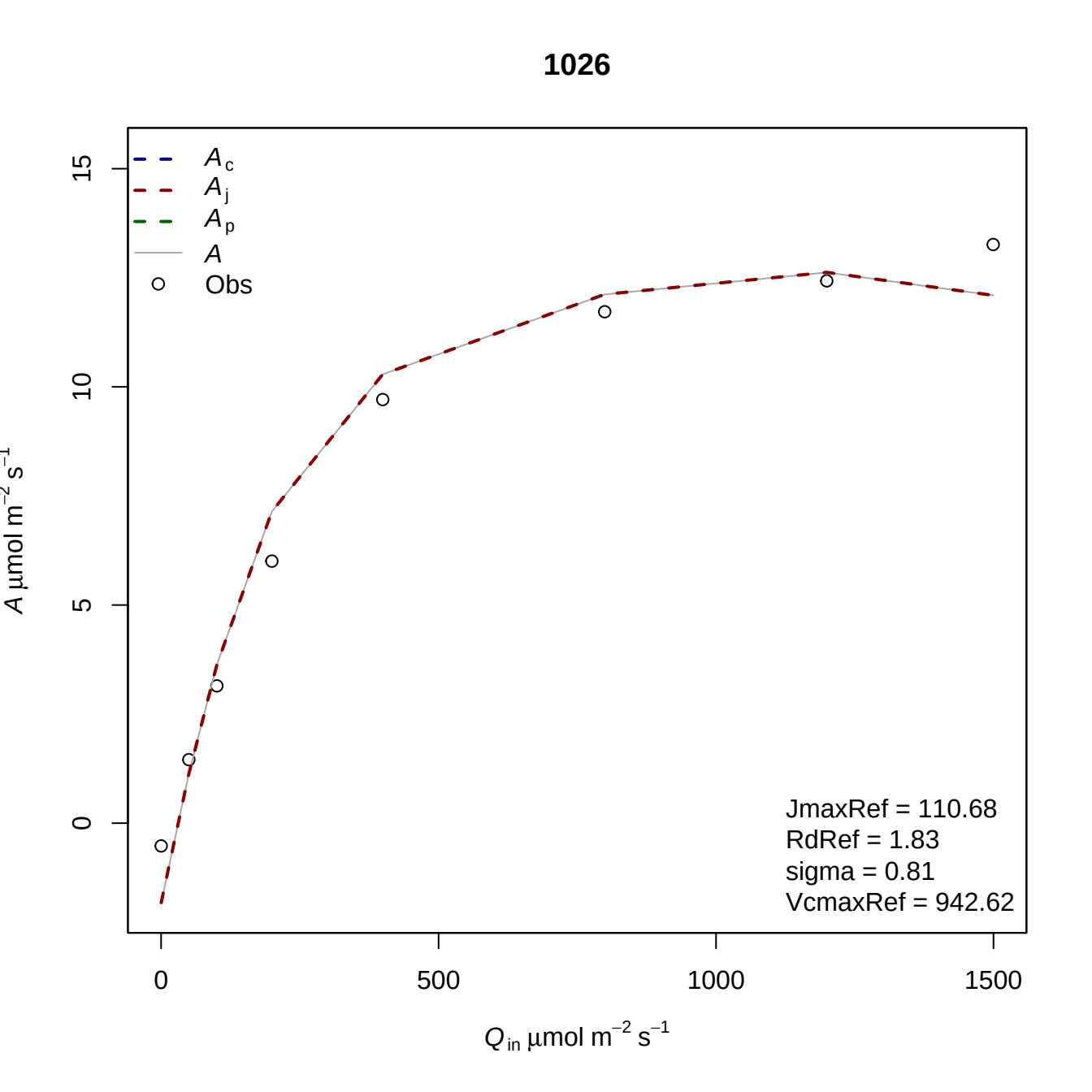

1019
A_c
A_j
A_p
A
Obs
JmaxRef = 85.33
RdRef = 1.76
sigma = 0.47
VcmaxRef = 497.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

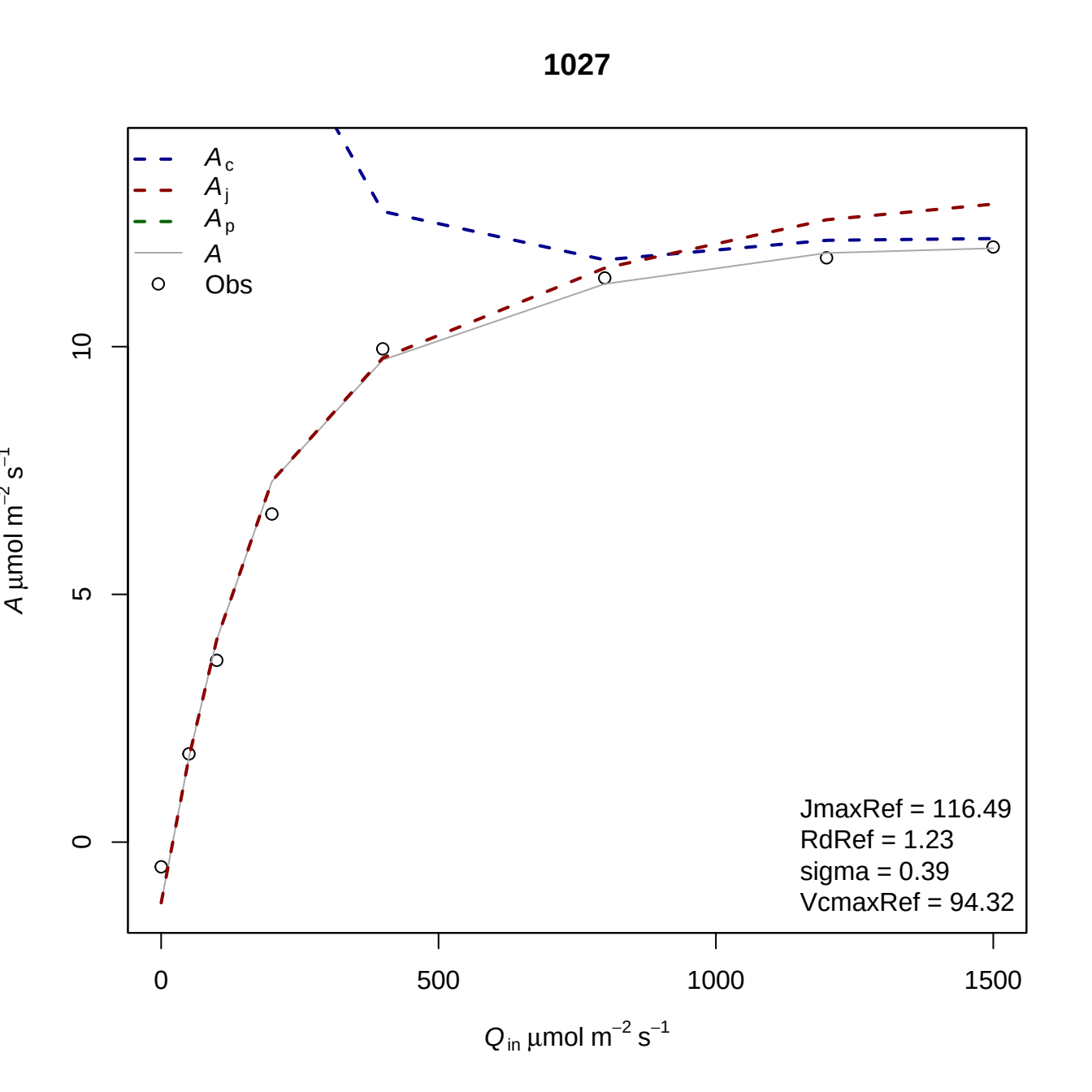

1020
A_c
A_j
A_p
A
Obs
JmaxRef = 109.4
RdRef = 1.34
sigma = 0.27
VcmaxRef = 78.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

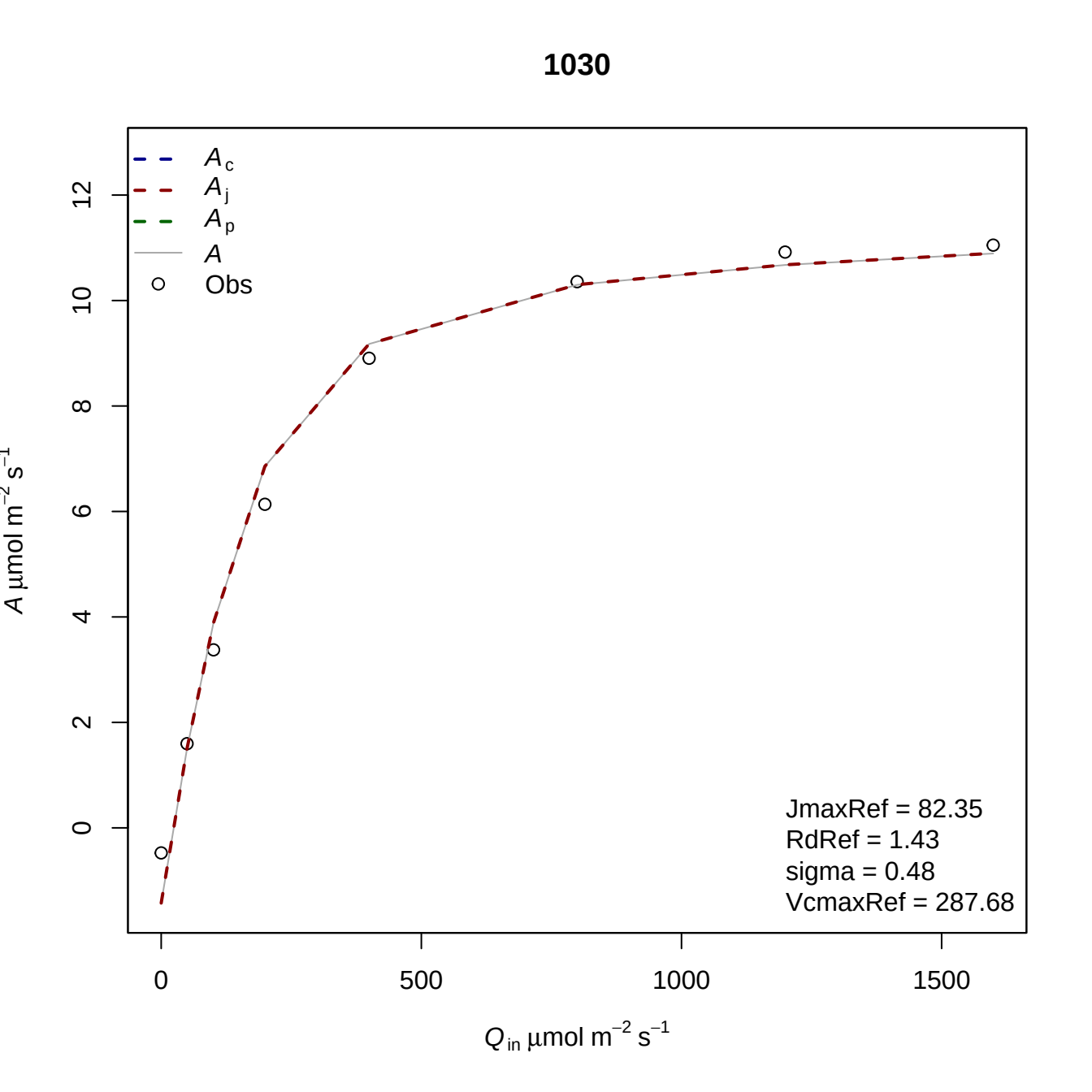

1021
A_c
A_j
A_p
A
Obs
JmaxRef = 122.04
RdRef = 1.58
sigma = 0.3
VcmaxRef = 91.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

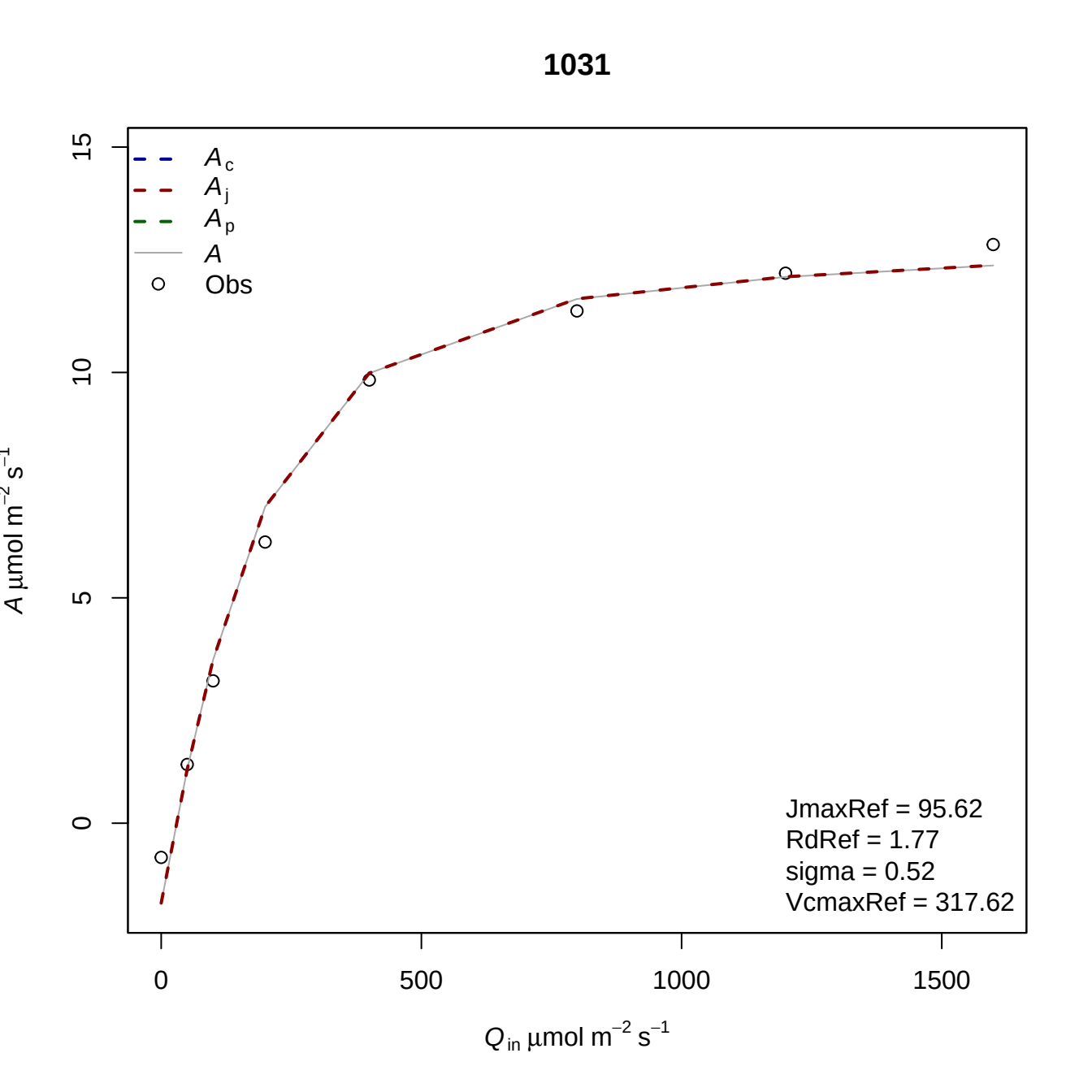

1024
A_c
A_j
A_p
A
Obs
JmaxRef = 83.22
RdRef = 1.81
sigma = 0.5
VcmaxRef = 239.97
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

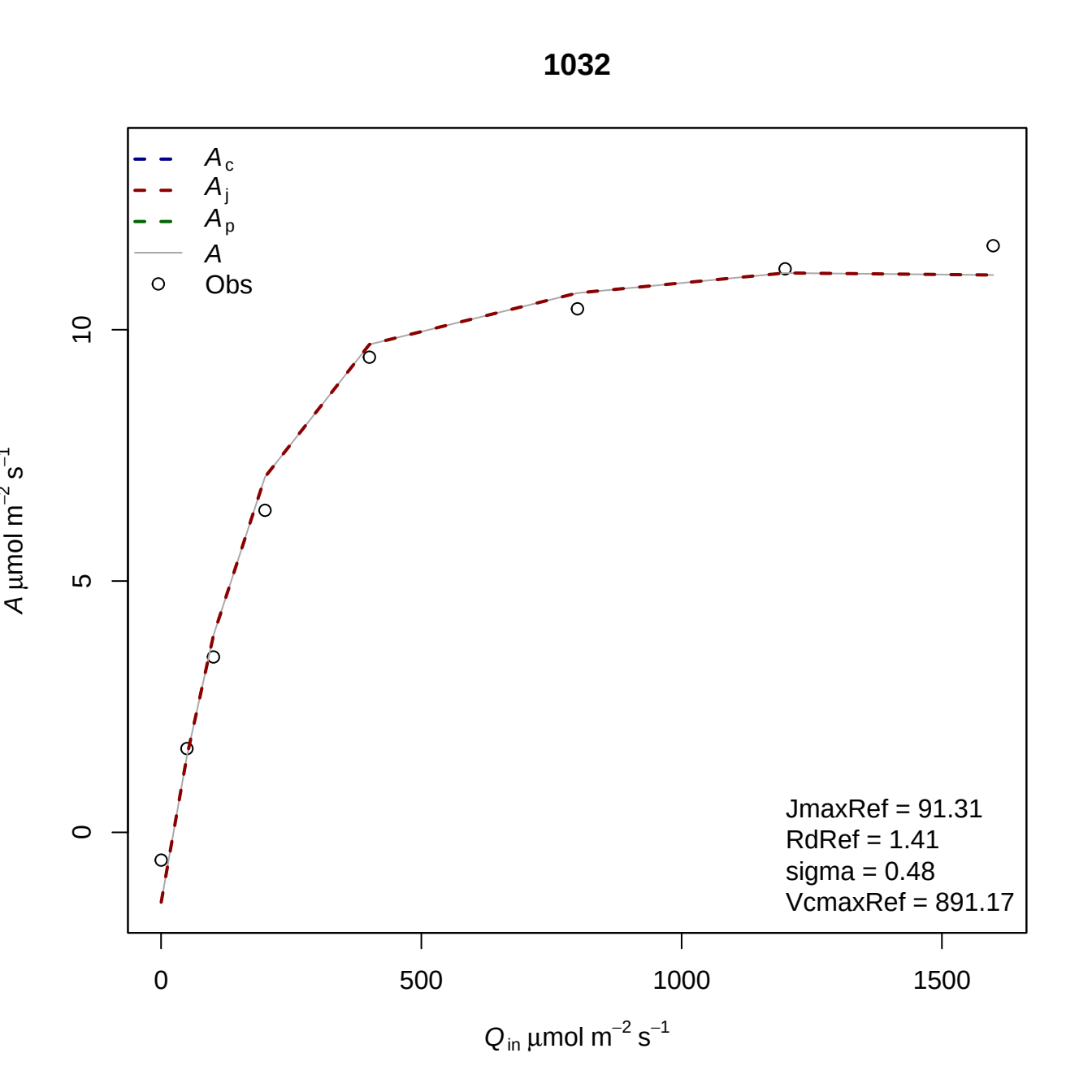

1025
A_c
A_j
A_p
A
Obs
JmaxRef = 106.77
RdRef = 2.12
sigma = 0.79
VcmaxRef = 449.58
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

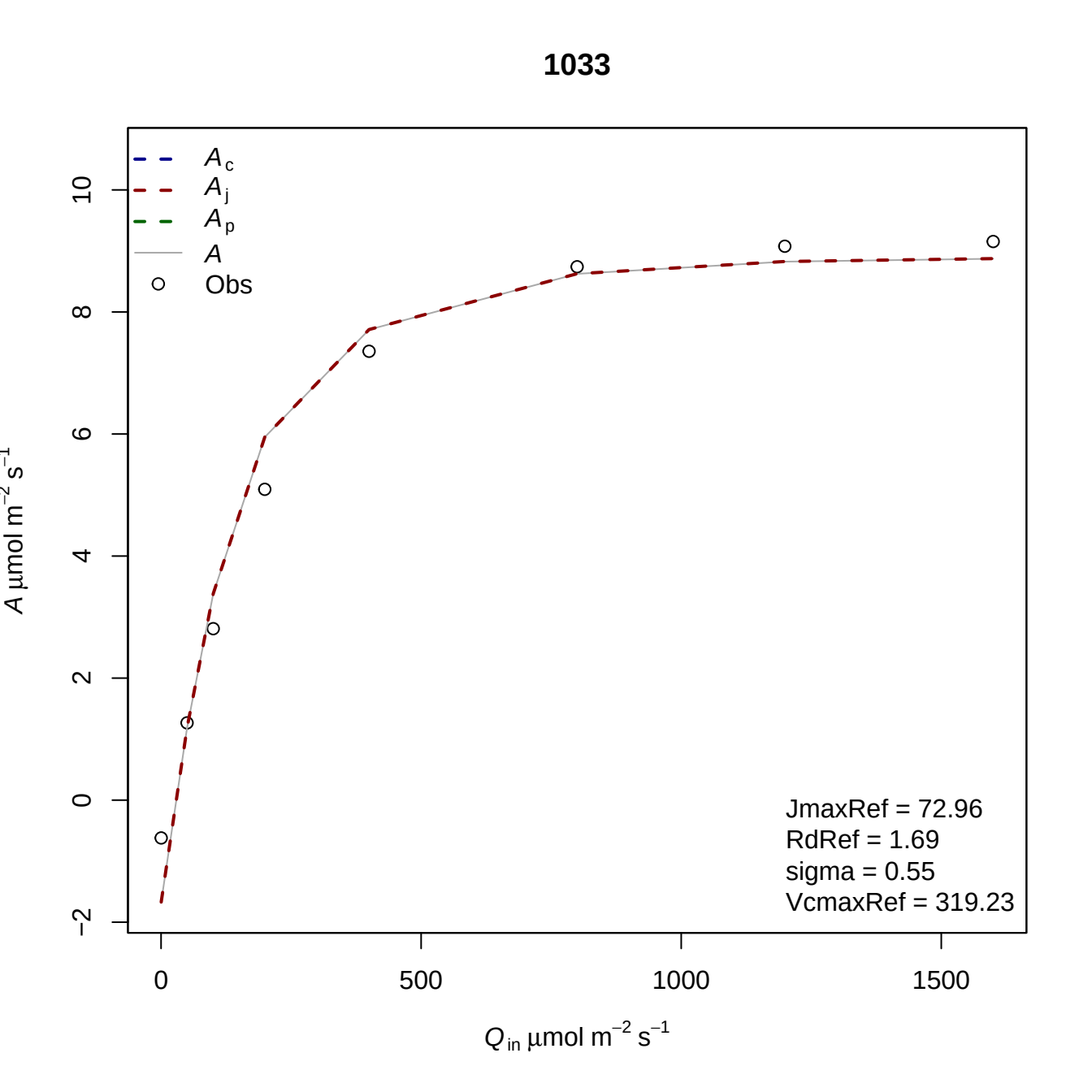

1026
A_c
A_j
A_p
A
Obs
JmaxRef = 110.68
RdRef = 1.83
sigma = 0.81
VcmaxRef = 942.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

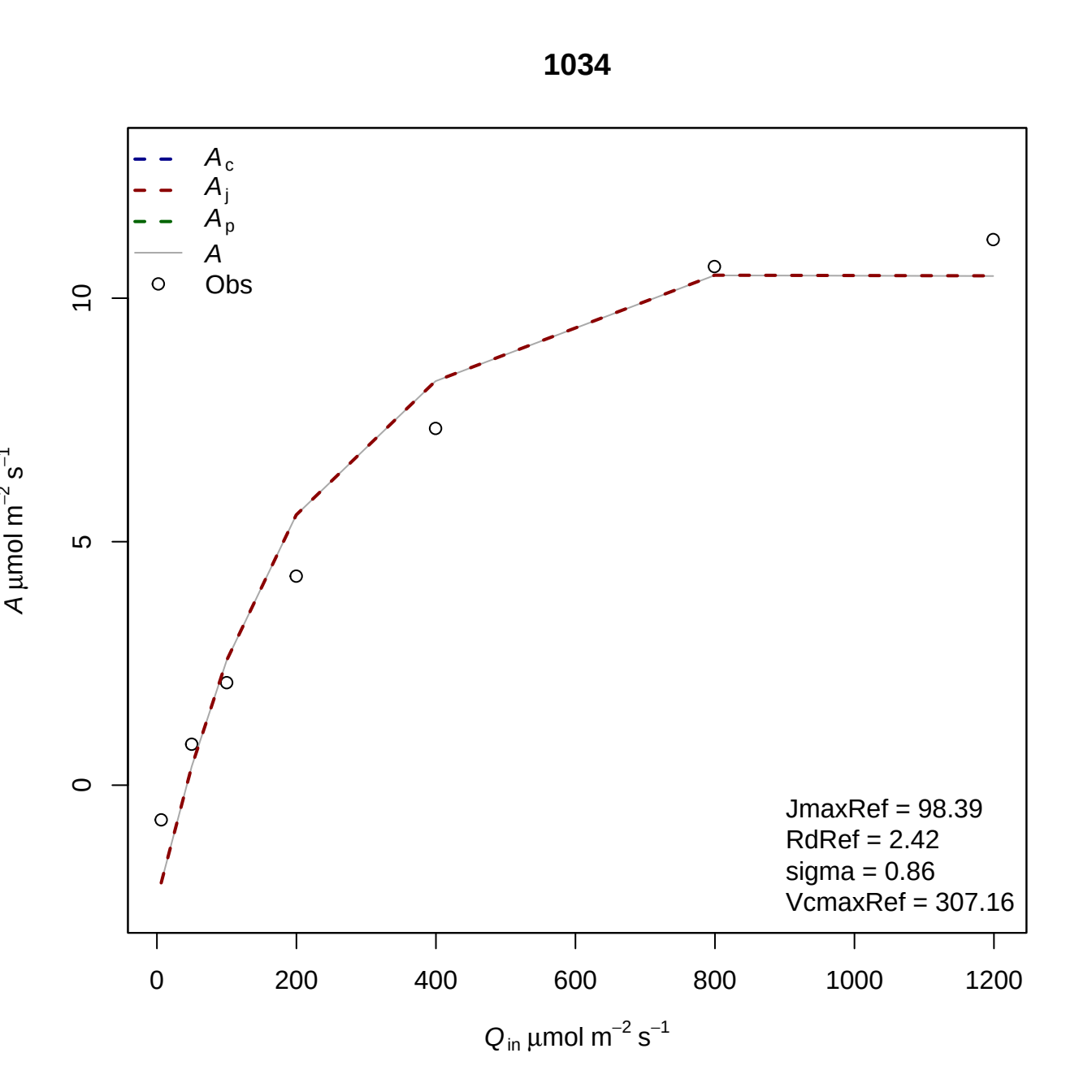

1027
A_c
A_j
A_p
A
Obs
JmaxRef = 116.49
RdRef = 1.23
sigma = 0.39
VcmaxRef = 94.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

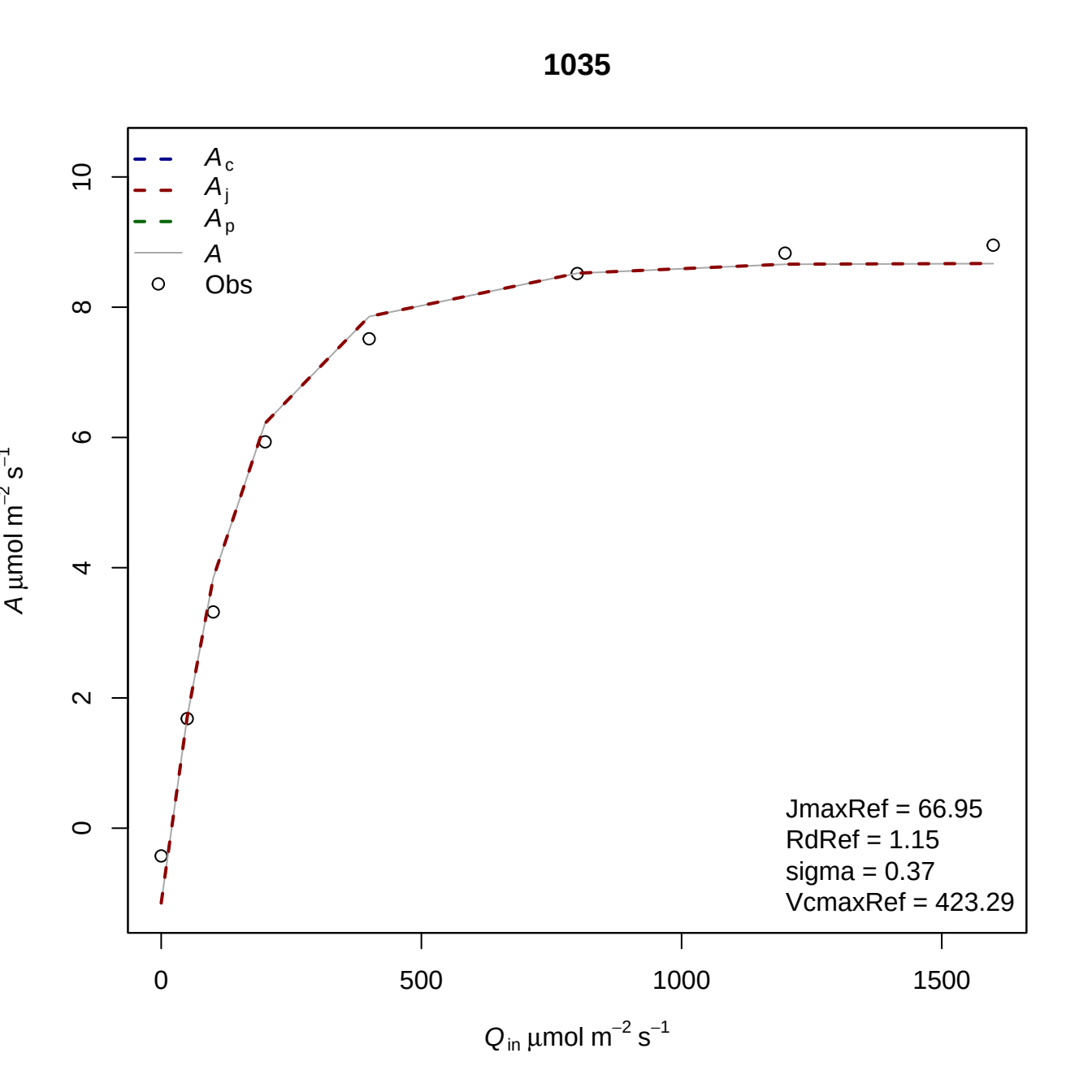

1030
A_c
A_j
A_p
A
Obs
JmaxRef = 82.35
RdRef = 1.43
sigma = 0.48
VcmaxRef = 287.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

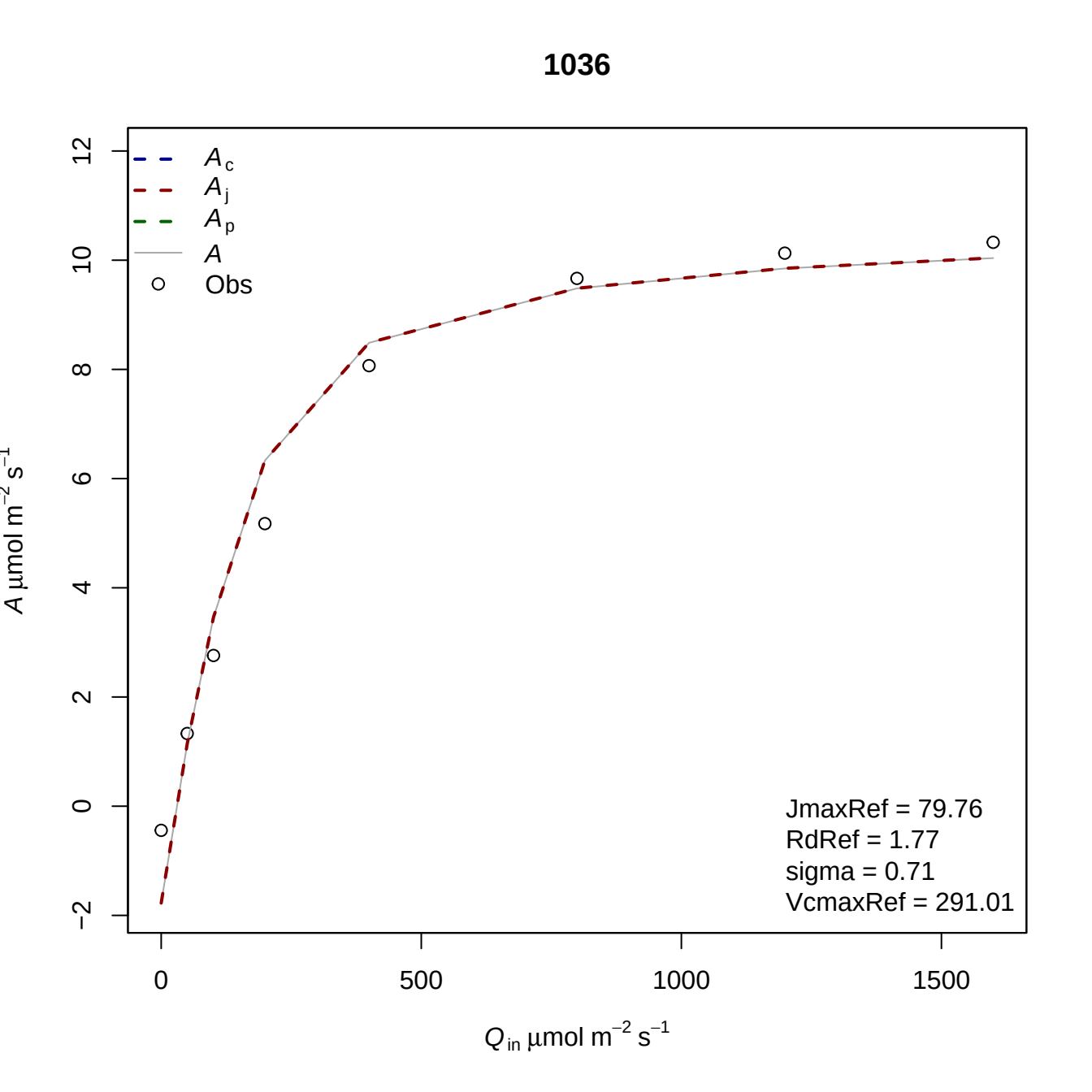

1031
A_c
A_j
A_p
A
Obs
JmaxRef = 95.62
RdRef = 1.77
sigma = 0.52
VcmaxRef = 317.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

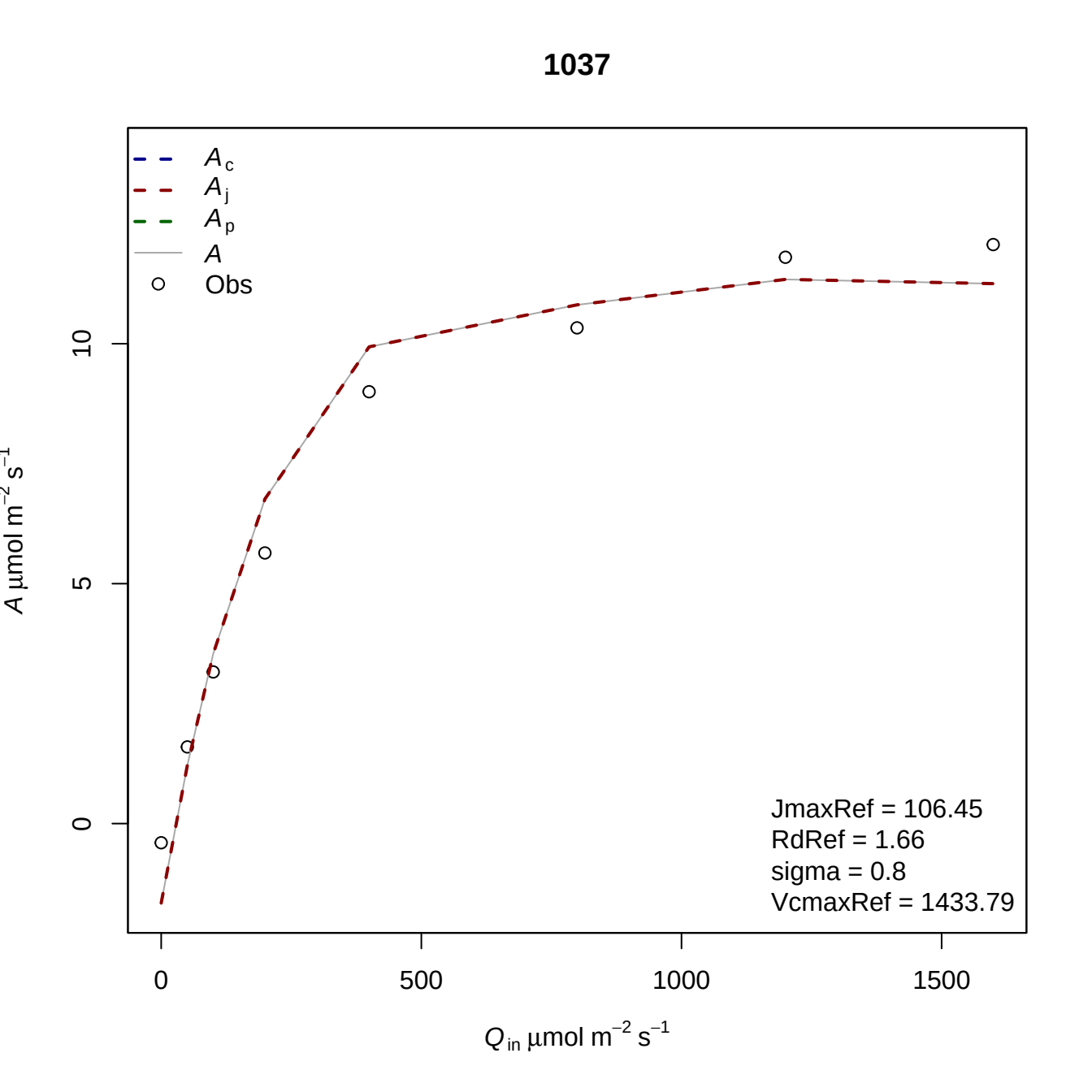

1032
A_c
A_j
A_p
A
Obs
JmaxRef = 91.31
RdRef = 1.41
sigma = 0.48
VcmaxRef = 891.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

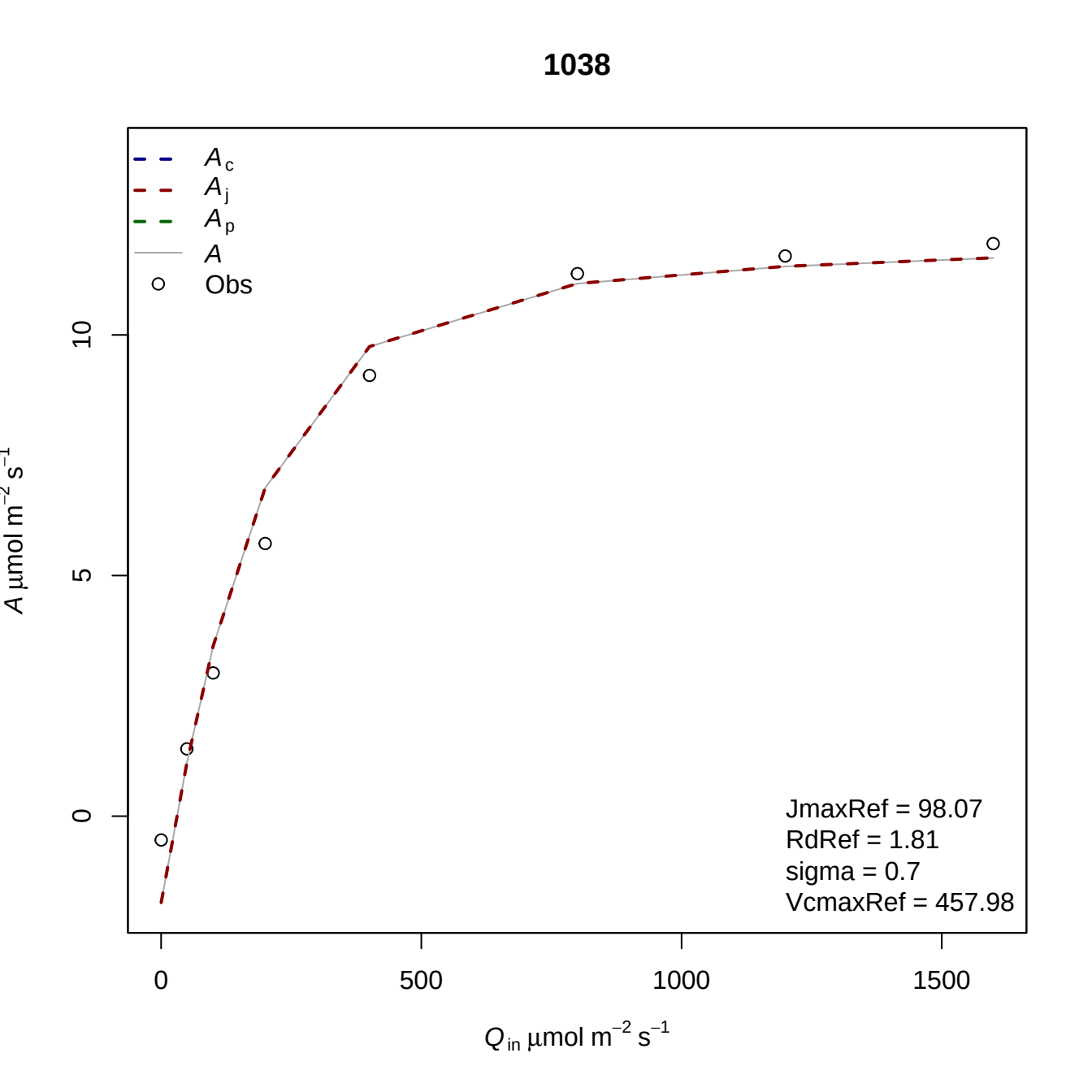

1033
A_c
A_j
A_p
A
Obs
JmaxRef = 72.96
RdRef = 1.69
sigma = 0.55
VcmaxRef = 319.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

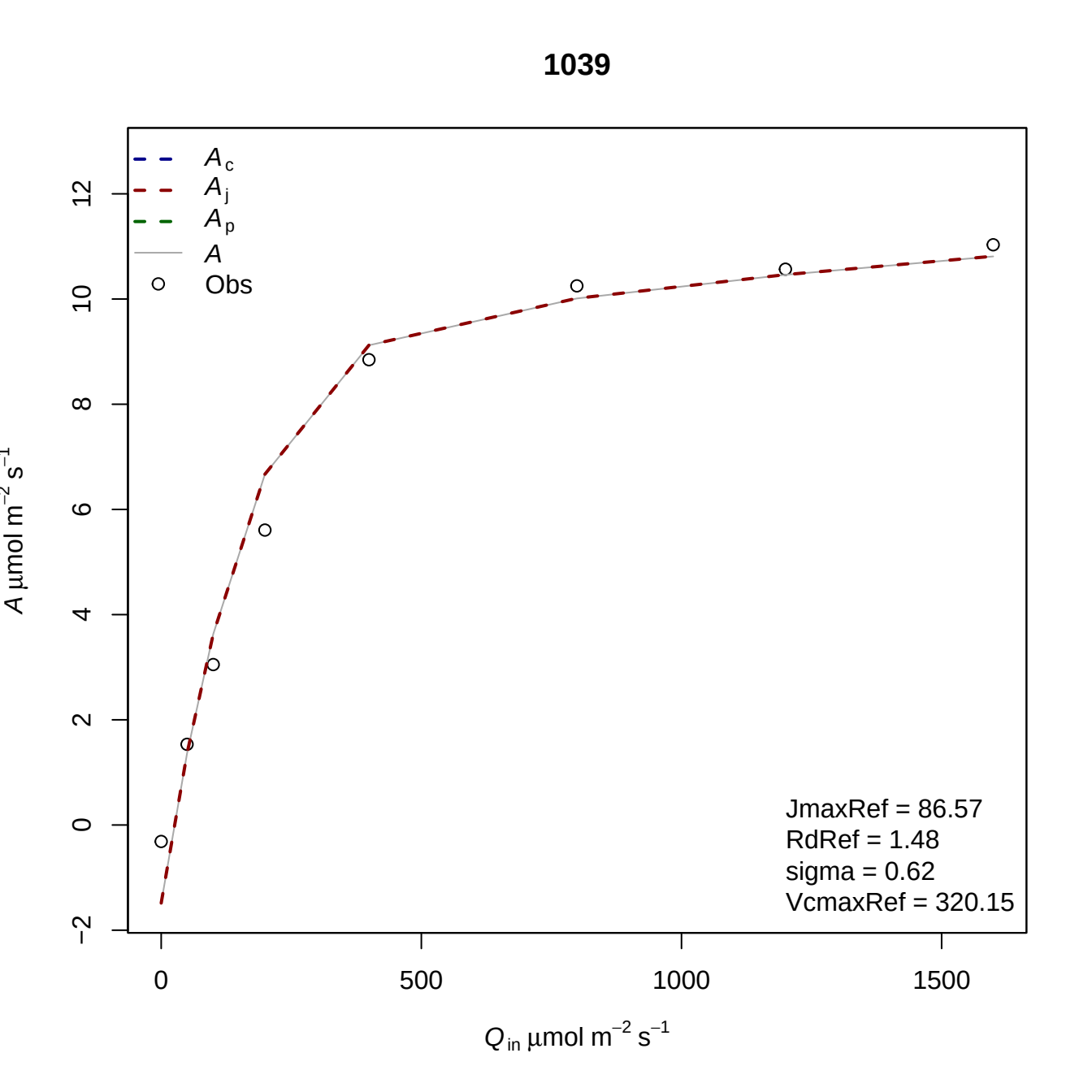

1034
A_c
A_j
A_p
A
Obs
JmaxRef = 98.39
RdRef = 2.42
sigma = 0.86
VcmaxRef = 307.16
A µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

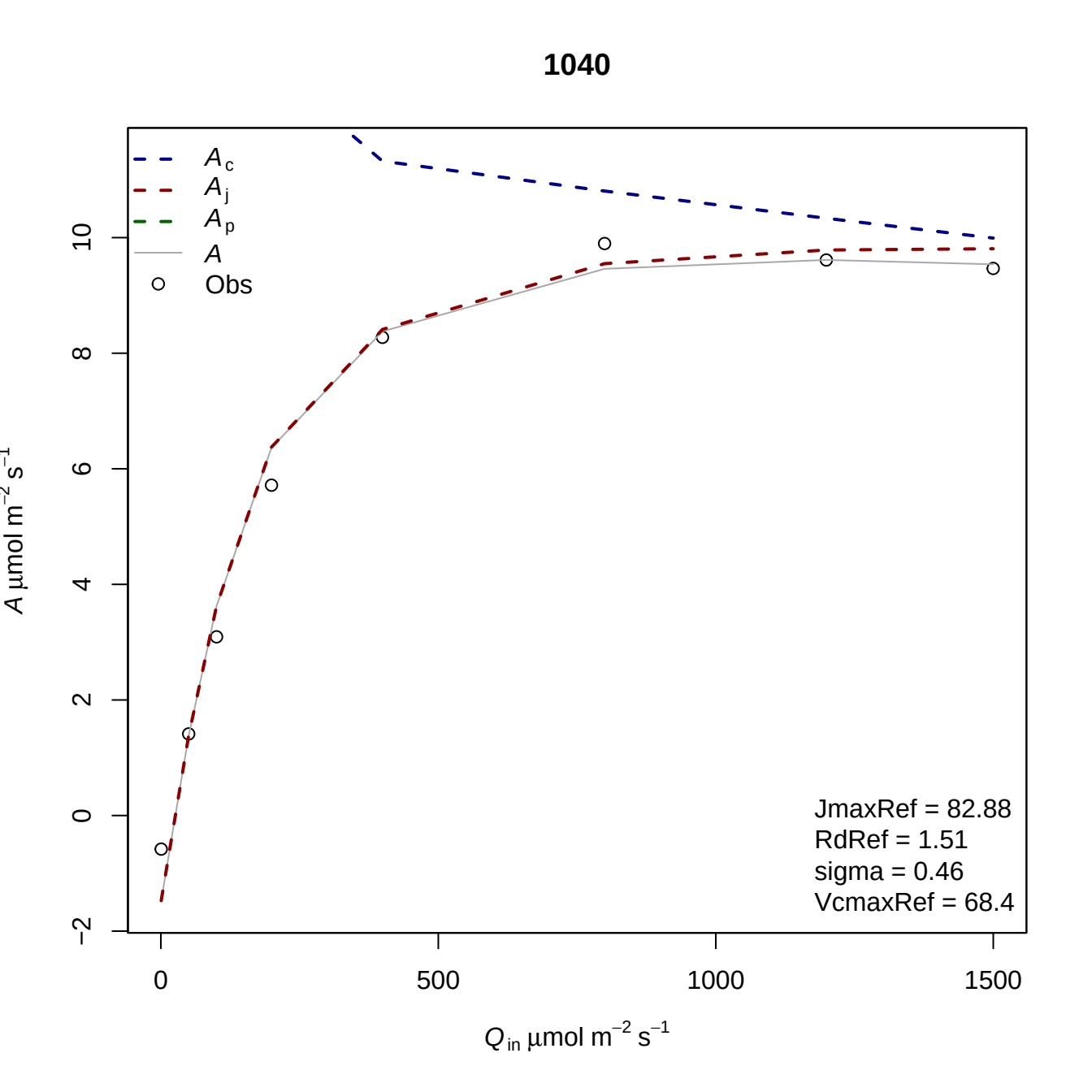

1035
A_c
A_j
A_p
A
Obs
JmaxRef = 66.95
RdRef = 1.15
sigma = 0.37
VcmaxRef = 423.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

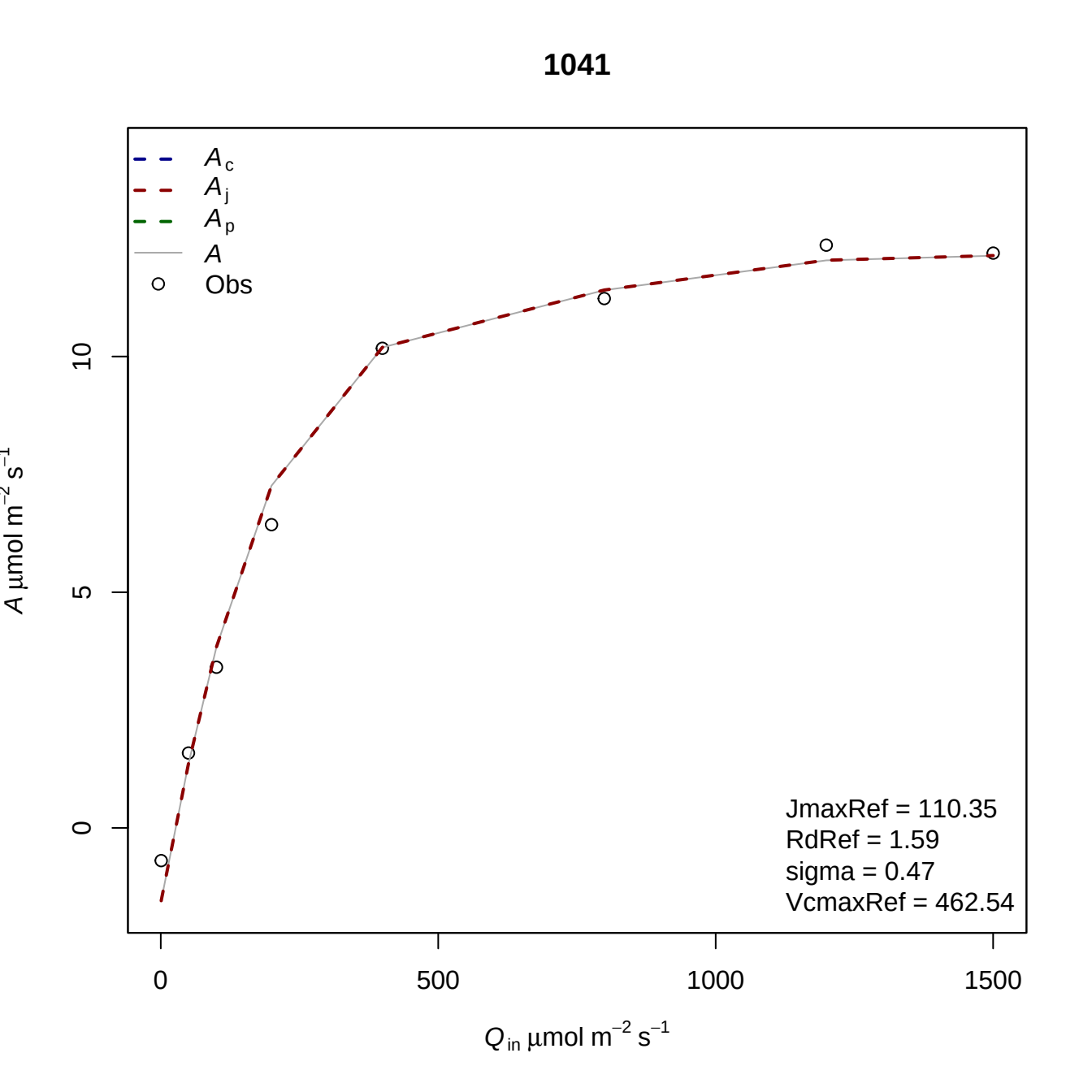

1036
A_c
A_j
A_p
A
Obs
JmaxRef = 79.76
RdRef = 1.77
sigma = 0.71
VcmaxRef = 291.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

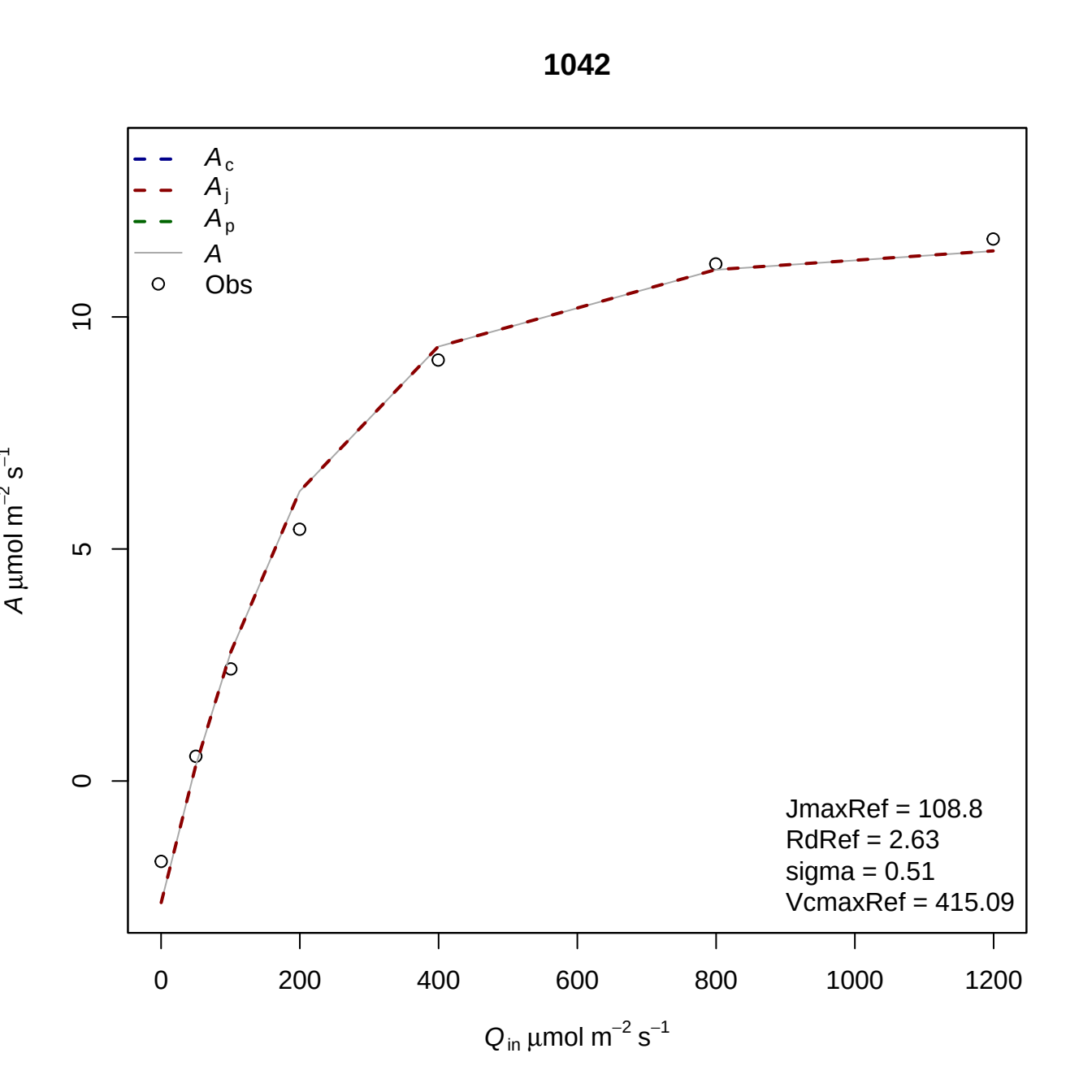

1037
A_c
A_j
A_p
A
Obs
JmaxRef = 106.45
RdRef = 1.66
sigma = 0.8
VcmaxRef = 1433.79
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

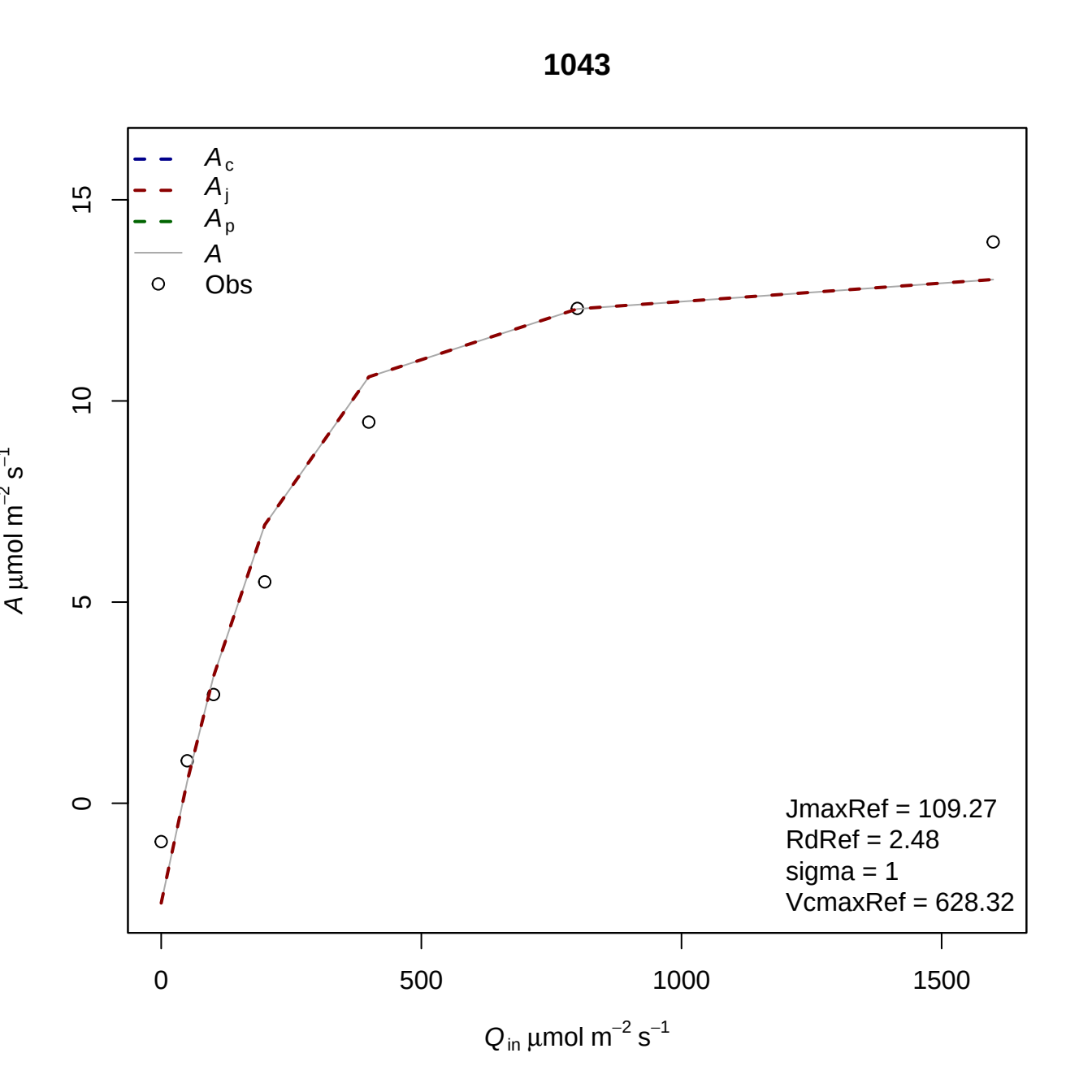

1038
A_c
A_j
A_p
A
Obs
JmaxRef = 98.07
RdRef = 1.81
sigma = 0.7
VcmaxRef = 457.98
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

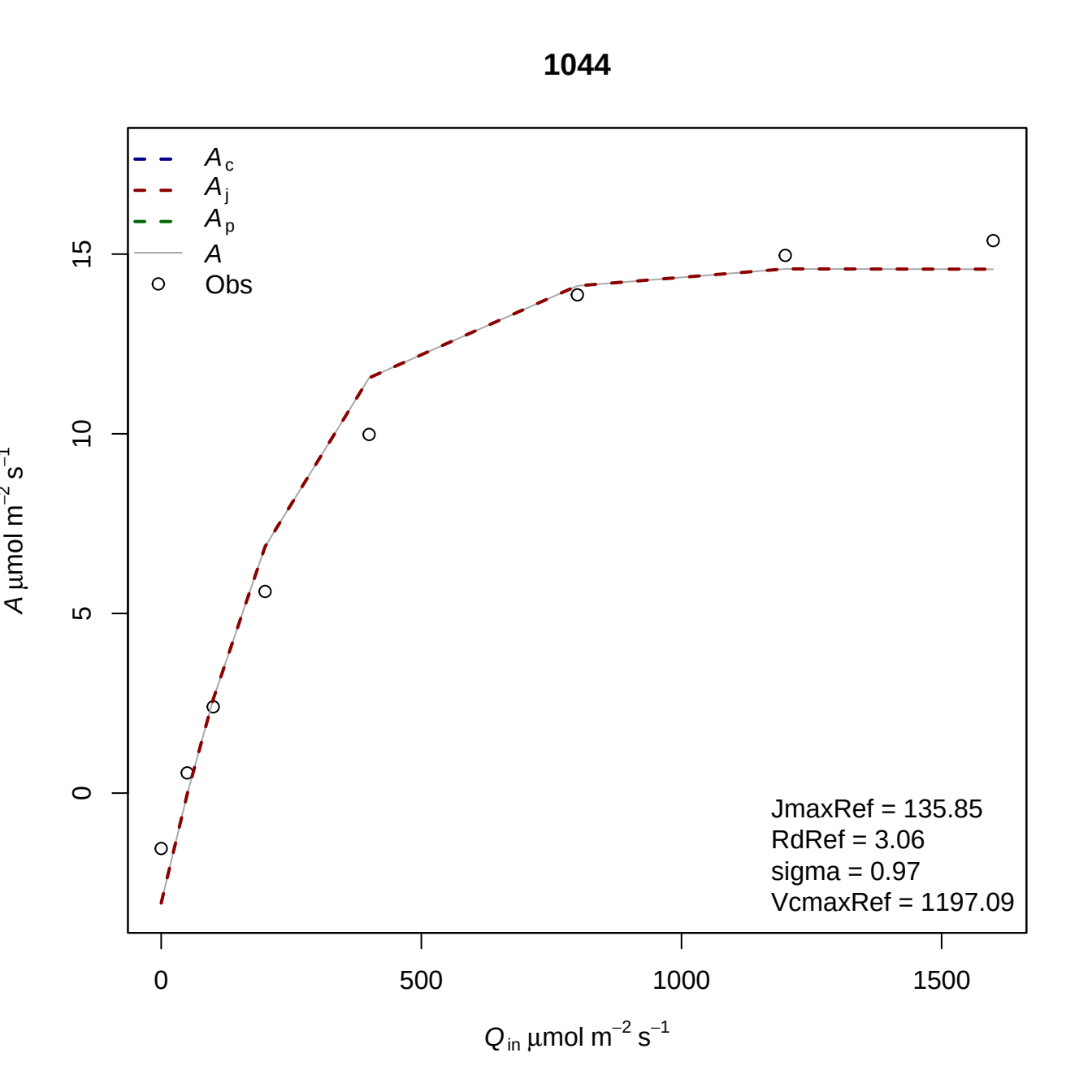

1039
A_c
A_j
A_p
A
Obs
JmaxRef = 86.57
RdRef = 1.48
sigma = 0.62
VcmaxRef = 320.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

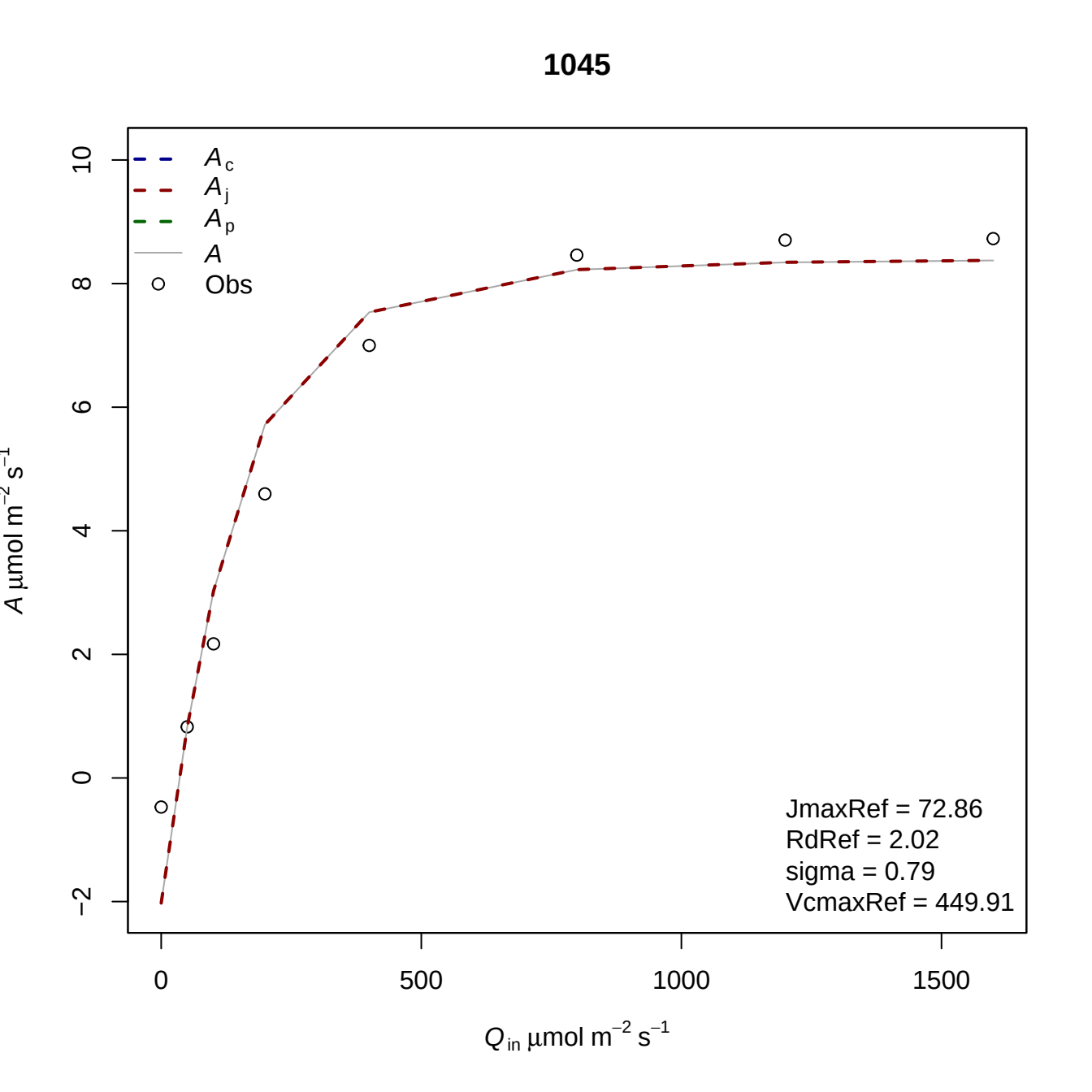

1040
A_c
A_j
A_p
A
Obs
JmaxRef = 82.88
RdRef = 1.51
sigma = 0.46
VcmaxRef = 68.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

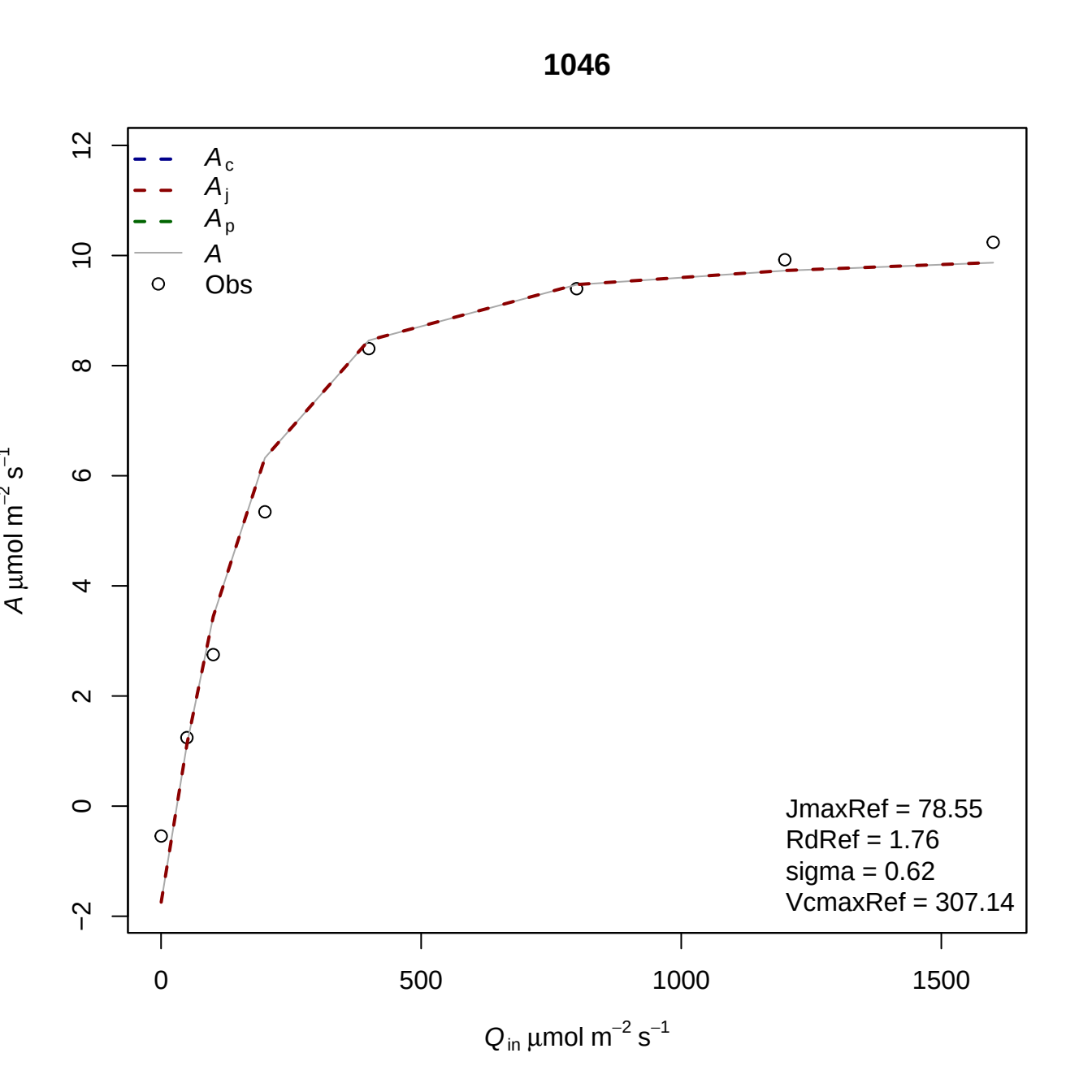

1041
A_c
A_j
A_p
A
Obs
JmaxRef = 110.35
RdRef = 1.59
sigma = 0.47
VcmaxRef = 462.54
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

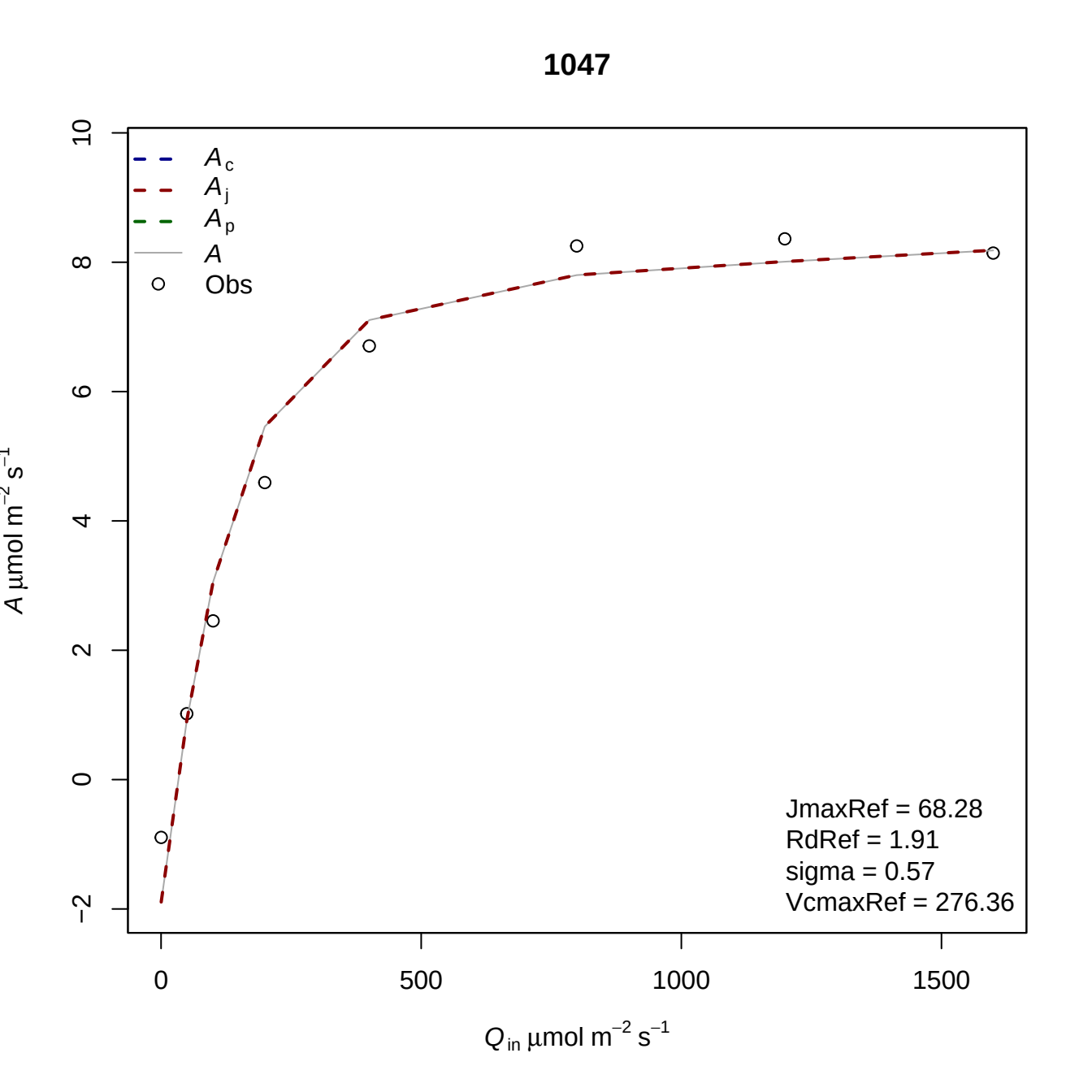

1042
A_c
A_j
A_p
A
Obs
JmaxRef = 108.8
RdRef = 2.63
sigma = 0.51
VcmaxRef = 415.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

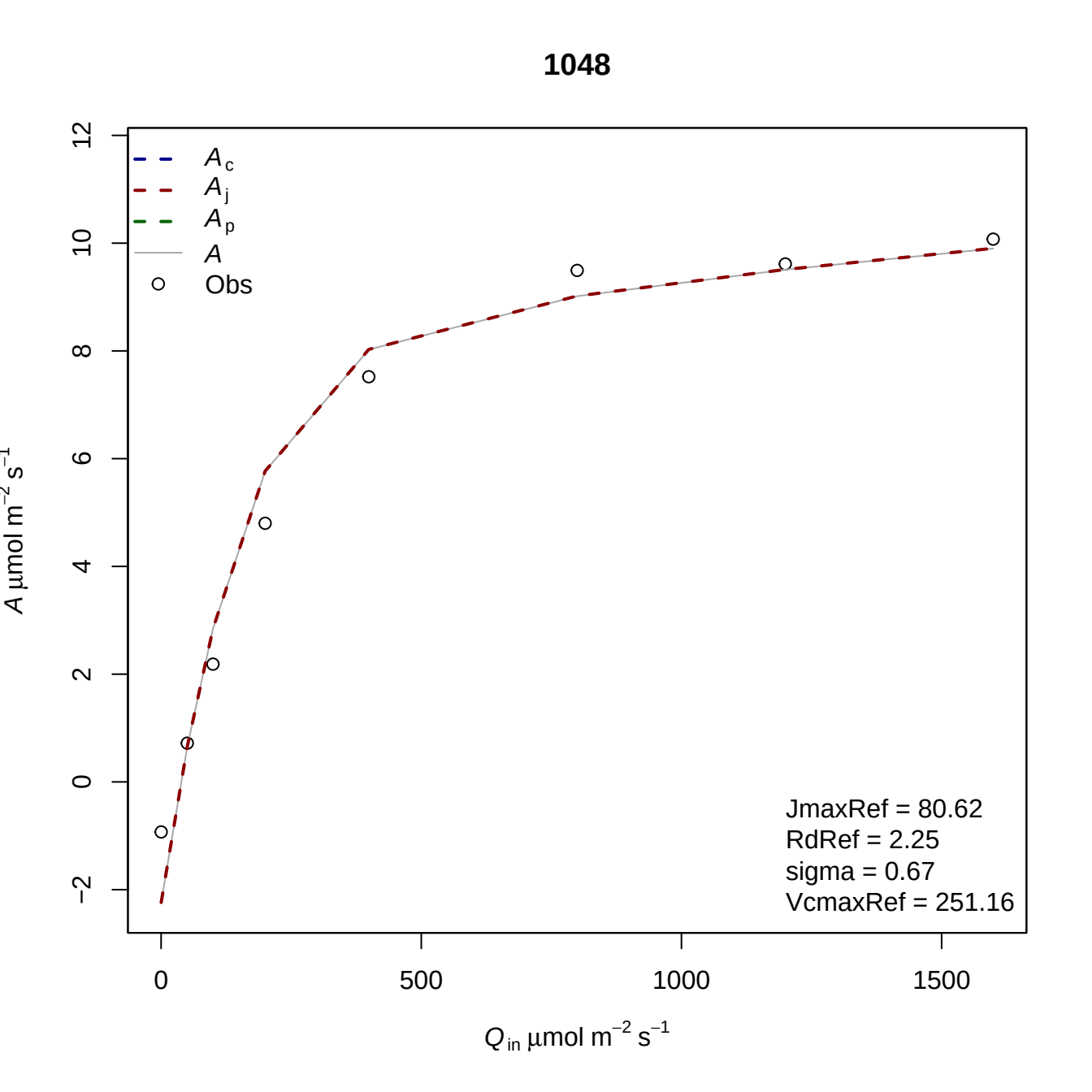

1043
A_c
A_j
A_p
A
Obs
JmaxRef = 109.27
RdRef = 2.48
sigma = 1
VcmaxRef = 628.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

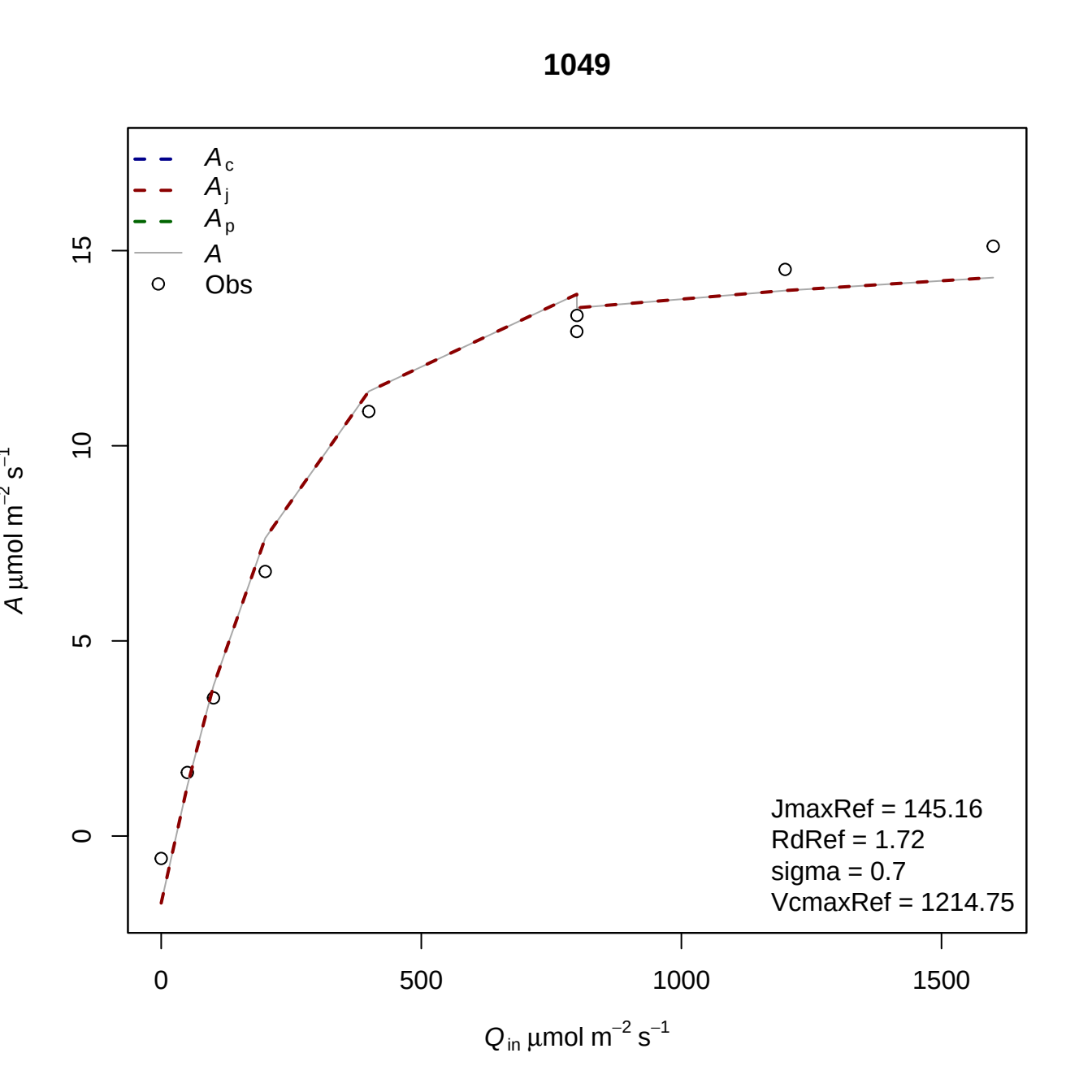

1044
A_c
A_j
A_p
A
Obs
JmaxRef = 135.85
RdRef = 3.06
sigma = 0.97
VcmaxRef = 1197.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

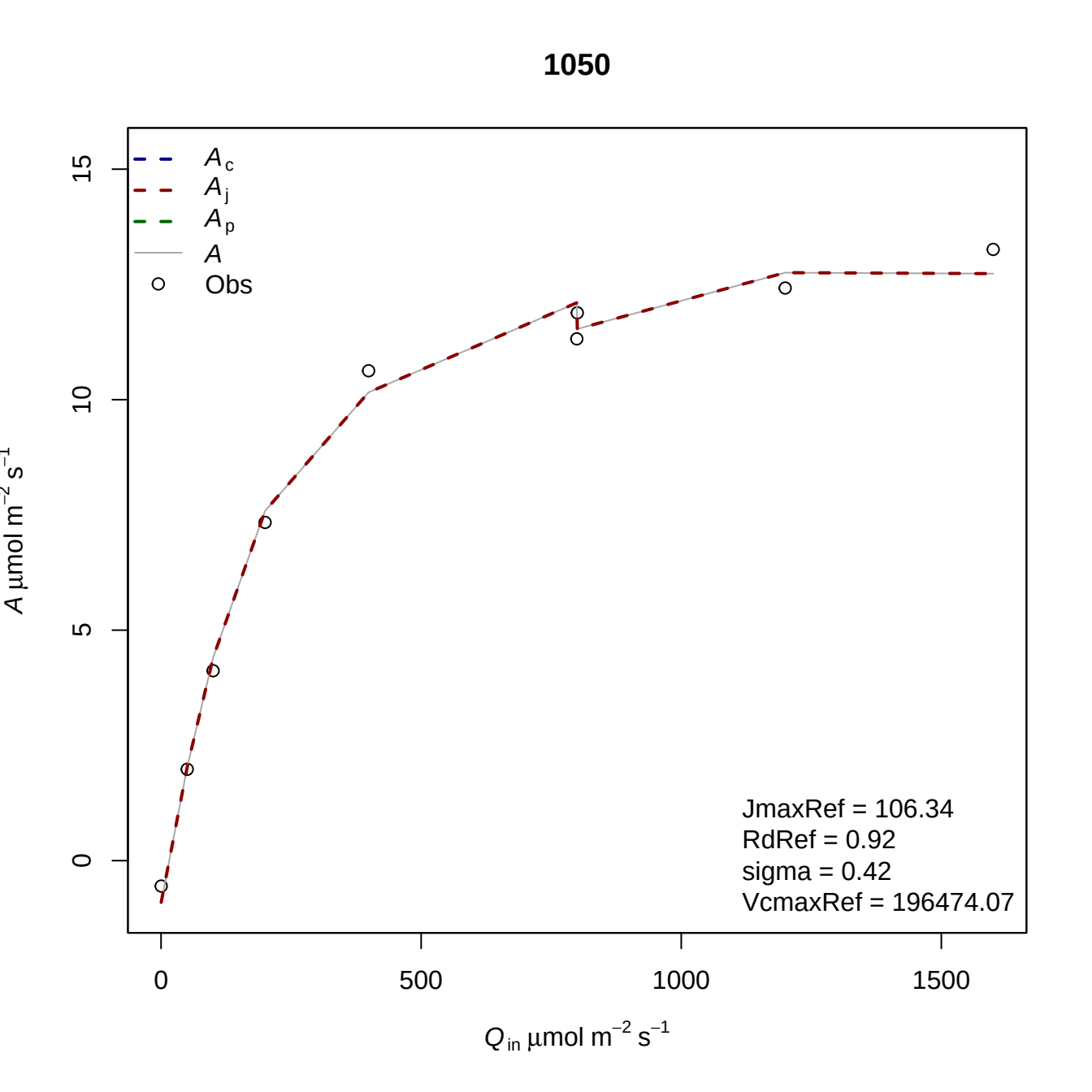

1045
A_c
A_j
A_p
A
Obs
JmaxRef = 72.86
RdRef = 2.02
sigma = 0.79
VcmaxRef = 449.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

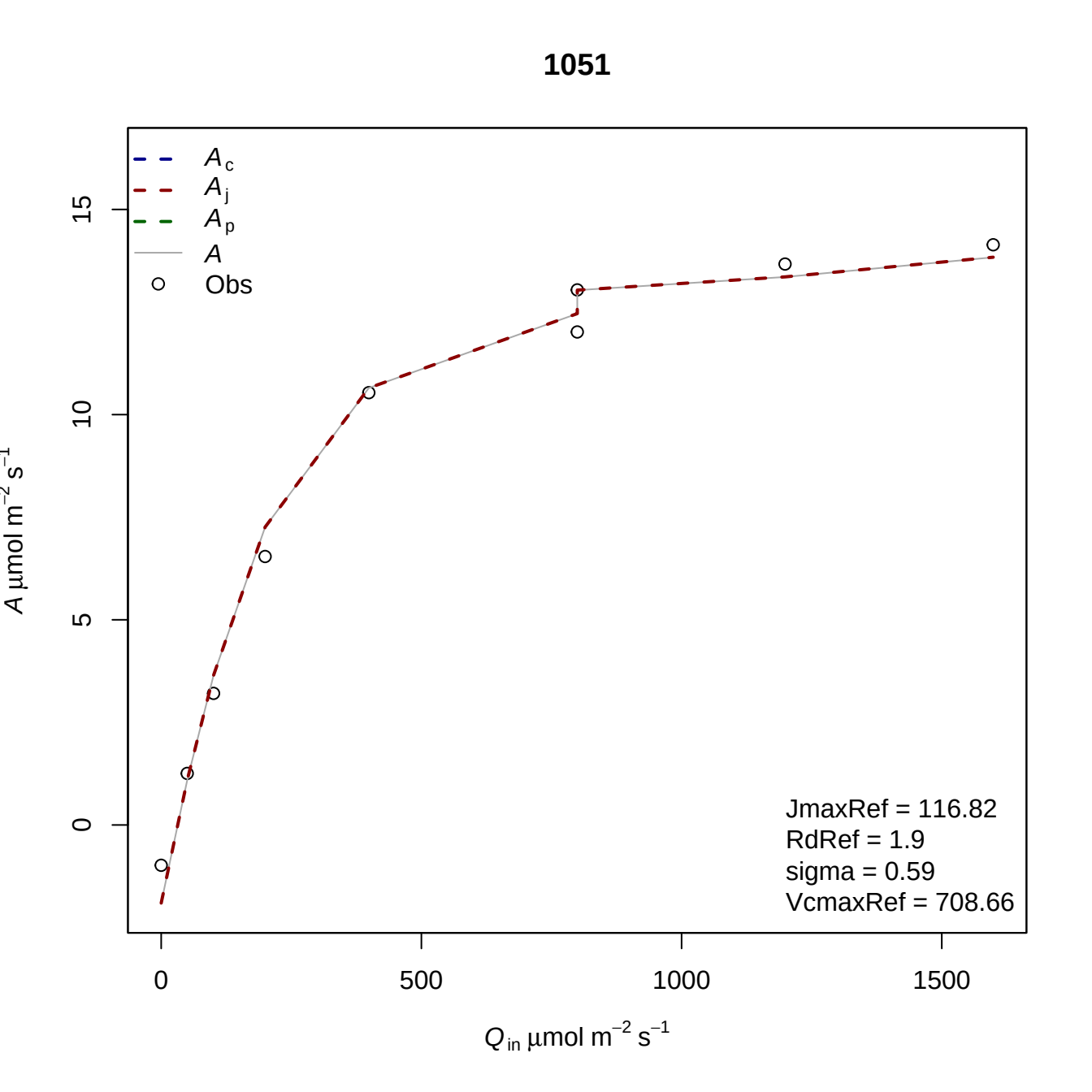

1046
A_c
A_j
A_p
A
Obs
JmaxRef = 78.55
RdRef = 1.76
sigma = 0.62
VcmaxRef = 307.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

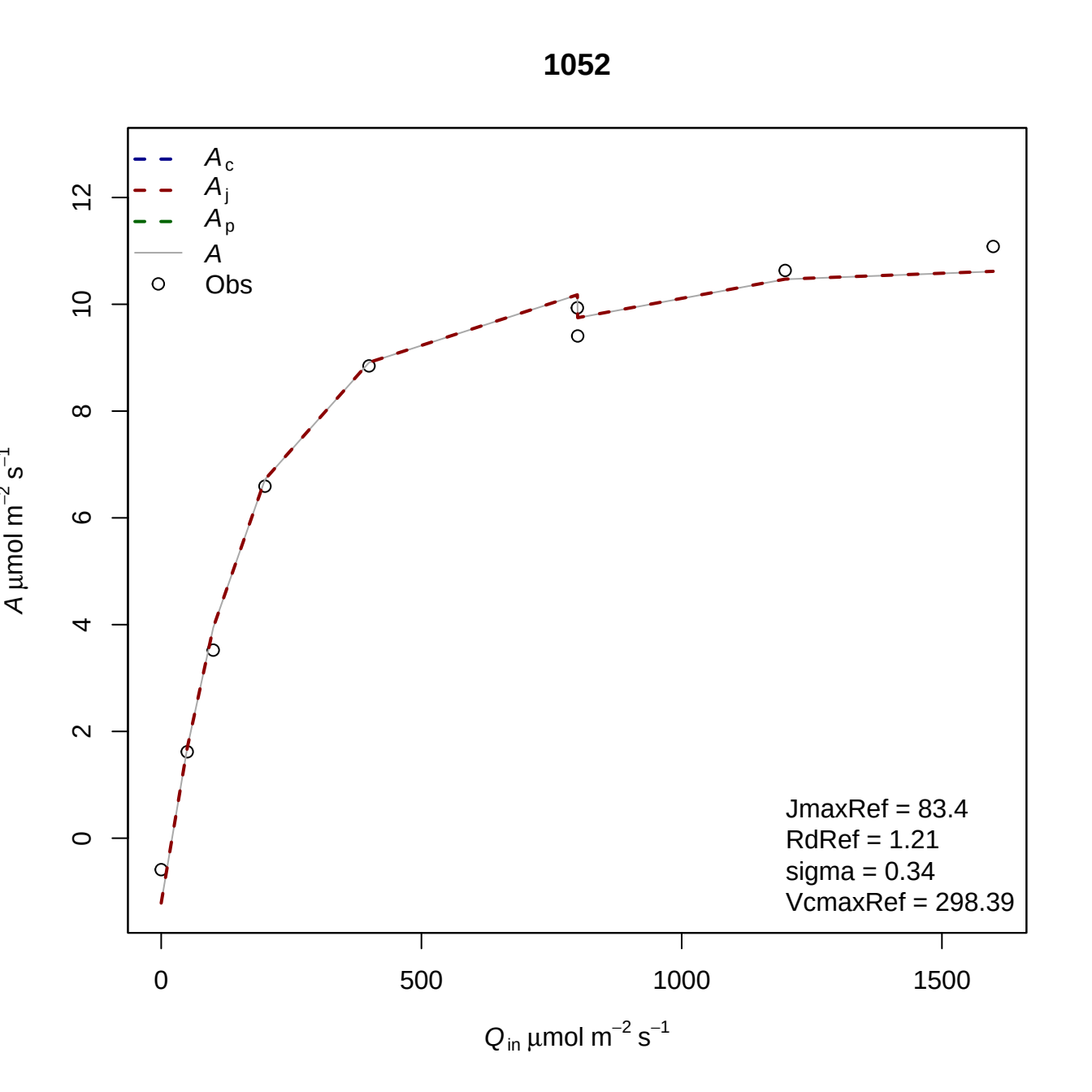

1047
A_c
A_j
A_p
A
Obs
JmaxRef = 68.28
RdRef = 1.91
sigma = 0.57
VcmaxRef = 276.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

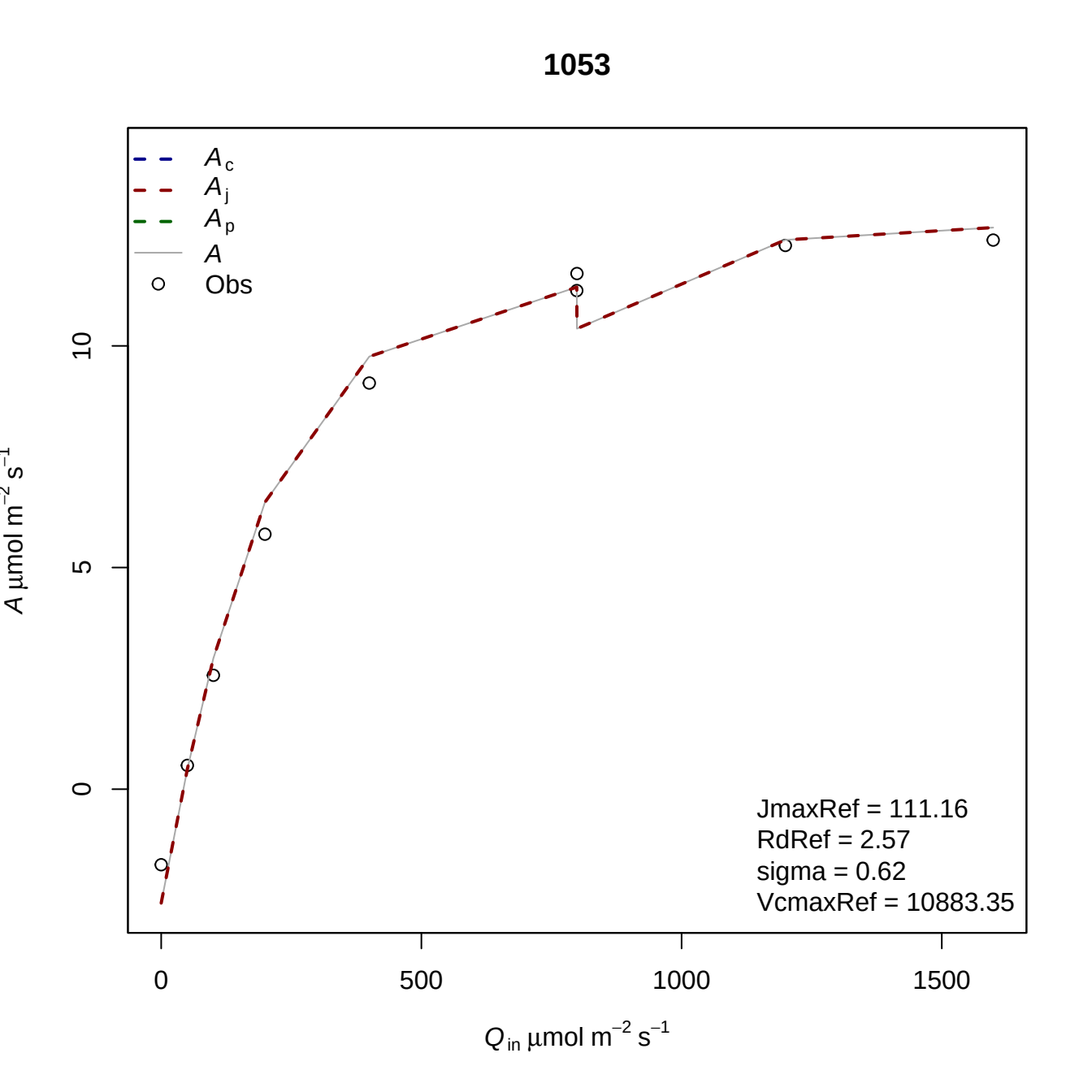

1048
A_c
A_j
A_p
A
Obs
JmaxRef = 80.62
RdRef = 2.25
sigma = 0.67
VcmaxRef = 251.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

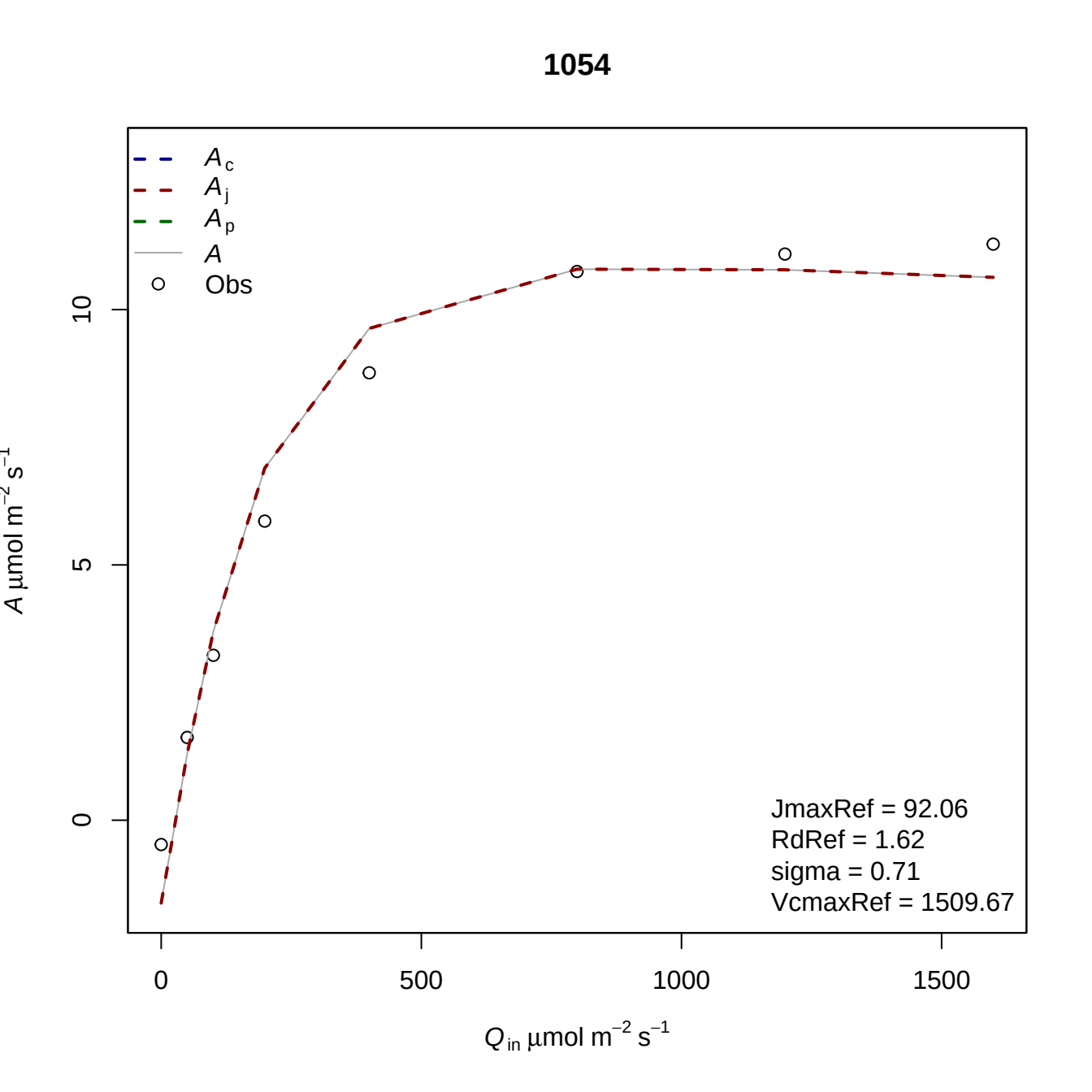

1049
A_c
A_j
A_p
A
Obs
JmaxRef = 145.16
RdRef = 1.72
sigma = 0.7
VcmaxRef = 1214.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

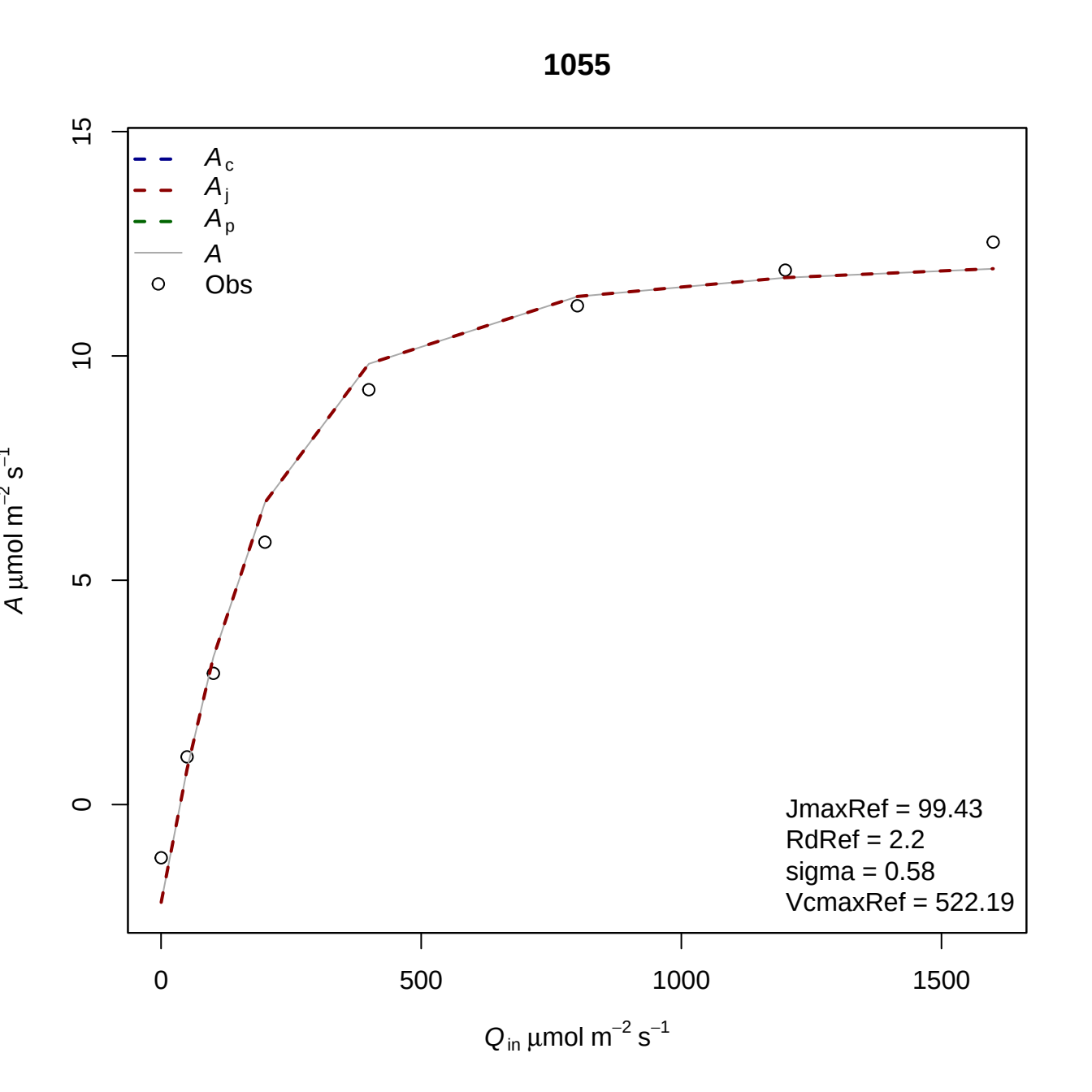

1050
A_c
A_j
A_p
A
Obs
JmaxRef = 106.34
RdRef = 0.92
sigma = 0.42
VcmaxRef = 196474.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

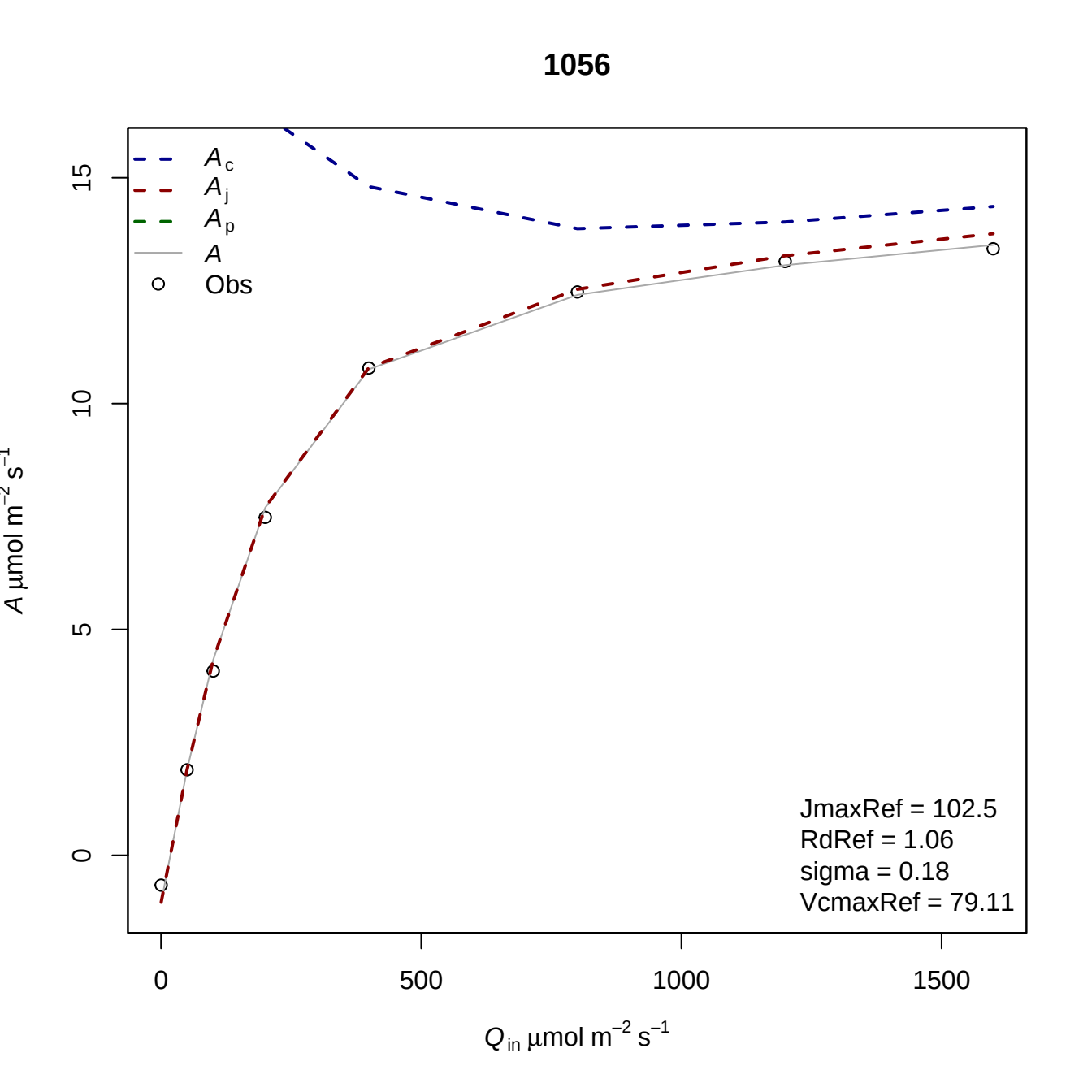

1051
A_c
A_j
A_p
A
Obs
JmaxRef = 116.82
RdRef = 1.9
sigma = 0.59
VcmaxRef = 708.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

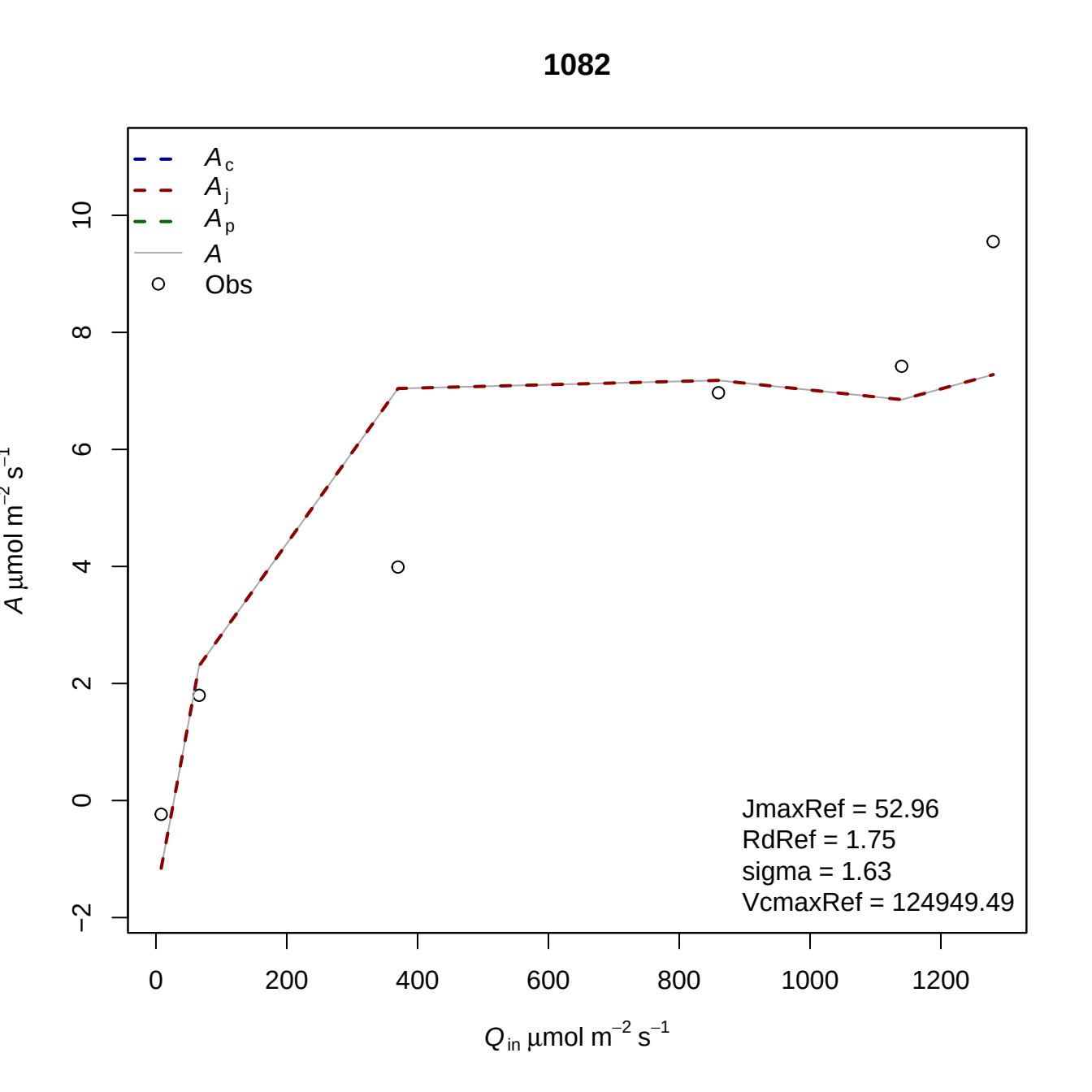

1052
A_c
A_j
A_p
A
Obs
JmaxRef = 83.4
RdRef = 1.21
sigma = 0.34
VcmaxRef = 298.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

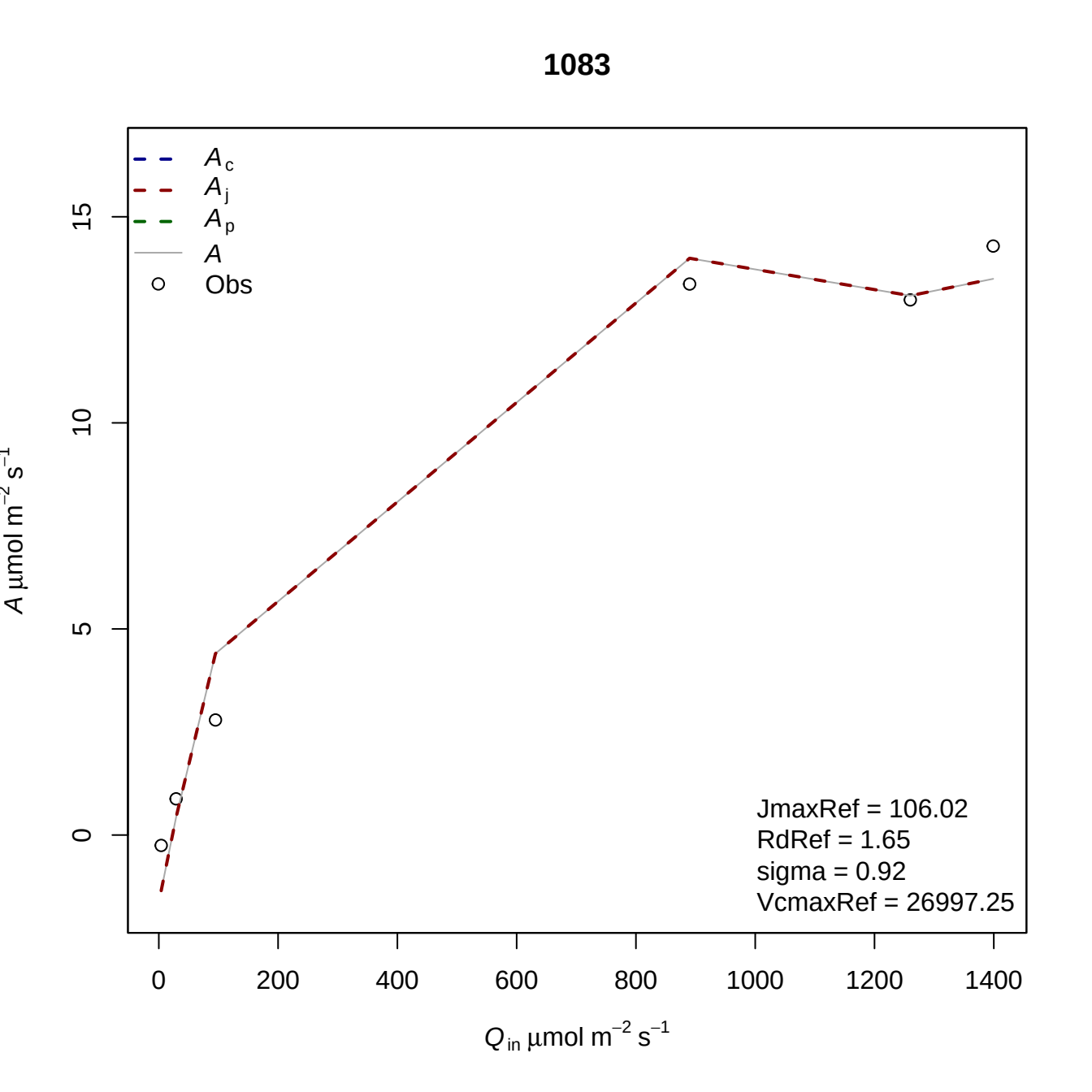

1053
A_c
A_j
A_p
A
Obs
JmaxRef = 111.16
RdRef = 2.57
sigma = 0.62
VcmaxRef = 10883.35
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

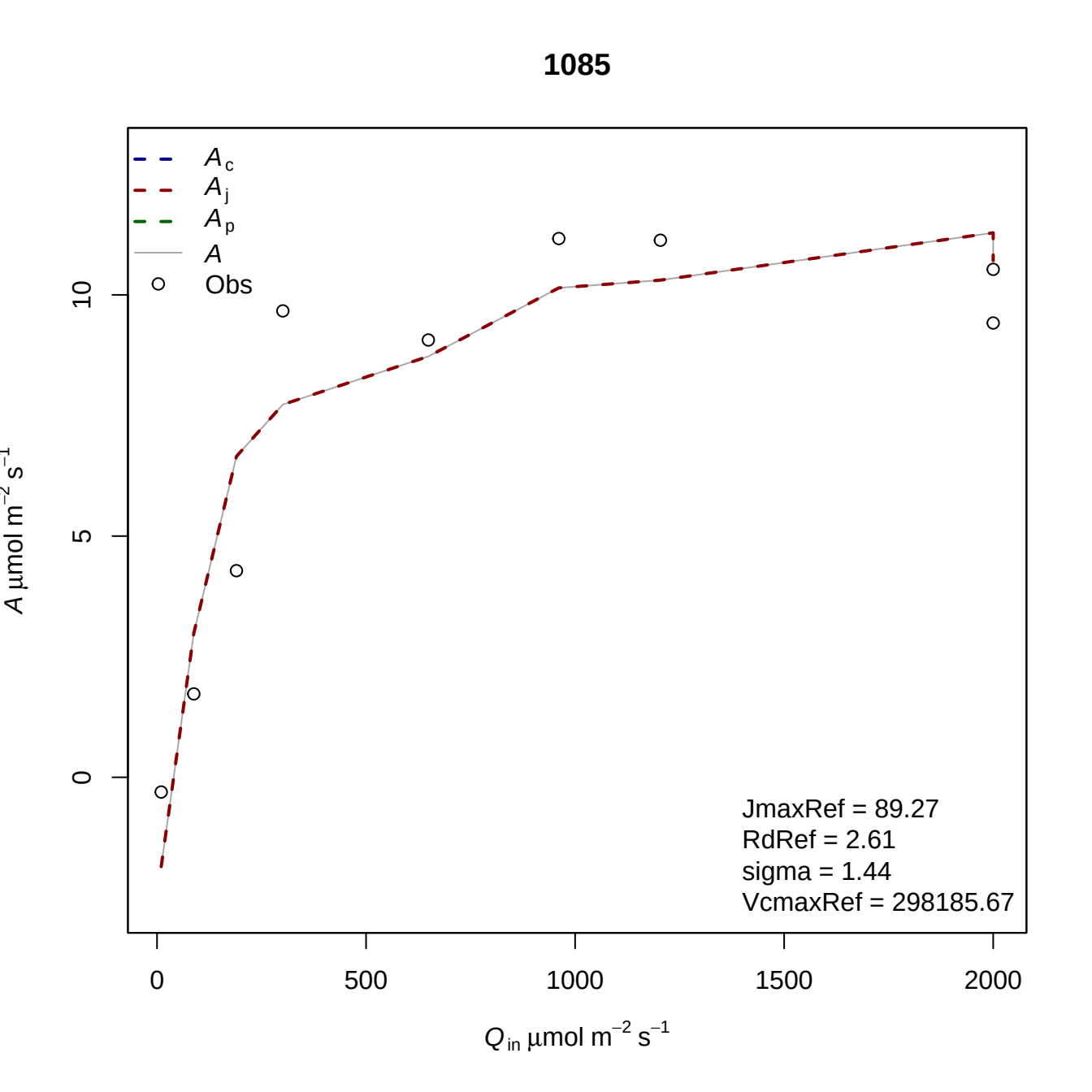

1054
A_c
A_j
A_p
A
Obs
JmaxRef = 92.06
RdRef = 1.62
sigma = 0.71
VcmaxRef = 1509.67
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

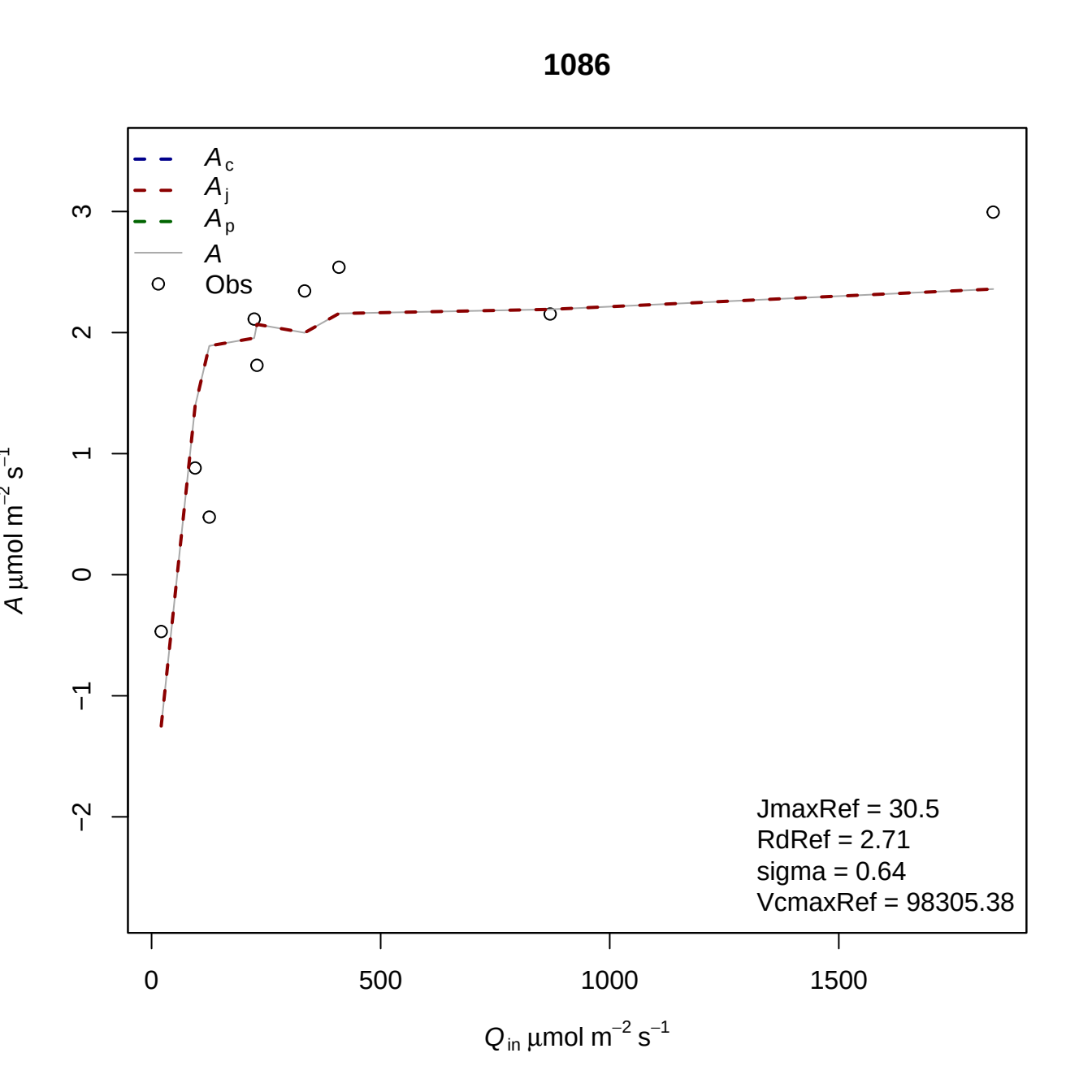

1055
A_c
A_j
A_p
A
Obs
JmaxRef = 99.43
RdRef = 2.2
sigma = 0.58
VcmaxRef = 522.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

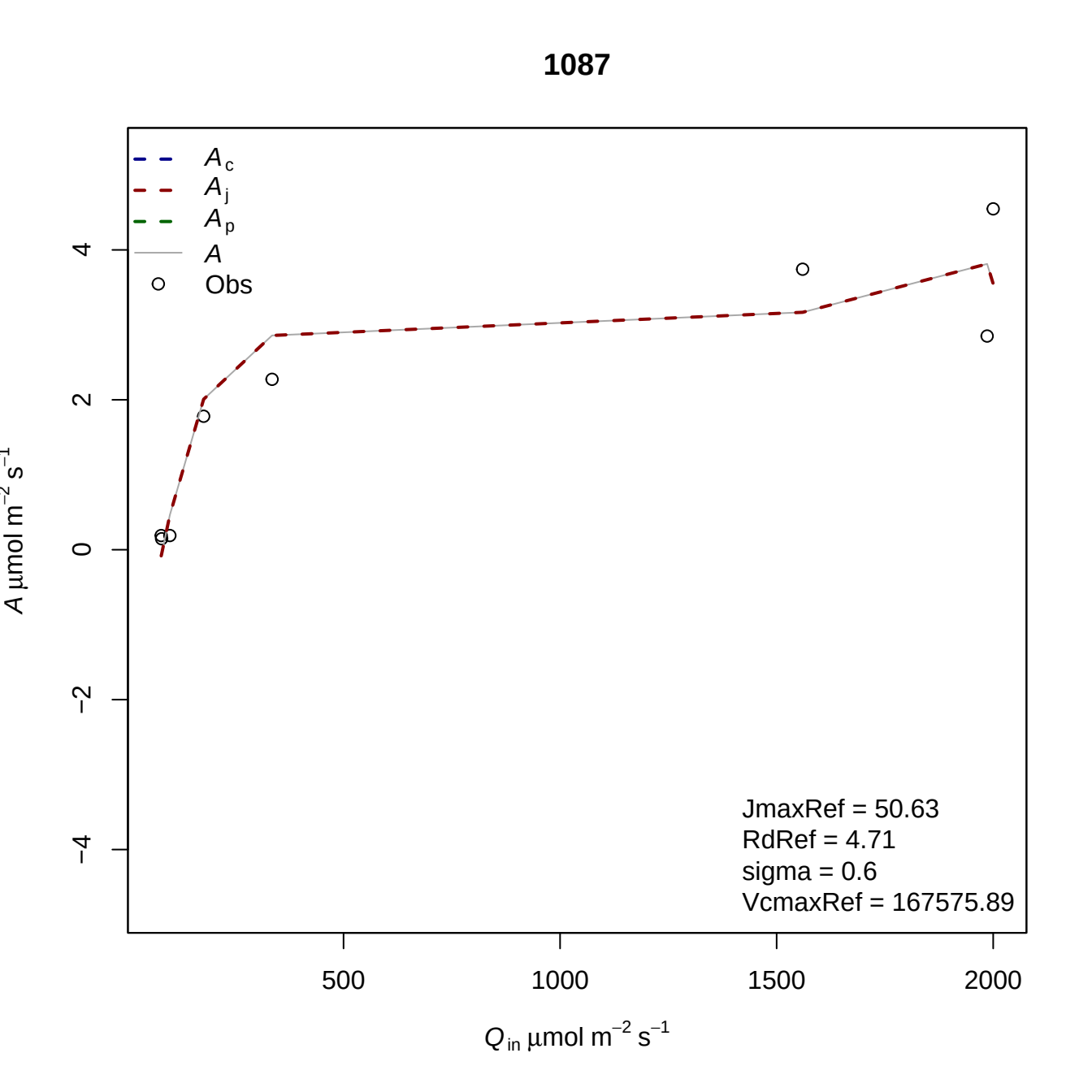

1056
A_c
A_j
A_p
A
Obs
JmaxRef = 102.5
RdRef = 1.06
sigma = 0.18
VcmaxRef = 79.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

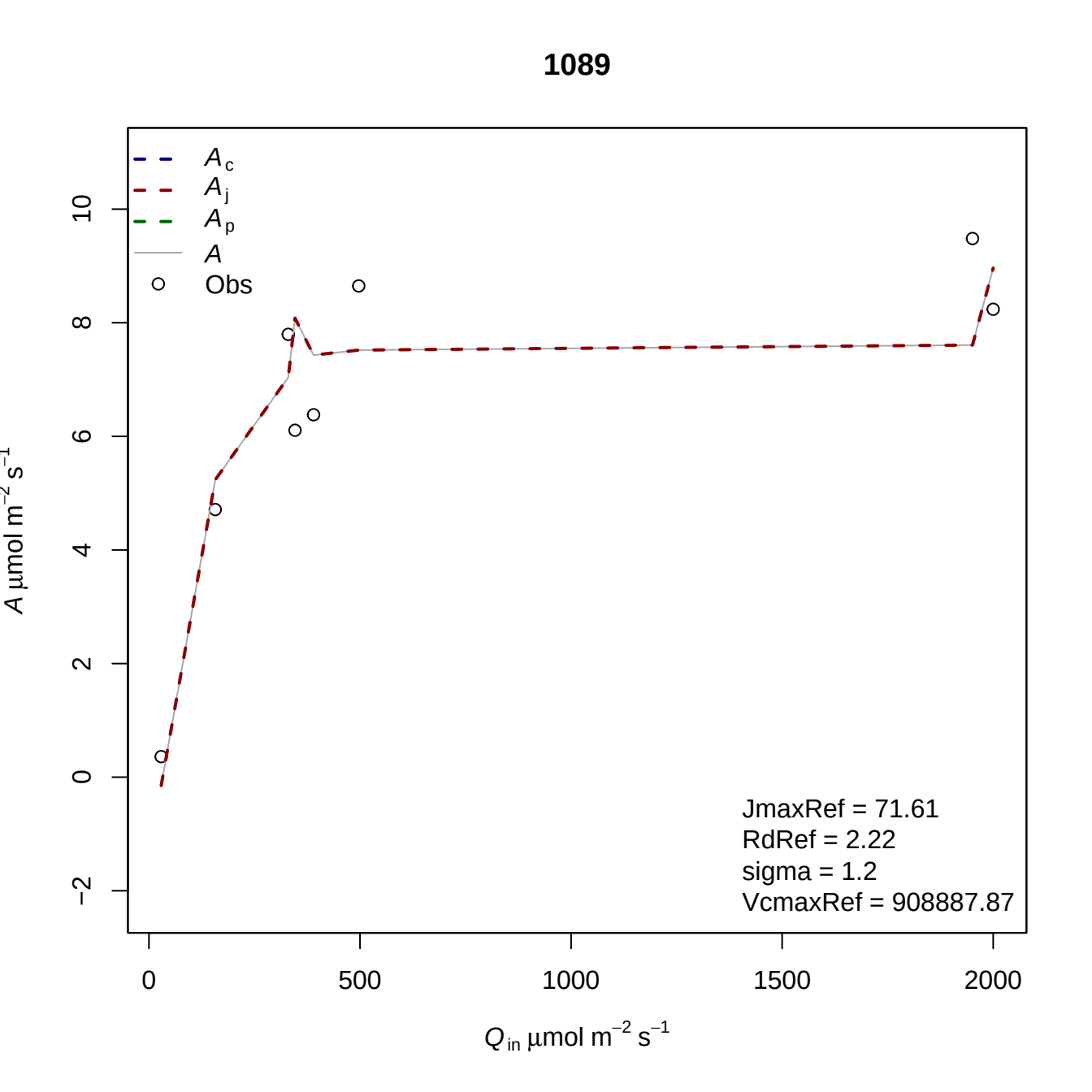

1082
A_c
A_j
A_p
A
Obs
JmaxRef = 52.96
RdRef = 1.75
sigma = 1.63
VcmaxRef = 124949.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

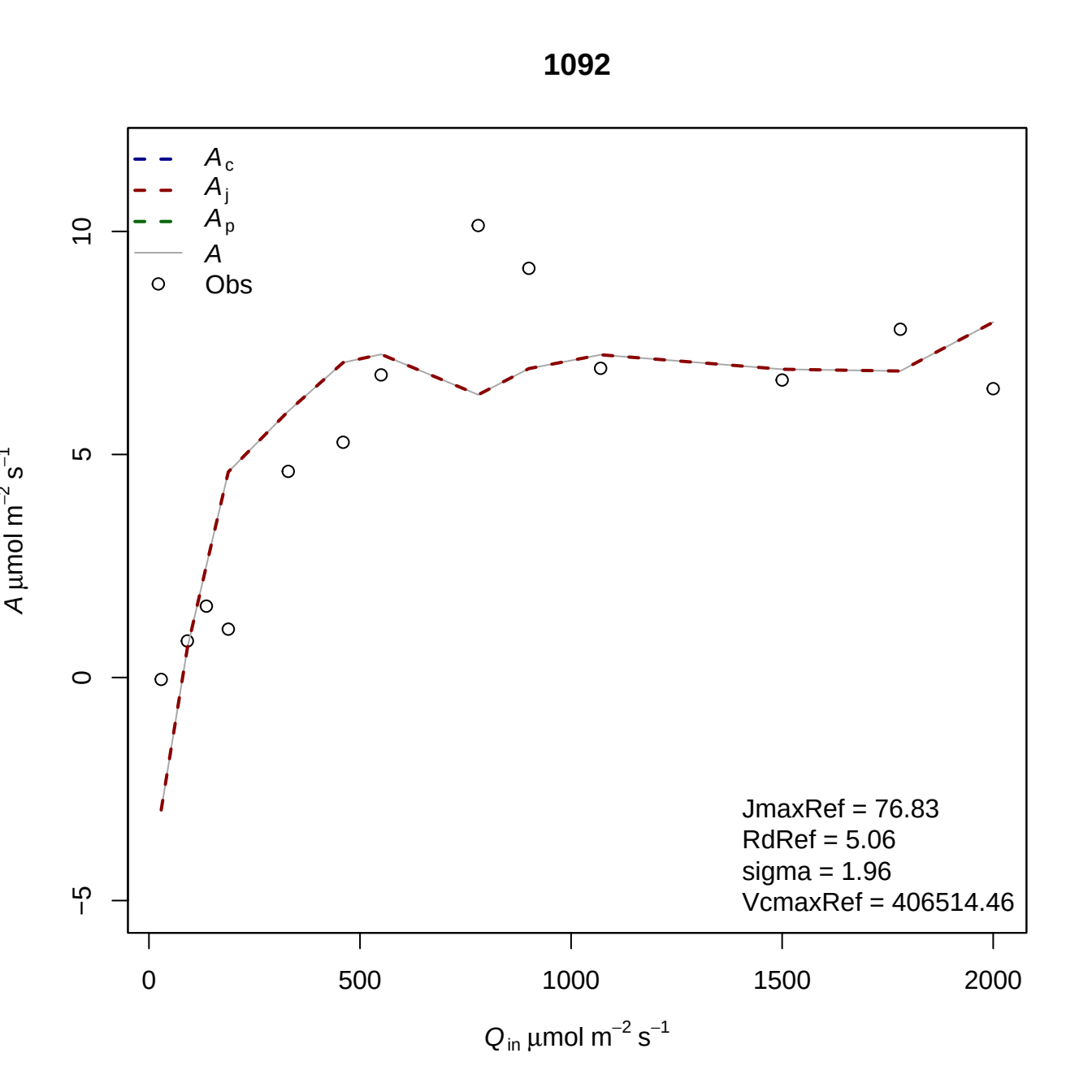

1083
A_c
A_j
A_p
A
Obs
JmaxRef = 106.02
RdRef = 1.65
sigma = 0.92
VcmaxRef = 26997.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

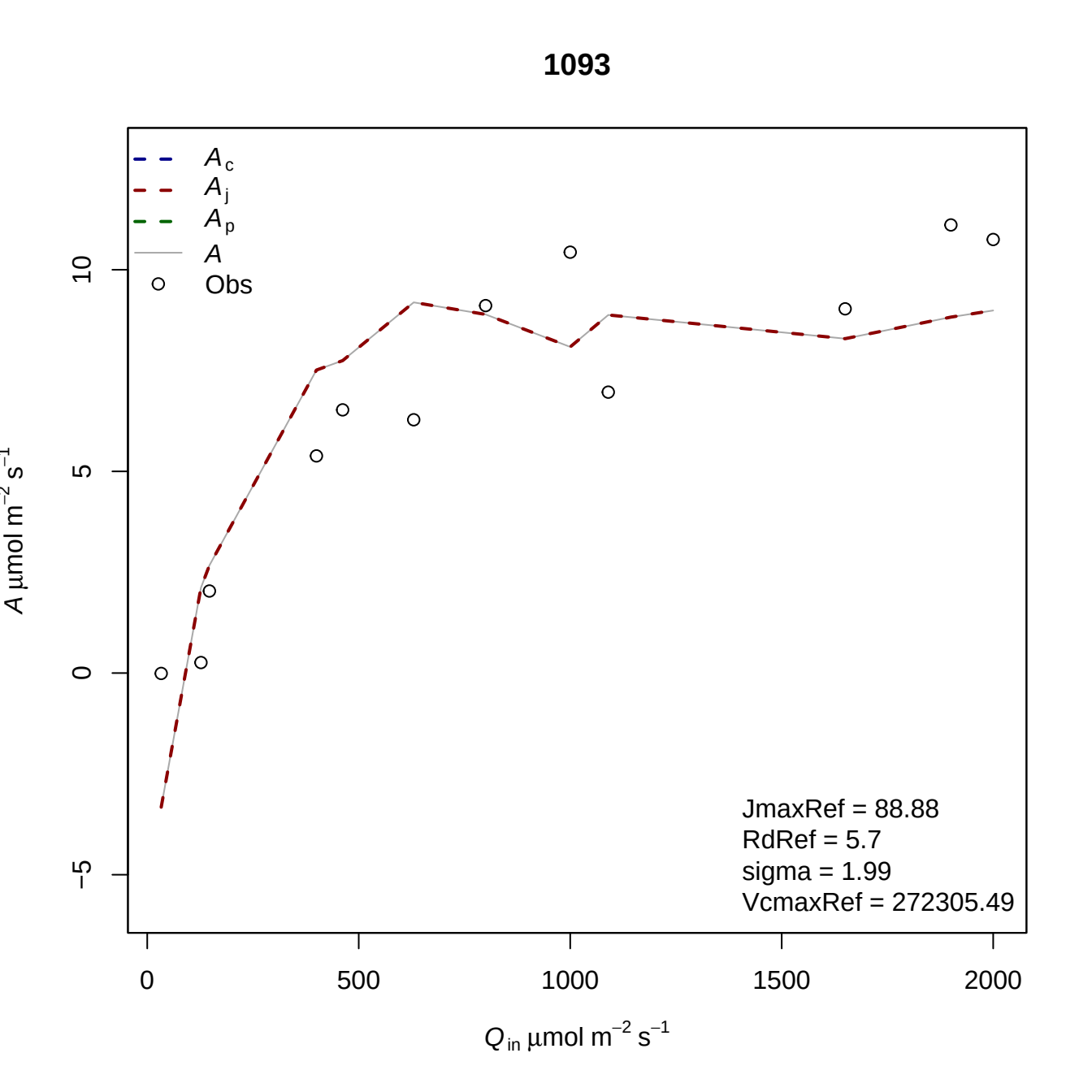

1085
A_c
A_j
A_p
A
Obs
JmaxRef = 89.27
RdRef = 2.61
sigma = 1.44
VcmaxRef = 298185.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

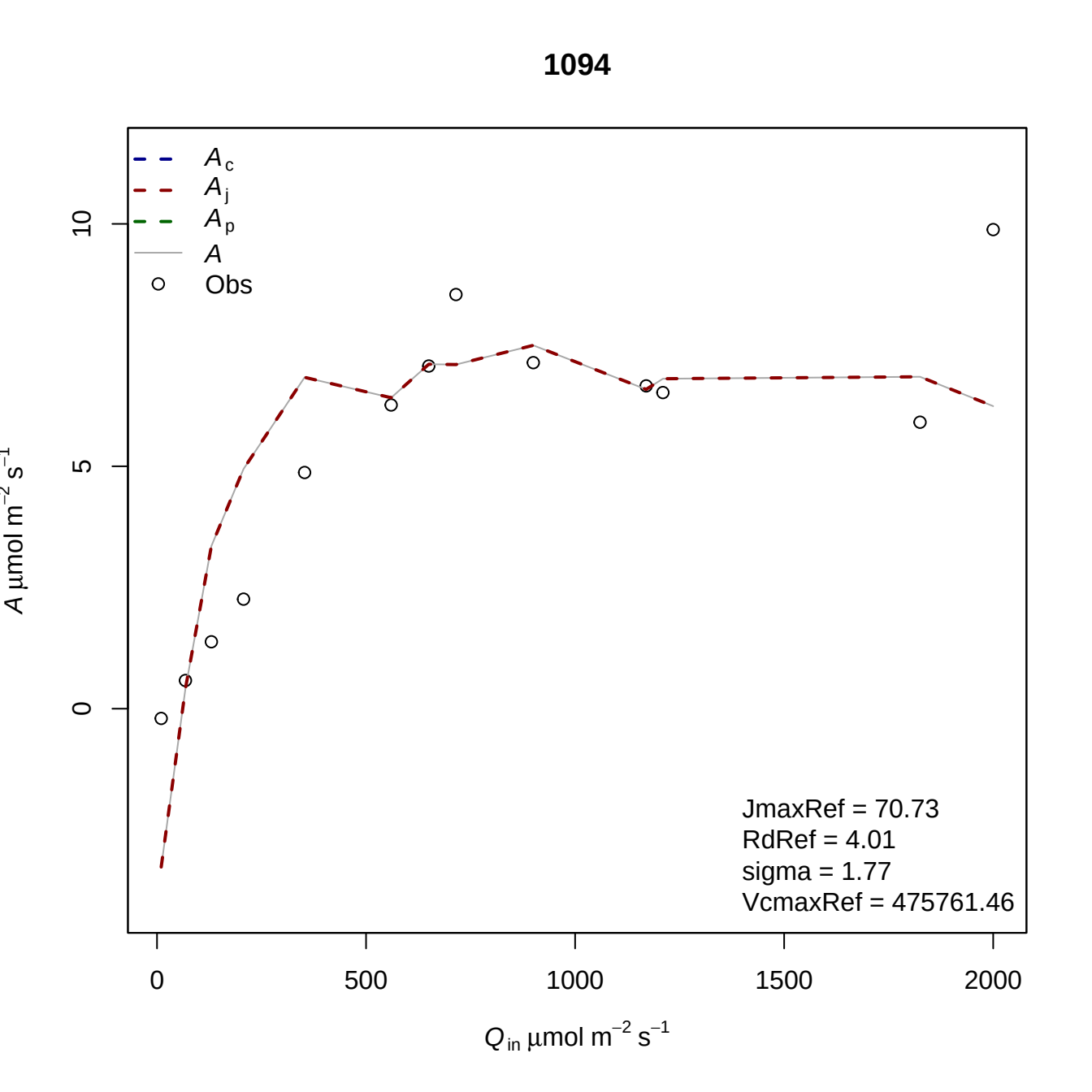

1086
A_c
A_j
A_p
A
Obs
JmaxRef = 30.5
RdRef = 2.71
sigma = 0.64
VcmaxRef = 98305.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
500
1000
1500

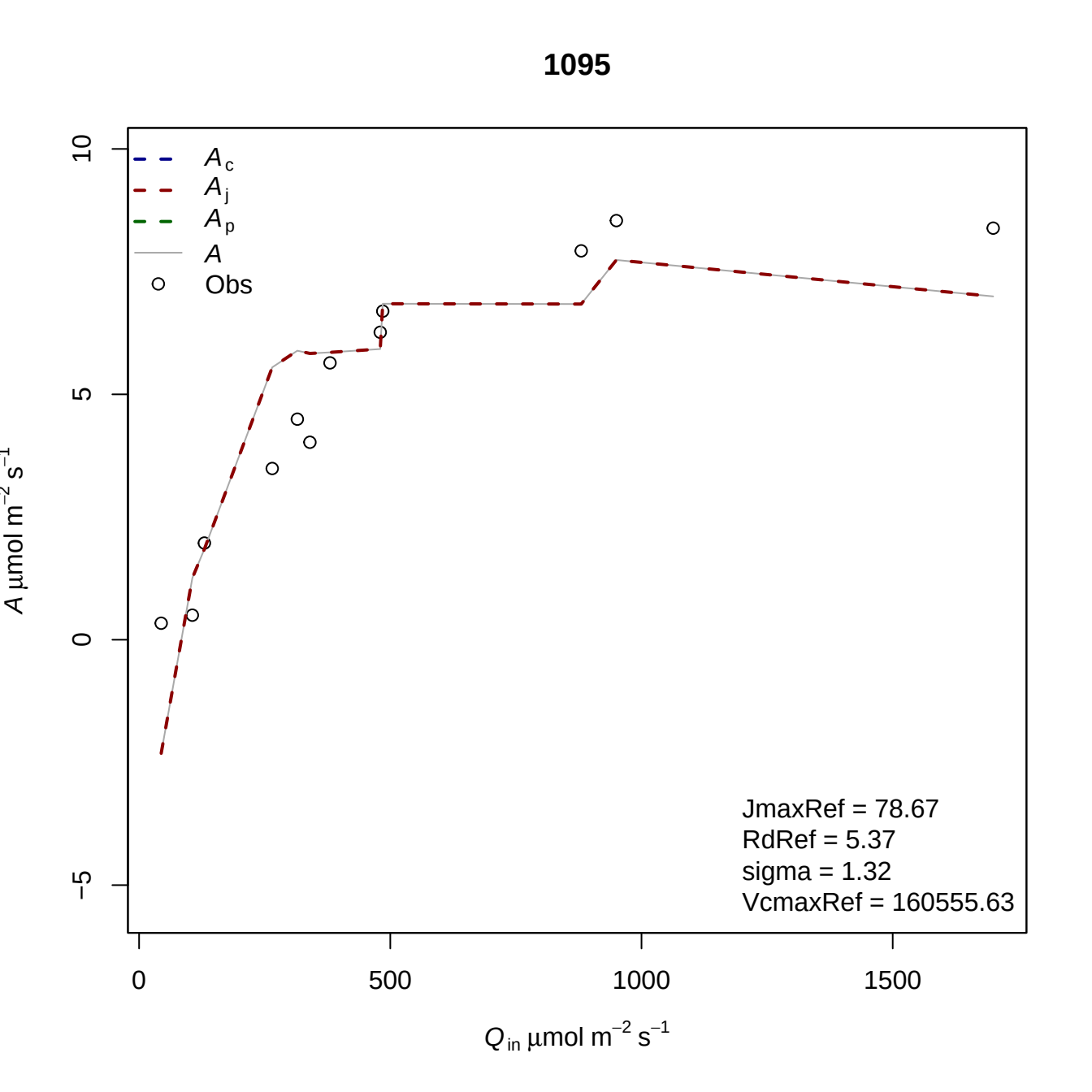

1087
A_c
A_j
A_p
A
Obs
JmaxRef = 50.63
RdRef = 4.71
sigma = 0.6
VcmaxRef = 167575.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
500
1000
1500
2000

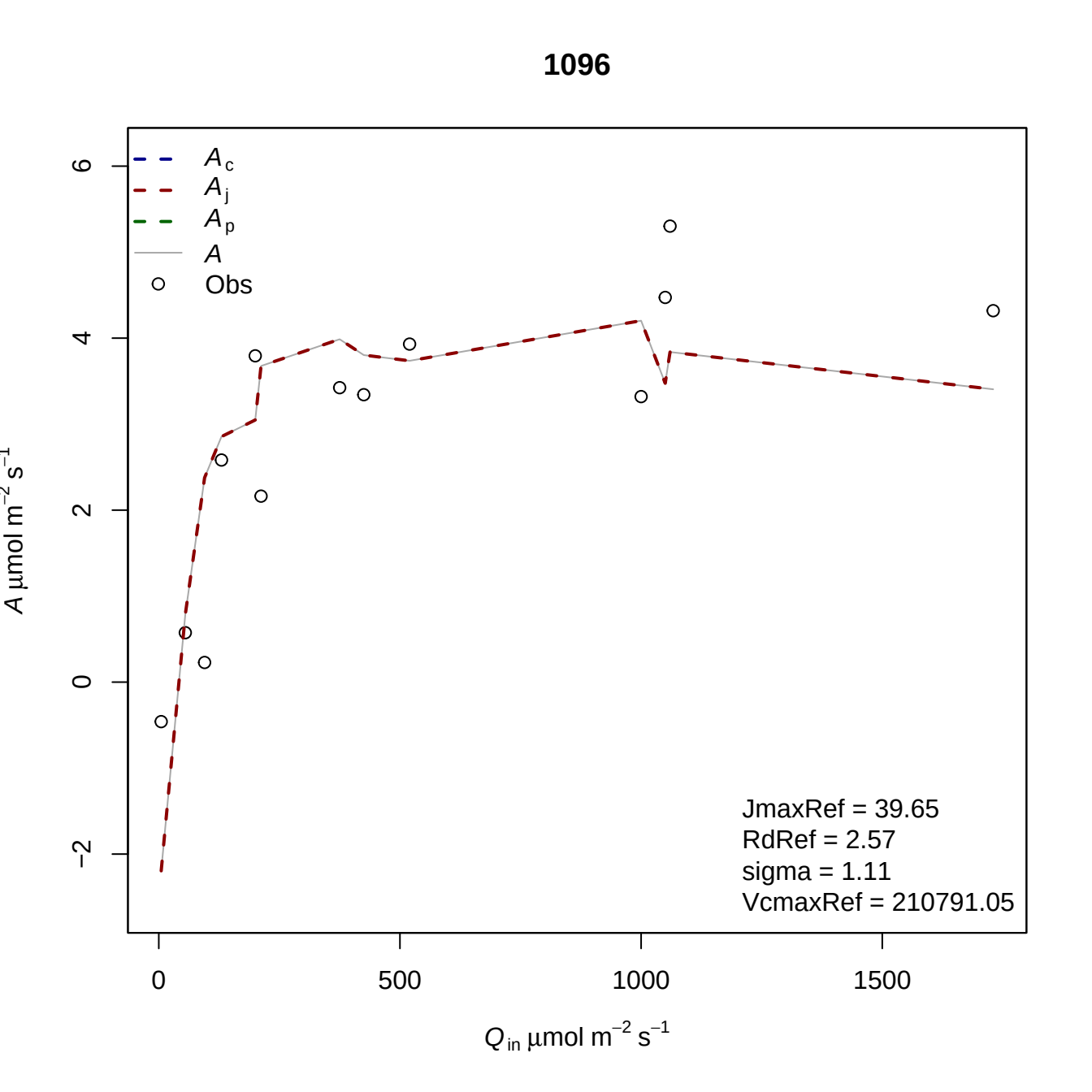

1089
A_c
A_j
A_p
A
Obs
JmaxRef = 71.61
RdRef = 2.22
sigma = 1.2
VcmaxRef = 908887.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

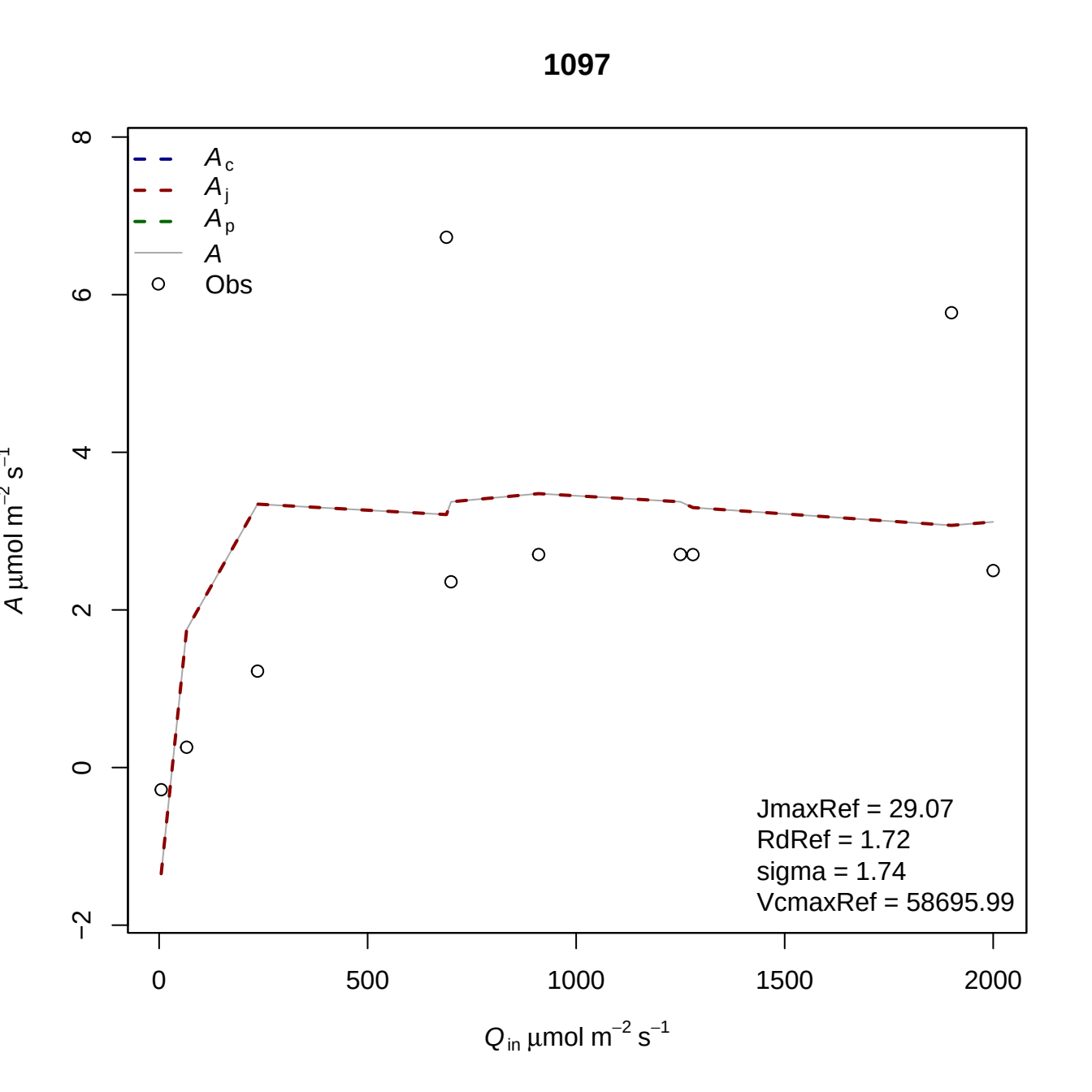

1092
A_c
A_j
A_p
A
Obs
JmaxRef = 76.83
RdRef = 5.06
sigma = 1.96
VcmaxRef = 406514.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

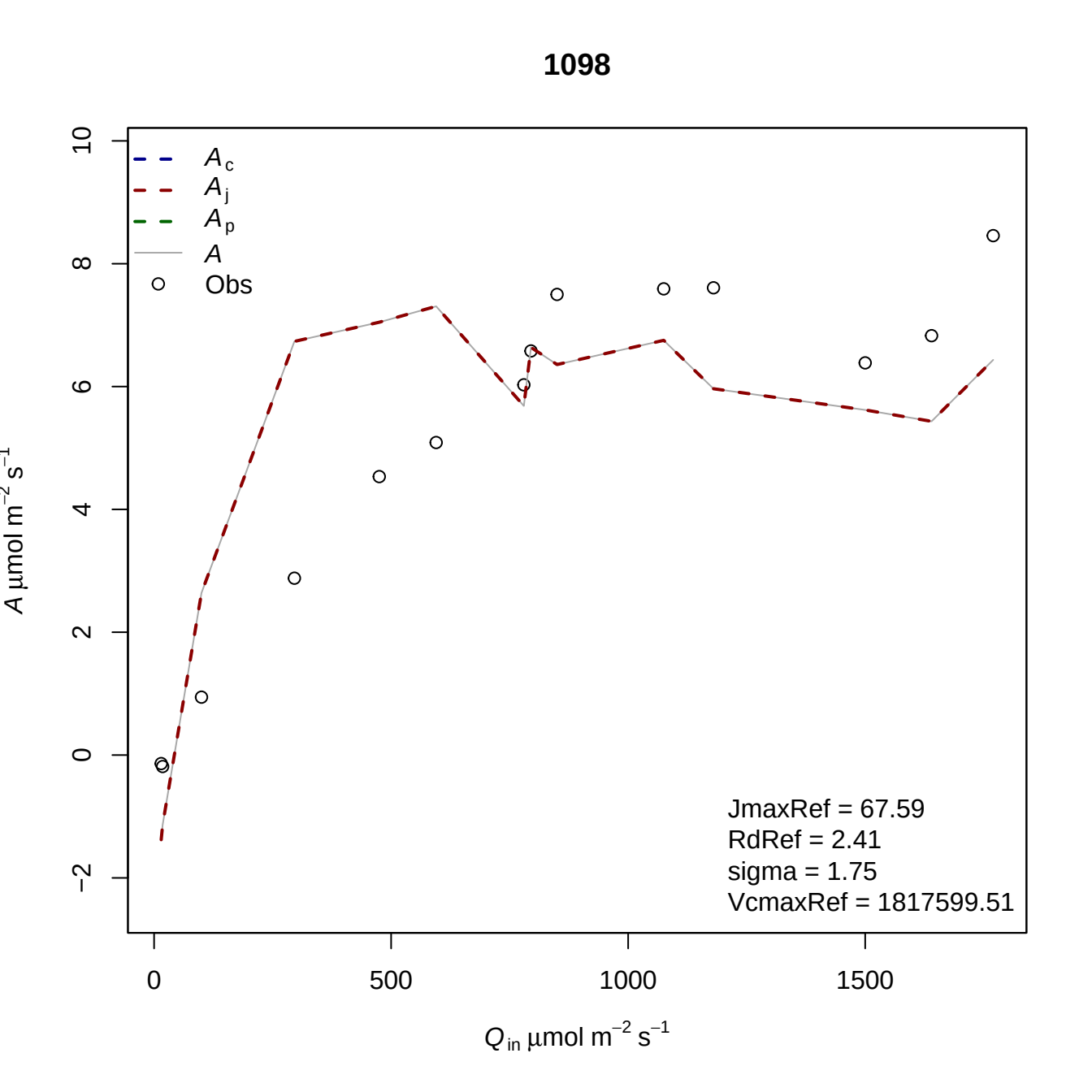

1093
A_c
A_j
A_p
A
Obs
JmaxRef = 88.88
RdRef = 5.7
sigma = 1.99
VcmaxRef = 272305.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

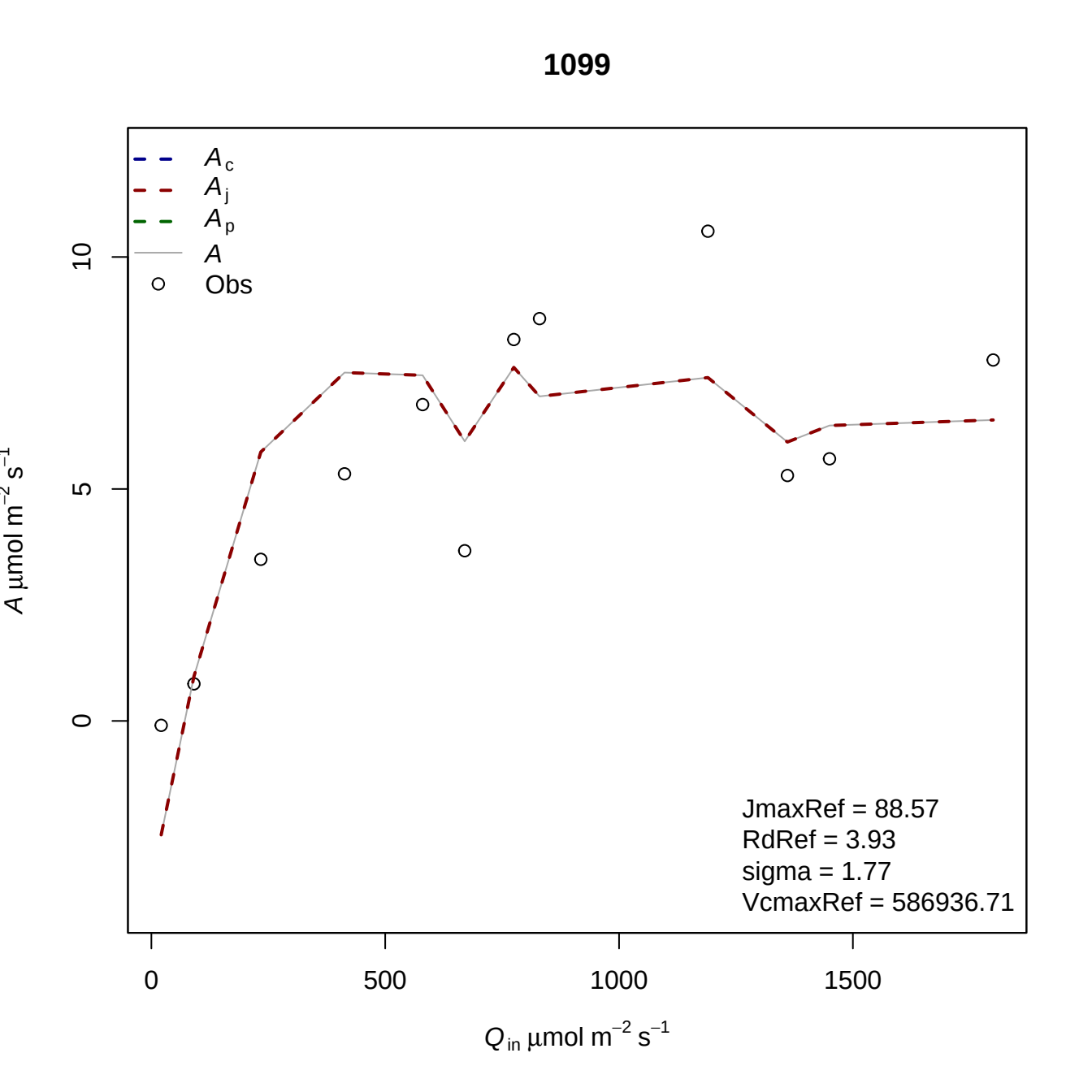

1094
A_c
A_j
A_p
A
Obs
JmaxRef = 70.73
RdRef = 4.01
sigma = 1.77
VcmaxRef = 475761.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

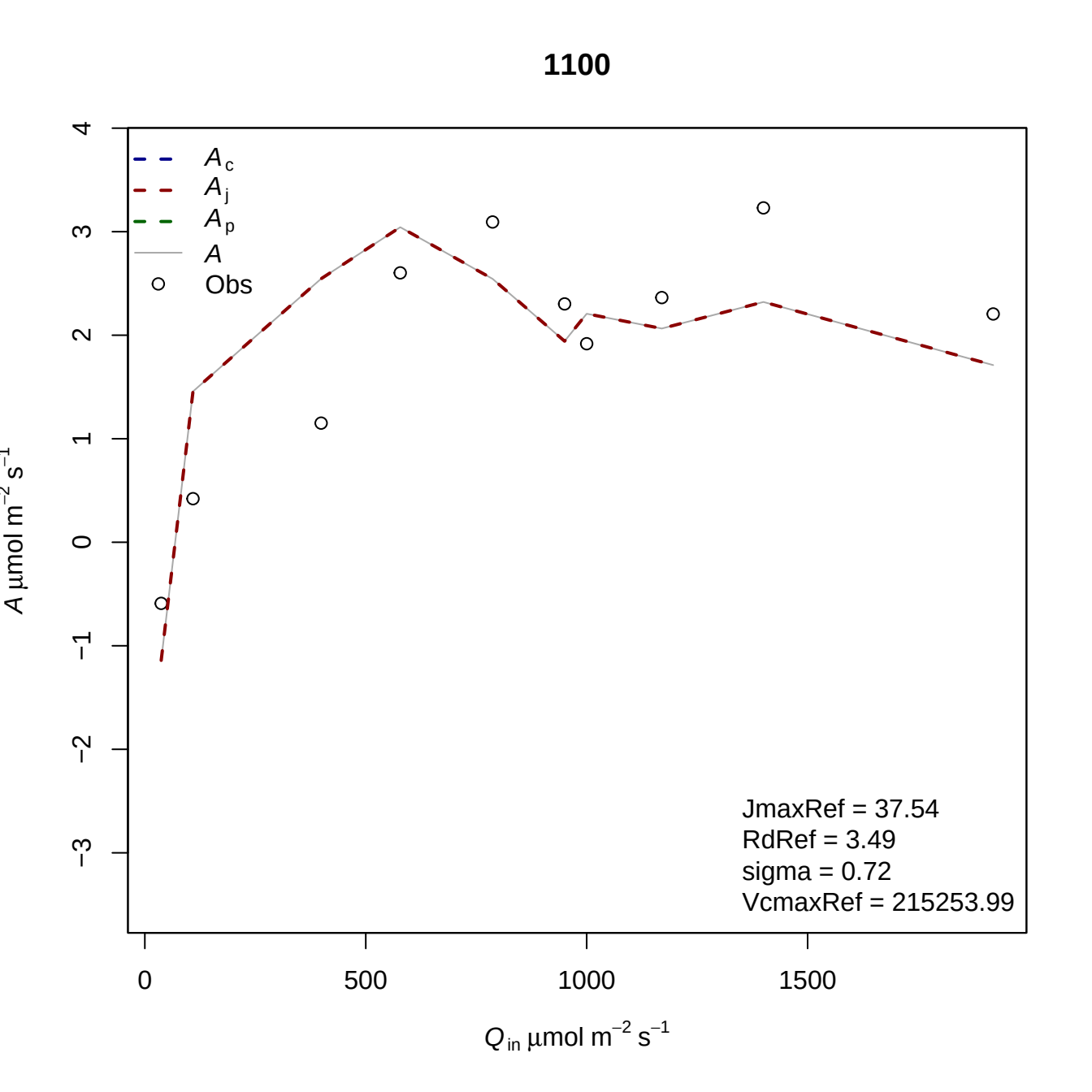

1095
A_c
A_j
A_p
A
Obs
JmaxRef = 78.67
RdRef = 5.37
sigma = 1.32
VcmaxRef = 160555.63
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

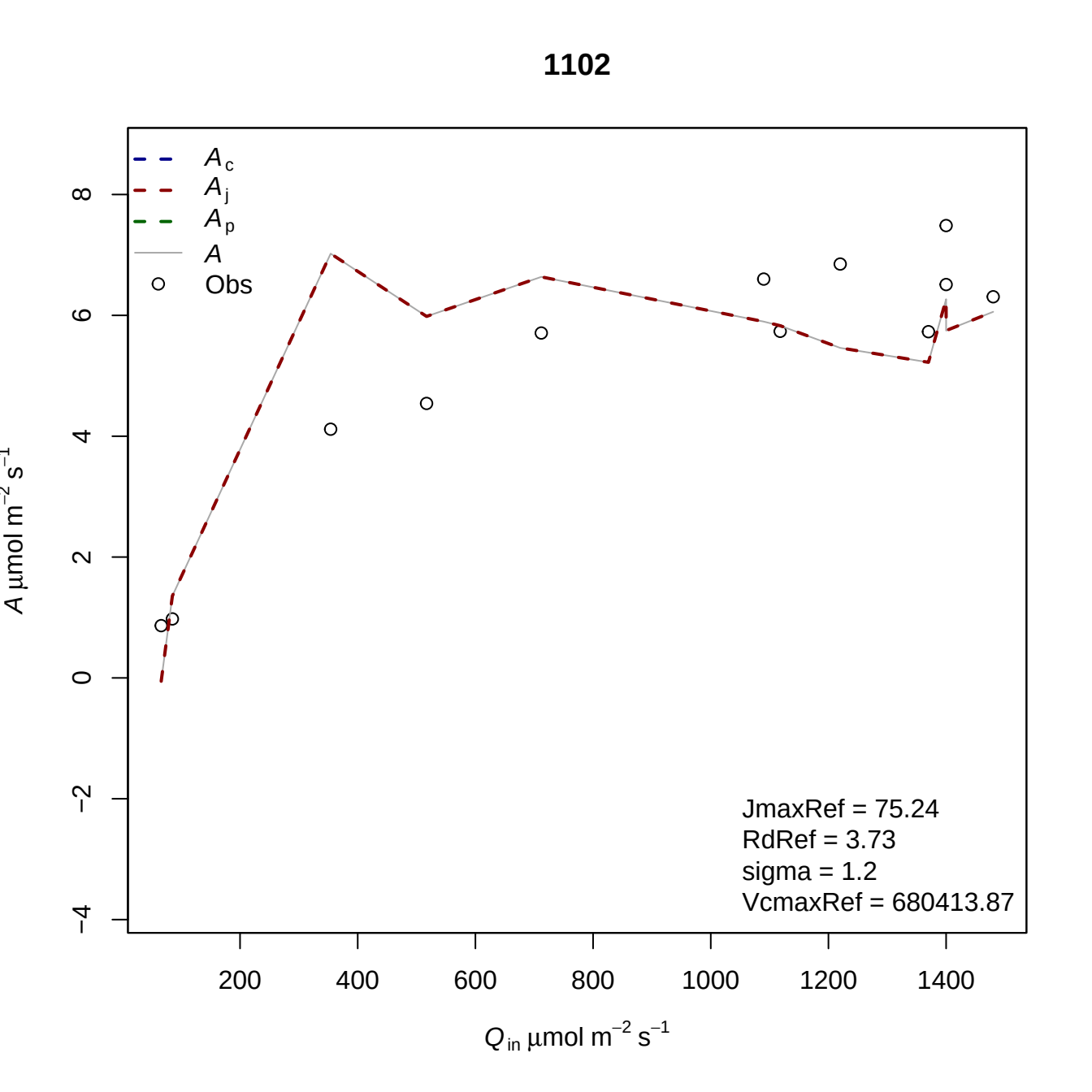

1096
A_c
A_j
A_p
A
Obs
JmaxRef = 39.65
RdRef = 2.57
sigma = 1.11
VcmaxRef = 210791.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

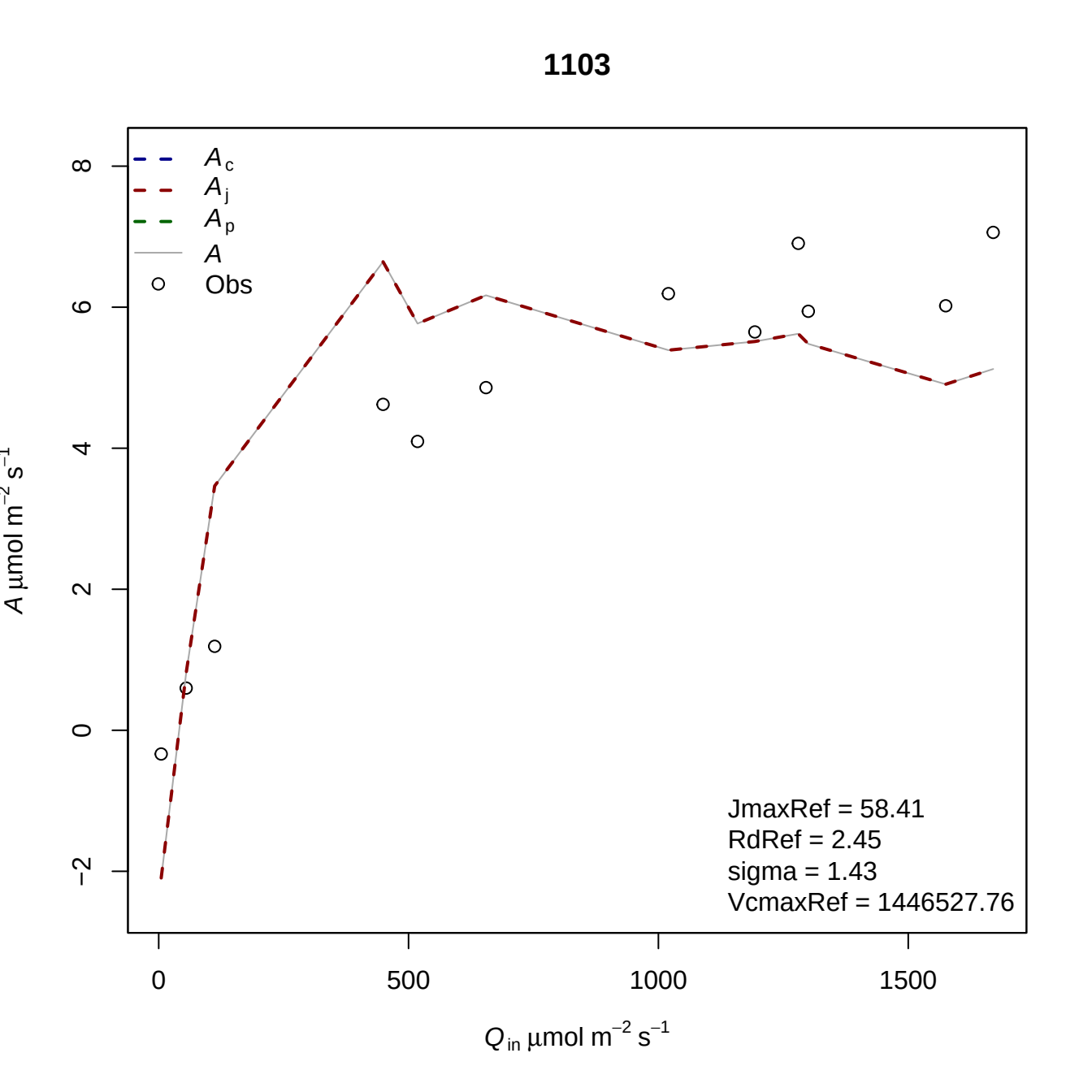

1097
A_c
A_j
A_p
A
Obs
JmaxRef = 29.07
RdRef = 1.72
sigma = 1.74
VcmaxRef = 58695.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

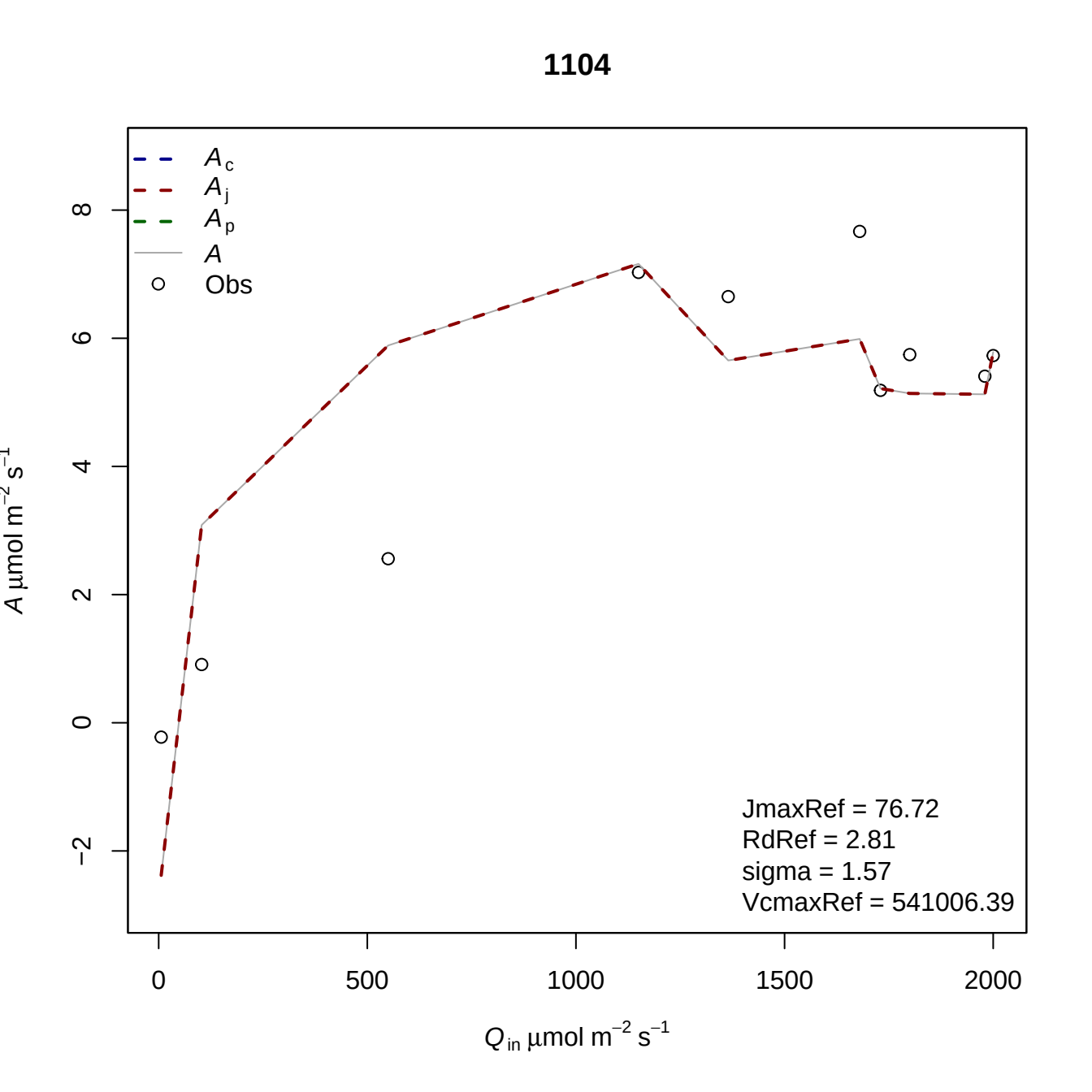

1098
A_c
A_j
A_p
A
Obs
JmaxRef = 67.59
RdRef = 2.41
sigma = 1.75
VcmaxRef = 1817599.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

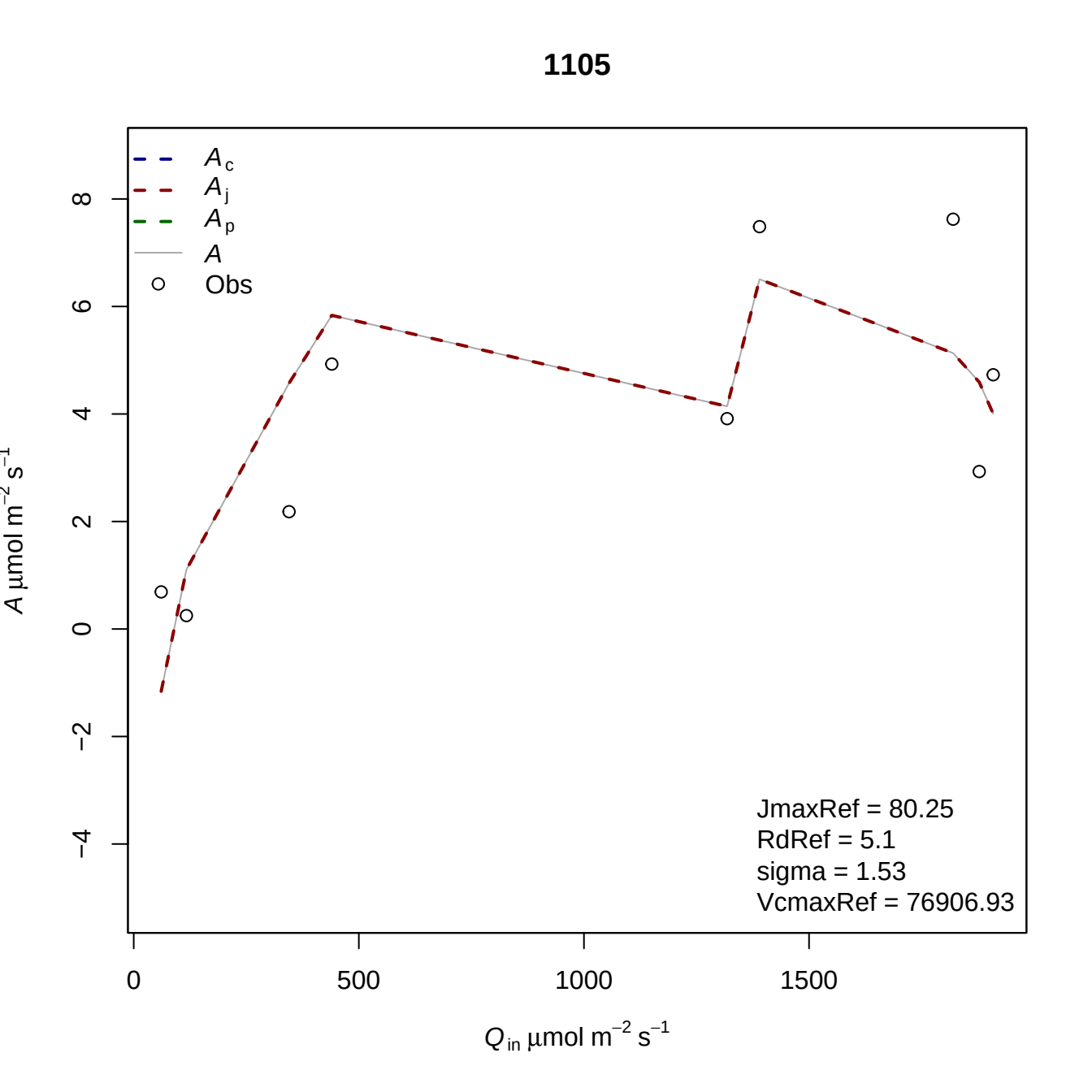

1099
A_c
A_j
A_p
A
Obs
JmaxRef = 88.57
RdRef = 3.93
sigma = 1.77
VcmaxRef = 586936.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

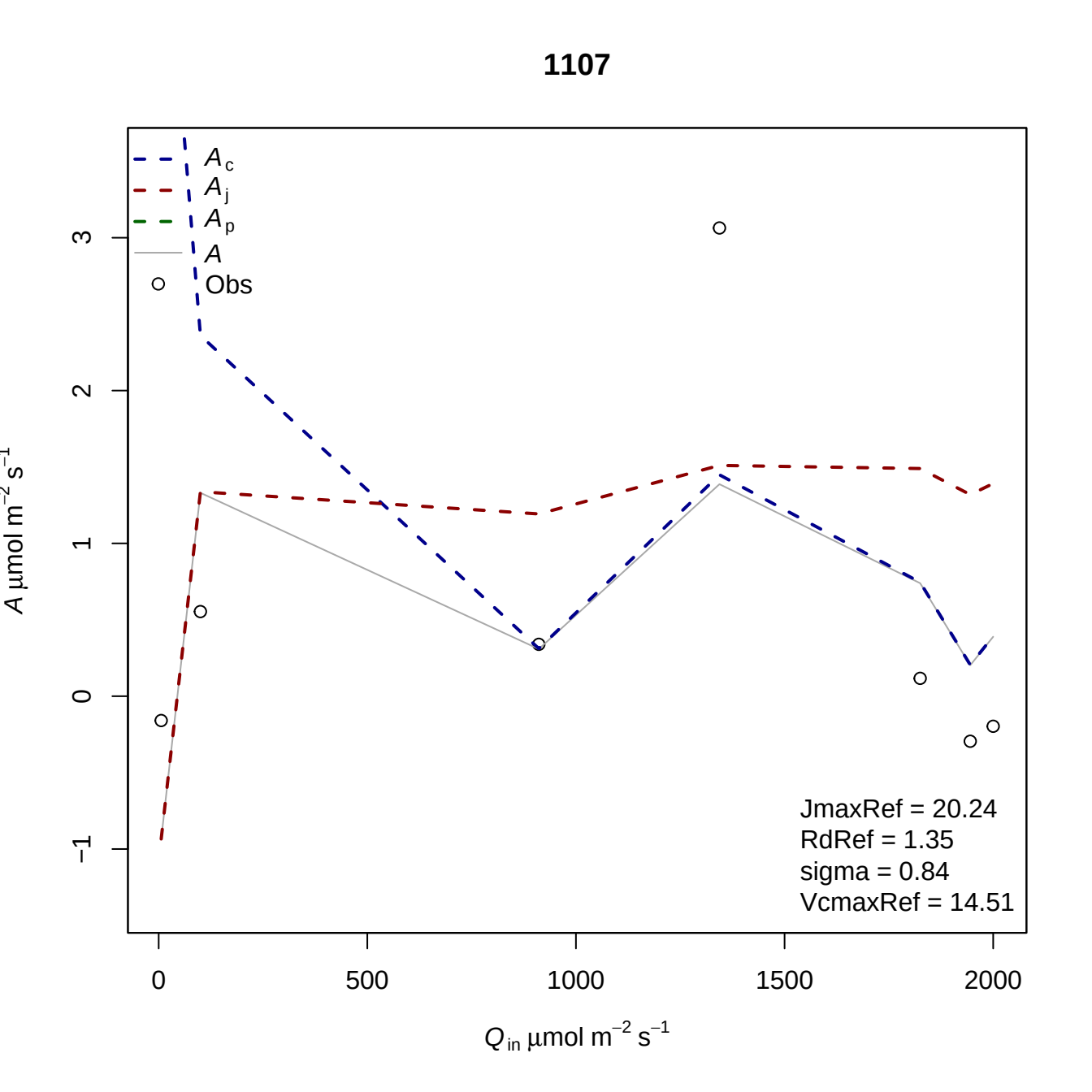

1100
A_c
A_j
A_p
A
Obs
JmaxRef = 37.54
RdRef = 3.49
sigma = 0.72
VcmaxRef = 215253.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500

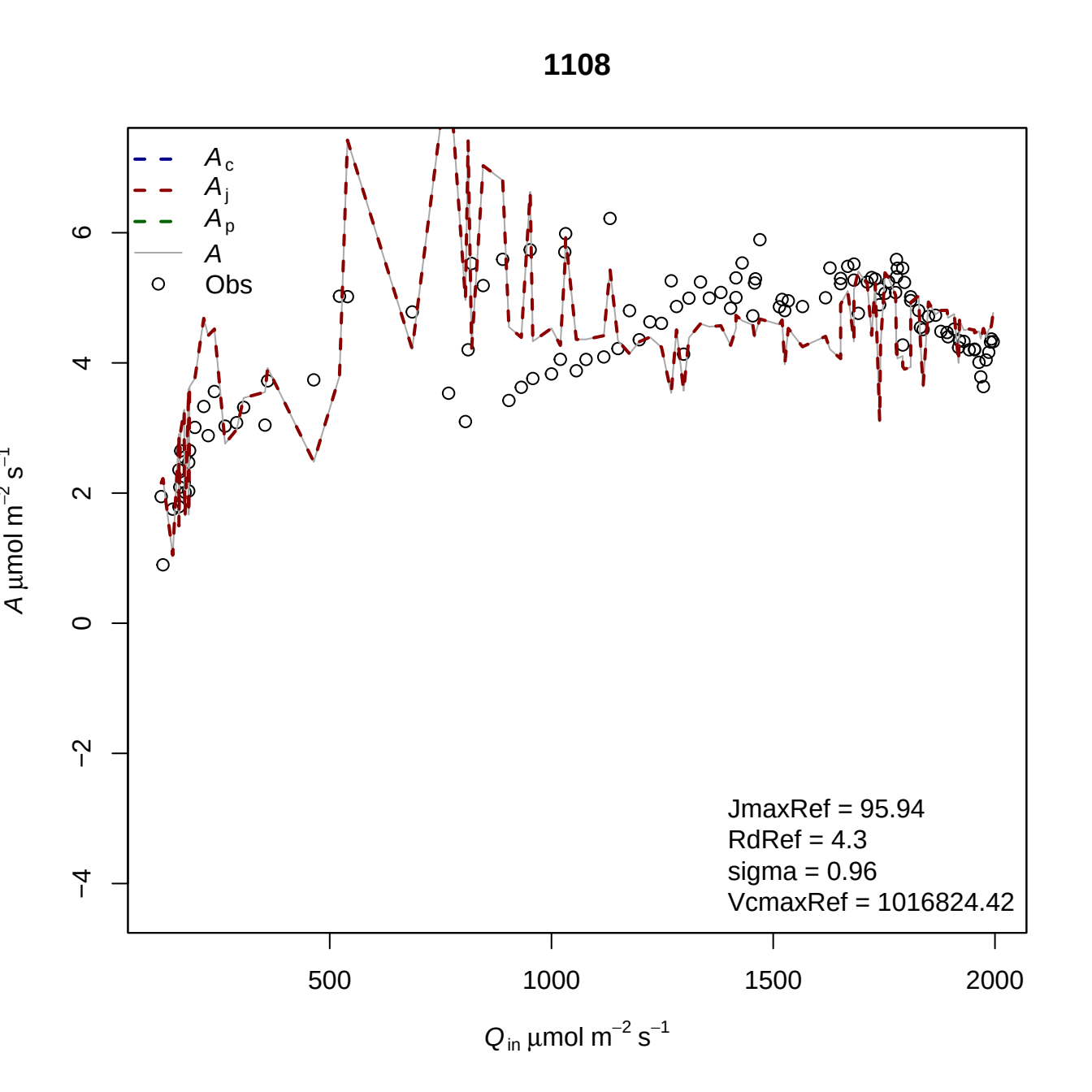

1102
A_c
A_j
A_p
A
Obs
JmaxRef = 75.24
RdRef = 3.73
sigma = 1.2
VcmaxRef = 680413.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

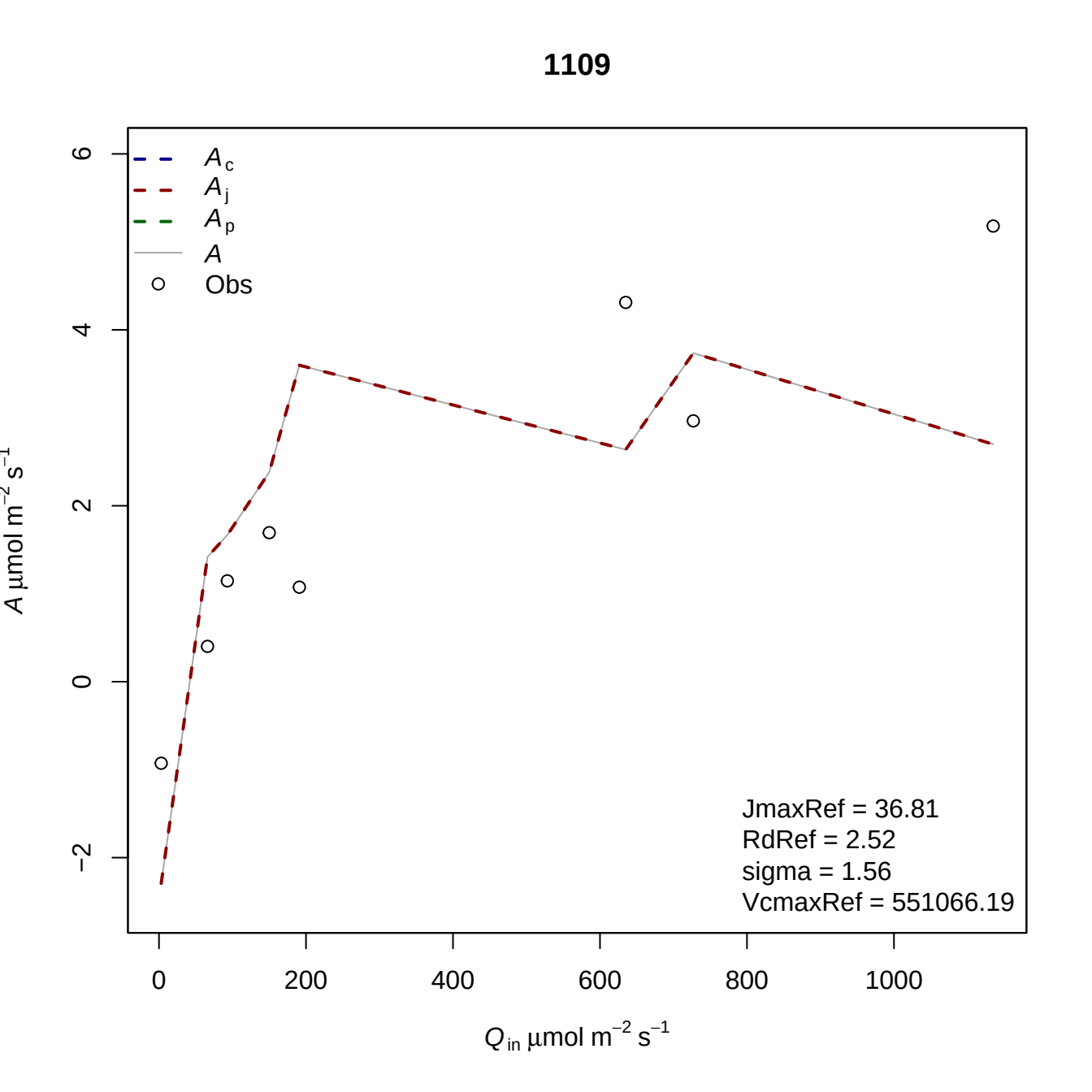

1103
A_c
A_j
A_p
A
Obs
JmaxRef = 58.41
RdRef = 2.45
sigma = 1.43
VcmaxRef = 1446527.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

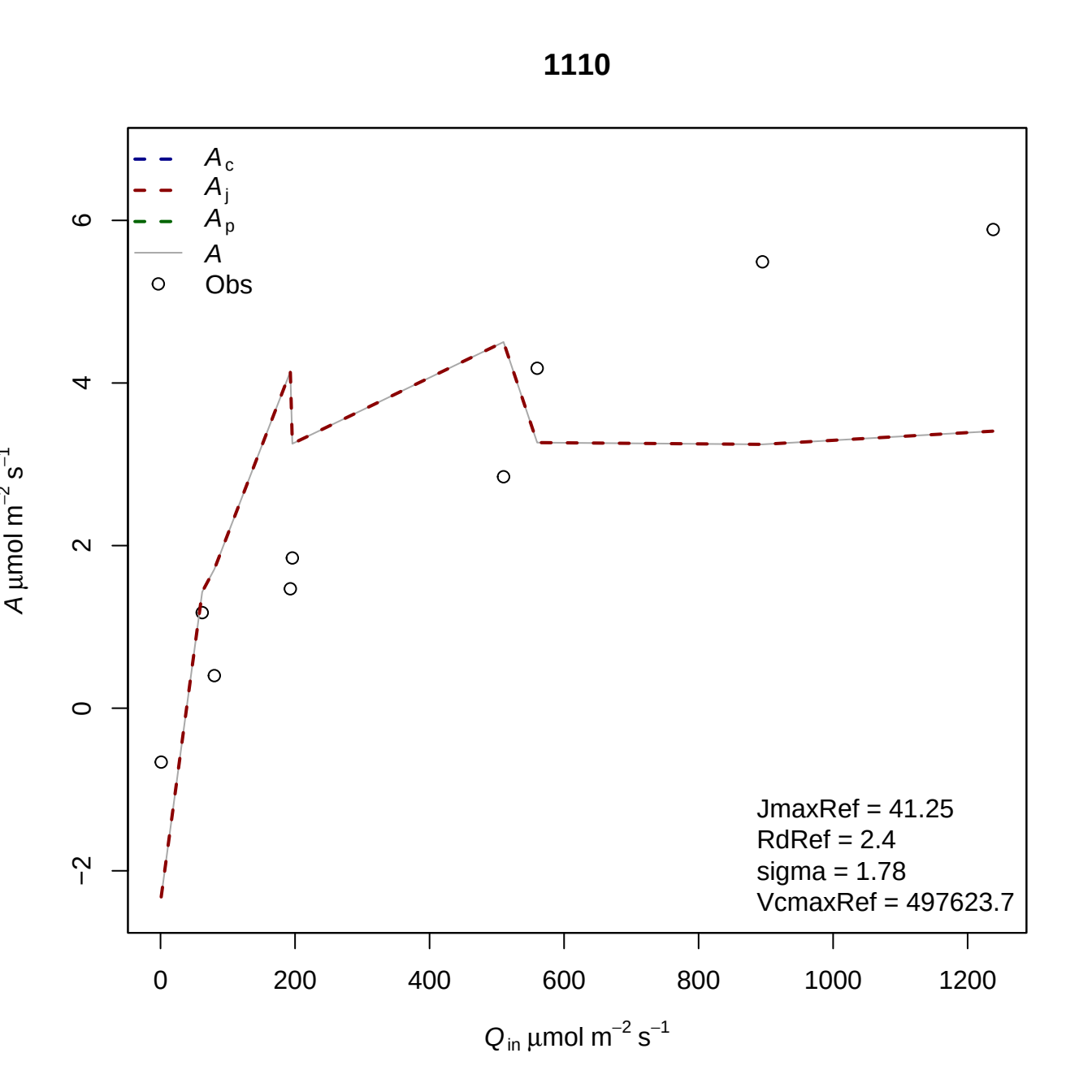

1104
A_c
A_j
A_p
A
Obs
JmaxRef = 76.72
RdRef = 2.81
sigma = 1.57
VcmaxRef = 541006.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

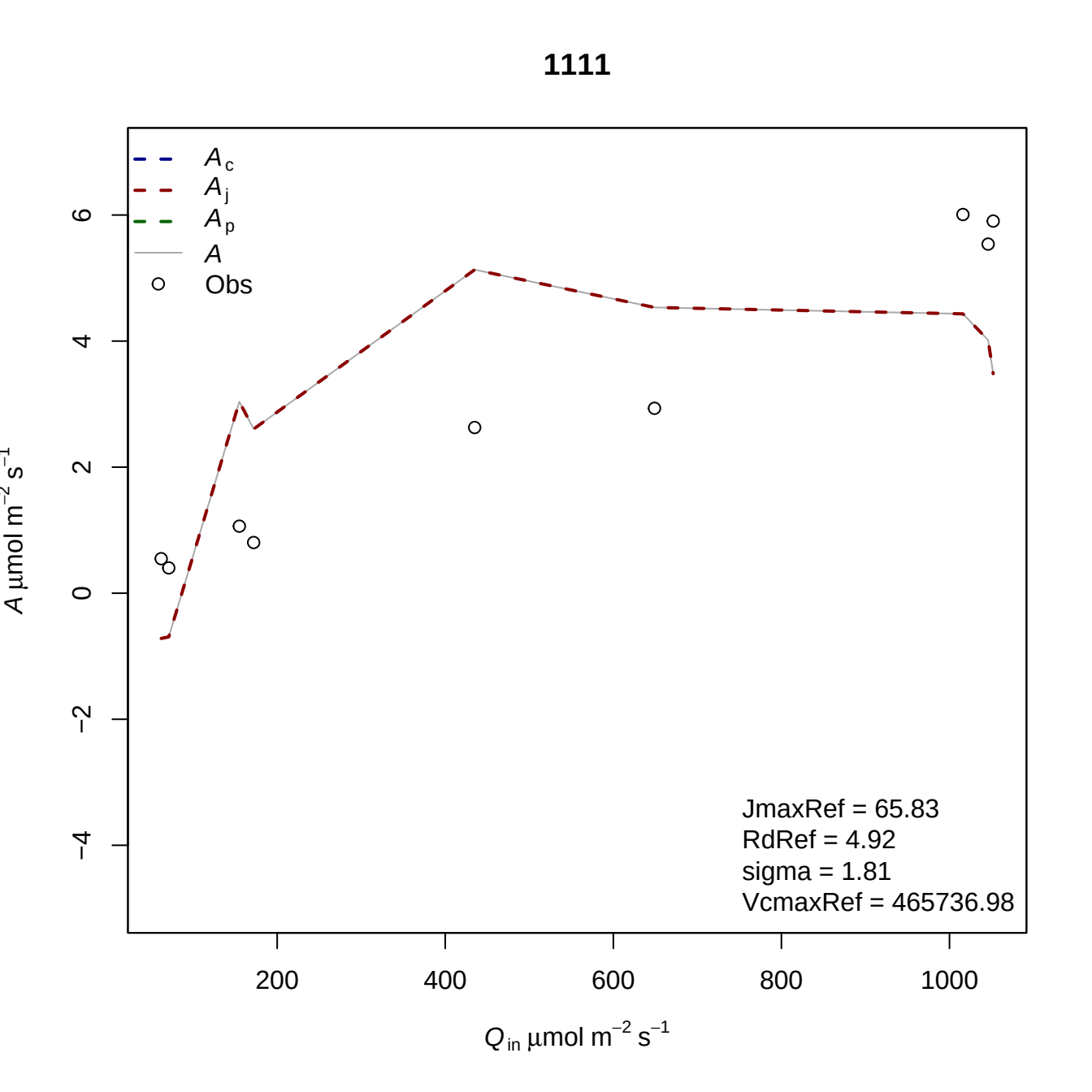

1105
A_c
A_j
A_p
A
Obs
JmaxRef = 80.25
RdRef = 5.1
sigma = 1.53
VcmaxRef = 76906.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

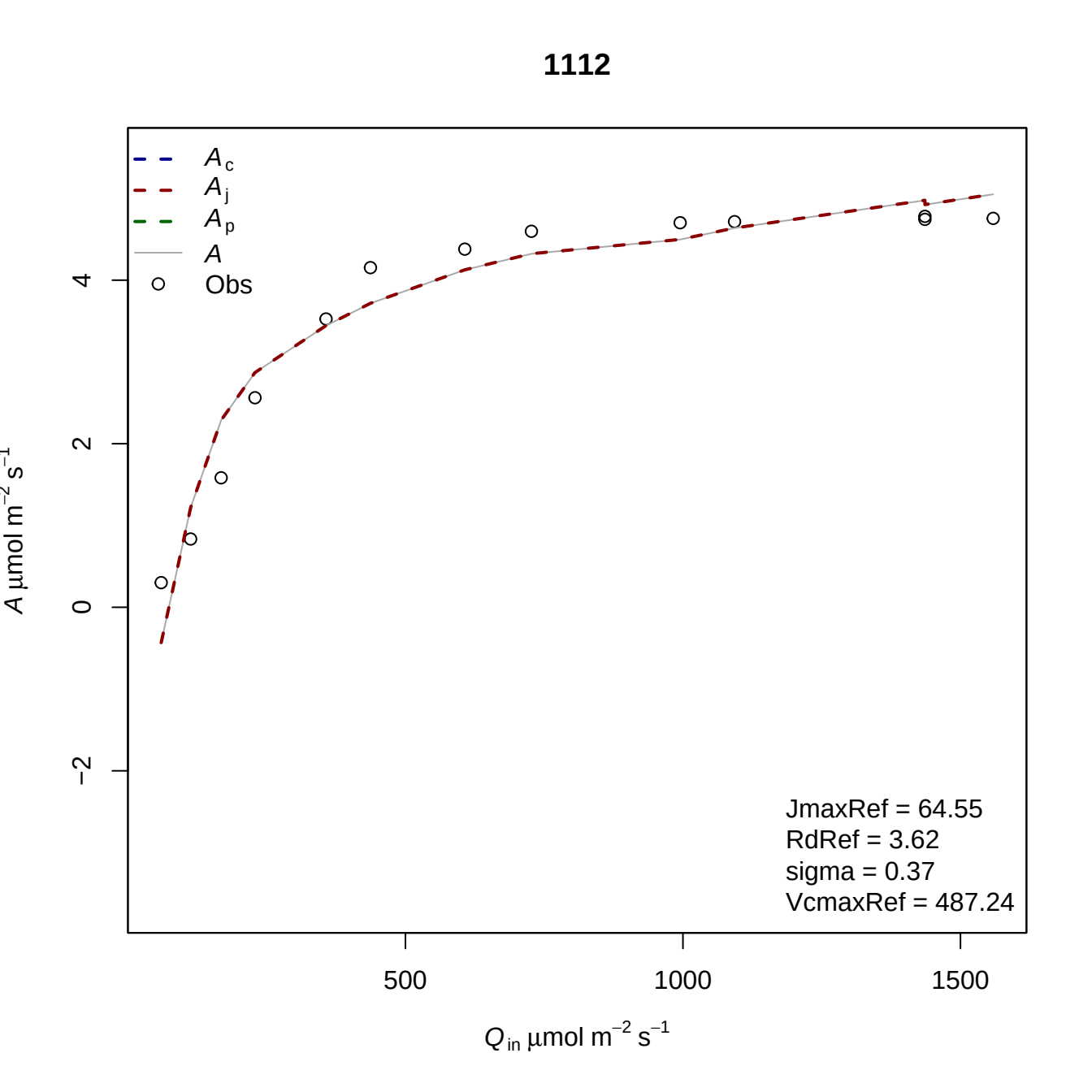

1107
A_c
A_j
A_p
A
Obs
JmaxRef = 20.24
RdRef = 1.35
sigma = 0.84
VcmaxRef = 14.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

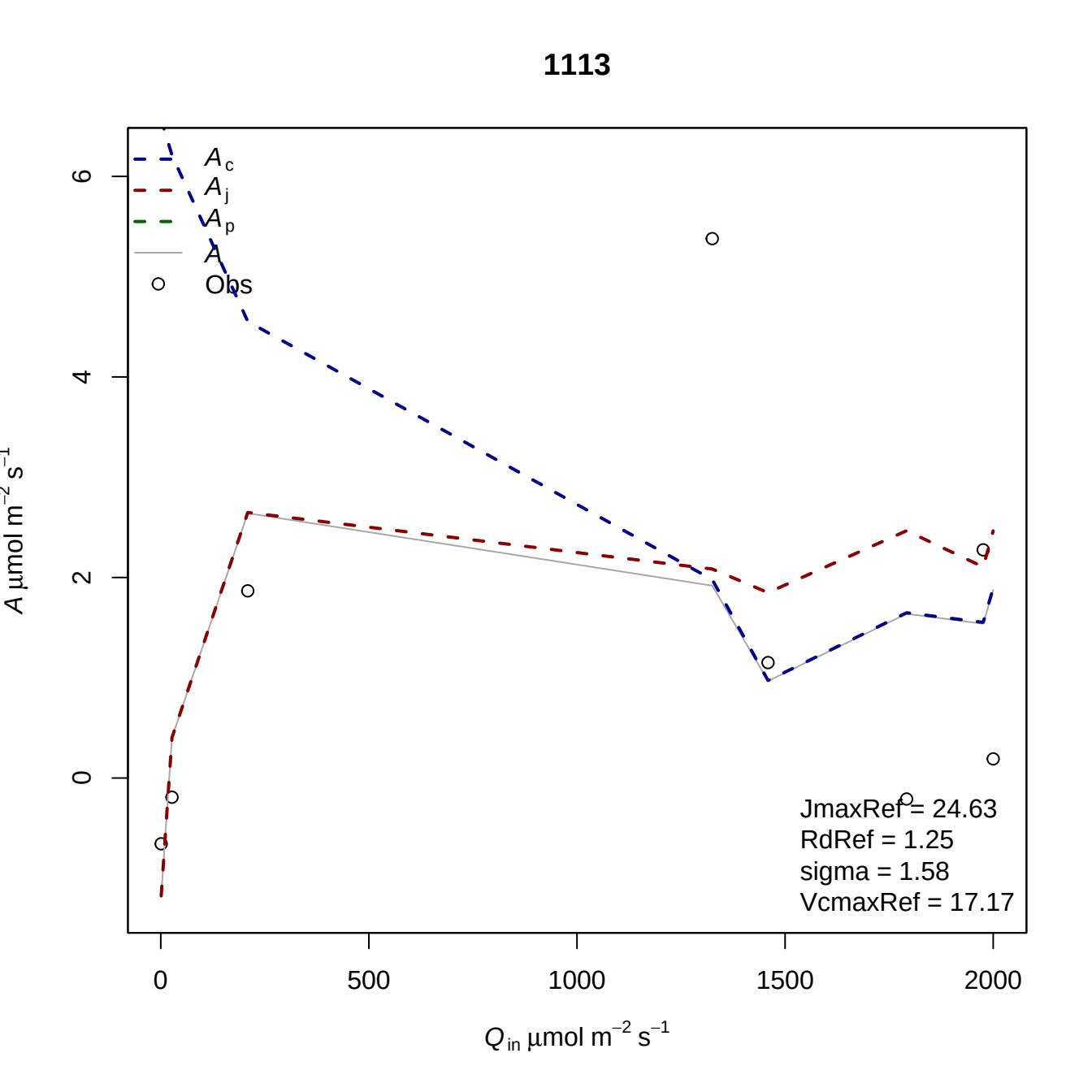

1108
A_c
A_j
A_p
A
Obs
JmaxRef = 95.94
RdRef = 4.3
sigma = 0.96
VcmaxRef = 1016824.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
500
1000
1500
2000

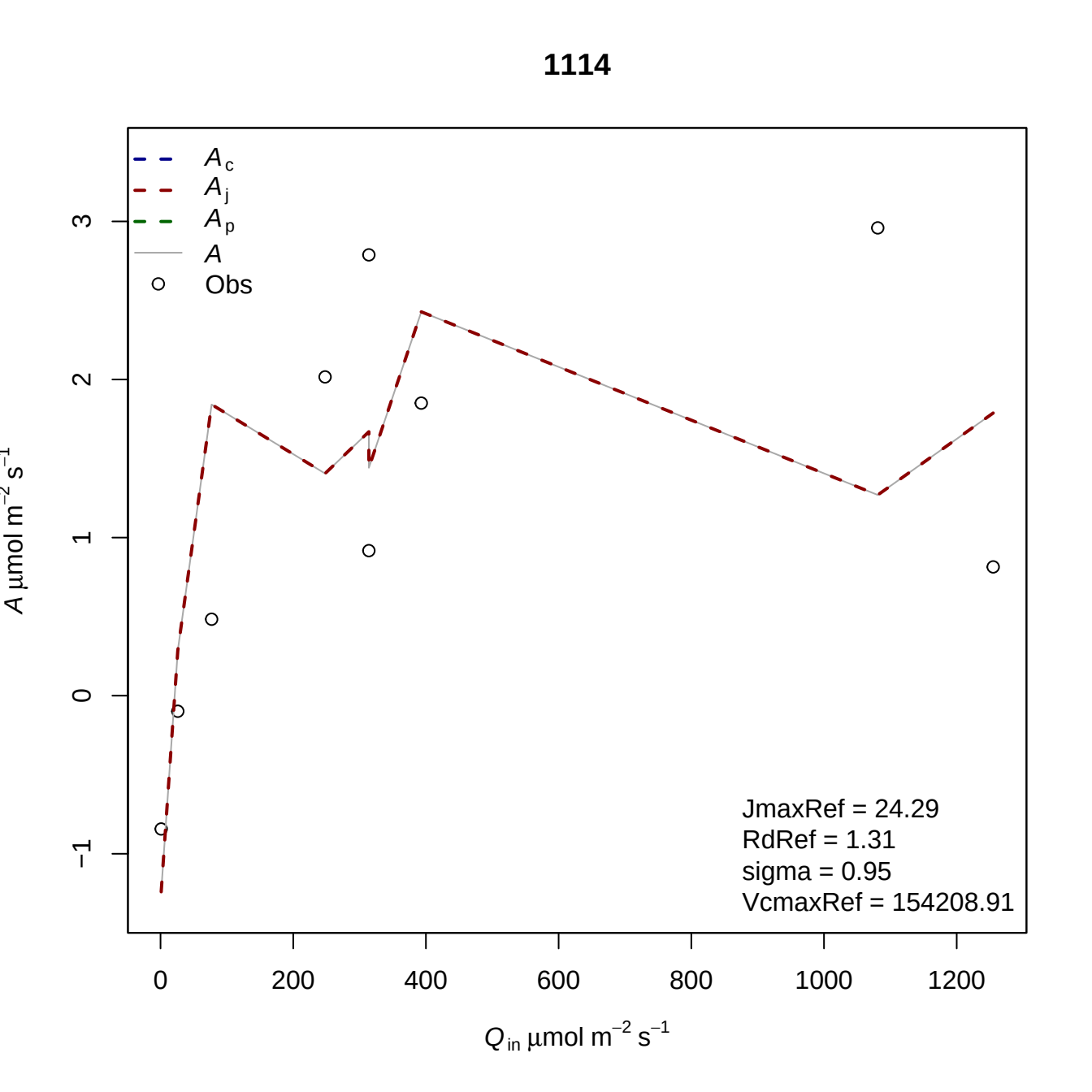

1109
A_c
A_j
A_p
A
Obs
JmaxRef = 36.81
RdRef = 2.52
sigma = 1.56
VcmaxRef = 551066.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

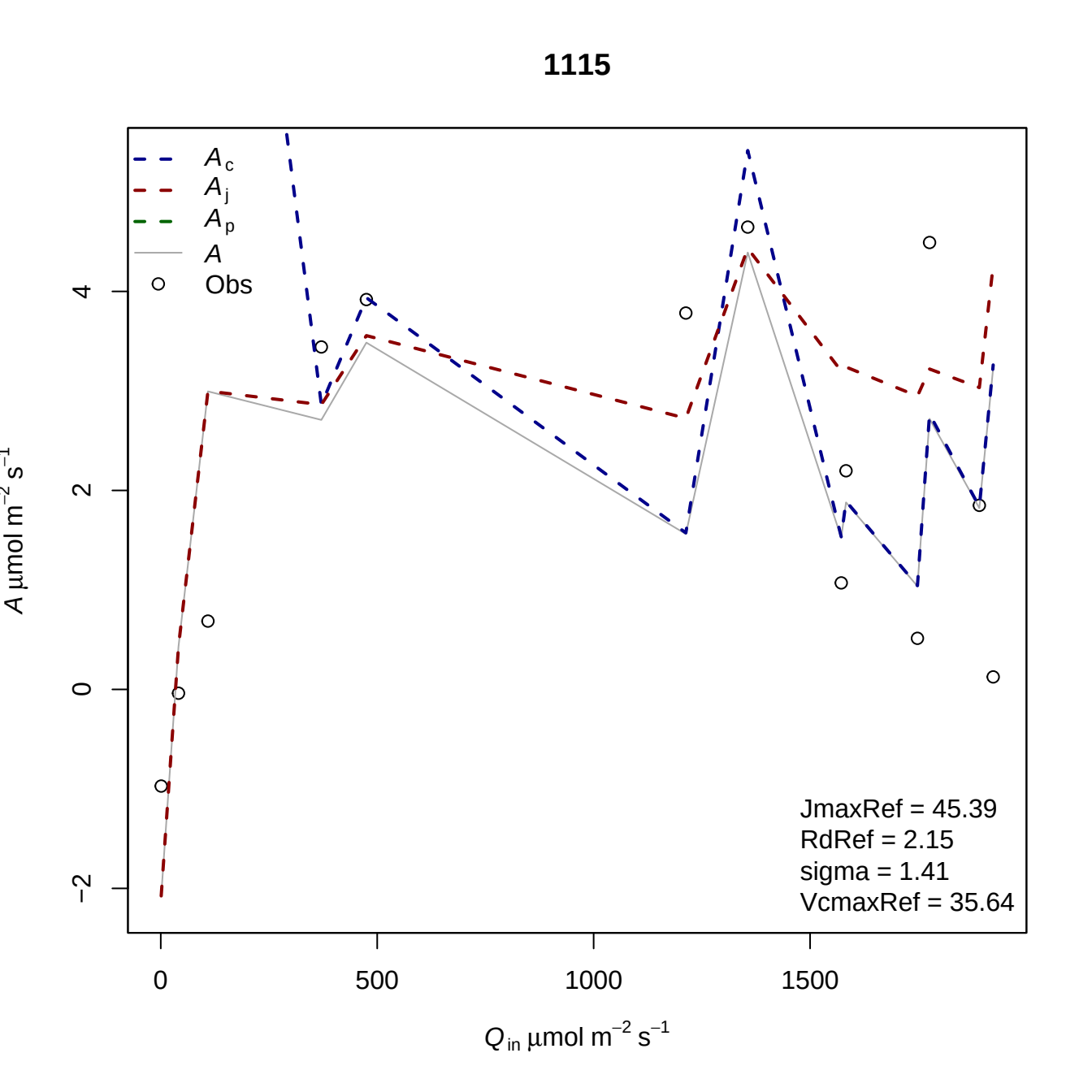

1110
A_c
A_j
A_p
A
Obs
JmaxRef = 41.25
RdRef = 2.4
sigma = 1.78
VcmaxRef = 497623.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

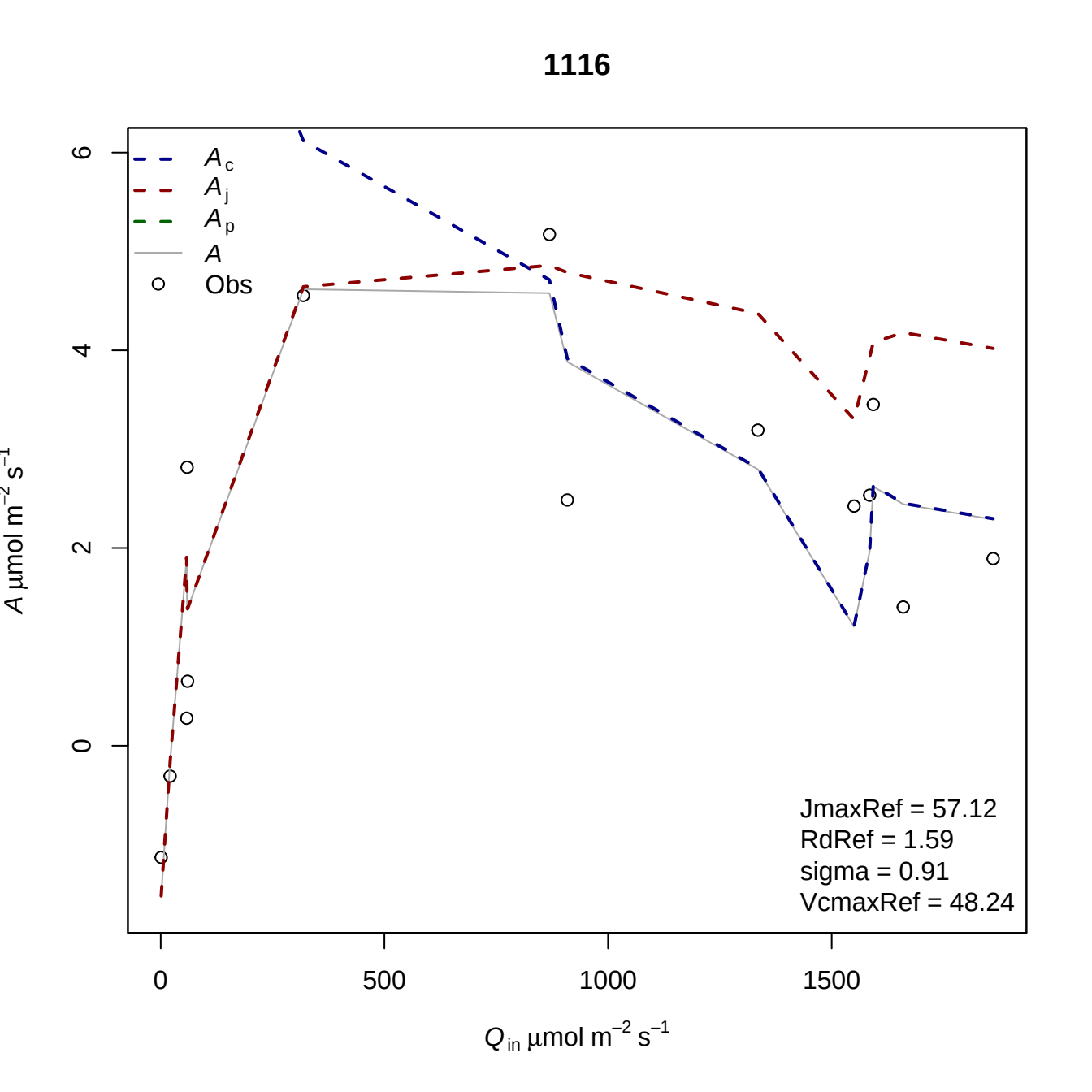

1111
A_c
A_j
A_p
A
Obs
JmaxRef = 65.83
RdRef = 4.92
sigma = 1.81
VcmaxRef = 465736.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
200
400
600
800
1000

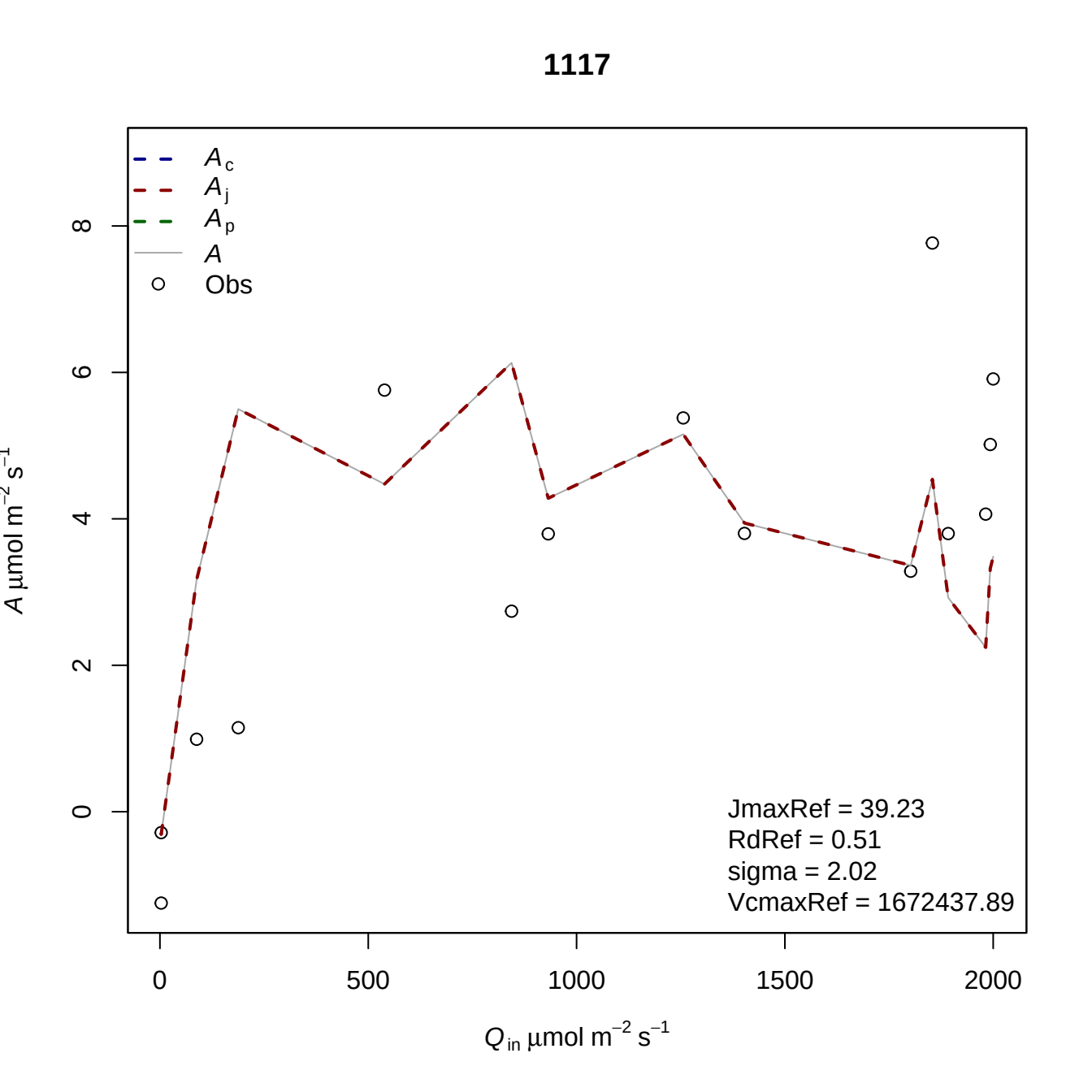

1112
A_c
A_j
A_p
A
Obs
JmaxRef = 64.55
RdRef = 3.62
sigma = 0.37
VcmaxRef = 487.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
500
1000
1500

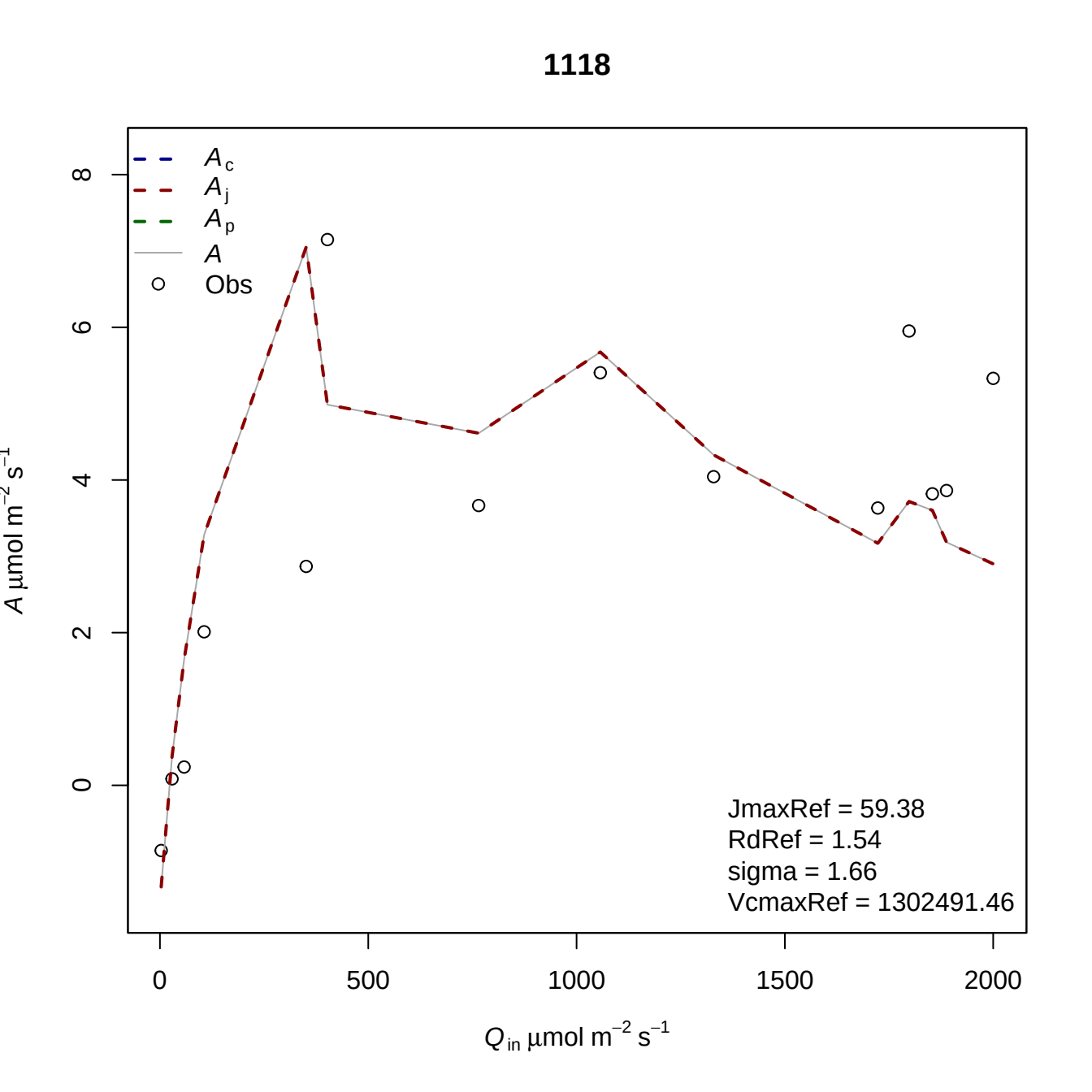

1113
A_c
A_j
A_p
A
Obs
JmaxRef = 24.63
RdRef = 1.25
sigma = 1.58
VcmaxRef = 17.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

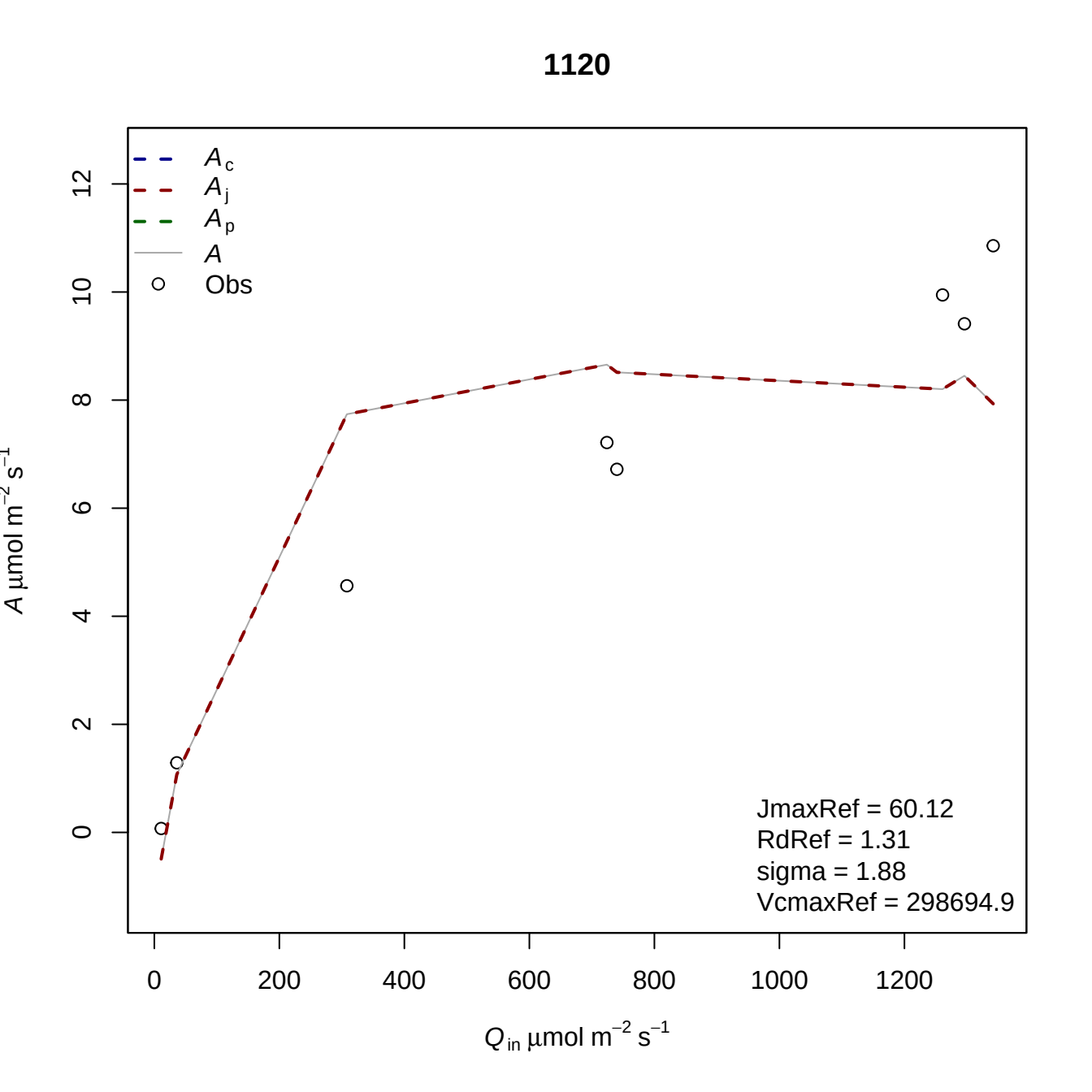

1114
A_c
A_j
A_p
A
Obs
JmaxRef = 24.29
RdRef = 1.31
sigma = 0.95
VcmaxRef = 154208.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200

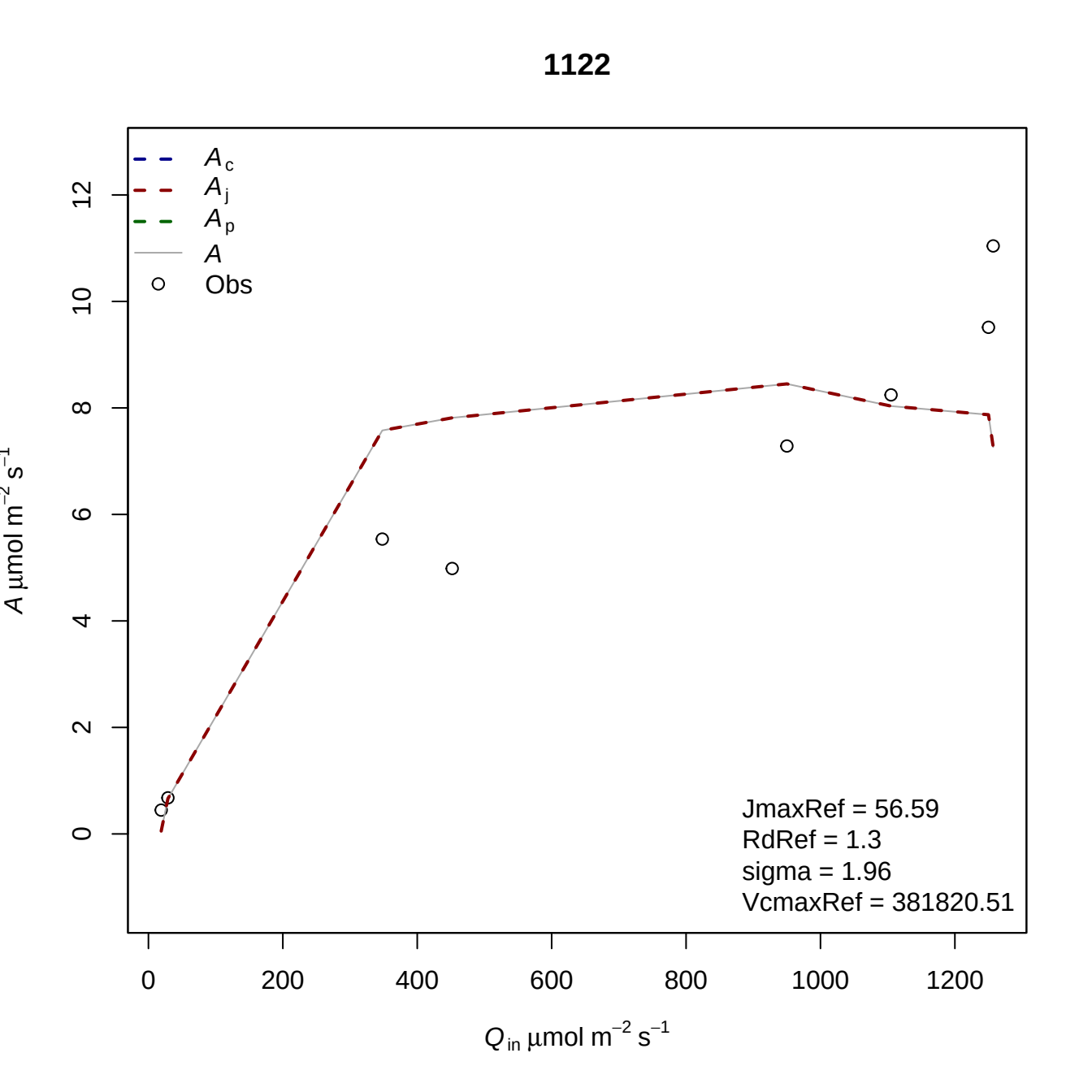

1115
A_c
A_j
A_p
A
Obs
JmaxRef = 45.39
RdRef = 2.15
sigma = 1.41
VcmaxRef = 35.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

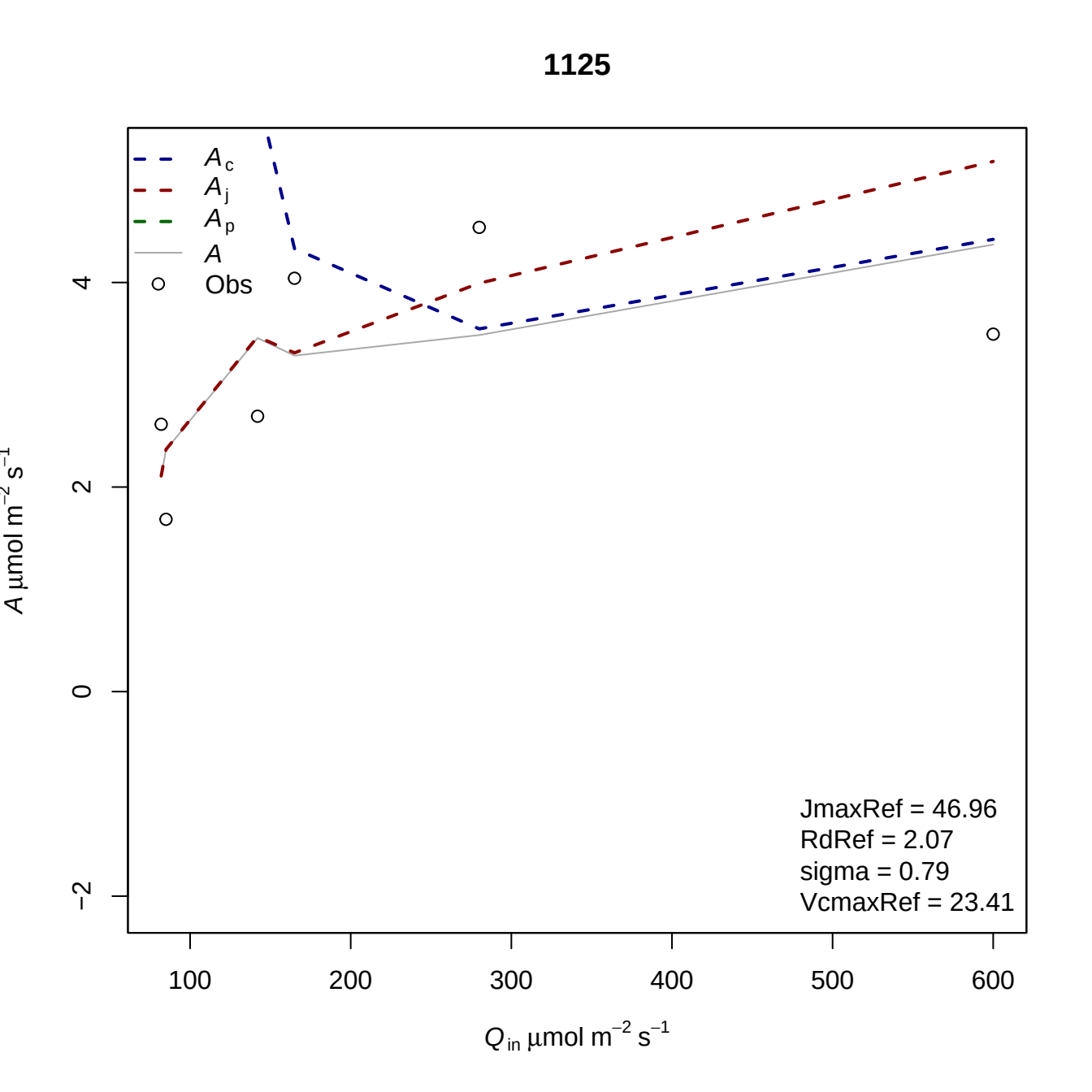

1116
A_c
A_j
A_p
A
Obs
JmaxRef = 57.12
RdRef = 1.59
sigma = 0.91
VcmaxRef = 48.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

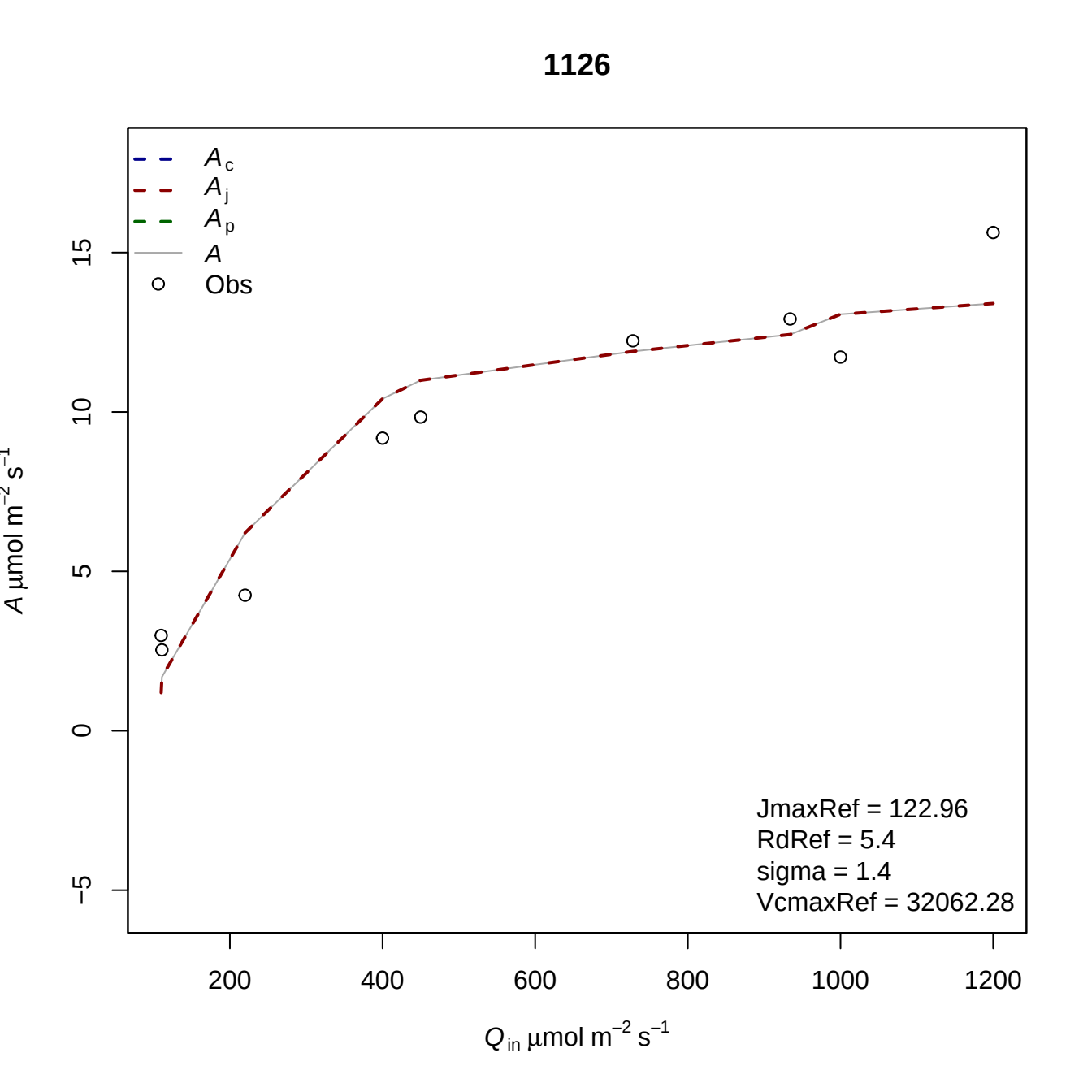

1117
A_c
A_j
A_p
A
Obs
JmaxRef = 39.23
RdRef = 0.51
sigma = 2.02
VcmaxRef = 1672437.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

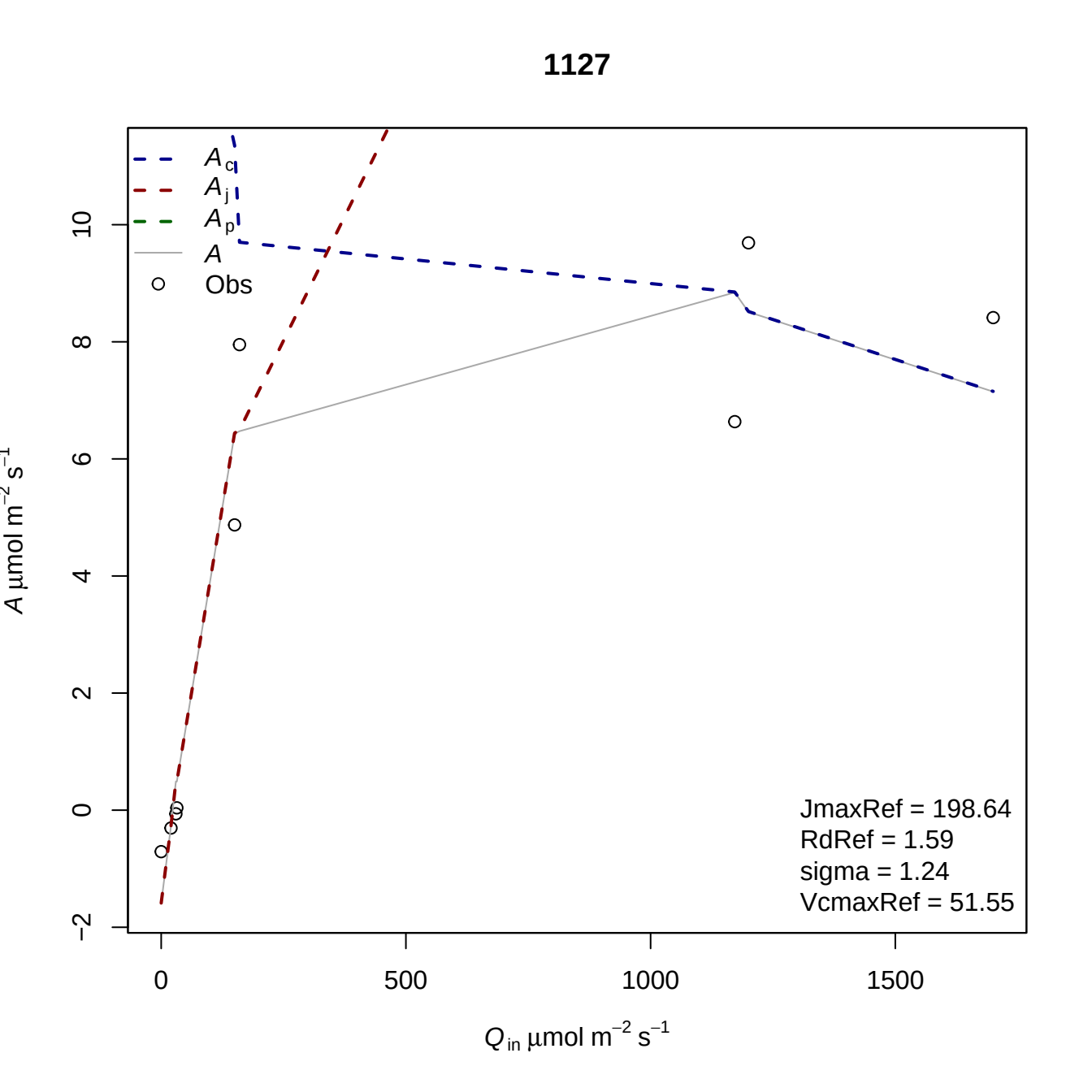

1118
A_c
A_j
A_p
A
Obs
JmaxRef = 59.38
RdRef = 1.54
sigma = 1.66
VcmaxRef = 1302491.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

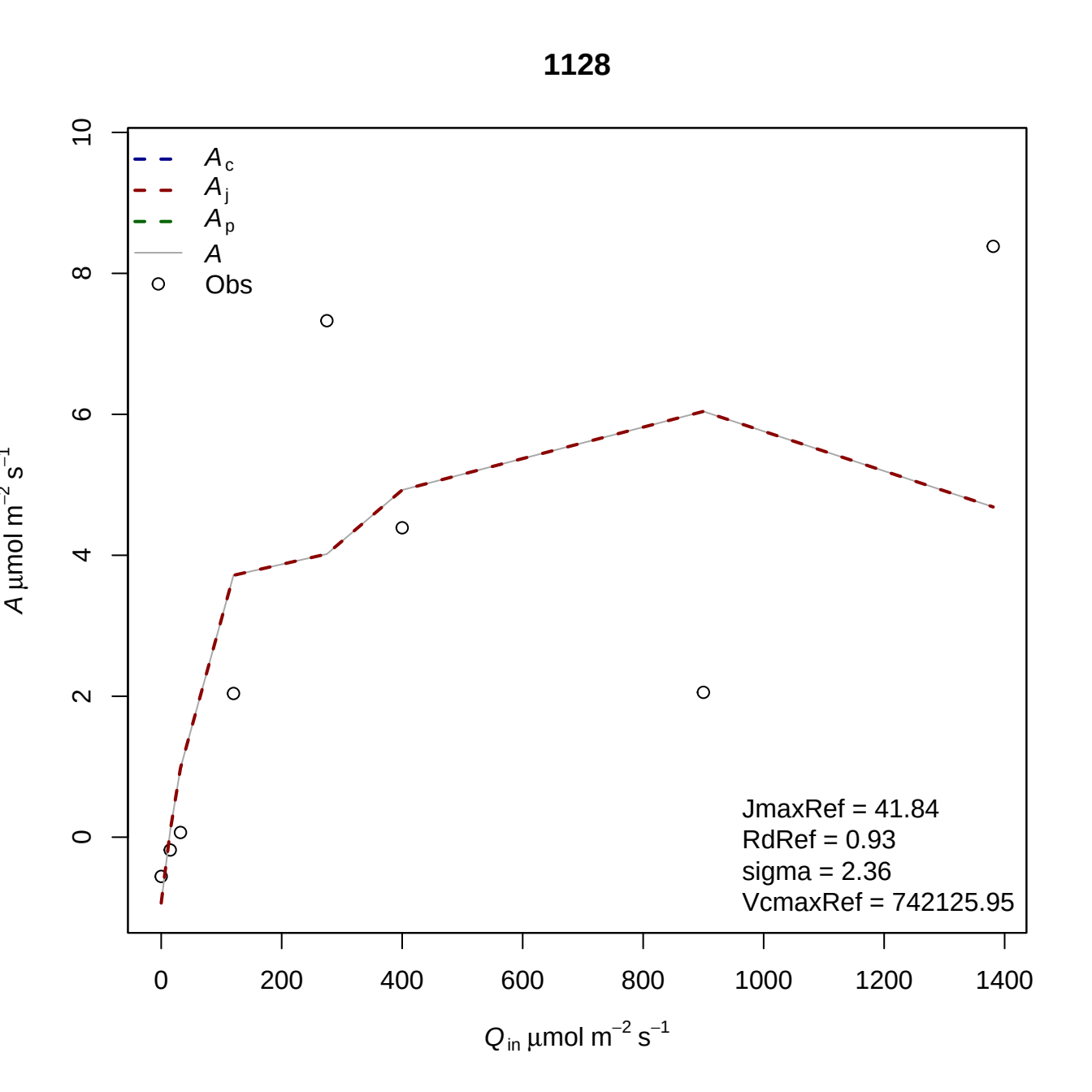

1120
A_c
A_j
A_p
A
Obs
JmaxRef = 60.12
RdRef = 1.31
sigma = 1.88
VcmaxRef = 298694.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

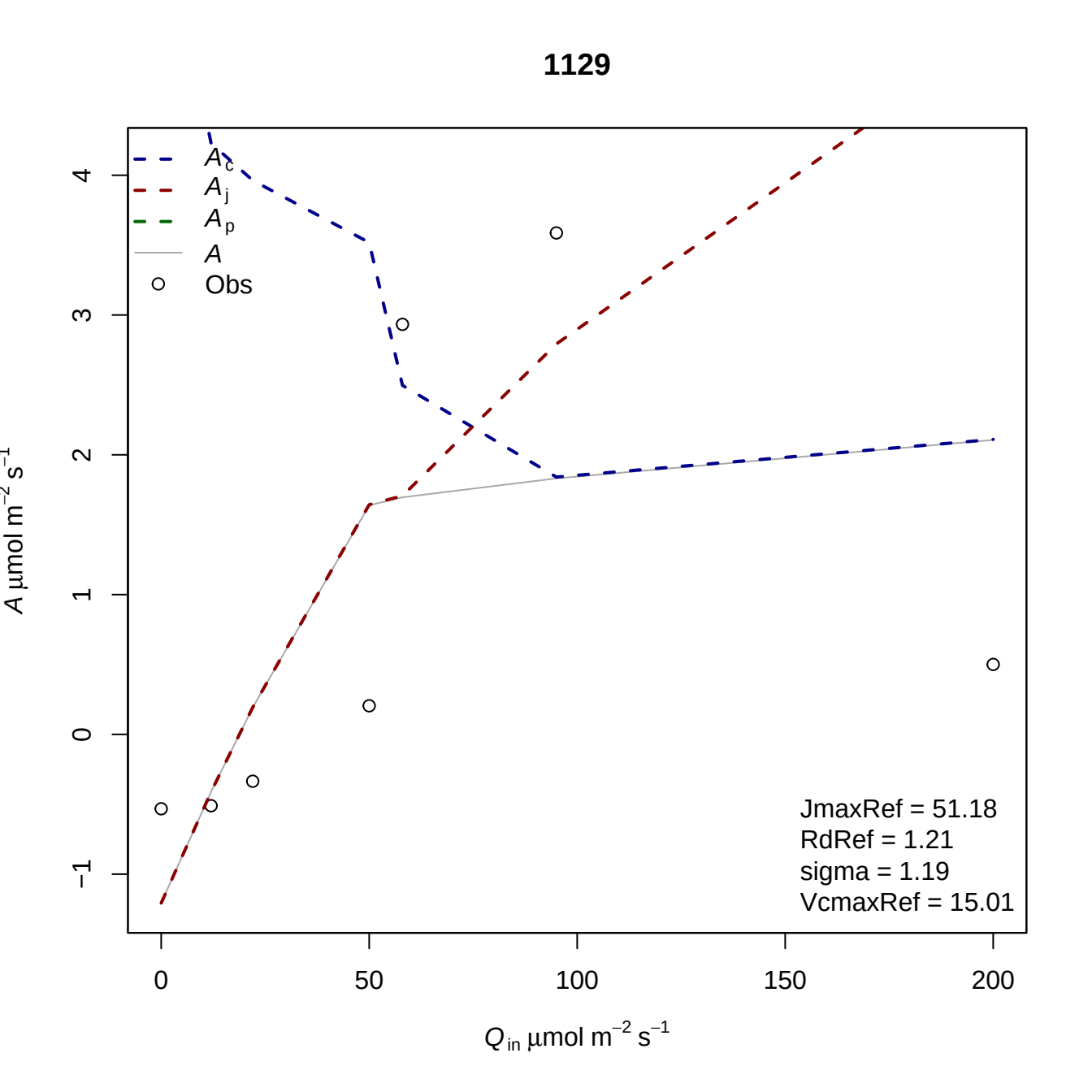

1122
A_c
A_j
A_p
A
Obs
JmaxRef = 56.59
RdRef = 1.3
sigma = 1.96
VcmaxRef = 381820.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

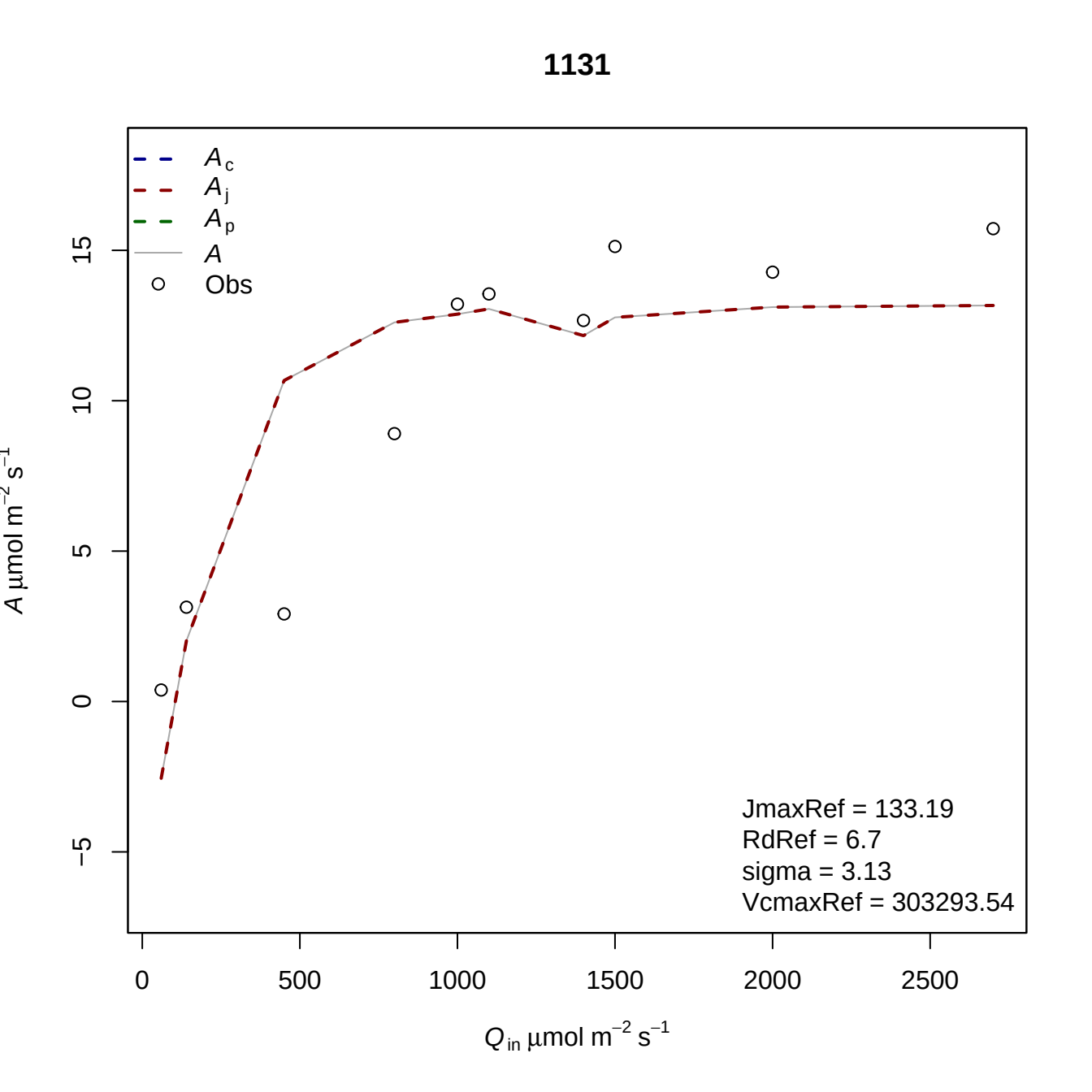

1125
A_c
A_j
A_p
A
Obs
JmaxRef = 46.96
RdRef = 2.07
sigma = 0.79
VcmaxRef = 23.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
100
200
300
400
500
600

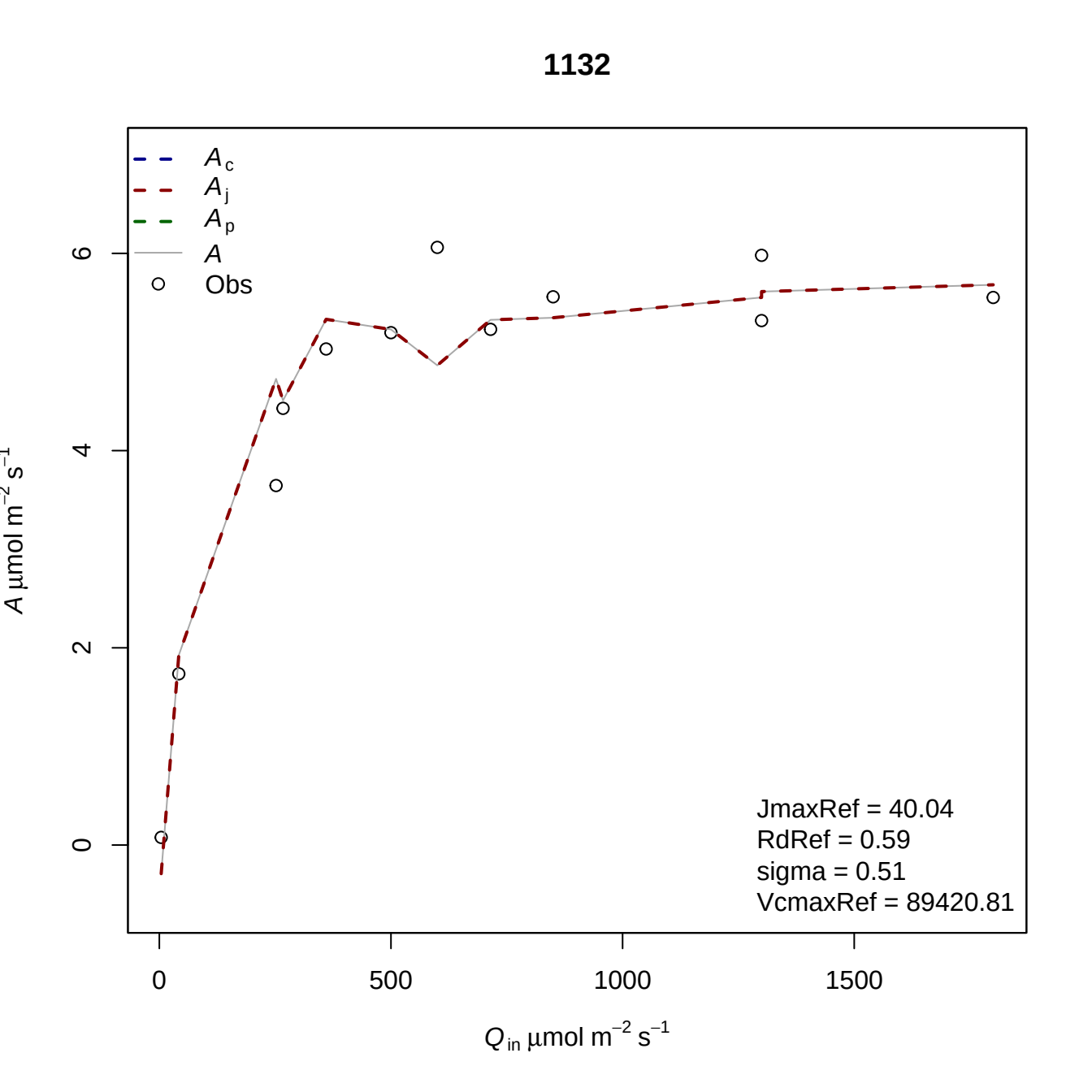

1126
A_c
A_j
A_p
A
Obs
JmaxRef = 122.96
RdRef = 5.4
sigma = 1.4
VcmaxRef = 32062.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
200
400
600
800
1000
1200

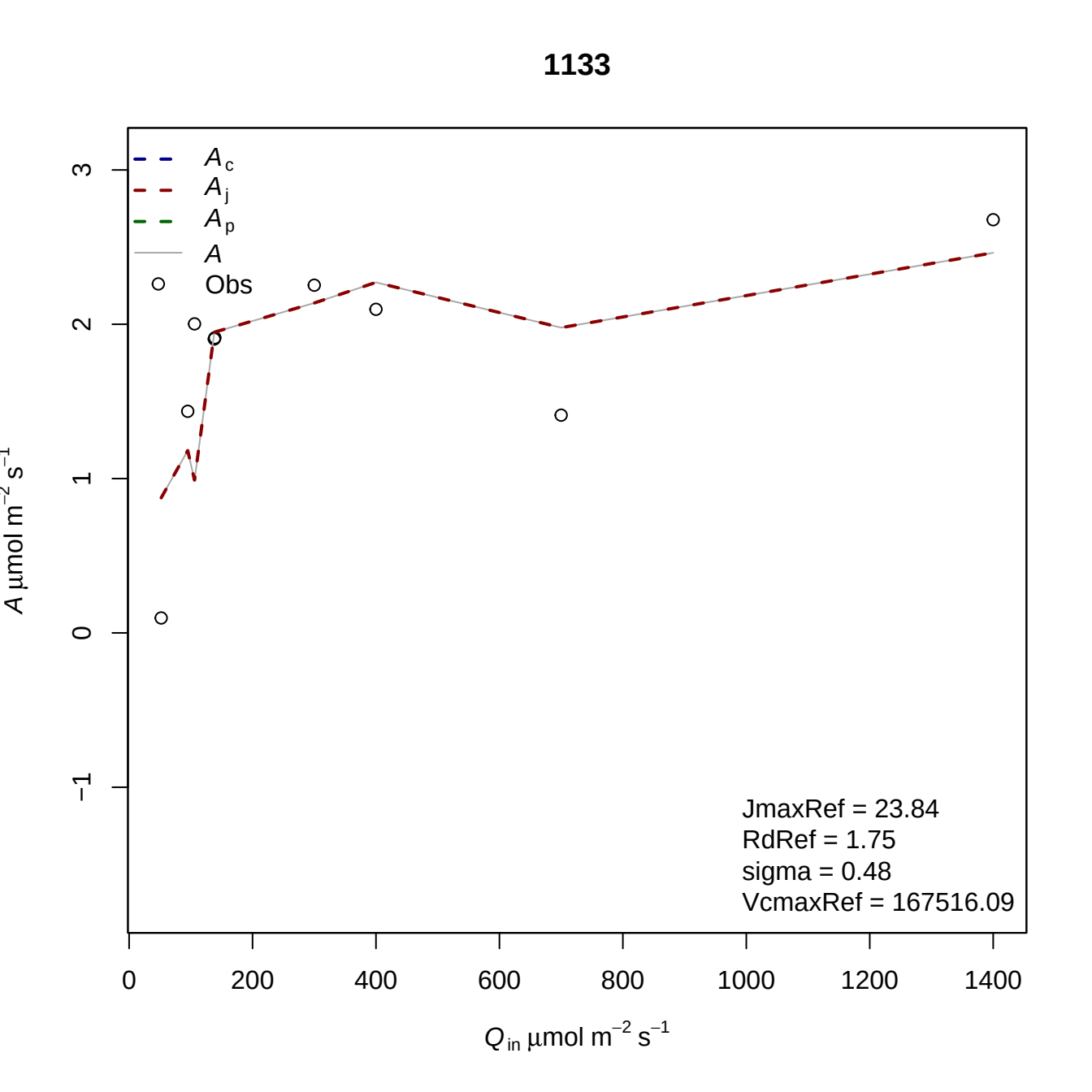

1127
A_c
A_j
A_p
A
Obs
JmaxRef = 198.64
RdRef = 1.59
sigma = 1.24
VcmaxRef = 51.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

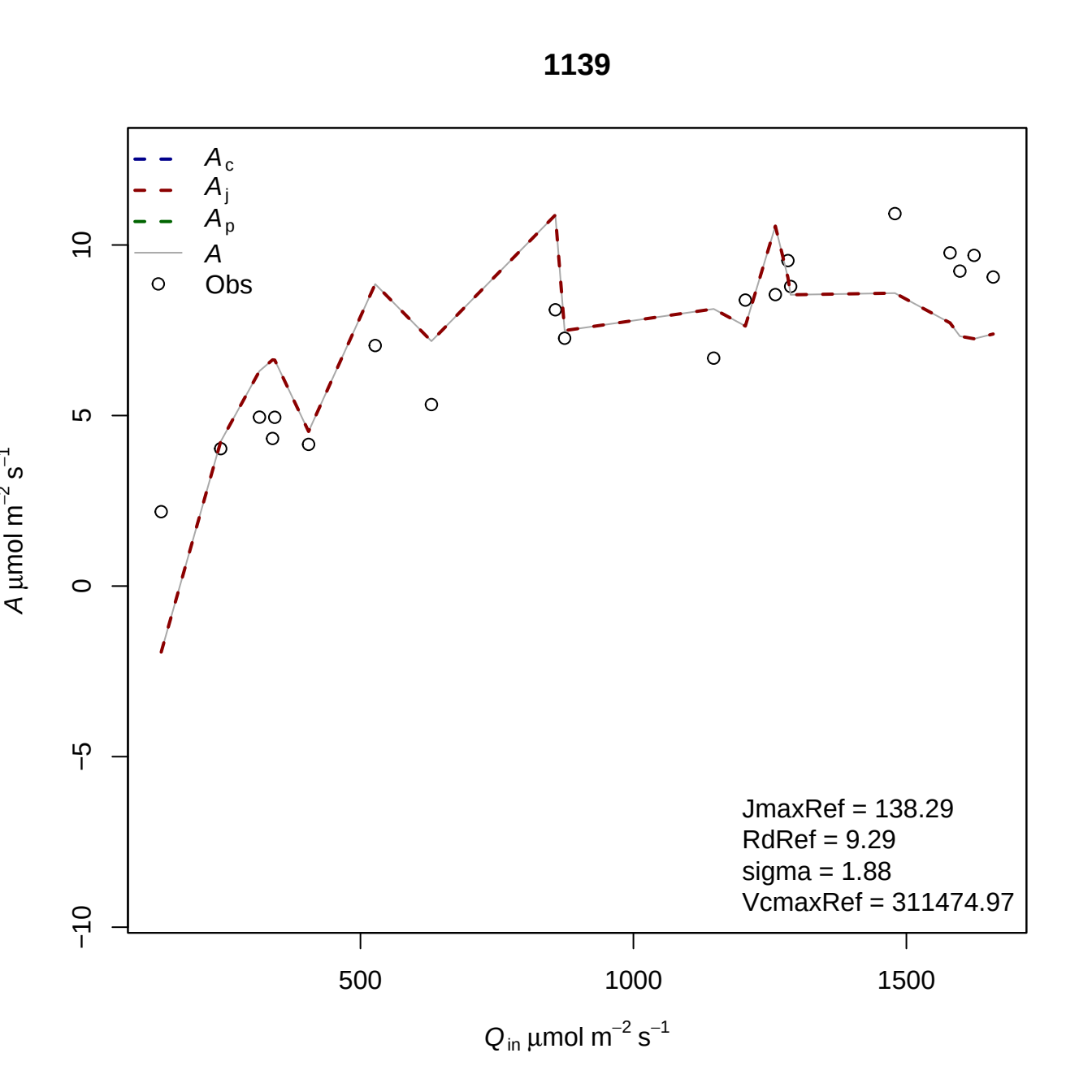

1128
A_c
A_j
A_p
A
Obs
JmaxRef = 41.84
RdRef = 0.93
sigma = 2.36
VcmaxRef = 742125.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

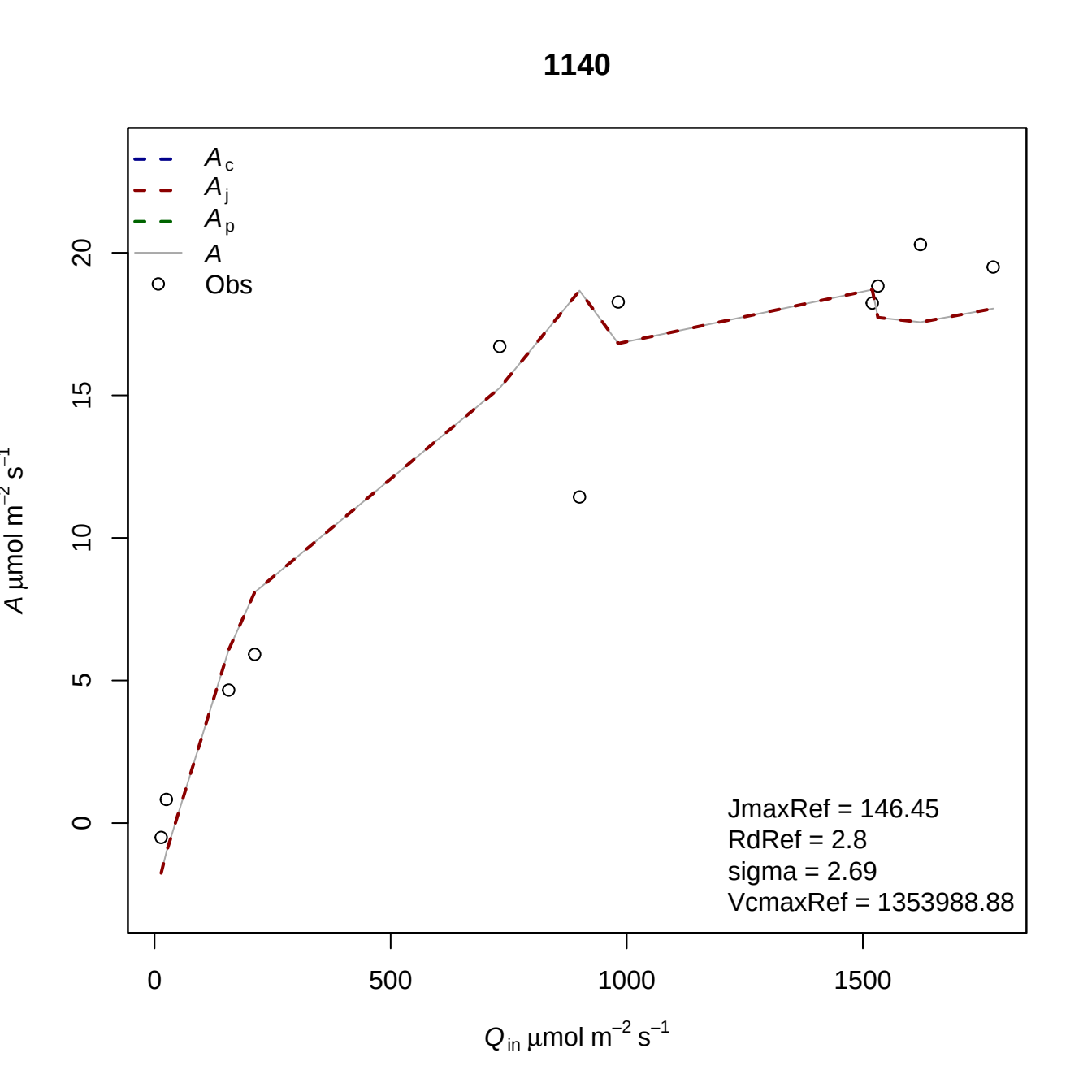

1129
A_c
A_j
A_p
A
Obs
JmaxRef = 51.18
RdRef = 1.21
sigma = 1.19
VcmaxRef = 15.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
50
100
150
200

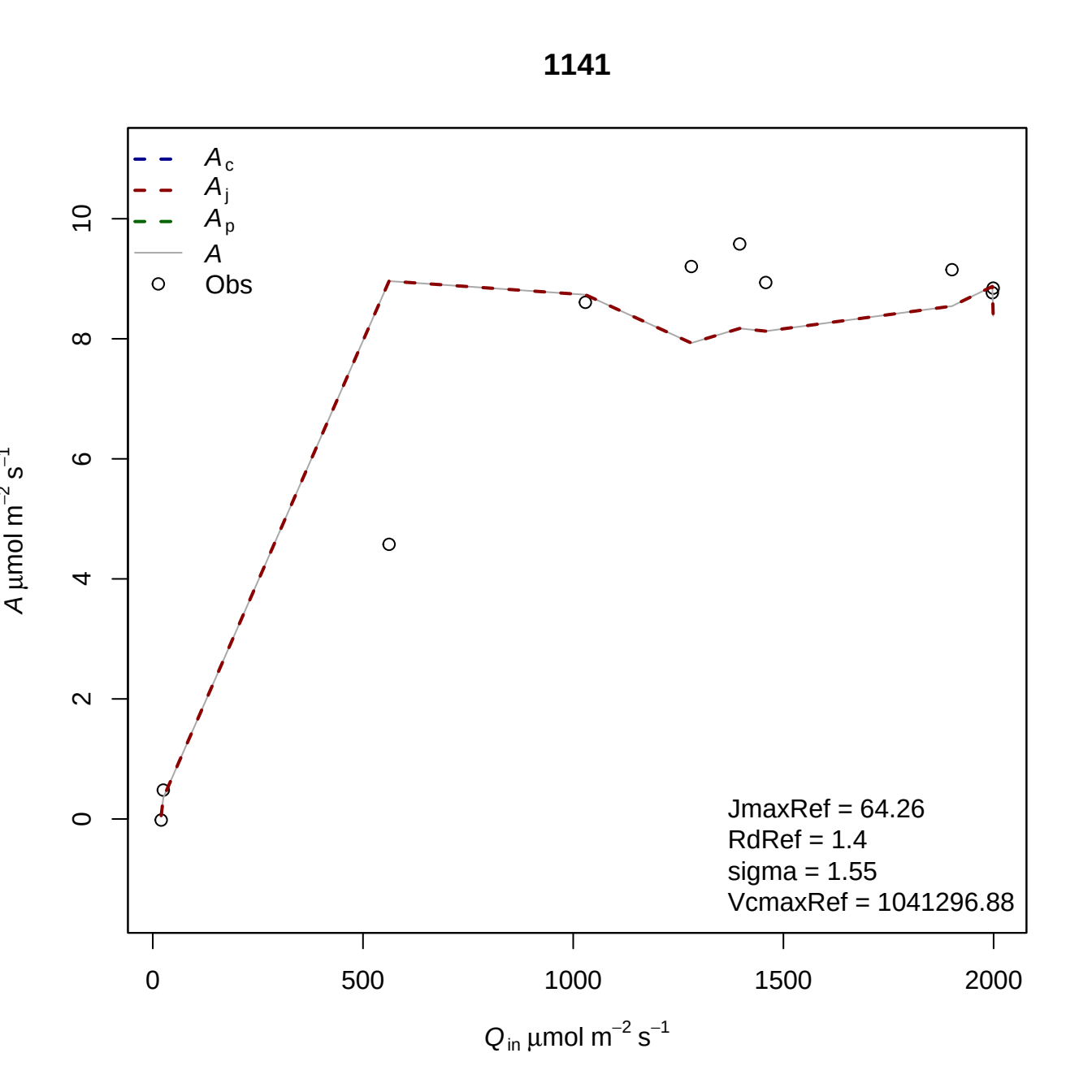

1131
A_c
A_j
A_p
A
Obs
JmaxRef = 133.19
RdRef = 6.7
sigma = 3.13
VcmaxRef = 303293.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000
2500

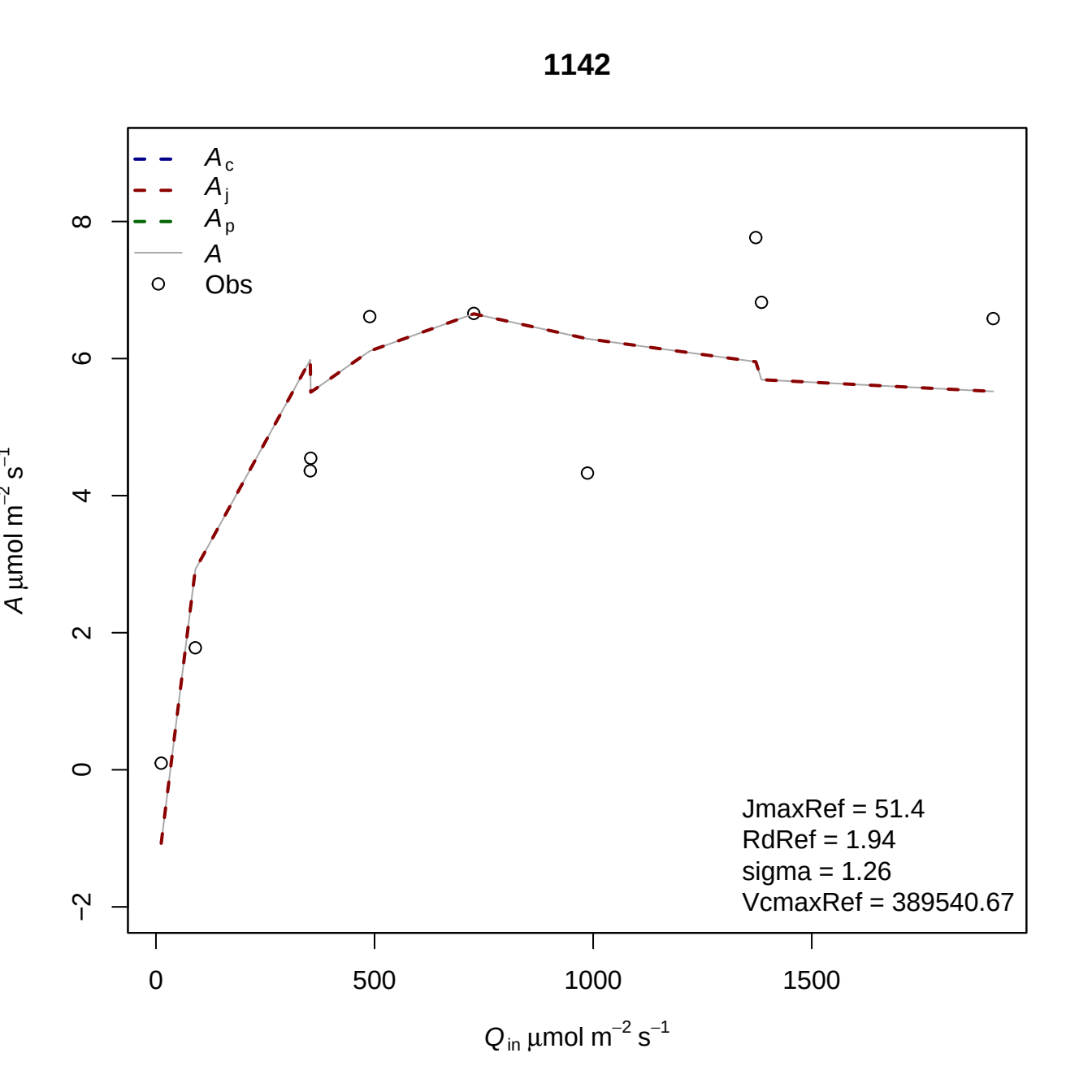

1132
A_c
A_j
A_p
A
Obs
JmaxRef = 40.04
RdRef = 0.59
sigma = 0.51
VcmaxRef = 89420.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

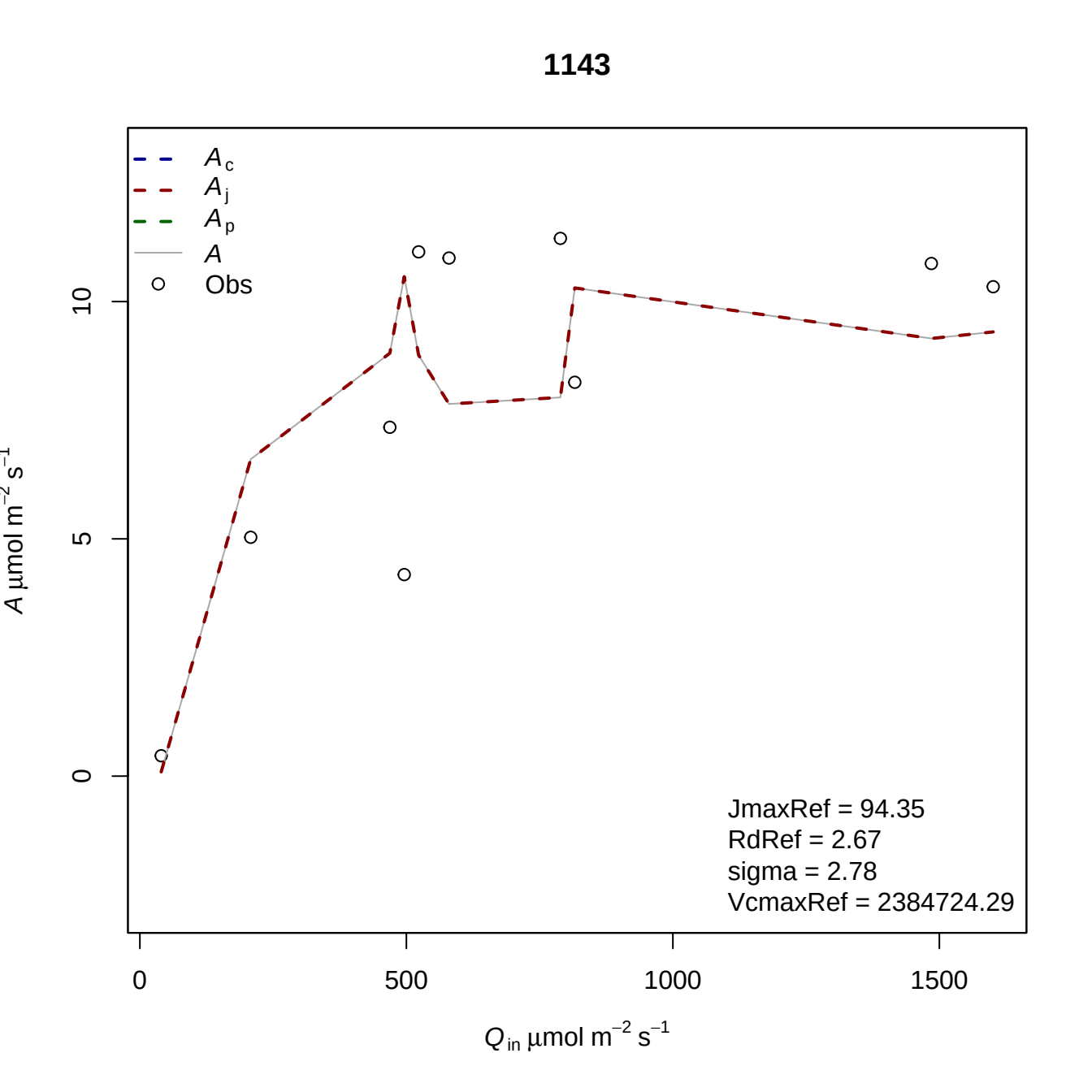

1133
A_c
A_j
A_p
A
Obs
JmaxRef = 23.84
RdRef = 1.75
sigma = 0.48
VcmaxRef = 167516.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
200
400
600
800
1000
1200
1400

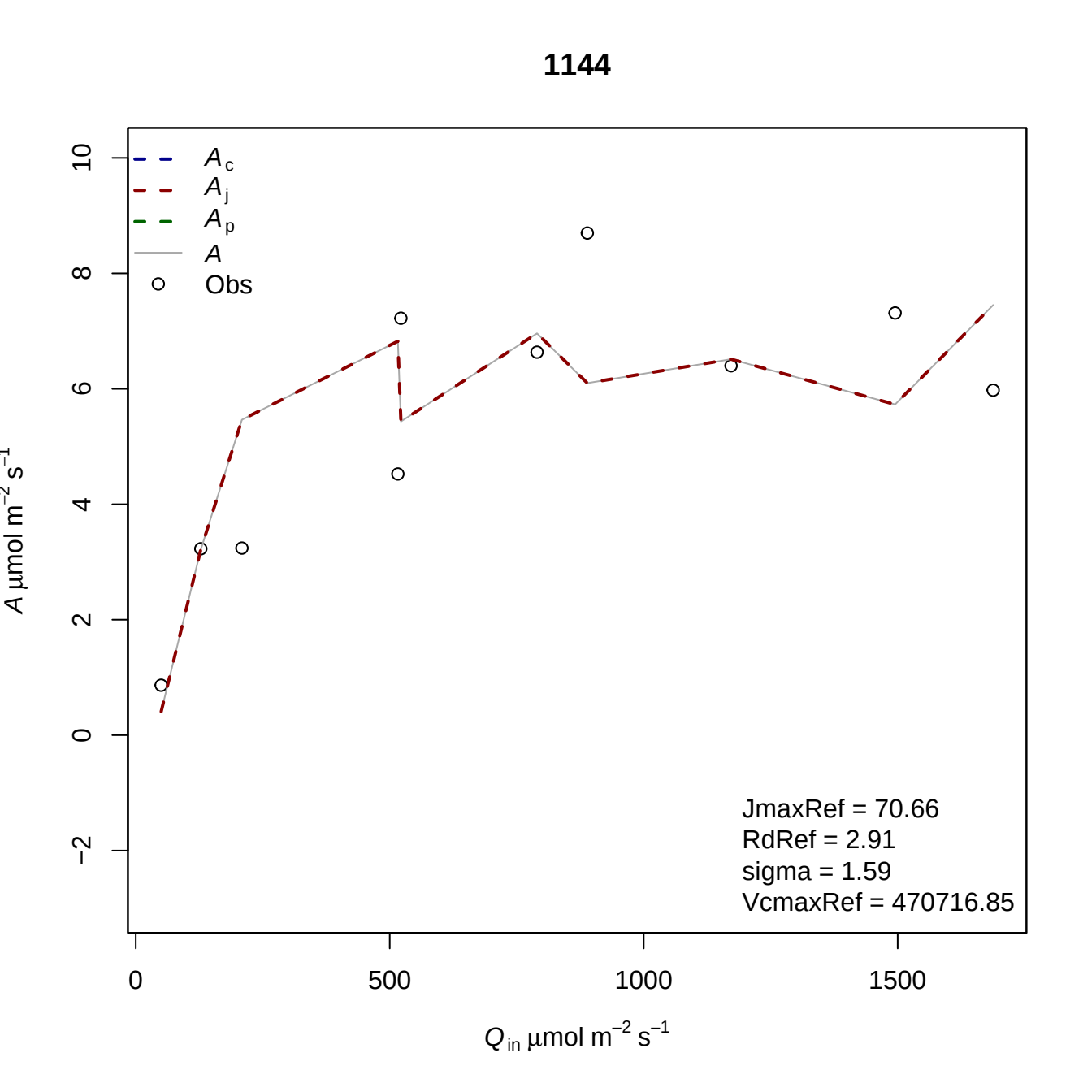

1139
A_c
A_j
A_p
A
Obs
JmaxRef = 138.29
RdRef = 9.29
sigma = 1.88
VcmaxRef = 311474.97
10
5
0
−5
−10
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

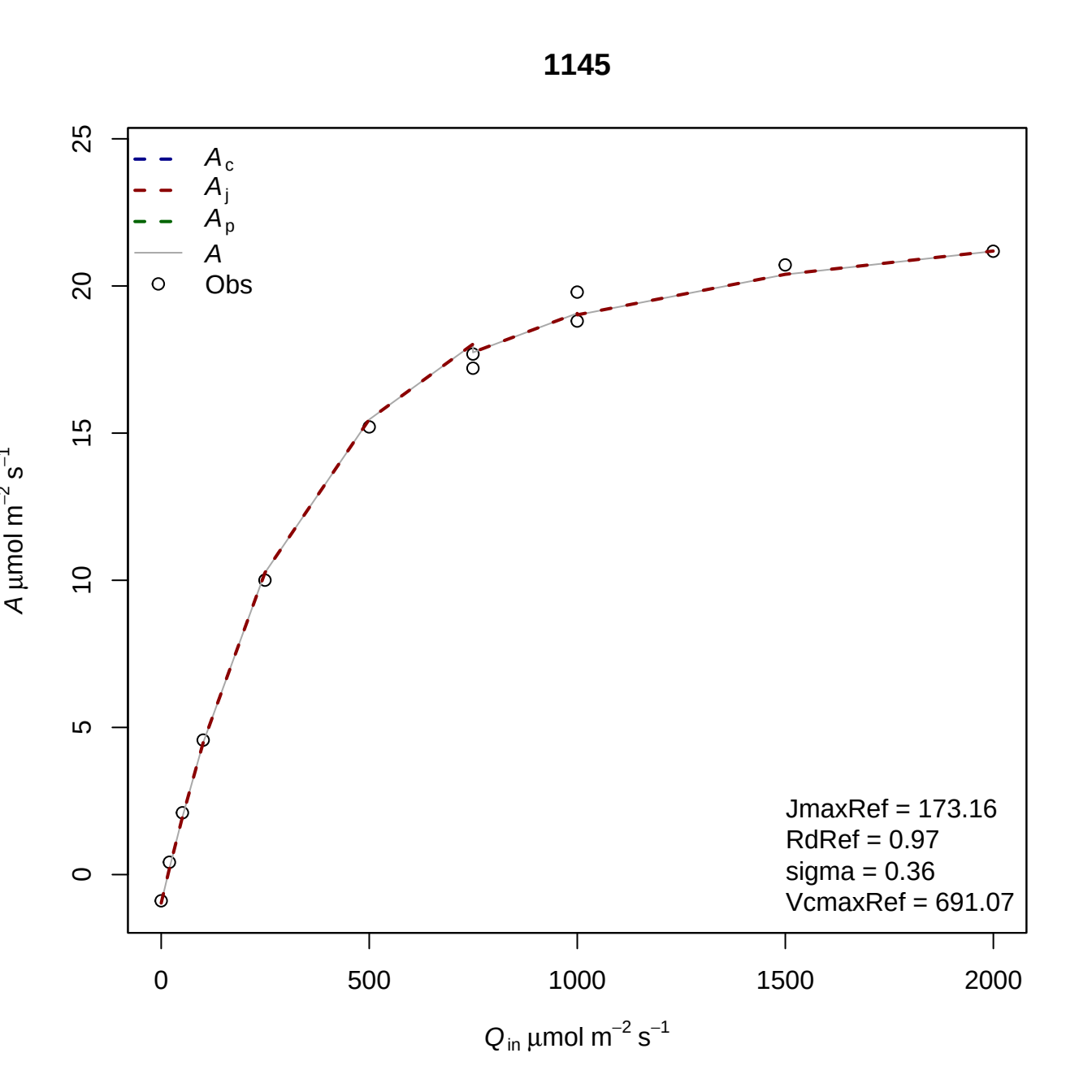

1140
A_c
A_j
A_p
A
Obs
JmaxRef = 146.45
RdRef = 2.8
sigma = 2.69
VcmaxRef = 1353988.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

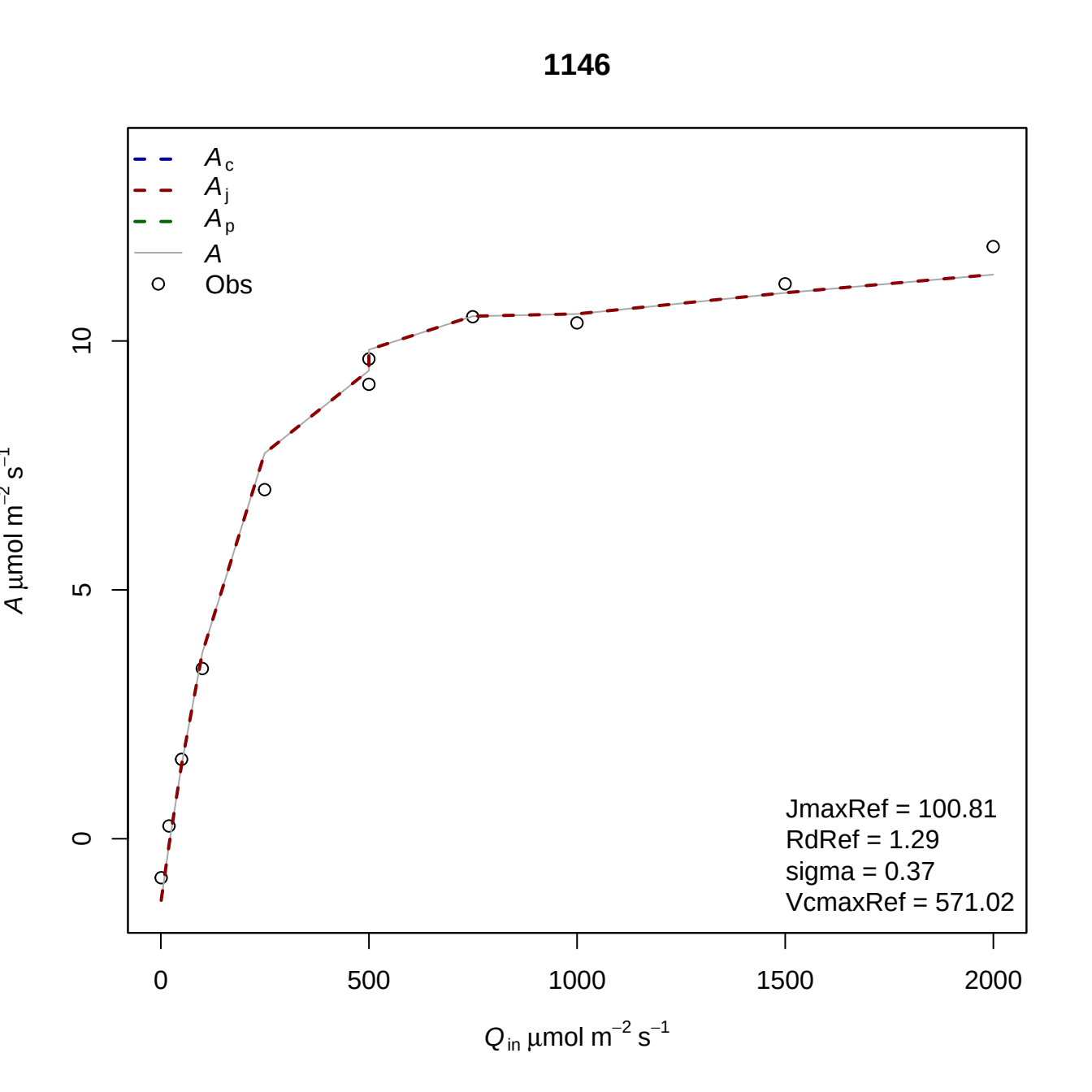

1141
A_c
A_j
A_p
A
Obs
JmaxRef = 64.26
RdRef = 1.4
sigma = 1.55
VcmaxRef = 1041296.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

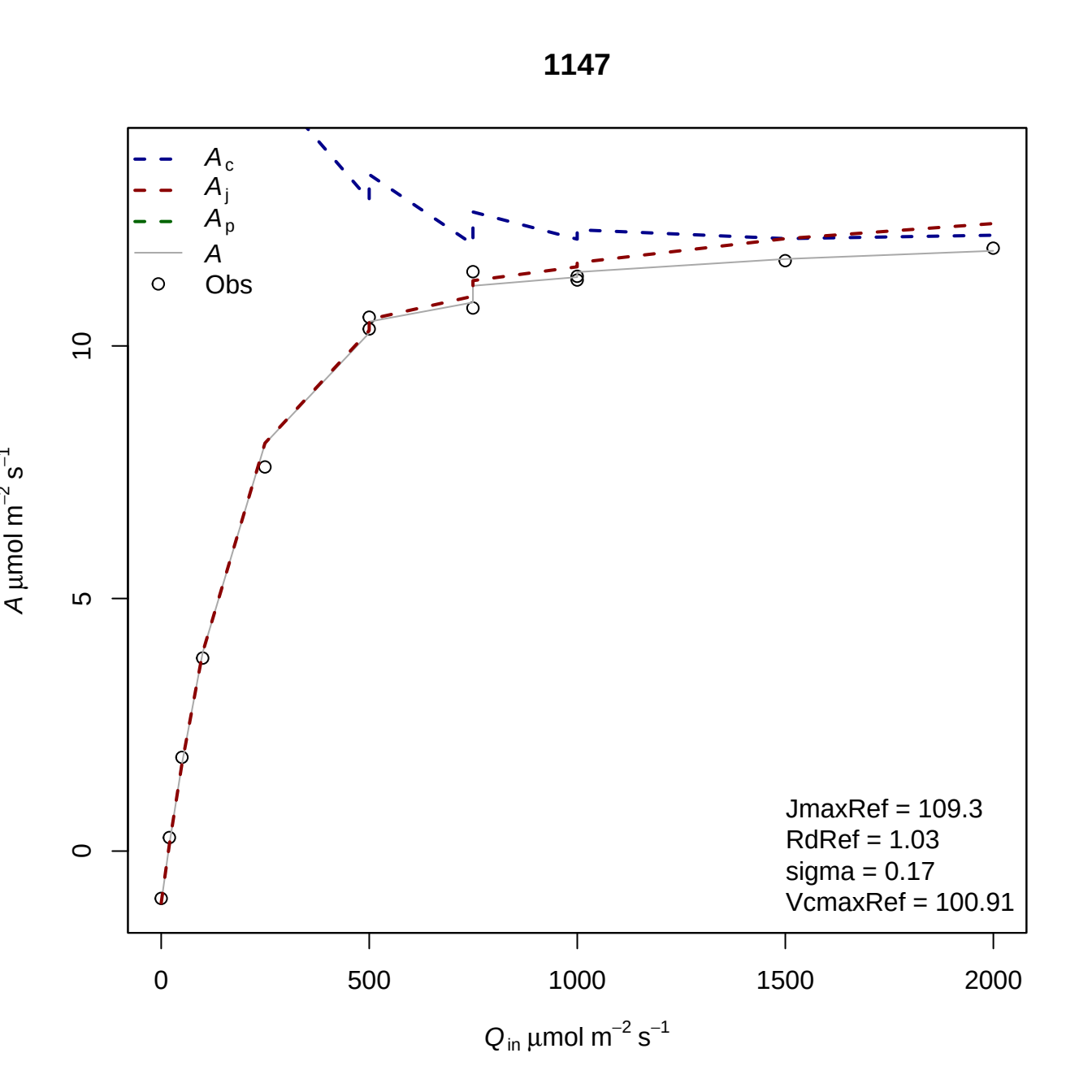

1142
A_c
A_j
A_p
A
Obs
JmaxRef = 51.4
RdRef = 1.94
sigma = 1.26
VcmaxRef = 389540.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

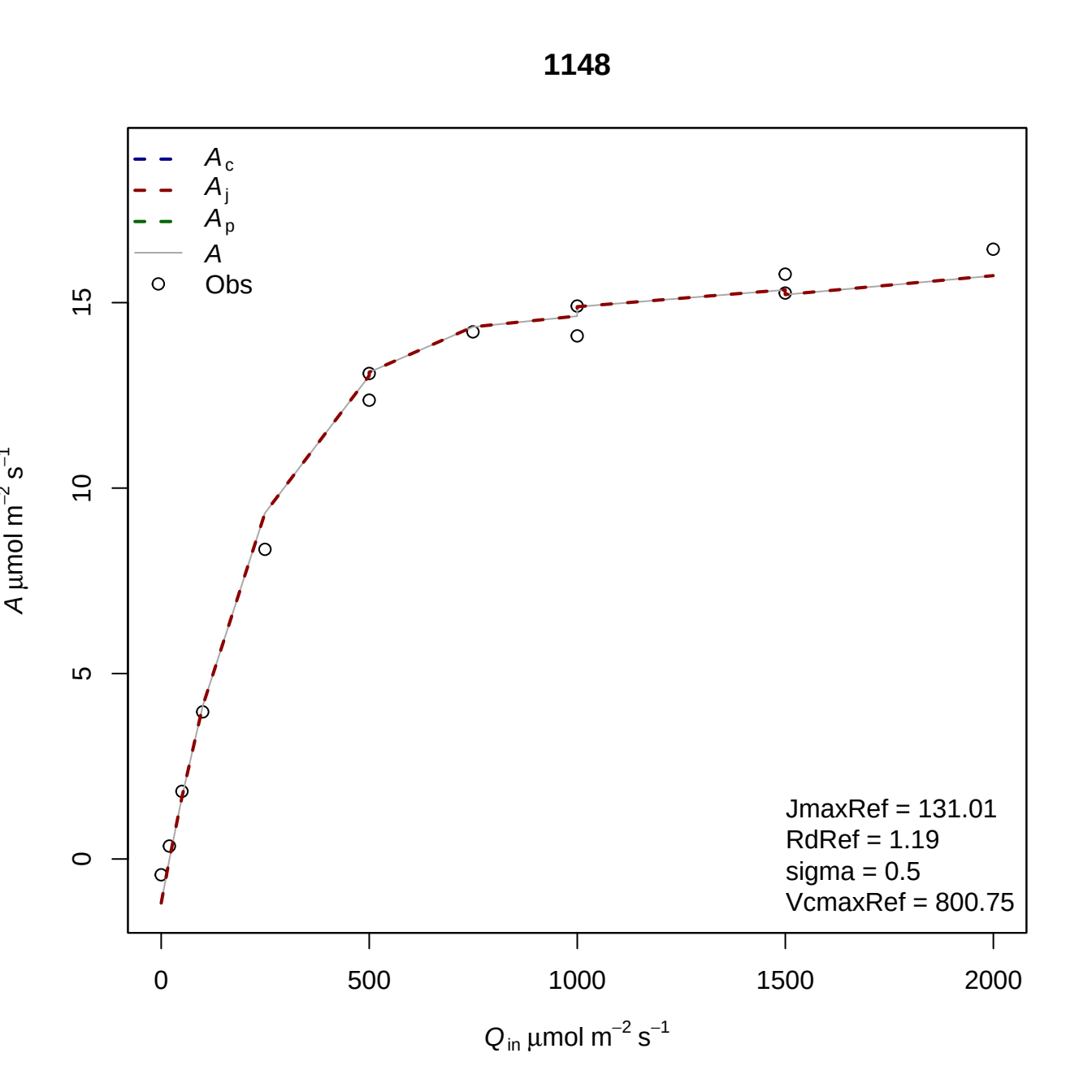

1143
A_c
A_j
A_p
A
Obs
JmaxRef = 94.35
RdRef = 2.67
sigma = 2.78
VcmaxRef = 2384724.29
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

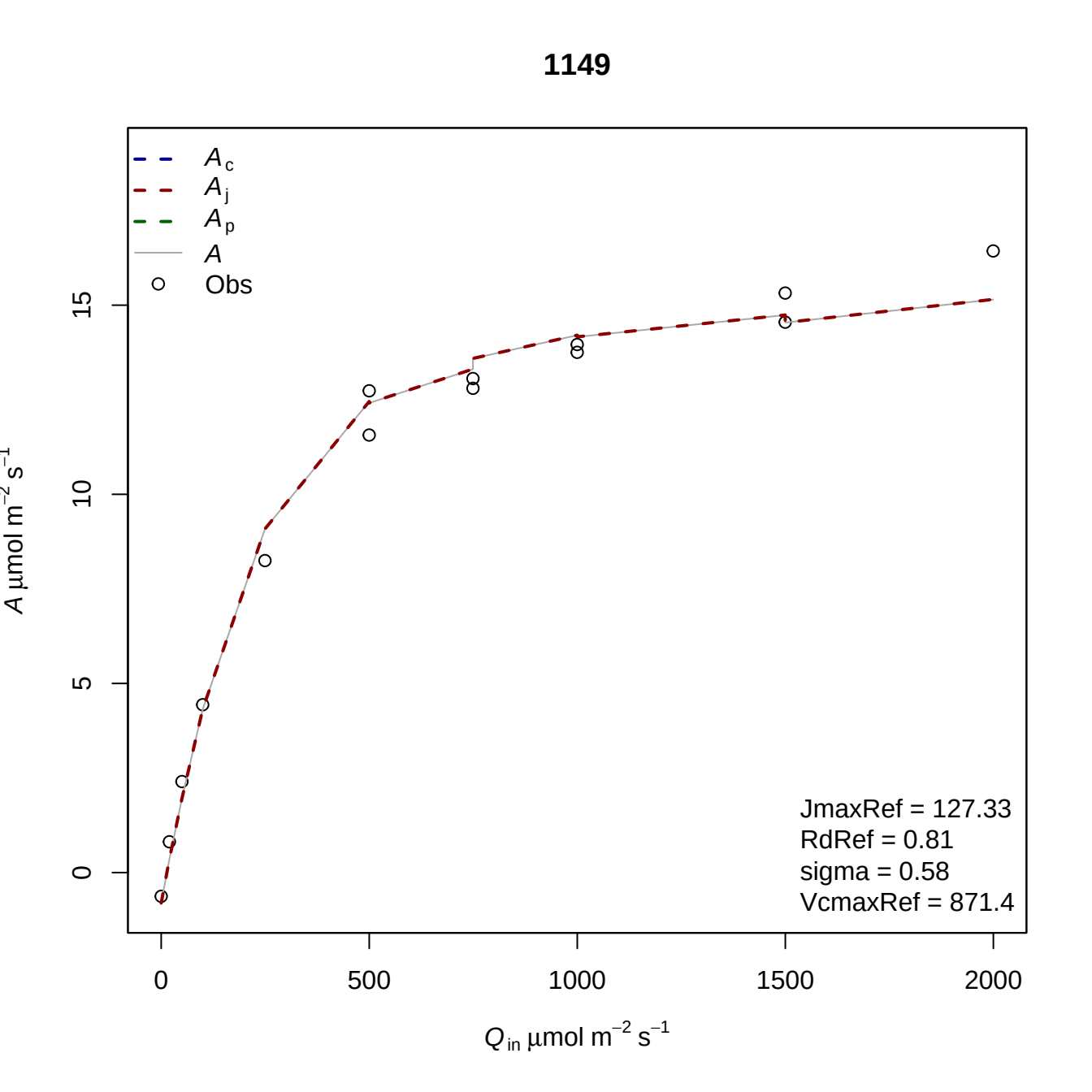

1144
A_c
A_j
A_p
A
Obs
JmaxRef = 70.66
RdRef = 2.91
sigma = 1.59
VcmaxRef = 470716.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

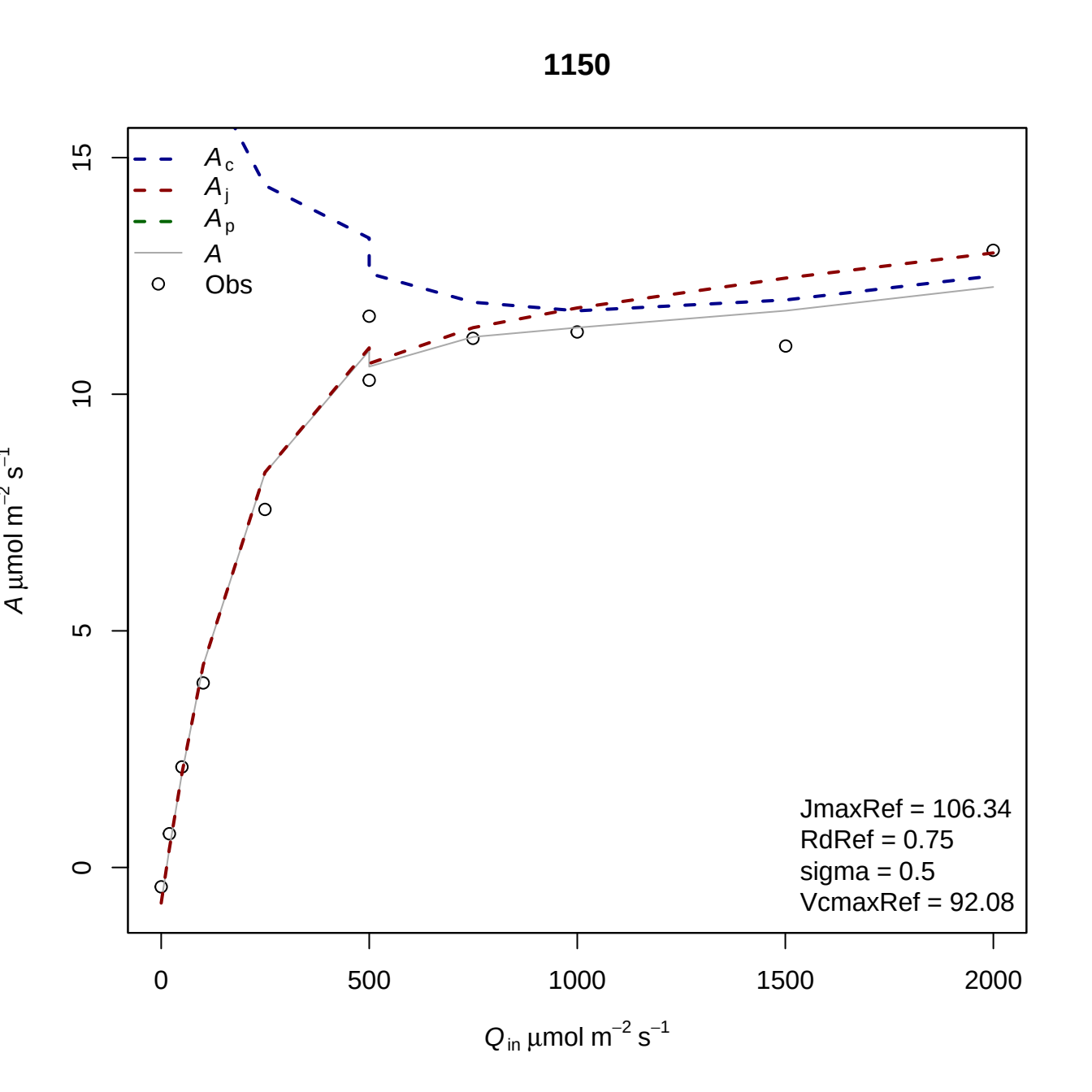

1145
A_c
A_j
A_p
A
Obs
JmaxRef = 173.16
RdRef = 0.97
sigma = 0.36
VcmaxRef = 691.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

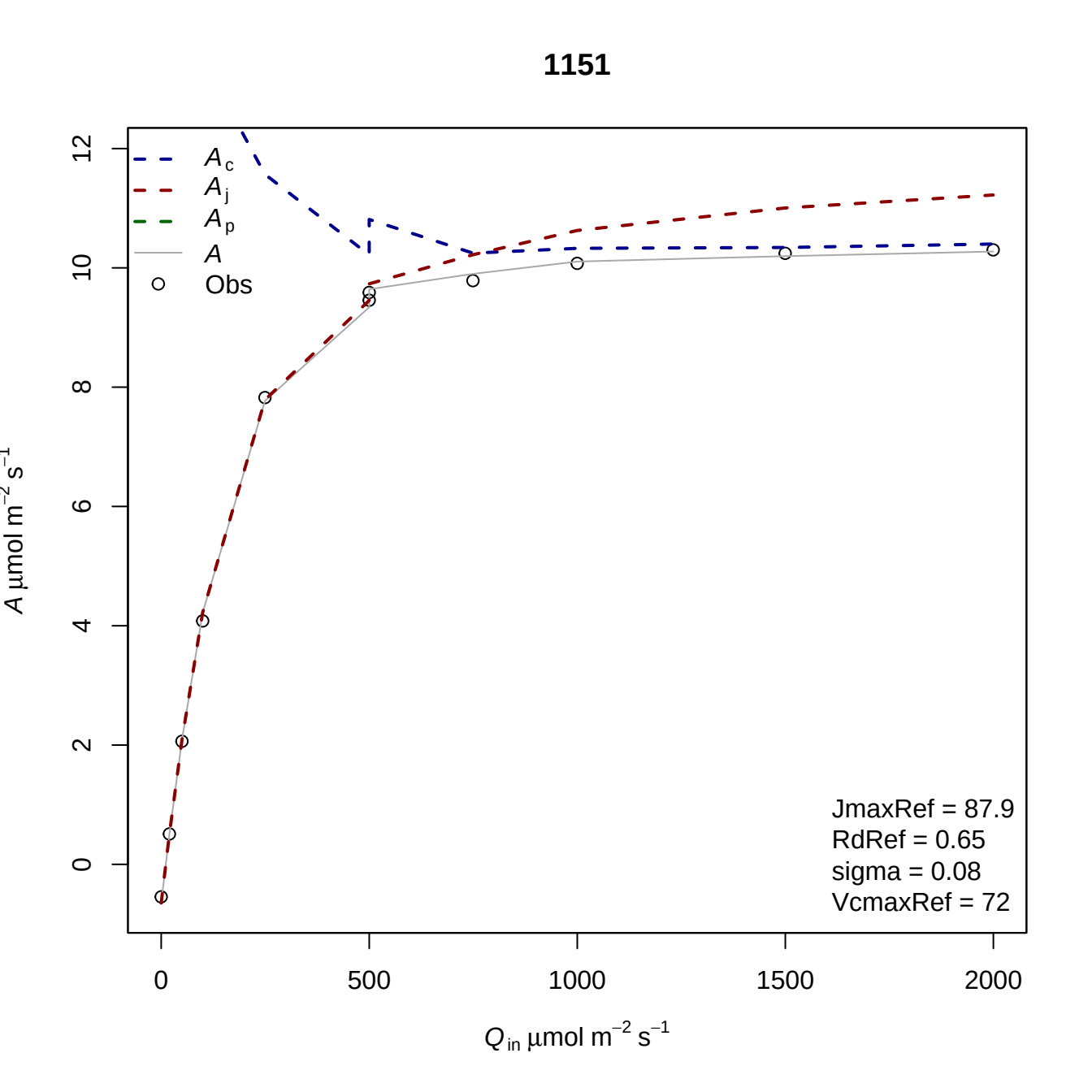

1146
A_c
A_j
A_p
A
Obs
JmaxRef = 100.81
RdRef = 1.29
sigma = 0.37
VcmaxRef = 571.02
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

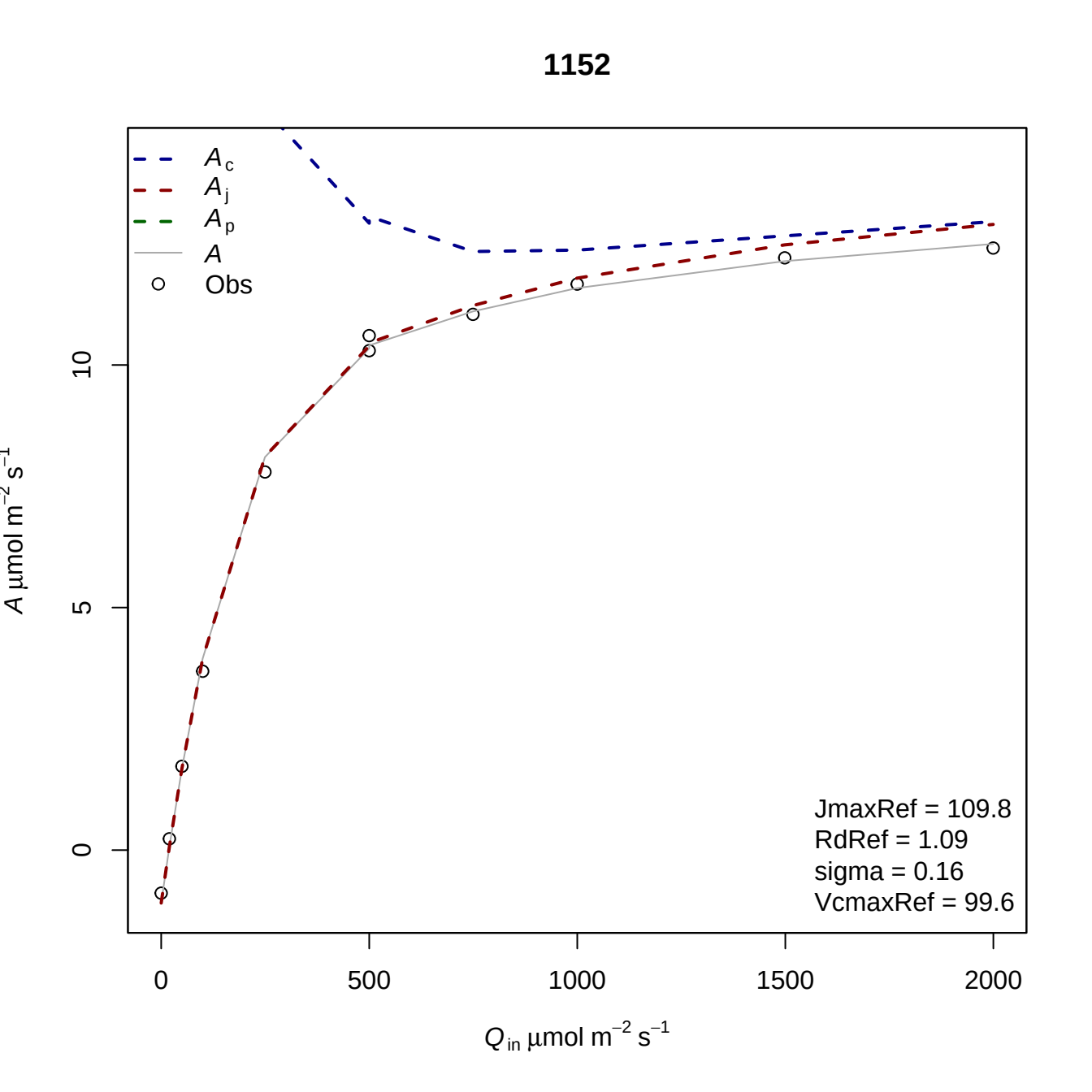

1147
A_c
A_j
A_p
A
Obs
JmaxRef = 109.3
RdRef = 1.03
sigma = 0.17
VcmaxRef = 100.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

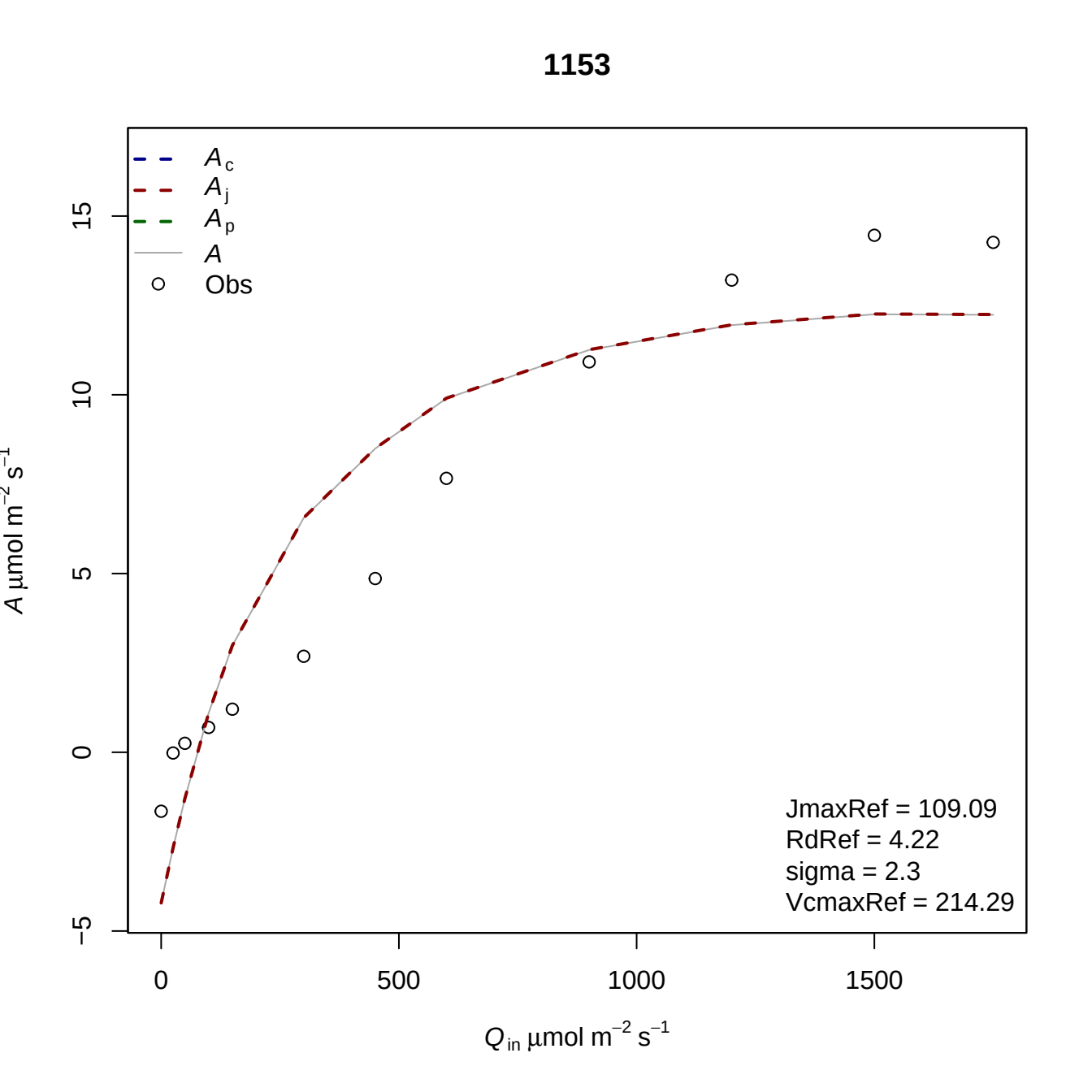

1148
A_c
A_j
A_p
A
Obs
JmaxRef = 131.01
RdRef = 1.19
sigma = 0.5
VcmaxRef = 800.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

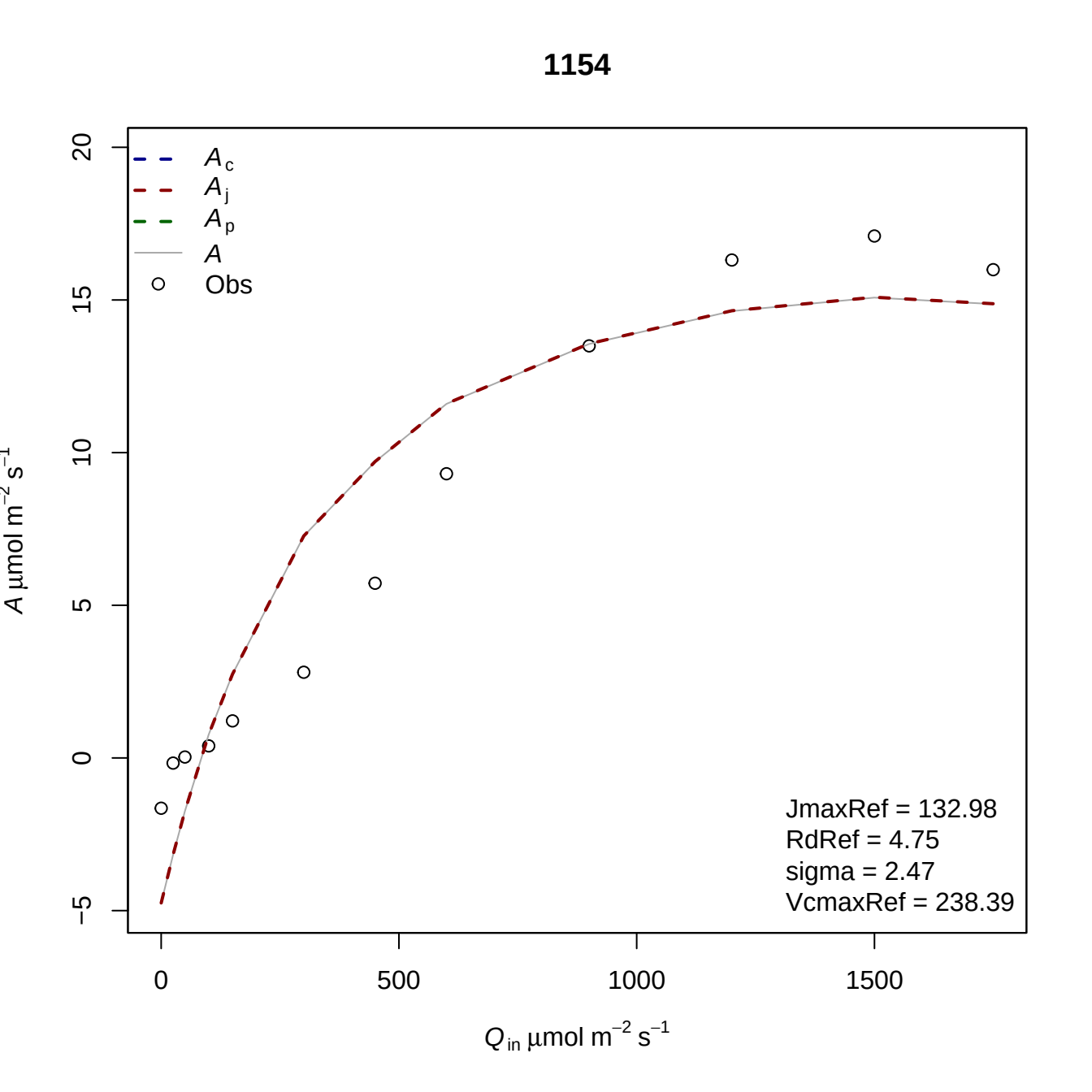

1149
A_c
A_j
A_p
A
Obs
JmaxRef = 127.33
RdRef = 0.81
sigma = 0.58
VcmaxRef = 871.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

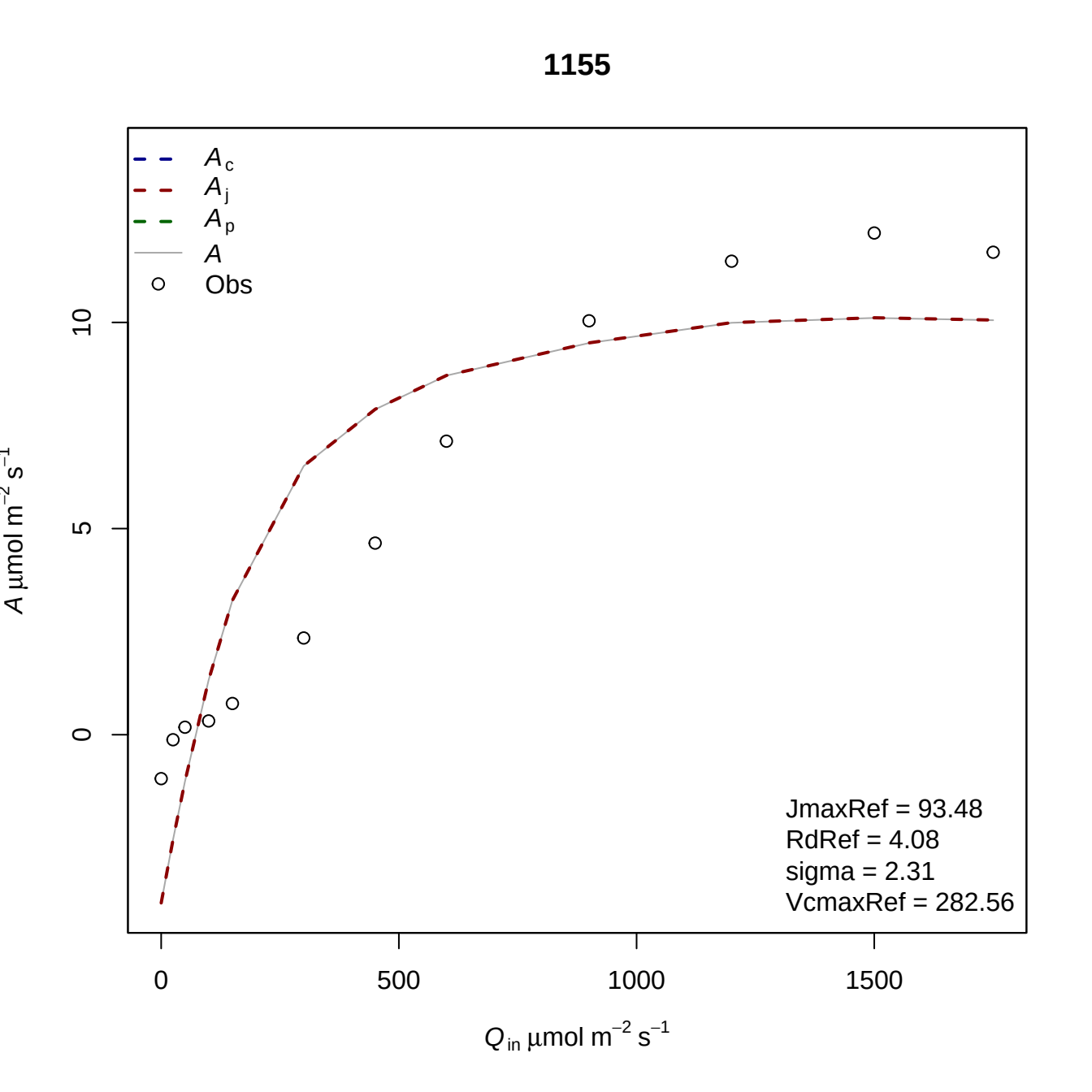

1150
A_c
A_j
A_p
A
Obs
JmaxRef = 106.34
RdRef = 0.75
sigma = 0.5
VcmaxRef = 92.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

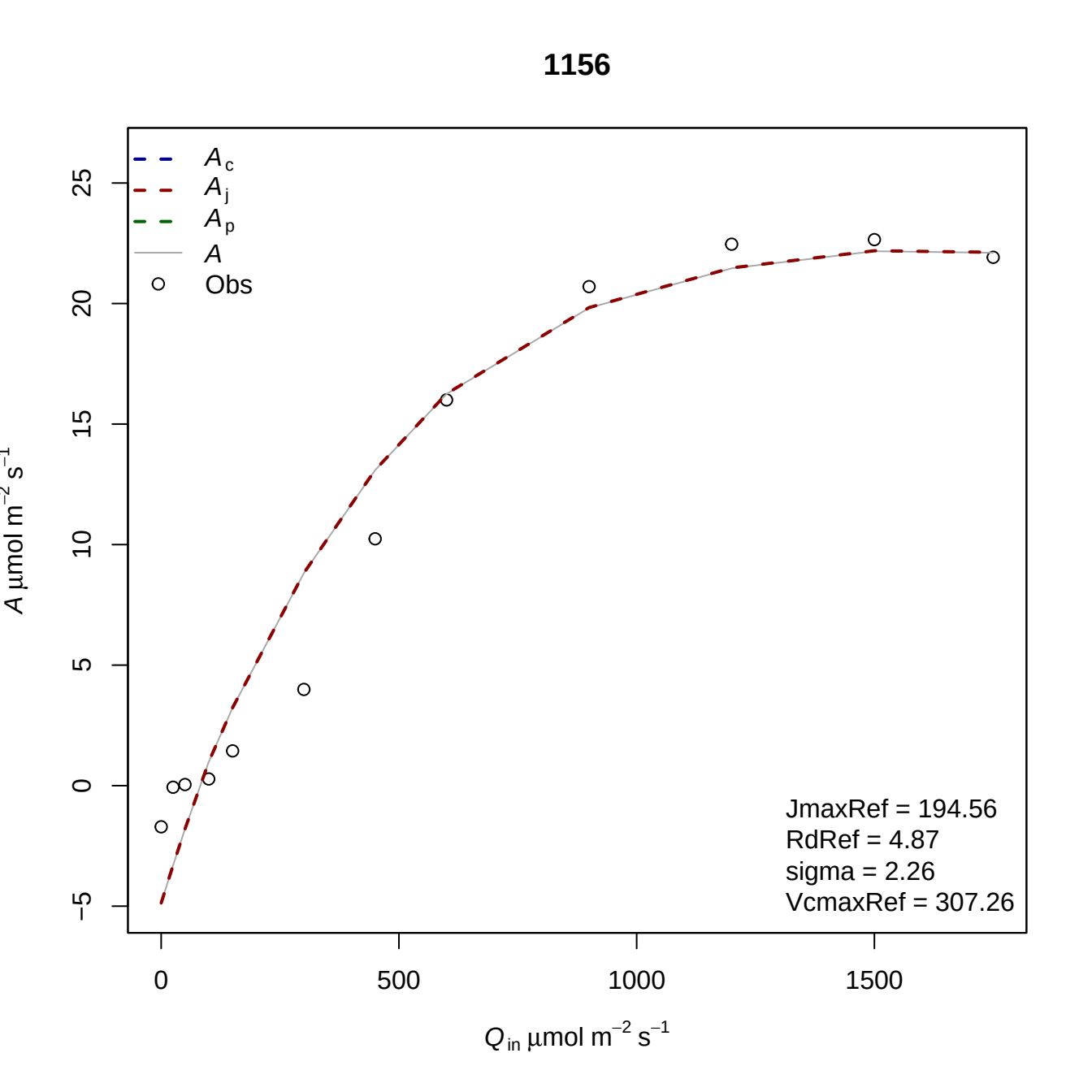

1151
A_c
A_j
A_p
A
Obs
JmaxRef = 87.9
RdRef = 0.65
sigma = 0.08
VcmaxRef = 72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

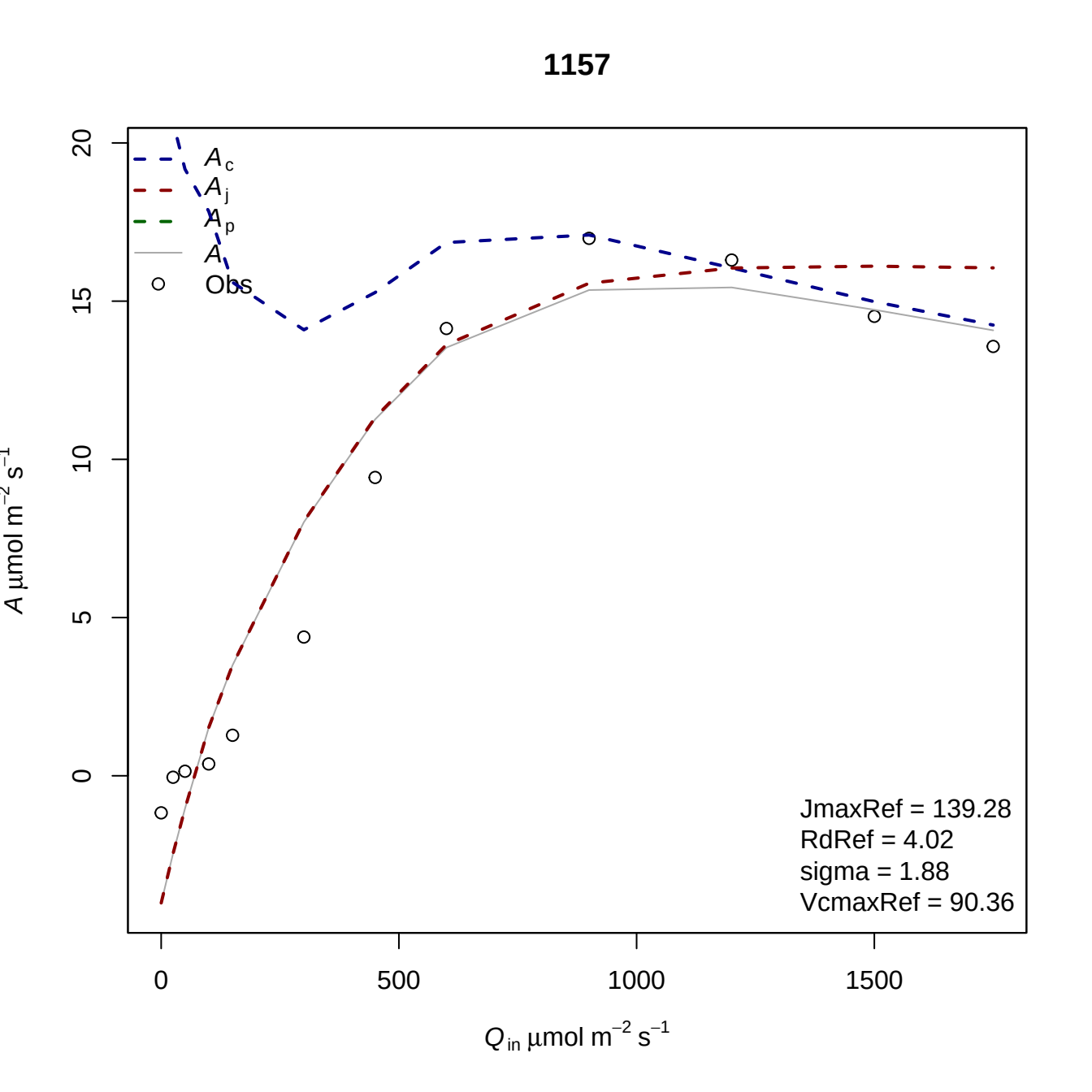

1152
A_c
A_j
A_p
A
Obs
JmaxRef = 109.8
RdRef = 1.09
sigma = 0.16
VcmaxRef = 99.6
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

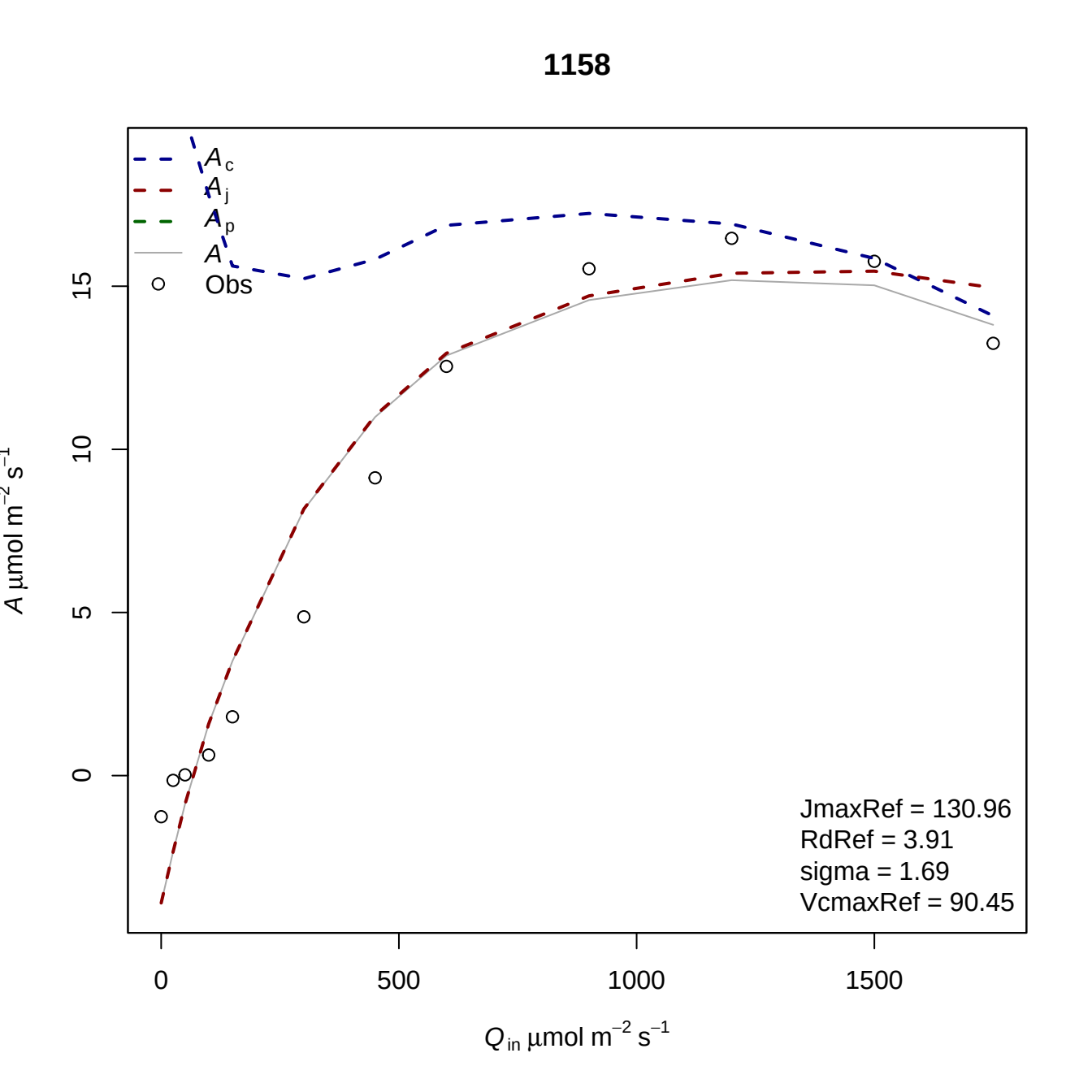

1153
A_c
A_j
A_p
A
Obs
JmaxRef = 109.09
RdRef = 4.22
sigma = 2.3
VcmaxRef = 214.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500

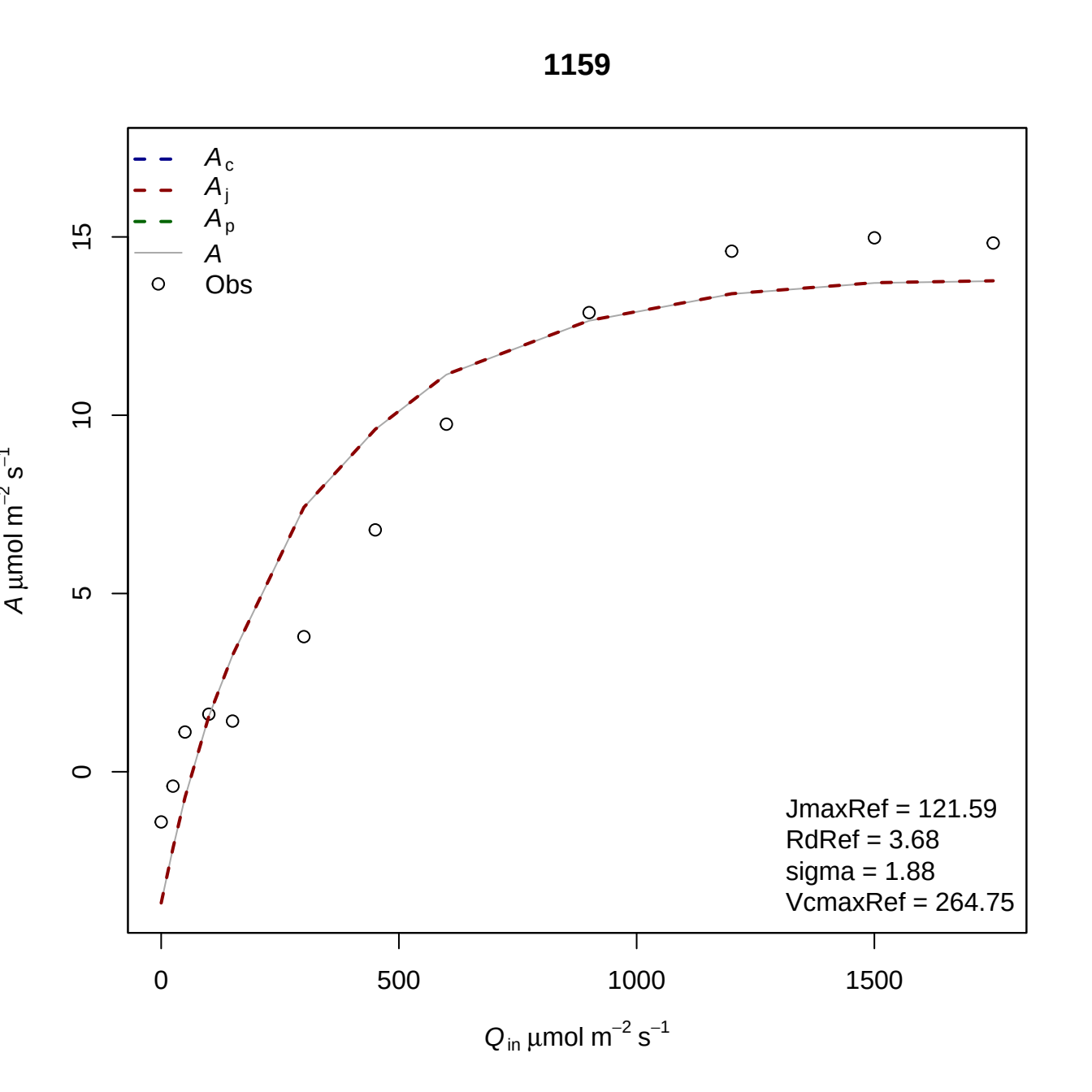

1154
A_c
A_j
A_p
A
Obs
JmaxRef = 132.98
RdRef = 4.75
sigma = 2.47
VcmaxRef = 238.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500

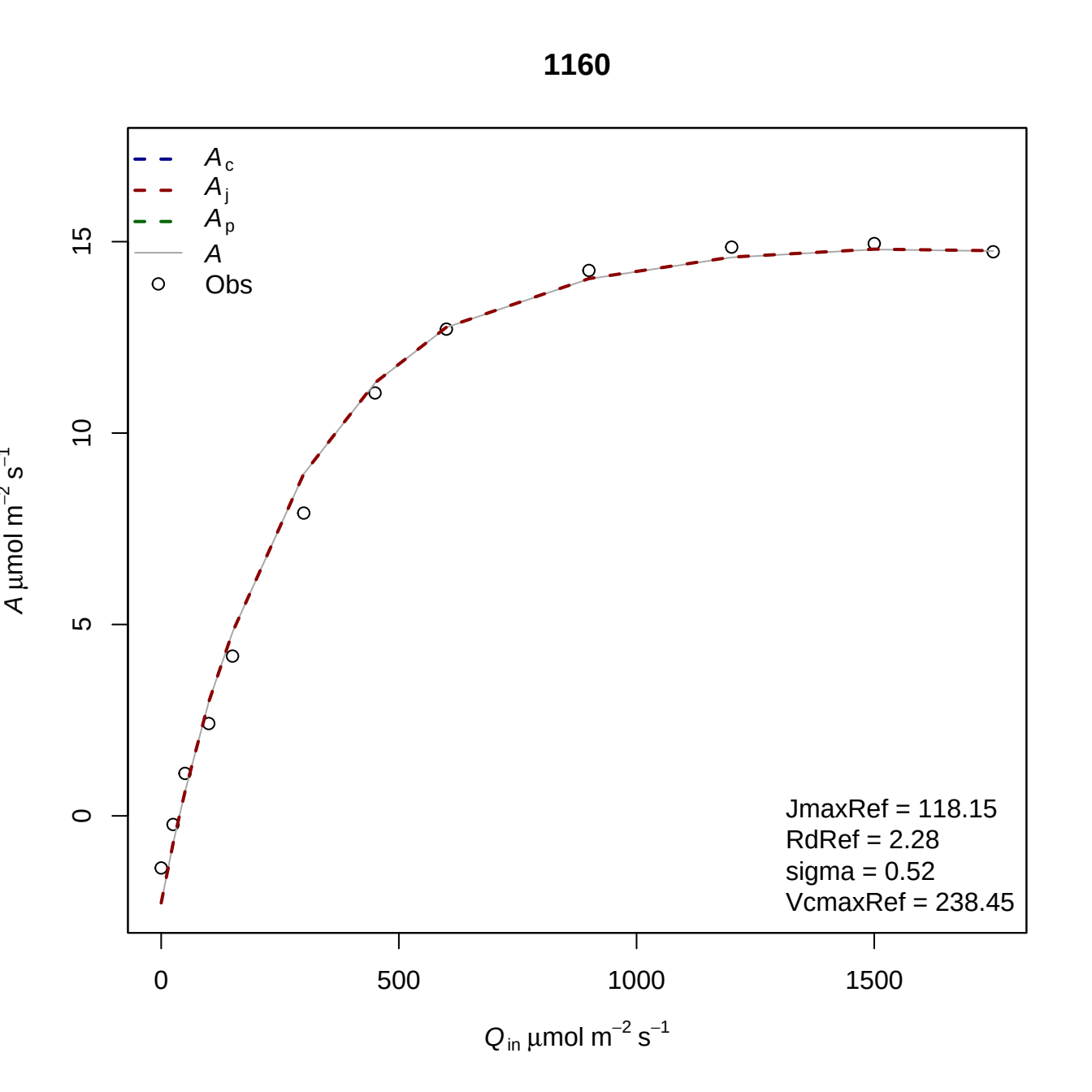

1155
A_c
A_j
A_p
A
Obs
JmaxRef = 93.48
RdRef = 4.08
sigma = 2.31
VcmaxRef = 282.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

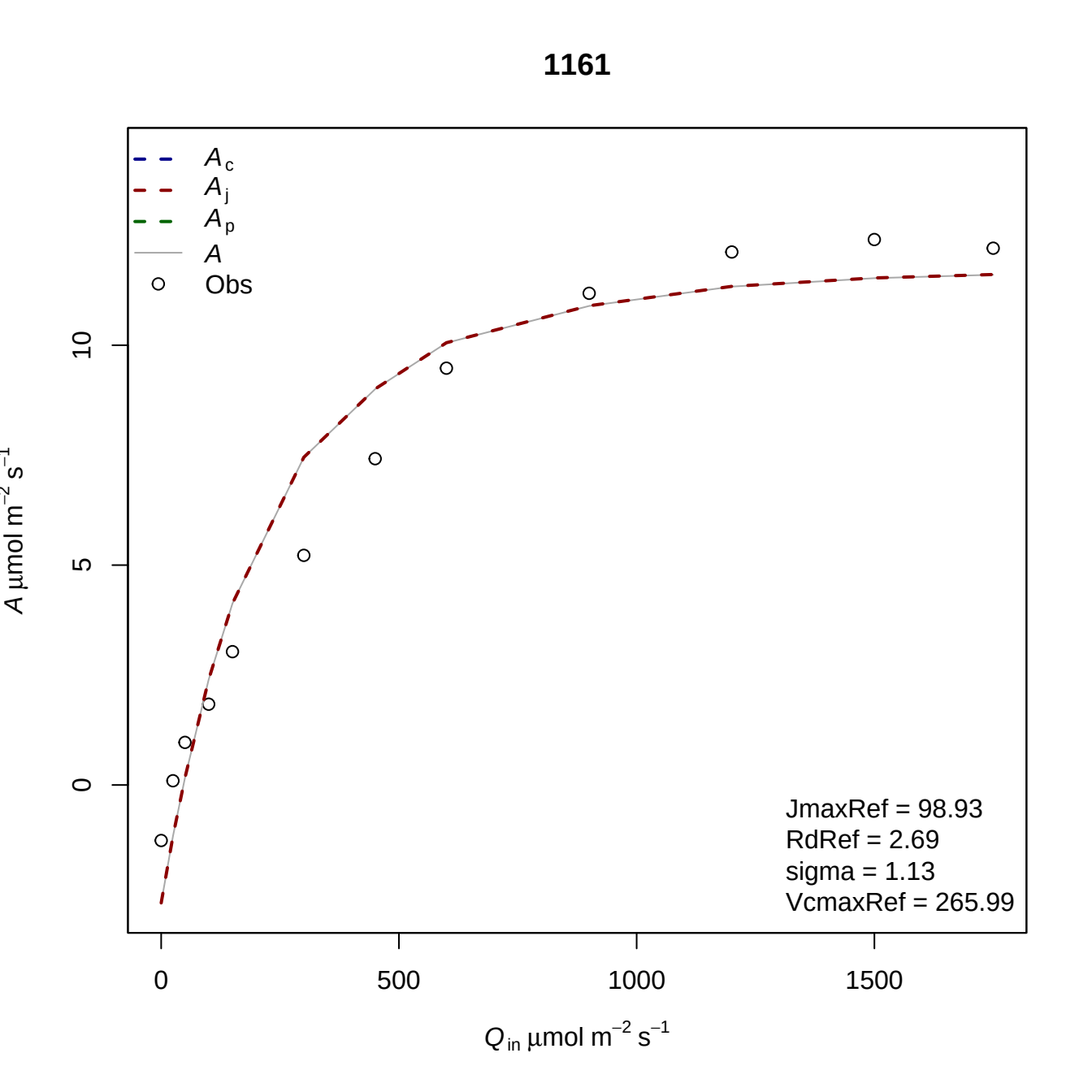

1156
A_c
A_j
A_p
A
Obs
JmaxRef = 194.56
RdRef = 4.87
sigma = 2.26
VcmaxRef = 307.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500

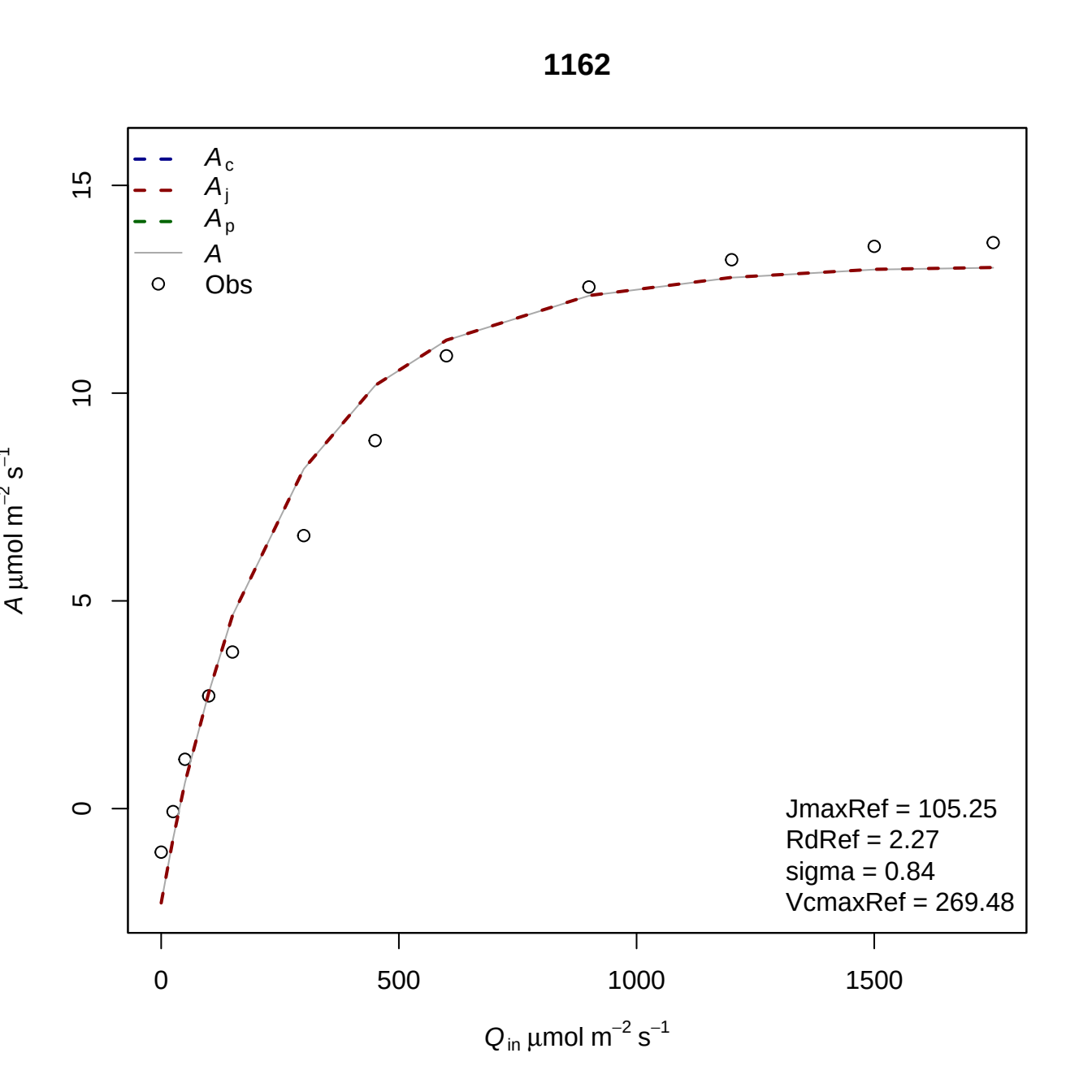

1157
A_c
A_j
A_p
A
Obs
JmaxRef = 139.28
RdRef = 4.02
sigma = 1.88
VcmaxRef = 90.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

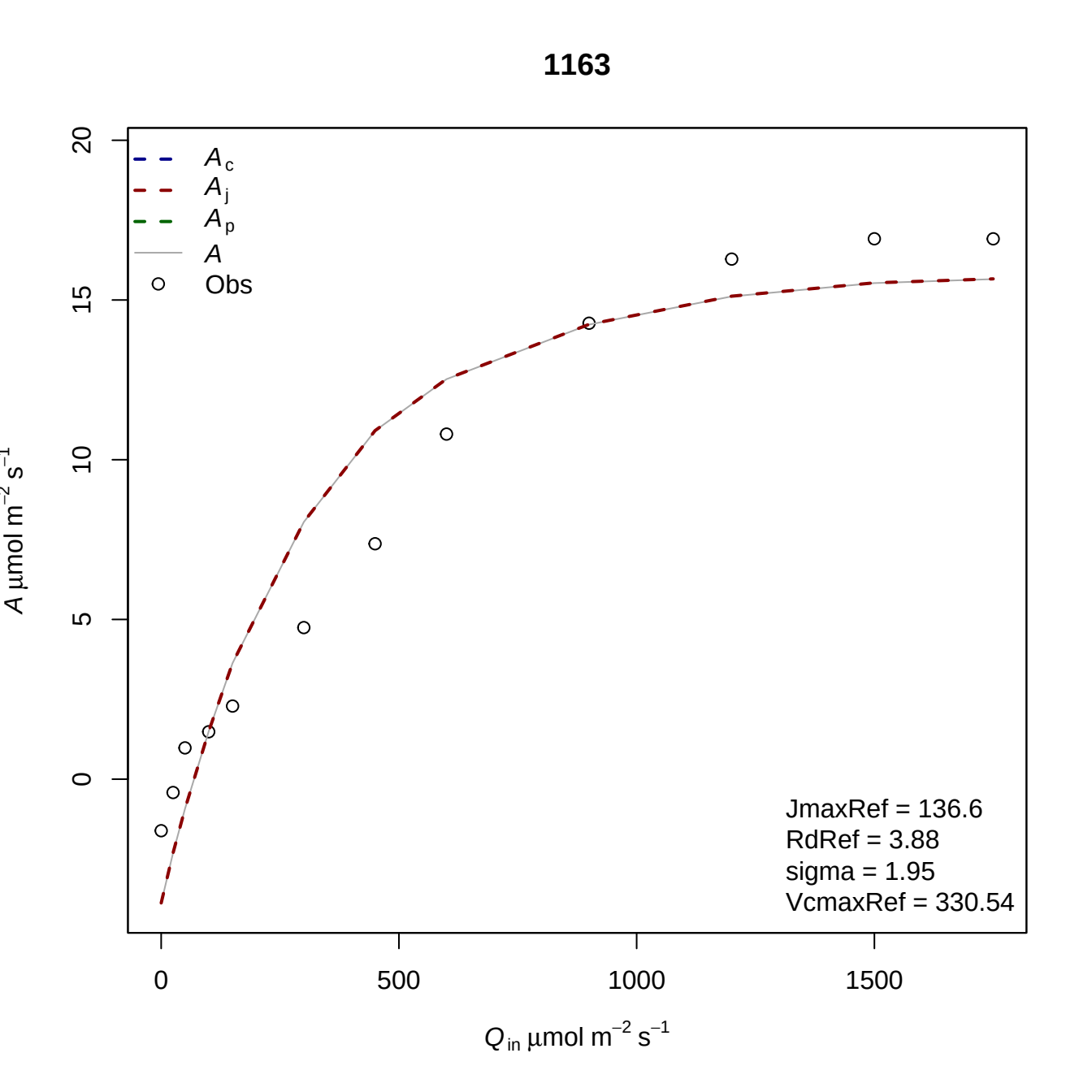

1158
A_c
A_j
A_p
A
Obs
JmaxRef = 130.96
RdRef = 3.91
sigma = 1.69
VcmaxRef = 90.45
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

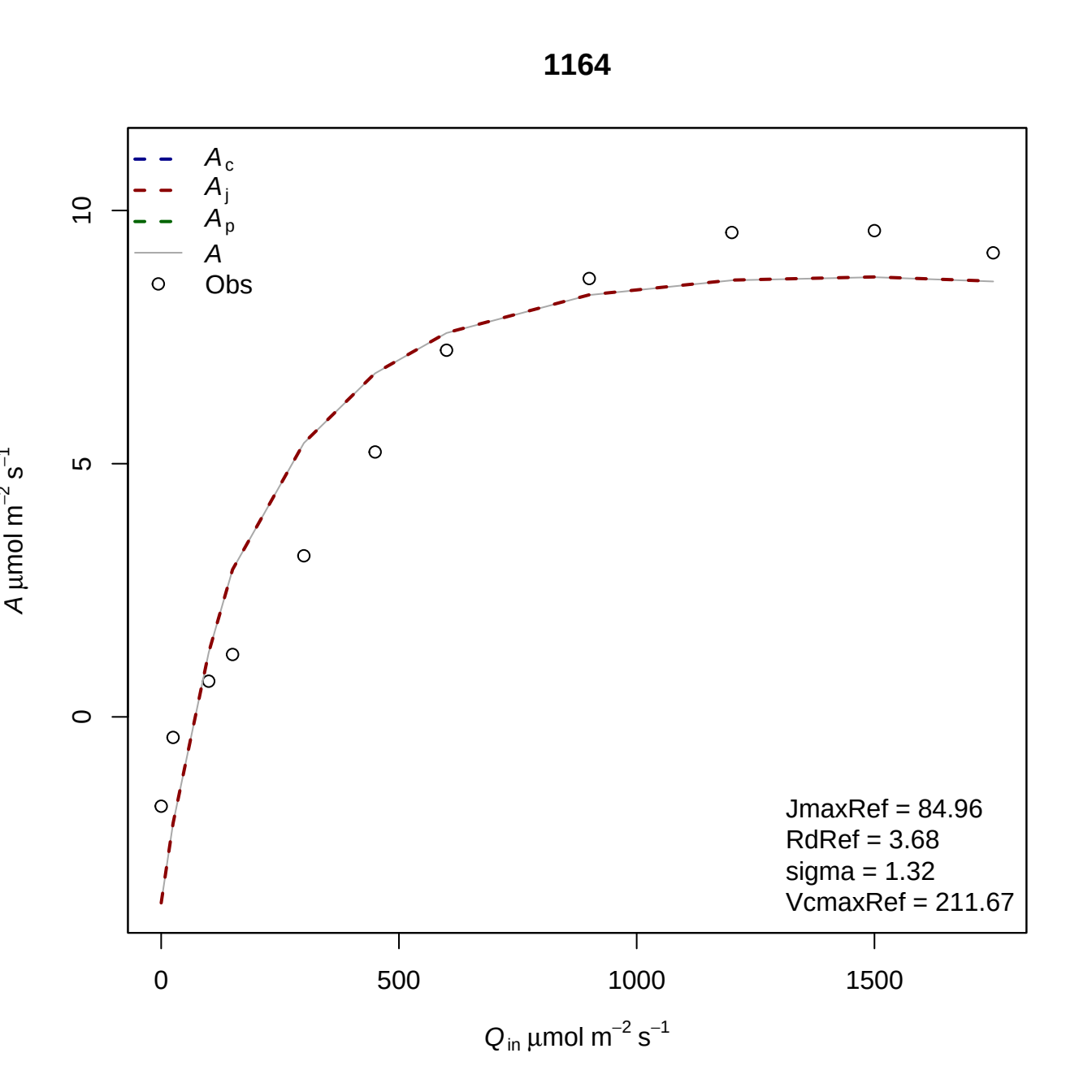

1159
A_c
A_j
A_p
A
Obs
JmaxRef = 121.59
RdRef = 3.68
sigma = 1.88
VcmaxRef = 264.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

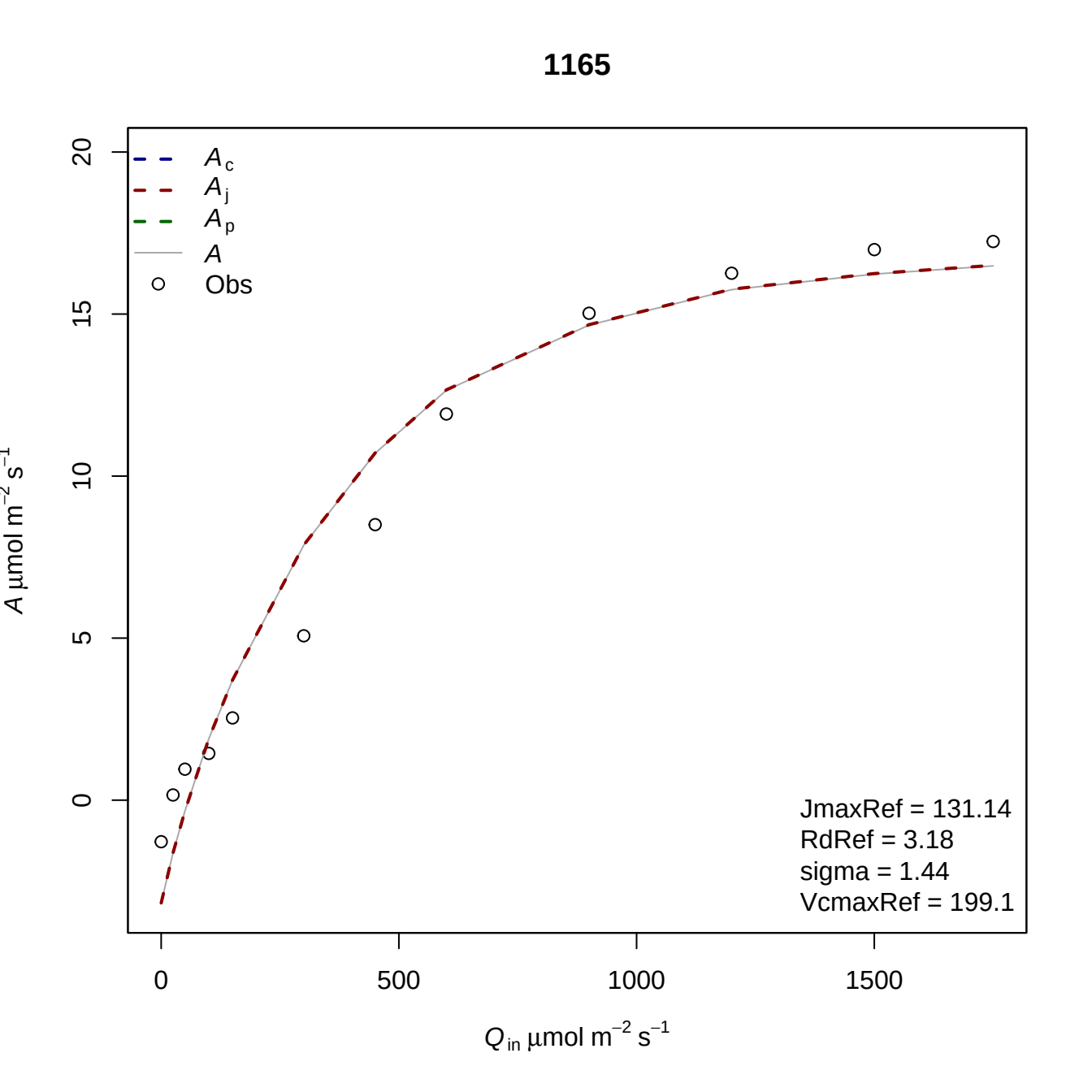

1160
A_c
A_j
A_p
A
Obs
JmaxRef = 118.15
RdRef = 2.28
sigma = 0.52
VcmaxRef = 238.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

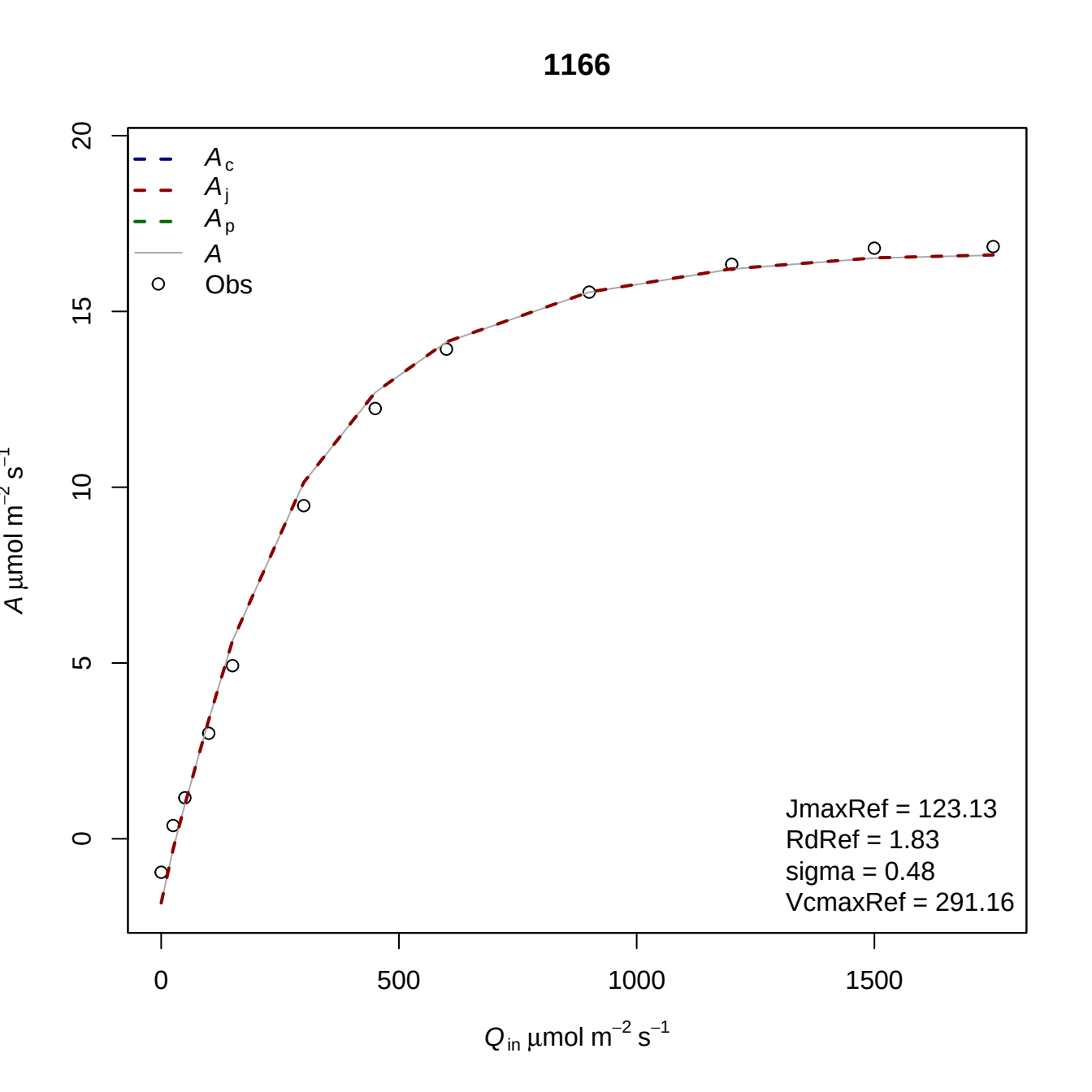

1161
A_c
A_j
A_p
A
Obs
JmaxRef = 98.93
RdRef = 2.69
sigma = 1.13
VcmaxRef = 265.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

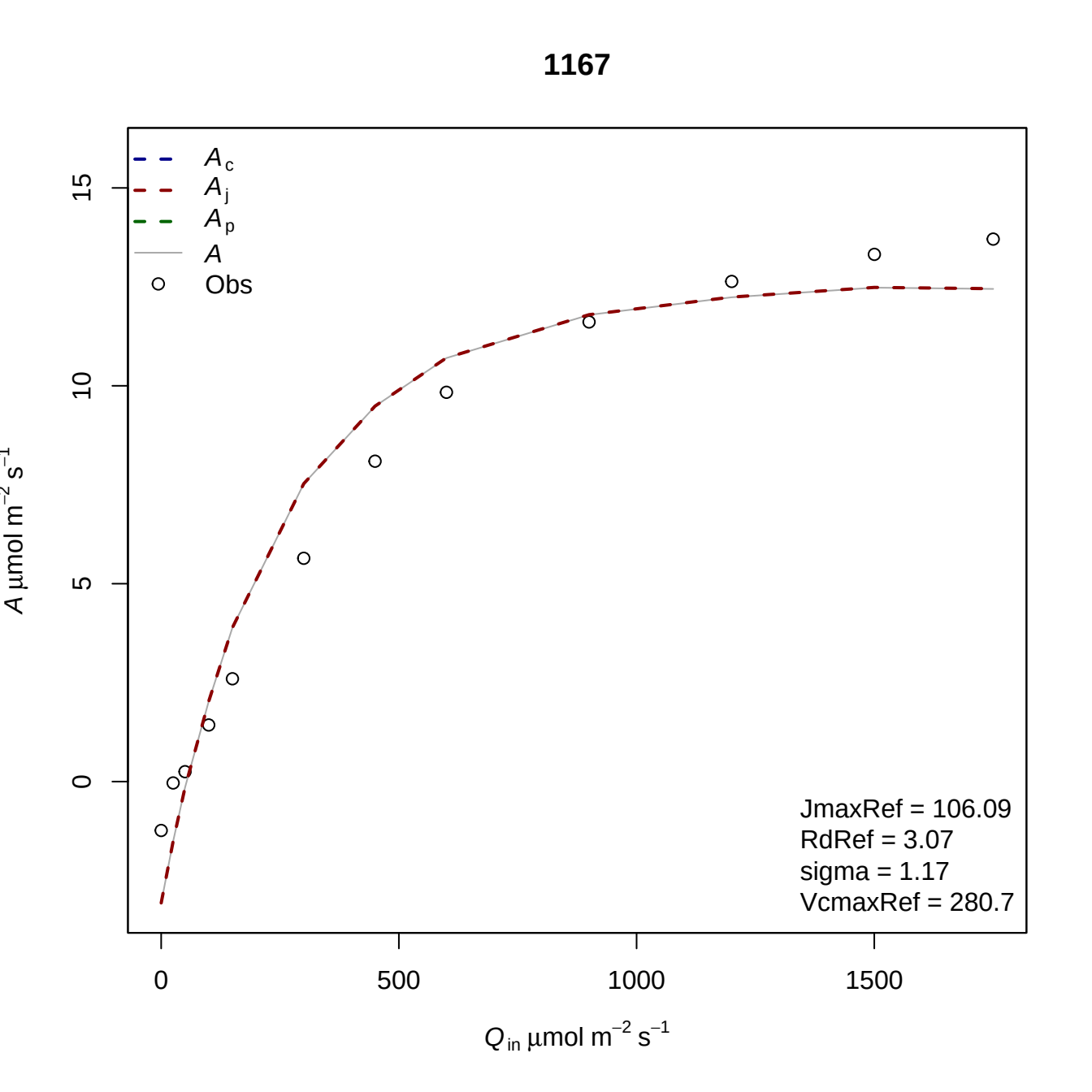

1162
A_c
A_j
A_p
A
Obs
JmaxRef = 105.25
RdRef = 2.27
sigma = 0.84
VcmaxRef = 269.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

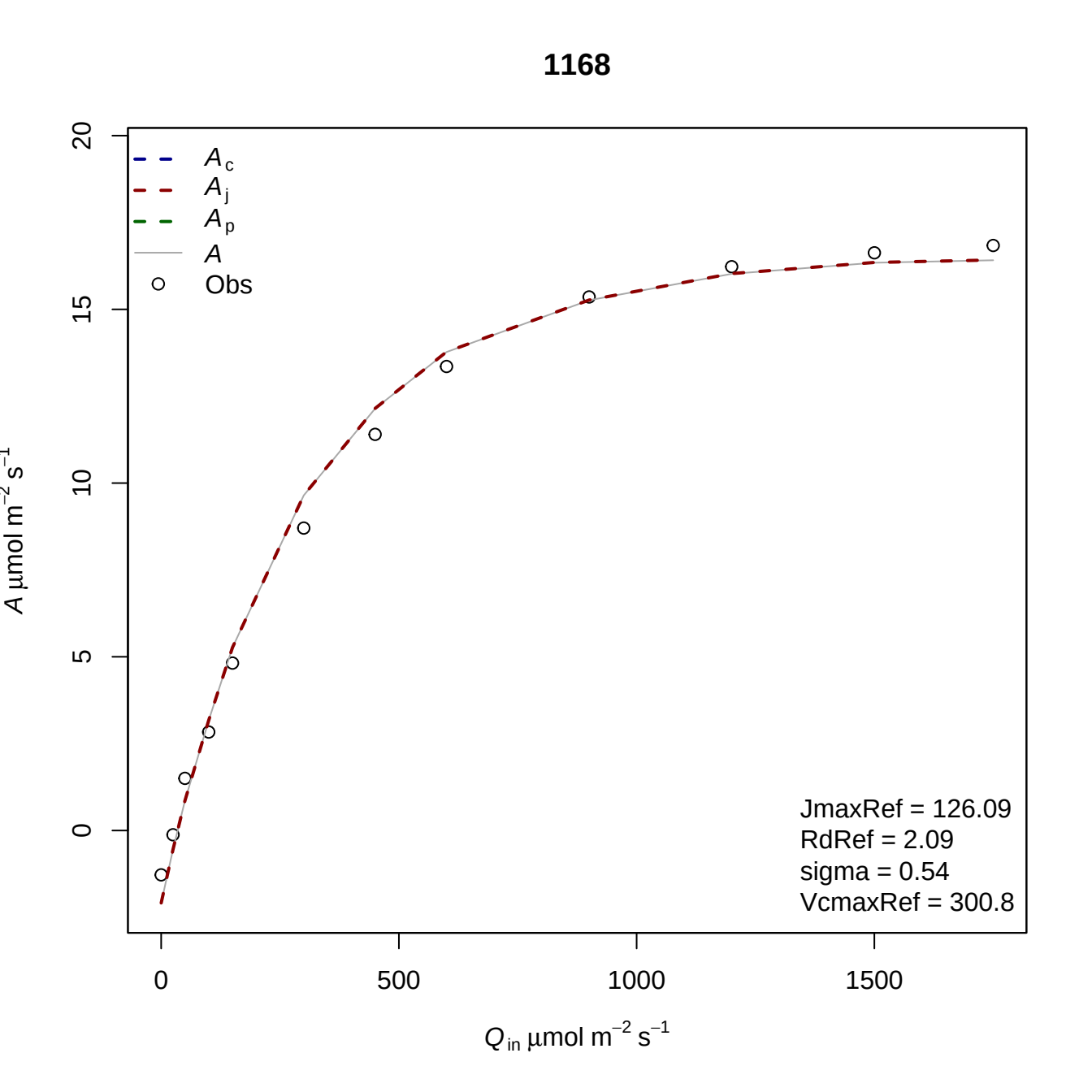

1163
A_c
A_j
A_p
A
Obs
JmaxRef = 136.6
RdRef = 3.88
sigma = 1.95
VcmaxRef = 330.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

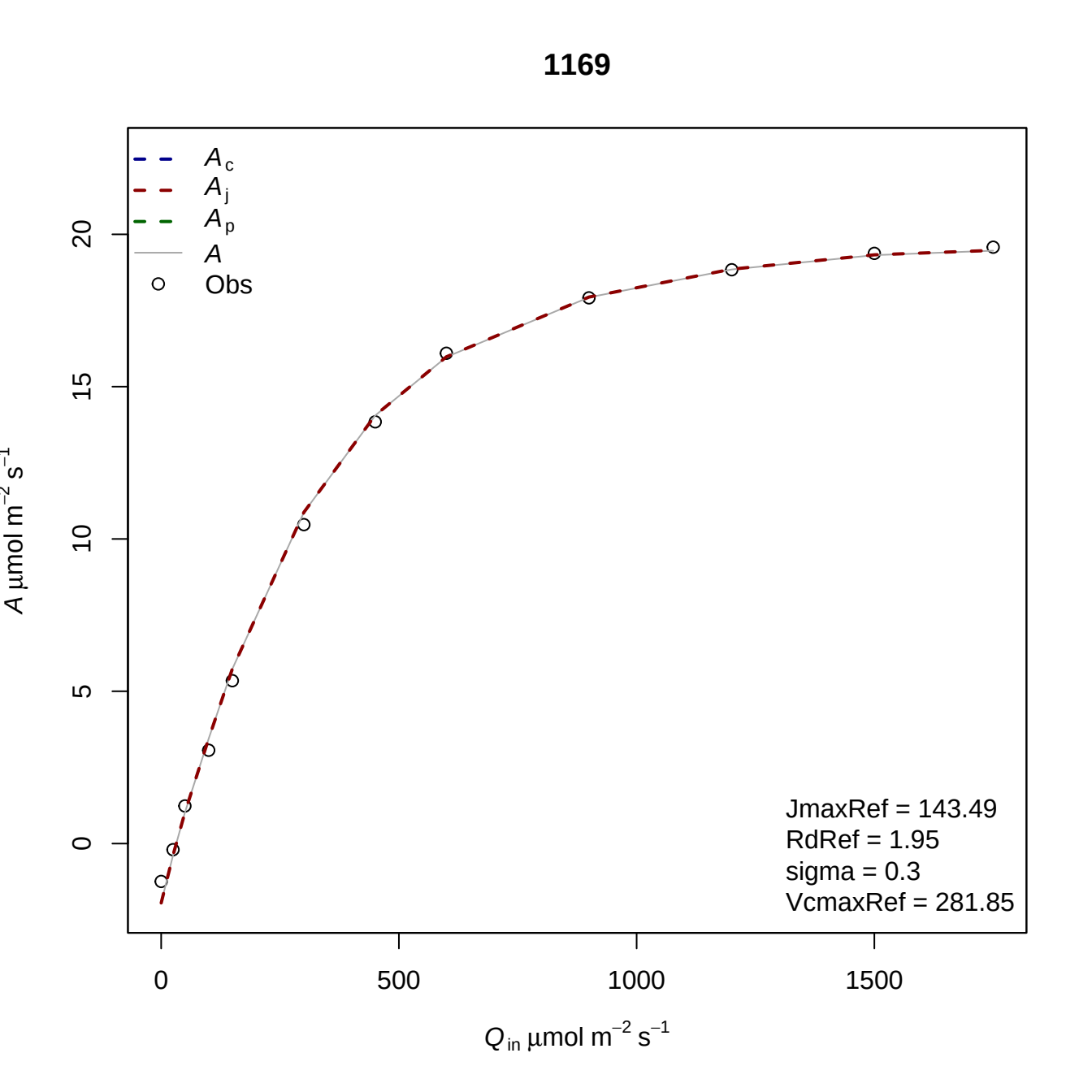

1164
A_c
A_j
A_p
A
Obs
JmaxRef = 84.96
RdRef = 3.68
sigma = 1.32
VcmaxRef = 211.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

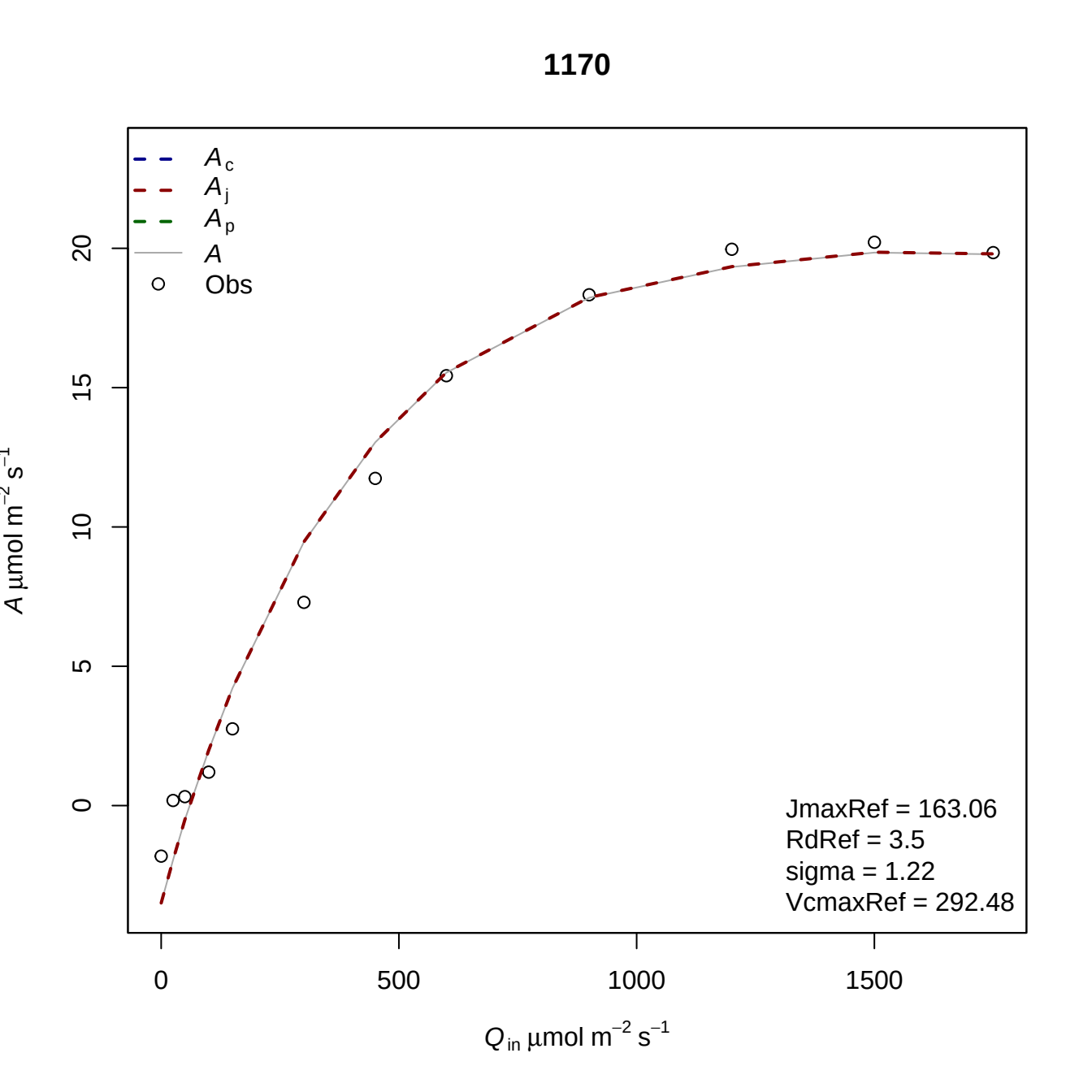

1165
A_c
A_j
A_p
A
Obs
JmaxRef = 131.14
RdRef = 3.18
sigma = 1.44
VcmaxRef = 199.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

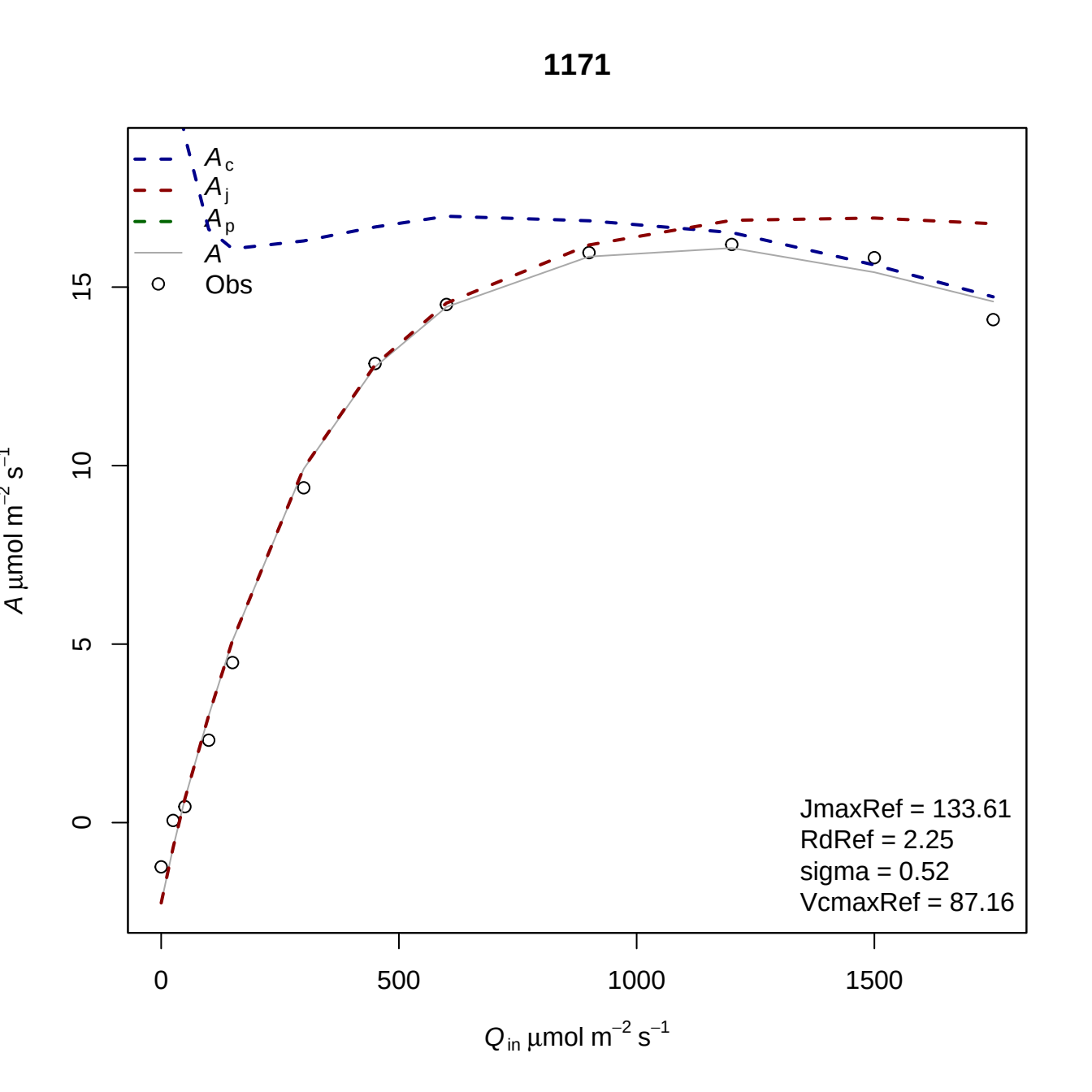

1166
A_c
A_j
A_p
A
Obs
JmaxRef = 123.13
RdRef = 1.83
sigma = 0.48
VcmaxRef = 291.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

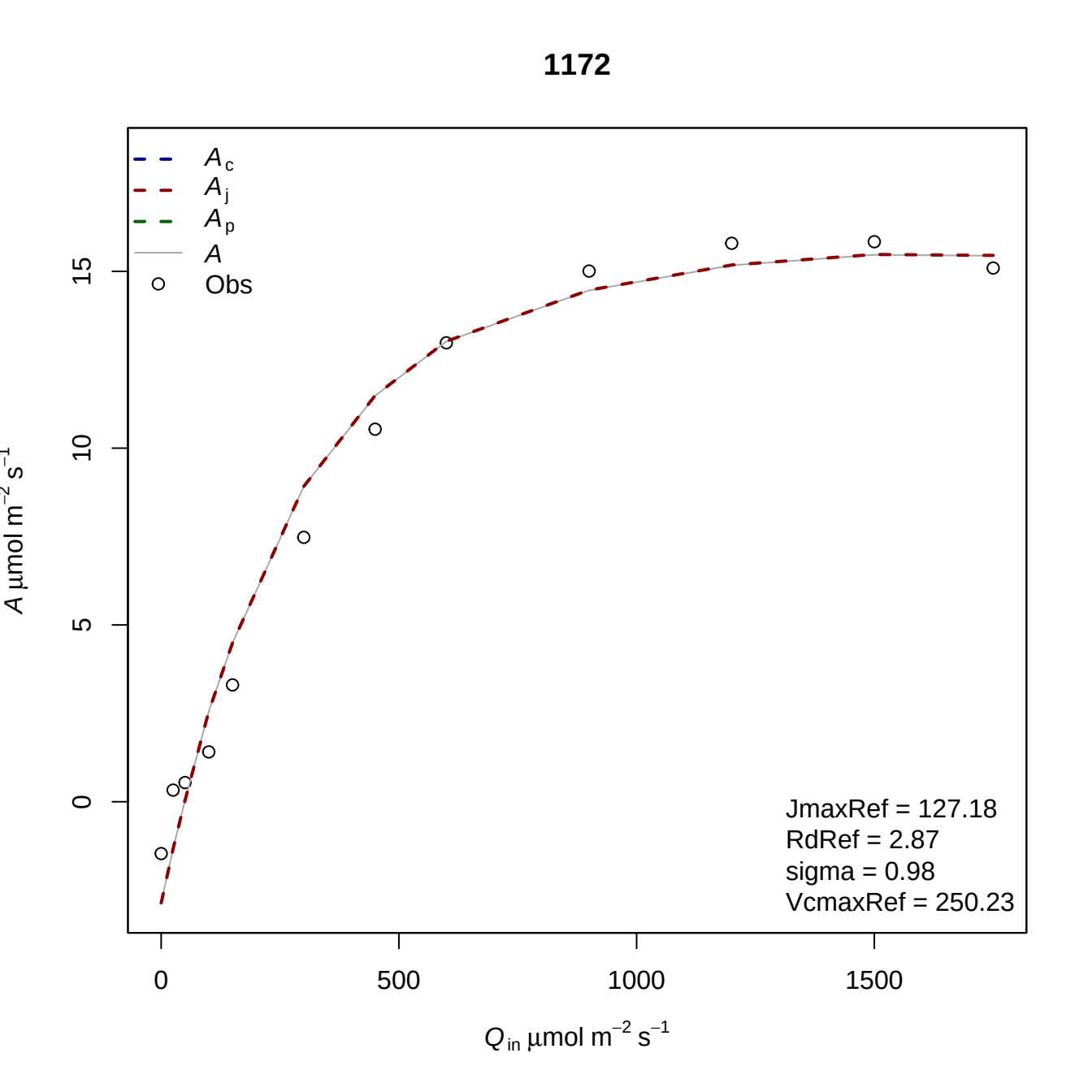

1167
A_c
A_j
A_p
A
Obs
JmaxRef = 106.09
RdRef = 3.07
sigma = 1.17
VcmaxRef = 280.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

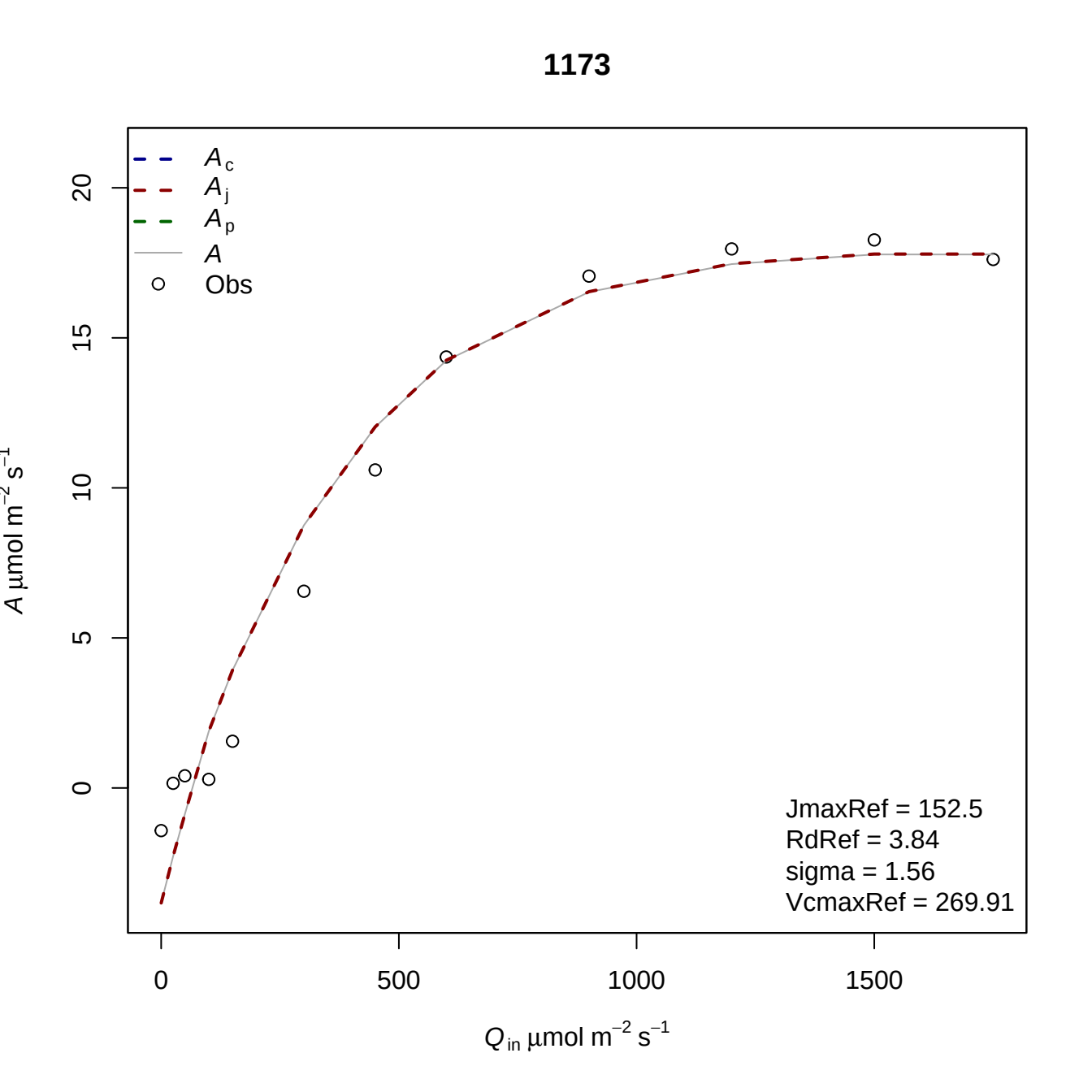

1168
A_c
A_j
A_p
A
Obs
JmaxRef = 126.09
RdRef = 2.09
sigma = 0.54
VcmaxRef = 300.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

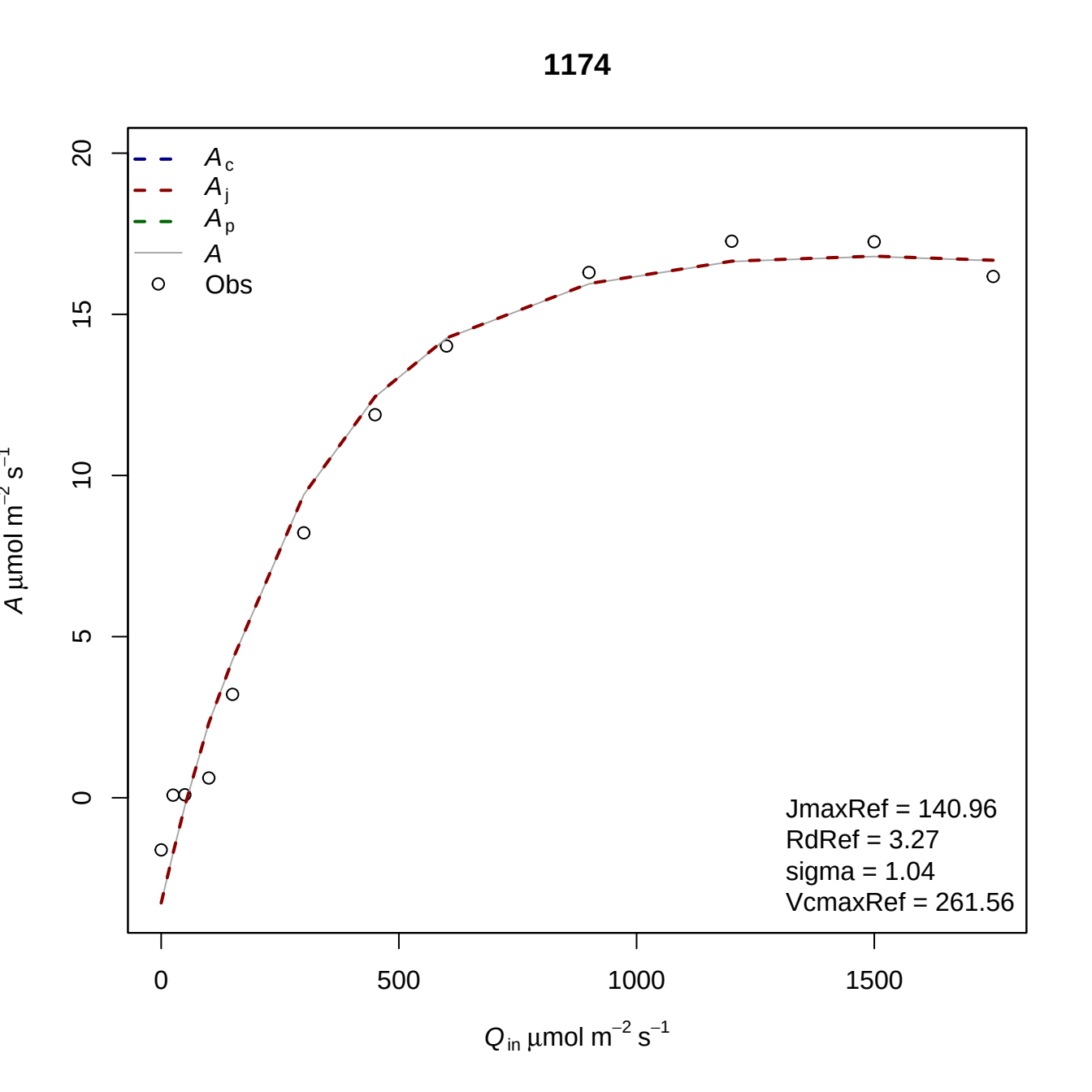

1169
A_c
A_j
A_p
A
Obs
JmaxRef = 143.49
RdRef = 1.95
sigma = 0.3
VcmaxRef = 281.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

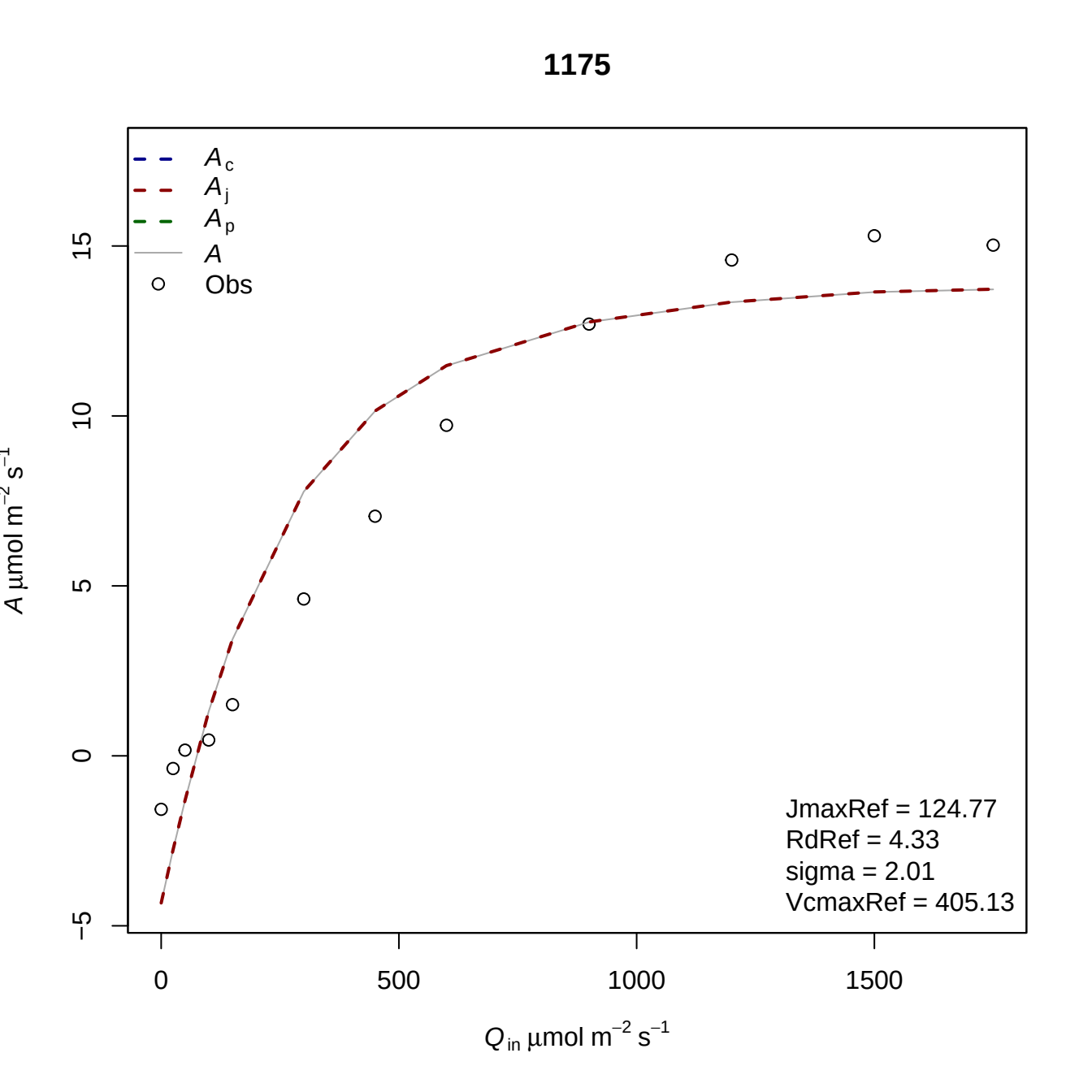

1170
A_c
A_j
A_p
A
Obs
JmaxRef = 163.06
RdRef = 3.5
sigma = 1.22
VcmaxRef = 292.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

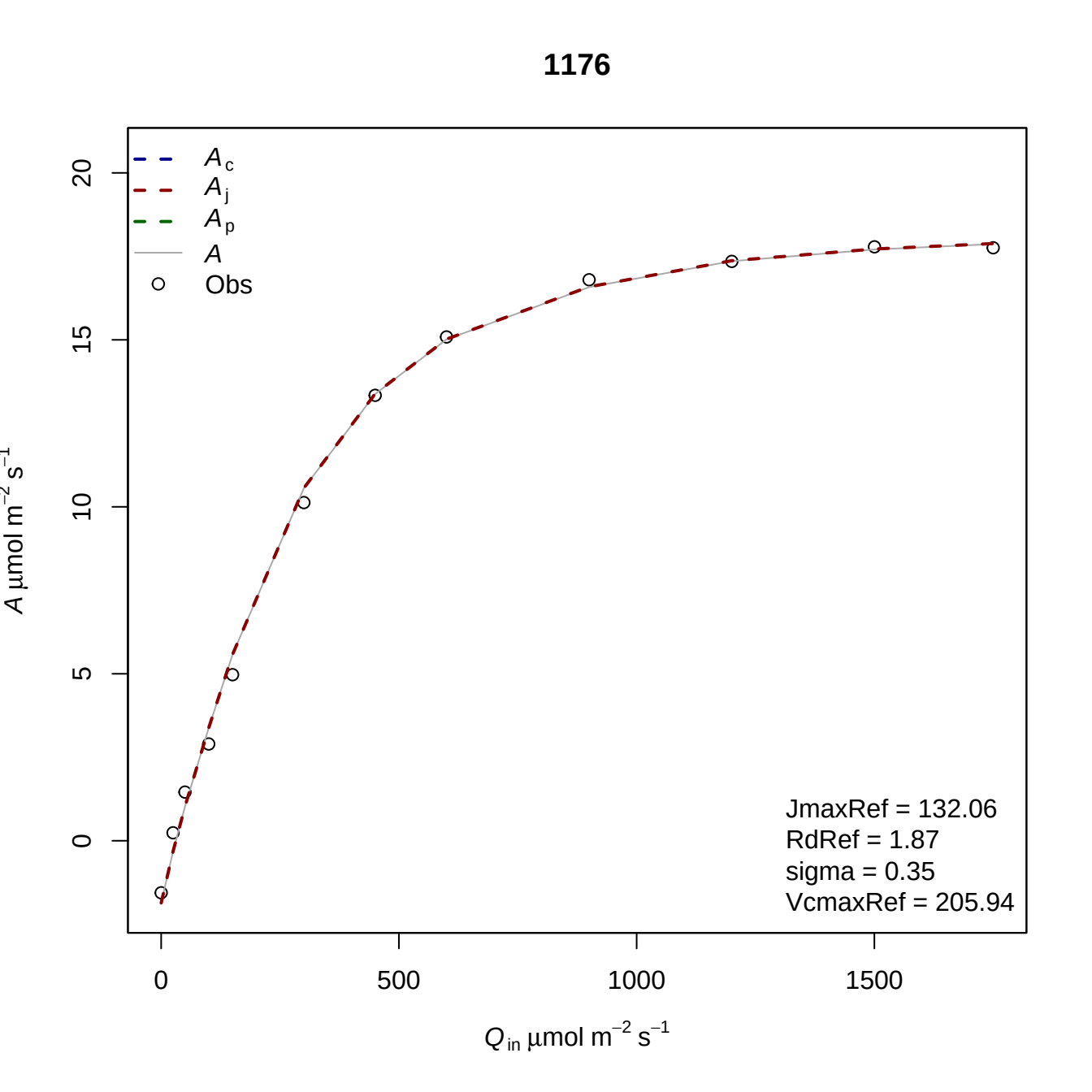

1171
A_c
A_j
A_p
A
Obs
JmaxRef = 133.61
RdRef = 2.25
sigma = 0.52
VcmaxRef = 87.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

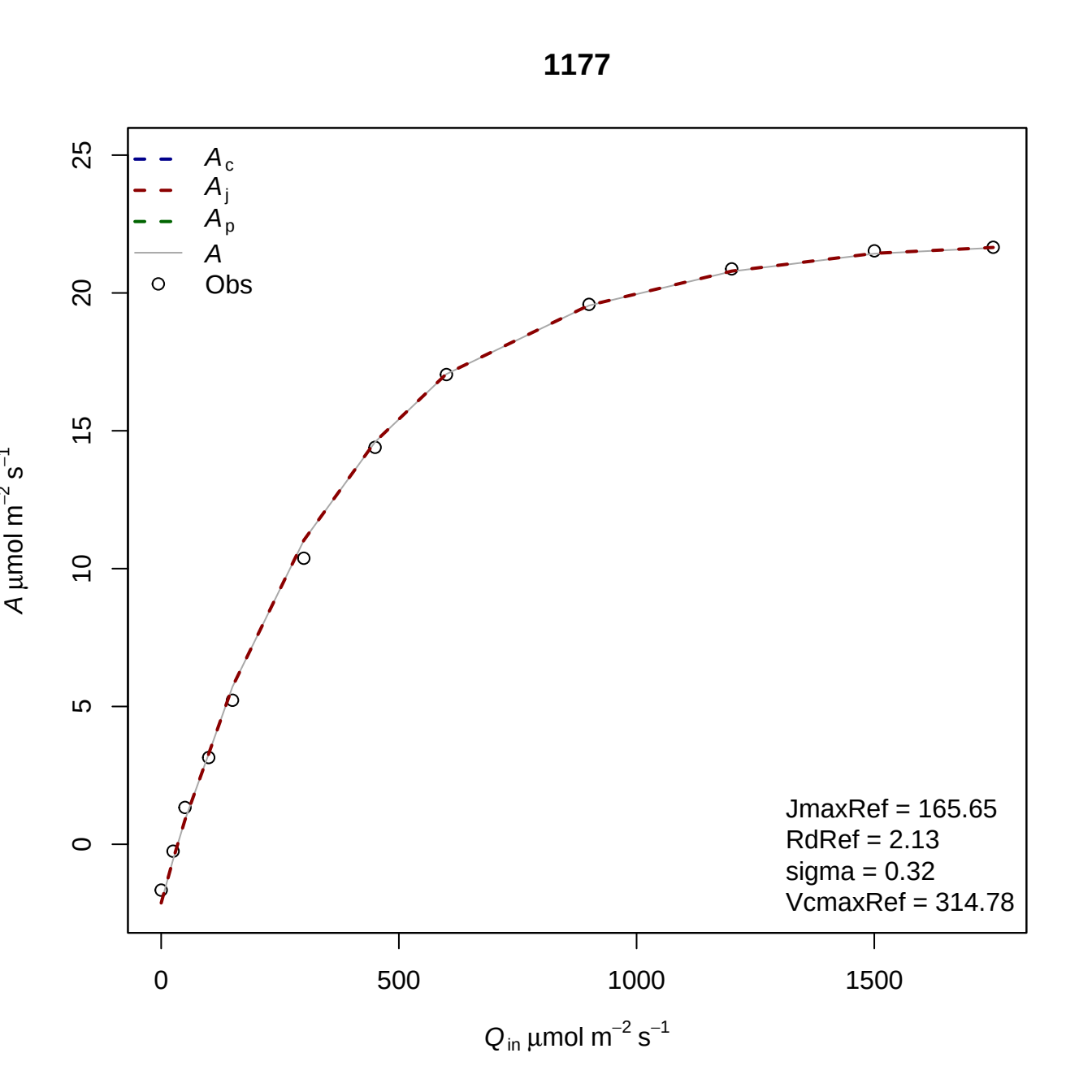

1172
A_c
A_j
A_p
A
Obs
JmaxRef = 127.18
RdRef = 2.87
sigma = 0.98
VcmaxRef = 250.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

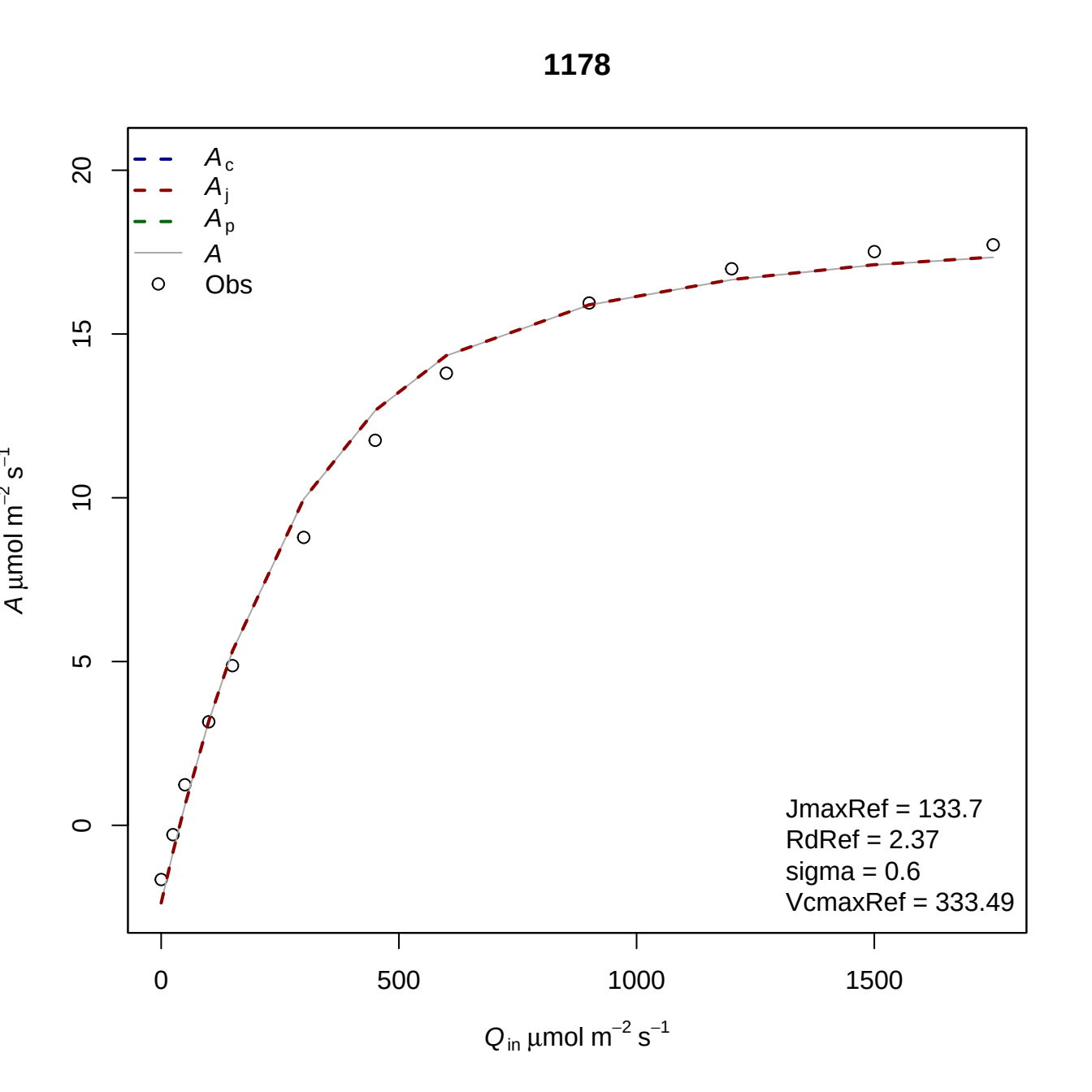

1173
A_c
A_j
A_p
A
Obs
JmaxRef = 152.5
RdRef = 3.84
sigma = 1.56
VcmaxRef = 269.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

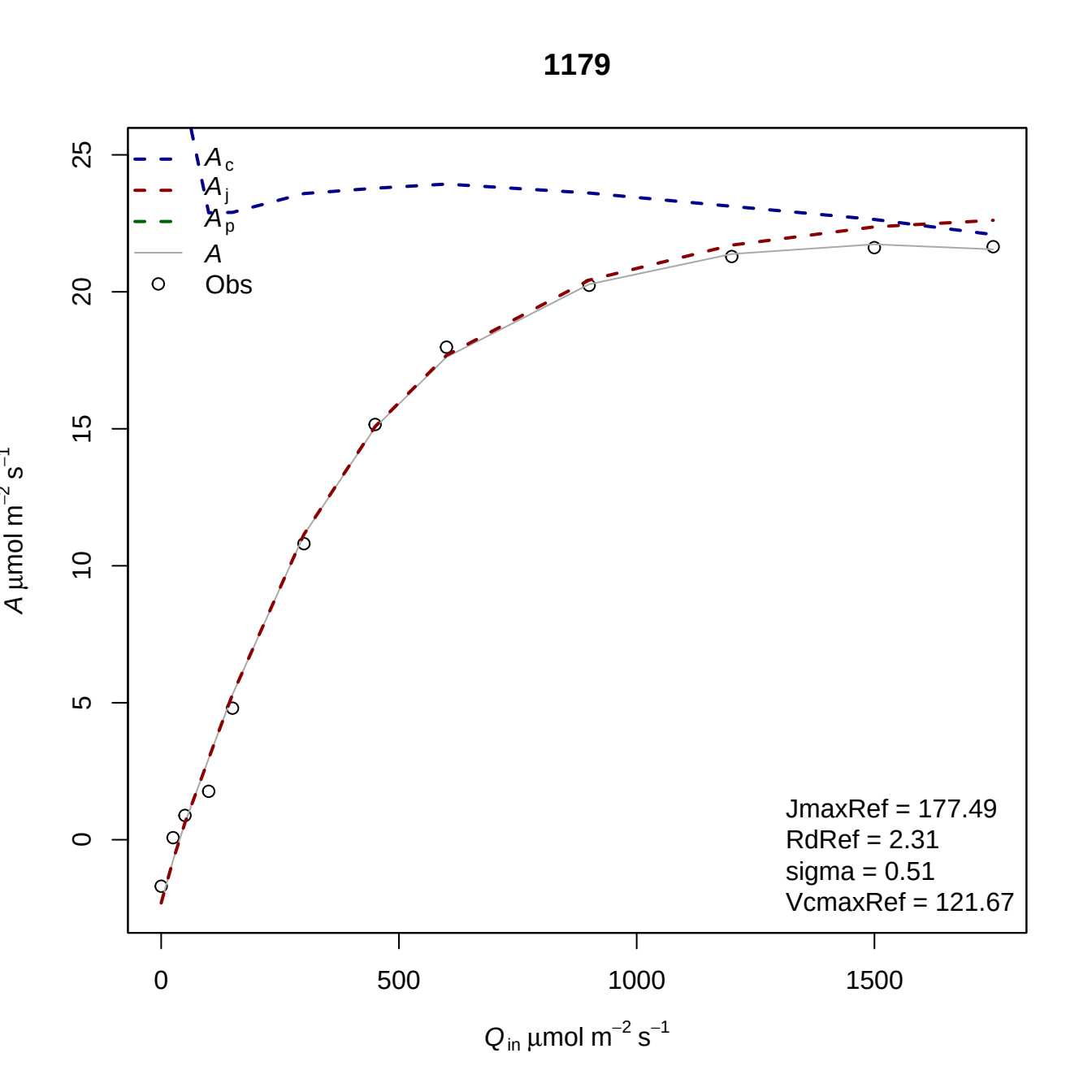

1174
A_c
A_j
A_p
A
Obs
JmaxRef = 140.96
RdRef = 3.27
sigma = 1.04
VcmaxRef = 261.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

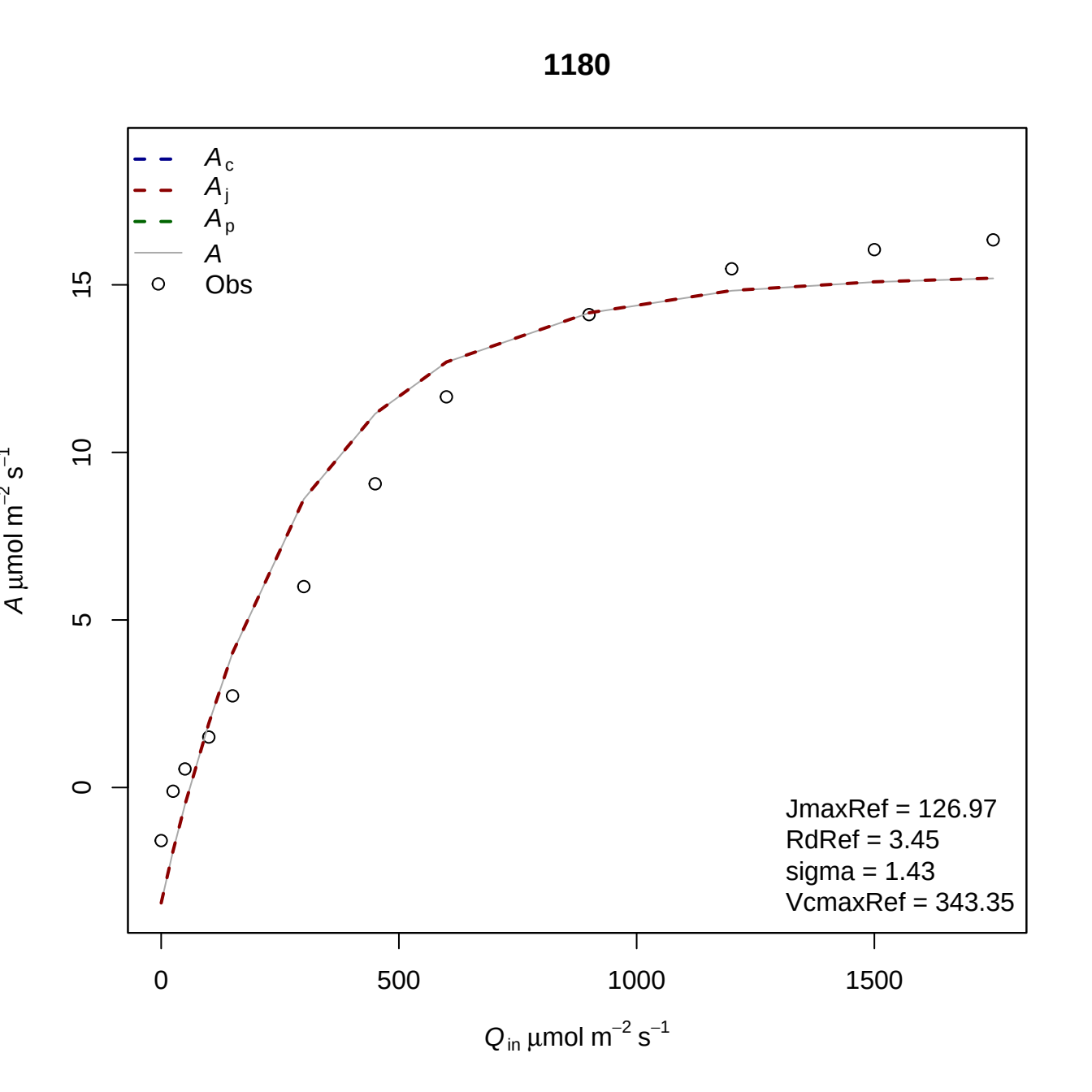

1175
A_c
A_j
A_p
A
Obs
JmaxRef = 124.77
RdRef = 4.33
sigma = 2.01
VcmaxRef = 405.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500

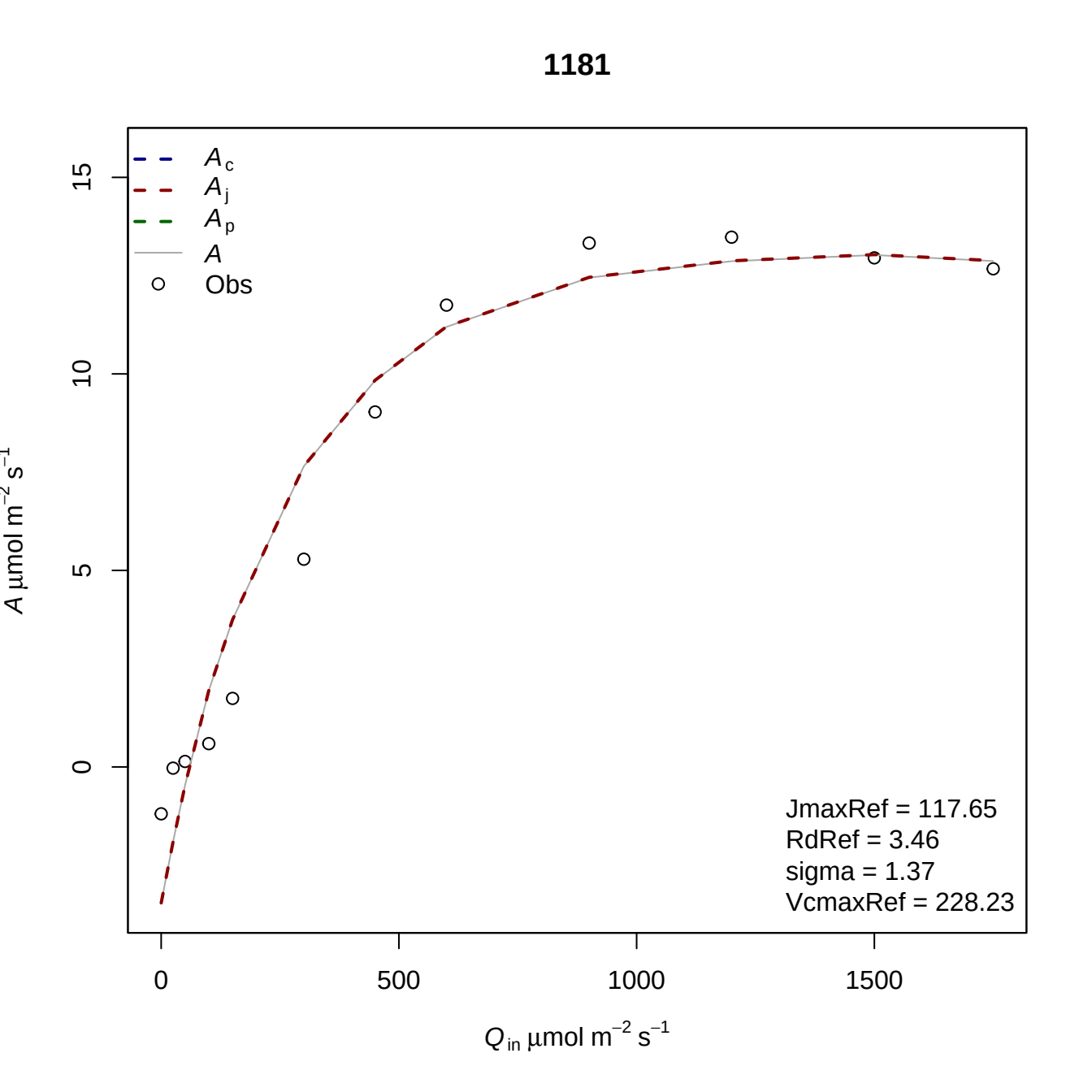

1176
A_c
A_j
A_p
A
Obs
JmaxRef = 132.06
RdRef = 1.87
sigma = 0.35
VcmaxRef = 205.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

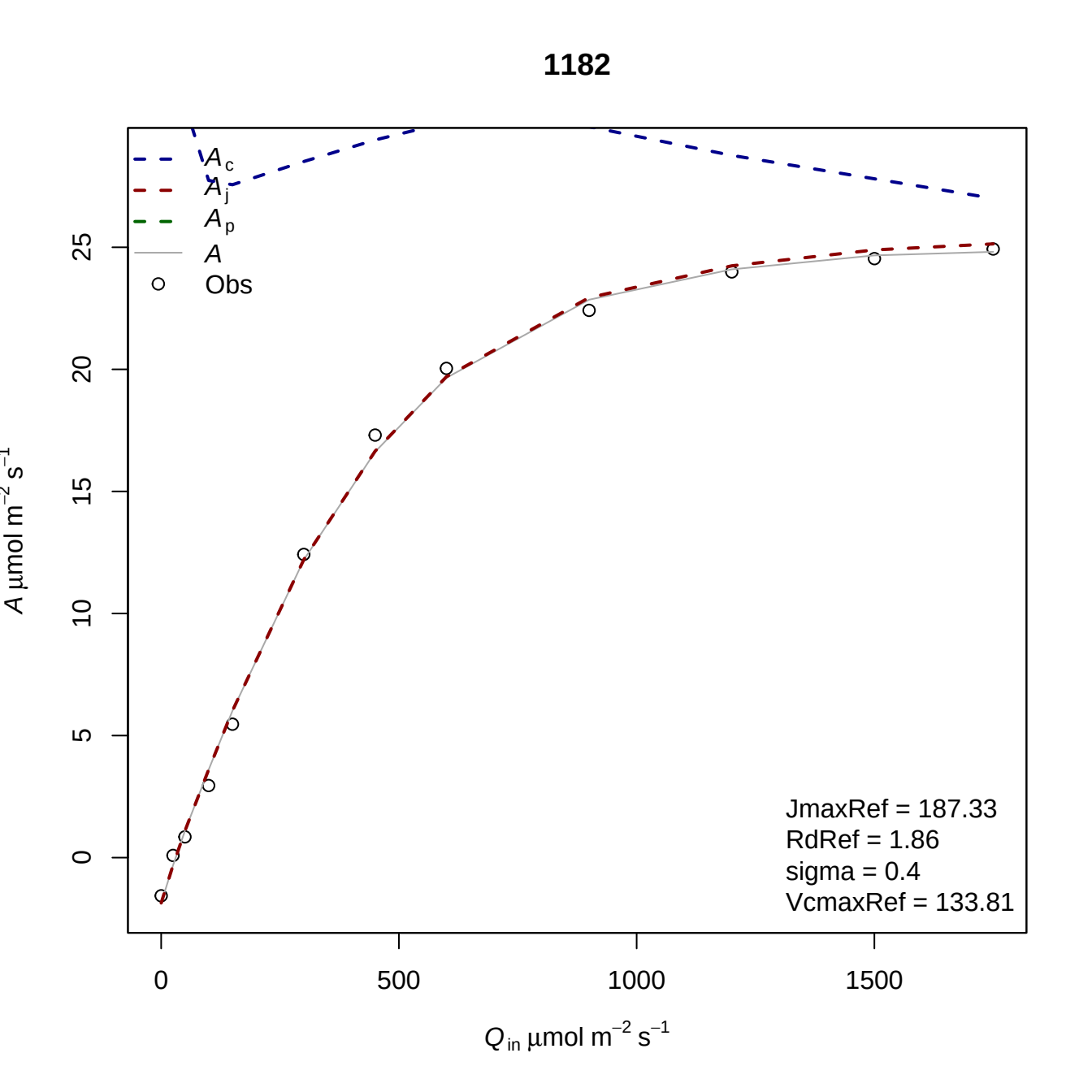

1177
A_c
A_j
A_p
A
Obs
JmaxRef = 165.65
RdRef = 2.13
sigma = 0.32
VcmaxRef = 314.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

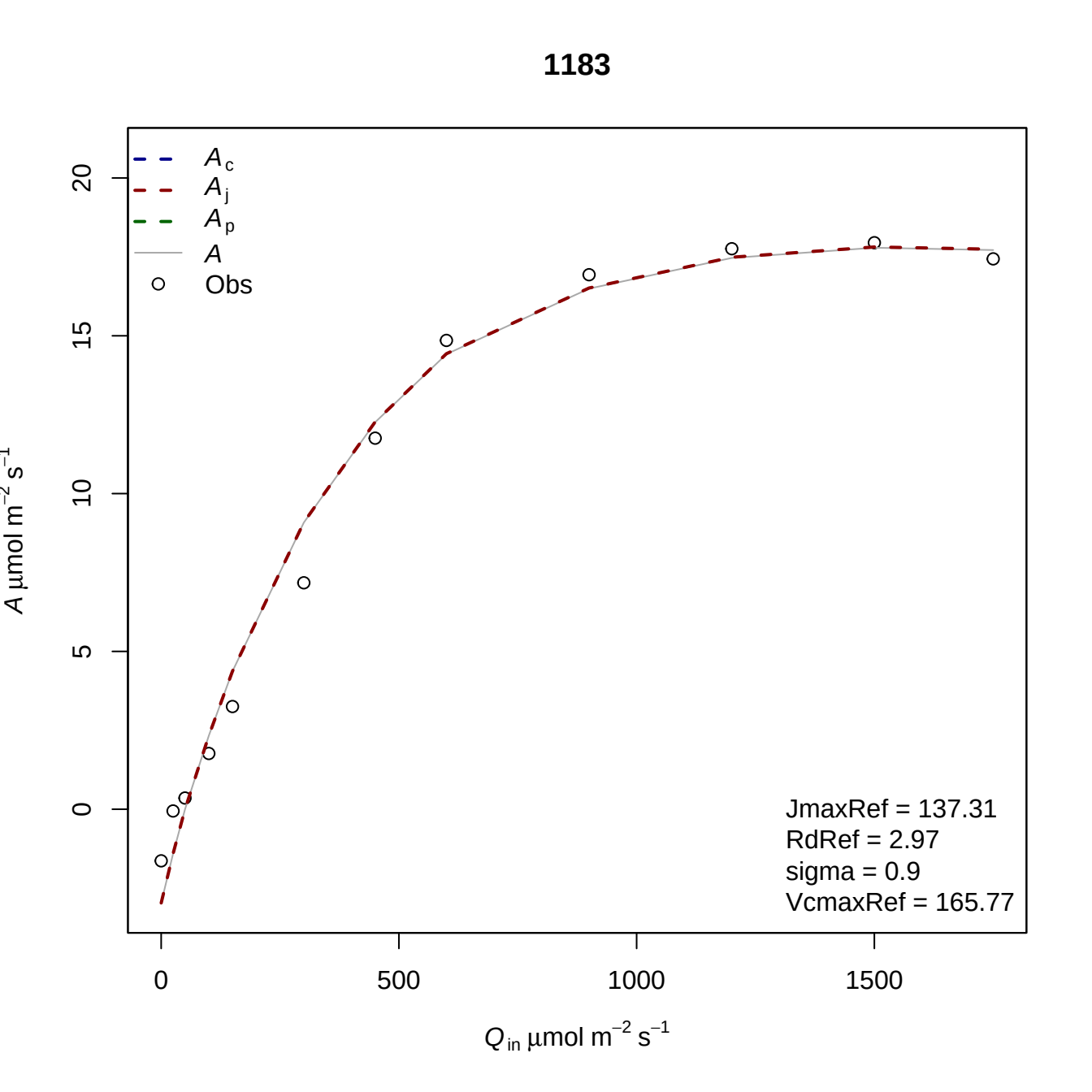

1178
A_c
A_j
A_p
A
Obs
JmaxRef = 133.7
RdRef = 2.37
sigma = 0.6
VcmaxRef = 333.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

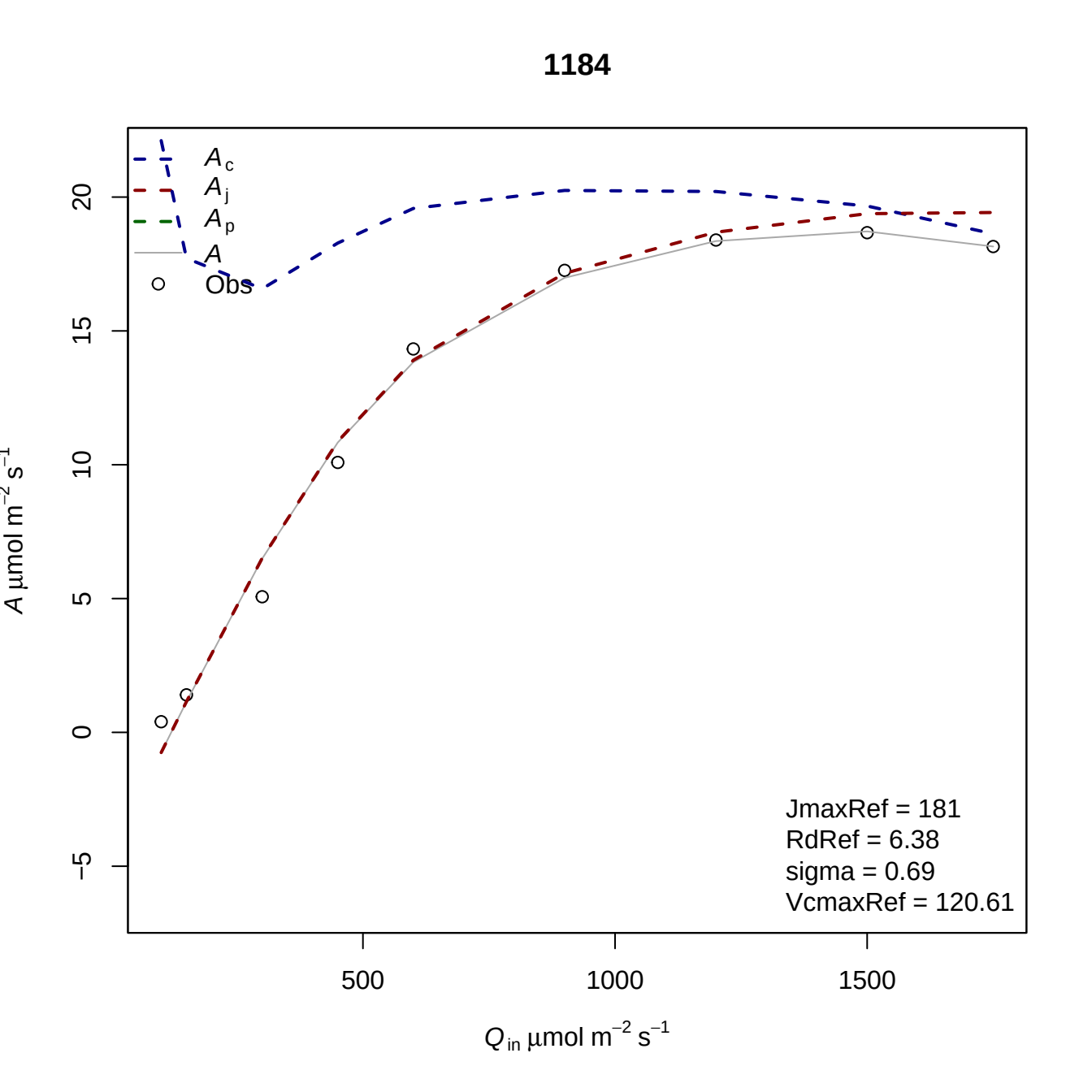

1179
A_c
A_j
A_p
A
Obs
JmaxRef = 177.49
RdRef = 2.31
sigma = 0.51
VcmaxRef = 121.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

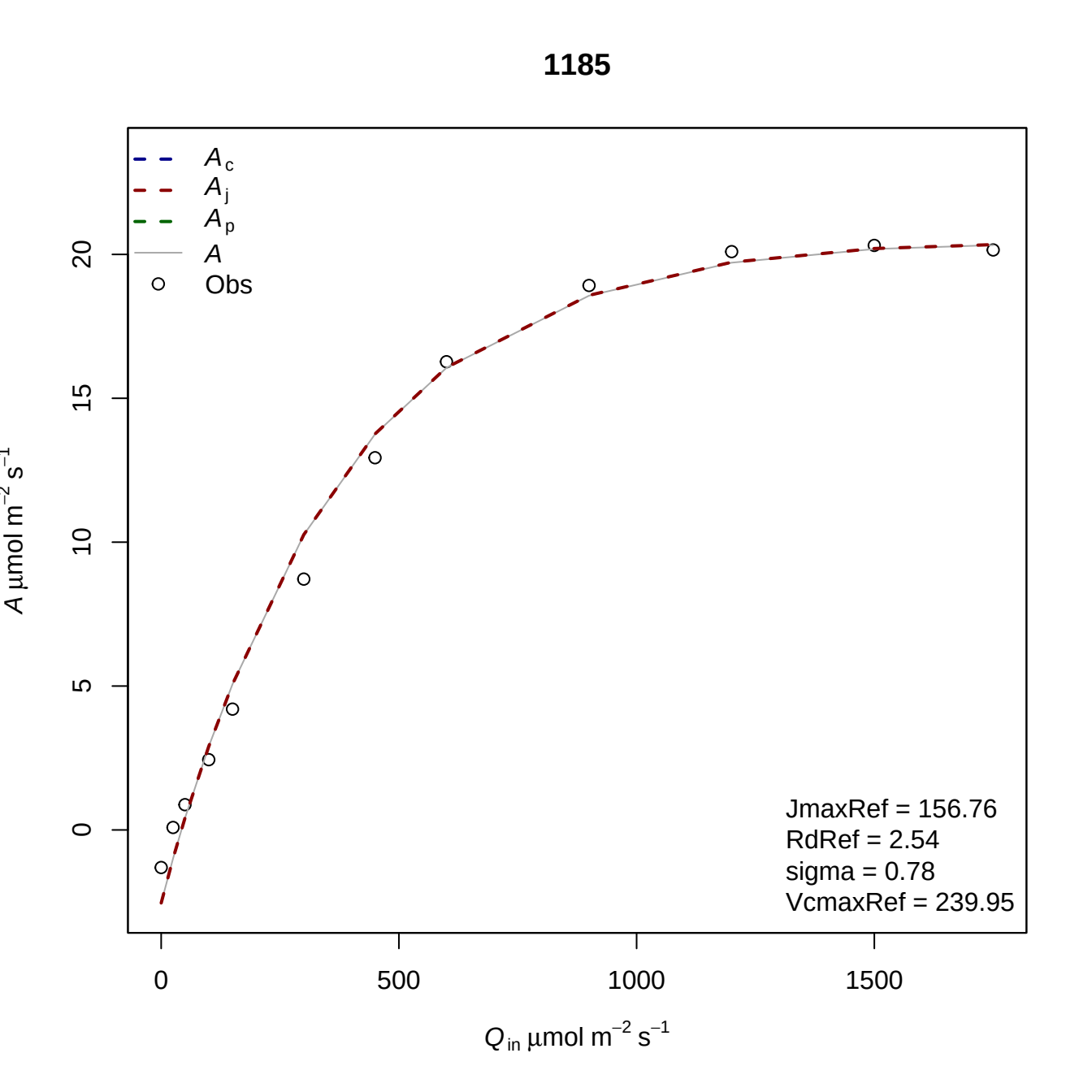

1180
A_c
A_j
A_p
A
Obs
JmaxRef = 126.97
RdRef = 3.45
sigma = 1.43
VcmaxRef = 343.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

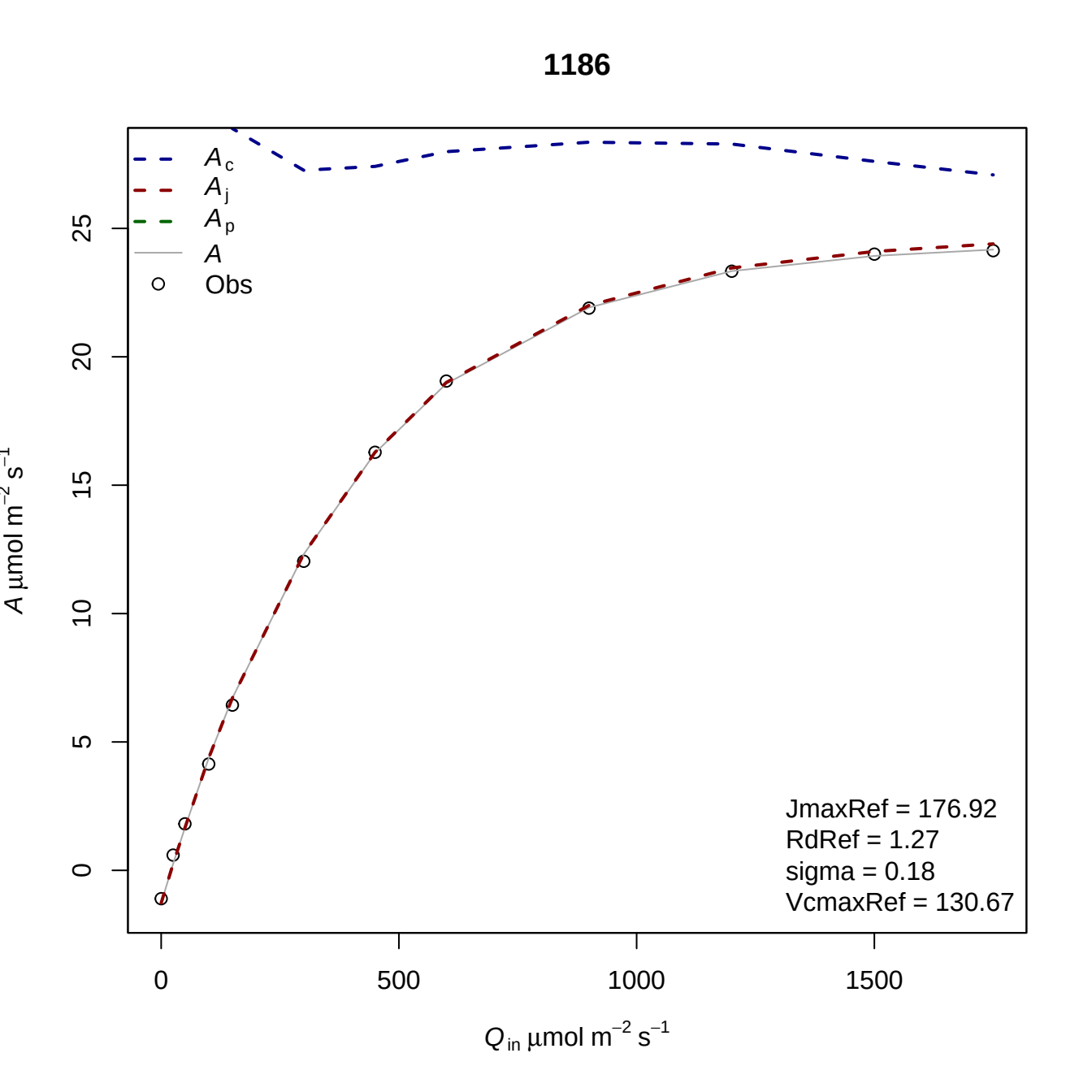

1181
A_c
A_j
A_p
A
Obs
JmaxRef = 117.65
RdRef = 3.46
sigma = 1.37
VcmaxRef = 228.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

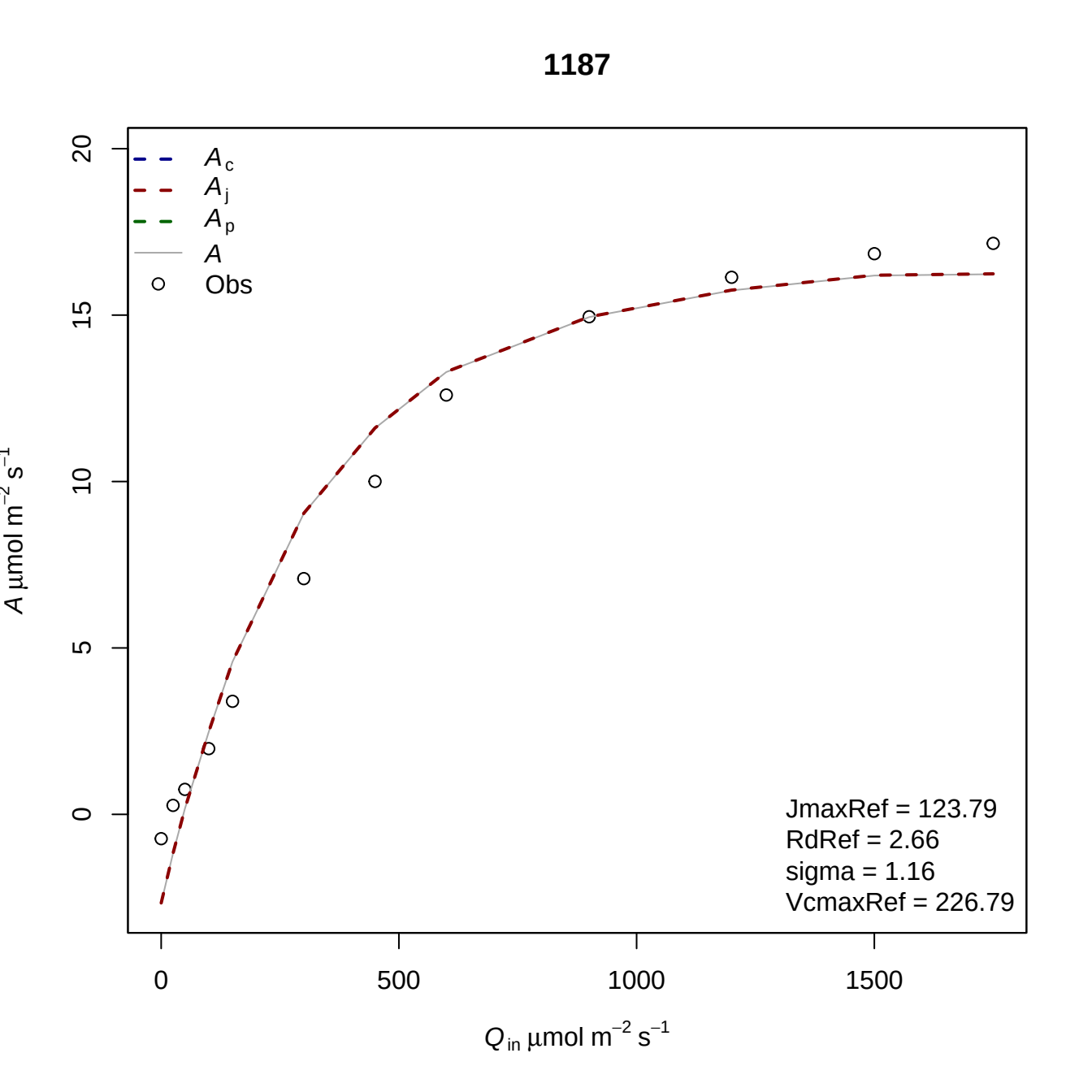

1182
A_c
A_j
A_p
A
Obs
JmaxRef = 187.33
RdRef = 1.86
sigma = 0.4
VcmaxRef = 133.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

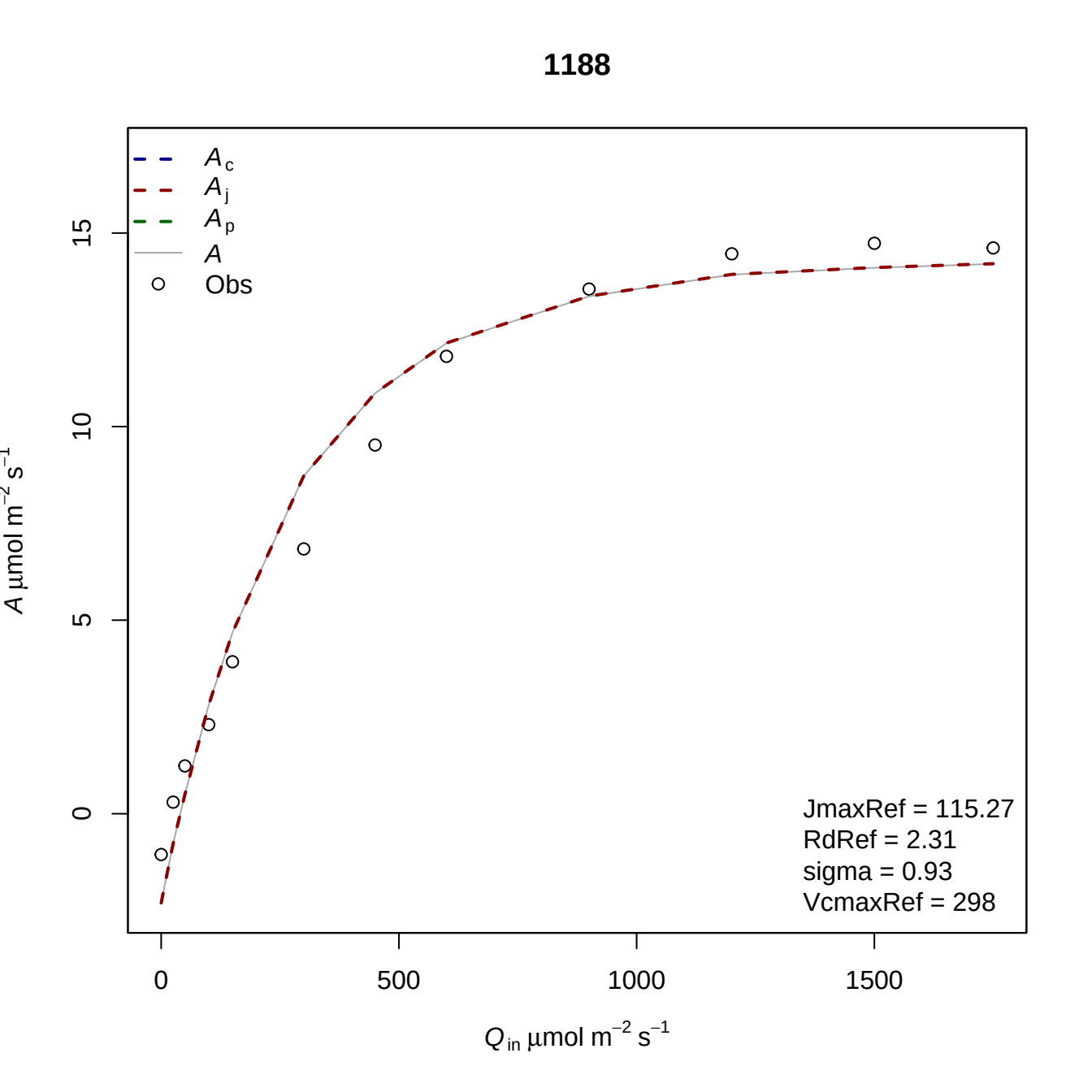

1183
A_c
A_j
A_p
A
Obs
JmaxRef = 137.31
RdRef = 2.97
sigma = 0.9
VcmaxRef = 165.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

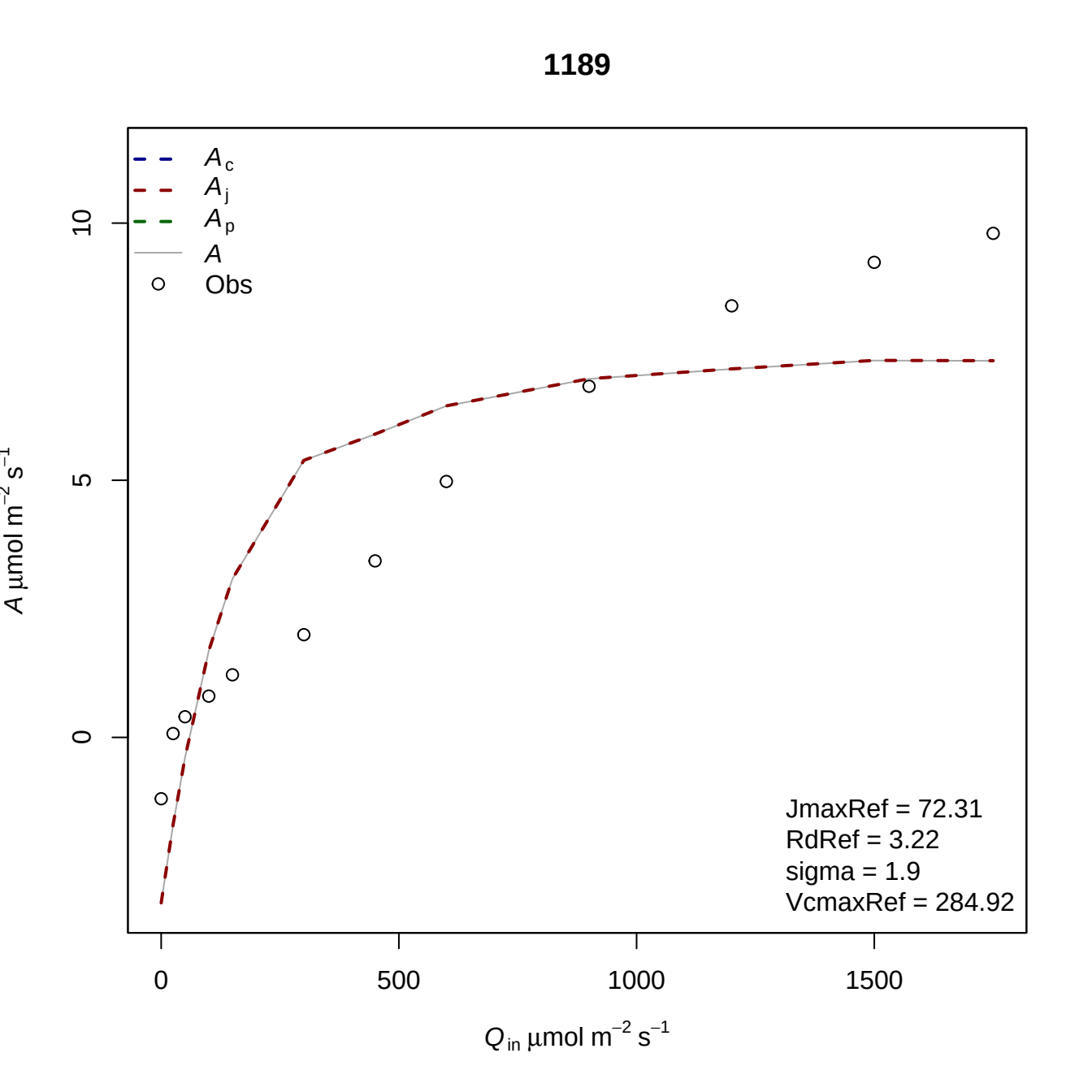

1184
A_c
A_j
A_p
A
Obs
JmaxRef = 181
RdRef = 6.38
sigma = 0.69
VcmaxRef = 120.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
500
1000
1500

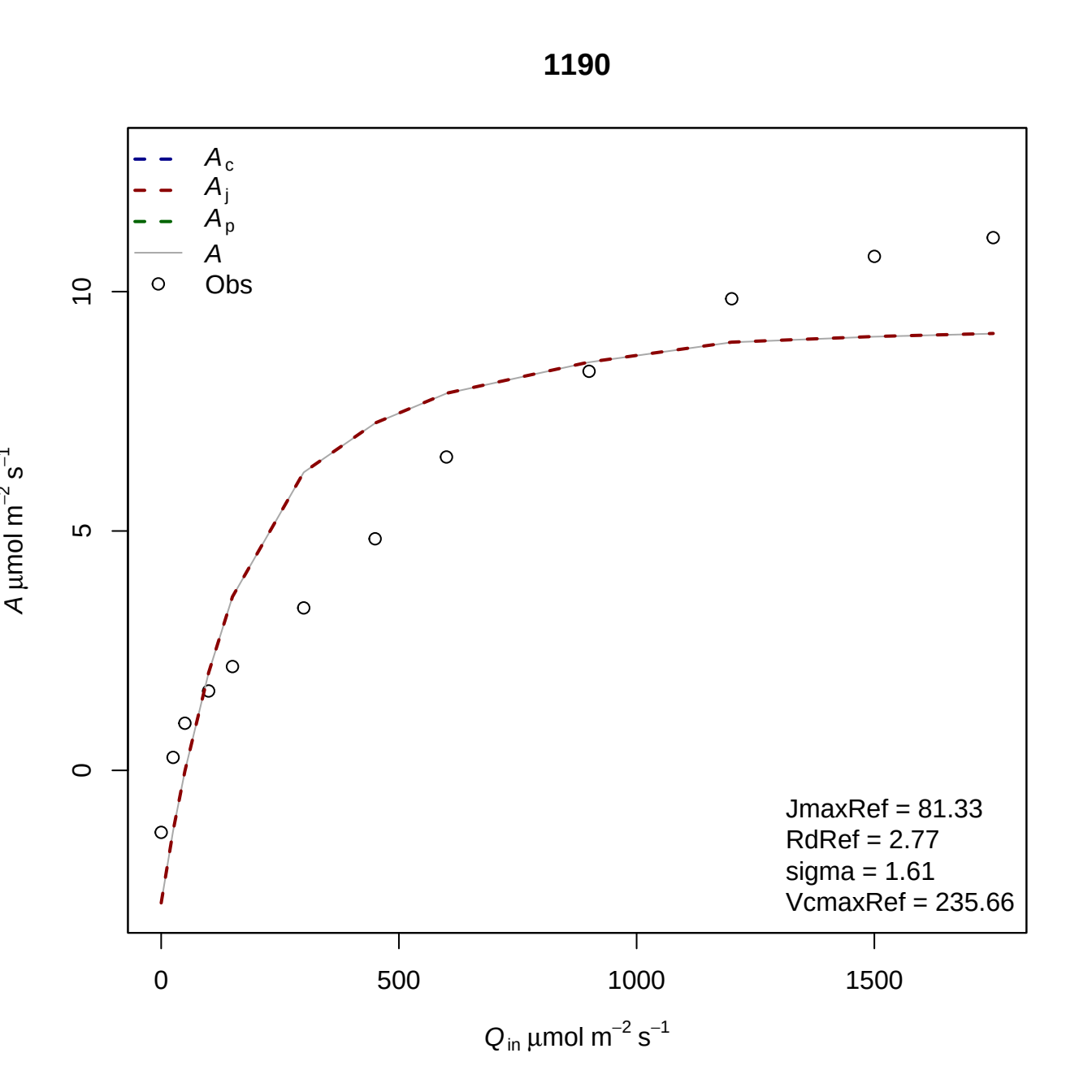

1185
A_c
A_j
A_p
A
Obs
JmaxRef = 156.76
RdRef = 2.54
sigma = 0.78
VcmaxRef = 239.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

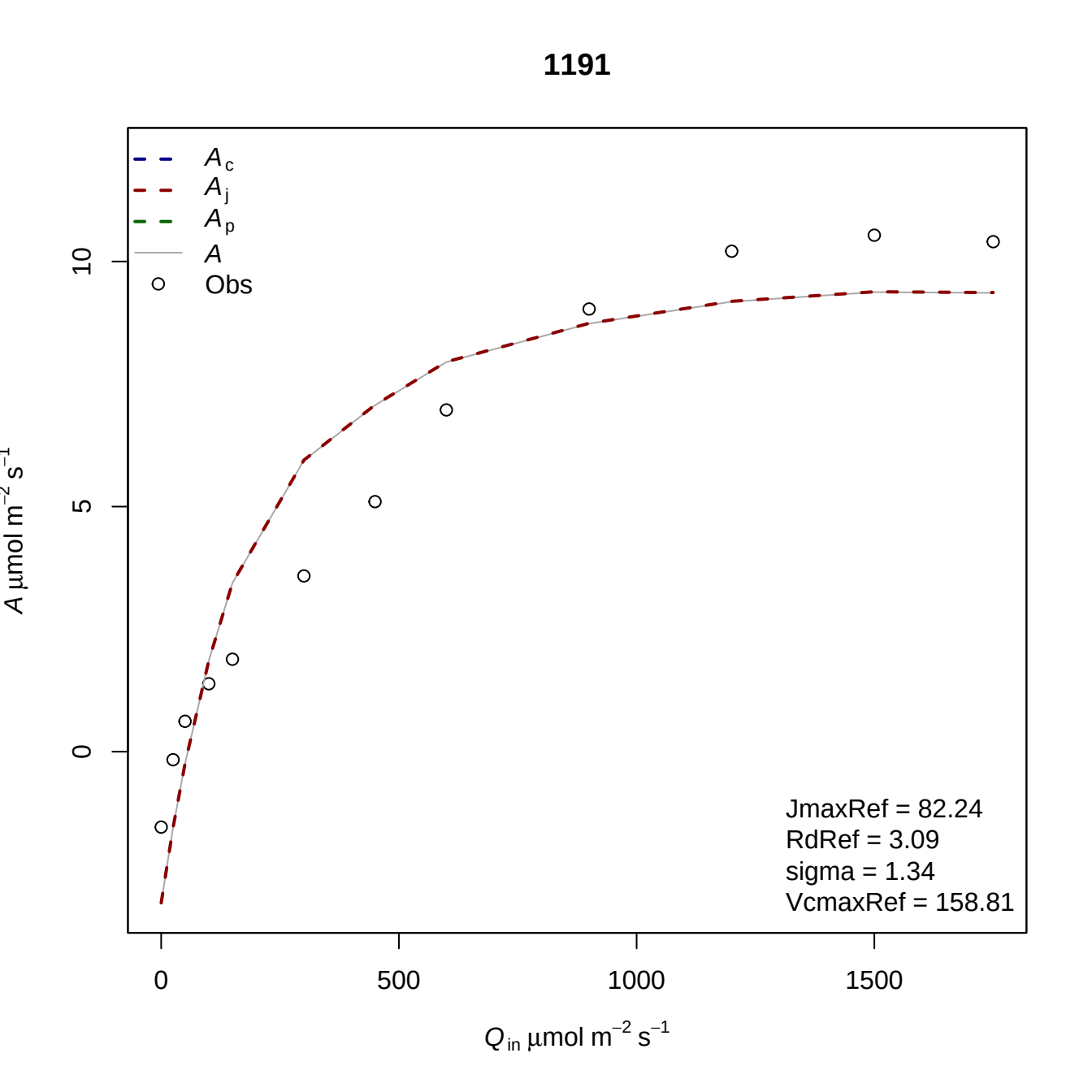

1186
A_c
A_j
A_p
A
Obs
JmaxRef = 176.92
RdRef = 1.27
sigma = 0.18
VcmaxRef = 130.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

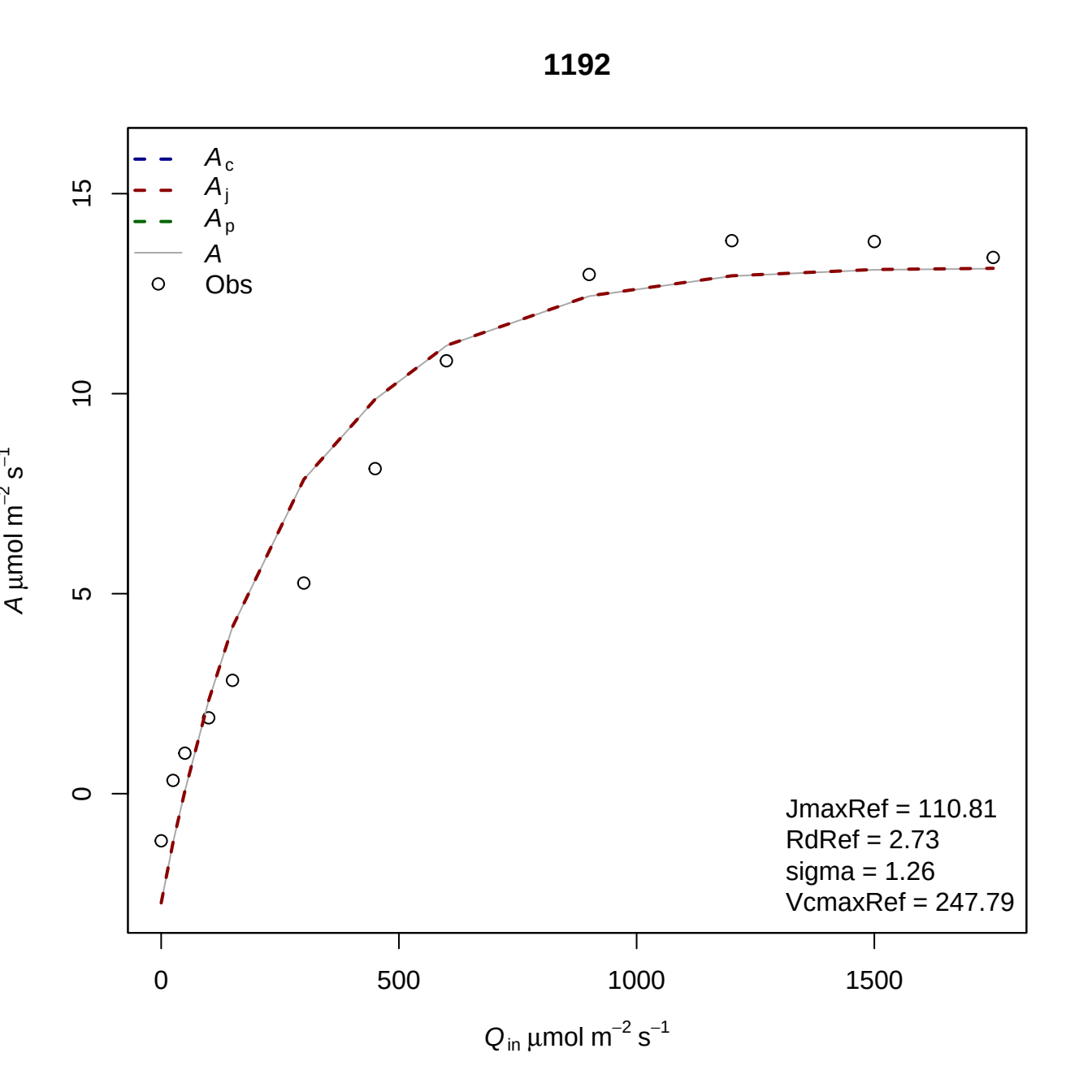

1187
A_c
A_j
A_p
A
Obs
JmaxRef = 123.79
RdRef = 2.66
sigma = 1.16
VcmaxRef = 226.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

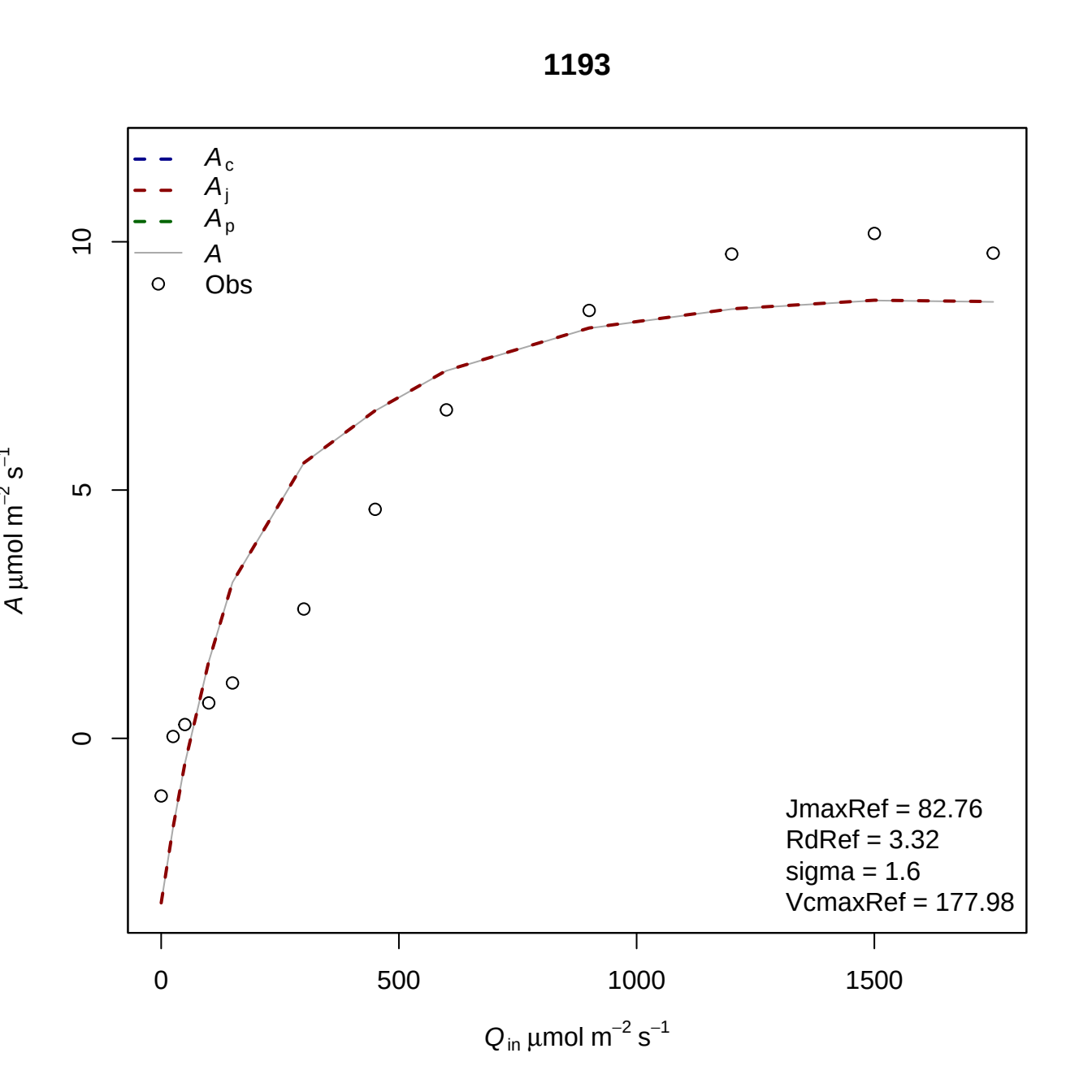

1188
A_c
A_j
A_p
A
Obs
JmaxRef = 115.27
RdRef = 2.31
sigma = 0.93
VcmaxRef = 298
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

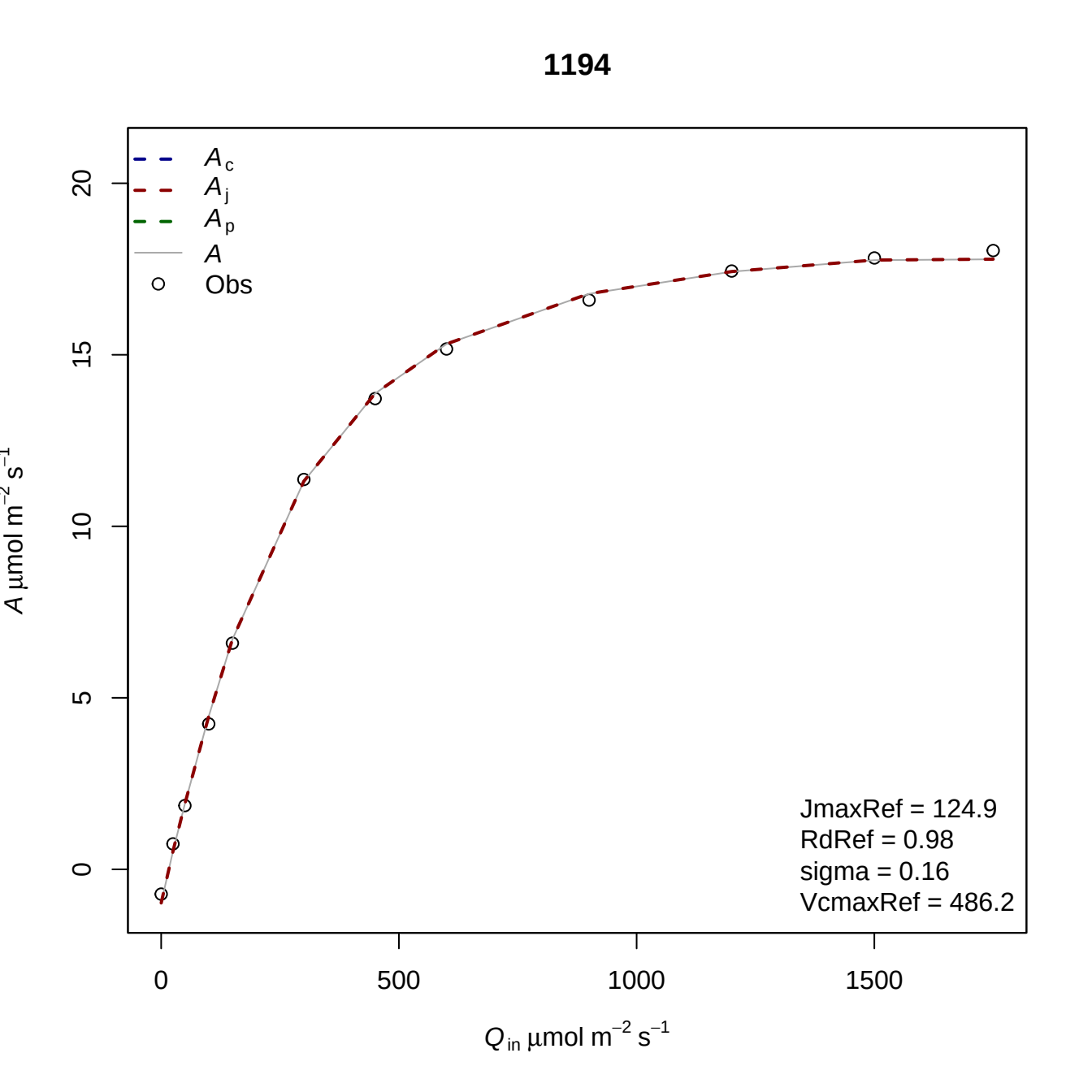

1189
A_c
A_j
A_p
A
Obs
JmaxRef = 72.31
RdRef = 3.22
sigma = 1.9
VcmaxRef = 284.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

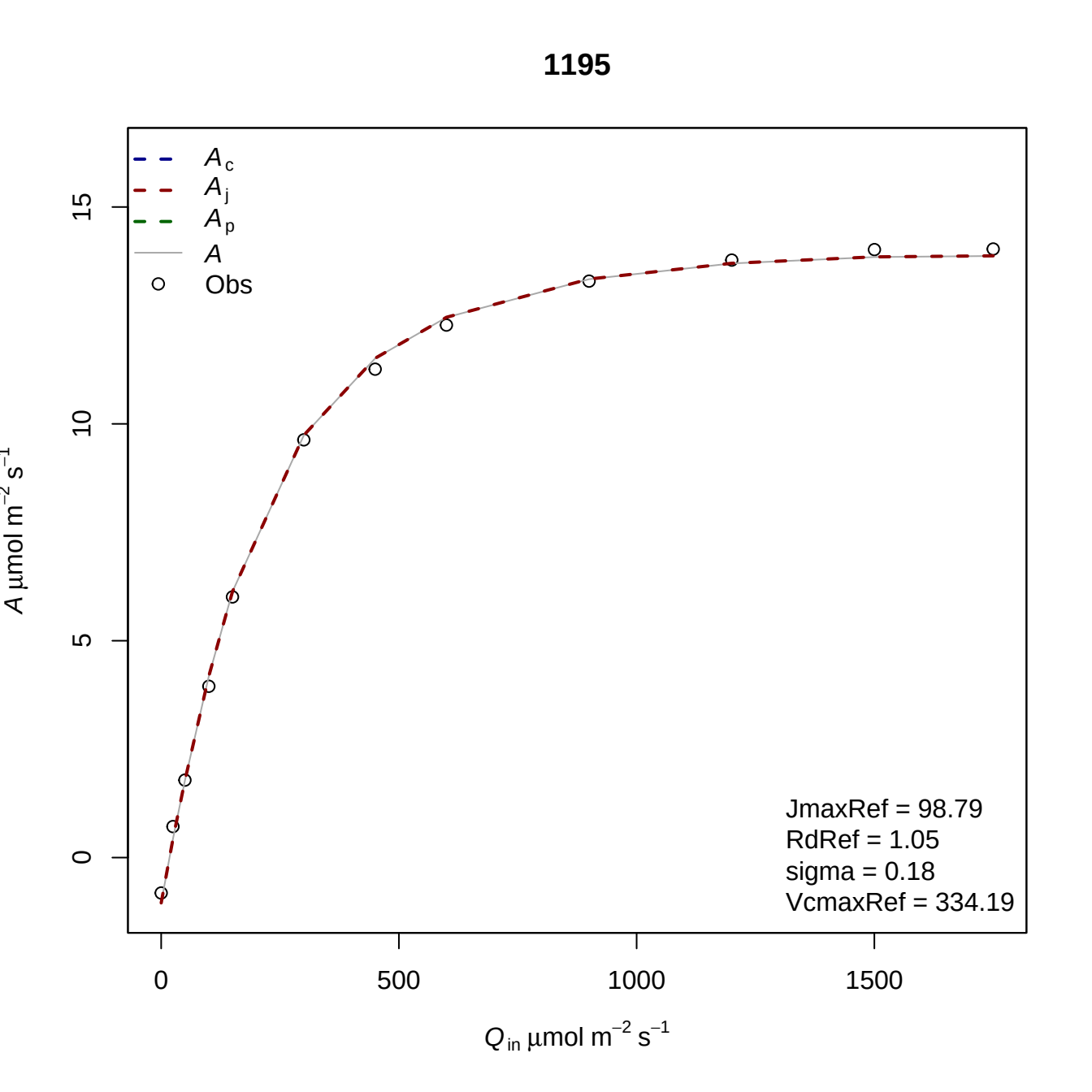

1190
A_c
A_j
A_p
A
Obs
JmaxRef = 81.33
RdRef = 2.77
sigma = 1.61
VcmaxRef = 235.66
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

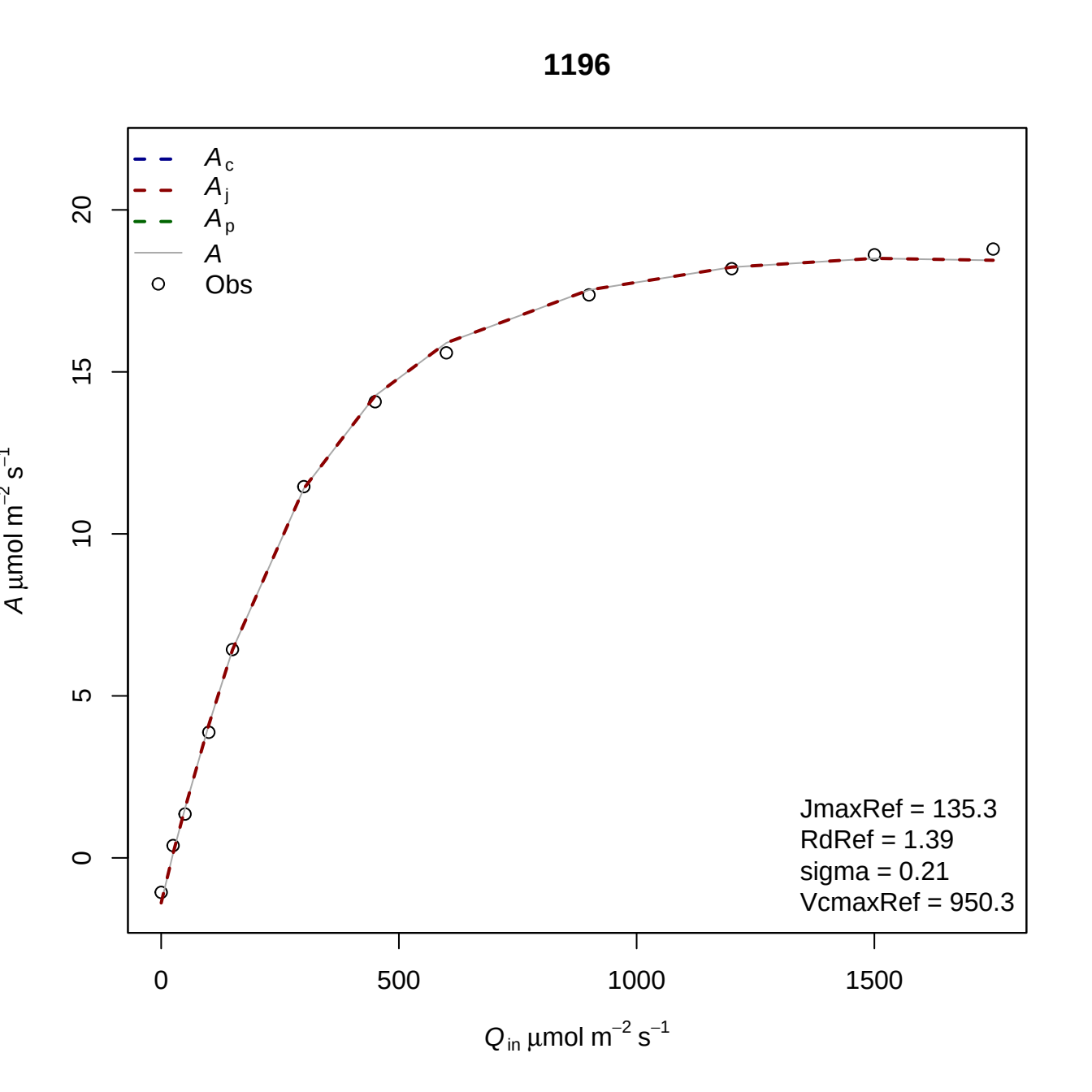

1191
A_c
A_j
A_p
A
Obs
JmaxRef = 82.24
RdRef = 3.09
sigma = 1.34
VcmaxRef = 158.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

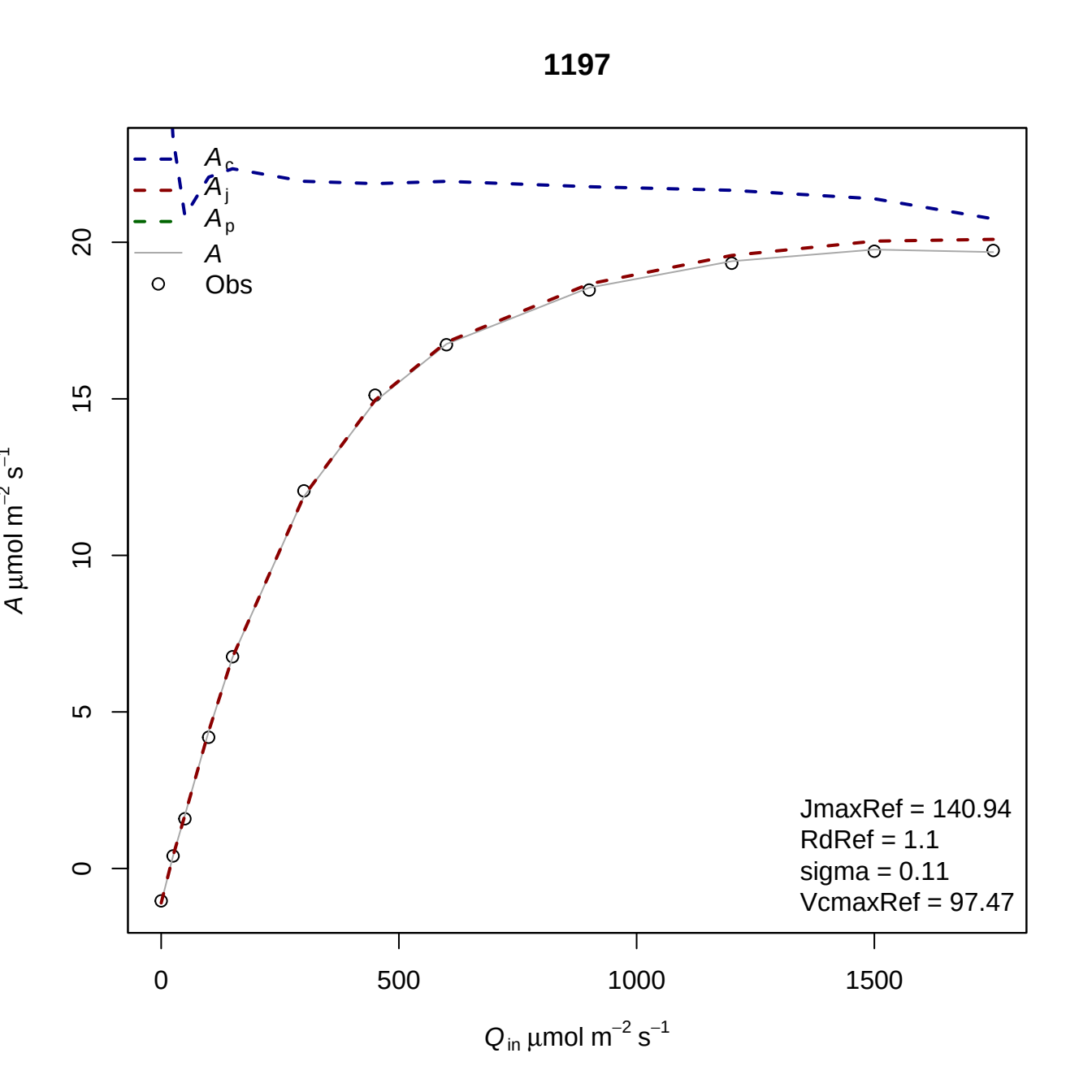

1192
A_c
A_j
A_p
A
Obs
JmaxRef = 110.81
RdRef = 2.73
sigma = 1.26
VcmaxRef = 247.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

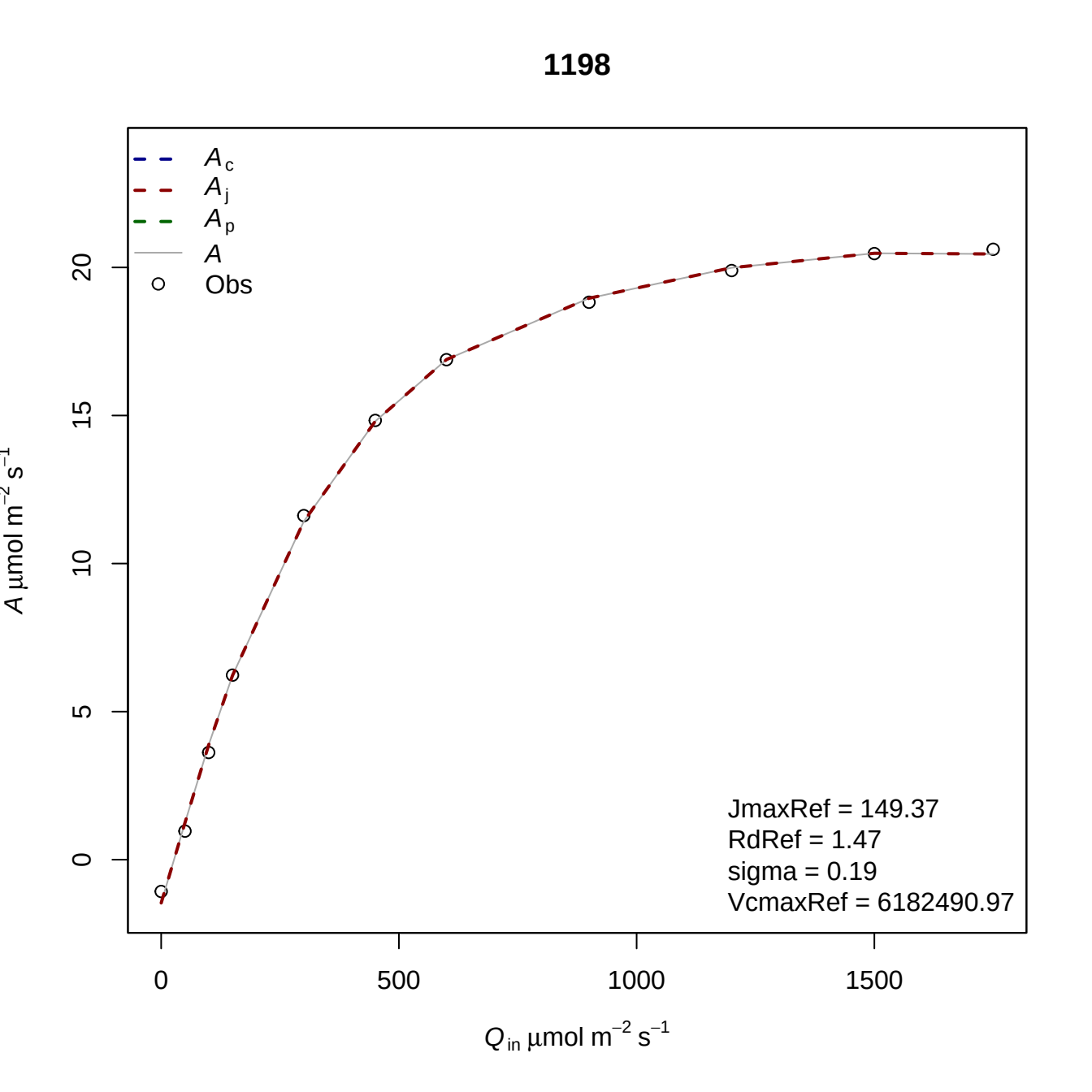

1193
A_c
A_j
A_p
A
Obs
JmaxRef = 82.76
RdRef = 3.32
sigma = 1.6
VcmaxRef = 177.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

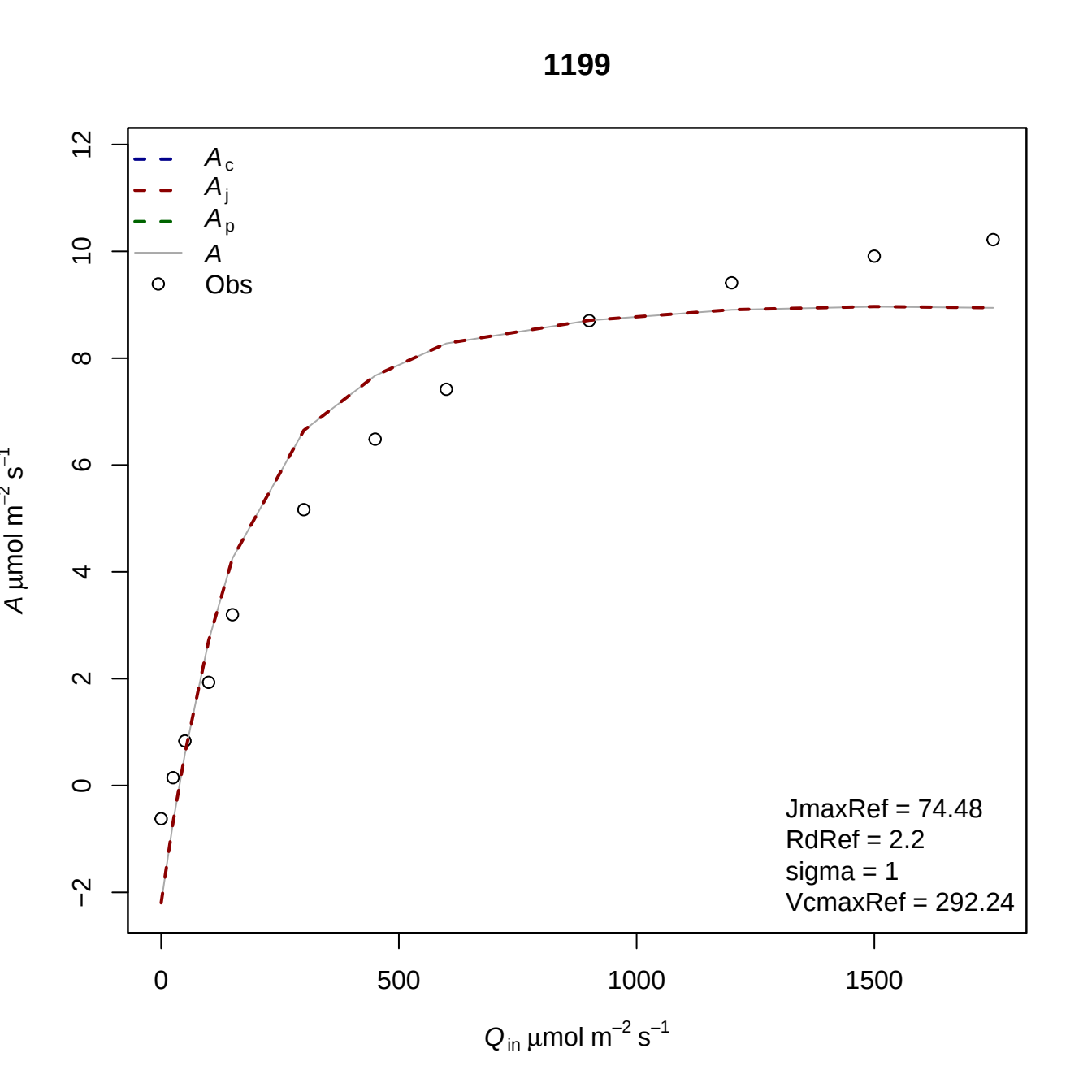

1194
A_c
A_j
A_p
A
Obs
JmaxRef = 124.9
RdRef = 0.98
sigma = 0.16
VcmaxRef = 486.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

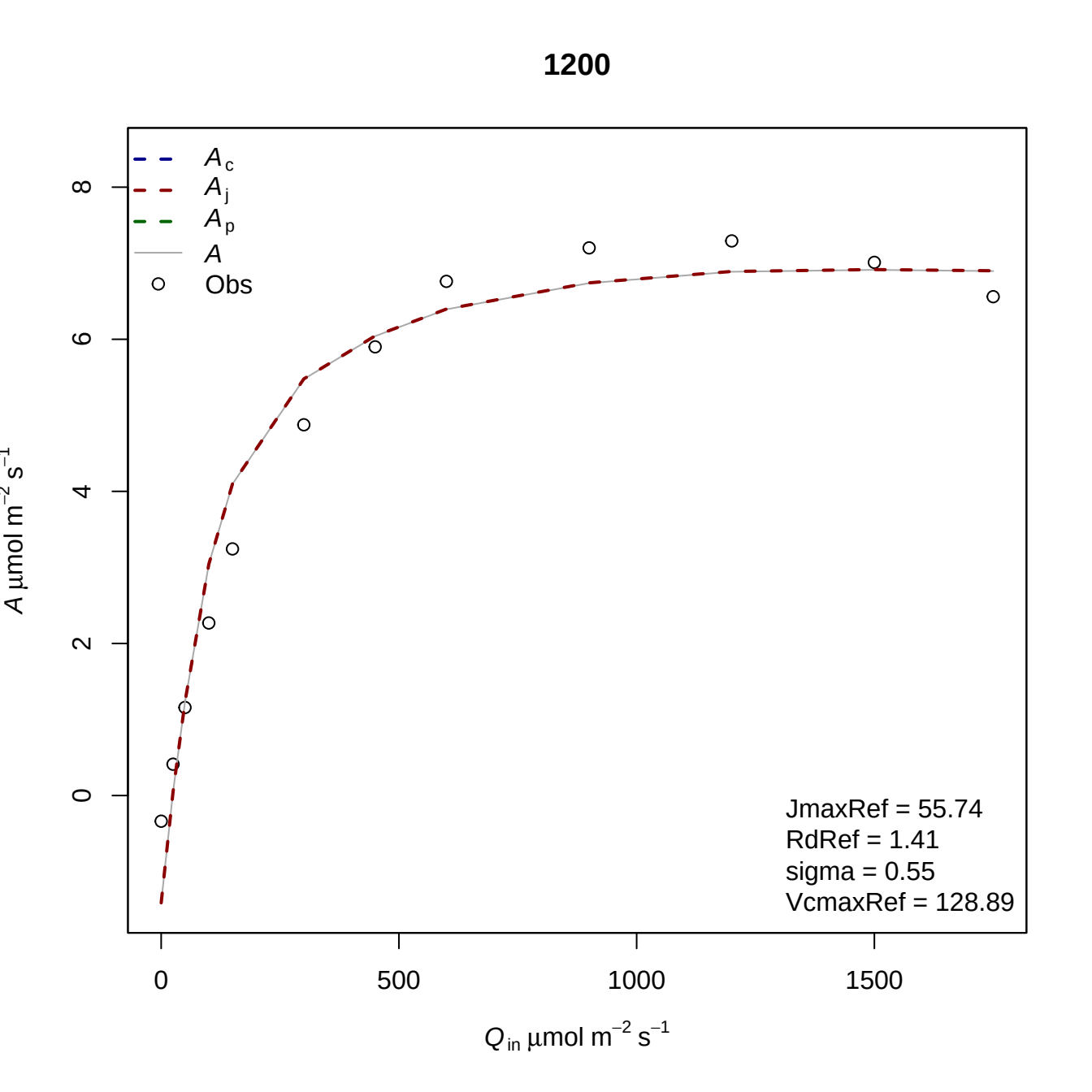

1195
A_c
A_j
A_p
A
Obs
JmaxRef = 98.79
RdRef = 1.05
sigma = 0.18
VcmaxRef = 334.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

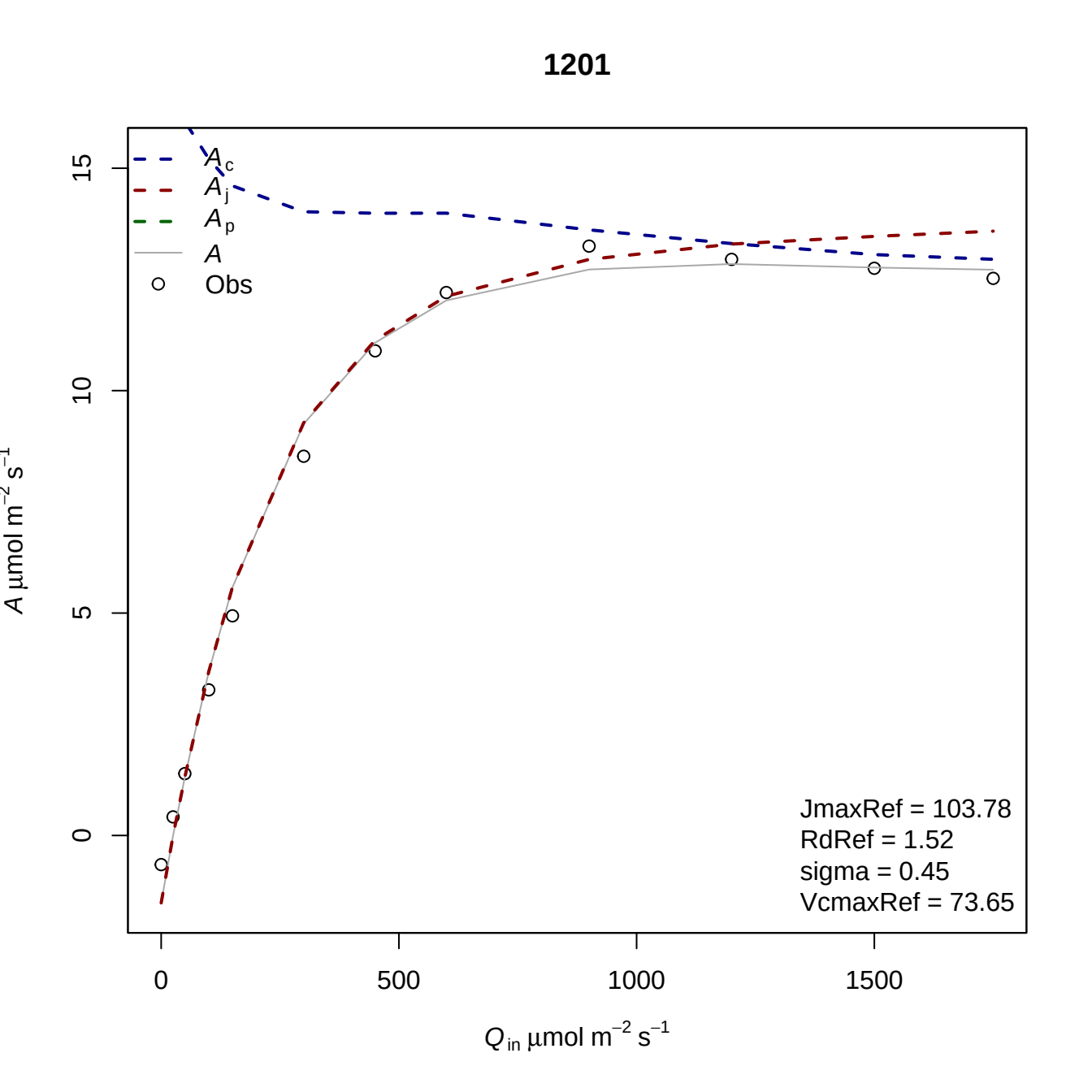

1196
A_c
A_j
A_p
A
Obs
JmaxRef = 135.3
RdRef = 1.39
sigma = 0.21
VcmaxRef = 950.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

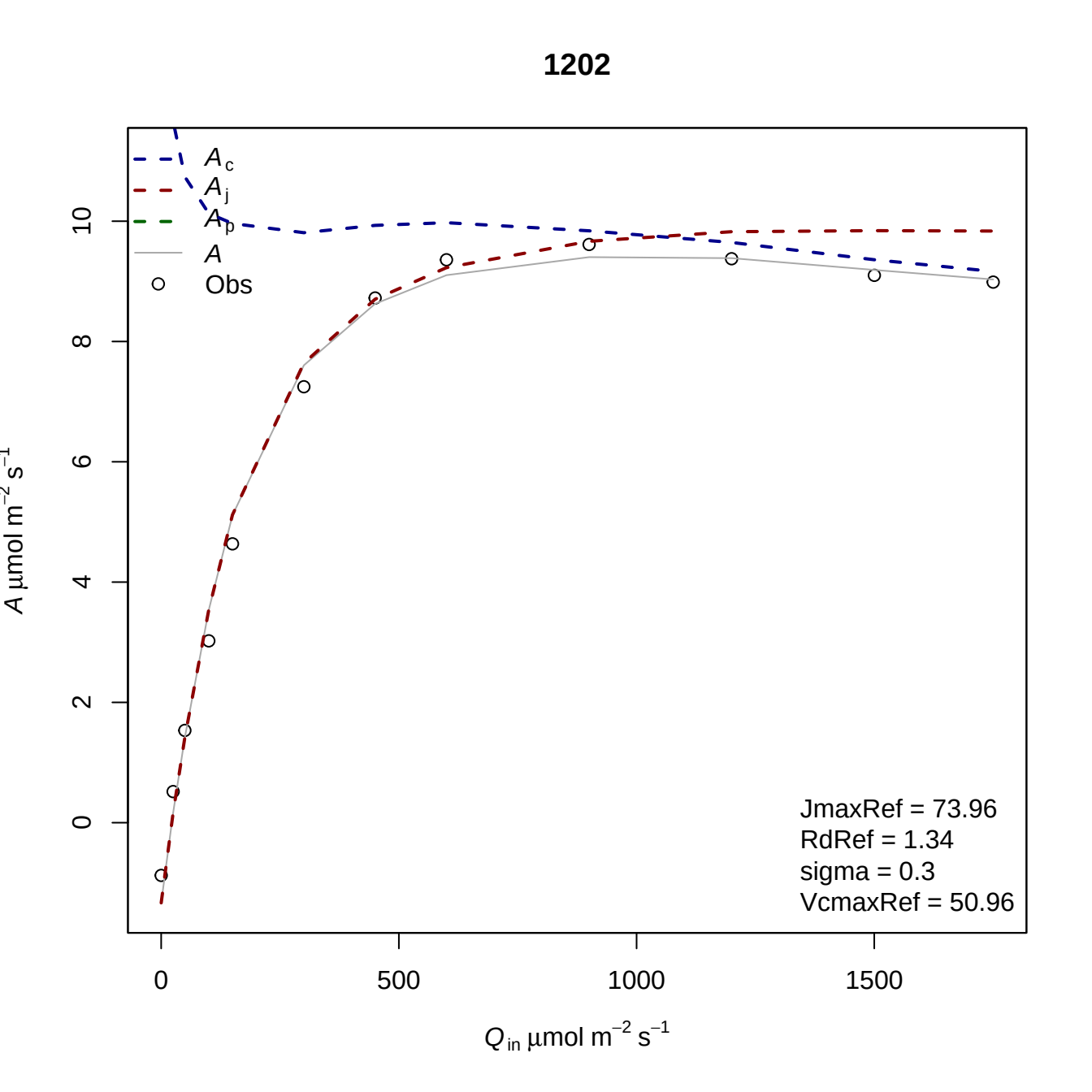

1197
A_c
A_j
A_p
A
Obs
JmaxRef = 140.94
RdRef = 1.1
sigma = 0.11
VcmaxRef = 97.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

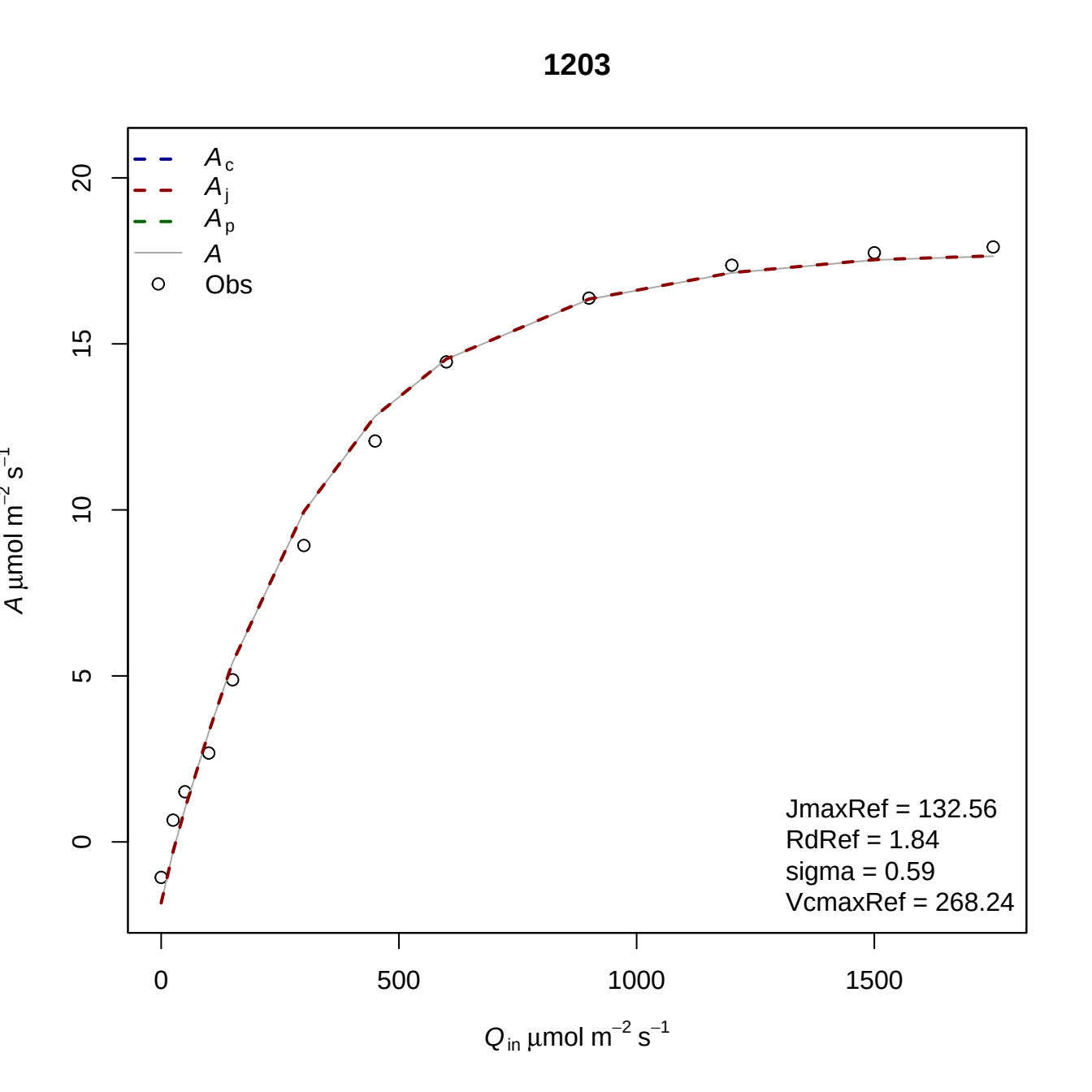

1198
A_c
A_j
A_p
A
Obs
JmaxRef = 149.37
RdRef = 1.47
sigma = 0.19
VcmaxRef = 6182490.97
A µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

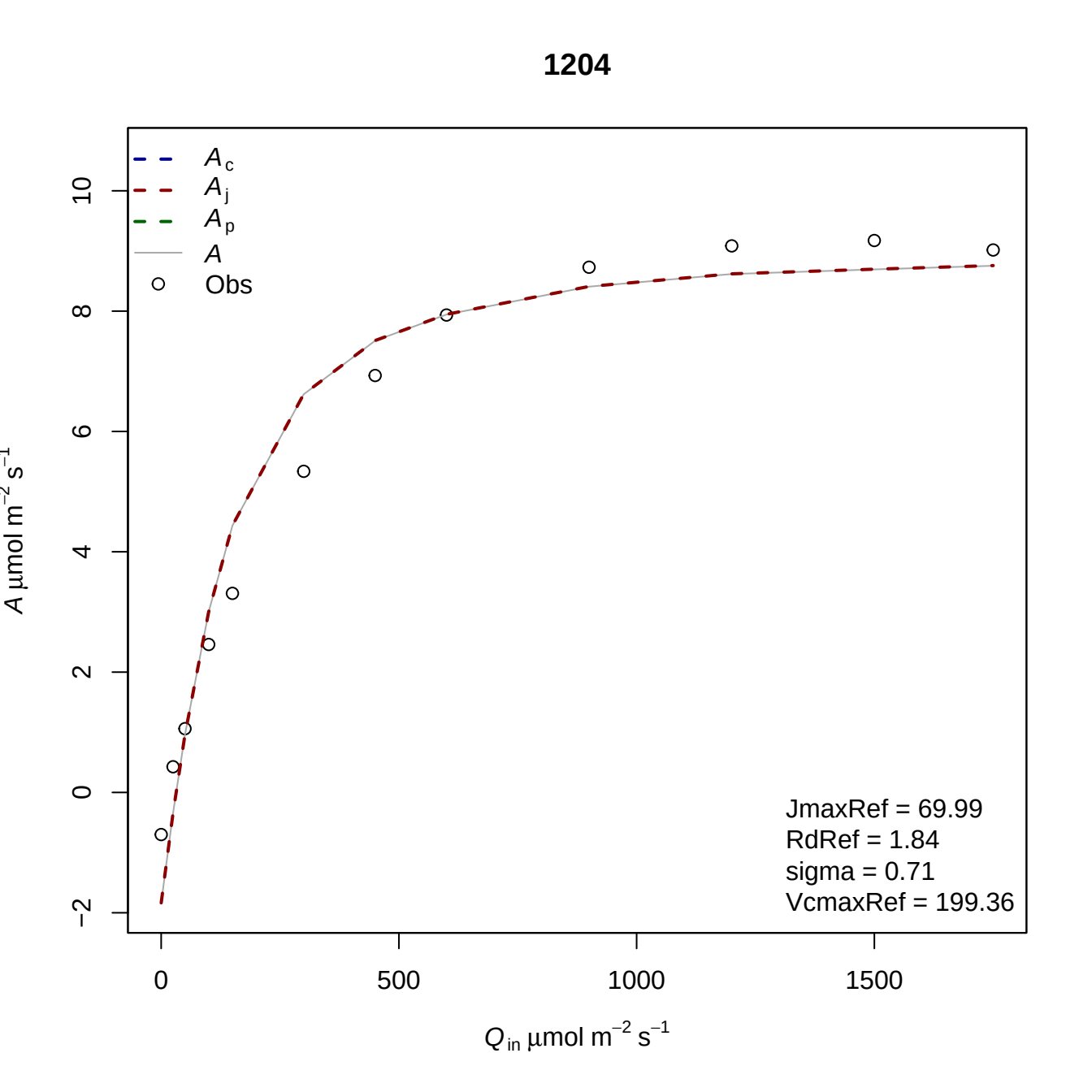

1199
A_c
A_j
A_p
A
Obs
JmaxRef = 74.48
RdRef = 2.2
sigma = 1
VcmaxRef = 292.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

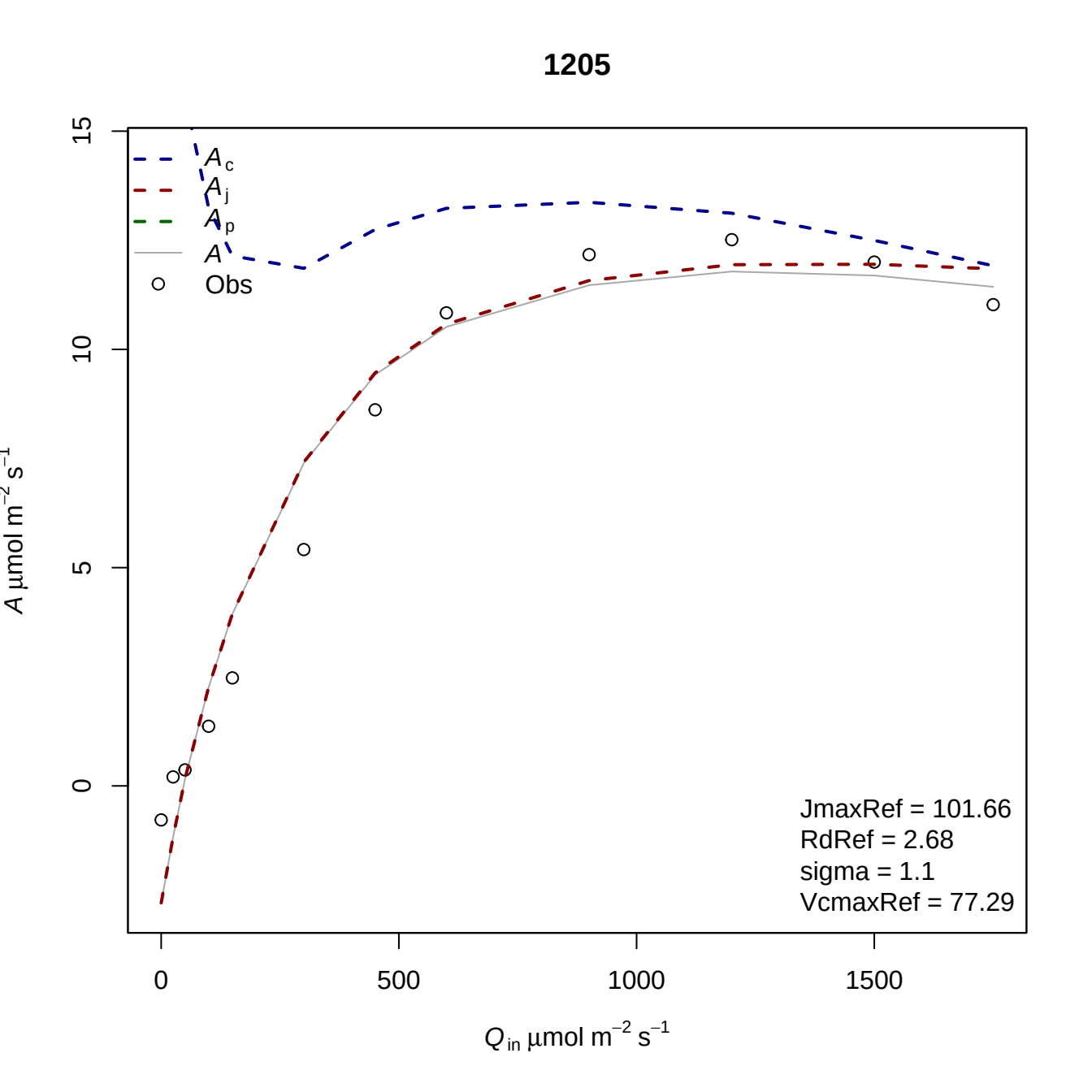

1200
A_c
A_j
A_p
A
Obs
JmaxRef = 55.74
RdRef = 1.41
sigma = 0.55
VcmaxRef = 128.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

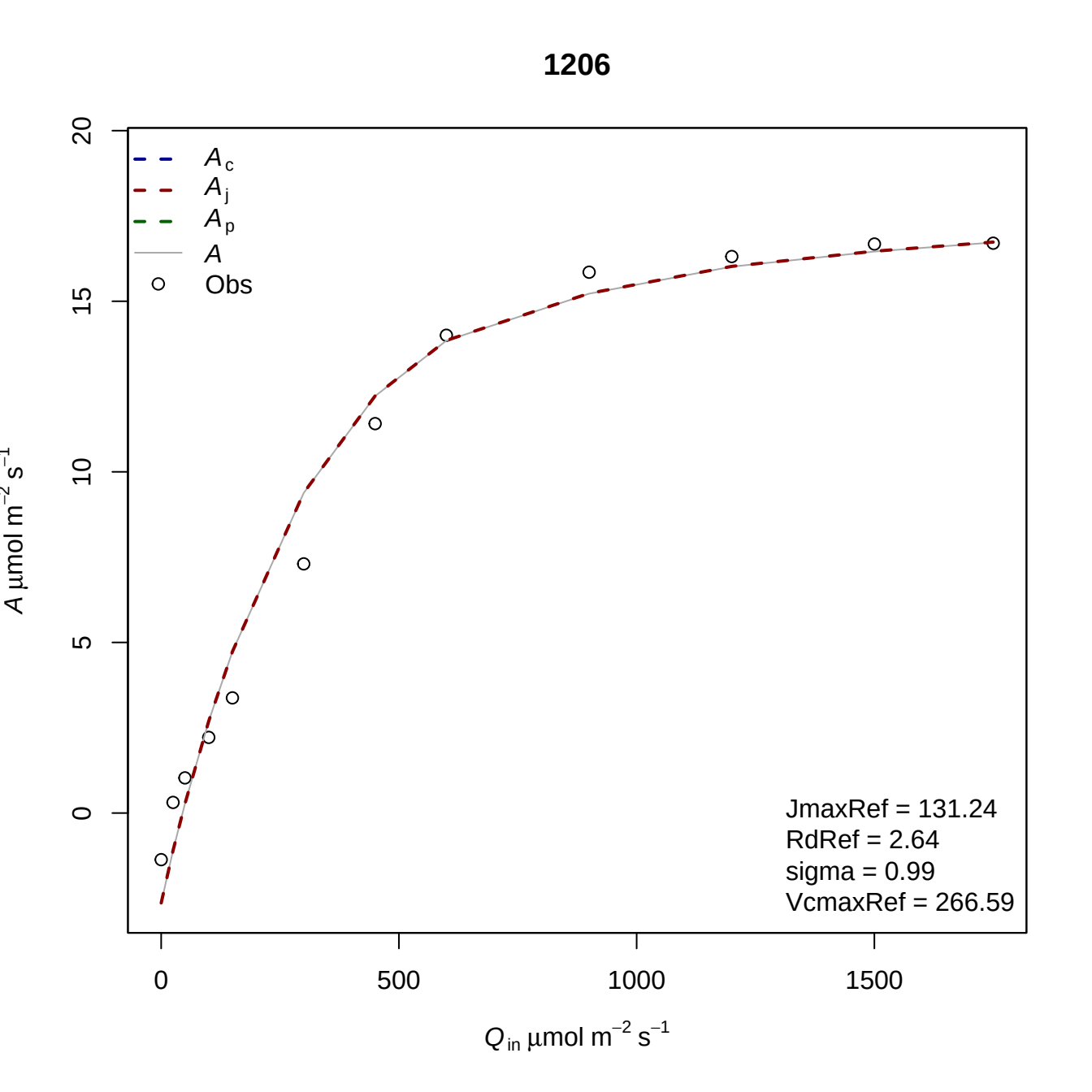

1201
A_c
A_j
A_p
A
Obs
JmaxRef = 103.78
RdRef = 1.52
sigma = 0.45
VcmaxRef = 73.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

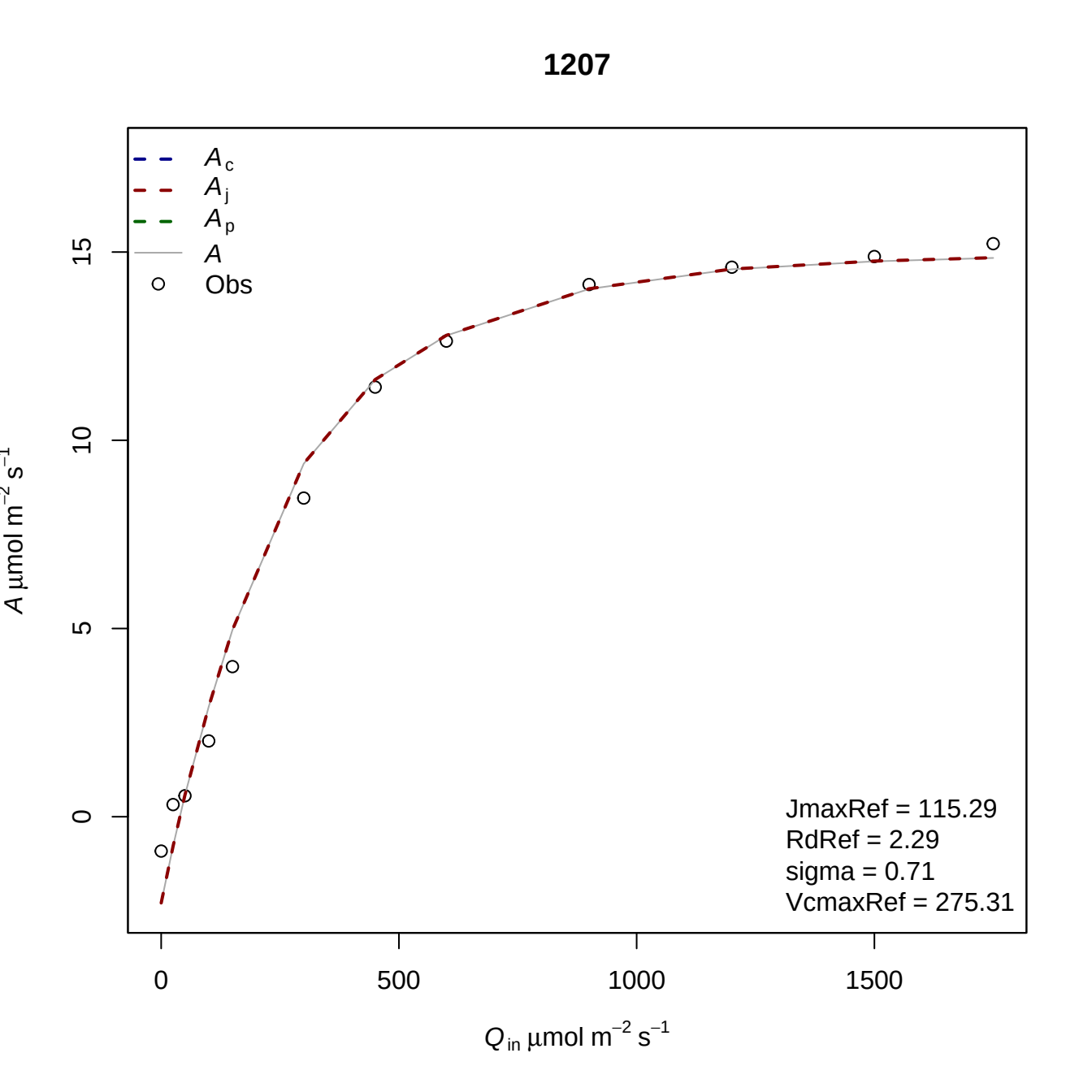

1202
A_c
A_j
A_p
A
Obs
JmaxRef = 73.96
RdRef = 1.34
sigma = 0.3
VcmaxRef = 50.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

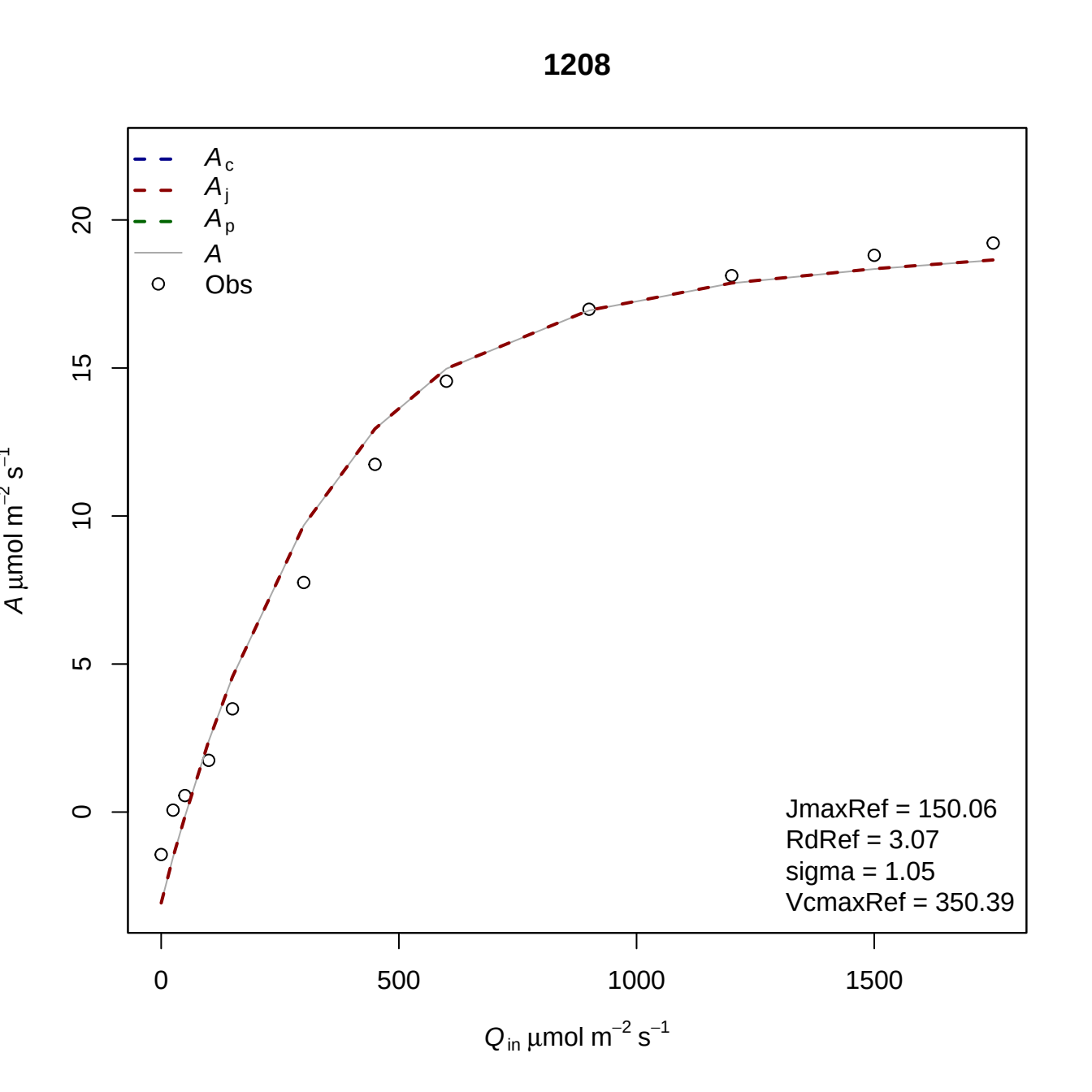

1203
A_c
A_j
A_p
A
Obs
JmaxRef = 132.56
RdRef = 1.84
sigma = 0.59
VcmaxRef = 268.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

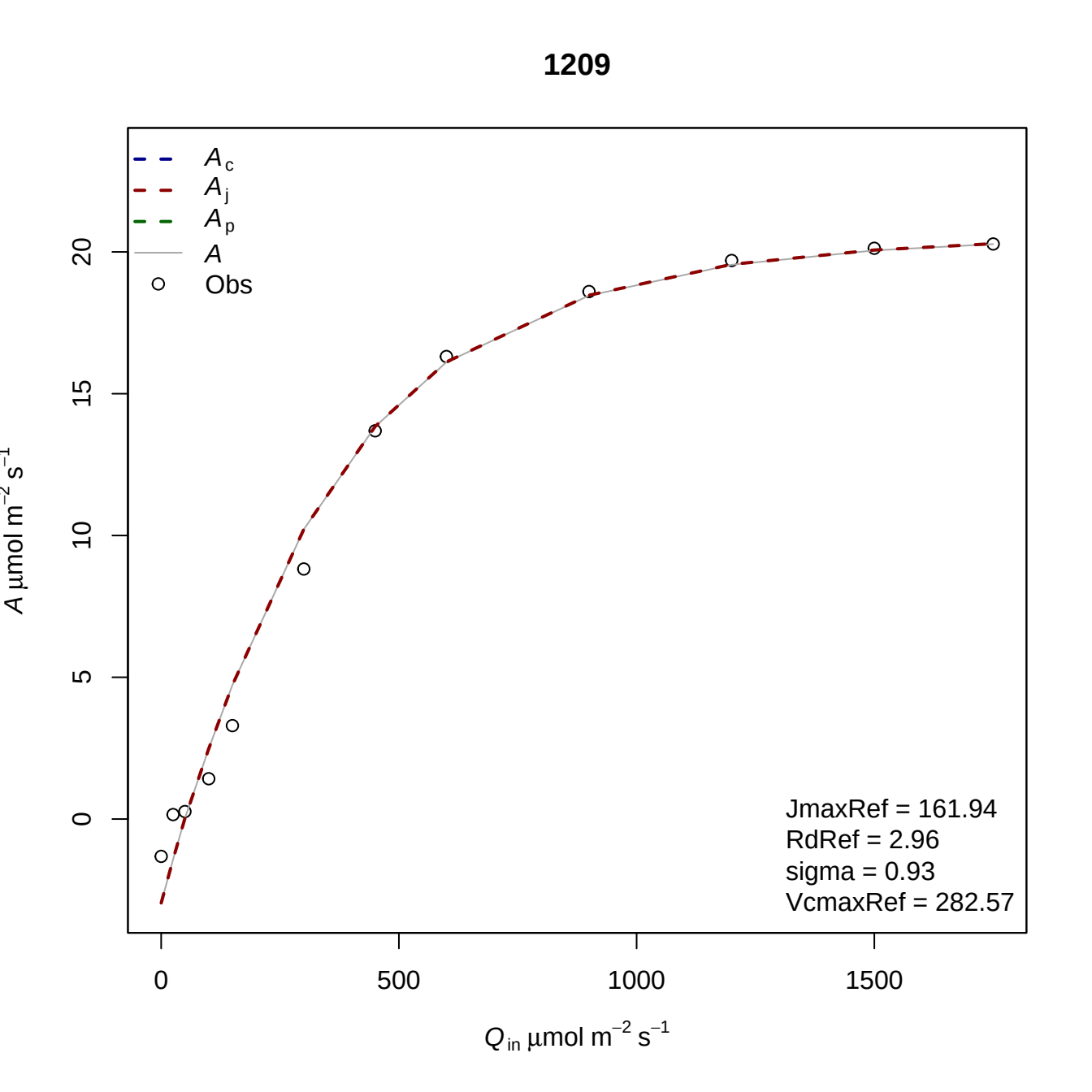

1204
A_c
A_j
A_p
A
Obs
JmaxRef = 69.99
RdRef = 1.84
sigma = 0.71
VcmaxRef = 199.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

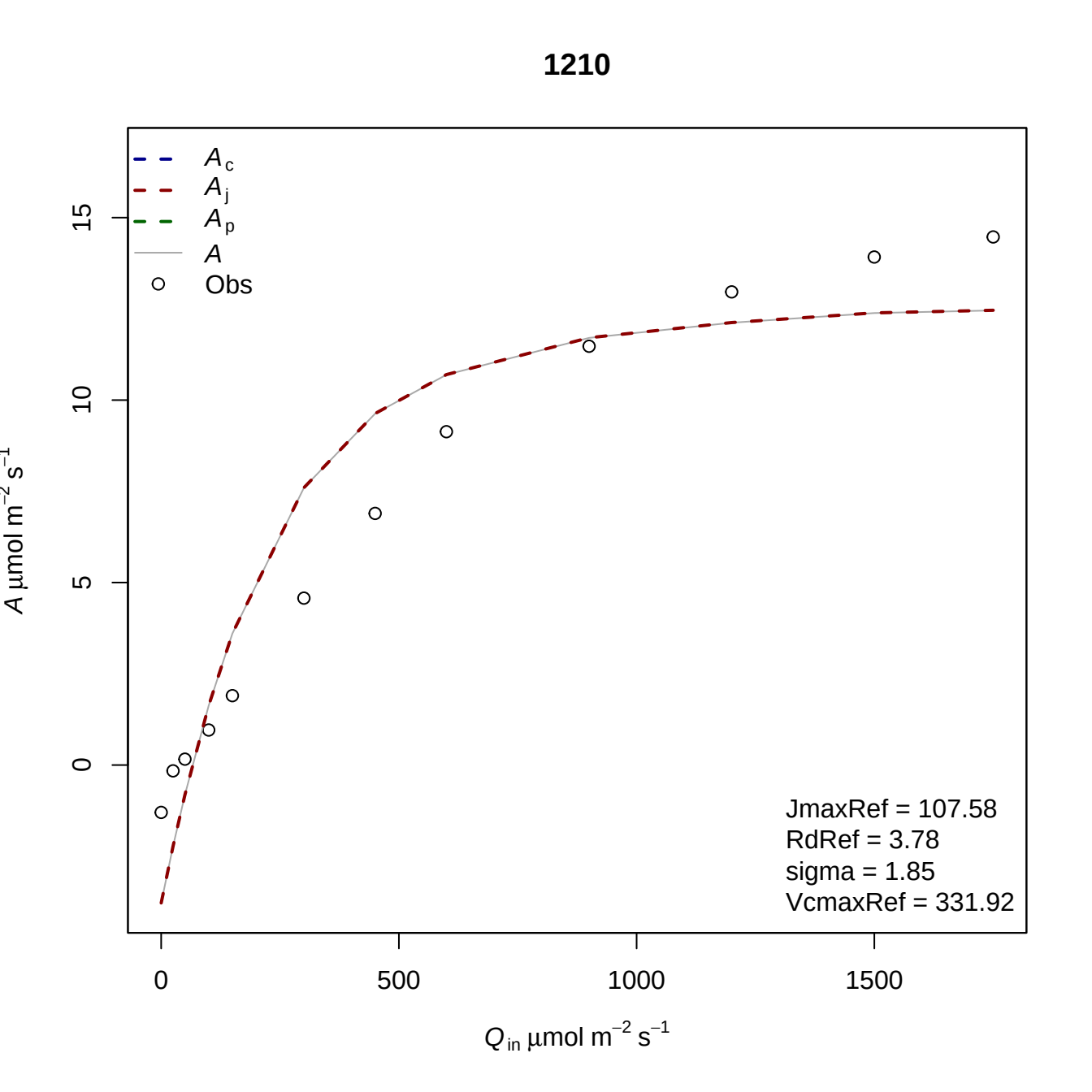

1205
A_c
A_j
A_p
A
Obs
JmaxRef = 101.66
RdRef = 2.68
sigma = 1.1
VcmaxRef = 77.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

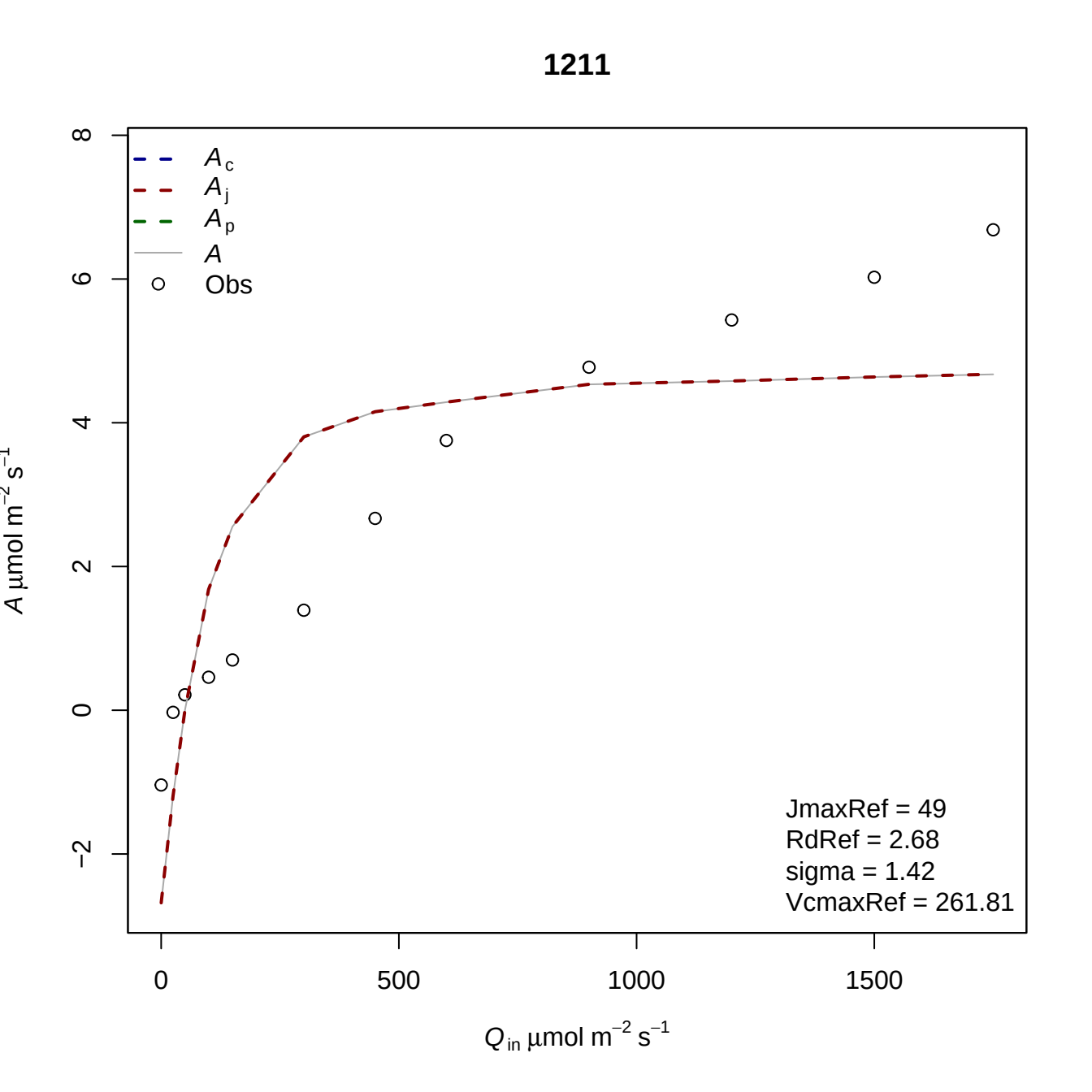

1206
A_c
A_j
A_p
A
Obs
JmaxRef = 131.24
RdRef = 2.64
sigma = 0.99
VcmaxRef = 266.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

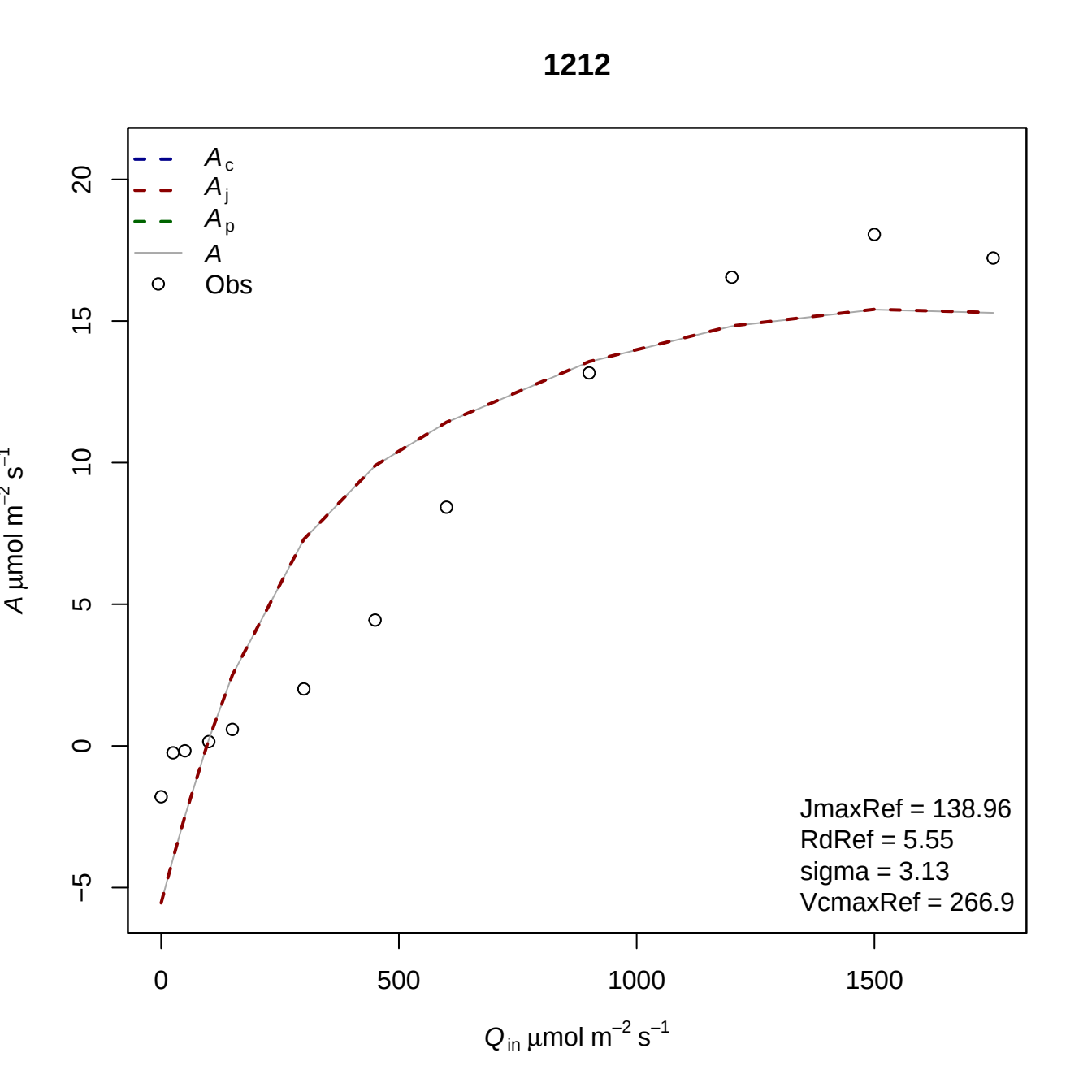

1207
A_c
A_j
A_p
A
Obs
JmaxRef = 115.29
RdRef = 2.29
sigma = 0.71
VcmaxRef = 275.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

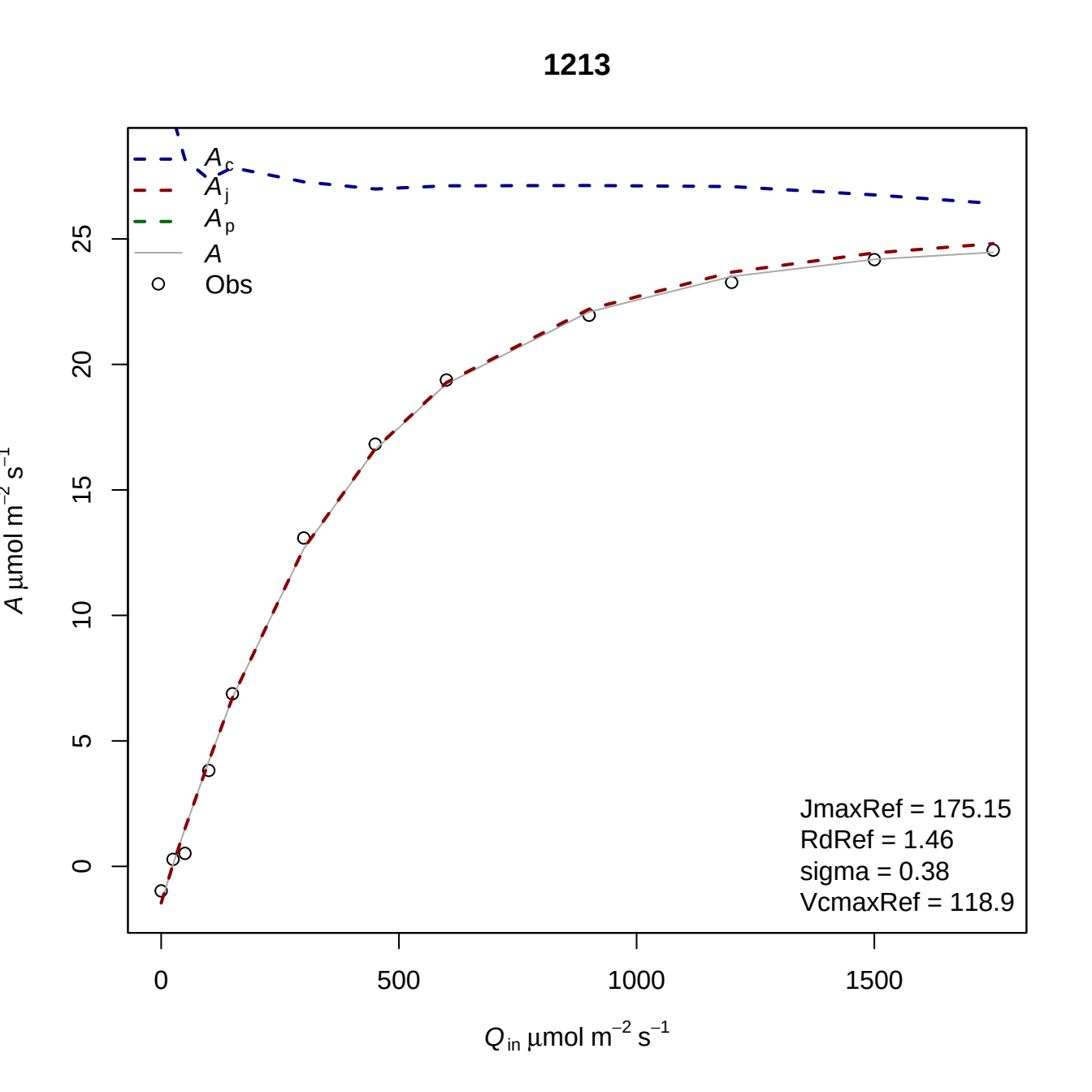

1208
A_c
A_j
A_p
A
Obs
JmaxRef = 150.06
RdRef = 3.07
sigma = 1.05
VcmaxRef = 350.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

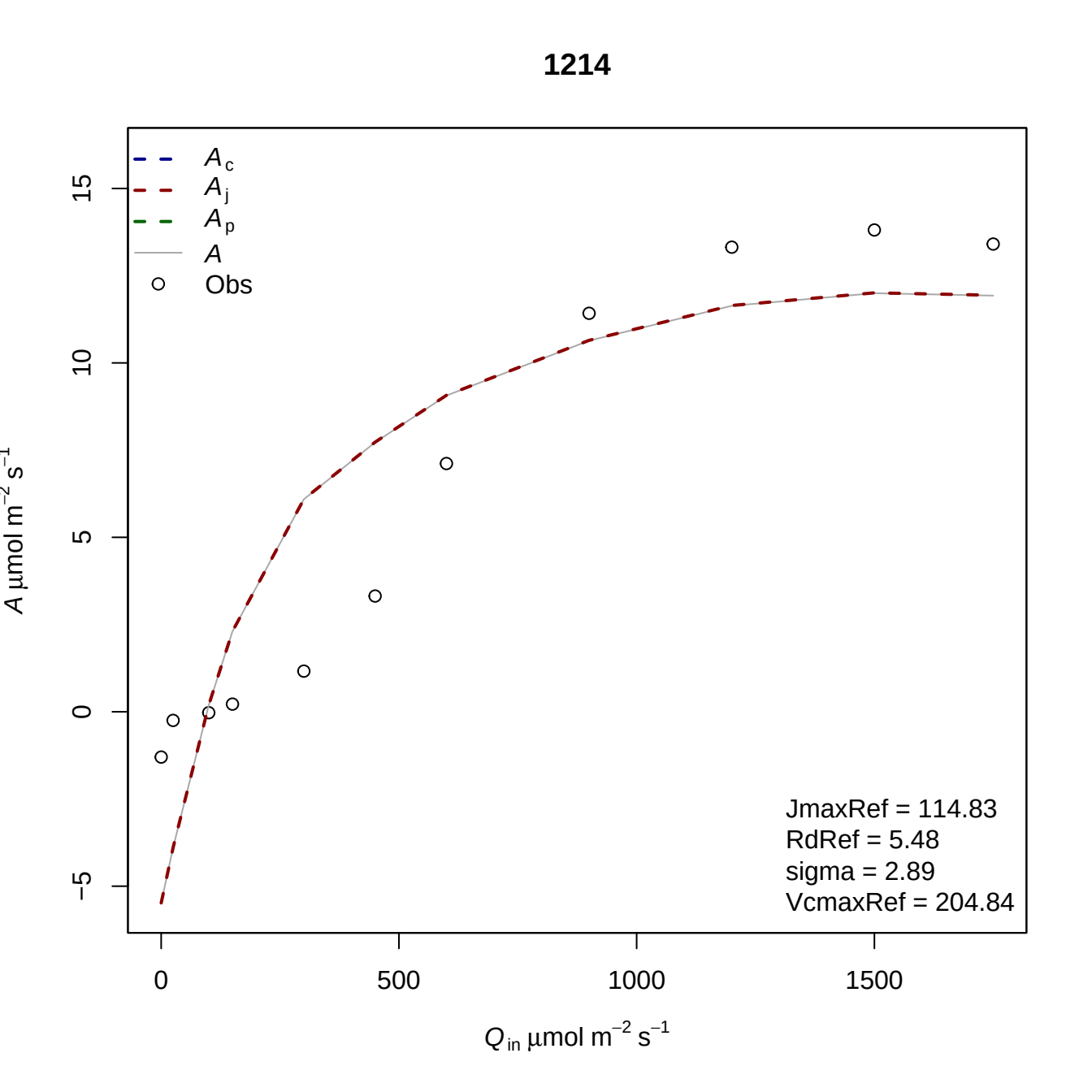

1209
A_c
A_j
A_p
A
Obs
JmaxRef = 161.94
RdRef = 2.96
sigma = 0.93
VcmaxRef = 282.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

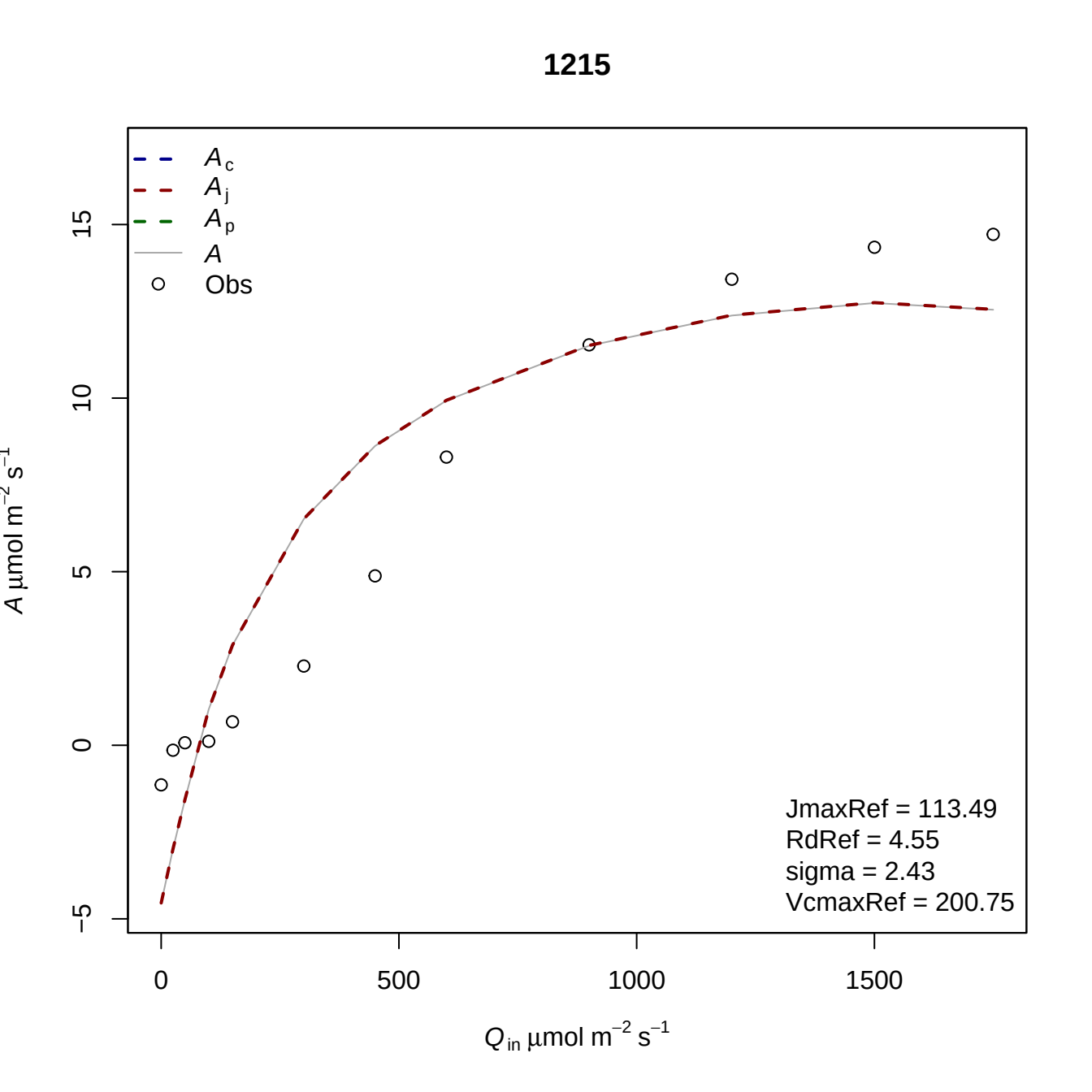

1210
A_c
A_j
A_p
A
Obs
JmaxRef = 107.58
RdRef = 3.78
sigma = 1.85
VcmaxRef = 331.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

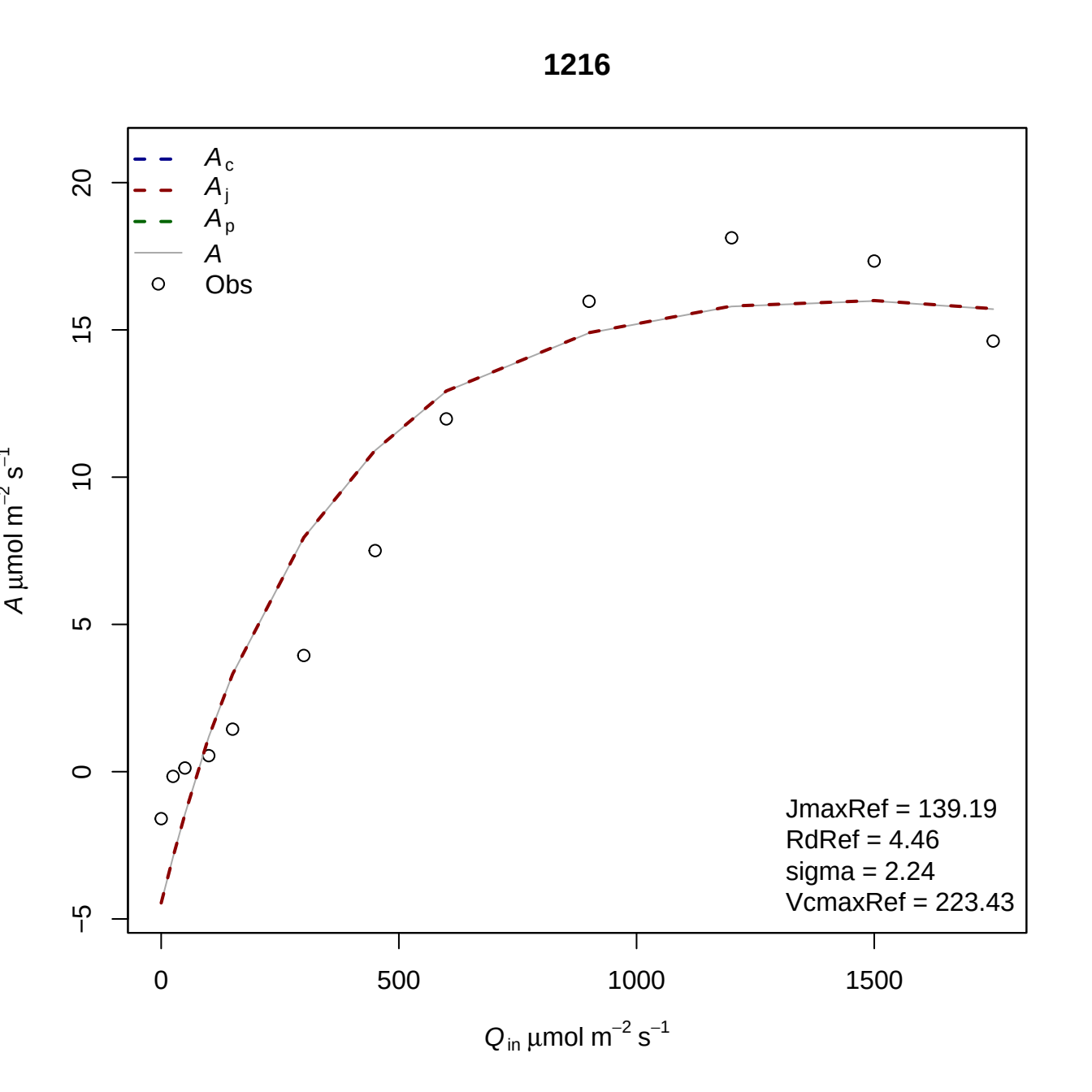

1211
A_c
A_j
A_p
A
Obs
JmaxRef = 49
RdRef = 2.68
sigma = 1.42
VcmaxRef = 261.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

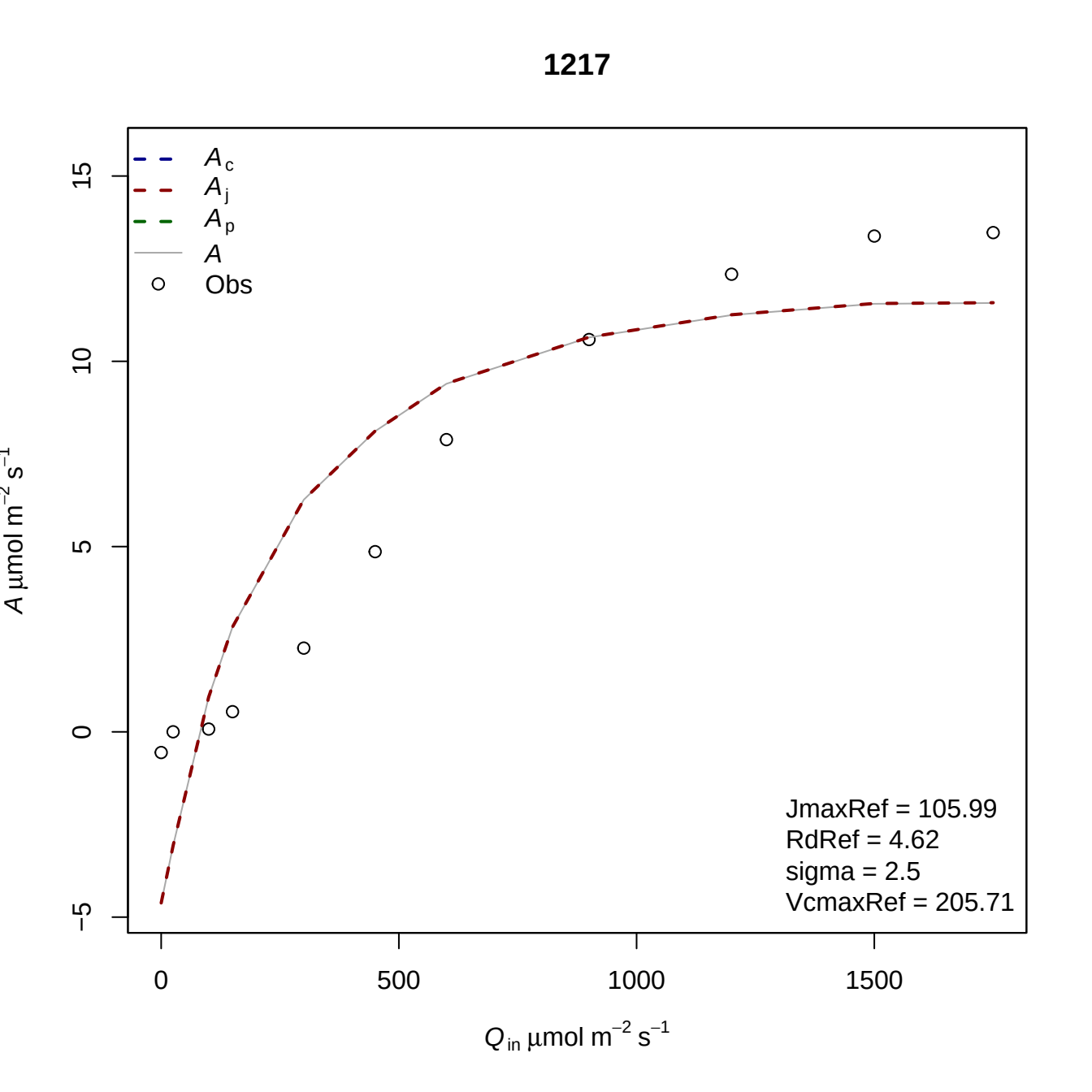

1212
A_c
A_j
A_p
A
Obs
JmaxRef = 138.96
RdRef = 5.55
sigma = 3.13
VcmaxRef = 266.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500

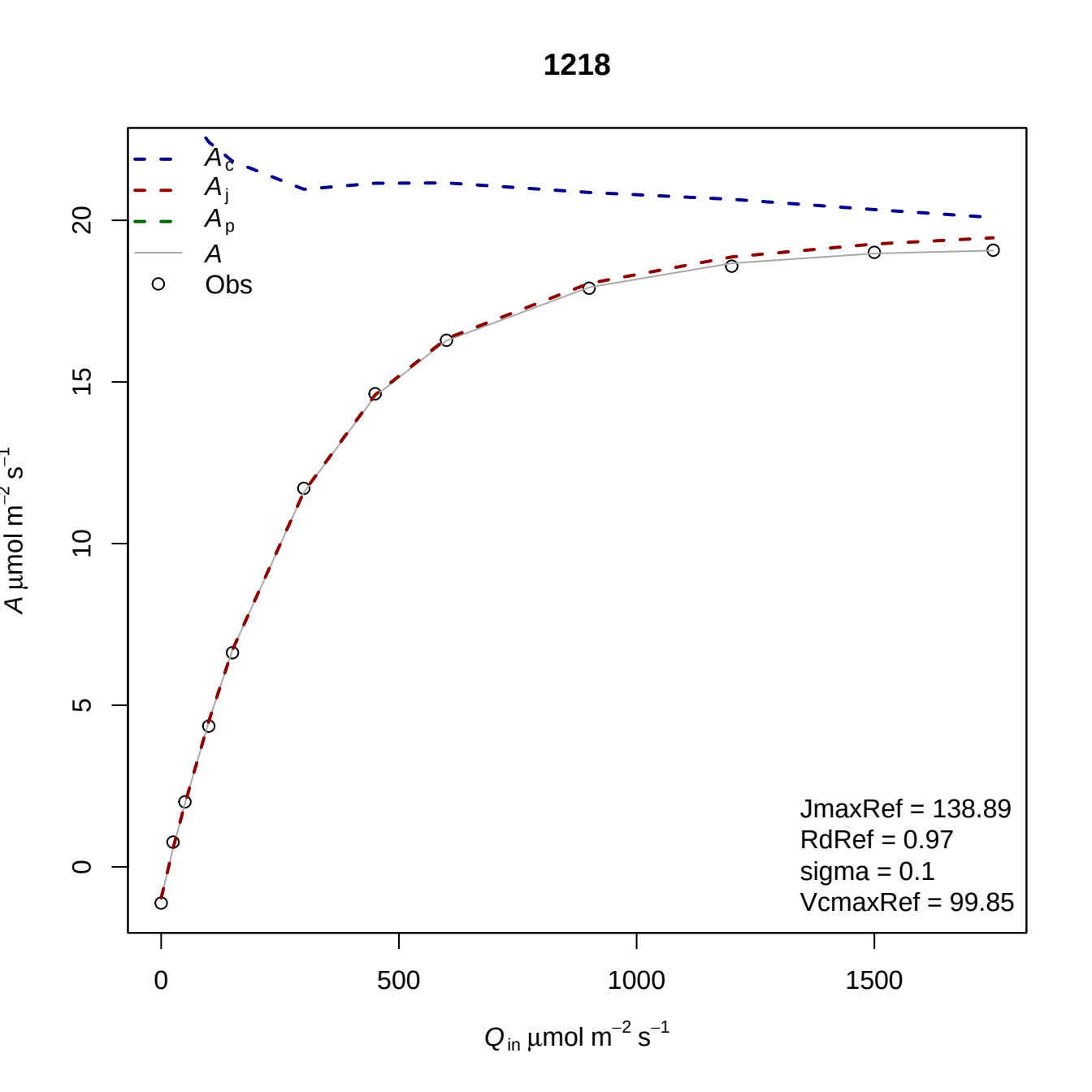

1213
A_c
A_j
A_p
A
Obs
JmaxRef = 175.15
RdRef = 1.46
sigma = 0.38
VcmaxRef = 118.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

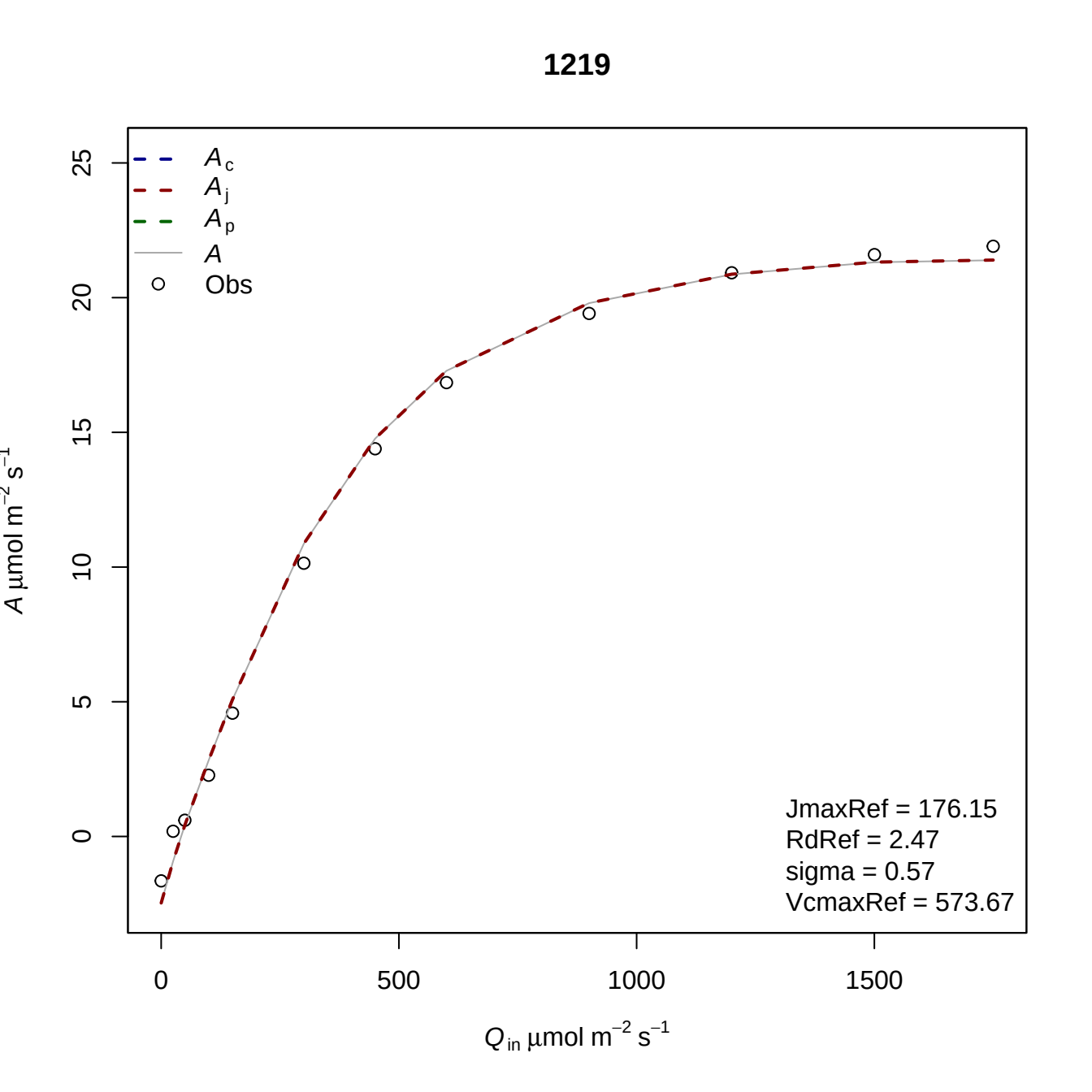

1214
A_c
A_j
A_p
A
Obs
JmaxRef = 114.83
RdRef = 5.48
sigma = 2.89
VcmaxRef = 204.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500

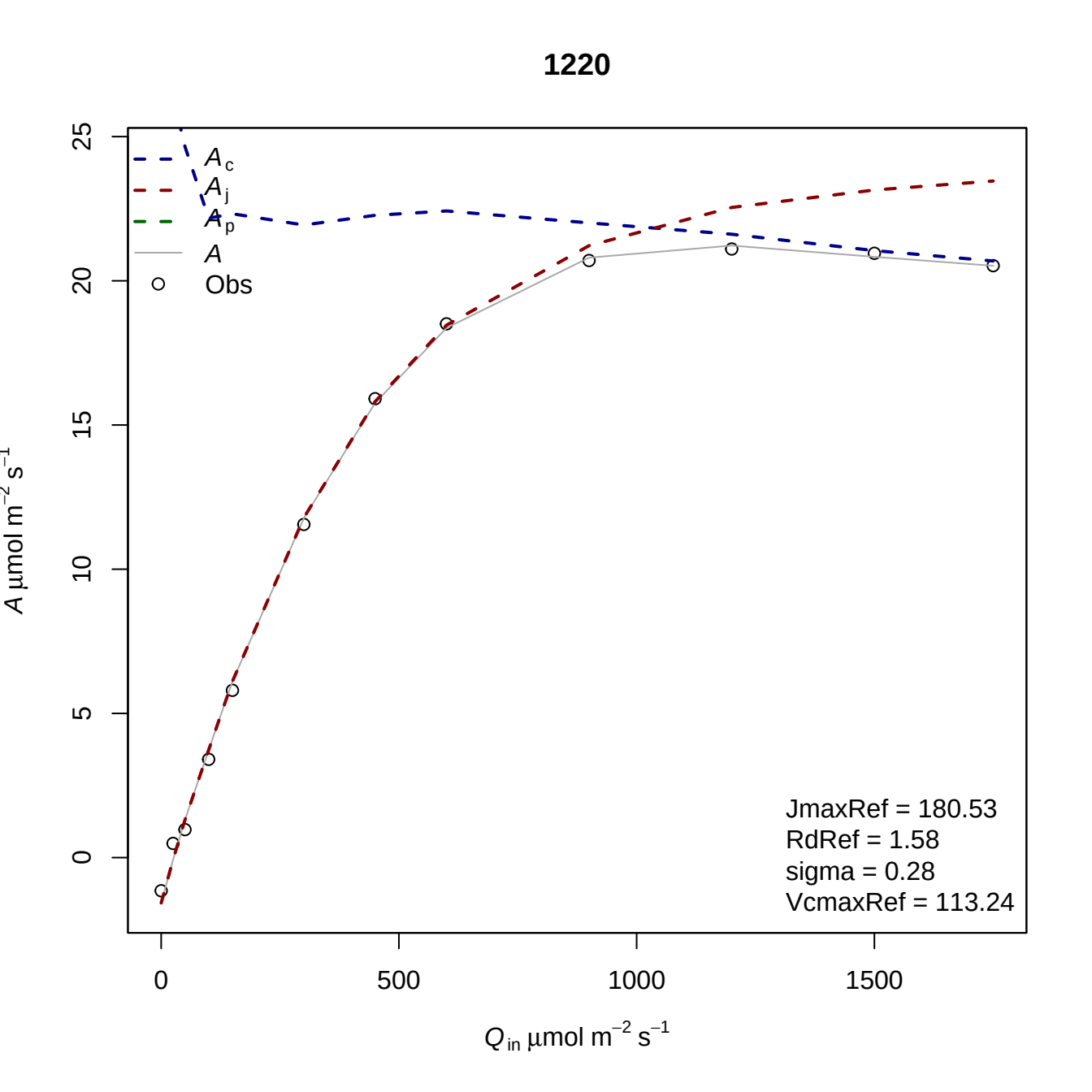

1215
A_c
A_j
A_p
A
Obs
JmaxRef = 113.49
RdRef = 4.55
sigma = 2.43
VcmaxRef = 200.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500

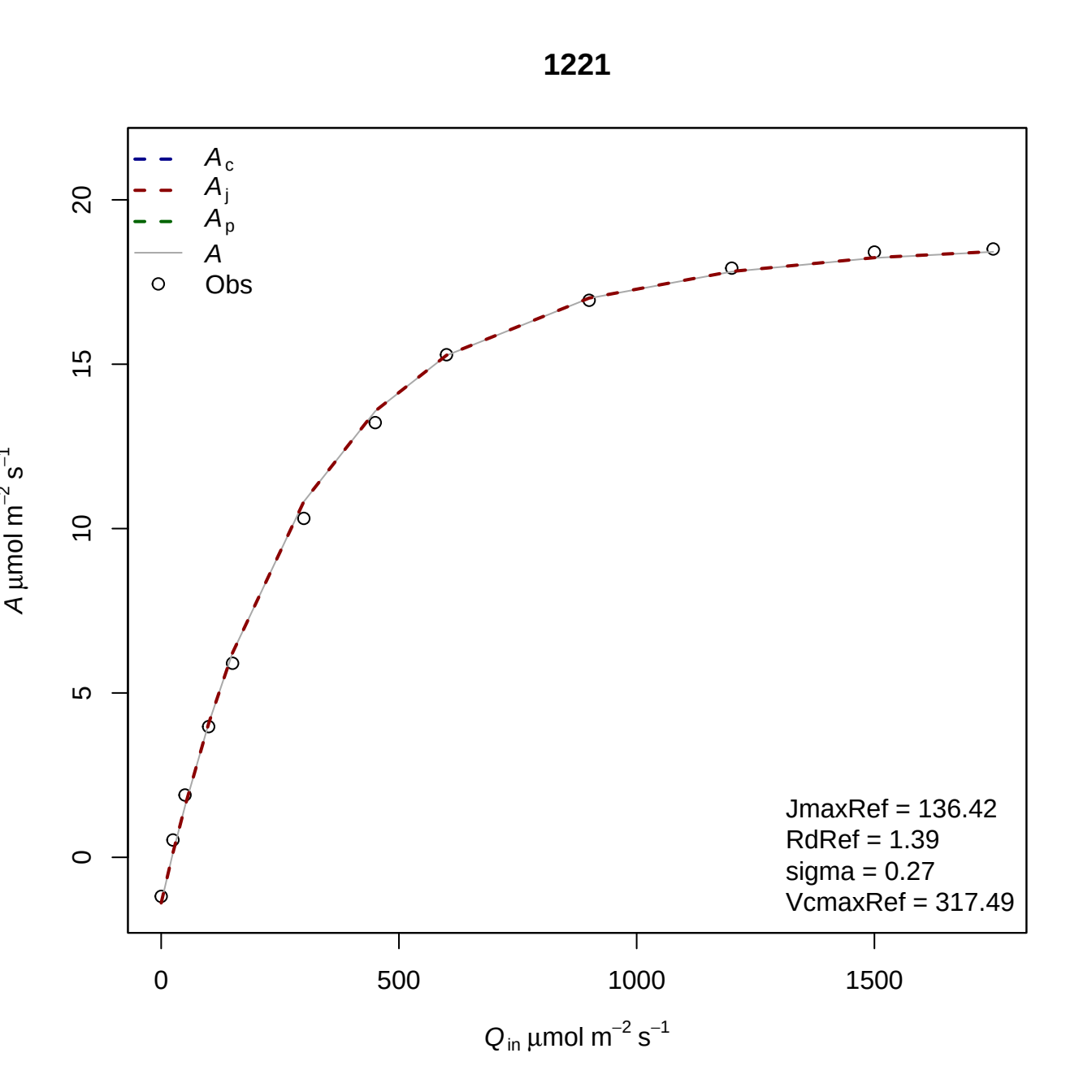

1216
A_c
A_j
A_p
A
Obs
JmaxRef = 139.19
RdRef = 4.46
sigma = 2.24
VcmaxRef = 223.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500

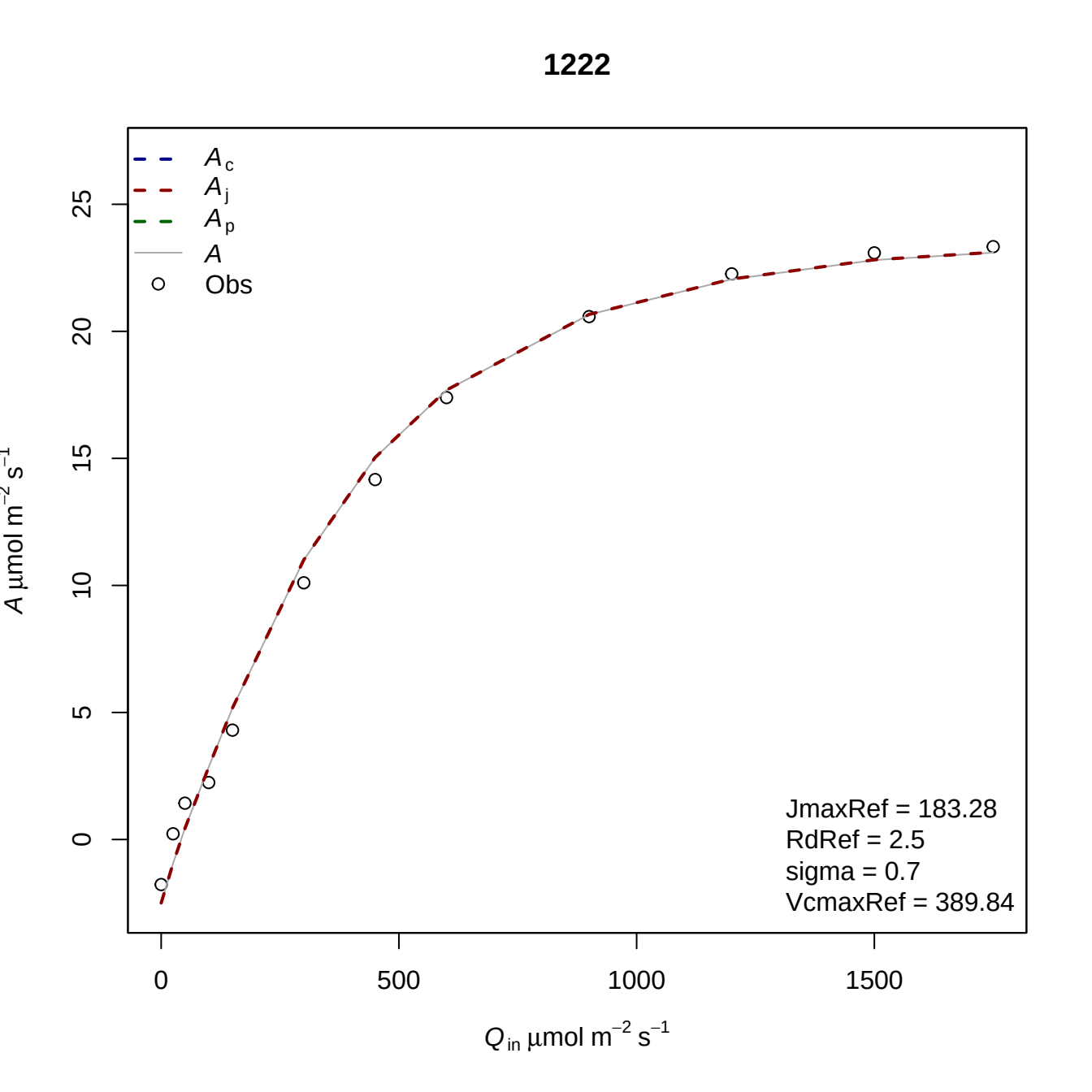

1217
A_c
A_j
A_p
A
Obs
JmaxRef = 105.99
RdRef = 4.62
sigma = 2.5
VcmaxRef = 205.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500

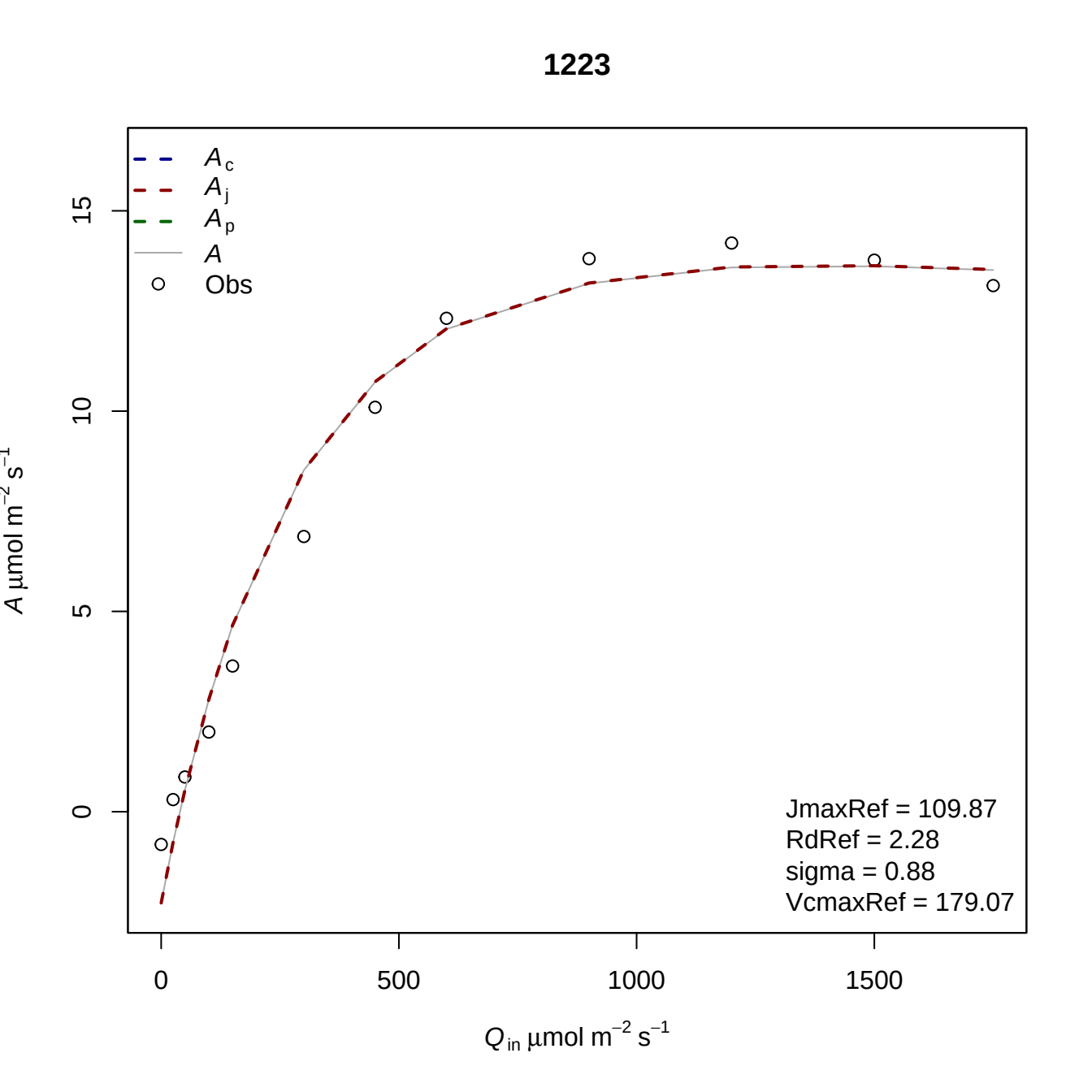

1218
A_c
A_j
A_p
A
Obs
JmaxRef = 138.89
RdRef = 0.97
sigma = 0.1
VcmaxRef = 99.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

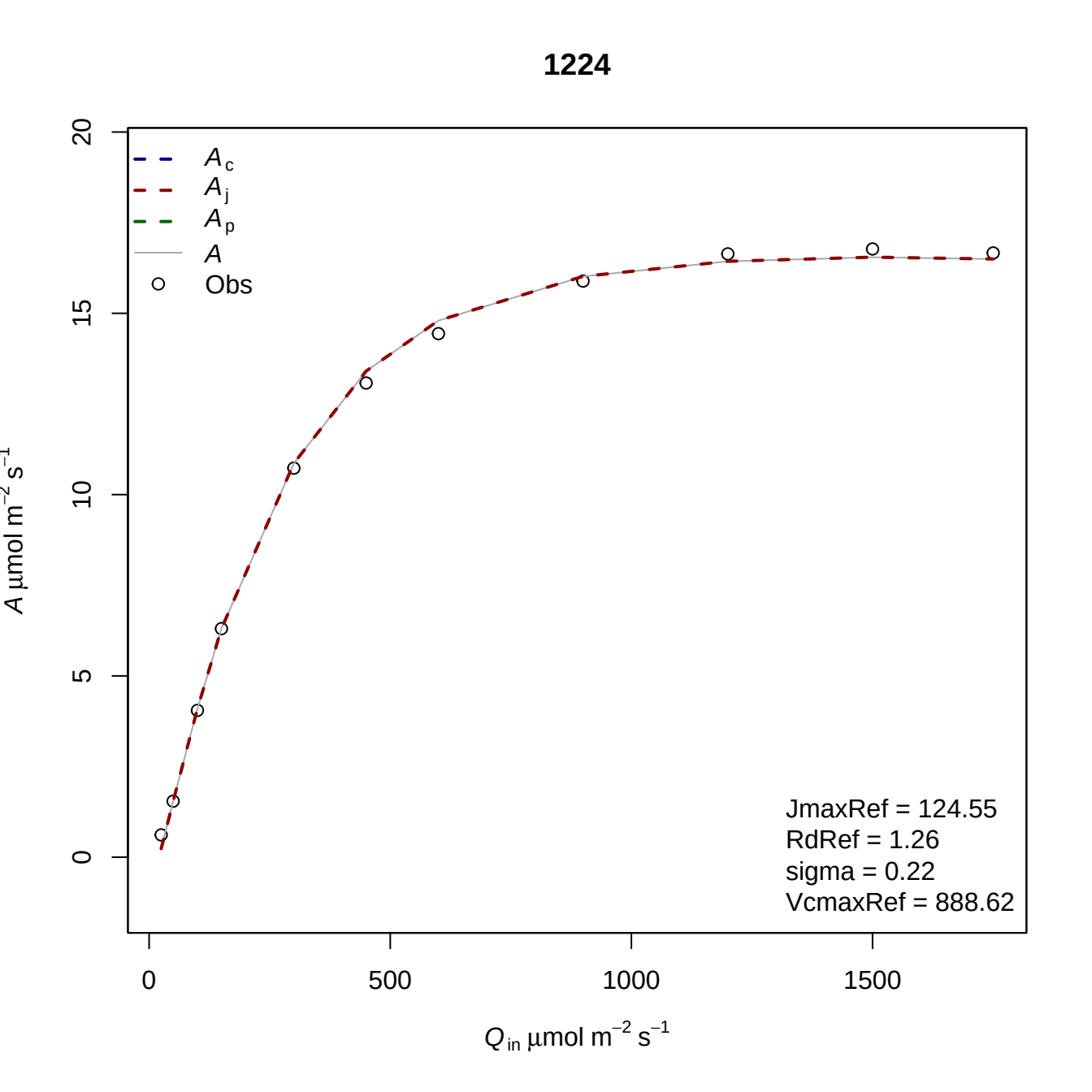

1219
A_c
A_j
A_p
A
Obs
JmaxRef = 176.15
RdRef = 2.47
sigma = 0.57
VcmaxRef = 573.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

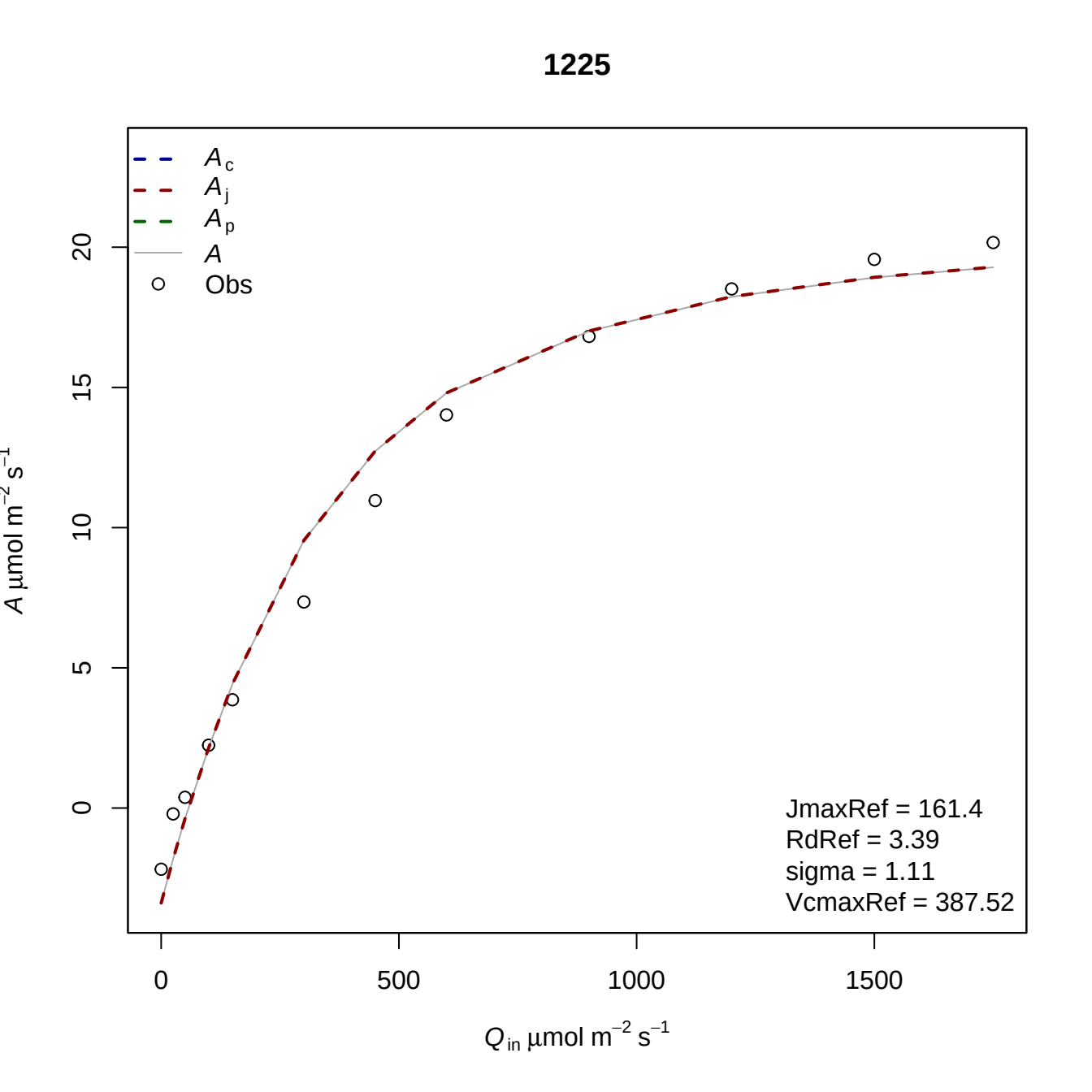

1220
A_c
A_j
A_p
A
Obs
JmaxRef = 180.53
RdRef = 1.58
sigma = 0.28
VcmaxRef = 113.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

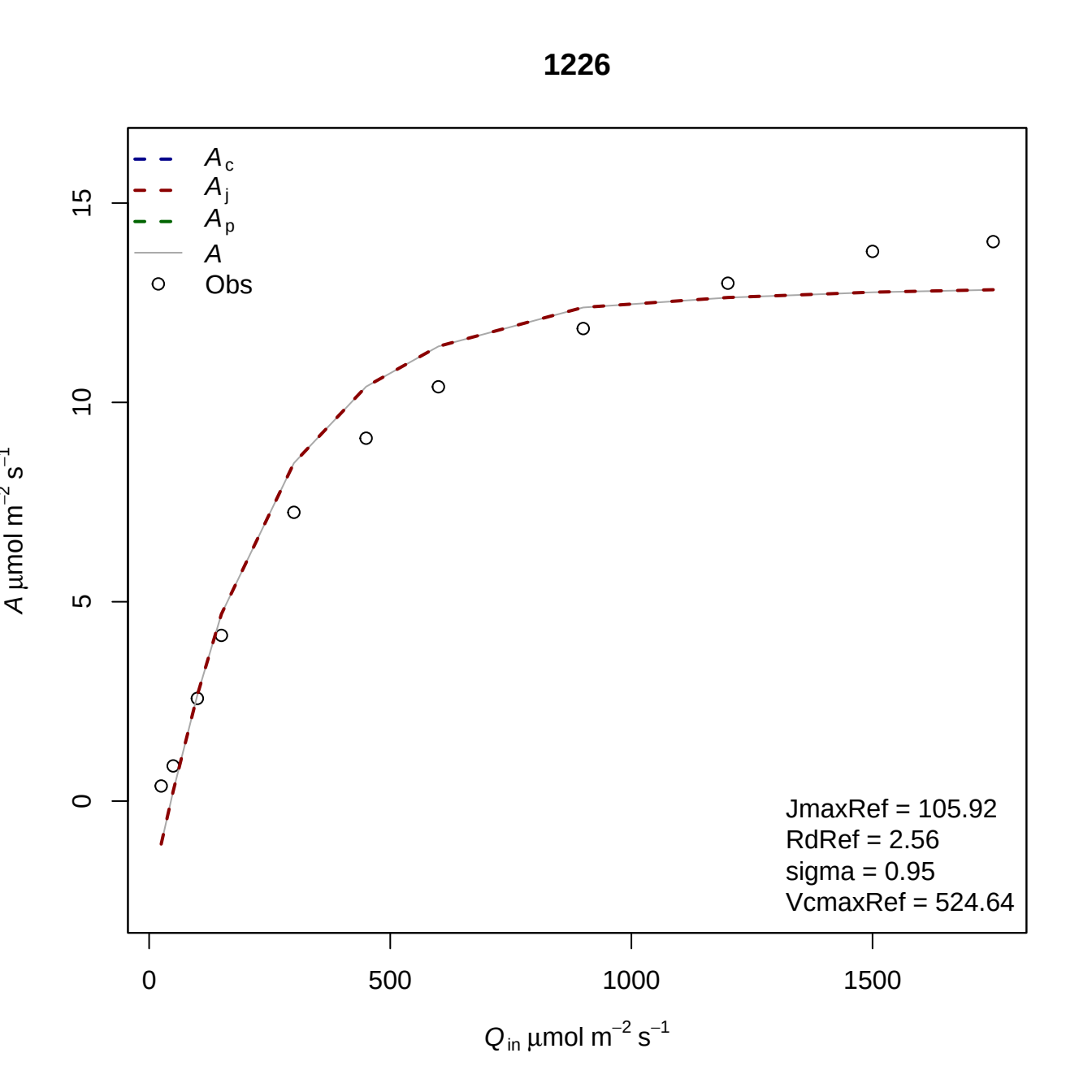

1221
A_c
A_j
A_p
A
Obs
JmaxRef = 136.42
RdRef = 1.39
sigma = 0.27
VcmaxRef = 317.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

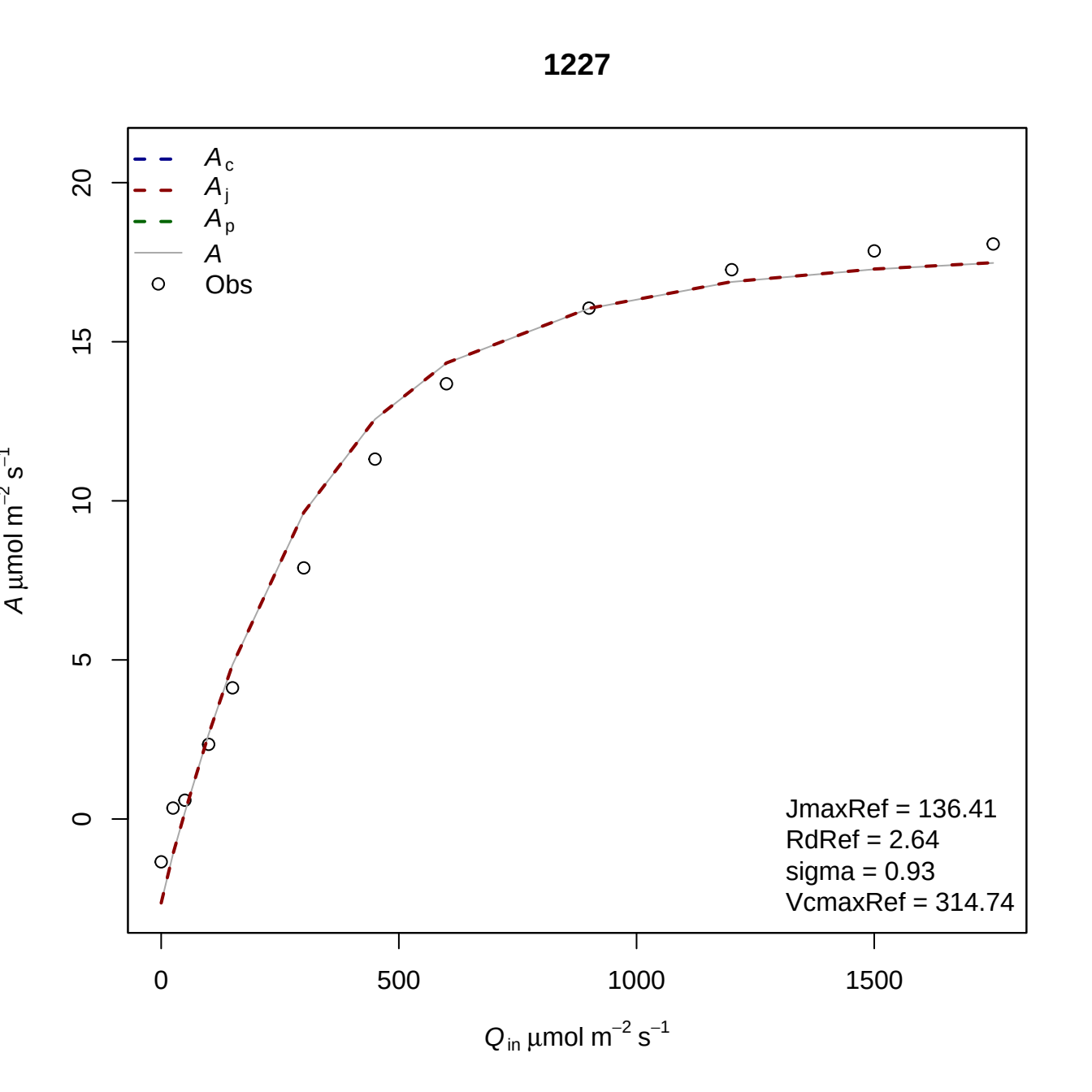

1222
A_c
A_j
A_p
A
Obs
JmaxRef = 183.28
RdRef = 2.5
sigma = 0.7
VcmaxRef = 389.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

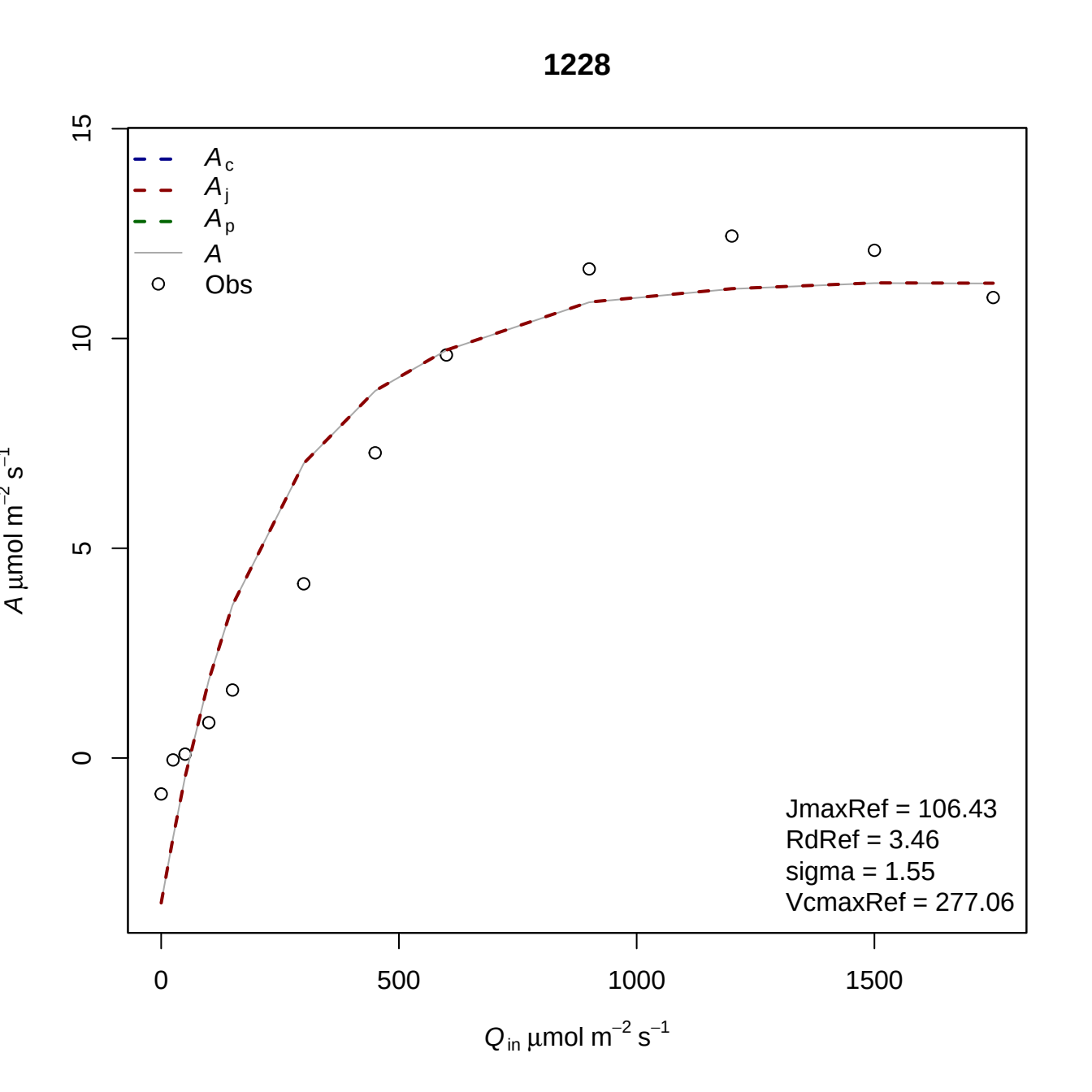

1223
A_c
A_j
A_p
A
Obs
JmaxRef = 109.87
RdRef = 2.28
sigma = 0.88
VcmaxRef = 179.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

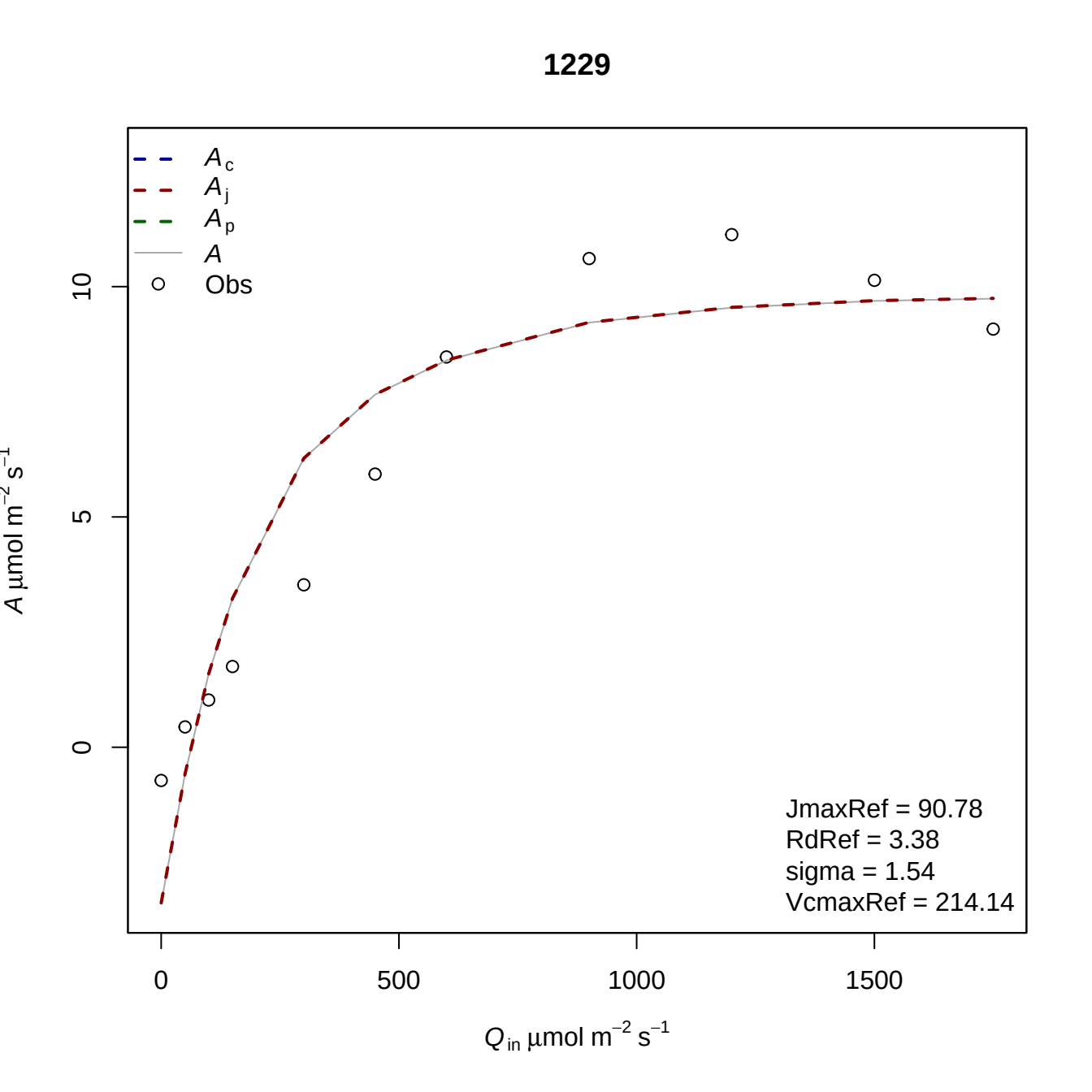

1224
A_c
A_j
A_p
A
Obs
JmaxRef = 124.55
RdRef = 1.26
sigma = 0.22
VcmaxRef = 888.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

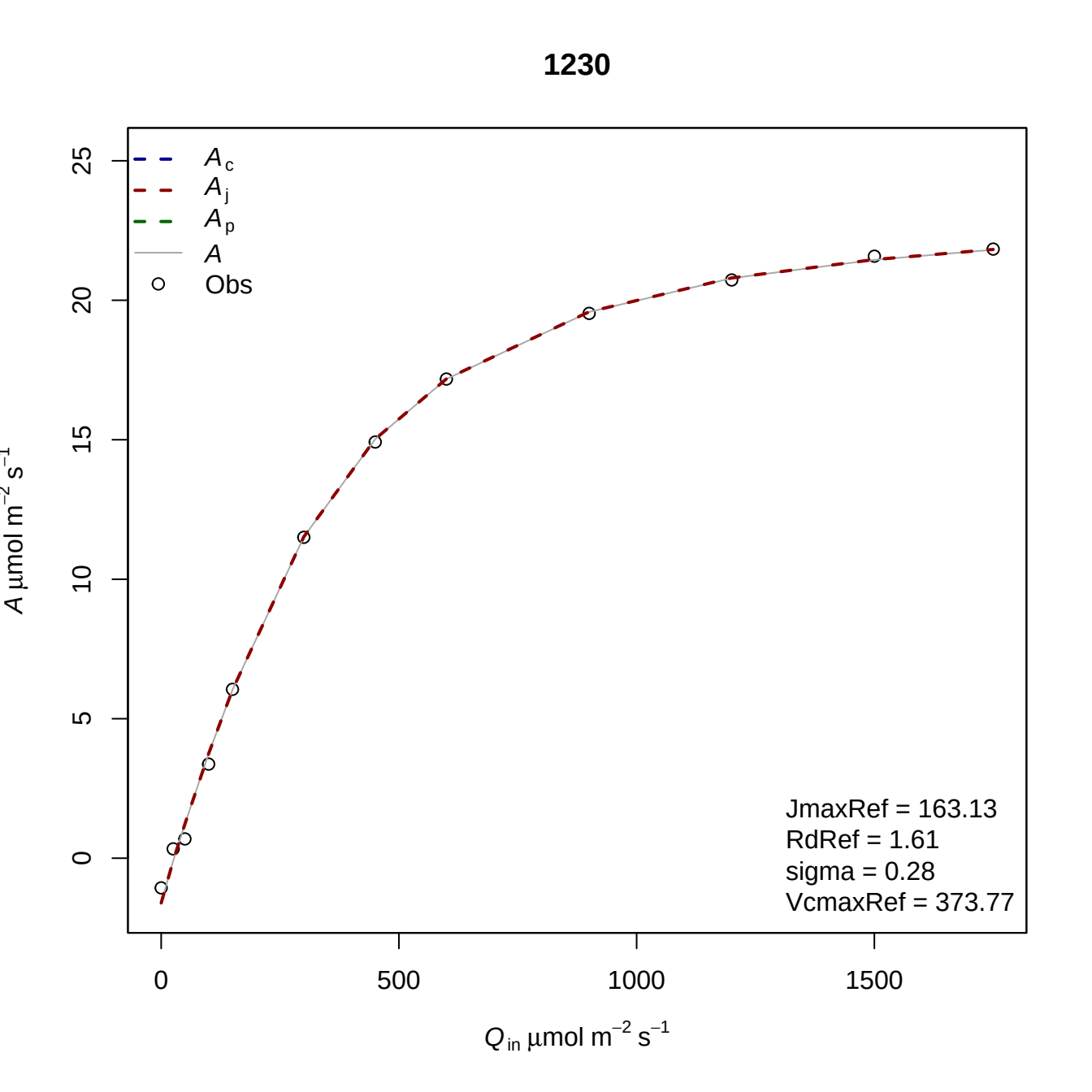

1225
A_c
A_j
A_p
A
Obs
JmaxRef = 161.4
RdRef = 3.39
sigma = 1.11
VcmaxRef = 387.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

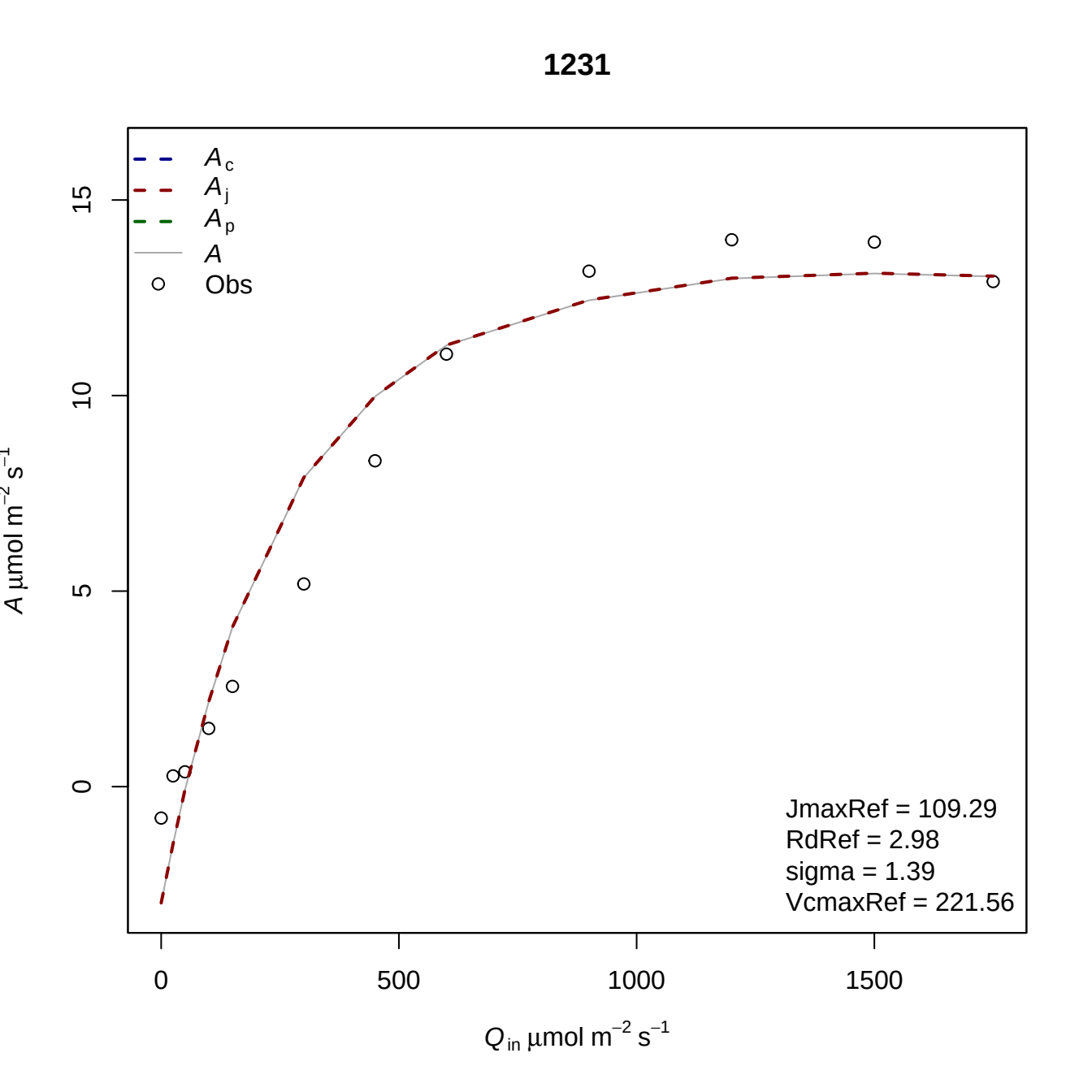

1226
A_c
A_j
A_p
A
Obs
JmaxRef = 105.92
RdRef = 2.56
sigma = 0.95
VcmaxRef = 524.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

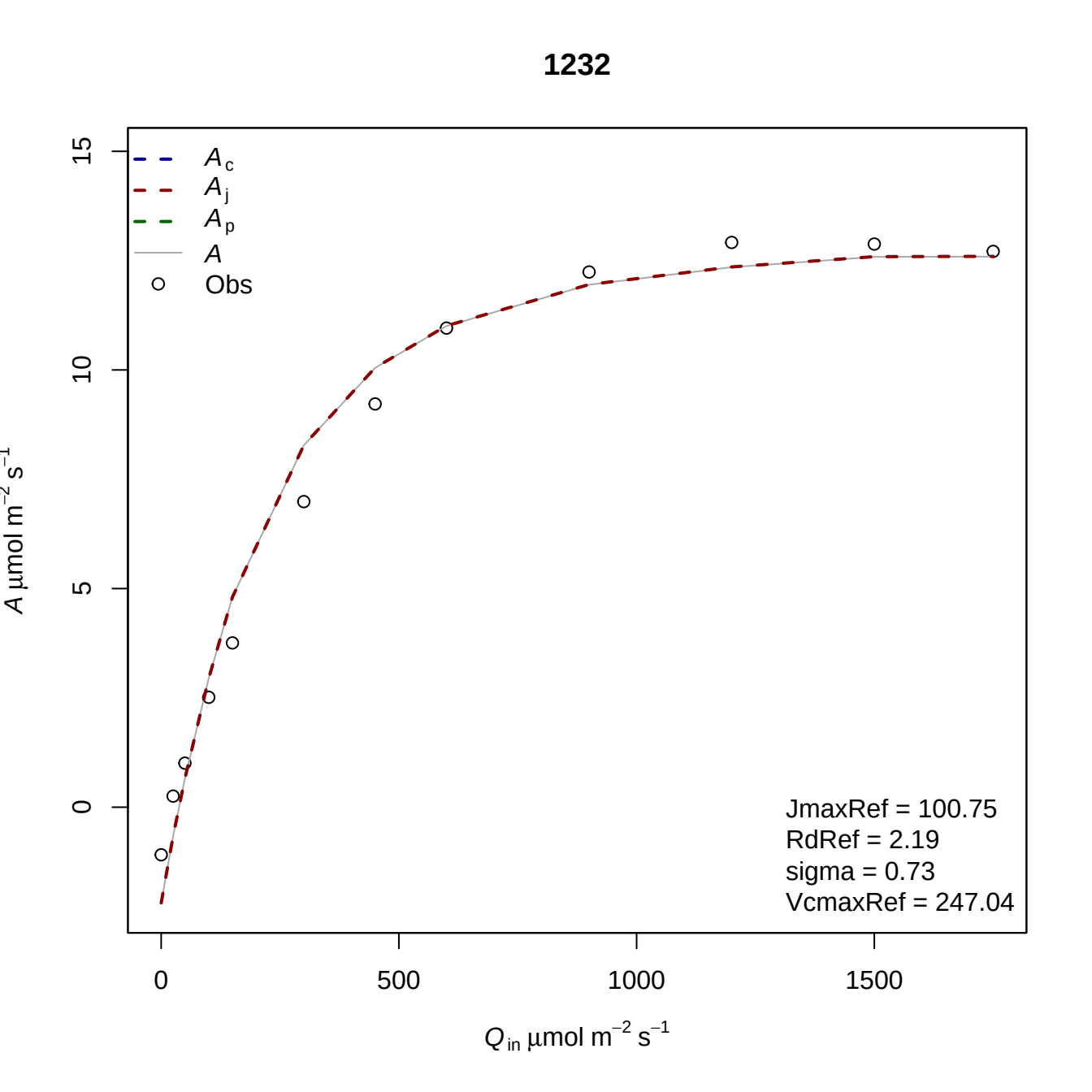

1227
A_c
A_j
A_p
A
Obs
JmaxRef = 136.41
RdRef = 2.64
sigma = 0.93
VcmaxRef = 314.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

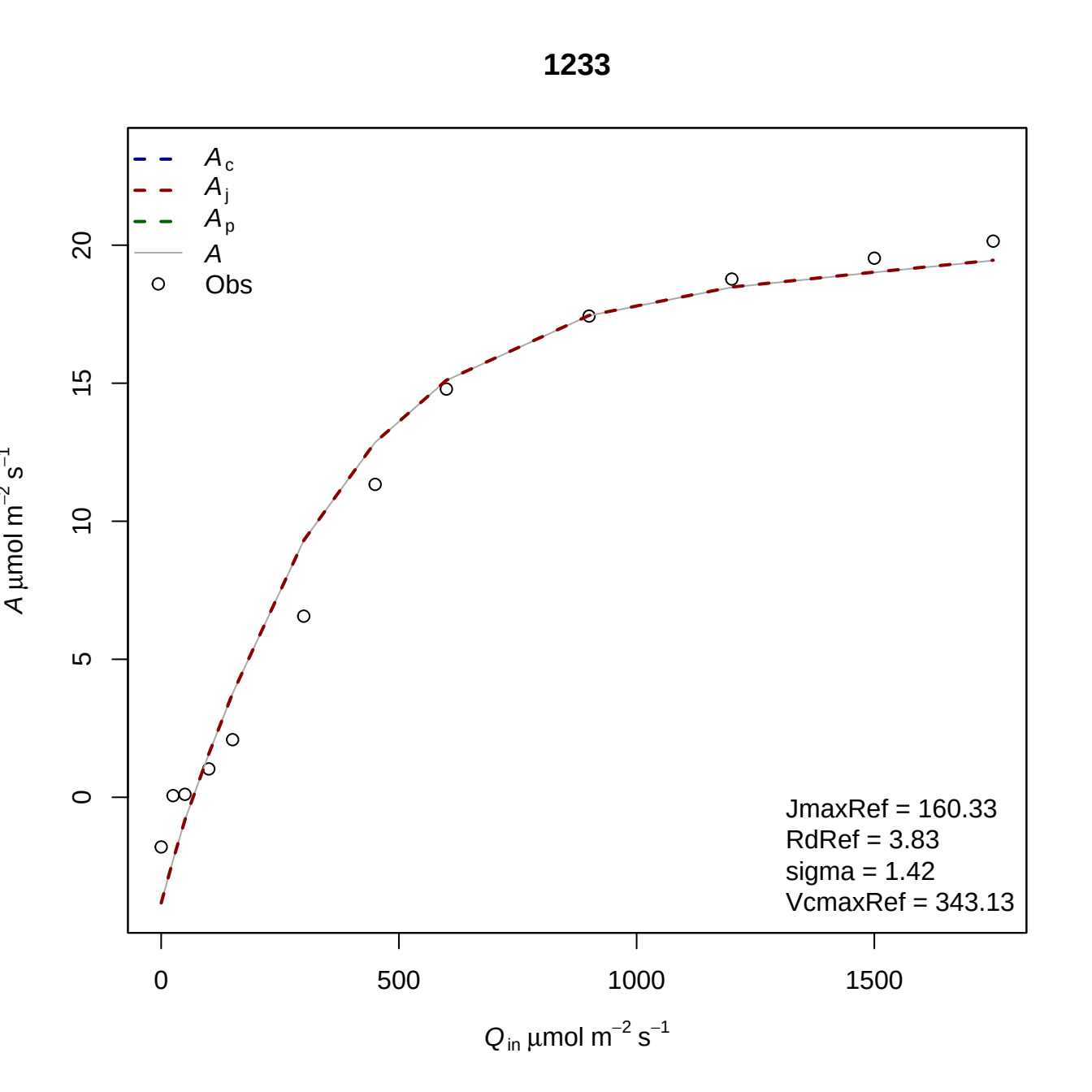

1228
A_c
A_j
A_p
A
Obs
JmaxRef = 106.43
RdRef = 3.46
sigma = 1.55
VcmaxRef = 277.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

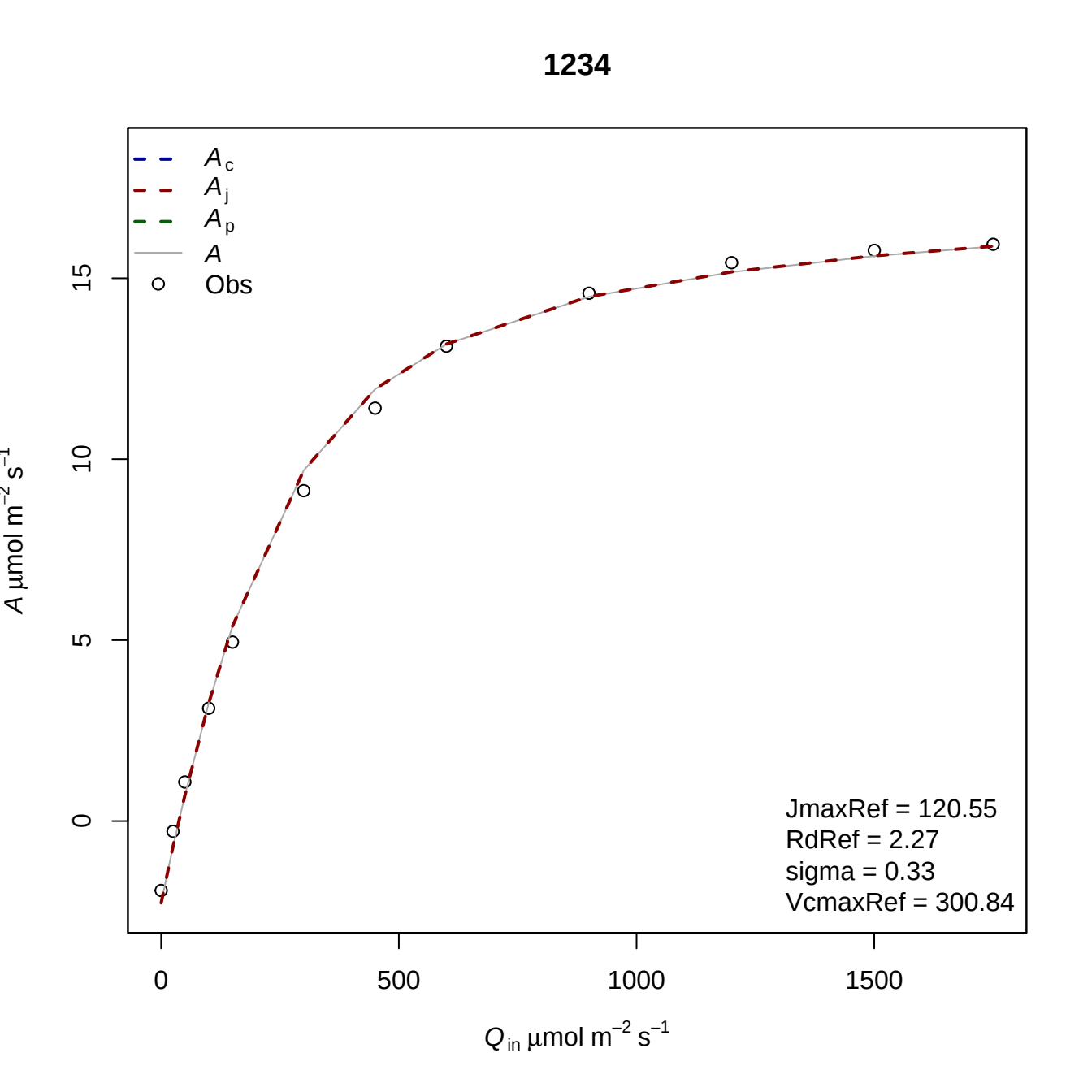

1229
A_c
A_j
A_p
A
Obs
JmaxRef = 90.78
RdRef = 3.38
sigma = 1.54
VcmaxRef = 214.14
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

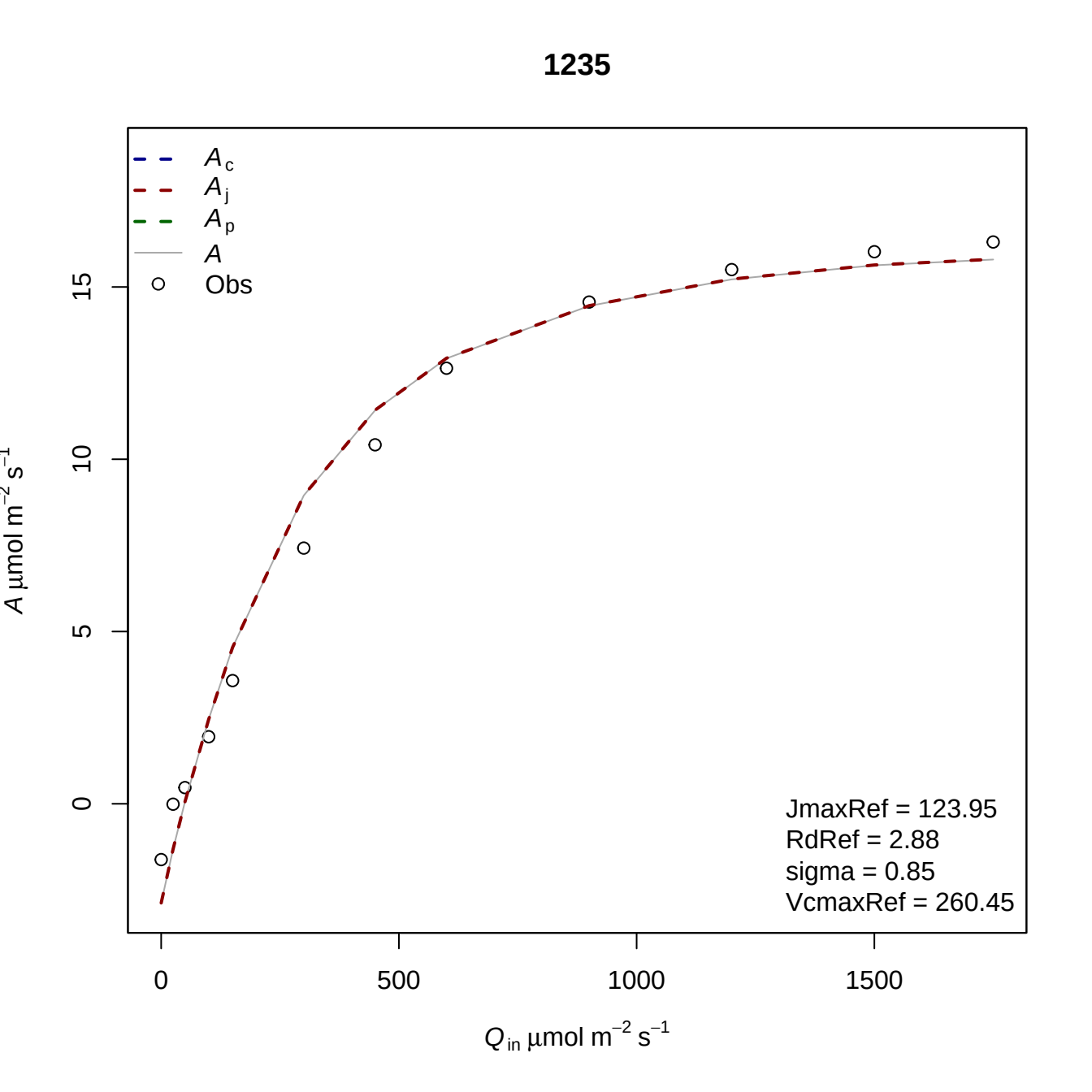

1230
A_c
A_j
A_p
A
Obs
JmaxRef = 163.13
RdRef = 1.61
sigma = 0.28
VcmaxRef = 373.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

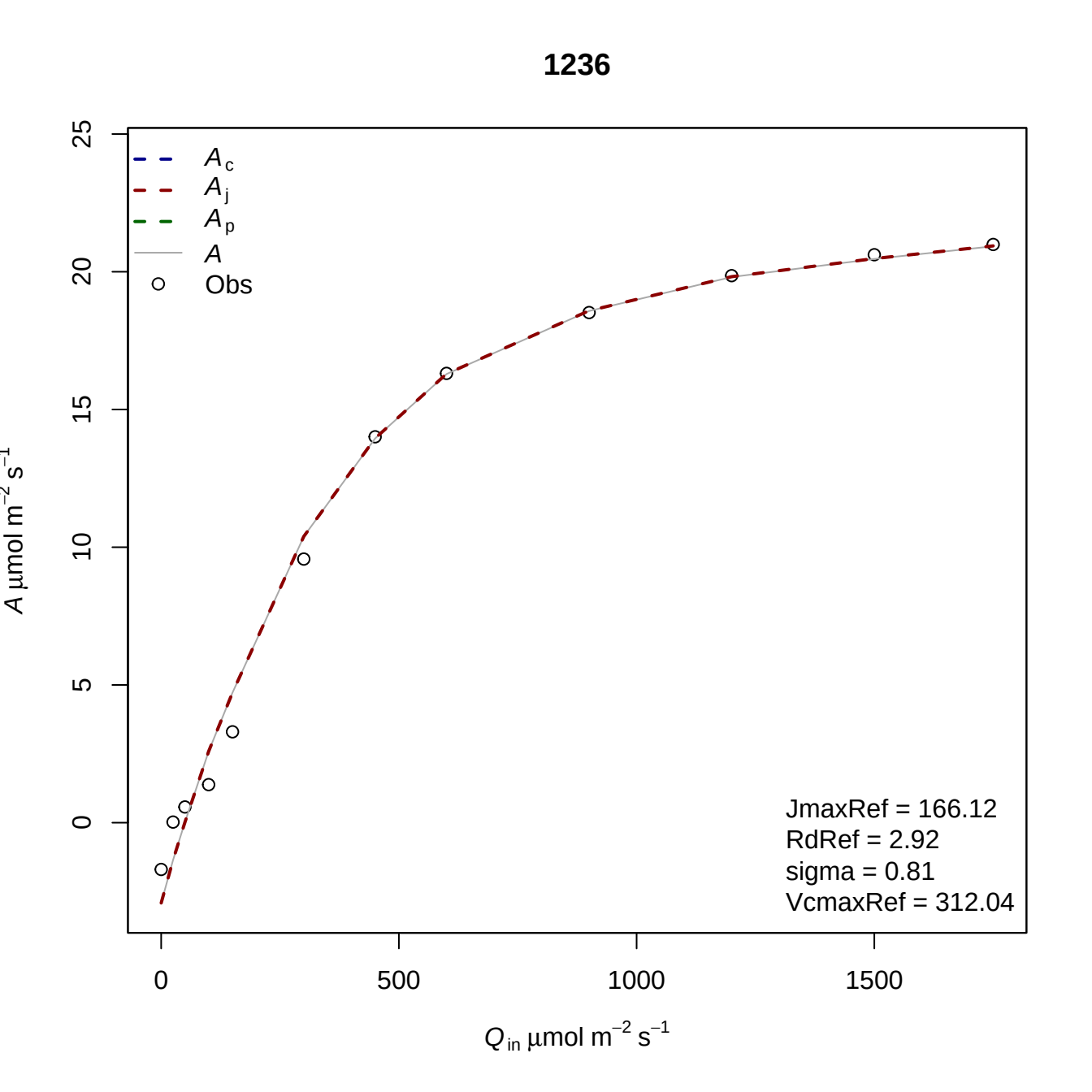

1231
A_c
A_j
A_p
A
Obs
JmaxRef = 109.29
RdRef = 2.98
sigma = 1.39
VcmaxRef = 221.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

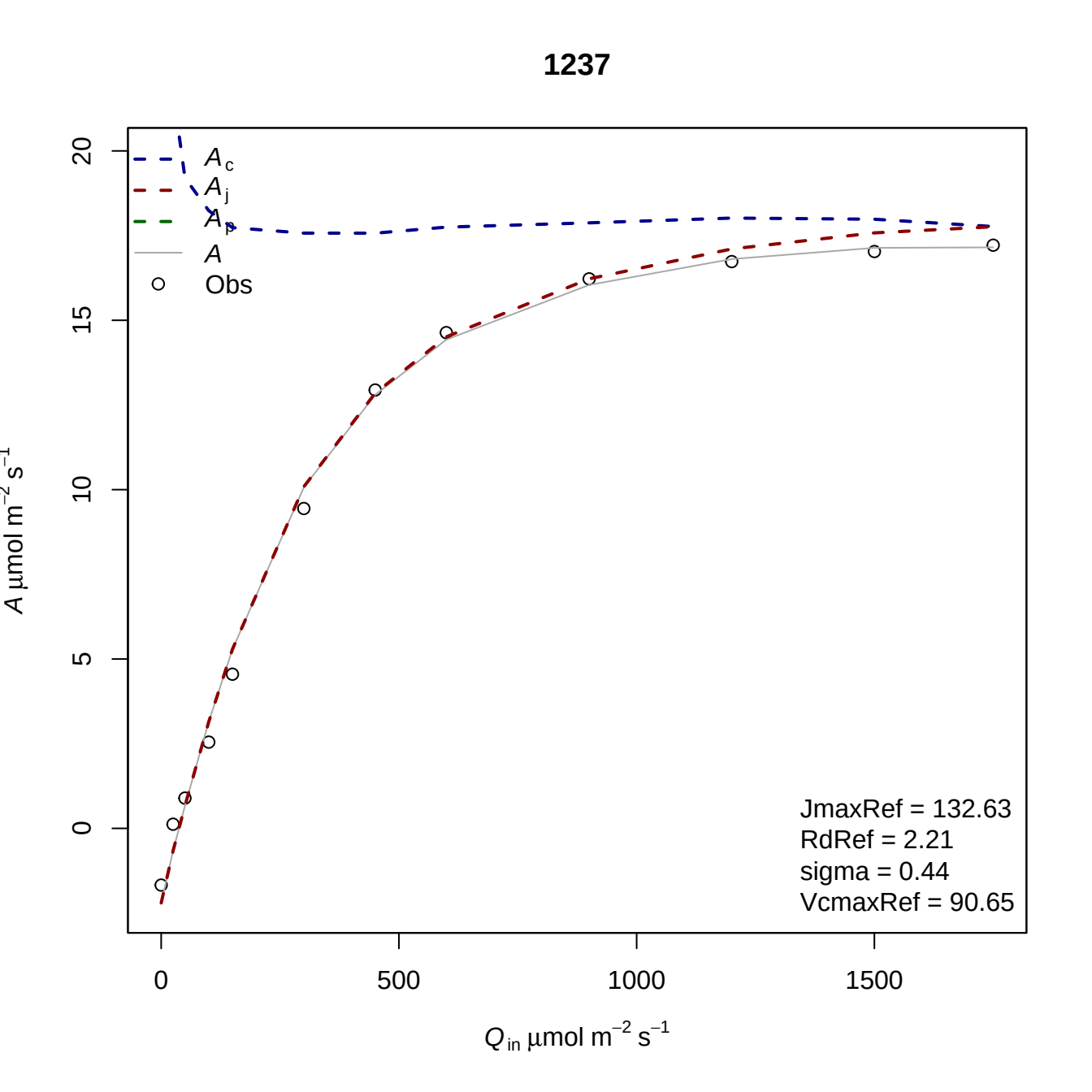

1232
A_c
A_j
A_p
A
Obs
JmaxRef = 100.75
RdRef = 2.19
sigma = 0.73
VcmaxRef = 247.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

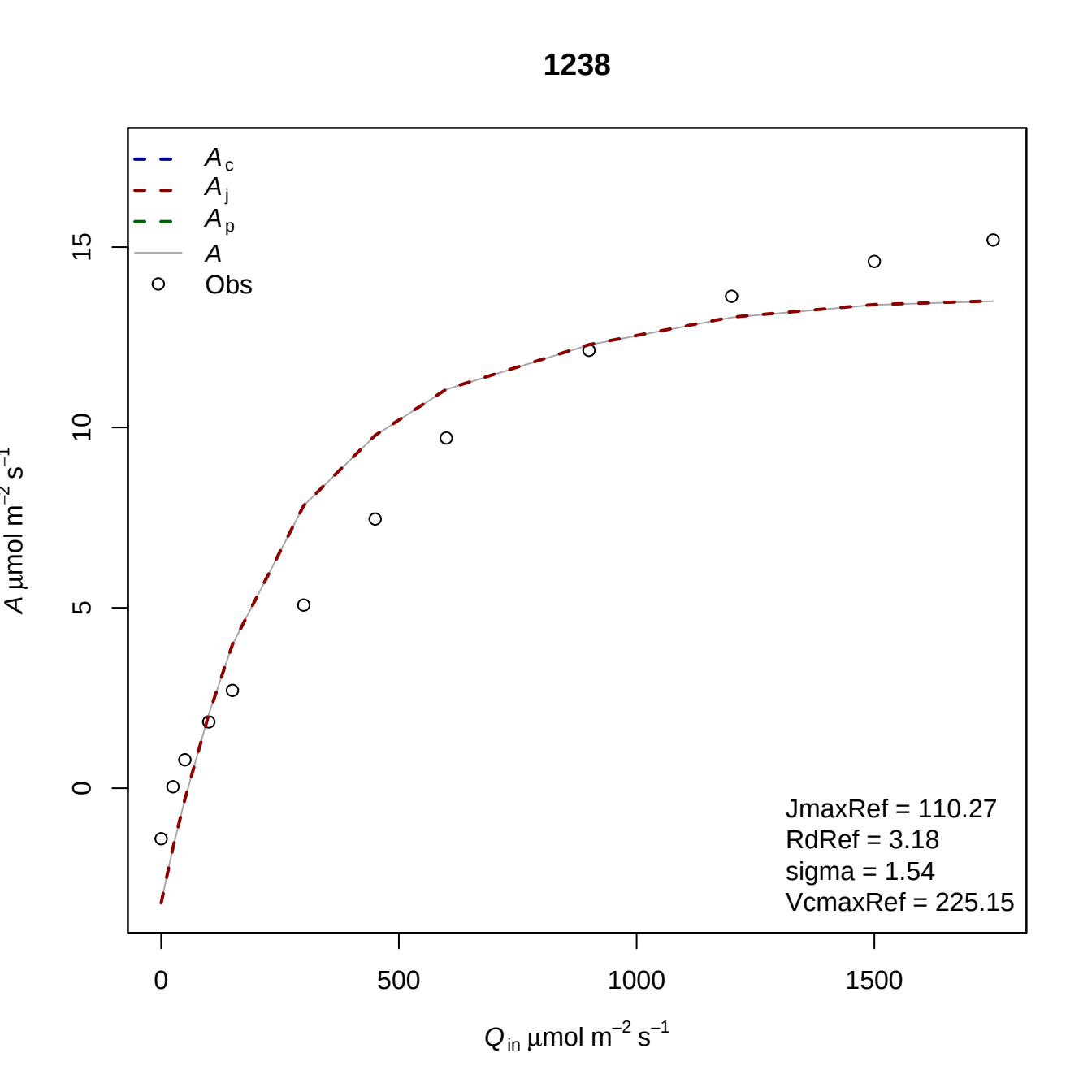

1233
A_c
A_j
A_p
A
Obs
JmaxRef = 160.33
RdRef = 3.83
sigma = 1.42
VcmaxRef = 343.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

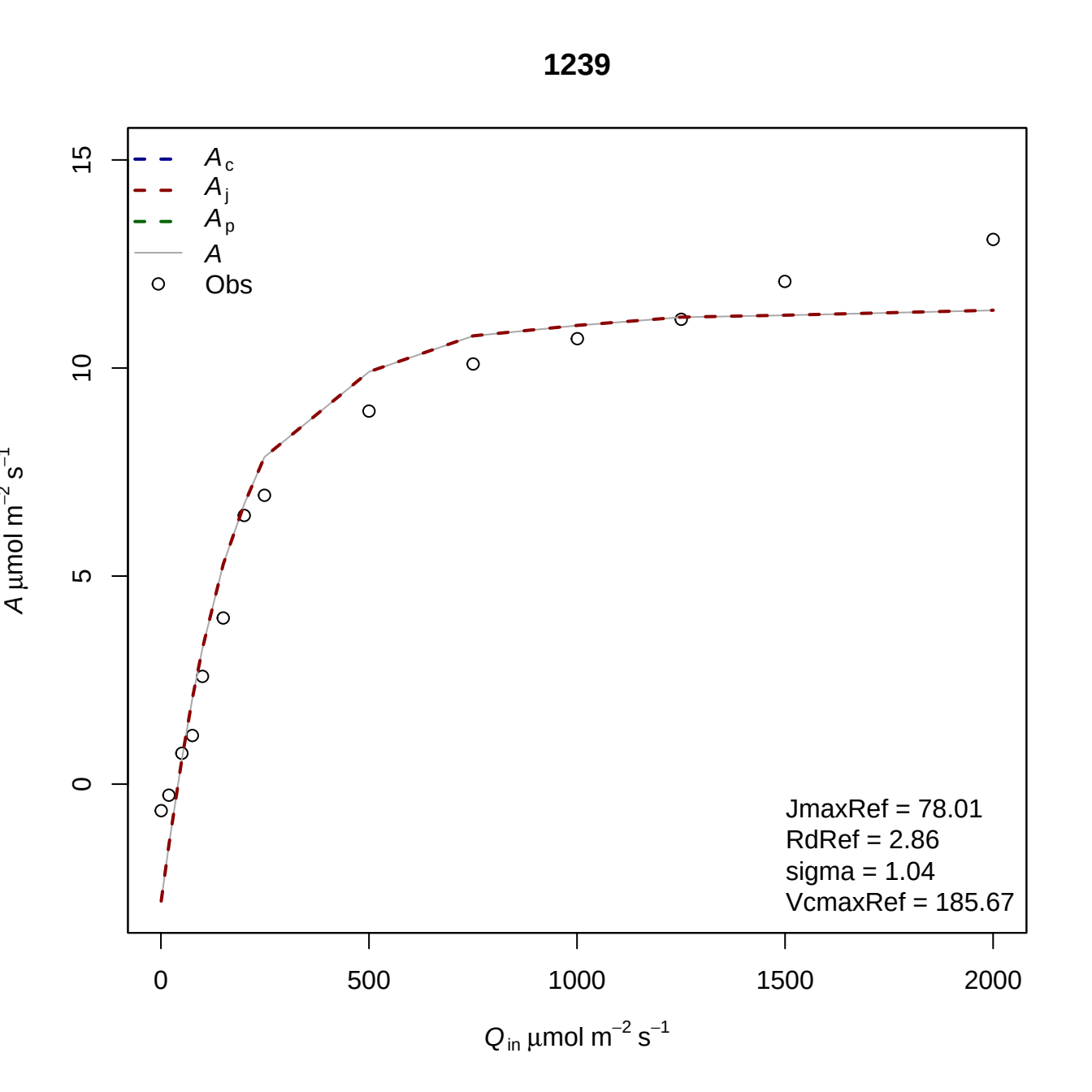

1234
A_c
A_j
A_p
A
Obs
JmaxRef = 120.55
RdRef = 2.27
sigma = 0.33
VcmaxRef = 300.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

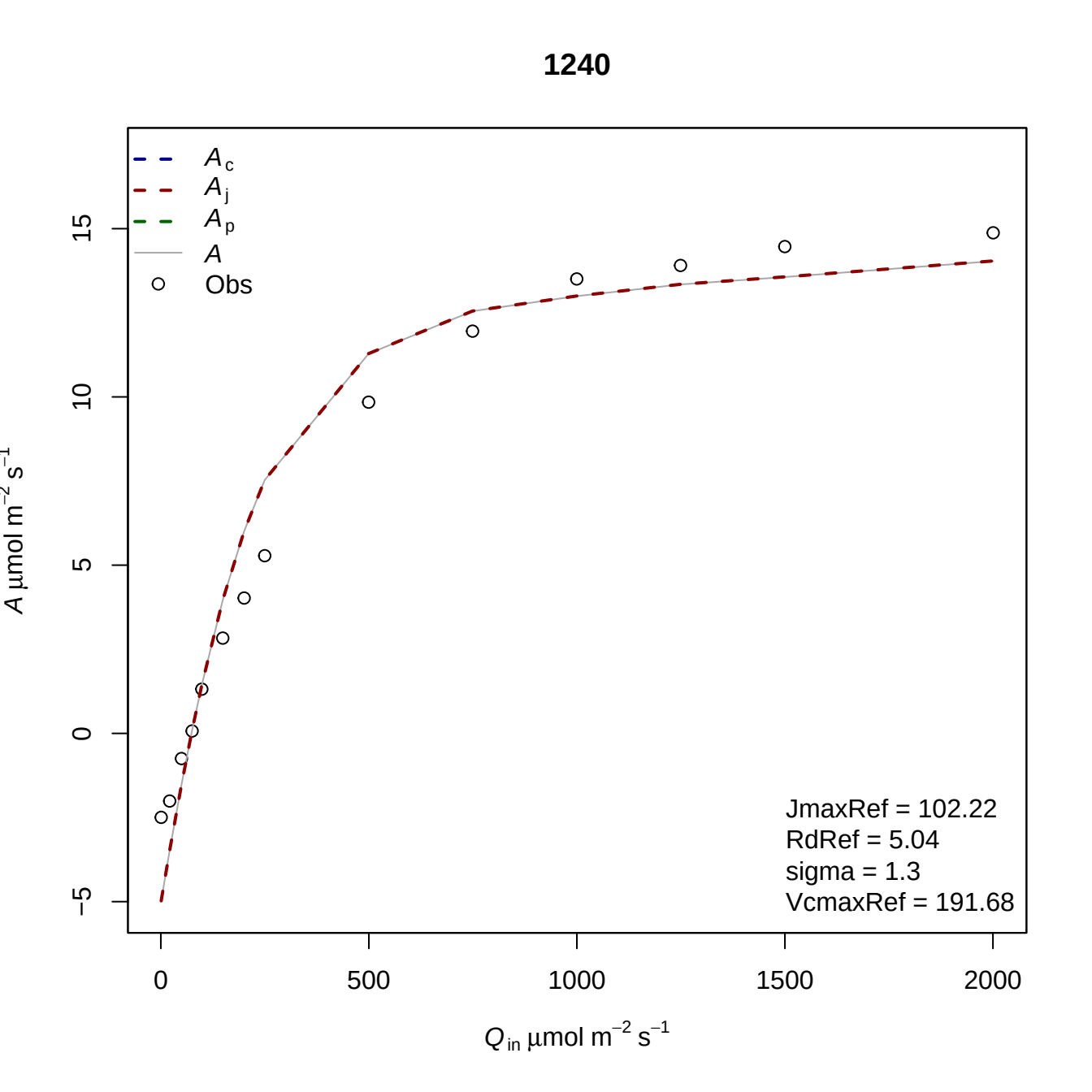

1235
A_c
A_j
A_p
A
Obs
JmaxRef = 123.95
RdRef = 2.88
sigma = 0.85
VcmaxRef = 260.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

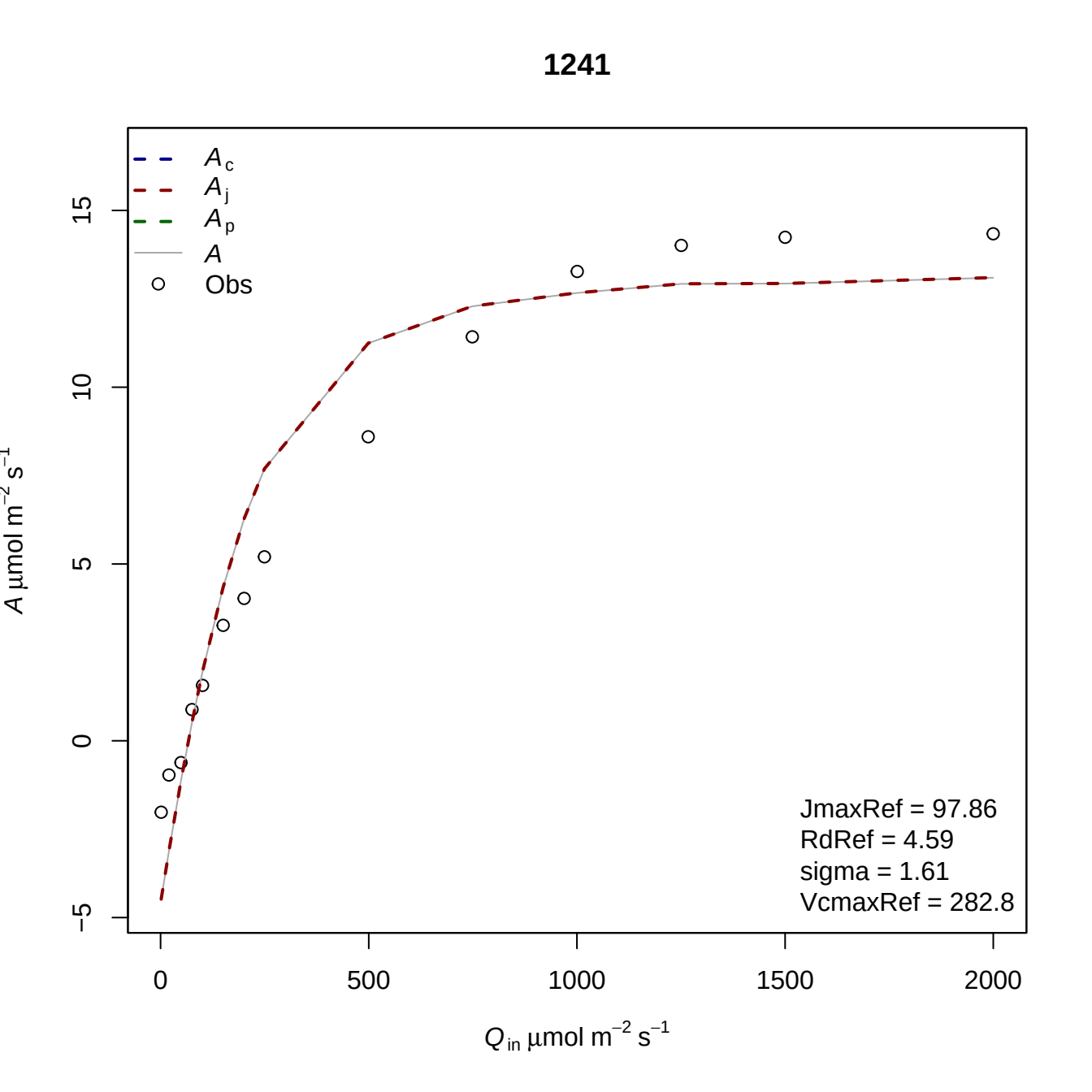

1236
A_c
A_j
A_p
A
Obs
JmaxRef = 166.12
RdRef = 2.92
sigma = 0.81
VcmaxRef = 312.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

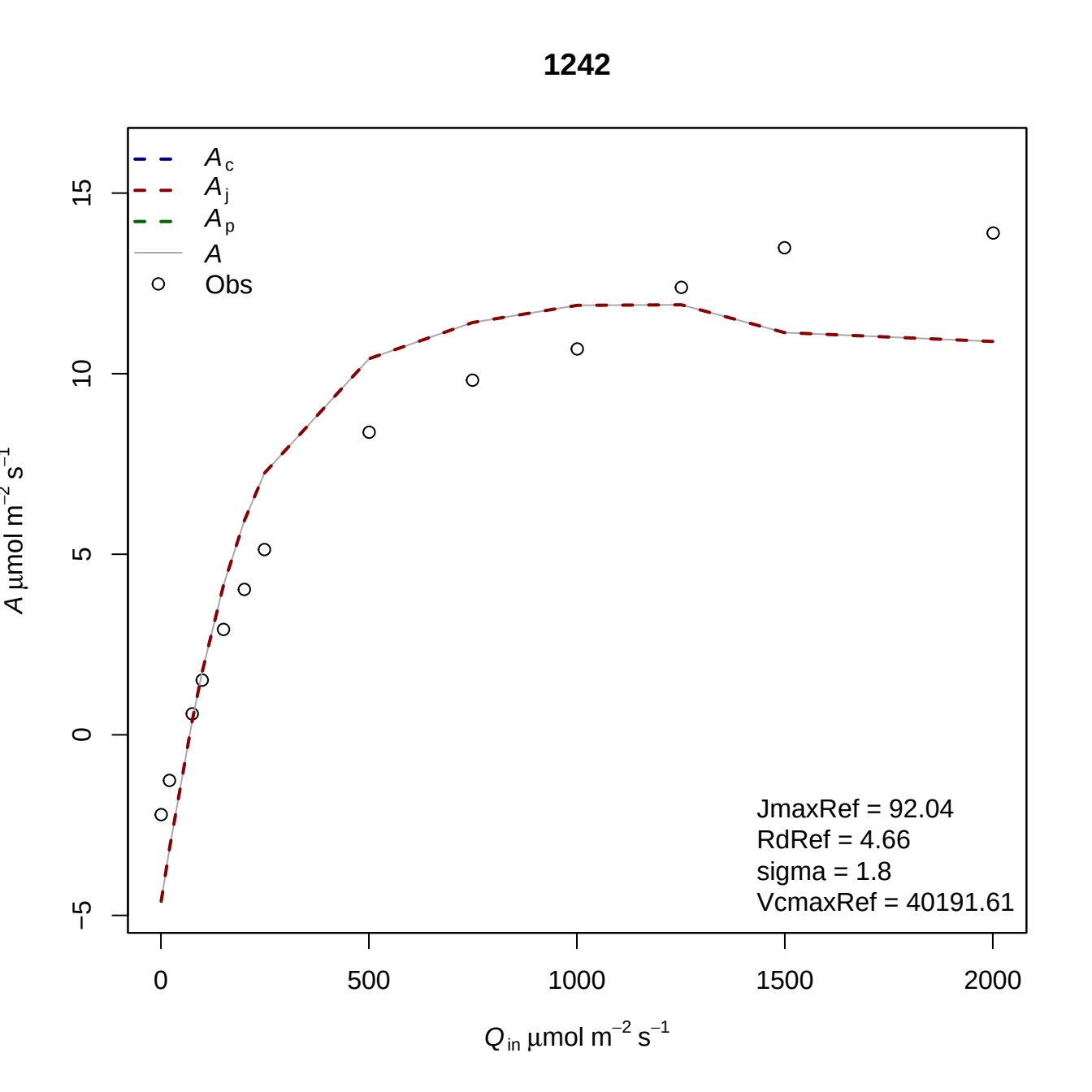

1237
A_c
A_j
A_p
A
Obs
JmaxRef = 132.63
RdRef = 2.21
sigma = 0.44
VcmaxRef = 90.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

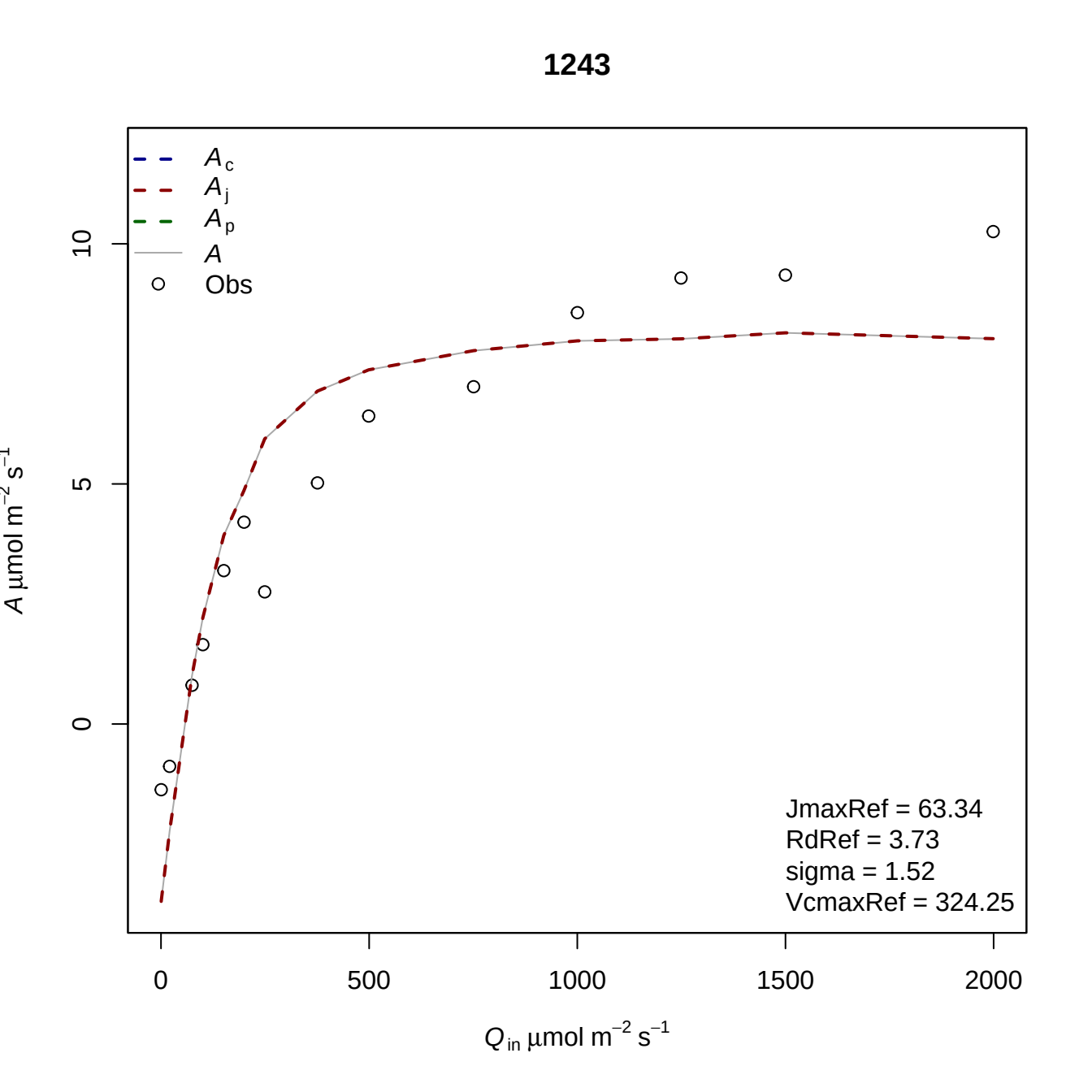

1238
A_c
A_j
A_p
A
Obs
JmaxRef = 110.27
RdRef = 3.18
sigma = 1.54
VcmaxRef = 225.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

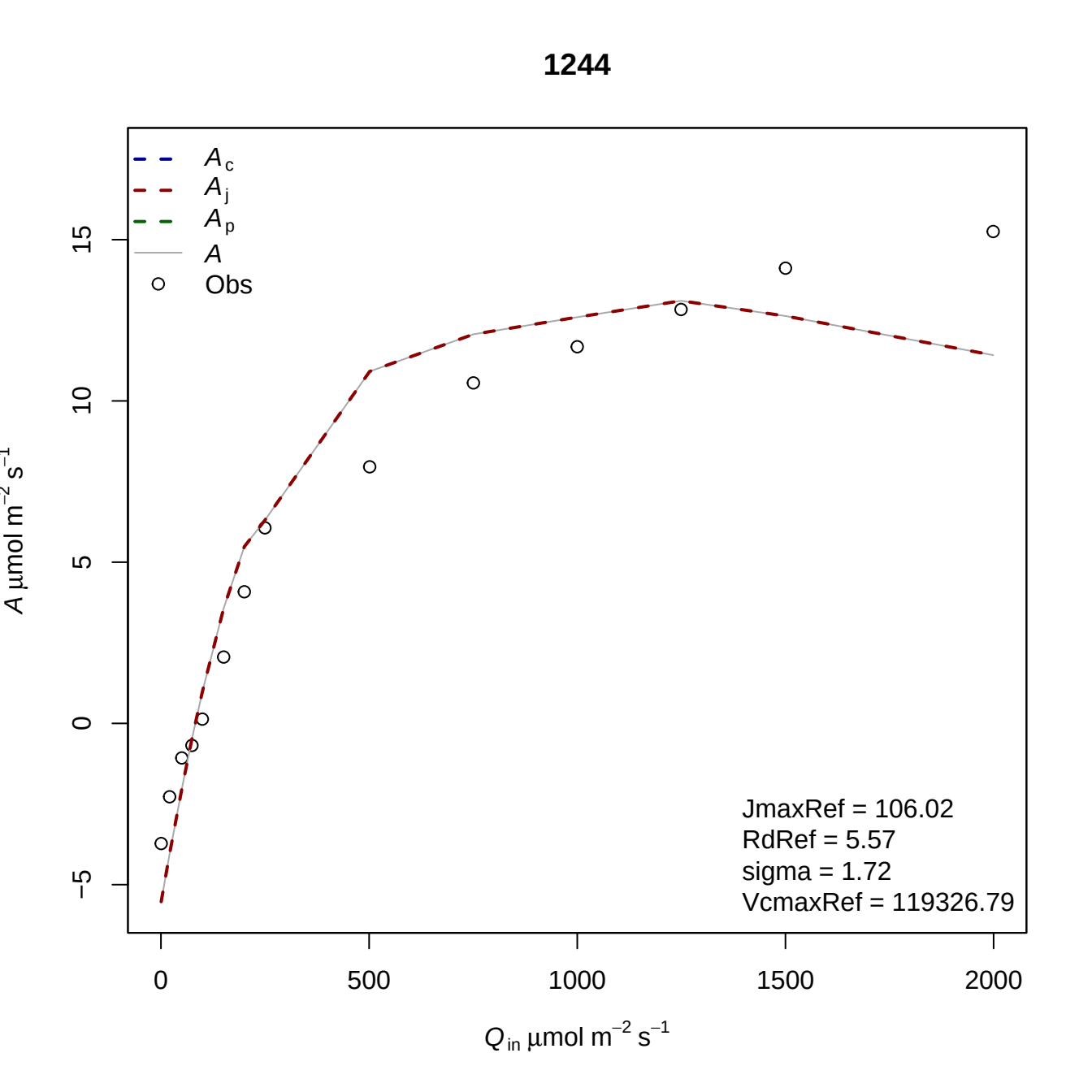

1239
A_c
A_j
A_p
A
Obs
JmaxRef = 78.01
RdRef = 2.86
sigma = 1.04
VcmaxRef = 185.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

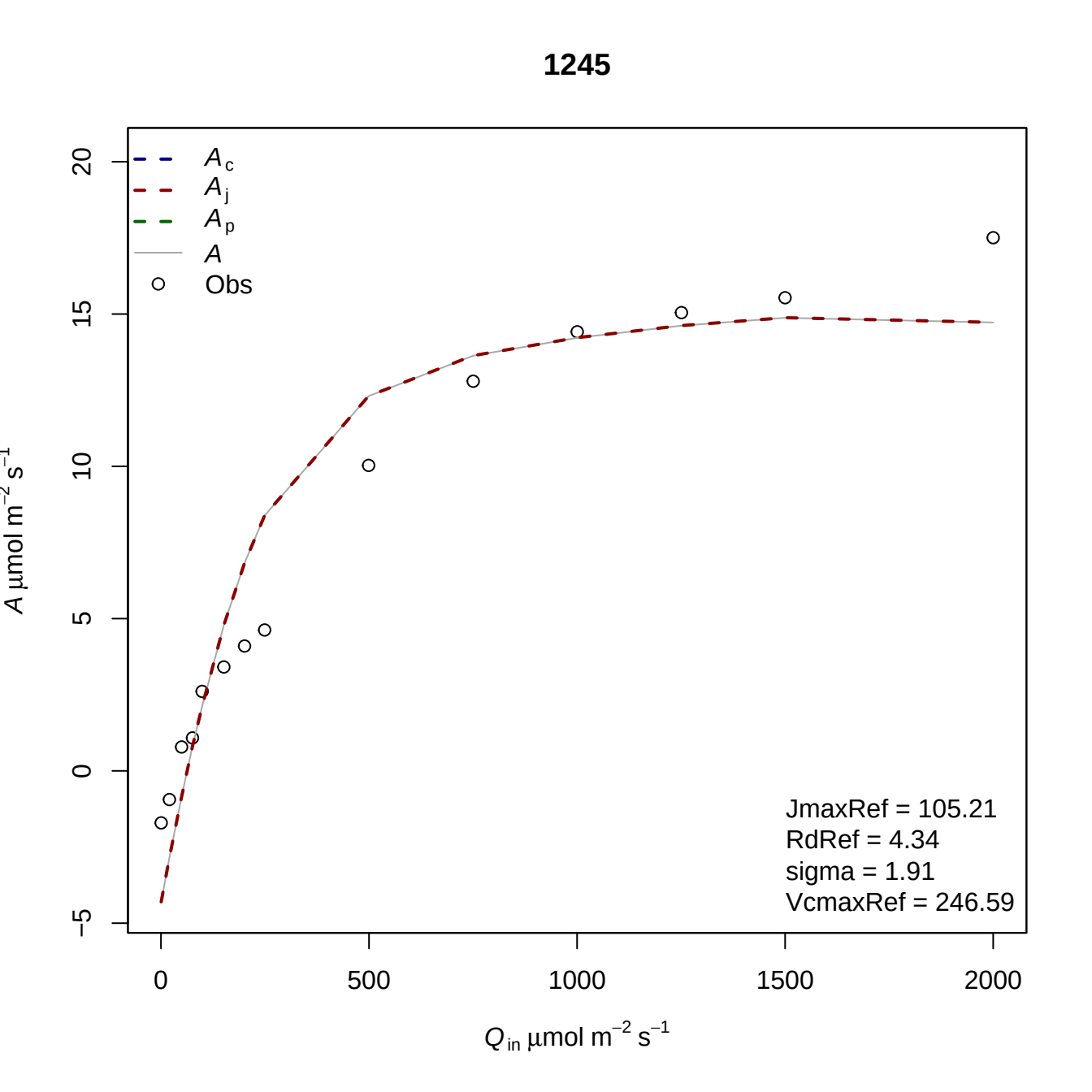

1240
A_c
A_j
A_p
A
Obs
JmaxRef = 102.22
RdRef = 5.04
sigma = 1.3
VcmaxRef = 191.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

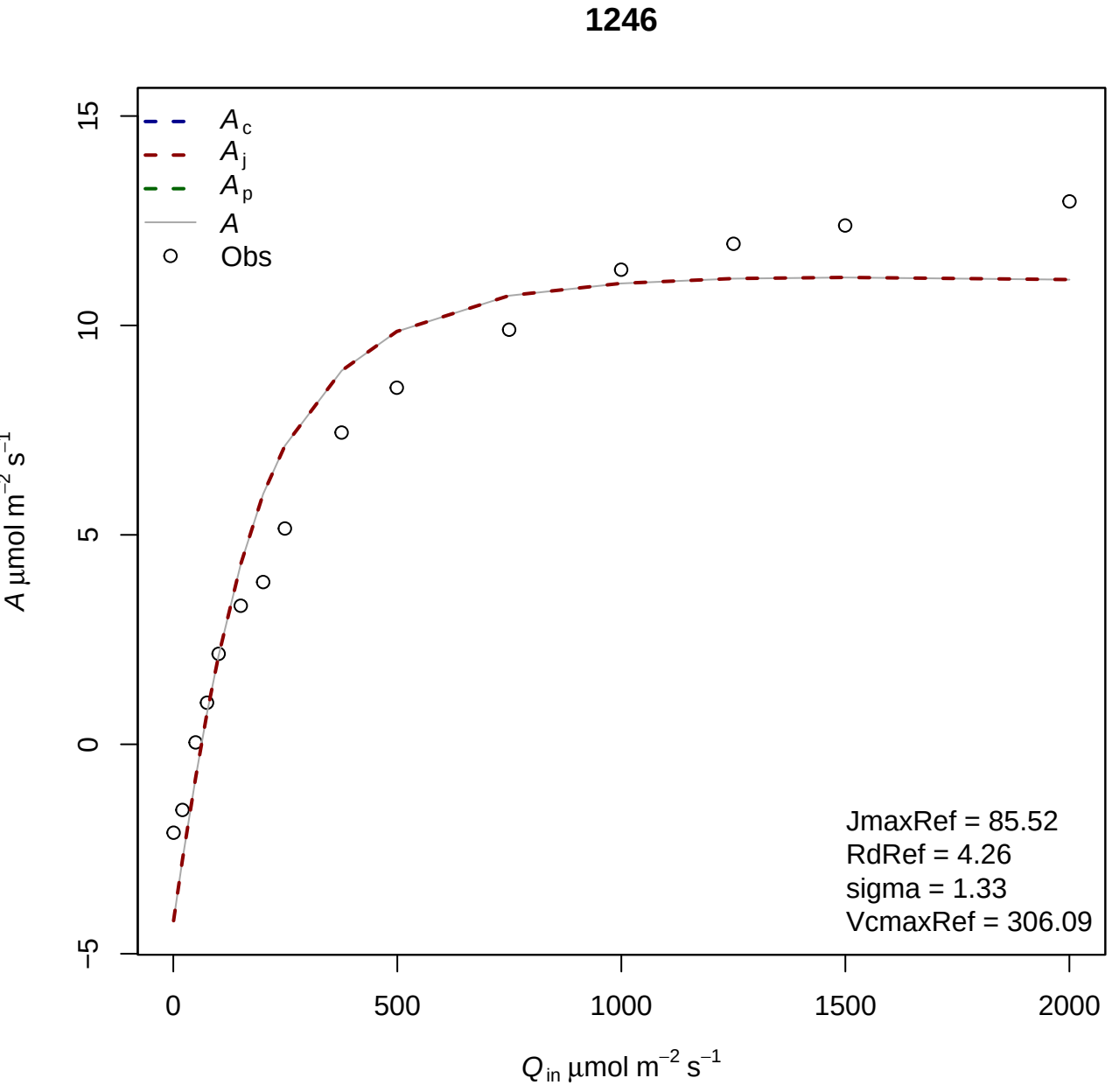

1241
A_c
A_j
A_p
A
Obs
JmaxRef = 97.86
RdRef = 4.59
sigma = 1.61
VcmaxRef = 282.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

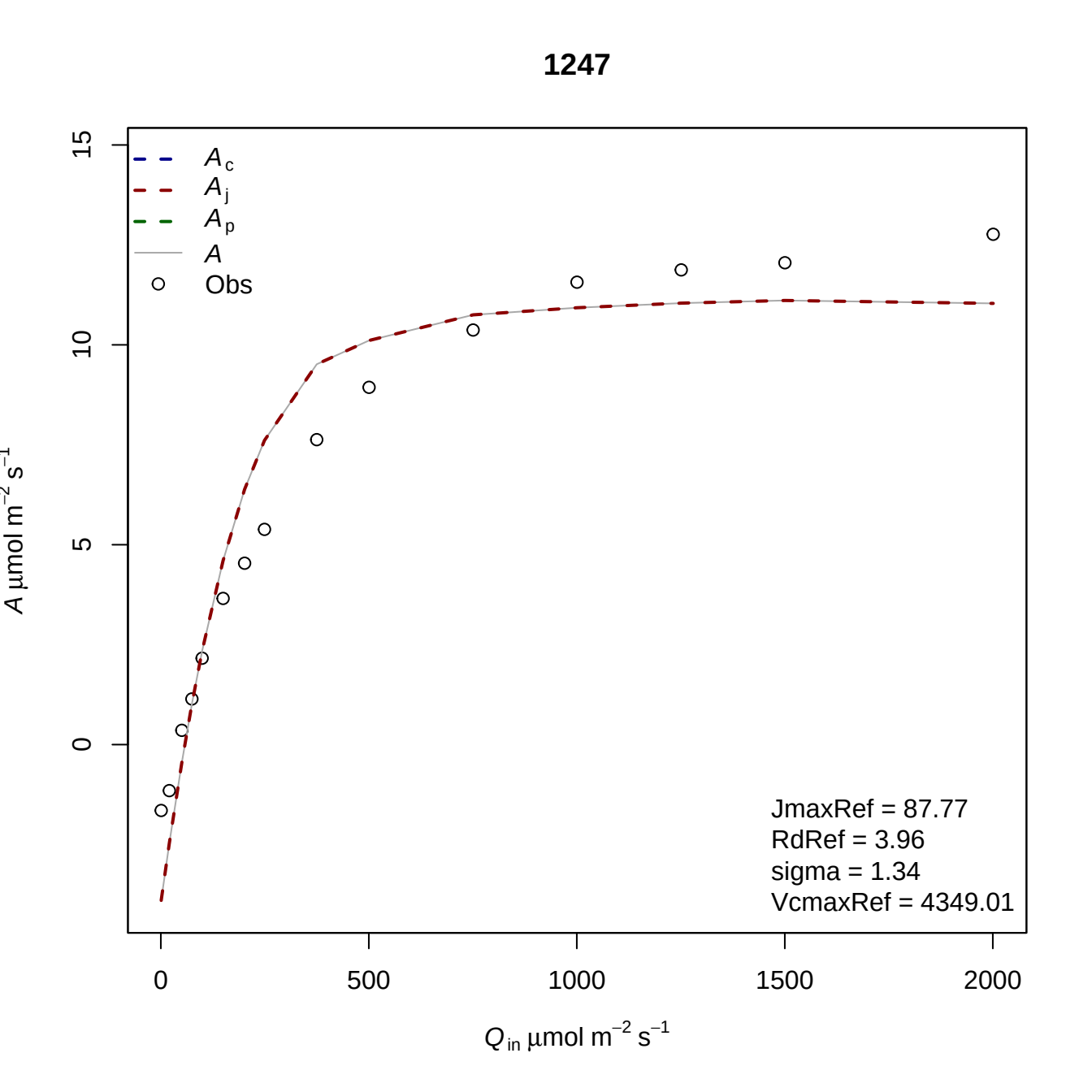

1242
A_c
A_j
A_p
A
Obs
JmaxRef = 92.04
RdRef = 4.66
sigma = 1.8
VcmaxRef = 40191.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

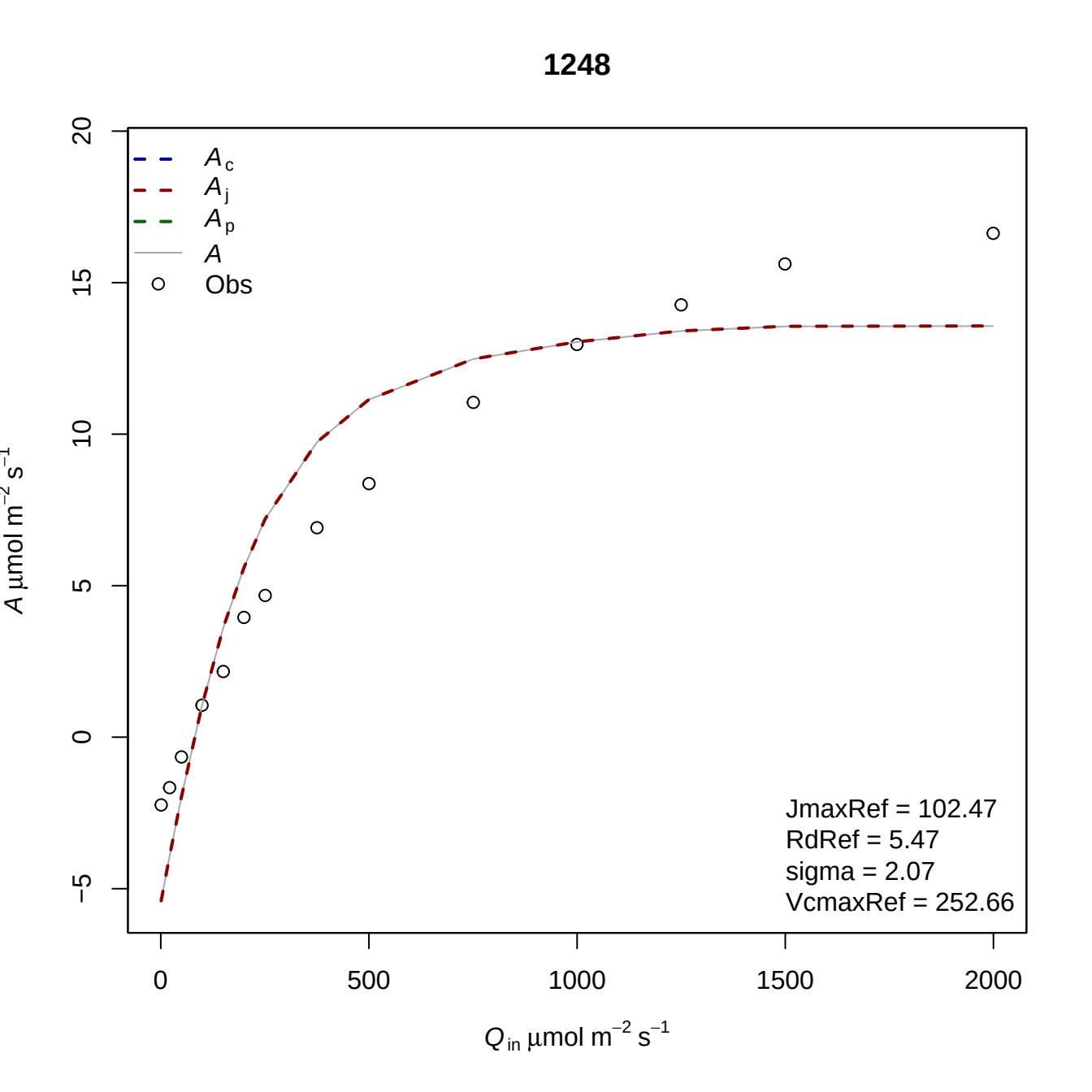

1243
A_c
A_j
A_p
A
Obs
JmaxRef = 63.34
RdRef = 3.73
sigma = 1.52
VcmaxRef = 324.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

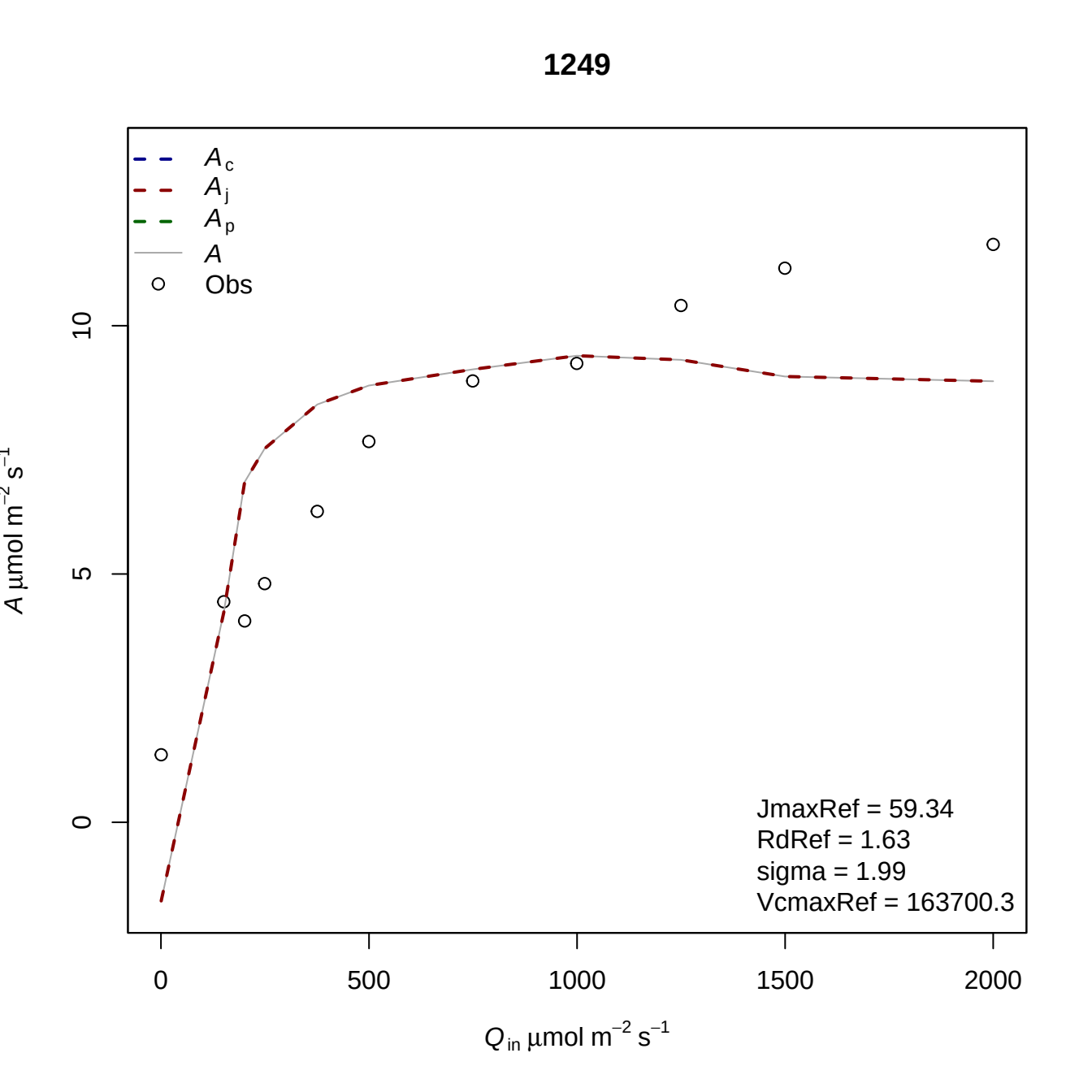

1244
A_c
A_j
A_p
A
Obs
JmaxRef = 106.02
RdRef = 5.57
sigma = 1.72
VcmaxRef = 119326.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

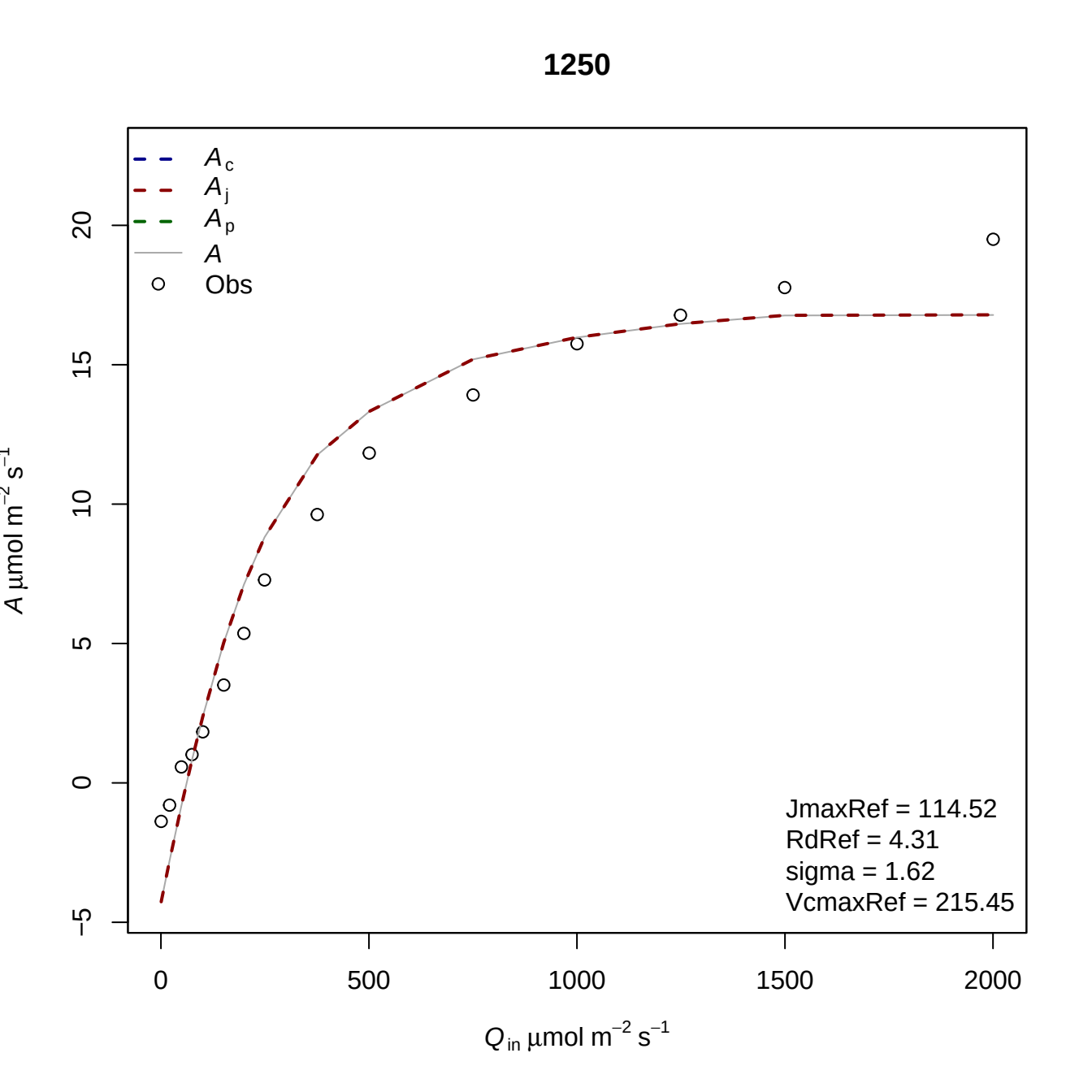

1245
A_c
A_j
A_p
A
Obs
JmaxRef = 105.21
RdRef = 4.34
sigma = 1.91
VcmaxRef = 246.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

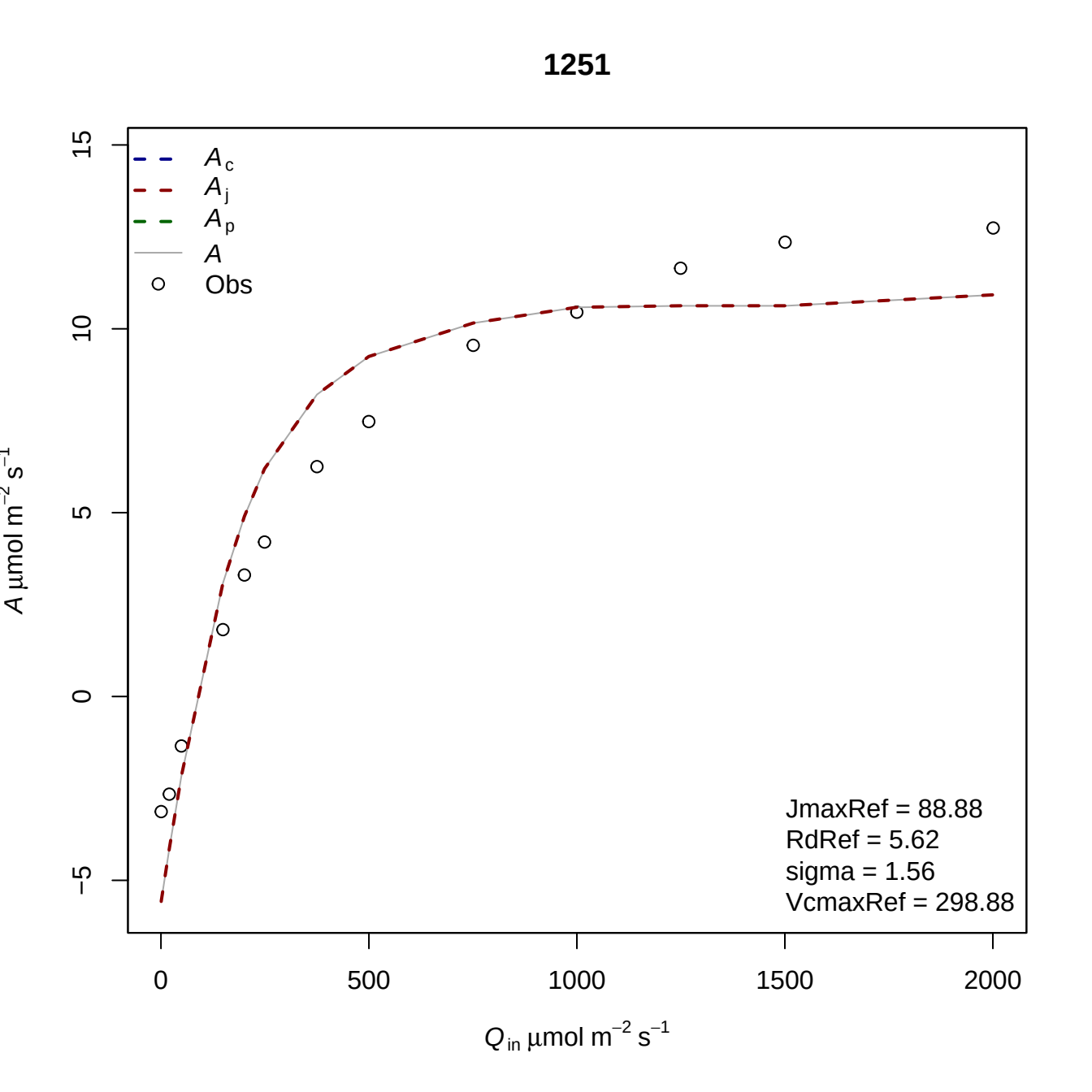

1246
A_c
A_j
A_p
A
Obs
JmaxRef = 85.52
RdRef = 4.26
sigma = 1.33
VcmaxRef = 306.09
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

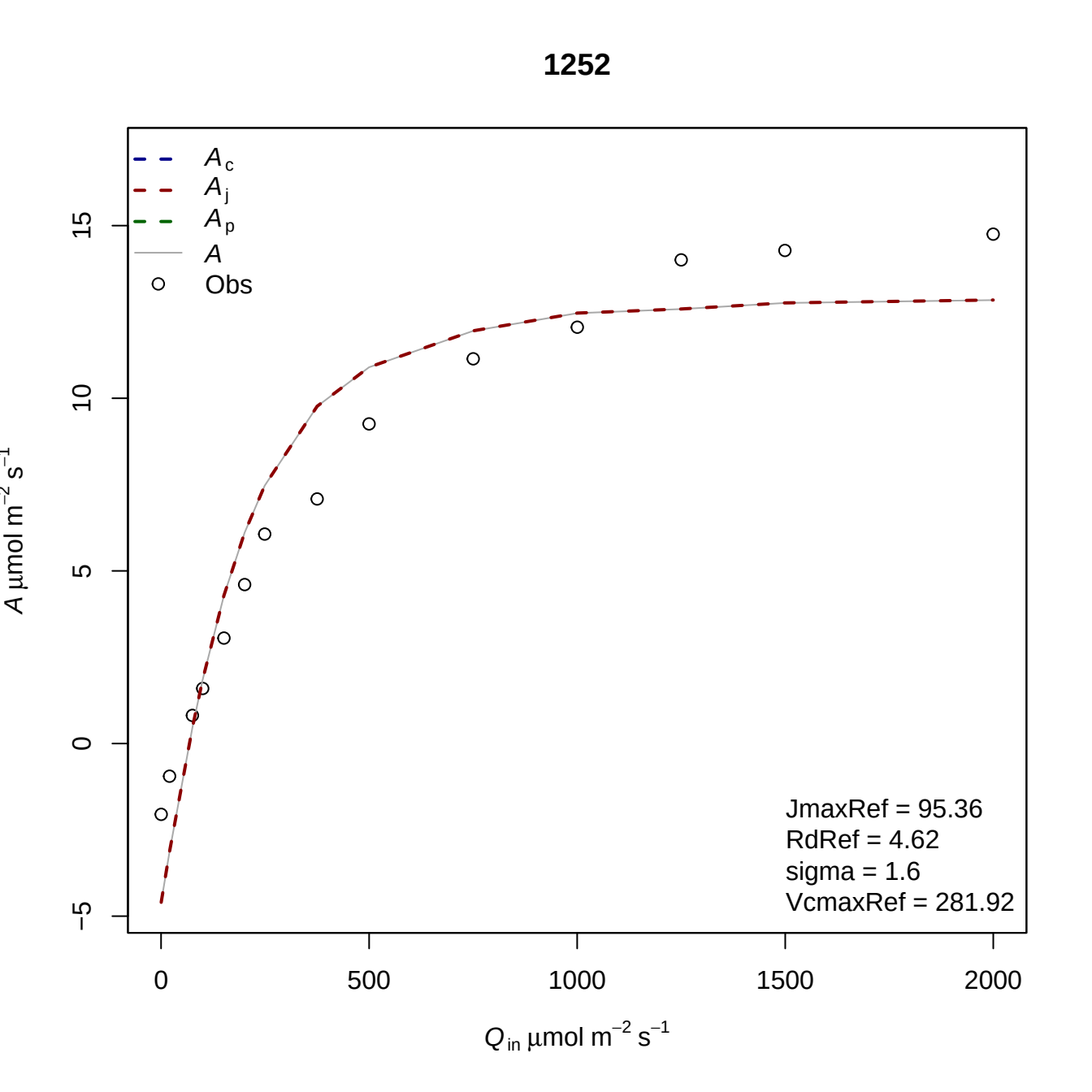

1247
A_c
A_j
A_p
A
Obs
JmaxRef = 87.77
RdRef = 3.96
sigma = 1.34
VcmaxRef = 4349.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

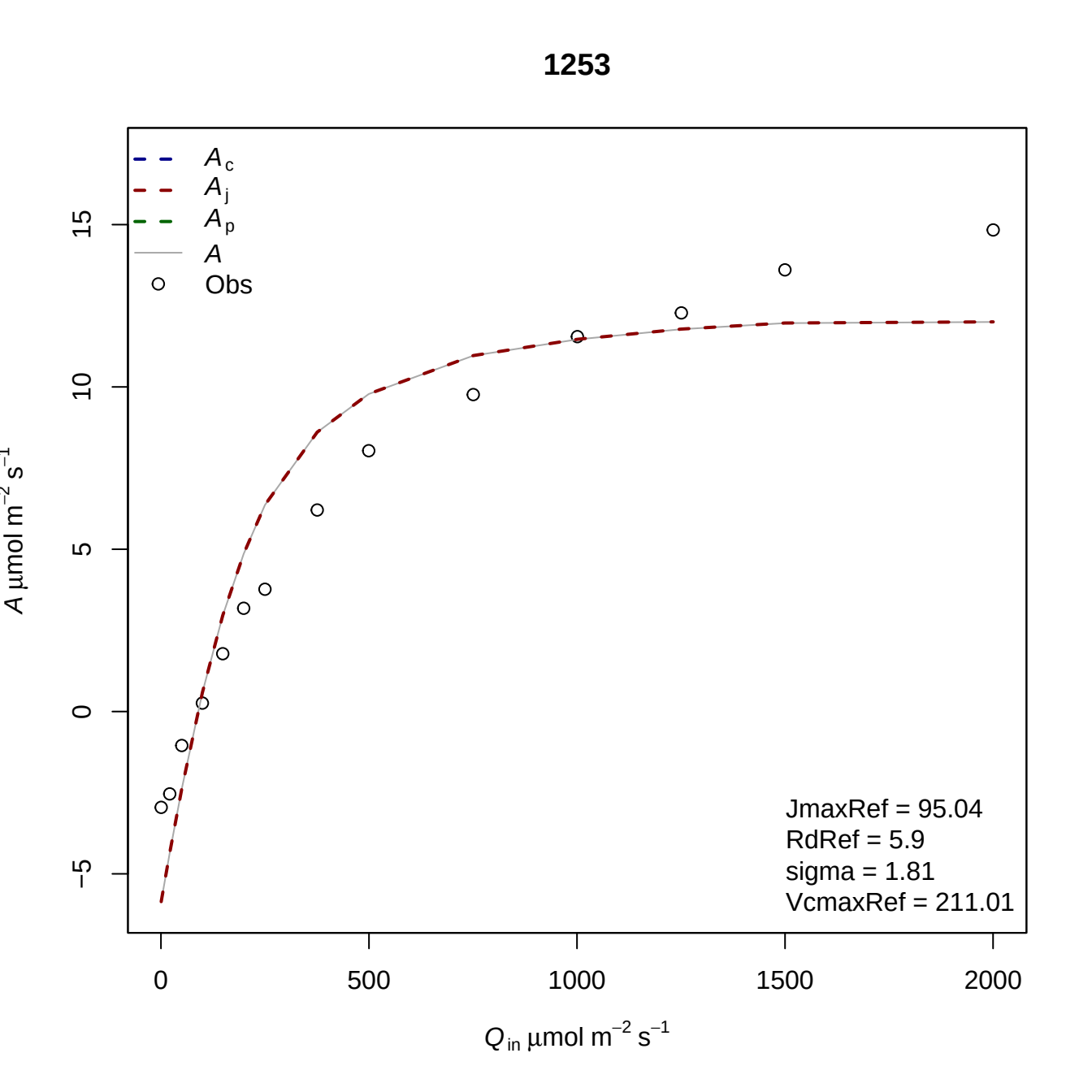

1248
A_c
A_j
A_p
A
Obs
JmaxRef = 102.47
RdRef = 5.47
sigma = 2.07
VcmaxRef = 252.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

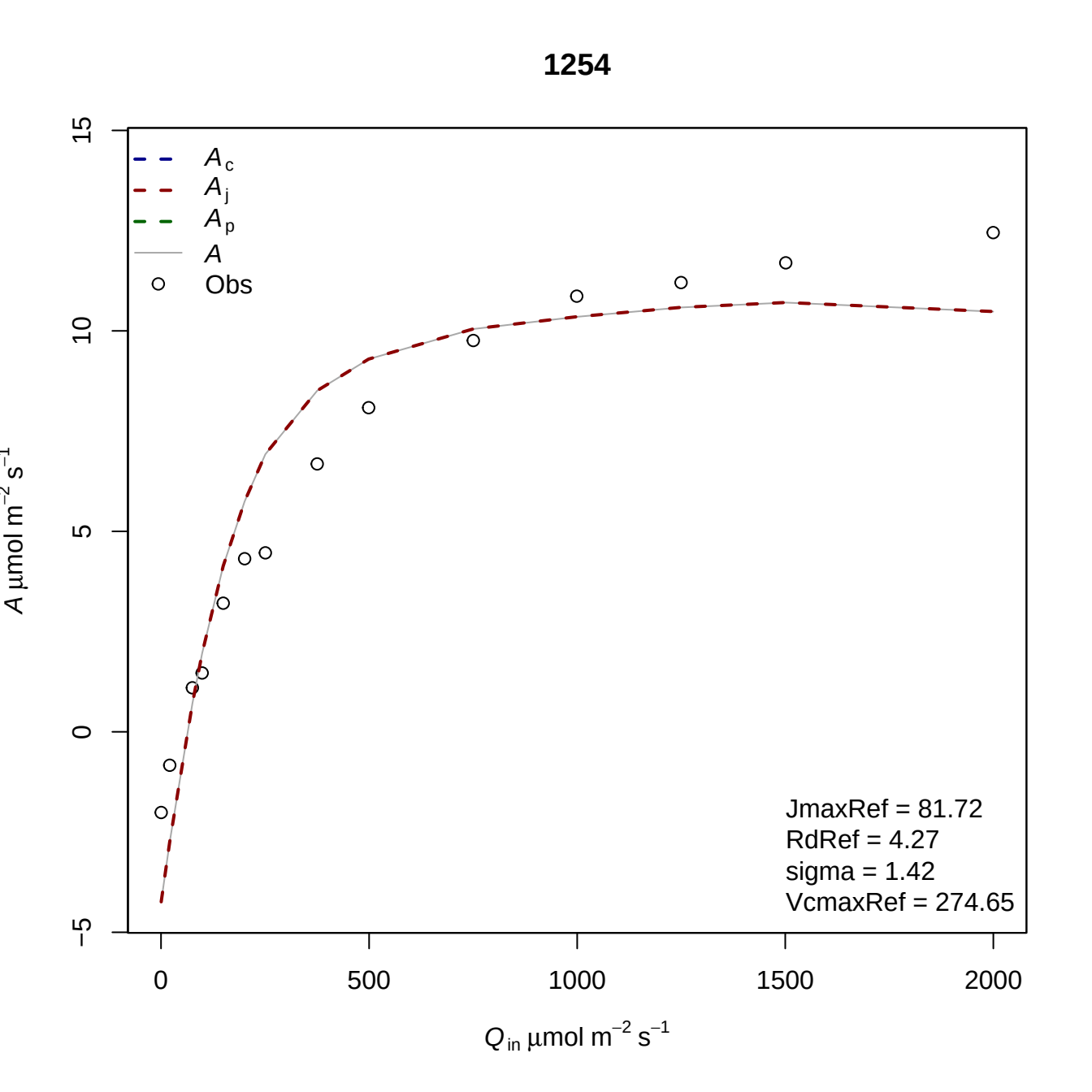

1249
A_c
A_j
A_p
A
Obs
JmaxRef = 59.34
RdRef = 1.63
sigma = 1.99
VcmaxRef = 163700.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

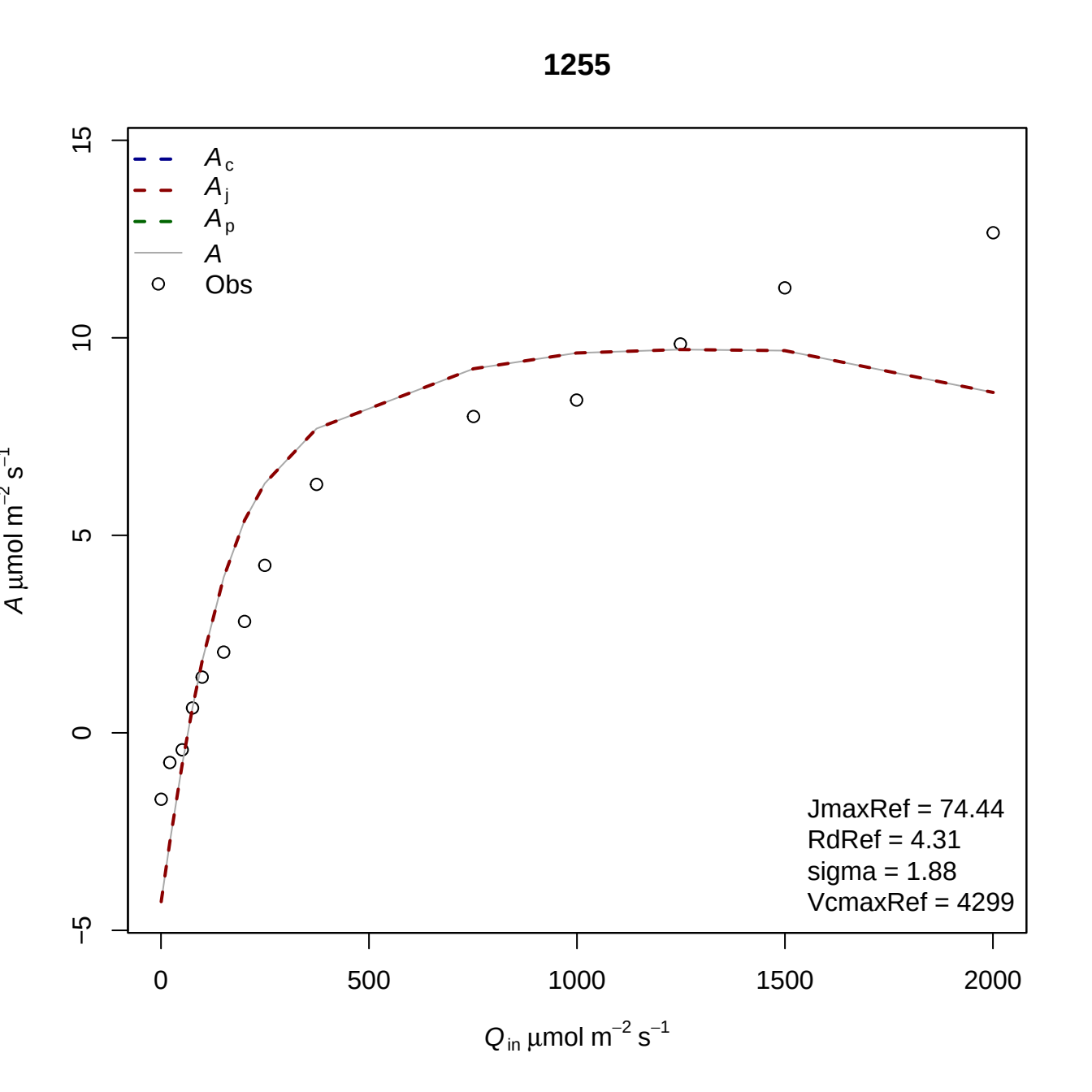

1250
A_c
A_j
A_p
A
Obs
JmaxRef = 114.52
RdRef = 4.31
sigma = 1.62
VcmaxRef = 215.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

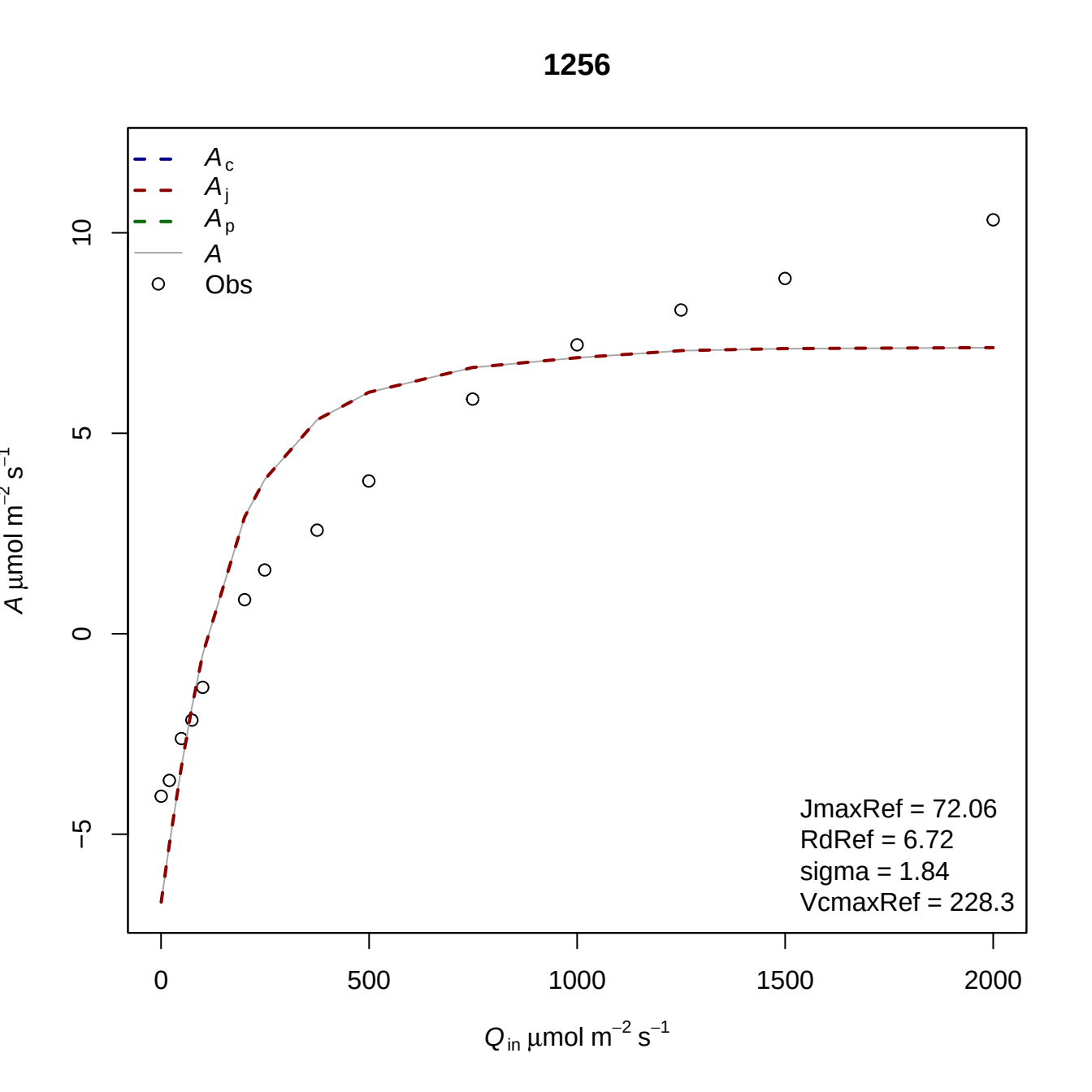

1251
A_c
A_j
A_p
A
Obs
JmaxRef = 88.88
RdRef = 5.62
sigma = 1.56
VcmaxRef = 298.88
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

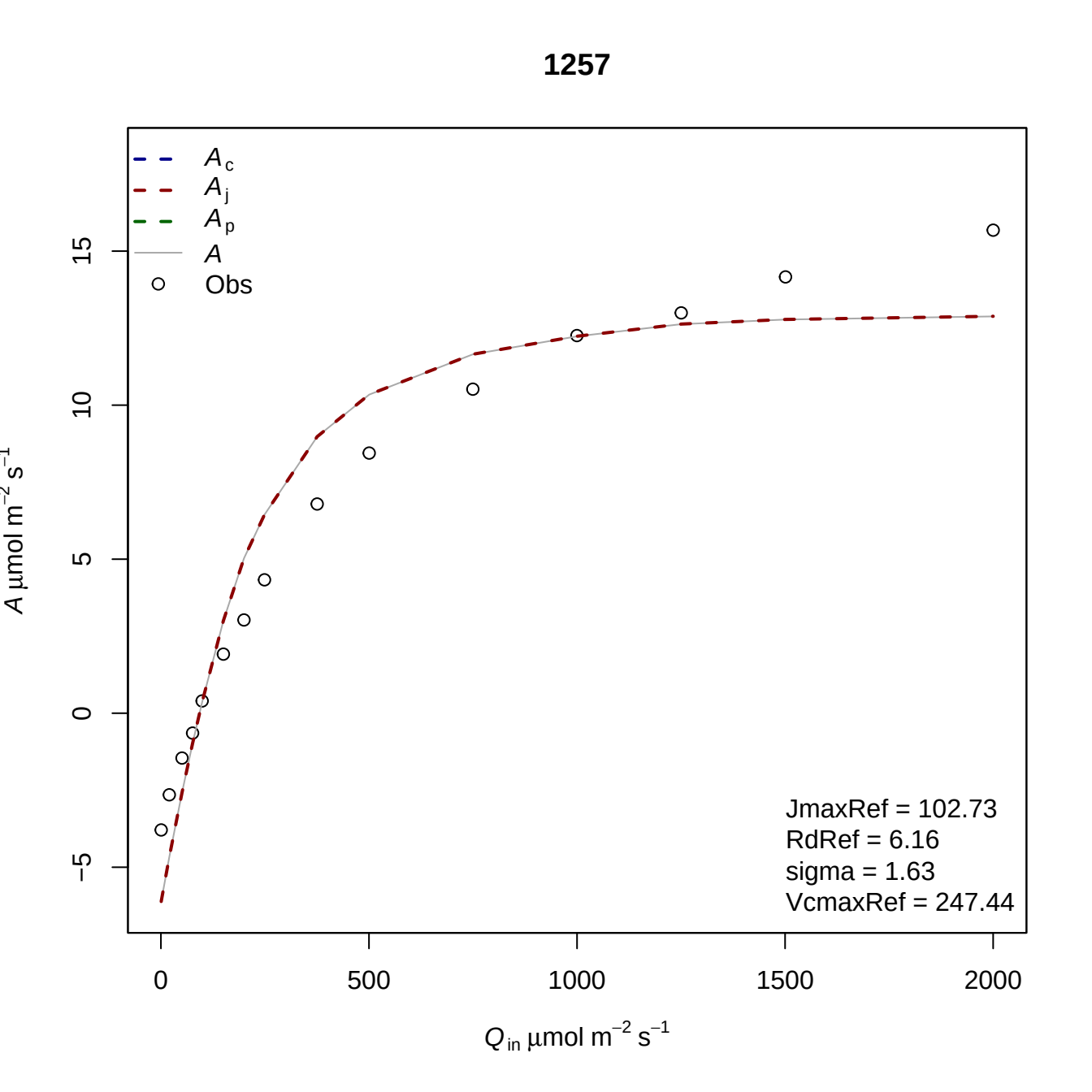

1252
A_c
A_j
A_p
A
Obs
JmaxRef = 95.36
RdRef = 4.62
sigma = 1.6
VcmaxRef = 281.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

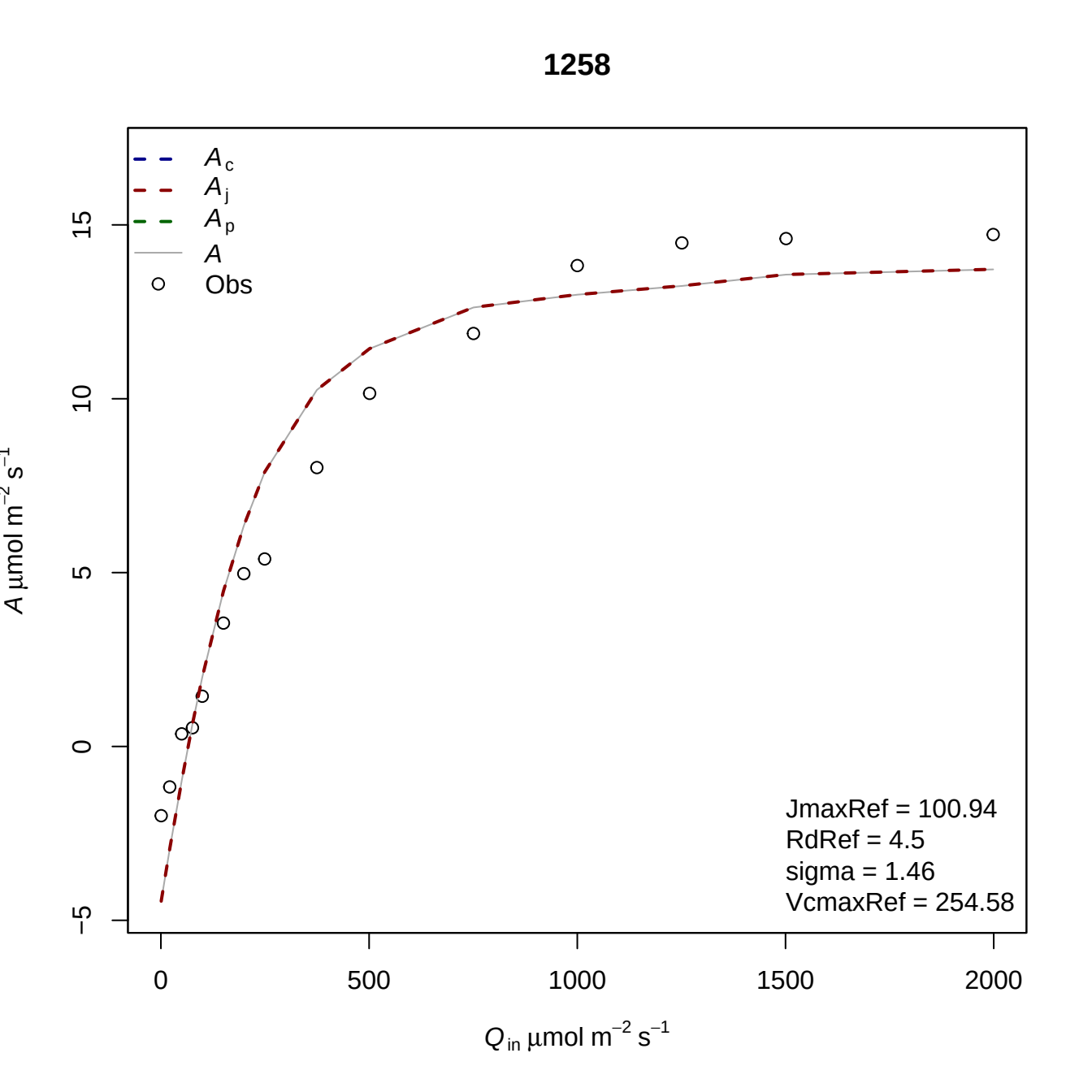

1253
A_c
A_j
A_p
A
Obs
JmaxRef = 95.04
RdRef = 5.9
sigma = 1.81
VcmaxRef = 211.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

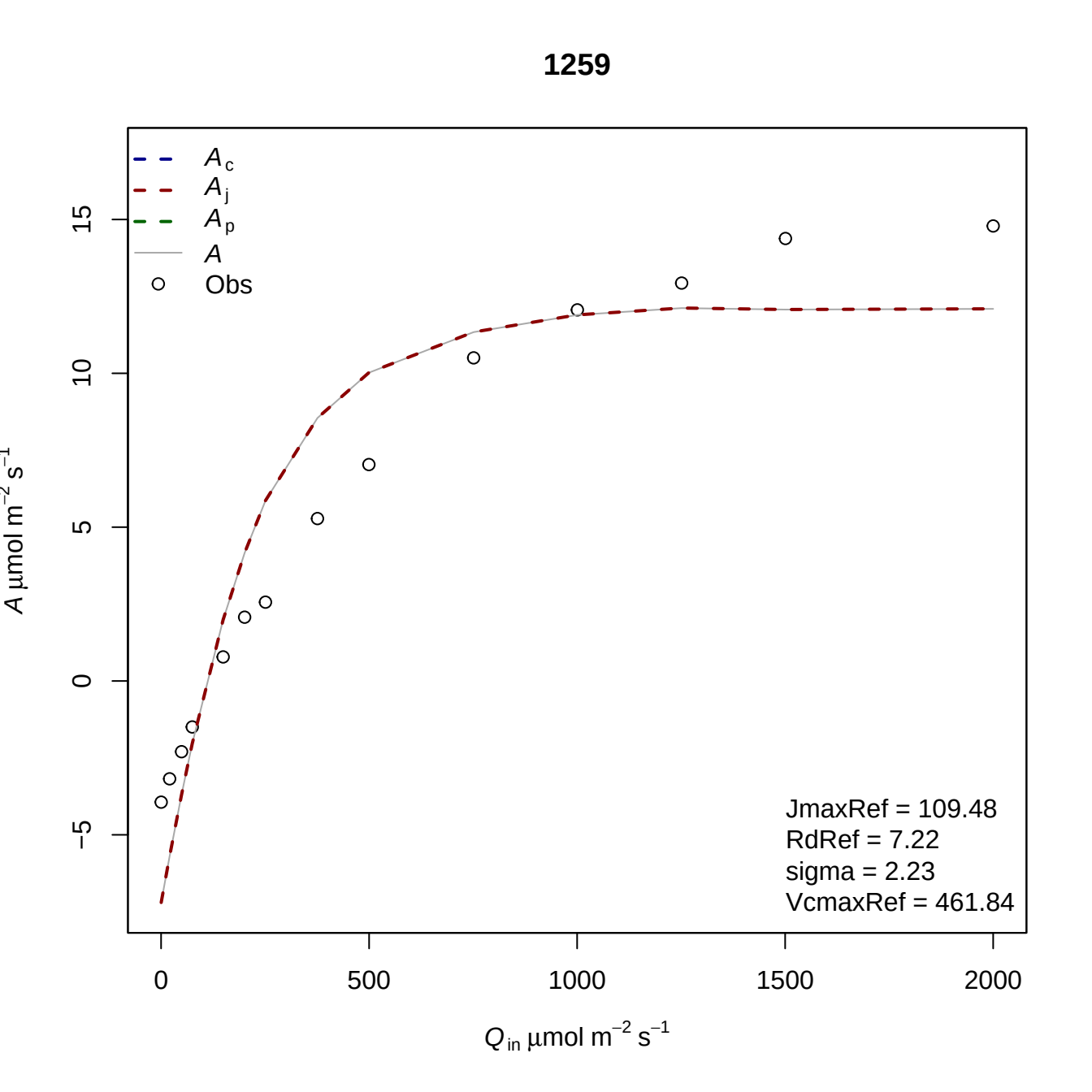

1254
A_c
A_j
A_p
A
Obs
JmaxRef = 81.72
RdRef = 4.27
sigma = 1.42
VcmaxRef = 274.65
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

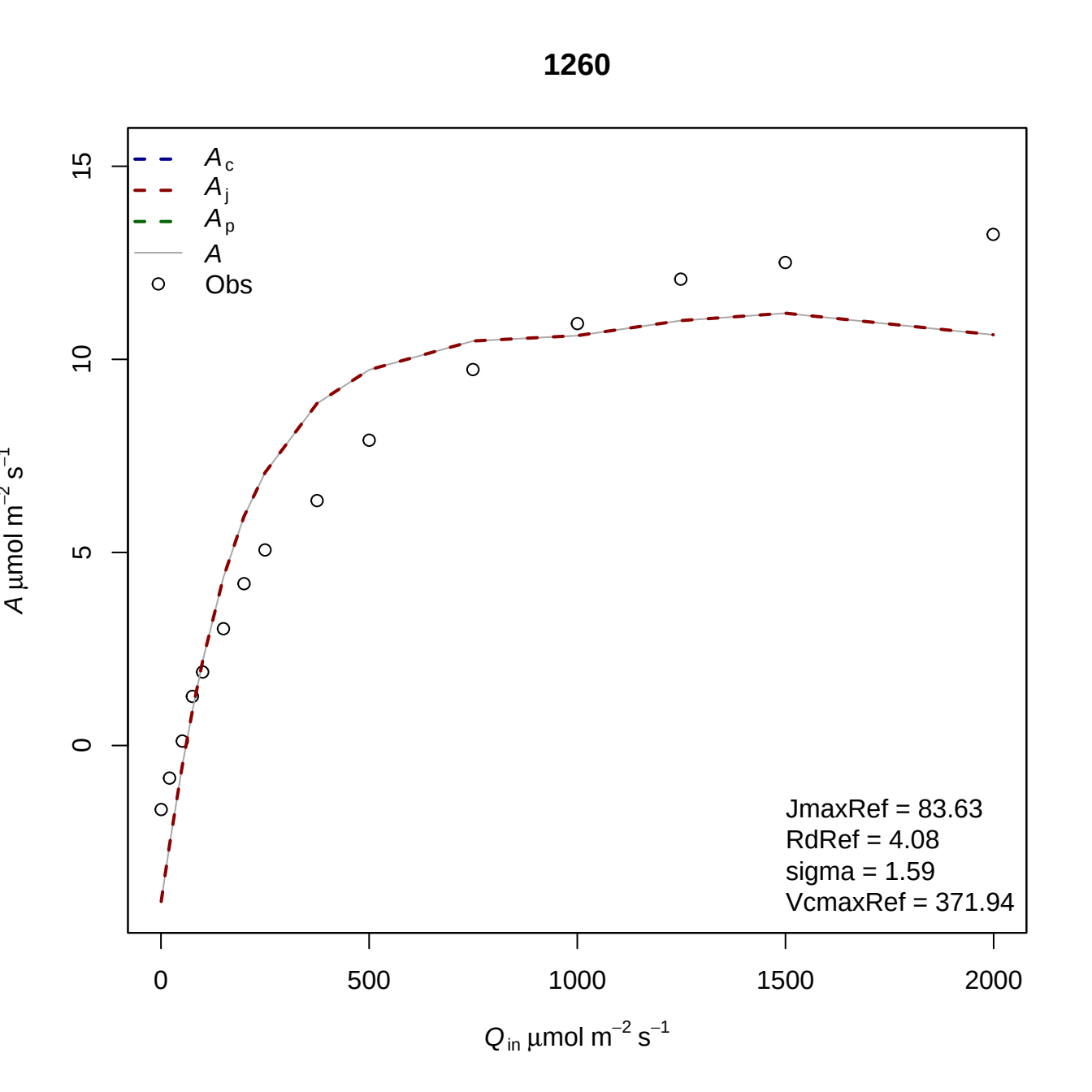

1255
A_c
A_j
A_p
A
Obs
JmaxRef = 74.44
RdRef = 4.31
sigma = 1.88
VcmaxRef = 4299
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

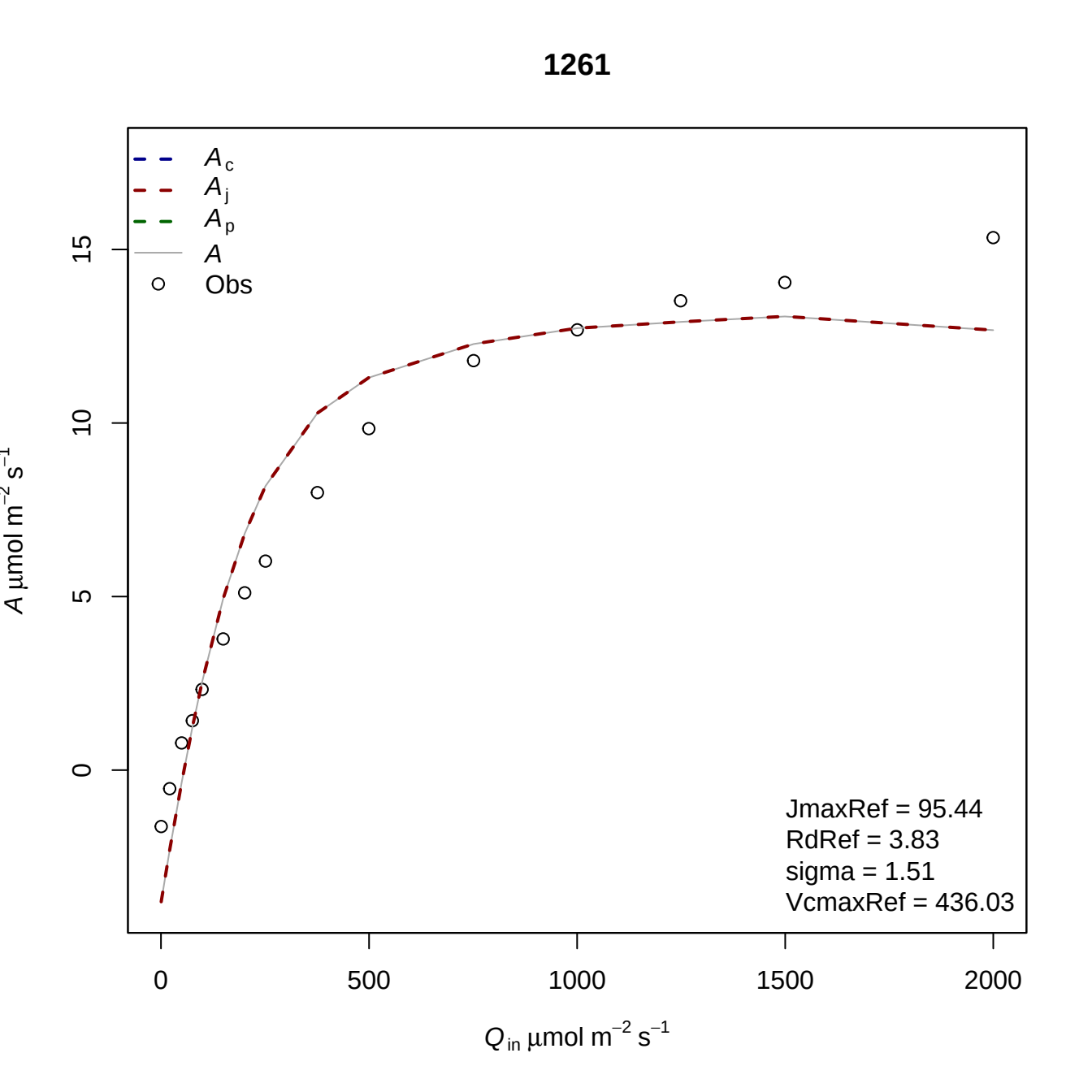

1256
A_c
A_j
A_p
A
Obs
JmaxRef = 72.06
RdRef = 6.72
sigma = 1.84
VcmaxRef = 228.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

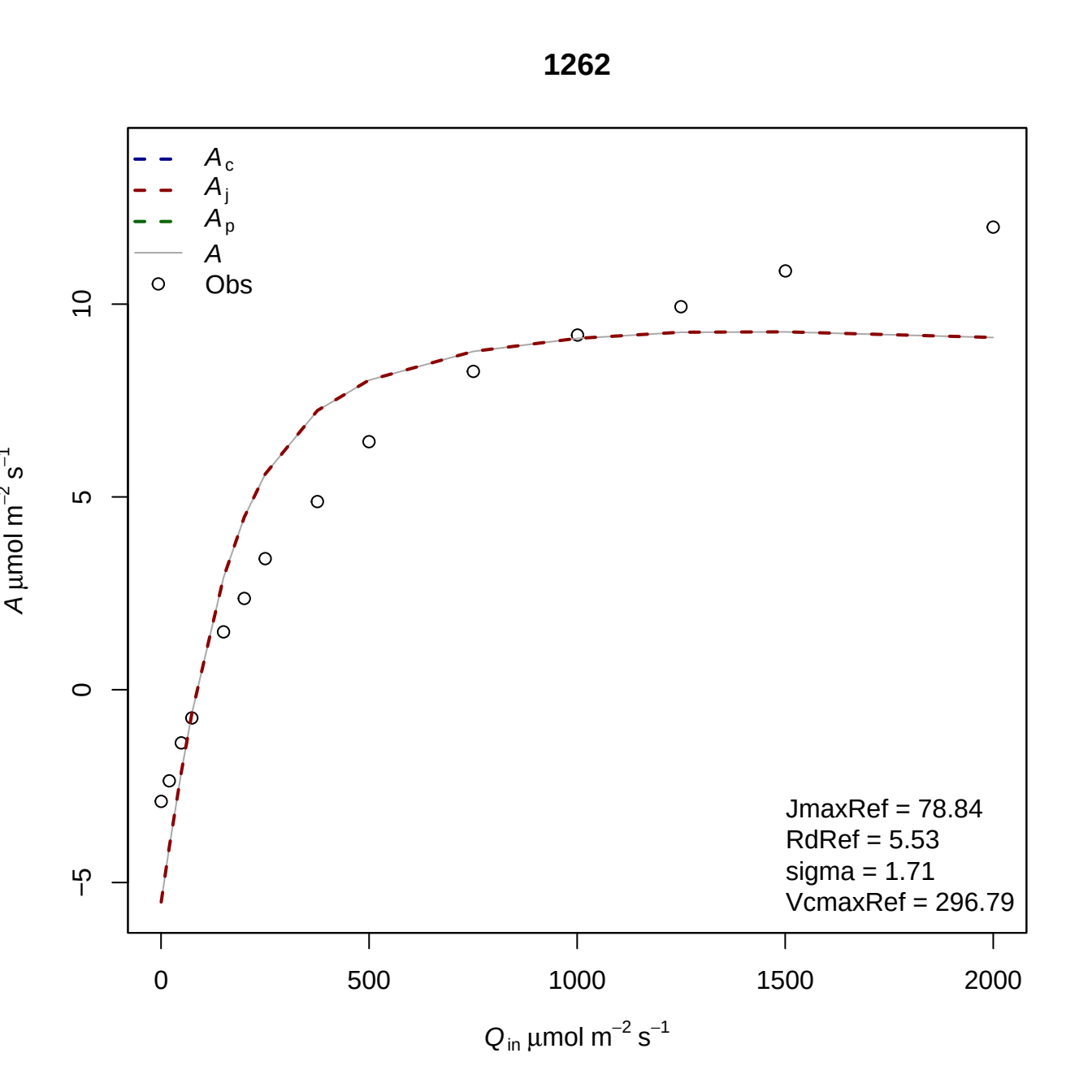

1257
A_c
A_j
A_p
A
Obs
JmaxRef = 102.73
RdRef = 6.16
sigma = 1.63
VcmaxRef = 247.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

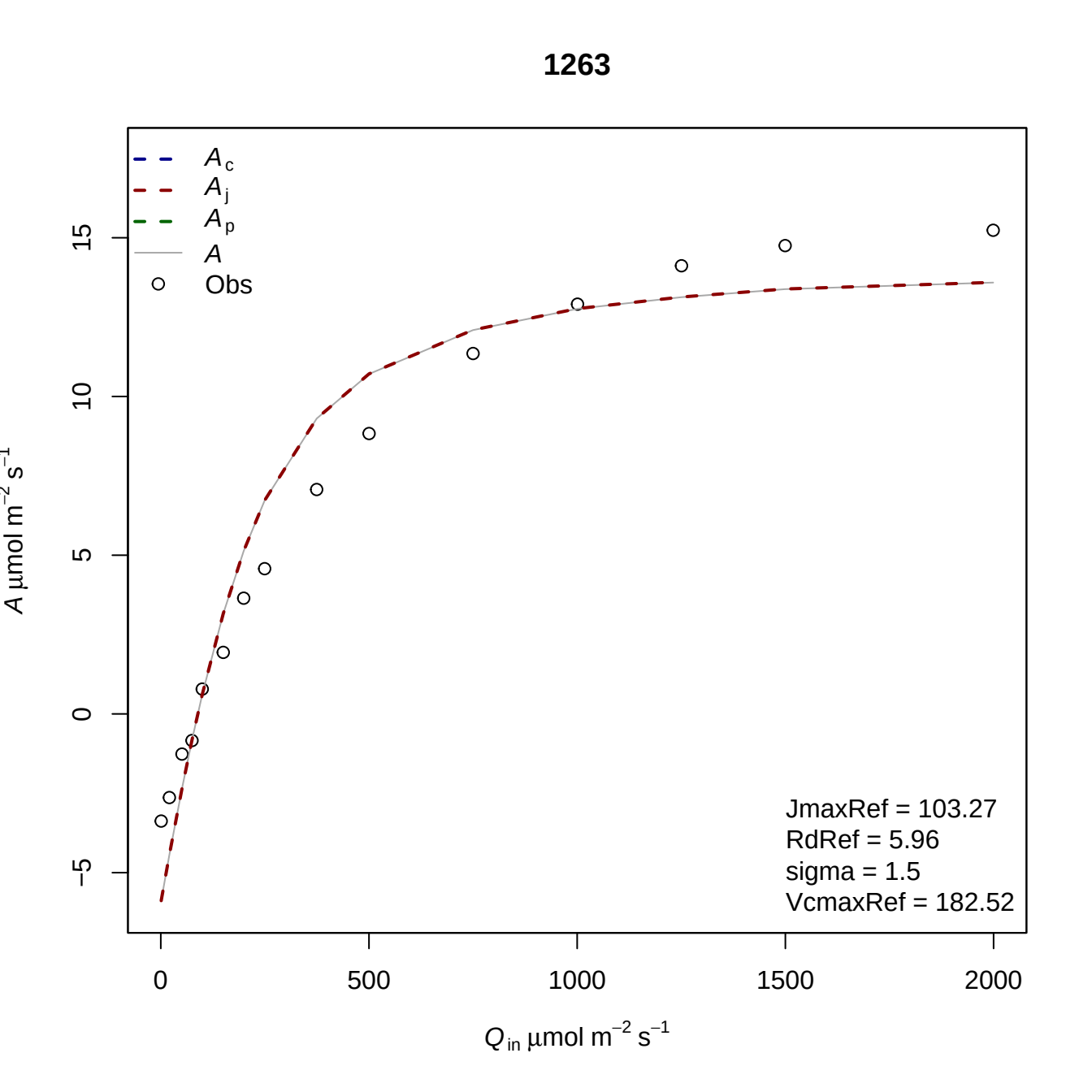

1258
A_c
A_j
A_p
A
Obs
JmaxRef = 100.94
RdRef = 4.5
sigma = 1.46
VcmaxRef = 254.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

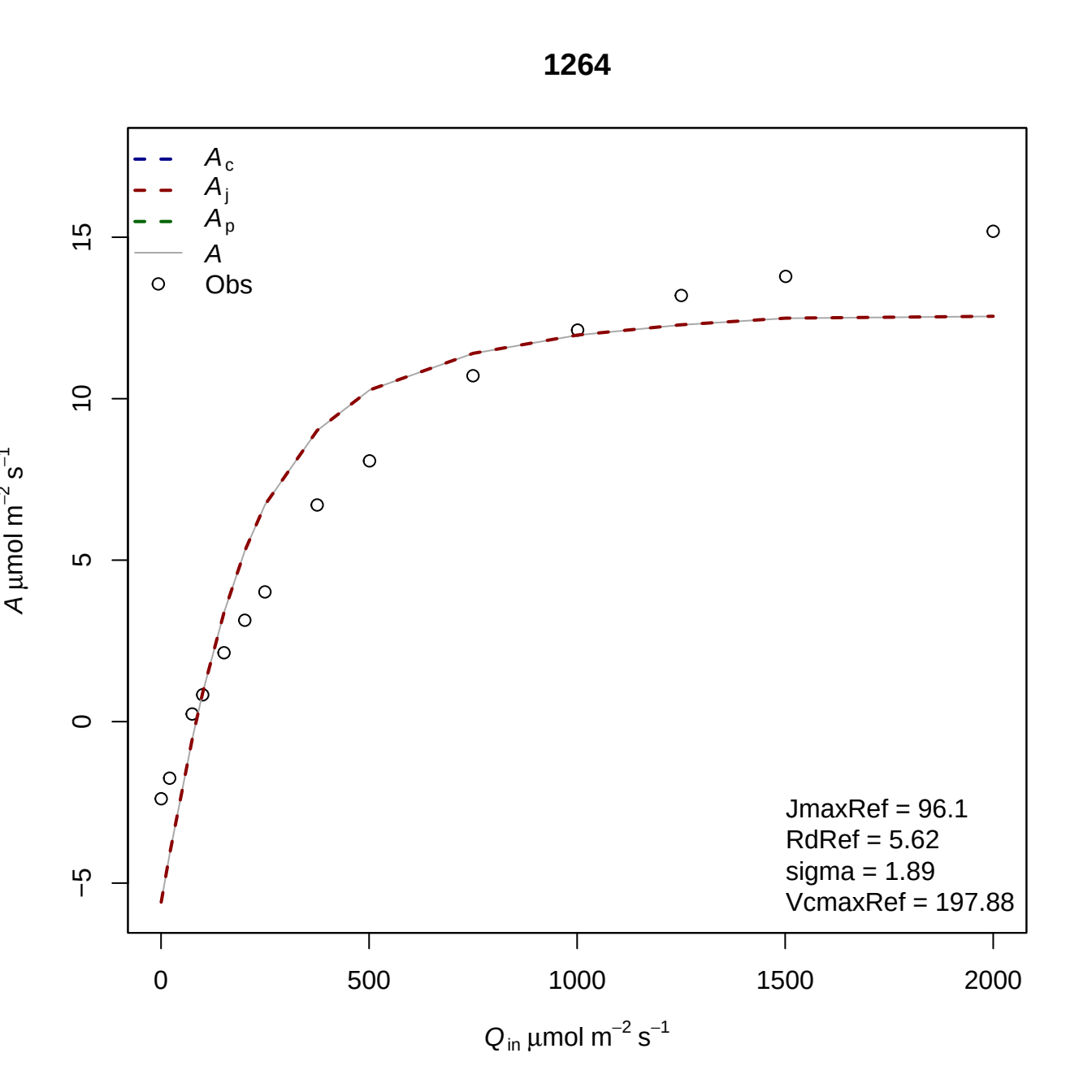

1259
A_c
A_j
A_p
A
Obs
JmaxRef = 109.48
RdRef = 7.22
sigma = 2.23
VcmaxRef = 461.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

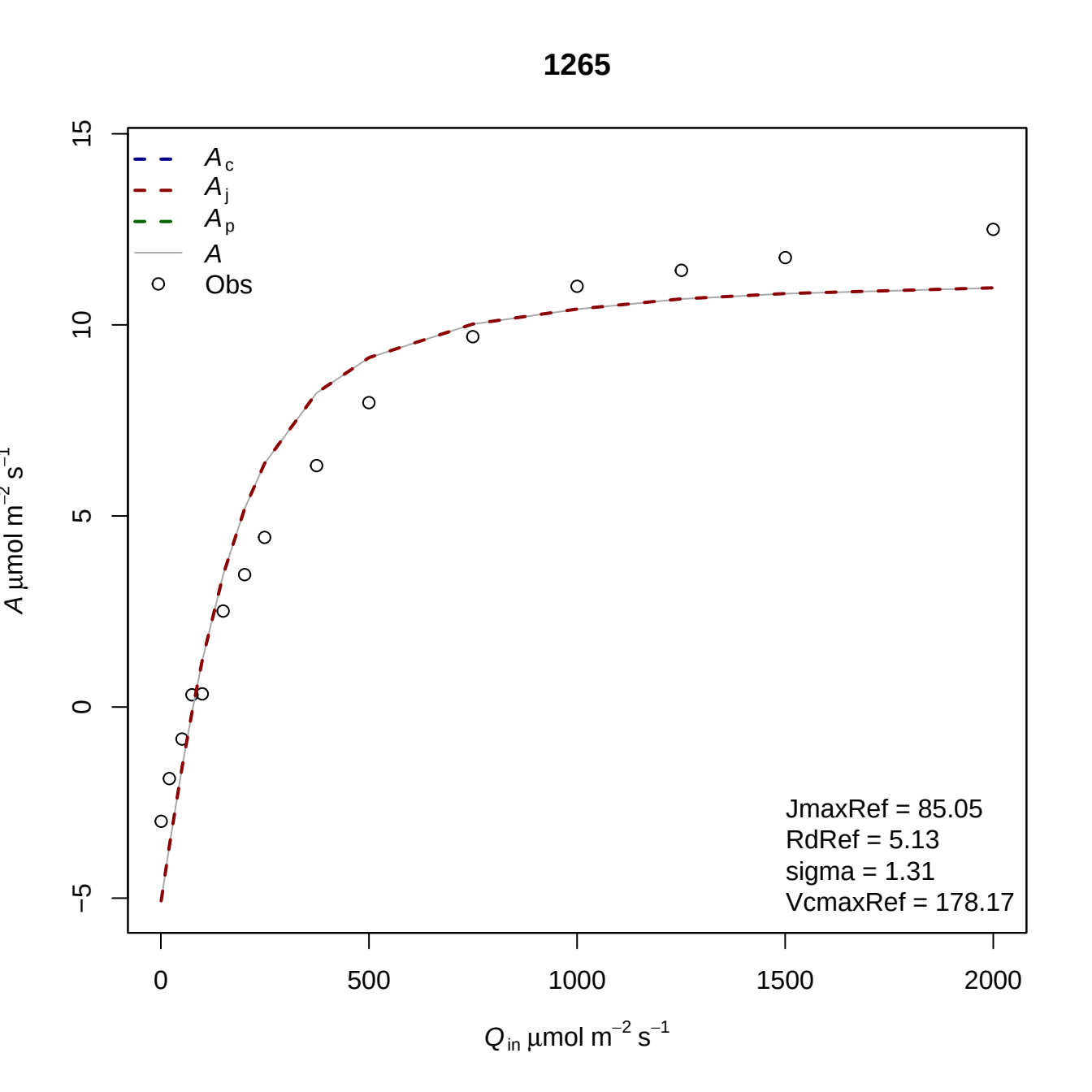

1260
A_c
A_j
A_p
A
Obs
JmaxRef = 83.63
RdRef = 4.08
sigma = 1.59
VcmaxRef = 371.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

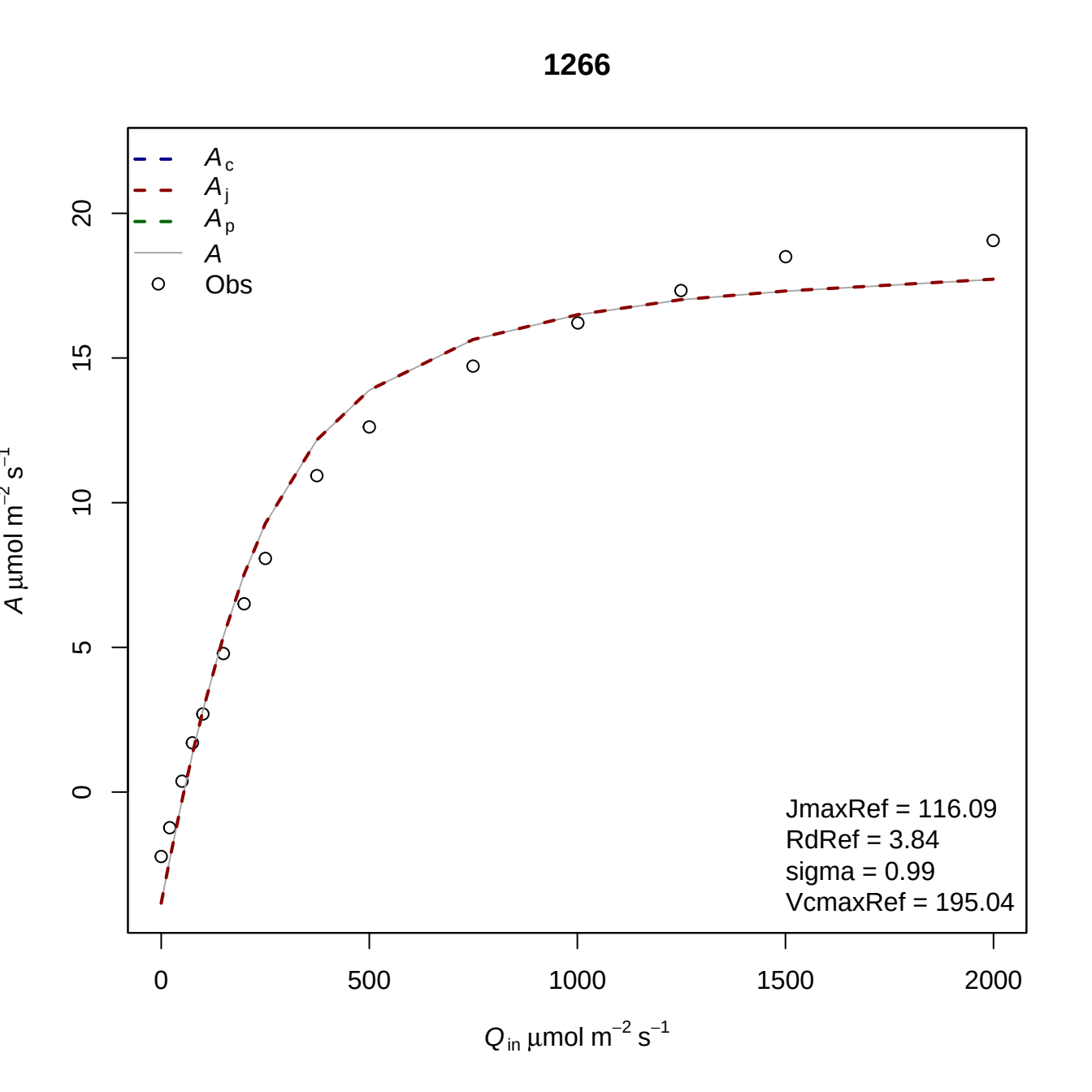

1261
A_c
A_j
A_p
A
Obs
JmaxRef = 95.44
RdRef = 3.83
sigma = 1.51
VcmaxRef = 436.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

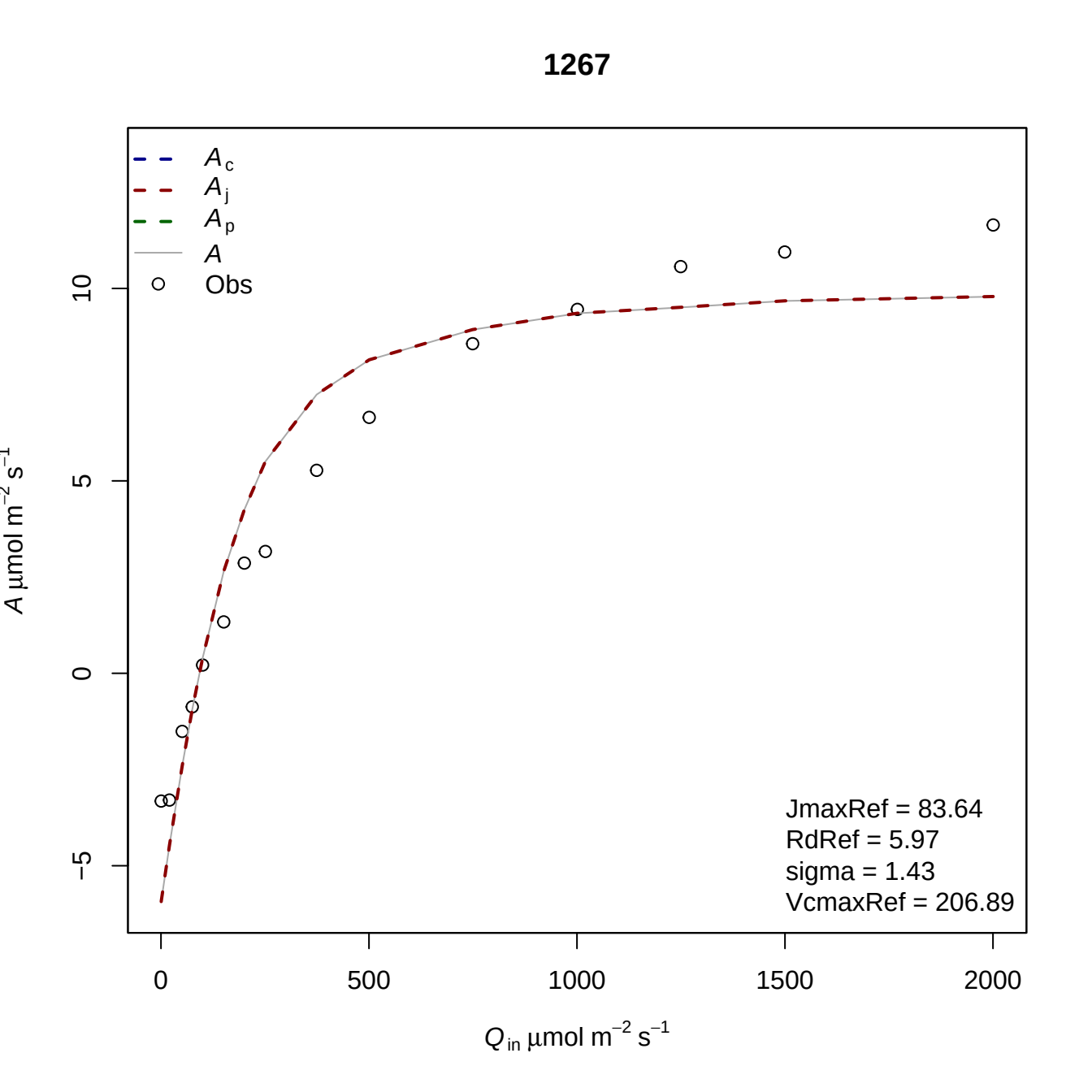

1262
A_c
A_j
A_p
A
Obs
JmaxRef = 78.84
RdRef = 5.53
sigma = 1.71
VcmaxRef = 296.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

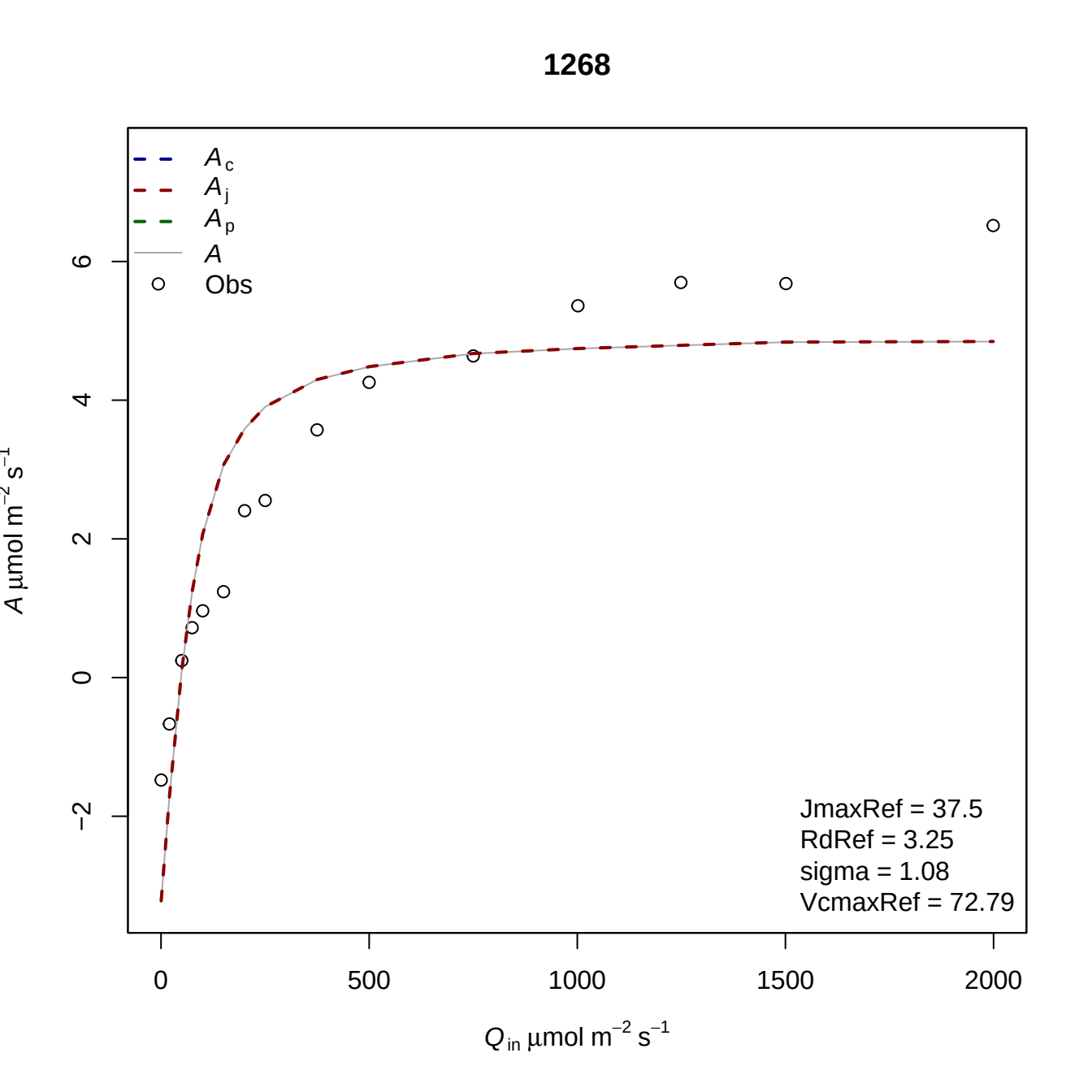

1263
A_c
A_j
A_p
A
Obs
JmaxRef = 103.27
RdRef = 5.96
sigma = 1.5
VcmaxRef = 182.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

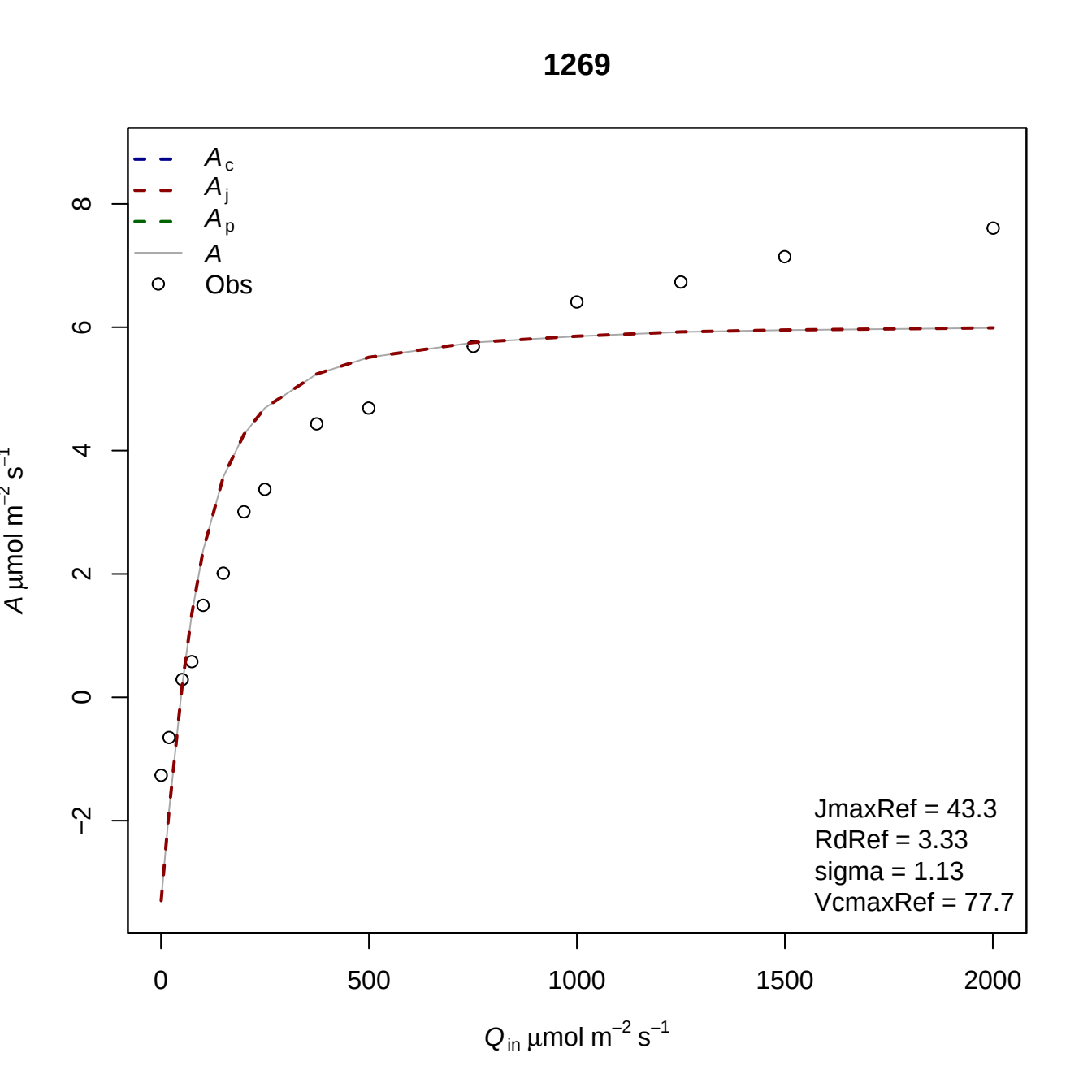

1264
A_c
A_j
A_p
A
Obs
JmaxRef = 96.1
RdRef = 5.62
sigma = 1.89
VcmaxRef = 197.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

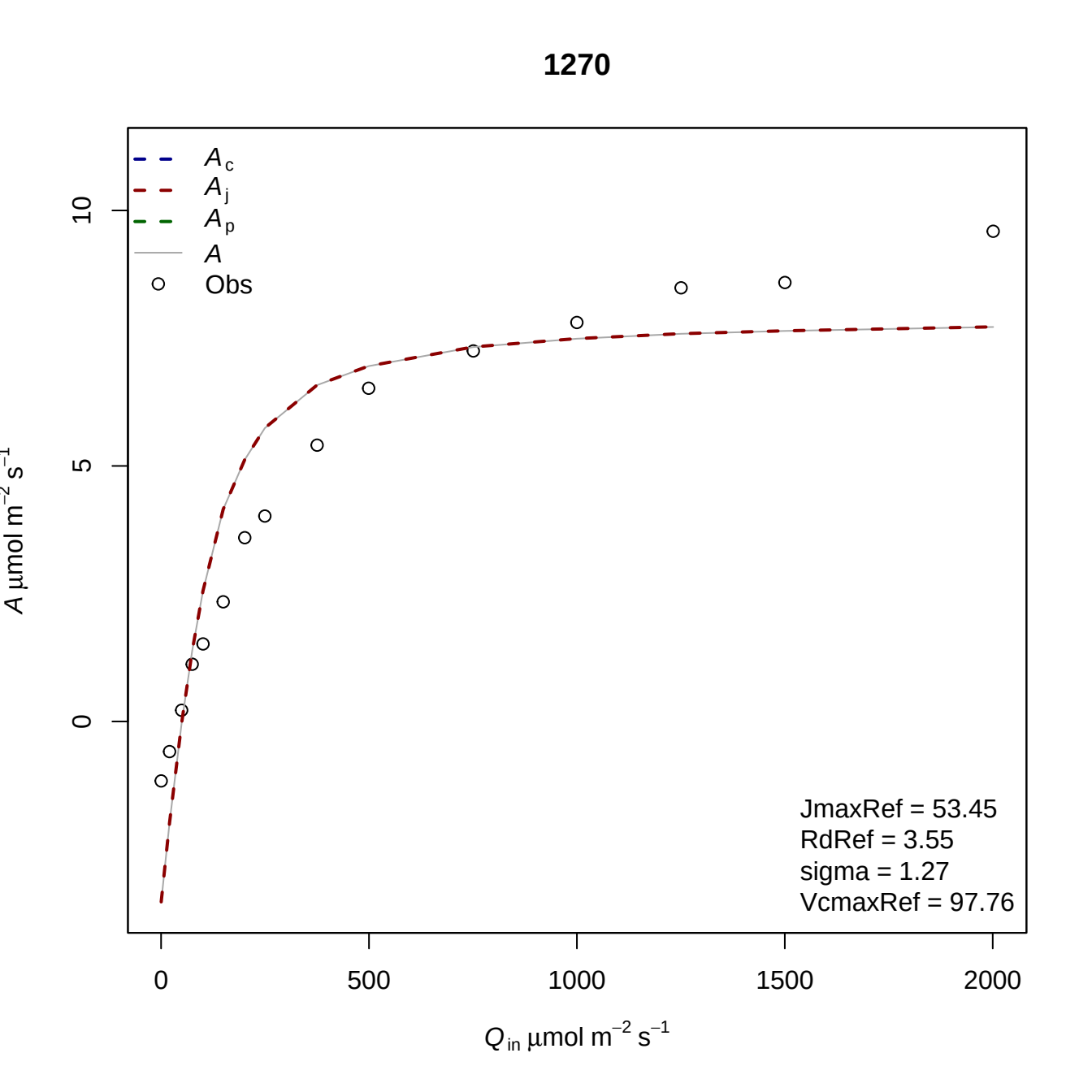

1265
A_c
A_j
A_p
A
Obs
JmaxRef = 85.05
RdRef = 5.13
sigma = 1.31
VcmaxRef = 178.17
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

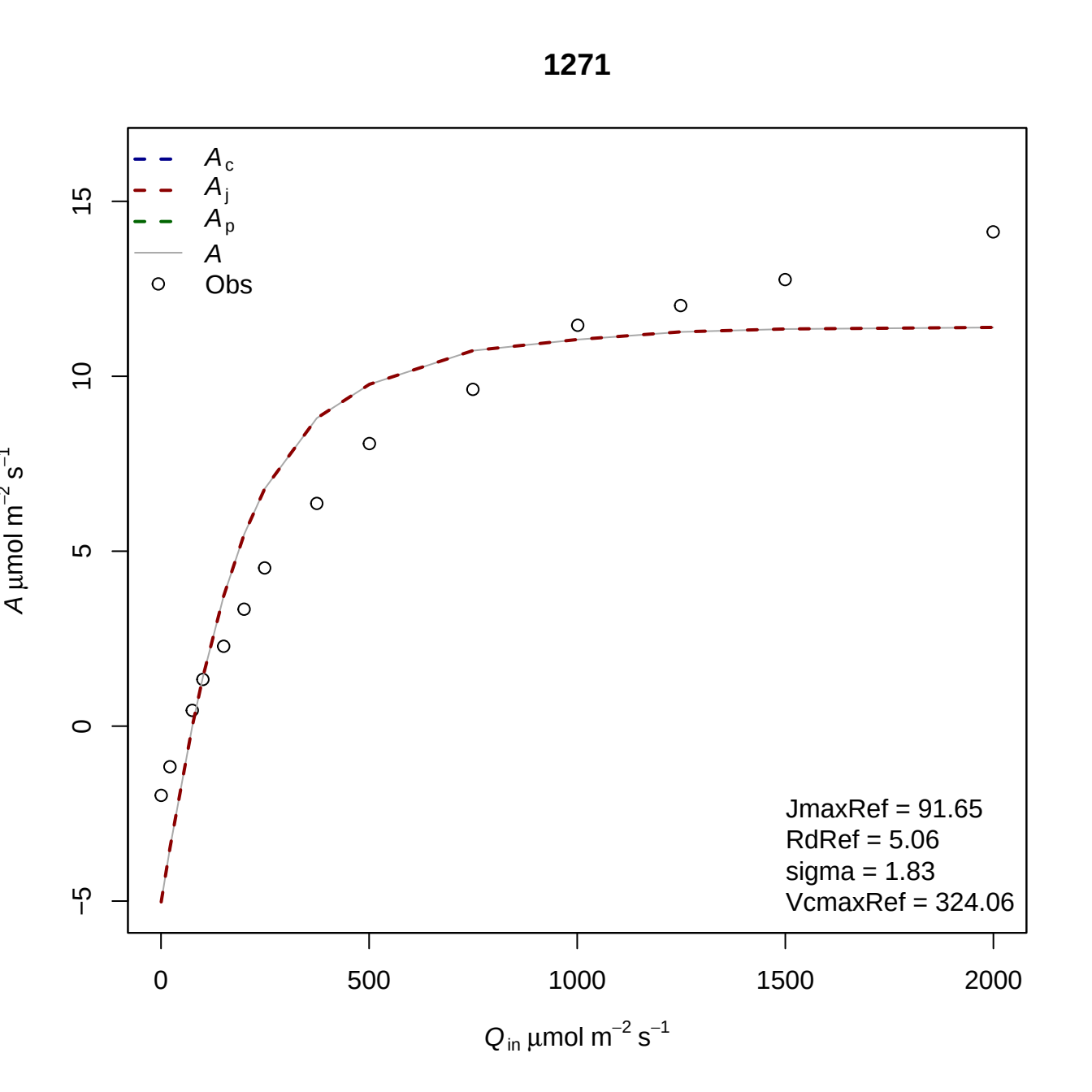

1266
A_c
A_j
A_p
A
Obs
JmaxRef = 116.09
RdRef = 3.84
sigma = 0.99
VcmaxRef = 195.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

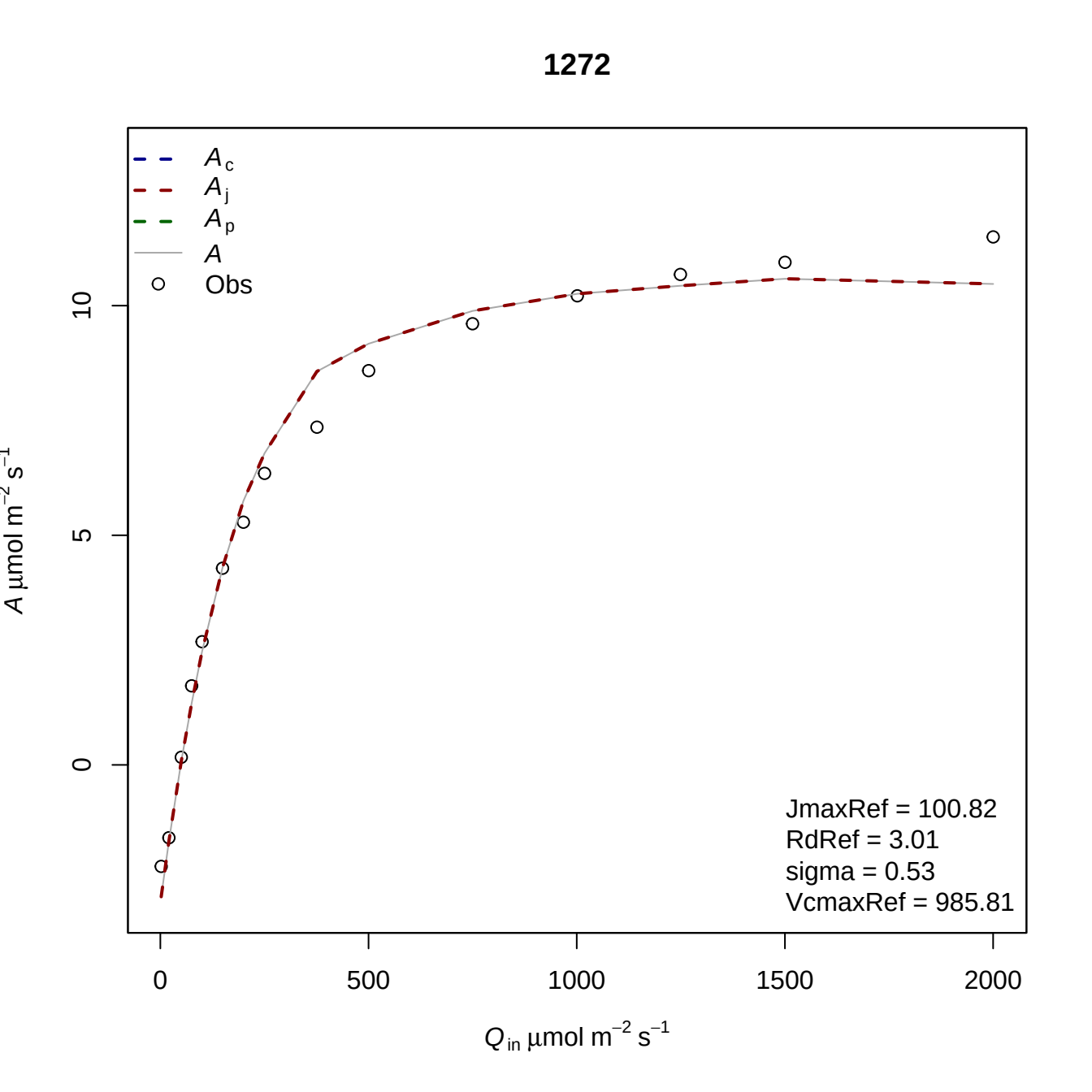

1267
A_c
A_j
A_p
A
Obs
JmaxRef = 83.64
RdRef = 5.97
sigma = 1.43
VcmaxRef = 206.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

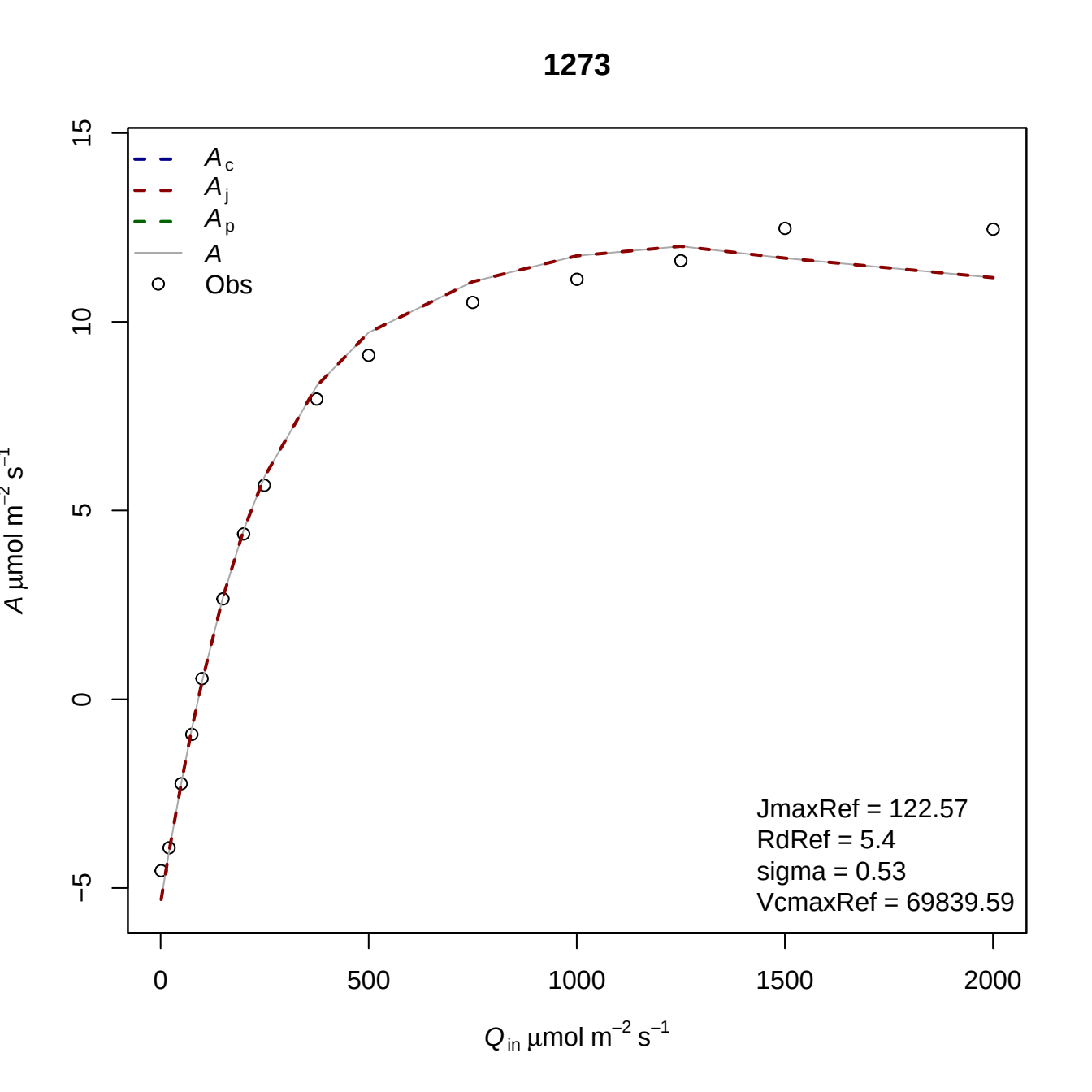

1268
A_c
A_j
A_p
A
Obs
JmaxRef = 37.5
RdRef = 3.25
sigma = 1.08
VcmaxRef = 72.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

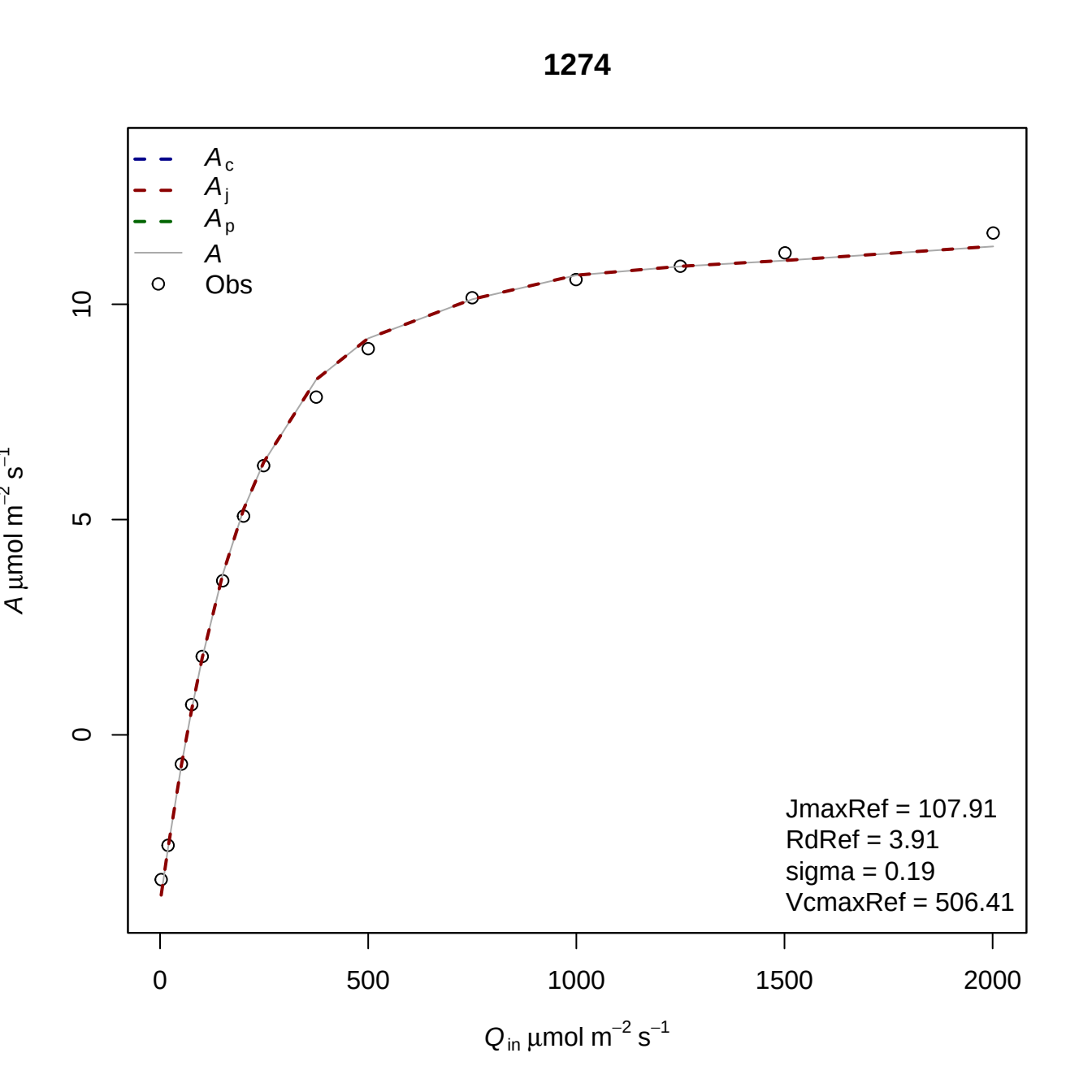

1269
A_c
A_j
A_p
A
Obs
JmaxRef = 43.3
RdRef = 3.33
sigma = 1.13
VcmaxRef = 77.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

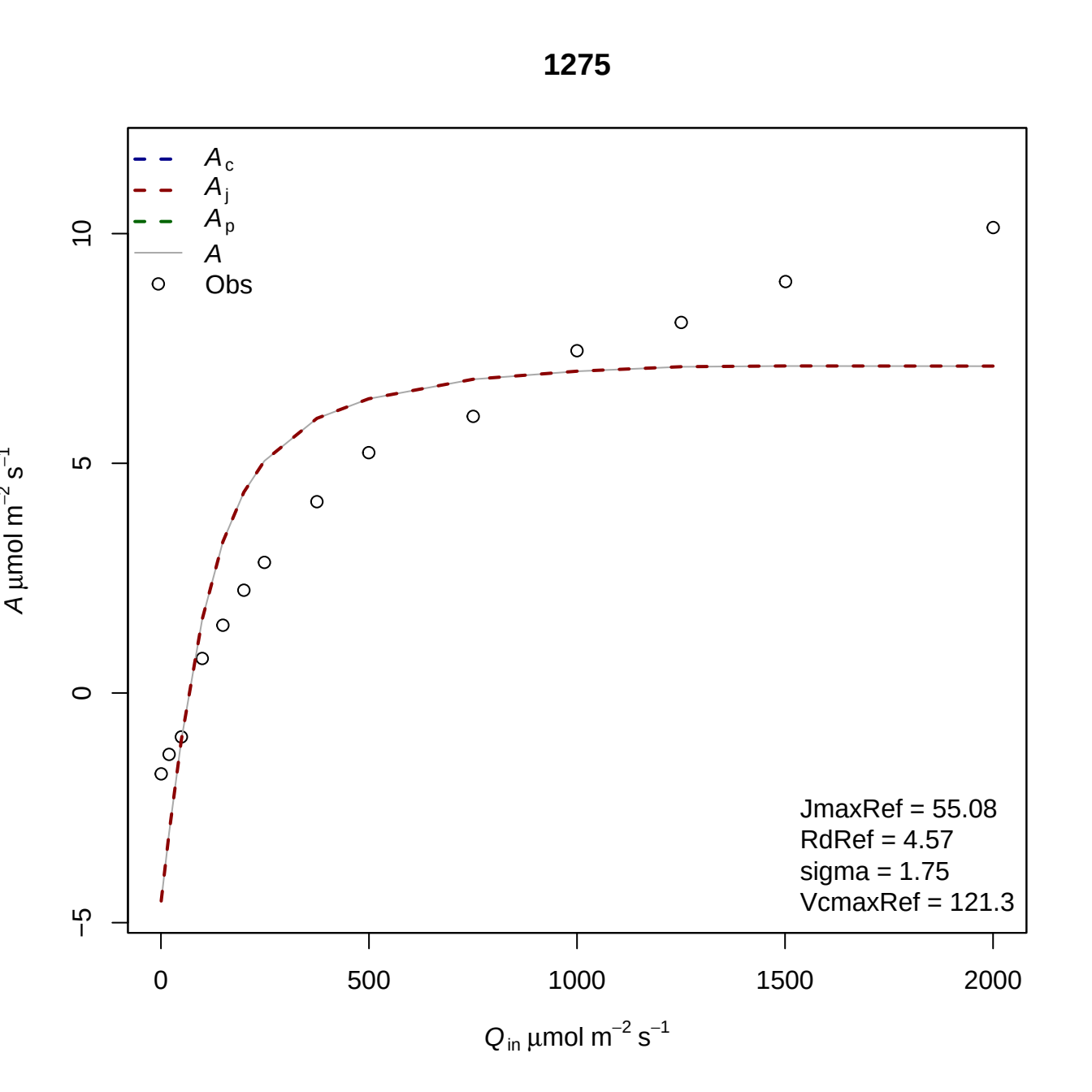

1270
A_c
A_j
A_p
A
Obs
JmaxRef = 53.45
RdRef = 3.55
sigma = 1.27
VcmaxRef = 97.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

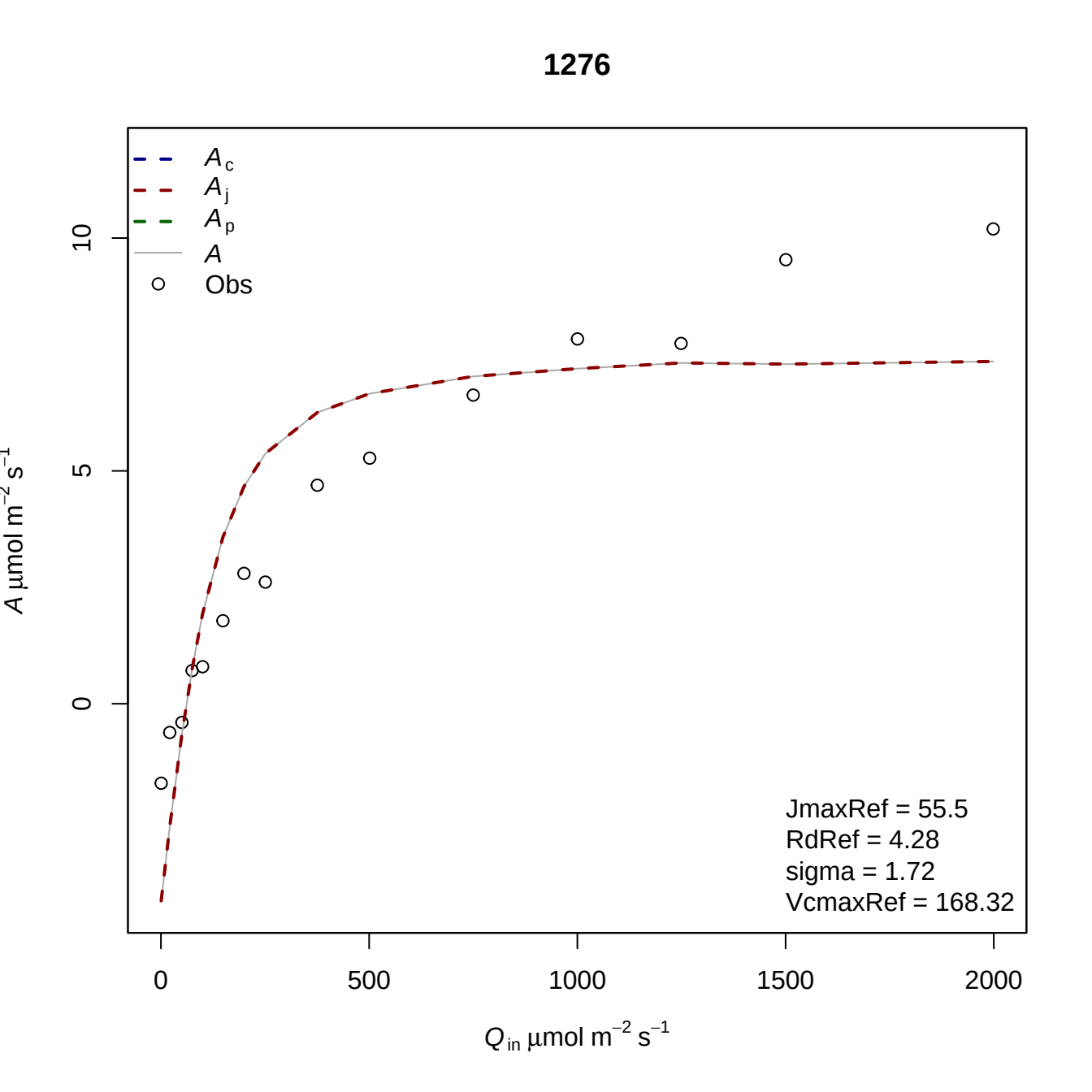

1271
A_c
A_j
A_p
A
Obs
JmaxRef = 91.65
RdRef = 5.06
sigma = 1.83
VcmaxRef = 324.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

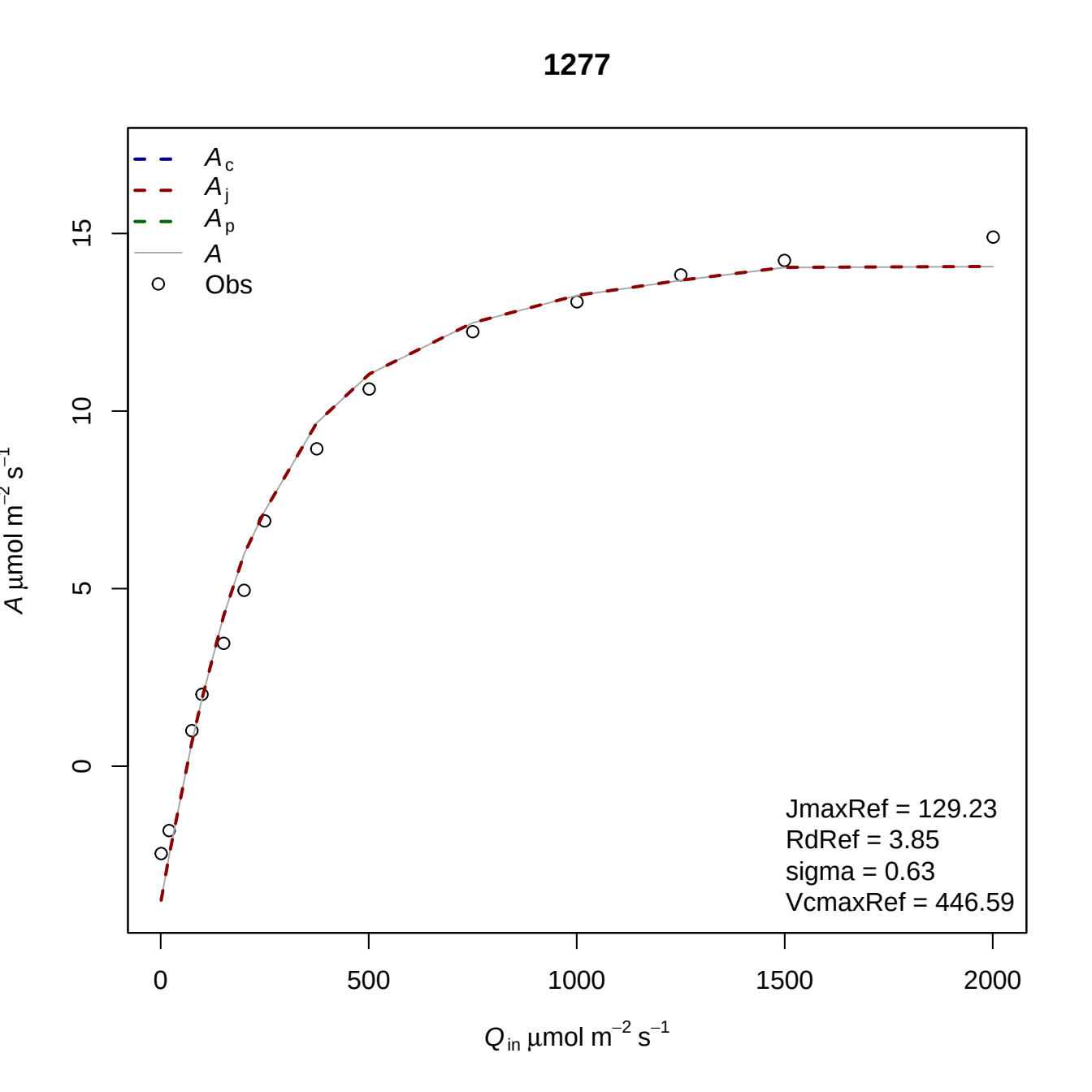

1272
A_c
A_j
A_p
A
Obs
JmaxRef = 100.82
RdRef = 3.01
sigma = 0.53
VcmaxRef = 985.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

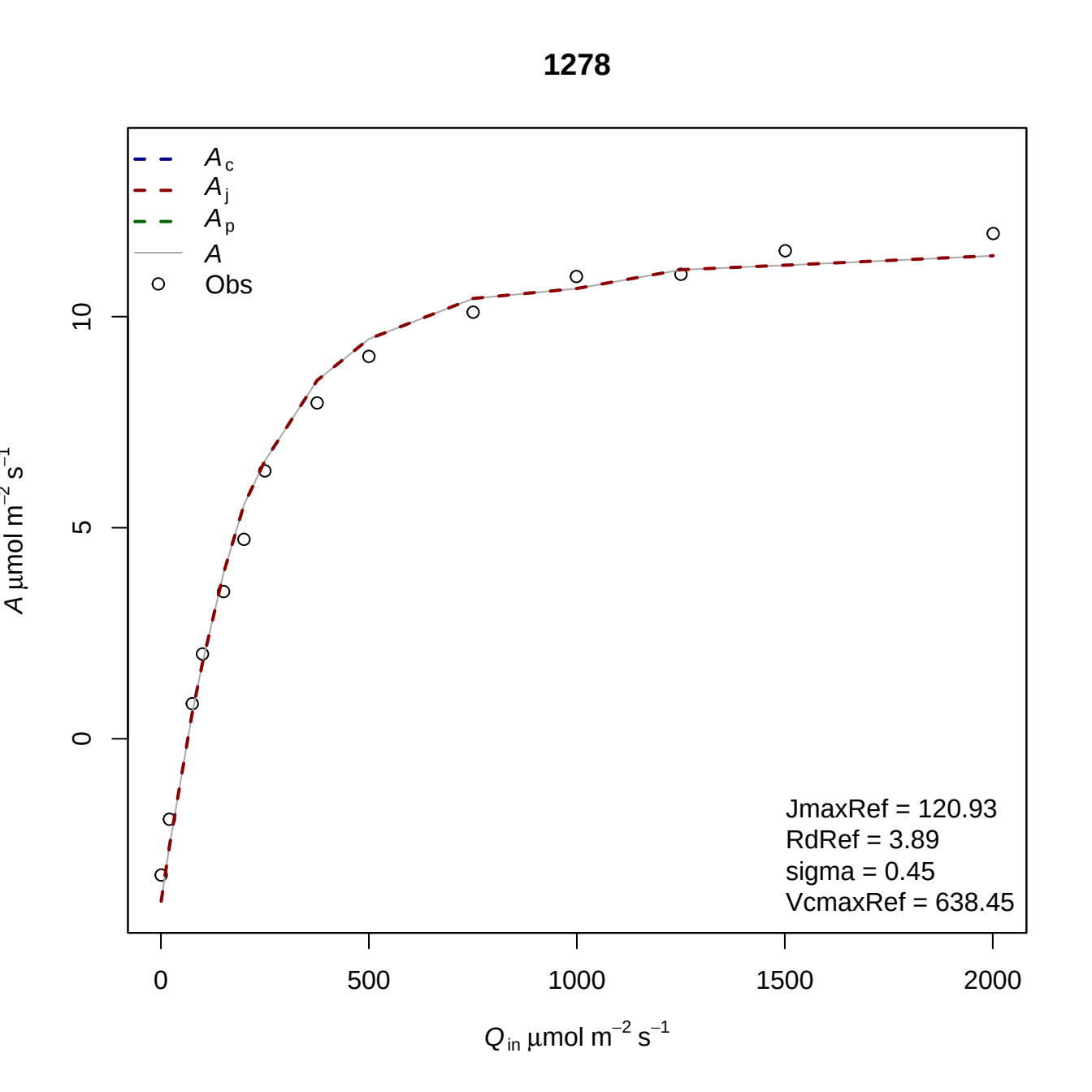

1273
A_c
A_j
A_p
A
Obs
JmaxRef = 122.57
RdRef = 5.4
sigma = 0.53
VcmaxRef = 69839.59
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

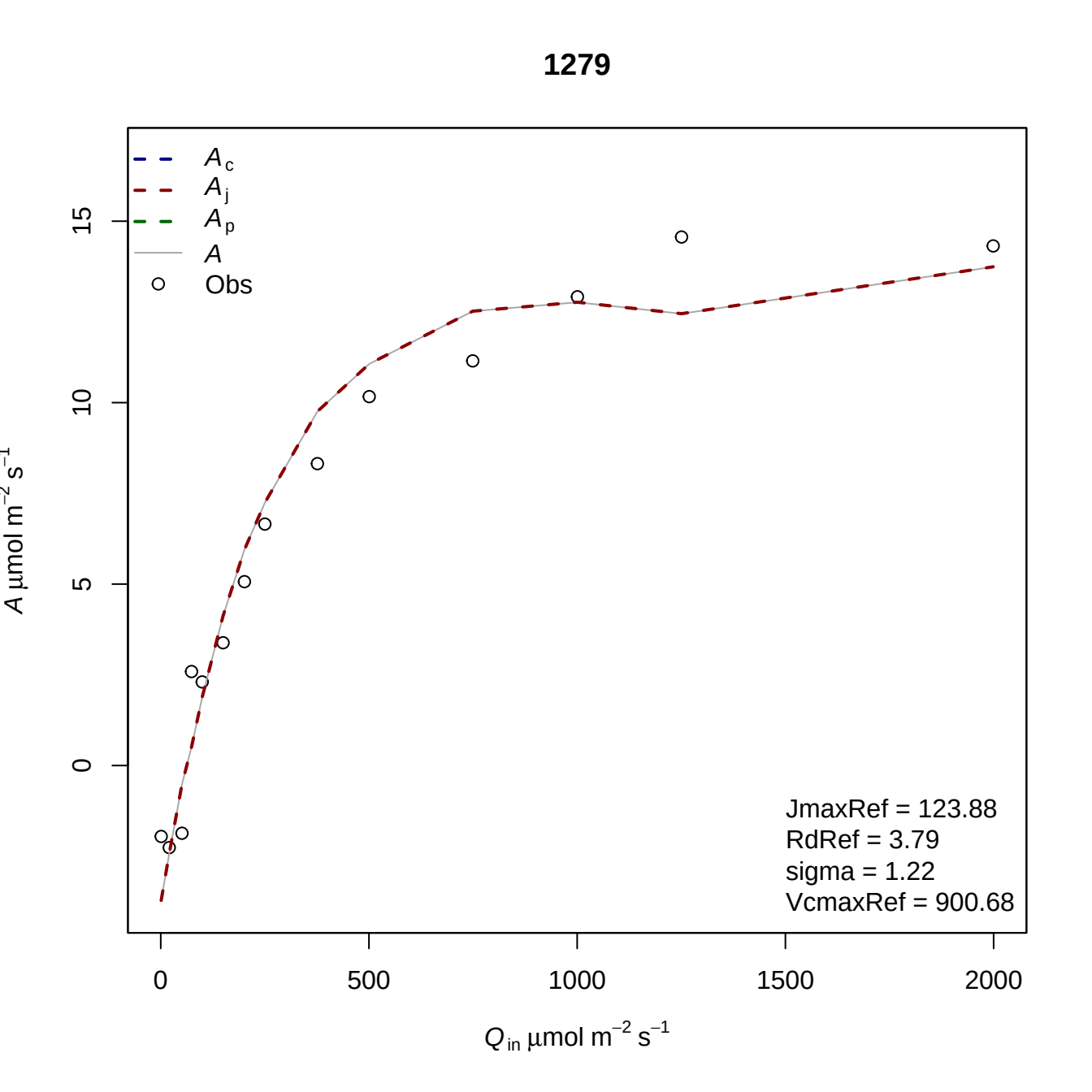

1274
A_c
A_j
A_p
A
Obs
JmaxRef = 107.91
RdRef = 3.91
sigma = 0.19
VcmaxRef = 506.41
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

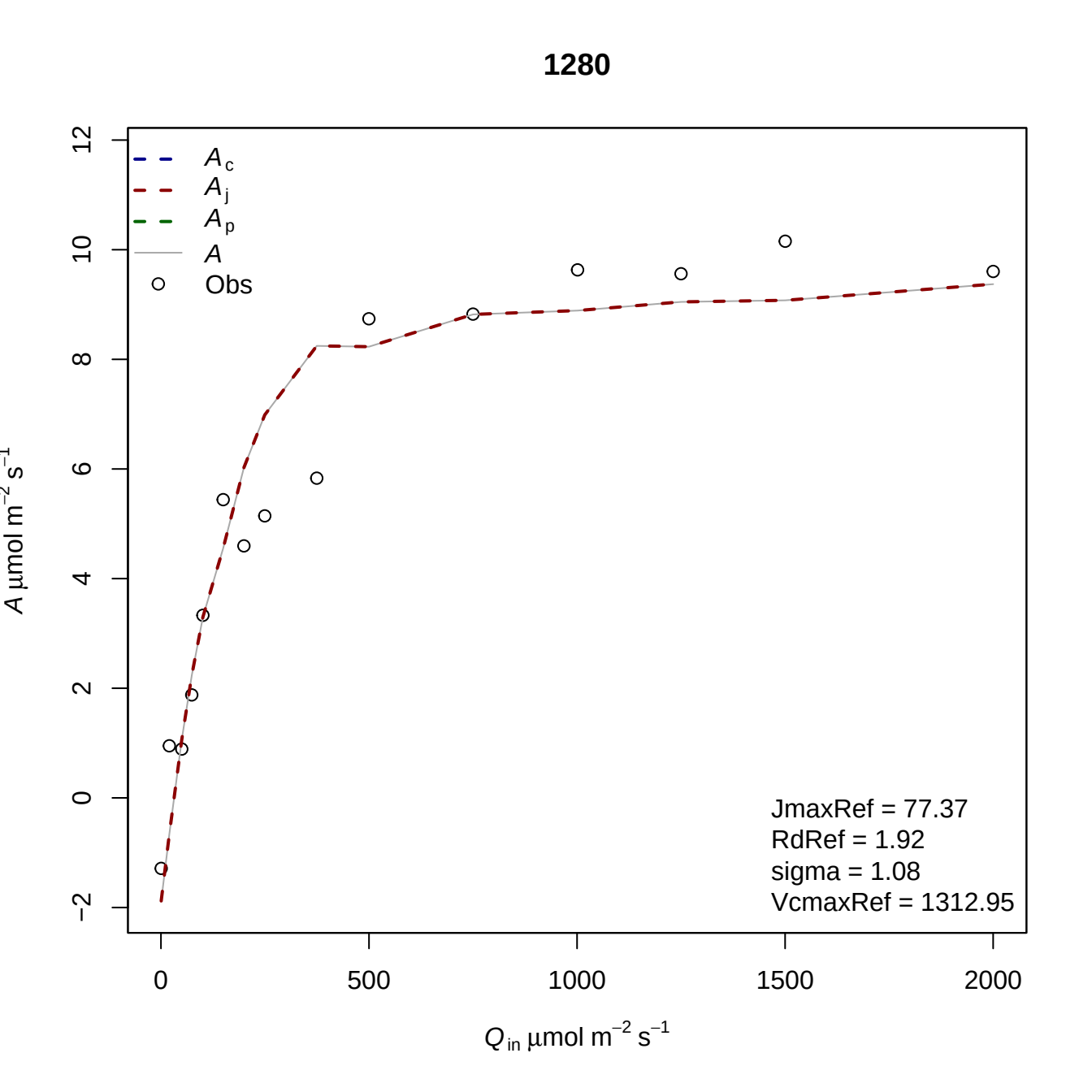

1275
A_c
A_j
A_p
A
Obs
JmaxRef = 55.08
RdRef = 4.57
sigma = 1.75
VcmaxRef = 121.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

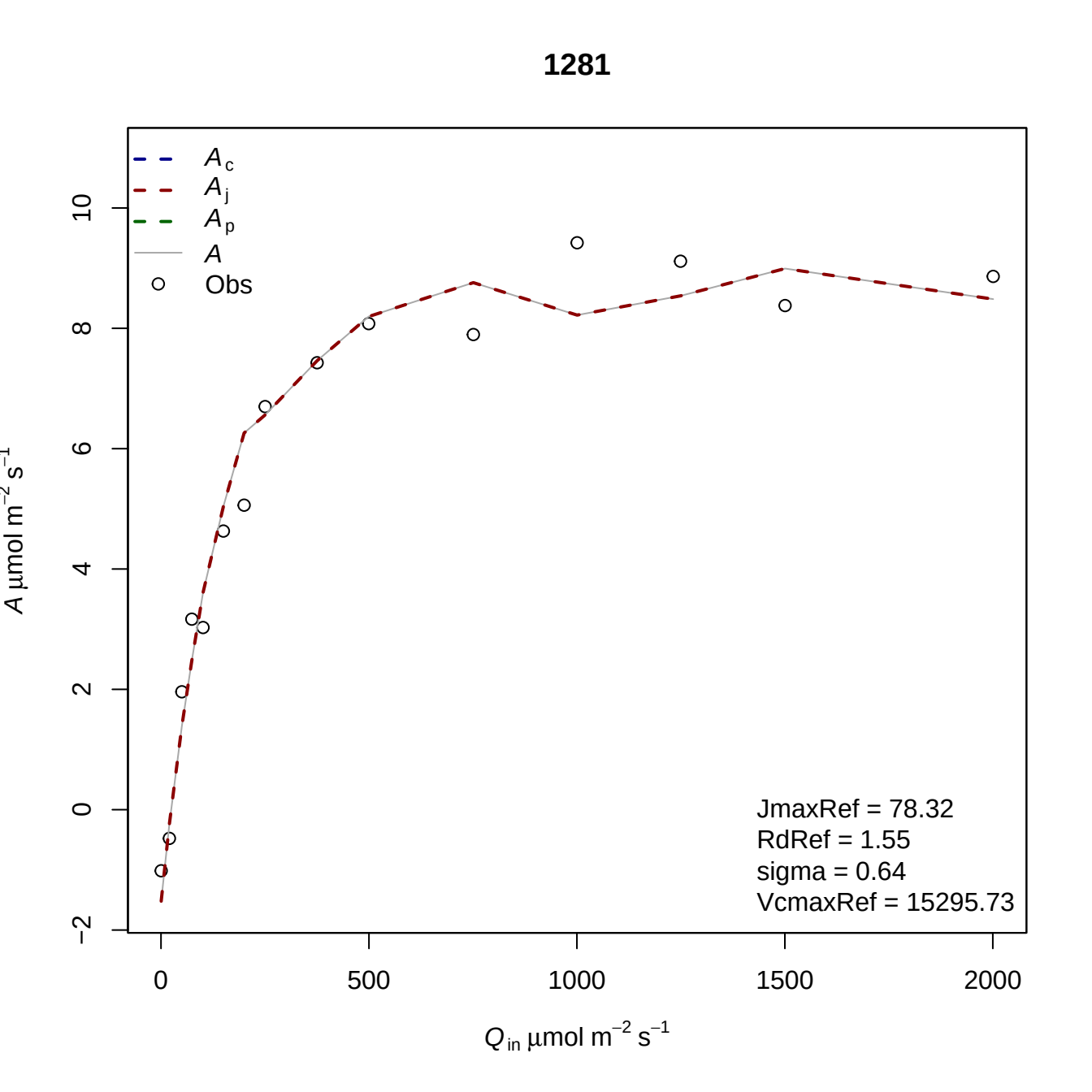

1276
A_c
A_j
A_p
A
Obs
JmaxRef = 55.5
RdRef = 4.28
sigma = 1.72
VcmaxRef = 168.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

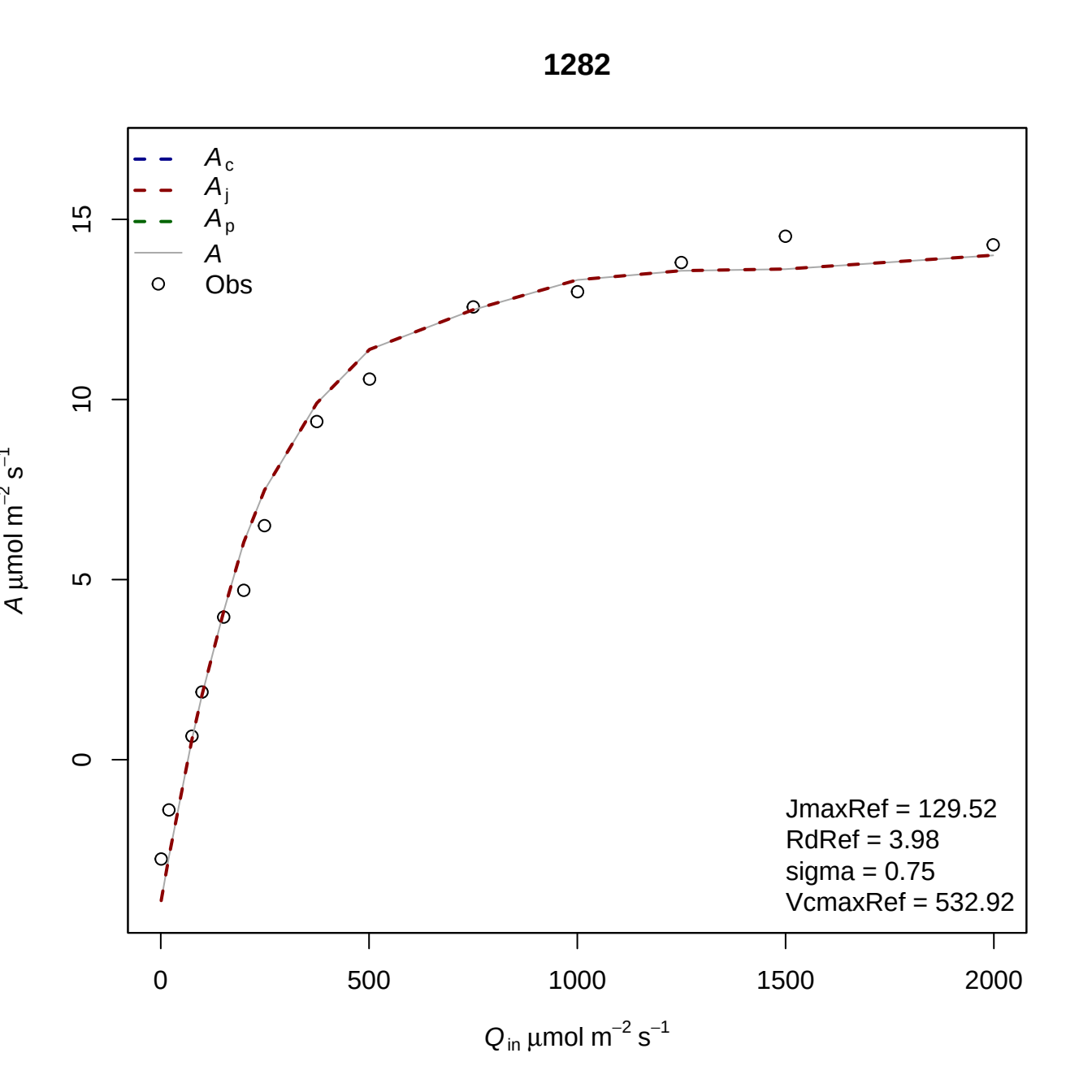

1277
A_c
A_j
A_p
A
Obs
JmaxRef = 129.23
RdRef = 3.85
sigma = 0.63
VcmaxRef = 446.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

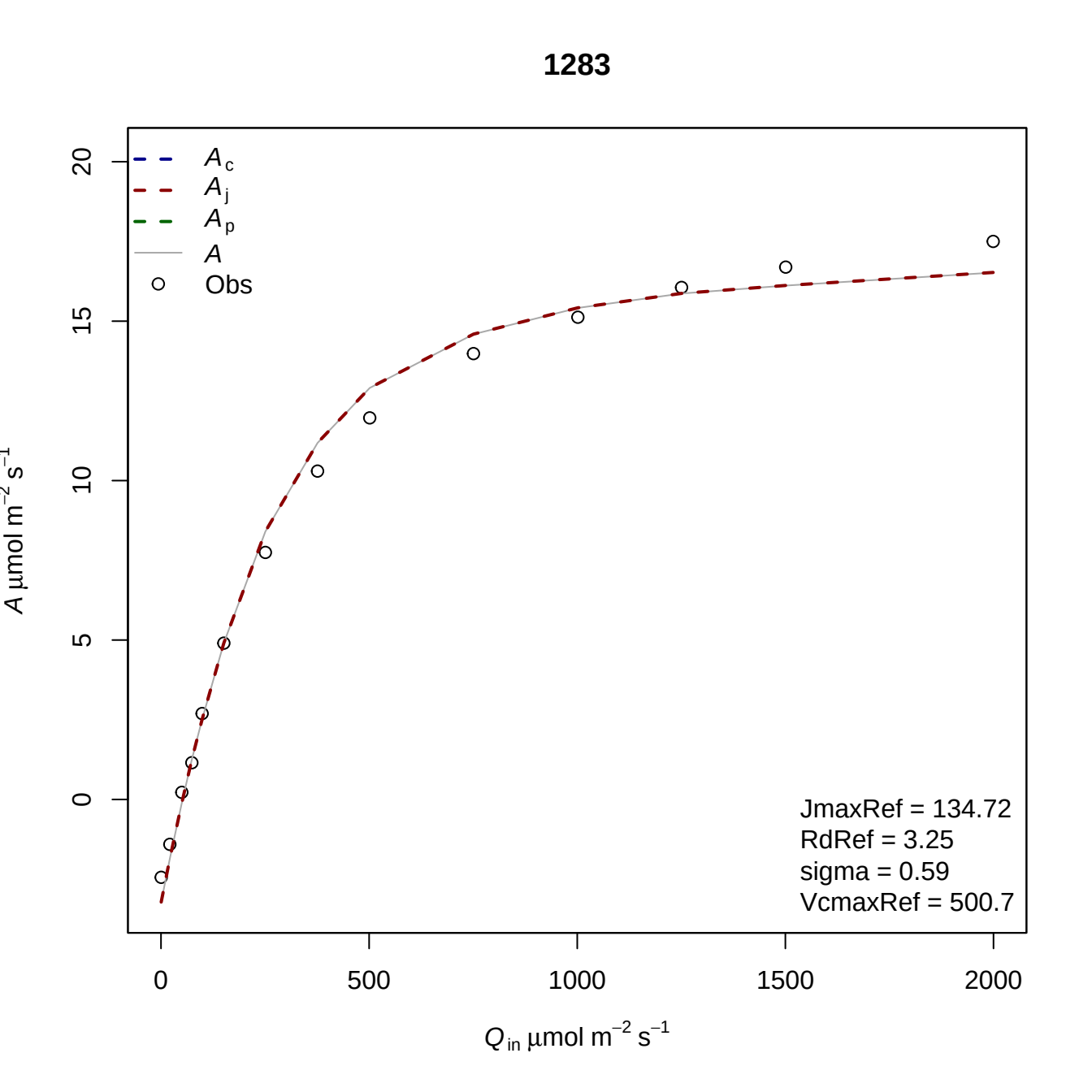

1278
A_c
A_j
A_p
A
Obs
JmaxRef = 120.93
RdRef = 3.89
sigma = 0.45
VcmaxRef = 638.45
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

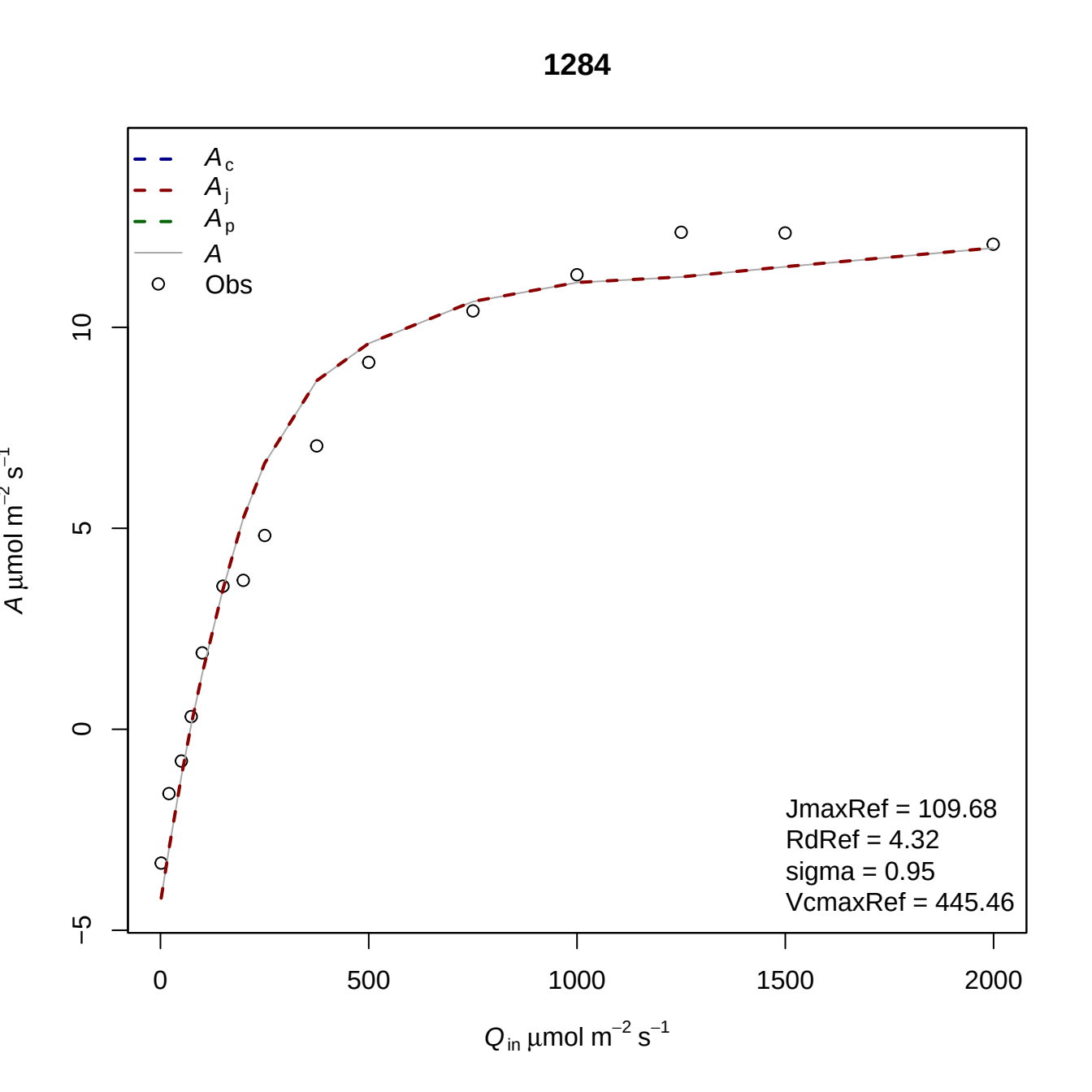

1279
A_c
A_j
A_p
A
Obs
JmaxRef = 123.88
RdRef = 3.79
sigma = 1.22
VcmaxRef = 900.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

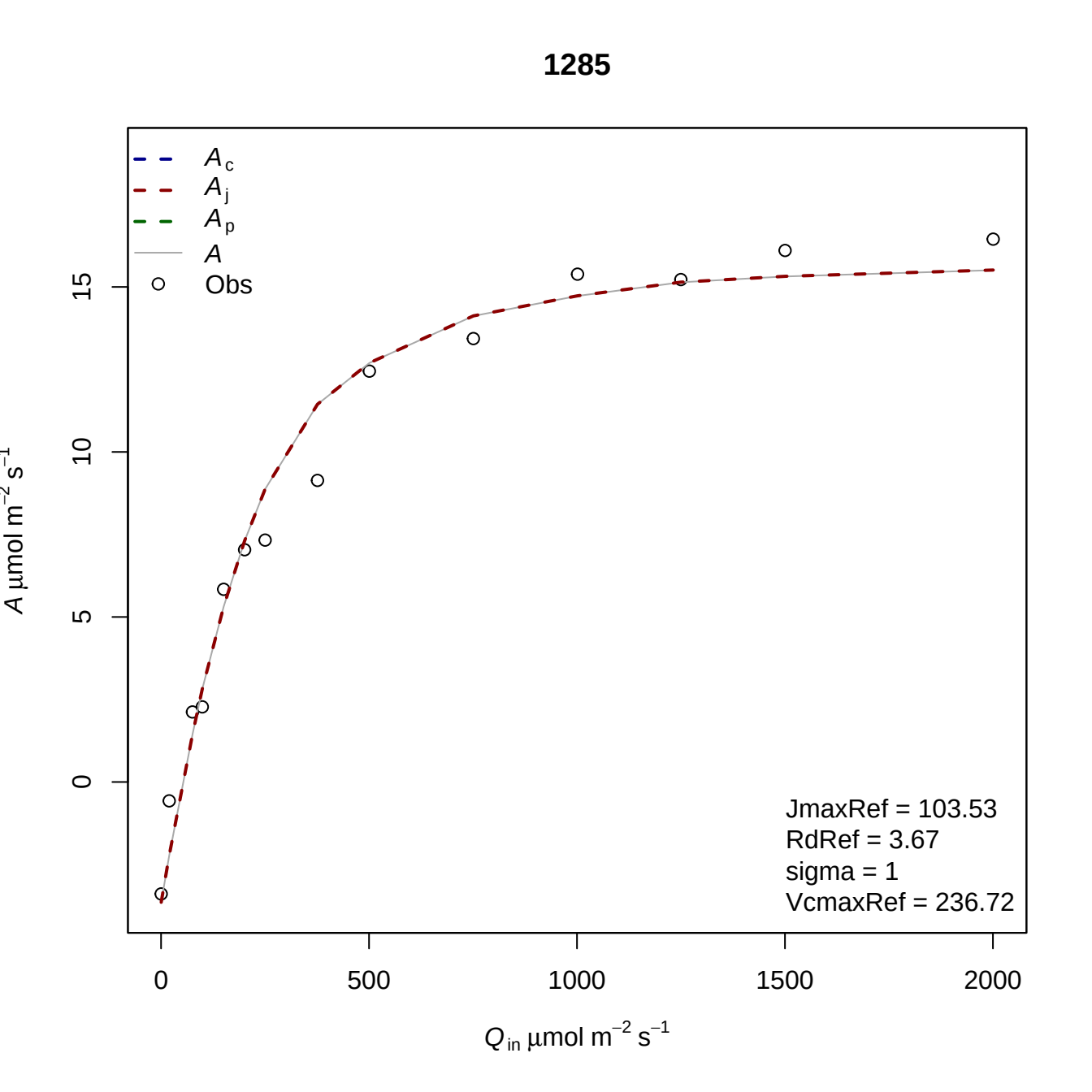

1280
A_c
A_j
A_p
A
Obs
JmaxRef = 77.37
RdRef = 1.92
sigma = 1.08
VcmaxRef = 1312.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

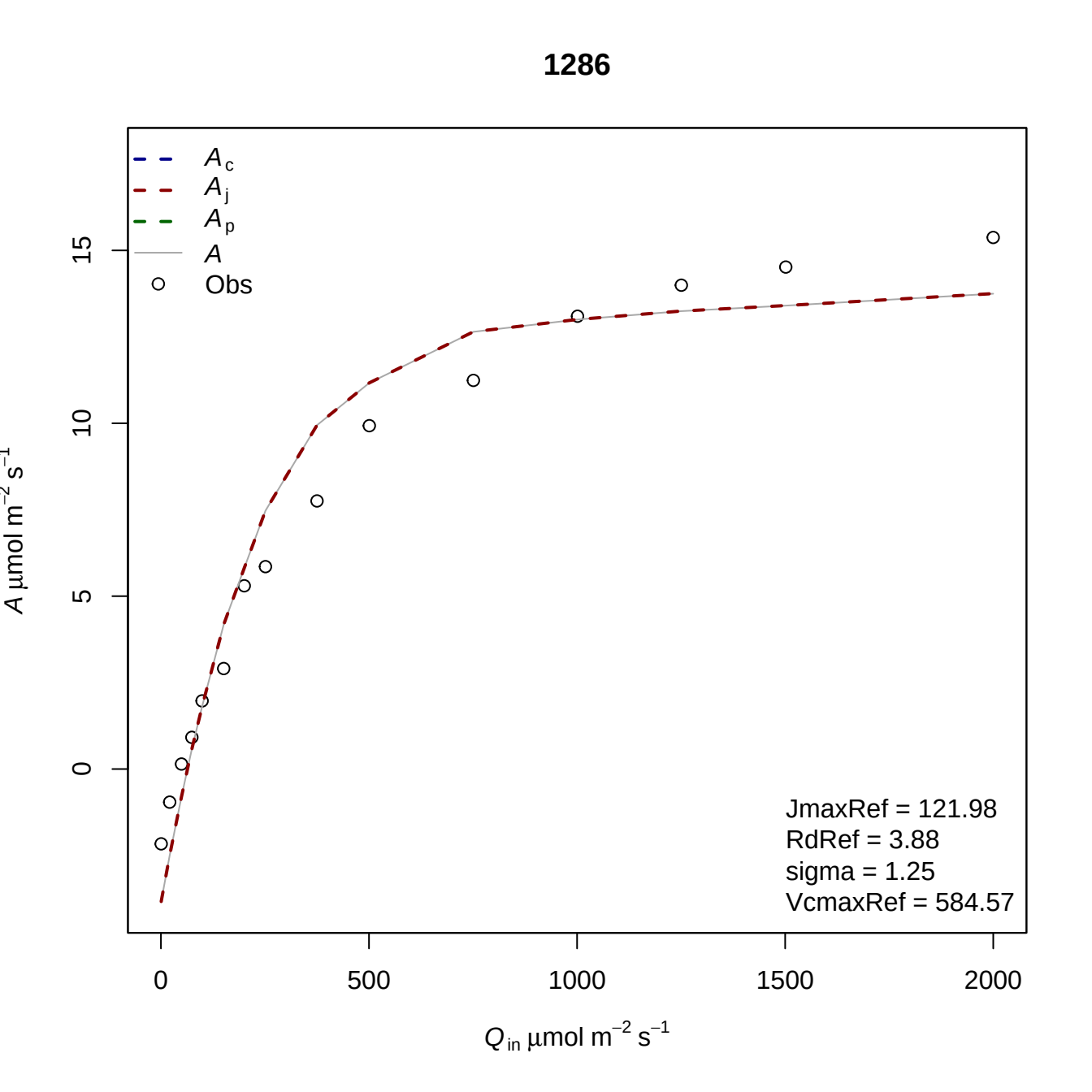

1281
A_c
A_j
A_p
A
Obs
JmaxRef = 78.32
RdRef = 1.55
sigma = 0.64
VcmaxRef = 15295.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

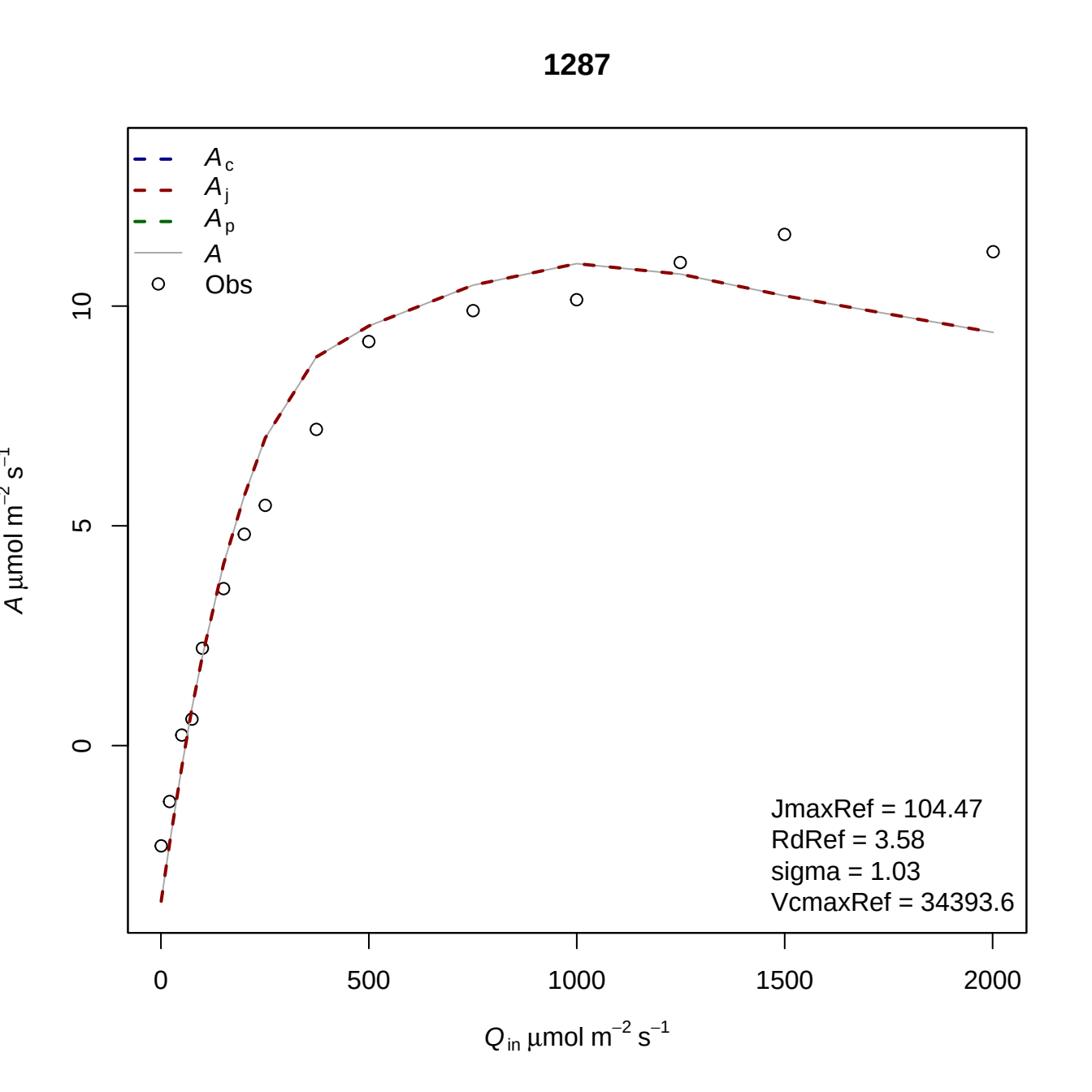

1282
A_c
A_j
A_p
A
Obs
JmaxRef = 129.52
RdRef = 3.98
sigma = 0.75
VcmaxRef = 532.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

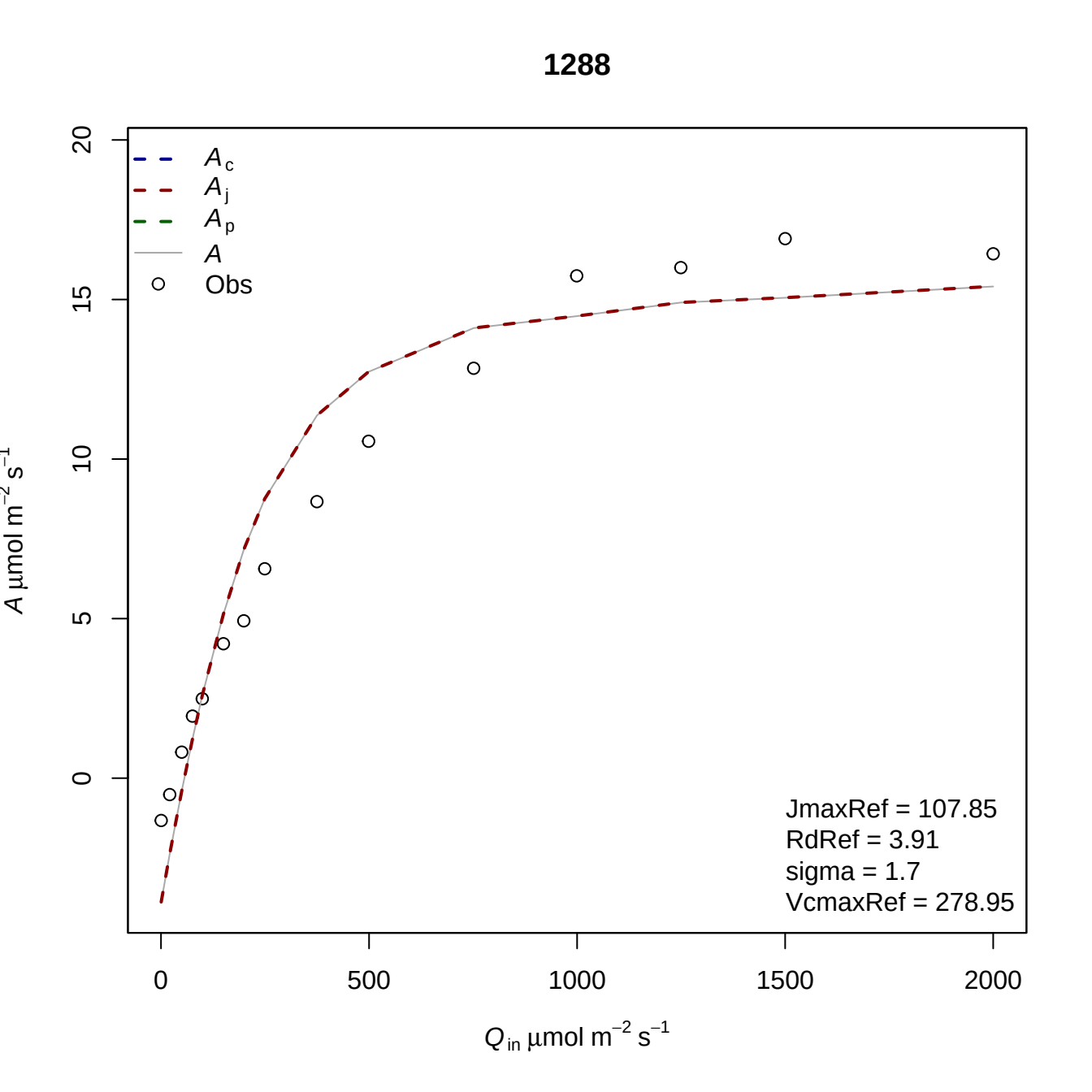

1283
A_c
A_j
A_p
A
Obs
JmaxRef = 134.72
RdRef = 3.25
sigma = 0.59
VcmaxRef = 500.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

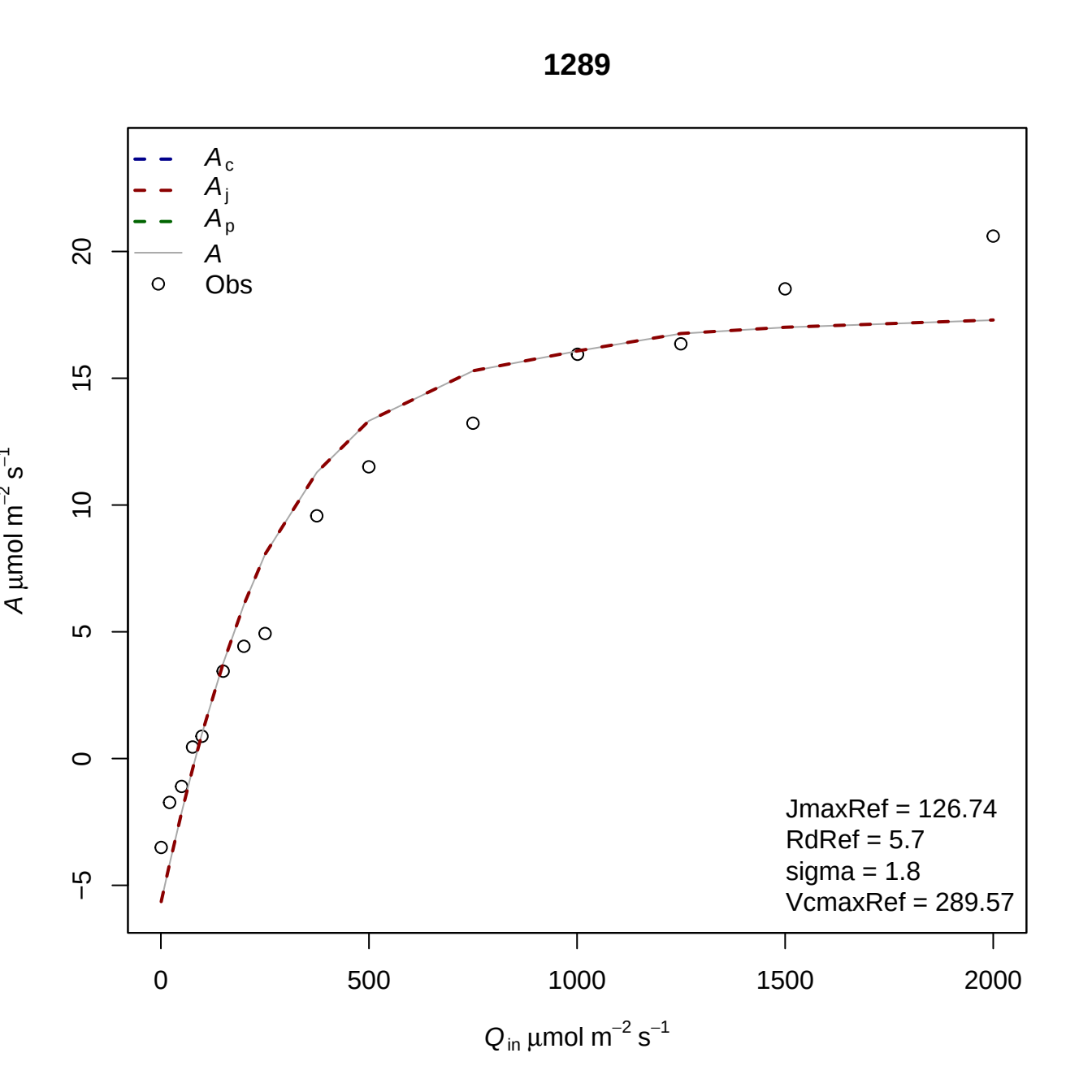

1284
A_c
A_j
A_p
A
Obs
JmaxRef = 109.68
RdRef = 4.32
sigma = 0.95
VcmaxRef = 445.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

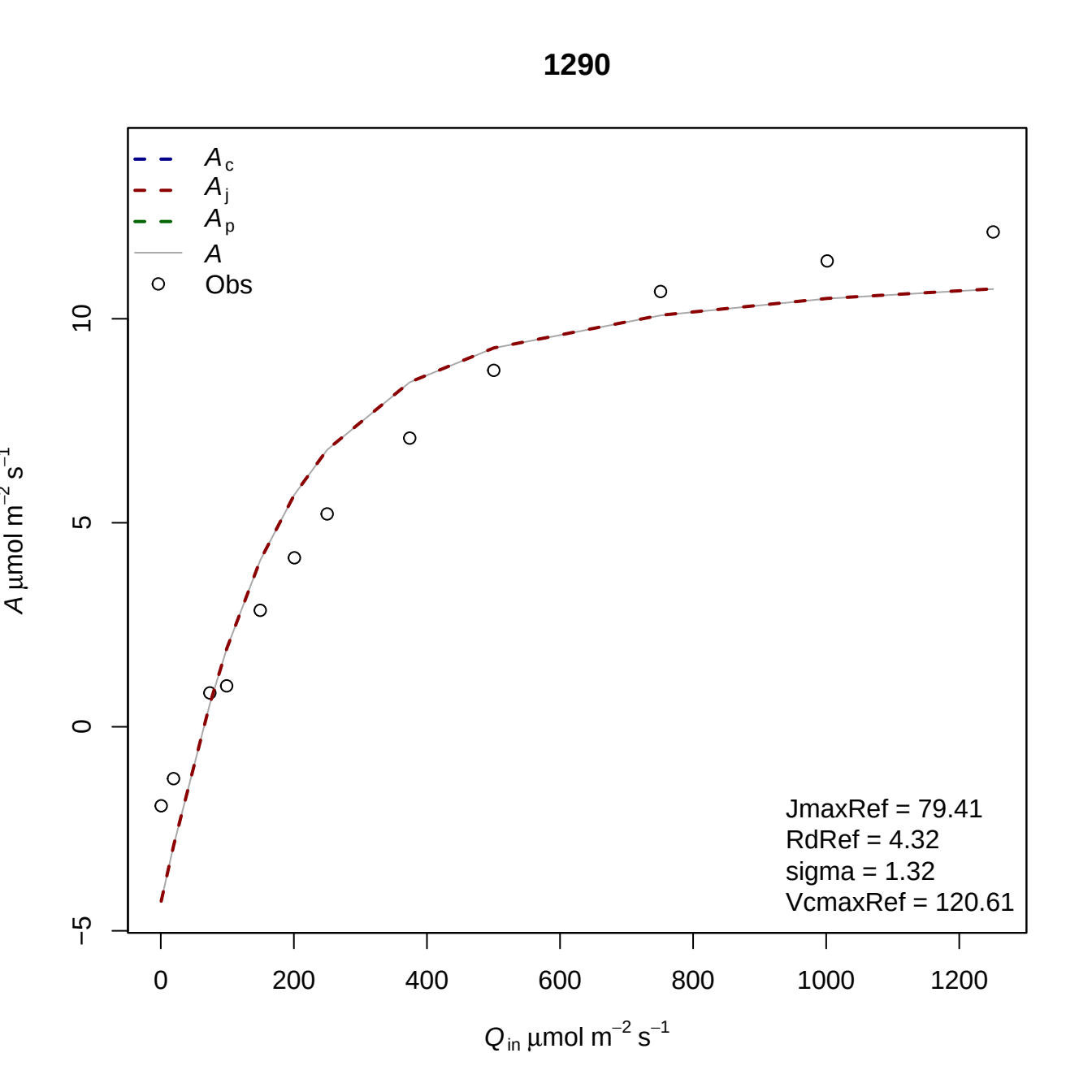

1285
A_c
A_j
A_p
A
Obs
JmaxRef = 103.53
RdRef = 3.67
sigma = 1
VcmaxRef = 236.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

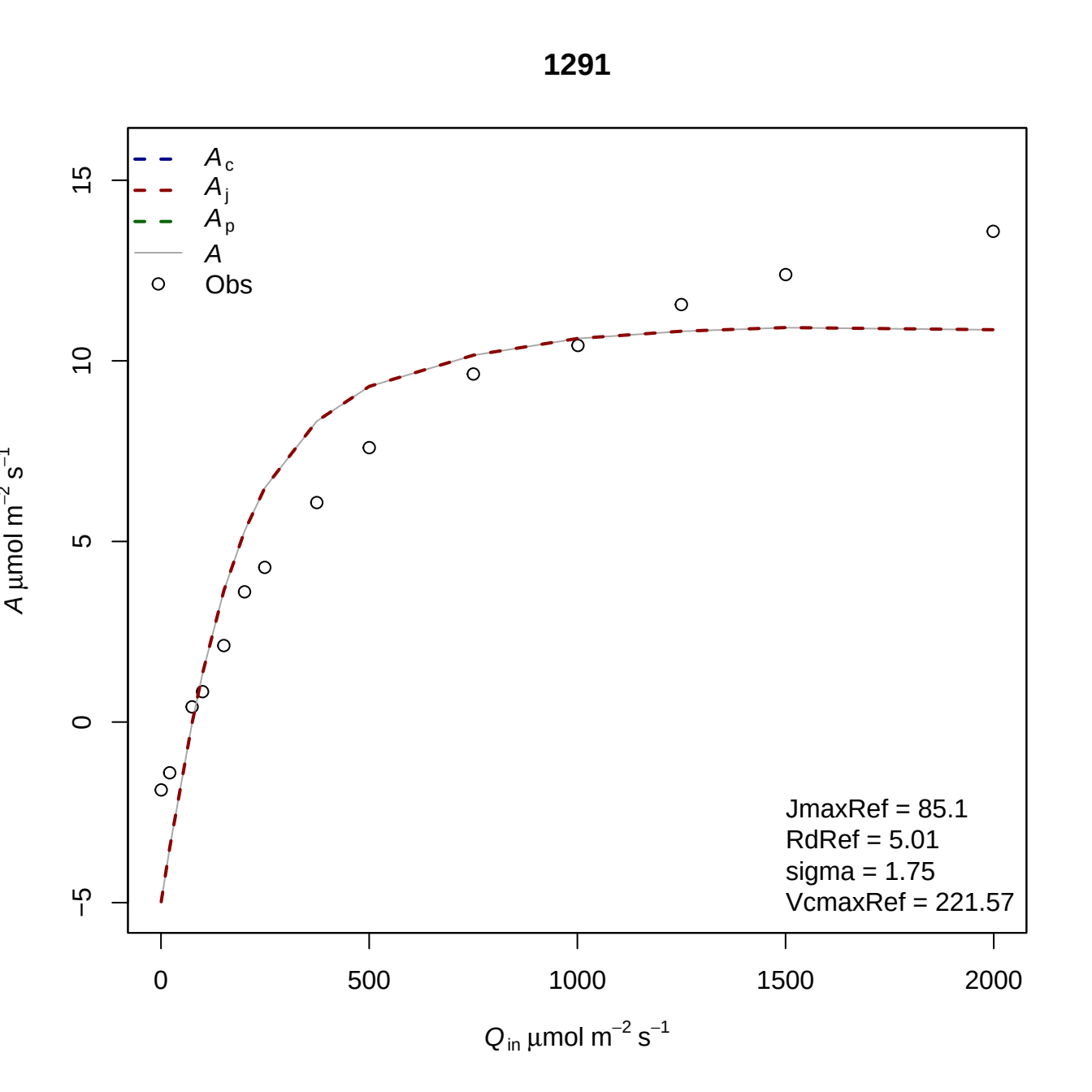

1286
A_c
A_j
A_p
A
Obs
JmaxRef = 121.98
RdRef = 3.88
sigma = 1.25
VcmaxRef = 584.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

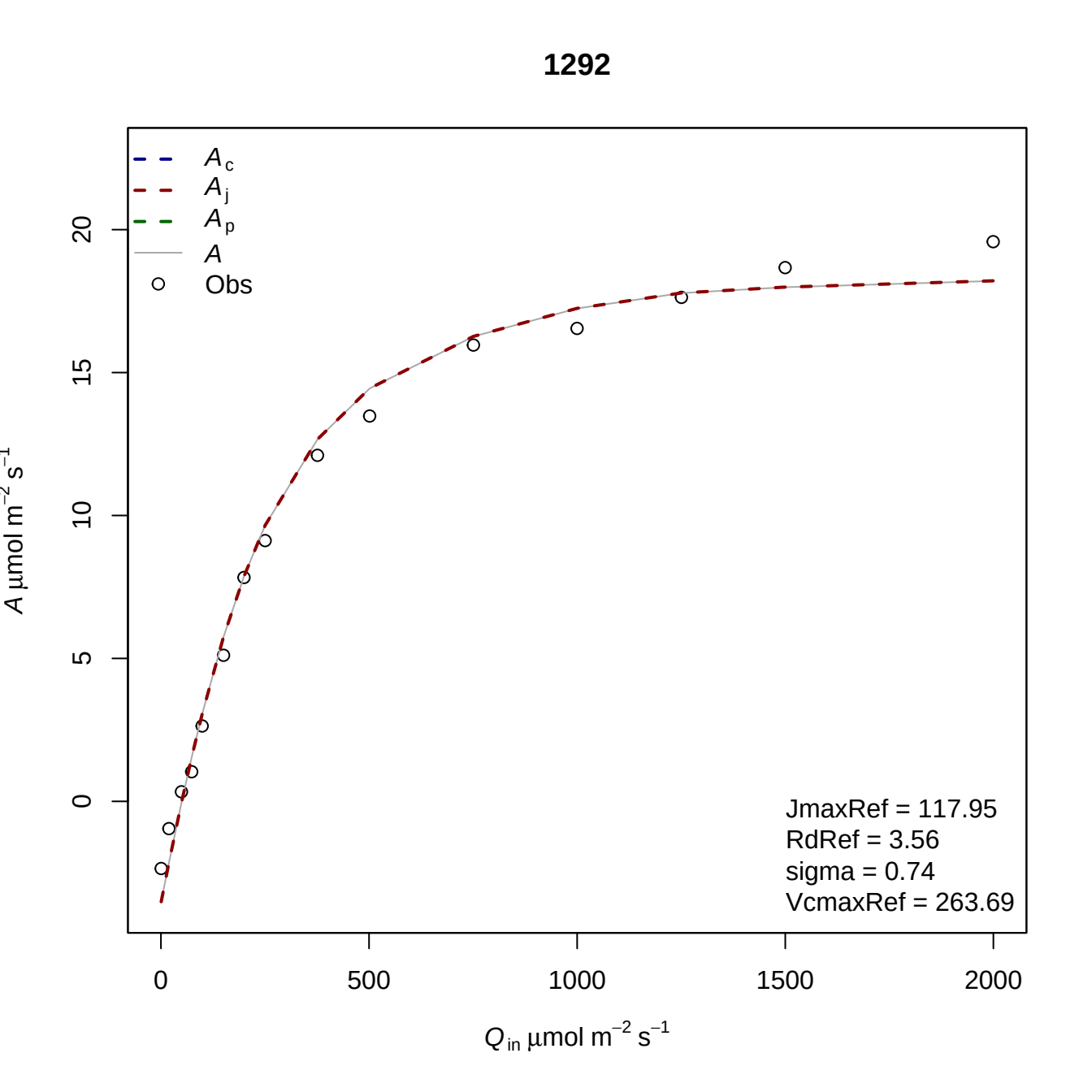

1287
A_c
A_j
A_p
A
Obs
JmaxRef = 104.47
RdRef = 3.58
sigma = 1.03
VcmaxRef = 34393.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

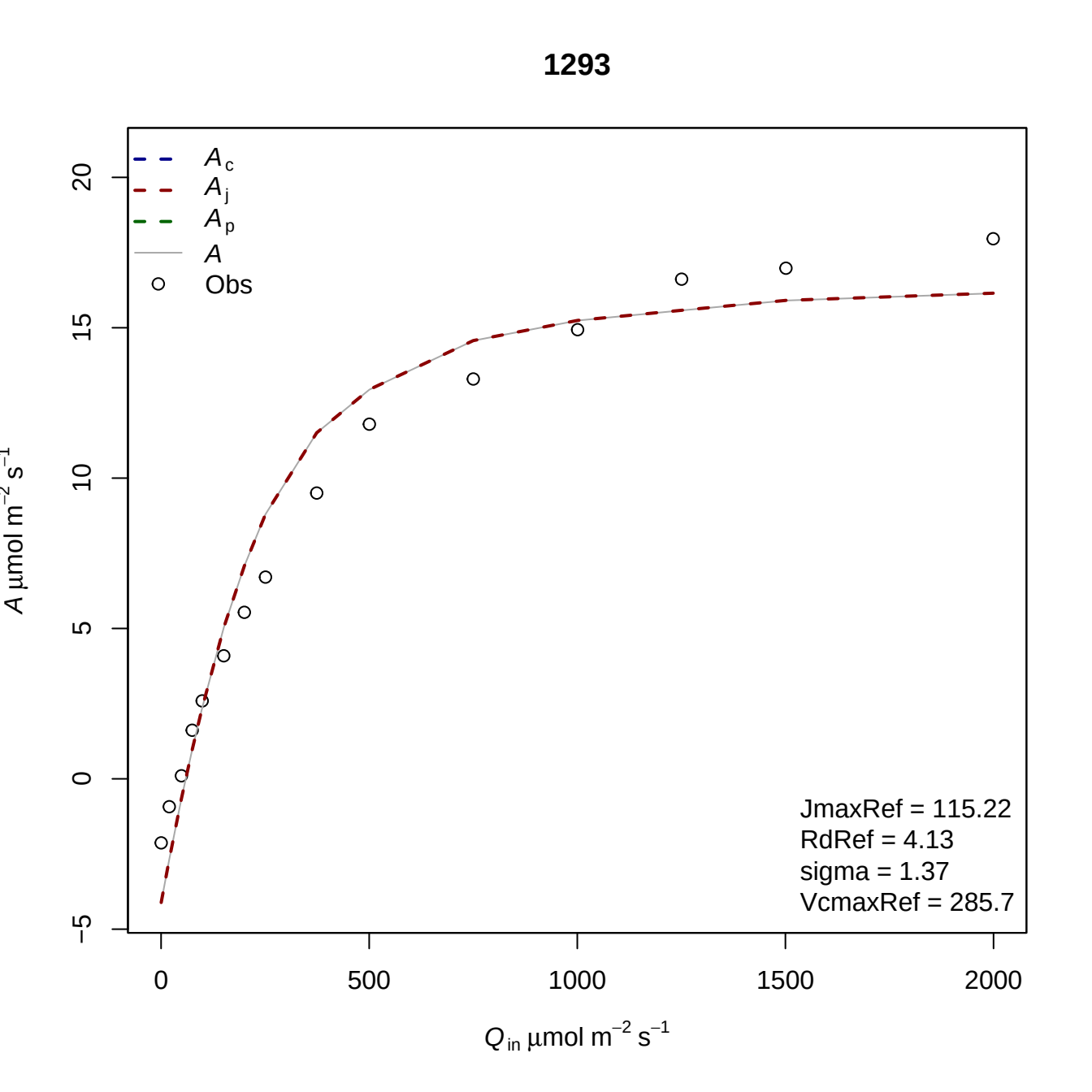

1288
A_c
A_j
A_p
A
Obs
JmaxRef = 107.85
RdRef = 3.91
sigma = 1.7
VcmaxRef = 278.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

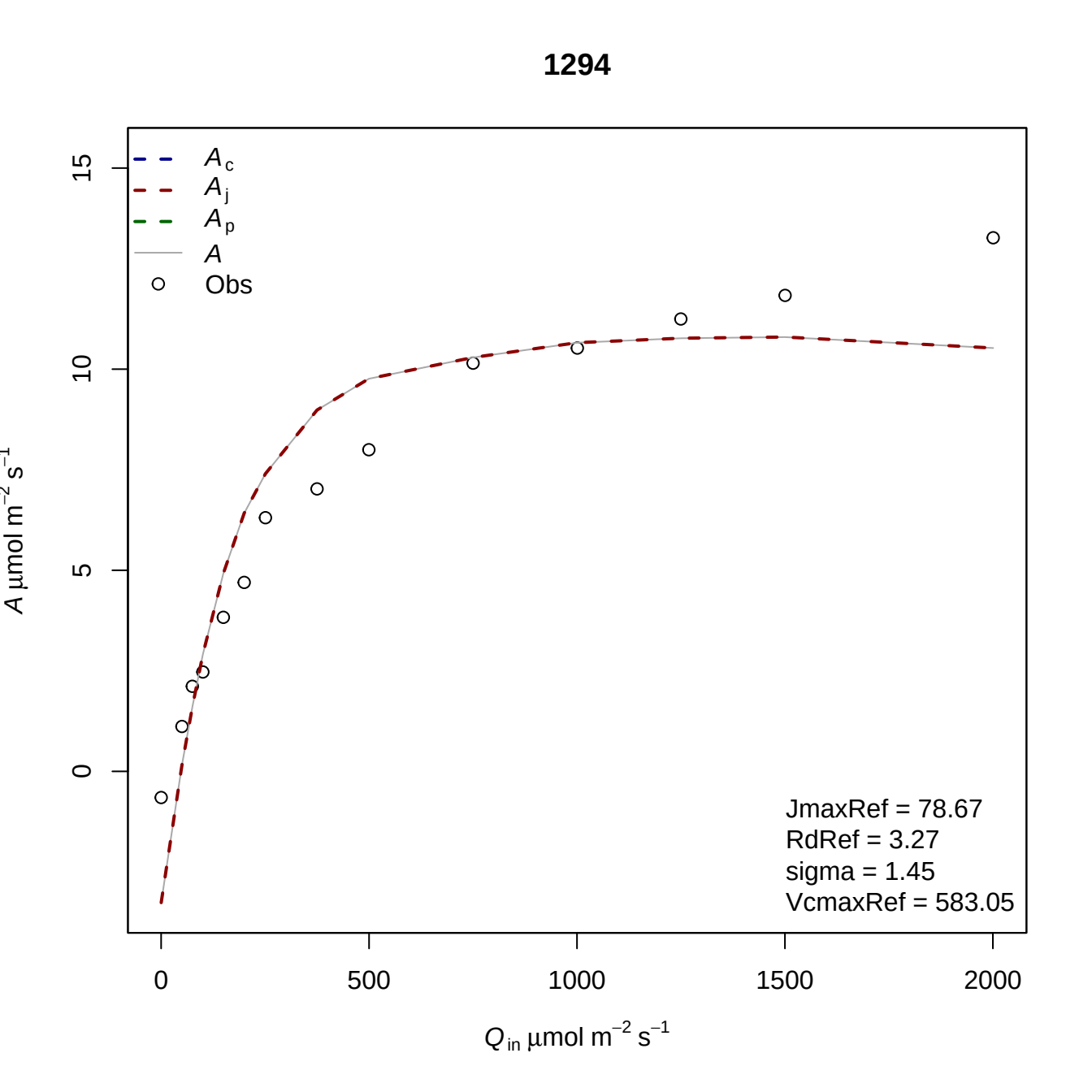

1289
A_c
A_j
A_p
A
Obs
JmaxRef = 126.74
RdRef = 5.7
sigma = 1.8
VcmaxRef = 289.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

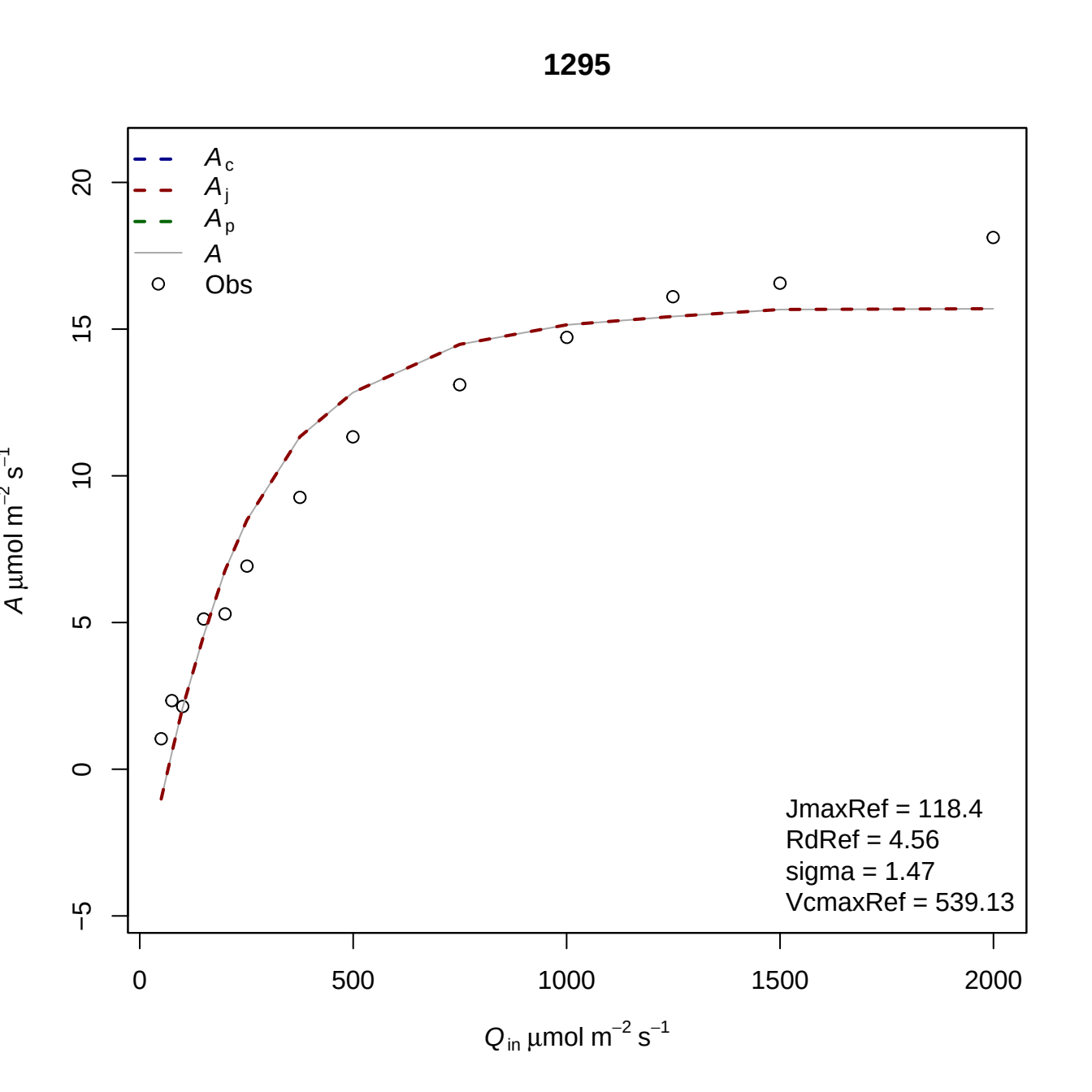

1290
A_c
A_j
A_p
A
Obs
JmaxRef = 79.41
RdRef = 4.32
sigma = 1.32
VcmaxRef = 120.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
200
400
600
800
1000
1200

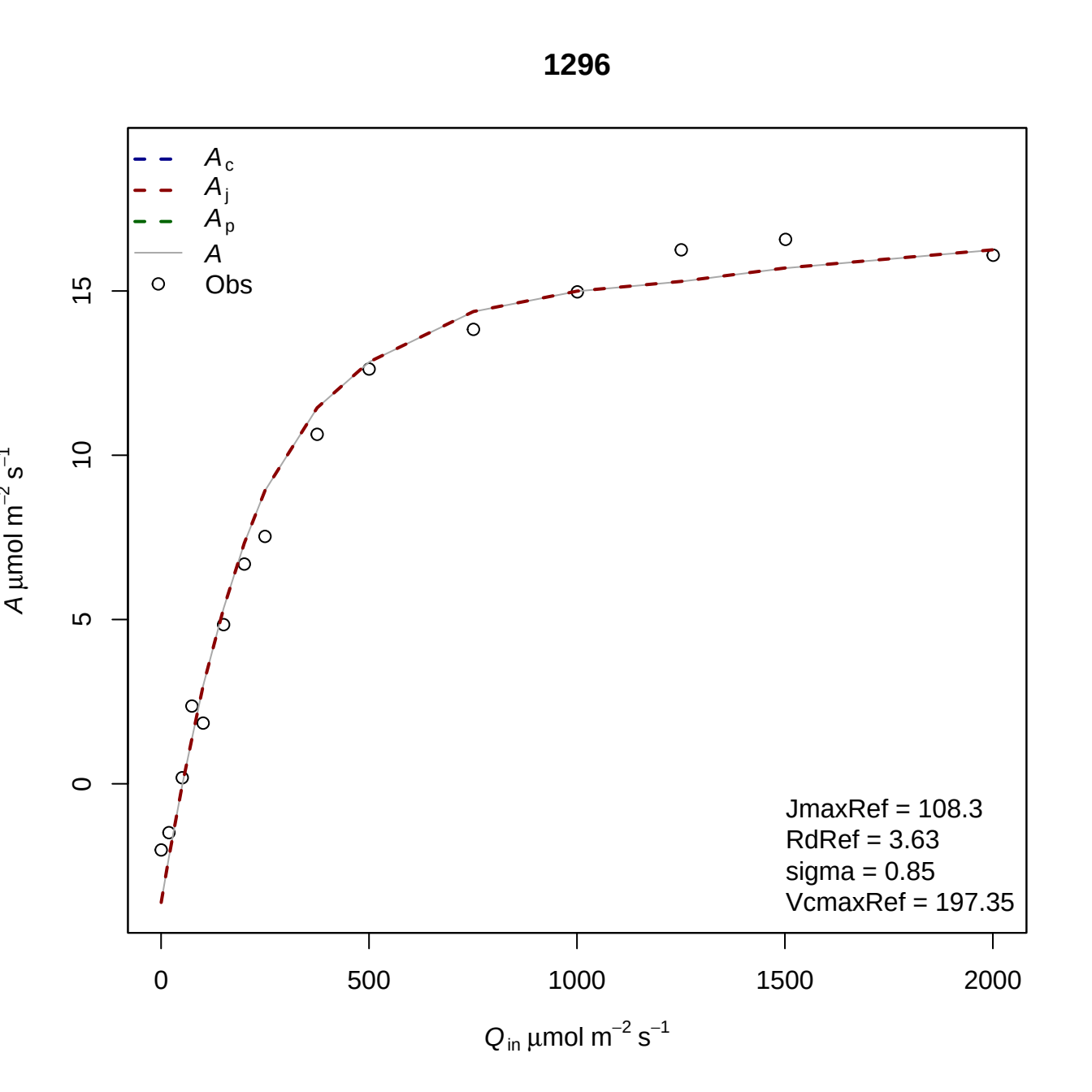

1291
A_c
A_j
A_p
A
Obs
JmaxRef = 85.1
RdRef = 5.01
sigma = 1.75
VcmaxRef = 221.57
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

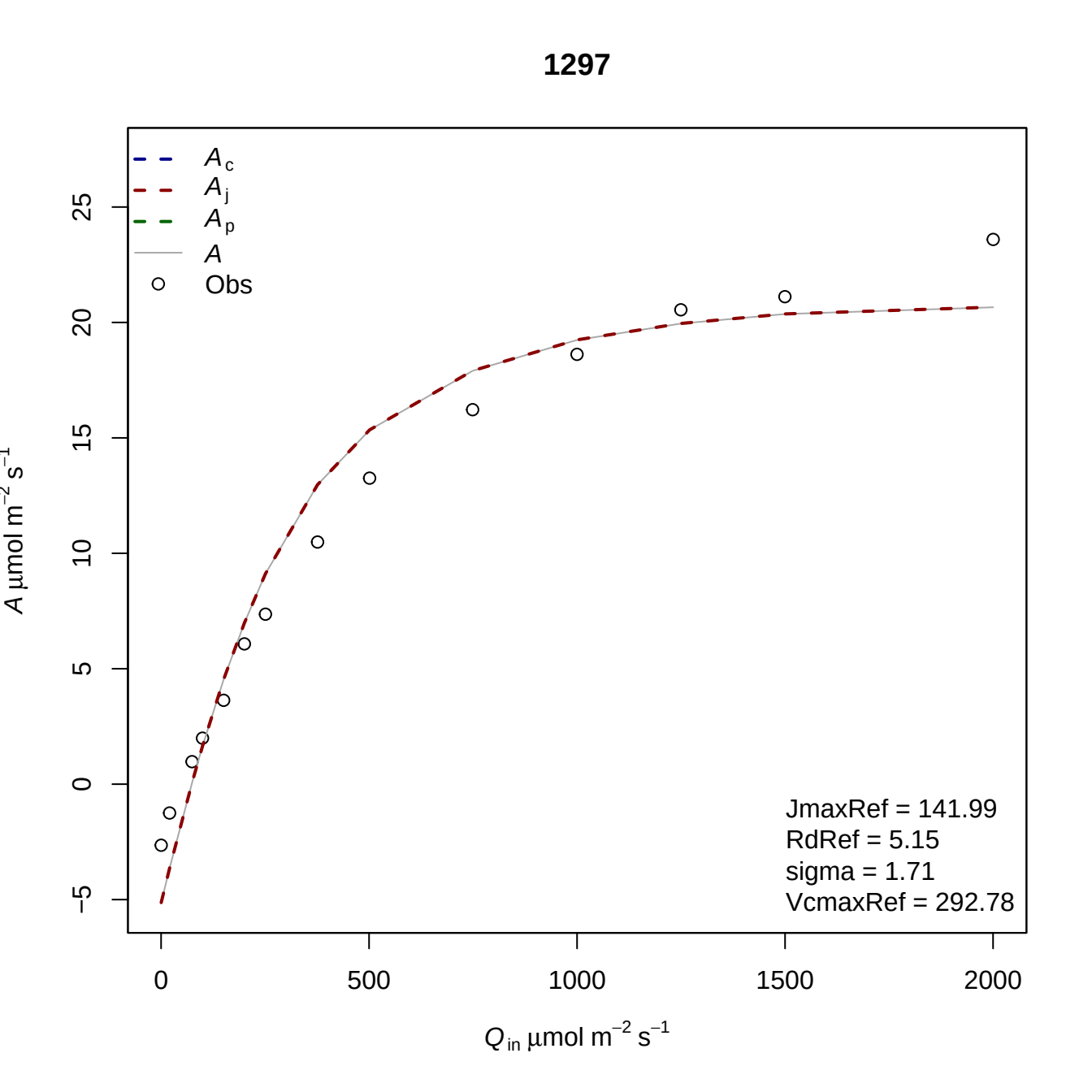

1292
A_c
A_j
A_p
A
Obs
JmaxRef = 117.95
RdRef = 3.56
sigma = 0.74
VcmaxRef = 263.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

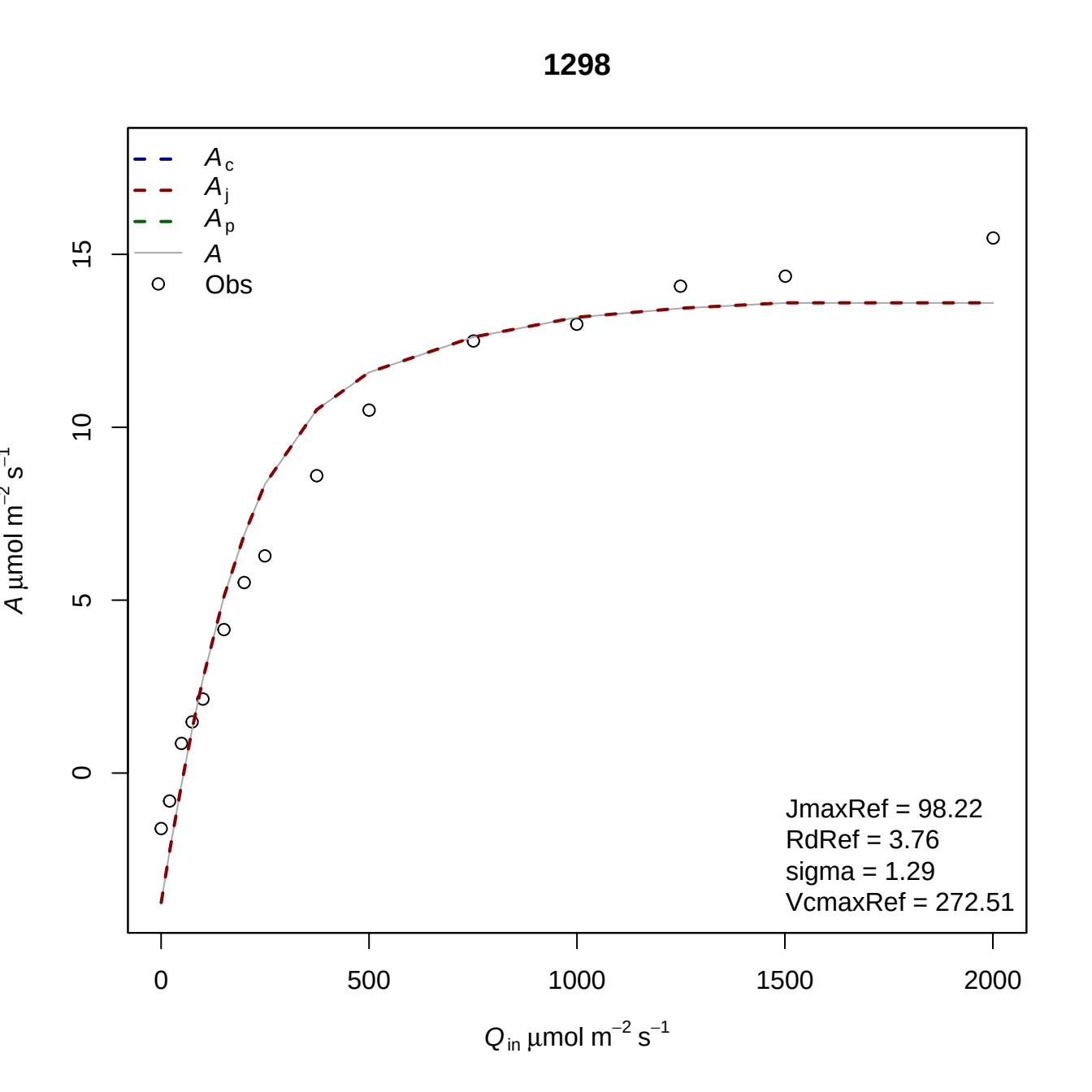

1293
A_c
A_j
A_p
A
Obs
JmaxRef = 115.22
RdRef = 4.13
sigma = 1.37
VcmaxRef = 285.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

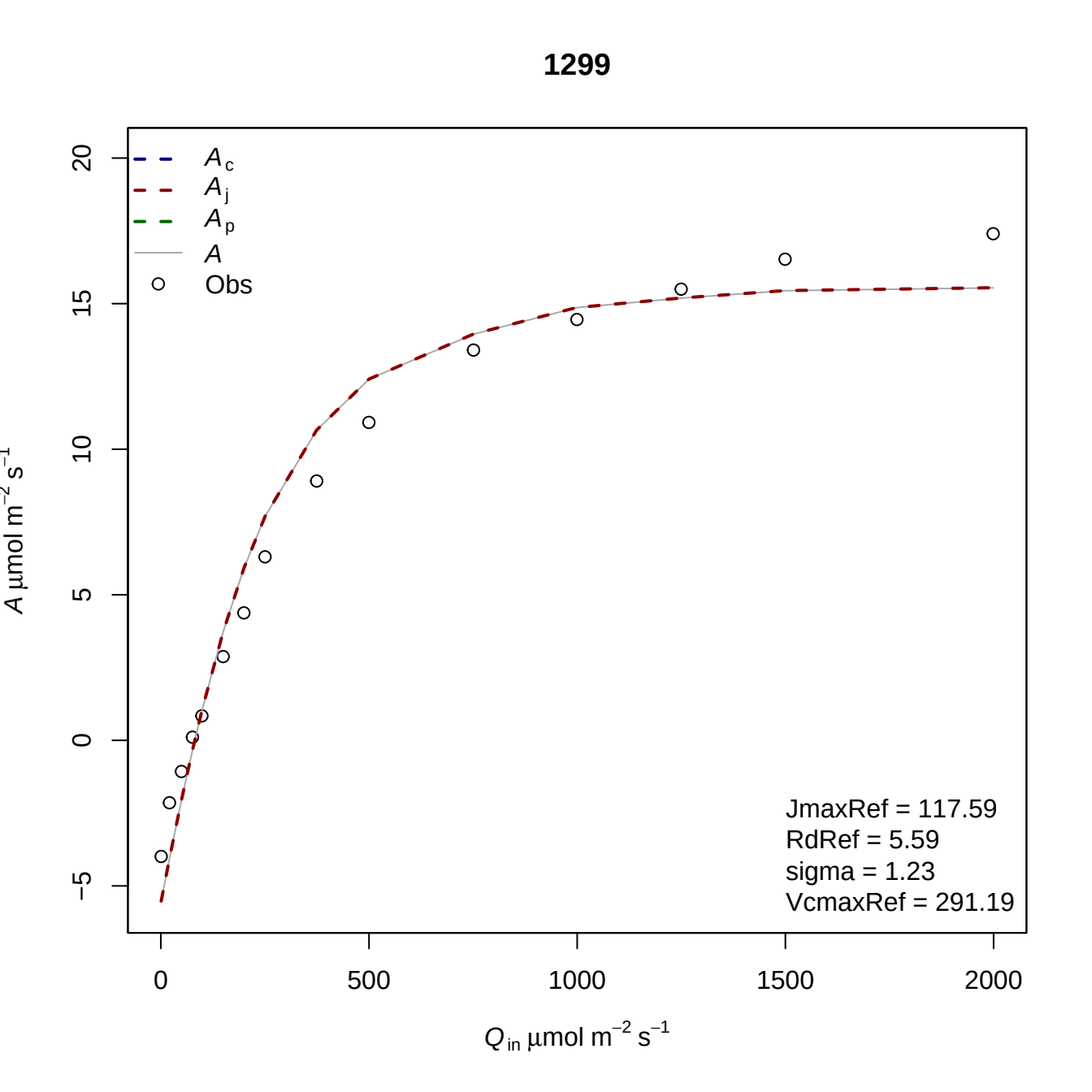

1294
A_c
A_j
A_p
A
Obs
JmaxRef = 78.67
RdRef = 3.27
sigma = 1.45
VcmaxRef = 583.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

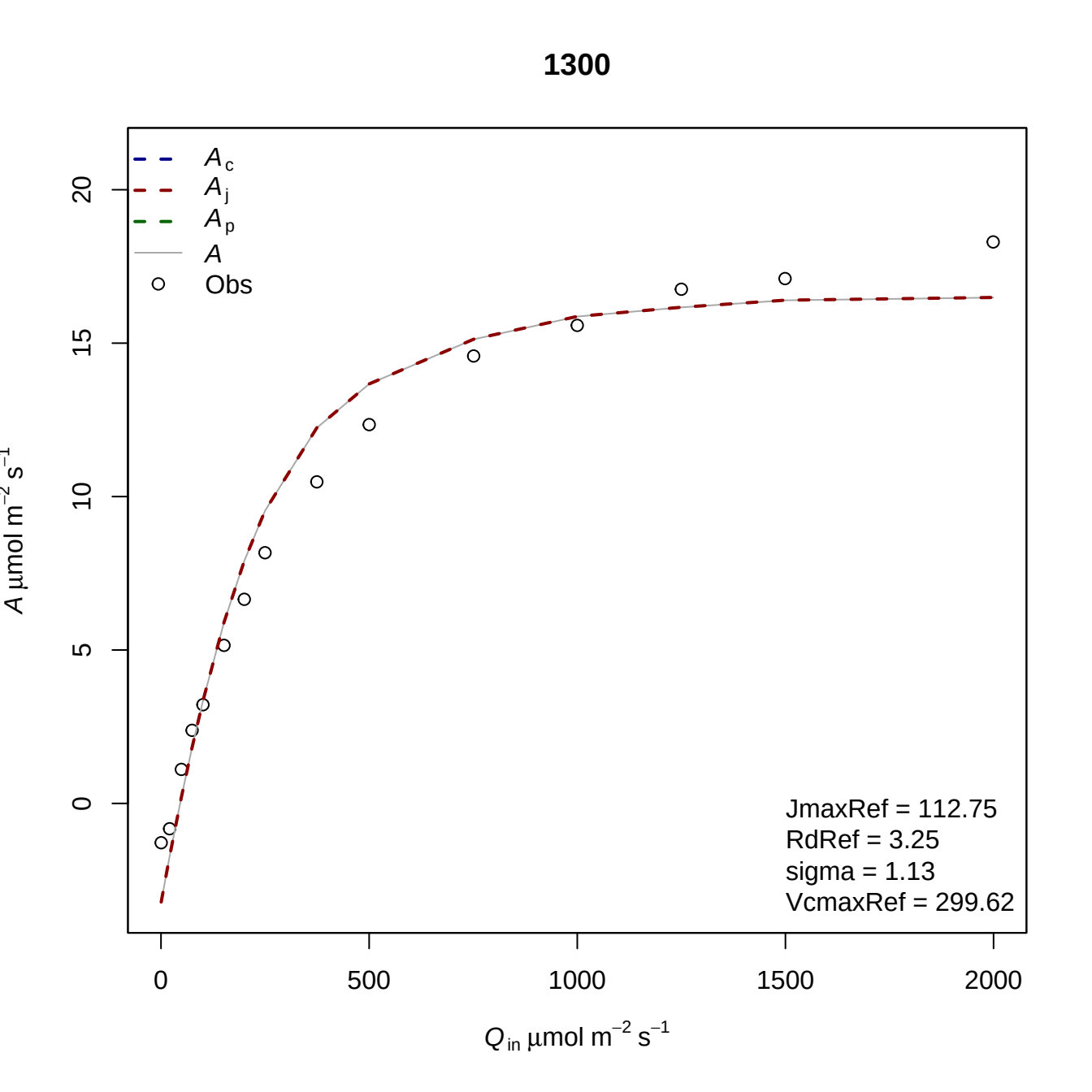

1295
A_c
A_j
A_p
A
Obs
JmaxRef = 118.4
RdRef = 4.56
sigma = 1.47
VcmaxRef = 539.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

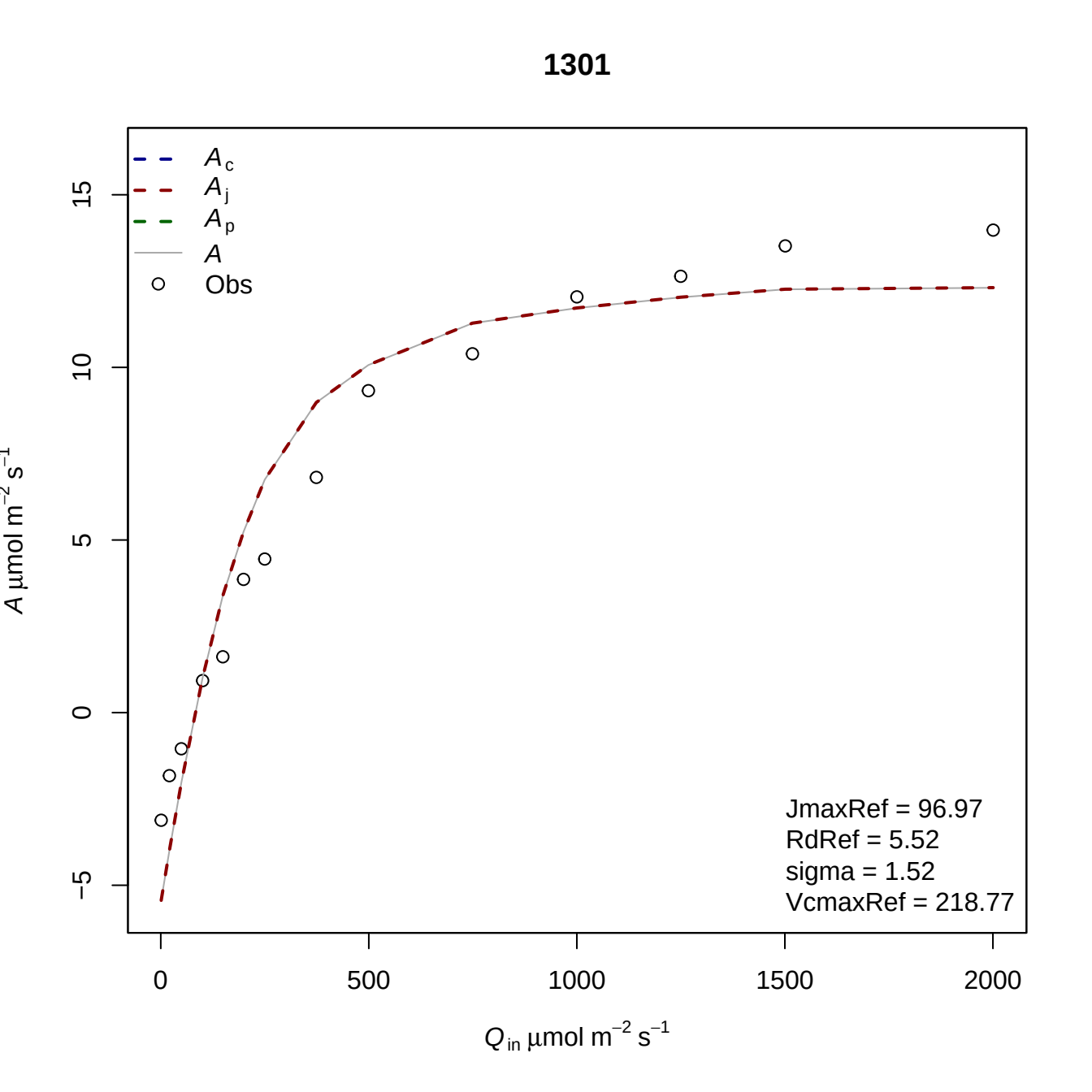

1296
A_c
A_j
A_p
A
Obs
JmaxRef = 108.3
RdRef = 3.63
sigma = 0.85
VcmaxRef = 197.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

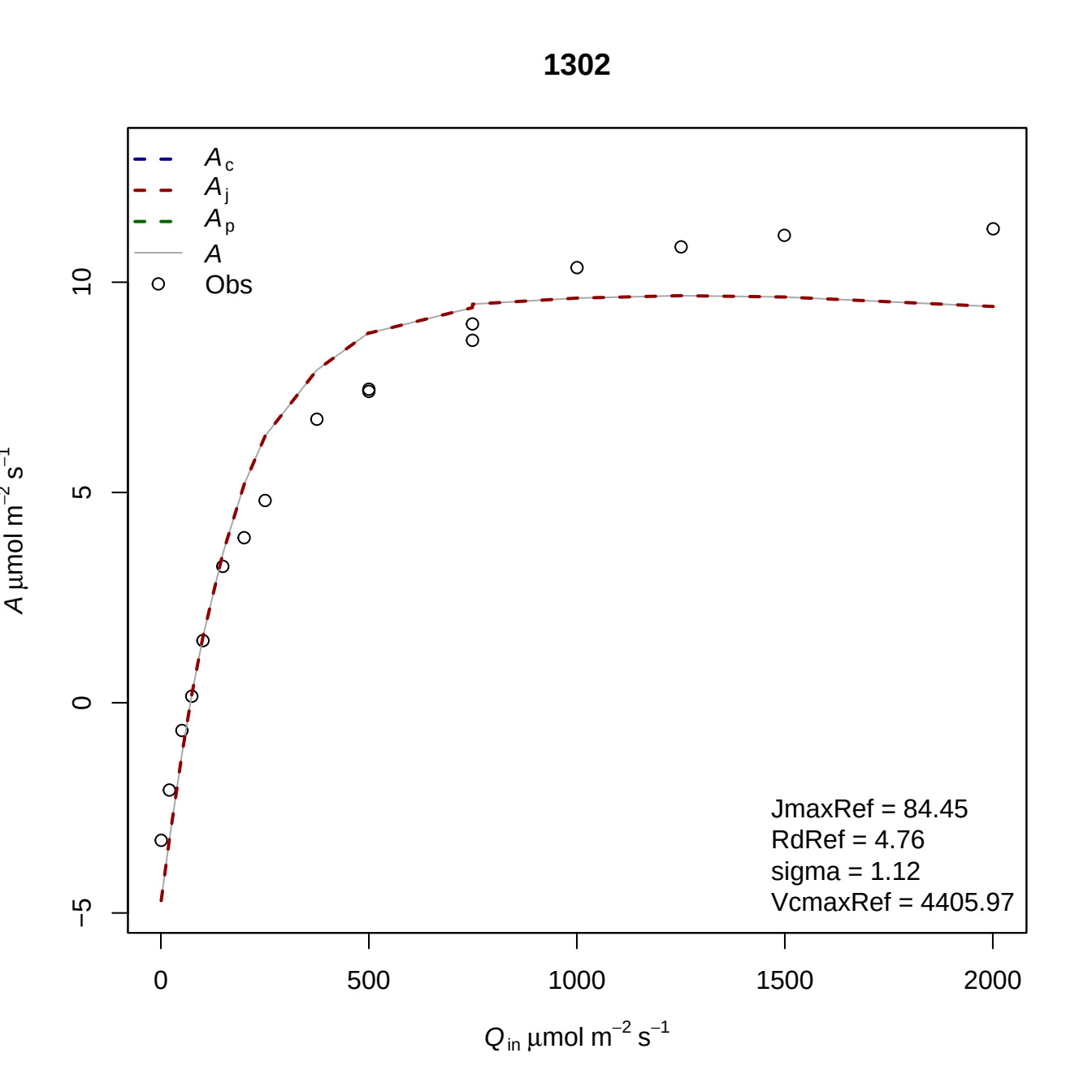

1297
A_c
A_j
A_p
A
Obs
JmaxRef = 141.99
RdRef = 5.15
sigma = 1.71
VcmaxRef = 292.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500
2000

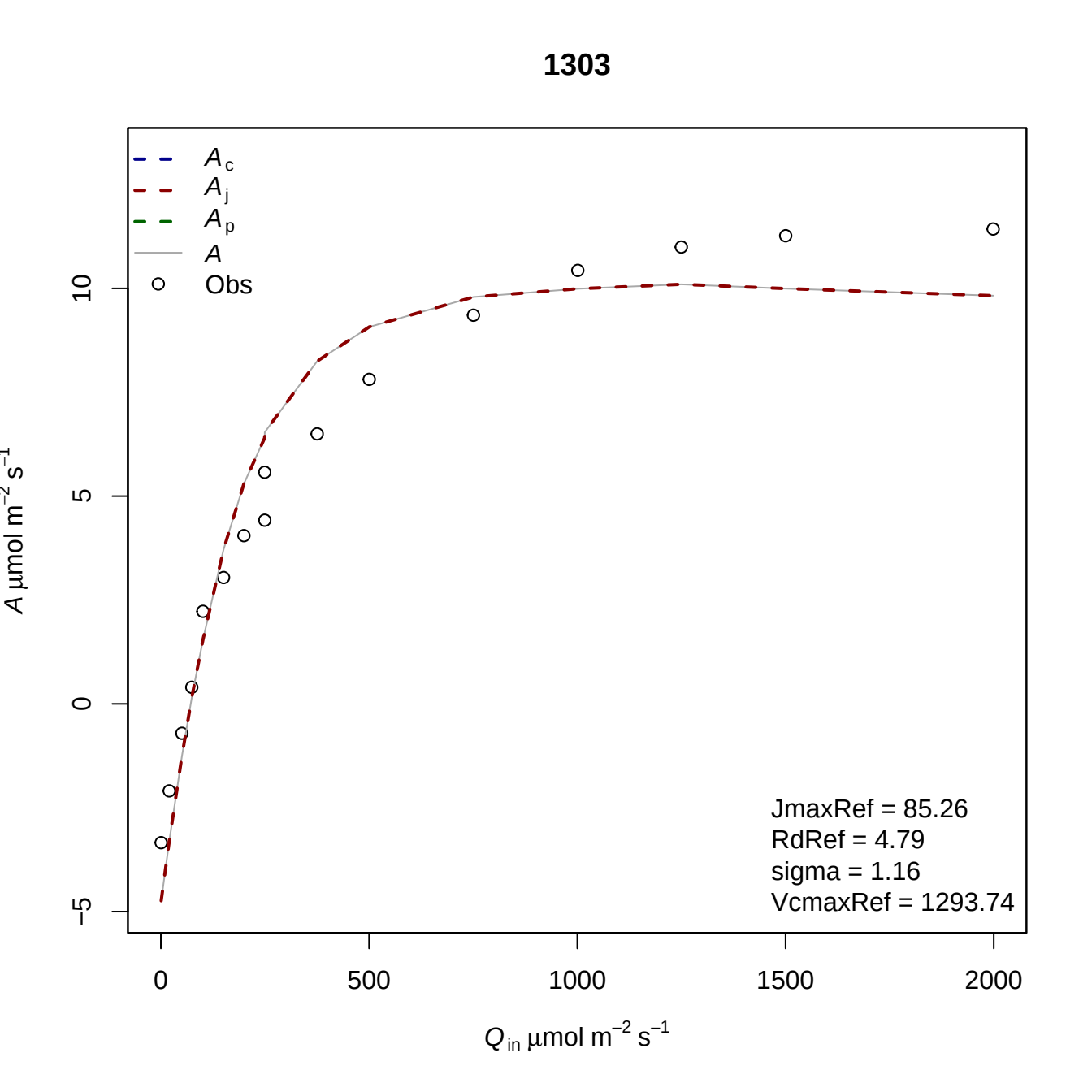

1298
A_c
A_j
A_p
A
Obs
JmaxRef = 98.22
RdRef = 3.76
sigma = 1.29
VcmaxRef = 272.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

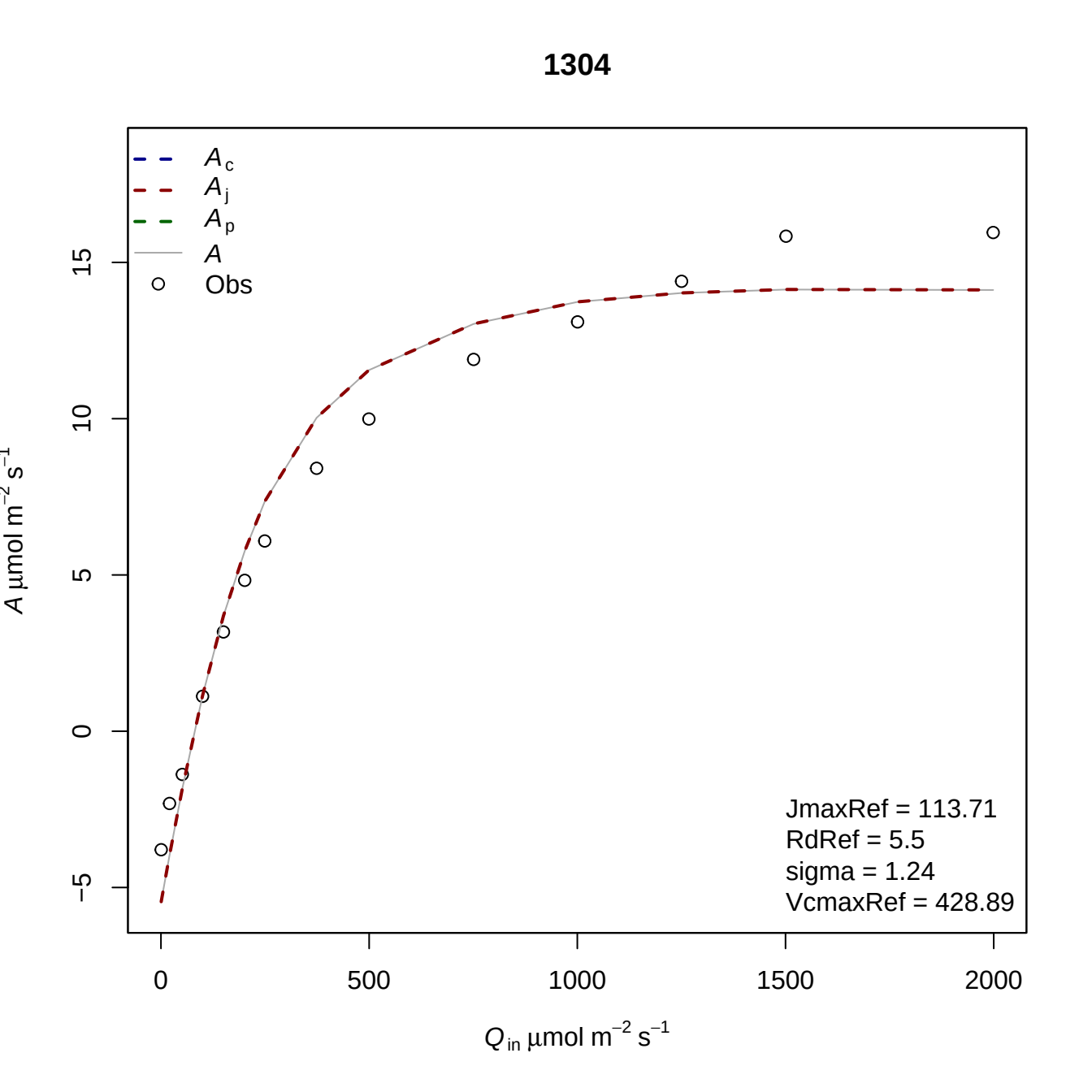

1299
A_c
A_j
A_p
A
Obs
JmaxRef = 117.59
RdRef = 5.59
sigma = 1.23
VcmaxRef = 291.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

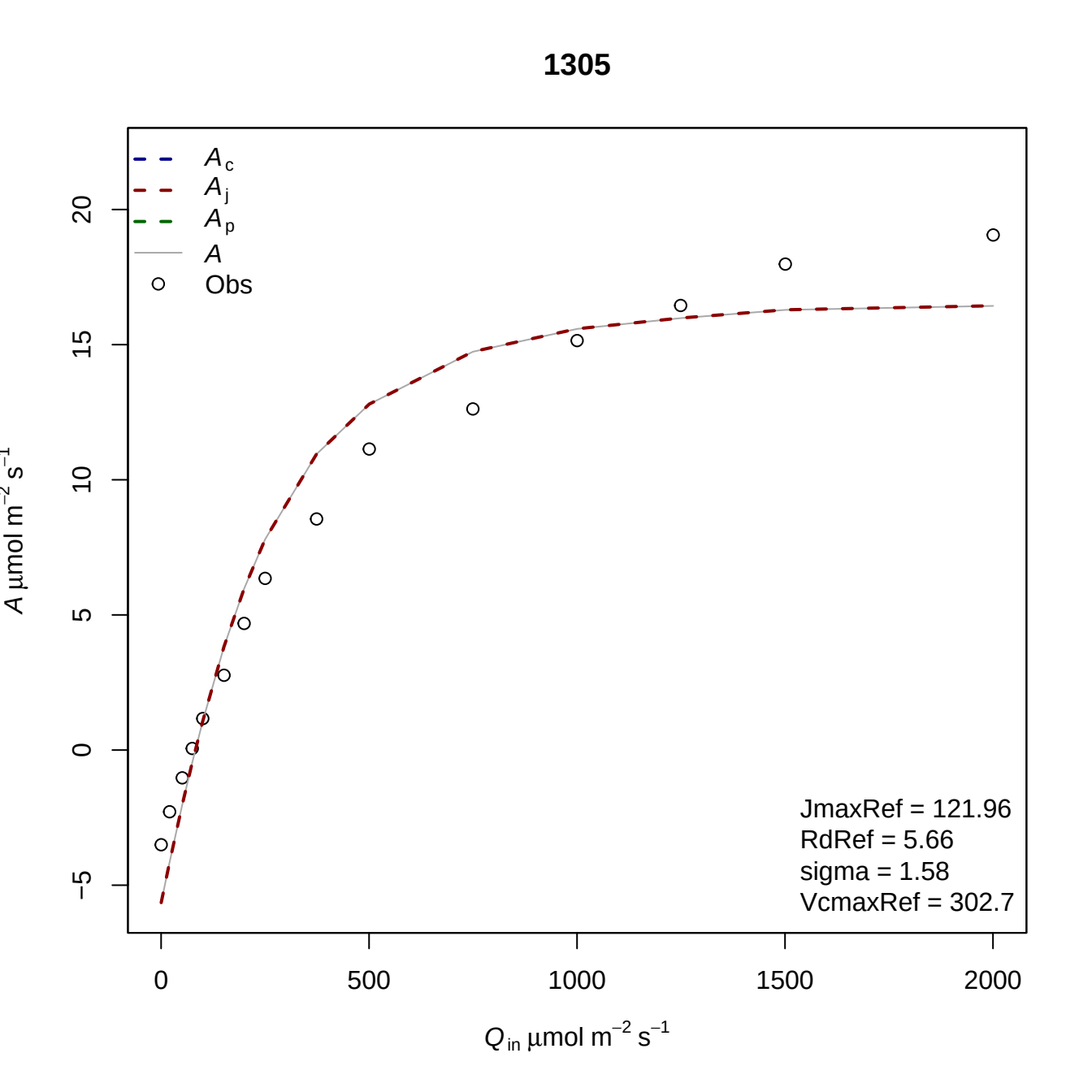

1300
A_c
A_j
A_p
A
Obs
JmaxRef = 112.75
RdRef = 3.25
sigma = 1.13
VcmaxRef = 299.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

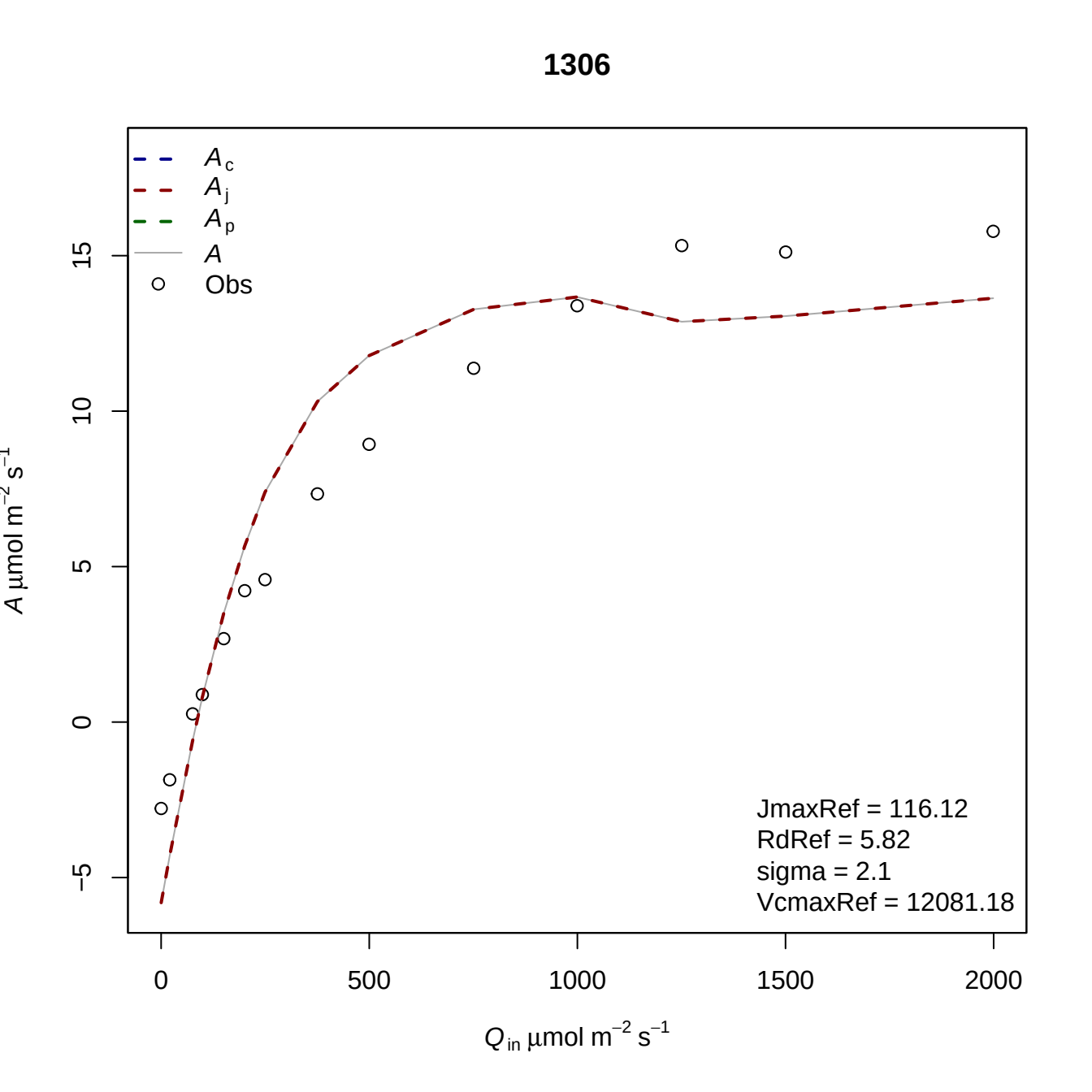

1301
A_c
A_j
A_p
A
Obs
JmaxRef = 96.97
RdRef = 5.52
sigma = 1.52
VcmaxRef = 218.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

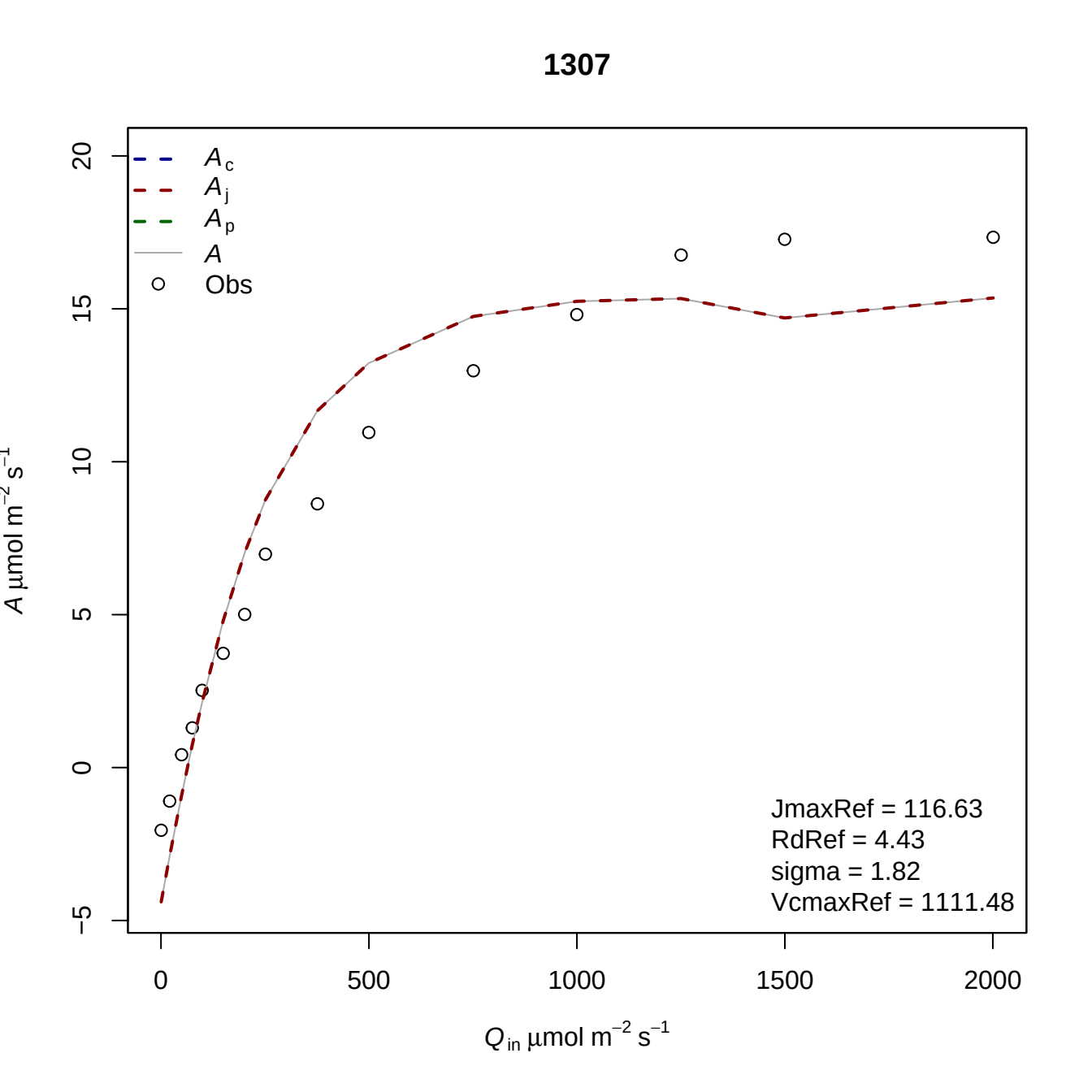

1302
A_c
A_j
A_p
A
Obs
JmaxRef = 84.45
RdRef = 4.76
sigma = 1.12
VcmaxRef = 4405.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

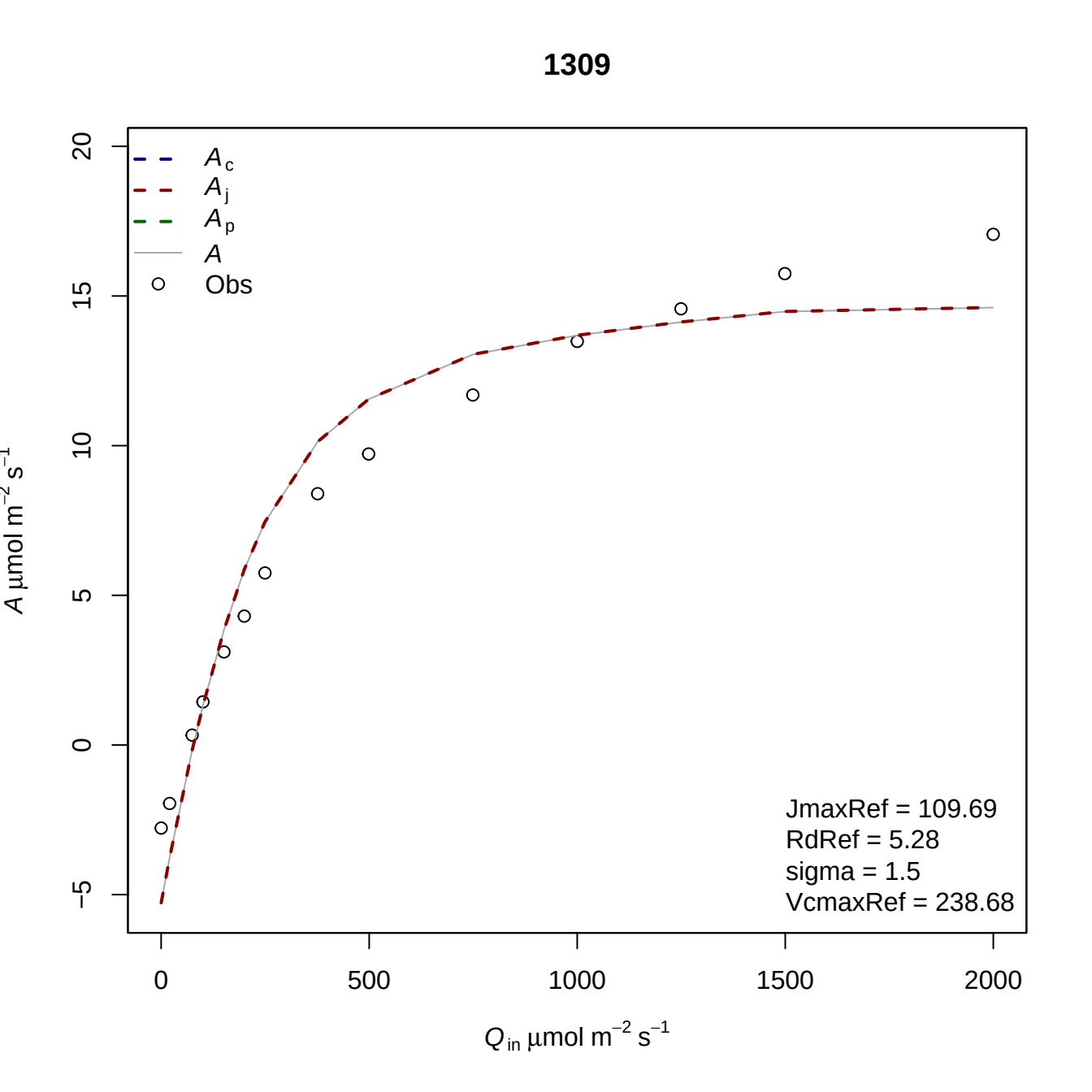

1303
A_c
A_j
A_p
A
Obs
JmaxRef = 85.26
RdRef = 4.79
sigma = 1.16
VcmaxRef = 1293.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

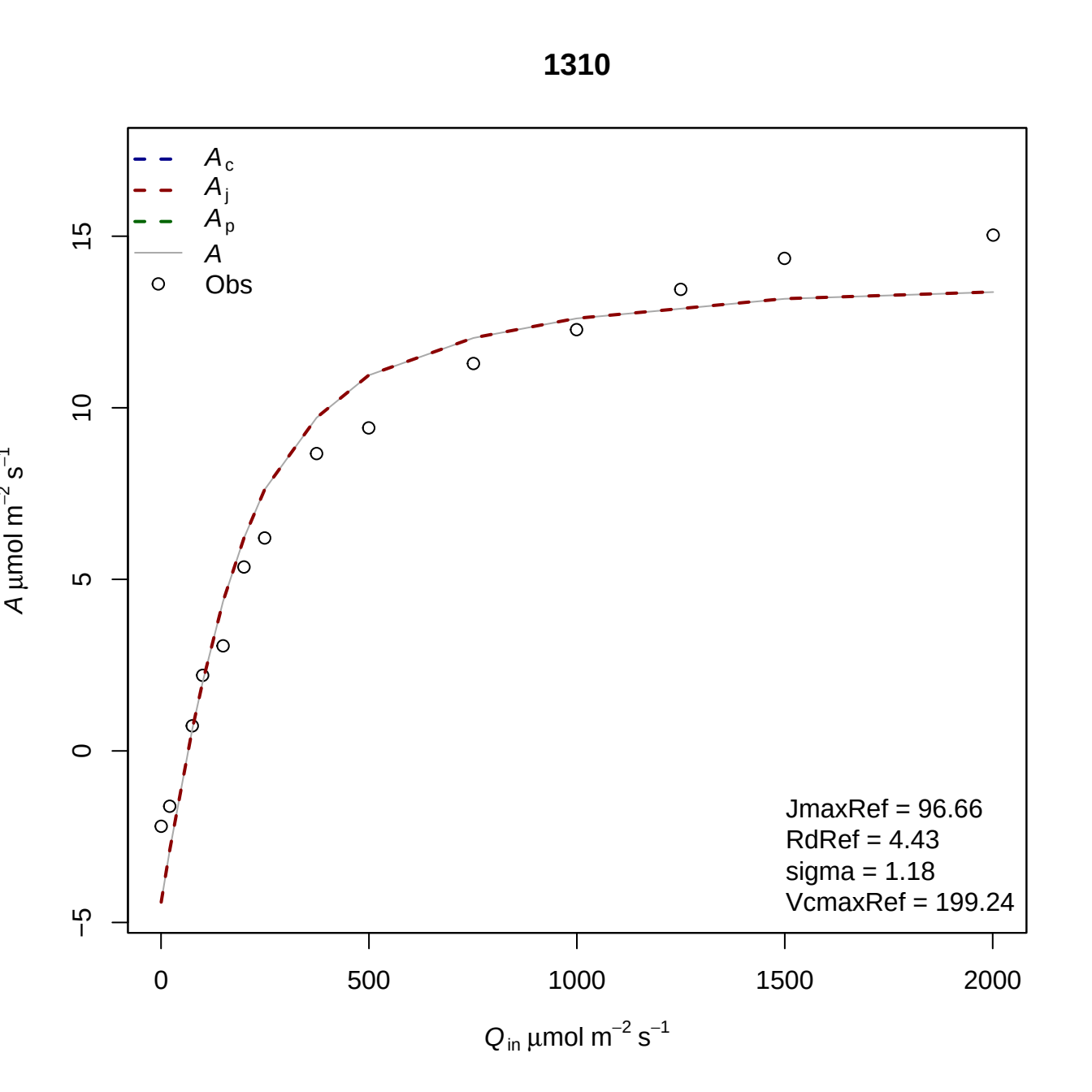

1304
A_c
A_j
A_p
A
Obs
JmaxRef = 113.71
RdRef = 5.5
sigma = 1.24
VcmaxRef = 428.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

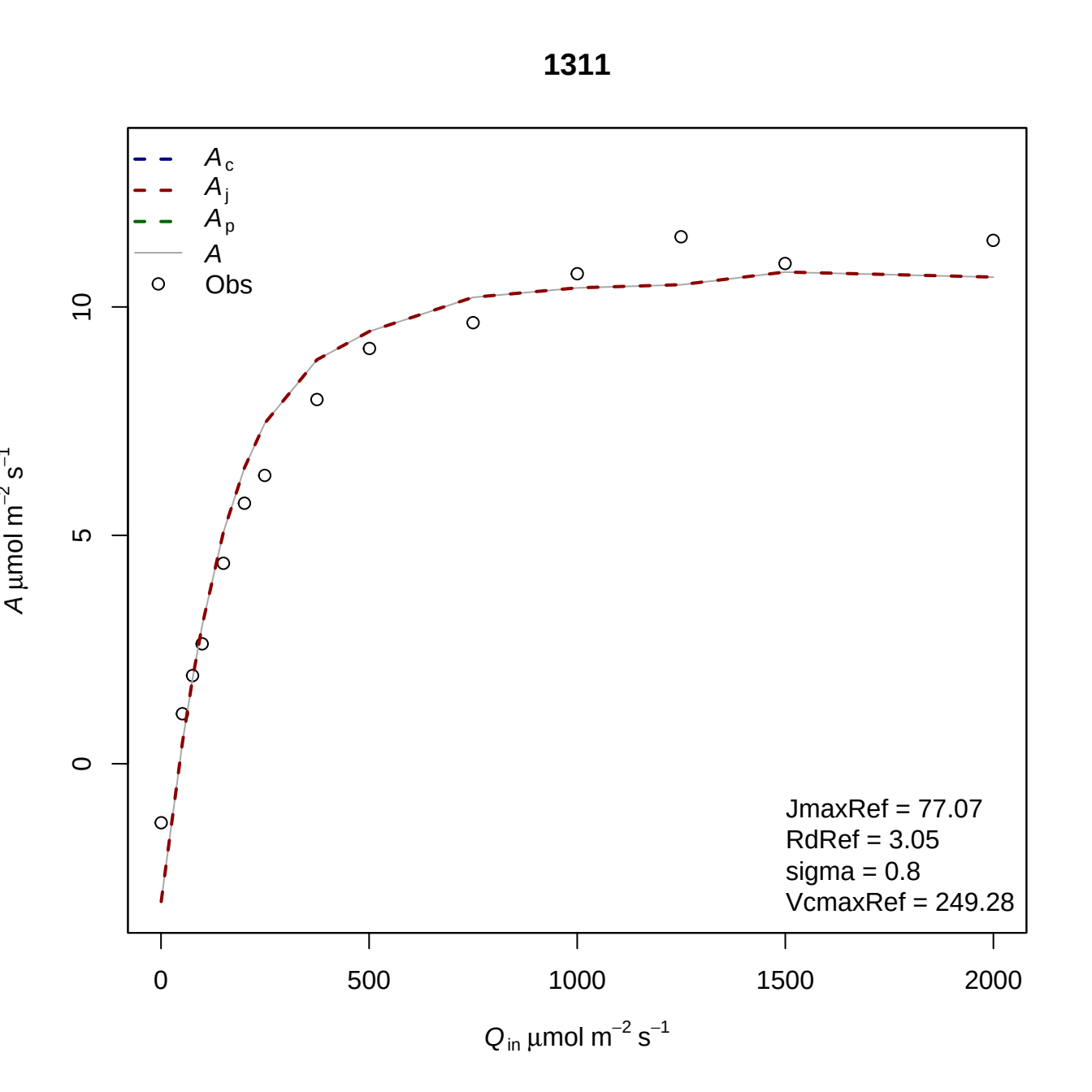

1305
A_c
A_j
A_p
A
Obs
JmaxRef = 121.96
RdRef = 5.66
sigma = 1.58
VcmaxRef = 302.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

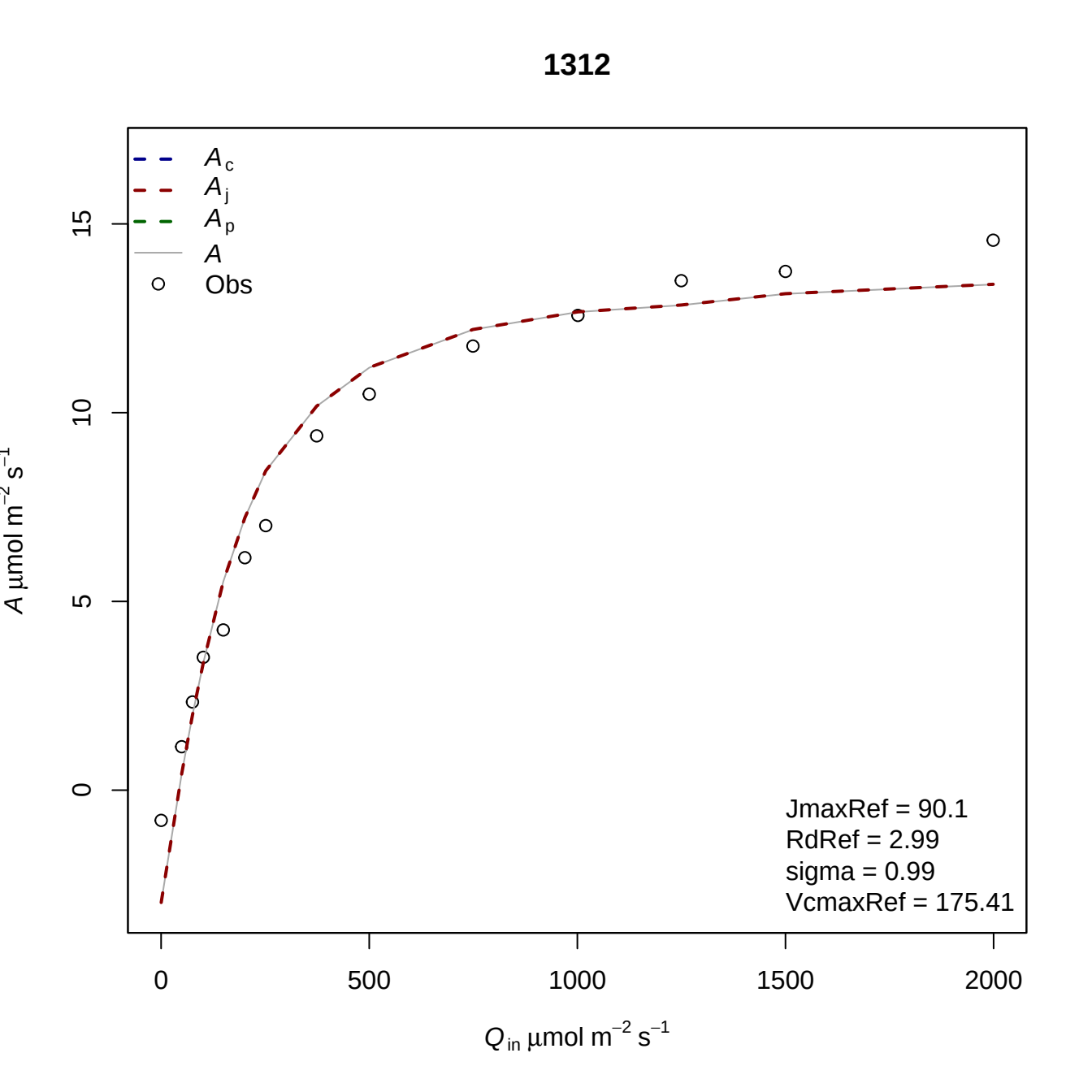

1306
A_c
A_j
A_p
A
Obs
JmaxRef = 116.12
RdRef = 5.82
sigma = 2.1
VcmaxRef = 12081.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

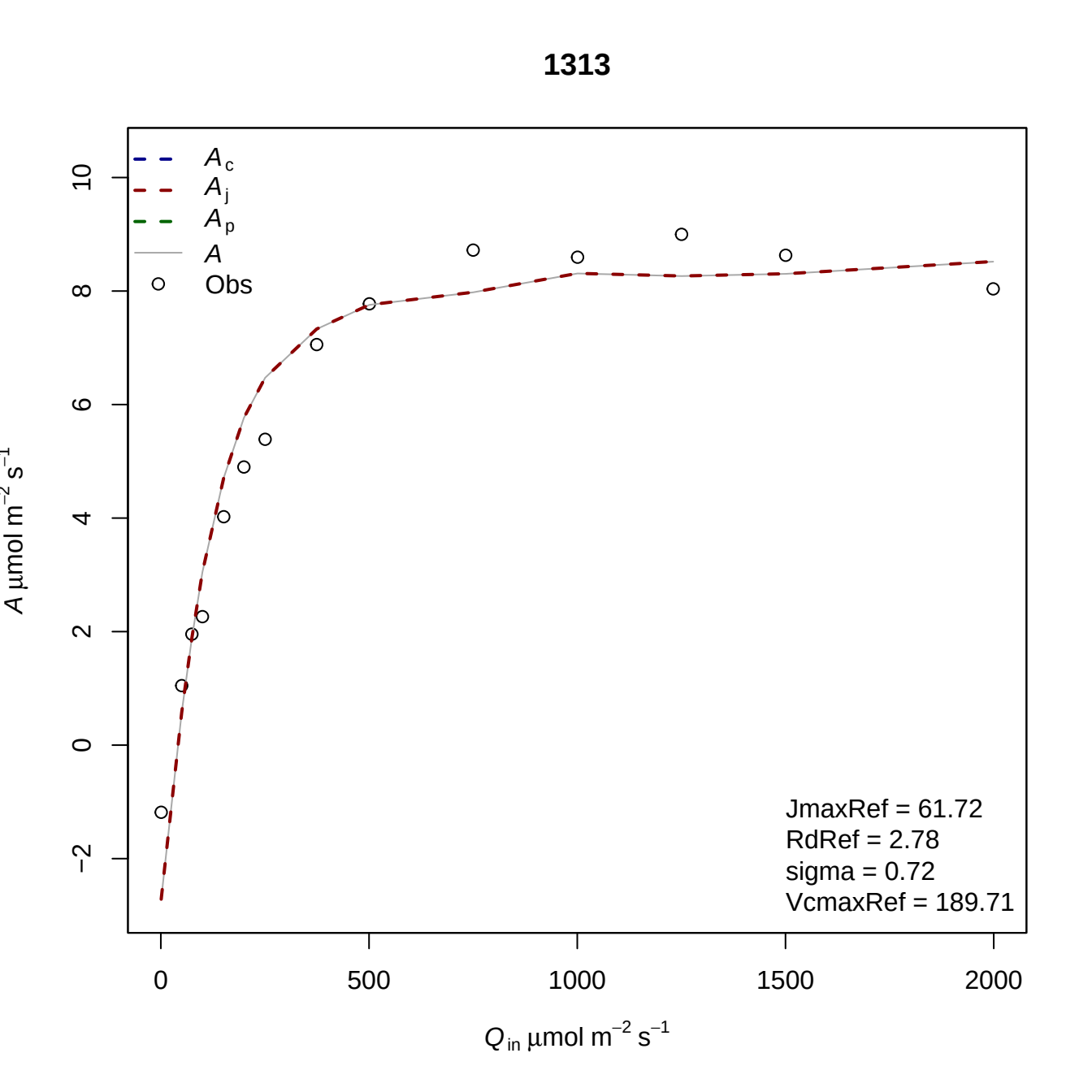

1307
A_c
A_j
A_p
A
Obs
JmaxRef = 116.63
RdRef = 4.43
sigma = 1.82
VcmaxRef = 1111.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

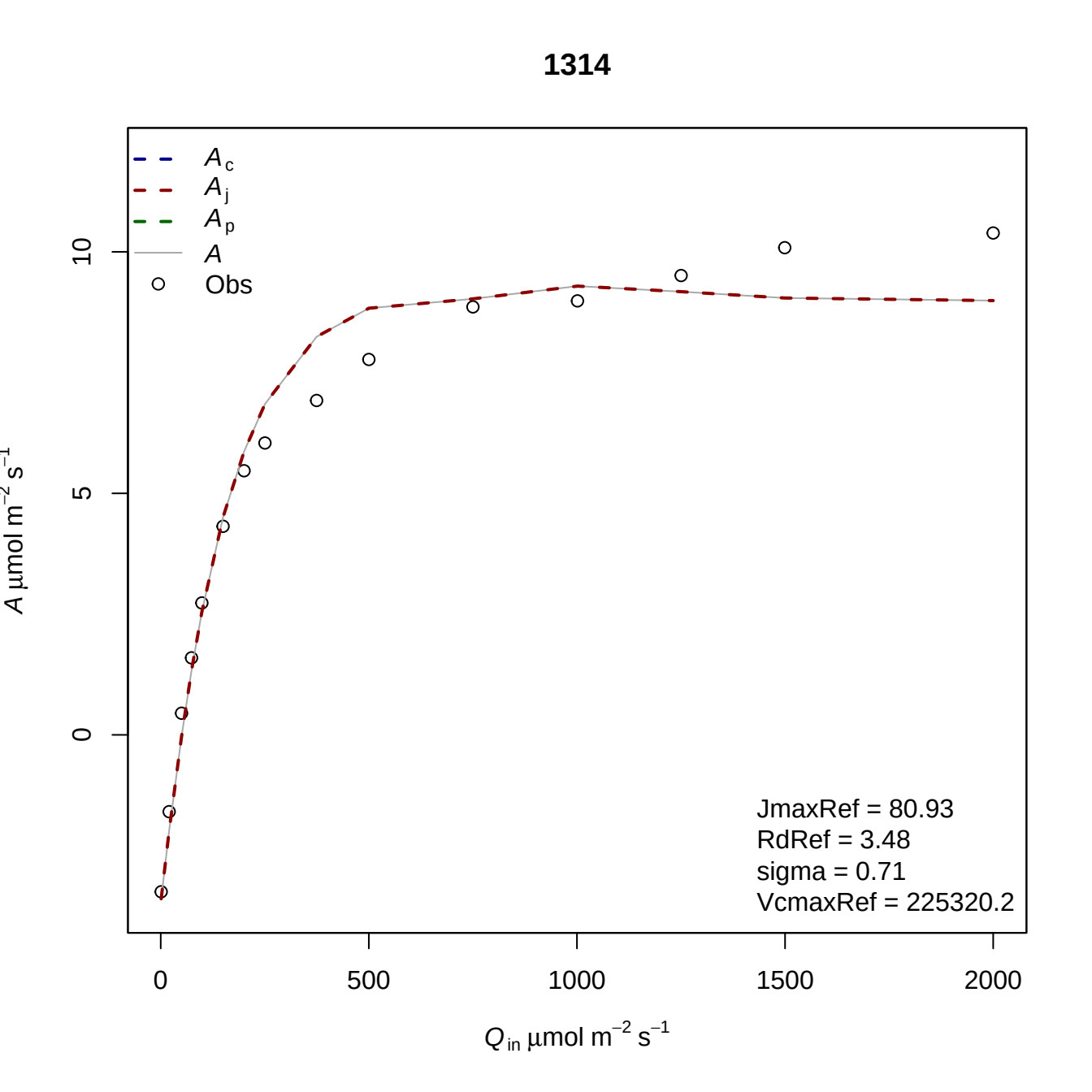

1309
A_c
A_j
A_p
A
Obs
JmaxRef = 109.69
RdRef = 5.28
sigma = 1.5
VcmaxRef = 238.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

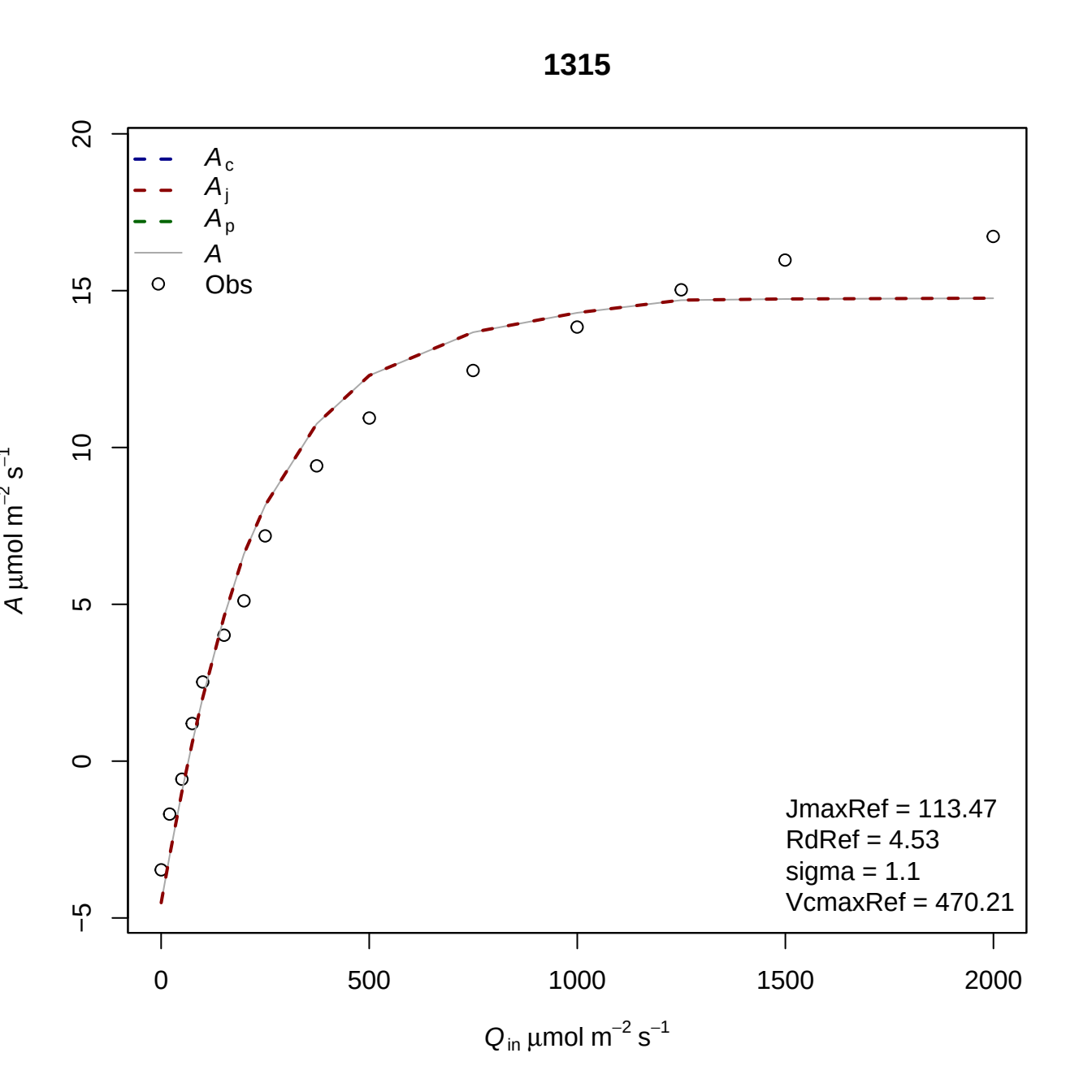

1310
A_c
A_j
A_p
A
Obs
JmaxRef = 96.66
RdRef = 4.43
sigma = 1.18
VcmaxRef = 199.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

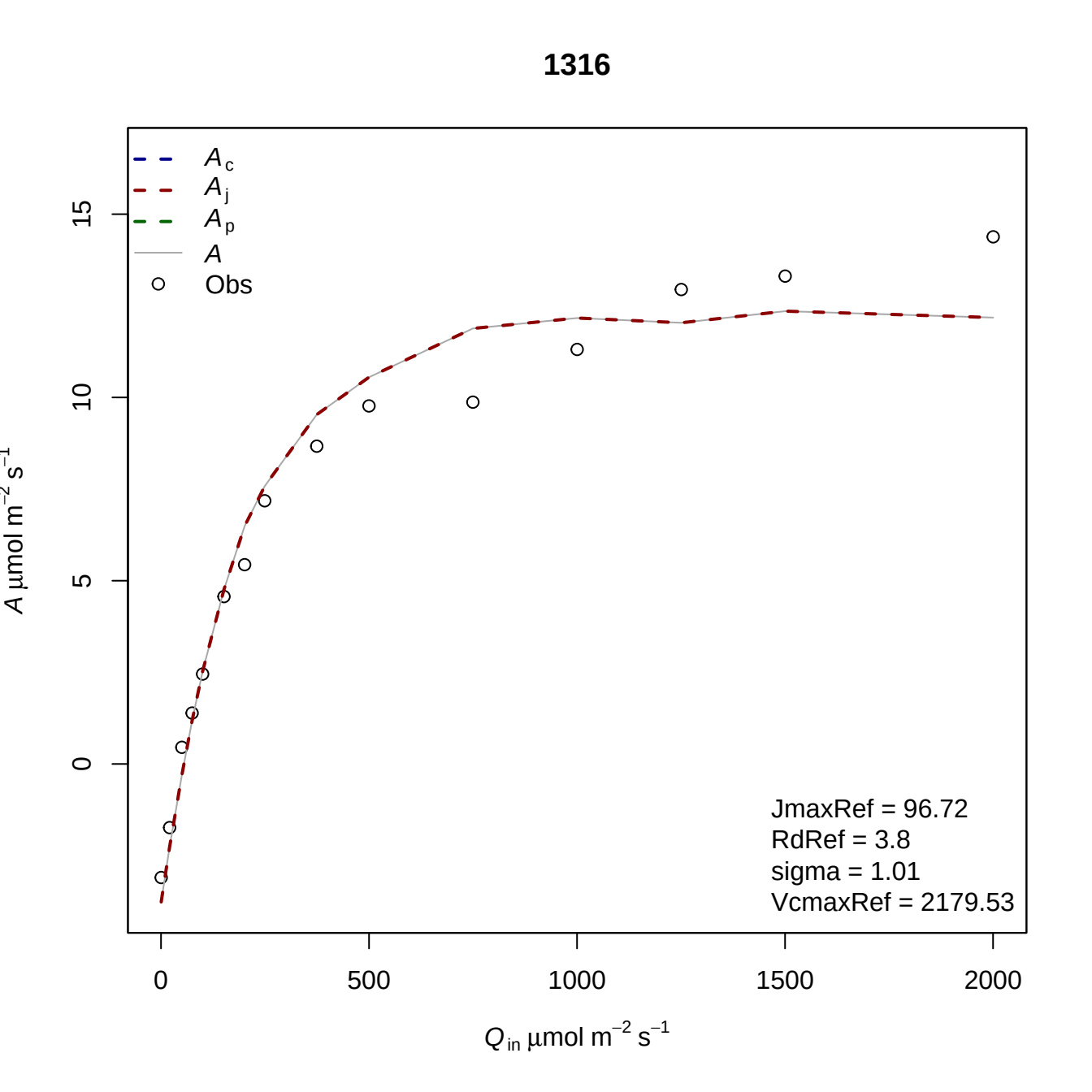

1311
A_c
A_j
A_p
A
Obs
JmaxRef = 77.07
RdRef = 3.05
sigma = 0.8
VcmaxRef = 249.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

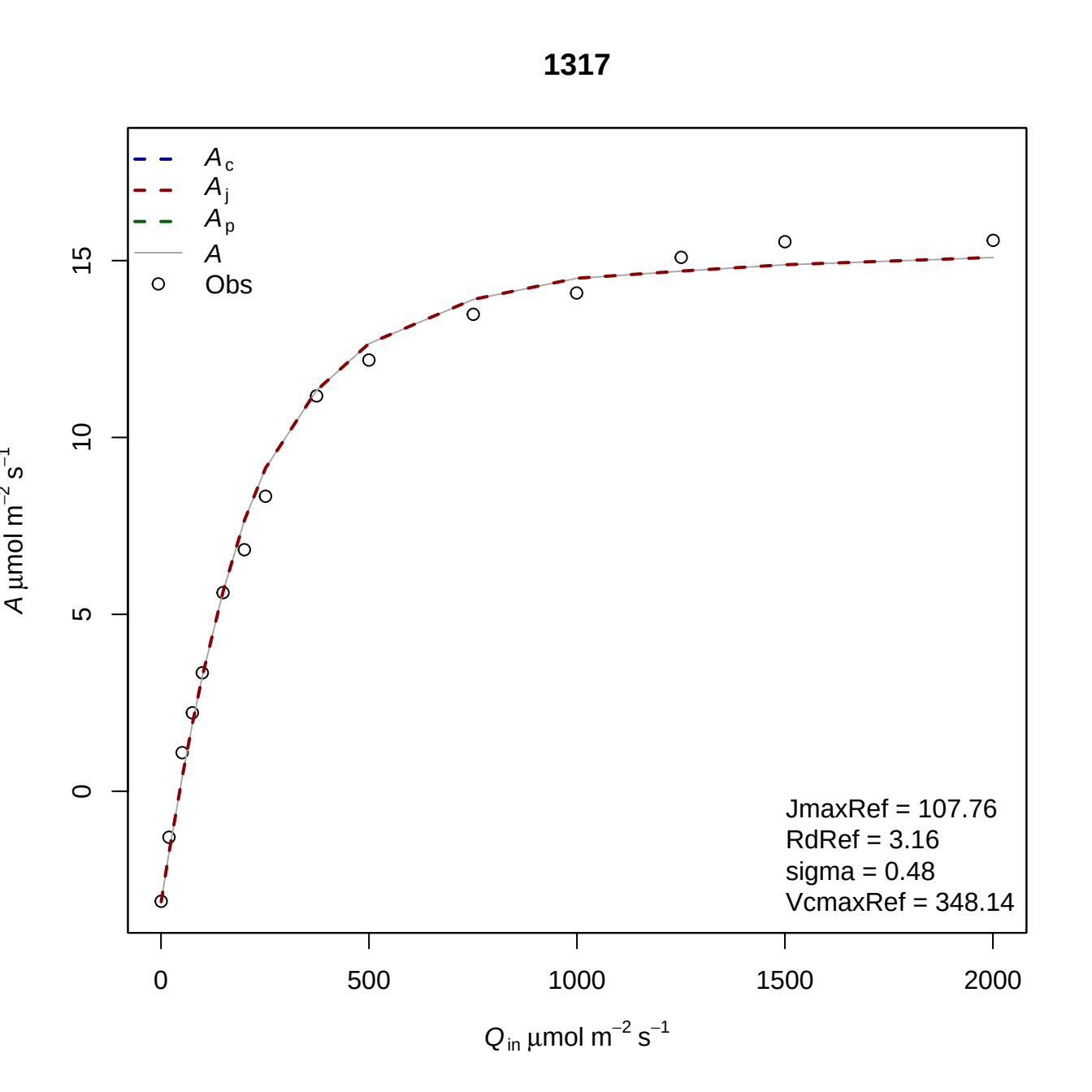

1312
A_c
A_j
A_p
A
Obs
JmaxRef = 90.1
RdRef = 2.99
sigma = 0.99
VcmaxRef = 175.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

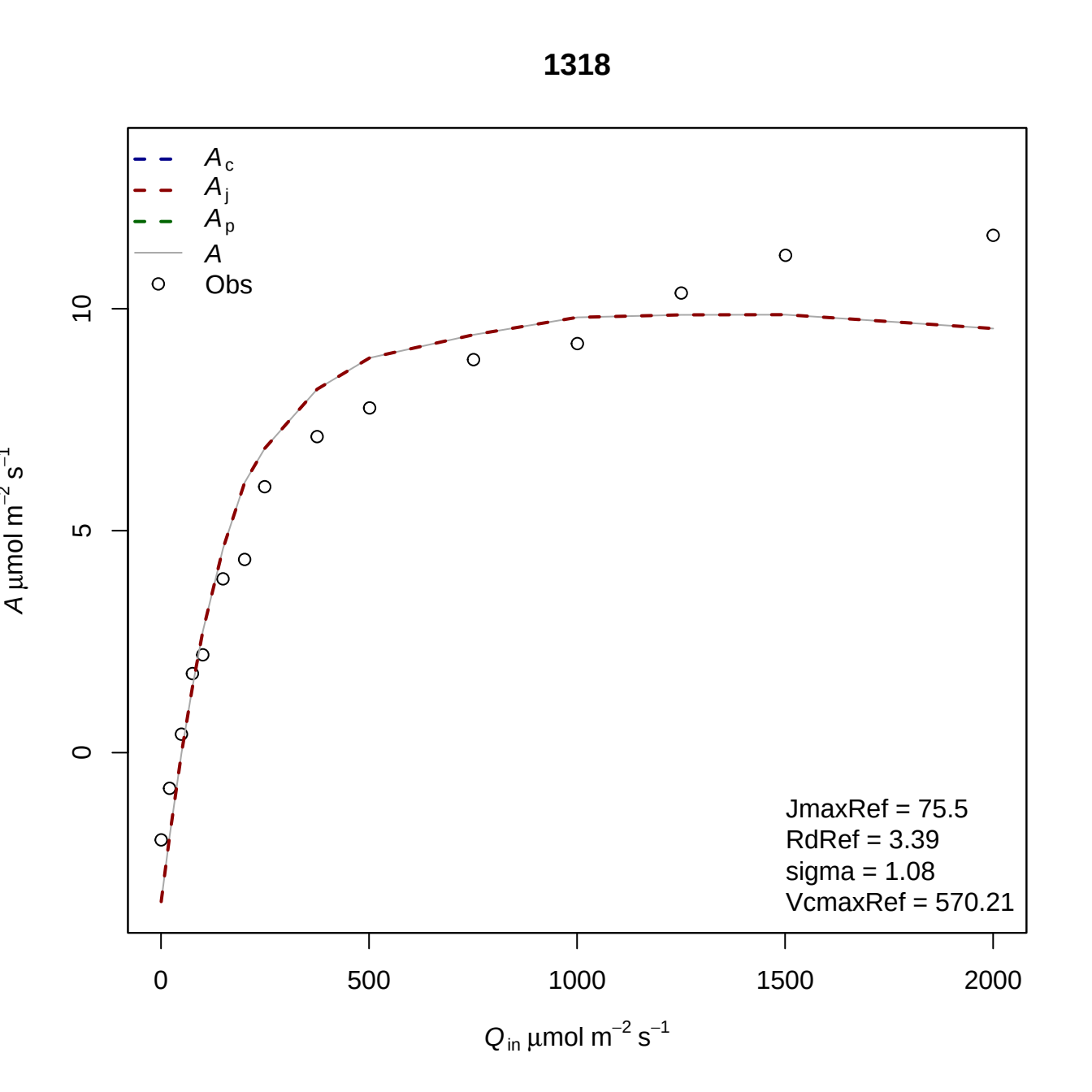

1313
A_c
A_j
A_p
A
Obs
JmaxRef = 61.72
RdRef = 2.78
sigma = 0.72
VcmaxRef = 189.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

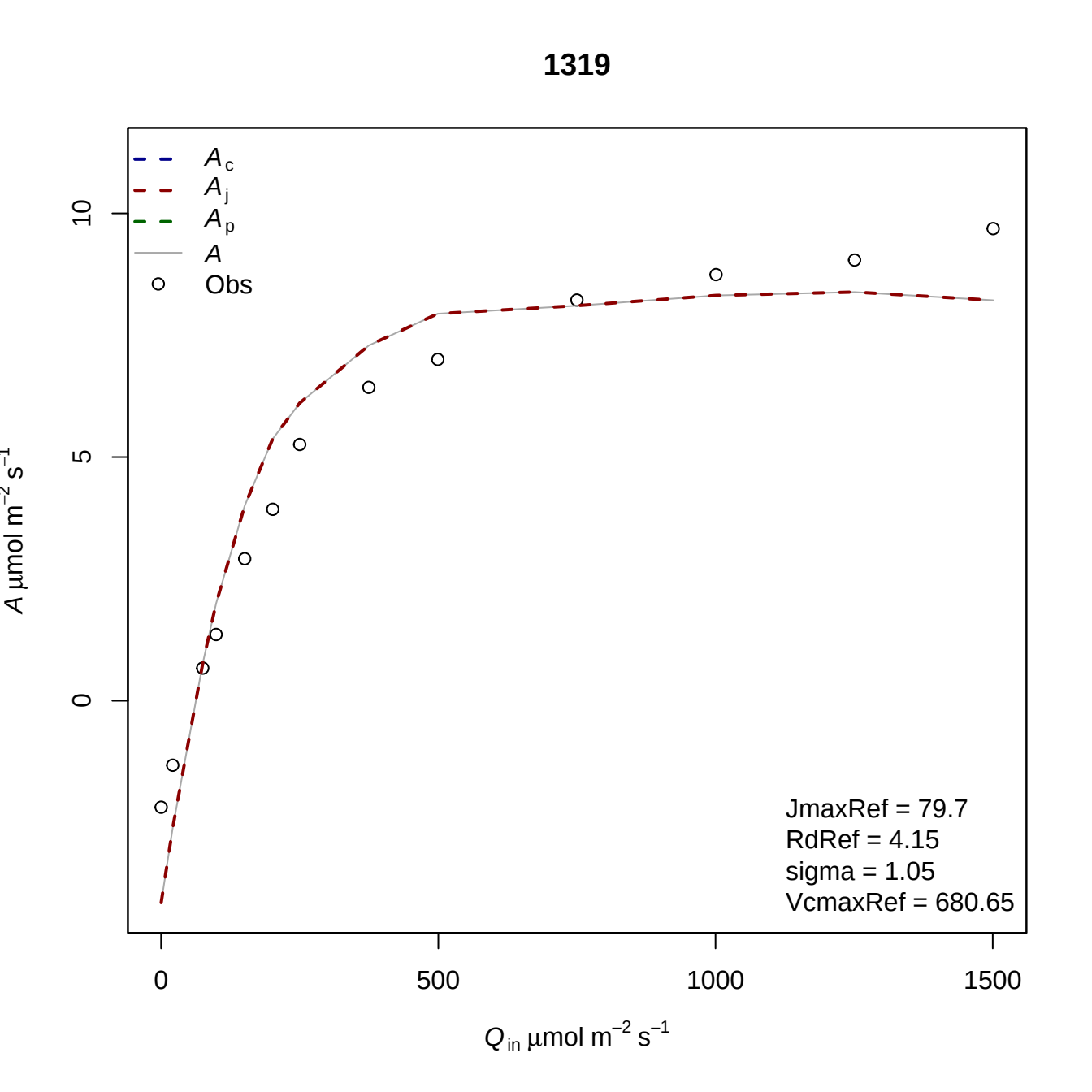

1314
A_c
A_j
A_p
A
Obs
JmaxRef = 80.93
RdRef = 3.48
sigma = 0.71
VcmaxRef = 225320.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

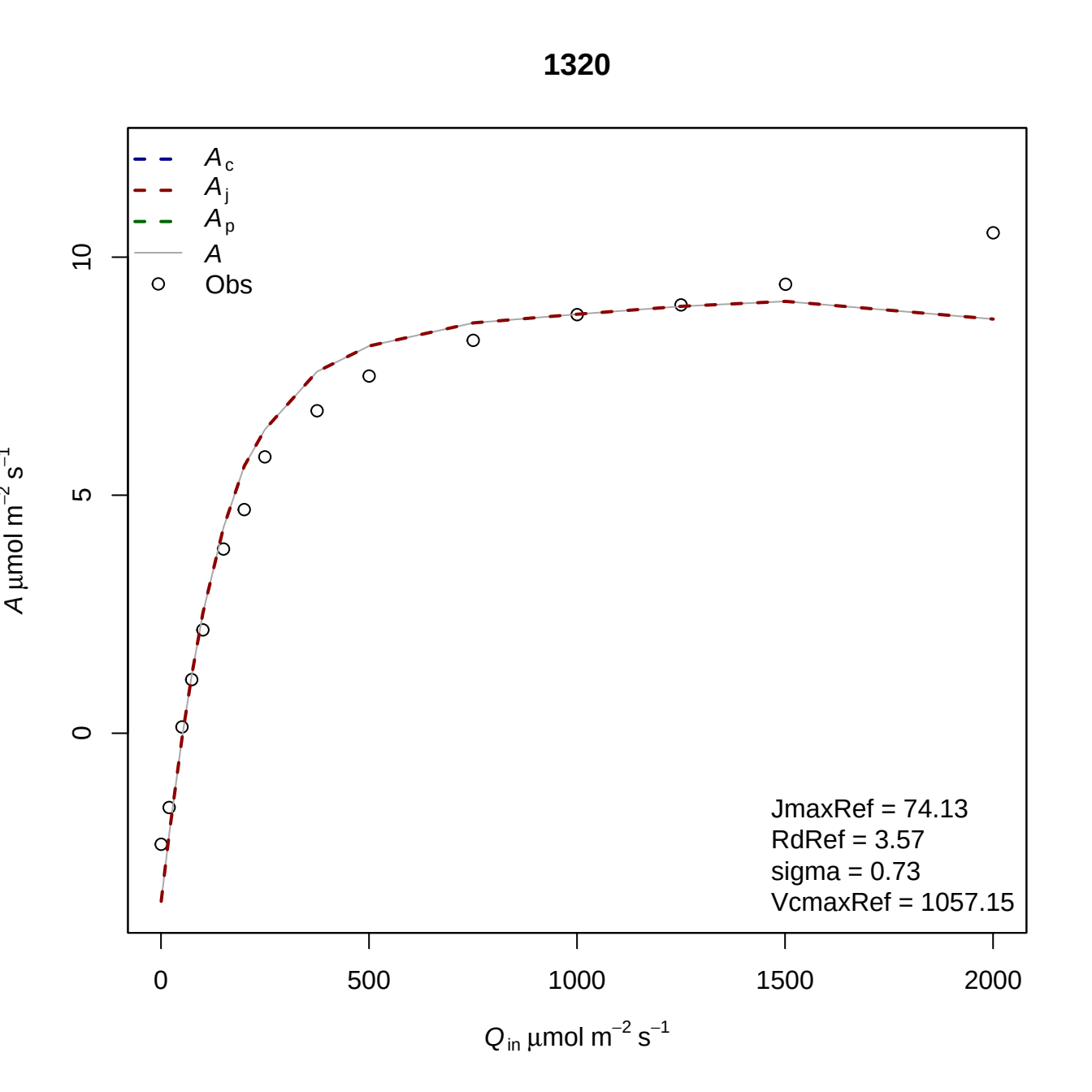

1315
A_c
A_j
A_p
A
Obs
JmaxRef = 113.47
RdRef = 4.53
sigma = 1.1
VcmaxRef = 470.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

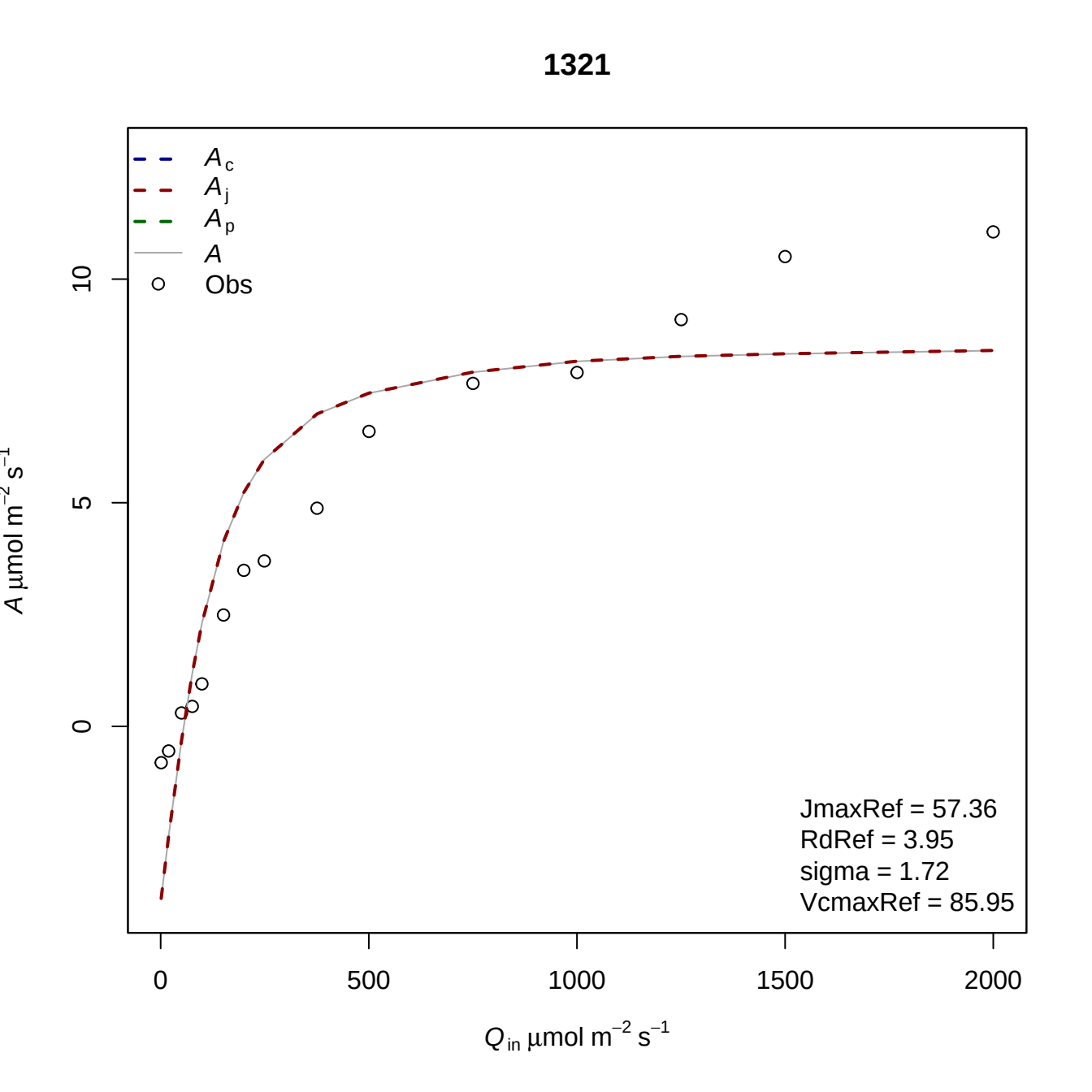

1316
A_c
A_j
A_p
A
Obs
JmaxRef = 96.72
RdRef = 3.8
sigma = 1.01
VcmaxRef = 2179.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

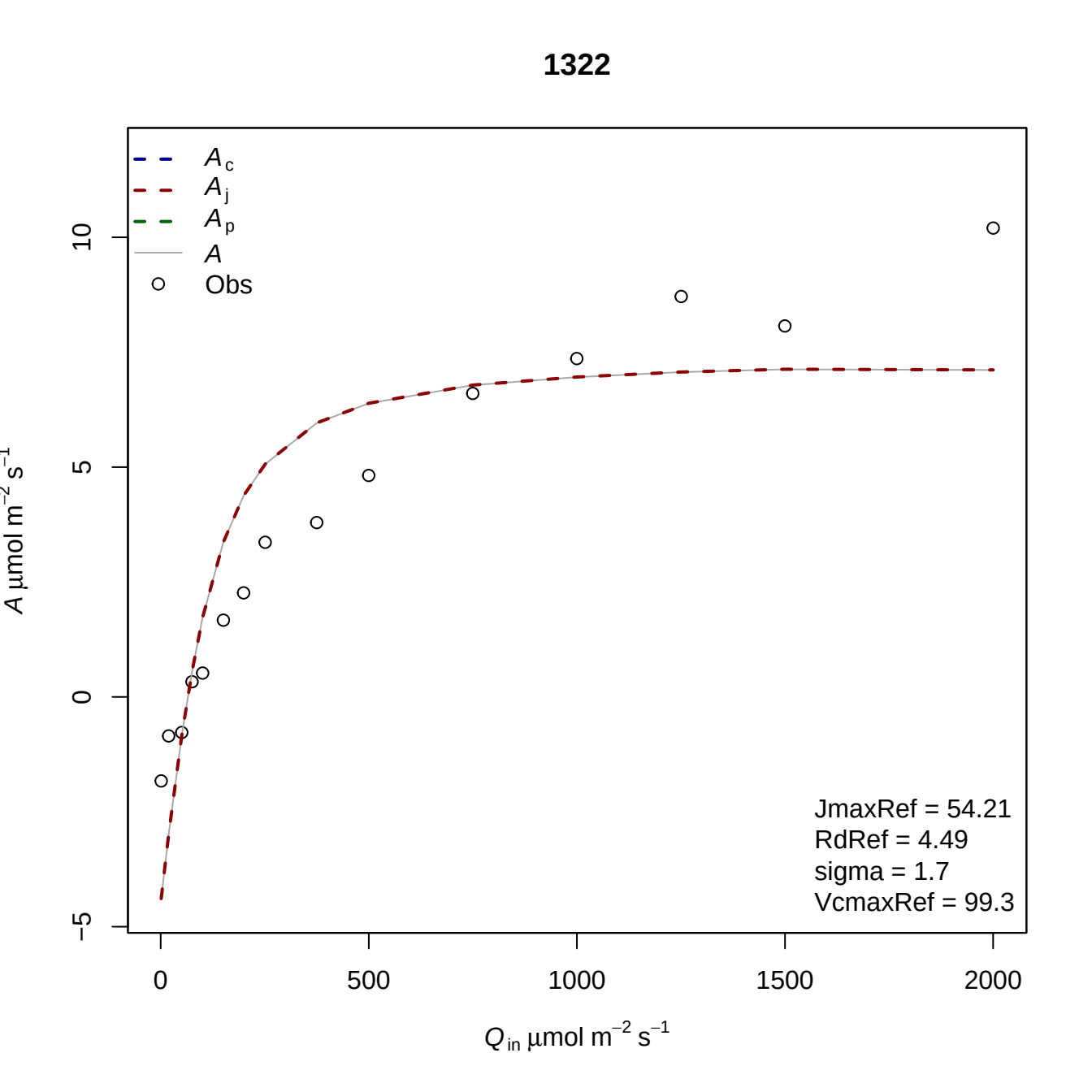

1317
A_c
A_j
A_p
A
Obs
JmaxRef = 107.76
RdRef = 3.16
sigma = 0.48
VcmaxRef = 348.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

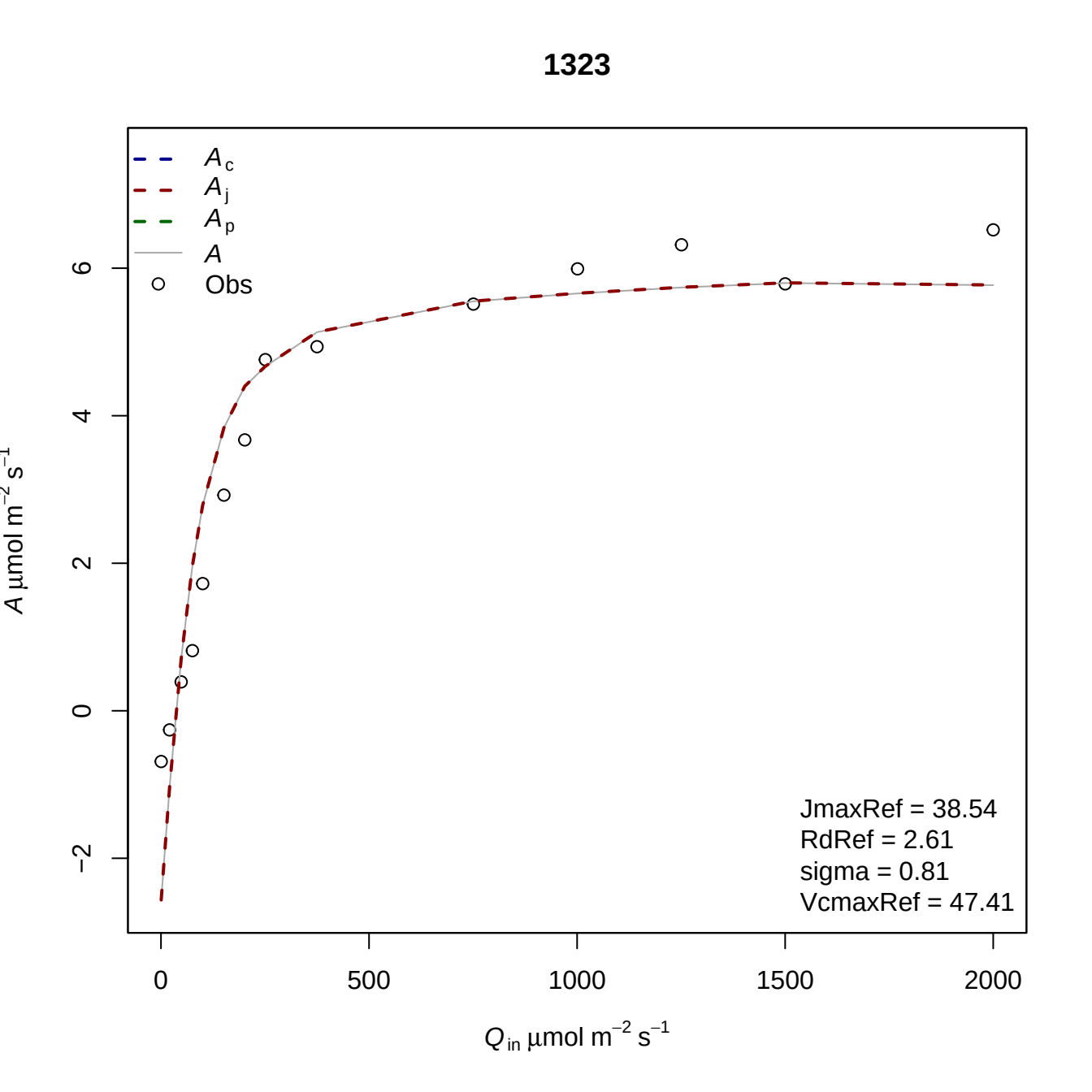

1318
A_c
A_j
A_p
A
Obs
JmaxRef = 75.5
RdRef = 3.39
sigma = 1.08
VcmaxRef = 570.21
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

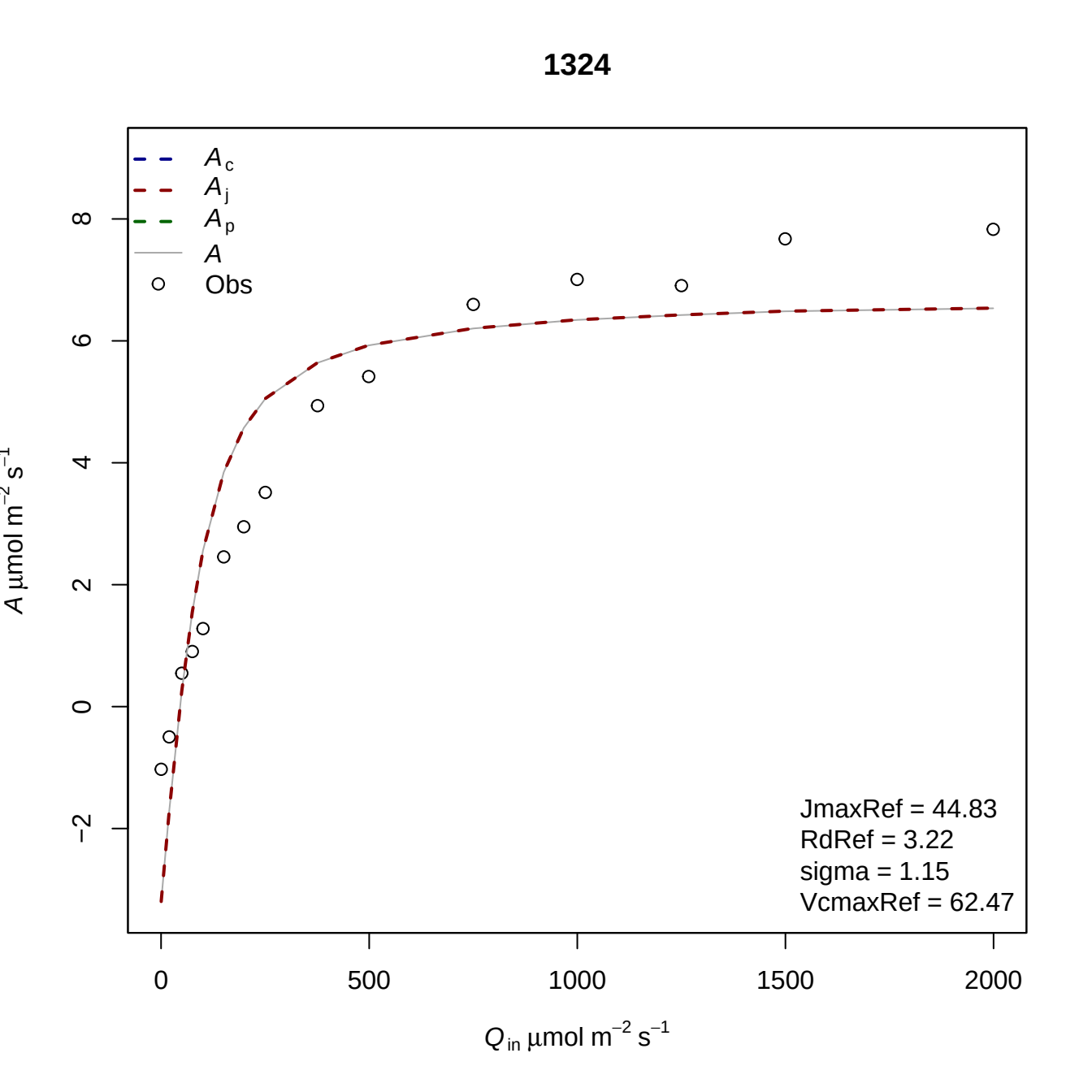

1319
A_c
A_j
A_p
A
Obs
JmaxRef = 79.7
RdRef = 4.15
sigma = 1.05
VcmaxRef = 680.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

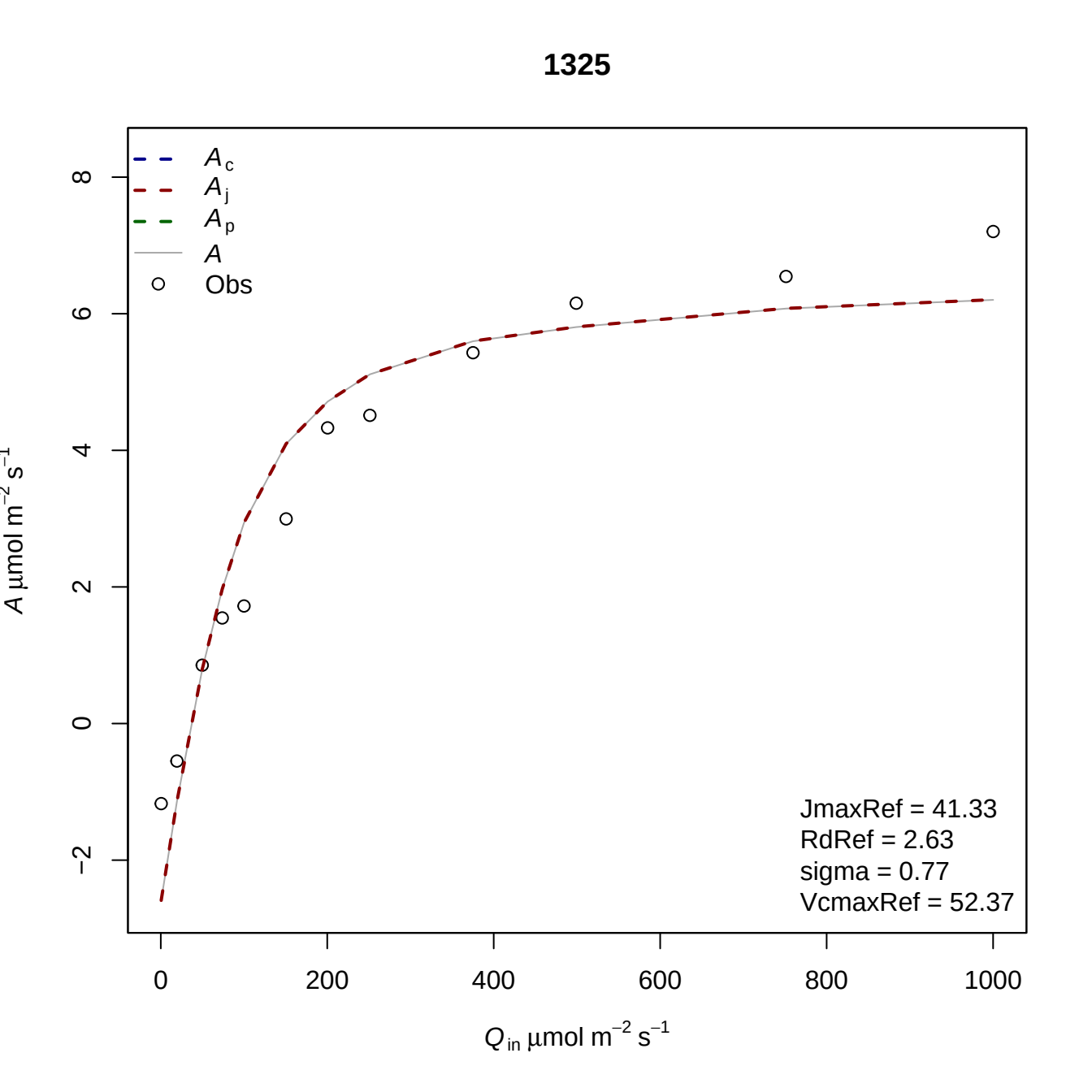

1320
A_c
A_j
A_p
A
Obs
JmaxRef = 74.13
RdRef = 3.57
sigma = 0.73
VcmaxRef = 1057.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

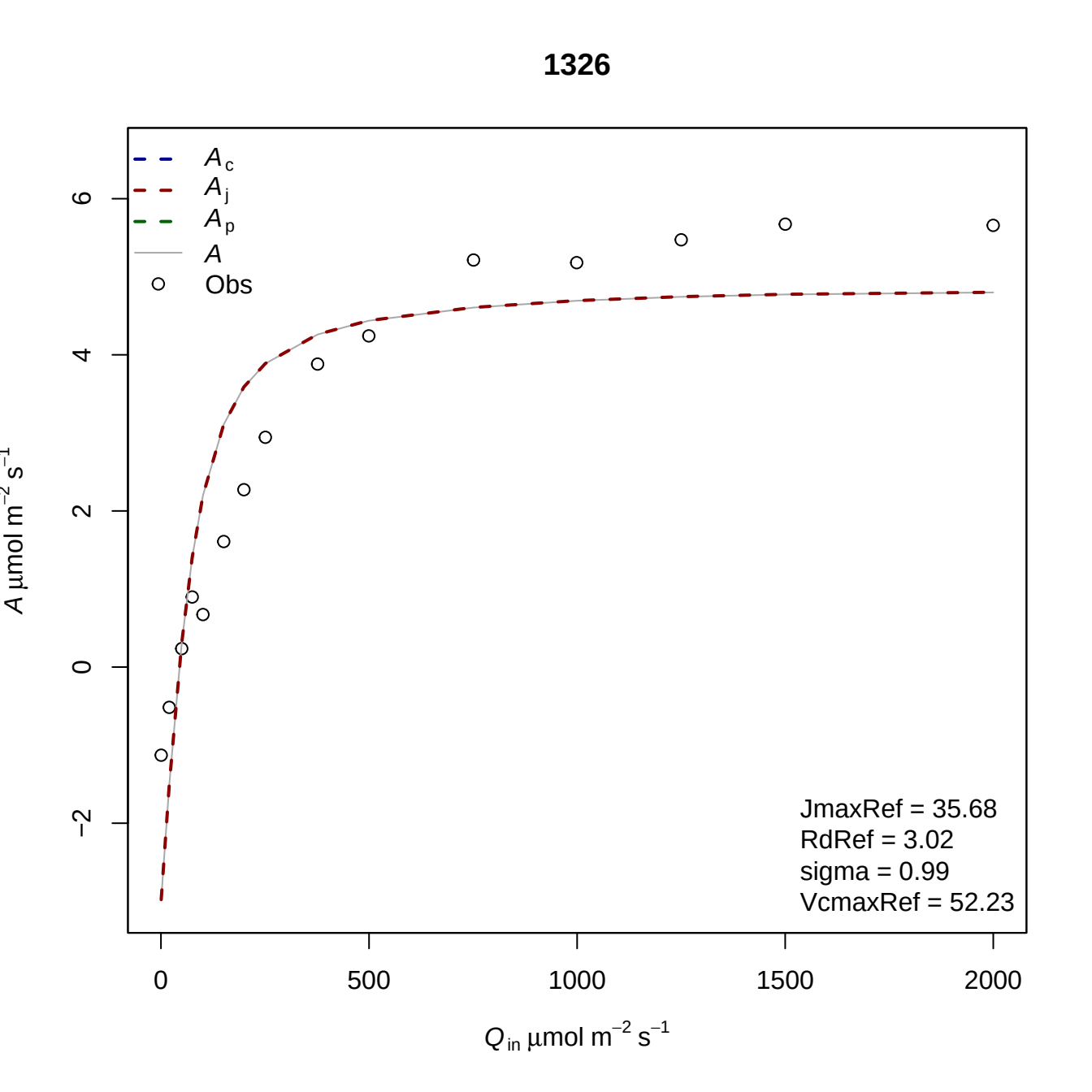

1321
A_c
A_j
A_p
A
Obs
JmaxRef = 57.36
RdRef = 3.95
sigma = 1.72
VcmaxRef = 85.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

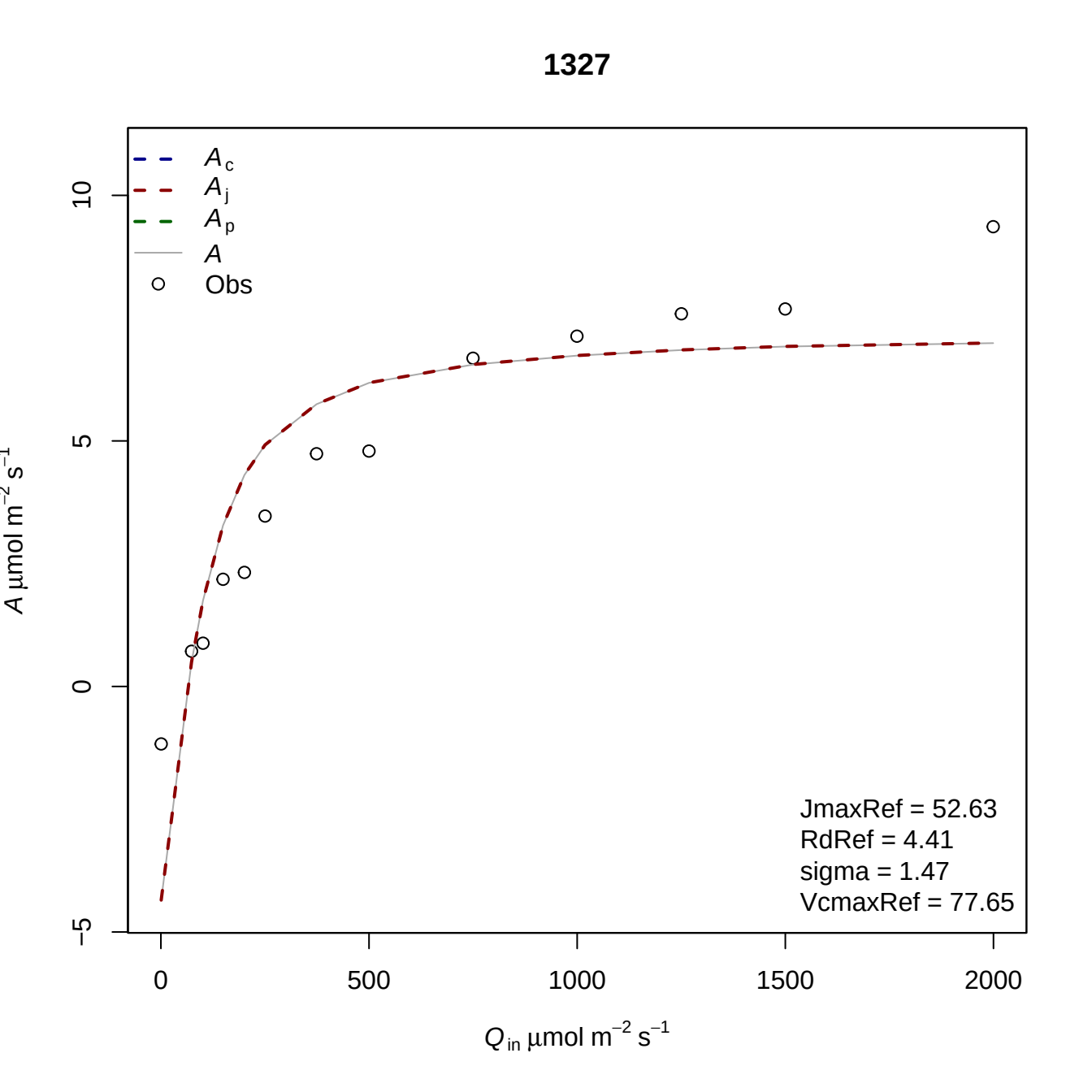

1322
A_c
A_j
A_p
A
Obs
JmaxRef = 54.21
RdRef = 4.49
sigma = 1.7
VcmaxRef = 99.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

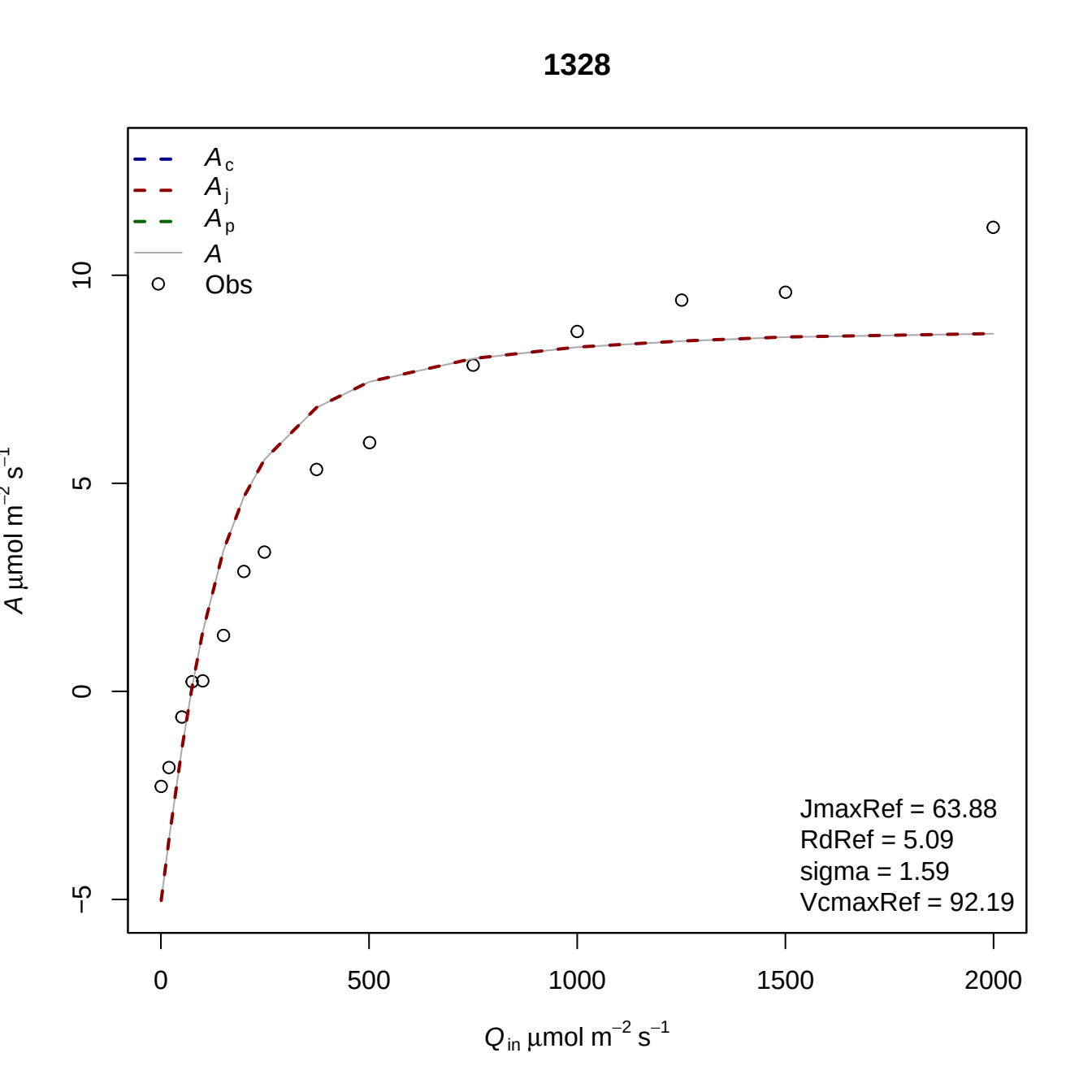

1323
A_c
A_j
A_p
A
Obs
JmaxRef = 38.54
RdRef = 2.61
sigma = 0.81
VcmaxRef = 47.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

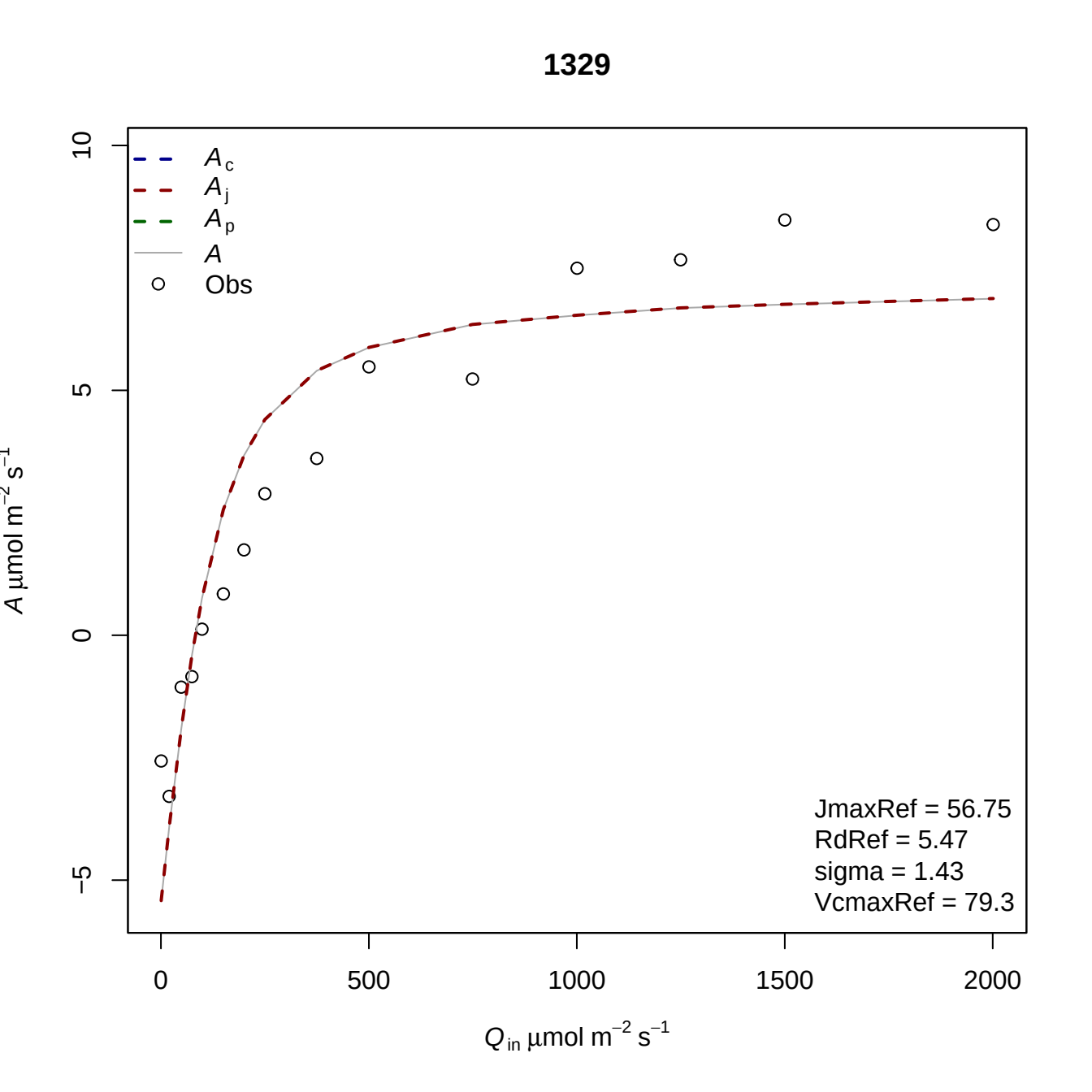

1324
A_c
A_j
A_p
A
Obs
JmaxRef = 44.83
RdRef = 3.22
sigma = 1.15
VcmaxRef = 62.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

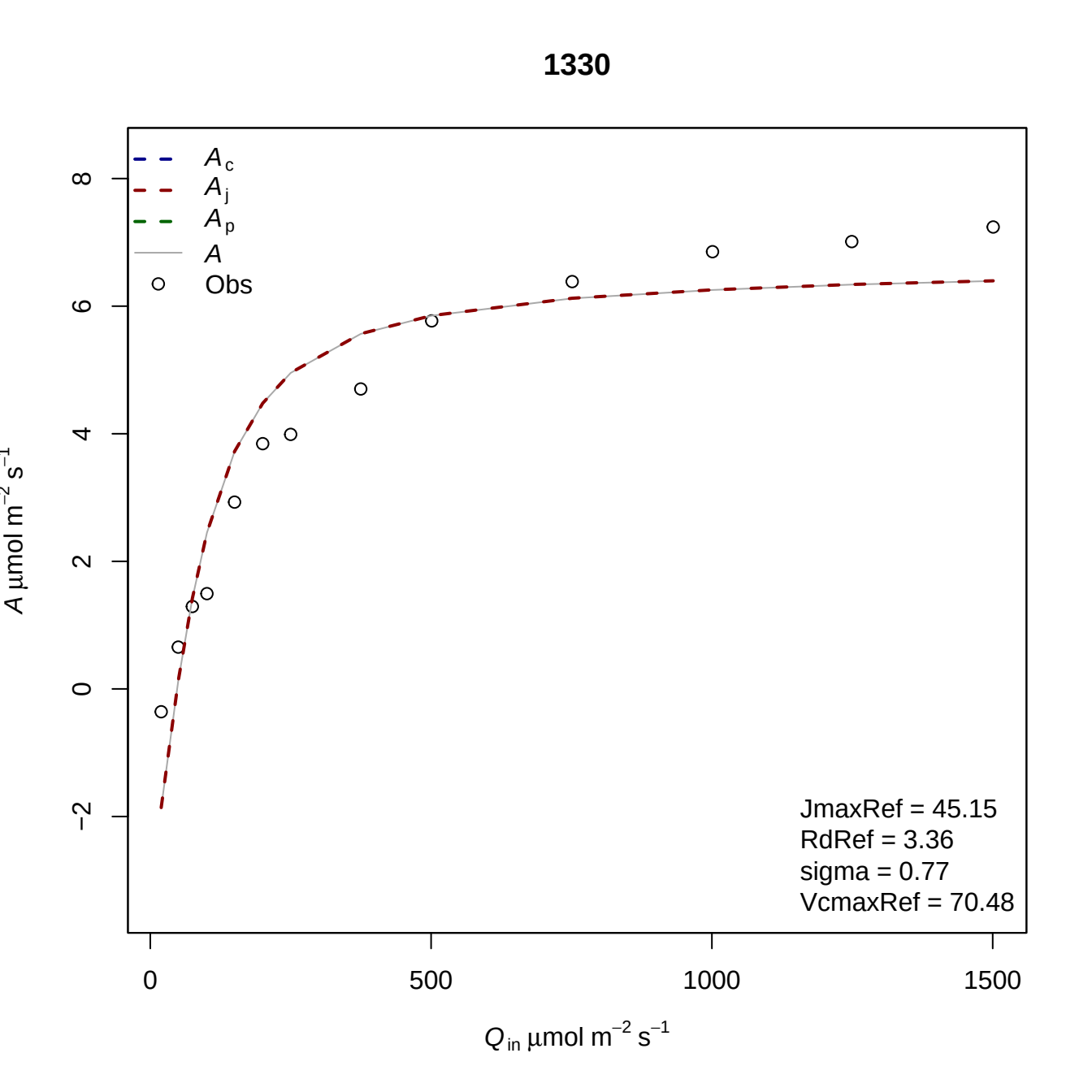

1325
A_c
A_j
A_p
A
Obs
JmaxRef = 41.33
RdRef = 2.63
sigma = 0.77
VcmaxRef = 52.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
200
400
600
800
1000

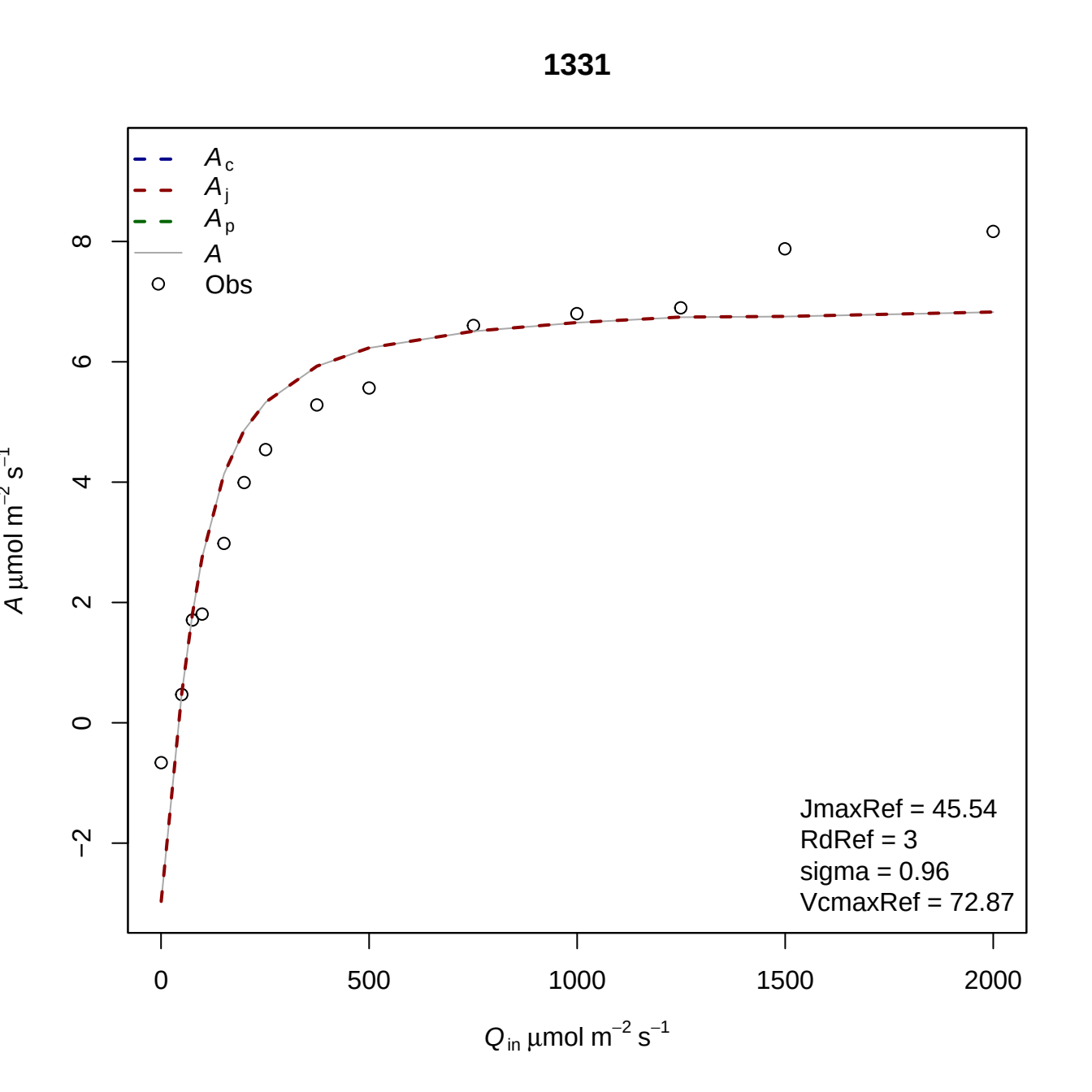

1326
A_c
A_j
A_p
A
Obs
JmaxRef = 35.68
RdRef = 3.02
sigma = 0.99
VcmaxRef = 52.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

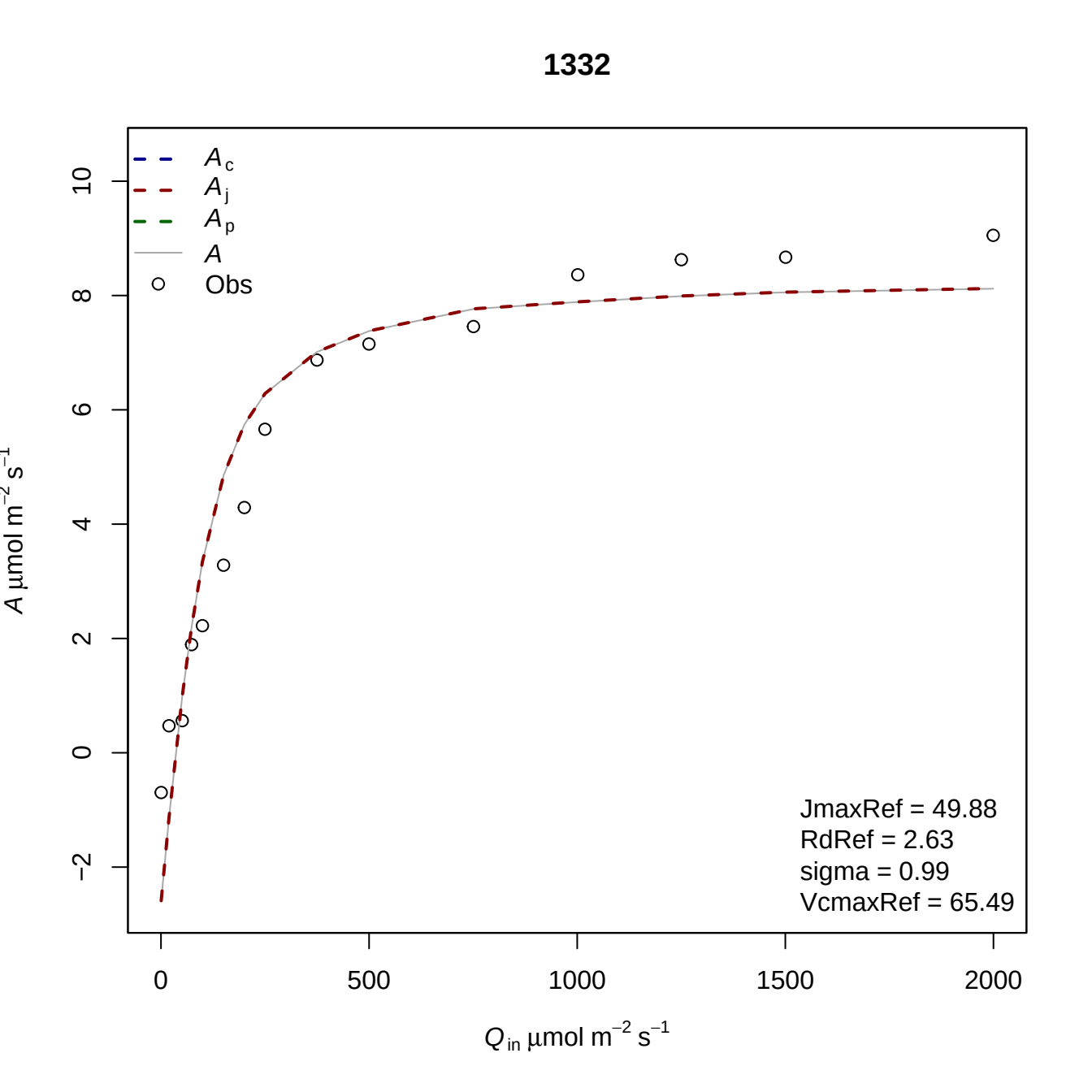

1327
A_c
A_j
A_p
A
Obs
JmaxRef = 52.63
RdRef = 4.41
sigma = 1.47
VcmaxRef = 77.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

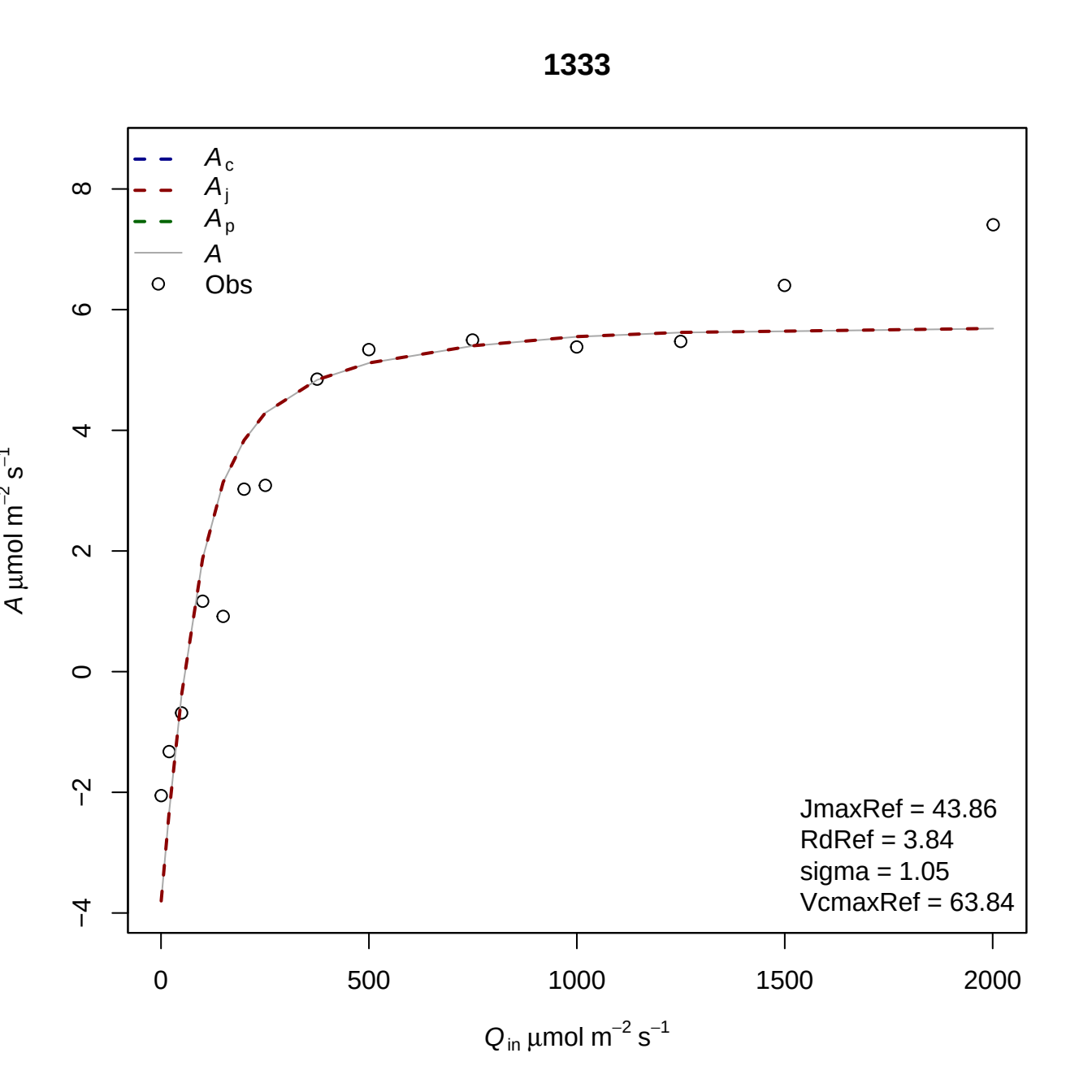

1328
A_c
A_j
A_p
A
Obs
JmaxRef = 63.88
RdRef = 5.09
sigma = 1.59
VcmaxRef = 92.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

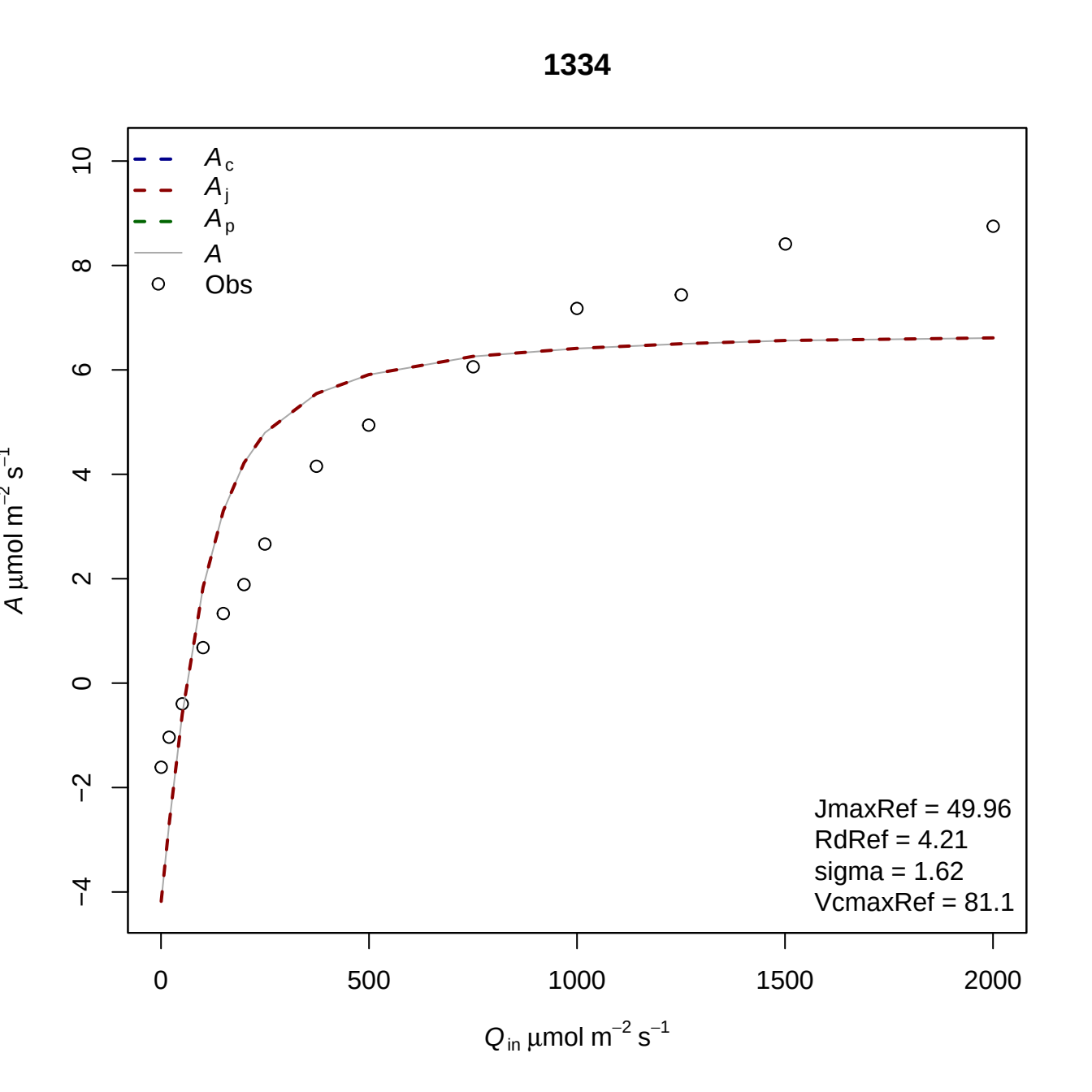

1329
A_c
A_j
A_p
A
Obs
JmaxRef = 56.75
RdRef = 5.47
sigma = 1.43
VcmaxRef = 79.3
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

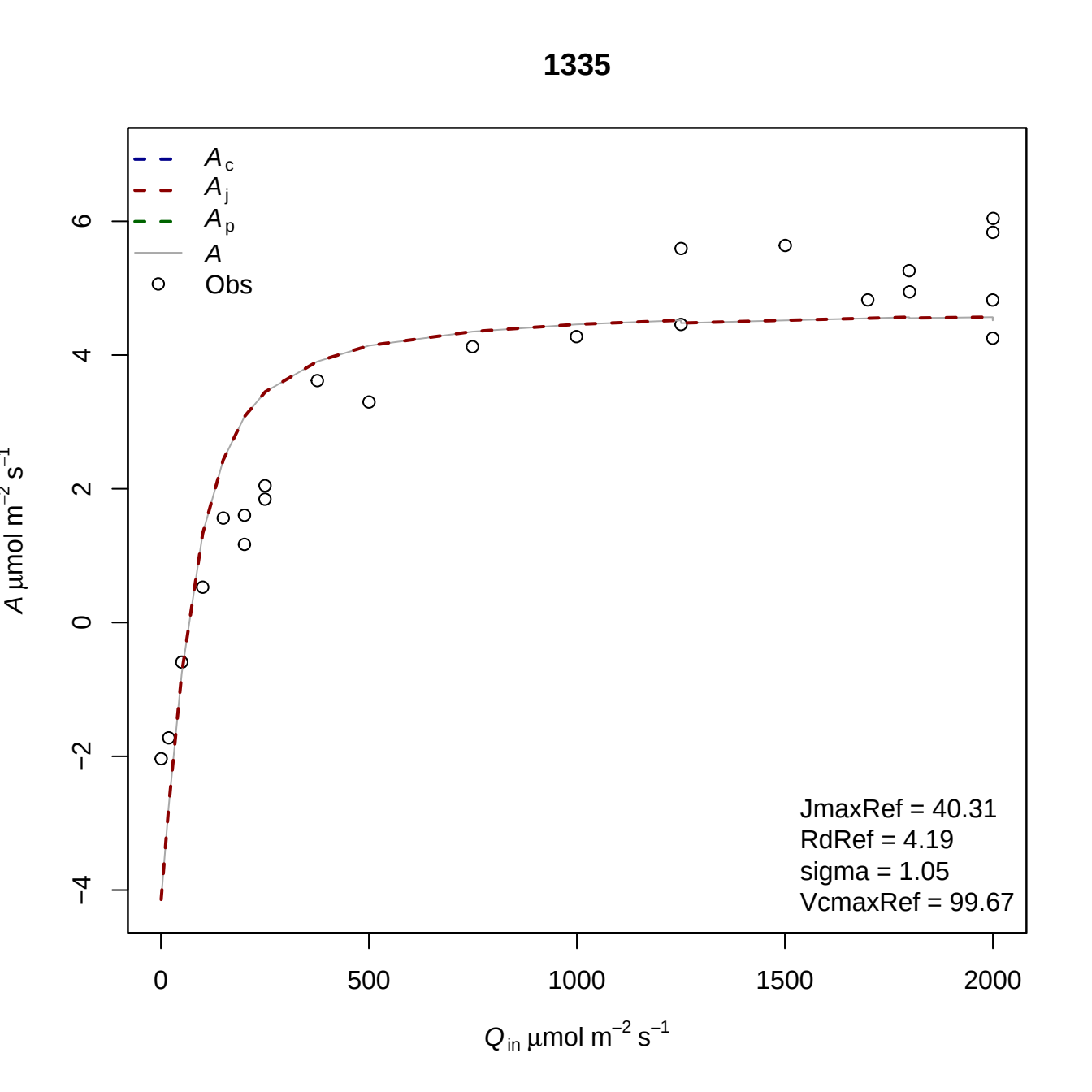

1330
A_c
A_j
A_p
A
Obs
JmaxRef = 45.15
RdRef = 3.36
sigma = 0.77
VcmaxRef = 70.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

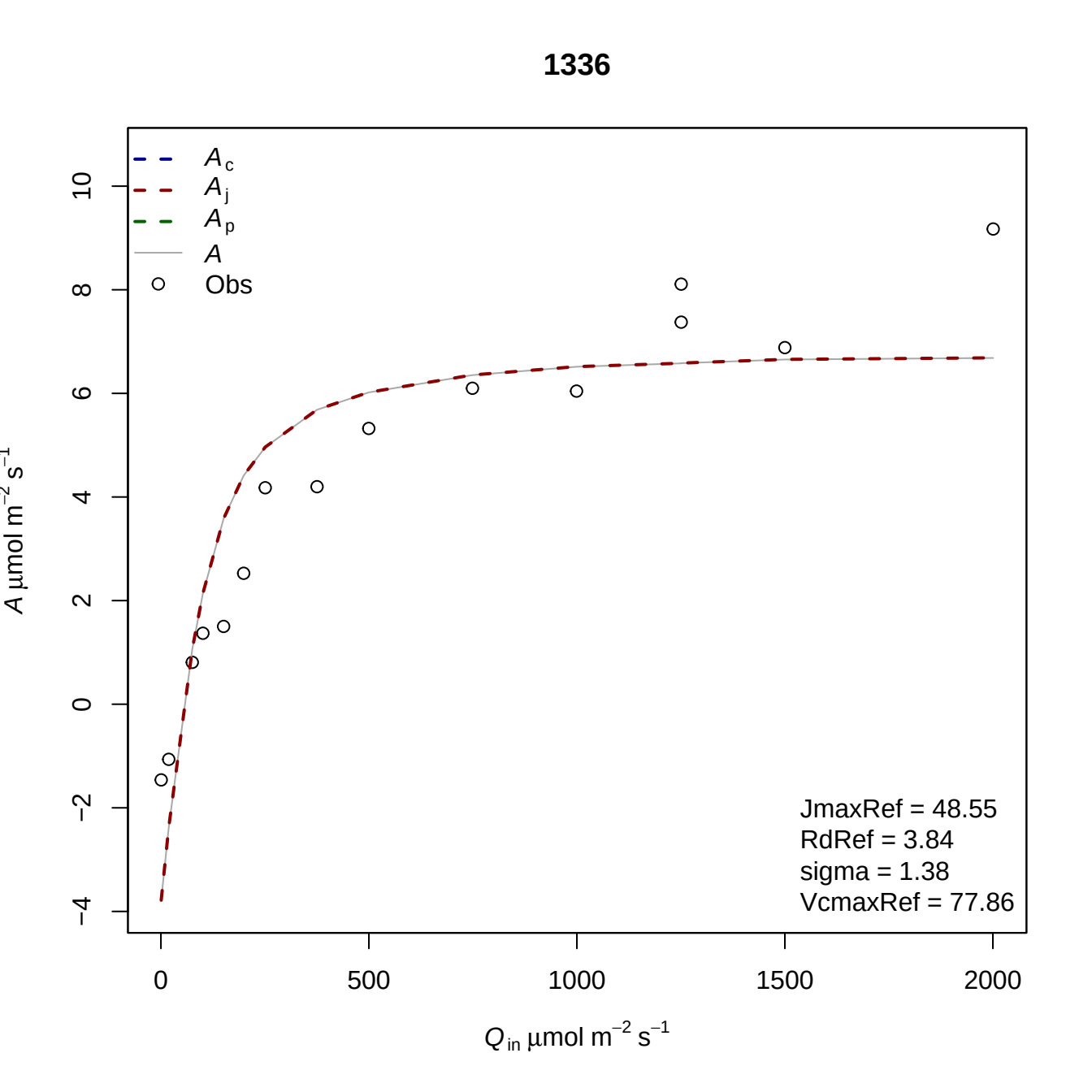

1331
A_c
A_j
A_p
A
Obs
JmaxRef = 45.54
RdRef = 3
sigma = 0.96
VcmaxRef = 72.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

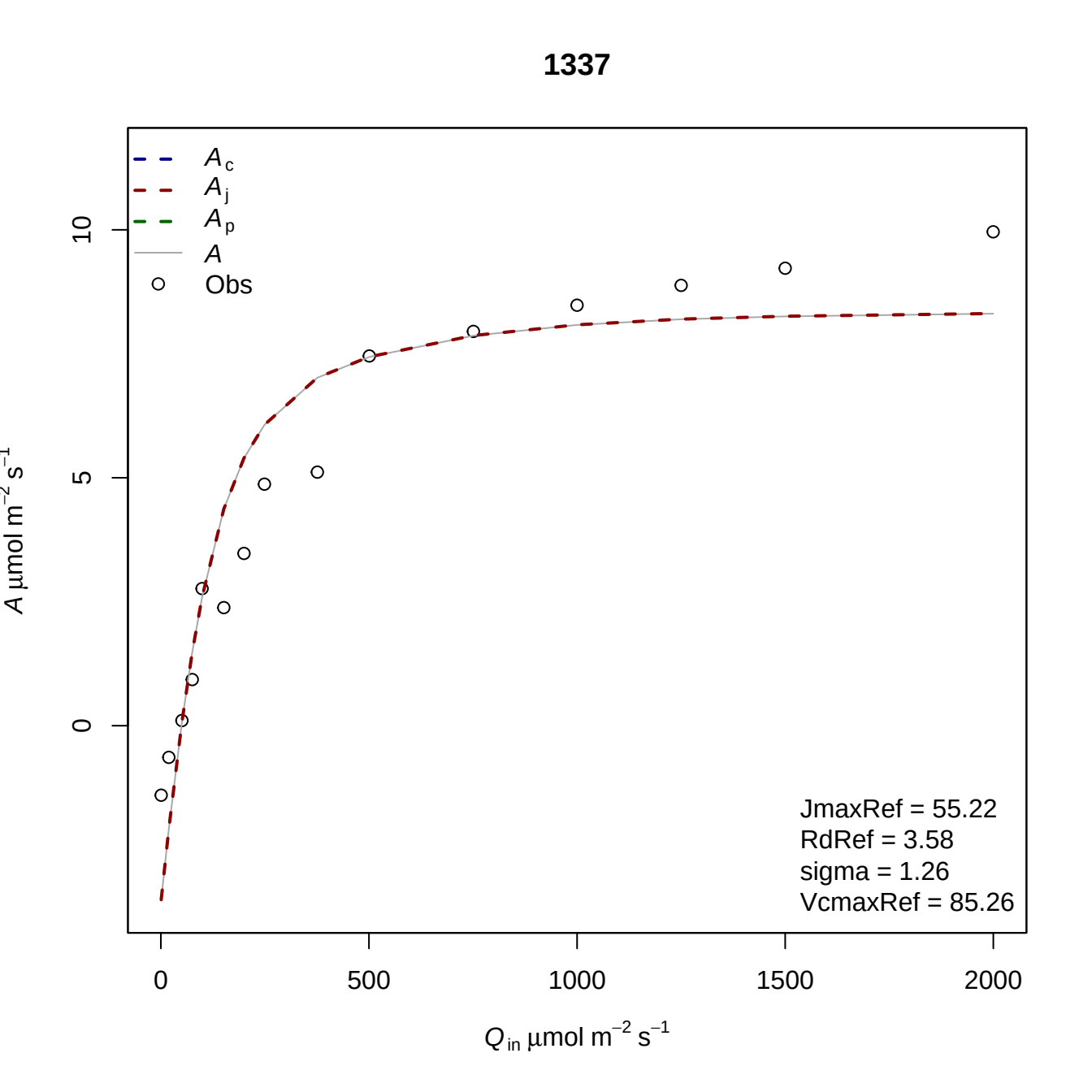

1332
A_c
A_j
A_p
A
Obs
JmaxRef = 49.88
RdRef = 2.63
sigma = 0.99
VcmaxRef = 65.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

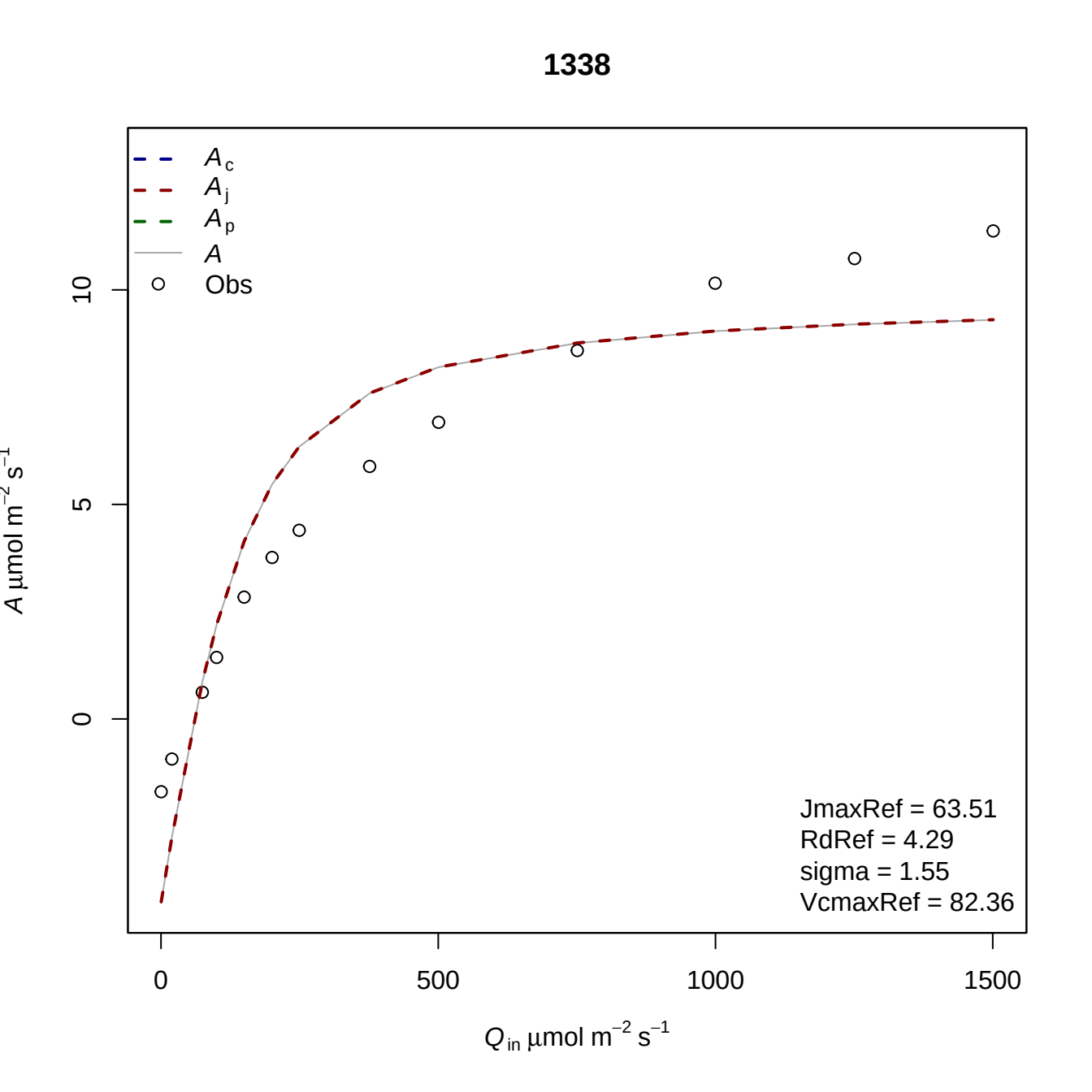

1333
A_c
A_j
A_p
A
Obs
JmaxRef = 43.86
RdRef = 3.84
sigma = 1.05
VcmaxRef = 63.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

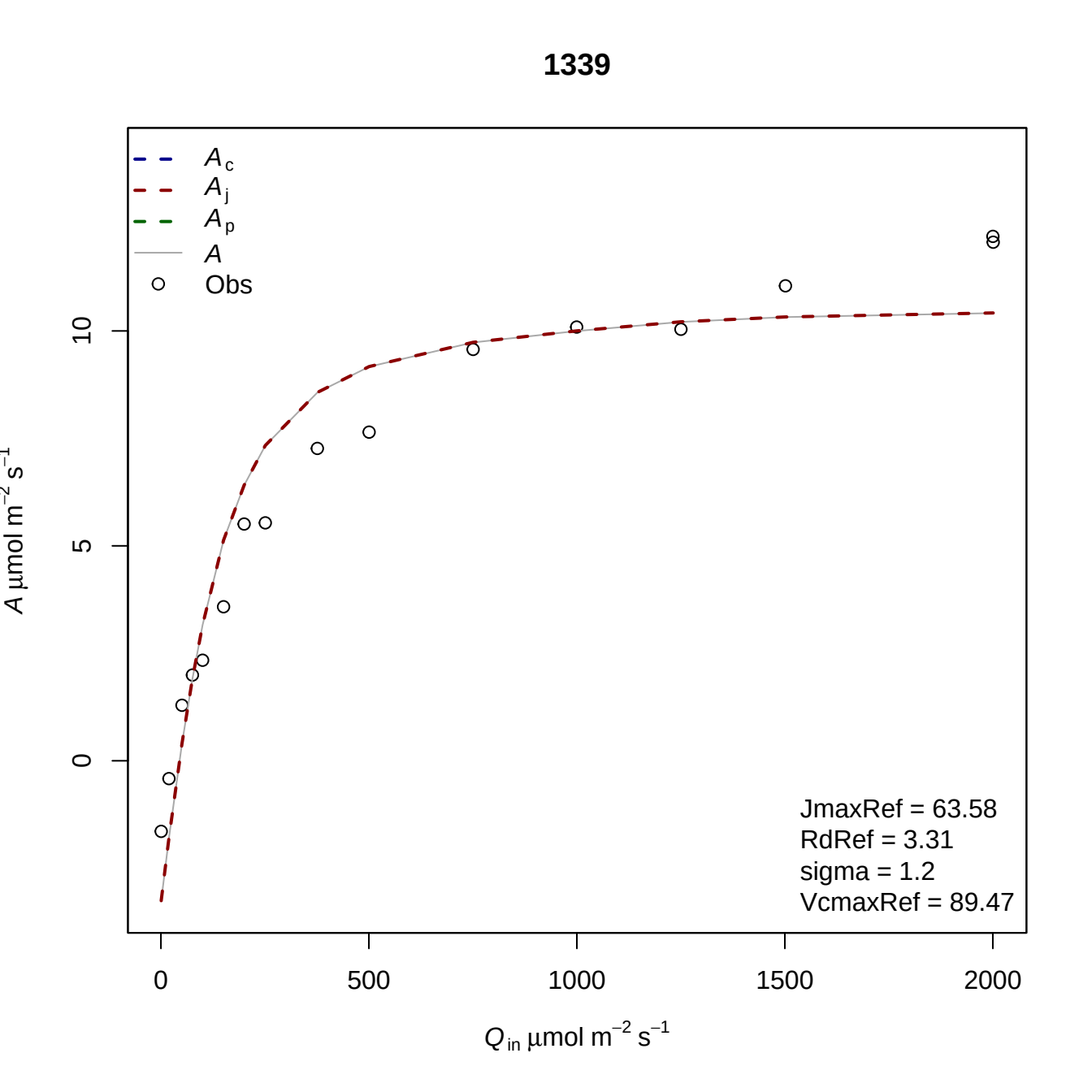

1334
A_c
A_j
A_p
A
Obs
JmaxRef = 49.96
RdRef = 4.21
sigma = 1.62
VcmaxRef = 81.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

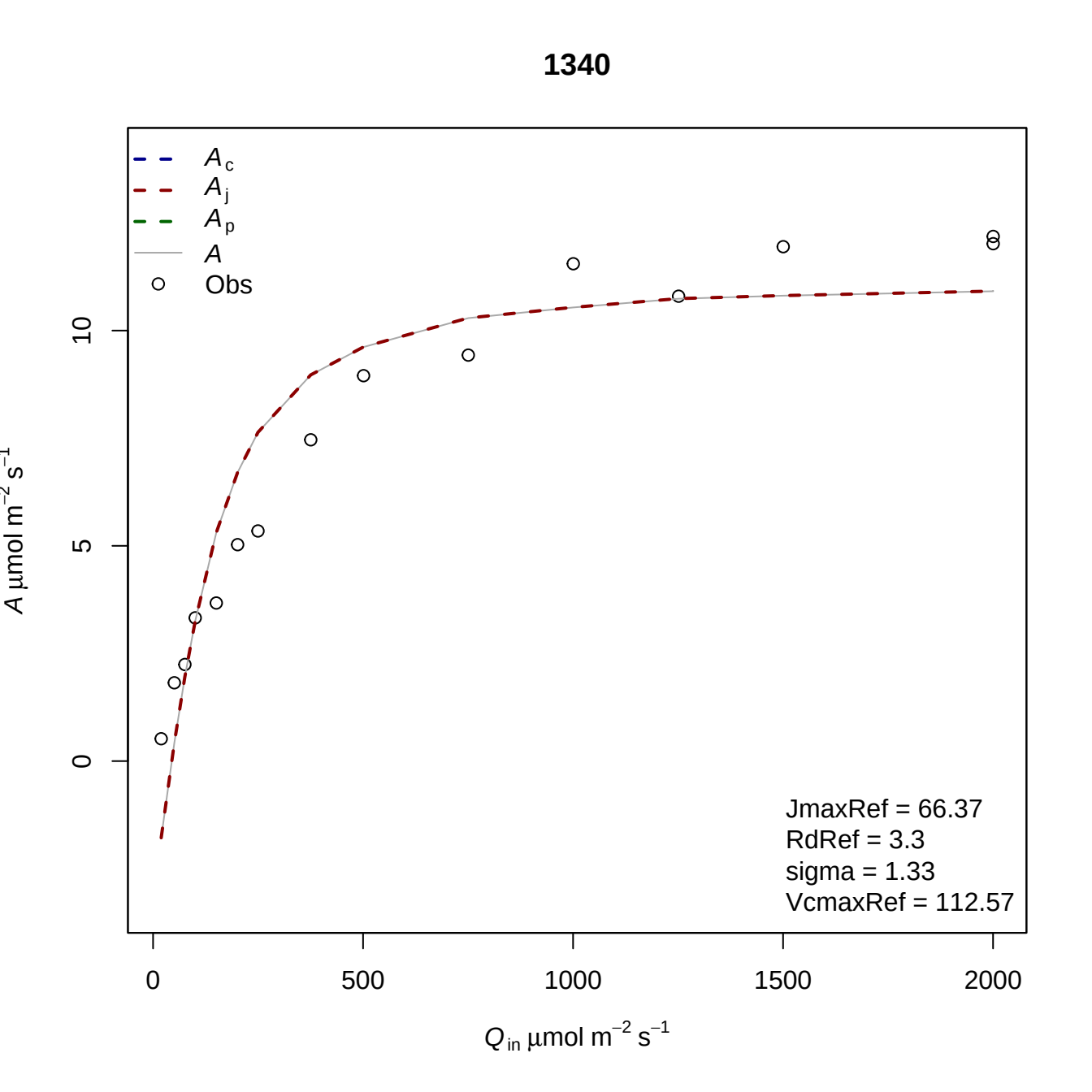

1335
A_c
A_j
A_p
A
Obs
JmaxRef = 40.31
RdRef = 4.19
sigma = 1.05
VcmaxRef = 99.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

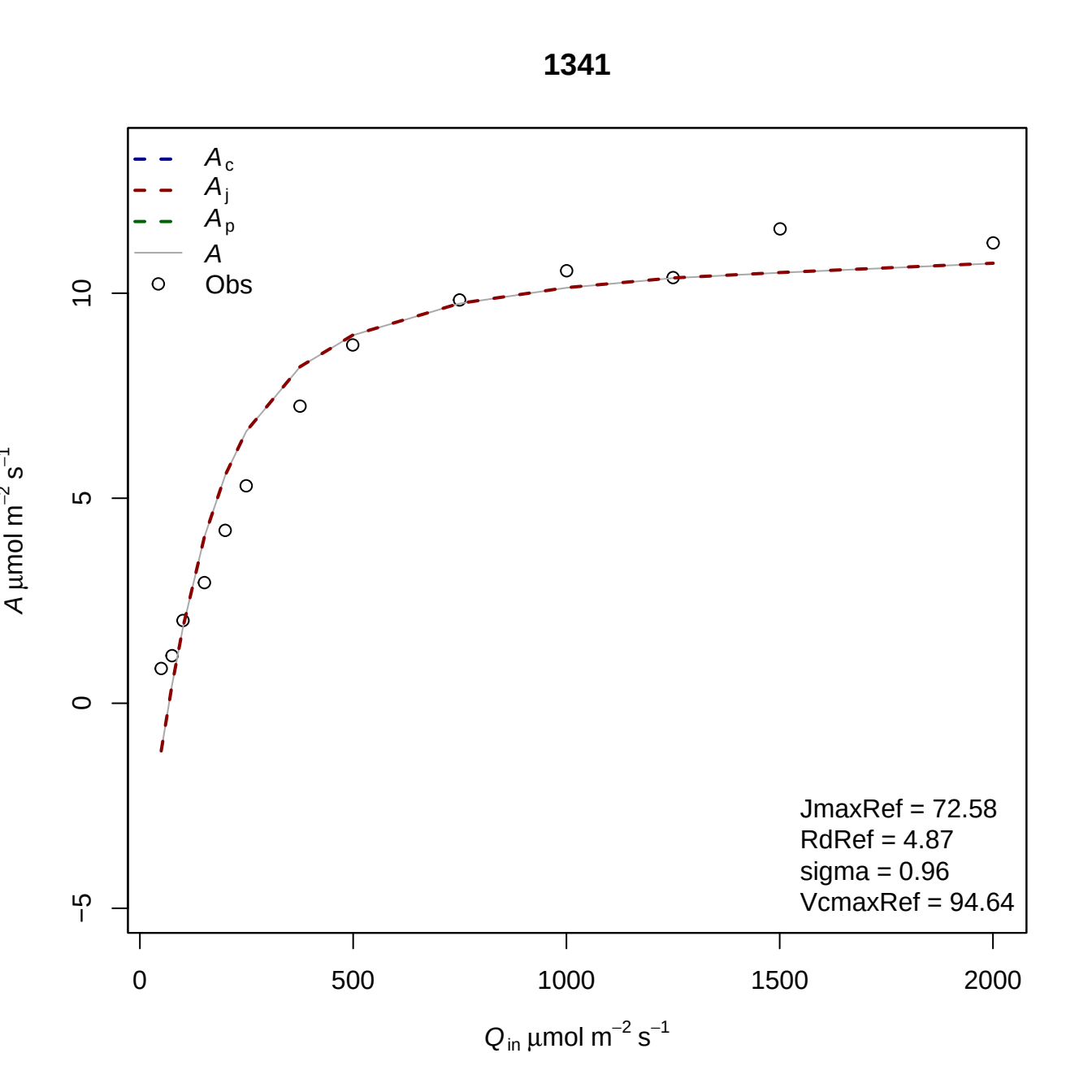

1336
A_c
A_j
A_p
A
Obs
JmaxRef = 48.55
RdRef = 3.84
sigma = 1.38
VcmaxRef = 77.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

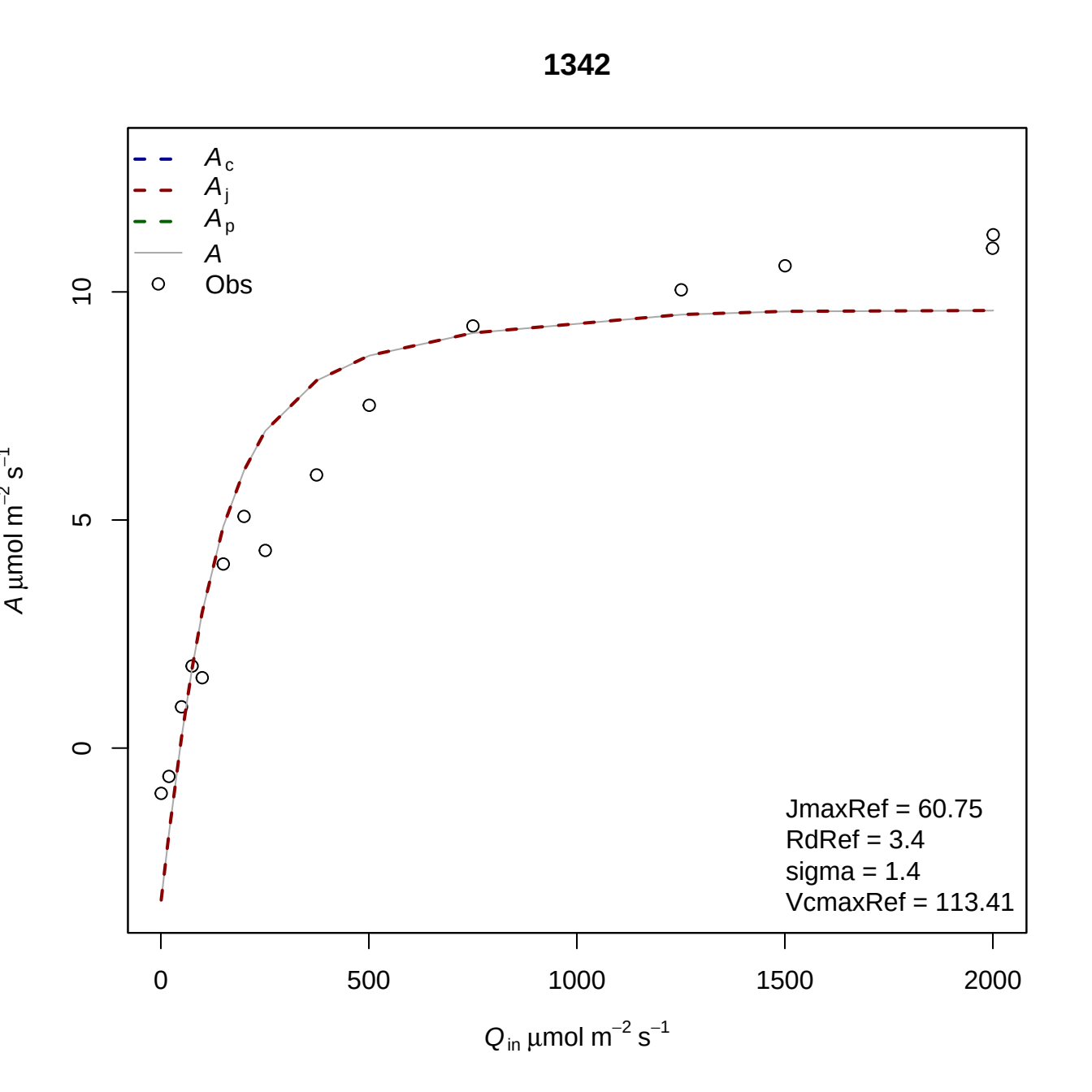

1337
A_c
A_j
A_p
A
Obs
JmaxRef = 55.22
RdRef = 3.58
sigma = 1.26
VcmaxRef = 85.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

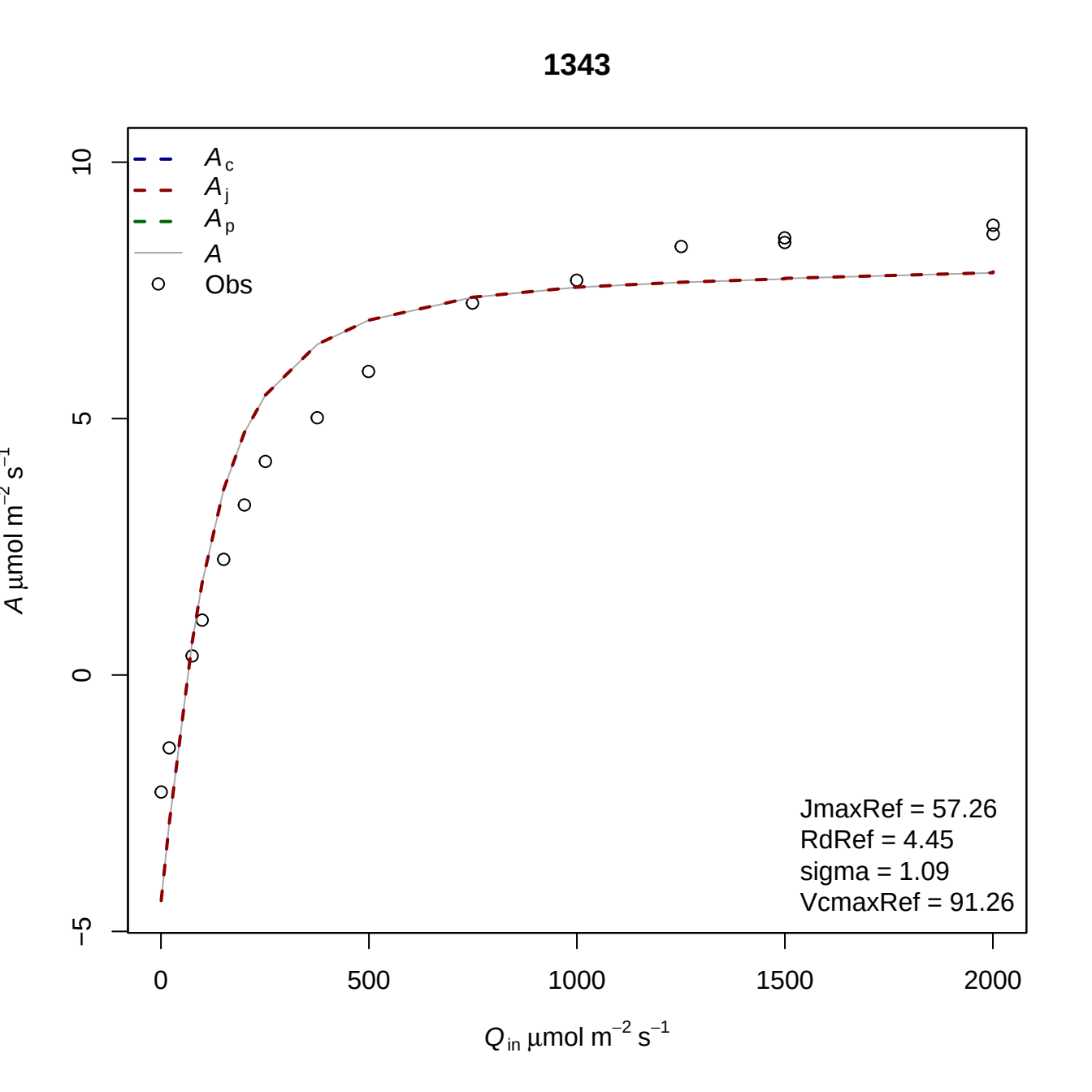

1338
A_c
A_j
A_p
A
Obs
JmaxRef = 63.51
RdRef = 4.29
sigma = 1.55
VcmaxRef = 82.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

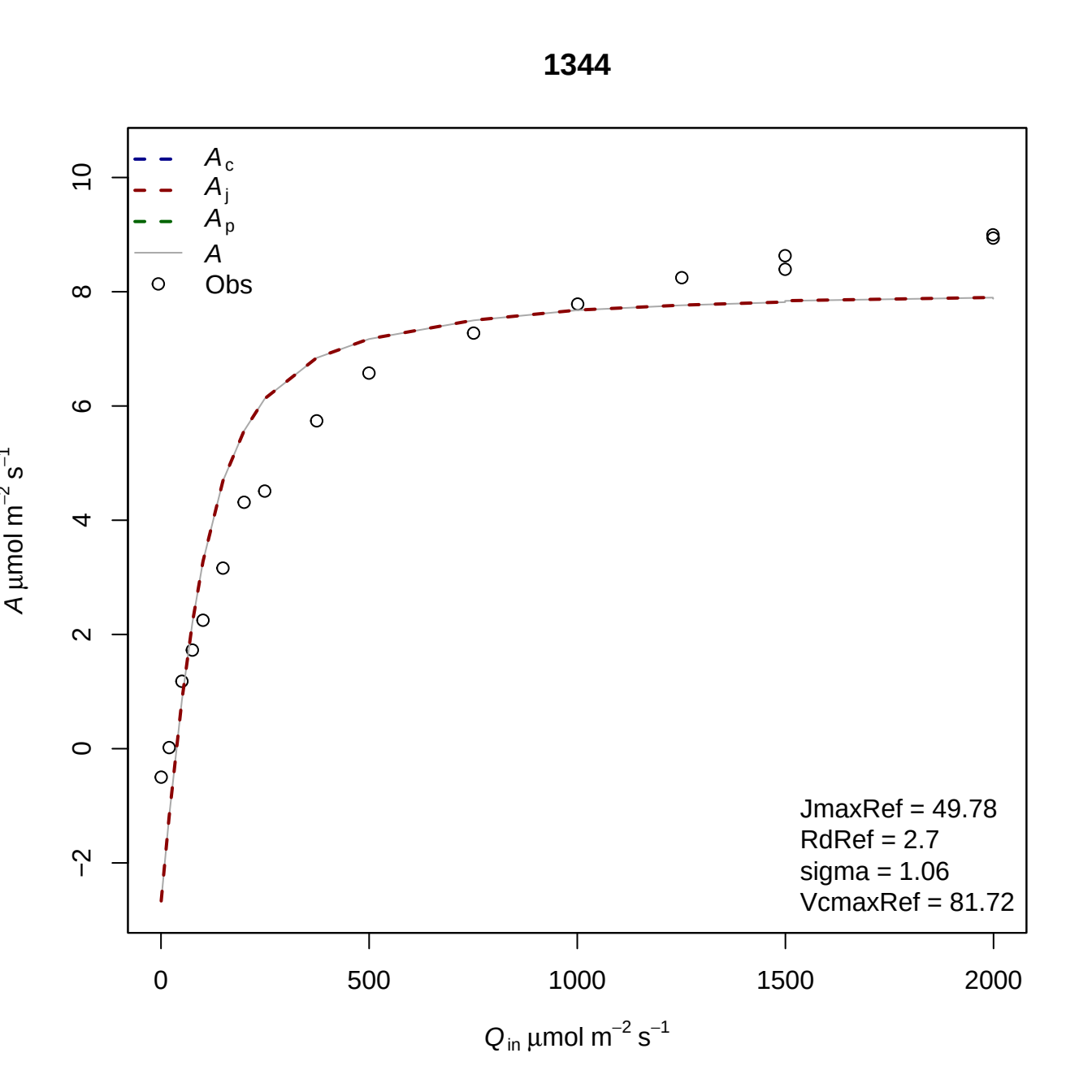

1339
A_c
A_j
A_p
A
Obs
JmaxRef = 63.58
RdRef = 3.31
sigma = 1.2
VcmaxRef = 89.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

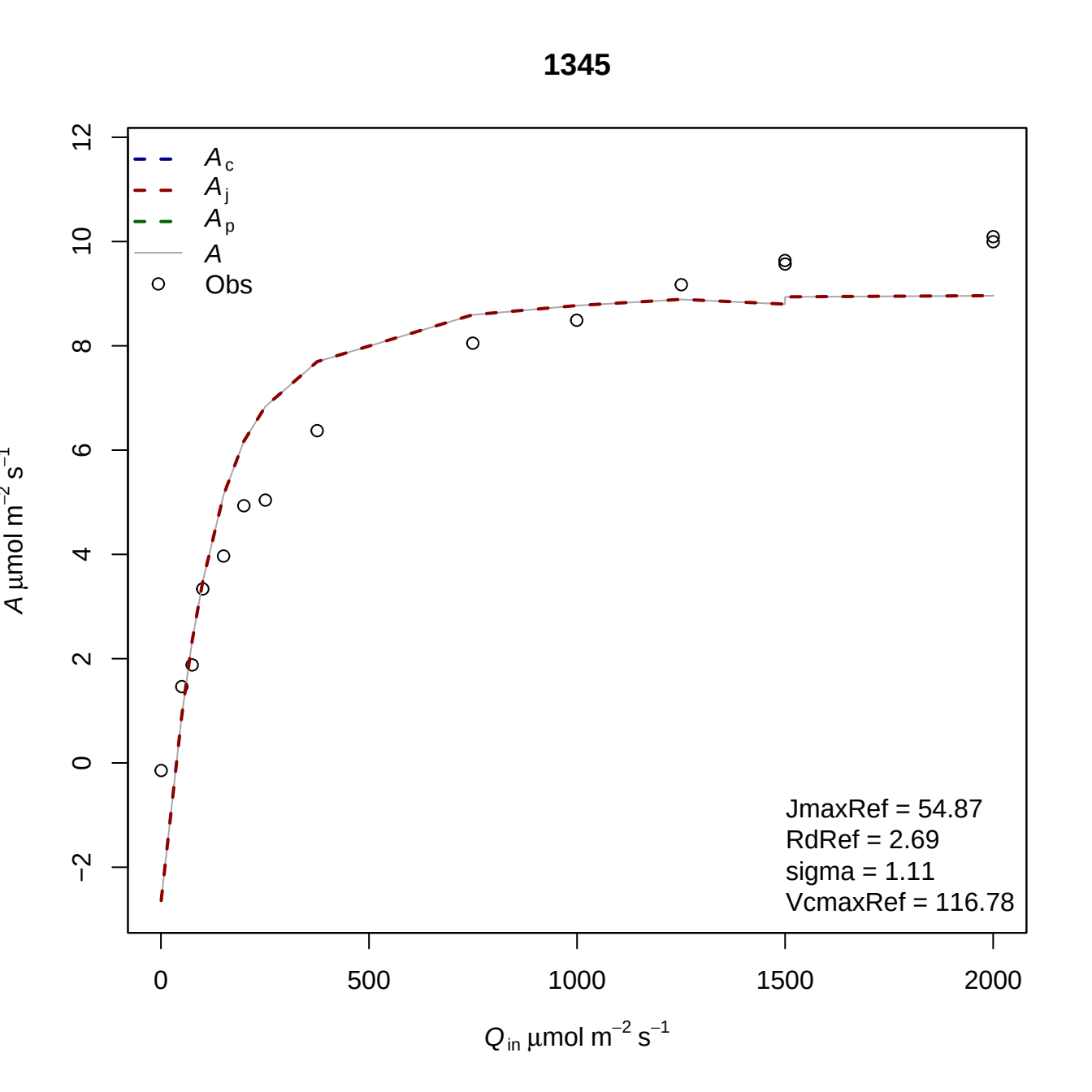

1340
A_c
A_j
A_p
A
Obs
JmaxRef = 66.37
RdRef = 3.3
sigma = 1.33
VcmaxRef = 112.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

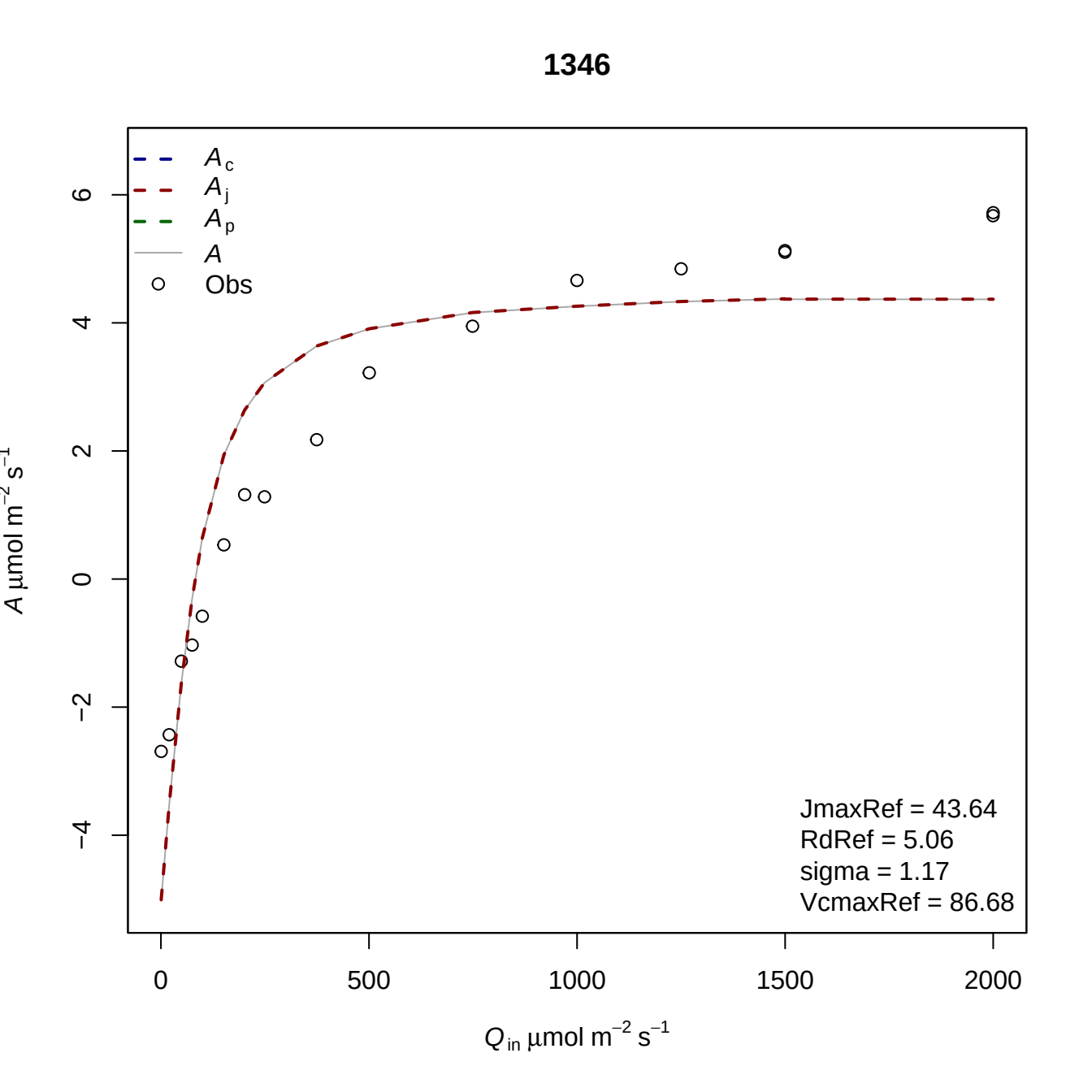

1341
A_c
A_j
A_p
A
Obs
JmaxRef = 72.58
RdRef = 4.87
sigma = 0.96
VcmaxRef = 94.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

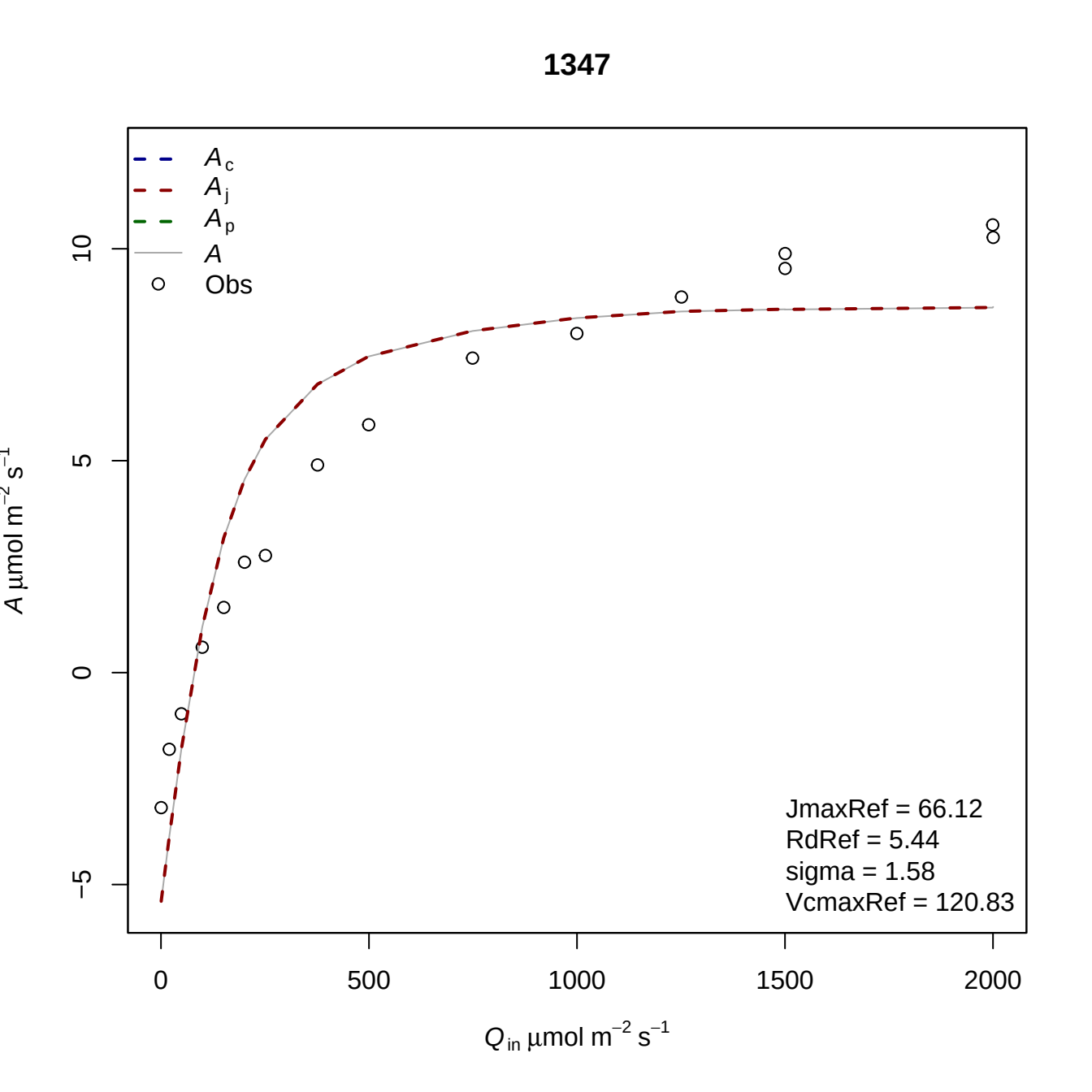

1342
A_c
A_j
A_p
A
Obs
JmaxRef = 60.75
RdRef = 3.4
sigma = 1.4
VcmaxRef = 113.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

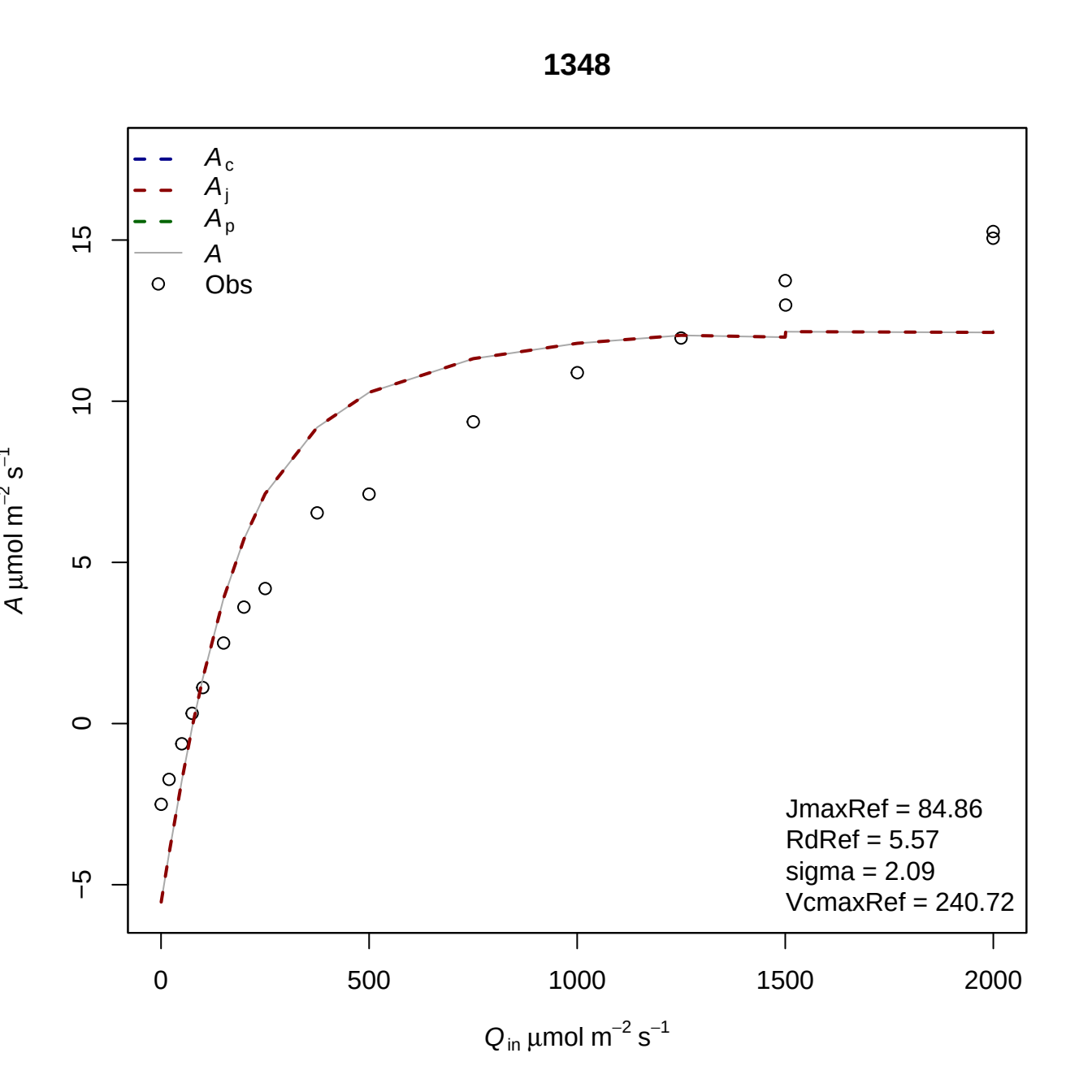

1343
A_c
A_j
A_p
A
Obs
JmaxRef = 57.26
RdRef = 4.45
sigma = 1.09
VcmaxRef = 91.26
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

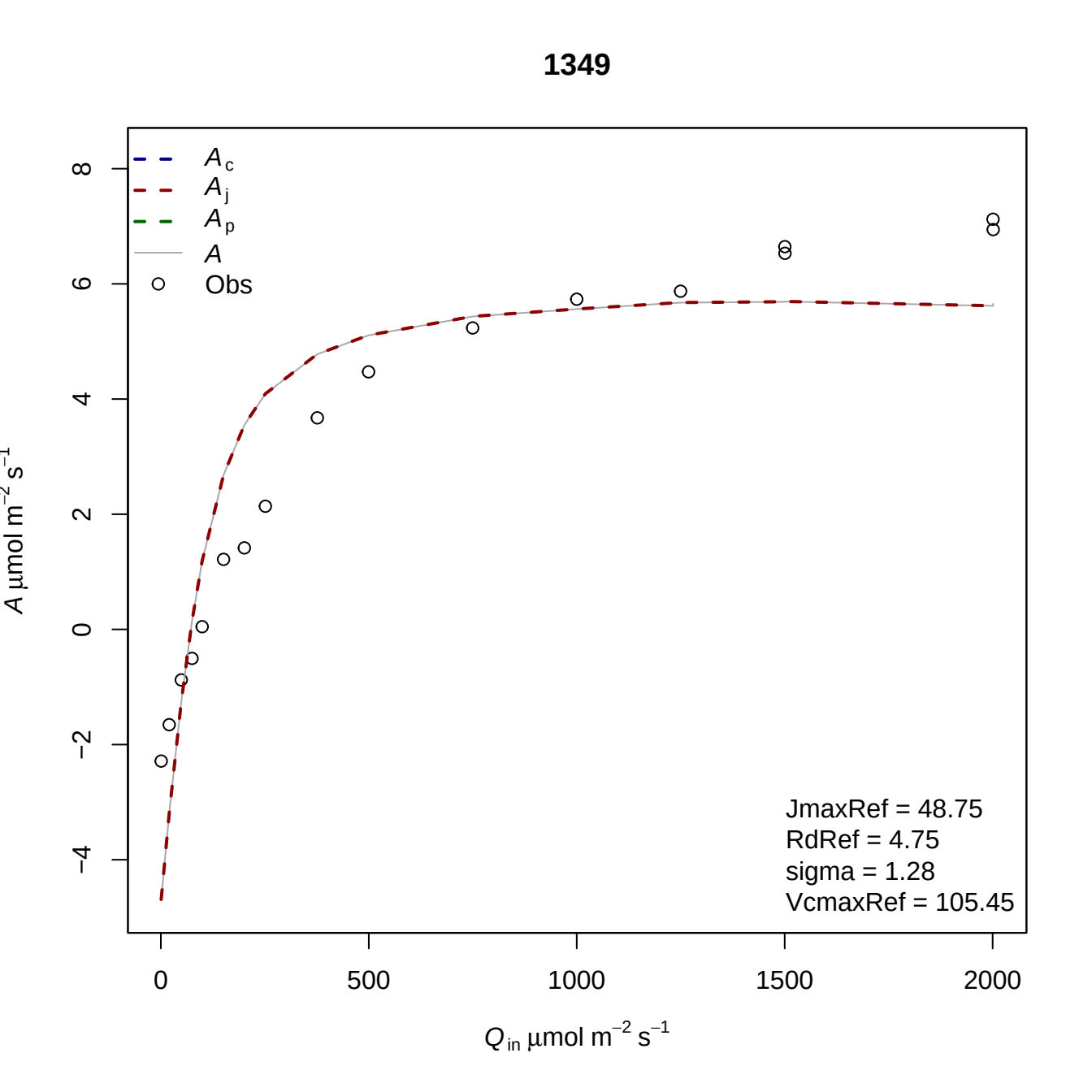

1344
A_c
A_j
A_p
A
Obs
JmaxRef = 49.78
RdRef = 2.7
sigma = 1.06
VcmaxRef = 81.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

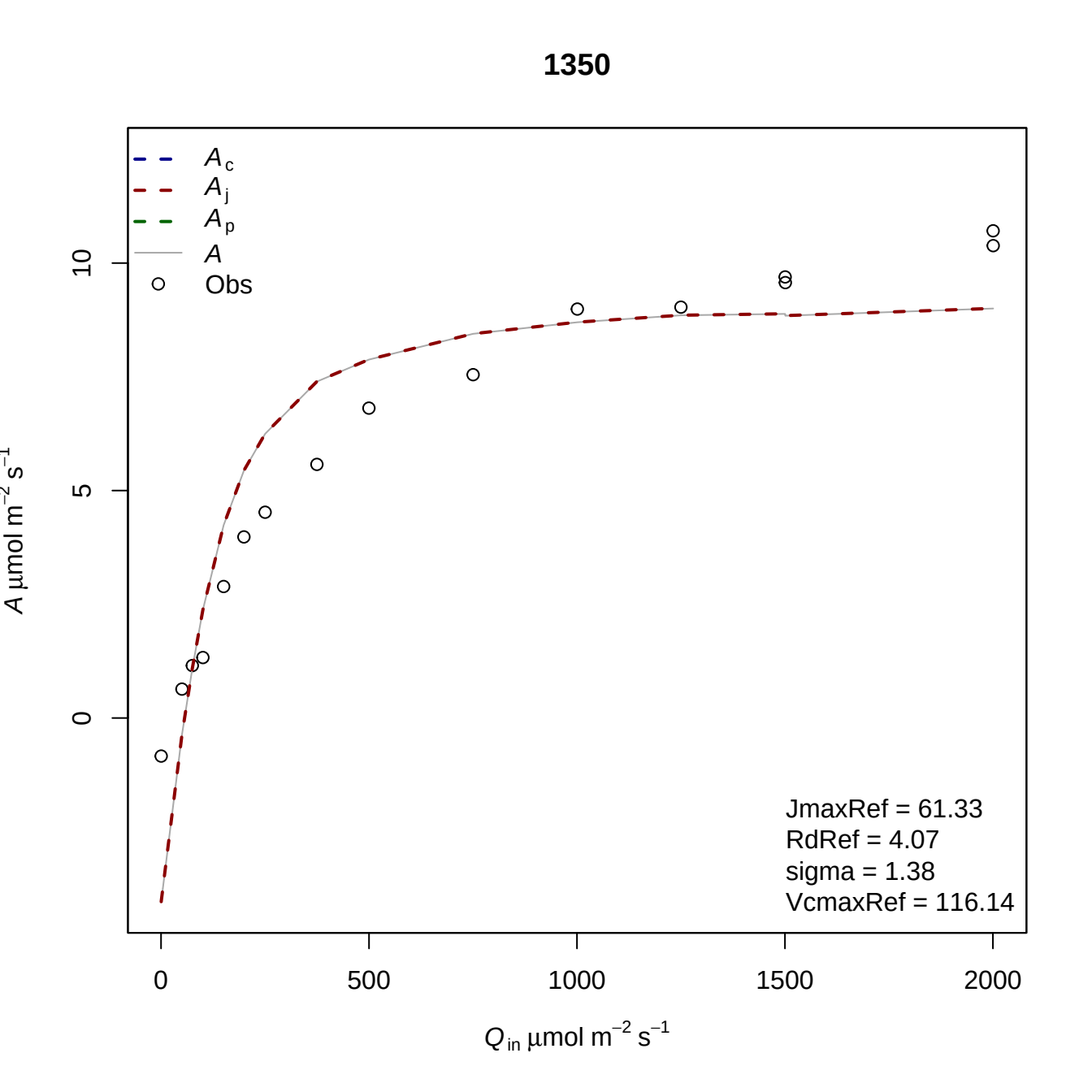

1345
A_c
A_j
A_p
A
Obs
JmaxRef = 54.87
RdRef = 2.69
sigma = 1.11
VcmaxRef = 116.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

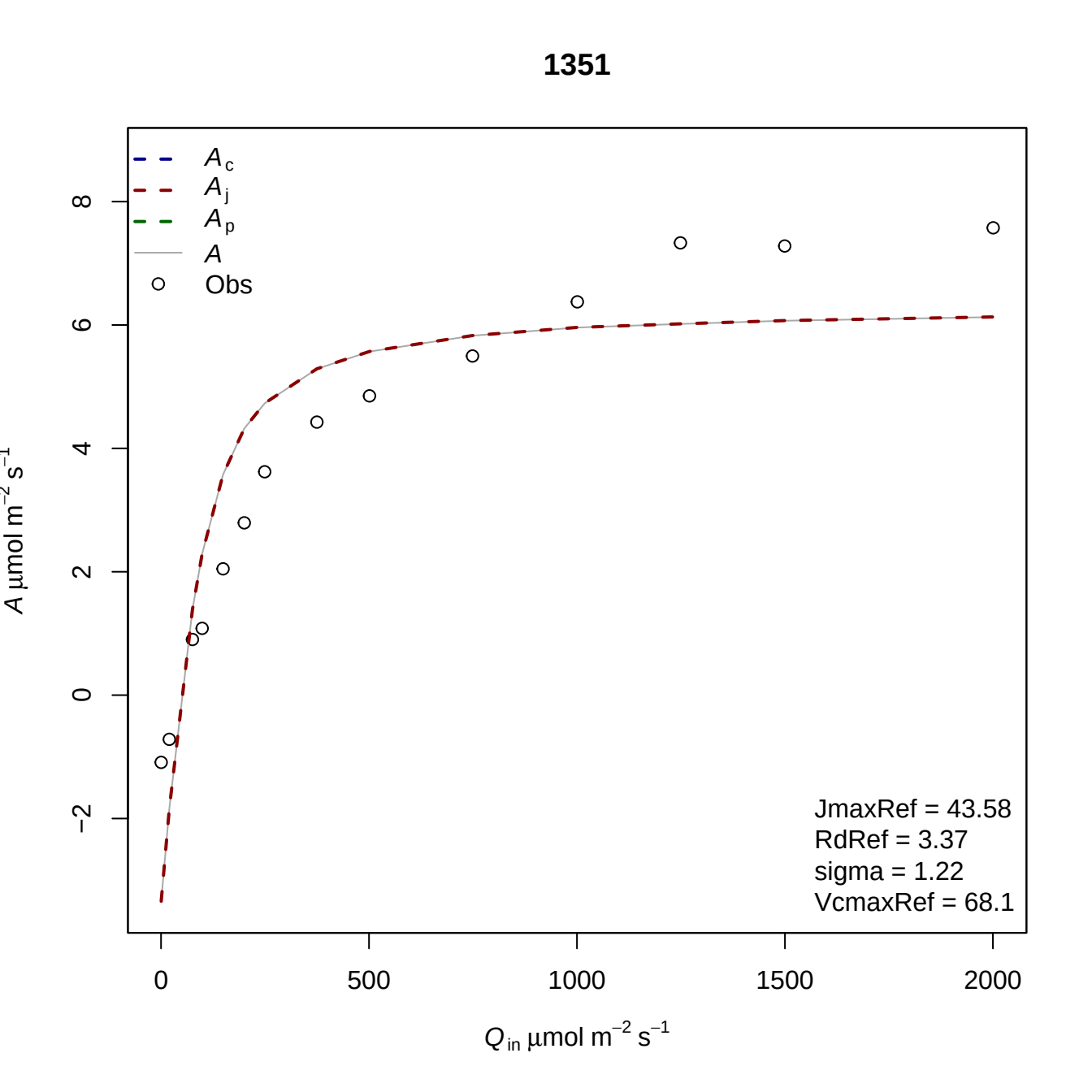

1346
A_c
A_j
A_p
A
Obs
JmaxRef = 43.64
RdRef = 5.06
sigma = 1.17
VcmaxRef = 86.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

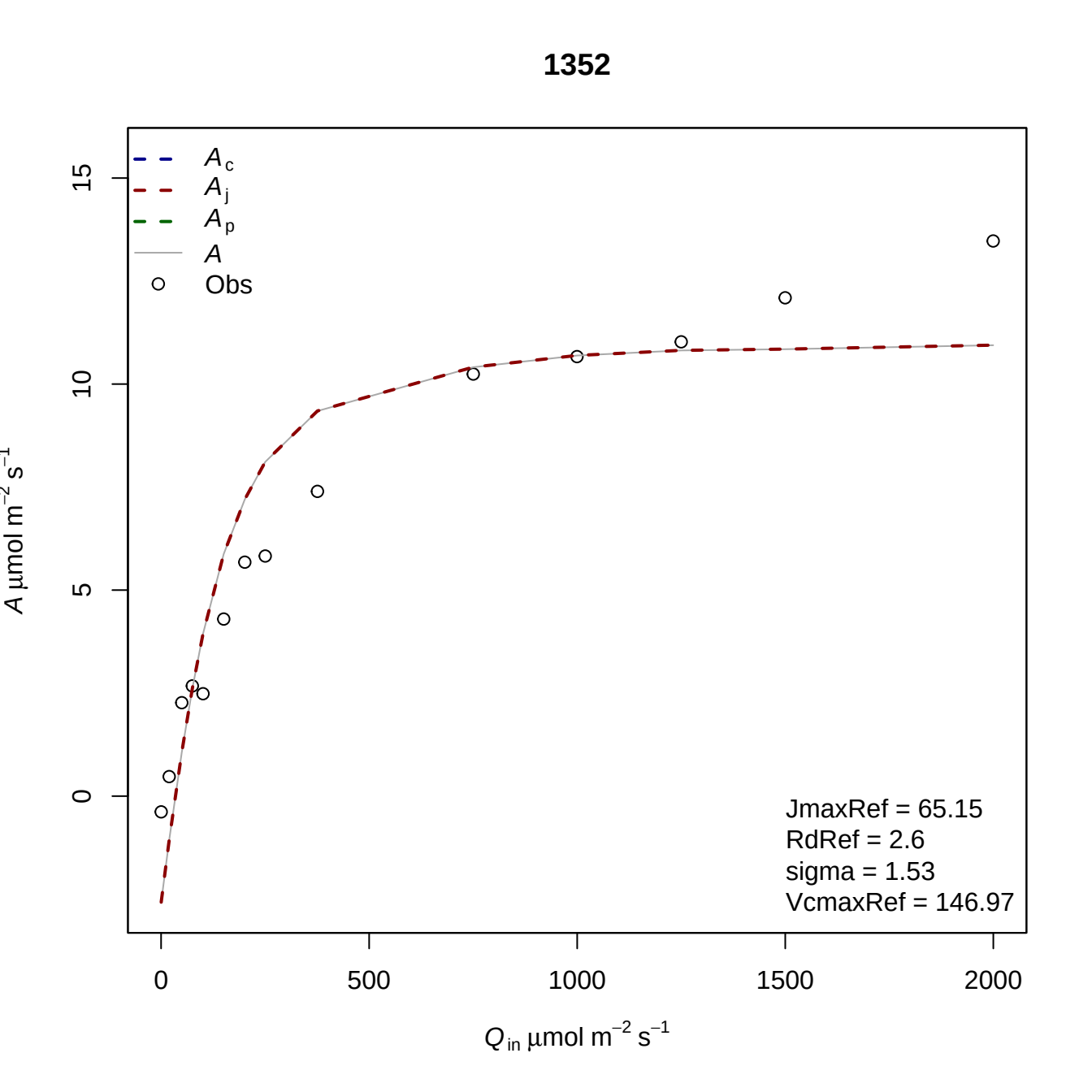

1347
A_c
A_j
A_p
A
Obs
JmaxRef = 66.12
RdRef = 5.44
sigma = 1.58
VcmaxRef = 120.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

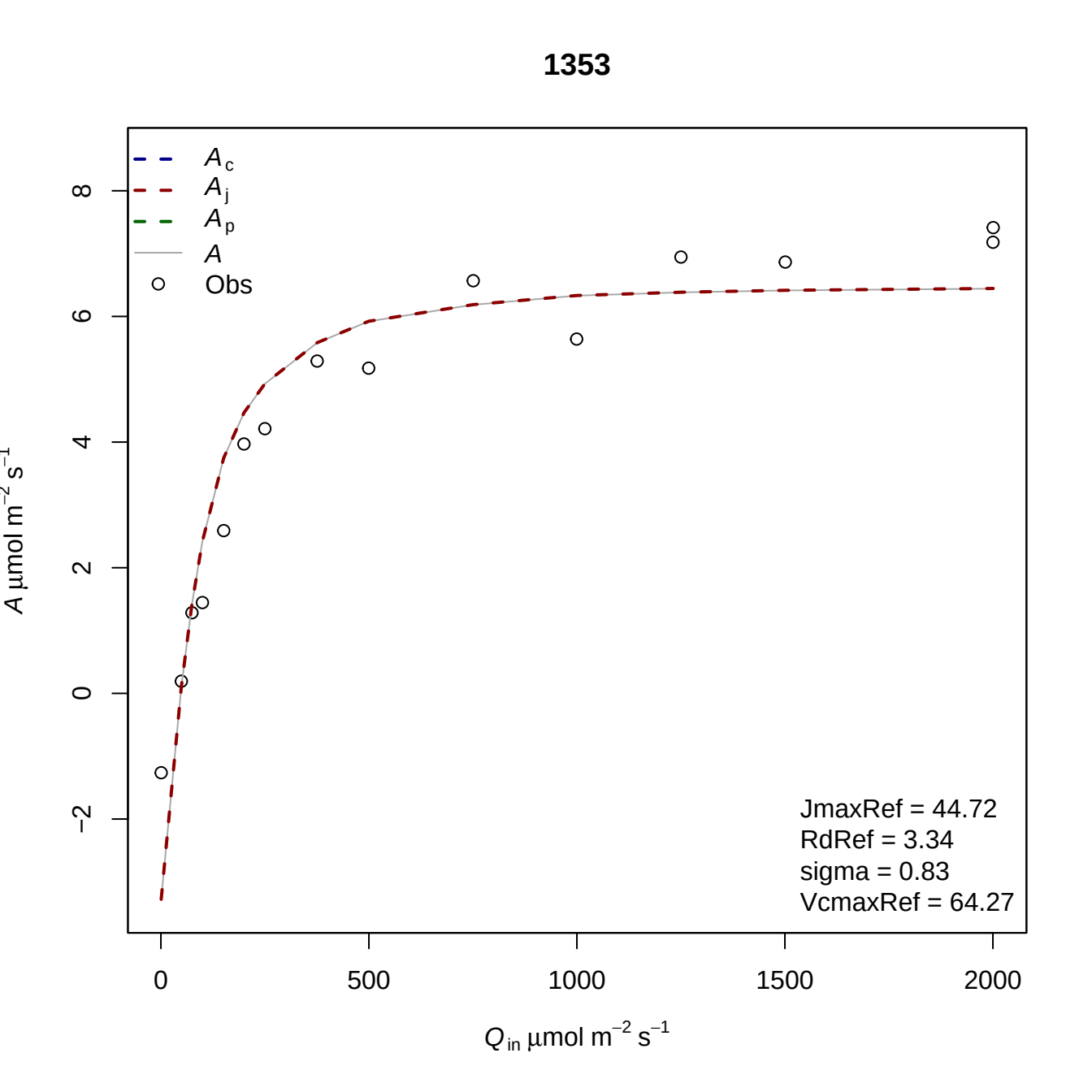

1348
A_c
A_j
A_p
A
Obs
JmaxRef = 84.86
RdRef = 5.57
sigma = 2.09
VcmaxRef = 240.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

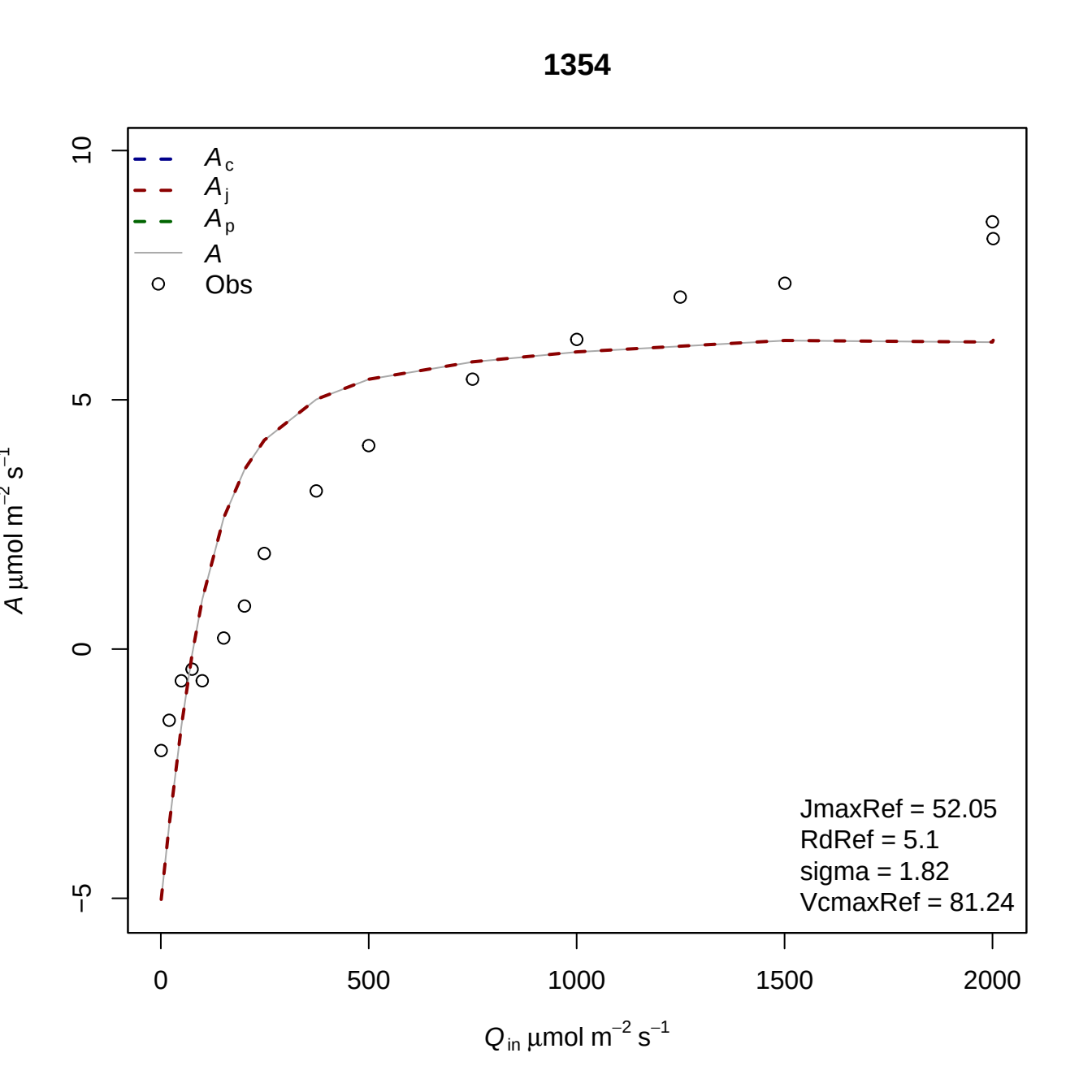

1349
A_c
A_j
A_p
A
Obs
JmaxRef = 48.75
RdRef = 4.75
sigma = 1.28
VcmaxRef = 105.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

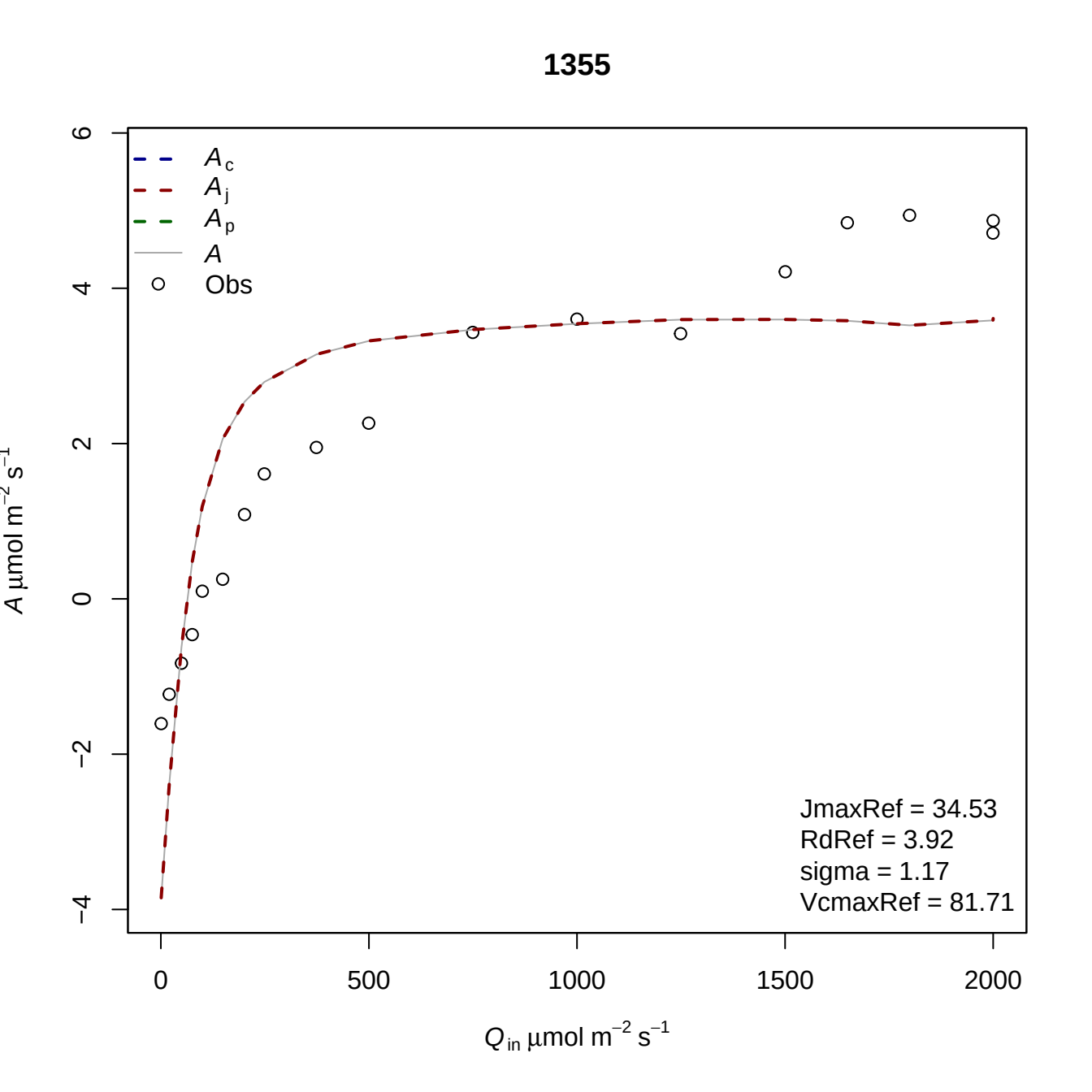

1350
A_c
A_j
A_p
A
Obs
JmaxRef = 61.33
RdRef = 4.07
sigma = 1.38
VcmaxRef = 116.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

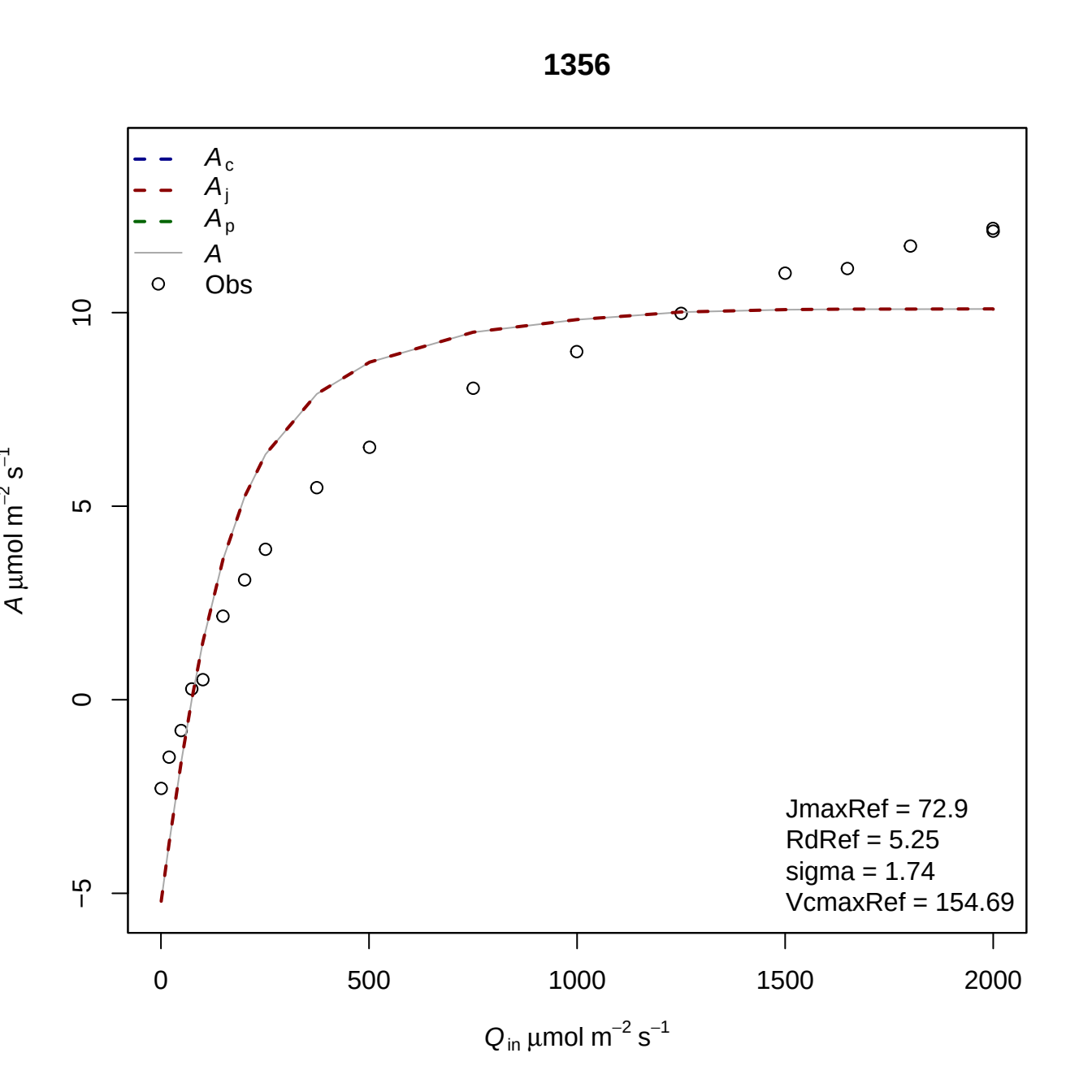

1351
A_c
A_j
A_p
A
Obs
JmaxRef = 43.58
RdRef = 3.37
sigma = 1.22
VcmaxRef = 68.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

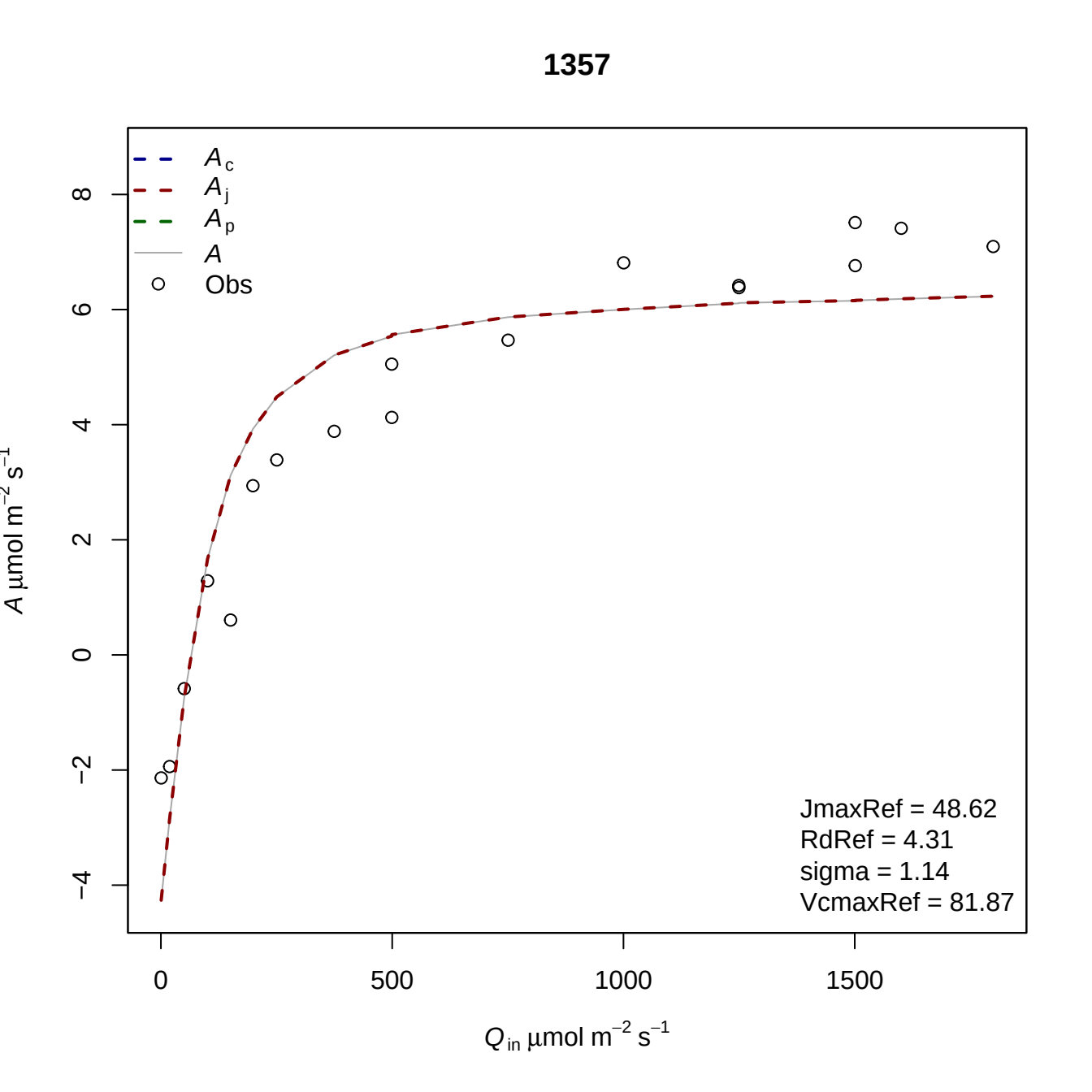

1352
A_c
A_j
A_p
A
Obs
JmaxRef = 65.15
RdRef = 2.6
sigma = 1.53
VcmaxRef = 146.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

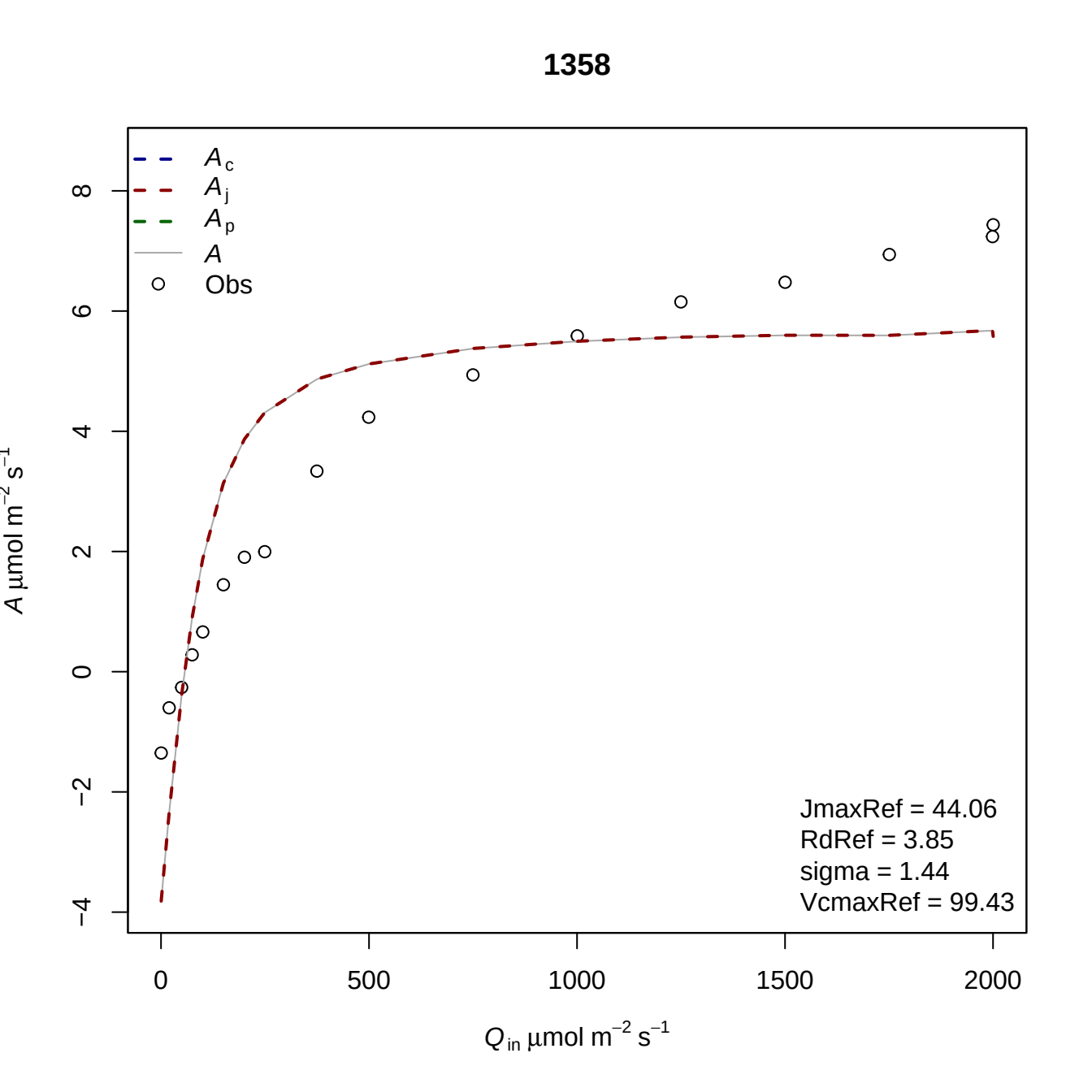

1353
A_c
A_j
A_p
A
Obs
JmaxRef = 44.72
RdRef = 3.34
sigma = 0.83
VcmaxRef = 64.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

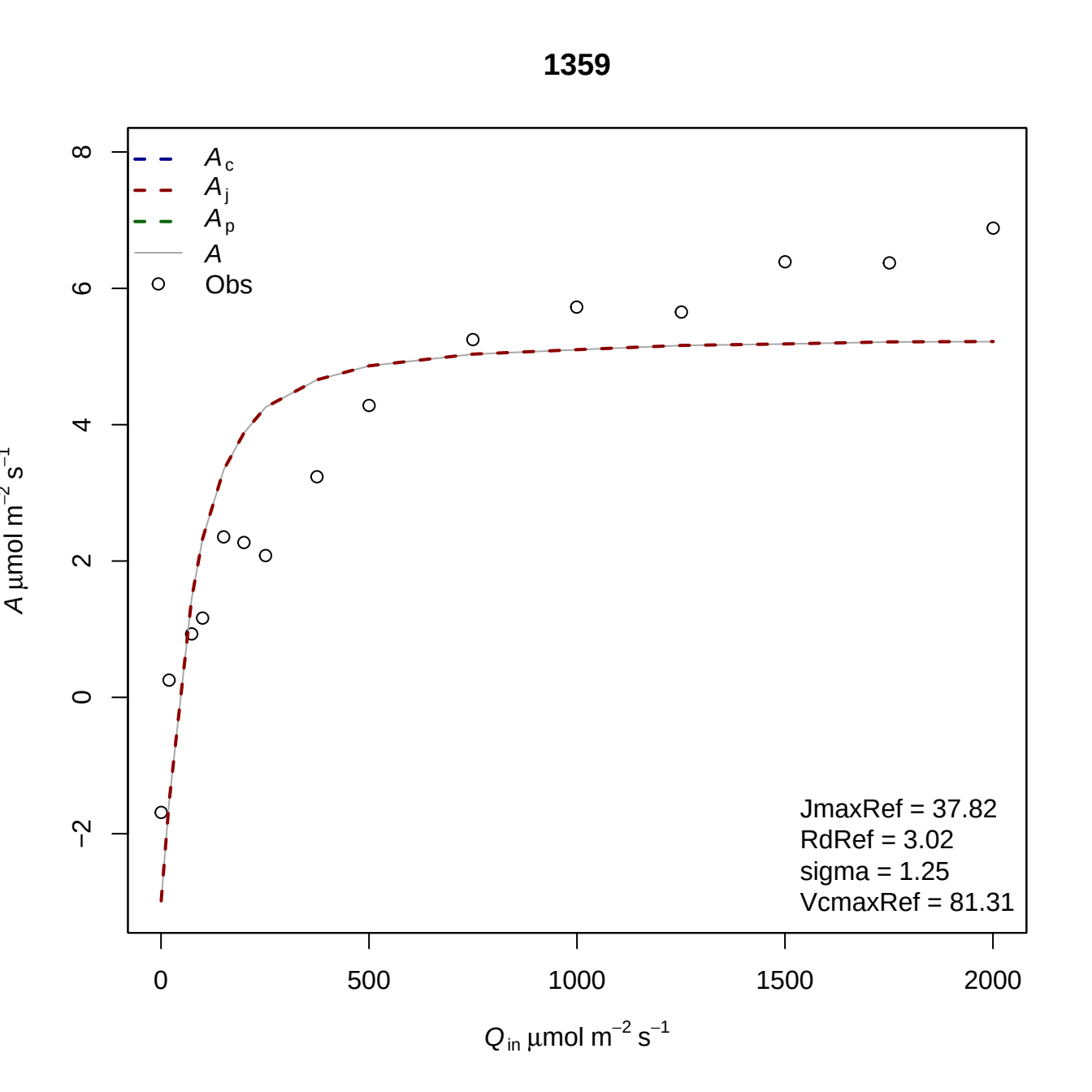

1354
A_c
A_j
A_p
A
Obs
JmaxRef = 52.05
RdRef = 5.1
sigma = 1.82
VcmaxRef = 81.24
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

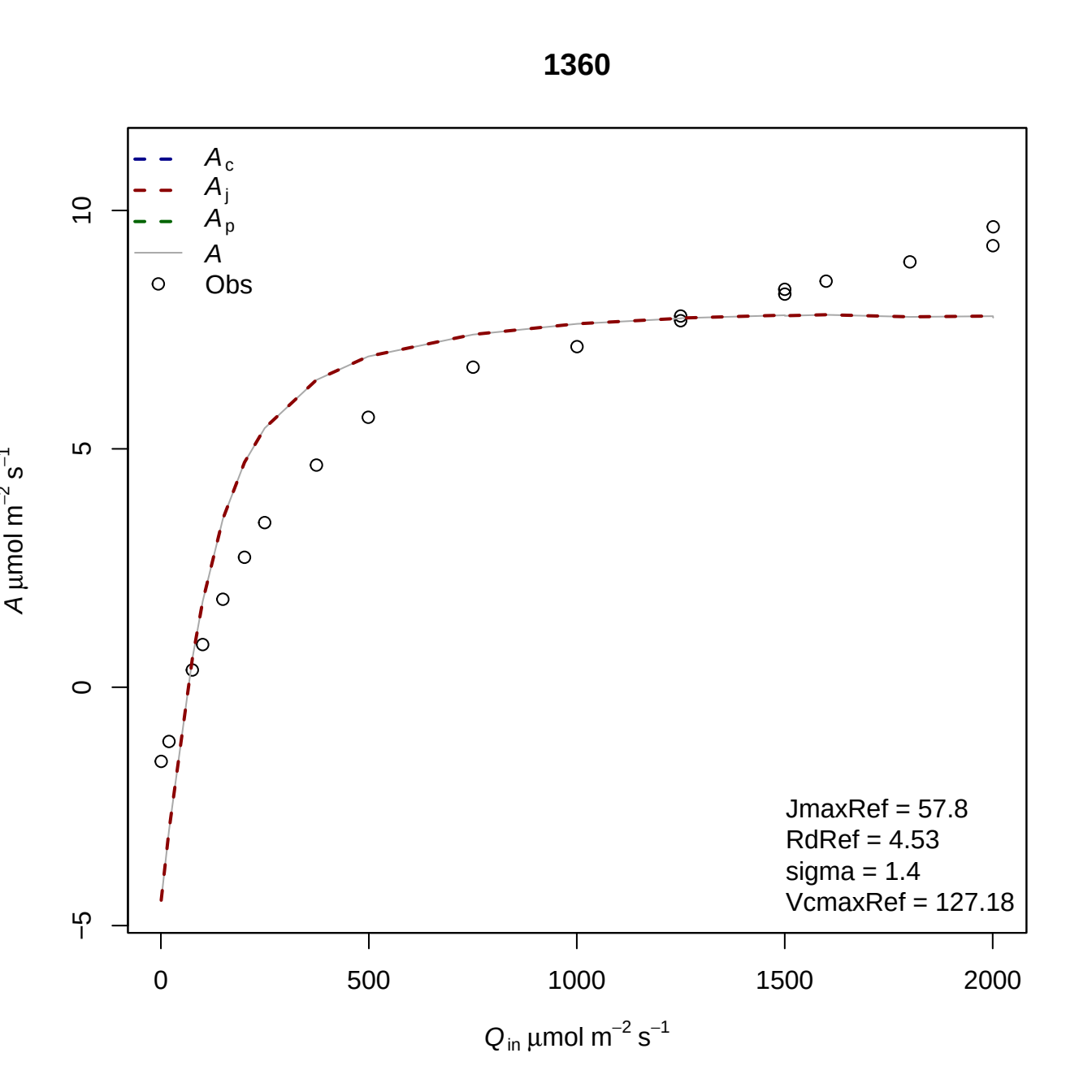

1355
A_c
A_j
A_p
A
Obs
JmaxRef = 34.53
RdRef = 3.92
sigma = 1.17
VcmaxRef = 81.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

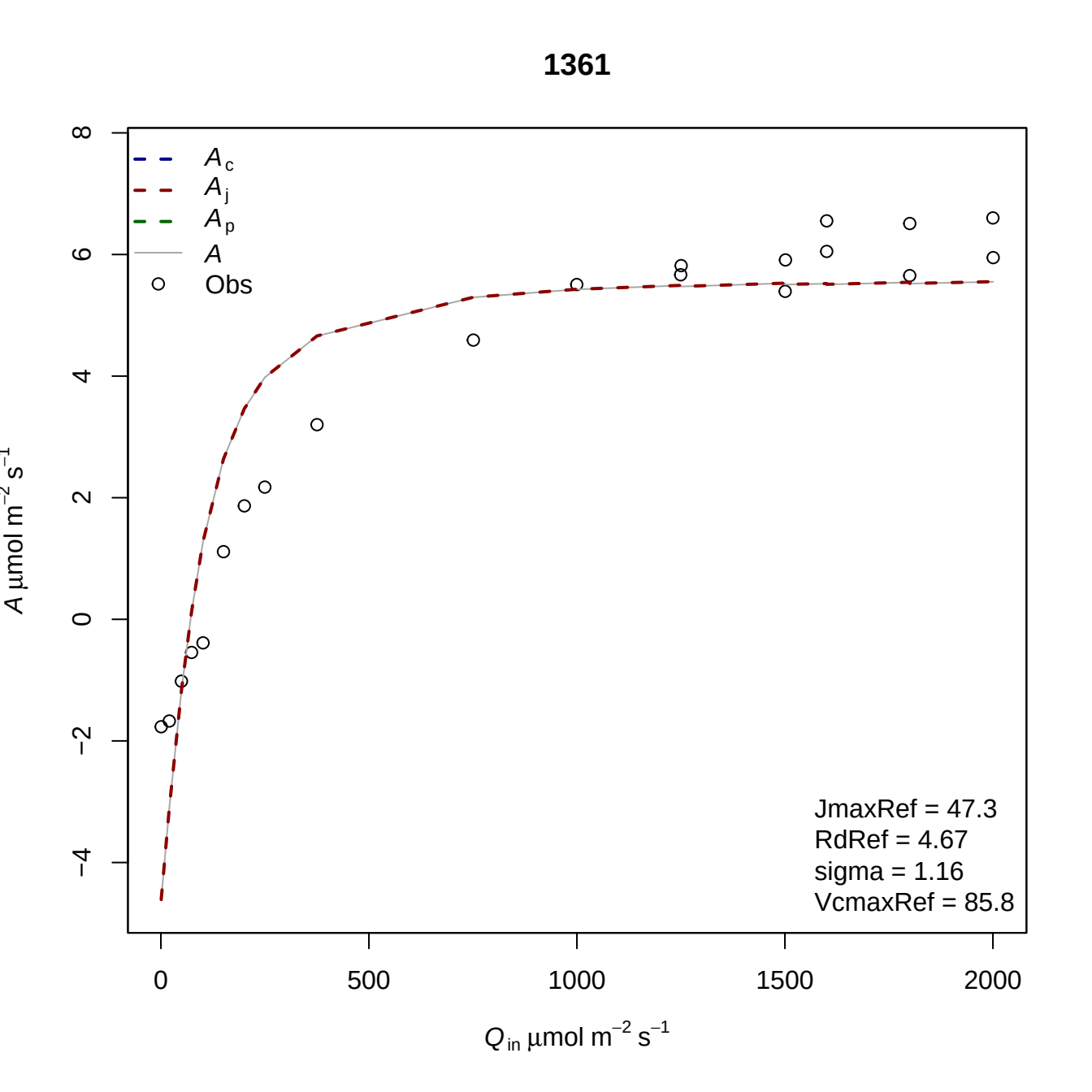

1356
A_c
A_j
A_p
A
Obs
JmaxRef = 72.9
RdRef = 5.25
sigma = 1.74
VcmaxRef = 154.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

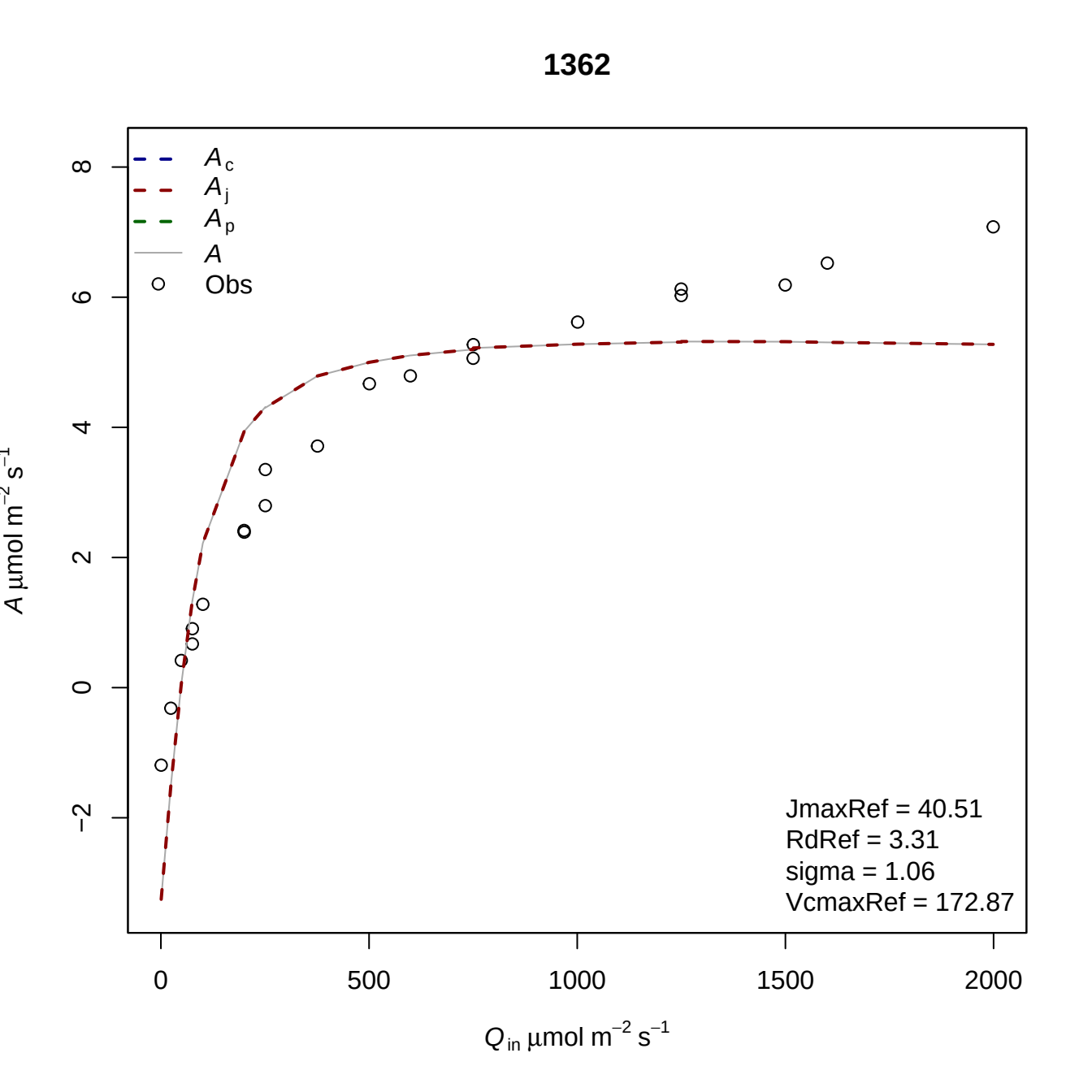

1357
A_c
A_j
A_p
A
Obs
JmaxRef = 48.62
RdRef = 4.31
sigma = 1.14
VcmaxRef = 81.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

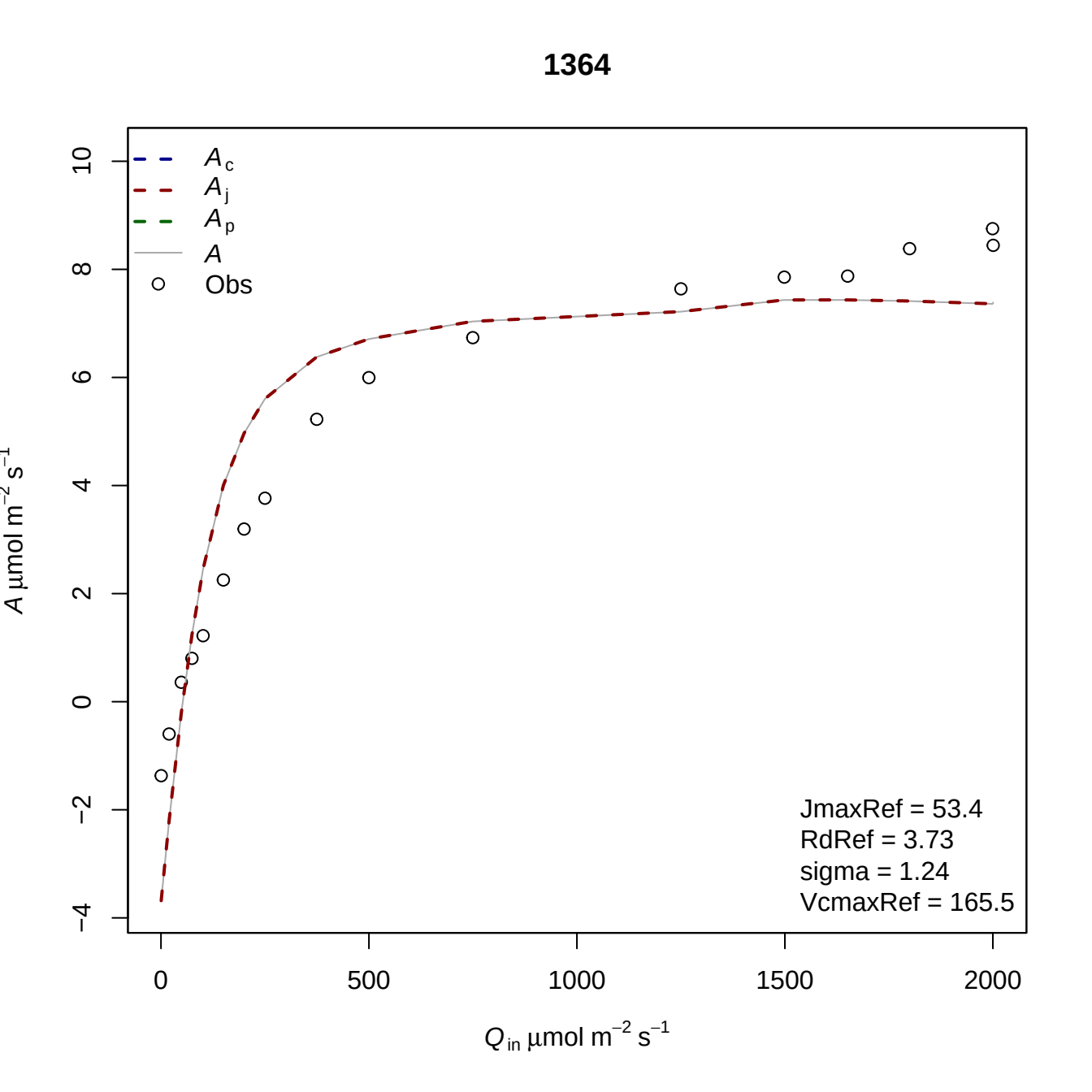

1358
A_c
A_j
A_p
A
Obs
JmaxRef = 44.06
RdRef = 3.85
sigma = 1.44
VcmaxRef = 99.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

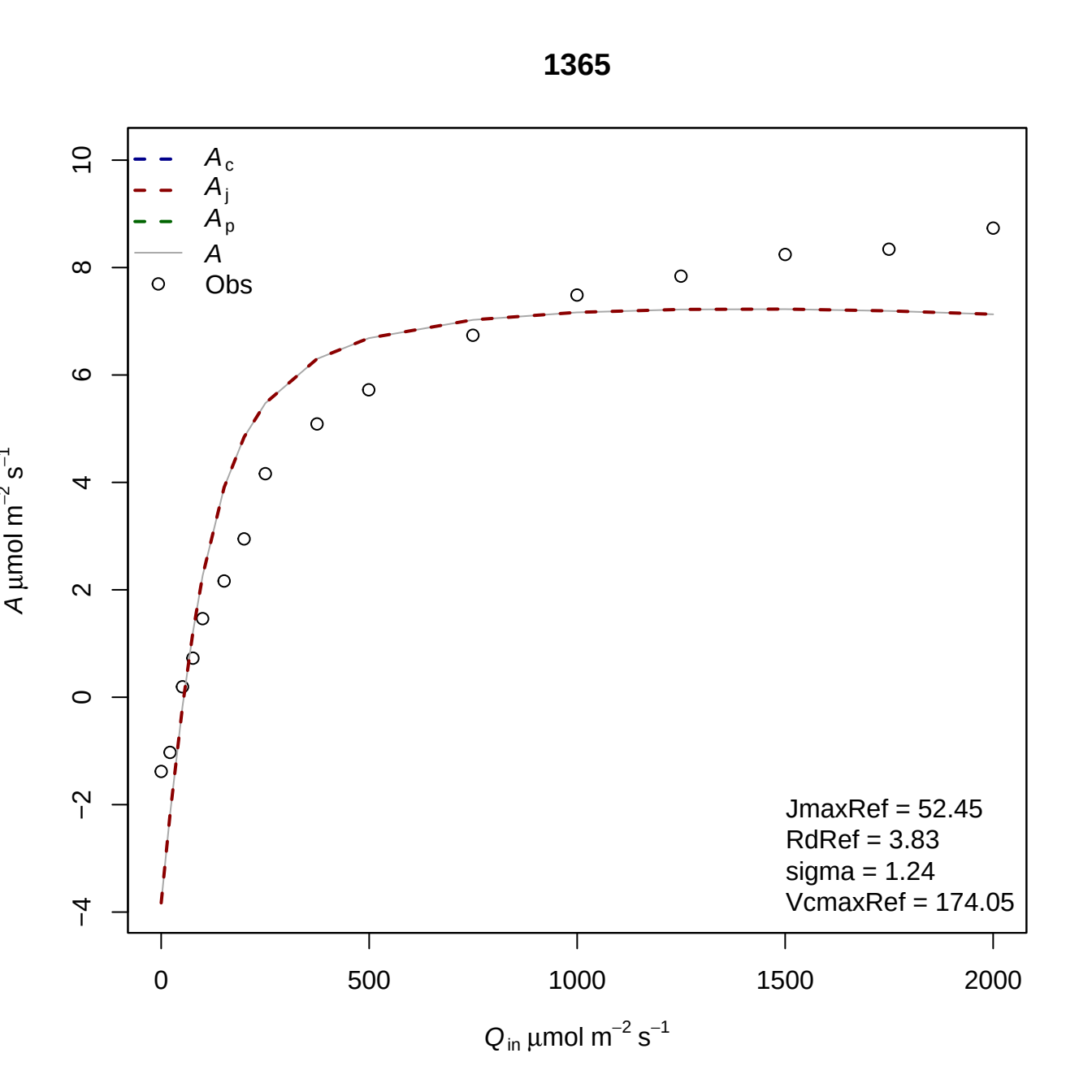

1359
A_c
A_j
A_p
A
Obs
JmaxRef = 37.82
RdRef = 3.02
sigma = 1.25
VcmaxRef = 81.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

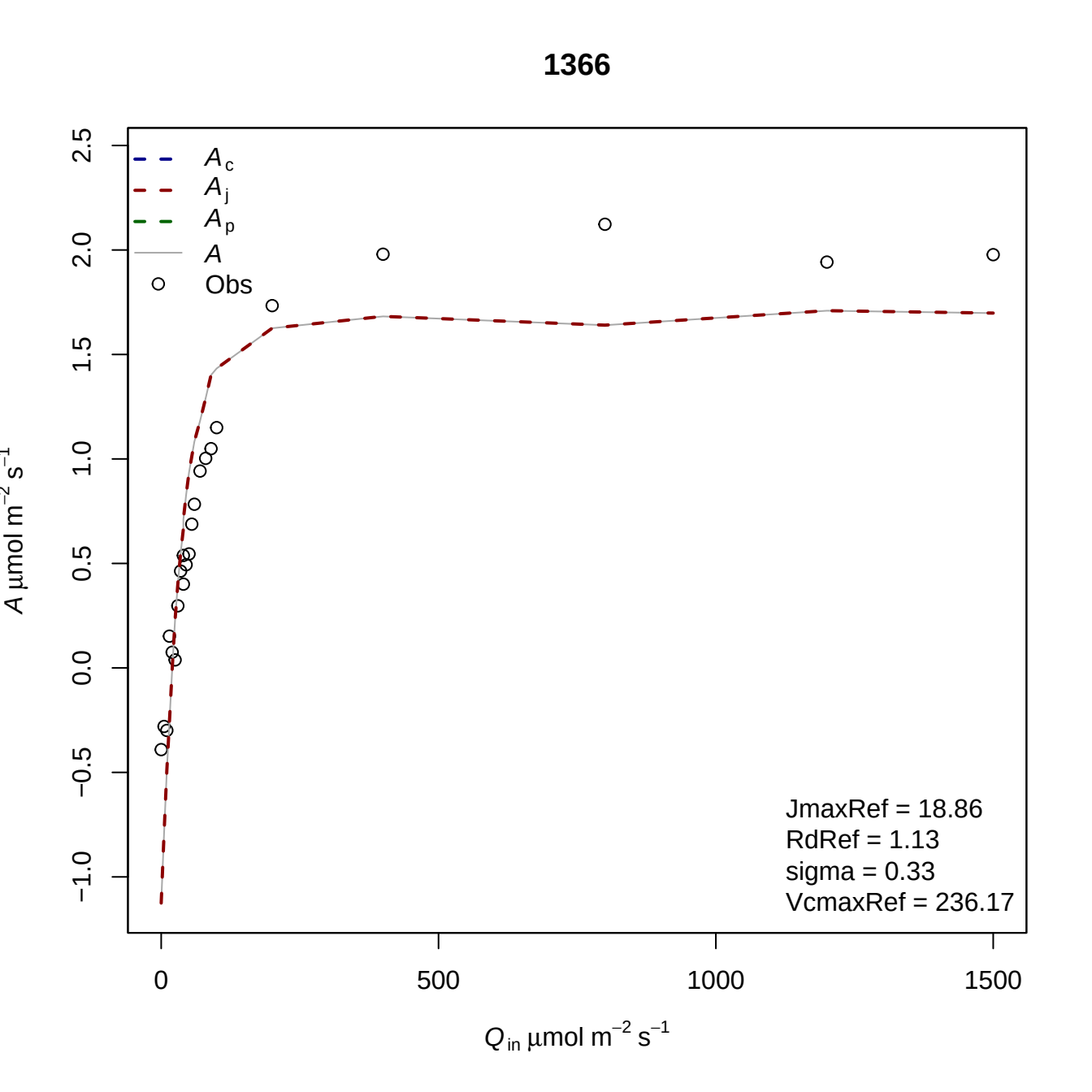

1360
A_c
A_j
A_p
A
Obs
JmaxRef = 57.8
RdRef = 4.53
sigma = 1.4
VcmaxRef = 127.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

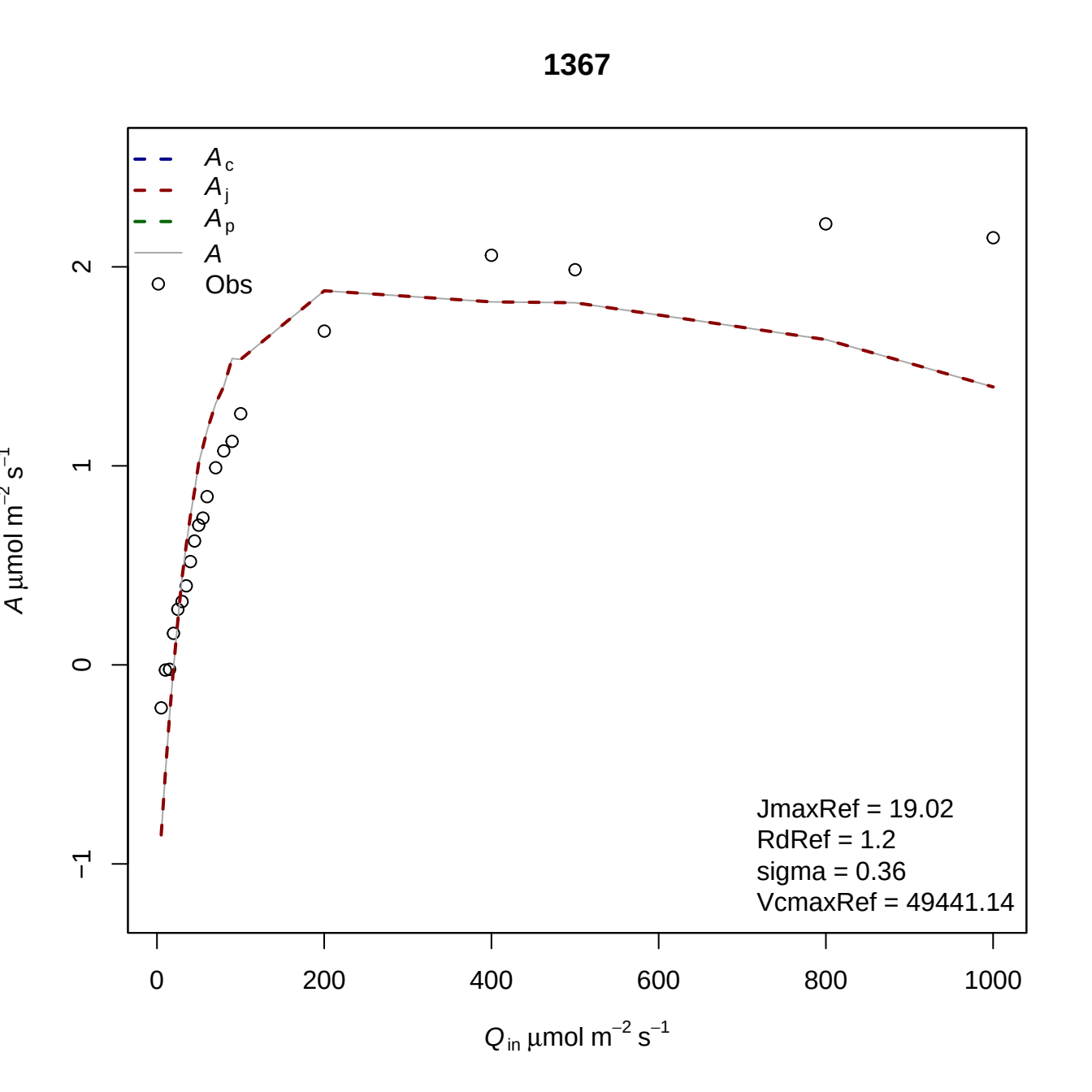

1361
A_c
A_j
A_p
A
Obs
JmaxRef = 47.3
RdRef = 4.67
sigma = 1.16
VcmaxRef = 85.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

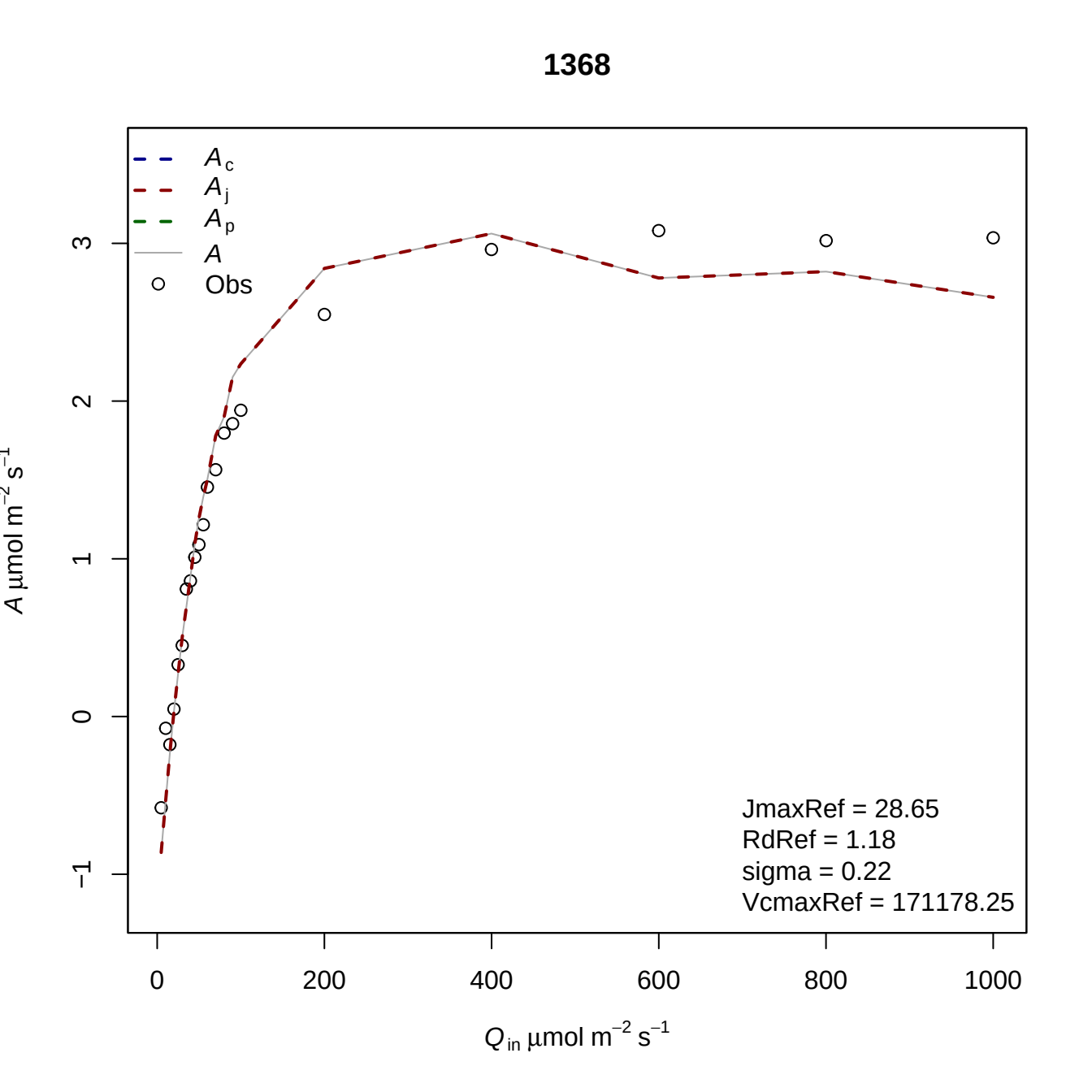

1362
A_c
A_j
A_p
A
Obs
JmaxRef = 40.51
RdRef = 3.31
sigma = 1.06
VcmaxRef = 172.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

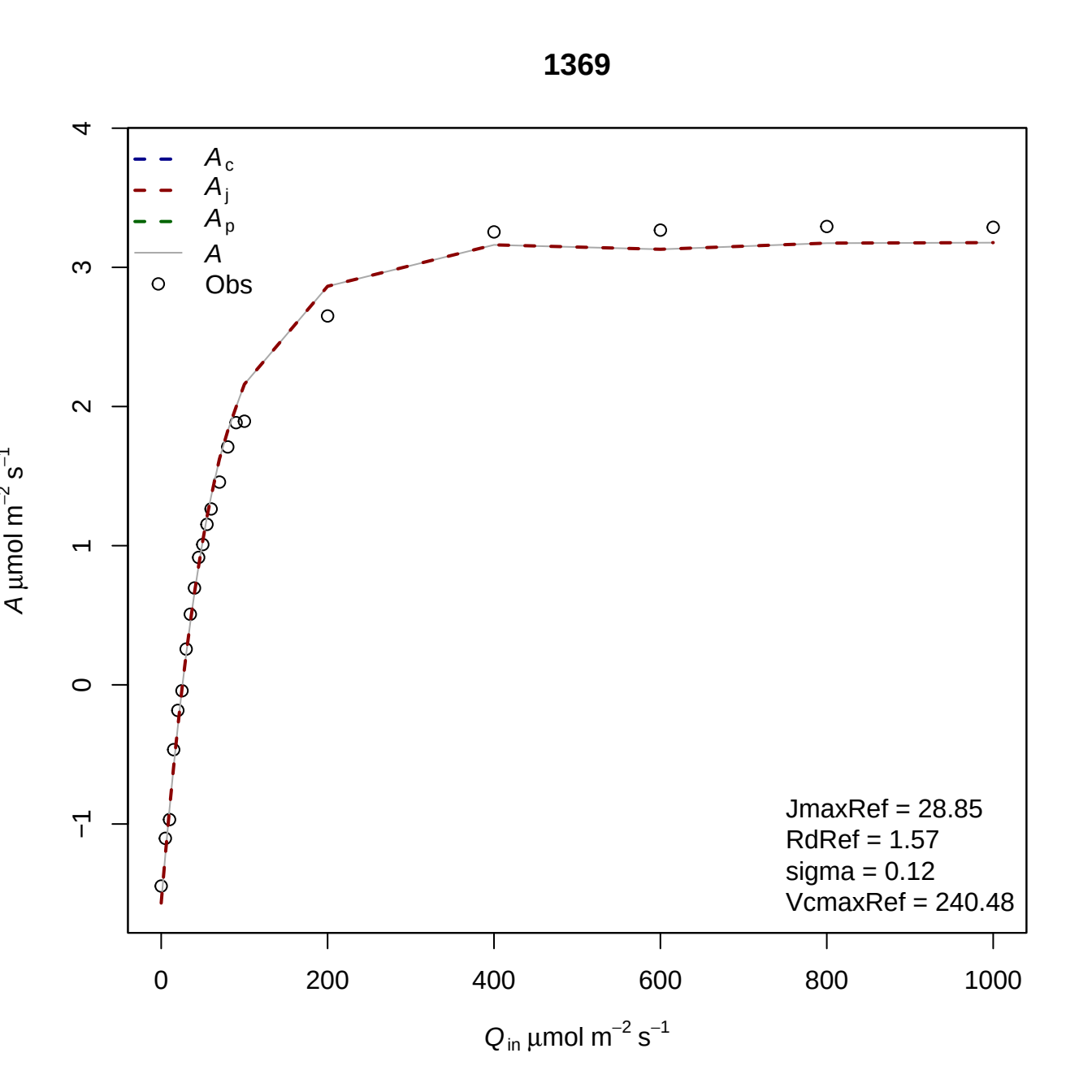

1364
A_c
A_j
A_p
A
Obs
JmaxRef = 53.4
RdRef = 3.73
sigma = 1.24
VcmaxRef = 165.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

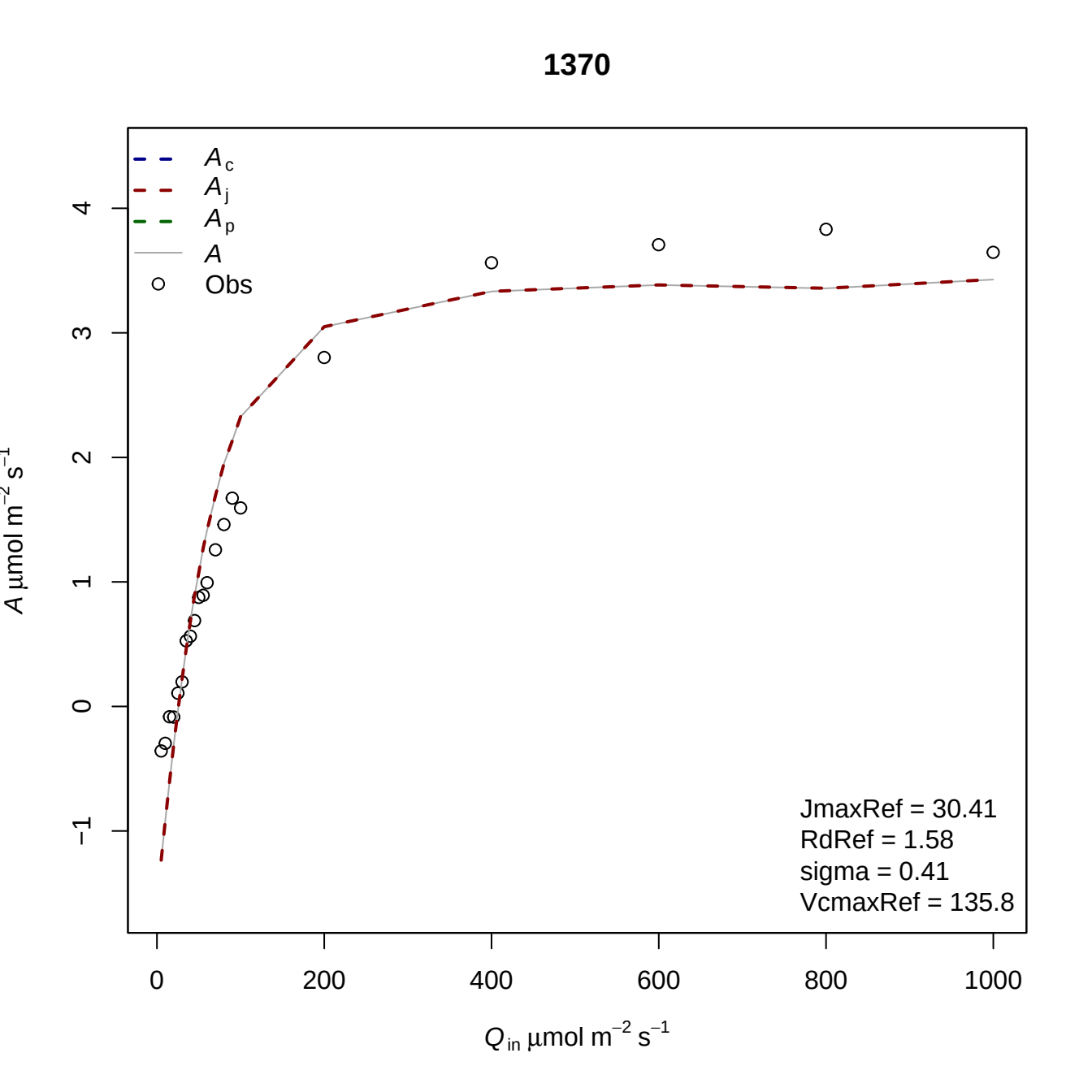

1365
A_c
A_j
A_p
A
Obs
JmaxRef = 52.45
RdRef = 3.83
sigma = 1.24
VcmaxRef = 174.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

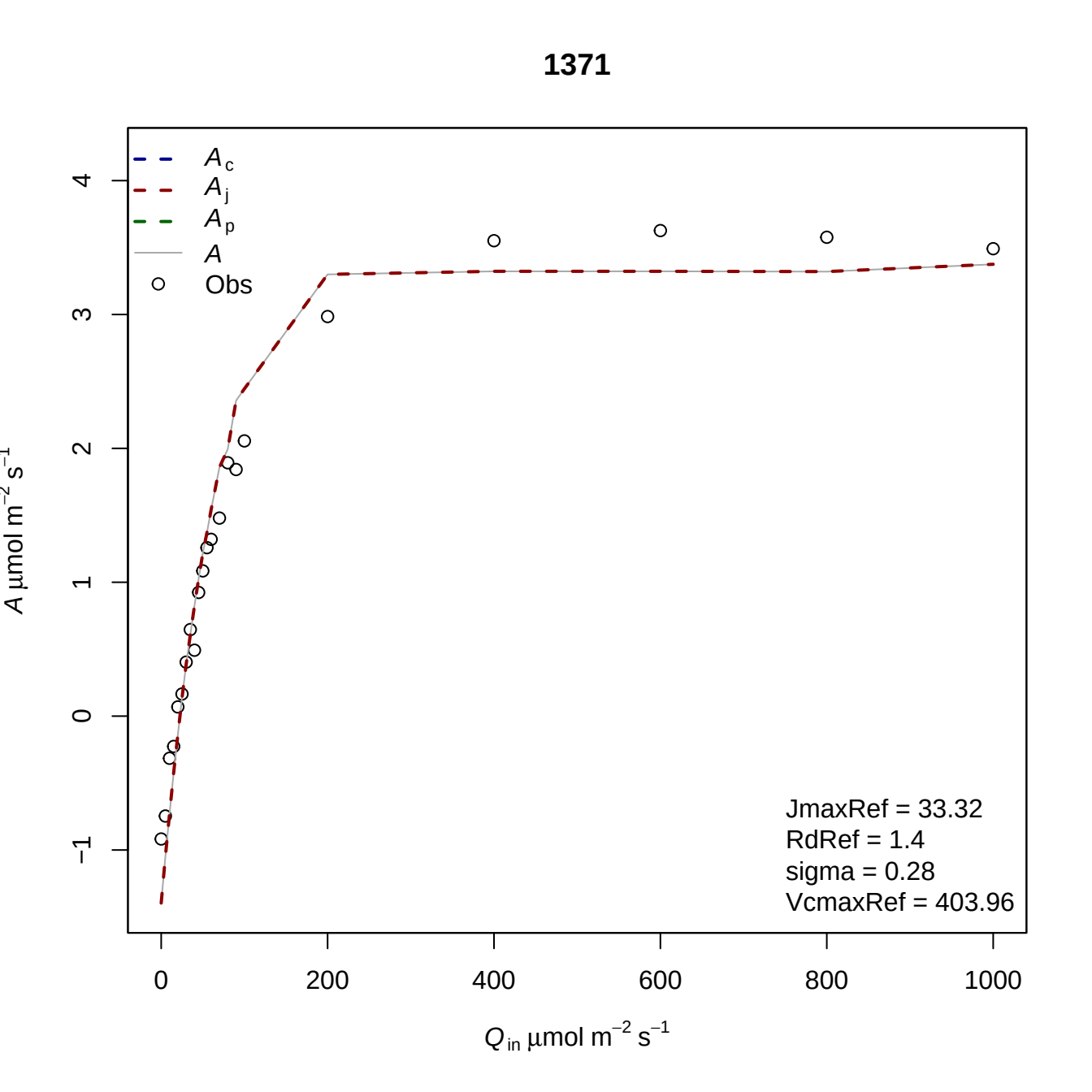

1366
A_c
A_j
A_p
A
Obs
JmaxRef = 18.86
RdRef = 1.13
sigma = 0.33
VcmaxRef = 236.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500

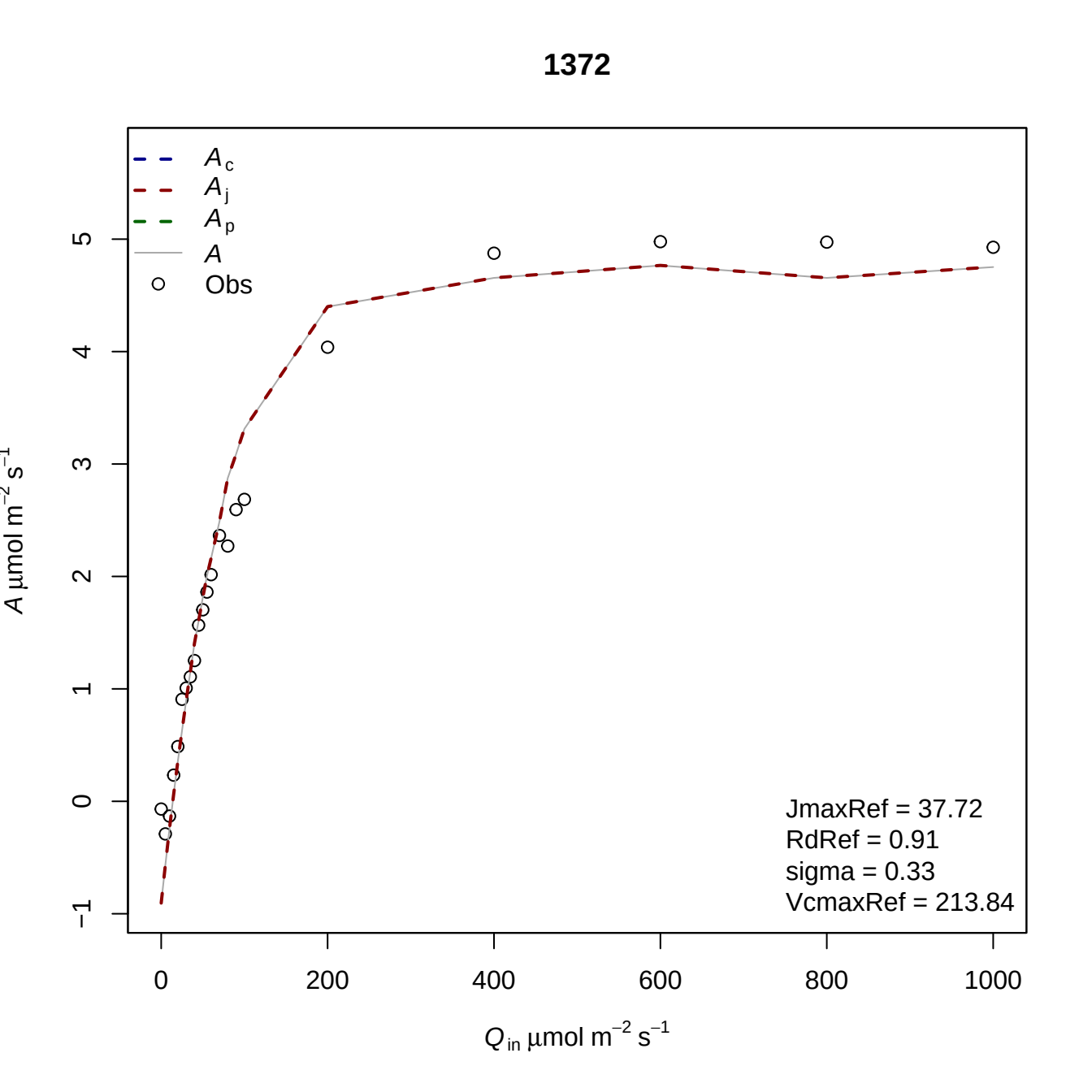

1367
A_c
A_j
A_p
A
Obs
JmaxRef = 19.02
RdRef = 1.2
sigma = 0.36
VcmaxRef = 49441.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
200
400
600
800
1000

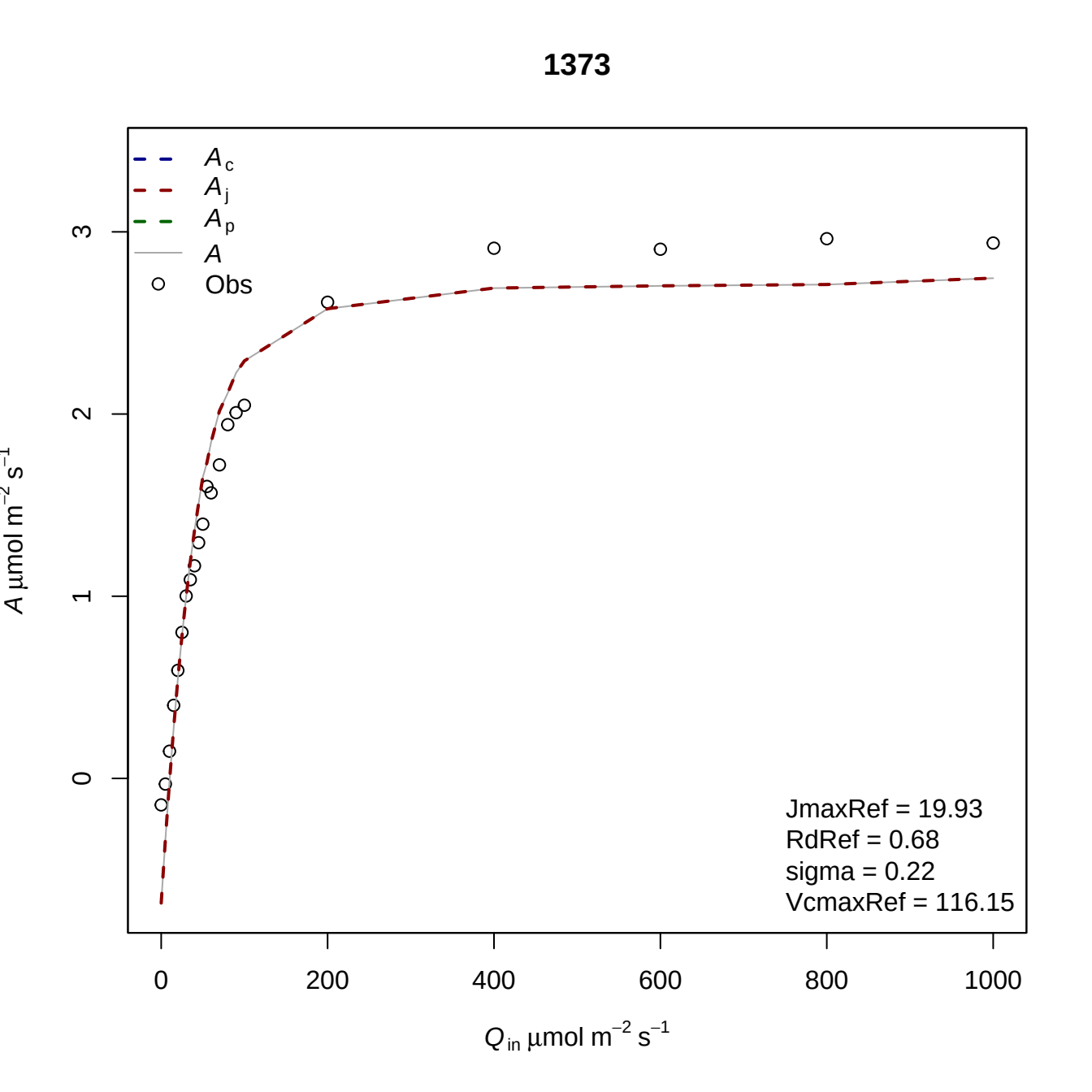

1368
A_c
A_j
A_p
A
Obs
JmaxRef = 28.65
RdRef = 1.18
sigma = 0.22
VcmaxRef = 171178.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

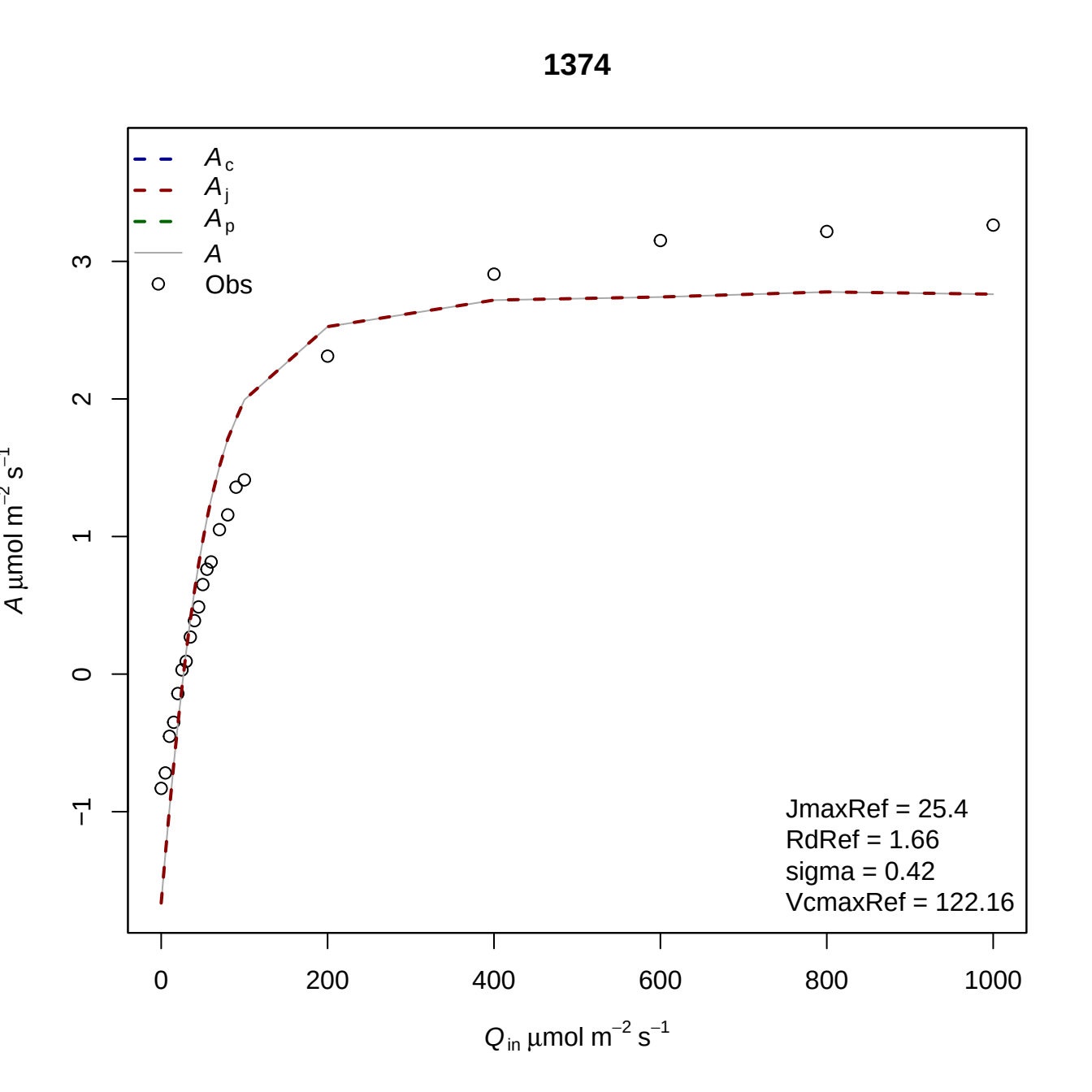

1369
A_c
A_j
A_p
A
Obs
JmaxRef = 28.85
RdRef = 1.57
sigma = 0.12
VcmaxRef = 240.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

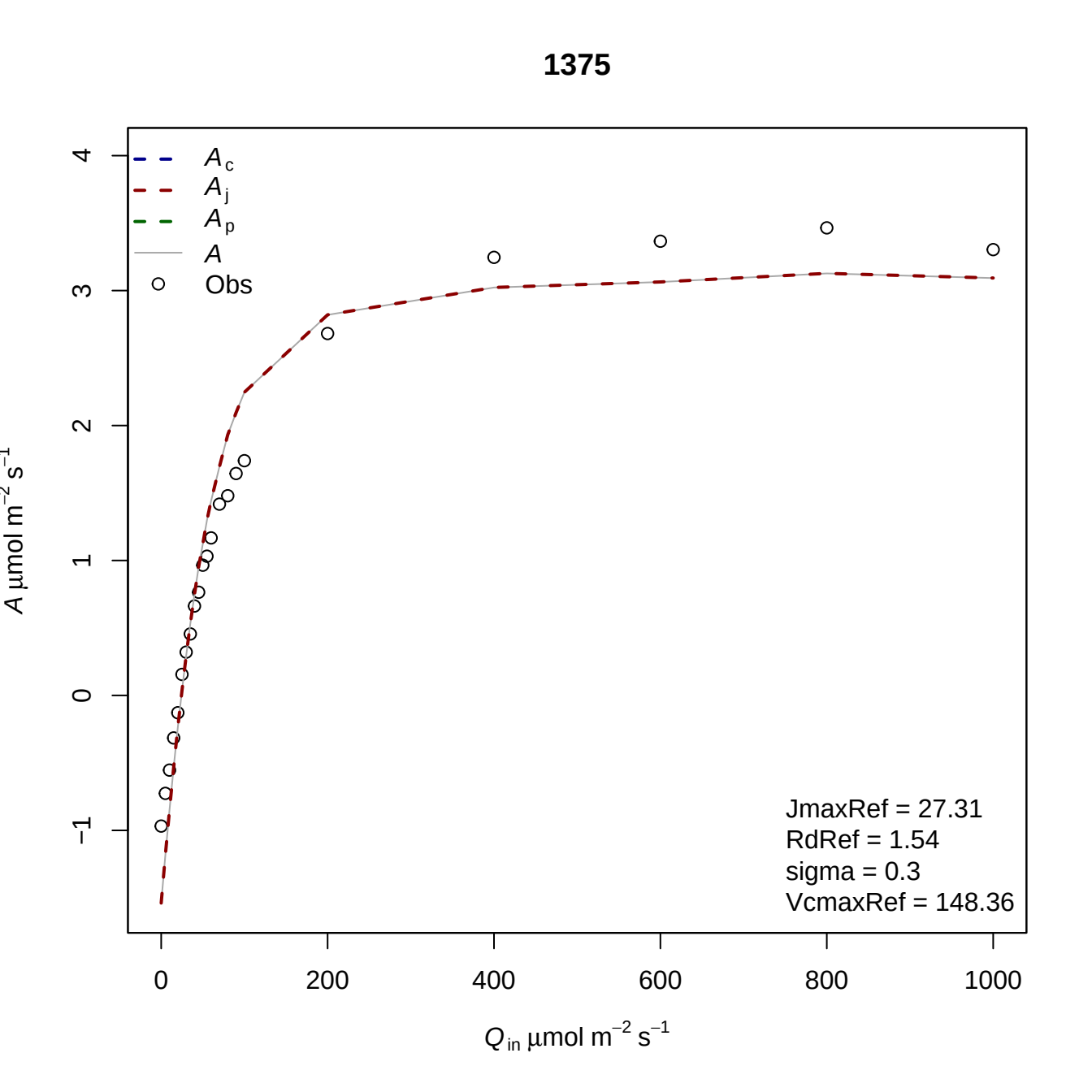

1370
A_c
A_j
A_p
A
Obs
JmaxRef = 30.41
RdRef = 1.58
sigma = 0.41
VcmaxRef = 135.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

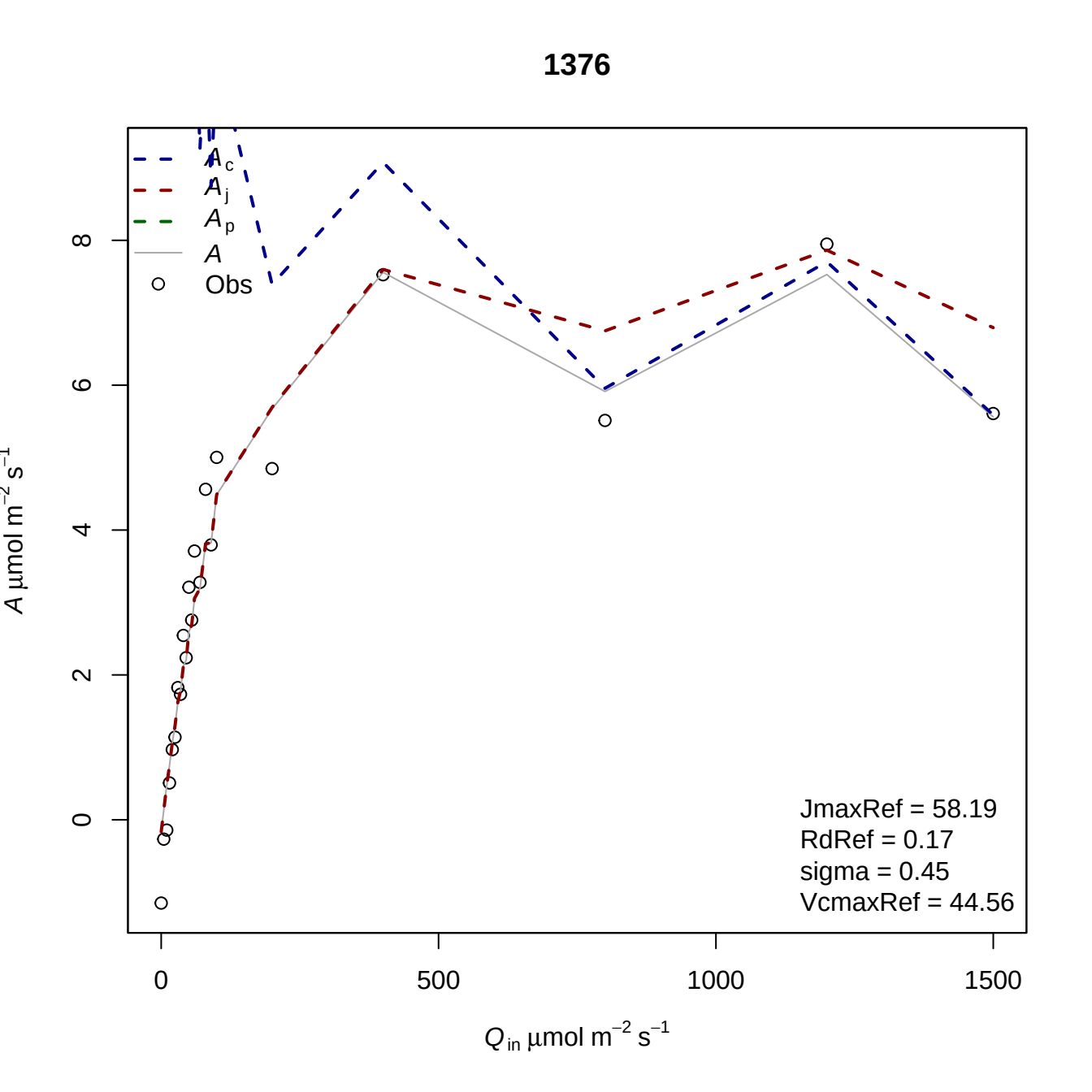

1371
A_c
A_j
A_p
A
Obs
JmaxRef = 33.32
RdRef = 1.4
sigma = 0.28
VcmaxRef = 403.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

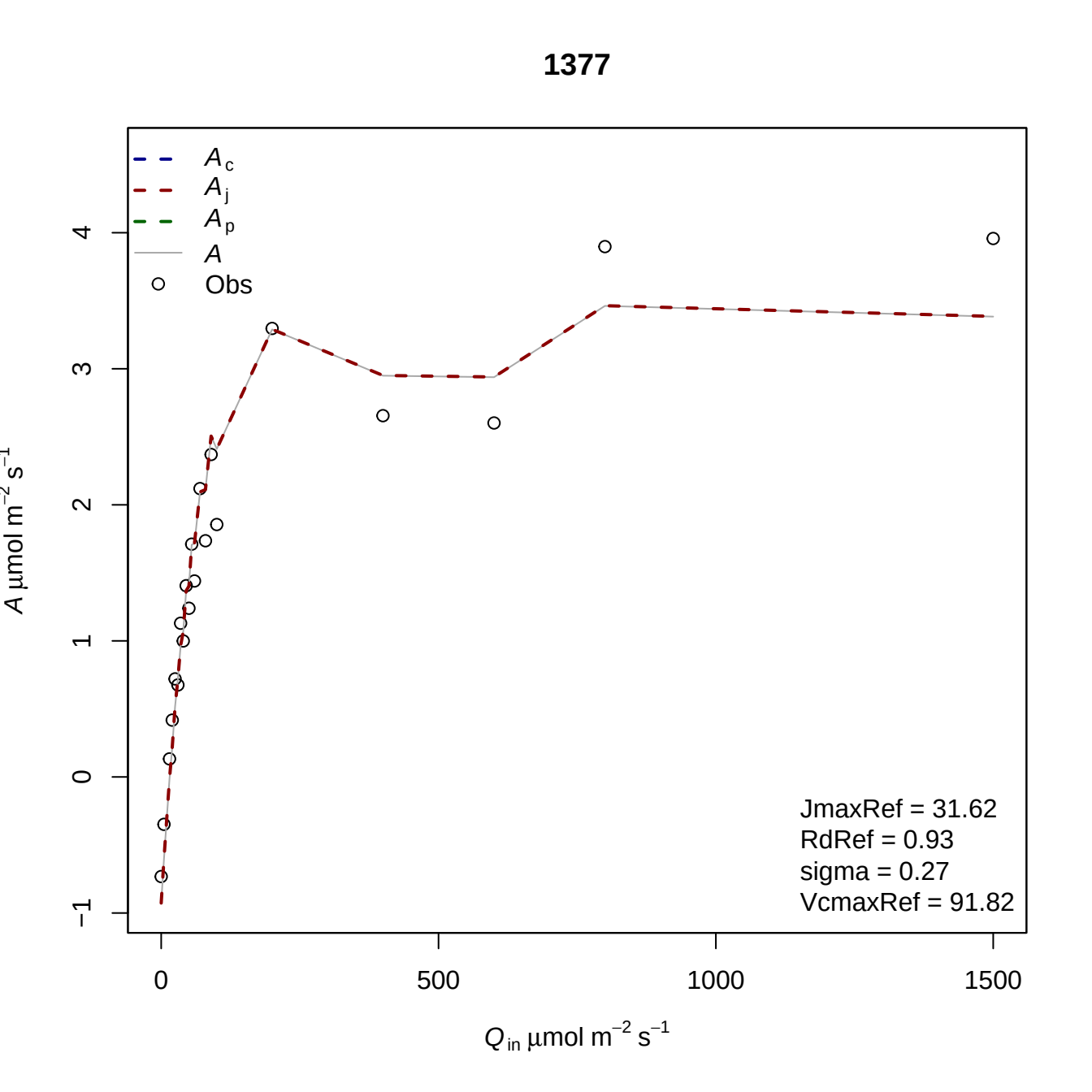

1372
A_c
A_j
A_p
A
Obs
JmaxRef = 37.72
RdRef = 0.91
sigma = 0.33
VcmaxRef = 213.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000

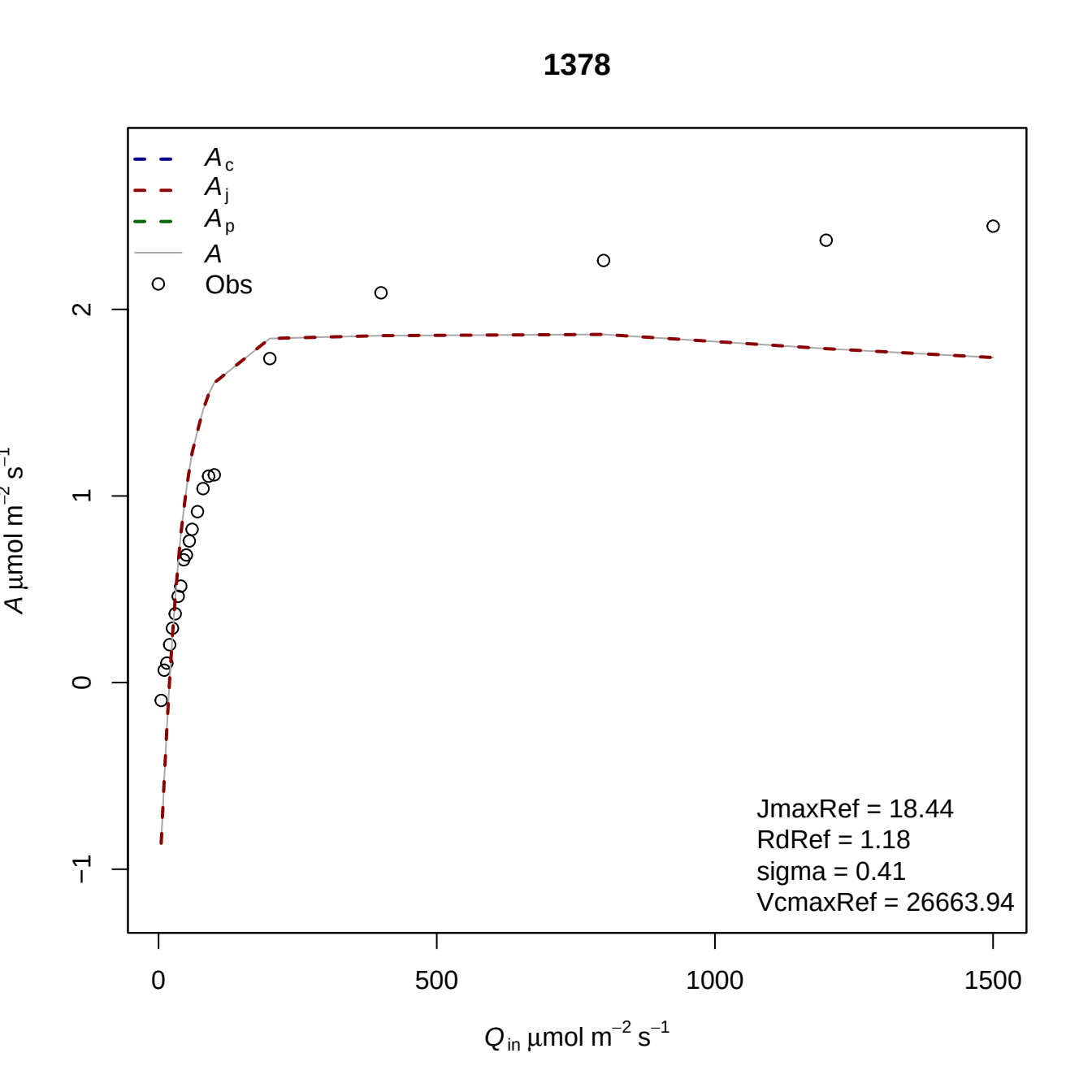

1373
A_c
A_j
A_p
A
Obs
JmaxRef = 19.93
RdRef = 0.68
sigma = 0.22
VcmaxRef = 116.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

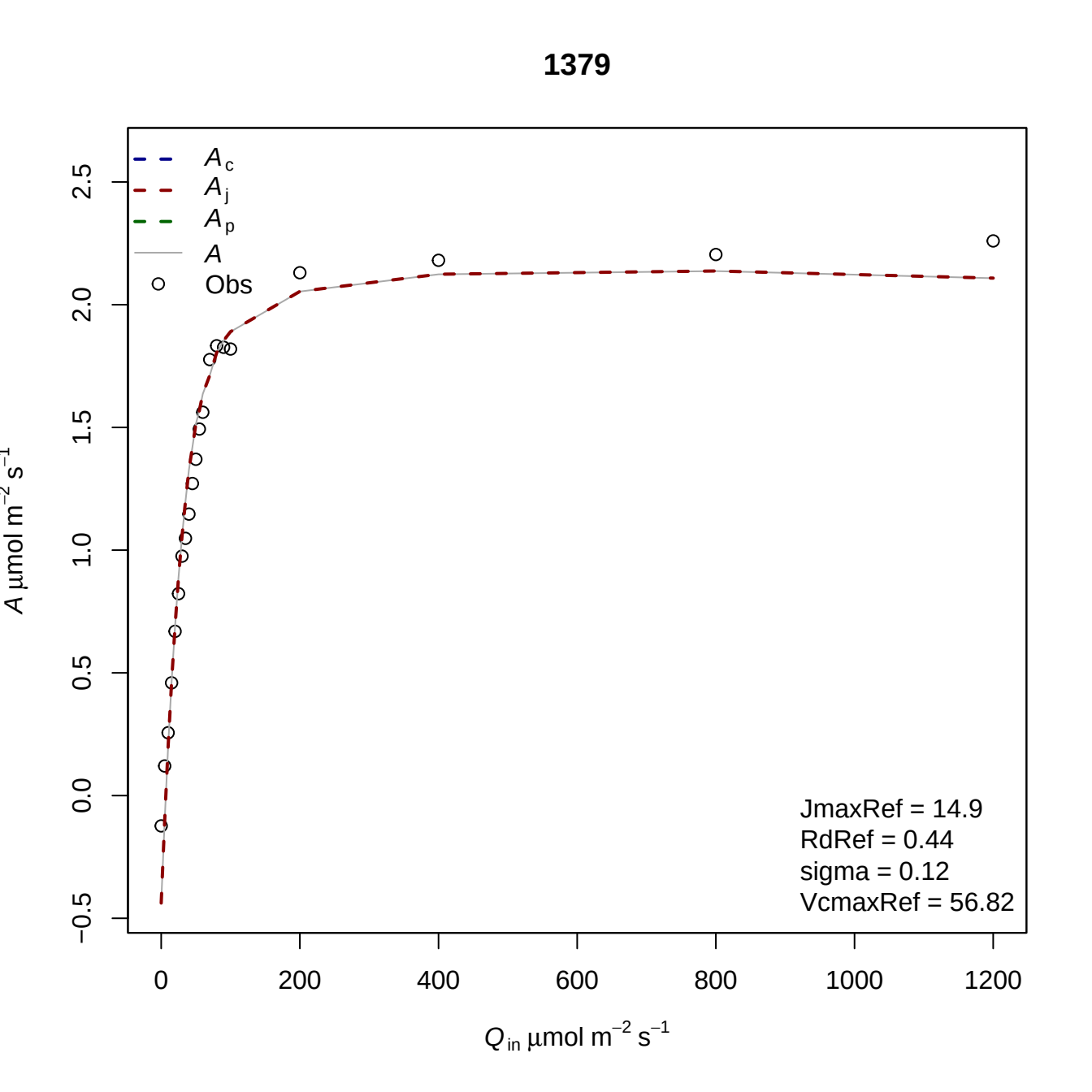

1374
A_c
A_j
A_p
A
Obs
JmaxRef = 25.4
RdRef = 1.66
sigma = 0.42
VcmaxRef = 122.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

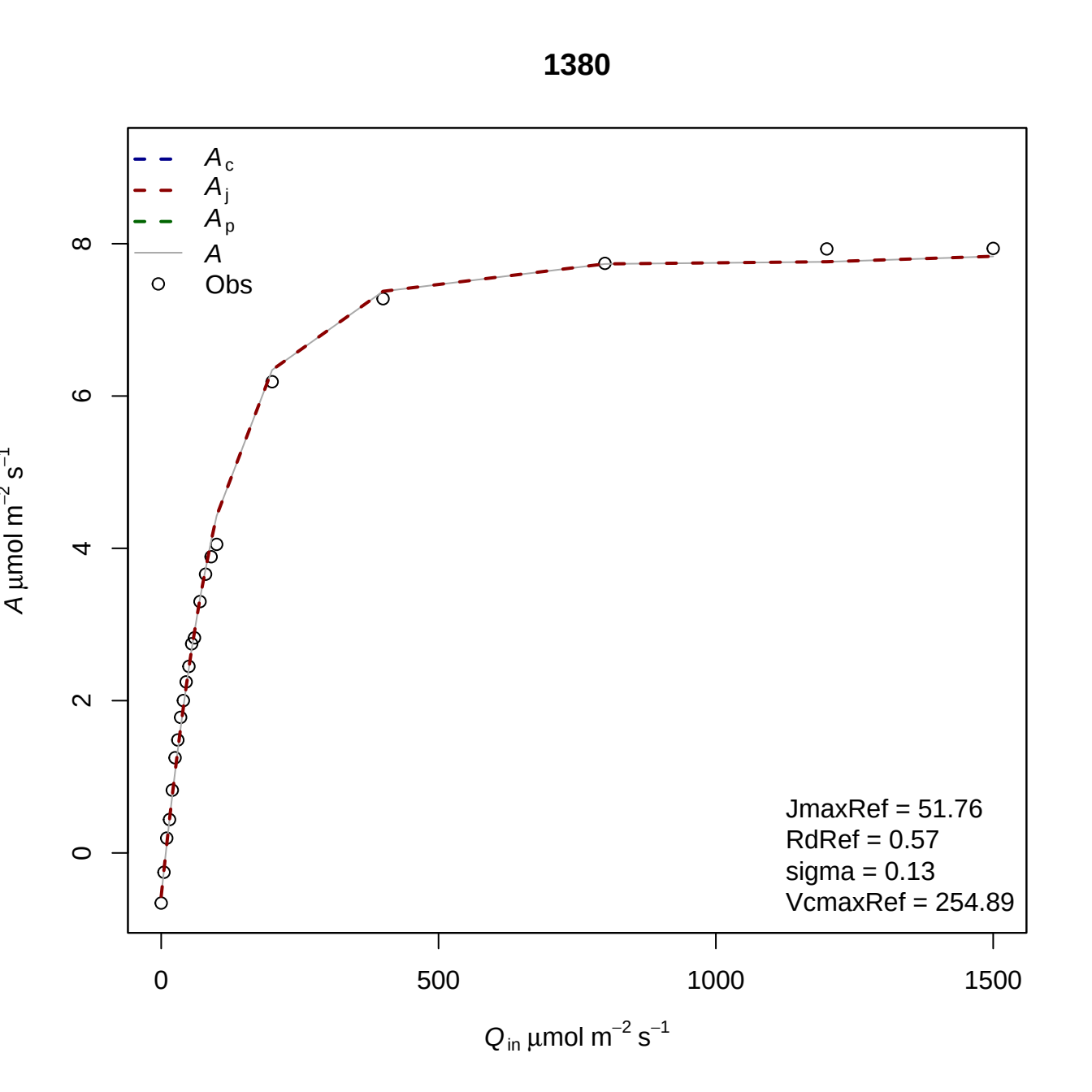

1375
A_c
A_j
A_p
A
Obs
JmaxRef = 27.31
RdRef = 1.54
sigma = 0.3
VcmaxRef = 148.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

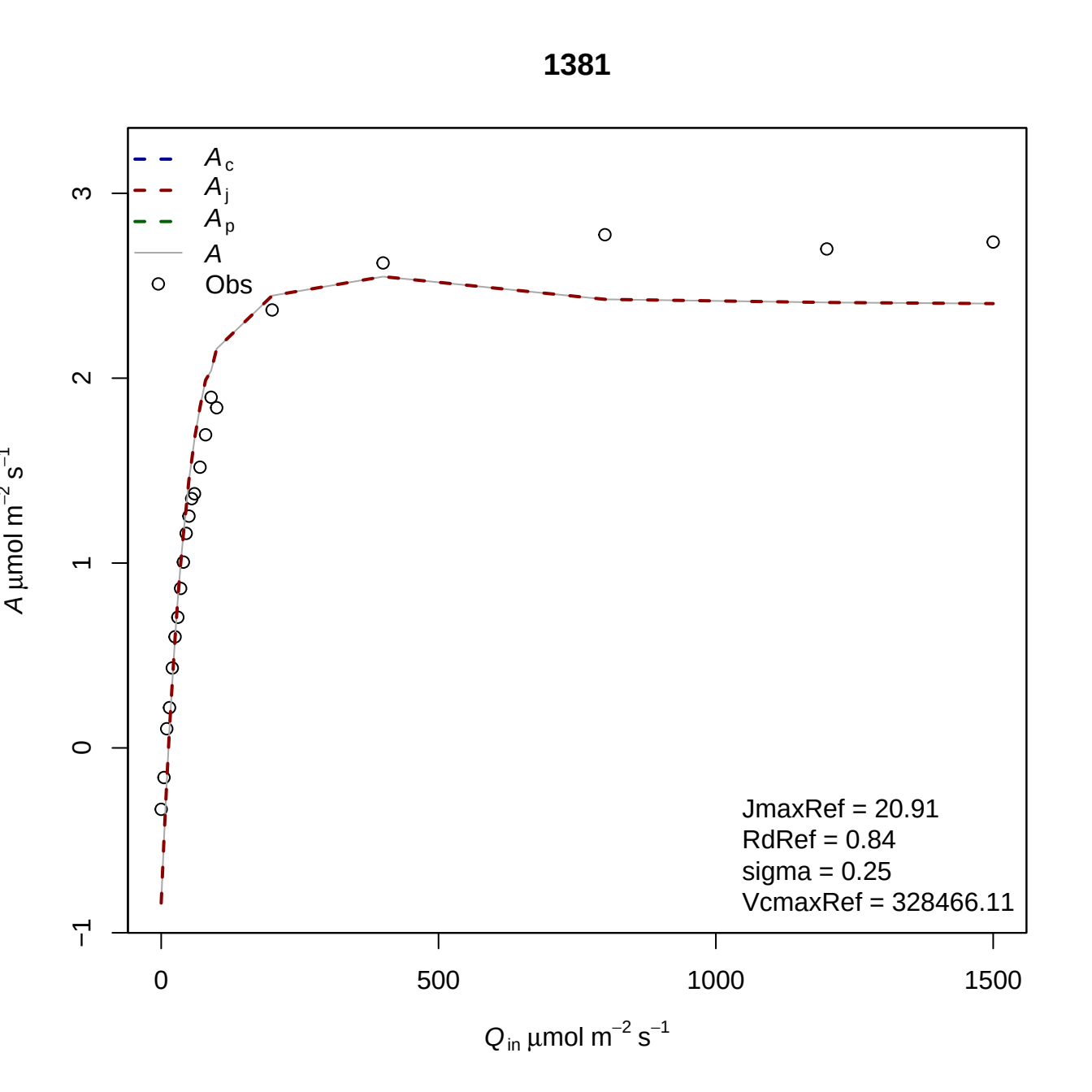

1376
A_c
A_j
A_p
A
Obs
JmaxRef = 58.19
RdRef = 0.17
sigma = 0.45
VcmaxRef = 44.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

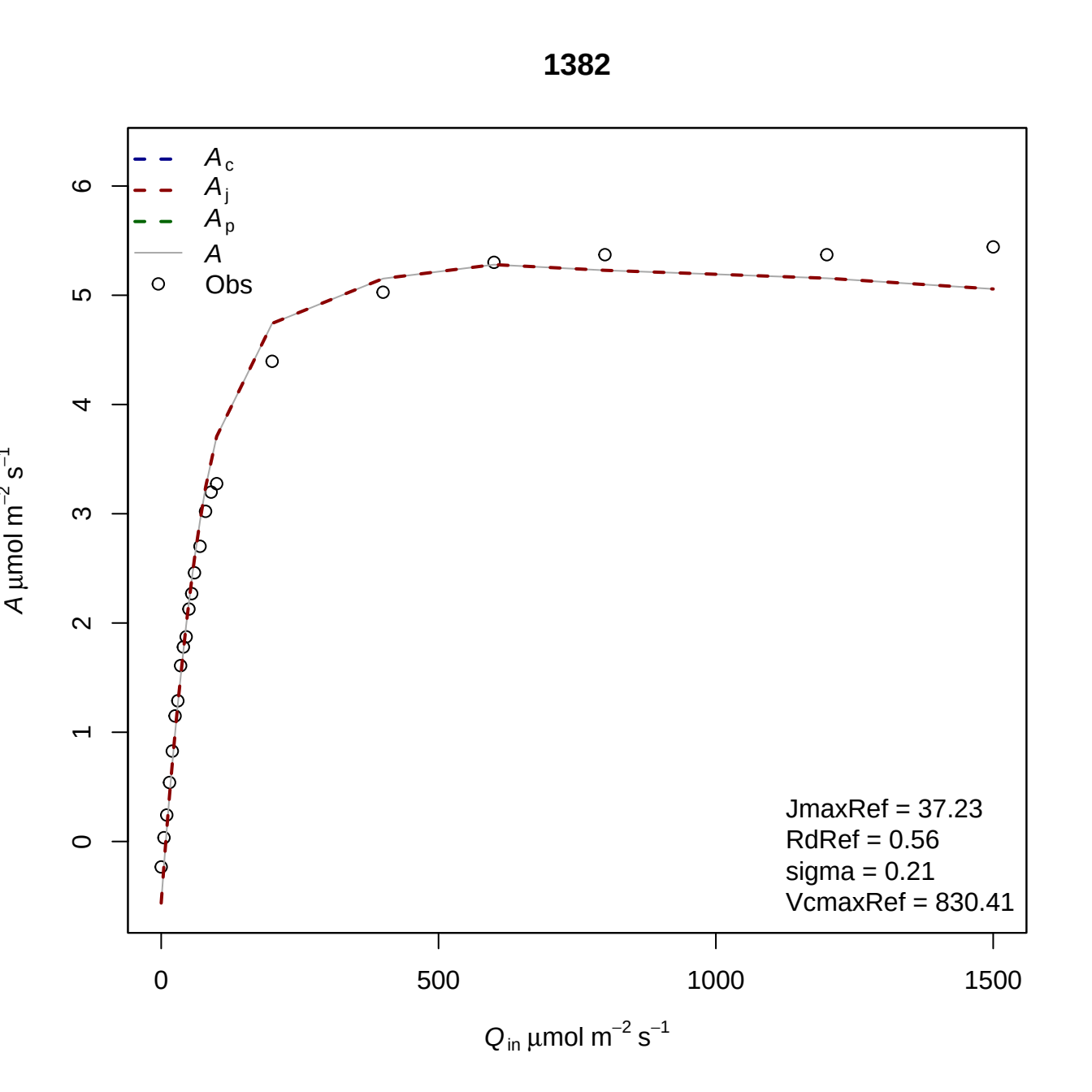

1377
A_c
A_j
A_p
A
Obs
JmaxRef = 31.62
RdRef = 0.93
sigma = 0.27
VcmaxRef = 91.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

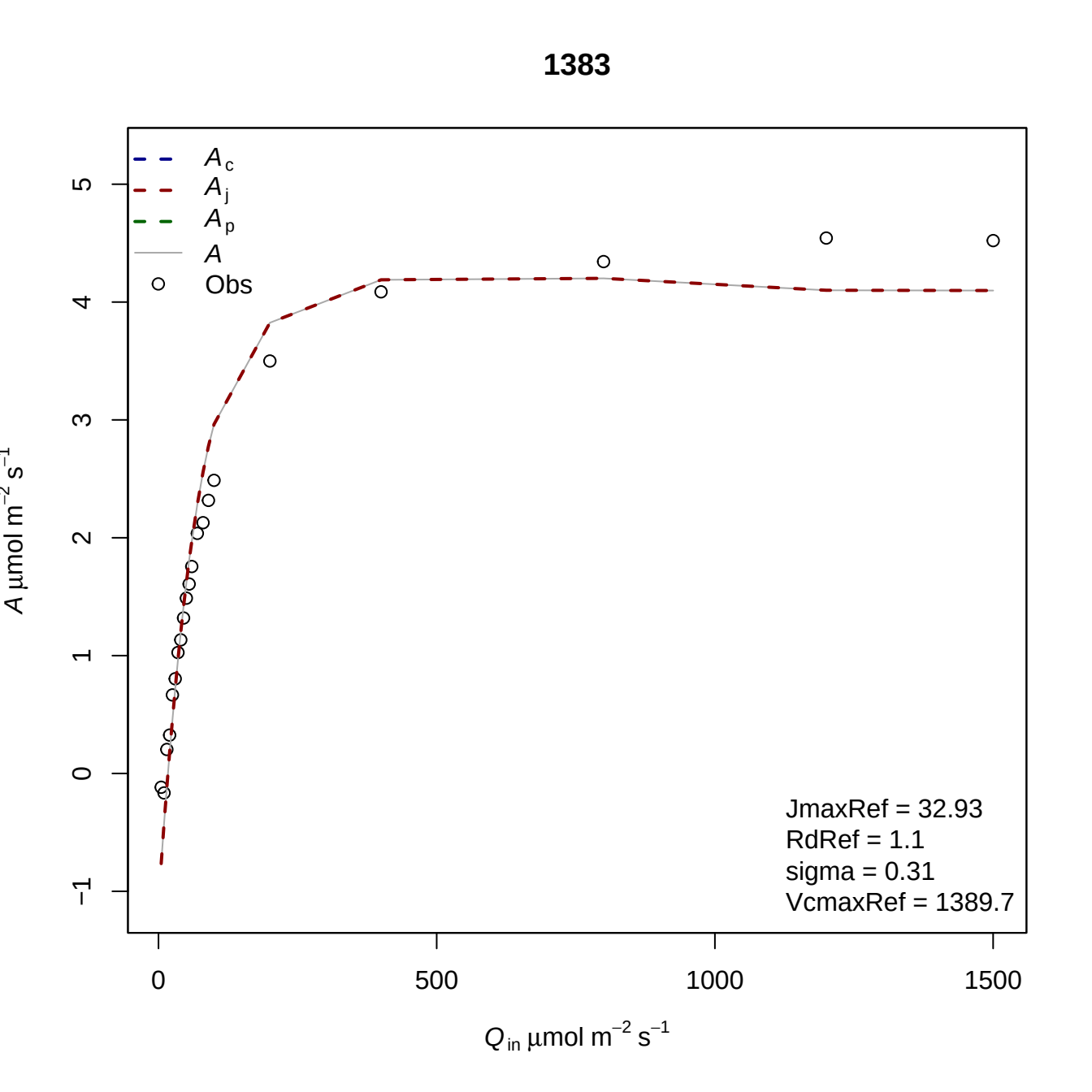

1378
A_c
A_j
A_p
A
Obs
JmaxRef = 18.44
RdRef = 1.18
sigma = 0.41
VcmaxRef = 26663.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

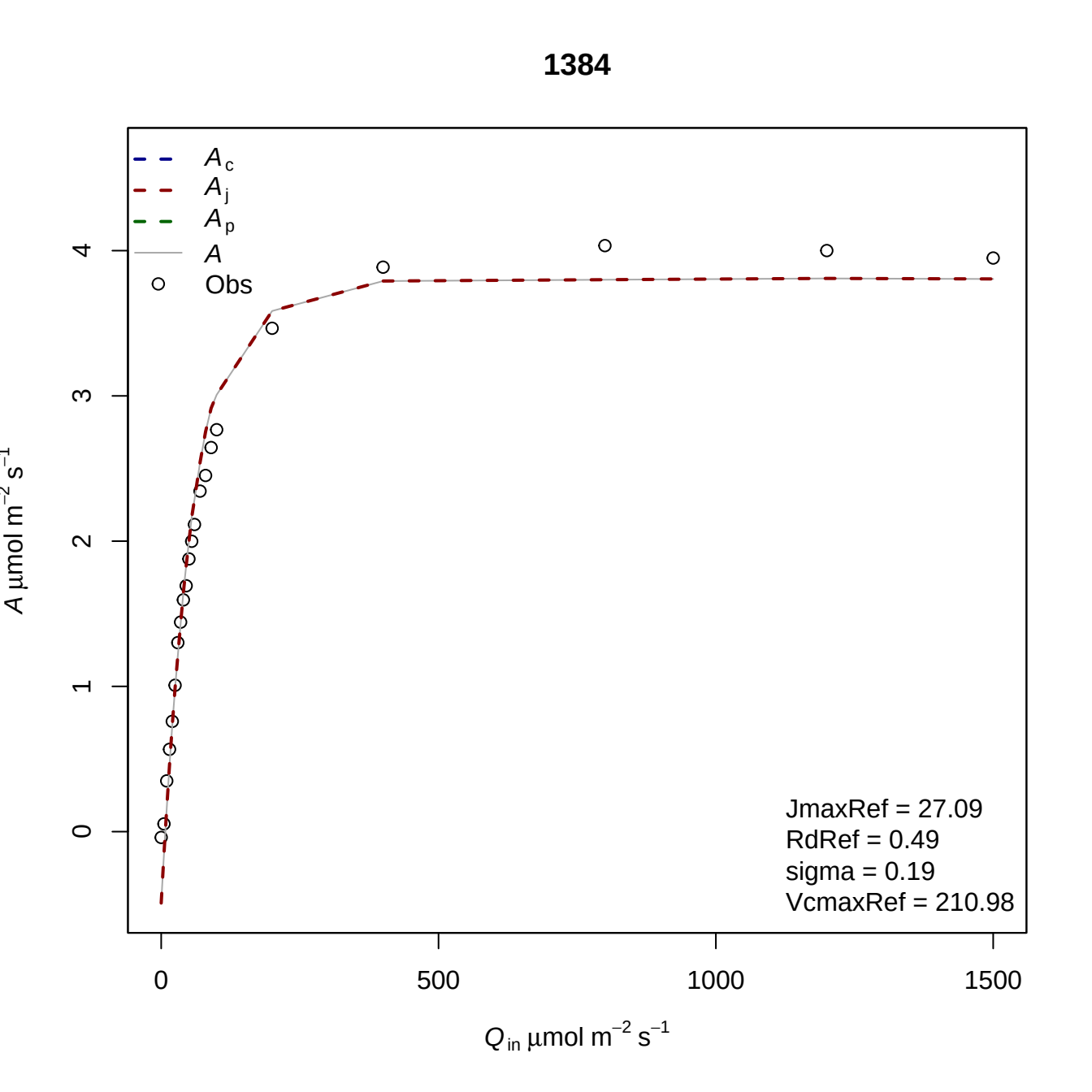

1379
A_c
A_j
A_p
A
Obs
JmaxRef = 14.9
RdRef = 0.44
sigma = 0.12
VcmaxRef = 56.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
200
400
600
800
1000
1200

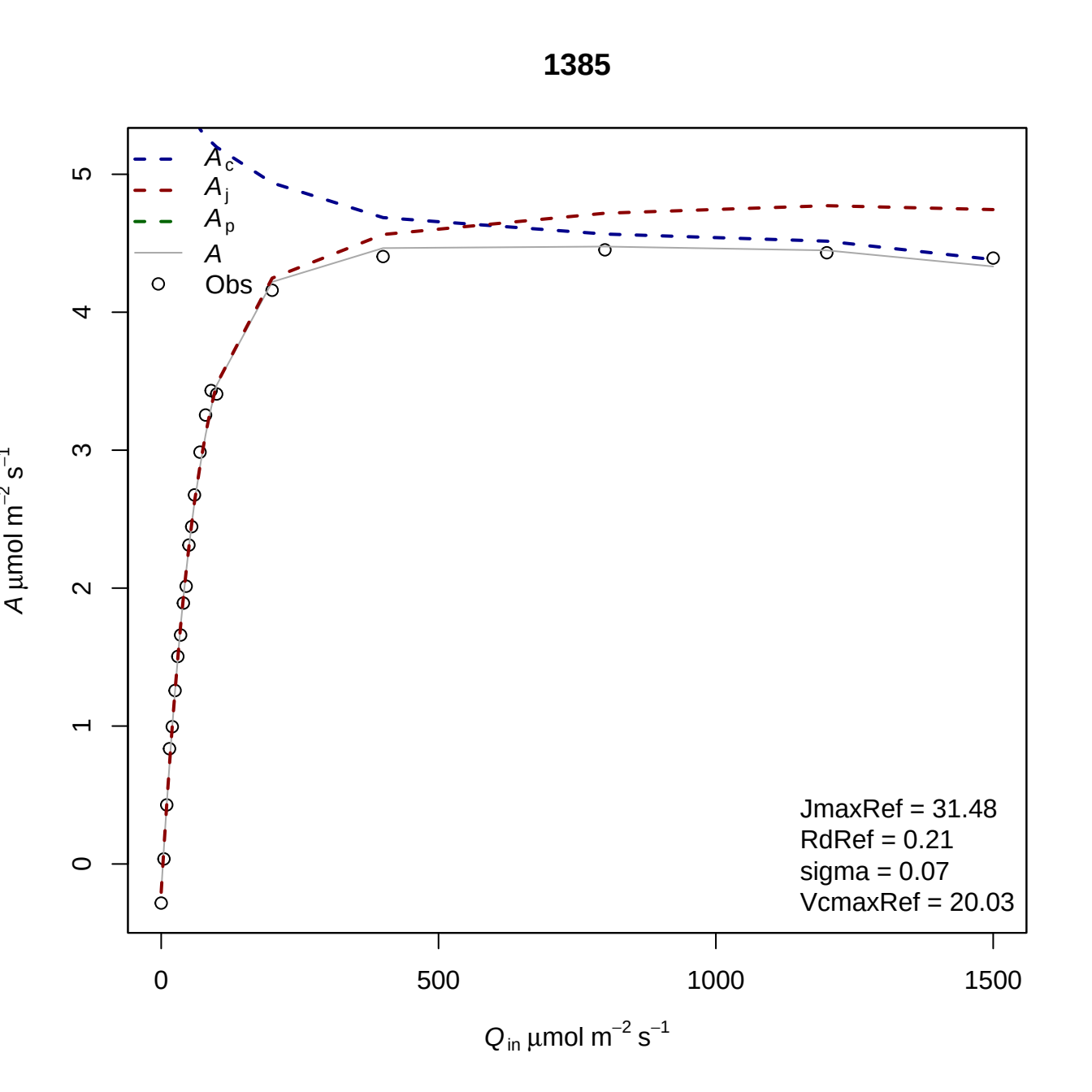

1380
A_c
A_j
A_p
A
Obs
JmaxRef = 51.76
RdRef = 0.57
sigma = 0.13
VcmaxRef = 254.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

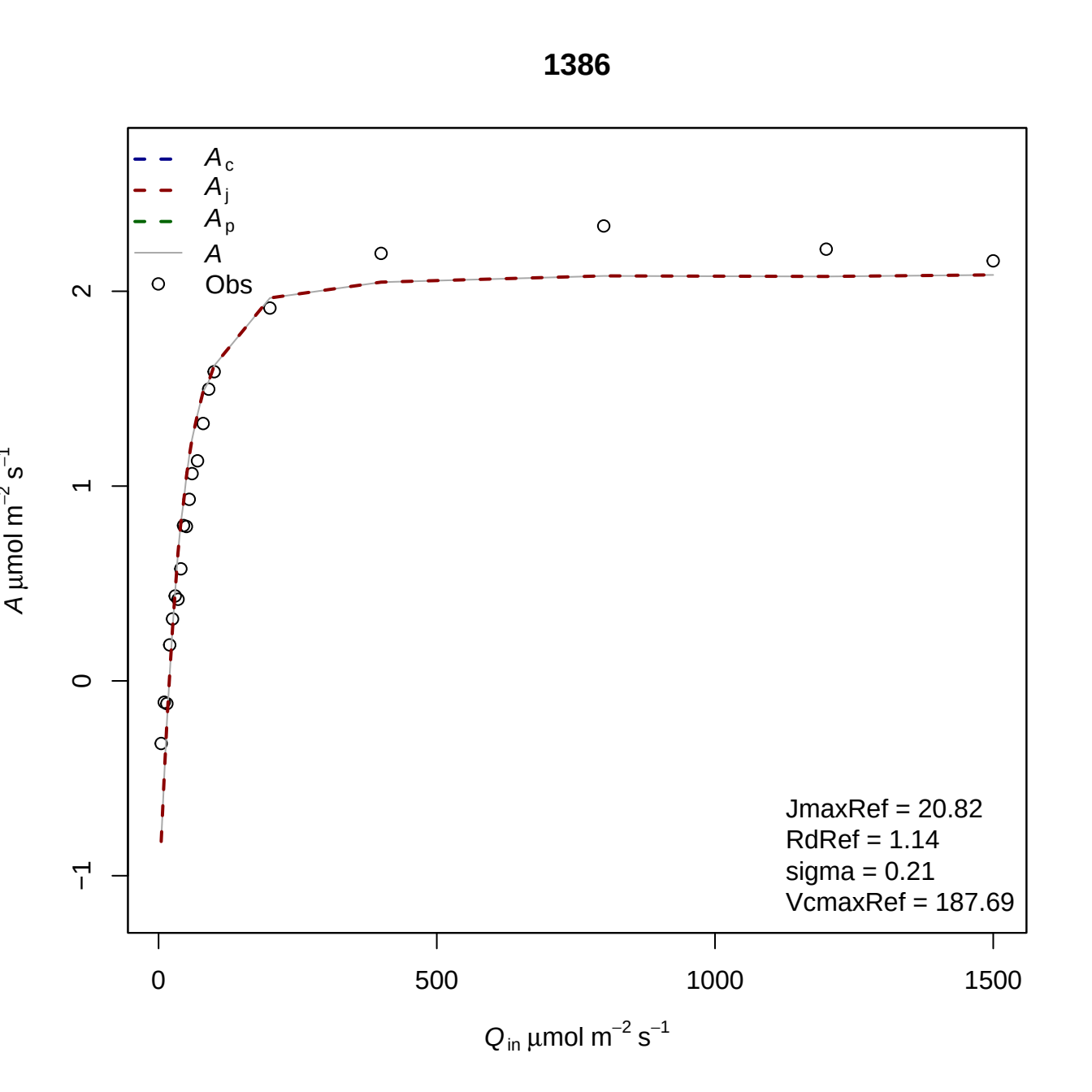

1381
A_c
A_j
A_p
A
Obs
JmaxRef = 20.91
RdRef = 0.84
sigma = 0.25
VcmaxRef = 328466.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

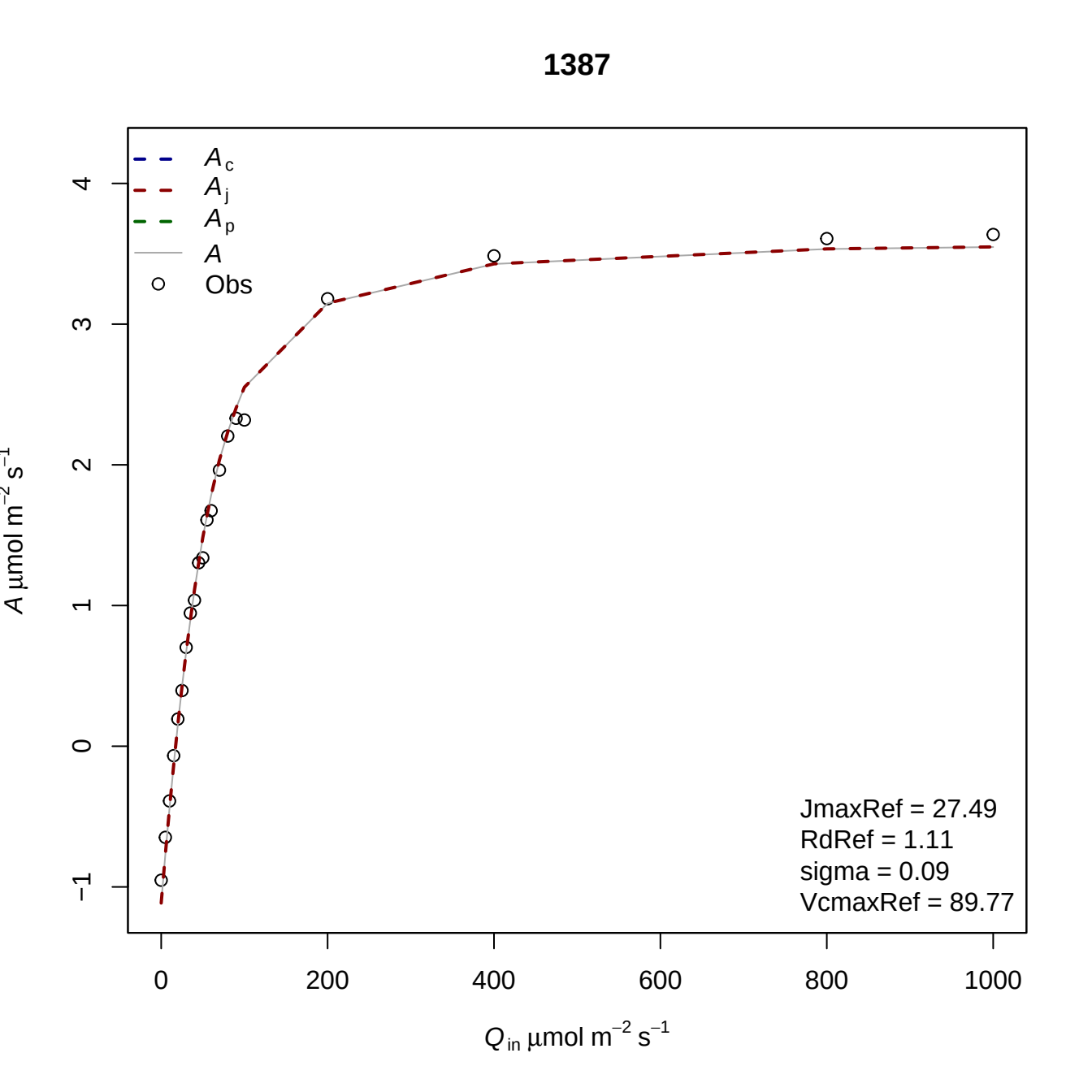

1382
A_c
A_j
A_p
A
Obs
JmaxRef = 37.23
RdRef = 0.56
sigma = 0.21
VcmaxRef = 830.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

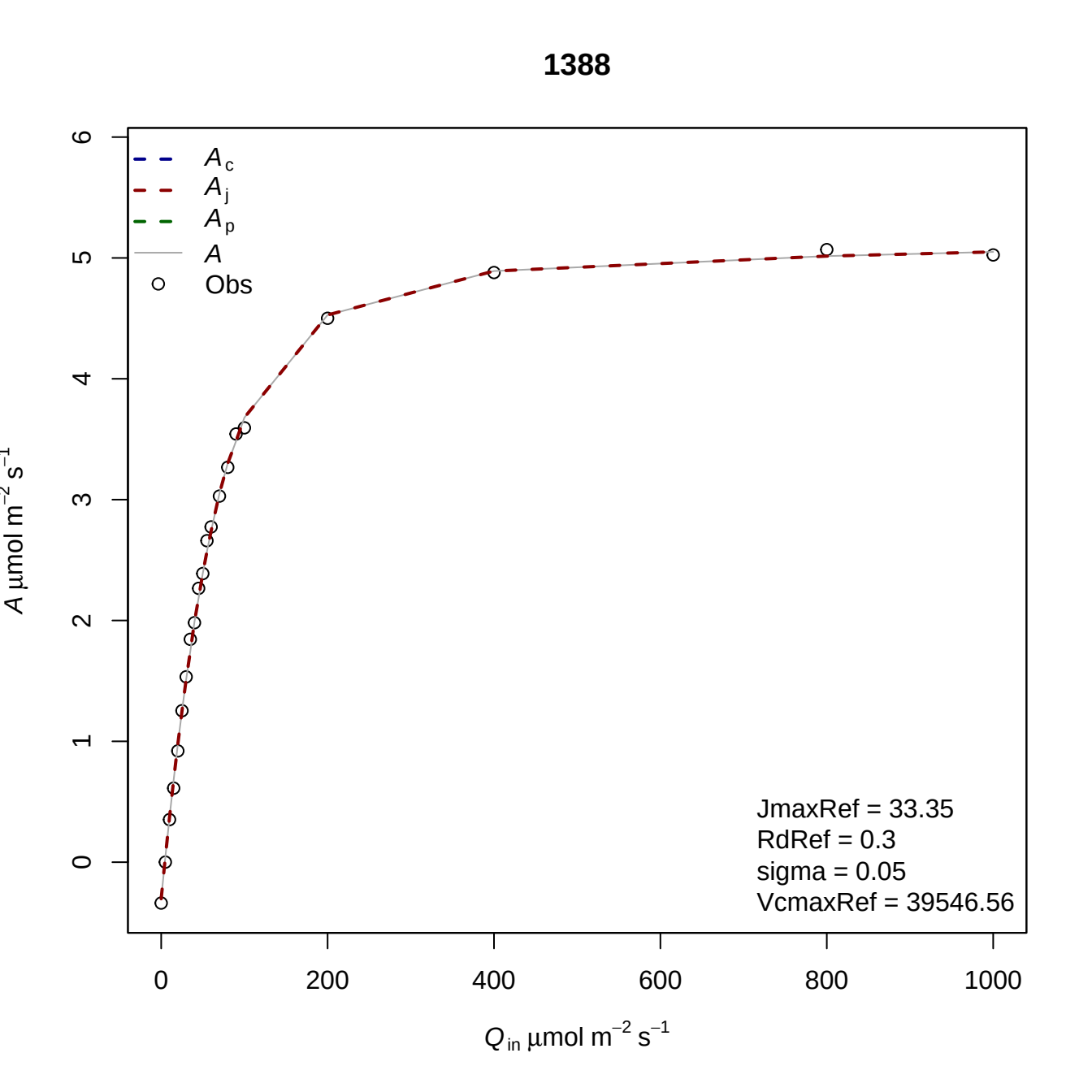

1383
A_c
A_j
A_p
A
Obs
JmaxRef = 32.93
RdRef = 1.1
sigma = 0.31
VcmaxRef = 1389.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

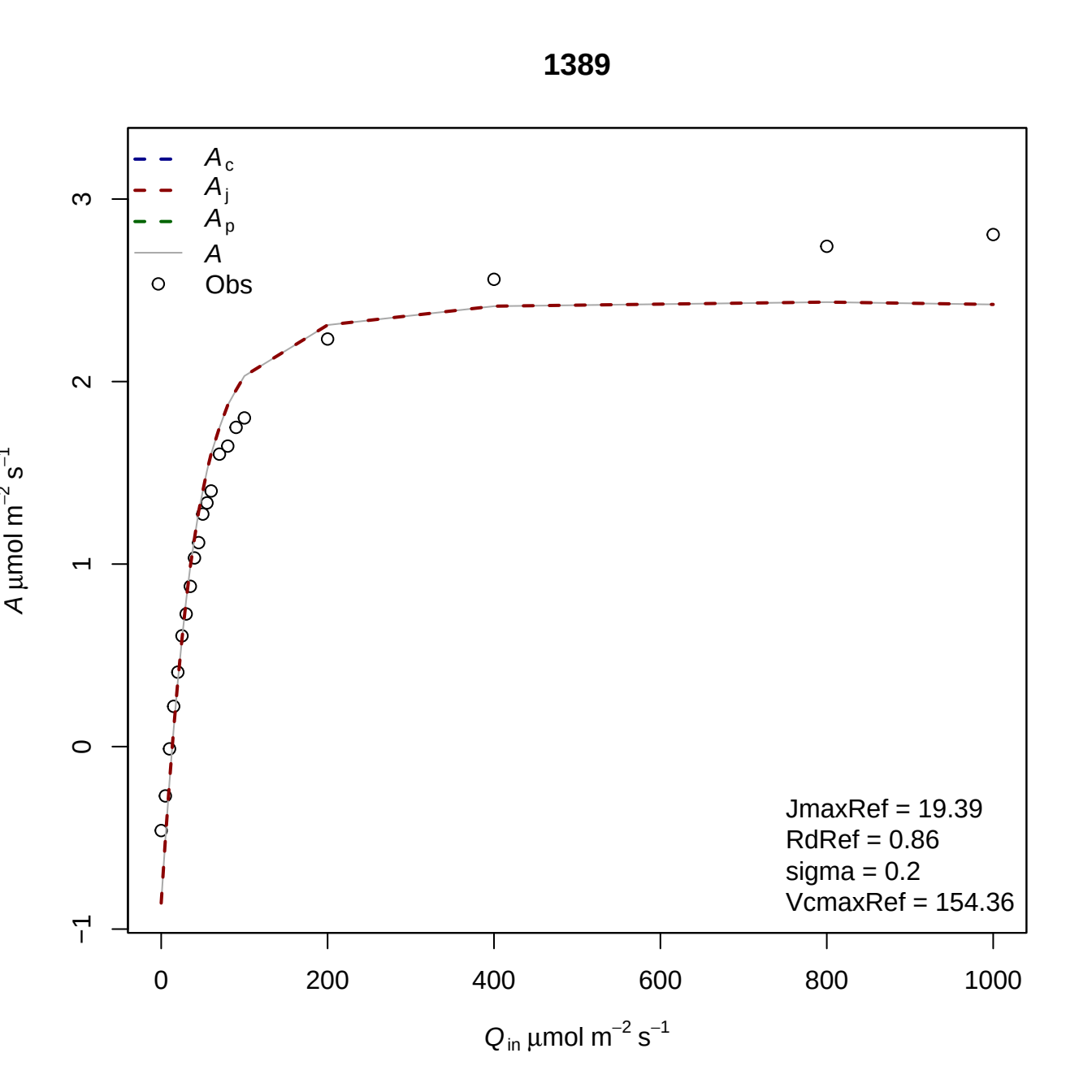

1384
A_c
A_j
A_p
A
Obs
JmaxRef = 27.09
RdRef = 0.49
sigma = 0.19
VcmaxRef = 210.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

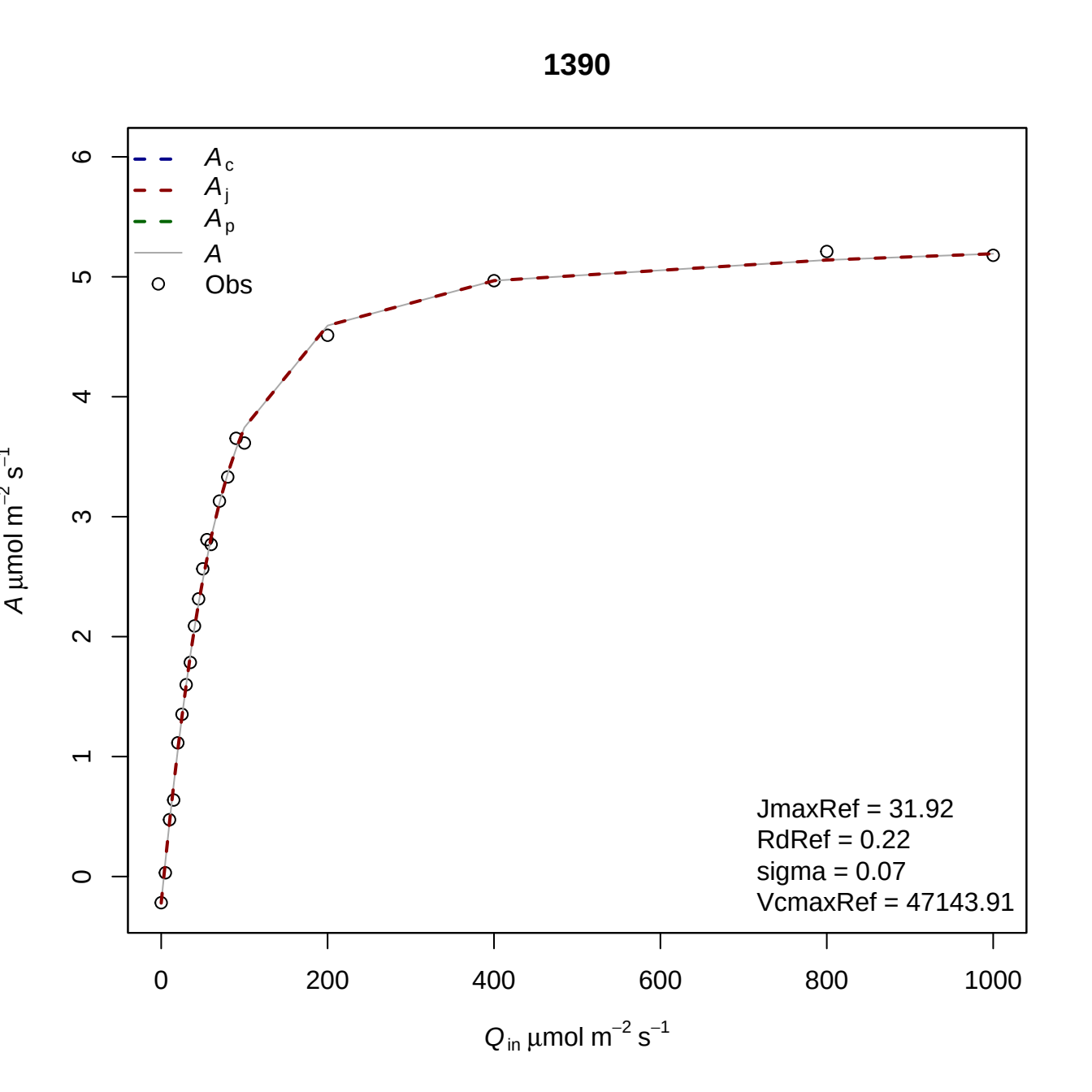

1385
A_c
A_j
A_p
A
Obs
JmaxRef = 31.48
RdRef = 0.21
sigma = 0.07
VcmaxRef = 20.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

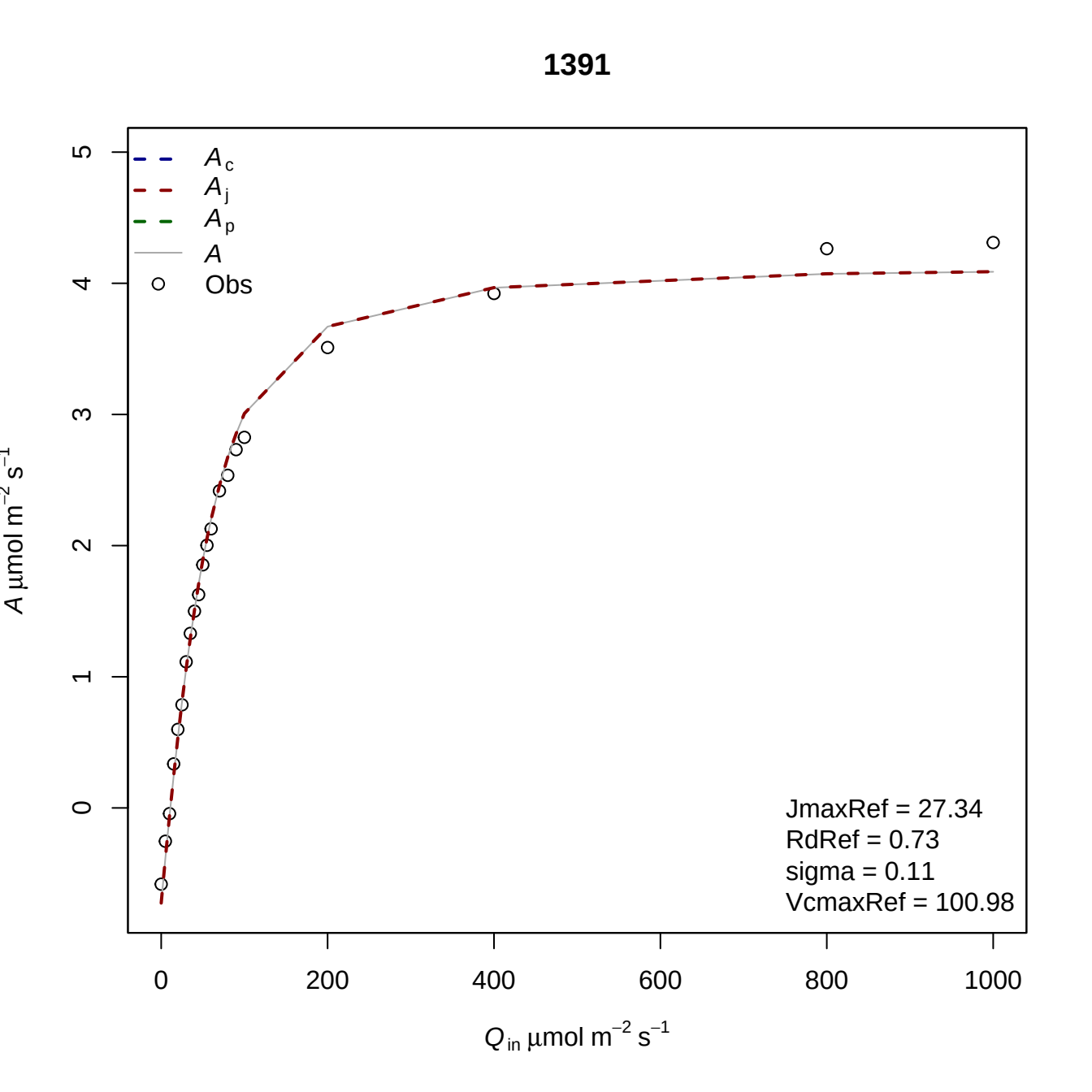

1386
A_c
A_j
A_p
A
Obs
JmaxRef = 20.82
RdRef = 1.14
sigma = 0.21
VcmaxRef = 187.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

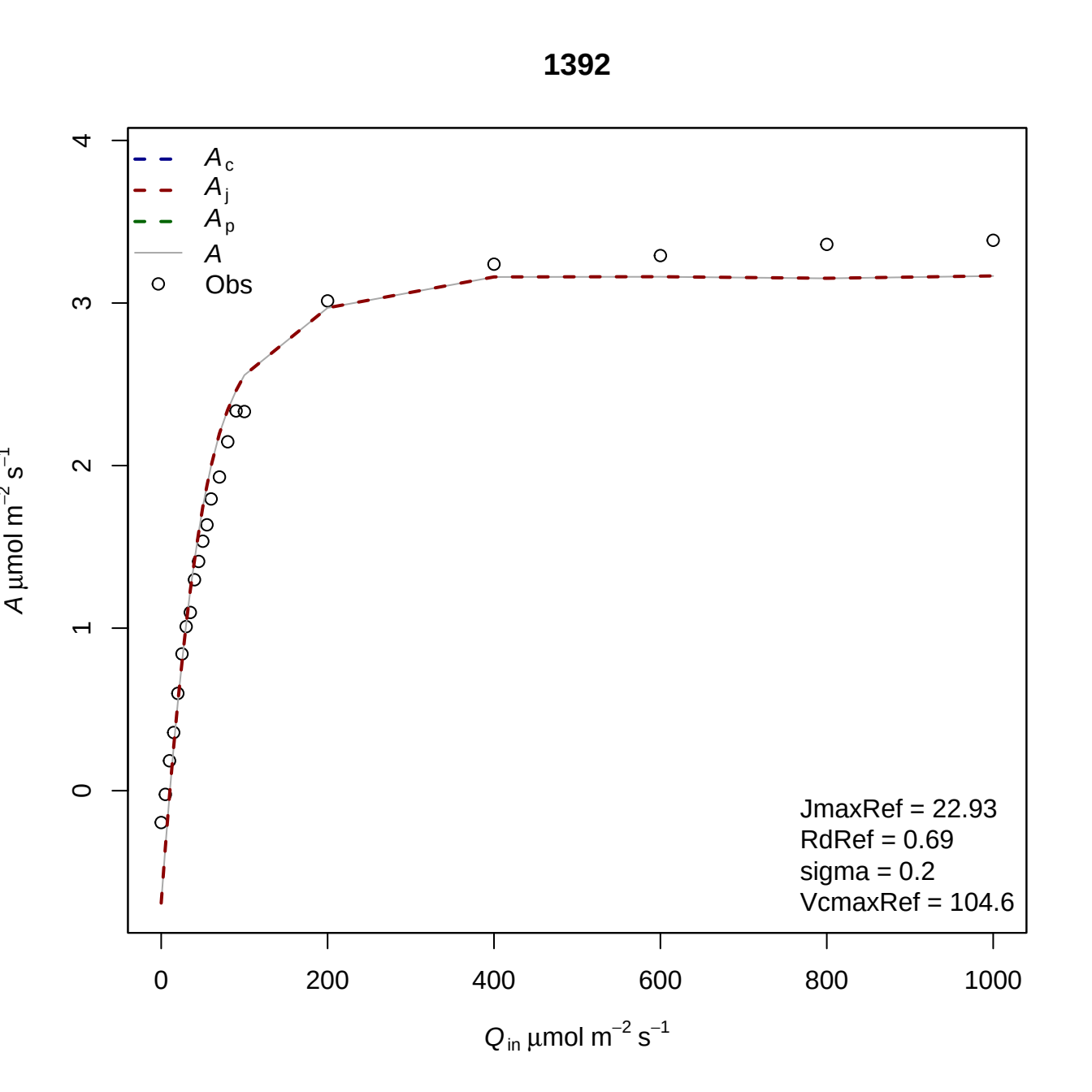

1387
A_c
A_j
A_p
A
Obs
JmaxRef = 27.49
RdRef = 1.11
sigma = 0.09
VcmaxRef = 89.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

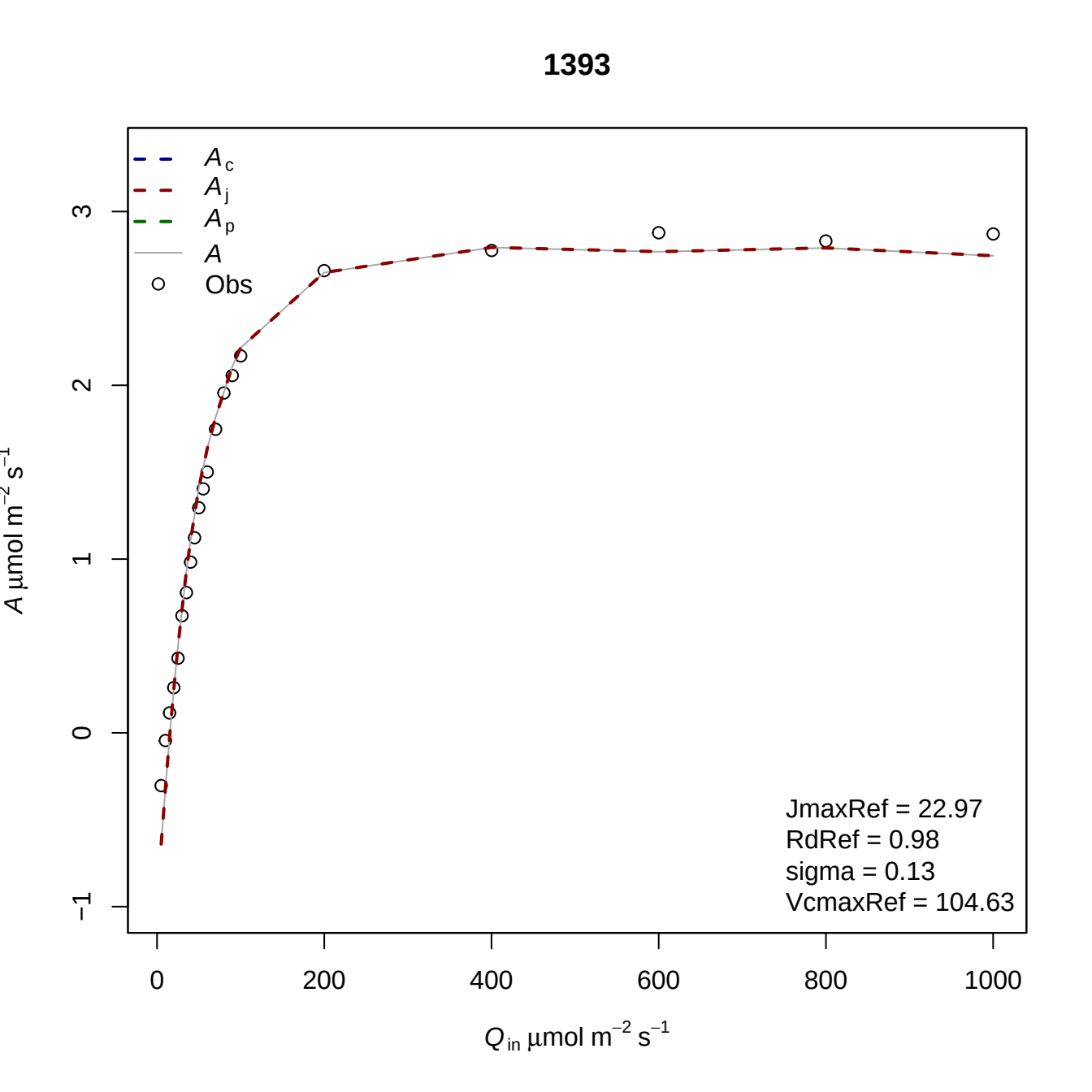

1388
A_c
A_j
A_p
A
Obs
JmaxRef = 33.35
RdRef = 0.3
sigma = 0.05
VcmaxRef = 39546.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

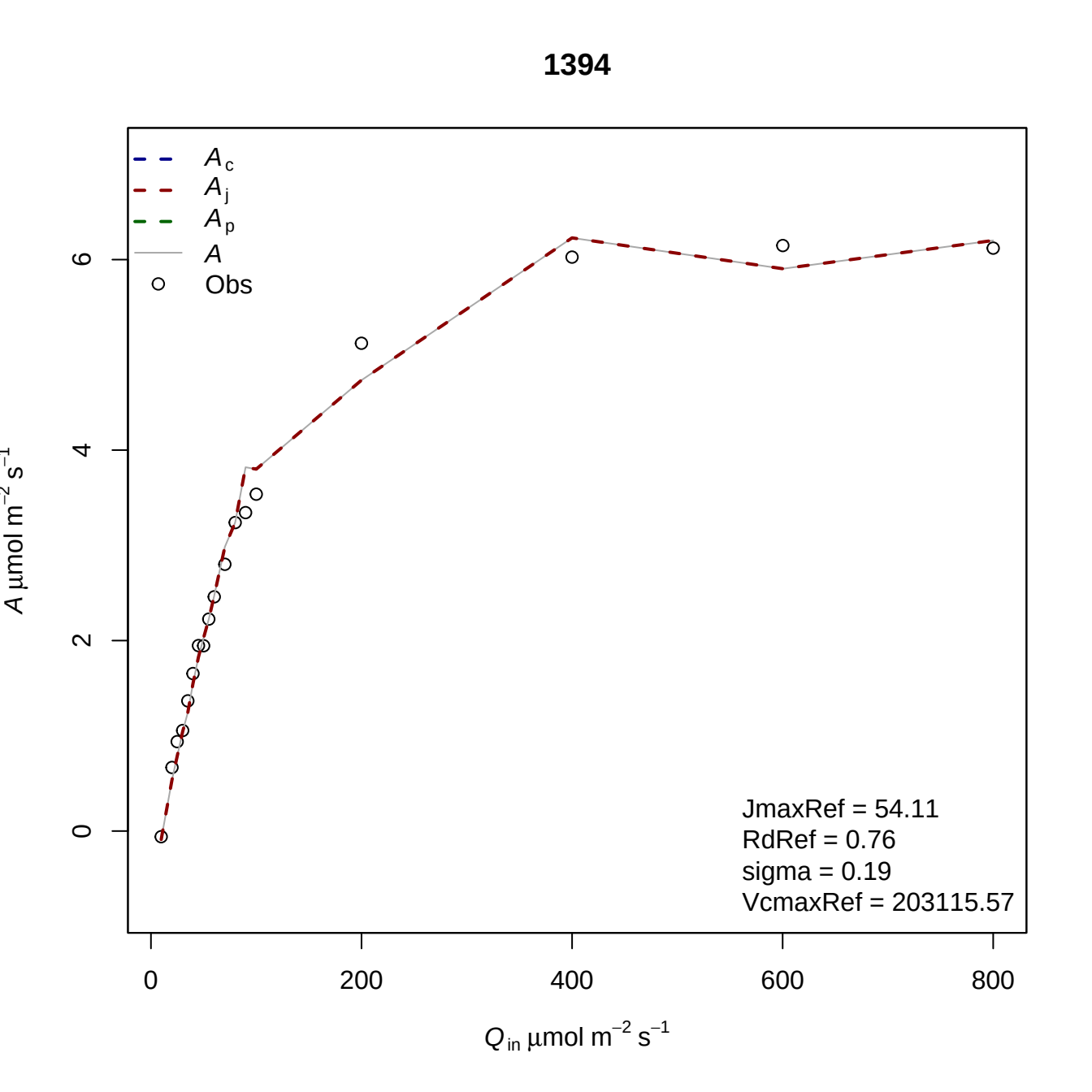

1389
A_c
A_j
A_p
A
Obs
JmaxRef = 19.39
RdRef = 0.86
sigma = 0.2
VcmaxRef = 154.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

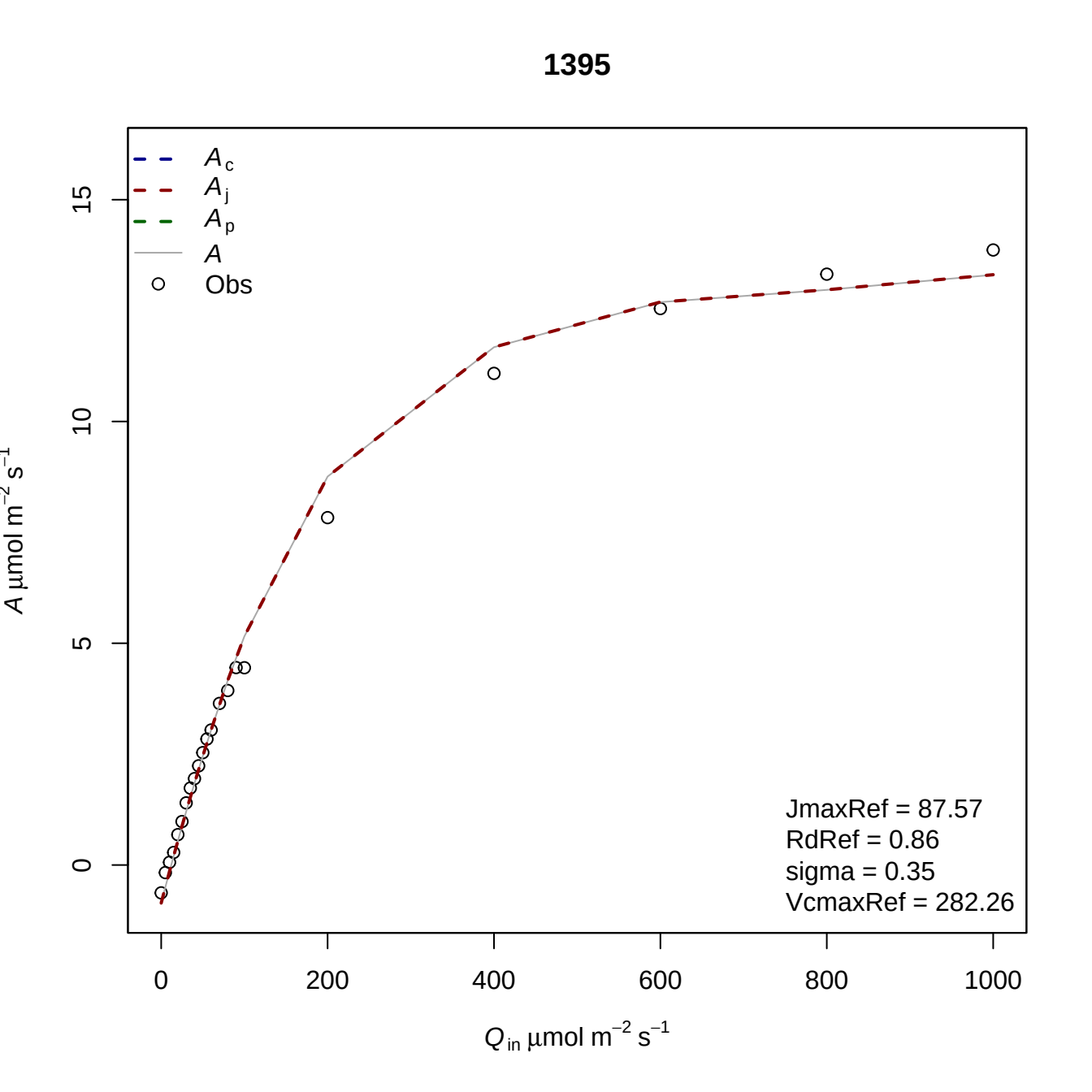

1390
A_c
A_j
A_p
A
Obs
JmaxRef = 31.92
RdRef = 0.22
sigma = 0.07
VcmaxRef = 47143.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

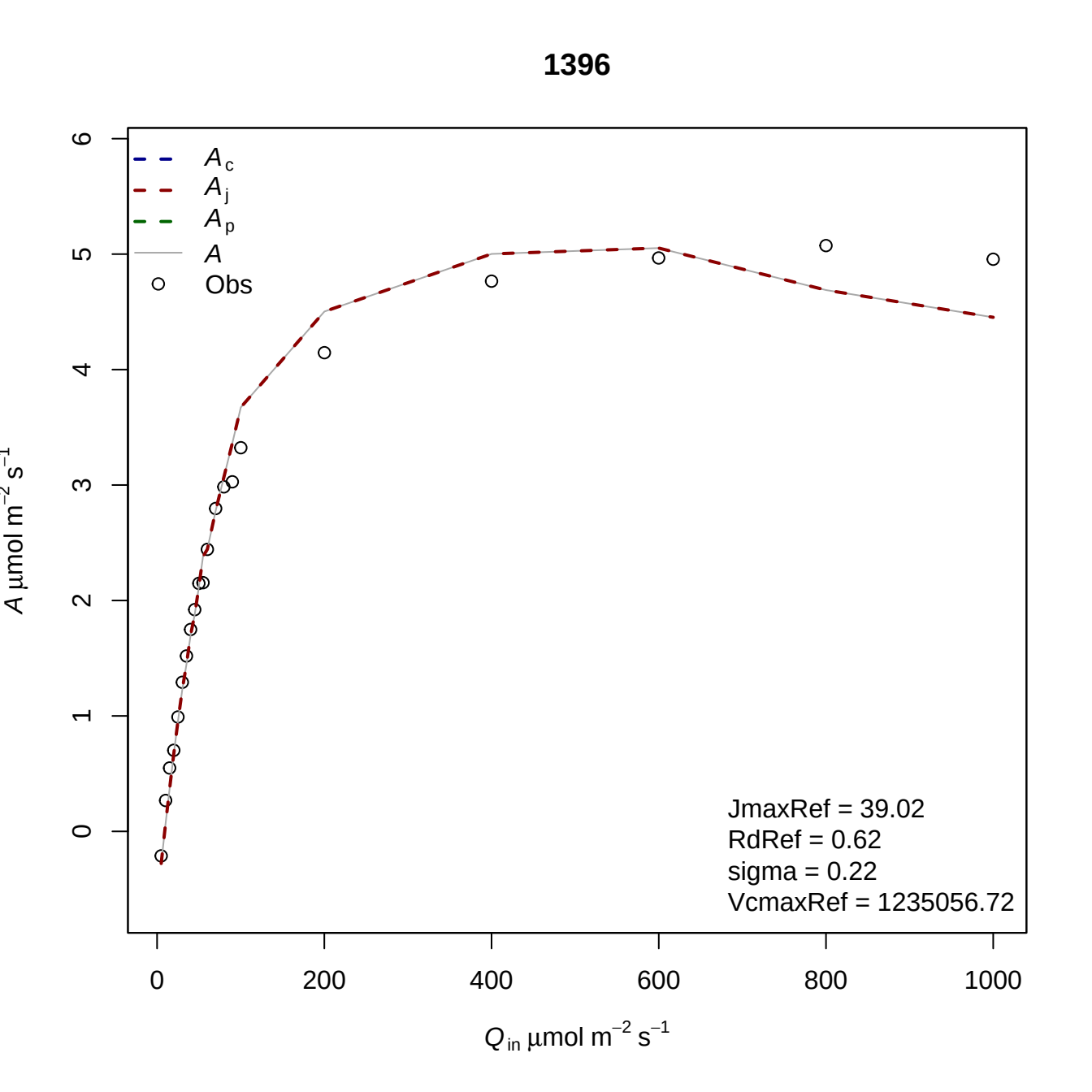

1391
A_c
A_j
A_p
A
Obs
JmaxRef = 27.34
RdRef = 0.73
sigma = 0.11
VcmaxRef = 100.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000

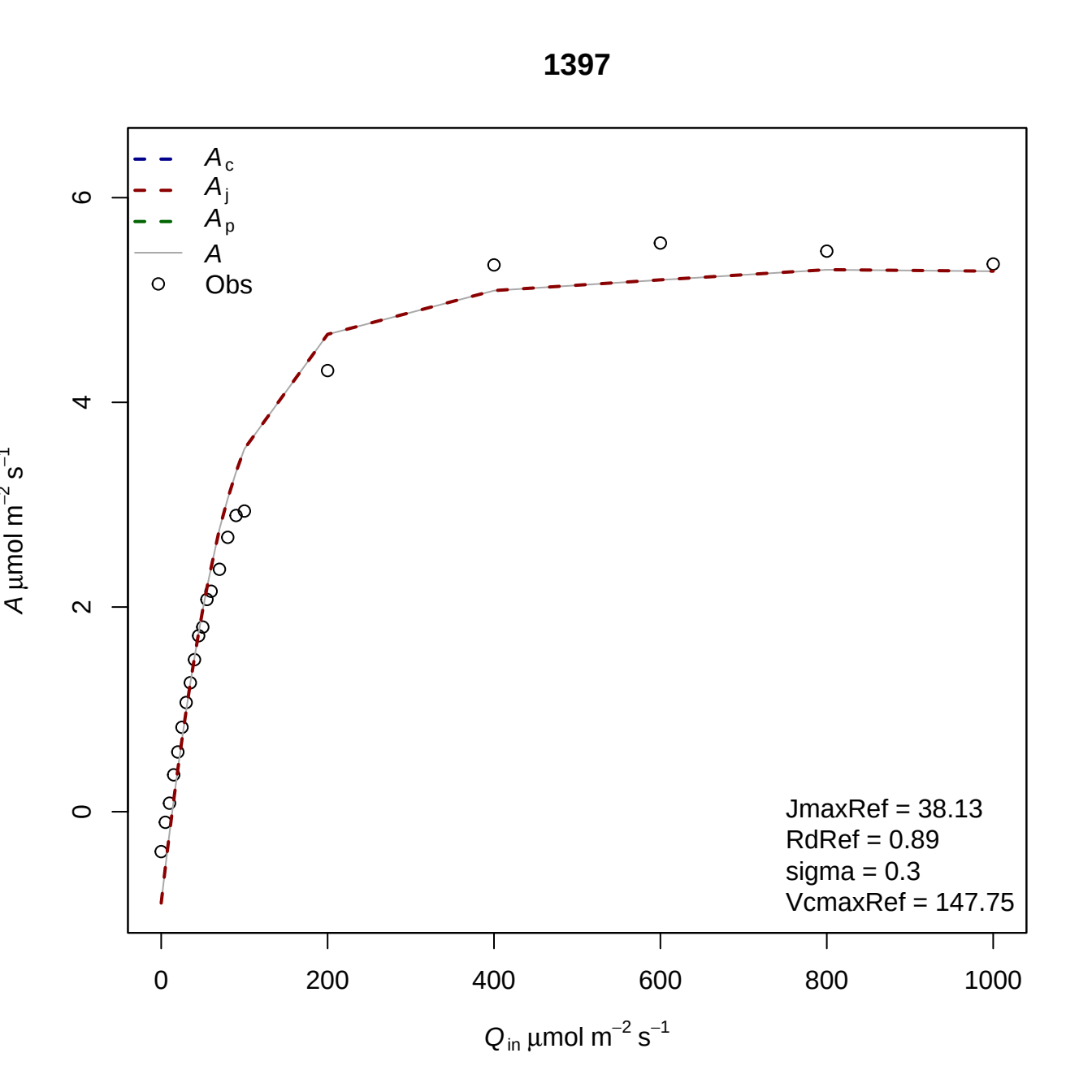

1392
A_c
A_j
A_p
A
Obs
JmaxRef = 22.93
RdRef = 0.69
sigma = 0.2
VcmaxRef = 104.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
200
400
600
800
1000

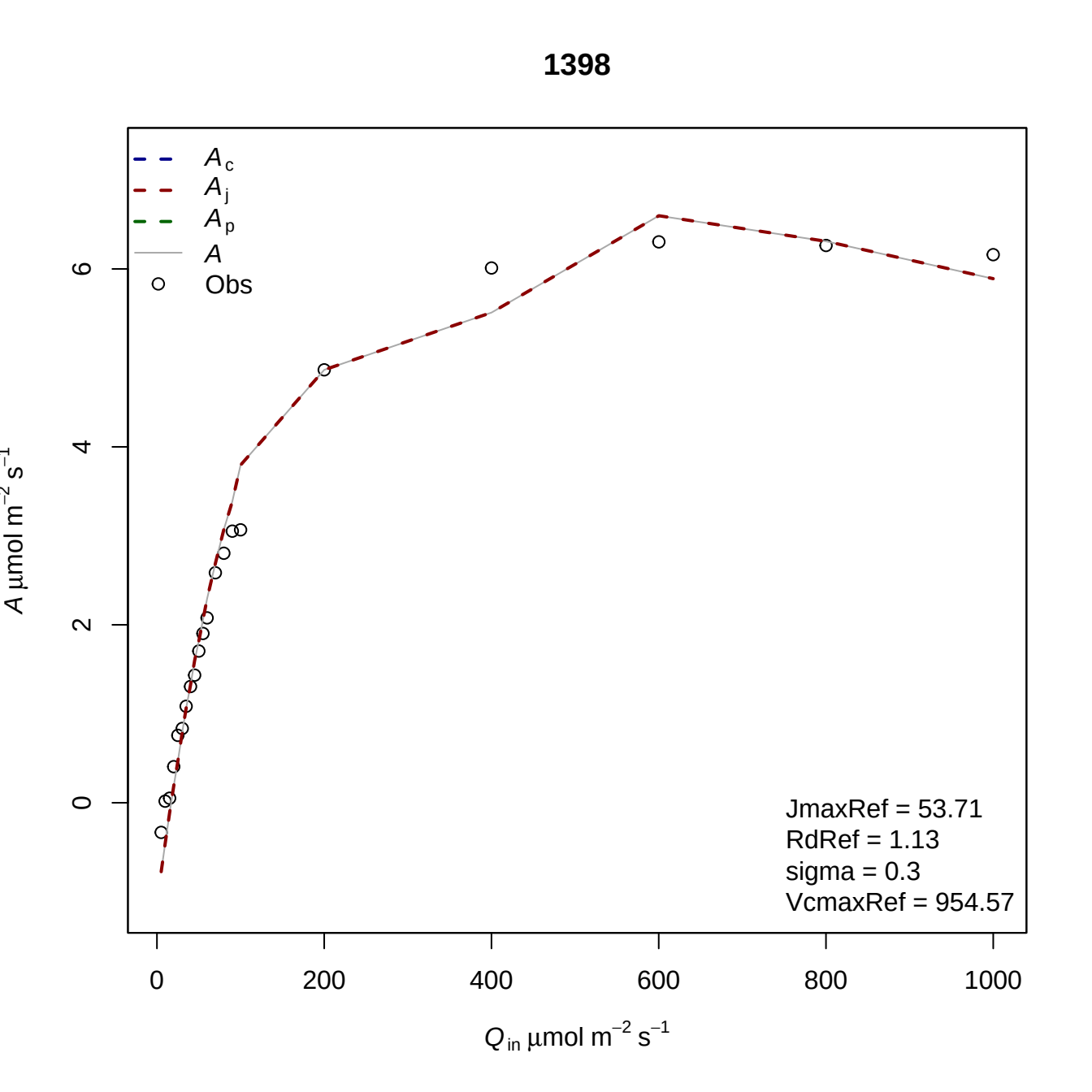

1393
A_c
A_j
A_p
A
Obs
JmaxRef = 22.97
RdRef = 0.98
sigma = 0.13
VcmaxRef = 104.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

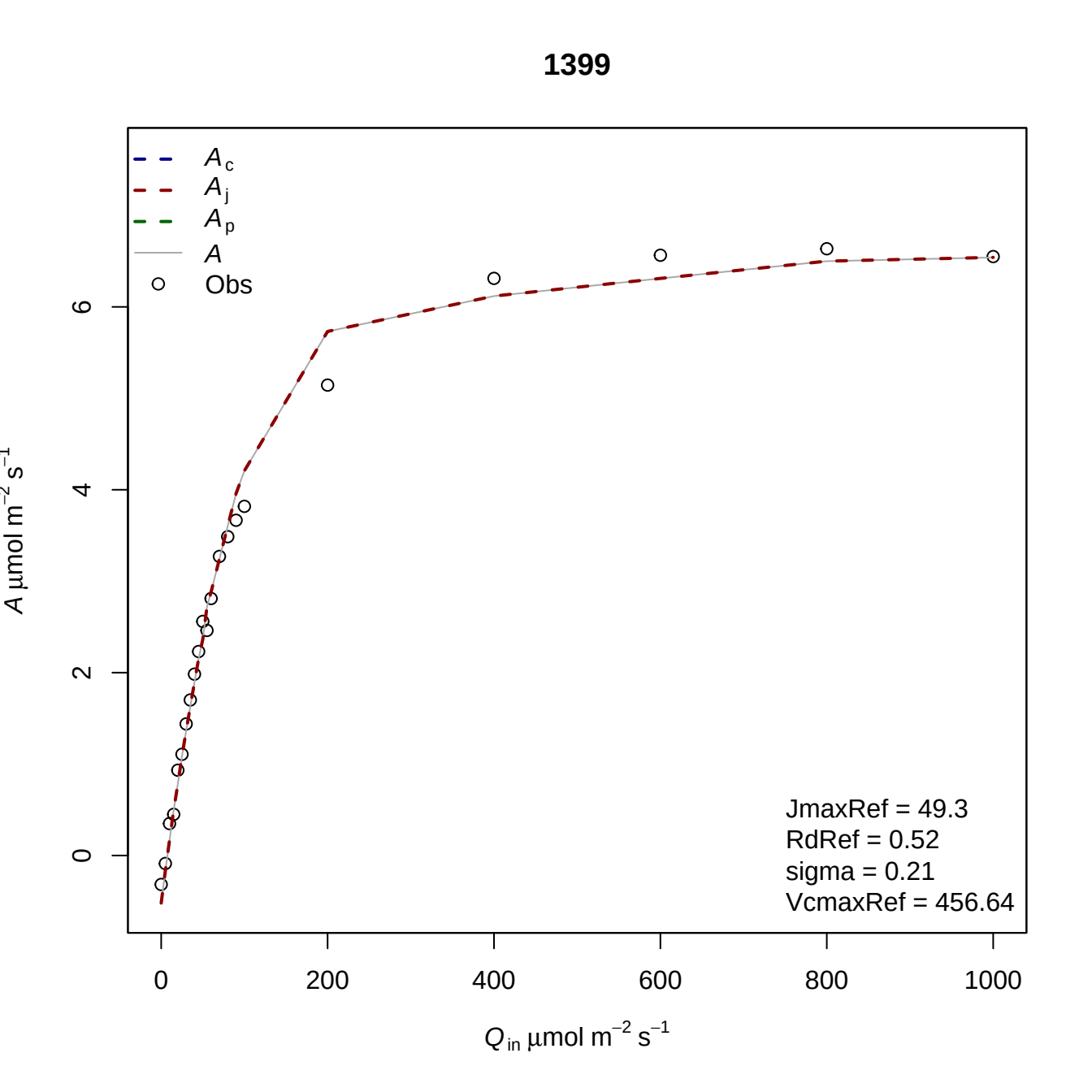

1394
A_c
A_j
A_p
A
Obs
JmaxRef = 54.11
RdRef = 0.76
sigma = 0.19
VcmaxRef = 203115.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800

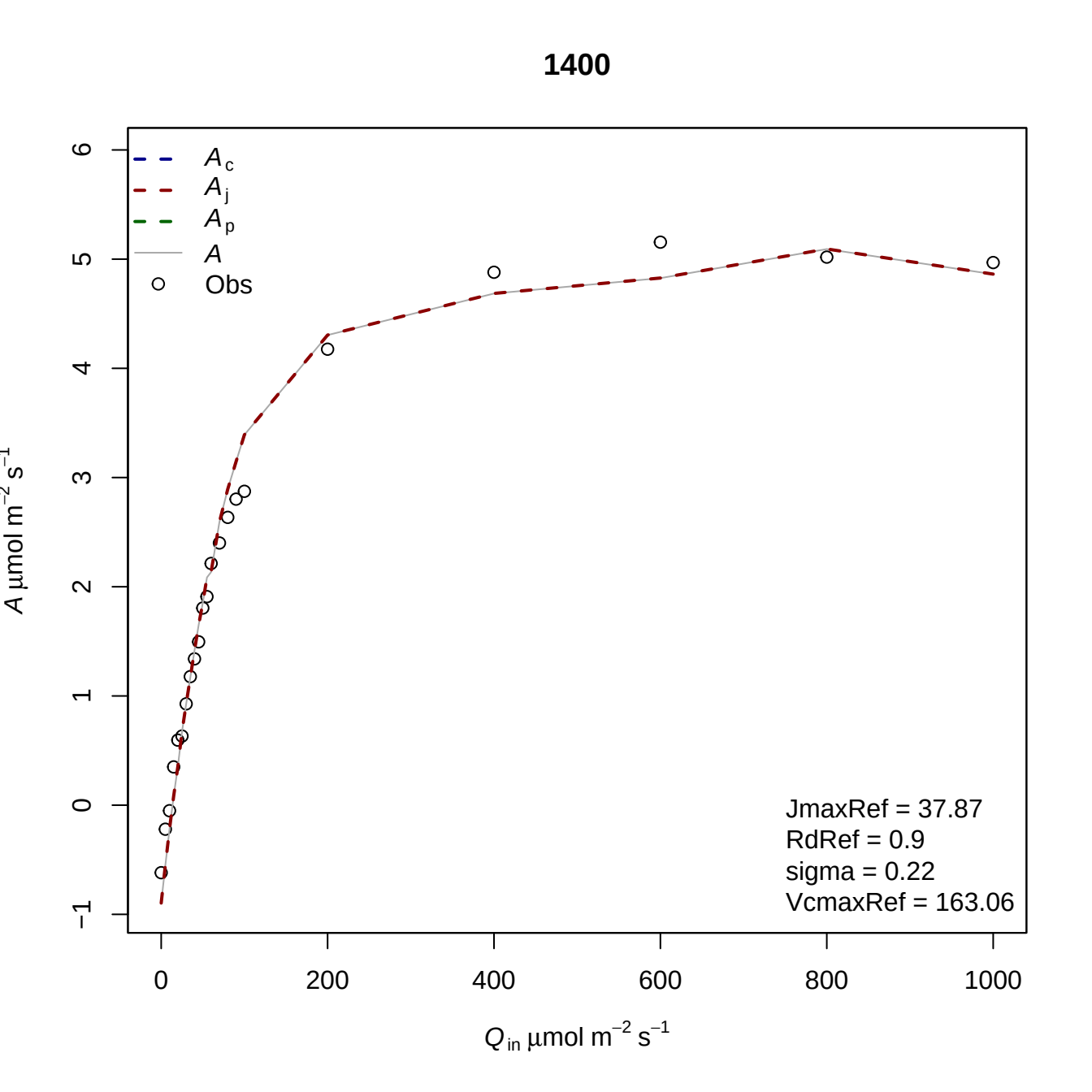

1395
A_c
A_j
A_p
A
Obs
JmaxRef = 87.57
RdRef = 0.86
sigma = 0.35
VcmaxRef = 282.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000

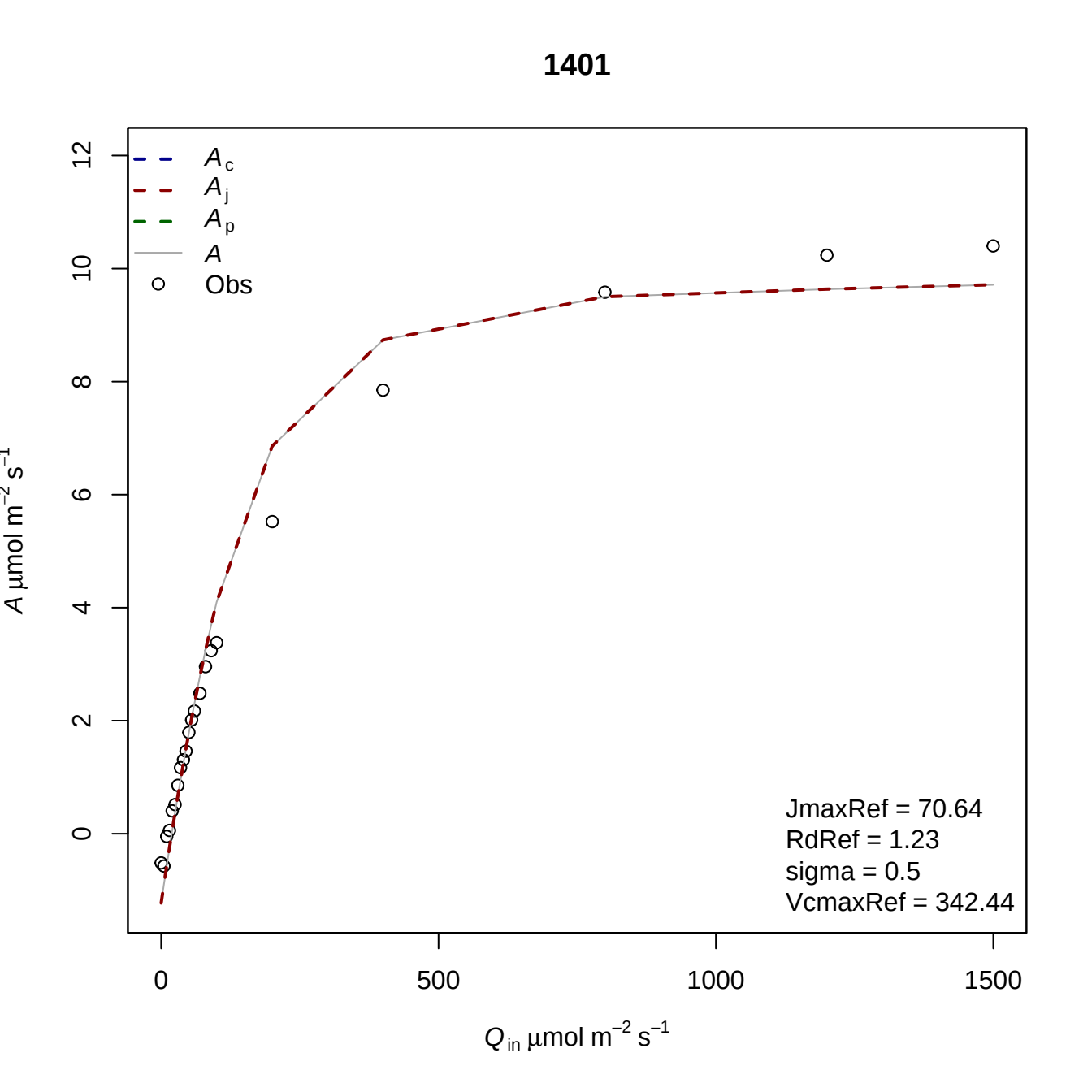

1396
A_c
A_j
A_p
A
Obs
JmaxRef = 39.02
RdRef = 0.62
sigma = 0.22
VcmaxRef = 1235056.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

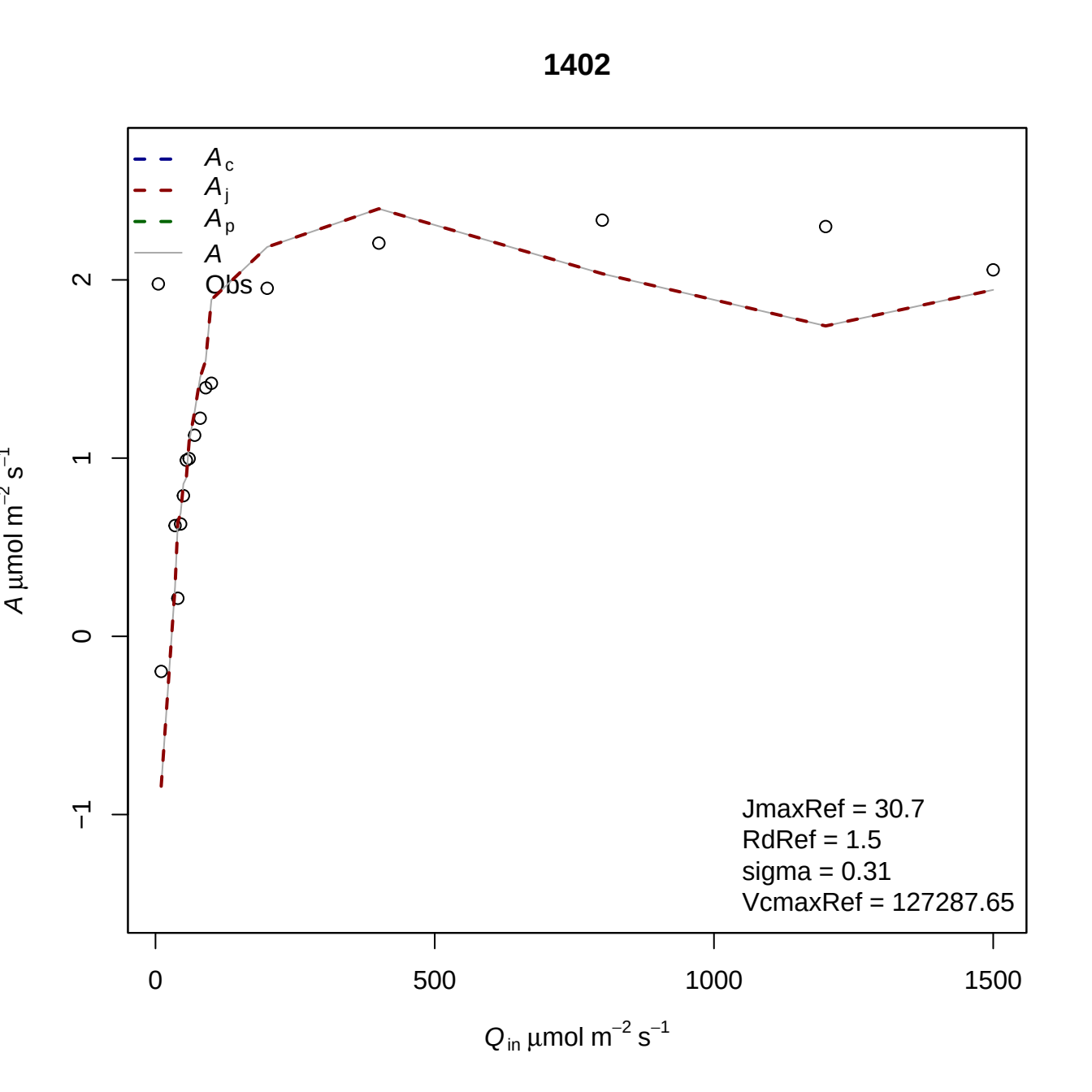

1397
A_c
A_j
A_p
A
Obs
JmaxRef = 38.13
RdRef = 0.89
sigma = 0.3
VcmaxRef = 147.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

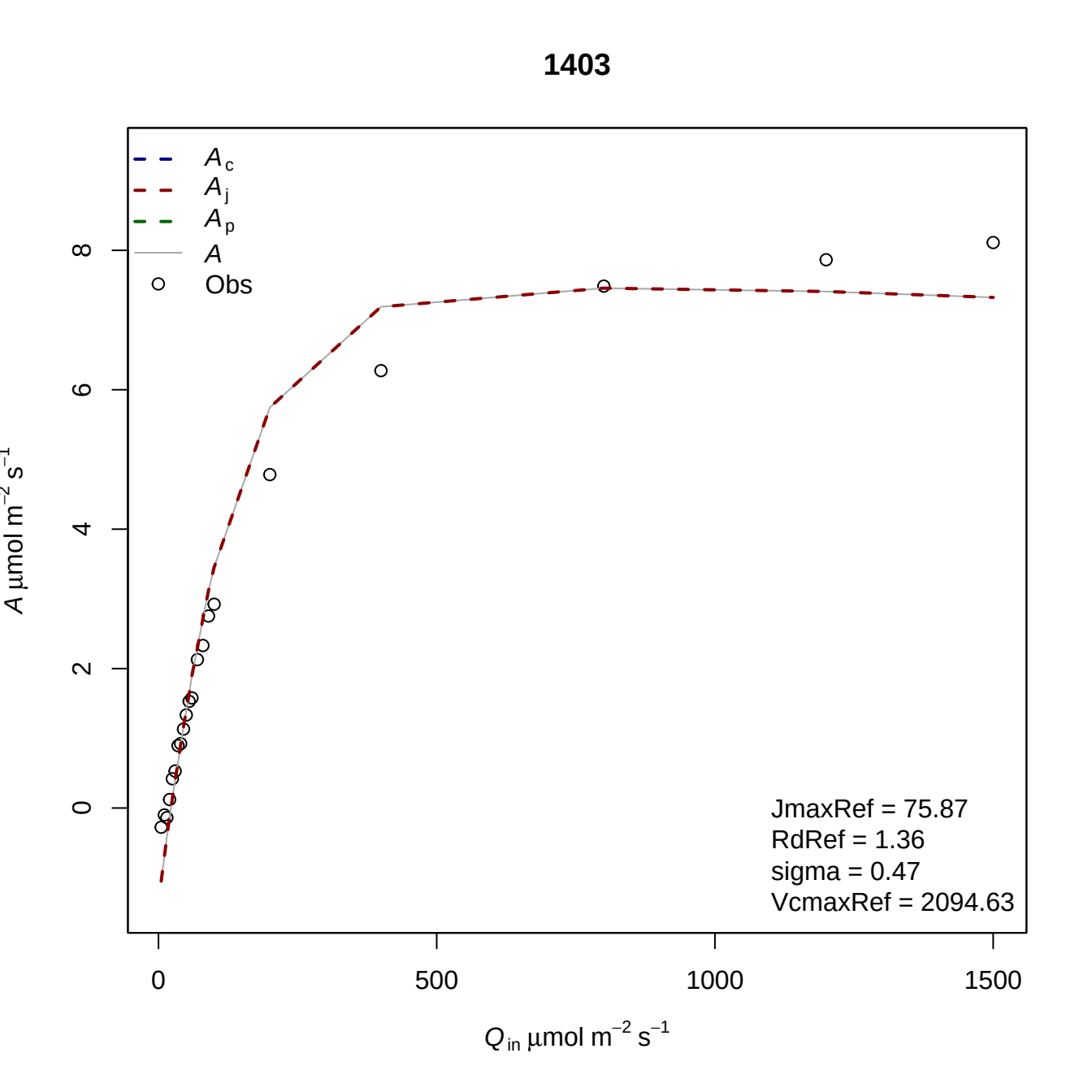

1398
A_c
A_j
A_p
A
Obs
JmaxRef = 53.71
RdRef = 1.13
sigma = 0.3
VcmaxRef = 954.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

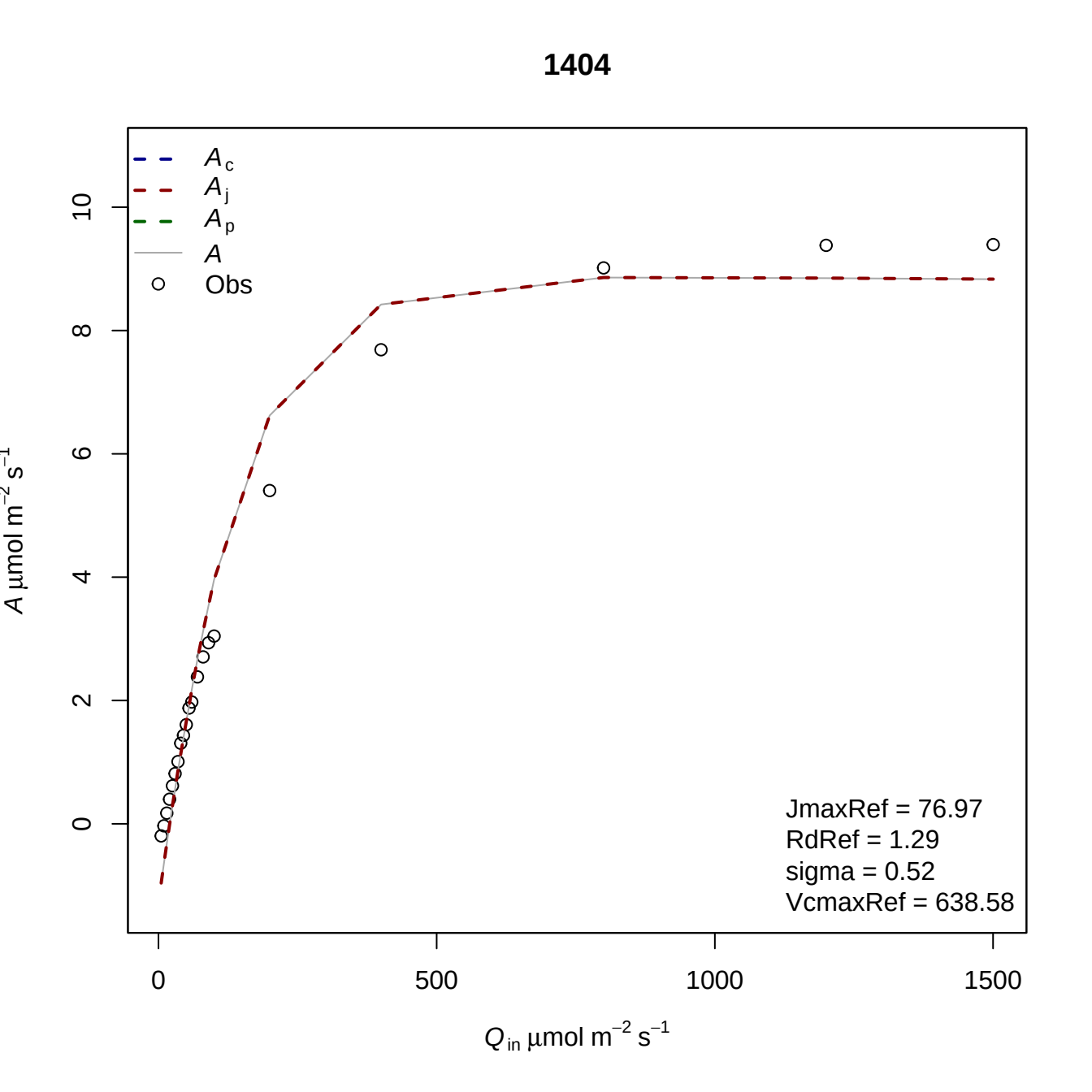

1399
A_c
A_j
A_p
A
Obs
JmaxRef = 49.3
RdRef = 0.52
sigma = 0.21
VcmaxRef = 456.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

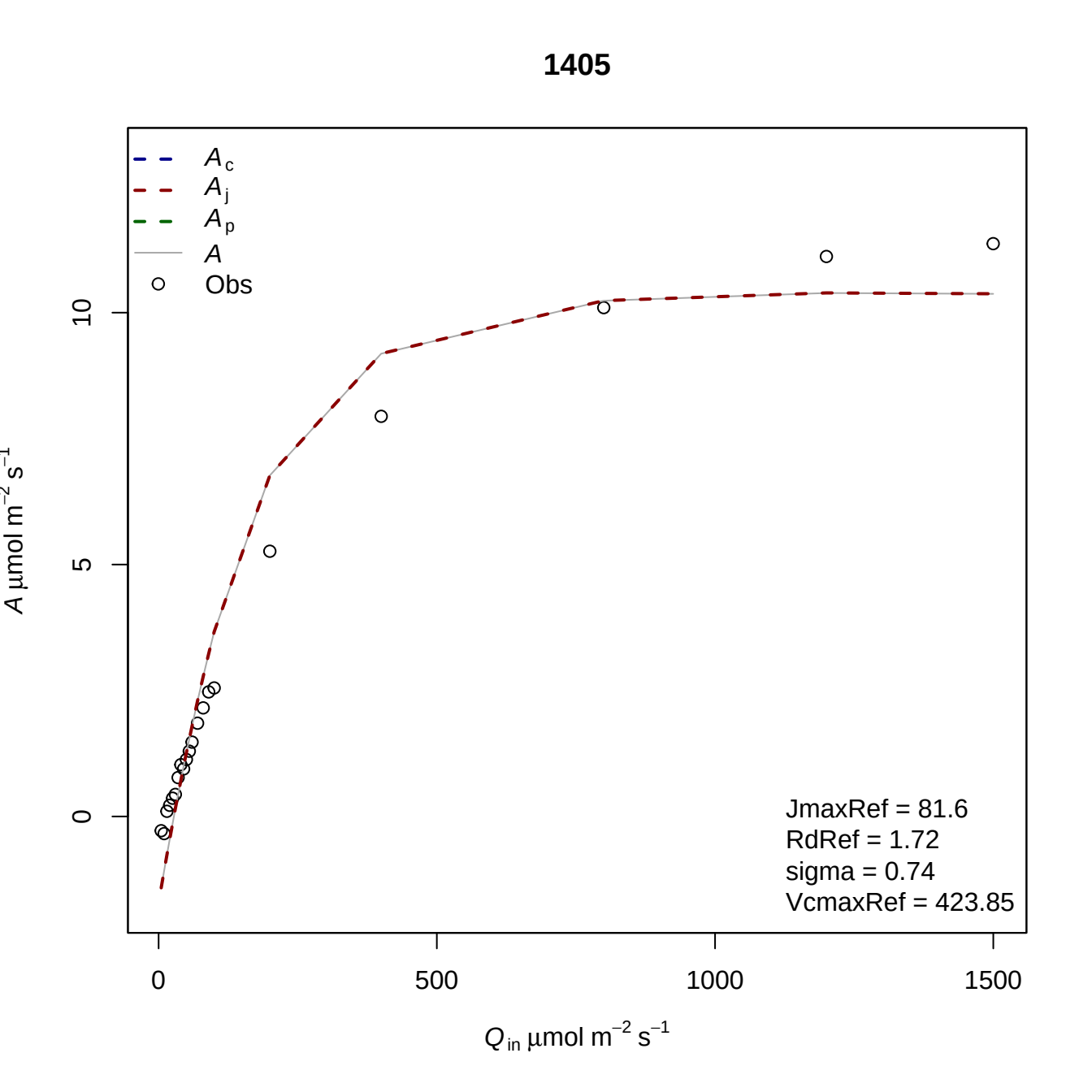

1400
A_c
A_j
A_p
A
Obs
JmaxRef = 37.87
RdRef = 0.9
sigma = 0.22
VcmaxRef = 163.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
200
400
600
800
1000

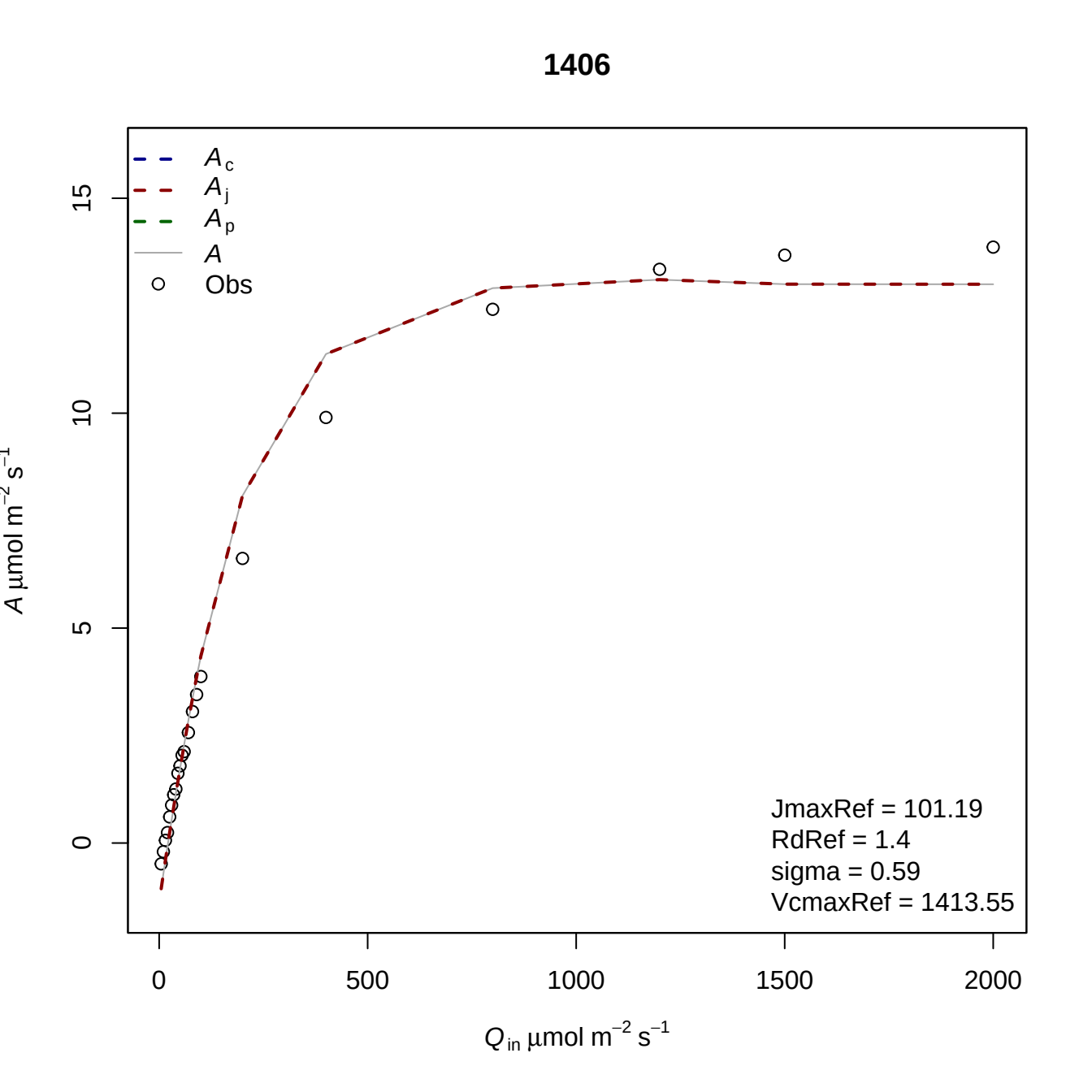

1401
A_c
A_j
A_p
A
Obs
JmaxRef = 70.64
RdRef = 1.23
sigma = 0.5
VcmaxRef = 342.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

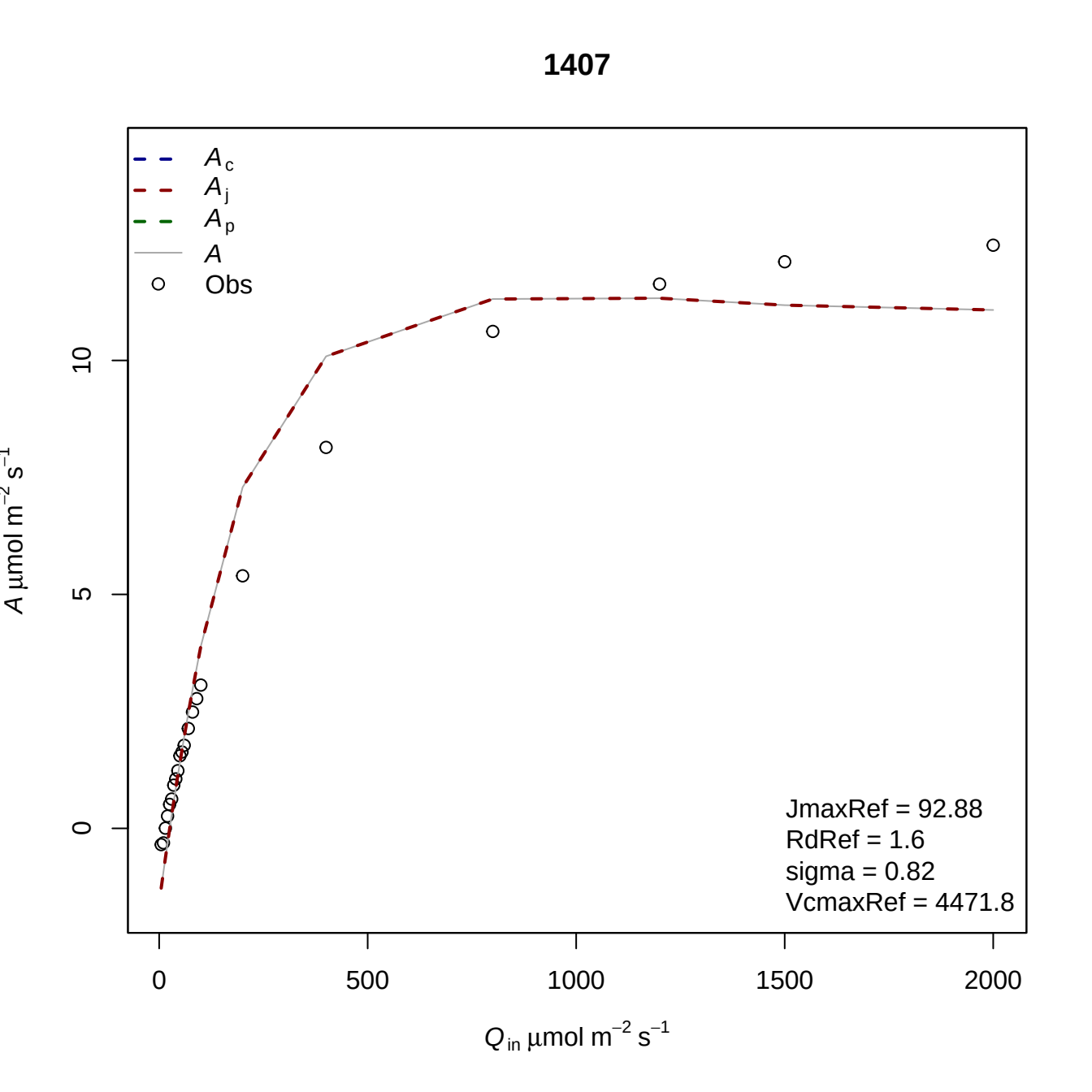

1402
A_c
A_j
A_p
A
Obs
JmaxRef = 30.7
RdRef = 1.5
sigma = 0.31
VcmaxRef = 127287.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

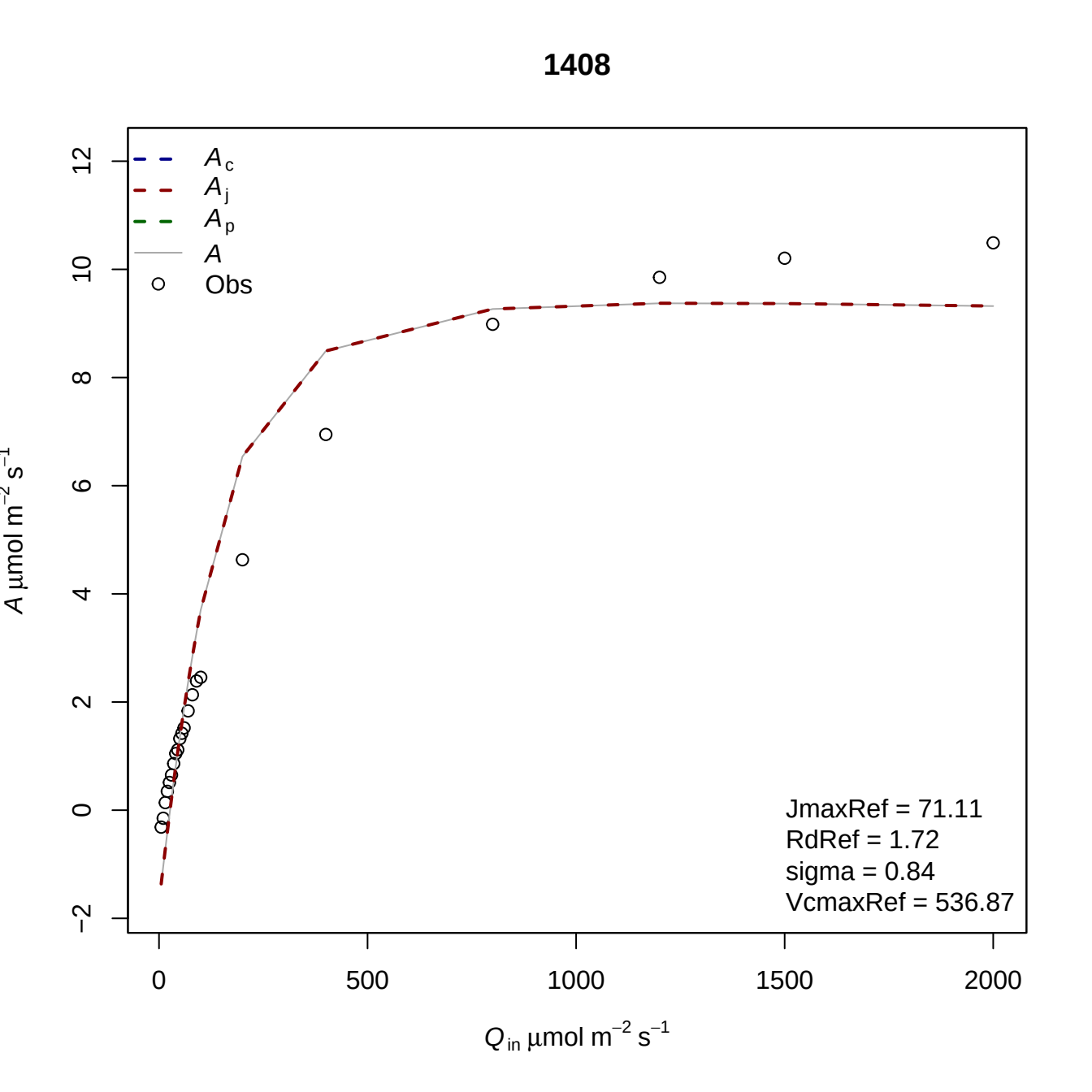

1403
A_c
A_j
A_p
A
Obs
JmaxRef = 75.87
RdRef = 1.36
sigma = 0.47
VcmaxRef = 2094.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

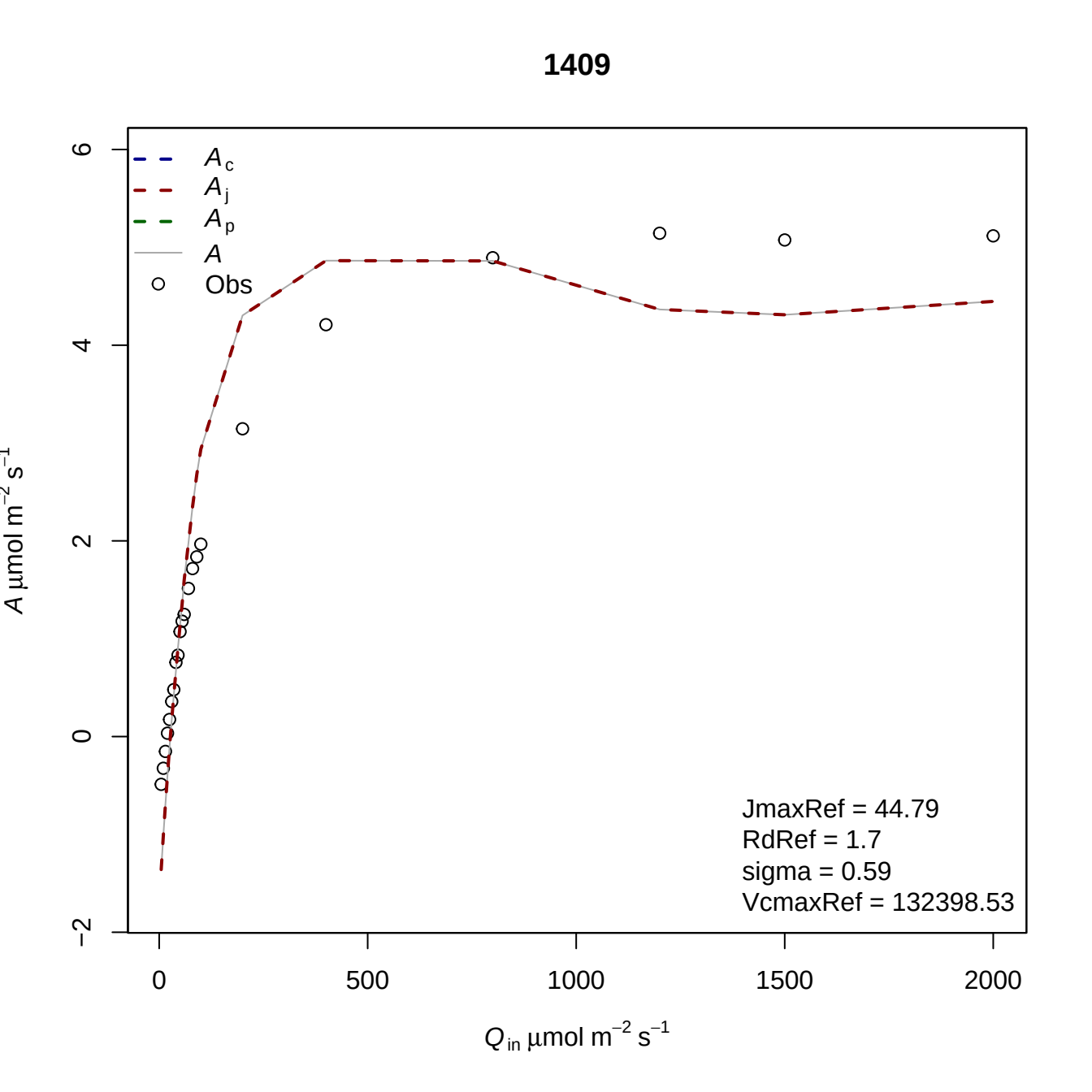

1404
A_c
A_j
A_p
A
Obs
JmaxRef = 76.97
RdRef = 1.29
sigma = 0.52
VcmaxRef = 638.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

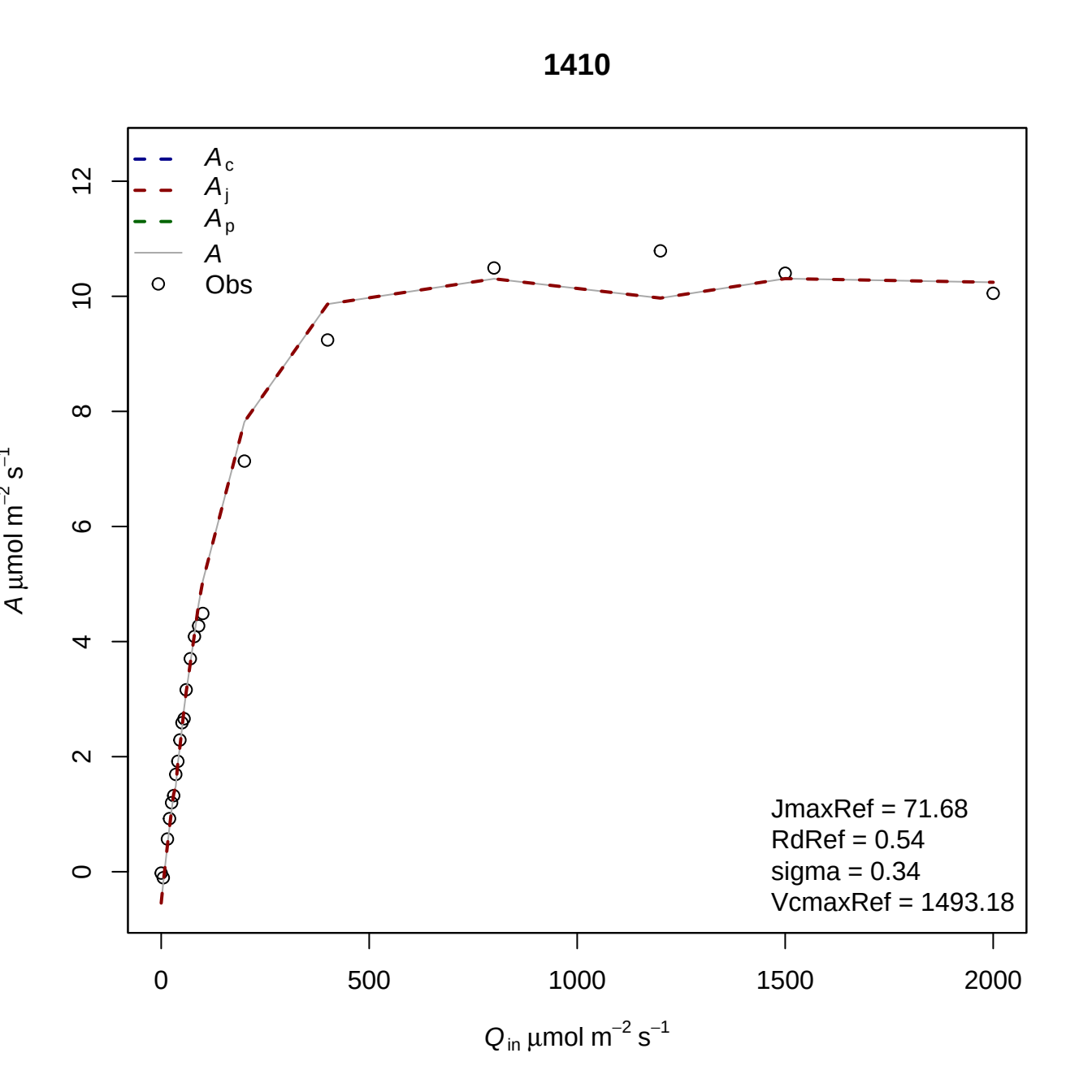

1405
A_c
A_j
A_p
A
Obs
JmaxRef = 81.6
RdRef = 1.72
sigma = 0.74
VcmaxRef = 423.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

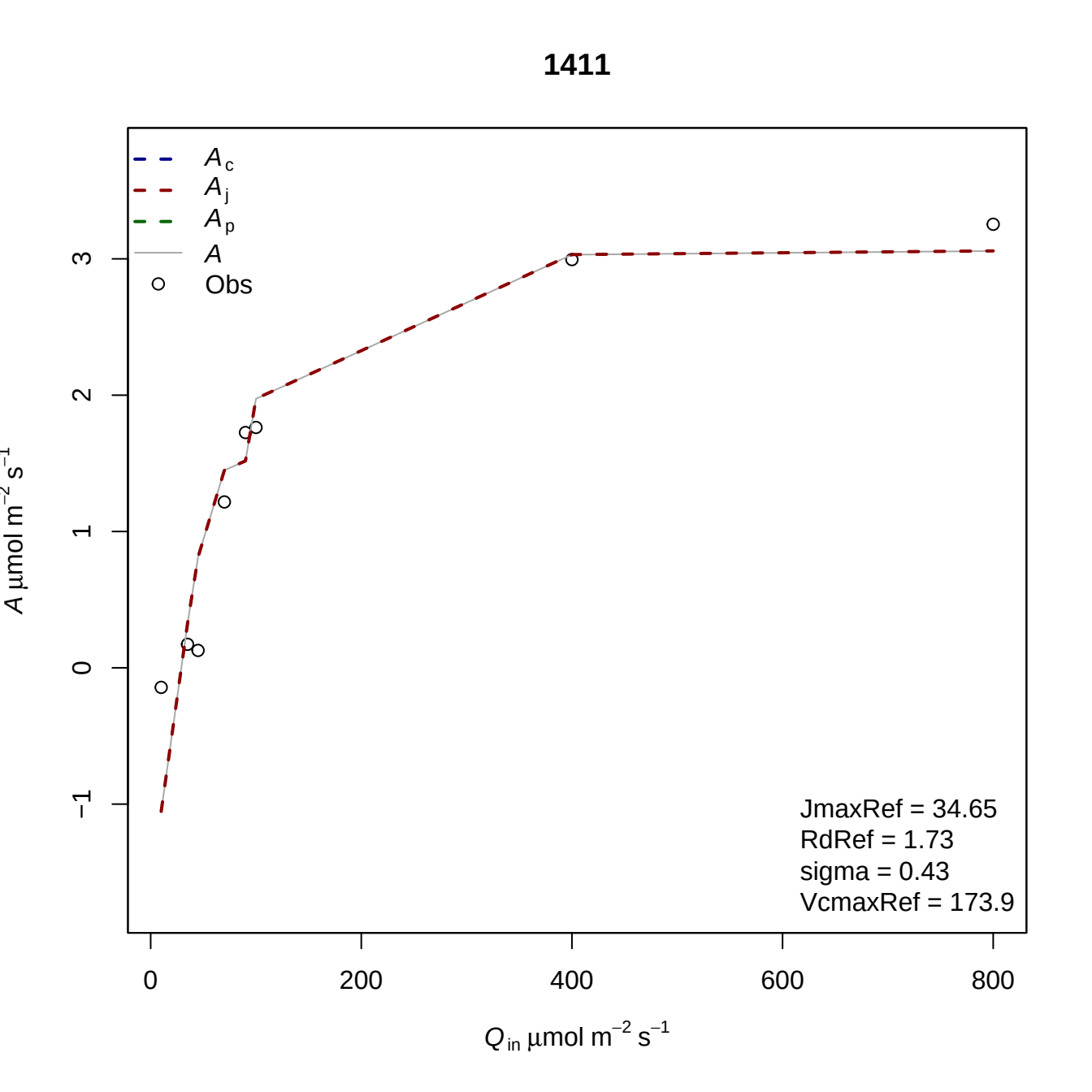

1406
A_c
A_j
A_p
A
Obs
JmaxRef = 101.19
RdRef = 1.4
sigma = 0.59
VcmaxRef = 1413.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

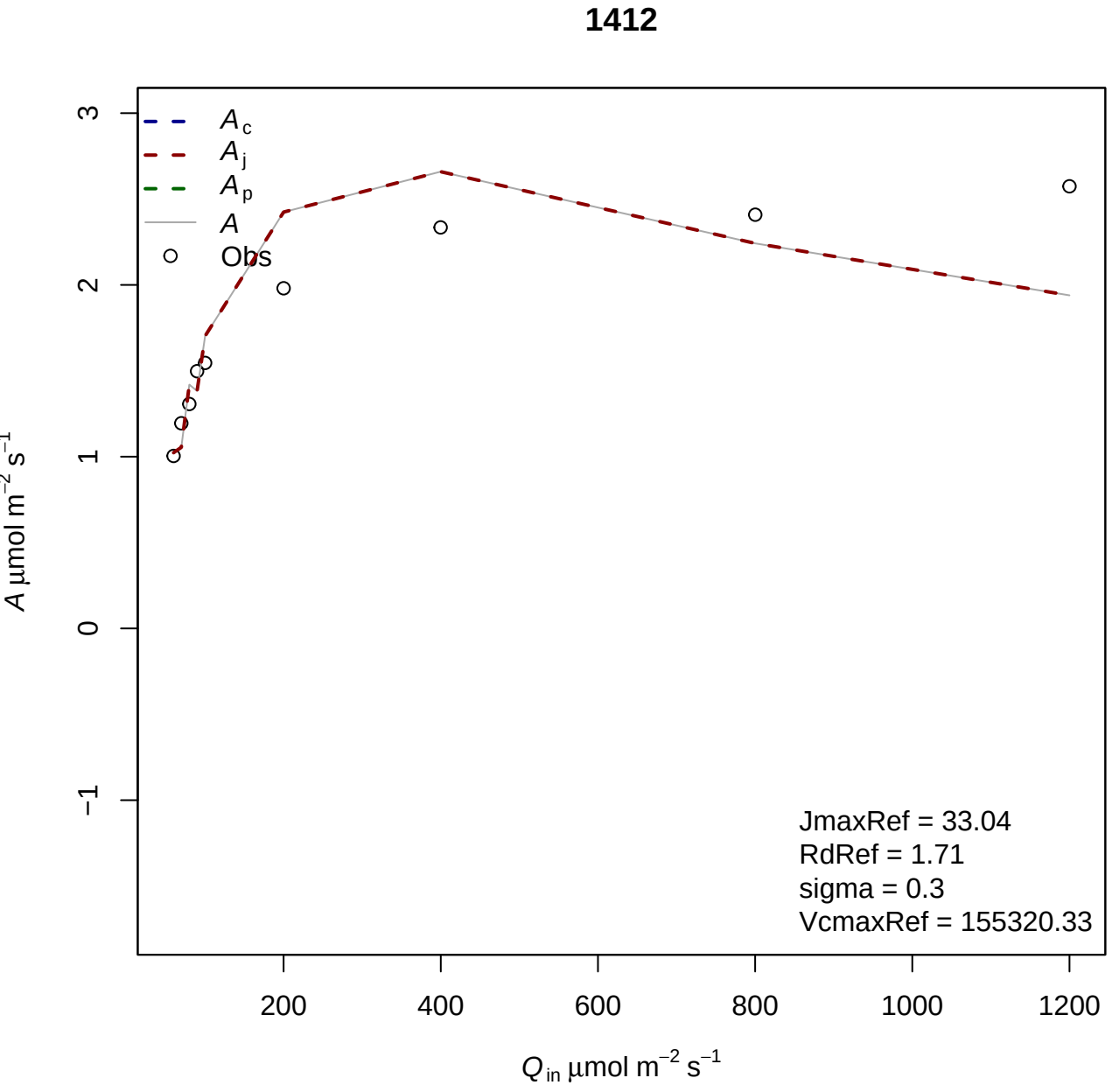

1407
A_c
A_j
A_p
A
Obs
JmaxRef = 92.88
RdRef = 1.6
sigma = 0.82
VcmaxRef = 4471.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

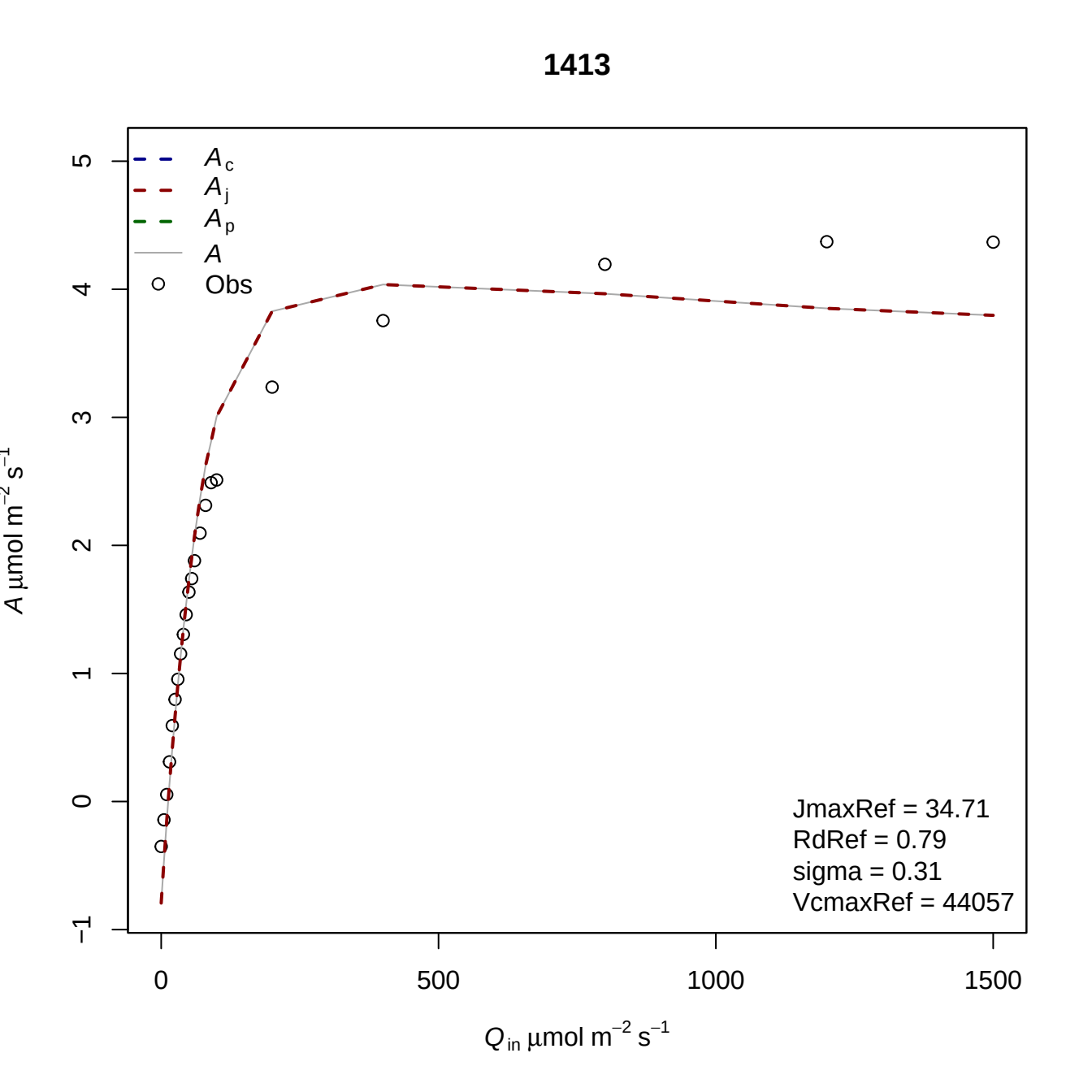

1408
A_c
A_j
A_p
A
Obs
JmaxRef = 71.11
RdRef = 1.72
sigma = 0.84
VcmaxRef = 536.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

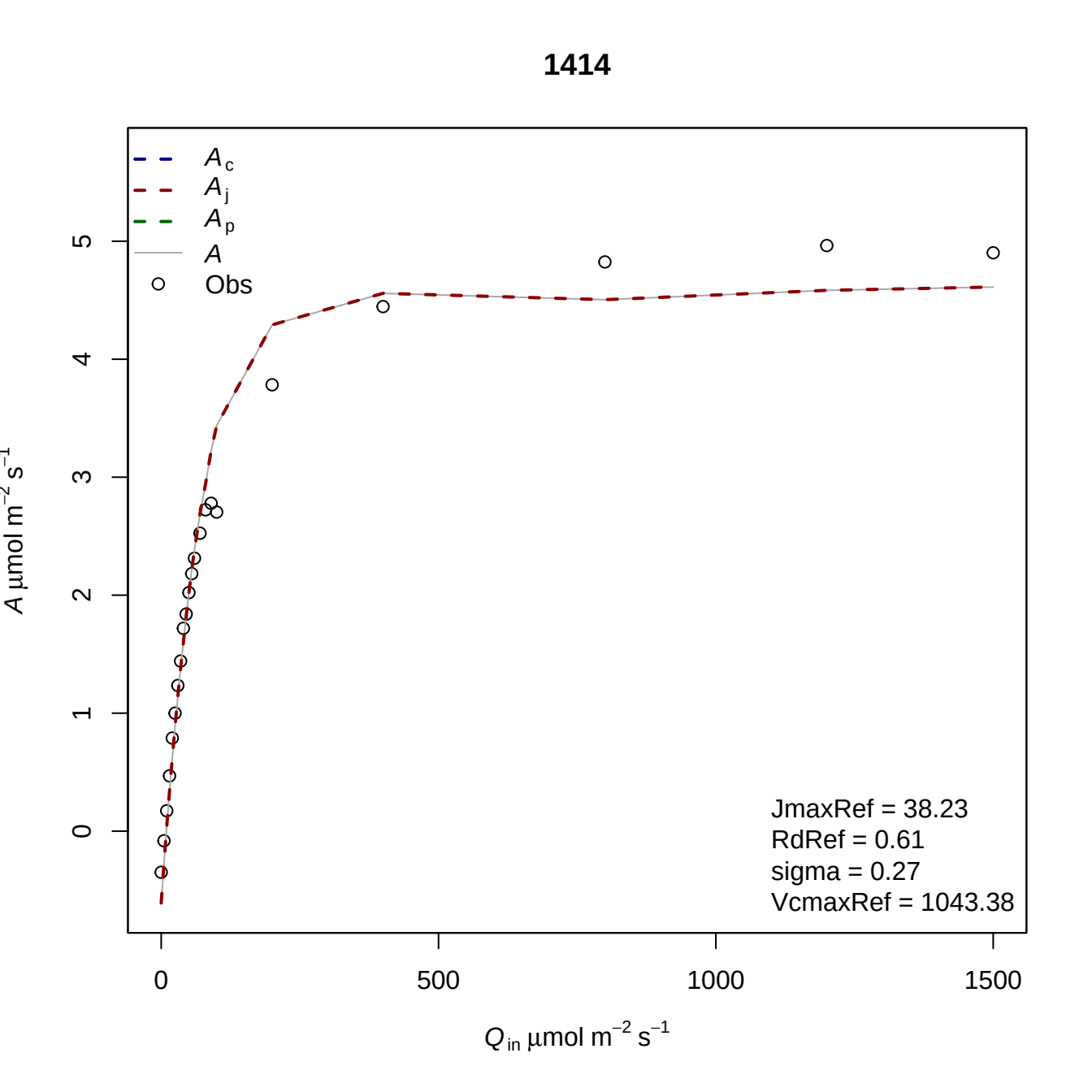

1409
A_c
A_j
A_p
A
Obs
JmaxRef = 44.79
RdRef = 1.7
sigma = 0.59
VcmaxRef = 132398.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

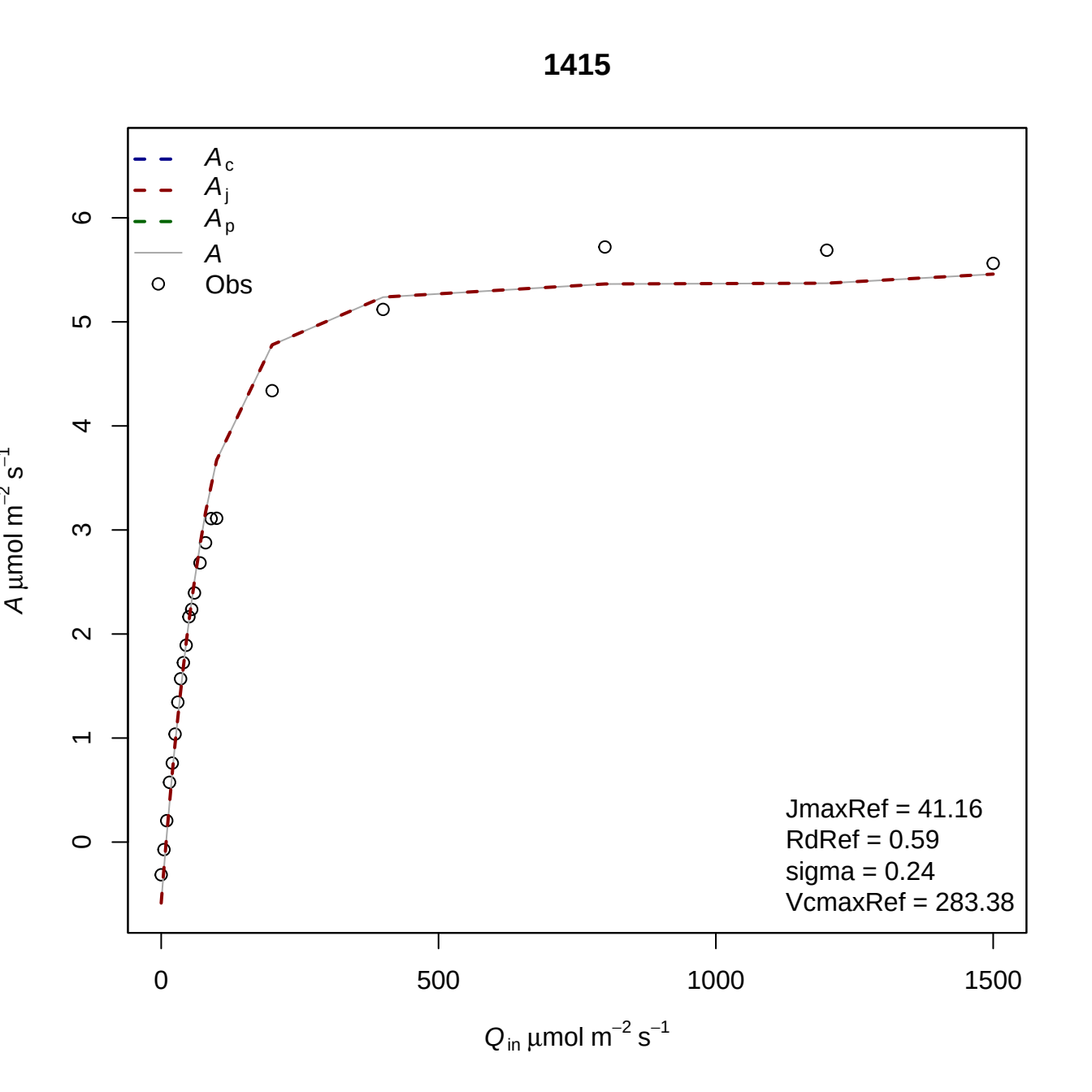

1410
A_c
A_j
A_p
A
Obs
JmaxRef = 71.68
RdRef = 0.54
sigma = 0.34
VcmaxRef = 1493.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

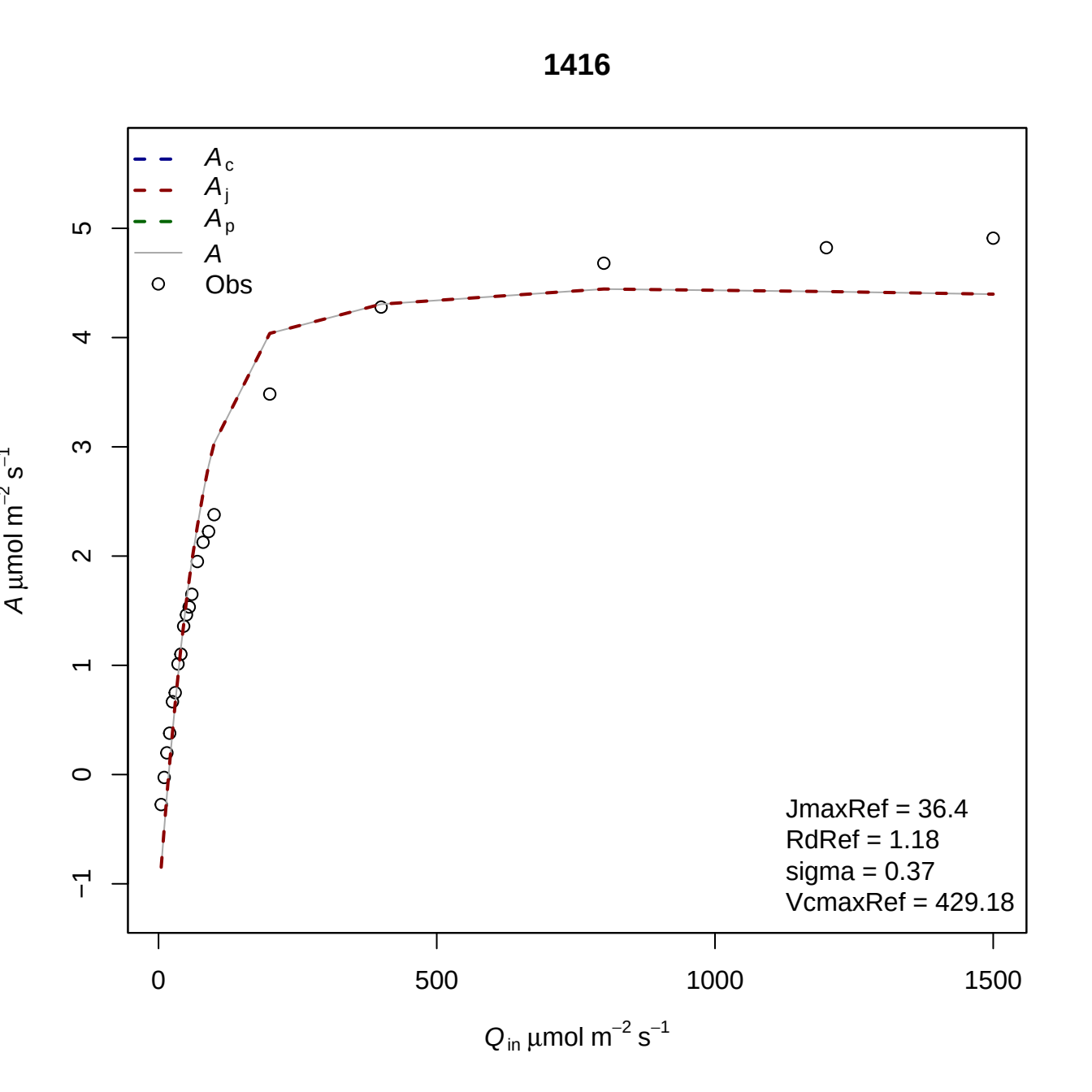

1411
A_c
A_j
A_p
A
Obs
JmaxRef = 34.65
RdRef = 1.73
sigma = 0.43
VcmaxRef = 173.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800

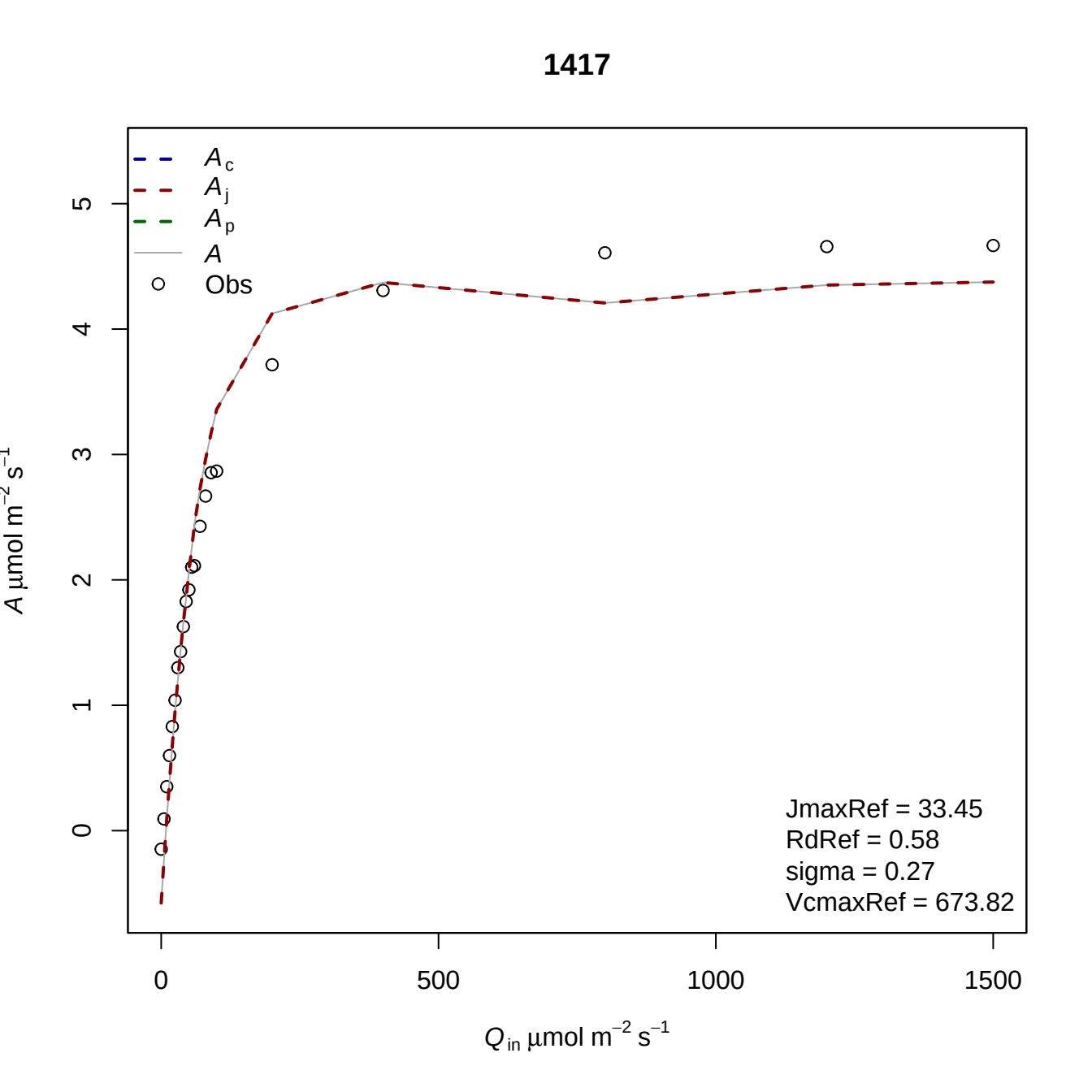

1412
A_c
A_j
A_p
A
Obs
3
2
1
0
−1
A µmol m^{-2} s^{-1}
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 33.04
RdRef = 1.71
sigma = 0.3
VcmaxRef = 155320.33

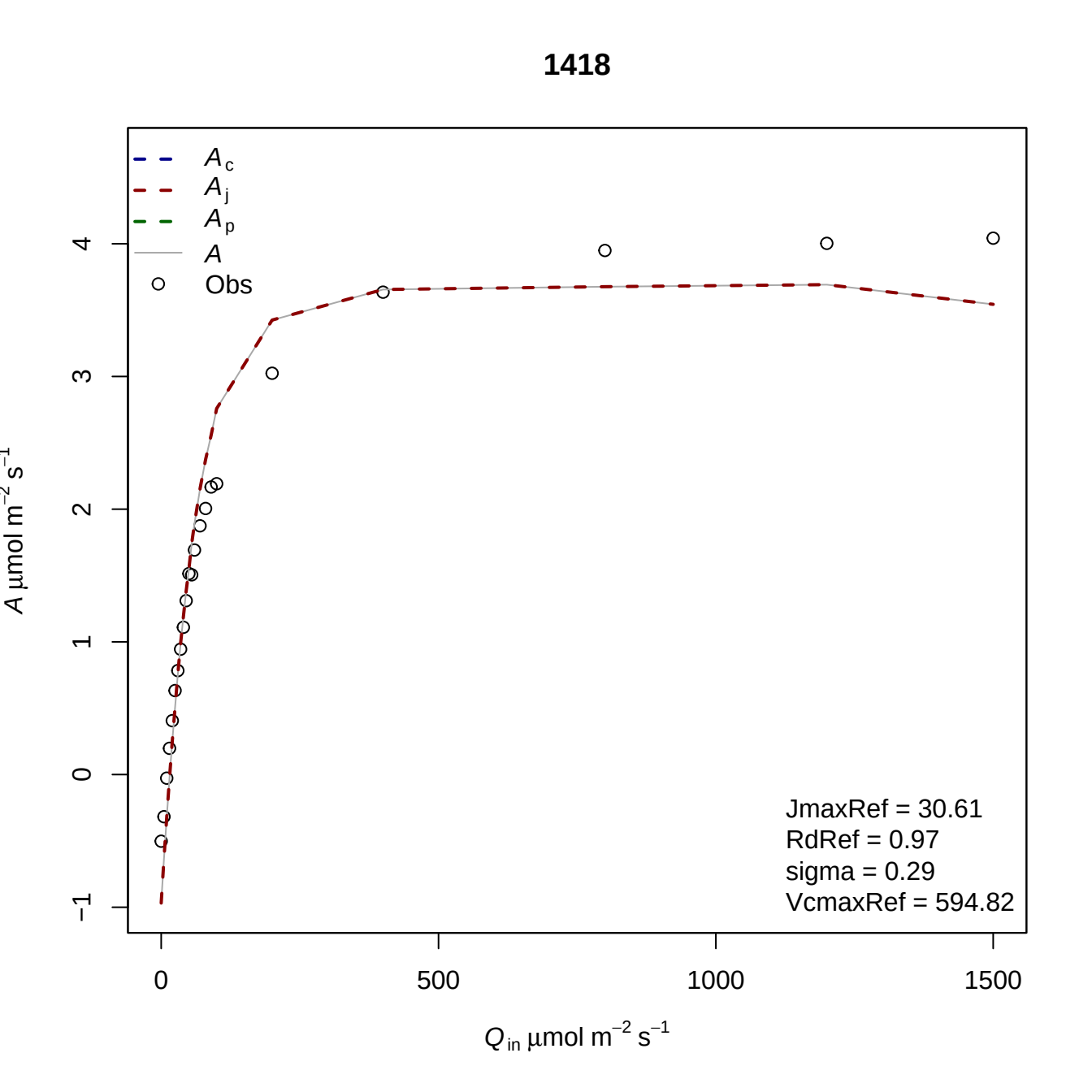

1413
A_c
A_j
A_p
A
Obs
JmaxRef = 34.71
RdRef = 0.79
sigma = 0.31
VcmaxRef = 44057
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

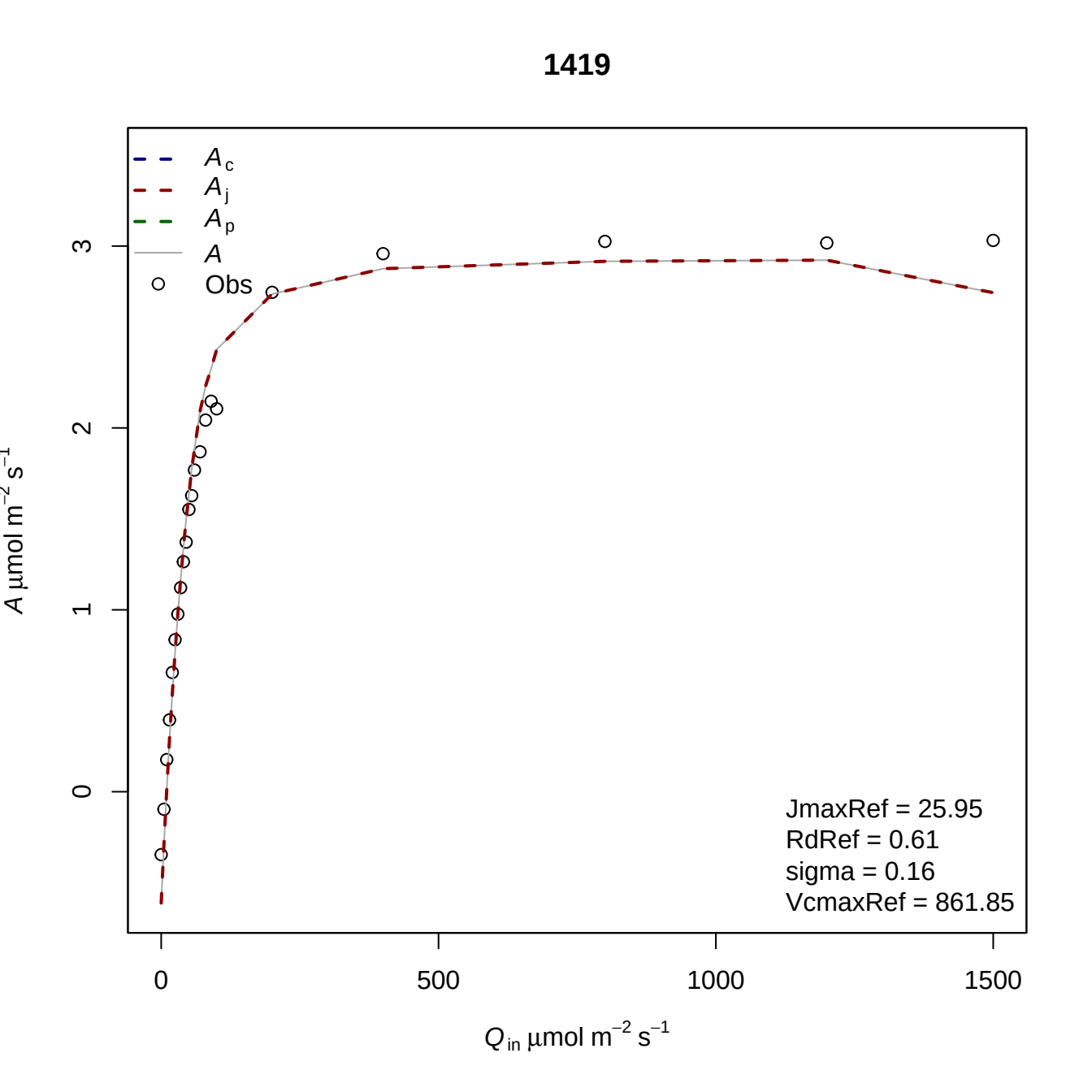

1414
A_c
A_j
A_p
A
Obs
JmaxRef = 38.23
RdRef = 0.61
sigma = 0.27
VcmaxRef = 1043.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

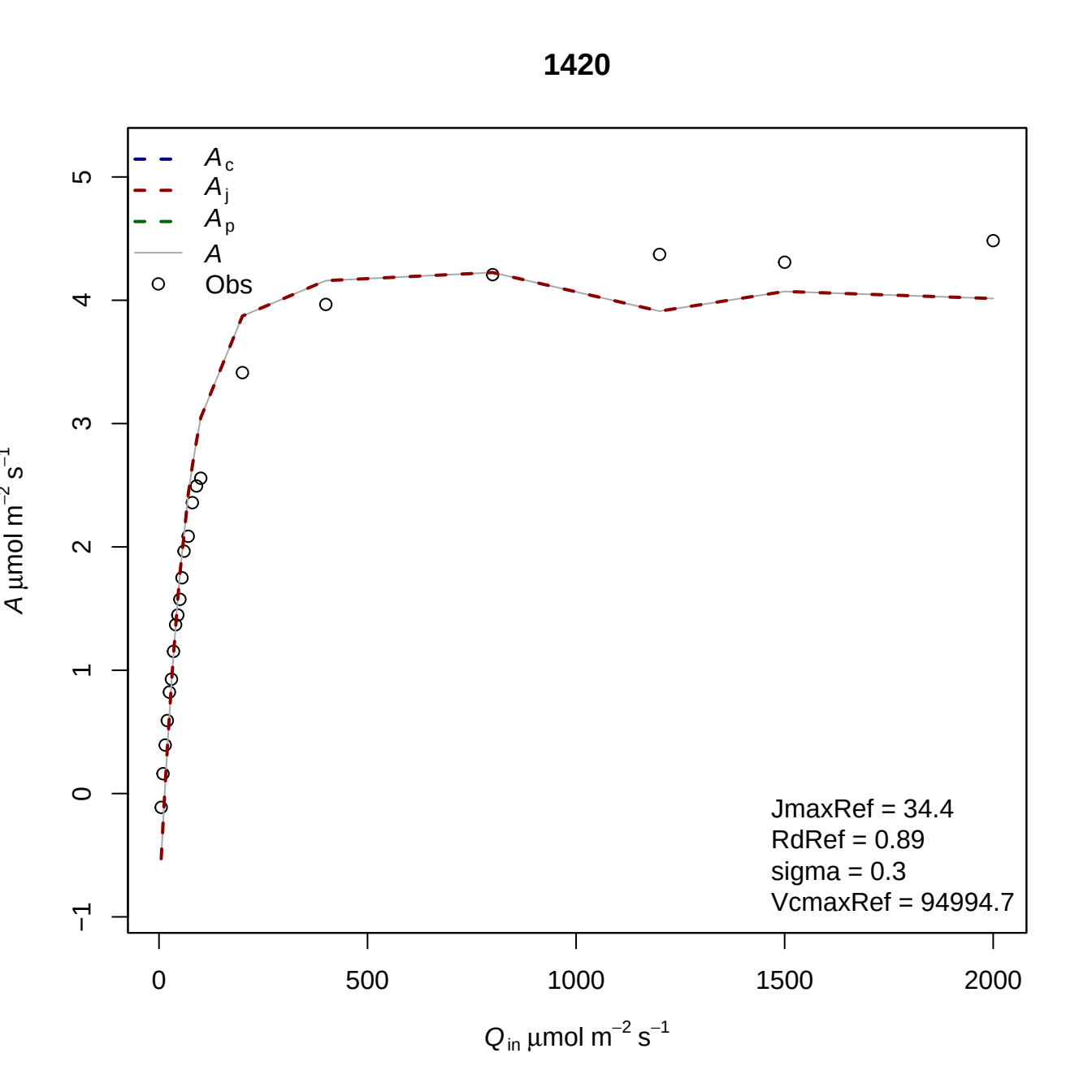

1415
A_c
A_j
A_p
A
Obs
JmaxRef = 41.16
RdRef = 0.59
sigma = 0.24
VcmaxRef = 283.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

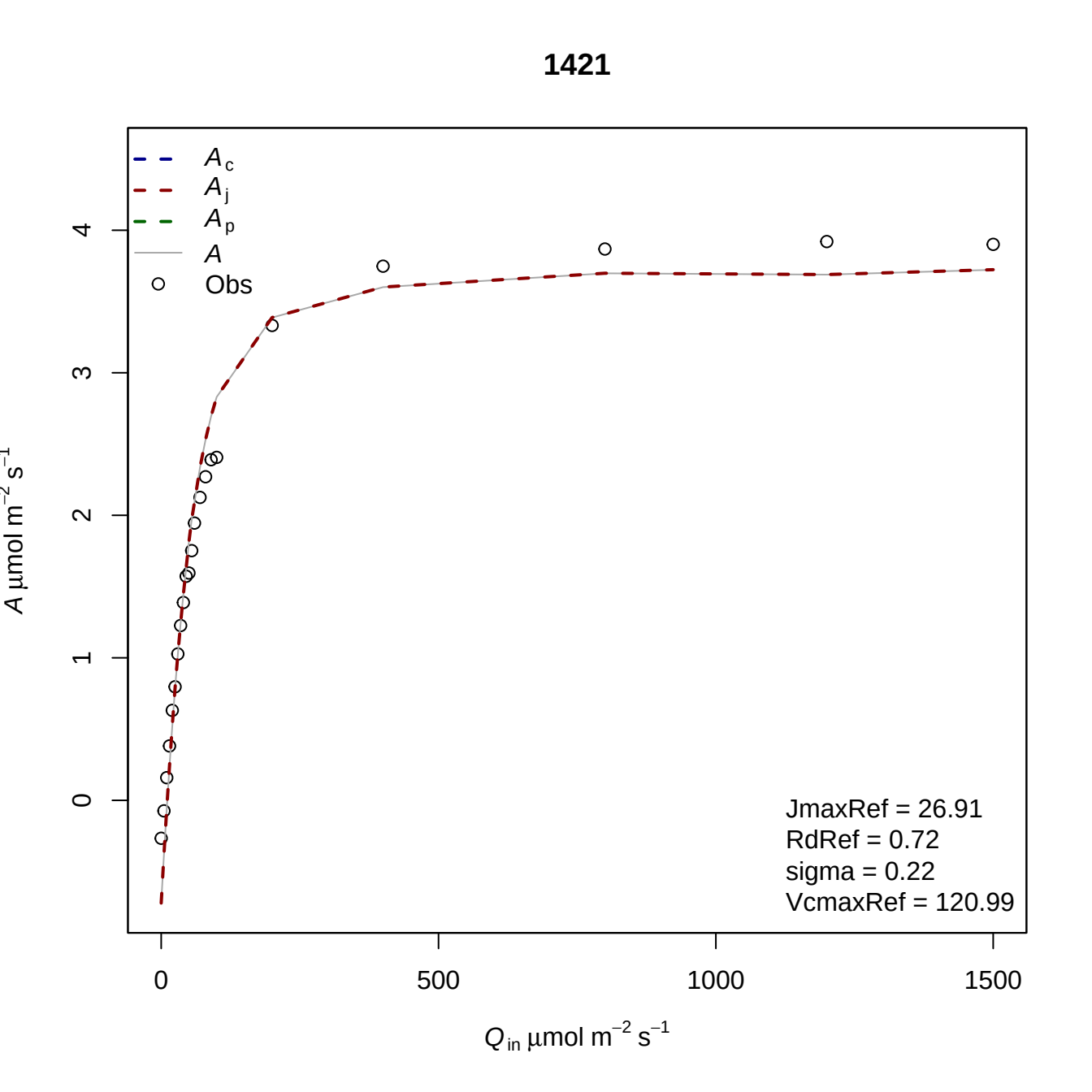

1416
A_c
A_j
A_p
A
Obs
JmaxRef = 36.4
RdRef = 1.18
sigma = 0.37
VcmaxRef = 429.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

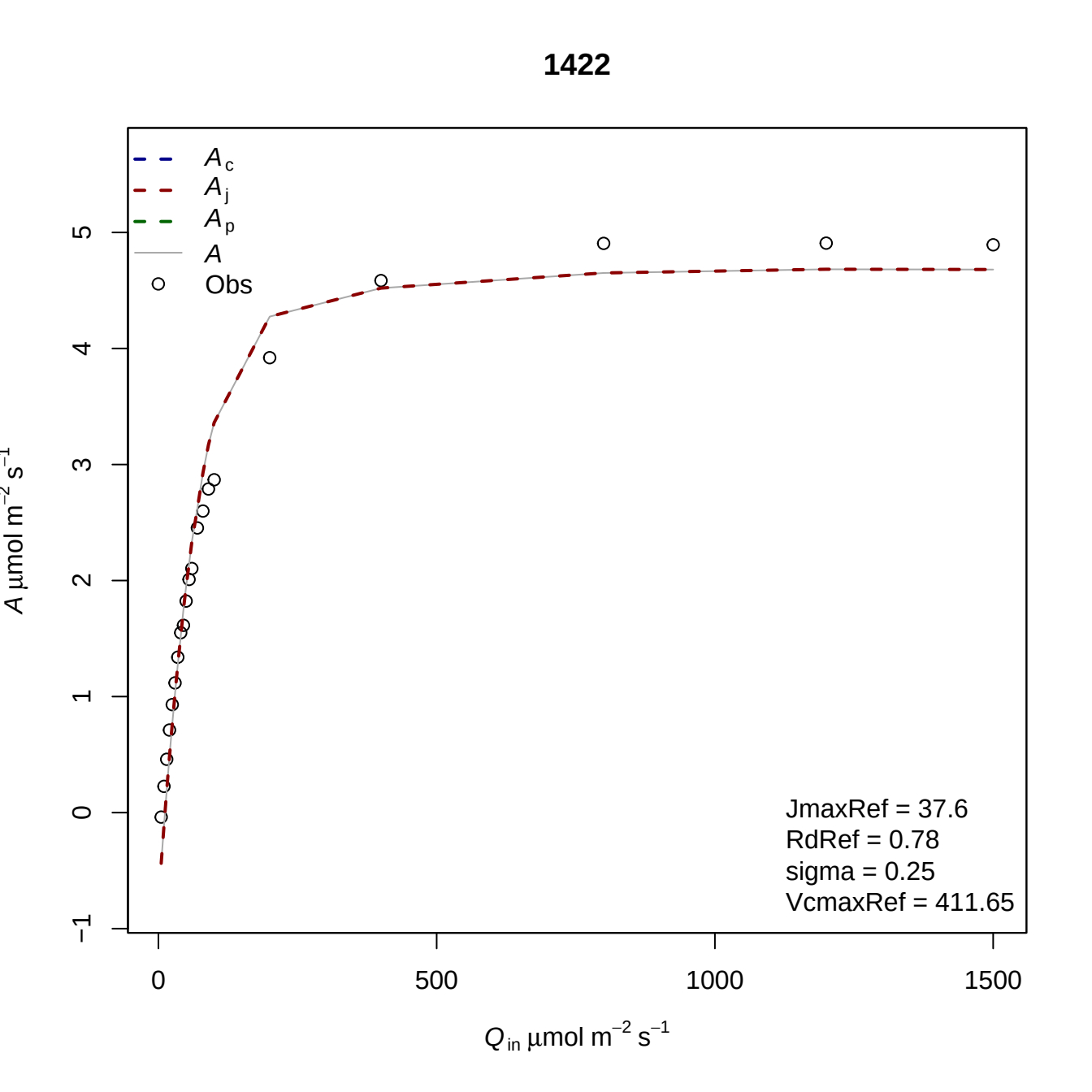

1417
A_c
A_j
A_p
A
Obs
JmaxRef = 33.45
RdRef = 0.58
sigma = 0.27
VcmaxRef = 673.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

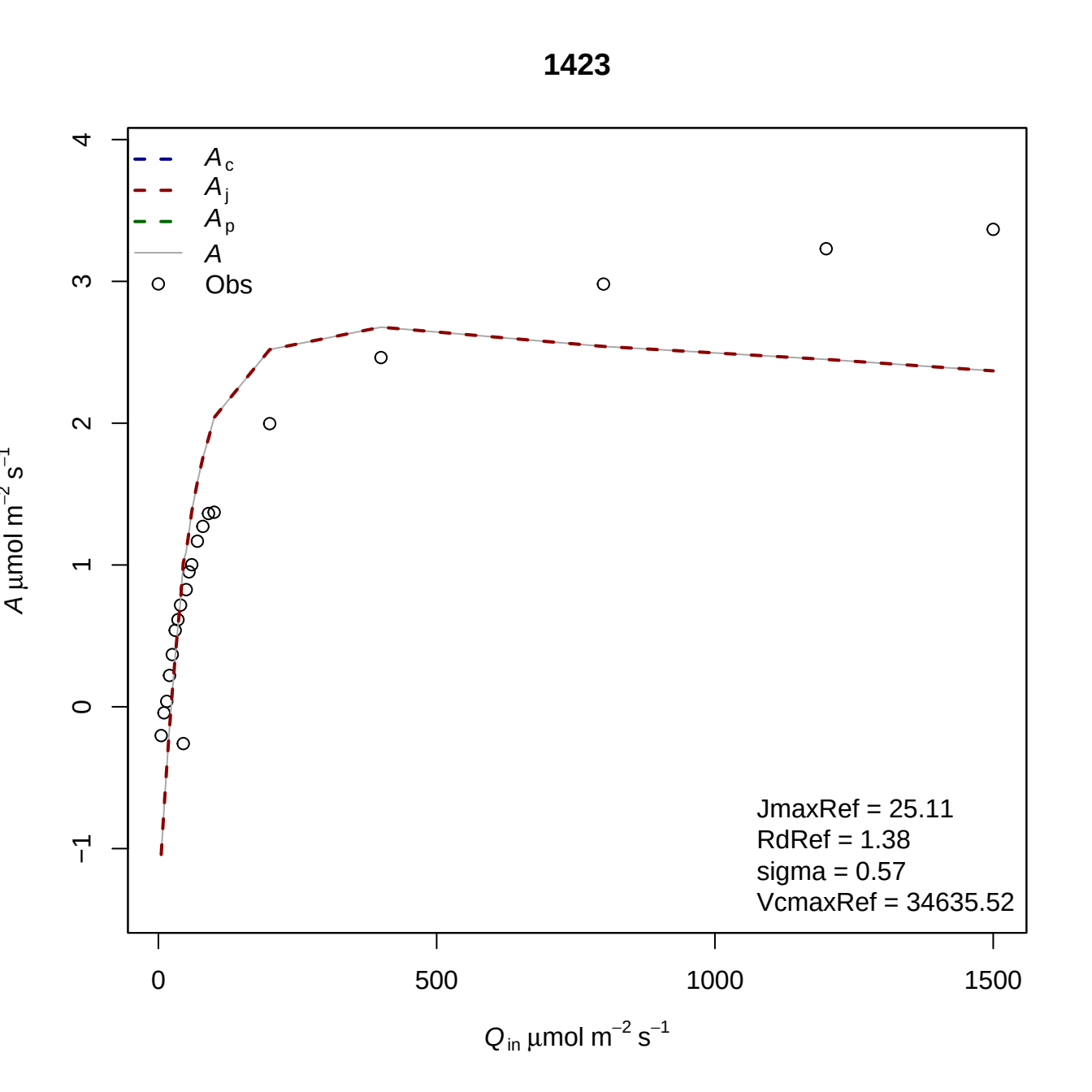

1418
A_c
A_j
A_p
A
Obs
JmaxRef = 30.61
RdRef = 0.97
sigma = 0.29
VcmaxRef = 594.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

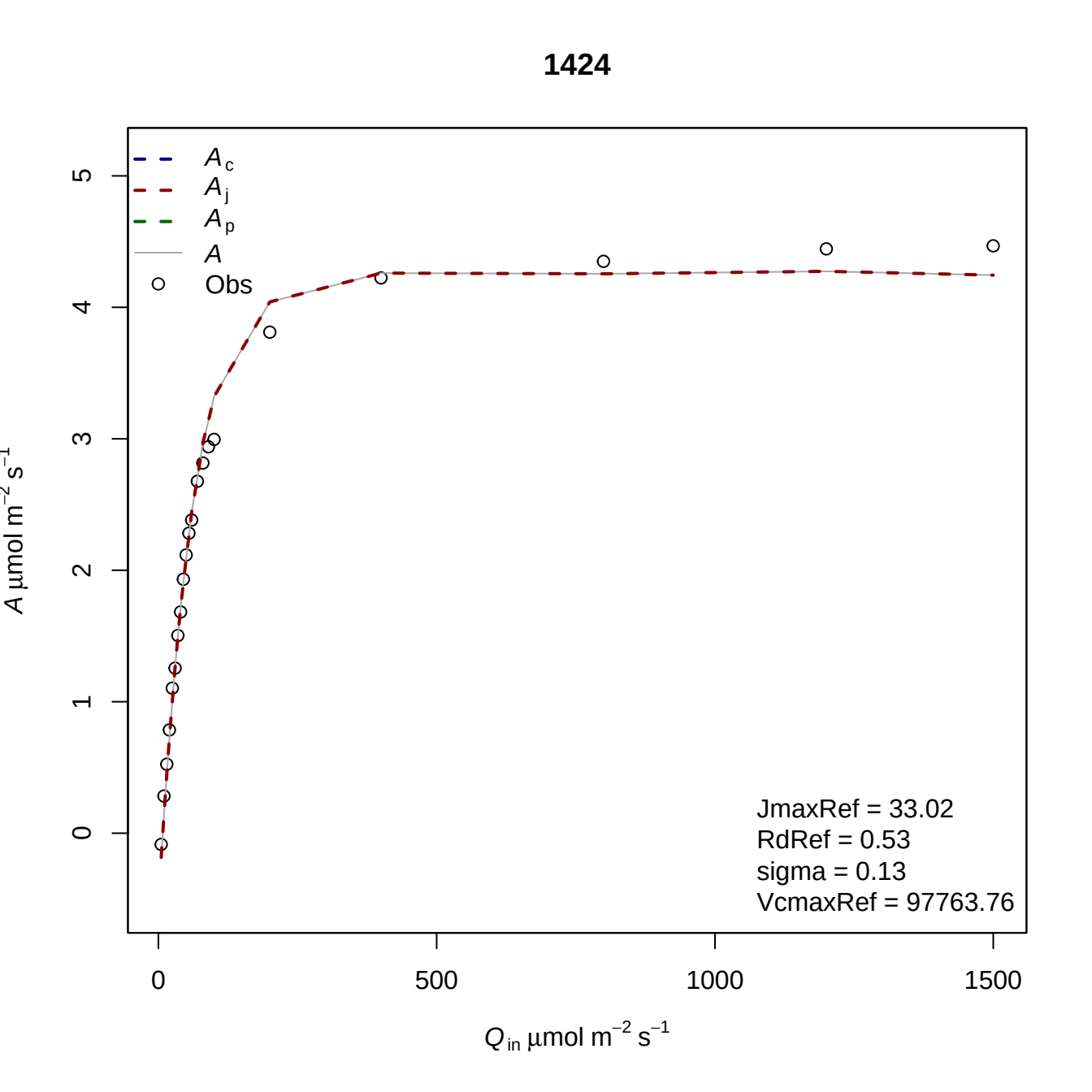

1419
A_c
A_j
A_p
A
Obs
JmaxRef = 25.95
RdRef = 0.61
sigma = 0.16
VcmaxRef = 861.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

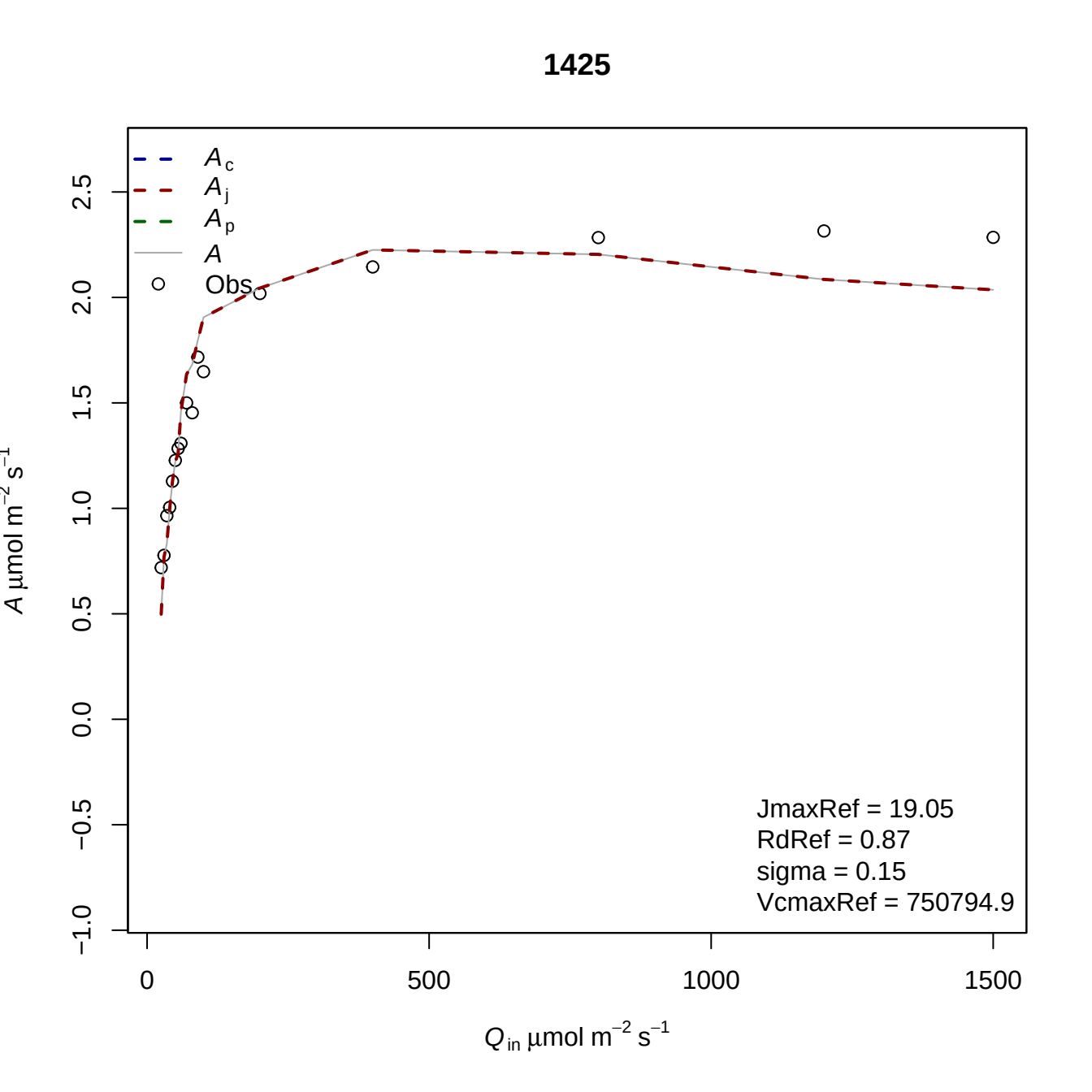

1420
A_c
A_j
A_p
A
Obs
JmaxRef = 34.4
RdRef = 0.89
sigma = 0.3
VcmaxRef = 94994.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

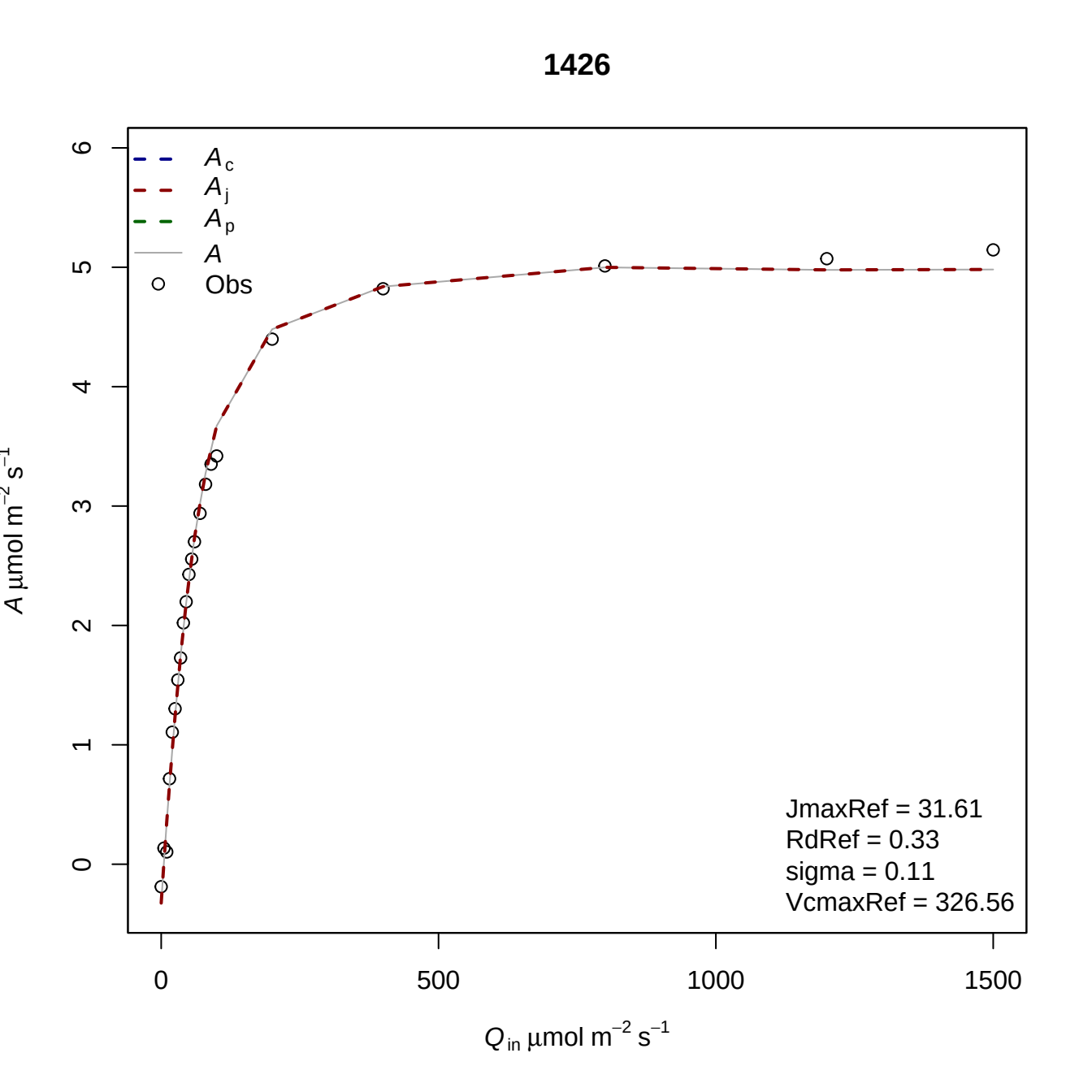

1421
A_c
A_j
A_p
A
Obs
JmaxRef = 26.91
RdRef = 0.72
sigma = 0.22
VcmaxRef = 120.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

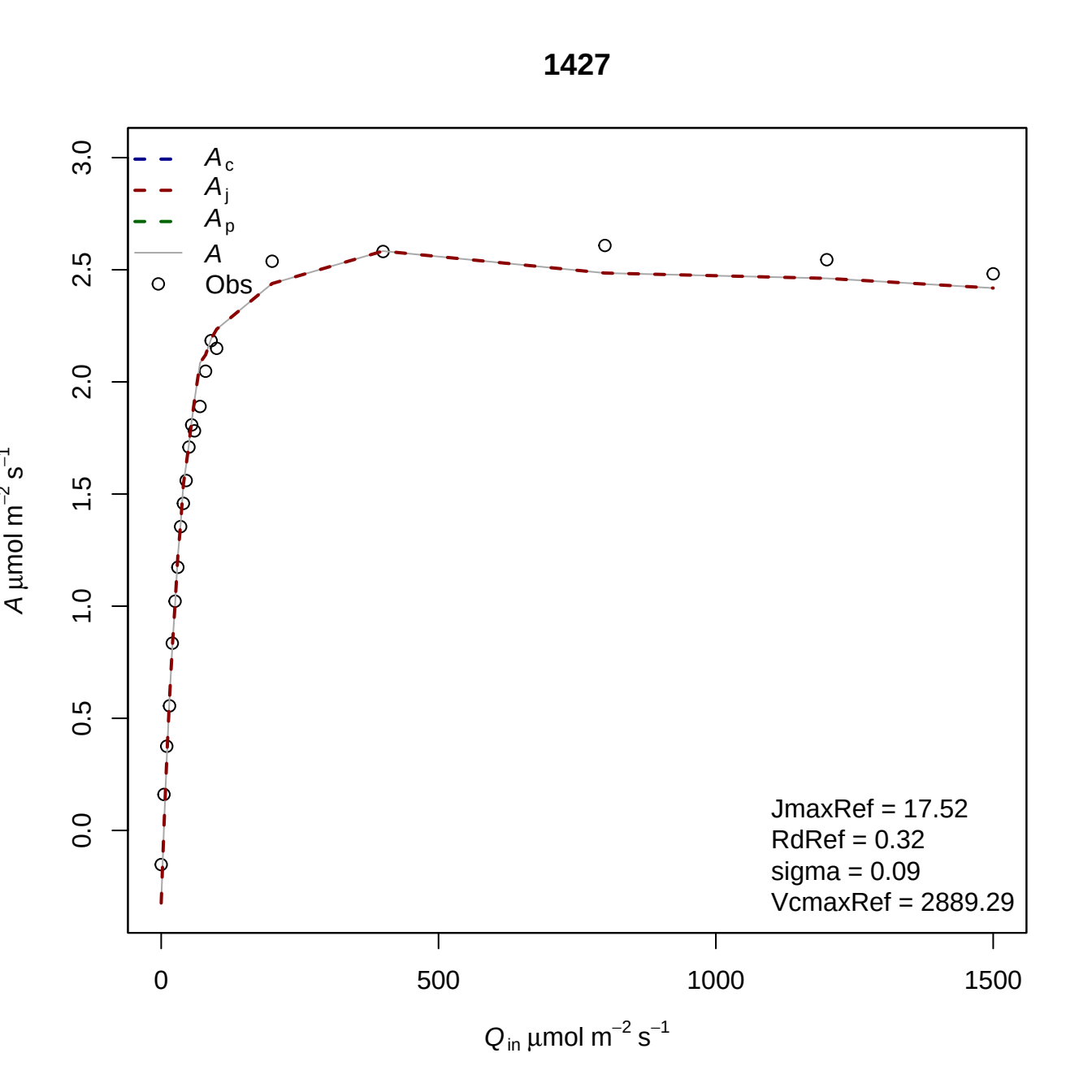

1422
A_c
A_j
A_p
A
Obs
JmaxRef = 37.6
RdRef = 0.78
sigma = 0.25
VcmaxRef = 411.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

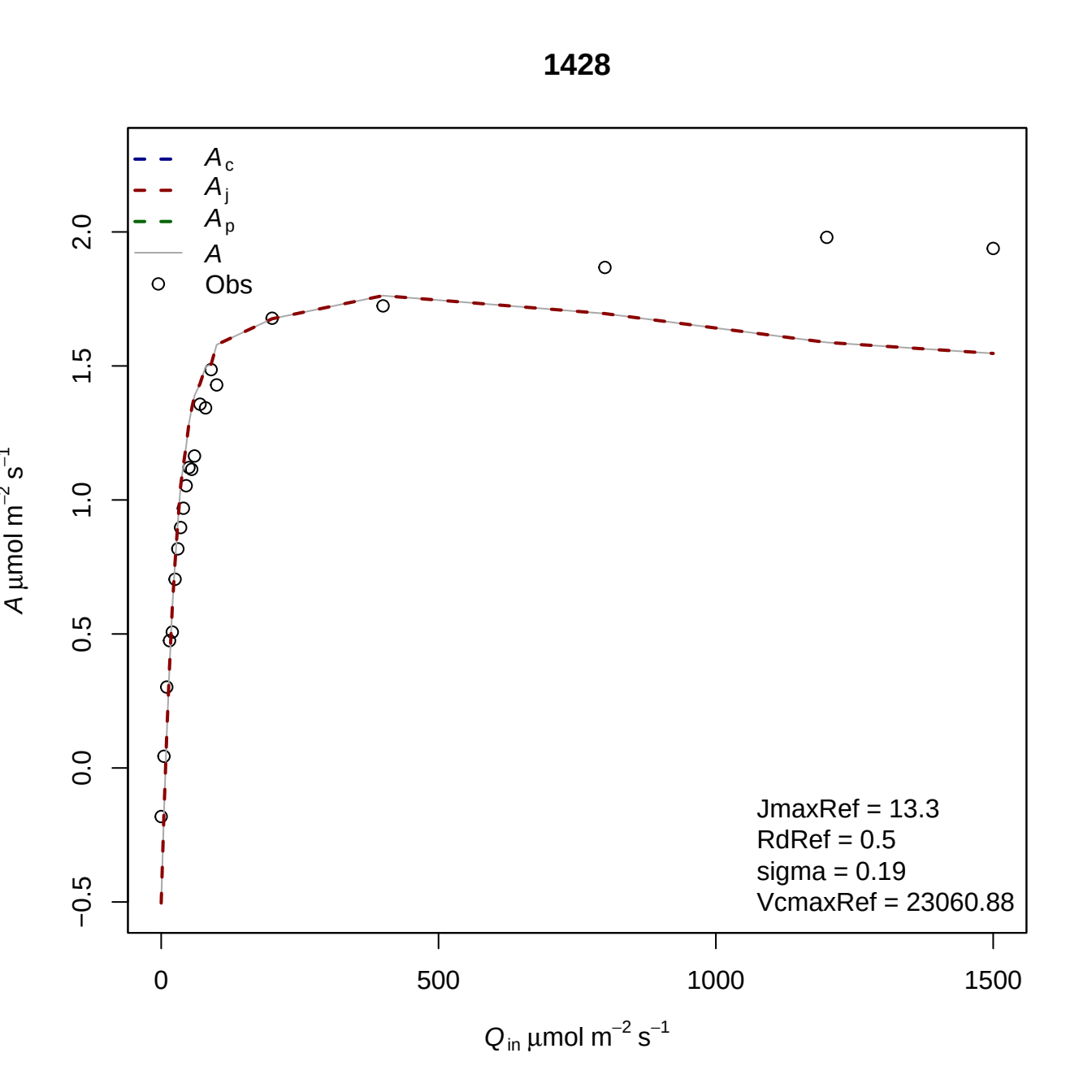

1423
A_c
A_j
A_p
A
Obs
JmaxRef = 25.11
RdRef = 1.38
sigma = 0.57
VcmaxRef = 34635.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

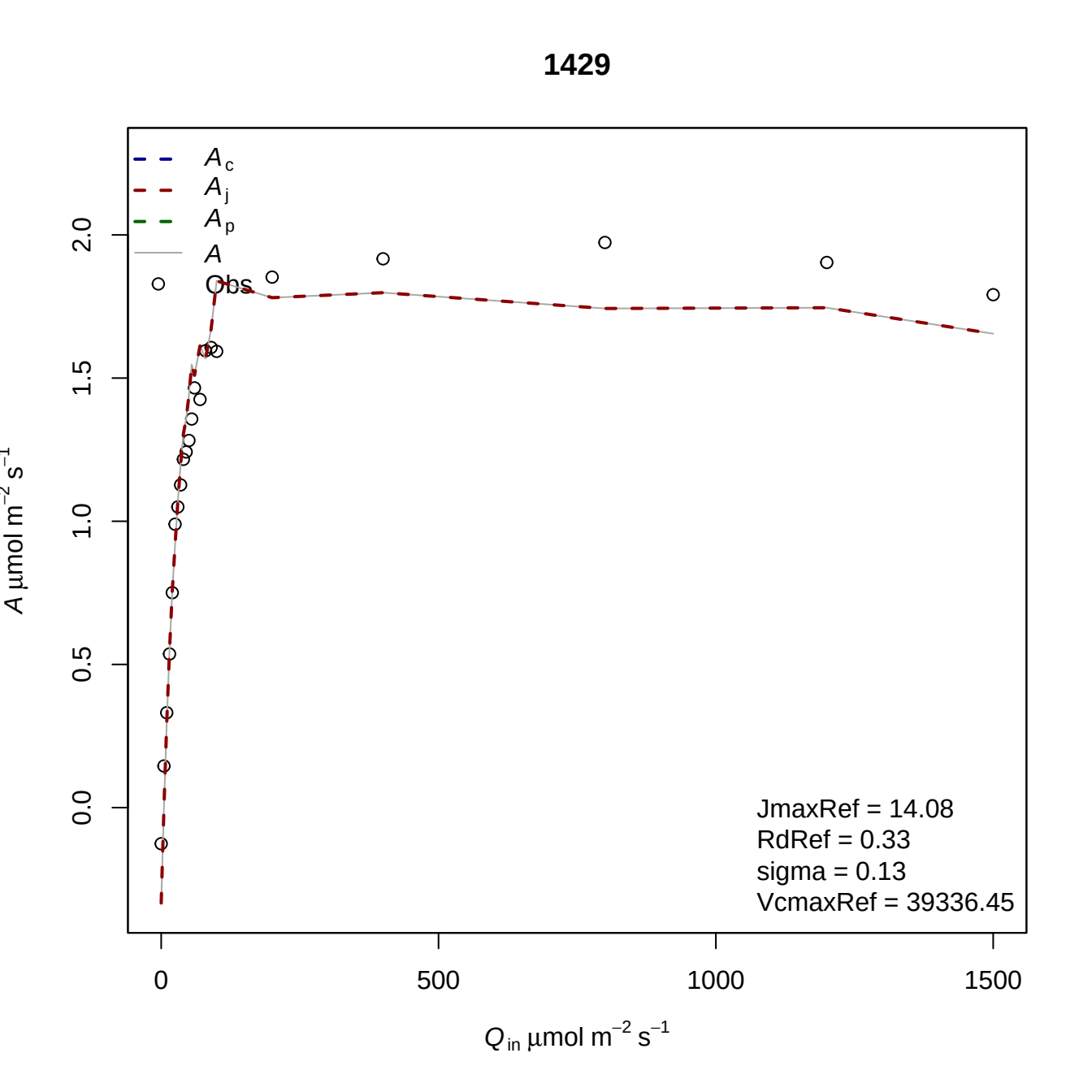

1424
A_c
A_j
A_p
A
Obs
JmaxRef = 33.02
RdRef = 0.53
sigma = 0.13
VcmaxRef = 97763.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

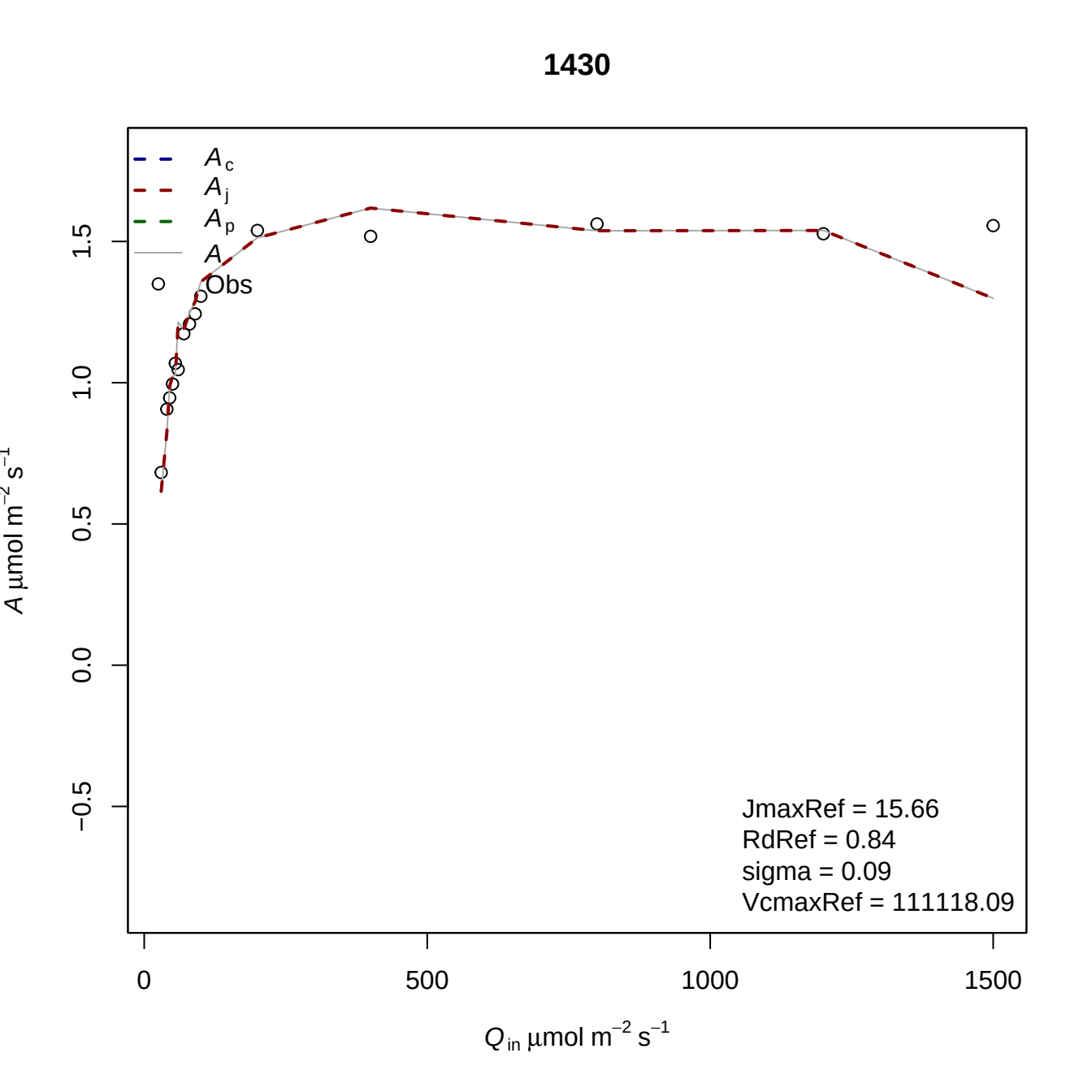

1425
A_c
A_j
A_p
A
Obs
JmaxRef = 19.05
RdRef = 0.87
sigma = 0.15
VcmaxRef = 750794.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500

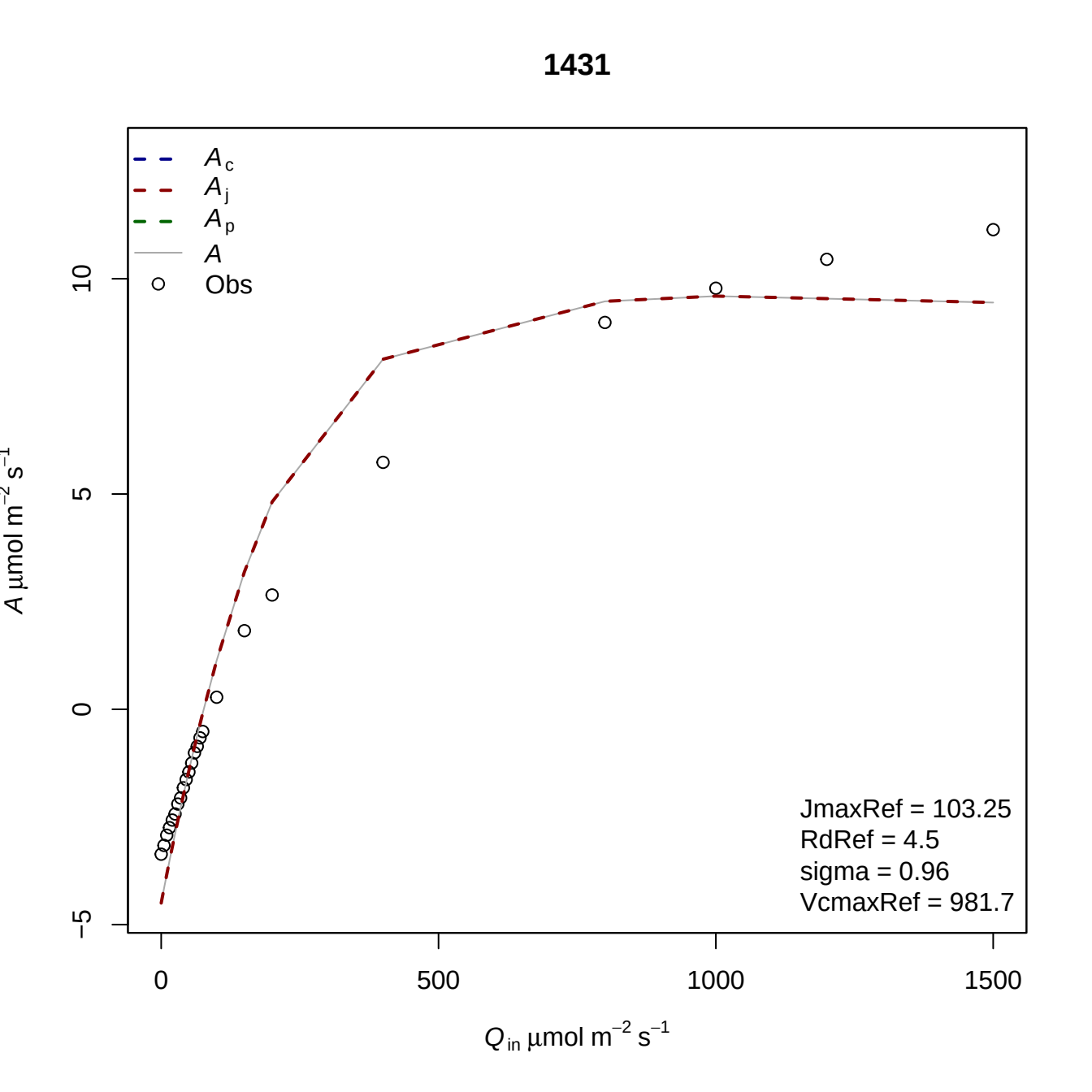

1426
A_c
A_j
A_p
A
Obs
JmaxRef = 31.61
RdRef = 0.33
sigma = 0.11
VcmaxRef = 326.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

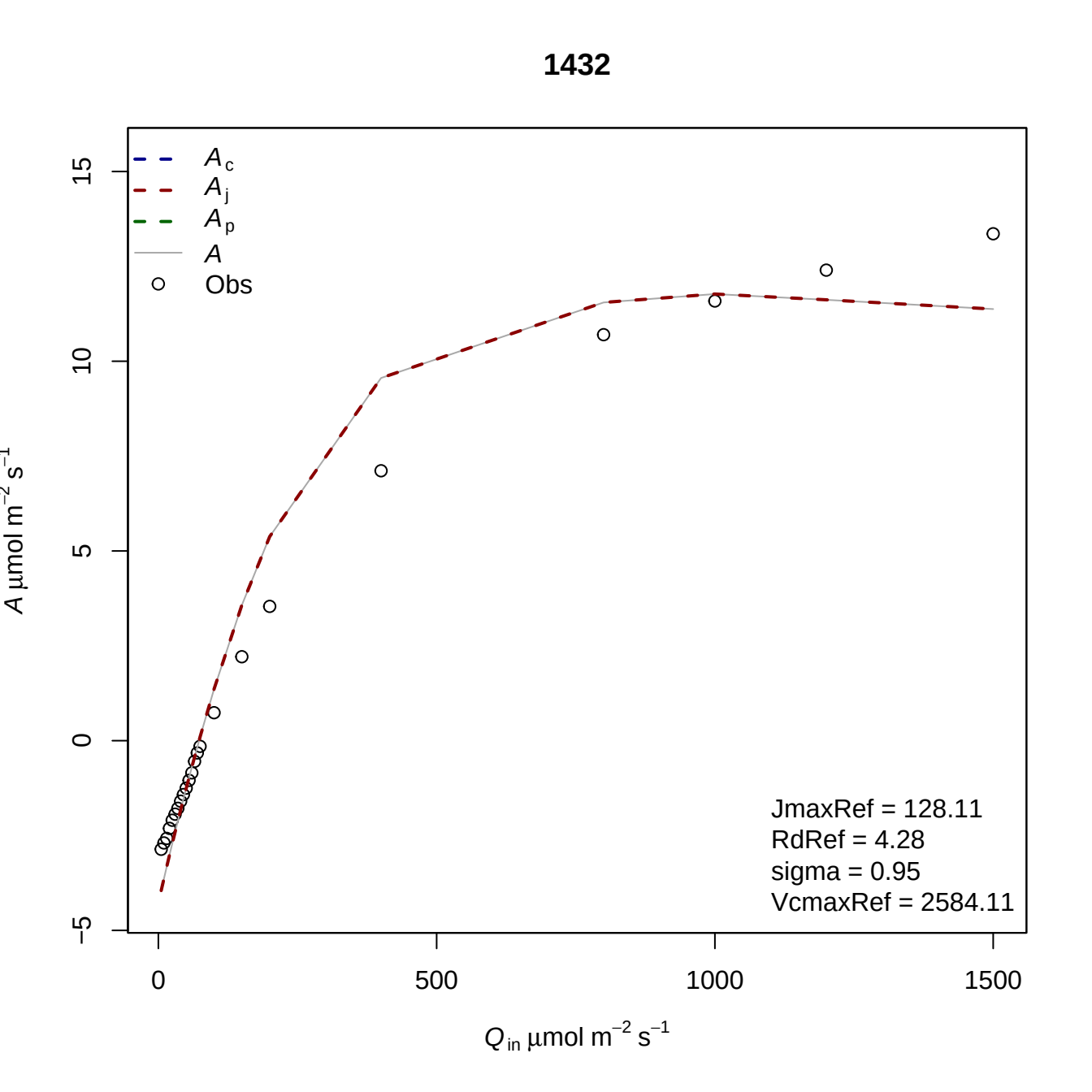

1427
A_c
A_j
A_p
A
Obs
JmaxRef = 17.52
RdRef = 0.32
sigma = 0.09
VcmaxRef = 2889.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500

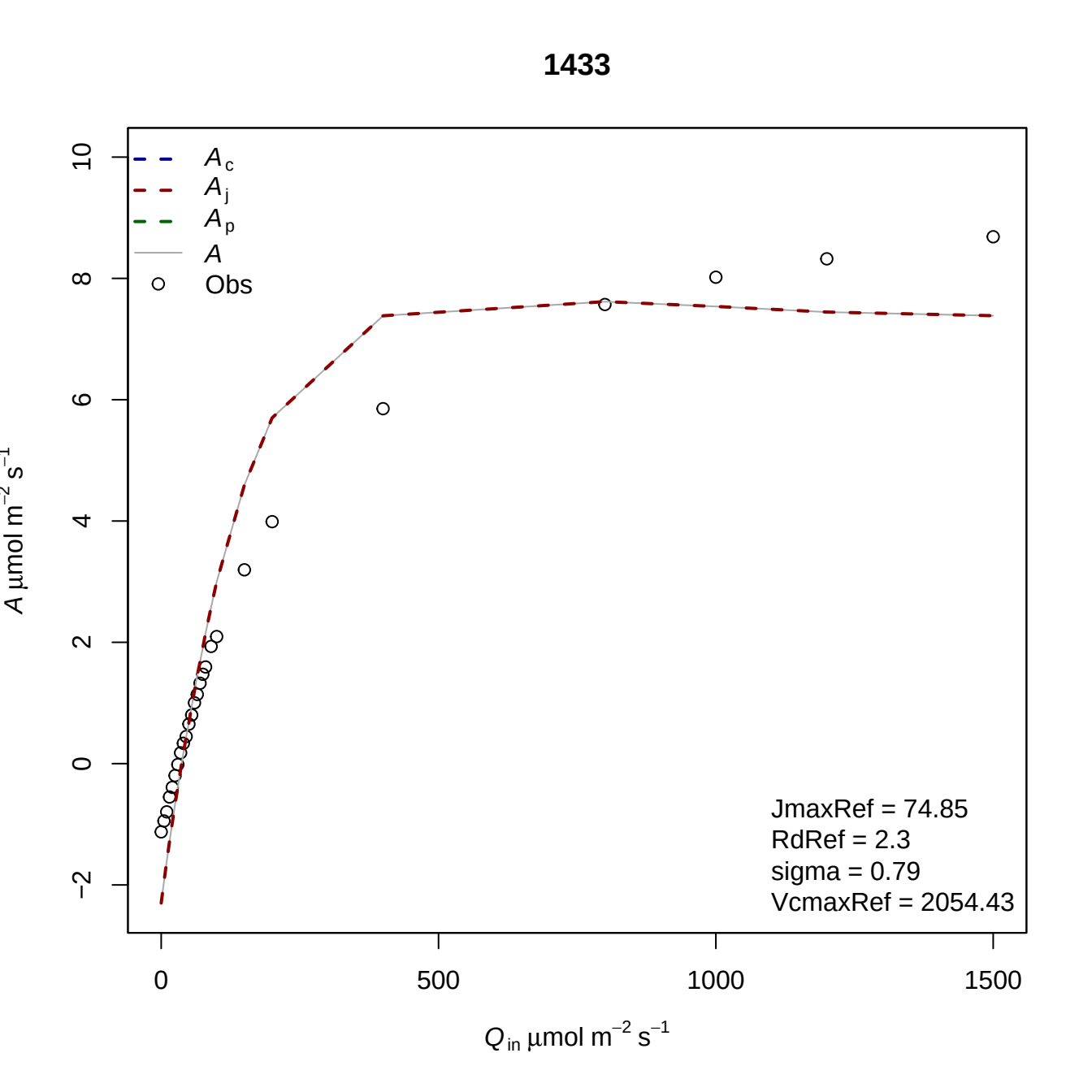

1428
A_c
A_j
A_p
A
Obs
JmaxRef = 13.3
RdRef = 0.5
sigma = 0.19
VcmaxRef = 23060.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500

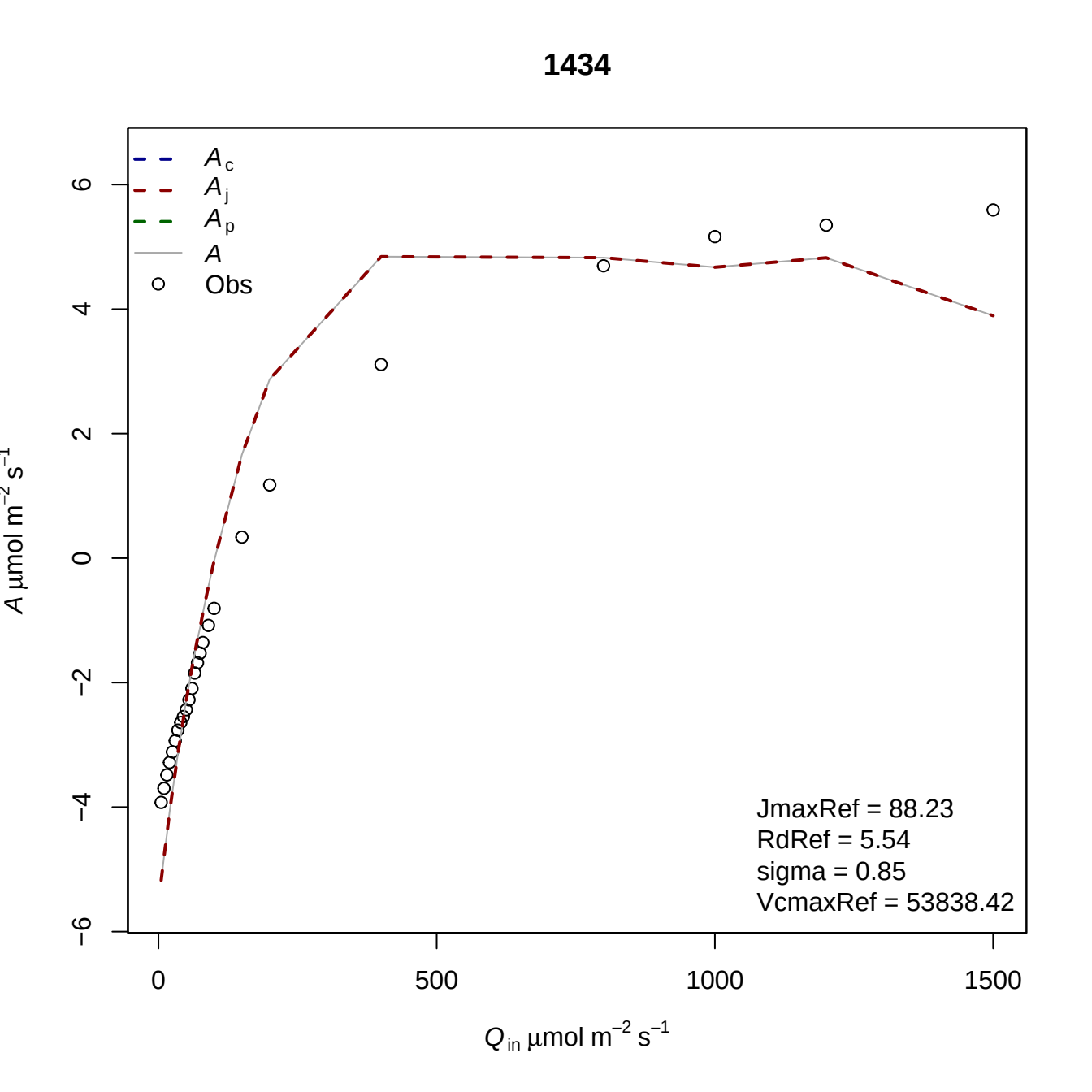

1429
A_c
A_j
A_p
A
Obs
JmaxRef = 14.08
RdRef = 0.33
sigma = 0.13
VcmaxRef = 39336.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500

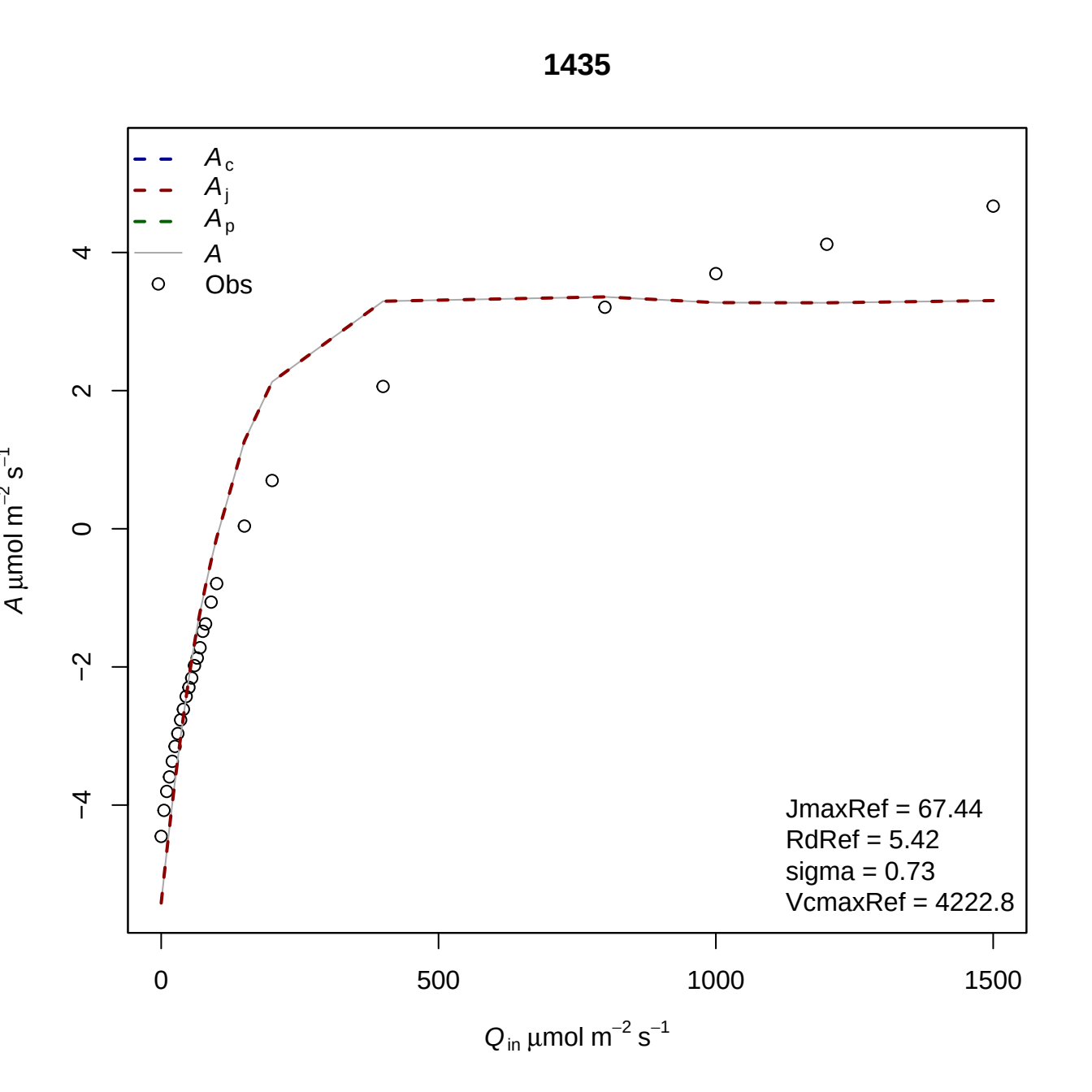

1430
A_c
A_j
A_p
A
Obs
JmaxRef = 15.66
RdRef = 0.84
sigma = 0.09
VcmaxRef = 111118.09
A µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

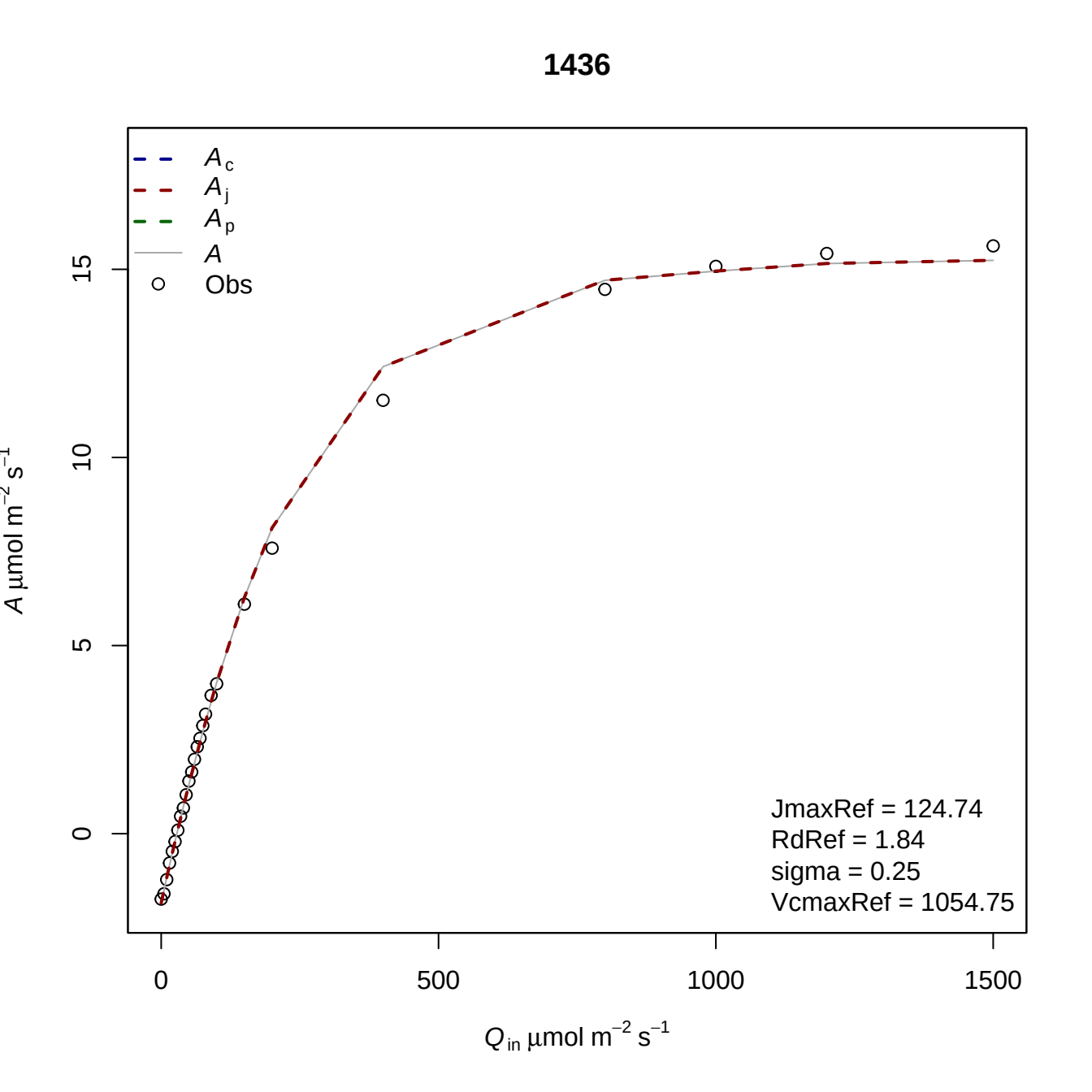

1431
A_c
A_j
A_p
A
Obs
JmaxRef = 103.25
RdRef = 4.5
sigma = 0.96
VcmaxRef = 981.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

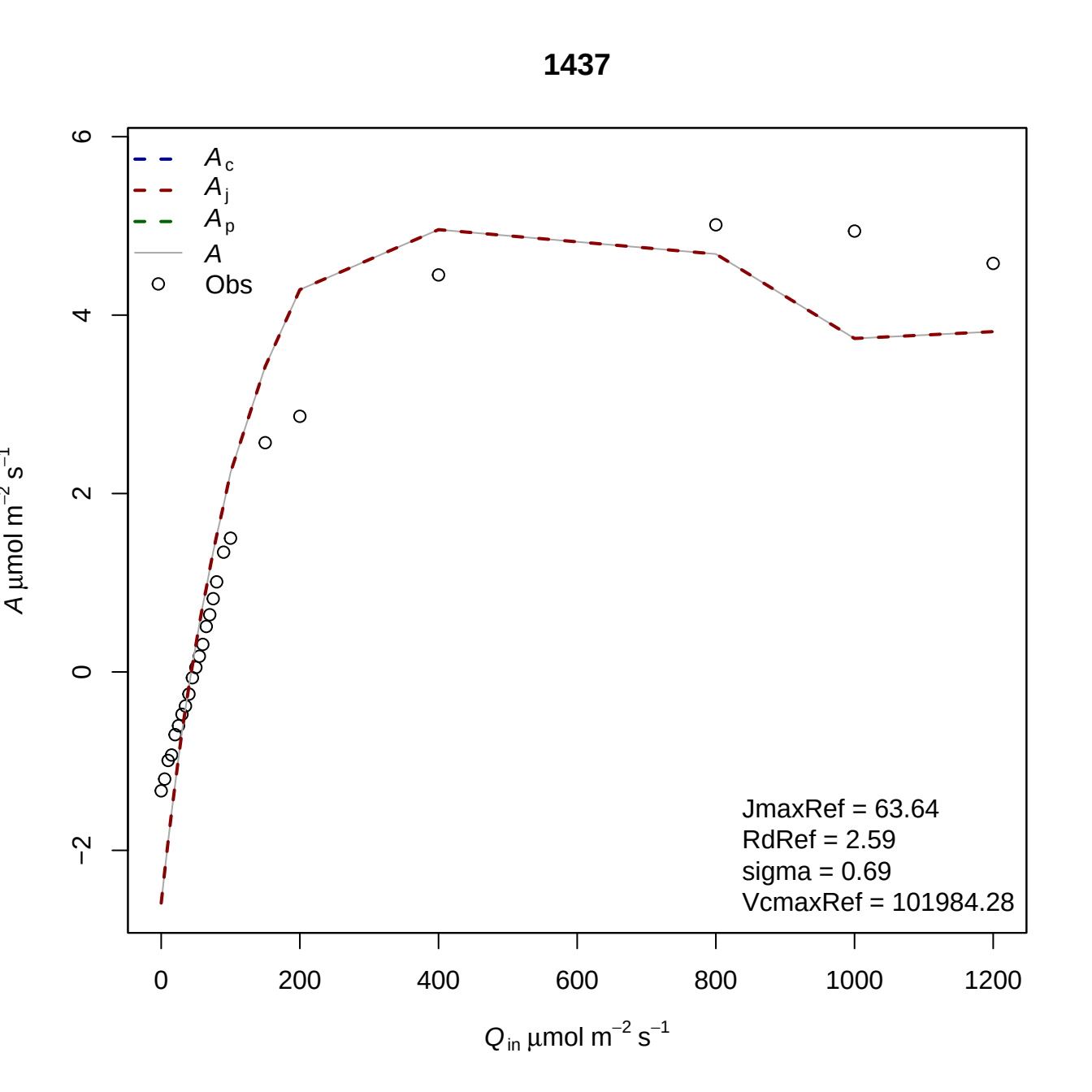

1432
A_c
A_j
A_p
A
Obs
JmaxRef = 128.11
RdRef = 4.28
sigma = 0.95
VcmaxRef = 2584.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500

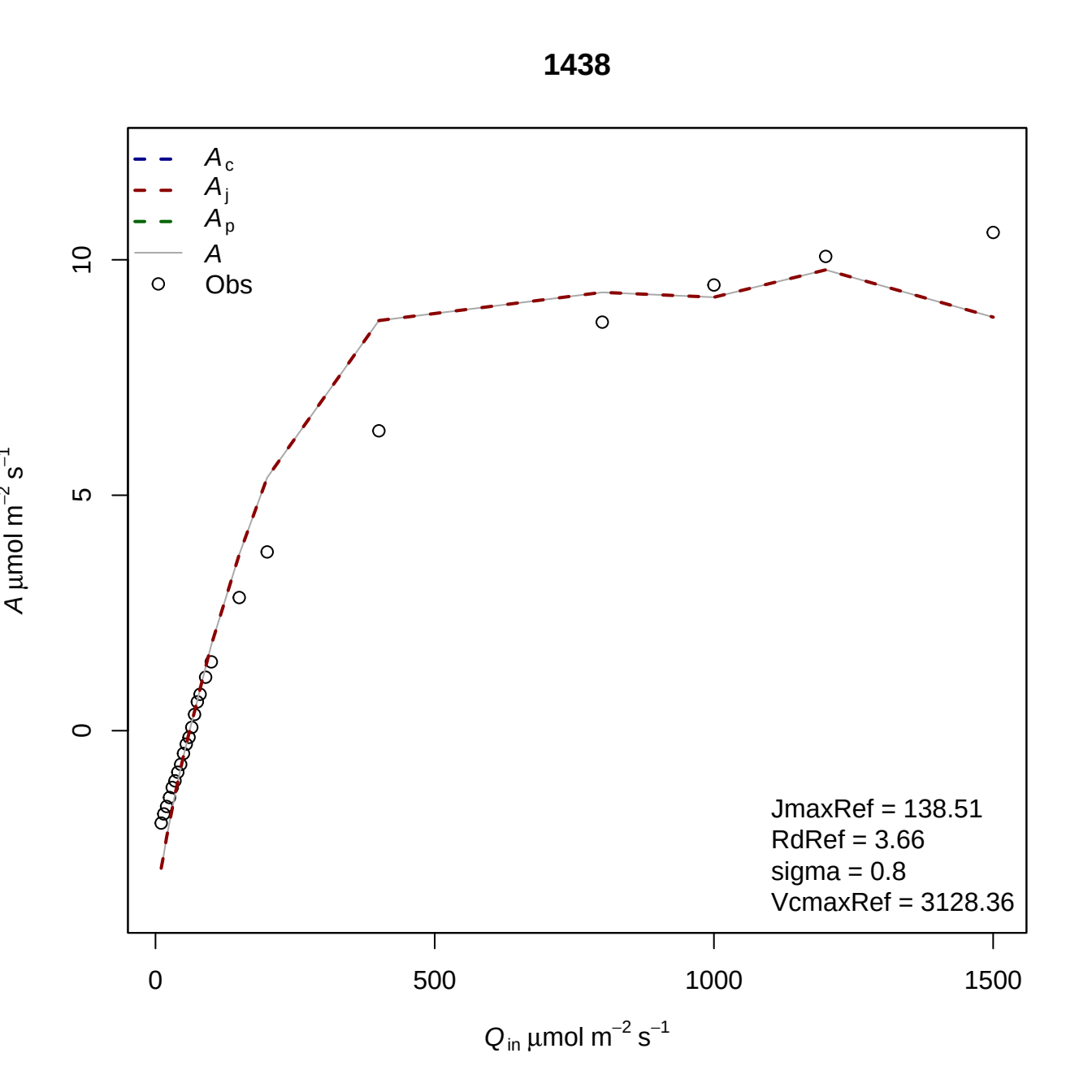

1433
A_c
A_j
A_p
A
Obs
JmaxRef = 74.85
RdRef = 2.3
sigma = 0.79
VcmaxRef = 2054.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

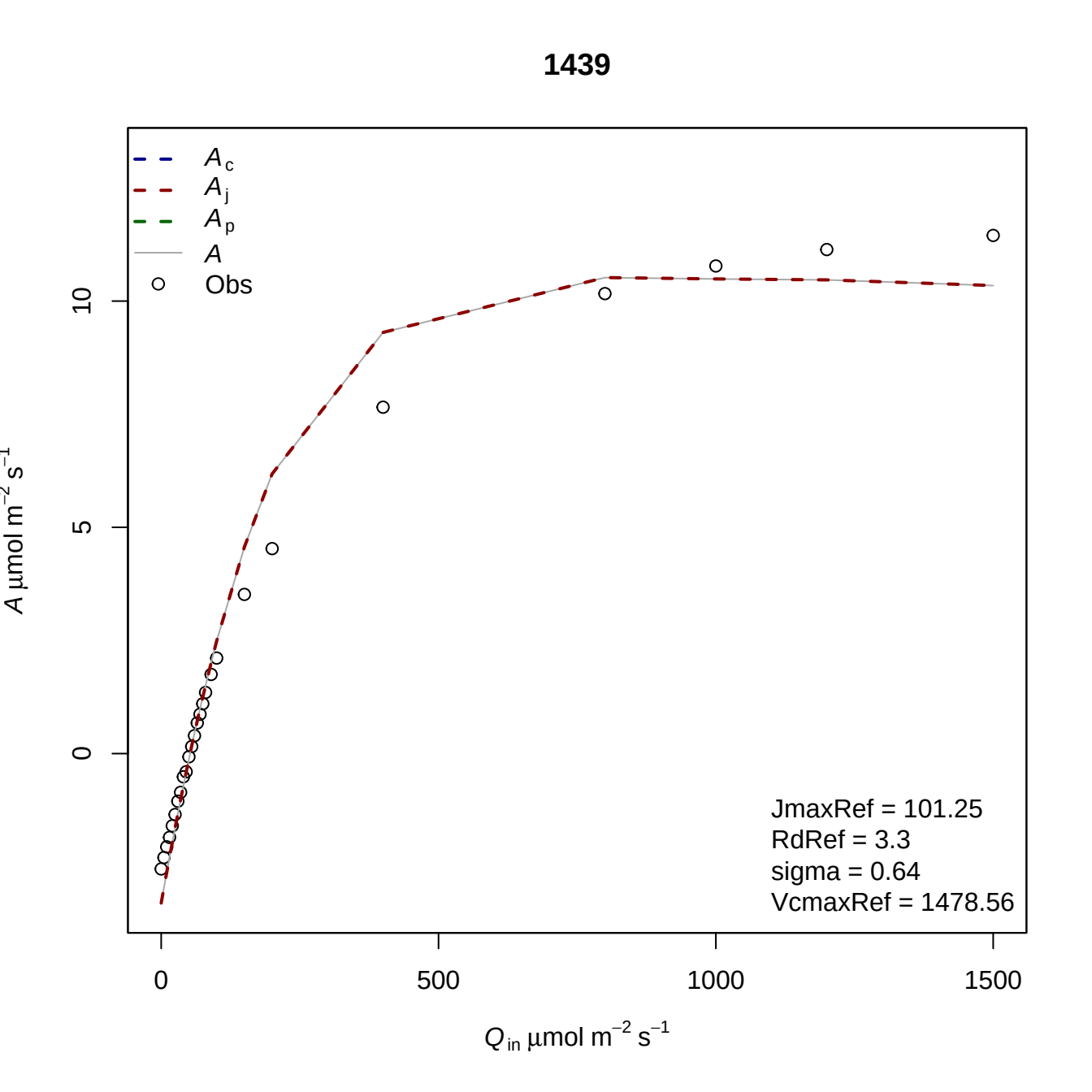

1434
A_c
A_j
A_p
A
Obs
JmaxRef = 88.23
RdRef = 5.54
sigma = 0.85
VcmaxRef = 53838.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
−6
0
500
1000
1500

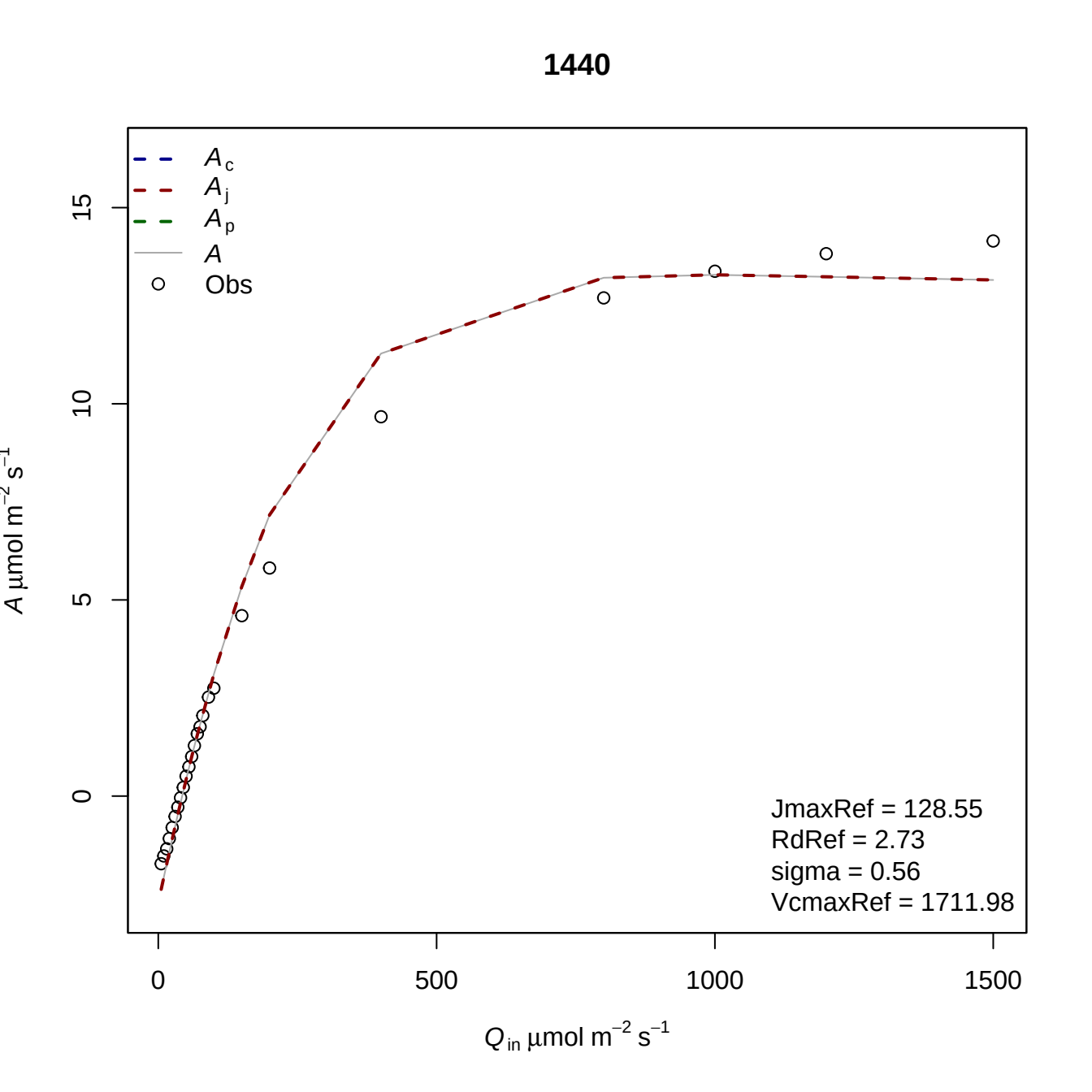

1435
A_c
A_j
A_p
A
Obs
JmaxRef = 67.44
RdRef = 5.42
sigma = 0.73
VcmaxRef = 4222.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

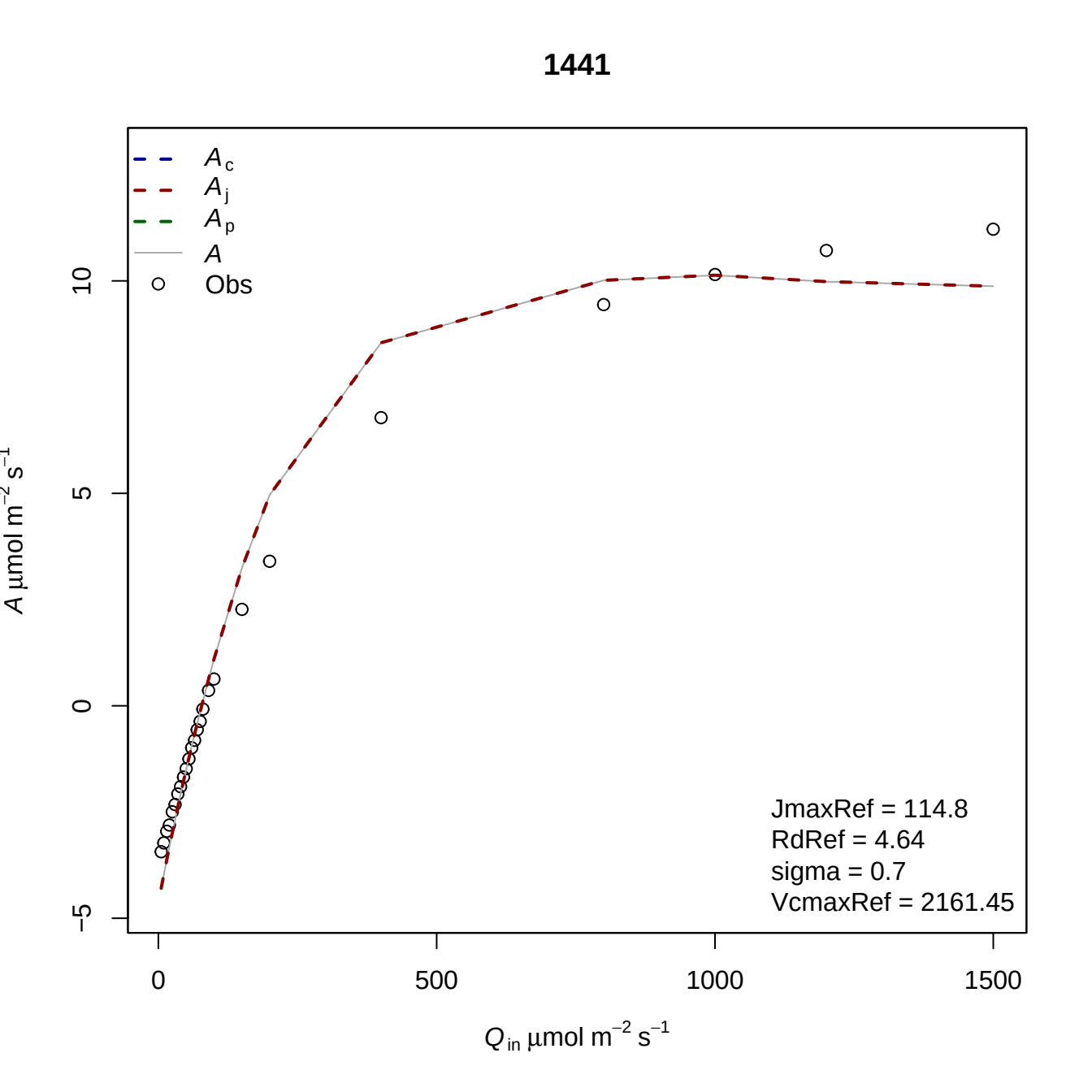

1436
A_c
A_j
A_p
A
Obs
JmaxRef = 124.74
RdRef = 1.84
sigma = 0.25
VcmaxRef = 1054.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

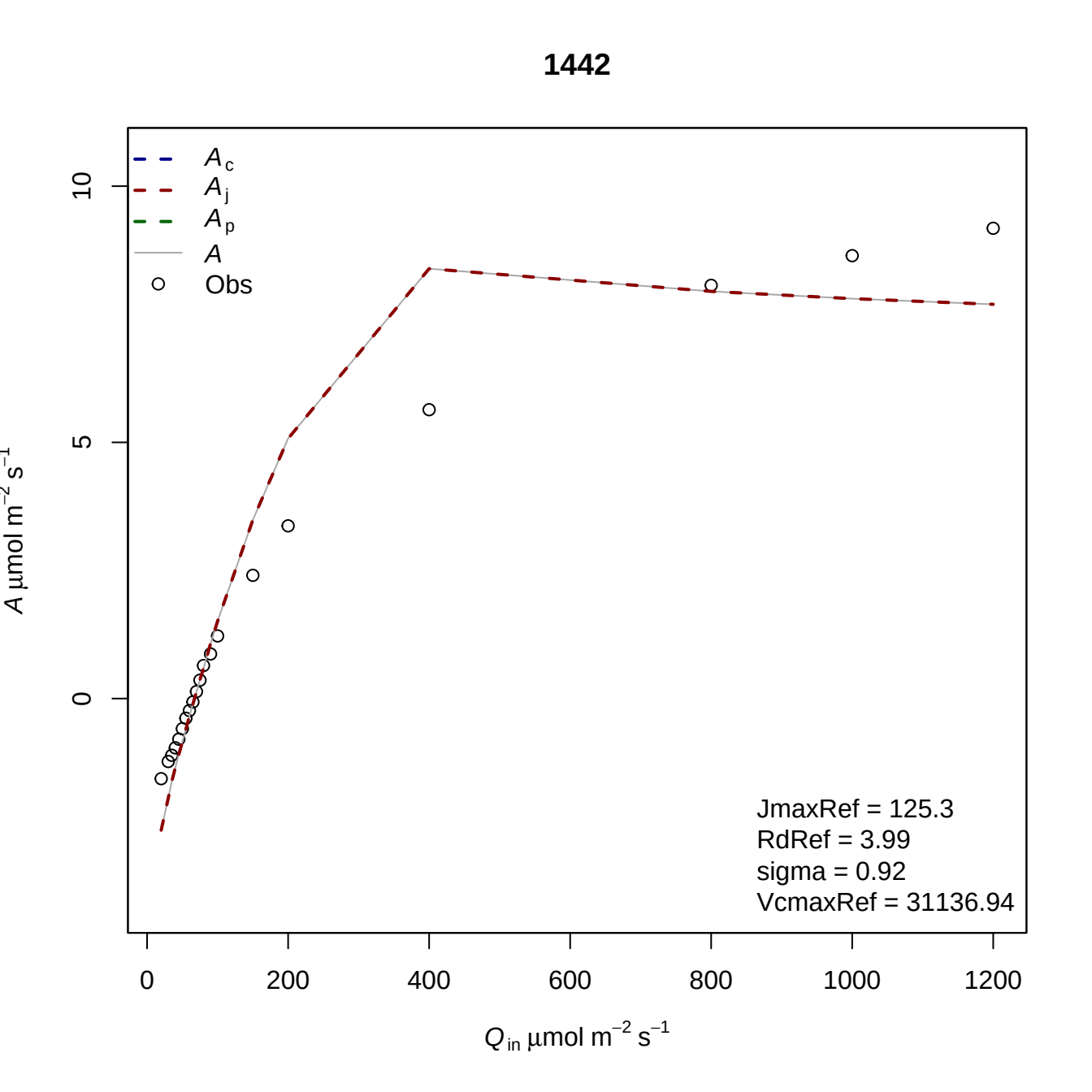

1437
A_c
A_j
A_p
A
Obs
JmaxRef = 63.64
RdRef = 2.59
sigma = 0.69
VcmaxRef = 101984.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

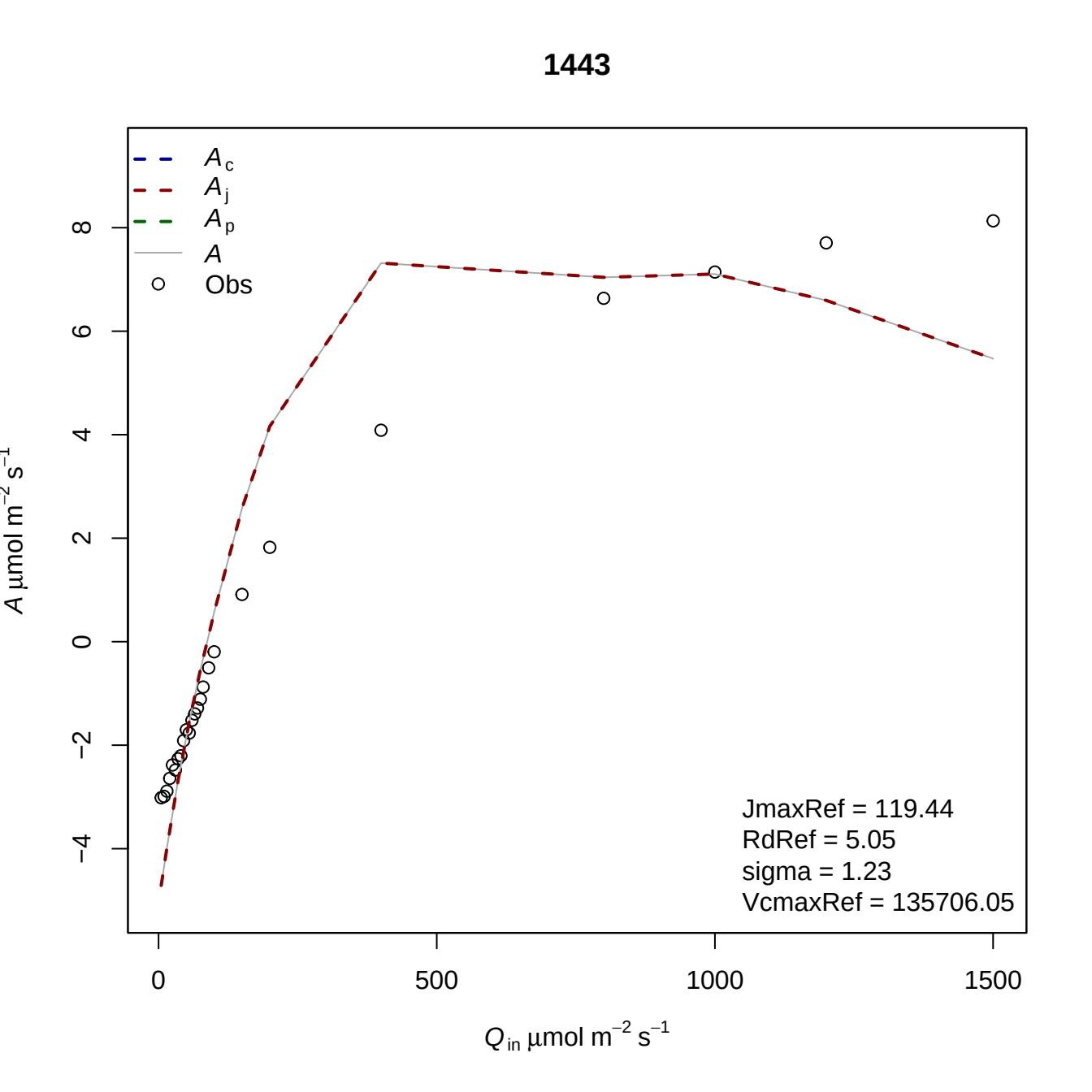

1438
A_c
A_j
A_p
A
Obs
JmaxRef = 138.51
RdRef = 3.66
sigma = 0.8
VcmaxRef = 3128.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

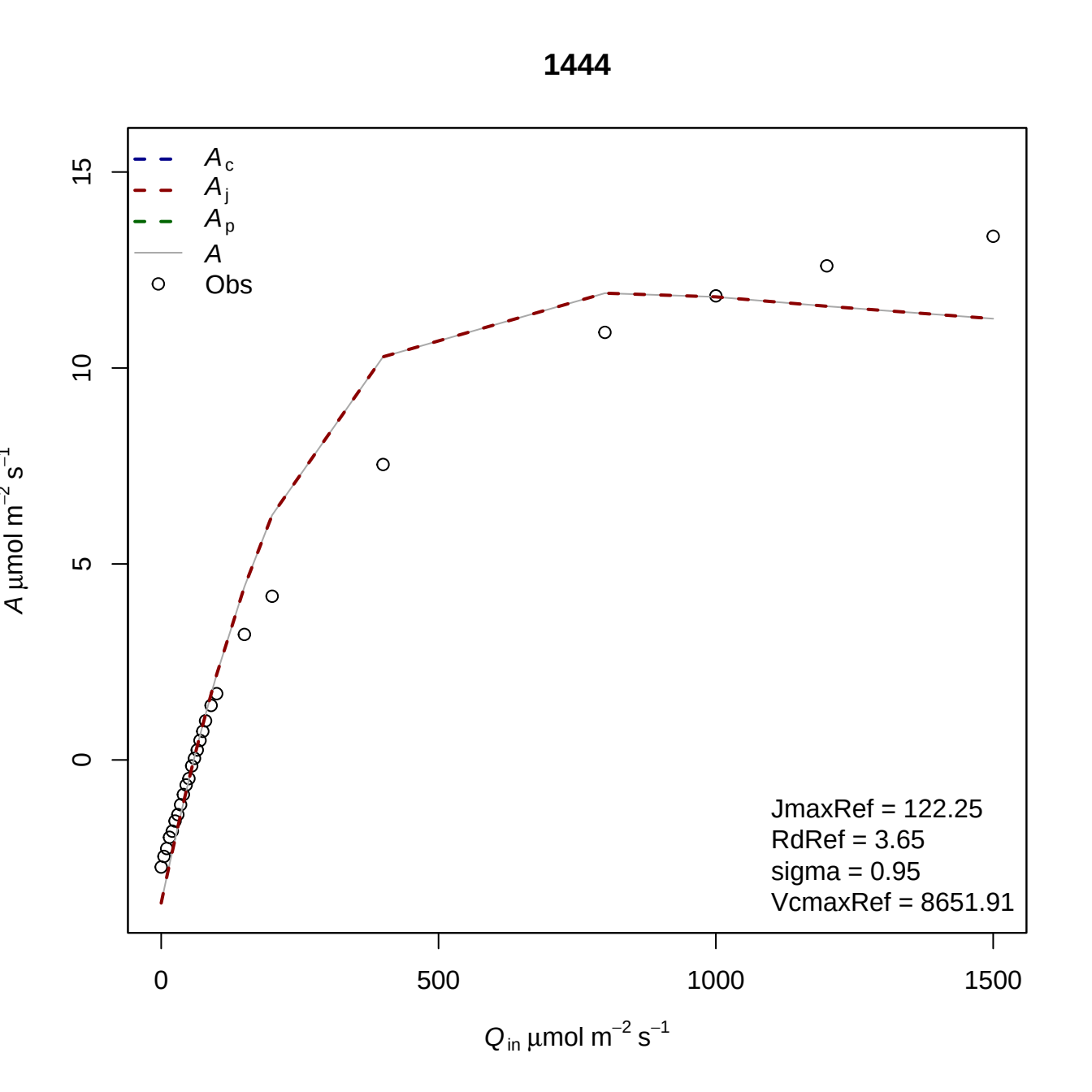

1439
A_c
A_j
A_p
A
Obs
JmaxRef = 101.25
RdRef = 3.3
sigma = 0.64
VcmaxRef = 1478.56
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

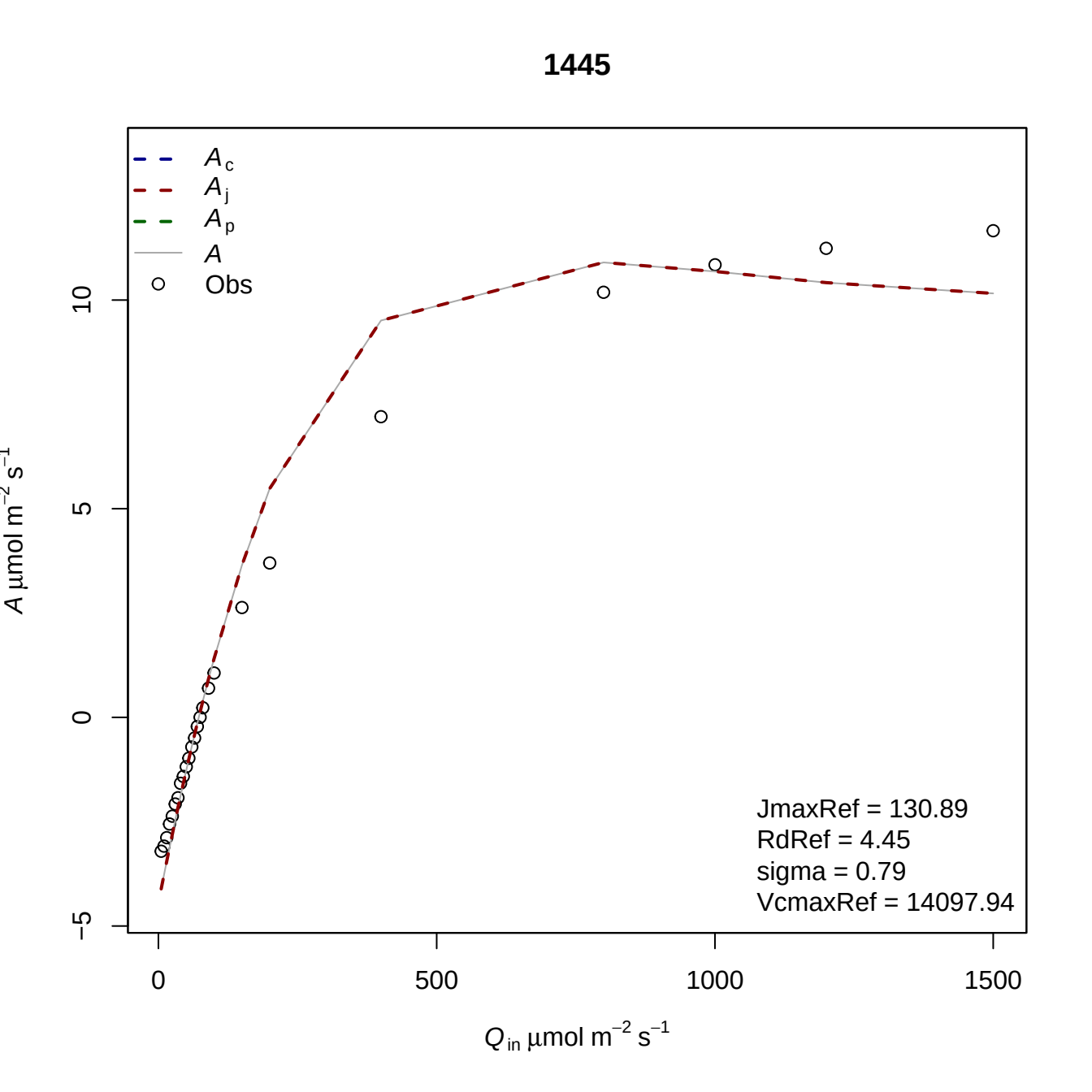

1440
A_c
A_j
A_p
A
Obs
JmaxRef = 128.55
RdRef = 2.73
sigma = 0.56
VcmaxRef = 1711.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

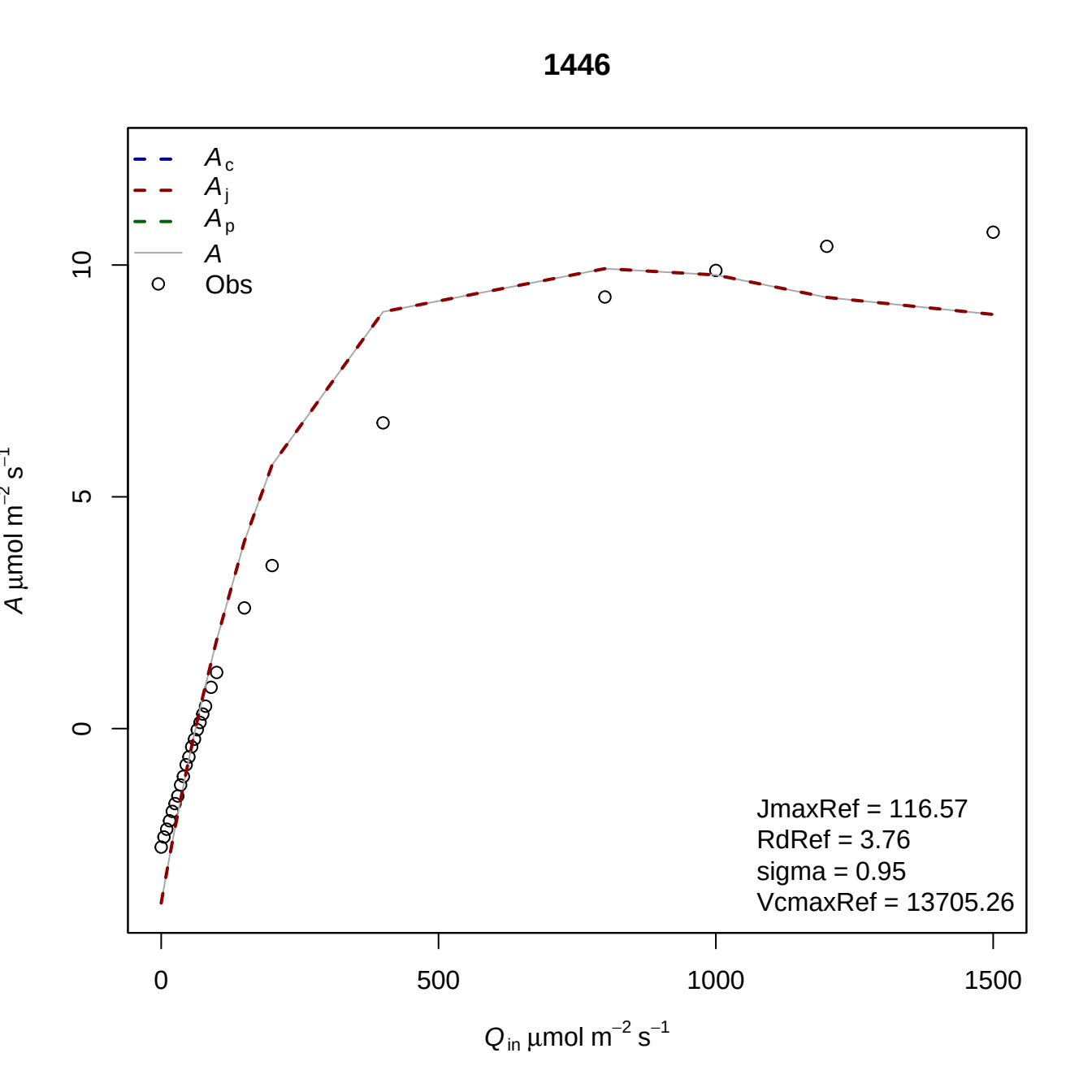

1441
A_c
A_j
A_p
A
Obs
JmaxRef = 114.8
RdRef = 4.64
sigma = 0.7
VcmaxRef = 2161.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

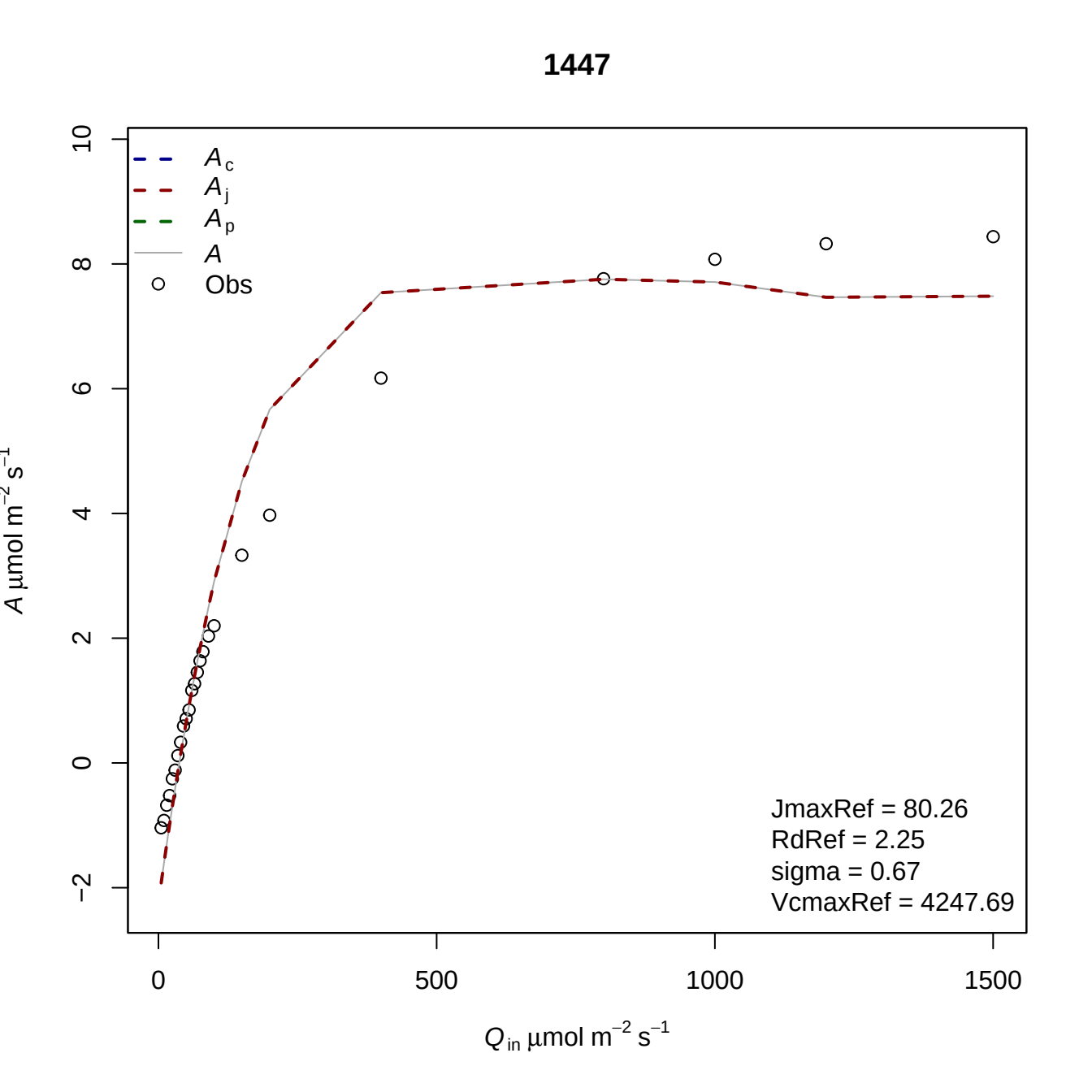

1442
A_c
A_j
A_p
A
Obs
JmaxRef = 125.3
RdRef = 3.99
sigma = 0.92
VcmaxRef = 31136.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

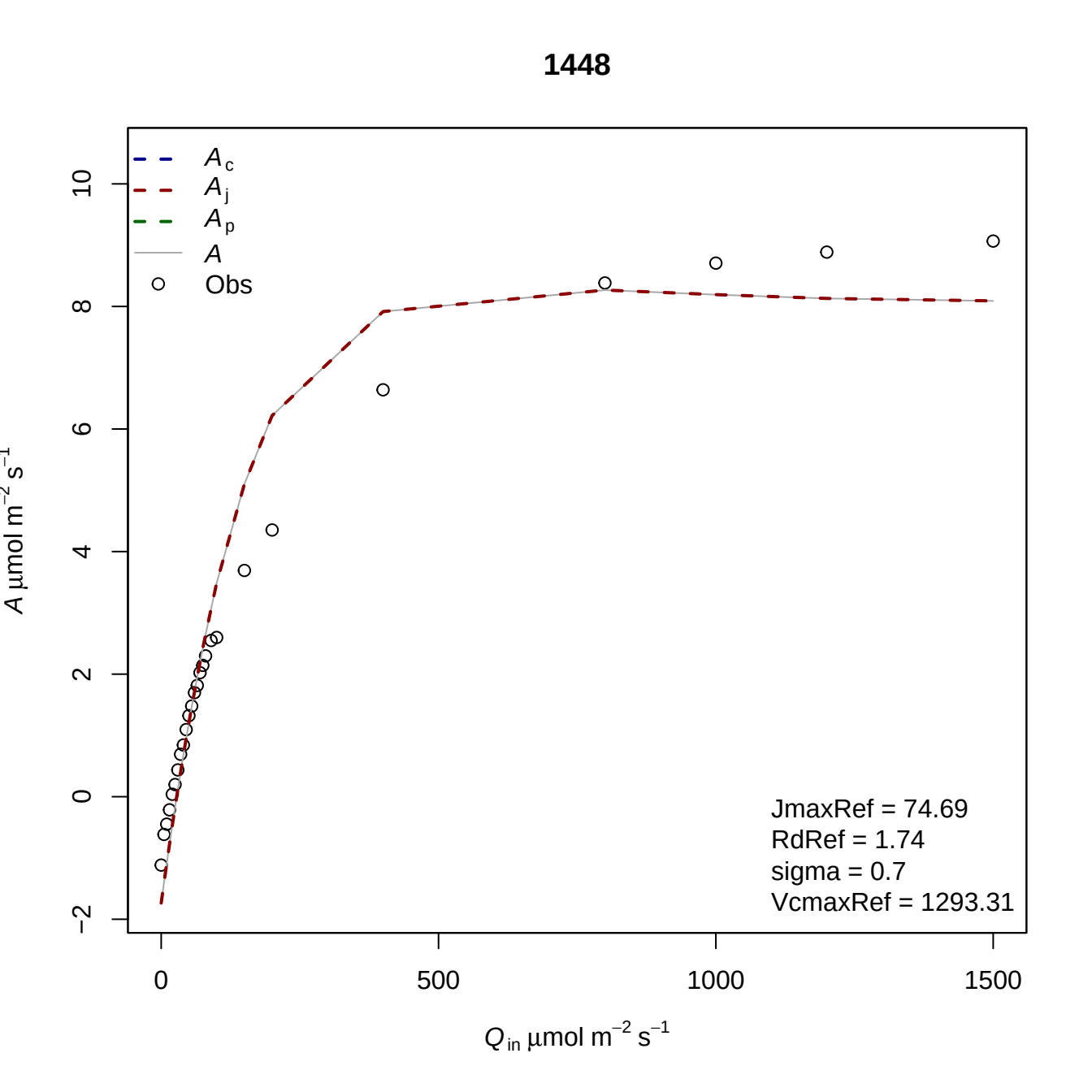

1443
A_c
A_j
A_p
A
Obs
JmaxRef = 119.44
RdRef = 5.05
sigma = 1.23
VcmaxRef = 135706.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

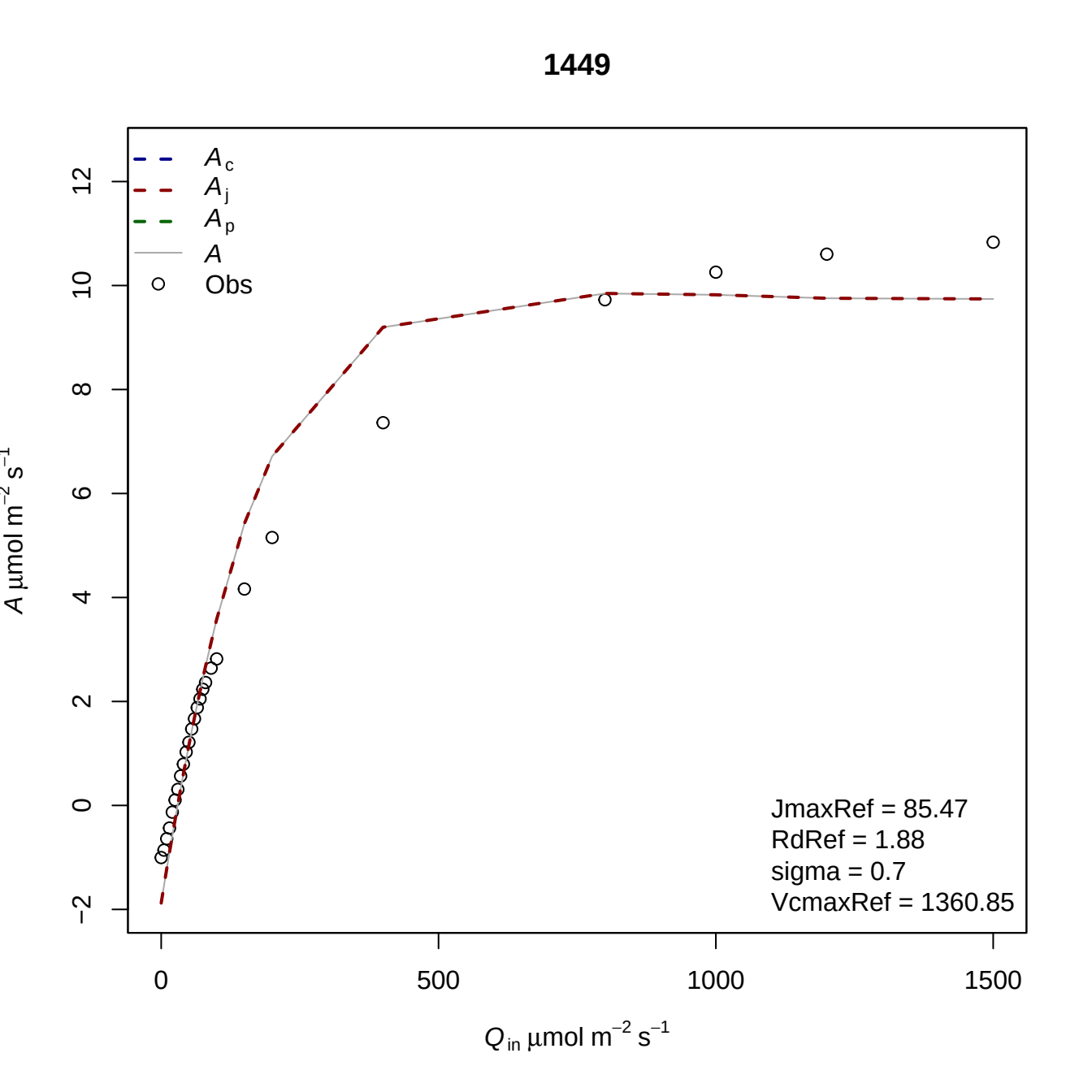

1444
A_c
A_j
A_p
A
Obs
JmaxRef = 122.25
RdRef = 3.65
sigma = 0.95
VcmaxRef = 8651.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

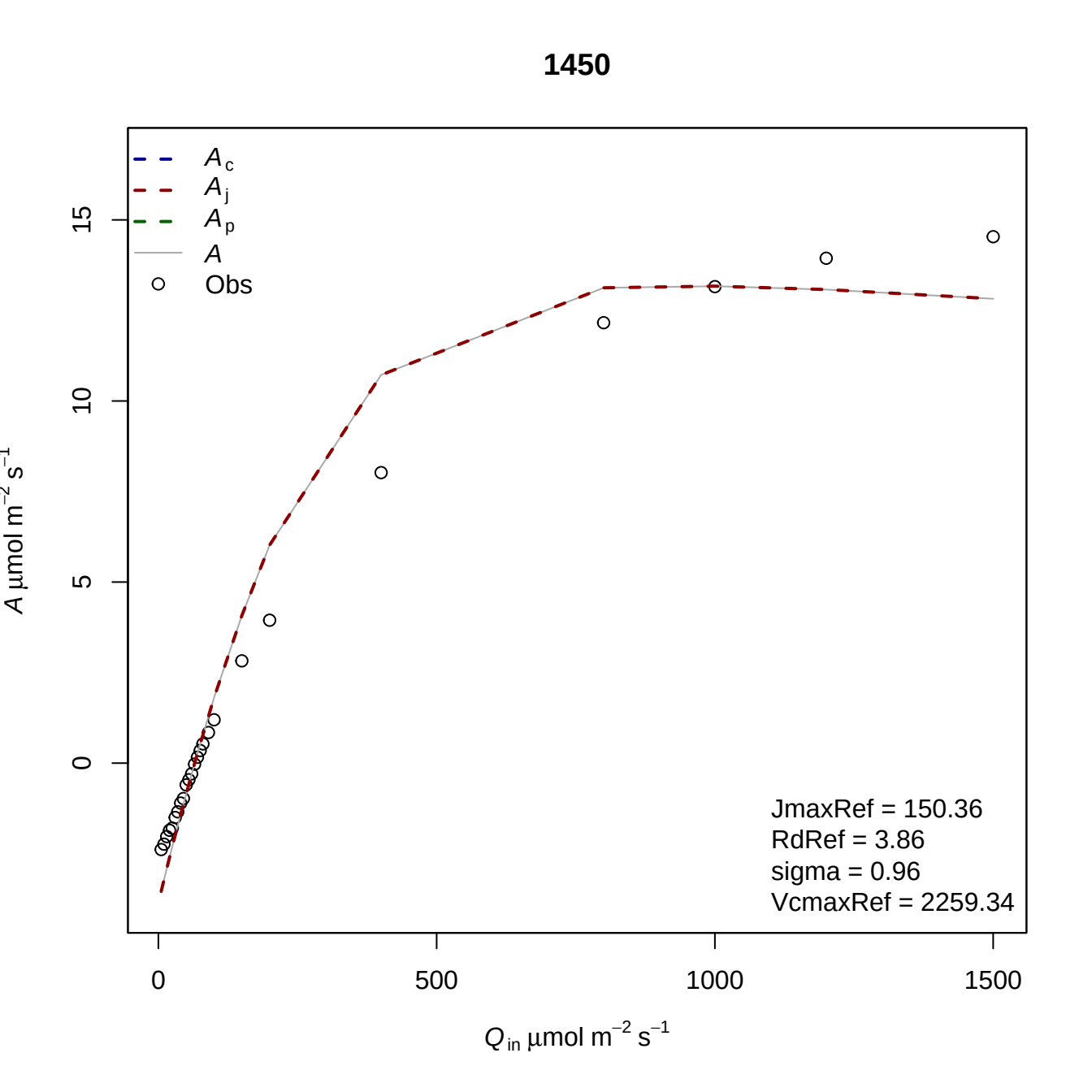

1445
A_c
A_j
A_p
A
Obs
JmaxRef = 130.89
RdRef = 4.45
sigma = 0.79
VcmaxRef = 14097.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

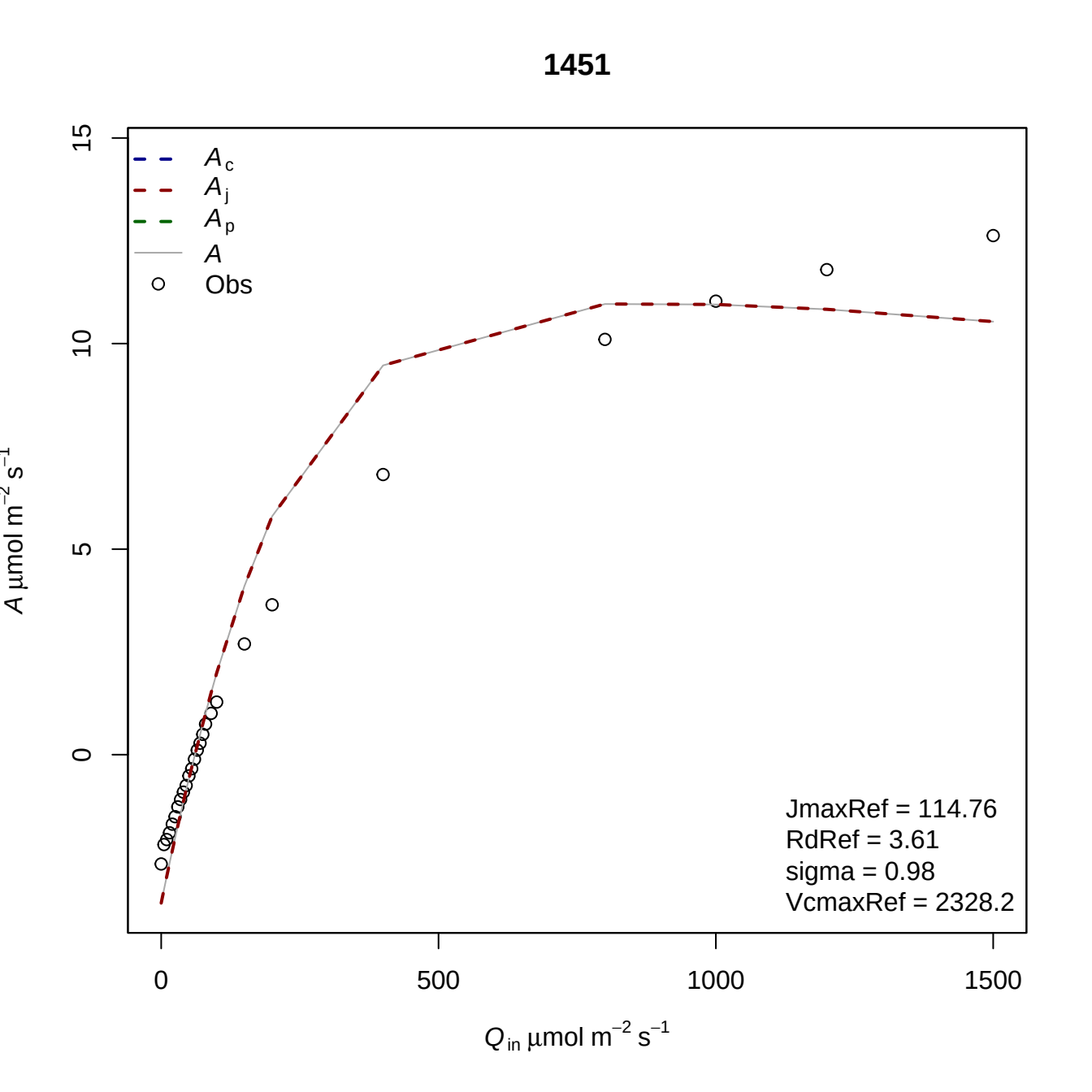

1446
A_c
A_j
A_p
A
Obs
JmaxRef = 116.57
RdRef = 3.76
sigma = 0.95
VcmaxRef = 13705.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

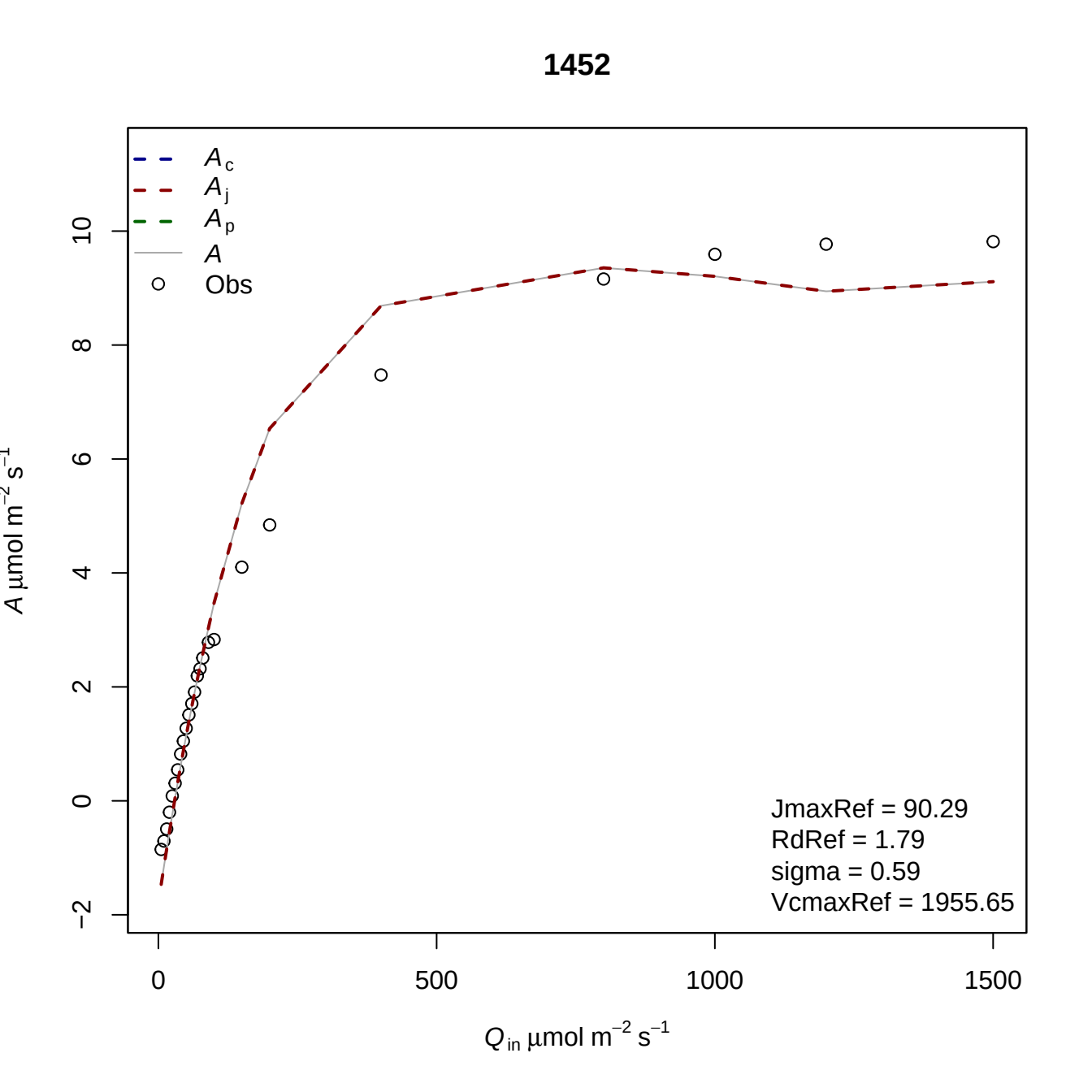

1447
A_c
A_j
A_p
A
Obs
JmaxRef = 80.26
RdRef = 2.25
sigma = 0.67
VcmaxRef = 4247.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

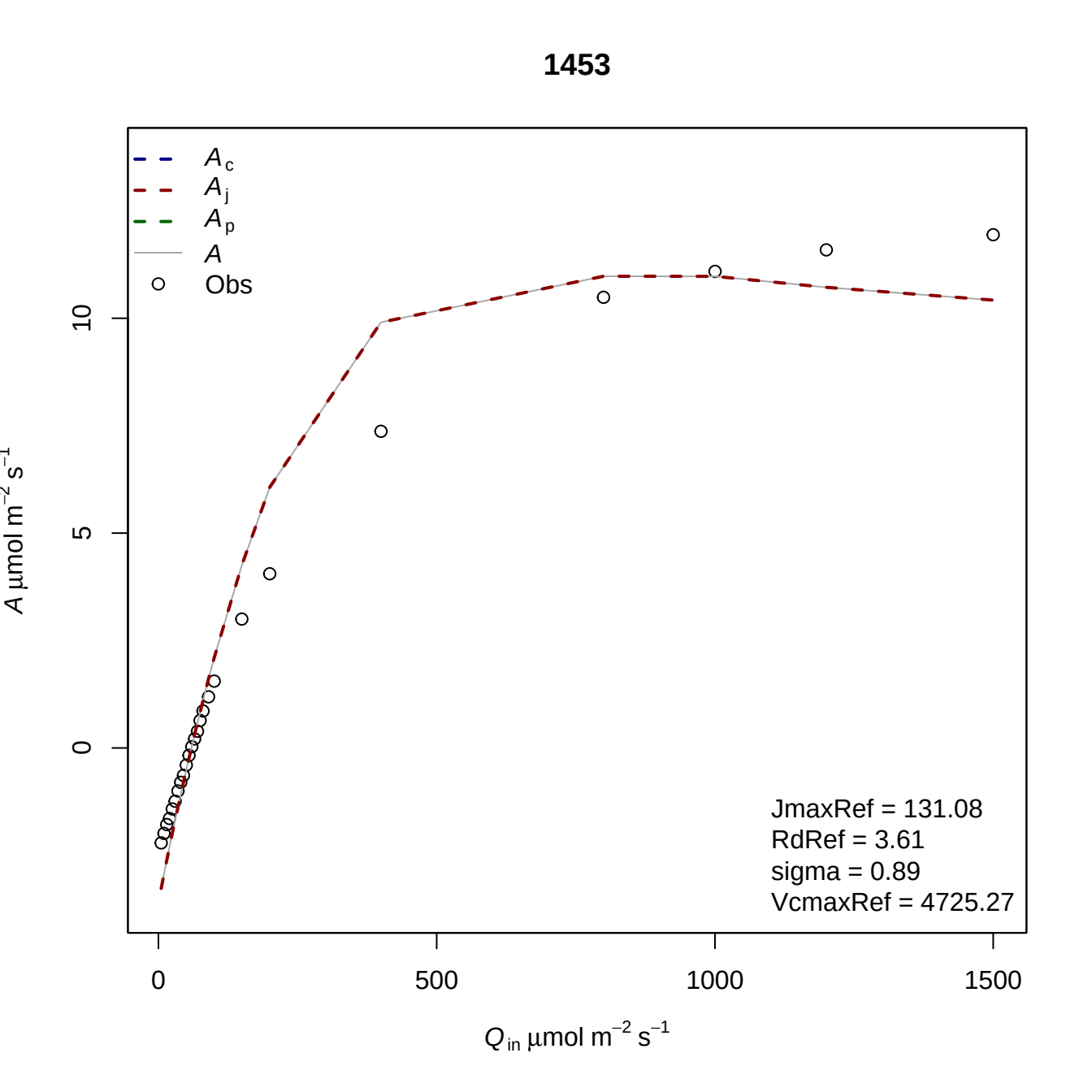

1448
A_c
A_j
A_p
A
Obs
JmaxRef = 74.69
RdRef = 1.74
sigma = 0.7
VcmaxRef = 1293.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

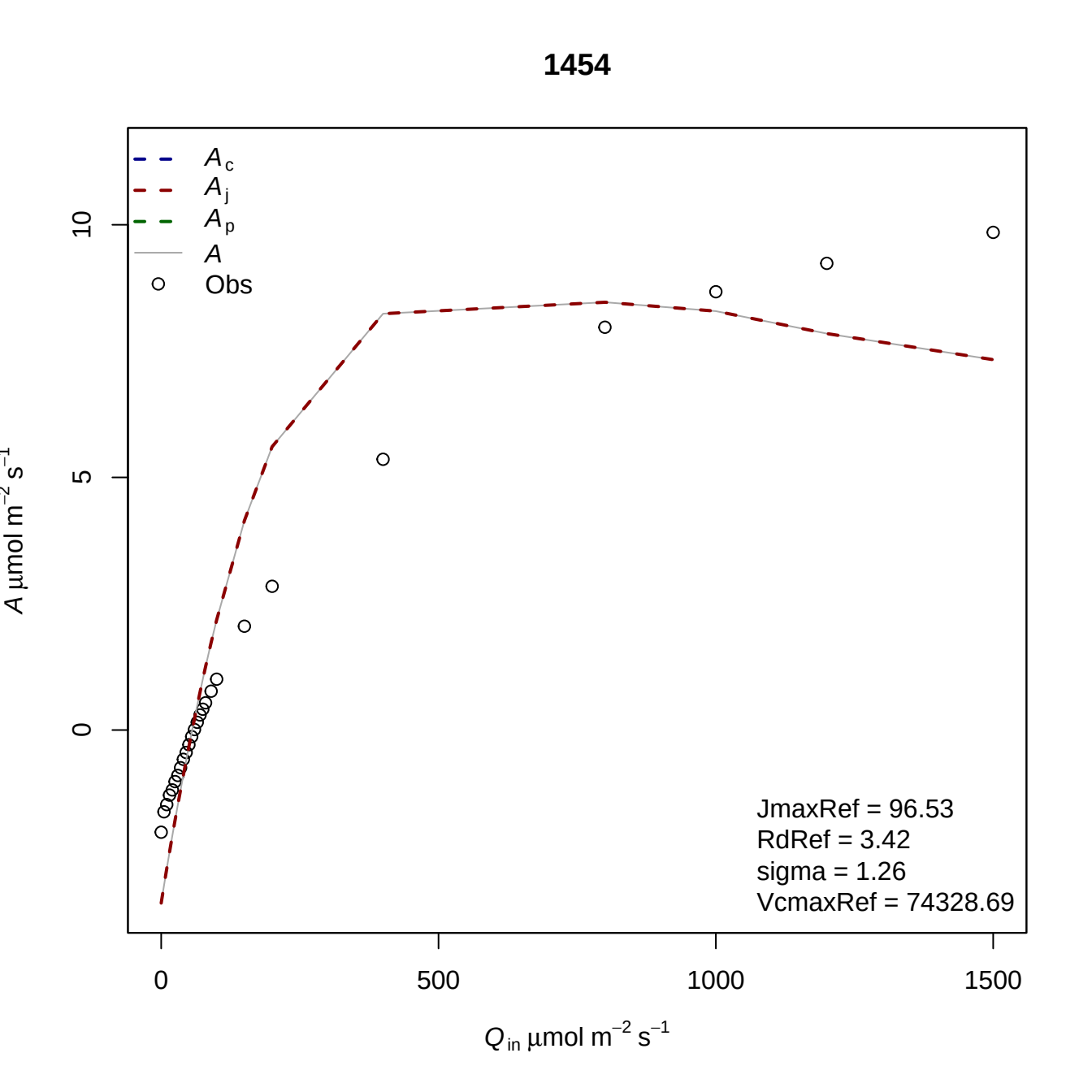

1449
A_c
A_j
A_p
A
Obs
JmaxRef = 85.47
RdRef = 1.88
sigma = 0.7
VcmaxRef = 1360.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

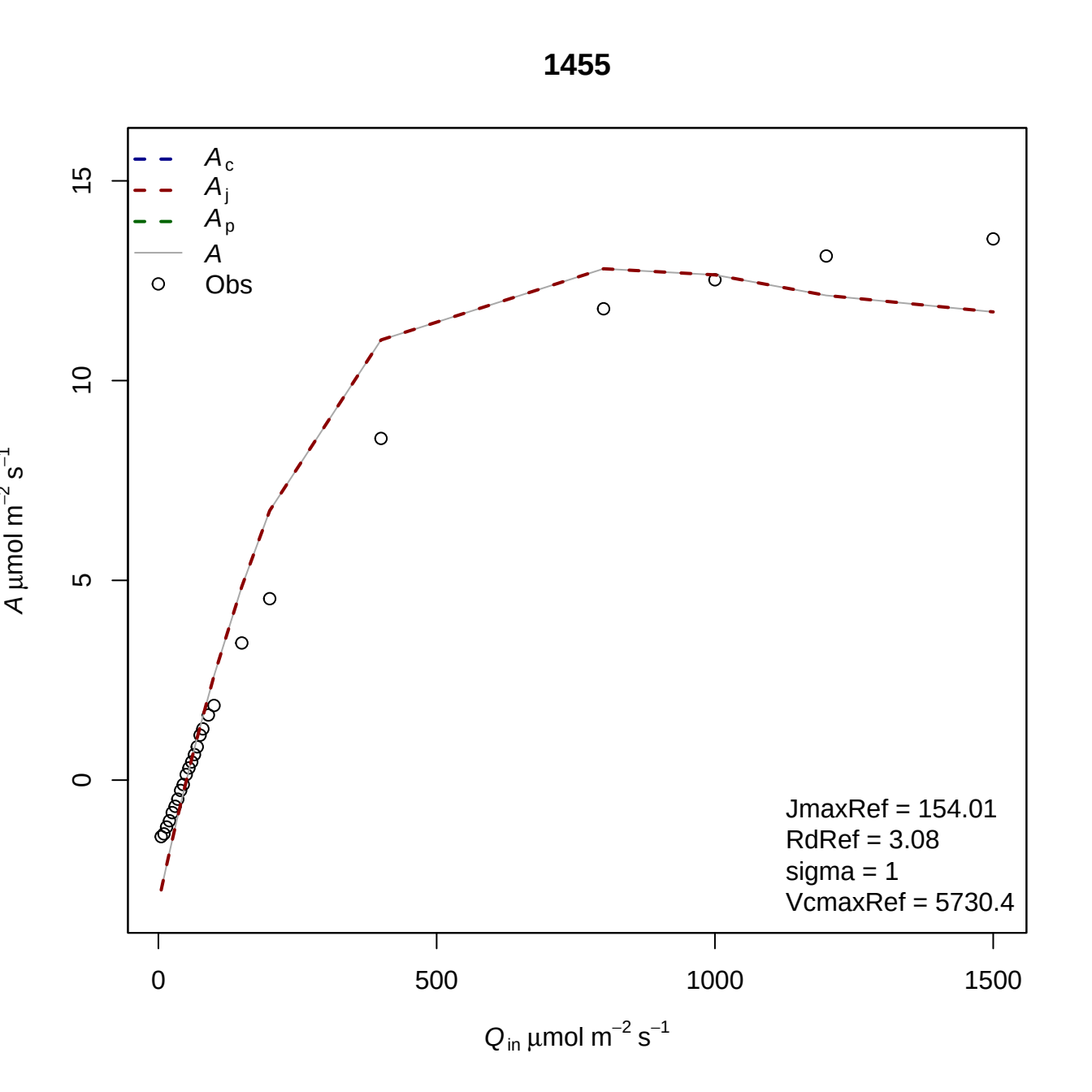

1450
A_c
A_j
A_p
A
Obs
JmaxRef = 150.36
RdRef = 3.86
sigma = 0.96
VcmaxRef = 2259.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

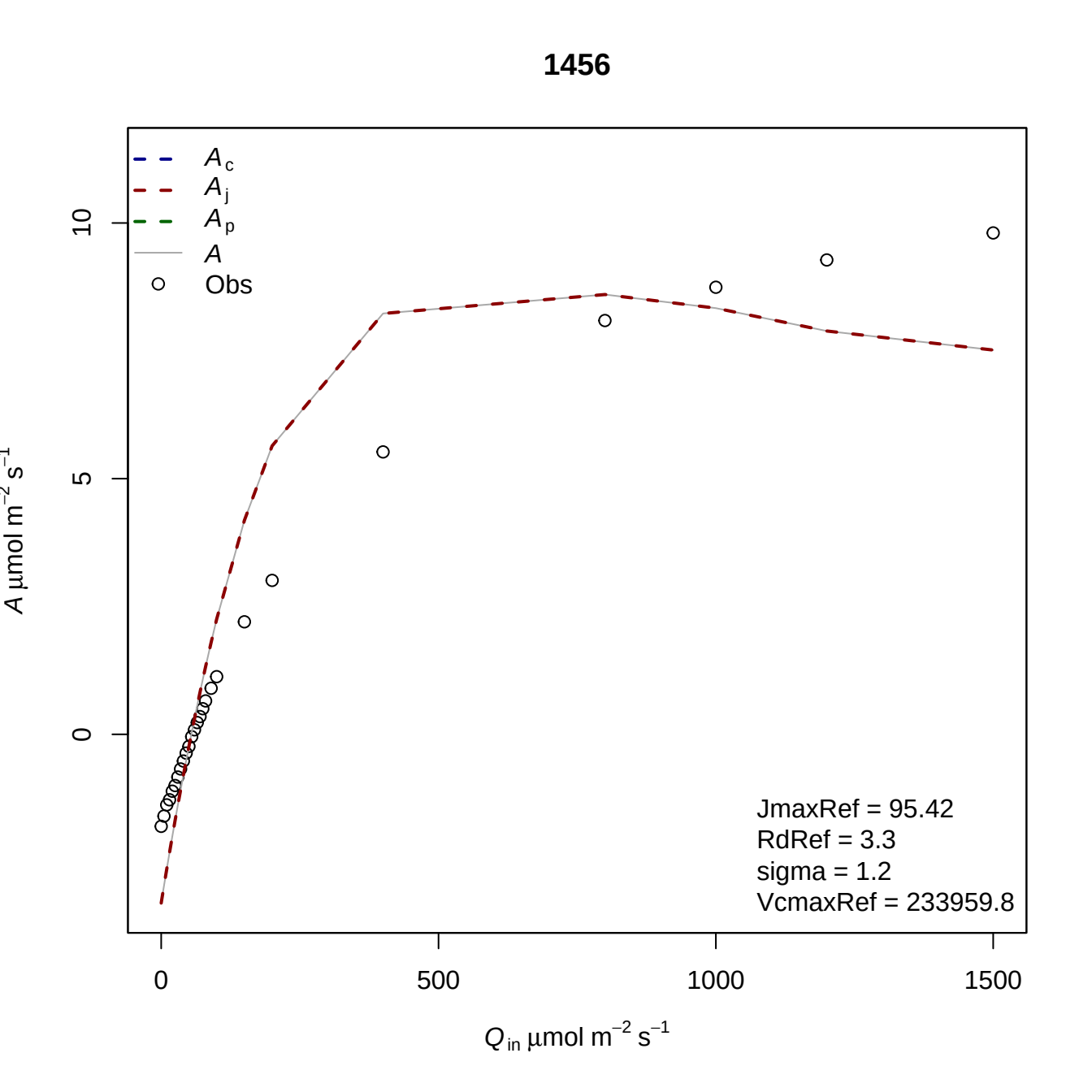

1451
A_c
A_j
A_p
A
Obs
JmaxRef = 114.76
RdRef = 3.61
sigma = 0.98
VcmaxRef = 2328.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

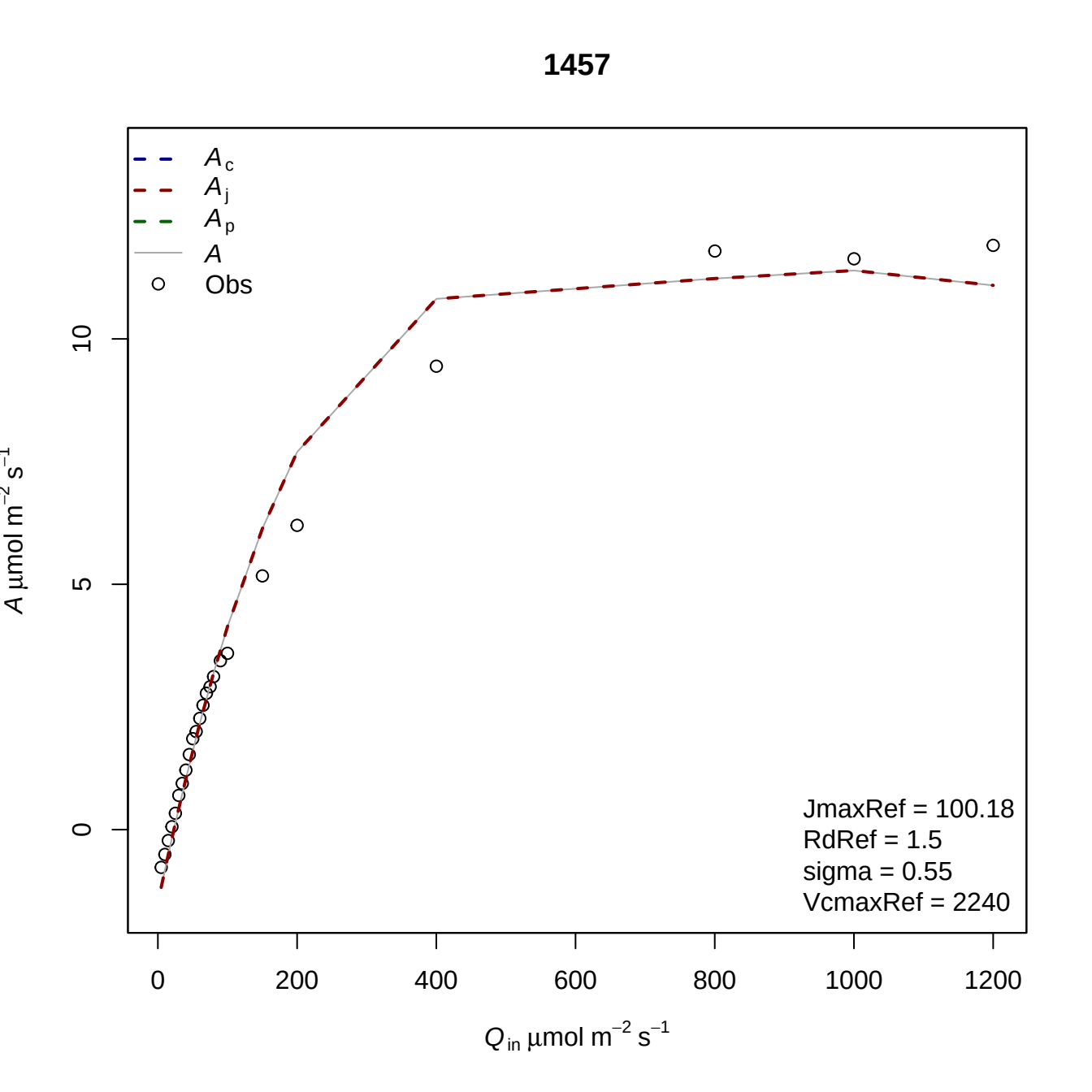

1452
A_c
A_j
A_p
A
Obs
JmaxRef = 90.29
RdRef = 1.79
sigma = 0.59
VcmaxRef = 1955.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

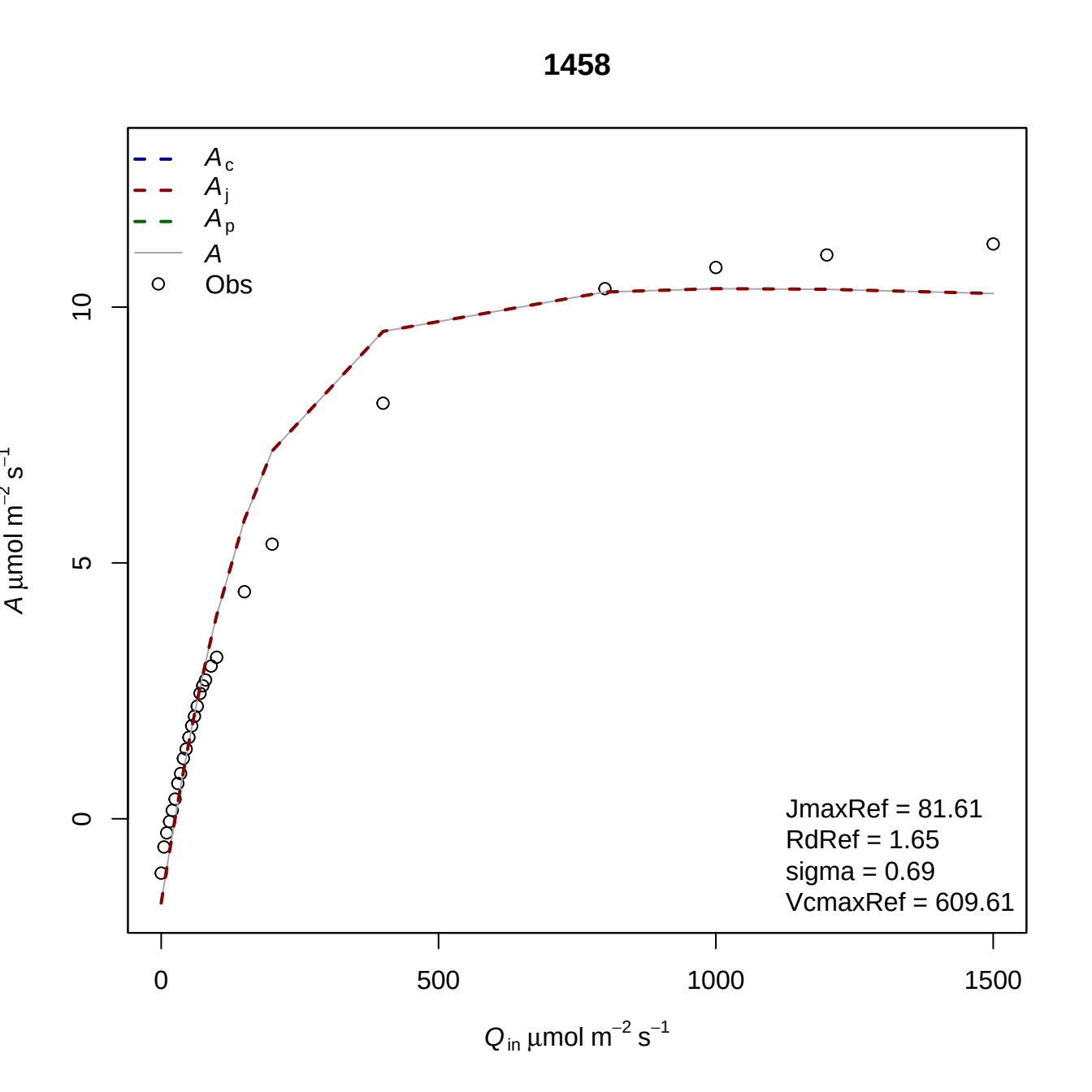

1453
A_c
A_j
A_p
A
Obs
JmaxRef = 131.08
RdRef = 3.61
sigma = 0.89
VcmaxRef = 4725.27
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

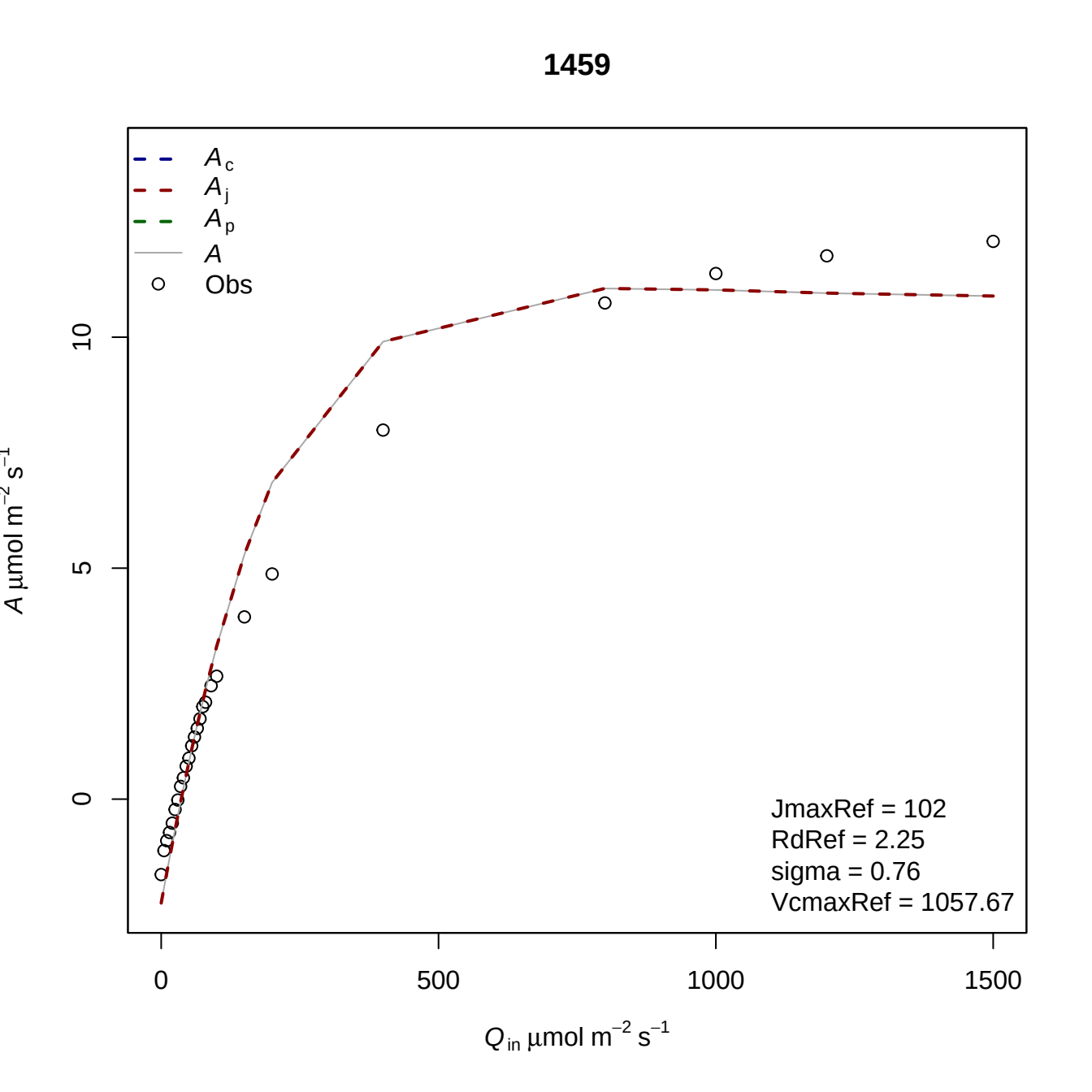

1454
A_c
A_j
A_p
A
Obs
JmaxRef = 96.53
RdRef = 3.42
sigma = 1.26
VcmaxRef = 74328.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

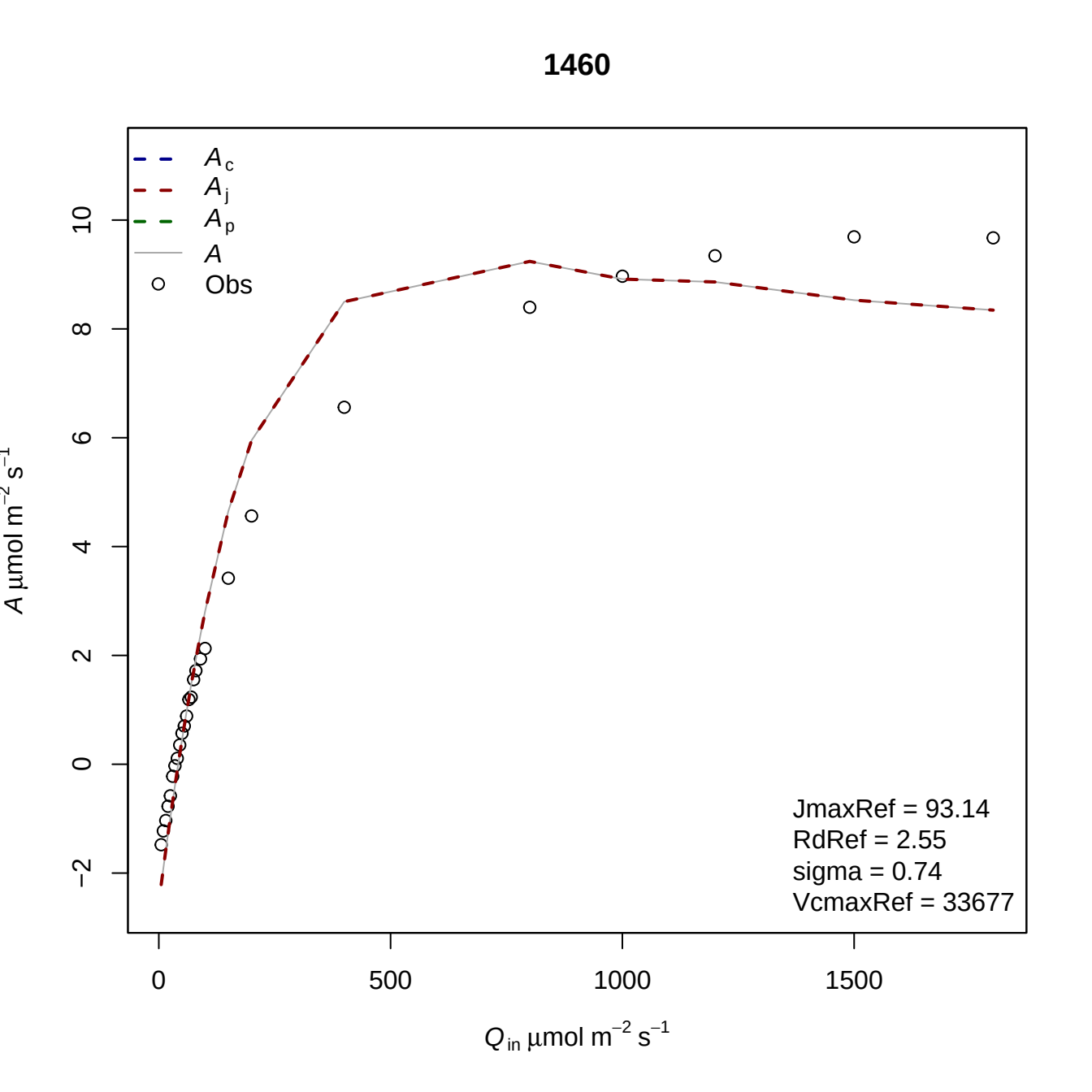

1455
A_c
A_j
A_p
A
Obs
JmaxRef = 154.01
RdRef = 3.08
sigma = 1
VcmaxRef = 5730.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

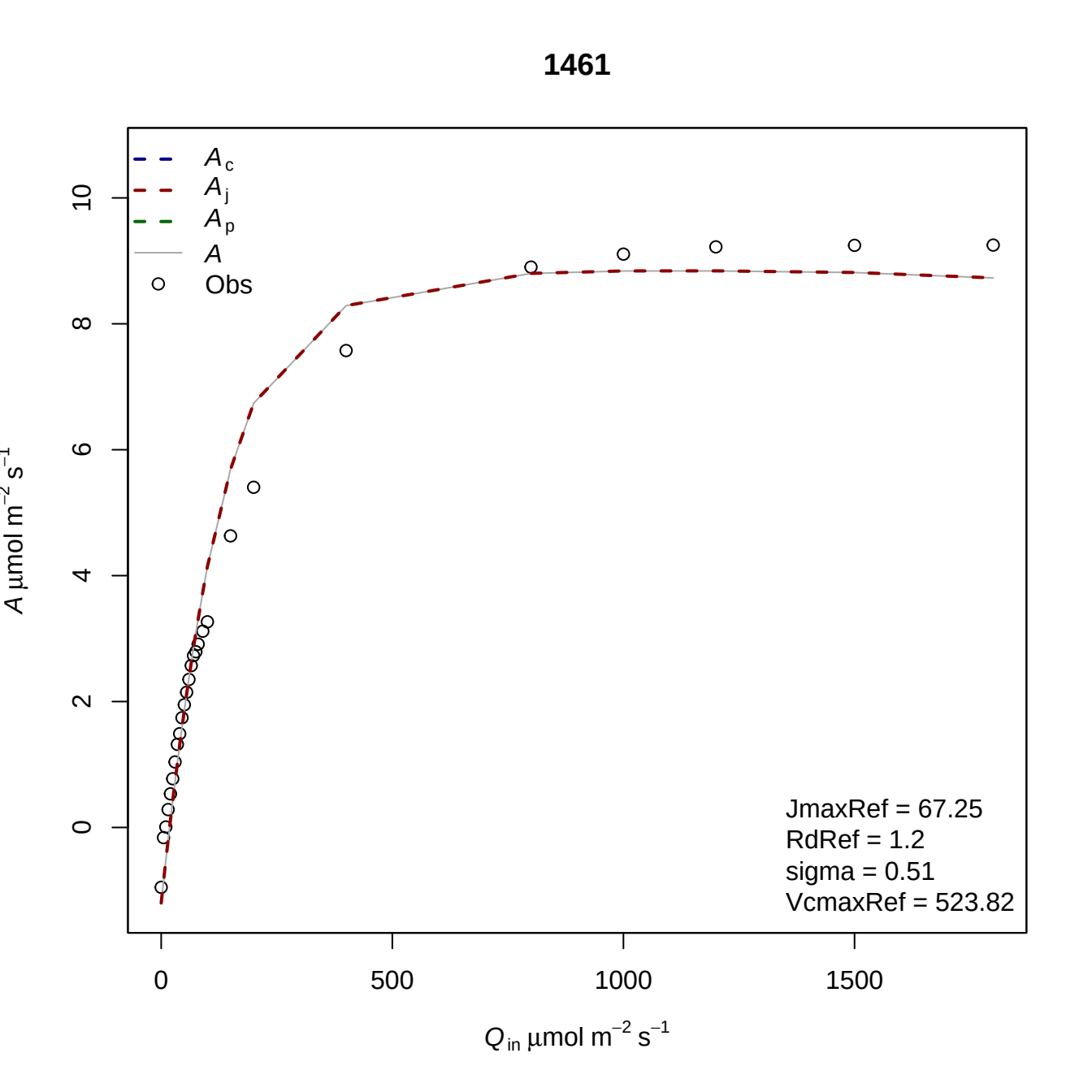

1456
A_c
A_j
A_p
A
Obs
JmaxRef = 95.42
RdRef = 3.3
sigma = 1.2
VcmaxRef = 233959.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

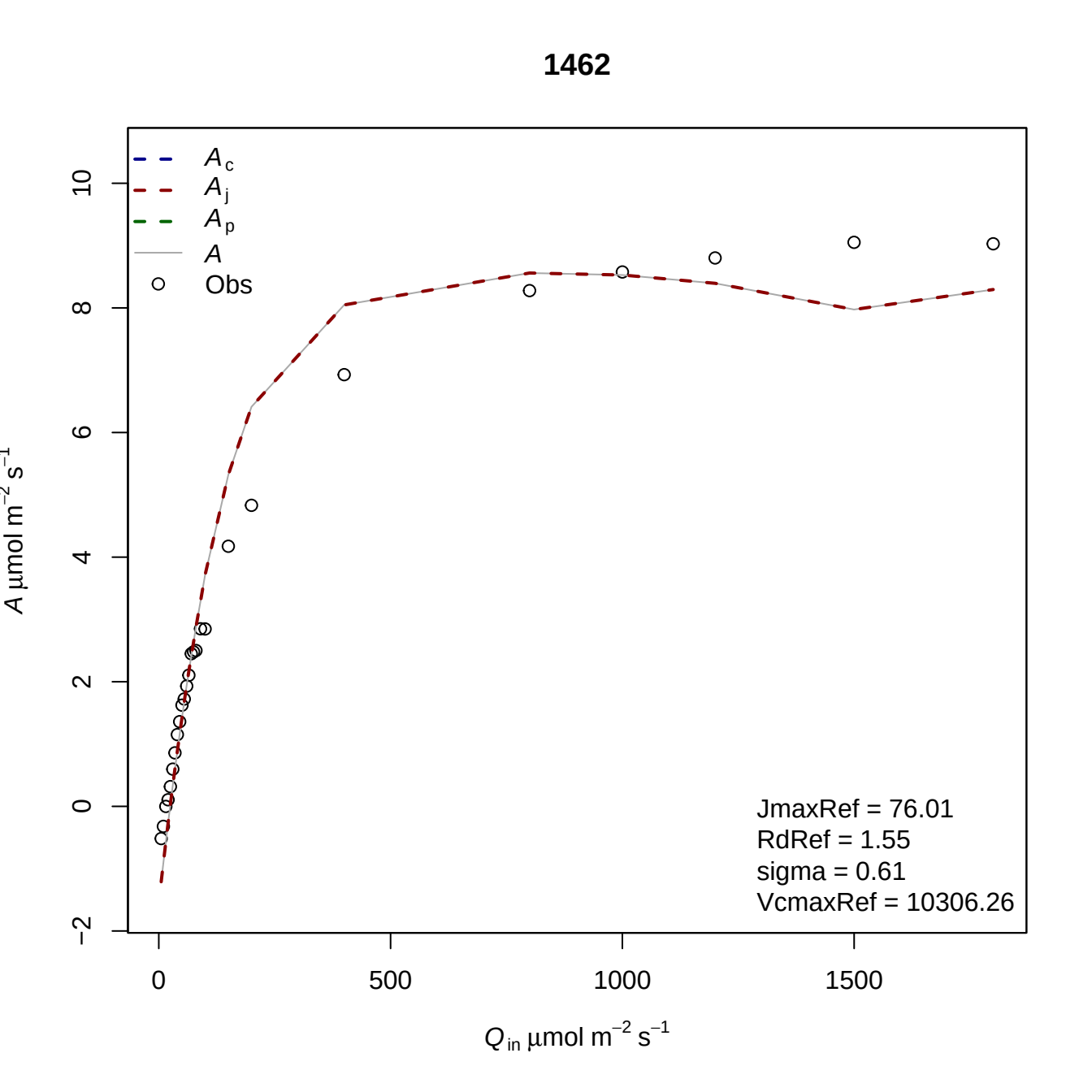

1457
A_c
A_j
A_p
A
Obs
JmaxRef = 100.18
RdRef = 1.5
sigma = 0.55
VcmaxRef = 2240
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

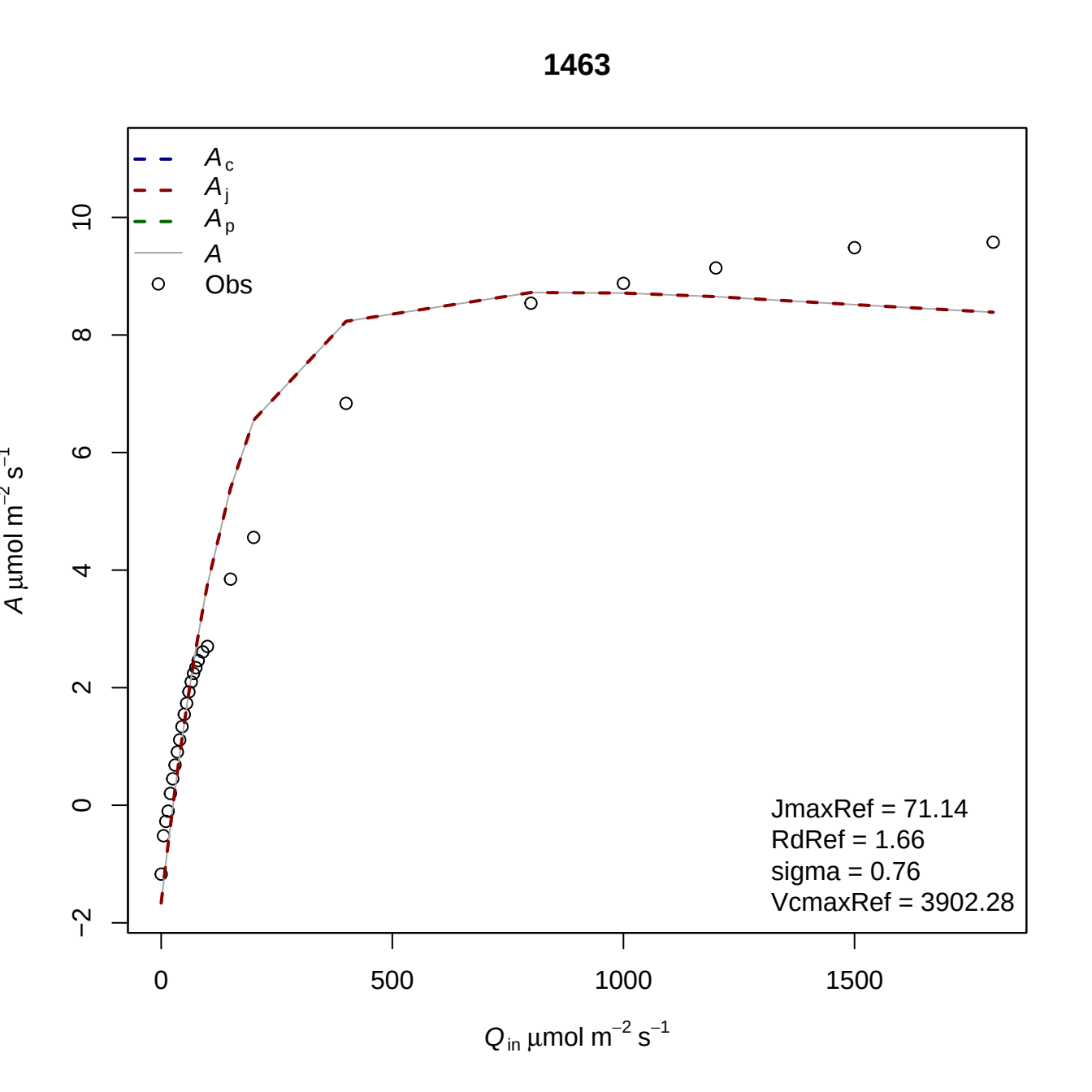

1458
A_c
A_j
A_p
A
Obs
JmaxRef = 81.61
RdRef = 1.65
sigma = 0.69
VcmaxRef = 609.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

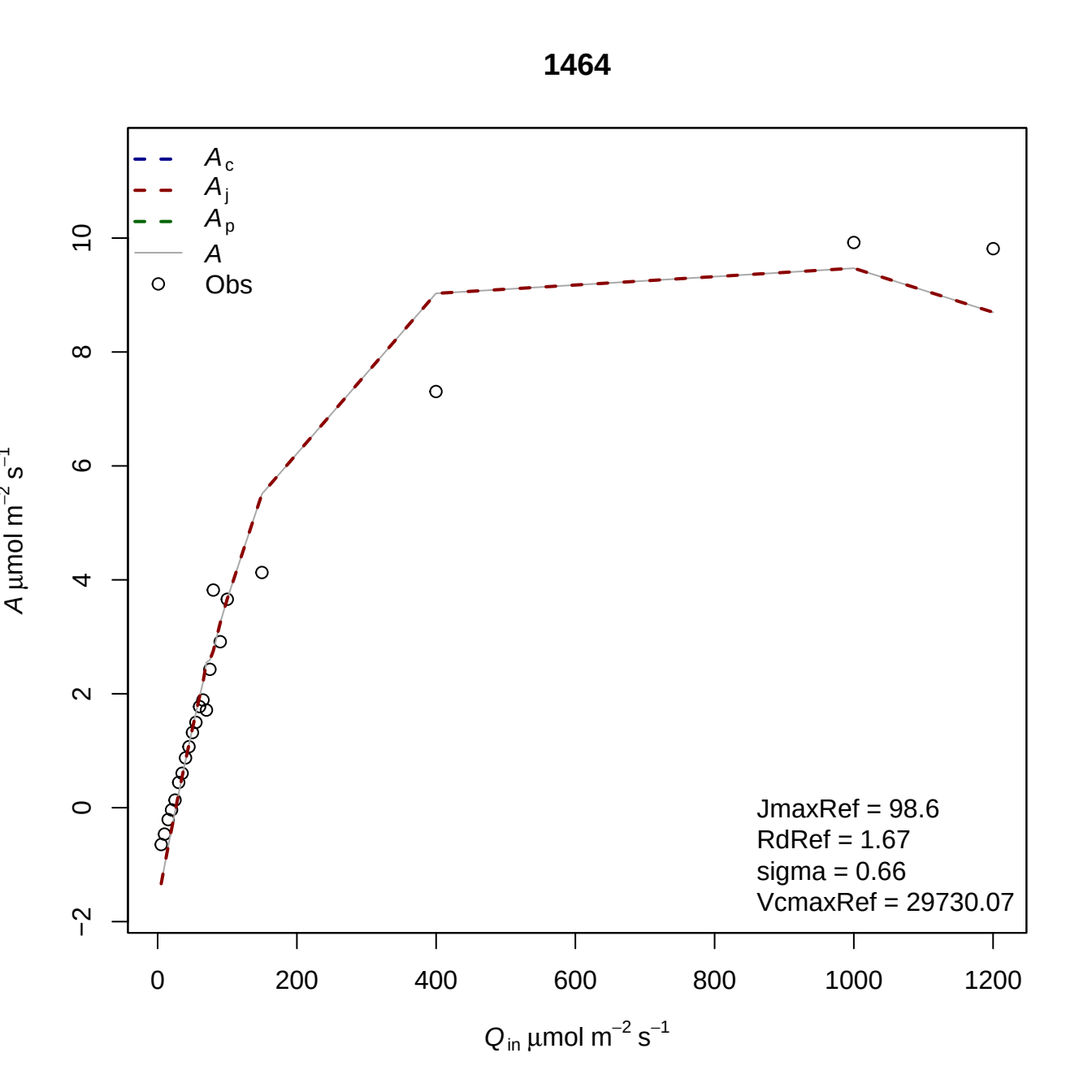

1459
A_c
A_j
A_p
A
Obs
JmaxRef = 102
RdRef = 2.25
sigma = 0.76
VcmaxRef = 1057.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

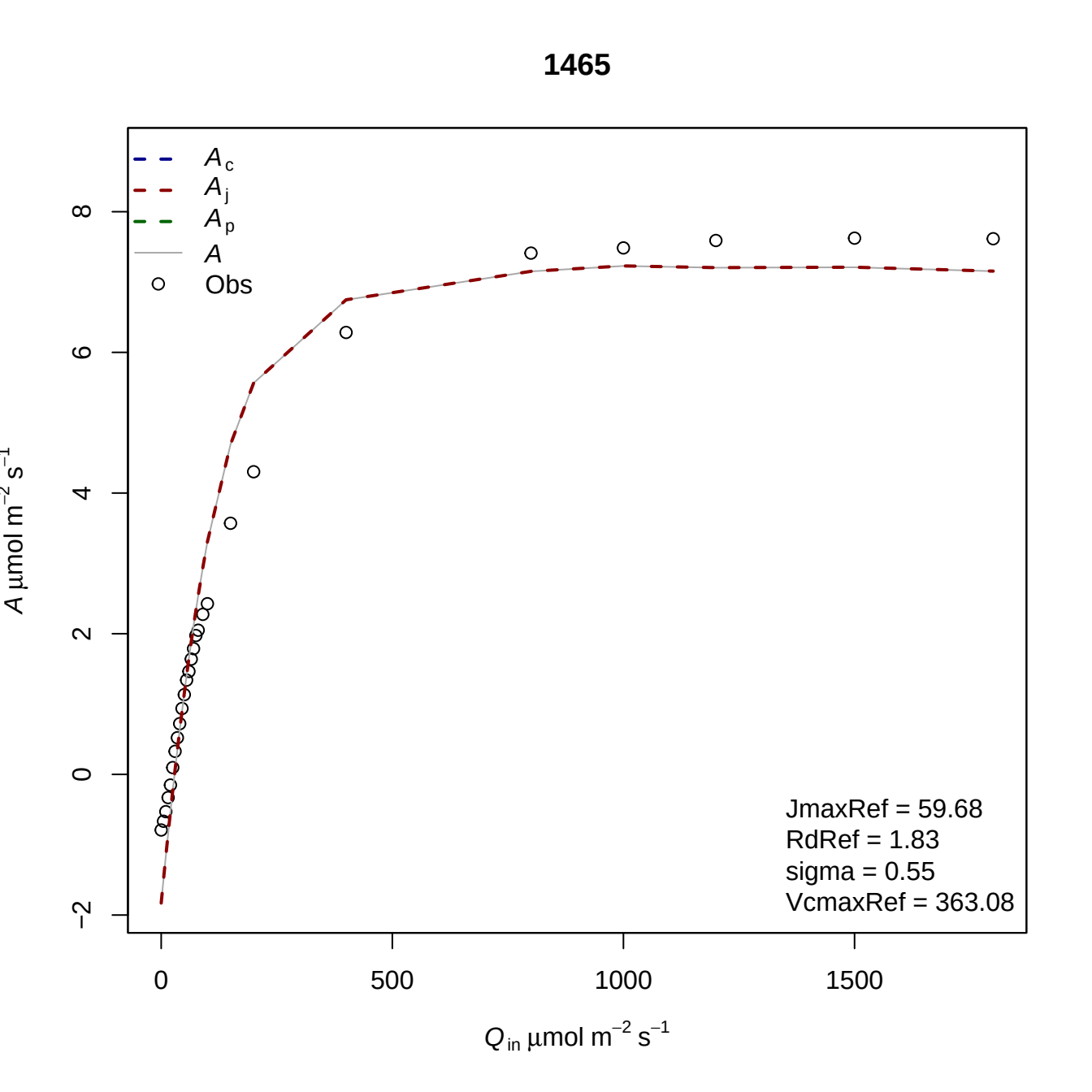

1460
A_c
A_j
A_p
A
Obs
JmaxRef = 93.14
RdRef = 2.55
sigma = 0.74
VcmaxRef = 33677
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

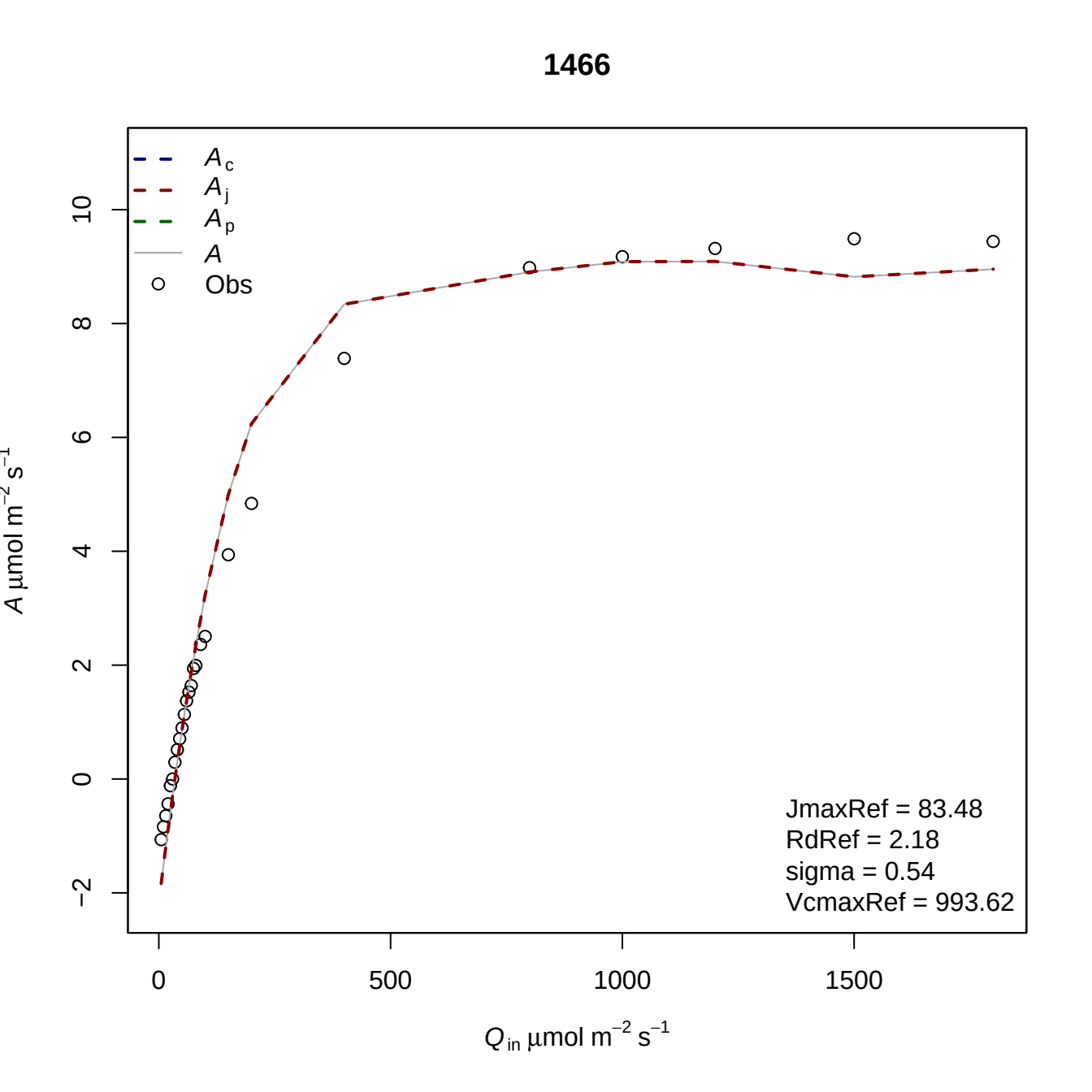

1461
A_c
A_j
A_p
A
Obs
JmaxRef = 67.25
RdRef = 1.2
sigma = 0.51
VcmaxRef = 523.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

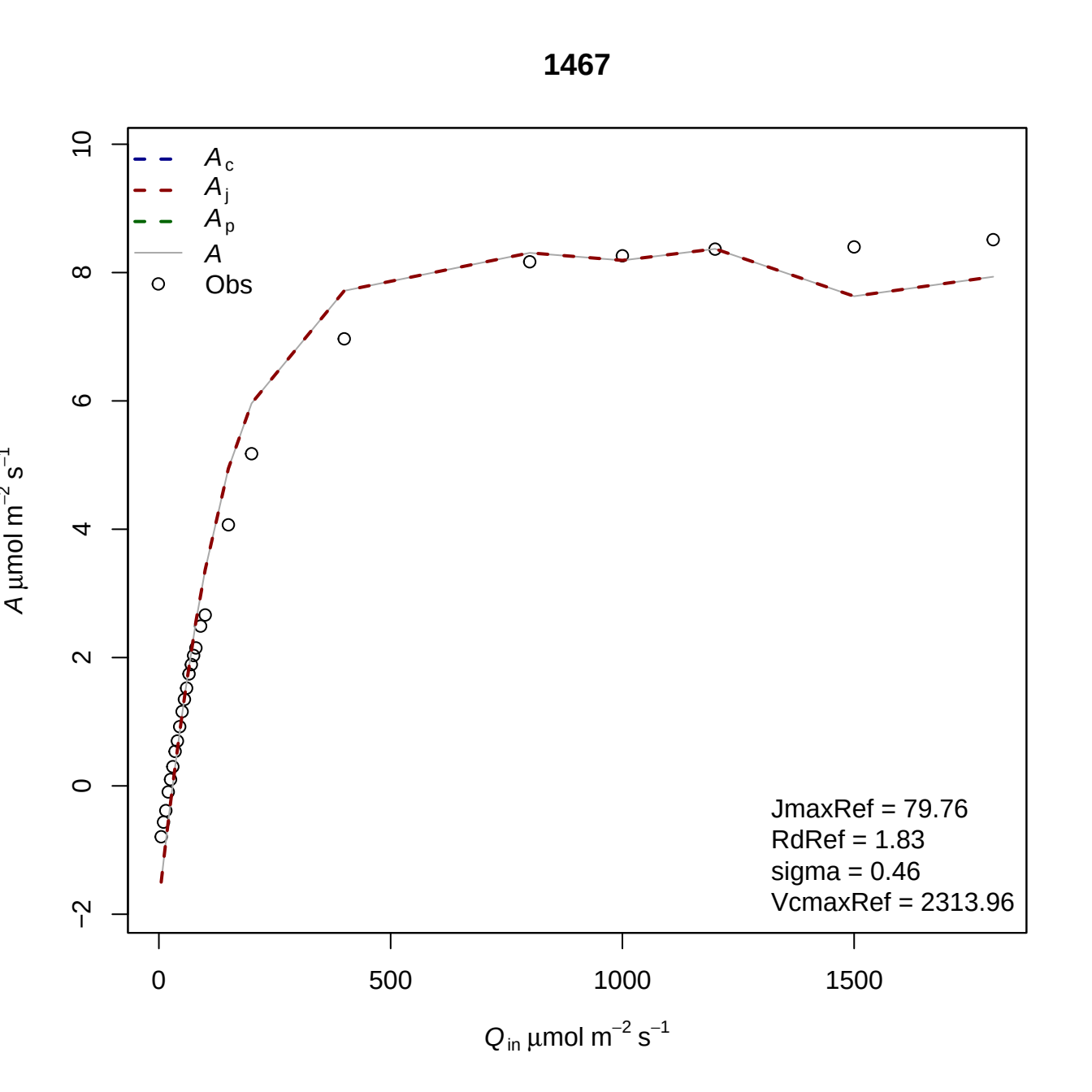

1462
A_c
A_j
A_p
A
Obs
JmaxRef = 76.01
RdRef = 1.55
sigma = 0.61
VcmaxRef = 10306.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

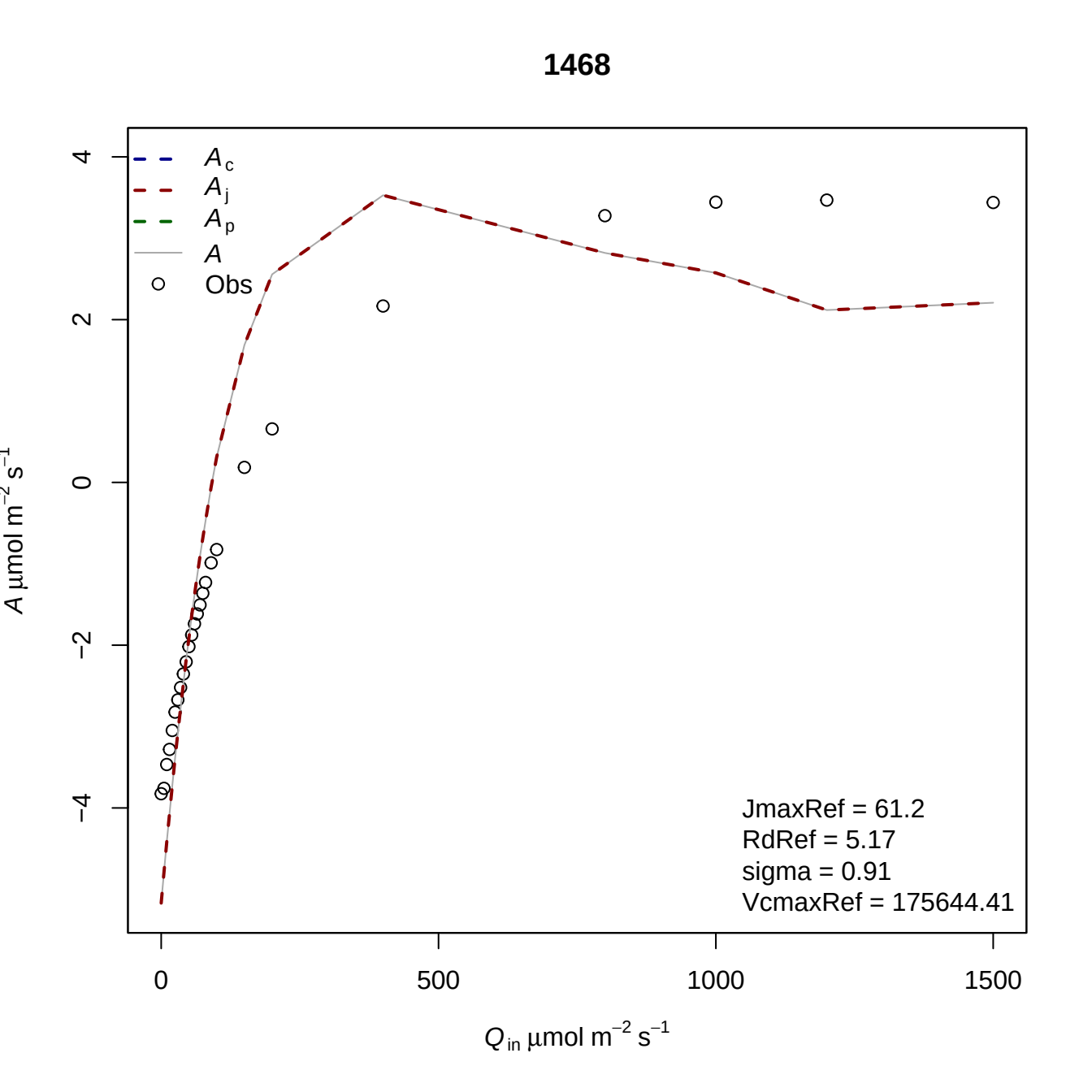

1463
A_c
A_j
A_p
A
Obs
JmaxRef = 71.14
RdRef = 1.66
sigma = 0.76
VcmaxRef = 3902.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

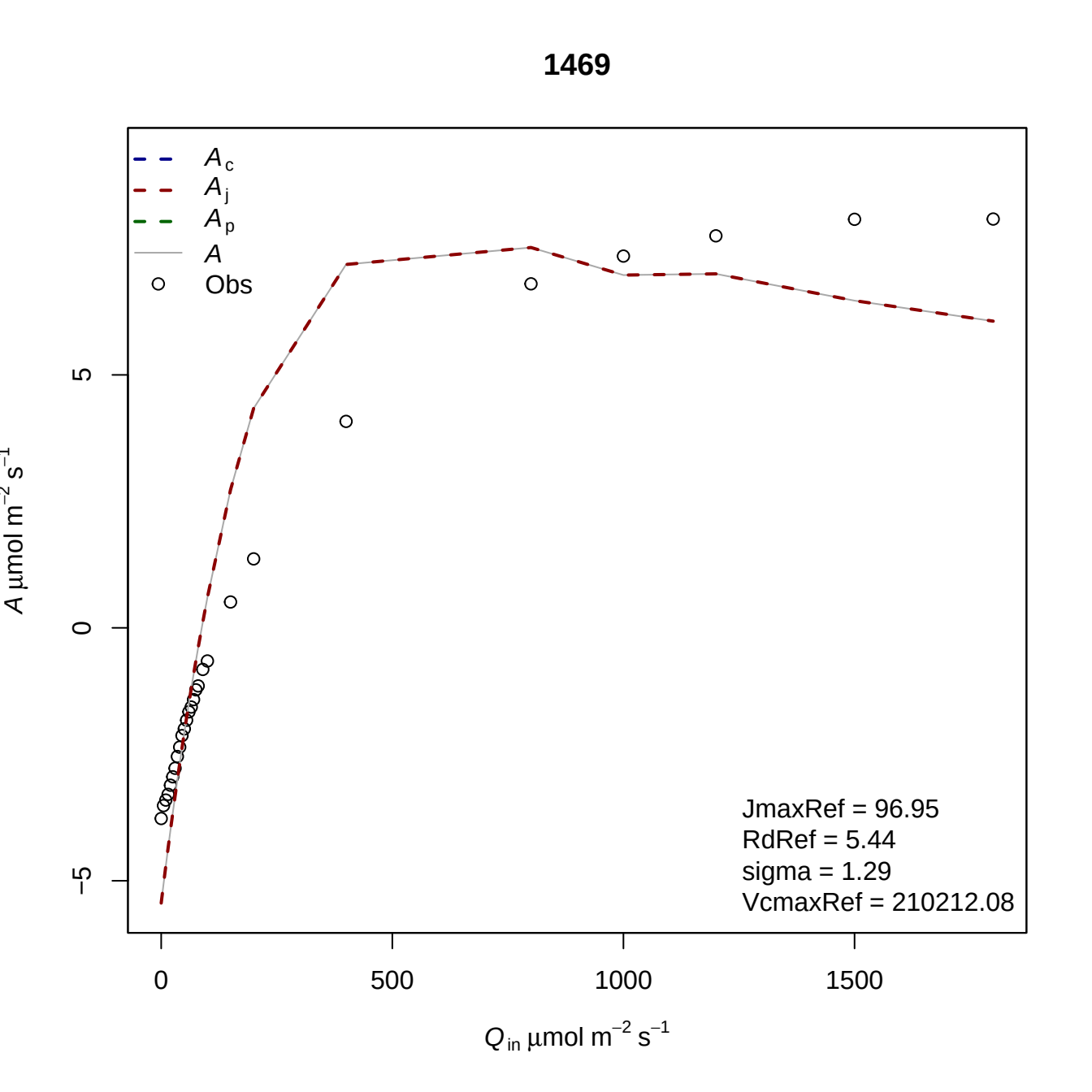

1464
A_c
A_j
A_p
A
Obs
JmaxRef = 98.6
RdRef = 1.67
sigma = 0.66
VcmaxRef = 29730.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

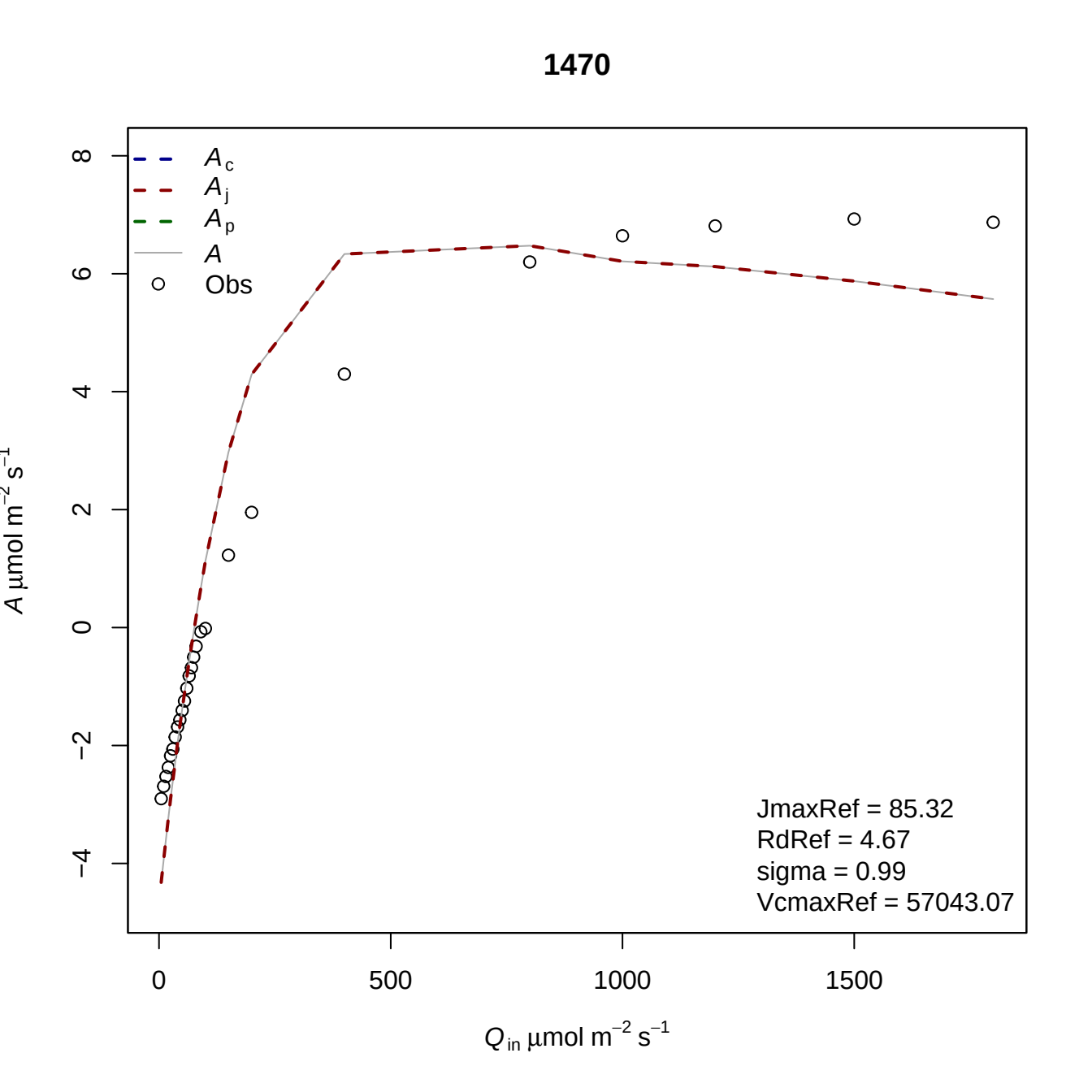

1465
A_c
A_j
A_p
A
Obs
JmaxRef = 59.68
RdRef = 1.83
sigma = 0.55
VcmaxRef = 363.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

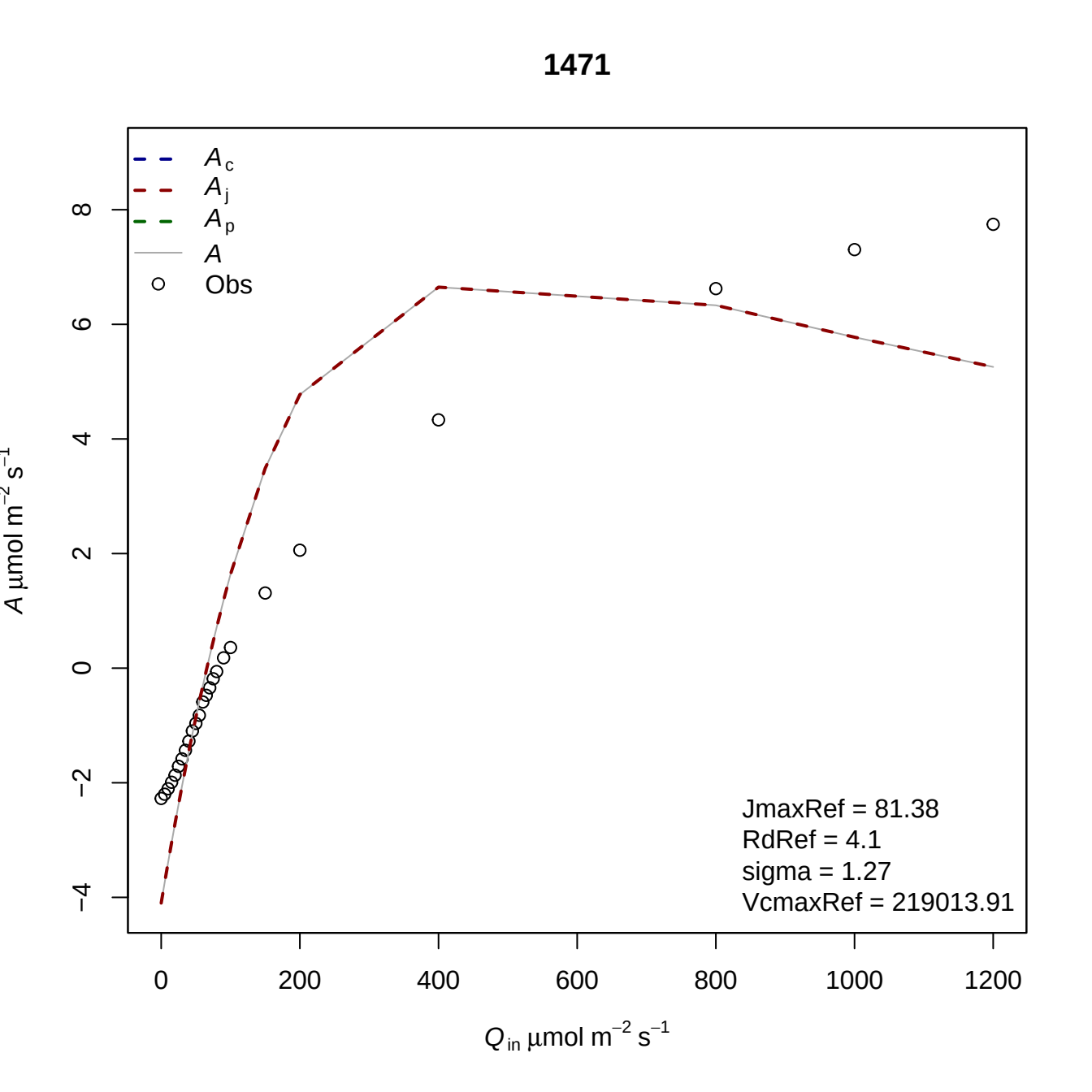

1466
A_c
A_j
A_p
A
Obs
JmaxRef = 83.48
RdRef = 2.18
sigma = 0.54
VcmaxRef = 993.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

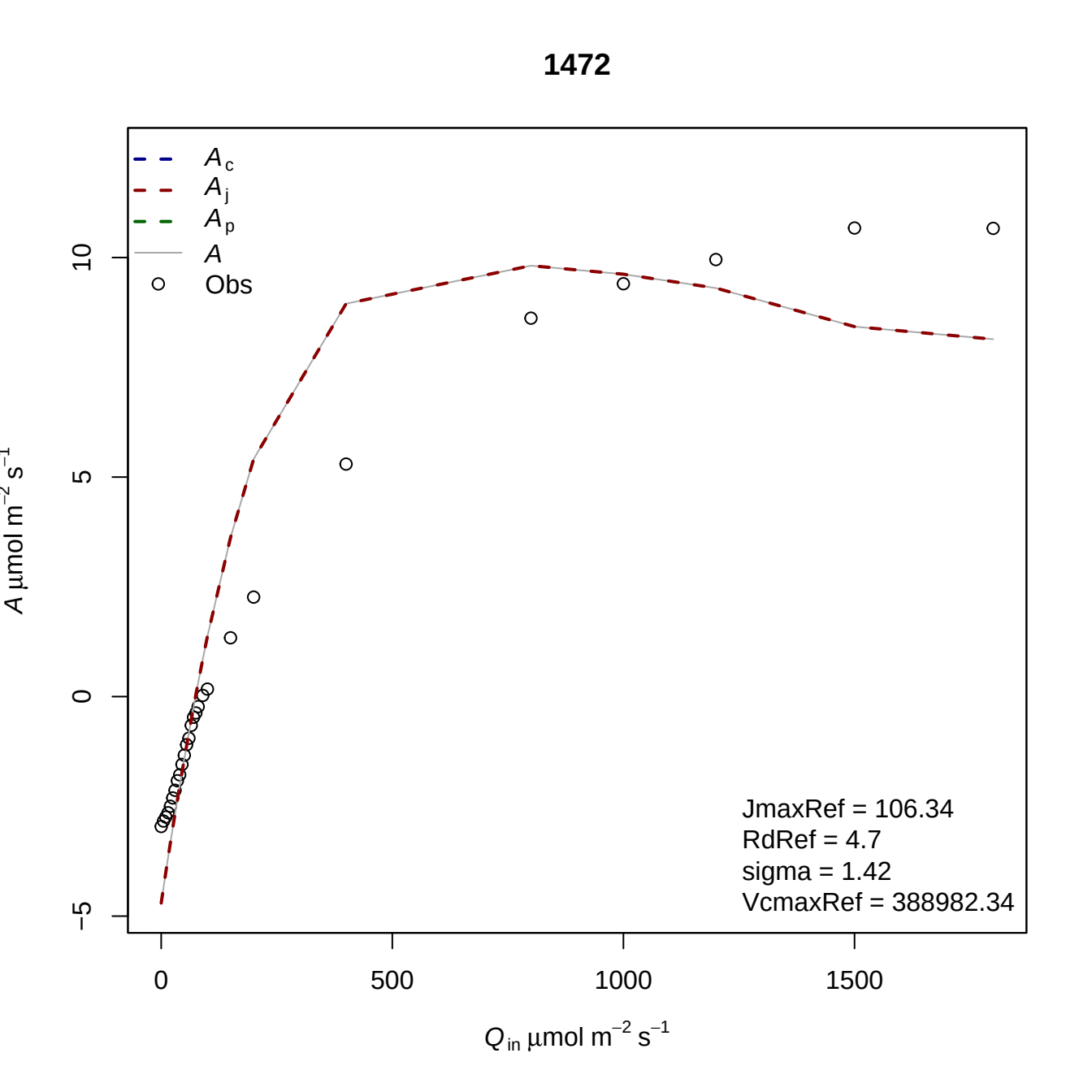

1467
A_c
A_j
A_p
A
Obs
JmaxRef = 79.76
RdRef = 1.83
sigma = 0.46
VcmaxRef = 2313.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

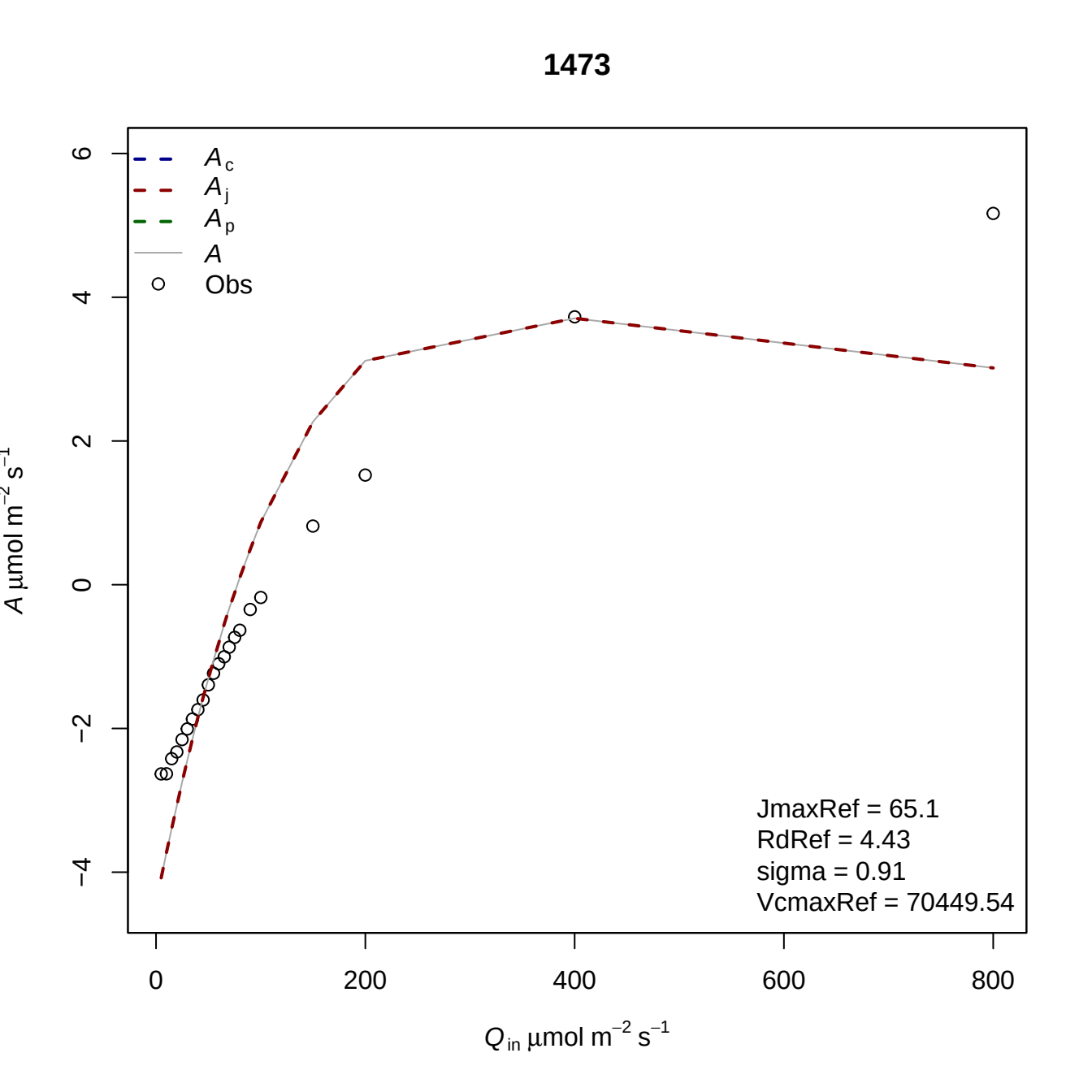

1468
A_c
A_j
A_p
A
Obs
JmaxRef = 61.2
RdRef = 5.17
sigma = 0.91
VcmaxRef = 175644.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

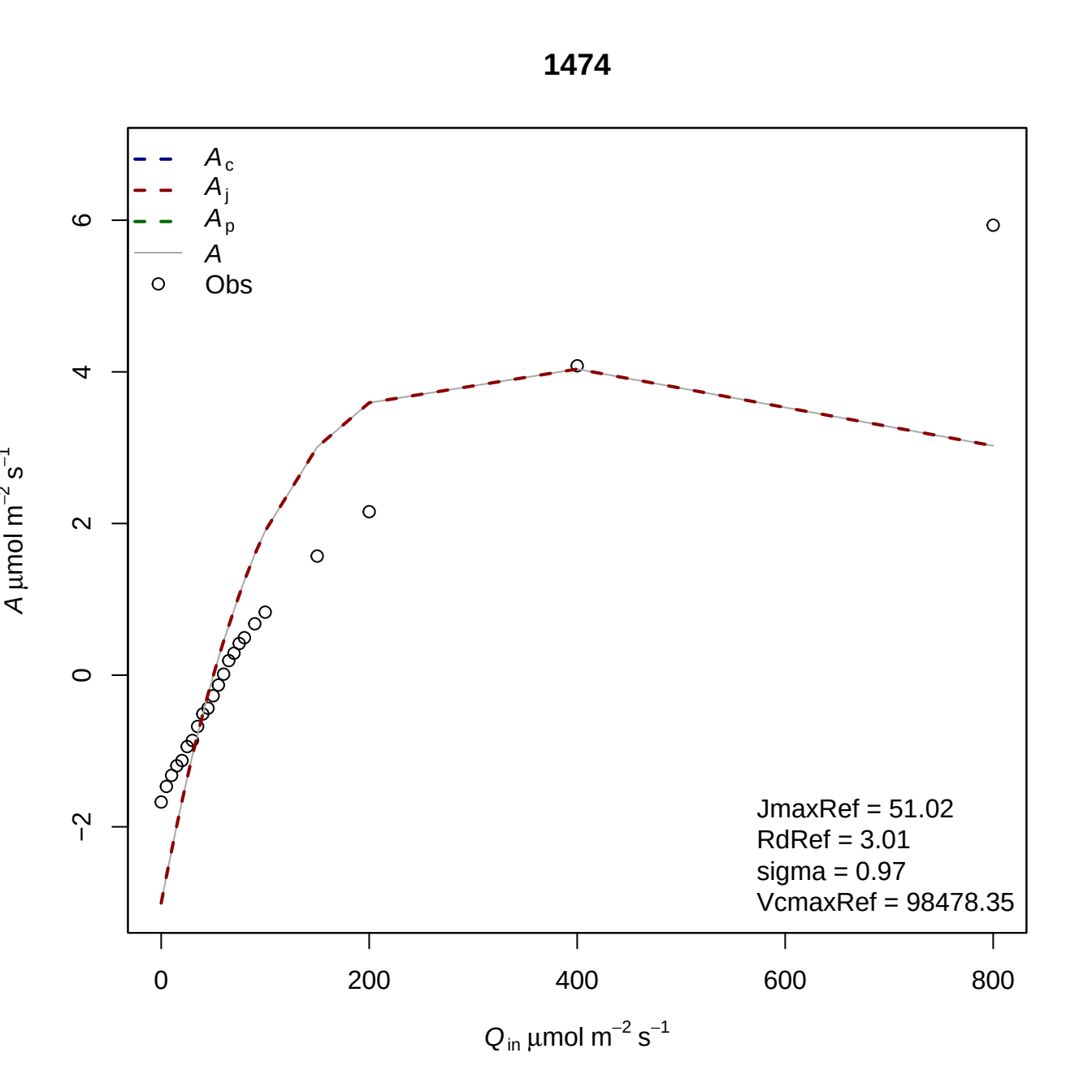

1469
A_c
A_j
A_p
A
Obs
JmaxRef = 96.95
RdRef = 5.44
sigma = 1.29
VcmaxRef = 210212.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
0
500
1000
1500

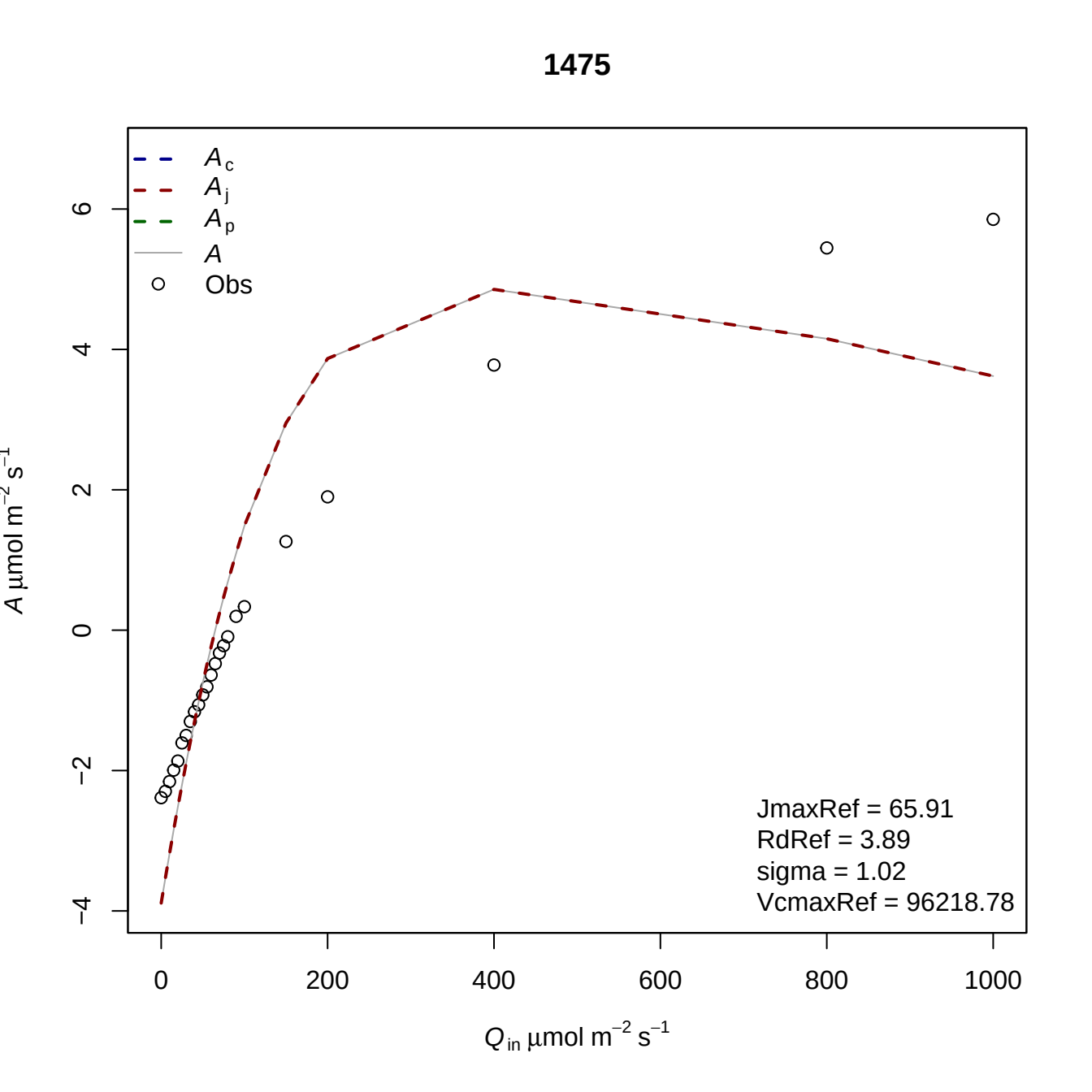

1470
A_c
A_j
A_p
A
Obs
JmaxRef = 85.32
RdRef = 4.67
sigma = 0.99
VcmaxRef = 57043.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

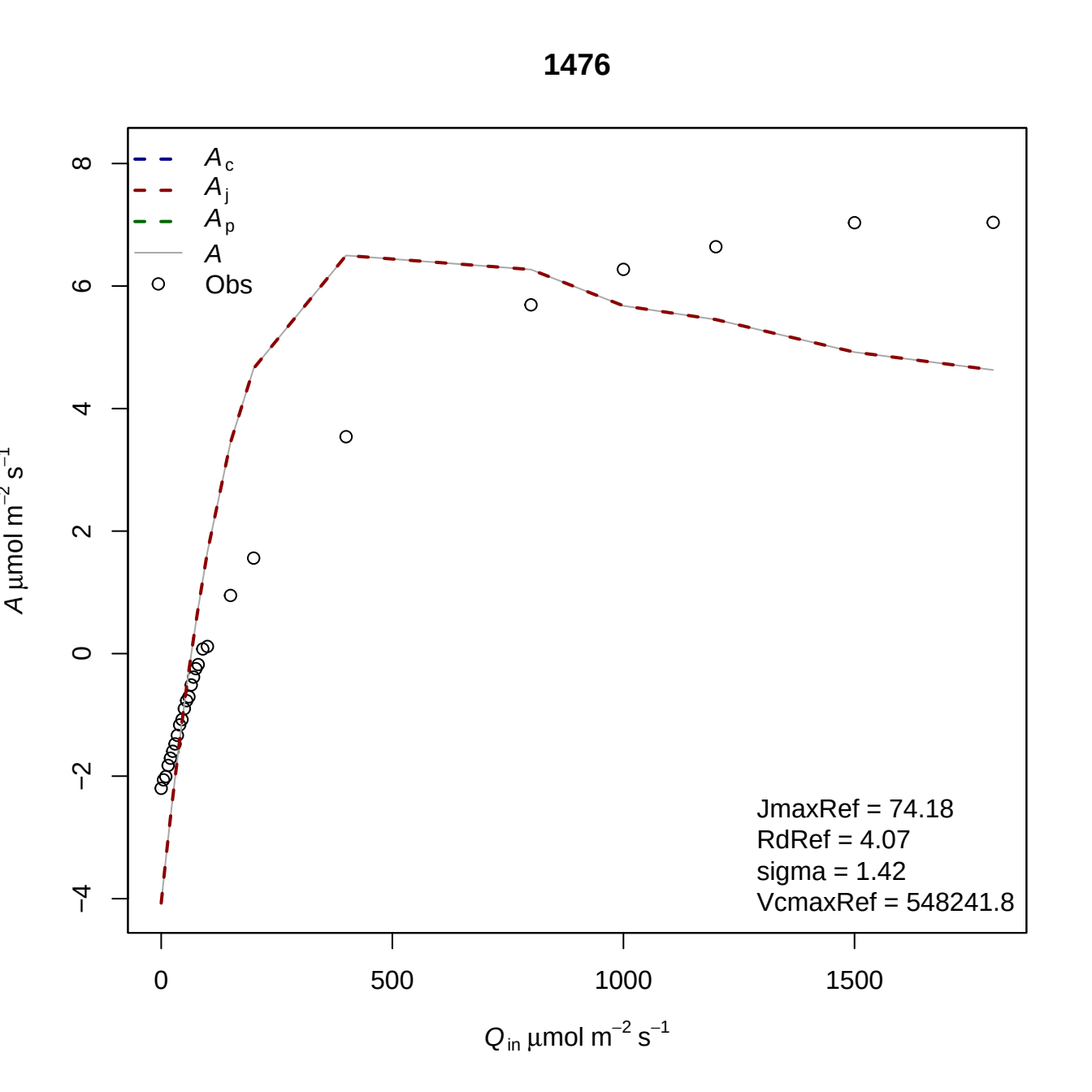

1471
A_c
A_j
A_p
A
Obs
JmaxRef = 81.38
RdRef = 4.1
sigma = 1.27
VcmaxRef = 219013.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

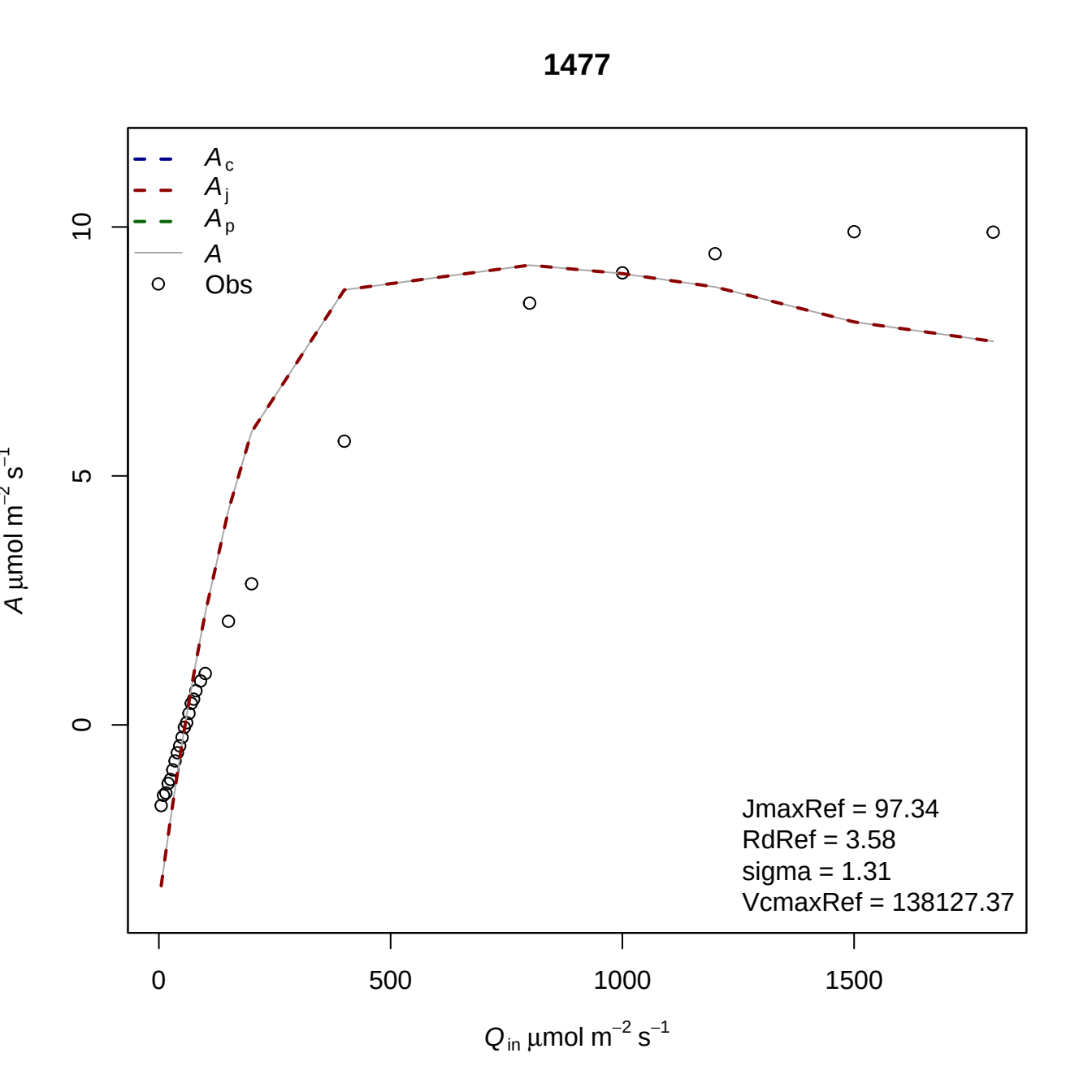

1472
A_c
A_j
A_p
A
Obs
JmaxRef = 106.34
RdRef = 4.7
sigma = 1.42
VcmaxRef = 388982.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500

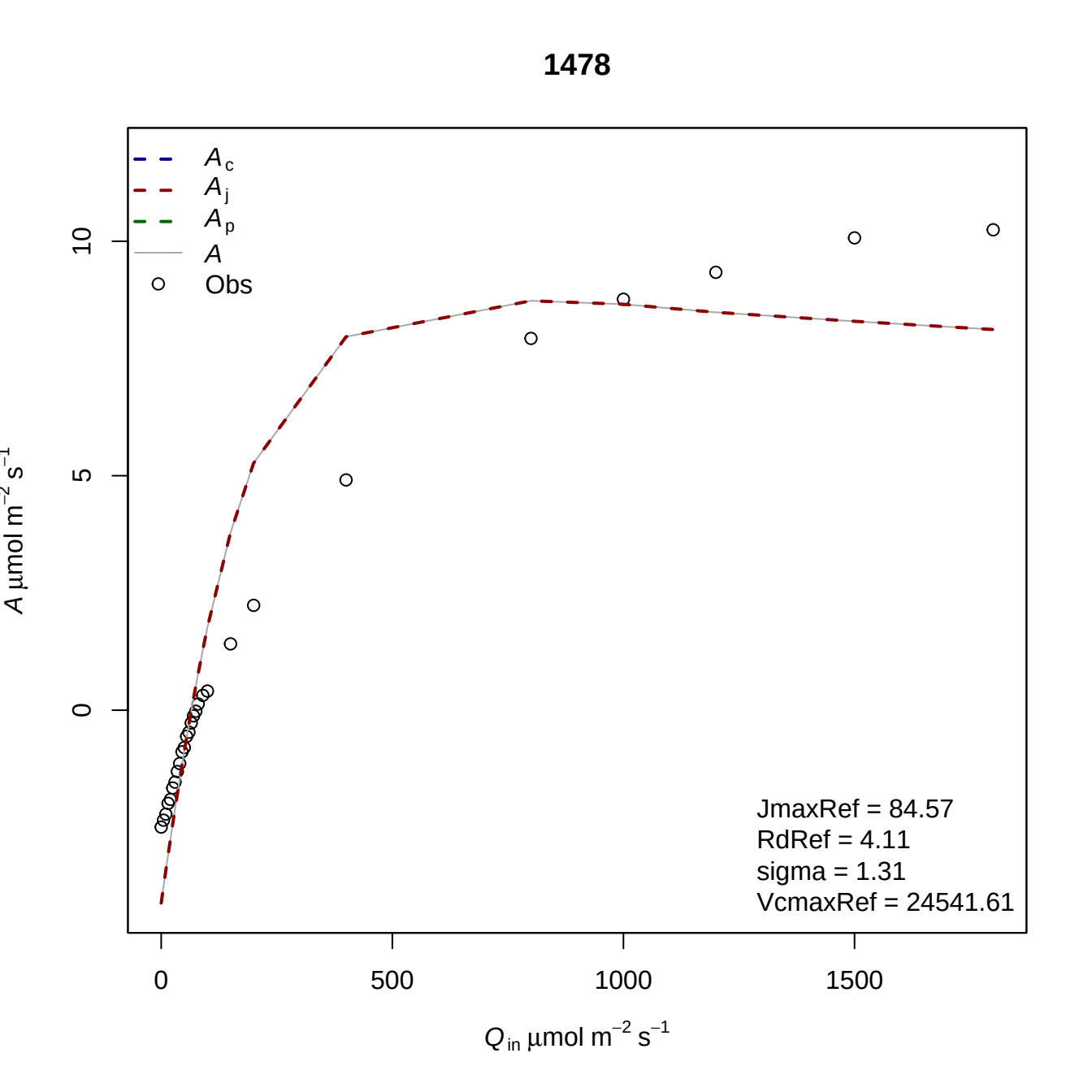

1473
A_c
A_j
A_p
A
Obs
JmaxRef = 65.1
RdRef = 4.43
sigma = 0.91
VcmaxRef = 70449.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800

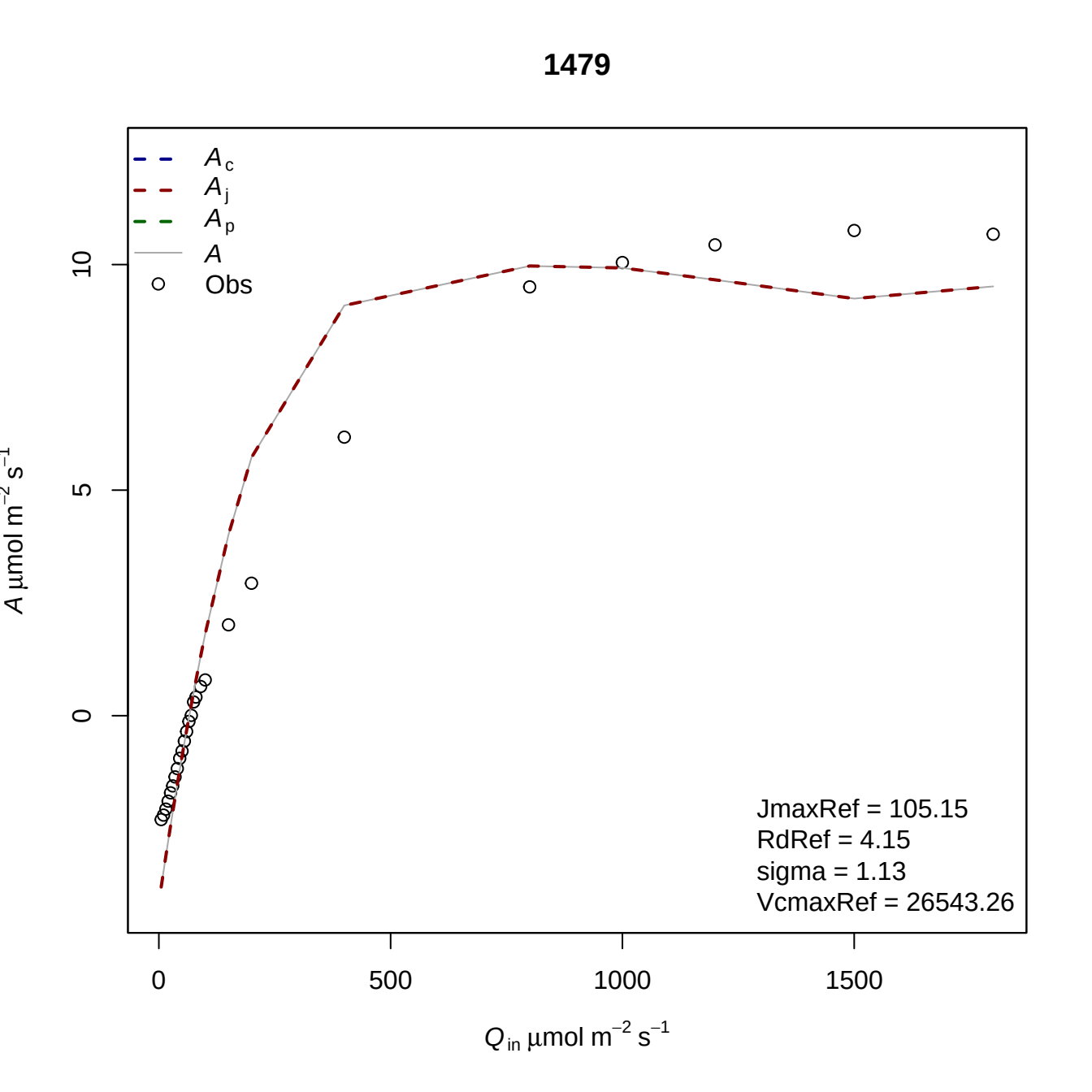

1474
A_c
A_j
A_p
A
Obs
JmaxRef = 51.02
RdRef = 3.01
sigma = 0.97
VcmaxRef = 98478.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

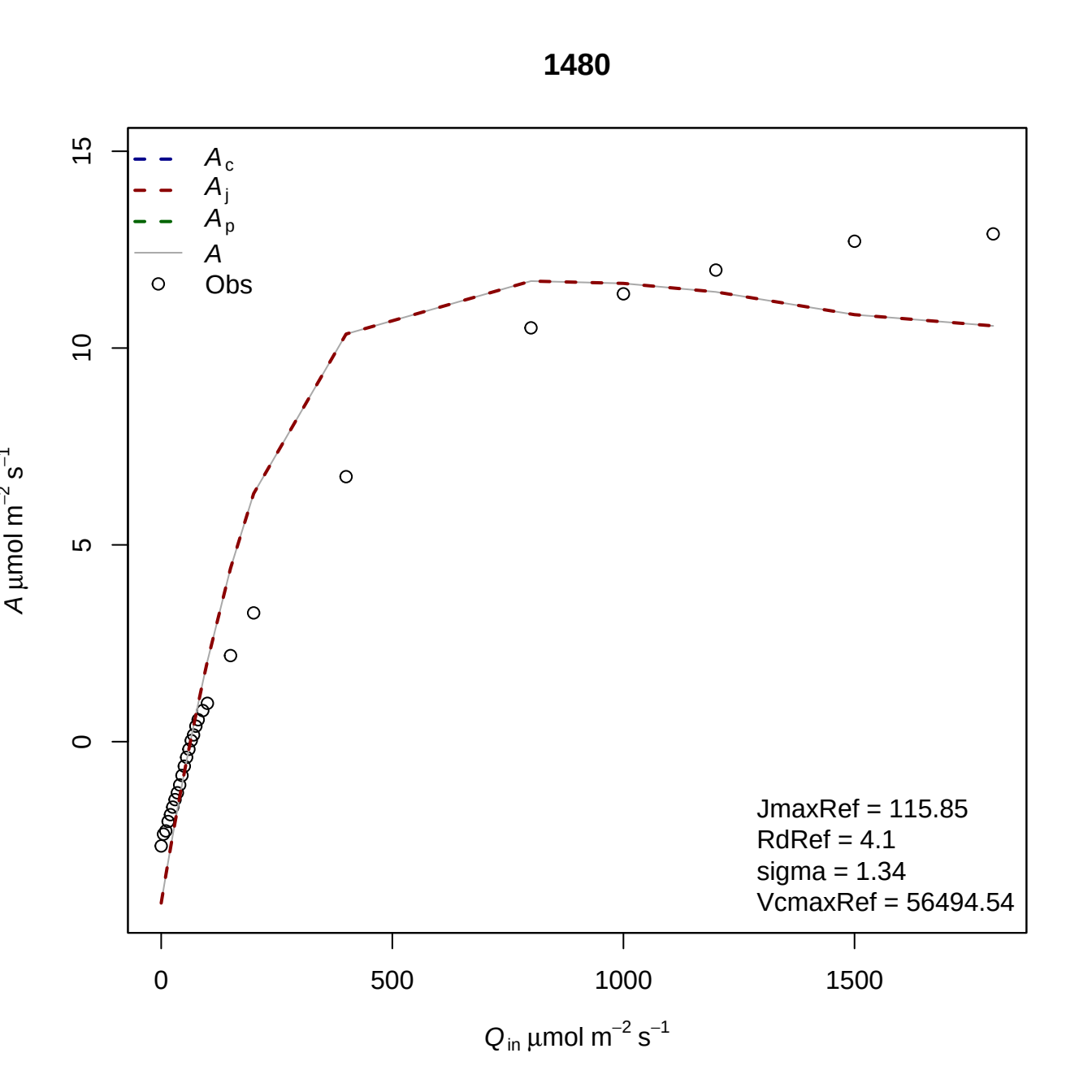

1475
A_c
A_j
A_p
A
Obs
JmaxRef = 65.91
RdRef = 3.89
sigma = 1.02
VcmaxRef = 96218.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800
1000

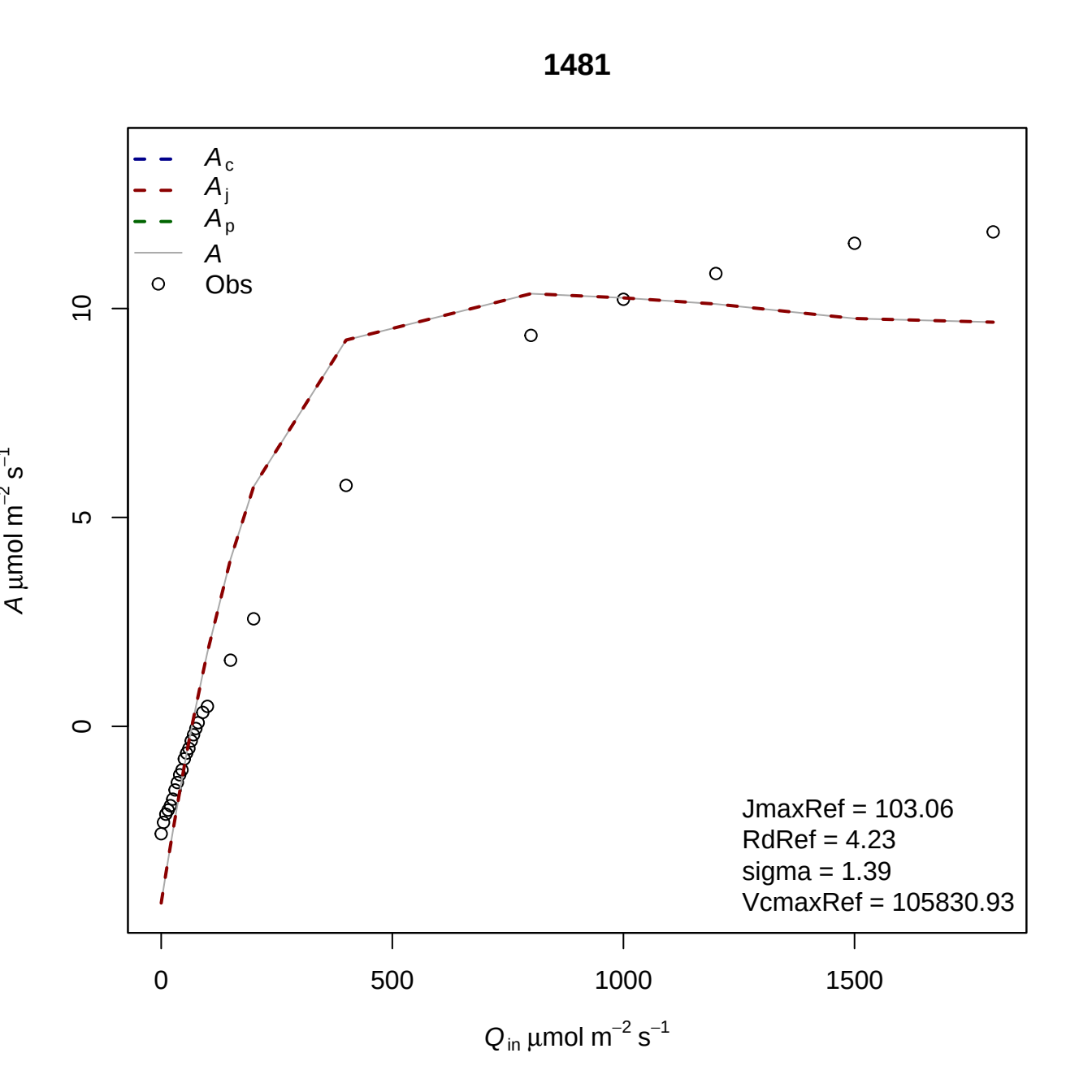

1476
A_c
A_j
A_p
A
Obs
JmaxRef = 74.18
RdRef = 4.07
sigma = 1.42
VcmaxRef = 548241.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

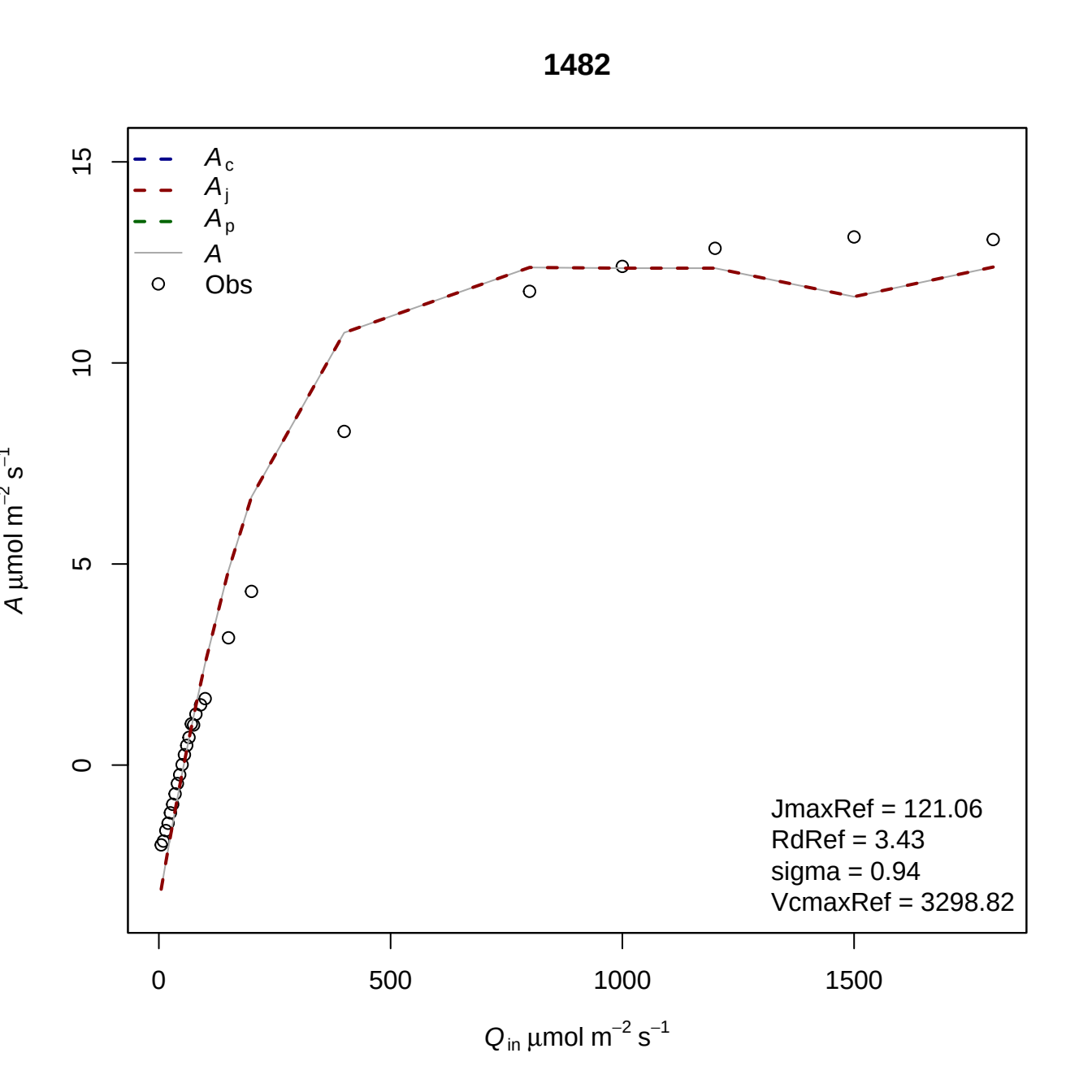

1477
A_c
A_j
A_p
A
Obs
JmaxRef = 97.34
RdRef = 3.58
sigma = 1.31
VcmaxRef = 138127.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

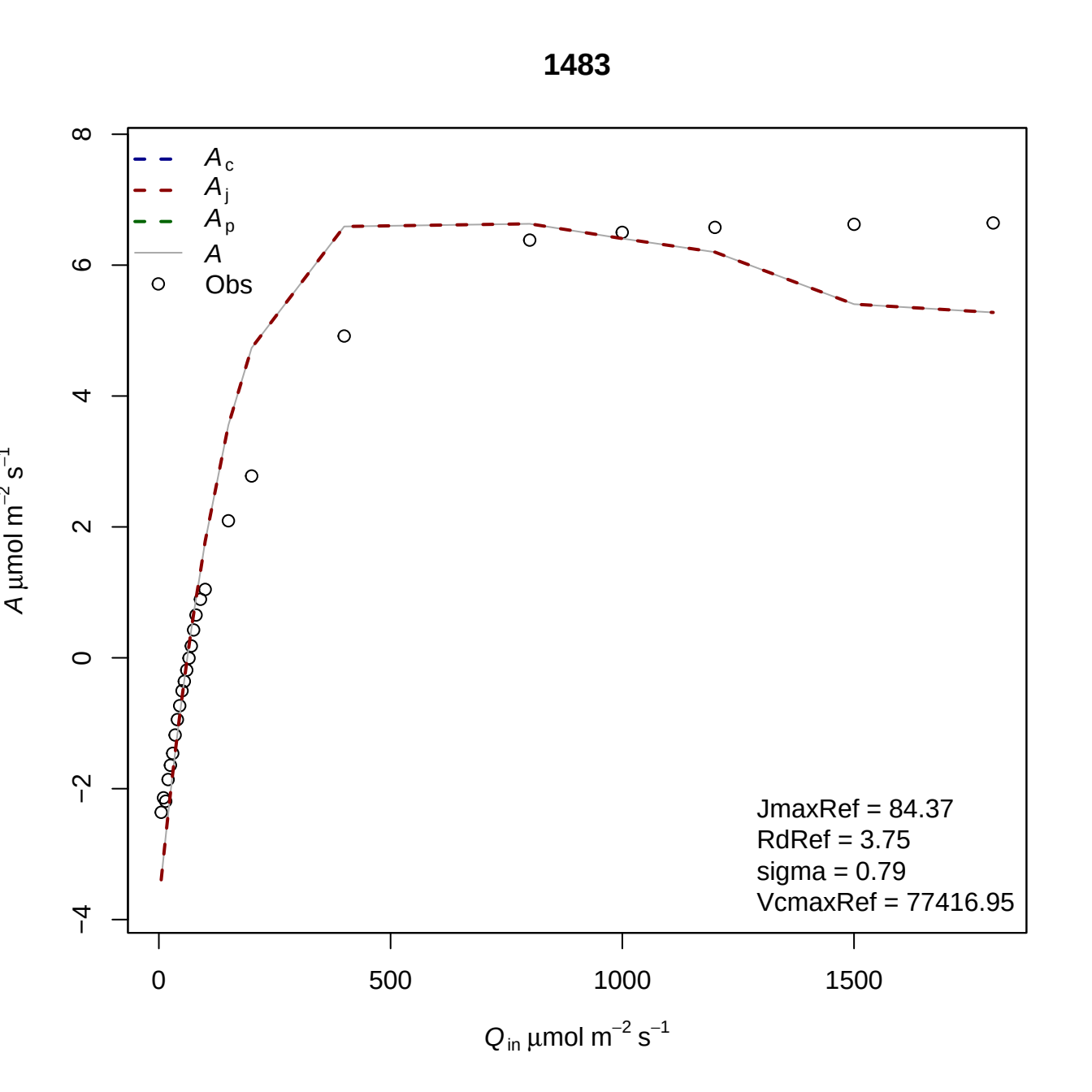

1478
A_c
A_j
A_p
A
Obs
JmaxRef = 84.57
RdRef = 4.11
sigma = 1.31
VcmaxRef = 24541.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

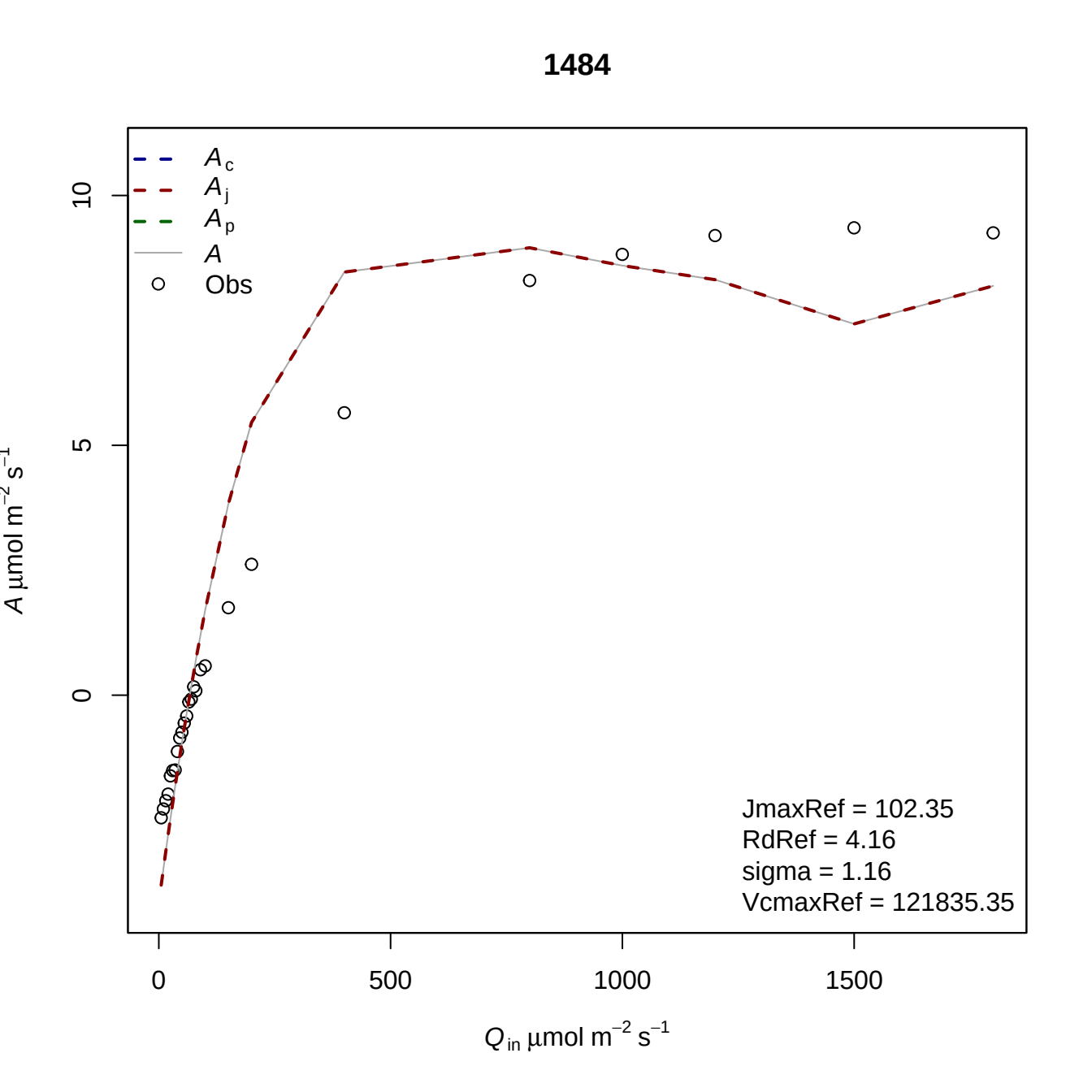

1479
A_c
A_j
A_p
A
Obs
JmaxRef = 105.15
RdRef = 4.15
sigma = 1.13
VcmaxRef = 26543.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

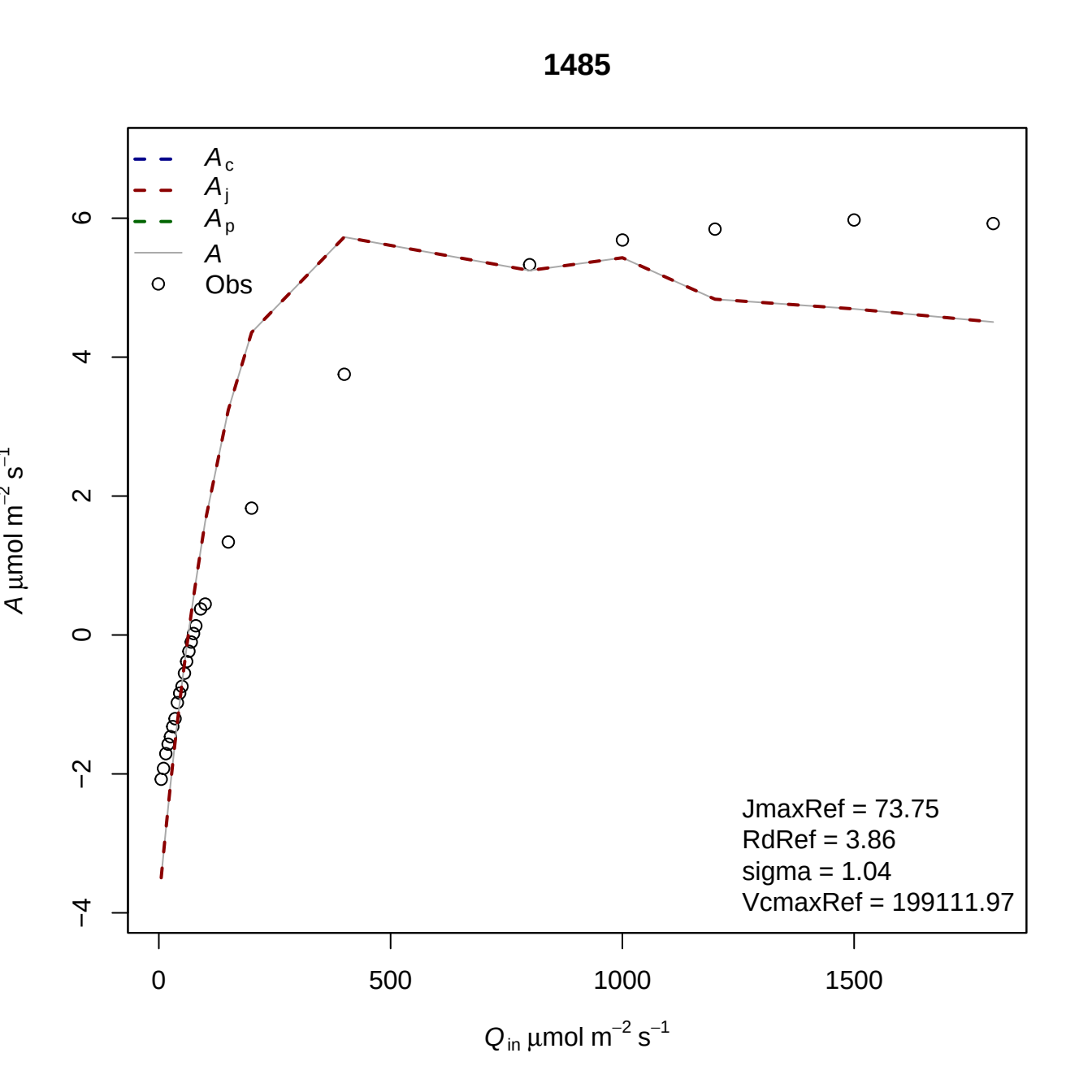

1480
A_c
A_j
A_p
A
Obs
JmaxRef = 115.85
RdRef = 4.1
sigma = 1.34
VcmaxRef = 56494.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

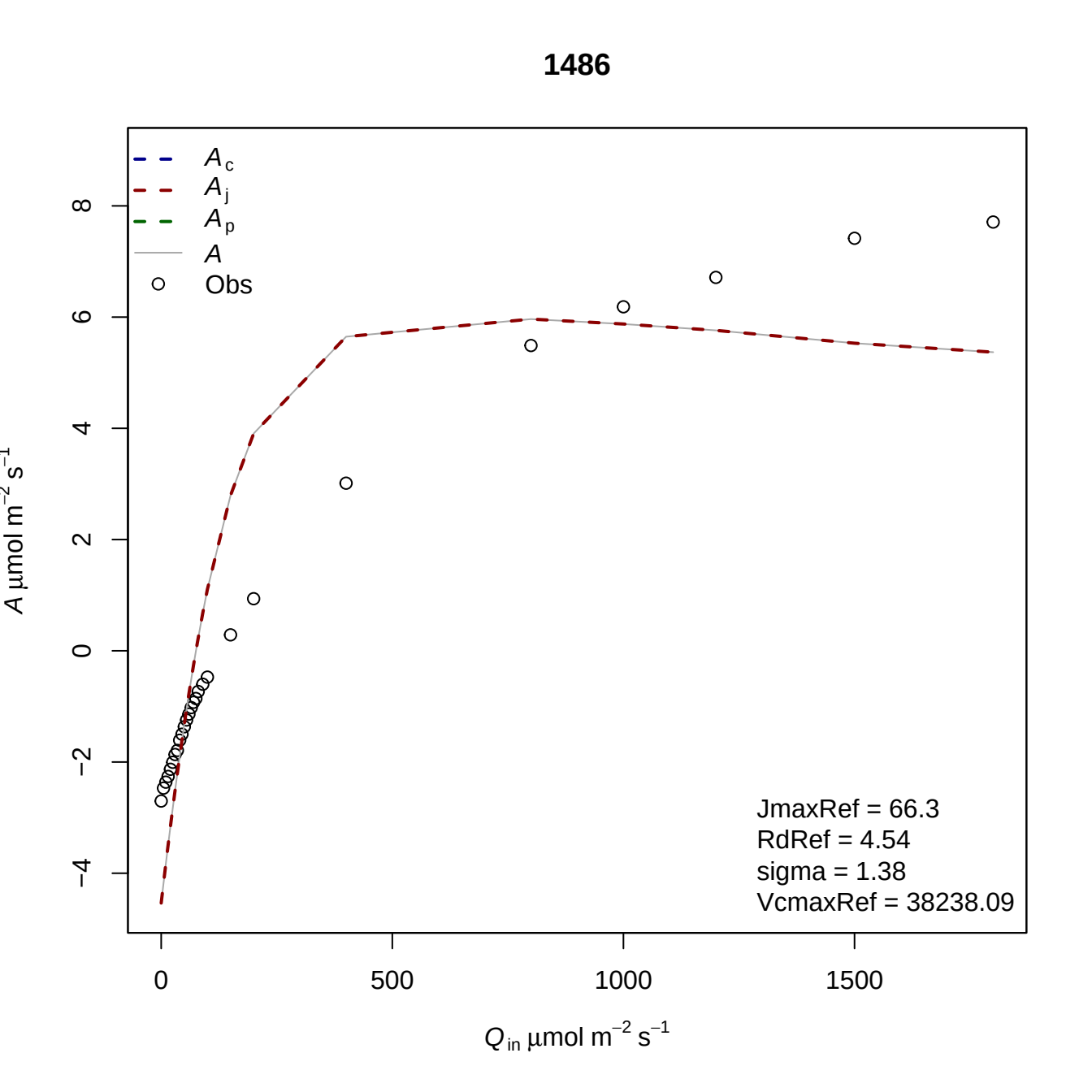

1481
A_c
A_j
A_p
A
Obs
JmaxRef = 103.06
RdRef = 4.23
sigma = 1.39
VcmaxRef = 105830.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

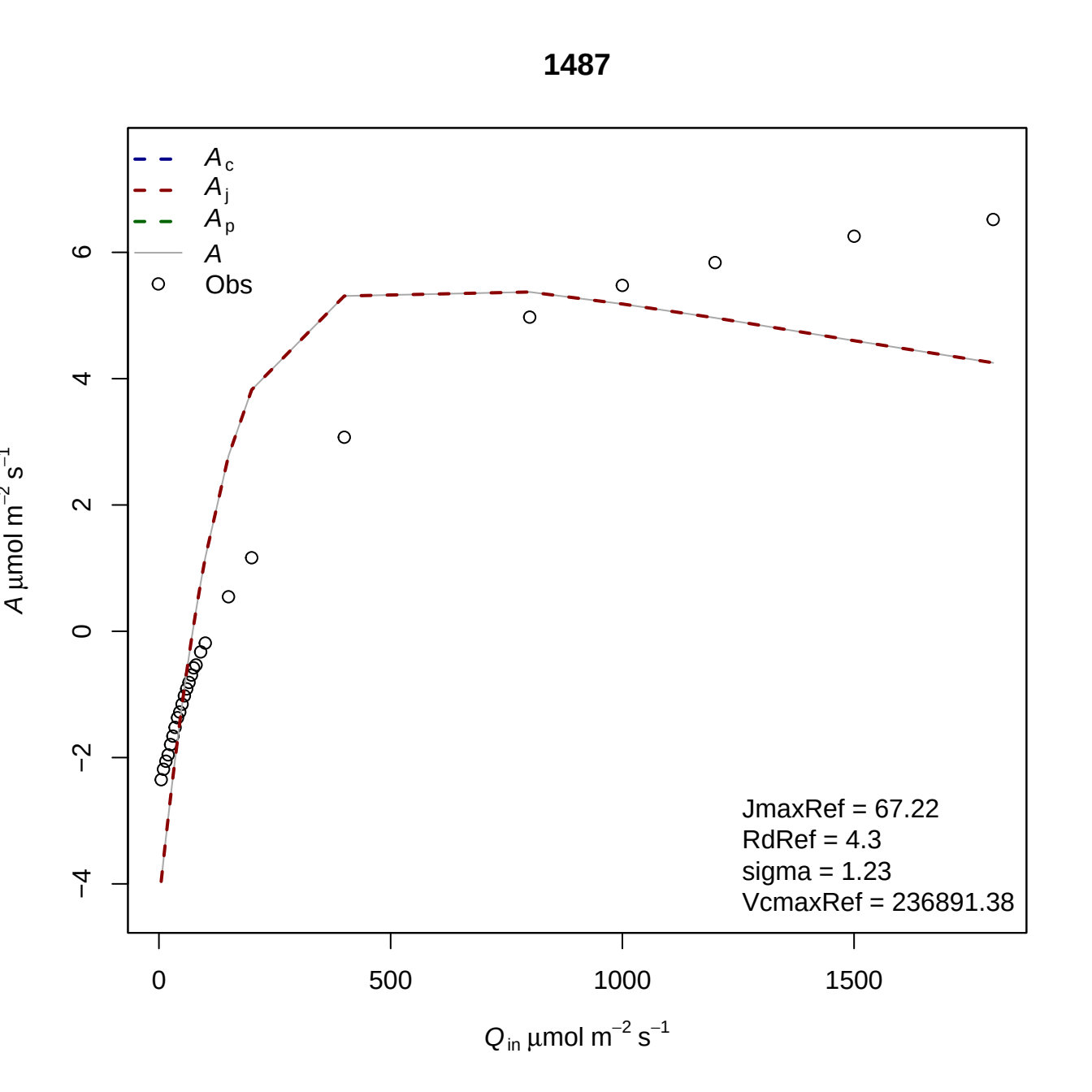

1482
A_c
A_j
A_p
A
Obs
JmaxRef = 121.06
RdRef = 3.43
sigma = 0.94
VcmaxRef = 3298.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

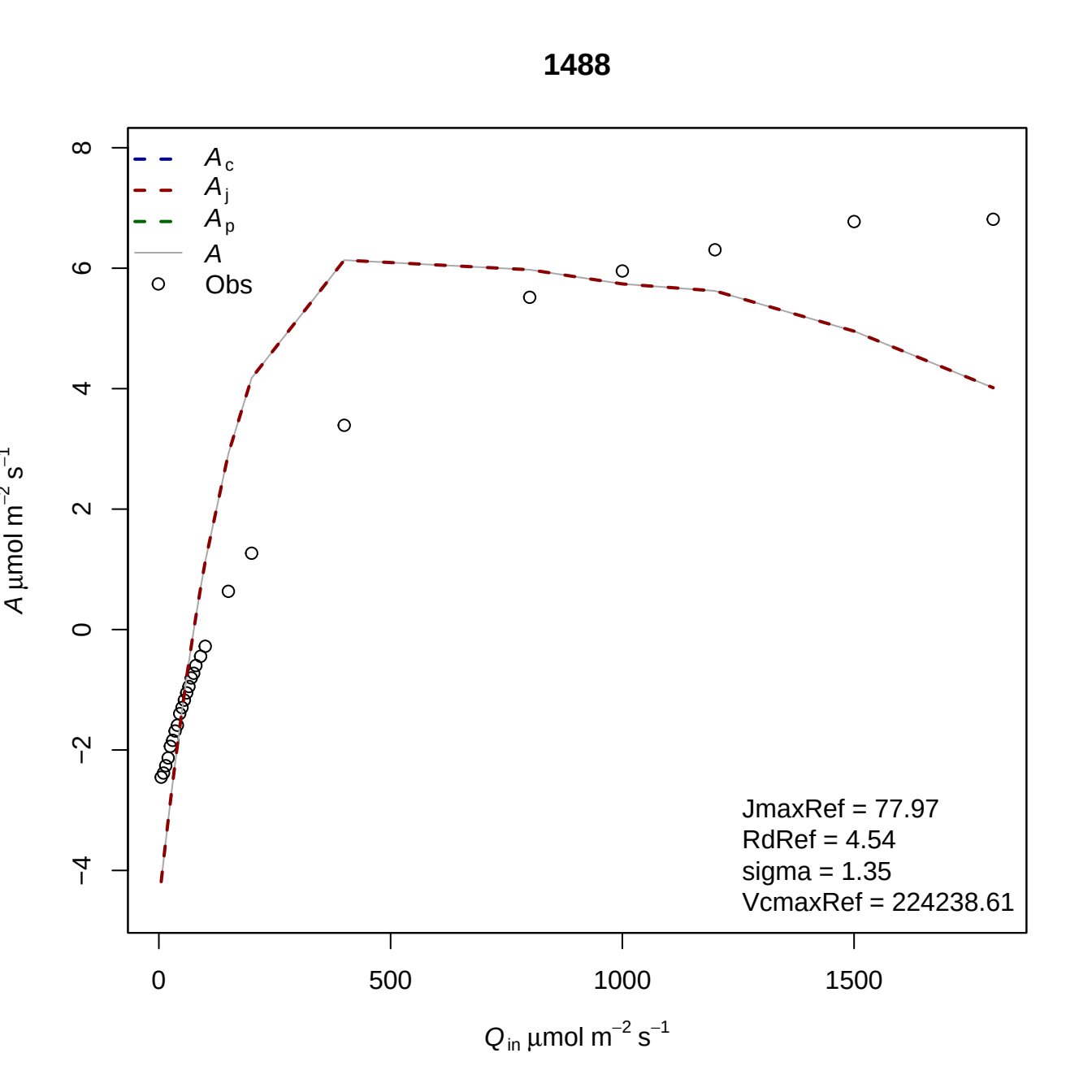

1483
A_c
A_j
A_p
A
Obs
JmaxRef = 84.37
RdRef = 3.75
sigma = 0.79
VcmaxRef = 77416.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

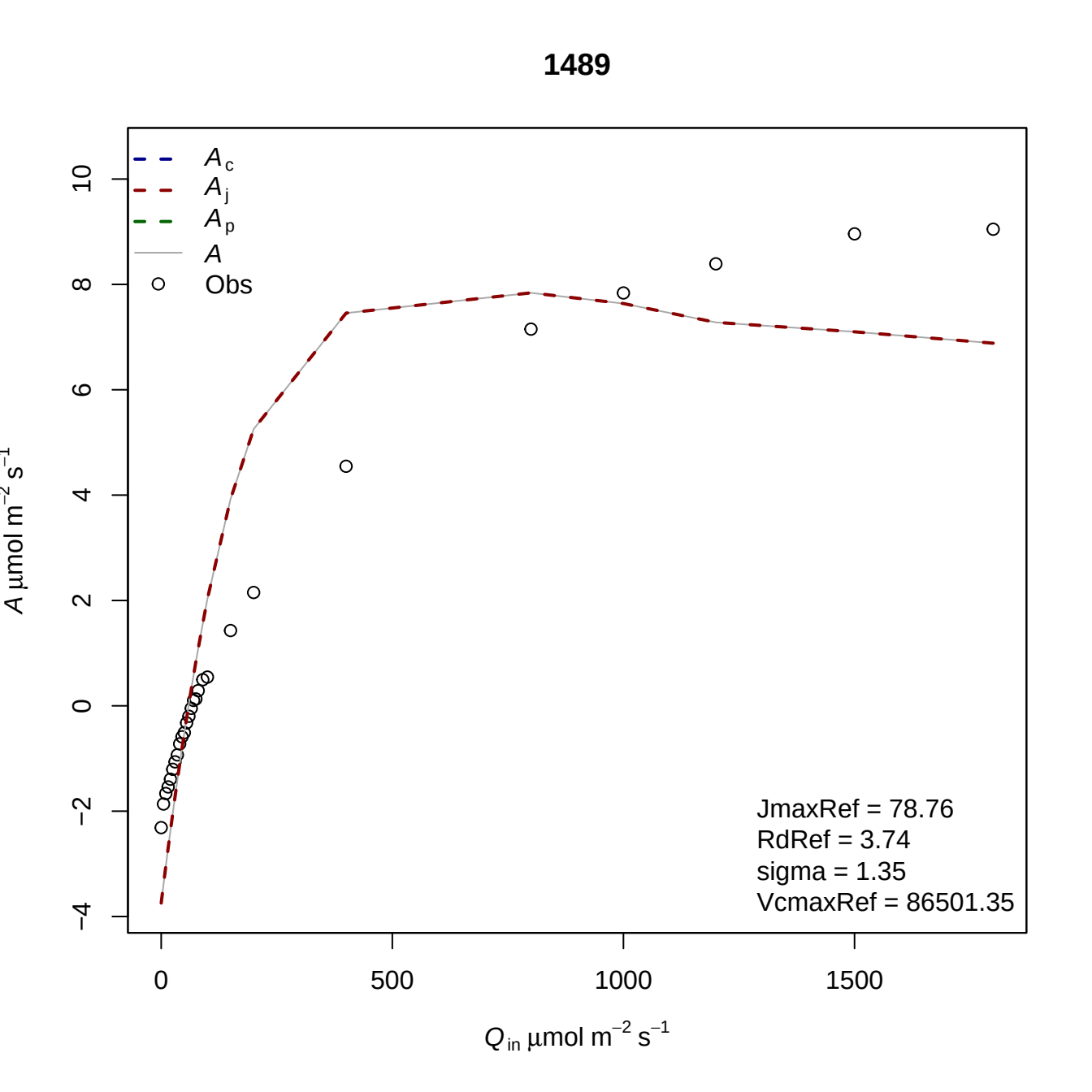

1484
A_c
A_j
A_p
A
Obs
JmaxRef = 102.35
RdRef = 4.16
sigma = 1.16
VcmaxRef = 121835.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

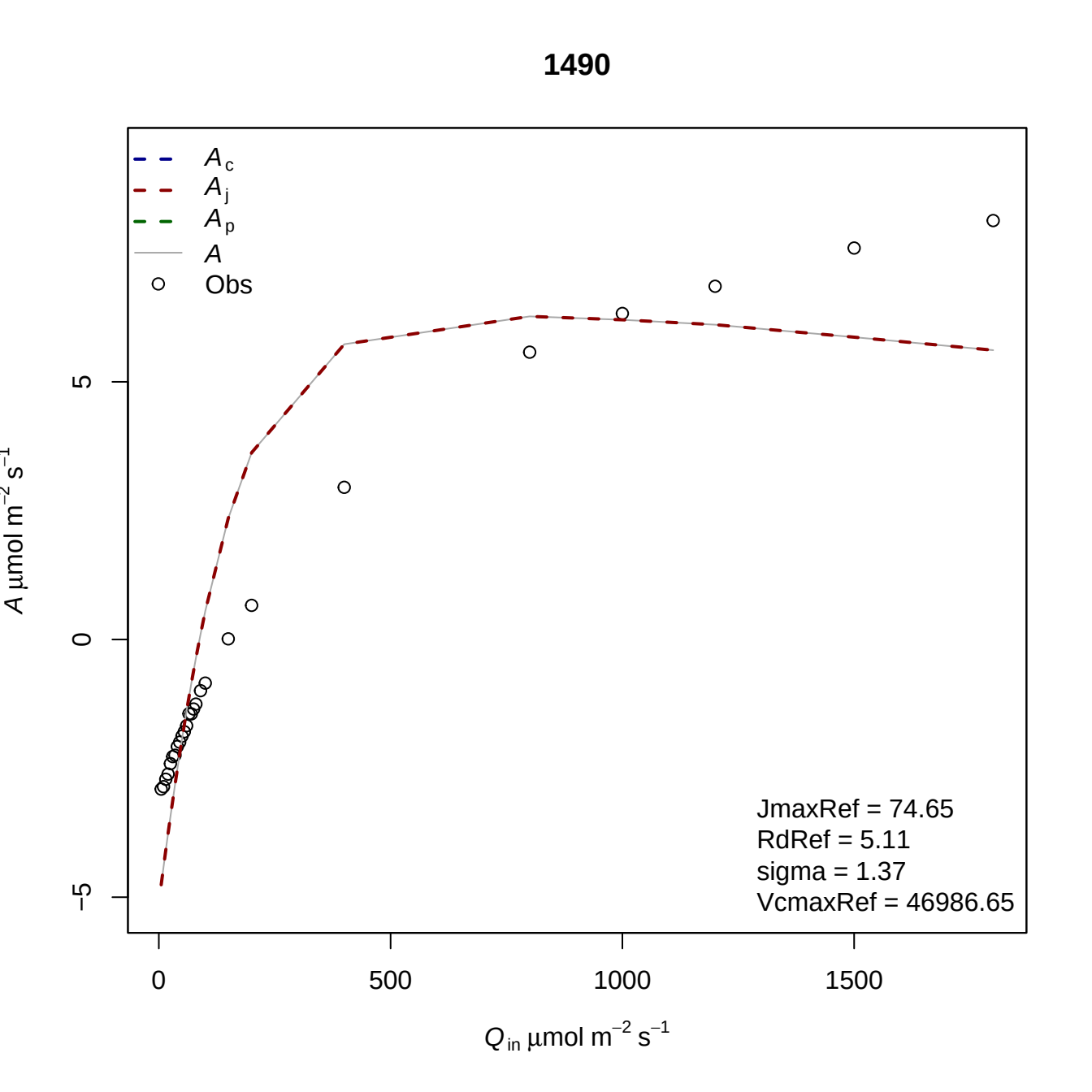

1485
A_c
A_j
A_p
A
Obs
JmaxRef = 73.75
RdRef = 3.86
sigma = 1.04
VcmaxRef = 199111.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

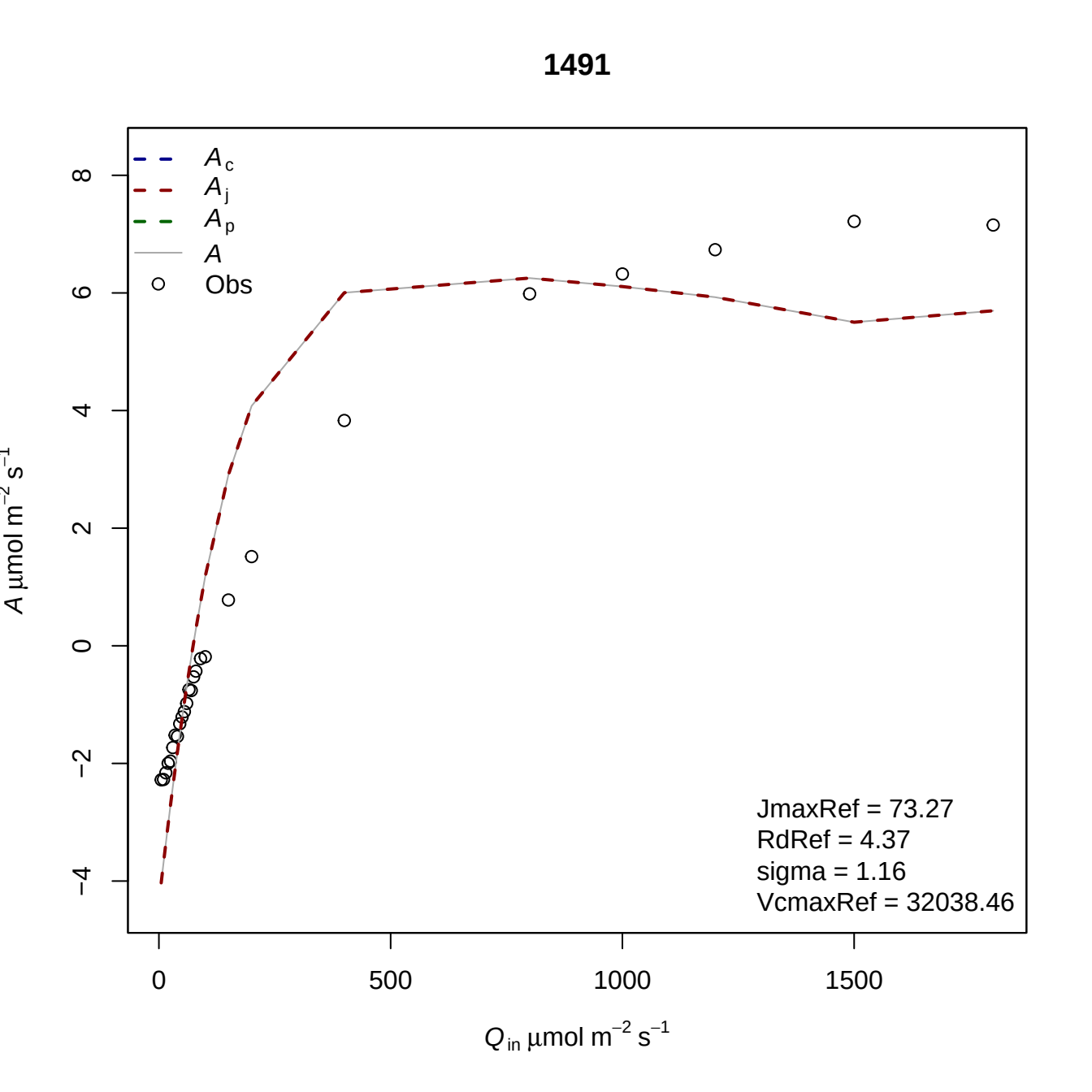

1486
A_c
A_j
A_p
A
Obs
JmaxRef = 66.3
RdRef = 4.54
sigma = 1.38
VcmaxRef = 38238.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

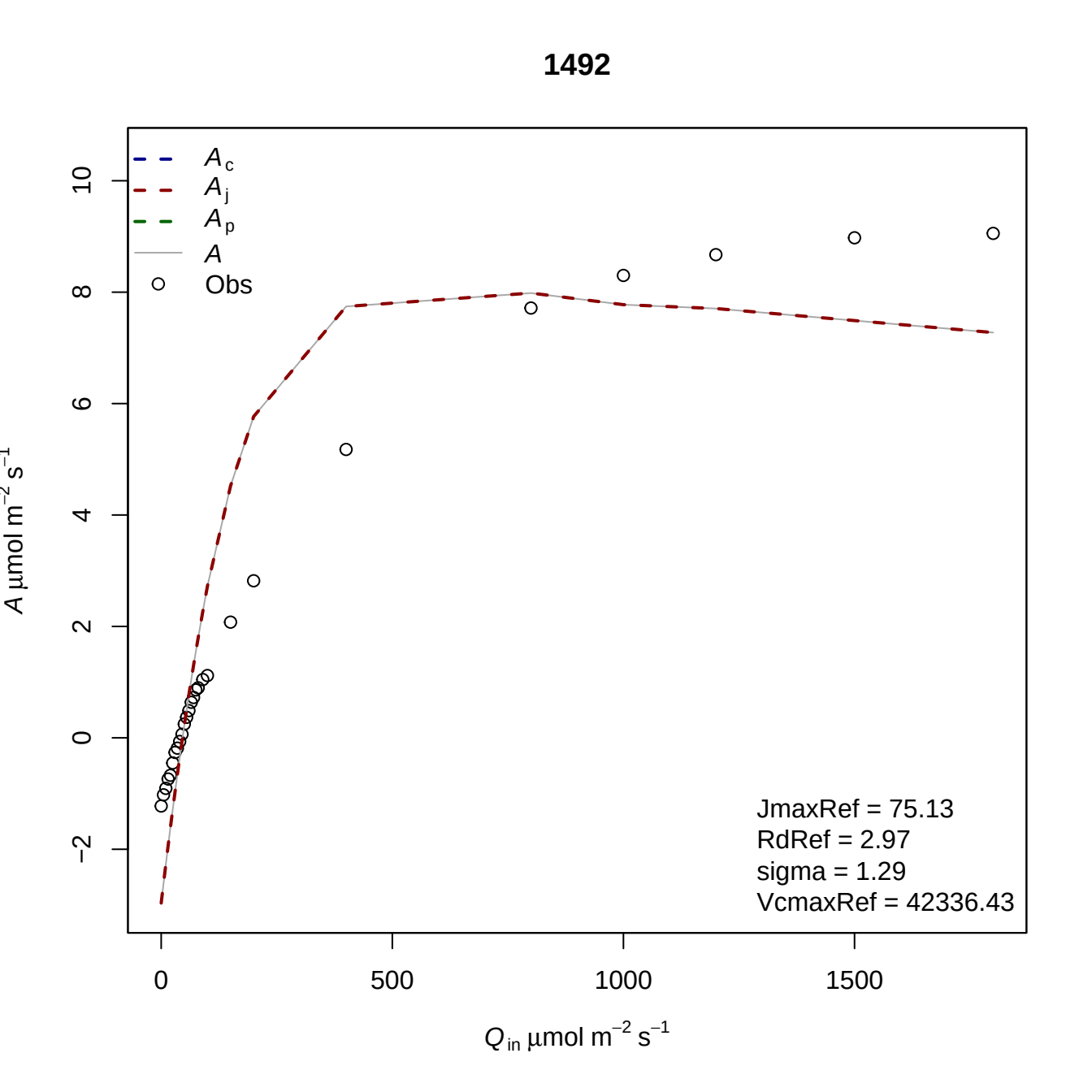

1487
A_c
A_j
A_p
A
Obs
JmaxRef = 67.22
RdRef = 4.3
sigma = 1.23
VcmaxRef = 236891.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

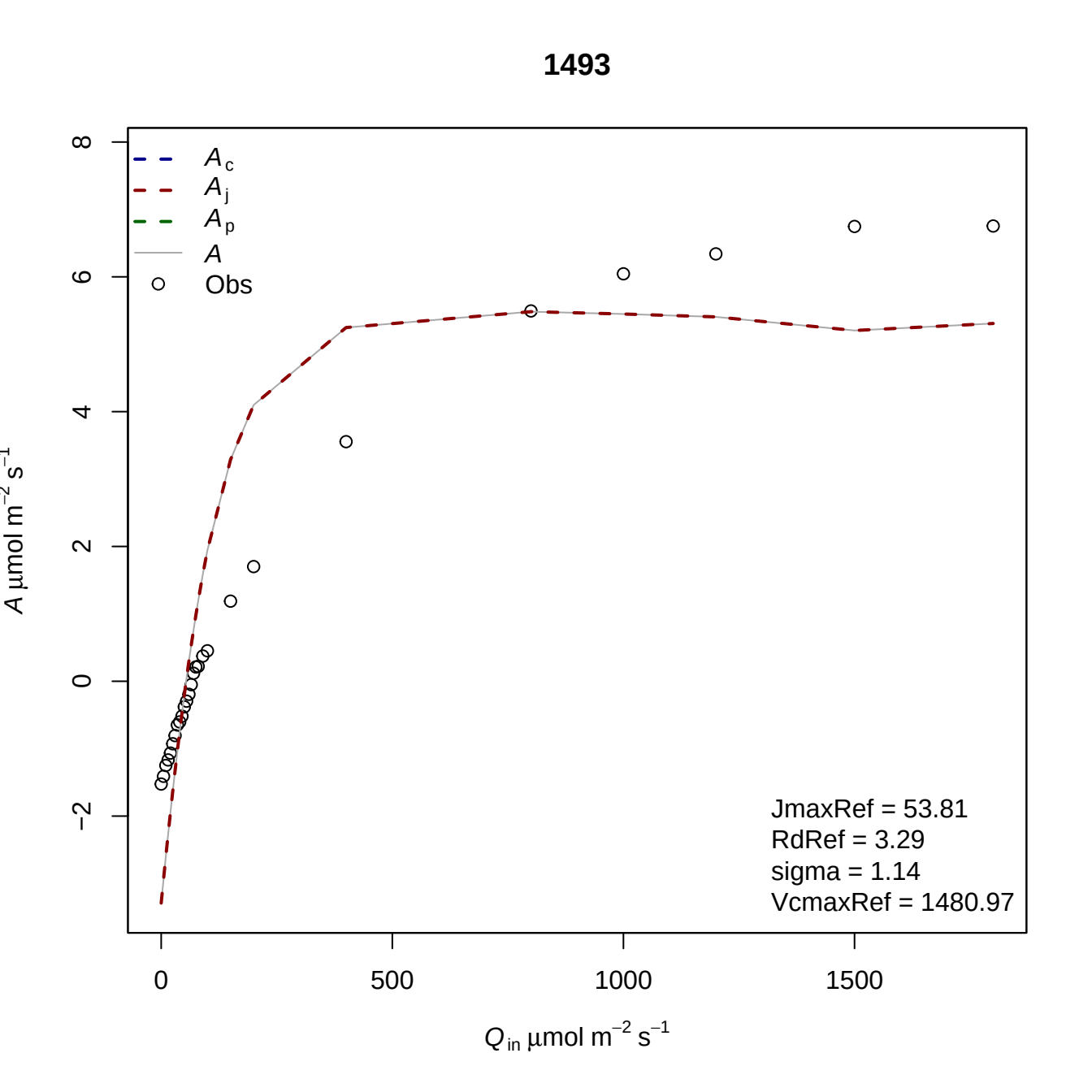

1488
A_c
A_j
A_p
A
Obs
JmaxRef = 77.97
RdRef = 4.54
sigma = 1.35
VcmaxRef = 224238.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

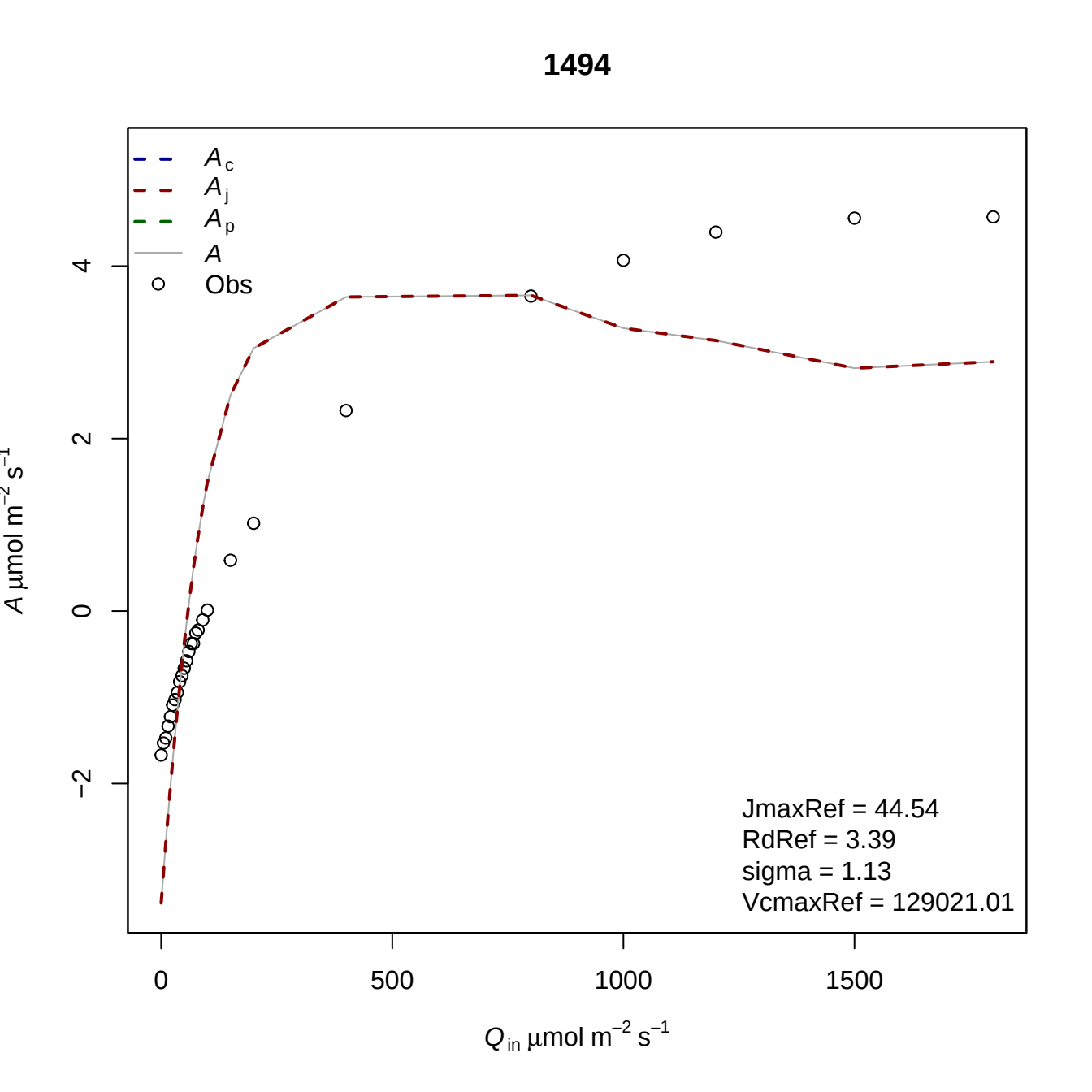

1489
A_c
A_j
A_p
A
Obs
JmaxRef = 78.76
RdRef = 3.74
sigma = 1.35
VcmaxRef = 86501.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

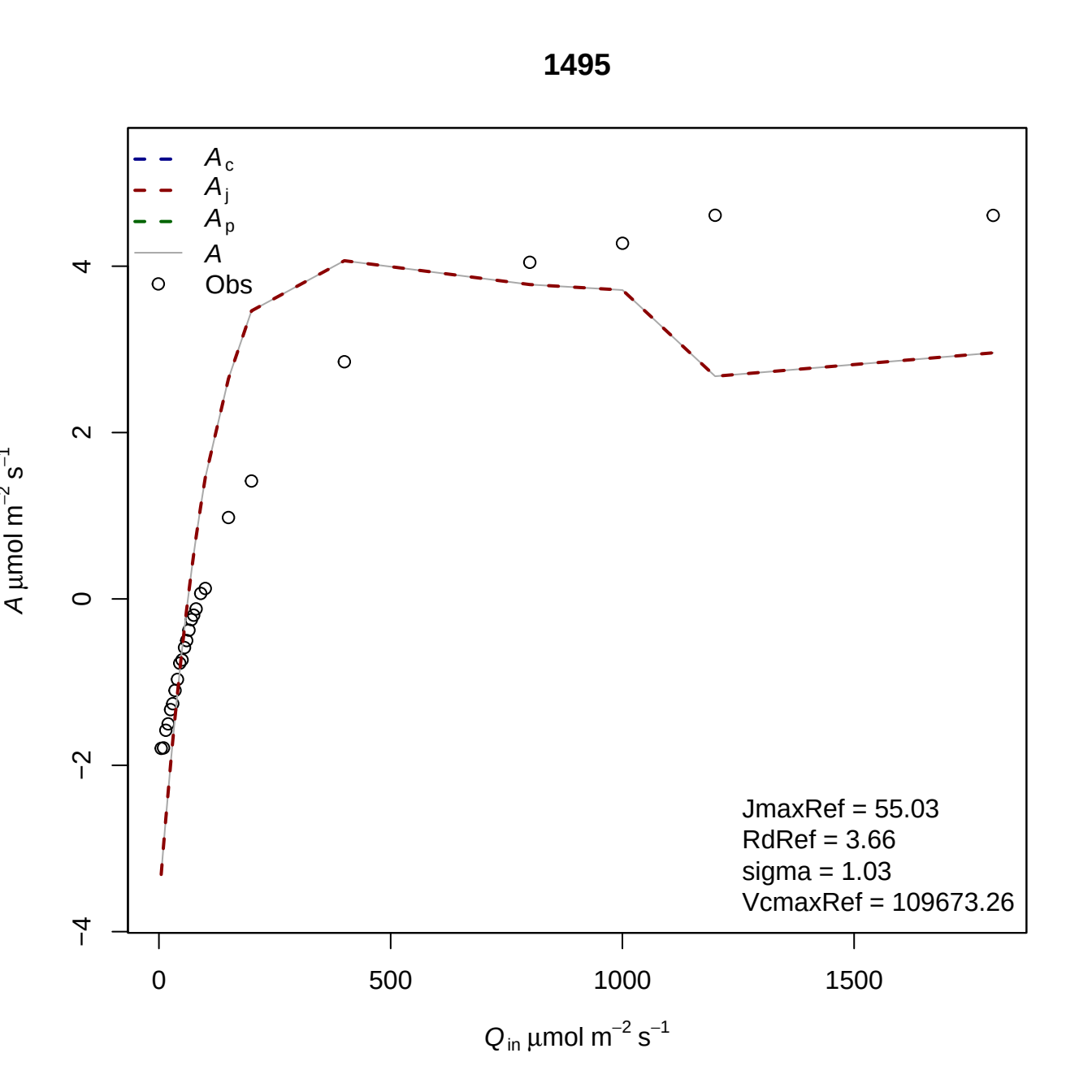

1490
A_c
A_j
A_p
A
Obs
JmaxRef = 74.65
RdRef = 5.11
sigma = 1.37
VcmaxRef = 46986.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
0
500
1000
1500

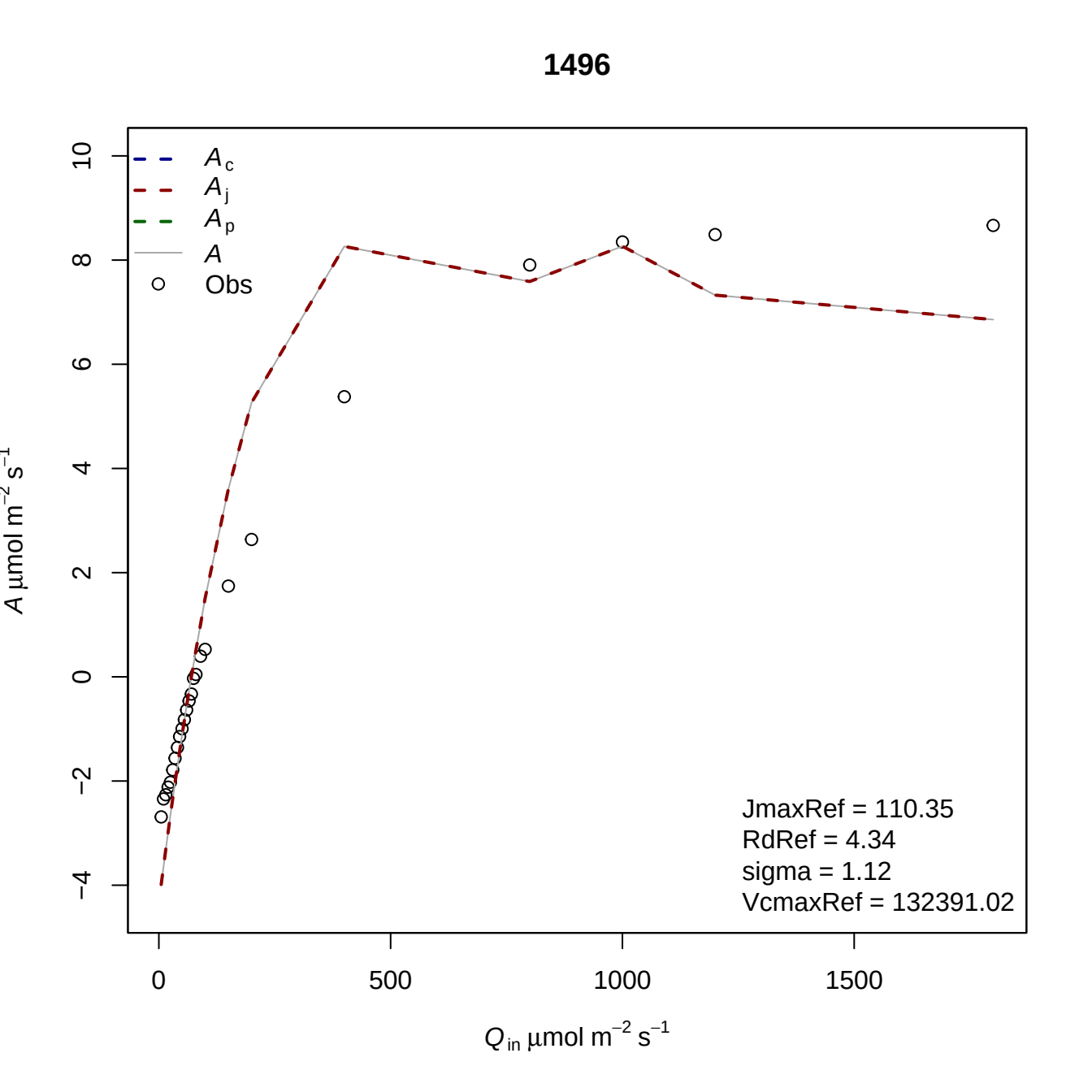

1491
A_c
A_j
A_p
A
Obs
JmaxRef = 73.27
RdRef = 4.37
sigma = 1.16
VcmaxRef = 32038.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

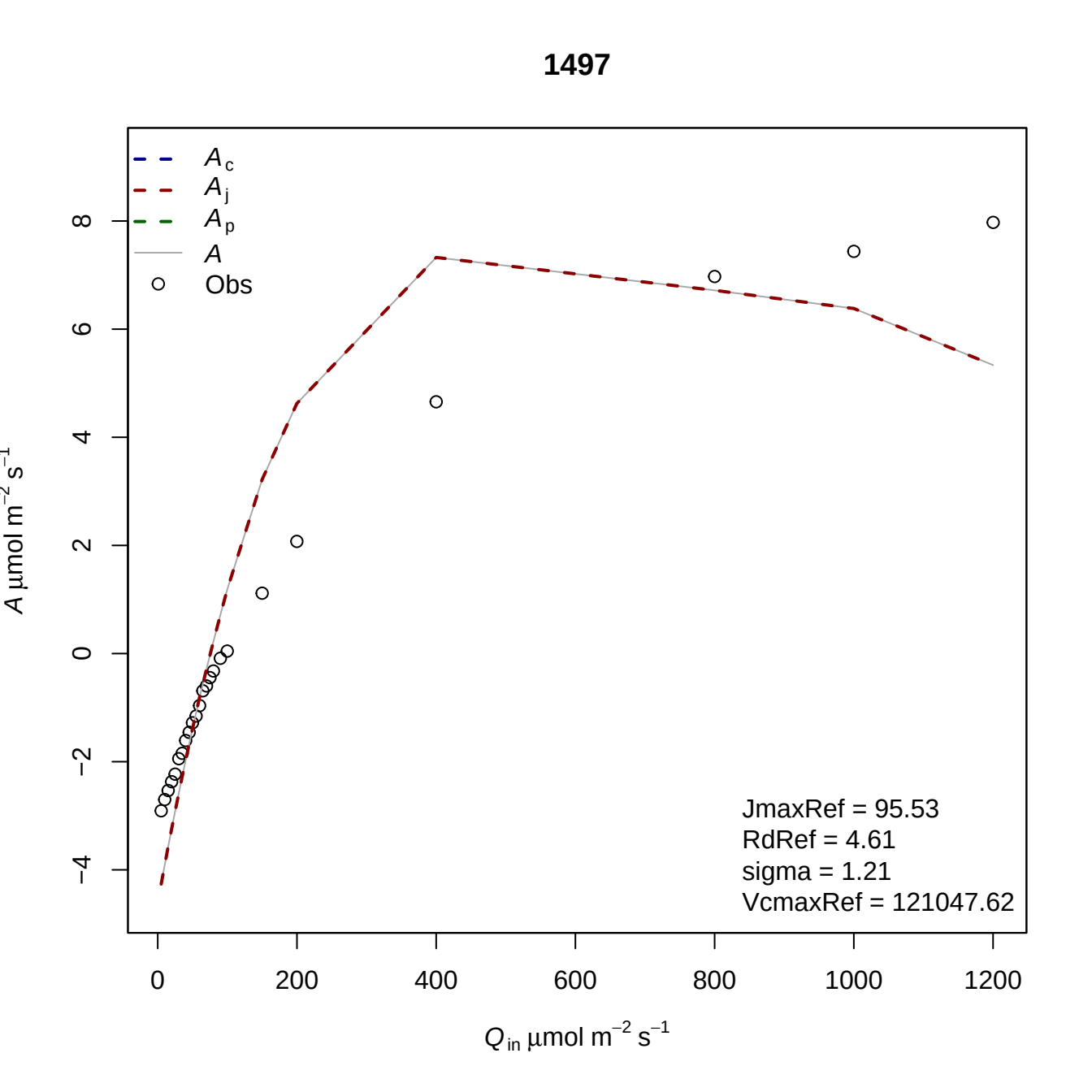

1492
A_c
A_j
A_p
A
Obs
JmaxRef = 75.13
RdRef = 2.97
sigma = 1.29
VcmaxRef = 42336.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

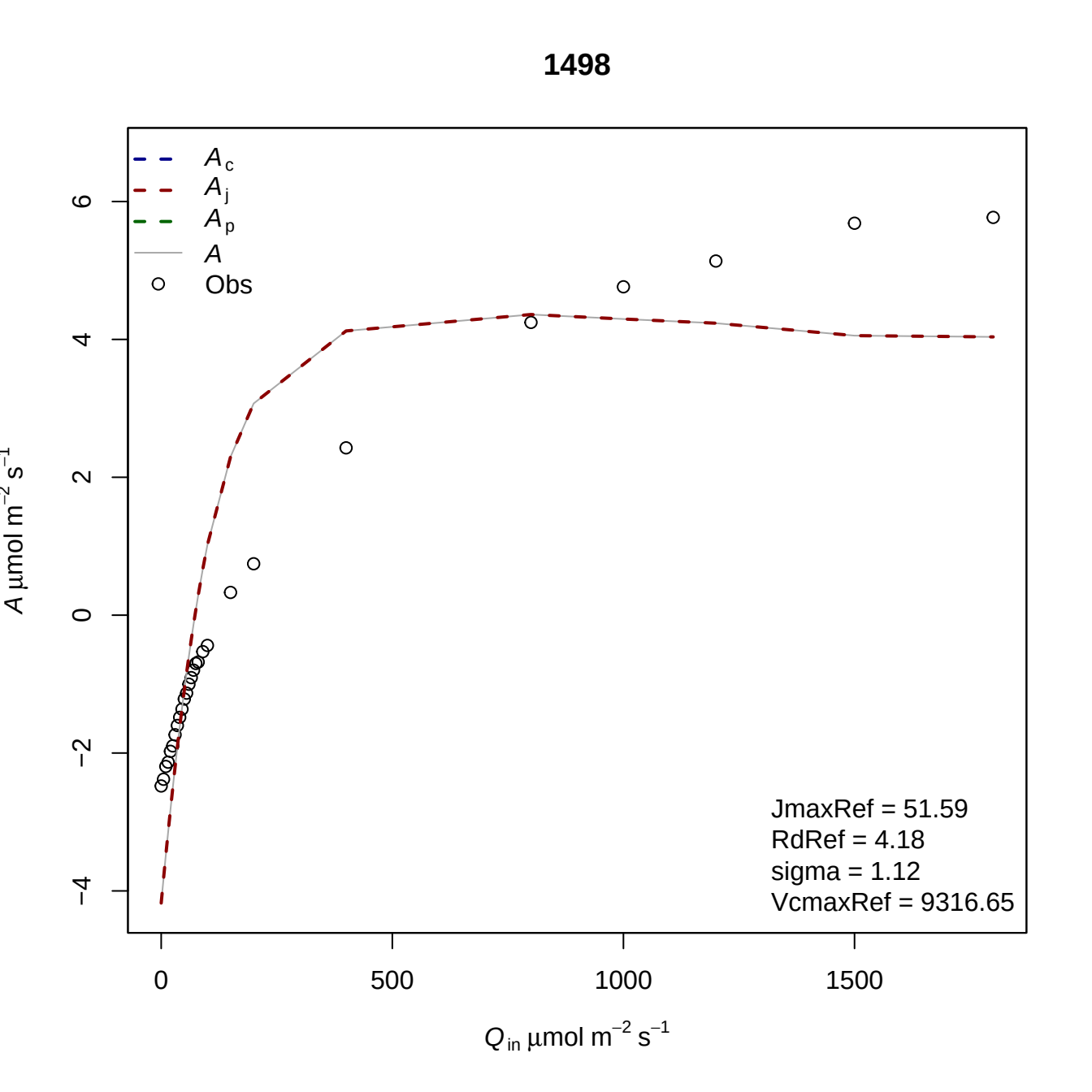

1493
A_c
A_j
A_p
A
Obs
JmaxRef = 53.81
RdRef = 3.29
sigma = 1.14
VcmaxRef = 1480.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

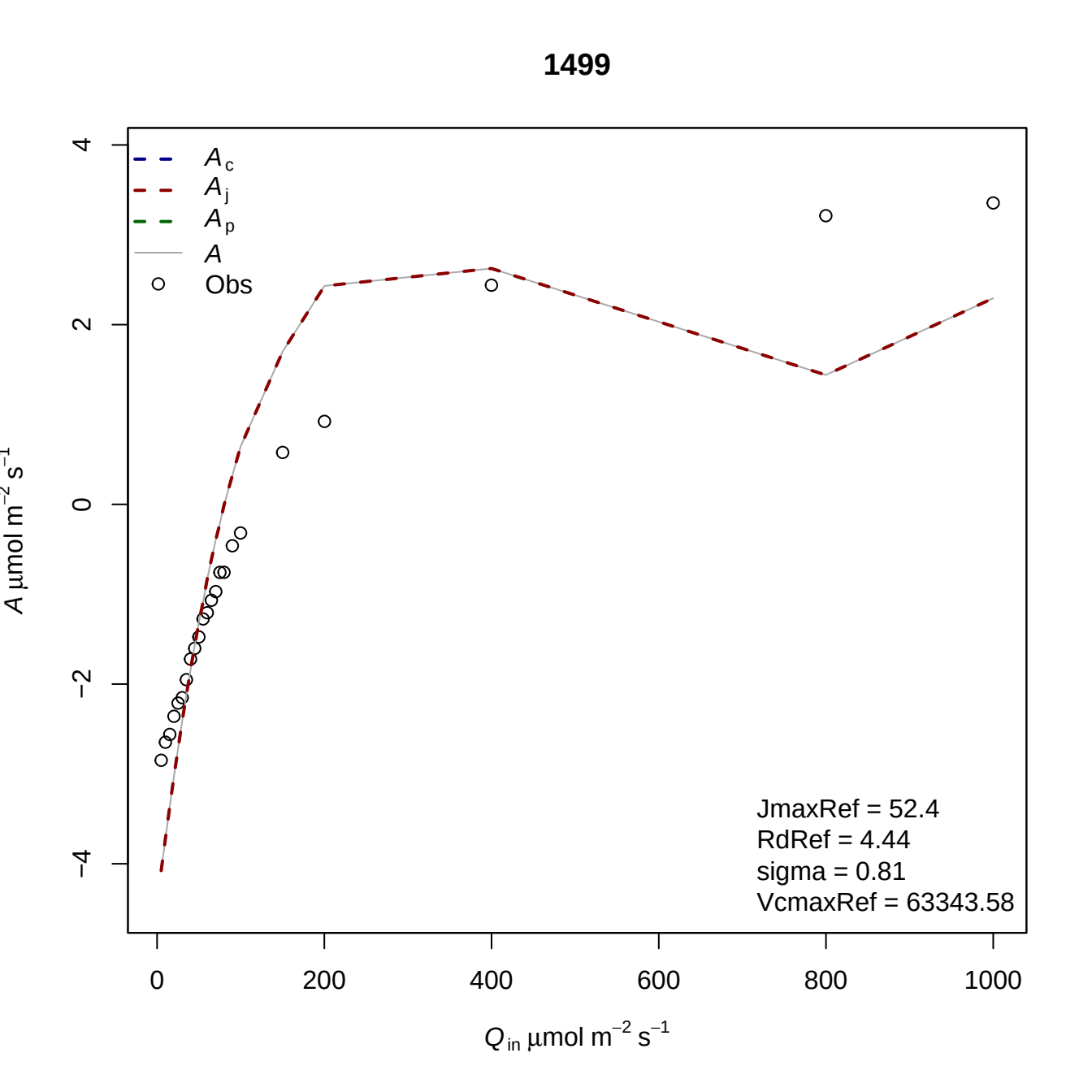

1494
A_c
A_j
A_p
A
Obs
JmaxRef = 44.54
RdRef = 3.39
sigma = 1.13
VcmaxRef = 129021.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

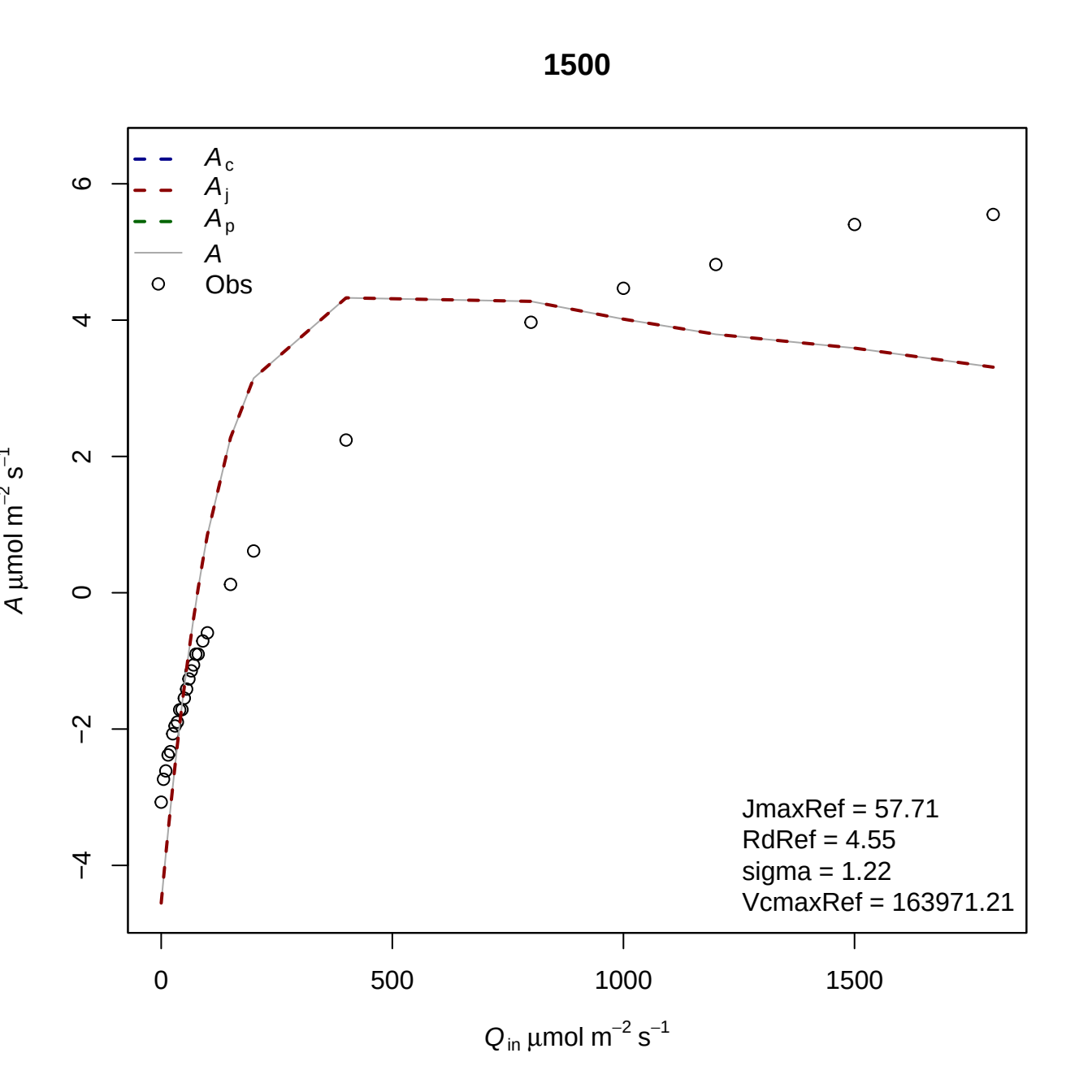

1495
A_c
A_j
A_p
A
Obs
JmaxRef = 55.03
RdRef = 3.66
sigma = 1.03
VcmaxRef = 109673.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

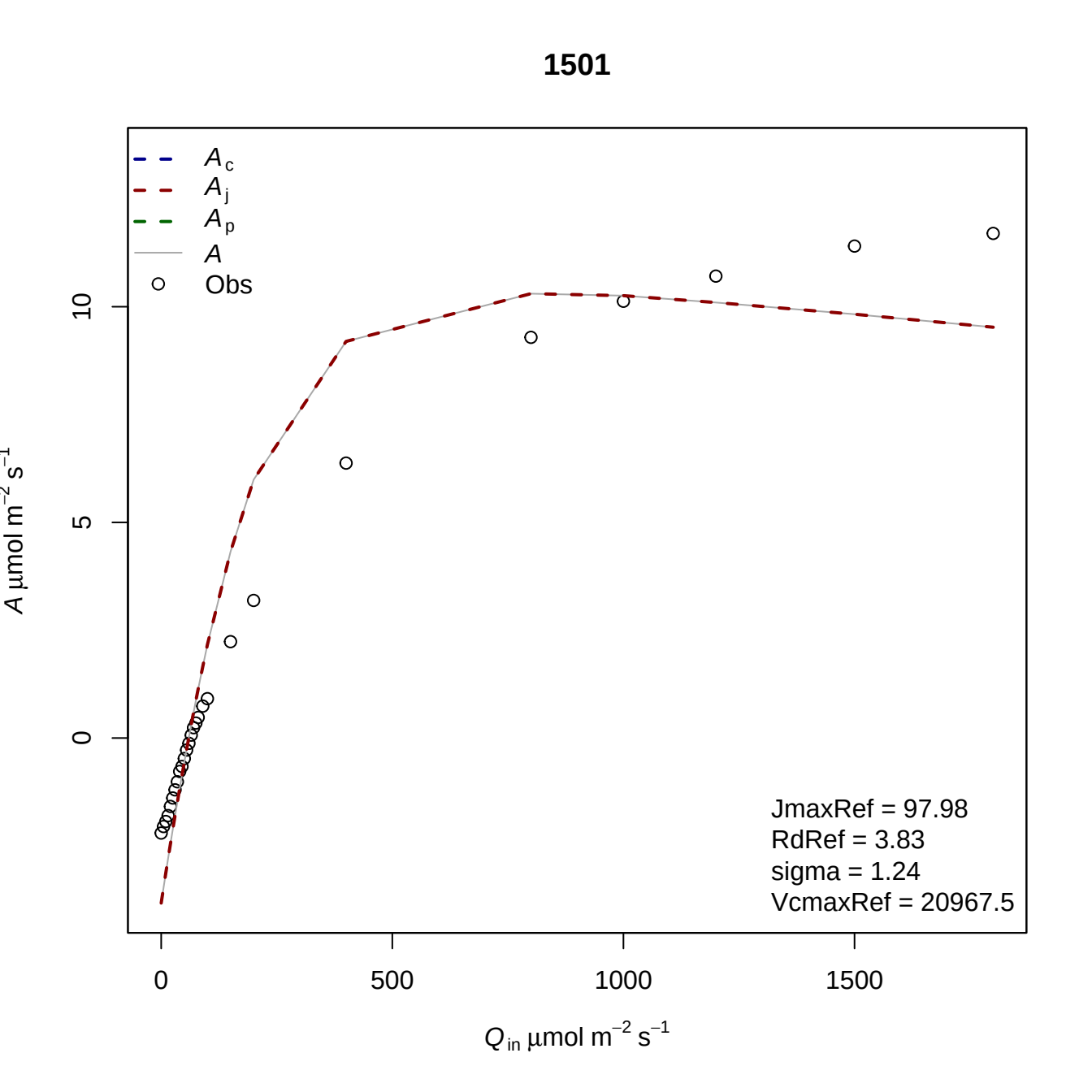

1496
A_c
A_j
A_p
A
Obs
JmaxRef = 110.35
RdRef = 4.34
sigma = 1.12
VcmaxRef = 132391.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

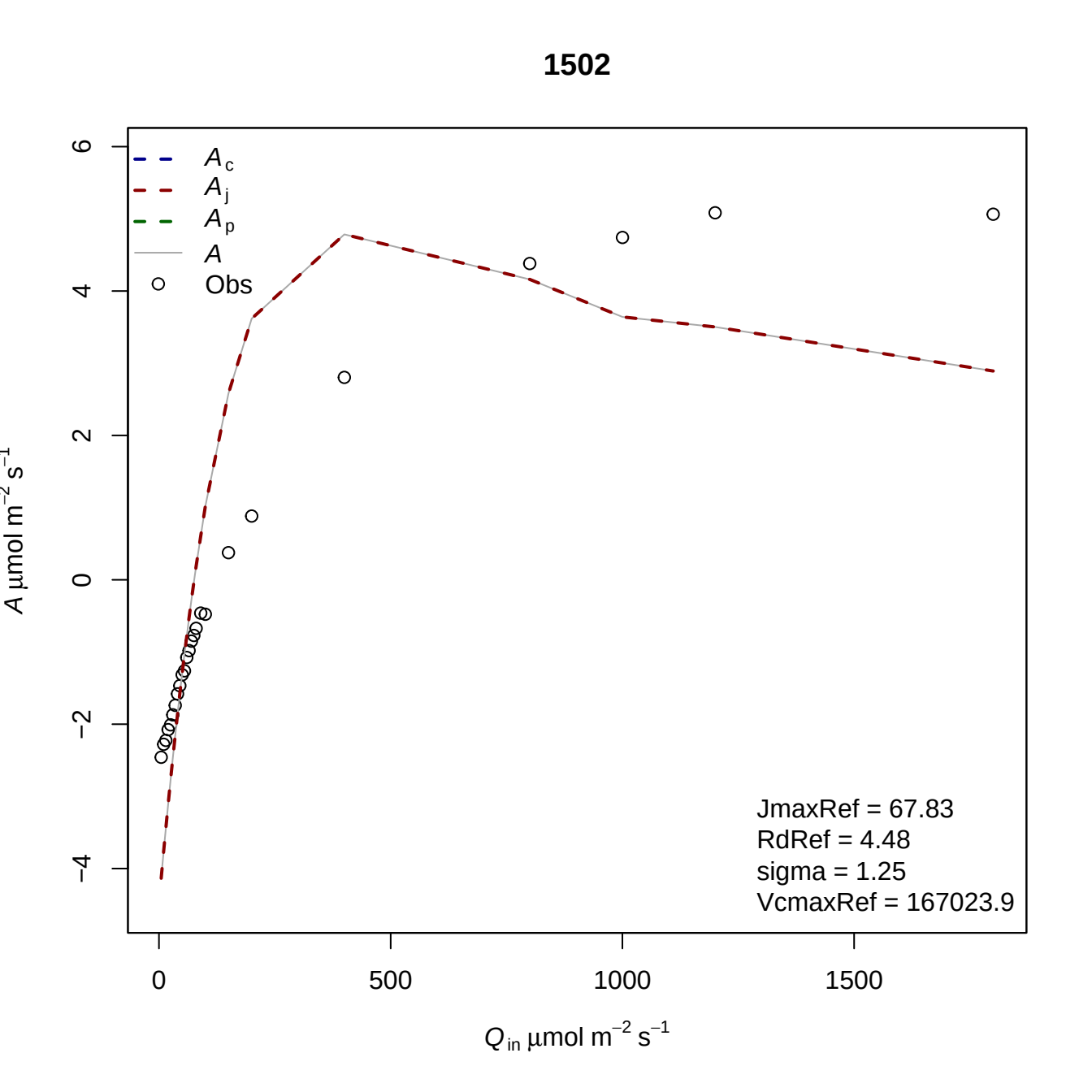

1497
A_c
A_j
A_p
A
Obs
JmaxRef = 95.53
RdRef = 4.61
sigma = 1.21
VcmaxRef = 121047.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

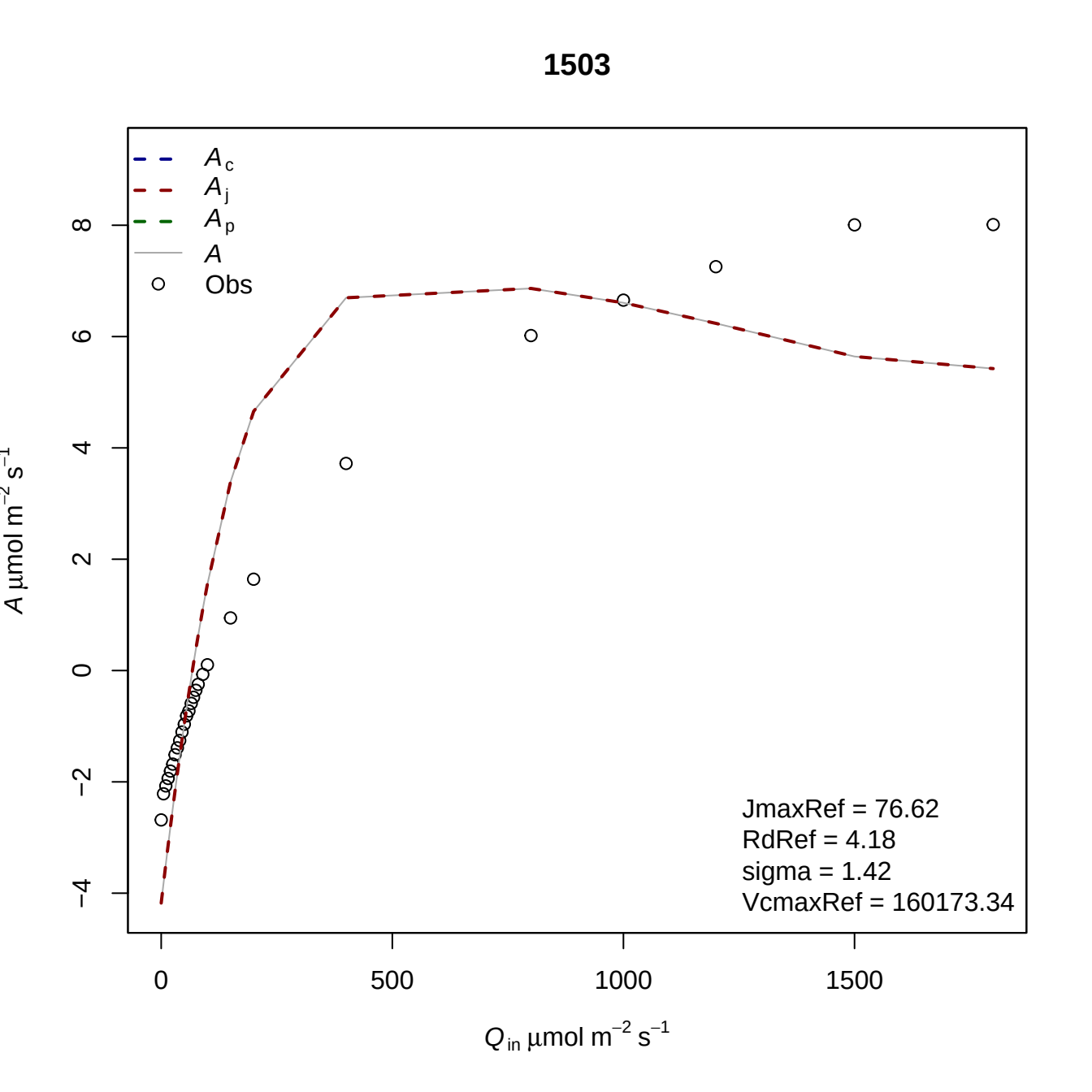

1498
A_c
A_j
A_p
A
Obs
JmaxRef = 51.59
RdRef = 4.18
sigma = 1.12
VcmaxRef = 9316.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

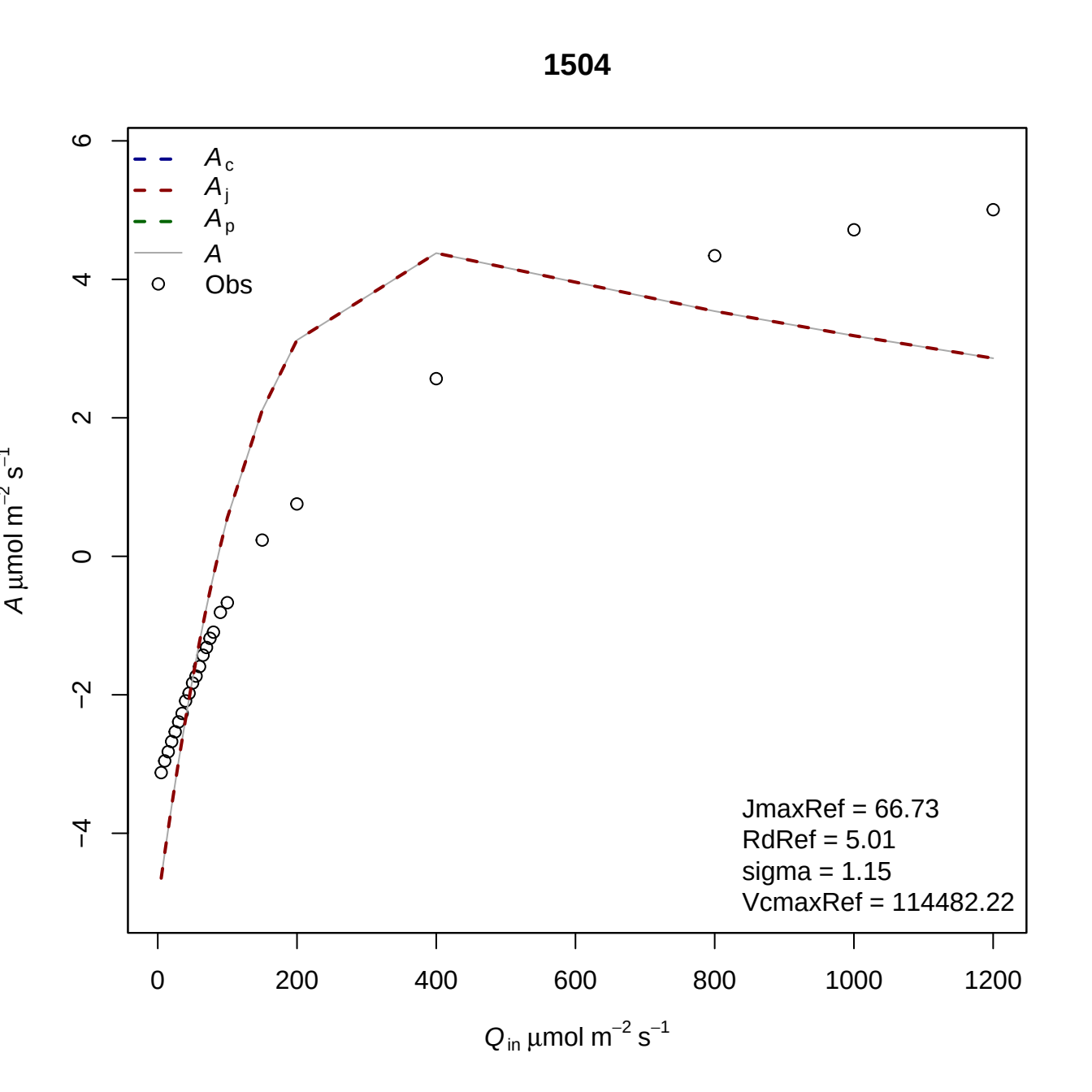

1499
A_c
A_j
A_p
A
Obs
JmaxRef = 52.4
RdRef = 4.44
sigma = 0.81
VcmaxRef = 63343.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
200
400
600
800
1000

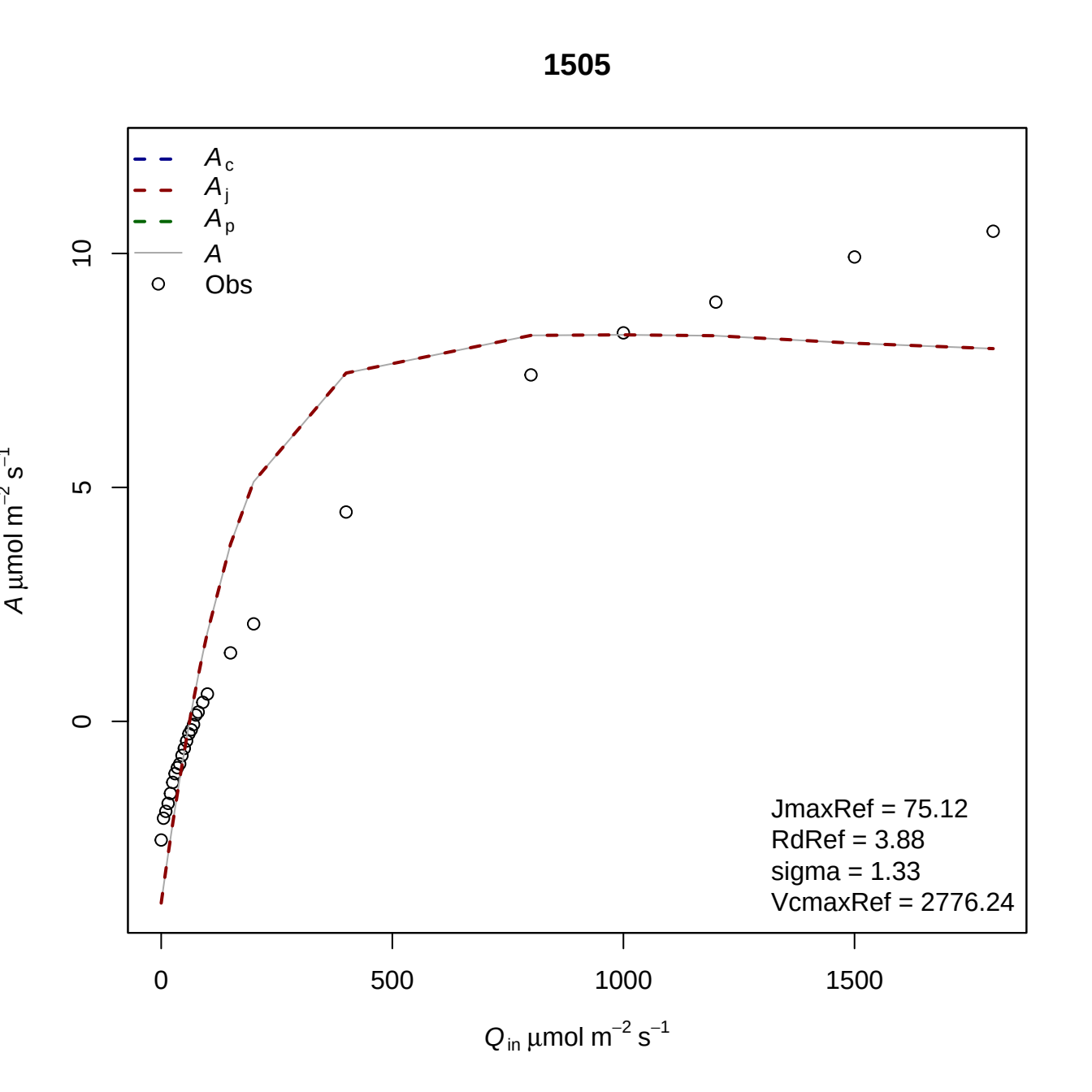

1500
A_c
A_j
A_p
A
Obs
JmaxRef = 57.71
RdRef = 4.55
sigma = 1.22
VcmaxRef = 163971.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

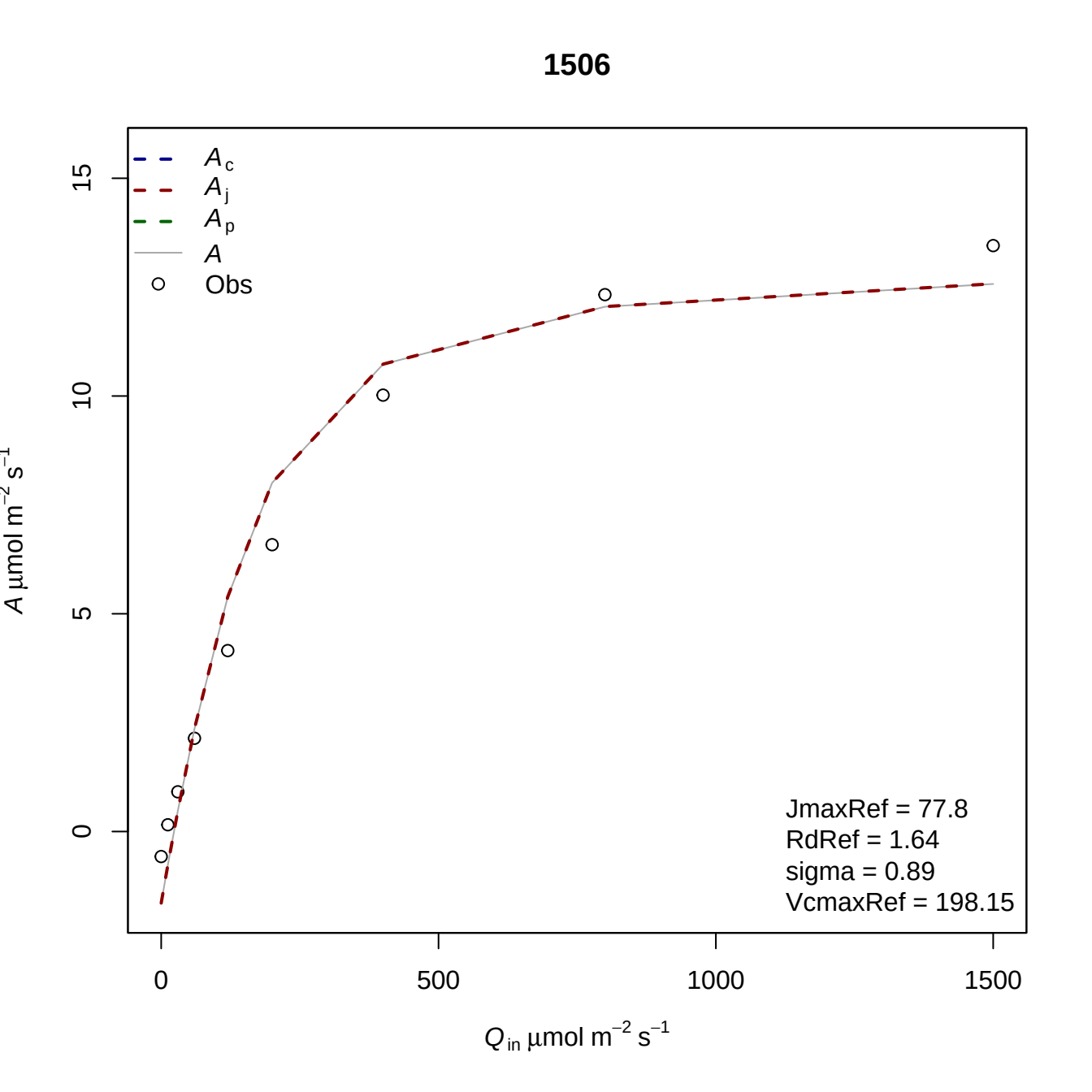

1501
A_c
A_j
A_p
A
Obs
JmaxRef = 97.98
RdRef = 3.83
sigma = 1.24
VcmaxRef = 20967.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

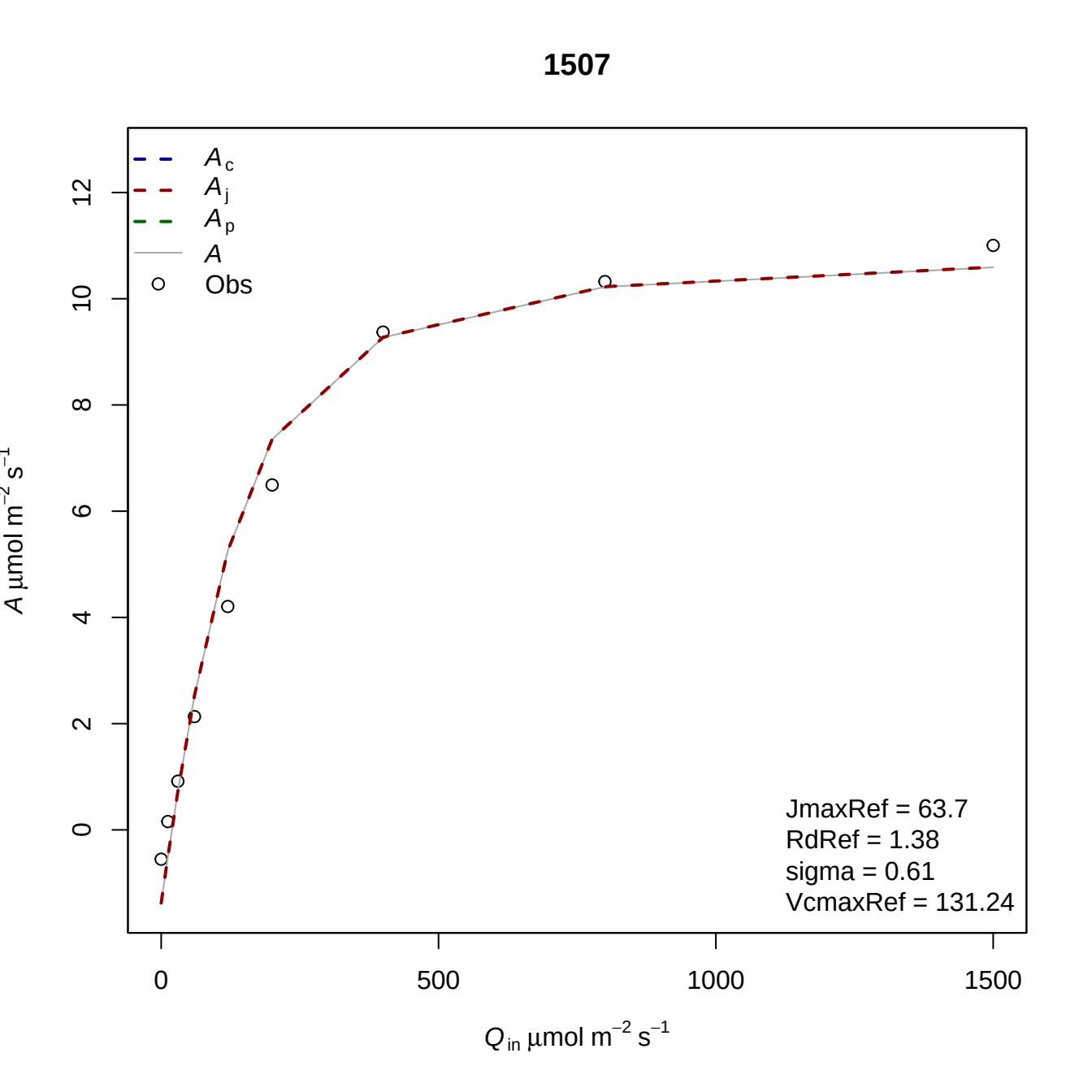

1502
A_c
A_j
A_p
A
Obs
JmaxRef = 67.83
RdRef = 4.48
sigma = 1.25
VcmaxRef = 167023.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

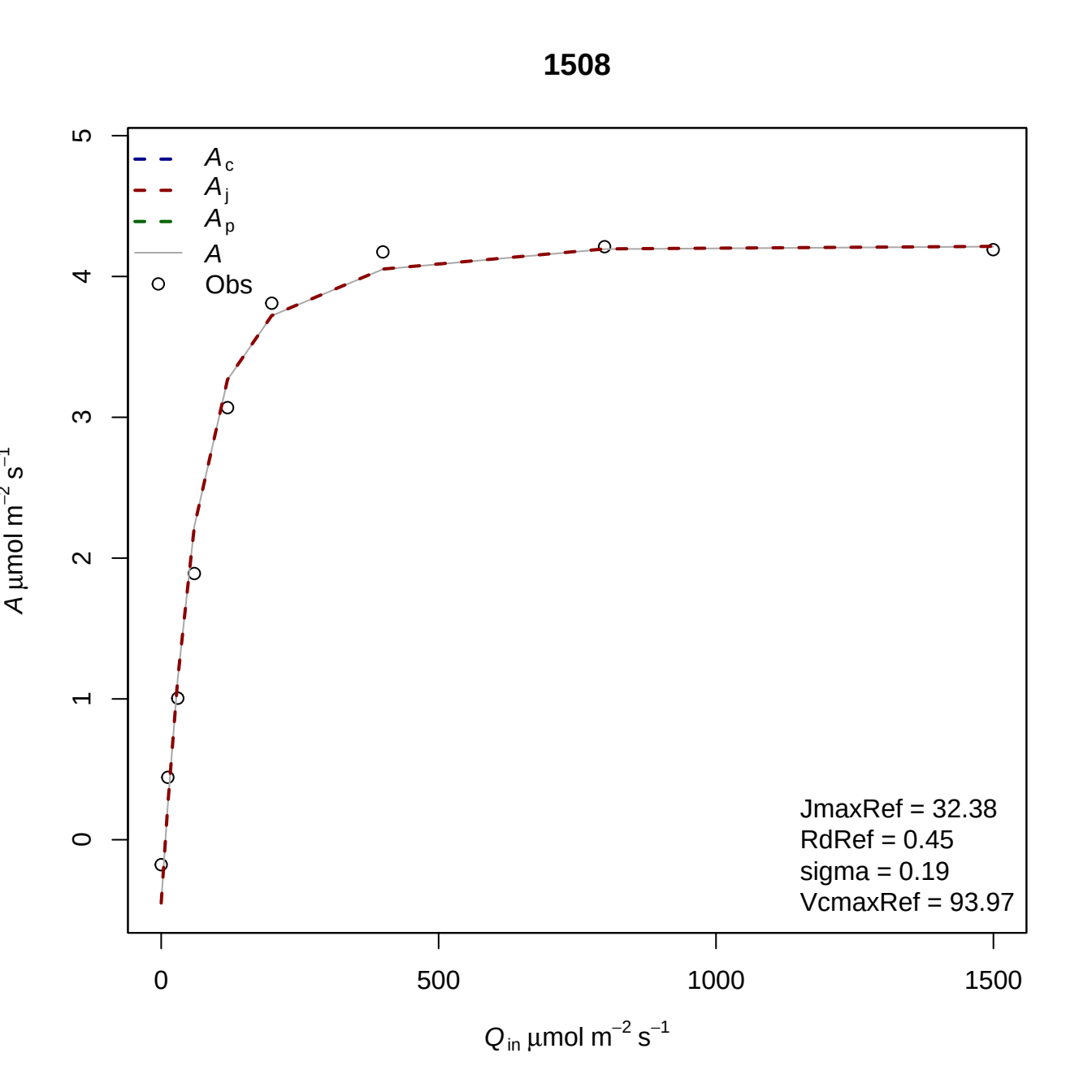

1503
A_c
A_j
A_p
A
Obs
JmaxRef = 76.62
RdRef = 4.18
sigma = 1.42
VcmaxRef = 160173.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

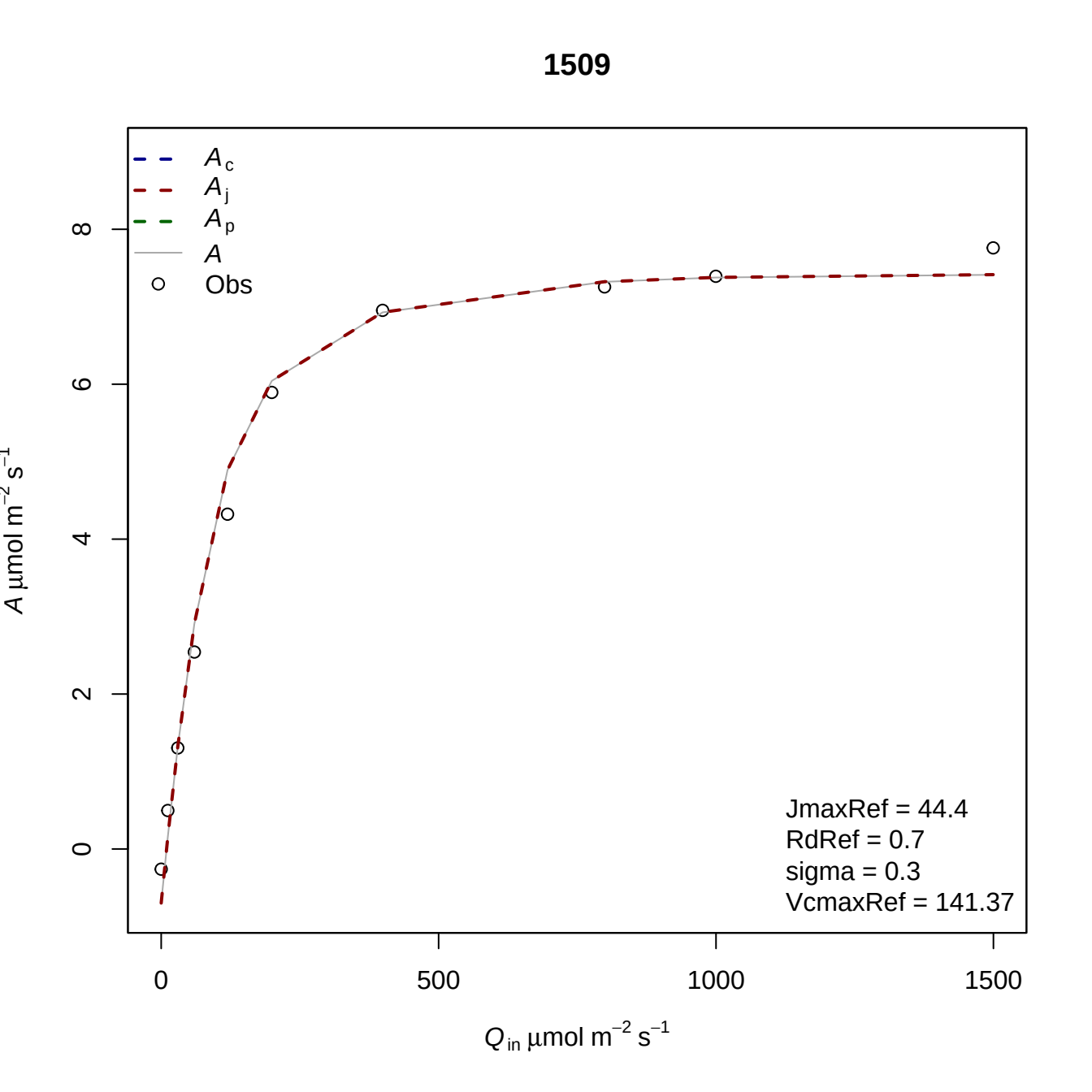

1504
A_c
A_j
A_p
A
Obs
JmaxRef = 66.73
RdRef = 5.01
sigma = 1.15
VcmaxRef = 114482.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

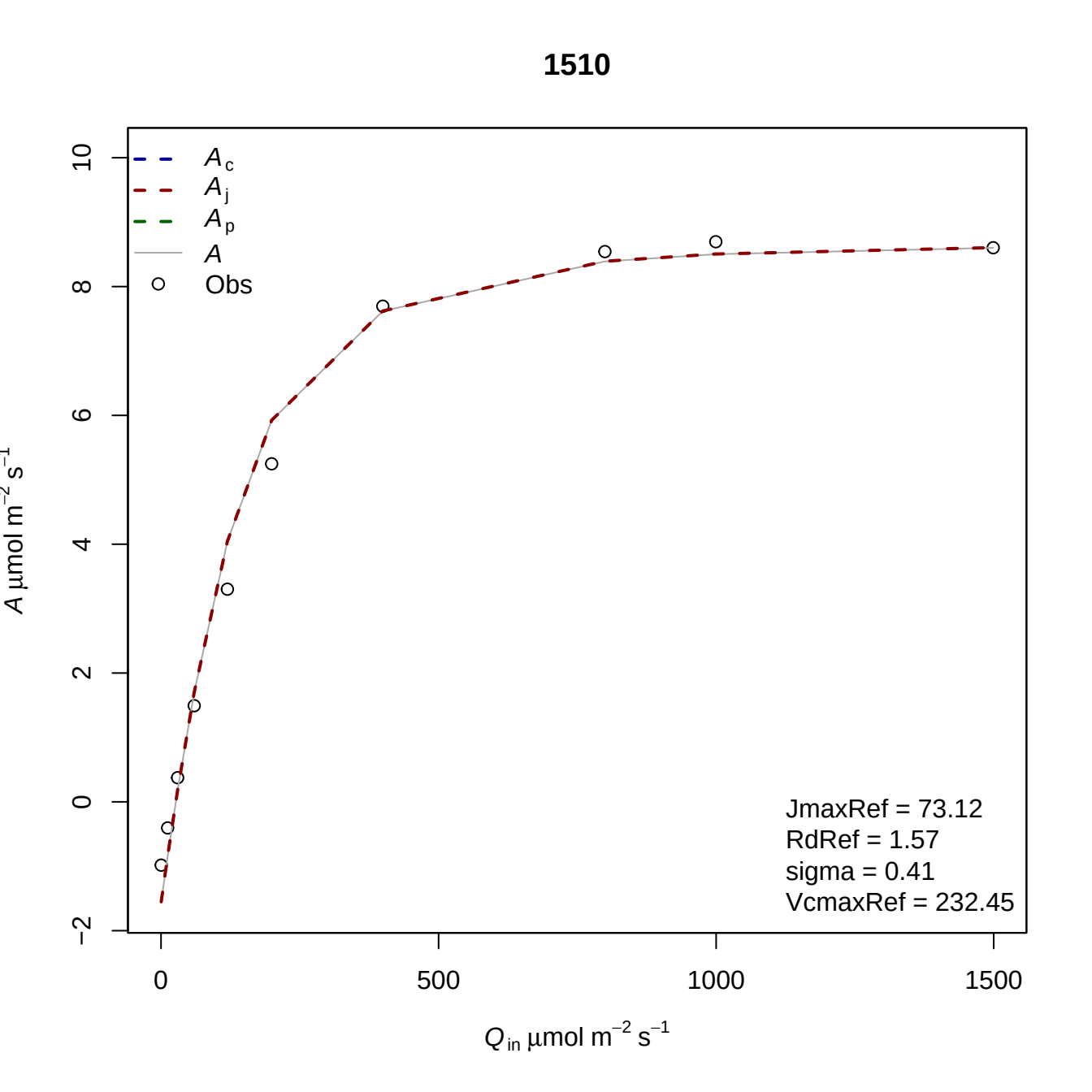

1505
A_c
A_j
A_p
A
Obs
JmaxRef = 75.12
RdRef = 3.88
sigma = 1.33
VcmaxRef = 2776.24
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

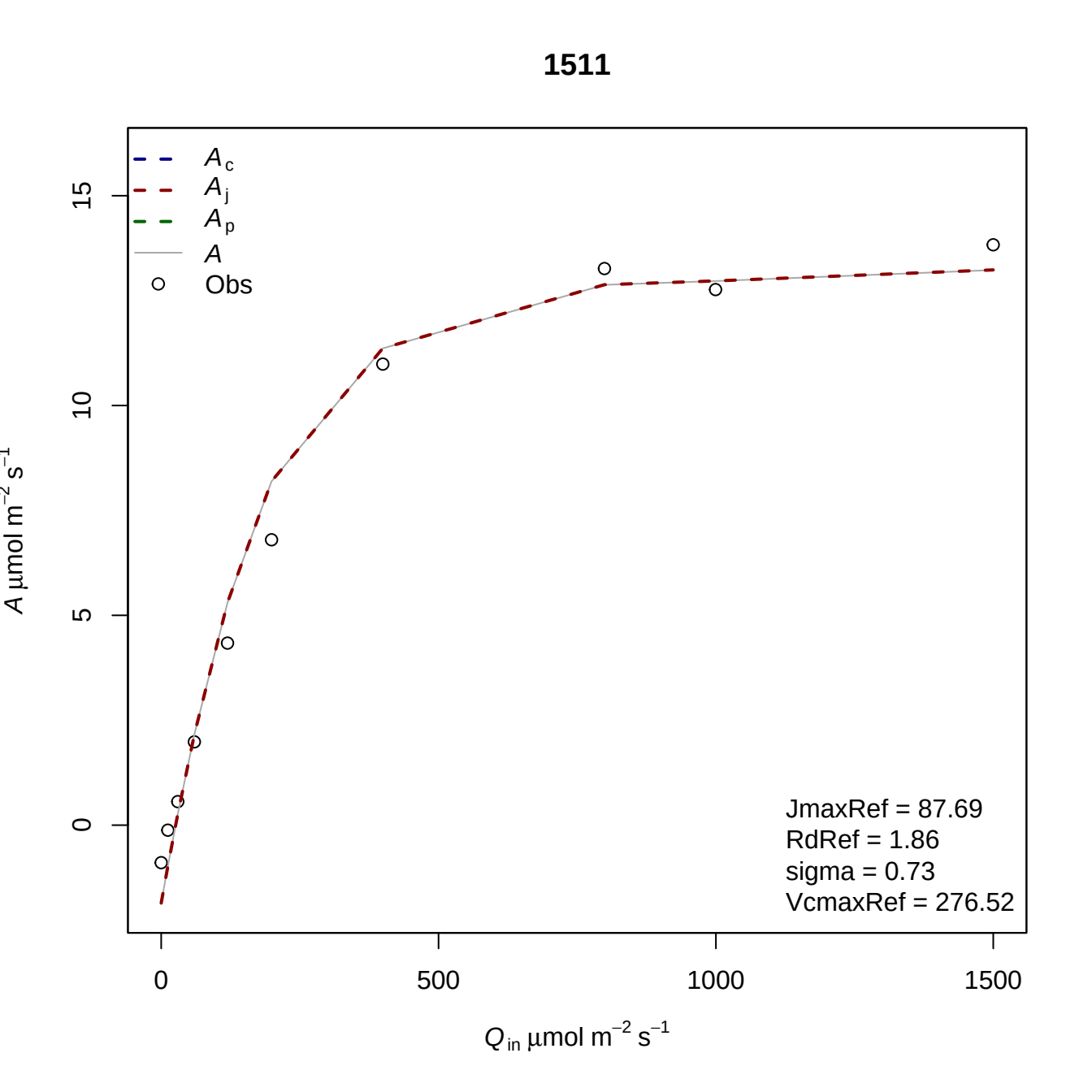

1506
A_c
A_j
A_p
A
Obs
JmaxRef = 77.8
RdRef = 1.64
sigma = 0.89
VcmaxRef = 198.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

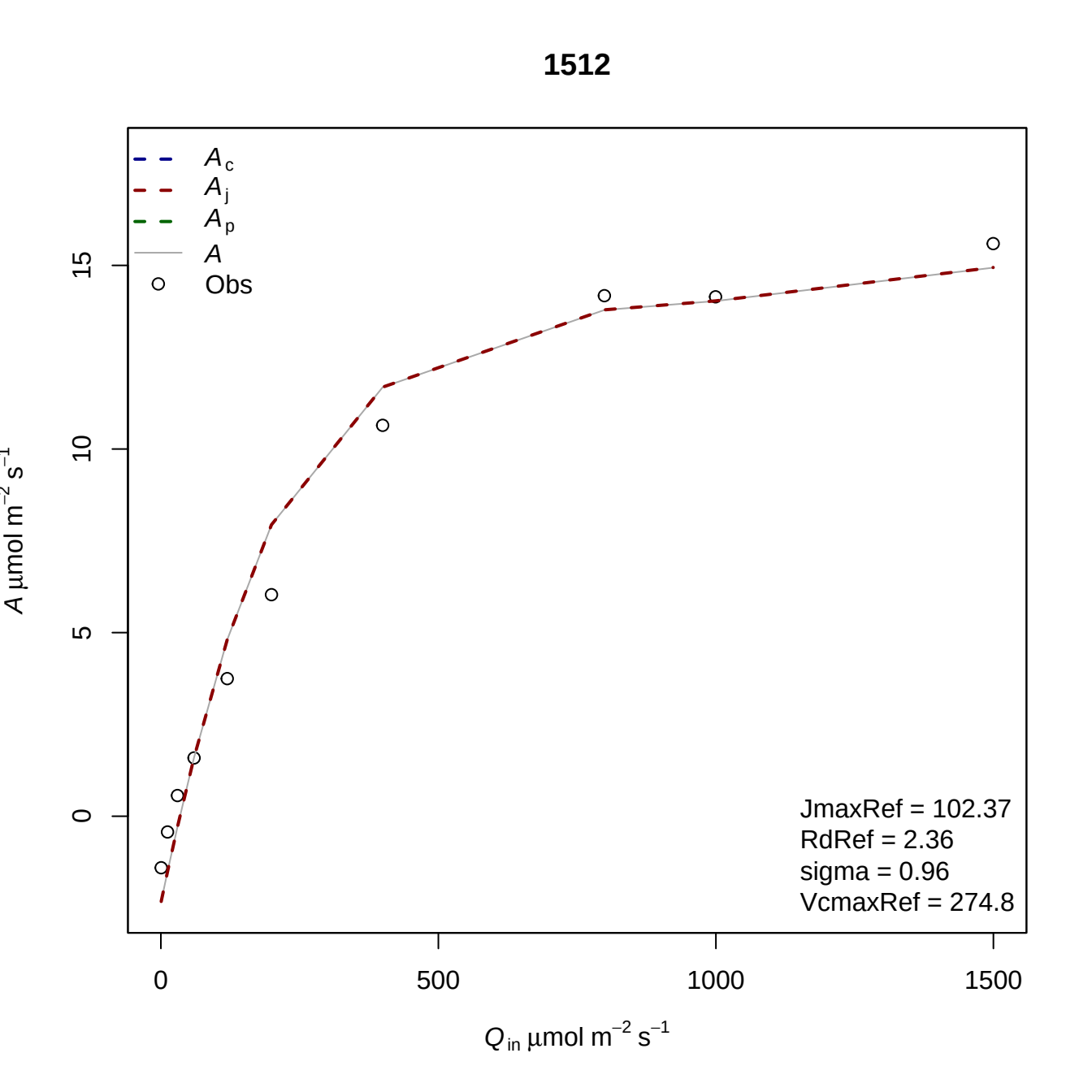

1507
A_c
A_j
A_p
A
Obs
JmaxRef = 63.7
RdRef = 1.38
sigma = 0.61
VcmaxRef = 131.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

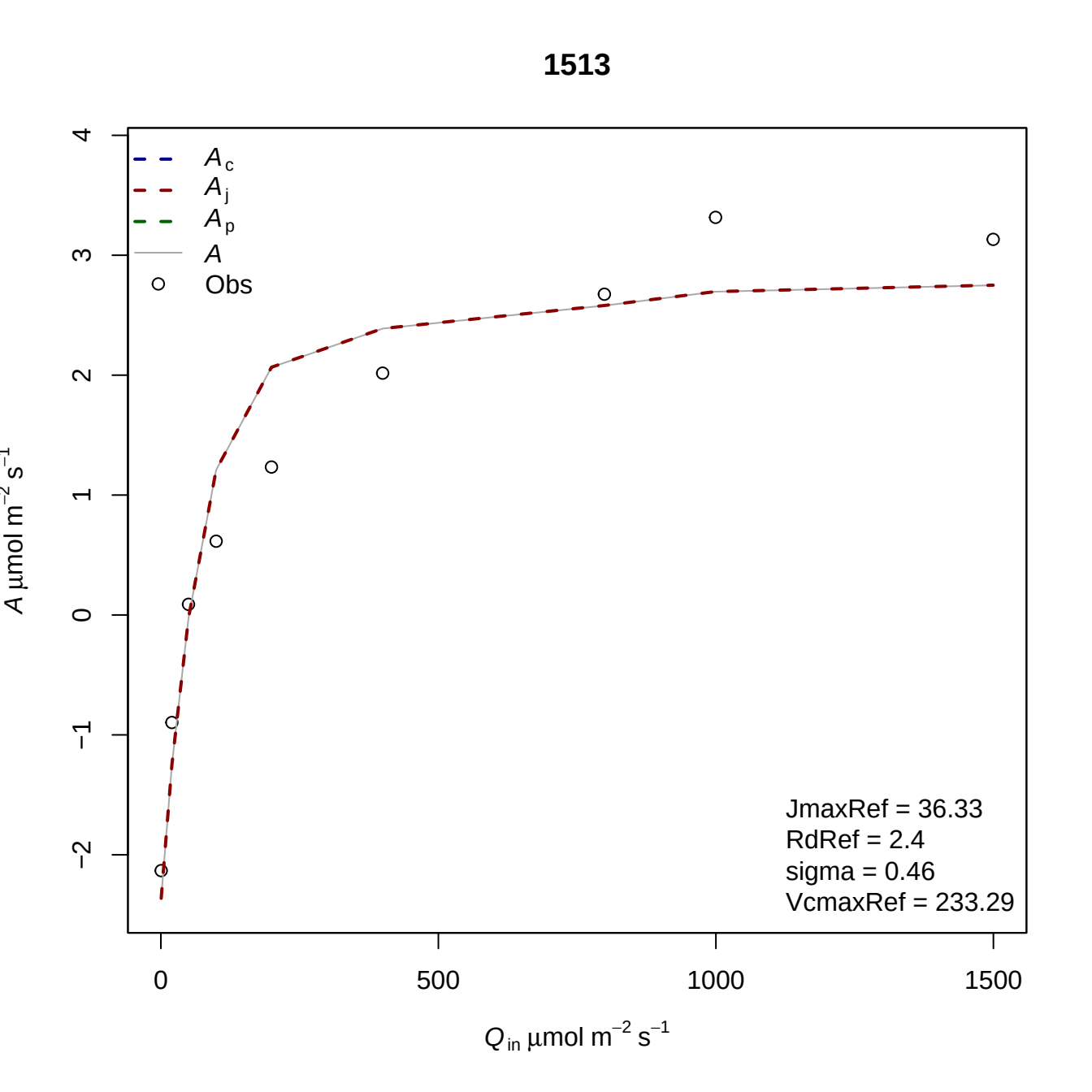

1508
A_c
A_j
A_p
A
Obs
JmaxRef = 32.38
RdRef = 0.45
sigma = 0.19
VcmaxRef = 93.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

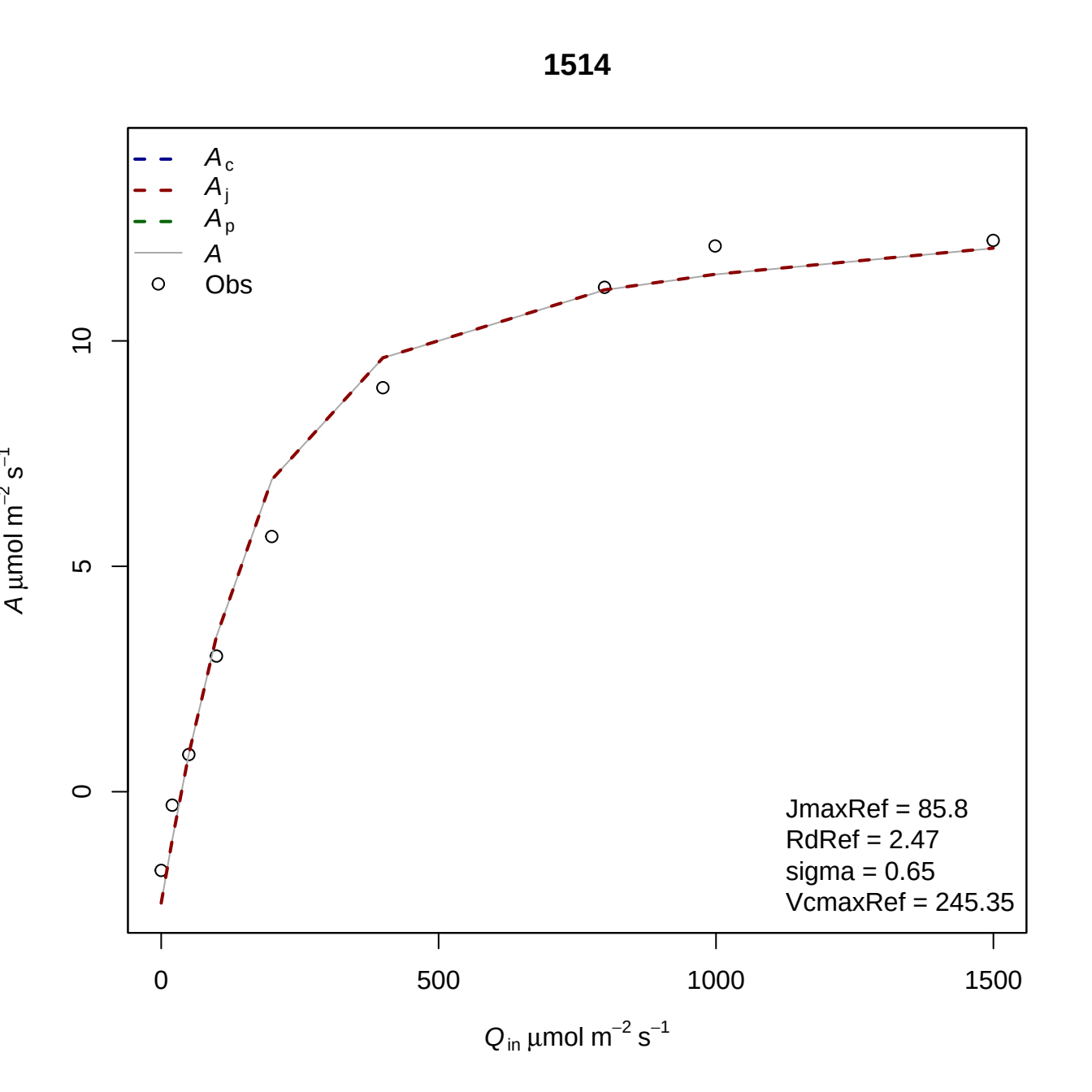

1509
A_c
A_j
A_p
A
Obs
JmaxRef = 44.4
RdRef = 0.7
sigma = 0.3
VcmaxRef = 141.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

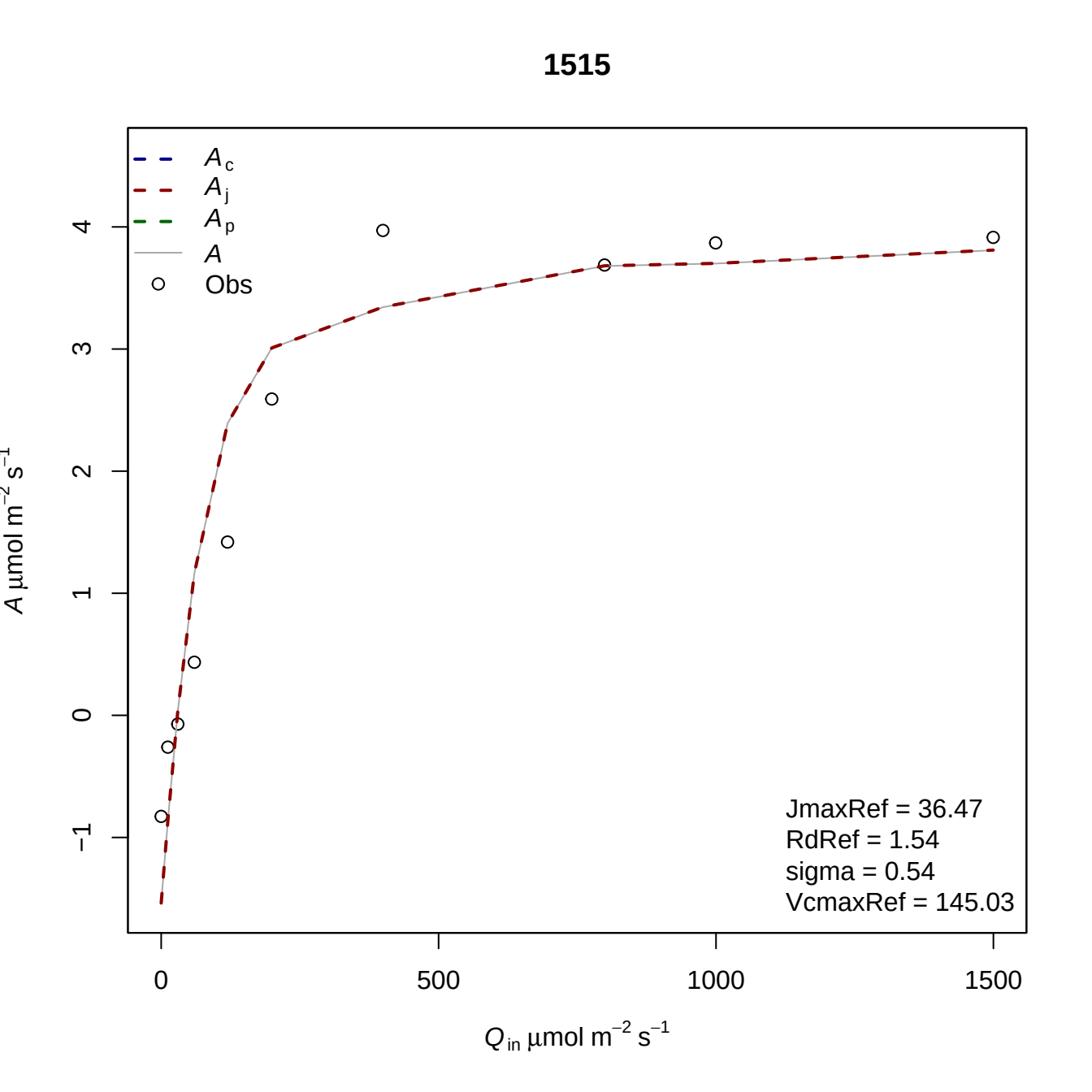

1510
A_c
A_j
A_p
A
Obs
JmaxRef = 73.12
RdRef = 1.57
sigma = 0.41
VcmaxRef = 232.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

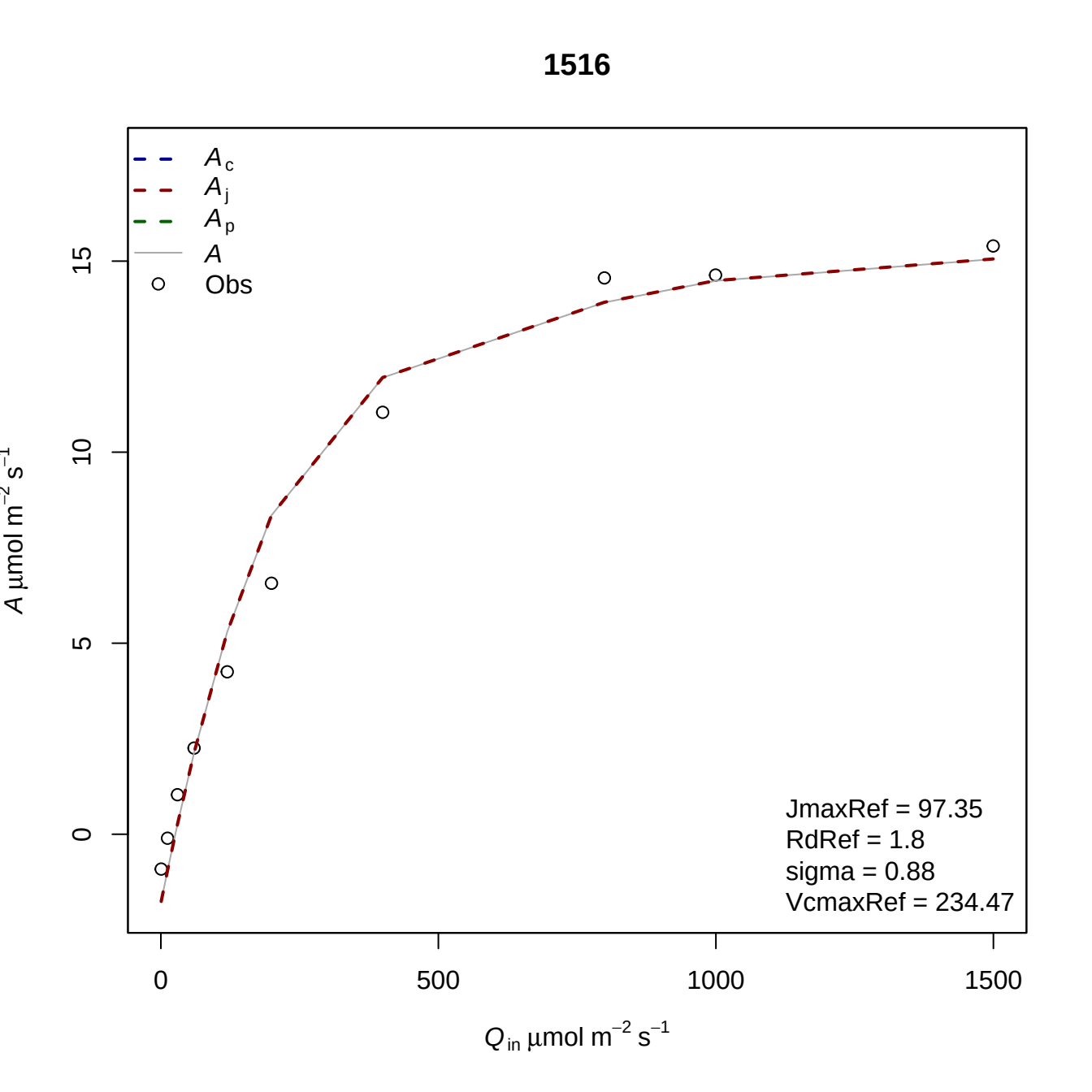

1511
A_c
A_j
A_p
A
Obs
JmaxRef = 87.69
RdRef = 1.86
sigma = 0.73
VcmaxRef = 276.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

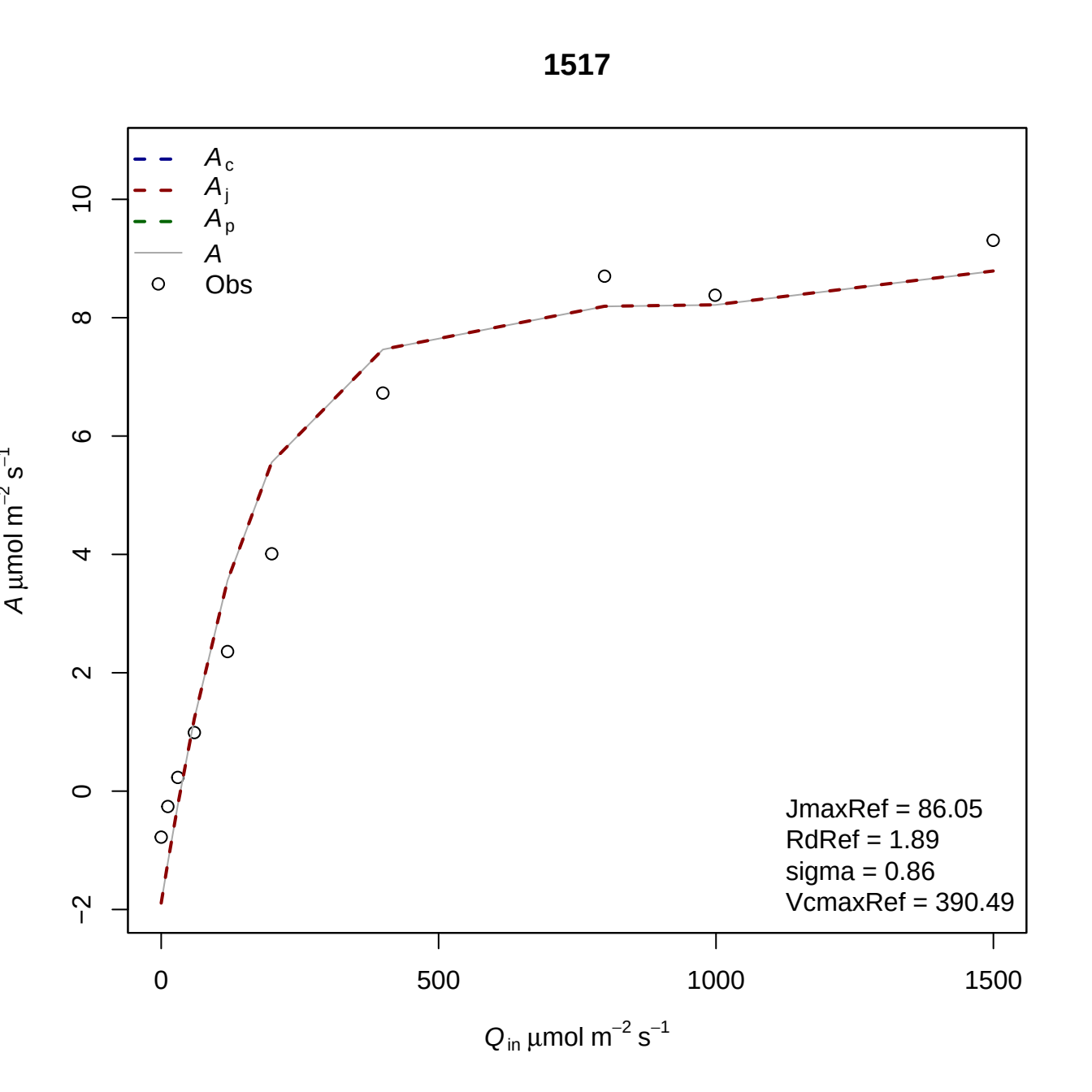

1512
A_c
A_j
A_p
A
Obs
JmaxRef = 102.37
RdRef = 2.36
sigma = 0.96
VcmaxRef = 274.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

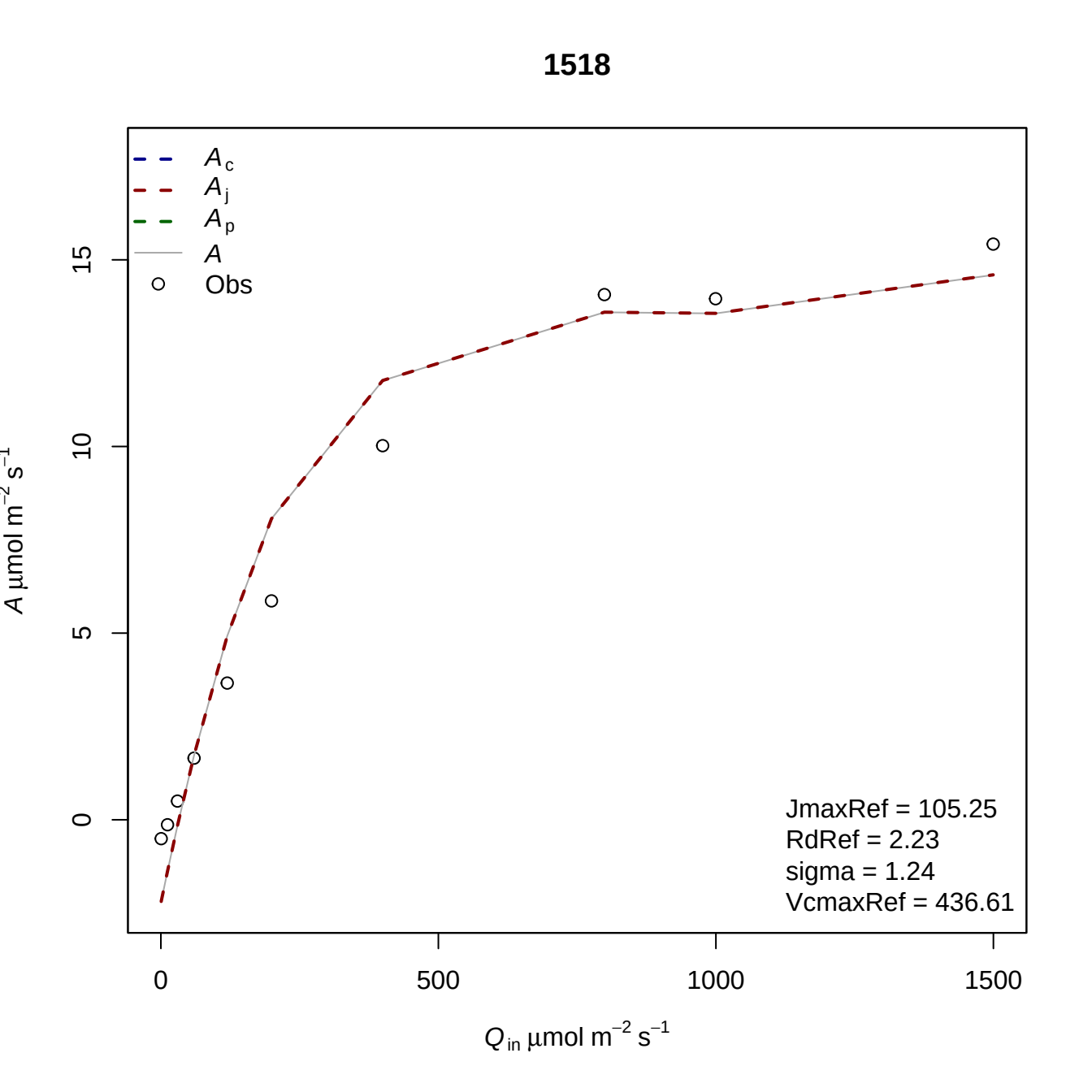

1513
A_c
A_j
A_p
A
Obs
JmaxRef = 36.33
RdRef = 2.4
sigma = 0.46
VcmaxRef = 233.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

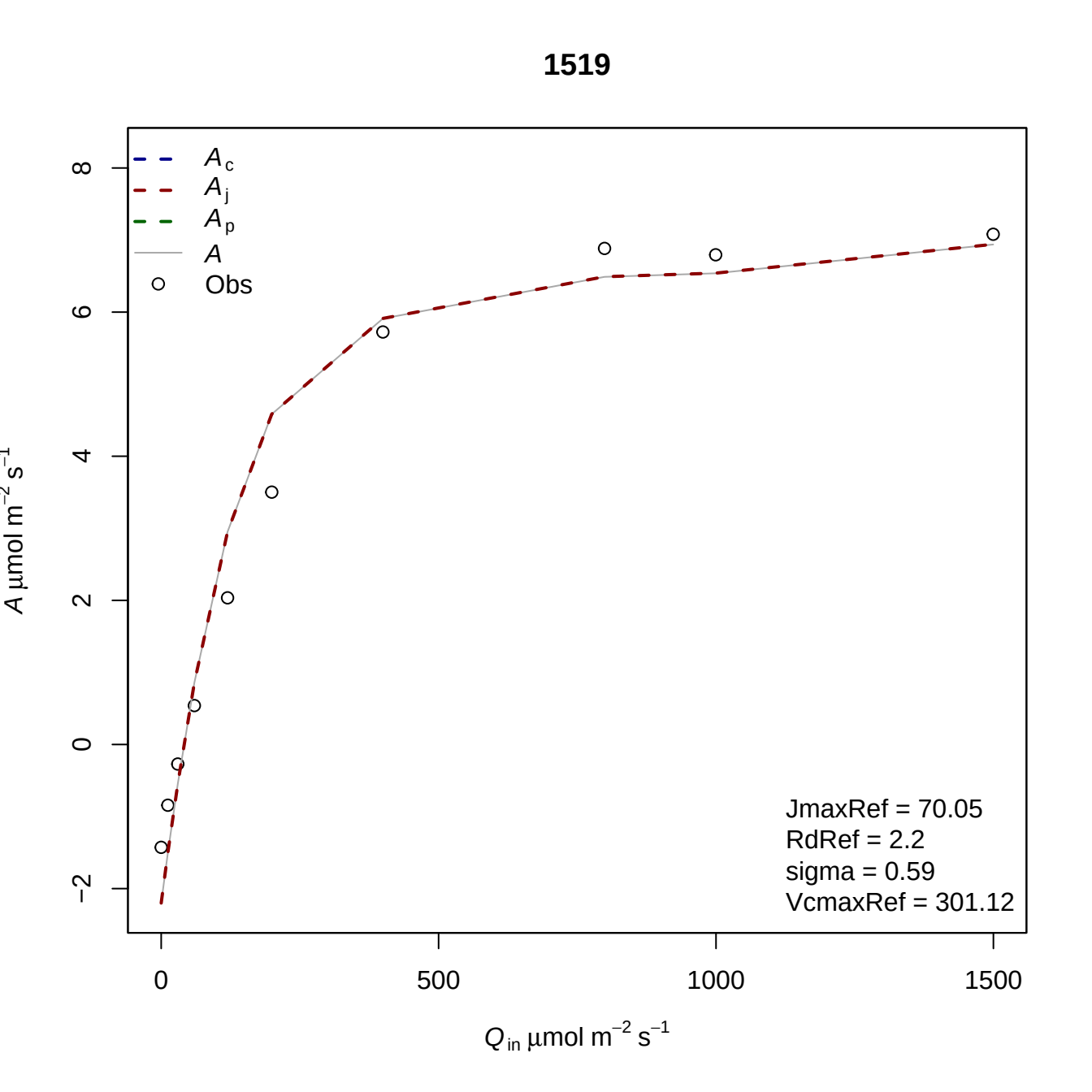

1514
A_c
A_j
A_p
A
Obs
JmaxRef = 85.8
RdRef = 2.47
sigma = 0.65
VcmaxRef = 245.35
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

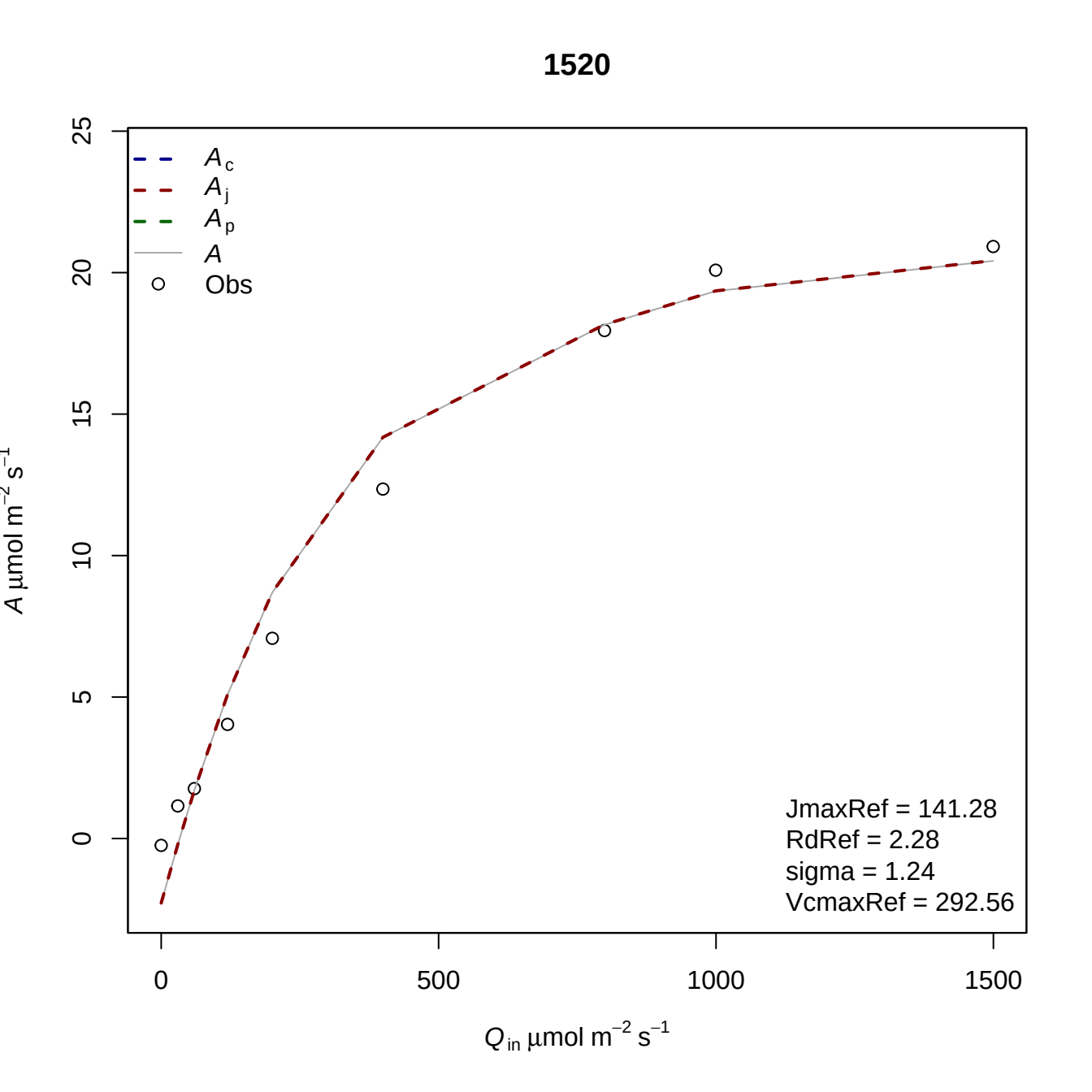

1515
A_c
A_j
A_p
A
Obs
JmaxRef = 36.47
RdRef = 1.54
sigma = 0.54
VcmaxRef = 145.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

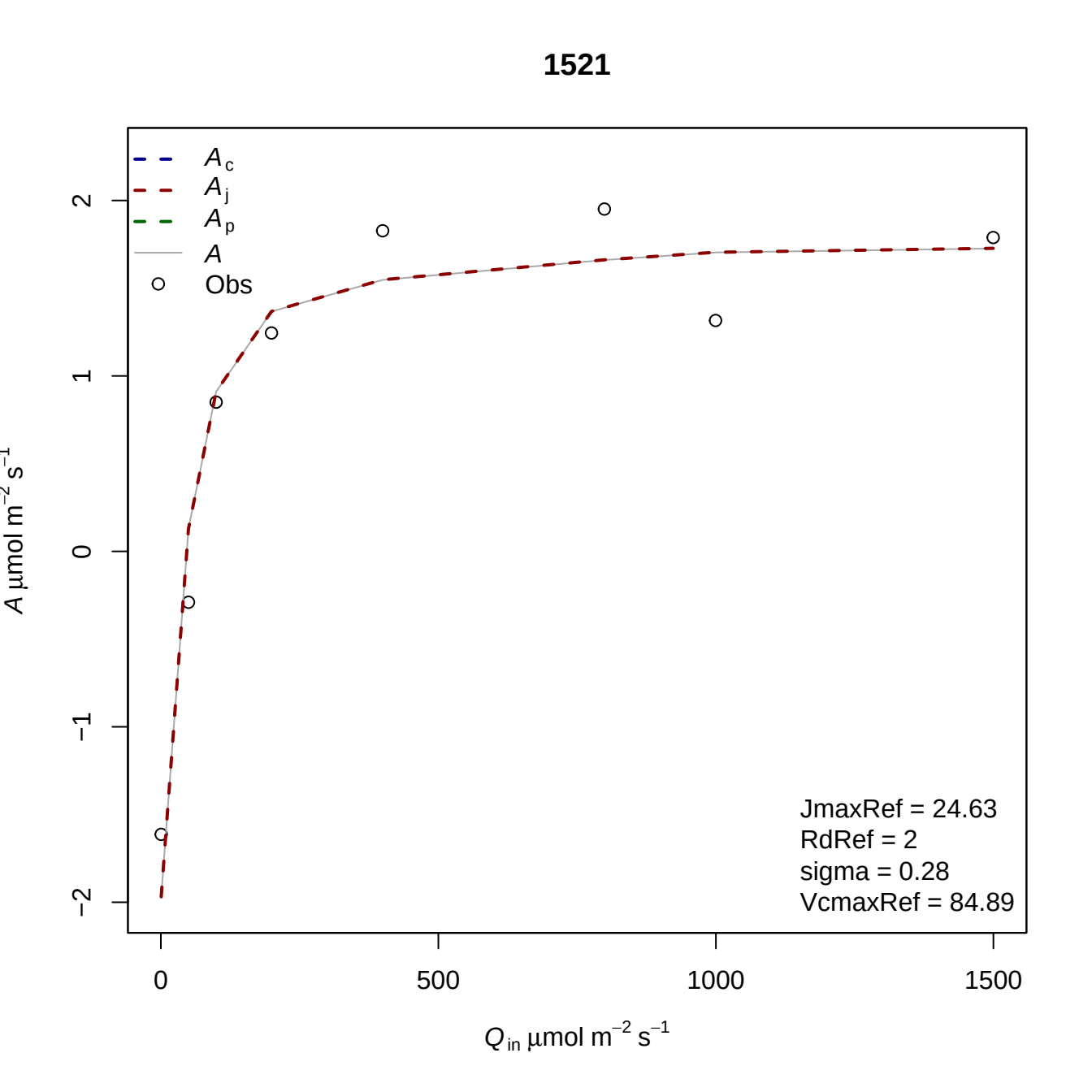

1516
A_c
A_j
A_p
A
Obs
JmaxRef = 97.35
RdRef = 1.8
sigma = 0.88
VcmaxRef = 234.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

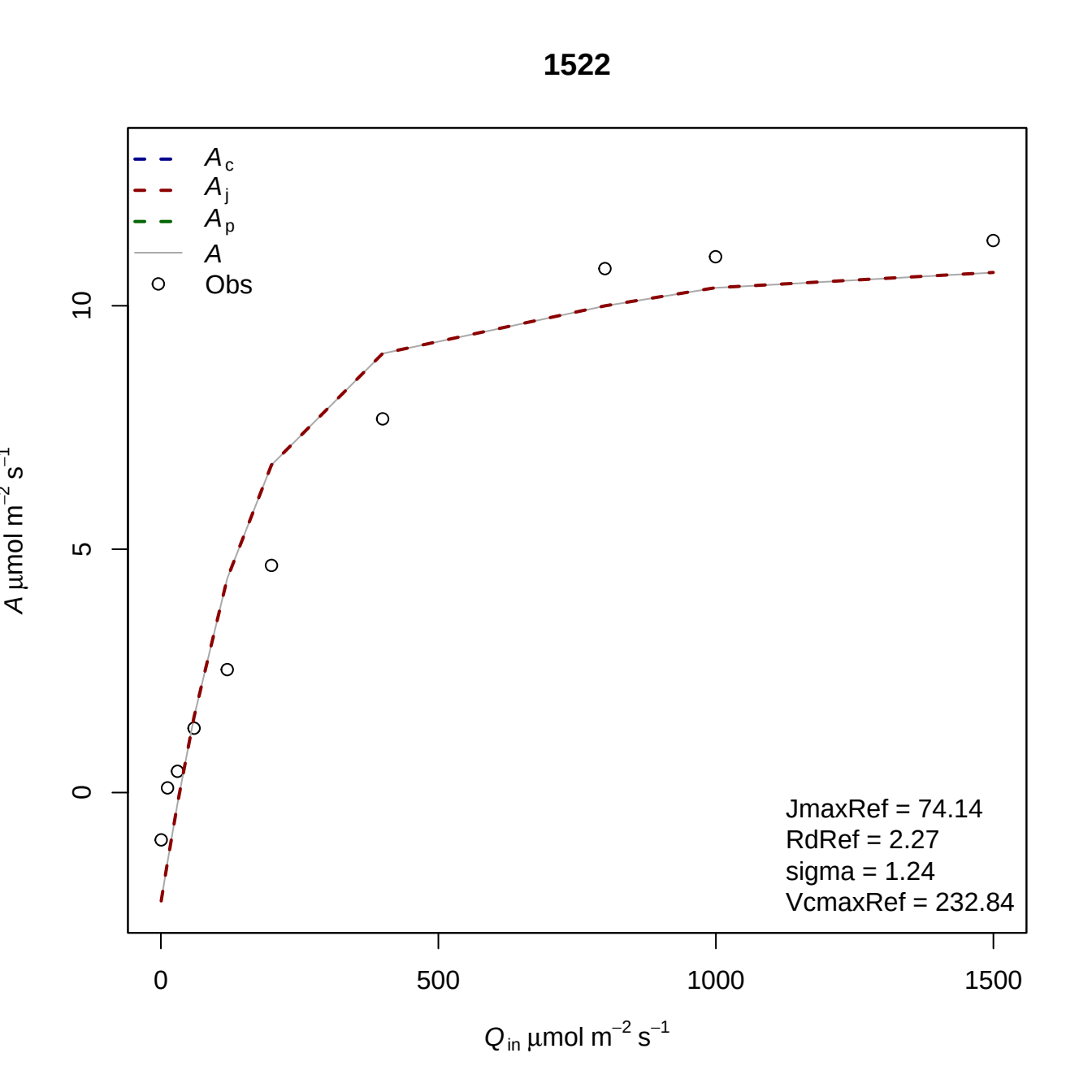

1517
A_c
A_j
A_p
A
Obs
JmaxRef = 86.05
RdRef = 1.89
sigma = 0.86
VcmaxRef = 390.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

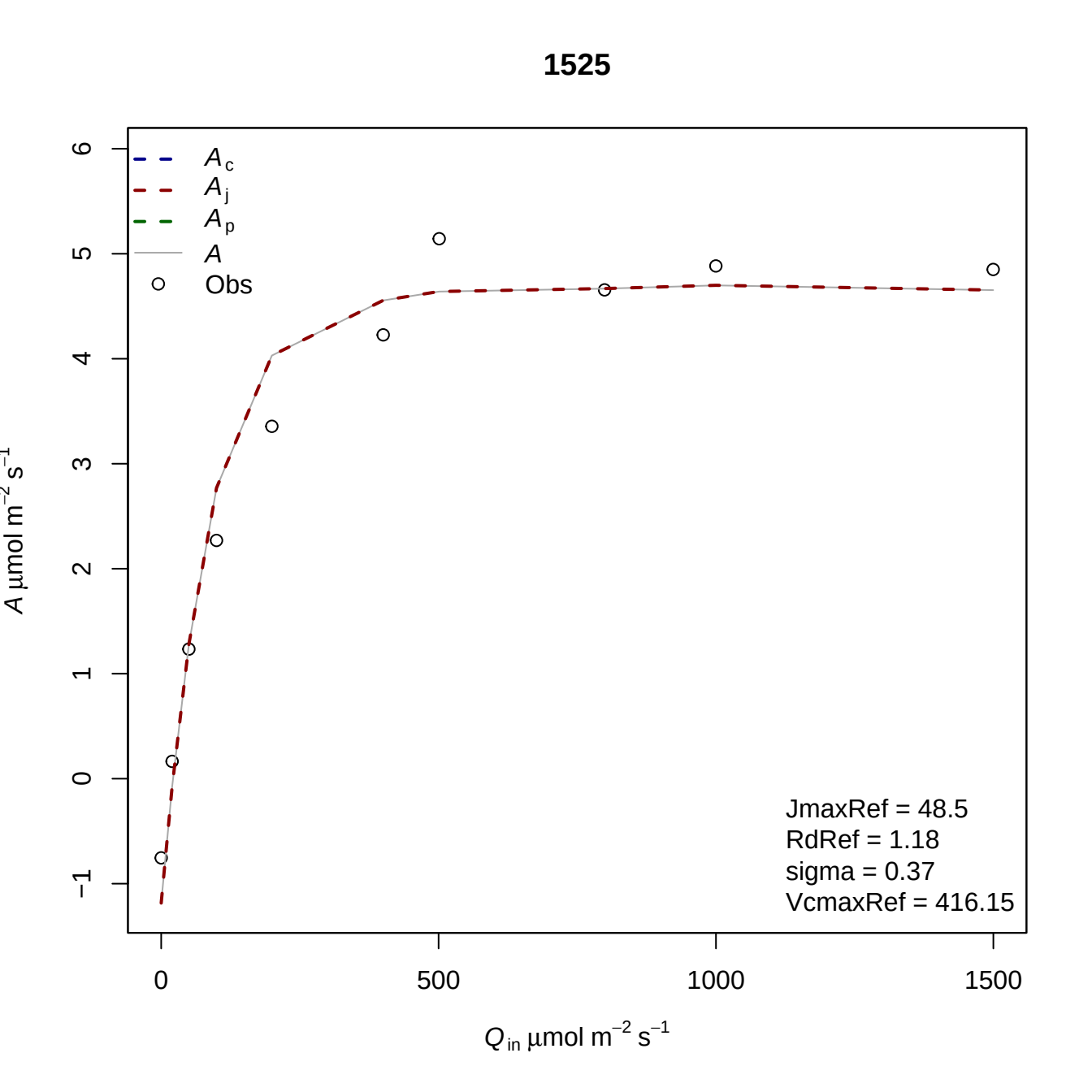

1518
A_c
A_j
A_p
A
Obs
JmaxRef = 105.25
RdRef = 2.23
sigma = 1.24
VcmaxRef = 436.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

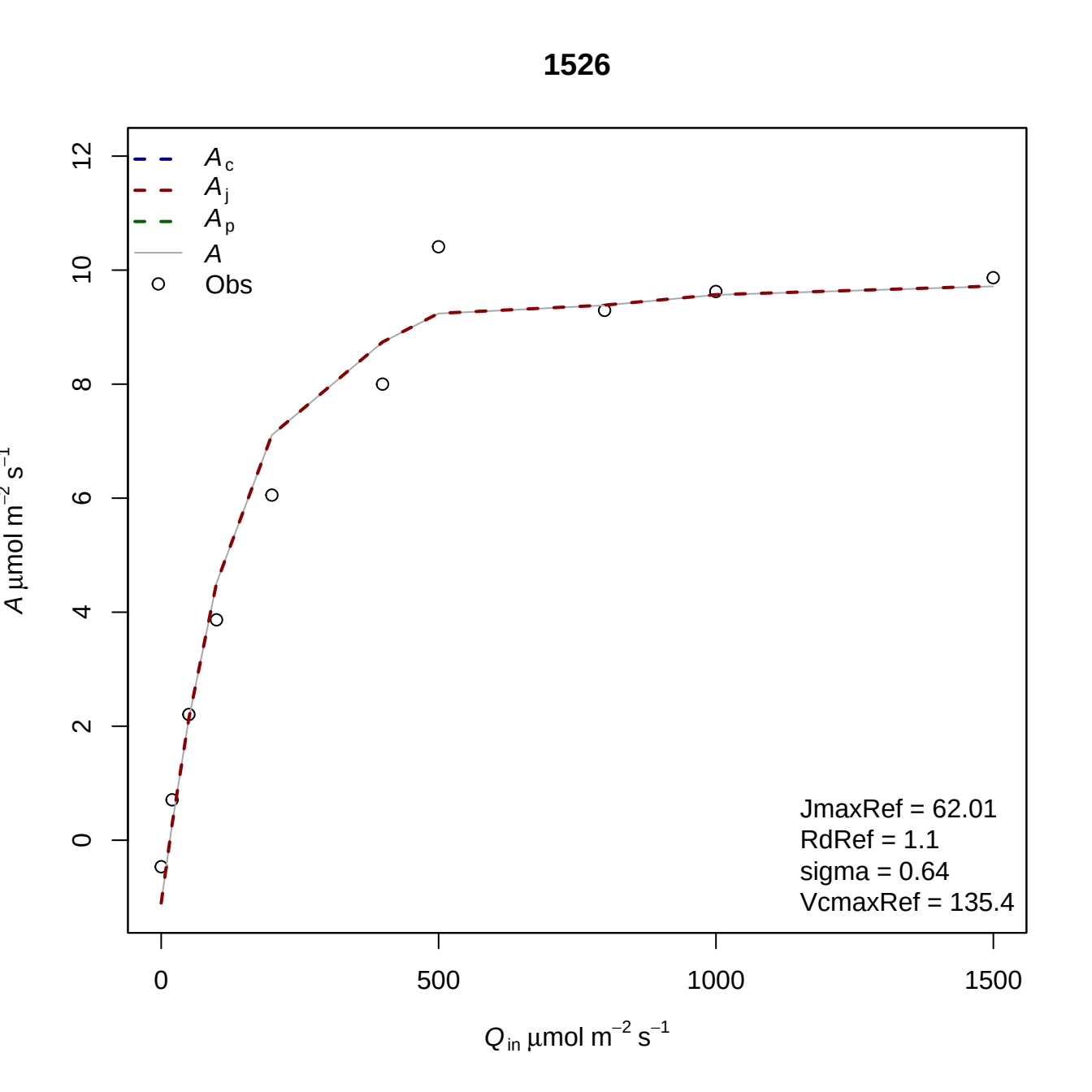

1519
A_c
A_j
A_p
A
Obs
JmaxRef = 70.05
RdRef = 2.2
sigma = 0.59
VcmaxRef = 301.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

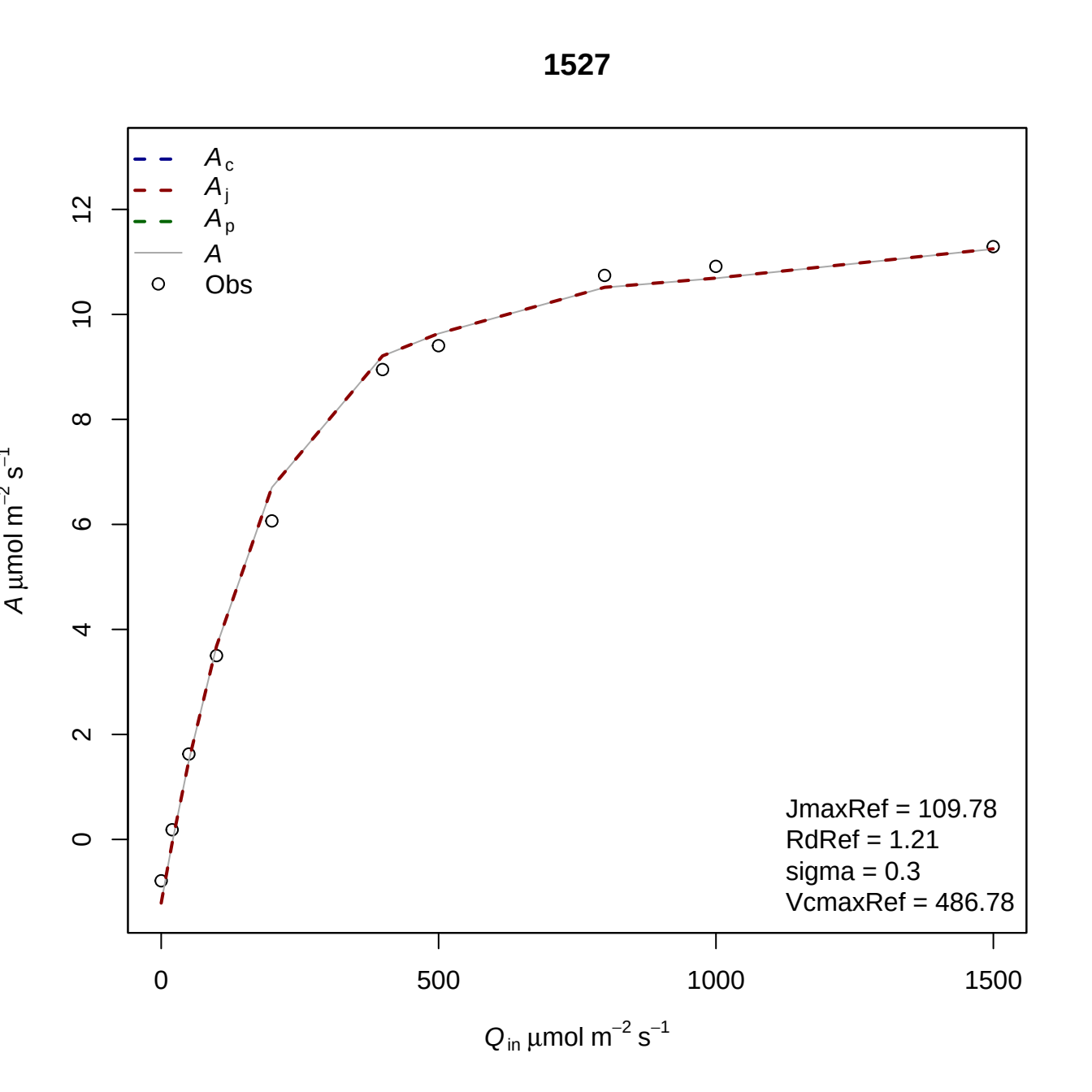

1520
A_c
A_j
A_p
A
Obs
JmaxRef = 141.28
RdRef = 2.28
sigma = 1.24
VcmaxRef = 292.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

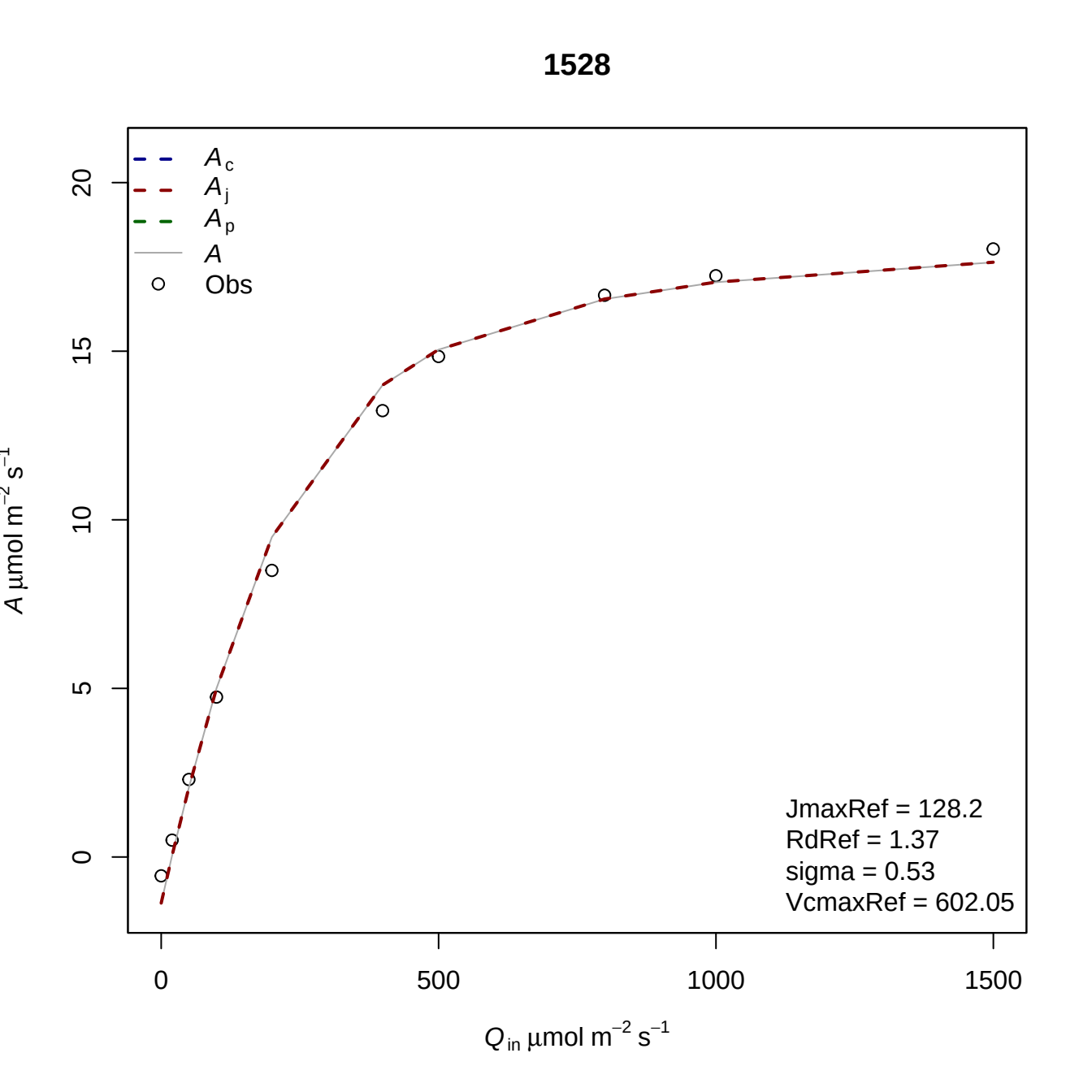

1521
A_c
A_j
A_p
A
Obs
JmaxRef = 24.63
RdRef = 2
sigma = 0.28
VcmaxRef = 84.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
500
1000
1500

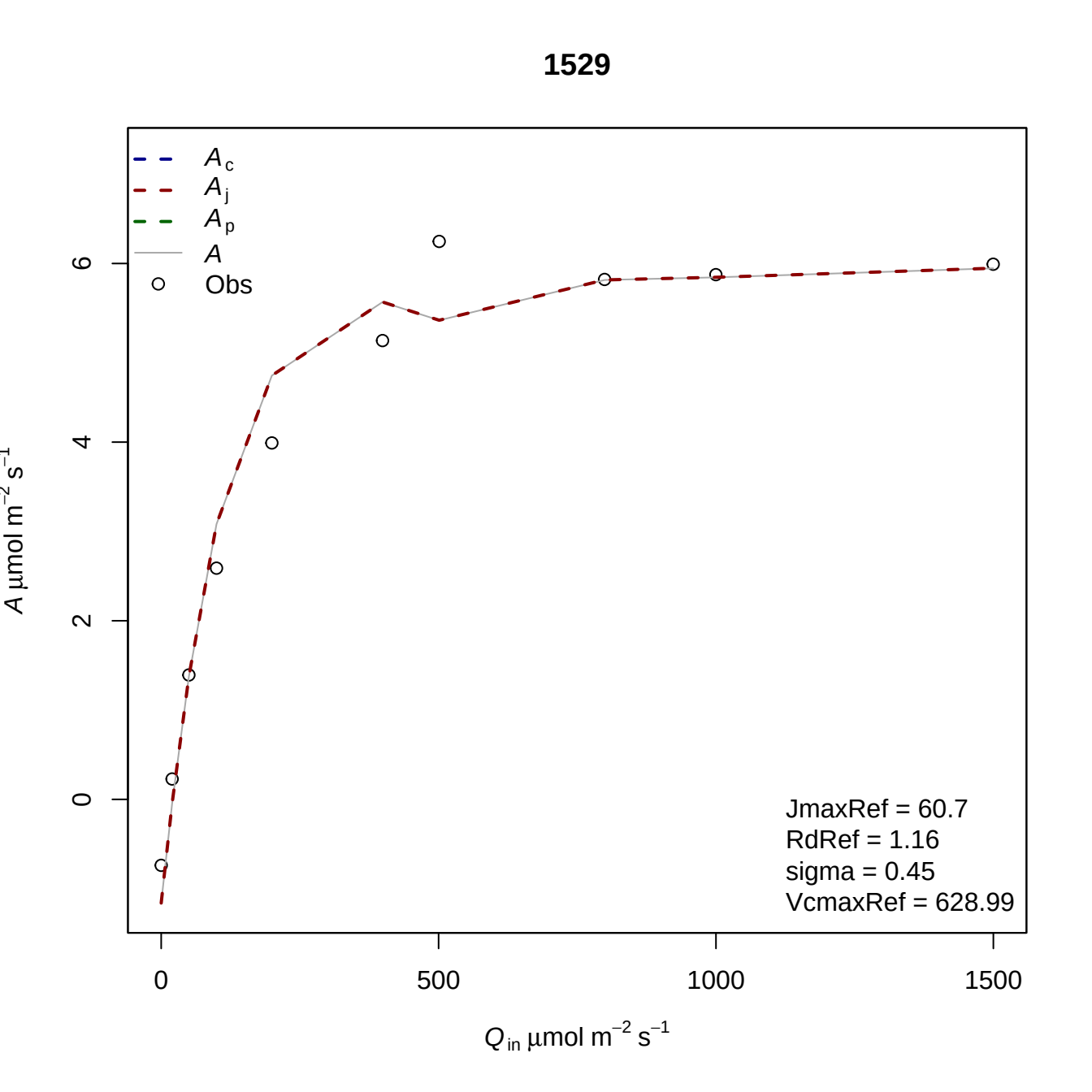

1522
A_c
A_j
A_p
A
Obs
JmaxRef = 74.14
RdRef = 2.27
sigma = 1.24
VcmaxRef = 232.84
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

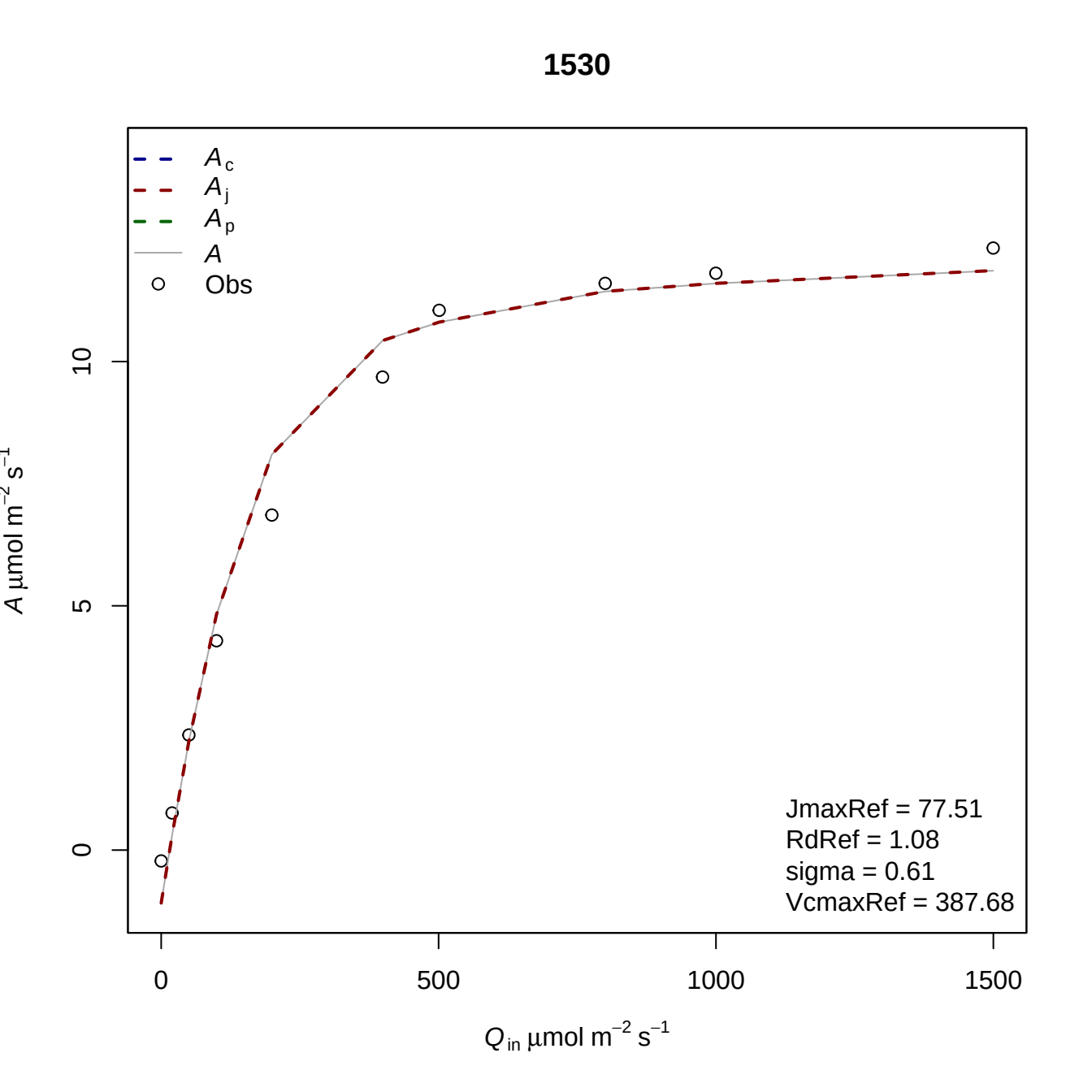

1525
A_c
A_j
A_p
A
Obs
JmaxRef = 48.5
RdRef = 1.18
sigma = 0.37
VcmaxRef = 416.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

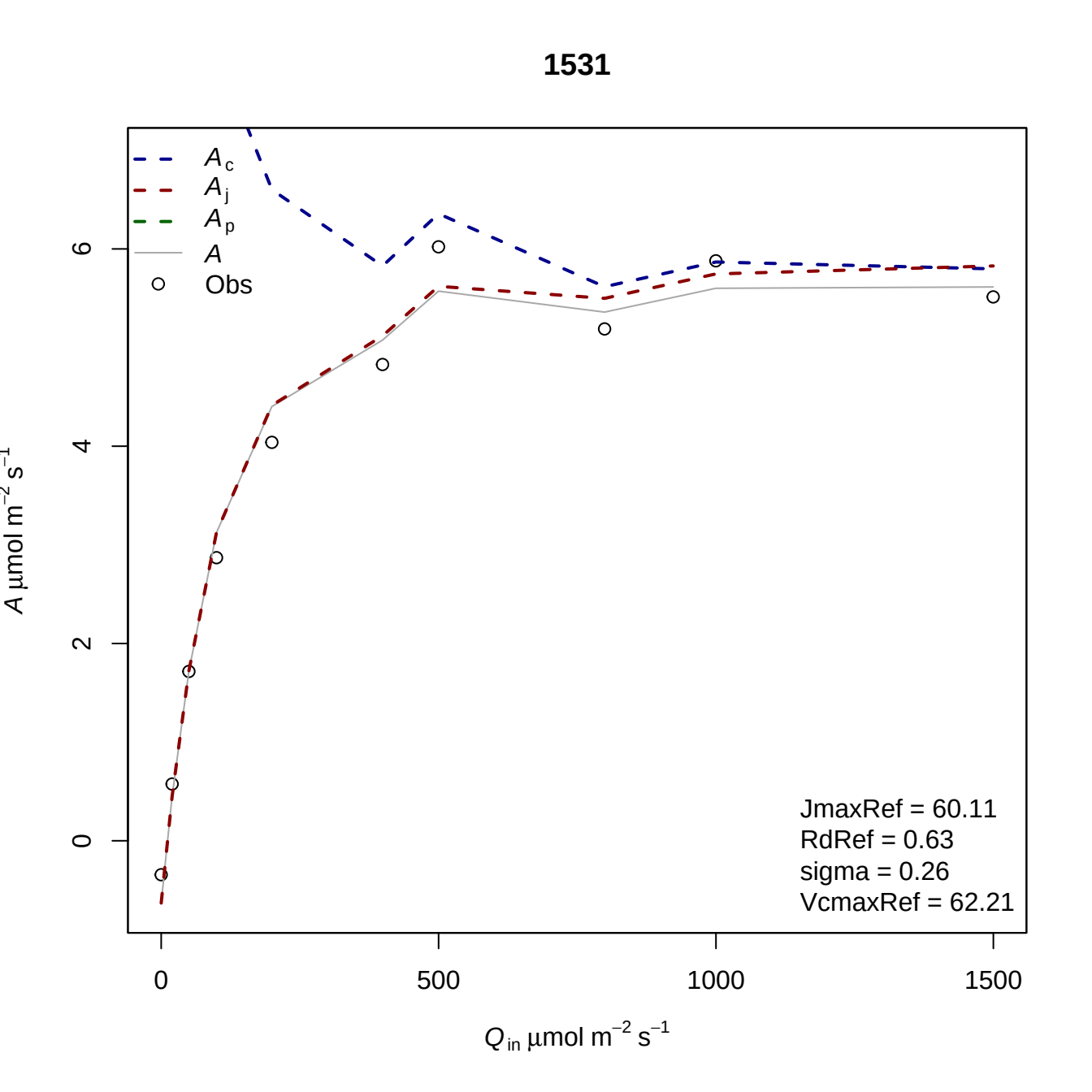

1526
A_c
A_j
A_p
A
Obs
JmaxRef = 62.01
RdRef = 1.1
sigma = 0.64
VcmaxRef = 135.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

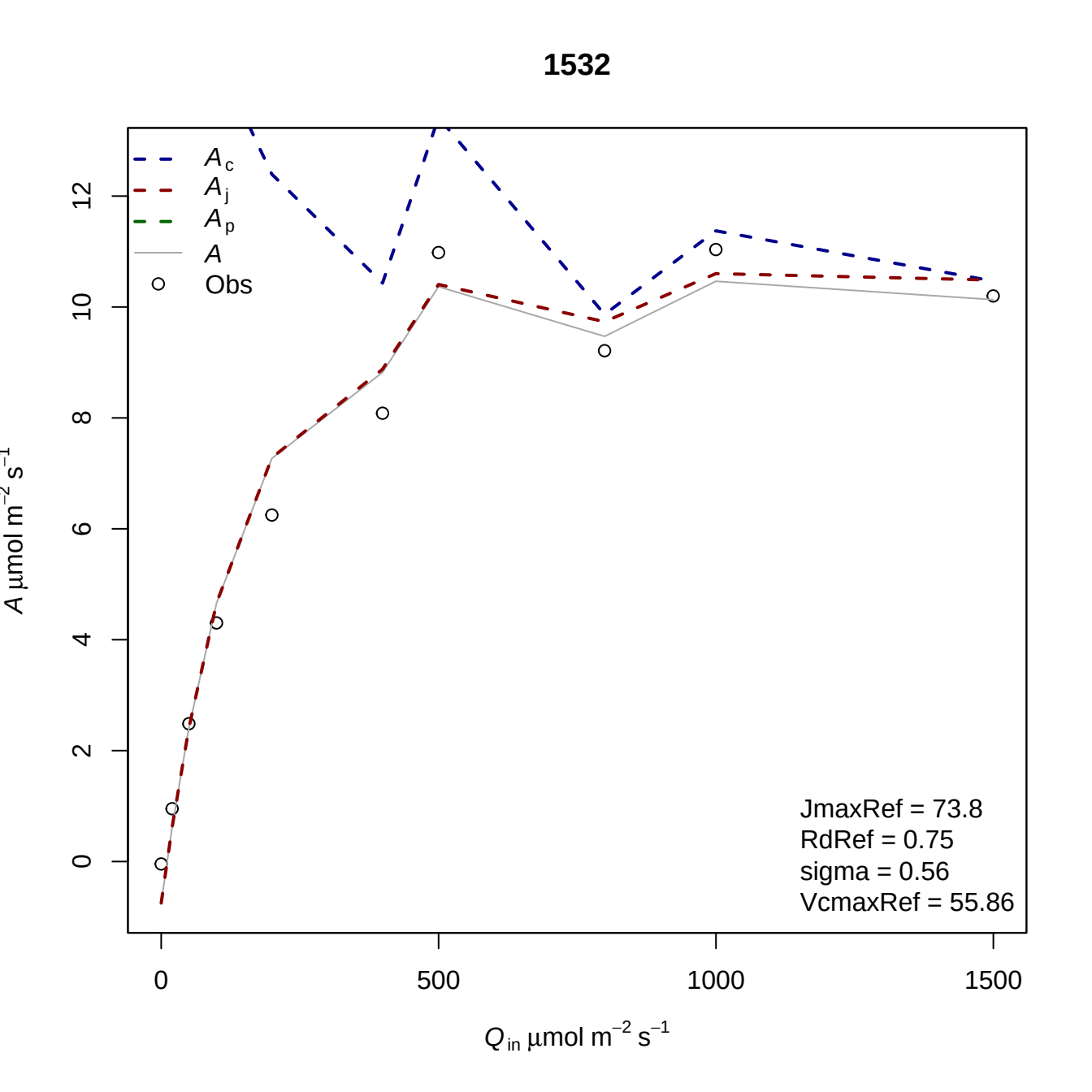

1527
A_c
A_j
A_p
A
Obs
JmaxRef = 109.78
RdRef = 1.21
sigma = 0.3
VcmaxRef = 486.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

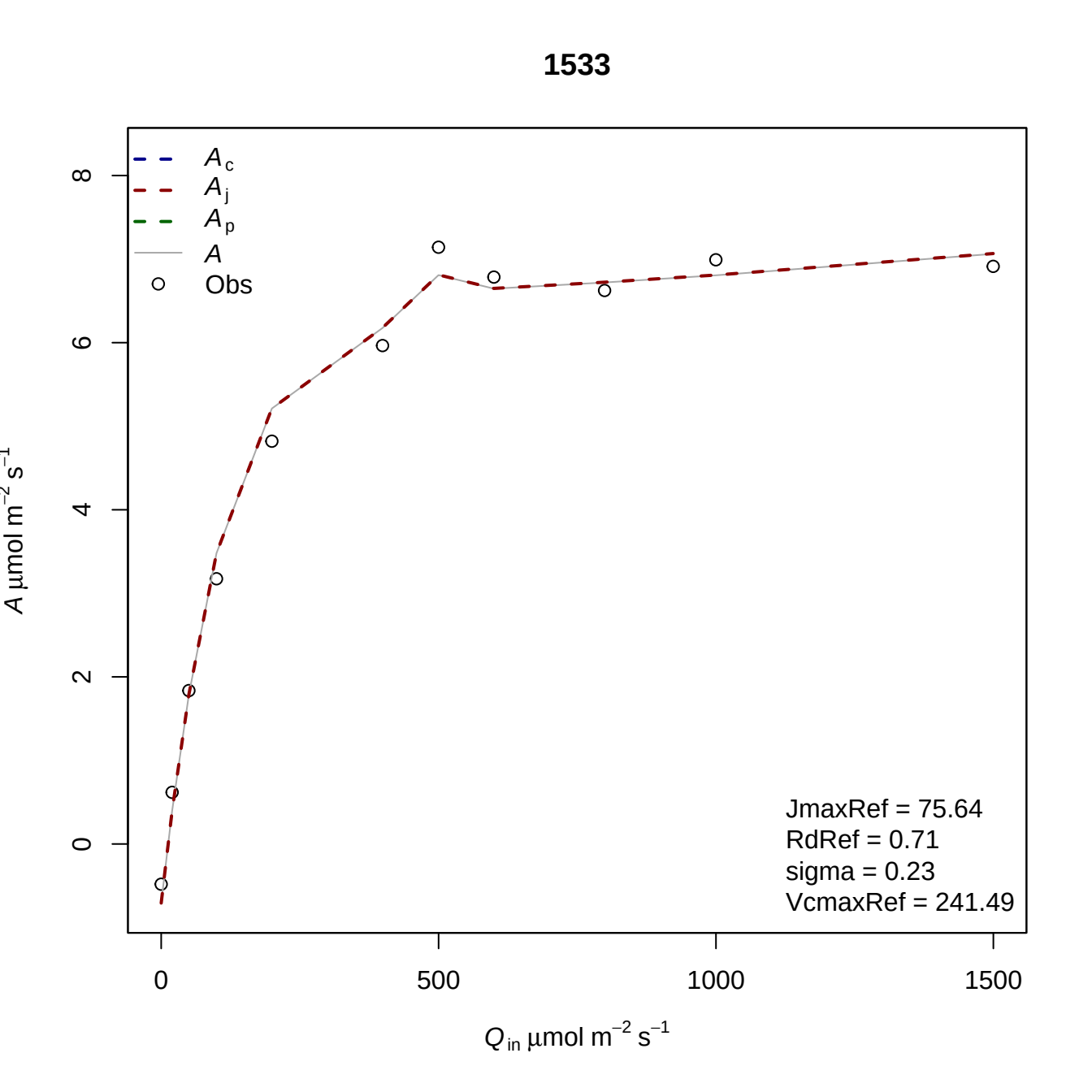

1528
A_c
A_j
A_p
A
Obs
JmaxRef = 128.2
RdRef = 1.37
sigma = 0.53
VcmaxRef = 602.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

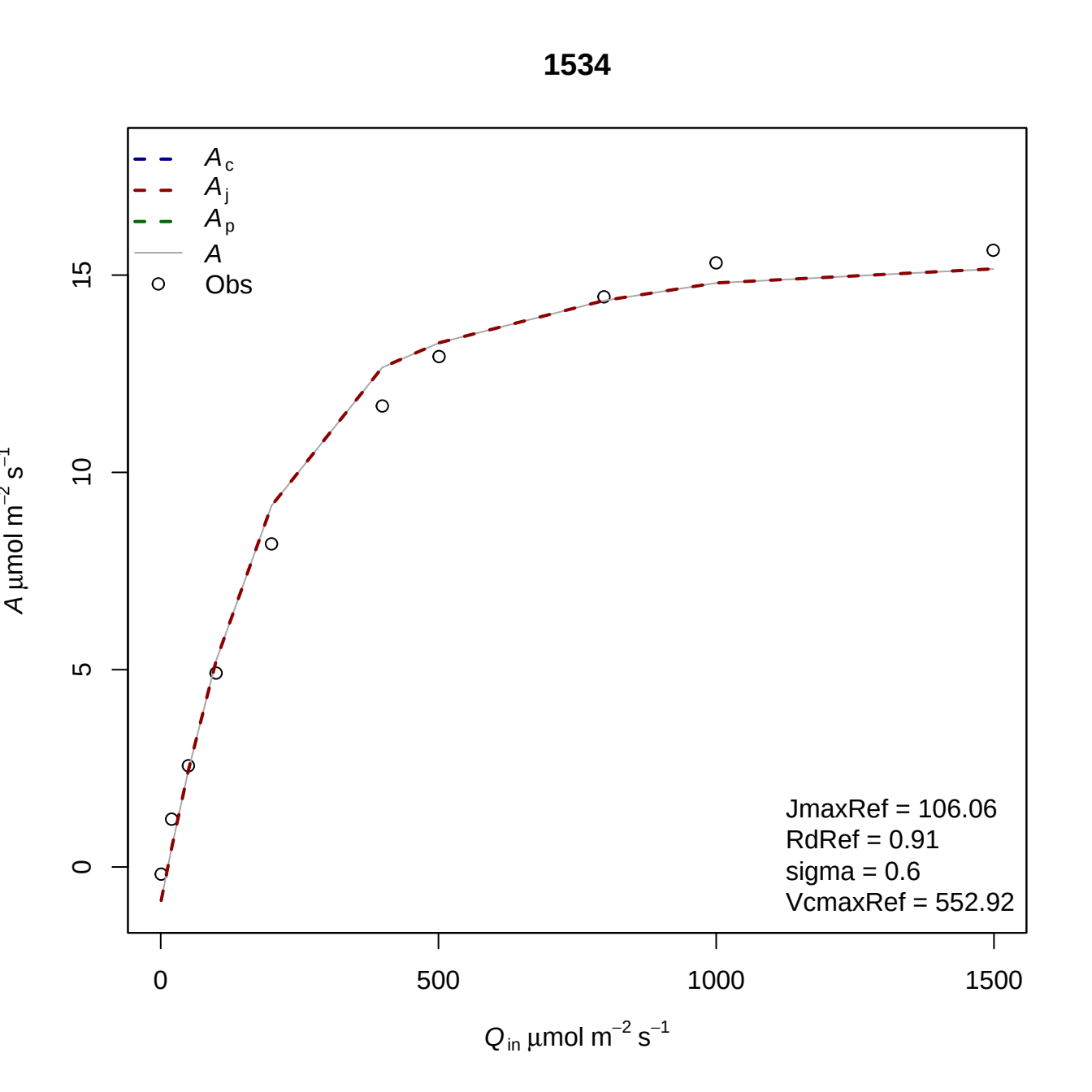

1529
A_c
A_j
A_p
A
Obs
JmaxRef = 60.7
RdRef = 1.16
sigma = 0.45
VcmaxRef = 628.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

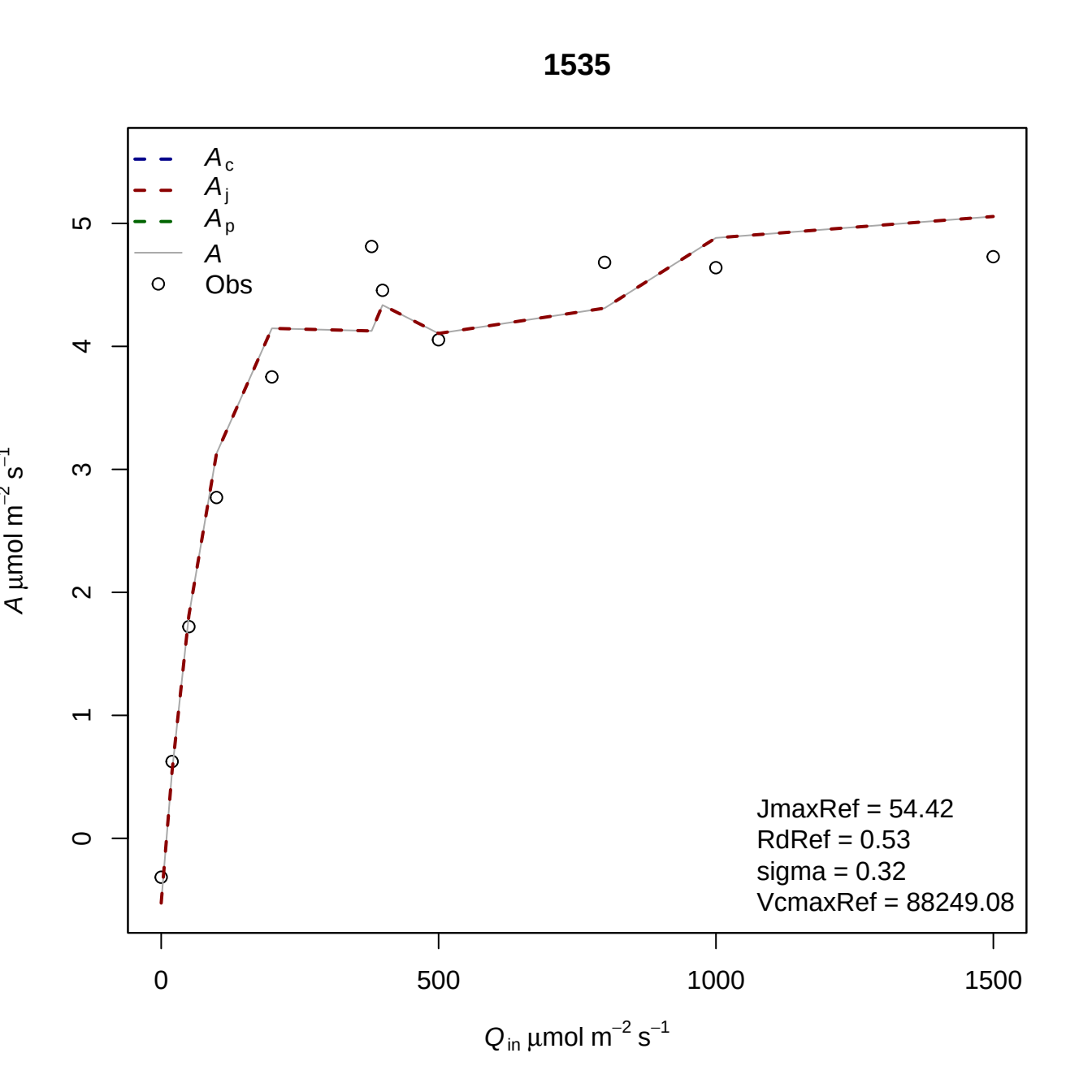

1530
A_c
A_j
A_p
A
Obs
JmaxRef = 77.51
RdRef = 1.08
sigma = 0.61
VcmaxRef = 387.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

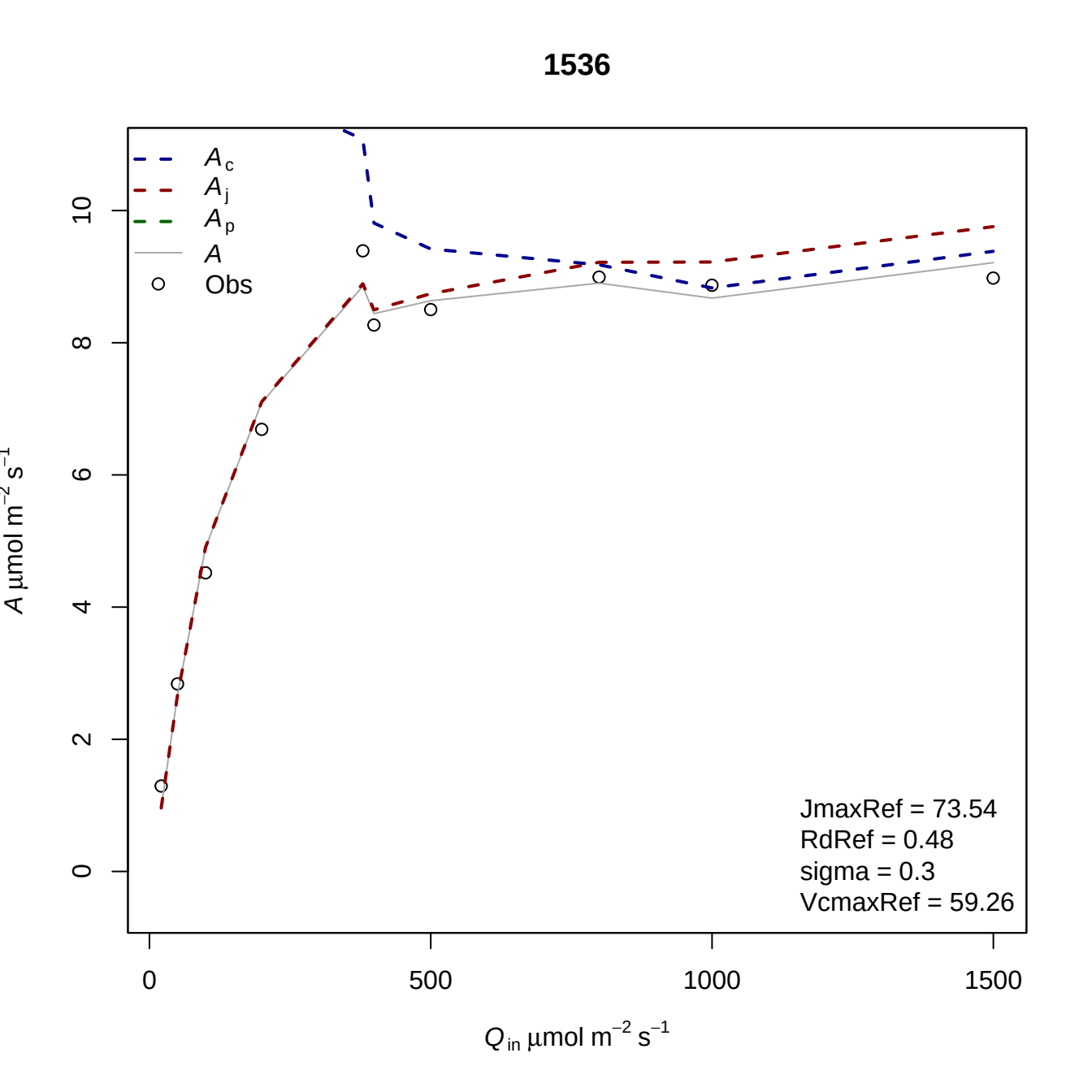

1531
A_c
A_j
A_p
A
Obs
JmaxRef = 60.11
RdRef = 0.63
sigma = 0.26
VcmaxRef = 62.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

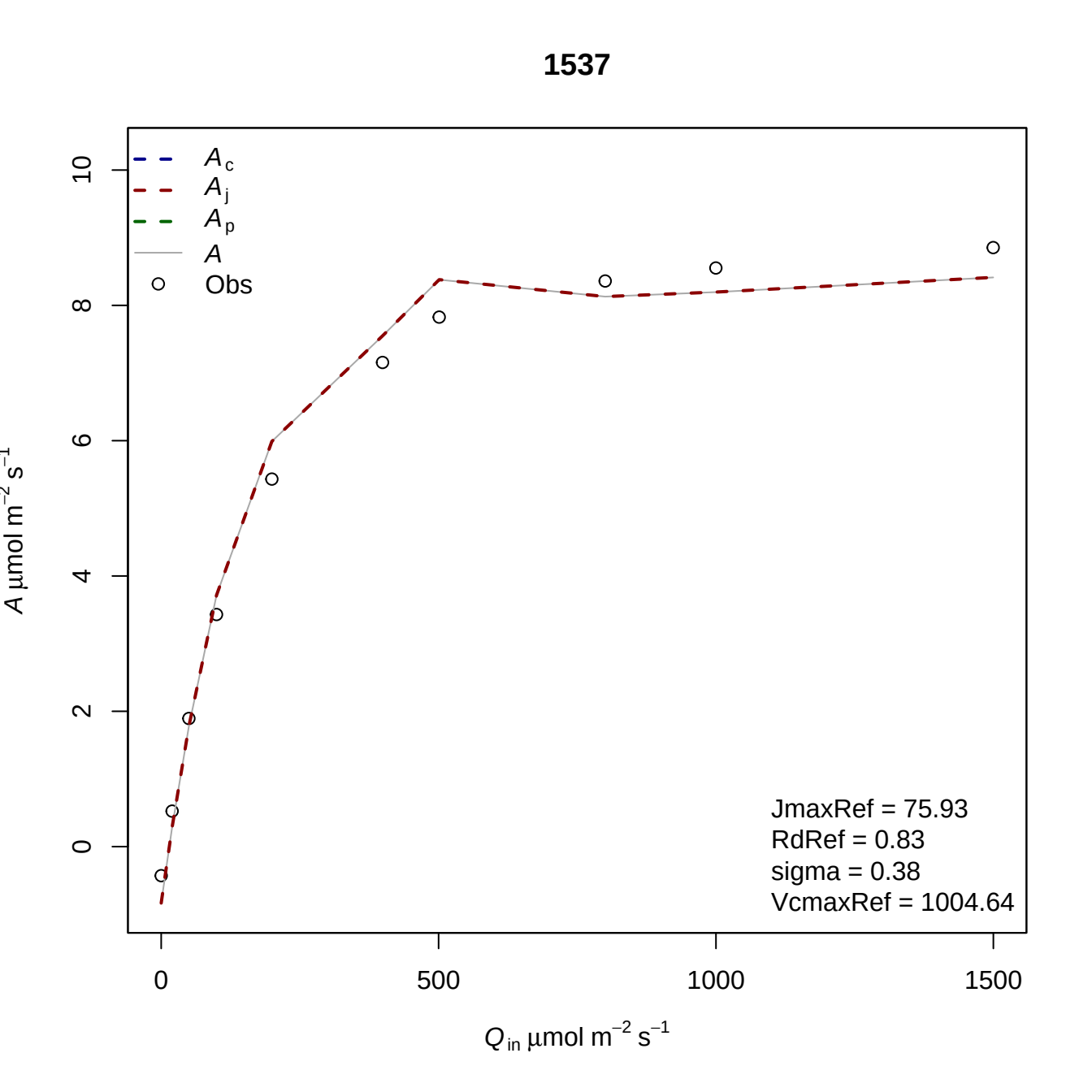

1532
A_c
A_j
A_p
A
Obs
JmaxRef = 73.8
RdRef = 0.75
sigma = 0.56
VcmaxRef = 55.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

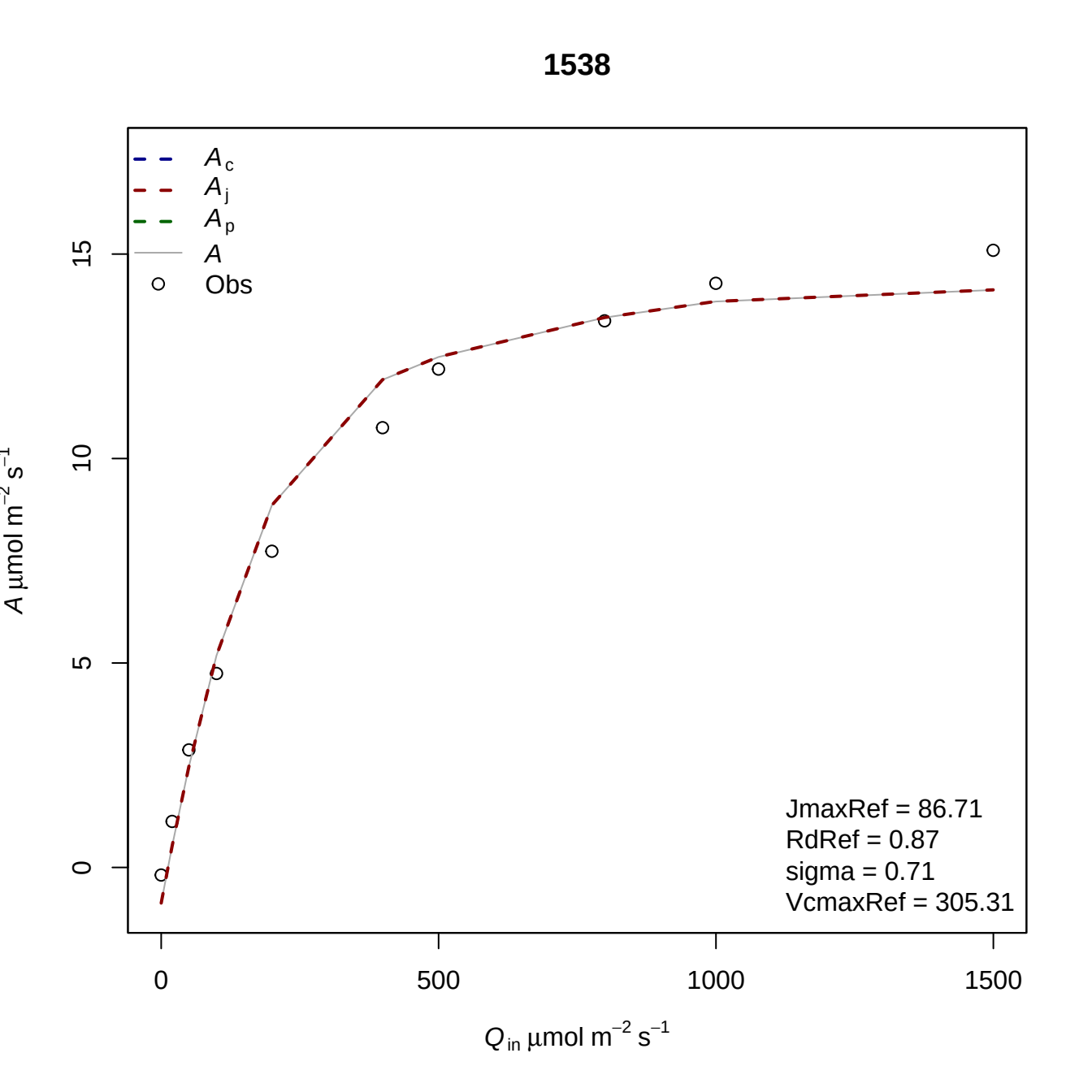

1533
A_c
A_j
A_p
A
Obs
JmaxRef = 75.64
RdRef = 0.71
sigma = 0.23
VcmaxRef = 241.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

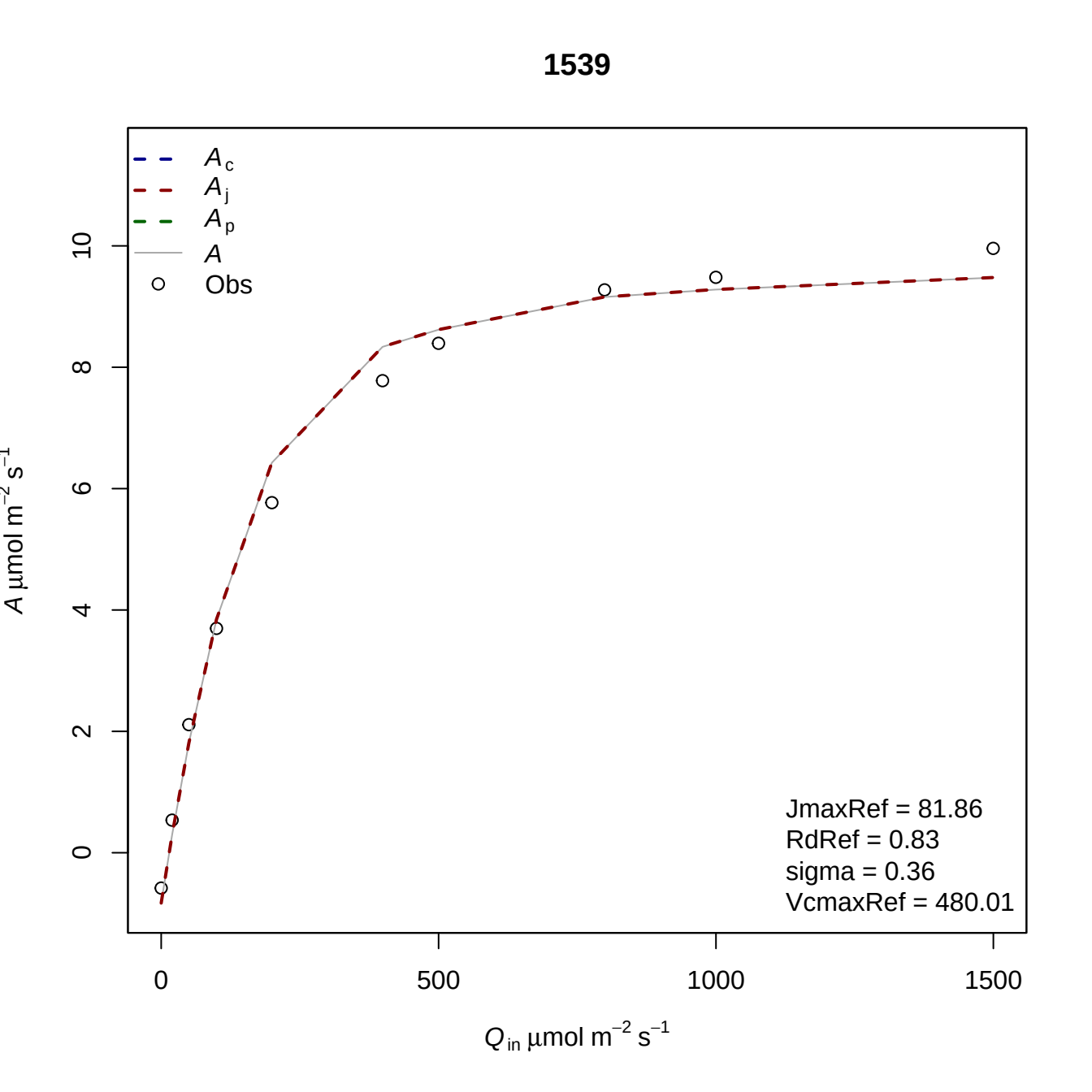

1534
A_c
A_j
A_p
A
Obs
JmaxRef = 106.06
RdRef = 0.91
sigma = 0.6
VcmaxRef = 552.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

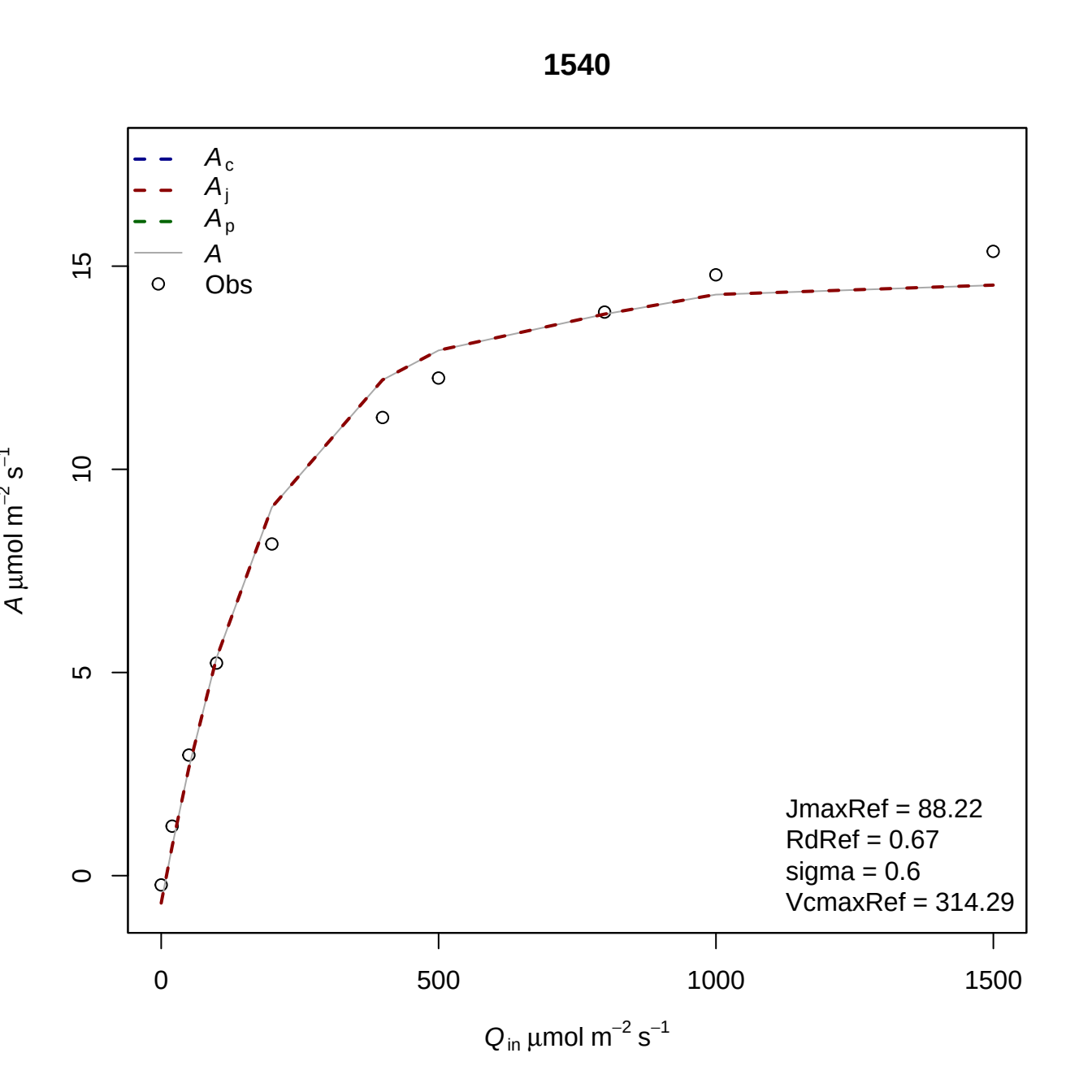

1535
A_c
A_j
A_p
A
Obs
JmaxRef = 54.42
RdRef = 0.53
sigma = 0.32
VcmaxRef = 88249.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

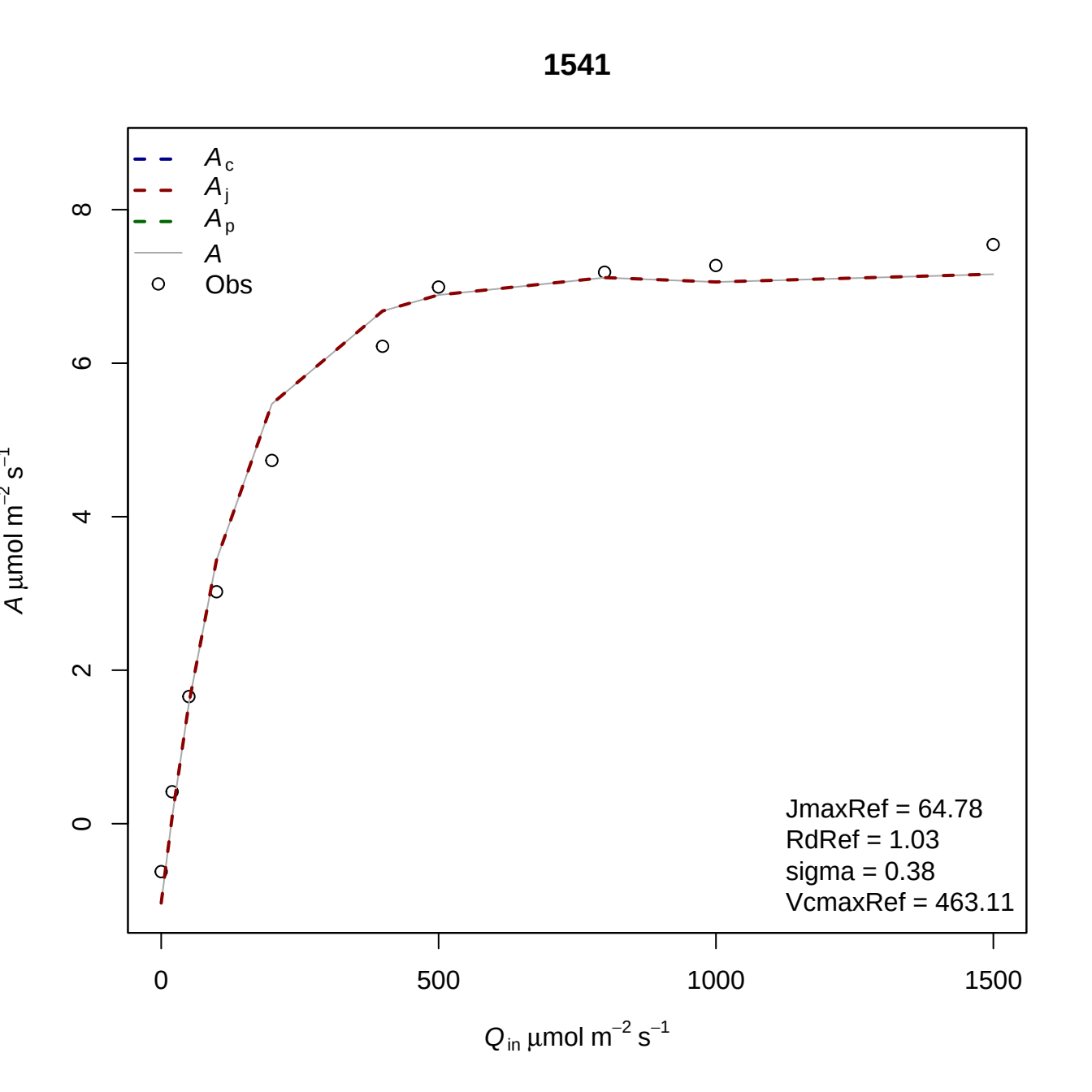

1536
A_c
A_j
A_p
A
Obs
JmaxRef = 73.54
RdRef = 0.48
sigma = 0.3
VcmaxRef = 59.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

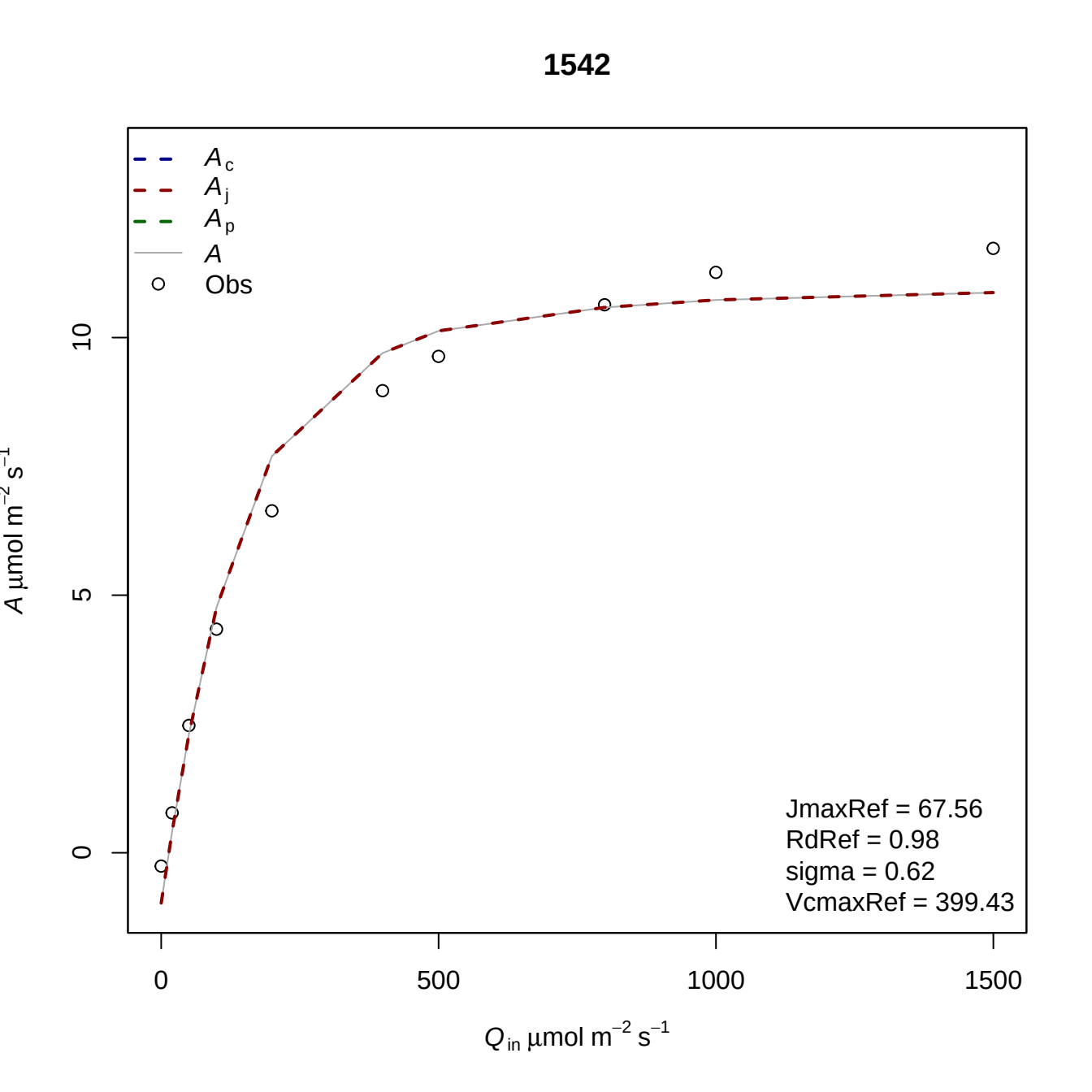

1537
A_c
A_j
A_p
A
Obs
JmaxRef = 75.93
RdRef = 0.83
sigma = 0.38
VcmaxRef = 1004.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

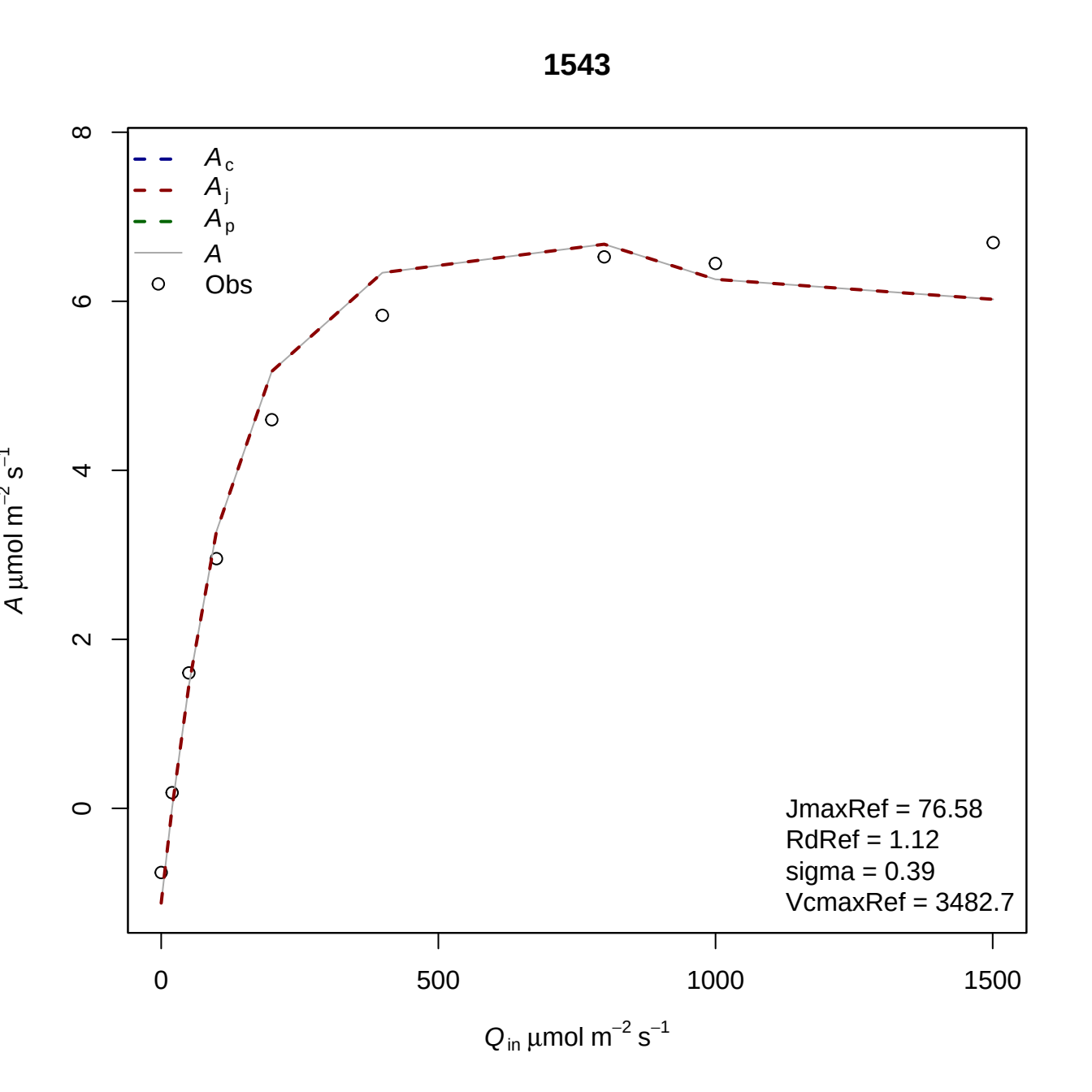

1538
A_c
A_j
A_p
A
Obs
JmaxRef = 86.71
RdRef = 0.87
sigma = 0.71
VcmaxRef = 305.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

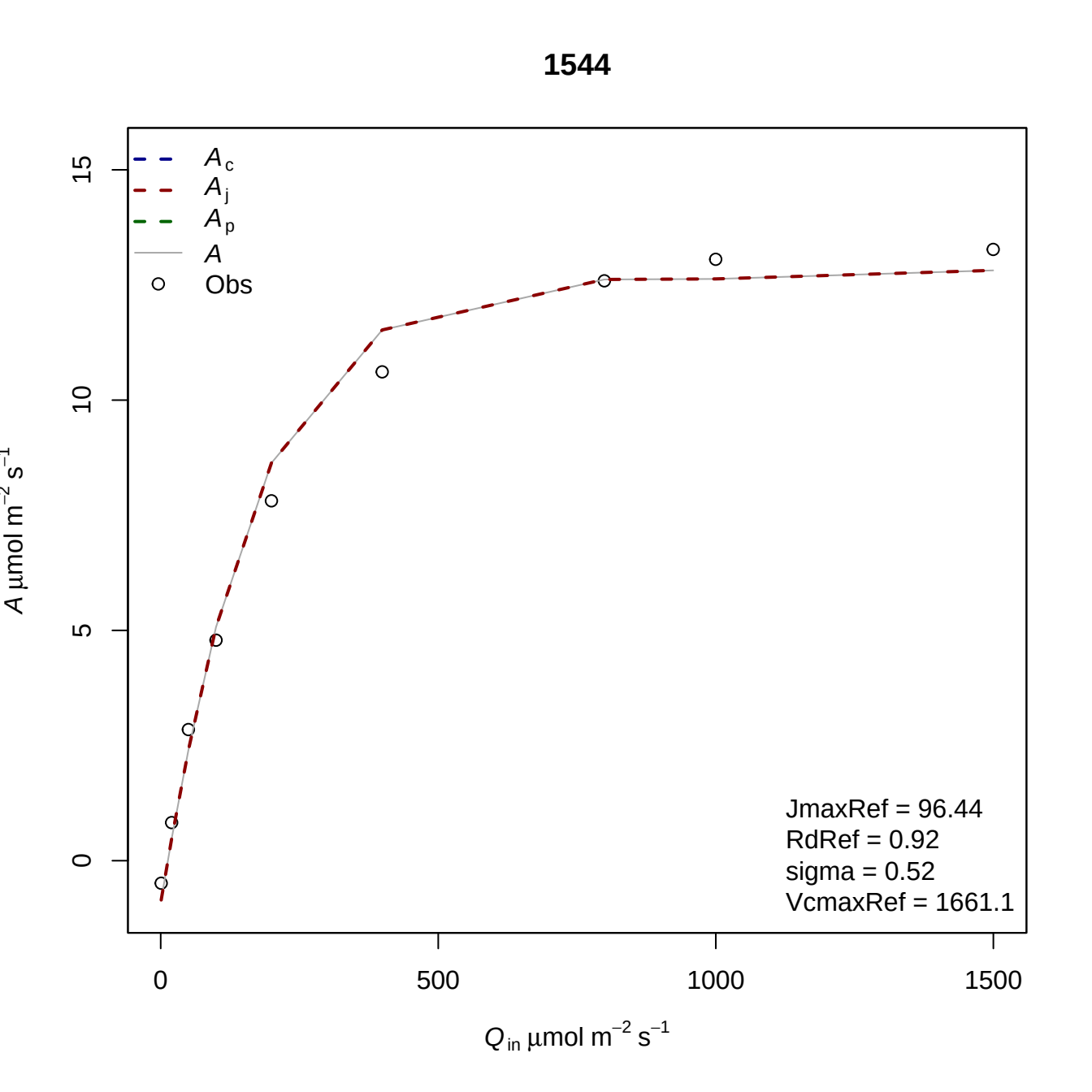

1539
A_c
A_j
A_p
A
Obs
JmaxRef = 81.86
RdRef = 0.83
sigma = 0.36
VcmaxRef = 480.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

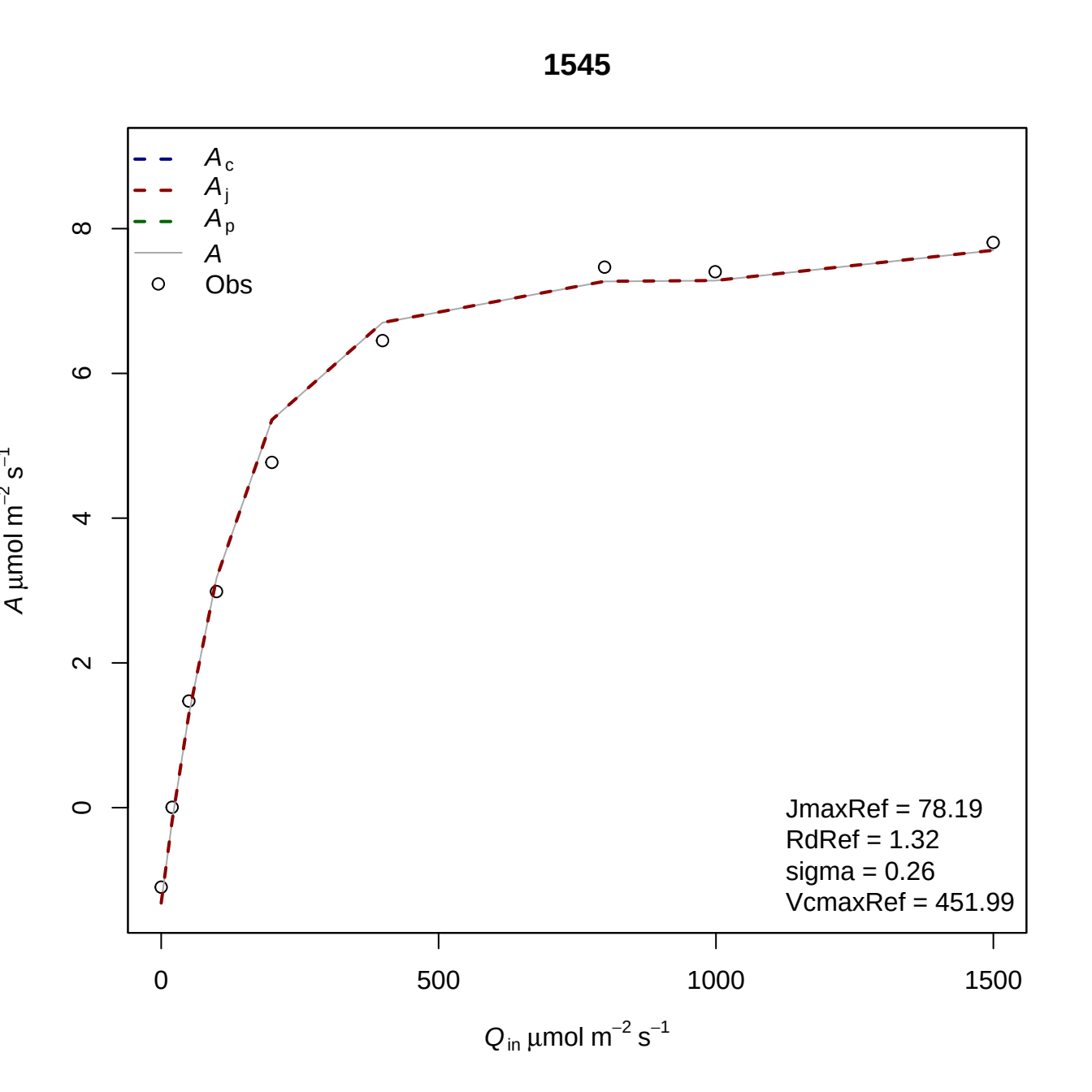

1540
A_c
A_j
A_p
A
Obs
JmaxRef = 88.22
RdRef = 0.67
sigma = 0.6
VcmaxRef = 314.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

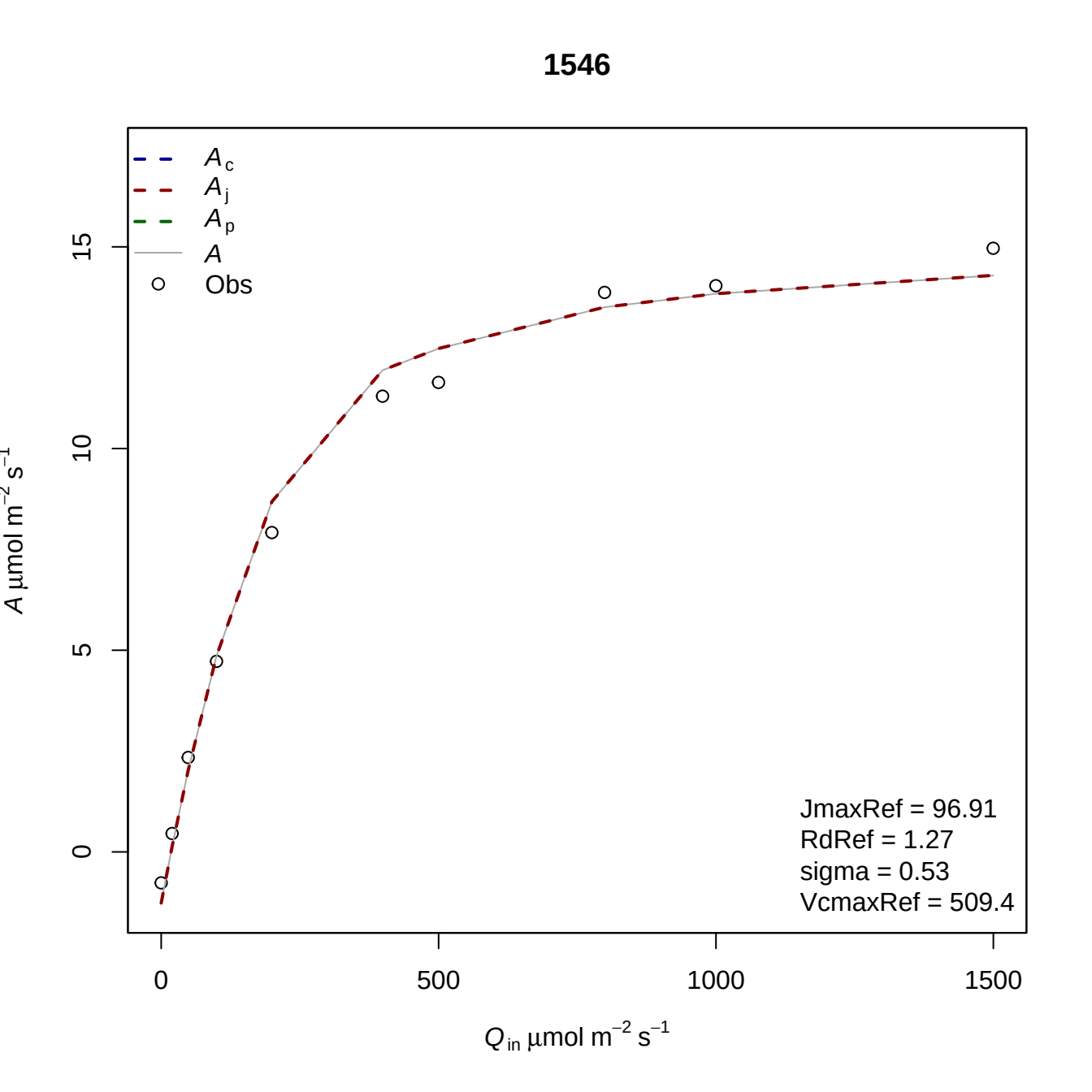

1541
A_c
A_j
A_p
A
Obs
JmaxRef = 64.78
RdRef = 1.03
sigma = 0.38
VcmaxRef = 463.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

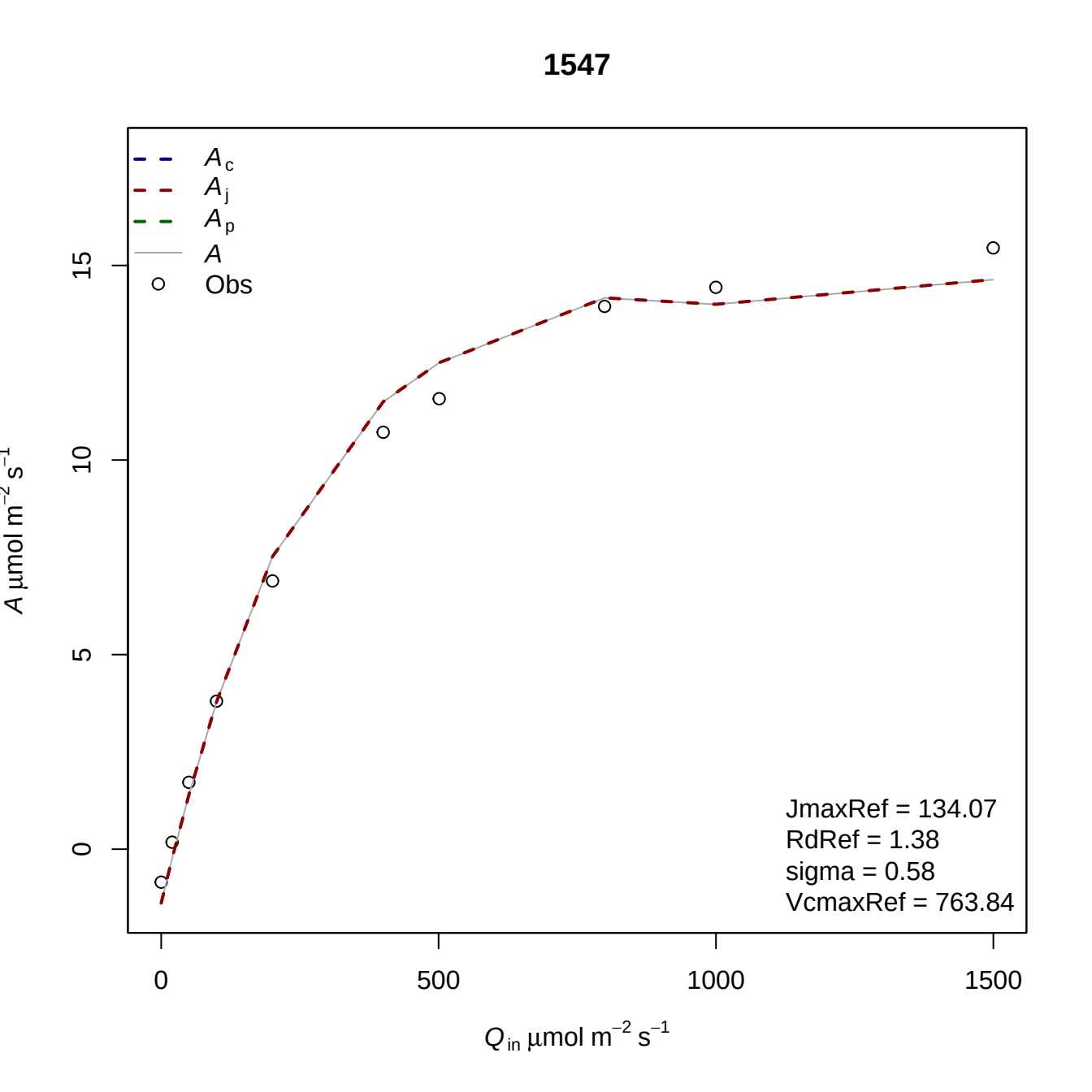

1542
A_c
A_j
A_p
A
Obs
JmaxRef = 67.56
RdRef = 0.98
sigma = 0.62
VcmaxRef = 399.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

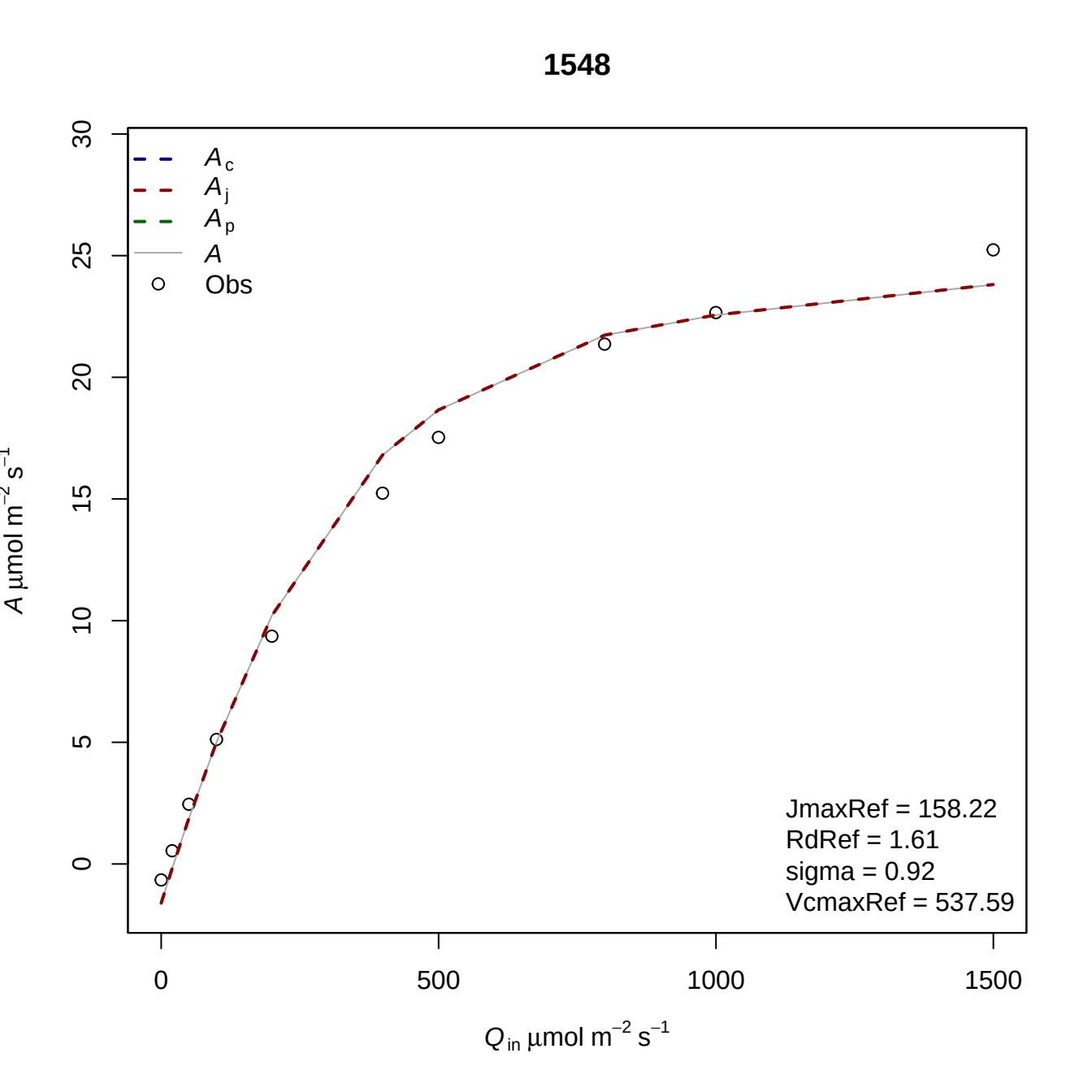

1543
A_c
A_j
A_p
A
Obs
JmaxRef = 76.58
RdRef = 1.12
sigma = 0.39
VcmaxRef = 3482.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

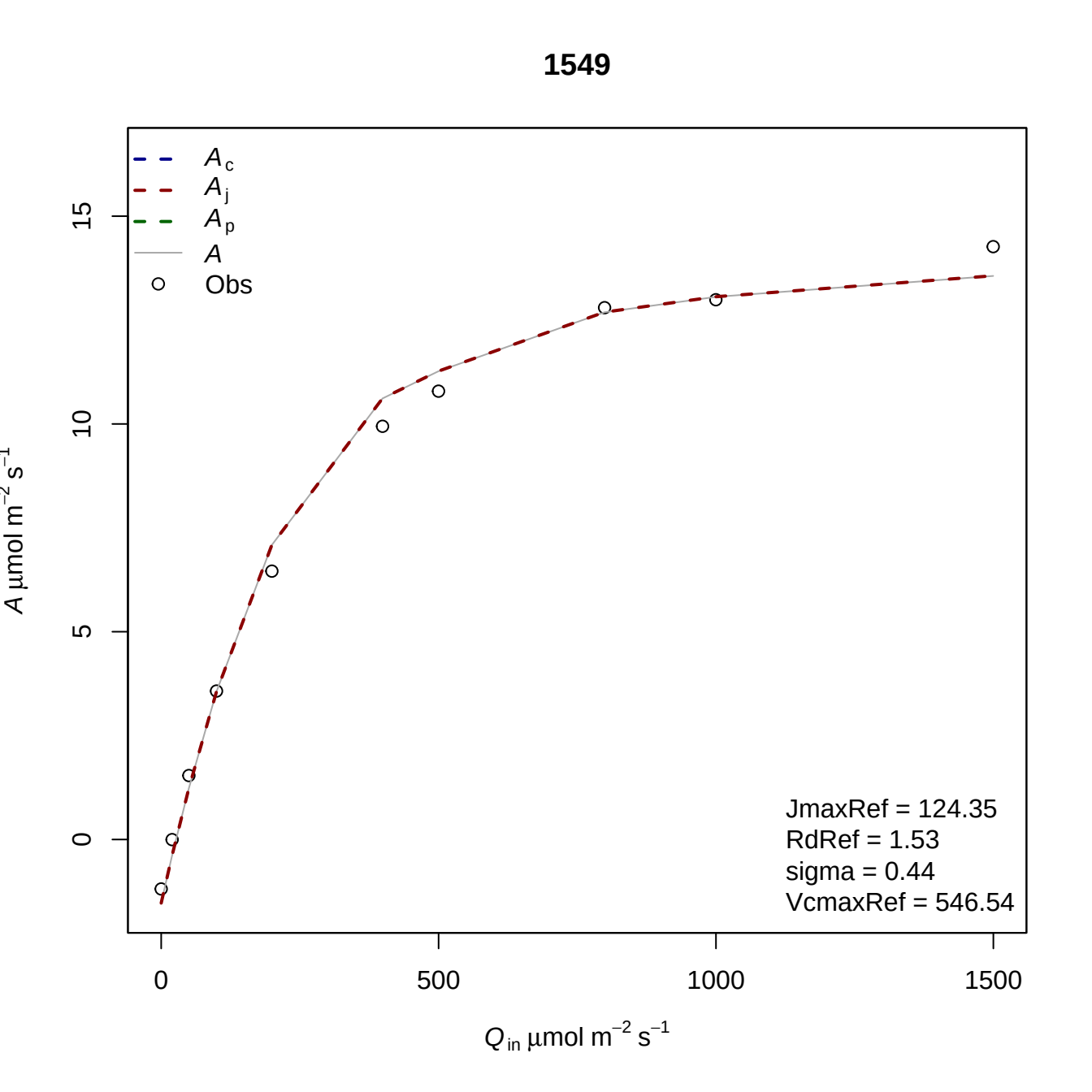

1544
A_c
A_j
A_p
A
Obs
JmaxRef = 96.44
RdRef = 0.92
sigma = 0.52
VcmaxRef = 1661.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

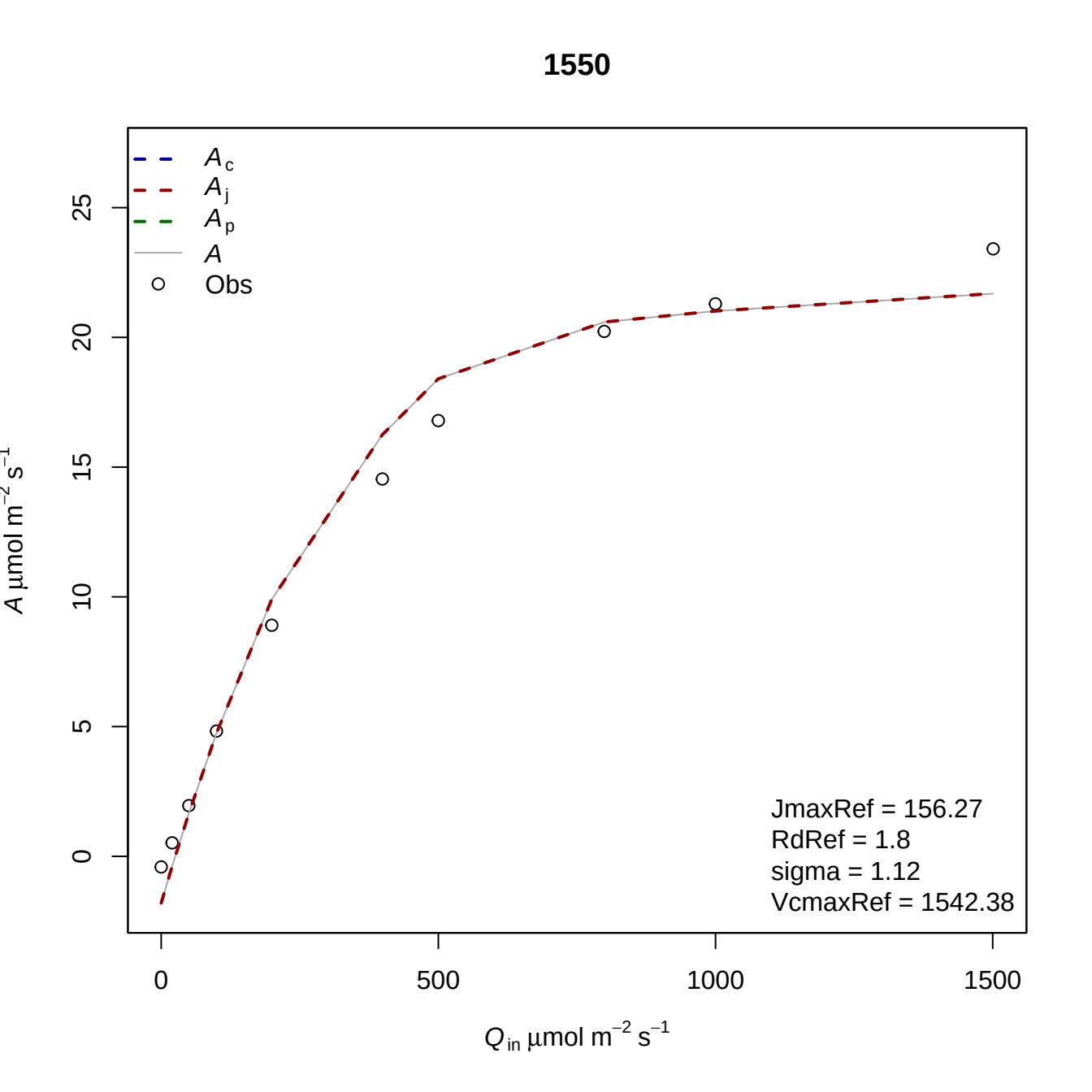

1545
A_c
A_j
A_p
A
Obs
JmaxRef = 78.19
RdRef = 1.32
sigma = 0.26
VcmaxRef = 451.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

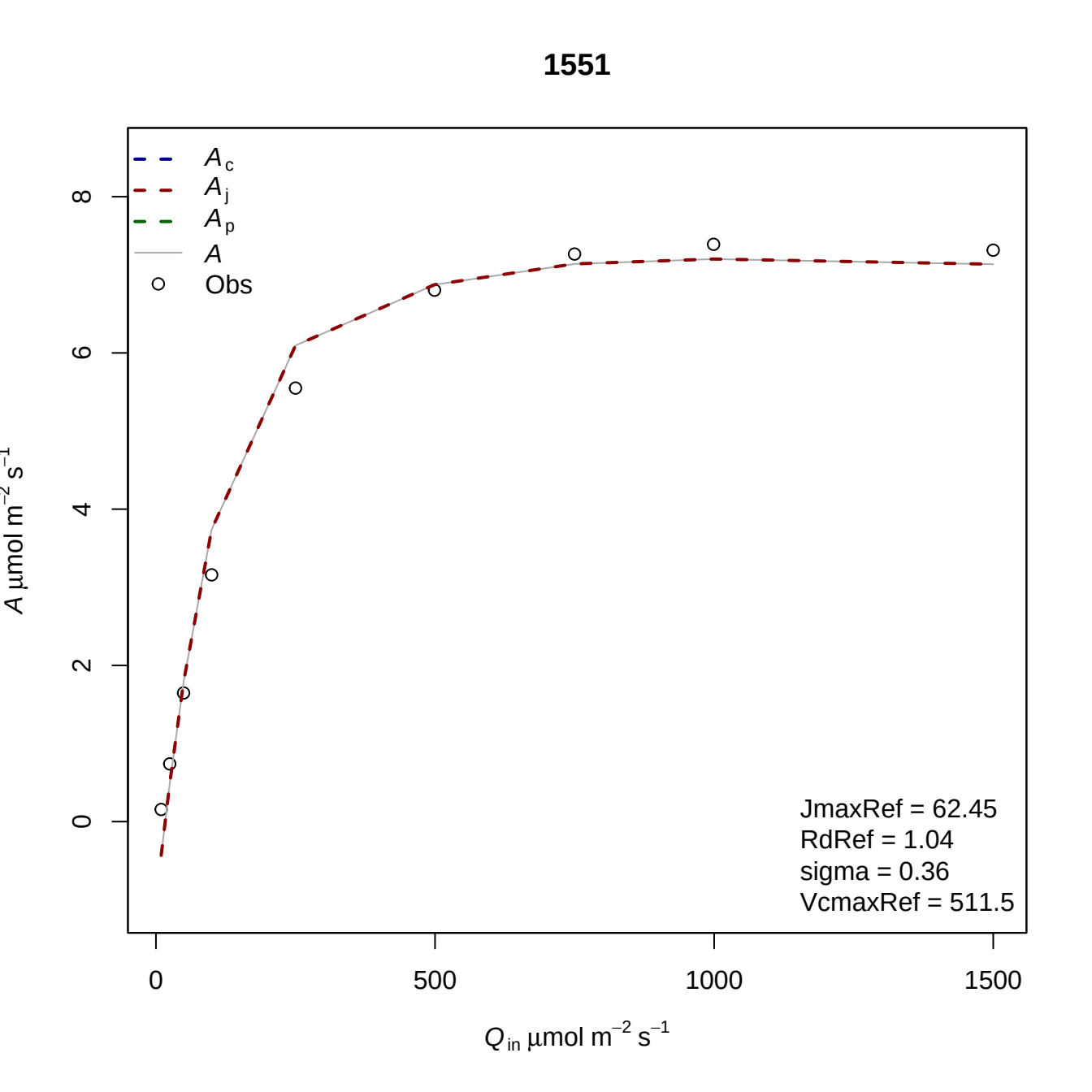

1546
A_c
A_j
A_p
A
Obs
JmaxRef = 96.91
RdRef = 1.27
sigma = 0.53
VcmaxRef = 509.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

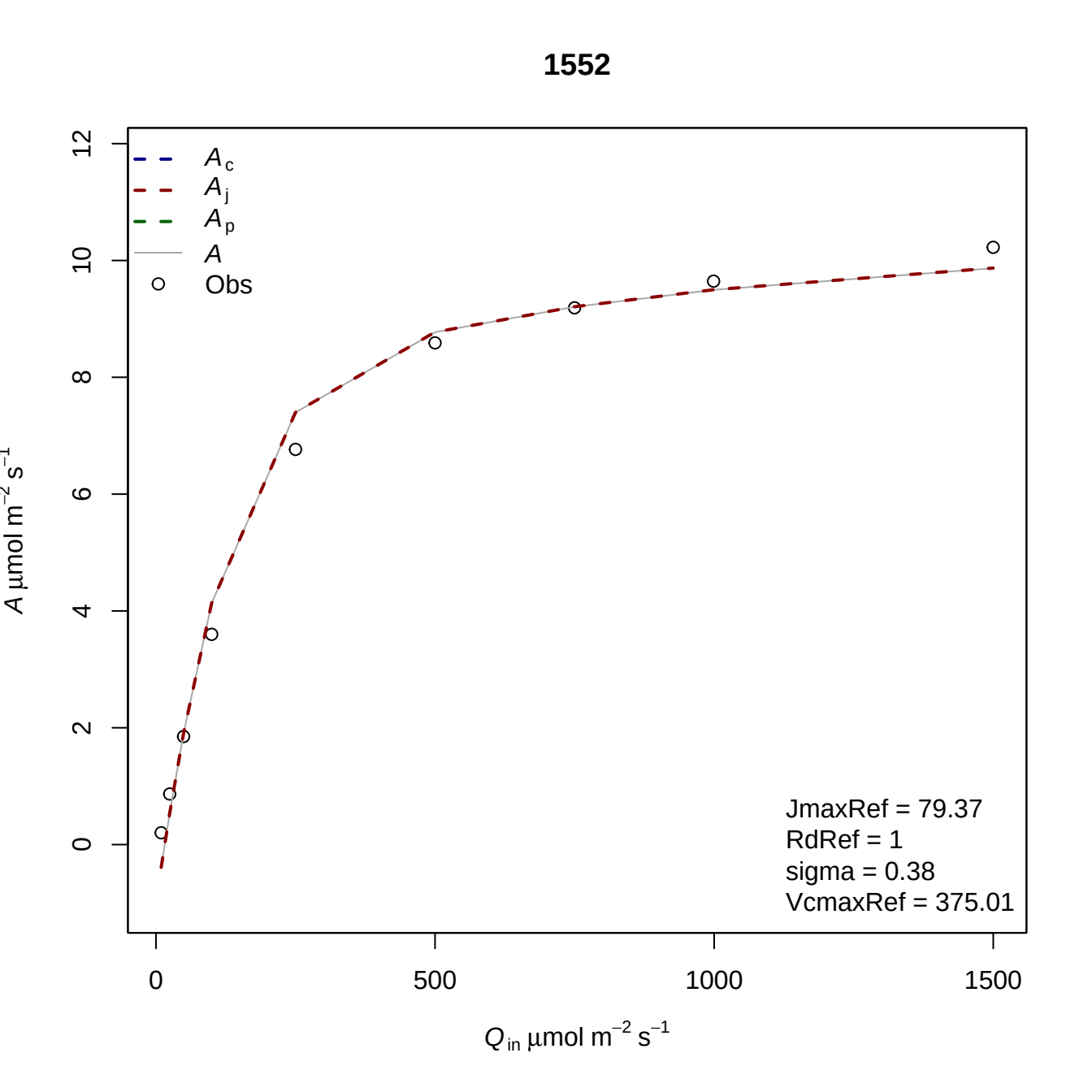

1547
A_c
A_j
A_p
A
Obs
JmaxRef = 134.07
RdRef = 1.38
sigma = 0.58
VcmaxRef = 763.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

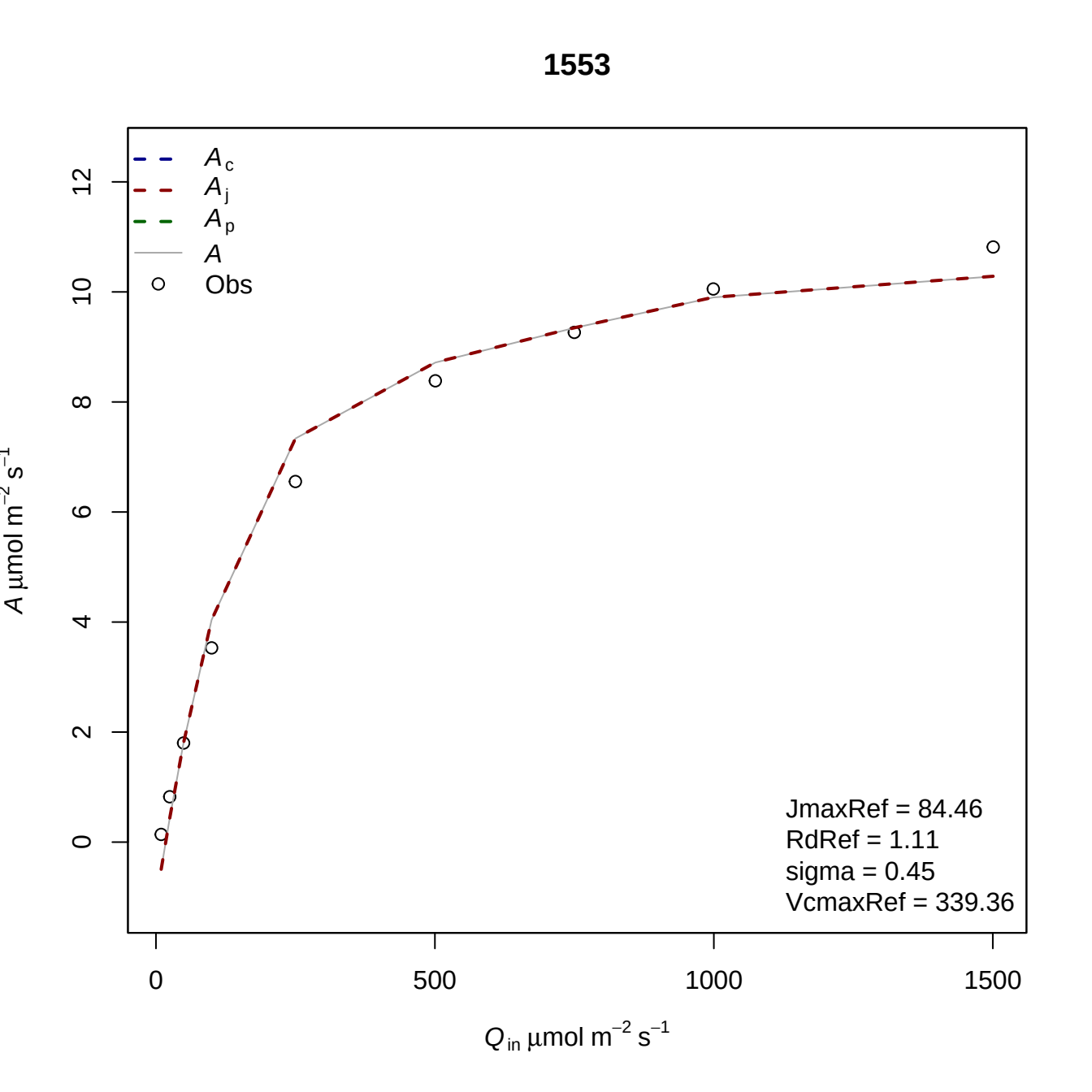

1548
A_c
A_j
A_p
A
Obs
JmaxRef = 158.22
RdRef = 1.61
sigma = 0.92
VcmaxRef = 537.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500

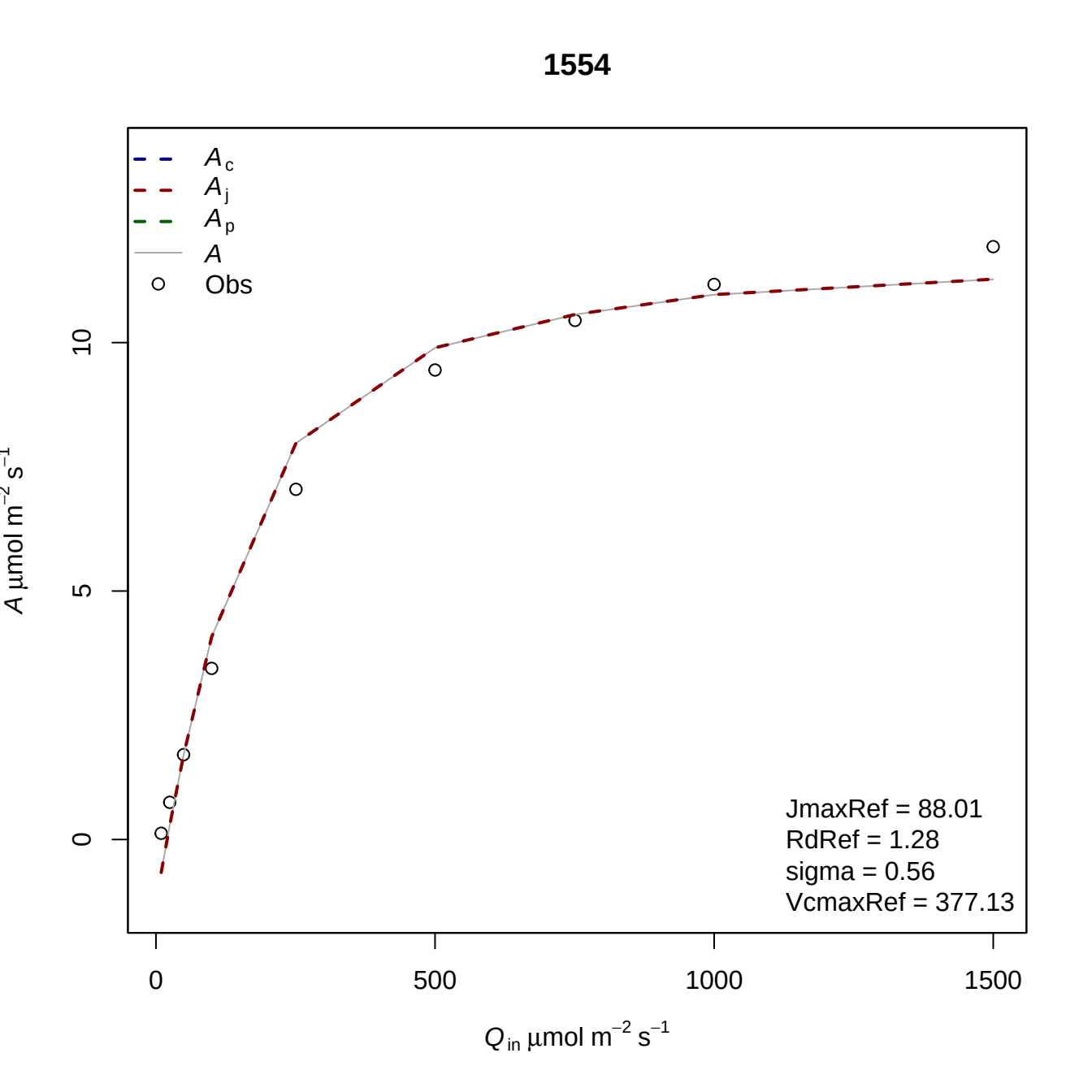

1549
A_c
A_j
A_p
A
Obs
JmaxRef = 124.35
RdRef = 1.53
sigma = 0.44
VcmaxRef = 546.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

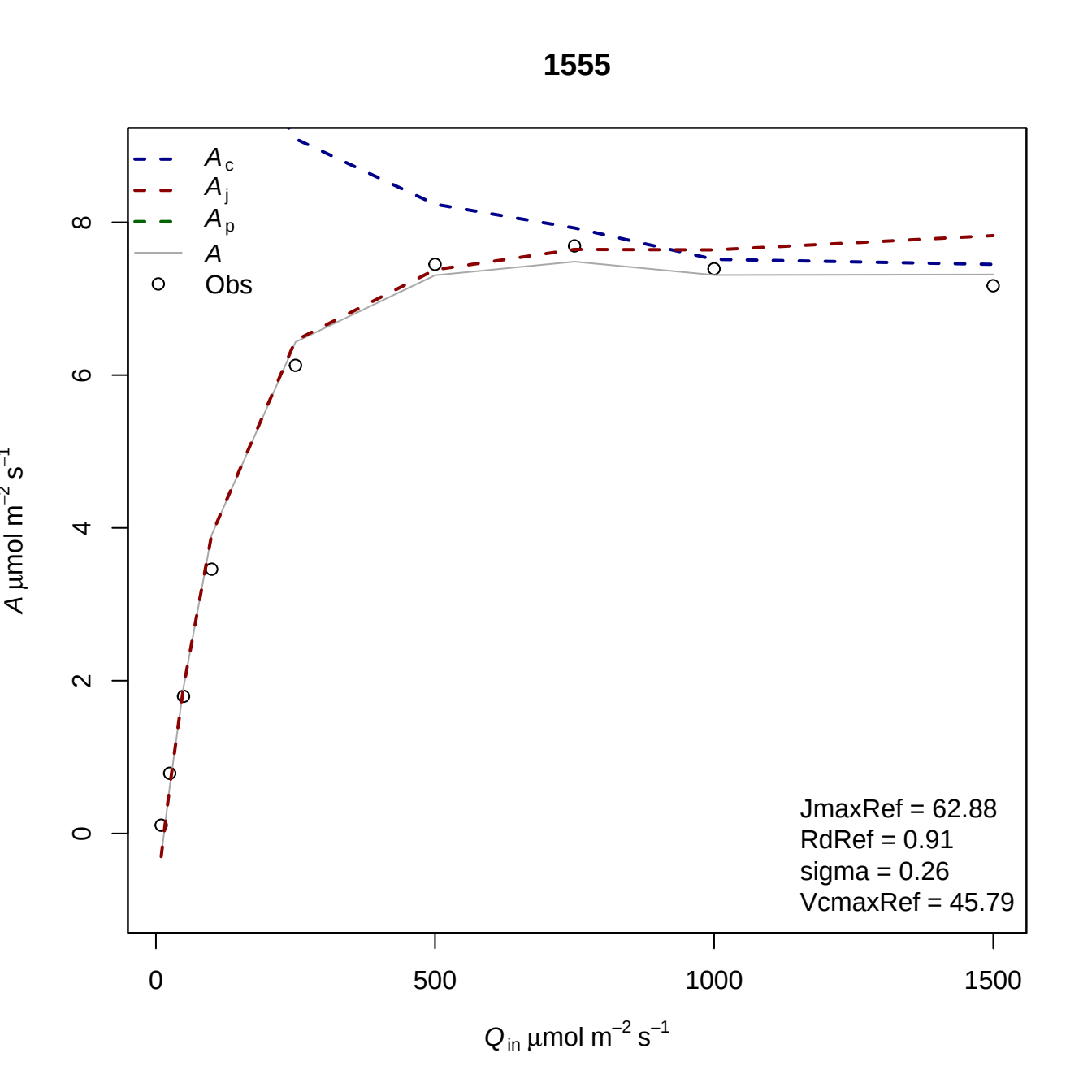

1550
A_c
A_j
A_p
A
Obs
JmaxRef = 156.27
RdRef = 1.8
sigma = 1.12
VcmaxRef = 1542.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

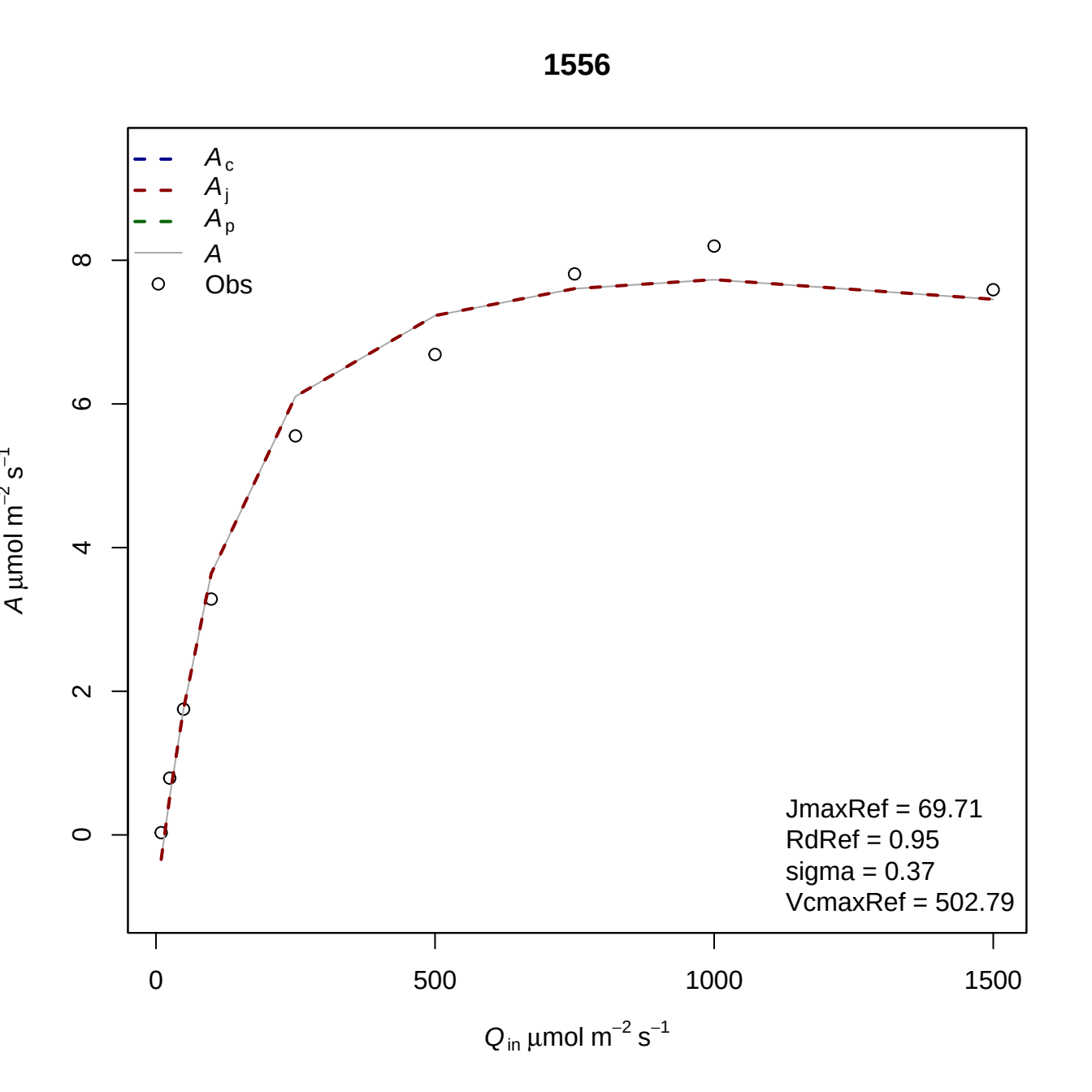

1551
A_c
A_j
A_p
A
Obs
JmaxRef = 62.45
RdRef = 1.04
sigma = 0.36
VcmaxRef = 511.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

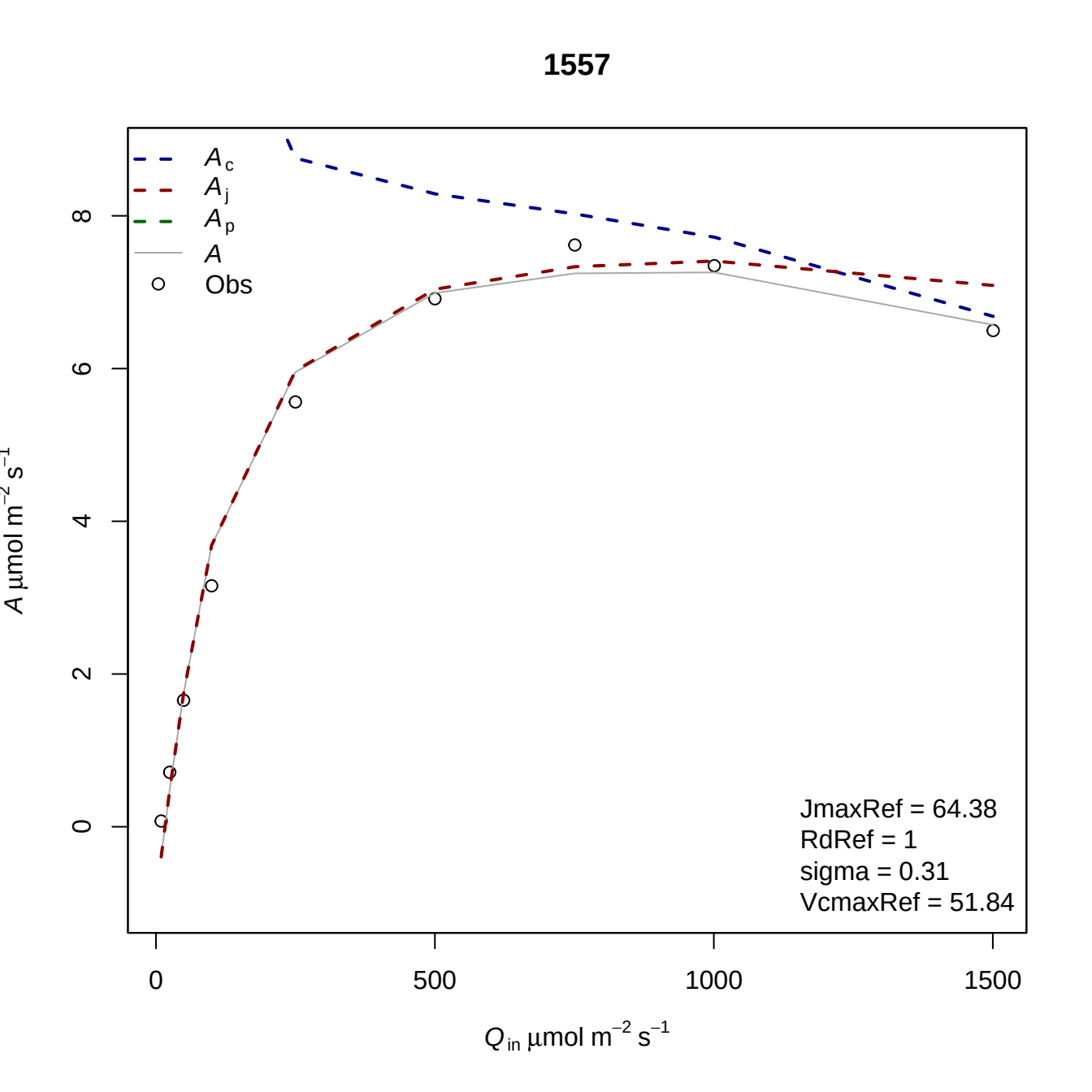

1552
A_c
A_j
A_p
A
Obs
JmaxRef = 79.37
RdRef = 1
sigma = 0.38
VcmaxRef = 375.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

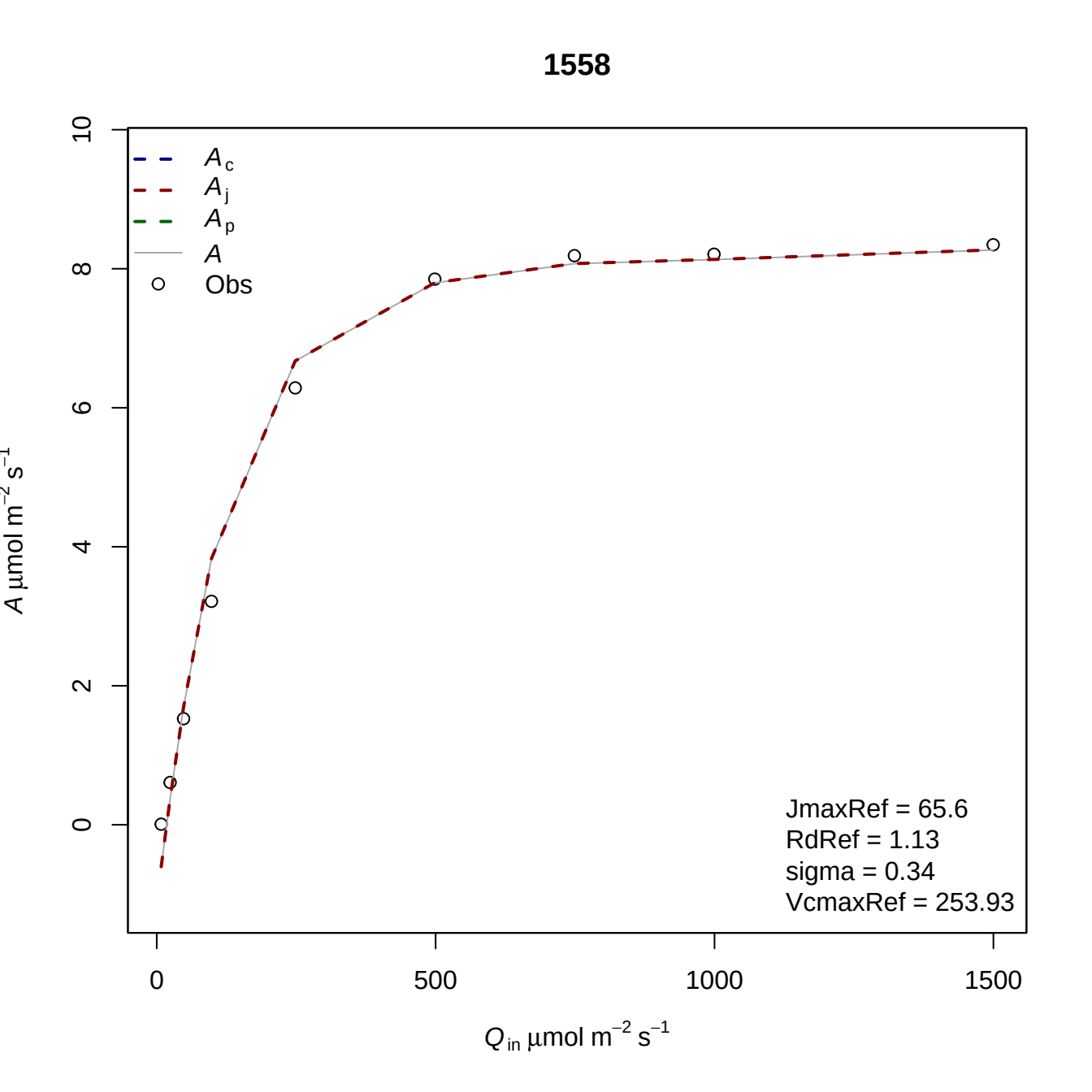

1553
A_c
A_j
A_p
A
Obs
JmaxRef = 84.46
RdRef = 1.11
sigma = 0.45
VcmaxRef = 339.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

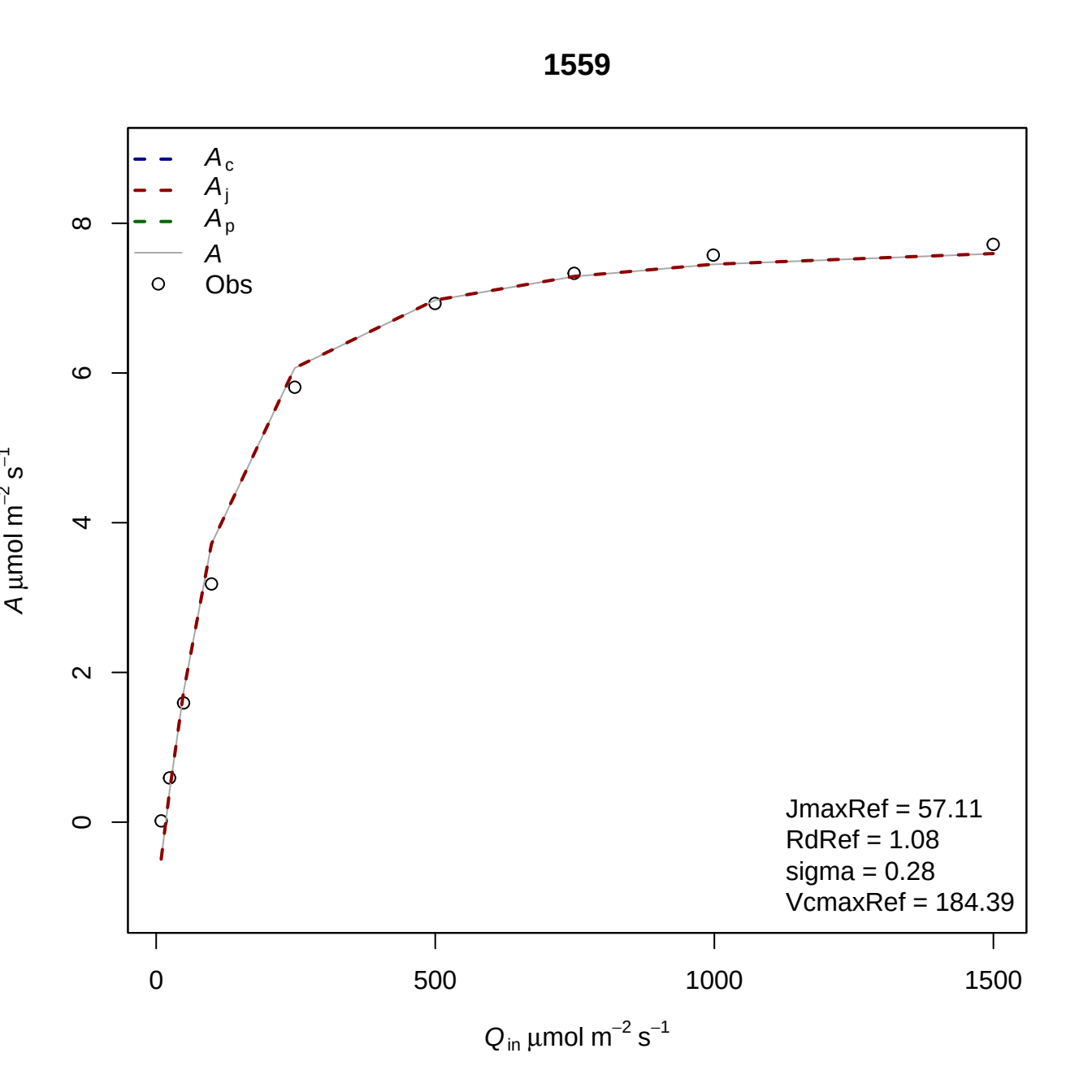

1554
A_c
A_j
A_p
A
Obs
JmaxRef = 88.01
RdRef = 1.28
sigma = 0.56
VcmaxRef = 377.13
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

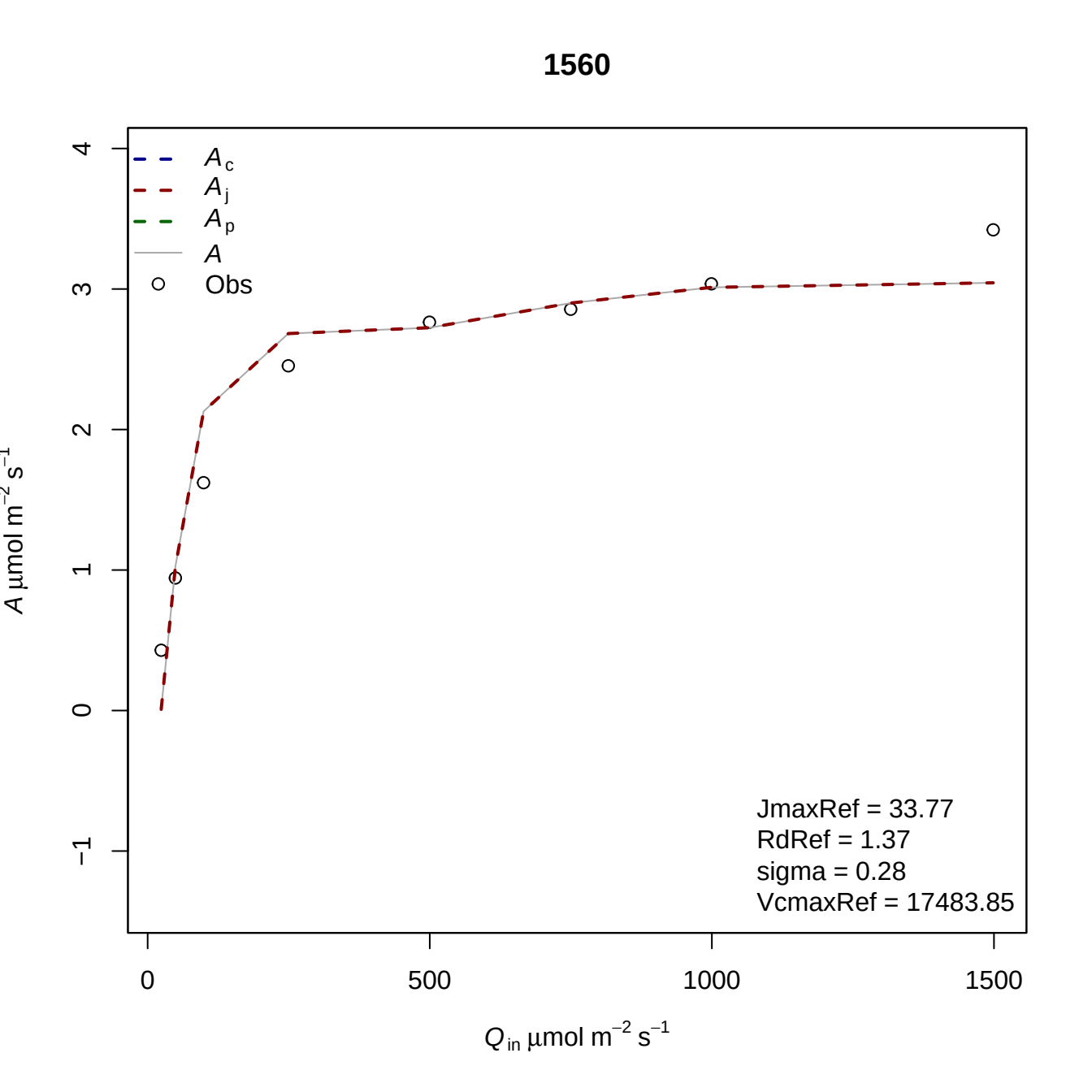

1555
A_c
A_j
A_p
A
Obs
JmaxRef = 62.88
RdRef = 0.91
sigma = 0.26
VcmaxRef = 45.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

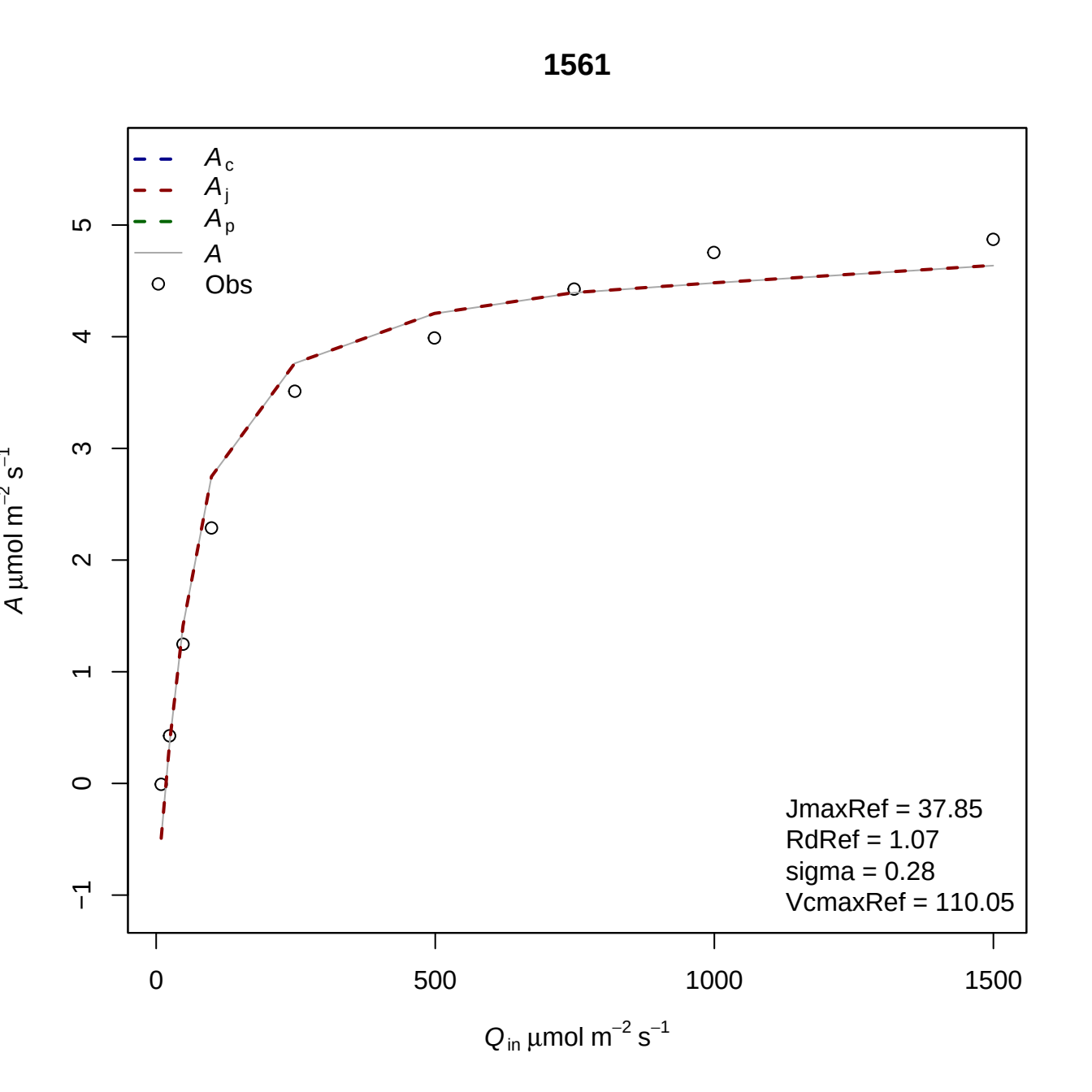

1556
A_c
A_j
A_p
A
Obs
JmaxRef = 69.71
RdRef = 0.95
sigma = 0.37
VcmaxRef = 502.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

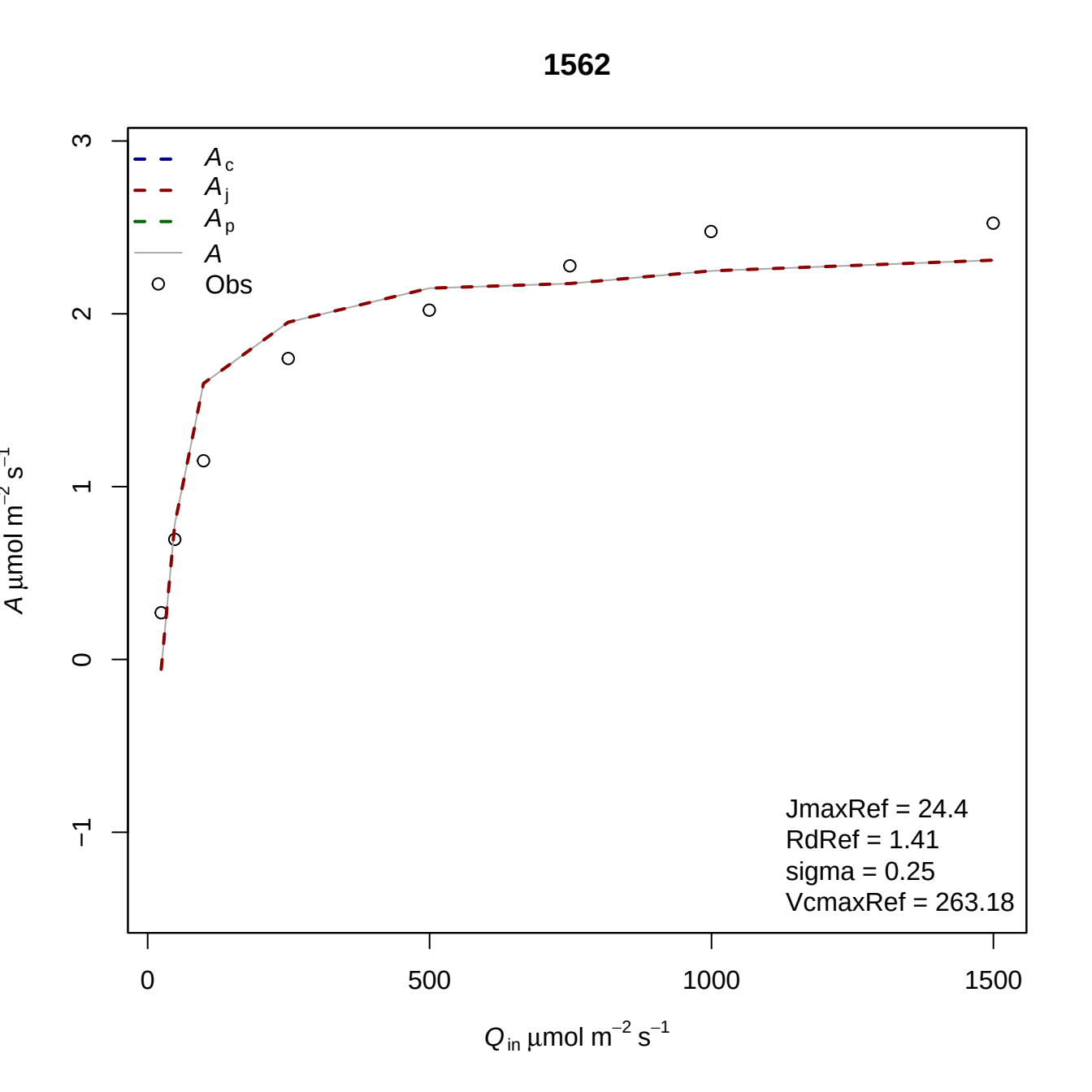

1557
A_c
A_j
A_p
A
Obs
JmaxRef = 64.38
RdRef = 1
sigma = 0.31
VcmaxRef = 51.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

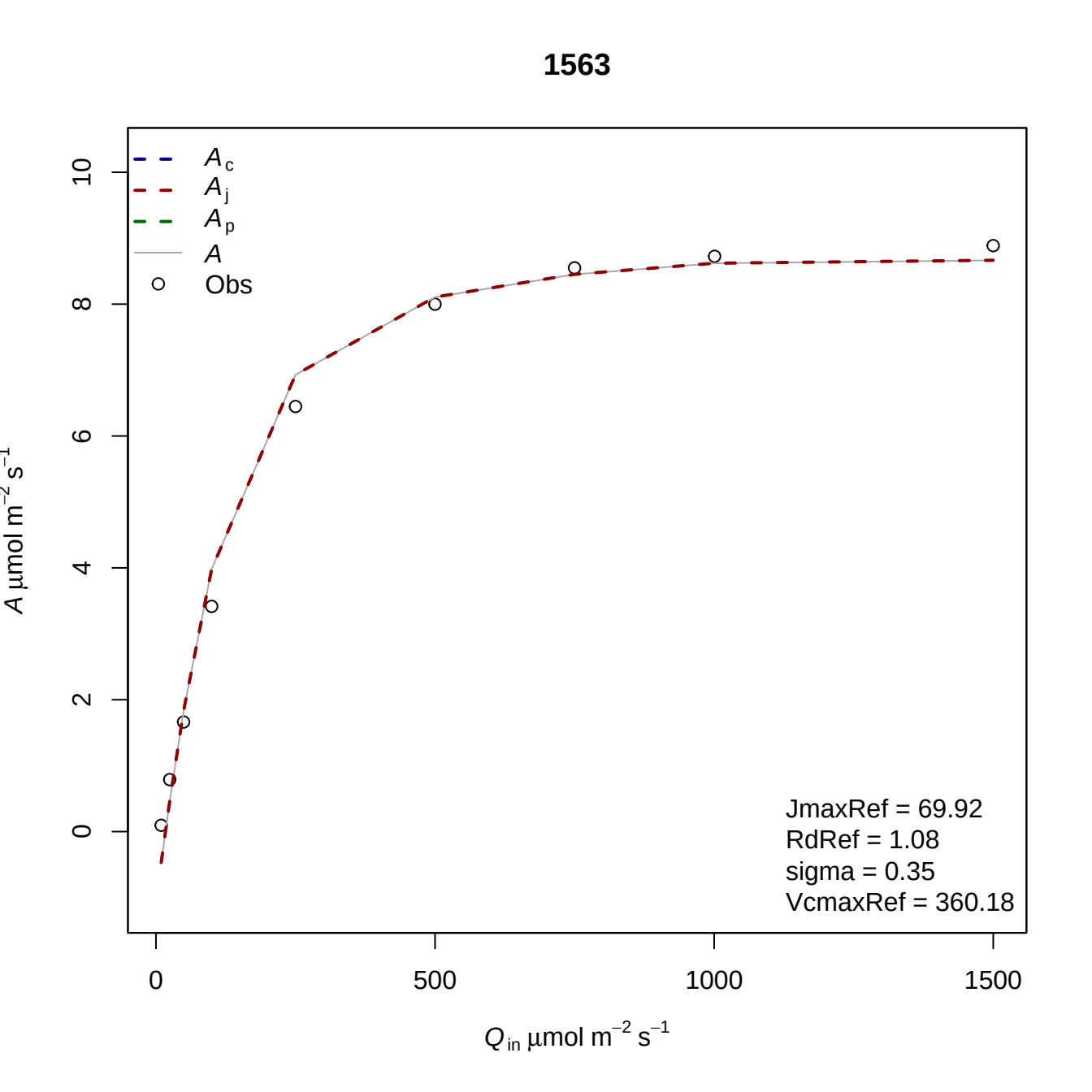

1558
A_c
A_j
A_p
A
Obs
JmaxRef = 65.6
RdRef = 1.13
sigma = 0.34
VcmaxRef = 253.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

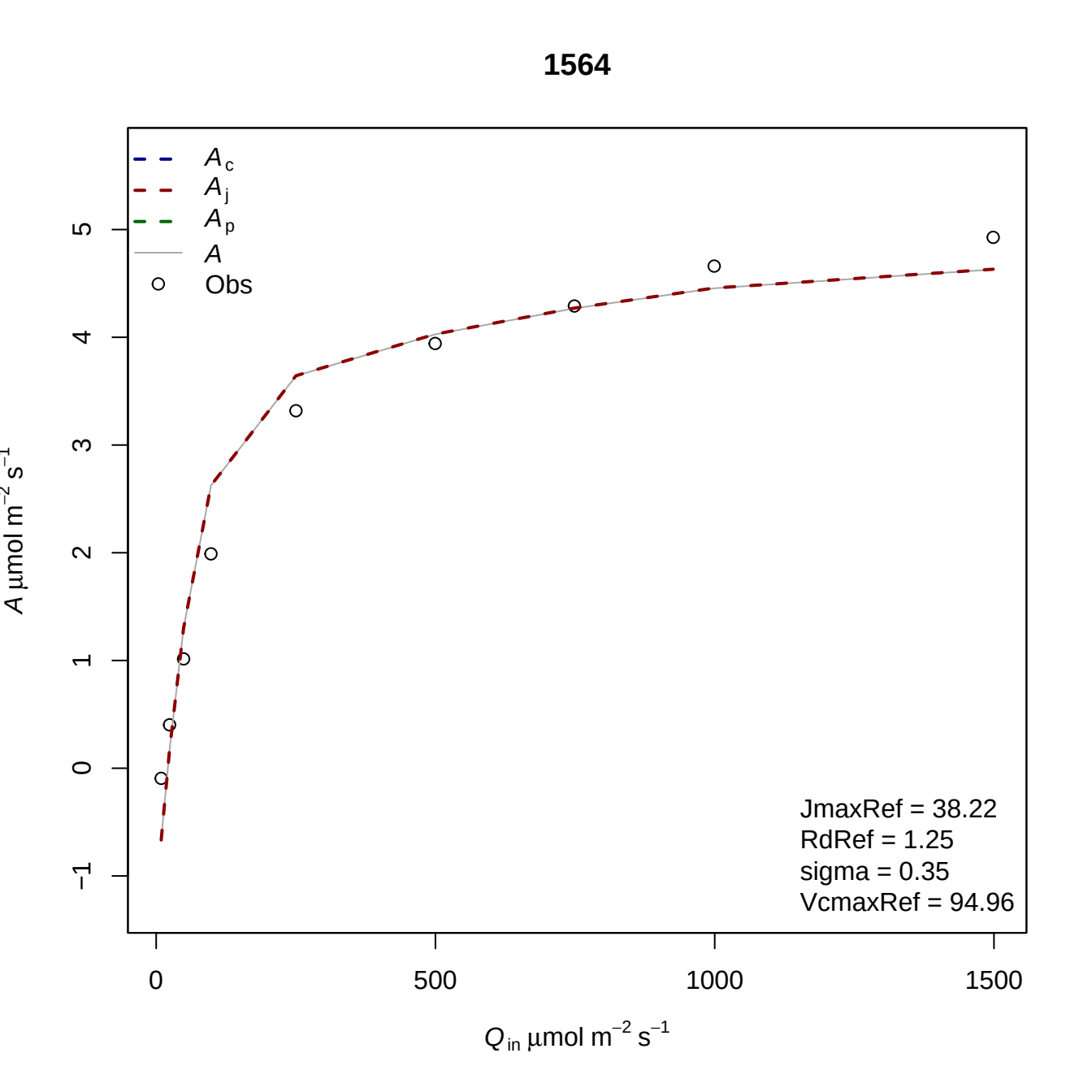

1559
A_c
A_j
A_p
A
Obs
JmaxRef = 57.11
RdRef = 1.08
sigma = 0.28
VcmaxRef = 184.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

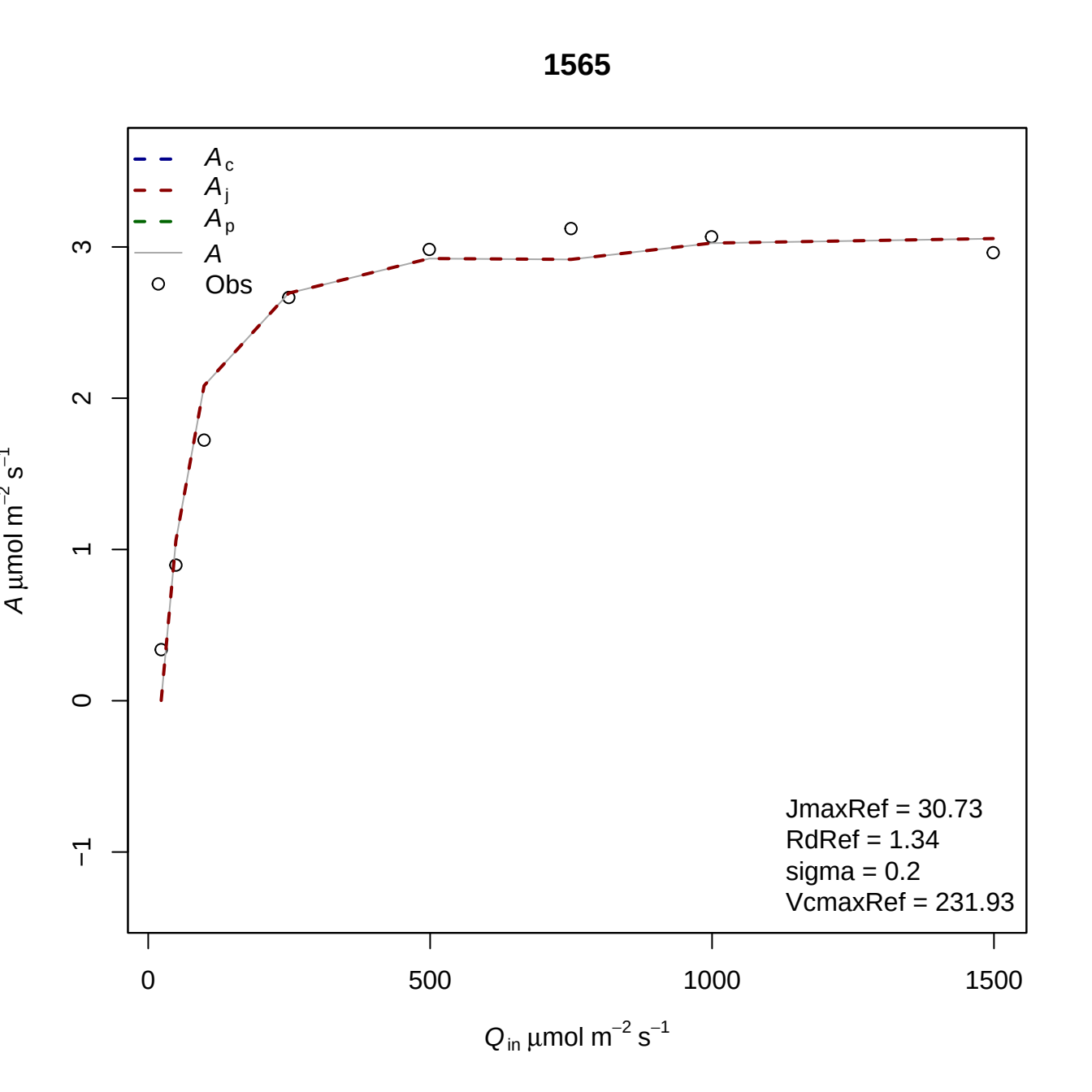

1560
A_c
A_j
A_p
A
Obs
JmaxRef = 33.77
RdRef = 1.37
sigma = 0.28
VcmaxRef = 17483.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
500
1000
1500

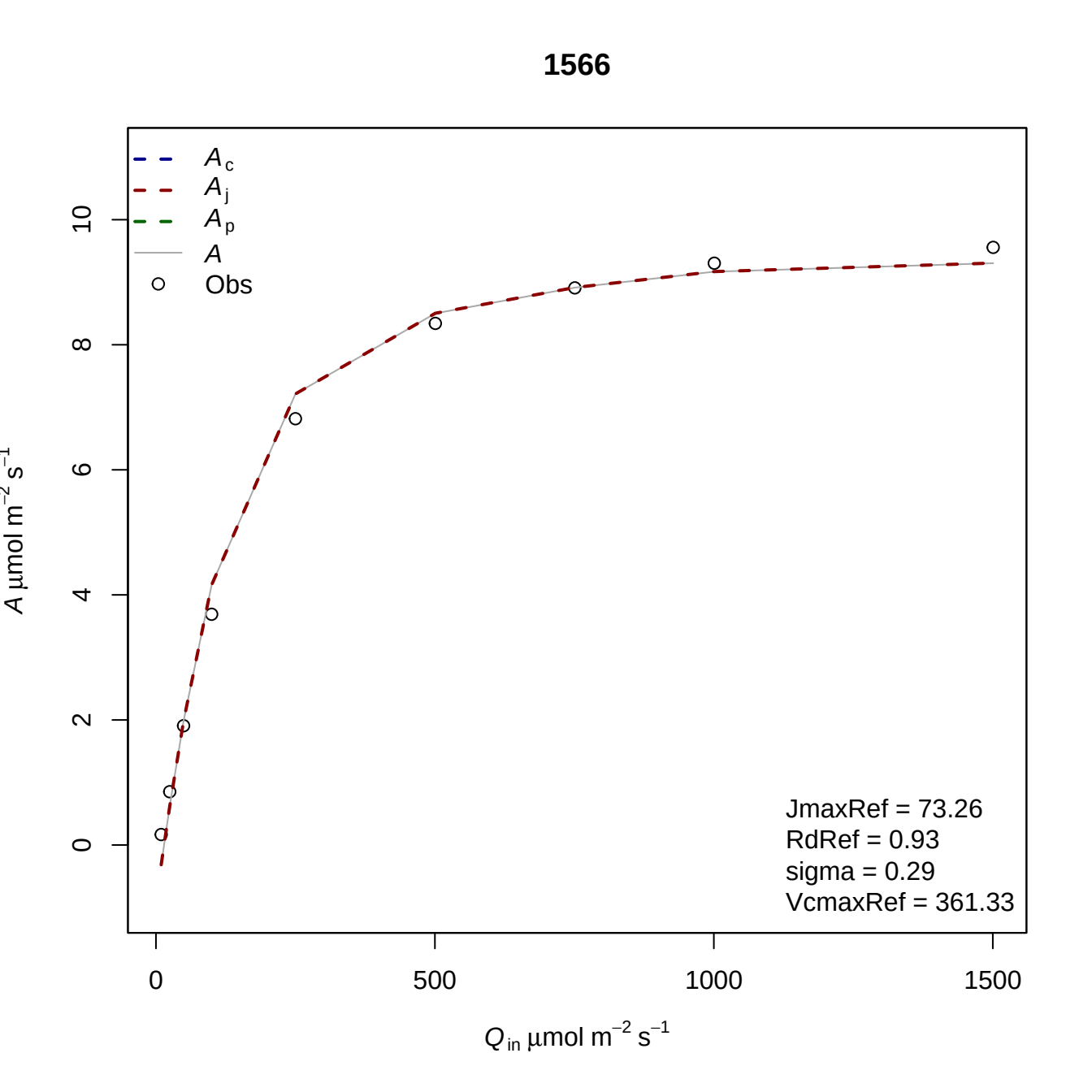

1561
A_c
A_j
A_p
A
Obs
JmaxRef = 37.85
RdRef = 1.07
sigma = 0.28
VcmaxRef = 110.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

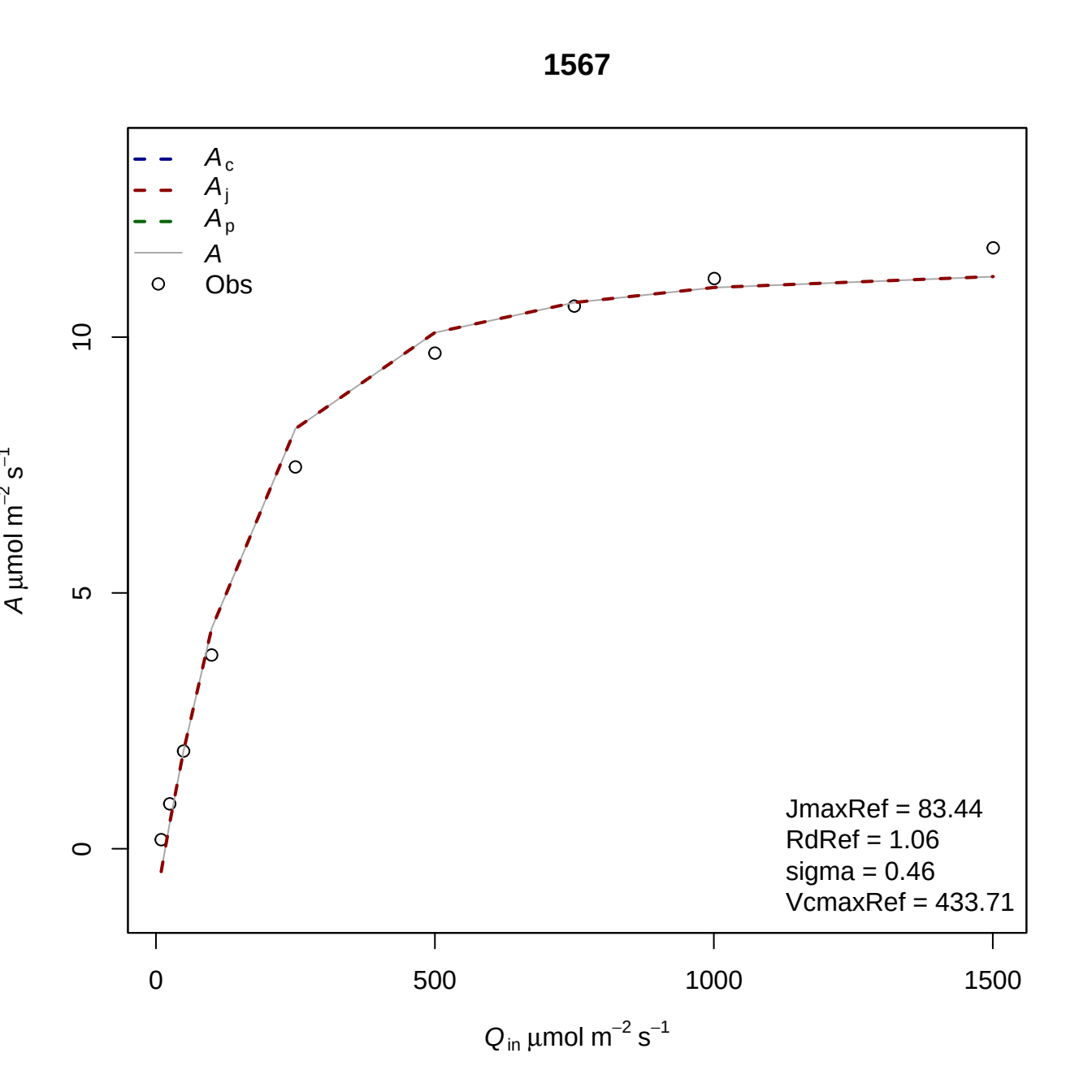

1562
A_c
A_j
A_p
A
Obs
JmaxRef = 24.4
RdRef = 1.41
sigma = 0.25
VcmaxRef = 263.18
3
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

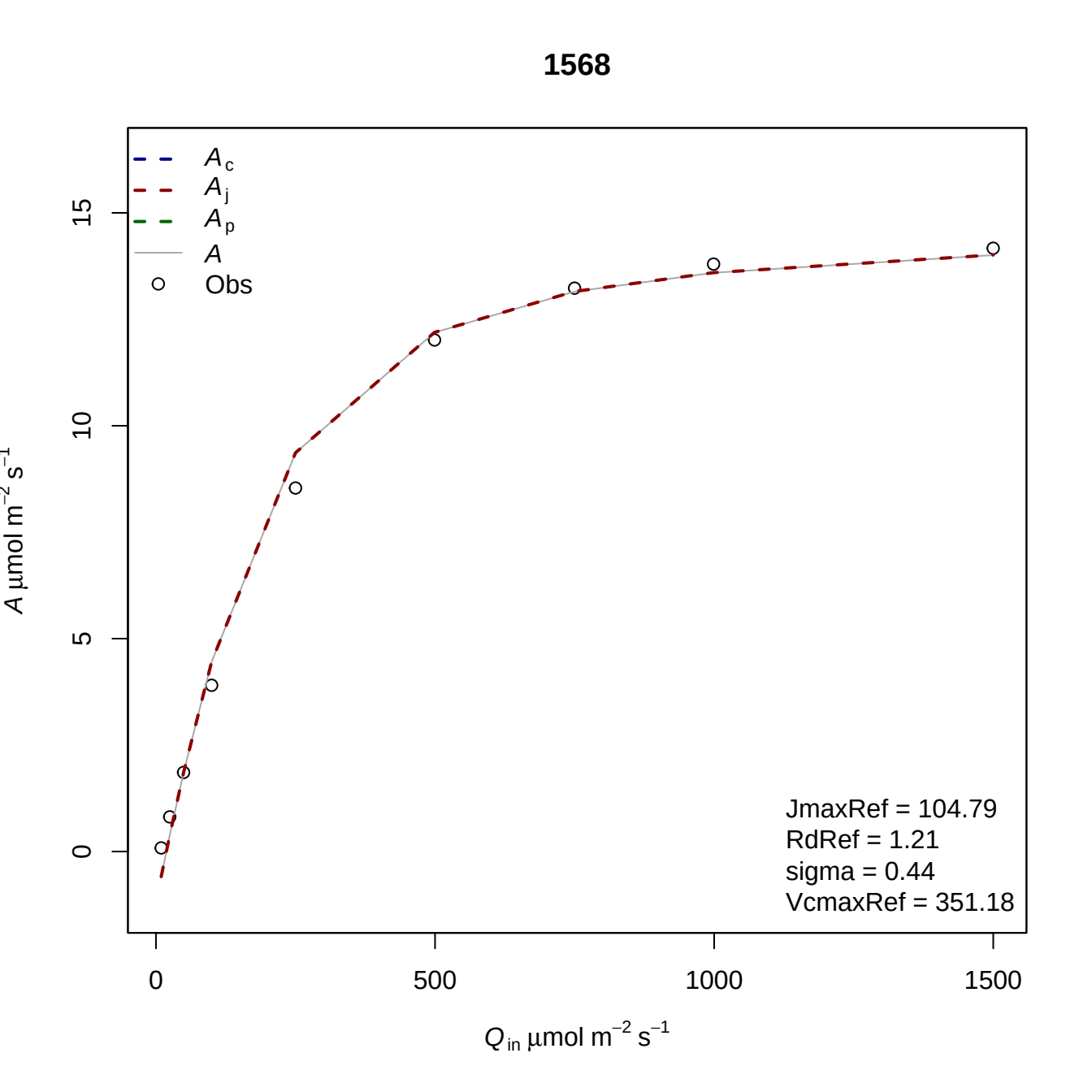

1563
A_c
A_j
A_p
A
Obs
JmaxRef = 69.92
RdRef = 1.08
sigma = 0.35
VcmaxRef = 360.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

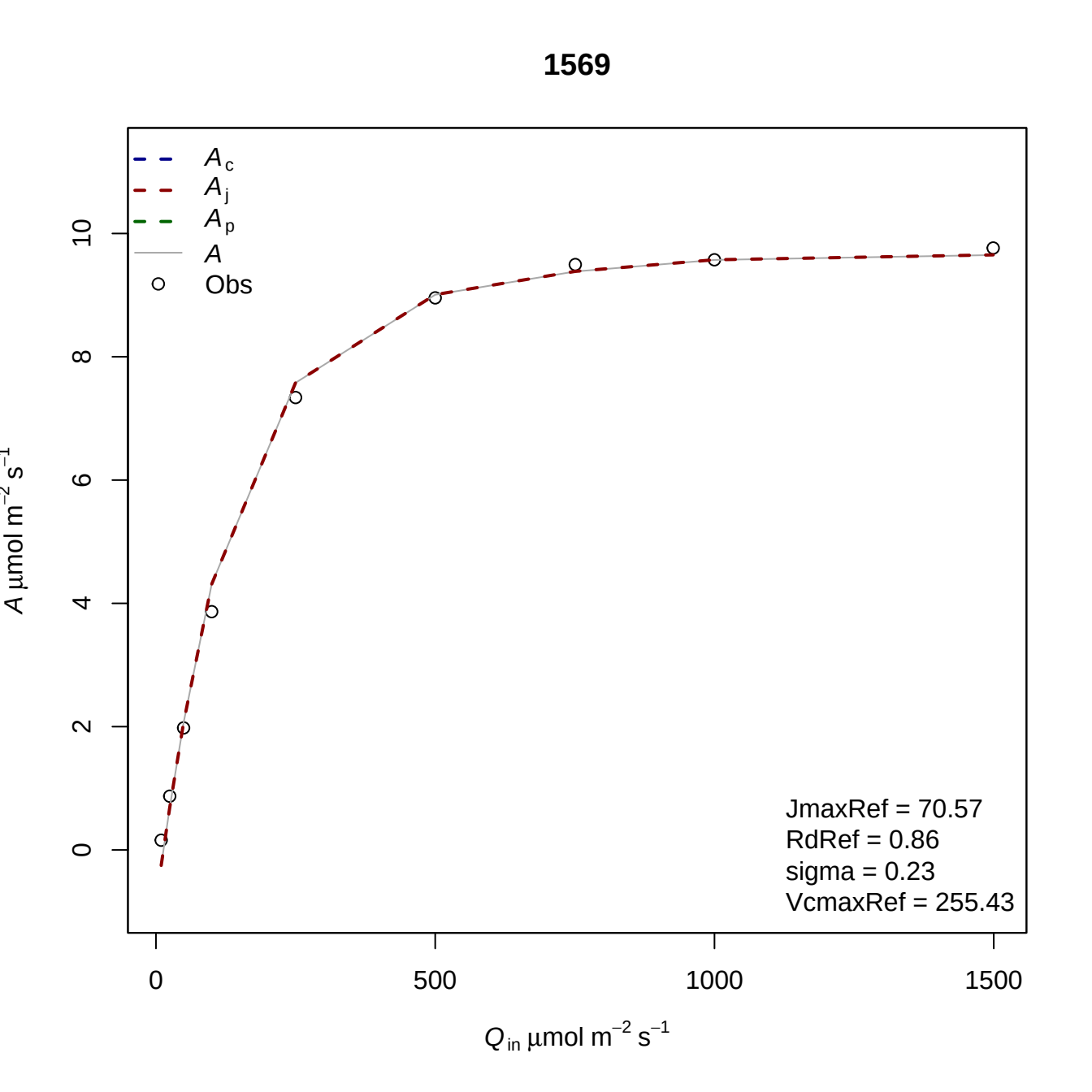

1564
A_c
A_j
A_p
A
Obs
JmaxRef = 38.22
RdRef = 1.25
sigma = 0.35
VcmaxRef = 94.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

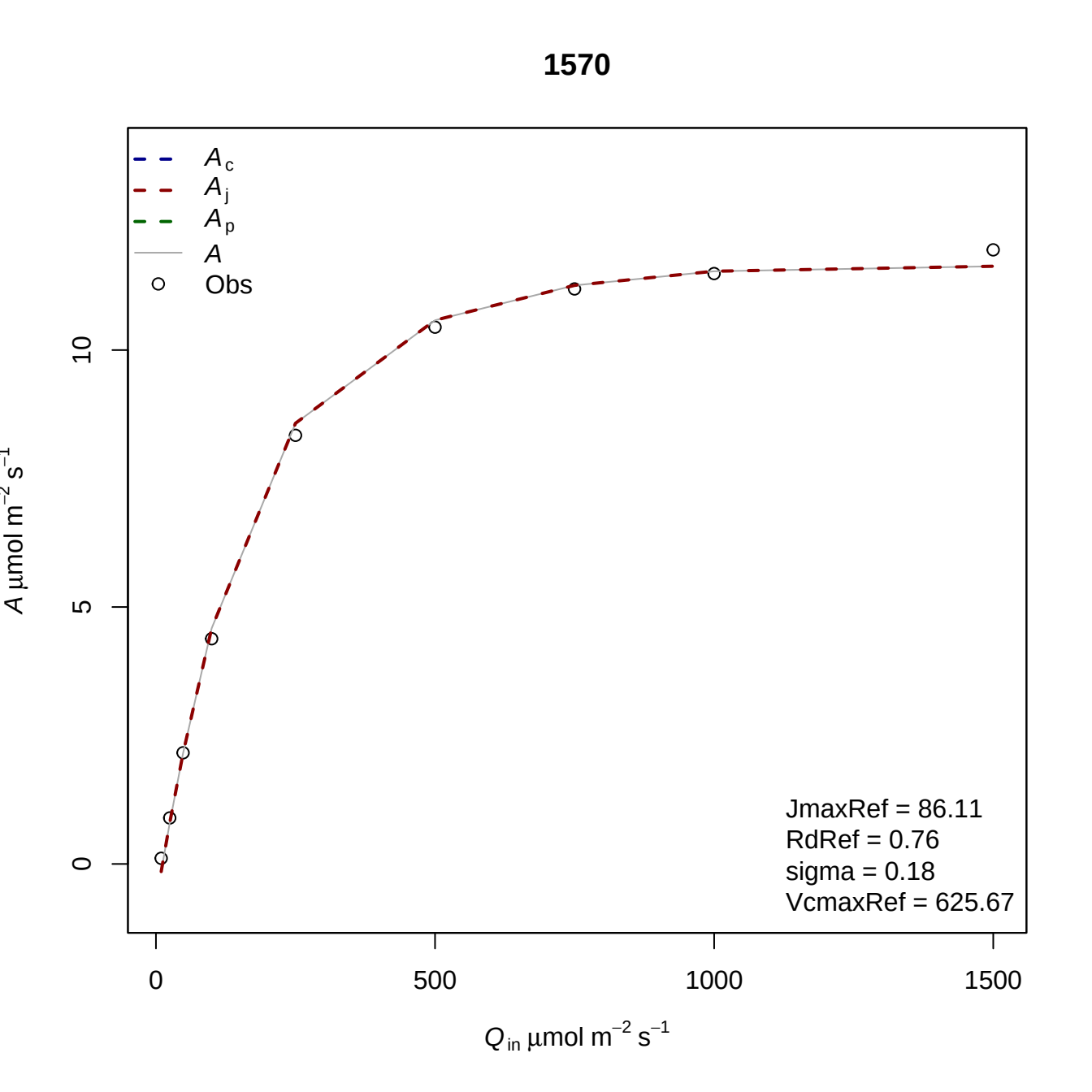

1565
A_c
A_j
A_p
A
Obs
JmaxRef = 30.73
RdRef = 1.34
sigma = 0.2
VcmaxRef = 231.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

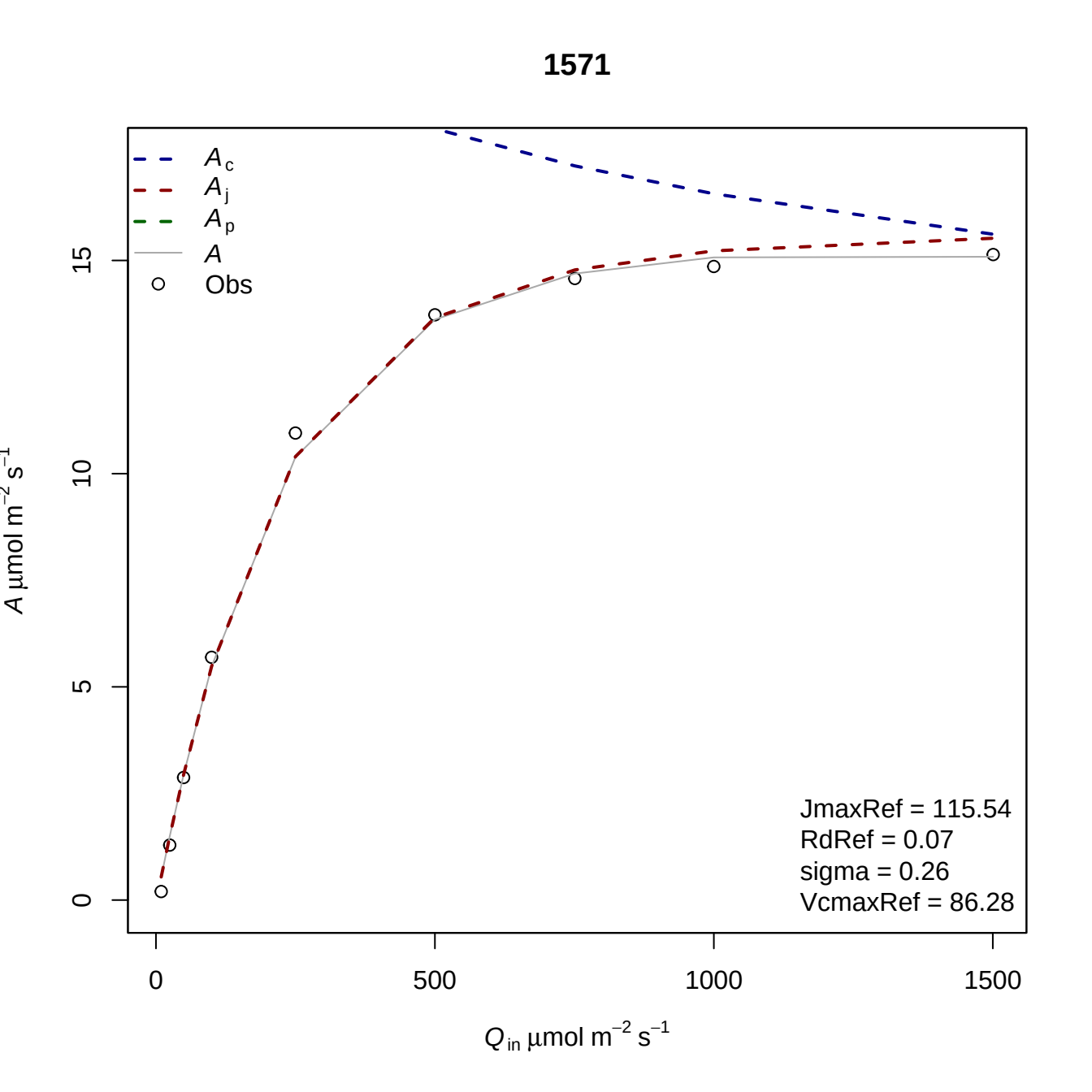

1566
A_c
A_j
A_p
A
Obs
JmaxRef = 73.26
RdRef = 0.93
sigma = 0.29
VcmaxRef = 361.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

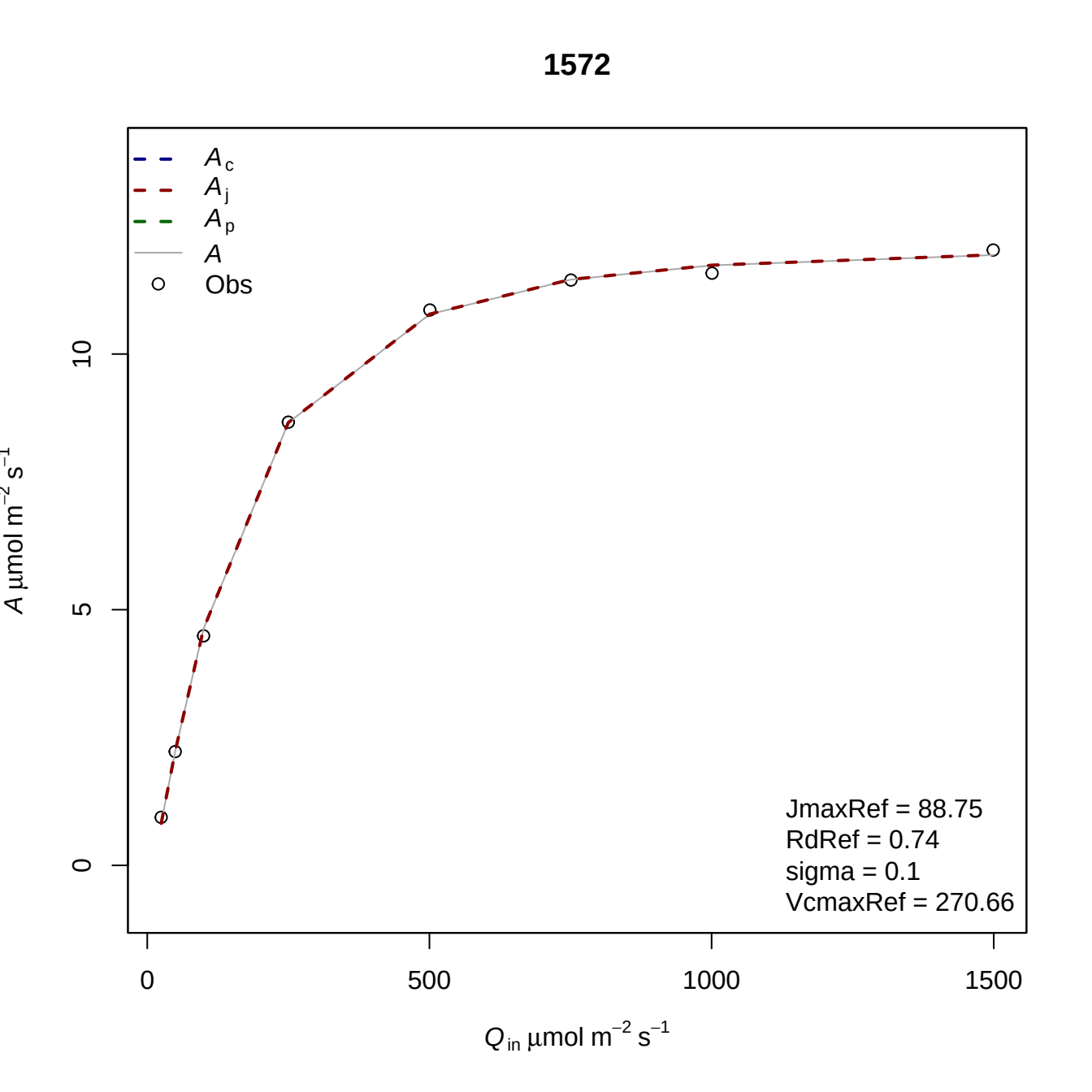

1567
A_c
A_j
A_p
A
Obs
JmaxRef = 83.44
RdRef = 1.06
sigma = 0.46
VcmaxRef = 433.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

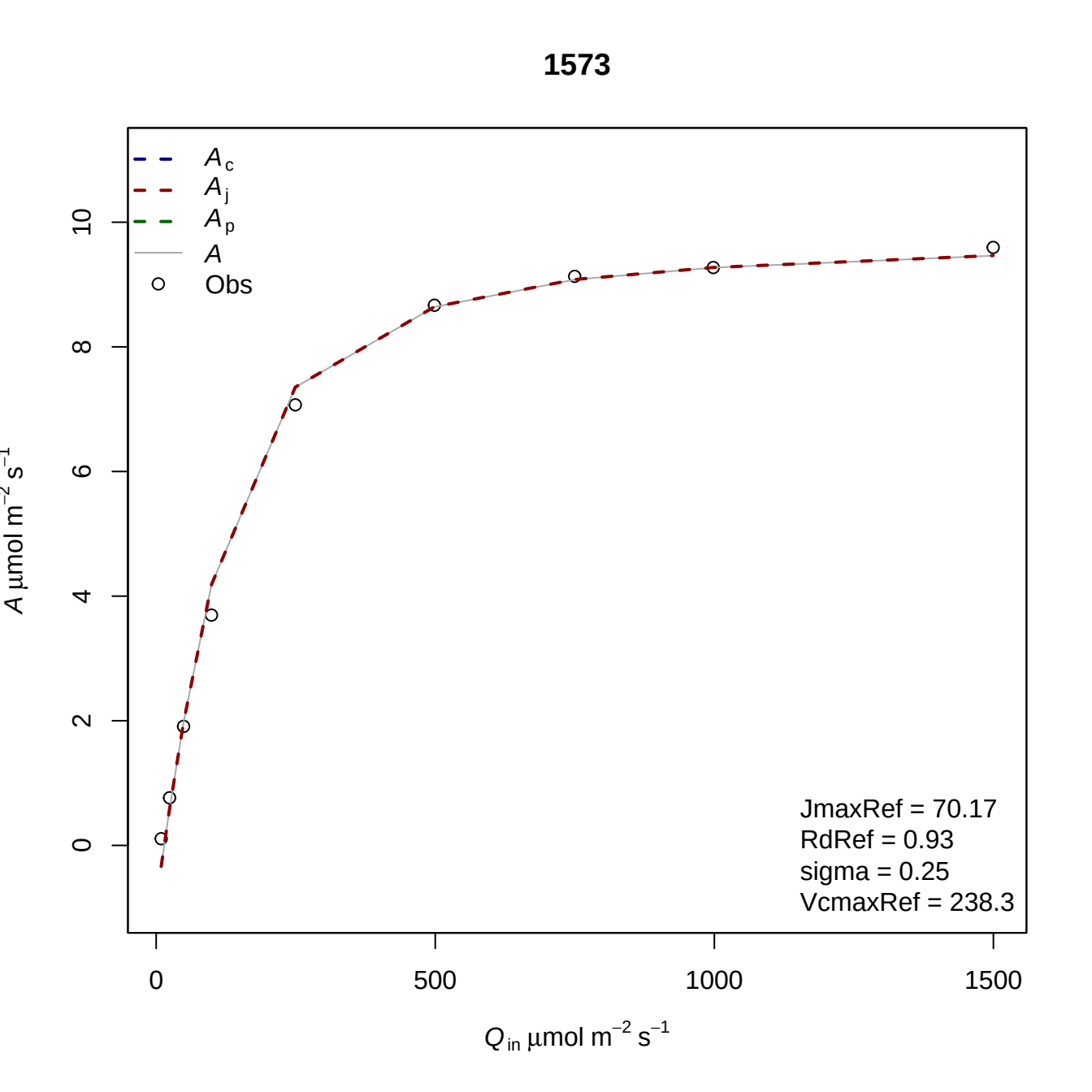

1568
A_c
A_j
A_p
A
Obs
JmaxRef = 104.79
RdRef = 1.21
sigma = 0.44
VcmaxRef = 351.18
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

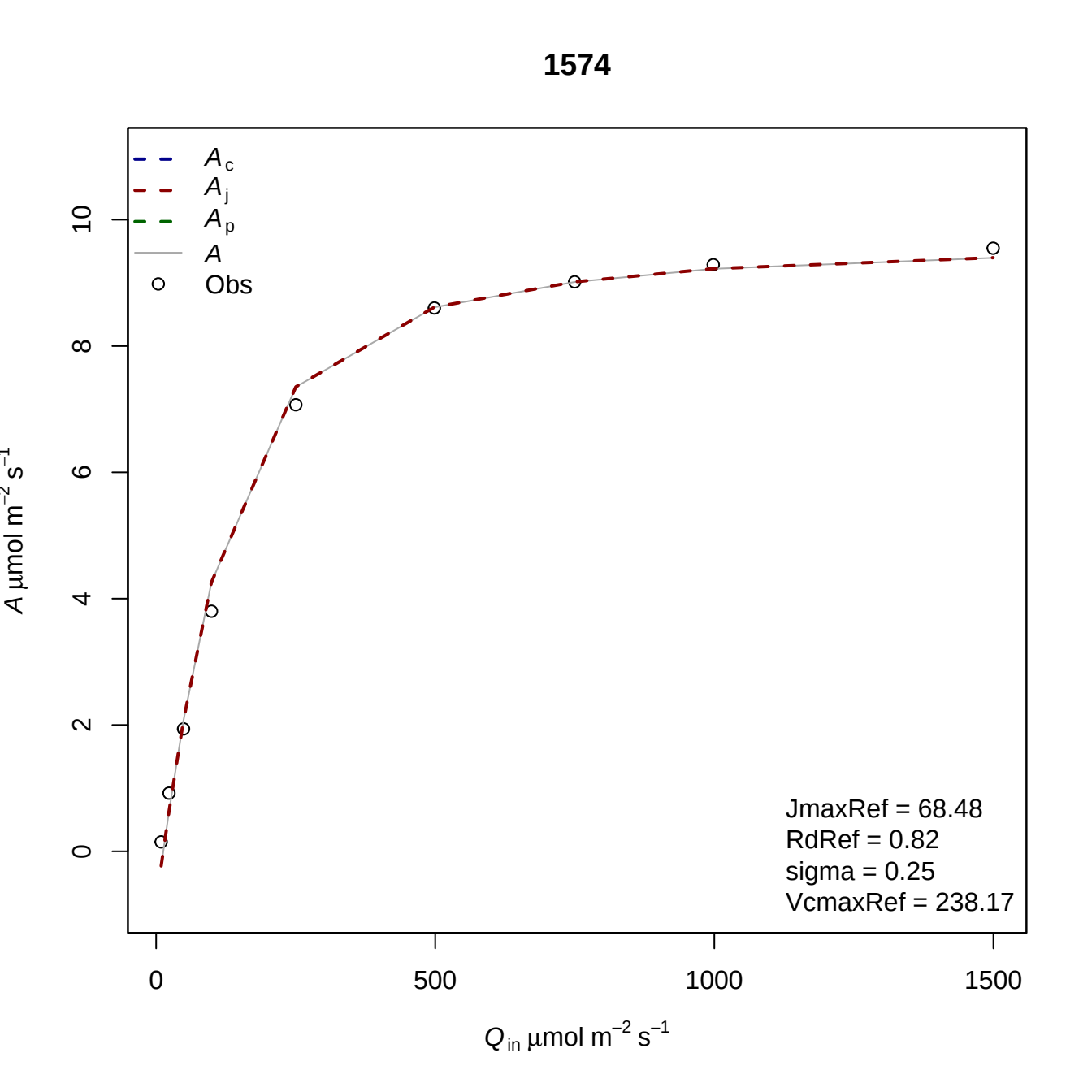

1569
A_c
A_j
A_p
A
Obs
JmaxRef = 70.57
RdRef = 0.86
sigma = 0.23
VcmaxRef = 255.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

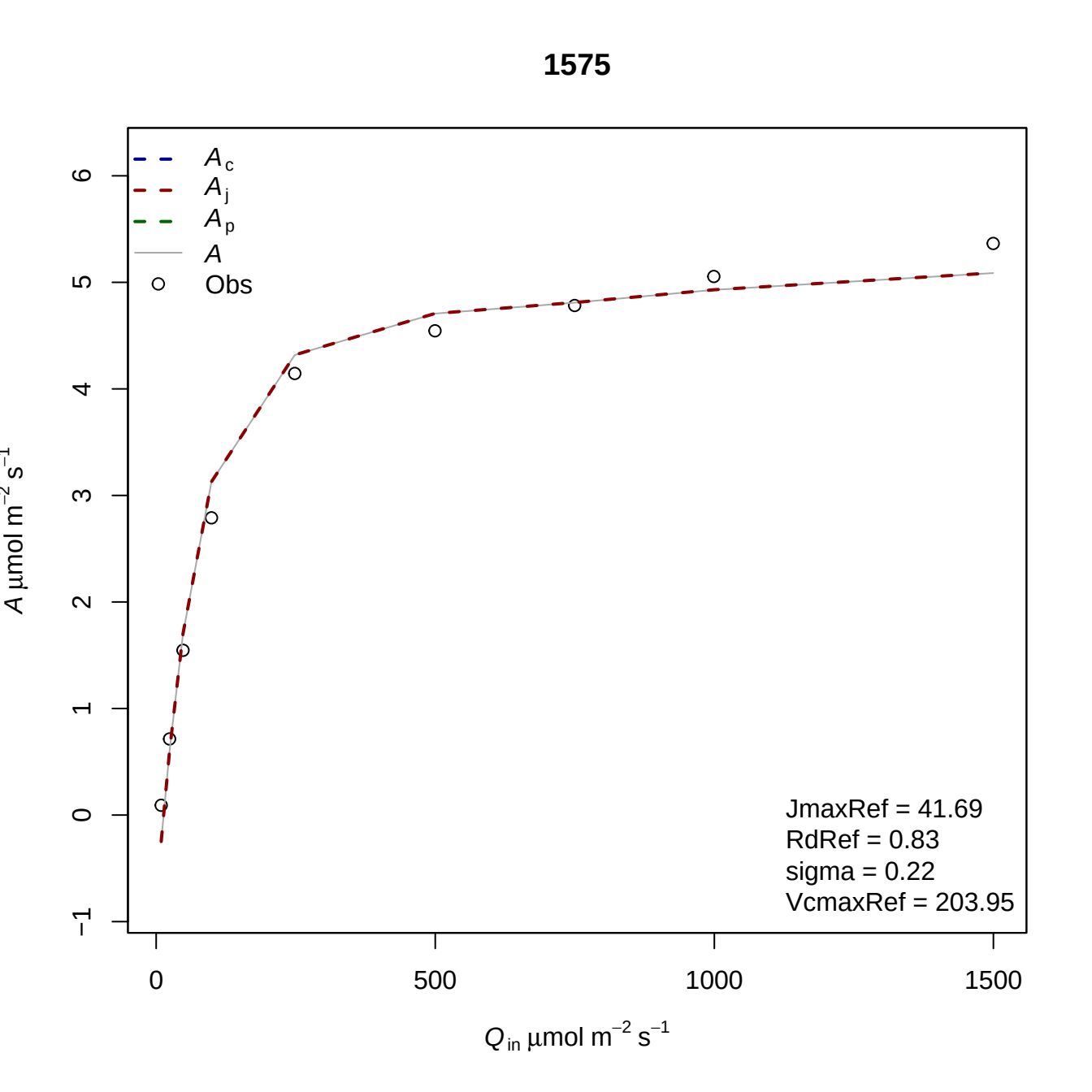

1570
A_c
A_j
A_p
A
Obs
JmaxRef = 86.11
RdRef = 0.76
sigma = 0.18
VcmaxRef = 625.67
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

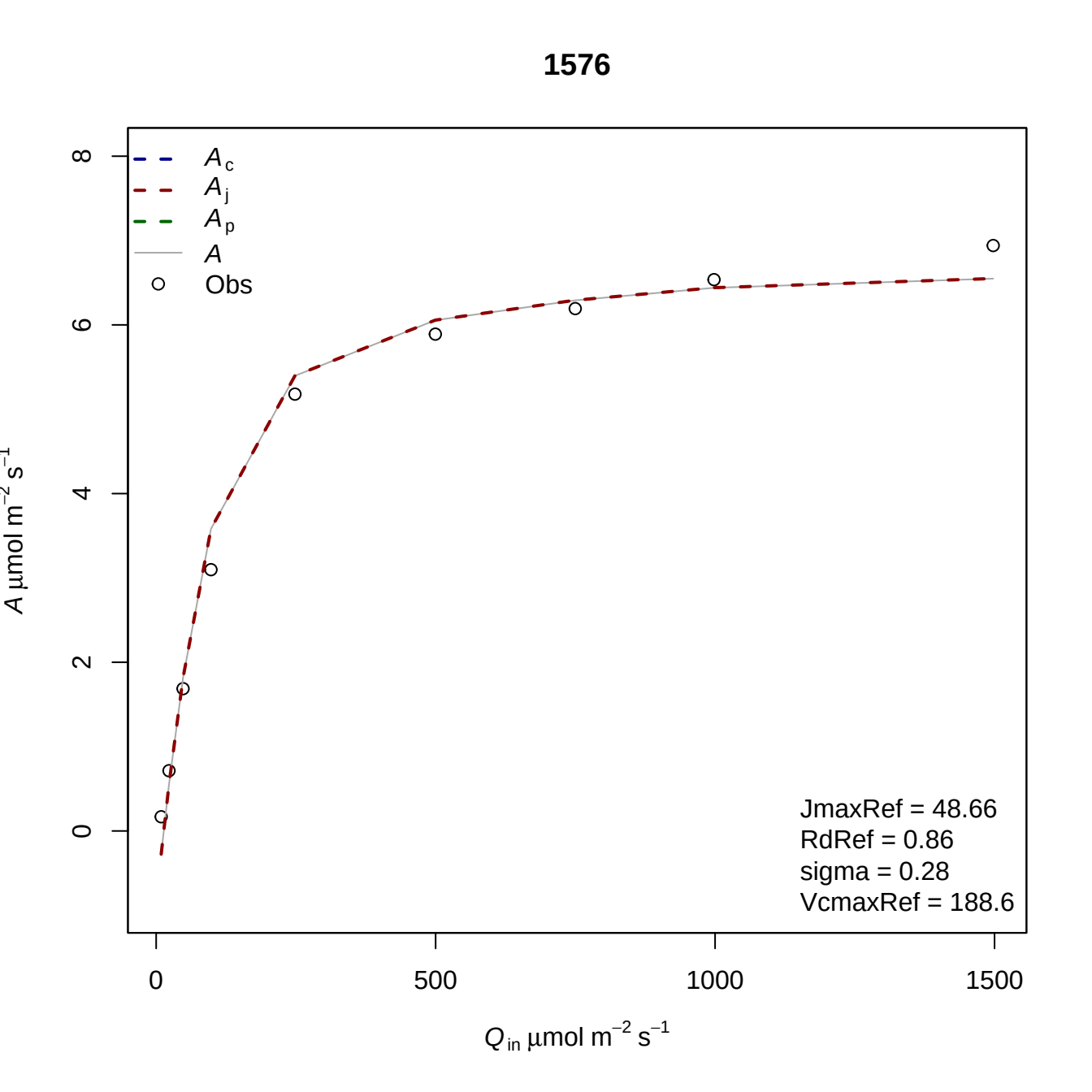

1571
A_c
A_j
A_p
A
Obs
JmaxRef = 115.54
RdRef = 0.07
sigma = 0.26
VcmaxRef = 86.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

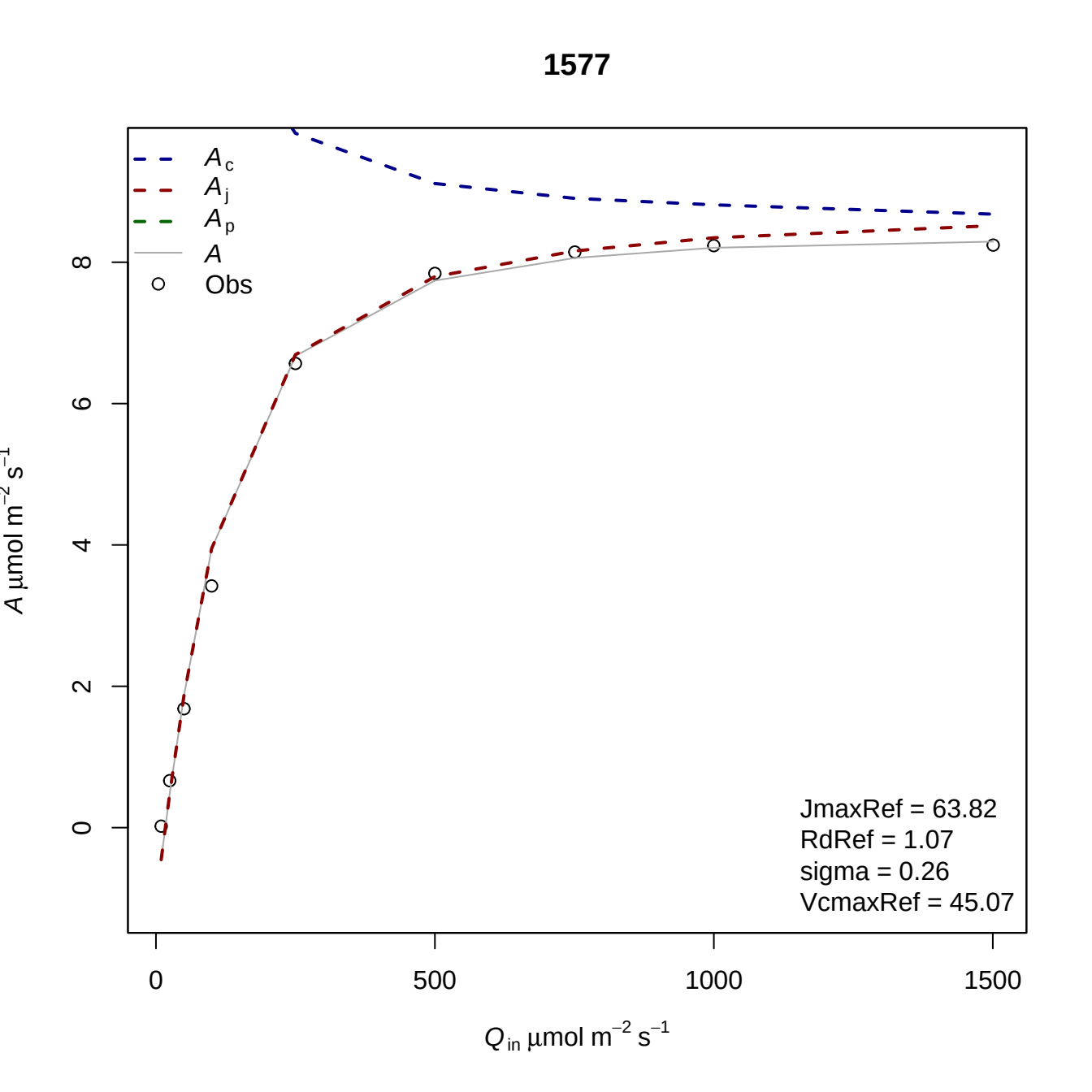

1572
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 88.75
RdRef = 0.74
sigma = 0.1
VcmaxRef = 270.66

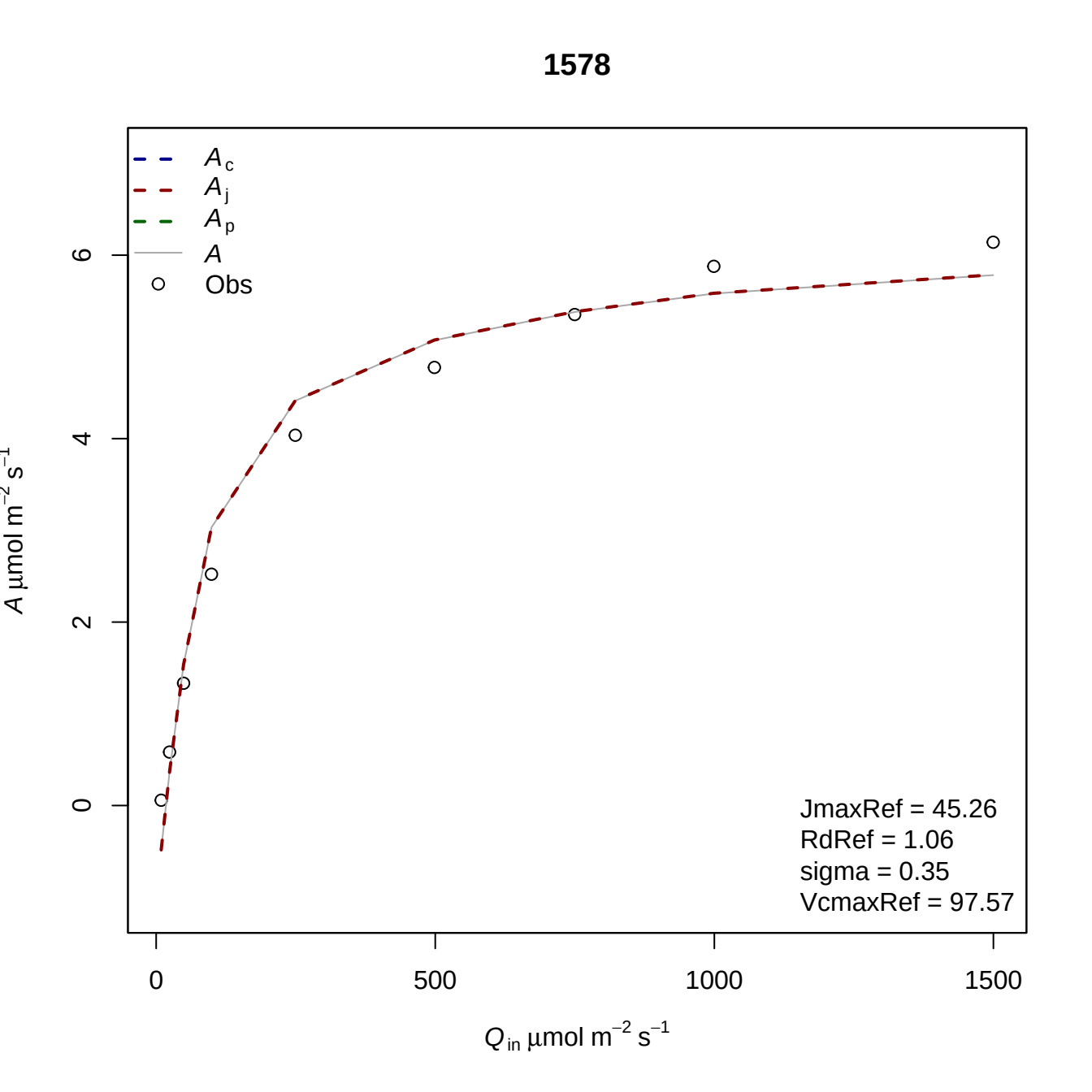

1573
A_c
A_j
A_p
A
Obs
JmaxRef = 70.17
RdRef = 0.93
sigma = 0.25
VcmaxRef = 238.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

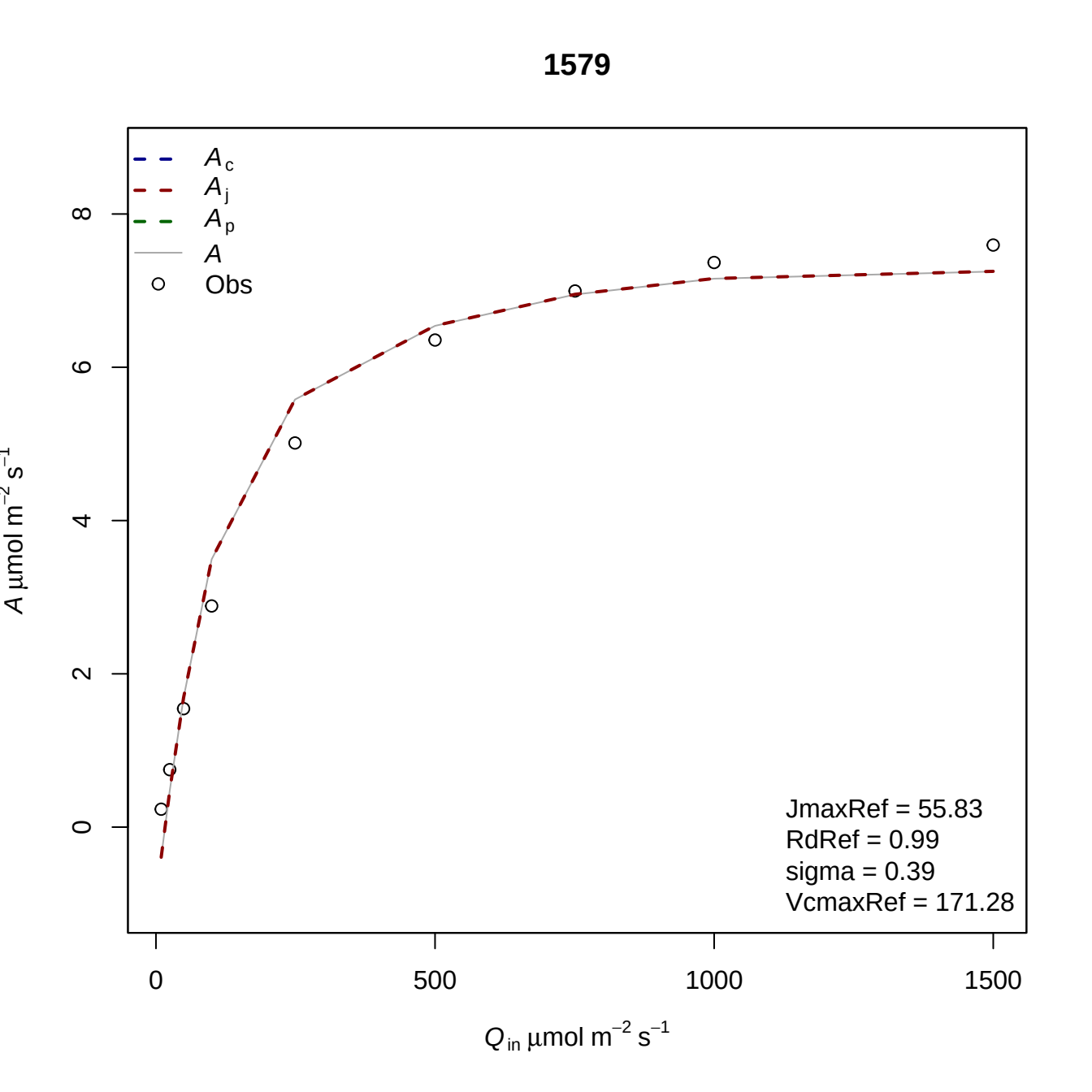

1574
A_c
A_j
A_p
A
Obs
JmaxRef = 68.48
RdRef = 0.82
sigma = 0.25
VcmaxRef = 238.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

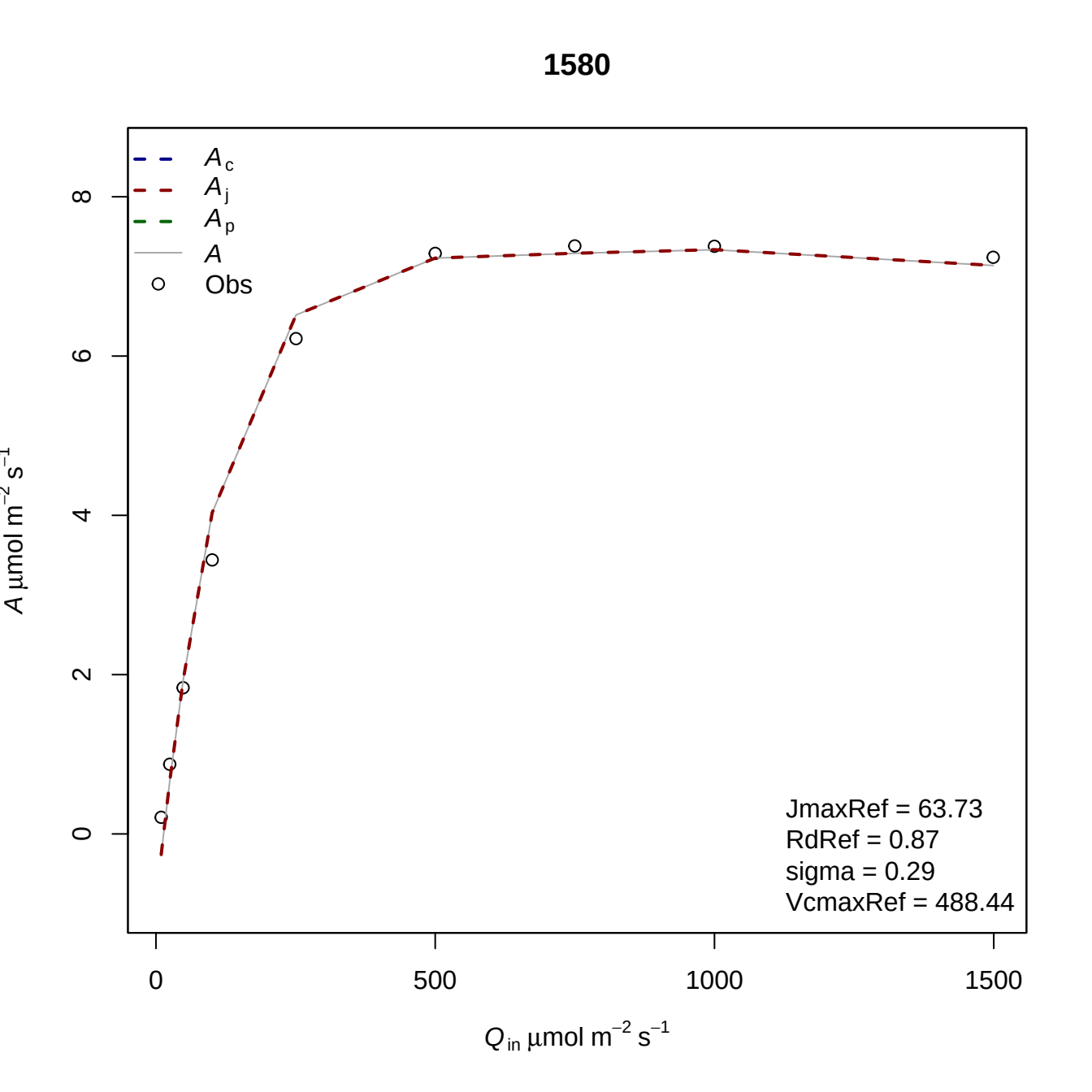

1575
A_c
A_j
A_p
A
Obs
JmaxRef = 41.69
RdRef = 0.83
sigma = 0.22
VcmaxRef = 203.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

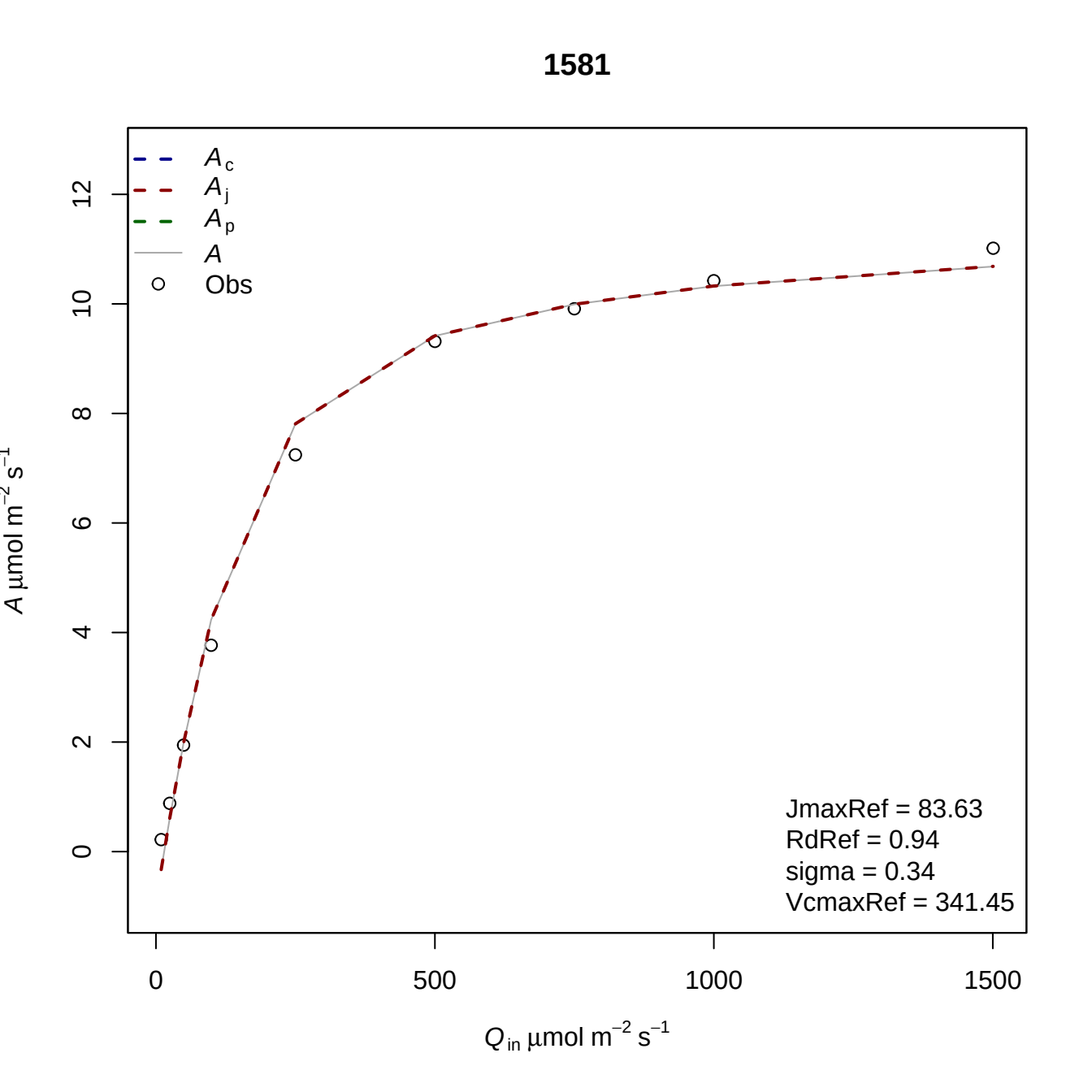

1576
A_c
A_j
A_p
A
Obs
JmaxRef = 48.66
RdRef = 0.86
sigma = 0.28
VcmaxRef = 188.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

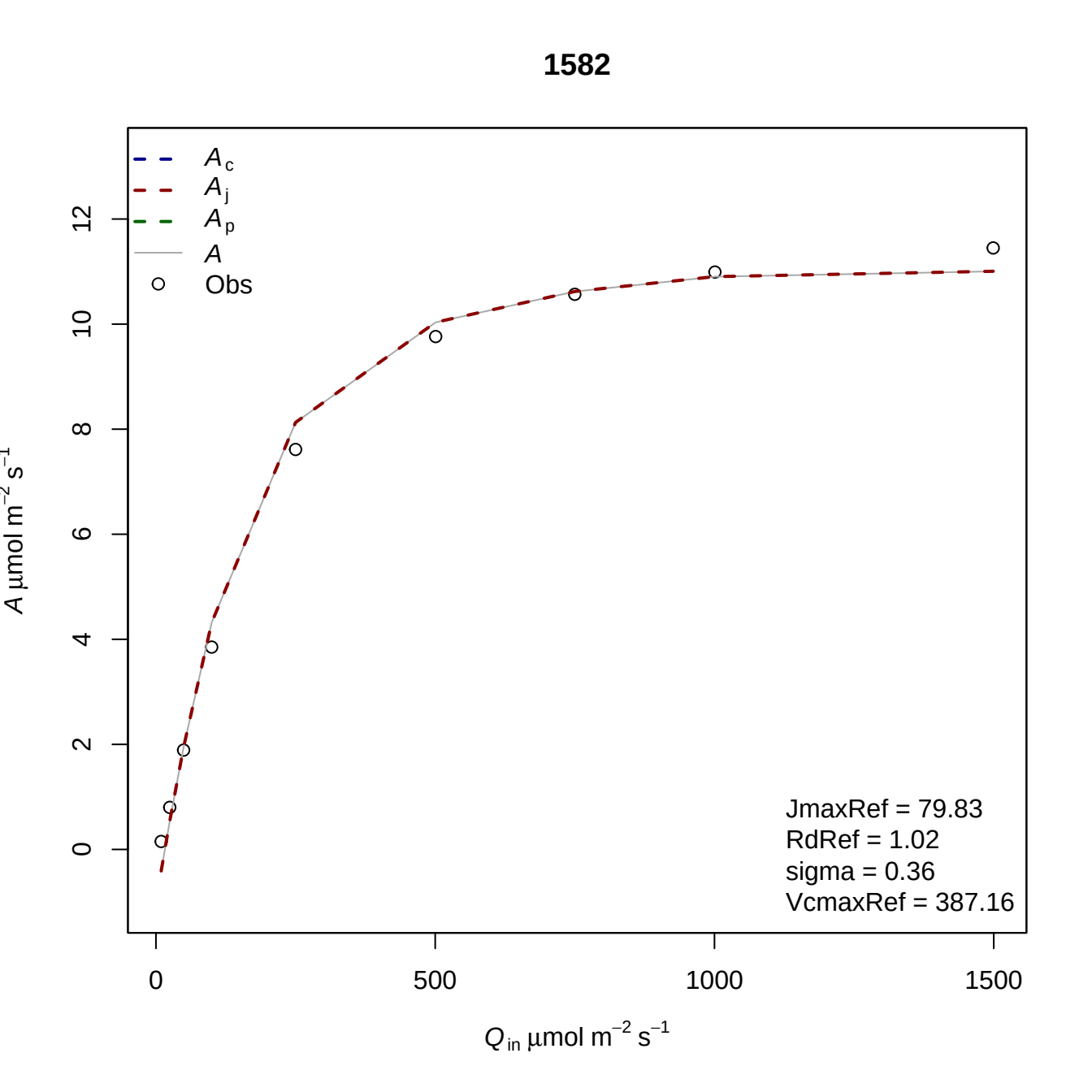

1577
A_c
A_j
A_p
A
Obs
JmaxRef = 63.82
RdRef = 1.07
sigma = 0.26
VcmaxRef = 45.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

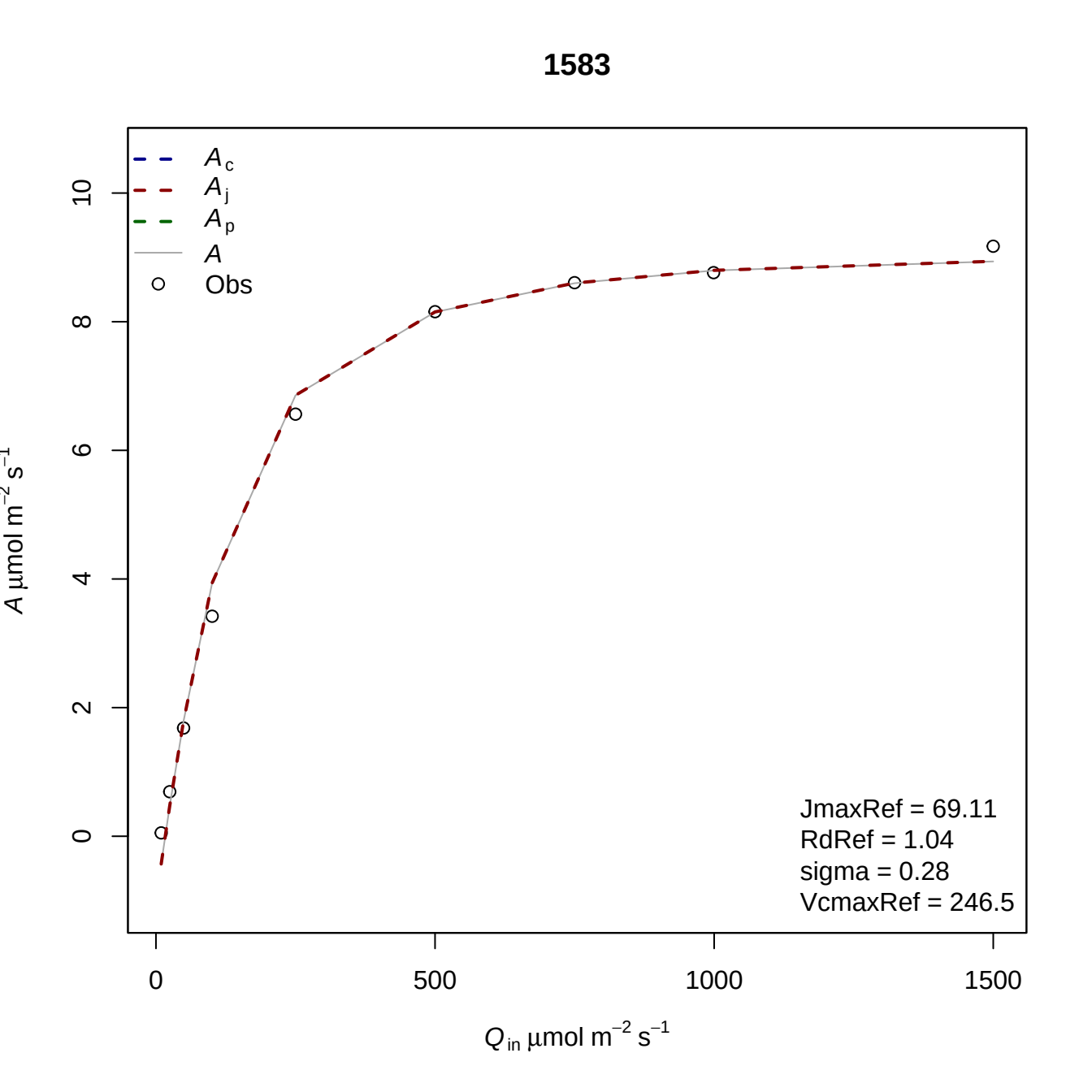

1578
A_c
A_j
A_p
A
Obs
JmaxRef = 45.26
RdRef = 1.06
sigma = 0.35
VcmaxRef = 97.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

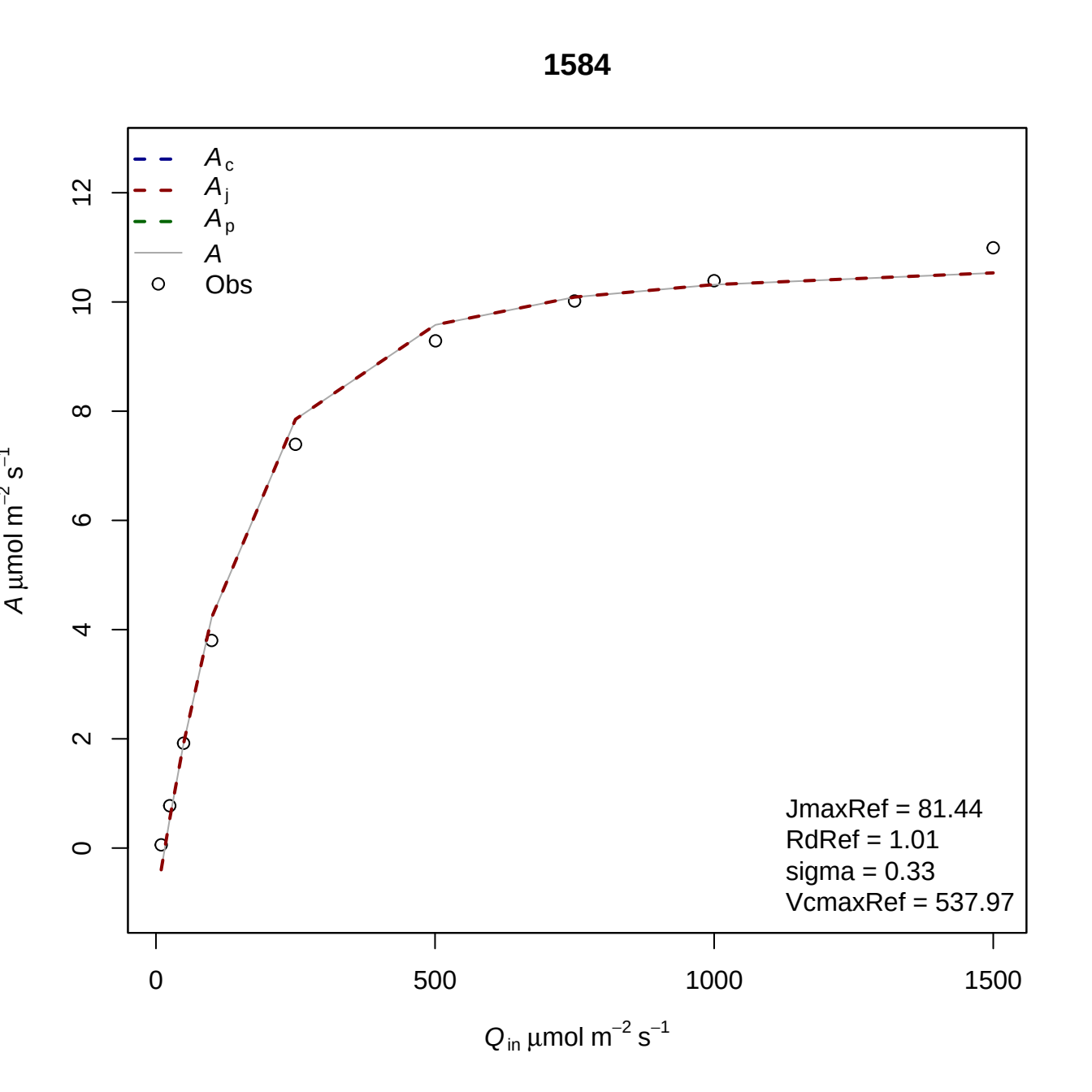

1579
A_c
A_j
A_p
A
Obs
JmaxRef = 55.83
RdRef = 0.99
sigma = 0.39
VcmaxRef = 171.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

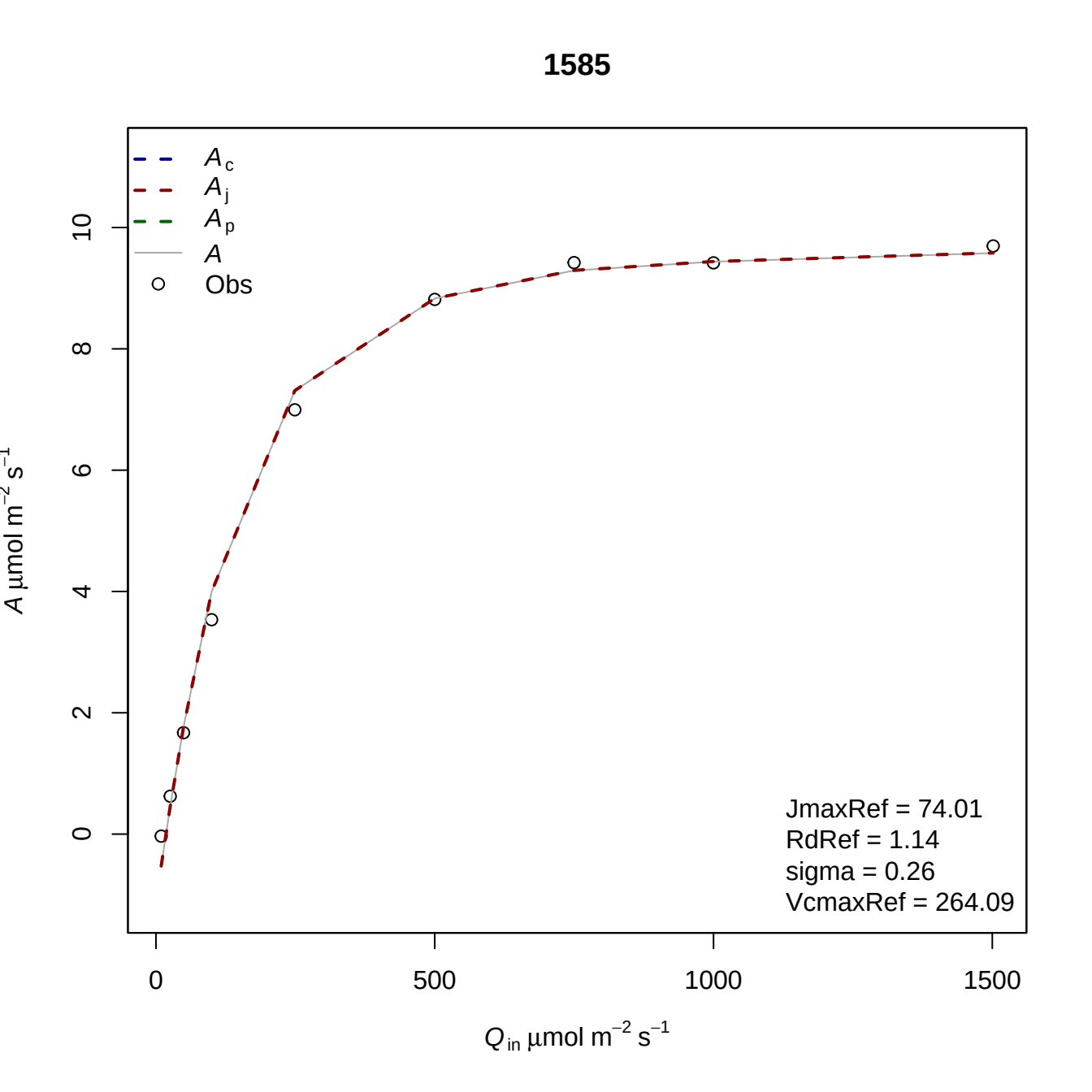

1580
A_c
A_j
A_p
A
Obs
JmaxRef = 63.73
RdRef = 0.87
sigma = 0.29
VcmaxRef = 488.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

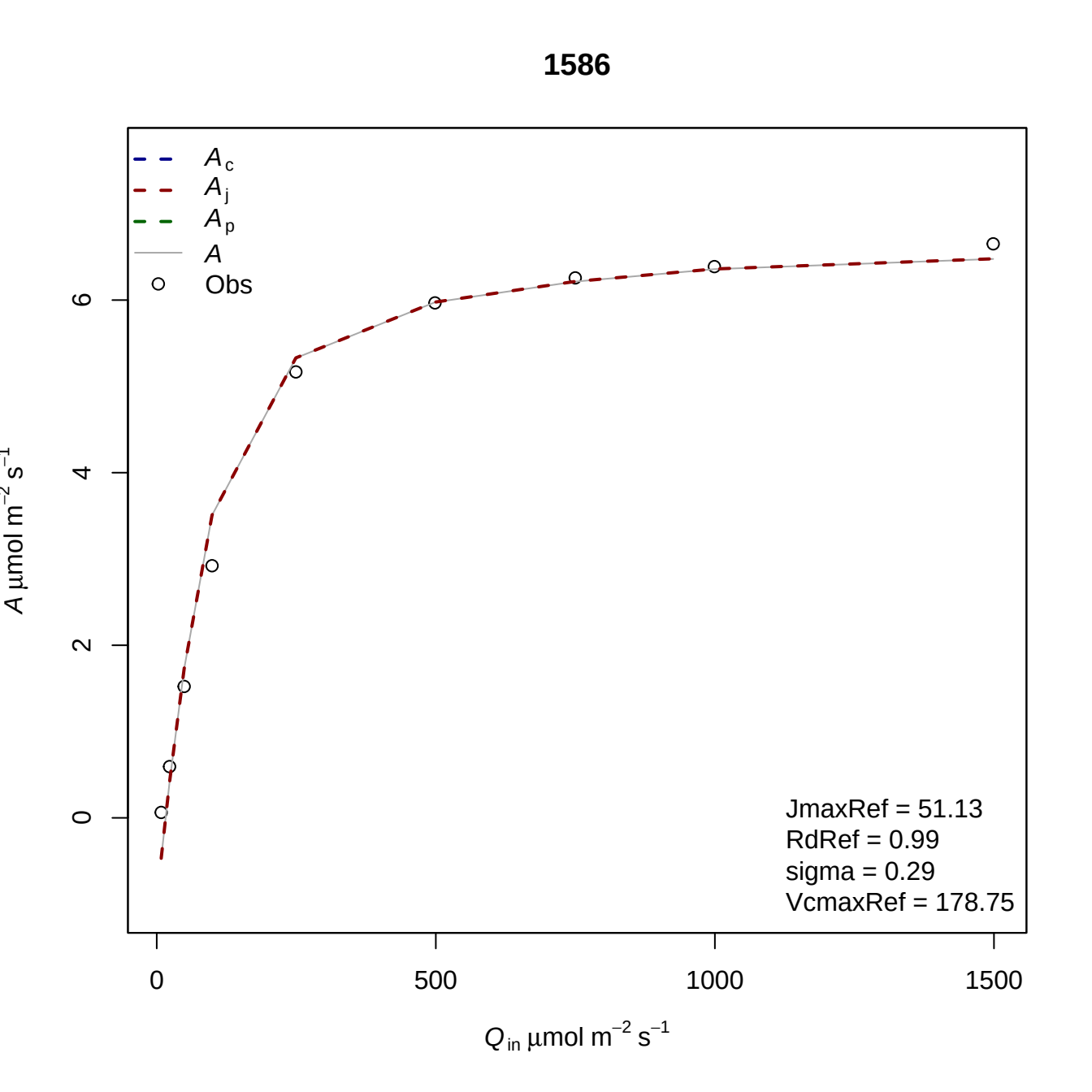

1581
A_c
A_j
A_p
A
Obs
JmaxRef = 83.63
RdRef = 0.94
sigma = 0.34
VcmaxRef = 341.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

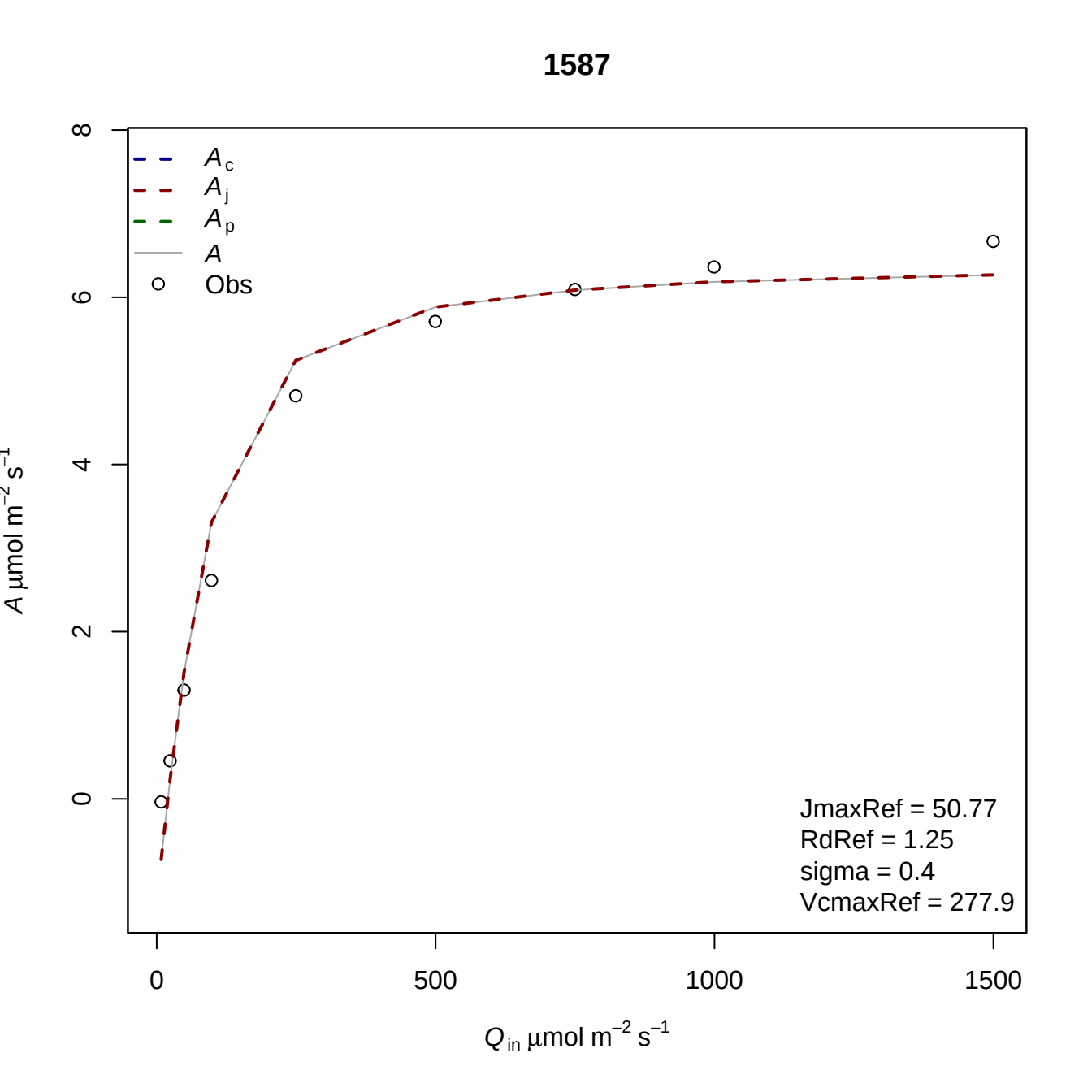

1582
A_c
A_j
A_p
A
Obs
JmaxRef = 79.83
RdRef = 1.02
sigma = 0.36
VcmaxRef = 387.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

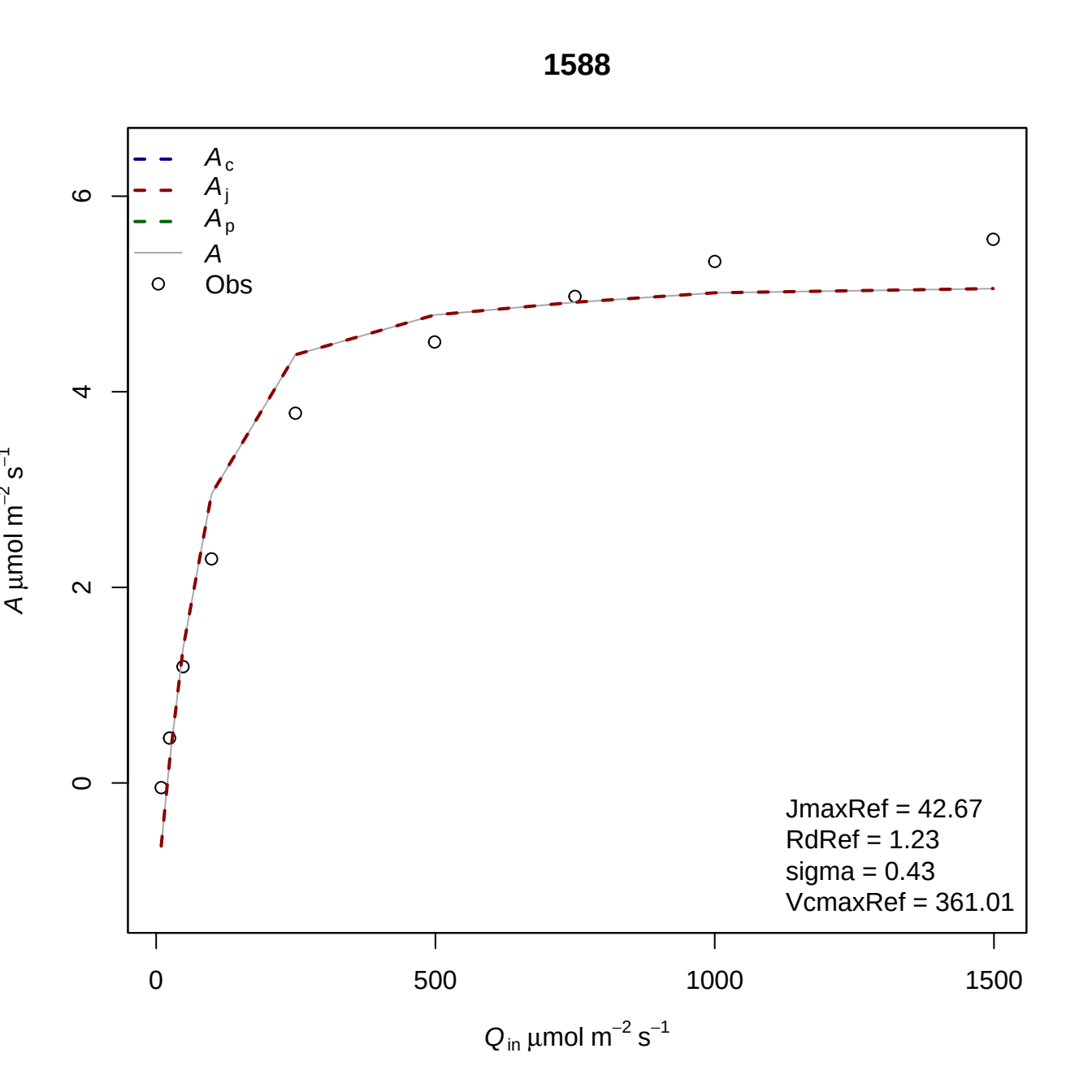

1583
A_c
A_j
A_p
A
Obs
JmaxRef = 69.11
RdRef = 1.04
sigma = 0.28
VcmaxRef = 246.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

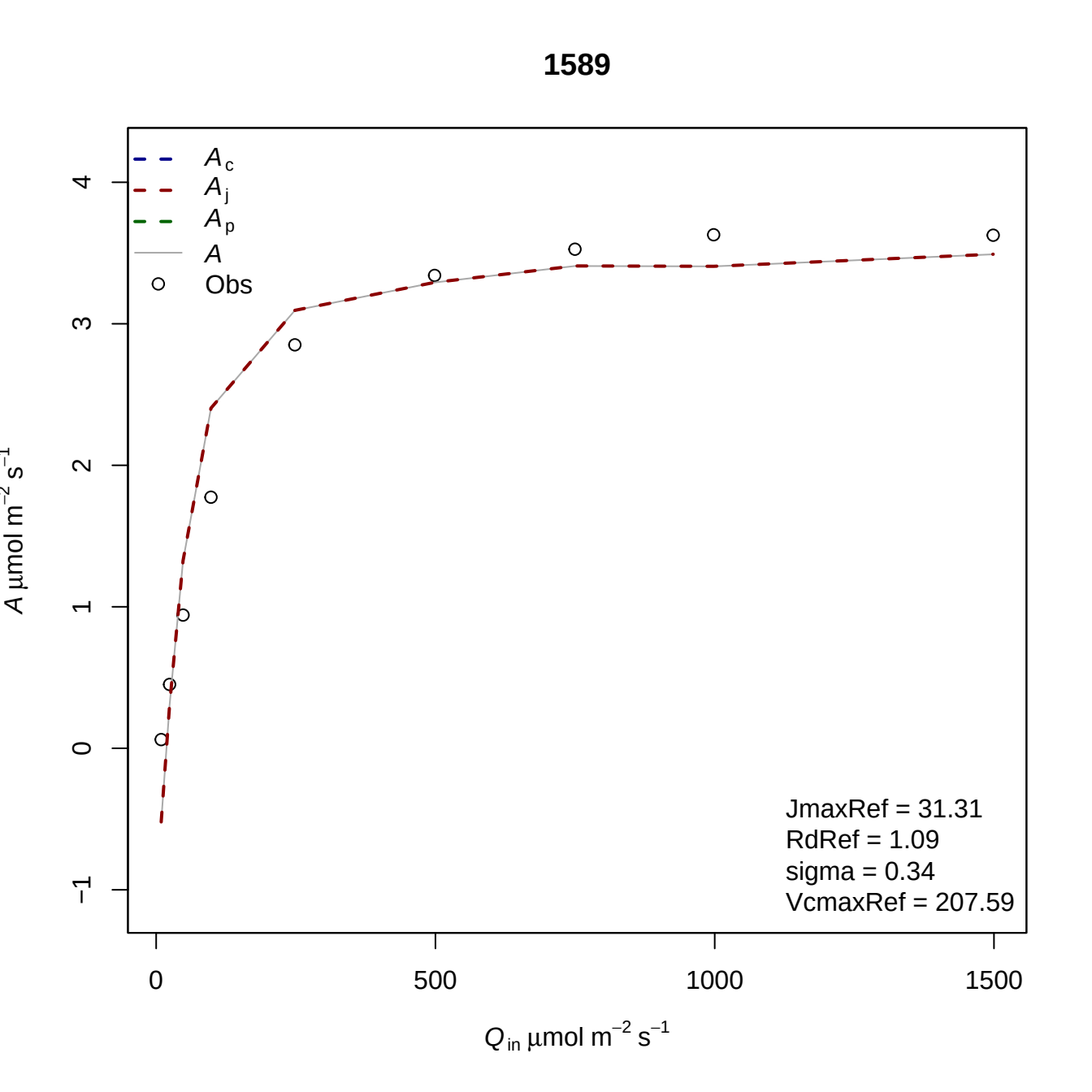

1584
A_c
A_j
A_p
A
Obs
JmaxRef = 81.44
RdRef = 1.01
sigma = 0.33
VcmaxRef = 537.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

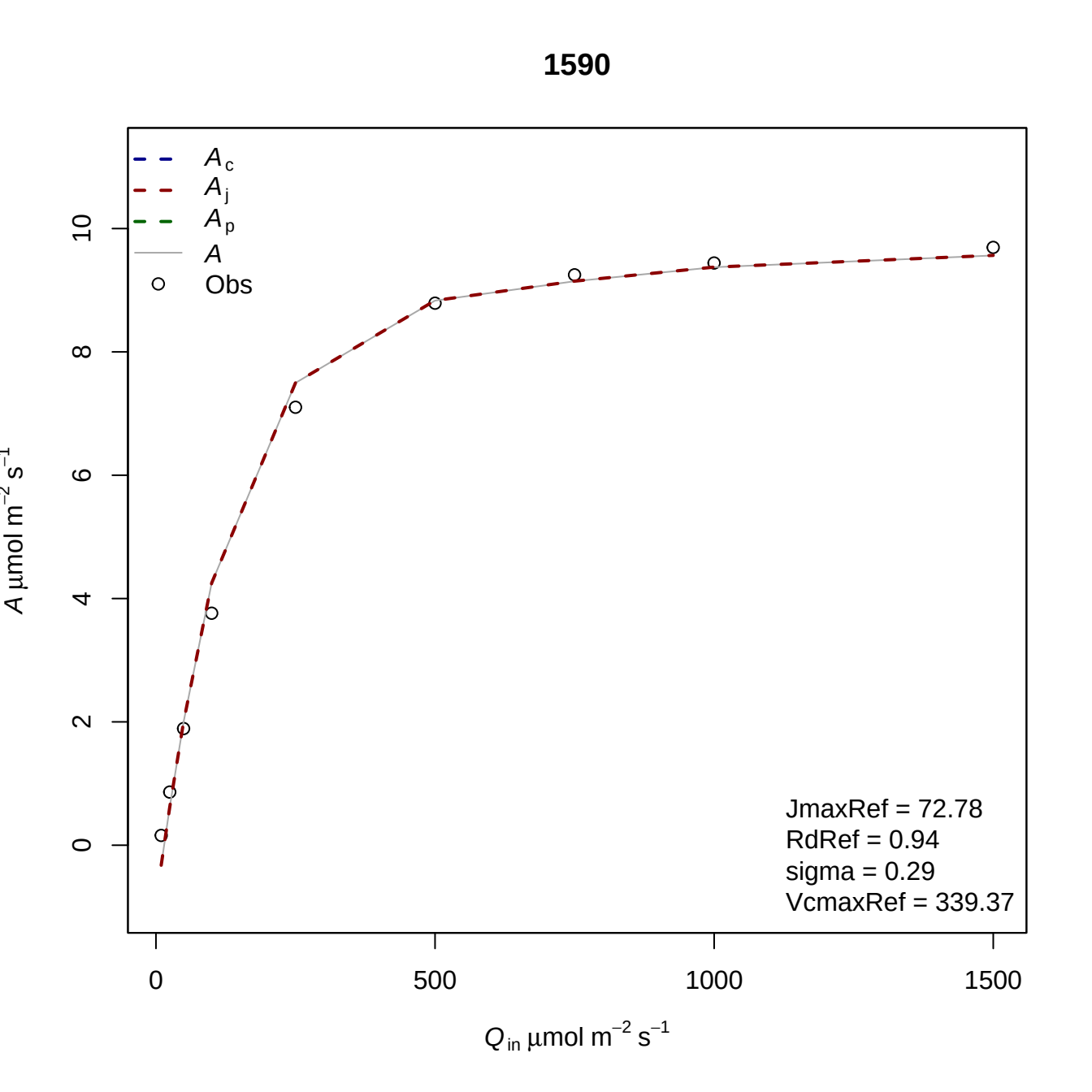

1585
A_c
A_j
A_p
A
Obs
JmaxRef = 74.01
RdRef = 1.14
sigma = 0.26
VcmaxRef = 264.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

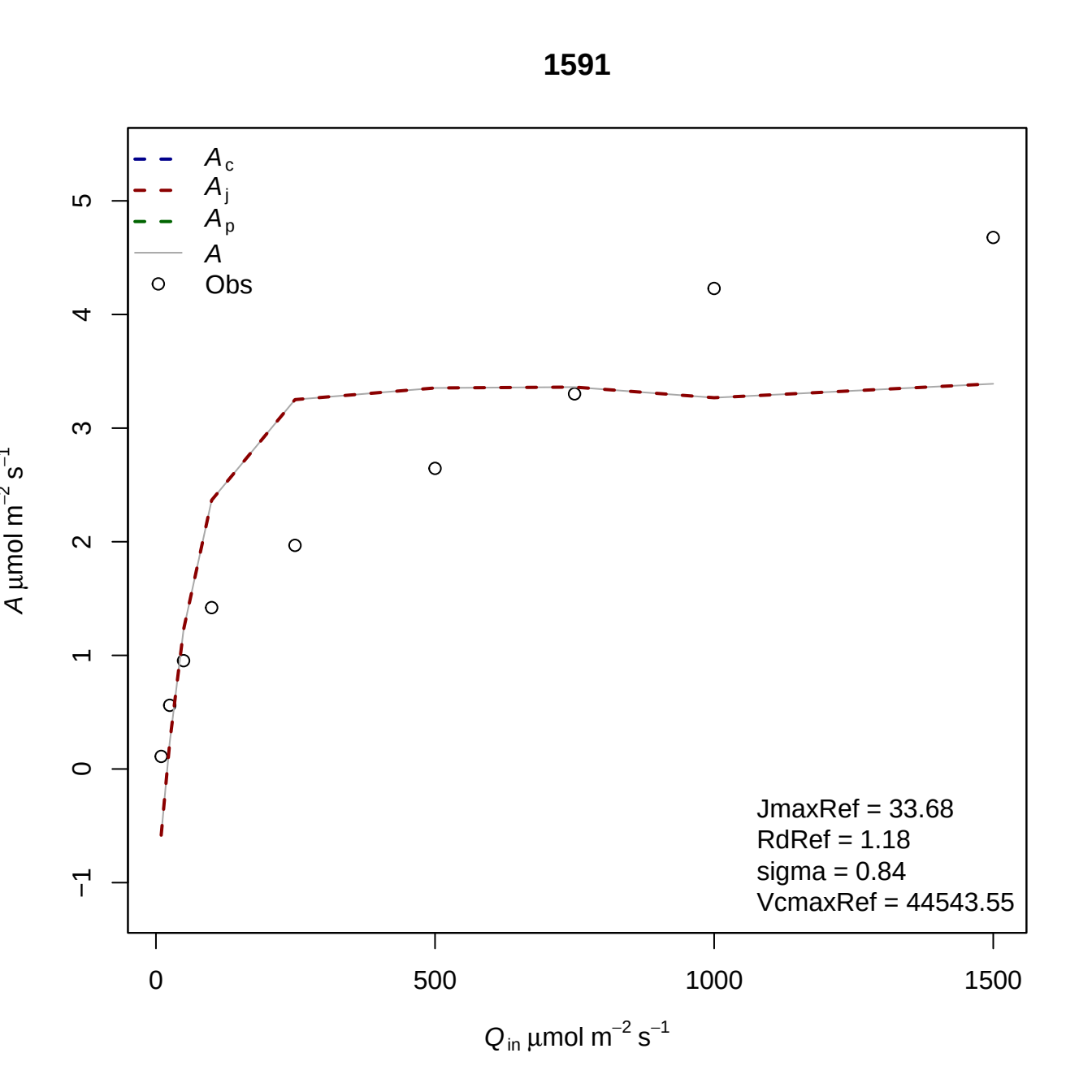

1586
A_c
A_j
A_p
A
Obs
JmaxRef = 51.13
RdRef = 0.99
sigma = 0.29
VcmaxRef = 178.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

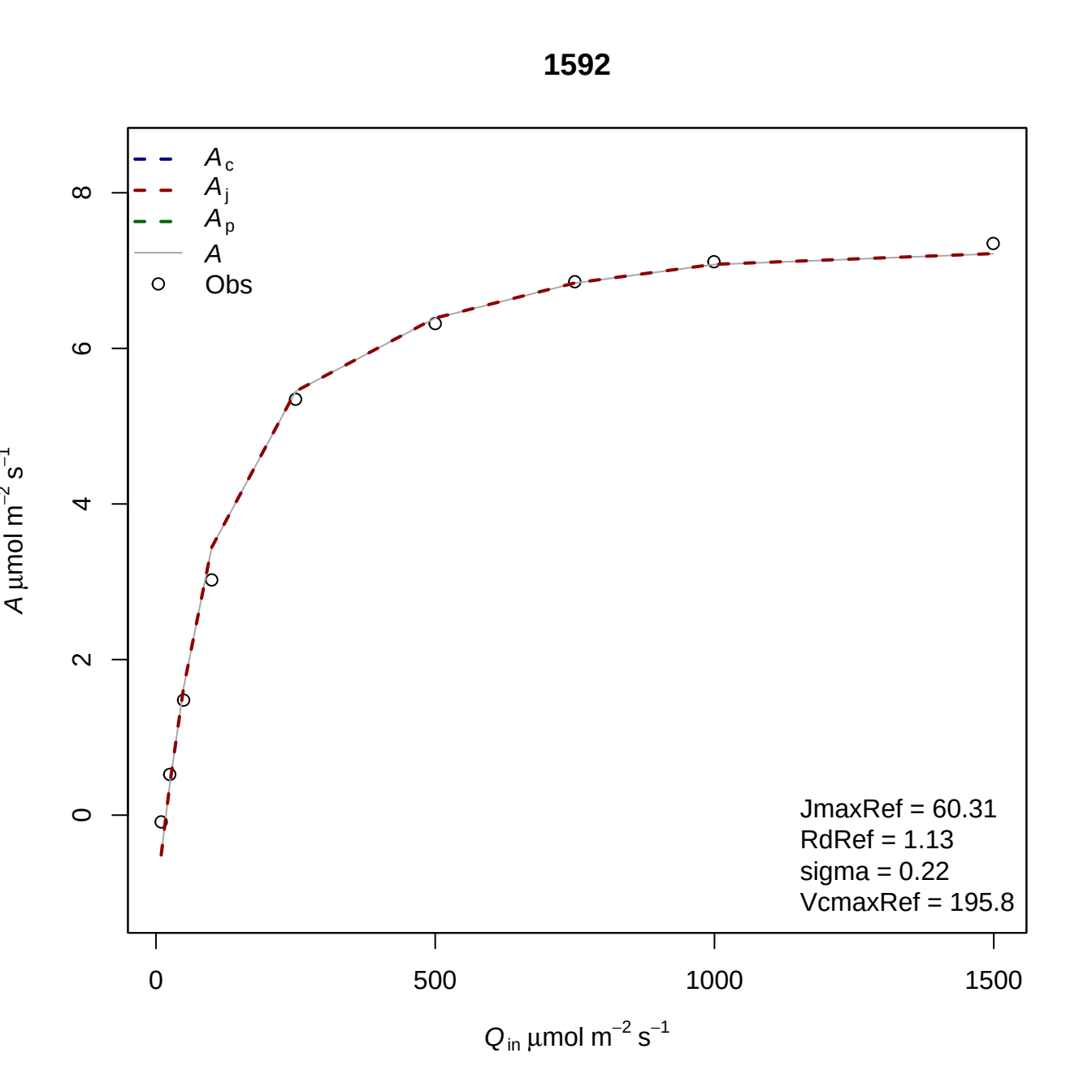

1587
A_c
A_j
A_p
A
Obs
JmaxRef = 50.77
RdRef = 1.25
sigma = 0.4
VcmaxRef = 277.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

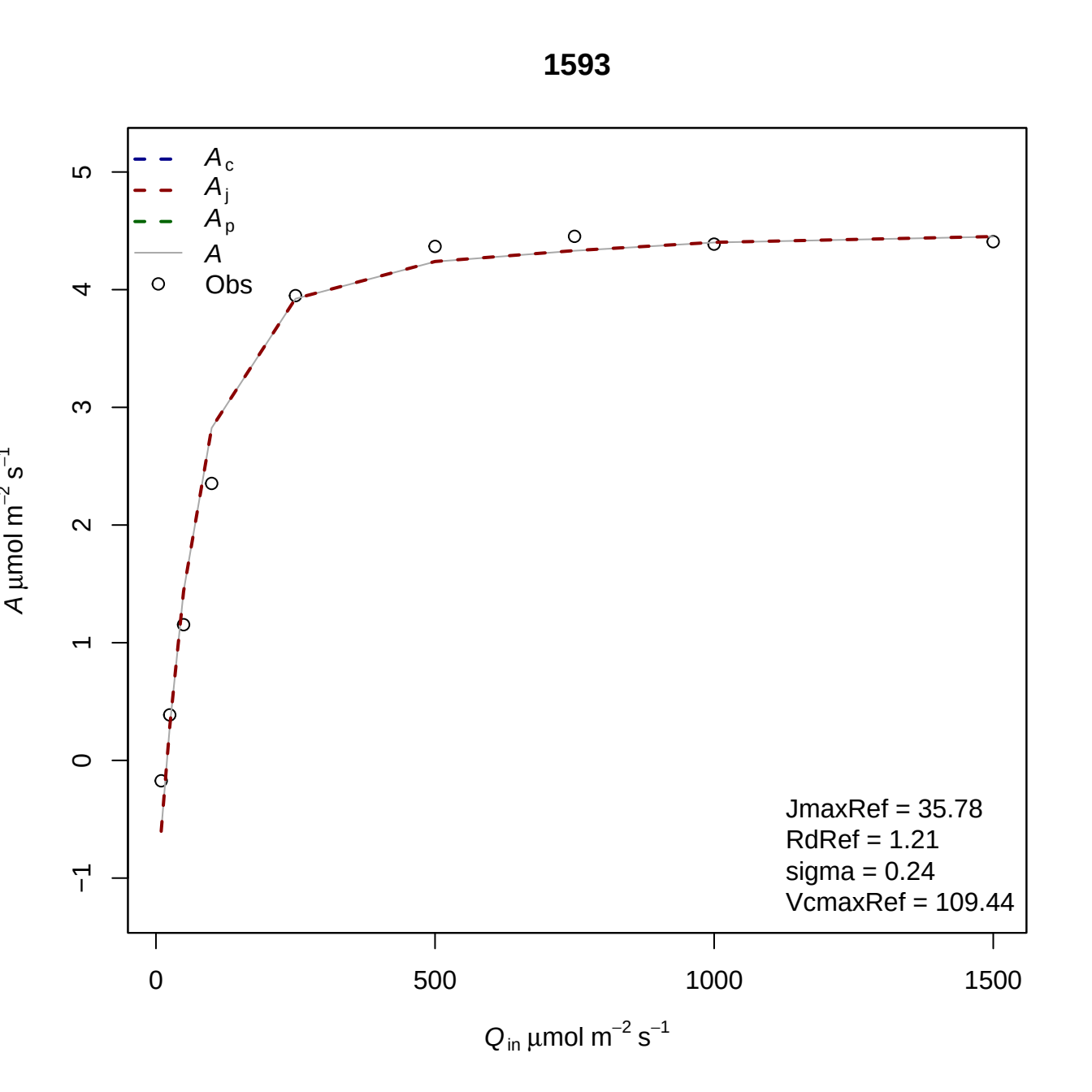

1588
A_c
A_j
A_p
A
Obs
JmaxRef = 42.67
RdRef = 1.23
sigma = 0.43
VcmaxRef = 361.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

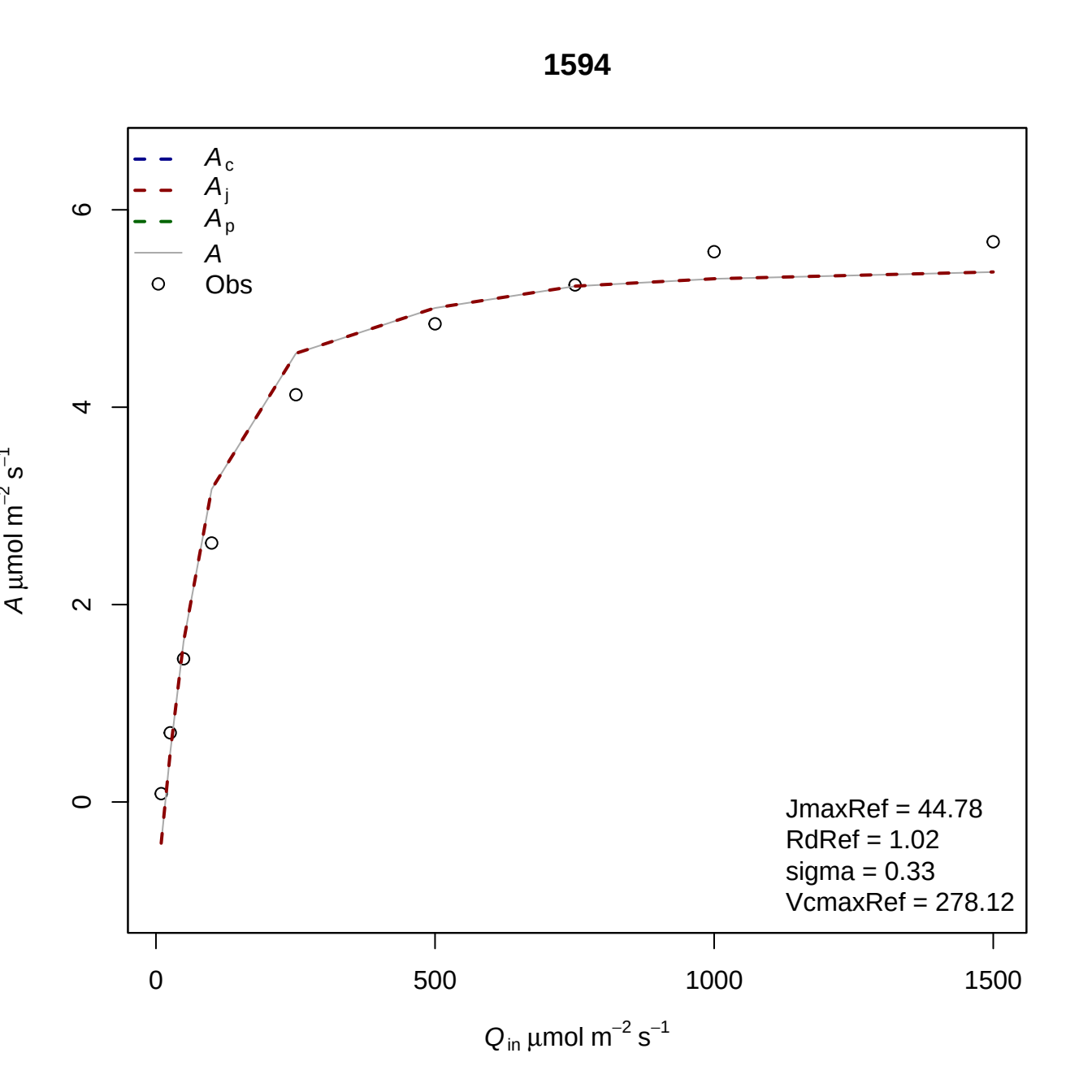

1589
A_c
A_j
A_p
A
Obs
JmaxRef = 31.31
RdRef = 1.09
sigma = 0.34
VcmaxRef = 207.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

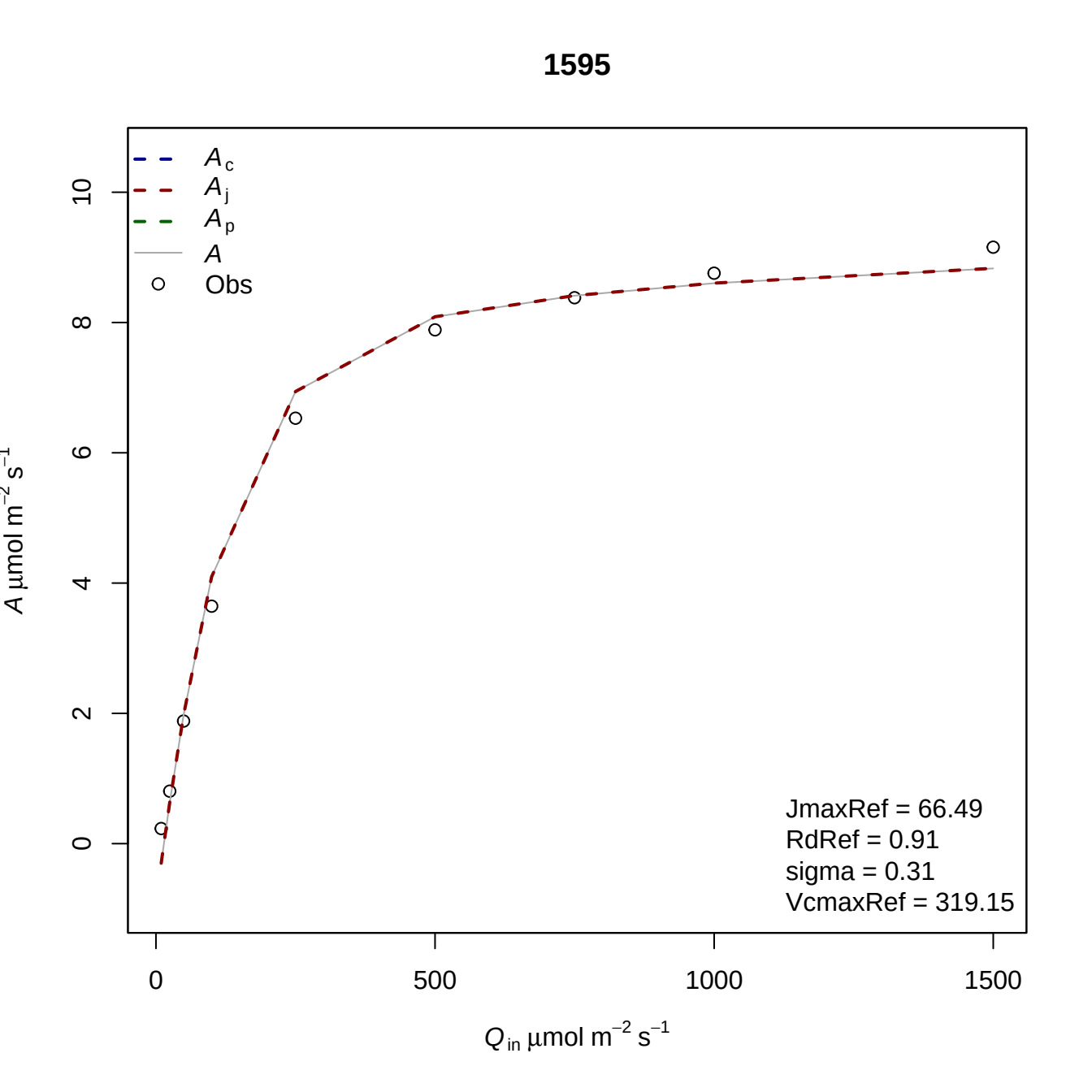

1590
A_c
A_j
A_p
A
Obs
JmaxRef = 72.78
RdRef = 0.94
sigma = 0.29
VcmaxRef = 339.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

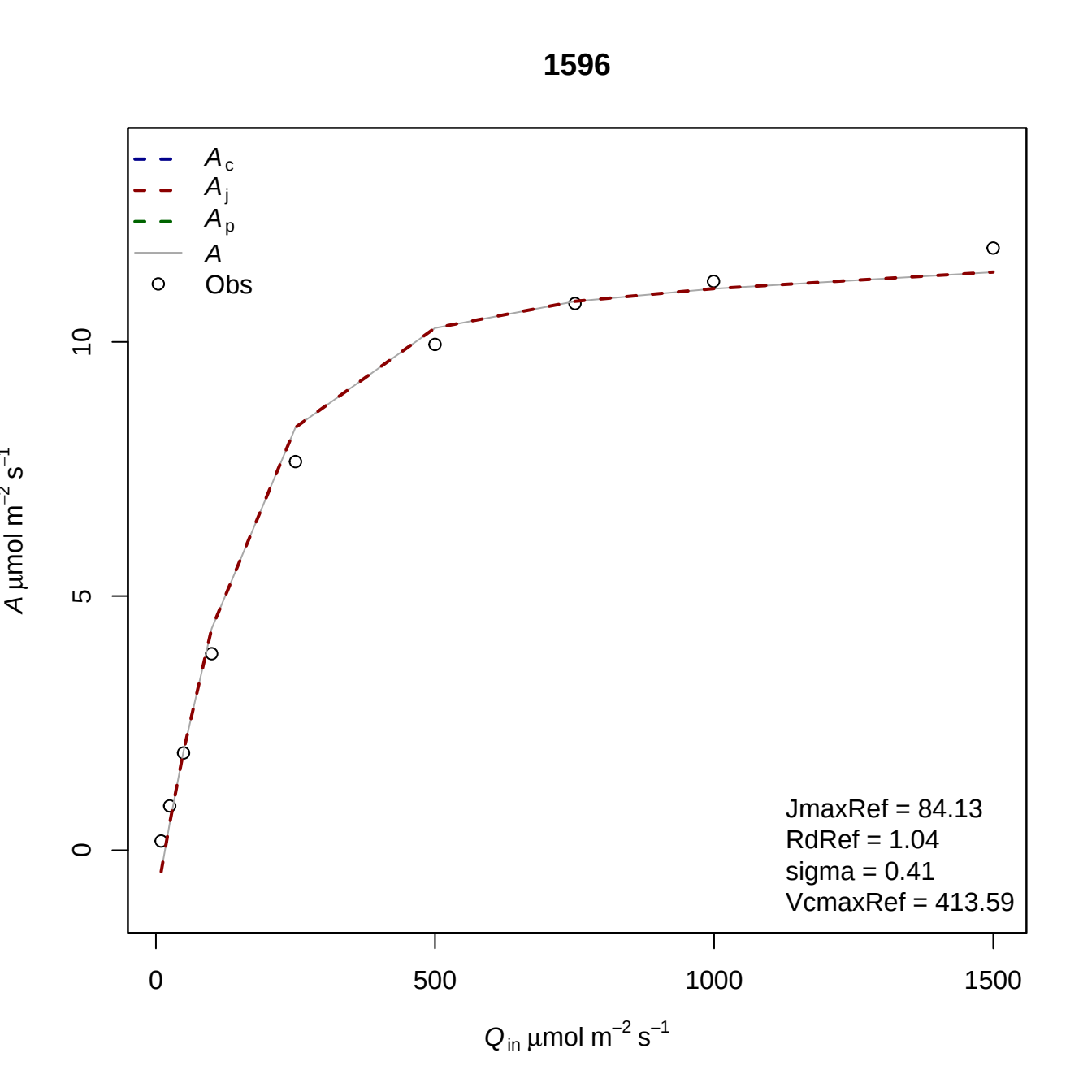

1591
A_c
A_j
A_p
A
Obs
JmaxRef = 33.68
RdRef = 1.18
sigma = 0.84
VcmaxRef = 44543.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

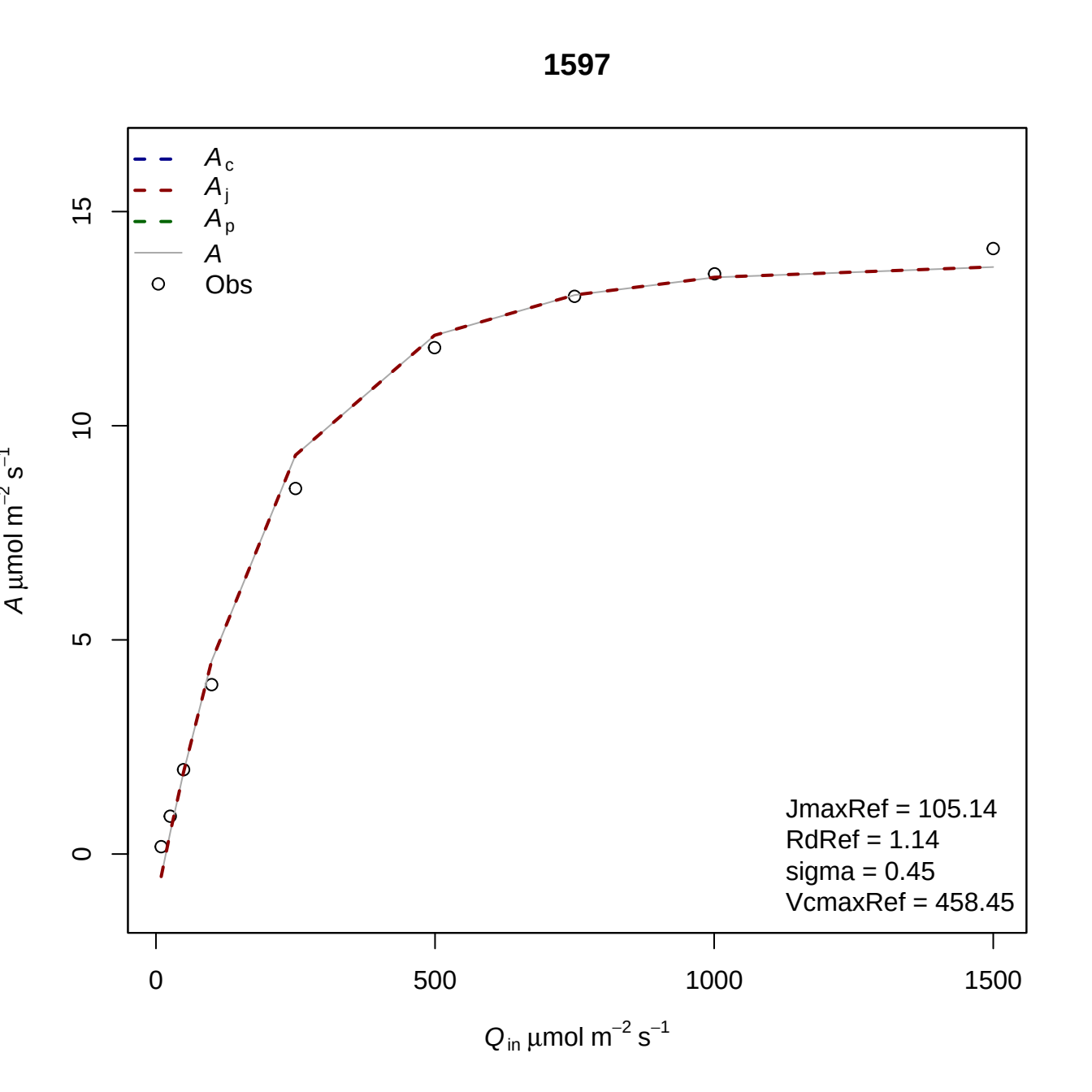

1592
A_c
A_j
A_p
A
Obs
JmaxRef = 60.31
RdRef = 1.13
sigma = 0.22
VcmaxRef = 195.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

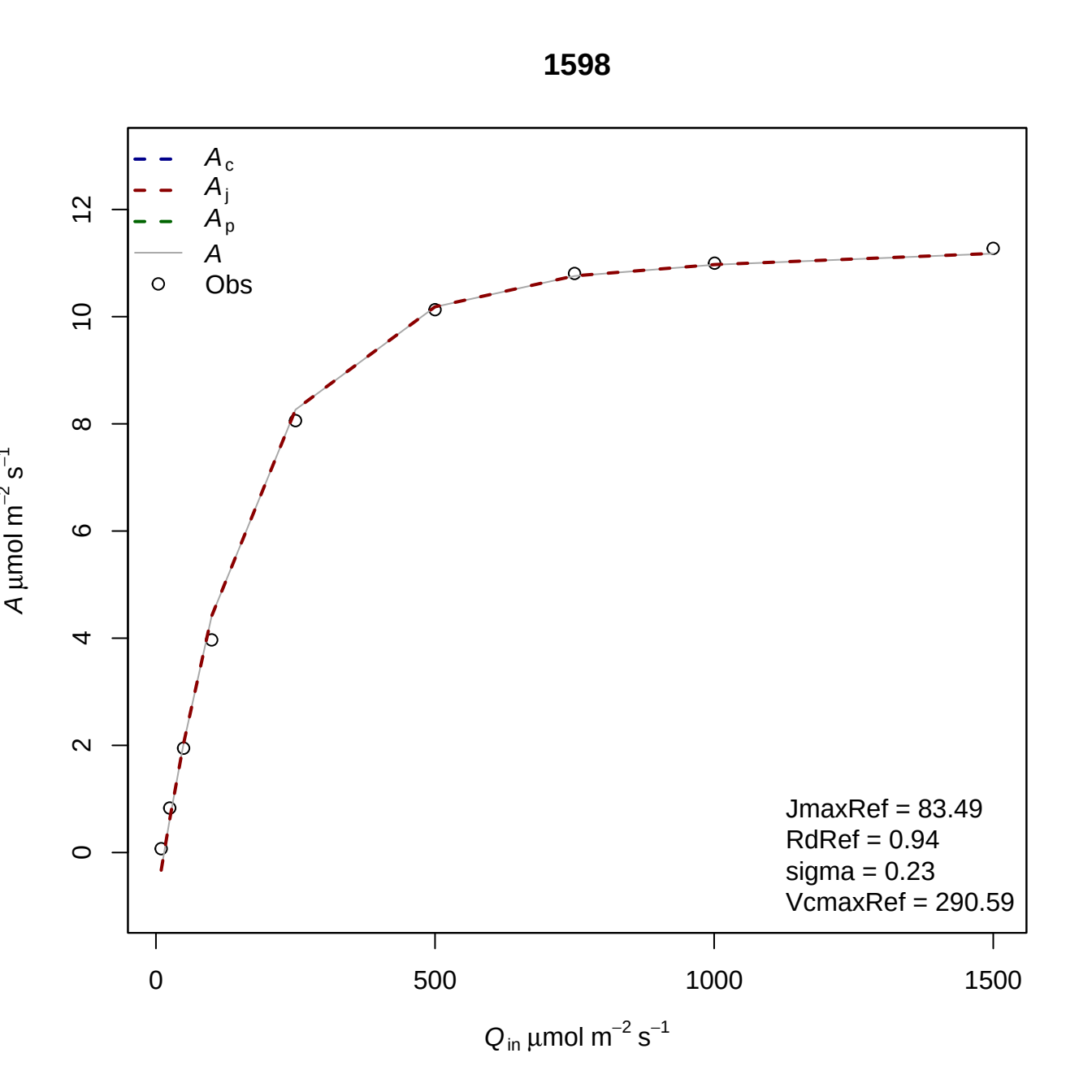

1593
A_c
A_j
A_p
A
Obs
JmaxRef = 35.78
RdRef = 1.21
sigma = 0.24
VcmaxRef = 109.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

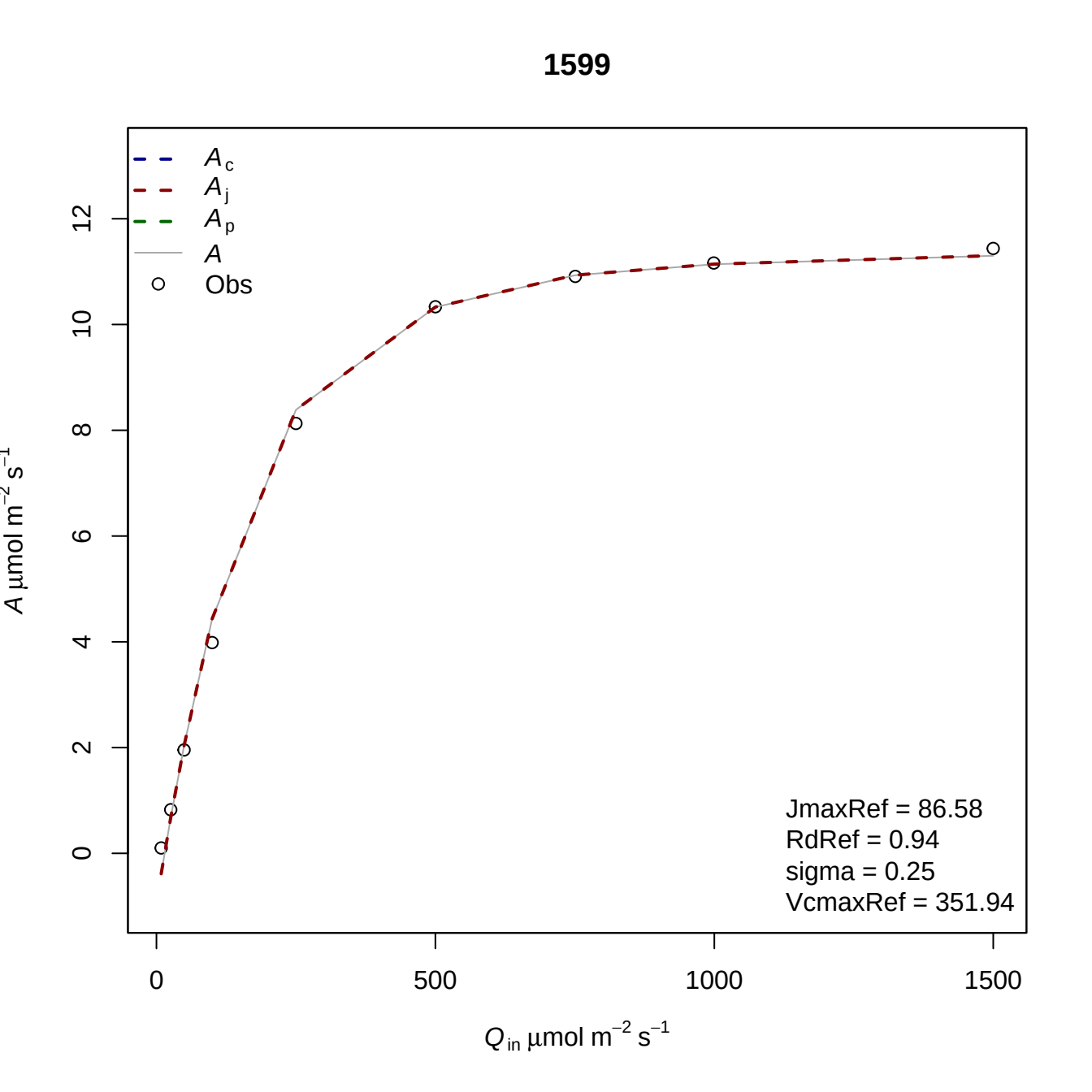

1594
A_c
A_j
A_p
A
Obs
JmaxRef = 44.78
RdRef = 1.02
sigma = 0.33
VcmaxRef = 278.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

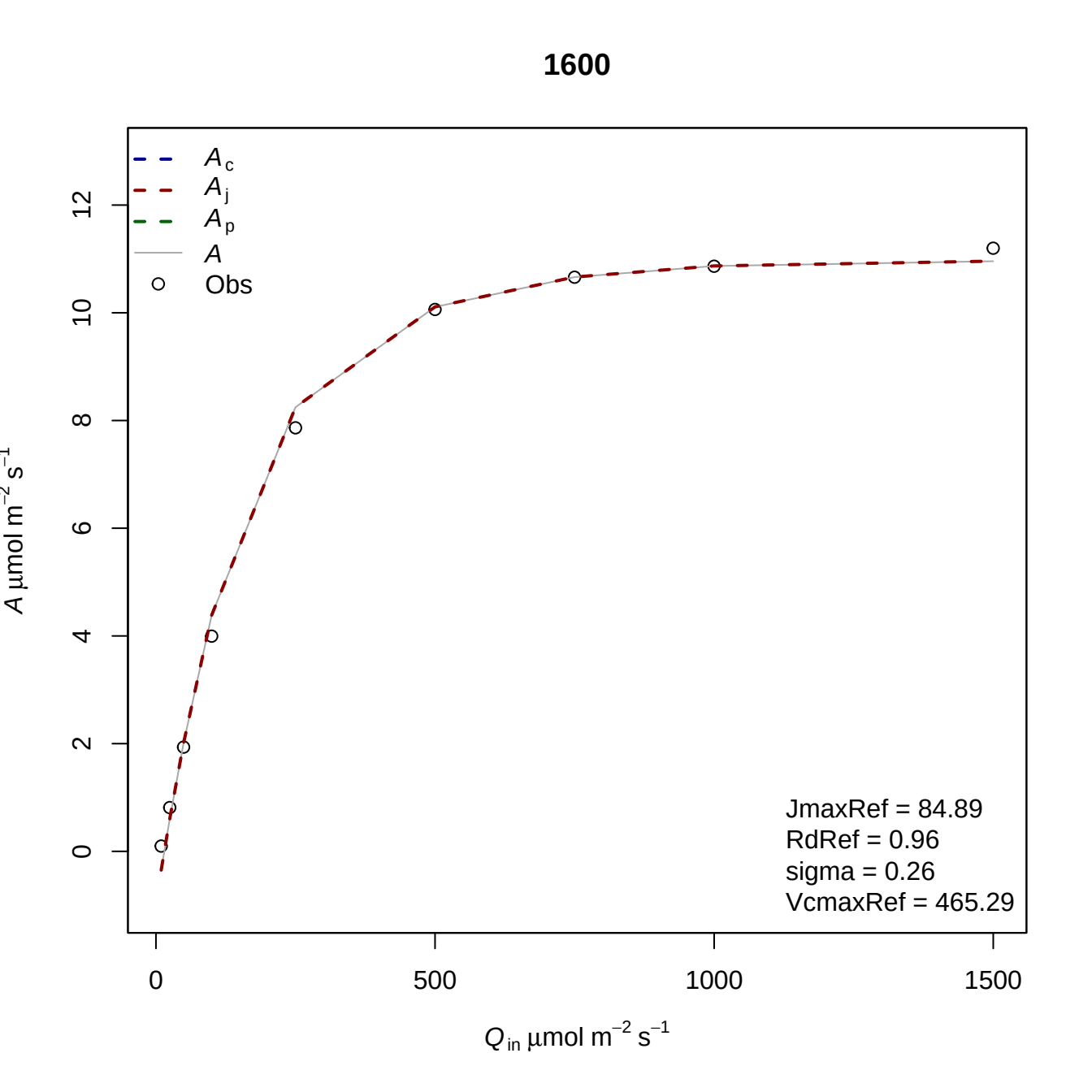

1595
A_c
A_j
A_p
A
Obs
JmaxRef = 66.49
RdRef = 0.91
sigma = 0.31
VcmaxRef = 319.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

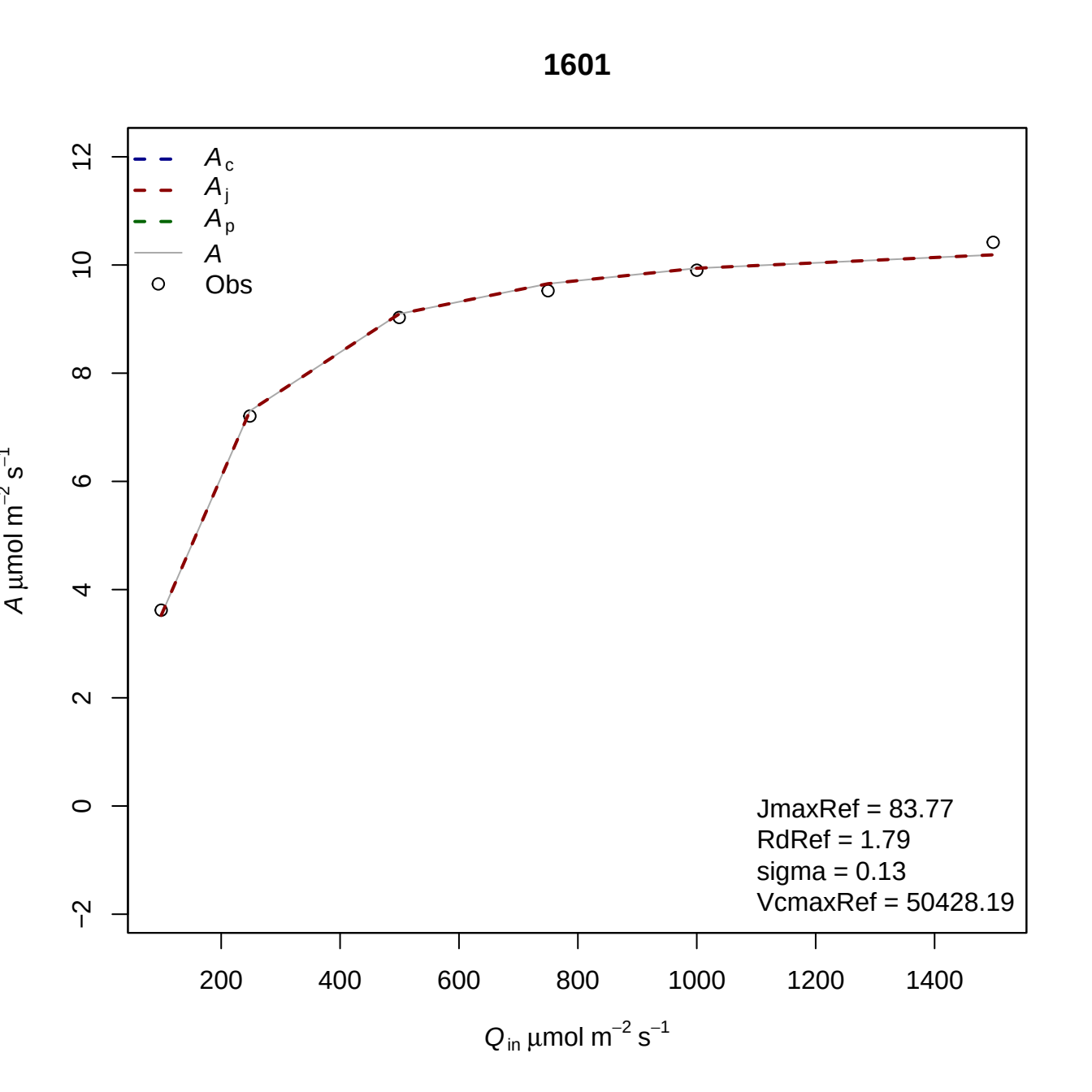

1596
A_c
A_j
A_p
A
Obs
JmaxRef = 84.13
RdRef = 1.04
sigma = 0.41
VcmaxRef = 413.59
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

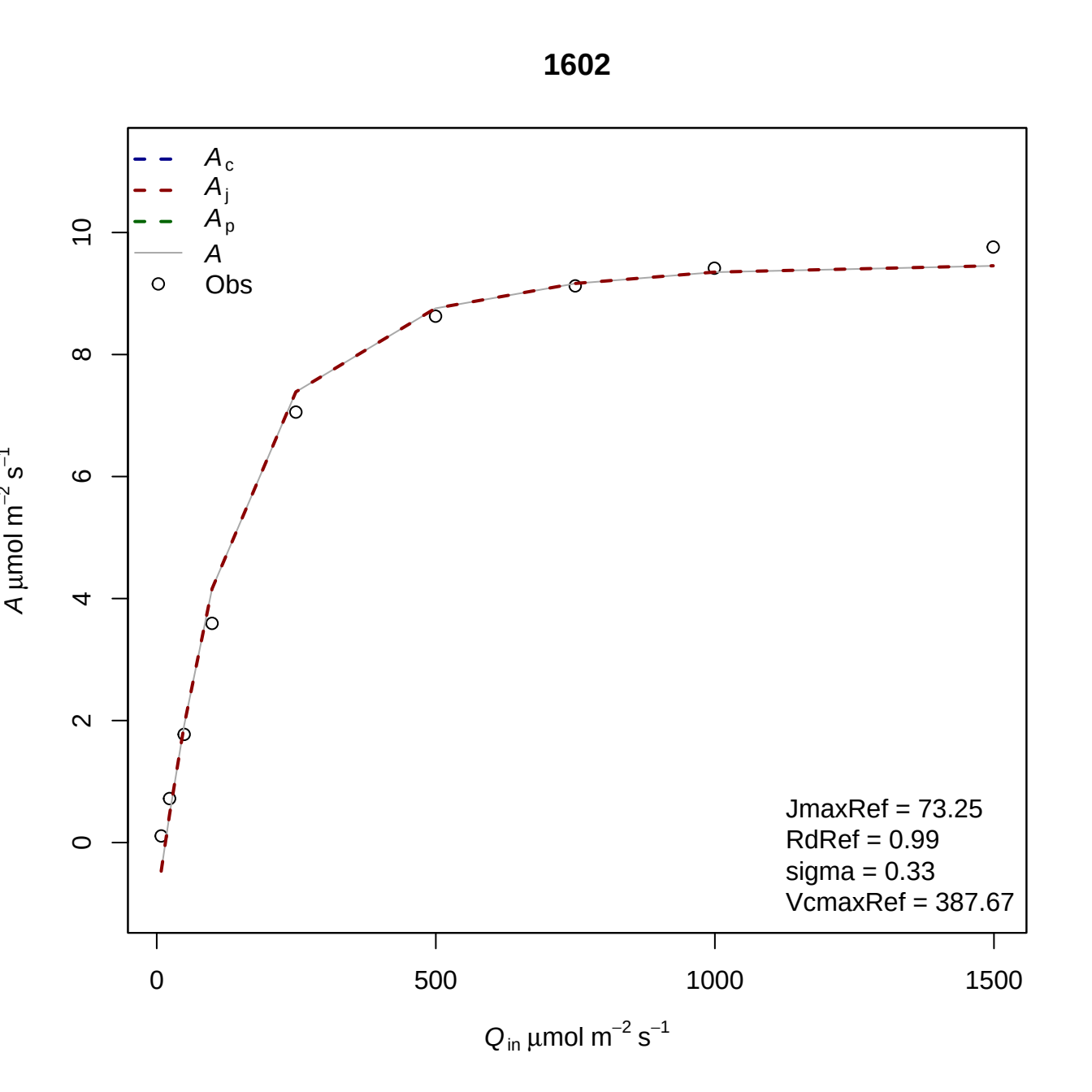

1597
A_c
A_j
A_p
A
Obs
JmaxRef = 105.14
RdRef = 1.14
sigma = 0.45
VcmaxRef = 458.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

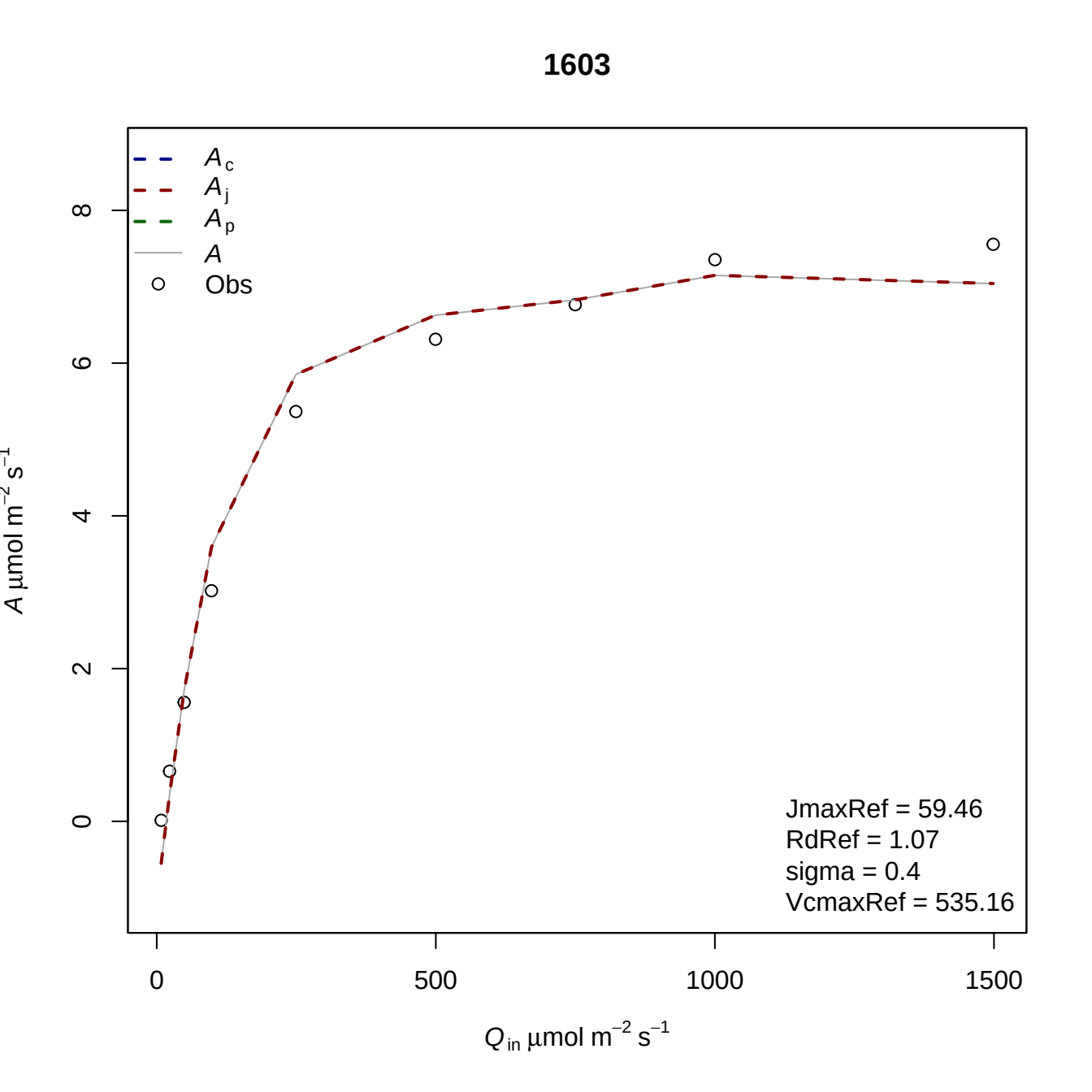

1598
A_c
A_j
A_p
A
Obs
JmaxRef = 83.49
RdRef = 0.94
sigma = 0.23
VcmaxRef = 290.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

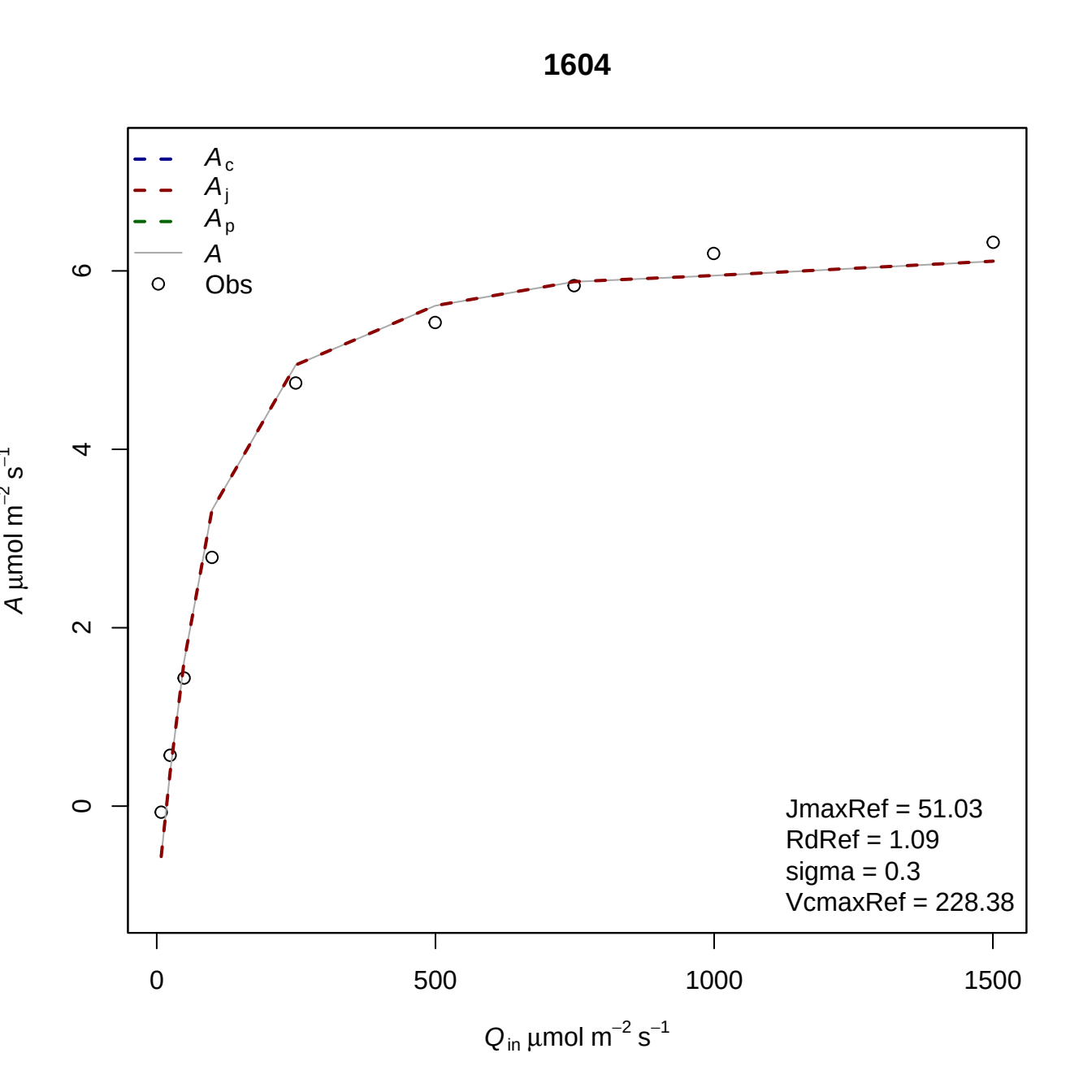

1599
A_c
A_j
A_p
A
Obs
JmaxRef = 86.58
RdRef = 0.94
sigma = 0.25
VcmaxRef = 351.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

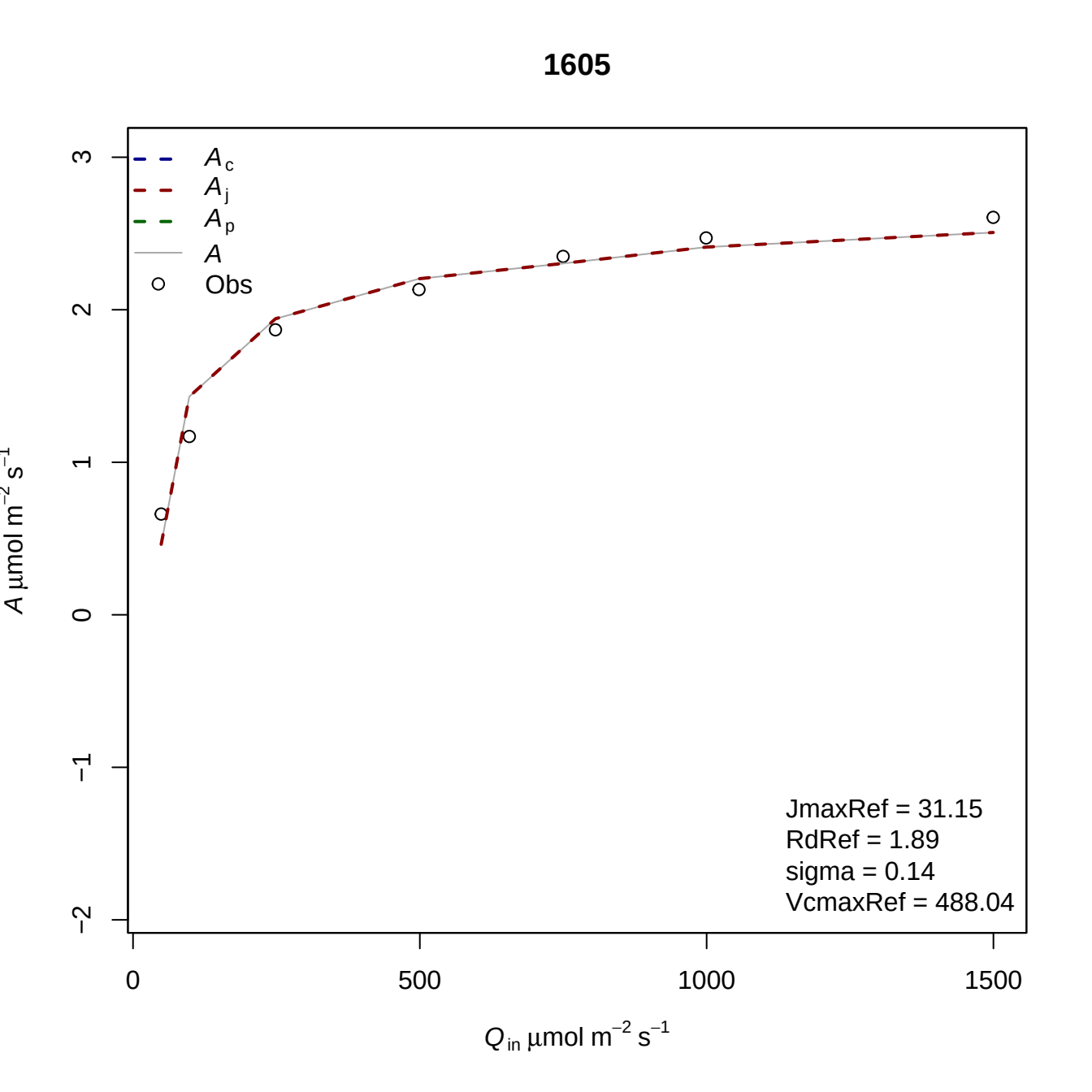

1600
A_c
A_j
A_p
A
Obs
JmaxRef = 84.89
RdRef = 0.96
sigma = 0.26
VcmaxRef = 465.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

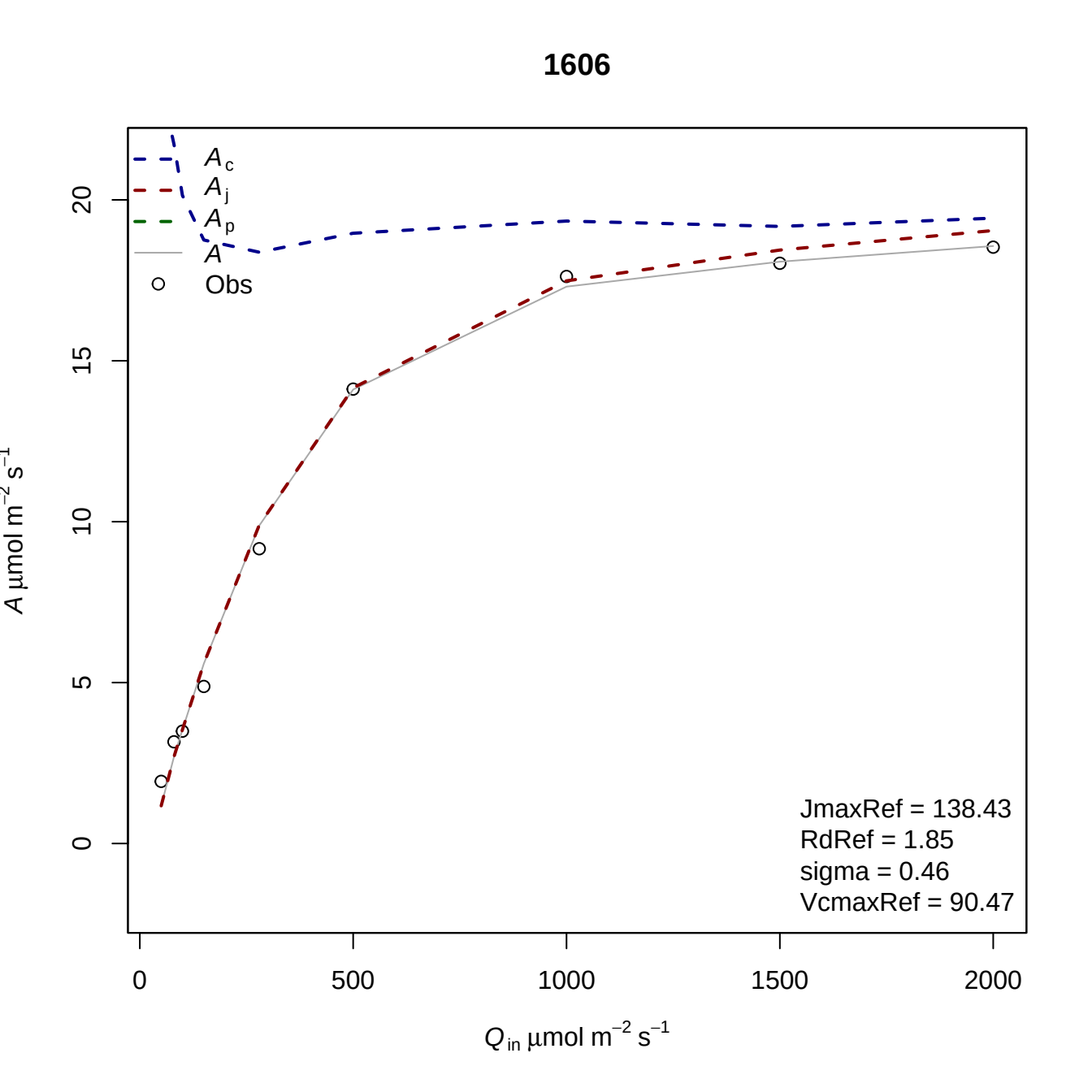

1601
A_c
A_j
A_p
A
Obs
JmaxRef = 83.77
RdRef = 1.79
sigma = 0.13
VcmaxRef = 50428.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400

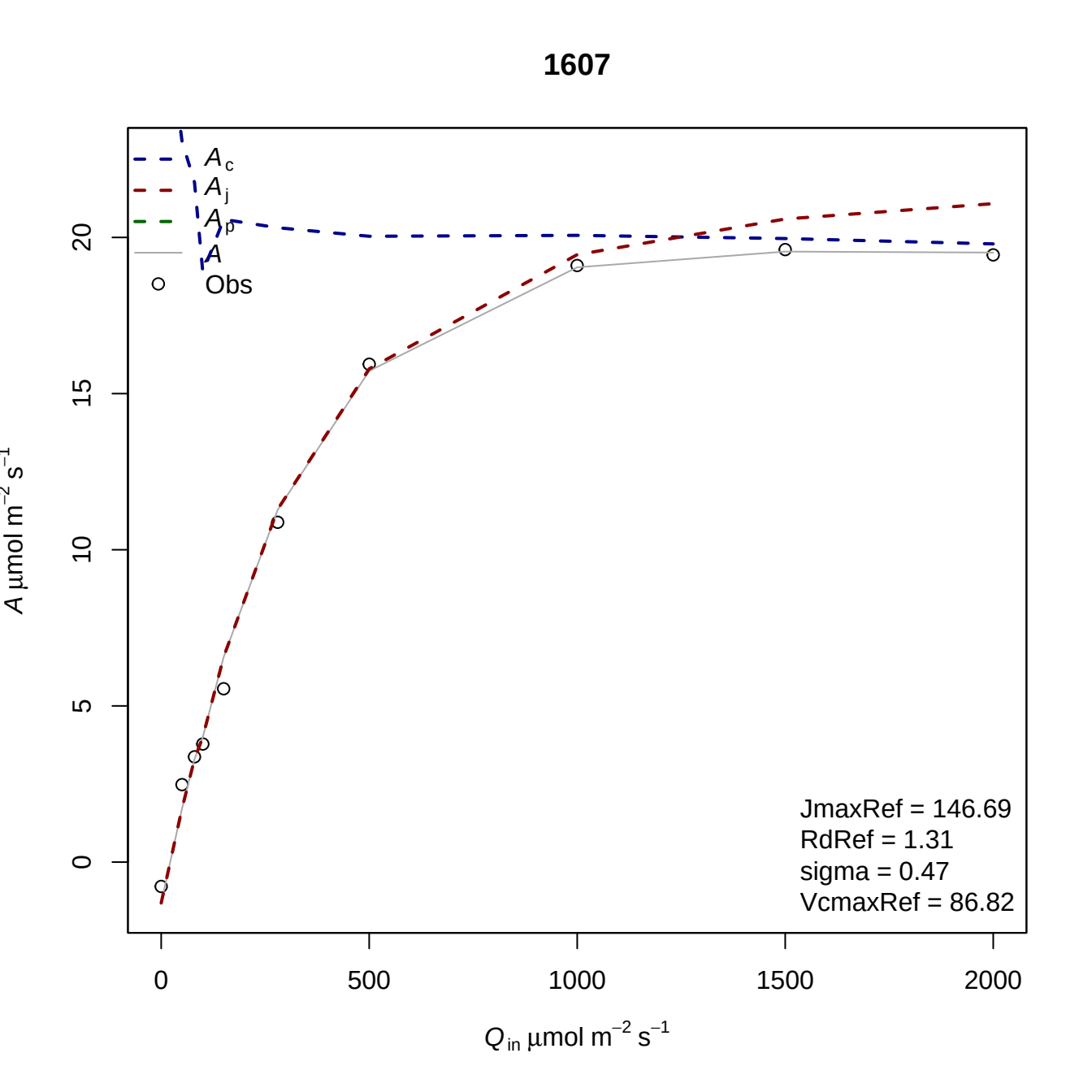

1602
A_c
A_j
A_p
A
Obs
JmaxRef = 73.25
RdRef = 0.99
sigma = 0.33
VcmaxRef = 387.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

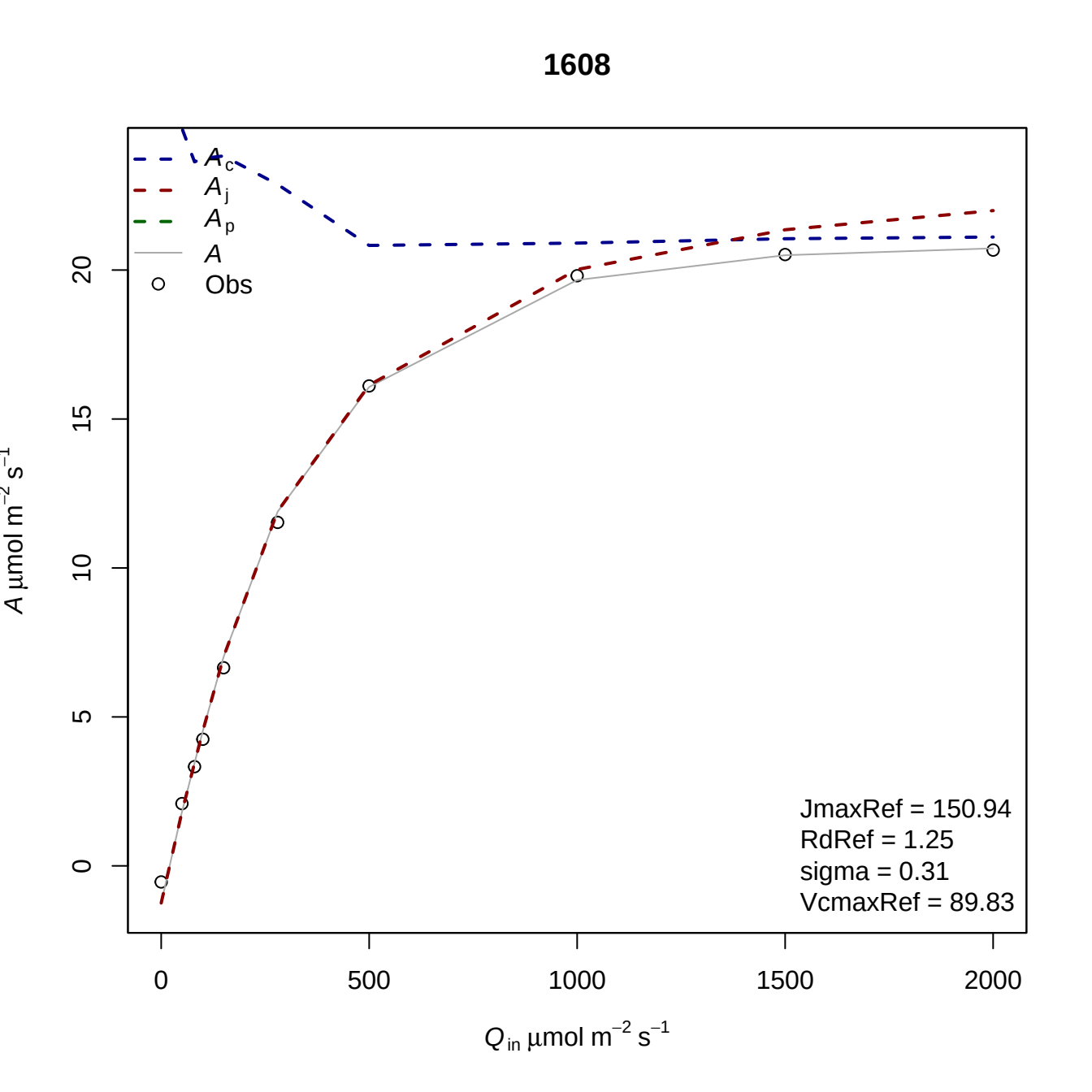

1603
A_c
A_j
A_p
A
Obs
JmaxRef = 59.46
RdRef = 1.07
sigma = 0.4
VcmaxRef = 535.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

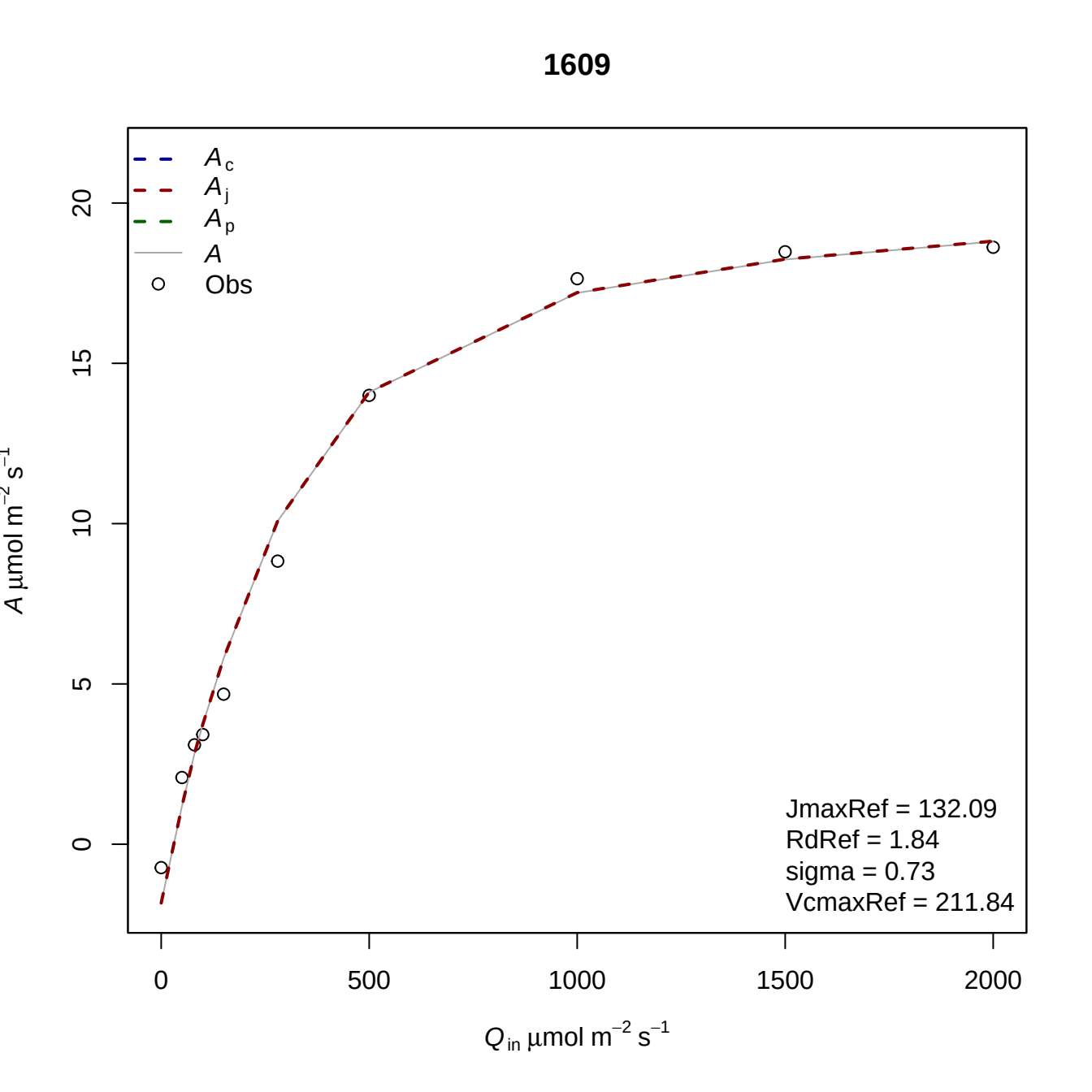

1604
A_c
A_j
A_p
A
Obs
JmaxRef = 51.03
RdRef = 1.09
sigma = 0.3
VcmaxRef = 228.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

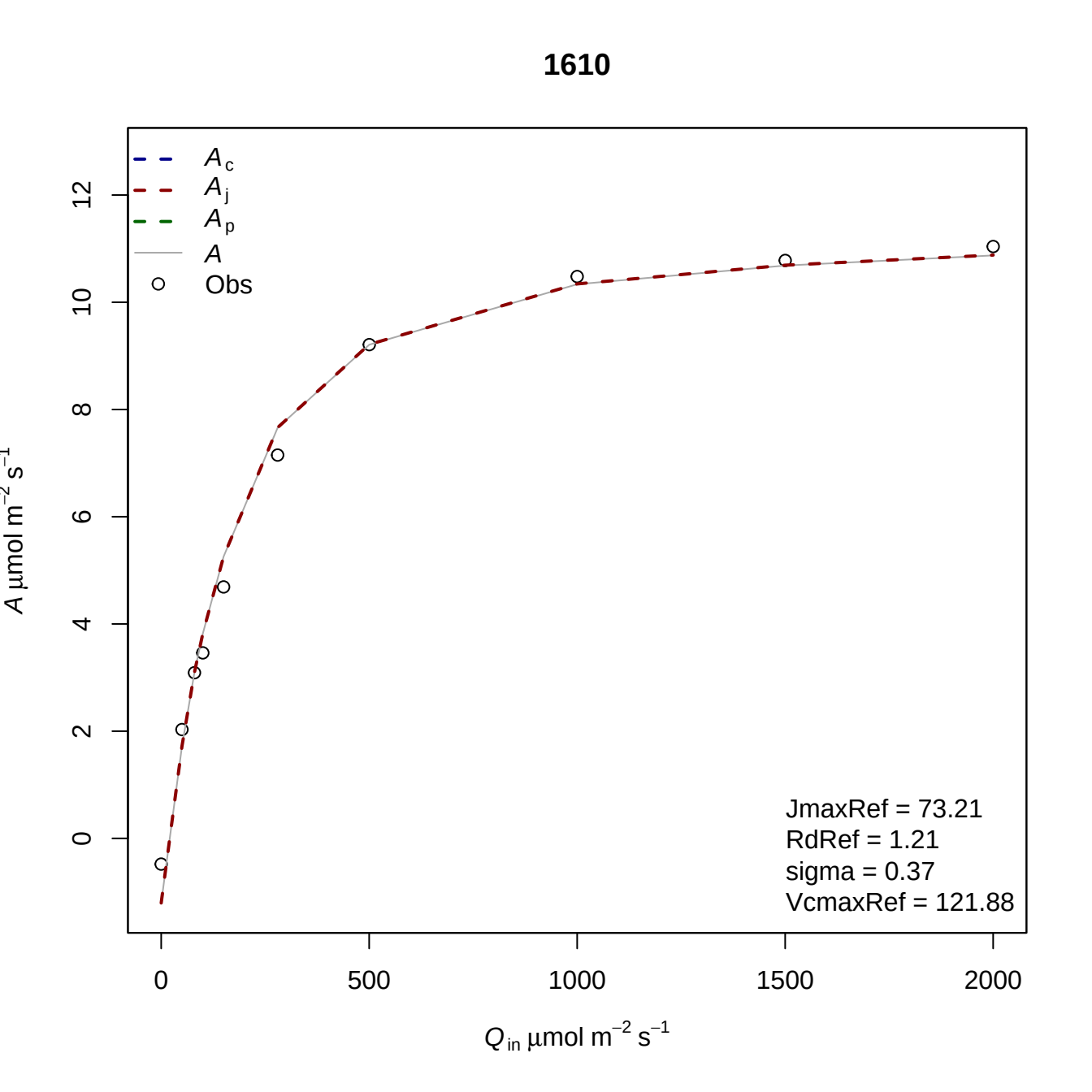

1605
A_c
A_j
A_p
A
Obs
JmaxRef = 31.15
RdRef = 1.89
sigma = 0.14
VcmaxRef = 488.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
500
1000
1500

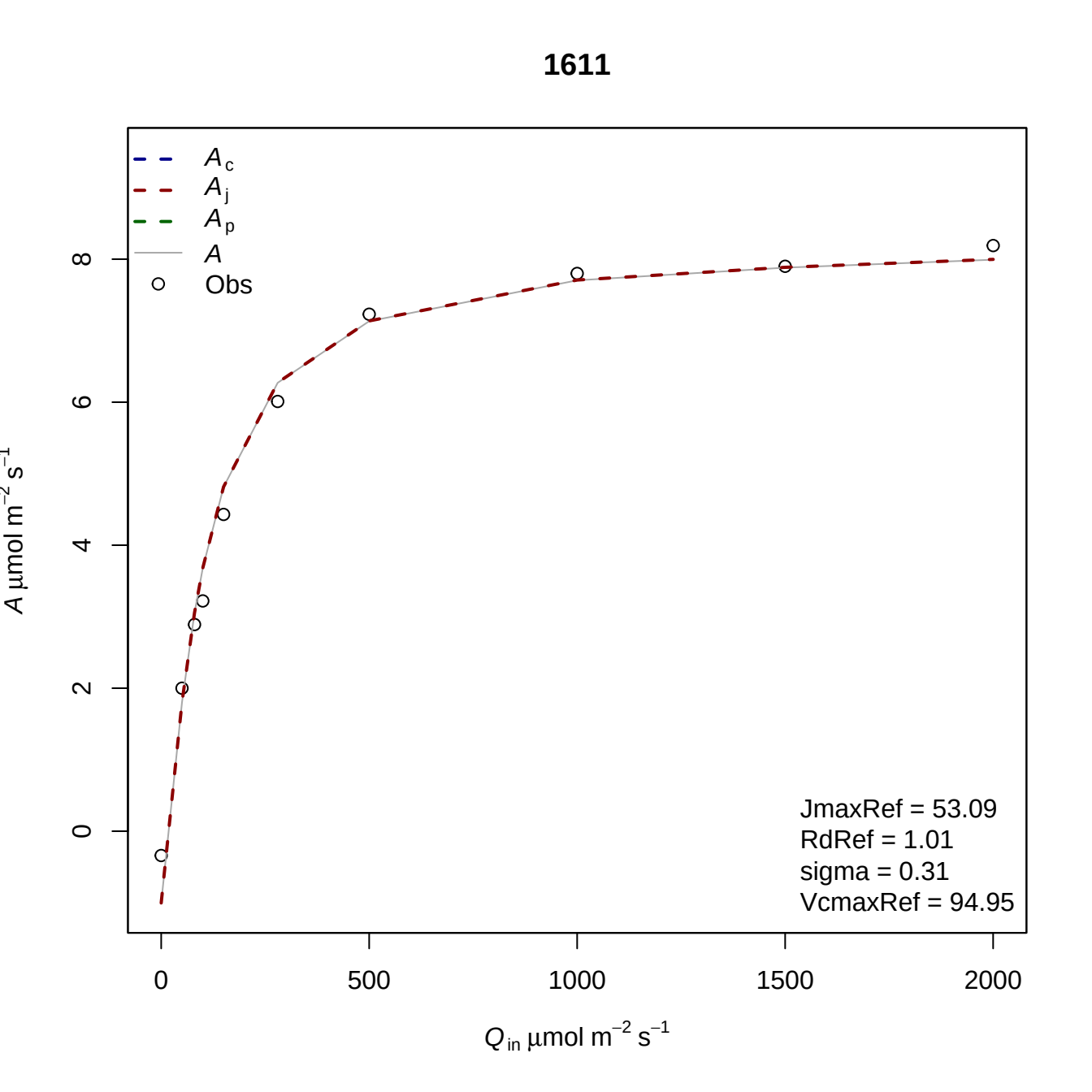

1606
A_c
A_j
A_p
A
Obs
JmaxRef = 138.43
RdRef = 1.85
sigma = 0.46
VcmaxRef = 90.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

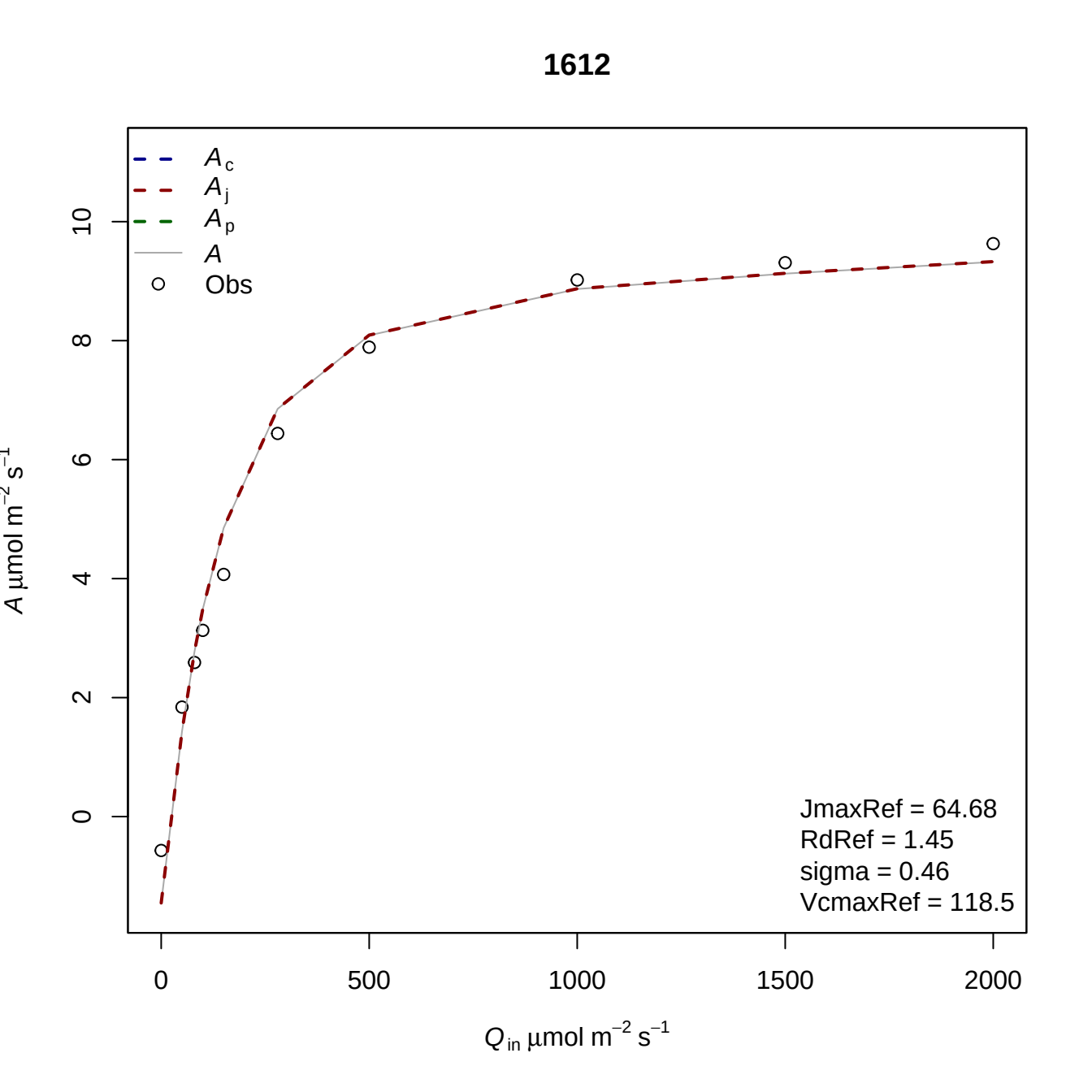

1607
A_c
A_j
A_p
A
Obs
JmaxRef = 146.69
RdRef = 1.31
sigma = 0.47
VcmaxRef = 86.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

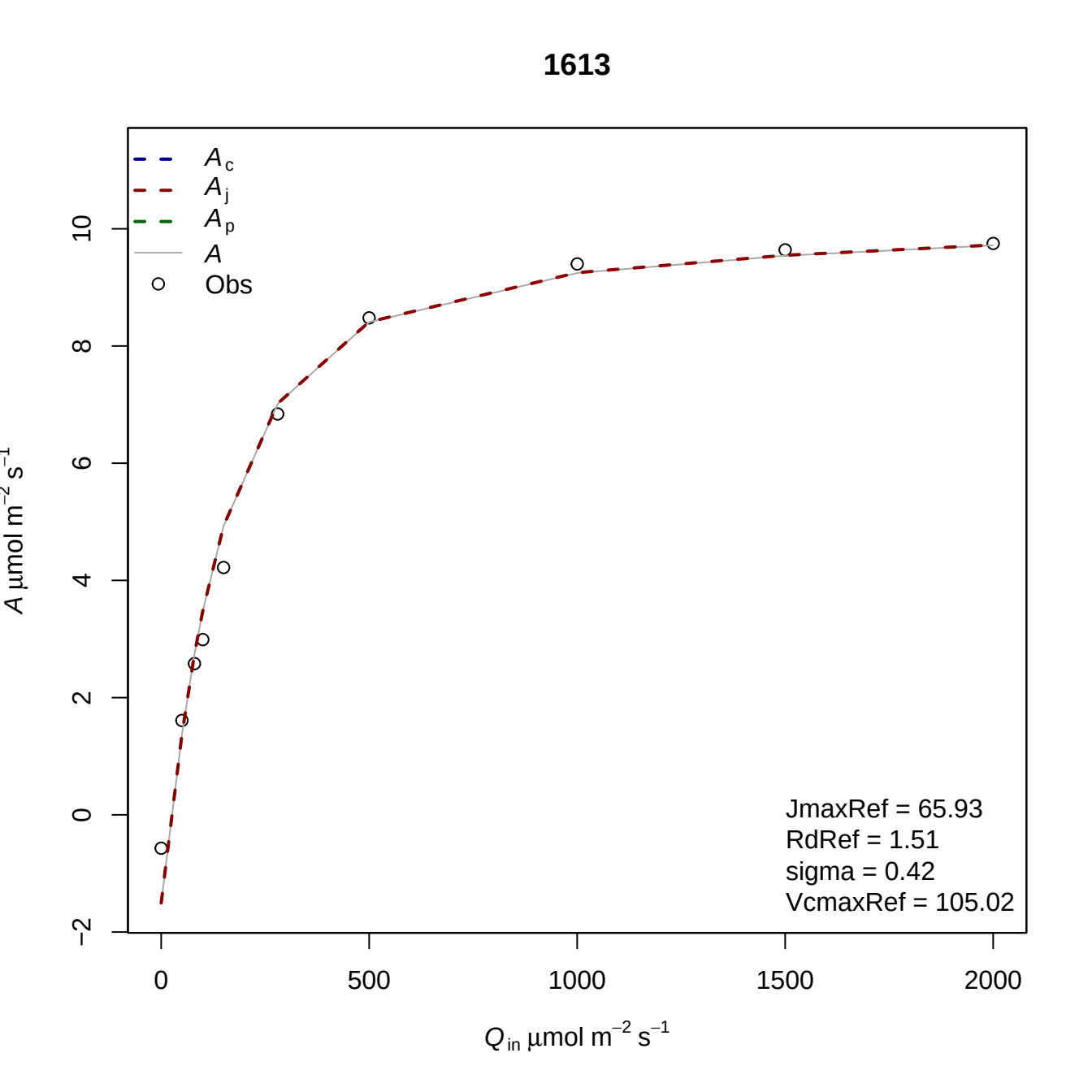

1608
A_c
A_j
A_p
A
Obs
JmaxRef = 150.94
RdRef = 1.25
sigma = 0.31
VcmaxRef = 89.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

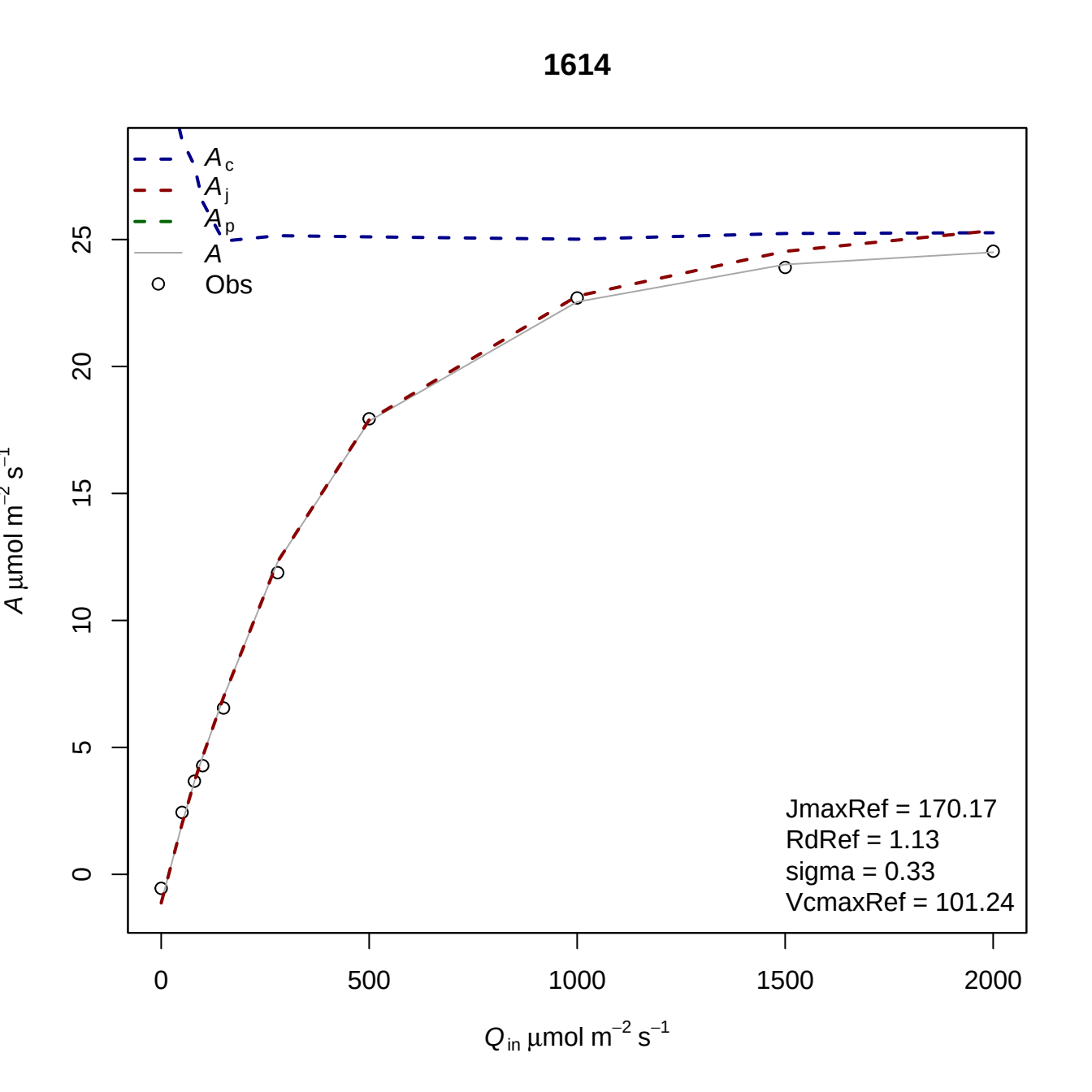

1609
A_c
A_j
A_p
A
Obs
JmaxRef = 132.09
RdRef = 1.84
sigma = 0.73
VcmaxRef = 211.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

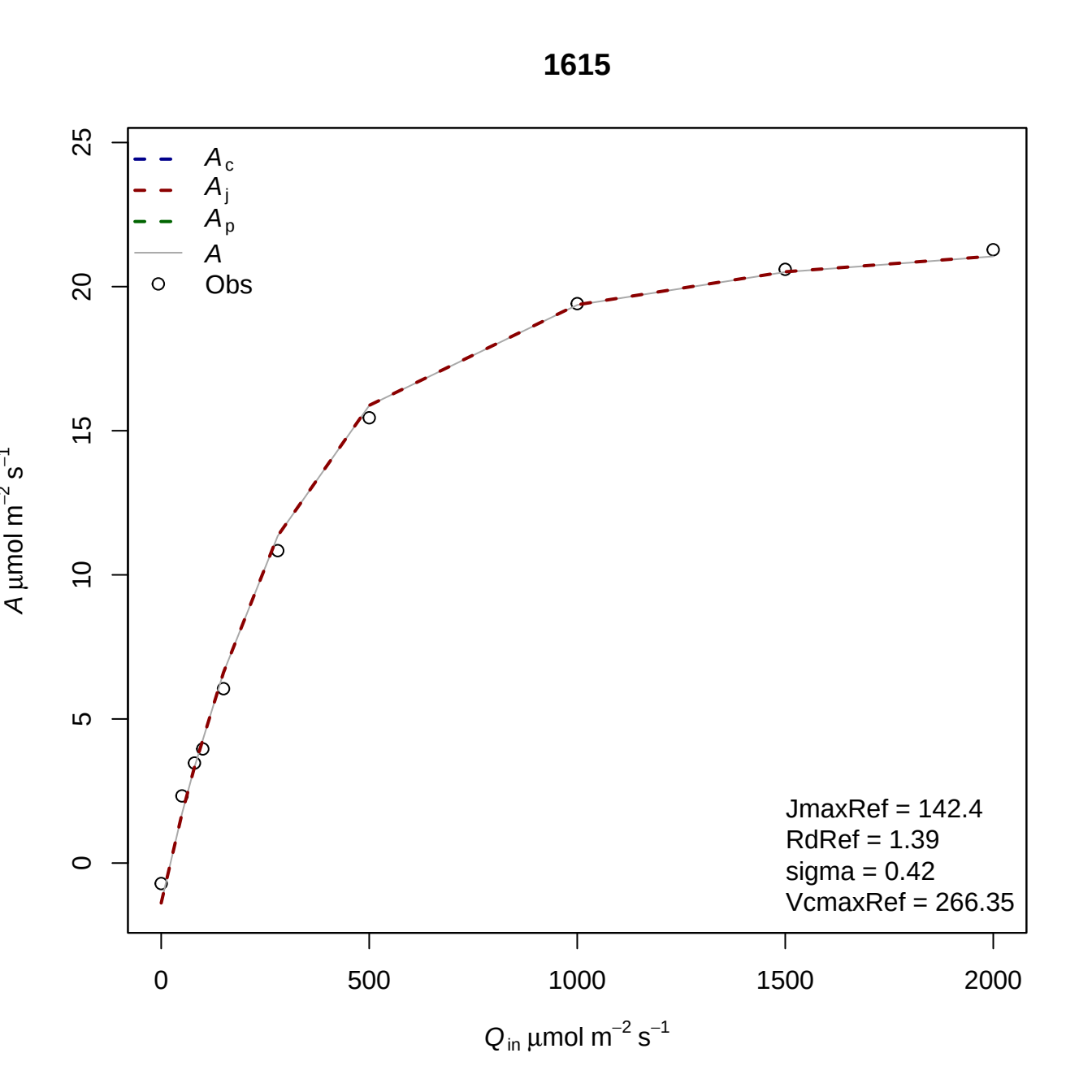

1610
A_c
A_j
A_p
A
Obs
JmaxRef = 73.21
RdRef = 1.21
sigma = 0.37
VcmaxRef = 121.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

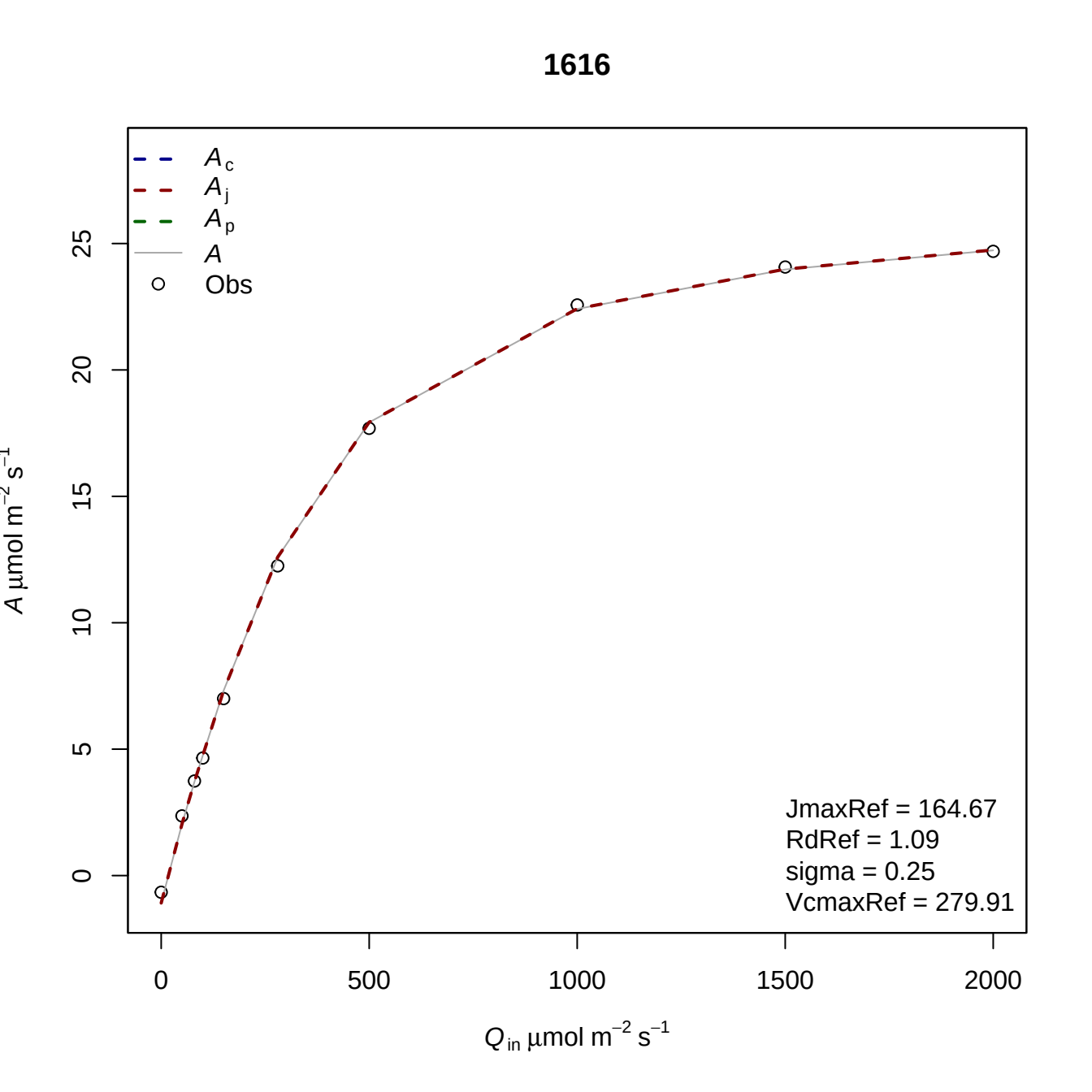

1611
A_c
A_j
A_p
A
Obs
JmaxRef = 53.09
RdRef = 1.01
sigma = 0.31
VcmaxRef = 94.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

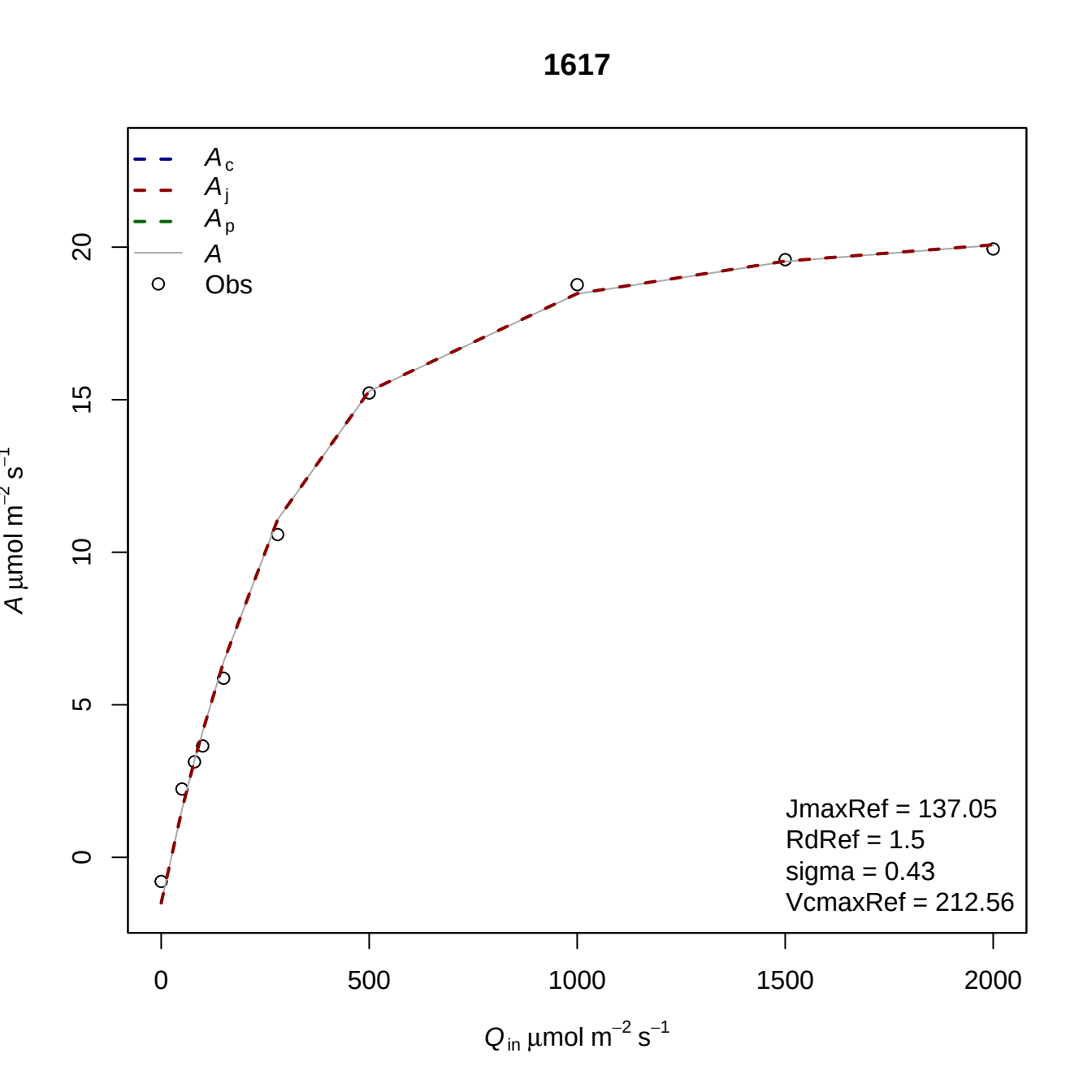

1612
A_c
A_j
A_p
A
Obs
JmaxRef = 64.68
RdRef = 1.45
sigma = 0.46
VcmaxRef = 118.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

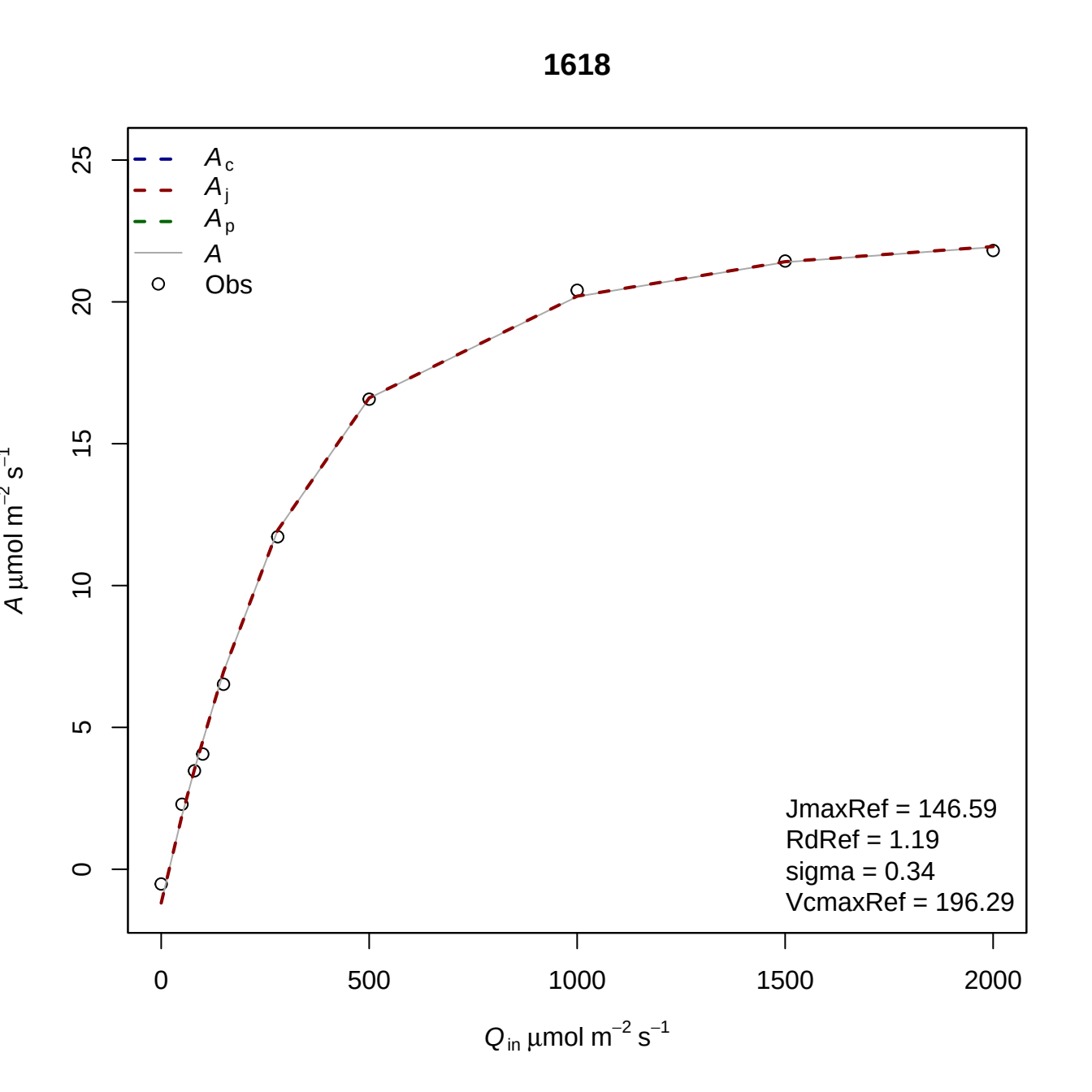

1613
A_c
A_j
A_p
A
Obs
JmaxRef = 65.93
RdRef = 1.51
sigma = 0.42
VcmaxRef = 105.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

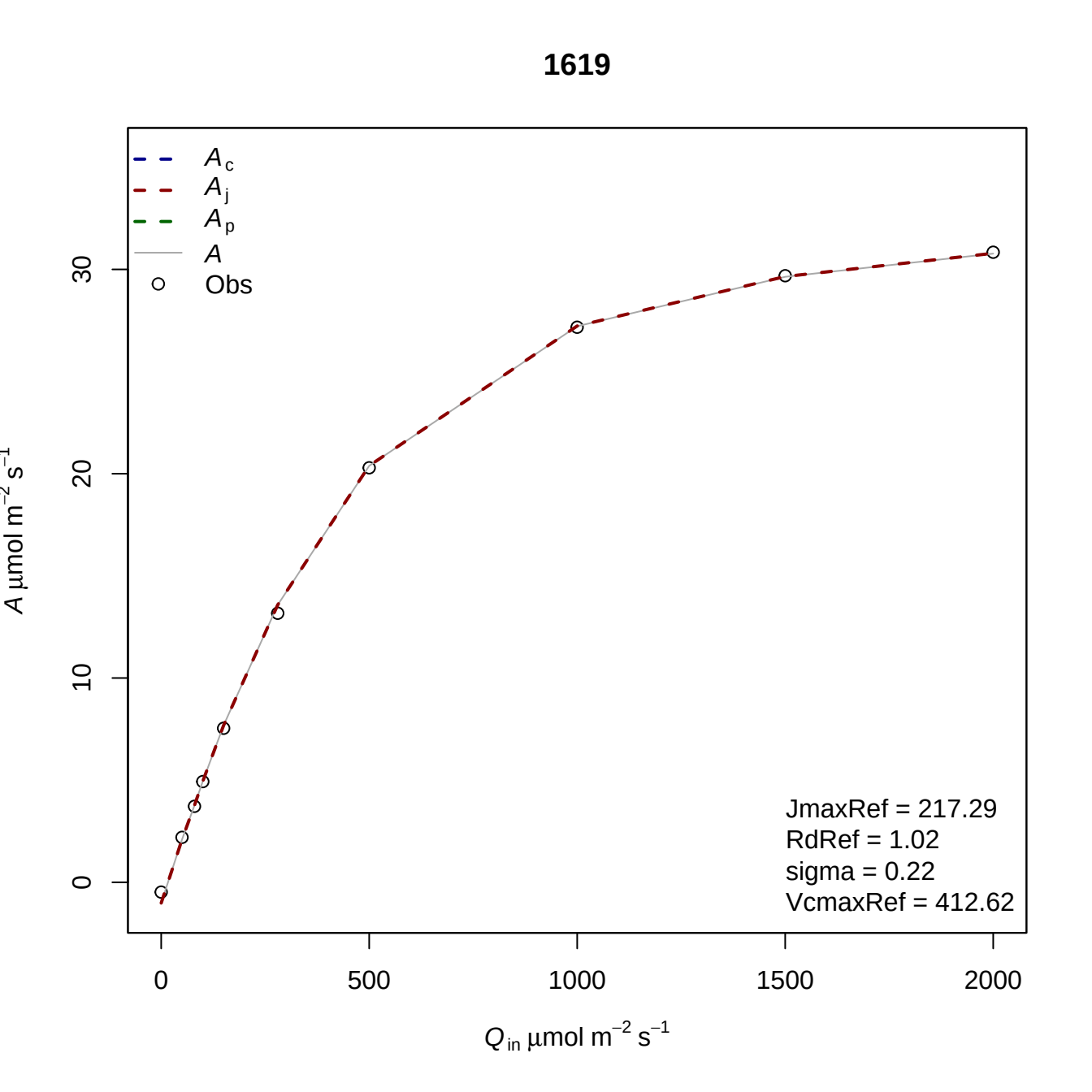

1614
A_c
A_j
A_p
A
Obs
JmaxRef = 170.17
RdRef = 1.13
sigma = 0.33
VcmaxRef = 101.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

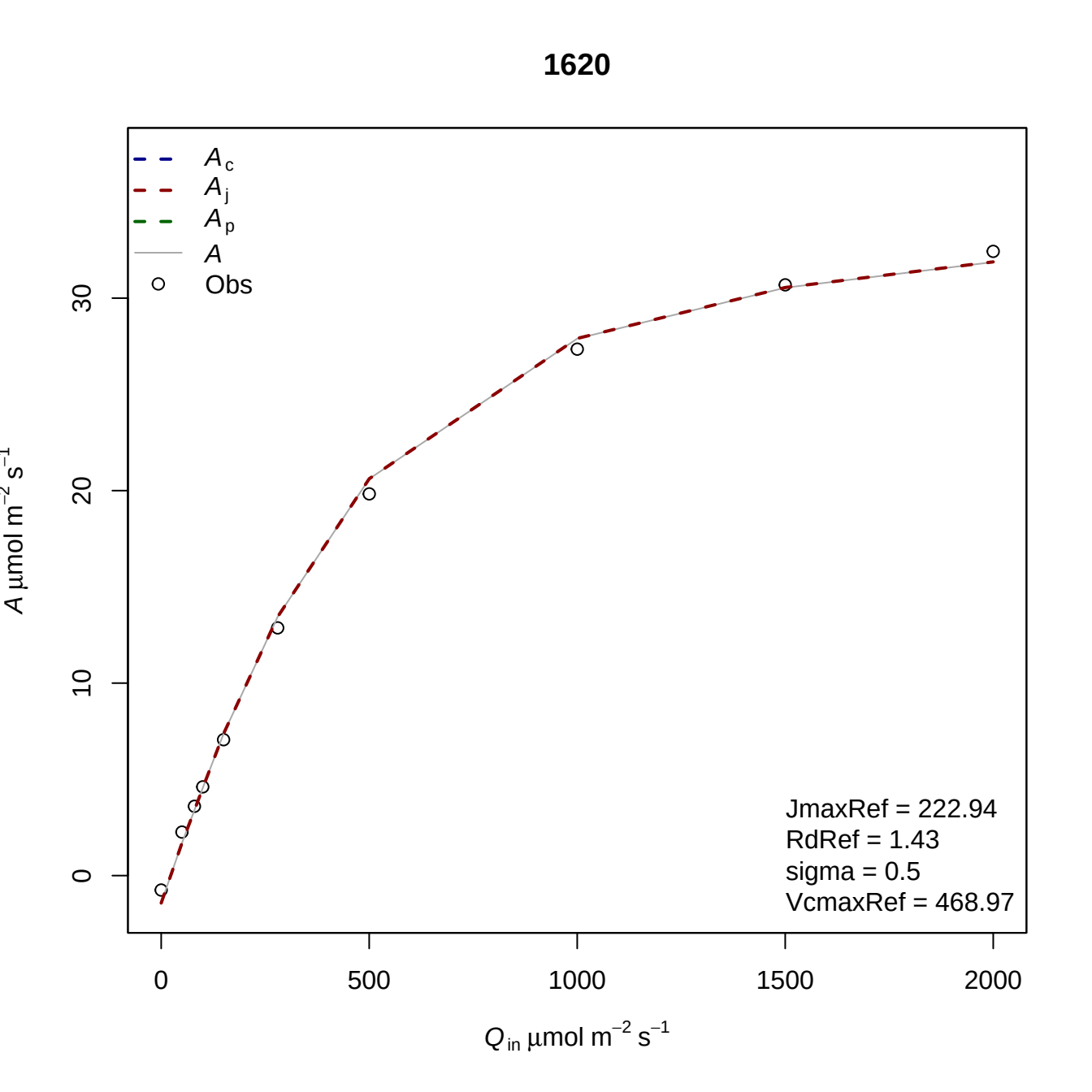

1615
A_c
A_j
A_p
A
Obs
JmaxRef = 142.4
RdRef = 1.39
sigma = 0.42
VcmaxRef = 266.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

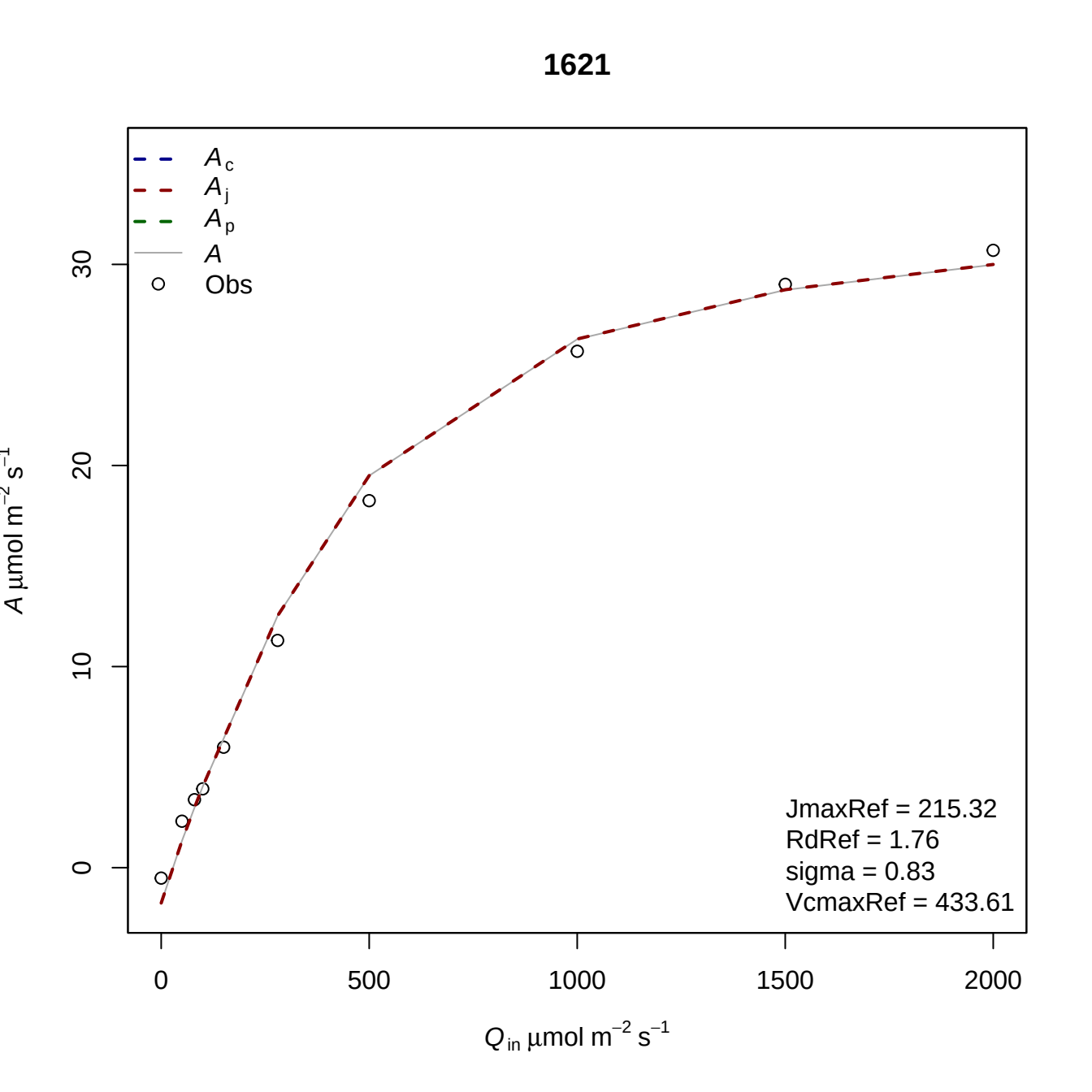

1616
A_c
A_j
A_p
A
Obs
JmaxRef = 164.67
RdRef = 1.09
sigma = 0.25
VcmaxRef = 279.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

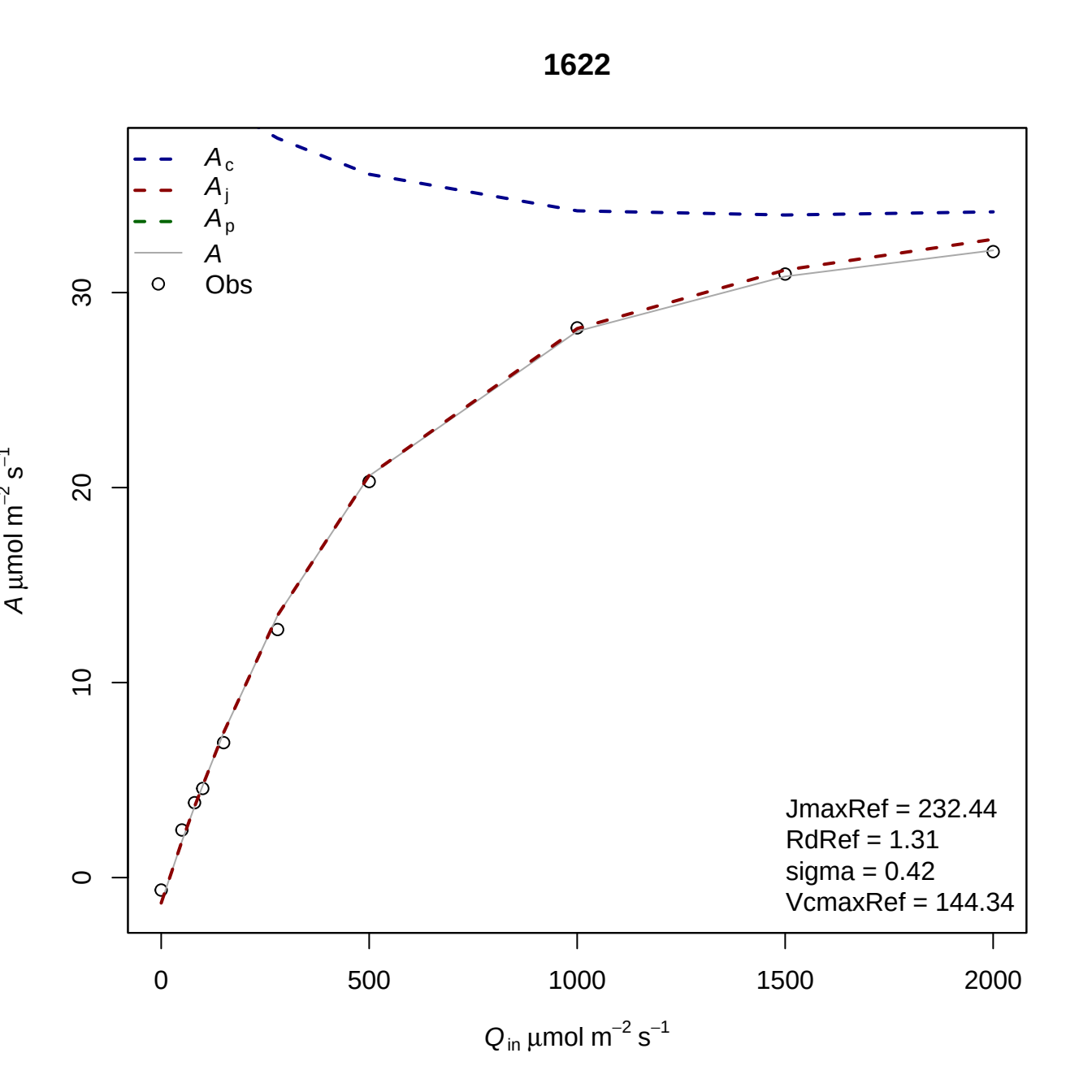

1617
A_c
A_j
A_p
A
Obs
JmaxRef = 137.05
RdRef = 1.5
sigma = 0.43
VcmaxRef = 212.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

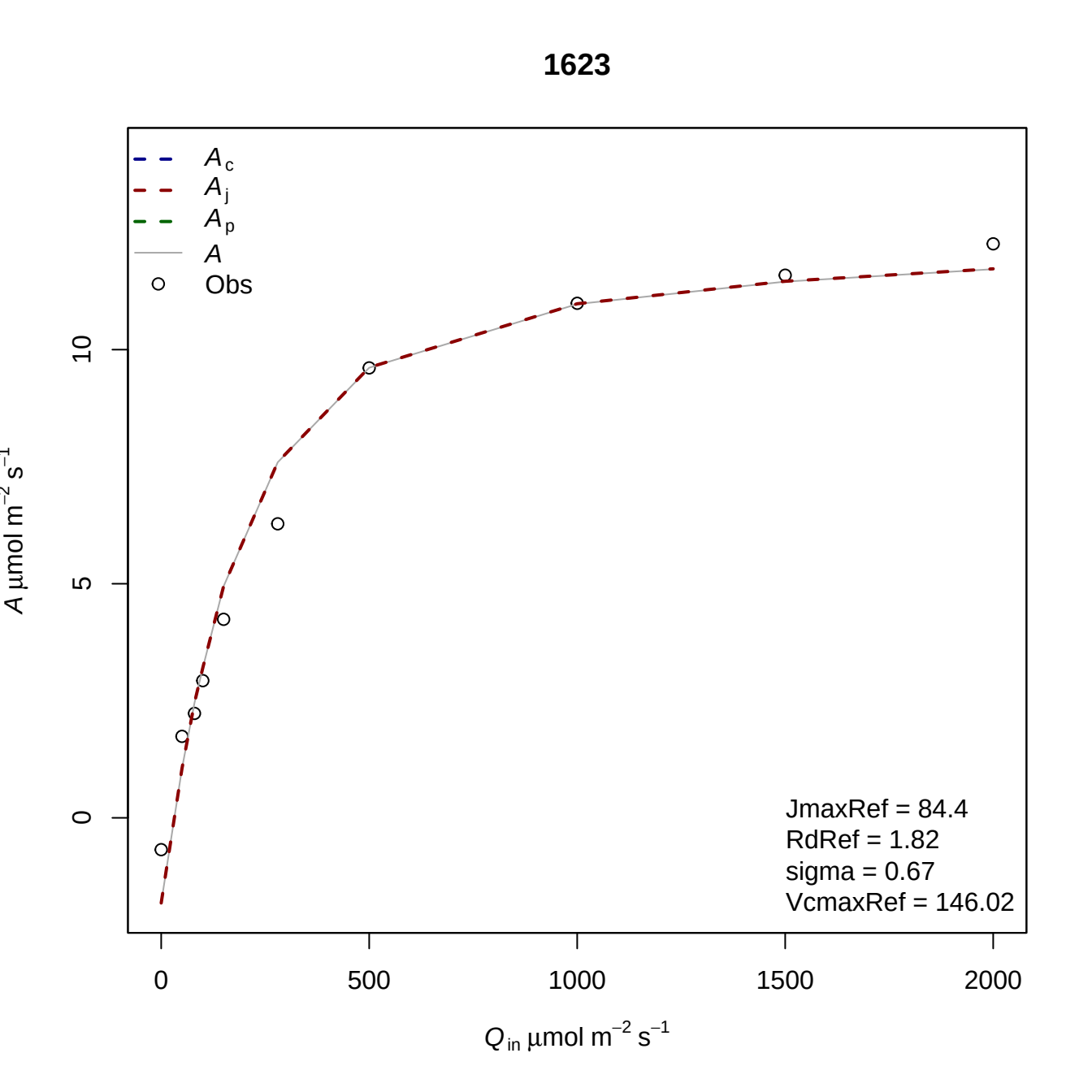

1618
A_c
A_j
A_p
A
Obs
JmaxRef = 146.59
RdRef = 1.19
sigma = 0.34
VcmaxRef = 196.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

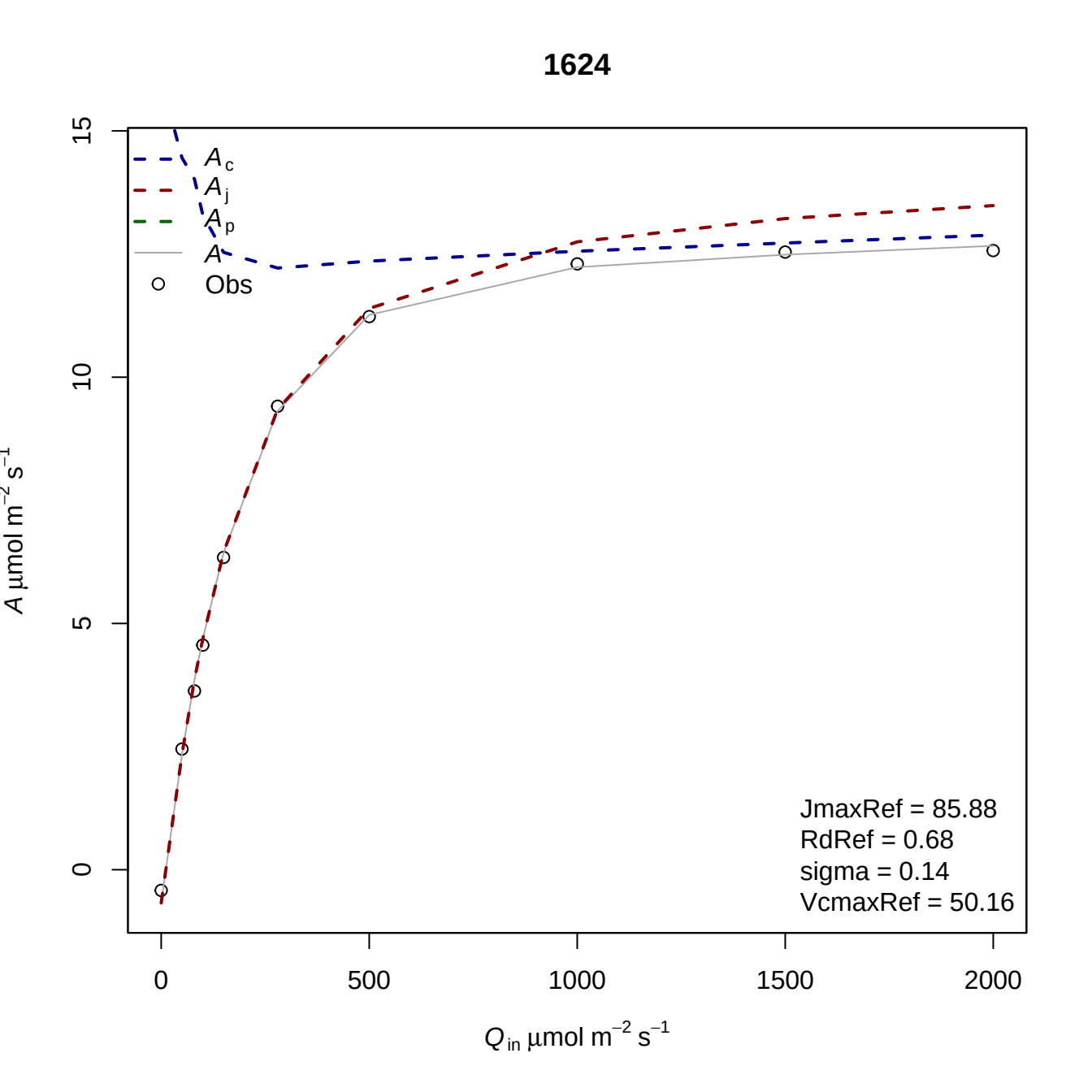

1619
A_c
A_j
A_p
A
Obs
JmaxRef = 217.29
RdRef = 1.02
sigma = 0.22
VcmaxRef = 412.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

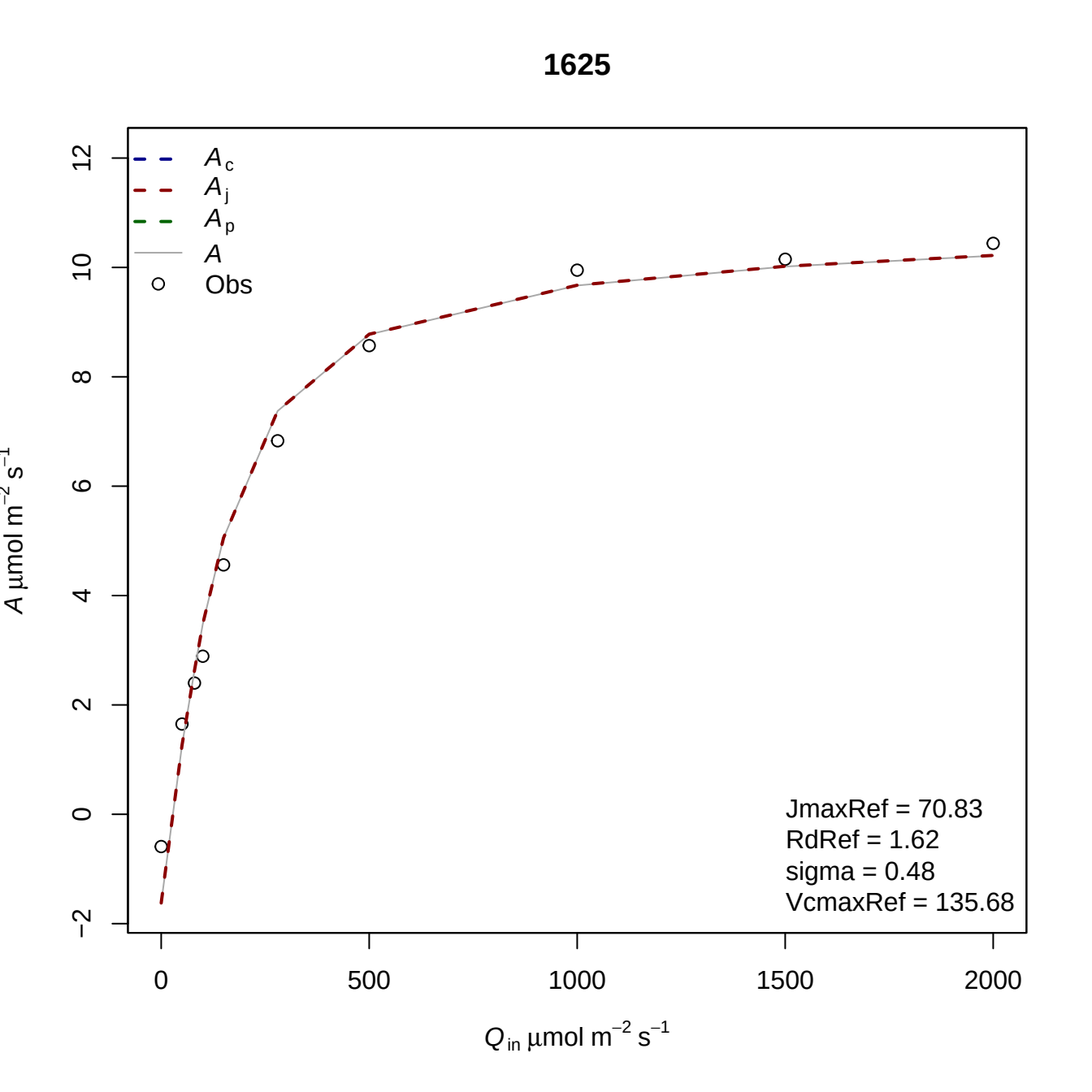

1620
A_c
A_j
A_p
A
Obs
JmaxRef = 222.94
RdRef = 1.43
sigma = 0.5
VcmaxRef = 468.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

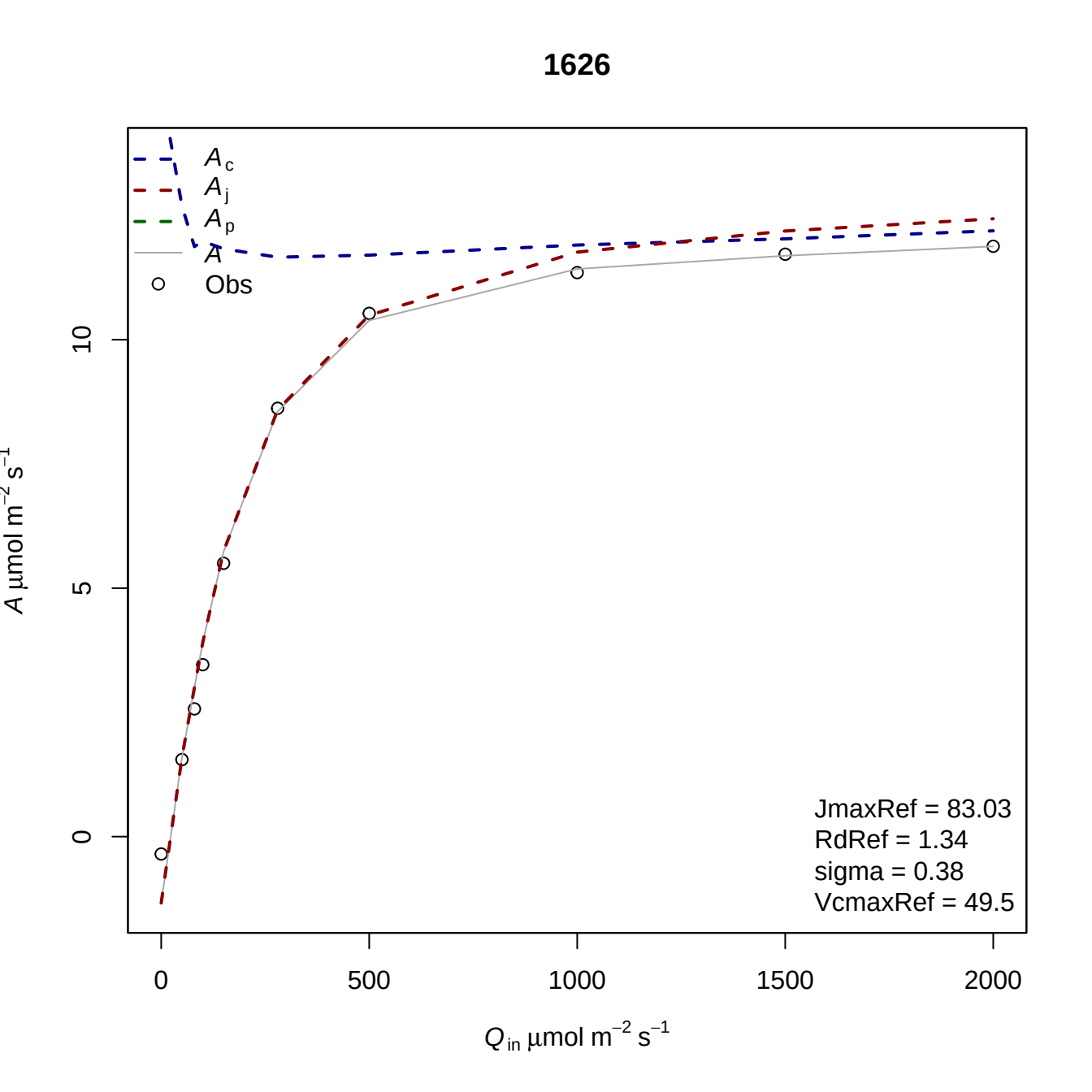

1621
A_c
A_j
A_p
A
Obs
JmaxRef = 215.32
RdRef = 1.76
sigma = 0.83
VcmaxRef = 433.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

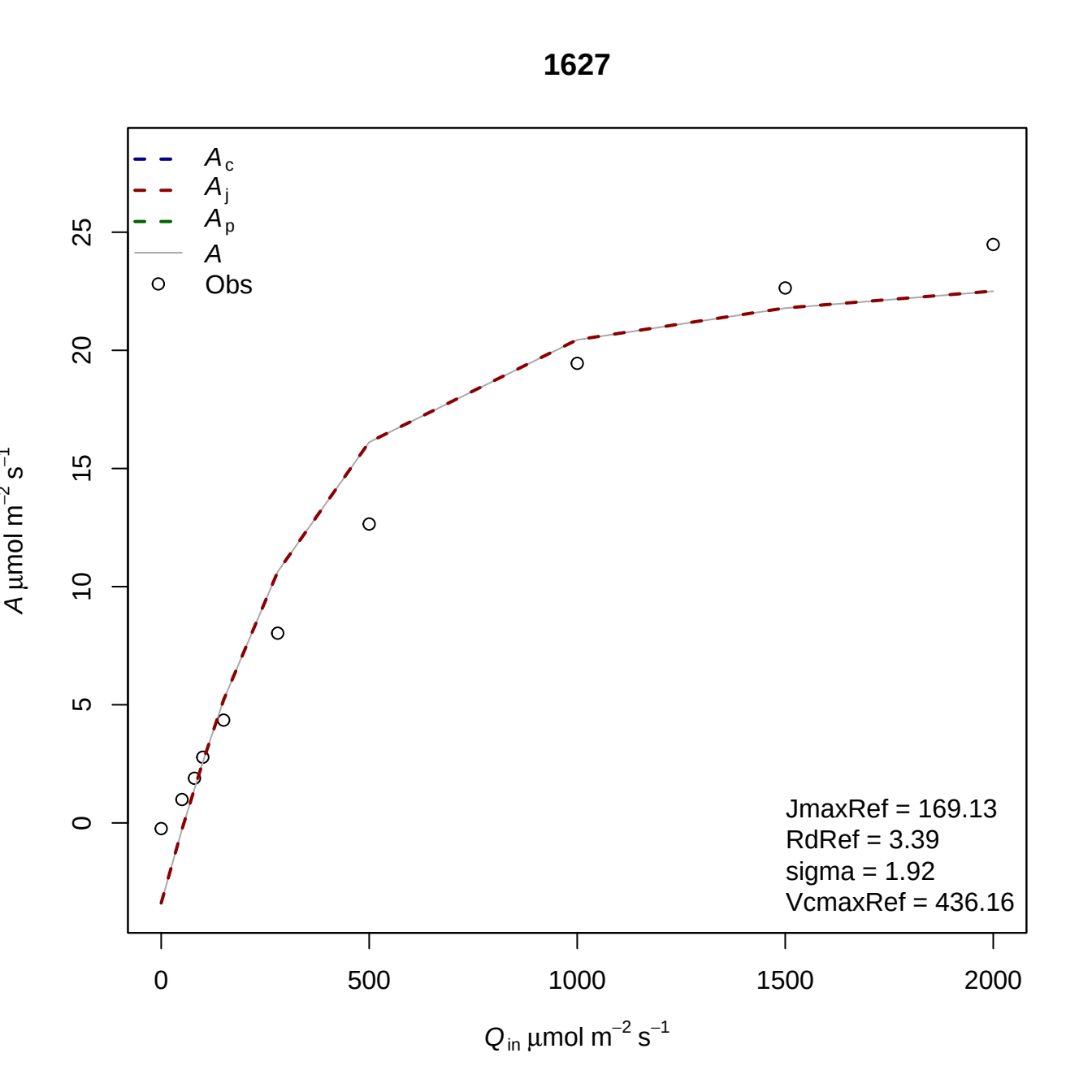

1622
A_c
A_j
A_p
A
Obs
JmaxRef = 232.44
RdRef = 1.31
sigma = 0.42
VcmaxRef = 144.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

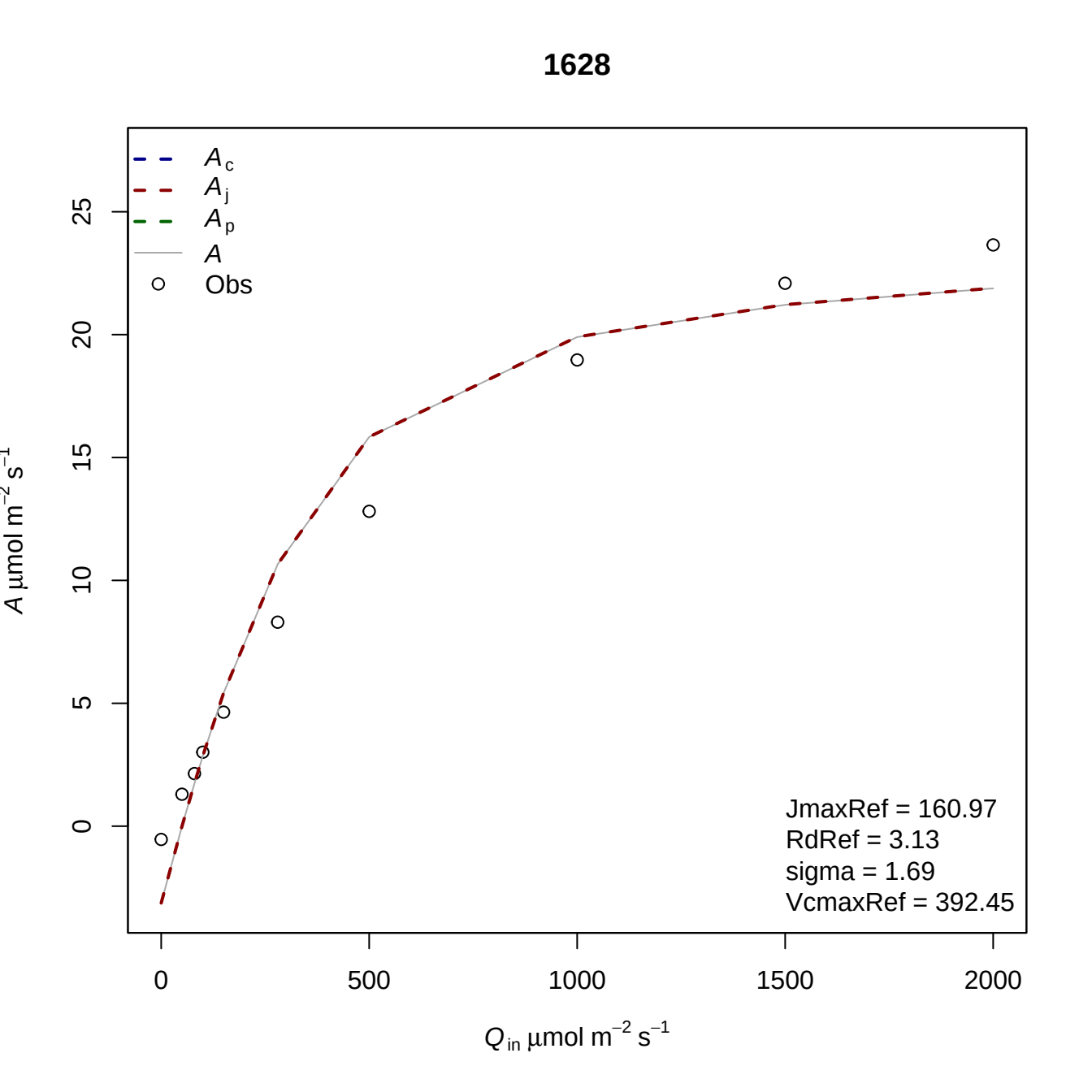

1623
A_c
A_j
A_p
A
Obs
JmaxRef = 84.4
RdRef = 1.82
sigma = 0.67
VcmaxRef = 146.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

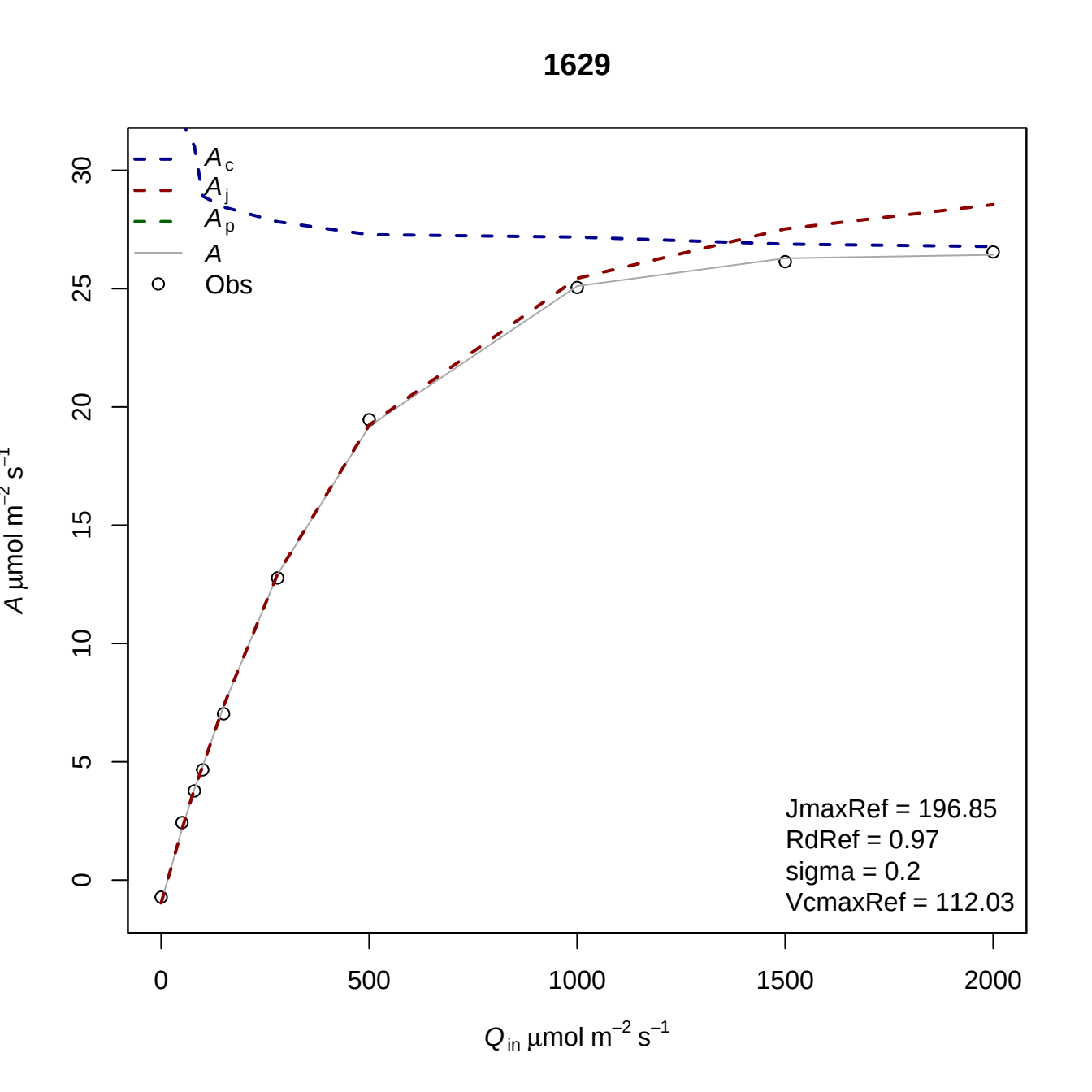

1624
A_c
A_j
A_p
A
Obs
JmaxRef = 85.88
RdRef = 0.68
sigma = 0.14
VcmaxRef = 50.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

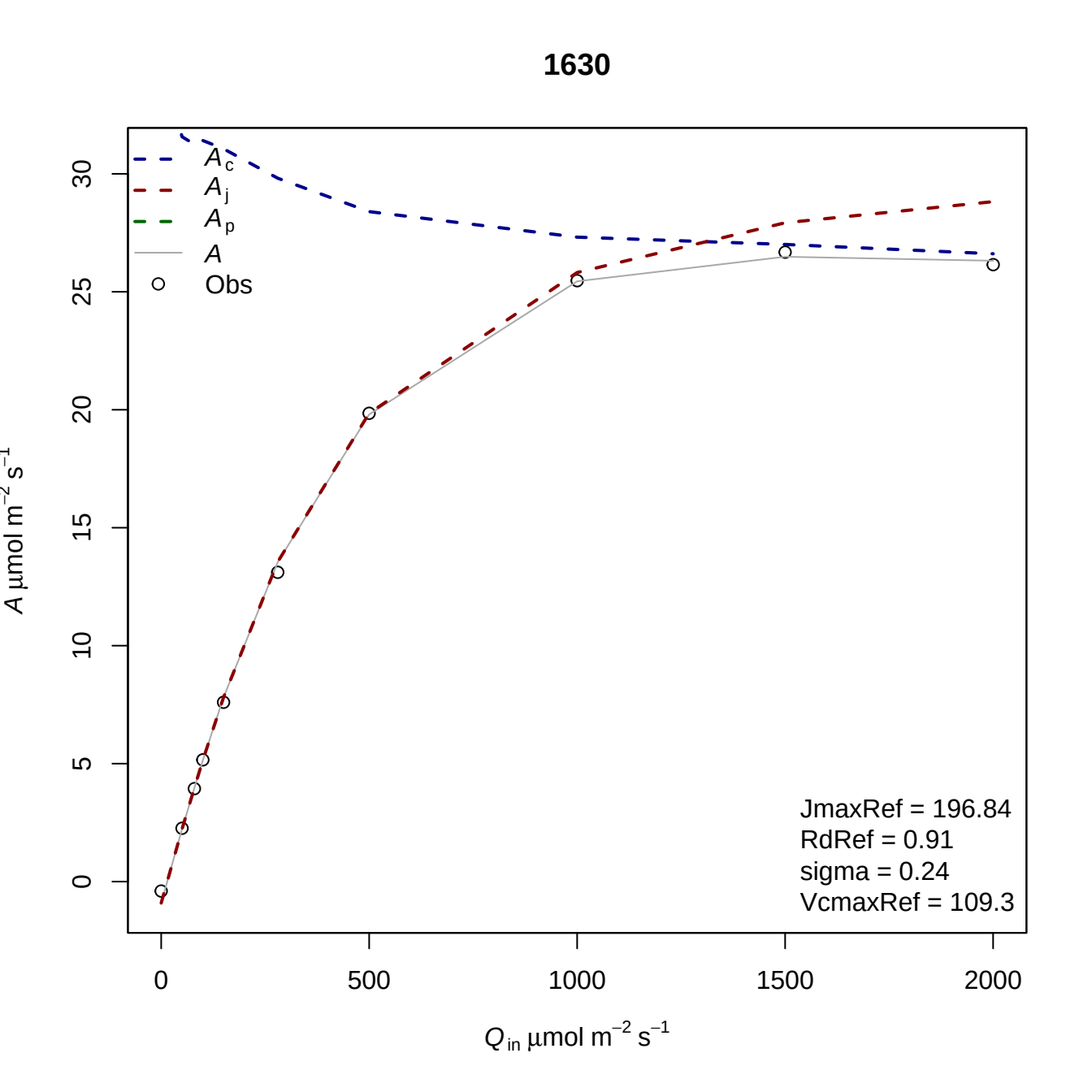

1625
A_c
A_j
A_p
A
Obs
JmaxRef = 70.83
RdRef = 1.62
sigma = 0.48
VcmaxRef = 135.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

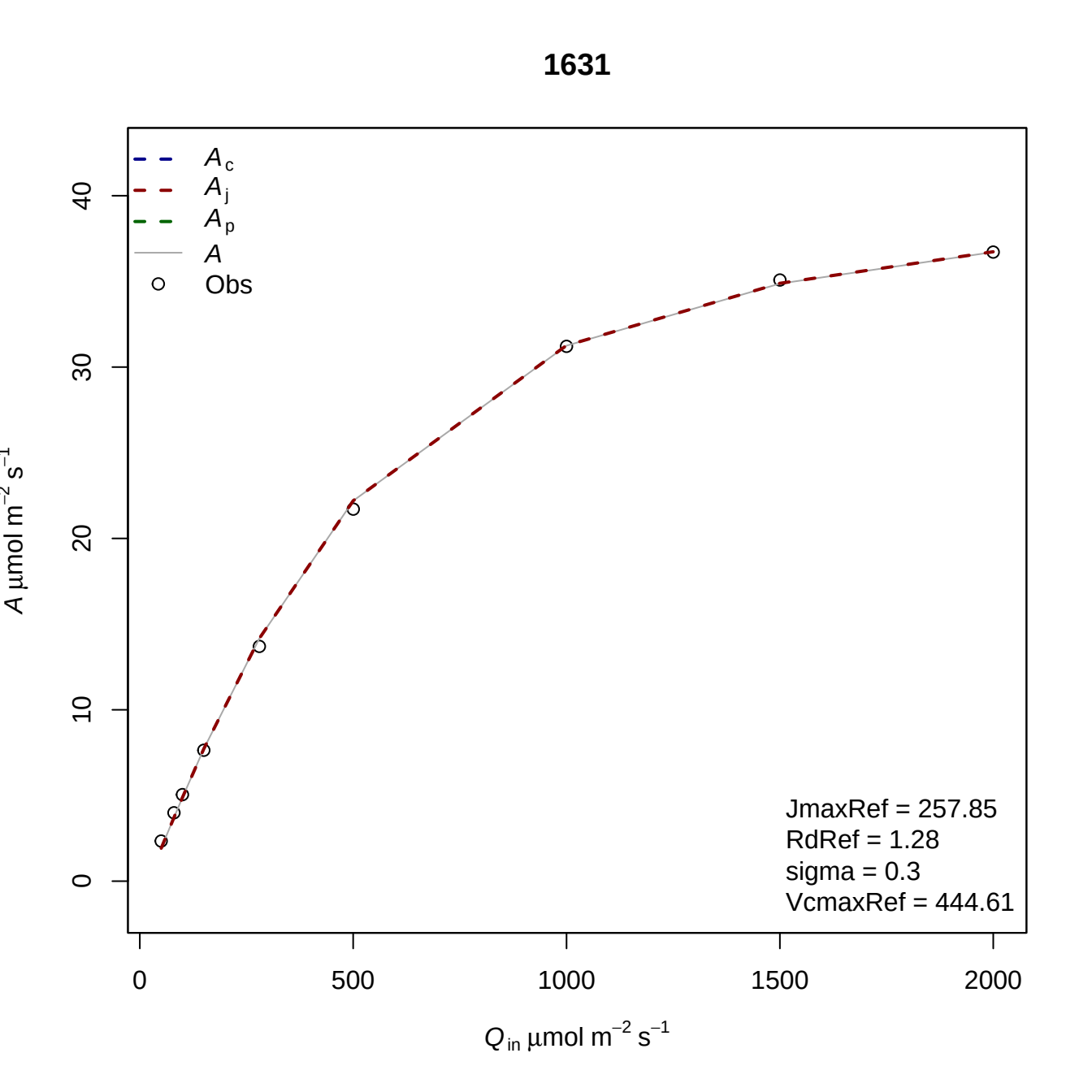

1626
A_c
A_j
A_p
A
Obs
JmaxRef = 83.03
RdRef = 1.34
sigma = 0.38
VcmaxRef = 49.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

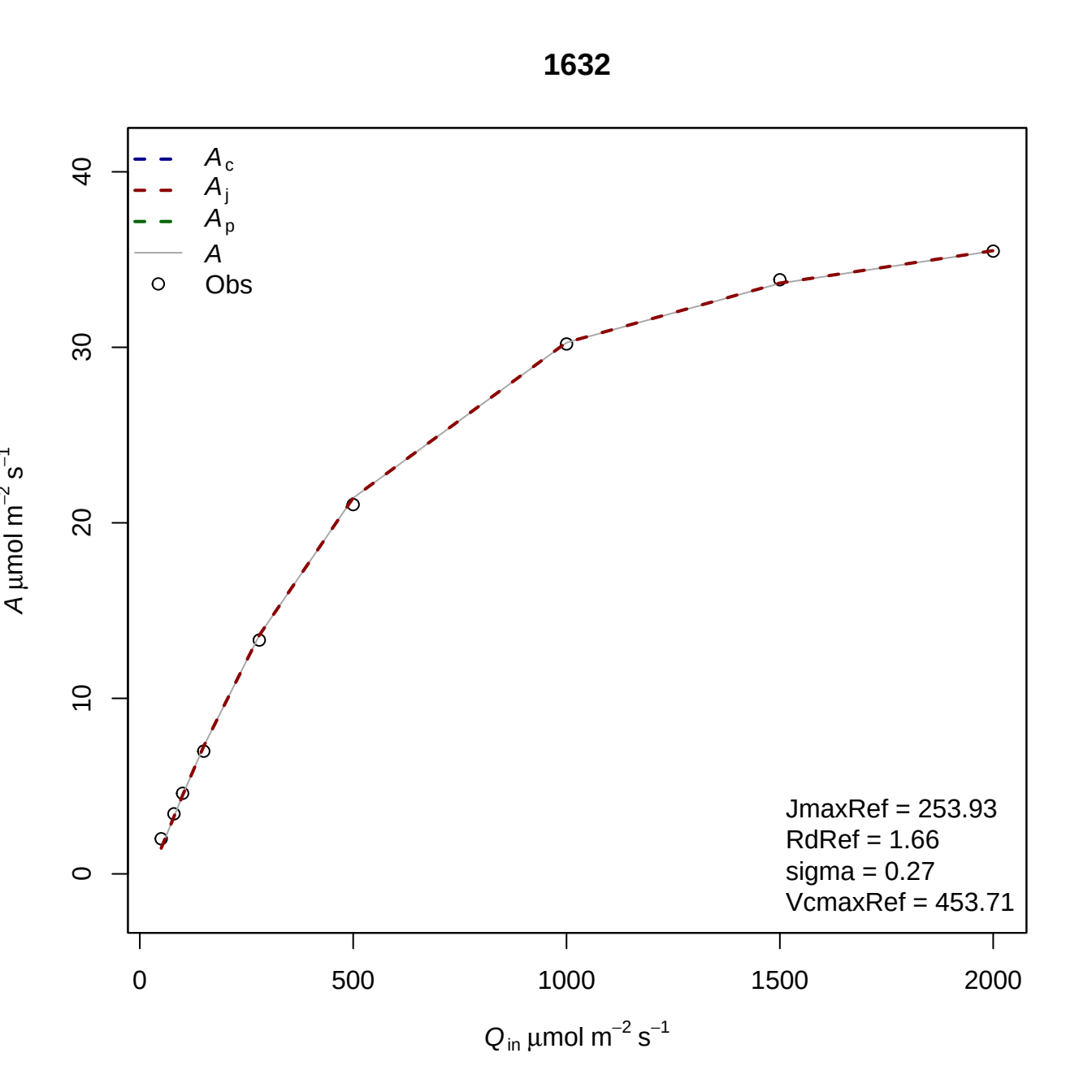

1627
A_c
A_j
A_p
A
Obs
JmaxRef = 169.13
RdRef = 3.39
sigma = 1.92
VcmaxRef = 436.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

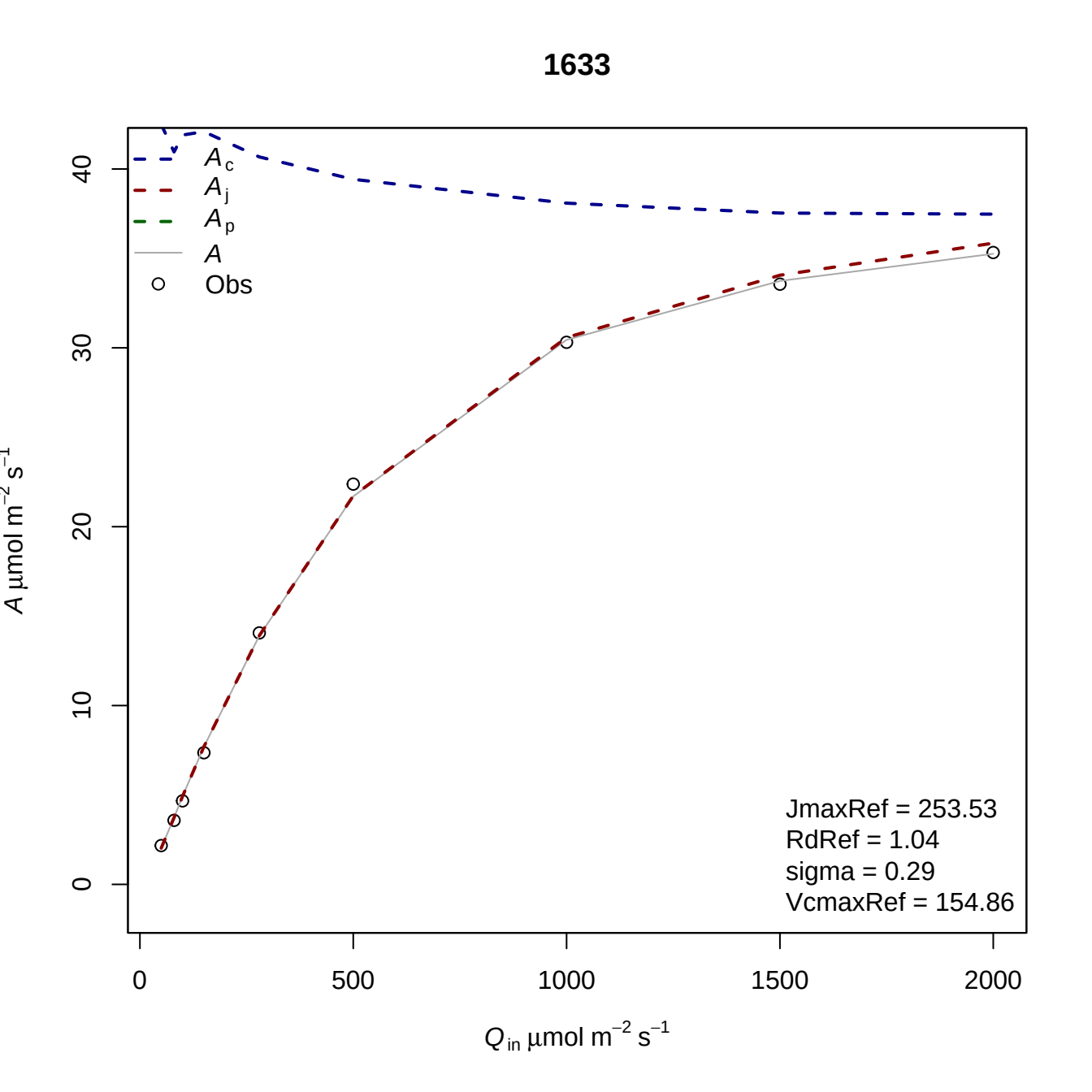

1628
A_c
A_j
A_p
A
Obs
JmaxRef = 160.97
RdRef = 3.13
sigma = 1.69
VcmaxRef = 392.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

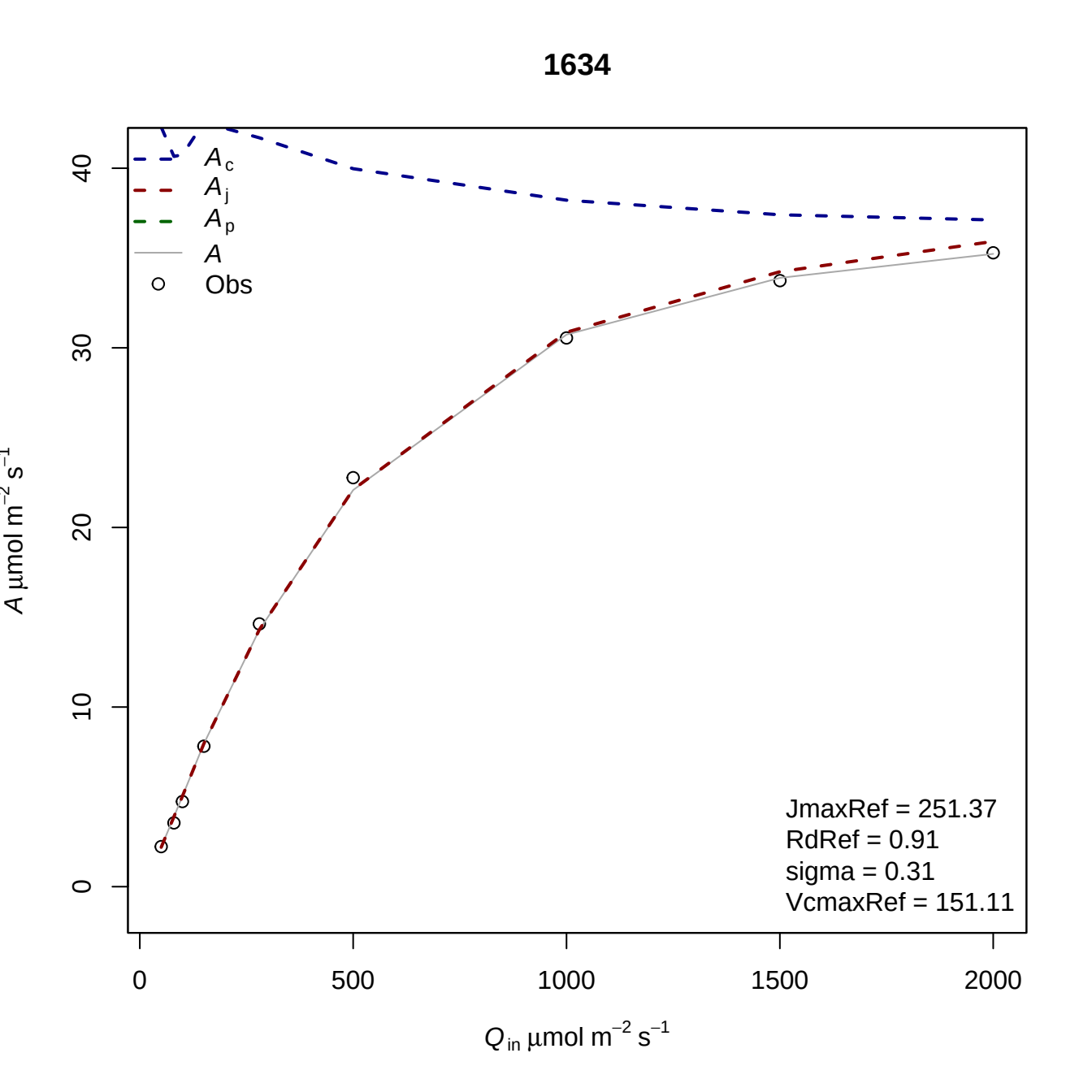

1629
A_c
A_j
A_p
A
Obs
JmaxRef = 196.85
RdRef = 0.97
sigma = 0.2
VcmaxRef = 112.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

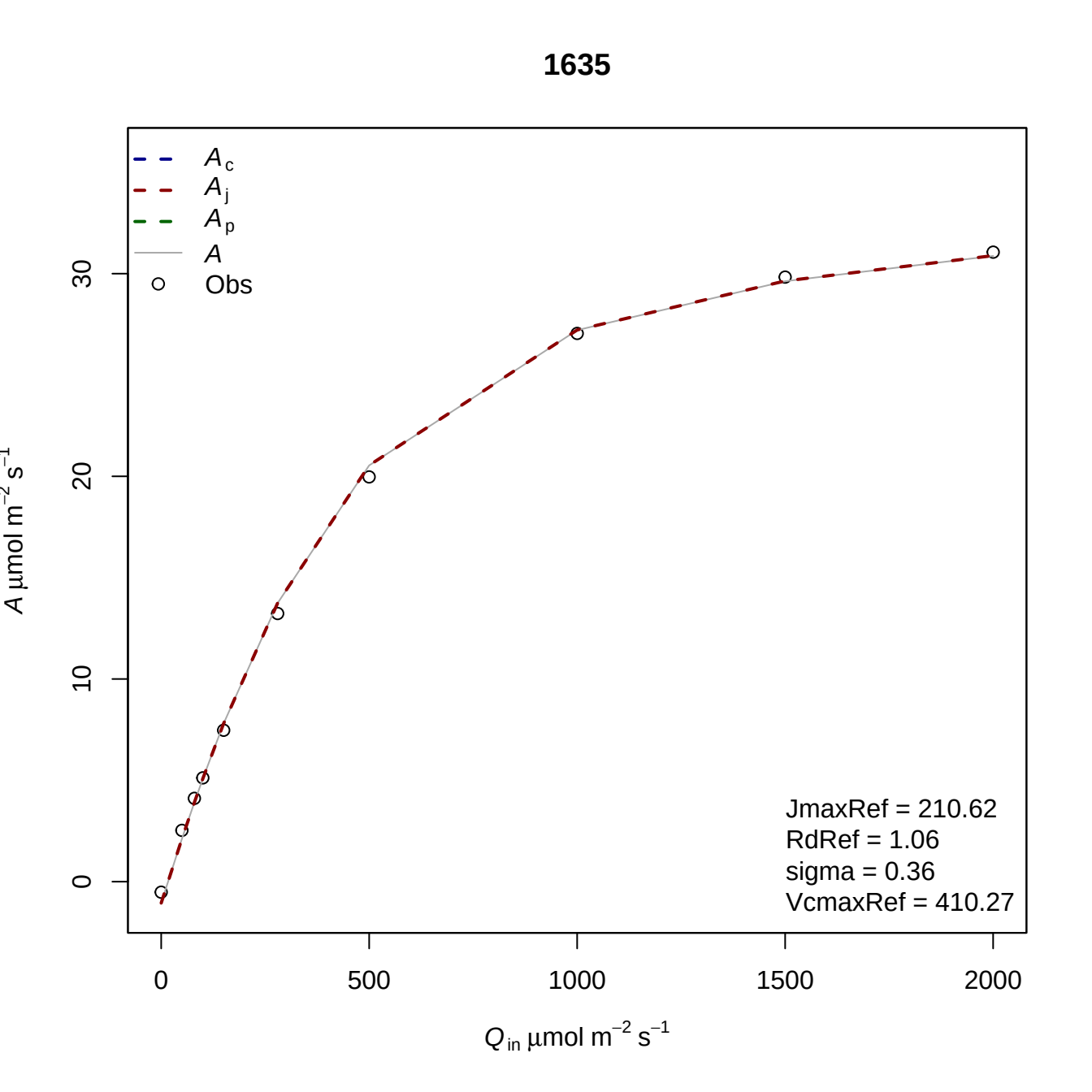

1630
A_c
A_j
A_p
A
Obs
JmaxRef = 196.84
RdRef = 0.91
sigma = 0.24
VcmaxRef = 109.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

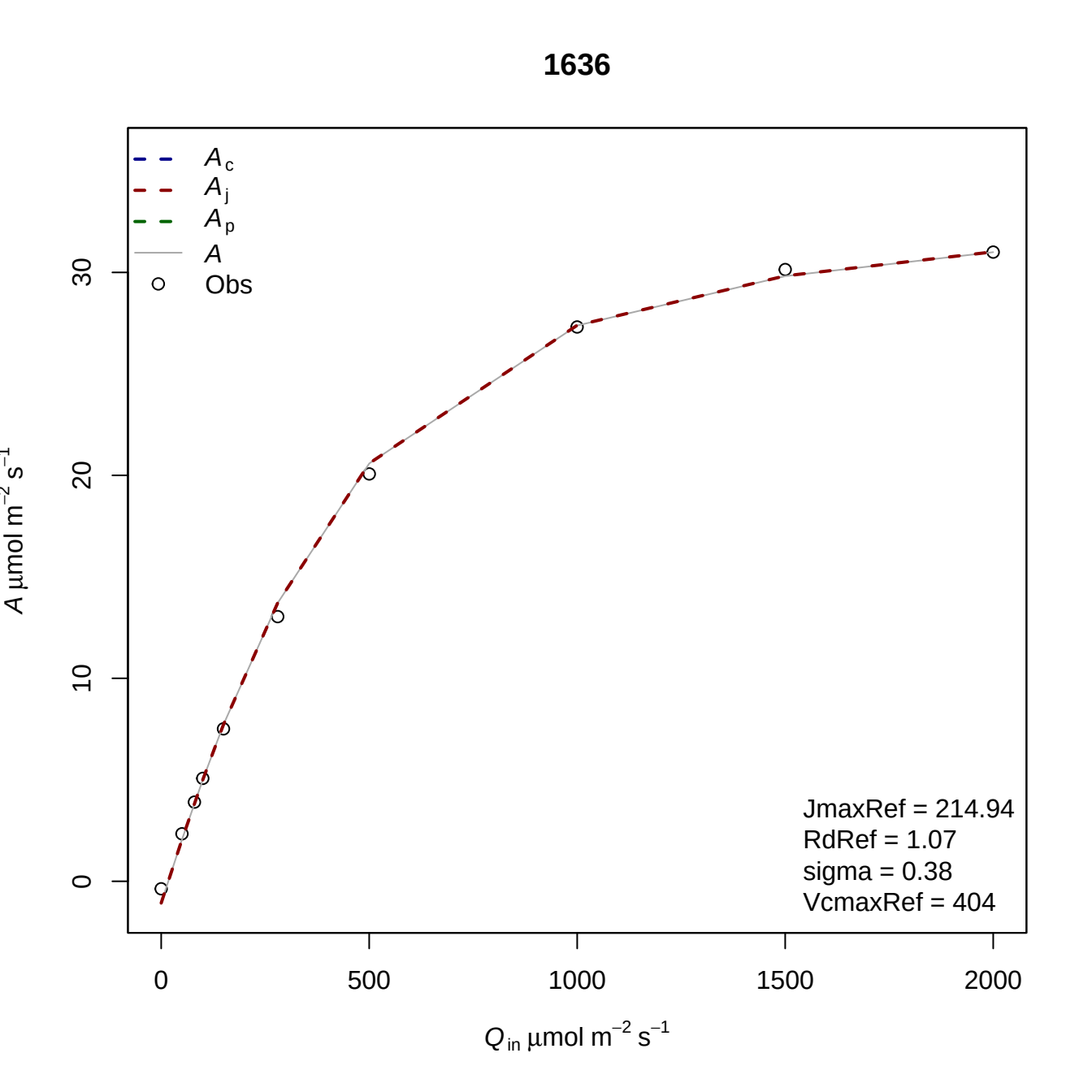

1631
A_c
A_j
A_p
A
Obs
JmaxRef = 257.85
RdRef = 1.28
sigma = 0.3
VcmaxRef = 444.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

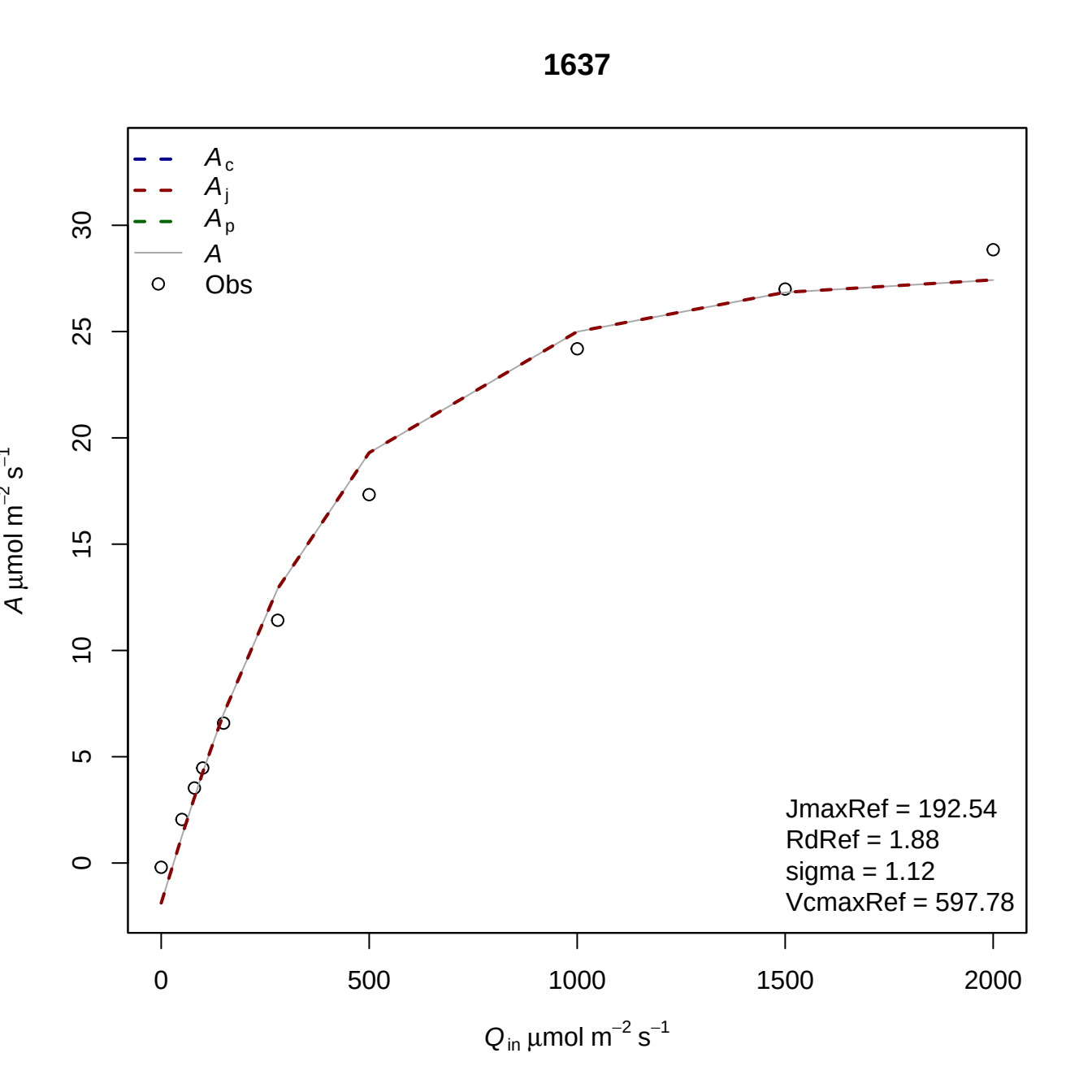

1632
A_c
A_j
A_p
A
Obs
JmaxRef = 253.93
RdRef = 1.66
sigma = 0.27
VcmaxRef = 453.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

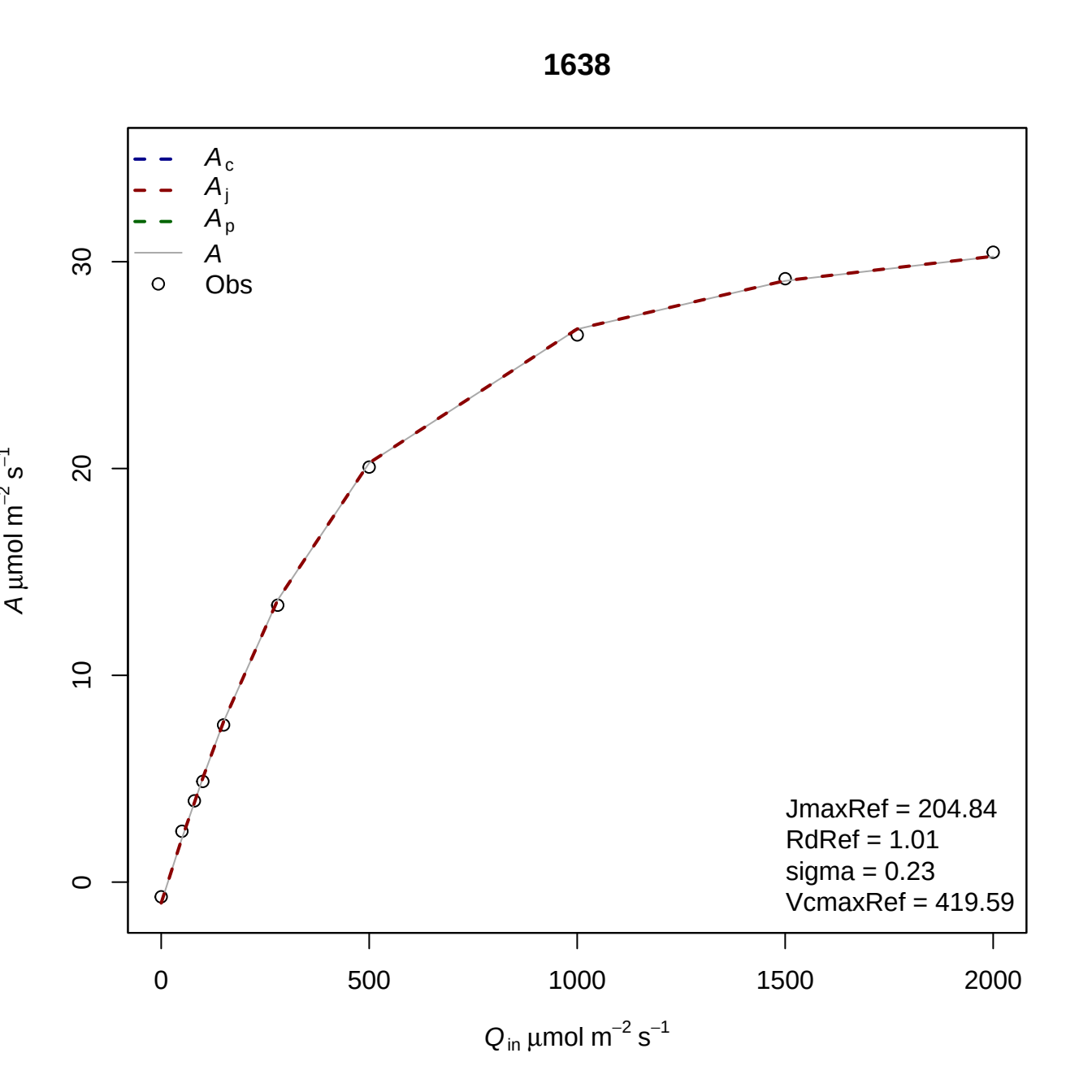

1633
A_c
A_j
A_p
A
Obs
JmaxRef = 253.53
RdRef = 1.04
sigma = 0.29
VcmaxRef = 154.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

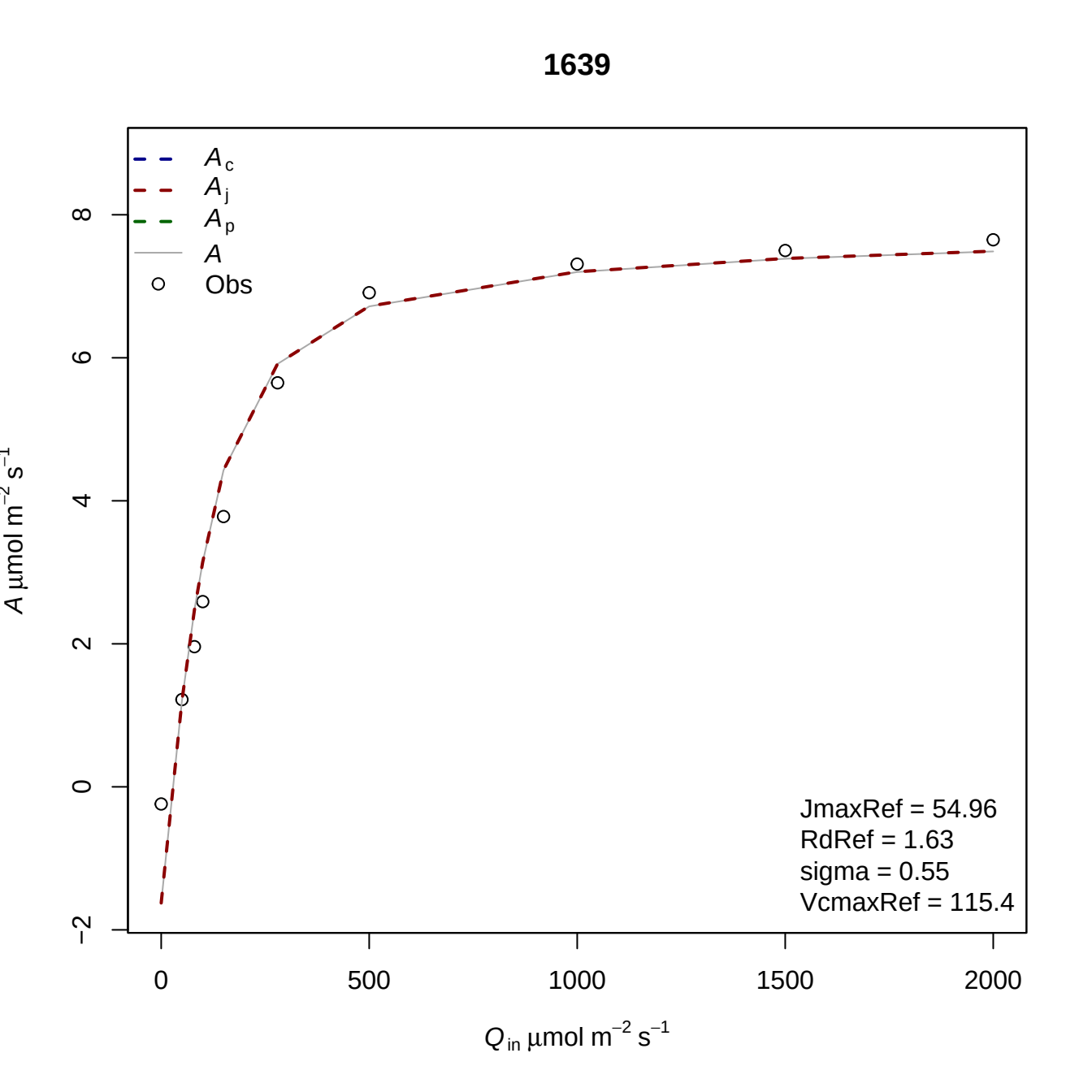

1634
A_c
A_j
A_p
A
Obs
JmaxRef = 251.37
RdRef = 0.91
sigma = 0.31
VcmaxRef = 151.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

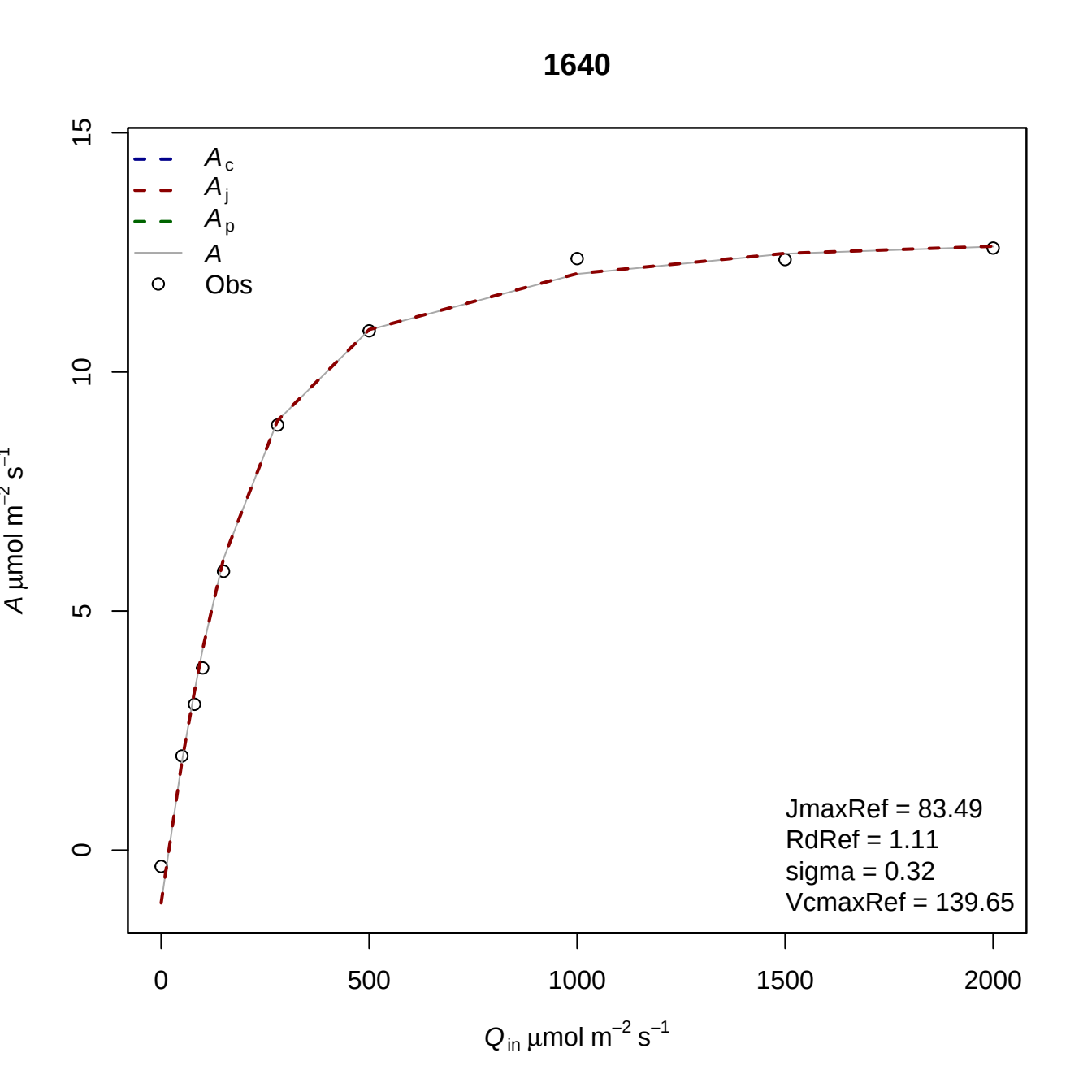

1635
A_c
A_j
A_p
A
Obs
JmaxRef = 210.62
RdRef = 1.06
sigma = 0.36
VcmaxRef = 410.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

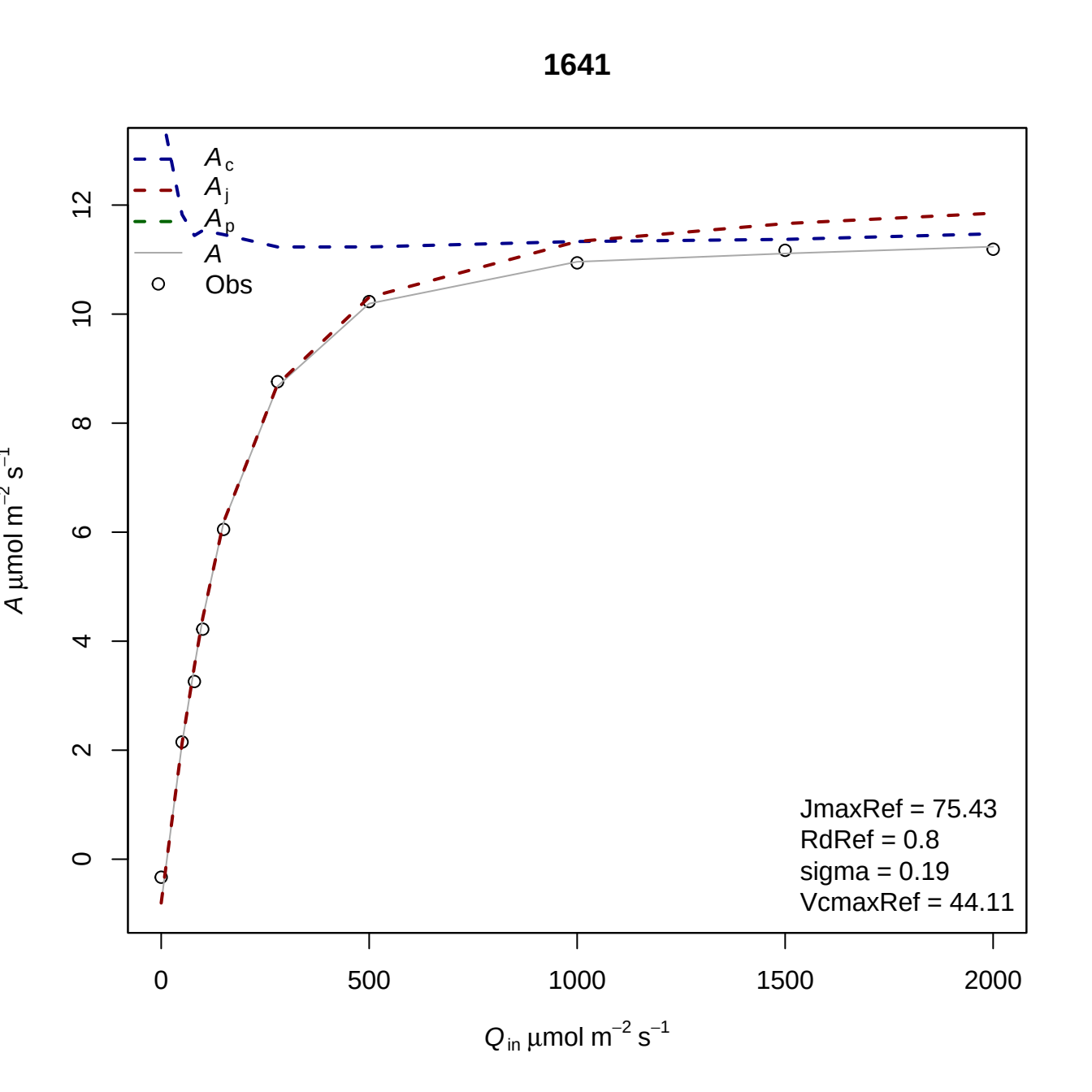

1636
A_c
A_j
A_p
A
Obs
JmaxRef = 214.94
RdRef = 1.07
sigma = 0.38
VcmaxRef = 404
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

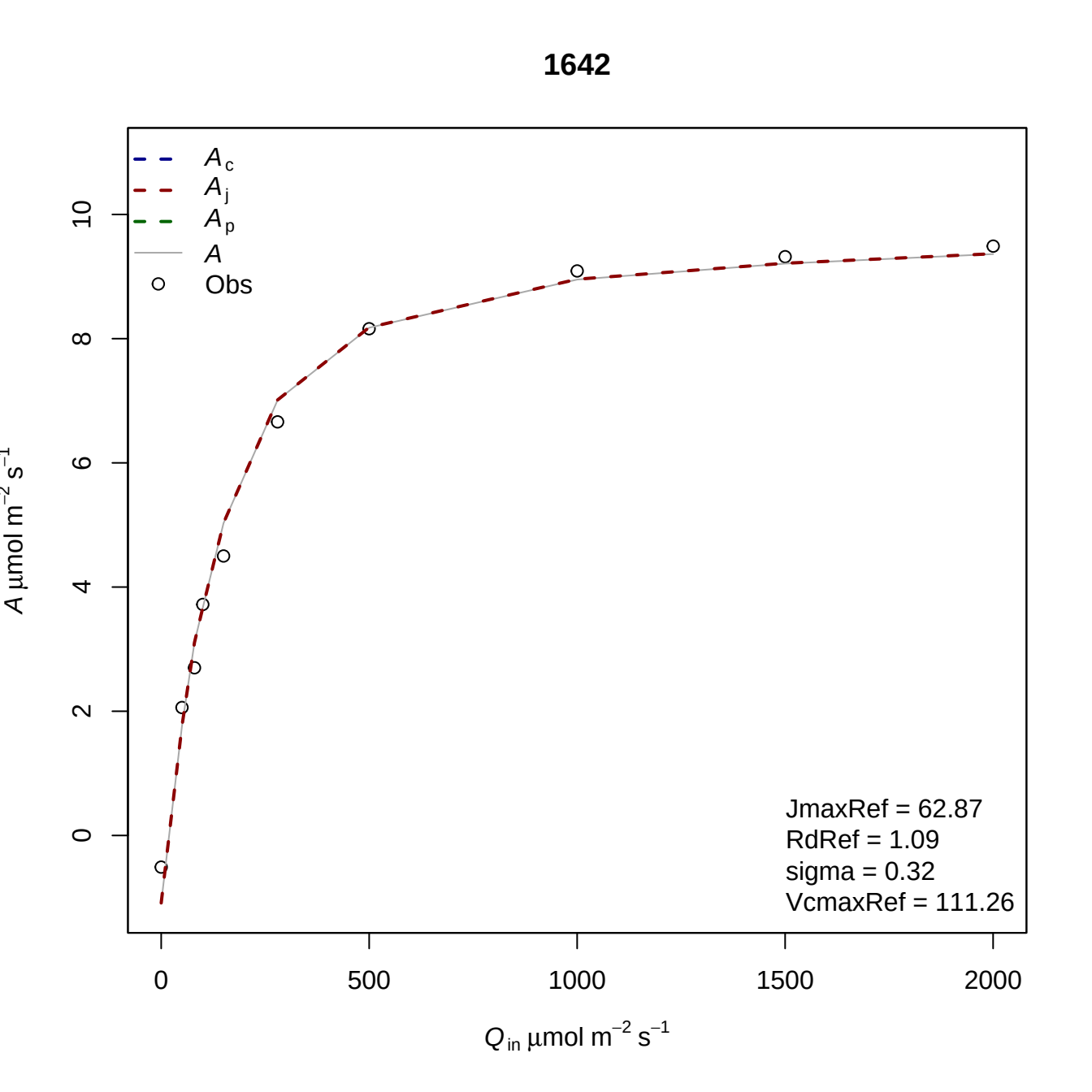

1637
A_c
A_j
A_p
A
Obs
JmaxRef = 192.54
RdRef = 1.88
sigma = 1.12
VcmaxRef = 597.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

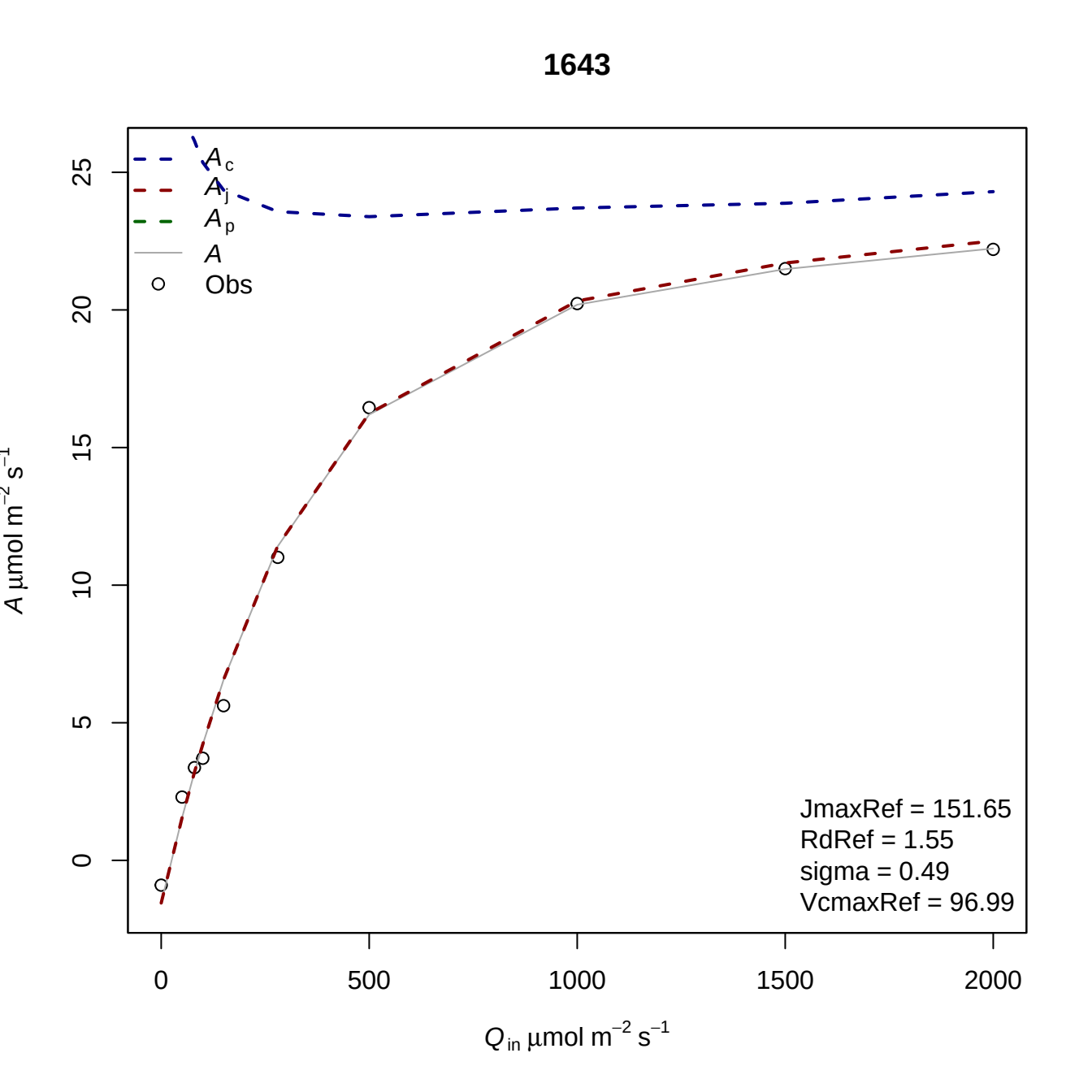

1638
A_c
A_j
A_p
A
Obs
JmaxRef = 204.84
RdRef = 1.01
sigma = 0.23
VcmaxRef = 419.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

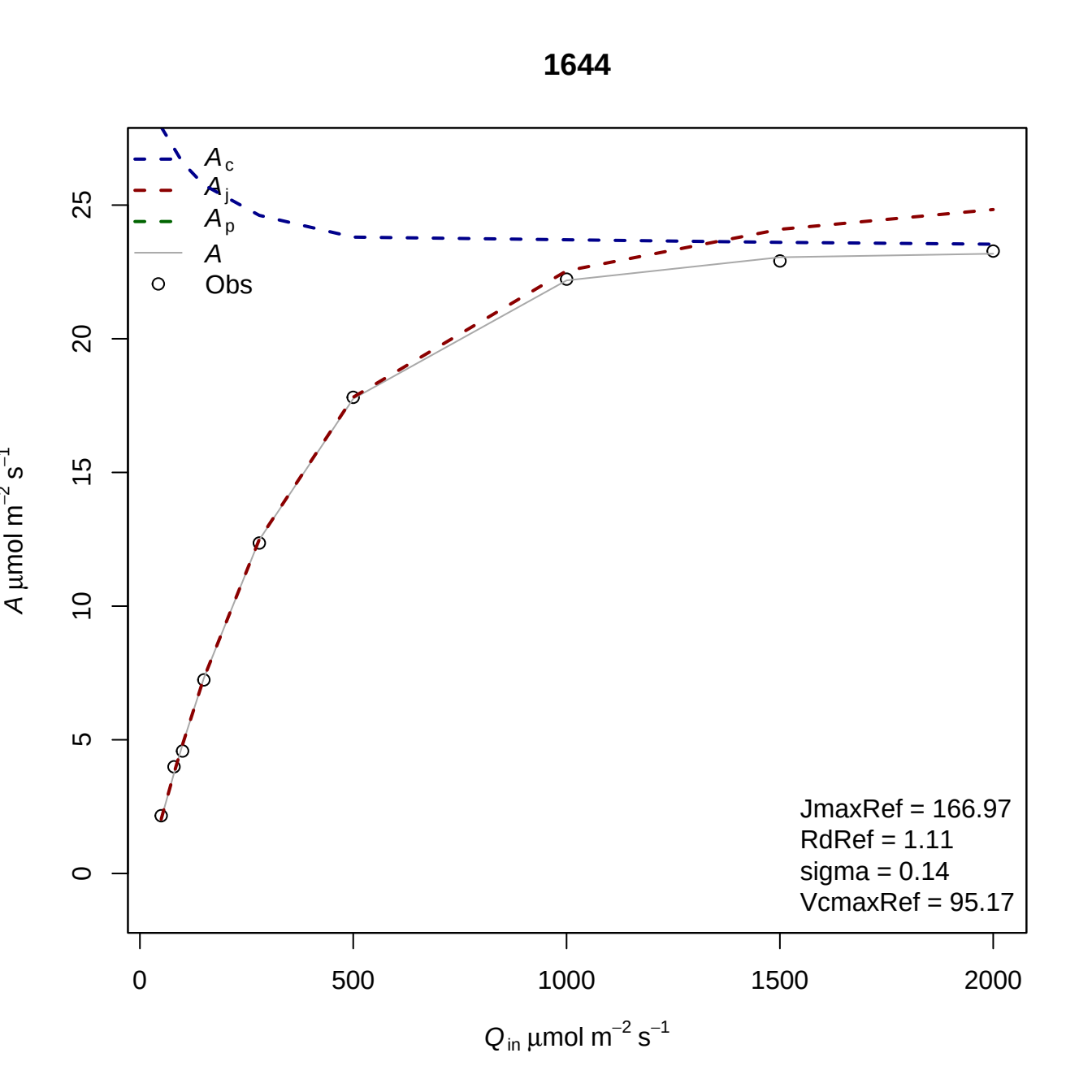

1639
A_c
A_j
A_p
A
Obs
JmaxRef = 54.96
RdRef = 1.63
sigma = 0.55
VcmaxRef = 115.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

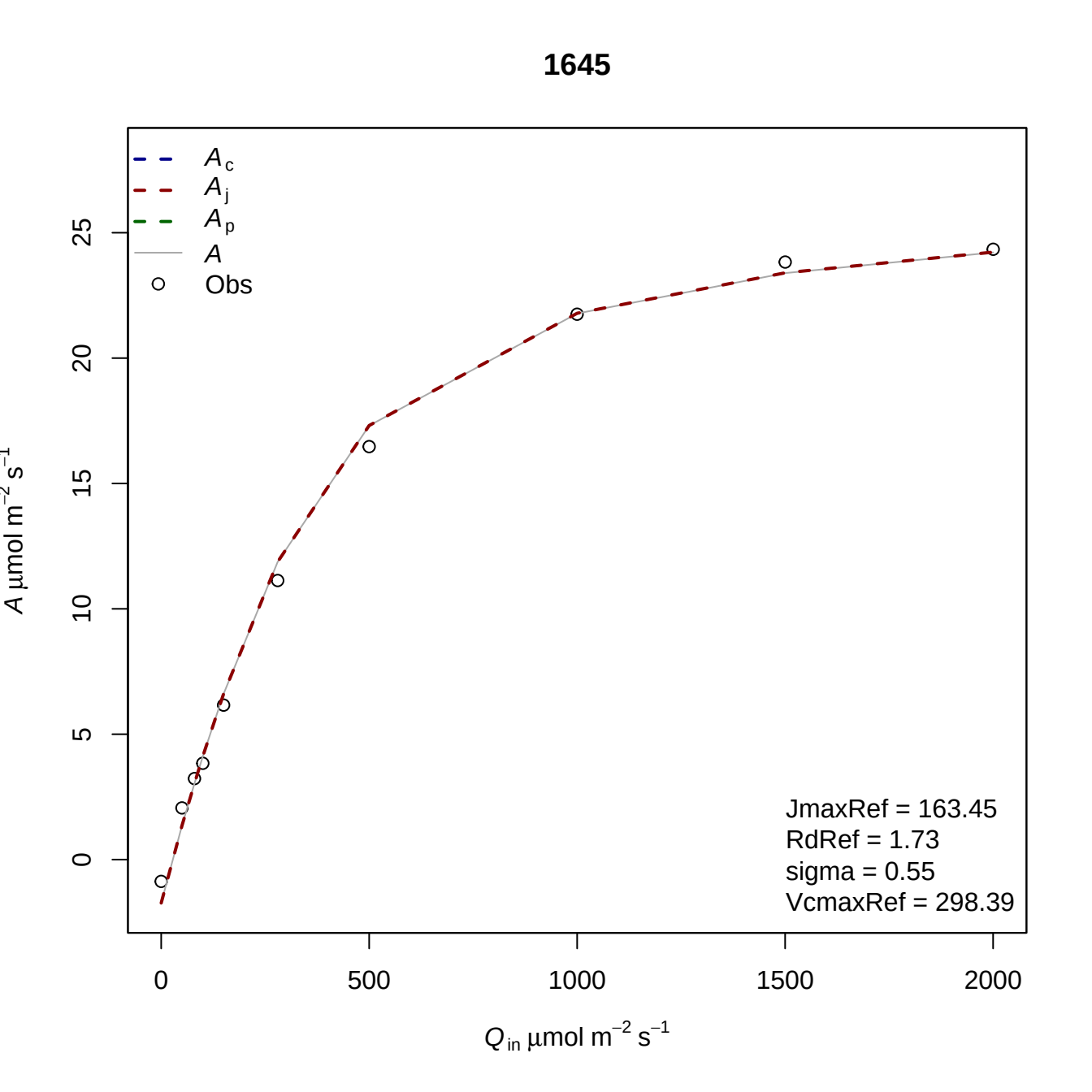

1640
A_c
A_j
A_p
A
Obs
JmaxRef = 83.49
RdRef = 1.11
sigma = 0.32
VcmaxRef = 139.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

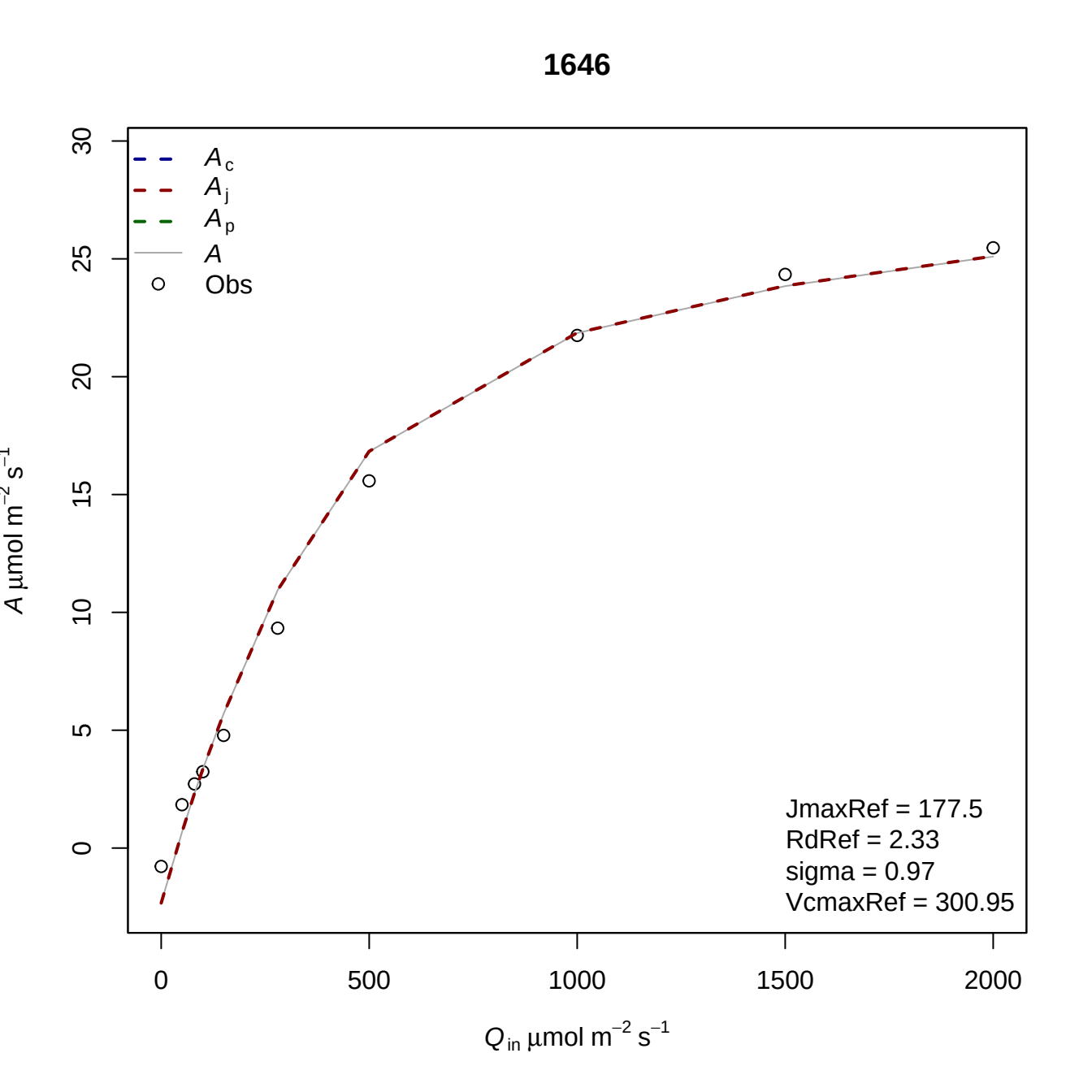

1641
A_c
A_j
A_p
A
Obs
JmaxRef = 75.43
RdRef = 0.8
sigma = 0.19
VcmaxRef = 44.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

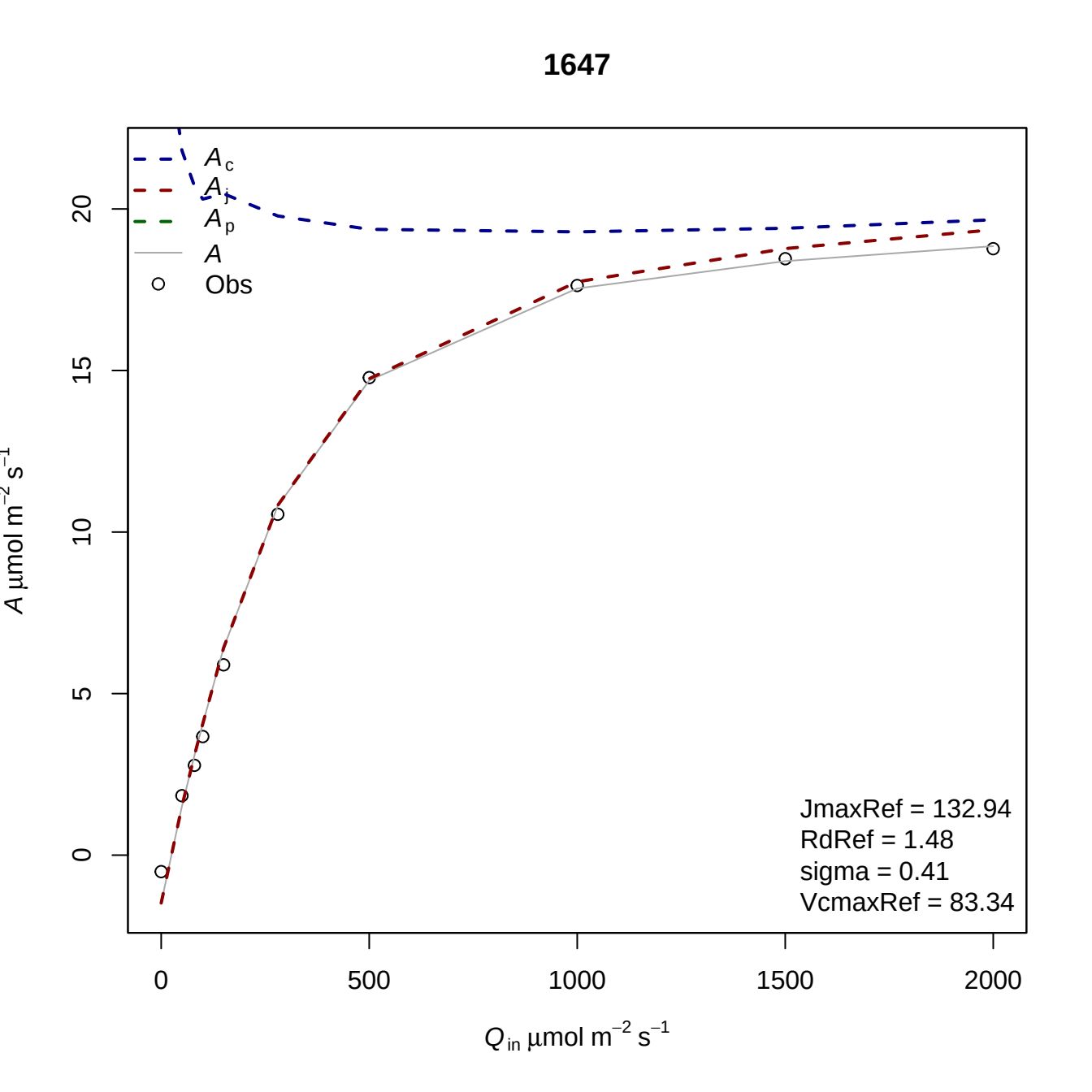

1642
A_c
A_j
A_p
A
Obs
JmaxRef = 62.87
RdRef = 1.09
sigma = 0.32
VcmaxRef = 111.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

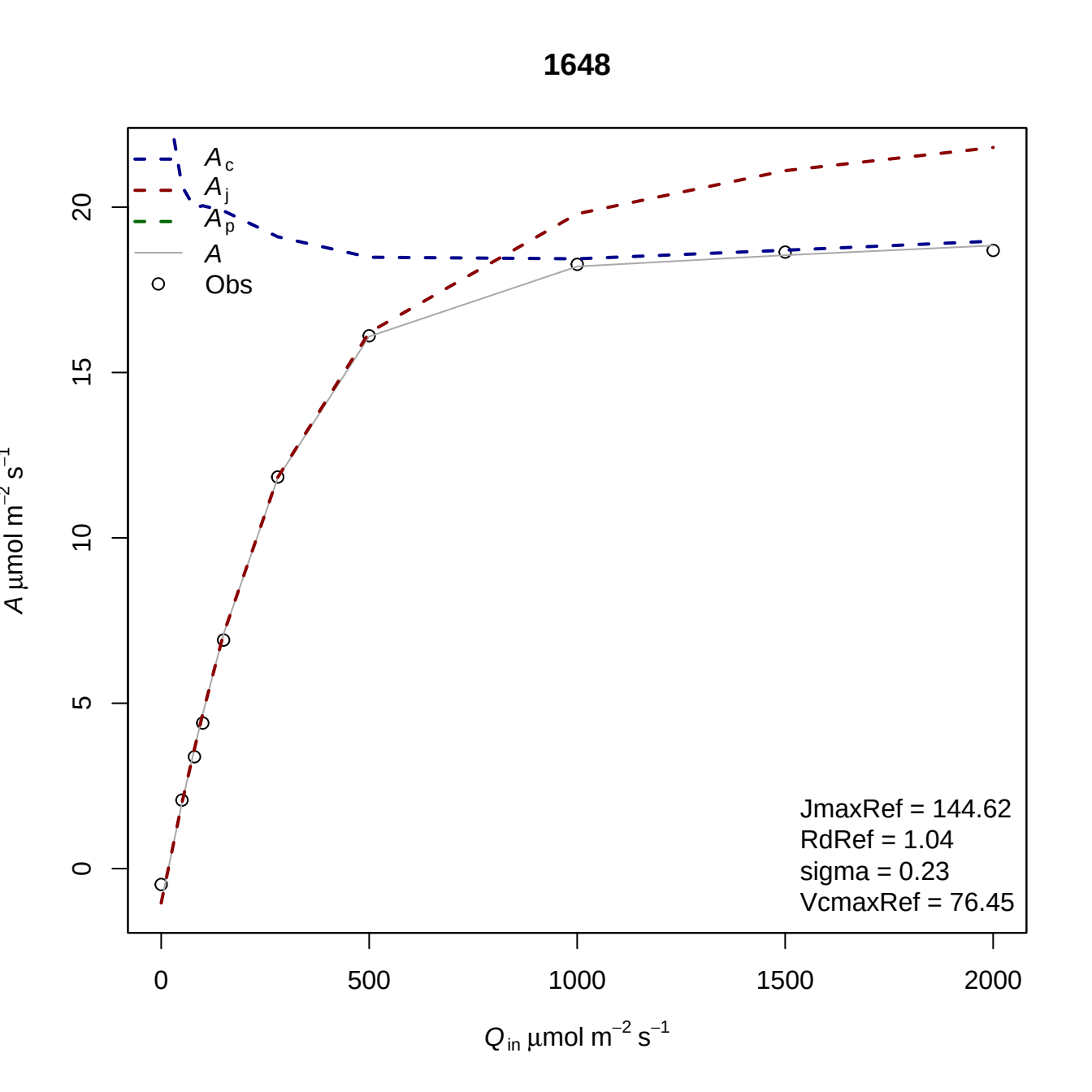

1643
A_c
A_j
A_p
A
Obs
JmaxRef = 151.65
RdRef = 1.55
sigma = 0.49
VcmaxRef = 96.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

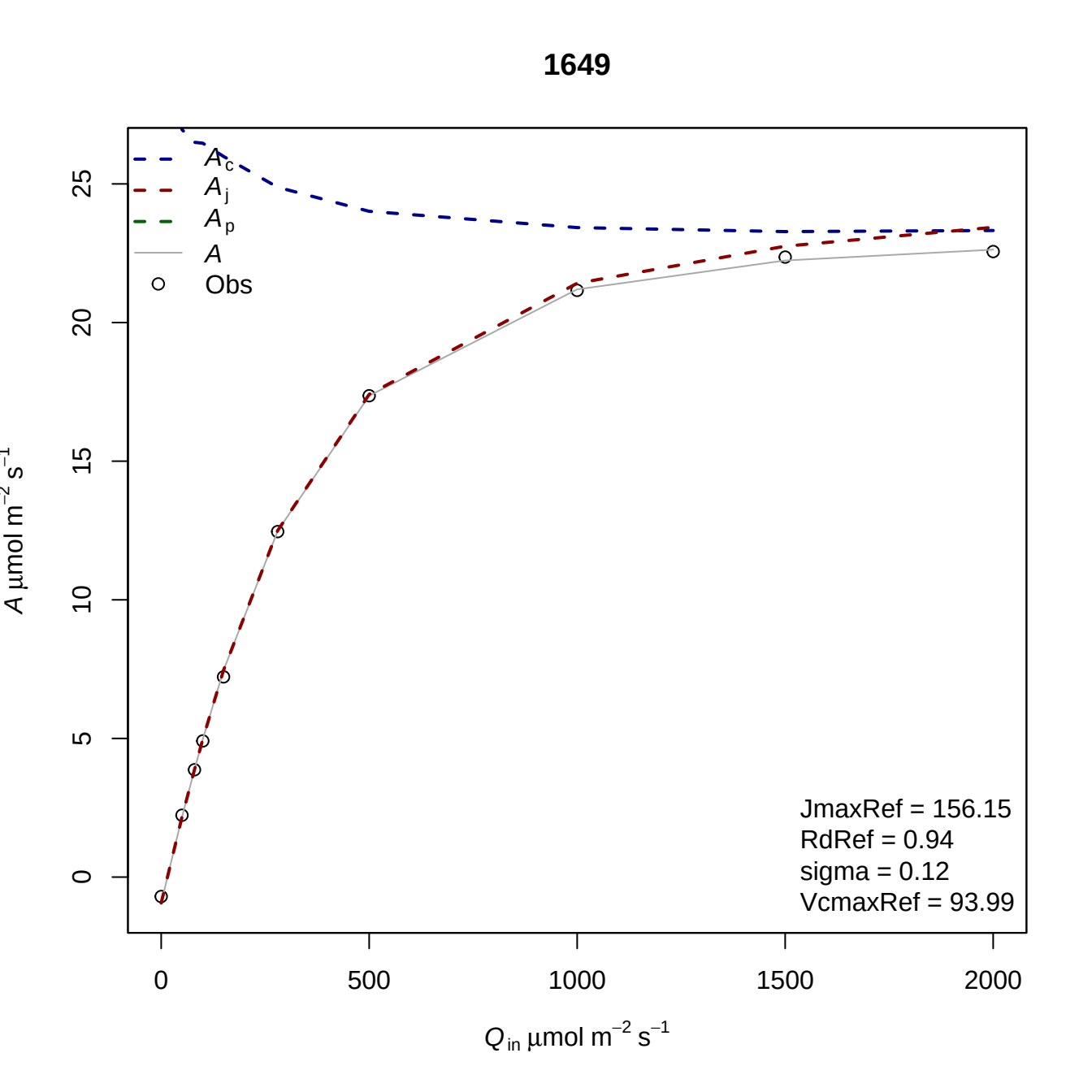

1644
A_c
A_j
A_p
A
Obs
JmaxRef = 166.97
RdRef = 1.11
sigma = 0.14
VcmaxRef = 95.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

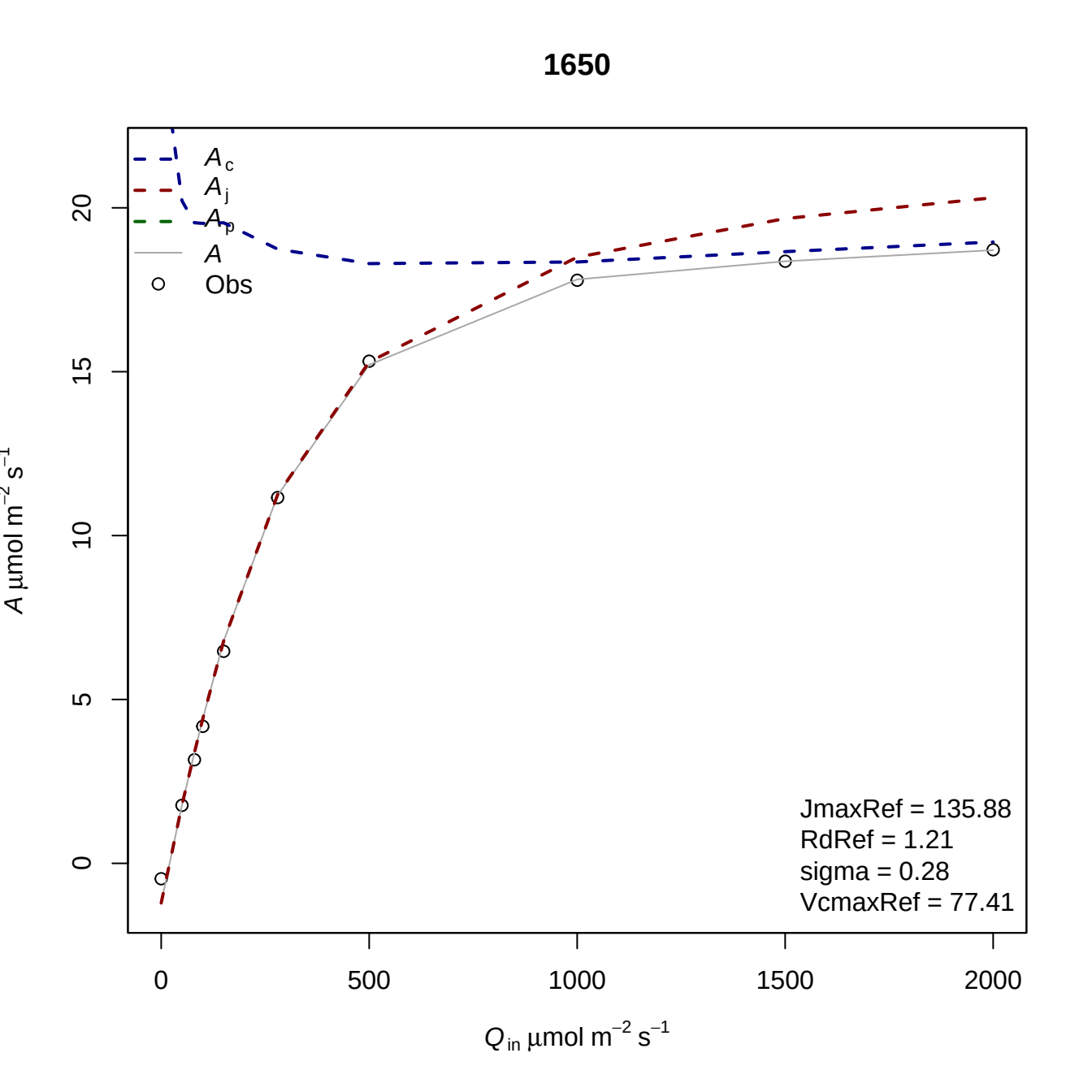

1645
A_c
A_j
A_p
A
Obs
JmaxRef = 163.45
RdRef = 1.73
sigma = 0.55
VcmaxRef = 298.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

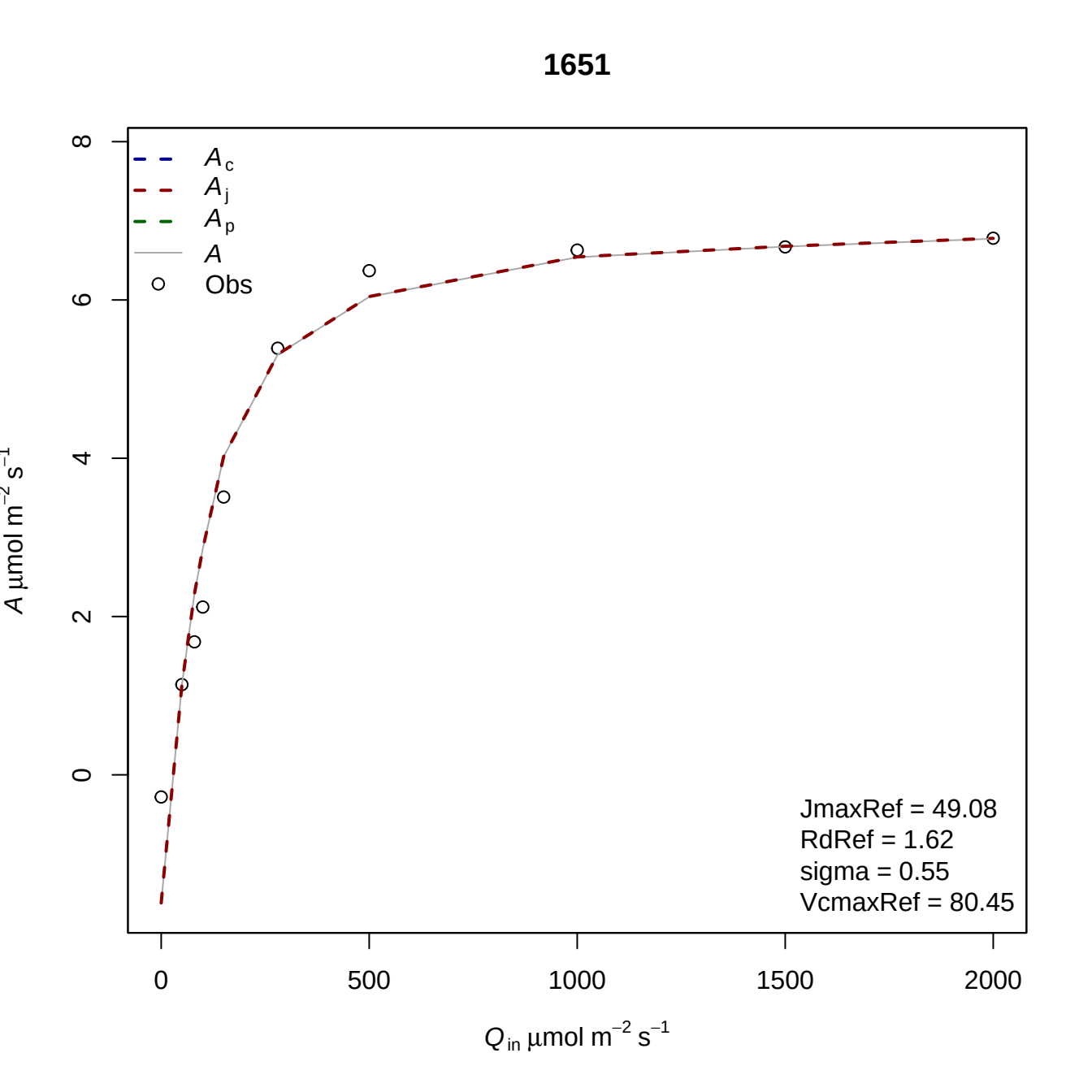

1646
A_c
A_j
A_p
A
Obs
JmaxRef = 177.5
RdRef = 2.33
sigma = 0.97
VcmaxRef = 300.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

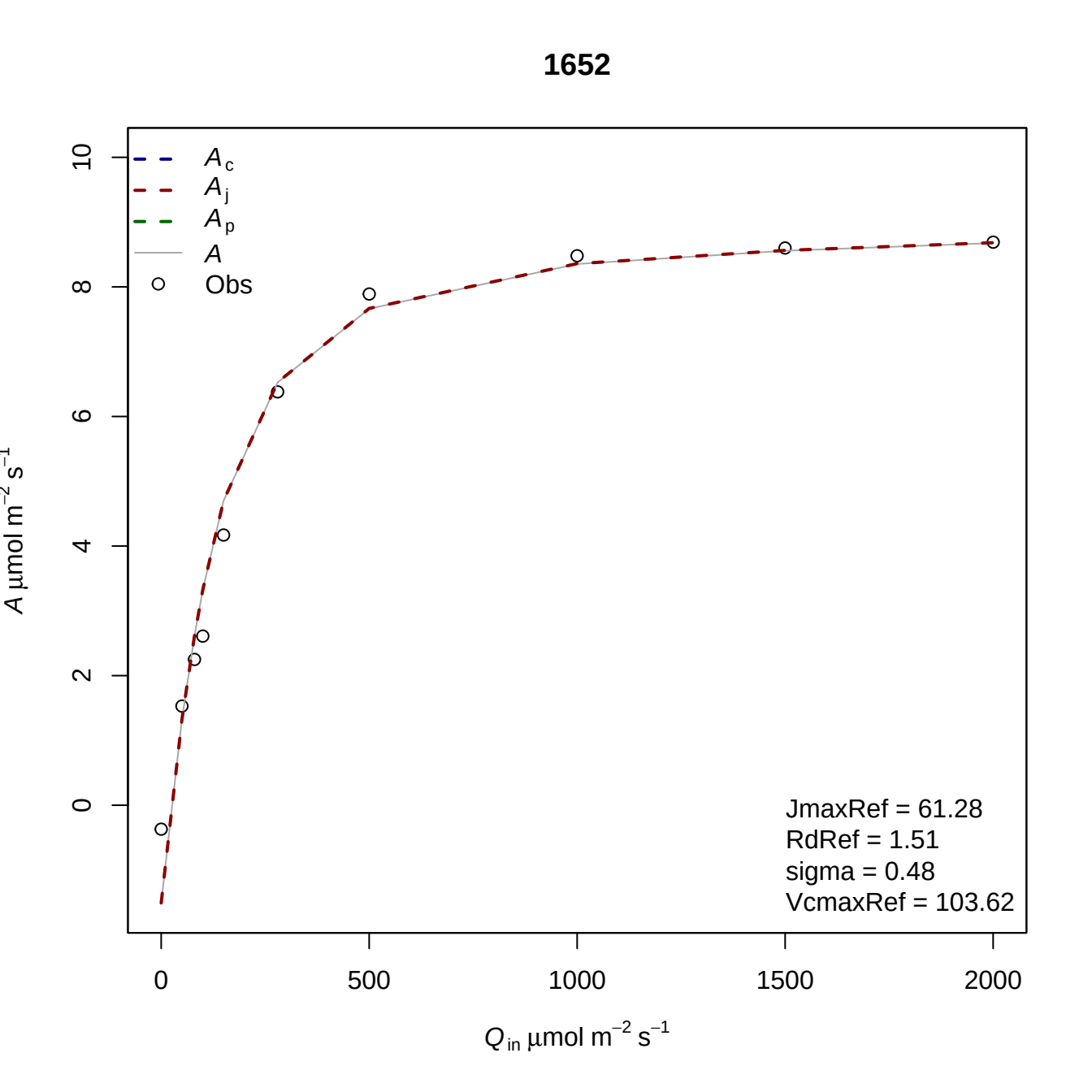

1647
A_c
A_j
A_p
A
Obs
JmaxRef = 132.94
RdRef = 1.48
sigma = 0.41
VcmaxRef = 83.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

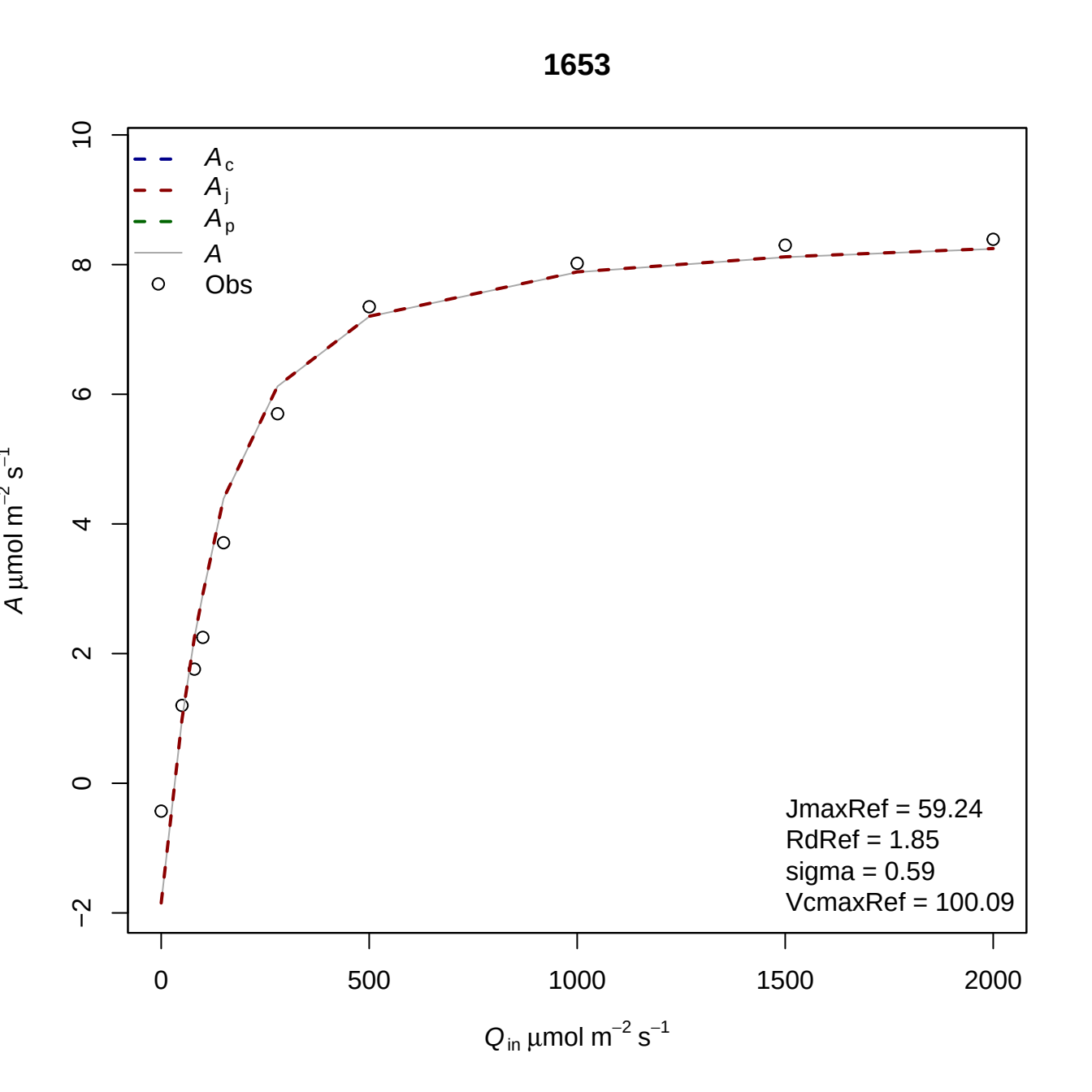

1648
A_c
A_j
A_p
A
Obs
JmaxRef = 144.62
RdRef = 1.04
sigma = 0.23
VcmaxRef = 76.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

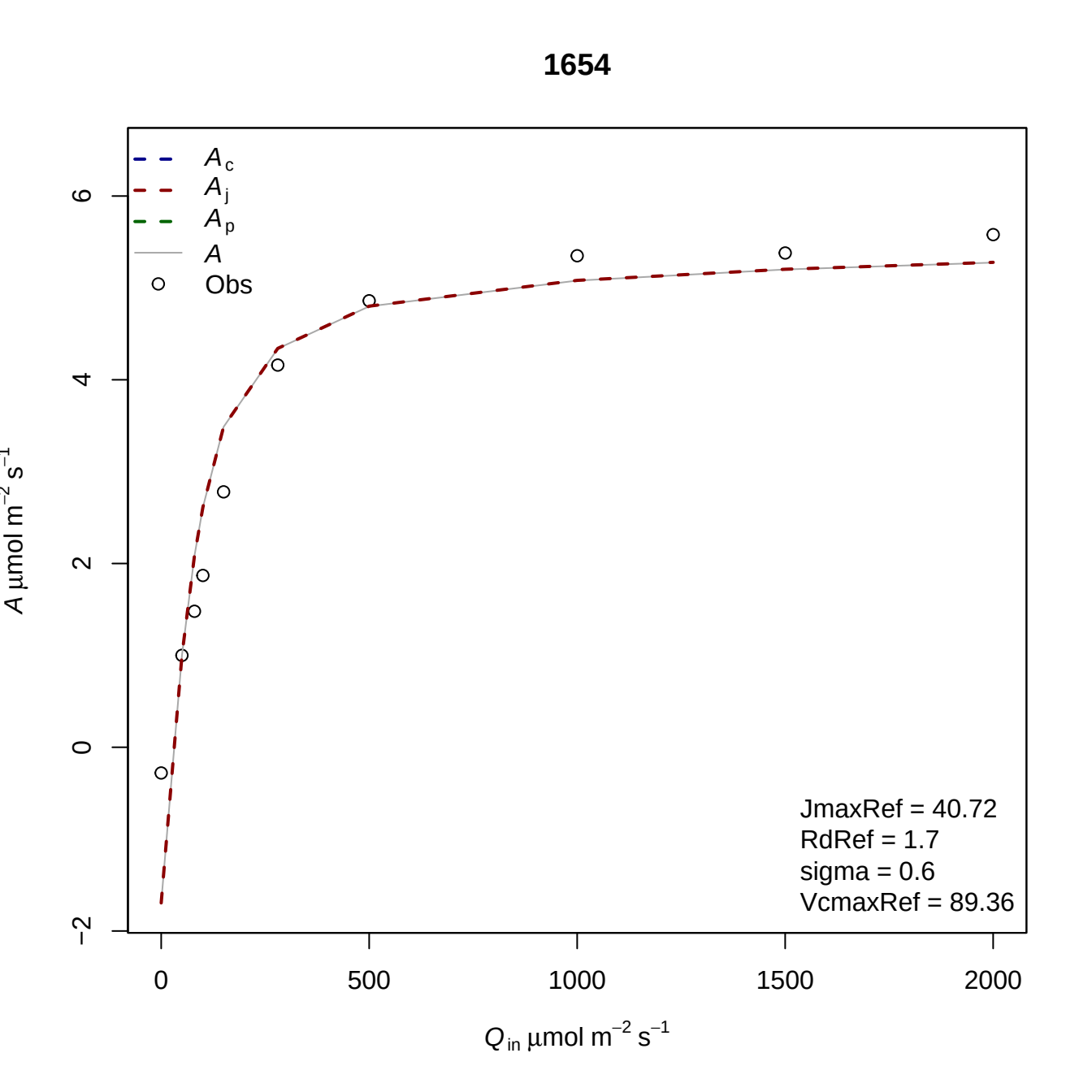

1649
A_c
A_j
A_p
A
Obs
JmaxRef = 156.15
RdRef = 0.94
sigma = 0.12
VcmaxRef = 93.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

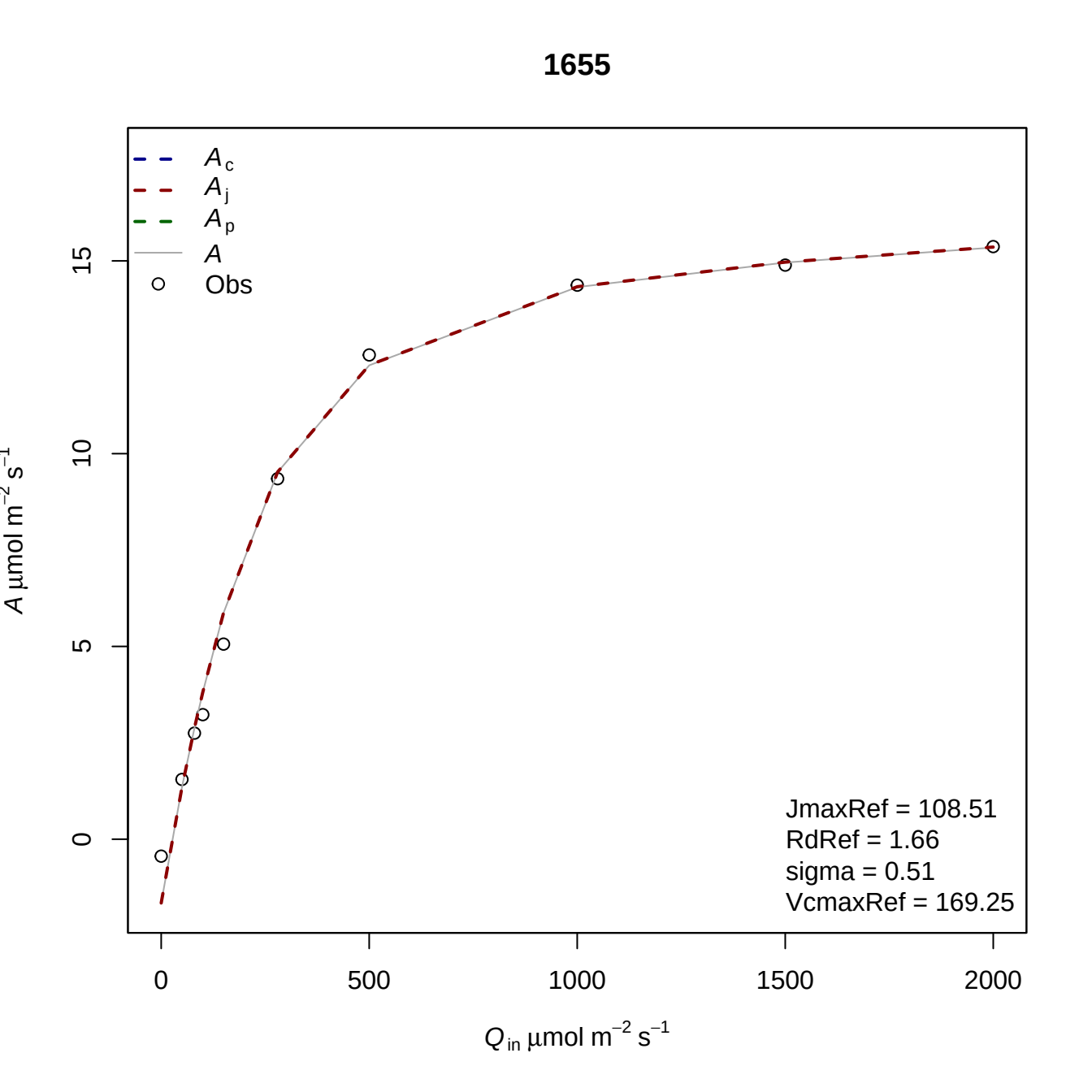

1650
A_c
A_j
A_p
A
Obs
JmaxRef = 135.88
RdRef = 1.21
sigma = 0.28
VcmaxRef = 77.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

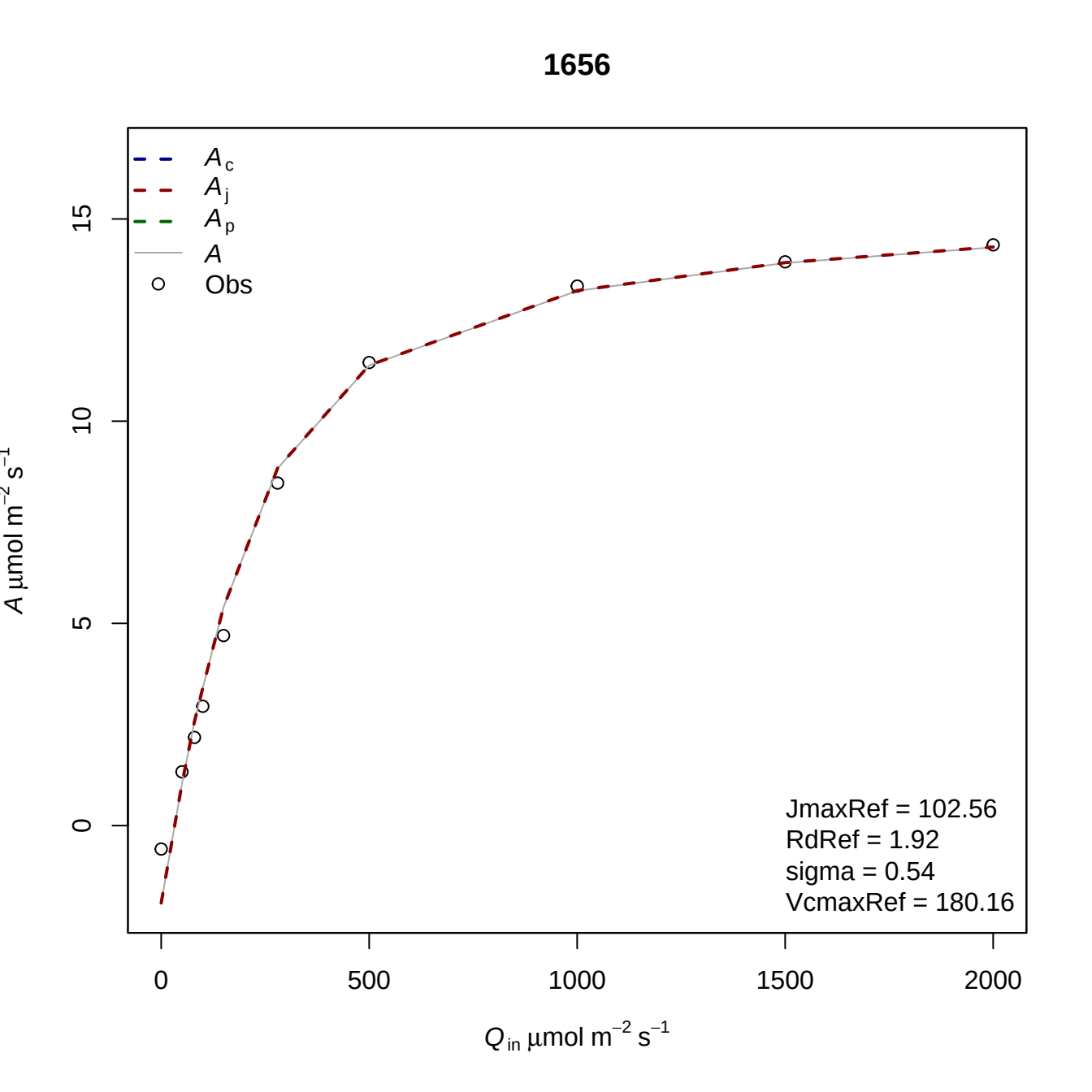

1651
A_c
A_j
A_p
A
Obs
JmaxRef = 49.08
RdRef = 1.62
sigma = 0.55
VcmaxRef = 80.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

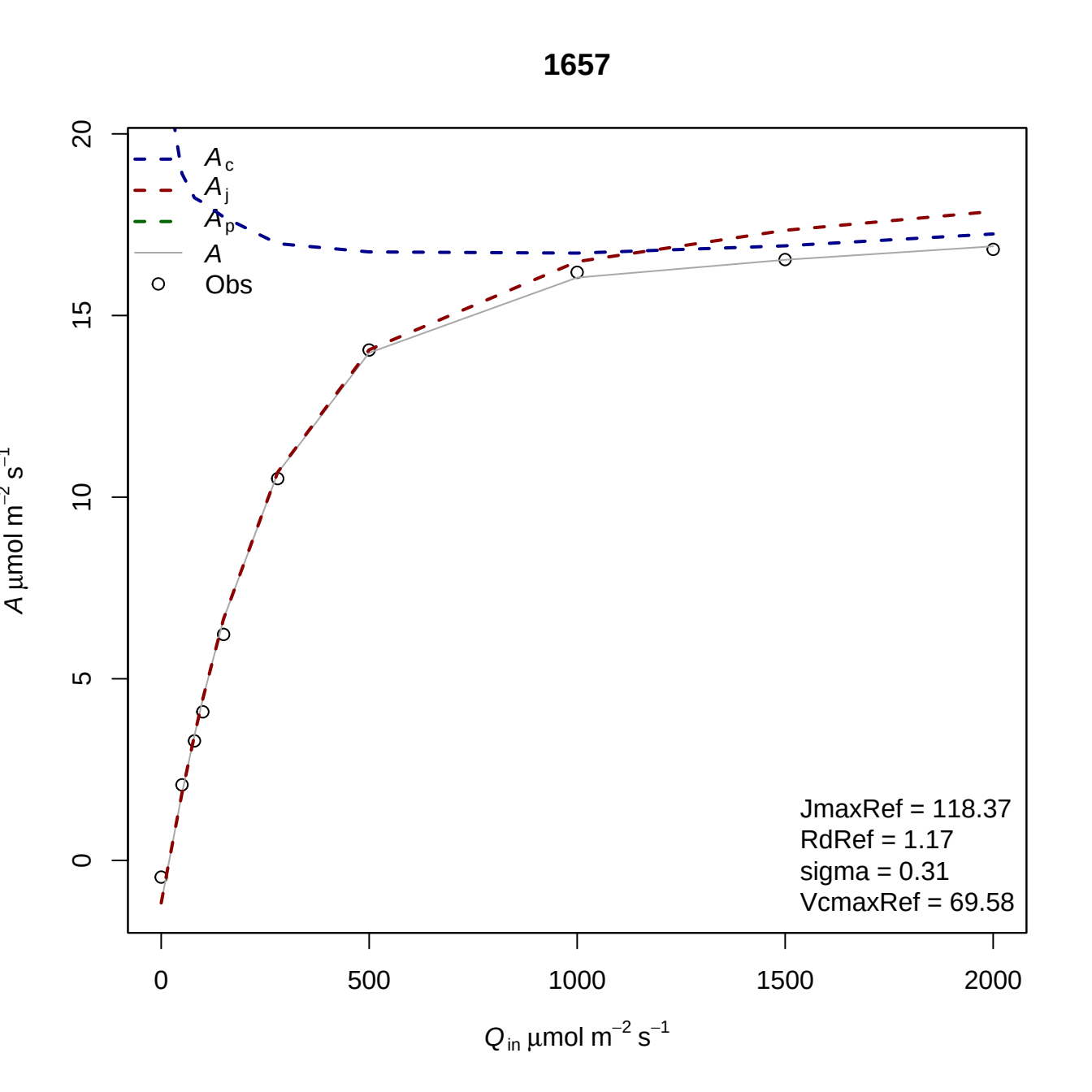

1652
A_c
A_j
A_p
A
Obs
JmaxRef = 61.28
RdRef = 1.51
sigma = 0.48
VcmaxRef = 103.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

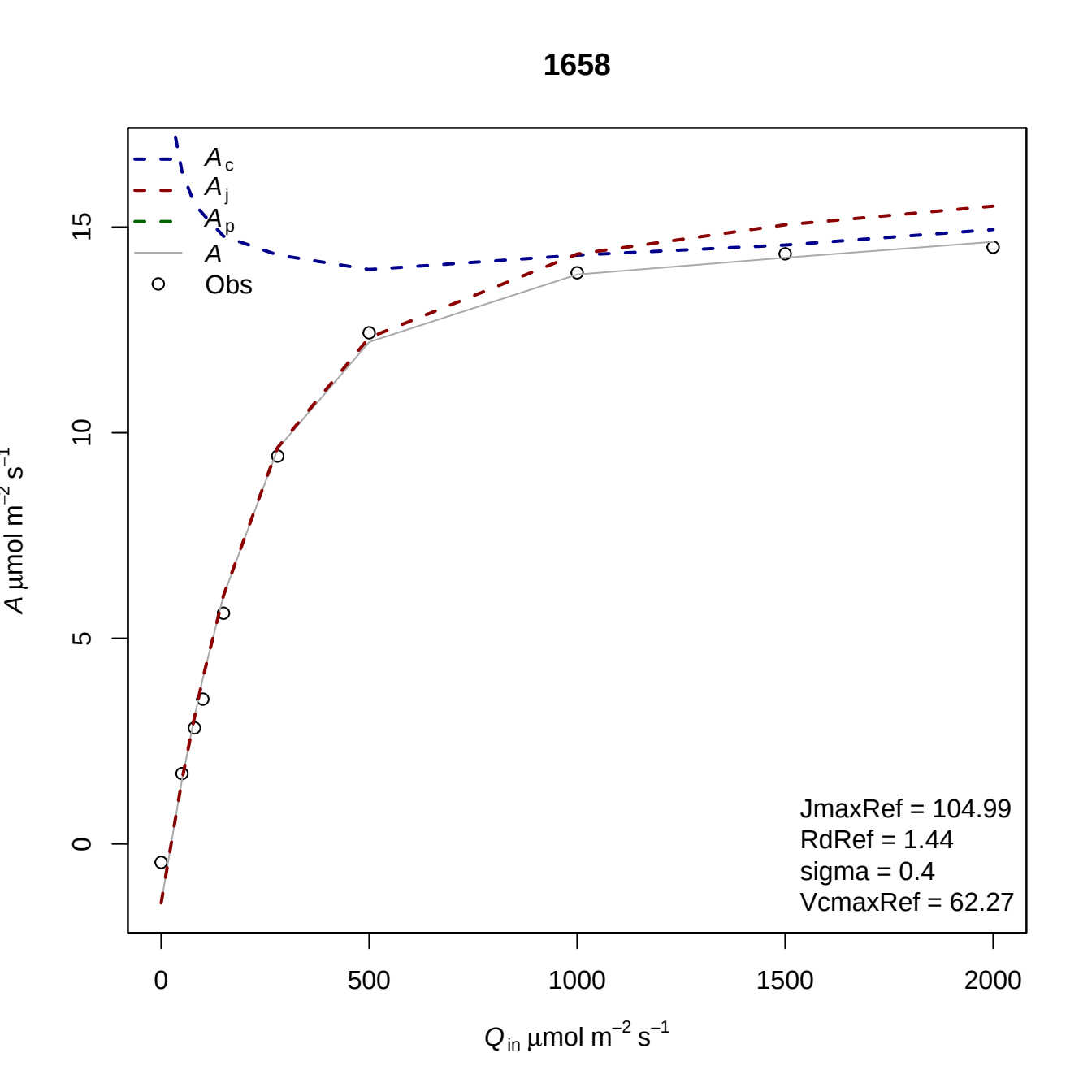

1653
A_c
A_j
A_p
A
Obs
JmaxRef = 59.24
RdRef = 1.85
sigma = 0.59
VcmaxRef = 100.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

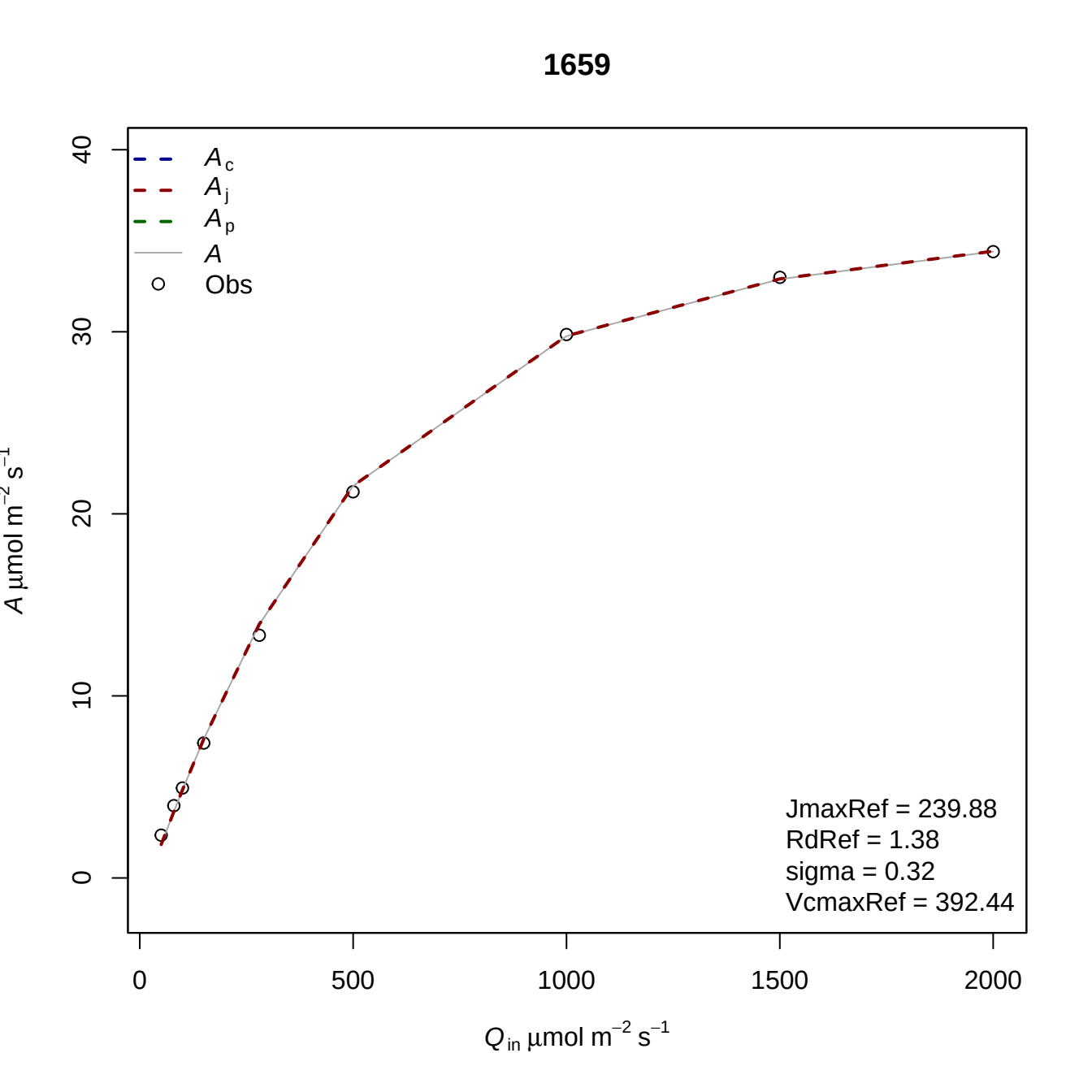

1654
A_c
A_j
A_p
A
Obs
JmaxRef = 40.72
RdRef = 1.7
sigma = 0.6
VcmaxRef = 89.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

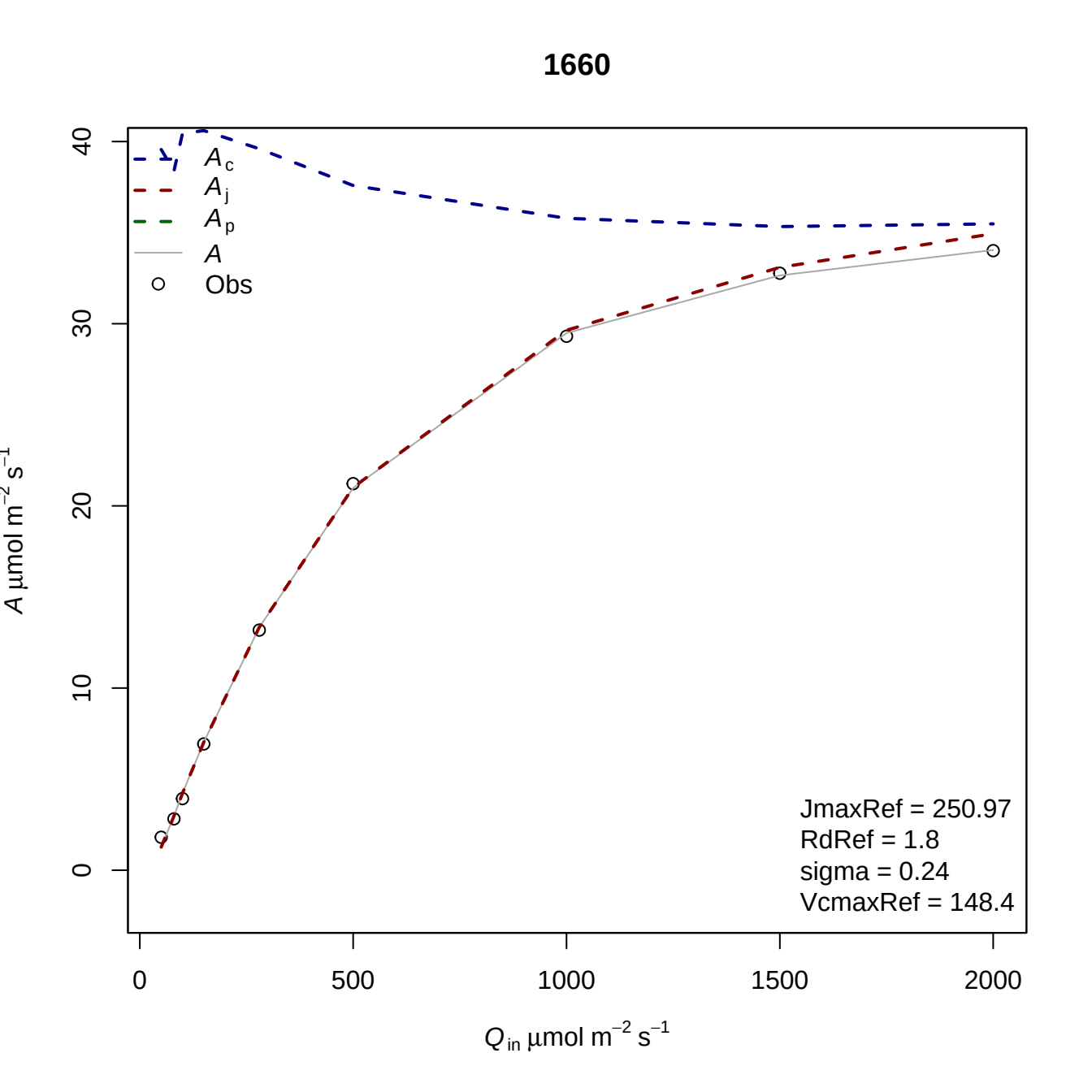

1655
A_c
A_j
A_p
A
Obs
JmaxRef = 108.51
RdRef = 1.66
sigma = 0.51
VcmaxRef = 169.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

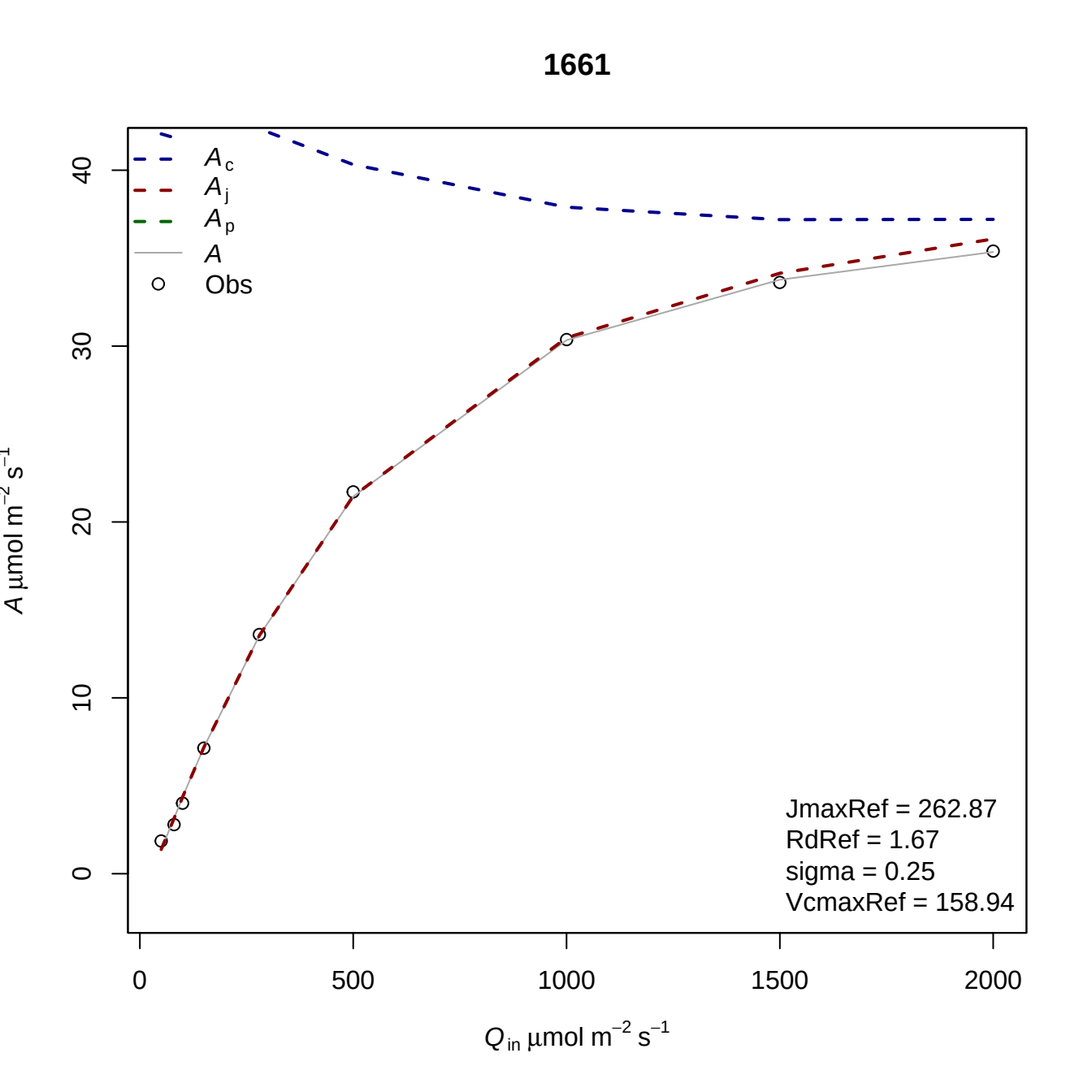

1656
A_c
A_j
A_p
A
Obs
JmaxRef = 102.56
RdRef = 1.92
sigma = 0.54
VcmaxRef = 180.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

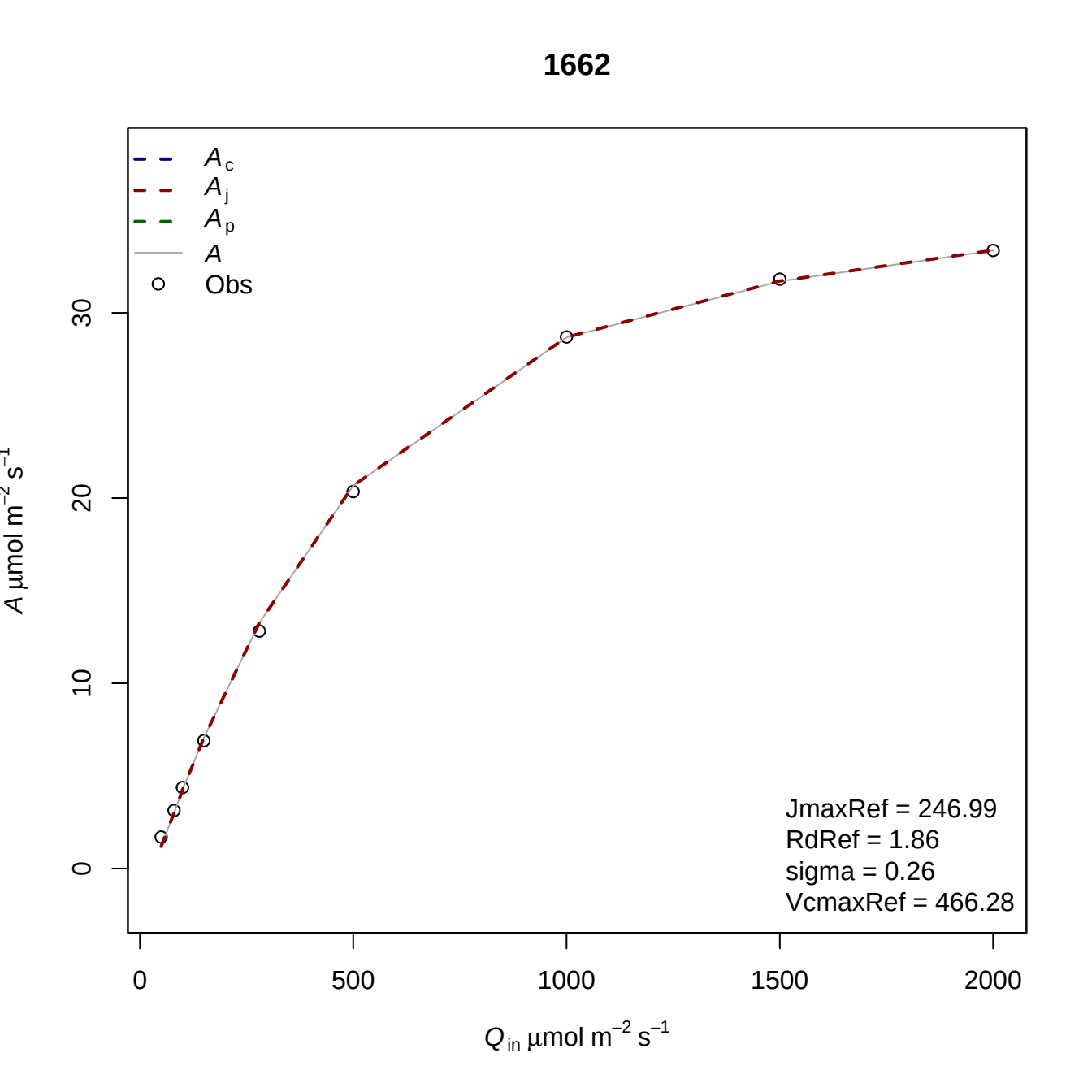

1657
A_c
A_j
A_p
A
Obs
JmaxRef = 118.37
RdRef = 1.17
sigma = 0.31
VcmaxRef = 69.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000


1658
A_c
A_j
A_p
A
Obs
JmaxRef = 104.99
RdRef = 1.44
sigma = 0.4
VcmaxRef = 62.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000


1659
A_c
A_j
A_p
A
Obs
JmaxRef = 239.88
RdRef = 1.38
sigma = 0.32
VcmaxRef = 392.44
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}


1660
A_c
A_j
A_p
A
Obs
JmaxRef = 250.97
RdRef = 1.8
sigma = 0.24
VcmaxRef = 148.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000


1661
A_c
A_j
A_p
A
Obs
JmaxRef = 262.87
RdRef = 1.67
sigma = 0.25
VcmaxRef = 158.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000


1662
A_c
A_j
A_p
A
Obs
JmaxRef = 246.99
RdRef = 1.86
sigma = 0.26
VcmaxRef = 466.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000